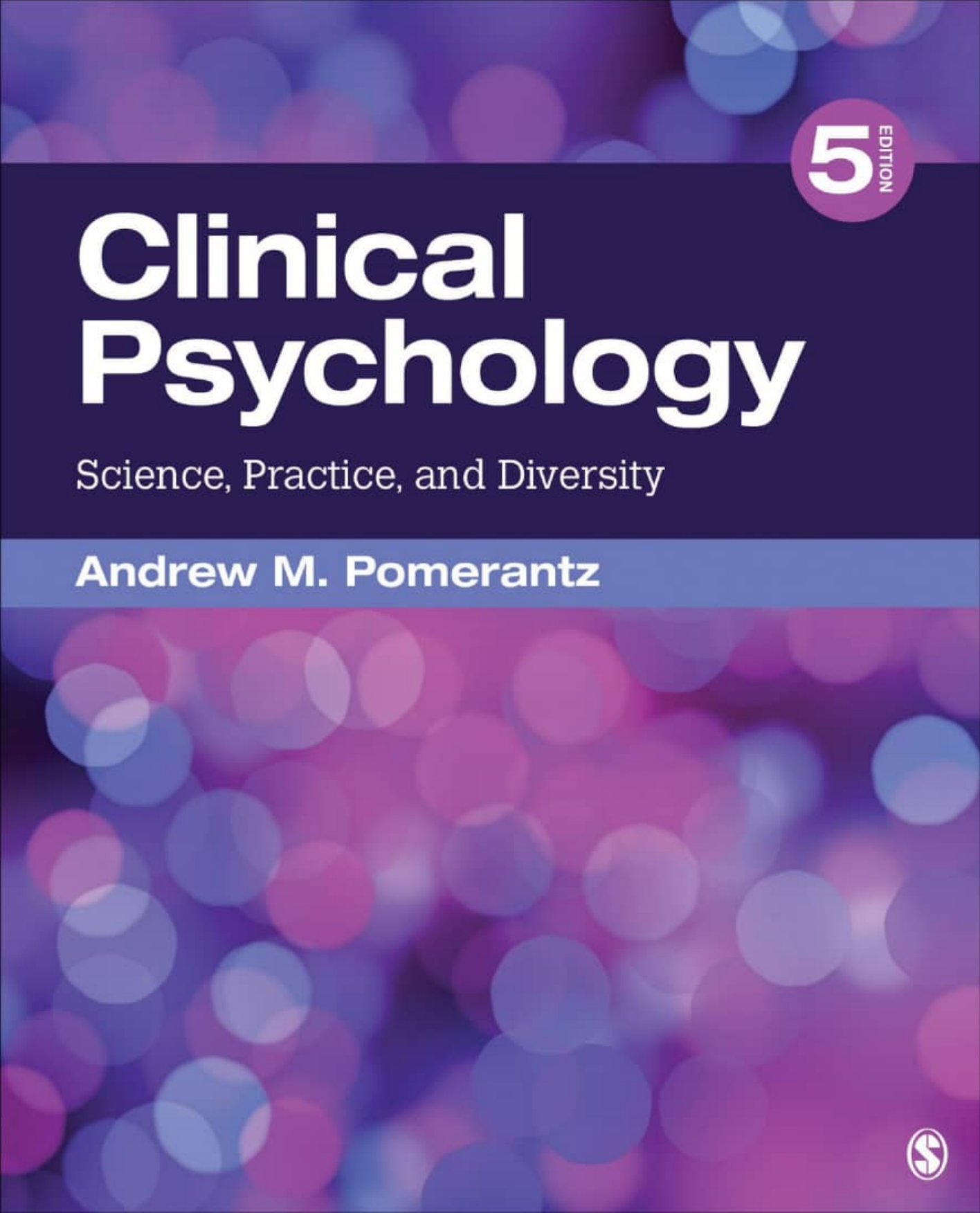
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Science, Practice, and Diversity

Fifth Edition

Andrew M. Pomerantz

Southern Illinois University Edwardsville

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PREFACE

Clinical psychology is an increasingly expansive field. This textbook is my attempt to introduce its students to the vast range of issues it encompasses. My rationale for creating this textbook was multifaceted:

- *To provide a balanced approach to clinical psychology*. There is no shortage of diverse points of view and ongoing debates within clinical psychology, and I believe that the ideal way to introduce students to the spectrum of opinions represented by clinical psychologists is to maximize the even-handedness of the text.
- *To promote cultural competence*. Clinical psychologists must appreciate cultural and diversity-related factors in all their professional activities, and as students familiarize themselves with the field, cultural sensitivity should be woven into their lessons.
- *To offer many illustrative examples*, including clinical applications for clinically relevant topics.
- *To write in a scholarly yet clear and accessible style and include up-to-date information*. Along the same lines, the text covers a distinctly broad range of topics while maintaining an adequate degree of depth.

Previous editions of this text were enthusiastically received by both instructors and students. We were pleased to learn that they found numerous components of the book—its emphasis on issues of culture, its unique pedagogical features, its scholarly yet readable style, its many clinical examples, its balanced approach, and its ancillary package, among others—beneficial to student learning.

WHAT'S THE SAME IN THE FIFTH EDITION

This edition retains all the strengths of the previous edition of the book:

- *Considering Culture* boxes appear in almost every chapter. These boxes highlight multicultural aspects of the various topics covered throughout the book. Along with the discussions of culture integrated throughout the text, these boxes encourage the student to appreciate culturally relevant issues surrounding research, psychotherapy, assessment, and other topics.
- Chapter 4 (Diversity and Cultural Issues in Clinical Psychology) is devoted entirely to topics related to multiculturalism.
- Interviews with nine renowned experts in multicultural clinical work appear in Chapter 4 and on the companion website. Each of these experts—Melba

Vasquez, Frederick Leong, Joseph E. Trimble, Robert L. Williams, Monica McGoldrick, Nadya A. Faoud, Karen Haboush, Kathleen J. Bieschke, and Lewis Z. Schlosser—discusses cultural competence with a specific cultural group based on ethnicity, religion, gender, or sexual orientation.

- *In My Practice* text boxes appear in most chapters (including all chapters focusing on individual psychotherapy, as well as the chapters focusing on interviewing, diagnosis, ethics, group and family therapy, clinical child psychology, and cultural issues), in which textbook author Andrew Pomerantz shares cases and stories directly from his own clinical psychology practice to illustrate key concepts. These text boxes are accompanied by whiteboard videos, which bring the cases and stories to life for students. In all of the *In My Practice* text boxes and accompanying whiteboard videos, information that might identify the client has been disguised, altered, or omitted to protect client confidentiality.
- *Key Journals* sections, in the end-of-chapter material for every chapter, list important journals to which students can turn for more information on chapter-related topics.
- *Metaphorically Speaking* boxes appear in almost every chapter. These boxes use metaphors to teach students about novel concepts by drawing parallels to concepts with which they are already familiar.
- *Denise in _____ Psychotherapy* boxes appear in all the chapters relevant to psychotherapy. "Denise" is a fictional therapy client created exclusively for this textbook. She is introduced at the end of Chapter 11, and the *Denise* boxes that appear at the end of each of the subsequent chapters illustrate how she would be treated according to psychodynamic, humanistic, behavioral, cognitive, and group therapy approaches.
- Chapter 3 (Current Controversies and Directions in Clinical Psychology) is devoted entirely to contemporary issues such as prescription privileges, evidence-based practice, payment methods, and technological advances.
- Chapter 5 (Ethical and Professional Issues in Clinical Psychology) is devoted entirely to ethical and professional issues in clinical psychology, including confidentiality, multiple relationships, and more.
- Chapters 12 through 16 are each devoted entirely to a particular approach or modality of psychotherapy (i.e., psychodynamic, humanistic, behavioral, cognitive/mindfulness-based, and group/family), and each contains coverage of contemporary versions and variants of these therapies.
- Icons (three to five per chapter) in the margin refer students to web-based resources (e.g., videos, websites, articles), accessible through the student study site, that have been carefully selected to enhance learning of key concepts.
- Chapter 7 (Diagnosis and Classification Issues: *DSM-5* and More) and other chapters contain detailed descriptions of what changed in *DSM-5*, including
    - new features (e.g., Arabic rather than Roman numeral system to facilitate the transition to a "living document"),

- new disorders (e.g., disruptive mood dysregulation disorder, mild neurocognitive disorder, binge eating disorder), and
- adjusted criteria for existing disorders (e.g., autism spectrum disorder encompassing autism and Asperger's disorder, revision of the bereavement exclusion criteria for major depressive episodes, more inclusive criteria for bulimia nervosa and attention-deficit/hyperactivity disorder).

- Coverage of what didn't change in *DSM-5* (changes that were considered but rejected).
- Coverage of how *DSM-5* was made, including
  - steps in the revision process,
  - controversies that arose during the process,
  - criticisms by prominent authors about the process, and
  - forces that may have influenced the process.
- Possible consequences of *DSM-5*, including
  - continued broadening of the scope of mental illness,
  - treatment issues (e.g., insurance coverage, psychotherapy, drug treatment), and
  - other implications (e.g., legal issues, disability claims).

WHAT'S NEW IN THE FIFTH EDITION

This edition includes numerous important enhancements and updates:

- More than 320 new references, including more than 110 published in 2018/2019 and more than 160 published in 2016/2017 ensuring extremely current coverage across all chapters.
- A new end-of-chapter feature, *Looking Toward Graduate School*, which serves as a bridge between the topics in a particular chapter and graduate programs that offer further training or experience in those topics. Specifically, this feature provides students interested in applying to graduate school with relevant index terms they can find in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018). Students can use these terms to find specific graduate programs that offer clinical opportunities, research experiences, and concentrations/tracks in topics within the chapter. This process should help students appreciate the vast range of experiences in various graduate programs and give them a practical head start toward finding programs that best match their interests.
- New videos, created by and featuring the author, on ethics topics including confidentiality and multiple relationships. These videos, professionally produced by Sage Knowledge, feature the author playing the role of the clinician (and narrator) in simulated clinical cases in which actors play the roles of clients.

- Chapter 1 (Clinical Psychology: Definition and Training)
  - New coverage of research on what PhD and PsyD graduate programs value among application materials
  - New coverage of marriage and family therapists (as an additional mental health profession similar to clinical psychology)
  - New coverage of states' decisions since 2008 to drop the postdoc requirement for licensure
  - New coverage of research on trends across doctoral training in clinical psychology
  - Update of Table 1.2 to include new examples of doctoral programs following various training models
  - Minor update on the origins of the PsyD degree
  - Minor update on the internship shortage/crisis
  - Minor update on master's level training
  - Minor update on counseling psychologists
  - Minor update on school psychologists
  - Minor update on professional counselors
- Chapter 2 (Evolution of Clinical Psychology)
  - Minor update on the number of clinical and counseling psychologists working in the Department of Veterans Affairs
- Chapter 3 (Current Controversies and Directions in Clinical Psychology)
  - Revised language from *cybertherapy* to *telepsychology* to match APA usage
  - New coverage regarding prescription privileges, including states that have approved it, number of psychologists who have obtained it, attitudes of medical professionals toward prescribing psychologists, and timing of prescription training
  - New coverage regarding evidence-based treatment, including the three-legged stool model, research on the importance of accommodating clients' preferences, APA Practice Guidelines as an additional source for therapists to learn "what works" for particular disorders, and flexibility-within-fidelity regarding therapy manuals
  - New coverage regarding telepsychology suggestions, specifically regarding the disposal of technologies and data, practicing across state lines, and choosing apps and other technologies carefully
- Chapter 4 (Diversity and Cultural Issues in Clinical Psychology)
  - Added the word *Diversity* to the chapter title to broaden the focus
  - Added "diversity" terminology throughout the chapter
  - Updated data about the population of immigrants in the United States
  - Added brief mention of connection between cultural competence and lower therapy drop-out rates

- New coverage of the effects of microaggressions on clients
- New coverage of research showing potentially harmful effects when cultural adaptations of therapies are not considered
- New coverage of language-based differences in symptom reports in interviews
- Added brief mention of increased research on cultural humility

- Chapter 5 (Ethical and Professional Issues in Clinical Psychology)
  - Added "Professional Issues" to title to more accurately reflect the scope of the chapter
  - Added links to four videos featuring narration by the author and clinical vignettes in which clients (played by actors) and a therapist (played by the author) enact a variety of ethically challenging situations involving multiple relationship and confidentiality issues
  - Supplemented the history of the *Tarasoff* case, including a better explanation of "duty to warn" and "duty to protect"
  - New coverage of difference in duty to warn and duty to protect laws in various states
  - Supplemented the list of APA Professional Guidelines regarding the treatment of diverse groups
  - New coverage of research on burnout rates among psychologists and the consequences of psychologist burnout for therapy clients
  - New coverage of self-care, including research on the rates at which psychologists seek their own therapy
  - New coverage, in the section on Ethics in Small Communities, of the issue of clients becoming aware of the psychologist's own personal information
  - Minor updates to the Metaphorically Speaking box about the "Six Degrees of Kevin Bacon" game and multiple relationships for clinical psychologists in small communities

- Chapter 6 (Conducting Research in Clinical Psychology)
  - Added a new subsection, Psychological Disorders, under the heading Why Do Clinical Psychologists Do Research?, including several recent specific examples
  - Added coverage of dissemination strategies as a research topic
  - Added two new examples of research on assessment methods
  - Added a new example of research on professional issues
  - Added a new example of research on teaching and training issues
  - Clarified the coverage of randomized clinical trials
  - Added an example of correlational research
  - Added brief coverage of the alternative treatments design (variation of ABAB design)
  - Added examples of ethical issues in clinical psychology research

- Chapter 7 (Diagnosis and Classification Issues: *DSM-5* and More)
  - Added update regarding the use of Arabic, rather than Roman, numerals in the title of *DSM-5*
  - Added full name of *International Classification of Diseases* (rather than only *ICD* abbreviation) for clarity
  - Added coverage of upcoding and related research in the subsection on controversial cutoffs in *DSM*
- Chapter 8 (The Clinical Interview)
  - Added a subsection, Observing Client Behaviors, under the heading Specific Behaviors
  - Slightly edited the primary example of an open-ended interview question to better distinguish it from closed-ended interview questions
  - Added example and a clinical example about judicious decisions to use clarification during interviewing
  - Added coverage of no-suicide contracts, including research on their effects
- Chapter 9 (Intellectual and Neuropsychological Assessment)
  - Added coverage, including research, of Q-interactive system of administering and scoring Wechsler tests
  - Added coverage of the NEPSY-II neuropsychological battery
  - Added mention of additional intelligence tests, including Woodcock-Johnson and Kaufman tests
  - Added brief explanation that intelligence tests and achievement tests by same company are typically paired and co-normed
  - Added mention of the Wechsler Nonverbal Scale of Ability in the section on addressing cultural fairness in intelligence testing
  - Slightly edited title and content of Metaphorically Speaking box to broaden and update it
  - Added coverage of evolution of neuropsychological testing from fixed-battery phase to flexible-battery phase
  - Added coverage of neuropsychological assessment anxiety
- Chapter 10 (Personality Assessment and Behavioral Assessment)
  - Added coverage of Personality Assessment Inventory
  - Removed coverage of California Psychological Inventory
  - Updated coverage of the NEO-PI-R to NEO-PI-3
  - Added brief coverage of the research on relative popularity of the MMPI-2 and MMPI-2-RF
  - Added brief mentions of the Beck Anxiety Inventory and Beck Hopelessness Scale in the section on the Beck Depression Inventory

- Added brief mention of the use of the term *performance-based test* in place of *projective test*

- Chapter 11 (General Issues in Psychotherapy)
  - Added coverage of recent research on the components of the therapeutic relationship that are demonstrably effective regarding therapy outcome
  - Added coverage of research on graduate training related to the therapeutic relationship/working alliance
  - Added coverage of APA Clinical Practice Guidelines and APA Guidelines for Practitioners to Box 11.4: Considering Culture
  - Added coverage of research on stages of change and therapy outcome
  - Added coverage of the assimilative approach to psychotherapy (as an alternative to the eclectic and integrative approaches)
  - Briefly clarified Eysenck's early findings on the outcome of psychotherapy
  - Minor edit to the definitions and explanations of efficacy and effectiveness
  - Briefly added explanation of opposition to the dodo bird verdict
- Chapter 12 (Psychodynamic Psychotherapy)
  - Added coverage of research on clients doing online searches of their therapists and its impact on the "blank screen" role preferred for transference
  - Added brief coverage of a research treatment outcome study involving insight via transference
  - Added a note that free association is rare, even in psychotherapy
- Chapter 13 (Humanistic Psychotherapy)
  - Added a seven-point summary of techniques commonly used in positive psychology interventions
  - Added a couple of health-related items to the list of psychological problems that motivational interviewing has evidence for improving
  - Added brief coverage, in the section on positive psychology, of research indicating that happiness is a protective factor against mental disorders
  - Added brief coverage of research on the relative acceptability of positive psychology interventions compared with cognitive-behavioral therapy
  - Added coverage of research on the benefits of humanistic therapy both overall and with specific disorders and problems
- Chapter 14 (Behavior Therapy)
  - Added coverage of research on the applicability of behavioral activation with disorders beyond depression, and on the idea that behavioral activation may be a common mechanism of change across disorders and therapy approaches
  - Added brief clarification regarding the disorders for which exposure therapies are effective, and the fact that exposure and response prevention is the treatment of choice for obsessive-compulsive disorder

- Chapter 15 (Cognitive Psychotherapy and Mindfulness-Based Therapies)
  - Added a brief note about the connection between mindfulness and Zen Buddhism
  - Added brief coverage of research suggesting that mindfulness can have therapeutic benefits as a supplement to more traditional treatments
  - Added explanation of the dialectics on which therapists often focus in dialectical behavior therapy (DBT)
  - Added coverage of research on the efficacy of Albert Ellis's Rational Emotive Behavior Therapy (REBT) specifically, and on the negative correlation between rational beliefs and psychological distress.
  - Added brief coverage of the results of meta-analyses of mindfulness-based therapies for specific disorders, including eating disorders and psychosis
- Chapter 16 (Group and Family Therapy)
  - Added, in the section on confidentiality issues in group therapy, brief mention of social media as a potential method of clients sharing confidential information about other clients, and of confidentiality contracts as a possible method of minimizing such confidentiality violations
  - Added brief coverage of the efforts to create relational diagnostic alternatives to *DSM* since the 1980s
  - Added coverage of the research on financial benefits, in terms of spending of taxpayer dollars, for multisystemic family therapy
- Chapter 17 (Clinical Child and Adolescent Psychology)
  - Added coverage of "youth culture" as a cultural variable to which psychologists should be sensitive, and related research on the correlation between adolescent clients' ratings of their therapists' social media competency and the therapeutic alliance
  - Added coverage of problem-solving strategies and affective education strategies within the discussion of cognitive-behavioral therapy (CBT) approaches and self-instructional training
  - Added research on play therapy outcome studies, including their methodological limitations
  - Added research on a five-decade meta-analysis on overall efficacy of psychotherapy with children and adults
  - Added updated coverage of the most frequently used assessment techniques for children and adolescents
  - Added a brief note about how the inferential, rather than empirically-based, method of interpreting drawing techniques brings their reliability and validity into question
- Chapter 18 (Health Psychology)
  - Added coverage of the tend-and-befriend response to stress as an alternative to fight-or-flight

- Added coverage of emotion-focused coping as an alternative to problem-focused coping
- Added a bullet-point list of components of successful weight loss strategies
- Added coverage of the STAR approach to smoking cessation

- Chapter 19 (Forensic Psychology)
  - Added coverage of the use of Internet searches and social media information as part of forensic assessments
  - Added coverage of the fact that the approach to criminal responsibility in not guilty by reason of insanity (NGRI) cases differs across diverse countries
  - Added coverage of research on rehospitalization outcomes of NGRI release evaluations with agreement or disagreement among evaluators
  - Added brief coverage of research on linguistic issues, including the use of translators and interpreters, in forensic assessment in the Considering Culture box
  - Added brief mention of the fact that a small number of forensic psychologists are conducting evaluations to determine the need for emotional support animals
  - Removed coverage of Psychology and the Nuremberg Ethic

A CHAPTER-BY-CHAPTER OVERVIEW

The textbook begins with a definition of clinical psychology, a consideration of how clinical psychologists are trained, and a survey of the professional activities and settings of clinical psychologists (Chapter 1). Chapter 2 considers the rich history of clinical psychology, and Chapter 3 highlights the current controversies and directions that characterize the field. Chapter 4 discusses the diversity-related cultural issues relevant to clinical psychology. Chapter 5 offers detailed analysis of some of the most important ethical and professional issues for clinical psychologists, including confidentiality and multiple relationships. Chapter 6 focuses on research and describes both how and why clinical psychologists conduct it.

Chapter 7 marks the beginning of the Assessment part of the textbook, and it focuses on issues of diagnosis and classification of disorders, with special attention paid to *DSM-5*. Chapter 8 spotlights the clinical interview. Chapter 9 outlines intellectual and neuropsychological assessment, and Chapter 10 focuses on personality and behavioral assessment.

Part III, on psychotherapy, begins with Chapter 11, which provides an overview of general psychotherapy issues, such as efficacy, effectiveness, and the commonality of various psychotherapy approaches. Chapters 12 through 15 each focus on a single approach to individual psychotherapy: psychodynamic (Chapter 12), humanistic (Chapter 13), behavioral (Chapter 14), and cognitive/mindfulness-based (Chapter 15). Chapter 16 is separated into two parts, one covering group therapy and the other family therapy.

Special topics are featured in the last three chapters. Chapter 17 discusses clinical child and adolescent psychology, including assessment and psychotherapy topics. The final two chapters cover growing specialty areas among clinical psychologists: health psychology (Chapter 18) and forensic psychology (Chapter 19).

SUPPLEMENTS FOR STUDENTS AND INSTRUCTORS

Student Study Site: edge.sagepub/pomerantz5e

This open-access student study site provides a variety of additional resources to build on students' understanding of the book content and extend their learning beyond the classroom. Students will have access to the following resources:

- Each chapter in the text is accompanied by **self-quizzes**, which include 10 to 15 true/false and multiple-choice questions for students to independently assess their progress in learning course material.
- **eFlashcards** reinforce student understanding and learning of key terms and concepts that are outlined in the book.
- Fictional vignettes in the form of **sample case studies** allow students the opportunity to apply therapeutic principles introduced in key chapters.
- **Culture expert interviews** with renowned experts in multicultural issues discuss psychotherapy, assessment, and training regarding specific cultural groups, including some based on ethnicity, religion, gender, and sexual orientation.
- **SAGE journal articles** provide access to recent, relevant full-text articles from SAGE's leading research journals. Each article includes discussion questions to focus and guide student interpretation.
- Carefully selected **web resources** feature relevant content for use in independent and classroom-based exploration of key topics.
- **Mock assessment data** provide realistic assessment profiles to invite in-depth consideration of fictional clients.

Instructor Teaching Site: edge.sagepub/pomerantz5e

A password-protected instructor teaching site offers the following resources for each chapter:

- An updated **test bank** available in Microsoft Word offers a diverse set of test questions and answers to aid instructors in assessing students' progress and understanding.

- **PowerPoint presentations** are designed to assist with lecture and review highlighted essential content, features, and artwork from the book.
- Classroom activities and discussion questions are provided to reinforce active learning.

Margin Icons

Icons appearing in the margin of the text will direct you to corresponding links on the open-access study site. These additional media include video, audio, and web links that elaborate on key concepts within the chapter.

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This book was undoubtedly a team effort. Many, many people facilitated or contributed to it in meaningful ways. For what they offered before or during the process of this book's creation, I am immeasurably appreciative. I wish to sincerely thank

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PART I

INTRODUCING CLINICAL PSYCHOLOGY

CHAPTER 1 • **Clinical Psychology: Definition and Training**

CHAPTER 2 • **Evolution of Clinical Psychology**

CHAPTER 3 • **Current Controversies and Directions in Clinical Psychology**

CHAPTER 4 • **Diversity and Cultural Issues in Clinical Psychology**

CHAPTER 5 • **Ethical and Professional Issues in Clinical Psychology**

CHAPTER 6 • **Conducting Research in Clinical Psychology**

1

CLINICAL PSYCHOLOGY

Definition and Training

What Is Clinical Psychology? 3
- *Original Definition* 3
- *More Recent Definitions* 4

Education and Training in Clinical Psychology 5
- *Balancing Practice and Science: The Scientist-Practitioner (Boulder) Model* 6
- *Leaning Toward Practice: The Practitioner-Scholar (Vail) Model* 6
- *Leaning Toward Science: The Clinical Scientist Model* 8
- *Getting In: What Do Graduate Programs Prefer?* 13
- *Internships: Predoc and Postdoc* 16
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- *Where Do Clinical Psychologists Work?* 18
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How Are Clinical Psychologists Different From . . . 19
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Welcome to clinical psychology! Throughout this book, you'll learn quite a bit about this field: history and current controversies, interviewing and psychological assessment methods, and psychotherapy approaches. Let's start by defining clinical psychology.

WHAT IS CLINICAL PSYCHOLOGY?

Original Definition

The term **clinical psychology** was first used in print by **Lightner Witmer** in 1907. Witmer was also the first to operate a psychological clinic (L. T. Benjamin, 1996, 2005). More about Witmer's pioneering contributions will appear in Chapter 2, but for now,

Learning Objectives

1.1 Describe the evolution of the definition of clinical psychology from the early 1900s to present.

1.2 Paraphrase the definition of clinical psychology provided by Division 12 of the American Psychological Association.

1.3 Compare the training foci of the scientist-practitioner (Boulder), practitioner-scholar (Vail), and clinical scientist models of training.

1.4 Discuss perceived advantages and limitations of the scientist-practitioner (Boulder), practitioner-scholar (Vail), and clinical scientist models of training.

1.5 Recount key applicant characteristics preferred by clinical psychology graduate programs.

1.6 Summarize current professional activities and employment settings of clinical psychologists in the United States.

1.7 Differentiate clinical psychologists from related professions, including counseling psychologists, psychiatrists, social workers, school psychologists, professional counselors, and marriage and family therapists.

let's consider how he chose to define his emerging field. Witmer envisioned clinical psychology as a discipline with similarities to a variety of other fields, specifically medicine, education, and sociology. A clinical psychologist, therefore, was a person whose work with others involved aspects of treatment, education, and interpersonal issues. At his clinic, the first clients were children with behavioral or educational problems. However, even in his earliest writings, Witmer (1907) foresaw clinical psychology as applicable to people of all ages and with a variety of presenting problems.

More Recent Definitions

Defining clinical psychology is a greater challenge today than it was in Witmer's time. The field has witnessed such tremendous growth in a wide variety of directions that most simple, concise definitions fall short of capturing the field in its entirety. As a group, contemporary clinical psychologists do *many* different things, with *many* different goals, for *many* different people.

Some have tried to offer "quick" definitions of clinical psychology to provide a snapshot of what our field entails. For example, according to various introductory psychology textbooks and dictionaries of psychology, clinical psychology is essentially the branch of psychology that studies, assesses, and treats people with psychological problems or disorders (e.g., Myers, 2013, VandenBos, 2007). Such a definition sounds reasonable enough, but it is not without its shortcomings. It doesn't portray all that clinical psychologists do, how they do it, or who they do it for.

An accurate, comprehensive, contemporary definition of clinical psychology would need to be more inclusive and descriptive. The **Division of Clinical Psychology (Division 12)** of the **American Psychological Association (APA)** defines clinical psychology as follows:

> The field of Clinical Psychology integrates science, theory, and practice to understand, predict, and alleviate maladjustment, disability, and discomfort as well as to promote human adaptation, adjustment, and personal development. Clinical Psychology focuses on the intellectual, emotional, biological, psychological, social, and behavioral aspects of human functioning across the life span, in varying cultures, and at all socioeconomic levels. (APA, 2012a)

The sheer breadth of this definition reflects the rich and varied growth that the field has seen in the century since

Witmer originally identified it. (As Norcross & Sayette, 2016, put it, "Perhaps the safest observation about clinical psychology is that both the field and its practitioners continue to outgrow the classic definitions" [p. 1].) Certainly, its authors do not intend to suggest that each clinical psychologist spends equal time on each component of that definition. But, collectively, the work of clinical psychologists does indeed encompass such a wide range. For the purposes of this textbook, a similarly broad but somewhat more succinct definition will suffice: Clinical psychology involves rigorous study and applied practice directed toward understanding and improving the psychological facets of the human experience, including but not limited to issues or problems of behavior, emotions, or intellect.

Web Link 1.1
APA Division 12

EDUCATION AND TRAINING IN CLINICAL PSYCHOLOGY

In addition to looking at explicit definitions such as those listed earlier, we can infer what clinical psychology is by learning how clinical psychologists are educated and trained. The basic components of clinical psychology training are common across programs and are well established (Vaughn, 2006). The aspiring clinical psychologist must obtain a doctoral degree in clinical psychology, about 3,000 of which are awarded each year (Norcross & Sayette, 2016). Most students enter a doctoral program with only a bachelor's degree, but some enter with a master's degree. Often, that master's degree was earned from a "terminal" master's program in clinical psychology (meaning that their program ends at the master's level). Some graduates of such master's programs go on to earn doctoral degrees, while others enter the work force in some capacity (Campbell, Worrell, Dailey, & Brown, 2018; Pomerantz & Murphy, 2016).

For those entering with a bachelor's degree, training typically consists of at least 4 years of intensive, full-time coursework, followed by a 1-year, full-time predoctoral internship. Required coursework includes courses on psychotherapy, assessment, statistics, research design and methodology, biological bases of behavior, cognitive-affective bases of behavior, social bases of behavior, individual differences, and other subjects. A master's thesis and doctoral dissertation are also commonly required, as is a practicum in which students start to accumulate supervised experience doing clinical work. When the on-campus course responsibilities are complete, students move on to the predoctoral internship, in which they take on greater clinical responsibilities and obtain supervised experience on a full-time basis. This predoctoral internship, along with the postdoctoral internship that occurs after the degree is obtained, is described in more detail later in this chapter.

Beyond these basic requirements, especially in recent decades, there is no single way by which someone becomes a clinical psychologist. Instead, there are many paths to the profession. One indication of these many paths is the multitude of specialty tracks within clinical psychology doctoral programs. Indeed, more than half of APA-accredited doctoral programs in clinical psychology offer (but may not require) training within a specialty track. The most common specialty areas are clinical child, clinical health, forensic, family, and clinical neuropsychology (K. M. Perry & Boccaccini, 2009). (Each of these specialty areas receives attention in a later chapter of this book.) Another indication

of the many paths to the profession of clinical psychology is the coexistence of three distinct models of training currently used by various graduate programs: the scientist-practitioner (Boulder) model, the practitioner-scholar (Vail) model, and the clinical scientist model (Routh, 2015a). Let's consider each of these in detail.

Balancing Practice and Science: The Scientist-Practitioner (Boulder) Model

In 1949, the first conference on graduate training in clinical psychology was held in Boulder, Colorado. At this conference, training directors from around the country reached an important consensus: Training in clinical psychology should jointly emphasize both practice and research. In other words, to become a clinical psychologist, graduate students would need to receive training and display competence in the application of clinical methods (assessment, psychotherapy, etc.) *and* the research methods necessary to study and evaluate the field scientifically (Grus, 2016; N. L. Johnson & Baker, 2015; Klonoff, 2011, 2016). Those at the conference also agreed that coursework should reflect this dual emphasis, with classes in statistics and research methods as well as classes in psychotherapy and assessment. Likewise, expectations for the more independent aspects of graduate training would also reflect the dual emphasis: Graduate students would (under supervision) conduct both clinical work and their own empirical research (thesis and dissertation). These graduate programs would continue to be housed in departments of psychology at universities, and graduates would be awarded the PhD degree. The term **scientist-practitioner model** was used to label this two-pronged approach to training (McFall, 2006; Norcross & Sayette, 2016).

For decades, the scientist-practitioner—or the **Boulder model**—approach to clinical psychology training unquestionably dominated the field (Klonoff, 2011, 2016). In fact, more programs still subscribe to the Boulder model than to any other. However, as time passed, developments took place that produced a wider range of options in clinical psychology training. The pendulum did not remain stationary at its midpoint between practice and research; instead, it swung toward one extreme and then toward the other.

Leaning Toward Practice: The Practitioner-Scholar (Vail) Model

In 1973, another conference on clinical psychology training was held in Colorado—this time, in the city of Vail (Grus, 2016, Klonoff, 2016). In the years preceding this conference, some discontent had arisen regarding the Boulder, or scientist-practitioner, model of training. In effect, many current and aspiring clinical psychologists had been asking, "Why do I need such extensive training as a scientist when my goal is simply to practice?" After all, only a minority of clinical psychologists were entering academia or otherwise conducting research as a primary professional task. Clinical practice was the more popular career choice (Boneau & Cuca, 1974; McConnell, 1984; Stricker, 2011), and many would-be clinical psychologists sought a doctoral-level degree with less extensive training in research and more extensive training in the development of applied clinical skills. Additionally, some within the profession were questioning whether the quality

and quantity of practitioners was sufficient to serve the population (Stricker, 2016). So the **practitioner-scholar model** of training was born, along with a new type of doctoral degree, the **PsyD** (Foley & McNeil, 2015; Routh, 2015b; Stricker & Lally, 2015). Since the 1970s, graduate programs offering the PsyD degree have proliferated. In fact, in the 1988 to 2001 time period alone, the number of PsyD degrees awarded increased by more than 160% (McFall, 2006). Compared with PhD programs, these programs typically offer more coursework directly related to practice and fewer courses related to research and statistics (Norcross, Sayette, Mayne, Karg, & Turkson, 1998). See Box 1.1 for a point-by-point comparison of PhD and PsyD models of training.

BOX 1.1

Comparing PhD Programs With PsyD Programs

Quite a bit of variation exists between PhD programs, just as it does between PsyD programs (Gardner, 2015). However, a few overall trends distinguish one degree from the other. ***In general, compared with PhD programs, PsyD programs tend to***

- place less emphasis on research-related aspects of training and more emphasis on clinically relevant aspects of training;
- accept and enroll a much larger percentage and number of applicants;
- be housed in freestanding, independent (or university-affiliated) "professional schools," as opposed to departments of psychology in universities;
- accept students with lower Graduate Record Examination (GRE) scores and undergraduate grade point averages (GPAs);
- offer significantly less funding to enrolled students in the form of graduate assistantships, fellowships, tuition remission, and so on;
- accept and enroll a higher percentage of students who have already earned a master's degree;
- have lower rates of success placing their students in APA-accredited predoctoral internships;
- produce graduates who score lower on the national licensing exam (EPPP);
- graduate students in a briefer time period (about 1.5 years sooner);
- graduate students who pursue practice-related careers rather than academic or research-related careers; and
- have at least a slightly higher percentage of faculty members who subscribe to psychodynamic approaches, as opposed to cognitive-behavioral approaches.

Sources: Gaddy, Charlot-Swilley, Nelson, & Reich (1995); Klonoff (2011); Mayne, Norcross, & Sayette (1994); McFall (2006); Norcross & Castle (2002); Norcross & Sayette (2016); Norcross et al., (1998).

The growth of the PsyD (or practitioner-scholar or **Vail model**) approach to training in clinical psychology has influenced the field tremendously. Of course, before the emergence of the PsyD, the PhD was the only doctoral degree for clinical psychology. But, currently, more than half the doctoral degrees being awarded in the field are PsyD degrees (Norcross, Kohout, & Wicherski, 2005). The number of PsyD programs is actually quite small in comparison with the number of PhD programs—about 80 versus about 250—but the typical PsyD program accepts and graduates a much larger number of students than does the typical PhD program, so the number of people graduating with each degree is about the same (roughly 1,500 each; Klonoff, 2011; Norcross & Sayette, 2016; Stricker, 2011).

Table 1.1, which features data from a large-scale survey of graduate programs (J. M. Graham & Kim, 2011), offers more detailed findings regarding the general trends listed in Box 1.1.

Leaning Toward Science: The Clinical Scientist Model

After the advent of the balanced Boulder model in the late 1940s and the subsequent emergence of the practice-focused Vail model in the 1970s, the more empirically minded members of the clinical psychology profession began a campaign for a strongly research-oriented model of training.

Table 1.1 Comparison of PsyD and PhD Programs in Clinical Psychology

Variable	PsyD	PhD
Mean GRE (Verbal + Quantitative) score of admitted students*	1116	1256
Mean undergraduate GPA	3.4	3.6
Percentage of students receiving at least partial tuition remission or assistantship	13.9	78.4
Number of students in incoming class	37.4	9.7
Percentage of applicants attending	26.3	7.4
Percentage successfully placed in APA-accredited predoctoral internships	66.0	92.8

Source: J. M. Graham & Kim (2011).

**GRE scores reported were based on the previous GRE scale. Estimated conversions to the current GRE scale are 303 for PsyD and 312 for PhD, based on data provided at https://www.ets.org/s/gre/pdf/concordance_information.pdf.*

Indeed, in the 1990s, a movement toward increased empiricism took place among numerous graduate programs and prominent individuals involved in clinical psychology training. In essence, the leaders of this movement argued that science should be the bedrock of clinical psychology. They sought and created a model of training—the **clinical scientist model**—that stressed the scientific side of clinical psychology more strongly than did the Boulder model (McFall, 2006; McFall, Treat, & Simons, 2015). Unlike those who created the Vail model in the 1970s, the leaders of the clinical scientist movement have not suggested that graduates of their program should receive an entirely different degree—they still award the PhD, just as Boulder model graduate programs do. However, a PhD from a clinical scientist program implies a very strong emphasis on the scientific method and evidence-based clinical methods (R. W. Levenson, 2014; Onken Carroll, Shoham, Cuthbert, & Riddle, 2014; Shoham et al., 2014).

Two defining events highlight the initial steps of this movement. In 1991, **Richard McFall**, at the time a professor of psychology at Indiana University, published an article that served as a rallying call for the clinical scientist movement (Treat & Bootzin, 2015). In this article, "Manifesto for a Science of Clinical Psychology," McFall (1991) argued that "scientific clinical psychology is the only legitimate and acceptable form of clinical psychology . . . after all, what is the alternative? . . . Does anyone seriously believe that a reliance on intuition and other unscientific methods is going to hasten advances in knowledge?" (pp. 76–77).

A few years later, a conference of prominent leaders of select clinical psychology graduate programs took place at Indiana University. The purpose of the conference was to unite in an effort to promote clinical science. From this conference, the **Academy of Psychological Clinical Science** was founded. McFall served as its president for the first several years of its existence, and as time has passed, an increasing number of graduate programs and internships have become members. The programs in this academy still represent a minority of all graduate programs in clinical psychology, but among the members are many prominent and influential programs and individuals (Academy of Psychological Clinical Science, 2009; Fowles, 2015; Klonoff, 2016; McFall et al., 2015).

Considering the discrepancies between the three models of training available today—the traditional, middle-of-the-road Boulder model; the Vail model, emphasizing clinical skills; and the clinical scientist model, emphasizing empiricism—the experience of clinical psychology graduate students varies widely from one program to the next. In fact, it's no surprise that in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018), a valuable resource used by many applicants to learn about specific graduate programs in clinical psychology, the first piece of information listed about each program is that program's self-rating on a 7-point scale from "practice oriented" to "research oriented." Moreover, it's no surprise that applicants can find programs at both extremes and everywhere in between. Table 1.2 shows examples of specific graduate programs representing each of the three primary training models (scientist-practitioner, practitioner-scholar, and clinical scientist), including quotes from the programs' websites that reflect their approach to training.

Just as training in clinical psychology has changed dramatically throughout its history, it continues to change today and promises to change further in the future (Grus, 2011). One study examined the broad trends in training in clinical psychology since the early 1990s. It found that there has been a shift in the theoretical orientation

Table 1.2 Sample Information Regarding Specific Graduate Programs in Clinical Psychology

Graduate Program	Training Model	Degree Awarded	Clinical/ Research Rating	Self-Description on Program Website
Indiana University	Clinical scientist	PhD	7	"Our Clinical Science program is not for everyone. If your primary interests include becoming a practitioner or service provider, you are encouraged to apply elsewhere. . . . We believe that applicants with a clear preference for careers as clinical scientists will find Indiana's program challenging and rewarding."
Northwestern University	Clinical scientist	PhD	7	"The Clinical Psychology Program . . . is designed to train students for primary careers in research and teaching in clinical psychology. . . . The major emphasis of the program is clinical research and research methods."
University of California, Los Angeles	Clinical scientist	PhD	7	"UCLA's Clinical Psychology program . . . aims to produce future faculty, researchers, and leaders in clinical science . . . The curriculum is designed to produce scientifically-minded scholars."
Virginia Tech	Clinical scientist	PhD	7	"Our program's mission is to advance clinical science. . . . Our program's emphasis on the term *science* underscores its commitment to empirical approaches. . . . Our program is most suited to students who are interested in pursuing science-, research-, academic-, or training-related careers. Students whose primary interest and career goal is solely the direct practice of psychology would likely be more satisfied in a different training program."

Graduate Program	Training Model	Degree Awarded	Clinical/ Research Rating	Self-Description on Program Website
Central Michigan University	Boulder/ scientist-practitioner	PhD	4	"The Clinical Psychology program at CMU maintains a balance between training in science and practice. . . . Upon graduation, students are prepared to pursue clinical or research careers."
University of Alabama	Boulder/ scientist-practitioner	PhD	4	"Graduates function in a variety of settings as teachers, researchers, and providers of clinical services, and policy makers. . . . The program emphasizes the integration of scientific knowledge and the professional knowledge and skills needed to function as a clinical psychologist in academic, research, or clinical settings."
Saint Louis University	Boulder/ scientist-practitioner	PhD	4	"The mission of the clinical psychology graduate program is to educate and train students broadly in the science and the practice of clinical psychology."
DePaul University	Boulder/ scientist-practitioner	PhD	4	"[Our] program emphasizes training in both scientific research and practice."
University of Denver	Vail/ practitioner-scholar	PsyD	2	"Historically, our program has emphasized the importance of clinical training. . . . The mission of the PsyD program is to train competent doctoral level practitioner/scholars who . . . have a functional mastery of psychological assessment and intervention, and can apply this knowledge and skill in a wide range of settings."
Chicago School of Professional Psychology	Vail/ practitioner-scholar	PsyD	2	"Guided by our practitioner-scholar faculty, our clinical psychology doctoral graduates are well prepared to provide assessment, intervention, and consultation to meet the needs of diverse populations."

(Continued)

Table 1.2 (Continued)

Graduate Program	Training Model	Degree Awarded	Clinical/ Research Rating	Self-Description on Program Website
Alliant University, San Diego	Vail/ practitioner-scholar	PsyD	2	"[The program] emphasizes the applications of theory and research to clinical practice. The program develops competent professional clinical psychologists skilled in delivering a variety of clinical services to diverse populations in varied settings."

Sources: Indiana University, https://psych.indiana.edu/graduate/degree-programs/clinical-science/index.html; Northwestern University, http://www.wcas.northwestern.edu/psych/program_areas/clinical/; University of California, Los Angeles, https://www.psych.ucla.edu/graduate/areas-of-study/clinical-psychology; Virginia Tech, http://www.psyc.vt.edu/graduate/clinical; Central Michigan University, https://www.cmich.edu/colleges/cgs/programs/Pages/clinical-psychology-phd.aspx; University of Alabama, https://psychology.ua.edu/graduate-studies/clinical-psychology/; St. Louis University, http://www.slu.edu/x13071.xml; DePaul University, https://csh.depaul.edu/academics/psychology/graduate/clinical-psychology-ma-phd/pages/default.aspx; University of Denver, https://www.du.edu/gspp/programs/psyd-clinical/; Chicago School of Professional Psychology, http://www.thechicagoschool.edu/content.cfm/about; Alliant University, San Diego, https://www.alliant.edu/cspp/programs-degrees/clinical-psychology/psyd-san-diego/

Note: Clinical/research ratings by directors of each graduate program, as reported in Sayette and Norcross (2018). Ratings range from 1 ("practice oriented") to 7 ("research oriented"), with 4 representing "equal emphasis."

of faculty toward cognitive/cognitive behavioral and away from psychodynamic/psychoanalytic, as well as a considerable increase in the percentage of doctoral students who are female and members of racial or ethnic minorities (Norcross, Sayette, & Pomerantz, 2018).

Undoubtedly, technology is increasingly influential in the training of clinical psychologists. For an increasing number of students, learning psychotherapy or assessment techniques involves the use of webcams and other computer-based methods that allow supervisors to view, either live or recorded, students trying to apply what they have learned in class (Barnett, 2011; Manring, Greenberg, Gregory, & Gallinger, 2011; Wolf, 2011). (We will discuss technology, including telepsychology, in more detail in Chapter 3.) Another growing emphasis in training is specific *competencies*, or outcome-based skills the students must be able to demonstrate. Emphasizing competencies ensures that the students who graduate from clinical psychology programs not only will have earned good grades on exams, papers, and other academic tasks but also will be able to apply what they have learned. Specific competencies that may be required of students could center on intervention (therapy), assessment, research, consultation/collaboration, supervision/teaching, ethics, cultural diversity, and management/administration (Barlow & Carl, 2011; N. J. Kaslow & Graves, 2015; Peterson, Peterson, Abrams, Stricker, & Ducheny, 2010).

Web Link 1.2
Manifesto for a Science of Clinical Psychology

Web Link 1.3
Academy of Psychological Clinical Science

Getting In: What Do Graduate Programs Prefer?

The *Insider's Guide* mentioned earlier (Sayette & Norcross, 2018) is one of several resources to educate and advise aspiring clinical psychology graduate students. Others include *Graduate Study in Psychology* (APA, 2012b) and *Getting In: A Step-by-Step Plan for Gaining Admission to Graduate School in Psychology* (APA, 2007). Getting into a graduate program in clinical psychology is no easy task: Admission rates are competitive, and the application process is demanding. On average, APA-accredited PhD programs admit only 7% to 16% of their applicants, and APA-accredited PsyD programs admit about 40% to 50% of their applicants (Sayette & Norcross, 2018). Knowing how to prepare, especially early in the process, can provide an applicant with significant advantages. Among the suggestions offered by resources such as those listed here are the following:

- *Know your professional options.* Numerous roads lead to the clinical psychologist title; moreover, numerous professions overlap with clinical psychology in terms of professional activities. Researching these options will allow for more informed decisions and better matches between applicants and graduate programs.

- *Take, and earn high grades in, the appropriate undergraduate courses.* Graduate programs want trainees whose undergraduate programs maximize their chances of succeeding at the graduate level. Among the most commonly required or recommended courses are statistics, research/experimental methods, psychopathology, biopsychology, and personality (Norcross & Sayette, 2016; Norcross, Sayette, Stratigis, & Zimmerman, 2014). Choose electives carefully, too—classes that have direct clinical relevance, including field studies or internships, are often seen favorably (Mayne et al., 1994).

- *Get to know your professors.* Letters of recommendation are among the most important factors in clinical psychology graduate admissions decisions (Norcross, Hanych, & Terranova, 1996). Professors (and, to some extent, supervisors in clinical or research positions) can be ideal writers of such letters—assuming the professor actually knows the student. The better you know the professor, the more substantial your professor's letter can be. For example, a professor may be able to write a brief, vaguely complimentary letter for a quiet student who earned an A in a large lecture course. But the professor would be able to write a much more meaningful, persuasive, and effective letter for the same student if the two of them had developed a strong working relationship through research, advising, or other professional activities.

- *Get research experience.* Your experience in a research methods class is valuable, but it won't distinguish you from most other applicants. Conducting research with a professor affords you additional experience with the empirical process, as well as a chance to learn about a specialized body of knowledge and develop a working relationship with the professor (as described in the previous bulleted point). If your contribution is significant enough, this research experience could also yield a publication or presentation on which you are listed as an author, which will further enhance your application file. In some cases, professors seek assistants for ongoing projects they have designed. In others, the undergraduate student may approach the faculty member with an original

idea for an independent study. Regardless of the arrangement, conducting research at the undergraduate level improves an applicant's chances of getting into and succeeding in a graduate program.

- *Get clinically relevant experience.* For undergraduates, the options for direct clinical experience (therapy, counseling, interviewing, testing, etc.) are understandably limited. Even for those who have earned a bachelor's degree and are considering returning to school at the graduate level, clinical positions may be hard to find. However, quite a few settings may offer exposure to the kinds of clients, professionals, and issues that are central to clinical psychology. These settings include community mental health centers, inpatient psychiatric centers, crisis hotlines, alternative schools, camps for children with behavioral or emotional issues, and others. Whether the clinical experience takes the form of an internship or practicum (for which course credit is earned), a paid job, or a volunteer position, it can provide firsthand knowledge about selected aspects of the field, and it demonstrates to admissions committees that you are serious and well informed about clinical psychology.

- *Maximize your GRE score.* Along with undergraduate GPA, scores on the GRE are key determinants of admission to graduate programs. Appropriately preparing for this test—by learning what scores your preferred programs seek, studying for the test either informally or through a review course, taking practice exams, and retaking it as necessary—can boost your odds of admission.

- *Select graduate programs wisely.* Getting in is certainly important, but getting into a program that proves to be a bad match benefits neither the student nor the program. It is best to learn as much as possible about potential programs: What is the model of training (Boulder, Vail, or clinical scientist)? To what clinical orientations does the faculty subscribe? What areas of specialization do the faculty members represent? What clinical opportunities are available? Of course, your own preferences or constraints—geography, finances, family—deserve consideration as well.

- *Write effective personal statements.* In addition to the many other items in your application file, graduate programs will require you to write a personal statement (or goal statement). This is your opportunity to discuss career aspirations as well as your research and clinical interests—all of which should fit well with the program to which you are applying. It is also a chance to explain in more detail information that may have appeared only briefly on a résumé or vita, such as clinical experiences or research with an undergraduate professor. Make sure your writing ability appears strong and that you don't make the statement overly personal or revealing.

- *Prepare well for admissions interviews.* Most doctoral programs invite high-ranking applicants for an in-person interview. These interviews are a wonderful opportunity for professors in the program to get to know you and for you to get to know the program. Arrive (professionally dressed, of course) with a strong understanding of the program and your interest in it. The more specific, the better: Interest in particular professors' research concentrations, for example, makes a better impression than the fact that the program has a strong reputation. Box 1.2 lists some of the questions you should be prepared to answer. And don't forget to develop a list of your own questions—good questions can solicit more detailed information than you were able to find on the program's website and can impress interviewers in the process.

- *Consider your long-term goals.* Down the road, do you see yourself as a clinician or a researcher? Have you firmly determined your own theoretical orientation already, or do you seek a program that will expose you to a variety? What specific areas of clinical or scientific work are most interesting to you? How much financial debt are you willing to incur? Thinking ahead about these and other questions can increase the likelihood that you will find yourself at a graduate program at which you thrive and that sets you up for a fulfilling career.

- In addition to the previously mentioned general points of advice, it's a good idea for future applicants to search the literature for studies on how graduate programs handle their admissions processes. For example, one study surveyed the directors of clinical psychology doctoral programs about the importance they place on various components of the application process (Littleford, Buxton, Bucher, Simon-Dack, & Yang, 2018). These researchers found that both PhD and PsyD programs placed high value on several core elements: undergraduate GPA, GRE scores, letters of recommendation, personal statement, curriculum vitae, and interview performance. There were some differences between PhD and PsyD programs: Compared with PsyD programs, PhD programs placed much greater importance on the student–mentor research match (making sure that an incoming student has research interests that closely correspond to those of a faculty member) and research assistance experience. Also, compared with PhD programs, PsyD programs were more likely to offer admission to applicants who already had a master's degree.

BOX 1.2

Interview Questions to Anticipate

There is no formula for the kinds of questions that interviewers might ask an applicant to a clinical psychology program, but these questions are especially common. Whether they ask these particular questions or not, you enhance your chances of finding a graduate program that truly fits your interests by giving them serious consideration.

- Why do you want to be a clinical psychologist?
- What attracts you to our graduate program specifically?
- What are your research interests?
- What approach(es) to psychotherapy do you prefer?
- Which of our faculty members would you like to work with?
- What are your long-term career goals? If you were a student in our program, what would you like to do after you graduated?

Source: Adapted from Norcross & Sayette (2016).

Internships: Predoc and Postdoc

All clinical psychology doctoral programs culminate in the **predoctoral internship** (N. J. Kaslow & Webb, 2011; McCutcheon & Sannes, 2015). Typically, this internship consists of a full year of supervised clinical experience in an applied setting—a psychiatric hospital, a Veterans Affairs medical center, a university counseling center, a community mental health center, a medical school, or another agency where clinical psychologists work (R. R. Baker & Pickren, 2011). As implied by the term *predoctoral*, this internship year takes place before the PhD or the PsyD is awarded. (Along with completion of the dissertation, it is likely to be one of the final hurdles.) It is generally considered a year of transition, a sort of advanced apprenticeship in which the individual begins to outgrow the role of "student" and grow into the role of "professional." In some settings, it is also an opportunity to gain more specialized training than may have been available in graduate school so far. Many internships are accredited by the APA; those that are not may be looked on less favorably by state licensing boards.

The process of applying for a predoctoral internship can feel a lot like the process of applying to graduate school some years earlier. It often involves researching various internships, applying to many, traveling for interviews, ranking preferences, anxiously awaiting feedback, and relocating to a new geographic area. Some students apply to 20 or more internship sites (Keilin, 2000), but 10 to 15 may be more reasonable and equally effective (Keilin & Constantine, 2001). Adding stress to the situation is the fact that in some years, the number of graduate students seeking predoctoral internships has either approached or exceeded the number of available slots (N. J. Kaslow & Webb, 2011; Keilin, Thorn, Rodolfa, Constantine, & Kaslow, 2000; Klonoff, 2016). In fact, the shortage worsened considerably after roughly 2002, as the number of students who applied but were not successfully placed at an internship increased dramatically, to hundreds per year (Dingfelder, 2012; Hatcher, 2014). That number has since decreased, but the issue remains, at least to some extent (Doran & Cimbora, 2016; Norcross et al., 2018). Numerous causes for the internship shortage (also called an imbalance or crisis by some) have been proposed, but the factor receiving the most attention is the drastic increase in the number of PsyD applicants without a corresponding increase in the total number of internship placements (McCutcheon & Sannes, 2015). The internship application process can certainly generate stress for applicants and feel a bit like a game of musical chairs, but numerous strategies to improve the current situation are under consideration (Hatcher, 2014; C. Webb & Hill, 2016). Applicants are generally successful in finding an internship position—especially if they don't overly restrict themselves in terms of the number of applications or geographic range.

Beyond the predoctoral internship and the doctoral degree that follows, most states require a **postdoctoral internship** (or postdoc) for licensure as a psychologist. The postdoc typically lasts 1 to 2 years (Vaughn, 2006), and it is essentially a step up from the predoctoral internship. Postdocs take on more responsibilities than they did as predoctoral interns, but they remain under supervision. Like the predoctoral internship, the postdoc often provides an opportunity for specialized training. After postdoctoral interns accumulate the required number of supervised hours (and pass the applicable licensing exams), they can become licensed to practice independently. Some clinical psychologists obtain postdoc positions that are explicitly designed from the start to meet licensing requirements for a particular state; sometimes, such positions are continuations

Web Link 1.4
Association of State and Provincial Psychology Boards

of predoctoral internship experiences. Other clinical psychologists may obtain an entry-level position with an agency and tailor it to meet postdoctoral requirements for licensure (Hatcher & Buchtel, 2016; McQuaid & McCutcheon, 2018; Silberbogen, Aosved, Cross, Cox, & Felleman, 2018).

Since 2008, a small but growing number of states have decided to drop the requirement of a postdoc for licensure. The rationale behind this decision involves several factors: Current graduate training (including the predoctoral internship) provides far more clinical experience than it did years ago when postdoc requirements were put into place; employers often have difficulties funding postdoc positions because third-party payers (e.g., health insurance companies) who pay the bills for many clients can refuse to pay for services by someone who is not yet licensed; postdoc positions can be difficult to find, especially in rural areas; and some states have a shortage of clinical psychologists to serve their residents, so making licensure easier to obtain is a win-win for the psychologists and the public (Boon, Lutz, & Marburger, 2015; Calkins, 2018).

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Photos 1.1, 1.2, and 1.3
Clinical psychologists work in a variety of settings, including private offices, universities, and hospitals.

Getting Licensed

Once all the training requirements are met—graduate coursework, predoctoral internship, postdoctoral internship—**licensure** appears on the horizon. Becoming licensed gives professionals the right to identify as members of the profession—to present themselves as psychologists (or clinical psychologists—the terminology, as well as licensing requirements in general, differs from state to state). It also authorizes the psychologist to practice independently (APA, 2007; J. E. Hall, 2015; Schaffer, DeMers, & Rodolfa, 2011; Siegel & DeMers, 2016).

But you won't be handed a license when you get your doctoral degree or when you finish your postdoc. Becoming licensed also requires passing licensure exams—typically, the **Examination for Professional Practice in Psychology (EPPP)** and a state-specific exam on laws and ethics. The EPPP is a standardized multiple-choice exam on a broad range of psychology topics; all U.S. states and most provinces of Canada establish a minimum score for licensure (Rehm & Lipkins, 2006; Schaffer et al., 2011). The state exams vary, of course, according to state regulations but tend to center on legal issues relevant to the practice of psychology in the state in question. The state exams may be written or oral.

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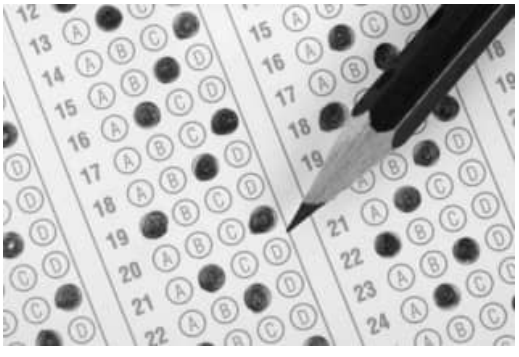

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Photo 1.4, 1.5, and 1.6 Clinical psychologists' professional activities include psychotherapy, assessment, and education.

Once licensed, clinical psychologists in many states must accumulate **continuing education units (CEUs)** to renew the license from year to year (G. J. Neimeyer & Taylor, 2011). In various states, psychologists can meet these ongoing requirements in a number of ways—by attending workshops, taking courses, undergoing additional specialized training, passing exams on selected professional reading material, and the like (Babeva & Davison, 2015; J. M. Taylor & Neimeyer, 2016). The purpose of requiring CEUs is to ensure that clinical psychologists stay up to date on developments in the field, with the intention of maintaining or improving the standard of care they can provide to clients.

PROFESSIONAL ACTIVITIES AND EMPLOYMENT SETTINGS

Where Do Clinical Psychologists Work?

The short answer is that clinical psychologists work in a wide variety of settings but that private practice is the most common. In fact, this answer is evident not only from a survey of clinical psychologists conducted in the 2000s but also according to similar surveys conducted in the 1980s and 1990s (Norcross & Karpiak, 2012; Norcross & Rogan, 2013).

Since the 1980s, private practice has been the primary employment site of 30% to 41% of clinical psychologists. The second-place finisher in each survey during that time has been the university psychology department, but that number has not exceeded 19%. Between 2% and 9% of clinical psychologists have listed each of the following as their primary work setting: psychiatric hospitals, general hospitals, community mental health centers, medical schools, and Veterans Affairs medical centers. Interestingly, the third-place finisher (after private practice and university psychology department) in each survey since the 1980s has been the "other" category; for example, in 2003, 15% of psychologists listed "other," writing in diverse settings such as government agency, public schools, substance abuse center, corporation, and university counseling center. It is clear that although private practice remains a common destination, clinical

psychologists are finding employment across an expanding range of settings (Borel, 2015; Norcross & Karpiak, 2012; Norcross, Karpiak, & Santoro, 2005; Vasquez & Kelly, 2016).

What Do Clinical Psychologists Do?

Again, the short answer first: Clinical psychologists are engaged in an enormous range of professional activities, but psychotherapy is foremost. As is the case with employment settings, this finding is as true today and has been for decades—at least since the 1970s (Norcross & Karpiak, 2012).

Since 1973, the number of clinical psychologists reporting that they are involved in psychotherapy has always outranked that of any other professional activity and has ranged from 76% to 87%. Moreover, when asked what percentage of their time they spend in each activity, clinical psychologists have reported that they spend between 31% and 37% of their time conducting psychotherapy—a percentage more than double that of any other activity. Of those who practice psychotherapy, individual therapy occupies the largest percentage of their therapy time (76%), with group, family, and couples therapy far behind (6% to 9% each; Norcross & Karpiak, 2012).

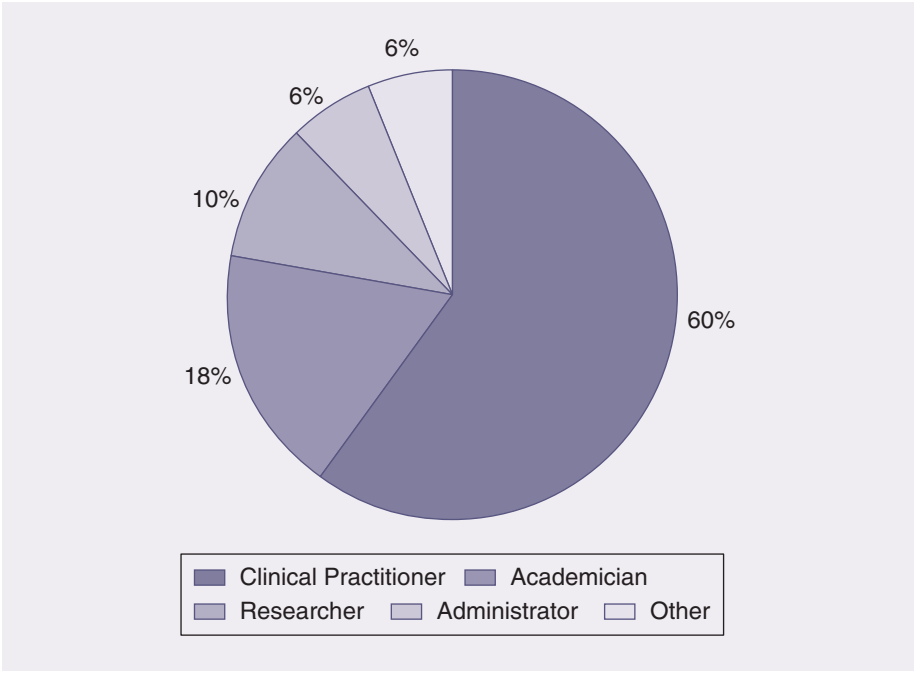
Of course, a sizable number of psychologists—more than half—have also reported that they are at least somewhat involved in each of the following activities: diagnosis/assessment, teaching, supervision, research/writing, consultation, and administration. Of these, diagnosis and assessment generally occupy more of clinical psychologists' time than do the others. Overall, it is evident that "clinical psychologists are involved in multiple professional pursuits across varied employment sites" (Norcross, Karpiak, & Santoro, 2005, p. 1474). In fact, more than half of clinical psychologists hold at least two professional positions (Norcross & Sayette, 2016). Figure 1.1 illustrates the professional self-views of clinical psychologists.

HOW ARE CLINICAL PSYCHOLOGISTS DIFFERENT FROM . . .

Counseling Psychologists

There may have been a time when counseling psychology and clinical psychology were quite distinct, but today, there is significant overlap between these two professions. Historically, they have differed primarily in terms of their clients' characteristics: Clinical psychologists were more likely to work with seriously disturbed individuals, whereas **counseling psychologists** were more likely to work with ("counsel") less pathological clients. But today, many clinical and counseling psychologists see the same types of clients, sometimes as colleagues working side by side. These two fields are also similar in that their graduate students occupy the same internship sites, often earn the same degree (the PhD), and obtain the same licensure status (Norcross, 2000). In fact, the two professions share so much common ground

Figure 1.1 Professional Self-Views of Clinical Psychologists



Source: Norcross, Karpiak, & Santoro (2005), Figure 3.

that it is entirely possible for a client who seeks the services of a psychologist with a doctoral degree behind their name never to know whether that doctoral degree is in clinical or counseling psychology.

A few meaningful differences, however, remain between clinical and counseling psychologists. Compared with counseling psychologists, clinical psychologists still tend to work with more seriously disturbed populations and, correspondingly, tend to work and complete internships more often in settings such as hospitals and inpatient psychiatric units. And compared with clinical psychologists, counseling psychologists still tend to work with less seriously disturbed populations and, correspondingly, tend to work and complete internships more often in university counseling centers (Eisman & Wright, 2016; Gaddy et al., 1995). Some differences in theoretical orientation are also evident: Both fields endorse the eclectic orientation more than any other, but clinical psychologists tend to endorse behaviorism more strongly, and counseling psychologists tend to endorse humanistic/client-centered approaches more strongly. Additionally, counseling psychologists tend to be more interested in vocational testing and career counseling, whereas clinical psychologists tend to be more interested in applications of psychology to medical settings (Banyasz & Baker, 2015; Norcross & Sayette, 2016).

Psychiatrists

Unlike clinical (or counseling) psychologists, **psychiatrists** go to medical school and are licensed as physicians. (In fact, their specialized training in psychiatry doesn't begin until well into their training; the first several years are often identical to that of other types of physicians.) As physicians, they are allowed to prescribe medication. Until recently, psychologists could not prescribe medication, but as described in Chapter 3, psychologists have rallied in recent years to obtain prescription privileges and have earned important victories in a small number of states.

The difference between psychiatrists and clinical psychologists is more than just the ability to prescribe medication. The two professions fundamentally differ in their understanding of and approach to behavioral or emotional problems. Clinical psychologists are certainly trained to appreciate the biological aspects of their clients' problems, but psychiatrists' training emphasizes biology to such an extent that disorders—depression, anxiety disorders, attention-deficit/hyperactivity disorder (ADHD), borderline personality disorder, and so on—are viewed first and foremost as physiological abnormalities of the brain. So, to fix the problem, psychiatrists tend to fix the brain by prescribing medication (Noll, 2015). This is not to imply that psychiatrists don't respect "talking cures" such as psychotherapy or counseling, but they favor medication more than they used to (G. Harris, 2011; Manninen, 2006). For clinical psychologists, the biological aspects of clients' problems may not be their defining characteristic; nor is pharmacology the first line of defense. Instead, clinical psychologists view clients' problems as behavioral, cognitive, emotional—still stemming from brain activity, of course, but amenable to change via nonpharmacological methods.

BOX 1.3

In My Practice . . .

In my practice, many of my clients don't know the difference between a psychologist and a psychiatrist. Of course, there are those who do know the difference. Those who know often have some firsthand experience with the mental health field: Some have seen both psychologists and psychiatrists before, some have accompanied family members to appointments with both, and some are members of the mental health profession themselves. But many clients don't know the difference. These are not ignorant people at all. Some have earned impressive degrees, and many have struck me as brilliant in their own ways. They simply don't have this particular piece of information.

Their lack of knowledge has become apparent in a variety of ways. Once, after finishing a long clinical interview with a client, he looked at me expectantly. When I asked if he had a question for me, he asked impatiently, "So? What's the prescription, doc?" When I explained that I was a psychologist and not a psychiatrist, and that I didn't prescribe medication, he seemed completely shocked. Another time, a mother who was bringing her 12-year-old son to his first appointment asked if she could have a few minutes alone with me.

(Continued)

(Continued)

She delivered an impassioned speech detailing her stance against psychiatric medication for children and described how she couldn't sleep the previous night because she was so afraid that I would recommend medication for her strong-willed, possibly hyperactive boy. She seemed quite relieved when I explained who I was and what I did.

I have often wondered why so many people have such trouble separating psychologists from psychiatrists. The similarity of the words certainly doesn't help: They both start with "psych" and end with "ist." The media don't help either: So often in movies, TV shows, and news reports, the names of the mental health professions, especially psychologists and psychiatrists, get jumbled. I have come to the conclusion that part of my job when I speak with new or prospective clients is to make sure that they understand the difference between a psychologist and a psychiatrist. I consider it part of the informed consent process (which we will discuss in greater detail in Chapter 5)—an important element of what clients need to know in order to make an educated decision about whether to move forward with treatment.

To see this box come to life as a whiteboard video, please visit edge.sagepub/pomerantz5e.

Social Workers

Traditionally, **social workers** have focused their work on the interaction between an individual and the components of society that may contribute to or alleviate the individual's problems. They saw many of their clients' problems as products of social ills—racism, oppressive gender roles, poverty, abuse, and so on. They also helped their clients by connecting them with social services, such as welfare agencies, disability offices, or job-training sites (Fossum, Logeais, & Robiner, 2016). More than their counterparts in psychology or psychiatry, they were likely to get into the "nitty-gritty" of their clients' worlds by visiting their homes or workplaces, or by making contacts on their behalf with organizations that might prove beneficial. When they worked together with psychologists and psychiatrists (e.g., in institutions), they usually focused on issues such as arranging for clients to transition successfully to the community after leaving an inpatient unit by making sure that needs such as those for housing, employment, and outpatient mental health services were being met.

In more recent years, the social work profession has grown to encompass a wider range of activities, and the similarity of some social workers (especially those conducting therapy) to clinical psychologists has increased (Fossum et al., 2016; Wittenberg, 1997). The training of social workers, however, remains quite different from the training of clinical psychologists. They typically earn a master's degree rather than a doctorate, and although their training includes a strong emphasis on supervised fieldwork, it includes very little on research methods, psychological testing, or physiological psychology. Their theories of psychopathology and therapy continue to emphasize social and environmental factors.

School Psychologists

As the name implies, **school psychologists** usually work in schools, but some may work in other settings such as day-care centers or correctional facilities. Their primary function is to enhance the intellectual, emotional, social, and developmental lives of students. They frequently conduct psychological testing (especially intelligence and achievement tests) to determine diagnoses such as learning disorders and ADHD. They use or develop programs designed to meet the educational and emotional needs of students. They also consult with adults involved in students' lives—teachers, school administrators, school staff, parents—and are involved, to a limited degree, in direct counseling with students (Albers & Felt, 2015). In many states, the title of school psychologist requires only a master's degree rather than a doctorate (Eisman & Wright, 2016).

Professional Counselors

Professional counselors (often called licensed professional counselors, or LPCs) earn a master's (rather than a doctoral) degree and often complete their training within 2 years. They attend graduate programs in counseling or professional counseling, which should not be confused with doctoral programs in counseling psychology. These programs typically have rather high acceptance rates compared with programs in many similar professions. Professional counselors' work generally involves counseling of people with problems in living or mild mental illness (as opposed to serious mental illness). Their training, compared with that of clinical psychologists, typically includes very little emphasis on psychological testing or conducting research. Correspondingly, their training programs include few if any courses on these topics, focusing instead on providing services to clients. Increasingly, professional counselors are among the clinicians who serve wide varieties of clients in community agencies (Cohen-Filipic, 2015b; Norcross & Sayette, 2016), and they often enter private practice as well. They often specialize in such areas as career, school, addiction, couple/family, or college counseling. Every state has some version of professional counselor licensure, but the name may vary slightly, with common alternatives including mental health counselor, licensed professional mental health counselor, licensed clinical professional counselor, and licensed counselor of mental health (Cohen-Filipic, 2015b; Fossum et al., 2016).

Marriage and Family Therapists

Marriage and family therapists (MFTs) earn master's degrees. Their training focuses on working with couples and families, but sometimes they also see individuals struggling with issues related to their partners or families. About half of MFTs work in private practice, and most of the others practice in other types of work settings like clinics or agencies (Fossum et al., 2016). Compared to the training of clinical psychologists, the training of MFTs places relatively little emphasis on research and assessment.

CHAPTER SUMMARY

The scope of clinical psychology has expanded greatly since the inception of the field by Lightner Witmer near the turn of the 20th century. Currently, there are multiple paths to the profession, including three distinct approaches to training: the scientist-practitioner (Boulder) approach, with roughly equal emphasis on empiricism and practice; the practitioner-scholar (Vail) approach, with stronger emphasis on practice; and the clinical scientist approach, with stronger emphasis on empiricism. Gaining admission to a training program is a competitive endeavor. Knowledge of the professional training options, successful completion of the appropriate undergraduate courses, research experience, and clinical experience are among the factors that can distinguish an applicant and enhance chances for admission. The final steps of the training process for clinical psychologists are the predoctoral and postdoctoral internships, in which the trainee practices under supervision to transition into the full-fledged professional role. Licensure, which requires a passing grade on the EPPP as well as meeting state-specific requirements, allows clinical psychologists to practice independently. The most common work setting for clinical psychologists is private practice, but university psychology departments and hospitals of various types are also somewhat frequent. The most common professional activity for clinical psychologists is psychotherapy, but they also spend significant amounts of time in assessment, teaching, research, and supervision activities. The professional roles of counseling psychologists, psychiatrists, social workers, and school psychologists each overlap somewhat with the role of clinical psychologists, yet clinical psychology has always retained its own unique professional identity.

KEY TERMS AND NAMES

Academy of Psychological Clinical Science 9
American Psychological Association (APA) 4
Boulder model 6
clinical psychology 3
clinical scientist model 9
continuing education units (CEUs) 18
counseling psychologists 19
Division of Clinical Psychology (Division 12) 4
Examination for Professional Practice in Psychology (EPPP) 17
licensure 17
marriage and family therapists (MFTs) 23
Richard McFall 9
postdoctoral internship 16
practitioner-scholar model 7
predoctoral internship 16
professional counselors 23
psychiatrists 21
PsyD 7
school psychologists 23
scientist-practitioner model 6
social workers 22
Vail model 8
Lightner Witmer 3

CRITICAL THINKING QUESTIONS

1. Lightner Witmer originally defined clinical psychology as a discipline with similarities to medicine, education, and sociology. In your opinion, to what extent does contemporary clinical psychology remain similar to these fields?
2. Considering the trends in graduate training models observed recently, how popular do you expect the scientist-practitioner, practitioner-scholar, and clinical scientist models of training to be 10 years from now? What about 50 years from now?

3. What specific types of research or clinical experience do you think would be most valuable for an undergraduate who hopes to become a clinical psychologist?
4. In your opinion, to what extent should graduate programs use the GRE as an admission criterion for graduate school in clinical psychology?
5. In your opinion, how much continuing education should licensed clinical psychologists be required to undergo? What forms should this continuing education take (workshops, courses, readings, etc.)?

LOOKING TOWARD GRADUATE PROGRAMS

For undergraduates who are seriously considering graduate school, the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018) can be a valuable resource. The *Insider's Guide* is a book filled with hundreds of pages of detailed information about almost all doctoral programs in clinical and counseling psychology, provided by the programs themselves. Included among that detailed information are lists of special areas of focus for each program, in the form of *research areas* (e.g., faculty members doing research on specific topics, often with grants); *specialty clinics and practica sites* (where graduate students may be able to get targeted clinical experience); and *program concentrations and tracks*. The *Insider's Guide* conveniently provides three separate indices in which you can look up a specialty area and be guided to specific graduate programs (Appendix E for research areas, Appendix F for specialty clinics and practica sites, and Appendix G for program concentrations and tracks). Those indices are great resources if you want an easy way to find out which graduate programs focus on which areas of clinical or counseling psychology.

The following table is a list of index entries relevant to the topics in this chapter. Keep in mind a few things as you consider these listings: They are current as of the 2018/2019 edition of the *Insider's Guide* (which typically revises every two years); they may have changed as faculty members retire, take new jobs, or develop new interests; and a graduate program's own website may have more extensive or up-to-date information.

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Professional Issues	Career Counseling/Development	Miscellaneous/Other—Preparing Future Faculty
Supervision/Mentoring/Training	College/University Counseling/ Psychotherapy Centers	
Miscellaneous and Other—Licensure and Regulatory Boards	Private Practice	
Miscellaneous and Other—Psychotherapist's Development		

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

Annual Review of Clinical Psychology

http://www.annualreviews.org/journal/clinpsy

Clinical Psychology Review

http://www.journals.elsevier.com/clinical-psychology-review

Clinical Psychology & Psychotherapy

http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1099-0879

Journal of Clinical Psychology

http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1097-4679

Journal of Counseling Psychology

http://www.apa.org/pubs/journals/cou/index.aspx

Training and Education in Professional Psychology

http://www.apa.org/pubs/journals/tep/

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- Culture expert interviews
- In My Practice demonstration video
- Full-text SAGE journal articles
- Additional web resources
- Mock assessment data

2

EVOLUTION OF CLINICAL PSYCHOLOGY

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Philippe Pinel (1745–1826)	28
Eli Todd (1769–1833)	29
Dorothea Dix (1802–1887)	29
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Assessment of Personality	36
Psychotherapy	**37**
Development of the Profession	**40**

ORIGINS OF THE FIELD

Psychology, as we know, has a long, rich history, with roots winding back to some of the great thinkers of prior millennia, such as Socrates, Plato, and Aristotle (L. T. Benjamin, 2007; Ehrenwald, 1991). The history of *clinical* psychology is certainly rich as well, but it's also shorter than some might expect. Of course, in recent decades, the clinical specialization has enjoyed great popularity and notoriety among psychology professionals and the general public; in fact, today, when many people hear of a "psychologist," they immediately think of a "clinical psychologist" practicing psychotherapy or assessment. This assumption regarding psychology was inaccurate until at least the early 1900s. The discipline of clinical psychology simply didn't exist until around the turn of the 20th century, and it didn't rise to prominence for decades after that.

EARLY PIONEERS

Even before "clinical psychology" per se had been created, numerous influential individuals in various parts of the world were working to make positive changes in the lives

Learning Objectives

2.1 Summarize major contributions of early pioneers, such as William Tuke and Eli Todd, to mental health care from the 1700s to the late 1800s.

2.2 Describe Lightner Witmer's key contributions to the field of clinical psychology.

2.3 Recount the development of key diagnostic systems for mental illness from the 1800s to present.

2.4 Evaluate the process used to decide what will become a *DSM* disorder.

2.5 Identify important people and psychological instruments in the history of intellectual and personality assessment.

2.6 Explain the influence of war on the professional activities of clinical psychologists.

2.7 Discuss the growth of clinical psychology in the United States from the early 1900s to present.

of the mentally ill. Collectively, their accomplishments created a climate from which clinical psychology could emerge. Specifically, in the 1700s and 1800s, the mentally ill were generally viewed and treated much more unfavorably than they are today. In many parts of the world, including much of the Western Hemisphere, they were understood to be possessed by evil spirits. Or they were seen as deserving of their symptoms as a consequence of some reprehensible action or characteristic. They were frequently shunned by society and were "treated" in institutions that resembled prisons more than they did hospitals (Reisman, 1991).

During this time, numerous individuals of various professional backgrounds from Europe and North America assumed the challenge of improving the way people with psychological problems were regarded and treated. Through their efforts, the Western world eventually adopted a new, more humane, approach to the mentally ill, foretelling the emergence of clinical psychology as a formal discipline. The accomplishments of several of these pioneers are described below.

William Tuke (1732–1822)

In his homeland of England, **William Tuke** heard about the deplorable conditions in which the mentally ill lived. He visited asylums to get a firsthand look, and he was appalled by what he saw. Tuke devoted much of his life to improving these conditions. He raised funds to open the **York Retreat**, a residential treatment center where the mentally ill would always be cared for with kindness, dignity, and decency. (The simple act of labeling his facility a "retreat" suggests a fundamentally different approach to the mentally ill compared with the dominant approach at the time.) Patients received good food, frequent exercise, and friendly interactions with staff. The York Retreat became an example of humane treatment, and soon similar institutions opened throughout Europe and the United States. Long after his death, Tuke's family members continued to be involved in the York Retreat and the movement to improve treatment of mentally ill individuals (Reisman, 1991).

Web Link 2.1
William Tuke

Philippe Pinel (1745–1826)

What William Tuke was to England, **Philippe Pinel** was to France—a liberator of the mentally ill. Like Tuke, Pinel worked successfully to move mentally ill individuals out of dungeons in Paris, where they were held as inmates rather than treated as patients (Cautin, 2011; Charland, 2015; Ehrenwald, 1991). He went to great lengths to convince

his contemporaries and those with power in France that the mentally ill were not possessed by devils and that they deserved compassion and hope rather than maltreatment and scorn. He created new institutions in which patients were not kept in chains or beaten but, rather, were given healthy food and benevolent treatment. Particularly noteworthy, Pinel advocated for the staff to include in their treatment of each patient a case history, ongoing treatment notes, and an illness classification of some kind—components of care that suggested he was genuinely interested in improving these individuals rather than locking them away (Reisman, 1991).

From Pinel's *Treatise on Insanity* in 1806, we get a sense of his goal of empathy rather than cruelty for the mentally ill: "To rule [the mentally ill] with a rod of iron, as if to shorten the term of an existence considered miserable, is a system of superintendence, more distinguished for its convenience than for its humanity or success" (as quoted in Ehrenwald, 1991, p. 217). As Frances (2013a) describes Pinel, "our field couldn't possibly have a better father and role model. . . . Pinel liked his patients as people and treated them as if they were simply human. When given the choice of joining Napoleon as a personal physician or staying with his patients, he turned down Napoleon" (p. 57).

In Europe in the late 18th and early 19th centuries, society's views toward the mentally ill were undergoing significant change and "the voices of Pinel and Tuke were part of a growing chorus that sang of individual rights and social responsibility" (Reisman, 1991, p. 9).

Eli Todd (1769–1833)

Eli Todd made sure that the chorus of voices for humane treatment of the mentally ill was also heard on the other side of the Atlantic Ocean. Todd was a physician in Connecticut in 1800, a time when only three states had hospitals for the mentally ill. The burden for treating the mentally ill typically fell on their families, who often hid their mentally ill relatives out of shame and embarrassment. Todd had learned about Pinel's efforts in France, and he spread the word among his own medical colleagues in the United States. They supported Todd's ideals both ideologically and financially, such that Todd was able to raise funds to open **The Retreat** in Hartford, Connecticut, in 1824. Todd ensured that patients at The Retreat were always treated in a humane and dignified way. He and his staff emphasized patients' strengths rather than weaknesses, and they allowed patients to have significant input in their own treatment decisions. Similar institutions were soon opened in other U.S. states as leaders elsewhere learned of Todd's successful treatment of the mentally ill (Reisman, 1991).

Web Link 2.2
Eli Todd

Dorothea Dix (1802–1887)

Despite Todd's efforts, there were simply not enough hospitals in the United States to treat the mentally ill, and as a result, these individuals were too often sent to prisons or jails in an attempt to find any social institution that could house them. In 1841, **Dorothea Dix** was working as a Sunday school teacher in a jail in Boston, where she saw firsthand that many of the inmates were there as a result of mental illness or retardation rather than crime. Dix devoted the rest of her life to improving the lives and treatment

of the mentally ill. Typically, she would travel to a city, collect data on its treatment of the mentally ill, present her data to local community leaders, and persuade them to treat the mentally ill more humanely and adequately. She repeated this pattern again and again, in city after city, with remarkable success. Her efforts resulted in the establishment of more than 30 state institutions for the mentally ill throughout the United States (and even more in Europe and Asia), providing more decent, compassionate treatment for the mentally ill than they might have otherwise received (Reisman, 1991).

Tuke, Pinel, Todd, and Dix did not create clinical psychology. Their efforts do, however, represent a movement prevalent through much of the Western world in the 1700s and 1800s that promoted the fundamental message that people with mental illness deserve respect, understanding, and help rather than contempt, fear, and punishment. As this message gained power and acceptance, it created fertile ground in which someone—Lightner Witmer, most would argue—could plant the seed that would grow into clinical psychology (Cautin, 2011).

Web Link 2.3
Dorothea Dix

LIGHTNER WITMER AND THE CREATION OF CLINICAL PSYCHOLOGY

Lightner Witmer (1867–1956) was born in Philadelphia and earned an undergraduate degree in business at the University of Pennsylvania. He eventually received his doctorate in psychology in 1892 in Germany under Wilhelm Wundt, who many view as the founder of experimental psychology. He also studied under James McKeen Cattell, another pioneer of experimental psychology (Reisman, 1991; Routh, 2015c). At the time Witmer received his doctorate, psychology was essentially an academic discipline, a field of research. It had almost none of the applied functions that characterize the field today. In short, in the late 1800s, psychologists didn't *practice* psychology, they *studied* it (Vasquez & Kelly, 2016).

A major historical shift took place 4 years after Witmer received his doctoral degree when, in 1896, he founded the first psychological clinic at the University of Pennsylvania, where he had returned as a professor (Routh, 1996). Although it may be difficult to imagine from our contemporary perspective of the field, this was the first time that the science of psychology was systematically and intentionally applied to people's problems. At the 1896 convention of the American Psychological Association, Witmer (1897) spoke to his colleagues and fellow members about his clinic and encouraged them to open their own—to "throw light upon the problems that confront humanity" (p. 116)—but they were largely unenthusiastic (Reisman, 1991). Decades later, though, clinical settings would certainly proliferate. By 1914, there were about 20 psychological clinics in the United States, most of which were modeled on Witmer's (Schultz & Schultz, 2011). By 1935, the number had soared to more than 150, and an issue of Witmer's journal from that year was dedicated to a survey of activities taking place in these clinics, as well as some specific suggestions for the training of clinical psychologists (A. W. Brown, 1935).

In his clinic, Witmer and his associates worked with children whose problems arose in school settings and were related to learning or behavior (L. T. Benjamin, 2007; Routh, 2015c). They were referred by their schools, parents, physicians, or community

authorities (McReynolds, 1997). Witmer (1907) emphasized that clinical psychology could be applied to adults as well as children, or to problems that had nothing to do with school: "Indeed, the clinical method is applicable even to the so-called normal child. . . . Whether the subject be a child or an adult, the examination and treatment may be conducted and their results expressed in the terms of the clinical method" (p. 9).

In addition to establishing his clinic, Witmer also founded the first scholarly journal in the field (called ***The Psychological Clinic***) in 1907 (L. T. Benjamin, 2007). Witmer authored the first article, titled "Clinical Psychology," in the first issue. This article included the first known publication of the term *clinical psychology*, as well as a definition of the term and an explanation of the need for its existence and growth. The article began with a description of Witmer's (1907) innovation: "During the last ten years the laboratory of psychology at the University of Pennsylvania has conducted, under my direction, what I have called a 'psychological clinic'" (p. 1).

We discussed Witmer's original definition of clinical psychology in Chapter 1, but a consideration in this chapter of Witmer's historical significance gives us an opportunity to examine it further. First, it is worth noting that Witmer defined clinical psychology as related to medicine, education, and social work but stated that physicians, teachers, and social workers would not be qualified to practice clinical psychology. Instead, this new field represented a hybrid of these and other influences, requiring a specially trained professional who, of course, would work collaboratively with members of related fields. It is also interesting that Witmer's definition of clinical psychology is basically uninfluenced by Freud, whose ideas appeared throughout many fields at the time, and that psychotherapy, as it would come to be known, was not explicitly discussed at all. Finally, the treatments that Witmer discusses in his original definition aren't accompanied by any mention of a plan for empirically evaluating their effectiveness, which is a bit surprising given Witmer's graduate training as an experimental researcher (Routh, 1996; Witmer, 1907).

So, by the late 1800s, the work of Tuke, Pinel, Todd, Dix, and others had set the stage for the birth of clinical psychology. Witmer proudly announced its arrival, and although it would thrive in later years, "clinical psychology in the 1890s was a glimmer, a baby just catching its first breath, drawing its life from the new science of psychology" (Reisman, 1991, p. 44).

Public Domain

Photo 2.1
Lightner Witmer (1867–1956), who created the term *clinical psychology* and founded the first psychological clinic. How has the field changed since Witmer's era?

Web Link 2.4
Lightner Witmer

Web Link 2.5
First "Clinical Psychology" article

ASSESSMENT

Diagnostic Issues

Categorizing mental illness has been an issue central to clinical psychology since Witmer defined the field. Actually, the debate began long before Witmer entered the picture.

With so many mental disorders tossed around as common terminology today, it can be difficult to imagine a time when such labels didn't exist at all, at least not in any formal sense. In the 1800s in Europe, labeling systems for mental illness began to take shape in a very rudimentary way. Specifically, mental illnesses were often placed in one of two very broad categories: **neurosis** and **psychosis**. Neurotic individuals were thought to suffer from some psychiatric symptoms (including what we would now call anxiety and depression) but to maintain an intact grasp on reality. Psychotic individuals, on the other hand, demonstrated a break from reality in the form of hallucinations, delusions, or grossly disorganized thinking (Reisman, 1991).

Emil Kraepelin (1855–1926), considered the "father of descriptive psychiatry" (Reisman, 1991, p. 30), offered a different two-category system of mental illness. Kraepelin differentiated **exogenous disorders** (caused by external factors) from **endogenous disorders** (caused by internal factors) and suggested that exogenous disorders were the far more treatable type. Kraepelin also assigned names to specific examples of disorders in the broad exogenous or endogenous categories. For example, Kraepelin put forth the term **dementia praecox** to describe one endogenous disorder similar to what is now known as schizophrenia. Later, he also proposed terms such as *paranoia*, *manic depressive psychosis*, *involutional melancholia*, *cyclothymic personality*, and *autistic personality*—terms that had not yet been coined (Millon & Simonsen, 2010). Most of Kraepelin's specific terms have long since been replaced, but by offering such specific terminology, he set a precedent for the creation of diagnostic terms that eventually led to the *Diagnostic and Statistical Manual of Mental Disorders (DSM)* (Engstrom & Weber, 2015; Widiger & Mullins-Sweatt, 2008).

In the United States, long before the appearance of the first *DSM*, the original reason for categorizing mental disorders was to collect statistics on the population. In 1840, the U.S. Census Bureau included a single category—"idiocy/insanity"—for this purpose. In 1880, there were seven such categories, and soon the American Medical Association and the U.S. Army each made preliminary attempts at classifying mental illness (American Psychiatric Association, 2000).

The original *DSM* was published by the American Psychiatric Association in 1952, representing a more sophisticated attempt to define and organize mental diagnoses. A revision (*DSM-II*) followed in 1968, but it was not considered to be significantly different from the original. However, the next revision—*DSM-III*, which arrived in 1980—signified an entirely new way of thinking about mental disorders (Decker, 2013). Whereas *DSM* and *DSM-II* included somewhat vague descriptions of each disorder, *DSM-III* provided specific **diagnostic criteria**—lists indicating exactly what symptoms constitute each disorder. *DSM-III* also introduced a **multiaxial system**, a way of cataloguing problems of different kinds on different axes, which remained for multiple editions before being taken out of the most recent. The *DSM* has been revised several more times, with *DSM-III-R*, *DSM-IV*, and *DSM-IV-TR* appearing in 1987, 1994, and 2000, respectively. The current edition is *DSM-5*, which was published in May 2013 after significant anticipation and controversy. (*DSM-5* will be covered in more detail elsewhere, especially in Chapter 7.) Each edition varies from its predecessor to some extent, but the most drastic change occurred with the publication of *DSM-III* in 1980 (Lilienfeld & Landfield, 2008a; Widiger & Mullins-Sweatt, 2008).

Sheer size is certainly among the most notable differences between *DSM-II* and *DSM-III*. In fact, new *DSM*s typically include more disorders than the editions they

replace, with the jump between *DSM-II* and *DSM-III* being the largest. In the time between the original *DSM* (in 1952) and *DSM-IV* (in 1994), the number of disorders increased by more than 300% to a total of 368 distinct diagnoses covering an increasing scope of human behavior (Houts, 2002). The first two editions of the *DSM* were brief, spiral-bound books; the current *DSM* is 947 pages long.

What are the reasons for such an increase? On one hand, it is possible that in a relatively brief period of time, psychological science is accurately recognizing disorders that went unrecognized (or at least unlabeled) for centuries before, an explanation called "scientific discovery." On the other hand, it is also possible that psychology is making disorders out of some aspects of human experience that had previously been considered normal, an explanation called "social invention" (Houts, 2002). Ongoing debates have arisen regarding the relative truth of both of these explanations. Debates also continue regarding the expanding range of the *DSM*, in general, and the inclusion and exclusion of some specific disorders as well. Some of these debates call into question the extent to which factors other than empirical data drive the decision making of *DSM* authors, and the histories of certain disorders—some of which appear in the *DSM*, and some of which do not—have been offered as evidence (Caplan, 1995; Eriksen, 2005; Kutchins & Kirk, 1997). Box 2.1 illustrates some of the decisions represented in *DSM* revisions. Regardless of the outcomes or current status of diagnostic debates, recent *DSM* revisions represent important chapters of a long history of diagnostic labeling that characterizes clinical psychology.

BOX 2.1

Is It a *DSM* Disorder? Decisions to Include or Exclude Potential Disorders

Before each edition of the *DSM* is published, its authors oversee an extensive process during which they must, among other important tasks, decide whether or not to include certain experiences or sets of symptoms as official diagnoses. The implications of these classification decisions are quite significant for many people, including clients who may be assigned the diagnosis, mental health professionals who may treat them, health insurance companies who may pay for the treatment, and researchers who may investigate the issue.

Sometimes, *DSM* authors have decided to add an entirely new disorder to a revised edition. For example, borderline personality disorder, narcissistic personality disorder, and social phobia each appeared for the first time in *DSM-III* in 1980. At other times, *DSM* authors have, after serious consideration, decided not to include a proposed set of symptoms as an official disorder. For years prior to the publication of *DSM-IV, DSM* authors contemplated adding disorders tentatively named sadistic personality disorder and self-defeating personality disorder,

(Continued)

(Continued)

among others, but ultimately decided against it. On occasion, the *DSM* authors reverse a previous decision to include a disorder. Homosexuality was listed as a disorder in *DSM-I* and *DSM-II* but was excluded from *DSM-III* (and subsequent editions) after extensive controversy. *DSM-III* also omitted inadequate personality disorder and asthenic personality disorder, which were included in previous editions.

A section of the current edition of the *DSM* (*DSM*-5) lists numerous proposed disorders that were considered for inclusion but rejected by *DSM* authors. They are included as unofficial "proposed criteria sets" with the hope that additional research will inform future decisions to include or exclude them as official disorders. Included among these proposed disorders are the following (American Psychiatric Association, 2013):

- *Internet gaming disorder*, in which the person's Internet game-playing behavior causes clinically significant impairment or stress. The person is preoccupied with gaming, can't control their gaming behavior, experiences withdrawal when access is taken away, experiences relationship problems due to gaming, or has similar symptoms.
- *Attenuated psychosis syndrome,* in which the person experiences mild or brief delusions, hallucinations, or other psychotic phenomena. These are the same kinds of symptoms experienced by people with schizophrenia, but in attenuated psychosis syndrome, the symptoms would be more fleeting, less intense, and the person's perception of reality would remain largely intact. (*Attenuated* means reduced or lessened.)

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Photo 2.2
What distinguishes sadness, or other common but undesirable experiences, from mental illness?

- *Persistent complex bereavement,* in which the person experiences the death of a loved one and continues to be preoccupied with, yearn for, and feel intense sorrow about the person (among other symptoms) for over 12 months (or 6 months for children).
- *Nonsuicidal self-injury,* which can involve cutting, burning, or otherwise intentionally hurting one's own body without the intent to kill oneself, on 5 or more days in the last year. This behavior has long been recognized as a symptom of other disorders (especially borderline personality disorder), but with this proposed disorder, nonsuicidal self-injury would be a full-fledged disorder of its own.

There is no doubt that many individuals experience the phenomena described in these "proposed criteria sets." The question is whether these experiences should be categorized as forms of mental illness or understood to be part of the range of normal human experience. Whether these experiences will be classified as official disorders in a future edition of *DSM* depends, ultimately, on decisions made by the *DSM* authors.

Assessment of Intelligence

The emergence of the field of clinical psychology around the turn of the 20th century coincided with a dispute among psychology's pioneers about the nature of intelligence. **Edward Lee Thorndike** was among those who promoted the idea that each person possesses separate, independent intelligences, whereas Charles Spearman led a group of theorists who argued for the existence of "g," a general intelligence thought to overlap with many particular abilities (Reisman, 1991). The outcome of this dispute would profoundly influence how clinical psychologists assessed intellectual abilities, an activity that, more than psychotherapy or any other, characterized clinical psychology as a profession in the early years of the profession (L. T. Benjamin, 2007).

An important development in the history of intelligence testing arose in the early 1900s, when the French government sought help in determining which public school students should qualify for special services. In response to this request, **Alfred Binet** (along with Theodore Simon) created the first Binet-Simon scale in 1905 (Ferrand & Nicolas, 2015). This test yielded a single overall score, endorsing the concept of "g." It was the first to incorporate a comparison of mental age to chronological age as a measure of intelligence. This comparison, when expressed as a division problem, yielded the "intelligence quotient," or IQ. Binet's test grew in popularity and was eventually revised by Lewis Terman in 1937. Terman's revision was called the **Stanford-Binet Intelligence Scales**, the name by which the test is currently known (G. Goldstein, 2008; Reisman, 1991).

The standardization sample and the age range of test takers improved with each new version of Binet's test, but even after Terman revised it in 1937, it was still a child-focused measure of IQ. In 1939, David Wechsler filled the need for a test of intelligence designed specifically for adults with the publication of his **Wechsler-Bellevue** test. It quickly became popular among psychologists working with adults, and its more recent revisions remain popular today. Since its creation, Wechsler's adult intelligence scale has been revised and restandardized numerous times: the **Wechsler Adult Intelligence Scale (WAIS)** in 1955, the WAIS-R in 1981, the WAIS-III in 1997, and the WAIS-IV in 2008 (G. Goldstein, 2008; Reisman, 1991; Wasserman & Kaufman, 2015).

In 1949, Wechsler released a children's version of his intelligence test (a more direct competitor for the Stanford-Binet), which he called the **Wechsler Intelligence Scale for Children (WISC)**. The WISC distinguished itself from the Stanford-Binet by the inclusion of specific subtests as well as verbal and performance scales (in addition to overall IQ). The WISC has been revised and restandardized several times: the WISC-R in 1974, the WISC-III in 1991, the WISC-IV in 2003, and the WISC-V in 2014. In 1967, Wechsler added an intelligence test designed for very young children called the **Wechsler Preschool and Primary Scale of Intelligence (WPPSI)**. The WPPSI was revised in 1989 (WPPSI-R), 2002 (WPPSI-III), and 2012 (WPPSI-IV).

Many other measures of child and adult intelligence have appeared during the time of the Stanford-Binet and the Wechsler tests, but more than any others, these two have established themselves as standards in the field. They have also established themselves as competitors in the marketplace of psychological assessment, and recent revisions in one have at times represented responses to successful strides made by the other.

Assessment of Personality

The term *mental test* was first used by James McKeen Cattell in 1890 in an article titled "Mental Tests and Measurements" (Cattell, 1890). At that time, the term was used to refer to basic tests of abilities such as reaction time, memory, and sensation/perception. Soon, though, the term encompassed a wider range of measures, including not only the intelligence tests described above but also tests of personality characteristics (Butcher, 2010).

The first two decades of the 20th century witnessed some of the earliest attempts to measure personality attributes empirically, but few had significant impact. In 1921, however, **Hermann Rorschach** published a test that had significant impact for many years to come. Rorschach, a Swiss psychiatrist, released his now-famous set of 10 inkblots, which rose quickly in popularity (despite the fact that in the early years, several different competing Rorschach scoring systems existed). As a *projective personality test*, the *Rorschach Inkblot Method* was based on the assumption that people will "project" their personalities onto ambiguous or vague stimuli; hence, the way individuals perceive and make sense of the blots corresponds to the way they perceive and make sense of the world around them. Psychodynamic practitioners, who dominated during the early and mid-1900s, found such tests especially compatible with their clinical approach to clients.

The success of Rorschach's test was followed by a number of other projective techniques (Reisman, 1991). For example, **Christiana Morgan** and **Henry Murray** published the Thematic Apperception Test (TAT) in 1935. The TAT was similar to the Rorschach in that the test taker responded to cards featuring ambiguous stimuli. However, instead of inkblots, the TAT cards depicted people in scenes or situations that could be interpreted in a wide variety of ways. Instead of identifying objects in the card (as they might with Rorschach's inkblots), clients were asked to tell stories to go along with the interpersonal situations presented in the TAT cards. Again, their responses were thought to reflect personality characteristics. Other projective techniques that appeared in the aftermath of the Rorschach included the Draw-a-Person test, in which psychologists infer personality characteristics from clients' drawings of human figures, and Julian Rotter's Incomplete Sentence Blank (Rotter & Rafferty, 1950), in which psychologists assess personality by examining the ways clients finish sentence stems. Like the Rorschach, these and other projective personality tests have certainly enjoyed some degree of popularity, but their popularity has declined in recent decades as questions about their reliability and validity have accumulated.

Objective personality tests appeared soon after projectives, offering a very different (and, in many cases, more scientifically sound) method of assessing personality. Typically, these tests were pencil-and-paper instruments for which clients answered multiple-choice or true/false questions about themselves, their experiences, or their preferences. Scoring and interpretation were typically more straightforward than for projective tests. Some objective tests focused on specific aspects of personality, whereas others aimed to provide a more comprehensive overview of an individual's personality. The **Minnesota Multiphasic Personality Inventory (MMPI)**, written by **Starke Hathaway** and **J. C. McKinley**, is perhaps the best example of a comprehensive personality measure. When it was originally published in 1943, it consisted of 550 true/false statements. Test takers' patterns of responses were compared with those of groups in the standardization sample who represented many diagnostic categories. Not only could this test help a psychologist categorize a client through use of its clinical scales, it also used validity scales

to assess the test taker's approach to the test. In other words, the MMPI had a built-in system to detect random responding or intentionally misleading responses. The MMPI became very popular, and by 1959, there were more than 200 separate scales consisting of combinations of MMPI items (Reisman, 1991).

In 1989, the Minnesota Multiphasic Personality Inventory-2 (MMPI-2) was released. Its norms were more appropriate than those of the original MMPI, especially in terms of including minorities and individuals from various regions of the country in the standardization sample. It also eliminated some of the outdated or confusing language from the original test. An adolescent version of the test (the Minnesota Multiphasic Personality Inventory-Adolescent [MMPI-A]) followed in 1992. All versions of the MMPI have featured hallmarks of high-quality objective personality tests: easy administration and scoring, demonstrable reliability and validity, and clinical utility.

Other objective tests have come and gone but none with the lasting impact or research base of the MMPI. The NEO Personality Inventory (NEO-PI), for example, and its successors, the NEO-PI-R and NEO-PI-3, have risen to some degree of prominence in more recent decades as a personality measure less geared toward psychopathology than is the MMPI (Costa & McCrae, 1985, 1992; McCrae, Costa, & Martin, 2005). Rather than diagnostic categories, its scales are based on universal personality characteristics common to all individuals. Instruments measuring more specific states or traits have also appeared, including the Beck Depression Inventory (now in its second edition) and the Beck Anxiety Inventory (A. T. Beck & Steer, 1993; A. T. Beck, Steer, & Brown, 1996).

In recent decades, personality assessment tools have been used for an increasingly wider range of purposes, including job screenings and forensic purposes (e.g., child custody evaluations; Butcher, 2010). These uses have often generated significant controversy and highlight the importance of validity and reliability in such tests—topics we cover in more detail in Chapter 6.

PSYCHOTHERAPY

Psychotherapy is the primary activity of clinical psychologists today, but that hasn't always been the case. In fact, in 1930—more than a quarter century after Witmer founded the field—almost every clinical psychologist worked in academia (rather than as a practitioner), and it wasn't until the 1940s or 1950s that psychotherapy played a significant role in the history of clinical psychology (L. T. Benjamin, 2005; Farreras, 2016; Humphreys, 1996; McFall, 2006; Vasquez & Kelly, 2016; Wertheimer, 2000). In fact, in the first half of the 1900s, psychological testing was familiar territory for clinical psychologists, "but it was important that they knew their place in a field dominated by medical practitioners . . . a strategy for treatment, and treatment—those were in the job description of the physician, not the psychologist" (L. T. Benjamin, 2007, p. 163). Without the demand created by the psychological consequences of World War II on U.S. soldiers, psychotherapy might have remained an uncommon activity of clinical psychologists even longer (Bazar, 2015b, 2015c; L. T. Benjamin, 2007; Farreras, Routh, & Cautin, 2016). (See Box 2.2 for further exploration of the impact of war on the history of clinical psychology.)

In the mid-20th century, when psychotherapy rose to a more prominent place in clinical psychology, the **psychodynamic** approach to therapy dominated (Routh, 1996). With time, challengers to the psychodynamic approach emerged (Engel, 2008; Hollon & DiGiuseppe, 2011; Routh, 2011). In the 1950s and 1960s, for example, behaviorism surfaced as a fundamentally different approach to human beings and their behavioral or emotional problems. The **behavioral** approach emphasizes an empirical method, with problems and progress measured in observable, quantifiable terms. This emphasis was in part a reaction to the lack of empiricism evident in psychodynamic psychotherapy. **Humanistic** (or "client-centered") therapy also flourished in the 1960s, as Carl Rogers's relationship- and growth-oriented approach to therapy offered an alternative to both psychodynamic and behavioral approaches that many therapists and clients found attractive. The *family therapy* revolution took root in the 1950s, and as the 1960s and 1970s arrived, understanding mentally ill individuals as symptomatic of a flawed system had become a legitimate—and, by some clinicians, the preferred—therapeutic perspective.

Most recently, interest in **cognitive** therapy, with its emphasis on logical thinking as the foundation of psychological wellness, has intensified to the point that it has become the most popular singular orientation among clinical psychologists (excluding eclectic or integrative approaches; Engel, 2008; Norcross & Karpiak, 2012; O'Donohue, 2009). Apart from the rise of cognitive therapy, the most striking feature of the current therapy marketplace is the utter range of therapy approaches. To illustrate, modern graduate textbooks for psychotherapy courses typically include at least a dozen chapters on distinct approaches (e.g., Corey, 2009; Prochaska & Norcross, 2014), with each chapter representing a full spectrum of more specific variations.

In addition to the sequential rise of these therapy approaches, recent decades have witnessed a movement toward combining them, in either eclectic or integrative ways (Goldfried, Glass, & Arnkoff, 2011), as well as the tremendous influence of cultural competence on any and all such approaches (Comas-Díaz, 2011a, 2011b).

Many of these therapy approaches are covered in detail elsewhere in this book, but for now, the important point is that the plethora of therapy options currently available to clinical psychologists did not always exist. Rather, these methods have evolved over the history of the discipline, with each new therapy approach emerging from the context of—and often as a reaction against—the therapies that came before it.

BOX 2.2

The Influence of War on Clinical Psychology

It is difficult to overestimate the influence of war and its aftermath on the development of clinical psychology as a profession. Therapy, assessment, and training have all been shaped by the attempts of various governments and individuals to select soldiers and treat them after they have served their countries (R. R. Baker & Pickren, 2011; Bazar, 2015a, 2015c; L. T. Benjamin, 2007; Tryon, 2008). Numerous critical incidents in the history of clinical psychology can be directly tied to military factors:

- Robert Yerkes chaired the Committee on the Psychological

Examination of Recruits that created the Army Alpha and Beta intelligence tests during World War I. These tests, which were used to measure the intelligence of recruits and soldiers, are considered precursors to today's most widely used measures of intelligence (McGuire, 1994).

- David Wechsler's creation of the Wechsler-Bellevue, his first intelligence test, stemmed from his clinical experiences during World War I measuring intellectual capacities of military personnel. This test led to the WAIS, and ultimately the WISC and WPPSI, the revisions of which are currently the most widely used measures of adult and child intelligence in the United States (Boake, 2002; Reisman, 1991).
- In the aftermath of World War II, many U.S. veterans returned home with "shell shock," as it was called at the time, and other psychological effects of battle (L. T. Benjamin, 2005; Farreras et al., 2016; J. G. Miller, 1946). (*Posttraumatic stress disorder* later replaced *shell shock* as the accepted diagnostic label.) The U.S. government (specifically, the Veterans Administration—now the Department of Veterans Affairs) responded by requesting that the American Psychological Association formalize the training of clinical psychologists and provided significant funding to ensure the availability of such training opportunities. These efforts led to accreditation of clinical psychology doctoral training programs

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Photo 2.3
War has influenced the field of clinical psychology in numerous ways. For soldiers involved in direct combat, how might clinical psychologists be most helpful?

and ultimately to the scientist-practitioner (i.e., Boulder) model of training. This training model continues to dominate the field, and the strong relationship between Veterans Affairs and clinical psychology training, including a large number of internships at various levels, continues as well (R. R. Baker & Pickren, 2011; Humphreys, 1996; Kutchins & Kirk, 1997). By 2014, there were 4,700 psychologists (including both clinical psychologists and counseling psychologists) working within the Department of Veterans Affairs (Zeiss, Zeiss, & Carroll, 2016).

- The Nazi presence in Eastern Europe in the 1930s forced many influential figures in clinical psychology—most notably, Sigmund Freud and other Jewish psychodynamic leaders—to flee their home countries. This forced relocation facilitated the spread of their theories and clinical approaches to England and,

(Continued)

(Continued)

ultimately, to the United States (Reisman, 1991).

- Recent U.S. military events, including efforts in Iraq and Afghanistan, have illustrated the crucial role that clinical psychologists continue to play for soldiers and veterans (Lorber & Garcia, 2010; Maguen et al., 2010). In fact, Veterans Affairs is one of the country's largest providers of mental health services, with almost 1 million veterans receiving such services within a recent 1-year period (M. G. Hunt & Rosenheck, 2011).

Consider the most recent missions carried out by U.S. military personnel. How will the profession of clinical psychology be shaped by these activities? What needs will clinical psychologists be challenged to meet, and how do these needs compare to those stemming from early war-related activities? Besides military personnel themselves, who might benefit from the services of clinical psychologists in relation to these activities?

DEVELOPMENT OF THE PROFESSION

Just as clinical psychology's primary activities, such as psychotherapy and assessment, have evolved, the profession itself has progressed since its inception. Even in the earliest years, significant steps were evident. For example, in 1917, the American Association of Clinical Psychologists was founded, and in 1919 it transitioned into the Clinical Section of the American Psychological Association. In 1921, the Psychological Corporation was founded, foreshadowing the big business that was to become of psychological tests and measures of intelligence and personality.

In the 1940s, education and training in clinical psychology became more widespread and more standardized. The number of training sites increased dramatically, and the American Psychological Association began accrediting graduate programs that offered appropriate training experiences in therapy, assessment, and research. Veterans Affairs hospitals began their long-standing relationship with clinical psychology by funding graduate training and internships. And in 1949, the historic **Boulder conference** took place, at which training directors from around the country agreed that both practice and research were essential facets of PhD clinical psychology training (Cautin, 2011).

The 1950s produced more evidence that clinical psychology was a burgeoning profession. Therapy approaches proliferated, with new behavioral and humanistic/existential approaches rivaling established psychodynamic techniques. The extent to which psychotherapy did or didn't work also received increased attention, kick-started by Eysenck's (1952) critical analysis. The American Psychological Association also published the first edition of its ethical code in 1953, with significant discussion of clinical activities, reflecting a new level of professional establishment for clinical psychology (McFall, 2006; Tryon, 2008).

In the 1960s and 1970s, the profession of clinical psychology continued to diversify, successfully recruiting more females and minorities into the field. Clinical approaches continued to diversify as well, as behaviorism, humanism, and dozens of other approaches garnered large followings. The first PsyD programs appeared, offering graduate training options that emphasized applied clinical skills over research expertise. And signifying that psychotherapy was becoming a recognized part of American health care, insurance companies began to authorize payment for clinical psychologists' services just as they did for the services of many medical specialists.

In the 1980s, clinical psychologists enjoyed increased respect from the medical establishment as they gained hospital admitting privileges and Medicare payment privileges. Larger numbers of graduate training institutions continued to train larger numbers of new clinical psychologists, and the number of American Psychological Association members who were clinicians approached 50%. Psychotherapy burgeoned, especially in private practice settings, but the use of intelligence and personality testing decreased (Reisman, 1991). The growth of the profession continued through the 1990s and 2000s, as did the trend toward diversity in gender and ethnicity of those joining it (DeLeon, Kenkel, Garcia-Shelton, & VandenBos, 2011).

The size and scope of the field continues to grow, largely to meet the demand for psychotherapy services. In the late 1950s, only 14% of the U.S. population had ever received any kind of psychological treatment; by 2010, that number had climbed to 50%. Professional training options continue to multiply as well. Today's aspiring clinical psychologists have more choices than ever: the science/clinical balance of traditional PhD programs, PsyD programs emphasizing clinical skills, and more selected PhD programs that endorse the clinical scientist model of training and lean heavily toward the empirical side of the science/clinical continuum (Klonoff, 2011; McFall, 1991; Stricker, 2011). Numerous specializations, including forensic psychology and health psychology (illustrated by the inclusion of increasing numbers of clinical psychologists on primary health care teams), are flourishing (Goodheart & Rozensky, 2011). Empirical support for clinical techniques, prescription privileges, and new technologies (as described in Chapter 3) are among the other major professional developments in recent years.

To summarize the events described in this chapter, Box 2.3 presents a timeline of important events in the history of clinical psychology.

BOX 2.3

Timeline of Key Historical Events in Clinical Psychology

Year	Event
1796	William Tuke opens York Retreat in England
1801	Philippe Pinel publishes book on humane treatment of mentally ill (*Medico-Philosophical Treatise on Mental Alienation or Mania*)

(Continued)

(Continued)

Year	Event
1824	Eli Todd opens The Retreat in Hartford, Connecticut
1840	U.S. Census Bureau lists one category of mental disorder ("idiocy/insanity")
1841	Dorothea Dix encounters mentally ill in Boston prison, prompting extensive efforts for better treatment
1880	U.S. Census Bureau lists seven categories of mental disorders
1890	"Mental test" is used in print for the first time by Cattell
1892	Lightner Witmer earns his doctoral degree
1893	Emil Kraepelin proposes the early diagnostic category "dementia praecox"
1896	Lightner Witmer opens the first psychological clinic at the University of Pennsylvania
1905	Binet-Simon intelligence test is published in France
1907	Lightner Witmer founds the first professional journal of clinical psychology, *The Psychological Clinic*
1914	Psychology clinics proliferate; about 20 in operation
1916	Stanford-Binet Intelligence Test (as translated by Terman) is published in the United States
1917	American Association of Clinical Psychologists is founded
1921	Psychological Corporation is founded
1921	Rorschach inkblot technique published
1930s to 1950s	Psychoanalysis dominates psychotherapy
1935	Thematic Apperception Test published
1935	Psychology clinics proliferate further; more than 150 in operation
1939	Wechsler-Bellevue Intelligence Test published; first designed for adults
1943	Minnesota Multiphasic Personality Inventory (MMPI) published
1949	Wechsler Intelligence Test for Children (WISC) published
1949	Boulder conference held; yields scientist-practitioner training model
1950s to 1970s	Alternatives to psychoanalytic psychotherapy emerge (e.g., behaviorism, humanism, family/systems)

Year	Event
1952	Hans Eysenck publishes early, critical review of psychotherapy outcome
1952	*Diagnostic and Statistical Manual of Mental Disorders (DSM)* published
1953	American Psychological Association publishes first ethical code
1955	Wechsler Adult Intelligence Scale (WAIS) published
1967	Wechsler Preschool and Primary Scale of Intelligence (WPPSI) published
1968	*DSM-II* published
1973	Vail conference held; yields PsyD and practitioner-scholar training model
1974	WISC-R published
1980s to 2000s	Cognitive psychotherapy rises in prominence
1980	*DSM-III* published; specific diagnostic criteria and multiple axes appear
1981	WAIS-R published
1985	NEO Personality Inventory (NEO-PI) published
1987	*DSM-III-R* published
1989	MMPI-2 published
1989	WPPSI-R published
1991	Richard McFall publishes "manifesto"; yields "clinical scientist" training model
1991	WISC-III published
1992	MMPI-A published
1994	*DSM-IV* published
1997	WAIS-III published
1999	NEO-PI-R published
2000	*DSM-IV-R* published
2002	WPPSI-III published
2003	WISC-IV published
2008	WAIS-IV published
2008	MMPI-2-RF published
2013	*DSM-5* published

CHAPTER SUMMARY

The roots of clinical psychology can be traced to pioneering efforts in the late 1700s and 1800s by William Tuke, Philippe Pinel, Eli Todd, Dorothea Dix, and others to treat the mentally ill in a humane rather than punitive way. The field of clinical psychology was formally founded by Lightner Witmer, who founded the first psychological clinic at the University of Pennsylvania in 1896 and created the first professional journal devoted to clinical psychology in 1907. Early efforts to diagnose mental problems were quite rudimentary, but the work of Emil Kraepelin and others eventually led to more sophisticated diagnostic classification systems, culminating in the current edition of the *DSM* (*DSM-5*), which defines hundreds of disorders according to specific diagnostic criteria. The assessment of intelligence has evolved from the earliest work of Alfred Binet, David Wechsler, and others to the current editions of their tests, such as the Stanford-Binet and the WAIS, WISC, and WPPSI. Early attempts to assess personality were primarily projective tests, such as the Rorschach Inkblot Method and the TAT. Those tests were soon followed by objective personality tests such as the MMPI, many of which have achieved high levels of reliability and validity. Although psychotherapy is currently the dominant professional activity of clinical psychologists, it was relatively uncommon until the 1940s and 1950s. At that time, the psychodynamic approach to therapy prevailed, but behaviorism and humanism rose to popularity in the decades that followed. Currently, the cognitive approach is the most popular single-school therapy approach, and the number of distinct approaches to therapy continues to proliferate. As a profession, clinical psychology continues to evolve in many ways, including a diversification of its members and its graduate training options.

KEY TERMS AND NAMES

behavioral 38
Alfred Binet 35
Boulder conference 40
cognitive 38
dementia praecox 32
diagnostic criteria 32
Dorothea Dix 29
endogenous disorders 32
exogenous disorders 32
Starke Hathaway 36
humanistic 38
Emil Kraepelin 32
J. C. McKinley 36
Minnesota Multiphasic Personality Inventory (MMPI) 36
Christiana Morgan 36
multiaxial system 32
Henry Murray 36
neurosis 31
Philippe Pinel 28
psychodynamic 38
The Psychological Clinic 31
psychosis 31
The Retreat 29
Hermann Rorschach 36
Stanford-Binet Intelligence Scales 35
Edward Lee Thorndike 35
Eli Todd 29
William Tuke 28
Wechsler Adult Intelligence Scale (WAIS) 35
Wechsler Intelligence Scale for Children (WISC) 35
Wechsler Preschool and Primary Scale of Intelligence (WPPSI) 35
Wechsler-Bellevue 35
Lightner Witmer 30
York Retreat 28

CRITICAL THINKING QUESTIONS

1. How essential were the contributions of William Tuke, Philippe Pinel, Eli Todd, and Dorothea Dix to the creation of the field of clinical psychology? Would the field exist today without their work?
2. Psychotherapy was not even mentioned in Lightner Witmer's original definition of clinical psychology, but in recent decades, psychotherapy has been the most common activity of clinical psychologists. In your opinion, what factors might have contributed to the rise in prominence of psychotherapy?
3. In your opinion, why has the number of disorders defined by successive editions of the *DSM* continued to increase?
4. In your opinion, which of the current proposed disorders (as listed in Box 2.1) should be included as official disorders in the next edition of the *DSM?* On what do you base your opinion?
5. To what extent would you expect a graduate program's model of training (e.g., scientist-practitioner, practitioner-scholar, or clinical scientist) to influence the types of personality assessment tools (e.g., projective or objective) it trains its students to use?

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

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http://www.apa.org/pubs/journals/hop/

Journal of the History of the Behavioral Sciences

http://www.wiley.com/WileyCDA/WileyTitle/productCd-JHBS.html

Journal of Clinical Psychology

http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1097-4679

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3

CURRENT CONTROVERSIES AND DIRECTIONS IN CLINICAL PSYCHOLOGY

Prescription Privileges 47
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Chapter 2 illustrated that at any given point in the history of clinical psychology, the field is defined by a few topical issues, challenges, and controversies. The present is certainly no exception. The issues facing the field today have an undeniable influence on clinical psychologists and those with whom they work, and the way these issues are resolved will shape the field for decades to come.

PRESCRIPTION PRIVILEGES

Historically, the ability to prescribe medication has been one of the primary distinctions between psychiatrists and psychologists. In fact, in the eyes of the general public, it is a defining difference between the professions (Balon, Martini, & Singareddy,

Learning Objectives

3.1 Articulate the arguments for and against clinical psychologists gaining prescription privileges.

3.2 Discuss the merits and limitations of evidence-based practice/manualized therapy.

3.3 Debate the overexpansion of mental disorders attributed to the *DSM*.

3.4 Understand the influence of third-party payers on the provision of psychological services.

3.5 Describe how technology can be used in the delivery of psychological services.

2004). However, in recent years, some clinical psychologists have actively pursued **prescription privileges** (Burns, Rey, & Burns, 2008; DeLeon, Kenkel, Gray, & Sammons, 2011; Heiby, 2015; McGrath, 2010; Sammons, 2016; Tryon, 2008). The roots of the movement were established in the 1980s, but in the 1990s and 2000s, it rose to the level of a high-profile, high-stakes debate. The American Psychological Association published numerous articles endorsing prescription privileges (e.g., American Psychological Association, 1996a) and offering suggestions for training of psychologists to become proficient in the knowledge necessary to prescribe safely and effectively (American Psychological Association, 1996b). In addition, several outspoken and prominent individuals have also promoted the movement toward prescribing, including **Patrick H. DeLeon**, a former president of the American Psychological Association (DeLeon & Wiggins, 1996); **Morgan T. Sammons**, a widely recognized expert on psychopharmacology and 1 of 10 psychologists who took part in the first experimental pilot program of psychologists prescribing medication (Dittman, 2003); and **Robert McGrath**, training director of the Psychopharmacology Postdoctoral Training Program in the School of Psychology at Fairleigh Dickinson University and former president of the **American Society for the Advancement of Pharmacotherapy (American Psychological Association, Division 55)**.

The prescription privilege movement scored notable victories when two states—New Mexico and Louisiana—became the first to grant prescription privileges to appropriately trained psychologists in 2002 and 2004, respectively. Illinois followed in 2014, as did Iowa in 2016 and Idaho in 2017 (DeAngelis, 2017). Many other states have given serious consideration to similar legislation in recent years.

Web Link 3.1
Video of group advocating prescription privileges

Web Link 3.2
Prescription privileges in New Mexico

Other significant steps in the movement toward prescription privileges include the creation of the aforementioned APA Division 55 in 2000, as well as the psychopharmacology training programs available for psychologists in the U.S. military (Sammons, 2011). Nonetheless, the issue remains hotly debated, with significant numbers of clinical psychologists and worthy arguments on both sides (e.g., DeLeon, Dunivin, & Newman, 2002; Heiby, 2002; Robiner, Tumlin, & Tompkins, 2013; Sammons, 2016).

Why Clinical Psychologists Should Prescribe

- *Shortage of psychiatrists.* In some parts of the country, there simply aren't enough psychiatrists to serve the population adequately. Especially in some rural areas, there is a strikingly low ratio of professionals with the training and ability to prescribe psychoactive medications to the number of people who need them. In fact, when clinical psychologists successfully lobbied for prescription privileges in New Mexico and Louisiana, a cornerstone of their argument was the low number of psychiatrists per capita in many parts of these two states (Long, 2005). Underserved segments of society would benefit from a higher ratio of prescribers to patients.

- *Clinical psychologists are more expert than primary care physicians.* Although psychiatrists have specialized training in mental health issues, they aren't the only ones prescribing psychoactive medications. In fact, by some estimates, more than 80% of the prescriptions written for psychoactive medications come from primary care physicians (e.g., Cummings, 2007). When it comes to expertise in mental health problems, clinical psychologists' training is more extensive and specialized than physicians'; therefore, clinical psychologists could be better able to diagnose problems correctly and select effective medications.

- *Other nonphysician professionals already have prescription privileges.* Dentists, podiatrists, optometrists, and advanced practice nurses are among the professionals who are not physicians but have some rights to prescribe medication to their patients. Their success in this activity sets a precedent for specially trained clinical psychologists to do the same. Especially when we consider that general practitioner physicians—whose specific training in psychological issues is limited—currently prescribe a high proportion of psychoactive medication, it seems reasonable to allow clinical psychologists to use their specialized expertise for the purpose of prescribing.

- *Convenience for clients.* Antonio is a 9-year-old boy with attention-deficit/hyperactivity disorder (ADHD). Angela is a 38-year-old woman with major depression. Antonio and Angela could benefit from both nonpharmacological interventions (psychotherapy, counseling, etc.) and prescription medications. Without prescription privileges, clinical psychologists can provide the therapy, but they cannot provide the medication. The result is that Antonio (and his parents) and Angela will both need to be referred to a physician, such as a psychiatrist, to be evaluated for medication. Of course, this increases the time and money that these clients must spend on appointments. In addition, it requires the two busy mental health professionals to communicate consistently with each other about their shared clients, and a misunderstanding between the two could result in complications for the clients. With prescription privileges, Antonio and Angela could get both their therapy and their medication from the same source—the clinical psychologist. From the client's perspective, treatment is streamlined, saving both time and money. And the risk of problems due to miscommunication between professionals is eliminated.

- *Professional autonomy.* With prescription privileges, clinical psychologists can feel capable of independently providing a wider range of services to their clients. Without them, they may feel restricted in what they can accomplish for their clients. Of course, clinical psychologists should always strive to work collaboratively with other professionals involved in their clients' treatment. But at the same time, with prescription privileges, their ability to treat the physical and psychological aspects of their clients' difficulties autonomously, without relying on psychiatrists or other physicians, is greatly increased.

- *Professional identification.* In the eyes of the general public, psychologists may be difficult to distinguish from other nonprescribing therapists or counselors such as licensed professional counselors, social workers, and the like. The ability to prescribe immediately sets psychologists apart from—and, many would argue, above—these other professions.

- *Evolution of the profession.* Clinical psychology has undergone many significant changes in its brief history. It has incorporated many treatment techniques (e.g., therapy approaches) that were initially unfamiliar, and in the process, the profession has

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Photo 3.1
In your opinion, should clinical psychologists have prescription privileges? Why or why not?

thrived. Embracing prescription privileges is seen by many as the next logical step in the progression. To stand in its way, some argue, is to impede the evolution of the field (DeLeon et al., 2002). Prescriptive authority could open multiple doors to professional opportunity for clinical psychologists, from direct pharmaceutical treatment of clients to consultation with physicians about psychoactive medications for their patients (Burns et al., 2008).

- *Revenue for the profession.* The profession and its members stand to benefit financially from prescription privileges as well (Cummings, 2007). The potential for increased income as a result of prescription practices may offset some of the salary decreases reported by psychologists in recent decades, including those occurring as a result of the impact of managed care (as reported by M. J. Murphy, DeBernardo, & Shoemaker, 1998). In fact, strong opposition to the prescription privilege movement has emerged from psychiatrist organizations, whose members stand to lose business if psychologists gain the ability to prescribe. And the scope of this business should not be underestimated: During the 1990s, the percentage of people in treatment for depression who received an antidepressant drug doubled, from 37% to 74%, and in 2007, antidepressants were the most commonly prescribed category of drug in the United States (Sammons, 2011).

Why Clinical Psychologists Should Not Prescribe

- *Training issues.* What kind of education should clinical psychologists receive before they are licensed to prescribe? What should it cover? Who should teach them? When should it take place? (Early in graduate school? During the predoctoral internship? As specialized training after the doctoral degree?) All these questions complicate the pursuit of prescription privileges. Some have argued that for a comprehensive understanding of everything involved in a prescription decision, prescribers need something close to full-fledged medical school training. Only in this way could they appreciate the potential impact of a drug on the multiple systems of the body, the possibility of drug interactions, and all other medical factors (Griffiths, 2001; Robiner et al., 2002). Others have argued that with far less training, clinical psychologists could gain a basic competence in psychopharmacology (R. J. Resnick & Norcross, 2002; Sammons, Sexton, & Meredith, 1996). Most proposals fall somewhere between these two extremes, but the scope of pharmacology training for clinical psychologists is not entirely resolved. In fact, debates about current training standards are ongoing, with some arguing that pharmacologically trained psychologists are better trained in psychoactive medications than are the physicians and nurses who prescribe them (Muse & McGrath, 2010) and others strongly disagreeing, labeling psychologists' training substandard (Heiby, 2010). Some have even promoted the notion that clinical psychologists should receive some training in psychopharmacology even if they don't plan to prescribe, because without such training, they are unable to communicate effectively with medical professionals with whom

they share clients (Julien, 2011). Moreover, the pragmatics of pharmacological training remain uncertain. If such training were added to existing graduate programs, it might extend them by many semesters. (Currently, the only state with a plan for training at the predoctoral level is Illinois; the other states that have approved prescription privileges all require that the psychologist already be licensed before pursuing additional training [DeAngelis, 2017].) And many graduate programs in clinical psychology currently lack faculty with the expertise to teach these courses. It is noteworthy that only a small fraction of psychologists in New Mexico and Louisiana have actually obtained prescription privileges (as of 2015, the total was approximately 160)—a fact probably related to training requirements that extend for long periods of time (DeAngelis, 2017; Heiby, 2015; Linda & McGrath, 2017). A survey of medical colleagues of those prescribing psychologists found that they had extremely positive impressions. Specifically, the medical colleagues strongly believe that prescribing psychologists have adequate training, make prescription decisions safely, and know when to refer to other specialists (Linda & McGrath, 2017).

- *Threats to psychotherapy.* If clinical psychologists can prescribe, what will become of psychotherapy? Some have wondered if we will see a drift within the profession from "talk therapy" to pharmacological intervention. Clients may come to expect medication from clinical psychologists, and clinical psychologists may discover that prescribing is more profitable than therapy. The way psychologists understand and intervene with their clients may fundamentally shift from an appreciation of behavioral, cognitive, or emotional processes to symptom reduction via pharmacology. Some have pointed out that the profession of psychiatry has witnessed a drift of this sort and that clinical psychology could lose something of its essence if it does the same (McGrath, 2004; McGrath et al., 2004). As Cummings (2007) put it,

> Undoubtedly, the acquisition of prescription authority . . . would significantly expand the economic base of psychological practice. When that day comes, it remains to be seen, however, whether they abandon the hard work of psychotherapy for the expediency of the prescription pad. (p. 175)

- *Identity confusion.* Until all active psychologists prescribe (which, if it occurs at all, is certainly decades away), an identity crisis could emerge within the clinical psychology profession. Some clinical psychologists will prescribe, whereas others won't. Some may have been trained during graduate school; others may have returned for specialized training long after they gained their doctoral degrees. Without an effective effort to keep the public educated about our profession, a client referred to a particular clinical psychologist may feel justifiably puzzled about whether prescription medication might be part of the treatment program.

- *The potential influence of the pharmaceutical industry.* Reports of drug companies' attempts to increase profit—by offering gifts to prescribers, funding research, and controlling the publication of research results—have become widespread (e.g., Healy, 2004; C. Lane, 2007). Some opponents of the prescription movement have expressed concern that if psychologists prescribe, they will inevitably find themselves targeted by the pharmaceutical industry and will be pressured to consider factors other than client welfare when making prescription decisions. Some psychologists have gone so far as to call for

Web Link 3.3
Psychiatry turns to drug therapy

formal professional guidelines to address how psychologists should interact with drug companies (McGrath et al., 2004).

EVIDENCE-BASED PRACTICE/MANUALIZED THERAPY

Like the prescription privilege movement, the movement in favor of **evidence-based practice** and **manualized therapy** has intensified in recent decades, as has the reaction to it. In fact, "few topics in mental health are as incendiary, consequential, and timely as evidence-based practices" (Norcross, Beutler, & Levant, 2006, p. 3; see also Gaudiano & Miller, 2013; Ollendick & King, 2012; Truijens, Zühlke-van Hulzen, & Vanheule, 2019). Before we consider both sides of the argument, let's consider a brief history of the factors that have led to the debate.

For many decades, researchers in the field of psychotherapy have sought to answer questions about its benefits, or how well therapy "works." Hans Eysenck's (1952) bold statement that therapy had no proof of positive outcome sparked much research on the topic, and this early wave of research basically proved Eysenck wrong—psychotherapy was found to be quite efficacious (M. L. Smith & Glass, 1977; M. L. Smith, Glass, & Miller, 1980). However, much of this research was specific neither to any particular type of psychotherapy nor to any particular disorder; it simply supported the idea that therapy of any kind was more helpful than no therapy for people with any disorder. Subsequent steps in this line of research addressed more specific, targeted questions. Rather than asking broadly, "Does therapy work?" researchers began asking more specifically, "Which forms of therapy work best for which disorders?"

Take an insider's perspective on the methodological differences between the more general and more specific types of therapy outcome studies. Imagine that you are one of the therapists involved in an early study on psychotherapy efficacy, perhaps in the 1970s. You and the other therapists are each assigned a certain number of clients and are instructed simply to provide therapy to them. The instructions are no more specific or detailed than that. As a result, the therapy done in this study varies widely from one client to the next and from one therapist to the next. This lack of uniformity between therapies is not a problem, though, because the aim of the study is simply to compare those receiving therapy of any kind with similar individuals not receiving therapy (the control group).

But this lack of uniformity between therapies would be a major problem in the type of therapy outcome study more common in recent years, such as a more targeted study on the effects of a specific form of therapy. In fact, if you designed this type of study, you would want to make absolutely sure that the therapy was somewhat uniform across all clients, even if different therapists provided it. After all, defining the therapy in exact terms is a prerequisite to demonstrating how effective that particular type of therapy is and offering a description of it to others. So, rather than broadly instructing the study's therapists to practice therapy, or providing a vague description of the category of therapy to practice, you would provide much more detailed instructions. In short, you would provide a **therapy manual**. One purpose of this manual is to keep variability among

therapists to a minimum, such that if the technique proves effective, it can be shared with others in exact, unambiguous terms.

Since the 1980s, publications of this type of outcome research—how well a manualized therapy works on a particular disorder—have proliferated. As examples, researchers have tested exposure and response prevention, a specific form of behavior therapy, for the treatment of obsessive-compulsive disorder; dialectical behavior therapy (DBT) for the treatment of borderline personality disorder; and specific cognitive therapy techniques for the treatment of depression (Woody & Sanderson, 1998). In each of these cases, the specific therapy proved effective in the treatment of the target disorder, and the authors published their results in professional journals. As such published reports accumulated, Division 12 (Clinical Psychology) of the American Psychological Association created a task force to compile them into a list to serve as a reference for therapists who sought the most proven therapies for particular disorders. The therapies on this list were originally called "empirically validated" treatments, but the terminology soon changed to "empirically supported" treatments and eventually to "evidence-based practice" (e.g., Kazdin & Weisz, 2003; La Roche & Christopher, 2009). This change in terminology is important because the current term, *evidence-based practice*, incorporates not only the particular treatment itself but also factors related to the people providing and receiving it. More specifically, evidence-based practice is defined as "the integration of the best available research with clinical expertise in the context of patient characteristics, culture, and preferences" (APA Presidential Task Force on Evidence-Based Practice, 2006, p. 273). This three-part definition is often called a three-legged "stool," with each leg representing an essential part of what supports psychotherapy. So research (often in the form of randomized clinical trials of manualized therapies) is just one leg of the stool; the other legs focus on the clinician's knowledge and capability regarding the treatment and the qualities of the client (including culture and personal preferences) that could influence how effective the treatment will be (Barlow, Bullis, Comer, & Ametaj, 2013; Goldfried, 2013; Norcross, Freedheim, & VandenBos, 2011; Spring & Neville, 2011). Research on the leg involving the client's preferences shows that it is quite important (E. Murphy & Hankerson, 2018). A meta-analysis of 53 studies on client preference that considered therapy outcomes of over 16,000 clients found that preference accommodation was associated with fewer dropouts and more positive treatment outcome, regardless of what the specific preference was (Swift, Callahan, Cooper, & Parkin, 2018).

The first list of empirically validated treatments was published in 1995, and updates appeared in 1996 and 1998 (Chambless et al., 1996, 1998; Task Force on Promotion and Dissemination of Psychological Procedures, 1995). A landmark book titled *A Guide to Treatments That Work* was also published around this time, with more detailed information about these therapies (Nathan & Gorman, 1998). In the years that have followed, numerous books of this type, offering chapter-by-chapter guides to "what works" for which disorders, have emerged (e.g., Carr, 2008; Norcross, Hogan, Koocher, & Maggio, 2017; A. Roth & Fonagy, 2005). Numerous websites now provide continuously updated lists of treatments that work to both therapists and prospective clients. One example is psychologicaltreatments.org, a site maintained by Division 12 (Clinical Psychology) of the American Psychological Association, where visitors can search by disorder or treatment to find detailed information about research support that a particular treatment has for a particular disorder. Division 53 (Clinical Child and Adolescent Psychology) offers a

BOX 3.1

Metaphorically Speaking: Evidence-Based Treatment Manuals and Teaching Manuals

Think about your high school chemistry teacher. Specifically, think about the way that teacher taught chemistry—the general style or approach he or she had, the lectures, discussions, lab experiments, homework, exams, and so on. The material was basically the same for all of us, but we can be sure that our teachers' presentation styles varied widely.

Which of their methods works best? With so many different techniques in practice, it seems reasonable to wonder which particular way of teaching high school chemistry is most effective. To begin to answer this question empirically, we would need to define a few things up front. For example, we'd have to define the teaching techniques very specifically. We couldn't simply compare, say, a "lecture-based" style with a "discussion-based" style, or a "laid-back" approach with a "high-pressure" approach—these descriptions are too vague. Instead, we would need to instruct chemistry teachers *exactly* how to teach—in other words, give them a manual with detailed instructions—to determine exactly what works and what doesn't.

We'd also have to define up front the outcomes, or the results, of the teaching methods we're examining. From our empirical point of view, objective and measurable results would be best. Scores on standardized chemistry exams might work well, for example. We certainly would want to avoid more subjective outcome measures such as, say, students' level of interest in chemistry or the degree to which the course inspired a general passion for learning—these are simply too hard to measure reliably and validly.

Once we have our methods and outcome measures defined, we could run our "chemistry teaching experiment." If the results

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demonstrate that a particular approach to teaching high school chemistry was in fact effective (especially if these results were obtained in multiple studies), we could share our results with high school chemistry teachers around the world. They, in turn, might adopt our "proven" method rather than use their own personal or idiosyncratic methods, which lack any such empirical support. Eventually, it is possible that all high school chemistry teachers would teach chemistry the same way, using the one method that has the most empirical evidence.

Like teachers, psychologists all have their own unique methods when practicing psychotherapy. But which methods are best for which disorders? Hundreds of empirical studies by clinical psychology researchers have addressed this question, using an approach similar to the one we considered above for teaching high school chemistry. First, the researchers defined the therapy in specific terms in a therapy manual and instructed therapists to follow it closely. They also defined therapy outcome in measurable, objective terms. Then they tested their manualized therapy with clients diagnosed with a particular disorder—for example, depression.

When the results were positive, the researchers shared their results with other psychologists in journal articles, book chapters, or conference presentations. These psychologists, in turn, have the option of adopting this "proven" method with their own clients, rather than using their own personal or idiosyncratic methods, which lack any such empirical support. Eventually, it is possible that all psychologists would treat depressed clients the same way, using the one method that has the most empirical support.

Champions of evidence-based treatment manuals see this movement as a significant step forward for the science of clinical psychology. In fact, some have argued that psychologists should use only therapies that have demonstrated empirical support, and that to shun these approaches in favor of other approaches that lack empirical support is professionally irresponsible (e.g., Lilienfeld, Lynn, Ruscio, & Beyerstein, 2010; Lohr, Fowler, & Lilienfeld, 2002). On the other hand, some psychologists feel strongly that the push toward evidence-based treatments and policies that require their use overly restricts psychologists from customizing/personalizing treatments for particular individuals, or is based on flawed scientific methods (e.g., Rosenfeld, 2009; Silverman, 1996), or that the problems clients bring to therapists are too idiosyncratic and span too wide a range to ever be captured by a catalogue of treatment manuals (e.g., Zedlow, 2009).

Again, think about your high school chemistry teacher and the thousands of others who teach the same material. If a single, manualized method of teaching high school chemistry had more empirical support than any other, should all high school chemistry teachers use it? If your chemistry teacher had taught "by the manual" in this way, how would your experience as a chemistry student have changed? Those of us whose chemistry teachers were incompetent may have learned more, but those of us whose chemistry teachers were uniquely innovative and effective may have learned less. The same questions, certainly, apply to the issue of evidence-based practice/manualized psychotherapies.

similar site for treatments for kids: effectivechildtherapy.org. Additional sources of "what works" have come in the form of APA Practice Guidelines, an effort that began in 2010 and has thus far produced guidelines for specific clinical problems, including posttraumatic stress disorder (PTSD) and obesity/overweight in children, available to professionals and the public at apa.org/about/offices/directorates/guidelines/clinical-practice.aspx (DeAngelis, 2018).

Such lists of "therapies that work" have been celebrated by some as a major step forward for clinical psychology, legitimizing it as a scientific enterprise. Others have voiced concern or disapproval of the implications of evidence-based, manualized therapy.

Web Link 3.4
Evidence-based practice

Advantages of Evidence-Based Practice/Manualized Therapy

- *Scientific legitimacy.* Before the emergence of manualized therapies with empirical support, clinical psychology might have been described by empirically minded critics as a "cottage industry" of sorts. In other words, each psychologist provided

his or her own unique, "homespun" form of psychotherapy for a particular disorder. A depressed client who saw Dr. A might have received Dr. A's idiosyncratic brand of therapy, but the same client might receive an entirely different brand of therapy from Dr. B, with the differences in treatment stemming from differences in training or the psychologist's professional preference. Such inconsistency hardly seems appropriate for a discipline that calls itself a science. A more scientific, empirical approach to the treatment of depression, or any other disorder, requires that the discipline of clinical psychology determine a beneficial treatment for the disorder and that members of the discipline uniformly practice that treatment. We expect such standards of treatment from medical doctors treating physical diseases, and if clinical psychology subscribes to the medical model of diagnosis and treatment, the same expectation should apply to our field as well.

- *Establishing minimal levels of competence.* Inevitably, if all clinical psychologists are allowed to practice their own unique brands of therapy, a few of them will be ineffective or even harmful to clients. However, as manualized, evidence-based treatments are disseminated and used by clinical psychologists, their presence ensures that a potentially incompetent or detrimental therapist will be educated in treatments with demonstrated effectiveness. Moreover, as these treatments evolve into professional standards, psychologists will be obligated to follow them to some degree. As a result, the public will receive a more consistent, proven brand of treatment for each disorder, and psychologists can be held to a greater standard of accountability (Sanderson, 2003).

- *Training improvements.* Just as therapy in a "cottage industry" can vary widely among psychologists, the training that psychologists receive can also fluctuate widely among graduate programs. As the list of manualized, evidence-based therapies grows, they can be incorporated into graduate programs. In fact, when the American Psychological Association makes accreditation decisions about graduate training programs, it does consider criteria related to training in evidence-based treatments (Lyddon & Chatkoff, 2001). The result is that upcoming generations of psychologists will have been educated in the therapies that have empirical data supporting their success with particular disorders. Some programs may emphasize evidence-based treatments more than others (Hebert, 2002), but all programs can benefit to some extent from their inclusion (Hofmann, 2013).

- *Decreased reliance on clinical judgment.* Clinical judgment can be susceptible to bias and, as a result, quite flawed (Dawes, Faust, & Meehl, 1989; Garb, 2005; C. A. Webb & Keeley, 2015). When subjective, personal judgment of this type is applied to therapy decisions, the outcome of therapy can be compromised. Vrieze and Grove (2009) surveyed clinical psychologists about the way they make decisions in their practices and found that they did, in fact, rely much more often on their own clinical judgment than on mechanical or actuarial judgment based on empirical evidence. To the extent that manualized therapies lessen the reliance on clinical judgment and replace these components of therapy with evidence-based techniques, outcome may be enhanced (Lilienfeld et al., 2010).

Disadvantages of Evidence-Based Practice/Manualized Therapy

- *Threats to the psychotherapy relationship.* Although some of "what works" in therapy is attributable to specific techniques that therapists employ, a greater proportion of therapy's success is due to the quality of the therapeutic relationship (or "alliance") between the therapist and the client (Norcross & Wampold, 2011a; Rosenfeld, 2009; Wampold & Imel, 2015). Therapy manuals typically don't emphasize this relationship; instead, they tend to emphasize technique. In other words, they generally overlook "how" therapists relate to their clients in favor of "what" therapists do with (or to) their clients. A therapist who operates as a technician carrying out mechanical, predetermined methods can do a disservice to clients who seek a meaningful human connection (Sommers-Flanagan & Sommers-Flanagan, 2009). In fact, some psychologists argue that it's the relationship, not the technique, that should be evidence-based (Kazantzis, Cronin, Norton, Lai, & Hofmann, 2015). Along those lines, soon after proponents of the manualized treatment movement published their landmark book *A Guide to Treatments That Work* (Nathan & Gorman, 1998), another group of psychologists responded with contrasting viewpoints in a book fittingly titled *Psychotherapy Relationships That Work* (Norcross, 2002), insisting that the therapist–client relationship should not be neglected but should be recognized and studied as a focal point of what makes therapy work.

- *Diagnostic complications.* Each evidence-based treatment manual targets a particular disorder or issue. Such specificity is at the heart of testing specific treatments for specific disorders. In fact, when manualized treatments are tested in clinical trials, the clients who are allowed into the study are those with the target problem—for example, panic disorder—and without other complicating factors. In other words, researchers typically strive to test their manualized therapies on clients with "textbook cases" of a disorder. But that's not necessarily how clients present themselves in the real world of clinical psychology. In a private practice, a community mental health center, or a hospital, a client with panic disorder might also have something else—another anxiety disorder, a mood disorder, a personality disorder, or cognitive limitations, perhaps. This comorbidity means that the therapy that worked on clients with "clean" (i.e., uncomplicated) disorders in a treatment outcome study may not work as well on clients with more "messy" diagnostic features who commonly seek therapy from clinical psychologists in real-world settings (Angold, Costello, & Erkanli, 1999; Kessler, 1994; Zedlow, 2009).

- *Restrictions on practice.* To some, the evidence-based practice/manualized therapy movement has suggested that the only therapies worth practicing are those with empirical support (Lohr et al., 2002). In other words, if a therapy is not on the list of evidence-based treatments, it is unfounded and should be avoided. That is, "narrowly interpreted, evidence-based practice has become a rationale for mandating particular treatments" (Wampold, 2009, p. ix). In fact, some psychologists have used the term *malpractice* in reference to the act of using a therapy that lacks empirical support (Nathan & Gorman, 1998). In the "advantages" section above, a decreased reliance on personal, clinical judgment was praised, but if empirical support trumps personal decision making completely, the therapist's role may include little more than the routine application

of prescribed techniques. Such a job description may prove unsatisfying for many psychologists who seek autonomy and who rely on clinical creativity to customize their treatments, and it may not justify the extensive training they have obtained (Beutler, Kim, Davison, & Karno, 1996; Buscemi & Spring, 2015; Lebow, 2006; Rosenfeld, 2009). Also, because many treatment manuals emphasize brief (inexpensive) treatments, managed-care and health insurance companies can use empirical support to argue that psychologists should practice them exclusively, further limiting psychologists' autonomy (Seligman & Levant, 1998). Proponents of manualized therapy often respond that manuals get a "bad rap" of sorts:

> We believe that manuals are mischaracterized when they are described as rigid . . . specifying the components of the therapy does not have to deprive a therapy of its lifeblood. At best, it can help everyone involved come to understand what that lifeblood actually is. (Spokas, Rodebaugh, & Heimburg, 2008, p. 322)

Indeed, within the larger debate over manualized therapy, a debate exists regarding the extent to which strict adherence to manuals produces better outcomes (Asnaani, Gallagher, & Foa, 2018; Norcross, Beutler, & Levant, 2006; Stirman & Comer, 2018). Some studies have actually found that, compared with therapists who follow manuals rigidly, therapists who demonstrate flexibility while using manuals are more successful in terms of engaging clients in therapy and ultimately producing better outcomes (e.g., B. C. Chu & Kendall, 2009). The option of flexibility, as opposed to required, rigid adherence, also helps the clinicians themselves accept manualized therapies as a means of practice (Forehand, Dorsey, Jones, Long, & McMahon, 2010). The expression "flexibility within fidelity" is often used to describe this strategy of following a manual while adapting it as necessary (Kendall & Frank, 2018; Sanetti, Collier-Meek, & Fallon, 2016). Those adaptations may stem from the client's cultural background, personal preferences, or unique combination of symptoms or problems (Norcross & Wampold, 2018; Stirman, Gamarra, Bartlett, Calloway, & Gutner, 2017). J. F. Curry (2009) offered an intriguing metaphor for this balance between rigidity and flexibility regarding therapy manuals by referring to training in musical instruments: It requires the learning of standard techniques but also allows for (or even encourages) improvisation.

- *Debatable criteria for empirical evidence.* What should it take for a manualized therapy to make the "empirically supported" or "evidence-based" list? Some have argued that the current criteria are questionable, or that they do not account for failed trials of a treatment (Garfield, 1996; Tolin, McKay, Forman, Klonsky, & Thombs, 2015). Others have argued that the criteria are biased in favor of more empirically oriented therapies (e.g., behavioral and cognitive), while therapies that produce less easily quantifiable results (e.g., psychodynamic or humanistic) are shut out (Lebow, 2006). Behavioral and cognitive therapies do dominate the lists of evidence-based therapies, although other orientations are also represented to a more limited extent (Norcross, Beutler, et al., 2006). Some have attempted to address this issue by identifying treatments at the other end of the spectrum—those that are likely *not* to work, as indicated by surveys of psychotherapy experts (Norcross, Koocher, Fala, & Wexler, 2010; Norcross, Koocher, et al., 2006). For

example, such a survey of experts in child psychotherapy identified certain treatments, including past-life regression therapies, rebirthing therapies, and crystal healing, among the most discredited in the field (Koocher, McMann, Stout, & Norcross, 2015).

Despite the ongoing heated debate, evidence-based practice need not be an "either/or" issue. Messer (2004) argues, "As practitioners, we cannot manage without nomothetic *and* idiographic data, findings based on quantitative *and* qualitative method, and a mixture of scientific *and* humanistic outlooks, which are psychology's dual heritage" (p. 586).

OVEREXPANSION OF MENTAL DISORDERS

As we discussed in Chapter 2 (and will discuss further in Chapter 7), the size and scope of the *Diagnostic and Statistical Manual* (*DSM*) has vastly increased since its inception in the 1950s. Correspondingly, the number of people with mental disorders has climbed: Half of the U.S. population is diagnosable at some point in their lifetime, and 11% of the population is currently taking antidepressant medication (Paris, 2013a). This climb in mental disorder rates goes by many names: overdiagnosis, diagnostic expansion, diagnostic inflation, diagnostic creep, medicalization of everyday problems, false positives, and in severe cases, false epidemics (Frances, 2013a; Pierre, 2013).

The authors of the *DSM*, including the current *DSM-5* published in 2013, spend significant time and energy considering every proposed disorder. They review relevant research and solicit feedback from practicing professionals and the public before they decide to include any new disorder or change the criteria for an existing disorder. Among other concerns, they want to minimize the chances that people struggling with mental illness fall through the cracks, so they want to create a diagnostic manual that captures all of them. Obviously, they believe that any disorder that "makes the cut" to appear on the pages of the *DSM* truly deserves to be there. But their decisions have caused significant controversy among those who believe that many of today's disorders actually describe normal life experiences—unfortunate or unpleasant experiences, certainly, but nothing that warrants a label of mental disorder. This was particularly true with *DSM-5*, which introduced numerous new disorders and which changed the criteria for some existing disorders in such a way that they would include more people (to be covered in more detail in Chapter 7; Paris, 2013b). Among the criticisms about overdiagnosis that were offered about *DSM-5* were these:

- "There has been no real epidemic of mental illness, just a much looser definition of sickness, making it harder for people to be considered well. The people remain the same; the diagnostic labels have changed and are too elastic. Problems that used to be an expected and tolerated part of life are now diagnosed and treated as mental disorder." (Frances, 2013a, p. 82)
- "The danger of DSM-5 ideology is that it extends the scope of mental disorder to a point where almost anyone can be diagnosed with one." (Paris, 2013a, p. 41)

- "The more that psychiatric diagnoses appear to encroach on the boundaries of normal behavior, the more psychiatry opens itself to criticisms that there is no validity to the concept of mental disorders (e.g., there's no such thing as mental illness—it's a 'myth')." (Pierre, 2013, p. 109)

Concerns about the expanding definition of mental illness were around long before *DSM-5* was published (Dobbs, 2013; Frances, 2013b; Horwitz & Wakefield, 2007). For example, excessive shyness that interfered with a person's life was once considered an unfortunate personality characteristic, but since 1980, it has been included in the *DSM* as social anxiety disorder (C. Barber, 2008; Horwitz & Wakefield, 2012). But with each edition of the *DSM*, the scope of mental illness has expanded, along with the controversy surrounding it. In fact, in a 2014 survey of more than 500 therapists from eight countries around the world, over 60% of participants identified at least one disorder that should be completely removed from *DSM*, and the most common reasons for their choice were unclear boundaries between the disorder and either another disorder or normalcy (Robles et al., 2014).

New Disorders and New Definitions of Old Disorders

This expansion of the scope of mental disorders happens in at least two ways: introduction of new disorders to capture experiences once considered normal and "lowering the bar" for existing disorders such that more people meet the criteria. As examples of new disorders, critics might point to premenstrual dysphoric disorder (a more disabling version of the symptoms of premenstrual syndrome) or binge eating disorder (out-of-control overeating at least once per week), both of which appear for the first time in *DSM-5* and which have the potential to describe large numbers of people. As examples of "lowering the bar," critics might point to changing the age by which symptoms of ADHD must appear from 7 to 12, or changing the required frequency of binges in bulimia nervosa from twice per week to once per week, both of which are also *DSM-5* innovations.

Whether the risk of overdiagnosis comes from a new disorder or a new definition of an old disorder, the consequences can be very real. A diagnosis can help some people with problems get treatment they need, but a diagnosis can also help some people with problems get treatment they *don't* need—medications that have harmful side effects or unnecessary therapy that undermines a person's ability to use her or his own coping skills and could have been offered to someone else. A mental illness diagnosis can have many other effects as well: It can affect a person's self-image via the stigma that some people attach to mental illness ("I'm mentally ill") and subsequently the person's self-efficacy and overall wellness; it can influence how health insurance companies consider the person as a potential enrollee; and it can affect how a court of law views the person in terms of guilt regarding a crime or suitability for child custody (Caplan, 2012; Frances, 2013a, 2013c).

Of course, our discussion of the controversy surrounding overdiagnosis must include not only the diagnoses themselves but also the way they are used in the real world. In other words, it hardly matters what the *DSM* authors label as a mental disorder, or how

they define it, if those who diagnose and treat clients ignore such information. There is at least some truth to the notion that practicing clinicians make diagnoses without detailed consideration of the precise definition of a mental disorder, and in some cases, they offer treatments whether or not a diagnosis has been made at all (G. Greenberg, 2013; Paris, 2013b). For example, 72% of people who receive a prescription for an antidepressant medication do not receive any mental health diagnosis (Mojtabai & Olfson, 2011). Other surveys of mental health professionals indicate that many rely more on professional experience, intuition, and "gut feelings" than on symptom checklists to guide diagnostic decisions (Mishara & Schwartz, 2013). In these cases, overdiagnosis is as likely to stem from decisions made by the mental health professional who sees the client as from the mental health professionals who wrote the diagnostic manual.

BOX 3.2

In My Practice . . .

In my practice, I started working with Jonathan, a 40-year-old man, in June 2013. This was about a month after *DSM*-5 was released, and the mental health field was still abuzz about the changes that the new edition of the manual introduced. An especially hot topic among my colleagues was the addition of new disorders (as described in this chapter and in more detail in Chapter 7). In our initial clinical interview, Jonathan explained his primary reason for seeking help: He wanted to kick his "bad habit" of "occasional overeating." He gave me some details: Most nights he didn't eat at all after dinner, but "once in a while" he would "pig out" on ice cream. The situation was usually the same—his wife was asleep or out of town, and he would sit by himself in front of the TV with the carton of ice cream (not a bowl with a reasonable serving of ice cream, but the whole carton) and eat "way too much" of it, even though he "wasn't really hungry" and later "wished he hadn't."

As Jonathan spoke, I thought about binge eating disorder. It was one of the new disorders in *DSM*-5, so it was fresh in my mind. Several of his comments brought to mind specific criteria for the diagnosis: eating far more food than most others would eat in similar circumstances, doing so with at least some frequency, and doing so alone. Other diagnostic questions occurred to me: Did his behavior meet the minimum threshold of once per week? Was the duration long enough (i.e., had he been doing this for at least 3 months)? When he said he "wished he hadn't," did he mean that his stomach hurt, or that he was emotionally upset? How fast did he eat? How much control did he feel like he had over the eating?

I wondered what would happen if I asked Jonathan those questions and discovered that he met the diagnostic criteria for binge eating disorder. In particular, I wondered how he would respond to feedback from me that I was diagnosing him with binge eating disorder—actually, that I was diagnosing him with anything. After all, this was a man who was high functioning in many ways. He performed well in his profession, maintained a happy marriage, and had many friends. He also used the words "bad habit" to describe his problem, and I would essentially be telling him that his bad

(Continued)

(Continued)

habit was in fact a mental disorder. I don't think he arrived in my office with a view of himself as mentally ill. How would he interpret my conclusion that he was? Would he self-stigmatize? Would he find relief in the fact that the experts had a label for his problem? I've had other clients react in one of those directions or the other, and in some cases both at the same time.

Ultimately, after I asked Jonathan the questions listed above (and many more), I came to the conclusion that he didn't qualify for the binge eating disorder diagnosis. The behavior happened only once or twice per month, and he felt like he had some control over it (he had stopped himself numerous times). But if his symptoms were slightly more severe, Jonathan would have qualified for binge eating disorder, a diagnosis that didn't have its own category until just a month prior to his visit to my office.

To see this box come to life as a whiteboard video, please visit edge.sagepub/pomerantz5e.

The Influence of the Pharmaceutical Industry

Increasingly, the overexpansion of mental health diagnoses has been connected, at least by some experts in the field, to the possible influence of the pharmaceutical industry. To be sure, big drug companies have a stake in the way mental disorders are defined, and to them, the broader the better. The more disorders there are, and the more they overlap with the unfortunate experiences of normal life, the more potential customers these companies have to target their advertising toward (C. Barber, 2008; Frances, 2013a; Paris, 2013b; Pierre, 2013; Sadler, 2013; Whitaker & Cosgrove, 2015).

As such, pharmaceutical companies might be pleased if those who write the *DSM*—the prominent research psychiatrists and other professionals who decide what's a mental disorder and what's not—were on their payrolls. A series of recent studies by Lisa Cosgrove and her colleagues found that a majority of them are. Specifically, Cosgrove, Krimsky, Vijayaraghavan, and Schneider (2006) found that of the 170 panel members of *DSM-IV,* 95 of them, or 57%, had financial ties to the major pharmaceutical companies. Interestingly, in the Work Groups, or teams of DSM authors with expertise in particular areas, for mood disorders (e.g., major depressive disorder, bipolar disorder) and psychotic disorders (e.g., schizophrenia), disorders for which medication is extremely common, the percentage was 100%. The percentages were also high for other Work Groups in which medication is common, such as anxiety disorders (81%), eating disorders (83%), and childhood disorders (62%), but lower for Work Groups in which medication is uncommon, such as substance-related disorders (17%). The most frequent types of financial ties were research funding, consultant fees, and speaking fees.

In 2012, Cosgrove and Krimsky repeated their research for those working on *DSM-5*. The results indicated that the relationship between industry and authors remained. Overall, the percentage with financial ties to the pharmaceutical companies was 69%. In terms of specific Work Groups, psychotic disorders was 83%, mood disorders was 67%, anxiety disorders was 57%, eating disorders was 50%, and ADHD and disruptive

behavior disorders (which covered much of the same ground as childhood disorders in *DSM-IV*) was 78%. Cosgrove and Wheeler (2013) have since argued that the pharmaceutical industry is trying to "colonize" psychiatry—that is, attempting to control the mental health field, beginning with a deep connection between its core diagnostic manual and their financial interests.

Do financial connections with drug companies really influence mental health professionals? Couldn't the *DSM* authors make whatever decisions they thought were right, with no regard for any payment they may have received or may soon receive from a particular drug company? One study suggests that the power of such a financial arrangement can be remarkable. Carey and Harris (2008) studied the prescribing habits of psychiatrists in Minnesota and found that those who had received at least $5,000 from pharmaceutical companies had written three times as many prescriptions for antipsychotic medication as doctors who had received none. Of course, clinical psychologists are not going to be affected in a major way by prescription rates; as explained earlier in this chapter, only a small minority have the ability to prescribe at all at this point. But clinical psychologists (and most other mental health professionals) use the *DSM*, and the notion that its disorders were defined by people under the influence of financial relationships with pharmaceutical companies whose profits may depend on their decisions is troubling to many.

It is important to note that those in charge of *DSM-5* did place some limits on its authors' financial links to drug companies, in terms of the amount of companies' stock they could own, and the amount of payment they received from the companies; however, neither of these was required to be zero (Greenberg, 2013).

In Chapter 7, we will explore possible reasons for overexpansion of mental disorders and other issues involving the *DSM-5* in greater detail.

PAYMENT METHODS: THIRD-PARTY PAYMENT VERSUS SELF-PAYMENT

In the earliest days of psychotherapy, clients paid directly out of pocket. With time (and significant effort by professional psychologist organizations), health insurance companies increasingly recognized the worth of clinical psychologists' practices and included them in their coverage. So today, although some clients still pay for therapy on their own, many use their **health insurance/managed care** benefits to pay for therapy, at least partially. The presence of this **third-party payer** in the therapy relationship has numerous consequences (Reich & Kolbasovsky, 2006). Certainly, managed-care and insurance benefits bring therapy to many individuals who might not otherwise be able to pay for it. However, the companies who control these benefits are concerned about their financial bottom line as well as the health of their members, and at times, their priorities can strongly affect the work of clinical psychologists.

Effect on Therapy

It would be ideal if clients received the same treatment from psychologists regardless of how they pay, but a growing body of research suggests that this may not be true. Instead,

it appears that managed care exerts quite an influence on the day-to-day practices of clinical psychologists. According to a survey by M. J. Murphy et al. (1998), psychologists in private practice describe managed care as having a negative impact on their practices and, more specifically, on the quality of therapy they provide. Most of these psychologists reported that their practices were affected by managed care, and, in general, they portrayed managed-care companies as exercising too much control over clinical decisions. Furthermore, their responses indicated that the managed-care companies' emphasis on financial concerns often made it difficult for them to provide appropriate, ethical psychological services. Confidentiality was specifically noted as an ethical concern. Psychologists can control firsthand the confidentiality of their own private files but not the clinical information they have been required to share with the insurance company. Additional surveys of psychologists have similarly concluded that most psychologists are affected by managed care and that the influences on their practices are generally negative (Rothbaum, Bernstein, Haller, Phelps, & Kohout, 1998; Rothstein, Haller, & Bernstein, 2000). From the psychologists' point of view, the downside of working with managed-care companies can have many facets: lower pay than from clients who pay directly; taking time away from direct clinical work to spend on paperwork, phone calls, and other interaction with the managed-care company; denial of care that the psychologist believes is necessary; and numerous other frustrations (Reich & Kolbasovsky, 2006).

Should clients be informed about the negative influence of managed care? Put yourself in the client's role: If you were seeking therapy and were considering using your insurance benefits or possibly paying out of pocket, would you want to know how your method of payment might influence your treatment? Of course, psychologists are required by the ethical code (American Psychological Association, 2002) to inform therapy clients about the therapy process as early as possible in the process, but the guidelines are vague about exactly what information they should include. An empirical study on the topic indicated that when they learned about psychologists' negative reactions to managed care, many individuals thought more negatively about therapy than they had before. These individuals also felt entitled to have this information before beginning therapy (Pomerantz, 2000).

Of course, paying for psychotherapy without using insurance or managed-care benefits has its own drawbacks. First and foremost, many individuals would struggle to pay for therapy out of pocket, at least without reduced fees or services from low-cost community clinics. For a large segment of the population, self-pay is simply an unaffordable option (and the health insurance/managed-care option is what makes therapy attainable). For those who can afford it, self-pay therapy does allow the therapist and client to make important decisions—such as establishing the goals of therapy, agreeing on a treatment method, and determining when therapy should end—without the intervention of a third party with a financial interest.

Effect on Diagnosis

Although it should not, how clients pay for therapy not only influences therapy but also influences the diagnostic process. It is worth noting first that most health insurance and managed-care companies require a *Diagnostic and Statistical Manual of Mental Disorders* diagnosis for treatment (Ackley, 1997; Chambliss, 2000; Kutchins & Kirk, 1997).

Typically, they will not pay for the treatment of issues that a client brings to therapy if those issues do not qualify for a diagnosis. Thus, clients whose symptoms are not severe enough to be diagnosable may find that treatment will not be covered.

A few recent surveys have suggested that psychologists diagnose clients differently depending on how the clients pay for therapy. When psychologists considered clients with mild symptoms of depression or anxiety, they were much more likely to assign a diagnosis when these clients paid via managed care rather than out of pocket. The specific choice of diagnosis depended somewhat on payment method as well (Kielbasa, Pomerantz, Krohn, & Sullivan, 2004). Additional studies have suggested that for a wider range of problems, including symptoms of inattention/hyperactivity and social phobia (Lowe, Pomerantz, & Pettibone, 2007) or for symptoms that are clearly below diagnosable levels (Pomerantz & Segrist, 2006), psychologists' diagnostic decisions depend on whether the client or the client's insurance company pays for therapy.

THE INFLUENCE OF TECHNOLOGY: TELEPSYCHOLOGY AND MORE

Like many other professions, especially in the health care field, clinical psychology has been significantly affected by technological advances in recent years (Barnett, 2018; A. Z. Barrera, Stanley, & Kelman, 2017; L. F. Campbell & Norcross, 2018; Dimeff, Paves, Skutch, & Woodcock, 2011; Kraus, Zack, & Stricker, 2004; Marks & Cavanagh, 2009). Particularly groundbreaking—and controversial—is the use of technological tools in the direct delivery of psychological services. Contemporary clinical psychologists can perform assessments and treatments via computer or smartphone as a supplement to, or instead of, traditional in-person meetings with clients (Eonta et al., 2011). This use of technology, and particularly the Internet, by clinical psychologists often goes by the name **telepsychology** but is also called telehealth, telemental health, and cybertherapy, among other labels (Luxton, Nelson, & Maheu., 2016; Mohr, 2009; Yuen, Goetter, Herbert, & Forman, 2012). Telepsychology and other recent applications of technology in clinical psychology and related professions have generated both enthusiasm and skepticism (e.g., K. D. Baker & Ray, 2011; Glueckauf et al., 2018; Perle et al., 2013; Tao, 2014). Let's explore various facets of this issue, including specific uses of technology, how well it works, and professional issues surrounding it.

Applications of Technology in Clinical Psychology

Today, technologically savvy clinical psychologists can use

- videoconferencing (similar to Skype or FaceTime) to interview or treat a client;
- e-mail or text (in chat-room or one-on-one formats) to provide psychotherapy to a client;
- interactive Internet sites to educate the public by responding to questions about mental health concerns;

- online psychotherapy programs to diagnose and treat specific conditions;
- virtual-reality techniques in which clients undergo therapeutic experiences, such as virtual exposure to feared objects;
- computer-based self-instructional programs designed as specific components of a treatment that is otherwise provided face to face; and
- apps and biofeedback sensors on handheld devices (e.g., smartphones, tablets) to monitor clients and interact with them on a regular or random basis between meetings with the psychologist (as described in Campbell, Millán, & Martin, 2018; Dimeff et al., 2011; Hsiung, 2002; J. H. L. Lui, Marcus, & Barry, 2017; Luxton et al., 2016; Marks & Cavanagh, 2009; Nitzburg & Farber, 2019; Riper & Cuijpers, 2016; Yuen et al., 2012).

Web Link 3.5
Psychotherapy via videoconference

The applications of emerging technologies represent an exciting new horizon to many clinical psychologists. They promise to provide services to populations that have been underserved by clinical psychologists, including people living in poverty, in rural areas, or in war-torn/violent regions (A. Z. Barrera et al., 2017; Duncan, Velasquez, & Nelson, 2014; Gale, 2015; Kraus, 2004; Luxton et al., 2016; E. L. Nelson & Bui, 2010; Reger & Gahm, 2009). Maheu, Pulier, Wilhelm, McMenamin, and Brown-Connolly (2005) summarize the benefits of these technologies as "accessibility, affordability, anonymity, acceptability, and adaptability" (p. 10).

How Well Do Telepsychology and Other Applications of Technology Work?

Because telepsychology and other applications of technology are recent developments, the amount of data on their benefits to clients is still growing, and the methodology used to collect the data may be imperfect to some extent (Richardson, Frueh, Grubaugh, Egede, & Elhai, 2009). But the data so far are promising: Reviews and meta-analyses suggest that telepsychology works as well as more traditional, face-to-face forms of psychotherapy for a variety of disorders (Andersson, 2015a, 2015b; Andersson, Cuijpers, Carlbring, Riper, & Hedman, 2014; Barak, Hen, Boniel-Nissim, & Shapira, 2008; Ebert et al., 2015; Emmelkamp, 2011; Kraus, 2011; Pennant et al., 2015; D. Richards & Richardson, 2012; D. Richards & Viganó, 2013; Turner & Casey, 2014). Some studies have reached this conclusion regarding particular treatments for particular disorders. Consider this sampling of outcome studies of therapies utilizing technology: Several studies have found that cognitive-behavioral treatment of anxiety disorders was equally effective whether the treatment was delivered in person or via computer (Dear et al., 2015; Reger & Gahm, 2009; Spence et al., 2011; Titov et al., 2015). A meta-analysis of Internet-based therapy for PTSD produced similarly favorable results (Kuester, Niemeyer, & Knaevelsrud, 2016). The use of a smartphone app designed to deliver attention-bias modification (a treatment described in more detail in Chapter 15) significantly reduced stress and anxiety in highly anxious people (Dennis & O'Toole, 2014). Adding numerous technological components—like videoconferencing, Internet-delivered treatment elements,

and texting between therapist and client—enhanced the effectiveness of cognitive-behavioral treatment of binge eating disorder and bulimia (Shingleton et al., 2016; Shingleton, Richards, & Thompson-Brenner, 2013). An Internet-based intervention designed to increase optimism reduced depression in people high in pessimism (Sergeant & Mongrain, 2014). Internet-guided programs designed to improve mood were effective in decreasing depression symptoms (Eells, Barrett, Wright, & Thase, 2014). A review and meta-analysis of videoconference therapy found that although the working alliance (essentially, the quality of the client–therapist relationship) was slightly weaker in videoconference therapy than in-person therapy, the reduction of symptoms was about equal (Norwood, Moghaddam, Malins, & Sabin-Farrell, 2018).

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Photo 3.2 Increasingly, clinical psychologists are using the Internet and other forms of technology to conduct clinical activities. In your opinion, what are the potential advantages and disadvantages of this increase?

Emerging Professional Issues

As clinical psychologists embrace new technologies and incorporate them into their practices, professional issues—including both the ethical and the practical—continue to arise (Koocher, 2009). In response, the profession is making efforts to ensure that the services provided are safe, effective, and ethical (D. C. Baker & Bufka, 2011). The American Psychological Association (2002) ethical code includes several updates designed to address the increasing use of new technological tools. Some experts in the field have also proposed ethical guidelines for psychologists practicing online, via videoconferencing, or through other similar means (e.g., Barnett & Kolmes, 2016; Hsiung, 2002; Ragusea & VandeCreek, 2003).

Like the technologies themselves, guidelines regarding their use are continuously in the works. Some of these guidelines are simply extensions of the guidelines that govern traditional, face-to-face practice, whereas others are quite distinct. As the technology-based practice of clinical psychology evolves, those providing it should follow some fundamental suggestions (adapted from Barnett & Scheetz, 2003; Barnwell & Larsen, 2018; Barnwell, McCann, & McCutcheon, 2018; DeMers, Harris, & Baker, 2018; Larsen & Juntunen, 2018; Larson, 2018; Lustgarten & Elhai, 2018; Luxton et al., 2016; Novotney, 2018; Poole & Crow, 2018; Ragusea, 2012; Rummell & Joyce, 2010):

- Obtain informed consent from clients about the services they may receive, the technologies that may be used to provide them, and the confidentiality of the communication.
- Know and follow any applicable laws and guidelines on telehealth and telemedicine. This includes laws and guidelines related to practicing across state lines—for example, when either the client or the therapist has traveled outside the state in which the therapist is licensed. Different states have different rules regarding practice in such situations.

- Know and follow the most recent version of the APA ethical code, especially the portions that address technological issues. Similarly, be familiar with the APA Guidelines for the Practice of Telepsychology (APA, 2013).
- Ensure client confidentiality as much as possible by using encryption or similar methods. Keep updated on ways clinical information could be accessed by hackers and techniques for stopping them. Avoid public WiFi hotspots, and consider using a VPN.
- Appreciate how issues of diversity and culture may be involved. As technological tools replace face-to-face meetings, psychologists may need to make special efforts to assess the cultural backgrounds of the clients they serve.
- Do not practice outside the scope of your expertise. Merely having a license may not be enough. Advanced training—either clinical or technological—may be necessary to use a particular technique.
- Be knowledgeable about emergency resources in any community from which your clients may seek services. A crisis related to suicide or psychosis, for example, may require an immediate face-to-face intervention that the psychologist is simply too far away to provide.
- Choose technologies carefully. For example, when choosing an app for clients to use on their smartphones, consider whether the app was developed and tested by qualified mental health professionals, how widely available the app is (on different types of operating systems), how long the app has been available (often a sign of its stability), and the extent to which you will be able to troubleshoot if a client asks you questions about its use.
- Be careful about how you dispose of data and technologies you use, including old phones, tablets, computers, and passwords. Similarly, make sure all of your devices are password-protected and guard against their theft or loss.
- Stay abreast of changes to the laws, ethical standards, guidelines, and technology relevant to your practice.

As this list suggests, numerous specific ethical issues have arisen as the use of technology in clinical psychology has expanded (C. B. Fisher & Fried, 2003; Koocher, 2009; Kraus, 2004; Naglieri et al., 2004; Rummell & Joyce, 2010). These issues begin even before the assessment or therapy does, as the very identity of the client may be questionable. That is, how can the psychologist be sure that the client is in fact the person agreeing to the informed consent statement, responding to assessment items, or providing comments during online therapy? Even if identity is confirmed, the psychologist must be concerned with maintaining confidentiality across electronic transmission, making appropriate interpretations in the absence of ability to observe nonverbal cues that would be present in a face-to-face meeting, and remaining competent regarding not only clinical but technical skills.

In spite of these ethical and pragmatic hurdles and a very short history, evidence is beginning to indicate that telepsychology can be quite successful, and it is certainly

becoming more widespread. However, it is important to remember that the effectiveness of a telepsychology treatment can depend on a number of factors (adapted from Marks & Cavanagh, 2009):

- Which telepsychology is being used for which disorder?
- On what device and via what means is the telepsychology delivered—computer, smartphone, e-mail, text, videoconference, or something else?
- In what setting is the telepsychology being delivered—home, clinic, school, public setting (library, café, etc.), or somewhere else?
- How did clients find the telepsychology? Did they receive a specific referral from a knowledgeable source or stumble across it on the Internet?
- Does the technology have live human support, and is that support monitored for quality?

Although many therapists remain wary of telepsychology and do not utilize it at all (McMinn, Bearse, Heyne, Smithberger, & Erb, 2011), the practice of technologically based therapy and diagnosis is undeniably on the rise and will probably remain so as technology improves and becomes more widespread. As telepsychology progresses, the current body of research on its effectiveness will certainly grow as well (Maheu et al., 2005), and the guidelines and training for its use will strengthen. Among the more interesting questions to be addressed will be the effect of technological tools on the client–therapist alliance (Rummell & Joyce, 2010). As we will see in later chapters, a strong professional relationship between client and therapist is crucial to any successful therapy. To the extent that therapy via e-mail, text, website, or videoconference can maintain or enhance that relationship, it stands to benefit the client.

CHAPTER SUMMARY

Several current issues and controversies dominate the contemporary field of clinical psychology. As a primary example, the issue of prescription privileges for clinical psychologists has emerged as a significant development in the profession. Those in favor of prescription privileges cite numerous justifications, including the shortage of psychiatrists in many geographic areas, other nonphysician professions that have obtained prescription privileges, convenience for clients, and autonomy for clinical psychologists. Opponents of prescription privileges cite difficulties related to training, medication as a threat to the practice of psychotherapy, identity confusion with the profession of clinical psychology, and the potentially negative influence of the pharmaceutical industry. The issue of evidence-based practice/manualized therapy has also stirred debate in recent years. Proponents point out that these treatments enhance the scientific legitimacy of psychotherapy, help establish standards of competence, and decrease reliance on therapists' clinical judgment. Those who oppose evidence-based practice/manualized therapy emphasize that these

therapies can threaten the therapeutic relationship, can overly restrict practice options, and may be based on empirical standards or specifically selected client populations that limit applicability to the broad field of psychotherapy. Overdiagnosis—also known as diagnostic inflation or the medicalization of everyday problems—has stirred significant controversy with each new edition of the *DSM*, especially *DSM-5*. Although the *DSM* authors follow a thorough process when deciding whether to label an experience as a mental disorder, critics argue that too often, they have included experiences that are unfortunate parts of normal life or have used criteria that set the bar so low that too many people qualify for the disorder. Consequences of an overly broad or loose definition of mental illness can include unnecessary treatment, stigma, and legal implications. How a client pays for psychotherapy—specifically, whether a managed-care/health insurance company pays the bill or the client pays independently—appears to have an impact on clinical psychologists and their work with clients. Especially in the private practice field, psychologists report that their practices have been negatively affected by the increased involvement of managed-care companies. Evidence suggests that psychologists' diagnostic decisions about clients can be significantly influenced by the presence or absence of a health insurance/managed-care company paying the bill. Recent technological innovations have expanded the range of tools that clinical psychologists can use to treat clients. Telepsychology of various kinds, including videoconferencing, online therapies, e-mail- or text-based therapeutic interventions, virtual-reality technology, and other forms of computer-aided treatment have all been successfully used. Initial outcome data are promising, but outcome may depend on a wide variety of factors related to the intervention, the delivery mode, and the setting in which it is used. Along with these new technologies have come corresponding ethical and professional issues, including informed consent, laws and ethics, confidentiality, and effectively managing client emergencies from a distance.

KEY TERMS AND NAMES

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evidence-based practice 52
health insurance/managed care 63
manualized therapy 52
Robert McGrath 48
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telepsychology 65
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CRITICAL THINKING QUESTIONS

1. In your opinion, should clinical psychologists have prescription privileges? Why or why not?
2. In your opinion, is mental illness currently overdiagnosed?
3. In your opinion, to what extent should the use of evidence-based practice be required for clinical psychologists?
4. In your opinion, how much should clinical psychologists tell new clients about the impact that payment method (e.g., managed care/health insurance vs. out of pocket) might have on their diagnosis or treatment?
5. In your opinion, what are the most important advantages and disadvantages of recent forms of technology on the practice of clinical psychology?

LOOKING TOWARD GRADUATE PROGRAMS

The following table is a list of index entries in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018) relevant to the topics in this chapter (see page 25 in Chapter 1 for more information about the *Insider's Guide*). Keep in mind a few things as you consider these listings: They are current as of the 2018/2019 edition of the *Insider's Guide* (which typically revises every two years); they may have changed as faculty members retire, take new jobs, or develop new interests; and a graduate program's own website may have more extensive or up-to-date information.

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Evidence-Based/Empirically Supported Treatments	Evidence-Based/Empirically Supported Treatments	Miscellaneous/Other—Evidence-Based Practice
Psychopharmacology	Technology	Miscellaneous/Other—Telepsychology
Technology and Practice		

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

Clinical Psychological Science

http://www.psychologicalscience.org/index.php/publications/journals/clinical

Clinical Psychology: Science and Practice

http://onlinelibrary.wiley.com/journal/10.1111/(ISSN)1468-2850

Evidence-Based Mental Health

http://ebmh.bmj.com/

Journal of Technology in Human Services

http://www.tandfonline.com/loi/wths20#.V08Y3vkrLX5

Professional Psychology: Research and Practice

http://www.apa.org/pubs/journals/pro/index.aspx

Psychotherapy

http://www.apa.org/pubs/journals/pst/index.aspx

Psychotherapy Research

http://www.tandfonline.com/loi/tpsr20#.V08V2fkrLX4

The Tablet (Newsletter of the American Society for the Advancement of Pharmacotherapy, APA Division 55)

http://www.apadivisions.org/division-55/publications/tablet/index.aspx

STUDENT STUDY SITE RESOURCES

SAGE edge™

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- Self-quizzes
- eFlashcards
- Culture expert interviews
- In My Practice demonstration video
- Full-text SAGE journal articles
- Additional web resources
- Mock assessment data

4

DIVERSITY AND CULTURAL ISSUES IN CLINICAL PSYCHOLOGY

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THE RISE OF MULTICULTURALISM IN CLINICAL PSYCHOLOGY

The Diversification of the U.S. Population

Cultural diversity has historically been a hallmark of the U.S. population, but in recent years, the country has become *much* more diverse. The number of people of minority ethnicities, as well as the proportion of the U.S. population they represent, has increased dramatically. For example, in a single decade (1990–2000), the Asian American/Pacific Islander population and the Latino/Latina/Hispanic population each grew by about 50% (U.S. Census Bureau, 2001). As of 2016, there were over 43 million immigrants in the United States, representing over 13% of the population (U.S. Census Bureau, 2017).

Learning Objectives

4.1 Explain what it means to describe multiculturalism as the "fourth force" in clinical psychology.

4.2 List professional efforts in clinical psychology that demonstrate the field's current focus on multiculturalism.

4.3 Speculate how cultural competence could be demonstrated with psychotherapy clients of various backgrounds.

4.4 Compare major perspectives on the similarities and differences among people.

4.5 Define "culture" in the context of clinical psychology.

4.6 Describe methods for training clinical psychologists in cultural issues.

One in five U.S. schoolchildren speaks a language other than English at home (Roberts, 2004). And by 2050, about half the country's population will identify at least partially as African American, American Indian, Asian American, or Latino/Latina (U.S. Census Bureau, 2008).

In certain parts of the United States, the increasing diversity is especially pronounced. In Miami, for example, over two thirds of the population is Latino/Latina/Hispanic (U.S. Census Bureau, 2018b). In San Francisco, individuals of Asian descent represent about one third of the population (U.S. Census Bureau, 2018c). And about three quarters of the population of Detroit and half the population of Washington, D.C., are African American (U.S. Census Bureau, 2018a, 2018d).

Clinical psychologists have recognized that the people who might seek their professional services represent a growing variety of cultural backgrounds. As individuals and as a profession, clinical psychologists are making efforts to address issues of culture sensitively and competently (Comas-Díaz, 2011a, 2011b, 2012; M. L. Melton, 2018; Suzuki & Wilton, 2016). As stated by McGoldrick, Giordano, and Garcia-Preto (2005b), "We must incorporate cultural acknowledgment into our theories and into our therapies, so that clients not of the dominant culture will not have to feel lost, displaced, or mystified" (p. 4).

Multiculturalism as the "Fourth Force"

The impact of issues of culture and diversity on mental health professionals in recent years has been so extensive that some authors have identified **multiculturalism** as a defining issue of the current era of psychology. For example, Pedersen (1990, 1999, 2008) has put forth the argument that in the evolution of the clinical/counseling field, multiculturalism represents the "fourth force." With this label, multiculturalism takes its place with the three previous movements that have been broadly recognized as dominant paradigms in their respective eras: psychoanalysis as the first force, behaviorism as the second force, and humanism/person-centered psychology as the third force (Bugental, 1964). Multiculturalism, then, stands as a major pervasive influence on the work of contemporary clinical psychologists (Comas-Díaz & Brown, 2016; Gelso, 2011). It represents a fundamental change of emphasis but one unlike the previous three in terms of its method of impact. Whereas behaviorism and humanism emerged as challenges to the incumbent first force of psychoanalysis, multiculturalism does not necessarily aim to dethrone any of the first three forces. Instead, it enhances and strengthens existing models by infusing them with sensitivity and awareness of how they can be best applied to individuals of various cultural backgrounds (Mio, Barker-Hackett, & Tumambing, 2006).

Web Link 4.1
Cultural diversity in the United States

One reason cultural diversity is such a powerful force in the clinical and counseling fields is that it shapes the way the client understands the very problem for which he or she is seeking help. This understanding—this worldview, applied to psychological problems—is what the therapist should appreciate as he or she devises an approach to helping the client. Comas-Díaz (2011a) encourages therapists to directly assess clients' understanding of their own psychological problem by asking them these questions:

- What do you call your problem (or illness or distress)?
- What do you think your problem does to you?
- What do you think the natural cause of your problem is?
- Why do you think this problem has occurred?
- How do you think this problem should be treated?
- How do you want me to help you?
- Who else (e.g., family, friends, religious leader) do you turn to for help?
- Who (e.g., family, friends, religious leader) should be involved in decision making about this problem? (Adapted from p. 875)

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Photo 4.1 Cultural diversity is a hallmark of the U.S. population, especially in certain geographical areas. How has the profession of clinical psychology emphasized the importance of cultural diversity in recent years?

Recent Professional Efforts to Emphasize Issues of Diversity and Culture

Clinical psychology and related professions have addressed the issue of cultural diversity in many tangible ways. In the 1970s, efforts toward educating therapists and therapists-in-training on the importance of race and ethnicity were in their early stages. These efforts expanded through the 1980s and by the 1990s were much more widespread and comprehensive in terms of the variables, beyond ethnicity and race, around which culture might be defined (J. E. Harris, 2012). In recent years, a plethora of efforts reflects the increasing importance of culture within clinical psychology.

Journals and Books

Numerous publications on diversity and cultural issues in mental health have appeared in recent years. Scholarly journals in clinical psychology have increasingly included articles on cultural and diversity topics, and some psychology journals are devoted entirely to such issues (Koydemir & Essau, 2015). (See Table 4.1 for examples.) In addition, a wide variety of books now offer education and guidance to psychologists working

Table 4.1 Scholarly Journals Relevant to Diversity and Multicultural Issues in Clinical Psychology

Among others, these journals focus on issues of culture and commonly include articles of clinical relevance:	
Asian American Journal of Psychology	*Journal of Latinx Psychology*
Cultural Diversity and Ethnic Minority Psychology	*Journal of Multicultural Counseling and Development*
Hispanic Journal of Behavioral Sciences	*Journal of Rural Mental Health*
Journal of Black Psychology	*Psychoanalysis, Culture, and Society*
Journal of Cross-Cultural Psychology	*Psychology of Men and Masculinity*
Journal of Cultural Diversity	*Psychology of Sexual Orientation and Gender Diversity*
Journal of Gender, Culture, and Health	
Among others, these journals focus on clinically relevant issues and commonly feature articles emphasizing culture or diversity:	
Clinical Psychology: Science and Practice	*Journal of Counseling Psychology*
Counseling Psychologist	*Journal of Marital and Family Therapy*
Journal of Abnormal Psychology	*Journal of Mental Health Counseling*
Journal of Clinical Psychology	*Professional Psychology: Research and Practice*
Journal of Consulting and Clinical Psychology	*Psychotherapy: Theory, Research, Practice, Training*

with diverse populations. Some of these books focus on a single population, such as *Working With Asian Americans* (E. Lee, 1997), *Psychotherapy With Women* (Mirkin, Suyemoto, & Okun, 2005), or *Counseling Muslims* (Ahmed & Amer, 2012), whereas others compile chapters on many different populations, such as *Ethnicity and Family Therapy* (McGoldrick, Giordano, & Garcia-Preto, 2005a) or *Counseling Diverse Populations* (Atkinson & Hackett, 2003). Collectively, these publications represent a wealth of cultural knowledge for contemporary clinical psychologists, and their increasing presence acknowledges the importance of the topic.

Emergence of American Psychological Association Divisions

Within the American Psychological Association, new divisions arise when a subset of members recognizes a need to study or examine a specific topic in depth. Among the divisions created most recently, many focus on diversity or cultural issues, including the following:

- Division 35—Society for the Psychology of Women
- Division 36—Psychology of Religion
- Division 44—Society for the Psychological Study of Lesbian, Gay, and Bisexual Issues
- Division 45—Society for the Study of Ethnic Minority Issues
- Division 51—Society for the Psychological Study of Men and Masculinity

American Psychological Association Ethical Code

Numerous specific standards and principles in the most recent edition of the American Psychological Association (2002) ethical code compel psychologists to work with cultural sensitivity and competence. Their inclusion as standards makes it clear that awareness of diversity issues is a requirement rather than merely an aspiration for ethical psychologists. See Table 4.2 for a list of specific ethical standards and principles that relate to multiculturalism.

Web Link 4.2
Cultural Diversity & Ethnic Minority Psychology

Table 4.2 Selected Excerpts From the American Psychological Association's (2002) "Ethical Principles of Psychologists and Code of Conduct" Relating to Multiculturalism or Diversity

- Principle E: Respect for People's Rights and Dignity

Psychologists respect the dignity and worth of all people, and the rights of individuals to privacy, confidentiality, and self-determination. Psychologists are aware that special safeguards may be necessary to protect the rights and welfare of persons or communities whose vulnerabilities impair autonomous decision making. Psychologists are aware of and respect cultural, individual, and role differences, including those based on age, gender, gender identity, race, ethnicity, culture, national origin, religion, sexual orientation, disability, language, and socioeconomic status and consider these factors when working with members of such groups. Psychologists try to eliminate the effect on their work of biases based on those factors, and they do not knowingly participate in or condone activities of others based on such prejudices.

- Standard 2.01 Boundaries of Competence

(b) Where scientific or professional knowledge in the discipline of psychology establishes that an understanding of factors associated with age, gender, gender identity, race, ethnicity, culture, national origin, religion, sexual orientation, disability, language, or socioeconomic status is essential for effective implementation of their services or research, psychologists have or obtain the training, experience, consultation, or supervision necessary to ensure the competence of their services, or they make appropriate referrals, except as provided in Standard 2.02, Providing Services in Emergencies.

- Standard 3.01 Unfair Discrimination

In their work-related activities, psychologists do not engage in unfair discrimination based on age, gender, gender identity, race, ethnicity, culture, national origin, religion, sexual orientation, disability, socioeconomic status, or any basis proscribed by law.

(Continued)

Table 4.2 (Continued)

- Standard 9.06 Interpreting Assessment Results

When interpreting assessment results, including automated interpretations, psychologists take into account the purpose of the assessment as well as the various test factors, test-taking abilities, and other characteristics of the person being assessed, such as situational, personal, linguistic, and cultural differences, that might affect psychologists' judgments or reduce the accuracy of their interpretations. They indicate any significant limitations of their interpretations. (See also Standards 2.01b and c, Boundaries of Competence, and 3.01, Unfair Discrimination.)

Source: American Psychological Association (2002), pp. 1060–1073.

American Psychological Association Accreditation Standards

When the American Psychological Association decides whether to give its "seal of approval"—in other words, accreditation—to a graduate program in psychology, cultural diversity is a primary focus. In the American Psychological Association (2005) standards of accreditation, "Cultural and Individual Differences and Diversity" is one of the eight domains that an educational program must address adequately to be accredited. This requirement applies to doctoral programs, predoctoral internships, and postdoctoral internships seeking accreditation. Specifically, the accreditation standards for doctoral programs list criteria such as (1) including people of diverse backgrounds among students and faculty and (2) educating students about the role of culture in the science and practice of professional psychology.

DSM Efforts Toward Multiculturalism

The authors of the fifth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (*DSM-5*) state in the Introduction that "key aspects of culture relevant to diagnostic classification and assessment have been considered in the development of *DSM-5*" (American Psychiatric Association, 2013, p. 14). In addition to information on cultural variation embedded in the descriptions of specific disorders, it provides more general guidance for clinicians to help with overall cultural competence. For example, it offers an "Outline for Cultural Formulation," which instructs clinicians in various aspects of culture to assess in clients, such as cultural identity, cultural conceptualization of distress, and cultural features of the relationship with the mental health professional. It also offers a "Cultural Formulation Interview," a series of 16 specific questions that can guide a clinician toward a culturally informed interview. All of these efforts support the growing body of research suggesting that culture influences the experience or expression of a variety of psychological problems, including anxiety disorders, eating disorders, substance use, and many others (e.g., Chentsova-Dutton & Tsai, 2007).

Another effort toward cultural awareness incorporated into *DSM-5* is a glossary listing **cultural concepts of distress** (many of which were called "culture-bound

syndromes" in previous *DSM* editions; Alarcón, 2015). The glossary of cultural concepts of distress includes nine terms that represent psychological problems observed in groups from various parts of the world. Some are described as similar to a *DSM-5* disorder, but others bear little resemblance. Examples include *taijin kyofusho*, in which a person anxiously avoids interpersonal situations because they believe that their appearance, actions, or odor will offend other people (found in Japanese and some other cultures); *susto*, in which a frightening event is thought to cause the soul to leave the body, resulting in depressive symptoms (found in some Latino/Latina/Hispanic cultures); and *maladi moun*, in which one person can "send" psychological problems like depression and psychosis to another, usually as a result of envy or hatred toward the other person's success (found in some Haitian communities; similar experiences called the "evil eye" are more common in other parts of the world). Although some authors (McGoldrick et al., 2005b) have pointed out that a few of the official disorders included in *DSM* might in fact be better described as culturally bound to U.S. or North American cultures (e.g., eating disorders, as discussed in Chapter 7), the list of culture-bound syndromes in the *DSM* nonetheless signifies an increase in the profession's recognition of multicultural issues.

Revisions of Prominent Assessment Methods

Several prominent assessment tools used by clinical psychologists have been revised in recent years with the specific intent of making them more appropriate and serviceable to diverse clients (Braje & Hall, 2015). The Minnesota Multiphasic Personality Inventory (MMPI), an especially popular and well-respected personality test, underwent a major overhaul in the 1980s, resulting in the publication of the MMPI-2 in 1989. Compared with the original MMPI, the normative scores for the MMPI-2 were based on population samples much more representative of the cultural diversity of the U.S. population (D. S. Nichols, 2001). (Chapter 10 includes much more information on the MMPI and MMPI-2.) Other examples include the adult and child versions of the Wechsler tests of intelligence (e.g., Wechsler Adult Intelligence Scale, Wechsler Intelligence Scale for Children), which are among the most widely used and highly esteemed in their respective categories. As these tests have been revised in recent years, their authors have made efforts to create instruments that minimize cultural bias and maximize cultural inclusion (D. P. Flanagan & Kaufman, 2009). (Chapter 9 includes much more information on the Wechsler tests.)

Web Link 4.3
National Center for Cultural Competence

CULTURAL COMPETENCE

What Is Cultural Competence?

Clinical psychologists should strive for **cultural competence** (Comas-Díaz & Brown, 2016; J. R. Graham & Roemer, 2015; Vasquez, 2010). Indeed, when clients perceive their therapists as culturally competent, they are more likely to form strong working relationships with them, which leads to fewer therapy drop-outs and better therapy outcomes (Owen, Tao, Leach, & Rodolfa, 2011; Soto, Smith, Griner, Domenech Rodríguez, & Bernal, 2018).

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Photo 4.2
Cultural self-awareness, or an appreciation of the clinical psychologist's own unique cultural viewpoint, is a key element of cultural competence.

But what exactly does cultural competence involve? According to D. W. Sue and Sue (2008),

> Multicultural counseling competence is defined as the counselor's acquisition of awareness, knowledge, and skills needed to function effectively in a pluralistic democratic society . . . and on an organizational/societal level, advocating effectively to develop new theories, practices, policies, and organizational structures that are more responsive to all groups. (p. 46)

A key phrase in D. W. Sue and Sue's (2008) definition of cultural competence is "awareness, knowledge, and skills"; these are the three primary components to multicultural competence as applied to clinical/counseling work. Let's examine each one in detail.

Cultural Self-Awareness

Web Link 4.4
Cultural competence & cultural self-awareness

Cultural competence begins with learning about one's own culture—not only basic facts such as where one's parents or ancestors came from but also the values, assumptions, and biases that one has developed as a result of all cultural influences (J. E. Harris, 2012). When a clinical psychologist attains **cultural self-awareness**—that is, comes to understand that their viewpoint is (like everyone's) unique and idiosyncratic—several conclusions are within reach (Fouad & Arredondo, 2007). For example, the psychologist may adopt a viewpoint toward clients that is less egocentric, more humble, and more appreciative of the varying experiences of life (Mosher et al., 2017). Also, the psychologist may come to recognize that differences between people are not necessarily deficiencies, especially if the difference demonstrated by the client is common or valued in the client's own cultural group. Rather than glossing over differences between themselves and their clients, psychologists should explore their own personal reactions to these differences and address any discomfort they may initially feel about them (B. Greene, 2007).

Of course, the process of cultural self-awareness can be difficult or unpleasant for psychologists, because it may require admitting and coming to terms with some undesirable "isms"—racism, sexism, heterosexism, classism, ethnocentrism, or similar prejudicial or discriminatory belief systems that we'd rather pretend we don't have. But by examining them and exposing them to ourselves, we can take steps toward minimizing them and the negative impact they might have on our clients (Vasquez, 2010).

Cultural self-awareness is important regardless of the psychologist's own cultural background. Whether a member of a majority or minority, the psychologist will inevitably encounter clients whose cultural backgrounds differ—sometimes slightly, sometimes considerably—from their own. Thankfully, scholars in our field are paying increasing attention to the cultural status of the therapist (e.g., Gelso, 2010; Mirsalimi, 2010; Nezu, 2010).

BOX 4.1

In My Practice . . .

In my practice, self-awareness of my own cultural background has always been important. It consistently reminds me that my way is just one way of understanding the world and helps me appreciate the validity of the perspectives that diverse clients hold.

With regard to my own cultural background, one particular challenge involves disclosing it to clients. The questions that I consider are the same that most other clinical psychologists (and other mental health professionals) consider: How much of my cultural background should my clients know? Are there certain characteristics about myself that I should voluntarily share with every client? Are there certain characteristics about myself that I should willingly share if the client requests it? Are there certain characteristics about myself that no client ever needs to know?

Clients may come to certain conclusions about me (whether right or wrong) based on what they see when they sit in my office: my race/ethnicity, my approximate age, my gender. But other variables may be tougher for them to guess: my religion, my ethnicity, my sexual orientation, my socioeconomic status, where I was raised, and more. Usually, when clients ask me to fill in those blanks, their underlying concern is whether I have the experience necessary to fully understand or empathize with them—in other words, whether I'll "get" them. A teen client who eventually came out asked me directly in an early session about my sexual orientation, later explaining that he thought a gay therapist would better recognize the issues he had around telling his family about his sexual orientation. When an 80-year-old client called me to inquire about my services, her first question was how old I was, because she was concerned that a young therapist couldn't grasp how it felt to experience the illnesses, cognitive declines, and deaths of friends that she had endured in recent years. A 21-year-old woman who was considering an abortion asked me about my religion, hoping that if mine matched hers I would have a more immediate appreciation of the issues with which she wrestled.

Over the course of my career, I have leaned toward keeping my own cultural background private, but in selected situations (including some of the above), when I thought it would benefit the client, I selectively disclosed.

In your opinion, should therapists disclose this kind of information about themselves? Why or why not? Does a shared cultural background in fact make a therapist better able to "get" a client, and does that lead to better assessment or treatment?

To see this box come to life as a whiteboard video, please visit edge.sagepub/pomerantz5e.

Knowledge of Diverse Cultures

To know one's own culture is a good first step for the psychologist, but it won't amount to much unless the psychologist also possesses information about the client's cultural groups. Simply put, the psychologist should know as much as possible about the client's culture. Efforts in this direction should be continual—learning through reading, direct experiences, relationships with people in various cultures, and other means. Of course, therapists can't know everything about every culture that might be represented by a client in a country as diverse as the United States. In fact, acknowledging cultural differences with clients can be a good idea, and asking a client to explain the meaning or importance of a particular experience from their point of view can ensure a more

BOX 4.2

Considering Culture

Interviews With Multicultural Experts: Cultural Competence With Clients From Specific Cultures

Knowledge of diverse cultures is one of the core elements of cultural competence. Here, we ask nine highly respected experts in multicultural mental health to provide their thoughts about what clinical psychologists should know about particular cultural groups. The cultural groups they discuss span ethnicity, gender, religion, and sexual orientation.

This box is just a start! It contains brief excerpts of the experts' responses, but this textbook's companion website (edge.sagepub/pomerantz5e) contains the experts' full biographies as well as their full-length responses to these five questions about clinical work with their cultural groups:

1. In general, why is it important for clinical psychologists to be culturally competent when working with members of this culture?
2. What can clinical psychologists (or students in training) do to enhance their cultural competence with members of this culture?
3. What specific considerations should clinical psychologists keep in mind when conducting assessment (interviewing, intelligence testing, personality testing, etc.) and diagnosis with members of this culture?
4. What specific considerations should clinical psychologists keep in mind when conducting psychotherapy with members of this culture?
5. Any other thoughts about culturally competent practice with members of this culture?

Latino/Latina Clients

Dr. Melba Vasquez is the cofounder of the Society for the Psychological Study of Ethnic Minority Issues (American Psychological Association Division 45), the first Latina member-at-large on the board of directors of the American Psychological Association, and 2011 president of the American Psychological Association. She sees clients in private practice in Texas.

- "Latino/a cultural factors for assessment may include relevant generational history (e.g., number of generations in the country, manner of coming to the country); citizenship or residency status (e.g., number of years in the country, parental history of migration, refugee flight, or immigration); fluency in 'standard' English or other language; extent of family support or disintegration of family; availability of community resources; level of education; change in social status as a result of coming to this country (for immigrant or refugee); work history; and level of stress related to acculturation and/or oppression."
- "The demographic changes in the world and in this country have significant implications for counseling and clinical psychologists. Although Latinos/Latinas typically present with similar problems, relative to other clients, variations in conceptualizations and interventions may be important in providing effective services."

Asian and Asian American Clients

Dr. Frederick Leong is a professor of psychology at Michigan State University, director of the Consortium for Multicultural Psychology Research, and founding editor of the *Asian American Journal of Psychology.*

1. "Intra-group heterogeneity is particularly important to recognize when it comes to a group such as Asian Americans given that this population comprises approximately 43 different ethnic groups with over 100 languages and dialects represented."
2. "For many Asian Americans, constancy and equilibrium, duty, obligation and appearance of harmonious relations are important in their family relations. In addition, Asian families tend to emphasize connectedness of the family, while European Americans tend to prioritize separateness and clear boundaries in relationships due to the two groups' value differences. It has become well known that Asian Americans tend to be more collectivistic in cultural orientation while European Americans tend to be individualistic."

American Indian/Alaska Native Clients

Dr. Joseph E. Trimble is a Distinguished University Professor and member of the Department of Psychology and a research associate in the Center for Cross-Cultural Research at Western Washington University. In 1994, he received a Lifetime Distinguished Career Award from American Psychological Association Division 45 for his research and dedication to cross-cultural and ethnic psychology. He also won the American Psychological Foundation's 2017 Gold Medal Award for Life Achievement in Psychology in the Public Interest.

- "Providers of traditional helping services in Indian communities most likely exemplify empathy, genuineness, availability, respect, warmth, congruence, and concreteness, characteristics that are likely to be effective in any therapeutic treatment setting, regardless of the provider's theoretical orientation or counseling style. Effective counseling with Indians begins when a counselor carefully internalizes and uses these basic characteristics in counseling settings."
- "A constant theme occurs repeatedly in the Indian and Native counseling literature—counselors of Indian and Native clients must be adaptive and flexible in their personal orientations and in their use of conventional counseling techniques."

African American Clients

Dr. Robert L. Williams was a founding member of the National Association of Black Psychologists in 1968. From 1970 to 1992, he served as professor of psychology and African and African-American studies at Washington University in St. Louis. He is currently professor emeritus of the departments of Psychological & Brain Sciences and African and Afro-American Studies at Washington University in St. Louis.

- "The first step in gaining cultural competence is self-knowledge. Psychologists need to become aware of their own racial scripts and beliefs, and how these might

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affect the way they conduct therapy. Racial scripts are programmed messages from parents to children about African Americans of which the children, even when they have become adults, are not fully aware. They can have a powerful influence, so it's important for psychologists to recognize their own racial scripts and alter them to meet the reality of the African American community."

- "Perhaps most importantly, psychologists working with clients of diverse backgrounds, including African American clients, need to know and accept that a *cultural difference is not a deficiency.* In other words, differences between the cultures exist, but they need not suggest that one is better or worse than the other."

Irish American Clients

Dr. Monica McGoldrick is the director of the Multicultural Family Institute of New Jersey and an adjunct associate professor of clinical psychiatry at UMDNJ–Robert Wood Johnson Medical School. Her many books include *Ethnicity and Family Therapy* (McGoldrick et al., 2005a) and *The Expanding Family Life Cycle* (McGoldrick, Carter, & Garcia-Preto, 2016).

- "Traditionally (and I'm overgeneralizing a bit to make the point) the Irish, although they were big talkers, were not big talkers about emotional issues. In fact, they seemed afraid of emotional issues. So, if you asked them the kinds of questions that would, say, come from a Freudian perspective (which is still very strong in psychology), they would probably look much more pathological than they really are, because they would be extremely uncomfortable with questions about their inner feelings, especially negative feelings or sexual feelings."
- "The Irish have a long history in which they learned how to keep their emotional process under wraps, and often the church encouraged certain attitudes which led them to feel guilty about some feelings that wouldn't even be an issue in other cultural groups."

Female Clients

Dr. Nadya A. Fouad is Mary and Ted Kellner Endowed Chair of Educational Psychology and Distinguished Professor of Counseling Psychology, School & Clinical Mental Health Counseling in the Department of Educational Psychology at the University of Wisconsin–Milwaukee. She served as cochair of the writing team for the American Psychological Association's *Multicultural Guidelines on Education, Training, Research, Practice, and Organizational Change.* She is also the recipient of the Leona Tyler Award for Lifetime Achievement in Counseling Psychology from APA Division 17.

- "It's important to understand how a client's gender is interwoven in his or her culture or ethnicity. For example, a traditional Hispanic woman facing a decision like moving across the country to go to college and moving away from her family might experience that decision in a unique way, and

differently from a man in the same situation. If a client senses that the psychologist isn't taking these kinds of issues into account, the client might not return at all."

- "Be aware of your own biases with regard to gender. For example, do you have a bias toward women being in a particular role, either traditional or nontraditional? Do you have a bias toward one type of relationship in which women should be?"

Middle Eastern Clients

Dr. Karen Haboush (1961–2016) was a clinical associate professor in the Department of Applied Psychology at Rutgers University and maintained a private practice as well. She published articles and chapters on culturally competent practice with children and families of Middle Eastern descent.

- "Because the popular media often presents predominantly negative images of Middle Easterners (i.e., terrorists, religious extremists), psychologists may unwittingly internalize these images, which subsequently influences their clinical work. As with all ethnic groups, the first step in developing cultural competence is for psychologists to examine their own attitudes and knowledge."
- "In Middle Eastern culture, the welfare of the family has much greater significance than individual autonomy and independence. . . . This makes Middle Eastern culture quite different from the prevailing emphasis on individual achievement, which is more characteristic of North American and European countries. Of course, great variability exists across cultures and countries, but generally speaking, a collectivist emphasis on the well-being of the family tends to characterize the Middle East."

Lesbian/Gay/Bisexual/ Transgender (LGBT) Clients

Dr. Kathleen J. Bieschke is a professor of counseling psychology at Pennsylvania State University. Dr. Bieschke has written extensively about the delivery of affirmative counseling and psychotherapy to LGBT clients. Dr. Bieschke is a coeditor of the *Handbook of Counseling and Psychotherapy With Lesbian, Gay, Bisexual, and Transgender Clients* (Bieschke, Perez, & DeBord, 2007) and the *Handbook of Sexual Orientation and Gender Diversity in Counseling and Psychotherapy* (DeBord, Fischer, Bieschke, & Perez, 2017).

- "Attention must be paid to the development of a productive therapeutic relationship. This is particularly true when working with sexual minority clients, as LGBT individuals have learned to carefully assess the extent to which therapists are affirmative. There are myriad ways in which therapists can convey affirmation to clients. For example, having a symbol in one's office indicating knowledge of the LGBT community (e.g., a small pink triangle) or using language that is inclusive (e.g., not using gender-specific pronouns when referring to sexual partners) can provide LGBT clients with concrete evidence of

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a therapist's openness to sexual minorities."

- "A particularly fruitful strategy is getting to know someone who identifies as a sexual minority or as a strong ally; relationships such as these can shatter stereotypes and assumptions."

Jewish American Clients

Dr. Lewis Z. Schlosser has worked as associate professor of counseling psychology at Seton Hall University and has practiced in counseling and forensic settings. He has published articles and book chapters on multicultural counseling and development, specifically the intersection of race, religion, and ethnicity; anti-Semitism; and psychotherapy for American Jews (e.g., Schlosser, 2006; Schlosser, Ali, Ackerman, & Dewey, 2009; Schlosser, Safran, Suson, Dettle, & Dewey, 2013).

- "Being Jewish is largely an invisible minority status. Psychologists might never know that they have a Jewish client in front of them unless the client discloses her or his identity. Because Jews have endured a long history of oppression, many American Jews will assess the safety of the current environment prior to disclosing their identity. A culturally competent psychologist will strive to foster an environment of safety so that the American Jewish client would feel comfortable disclosing her or his Jewish identity."
- "It is important to note that cultural competence is not assured simply by being Jewish. That is, we can't assume that Jews are going to be culturally competent with Jewish clients, as internalized anti-Semitism or other factors might be operating."

culturally sensitive understanding. But clients shouldn't bear too much of the burden of educating the psychologist; instead, the psychologist should aim to enter each session with sufficient knowledge of the client's cultural background.

Cultural knowledge should include not only the current lifestyle of the members of the cultural group but also the group's history, especially regarding major social and political issues. For example, the history of African Americans—including slavery, cruelty, exploitation, and overt and covert racism—can understandably affect the formation of a trusting relationship with a psychologist or with the mental health system more generally (Constantine, Redington, & Graham, 2009; Terrell, Taylor, Menzise, & Barett, 2009). A clinical psychologist who fails to recognize these historical realities and their potential impact on clients may form expectations or make interpretations that are culturally insensitive and jeopardize the therapeutic relationship (Shorter-Gooden, 2009).

Of course, the psychologist should not assume that every individual is typical of their cultural group. In other words, although a cultural group may have a collective tendency, an individual member of that group may vary greatly from that tendency. To assume that a member of a cultural group will exhibit all the characteristics common to that group is to prejudge. The individual would be better served by a psychologist who appreciates the cultural group norms but also appreciates the **heterogeneity** inherent in every culture. See Box 4.3 for further discussion of heterogeneity within a culture.

Some of the heterogeneity within a culture stems from differences in **acculturation** (J. Chu, Leino, Pflum, & Sue, 2017; Organista, Marin, & Chun, 2010; Rivera, 2010). That is, when people find themselves in a new cultural environment, they may respond in a variety of ways, especially with regard to adopting elements of the new culture or retaining elements of their original culture. Four separate acculturation strategies have been identified (Berry, 2003; Rivera, 2008): *assimilation*, in which the individual adopts much of the new culture and abandons much of the original; *separation*, in which the individual rejects much of the new culture and retains much of the original; *marginalization*, in which the individual rejects both the new and the original culture; and *integration*, in which the individual adopts much of the new culture and retains much of the original. As these four strategies illustrate, individuals can combine an appreciation of their new and original cultures in many ways, resulting in remarkable diversity within any cultural group. So simply knowing that Hajra immigrated to the United States from Bosnia 15 years ago leaves much unknown about her cultural identity. To what extent has she embraced mainstream U.S. culture? To what extent has she carried on her Bosnian cultural beliefs and traditions? Culturally competent clinical psychologists strive to learn their clients' acculturation strategies in an effort to understand more completely their unique ways of life (Shin & Muñoz, 2009).

BOX 4.3

Metaphorically Speaking

If You've Seen Yao Ming, You Understand Heterogeneity Within a Culture

The average height of a Chinese man is about 5 ft. 7 in.—about 2 to 3 in. shorter than the average height of men from the United States. So if you are a clinical psychologist and you know that the new client in your waiting room is a Chinese man, you might expect to see someone around 5 ft. 7 in. tall. But when you open the waiting room door and Yao Ming stands up—all 7 ft. 6 in. of him—you understand right away that your client is a huge exception to the rule.

Yao Ming is a Chinese man and was a star basketball player who entered the NBA Hall of Fame in 2016. His height illustrates a powerful lesson about cultural competence: Although members of a culture *as a group* may tend toward certain norms, any *individual* within that group may fall far from that norm.

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Photo 4.3
Yao Ming's extraordinary height illustrates that the characteristics of an individual do not necessarily follow cultural trends. Regarding their clients' psychological characteristics, culturally competent clinical psychologists appreciate the exceptions as well as the rules.

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Clinical psychologists should aspire to understand the norms of the cultures with which they work, but if they rigidly assume that every person in that culture fits those norms, they are guilty of unfair and often inaccurate prejudice. To some extent, generalizing is inevitable when discussing cultural groups (McGoldrick et al., 2005a), but our generalizations should be "guidelines for our behaviors, to be tentatively applied in new situations, and they should be open to change and challenge. It is exactly at this stage that generalizations remain generalizations or become stereotypes" (D. W. Sue & Sue, 2008, p. 154).

Actually, your statistics course probably included some basic concepts that can help illustrate this point: measures of central tendency, such as the mean, and measures of variability, such as range or standard deviation. Any group of numbers will yield a mean, but a quick glance at the scatterplot shows that there is some variation around that mean, and in some cases the outliers are quite extreme. The client in the psychologist's office represents one of those points on the scatterplot, and although it is important to appreciate the central tendencies of that group, it is equally important to recall that the client might be an outlier.

Consider another central tendency of Chinese and many other Asian cultures—the emphasis on collectivism over individualism. In contrast to the tendencies in European American cultures, members of Asian cultures tend to value the welfare of the family or group over their own welfare as individuals. Numerous authors on the subject have indicated that members of Asian cultures assign great significance to harmony, interdependence, respect, and loyalty in close relationships, and they will often forgo individual self-directed accomplishments as a result (e.g., Dana, 1993; E. Lee & Mock, 2005; Shibusawa, 2005). It is important for clinical psychologists to be aware of this tendency toward collectivism among this group but simultaneously be aware that any particular client may be an exception to the rule. Just as Yao Ming stands almost 2 ft. higher than the average Chinese man, the Chinese client in the waiting room may hold very individualistic—rather than collectivistic—values. Although this client's status as a cultural outlier is less obviously visible than Yao Ming's height, it is nonetheless the responsibility of clinical psychologists to be perceptive and responsive to the existence of such atypical cultural members.

Culturally Appropriate Clinical Skills

Once the psychologist has attained cultural knowledge of self and clients, the next step is to develop suitable strategies for assessment and treatment. In other words, the approaches and techniques that a psychologist uses to improve a client's life should be consistent with the values and life experience of that client (Hall, Hong, Zane, & Meyer, 2011; Hwang, 2011; Toporek, 2012). "Talk therapy" may work well for many, but for members of some cultural groups, it may be a bad fit. Similarly, clients from some cultures may place great value on "insight" into their psychological problems obtained over many months, but clients from other cultures may respond much more positively to action-oriented therapies with a short-term focus. Other common features of traditional psychotherapy, including verbal self-disclosure of personal problems and 50-minute

sessions in an office building, may not be entirely compatible with clients from certain cultural backgrounds (Comas-Díaz, 2012).

One essential culturally appropriate clinical skill that is receiving more attention in recent years involves microaggressions. **Microaggressions** are comments or actions made in a cross-cultural context that convey prejudicial, negative, or stereotypical beliefs and may suggest dominance or superiority of one group over another (Fouad & Arredondo, 2007; D. W. Sue, 2010; D. W. Sue et al., 2007). Often, they are "little things" that one person may say to another without any intention of hostility or any awareness that the comments might be invalidating or insulting—but, in fact, they are. Microaggressions often center on ethnicity or race (e.g., A. J. Franklin, 2007; D. W. Sue, Capodilupo, & Holder, 2008) but can involve any number of differences between people, such as age, gender, socioeconomic status, religion/spirituality, or sexual orientation. A review of research on the effects of microaggressions finds that they cause distress and can contribute to serious psychological problems (P. P. Lui & Quezada, 2019). Unfortunately, clinical psychologists (and other mental health professionals) are not immune from committing microaggressions against their clients, and those microaggressions can cause damage to the progress a client might otherwise make (Owen et al., 2018; Spengler, Miller, & Spengler, 2016). As an example, consider a psychologist who, during an initial interview with a 19-year-old male college student, asks, "Do you have a girlfriend?" The "girlfriend" question might communicate an assumption on the psychologist's part that being straight is the norm or what is expected or "right." Especially if the client is gay or bisexual, such a question might have negative consequences in terms of forming a therapeutic relationship in which the client feels valued and accepted. Or if a psychologist meets with a 7-year-old therapy client on December 27 and asks, "So what did Santa bring you?" the child may feel devalued if he or she is Muslim, Jewish, or Buddhist, or doesn't celebrate Christmas for any other reason. The best way for psychologists to avoid microaggressions is to examine the thoughts and beliefs that underlie them, which can result in greater humility and self-awareness for the psychologist (Vasquez, 2010).

Recent efforts toward the attainment of culturally appropriate clinical skills have emphasized the notion of *cultural adaptation* of treatments with empirical evidence to support them (Bernal, Jiménez-Chafey, & Domenech Rodríguez, 2009; Smith, Domenech Rodríguez, & Bernal, 2011). In other words, now that clinical psychology has generated lists of treatments that work, an important subsequent step is to determine how those treatments might need to be adapted for members of diverse cultures (Castro, Barrera, & Steiker, 2010; J. R. Graham & Roemer, 2015; Mulvaney-Day, Earl, Diaz-Linhart, & Alegría, 2011). (After all, many of the studies that generated the empirical support for evidence-based treatments were conducted on clients whose collective cultural range was very narrow.) For example, La Roche, Batista, and D'Angelo (2011) examined a large number of guided imagery scripts—the instructions that psychologists read or record for their clients when they are trying to induce relaxation. Often, these scripts include statements such as "Imagine yourself alone on a calm beach" or "Picture yourself in a beautiful meadow." Such situations are remarkably "solo"—in other words, the idyllic setting involves the client alone. In many ethnicities, scenes that invoke feelings of togetherness or connectedness with other people might better capture relaxation or happiness—something along the lines of "Imagine yourself among people who are positive and make you feel good about yourself." La Roche et al. found that guided imagery scripts emphasize a "solo" (or idiocentric) orientation rather than a "together"

Web Link 4.5
Microaggressions

(or allocentric) orientation, which could prove inconsistent with the cultural values of many clients. They recommend that clinicians who use such techniques develop a variety of guided imagery scripts, including some that are allocentric, rather than imposing idiocentric scripts on all clients. This kind of adaptation—in which clinicians consider how diverse clients might respond differently to the standard (often evidence-based) treatment, versus one that has been customized for them—is on the rise (Gelso, 2011; G. C. N. Hall & Ibaraki, 2016; Hwang, 2011). Another study shows the importance of cultural adaptation of therapies by showing what can happen when such adaptation does not happen: Expressive writing, a technique that has produced positive therapy outcomes for many clients struggling with posttraumatic symptoms (in studies that featured mostly White clients) unexpectedly worsened symptoms of a group of about 100 Chinese American women (Gallagher, Long, Tsai, Stanton, & Lu, 2018). Meta-analyses suggest that overall, cultural adaptations to evidence-based therapies produce positive therapy outcomes (Soto et al., 2018).

Another clinically appropriate skill to consider, as basic as it may sound, is simply the language in which the therapist talks with the client. For clients who speak more than one language, this choice can have important consequences. One study focused on the assessment of bilingual individuals with symptoms of schizophrenia and schizoaffective disorder. Those who were interviewed in their first language (their "mother tongue") generally reported a higher number and severity of symptoms than those who were interviewed in their second ("acquired") language (C. A. Brown & Weisman de Mamani, 2017). Of course, no therapist is required to speak every language, but a skillful therapist might either keep such language effects in mind during their interactions with a client, or be aware of other available mental health resources (including therapists who speak other languages) and be capable of referring clients when appropriate.

ARE WE ALL ALIKE? OR ALL DIFFERENT?

The discussion of cultural issues brings up some important, fundamental questions about human beings that are applicable to psychologists and the clients they see. To what extent are all people—and the experiences and problems they bring to therapy—similar? And to what extent might they differ from one another?

Etic Versus Emic Perspective

Dana (1993) describes two distinct perspectives that psychologists have used during the history of the profession. The first, known as the **etic** perspective, emphasizes the similarities between all people. It assumes universality among all people and generally does not attach importance to differences among cultural groups. This perspective was more dominant in the early days of psychology, when most of the people teaching and practicing psychology were male, of European descent, and of middle-class or higher socioeconomic standing. Generally, their viewpoint was put forth as the normative viewpoint on issues such as defining psychological health, identifying and labeling psychological disorders, and developing therapy approaches.

The **emic** perspective differs from the etic perspective in that it recognizes and emphasizes culture-specific norms. A psychologist employing the emic perspective—which has grown in prominence along with the rise in multiculturalism—considers a client's behaviors, thoughts, and feelings within the context of the client's own culture rather than imposing norms of another culture on the client (Koydemir & Essau, 2015). Compared with the etic perspective, the emic perspective allows psychologists more opportunity to appreciate and understand how the client might be viewed by members of their own cultural group. In short, the emic approach stresses that individuals from various cultural groups "must be understood on their own terms" (Dana, 1993, p. 21).

As a side note, Dana (1993) mentions that the terms *etic* and *emic* were derived from the field of linguistics and, specifically, from the terms *phonetic* and *phonemic*. Historically, linguists have used the term *phonetics* for sounds that are common to all languages and the term *phonemics* for sounds that are specific to a particular language (Dana, 1993; Pike, 1967). The distinction between the way the two terms were used in linguistics—essentially, emphasizing *universality* versus *culture specificity*—remains in the way the terms *etic* and *emic* are currently used in psychology.

Tripartite Model of Personal Identity

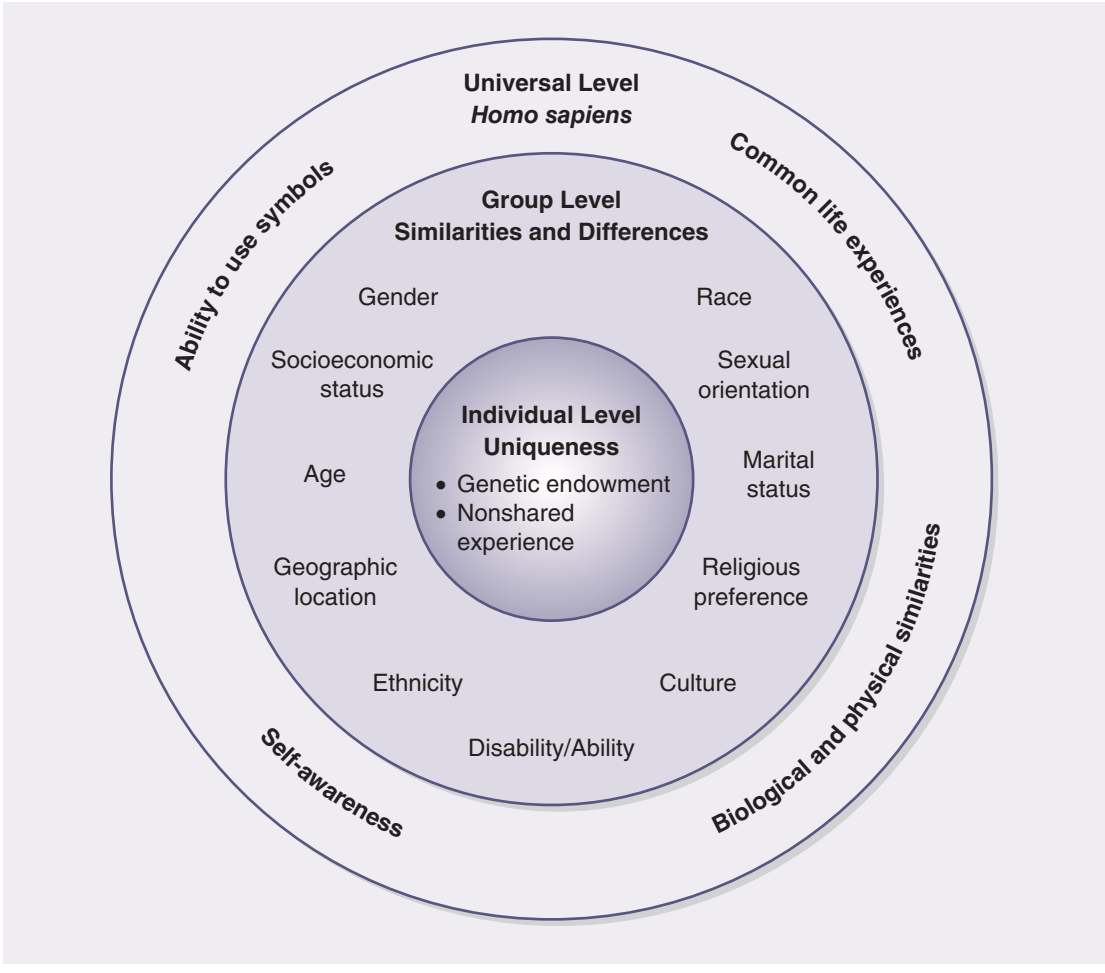
If the etic and emic perspectives represent two opposite viewpoints, perhaps it would be beneficial to consider a continuum that includes not only these two extremes but also some middle ground. D. W. Sue and Sue (2008) offer a three-level model called the **tripartite model of personal identity** in which all levels hold some degree of importance.

One level in this model is the **individual level.** Here, the premise is that "all individuals are, in some respects, like no other individuals." A second level is the **group level,** where the premise is that "all individuals are, in some respects, like some other individuals." The final level is the **universal level,** based on the premise that "all individuals are, in some respects, like all other individuals" (D. W. Sue & Sue, 2008, pp. 38–39). A psychologist who can appreciate a client on all three levels will be able to recognize characteristics that are entirely unique to the client, others that are common within the client's cultural group, and still others that are common to everyone. D. W. Sue and Sue (2008) argue that appreciation of all three levels is indeed the goal but that the group level has been overlooked traditionally in psychology, especially when the group is a minority culture, so psychologists may need to make more deliberate efforts in that direction. See Figure 4.1 for a visual representation of the tripartite model of personal identity.

WHAT CONSTITUTES A CULTURE?

When someone inquires about your cultural background, which of your characteristics come to mind? Many would list their race and ethnicity, and others would include a multitude of additional characteristics. If clinical psychologists are to function in a culturally competent and sensitive way, it makes sense to consider what exactly we might mean when we say "culture."

Figure 4.1 Tripartite Model of Personal Identity



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Narrow Versus Broad Definitions

Those who argue for a more narrow definition of culture typically point to ethnicity and race as the defining characteristics. Indeed, many books and articles on the topic of culture focus exclusively on issues of race or ethnicity. According to some who endorse this perspective, the inclusion of other variables as "cultural" would unfairly deemphasize the socially, politically, and personally important characteristics of race and ethnicity (Mio et al., 2006).

On the other hand, some argue that culture can be defined by a much broader range of variables, including "any and all potentially salient ethnographic, demographic, status,

or affiliation identities" (Pedersen, 1999, p. 3), or that culture can be composed of "any group that shares a theme or issues(s)" (D. W. Sue, Ivey, & Pedersen, 1996, p. 16). Others have stated that ethnicity may be the primary determinant of culture but not the only one: Further factors can include socioeconomic status, gender, geography/region, age, sexual orientation, religion/spirituality, disability/ability status, immigrant/refugee status, and political affiliation, among others (Alessi, Dillon, & Kim, 2015; Artman & Daniels, 2010; Consoli, Wang, Delucio, & Yakushko, 2017; Goldblum, Pflum, Skinta, Evans, & Balsam, 2017; Hope & Chappell, 2015; Liang & Molenaar, 2017; Lyons, Bieschke, Dendy, Worthington, & Georgemiller, 2010; McGoldrick et al., 2005a; McKitrick & Li, 2008; Moradi & Budge, 2018; Pepping, Lyons, & Morris, 2018; Robinson-Wood, 2009; Sewell, 2009; Shipherd, 2015; Vasquez & Vasquez, 2017).

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Photo 4.4
In your opinion, what variables should clinical psychologists consider when they appreciate their clients' cultural backgrounds?

Whether or not the characteristics beyond race and ethnicity are universally accepted as components of culture per se, quite a few books, chapters, and articles have been written with the intent of making psychotherapists and counselors more sensitive to these characteristics. For example, therapists can educate themselves about LGBT clients (Bieschke et al., 2007; Diamond, Butterworth, & Savin-Williams, 2011; Russell & Fish, 2016); clients with disabilities (Kosciulek, 2003); deaf clients (Leigh, 2010); elderly clients (Qualls, 2003); rural clients (Rainer, 2010; Smalley et al., 2010); low-income clients (Acosta, Yamamoto, & Evans, 1982); Appalachian clients (Harper, 1996); Orthodox Jewish clients (Mirkin & Okun, 2005); and many, many other specific groups whose defining characteristic is neither race nor ethnicity. Indeed, as time goes by, the range of variables that clinical psychologists include when they consider culture continues to broaden (L. S. Brown, 2011; Green, Callands, Radcliffe, Luebbe, & Klonoff, 2009).

Additionally, some subsections of society—**subcultures,** if you will—may be especially relevant for certain clients. As an example, consider a psychologist working in a prison setting or with a former prison inmate who was recently released. If there is such a thing as "prison culture"—a shared lifestyle with its own unique norms, expectations, values, and so on—it would probably be wise for the psychologist to consider it in addition to variables such as race, ethnicity, gender, and others. Likewise, a psychologist working with military personnel should have some appreciation of how military culture differs from civilian culture (Atuel & Castro, 2018). Even adolescents have been identified as having a culture of their own, with shared values around the importance of technology, peer relationships, and independence from parents (Nelson & Nelson, 2010). Many other "subcultures" based on specific work settings, living communities, or other variables may represent enough of an influence on the life experiences of clients to justify tailoring the treatment to best fit them (Arredondo et al., 1996; Truscott, 2010).

Interacting Cultural Variables

When we consider how many variables might contribute to culture, it's hard to avoid the conclusion that for any individual with whom a psychologist works, culture might be

multifactorial. In other words, lots of cultural variables may interact in unique ways to shape the life experience of a client. Of course, ethnicity and race may be most important for certain clients. But other variables might play significant roles as well.

Consider, for example, Esteban and Maria, two clients who share similar Latino/Latina/Hispanic ethnic backgrounds. In the spirit of cultural competence, their respective therapists should appreciate their ethnicity, but perhaps the cultural considerations should incorporate other factors as well. If Esteban is a 28-year-old gay man living an upper-middle-class life in Los Angeles, while Maria is a 66-year-old heterosexual woman living a lower-class life in a small town in rural West Virginia, the cultures of their day-to-day lives are probably quite different in spite of similar ethnic heritage. Indeed, if they visited each other's homes, they might find themselves living in very different worlds. Culturally competent therapists would certainly appreciate their ethnicity, but such therapists would also consider the way that other variables in their lives interact with their ethnicity to create a unique set of cultural circumstances (Arredondo et al., 1996; Truscott, 2010).

TRAINING PSYCHOLOGISTS IN ISSUES OF DIVERSITY AND CULTURE

With the increasing emphasis on multiculturalism in clinical psychology comes an increased responsibility to train psychologists to become culturally sensitive and competent. Graduate program directors, professors, and providers of continuing education share this challenge, along with psychologists striving to educate themselves.

Educational Alternatives

What are the best methods for training clinical psychologists in multicultural issues? Graduate programs have tried a variety of approaches (Benuto, Casas, & O'Donohue, 2018; Tormala, Patel, Soukup, & Clarke, 2018). Often, graduate programs include one or more courses specifically designed to address culture. In addition, some graduate programs may weave cultural training into all the educational experiences of the graduate student. Courses in psychotherapy, assessment, and research, as well as practicum training, can be designed to incorporate issues of culture. This way, issues of culture are not considered a specialized topic to be examined in isolation but a factor relevant to all professional activities of the clinical psychologist.

Another, less traditional approach to training in cultural issues emphasizes real-world experience with individuals of diverse cultures. Supporters of this approach contend that reading about a different culture in a book, or discussing a different culture in class, is no substitute for immersing oneself in that culture to some extent. Thus, through experiences that are professional (such as clinical work or research projects incorporating diverse clients, participants, colleagues, or supervisors) or personal in nature, some training programs promote learning about cultural groups by interacting directly with their members (Center for Multicultural Human Services, 2006; Magyar-Moe et al., 2005).

No single "best method" or consensus has emerged for training psychologists to be culturally competent. However, leaders in this field have begun to identify essential

components for graduate training programs. For example, Fouad and Arredondo (2006, as cited in Fouad, 2006) identify seven specific "critical elements of a multiculturally infused psychology curriculum" that they believe will improve the training of psychologists working as practitioners, teachers, or researchers. According to these authors, graduate training programs should

1. explicitly state a commitment to diversity;
2. actively make an effort to recruit graduate students from diverse populations;
3. actively make an effort to recruit and retain a diverse faculty;
4. make efforts to make the admissions process fair and equitable;
5. ensure that students gain awareness of their own cultural values and biases, knowledge of other groups, and skills to work with diverse populations;
6. examine all courses for an infusion of a culture-centered approach throughout the curriculum; and
7. evaluate students on their cultural competence on a regular basis. (Adapted from Fouad, 2006, pp. 7–9)

Regardless of the methods used to train clinical psychologists to be culturally competent, an essential ingredient is that the psychologist (or trainee) reaches a deeper appreciation of their own cultural identity. Hardy and Laszloffy (1992) describe numerous ways in which self-knowledge can be examined during training, such as in-class discussions, in-class presentations, self-guided assignments, and assigned discussions with one's own family of origin. In the end, the ability to relate to clients of diverse cultures may depend not only on information obtained through courses and assignments but also on an attitude of "respect, curiosity, and especially humility" (McGoldrick et al., 2005b, p. 6). Cultural humility, in fact, has been explored by numerous researchers in recent years and linked to better outcomes for therapy clients (D. E. Davis et al., 2018; Mosher et al., 2017).

Measuring the Outcome of Culture-Based Training Efforts

Let us not forget that psychology is a science, and as such we take a keen interest in measuring the outcome of our efforts to increase cultural sensitivity and competence of psychologists. But consider some of the difficult methodological questions:

- How should we reliably and validly measure the outcome of culture-based training efforts?
- How can we reliably and validly establish a baseline for the level of cultural competence of a psychologist or trainee before the training takes place?
- When we assess the cultural competence component of psychotherapy, whose opinion should we seek? The client's? The psychologist's? The supervisor's? Another interested party's?

- How can we make a causal connection between particular culture-based training efforts and particular outcomes? How can we be sure that confounding or unexamined variables aren't responsible for the outcomes we observe?

At the moment, measuring the outcome of culture-based efforts is at a very early stage of empirical investigation, as researchers grapple with issues such as those suggested by the questions above. There is some evidence to suggest that psychologists are learning the ideals of cultural competence but are not always implementing them as often or as comprehensively as they know they should. In other words, there may be a gap between what psychologists "practice" and what they "preach" regarding multicultural competence (Hansen et al., 2006).

On a more positive note, efforts promoting multiculturalism are clearly resulting in some needed improvements related to clinical and research activities of clinical psychologists. In 2003, D. W. Sue and Sue lamented the fact that evidence-based treatments very rarely incorporate significant numbers of minority clients in their research trials, so despite the growing number of evidence-based treatments, these treatments may not be applicable to diverse populations. Only 2 years later, however, Muñoz and Mendelson (2005) provided one of the first reports of a study attempting to establish empirical evidence for a treatment with a specific minority population. This report outlined the development and empirical evaluation of prevention and treatment manuals for depression and other mental health problems designed for San Francisco's low-income ethnic minority populations at San Francisco General Hospital. The authors noted many promising evaluations of these culture-specific manuals and concluded that "certain psychological theories describe universal aspects of human behavior and can thus profitably inform core therapeutic strategies. However, the effective clinical application of such strategies requires group-specific knowledge and cultural adaptation to increase the likelihood of positive outcomes" (p. 797). Additional research (M. Barrera & Castro, 2006; Lau, 2006; Spilka & Dobson, 2015) has significantly advanced the efforts to adapt evidence-based practices for specific cultural groups. In fact, as suggested earlier in the chapter, reviews of studies that examine the outcomes of evidence-based practices that have been tailored for a particular culture generally find that they work more effectively than nontailored versions (Huey, Tilley, Jones, & Smith, 2014; Soto et al., 2018).

AN EXAMPLE OF CULTURE INFLUENCING THE CLINICAL CONTEXT: THE PARENT–CHILD RELATIONSHIP

Let's consider the cultural issues related to a specific aspect of clients' lives that might be involved in the assessment or treatment of a wide variety of individuals and families: the relationship between parents and their children. Perhaps the first thing that the psychologists should recall is that their own expectations regarding parent–child relationships are probably influenced by their own culture and that those expectations don't hold true for everyone. For example, in some cultures (e.g., British), parenting that produces a child who grows up, moves out, and lives an independent life is usually considered successful. But families of other cultural backgrounds (e.g., Italian) usually prefer that their children

stay geographically and emotionally close even after they reach adulthood. Some cultures (e.g., Chinese) tend to insist that children obey parental authority without discussion or negotiation. But families of other cultural backgrounds (e.g., Jewish) tend to create a home life in which open discussion of feelings, including children disagreeing or arguing with parents, is tolerated or even encouraged (McGoldrick et al., 2005a). (Of course, there is heterogeneity within any of these groups.)

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Photo 4.5 Norms or expectations for the parent–child relationship can differ across cultures. How is this fact important to clinical psychologists who work with adults, children, or families?

It is essential that the clinical psychologist seeing an individual or family such as those described above attain the multicultural competence to consider these varying norms and implement a form of treatment that is consistent with them. Especially in the United States, where diversity is extensive and on the rise, clinical psychologists are likely to work with clients from a wide variety of backgrounds. And, as in this example, the cultural issue may serve as a backdrop to any number of presenting problems, including mood disorders, disruptive behavior disorders, relationship problems, and many others. A culturally sensitive appreciation of the Italian family, for example, might include exploration of parents' depressive feelings about a 25-year-old daughter whose successful career has served to separate her from them. But if the family were British, an exploration of depressive feelings in the parents might strike the clients as off the mark. Similarly, if an 11-year-old boy had a heated argument with his parents because he didn't want to take the piano lessons they had arranged, a culturally sensitive response might depend on whether the family were Chinese, Jewish, or of another cultural background.

CHAPTER SUMMARY

As the U.S. population has become increasingly diverse, multiculturalism has risen to prominence in clinical psychology. Evidence of its growing influence includes books and articles on diversity and multiculturalism; revisions to the *DSM*, including the inclusion of cultural concepts of distress; the creation of culturally relevant American Psychological Association divisions; and the addition of ethical standards directly related to culture. Cultural competence, for which all clinical psychologists should strive, involves cultural self-awareness, knowledge of diverse cultures, and culturally appropriate clinical skills. Knowledge of cultural norms should be accompanied by an appreciation of the heterogeneity of that culture and the likelihood that an individual may vary from some cultural norms, especially when acculturation strategies are considered. The tripartite model of personal identity suggests that an individual can be understood as an entirely unique person, similar to members of a cultural group, or similar to all human beings. Cultures are often defined by ethnicity or race, but numerous other variables may also constitute culture, such as gender, religion/spirituality, disability status, socioeconomic status, age, and sexual orientation. Training efforts intended to increase cultural sensitivity and competence among clinical psychologists and trainees include traditional coursework as well as direct interaction with members of diverse cultures.

KEY TERMS AND NAMES

acculturation 87
cultural competence 79
cultural concepts of distress 78
cultural diversity 73
cultural self-awareness 80
emic 91
etic 90
group level 91
heterogeneity 87
individual level 91
microaggressions 89
multiculturalism 74
subcultures 93
tripartite model of personal identity 91
universal level 91

CRITICAL THINKING QUESTIONS

1. In your opinion, how important is the issue of cultural self-awareness to clinical psychologists? What is the best way to increase cultural self-awareness among current members of the profession?
2. In your opinion, which level of the tripartite model of personal identity (individual level, group level, or universal level) is most important in the conceptualization of clients?
3. What are the pros and cons of defining culture in a narrow versus broad way?
4. If you were a client, how important would it be to you that your clinical psychologist had received training in cultural issues? Which methods of training would you expect to contribute most to your clinical psychologist's cultural competence?
5. Considering the discussion in Box 4.3 about heterogeneity within a culture, can you think of a cultural group to which you belong but within which you represent an exception to a cultural tendency?

LOOKING TOWARD GRADUATE PROGRAMS

The following table is a list of index entries in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018) relevant to the topics in this chapter (see page 25 in Chapter 1 for more information about the *Insider's Guide*). Keep in mind a few things as you consider these listings: They are current as of the 2018/2019 edition of the *Insider's Guide* (which typically revises every two years); they may have changed as faculty members retire, take new jobs, or develop new interests; and a graduate program's own website may have more extensive or up-to-date information.

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Acculturation	Aging/Gerontology	Gender Studies
African American Studies	Correctional Psychology/Prisons	Latino Psychology/Spanish Bilingual
Aging/Gerontology/Adult Development	Immigrant/Refugee Populations	Multicultural/Cross-Cultural/ Diversity
Asian Studies	Lesbian/Gay/Bisexual/ Transgender	

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Deafness/Hard of Hearing	Men's Issues	Geropsychology/Aging
Discrimination	Military/Armed Forces	Miscellaneous/Other—Global Mental Health
Gender Roles/Sex Differences	Minority/Cross-Cultural/ Multicultural	Miscellaneous/Other—LGBTQ
Hispanic Studies	Native American Health Services	Miscellaneous/Other—Rural Integrated Primary Care
Immigration	Religion/Spirituality	Miscellaneous/Other—Rural Mental Health
Indigenous/Native American	Rural Mental Health/Psychology	Miscellaneous/Other—Rural/ Multicultural Psychology
Lesbian/Gay/Bisexual/ Transgender	Spanish-Speaking Clients	Miscellaneous/Other—Women and Diversity Studies
Men's Issues	Underserved Populations	
Minority/Cross-Cultural/ Diversity	Women's Issues	
Poverty	Miscellaneous—Deaf and Hard-of-Hearing	
Religion/Spirituality	Miscellaneous—Indian Health Services	
Rural Mental Health	Miscellaneous—Inner-City Populations	
Veteran/Military Issues	Miscellaneous—Low Income	
Women's Studies/Feminism	Miscellaneous—Socioeconomic Disadvantage	
Miscellaneous and Other—American Indian		
Miscellaneous and Other—Cultural Adaptations		
Miscellaneous and Other—Cultural Humility		
Miscellaneous and Other—Filipino American Psychology		
Miscellaneous and Other—Intersectionality; Microaggressions		
Miscellaneous and Other—Interventions With Refugees		
Miscellaneous and Other—Microaggressions		
Miscellaneous and Other—Muslim Mental Health		
Miscellaneous and Other—Stereotyping and Prejudice		

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

Asian American Journal of Psychology
http://www.apa.org/pubs/journals/aap/index.aspx

Cultural Diversity & Ethnic Minority Psychology
http://www.apa.org/pubs/journals/cdp/index.aspx

Journal of Black Psychology
http://jbp.sagepub.com/

Journal of Latinx Psychology
http://www.apa.org/pubs/journals/lat/index.aspx

Psychological Services
http://www.apa.org/pubs/journals/ser/index.aspx

Psychology and Aging
http://www.apa.org/pubs/journals/pag/index.aspx

Psychology of Sexual Orientation and Gender Diversity
http://www.apa.org/pubs/journals/sgd/index.aspx

STUDENT STUDY SITE RESOURCES

SAGE edge™

Visit the study site at **edge.sagepub/pomerantz5e** for these additional learning tools:

- Self-quizzes
- eFlashcards
- Culture expert interviews
- In My Practice demonstration video
- Full-text SAGE journal articles
- Additional web resources
- Mock assessment data

5

ETHICAL AND PROFESSIONAL ISSUES IN CLINICAL PSYCHOLOGY

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What Makes Multiple Relationships Unethical? 114
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Clinical psychologists are obligated to behave ethically in all their professional activities (Hummel, Bizar-Stanton, Packman, & Koocher, 2017; Pope & Vasquez, 2016). As such, discussion of ethical issues occurs at numerous points in this textbook. For example, among our considerations of diversity and culture, we discuss a subset of ethical standards most relevant to cultural sensitivity and competence. In another chapter, we discuss the ethical standards most relevant to research. In this chapter, we focus entirely on ethical issues as they apply to the wide spectrum of clinical psychology.

AMERICAN PSYCHOLOGICAL ASSOCIATION'S CODE OF ETHICS

The American Psychological Association published its first **code of ethics** in 1953. The appearance of the initial code corresponded with the rise of professional psychology

Learning Objectives

5.1 Describe the five ethical principles and 10 ethical standards of the American Psychological Association's code of ethics.

5.2 Distinguish between the aspirational and enforceable components of the American Psychological Association's code of ethics.

5.3 Explain the limits of confidentiality for clients receiving services from a clinical psychologist.

5.4 Differentiate between ethical and unethical multiple relationships for clinical psychologists.

5.5 Determine key ethical considerations for psychological assessment and research.

5.6 Propose ways that a clinical psychologist could ethically navigate contemporary ethical issues that have arisen in the profession.

around that time period. Subsequently, nine revised editions of the ethical code have been published, including the most recent edition published in 2002 (Behnke & Jones, 2012; Koocher & Campbell, 2016). (Two amendments were added in 2010, emphasizing the fact that psychologists cannot use particular ethical standards to justify or defend the violation of human rights. These amendments relate to the disclosure that high-ranking members of the American Psychological Association had worked with the U.S. Department of Defense in their interrogation efforts, described by some as torture, of detainees during the war on terror [Bohannon, 2015; C. B. Fisher, 2017; Pope & Vasquez, 2016].) Since its inception, the code has applied not only to clinical psychologists but also to psychologists of all specialties. Some of its guidelines are especially relevant to the most common professional activities of clinical psychologists, such as therapy, assessment, research, and teaching.

Aspirational and Enforceable

The current American Psychological Association (2002) ethical code features two distinct sections: General Principles and Ethical Standards (Vasquez, 2015). Each of these sections steers psychologists toward ethical behavior in a different way. The items in the **General Principles** section are **aspirational**. In other words, they describe an ideal level of ethical functioning or how psychologists should strive to conduct themselves. They don't include specific definitions of ethical violations; instead, they offer more broad descriptions of exemplary ethical behavior. There are five general principles, and in Table 5.1, each appears alongside a selected sentence cited from the lengthier description included in the ethical code.

In contrast to the General Principles section, the **Ethical Standards** section of the ethical code includes **enforceable** rules of conduct. Thus, if a psychologist is found guilty of an ethical violation, it is a standard (not a principle) that has been violated. These standards are written broadly enough to cover the great range of activities in which psychologists engage, but they are nonetheless more specific than the general principles. Although each general principle could apply to almost any task a psychologist performs, each ethical standard typically applies to a more targeted aspect of professional activity. The ethical standards are divided into 10 categories (listed in Table 5.2), and, collectively, these 10 categories include 89 individual standards.

Web Link 5.1
Code of ethics

Web Link 5.2
Comparison of 1992 & 2002 ethical codes

In this chapter, we zoom in on some of the standards most relevant to clinical psychologists. Throughout our discussion, it's important to remember that the ethical code should be understood not only as a list of rules to follow and mistakes to avoid but also as a source of inspiration for ethical behavior of the highest order. Knapp and VandeCreek (2006) describe these two approaches to ethics as "remedial" and "positive"

Table 5.1 American Psychological Association's General Ethical Principles

Ethical Principle	Sample Sentence From Description in Ethical Code
Beneficence and Nonmaleficence	"Psychologists strive to benefit those with whom they work and take care to do no harm."
Fidelity and Responsibility	"Psychologists establish relationships of trust with those with whom they work. They are aware of their professional and scientific responsibilities to society and to the specific communities in which they work."
Integrity	"Psychologists seek to promote accuracy, honesty, and truthfulness in the science, teaching, and practice of psychology."
Justice	"Psychologists recognize that fairness and justice entitle all persons to access and benefit from the contributions of psychology and to equal quality in the processes, procedures, and services being conducted by psychologists."
Respect for People's Rights and Dignity	"Psychologists respect the dignity and worth of all people, and the rights of individuals to privacy, confidentiality, and self-determination."

Source: American Psychological Association (2002), 1060–1073.

Table 5.2 Categories of American Psychological Association Ethical Standards

1. Resolving Ethical Issues
2. Competence
3. Human Relations
4. Privacy and Confidentiality
5. Advertising and Other Public Statements
6. Record Keeping and Fees
7. Education and Training
8. Research and Publication
9. Assessment
10. Therapy

Source: American Psychological Association (2002), 1060–1073.

ethics, respectively. A remedial approach to ethics would involve doing just enough to avoid any trouble that might come from a violation of ethical standards, but a positive approach to ethics would involve making every effort to ensure that one's professional behavior was as consistent with ethical principles as possible. As an example, Knapp and VandeCreek consider the ethical obligation of competence (which we discuss in more detail later in this chapter). Psychologists with a remedial approach to ethics might do the bare minimum to make themselves competent for a particular activity (e.g., taking courses, getting supervision), but psychologists with a positive approach to ethics will strive to become as competent as possible (e.g., additional courses, extra supervision, self-study, self-care).

Ethical Decision Making

When any ethical issue arises, a clinical psychologist should be equipped with a process by which to make the most ethical decision possible. The American Psychological Association's (2002) ethical code does not offer any such decision-making models per se, but such models have been recommended by a number of experts in the field (e.g., Knapp & VandeCreek, 2006; Koocher & Keith-Spiegel, 2008; Pope, 2011; Treppa, 1998). One such expert is **Celia Fisher**, who served as chair of the American Psychological Association's Ethics Code Task Force, the committee responsible for creating the 2002 revision of the ethical code. In her book *Decoding the Ethics Code*, Fisher (2017) proposes a model for **ethical decision making**. In slightly adapted language, that model is presented here:

1. Prior to any ethical dilemma arising, make a commitment to doing what is ethically appropriate.
2. Become familiar with the American Psychological Association's ethical code.
3. Consult any law or professional guidelines relevant to the situation at hand.
4. Try to understand the perspectives of various parties affected by the actions you may take. Consult with colleagues (always protecting confidentiality) for additional input and discussion.
5. Generate and evaluate your alternatives.
6. Select and implement the course of action that seems most ethically appropriate.
7. Monitor and evaluate the effectiveness of your course of action.
8. Modify and continue to evaluate the ethical plan as necessary.

Before turning to any step-by-step ethical decision-making model, psychologists best prepare themselves to deal with ethical dilemmas by becoming generally ethical people with sound values. In other words, functioning as an ethical psychologist should not be a rote exercise in which a predetermined model for decision making is the sole guide. While such models can certainly enhance the psychologist's chances of making the most ethical decision, they work best when utilized by a person who has already examined his or her own values and aligned them with the ethics of the profession (Tjeltveit & Gottlieb, 2010).

Psychologists' Ethical Beliefs

The American Psychological Association's (2002) ethical code may instruct psychologists on how to conduct themselves ethically, but what do psychologists actually believe about the ethicality of various behaviors they might perform? In other words, as a group, what ethical beliefs do psychologists hold? This question, especially as it applies to psychotherapy-related behaviors, was addressed in a large-scale survey of American Psychological Association members (Pope, Tabachnick, & Keith-Spiegel, 1987). In this study, more than 450 members of Division 29 (Psychotherapy) of the American Psychological Association rated the ethicality of 83 separate behaviors that a psychologist might perform toward, with, or in response to a client. Results indicated that a few behaviors—for example, sex with clients or former clients, socializing with current clients, and disclosing confidential information without cause or permission—are viewed as blatantly unethical. In contrast, a few other behaviors—for example, shaking hands with clients, addressing clients by first name, and breaking confidentiality if clients are suicidal or homicidal—are viewed as unquestionably ethical. But most of the 83 behaviors fell in the gray area between ethical and unethical, illustrating both the challenges that psychologists face in making wise judgments regarding ethical issues and the importance of a sound model of ethical decision making (Cottone, 2012).

In the years since the 1987 study by Pope et al., other researchers have used similar methodologies to further examine the ethical beliefs of psychologists. One study found that psychologists' ethical beliefs may vary according to the point in time or the region of the country in which they are collected (Tubbs & Pomerantz, 2001), whereas others found that psychologists' ethical beliefs may vary according to the gender or age of the clients toward whom the behaviors may be directed (Pomerantz, 2012a; Pomerantz & Pettibone, 2005). Thus, although the American Psychological Association's (2002) ethical code serves as a guiding force, the beliefs psychologists actually hold, which correspond strongly with the behaviors they act out (Pope et al., 1987), may be subject to other influences.

CONFIDENTIALITY

One of the characteristics most closely associated with the ethical practice of clinical psychology is **confidentiality** (M. A. Fisher, 2012). In fact, confidentiality is specifically mentioned among the general principles (in Principle E: Respect for People's Rights and Dignity) and in numerous specific ethical standards—including Standard 4.01, "Maintaining Confidentiality," which begins, "Psychologists have a primary obligation and take reasonable precautions to protect confidential information" (American Psychological Association, 2002, p. 1066).

There is good reason for the emphasis on confidentiality in the profession of psychology: Our profession is entrusted by the public to provide professional services without sharing the private, personal details offered in the process. However, the public may be unaware of the fact that confidentiality is not absolute. Although most people outside of the mental health profession may assume that psychologists hold all information confidential (D. J. Miller & Thelen, 1986), the truth is that situations arise in which

psychologists are obligated to break confidentiality. Many such situations have been defined by court cases, including the well-known case involving the death of Tatiana Tarasoff.

Web Link 5.3
Tarasoff case

Tarasoff and the Duty to Warn

In 1969, Prosenjit Poddar was a student at the University of California at Berkeley. He became romantically interested in Tatiana Tarasoff, and when their relationship did not advance as he hoped, his mental state worsened and he sought psychotherapy at the university counseling center from a psychologist, Dr. Lawrence Moore. During a session in August 1969, Poddar told Dr. Moore that he intended to kill Tarasoff. Dr. Moore believed that Poddar's comment was credible, so he broke therapist–client confidentiality and contacted campus police. The campus police interviewed Poddar but did not hold him, because he promised to avoid Tarasoff and seemed rational at the time of the interview. Poddar never returned to therapy. Two months later, on October 27, 1969, Poddar killed Tarasoff by stabbing and shooting her. Tarasoff's parents later sued Dr. Moore and the others involved in the case for wrongful death. The court found that the psychologist was liable for failure to warn Tarasoff of the danger (Knapp & VandeCreek, 2006). Two years later, the court reheard the case and slightly revised their verdict, from a duty to warn to a duty to protect (which could involve something other than a direct warning by the therapist); however, the "duty to warn" language is still commonly used (Bucker & Firestone, 2000; DeMers & Siegel, 2016; Foellmi & Rosenfeld, 2015). Both "duty to warn" or "duty to protect" decisions suggest that the court saw Dr. Moore's actions as insufficient. Although he broke confidentiality and contacted campus police, the court determined that he should have made additional efforts to ensure that Tarasoff knew that she was in harm's way.

From the ***Tarasoff* case**, the **duty to warn** (and duty to protect) was born. Since the *Tarasoff* case set the legal precedent, clinical psychologists (and other therapists) have understood that there are limits to their confidentiality agreements with clients and that they have a duty to warn or protect people toward whom their clients make credible, serious threats. As stated in the first *Tarasoff* ruling, "The confidential character of patient-psychotherapist communications must yield to the extent to which disclosure is essential to avert danger to others. The protective privilege ends where the public peril begins" (*Tarasoff v. The Regents of the University of California*, 1974, p. 561, as cited in Tribbensee & Claiborn, 2003, p. 287).

Certainly, the rationale behind the duty to warn is clear: In the Tarasoff situation, a young woman's life could have been spared if a warning had been issued. And if we consider more recent tragedies—the mass shootings in Las Vegas and Newtown, Connecticut, or the September 11 terrorist attacks, for example—we can see how, if the perpetrators had disclosed their homicidal plans to therapists, the duty to warn could have prevented large-scale "public peril" (Pope, 2011). In its application, however, the duty-to-warn/protect issue is fraught with difficult questions for the clinical psychologists expected to uphold it. For example, how accurately can clinical psychologists assess the credibility of their clients' threatening statements or their intent to follow through with them? What kinds of threats merit warnings—only blatant life-or-death threats, or other kinds of harm, such as drunk driving or intimate partner violence as

well (Guedj, Sastre, Mullet, & Sorum, 2009; Welfel, Werth, & Benjamin, 2012)? At what point in therapy and to what extent should psychologists prioritize the protection of potential victims over the treatment of their clients (Bersoff, 1976; Knapp & VandeCreek, 2006; Tribbensee & Claiborn, 2003)?

In recent decades, clinical psychologists have faced a version of the duty-to-warn decision regarding clients with HIV/AIDS (Chenneville, 2000; Peter, 1998). To exemplify this dilemma, imagine that Paul, a 30-year-old, single, HIV-positive client, is seeing Dr. Reed, a clinical psychologist, for depressive symptoms. During the course of their conversations, Paul mentions to Dr. Reed that he is sexually active, that he sometimes engages in sexual activity that could put his partner at risk, and that he has withheld from his partner that he is HIV positive. Dr. Reed immediately faces numerous challenging questions: Does Paul's behavior constitute a threat worthy of a warning to the potential victim? How does it compare with Poddar's threat to kill Tarasoff? If Dr. Reed breaks confidentiality to warn a potential victim, what effect will that have on the therapeutic alliance with Paul and, ultimately, on Paul's well-being? To whom does Dr. Reed have a primary obligation—Paul or someone whom Paul might endanger? Unfortunately for clinical psychologists, there are no easy answers to questions of this type.

Video Link 5.1
Professional Issues and Ethics in Counseling & Psychotherapy: Threats of Violence

Adding to the confusion for many psychologists is the fact that different states have adopted different interpretations of the rules and precedents created by the *Tarasoff* case and others like it. For example, some states require the breaking of confidentiality in situations where other states allow (but don't require) it (G. A. H. Benjamin, Kent, & Sirikantraporn, 2009). States also differ on their definitions of a warning and what should trigger it. Some states' rules even mention that the protecting the client could take the form of intensified treatment or hospitalization of the client (without notifying the client), while other states' rules make no such mention (G. A. H. Benjamin & Beck, 2018).

The video available through Video Link 5.1 features the textbook author (with an actor playing the role of the client) navigating and commenting on numerous clinical situations in which questions related to breaking confidentiality could arise, including the client threatening (to varying degrees of credibility and specificity) different kinds of harm to another person.

When the Client Is a Child or Adolescent

More confidentiality-related challenges arise when clinical psychologists provide services to minors. One particular challenge centers on the fact that for many children and adolescents, the establishment of a close, trusting relationship with a clinical psychologist depends on the extent to which the psychologist reveals details of one-on-one conversations with the child's parents (Knapp & VandeCreek, 2006; Richards, 2003; Tribbensee & Claiborn, 2003). Simply put, kids might choose to withhold rather than discuss important personal issues if they know that their psychologists will subsequently share the information with the kids' parents. Of course, as legal guardians, parents are ultimately entitled to be informed about their children's progress in therapy; moreover, it's effective clinical practice to keep parents actively engaged in a child's therapy.

Sometimes, clinical psychologists can discuss this confidentiality dilemma openly with child clients and their parents, and a mutually agreeable arrangement can be reached (Knapp & VandeCreek, 2006; Richards, 2003). In other words, "therapists

BOX 5.1

In My Practice . . .

In my practice, I once had a client named Sophie, who caused me to think quite a bit about *Tarasoff* related issues. Sophie, who was 26 years old, had generalized anxiety disorder. Throughout our work together, Sophie shared lots of worries about her day-to-day life. She worried unnecessarily about her health, her friendships, her money, her job, and much more.

One day, she came in with a very specific worry: "You know how I told you the walls in my apartment building are so thin that I can literally hear what my neighbors are saying sometimes? Well, last night the guy next door was talking on the phone and I'm pretty sure I heard him say something about robbing someone. It sounded like he was making a plan with the other person on the phone. I heard him say things like, 'I know for sure that she won't be home all day Sunday. Let's do it then. She won't even know the stuff is missing. We'll totally get away with it. And if I'm wrong, and she is home, we'll just deal with it.'"

Sophie went on to share her anxiety about living next to someone who might be a criminal, and I did my best to help her deal with those worries. But in the back of my mind, and after her session ended, I considered the *Tarasoff* case. Of course, Sophie's situation had neither the clarity nor the high stakes of the actual *Tarasoff* case, in which the client explicitly stated his intentions, those intentions were life-threatening, and the person whose life was threatened was clearly identified. And in the end (spoiler alert), I did not break confidentiality in Sophie's case. In fact, I didn't give it very serious consideration. But many *Tarasoff*-related questions did occur to me. How dangerous does the potential behavior need to be to trigger a duty to warn? Do only life-or-death situations count, or could other forms of victimization (robbery, assault, kidnapping, identity theft, etc.) count as well under certain circumstances? How much should the credibility of a client's story be affected by the client's clinical issues—in this case, Sophie's tendency to worry about things unnecessarily? How much credence should be given to a client's secondhand reports—not dangerous behaviors they plan to do themselves, but dangerous behaviors they believe others might do? Once a client triggers thoughts in a psychologist's mind of a possible duty to warn, how much responsibility does the psychologist have to "dig" for more information beyond what the client has volunteered, which (as in Sophie's case) is often incomplete and unclear? When do psychologists' obligations to potential victims outweigh psychologists' obligations to their clients? These are the kinds of questions that make *Tarasoff*-related decisions especially challenging.

To see this box come to life as a whiteboard video, please visit edge.sagepub/pomerantz5e.

working with minor clients often negotiate an arrangement with their clients and their parents whereby the parents will not expect to be informed of what the client discloses in therapy except under specified conditions" (Tribbensee & Claiborn, 2003, p. 297). Those specified conditions should, at the very least, include any situation in which children might be harmed by the behavior of others or themselves, as well as general information about the child's psychological condition or progress in therapy. Even if the clinical psychologist offers the rationale that such an agreement would benefit the

child's well-being, parents may refuse it, and the acceptability of such an arrangement may depend on cultural variables involving parent–child relationships. Koocher and Daniel (2012) offer this script as a prototype of what the psychologist may say to an adolescent client and his or her parent(s) together at the outset of therapy:

> Psychotherapy works best when people have confidence in the privacy of their conversations. At the same time, parents do want to feel confident about their child's well-being and safety. Since parents were once teenagers, you certainly know that an adolescent may want to use therapy to talk about sex, alcohol, smoking, or other activities that parents may not approve of. Let's talk about how we can assure your child of confidentiality so s/he can talk openly about what's on her/his mind and at the same time assure your parents about your safety. (p. 12)

When parents do accept such an arrangement, they place quite a bit of trust in the psychologist's judgment. Consider some of the behaviors in which minor clients might engage: smoking, drinking, sex, drug use, crime, and "cutting" (self-mutilation), to name a few. At what point or under what conditions do these behaviors constitute harm that merits the psychologist's informing the child's parents? To what extent is the child's age a factor?

As a clinical example, consider Danica, a 17-year-old girl seeing Dr. Terry, a clinical psychologist. Danica's parents believe that Danica deserves some confidentiality with Dr. Terry, and they agree that Dr. Terry need not repeat the full contents of their sessions; however, they understandably insist that they be informed of any harm or danger that Danica may experience. As the sessions progress, the therapeutic relationship strengthens, and Danica begins to reveal to Dr. Terry details of her life about which Danica's parents are unaware. These details include the fact that Danica drinks alcohol about once a week (but does not get drunk), that she intentionally cut her forearm with a razor blade once a few months ago, and that one night she was a passenger in a car driven by a friend who may have been high. Do any of these behaviors or situations call for Dr. Terry to inform Danica's parents? If not, how much would the behaviors have to intensify before they did? What consequences might Dr. Terry be able to expect if she did or did not tell Danica's parents? Would the answers differ if Danica was 14, or 11, or 8 years old?

A separate confidentiality issue for clients who are minors involves **child abuse**. Every state has laws requiring mental health professionals to break confidentiality to report known or suspected child abuse (Benjamin & Beck, 2018; Kenny et al., 2017; Knapp & VandeCreek, 2006; Koocher & Daniel, 2012; Tribbensee & Claiborn, 2003). (Many states have similar laws pertaining to abuse of vulnerable adults as well.) The rationale behind such laws is similar to the rationale behind the *Tarasoff* ruling; namely, some situations demand that the clinical psychologist's primary responsibility shift to the immediate prevention of harm. And, like duty-to-warn situations, child abuse situations often require the clinical psychologist to make difficult judgment calls. It can be exceedingly challenging to determine with confidence whether child abuse is likely to have taken place, especially with children who may not be entirely forthcoming, who may exaggerate claims against their parents, or whose communication skills may be limited. The clinical psychologist's overarching goal of treatment may remain simple—the well-being of the child—but in cases in which child abuse is suspected, determining the means to attain this goal can become complex.

BOX 5.2

Considering Culture

Confidentiality, Ethnicity, and Family

Clients from diverse cultural groups, along with their relatives, may have very different beliefs regarding confidentiality in mental health (e.g., McGoldrick, Giordano, & Garcia-Preto, 2005b). An informative book on this topic, *Ethics, Culture, and Psychiatry* (Okasha, Arboleda-Florez, & Sartorius, 2000), focuses on such cultural differences regarding confidentiality and other ethical issues. It includes separate chapters on many specific groups, including African, Indian, Japanese, Latin American, German, West Mediterranean, and Chinese cultures. When directly compared with each other, two of these chapters—on Arab and Scandinavian culture—illustrate quite a contrast in beliefs regarding confidentiality. Regarding Arab culture, Okasha (2000) states the following:

- "In Arab culture, issues of illness are dealt with as family matters. Whether a patient is hospitalized, for example . . . is dependent not on what the patient wants himself or herself but on the estimation, need, or wish of the extended family." (pp. 24–25)
- "In the Arab region, a person may actually change doctors . . . if the doctor persists in considering the patient the only decision maker." (p. 26)
- "In Arab culture, the norm is to convey the [diagnostic] information to the family first and then leave it up almost entirely to the family to decide whether to inform the patient." (p. 27)

Regarding Scandinavian culture, Kastrup (2000) offers these descriptions:

- "The key role of patient autonomy has been increasingly emphasized in recent years. A patient has the right to make decisions about his or her treatment without interference from family." (p. 79)
- "Emphasis on autonomy is part of Scandinavian culture, and . . . lack of autonomy is considered the greatest unhappiness of the modern person. . . . The ideal is to be able . . . to live independently, without the need for support from others." (p. 79)

It is clear that Arab clients and Scandinavian clients—and possibly their family members as well—may enter into a working relationship with a clinical psychologist holding very different assumptions about confidentiality. In one culture, it is assumed that family members will be included, whereas in the other, it is assumed that family members will be excluded. Other cultures will certainly have their own unique assumptions about confidentiality as well.

In culturally diverse countries such as the United States, a clinical psychologist could see both Arab clients and Scandinavian clients. Even if these clients have similar diagnoses, clinical psychologists may find themselves working in dissimilar ways, especially in terms of the involvement of family members. To be ethical, of course, clinical psychologists must adhere to the American Psychological Association's (2002) code of ethics, yet to be culturally competent, they must also recognize that clients of various cultures will bring their own ethical values as well. If you were a clinical

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Photos 5.1 and 5.2
Assumptions about confidentiality may vary between cultural groups. In your opinion, how should clinical psychologists address these assumptions?

psychologist, how would you handle apparent conflicts between the ethical code and clients' cultural norms? Besides confidentiality, in what other areas of ethics might culture play a role in clients' expectations about clinical psychologists' behavior?

As a final note on confidentiality, it is important to note the difference between legal standards and ethical standards. Although they may differ in rare situations, the directives for clinical psychologists listed in this section—duty to warn in *Tarasoff*-like situations and required reporting of suspected child abuse—often represent both legal standards and ethical standards. That is, state law typically requires such behavior by clinical psychologists, and the American Psychological Association's (2002) ethical code includes standards that are consistent with these laws. In fact, one ethical standard (4.02, "Discussing the Limits of Confidentiality") specifically instructs clinical psychologists to "discuss . . . the relevant limits of confidentiality" with clients (p. 1066). Such a discussion is a key component of the informed consent process to which we now turn our attention.

INFORMED CONSENT

You may have been exposed to the notion of **informed consent** through psychological research. If you have participated in a psychological study, you probably received written information about the study first, and only after you provided consent by signing your name did the research begin. Research is certainly an important application of the ethical standards involving informed consent, but it is not the only one. Assessment and therapy also require informed consent according to the ethical code (American Psychological Association, 2002). Actually, in any professional activity conducted by psychologists, informed consent is an essential process. It ensures the person with whom the psychologist is working the opportunity to become knowledgeable about the activities

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Photo 5.3
In your opinion, to what extent should clinical psychologists share the content of a child's sessions with the child's parent(s)?

in which they may participate, and it facilitates an educated decision. Moreover, it affords individuals the opportunity to refuse to consent if they so choose.

Regarding research, Standard 8.02 ("Informed Consent to Research") of the American Psychological Association's (2002) ethical code instructs psychologists to inform prospective participants about numerous aspects of the study, including its purpose, procedures, and length of time it may require; any predictable risks or adverse effects; incentives for participation; and the right to decline or withdraw from participation. If the study is an investigation of a treatment method, psychologists should also inform clients that the treatment is experimental in nature, that some clients may be assigned to groups that receive no treatment (control groups), and of available alternative treatments outside the scope of the current study.

It is also necessary to obtain informed consent for psychological assessments. According to Standard 9.03 ("Informed Consent in Assessments" [American Psychological Association, 2002]), psychologists should offer information about the nature and purpose of the assessment; any relevant fees; the involvement of other parties, if any; and limits of confidentiality (e.g., duty-to-warn or child abuse situations).

Psychotherapy requires an informed consent process as well. Ethical Standard 10.01 ("Informed Consent to Therapy") explains that

> psychologists inform clients/patients as early as is feasible in the therapeutic relationship about the nature and anticipated course of therapy, fees, involvement of third parties, and limits of confidentiality and provide sufficient opportunity for the client/patient to ask questions and receive answers. (American Psychological Association, 2002, p. 1072)

Several phrases of this ethical standard highlight the fact that therapy stands apart from other professional activities of clinical psychologists. For one, the phrase "as early as is feasible" suggests that there may be different points at which information could be presented to clients. In fact, a survey of psychologists providing therapy found that in general, they are comfortable providing some generic information, such as payment and confidentiality policies, at the outset of therapy, but more specific information, such as the length, goals, and substance of psychotherapy, requires more time for therapists to get to know their clients. As a result, informed consent to therapy—unlike informed consent to research or assessment—may be best understood as an ongoing process rather than a one-time event (Pomerantz, 2005). Another noteworthy phrase in Standard 10.01 is "the involvement of third parties"; indeed, it has been a subject of debate exactly what or how much to tell clients about managed care's influence on the therapy process (e.g., J. Cohen, Marecek, & Gillham, 2006; Huber, 1997; Pomerantz, 2000). Finally, the phrase "provide sufficient opportunity for the client/patient to ask questions and receive answers" has been addressed by the publication of lists of questions that psychologists could offer to clients at the outset of therapy or at any other relevant point (e.g., J. M. Murphy &

Pomerantz, 2016; Pomerantz & Handelsman, 2004). From this list, clients could choose the questions in which they have interest, some of which may not have occurred to them on their own.

Especially in psychotherapy, the informed consent process presents the clinical psychologist the chance to begin to establish a collaborative relationship with the client (Pomerantz, 2012b, 2015). As we examine in more detail in later chapters, this kind of relationship is central to the success of psychotherapy of all kinds. Thus, it can be beneficial for clinical psychologists to invite clients to participate actively in the informed consent process and genuinely to join in the decision-making process regarding their treatment plan (Knapp & VandeCreek, 2006; Pomerantz & Handelsman, 2004).

BOUNDARIES AND MULTIPLE RELATIONSHIPS

In general, it can be problematic for clinical psychologists to know someone professionally—as, say, a therapy client or student—and also to know that person in another way—as, say, a friend, business partner, or romantic partner. The term used to describe such situations is **multiple relationships** (although the term *dual relationships* has also been used). It would be nice to state that psychologists never engage in such relationships, but such a claim would be false (Borys & Pope, 1989); in fact, a significant portion of complaints to the American Psychological Association Ethics Committee involve "incidents of blurred boundaries" (Schank, Slater, Banerjee-Stevens, & Skovholt, 2003, p. 182).

Defining Multiple Relationships

Web Link 5.4
Multiple relationships

Ethical Standard 3.05a (American Psychological Association, 2002) states that a multiple relationship

> occurs when a psychologist is in a professional role with a person and (1) at the same time is in another role with the same person, (2) at the same time is in a relationship with a person closely associated with or related to the person with whom the psychologist has the professional relationship, or (3) promises to enter into another relationship in the future with the person or a person closely associated with or related to the person. (p. 1065)

So multiple relationships can form not only when a psychologist knows one person both professionally and nonprofessionally but also when a psychologist has a relationship with someone "closely associated with or related to" someone the psychologist knows professionally. As an example, if Monique, a 36-year-old woman, is the therapy client of Dr. Davis, a clinical psychologist, Dr. Davis would be forming a multiple relationship if she became Monique's friend, business partner, or romantic partner. Additionally, Dr. Davis would be forming a multiple relationship if she became similarly involved with Monique's romantic partner, sibling, or close friend. For clinical psychologists

considering this ethical standard, defining "closely associated or related" can be a challenge, especially considering the "six degrees of separation" nature of our cities and communities. For example, would Dr. Davis be forming a multiple relationship if she dated Monique's second cousin, went into business with Monique's neighbor, or became a friend of Monique's coworker?

As indicated by the examples above, multiple relationships can take many forms (Sommers-Flanagan, 2012). Perhaps the most blatant and damaging are **sexual multiple relationships**, in which the clinical psychologist becomes a sexual partner of the client. The American Psychological Association's (2002) code of ethics offers a direct and inflexible standard about such behavior: "Psychologists do not engage in sexual intimacies with current therapy clients/patients" (Standard 10.05, p. 1073). Such behavior represents a fundamental breach of the healthy therapist–client relationship and often results in significant psychological or emotional damage for the client (Pope, 1994; Sonne, 2012). Psychologists are human, of course, and they may experience feelings of attraction toward a client from time to time. Surveys of psychologists and other therapists suggest that such feelings do in fact occur (Sonne & Jochai, 2014). The important issue is how psychologists deal with such feelings: preferably by discussing them in consultation with other professionals or perhaps their own therapist, but certainly not by acting on them (Barnett, 2014; Gelso, Pérez Rojas, & Marmarosh, 2014).

The nonprofessional involvement between client and clinical psychologist need not be sexual to constitute a multiple relationship or to cause the client harm. Indeed, psychologists may have opportunities to engage in a wide variety of **nonsexual multiple relationships**: friendships, business or financial relationships, coworker or supervisory relationships, affiliations through religious activities, and many others (Anderson & Kitchener, 1996; Zur, 2007). An essential task for clinical psychologists is to appropriately recognize the overlapping nature of these relationships as well as their potential to cause problems for the client.

What Makes Multiple Relationships Unethical?

Not every multiple relationship is, by definition, unethical. To help identify the specific elements of multiple relationships that characterize them as unethical, we again turn to Ethical Standard 3.05a:

> A psychologist refrains from entering into a multiple relationship if the multiple relationship could reasonably be expected to impair the psychologist's objectivity, competence, or effectiveness in performing his or her functions as a psychologist, or otherwise risks exploitation or harm to the person with whom the professional relationship exists. Multiple relationships that would not reasonably be expected to cause impairment or risk exploitation or harm are not unethical. (American Psychological Association, 2002, p. 1065)

As this standard indicates, there are essentially two criteria for impropriety in a multiple relationship. The first involves impairment in the psychologist; if the dual role with the client makes it difficult for the psychologist to remain objective, competent, or effective, then it should be avoided. The second involves exploitation or harm to

the client. Psychologists must always remember that the therapist–client relationship is characterized by unequal power, such that the therapist's role involves more authority and the client's role involves more vulnerability, especially as a consequence of some clients' presenting problems (Pope, 1994; Schank et al., 2003). Thus, ethical psychologists remain vigilant about exploiting or harming clients by clouding or crossing the boundary between professional and nonprofessional relationships. Above all, the client's well-being, not the psychologist's own needs, must remain the overriding concern.

As the last line of the standard above indicates, it is possible to engage in a multiple relationship that is neither impairing to the psychologist nor exploitive or harmful to the client. (And in some settings, such as small communities, such multiple relationships may be difficult to avoid. We discuss this in more detail later in this chapter.) However, multiple relationships can be ethically treacherous territory, and clinical psychologists owe it to their clients and themselves to ponder such relationships with caution and foresight. Sometimes, major violations of the ethical standard of multiple relationships are preceded by "a slow process of boundary erosion" (Schank et al., 2003, p. 183). That is, a clinical psychologist may engage in some seemingly harmless, innocuous behavior that doesn't exactly fall within the professional relationship—labeled by some as a "boundary crossing" (Gabbard, 2009b; Zur, 2007)—and although this behavior is not itself grossly unethical, it can set the stage for future behavior that is. These harmful behaviors are often called "boundary violations" and can cause serious harm to clients, regardless of their initial intentions (Cohen-Filipic, 2015a; Gutheil & Brodsky, 2008; Zur, 2009).

As an example of an ethical "slippery slope" of this type, consider Dr. Greene, a clinical psychologist in private practice. Dr. Greene finishes a therapy session with Annie, a 20-year-old college student, and soon after the session, Dr. Greene walks to his car in the parking lot. On the way, he sees Annie unsuccessfully trying to start her car. He offers her a ride to class, and she accepts. As they drive and chat, Annie realizes that she left her backpack in her car, so Dr. Greene lends her some paper and pens from his briefcase so she will be able to take notes in class. Dr. Greene drops off Annie and doesn't give his actions a second thought; after all, he was merely being helpful. However, his actions set a precedent with Annie that a certain amount of nonprofessional interaction is acceptable. Soon, their out-of-therapy relationship may involve socializing or dating, which would undoubtedly constitute an unethical circumstance in which Annie could eventually be exploited or harmed. Although such "boundary erosion" is not inevitable (Gottlieb & Younggren, 2009), minor boundary infractions can foster the process. As such, clinical psychologists should give careful thought to certain actions—receiving or giving gifts, sharing food or drink, self-disclosing their own thoughts and feelings, borrowing or lending objects, hugging—that may be expected and normal within most interpersonal relationships but may prove detrimental in the clinical relationship (Gabbard, 2009b; Gutheil & Brodsky, 2008; Zur, 2009).

The videos available through the Video Links feature the textbook author (with actors playing the roles of clients) navigating and commenting on numerous clinical situations in which questions related to multiple relationships could arise, including the client giving a gift to the therapist, the client requesting various kinds of interactions outside of therapy, and the therapist self-disclosing information.

Video Link 5.2
Professional Issues and Ethics in Counseling & Psychotherapy: Gift Giving

Video Link 5.3
Professional Issues and Ethics in Counseling & Psychotherapy: Multiple Relationships

Video Link 5.4
Professional Issues and Ethics in Counseling & Psychotherapy: Therapist's Self-Disclosure

COMPETENCE

The American Psychological Association's (2002) code of ethics devotes an entire section of ethical standards to the topic of **competence**. In general, competent clinical psychologists are those who are sufficiently capable, skilled, experienced, and expert to adequately complete the professional tasks they undertake (Nagy, 2012).

One specific ethical standard in the section on competence (2.01a) addresses the **boundaries of competence**: "Psychologists provide services, teach, and conduct research with populations and in areas only within the boundaries of their competence, based on their education, training, supervised experience, consultation, study, or professional experience" (American Psychological Association, 2002, p. 1063).

An important implication of this standard is that having a doctoral degree or a license in psychology does not automatically make a psychologist competent for all professional activities. Instead, the psychologist must be specifically competent for the task at hand (Spotts-De Lazzer & Muhlheim, 2016). As an example, consider Dr. Kumar, a clinical psychologist who attended a doctoral training program in which she specialized in child clinical psychology. All her graduate coursework in psychological testing focused on tests appropriate for children, and in her practice, she commonly uses such tests. Dr. Kumar receives a call from Rick, an adult seeking an intelligence test for himself. Although Dr. Kumar has extensive training and experience with children's intelligence tests, she lacks training and experience with the adult versions of these tests. Rather than reasoning, "I'm a licensed clinical psychologist, and clinical psychologists give these kinds of tests, so this is within the scope of my practice," Dr. Kumar takes a more responsible, ethical approach. She understands that she has two options: Become adequately competent (through courses, readings, supervision, etc.) before testing adults such as Rick, or refer adults to another clinical psychologist with more suitable competence.

Psychologists not only need to become competent, but they must also remain competent: "Psychologists undertake ongoing efforts to develop and maintain their competence" (Standard 2.03, American Psychological Association, 2002, p. 1064). This standard is consistent with the **continuing education** regulations of many state licensing boards. That is, to be eligible to renew their licenses, psychologists in many states must attend lectures, participate in workshops, complete readings, or demonstrate in some other way that they are sharpening their professional skills and keeping their knowledge of the field current.

Among the many aspects of competence that clinical psychologists must demonstrate is *cultural competence* (as discussed extensively in the previous chapter). Ethical Standard 2.01b (American Psychological Association, 2002) states that when

> an understanding of factors associated with age, gender, gender identity, race, ethnicity, culture, national origin, religion, sexual orientation, disability, language, or socioeconomic status is essential for effective implementation of their services or research, psychologists have or obtain the training, experience, consultation, or supervision necessary to ensure the competence of their services. (pp. 1063–1064)

Ethical psychologists do not assume a "one-size-fits-all" approach to their professional work. Instead, they realize that clients differ in important ways, and they ensure that

they have the competence to choose or customize services to suit culturally diverse clients (Salter & Salter, 2012). Such competence can be obtained in many ways, including through coursework, direct experience, and efforts to increase one's own self-awareness. Professional guidelines sponsored by the American Psychological Association, such as "Multicultural Guidelines: An Ecological Approach to Context, Identity, and Intersectionality" (American Psychological Association, 2017), "Guidelines for Psychological Practice With Transgender and Gender Nonconforming People" (American Psychological Association, 2015), "Guidelines for Psychotherapy With Lesbian, Gay, and Bisexual Clients" (Division 44, 2000), and "Guidelines for Providers of Psychological Services to Ethnic, Linguistic, and Culturally Diverse Populations" (American Psychological Association, 1993), can also be important contributors to cultural competence for clinical psychologists.

It is important to note that ethical violations involving cultural incompetence (e.g., actions reflecting racism or sexism) are viewed just as negatively by nonprofessionals as other kinds of ethical violations, such as confidentiality violations and multiple relationships (D. L. Brown & Pomerantz, 2011). In other words, cultural competence is not only a wise clinical strategy; it is an essential component of the ethical practice of clinical psychology that can lead to detrimental consequences for clients when violated (Gallardo, Johnson, Parham, & Carter, 2009).

The American Psychological Association's (2002) code of ethics also recognizes that psychologists' own personal problems can lessen their competence: "When psychologists become aware of personal problems that may interfere with their performing work-related duties adequately, they take appropriate measures, such as obtaining professional consultation or assistance, and determine whether they should limit, suspend, or terminate their work-related duties" (Standard 2.06, p. 1064). Of course, personal problems that impede psychologists' performance can stem from any aspect of their personal or professional lives (Barnett, 2008). On the professional side, the phenomenon of burnout among clinical psychologists has been recognized in recent decades (e.g., Grosch & Olsen, 1995; Morrissette, 2004). **Burnout** refers to a state of exhaustion that relates to engaging continually in emotionally demanding work that exceeds the normal stresses or psychological "wear and tear" of the job (Pines & Aronson, 1988). Due to the nature of the work they often perform, clinical psychologists can find themselves quite vulnerable to burnout. In one study of more than 500 licensed psychologists practicing therapy (Ackerley, Burnell, Holder, & Kurdek, 1988), more than one third reported that they had experienced high levels of some aspects of burnout, especially emotional exhaustion. In this study, the factors that increased a psychologist's susceptibility to burnout included feeling overcommitted to clients, having a low sense of control over the therapy, and earning a relatively low salary. A more recent study confirmed that overinvolvement with clients correlates strongly with burnout, particularly in the form of emotional exhaustion (J. Lee, Lim, Yang, & Lee, 2011). Another study of more than 600 clinical psychologists found that about two thirds had experienced mental health problems themselves, with the most common being depression, anxiety, and eating disorders (Tay, Alcock, & Scior, 2018). Other research suggests that burnout is more common among therapists who repeatedly think about frustrating aspects of their work, therapists with relatively little experience, and therapists who have high levels of neuroticism and perfectionism, but burnout is less common among therapists who believe they have sufficient social support (Rzeszutek & Schier, 2014; Simpson et al., 2019). The impairment associated with burnout can definitely translate into less effective treatment. One study

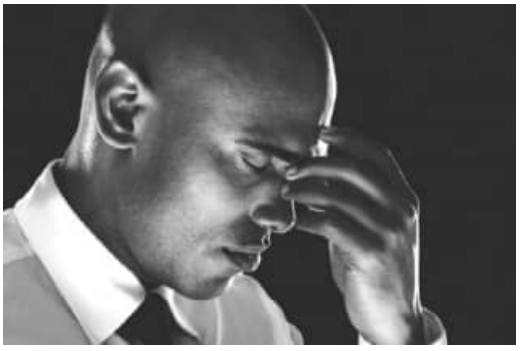

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Photo 5.4
Burnout can have a negative influence on a clinical psychologist's competence. If you were a clinical psychologist, what steps would you take to avoid burnout?

found a strong correlation between higher rates of therapist burnout and lower therapy outcomes in clients with depression and anxiety, mostly because of disengagement by the therapists (Delgadillo, Saxon, & Barkham, 2018).

Clearly, burnout and other factors can contribute to a level of impairment—in the form of depression, substance use, or other manifestations—that directly interferes with clinical work (Tamura, 2012; B. E. Williams, Pomerantz, Pettibone, & Segrist, 2010). So, as the ethical standard suggests, psychologists should take action to prevent or minimize their own impairment, including professional burnout. Such actions can include varying one's work responsibilities, keeping one's expectations reasonable, consulting with other professionals, maintaining a balanced and healthy personal life, or seeking their own psychotherapy as necessary (Barnett, 2008; Grosch & Olsen, 1995; P. L. Smith & Moss, 2009). (About 80% of therapists, including 92% of psychologists, seek their own therapy at some point during their careers [Orlinsky, Schofield, Schroder, & Kazantzis, 2011; Rønnestad, Orlinsky, & Wiseman, 2016]). Such efforts fall under the heading of *self-care*, a priority that should be near the top of the list for psychologists (and graduate students) for their own sake and for the sake of their clients (Wise & Barnett, 2016; Zahniser, Rupert, & Dorociak, 2017). At the same time, it is important for psychologists to remain alert to signs that they are experiencing impairment—which, to some extent, is a universal experience for those who stay in the profession—and take appropriate action when such situations arise (Good, Khairallah, & Mintz, 2009). Collective efforts can also be helpful; that is, not only can each psychologist look out for herself or himself, but clinical psychologists can look out for one another, both informally (e.g., among colleagues) and formally via efforts by professional organizations to promote self-care among their members (Barnett & Cooper, 2009).

ETHICS IN CLINICAL ASSESSMENT

Many of the principles and standards discussed so far in this chapter relate to all clinical activities, but there are some that address assessment specifically. For example, the American Psychological Association's (2002) code of ethics obligates psychologists to select tests that are appropriate for the purpose of the assessment and the population being tested. **Test selection** should entail a number of factors, including the psychologist's competence; the client's culture, language, and age; and the test's reliability and validity. Additionally, psychologists must not select tests that have become obsolete or have been replaced by revised editions that are better suited to the assessment questions being addressed. Sometimes, psychologists find themselves in a position of constructing a new test rather than selecting from existing tests. Psychologists involved in test construction should do their best to establish adequate reliability and validity, minimize test bias, and accompany the test with a coherent, user-friendly test manual (Pomerantz & Sullivan, 2006).

Test security represents another specific area of focus of the American Psychological Association's (2002) ethical code. Psychologists should make efforts to protect the security and integrity of the test materials they use. In other words, psychologists should prevent the questions, items, and other stimuli included in psychological tests from entering the public domain (Bersoff, DeMatteo, & Foster, 2012). When psychologists allow test materials to be taken home by clients, photocopied, or posted on Internet sites, not only might they be violating copyright laws, but they might also be allowing prospective test takers inappropriate access to tests. This could lead to preparation or coaching for psychological tests, which could, in turn, produce invalid test results. As described by Knapp and VandeCreek (2006), depending on the test in question, such invalid test results could place a nongifted student in a school's gifted program, a psychologically unstable police officer on the streets, or a child in the custody of an emotionally unfit parent.

Although psychologists should keep test materials secure, the American Psychological Association's (2002) ethical code explains that they are generally obligated to release test data to clients on request. **Test data** refers to the raw data the client provided during the assessment—responses, answers, and other notes the psychologist may have made. Although previous editions of the ethical code instructed psychologists not to release test data to clients, the current edition instructs psychologists to release test data unless there is reason to believe that the data will be misused or will harm the client. This revision reflects the more global shift toward patient autonomy in the health care field (C. B. Fisher, 2017).

ETHICS IN CLINICAL RESEARCH

The American Psychological Association's code of ethics includes numerous standards that apply to research of all kinds, including clinical research. So, just like psychologists from other specialty areas, clinical psychologists who conduct research are ethically obligated to minimize harm to participants, steer clear of plagiarism, and avoid fabrication of data, among other things (C. B. Fisher & Vacanti-Shova, 2012). Here, we will focus our discussion on one issue particularly relevant to an essential area of clinical research: *efficacy* of psychotherapy.

When clinical psychologists conduct empirical studies to measure how well a particular therapy works, they typically conduct the therapy in question with one group of participants, whereas a second group does not receive this therapy. What should the second group receive? This is a question with important ethical implications (Imber et al., 1986; Lindsey, 1984; Saks, Jeste, Granholm, Palmer, & Schneiderman, 2002). Although studies of this type may ultimately benefit many clients via the identification of evidence-based treatments, psychologists should be careful not to mistreat or harm some of their clients/research participants in the process.

Most commonly, the participants in therapy efficacy studies who don't receive the treatment being studied are placed in one of three conditions: no treatment (often called a "wait-list control" group), a placebo treatment (some kind of interpersonal interaction with a professional but with presumably therapeutic techniques deliberately omitted), or an alternate treatment (the efficacy of which may be unknown; Bjornsson, 2011;

Lohr & Adams, 2015). Is it ethical to provide any of these options to people who have psychological problems and have chosen, presumably with the hope of improvement, to participate in a study on its treatment? Of course, it is essential to inform participants before they consent to the study that some of them may not receive the treatment being studied or any treatment at all. Even if participants agree to this arrangement, the ethicality of the treatment they receive throughout a study of this type has been questioned (e.g., Arean & Alvidrez, 2002; Street & Luoma, 2002). Indeed, it is a significant ethical challenge for clinical psychologists to determine empirically the efficacy of their therapies without unduly exploiting or failing to help some of the participants while doing so (Saks et al., 2002).

Web Link 5.5
Burnout among therapists

CONTEMPORARY ETHICAL ISSUES

Managed Care and Ethics

As we discuss in another chapter, managed care exerts a strong influence on the current practice of clinical psychology. Among the new challenges it has presented are numerous ethical issues.

To begin, managed-care companies can put clinical psychologists in a position of divided loyalty. Although psychologists are ethically committed to "strive to benefit" and "safeguard the welfare" of their clients (American Psychological Association, 2002, p. 1062), they may be professionally pressured to minimize the services they provide to limit the cost of mental health care. In other words, clinical psychologists may find themselves in a tug-of-war between the managed-care companies' profits and their clients' psychological well-being (Alarcón, 2000; Wilcoxon, Remley, Gladding, & Huber, 2007). Moreover, to the extent that clients can perceive that the clinical psychologist has divided loyalties, the therapeutic relationship may suffer (L. J. Haas & Cummings, 1991).

To deal with this issue, the clinical psychologist could tell the client about loyalties to managed-care companies—and much more about managed care for that matter—during the informed consent process. Indeed, there are many aspects of managed mental health care that could be included in the informed consent process. For example, managed-care plans typically require that a *Diagnostic and Statistical Manual of Mental Disorders* (*DSM*) diagnosis be assigned to a client to qualify for payment (Ackley, 1997; Chambliss, 2000). Also, managed-care companies typically require that the clinical psychologist share at least some clinical information about the problem or its treatment with the company. To what extent should clinical psychologists discuss this and other facts of managed-care therapy with clients at the outset of therapy? Although the question is important, the answer is unclear (Acuff et al., 1999). Little empirical research has been conducted on the question, but some studies have suggested that information on managed care strongly impacts attitudes toward therapy (e.g., Pomerantz, 2000).

The aforementioned requirement that clients must be diagnosed with a *DSM* disorder for their managed-care companies to pay for treatment can present another ethical dilemma for clinical psychologists. If a client is struggling with a problem that does not meet the criteria for any *DSM* disorder, or if a family or couple has problems that don't stem directly from an identifiable disorder in one person, the psychologist can face

pressure to falsely assign a diagnosis to ensure that the managed-care company will pay the bill (Kielbasa et al., 2004; Pomerantz & Segrist, 2006; Wilcoxon et al., 2007). Or the psychologist may be tempted to "upcode," or assign a more serious diagnosis than the client's symptoms actually merit, to increase the number of sessions or amount of money the managed-care company will devote to treatment (C. B. Fisher, 2017; Parry, Furber, & Allison, 2009). Even if the psychologist's motivation is the client's welfare, such actions are unethical and can constitute the illegal act of insurance fraud as well (R. Miranda & Marx, 2003).

Technology and Ethics

Along with managed care, technological advances have led to changes in the practice of clinical psychology in recent years, and new ethical challenges have also arisen. For example, a quick Internet search will yield a wide array of so-called psychological tests of one kind or another, claiming to measure intelligence, personality, and other variables. Many of these tests have questionable validity or reliability, and the feedback they provide may be inaccurate and distressing to clients. Any clinical psychologist who creates or uses unscientifically sound tests of this type may be engaging in unethical practice (Buchanan, 2002). And online assessment raises many more ethical questions (e.g., Knapp & VandeCreek, 2006; Naglieri et al., 2004): As the client is taking the test, are the testing conditions standardized? Will the client keep the test materials secure? Is the client distracted by other tasks? Is the client actually the person completing the test? Are there important behavioral observations the psychologist might be missing because of the remote nature of the assessment?

Some of these concerns apply to online therapy practices as well (e.g., D. E. Shapiro & Schulman, 1996). When therapy is done via computer, the clinical psychologist and client may not be able to fully appreciate all aspects of communication (e.g., nonverbals). Moreover, online therapy gives rise to concerns about confidentiality and client identity that don't exist when the clinical psychologist works with the client in person (C. B. Fisher & Fried, 2003; Kraus, 2004).

Ethics in Small Communities

Although they have become increasingly acknowledged and researched in recent decades, the ethical challenges unique to small communities are certainly not a recent development. Clinical psychologists who work and live in small communities have always experienced these challenges (Werth, Hastings, & Riding-Malon, 2010). Rural areas and small towns may be the most obvious examples of small communities, but there are many others as well. Even within large cities, clinical psychologists can find themselves living and working in small communities defined by ethnicity, religion, or sexual orientation, or on military bases, at small colleges, or in similar settings (Hargrove, 1986; Schank, Helbok, Haldeman, & Gallardo, 2010; Schank & Skovholt, 1997, 2006; Schank et al., 2003).

Multiple relationships are a distinctively difficult ethical issue for clinical psychologists in small communities. In fact, "nonsexual overlapping relationships are not a matter of *if* as much as *when* in the daily lives of many small- and contained-community

psychologists" (Schank et al., 2003, p. 191). Unlike clinical psychologists in larger communities, those in small communities may not be able to live in one population and practice in another, so keeping personal and professional aspects of their lives entirely separate may prove impossible. Consider Dr. Villanueva, the only clinical psychologist in a remote town with a population of 1,500. Any of Dr. Villanueva's activities—shopping at the grocery store, working out at the gym, visiting a dentist—might require her to interact with a client or former client. And when we recall that multiple relationships can involve those close to clients (e.g., family, friends) in addition to clients themselves, the likelihood of such interactions increases tremendously. Dr. Villanueva may be the only qualified mental health professional in the community, so for her, referring clients to others may not be a viable option. Further complicating the situation is the fact that clients may know more about Dr. Villanueva than is typical in larger communities (Juntunen, Quincer, & Unsworth, 2018). For example, Dr. Villanueva may be working with a married couple who are considering divorce. If Dr. Villanueva gets a divorce herself and the couple finds out about it—more likely in a small community than in a large one—the couple may lose confidence in her ability to help them, whether such a loss of confidence is warranted or not.

For clinical psychologists such as Dr. Villanueva, it is wise to discuss the issue of multiple relationships with clients at the outset of psychological services as part of the informed consent process. Although multiple relationships may be somewhat inevitable, educating clients about the complications they can cause, as well as the psychologist's ethical obligations, can clarify boundaries and prevent misunderstandings. In addition, clinical psychologists in small communities should do their best to live a healthy, well-balanced personal life to ensure that they don't find themselves inappropriately leaning on clients to meet their own personal needs. And even if some degree of overlap is inevitable, clinical psychologists in small communities must nonetheless make every effort to avoid the impaired judgment and client exploitation that can make multiple relationships unethical and harmful (Curtin & Hargrove, 2010).

BOX 5.3

Metaphorically Speaking

If You've Played the "Six Degrees of Kevin Bacon" Game, You Understand Multiple Relationships in Small Communities

You may be familiar with the idea of "six degrees of separation," which suggests that any person on earth can be connected to any other person on earth within just six steps (in a "friend of a friend" kind of way). But have you ever played the "six degrees of Kevin Bacon" version?

It's nearly impossible to find a Hollywood actor or actress without a connection to Kevin Bacon. Hundreds have direct connections, meaning that they have costarred in films with him. Thousands more are indirectly connected to him, often through a surprisingly small number of links. Take Keira Knightley, for example. She's just two links away from Kevin Bacon—she appeared in *Love Actually* with an actress who appeared in *Wild Things* with Kevin Bacon. Dwayne "The Rock" Johnson? Melissa McCarthy? Samuel L. Jackson? Seth Rogen?

Meryl Streep?—all just a couple of steps away from Kevin Bacon. (Check www.oracleofbacon.org for more Kevin Bacon connections.)

The experience of clinical psychologists living and working in small communities might be a bit like Kevin Bacon's experience at a Hollywood party. For Bacon, everyone at the party could be a friend or the friend of a friend; for the clinical psychologist, everyone in the community could be a client or a client's friend, partner, brother, sister, parent, child, and so on. There are a few differences, however. For one, Kevin Bacon can skip the party if he chooses to avoid seeing people to whom he is connected, but that's not so easy for the clinical psychologist, whose personal life requires regular contact with community members. Also, Kevin Bacon doesn't need to be wary of ethical pitfalls as the clinical psychologist does; if he blurs relationship boundaries at the party, it might be called "networking," but if a clinical psychologist does so in the small community, it might result in confusion, reduced objectivity, or even exploitation.

In larger communities, clinical psychologists may have more options to avoid multiple-relationship predicaments. They might be able to find accountants, physicians, yoga instructors, or basketball coaches for their kids who have no connections to their clients. But in smaller communities, the degrees of separation are far fewer, so multiple relationships can be unavoidable. It is possible to manage this situation successfully, however, and the American Psychological Association's (2002) ethical code explicitly states that multiple relationships need not be unethical. An essential strategy for clinical psychologists in small communities includes communication with clients early in the professional relationship about the "ground rules" regarding multiple relationships and always steering clear of the most toxic elements of multiple relationships, such as exploitation and harm of clients.

CHAPTER SUMMARY

The American Psychological Association's code of ethics was originally created in 1953 and was most recently fully revised in 2002. The current code includes both aspirational principles and enforceable standards. In addition to a thorough knowledge of the code, a psychologist's ethical decision-making process should include a thoughtful consideration of the perspectives of all parties involved in the situation, consultation with trusted colleagues, and careful evaluation of all alternatives. Confidentiality is a particularly significant ethical issue for clinical psychologists. Although some clients may mistakenly believe that confidentiality is absolute, some situations necessitate a breach of confidentiality. For example, the ruling in the *Tarasoff* case assigns psychologists the duty to warn potential victims of harm, and state laws require psychologists to report ongoing child abuse. Policies regarding confidentiality and other aspects of psychological services should be addressed in the informed consent process. Unlike informed consent for research, which is typically a single event at the outset, informed consent for psychotherapy may be better understood as a process that continues over time as the psychologist is increasingly able to provide information specific to each client. Multiple relationships, in which a psychologist knows a client both professionally and nonprofessionally (or has a relationship with someone close to the client), can constitute ethical violations when they impair the psychologist's objectivity, competence, or

effectiveness, or when they exploit or harm the client. Not all multiple relationships are unethical, however, and in small communities, some degree of overlapping relationships may be unavoidable. Regarding competence, psychologists should (1) ensure that their professional activities match their training and expertise, (2) make efforts to avoid any negative impact of burnout or personal problems on their work, and (3) attain cultural competence for a diverse range of clients. Some contemporary developments have created additional ethical concerns for psychologists, including diagnostic and informed consent issues related to managed care and numerous ethical issues related to online assessment and therapy.

KEY TERMS AND NAMES

aspirational 102
boundaries of competence 116
burnout 117
child abuse 109
code of ethics 101
competence 116
confidentiality 105
continuing education 116
duty to warn 106
enforceable 102
ethical decision making 104
Ethical Standards 102
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informed consent 111
multiple relationships 113
nonsexual multiple relationships 114
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test selection 118

CRITICAL THINKING QUESTIONS

1. What is the primary difference between the aspirational and enforceable sections of the American Psychological Association's (2002) code of ethics? Which would you expect to have the most direct influence on the professional behavior of a clinical psychologist?
2. What conclusions do you draw from the research findings that psychologists' ethical beliefs may vary across time, client gender, client age, or other variables?
3. Provide separate lists of examples of client threats that, in your opinion, do and do not invoke the "duty to warn" as established by the *Tarasoff* case.
4. In your opinion, what information is most essential to informed consent for psychotherapy?
5. If you were a clinical psychologist, what efforts would you make to ensure that you did not experience personal problems or burnout that impaired your work?

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

Journal of Clinical Psychology
http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1097-4679

Journal of Contemporary Psychotherapy
http://link.springer.com/journal/10879

Ethics & Behavior
http://www.tandfonline.com/loi/hebh20#.V08aRfkrLX4

Professional Psychology: Research and Practice
http://www.apa.org/pubs/journals/pro/index.aspx

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- In My Practice demonstration video
- Full-text SAGE journal articles
- Additional web resources
- Mock assessment data

6

CONDUCTING RESEARCH IN CLINICAL PSYCHOLOGY

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Ethical Issues in Research in Clinical Psychology 145

Like the other specialty areas in psychology, clinical psychology addresses its fundamental questions via scientific study and empirical research. Of course, many of the research methods used by clinical psychologists overlap with those used by other psychologists and scientists (and with those that may have been covered in other courses you have taken by this point). The goal of this chapter is to place emphasis on how *clinical* psychologists conduct research, which involves the introduction of some research-related concepts as well as the reconsideration of others in this specialized context.

WHY DO CLINICAL PSYCHOLOGISTS DO RESEARCH?

Psychological Disorders

Perhaps the most fundamental reason clinical psychologists do research is to gain knowledge about psychological disorders (Zvolensky, Forsyth, & Johnson, 2013).

Learning Objectives

6.1 Describe the reasons why clinical psychologists perform research.

6.2 Explain the experimental method as used by clinical psychologists.

6.3 Differentiate among various experimental designs employed by clinical psychologists (e.g., between-group, within-group).

6.4 Speculate on the impact of technology on future clinical psychology research.

6.5 Indicate ethical issues that can arise in clinical psychology research.

Remember—there is no doctoral degree in psychology in "abnormal psychology" or "psychopathology" per se, so psychologists who choose to conduct basic research on psychological disorders are typically *clinical* psychologists. Other reasons for research that we will cover as we move through this section may be more applied (e.g., focusing on practice activities like therapy or assessment), but research on the psychological disorders is more basic, establishing a foundation for the field. A few examples:

- Is social anxiety disorder experienced differently by men and women? A study found that women are more likely to have the disorder and typically have more severe symptoms and a higher level of distress related to it. Women are more likely to have comorbid phobias, generalized anxiety disorder, and posttraumatic stress disorder (PTSD), whereas men are more likely to have comorbid substance abuse disorder and conduct disorder (Asher & Aderka, 2018).
- Do specific symptoms of PTSD predict suicidal behavior in military personnel more than others? A study found that hyperarousal symptoms (e.g., agitation) do predict suicidal behavior to a significantly greater extent than other symptoms of the disorder (I. H. Stanley, Rogers, Hanson, Gutierrez, & Joiner, 2019).
- People with borderline personality disorder often overreact to the possibility of interpersonal rejection, but is it possible that they also experience poor connectedness with others when they are being accepted and included? One study found that people with the disorder (compared to people without it) did in fact respond more negatively to positive social feedback—for example, they were less cooperative (Liebke et al., 2018).

Treatment Outcome

Another important reason clinical psychologists conduct research is to determine how well their therapies work (M. J. Lambert & Ogles, 2016; Spokas et al., 2008). Research on **treatment outcome** has a rather extensive history, with Hans Eysenck's (1952) unfavorable review of psychotherapy outcome serving as a landmark study and a catalyst for additional investigations on the subject. In the decades subsequent to Eysenck's review, researchers established that psychotherapy is indeed effective and ultimately moved on to exploring which particular therapies are most successful at treating which particular disorders (a topic covered in more detail in Chapter 11). Today, professional journals such as *Professional Psychology: Research & Practice; Psychotherapy: Theory, Research, Practice, Training;* and *Clinical Psychology: Science and Practice* frequently feature treatment outcome studies.

BOX 6.1

Measuring Therapy Outcome: Essential Questions

Clinical psychologists who investigate therapy outcome face many fundamental questions when designing their studies. The choices they make can have a profound influence on the data they collect and the conclusions they draw.

How should psychotherapy outcome be measured?

Questionnaires are an efficient, convenient way to evaluate therapy outcome. In fact, some of the most commonly used and widely accepted measures of therapy outcome are written surveys of some type. However, other possibilities certainly exist, including interviews and direct observation of the client in or out of therapy sessions. If questionnaires are the chosen method, more specific "how" questions arise, such as the length and format of the questionnaire.

What should be measured when assessing psychotherapy outcome?

Specifically, what aspects of a client do we hope to change via psychotherapy? Given the enormous breadth of problems for which people seek psychotherapy and the vast array of therapy techniques and approaches available, the answer can vary greatly: observable behavior, thought patterns or cognitive processes, emotional responses, interpersonal tendencies, awareness of internal mental processes, and academic or professional outcomes, among others. Researchers who conduct empirical inquiries about therapy outcome must consider which of these outcomes is most important for the purpose of their particular study and design their outcome measure accordingly.

When should psychotherapy outcome be measured?

The possible answers to this question vary widely as well. If the emphasis is evaluative (i.e., to determine after the fact how well the therapy worked), measurements could occur immediately after therapy ends or as "follow-up" efforts after a week, a month, a year, or any other chosen time period has elapsed. If the emphasis is formative (i.e., to determine how therapy is working while it is still happening, with the intent of making immediate adjustments), measurements could take place at each session or on a less frequent basis throughout the course of therapy. Of course, researchers could combine these approaches as well.

Who should rate psychotherapy outcome?

There are numerous parties who may have an interest and some knowledge regarding how psychotherapy turns out for a particular client. The client is an obvious source of information, but the reliability and validity of clients' opinions of their own therapy can be questionable as a result of their biases, expectations, and, in some cases, the very psychological issues that prompted the therapy. (For example, a depressed client who consistently focuses on the negative aspects of his life while ignoring the positive aspects may apply this negative bias toward the question of therapy outcome as well.) Moreover, if therapy involves multiple individuals (as in couples therapy or family therapy), different people may have very different opinions. Psychotherapists can provide their own perspectives, but these are subject to biases and other threats to objectivity as well. In addition to the client and the therapist, other parties can have a justified interest in therapy outcome, including spouses, family members, employers, law enforcement and judicial institutions, managed-care/insurance companies, and society as a whole (Strupp, 1996; Strupp & Hadley, 1977).

Efficacy Versus Effectiveness

It is important to distinguish studies of psychotherapy outcome that measure efficacy from those studies that measure effectiveness (Spokas et al., 2008; Truax & Thomas, 2003). *Efficacy* refers to the success of a particular therapy in a controlled study conducted with clients who were chosen according to particular study criteria. In short, the efficacy of a form of therapy is how well it works "in the lab," where it is practiced according to manualized methods and where outcome for treated individuals is compared via clinical trial with outcome for individuals who receive alternate or no treatment (Chambless et al., 1998; Garske & Anderson, 2003). Participants are often selected on the basis of meeting strict (e.g., diagnostic) criteria and are randomly assigned to either a treatment group or a control group (Durand & Wang, 2011). (In many cases, more than half of those interested in being participants are excluded. Those who are included are typically less severe, more likely to have comorbid diagnoses, and less likely to be from minority populations than are comparable clients in actual clinical settings [Rosqvist, Thomas, & Truax, 2011].) As an example, cognitive-behavior therapy has been listed as a treatment with demonstrated efficacy for panic disorder (Chambless et al., 1998) as a result of efficacy studies conducted with carefully selected clients in a controlled, quasi-experimental setting (e.g., Barlow, Craske, Cerny, & Klosko, 1989; D. M. Clark et al., 1994).

Effectiveness, on the other hand, refers to the success of a therapy in actual clinical settings in which client problems span a wider range, and clients are not chosen as a result of meeting certain diagnostic criteria. In short, the effectiveness of psychotherapy is how well it works "in the real world," or how well it translates from the lab to the clinics, agencies, hospitals, private practices, and other settings where clinical psychologists conduct therapy (Chambless et al., 1998; Garske & Anderson, 2003). For example, the November 1995 issue of *Consumer Reports* magazine featured a survey of its readers who had received psychotherapy in the early 1990s (Seligman, 1995). Overall, they provided a positive view of their psychotherapy experiences: More than 90% reported that therapy was beneficial; the type of therapy, the profession of the licensed provider, and the presence of medication were generally unimportant; and longer therapy tended to produce greater benefit. Additional findings were reported as well, but the essential point here is that this study provides data on the effectiveness, not the efficacy, of psychotherapy. This was not a study that was designed and controlled by researchers from the outset. An experimental or quasi-experimental design was not used, nor was an untreated control group included. Although we can be sure that these results are valid in the "real world," the lack of scientific rigor leaves us unsure that the therapy is undoubtedly the factor that produced the outcome. Issues of effectiveness and efficacy are considered further in Chapter 11.

Web Link 6.1
Article about *Consumer Reports* effectiveness study

When measuring the outcome of therapy in either an efficacy or an effectiveness study, researchers must be careful to distinguish *statistical significance* from *clinical significance* when interpreting their results. Simply put, statistical tests can, at times, indicate that a significant difference exists between two groups (e.g., those who received the treatment and those who did not). But a statistically significant difference doesn't necessarily mean that the two groups differ much in real-world terms (Kendall, Comer, & Chow, 2013). For example, consider a study in which the two groups both start therapy with average ratings of 80 on a 100-point depression scale—in other words, both groups are markedly depressed. After one group receives Therapy A, that group's average drops to

75, while the control group's average stays at 80. Especially if the number of participants in each group is high, that 5-point difference could turn up as significant according to a statistical test. But is that jump—a mere 5 points on a 100-point scale—clinically significant? Does it make a meaningful difference in the quality of the participants' lives? Probably not. For this reason, both statistical significance, which is assessed quantitatively, and clinical (or real-world) significance, which is measured more qualitatively, should be taken into account when interpreting the results of a therapy outcome study (Conner, 2010).

Internal Versus External Validity

The distinction between efficacy and effectiveness exemplifies how the fundamental research concepts of internal and external validity are applicable to clinical psychology. **Internal validity** refers to the extent to which the change in the dependent variable is due solely to the change in the independent variable. On the other hand, **external validity** refers to the generalizability of the result—to what extent is the same finding valid for different settings and populations?

When psychotherapy outcome researchers design an efficacy study, they aim to minimize any element of the study that may bring into question the causal connection between the treatment and the outcome (such as variability in client diagnosis or in therapist methods). This bolsters the study's internal validity. However, even in the most internally valid study, the extent to which its results are replicable in other settings with other clients—in other words, the generalizability or external validity—is not guaranteed. In fact, the more "controlled" and internally valid an efficacy study is, the more dissimilar it can become from therapy in the "real world," thus undermining external validity (Lebow & Jenkins, 2018). Efficacy studies are often respected for their internal validity but discounted for their external validity, whereas the opposite is often true for effectiveness studies (Scotti, Morris, Stacom, & Cohen, 2011; Spokas et al., 2008).

Another psychotherapy-related type of research that clinical psychologists do focuses on *dissemination strategies*—in other words, efforts to promote therapies that have demonstrated effectiveness in treatment studies to large numbers of therapists. Discovering through a research study that a particular type of therapy works is great, but the widespread use of that therapy depends on how broadly and persuasively those results are shared among people who conduct therapy and might choose to use it. Results of dissemination strategies show that therapists in clinical (as opposed to research) settings are often less than eager to accept and use evidence-based therapies and, in some cases, are quite resistant (Beidas, Mehta, Atkins, Solomon, & Merz, 2013; Stewart, Chambless, & Stirman, 2018). Some researchers have suggested that in order to increase the uptake of evidence-based therapies, thereby increasing the number of clients who receive them, dissemination strategies should shift from what has not worked especially well (e.g., offering lists of evidence-based treatments, providing guidelines according to disorder, trusting therapists to read and implement research themselves) to what might work in the future (e.g., directly advertising to consumers/clients about evidence-based therapies, requiring use of evidence-based therapies for licensure or payment, testing clinicians on competence with evidence-based therapies, changing accreditation standards of graduate programs to more strongly emphasize evidence-based therapies, and even threatening lawsuits) (Forman, Gaudiano, & Herbert, 2016).

BOX 6.2

Considering Culture

Treatments That Work, but for Whom?

When researchers investigate the efficacy of a particular form of therapy, they select clients who meet the criteria of interest for that study. For example, if the treatment targets generalized anxiety disorder (GAD), the researchers will select clients who meet the official diagnostic criteria of GAD and whose diagnostic profiles are free of other complicating factors or comorbidities.

What if the resulting population of selected clients is culturally limited? To what extent can the study's results be generalized to individuals from a broader range of diverse groups? Some have argued that the outcome studies that have formed the empirical basis of "what works" in psychotherapy lack minority participants and, moreover, that their conclusions regarding efficacy are not necessarily applicable to those who differ from the studies' participants in terms of ethnicity, gender, or other variables (e.g., Bernal & Scharrodel-Rio, 2001; Leong & Kalibatseva, 2013; Miranda, Azocar, Organista, Muñoz, & Lieberman, 1996; D. W. Sue & Sue, 2008; S. Sue, Cheng, & Sue, 2011). Muñoz and Mendelson (2005) stated, "Clinical trials have seldom included adequate samples of people of color. . . . Indeed, most empirically supported therapies were evaluated with samples drawn primarily from a White, middle class, English-speaking, female population" (p. 790). In many cases, the characteristics of the participants in clinical trials are not even specified when the results are shared (Callahan, Heath, Aubuchon-Endsley, Collins, & Herbert, 2013).

It is encouraging that numerous investigations of evidence-based treatments have found that many do, in fact, work as well with ethnic minority clients as with the majority culture. For example, Miranda et al. (2005) found that Latino/Latina clients and African American clients who have received evidence-based therapy for major depression have experienced results similar to those of white clients. However, studies of this type are limited, particularly for certain ethnic groups such as Asian American and American Indian populations. If additional research establishes more firmly that "what works" works for all—or if it determines that such equality requires certain adaptations for certain populations—the related problem of cultural disparities in access to care can receive more attention. That is, ethnic minorities simply receive less psychotherapy than those in the majority (Snowden & Yamada, 2005). So clinical psychologists should ensure not only that they provide members of diverse groups therapy that works for them specifically but, more fundamentally, that these clients have access to their services in the first place.

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Photo 6.1
When clinical psychologists conduct research on therapy outcome, how can they ensure that their results will apply to diverse clients?

Imagine that you are a clinical psychologist and you learned that the GAD study described above, as well as others examining

the same form of therapy, concluded that this therapy was efficacious. If the sample in these studies was predominantly white, middle class, English speaking, and female, how confident would you be in its potential for success with your clients who don't match these characteristics? If you were a client who did not match these characteristics, how would you feel about a clinical psychologist choosing this form of therapy for you?

Assessment Methods

Clinical psychologists conduct research to evaluate and improve the assessment methods they use with clients. Such studies may involve the development, validation, or expanded use of new instruments; the establishment of normative data for specific populations on an assessment tool; a comparison of multiple assessment tools to one another; or other research questions (Geisenger, 2016). Numerous professional journals, including *Assessment*, *Psychological Assessment*, and *Journal of Personality Assessment*, are dedicated primarily to empirical studies of assessment issues in clinical psychology.

Here are a few examples from the spectrum of research being conducted in this area.

- Can an Internet-based, self-administered version of a clinical interview match an in-person interview with a live clinician in terms of diagnosing attention-deficit/hyperactivity disorder (ADHD) in children? Researchers examined a new instrument of this type, the Diagnostic Interview Schedule for Children—Version 4 (DISC-IV), and found that, indeed, the correspondence between the two types of interviews (conducted with parents of the child in question) was extremely high (Steenhuis, Serra, Minderaa, & Hartman, 2009). The researchers acknowledge that further research is necessary, but this study is a promising first step toward establishing a potentially cost-effective, efficient, and convenient form of assessment.

- Speaking of ADHD, do men and women (as parents or teachers) rate boys' ADHD symptoms differently on rating scales? One study found that they do: Specifically, female teachers and parents provided consistently higher ratings of the boys' symptoms than male teachers and parents, in a way that would result in a significant difference in the number of boys who would be considered for the diagnosis (Anastopoulos et al., 2018).

- When psychologists assess depressive symptoms, does the mode of assessment—a self-report questionnaire versus an in-person interview—make a difference? One study found that among adults who were young and better educated, the answer was yes: Their reports of depressive symptoms were lower in the interview than on the questionnaire (Levin-Aspenson & Watson, 2018).

- Researchers conducted a study to compare English and Spanish versions of a commonly used written self-report measure of depression (Wiebe & Penley, 2005). Titled "A Psychometric Comparison of the Beck Depression Inventory-II in English and Spanish," the study concluded that the reliability and validity of the Spanish version of the test was comparable to that of the original English version.

• The Minnesota Multiphasic Personality Inventory-2 (MMPI-2) is the most widely used and psychometrically sound objective personality test, and one of its strengths is its ability to detect "faking" by the test taker. Hahn (2005) conducted a study to examine whether the scales that detect faking in North American, English-speaking clients are also effective in the translation of the MMPI-2 for Korean clients. Results indicated that the scales that identify "fake-good" and "fake-bad" test takers were equally adequate in the North American and Korean versions of the test.

Diagnostic Issues

Clinical psychologists conduct research to explore issues of diagnosis and categorization regarding psychological problems. Such studies may examine the validity or reliability of existing or proposed diagnostic constructs, the relationships between disorders, the prevalence or course of disorders, or numerous related topics. Such studies often appear in *Clinical Psychology: Research and Practice*, *Journal of Abnormal Psychology*, *Journal of Consulting and Clinical Psychology*, and other journals. A few recent examples include the following:

• Paranoid personality disorder is a commonly diagnosed personality disorder, but is paranoid personality best understood as a categorical or a dimensional characteristic? In other words, is paranoid personality something an individual "has" or "doesn't have," or is it something we all have to varying degrees? Edens, Marcus, and Morey (2009) empirically examined this question by reviewing hundreds of structured interviews and self-report questionnaires of individuals assessed for personality disorder characteristics. Their primary finding was that paranoid personality has a dimensional, rather than categorical, underlying structure. Research of this type can have an influence on the way psychological disorders are defined in future editions of the *Diagnostic and Statistical Manual of Mental Disorders (DSM)*.

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Photo 6.2 Professional issues—for example, the frequency and effect of discussing cultural differences directly with clients—are common topics of research within clinical psychology.

• When clinical psychologists assess children for ADHD or oppositional defiant disorder (ODD), should they use gender-specific norms? In other words, should girls be compared only with girls and boys be compared only with boys? Researchers recently examined this question using teacher ratings of about 1,500 elementary school students (Waschbusch & King, 2006). Results indicated that when gender-specific norms were used, a small group of girls were identified who had significantly elevated ADHD and ODD levels but were not diagnosable by standard *DSM* criteria.

• Lilenfeld, Wonderlich, Riso, Crosby, and Mitchell (2006) conducted a review of empirical studies on the relationship between eating disorders and personality disorders. Among other conclusions, the authors found that anorexia nervosa overlaps significantly with obsessive-compulsive personality disorder and that individuals predisposed to one of these may be similarly predisposed toward the other.

- Personality disorders are often assumed to be stable over time. Researchers recently tested this assumption by assessing personality disorder characteristics in the same clients five times over a 10-year period (Durbin & Klein, 2006). According to this longitudinal study, stability of personality disorder traits was not as strong as often assumed and was comparable with some Axis I disorders, such as anxiety disorders.

Professional Issues

Clinical psychologists also examine elements of their own profession through empirical research. They study clinical psychologists' activities, beliefs, and practices, among other aspects of their professional lives. *Professional Psychology: Research & Practice; Psychotherapy: Theory, Research, Practice, Training;* and *Journal of Clinical Psychology* are among the journals that feature such research. Some recent studies of this type include the following:

- Do the ethical dilemmas and problems psychologists encounter depend on whether they work in rural or urban communities? Helbok, Marinelli, and Walls (2006) conducted a survey to explore this question and discovered some significant differences. Among other findings, results suggested that small-town/rural psychologists were more likely than their urban counterparts to encounter multiple relationships with clients (i.e., to find themselves interacting with clients in another context besides the clinical work), and small-town/rural psychologists were also less likely to have access to professional resources such as medication prescribers and specialized services.

- How do psychologists compare to other mental health professionals (social workers, counselors, and marriage and family therapists) in terms of integrating client religion into therapy? One study found that the professions were generally similar in terms of attitude toward doing so, but that marriage and family therapists actually did so more frequently (Oxhandler & Parrish, 2018).

- Maxie, Arnold, and Stephenson (2006) surveyed licensed psychologists to assess the extent to which they discuss ethnic and racial differences directly with psychotherapy clients. The researchers found that while most psychologists have had such discussions with some clients in the past, they don't happen as consistently as might be expected or hoped. The psychologists reported that when they took place, these discussions facilitated the therapy process.

- Stevanovic and Rupert (2009) empirically investigated a simple question: "How do the professional lives of psychologists influence their personal and family lives?" (p. 62). They surveyed almost 500 psychologists and found evidence for both positive and negative "spillover." That is, psychologists reported that their work lives provided both enhancement and stress to their personal lives. Good news: Positive spillover, which often stemmed from a sense of accomplishment at work, was more common than negative spillover, which often stemmed from work-related emotional exhaustion.

- What attitudes do contemporary psychologists hold toward gay, lesbian, and bisexual lifestyles? Kilgore, Sideman, Amin, Baca, and Bohanske (2005) examined this question via a survey of psychologists. They found that psychologists are more likely to view such lifestyles as acceptable and nonpathological now than they have been in

the past. They also found that compared with male psychologists, female psychologists held more positive beliefs about gay, lesbian, and bisexual lifestyles and gay-affirmative therapy.

Teaching and Training Issues

Clinical psychologists also pursue research questions related to how to educate those entering the profession. Areas of study include training philosophies, specific coursework, opportunities for specialized training, and the outcome of particular training efforts. *Training and Education in Professional Psychology*, *Teaching of Psychology*, and *Journal of Clinical Psychology* are among the journals that often publish articles of this type. A small sample of recent articles in this area includes the following:

- When training graduate students in assessment, teaching them to write effective reports can be particularly challenging, with a variety of different styles of report writing available for the instructor to emphasize. Pelco, Ward, Coleman, and Young (2009) conducted an interesting study in which sample reports of various types were provided to elementary school teachers (who often receive such reports about their students), and the teachers were asked to compare the effectiveness of the reports. Results indicated that teachers preferred a report in which the results of different psychological tests were organized by theme, rather than one in which the results were reported test by test. They also gave higher ratings to the most "readable" report, as opposed to reports with higher-level professional terminology. Results of studies such as this can guide graduate programs in providing instruction regarding psychological report writing for particular audiences.

- To what extent do graduate programs in clinical psychology train students to manage potentially violent clients? Researchers examined this question by surveying graduate students regarding their exposure to violent clients and the training they had received in dealing with them (Gately & Stabb, 2005). Results indicated that, in general, graduate students encountered violent clients infrequently and that they felt their training to handle such clients was lacking. The researchers noted a correlation between trainees' perception of training to manage violent clients and their confidence level in handling such situations.

- Both clinical psychology and counseling psychology graduate students complete predoctoral internships. How is the placement process—applying and obtaining the internship—similar and different between these two types of training programs? Neimeyer, Rice, and Keilin (2009) studied thousands of placement experiences of graduate students and found that each type of program "matched" at higher rates with certain types of internship sites. Specifically, clinical psychologist students had a higher match rate with medical facilities, whereas counseling psychology students had a higher match rate with college counseling centers.

- At predoctoral and postdoctoral internship sites, which personality assessment techniques are most commonly included in the training experience? One study found that objective tests like the Minnesota Multiphasic Personality Inventory (MMPI),

Personality Assessment Inventory (PAI), and Millon Clinical Multiaxial Inventory (MCMI) were much more commonly taught than projective tests like the Rorschach, storytelling tests, and sentence completion tests (Stedman, McGeary, & Essery, 2018).

HOW DO CLINICAL PSYCHOLOGISTS DO RESEARCH?

The Experimental Method

Like other psychologists (and, more generally, scientists), clinical psychologists adhere to the **experimental method** whenever possible in their research efforts. In general, this method involves a number of discrete, sequential steps. It begins with an observation of events. Second, the clinical psychologist develops a hypothesis to explain the observed events. Included in this hypothesis development is the definition of independent and dependent variables. **Independent variables** are those variables in the study that are manipulated by the experimenter, whereas **dependent variables** are those variables that are expected to change as a result of changes in the independent variables. In short, it is hypothesized that the dependent variables "depend" on the independent variables. After the hypothesis is developed, the third step, empirical testing of the hypothesis, ensues. The fourth step takes place after this testing has been completed and involves altering the hypothesis to match the results and interpretations obtained during empirical testing (Greenhoot, 2003; Moyer & Gross, 2011).

In clinical psychology, experimental studies often take the form of **randomized clinical trials (RCTs)** (Machado & Beutler, 2017). (A note on terminology—in the clinical psychology literature, there is some overlap between the use of "randomized clinical trial" and "randomized control trial," both of which are abbreviated "RCT." They refer to very similar studies that involve random assignment of participants to either an experimental or control group.) In an RCT, researchers test the outcome of a particular, manualized therapy on a particular diagnosis. An RCT begins with recruitment and selection of participants who meet the criteria for the diagnosis. Then, each participant is randomly assigned to a group that receives the therapy (the experimental group) or another group that does not receive the therapy (the control group). Participants in the control group may receive no treatment, be placed on a waitlist, receive a placebo or nonspecific treatment, or receive a "standard" or "routine" type of therapy. In both groups, the target variables (typically objective measures of symptoms, like frequency of panic attacks or self-reported depression symptoms) are assessed at the outset as a baseline and at the end of the study (and sometimes months later as a follow-up), and the comparison between the two groups indicates the degree to which the treatment was helpful. RCTs have been lauded by some for raising the level of scientific rigor in therapy outcome studies. They maximize internal validity by ensuring that improvements in participants receiving the therapy are indeed due to the therapy rather than extraneous factors (Kendall et al., 2013). On the other hand, RCTs have been criticized for oversimplifying clinical problems by reducing them to objective and easily measured symptoms, and for producing results that may not translate to "real-world" settings where clients

may not match the clients selected for the study (Comer & Kendall, 2013; Goldfried, 2013; Hollon, 2015; Westen, Novotny, & Thompson-Brenner, 2004). The coverage of evidence-based therapy/manualized therapy as a controversy in Chapter 3 explores this issue in greater depth.

Quasi-Experiments

Research in clinical psychology often involves variables that the researcher is not entirely able to control. Ethical, practical, or other constraints often limit the researcher's ability to assign people randomly to certain conditions, make particular manipulations, or otherwise experimentally test certain hypotheses. In such cases, clinical psychologists may use a **quasi-experimental design** rather than a true experimental design. Quasi-experimental designs are less scientifically sound than experimental designs; nonetheless, they are frequently used in clinical psychology and have often yielded very meaningful and important results (Greenhoot, 2003; Moyer & Gross, 2011; Steiner & Kim, 2015).

As an example, consider a group of clinical psychology researchers who are interested in examining the outcome of a particular therapy for specific phobias. If the researchers were designing a true experiment, they would need to take identical individuals, induce in them identical fears, and then assign them to different conditions (perhaps treatment vs. no treatment, or treatment vs. alternate treatment) to measure the influence of the treatment. Obviously, such a method is implausible and unjustifiable for both practical and ethical reasons. Instead, the researchers may choose to use a quasi-experimental design, acknowledging that the individuals in the two comparison groups will necessarily have phobias of different objects and different intensities and that they will differ in many other ways as well (age, personality characteristics, ethnicity, etc.) even if the researchers make attempts to "match" the groups as much as possible. Such a study may not be as internally valid as a true experiment, but especially when many such studies converge on the same conclusions, they can provide important data.

Between-Group Versus Within-Group Designs

When clinical psychologists conduct a study with a **between-group design**, participants in different conditions receive entirely different treatments. Often, such studies involve two conditions or groups: One (the **experimental condition**) receives the experimental treatment, whereas the other (the **control group**) does not receive the experimental treatment (Moyer & Gross, 2011). The control group can take a number of forms, including no treatment at all (and no expectation for treatment), being placed on a wait list for treatment (such that treatment is expected later), receiving a placebo (in which the client receives a form of intervention that lacks the "active ingredient" of the experimental treatment), or receiving treatment-as-usual (another treatment already established as beneficial or typical for the disorder; Comer & Kendall, 2013; Kazdin, 2015). As an example of a between-group design, consider a clinical psychology

researcher who examines the relationship between physical exercise and depression. After taking a baseline measure of depression of all participants at the outset of the study, the researcher might assign some participants to Condition A, in which they follow an exercise regimen over 8 weeks, and other participants to Condition B, in which they do not follow the exercise regimen. The crucial question is whether changes in depression levels from pretest to posttest depend on whether a participant is assigned to Condition A or Condition B.

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Photo 6.3 A clinical psychologist conducting research on any topic—for example, the relationship between physical exercise and depression—makes numerous important choices regarding the design of the study.

By contrast, a **within-group design** involves comparisons of participants in a single condition to themselves at various points in time. Essentially, all participants experience the same condition or conditions (Moyer & Gross, 2011). For example, the same empirical issue—the relationship between physical exercise and depression—could be examined via a within-group design if the researcher introduced to all participants an exercise regimen lasting 8 weeks. The researcher could compare depression levels each week from start to finish. The crucial question is whether changes in depression levels depend on the point in time when depression is assessed.

At times, clinical psychology researchers will combine aspects of between-group and within-group designs, creating a **mixed-group design** (Moyer & Gross, 2011). As an example, researchers could merge the approaches to studying exercise and depression described in the two previous paragraphs by assigning participants to two conditions (exercise regimen vs. no exercise regimen) *and* by measuring changes of all participants on a weekly basis. This way, at the end of the study, the researchers would be able to compare the two groups with each other, and they would also be able to compare each group with itself at various points in time.

Analogue Designs

When clinical psychologists seek to study clinical populations or situations but are unable to access them adequately, they may conduct a study with an **analogue design**. A study of this type involves an approximation of the target client or situation as a substitute for the "real thing." At times, it can involve using participants whose characteristics resemble (but don't exactly match) those of the target population, or asking participants to remember or imagine themselves in a certain situation (McNeil & Hayes, 2015).

Consider again the physical exercise/depression research topic. If the researchers are primarily interested in the relationship of physical exercise on severely depressed individuals but cannot access this population in sufficient numbers, they might access a more available population, such as undergraduates at their university whose depression levels are mild to moderate according to a self-report depression questionnaire. Of course, if the researchers conduct the study with this analogue population and find that mood improved as exercise increased, they should not definitively state that this result is similarly applicable to the severely depressed.

Correlational Methods

Clinical psychology researchers often conduct a correlational study, especially when neither an experiment nor a quasi-experiment is plausible. Studies utilizing **correlational methods** examine the relationship that exists between two or more variables (G. Goldstein, 2011). Unlike experiments and quasi-experiments, correlational studies do not identify variables as either independent or dependent. Similarly, causality is not implied by the results of correlational studies. In other words, although a correlational study can conclude that one variable predicts or associates with another, it cannot conclude that one variable causes changes in another (MacDonald, Wong, & Dionne, 2015). Sometimes, correlational studies constitute important early steps in an evolution of research on a particular topic, leading eventually to more experimental or quasi-experimental studies.

Again, consider the research topic of the relationship between physical exercise and depression. If a correlational study produces results that include a negative correlation between these two variables (increased exercise is associated with decreased depression), numerous causal questions arise. Does increased exercise cause mood to improve? Does an improvement in mood cause exercise to increase? Do both of these changes—increased exercise and improved mood—cause each other in a reciprocal manner? Could both of these observed changes be by-products of some other cause that was outside the scope of this study? Although the link between the variables can be determined, none of these questions can be answered definitively by a study with a correlational design.

Statistically, correlations between two variables can range from +1 to –1. Correlations approaching +1 are positive correlations, suggesting that the two variables vary together—as A increases, B increases; as A decreases, B decreases. Correlations approaching –1 are negative correlations, suggesting that the two variables vary in an inverse way—as A increases, B decreases; as A decreases, B increases. Correlations near 0 are neither strongly positive nor negative and suggest that the two variables vary independently of each other.

In one correlational study, researchers examined therapy outcomes of more than 2,000 clients over a 5-year period who had taken sick leave from their jobs because of anxiety, depression, or similar disorders. They found a significant positive correlation (.5) between the amount of time clients wait to start therapy and the amount of time clients stay on sick leave—in other words, those who start therapy sooner get back to work sooner, and those who start therapy later get back to work later (Alonso, Marco, & Andani, 2018). (It may seem like one of these variables could cause the other, but correlation does not necessarily imply cause. Either of these variables could have influenced the other, or a third variable could have influenced them both.)

Case Studies

Unlike the methods described thus far that involve larger numbers of participants, **case studies** involve a thorough and detailed examination of one person or situation. Typically, they include descriptive observations of an individual's behavior and an attempt by the researcher to interpret it. The researcher may speculate about how explanations about the target individual may also apply to others who are similar in

some way. Also, case studies can stimulate more systematic research, and together, these forms of research can converge on important findings (Davison & Lazarus, 2007; Fishman, 2016b; Kazdin, 2011; Kratochwill, Mission, & Hagermoser, 2015). For example, in the mid-1960s, when obsessive-compulsive disorder was considered essentially untreatable, Dr. Victor Meyer published a series of case studies in which he described his successful treatment of patients with obsessive-compulsive disorder that emphasized exposure to the stimulus that caused anxiety. If the patient obsessed about dirt and compulsively washed her hands, Dr. Meyer had her touch dirty things, prevented her from washing her hands, and pointed out that nothing catastrophic actually took place. Meyer's case studies are widely considered the seed that grew into exposure and response prevention, the current treatment of choice for obsessive-compulsive disorder that gained its status through many larger-scale, controlled research studies (McKay, 2015; V. Meyer, 1966).

Case studies tend to be held in high regard by researchers who prefer the **idiographic approach**—emphasizing or revealing the unique qualities of each person—to the **nomothetic approach**—determining similarities or common qualities among people. Sometimes, case studies are qualitative in nature, such as Freud's descriptions of "rat man" and "wolf man" (as cited in Gay, 1995), Watson's descriptions of Little Albert (J. B. Watson & Rayner, 1920), and many other classic studies in clinical psychology.

Web Link 6.2
Debate between the idiographic and nomothetic approaches

Although case studies are not experiments, they can feature some experimental qualities. For example, a single-subject design study focuses on one person at various points in time under changing conditions. It is similar to the within-group design discussed previously, but the "group" consists of only one person. Commonly, such studies use some variation of an **ABAB design** (also known as a *reversal* design), in which a treatment is alternately applied and removed (with *A* and *B* each representing the presence or absence of the treatment). A variation of this design, called the *alternating treatments* design, involves two treatments alternating rather than a single treatment alternating with no treatment (Hurst & Jones, 2016). Although case studies such as these may have limited generalizability on their own, they can be a relatively inexpensive and convenient way to do research with direct clinical relevance (K. A. Freeman & Eagle, 2011; Photos, Michel, & Nock, 2008).

Meta-Analysis

Among the most powerful tools used by clinical psychology researchers is the **meta-analysis**. A meta-analysis is a statistical method of combining results of separate studies (translated into effect sizes) to create a summation (or, statistically, an overall effect size) of the findings (Hedges & Citkowicz, 2015). As its label implies, a meta-analysis is a study of studies, a quantitative analysis in which the full results of previous studies each represent a small part of a larger pool of data (Durlak, 2003; Thomas & Michael, 2011). So, although it is informative to read the results of a single study on, say, the benefits of psychotherapy, it can be far more enlightening to read a meta-analysis on the topic that encompasses many comparable studies (e.g., M. L. Smith et al., 1980; Wampold et al., 1997).

Web Link 6.3
History of meta-analysis in psychotherapy

Meta-analysis is a sophisticated statistical technique, and meta-analysts must ensure that their method is scientifically sound (Field, 2013; Del Re & Flückiger, 2016).

BOX 6.3

Metaphorically Speaking

If You Read Movie Reviews, You Understand Meta-Analysis

A bad review of a new movie will make you think twice about seeing it. But 10 or 20 bad reviews of the same movie, especially if they come from different sources, will definitely keep you away from it. We can't put too much trust in any single review, but when multiple reviews agree (or disagree) about a movie, we feel more confident in allowing our impression to be influenced by their collective opinions. In an informal way, we combine the separate impressions of the various reviewers into a single, overall impression. Stated another way, as we read each new review, we meld the "effect" the movie had on each reviewer into an overall, cumulative "effect."

Meta-analyses are a conceptually similar but methodologically more rigorous way of combining effects. In psychology meta-analysis, the melding happens quantitatively rather than qualitatively. Researchers collect studies that have investigated a similar topic and convert the results of each into an effect size, which provides a common numerical language to make the findings comparable and compatible, and then statistically merge them to form one overall effect size to encapsulate the multitude of findings.

The popular websites Rotten Tomatoes and Metacritic resemble meta-analyses. They each use the content of individual movie reviews to come up with a single statistic to represent the overall quality of a movie. (On Rotten Tomatoes, this is the "Tomatometer," and on Metacritic, it's the "Metascore.") With a glance at these single numbers, readers can learn not only what a particular reviewer thought about a movie but what many other reviewers thought of it as well.

Meta-analysts do the same thing (although using a more sophisticated statistical process) with the outcomes of studies on the same variable. For example, a meta-analysis of cognitive therapy for generalized anxiety disorder combined the results of 15 studies that examined this topic and found that, overall, cognitive therapy was much more effective than no therapy and at least slightly more effective than other kinds of therapy (Hanrahan, Field, Jones, & Davey, 2013). Another meta-analysis combined the results of 18 studies on remote cognitive-behavioral treatment of obsessive-compulsive disorder (through videoconference and other media) and found that the remote treatments were much more effective than control conditions and equally effective to face-to-face treatments (Wootton, 2016).

Whether evaluating the "effects" of movies or psychology research, rather than relying on a single study (or review), it is wise to rely on the meta-analysis, which provides an overview of the "effect size" of a treatment (or movie) across multiple research settings (or movie reviewers).

According to Durlak (2003), meta-analysts in clinical psychology should incorporate the following five steps:

1. Formulate the research question, including reasonable, important, testable hypotheses.

2. Obtain a representative study sample. Determine and explain the criteria by which individual studies will be included or excluded. Consider unpublished reports as well as those that have been published.

3. Obtain information from individual studies. Take into account the specific methodology of each individual study and ensure that reliability is adequate when coding previous findings.

4. Conduct appropriate analyses. When combining effect sizes from individual studies, assign appropriate weights to each, such that larger-scale studies "count" more than smaller-scale studies.

5. Reach conclusions and offer suggestions for future research.

Box 6.3 offers further discussion of meta-analysis in clinical psychology.

Cross-Sectional Versus Longitudinal Designs

Cross-sectional designs assess or compare a participant or group of participants at one particular point in time. By contrast, **longitudinal designs** emphasize changes across time, often making within-group comparisons from one point in time to another (A. G. C. Wright & Markon, 2016). Cross-sectional designs tend to be easier and more efficient, whereas longitudinal designs require much longer periods of time. However, cross-sectional studies may not provide valid approximations for changes that take place or evolve over time, whereas longitudinal studies may have a unique ability to do so.

Reconsider the research topic of the relationship between physical exercise and depression. A longitudinal study of this topic might be quite different from any variation we have considered previously. For example, rather than focusing on an 8-week time period, the researchers could extend the scope of the study to 10 years. They could follow the participants to determine whether the relationship of the two variables remains the same or changes over that time. They could consider such issues as long-term adherence to the exercise regimen, the effect of physical injury and aging, lifestyle changes, spontaneous remission of depression, recurrence of depression, and other issues that would have little opportunity to arise in a cross-sectional study.

Actual longitudinal studies are often much longer than 10 years. For example, Kamen, Cosgrove, McKellar, Cronkite, and Moos (2011) followed individuals for 23 years to assess the relationship between family support and depression, and found that those with higher family support at the beginning (and throughout) predicted lower levels of depressive symptoms. In a 15-year study, Lönnqvist, Verkasalo, Mäkinen, and Henriksson (2009) found that neuroticism among 20-year-olds predicted the likelihood of mental disorders and low self-esteem among the same individuals at age 35. Such results would simply be unattainable by any method other than longitudinal research.

Use of Technology in Clinical Psychology Research

Increasingly, clinical psychologists are involving new forms of technology in their research. Sometimes, they use the technology to collect data (Trull & Ebner-Priemer, 2014). For example, one group of researchers examined the extent to which negative thoughts in the evening impaired sleep and decreased mood the next day. They measured negative thoughts and mood by sending e-mails to the smartphones of participants at eight randomly selected times per day. Each e-mail provided a link to a website at which participants immediately completed a brief survey about either their negative thoughts or their mood. Sleep quality was measured by actigraphs (similar to a Fitbit tracker) worn on the participants' wrists each night. This high-tech method of data collection—often called *experience sampling*, for its ability to tap into participants' behavior in real time—produced results indicating that negative thoughts did in fact significantly correlate with reduced sleep quality and poorer mood the next day (Takano, Sakamoto, & Tanno, 2014).

Another way that clinical psychologists use technology to collect data is by using Amazon Mechanical Turk, or MTurk, to recruit participants. MTurk is an Internet-based crowdsourcing marketplace run by the Amazon corporation that matches researchers with willing participants. Clinical psychologists are increasingly turning to MTurk as an alternative to undergraduate psychology students or other available pools of participants. MTurk has been available since 2007, but its use has increased dramatically since 2011. There are hundreds of thousands of people registered as MTurk "workers" (potential participants, essentially), most of whom live in the United States or India. Here's how it works: The researcher posts a description of the study, including criteria for eligibility (age, gender, employment status, etc.) and the amount of payment available. Interested and eligible MTurk workers select the study and complete their participation online. The researcher downloads the data and moves forward with the study. Research on U.S. MTurk participants suggests that, as a group, they differ slightly from the U.S. general public: They are slightly younger, better educated, more introverted, and more likely to be European American or Asian American than other ethnicities, for example. Even with these characteristics, the MTurk population is at least as representative as other commonly used samples, such as college students or samples drawn from specific local communities (Chandler & Shapiro, 2016; D. N. Shapiro, Chandler, & Mueller, 2013). Psychologists have used MTurk to study a wide range of topics such as perceptions of mental disorders, attitudes toward therapy, prevalence of specific psychological symptoms, and other clinically relevant topics. For example, one group of researchers recruited several hundred MTurk participants to examine how biological explanations for the cause of generalized anxiety disorder influenced their expectations to overcome the disorder. They found that participants who received the biological explanation expected the disorder to last significantly longer than those who did not (Lebowitz, Pyun, & Ahn, 2014).

Additionally, clinical psychologists have involved technology in their research by focusing on the technology as a clinical intervention. In other words, they have examined how well a particular use of technology works as a component of therapy. For example, researchers studied the benefits of a smartphone app designed specifically for the treatment of alcohol use disorder and found that it worked about as well as alternate treatments (Gonzalez & Dulin, 2015). Another group of researchers examined the outcome

of a parent training program for kids with disruptive behavior disorders in which all parents received the face-to-face training, but only half got an additional package of "technology enhancements" delivered to their smartphones, including videoconferencing, instructional videos, and brief daily surveys and reminders. They found that the technology enhancements improved both parental engagement in treatment and children's behavior change (Jones et al., 2014). One study found that sending motivational texts to clients' smartphones improved the benefits of therapy for eating disorders, and another found that sending texts to clients shortly before their appointments significantly increased attendance in therapy (Delgadillo, Moreea, Murphy, Ali, & Swift, 2015; Shingleton et al., 2016).

ETHICAL ISSUES IN RESEARCH IN CLINICAL PSYCHOLOGY

Of course, clinical psychology researchers must adhere to ethical standards when conducting research. In the ethics chapter of this textbook, we discuss in detail the ethical issues involved in psychotherapy outcome studies in which therapy is withheld from some of the participants (e.g., Saks et al., 2002; Street & Luoma, 2002). Those issues include the use of a control condition in an RCT (e.g., no-treatment control or waitlist control) that prevents the participant from receiving help for their psychological disorder. Other ethical issues that can arise in clinical psychology research include a participant in a study involving assessment receiving feedback that upsets them (e.g., they are diagnosable with a disorder), or a participant in a study that involves a clinical interview reveals suicidal intentions or ongoing child abuse but the researcher does not respond adequately in terms of breaking confidentiality (Koocher, 2013).

The American Psychological Association's (2002) code of ethics includes many specific standards that apply to clinical psychologists conducting research (C. Miller & Williams, 2011). Box 6.4 offers some of the most relevant of these standards, additional discussion of which appears in Chapter 5.

BOX 6.4

Key American Psychological Association Ethical Standards Related to Research in Clinical Psychology

- Researchers should obtain informed consent from prospective participants. The information they present should include the purpose and duration of participation as well as the participants' right to decline or withdraw. (Standard 8.02)
- Researchers should not coerce participation by offering excessive or inappropriate rewards (financial or otherwise). (Standard 8.06)
- Researchers should avoid using deception in their studies unless the

(Continued)

(Continued)

prospective value of the research justifies its use and nondeceptive methods are not feasible. If deception is used, the researcher should explain it fully to participants as soon as possible and allow them to withdraw their data if they choose. (Standard 8.07)

- If researchers discover that research procedures have harmed a participant, they should take reasonable steps to minimize the harm. (Standard 8.08)
- Researchers should never fabricate or falsify data. (Standard 8.10)
- Researchers should list themselves as authors of a study only when their contributions are justifiably substantial and the order of authorship reflects the relative contributions of the individuals involved. (Standard 8.12)
- Researchers should share their data with other competent researchers who intend to reanalyze it for verification. (Standard 8.14)

Source: Adapted from the American Psychological Association's (2002) Code of Ethics.

CHAPTER SUMMARY

Clinical psychologists conduct empirical research for a number of purposes. For example, empirical studies examine the outcome of treatment, in terms of both efficacy (how well a therapy works under controlled conditions with selected clients) and effectiveness (how well a therapy works in "real-world" clinical settings with a wider variety of clients). Clinical psychologists similarly conduct empirical research to study assessment methods, diagnostic issues, professional issues, and training issues. Often, this research follows the experimental or quasi-experimental method, in which the impact of an independent variable on a dependent variable is measured. In other cases, a correlational design is employed to establish a noncausal link between two variables. Between-group designs typically involve a comparison of a control (or no-treatment, or placebo) group to an experimental group, whereas within-group designs typically involve a comparison of the same group to itself at different points in time. Case studies have limited generalizability but can provide important clinical information or lead to more large-scale research. When it is impractical to study a clinical phenomenon as it happens, researchers can conduct an analogue study, in which the situation is simulated in the lab. Meta-analyses enable researchers to statistically combine the results of individual studies regarding the same research question, producing a single overall estimate of effect size. Increasingly, research in clinical psychology incorporates technology, either as a means to collect data (such as experience sampling by smartphone or finding participants via MTurk) or as the subject of outcome studies (such as the impact of technology utilized as part of therapy).

KEY TERMS AND NAMES

ABAB design 141
analogue design 139
between-group design 138
case studies 140
control group 138
correlational methods 140
cross-sectional designs 143
dependent variables 137
experimental condition 138
experimental method 137
external validity 131
idiographic approach 141
independent variables 137
internal validity 131
longitudinal designs 143
meta-analysis 141
mixed-group design 139
nomothetic approach 141
quasi-experimental design 138
randomized clinical trials (RCTs) 137
treatment outcome 128
within-group design 139

CRITICAL THINKING QUESTIONS

1. What is the difference between efficacy and effectiveness? Why is this distinction important to clinical psychologists involved in research and practice?
2. What can clinical psychologists do to maximize the generalizability of an analogue study to a broader population?
3. In your opinion, how important should case studies be in the establishment of a treatment method as valid?
4. Considering the questions raised in Box 6.1, when, in your opinion, should the outcome of psychotherapy be measured?
5. Considering the questions raised in Box 6.1, who, in your opinion, should rate the outcome of psychotherapy?

LOOKING TOWARD GRADUATE PROGRAMS

The following table is a list of index entries in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018) relevant to the topics in this chapter (see page 25 in Chapter 1 for more information about the *Insider's Guide*). Keep in mind a few things as you consider these listings: They are current as of the 2018/2019 edition of the *Insider's Guide* (which typically revises every two years); they may have changed as faculty members retire, take new jobs, or develop new interests; and a graduate program's own website may have more extensive or up-to-date information.

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Dissemination/Implementation/Translation Methodology Psychometrics/Measurement Statistics	Dissemination	Quantitative/Statistics Miscellaneous/Other—Applied Methodology Miscellaneous/Other—Experimental Psychopathology Miscellaneous/Other—Research Methods/Statistics

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

Clinical Psychological Science

http://www.psychologicalscience.org/index.php/publications/journals/clinical

Clinical Psychology: Science and Practice

http://onlinelibrary.wiley.com/journal/10.1111/(ISSN)1468-2850

Journal of Clinical Psychology

http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1097-4679

Journal of Consulting and Clinical Psychology

http://www.apa.org/pubs/journals/ccp/index.aspx

Professional Psychology: Research and Practice

http://www.apa.org/pubs/journals/pro/index.aspx

Psychotherapy Research

http://www.tandfonline.com/loi/tpsr20#.V08V2fkrLX4

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- Full-text SAGE journal articles
- Additional web resources
- Mock assessment data

PART II

ASSESSMENT

CHAPTER 7 • **Diagnosis and Classification Issues: *DSM-5* and More**

CHAPTER 8 • **The Clinical Interview**

CHAPTER 9 • **Intellectual and Neuropsychological Assessment**

CHAPTER 10 • **Personality Assessment and Behavioral Assessment**

7

DIAGNOSIS AND CLASSIFICATION ISSUES

DSM-5 and More

Defining Normality and Abnormality 151
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- *Before the* DSM 156
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- DSM—*More Recent Editions* (III, III-R, IV, *and* IV-TR) 158
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- *Criticisms of the* DSM 168

Alternative Directions in Diagnosis and Classification 174

DEFINING NORMALITY AND ABNORMALITY

Much of the practice and research conducted by clinical psychologists focuses on **abnormality**, also known as mental disorders, psychiatric diagnoses, or, more broadly, psychopathology. Through their training and their professional activities, clinical psychologists become very familiar with the definitions of various forms of abnormal behavior and the ways it differs from normal behavior. But before these diagnostic categories are put to use by clinicians and scholars, they must be defined.

What Defines Abnormality?

Over time, mental health professionals have put forth a variety of answers to the question of what makes behavior abnormal. These answers have included criteria such as personal distress to the individual (as in severe depression or panic disorder), deviance from cultural norms (as in many cases of schizophrenia), statistical infrequency (as in rarer disorders such as dissociative identity disorder), and impaired social functioning (as in social phobia and, in a more dangerous way, antisocial personality disorder).

Learning Objectives

7.1 Compare major definitions of abnormality proposed in the field of clinical psychology.

7.2 Describe the evolution of the *DSM* from its first edition to the current *DSM*-5.

7.3 List noteworthy changes that were implemented in *DSM*-5.

7.4 Evaluate criticisms of *DSM*-5.

7.5 Explain how the dimensional approach differs from the categorical approach of diagnosis and classification used by the *DSM*.

In the 1990s, Jerome Wakefield, a renowned scholar in the field of abnormal psychology, offered a theory that put forth a more simplified (in comparison with the multifaceted criteria above) definition of mental disorders (Wakefield, 1992, 1999). His theory has generated enough support to become quite prominent in recent years. Wakefield (1992) explains his **harmful dysfunction theory** of mental disorders in the following way:

> I argue that a disorder is a harmful dysfunction, wherein harmful is a value term based on social norms, and dysfunction is a scientific term referring to the failure of a mental mechanism to perform a natural function for which it was designed by evolution. Thus, the concept of disorder combines value and scientific components. (p. 373)

The harmful dysfunction theory proposes that in our efforts to determine what is abnormal, we consider both scientific (e.g., evolutionary) data and the social values in the context of which the behavior takes place. As such, it can account for a wide range of behavior that clinical psychologists have traditionally labeled as psychopathological according to the multiple criteria listed above.

Who Defines Abnormality?

Wakefield's definition of abnormality, along with other definitions, continues to be debated by academics and researchers in the field (e.g., Lilienfeld & Marino, 1999; Wakefield, 2010). However, clinical psychologists have certainly not waited for a resolution of this scholarly debate before assigning, treating, and studying disorders. They use disorders—as defined by the ***Diagnostic and Statistical Manual of Mental Disorders (DSM)***, the prevailing diagnostic guide for mental health professionals—every day as they perform assessments, conduct therapy, and design and execute research studies.

In the introductory pages of the latest version of the *DSM* (*DSM-5;* American Psychiatric Association, 2013), its authors offer a broad definition of mental disorder. It is not entirely dissimilar to Wakefield's harmful dysfunction theory (Spitzer, 1999), yet it also incorporates aspects of the other criteria discussed above.

In *DSM-5*, mental disorder is defined as a "clinically significant disturbance" in "cognition, emotion regulation, or behavior" that indicates a "dysfunction" in "mental functioning" that is "usually associated with significant distress or disability" in work, relationships, or other areas of functioning (American Psychiatric Association, 2013, p. 20). In addition, the definition states that expectable reactions to common stressors are not mental disorders.

Who created this definition, as well as the specific diagnostic categories that fill *DSM?* Many people played significant roles, but the most significant were those on the

BOX 7.1

Considering Culture

Typical but Abnormal?

At one time or another, many of us have used the "everybody else is doing it" explanation to rationalize our aberrant behavior. Drivers who speed, kids who steal candy, partiers who drink too much, citizens who cheat on their taxes: Any of them might argue, "I'm not the only one," and they'd be right. Should the commonality of a behavior affect the way we evaluate that behavior? If we tweak the question to consider abnormal (but not necessarily unlawful or unruly) behavior, does behavior become normal—and possibly, by extension, healthy or acceptable—if lots of people engage in it? Or is it possible that abnormal is abnormal—and possibly, by extension, unhealthy or unacceptable—no matter how many people do it?

These questions are particularly relevant to the definition of mental illness, especially when we take issues of culture and diversity into account. Thomas A. Widiger and Stephanie Mullins-Sweatt (2008) considered the issue and came to this conclusion: "Simply because a behavior pattern is valued, accepted, encouraged, or even statistically normative within a particular culture does not necessarily mean it is conducive to healthy psychological functioning" (p. 360). This statement suggests that it is possible for a behavior to be quite common, even conventional, within a culture yet pathological at the same time. They question whether numerous behavior patterns—extensive and meticulous rituals among some religious groups, patterns of interpersonal submission in some Asian cultures, and the practice of remaining house-bound among some Muslim women in certain parts of the world—might actually represent disordered behavior (obsessive-compulsive disorder, dependent personality disorder, and agoraphobia, respectively) despite their prevalence within the culture.

Do you agree with the idea that a behavior can be pathological within a particular culture even if it is common or typical within that culture? Why or why not? If so, can you think of examples in other cultures and in your own culture as well? And, importantly, who should determine the definitions of universal psychological wellness or disorder?

Task Force for each edition of the *DSM* (American Psychiatric Association, 2013). This group consisted largely of leading researchers in various specialty areas within psychopathology who were selected for their scholarship and expertise in their respective fields. It is noteworthy that this task force consisted primarily of psychiatrists, and a relatively small number of psychologists and other mental health professionals were included. Moreover, the *DSM-5* and all previous editions of the *DSM* have been published by the American *Psychiatric* Association (as opposed to the American *Psychological* Association). Thus, although the *DSM* has been used extensively by clinical psychologists and a wide range of other nonmedical mental health professionals (social workers, counselors, etc.), the authors who have had the most significant impact on its contents are medical doctors. So it should come as no surprise that the *DSM* reflects a **medical model of psychopathology** in which each disorder is an entity defined categorically and features a list

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Photo 7.1 Distinguishing normality from abnormality has been the focus of many theories offered throughout the history of clinical psychology. In your opinion, what are the most important distinguishing factors?

of specific symptoms (Blaney, 2015; Deacon, 2013). (We discuss these aspects of *DSM* disorders in more detail later in this chapter.)

Besides their profession, what else do we know about the primary authors of the *DSM?* The first edition of the *DSM*, published in 1952, was created by the foremost mental health experts of the time, who were almost exclusively white, male, trained in psychiatry, at least middle age, and at least middle class. Especially with the most recent revisions of the *DSM*, deliberate efforts have been made to include more diversity among the contributors. In spite of this forward progress, some have remained critical: "The designers of the *DSM-III* and *DSM-III-R* (and to a lesser extent the fourth edition and the text revision) were still predominantly senior White male psychiatrists who embedded the document with their biases" (Malik & Beutler, 2002, pp. 5–6). As we will explore later in this chapter, the culture and values of those who define mental disorders can play an influential role in the definitions they produce.

Why Is the Definition of Abnormality Important?

The process by which disorders are defined is much more than an academic exercise for scholars to debate. Instead, this process and the definitions it yields have very real consequences for professionals and nonprofessionals alike (Kinghorn, 2013; Widiger & Mullins-Sweatt, 2008).

As an example, consider attenuated psychosis syndrome. Currently, attenuated psychosis syndrome is not an official diagnostic category. Instead, it is listed as a **proposed criteria set** in the "Emerging Measures and Models" section of *DSM-5* (American Psychiatric Association, 2013; Fusar-Poli, Carpenter, Woods, & McGlashan, 2014). This section describes conditions that *DSM* authors decided to leave out of the list of "official" disorders, at least for now, but to list as "unofficial" conditions for the purpose of inspiring clinicians and researchers to study them more. Attenuated psychosis syndrome (as described earlier in this chapter) is a bit like a "light" version of schizophrenia ("attenuated" means reduced or lessened). Its symptoms include delusions, hallucinations, and disorganized speech that are not severe or long-lasting. Its description also mentions that the person's "reality testing"—her or his ability to stay in touch with the same reality that the rest of us experience—remains relatively intact. The symptoms must be present only once per week within the last month but must be distressing or disabling.

Importance for Professionals

By listing attenuated psychosis syndrome as a proposed criteria set, the *DSM* authors have facilitated the study of attenuated psychosis syndrome by researchers and its consideration by clinicians. If attenuated psychosis syndrome becomes an official diagnosis, we will undoubtedly see an increase in both of these activities. Additionally, if attenuated psychosis syndrome becomes official, people will be diagnosed with it and will be

conceptualized (by professionals and themselves) as having this form of mental illness. On the other hand, if attenuated psychosis syndrome had never appeared in any form in any edition of *DSM*, it is less likely that researchers would study it or clinicians would add it to their professional vocabulary. And the same people who would have received the diagnosis would be viewed as slightly odd or eccentric but not mentally ill. Thus, the presence or absence of a diagnostic label for a particular human experience has a powerful impact on the attention it receives from clinical psychologists.

Importance for Clients

For clients, future decisions by *DSM* authors about the status of attenuated psychosis syndrome may carry special significance. Some clients could experience beneficial consequences of being diagnosed with attenuated psychosis syndrome. Consider Lucinda, a woman whose experience over the past few months meets the criteria for attenuated psychosis syndrome. If this disorder was available as an official label, the label could help Lucinda identify and demystify an otherwise nameless experience; feel as though she shares a recognized problem with others (rather than feeling like the only one with the problem); acknowledge the significance of her experience with family, friends, and employers; and gain access to treatment that might have been unavailable without a diagnosis (especially if she uses health insurance to pay [Eriksen & Kress, 2005; Langenbucher & Nathan, 2006]). On the other hand, Lucinda could also experience harmful consequences from being diagnosed with attenuated psychosis syndrome. The label could carry a stigma that damages her self-image; lead to stereotyping by individuals who know her or work with her; and even have an effect on the outcome of legal issues she may encounter, such as child custody cases, sentencing decisions, and fitness-to-stand-trial decisions (Butcher, Mineka, & Hooley, 2007; Eriksen & Kress, 2005).

Thus, the decisions *DSM* authors make when defining abnormality, both as a broad concept and as specific diagnostic categories, profoundly influence many aspects of clients' lives.

BOX 7.2

In My Practice . . .

In my practice, quite a few of my clients have revealed misconceptions about *DSM* diagnoses. Helping them overcome those misconceptions became a crucial component of the early stages of successful therapy.

In one case, Teresa, a 35-year-old woman, sought therapy after considering it for a long time. What "pushed her over the edge," she explained, was the fact that her 11-year-old son, Thomas, had recently been diagnosed with major depression by his pediatrician. Teresa held the mistaken belief that *DSM* diagnoses were permanent. So, in Teresa's mind, Thomas would never overcome his depression. He would struggle with it for the rest of his life. The thought of her "sweet little boy" condemned to a life of sadness caused Teresa so much despair that she ultimately feared that she, too, would become irreversibly depressed. A primary goal for my work with Teresa was to

(Continued)

(Continued)

help her understand that many *DSM* diagnoses, including major depression, were unlikely to be permanent conditions, especially with treatment (which Thomas was receiving).

Another case involved Julian, a 30-year-old man. Julian was overly concerned with germs and cleanliness. He showered several times a day and washed his hands about 20 times per day. These behaviors didn't prevent him from living his life—he had a long-term romantic relationship, performed well at work, and had an active social life—but there were at least occasional moments when his need to wash got in the way. For example, when his friends threw him a surprise birthday party at a restaurant, Julian needed to leave the party early to run home and shower. Another day, he missed much of his sister's wedding because he felt compelled to wash his hands twice during the ceremony. After a thorough assessment, I told Julian that I believed his problems were best diagnosed as a mild case of obsessive-compulsive disorder. That's when it became evident that Julian held the misconception that *DSM* diagnoses inevitably worsened over time. As soon as the words "obsessive-compulsive disorder" left my mouth, Julian's face went blank. He explained that he "knows what happens" to people with OCD: "They keep getting worse and worse until their symptoms totally take over. Pretty soon, I'm gonna be a prisoner in my own house. I won't be able to go out, work, or do anything. I'll turn into a total germophobe who washes himself all day every day." I immediately understood the urgency of correcting Julian's misconception. I told him emphatically that OCD does not inevitably get worse. I added that as his psychologist, I had a lot of confidence that with treatment (as he was admirably seeking), his OCD symptoms would actually improve significantly. It took Julian a while to believe me, but once he did, his treatment was on track.

To see this box come to life as a whiteboard video, please visit edge.sagepub/pomerantz5e.

DIAGNOSIS AND CLASSIFICATION OF MENTAL DISORDERS: A BRIEF HISTORY

Before the *DSM*

Abnormal behavior garnered attention long, long before the first version of the *DSM* appeared. Discussion of abnormal behavior appears in the writings of ancient Chinese, Hebrew, Egyptian, Greek, and Roman societies (Butcher et al., 2007; Millon & Simonsen, 2010). Hippocrates (460–377 BCE) wrote extensively about abnormality, but unlike most of his predecessors, he did not offer supernatural explanations such as possession by demons or gods. Instead, his theories of abnormality emphasized natural causes. Specifically, he pointed to an imbalance of bodily fluids (blood, phlegm, black bile, and yellow bile) as the underlying reason for various forms of mental illness (Blashfield, 1991; Butcher et al., 2007). Although his specific theory is viewed by contemporary mental health experts as outdated, Hippocrates' accent on natural causes of psychopathology was a significant early step toward more current definitions.

If we fast-forward from Hippocrates to the 19th century, we find an era when many cities in Europe and the United States were establishing asylums for the treatment of the mentally ill (as opposed to the imprisonment or abuse they might have suffered previously). In these inpatient treatment settings, mental health professionals had the opportunity to observe individuals with mental disorders for extended periods of time. In many such settings, a by-product of this extended care was a list of categories into which clients could be organized. One example occurred in France, where Philippe Pinel (discussed in more detail in Chapter 2) proposed specific categories such as melancholia, mania, and dementia, among others. Eventually, the staff of some of these institutions shared their idiosyncratic diagnostic systems with one another, and more common terminology evolved (Langenbucher & Nathan, 2006).

Around 1900, more important steps were taken toward the eventual *DSM* system that we currently use (Widiger & Mullins-Sweatt, 2008). Emil Kraepelin (also discussed in Chapter 2) labeled specific categories such as manic-depressive psychosis and dementia praecox (roughly equivalent to bipolar disorder and schizophrenia, respectively; Millon & Simonsen, 2010). These and other contributions by Kraepelin have resulted in his reputation as a founding father of the current diagnostic system (Langenbucher & Nathan, 2006). During the late 1800s and early 1900s, the primary purpose of diagnostic categories was the collection of statistical and census data. Later, in the mid-1900s, the U.S. Army and Veterans Administration (now Veterans Affairs) developed their own early categorization system in an effort to facilitate the diagnosis and treatment of soldiers returning from World War II (American Psychiatric Association, 2000; Office of the Surgeon General, Army Service Forces, 1946). This military categorization system was quite different from the most recent editions of the *DSM*, but it actually had significant influence on the creation of the first edition of the *DSM*, which appeared less than a decade later (Langenbucher & Nathan, 2006).

DSM—Earlier Editions (*I* and *II*)

DSM-I was published by the American Psychiatric Association in 1952. ***DSM-II*** followed as a revision in 1968. These two editions of the *DSM* were quite similar to each other, but as a pair, they were quite different from all the *DSM* editions subsequently published (Lilienfeld & Landfield, 2008a). *DSM-I* and *DSM-II* contained only three broad categories of disorders: psychoses (which would contain today's schizophrenia), neuroses (which would contain today's major depression, bipolar disorder, and anxiety disorders), and character disorders (which would contain today's personality disorders; Blashfield, Flanagan, & Raley, 2010; Horwitz, 2015).

It is particularly noteworthy that the definitions of disorders in *DSM-I* and *DSM-II* were not scientifically or empirically based. Instead, they represented "the accumulated clinical wisdom of the small number of senior academic psychiatrists who staffed the *DSM* task forces" (Langenbucher & Nathan, 2006, p. 5). Most of these psychiatrists were psychoanalytic in orientation, and the language of the first two *DSM* editions reflected the psychoanalytic approach to understanding people and their problems. Additionally, the descriptions of individual disorders in *DSM-I* and *DSM-II* were not lists of specific symptoms or criteria; instead, they were simply prose, typically one paragraph per disorder, offering relatively vague descriptions of clinical conditions. As a result, the diagnostic categories in *DSM-I* and *DSM-II* had very limited generalizability or utility for clinicians in practice at the time (Woo & Keatinge, 2008).

DSM—More Recent Editions (*III*, *III-R*, *IV*, and *IV-TR*)

DSM-III, published in 1980, was very dissimilar from *DSM-I* and *DSM-II* (Widiger & Mullins-Sweatt, 2008; Widiger & Trull, 2007). In comparison to *DSM-I* and *DSM-II*, it reflected an approach to defining mental disorders that differed substantially in some important ways (Blashfield et al., 2010; Decker, 2013; Sprock, 2015; Whooley & Horwitz, 2013):

- It relied to a much greater extent on empirical data to determine which disorders to include and how to define them.
- It used specific diagnostic criteria to define disorders. Whereas the *DSM-III* retained some descriptive paragraphs (and in fact augmented them for most disorders), these paragraphs were followed by specific criteria—checklists, basically—that delineated in much greater detail the symptoms that must be present for an individual to qualify for a diagnosis.
- It dropped any allegiance to a particular theory of therapy or psychopathology. As a result, the psychoanalytic language of previous editions was replaced by terminology that reflected no single school of thought.
- It introduced the **multiaxial assessment** system that remained in *DSM* through the next several editions but was dropped in *DSM-5*. When multiaxial assessment was in place, the psychiatric problems were described on each of five distinct axes. Axis I included disorders thought to be more episodic (likely to have beginning and ending points), and Axis II included disorders thought to be more stable or long-lasting. Axes III and IV offered clinicians a place to list medical conditions and psychosocial/environmental problems, respectively, relevant to the mental health issues at hand. And Axis V, known as the Global Assessment of Functioning (GAF) scale, provided clinicians an opportunity to place the client on a 100-point continuum describing the overall level of functioning.

An immediately noticeable feature of *DSM-III* was its size: It was a significantly longer, more expansive diagnostic manual than its predecessors. For the disorders that had appeared in earlier editions, *DSM-III* offered extended descriptions and added lists of specific criteria. In addition, it included many new disorders—265 disorders in total, as compared with 182 in *DSM-II* and 106 in *DSM-I*. As a result of all these factors, *DSM-III* contained more than three times as many pages as *DSM-II* (Houts, 2002). Subsequent revisions to the *DSM*—***DSM-III-R*** (American Psychiatric Association, 1987), ***DSM-IV*** (American Psychiatric Association, 1994), and ***DSM-IV-TR*** (American Psychiatric Association, 2000)—retained the major quantitative and qualitative changes instituted by *DSM-III* in 1980 (Widiger & Crego, 2015). *DSM-5* retained many of these changes as well, although it also features some significant changes of its own (to be discussed in detail later in this chapter).

Table 7.1 offers an example of a specific disorder—generalized anxiety disorder, or "anxiety neurosis," as it was previously described—defined at an early stage of *DSM*

Table 7.1 Anxiety Neurosis/Generalized Anxiety Disorder as Defined in *DSM-II* and *DSM-5*

***DSM-II* (American Psychiatric Association, 1968): 300.0 Anxiety Neurosis**	***DSM-5* (American Psychiatric Association, 2013): 300.02 Generalized Anxiety Disorder**
This neurosis is characterized by anxious over-concern extending to panic and frequently association with somatic symptoms. Unlike Phobic neurosis, anxiety may occur under any circumstances and is not restricted to specific situations or objects. This disorder must be distinguished from normal apprehension or fear, which occurs in realistically dangerous situations.	Excessive anxiety and worry (apprehensive expectation), occurring more days than not for at least 6 months, about a number of events or activities (such as work or school performance). The person finds it difficult to control the worry. The anxiety and worry are associated with three (or more) of the following six symptoms (with at least some symptoms present for more days than not for the past 6 months). Note: Only one item is required in children. 1. restlessness or feeling keyed up or on edge 2. being easily fatigued 3. difficulty concentrating or mind going blank 4. irritability 5. muscle tension 6. sleep disturbance (difficulty falling or staying asleep, or restless unsatisfying sleep) The anxiety, worry, or physical symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning. The disturbance is not attributable to the physiological effects of a substance (e.g., a drug of abuse, a medication) or another medical condition (e.g., hyperthyroidism). The disturbance is not better explained by another mental disorder (e.g., anxiety or worry about having panic attacks in panic disorder, negative evaluation in social anxiety disorder [social phobia], contamination or other obsessions in obsessive-compulsive disorder, separation from attachment figures in separation anxiety disorder, reminders of traumatic events in posttraumatic stress disorder, gaining weight in anorexia nervosa, physical complaints in somatic symptom disorder, perceived appearance flaws in body dysmorphic disorder, having a serious illness in illness anxiety disorder, or the content of delusional beliefs in schizophrenia or delusional disorder).

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history (*DSM-II;* American Psychiatric Association, 1968) and at the most recent stage (*DSM-5;* American Psychiatric Association, 2013). In contrast to the current definition, several features of the *DSM-II* definition are noteworthy, including the psychoanalytically derived term *neurosis* in the title of the disorder and the use of a brief descriptive paragraph rather than the more detailed checklist of specific criteria.

DSM-5: The Current Edition

In May 2013, ***DSM-5*** was published. It was the first substantial revision of the manual in about 20 years, and it was a massive effort. Led by two prominent mental health researchers, David Kupfer and Darrel Regier, it involved hundreds of experts from over a dozen countries contributing their time and expertise over a 12-year period that was particularly intensive in the last half-dozen years before its release (American Psychiatric Association, 2013; Kupfer, Kuhl, & Regier, 2013; Marty & Segal, 2015; Paris, 2013b, 2015a; Regier, Kuhl, & Kupfer, 2013).

Many steps were involved in this huge undertaking. Early on, a task force—essentially, a committee of prominent researchers in various areas of mental disorders—was created. The members of this task force then led work groups, each of which focused on a particular area of mental disorders (e.g., the Eating Disorders Work Group, the Psychotic Disorders Work Group). These Work Groups reviewed the disorders listed in the previous *DSM* and considered proposals for revision, including ideas for adding, eliminating, combining, splitting, or revising the definitions of disorders. The leaders also created a Scientific Review Committee of experts separate from the Work Groups whose job was to make sure that there was sufficient scientific evidence to support the changes proposed by the Work Groups (Kendler, 2013). Later in the process, they conducted field trials for their proposed changes—in other words, they asked practicing clinicians to try using them with clients, with the intention of determining how reliable and clinically useful they were (American Psychiatric Association, 2013; Clarke et al., 2013; Regier, Narrow, et al., 2013; Widiger & Glover, 2015).

Beginning in 2010, the *DSM-5* authors maintained a website (dsm5.org) on which they communicated to the public about their progress, including posts of the drafts of proposed changes. The website also solicited comments from interested parties, whether professionals, clients, mental health advocacy groups, or any member of the public. And they received plenty—over 13,000 comments through the website, plus another 12,000 in other forms such as e-mail and letters (American Psychiatric Association, 2013; Jeste, Lieberman, Scully, & Kupfer, 2012; D. Porter, 2013).

Throughout the process, the *DSM-5* authors tried to coordinate their efforts with those of the World Health Organization (WHO), which publishes the *International Classification of Diseases* (*ICD*). *ICD* is the primary way that diseases—both mental disorders and all other health-related problems—are coded and categorized in many countries outside the United States (S. C. Evans & Roberts, 2015). Greater consistency between *DSM* and *ICD* will allow better communication between countries and improved research design (American Psychiatric Association, 2013; Frances, 2013a; Frances & Nardo, 2013; Paris, 2015a; Regier, Kuhl, et al., 2013). (China also has its own system of classifying mental disorders, the *Chinese Classification of Mental Disorders*.)

Changes *DSM-5* Did Not Make

The authors of *DSM-5* made plenty of changes, but before we review them, let's review the changes they considered but ultimately did *not* make. These are important because they illustrate decision-making processes used by the *DSM-5* authors, the fact that not every proposal is adopted, and changes that may be reconsidered for future editions of the manual.

- Initially, the authors of *DSM-5* considered significantly overhauling the manual to emphasize neuropsychology, or the biological roots, of mental disorders. This would have been a significant paradigm shift in which the fundamental way we define mental disorders changes from descriptions of behavioral symptoms to biological evidence—in much the same way that many areas of medicine use blood tests, biopsies, x-rays, PET scans, gene analyses, or other lab results to determine whether a person has a disease. However, it became evident to the *DSM* authors that mental health lags behind many other specialties in this regard. In other words, although there are many mental disorders that involve biological factors, those disorders lack definitive, reliable "biological markers"—the kinds of things that indicate that a person "tests positive" or "tests negative" in a conclusive way. Such biological markers may become known in time as neuroscience advances, but for now, research has simply not yet uncovered them clearly enough to use them as diagnostic tools (Paris, 2013b, 2015a; Pierre, 2013; Stringaris, 2013). As one author put it, "*DSM-5* has been written for 2013, not for 2063 or 2113," at which time the connections between mental disorders and their neurological or biological underpinnings may be sufficiently strong (Paris, 2013a, p. 41).

- Another across-the-board change that the *DSM-5* authors considered was a shift toward a dimensional definition of mental disorders (Mullins-Sweatt, Samuel, & Helle, 2017; Regier, Narrow, Kuhl, & Kupfer, 2009; Widiger & Mullins-Sweatt, 2016). This issue is discussed in more detail later in this chapter, but essentially, it involves viewing disorders not strictly in a categorical or "yes or no" way, but along a continuum. In other words, rather than describing a client only as either "having" or "not having" major depressive disorder, the clinician could rate the client's depression symptoms (or another underlying characteristic) on a scale (e.g., a range of 1–10; Whooley & Horwitz, 2013). Ultimately, this idea was rejected as too premature and overly complicated to justify such a comprehensive renovation of the manual, but the *DSM-5* authors suggest that it may be seriously considered for future editions (American Psychiatric Association, 2013; De Fruyt et al., 2013; Gore & Widiger, 2013; J. Phillips, 2013).

- A dimensional approach was also seriously considered for a particular subset of mental disorders: personality disorders. As described later in this chapter, a significant body of research has accumulated to suggest that personality disorders fit best with the idea of dimensional, as opposed to strictly categorical, conceptualization. The *DSM-5* authors proposed a specific way of understanding personality disorders dimensionally, but the proposal was rejected as being too complex and not clinically useful enough. Although it is not in effect, that proposal is included in a later section of *DSM-5* called "Emerging Measures and Models" with the hope that researchers will study and possibly revise it for future consideration (American Psychiatric Association, 2013; Black, 2013; Hopwood, Schade, Krueger, Wright, & Markon, 2013; Skodol et al., 2013).

- Speaking of personality disorders, the *DSM-5* authors considered removing 5 of the 10 personality disorders previously included in that section, a change that would have significantly reshaped that category. The five that were on the chopping block at one point were paranoid, schizoid, histrionic, dependent, and narcissistic personality disorders. The *DSM-5* authors ultimately decided to retain all of them (American Psychiatric Association, 2011; Pull, 2013).

- There were numerous proposals for specific new disorders that were considered but rejected. Many of these appear in the "Emerging Measures and Models" section as "proposed criteria sets" to enable researchers to conduct studies that will help determine whether they "make the cut" for future editions of *DSM*. Among them are the following:
    - *Attenuated psychosis syndrome*, which features the hallucinations, delusions, and disorganized speech characteristic of schizophrenia but in much less intense and more fleeting forms, and in which the person doesn't lose touch with reality in a pervasive way (American Psychiatric Association, 2013; Frances, 2013a; Tsuang et al., 2013; Wakefield, 2013a)
    - *Mixed anxiety-depressive disorder*, which features some symptoms of anxiety, some symptoms of depression, but not enough of either to qualify for any existing disorder (such as generalized anxiety disorder or major depressive disorder; American Psychiatric Association, 2013; Frances, 2012a)
    - *Internet gaming disorder*, which features excessive and disruptive Internet game-playing behavior (Bean, Nielsen, van Rooij, & Ferguson, 2017; King & Delfabbro, 2014a, 2014b). To a lesser extent, the *DSM-5* authors also considered other disorders based on various nonsubstance addictive-related behaviors such as shopping, exercise, work, and sex (American Psychiatric Association, 2013; Frances, 2013a; G. Greenberg, 2013; Kafka, 2013; King & Delfabbro, 2013; Petry & O'Brien, 2013; Wakefield, 2013a).

New Features in *DSM-5*

Some of the most significant changes in *DSM-5* do not focus on specific disorders but on the way the entire manual is organized or presented. Among them are the following:

- The title of the manual is not *DSM-V*, but *DSM-5*. That is, the authors deliberately shifted away from the traditional Roman numerals used in previous editions (e.g., *DSM-II*, *DSM-III*, *DSM-IV*) and toward Arabic numerals instead. The reason for this shift is to enable more frequent minor updates that will be named just as changes to computer operating systems and applications are often named: *DSM-5.1*, *DSM-5.2*, et cetera. Thus, this naming change is not merely superficial. It suggests that *DSM* is a "living document" that, in the future, will be more quick to respond to new research that improves our understanding of mental disorders. No longer will we have to stick with a single, static version of *DSM* for a decade or two until the time arrives for a major revamp; instead, smaller-scale changes can be made more continuously. (As of early 2019, no 5.1 or otherwise updated version of *DSM* had yet been released.) The "Emerging Measures and Models" section of the manual (mentioned earlier) goes hand in hand with this notion of a living document, as it prompts researchers and clinicians

to consider conditions that have not yet been officially included in *DSM* but may, after more attention, be included in future editions (Kraemer, 2013; Moran, 2013a; Paris, 2013b, 2015a; Stringaris, 2013; Wakefield, 2013a; Whooley & Horwitz, 2013).

- The multiaxial assessment system—a central feature of *DSM* since its introduction in *DSM-III* in 1980—was dropped altogether from *DSM-5*. This removal brings a number of important changes to the way clinicians diagnose clients. For example, the tradition of separate axes for disorders that tend to persist long-term (such as developmental disorders and personality disorders, formerly on Axis II) from disorders that tend to be more short-term or episodic (such as major depression, formerly on Axis I) is now gone. This could result in different conceptualizations of these disorders by researchers, clinicians, clients, or third-party payers in the future. Also, Axis V, the Global Assessment of Functioning (GAF) scale, is now eliminated, so there is no longer a single numeric scale on which clinicians can describe their clients' level of functioning across all disorders (although a questionnaire that clients complete themselves that the World Health Organization has created is included for future consideration in the "Emerging Measures and Models" section; American Psychiatric Association, 2013; Kupfer et al., 2013; Paris, 2015a; Wakefield, 2013a).

New Disorders in *DSM-5*

DSM-5 introduced a number of new disorders—not merely revisions or regroupings of existing disorders (as we'll see in the next section) but disorders that, at least to some extent, cover problems that were not covered by any disorders in the previous edition of the manual. Among them are the following:

- *Premenstrual dysphoric disorder* (PMDD; discussed in more detail in a box later in this chapter), which is essentially a severe version of premenstrual syndrome (PMS), including a combination of at least five emotional and physical symptoms occurring in most menstrual cycles during the last year that cause clinically significant distress or interfere with work, school, social life, or relationships with others (American Psychiatric Association, 2013; Paris, 2013b; Regier, Kuhl, et al., 2013; Wakefield, 2013a).

- *Disruptive mood dysregulation disorder* (DMDD), which is essentially frequent temper tantrums in children 6 to 18 years old (at least three tantrums per week over the course of a year) that are clearly below the expected level of maturity and occur in at least two settings (e.g., home, school, or with friends) along with irritable or angry mood between the temper tantrums. The creation of this new diagnosis was prompted by the drastic increase in the diagnosis (and possible overdiagnosis and overmedication) of bipolar disorder in children in recent decades (American Psychiatric Association, 2013; Copeland, Angold, Costello, & Egger, 2013; Frances & Batstra, 2013; Pierre, 2013).

- *Binge eating disorder* (BED), which resembles the part of bulimia nervosa in which the person overindulges in food but lacks the part in which the person tries to subtract the calories through compensatory behaviors like excessive exercise. Binges must take place at least once per week for 3 months and be accompanied by a lack of control over the eating as well as other symptoms like rapid eating, eating until overly full, eating alone to avoid embarrassment, and feelings of guilt or depression afterward (American

Psychiatric Association, 2013; Cassin & David, 2015; Moran, 2013b; Ornstein et al., 2013; Stice, Marti, & Rohde, 2013).

- *Mild neurocognitive disorder* (mild NCD), which is essentially a less intense version of major neurocognitive problems like dementia and amnesia. It requires modest decline in such cognitive functions as memory, language use, attention, or executive function, but nothing serious enough that it interferes with the ability to live independently (American Psychiatric Association, 2013; Blazer, 2013; Frances, 2013a).

- *Somatic symptom disorder* (SSD), which involves a combination of at least one significantly disruptive bodily (somatic) symptom with excessive focus on that symptom (or symptoms) that involves perceiving it as more serious than it really is, experiencing high anxiety about it, or devoting excessive time and energy to it (American Psychiatric Association, 2013; Frances, 2012b, 2013d; Frances & Chapman, 2013; Rief & Martin, 2014).

- *Hoarding disorder*, in which the person has continuing difficulty discarding possessions no matter how objectively worthless they are and, as a result, lives in a congested or cluttered home and experiences impairment in important areas such as work, socialization, or safety. In past *DSM*s, the diagnosis of obsessive-compulsive disorder may have been considered for hoarders, but their behavior often matched OCD criteria imperfectly, and with *DSM-5*, the criteria for hoarding are now distinct (American Psychiatric Association, 2013; G. Greenberg, 2013; Regier, Kuhl, et al., 2013).

Revised Disorders in *DSM-5*

In some cases, changes in *DSM-5* involved established disorders being revised in some way—diagnostic criteria were modified, disorders were combined, or age limits were adjusted. Among the revisions:

- The "bereavement exclusion" formerly included in the diagnostic criteria for major depressive episode was dropped. To explain, previous editions of *DSM* featured a statement that major depression could not be diagnosed in a person who was mourning (or bereaving) the death of a loved one during the first two months following the death. The rationale for the exclusion was that the sadness that commonly comes with such loss should not be confused with the mental disorder of major depression. The decision to drop this statement in *DSM-5* means that now the diagnosis *can* be given to people who lost a loved one within the past 2 months, but only if the clinician determines that the symptoms (sadness, changes in sleeping and eating, etc.) exceed expectations based on the person's own history and culture. The rationale for dropping the exclusion was to make sure that people in mourning who are indeed experiencing abnormal levels of depressive symptoms will be recognized, diagnosed, and promptly treated before things get even more dire (American Psychiatric Association, 2013; Fox & Jones, 2013; Parker, 2013; R. Porter, Mulder, & Lacey, 2013; Wakefield, 2013b).

- The *DSM-IV* diagnoses of autistic disorder, Asperger's disorder, and related developmental disorders were combined into a single *DSM-5* diagnosis: *autism spectrum disorder*. The reason for consolidating these disorders is that, according to *DSM-5* authors, they represent various points on the same spectrum of impairment, defined

by social communication problems and restrictive or repetitive behaviors and interests (American Psychiatric Association, 2013; Kent et al., 2013; Kupfer et al., 2013; Mayes, Black, & Tierney, 2013; Pina-Camacho et al., 2013; Regier, Kuhl, et al., 2013). In other words, they are now seen as mild, moderate, or severe versions of the same problem.

- In the criteria for attention-deficit/hyperactivity disorder (ADHD), the age at which symptoms must first appear was changed from 7 to 12 years old, and the number of symptoms required for the diagnosis to apply to adults was specified as five (as opposed to six for kids; American Psychiatric Association, 2013; Frances, 2013a; Paris, 2013a; Wakefield, 2013a).

- In the criteria for bulimia nervosa, the frequency of binge eating required for the disorder was dropped from twice per week to once per week. In the diagnosis of anorexia nervosa, the requirement that menstrual periods stop has been omitted, and the definition of low body weight has been changed from a numeric definition (less than 85% of expected body weight) to a less specific description that takes into account age, sex, development, and physical health (American Psychiatric Association, 2013; Moran, 2013b; Ornstein et al., 2013).

- The two separate *DSM-IV* diagnoses of substance abuse and substance dependence have been combined into a single diagnosis: *substance use disorder*. Tolerance and withdrawal, which had been solely linked to substance dependence in *DSM-IV* (and confused with addiction), were not in fact solely experienced by those with substance dependence but also by people who use substances in various capacities (American Psychiatric Association, 2013; Compton, Dawson, Goldstein, & Grant, 2013; Hasin et al., 2013).

- Mental retardation was renamed *intellectual disability (intellectual development disorder)*, and learning disabilities in reading, math, and writing were combined into a single diagnosis with a new name: *specific learning disorder* (American Psychiatric Association, 2013; Berkule, 2015; Schalock, 2015).

- Obsessive-compulsive disorder was removed from the Anxiety Disorders category and placed into its own new category, Obsessive-Compulsive and Related Disorders, which also includes trichotillomania (hair-pulling), excoriation (skin-picking), and body dysmorphic disorder (Abramowitz & Jacoby, 2014, 2015; American Psychiatric Association, 2013).

- The category of Mood Disorders was split into two: Depressive Disorders (in which mood is singularly sad) and Bipolar and Related Disorders (in which mood alternates between sadness and mania; American Psychiatric Association, 2013; Gotlib & LeMoult, 2014).

Controversy Surrounding *DSM-5*

DSM-5 arrived in May 2013 amid controversy that had already been swirling for many months (Cooper, 2013). The controversy extended well beyond academic journals and professional conferences into popular media such as magazines, newspapers, books, television, and websites geared toward the general public. Coverage of controversy extended well past the United States into Europe, Australia, Asia, and South America (Frances, 2013e).

Members of multiple *DSM-5* Work Groups quit in the middle of the revision, publicly casting doubt on the process used to create the *DSM-5* (Frances, 2012c; G. Greenberg, 2013). Letters of protest came from leaders of multiple mental health organizations, including Division 32 (Society of Humanistic Psychology) of the American Psychological Association, the American Counseling Association, and the British Psychological Society (G. Greenberg, 2013; Whooley & Horwitz, 2013). Numerous mental health professionals called for a boycott of *DSM-5* (Caccavale, 2013; Frances, 2013g).

The majority of the commentary surrounding *DSM-5* was critical, and the most vocal critic was Allen Frances (2012a, 2012b, 2013a, 2013b, 2013c, 2013d, 2013e, 2013f, 2013g). It is important to recognize that Frances was the chair of the Task Force for *DSM-IV*—in other words, he led the previous revision of the manual that was published in 1994, and as such he had direct experience with the enormous challenge and consequences of producing a new *DSM*. Frances had actually been retired for almost a decade, with no intention of being involved in *DSM-5* at all, until he found himself at a cocktail party at the annual meeting of the American Psychiatric Association in 2009 and was pulled into the debate about *DSM-5* by colleagues who informed him about the revision process (Frances, 2013a). Since that time, Frances has put forth a steady stream of articles, blog posts, radio and TV appearances, and even a full book on the flaws and failings of *DSM-5*. His tone has been consistently critical and condemning, and in many cases, he has been remarkably outspoken. For example:

- "This is the saddest moment in my 45-year career of studying, practicing, and teaching psychiatry. The Board of Trustees of the American Psychiatric Association has given its final approval to a deeply flawed *DSM-5* containing many changes that seem clearly unsafe and scientifically unsound. . . . Our patients deserve better, society deserves better, and the mental health professions deserve better." (Frances, 2012b)

- "With the *DSM-5*, patients worried about having a medical illness will often be diagnosed with somatic symptom disorder, normal grief will be misidentified as major depressive disorder, the forgetfulness of old age will be confused with mild neurocognitive disorder, temper tantrums will be labeled disruptive mood dysregulation disorder, overeating will become binge eating disorder, and the already overused diagnosis of attention-deficit disorder will be even easier to apply." (Frances, 2013i)

- "[I have offered many] warnings about the risks that *DSM-5* will mislabel normal people, promote diagnostic inflation, and encourage inappropriate medication use. . . . Many other individuals, mental health organizations, professional journals, and the press have loudly sounded the very same alarm. We have had some positive impact . . . but overall we failed. *DSM-5* pushes psychiatric diagnosis in the wrong direction." (Frances, 2013a, p. xviii)

- "My advice . . . is to ignore *DSM-5*. . . . It is not well done. It is not safe. Don't buy it. Don't use it. Don't teach it." (Frances, 2013f)

Frances is certainly not alone. Although few were quite so blunt, many others offered harsh criticism of many aspects of *DSM-5* (e.g., G. Greenberg, 2013; Paris, 2013a, 2015a). What, specifically, did they criticize?

- *Diagnostic overexpansion*. The primary criticism of *DSM-5*, which is a continuation of a complaint of recent *DSM*s, is that its diagnoses cover too much of normal life—in other words, too often it takes difficult or inopportune life experiences and labels them as mental illnesses (Paris, 2015b; Wakefield, 2013a, 2016). (The history of this criticism is covered later in this chapter, and the current version of the controversy is covered in more detail in Chapter 3.) These critics argue that prior to *DSM-5*, a woman with severe PMS, an aging adult whose memory isn't quite what it used to be, an adult who binges on food once a week and feels guilty about it, and a child who throws temper tantrums were experiencing normal life struggles, but with the publication of *DSM-5*, they run the risk of being labeled mentally ill, with premenstrual dysphoric disorder, mild neurocognitive disorder, binge eating disorder, and disruptive mood dysregulation disorder, respectively. These critics also point out that the addition of these disorders in *DSM-5* was often done in spite of questionable research evidence (Frances, 2013a; Wakefield, 2013a). They further emphasize the likelihood that the revisions in diagnostic criteria to existing disorders—the removal of the bereavement exception for major depressive episode, the older age by which ADHD symptoms may now start, the lower frequency requirement for binge eating for bulimia—will result in higher rates of these disorders (Fox & Jones, 2013; Parker, 2013; Wakefield, 2013c). The only change likely to go against this trend is the consolidation of autistic disorder, Asperger's disorder, and similar developmental disorders into the single autism spectrum disorder (Dobbs, 2013; Frances, 2012b; Mayes et al., 2013). A possible factor in this overexpansion—the pharmaceutical industry—is discussed in more detail in Chapter 3.

- *Transparency of the revision process*. Although the *DSM-5* authors maintained a website on which they shared information throughout the revision process, including proposals for changes to the manual, some critics argued that they were vague and selective about what they shared, that too many of their ideas and decisions were eventually made behind closed doors, and that the members of the Work Groups were required to sign confidentiality agreements to keep their processes out of public awareness (Cosgrove & Wheeler, 2013; Frances, 2012c, 2012e; Paris, 2013a).

- *Membership of the work groups*. The decisions made by *DSM* authors depends, at least to some extent, on who those authors are. Those who were invited into the *DSM-5* revision process were, predominantly, researchers. Undoubtedly, they understand the disorders in their area of expertise in terms of designing and conducting empirical studies, but some of them do not practice at all, and those who do may only do so minimally, so their ability to assess the impact of *DSM* changes on full-time clinicians practicing in real-world clinics, hospitals, and private practices may have been lacking (Paris, 2013a, 2015a; Whooley & Horwitz, 2013). That is, the clinical utility of *DSM-5* may suffer from a lack of clinical practice experience among its authors (Mullins-Sweatt, Lengel, & DeShong, 2016). Supporting this argument is a study in which more than 500 practicing clinicians were given the opportunity to organize *DSM* disorders into categories. They consistently categorized the disorders differently from the *DSM*, but similar to each other (Reed et al., 2013). On the other hand, *DSM-5* also received criticism from the director of the National Institute of Mental Health (NIMH), a leading funding agency for mental health research, in which he announced that NIMH will be developing its own Research Domain Criteria (RDoC) by which to define mental health problems for research purposes. In the meantime,

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Photo 7.2
The definitions of mental disorders, including the anxiety-based disorders, have changed as the *DSM* has been revised. In your opinion, what are the pros and cons of the changes?

NIMH will show preference to research proposals that consider mental health issues across *DSM-5* categories rather than within them (Insel, 2013; Lilienfeld & Treadway, 2016; Sanislow, Quinn, & Sypher, 2015). Also related to the membership of the Work Groups is the fact that some of these members have "pet" diagnostic proposals of their own, and a Work Group member who promoted his or her pet proposal consistently and effectively had a better chance of a successful negotiation, resulting in *DSM-5* change, than someone who wasn't invited into the Work Group in the first place (Frances, 2013a; Shorter, 2013).

- *Field trial problems.* As mentioned earlier, the authors of *DSM-5* ran field trials in which they tested the reliability of their new diagnoses with clinicians in real-world clinical settings. The problem, according to critics, is that some of the reliability ratings that these field tests produced were too low. In other words, these changes didn't yield sufficiently consistent diagnoses across clinicians. A second stage of field trials, in which some of these issues could have been corrected, was cancelled (Frances, 2012d, 2013a).

- *Price.* The original list price for *DSM-5* was $199 hardback, $149 paperback, and $149 e-book. By comparison, when *DSM-IV* was released in 1994, it cost $65 (Frances, 2013h). Some critics view this as a steep price for a book considered an essential reference for all mental health professionals, all students of clinical psychology, psychiatry, and related fields, and many health professionals more broadly. (It is worth noting that much of the *International Classification of Diseases* [*ICD*], the manual of mental disorders used in other parts of the world outside the United States, is available for free on the Internet [Reed, Robles, & Dominguez-Martinez, 2016].) In fact, since the success of *DSM-III* in the 1980s, *DSM* proceeds have provided significant funding to its publisher, the American Psychiatric Association. Sadler (2013) reports that according to American Psychiatric Association treasurer reports, *DSM-IV* brought in between 5 and 6 million dollars per year between 2005 and 2011—years well after its initial publication, when sales might have been even higher. These critics and others have questioned the extent to which profits might influence the price point of the book as well as the decisions made in the process of creating it (Blashfield, Keeley, Flanagan, & Miles, 2014; Frances, 2012f).

Criticisms of the *DSM*

The most recent editions of *DSM* have been widely used by all mental health professions, and they undeniably represent improvements over their predecessors in some important ways. Strengths of recent editions include their emphasis on empirical research, the use of explicit diagnostic criteria, interclinician reliability, and atheoretical language (e.g., Klerman, 1984; Malik & Beutler, 2002; Matarazzo, 1983). Additionally, the *DSM* has facilitated communication between researchers and clinicians by providing a common professional language that has become very widely accepted (Lilienfeld & Landfield, 2008a; M. Wilson, 1993). However, the recent *DSM*s have received significant criticism as well. We touched on some of this criticism in our discussion of *DSM-5* earlier; here, we consider those topics in more detail and other topics as well.

Web Link 7.1
Open letter of concern about proposed *DSM* changes

BOX 7.3

Considering Culture

Are Eating Disorders Culturally Specific?

DSM-5 (American Psychiatric Association, 2013) includes a Glossary of Cultural Concepts of Distress, which lists syndromes and experiences that are relatively unique to particular cultures. Do anorexia and bulimia belong on this list?

Currently, of course, anorexia and bulimia are included among the official *DSM* disorders rather than among the cultural concepts of distress, most of which originate outside the United States. But there is reason to question whether these disorders are culturally specific—after all, they are far more prevalent among females in industrialized, Western societies, especially the United States, than in other cultures and locations. Pamela K. Keel and Kelly L. Klump (2003) conducted an extensive study of this question, examining the incidence rates of anorexia and bulimia in Western and non-Western parts of the world, as well as the history of these two disorders before they were officially defined and labeled. In their review of previous studies on the subject, Keel and Klump stated that some researchers had put forth the argument that these eating disorders were limited to Western culture and were therefore culture-bound. Other researchers emphasized that factors besides culture, including genetics, were the primary causes of these disorders. Ultimately, Keel and Klump concluded that although anorexia did not meet their definition of a culture-bound condition, bulimia did. Among the evidence they cited was a study of eating disorders in Fiji where bulimic behaviors (especially self-induced vomiting and other compensatory behaviors) were virtually nonexistent until immediately following the introduction of television—which glamorizes Hollywood's thinness-based definitions of female ideal body image—at which point these behaviors increased suddenly and dramatically (A. E. Becker, Burwell, Gilman, Herzog, & Hamburg, 2002).

The questions raised by Keel and Klump's (2003) study are certainly important to clinical psychology. What separates a local, indigenous, "folk" syndrome from a bona fide, official disorder? Culture can certainly shape psychological problems, but can each culture create entirely unique disorders that are unrelated to disorders elsewhere? And, ultimately, to what extent should a clinical psychologist's treatment of a client depend on whether the client's problem is defined as culturally bound or more universal?

Breadth of Coverage

Recent editions of the *DSM* include many more disorders than editions published a relatively short time earlier. Some have argued that this expansion has been too rapid and that the result is a list of mental disorders including some experiences that should not be categorized as forms of mental illness (e.g., Burr & Butt, 2000; Langenbucher & Nathan, 2006; Pilgrim & Bentall, 1999). Houts (2002) points out that many of the newer disorders are not entirely "mental" disorders, including some disorders with physical factors such as the sexual disorders, substance-related disorders, and sleep disorders. Kutchins and Kirk (1997) and Caplan (1995) make the case that the *DSM* includes an increasing number of disorders that are better understood as problems in day-to-day life than as diagnosable mental illnesses—an argument not dissimilar to the one made decades ago

Web Link 7.2
DSM criticism

by Thomas Szasz (1961, 1970), when the diagnostic manual was nowhere near as wide ranging. A potential risk of expanding the range of pathology this way is that greater numbers of people may face stigma associated with a diagnostic label, either from others or as self-stigma. Or the concept of mental illness may be spread too thin, which would solve the stigmatization problem but could result in the trivialization of any form of mental illness, such that they are not taken as seriously as they should be (Hinshaw & Stier, 2008). In addition, as psychological diagnoses multiply, the chances that they will overlap with one another increase, leading to a likelihood of comorbidity (diagnosis of more than one disorder) that some argue is excessively high (e.g., Kendall & Drabick, 2010).

Controversial Cutoffs

One of the essential differences between earlier and later editions of *DSM* is the presence of lists of specific symptoms and, moreover, specific cutoffs regarding those lists of symptoms. For example, the current diagnostic criteria for major depressive disorder include nine specific symptoms, at least five of which must be present for at least a 2-week period. Why a minimum of five symptoms rather than three, seven, or some other number? And why 2 weeks rather than 1 week, 1 month, or some other duration? Some have argued that these cutoffs have been arbitrarily or subjectively chosen by *DSM* authors and that, historically, the consensus of the *DSM* authors (as opposed to empirical data) has played a significant role in these cutoff decisions (Barlow & Durand, 2005; Widiger & Mullins-Sweatt, 2008). Similarly, many *DSM* diagnoses include criteria intended to establish a level of severity that must be met in order for the diagnosis to be assigned—statements that the symptoms must cause significant distress or impairment in various areas of the client's life, including interpersonal relationships or functioning at work/school. What, exactly, constitutes significant distress or impairment? And how are mental health professionals trained to make such a judgment? Such questions remain problematic in the controversy over cutoffs that separate normal from abnormal functioning (Narrow & Kuhl, 2011; Spitzer & Wakefield, 1999).

One reason why the cutoff for any disorder is especially relevant is that third-party payers (e.g., health insurance companies) typically pay for treatment only if it is for a diagnosed disorder. As a result, "upcoding" can become an ethical issue for psychologists who want to ensure that their clients receive their health insurance benefits even if they do not technically qualify for a *DSM* category. Upcoding, essentially, is falsely reporting that a client has a *DSM* diagnosis when in fact they fall short of the criteria (or reporting that a client has a more severe diagnosis than they warrant). For example, if a client is somewhat anxious but the client's symptoms are not sufficient in quantity, severity, or duration to qualify for generalized anxiety disorder, a psychologist may consider diagnosing the client with the disorder anyway. Upcoding, of course, can constitute insurance fraud and a violation of professional ethics (Cartwright, Lasser, & Gottlieb, 2017; Fried, 2018; Peck & Scheffler, 2002). Nonetheless, surveys of psychologists in which they indicate whether and how they would diagnose clients who are near or below the cutoff for a *DSM* diagnosis suggest that upcoding is not uncommon, especially when the psychologist knows that the client is using health insurance to pay (Lowe et al., 2007; Pomerantz & Segrist, 2006).

Cultural Issues

When *DSM-IV* arrived in 1994, it signaled important advances in the consideration of cultural issues regarding mental disorders. *DSM-5* continues many of those advances. As described in Chapter 4, *DSM-5* includes an "Outline for Cultural Formulation" and the accompanying "Cultural Formulation Interview" designed to help clinicians assess in a culturally competent way. It also includes a glossary of cultural concepts of distress, or terms used by various cultural groups to describe specific psychological conditions. In addition, for many disorders, the text describing the disorder now includes comments on cultural variations, such as the way a disorder may be expressed differently in diverse cultures. For example, in the paragraphs describing schizophrenia, *DSM-5* contains several paragraphs on cultural issues, including such statements as "Ideas that appear to be delusional in one culture (e.g., witchcraft) may be commonly held in another" (American Psychiatric Association, 2013, p. 103).

Despite this considerable progress toward improving cultural sensitivity, the recent editions of *DSM* have received pointed criticism for shortcomings related to culture and diversity (e.g., Mezzich, Kleinman, Fabrega, & Parron, 1996). For example, although more ethnic minorities and women were included among the authors of recent *DSM*s than earlier editions, these authors weren't creating a new *DSM* from scratch. Instead, they were revising an established manual that was originally created by a group of authors that was overwhelmingly white and male. So although their input was valued and incorporated, its impact was limited to revisions of previously confirmed *DSM* decisions rather than a more substantive renovation (Velasquez, Johnson, & Brown-Cheatham, 1993).

Also, although the recent revisions of the *DSM* have been praised for basing decisions on empirical data, critics have questioned the extent to which culturally diverse populations are included among the participants in those empirical studies. In fact, some detractors of the *DSM* have stated that very few of the empirical studies considered by the authors of the current *DSM* have focused sufficiently on ethnic minorities, which suggests that the *DSM* still may not reflect minority experiences (e.g., Markus & Kitayama, 1991; Mezzich et al., 1996). The implicit values of Western culture have been the focus of some *DSM* critics as well. They argue that these values are embedded in the *DSM* and that they don't encompass the values of many people from non-Western societies. Specifically, Eriksen and Kress (2005) explain that in comparison with the traditional Western view, the majority of the world's population believes that the boundaries between an individual and the people or environment surrounding them are more permeable and that interdependence between individuals, rather than striving for independence and self-sufficiency, is highly valued. The *DSM* conceptualizations of some disorders (such as some psychotic disorders, some personality disorders, and a number of others) may presume a Western worldview (Lewis-Fernandez & Kleinman, 1994).

Gender Bias

Some disorders are diagnosed far more often in males: alcohol use disorder, conduct disorder, ADHD, and antisocial personality disorder, to name a few. Other disorders are diagnosed far more often in females: major depression, many anxiety disorders, eating disorders, borderline personality disorder, histrionic personality disorder, and others (American Psychiatric Association, 2013; Cosgrove & Riddle, 2004). Critics of the

BOX 7.4

Premenstrual Dysphoric Disorder

One of the most notable new disorders listed in *DSM-5* is premenstrual dysphoric disorder (PMDD). It falls within the Depressive Disorders category, along with major depressive disorder and other disorders that center around sad or irritable mood. According to its description, PMDD should not be equated with premenstrual syndrome (PMS), which is more common and less severe than PMDD. Instead, PMDD requires either "clinically significant distress" or "interference with work, school, usual social activities, or relationships with others" (American Psychiatric Association, 2013, p. 172). The diagnosis of PMDD requires at least five symptoms from a list of 11 occurring in most menstrual cycles of the past year during the week before the menstrual period, at least one of which is mood-related (e.g., mood swings, irritability, sadness) and at least one of which is behavioral or physical (e.g., trouble concentrating, lack of energy, changes in eating or sleeping, breast tenderness or swelling, or pain in joints or muscles).

There is no question that some women experience severe premenstrual symptoms that meaningfully influence their lives. There is a question, however, as to whether that experience should be labeled a mental illness. Some benefits could certainly accompany the diagnosis of PMDD: It could legitimize the severity of the experience for women who receive the diagnosis, provide an explanation for their symptoms, elicit compassion from others, prevent misdiagnosis under another category, and enable treatment (Gallant & Hamilton, 1988). However, there may also be some drawbacks to the diagnosis of PMDD. Specifically, for a woman who has received the diagnostic label of PMDD and shared it with others, it may pose a significant challenge to convince other people (e.g., employers, friends, family) that she is not psychologically abnormal (Caplan, 1992). Moreover, the presence of a PMDD diagnosis in a woman's medical record could have an effect on legal cases in which she is involved. For example, in a custody dispute, a woman with a PMDD diagnosis may be at a distinct disadvantage compared with a woman with identical symptoms for which no diagnostic category exists (Caplan, 1992; Eriksen & Kress, 2005).

The idea of a PMDD diagnosis has generated controversy since it was first proposed in the 1980s (under the name "Late Luteal Phase Dysphoric Disorder" at the time) (Figert, 1996). It was considered for inclusion in previous editions of *DSM*, and listed in an appendix of conditions to be considered for the future in *DSM-IV* in 1994, but it became a full-fledged disorder in 2013 with the publication of *DSM-5*. Feminist groups in particular have opposed its inclusion as an official disorder in the *DSM*. They have emphasized that it represents a form of gender bias likely to overpathologize women, especially since there is no parallel diagnosis for men despite the fact that men also experience some degree of hormonal fluctuations. Additionally, they have pointed out that although premenstrual symptoms can be problematic, they are better understood as physical or gynecological problems rather than a form of mental illness, especially because they last a short time and subside on their own without treatment. Opponents of the proposed PMDD diagnosis have also criticized the quality and quantity of the research on PMDD, concluding that it does not substantiate PMDD as a distinct form of mental illness (Alevizou, Vousoura, & Leonardou, 2018; Caplan, 1992, 1995; Caplan & Cosgrove, 2004; Eriksen & Kress, 2005).

Many questions relevant to contemporary clinical psychologists arise from the

PMDD debate. Where should we draw the line between diagnosable mental illness and the unpleasant or problematic aspects of normal life? What quantity and quality of empirical research should be necessary to support the inclusion of a new disorder in the *DSM*?

Looking ahead, these questions apply not only to the new disorders included in *DSM*-5 but to its proposed criteria sets as well, in other words, the nonofficial conditions that are being considered for official status in a future edition of *DSM*.

current *DSM* have argued that some diagnostic categories are biased toward pathologizing one gender more than the other. Specifically, these diagnoses may represent exaggerations of socially encouraged gender roles, suggesting that society, rather than or in addition to the individual, plays a prominent role in their emergence (e.g., Caplan & Cosgrove, 2004; Kutchins & Kirk, 1997; Yonkers & Clarke, 2011). Some empirical studies have found that clinicians define mental health differently for males and females (e.g., Broverman, Broverman, Clarkson, Rosenkrantz, & Vogel, 1970; Ritchie, 1994; J. A. Sherman, 1980) and that clients of different genders with identical symptoms often receive different diagnoses from clinicians (D. Becker & Lamb, 1994). The controversy over **premenstrual dysphoric disorder (PMDD)**, which had been rejected from previous editions of *DSM* but is now included as a disorder in *DSM-5*, has renewed this criticism of *DSM* authors. See Box 7.4 for more on PMDD.

Web Link 7.3
Premenstrual dysphoric disorder

Nonempirical Influences

Authors of recent editions of the *DSM* have increasingly used empirical evidence to determine the diagnostic categories, and the manual is certainly more reliable as a result. However, nonempirical influences have intruded on the process to some extent, according to various commentators on the *DSM* process (e.g., Blashfield, 1991; Caplan, 1995; Kutchins & Kirk, 1997). At times, political wrangling and public opinion may have pressured *DSM* authors to make certain decisions. For example, once-proposed disorders such as masochistic personality disorder have been strongly and publicly opposed by political organizations, and decisions not to include them officially in the *DSM* may stem in part from this opposition. The changing status of homosexuality—which was an official disorder in *DSM-I* and *DSM-II*, was a disorder only if "ego-dystonic" (in other words, if it caused distress) in *DSM-III*, and has been absent from the manual since *DSM-III-R*—also may reflect efforts during the 1970s and 1980s by organized groups to influence the mental health establishment as well as changing values in *DSM* authors and society more generally.

In addition to politics and public opinion, financial concerns may have played a role in some *DSM* decisions, according to some (e.g., Blashfield, 1991; Cosgrove et al., 2006; Cosgrove & Vaswani, 2019; Kirmayer & Minas, 2000; Whitaker & Cosgrove, 2015). As increasing numbers of therapy clients paid via health insurance in the last several decades of the 20th century, clinicians found that, typically, the health insurance companies required a diagnosis for payment. As it happens, the number of *DSM* diagnoses

Web Link 7.4
DSM author ties to pharmaceutical companies

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Photo 7.3 Homosexuality was considered a disorder in earlier editions of the *DSM* but has not been considered a disorder in more recent editions.

expanded substantially around that time. Also, pharmaceutical companies sponsor a substantial number of research projects on various forms of psychopathology, including those that are proposed and under consideration for future inclusion in the *DSM*. As mentioned earlier in this chapter, quite a few *DSM* authors have financial ties to these companies (Cosgrove et al., 2006; Cosgrove & Krimsky, 2012). Of course, drug companies stand to boost profits when new diagnostic categories, and the new client "markets" that come along with them, are created.

Limitations on Objectivity

No matter how much the *DSM* authors emphasize empiricism in the process of defining mental disorders, there is a limit to their objectivity. In fact, the leading authors of the *DSM-IV* and *DSM-IV-TR* stated that "although based on empirical data, *DSM-IV* decisions were the results of expert consensus on how best to interpret the data" (Frances, First, & Pincus, 1995, p. 34). To some extent, the opinions of these experts can be influenced by changes in our society: "Throughout the manufacture of the *DSM*, people are making decisions and judgments in a social context. Whether or not a new set of behaviors warrants a diagnostic label depends on culturally varying judgments about what is clinically significant" (Houts, 2002, p. 53).

ALTERNATIVE DIRECTIONS IN DIAGNOSIS AND CLASSIFICATION

DSM has always offered a **categorical approach** to diagnosis. The word *categorical* refers to the basic view that an individual "has" or "does not have" the disorder—that is, the individual can be placed definitively in the "yes" or "no" category regarding a particular form of psychopathology. Popular and professional language reflects this categorical approach: "Does my child *have* ADHD?" "Some of my clients *have* bipolar disorder." "Michael *has* obsessive-compulsive disorder."

In recent years, and especially regarding some disorders (e.g., personality disorders), noncategorical approaches to psychopathology have received significant attention and empirical support. Specifically, the **dimensional approach** has been proposed by a number of researchers and clinicians (e.g., Blashfield, 1991; Costa & Widiger, 2001; Trull & Durrett, 2005; Widiger & Trull, 2007). According to a dimensional approach, the issue isn't the presence or absence of a disorder; instead, the issue is where on a continuum (or "dimension") a client's symptoms fall. As an example, consider Robert, a client with a strong tendency to avoid social situations because he fears criticism and rejection from others. A categorical system would require Robert's psychologist to determine whether Robert has a particular disorder—perhaps social phobia or avoidant personality disorder. A dimensional system wouldn't require a yes or no answer to the question. In place of

this dichotomous decision, Robert's psychologist would be asked to rate Robert on a continuum of anxious avoidance of social situations. In other words, rather than a "black-or-white" type of decision, Robert's psychologist would determine which "shade of gray" best describes Robert's symptoms.

The dimensions of psychological disorders may not be defined by traditional *DSM* categories. Instead, proponents of the dimensional approach to abnormality suggest that all of us—the normal and the abnormal—share the same fundamental characteristics but that we differ in the amounts of these characteristics that we each possess. What makes some of us abnormal is an unusually high or low level of one or more of these characteristics.

But what are these fundamental shared characteristics? The *five-factor model of personality* (also known as the "Big Five" model) is a leading candidate, at least for the personality disorders, and perhaps more broadly (Bagby & Whitaker, 2018; Costa & Widiger, 2001; Crego, Oltmanns, & Widiger, 2018; Gore & Widiger, 2013; Helle, DeShong, Baraldi, & Mullins-Sweatt, 2018; J. D. Miller, Lynam, Siedor, Crowe, & Campbell, 2018; Moorman & Samuel, 2018; D. Watson, Stasik, Ro, & Clark, 2013; Widiger & Mullins-Sweatt, 2009; Widiger & Trull, 2007). In other words, according to the dimensional approach to abnormality, each of our personalities contains the same five basic factors—neuroticism, extraversion, openness to experience, agreeableness, and conscientiousness. These five factors (or some variation of them) could constitute the dimensions on which clinical psychologists describe clients with personality problems. Thus, in the case of Robert (described earlier), instead of giving him a high rating on a dimension called "avoidant personality disorder" or "social phobia," the clinician might give the client a low rating on the dimension of extraversion. (As mentioned earlier in this chapter, the authors of *DSM-5* seriously considered a dimensional make-over of the personality disorders but ultimately decided against it, at least for now.)

Conceptualizing people's psychological problems the way *DSM* always has (i.e., categorically) certainly has many advantages. In fact, some have argued that categorical thinking is unavoidable and essential to our understanding and communication about mental disorders: "Just as water is basic to human existence, classification is fundamental to human cognition. As human beings, we are hardwired to think in categorical terms. Human cognition without categorization is unimaginable" (Blashfield & Burgess, 2007, p. 113). Perhaps most important, categories of mental illness facilitate efficient communication between professionals as well as nonprofessionals. If we instead conceptualized people's psychological problems dimensionally, communication would necessarily become more cumbersome. Moreover, generations of mental health professionals have been trained to understand psychopathology categorically, so it has become firmly established as the preferred model (Lilienfeld & Landfield, 2008a), and accumulating evidence suggests that some specific disorders, such as the psychotic disorders, may indeed exist categorically rather than dimensionally (Linscott & van Os, 2010). However, the dimensional model offers some interesting advantages as well. It allows for a more thorough description of clients. Also, rather than arbitrarily dividing a dimension into two categories, it allows for more detailed placement on that dimension. Although clinical psychologists currently work within the categorical *DSM* system, a growing body of research supports the possible incorporation of a dimensional model of psychopathology, perhaps in some kind of combination with categorical approaches and at least with

Web Link 7.5
Dimensional approach

the personality disorders (Achenbach, 2009b; Costa & Widiger, 2001; Maser et al., 2009; Simonsen, 2010; Skodol, 2010; Widiger & Gore, 2015; Widiger & Trull, 2007). Although this dimensional model was not used to revamp the personality disorders section of *DSM-5*, it may appear in some form in a future edition of the manual.

BOX 7.5

Metaphorically Speaking

If You've Eaten Chocolate Chip Cookies, You Understand the Dimensional Model of Psychopathology

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Chocolate chip cookies are usually made from the same short list of ingredients: flour, sugar, chocolate chips, butter, eggs, baking soda, vanilla. If you bite into a chocolate chip cookie and it tastes unusual, atypical, or even "abnormal," what's the reason? Perhaps it contains the same ingredients as other chocolate chip cookies but simply in different amounts—more sugar, less butter, fewer chocolate chips—than "normal" cookies. In other words, the difference between "abnormal" and "normal" cookies could be quantitative. Or maybe the "abnormal" cookie contains some entirely different ingredients that "normal" cookies simply don't contain—nutmeg, cinnamon, pepper, garlic, or something else. In this case, the difference is qualitative.

The proponents of the dimensional model of psychopathology believe that the same question of qualitative versus quantitative differences between normality and abnormality applies to people as well as to chocolate chip cookies. They believe that all our personalities are made of the same short list of "ingredients," or traits. In the past few decades, they have increasingly pointed to the Big Five—neuroticism, extraversion, openness, conscientiousness, and agreeableness—as characteristics that are common to all of us, including normal and abnormal individuals. According to the dimensional model, what is abnormal about abnormal people is the amount of one of these basic ingredients—extremely high neuroticism, extremely low extraversion, or extremely high openness, for example. In other words, abnormal people have the same "ingredients" as normal people but in significantly different amounts—a quantitative difference. In contrast, supporters of the categorical approach to psychopathology tend to believe that the difference between abnormal and normal people is more qualitative, suggesting that abnormal people have an "ingredient" that normal people simply don't have.

Certain forms of psychopathology tend to lend themselves to the dimensional model more than do others. Specifically, a growing body of research suggests that the personality disorders can be conceptualized quite well using the dimensional approach based on the Big Five personality factors (e.g., Widiger & Mullins-Sweatt, 2009; Widiger & Trull, 2007). This makes some intuitive sense: If human personality essentially consists of these five factors and we have a category of disorders called "personality disorders," it stands to reason that the personality disorders we have identified should

map onto the five factors of personality we have recognized. Empirical support for this theory has suggested several specific links: Paranoid personality disorder is characterized by very low agreeableness; histrionic personality disorder is characterized by very high extraversion; obsessive-compulsive personality disorder is characterized by very high conscientiousness; borderline personality disorder is characterized by very high neuroticism. Evidence linking the Big Five to other disorders, however, is less impressive (Costa & Widiger, 2001).

The debate regarding the dimensional and categorical models of psychopathology evokes important questions for clinical psychologists. Are people, in fact, all made of the same psychological "ingredients"? If so, should we understand the differences between abnormal and normal people as quantitative? Or are we made of entirely different "ingredients," and would it be more advantageous to understand our differences as qualitative? How might the answers to these questions influence the assessment and treatment that clinical psychologists conduct?

CHAPTER SUMMARY

Diagnosis is a primary task of clinical psychologists, but the process of defining mental disorders, and abnormality more generally, has been the subject of some debate. The *DSM* is generally considered the authoritative source of definitions for mental disorders, and it is widely accepted and used among mental health professionals. The *DSM* has been revised several times since its original publication by the American Psychiatric Association in 1952. The most recent edition, *DSM-5*, was published in 2013 and offers numerous new features, new disorders, and revised definitions of previously included disorders, many of which have elicited controversy and criticism. The *DSM* has always employed a medical model of psychopathology characterized by a categorical approach that describes disorders as either present or absent. An alternative, dimensional approach has been promoted in recent years, especially for the personality disorders, which would describe the extent to which clients have particular universal characteristics rather than whether or not they have a disorder. The *DSM* has become increasingly empirical and reliable as it has been revised, but it has also received criticism for pathologizing aspects of normal life, using arbitrary cutoffs regarding some diagnostic criteria, paying insufficient attention to issues of culture and diversity, and allowing nonempirical factors too much influence.

KEY TERMS AND NAMES

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proposed criteria set 154

CRITICAL THINKING QUESTIONS

1. In your opinion, what factors are most important in distinguishing abnormal from normal behavior?
2. In your opinion, who should define normal versus abnormal behavior?
3. How might the changes in the *DSM-5* influence both clinical psychologists and the clients they treat?
4. Of the criticisms of the current *DSM* described in this chapter, which seem most and least legitimate?
5. In your opinion, what are the strengths and weaknesses of the categorical and dimensional approaches to diagnosis?

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

American Journal of Psychiatry
http://ajp.psychiatryonline.org/

Depression and Anxiety
http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1520-6394

Journal of Abnormal Child Psychology
http://link.springer.com/journal/10802

Journal of Abnormal Psychology
http://www.apa.org/pubs/journals/abn/index.aspx

Journal of Clinical Psychology
http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1097-4679

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8

THE CLINICAL INTERVIEW

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This chapter focuses on the **clinical interview**, and it is the first of the series of chapters on techniques of **assessment**. Assessment is closely and uniquely associated with the identity of clinical psychology. None of the other mental health professionals—psychiatrists, social workers, professional counselors, nonclinical psychologists—incorporate assessment into their training and their work to the extent clinical psychologists do.

Any assessment technique used by a clinical psychologist should possess the qualities of validity, reliability, and clinical utility (Hogan & Tsushima, 2016; J. M. Wood et al., 2007). An assessment technique has **validity** to the extent that it measures what it claims to measure (Hood, 2015). If you've taken certain "intelligence" or "personality" tests, perhaps on the Internet or in popular magazines, and wondered, "Does this really measure what it says it measures?" you were wondering about validity. The technique has **reliability** to the extent that it yields consistent, repeatable results (Hogan, 2015). If its results depend on when, where, or how the technique was administered, its reliability might be questionable. Finally, assuming the technique is used in clinical settings, it should provide **clinical utility** as well. Even the most valid and reliable test ever created

Learning Objectives

8.1 Summarize key skills and behaviors for a psychologist conducting a clinical interview.

8.2 Define rapport as it relates to a clinical interview.

8.3 Explain major techniques that a clinician may use during a clinical interview.

8.4 Evaluate the potential impact of note taking during the clinical interview.

8.5 Describe characteristics of a clinician's office that may contribute to successful clinical interviews.

8.6 Differentiate among the types of clinical interviews.

8.7 Propose ways a psychologist could acknowledge cultural differences while interviewing clients from differing backgrounds.

wouldn't be helpful to a clinical psychologist if it didn't benefit the clinician and, ultimately, the client in some meaningful way (DeLao & Keeley, 2015). These terms, as well as subtypes of validity and reliability, are explained further in Table 8.1.

An element common to all kinds of psychological assessment is *feedback*. In other words, clinical psychologists provide their clients with the results of tests or interviews that have been conducted (American Psychological Association, 2002). This feedback can come in the form of a face-to-face meeting, a written report, or other forms. Psychologists learn to provide feedback during graduate training, but postgraduate internships and professional experience also shape the way they provide feedback to clients (K. T. Curry & Hanson, 2010). Most psychologists believe that clients find their feedback to be helpful and positive, even before any type of intervention (e.g., psychotherapy) is implemented (S. R. Smith, Wiggins, & Gorske, 2007).

Clinical psychologists conducting assessments can employ a wide variety of methods, including intelligence tests, achievement tests, neuropsychological tests, personality tests, and specialized measures for targeted variables. In fact, psychological testing, which we cover in the next chapters, has been called "one of the brightest jewels in the crown of clinical psychology" (J. M. Wood et al., 2007, p. 77). However, of all the available procedures, clinical psychologists rely most frequently on the clinical interview. The vast majority of clinical psychologists use interviews, and few assessments are conducted without an interview of some kind (Norcross, Hedges, & Castle, 2002; Watkins, Campbell, Nieberding, & Hallmark, 1995).

THE INTERVIEWER

The most pivotal element of a clinical interview is the person who conducts it. A skilled interviewer not only is a master of the technical and practical aspects of the interview but also demonstrates broad-based wisdom about the human interaction interviewing entails.

Web Link 8.1
Psychological assessment and assessment journals

General Skills

Before leaping into any interview, the interviewer should have acquired some general skills to serve as a foundation for interviewing in any context. These general skills focus on the interviewer's own frame of mind rather than any particular set of techniques. Sommers-Flanagan and Sommers-Flanagan (2009) describe several such requirements: quieting yourself, being self-aware, and developing positive working relationships.

Table 8.1 Defining Validity, Reliability, and Clinical Utility for Assessment Methods		
Validity	is the extent to which the assessment technique . . .	measures what it claims to measure.
Content validity	is the extent to which the assessment technique . . .	has content appropriate for what is being measured.
Convergent validity	is the extent to which the assessment technique . . .	correlates with other techniques that measure the same thing.
Discriminant validity	is the extent to which the assessment technique . . .	does not correlate with techniques that measure something else.
Reliability	is the extent to which the assessment technique . . .	yields consistent, repeatable results.
Test-retest reliability	is the extent to which the assessment technique . . .	yields similar results across multiple administrations at different times.
Interrater reliability	is the extent to which the assessment technique . . .	yields similar results across different administrators.
Internal reliability (e.g., split-half reliability)	is the extent to which the assessment technique . . .	consists of items that are consistent with one another.
Clinical utility	is the extent to which the assessment technique . . .	improves delivery of services or client outcome.

Source: Adapted from Wood et al. (2007).

The term **quieting yourself** does not simply mean that the interviewer shouldn't talk much during the interview (although rambling by the interviewer is usually not a good idea). Rather than the interviewer's speech, what should be quieted is the interviewer's internal, self-directed thinking pattern. To the extent that interviewers are preoccupied by their own thoughts, they will struggle in the fundamental task of listening to their clients. Simply put, the voice in the interviewer's own mind should not interrupt or drown out the voice of the client.

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Photo 8.1
The clinical interview is a core element of most assessments conducted by clinical psychologists. In your opinion, what are the most important skills and behaviors of the interviewer?

As an example, consider Joseph, a second-year graduate student in a clinical psychology program. Joseph has just begun his supervised practicum work in an outpatient mental health clinic, and clinical interviews are among his frequent activities at the site. As a novice interviewer, Joseph may find a number of self-directed questions repeatedly running through his mind while an interview takes place: "Was that a good question? Should I have asked something else? What does this client think of me? Do I come across as competent? Am I competent? What should I ask next?" A certain amount of self-reflection and thoughtfulness about the ongoing process of an interview is beneficial, but Joseph's extensive self-questioning may distract him from his client. As Joseph gains more experience and confidence through graduate school and later as a professional, his listening capacity may increase as he is increasingly able to quiet such internal questions.

Being self-aware is another important skill for the clinical interviewer. This self-awareness should not be confused with the excessive self-consciousness exemplified by Joseph in the previous example, which should be minimized. The type of self-awareness that should be maximized is the interviewer's ability to know how he or she tends to affect others interpersonally and how others tend to relate to him or her. To illustrate this point, imagine yourself as a new client in a clinical psychologist's waiting room for the first time. When the psychologist comes out to greet you, he looks and speaks exactly like Albert Einstein. Would his appearance and demeanor influence the way you responded to him during the interview? What if the psychologist resembled Kim Kardashian? Or LeBron James? Or Betty White? Every interviewer has a distinct combination of idiosyncratic qualities—looks, voice, mannerisms, body language, accents, and so on—that may evoke certain responses from clients. Skilled interviewers are cognizant of their unique traits, including their cultural assumptions as well as more superficial characteristics, and consider their effect on the interview process (Sommers-Flanagan & Sommers-Flanagan, 2009).

In addition to quieting themselves and knowing themselves, good interviewers appreciate the importance of **developing positive working relationships** with clients. In many clinical settings, an interview client is likely to become a psychotherapy client, so a good start to the working relationship benefits both parties. There is no formula for developing positive working relationships during an interview; however, attentive listening, appropriate empathy, genuine respect, and cultural sensitivity play significant roles. Positive working relationships are always a function of the interviewer's attitude as well as the interviewer's actions (Sommers-Flanagan & Sommers-Flanagan, 2009).

Specific Behaviors

When interviewers succeed in quieting themselves, knowing themselves, and developing good working relationships, they have laid the groundwork for conducting successful interviews. The next task is to master the tools of the trade, the specific behaviors characteristic of effective interviewers.

As mentioned earlier, a primary task of the interviewer is to listen. **Listening** may seem like a simple enough task, but it can be broken down into even more fundamental building blocks of **attending behaviors** (many described in Ivey, Ivey, & Zalaquett, 2010).

Eye Contact

Perhaps it's an obvious point, but interviewers must realize the connection between attentive listening and eye contact. Eye contact not only facilitates listening, but it also communicates listening. That is, when the client notices the interviewer's continuous, appropriate eye contact, the client feels heard. Of course, culture plays a significant role in the meaning of eye contact. In some cultures, eye contact that lasts too long or is too intense may communicate threat, seduction, or other messages that an interviewer would be wise to avoid. On the other hand, scarce, inconsistent eye contact may be viewed by some clients as inattentive or insulting. Like so many other aspects of the clinical interview, eye contact is a specific behavior that requires cultural knowledge and sensitivity on the part of the interviewer, both as the sender and receiver of eye contact.

Body Language

As with eye contact, culture can shape the connotations of **body language**. A few general guidelines for the interviewer include facing the client, appearing attentive, minimizing restlessness, and displaying appropriate facial expressions. But even these guidelines are subject to cultural interpretation, and a client may not feel understood if an interviewer unknowingly communicates disinterest through body language. Furthermore, the client's body language can be misinterpreted by an interviewer whose knowledge of the client's cultural background is deficient.

Vocal Qualities

Skilled interviewers have mastered the subtleties of the **vocal qualities** of language—not just the words but how those words sound to the client's ears. They use pitch, tone, volume, and fluctuation in their own voices to let clients know that their words and feelings are deeply appreciated. And these interviewers also attend closely to the vocal qualities of their clients.

Verbal Tracking

Effective interviewers are able to repeat key words and phrases back to their clients to assure the clients that they have been accurately heard. Interviewers don't do this in a

mechanical way; instead, they weave the clients' language into their own. In addition, interviewers skilled at **verbal tracking** monitor the train of thought implied by clients' patterns of statements and are thus able to shift topics smoothly rather than abruptly.

Referring to the Client by the Proper Name

It sounds simple enough, but using the client's name correctly is essential (Fontes, 2008). Inappropriately using nicknames or shortening names (e.g., calling Benjamin "Benji"), omitting a "middle" name that is in fact an essential part of the first name (e.g., calling John Paul "John"), or addressing a client by first name rather than a title followed by surname (e.g., addressing Ms. Washington as "Latrice") are presumptuous mistakes that, for some clients, can jeopardize the sense of comfort with the interviewer. Especially with clients of certain ethnicities or ages, the misuse of names in this way may be disrespectful and may be received as a microaggression (discussed in Chapter 4; D. W. Sue, Capodilupo, & Holder, 2008; D. W. Sue et al., 2007). The initial interview is an ideal opportunity to ask clients how they would prefer to be addressed and to confirm that it is being done correctly. This gesture in itself can be interpreted by clients as a sign of consideration.

Observing Client Behaviors

All of the previously described interviewer behaviors, and many other important decisions the interviewer makes, can be informed by *behavioral observations* of the client by the interviewer (Kamphaus & Dever, 2016; Leichtman, 2009). Typically, when psychologists write a report summarizing the results of an assessment (including the clinical interview), that report contains at least a brief section describing the behavior of the client during the process. That behavior can vary widely: Some clients are calm while others are nervous; some are easygoing while others are hostile; some stay on task while others stray in random directions; some are emotional while others are stoic; and so on. Observing client behaviors during the interview allows the psychologist to consider not only what the client said but how the client said it, and that "how" component can offer important information to the psychologist about responding during the interview and understanding the client.

The interviewer should carry out all these attending behaviors naturally and authentically (Ivey et al., 2010). Beginners may need to make deliberate attempts to incorporate such behaviors, but with experience, they should occur more effortlessly. As they do, clients will respond more positively.

Specific examples of the importance of cultural variables to listening during the clinical interview may help illustrate the previously mentioned points. Compared with traditional Western culture, Chinese culture tends to feature far less eye contact, especially between members of the opposite sex. Eye contact that falls within the normal range among Western individuals may communicate rudeness or sexual desire among Chinese people. Norms for physical touch also differ from one culture to another, with Asian cultures typically incorporating less than European Americans. People from diverse cultures

have also developed measurable differences in the physical distance or "personal space" they maintain. Individuals of Middle East and Latino/Latina cultures tend to stand or sit closer together, whereas North American and British people tend to maintain more space between themselves. Tendencies such as these represent essential knowledge for the culturally competent interviewer (Ivey et al., 2010; D. W. Sue & Sue, 2008).

COMPONENTS OF THE INTERVIEW

Although interviews may vary drastically according to the setting, purpose, and other factors, several components are universal to interviews: rapport, technique, and conclusions.

Rapport

Rapport refers to a positive, comfortable relationship between interviewer and client. When clients feel a strong sense of rapport with interviewers, they feel that the interviewers have "connected" with them and that the interviewers empathize with their issues. In this situation, clients tend to disclose more information and invest themselves further in the interview process than they might have otherwise.

How can an interviewer establish good rapport with clients? No foolproof technique exists, especially across the culturally, demographically, and diagnostically wide range of clients with whom a clinical psychologist might work. And if the goal is to make the client feel connected and sympathized with, there is no substitute for genuinely connecting and sympathizing with the client. However, there are some specific efforts interviewers can make to enhance the client's experience of rapport.

First, interviewers should make an effort to put the client at ease, especially early in the interview session. For some clients, the first visit to a clinical psychologist can be daunting. They may be fearful about the interview or may be under a misperception of what the session will entail. Engaging in small talk about innocuous topics ("Did you have any trouble finding the office?" or "I haven't been outside in a while—is it still cold out?") for a minute or two at the beginning of the interview can make an initially nervous client feel more comfortable (Othmer & Othmer, 1994). Of course, this small talk should not be excessive, nor should it distract from the purposes of the interview.

Second, interviewers can acknowledge the unique, unusual situation of the clinical interview. From the client's perspective, the interview experience is unlike any other interaction: They are expected to discuss private, difficult, often emotionally charged issues with a person they just met and know very little about. Letting clients know that you recognize their position and appreciate their willingness to participate communicates empathy and enhances rapport. Along the same lines, inviting clients to ask questions about the interview process provides them with a sense of knowledge and control, which can also improve their comfort level (Sommers-Flanagan & Sommers-Flanagan, 2009).

Third, interviewers can enhance rapport by noticing how the client uses language and then following the client's lead. Interviewers should pick up on the client's vocabulary and, as much as possible, speak in similar terms (Othmer & Othmer, 1994). For example,

if a client frequently uses visual imagery ("I can't see the forest for the trees," "That's a blind spot for me," etc.), interviewers should make an effort to use visual terminology as well. If they use more tactile, "touch" words ("It feels like everything's slipping through my fingers," "I'm losing my grip," etc.), interviewers should respond accordingly. When a client uses a metaphor, the interviewer can extend it. For example, consider Ian, a full-time undergraduate student with a demanding job, several extracurricular activities, and numerous family obligations. When he describes his anxiety and stress by saying, "It's like I'm juggling, and I can't keep all the balls in the air," the interviewer might establish better rapport by asking, "What if one of the balls fell?" than "What would happen if you failed in one of your responsibilities?" Clients whose interviewers literally "speak their language" tend to feel relaxed and understood.

Technique

If rapport is *how* an interviewer *is* with clients, **technique** is *what* an interviewer *does* with clients. These are the tools in the interviewer's toolbox, including questions, responses, and other specific actions.

BOX 8.1

Considering Culture

Communication Across Cultures

Members of some cultures tend to exhibit identifiably distinct communication behaviors. Interviewers should accept this concept generally, but they should also be aware of as many of these distinct communication tendencies as possible.

D. W. Sue and Sue (2008) identify some of these variables in communication style and describe how they differ across ethnicities. In terms of speaking volume, Asian Americans and Hispanic Americans tend to speak more softly than whites. Whites, in turn, tend to speak more quickly than some other cultural groups, such as Native Americans. African Americans tend to maintain more direct eye contact when speaking than when listening, whereas the opposite is true for whites. Native Americans often display an indirect gaze during conversation, whereas Asian Americans are inclined to avoid eye contact when speaking to people of higher status (including, possibly, the interviewer). And when we consider the amount of time before responding, African Americans and whites are likely to respond rather quickly, whereas Native Americans, Asian Americans, and Hispanic Americans may demonstrate a mild delay.

But ethnicity is not the only characteristic on which communication differences may center. Mio, Barker-Hackett, and Tumambing (2009) and J. Wood (1994, 1999) highlight the observable and common differences in speaking style between men and women. Men tend to use talk to assert themselves, promote their ideas, and achieve goals, whereas women are more likely to use talk to build rapport and sustain relationships. Women are apt to value self-disclosure for the closeness it brings; men are apt to believe that self-disclosure increases vulnerability. When men use speech to help others, the help is likely to take the form of

advice or problem solving. In contrast, when women use speech to help others, the help is likely to take the form of empathy or understanding of emotions.

Any or all of these differences could be among the cultural factors of interest to the clinical interviewer. Imagine, for example, that Dr. Ryan, a clinical psychologist, was to conduct clinical interviews of five clients: a Hispanic American woman, an African American man, an Asian American woman, a Native American man, and a white woman. How crucial is it for Dr. Ryan to have some appreciation of ethnicity- and gender-related communication tendencies? As each interview ends, how might Dr. Ryan's conclusions depend on this appreciation of communication tendencies?

Directive Versus Nondirective Styles

Interviewers who use a **directive** style get exactly the information they need by asking clients specifically for it. Directive questions tend to be targeted toward specific pieces of information, and client responses are typically brief, sometimes as short as a single word (e.g., "yes" or "no"). On the other hand, interviewers who use a **nondirective** style allow the client to determine the course of the interview. Without direction from the interviewer, a client may choose to spend a lot of time on some topics and none on others.

Certainly, both directive and nondirective approaches play an important role in interviewing. Direct questioning can provide crucial data that clients may not otherwise choose to discuss: important historical information ("How often have you attempted suicide in the past?" "Is there a history of schizophrenia in your family?"), the presence or absence of a particular symptom of a disorder ("How many hours per day do you typically sleep?"), frequency of behaviors ("How often have you had panic attacks?"), and duration of a problem ("How long has your son had problems with hyperactivity?"), among other information. For example, imagine Dr. Molina interviewing Raymond, a client whose presenting problem is depressive in nature. If Dr. Molina directly inquires about symptoms and duration—"When did you first notice these feelings of sadness? Has your weight changed during that time? Do you feel fatigued? Do you have trouble concentrating?"—Dr. Molina will be sure to gather information essential for diagnosis.

Indirect questioning, conversely, can provide crucial information that interviewers may not otherwise know to inquire about. If, instead of the direct questions listed in the previous paragraph, Dr. Molina asks Raymond, "Can you tell me more about your feelings of sadness?" Raymond's response may not specifically mention how long the symptoms have been present or refer to each individual criterion for major depression, dysthymia, or other related disorders. However, Dr. Molina's indirect question does allow Raymond to expand on anything he believes is essential, including symptoms, duration, background data, or other elements that may prove to be extremely relevant (J. Morrison, 2008).

Both directive and nondirective approaches have shortcomings as well, especially when an interviewer relies too heavily on either one. Sometimes, directive approaches can sacrifice rapport in favor of information. In other words, interviewers who are overly

directive may leave clients feeling as though they didn't have a chance to express themselves or explain what they thought was important. Although nondirective interviewing can facilitate rapport, it can fall short in terms of gathering specific information. Interviewers who are overly nondirective may finish an interview without specific data necessary for a valid diagnosis, conceptualization, or recommendation.

Perhaps the best strategy regarding directive and nondirective interviewing is one that involves balance and versatility (J. Morrison, 2008). At the outset of the interview, it may be wise to allow clients to speak freely to communicate what they think is most important. Even without direction from the interviewer, most clients will nonetheless address a number of direct questions that the interviewer might have asked. Later in the session, the interviewer can become more directive, asking the remaining targeted questions and soliciting more limited answers (e.g., yes/no, specific factual information) from clients.

Specific Interviewer Responses

Ultimately, interviewing technique consists of what the interviewer chooses to say. The interviewer's questions and comments can span a wide range and serve many purposes. There are numerous common categories of interviewer responses: open- and closed-ended questions, clarification, confrontation, paraphrasing, reflection of feeling, and summarizing (Othmer & Othmer, 1994; Sommers-Flanagan & Sommers-Flanagan, 2009).

Open- and Closed-Ended Questions

Whether a question is open- or closed-ended can have great impact on the information a client provides an interviewer. For example, consider a clinical psychologist interviewing Brianna, a 24-year-old outpatient client who mentioned in her initial phone call that she thinks she may be bulimic. An **open-ended question**, such as, "What more can you tell me about the eating problems you mentioned on the phone?" may take the interview in a different direction than a **closed-ended question**, such as, "How many times per week do you binge and purge?" "Which purging methods do you use—vomiting, exercise, nausea?" or "Have you been diagnosed with an eating disorder in the past?" Open-ended questions allow for individualized and spontaneous responses from clients. These responses tend to be relatively long, and although they may include a lot of information relevant to the client, they may lack details that are important to the clinical psychologist. Closed-ended questions allow for far less elaboration and self-expression by the client but yield quick and precise answers. Indeed, open-ended questions are the building blocks of the nondirective interviewing style, whereas the directive interviewing style typically consists of closed-ended questions.

Clarification

The purpose of a **clarification** question is to make sure the interviewer has an accurate understanding of the client's comments. Clarification questions not only enhance the interviewer's ability to "get it," they also communicate to the client that the interviewer is actively listening and processing what the client says. For example, with Brianna (the client with bulimic symptoms described earlier), the interviewer may at one point say,

BOX 8.2

Metaphorically Speaking

If You've Taken Multiple-Choice, True/False, and Essay Exams, You Understand Open- and Closed-Ended Interview Questions

As you are surely well aware by this point in your academic career, there is a big difference between multiple-choice/true-false (MC/TF) exams and essay exams. MC/TF questions usually target a specific piece of information, and students' responses are restricted (in most cases, to a single word or letter). There are usually more items on an MC/TF exam than on an essay exam, because each MC/TF question is brief and of limited scope. As a student, when you take an MC/TF exam, you are providing your teacher with the units of knowledge that the teacher has deemed important.

Essay exams are different in that they allow you to "show what you know" in a more unrestricted way. Although you may or may not provide the specific information that the teacher is looking for, you nonetheless choose what you think is most important in responding to the questions. The number of essay questions is typically small, since they allow for longer, more expansive answers. The scope of essay questions is typically broader than that of MC/TF questions.

Closed-ended interview questions are quite similar to MC/TF questions, whereas open-ended interview questions are quite similar to essay exam questions. Closed-ended questions (and the structured interviews in which they are prominently featured) are more targeted; elicit brief, pointed responses; and allow for less elaboration. They usually appear in long series, and they reflect the specific topics that the interviewer considers most relevant. Open-ended questions (and the unstructured interviews in which they tend to appear) are less focused and allow for more free-form responses. Clients determine the content of these responses according to the topics they believe are most relevant. There are usually a relatively small number of open-ended questions, since they can elicit lengthy responses.

One important point about this metaphor is that as the interviewer, your position is parallel to that of the teacher, not the student. In other words, as students, you are accustomed to answering test questions, but as interviewers, your task is to design test questions. In your opinion, what are the advantages and disadvantages from the teacher's perspective of MC/TF and essay exams? If you were the teacher, would one type of exam question (MC/TF or essay) tend to produce more valid assessments of your students' knowledge than the other? If you were an interviewer, would one type of interview question (open- or closed-ended) tend to produce more valid assessments of your clients than the other? What about combining the two types of questions?

"You mentioned that a few months ago you started exercising excessively after eating large amounts of food but that you've never made yourself vomit—do I have that right?" or "I want to make sure I'm understanding this correctly—did you mention that you've been struggling with eating-related issues for about 6 months?" Either of these questions would allow Brianna to confirm or correct the interviewer so that ultimately the client is correctly understood.

An important note about clarification: Sometimes, it's better to wait than to immediately demand clarification. Often, novice interviewers are so concerned about getting all the facts straight that they will interrupt the client at inopportune moments, particularly when the client is describing emotionally laden events. For example, imagine Dr. Palmer interviewing Austin, a client whose is seeking help for a phobia of driving that he developed after a serious car accident a few months ago. His work with Dr. Palmer starts with a clinical interview, and during the interview, Austin decides to describe the accident itself, in detail. It's one of the few times that Austin has described the accident to anyone, and doing so makes him very upset. As he starts to cry, Austin describes how he was making a left turn out of a gas station and saw two cars on the road: a red car coming from the left and a blue pickup truck coming from the right. Through more tears, he describes how he misjudged the distance and speed of the black pickup truck—but before he can finish that sentence and continue on to the moment of impact, Dr. Palmer interrupts him: "Wait—I thought the pickup truck was blue, but you just said black. What color was it?" By doing so, Dr. Palmer will get the information correct (Austin misspoke the second time—the pickup truck was in fact blue), but she risks coming across as insensitive to Austin's emotional experience. She may strike Austin as being more concerned about minor objective facts than his overall subjective experience. For a detail like this one, it might be wiser for Dr. Palmer to wait to clarify. Later, when Austin is less emotional, if it still seems important to know what color the pickup truck was, Dr. Palmer can choose to seek clarification. If it seems unimportant by that point, she can choose not to do so.

Confrontation

Interviewers use **confrontation** when they notice discrepancies or inconsistencies in a client's comments. Confrontations can be similar to clarifications, but they focus on apparently contradictory information provided by clients. For example, an interviewer might say to Brianna, "Earlier, you mentioned that you had been happy with your body and weight as a teenager, but then a few minutes ago you mentioned that during high school you felt fat in comparison with many of your friends. I'm a bit confused."

Paraphrasing

Unlike clarifying or confronting, paraphrasing is not prompted by the interviewer's need to resolve or clarify what a client has said. Instead, **paraphrasing** is used simply to assure clients that they are being accurately heard. When interviewers paraphrase, they typically restate the content of clients' comments, using similar language. So if Brianna mentions, "I only binge when I'm alone," the interviewer might immediately respond with a statement such as, "You only binge when no one else is around." A paraphrase usually doesn't break new ground; instead, it maintains the conversation by assuring the client that the interviewer is paying attention and comprehending.

Reflection of Feeling

Whereas a paraphrase echoes the client's words, a **reflection of feeling** echoes the client's emotions. Reflections of feeling are intended to make clients feel that their emotions

are recognized, even if their comments did not explicitly include labels of their feelings. For example, if Brianna's comment "I only binge when I'm alone" was delivered with a tone and body language that communicated shame—her hand covering her face, her voice quivering, and her eyes looking downward—the interviewer might respond with a statement such as "You don't want anyone to see you bingeing—do you feel embarrassed about it?" Unlike paraphrasing, reflecting a client's feelings often involves an inference by the interviewer about the emotions underlying the client's words.

Summarizing

At certain points during the interview—most often at the end—the interviewer may choose to summarize the client's comments. **Summarizing** usually involves tying together various topics that may have been discussed, connecting statements that may have been made at different points, and identifying themes that have recurred during the interview. Like many of the other responses described in this section, summarizing lets clients know that they have been understood but in a more comprehensive, integrative way than, say, paraphrasing single statements. After a full interview with Brianna, an interviewer may summarize by offering statements along the lines of "It seems as though you are acknowledging that your bingeing and purging have become significant problems in recent months, and although you've kept it to yourself and you may feel ashamed about it, you're willing to discuss it here with me and you want to work toward improving it." An accurate summarization conveys to the client that the interviewer has a good grasp on the "big picture."

Conclusions

Typically, clinical interviews involve a **conclusion** of some kind made by the interviewer. The conclusion can take a number of different forms, depending on the type of interview, the client's problem, the setting, or other factors. In some cases, the conclusion can be essentially similar to a summarization. Or the interviewer might be able to go a step further by providing an initial conceptualization of the client's problem that incorporates a greater degree of detail than a brief summarization statement. In some situations, the conclusion of the interview may consist of a specific diagnosis made by the interviewer on the basis of the client's response to questions about specific criteria. Or the conclusion may involve recommendations. These recommendations might include outpatient or inpatient treatment, further evaluation (by another psychologist, psychiatrist, or health professional), or any number of other options.

Web Link 8.2
Effects of note taking on clients

PRAGMATICS OF THE INTERVIEW

The interviewer encounters quite a few decisions long before the client arrives for the interview. Many of these decisions involve the setting in which the interview will take place and the professional behaviors the interviewer plans to use during the interview itself.

Note Taking

Should an interviewer take notes during an interview? There is little consensus about note taking among clinical psychologists. Some write profusely during interviews, typing furiously on laptops or tablets, or filling pages with handwritten observations of their clients and selected snippets of the conversation. Others choose to take no notes at all during the session, although they may write what they recall soon after the interview ends (Greenson & Jaffe, 2004; Hickling, Hickling, Sison, & Radetsky, 1984).

There are good reasons for taking notes. Written notes are certainly more reliable than the interviewer's memory. Many clients will expect the interviewer to take notes and may feel as though their words will soon be forgotten if the interviewer is not taking notes. On the other hand, there are also drawbacks to taking notes. The process of note taking can be a distraction, both for the interviewer, who may fail to notice important client behaviors while looking down to write, and for the client, who may feel that the interviewer's notebook is an obstacle to rapport. Some interviewers aim for a middle ground between no notes and excessive notes, such that they jot down the most essential highlights of an interview but prevent the note-taking process from disrupting the conversation or their attention toward the client. In some clinical situations, interviewers may be wise to explain their note-taking behavior to clients. In particular, a client-centered rationale for the note-taking behavior—"I'm taking notes because I want to make sure I have a good record of what you have told me" or "I don't take notes during interviews because I don't want anything to distract either of us from our conversation"—may enhance rapport between interviewer and client (Fontes, 2008; Sommers-Flanagan & Sommers-Flanagan, 2009).

A few empirical studies have examined the effects of note taking on important interview variables, with mixed results. In one study, researchers measured the quality of the relationship between interviewer and interviewee in three note-taking conditions: paper and pen, iPad, and computer. They found no differences across the three conditions (Wiarda, McMinn, Peterson, & Gregor, 2014). In another study, the quality of the relationship did depend on note-taking condition: It was significantly lower for interviewers who used computers versus interviewers who did not (D. C. Rosen, Nakash, & Alegría, 2016). It is likely that the effect of note taking is highly dependent on the particular situation, including exactly how a client interprets the use of any particular mode of note taking and exactly how an interviewer uses that mode.

Audio and Video Recordings

Written notes are not the only way of documenting an interview. Clinical psychologists may prefer to audio- or video-record the session. Unlike note taking, recording a client's interview requires that the interviewer obtain written permission from the client. While obviously providing a full record of the entire session, recordings can, with some clients, hinder openness and willingness to disclose information. As with note taking, an explanation of the rationale for the recording, as well as its intended use (e.g., review by the interviewer or a supervisor) and a date by which it will be erased or destroyed, is typically appreciated by the client.

BOX 8.3

In My Practice . . .

In my practice, I've had some interesting experiences regarding note taking during clinical interviews. Earlier in my career, I placed a high priority on the comprehensiveness of my notes. I tried to write down as much as I could as quickly as I could. This approach to note taking yielded interview notes that were lengthy and detailed but often proved costly in terms of rapport. One particularly outspoken client interrupted my scribbling and said, "Are you even listening?" I was surprised by the comment—I had indeed been listening intently and writing down every word—until I realized that when I looked up at her to answer her question, that was the first eye contact I had made with her in quite a while. I've had the opposite experience, too—listening intently without taking notes, prompting at least one client to ask, "How are you going to remember all of this? Don't you need to write it down?"

The middle ground is a nice place to settle—jotting down occasional notes on the most important material but spending more time looking at the client than at the notepad or laptop. However, even the middle ground has its disadvantages. I've had several clients who showed a keen awareness of exactly when I choose to write. One client who had been encouraged by her supervisor at work to seek therapy asked, "Why were you writing stuff down when I was talking about my job? Are you going to tell my boss?" A young adult client remarked, "I noticed you didn't write much down when I was talking about how my parents treated me when I was a kid. Does that mean that you don't focus on that kind of stuff in therapy?" A socially anxious client said, "I bet you thought I was crazy when I was talking about how nervous I get around other people. Is that what you were writing right then? That I seem crazy?"

I have always found it quite beneficial to explain to clients why I make the decisions I do about note taking during interviews, whether the purpose is to aid my memory later, pay attention to them in the moment, or some combination of reasons. Clients typically respect the explanation, and the focus then rightly shifts to the content of the interview itself rather than the note-taking process.

To see this box come to life as a whiteboard video, please visit edge.sagepub/pomerantz5e.

The Interview Room

What should the interview room look like? In truth, there is tremendous variability among the rooms in which clinical psychologists conduct interviews (D. Jackson, 2018). The size of the room, its furnishings, and its decor are among the features that may differ. As a general rule, "when choosing a room [for interviews], it is useful to strike a balance between professional formality and casual comfort" (Sommers-Flanagan & Sommers-Flanagan, 2009, p. 31). That is, the interview room should subtly convey the message to the client that the clinical interview is a professional activity but one in which warmth and comfort are high priorities.

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Photo 8.2
In your opinion, what are the most important reasons for and against note taking during clinical interviews?

As a specific example, consider seating arrangements, one of the many features of interview rooms that may vary. There are quite a few options: the traditional (and increasingly rare) psychoanalytic arrangement in which the client lies on a couch while the interviewer sits in a chair, interviewer and client sitting face-to-face, interviewer and client in chairs at an angle between 90 degrees and 180 degrees, and others. A table or desk may be positioned between the chairs, small side tables may accompany each chair, or the chairs may have no furniture between them at all. And depending on the setting, the chairs themselves may run the gamut of anything available at office- or home-furniture stores. As long as the interview room facilitates the fundamental goals of the interview, such as gathering information and building rapport in a private setting free of interruptions, its purpose is well served (Sommers-Flanagan & Sommers-Flanagan, 2009).

Devlin and Nasar (2012; see also Nasar & Devlin, 2011) have surveyed both therapists and nonprofessionals regarding the impressions that clinicians' offices can make on clients. They showed participants 30 different photographs of actual clinicians' offices—the rooms where interviews (and psychotherapy) take place—and asked them to rate the rooms in a variety of ways. Results indicated that the most preferred offices were the ones that were orderly rather than messy, and "soft" (with comfortable seats, decorative rugs, muted lighting, art, and plants) rather than "hard" (sparsely furnished, institutional, or overly modern). Additionally, more spacious, uncluttered offices were rated more positively than offices that were cramped and featured disordered piles of papers and tangles of electric cords. Although individual responses may certainly differ among clients, such studies indicate that decisions clinicians make about the rooms where they conduct interviews and other clinical services can be quite important.

When making decisions related to the decor of an office, clinical psychologists usually steer clear of overtly personal items such as family photos, souvenirs, and memorabilia. Such items might be inconsistent with the professional ambience that the interview room should exude, and, perhaps more important, they can influence the content of the interview in some cases. For example, consider Rochelle and Javon, a married couple seeking help from Dr. Warner for relationship problems due to infertility. A prominent photo of Dr. Warner's own infant son or daughter may evoke feelings in this couple that would alter the course of the interview. Rochelle or Javon may experience envy or resentment or (whether accurate or not) assume that Dr. Warner may be unable to empathize fully with their difficulties, and these feelings may impair the interview process.

Confidentiality

When clients enter the interview room, they may have inaccurate assumptions about the *confidentiality* of the ensuing discussion. Many people incorrectly assume that any session with a psychologist is absolutely confidential (D. J. Miller & Thelen, 1986), when in fact there are some situations that require the psychologist to break confidentiality. These

situations are typically defined by state law and often involve the psychologist discovering during a session that the client intends to seriously harm someone (self or others) or that ongoing child abuse is occurring. (These exceptions to confidentiality are discussed in greater detail in Chapter 5, "Ethical and Professional Issues in Clinical Psychology.") Some clients may make the opposite mistake, wrongly assuming that their relatives, supervisors, or others will have access to their interview records, and as a result may choose to disclose very little about their problems.

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Photo 8.3
The interview room itself—its decor, furnishings, style, size, etc.—can influence the client's level of comfort and ultimately the success of the interview. If you were a clinical psychologist, what priorities would you have regarding the design of your office?

To inform their clients about confidentiality, and especially to correct any misconceptions such as those described earlier, interviewers should routinely explain policies regarding confidentiality as early as possible. Such explanations should be consistent with state law and professional ethics and provided in writing, with ample opportunity for oral discussion offered as well. Psychologists who discuss confidentiality and its limits with interview clients demonstrate competent and ethical practice (American Psychological Association, 2002).

TYPES OF INTERVIEWS

Clinical interviews take on different forms according to the demands of the situation, which may depend on the setting, the client's presenting problem, and the issues the interview is intended to address. In actual practice, there are countless unique, idiosyncratic varieties of interviews, but most fall into a few broad categories: intake interviews, diagnostic interviews, mental status exams, and crisis interviews.

Intake Interviews

The purpose of the **intake interview** is essentially to determine whether to "intake" the client to the setting where the interview is taking place. In other words, the intake interview determines whether the client needs treatment; if so, what form of treatment is needed (inpatient, outpatient, specialized provider, etc.); and whether the current facility can provide that treatment or the client should be referred to a more suitable facility (Sommers-Flanagan, 2016).

Intake interviews typically involve detailed questioning about the presenting complaint. For example, consider Julia, a 45-year-old client who arrives for an intake interview at an outpatient community mental health center and describes "hearing voices." Dr. Epps, the clinical psychologist conducting the intake interview, is likely to ask questions about the frequency, intensity, and duration of Julia's psychotic symptoms, as well as symptoms of other conditions that Julia may exhibit. Dr. Epps would also closely observe Julia during the interview and note any relevant behaviors. Questions about Julia's personal history, especially the previous existence of this or other psychological or

psychiatric problems, would also be an important part of the intake interview. At the conclusion of the interview, Dr. Epps would use all information solicited to decide whether Julia's problems would be best treated at this community mental health center and, if so, recommend a particular mode or frequency of treatment. If not, Dr. Epps would provide Julia with referrals to other treatment providers, including inpatient or outpatient settings as appropriate.

Diagnostic Interviews

As the name indicates, the purpose of the **diagnostic interview** is to diagnose. At the end of a well-conducted diagnostic interview, the interviewer is able to confidently and accurately assign *Diagnostic and Statistical Manual of Mental Disorders* (*DSM*) diagnoses to the client's problems. When an interview yields a valid, specific diagnosis, the effectiveness of the recommendations and subsequent treatment may be increased.

If the purpose of the diagnostic interview is to produce a diagnosis, it would make sense for the diagnostic interview to include questions that relate to the criteria of *DSM* disorders. How directly should the questions reflect the diagnostic criteria? Some clinical psychologists believe that the questions should essentially replicate the *DSM* criteria; after all, if our profession follows a diagnostic manual with disorders defined by meticulous lists of symptoms, what better way to reach a diagnosis than to ask directly about each symptom? Other clinical psychologists believe that every question in a diagnostic interview need not be coupled with a specific *DSM* diagnostic criterion. These clinical psychologists prefer a more flexible interview style in which they choose or create questions as they move through the interview, and they use inference rather than absolute fact to make diagnostic decisions. In other words, some clinical psychologists prefer structured interviews, while others prefer unstructured interviews.

Structured Interviews Versus Unstructured Interviews

A **structured interview** is a predetermined, planned sequence of questions that an interviewer asks a client. Structured interviews are constructed for particular purposes, usually diagnostic. An **unstructured interview**, in contrast, involves no predetermined or planned questions. In unstructured interviews, interviewers improvise: They determine their questions on the spot, seeking information that they decide is relevant during the course of the interview (Maruish, 2008; Mueller & Segal, 2015; O'Brien & Tabaczynski, 2007; Sommers-Flanagan, Zeleke, & Hood, 2015).

Structured interviews possess a number of advantages over unstructured interviews, particularly from a scientific or empirical perspective (Sommers-Flanagan & Sommers-Flanagan, 2009; Villa & Reitman, 2007):

- Structured interviews produce a diagnosis based explicitly on *DSM* criteria, reducing reliance on subjective factors such as the interviewer's clinical judgment and inference, which can be biased or otherwise flawed.
- Structured interviews tend to be highly reliable, in that two interviewers using the same structured interview will come to the same diagnostic conclusions far

more often than two interviewers using unstructured interviews. Overall, they are more empirically sound than unstructured interviews.

- Structured interviews are standardized and typically uncomplicated in terms of administration.

On the other hand, structured interviews have numerous disadvantages as well (Sommers-Flanagan et al., 2015):

- The format of structured interviews is usually rigid, which can inhibit rapport and the client's opportunity to elaborate or explain as he or she wishes.
- Structured interviews typically don't allow for inquiries into important topics that may not be directly related to *DSM* criteria, such as relationship issues, personal history, and problems that fall below or between *DSM* diagnostic categories.
- Structured interviews often require a more comprehensive list of questions than is clinically necessary, which lengthens the interview.

In the early days of clinical psychology, structured interviews were uncommon, but in recent years they have become more accepted and in many settings preferred. There are a number of published structured interviews that focus on specific diagnostic issues, such as the Anxiety and Related Disorders Interview Schedule for *DSM-5* (T. A. Brown & Barlow, 2014). However, especially in recent years, the most prominent structured interview has been the more wide-ranging **Structured Clinical Interview for *DSM-5* Disorders** (**SCID**; First, 2015; First, Gibbon, Spitzer, Williams, & Benjamin, 1997; First, Spitzer, Gibbon, & Williams, 1997a, 1997c).

The SCID was created by some of the leading authors of recent editions of the *DSM*, and it is essentially a comprehensive list of questions that directly ask about the specific symptoms of the many disorders included in the *DSM*. For the most part, each question in the SCID has a one-to-one correspondence with a specific criterion of a *DSM* diagnosis. In fact, the SCID administration booklet is structured in a two-column format such that each *DSM* symptom appears parallel to a question specifically about it. As an example, one of the specific *DSM* criteria for obsessive-compulsive disorder is "recurrent and intrusive distressing recollections of the event, including images, thoughts, or perceptions"; the SCID, next to this criterion, includes the question, "Did you think about [TRAUMA] when you did not want to or did thoughts about [TRAUMA] come to you suddenly when you didn't want them to?" (First, Spitzer, Gibbon, & Williams, 1997b, p. 74). Like this one, most SCID questions are designed to elicit "yes" or "no" answers, and clients can be asked to elaborate briefly when they say "yes."

The SCID is modular, which means that interviewers can choose only those modules (or sections) of the SCID that are relevant for a particular clinical case and omit the other modules. For example, imagine that Dr. Rodriguez is interviewing Ethan, an adult client in an outpatient clinic. In an initial phone call to the clinic and on a brief waiting-room screening form, Ethan has indicated that his problem involves a severe phobia of dogs. Dr. Rodriguez may choose to administer the anxiety module of the SCID but

Web Link 8.3
SCID

to omit the modules that are irrelevant to Ethan's problems, such as the module on psychotic disorders. One benefit of this modular design is a reduction in the time needed to complete the SCID. Of course, if an interviewer prefers to administer the entire SCID, no module omission is required. When the assessment involves the possible presence of a personality disorder, a separate version of the SCID designed for that purpose (the SCID-PD) is necessary (First, 2015).

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Photo 8.4
The mental status exam is commonly used in hospitals and medical centers to quickly estimate a client's current level of functioning.

Whereas some clinical psychologists—namely, researchers and those clinicians who emphasize the empirical foundations of clinical psychology—use structured interviews such as the SCID regularly, others use unstructured interviews regularly. In actual practice, quite a few clinical psychologists blend these approaches to conduct an interview that may be labeled as a partially structured or **semistructured interview** (Maruish, 2008; Nock, Holmberg, Photos, & Michel, 2007). Such interviews may include unstructured segments, typically at the beginning of the interview, which allow clients to describe in their own words the current problem and any relevant history. (The SCID, actually, includes an "overview" section at the outset that is intended to fulfill this purpose.) After this unstructured segment, an interviewer may ask a succession of targeted questions to address specific diagnostic criteria.

Mental Status Exam

The **mental status exam** is employed most often in medical settings. Its primary purpose is to quickly assess how the client is functioning at the time of the evaluation. The mental status exam does not delve into the client's personal history, nor is it designed to determine a *DSM* diagnosis definitively. Instead, its yield is usually a brief paragraph that captures the psychological and cognitive processes of an individual "right now"—like a psychological snapshot (Lukas, 1993; J. Morrison, 2008; Sommers-Flanagan & Sommers-Flanagan, 2009; Strub & Black, 1977; Wiger & Mooney, 2015).

The format of the mental status exam is not completely standardized, so it may be administered differently by various health professionals. (Psychiatrists, in addition to clinical psychologists and other mental health professionals, often use the mental status exam.) Although specific questions and techniques may vary, the following main categories are typically covered (Sommers-Flanagan, 2016):

- Appearance
- Behavior/psychomotor activity
- Attitude toward interviewer
- Affect and mood

- Speech and thought
- Perceptual disturbances
- Orientation to person, place, and time
- Memory and intelligence
- Reliability, judgment, and insight

Because of its lack of standardization, two interviewers who use the mental status exam may ask different questions within the same category. For example, in the "memory and intelligence" category, the interviewer may assess memory by asking the client about schools attended or age differences with siblings and assess intelligence by asking the client to count backward by 7 from 100, to recall a brief series of numbers or words, or to indicate the direction traveled between two cities. The mental status exam is not intended as a meticulous, comprehensive diagnostic tool. Instead, it is intended for brief, flexible administration, primarily in hospitals and medical centers, requiring no manual or other accompanying materials.

Crisis Interviews

The **crisis interview** is a special type of clinical interview and can be uniquely challenging for the interviewer. Crisis interviews have purposes that extend beyond mere assessment. They are designed not only to assess a problem demanding urgent attention (most often, clients actively considering suicide or another act of harm toward self or others) but also to provide immediate and effective intervention for that problem (Sommers-Flanagan & Shaw, 2017). Crisis interviews can be conducted in person but also take place often on the telephone via suicide hotlines, crisis lines, and similar services.

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Photo 8.5 Crisis interviews are unique among clinical interviews because they typically involve not only assessment of the client's problems but immediate intervention as well.

Quickly establishing rapport and expressing empathy for a client in crisis, especially a suicidal client, are key components of this type of interview. Providing an immediate, legitimate alternative to suicide can enable the client to endure this period of very high distress and reach a later point in time when problems may feel less severe or solutions may be more viable. Sometimes, interviewers in these situations ask clients to sign or verbally agree to suicide prevention contracts. These contracts essentially require the client to contact the interviewer before committing any act of self-harm. Research on the use of contracts of this type (often called "no-suicide contracts") suggests that they are questionable in their effectiveness (McMyler & Pryjmachuk, 2008). In fact, in some cases they may have the unintended negative effect of communicating to the client that the psychologist is more concerned about protecting themselves from legal liability (by being able to provide documentation of their efforts in case suicidal behavior does take place) than the client's actual well-being (Range et al., 2002; B. Stanley & Brown, 2012). As an

alternative, some psychologists have suggested "commitment to treatment" statements, which essentially involve a promise to keep working toward improvement rather than a promise not to engage in suicidal behaviors (Rudd, Mandrusiak, & Joiner, 2006).

When interviewing an actively suicidal person, five specific issues should be assessed (adapted from Sommers-Flanagan & Sommers-Flanagan, 2009):

- *How depressed is the client?* Unrelenting, long-term depression and a lack of hope for the future indicate high risk.
- *Does the client have suicidal thoughts?* If such thoughts have occurred, it is important to inquire about their frequency and intensity.
- *Does the client have a suicide plan?* Some clients may have suicidal thoughts but no specific plan. If the client does have a plan, its feasibility (the client's access to the means of self-harm, such as a gun, pills, etc.), its lethality, and the presence of others (family, friends) who might prevent it are crucial factors.
- *How much self-control does the client currently appear to have?* Questions about similarly stressful periods in the client's past, or about moments when self-harm was previously contemplated, can provide indirect information about the client's self-control in moments of crisis.
- *Does the client have definite suicidal intentions?* Direct questions may be informative, but other indications such as giving away one's possessions, putting one's affairs in order, and notifying friends and family about suicide plans can also imply the client's intentions.

CULTURAL COMPONENTS

Appreciating the Cultural Context

All human behaviors, including the words and actions of a client during a clinical interview, take place in a cultural context. It is imperative that the interviewer retain this perspective when observing and listening to the client. Interviewers should possess sufficient knowledge about culture—their own as well as their clients'—to understand the meaning of interview material within the appropriate cultural context. Behaviors, thoughts, or emotions that might be viewed as abnormal or pathological by some cultural standards may in fact be normal according to others, and interviewers should be careful not to overpathologize by imposing their own cultural values. Clinical psychologists should make efforts, in interviews and other interactions with clients, to appreciate clients from a perspective that takes into account the clients' own cultures. Clinical psychologists also make adaptations to cultural expectations or norms when conducting the interview, such as perhaps including a bit more small talk with a member of a culture where its absence is likely to hinder rapport, or consulting with professionals who know the culture well (Sommers-Flanagan et al., 2015; Sommers-Flanagan, 2016).

As an example, consider Paula, a 60-year-old Jewish mother of two grown children, who is being interviewed by Dr. McMillan, a clinical psychologist whose own religious

background is Catholic. Paula explains her anxiety and depression as partially stemming from her son's recent announcement that his own 10-year-old son (Paula's grandson), unlike the generations before him, will not study Hebrew and become a bar mitzvah. Although the bar mitzvah (a traditional Jewish ritual symbolizing entry into adulthood that culminates in reading Hebrew from the Torah at age 13—or at age 12 for girls, for whom it is called a bat mitzvah) may not be a milestone for Dr. McMillan personally, it is vital that Dr. McMillan recognize its significance within Jewish culture and to Paula in particular. If Dr. McMillan fails to appreciate the significance of the bar or bat mitzvah in the Jewish community, Paula's problems may be misinterpreted or underestimated. And as a result, during her interview, Paula may feel misunderstood and rapport may be damaged.

Of course, the culturally competent interviewer should not assume that every person from a culture holds identical values. As a group, members of a culture may tend to act, believe, or feel a certain way, but tremendous variability often exists among individuals within that cultural group. Thus, in the previous example, Dr. McMillan should have some idea of the meaning that Jews attach to the bar or bat mitzvah but not rigidly assume that Paula's individual views are necessarily prototypical.

Religion, of course, is just one of many components of culture that may be important to a particular client (Grieger, 2008). Box 8.4 provides a broad list of questions interviewers can use to inquire about clients' cultural backgrounds (Fontes, 2008).

Acknowledging Cultural Differences

When cultural differences exist between interviewer and client—which happens quite frequently, especially if we define culture by ethnicity, gender, religion, age, or a number of other variables—it is often wise for the interviewer to acknowledge these differences openly. Open, respectful discussion of cultural variables can enhance rapport, increase the client's willingness to share information, and help the interviewer gain a more accurate understanding of the client's issues (Sommers-Flanagan & Sommers-Flanagan, 2009). Without it, the interview process—and the assessment process more broadly—can prove invalid and can lead to culturally insensitive treatment (Ridley, Tracy, Pruitt-Stephens, Wimsatt, & Beard, 2008).

In many cases, it is appropriate for the interviewer to inquire directly about cultural background to become more informed about the client's perspective. Of course, such questions (as listed in Box 8.4) should be asked sensitively rather than callously. Clients can be an enlightening source of information about their own cultural backgrounds; however, the interviewer should not lean on them too strongly. The purpose of the interview, after all, is not for the client to educate the interviewer. Interviewers should continually seek other sources of information about specific cultures—written, experiential, or otherwise—to increase their cultural awareness. In the earlier example of Paula, it would be preferable if Dr. McMillan came into the interview already having a sense of what a bar mitzvah is and what it typically means to Jewish people. If he made it clear to Paula that he had this basic knowledge but asked her about some of the finer points or some of her specific feelings about it, she might admire his efforts. On the other hand, if he entered the interview completely ignorant and insisted that Paula explain the basics to him—what a bar mitzvah is, when it takes place, what it means—she may perceive his questions as burdensome, and the interview may suffer as a result.

BOX 8.4

Interview Questions to Consider When Inquiring About the Cultural Backgrounds of Clients

1. Where were you born?
2. Who do you consider family?
3. What was the first language you learned to speak?
4. Tell me about the other language(s) you speak.
5. What language or languages are spoken in your home?
6. What is your religion? How observant are you in regard to practicing that religion?
7. What activities do you enjoy when you are not working?
8. How do you identify yourself culturally?
9. What aspects of [client's self-reported cultural background] are most important to you?
10. How would you describe your home and neighborhood?
11. Who do you usually turn to for help when facing a problem?
12. What are your goals for this interview today?

Source: Fontes (2008), p. 31.

CHAPTER SUMMARY

Like any assessment technique, the clinical interview should be reliable, valid, and clinically useful. The clinical interview is a core element of psychological assessment. Skillful interviewers are adept at quieting their internal self-talk to enhance attentiveness; being self-aware, particularly regarding the responses they tend to elicit from others; and developing positive working relationships. They establish rapport via a respectful, caring attitude and attend to subtle aspects of communication such as body language and vocal qualities in their clients. Skillful interviewers are also proficient with specific interviewing techniques, including open- and closed-ended questions, clarifications, confrontations, paraphrasing, reflection of feeling, summarizing, and forming conclusions. There are numerous pragmatic issues relevant to the clinical interview, including note taking, audio- or video-recording, the look and feel of the interview room, and confidentiality. It can be advantageous for the interviewer to discuss the rationale behind decisions regarding these pragmatic issues with clients. Clinical interviews take a variety of forms, but many can be categorized as intake interviews, diagnostic interviews, mental status exams, or crisis interviews. Structured clinical interviews such as the SCID tend to have strong reliability and validity but may sacrifice rapport; unstructured clinical interviews are more client directed and may facilitate rapport, but their reliability and validity are weaker. Semistructured interviews strike a balance between structured and unstructured approaches. Appreciation and acknowledgment of cultural issues enhance any clinical interview and should be a primary goal for all interviewers.

KEY TERMS AND NAMES

assessment 179
attending behaviors 183
being self-aware 182
body language 183
clarification 188
clinical interview 179
clinical utility 179
closed-ended question 188
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CRITICAL THINKING QUESTIONS

1. This chapter describes an interviewer's self-awareness as an important general skill. What steps should clinical psychologists take to become more self-aware?
2. Describe two different client–therapist pairings in which issues of culture might influence the meaning of eye contact or body language in different ways.
3. Considering the wide range of clinical problems that clients might bring to an interview, for which, in your opinion, would direct versus indirect questioning be best suited?
4. If you were a clinical psychologist, would you take notes during a clinical interview? Why or why not?
5. If you were a clinical psychologist, in what situations would you openly acknowledge cultural differences between yourself and an interview client?

LOOKING TOWARD GRADUATE PROGRAMS

The following table is a list of index entries in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018) relevant to the topics in this chapter (see page 25 in Chapter 1 for more information about the *Insider's Guide*). Keep in mind a few things as you consider these listings: They are current as of the 2018/2019 edition of the *Insider's Guide* (which typically revises every two years); they may have changed as faculty members retire, take new jobs, or develop new interests; and a graduate program's own website may have more extensive or up-to-date information.

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Assessment/Diagnosis	Assessment/Testing	Assessment/Testing

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

Assessment
http://asm.sagepub.com/

Journal of Clinical Psychology
http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1097-4679

Journal of Contemporary Psychotherapy
http://link.springer.com/journal/10879

Psychological Assessment
http://www.apa.org/pubs/journals/pas/index.aspx

STUDENT STUDY SITE RESOURCES

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- Self-quizzes
- eFlashcards
- Culture expert interviews
- In My Practice demonstration video
- Full-text SAGE journal articles
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Measuring cognitive performance has been a hallmark of clinical psychology since its origin (Wood, Garb, & Nezworski, 2007). Knowledge of a client's level of cognitive functioning, including both strengths and deficits, can help a clinical psychologist with diagnosis and treatment of many presenting problems. Some assessments, such as those for specific learning disorder (formerly known as learning disabilities), intellectual disability (also known as intellectual developmental disorder, and formerly known as mental retardation), or giftedness, focus on cognitive issues from the start. In other assessments, such as those focusing on mood disorders or disruptive behavior, cognitive tests can provide important contextual information.

This chapter focuses on three types of tests, each related to cognitive functioning in some way but each with a distinct purpose. **Intelligence tests** measure a client's intellectual abilities. **Achievement tests**, in contrast, measure what a client has accomplished with those intellectual abilities. **Neuropsychological tests** focus on issues of cognitive or brain dysfunction, including the effects of brain injuries and illnesses.

Learning Objectives

9.1 Compare major theories of intelligence from the early 1900s to present.

9.2 Describe how the Wechsler intelligence tests are used for intellectual assessment.

9.3 Differentiate between intelligence testing and achievement testing.

9.4 Explain the role of achievement testing in diagnosing specific learning disorder.

9.5 Define the purpose of neuropsychological testing.

9.6 List major neuropsychological batteries and instruments used by clinical neuropsychologists.

INTELLIGENCE TESTING

Classic Theories of Intelligence

The specific intelligence tests used by contemporary clinical psychologists are rooted in contrasting theories of intelligence. Each of these theories puts forth a different answer to the essential question: What is **intelligence**? Actually, clinical psychologists have never reached a consensus regarding the definition of intelligence. Experts in this area of clinical psychology have emphasized many abilities as central to intelligence: speed of mental processing, sensory capacity, abstract thinking, imagination, adaptability, capacity to learn through experience, memory, reasoning, and inhibition of instinct, to name a few (as summarized by Sternberg, 2000; Sternberg & Grigorenko, 2008; Wasserman & Tulsky, 2005). In the debate about defining intelligence, perhaps no specific issue has received as much attention as the singular versus plural nature of intelligence (Wasserman, 2018). In other words, is intelligence one thing or many things?

Charles Spearman: Intelligence Is One Thing

In the early 1900s, **Charles Spearman** proposed a theory: Intelligence is a singular characteristic. Spearman labeled this characteristic **"g"** for general intelligence and argued that it represented a person's global, overall intellectual ability. His theory was based on research in which he measured many different, specific capabilities of his participants, including academic abilities and sensory-discrimination tasks. The primary finding was a strong correlation between this wide range of abilities, suggesting that a single factor underlies them all. Spearman acknowledged that more specific abilities (**"s"**) existed, but he argued that they played a relatively minor role in intelligence. Essentially, according to Spearman, intelligence was one thing (Brody, 2000; Revelle, 2015).

Louis Thurstone: Intelligence Is Many Things

Louis Thurstone was among the first and the strongest opponents to Spearman's singular theory of intelligence. According to Thurstone, intelligence should not be understood as a single, unified ability but as numerous distinct abilities that have little relationship to one another. Among his other contributions in the first half of the 1900s, Thurstone was a pioneer of the statistical procedure called multiple factor analysis, which enabled him to identify underlying factors in a large data set. When he performed these statistical analyses on examinations of various intellectual abilities, he could have found one dominant factor underlying all abilities. Instead, he found several independent factors. These factors were given labels such as verbal comprehension, numerical ability, spatial reasoning, and memory. The specific names and number of the factors are less important,

however, than Thurstone's fundamental point: Intelligence is not one thing; it is many things (Brody, 2000). Thus, according to Thurstone, if you know how capable someone is regarding, say, mathematics, you cannot predict with confidence how capable that person is regarding, say, verbal skills.

Eventually, Spearman and Thurstone each acknowledged the validity of the other's arguments and came to somewhat of a compromise. They settled on a **hierarchical model of intelligence** in which specific abilities ("s") existed and were important, but they were all at least somewhat related to one another and to a global, overall, general intelligence ("g") (Willis, Dumont, & Kaufman, 2015a).

Copyright © Jupiterimages

Photo 9.1 Clinical psychologists use tests of intelligence, achievement, and neuropsychological functioning for many purposes, such as the assessment of a child exhibiting academic problems.

More Contemporary Theories of Intelligence

Other theories of intelligence have emerged since the debates between Spearman and Thurstone, most of which provide a fresh perspective on the singular versus plural issue (Brody, 2000; Davidson & Downing, 2000; Nisbett et al., 2012). For example, in the latter half of the 1900s, **James Cattell** proposed two separate intelligences: **fluid intelligence**—the ability to reason when faced with novel problems—and **crystallized intelligence**—the body of knowledge one has accumulated as a result of life experiences. With exactly two very broad categories of intelligence, Cattell's theory falls somewhere between Spearman's theory of a singular intelligence and Thurstone's theory of many intelligences. More recently, new hierarchical models of intelligence have emerged as well, including **John Carroll**'s (2005) **three-stratum theory of intelligence**, in which intelligence operates at three levels: a single "g" at the top, eight broad factors immediately beneath "g," and more than 60 highly specific abilities beneath these broad factors. This model, like most contemporary models of intelligence, not only acknowledges "g" but also recognizes that more specific abilities exist as well (Wasserman & Tulsky, 2005). As we see in this chapter, most contemporary intelligence tests mirror this view by producing a single overall score—one number to represent how intelligent a person is—in addition to a number of other scores representing more specific abilities. The Wechsler and Stanford-Binet intelligence tests are primary among these, and we will turn our attention to them soon. But it is worth mentioning that there are other intelligence tests besides the Wechsler and Stanford-Binet tests that are also widely used among clinical psychologists. Those tests include the Woodcock-Johnson tests, such as the Woodcock-Johnson IV Tests of Early Cognitive and Academic Development (for children aged 2.5–7) and the Woodcock-Johnson IV Tests of Cognitive Abilities (generally for clients aged 5–95), as well as the Kaufman Assessment Battery for Children-II (for children aged 3–18) (Drozdick, Singer, et al., 2018; Morrison, Singer, & Raiford, 2018; Schrank & Wendling, 2018a, 2018b). Each of these tests is paired with an achievement test created and typically co-normed by the same company (Woodcock-Johnson and Kaufman, respectively).

Web Link 9.1 John Carroll

BOX 9.1

Metaphorically Speaking

If You've Watched Multi-Sport Athletes, You Understand the Challenges of Defining and Assessing Intelligence

Michael Jordan is undoubtedly one of the greatest basketball players of all time. His five MVP awards and six NBA championships offer indisputable evidence. But if we broaden the question—is Michael Jordan a great *athlete?*—we can begin to appreciate some of the challenges in defining ability, whether athletic or intellectual.

In 1993, after 9 extremely successful years in the NBA, Jordan decided (to the surprise of many) to leave basketball and focus on baseball. He signed on with the Chicago White Sox and was assigned to the Birmingham Barons, a Double-A minor league team. His prowess did not easily transfer from the basketball court to the baseball diamond. Instead of the familiar highlights of acrobatic dunks and picturesque jump shots, sports fans saw Jordan flailing at curve balls and misjudging fly balls. His hitting and fielding improved slightly but not enough for him to advance any closer to the major leagues.

Should we expect that a person who excels at one sport will similarly excel at others? If so, perhaps each of us possesses a general athletic ability that influences our performance at any sport—a lot like Spearman's concept of a general intelligence ("g"). If not, perhaps each of us has a number of specific athletic abilities that are unrelated to one another—a lot like Thurstone's theory of specific ("s"), multiple intelligences.

Which theory does the example of Michael Jordan support? On the one hand, the contrast between Jordan's basketball excellence and baseball mediocrity supports the idea that our abilities are independent. Indeed, in the history of athletics, there have been very few people (e.g., Jim Thorpe, Bo Jackson, Danny Ainge, Deion Sanders) who have reached the highest level of multiple sports. On the other hand, there are quite a few athletes who have reached the highest level of one sport and had some success (perhaps at the college or minor league level) at another sport as well, including Russell Wilson (football and baseball), Tim Tebow (football and baseball), Jim Brown (football and lacrosse), Stephen Curry (basketball and golf), and Jackie Robinson (baseball, basketball, football, and track). Michael Jordan may belong on that list, too: Despite focusing exclusively on basketball for many years, Jordan was able, with minimal preparation, to step onto a minor league baseball team and at least hold his own. He may not have excelled, but he did achieve some success at a level of baseball—a minor league team just two steps away from the majors—that far exceeds the talents of the vast majority of players. This would suggest that he possesses some fundamental, general athletic ability that applies, to some extent, to any sport. Perhaps Michael Jordan's experiences best support the hierarchical models of ability, which acknowledge a general, across-the-board ability (of athletics or intelligence) as well as more specific talents (basketball vs. baseball, math vs. verbal skills) that are influenced by the general ability but may be somewhat independent as well.

In your personal experience, does athletic ability seem to be one thing or many things? Consider the athletic abilities of people you have known (or perhaps played with or against). Are the best at one sport typically among the best at other sports? What about intellectual ability? Among your friends and family, do people seem to have a singular, general intelligence that influences all their abilities, or many distinct intelligences that operate independently?

Wechsler Intelligence Tests

Since **David Wechsler**'s earliest attempts to measure intelligence in the early 1900s, the tests that bear his name have risen to prominence among clinical psychologists (Coalson, Raiford, Saklofske, & Weiss, 2010; Drozdick, Raiford, et al., 2018; G. Goldstein, 2008; Lichtenberger & Kaufman, 2009; Willis et al., 2015b). There are three separate Wechsler intelligence tests, each the most widely used among psychologists for its respective age range (Wright et al., 2017). The names and current editions of the Wechsler intelligence tests are as follows:

Web Link 9.2
David Wechsler

Web Link 9.3
Site for WAIS-IV

- **Wechsler Adult Intelligence Scale—Fourth Edition (WAIS-IV)**
- **Wechsler Intelligence Scale for Children—Fifth Edition (WISC-V)**
- **Wechsler Preschool and Primary Scale of Intelligence—Fourth Edition (WPPSI-IV)**

Table 9.1 provides a brief description for each of these Wechsler intelligence tests, as well as the other intelligence, achievement, and neuropsychological tests covered in detail in this chapter.

Table 9.1 At-a-Glance Information About the Tests Detailed in This Chapter

	Most Recent Edition	Year Published	Age Range (Years)
Intelligence Tests			
Wechsler Adult Intelligence Scale (WAIS)	IV	2008	16 to 90
Wechsler Intelligence Scale for Children (WISC)	V	2014	6 to 16
Wechsler Preschool and Primary Scale of Intelligence (WPPSI)	IV	2012	2 years and 6 months to 7 years and 3 months
Stanford-Binet Intelligence Scales	V	2003	2 to 85+
Universal Nonverbal Intelligence Test (UNIT)	2	2015	5 to 21
Achievement Tests			
Wechsler Individual Achievement Test (WIAT)	III	2009	4 to 50
Neuropsychological Tests			
Halstead-Reitan Neuropsychological Test Battery (HRB)	I	Varies by test	15+
Bender Visual-Motor Gestalt Test	II	2003	4+
NEPSY	II	2007	3 to 16

Collectively, the three Wechsler intelligence tests cover virtually the entire life span. They vary slightly from one another, as necessitated by the demands of measuring intelligence at different ages. And, in practical terms, they are indeed separate tests rather than slight variants of one another. (They are purchased as separate kits with separate manuals, answer sheets, materials, etc.) However, the three Wechsler intelligence tests share many fundamental characteristics:

- They yield a single **full-scale intelligence score,** four or five **index scores,** and about a dozen (give or take a few, depending on optional subtests chosen) specific **subtest scores.** Together, this collection of scores indicates that the Wechsler tests employ a hierarchical model of intelligence (as discussed earlier), in which the full-scale intelligence score reflects a general, global level of intelligence ("g") and the index/factor scores and subtest scores represent increasingly specific areas of ability ("s"). These scores allow clinical psychologists to focus broadly or narrowly when making interpretations regarding intellectual ability.

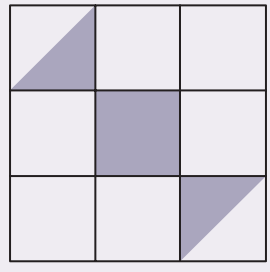
- They are administered one-on-one and face-to-face. In other words, the Wechsler tests cannot be administered to a group of examinees at the same time, nor are they entirely pencil-and-paper tests (e.g., multiple choice, true/false, essay) that examinees simply administer to themselves. Administration of the Wechsler intelligence tests is a structured interpersonal interaction requiring extensive training, typically received during graduate programs in clinical psychology (Raiford, Coalson, Saklofske, & Weiss, 2010).

- Each subtest is brief (lasting about 2–10 minutes) and consists of items that increase in difficulty as the subtest progresses. Most often, the subtests are designed such that examinees continue until they fail a predetermined number of consecutive items (or "max out," to state it informally). The nature of the tasks performed varies widely across the dozen or so subtests but includes both verbal and nonverbal tasks. (More detailed descriptions of the subtests follow.)

- Although each of the three Wechsler tests has a small number of unique subtests, most subtests appear in all three tests as either core or supplemental subtests. These common subtests are described in detail in Table 9.2.

- Originally, the Wechsler tests were designed with two categories of subtests: verbal and performance. ("Performance" was essentially equivalent to "nonverbal.") In more recent years, statistical tests including factor analyses have concluded that the subtests don't, in fact, cluster together into those two groups. Instead, they cluster together in four or five, rather than two, factors. Thus, the original verbal/performance split has been replaced by four or five factors, each receiving contributions from several subtests (Groth-Marnat, 2009; L. G. Weiss, Saklofske, Coalson, & Raiford, 2010). Specifically, the WAIS has four factors (listed below), and the WISC and WPPSI each have five factors (essentially, they share three with the WAIS, but the Perceptual Reasoning Index of the WAIS is divided into Visual Spatial Index and Fluid Reasoning Index).
    - **Verbal Comprehension Index**—a measure of verbal concept formation and verbal reasoning
    - **Perceptual Reasoning Index**—a measure of fluid reasoning, spatial processing, and visual-motor integration

Table 9.2 Subtests Common to Wechsler Tests of Intelligence

Vocabulary	Orally explain the meaning of a word	• What does "consistent" mean? • What is an "intersection"?
Similarities	Orally explain how two things or concepts are alike	• How are a door and a window alike? • How are success and failure alike?
Information	Orally answer questions focusing on specific items of general knowledge	• On what continent is Spain? • How many cents is a quarter worth?
Comprehension	Orally answer questions about general social principles and social situations	• Why is it important for people to show identification before being allowed to vote? • What are some advantages of using only the minimal amount of water necessary in our homes?
Block Design	Re-create a specific pattern or design of colored blocks	
Picture Completion	View a picture of a simple object or scene and identify the important part that is missing	
Matrix Reasoning	View an incomplete matrix and select the missing portion from multiple choices provided	
Coding	Using pencil and paper, repeatedly copy simple shapes/symbols in appropriate spaces according to a key provided	

(Continued)

Table 9.2 (Continued)

Symbol Search	Scan a group of visual shapes/symbols to determine if target shape(s)/symbol(s) appear in group	

- **Working Memory Index**—a measure of the capacity to store, transform, and recall incoming information and data in short-term memory
- **Processing Speed Index**—a measure of the ability to process simple or rote information rapidly and accurately

- They feature large, carefully collected sets of **normative data.** That is, the manual for each Wechsler test includes norms collected from about 2,000 people. These normative groups closely match recent U.S. Census data in terms of gender, age, race/ethnicity, and geographic region, among other variables (Zhu & Weiss, 2005). So when an examinee takes a Wechsler intelligence test, the examinee's performance is compared with the performance of a large, same-age sample of individuals representing a wide-scale national population.

- The full-scale and index scores generated by the Wechsler tests are "IQ" scores, meaning that they reflect an intelligence "quotient." This quotient is the result of a division problem in which the examinees' raw scores are compared with age-based expectations. For all Wechsler tests, mean IQ scores (full-scale and index) are 100, with a standard deviation of 15. For each subtest, a score of 10 is average, with a standard deviation of 3.

- The Wechsler intelligence tests share a general approach to interpretation of scores. Assessors are instructed to first consider the full-scale IQ score. Next, they move on to interpret each index score in relation to the others and then the pattern of subtest scores. Finally, they note more detailed aspects of the testing, such as observable patterns of behavior that may have contributed to scores or inconsistent performance within a single subtest. Essentially, these successive steps allow the assessor to grasp the "big picture" of the intellectual profile before delving into increasingly detailed levels of analysis (Groth-Marnat, 2009).

- The Wechsler intelligence tests are all backed by very impressive psychometric data. That is, a large number of empirical studies suggest that these tests have the characteristics that clinical psychologists should seek in any test: strong reliability and validity. They measure what they intend to measure, and they do so consistently (Canivez & Watkins, 2010; Groth-Marnat, 2009; Lichtenberger & Kaufman, 2004; Zhu & Weiss, 2005).

- The Wechsler intelligence tests—and most other intelligence tests, for that matter—are used for a wide range of clinical applications, including evaluations that focus on issues of intellectual disability (intellectual developmental disorder), developmental delays, giftedness, educational and vocational planning, school placement and qualification, and other targeted assessment questions (Zhu & Weiss, 2005). They can also be used to provide general intelligence information in broader contexts, including

a comprehensive assessment of a client whose presenting problems are more neuropsychological (e.g., Alzheimer's disease), emotional (e.g., mood disorders), or behavioral (e.g., attention-deficit/hyperactivity disorder [ADHD]).

- The Wechsler intelligence tests were among the first to become available on a digital platform as an alternative to the traditional pencil-and-paper format. Known as the "Q-interactive" system, this new method of administering and scoring Wechsler tests allows psychologists to use tablets (e.g., iPads) and promise to reduce both the overall length of the tests and the number of scoring errors that can happen when a person rather than a machine does the computations (S. W. Clark, Gulin, Heller, & Vrana, 2017). However, it also brings a number of challenges, including the need for psychologists to be sufficiently tech-savvy, clients' potential discomfort with the tablets, and financial expense for psychologists and graduate training programs (S. W. Clark et al., 2017; M. C. Roberts & Campbell, 2017). One study of graduate students trained to use the Q-interactive system found mixed reviews, with most preferring to learn the tests initially in pencil-and-paper format and only slightly more than half preferring to eventually switch to the Q-interactive format (Noland, 2017).

- Generally, the Wechsler intelligence tests have a number of notable strengths: They have impressive reliability and validity; they feature comprehensive and recent normative data; they cover an extremely wide age range; they provide full-scale, index, and subtest scores that have great clinical utility; and, at this point in their history, they are very familiar to most clinical psychologists. On the other hand, these tests have received criticism for some limitations: Some subtests may be culturally loaded or biased, the connection between the tests and day-to-day life (ecological validity) may be limited, and scoring can be complex or subjective on some subtests (Groth-Marnat, 2009; Zhu & Weiss, 2005). On the whole, however, the Wechsler tests are held in high esteem by clinical psychologists.

Stanford-Binet Intelligence Scales—Fifth Edition

The Stanford-Binet intelligence tests have a rich history in clinical psychology. The first editions of the Stanford-Binet intelligence test dominated the field in the early 1900s until the Wechsler tests began to provide competition. Although the Wechsler tests have taken a leading role in recent decades, the Stanford-Binet remains highly respected and offers an approach to assessing intelligence that is both similar to and different from that of the Wechsler tests (G. Goldstein, 2008).

Web Link 9.4
Site for SB5

The most recent revision of the Stanford-Binet, the **Stanford-Binet Intelligence Scales—Fifth Edition (SB5),** is like the Wechsler tests in many ways. It is administered face-to-face and one-on-one. It employs a hierarchical model of intelligence and therefore yields a singular measure of full-scale IQ (or "g"), five factor scores, and many more specific subtest scores. It features the same means (100) and standard deviations (15) as the Wechsler intelligence tests for its full-scale and factor scores. Its psychometric data, including reliability and validity, are similarly strong.

The SB5 differs from the Wechsler tests in some important ways, however. Rather than three separate tests for three different age ranges, the SB5 covers the entire life

span (ages 2–85+) as a single test. The normative sample is like those of the Wechsler tests in that it matches recent U.S. Census data on important variables, but it additionally includes normative data from individuals with specific relevant diagnoses, including learning problems, intellectual disability (intellectual developmental disorder), and ADHD. Its subtests include extensions at the low and high ends—in other words, a greater number of very easy and very difficult items—as an effort to more accurately assess people at the extremes, including those who may have intellectual disability or be gifted (Kamphaus & Kroncke, 2004).

Perhaps the most important difference between the SB5 and the Wechsler tests involves their specific factors and subtests. Whereas the Wechsler tests feature four or five factors, each of which yields an index score, the SB5 features exactly five, described briefly here:

- **Fluid Reasoning**—the ability to solve novel problems
- **Knowledge**—general information accumulated over time via personal experiences, including education, home, and environment
- **Quantitative Reasoning**—the ability to solve numerical problems
- **Visual-Spatial Processing**—the ability to analyze visually presented information, including relationships between objects, spatial orientation, assembling pieces to make a whole, and detecting visual patterns
- **Working Memory**—the ability to hold and transform information in short-term memory

Each of these five factors is measured both verbally and nonverbally, a deliberate innovation that is new to the SB5 and to IQ tests more generally. Each of these 10 areas (five factors, each measured both verbally and nonverbally) is assessed by one to three specific types of items (Kamphaus & Kroncke, 2004).

Overall, current editions of the Wechsler tests may have become more widely used than the current edition of the Stanford-Binet, but the SB5 continues to hold a similar position of respect and, in some settings, popularity as it has for the past century.

Additional Tests of Intelligence: Addressing Cultural Fairness

Especially in recent years, one of the primary criticisms of the Wechsler tests, the SB5, and other renowned IQ tests has centered on issues of **cultural fairness.** Specifically, these tests have been described as featuring numerous subtests, especially those relying on verbal skills, that place individuals from minority cultural groups at a disadvantage. In other words, to the extent that an intelligence test is based on culture-specific concepts, it may unfairly assess the intelligence of people of other cultures (Ortiz, Piazza, Ochoa, & Dynda, 2018). Certainly, the authors of prominent tests, including the Wechsler tests and the SB5, have made significant efforts to make recent editions of their tests less culturally biased or loaded. But entirely new tests have also emerged, with the explicit purpose of measuring IQ in a more culturally fair way.

A leading example of such a test is the **Universal Nonverbal Intelligence Test-2** (**UNIT-2**; Bracken & McCallum, 2009, 2015; McCallum & Bracken, 2005, 2018). Originally published in 1996 and revised in 2015, the UNIT-2 is a completely language-free test of intelligence. It requires no speaking or shared understanding of language between the person administering the test and the person taking it. Like the Wechsler and Stanford-Binet intelligence tests, the UNIT-2 is administered one-on-one and face-to-face, but rather than using verbal instructions, the examiner presents instructions via eight specific hand gestures taught in the test manual and demonstrated in an accompanying video. Additionally, the responses of the examinee all consist of either pointing with fingers or minor manipulation of objects with hands or fingers.

Web Link 9.5
Site for UNIT

The UNIT-2 is appropriate for clients aged 5 to 21 years and was normed on 1,800 people who match recent U.S. Census data in terms of age, sex, race, parent education, community size, geographic region, and ethnicity. A small body of psychometric data on the UNIT and UNIT-2 has been published and suggests generally acceptable reliability and validity (Bracken & McCallum, 2015).

The UNIT-2 consists of six subtests organized into a two-tiered model of intelligence. The two tiers are identified as Memory and Reasoning. The three subtests contributing to the memory tier are

- Object Memory, in which the examinee views a visual assortment of common objects for 5 seconds and then views a larger array and identifies the objects from the first array;
- Spatial Memory, in which the examinee recalls the placement of colored chips on a three-by-three or four-by-four grid; and
- Symbolic Memory, in which the examinee recalls and re-creates sequences of visually presented symbols.

The three subtests contributing to the reasoning tier are

- Cube Design, in which the examinee arranges colored blocks in a specific three-dimensional design;
- Mazes, in which the examinee completes traditional maze puzzles; and
- Analogic Reasoning, in which the examinee solves analogy problems that are presented visually rather than verbally.

The UNIT-2 is not without its shortcomings. It assesses a more limited range of abilities than more traditional IQ tests; it is appropriate only for young clients (no version for preschool children or adults older than 21 has yet been developed); and its psychometric data, although encouraging, is limited in quantity. Additionally, as a relative newcomer to the intelligence test field, it is nowhere near as well established or popular as the standard Wechsler or Stanford-Binet tests, so the structure and format of the UNIT-2 remain unfamiliar to many clinical psychologists. (And the Wechsler family of tests includes their own version of a nonverbal intelligence test: the Wechsler Nonverbal Scale of Ability. It includes only minimal spoken instructions at the outset and mostly pictorial directions for clients throughout the subtests [Naglieri & Otero, 2018].) Such limitations are common

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Photo 9.2 The Universal Nonverbal Intelligence Test-2 (UNIT-2) attempts to maximize cultural fairness by using tasks, such as mazes, that require no language skills. In your opinion, what are the strengths and weaknesses of language-free tests of intelligence?

to nonverbal or "culture-free" tests more generally (Ortiz & Dynda, 2005). However, the development and increasing acceptance of the UNIT-2 and similar tests represent a significant step forward in the culturally sensitive practice of intelligence assessment. Considering the large numbers of non-English-speaking people in the United States, clinical psychologists will increasingly require intelligence tests that are applicable and fair to clients of a broad cultural and linguistic range (Pieterse & Miller, 2010).

ACHIEVEMENT TESTING

Achievement Versus Intelligence

Clinical psychologists assess clients' intelligence, but with separate tests they also assess clients' achievement. What's the difference between intelligence and achievement? As we have discussed, intelligence refers to a person's cognitive capacity. In short, intelligence is what a person *can* accomplish intellectually. In contrast, **achievement** is what a person *has* accomplished, especially in the kinds of subjects that people learn in school, such as reading, spelling, writing, or math. Achievement tests typically produce age- or grade-equivalency scores as well as standard scores (D. Schneider & Mather, 2015).

Prior to *DSM-5*, the comparison of intelligence and achievement was a key factor in determining learning disabilities. In *DSM-5*, the terminology changed ("learning disability" is now called **specific learning disorder**), and the definition has changed as well, such that the primary comparison is between the person's achievement (as measured by achievement tests and performance at school or work) and expected levels of achievement for people of the same age (American Psychiatric Association, 2013; Hogan & Tsushima, 2016).

When clinical psychologists conduct tests of intelligence or achievement (or any other kind of assessment), they typically explain their findings in a written psychological report. Psychological reports are the formal means by which they communicate their findings to others. Depending on the case, these "others" can represent a wide range of individuals: clients, parents of child clients, teachers, lawyers, judges, physicians, or other mental health professionals, just to name a few. It is important that clinical psychologists not only write intelligible, accurate reports that address the reason for referral but that they also "know their audience" in order to customize their reports in terms of readability, selection of relevant information, and recommendations (Blais & Smith, 2008; Goldfinger & Pomerantz, 2010).

Achievement tests come in many varieties. Some measure a single area of achievement in detail, such as the KeyMath or the Gray Oral Reading achievement tests. Others are more comprehensive, featuring a wide range of subtests. Examples of these include the Woodcock-Johnson Tests of Achievement (WJ-ACH), the Kaufman Test of Educational Achievement (KTEA), and the Wechsler Individual Achievement Test (WIAT),

BOX 9.2

Considering Culture

Defining Intelligence Around the World

What is intelligence? It depends on the cultural values of those we ask. When we direct the question to people outside traditional Western culture, the answer sometimes features characteristics that are quite different from definitions that Spearman, Thurstone, Cattell, and Carroll have proposed (as summarized by Sternberg, 2000; Sternberg & Grigorenko, 2008).

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Photo 9.3
In various cultures around the world, intelligence is defined in different ways. In your opinion, are any aspects of intelligence universal?

- In some societies in Africa (e.g., Zambia, Mali, and Kenya), intelligence consists largely of skills that preserve harmony in interpersonal relationships, both between and within groups. Interviews with some residents of Zambia, for example, suggest that cooperation, deference, respect for elders, and acceptance of social responsibilities characterize intelligent people.
- In Zimbabwe, the word for intelligence—*ngware*—literally translates into a prudent and cautious approach to life and especially to social relationships.
- In some Asian cultures, the definition of intelligence also involves heavy doses of social responsibility and benevolence. More specifically, Taoist conceptions of intelligence highlight humility, independent (rather than conventional) standards of judgment, and thorough knowledge of self.
- The emphasis on social duties as central to intelligence appears in some Hispanic cultures as well. In fact, in a study of parents of schoolchildren in San Jose, California, in the 1990s, parents of Hispanic descent rated social competence as more closely related to intelligence than did parents of European descent (Okagaki & Sternberg, 1993).
- The Western emphasis on speed of mental processing is not shared by all ethnic groups. In fact, some ethnic groups may value depth of thought more highly than speed of thought and may look unfavorably or doubtfully on work performed very quickly.

This variety of defining characteristics of intelligence raises a number of important questions. Is the definition of intelligence completely dependent on cultural context, or are some aspects of intelligence universal? To what extent should intelligence tests reflect the alternate definitions of intelligence held around the world or around the United States? Where should we draw the line between personality traits and intelligence? And, as Spearman and Thurstone debated many years ago, is intelligence one thing or many things?

this last of which we consider below in more detail. Each of these achievement tests pairs with an intelligence test created and typically co-normed by the same company (Woodcock-Johnson, Kaufman, and Wechsler, respectively).

Wechsler Individual Achievement Test—Third Edition

The **Wechsler Individual Achievement Test—Third Edition (WIAT-III)** is a comprehensive achievement test for clients aged 4 to 50 years. Like the Wechsler intelligence tests, the WIAT-III is administered face-to-face and one-on-one.

The WIAT-III measures achievement in four broad areas: reading, math, written language, and oral language. Each of these broad areas is assessed by two to four subtests. The Oral Language Composite derives from two subtests: Listening Comprehension (paying attention to orally presented information and answering questions about it) and Oral Expression (using speech to repeat spoken material, create stories about presented pictures, provide directions, etc.). The Reading Composite derives from scores on such subtests as Word Reading (reading isolated words), Pseudoword Decoding (using phonetic skills to sound out nonsense words, such as *plore* or *tharch*), and Reading Comprehension (reading sentences or passages and answering questions about their content). The Mathematics Composite derives subtests such as Numerical Operations (written math problems) and Math Problem Solving (word problems, numerical patterns, statistics and probability questions, etc.). The Written Language Composite derives subtests such as Spelling (increasingly difficult words) and Sentence Composition, Essay Composition (constructing sentences, paragraphs, or essays as instructed; Lichtenberger & Breaux, 2010).

The WIAT-III yields standard scores on the same scale as most intelligence tests: a mean of 100 and a standard deviation of 15. It also yields age and grade equivalencies for each subtest. The test was standardized on 3,000 people who were chosen to match recent U.S. Census data in terms of sex, age, race/ethnicity, geographic region, and parent education level. Many of the individuals in the standardization sample also took the age-appropriate Wechsler intelligence scale, so the WIAT-III is "linked" to the Wechsler IQ tests, which enhances the validity of comparisons between these two types of tests. Overall, the reliability and validity data supporting the WIAT-III are quite strong (Lichtenberger & Breaux, 2010).

NEUROPSYCHOLOGICAL TESTING

Neuropsychological testing represents a specialized area of assessment within clinical psychology, typically practiced by clinical psychologists whose training includes extra training in neuropsychology during graduate school courses, the predoctoral internship, and the postdoctoral internship (Hill & Westervelt, 2015; Spencer & Adams, 2016). The intent of neuropsychological tests is to measure cognitive functioning or impairment of the brain and its specific components or structures. Medical procedures such as computed tomography (CT), magnetic resonance imaging (MRI), and positron emission tomography (PET) scans can show that parts of the brain look abnormal, but neuropsychological

tests can indicate how the brain is actually functioning. Such tests are especially useful for targeted assessment of problems that might result from a head injury, prolonged alcohol or drug use, or a degenerative brain illness. They can also be used to make a prognosis for improvement, plan rehabilitation, determine eligibility for accommodations at school or work, and establish a baseline of neuropsychological abilities to be used as a comparison at a later time (Dawson & Jacquin, 2015b; Golden, 2008; Hebben & Milberg, 2009; McCaffrey, Lynch, & Westervelt, 2011).

Some neuropsychological testing procedures are lengthy, comprehensive batteries that include a broad array of subtests. The patterns of scores on these subtests can go a long way toward pinpointing specific cognitive weaknesses. Other neuropsychological tests are much briefer and are typically used as screens for neuropsychological impairment rather than as full-fledged neuropsychological assessment tools. Over time, the field has evolved from a "fixed-battery phase" in which psychologists typically use the same standard set of tests for most clients to a more "flexible-battery phase" in which psychologists create a more customized battery by picking and choosing tests after considering the specific referral questions and areas of concern (Hale, Wilcox, & Reddy, 2016).

Psychologists who use neuropsychological tests have also become more increasingly aware of neuropsychological assessment anxiety, an experience in which clients' performance on these tests is influenced by the anxiety in the moment. One review of dozens of studies on the subject found that about two thirds of those studies identified a connection between performance and anxiety, with some areas of performance (e.g., attention, verbal memory, working memory) being especially vulnerable to decrease (Dorenkamp & Vik, 2018).

Let's consider several examples of full neuropsychological batteries and briefer neuropsychological measures.

Full Neuropsychological Batteries

The **Halstead-Reitan Neuropsychological Battery (HRB)** is a battery of eight standardized neuropsychological tests. It is suitable for clients of age 15 years and older, but alternate versions are available for younger clients. The HRB (also known as HRNB) is administered only as a whole battery; its components are not to be administered separately. Thus, it is a thorough (and rather lengthy) neuropsychological battery. Essentially, its primary purpose is to identify people with brain damage and, to the extent possible, provide detailed information or hypotheses about any brain damage identified, including specific cognitive impairments or physiological regions of the brain that may be deficient. The findings of the HRB can help in diagnosis and treatment of problems related to brain malfunction (Broshek & Barth, 2000).

Some of the eight tests in the HRB involve sight, whereas others involve hearing, touch, motor skills, and pencil-and-paper tasks. An example of one of the tests of the HRB is the Trail Making Test, which resembles the familiar "dot-to-dot" puzzles that children complete, but this test is timed, contains both numbers and letters, and produces a rather haphazard line instead of an identifiable figure or shape. A second example is the Category Test, in which a client sees a pattern of shapes and designs on a screen and presses the number key (1–4) that is suggested by the pattern. Clients hear a bell when correct and a buzzer when wrong, and their successes and failures are noted

by the examiner. The Finger Tapping (or Finger Oscillation) Test is a third example; in this test, clients tap a single typewriter key as rapidly as they can with the index finger for 10-second intervals. The number of taps they can produce, averaged across multiple attempts with each hand, estimates motor speed (G. Goldstein & Sanders, 2004; Reitan & Wolfson, 2004).

The examiner using the HRB compares a client's test scores with published norms to assess overall performance and also with each other to determine the client's own relative strengths and weaknesses. Interpretation of results can include detailed inferences about specific neuropsychological pathologies and the localization of problems in cognitive functioning (e.g., a particular lobe or hemisphere of the brain).

In the decades since it was introduced, the HRB has been evaluated more than any other comprehensive neuropsychological test. These studies suggest strongly that the HRB and each of its tests have been established as reliable and valid. Additional strengths of the HRB include its comprehensiveness and clinical usefulness. Drawbacks of the HRB include its length (and corresponding expense), inflexibility (as a fixed battery), and a limited overlap with real-life, day-to-day tasks (Broshek & Barth, 2000; Hebben & Milberg, 2009).

The HRB may be among the most commonly administered comprehensive neuropsychological batteries (Guilmette & Faust, 1991; Rabin, Paolillo, & Barr, 2016), but another long-standing, respected and popular alternative is the **Luria-Nebraska Neuropsychological Battery (LNNB).** The LNNB is similar to the HRB in that it is a wide-ranging test of neuropsychological functioning. It consists of 12 scales, with a similar range to that of the HRB. A primary difference is the LNNB's emphasis on qualitative data in addition to quantitative data. In other words, to a greater extent than the HRB, the LNNB relies on qualitative written comments from the examiner about the testing process. These comments describe what the examiner observed about the client, such as problems comprehending the test (e.g., confusion, poor attention), how or why the client is missing items (e.g., slow movement, sight or hearing problems, speech flaws), or unusual behaviors (e.g., inappropriate emotional reactions, hyperactivity, distraction). Another difference between the HRB and the LNNB is that the LNNB tends to be slightly briefer. As for the HRB, a strong body of psychometric data supports the LNNB's reliability and validity (Golden, 2004; Golden, Freshwater, & Vayalakkara, 2000; Hebben & Milberg, 2009).

The **NEPSY-II** is one of the few batteries of neuropsychological tests designed specifically for children. Specifically, it is suited for use in people between 3 and 16 years old. It is based on the same general principles put forth by the creator of the Luria-Nebraska test (Alexander Luria). It includes 32 separate subtests across 6 different categories: attention and executive functioning, language, memory and learning, sensorimotor, social perception, and visuospatial processing. Psychologists have flexibility in determining exactly how to use the various components of the test (Matthews & Davis, 2018).

Brief Neuropsychological Measures

The Bender-Gestalt test, originally published in 1938 and currently available as the **Bender Visual-Motor Gestalt Test—Second Edition (Bender-Gestalt-II),** is the most commonly used neuropsychological screen among clinical psychologists

(Watkins, Campbell, Nieberding, & Hallmark, 1995). The test is a straightforward copying task: The client is given a pencil, blank paper, and nine simple geometric designs (primarily made of combinations of circles, dots, lines, angles, and basic shapes) and is asked to copy each design as accurately as possible. It measures visuoconstructive abilities, which are also commonly known as perceptual-motor or visual-spatial skills (Lacks, 2000). The current edition is quite similar to the original Bender-Gestalt, but it offers memory tasks and additional stimuli.

This test is remarkably brief—on average, it takes only 6 minutes to administer (Lacks, 2000). It is appropriate for clients of any age above 3 years. For these reasons, clinical psychologists frequently include it as a quick "check" for neuropsychological problems. In other words, if it is already established that the client being evaluated has or is strongly suspected to have a neuropsychological problem, it is likely that the clinical psychologist will select a more comprehensive battery, such as the HRB or the LNNB, rather than the Bender-Gestalt. However, when the evaluation is for another purpose and the clinical psychologist simply wants a rapid appraisal of overall neuropsychological functioning, the Bender-Gestalt can be a good choice. Its results cannot specify locations of brain damage, but poor performance on the Bender-Gestalt can suggest brain damage in a diffuse way. As such, it can alert the clinical psychologist to the general presence of neuropsychological problems, and more thorough testing can subsequently be conducted. Poor performance on the Bender-Gestalt is indicated by a variety of errors that clients may make in copying the figures, including figures in which details are missing or notably inexact; figures that collide with each other on the page; inability to accurately "close" shapes such as circles or squares; disproportionate size of a figure or part of a figure; and angles in copied figures that do not match the angles in the originals (Lacks, 1999).

Another common neurological screen among clinical psychologists is the **Rey-Osterrieth Complex Figure Test,** originally published in 1941. This test is also a brief pencil-and-paper drawing task, but it involves only a single, more complex figure. The Rey-Osterrieth also features the use of pencils of different colors at various points in the test; this way, the examiner can trace the client's sequential approach to this complex copying task. It also includes a memory component, in which clients are asked 3 to 60 minutes after copying the form to reproduce it again from memory (Helmes, 2000; Lacks, 2000).

The **Repeatable Battery for the Assessment of Neuropsychological Status** (**RBANS;** Randolph, 1998) is a neuropsychological screen that focuses on a broader range of abilities than does either the Bender-Gestalt or the Rey-Osterrieth. The RBANS tests not only visuomotor abilities but also verbal skills, attention, and visual memory. It takes 20 to 30 minutes to complete and includes 12 subtests in 5 categories: immediate and delayed memory, visuospatial/constructional, language, attention, and delayed memory. The subtests involve such tasks as learning a list of 10 words presented orally, naming pictures of various objects, recalling an orally presented list of numbers, recalling a story told 20 minutes earlier, and (like the Bender-Gestalt and Rey-Osterrieth) copying of visual figures. The RBANS yields a total score as well as scores for each of the 12 subtests and 5 indices into which they are organized (Groth-Marnat, 2009; Groth-Marnat & Wright, 2016). A revision, called the RBANS Update, was published in 2012 and enables the test to be administered to children as young as 12 years old (Hale et al., 2016; Randolph, 2012).

Although it is used for a variety of other purposes as well, the **Wechsler Memory Scale—Fourth Edition** (**WMS-IV**; Wechsler, 2009a, 2009b) is frequently utilized for neuropsychological purposes. The WMS-IV is a memory test often used to assess individuals aged 16 to 90 who are suspected of having memory problems due to brain injury, dementia, substance abuse, or other factors. It assesses both visual and auditory memory across its seven subtests. It also assesses both immediate and delayed recall. One of the WMS-IV subtests, Logical Memory, involves the client hearing a story read aloud and then trying to recall the story both immediately and after a 20- to 30-minute delay. Another subtest, Visual Reproduction, involves presenting abstract visual figures to the client and asking him or her to reproduce them after they have been removed. For adults over 65 years of age, the WMS-IV offers a version that is shorter in duration, therefore minimizing the impact of fatigue on test results (Drozdick, Holdnack, & Hilsabeck, 2011; Drozdick, Raiford, et al., 2018; Holdnack & Drozdick, 2010).

A final note about neuropsychological testing: As with any other kind of assessment, it is essential for clinical psychologists to develop and use cultural competence when selecting, administering, scoring, and interpreting neuropsychological tests. Many neuropsychological instruments were standardized (either originally or currently) on European Americans, and their form or structure can emphasize European American values. When working with diverse clients, it is important to caution against overpathologizing and to consider (or consider developing) culture-specific norms whenever possible (Horton, 2008).

CHAPTER SUMMARY

Clinical psychologists use intelligence tests, achievement tests, and neuropsychological tests to assess various intellectual capacities of their clients. The most widely accepted intelligence tests endorse a hierarchical model of intelligence, as indicated by the fact that they yield a single overall intelligence score as well as more specific index, factor, or subtest scores. The Wechsler intelligence tests, including the WAIS-IV for adults, the WISC-V for school-age children, and the WPPSI-IV for preschoolers, rank among the most commonly used and psychometrically sound measures of intelligence. Each Wechsler test features about a dozen subtests grouped into four or five factors, each of which contributes to the full-scale IQ. The Stanford-Binet intelligence test, which covers the entire age range in a single version, is also frequently used, especially among clinical psychologists who may be testing clients at the extremes of the intelligence range. Concerns about the cultural fairness of prominent intelligence tests has fostered the development of less culturally dependent tests such as the UNIT-2, which does not depend on linguistic compatibility between examiner and client. Whereas intelligence tests measure abilities, achievement tests such as the WIAT-III measure accomplishments, particularly in core academic areas such as math and reading. Achievement tests are commonly used when assessing for specific learning disorder. Neuropsychological tests are intended to measure cognitive function and dysfunction and, in some cases, to localize impairment to a particular region of the brain. Some neuropsychological measures are lengthy and comprehensive, such as the HRB, LNNB, and NEPSY-II; others, such as the Bender-Gestalt-II, Rey-Osterrieth Complex Figure Test, and RBANS, are briefer screens for neuropsychological problems.

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cultural fairness 214
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CRITICAL THINKING QUESTIONS

1. In your opinion, what aspects of intelligence do contemporary intelligence tests overemphasize?
2. In your opinion, what aspects of intelligence do contemporary intelligence tests overlook?
3. To what extent do you agree with the idea that intelligence is a single ability ("g"), as opposed to separate unrelated abilities ("s")?
4. Which subtests of the Wechsler intelligence tests seem most culturally fair? Which seem most potentially culturally unfair?
5. Why is it important for clinical psychologists who do not specialize in neuropsychology to be competent in the use of brief neuropsychological screens?

LOOKING TOWARD GRADUATE PROGRAMS

The following table is a list of index entries in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018) relevant to the topics in this chapter (see page 25 in

Chapter 1 for more information about the *Insider's Guide*). Keep in mind a few things as you consider these listings: They are current as of the 2018/2019 edition of the *Insider's Guide* (which typically revises every two years); they may have changed as faculty members retire, take new jobs, or develop new interests; and a graduate program's own website may have more extensive or up-to-date information.

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Alzheimer's Disease	Assessment/Testing	Assessment/Testing
Assessment/Diagnosis	Neuroimaging	Neuropsychology
Brain Injury/Head Injury	Neuropsychology	Neuroscience
Neuroimaging/Functional Neuroimaging	Traumatic Brain Injury	
Neuropsychology	Miscellaneous—Alzheimer's Dementia	
Miscellaneous and Other—Clinical Neuroscience	Miscellaneous—Gifted Evaluation	
	Miscellaneous—Neurological Disorders	

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

Assessment
http://asm.sagepub.com/

The Clinical Neuropsychologist
http://www.tandfonline.com/loi/ntcn20#.V08kr_krLX4

Intelligence
http://www.journals.elsevier.com/intelligence

Neuropsychology
http://www.apa.org/pubs/journals/neu/index.aspx

Psychological Assessment
http://www.apa.org/pubs/journals/pas/index.aspx

STUDENT STUDY SITE RESOURCES

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- Self-quizzes
- eFlashcards
- Culture expert interviews
- In My Practice demonstration video
- Full-text SAGE journal articles
- Additional web resources
- Mock assessment data

10

PERSONALITY ASSESSMENT AND BEHAVIORAL ASSESSMENT

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Clinical psychologists have developed many different approaches to assessing clients' personality and behavior. But before we examine any specific tests or methods, there are three important, overarching themes to emphasize: multimethod assessment, evidence-based assessment, and culturally competent assessment.

MULTIMETHOD ASSESSMENT

No measure of personality or behavior is perfect. Some have excellent reliability, validity, and clinical utility, but even these have their limitations. For that reason, it is important for clinical psychologists not to rely exclusively on any single assessment method. Instead, personality is best assessed by using multiple methods, including tests of different types, interview data, observations, or other sources. Each method offers a unique perspective on the client, and often they converge on similar conclusions. To the extent that these conclusions are supported by multiple tests rather than only one, the clinical psychologist can assert them with confidence.

Learning Objectives

10.1 List the advantages of a multimethod assessment approach when conducting a psychological evaluation.

10.2 Propose how a psychologist conducting a personality assessment can demonstrate cultural competence.

10.3 Differentiate between objective and projective assessments of personality.

10.4 Identify objective personality tests commonly used by clinical psychologists.

10.5 Describe major projective personality tests used for psychological assessment.

10.6 Explain behavioral assessment methods used by clinical psychologists.

The advantages of **multimethod assessment** hold true even in less formal and professional settings. Consider a situation in which you are getting to know someone, such as in a new dating relationship. Your initial "assessment" of your partner's personality may take place on the first date—at a restaurant and a movie, perhaps. As the relationship continues, you'll have a chance to "assess" your partner's personality using other "methods"—you'll see his or her personality and behavior with family, with friends, at parties, at home, at work, at school, and so on. You may get a strong first impression on the first date, but you probably won't feel that you genuinely know and understand your partner's personality until you've "assessed" him or her in a variety of contexts, because each situation will reveal a different aspect of personality. In a more professional sense, the same applies to clinical psychologists "getting to know" a client via personality tests: Each method offers a unique perspective, and although some may be more enlightening than others, it is the integration of multiple methods that ultimately proves most informative.

EVIDENCE-BASED ASSESSMENT

In an earlier chapter, we discussed the recent advancement toward evidence-based treatments, or "treatments that work." The movement toward **evidence-based assessment**, resting on the same principle of "what works" empirically, isn't far behind. Clinical psychologists who practice evidence-based assessment select only those methods that have strong psychometrics, including reliability, validity, and clinical utility (each defined in Chapter 8; Therrien & Hunsley, 2015). Moreover, they select tests that have sufficient normative data and are sensitive to issues of diversity such as age, gender, race, and ethnicity. They typically target their assessment strategies toward a particular diagnosis or problem, such that "what works" for assessing attention-deficit/hyperactivity disorder (ADHD) might be a different set of assessment tools than "what works" for assessing panic disorder, schizophrenia, bulimia, PTSD, or any other clinical issue (Glynn & Mueser, 2018; Hunsley & Mash, 2008b; C. Johnston & Colalillo, 2018; Moshier, Parker-Guilbert, Marx, & Keane, 2018). For example, two methods—the Structured Clinical Interview for *DSM-5* (SCID, described in Chapter 8) and the Beck Depression Inventory-II (described later in this chapter)—are specifically identified as "highly recommended" instruments for diagnosing depression in adults (Persons & Fresco, 2008).

How do researchers determine "what works" when assessing particular clinical problems? In other words, what criteria should distinguish evidence-based assessment methods from those lacking evidence? This is one of the great challenges in

the early development of evidence-based assessment (Hunsley & Mash, 2008b, 2018). Of course, some of the strengths or weaknesses of an assessment method can be expressed quantitatively. For example, test-retest reliability, which measures the extent to which the method yields similar results at different points in time, is expressed as a correlation coefficient ranging from −1 to +1. In a landmark book on evidence-based assessment—*A Guide to Assessments That Work*—its editors define "adequate" test-retest reliability this way: "Preponderance of evidence indicates test-retest correlations of at least .70 over a period of several days to several weeks" (Hunsley & Mash, 2008a, p. 8). They go on to define "good" test-retest reliability as the same .70 level "over a period of several months" and "excellent" test-retest reliability as the same .70 level "over a period of a year or longer" (p. 8). These definitions seem reasonable enough, but they also point out that any criteria designed to delineate levels of evidence for assessment tools will incorporate some subjective decision making (e.g., Why .70 rather than .60 or .80? Why those specific time periods rather than shorter or longer periods?), along with objective data.

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Photo 10.1 Personality is best assessed by using multiple methods, including tests of different types, interview data, observations, or other sources.

Evidence-based assessment has gained quite a bit of momentum within the context of evidence-based practices in health care more broadly (Jensen-Doss, Walsh, & Ringle, 2018; Youngstrom et al., 2017). Some of its advocates hope it will influence not only current graduate students in clinical psychology but also experienced psychologists "who use assessment instruments because they learned them in graduate school, rather than because there is strong evidence that they work. Now, a different and better standard is available" (Nathan, 2008, p. xvii; see also Youngstrom & Van Meter, 2018). On the other hand, there are those who remind us that not every decision within a clinical assessment should be predetermined by the instructions of guidebooks or manuals. For example, in a discussion of clinical interviewing, Sommers-Flanagan and Sommers-Flanagan (2009) state that

> these manuals cannot address nuances in human interaction—nuances that sometimes elicit anger or tears—and these manuals cannot adequately guide clinicians on when to self-disclose, when to just listen, how to respond to client questions, or effective methods for establishing rapport with a living, breathing, culturally unique individual whom they want to help. (p. xi)

Undoubtedly, relying on assessment methods with demonstrable reliability, validity, and clinical utility for particular clinical problems is a professional step forward, especially if the alternative involves choosing methods on the basis of familiarity, popularity, or similarly unempirical factors. Contemporary clinical psychologists now face the challenge of integrating "what works" empirically with their own clinical judgment and their clients' needs as they make decisions about assessment.

CULTURALLY COMPETENT ASSESSMENT

Cultural competence is essential across all activities of clinical psychologists, especially personality assessment. Simply put, every culture has its own perception of "normal" and its own variations of "abnormal" as well. A personality assessment conducted without knowledge or sensitivity to these cultural specifics can be a dangerous thing; in fact, it has been labeled "cultural malpractice" by some (e.g., Dana, 2005). A primary danger lies in the possibility of **overpathologizing**—that is, viewing as abnormal that which is culturally normal. In other words, the clinical psychologist must appreciate the meaning of a behavior, thought, or feeling within the context of the client's culture, which may differ from the context of the psychologist's own culture.

As an example of overpathologizing, consider Duron, a 45-year-old African American client who was urged by his wife and children to seek psychotherapy. On the intake form Duron completed in the waiting room of Dr. Platt, a white clinical psychologist, he indicated that he was struggling with symptoms of depression and excessive alcohol use. But as Duron and Dr. Platt spoke during the initial session, Dr. Platt was struck by what he considered excessive suspiciousness. He noticed that Duron was somewhat reluctant to reveal details about himself, posed numerous questions about confidentiality, and asked, "Why do you need to know that?" several times in response to interview questions. Duron did not schedule a second appointment when the first appointment ended, and Dr. Platt interpreted this as a sign of distrust. By the end of the session, Dr. Platt was convinced that Duron's primary problem was not depression or alcohol abuse but paranoia. He subsequently administered personality tests to Duron and interpreted the results to support his hypothesis that Duron was paranoid. Of course, Dr. Platt's mistake throughout this assessment process was his lack of knowledge of African American cultural norms, especially as they relate to seeking psychological services from white therapists. In this situation, African Americans often exhibit a guardedness justified by many years of formal and informal racial prejudice, oppression, and betrayal (Boyd-Franklin, 1989; Hines & Boyd-Franklin, 2005; Maultsby, 1982). Through Dr. Platt's cultural insensitivity and ethnocentrism, he wrongly judged Duron's behavior as pathologically paranoid when, in fact, Duron's behavior was reasonable based on his cultural background.

In Duron's case, the source of the cultural bias was the clinician himself. During personality assessment, the bias can actually stem from a number of other sources as well, including the test and the service delivery method (Dana, 2005). For example, a test can be biased if it features language or structure that is culture specific rather than universally applicable. The test may also include norms that are not adequately culturally inclusive, resulting in a comparison of the client's scores with the scores of an inappropriate reference group (see Box 10.1). Additionally, the service delivery method (essentially, the way the clinician uses the test) can be biased by including interpersonal interaction or an approach to time and task that is inconsistent with some cultures. Whether the bias stems from the clinician, the test, or the service delivery method, the clinical psychologist is responsible for recognizing and minimizing it in order to conduct culturally competent assessments.

BOX 10.1

Considering Culture

Culture-Specific Norms for Personality Tests

Claudia is a Cuban American woman who has just completed the Minnesota Multiphasic Personality Inventory-2 (MMPI-2), a popular multiple-choice, pencil-and-paper personality test for adults (covered later in this chapter). The MMPI-2 yields a wide variety of scores on variables such as depression, anxiety, paranoia, schizophrenia, mania, and others. To score Claudia's test, her clinical psychologist will compare her responses with those of a "normative" group of people who took the test prior to its publication and whose scores form the basis of the norms tables in the MMPI-2 manual. What cultural characteristics should the people in the normative group have? In other words, with whose scores, specifically, should Claudia's scores be compared?

The possibilities range from very broad groups to very specific groups. Most broadly, the normative group could be a cultural mix representative of the U.S. population, and Claudia's scores could be compared with those of the entire group as a whole. On the other hand, a more narrowly defined normative group could be used, such as Cuban Americans or females. In fact, we could combine these variables and consider an even more culturally specific group: Cuban American females.

There would be advantages to ignoring the culture-specific norms and instead comparing Claudia's scores with those of the entire normative sample (which, for the MMPI-2, consists of 2,600 adults). The full normative sample is, by definition, larger than any cultural subset. Only a small fraction of those 2,600 adults were Cuban American females, and such a small group may not be statistically representative of Cuban American females more generally. Test developers and researchers could make efforts to establish sufficiently large normative groups specific to each culture, but this could prove excessively costly and time-consuming, especially if many cultural variables are considered in combination (Cuban American females of Claudia's age range and religion, for example).

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Photo 10.2
Should a client's scores on a personality test be compared with scores of the population as a whole or with scores of others who are culturally similar?

On the other hand, there would be advantages to comparing Claudia's scores with culture-specific norms. By doing so, we could ensure that Claudia's responses are being evaluated in a culturally sensitive manner. We could reduce the chances of overpathologizing her scores, a risk involved in comparing them with norms that do not reflect her own cultural background and values. Comparing Claudia's scores only with the scores of others

(Continued)

(Continued)

like her prevents an ethnocentric judgment of Claudia's mental health according to standards established by others culturally unlike her (J. M. Wood et al., 2007).

In actual practice, clinical psychologists compare client test data with both broad, heterogeneous normative groups and more specific, homogeneous normative groups. The MMPI-2 manual, for example, does not include ethnicity-specific norms, but it does include gender-specific norms, and in the years since the MMPI-2 was published, numerous researchers have established and published more culturally specific normative information for clinical psychologists to use (e.g., Butcher, Cheung, & Lim, 2003; Dana, 1995; Garrido & Velasquez, 2006; Kwan & Maestas, 2008; McNulty, Graham, Ben-Porath, & Stein, 1997; Velasquez, Garrido, Castellanos, & Burton, 2004). Even cultural groups as specific as the Old Order Amish have their own set of MMPI-2 norms (Knabb, Vogt, & Newgren, 2011). One solution to this quandary is to consider both broad and culture-specific normative data. Both can provide meaningful comparisons for a client's responses to a personality test.

OBJECTIVE PERSONALITY TESTS

Generally speaking, personality tests can be placed in one of two categories: objective and projective. **Objective personality tests** include unambiguous test items, offer clients a limited range of responses, and are objectively scored. Most often, the objective personality tests that clinical psychologists use are questionnaires that clients complete with pencil and paper (or in some cases, on a computer). They typically involve a series of direct, brief statements or questions and either true/false or multiple-choice response options in which clients indicate the extent to which the statement or question applies to them (Morey & Hopwood, 2008). By contrast, *projective* personality tests feature ambiguous stimuli and an open-ended range of client responses. They are based on the assumption that clients reveal their personalities by the way they make sense of vaguely defined objects or situations (S. R. Smith & Archer, 2008). Projective personality tests will be covered in more detail in a later section. For now, let's consider some of the most commonly used and reputable objective personality tests.

Minnesota Multiphasic Personality Inventory-2

The **Minnesota Multiphasic Personality Inventory-2 (MMPI-2)** is both the most popular and the most psychometrically sound objective personality test used by clinical psychologists (Butcher & Beutler, 2003; Camara, Nathan, & Puente, 2000; Frauenhoffer, Ross, Gfeller, Searight, & Piotrowski, 1998; C. V. Wright et al., 2017). It is used in many countries and cultures around the world and has been translated into dozens of languages (Butcher, Mosch, Tsai, & Nezami, 2006; J. R. Graham, 2012). The format of the MMPI-2 is simple: The client reads 567 self-descriptive sentences and, using a pencil-and-paper answer sheet, marks each sentence as either true or false as it applies to him or her. The items span a wide range of behavior, feelings, and attitudes. A few simulated MMPI-2 items appear in Box 10.2.

Web Link 10.1
MMPI-2 site

BOX 10.2

Simulated MMPI-2 Items

- I feel like I am low on energy much of the time.
- I often find myself in conflicts with people in authority.
- During times of stress, I usually experience an upset stomach.
- Most of the people I know have plans to hurt me in some way.
- I have visions of things that aren't real and that other people can't see.
- I enjoy going to social events with dozens of people.
- I often can't stop worrying about things even when there is no reason for the worry.

The MMPI-2 is a revision of the original MMPI, which was published in 1943. When Starke Hathaway and J. C. McKinley, the authors of the original MMPI, began their work in the 1930s, they sought an objective way to measure psychopathology. Numerous questionnaires of this type were available at that time, but none were based on a solid empirical foundation as the MMPI would be. When creating a personality test, it is relatively easy for an author to create a list of items that should, *theoretically*, elicit different responses from "normal" and "abnormal" people of various categories. Hathaway and McKinley chose to take on a greater challenge: to create a list of items that *empirically* elicit different responses from people in these normal and abnormal groups (Ben-Porath & Archer, 2008; J. R. Graham, 2012).

Hathaway and McKinley succeeded in creating such a list of items by using a method of test construction called **empirical criterion keying** (Arbisi, 2015). Essentially, this method involves identifying distinct groups of people, asking them all to respond to the same test items, and comparing responses between groups. If an item elicits different responses from one group than from another, it's a worthy item and should be included on the final version of the test. If the groups answer an item similarly, the item is discarded because it does not help categorize a client in one group or the other. When empirical criterion keying is used, it doesn't matter whether an item should, in theory, differentiate two groups; it matters only whether an item does, in actuality, differentiate two groups (Ruscio, 2015). As an example, consider the simulated item "I have visions of things that aren't real and that other people can't see." In theory, we may expect schizophrenics to endorse this item far more often than would nonschizophrenics. According to the empirical criterion keying method, this theoretical expectation is not enough. It must be supported by evidence that schizophrenics do, in fact, endorse this item far more often than do nonschizophrenics. According to the empirical criterion keying method, only if the two groups actually respond differently to the item should it be included on the final version of the test. See Box 10.3 for additional explanation of empirical criterion keying.

BOX 10.3

Metaphorically Speaking

If You've Shopped on Amazon.com, You Understand Empirical Criterion Keying

How do the people at Amazon.com (or most other online shopping sites) decide which items to suggest to customers? Actually, they use a strategy similar to empirical criterion keying. As you read through the description of an item on Amazon.com, the listing always includes the section, "Customers who bought this item also bought. . . ." The site uses its sales statistics to link certain items together: People who downloaded Drake's latest album also downloaded Nicki Manaj's, people who bought *Star Wars* DVDs also bought *Lord of the Rings* DVDs, and people who bought Dav Pilkey's newest books also bought Aaron Blabey's.

The important point is that these suggestions are based on empirical data—not theory, opinion, common sense, or any other nonempirical criteria. In other words, the people at Amazon.com could, if they chose, recommend to fans of *Star Wars* other DVDs that would seem, theoretically, to make sense. Rather than stating, "Customers who bought this item also bought . . . ," they could tell us, "Customers who like this item might also like. . . ." The "might also like" list would include DVDs similar in some way to *Star Wars*—other movies set in outer space, other movies directed by George Lucas, other movies featuring Harrison Ford, and so on.

Instead, the suggestions from Amazon.com are based entirely on connections among movies according to empirical sales data. Perhaps your intuition wouldn't have told you that customers who bought *Star Wars* DVDs also bought *Back to the Future* or *The Muppets,* but according to sales data, they did. None of these "also bought" movies is especially similar to *Star Wars,* but the sales figures nonetheless demonstrate empirically that they sell to the same audience. Appearance on the list of suggestions has nothing to do with a connection between movies in terms of actors, subject matter, or any other criteria besides empirical sales data.

In a similar way, the creators of the MMPI were blindly empirical when they constructed their tests. Previous designers of personality tests had primarily relied on items they believed would separate abnormal people from normal people. The MMPI's creators instead insisted that an item would be included in their test only if it demonstrated empirically that it produced different responses from people in different groups. It didn't matter whether the item *should* distinguish the groups according to intuition, logic, common sense, or clinical wisdom; what mattered was whether the empirical data showed that the item *did* in fact distinguish between groups. As a hypothetical example, if the item "I cry a lot" did not separate depressed people from nondepressed people, it would be eliminated from the test, even if in theory it seems like an obvious distinction between the two groups. And if a seemingly unrelated item such as, say, "I prefer apples to oranges," did separate depressed people from nondepressed people, it would be kept, despite its lack of a conceptual connection to depression.

What do you think of the "Customers who bought this item also bought . . ." suggestion lists on websites such as Amazon.com? What are the benefits and drawbacks to using an empirical criterion keying approach to create such suggestion lists? Similarly, how do you evaluate the empirical criterion keying approach to personality test construction? What are its benefits and drawbacks?

For Hathaway and McKinley, the distinct groups on which potential items were evaluated consisted of people who had been diagnosed with particular mental disorders (e.g., depressed, paranoid, schizophrenic, anxious, sociopathic, and hypochondriacal groups) and a group of "normals" who did not have a mental health diagnosis. Although they began the process with more than 1,000 potential items, only 550 were retained after the empirical criterion keying method was complete. The items appeared in the test in random order, but for scoring purposes, they were organized into groups related to 10 specific pathologies. Each of those 10 groups of items represented a **clinical scale**, and the higher a client scored on a particular scale, the greater the likelihood that he or she demonstrated that form of psychopathology. The 10 clinical scales remain the same in the MMPI-2 and are described in Table 10.1 (Ben-Porath & Archer, 2008).

Table 10.1 Clinical Scales of the MMPI, MMPI-2, and MMPI-A

Scale Number	Scale Name	Abbreviation	Description of High Scale Scores
1	Hypochondriasis	Hs	Somatic problems, excessive bodily concern, weakness, ailments, complaining and whining
2	Depression	D	Depressed, unhappy, low confidence, pessimistic
3	Hysteria	Hy	Vague medical reactions to stress, somatic symptoms, denial of conflict and anger
4	Psychopathic Deviate	Pd	Antisocial, rebellious, blaming others, poor consideration of consequences of actions
5	Masculinity-Femininity	Mf	Rejection of traditional gender roles, effeminate men, masculine women
6	Paranoia	Pa	Suspicious, guarded, hypersensitive, belief that others intend to harm
7	Psychasthenia	Pt	Anxious, nervous, tense, worrisome, obsessive
8	Schizophrenia	Sc	Psychotic, disorganized, or bizarre thought process, unconventional, hallucinations, delusions, alienated
9	Mania	Ma	Manic, elevated mood, energetic, overactive, accelerated movement and speech, flight of ideas
10	Social Introversion	Si	Introverted, shy, reserved, more comfortable alone than with others

Sources: Archer (1997) and Butcher (2011).

Some of the names of the clinical scales described in Table 10.1 may seem unfamiliar, but they actually refer to very familiar and common clinical issues. The unfamiliar terms were commonly used when the MMPI was originally created, and they were retained in the MMPI-2 to maintain continuity even though some have become outdated. For example, Scale 4, "Psychopathic Deviate," would probably be labeled "Antisocial" today. And the name for Scale 7, "Psychasthenia," is an obsolete word roughly equivalent to "Anxiety."

Another important feature introduced by the MMPI (and retained in the MMPI-2) was a way to assess clients' **test-taking attitudes**. Hathaway and McKinley realized that self-report instruments are vulnerable to insincere efforts by the client. Some clients may intentionally exaggerate their symptoms ("fake bad") to appear more impaired than they really are; others may intentionally minimize their symptoms ("fake good") to appear healthier than they really are. Still other clients may respond randomly without paying much attention to items at all. The MMPI and MMPI-2 include a number of items designed to "catch" these test-taking attitudes. When responses to these items are grouped together, they constitute the test's **validity scales**. These validity scales inform the clinical psychologist about the client's approach to the test and allow the psychologist to determine whether the test is valid and what kinds of adjustments might be appropriate during the process of interpreting the clinical scales. The MMPI and MMPI-2 contain three specific validity scales: L (*Lying*, suggesting "faking good"), K (*Defensiveness*, also suggesting "faking good"), and F (*Infrequency*, suggesting "faking bad").

As an example of the importance of validity scales, consider Tammy, a 24-year-old woman on trial for armed robbery. Her attorney has argued that Tammy is seriously mentally ill and that, as a result, she should be found not guilty by reason of insanity and placed in a psychiatric treatment center rather than imprisoned. The judge in Tammy's case orders Tammy to undergo a psychological examination, and Dr. Reed, the clinical psychologist in the case, uses the MMPI-2, among other tests. Dr. Reed finds that many of Tammy's MMPI-2 clinical scales, especially scales 2 (Depression) and 8 (Schizophrenia), are extremely high. Alone, this information would suggest that Tammy is experiencing significant psychopathology. However, Tammy's MMPI-2 validity scales (especially her F scale) strongly suggest that she "faked bad" on the test; that is, she exaggerated her symptoms in an attempt to look more pathological than she truly is. (Clinical psychologists often use the term *malingering* to describe this type of deliberate behavior.) With this information, Dr. Reed may determine that Tammy's MMPI-2 results are altogether invalid or may interpret the clinical scales with the knowledge that Tammy tended to overstate her problems.

The MMPI grew tremendously in popularity from the 1940s to the 1980s and was eventually replaced by the MMPI-2 in 1989. The revision process addressed several weaknesses that had become increasingly problematic for the original MMPI (Tarescavage & Ben-Porath, 2015). Foremost among these weaknesses was the inadequate normative sample of the original MMPI. The "normal" group with which the clinical groups were compared consisted of 724 individuals from Minnesota in the 1940s. (To be specific, they were visitors to the University of Minnesota Hospital.) This group was overwhelmingly rural and white, and as such, they were not at all an adequate microcosm of the U.S. population. So for the MMPI-2, normative data were solicited from a much larger and demographically diverse group. In all, the MMPI-2 normative group includes 2,600

people from seven states who closely matched U.S. Census data from the 1980s regarding age, ethnicity, and marital status (Butcher, 2011; R. L. Greene & Clopton, 2004). Other improvements included the removal or revision of some test items with outdated or awkward wording. Otherwise, the authors of the MMPI-2 sought to keep the revised version as similar as possible to the original MMPI to take advantage of the familiarity of the test to many clinical psychologists and the enormous body of literature that had accumulated regarding the MMPI.

Soon after the MMPI-2 was published, an alternate version of the MMPI was created for younger clients. Whereas the MMPI-2 is appropriate only for adults (18 years and older), the **Minnesota Multiphasic Personality Inventory-Adolescent (MMPI-A)** was designed for clients aged 14 to 18 years. It was published in 1992 and is very similar in administration, format, scoring, and interpretation to the MMPI-2. It is a true/false, pencil-and-paper test consisting of 478 items. Some of its items are shared with the MMPI-2, and some are original items targeting common teen issues such as school, family, substance use, and peer relations. It yields the same validity scales and clinical scales as the MMPI-2. It was normed on 2,500 adolescents chosen to match 1980 U.S. Census data on many important demographic variables (Archer, 1997; Baer & Rinaldo, 2004; Ben-Porath & Archer, 2008).

The MMPI-2 and MMPI-A yield the same 10 clinical scale scores that the original MMPI yielded. After considering the validity scores, clinical psychologists interpret the MMPI-2 or MMPI-A by considering the clinical scale scores most elevated above normal levels. In some cases, a single clinical scale score will stand out; in others, two or three clinical scale scores may be elevated. As shorthand, clinical psychologists often use the elevated scale numbers ("code-types") to refer to a client's profile, as in, "her MMPI-A profile is a 3/6" or "his MMPI-2 is a 2/4/8." Once these high clinical scales are identified, clinical psychologists turn to the empirical literature describing clients with that pattern of elevations. Numerous books and computer programs, many of which were authored by individuals involved in the creation of the MMPI, MMPI-2, or MMPI-A, supply psychologists with this interpretive information. In addition to the 10 clinical scales, a number of additional scales—known as **supplemental scales** and **content scales**—have been developed to measure other, often more specific aspects of personality and pathology (Butcher, 2011; Butcher & Beutler, 2003).

The MMPI-2 is extremely well supported by research on its psychometric characteristics. Thousands of studies examine some aspect of reliability or validity regarding the MMPI tests, and the results have been consistently positive. In short, the MMPI-2 is established as a reliable and valid test of personality and psychopathology, justifying its extensive use by clinical psychologists (R. L. Greene & Clopton, 2004). Although the research base is not as extensive, similar conclusions have been reached for the MMPI-A (Archer, 1997; Baer & Rinaldo, 2004).

In late 2008, a new, shorter version of the MMPI-2 was released. The **MMPI-2 Restructured Form** (**MMPI-2-RF**; Ben-Porath & Tellegen, 2008; Krishnamurthy & Meyer, 2016) contains only 388 of the 567 items on the MMPI-2. Its clinical scales, called the Restructured Clinical scales, are close to the 10 clinical scales in the other forms of the MMPI but different in important ways. Each scale includes fewer items, and each is more homogeneous or "tighter." In other words, each scale overlaps less with the others due to the removal of items that overlapped between the scales. Much

of this overlap stemmed from items that tap into the concept of demoralization, which developers of the MMPI-2-RF believed to be pervasive across many forms of psychopathology. So demoralization gets a scale of its own in the MMPI-2-RF, and the remaining clinical scales are a bit more streamlined and distinctive from one another. (Two of the original 10 clinical scales, scale 5 [Masculinity-Femininity] and scale 0 [Social Introversion], are omitted from the MMPI-2-RF.) The MMPI-2-RF also includes the new Higher-Order scales, which correspond with common 2-point codes from the MMPI-2 but are measured as single scales, and the new PSY-5 scales, which are related to personality disorders (Ben-Porath & Archer, 2008; Groth-Marnat, 2009). The MMPI-2-RF has been well received by some because of its brevity (compared to the MMPI-2) and ease of interpretation (J. R. Graham, 2012; R. L. Greene, 2011). However, the MMPI-2-RF also has a number of detractors, who raise questions about the quality of its norms and the reliability and validity of some of its scales (Butcher, 2010, 2011; Butcher & Williams, 2009; Tarescavage & Ben-Porath, 2015). The MMPI-2-RF has not entirely replaced the MMPI-2 in terms of usage by clinical psychologists. Instead, both versions are widely used. Researchers have made attempts to estimate the relative popularity of the MMPI-2 and MMPI-2-RF, with different results and explanations offered by different authors (Ben-Porath, 2017; Knauss, 2017; Lally & Williams, 2017; C. L. Williams & Lally, 2016, 2017).

The MMPI-2, MMPI-2-RF, and MMPI-A are currently used for a wide variety of purposes in a wide variety of settings. They are considered comprehensive tests of personality characteristics and psychopathology, and they can be helpful in forming *Diagnostic and Statistical Manual of Mental Disorders* (*DSM*) diagnoses and suggesting placement (e.g., inpatient vs. outpatient) and treatment. They are also used in numerous specialty areas of psychology, including forensic settings and personnel testing (Butcher, 2011).

An unusual but interesting use of the MMPI-2 has been described by Stephen Finn and his colleagues: *therapeutic assessment* (TA). As the name implies, TA involves the use of psychological testing—including feedback about the results—as a brief therapeutic intervention. Early in his career, Finn noticed that his assessment clients often seemed to benefit from the feedback session in which he explained the results of their testing. Despite the fact that the effect was unintended, Finn pursued specific methods by which to maximize it. He has focused on the MMPI-2 specifically and emphasizes a feedback session that is an interactive discussion between client and clinician rather than a unidirectional lecture. TA works most effectively when the assessment is voluntary (rather than required) and when the client can pose specific questions for the clinician to address during feedback (Finn & Kamphuis, 2006; Finn & Tonsager, 1992; M. L. Newman & Greenway, 1997). More broadly, the notion of psychological assessment as a therapeutic tool in and of itself has accumulated significant empirical support (Aschieri, Fantini, & Smith, 2016; Bornstein, 2016; Hanson & Poston, 2011; Poston & Hanson, 2010; Provenzi, Menichetti, Coin, & Aschieri, 2017) and has even been extended to assessments conducted via computer or smartphone (R. E. Smith et al., 2011).

The MMPI-2 and MMPI-A tests are not without limitations. They have been criticized for being too lengthy, requiring reading ability and prolonged attention beyond the capability of some clients, and being susceptible to "faking" by sophisticated clients who can outwit the validity scales. A final criticism of the MMPI tests focuses on their

emphasis on forms of psychopathology as the factors that make up personality. That is, the clinical scales of the MMPI tests describe a client's personality by describing the extent to which the client has various pathologies (depression, schizophrenia, etc.) as opposed to emphasizing other aspects of personality, such as normal traits or strengths.

Clinical psychologists use numerous other objective personality tests that are similar to the MMPI tests in some ways yet different in some important ways as well. Examples include the Personality Assessment Inventory; the Millon Clinical Multiaxial Inventory-IV, which emphasizes personality disorders; and the NEO Personality Inventory-3, which emphasizes normal personality traits.

Personality Assessment Inventory

The **Personality Assessment Inventory (PAI)** contains 344 items, each of which offers four responses: totally false, slightly true, mainly true, and very true. It is appropriate for clients 18 to 89 years old, but an adolescent version, the PAI-A, can be used for clients as young as 12. The PAI has a fourth-grade reading level, making it more intelligible than other broad-based objective personality tests. Like the MMPI, it includes multiple validity scales to detect the manner in which the client is taking the test. It includes 11 clinical scales, some that match with those of the MMPI but others that are more uniquely tied to specific diagnoses or problems, like Borderline Features, Antisocial Features, Anxiety-Related Disorders, Alcohol Problems, and Drug Problems. This direct correspondence to some (but not all) *DSM* disorders is considered a strength of the PAI (Groth-Marnat & Wright, 2016; Krishnamurthy & Meyer, 2016). The PAI was originally created in 1991 by Leslie Morey (Morey, 2003).

Millon Clinical Multiaxial Inventory-IV

The **Millon Clinical Multiaxial Inventory-IV (MCMI-IV)** is like the MMPI-2 and PAI in many ways: It is a comprehensive personality test in a self-report, pencil-and-paper, true/false format. The primary difference between the tests is the MCMI-IV's emphasis on personality disorders. Although it does feature scales for many clinical syndromes related to other disorders such as depression, anxiety, and posttraumatic stress, the MCMI-IV is notable for its many scales related to the personality disorders in *DSM*. In fact, the MCMI-IV features separate clinical scales corresponding to each of the 10 current personality disorders (e.g., antisocial, borderline, narcissistic, paranoid). It also includes clinical scales for other forms of personality pathology, many of which have been considered for inclusion as disorders in *DSM* but are currently omitted (e.g., self-defeating personality, negativistic/passive-aggressive personality, depressive personality, turbulent personality; Choca & Grossman, 2015; Craig, 2008; Retzlaff & Dunn, 2003). The MCMI-IV is designed for adults, but there are separate Millon personality inventories for adolescents and children, as well as another designed for use in medical situations in which the health care professional believes that personality may influence physical health (Millon & Millon, 2015).

The MCMI was originally created in 1977 by **Theodore Millon**, a widely recognized scholar on personality disorders. The current version, MCMI-IV, was published in

2015. The MCMI-IV consists of 195 true/false items. In addition to its clinical scales, it includes "modifier indices," which function similarly to the validity scales of the MMPI tests by assessing the test-taking attitude of the client. A version for adolescents, the Millon Adolescent Clinical Inventory (MACI), contains 160 items and is appropriate for clients aged 13 to 19 years old (Krishnamurthy & Meyer, 2016; J. L. Newman, Larsen, Cunningham, & Burkhart, 2015). Reliability and validity data for all editions of the MCMI are strong, suggesting it is a wise choice for clinical psychologists who seek a broad assessment of personality with an emphasis on personality disorders (Craig, 2008; Meagher, Grossman, & Millon, 2004; Pincus & Krueger, 2015).

Web Link 10.2
Theodore Millon

NEO Personality Inventory-3

As stated earlier, the MMPI (and MCMI and PAI) emphasizes the pathological aspects of personality by producing scores that indicate the extent to which the client is symptomatic of various disorders. The authors of the **NEO Personality Inventory-3 (NEO-PI-3)**, **Paul Costa** and **Robert McCrae**, sought to create a personality measure that assesses "normal" personality characteristics. The five characteristics measured by their test emerge from decades of research on normal personality, much of which was factor analytic in nature. In short, the authors of the NEO-PI-3 (who also put forth the corresponding **five-factor model of personality** or "Big Five") argue that the many words our language offers for describing personality traits "cluster" into five fundamental traits of personality that characterize everyone in varying degrees. These traits—Neuroticism, Extraversion, Openness, Agreeableness, and Conscientiousness—are the five primary scales yielded by the NEO-PI-3 and are described in Table 10.2. The

Table 10.2 Normal Personality Traits Assessed by the NEO-PI-3

Trait/Scale	Description of High Score	Description of Low Score
Neuroticism	Prone to emotional distress, negative affect, anxiety, sadness	Emotionally stable, even tempered, secure even under stressful conditions
Extraversion	Sociable, talkative, outgoing, prefer to be with others	Introverted, reserved, shy, prefer to be alone
Openness	Curious about novel ideas and values, imaginative, unconventional	Conventional, conservative, traditional, prefer familiar ideas and values
Agreeableness	Sympathetic, cooperative, accommodating, prefer to avoid conflict	Hardheaded, competitive, egocentric, uncooperative, unsympathetic
Conscientiousness	Organized, purposeful, disciplined, methodical, tend to make and carry out plans	Easygoing, spontaneous, disorganized, spur-of-the-moment, laid-back

Sources: Costa & McCrae (1992); Costa & Widiger (2001).

NEO-PI-3 also produces 30 "facet" scores (six facets within each of the five domains) to offer more specific descriptions of components within each trait (Costa & McCrae, 1992, 2008; McCrae & Costa, 2007; McCrae, Costa, & Martin, 2005; Simms, Williams, & Nus, 2017).

Web Link 10.3
NEO-PI-3 site

In terms of format, the NEO-PI-3 is comparable to the other objective personality tests described in this chapter. It is a 240-item, pencil-and-paper, self-report questionnaire. The items are short statements with multiple-choice responses ranging from "strongly agree" to "strongly disagree." The original NEO Personality Inventory was published in 1985, with a revised edition, NEO-PI-R, arriving in 1992, and the NEO-PI-3 in 2005. A short form of this test (with 60 items), the **NEO Five Factor Inventory (NEO-FFI)**, is also available but produces a less detailed profile. Psychometric data on the NEO-PI-3 suggest that it has very strong reliability and validity (Costa & McCrae, 1992). The main purpose of the revision that produced the NEO-PI-3 was to revise 37 items to become more readable and easily understood. The NEO-PI-3 has not replaced the NEO-PI-R in terms of popular usage, but it provides an alternative for clients as young as 12, or older clients whose reading skills may be relatively weak (Costa, McCrae, & Martin, 2008; McCrae & Costa, 2010; McCrae, Costa, & Martin, 2005).

The NEO-PI-3 has been criticized for its lack of validity scales. Unlike the MMPI tests and the MCMI-III, the NEO-PI-3 lacks any substantive measurement of the test taker's approach to the test, leaving it rather vulnerable to "faking" or inattention by clients. Clinicians have also commented on the test's limited clinical utility, especially regarding diagnosis. Although the NEO-PI-3 can indicate that an individual is extremely high or low in any of the five fundamental personality traits, such extreme scores do not readily translate into specific, diagnosable mental disorders. However, recent research suggests that some disorders, especially personality disorders, correlate strongly with particular combinations of high scores on NEO-PI-3 scales, which bolsters the clinical utility of the NEO-PI-3 for many psychologists (Costa & McCrae, 2008; Costa & Widiger, 2001).

Beck Depression Inventory-II

Comprehensive objective personality tests—such as the MMPI, PAI, MCMI-III, and the NEO-PI-3—provide clinical psychologists with a broad overview of personality and provide scores on a broad range of variables.

Web Link 10.4
BDI-II site

But sometimes, clinical psychologists prefer more targeted, noncomprehensive objective measures. These tests are typically briefer and focus exclusively on one characteristic, such as depression, anxiety, or eating disorders. The **Beck Depression Inventory-II (BDI-II)** is a widely respected and used example of this type of test (Piotrowski, 2018). (It appears in this section about objective personality tests even though it assesses only one characteristic rather than personality more fully.) Related instruments include the *Beck Hopelessness Scale* and the *Beck Anxiety Inventory*, each of which zooms in on the variable identified in its name. Here, we will focus on the BDI as an example of this type of test.

The BDI-II is a self-report, pencil-and-paper test that assesses depressive symptoms in adults and adolescents. The original BDI was created by Aaron Beck, a leader in cognitive therapy for depression and other disorders, in the 1960s (Rosner, 2015a);

the current revision was published in 1996. The BDI-II is brief—only 21 items, usually requiring a total of 5 to 10 minutes to complete. Each item is a set of four statements regarding a particular symptom of depression, listed in order of increasing severity. The clients choose the one sentence in each set that best describes their personal experience during the previous 2 weeks (a time period chosen to match *DSM* criteria). The 21 item scores are summed to produce a total score, which reflects the client's overall level of depression (Brantley, Dutton, & Wood, 2004; Quilty, Zhang, & Bagby, 2010). As a simulated sample BDI-II item, consider the following set of statements, of which the client would choose one:

- I never think about dying. (0 points)
- I occasionally think about dying. (1 point)
- I frequently think about dying. (2 points)
- I constantly think about dying. (3 points)

The BDI-II lacks validity scales, and of course its scope is much more limited than those of the other objective tests discussed in this chapter (Rosner, 2015b). However, its reliability and validity are strongly established (Brantley et al., 2004), and it is a frequent choice of psychologists seeking a quick, empirically sound answer to a specific assessment question regarding depression.

PROJECTIVE PERSONALITY TESTS

Objective personality tests are based on the assumption that personality is best assessed by directly asking people to describe themselves. **Projective personality tests** are based on a fundamentally different assumption: People will "project" their personalities if presented with unstructured, ambiguous stimuli and an unrestricted opportunity to respond (Tuber & Meehan, 2015). (The term *performance-based test* is sometimes used synonymously with *projective test* [Krishnamurthy & Meyer, 2016].) Imagine a group of people lying on the ground, looking up at the same series of vaguely shaped clouds in the sky. The way each person makes sense of the series of clouds implies something about that person's personality, according to supporters of projective personality tests. To extend this analogy, projective personality tests are similar to a series of "clouds" that psychologists display to clients. Clients are given freedom to make sense of these stimuli in any way they choose—they are not restricted to multiple-choice or true/false options. Clients' responses may be compared with those of others in a normative group, and psychologists will ultimately form hypotheses about the clients' personalities based on their responses.

The lack of objectivity, especially in scoring and interpreting, highlights the most frequently cited shortcoming of projective personality tests. Critics of projective tests stress that they are far too inferential to be empirically sound; that is, they rely too heavily on a psychologist's unique way of scoring and interpreting a client's responses (Hunsley, Lee, & Wood, 2003). In an objective test in which a client's responses consist of "true"

or "false," scoring is standardized and consistent. But in a projective test in which each client produces unique responses, scoring may vary across psychologists. Additionally, once items are scored, the process of assigning meaning to them is, according to many, less systematic and more idiosyncratic in projective tests. Indeed, the standing of projective tests has undoubtedly declined in popularity in recent decades, to the satisfaction of critics who claim that their reliability and validity are insufficient to justify psychologists using them or graduate programs teaching them (e.g., Lilienfeld, Wood, & Garb, 2000).

However, other experts in the field of projective personality testing have argued vigorously in favor of their use (G. J. Meyer & Viglione, 2008). They point to unique assets of projective tests, such as the fact that they aren't as "fake-able" as objective tests: "Given the research on the human capacity to create a façade of mental health, personality assessment using [projective] methods affords the clinical psychologist the means to bring to light a person's true emotional troubles" (Fowler & Groat, 2008, p. 491). They also point to empirical data, such as a review of 184 meta-analytic studies of the psychometric characteristics of personality tests that found the reliability and validity of common projectives such as the Rorschach Inkblot Method and the Thematic Apperception Test (TAT), when scored systematically and accurately, roughly equal to those of commonly used objective personality tests (G. J. Meyer, 2004). However, critics cast serious doubt on the methods and data used to support these conclusions. So the controversy continues. Although projective personality tests may not be as popular as they once were and some detractors decry their continued use, surveys of psychologists suggest that some, including the Rorschach, the TAT, and sentence completion tests, remain very much a part of contemporary clinical psychology (e.g., Watkins et al., 1995).

Rorschach Inkblot Method

In 1921, Hermann Rorschach created the **Rorschach Inkblot Method**. Rorschach was a Swiss psychiatrist who, as a child, played a game in which participants looked at vague blots of ink and said what they saw in the blots. As an adult, he decided to apply a similar method to his patients, with the hypothesis that their responses would reveal their personality characteristics (Weiner, 2004).

For his test, Rorschach created 10 inkblots, 5 with only black ink and the other 5 with multiple colors. Administration occurs in two phases. In the "response" or "free association" phase, the psychologist presents one inkblot card at a time, asks, "What might this be?" and writes down the client's responses verbatim. After the client has responded to all 10 cards, the "inquiry" phase begins, in which the psychologist reads the client's responses aloud and asks the client to describe exactly where in the inkblot each response was located and what features of the inkblot caused the client to offer that response. A simulated Rorschach inkblot appears in Photo 10.3.

When Rorschach published his inkblots in 1921, they were not accompanied by a scoring method. Rorschach died about a year later, and in his absence, numerous

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Photo 10.3 This inkblot closely resembles those used in the Rorschach Inkblot Method. In your opinion, how well can responses to inkblots reveal personality characteristics?

scoring systems were independently created and used in the United States and Europe. Ultimately, **John Exner** combined aspects of many scoring systems to create the **Comprehensive System**, which has become the most common method of scoring the Rorschach (Exner, 1986; J. M. Wood, Nezworski, Lilienfeld, & Garb, 2003). Exner's Comprehensive System includes normative data collected from thousands of children and adults; thus, the Rorschach can be used with clients across almost the entire life span.

Web Link 10.5
Hermann Rorschach Archives and Museum

When a clinical psychologist uses Exner's Comprehensive System, each Rorschach response is coded in many ways. The scoring of a Rorschach response has as much to do with the process by which the client made sense of the blot as the content of the client's perceptions. In other words, *how* the client perceives the blot is no less important than *what* the client sees. A small sample of the many variables examined by the Comprehensive System includes the following:

- *Location.* Does the response involve the whole inkblot, a large portion of it, or a small detail?
- *Determinants.* What aspect of the inkblot—its form, color, shading, and so on—caused the client to make a particular response?
- *Form Quality.* Is the response easily identifiable and conventional? Or is it unique or distorted?
- *Popularity.* Each card has a response or two that occurs relatively frequently. How often does the client offer these popular responses?
- *Content.* What kinds of objects appear with unusual frequency in the client's responses? People, animals, food, clothing, explosions, body parts, nature, or other categories of items?

After each response is scored, the scores are combined into a variety of indices, and the scores and indices are then interpreted. Although interpretive strategies vary somewhat and some may be more empirically based than others, a rule of thumb is that the way the client makes sense of the inkblots parallels the way the client makes sense of the world. Consider, for example, the Form Quality variable. Clients who consistently offer distorted, atypical perceptions of inkblots are thought to do similar things in life—misperceive situations, use distorted judgment, or make sense of things in an unconventional manner. Or consider the Location variable. A client whose responses almost always incorporate the whole inkblot may tend to focus on the "big picture" of life rather than "breaking it down" into smaller pieces or attending to details, whereas a client whose responses always involve very small parts of the inkblot may tend to focus excessively on details and overlook the "big picture" in day-to-day life.

Together, the multitude of variables yielded by the Comprehensive System shed light on many aspects of personality. However, the psychometric foundation underlying the Rorschach has, for some, cast doubt on the clinical conclusions it can offer. Supporters of the Rorschach point out that the reliability data regarding scoring, especially test-retest reliability, are quite strong according to some studies and that validity for the Rorschach has been measured at a comparable level to that of the MMPI tests (e.g., G. J. Meyer, 2004; Rose, Kaser-Boyd, & Maloney, 2001). However, many critics of the

Rorschach have pointed out some serious shortcomings: Numerous studies have found weak reliability and validity data; scoring and interpretation guidelines are complex, and psychologists do not always follow them as closely as they should; results can't often distinguish those who have a particular disorder from those who don't; moreover, norms for the Rorschach may be inadequate for some populations (as summarized by J. M. Wood et al., 2003; see also J. M. Wood et al., 2010). Amid the recent controversy regarding the scientific status of the Rorschach, some strong support has emerged. Specifically, meta-analyses have concluded that some variables in the Comprehensive System, especially those that measure psychoticism, psychological resources, and cognitive complexity, are quite valid (Mihura & Meyer, 2015a; Mihura, Meyer, Dumitrascu, & Bombel, 2013). These meta-analyses were so impressive that even vocal critics of the validity of the Rorschach published an article in which they endorsed the use of those variables in certain clinical and research settings (Mihura, Meyer, Bombel, & Dumitrascu, 2015; J. M. Wood et al., 2015). Those meta-analyses also inspired the creation of an entirely new scoring system, the Rorschach Performance Assessment System (R-PAS), which has stronger norms than the Comprehensive System (G. J. Meyer, Viglione, Mihura, Erard, & Erdberg, 2011; Mihura & Meyer, 2015a).

BOX 10.4

Simulated TAT Card With Simulated Responses

1. "She's really nervous about something. It's only breakfast time, but she's already anxious about something coming up during her day. Maybe she's worrying about a test at school, or some kids who might tease her, or the school bus getting into an accident, or something happening out of the blue that she hasn't even thought about yet. Not sure which, but she's definitely worried about something, and she's only gonna get more worried as the day goes on."
2. "She is a sad, sad girl. She's looking at that plate and thinking that it's empty, just like her life. Something good may appear on the plate soon, but that will only be temporary. It's just like her life—she wishes she could stay happy, but she just can't. No matter what, she ends up sad again."

© Kenney Mencher

Photo 10.4
Simulated TAT

3. "She's excited for a family dinner. Soon, her parents and brothers and sisters will be around the table, and there will be eating and laughing and fun. Then,

(Continued)

(Continued)

they will all help clean up and have a great evening. As she goes to bed, she'll be looking forward to another great day tomorrow."

4. "She's a guest at this house—her friend invited her to have dinner over. She's thinking about the jewelry she just stole from her friend's mother's bedroom. She snuck in there when no one was looking and grabbed some earrings and a necklace. Some people would be feeling guilty about it, but she doesn't—not at all. She actually gets a rush out of it."
5. "She just finished all of the food on her plate, and now she's thinking about making herself throw up. She definitely doesn't want to get fat. Her mom yells at her older sister all the time about being fat. She doesn't want her mom to criticize her like that, so she's trying hard to stay thin. Sometimes she doesn't eat much at all. When she does, she usually thinks about making herself throw up, and sometimes she actually does it."

Source: Kenney Mencher.

Thematic Apperception Test

The **Thematic Apperception Test (TAT)**, published by Henry Murray and Christiana Morgan, is similar to the Rorschach in that it involves presenting the client with a series of cards, each featuring an ambiguous stimulus (C. D. Morgan & Murray, 1935; Murray, 1943). Unlike the Rorschach cards, however, the TAT cards feature interpersonal scenes rather than inkblots. The client's task is to create a story to go along with each scene (Gieser & Stein, 1999; W. G. Morgan, 1999). They are asked to consider not only what is happening in the scene at the moment but also what happened before and what may happen after the scene. They are also asked to describe what the characters may be thinking and feeling (Bellak, 1993). Although the TAT is considered a global measure of personality by many, its strength may lie in its ability to measure interpersonal relationship tendencies (Ackerman, Fowler, & Clemence, 2008).

Box 10.4 features a simulated TAT card, along with a variety of simulated responses. What inferences would you make about the person telling each of the stories? Compare your interpretations to those of your classmates or friends—how consistent are the interpretations? (Note that the actual interpretation of TAT responses would emphasize recurring themes across multiple stories rather than an interpretation based on a single story.)

The TAT includes a total of 31 cards, but psychologists typically select their own subset of cards—often about 10 or so—to administer to a particular client. As the client tells stories aloud and the psychologist writes them down, the psychologist may ask questions during a client's story to solicit more information and can remind the client of the initial instructions as well. Henry Murray (1943) offered a scoring system for the TAT emphasizing "needs" of the main characters, "press" from the environment, and other variables. Others have offered additional formal scoring systems (e.g., Bellak,

1999; see also Abrams, 1999), but for the most part, formal TAT scoring systems have been "neglected or ignored" (Moretti & Rossini, 2004, p. 357). Currently, the TAT is often analyzed without formal scoring at all: "Most clinicians today seem to rely on their own impressionistic inferences," resulting in "idiosyncratic and inconsistent" use of the TAT (Moretti & Rossini, 2004, p. 357). Thus, TAT interpretation is frequently more art than science. A client's story may be interpreted one way by one psychologist but very differently by another.

As a result of this nonempirical scoring and interpretation procedure, the TAT is not a preferred test among clinical psychologists who insist on assessment methods supported by strong psychometric data (Holt, 1999). The validity and reliability of the TAT are less well established than those of other personality tests, largely because the scoring, interpretation, or administration is not uniform across psychologists (Mihura & Meyer, 2015b). Henry Murray (1943) himself stated, "The conclusions that are reached by an analysis of TAT stories must be regarded as good 'leads' or working hypotheses to be verified by other methods, rather than as proved facts" (p. 14)—a statement applicable to all assessment methods, especially those with debatable scientific status. Nonetheless, the use of the TAT continues, as well as the use of the related **Children's Apperception Test (CAT)** and **Senior Apperception Test (SAT**; Bellak, 1993; see also Teglasi, 2010), based on the hypothesis that "the stories we tell say something about who we are" (Moretti & Rossini, 2004, p. 357).

A final note on storytelling tests: the Tell-Me-a-Story (TEMAS) apperception test, designed for children and adolescents, is part of a newer generation of TAT-style tests that place greater emphasis on empirical scoring via normative data as well as cultural sensitivity (Malgady & Colon-Malgady, 2008). The TEMAS was developed specifically for Hispanic youth and was later expanded for use with African American and white youth—with images that feature culturally diverse people in scenes more typical of urban settings rather than the somewhat rural emphasis of the TAT (Teglasi, 2010). The TEMAS is described in more detail in Chapter 17.

Sentence Completion Tests

In **sentence completion tests**, the ambiguous stimuli are neither inkblots nor interpersonal scenes; instead, they are the beginnings of sentences. The assumption is that clients' personalities are revealed by the endings they add and the sentences they create. Although there are many projective tests using the sentence completion format, the **Rotter Incomplete Sentences Blank (RISB)** tests are by far the most widely known and commonly used (Sherry, Dahlen, & Holaday, 2004; Weis, 2015). The original RISB was published in 1950, with the most recent revised edition (including high school, college, and adult versions) appearing in 1992. The RISB tests include 40 written sentence "stems" referring to various aspects of the client's life. Each stem is followed by a blank space in which the client completes the sentence. Simulated sentence stems similar to those in the RISB include the following:

- I enjoy _____.
- It makes me furious _____.

- I feel very nervous _____.
- My proudest moment _____.
- My greatest weakness _____.

Like the TAT, the RISB includes a formal scoring system, but clinical psychologists may not use it regularly, and when they do, scoring is highly dependent on the clinical judgment of the psychologist. Thus, its scientific standing is questionable. However, the RISB can flesh out the information obtained through other tests, including objective personality tests (Sherry et al., 2004). For example, a clinical psychologist may conclude that a client is depressed after the client obtains a very high score on the Depression scale (scale 2) of the MMPI-2. But that scale score alone offers no explanation of the qualitative aspects of the depression or what the depression is "about." If the client answered the simulated RISB items above by stating that "I enjoy . . . very few things now that my spouse is gone," "It makes me furious . . . that life is so unfair," and "I feel very nervous . . . when I think about living alone," the clinical psychologist can hypothesize that the client's depression relates closely to the loss of this significant relationship.

BEHAVIORAL ASSESSMENT

Most clinical psychologists would consider all the objective and projective measures described so far in this chapter to be **traditional personality assessment** techniques. These traditional techniques share a few basic, implicit assumptions:

- Personality is a stable, internal construct. In other words, behavior is determined primarily by characteristics or dispositions "inside" the person.
- Assessing personality requires a high degree of inference. That is, clinical psychologists use the data provided by personality tests (e.g., MMPI-2 scale scores, Rorschach responses) to deduce or speculate about problem behaviors that clients may actually experience.
- Client behaviors are signs of deep-seated, underlying issues or problems, sometimes taking the form of *DSM* diagnoses.

Behavioral assessment challenges all these assumptions and offers a fundamentally different approach to assessment (Haynes, Kaholokula, & O'Brien, 2015; Heiby & Haynes, 2004; Ollendick, Alvarez, & Greene, 2004). According to behavioral assessment, client behaviors are not signs of underlying issues or problems; instead, those behaviors *are* the problems. Another way to state this is that the behavior a client demonstrates is a sample of the problem itself, not a sign of some deeper, underlying problem. For example, if Zach, a 9-year-old child, argues with teachers often at school, "arguing with teachers" is the problem. It is unnecessary and unwise, according to

behavioral assessors, to understand Zach's arguing as "symptomatic" of a deep-rooted issue, such as oppositional defiant disorder. So to assess the problem, assessment techniques should involve as little inference as possible. Rather than inkblots or questionnaires that may get at the problem indirectly, behavioral assessors would choose the most direct way to measure Zach's problem behavior: observation of Zach in his classroom. By rejecting the idea that Zach "has" some internal disorder or trait that underlies his arguing, behavioral assessors demonstrate their rejection of the more fundamental idea that enduring, internal "personality" characteristics cause all behavior. Instead, behavioral assessors argue that external, situational factors determine our behavior. As such, behavioral assessment is a uniquely empirical approach that capitalizes on "the use of valid, precise, and sensitive measures of well-defined behaviors and contemporaneous events to capture behavior-environment interactions" (Haynes & Kaholokula, 2008, p. 518) in which clinicians diagnose behavior problems rather than the clients who exhibit them.

Methods of Behavioral Assessment

The most essential technique in behavioral assessment is **behavioral observation** or the direct, systematic observation of a client's behavior in the natural environment (Bakeman & Haynes, 2015; Ollendick et al., 2004). Also known as **naturalistic observation**, this practice involves taking a direct sample of the problem at the site where it occurs (home, work, school, public places, etc.). The first step in behavioral observation involves identifying and operationally defining the problem behavior. This takes place via interviews, behavioral checklists, consultation with those who have observed the client (family members, coworkers, teachers, etc.), or self-monitoring by the client. Once the target behavior is identified and defined, systematic observation takes place. This process usually involves tallying the frequency, duration, or intensity of the target behavior across specified time periods—first as a baseline, and then at regular intervals to measure improvement as compared with that baseline. Such direct observation can provide a far more accurate assessment of problem behavior than merely asking the client to recall or summarize it verbally during an interview or on a questionnaire. When observation in the natural environment is not possible, behavioral assessors can arrange an analogue observation, in which they attempt to replicate the real-world setting in the clinic and observe the client's responses there (Haynes & Kaholokula, 2008).

Behavioral observation also typically includes keeping a record of the events that occur immediately before and after the target behavior. Depending on the client, the target behavior, and the setting, the record keeping can be completed by any number of individuals, including a teacher, a parent, a friend or family member, or the client. Documenting these events allows for clinical psychologists to understand the functionality of a particular behavior or how the behavior relates to the environment and contingencies that surround it (Heiby & Haynes, 2004; O'Brien, Haynes, & Kaholokula, 2016). As an example, let's consider 9-year-old Zach again. If Zach's clinical psychologist, Dr. Davis, was conducting a behavioral observation, Dr. Davis would visit Zach's classroom with an explicit definition of the arguing behavior and would systematically keep

track of its occurrence. However, Dr. Davis would also take note of the antecedents and consequences of Zach's behavior. Perhaps Zach's arguments are consistently preceded by a particular stimulus (e.g., a direct question from the teacher or an in-class math assignment) or followed by a particular consequence (e.g., prolonged attention from the teacher or removal from the room). These events surrounding Zach's behavior—the "if . . . , then . . ." contingencies that he has learned regarding the outcomes of his actions—can offer important information regarding the functionality of the target behavior (Cipani & Schock, 2007).

Traditional assessment methods such as interviews and questionnaires are not necessarily foreign to behavioral assessment, but if they are used, they emphasize the instrumental aspect of the behavior in question. That is, rather than having the aims of many traditional interviews and questionnaires—diagnostic categorization, for example—the interviews and questionnaires are used by behavioral assessors with the singular goal of identifying the function of the problem behavior. In fact, the terminology they use for these techniques—functional interview and functional questionnaires—illustrates this approach (Haynes & Kaholokula, 2008). If Dr. Davis conducted a functional interview with 9-year-old Zach (or his parents or his teacher), she would ask targeted questions about the antecedents and consequences of Zach's problematic arguing behavior. Of course, asking the client's opinion is no substitute for direct behavioral observation, because the client may lack insight or deliberately mislead the assessor about the sequence of events that typically takes place. However, functional interviews and functional questionnaires can provide important information to guide the behavioral observation. For example, Zach may mention that a consequence of his arguing behavior is that Thomas, the most popular boy in the class, roots him on by subtly smiling and nodding. If Thomas wasn't on Dr. Davis's radar before the functional interview, he will be after Zach mentions him.

Technology in Behavioral Assessment

Especially in recent years, computers and other forms of technology have been used productively by behavioral assessors. In Dr. Davis's behavioral observation of Zach, Dr. Davis could have used a laptop computer or smartphone to record her observations. Numerous software programs have emerged, allowing for simultaneous recording of multiple variables, instant graphing and statistical analysis of behavior patterns, and other advanced, time-saving features. In fact, clients themselves can benefit from the use of technology in behavioral assessment. For example, clients often conduct **self-monitoring**, as a way of either defining the target behavior or measuring changes in it over time. This self-monitoring has traditionally been done with pencil-and-paper journals, but computers and smartphones offer a method that can be more convenient (for e-mailing, texting, or downloading records) and private (if the device has a password-protection feature). Additionally, most of these devices have timer and alarm features, which can increase clients' compliance with self-monitoring schedules (Richard & Lauterbach, 2003).

CHAPTER SUMMARY

The personality assessment techniques traditionally used by clinical psychologists can generally be divided into objective and projective tests. Objective personality tests are typically pencil-and-paper questionnaires featuring unambiguous stimuli, a limited range of responses, and a self-report format. Some of these tests assess test takers' attitudes as well as clinically relevant variables, which aid in interpretation of test results. Examples of objective personality tests include the MMPI-2 and PAI, which focus on various types of psychopathology; the MCMI-IV, which focuses on personality disorders; the NEO-PI-3, which focuses on normal personality traits; and the BDI-II, which is a relatively brief, more targeted test assessing depression only. Projective personality tests involve ambiguous stimuli and an unlimited range of responses and are viewed by many as less psychometrically sound than objective tests. Examples of projective personality tests include the Rorschach Inkblot Method; the TAT, in which clients respond to interpersonal scenes rather than inkblots; and sentence completion tests, in which clients fill in the end of unfinished sentence stems. Unlike traditional clinical assessment involving objective or projective testing, behavioral assessment involves a more direct and less inferential approach to clients' problems. Behavioral assessors employ techniques such as behavioral observation and client self-monitoring with the belief that problem behavior is the clinical issue rather than a sign of a deeper underlying issue. Regardless of the techniques they choose when assessing personality or behavior, clinical psychologists should use multiple methods, consider evidence-based assessment methods for particular clinical problems, and strive for cultural competence.

KEY TERMS AND NAMES

Beck Depression Inventory-II (BDI-II) 239
behavioral assessment 246
behavioral observation 247
Children's Apperception Test (CAT) 245
clinical scale 233
Comprehensive System 242
content scales 235
Paul Costa 238
empirical criterion keying 231
evidence-based assessment 226
John Exner 242
five-factor model of personality 238
Robert McCrae 238
Theodore Millon 237
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Minnesota Multiphasic Personality Inventory-Adolescent (MMPI-A) 235
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validity scales 234

CRITICAL THINKING QUESTIONS

1. In your opinion, what are the most important strengths and weaknesses of projective personality tests?
2. In your opinion, what are the most important strengths and weaknesses of objective personality tests?
3. If you were designing an objective personality test, how would you go about creating a measure of test-taking attitudes (such as the validity scales of the MMPI-2)?
4. The NEO-PI-3 is founded on the idea that all people share the same five fundamental personality characteristics and that we differ only in the strength of these characteristics. Do you agree that all people share the same personality characteristics, or do you believe that some people have qualitatively different characteristics than others?
5. In your opinion, for what clinical problems is behavioral assessment best suited?

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

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http://asm.sagepub.com/

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http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1097-4679

Journal of Personality Assessment

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Journal of Research in Personality

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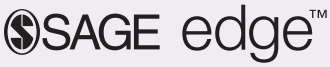

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PART III

PSYCHOTHERAPY

CHAPTER 11 • General Issues in Psychotherapy

CHAPTER 12 • Psychodynamic Psychotherapy

CHAPTER 13 • Humanistic Psychotherapy

CHAPTER 14 • Behavior Therapy

CHAPTER 15 • Cognitive Psychotherapy and Mindfulness-Based Therapies

CHAPTER 16 • Group and Family Therapy

11

GENERAL ISSUES IN PSYCHOTHERAPY

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With this chapter, we begin our exploration of **psychotherapy**, the most common professional activity of clinical psychologists (Norcross & Karpiak, 2012). Upcoming chapters offer details on specific approaches to psychotherapy, including psychoanalytic/psychodynamic, humanistic, behavioral, and cognitive techniques. But in this chapter, we consider some important "across-the-board" psychotherapy issues, including how well it works and how it has been practiced by clinical psychologists.

DOES PSYCHOTHERAPY WORK?

Questions about the outcome of psychotherapy have been prominent throughout the history of clinical psychology, and studies investigating these questions have taken many

Learning Objectives

11.1 Explain the tripartite model in the context of psychotherapy research.

11.2 Differentiate between efficacy and effectiveness psychotherapy research.

11.3 Summarize the findings of psychotherapy efficacy studies.

11.4 Summarize the findings of psychotherapy effectiveness studies.

11.5 Identify common factors that contribute to the success of psychotherapy.

11.6 List the most common types of psychotherapy practiced by clinical psychologists.

11.7 Distinguish between eclectic and integrative approaches to psychotherapy.

forms. Through the mid-1900s, most of the evidence offered in support of psychotherapy came in the form of anecdotes, testimonials, and case studies—essentially, subjective descriptions of individual clients' progress, usually written by the therapists themselves. Controlled, empirical studies of therapy outcome didn't appear regularly until the 1950s (Weissmark & Giacomo, 1998). **Hans Eysenck** (1952) published a historic study during this early period. Its claims were noteworthy and controversial: After reviewing some of the early empirical studies on psychotherapy outcome, Eysenck concluded that most clients got better without therapy and that, in general, psychotherapy was of little benefit. His scientific methods have since been criticized and his claims overturned, but Eysenck's allegation that therapy doesn't work inspired thousands of subsequent empirical studies on therapy outcome (Routh, 2011; Wampold, 2010b).

Many of those empirical studies on therapy outcome were conducted in the 1960s and 1970s. By the late 1970s and 1980s, meta-analyses began to appear. A *meta-analysis* statistically combines the results of many separate studies—in some cases, hundreds—to create numerical representations of the effects of psychotherapy as tested across massive numbers of settings, therapists, and clients. As we see in this chapter, these meta-analyses have yielded consistently positive results about how well psychotherapy works, so Eysenck's negative conclusion is now just a distant (and inaccurate) memory. But before exploring those results, let's consider just some of the methodological issues involved in a simple investigation of psychotherapy outcome.

Whom, When, and How Should Researchers Ask?

Designing and running an empirical study to measure the outcome of psychotherapy can be a complex task for researchers. One of the fundamental questions they must answer involves the fact that different people involved in the therapy may have different vantage points on the results of the therapy (Holmqvist, 2016). Whose opinion should researchers seek?

Hans Strupp, a legendary and pioneering psychotherapy researcher, identified three parties who have a stake in how well therapy works and who may have different opinions about what constitutes a successful therapy outcome. He and his colleagues labeled their theory the **tripartite model** (with *tripartite* literally meaning "three parties"; Stricker, 2012; Strupp, 1996; Strupp & Hadley, 1977; Strupp, Hadley, & Gomes-Schwartz, 1977). One party, of course, is the client. After all, clients are the ones whose lives are affected by therapy, and improving their lives in some meaningful way is presumably the focus of the therapy. Clients' opinions about therapy outcome are extremely valuable, but they can also be extremely biased. Some clients may be overly eager to see positive results, especially after investing significant time and money, and therefore overestimate therapy's

benefits. Other clients' opinions about psychotherapy may be negatively influenced by the very factors that brought them to therapy in the first place—for example, a depressed client who tends to interpret events in an unrealistically negative way may apply that kind of distorted thinking to his or her therapy. So a researcher may choose to turn to the therapist—the second party—as another source of feedback. The therapist typically has more experience in mental health issues than the client and may therefore have more reasonable expectations. However, therapists' views can be biased as well. Therapists witness only a fraction of clients' lives, and they may feel that negative evaluations reflect poorly on their own therapeutic skills. The third party identified by Strupp and his colleagues, society, can take the form of any outsider to the therapy process who has an interest in how therapy progresses. This can include the general public, the legal system, clients' family and friends, clients' employers, and, especially today, health insurance or managed-care companies that pay the psychotherapy bill. These third parties tend to bring a perspective that emphasizes the client's ability to perform expected duties in a stable, predictable, unproblematic way. Certainly, the question of whom to ask is crucial to the process of measuring therapy outcome. Thankfully, researchers need not choose a single perspective exclusively; they can and often do solicit multiple opinions.

Regardless of *whom* the researchers ask about therapy outcome, *when* should they ask? Perhaps the obvious answer is immediately after therapy ends. At that point, it would be reasonable to expect some improvement from the client's initial level of functioning. But how long should those benefits last? Is it reasonable to expect that therapy's benefits would continue to be evident 1 month, 6 months, 2 years, or 5 years down the road? And what about benefits before therapy ends? Should there be some evidence of improvement at various points in therapy or perhaps even at each session? Again, shrewd researchers may choose multiple answers to the question of *when*, either within or across psychotherapy studies. But it is evident that the time at which the researchers answer this question may influence the results they see (M. J. Lambert, 2011).

Finally, researchers have many options regarding *how* they measure the outcome of psychotherapy. If they choose to solicit opinions of an interested party, they can use questionnaires or interviews. Of course, the content and structure of these questionnaires or interviews will have an impact on the data they yield. Researchers may elect to use a more behavioral measure of therapy outcome instead. As an example, consider a researcher who seeks to determine the outcome of psychotherapy for a boy treated for attention-deficit/hyperactivity disorder. Rather than asking the boy, his therapist, his parents, or his teachers, a researcher could directly observe the boy at school or at home to determine if his behavior has changed since therapy began. As with the questions of *who* and *when*, the researcher's decision regarding *how* psychotherapy outcome is measured can shape the results of the investigation (Horvath, 2013; M. J. Lambert, 2013; Stiles, 2013; Wampold, 2013).

EFFICACY VERSUS EFFECTIVENESS OF PSYCHOTHERAPY

Empirical studies of psychotherapy outcome generally fall into two categories. The extent to which psychotherapy works in a controlled research study—that is, "in the

BOX 11.1

Considering Culture

Culture-Specific Expectations About Psychotherapy

Our discussion about psychotherapy outcome, especially from the client's perspective, must take into consideration the fact that clients from diverse cultures often hold widely varying expectations about the psychotherapy process. A culturally competent therapist will be able to attain a more successful psychotherapy outcome by appreciating the presumptions related to each client's cultural background.

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Photo 11.1
Individuals from diverse cultural backgrounds come to psychotherapy with very different expectations about the behaviors of clients and therapists.

As it is traditionally practiced in North America and the Western world, psychotherapy involves verbally discussing one's problems, focusing attention on them, and gaining greater understanding of or control over them. These basic processes clash

with cultural values common to some non-Western cultures whose members often prefer to "conceal" rather than "reveal" (Fontes, 2008; Toukmanian & Brouwers, 1998). Individuals from Eastern cultures often prefer to avoid, rather than confront, negative thoughts. Thus, the communication styles of clients with psychotherapists—in simplest terms, the extent to which they are comfortable self-disclosing about problematic behaviors, thoughts, and feelings—may depend significantly on their cultural values.

A client's willingness to self-disclose to a psychotherapist may also depend on the individualistic versus collectivistic nature of a client's culture. Western cultures tend to foster an individualistic way of life, emphasizing self-reliance and self-determination. In contrast, Eastern cultures tend to encourage collectivism, whereby the needs of the family/group and the relationships that bind the family/group take priority over individual goals. Thus, whereas a client of European descent may feel slightly uncomfortable discussing a personal problem or shortcoming, a client of Asian descent may feel a much stronger sense of shame about a similar problem or shortcoming because of his or her perception that the problem reflects on family members as well (D. W. Sue & Sue, 2008; Toukmanian & Brouwers, 1998).

Training psychotherapists to be culturally competent is the foundation on which they can build an appreciation of clients' expectations about psychotherapy. Like many others, Toukmanian and Brouwers (1998) recommend that training in psychotherapy emphasize the personal development of the therapist's attitude toward diverse cultures.

This training should encompass four levels: self-understanding (of one's own cultural values), listening to clients' cultural values, accepting clients' cultural values, and understanding clients' cultural values.

If you were the client, how important would it be for your psychotherapist to appreciate your expectations about psychotherapy that relate to your cultural background? Specifically, which values would be most important to recognize?

lab"—refers to its **efficacy**. Most recent studies of psychotherapy outcome are efficacy studies. They maximize internal validity—that is, the ability to draw conclusions about the cause–effect relationship between therapy and outcome—by controlling as many aspects of therapy as possible. Efficacy studies typically feature well-defined groups of patients, usually meeting diagnostic criteria for a chosen disorder (but no comorbid disorders); manualized treatment guidelines to minimize variability among therapists; and random assignment of participants to control and treatment groups (Jobe-Shields, Costello, Jackson, & Hanson, 2016; Nathan & Gorman, 2002; Spokas et al., 2008; Truax & Thomas, 2003).

In contrast, the extent to which psychotherapy works with actual clients treated by therapists in clinics, private practices, hospitals, and counseling centers—that is, "in the real world"—refers to its **effectiveness**. Effectiveness studies tend to include a wider range of clients, including those with complex and overlapping diagnostic profiles, or whose symptoms don't qualify for any specific disorder; allow for greater variability between therapists' methods than a manual would allow; and may or may not include a

control group for comparison to a treatment group. Effectiveness studies lack the internal validity of efficacy studies, because the researchers control and manipulate fewer variables. However, effectiveness studies typically have greater external validity than efficacy studies because their methods better match therapy that actually takes place in real clinical settings (Jobe-Shields et al., 2016; Nathan & Gorman, 2002; Spokas et al., 2008; Truax & Thomas, 2003).

Results of Efficacy Studies

Thousands of efficacy studies of psychotherapy have accumulated in recent decades; in fact, reviews and meta-analyses that serve as summaries of individual studies now number in the hundreds as well. Again and again, these research efforts yield the same affirmative conclusion: Psychotherapy works (L. F. Campbell, Norcross, Vasquez, & Kaslow, 2013; M. J. Lambert, 2011; Wampold, 2010a; Weiner, 2016). For example, a primary finding of a landmark meta-analysis of 475 psychotherapy efficacy studies was that the average effect size for psychotherapy was .85, indicating that "the average person who receives therapy is better off at the end of it than 80 percent of the persons who do not" (M. L. Smith et al., 1980, p. 87). More recent reviews and meta-analyses of therapy efficacy studies (e.g., Lambert & Simon, 2008; D. A. Shapiro & Shapiro, 1982), including some very large-scale mega-reviews of meta-analyses (e.g., Lipsey & Wilson, 1993; Luborsky et al., 2002) have confirmed these findings. Summarizing their recent comprehensive review of psychotherapy efficacy data, M. J. Lambert and Ogles (2004) stated, "The pervasive theme of this large body of psychotherapy research must remain the same—psychotherapy is beneficial. This consistent finding across thousands of studies and hundreds of meta-analyses is seemingly undebatable" (p. 148).

Not only does psychotherapy work, its benefits appear to endure over long periods of time, exceed placebo effects, and represent clinically (not just statistically) significant change in clients' well-being (e.g., M. J. Lambert & Ogles, 2004; M. L. Smith et al., 1980). It should be noted that psychotherapy is not a panacea—a small minority of therapy clients do appear to worsen during the therapy process (Striano, 1988; Strupp et al., 1977). On a similar note, some clients drop out of therapy prematurely, and others experience only short-lived benefits (Lebow, 2006). However, these negative effects clearly appear to be the exception rather than the rule.

In recent decades, efficacy studies have typically focused on how well a particular therapy works for a particular disorder. Collectively, those studies often produce overlapping results, such that very similar therapies work for different disorders (and those different disorders often have high rates of comorbidity). As a result, there has been a movement in recent years toward a *transdiagnostic approach* to efficacy studies, led by David Barlow and his colleagues (Barlow, Allen, & Choate, 2004; Moses & Barlow, 2006). According to the transdiagnostic approach, what needs treatment are not the superficial symptoms of a particular disorder, but the underlying pathology that causes those symptoms and the symptoms of related disorders. The main target of the transdiagnostic approach is the emotional disorders, a loose term intended to cover the disorders characterized by sadness (e.g., major depression) and anxiety (e.g., generalized anxiety disorder). These often-coexisting disorders share a core of negative affect or neuroticism, so it makes sense to target in therapy change of that core rather than change of

symptoms that manifest from it (Boswell, Thompson-Hollands, Farchione, & Barlow, 2013; Hong & Cheung, 2015). The *unified protocol* that Barlow and his colleagues recommend for this core of negative affect or neuroticism focuses on three basic elements: reducing the amount of negative thought in which clients engage, especially when they anticipate upcoming events; preventing unhealthy ways of avoiding unpleasant feelings; and encouraging behaviors that facilitate positive rather than negative feelings (Morris & Franklin, 2015). Data for the efficacy of the unified protocol for the emotional disorders is strong and growing (Bullis et al., 2015; Ellard, Fairholme, Boisseau, Farchione, & Barlow, 2010; Farchione et al., 2012; Ito et al., 2016).

Bridging the Gap Between Research and Practice

In spite of the overwhelming evidence offered by efficacy studies, their results are not always heeded by those who practice therapy in the real world, especially when their results point to a specific kind of therapy for a specific disorder (Boswell et al., 2011; Safran, Abreu, Ogilvie, & DeMaria, 2011; Stewart et al., 2018). There is an undeniable and unfortunate gap between those who conduct efficacy research on psychotherapy and those who practice it (Lilienfeld, Ritschel, Lynn, Cautin, & Latzman, 2015; Reese, Rosenfield, & Wilhelm, 2013; Teachman et al., 2012; Wolfe, 2012). Some might say the word *gap* is an understatement:

> In many ways, science and practice are two continents separated by a vast stretch of water. . . . Many books and articles discuss how to bridge the gap, and I expect the sheer volume and weight of these could serve as a physical bridge between the two metaphorical continents. (Kazdin, 2014, p. 6)

Depending on which side you ask, the gap is caused either by researchers who conduct studies that are artificial, contrived, and irrelevant to how clinicians really practice in their clinics, hospitals, and offices, or by practitioners who refuse to open their eyes to scientific data regarding the outcome of various therapies and who resist changing their practices accordingly. Regardless of the reason, this gap is problematic for the field of clinical psychology. Fortunately, efforts to recognize and bridge it have increased in recent years (e.g., Castonguay & Muran, 2015; Dattilio, Edwards, & Fishman, 2010; Littell, 2010).

One such effort focuses on attempts to understand how much reluctance therapists actually have to practicing according to efficacy research and specifically why that reluctance exists. To the question of how much, the answer is a lot (Stewart & Chambless, 2007). One survey of over 700 therapists asked what factors influenced their clinical decisions. It found that research such as efficacy studies had little influence over their choices, but their own clinical experience had a lot. This finding was true even for the many therapists who tend to use approaches consistent with therapies supported by many efficacy studies (i.e., cognitive and behavioral approaches; Gyani, Shafran, Myles, & Rose, 2014). Even clinicians who are themselves psychotherapy researchers don't always rank empirical research at the top of their list of factors by which to make clinical decisions (Safran et al., 2011). To the question of why, there are many answers. In some cases, therapists

simply place a higher value on their own intuition and judgments than on any data collected more methodically and scientifically, as in efficacy studies (Lilienfeld et al., 2013). In other cases, therapists see the research as irrelevant to their day-to-day practice: The clients in the studies differ in some important way from the clients they see in their own practices; the manuals limit their professional autonomy; the methodology of efficacy studies favors therapies with more measurable or "testable" outcomes, like behavioral and cognitive approaches, over other kinds of therapies; the manuals ignore or deemphasize the therapeutic alliance; and more (Herbert, 2015).

The reluctance of practitioners to adopt treatments supported by efficacy studies has created a significant challenge in terms of *dissemination* of these treatments. Indeed, dissemination of evidence-based treatments has become a major focus of researchers of psychotherapy outcome (Beidas & Kendall, 2010; Godley, Garner, Smith, Meyers, & Godley, 2011; McCloskey, 2011; Weisz, Ng, & Bearman, 2014). Larger organizations, including statewide professional associations, have also made efforts to increase dissemination within their territory (Hoagwood et al., 2014; Nakamura et al., 2014; Starin et al., 2014). Numerous strategies have been employed, with varying results. Stewart and Chambless (2010), for example, found that psychologists in private practice find case studies (i.e., detailed descriptions of a single client successfully treated) more persuasive than research reviews (i.e., presentations of data indicating that the treatment works). Stewart, Chambless, and Baron (2012) found that psychologists in private practice were more willing to undergo training in an empirically supported treatment if the training was brief (i.e., 3 hours) and inexpensive, as opposed to lengthy (i.e., 1 to 3 days) and expensive. Jensen-Doss, Hawley, Lopez, and Osterberg (2009) found that an organization that forces its clinicians to use evidence-based treatments, especially without any attempt to get clinicians "on board" first, can encounter significant resistance and resentment. S. J. Becker (2015) proposed that direct-to-consumer marketing (like when a pharmaceutical company advertises a medication directly to patients on television or magazine ads) could help dissemination efforts—that way, therapists would feel not only a "push" from researchers, but a "pull" from clients who have learned about what works for a particular disorder.

One dissemination strategy that has received a lot of attention and increasing collaboration between researchers and practitioners is a strategy often called *practice-oriented research* (Castonguay, Youn, Xiao, Muran, & Barber, 2015; Koerner & Castonguay, 2015). If researchers and therapists could feel like partners throughout the process—from the inception of a research idea to its execution to its conclusions—therapists would be more likely to accept and apply the results (Chambless, 2014; Ollendick, 2014; Weisz, Krumholz, Santucci, Thomassin, & Ng, 2015). *Practice-research networks* are one manifestation of this strategy, in which groups of therapists and researchers work as teammates in all phases of a research study (Castonguay & Muran, 2015; Youn, McAleavey, & Castonguay, 2015). In such an arrangement, researchers could ask practitioners how they should design their studies and, more generally, what kinds of challenges they face in their day-to-day practice (Goldfried et al., 2014). Researchers could also stay involved for the long haul—rather than just developing a treatment and handing it off to practitioners, they could train the practitioners and maintain collaboration with them as they implement the treatment and provide feedback about how well it works in their practices (Edmunds, Beidas, & Kendall, 2013).

Results of Effectiveness Studies

Effectiveness studies have not been conducted as frequently as efficacy studies, but those that have been conducted have generated similarly positive results. So, whereas efficacy studies indicate that psychotherapy works when tested in controlled settings, effectiveness studies indicate that psychotherapy works as it is commonly applied in realistic settings.

As an example of an effectiveness study, consider the investigation conducted by *Consumer Reports* magazine in 1995 (Dalrymple, Gaudiano, & Weinstock, 2015; "Mental Health," 1995; Seligman, 1995). The popular magazine—the same one that surveys its subscribers about their experiences with cars, DVD players, and laundry detergent—surveyed its many subscribers about their experiences with psychotherapy. The primary finding was that for the vast majority of respondents, psychotherapy had very positive, lasting effects. As stated by Seligman (1995),

> There were a number of clear-cut results, among them:
>
> Treatment by a mental health professional usually worked. Most respondents got a lot better.
>
> Averaged over all mental health professionals, of the 426 people who were feeling *very poor* when they began therapy, 87% were feeling *very good*, *good*, or at least *so-so* by the time of the survey. Of the 786 people who were feeling *fairly poor* at the outset, 92% were feeling *very good*, *good*, or at least *so-so* by the time of the survey. These findings converge with meta-analyses of efficacy (Lipsey & Wilson, 1993; Shapiro & Shapiro, 1982; Smith et al., 1980). (p. 968)

Like any effectiveness study, the *Consumer Reports* study is constrained by some troubling methodological questions (Seligman, 1995). For example, was there a sampling bias such that those whose therapy experience was successful were most likely to respond? Of the many clients who improved, how many would have improved without psychotherapy (i.e., in a control group)? How reliable and valid are clients' own self-reports about psychotherapy outcome, especially after months or years have elapsed? Nonetheless, effectiveness studies such as this one complement efficacy studies, and together they strongly support the benefits of psychotherapy. Two meta-analyses of effectiveness studies for cognitive-behavioral therapy—one combining the results of 71 effectiveness studies for anxiety disorders and another combining the results of 34 effectiveness studies for depression—found that the treatment was highly effective in both cases and fell only slightly below the levels of symptom relief found in efficacy studies (Hans & Hiller, 2013a, 2013b).

Alternate Ways to Measure Psychotherapy Outcome

Efficacy and effectiveness studies have been two primary ways of assessing how well therapy has worked, but there are others, both direct and indirect. For example, in recent years, researchers have paid increased attention to the neurobiological effects of

psychotherapy. Although this field of study is in the early stages, the initial conclusion is unmistakable: Psychotherapy changes the brain (Arden & Linford, 2009; Viamontes & Beitman, 2009). Studies using functional magnetic resonance imaging (fMRI) and positron emission tomography (PET) neuroimaging technologies have repeatedly found that successful therapy for particular disorders produces reliable changes in brain activity and structure. Among other findings, behavioral therapy for obsessive-compulsive disorder has been found to decrease metabolism in the caudate nucleus, behavior activation affects the dorsal striatum of depressed clients, and both cognitive-behavioral therapy and interpersonal psychotherapy have been found to decrease activity in dorsal frontal regions and increase activity in ventral frontal and subcortical regions (Dichter et al., 2009; Roffman, Marci, Glick, Dougherty, & Rauch, 2005). (All these types of therapy are described in more detail in later chapters.) Although medication studies have focused on neurological changes in the brain, such studies are relatively new for psychotherapy studies. As they accumulate, they will provide even further evidence of the effects of psychotherapy.

Another intriguing, but indirect, way to assess the outcome of psychotherapy is medical cost offset. The logic goes like this: Many people with mental or emotional problems will either seek medical (rather than psychological) assistance or put off treatment for so long that the problem worsens the individual's physical state, necessitating medical treatment. If they seek psychotherapy, will they reduce their medical costs? According to the available data, the answer appears to be a resounding yes (M. J. Lambert & Ogles, 2004). A review of more than 90 studies on this issue found that, on average, clients receiving therapy spent fewer days in the hospital and saw their medical costs reduced by 15.7%, while comparable clients in control groups spent more days in the hospital and saw their medical costs increased by 12.3% (Chiles, Lambert, & Hatch, 1999). Additionally, psychotherapy has been found to reduce the need for emergency room visits (Carr, 2008). Whether the medical bills are being paid by a health insurance company or by the clients themselves, the finding that psychotherapy is a good investment in both mental and physical health again points to its positive effects.

WHICH TYPE OF PSYCHOTHERAPY IS BEST?

Soon after the finding that psychotherapy works started to become an established fact, infighting began among the various orientations and approaches about which had the strongest empirical support (M. J. Lambert, 2011). Each claimed superiority over the others, and as the language in the following excerpt implies, the competition was quite fierce:

> Although there were many combatants—Freudians versus cognitivists versus humanists—the principals in this war were behaviorists and nonbehaviorists. These groups have called each other names and traded high sounding insults. But the issue was not over psychotherapy versus no psychotherapy, but brand A psychotherapy versus brand B psychotherapy. . . . Different forms of therapy

were viewed as adversaries, competitors, or contestants, and the arena of conflict was the controlled experiment. (M. L. Smith et al., 1980, pp. 2–3)

The "Dodo Bird Verdict" and Common Factors

Indeed, many empirical outcome studies throughout the latter half of the 1900s have pitted one form of therapy against another. The collective results of these studies have, again and again, yielded a result that surprised many in the field: a virtual tie. In other words, in the hundreds of empirical studies designed to compare the efficacy of one form of therapy with the efficacy of another, the typical result is that the competing therapies are found to work about equally well (M. J. Lambert, 2011; M. J. Lambert & Ogles, 2004; Norcross & Newman, 1992; M. L. Smith et al., 1980; Wampold, 2001, 2010a, 2010b; Weissmark & Giacomo, 1998). In one of the earliest review articles to reach this conclusion, the authors borrowed a line from the dodo bird in *Alice in Wonderland* who, after judging a race between many competitors, stated that "everybody has won and all must have prizes" (Luborsky, Singer, & Luborsky, 1975, p. 995).

How could the **"dodo bird verdict"** apply to psychotherapy outcome? The various forms of psychotherapy—psychoanalysis, humanism, cognitive, behavioral, and others—are indeed quite discrepant from one another, so how could they consistently produce such similar results? Most researchers explain this finding by pointing to **common factors** across all forms of psychotherapy (e.g., Huibers & Cuijpers, 2015; Stricker, 2010; Wampold, 2001). That is, although proponents of each school of therapy tout the unique and distinctive aspects of their own approaches, they all share some fundamental components as well. Actually, the notion that different therapies benefit from the same underlying mechanisms was suggested as early as the 1930s (Rosenzweig, 1936) and has been reiterated numerous times since then (e.g., Frank, 1961; Torrey, 1986). The difference in the most recent versions of this argument is that they are supported by extensive empirical data on psychotherapy outcome.

It is important to recognize that these common factors are not merely present; they are therapeutic (M. J. Lambert & Ogles, 2004; Wampold, 2010a, 2010b). They function as "active ingredients" in all forms of psychotherapy, which helps explain the comparable results of the various approaches: "All of the specific types of therapy achieve virtually equal—or insignificantly different—benefits because of a common core of curative processes" (Glass, 2001, p. ix). So what are the common factors that make up this common core of curative processes?

Therapeutic Relationship/Alliance

Of the many common factors for psychotherapy outcome that have been proposed, the one with the most evidence behind it is a strong relationship between therapist and client (Beitman & Manring, 2009; Crits-Christoph, Gibbons, Hamilton, Ring-Kurtz, & Gallop, 2011; Horvath, Del Re, Flückiger, & Symonds, 2011; Lebow, 2006; Norcross & Lambert, 2011a, 2011b, 2014). This relationship goes by multiple names: **therapeutic relationship**, **therapeutic alliance**, or **working alliance**. The word *alliance* is perhaps the most illustrative of the nature of this relationship—a coalition, a

partnership between two allies working in a trusting relationship toward a mutual goal. Research unequivocally indicates that whether the clients are adults, children, or families, and whether the therapy format is individual or group, the quality of the therapeutic relationship strongly contributes to psychotherapy outcome (Flückiger, Del Re, Wampold, Symonds, & Horvath, 2012; Horvath, 2015; Norcross & Wampold, 2011a, 2011b).

In fact, numerous studies have concluded that the therapeutic relationship is perhaps the most crucial single aspect of therapy. Specifically, researchers have argued that the quality of the therapeutic relationship is the best predictor of therapy outcome and that it accounts for more variability in therapy outcome than do the techniques specific to any given therapy approach (e.g., Beitman & Manring, 2009; Wampold, 2001, 2010b). The strength of the therapeutic relationship is especially important from the client's point of view; after all, the client's perception of this relationship is what facilitates positive change. Numerous surveys of clients and potential clients have found that they prefer a warm therapist with whom they can relate to a treatment with empirical support (Swan & Heesacker, 2013; Swift & Callahan, 2010).

It is also interesting to note that the quality of the therapeutic relationship is vital to therapy regardless of how much emphasis the therapist places on it. Some therapists (e.g., behaviorists) tend to deemphasize therapy relationships, others tend to pay it moderate attention (e.g., cognitive therapists), and others tend to focus more heavily on it (e.g., humanists and psychoanalysts). But through the eyes of the client, the therapeutic relationship remains a consistently vital component of psychotherapy (B. C. Chu, Skriner, & Zandberg, 2014; Norcross & Lambert, 2011a, 2011b). One study split hundreds of chronically depressed adults into groups receiving either brief supportive therapy or cognitive-behavioral therapy. The correlation between the client's rating of the therapy alliance and the outcome of therapy was strong for both groups, but it was higher for the cognitive-behavioral therapy group, which theoretically emphasizes relationship variables less than brief supportive therapy does (Arnow et al., 2013).

Having established the centrality of the therapeutic alliance, researchers have begun to investigate how, exactly, it contributes to successful psychotherapy. Kazdin (2007) brings up an interesting point: The connection between a good therapeutic alliance and client improvement is not necessarily a one-way street. In other words, in addition to the likelihood that a good alliance facilitates client improvement, it is also possible that as clients improve, they experience an enhanced relationship with their therapists. (Just imagine how you would rate your alliance with a physician if that physician cured you of a serious disease.) Of course, a reciprocal relationship between the two—an upward spiral of sorts—is also a possibility (Marker, Comer, Abramova, & Kendall, 2013; C. A. Webb et al., 2011; Zilcha-Mano, Dinger, McCarthy, & Barber, 2014). This issue highlights what you may have learned in a statistics course: A correlation between two variables does not necessarily indicate a definitive cause-and-effect relationship.

Researchers have also begun to break down the therapeutic relationship—that is, to look more closely at its specific components to determine exactly what makes a therapeutic relationship beneficial (Horvath et al., 2011; Orlinsky, 2010). Numerous elements have been identified—most emphatically, the therapist's ability to provide empathy and acceptance to the client (M. J. Lambert, 2011; Norcross, 2010). (These elements overlap significantly with cornerstones of humanistic therapy, which we will discuss in detail in a later chapter.) Empirical study of the therapeutic relationship remains in the early stages,

but as it continues, it promises to guide therapists toward establishing healing interpersonal relationships with their clients, regardless of the clients' presenting problems or the particular techniques chosen by the therapists (Norcross & Wampold, 2011a, 2011b).

More broadly, researchers have also conducted empirical research on specific components of the psychotherapy relationship that contribute to successful therapy. Norcross and Lambert (2018) summarize the findings with a list of relationship components they label as *demonstrably effective*:

- The alliance between therapist and client
- A sense of collaboration
- Goal consensus—the client and therapist agreeing on the purpose of therapy
- Cohesion—for group therapy, this is the equivalent of the alliance, but to the whole group
- Empathy
- Positive regard and affirmation
- Providing feedback to clients

So, in addition to empirical data supporting particular techniques that work, clinical psychologists also have data supporting particular kinds of therapeutic relationships that work. Rather than siding with either the techniques or the relationships, clinical psychologists can embrace both. In other words, these two components of therapy can complement each other to create the most beneficial experience for clients. As Castonguay and Beutler (2006) put it,

> One of the most salient controversies in the field of psychotherapy is whether client change is primarily due to the therapist's techniques or the quality of the therapeutic relationship. . . . This controversy reflects, more or less implicitly, an "either/or" assumption that is conceptually flawed and empirically untenable. The complexity of the process of change requires, at least in our view, a consideration of both technical and interpersonal factors. (p. 353)

One last note about the therapeutic relationship/alliance—researchers have also begun to examine the extent to which graduate programs teach their students about it. One survey of 84 doctoral programs found that most programs recognize how important the topic is, and may touch on the topic in some classes, but do not offer "systemic, evidence-based alliance training into their programs, despite believing that such systematic elements would contribute to ideal alliance-training practices" (Constantino, Morrison, Coyne, & Howard, 2017, p. 219).

Other Common Factors

The therapeutic relationship is not the only common factor that psychotherapy researchers have proposed. **Hope** (or positive expectations) has also received support as a common

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Photo 11.2
The relationship or "alliance" between therapist and client is a common factor for therapeutic success.

factor (Alavi & Sanderson, 2015; Constantino, Glass, Arnkoff, Ametrano, & Smith, 2011). Simply stated, therapists of all kinds provide hope or an optimism that things will begin to improve. Although the mechanism by which this improvement will take place may differ, the improvement may actually begin before any techniques, per se, have been applied. Anyone who has walked despairingly into a physician's office, or even a car repair shop, and received a confidently delivered message that the problem can be fixed understands the curative power of hope.

Attention may also be a common factor across psychotherapies. Also known as the Hawthorne effect (a name derived from classic organizational psychology studies in which factory workers' performance improved as a result of being observed), the attention the therapist and client direct toward the client's issues may represent a novel approach to the problem. That is, clients may have previously attempted to ignore problems that are ultimately addressed in therapy. Simply by openly acknowledging a problem and focusing on it with the therapist, a client may begin to experience improvement, even before formal intervention begins. In addition to the therapeutic relationship, hope, and attention, other common factors that researchers have proposed include reinforcement of novel behaviors, desensitization to threatening stimuli, confronting a problem, and skill training.

Is it possible that the common factors underlying all forms of psychotherapy occur in a predictable sequence? M. J. Lambert and Ogles (2004) put forth a **three-stage sequential model of common factors**, beginning with the "support factors" stage—common factors such as a strong therapist–client relationship, therapist warmth and acceptance, and trust. They label the second stage "learning factors," including such aspects as changing expectations about oneself, changes in thought patterns, corrective emotional experiences, and new insights. The third and final stage consists of "action factors," such as taking risks, facing fears, practicing and mastering new behaviors, and working through problems. In brief, this sequential model suggests that psychotherapists of all kinds help clients by moving them through three common steps: connecting with them and understanding their problems; facilitating change in their beliefs and attitudes about their problems; and, finally, encouraging new and more productive behaviors.

It is interesting to compare this three-step sequence with the more informal process of helping a friend who comes to you with a personal problem. Typically, we begin by communicating understanding and compassion, then move on to help them see their problems in a new light, and ultimately help them develop a strategy and take new action to address it. If we skip any of these steps or do them "out of order," the helping process may be hindered. By the same token, some friends (or clients) may need more time at certain stages and less time at others—perhaps more support and less action or vice versa. And cultural factors can play an important role in the value of each stage to the person seeking help, as members of some groups may tend to favor support, learning, or action.

BOX 11.2

Metaphorically Speaking

If You Use Toothpaste, You Understand Common Factors in Psychotherapy

Toothpaste companies spend a lot of time and money convincing us that, because of some "special" feature, their product is the best. Crest, Colgate, Aim, Gleem, Aquafresh—all make claims that they have something unique that sets their toothpaste apart. You've probably tried a few yourself: toothpastes with baking soda, with mouthwash, with sparkles, in a stand-up tube, in winter-fresh gel, and so on.

Ever read the list of ingredients on one of those tubes of toothpaste? Beneath the full list of ingredients, you'll see a separate, important category: "Active Ingredient." And that category has only one item listed: fluoride. That's true across all the brands, all the varieties. In other words, although the manufacturers and advertisers try to sell us on the unique features—extra ingredients, special flavors—what makes one toothpaste work is the same thing that makes its competitors work: fluoride. (That's why our dentists rarely mention a specific brand when they remind us to brush—as long as it has fluoride, any brand will prevent cavities about as well as the others.)

Decades of outcome studies have suggested that the same type of phenomenon has taken place in the psychotherapy field. Each "brand" of psychotherapy has promoted its unique features, those aspects that distinguish it and supposedly make it better than the other brands. But those claims are contradicted by the consistent result of many controlled, empirical psychotherapy outcome studies: Different forms of psychotherapy work about equally. Consequently, it makes sense to speculate about the underlying common ingredient—the "fluoride" of psychotherapy.

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So what is the "fluoride" of psychotherapy? At this point, the therapeutic relationship/alliance has emerged as the leading candidate. That is, a strong relationship between therapist and client has proven beneficial, regardless of whether the therapist in that relationship uses psychoanalytic, humanistic, cognitive, or behavioral techniques. Other common factors have garnered significant attention and support from psychotherapy researchers as well, including hope/optimism, attention, and insight (Wampold, Imel, Bhati, & Johnson-Jennings, 2007). Perhaps various forms of psychotherapy share not a single ingredient (like toothpastes share fluoride) but a "common core" of ingredients, in which some of the factors described here combine to help a client (Wampold, 2001, p. ix).

The fact that fluoride is the common active ingredient in all toothpastes doesn't make us indifferent about selecting one for ourselves. Just as you wouldn't choose a toothpaste that costs too much or tastes bad to you, you wouldn't choose a therapy that you found unjustifiably expensive or simply not palatable. Instead, with either toothpaste or therapy, your choice reflects your personal values and preferences about the means by which the active ingredients are delivered.

Reconsidering the Dodo Bird Verdict: Specific Treatments for Specific Disorders

Web Link 11.1
Dianne Chambless

The dodo bird verdict has not gone unchallenged. Although it is a widely accepted finding in the field that the various forms of psychotherapy are, in general, equally effective, some researchers have made the case that certain psychotherapies are, in fact, superior to others in the treatment of specific problems (e.g., Antony & Barlow, 2010; Carr, 2008; Chambless & Ollendick, 2001; Weiner, 2016). In fact, the existence and popularity of lists, books, and websites that offer pairings of "what works" between specific disorders and specific therapies suggests that for many, the assumption is that therapies are *not* equally effective.

Dianne Chambless, a prolific and highly respected psychotherapy researcher, has argued strongly against the idea that all psychotherapy approaches are equally efficacious (e.g., Chambless & Ollendick, 2001; Siev & Chambless, 2007). For instance, in her 2002 article (fittingly titled "Beware the Dodo Bird: Dangers of Overgeneralization"), Chambless points out that although empirical studies have compared many therapies with one another, there are many specific comparisons—certain therapies for certain disorders—that studies have not yet examined. Thus, it would be premature to conclude that all therapies are equal for the treatment of all disorders, even if equal efficacy has been the typical finding in studies so far. Additionally, as described in Chapter 3, Chambless is a champion of the movement toward manualized, evidence-based treatments and has led the task forces that established criteria for efficacious treatments for specific disorders and determined which therapies made that list (Chambless et al., 1996, 1998; Task Force on Promotion and Dissemination of Psychological Procedures, 1995).

The contention made by Chambless and others that the dodo bird verdict is inaccurate—and that, therefore, common factors should take a back seat to specific ingredients in each therapy technique—has itself been countered by other leading psychotherapy researchers (K. Elliott, Barker, & Hunsley, 2015; Norcross, 2002). For example, in their 2002 article (aptly titled "Let's Face Facts: Common Factors Are More Potent Than Specific Therapy Ingredients"), **Stanley Messer** and **Bruce Wampold** review the literature on therapy efficacy and conclude that "the preponderance of evidence points to the widespread operation of common factors such as therapist-client alliance . . . in determining treatment outcome" (p. 21). They further argue that the **prescriptive approach** to therapy—in which specific therapy techniques are viewed as the treatment of choice for specific disorders (e.g., Antony & Barlow, 2010)—should be replaced by an approach that more broadly emphasizes common factors, especially the therapeutic relationship. Wampold and his colleagues have gone so far as to argue that a common-factors approach to therapy is an evidence-based approach, despite the fact that it might not stem directly from a therapy manual for any particular disorder (Laska, Gurman, & Wampold, 2014; Laska & Wampold, 2014).

Others are quick to add that the debate over what's best for the client should not overlook what the client wants (Swift & Callahan, 2009; Swift, Callahan, & Vollmer, 2011). In other words, *client preferences* are important to consider for retention (keeping the client from dropping out of therapy), enhancement of the therapy relationship, and, ultimately, outcome. As an example, consider the client's preferred coping style (Beutler, Harwood, Kimpara, Verdirame, & Blau, 2011; Tompkins, Swift, & Callahan, 2013). Some

BOX 11.3

In My Practice . . .

In my practice, clients have displayed a range of reactions to the prescriptive approach to therapy. I'm referring specifically to situations in which a client has a definitive diagnosis and I tell the client that there is a clear-cut, evidence-based treatment of choice for that diagnosis. In many cases, the client's reaction is extremely positive. Consider Matthew, a 30-year-old man with obsessive-compulsive disorder. When I told Matthew that exposure and response prevention for OCD had received a tremendous amount of support from therapy outcome studies, he was thrilled. Actually, he was thrilled not only that outcome studies supported a particular type of therapy for his disorder, but also with the fact that such studies even happened at all. (Quite a few of my clients have expressed surprise at this: "You mean they actually test specific kinds of therapies? I assumed each therapist just used his or her own style.") Matthew started therapy confident that he was engaging in a treatment with evidence behind it, rather than something we were simply taking our chances with.

In other cases, the reactions from my clients about a prescriptive approach to therapy has been less positive, at least initially. Consider Lauren, a 25-year-old woman with generalized anxiety disorder. When I told Lauren that cognitive therapy for GAD had received a tremendous amount of support from therapy outcome studies, she actually seemed to balk at the idea. She had two primary concerns. One was that the therapy would be chosen *for* her, not *by* her. The other was that she would be treated in a "cookie-cutter" or "one size fits all" way—in other words, that she would receive exactly the same manual-based therapy as any other client with the same diagnosis, rather than receiving a more personalized treatment. I assured Lauren that her preferences would absolutely play a role in the type of therapy she received, and that we would make treatment decisions together (an approach consistent with the three-part model of evidence-based practice that we discussed in Chapter 3, involving not only research evidence but client and therapist factors as well). I also assured her that the use of a particular kind of therapy did not preclude personalization—in fact, rather than adhering blindly or absolutely to any manual, I would be customizing the process for her throughout. These explanations helped. But my experience with Lauren (and Matthew, and many other clients) reminds me that clients often enter therapy with little information, or plenty of misinformation, about how therapy decisions actually get made. Explaining this process to clients in an up-front way, and including an explanation of a prescriptive approach to therapy or any other way in which therapy outcome research can influence therapy choices, has generally proven beneficial in my interactions with clients.

To see this box come to life as a whiteboard video, please visit edge.sagepub/pomerantz5e.

clients may have an externalizing coping style, so they prefer to deal with problems by presuming that their origins are outside of themselves; as such, they may respond better to a therapy that emphasizes symptoms but deemphasizes their root causes. Other clients may have a more internalizing coping style—presuming that their problems originate from within themselves—and would therefore respond better to a therapy that emphasizes insight into root causes rather than just symptom reduction. Appreciating this client

preference could be the key to therapy's success; ignoring it could be the cause of its failure. A meta-analysis of 34 empirical studies of the impact of client preference on therapy revealed a powerful effect on whether clients completed therapy, how satisfied they were with therapy, and how much therapy benefited them (Lindhiem, Bennett, Trentacosta, & McLear, 2014).

Web Link 11.2
Bruce Wampold

Web Link 11.3
Role of science in psychotherapy

The debate over the dodo bird verdict—whether the benefits of psychotherapy are due primarily to ingredients shared across therapies or specific to certain therapies—carries on. In fact, *The Great Psychotherapy Debate*, an important book by Bruce Wampold and Zac Imel (2015), suggests that the quarrel is far from over. In the middle ground of this dispute, numerous researchers and clinicians have tried to form a compromise that recognizes the importance of both the specific treatment and more general, common effects (e.g., Norcross, 2005). A meta-analysis of treatment outcome studies since 1997 found some support for both sides of the argument: For targeted outcomes (the specific symptoms that a particular treatment aims to reduce), the particular techniques of cognitive-behavioral therapy were better predictors, but for general quality of life (not specific to any particular symptom), common factors were better predictors (Marcus, O'Connell, Norris, & Sawaqdeh, 2014). Some have pointed out that change in therapy can be attributable to *many* factors, including common factors, specific factors, and numerous others. For example, some call attention to the fact that in addition to the specific and common factors cited above, there are client characteristics, therapist characteristics, and problem characteristics (how severe, how chronic, etc.) that can affect the outcome of therapy (Beitman & Manring, 2009; Rosenfeld, 2009). Not to mention, extratherapeutic forces—that is, life events the client experiences while in therapy—can also have a powerful impact on the client's well-being at the end of therapy (Norcross, 2005; Rosenfeld, 2009; Wampold, 2001). Perhaps a wise resolution of this debate is to consider the full range of potential influences on therapy outcome, as suggested by Paul (2007):

> The treatment method, the therapist, the relationship, the client, and principles of change are vital contributors, and all must be studied. Comprehensive evidence-based practices will consider *all* of these determinants and their optimal combinations. "Common and specific effects" and "art and science" appear properly complementary, not as "either/or" dichotomies. (p. 141)

WHAT TYPES OF PSYCHOTHERAPY DO CLINICAL PSYCHOLOGISTS PRACTICE?

The Past and Present

Seven times since 1960, researchers have surveyed the Division of Clinical Psychology (Division 12) of the American Psychological Association to assess, among other things, the type or orientation of psychotherapy that its members practice. The most recent of these surveys incorporates responses from 549 clinical psychologists and includes a comparative review of the six previous surveys as well (Norcross & Karpiak, 2012). These results are summarized in Table 11.1.

BOX 11.4

Considering Culture

Are Evidence-Based Treatments Appropriate for Diverse Clients?

A form of psychotherapy becomes an evidence-based treatment when empirical studies demonstrate that it produces successful results with clients. But what if the clients from the study differ in important ways from clients who might receive the treatment in the real world? Can we expect similarly positive results with all clients?

Empirical studies of psychotherapy have done a poor job of including diverse populations in their clinical trials, according to some recent criticisms. Specifically, a series of entries to an important book titled *Evidence-Based Practices in Mental Health: Debate and Dialogue on the Fundamental Questions* (Norcross, Beutler, & Levant, 2006) argue strongly that studies examining the efficacy of manualized therapies have commonly neglected issues of

- ethnicity (S. Sue & Zane, 2006),
- gender (Levant & Silverstein, 2006),
- disability (Olkin & Taliaferro, 2006), and
- LGBTQ clients (L. S. Brown, 2006).

Regarding ethnicity, S. Sue and Zane (2006) report that despite the massive number of clinical studies evaluating the efficacy of specific therapies since 1986 (encompassing about 10,000 clients), the number of these studies that measured the efficacy of the treatment according to ethnicity or race is zero. Further, in about half of these studies, ethnicity information was not reported at all, and in most of the rest, very few minority

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Photo 11.3
How should clinical psychologists adapt evidence-based treatments for clients of minority groups—lesbians, for example?

clients were included. Regarding people with disabilities, Olkin and Taliaferro (2006) report that they "have been unable to locate any published materials on [evidence-based practices] and people with disabilities," which "fuels our concern that [evidence-based practice] will develop without due consideration of this minority group" (pp. 353–354). L. S. Brown (2006) and Levant and Silverstein (2006) offer comparable summaries regarding inattention to LGBT and gender issues, respectively.

Not only are clients from these diverse groups often omitted from clinical trials, but the authors of some clinical studies (and the therapy manuals they test) do not suggest specific adaptations to their treatment to better suit any such clients. Fortunately, very recent research has focused on exactly these kinds of adaptations. That is, there is a current movement among psychotherapy outcome researchers to describe how evidence-based therapies could be adapted to specific cultural populations and to empirically test these adaptations (Castro et al., 2010; Morales

(Continued)

(Continued)

& Norcross, 2010; T. B. Smith et al., 2011). For example, BigFoot and Schmidt (2010) successfully tested an adaptation of cognitive-behavioral therapy for American Indian and Alaska Native children with posttraumatic symptoms, and Aguilera, Garza, and Muñoz (2010) successfully tested an adaptation of cognitive-behavioral therapy for Spanish-speaking clients with depression. There is also a growing number of *Clinical Practice Guidelines* and *Guidelines for Practitioners* (both of which fall under the broader category of *Professional Practice Guidelines*) published by APA that offer advice to clinicians working with members of specific diverse groups. These include guidelines for practice with clients who have particular disorders or problems (e.g., PTSD) or who are members of particular groups (e.g., lesbian, gay, and bisexual clients; older adult clients; transgender and gender nonconforming clients; DeAngelis, 2018).

For a moment, imagine that you are a psychotherapy client. Your clinical psychologist informs you that a particular form of therapy has been shown in a series of studies to successfully treat the disorder with which you have been diagnosed. How important is it to you that the clients on whom the therapy was successfully tested may differ from you in important ways? If most of the clients in the successful clinical trials differed from you in terms of ethnicity, gender, disability status, or sexuality, would you feel confident that the therapy would work equally well for you? If you were the clinical psychologist, how comfortable would you be following a manual empirically supported by research conducted on clients who differ from the client in your office?

Table 11.1 Percentage of Clinical Psychologists Endorsing Leading Primary Theoretical Orientations Since 1960

Orientation	1960	1973	1981	1986	1995	2003	2010
Eclectic/Integrative	36	55	31	29	27	29	22
Cognitive	—	2	6	13	24	28	31
Psychodynamic/ Psychoanalytic	35	16	30	21	18	15	18
Behavioral	8	10	14	16	13	10	15
Humanistic/ Rogerian/ Existential/Gestalt	6	7	7	12	4	2	2

Source: Adapted from Norcross & Karpiak (2012); Norcross, Karpiak, & Santoro (2005).

Several observations and trends related to Table 11.1 are noteworthy:

- Eclectic/integrative therapy was the most commonly endorsed orientation in every survey summarized in Table 11.1 until 2010, when it fell to second place. That is, over the past half century, more psychologists have described themselves as "mutts" who blend multiple approaches or use an assortment of therapies than as "purebreds" who practice one type exclusively. Additional research (e.g., Cook, Biyanova, Elhai, Schnurr, & Coyne, 2010) has confirmed that combined orientations remain extremely common, not only among clinical psychologists but among psychotherapists more broadly. Therapists combine orientations in a wide variety of ways, but surveys suggest that the combination of cognitive and behavioral approaches (each of which is described in detail in a later chapter) is the most common (Hickman, Arnkoff, Glass, & Schottenbauer, 2009).

- Cognitive therapy has witnessed a remarkable rise in popularity, especially since the 1980s. Prior to the 1980s, the unpopularity of the cognitive approach suggested it was hardly worth including on these surveys, but by 2010, it had become by far the most commonly endorsed single-school approach and had also overtaken the eclectic/integrative approach. In surveys since 2010 in which the combined "cognitive-behavioral" option is available, that orientation has been so dominant that some authors have speculated about the possibility of it overtaking the field (Heatherington et al., 2012; Jaimes, Larose-Hébert, & Moreau, 2015; K. N. Levy & Anderson, 2013).

- Even among those therapists who endorse a singular orientation rather than calling themselves eclectic, the actual techniques they use often fall outside the boundaries of their singular orientation. For example, a therapist who calls herself behavioral may in fact use cognitive techniques from time to time, or another therapist who calls himself humanistic may in fact use psychodynamic techniques occasionally (Thoma & Cecero, 2009).

- The endorsement of psychodynamic/psychoanalytic therapy has declined significantly since 1960, when it far exceeded any other single approach and rivaled the eclectic/integrative approach in terms of popularity. In 2010, only 18% of clinical psychologists endorsed it as their primary orientation—about half the percentage reported in 1960. Nonetheless, psychodynamic/psychoanalytic therapy remains the second most commonly endorsed orientation among the single-school approaches.

The same survey (Norcross & Karpiak, 2012) offers a description of the formats, or modalities, most commonly used by clinical psychologists. Table 11.2 presents highlights of these results for clinical psychologists responding to the most recent edition of the survey.

Table 11.2 Involvement of Clinical Psychologists in Various Psychotherapy Formats

Psychotherapy Format	Percentage of Clinical Psychologists Who Practice It
Individual	98
Couples/Marital	48
Family	34
Group	20

Source: Adapted from Norcross, Karpiak, & Santoro (2005).

As Table 11.2 indicates, psychotherapy with individual clients dominates the professional activities of contemporary clinical psychologists (see also Cook et al., 2010). Almost all (98%) conduct some individual therapy. However, the other formats of therapy—couples/marital therapy, family therapy, and group therapy—are also practiced by sizable numbers of clinical psychologists.

Regardless of the type of therapy, it has become increasingly evident that clients enter therapy at various points regarding willingness to change. Simply put, some clients are quite ready to change when they enter therapy, whereas others are not. Psychotherapy researchers (e.g., Norcross, Krebs, & Prochaska, 2011) have developed and provided empirical support for a **stages of change model** to describe the various points where clients may fall in terms of readiness. Specifically, clients may enter therapy at one of these five stages:

Web Link 11.4
John Norcross discussing stages of change

- *Precontemplation stage*—no intention to change at all. These clients are largely unaware of their problems, and they may have been pressured to enter therapy by family or friends who are more aware than they are themselves.
- *Contemplation stage*—aware that a problem exists, considering doing something to address it, but not ready to commit to any real effort in that direction. These clients are often ambivalent and are not yet willing to give up the benefits of the behavior they recognize as somewhat problematic.
- *Preparation stage*—intending to take action within a short time (e.g., weeks, a month). These clients may be taking small steps but have not made significant or drastic change.
- *Action stage*—actively changing behavior and making notable efforts to overcome their problems. More than any other stage, this stage requires sustained effort and commitment to the therapeutic goals.
- *Maintenance stage*—preventing relapse and retaining the gains made during the action stage. This stage lasts indefinitely.

It is important for therapists to assess the stage of change clients are in when they seek therapy. It would be a mistake to assume that most are at the action stage; indeed, data suggest that the vast majority are at an earlier stage at the outset of therapy. Thus, initial goals should emphasize increasing clients' readiness to change rather than forcing them to change prematurely (Norcross, Krebs, & Prochaska, 2011). A meta-analysis on studies of the connection between stages of change and therapy outcome produces a clear result: The farther along the continuum the client is, the better the outcome will be, so it is important to customize treatment by "meeting the client where they are" and progressing one stage at a time (Krebs, Norcross, Nicholson, & Prochaska, 2018).

The Future

The practice of psychotherapy among clinical psychologists has certainly changed in the past half century. How might it change in the near future? Seventy

psychotherapy experts, including many editors of leading journals in the field, were surveyed about the trends they foresee for the near future (Norcross, Pfund, & Prochaska, 2013). The results included several provocative predictions, including a rise in the use of

- mindfulness-based approaches to therapy,
- cognitive and behavioral approaches to therapy,
- multicultural approaches to therapy,
- eclectic/integrative approaches to therapy,
- therapies involving the use of the Internet and other forms of technology, and
- evidence-based practice.

The survey also suggested that classic psychoanalysis will continue to decline in use soon. Of course, only time will tell if these predictions are accurate.

Eclectic and Integrative Approaches

Eclectic and integrative approaches to psychotherapy hold a unique position among clinical psychologists, as we have seen. Whereas the various single-school forms of therapy have risen and fallen in popularity over the years, eclectic/integrative therapy has remained at or near the top of the list in terms of popularity (Norcross, Hogan, Koocher, & Maggio, 2017; Norcross & Karpiak, 2012).

Although the terms are often linked and both involve multiple therapy approaches, an eclectic approach to therapy actually differs in important ways from an integrative approach. **Eclectic** therapy (also known as *technical eclecticism*) involves selecting the best treatment for a given client based on empirical data from studies of the treatment of similar clients (J. R. Gold, 1996; Norcross & Newman, 1992; Stricker, 2010). In other words, a truly eclectic therapist turns to the empirical literature as soon as the diagnosis is made and practices whatever technique the literature prescribes for that diagnosis. So if the empirical literature dictates it, an eclectic therapist might practice cognitive therapy with a 9:00 a.m. client with generalized anxiety disorder, behavioral therapy with a 10:00 a.m. client with a phobia, and so on.

An **integrative** approach to therapy, on the other hand, involves blending techniques in order to create an entirely new, hybrid form of therapy (Beitman & Manring, 2009; Norcross, 2005; Ziv-Beiman & Shahar, 2015). An integrative therapist may combine elements of psychoanalytic, cognitive, behavioral, humanistic, or other therapies into a personal therapy style applied to a wide range of clients.

In 1977, Paul Wachtel was one of the first to successfully integrate complementary (some might have said incompatible) approaches, namely, psychoanalysis and behavior therapy. Soon after, especially in the 1980s, integrative therapy grew into a full-fledged

BOX 11.5

Metaphorically Speaking

If You Know the Difference Between a Fruit Salad and a Smoothie, You Understand the Difference Between Eclectic and Integrative Psychotherapists

A fruit salad includes a variety of ingredients, but each bite brings only one flavor: The fork may stab a strawberry first, a blueberry next, and a pineapple chunk third. Each piece is pure, discrete, and easily distinguished from the others. But in a smoothie made of these ingredients, every sip includes the same combination of ingredients, and the taste of every sip reflects that unique blend. Mixed together, the ingredients create a distinct concoction with a taste wholly its own.

An eclectic approach to psychotherapy is a lot like a fruit salad. Eclectic therapists use a pure, discrete approach to therapy with each client, and they choose that approach according to empirical support. They allow empirical studies of psychotherapy efficacy to direct them toward the therapy most likely to succeed for a particular diagnosis. As described in Chapter 3, recent publications have supplied therapists with lists of the most empirically supported treatments for particular disorders (e.g., Chambless et al., 1998). An eclectic therapist would refer to such a list for each client separately. Because such lists prescribe very different forms of therapy for various disorders, eclectic therapists must be versatile enough to practice many techniques competently. Eclectic therapists have no loyalty to any particular approach to therapy; their loyalty is to the empirical data.

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Integrative therapy, in contrast, is more like a smoothie—a custom blend of ingredients that forms an original creation. This "hybrid" approach to therapy is often used across clients and across diagnoses. Thus, integrative therapists are less concerned with employing evidence-based, manualized techniques in their pure form; instead, they are concerned with synthesizing the best features of various theories of psychotherapy. Whereas an eclectic therapist's approach might contrast greatly from one client to the next (depending on empirical data for treatments of various disorders), an integrative therapist's approach might remain a bit more constant, just as in a smoothie—the flavor combination doesn't vary much from sip to sip.

As a clinical psychologist, which of these approaches to therapy—eclectic or integrative—would you prefer? If you were the client, which approach might you prefer that your clinical psychologist take? What are the pros and cons of each approach?

movement. One of the champions of this movement has been **John Norcross**, who explained that the psychotherapy integration movement grew out of "a dissatisfaction with single-school approaches and a concomitant desire to look across and beyond school boundaries to see what can be learned from other ways of thinking about psychotherapy and behavior change" (Norcross & Newman, 1992, p. 4). Norcross and Newman identify a number of factors that have fostered the popularity of integrative forms of psychotherapy, especially since the 1980s. Among the most important of these factors are two that we identify in this chapter: the lack of differential effectiveness among therapies and the recognition that common factors contribute significantly to therapy outcome.

More recently, an alternative to the eclectic and integrative approaches has emerged: an *assimilative* approach, in which the therapist maintains "a firm grounding in one system of psychotherapy, with a willingness to selectively incorporate (assimilate) practices and views from other systems" (Norcross, Goldfried, & Arigo, 2016, p. 306). The assimilative approach is not quite as blended as the integrative approach—a single approach, like cognitive or psychodynamic, clearly dominates—but the therapist is willing to bring in elements of other approaches in a limited way (Castonguay, Eubanks, et al., 2015).

Web Link 11.5
John Norcross

DENISE: A FICTIONAL CLIENT TO CONSIDER FROM MULTIPLE PERSPECTIVES

The upcoming chapters of this book focus on specific approaches to psychotherapy, including psychoanalytic, humanistic, behavioral, cognitive, and group. To illustrate each fully, we consider a therapy summary of a fictional client named Denise. (As this chapter indicates, if Denise were a real client seen by a real clinical psychologist, there is a good chance she would receive a combination or blend of treatments, but therapy summaries of the single-school approaches will nonetheless exemplify the elements of the various approaches.) As therapy summaries, these reports appear in the past tense, as if Denise has completed a full course of the featured treatment and the therapist has written a synopsis of the treatment that might be useful if Denise returns later or resumes therapy with another therapist.

As it is with any client, Denise's cultural background is an important aspect of her therapy. In creating this fictional client, many of her demographic characteristics were selected because they match descriptions of individuals most likely to seek psychotherapy (Vessey & Howard, 1993). Of course, culturally competent clinical psychologists may make adjustments to therapy for clients with similar presenting problems but different cultural characteristics.

A full description of Denise is presented in Box 11.6.

BOX 11.6

Denise: A Fictional Client to Consider From Multiple Perspectives

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Denise is a 30-year-old, single, straight, Caucasian woman who has lived her entire life in a large Midwestern city. She has no history of significant illness or injury and is in generally good health. Denise grew up as the fifth of six children in a middle-class family in which both parents worked full-time. She attended 2 years of college immediately after high school but then transferred to culinary school where she graduated near the top of her class. She currently lives alone and maintains an upper-middle-class lifestyle.

Denise has worked as the only chef in a small upscale restaurant for about 5 years. Denise has enjoyed her job very much. One of her favorite aspects of her job had always been coming out of the kitchen to ask customers how they were enjoying their meals, especially because the feedback she received from them was almost exclusively positive. The owner of the restaurant had allowed Denise to do this because he believed it added a personal touch to the dining experience. Another of her favorite aspects of the job had been the creative freedom she enjoyed in the kitchen. The owner had allowed her to create her own unique entrées and change them as frequently as she liked.

Recently, though, Denise's feelings toward her job have changed drastically. A new owner took over the restaurant, and the new owner firmly stated to Denise that her job was to stay in the kitchen preparing food and not talk with customers at all. The new owner has also provided Denise with a strict, predetermined, permanent menu that the owner alone created and that Denise must now follow. She finds this revised job description inconsistent with her own personal style, and she sorely misses both the praise she had become accustomed to receiving from customers and the opportunities to create her own dishes.

Since this change was implemented, Denise has been experiencing mild to moderate depressive symptoms, including sadness, loss of interest in daily activities, low energy, difficulty sleeping, and difficulty concentrating. She has HAD difficulty getting to work on time and preparing dishes in a timely and conscientious manner, and her exercise routine, which had been very regular, is now sporadic. Sometimes Denise suspects that the new owner may have implemented this new policy specifically to hurt her. She states that she wants to return to the way she felt before this happened, and she is concerned that if she cannot overcome this problem, her performance at work will suffer to the extent that she will lose her job.

CHAPTER SUMMARY

Psychotherapy is the most common professional activity of clinical psychologists. Following some published doubts about its efficacy in the 1950s, researchers in the subsequent decades amassed enormous amounts of empirical outcome data, much of which has been combined in meta-analyses, supporting the conclusion that for the vast majority of clients, psychotherapy works. Recent data suggest that it also reduces medical

costs and produces neurological changes in the brain. Outcome studies that have compared the efficacy of various approaches to therapy with one another have consistently resulted in a virtual tie, a finding that has been nicknamed the "dodo bird verdict." The presence of common factors across all forms of therapy, including a strong therapeutic relationship, hope, and attention, may underlie their equal efficacy. More recent and targeted outcome studies focusing on particular manualized therapies for particular disorders have given rise to further debate about the relative contributions of the common factors versus specific therapy techniques. All empirical psychotherapy outcome studies must address numerous methodological issues, including who, when, and how to ask about the outcome of therapy. Additionally, efficacy studies, which assess how well a therapy works "in the lab," should be understood in conjunction with effectiveness studies, which assess how well a therapy works "in the real world." A gap exists between psychotherapy researchers and practitioners, but dissemination of evidence-based therapies can be increased by various strategies, including practice-oriented research in which researchers and practitioners collaborate. Surveys indicate that since 1960, the eclectic/integrative therapy orientation has generally been most commonly endorsed by clinical psychologists, and among single-school approaches, the psychodynamic/psychoanalytic orientation has been on the decline, whereas the cognitive orientation has been on the rise.

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CRITICAL THINKING QUESTIONS

1. According to the tripartite model, parties other than the client and the therapist can have a meaningful perspective on the outcome of a client's psychotherapy. Specifically, which third parties might have the most valid perspective? Clients' partners, friends, kids, supervisors, coworkers, managed-care companies, or someone else?
2. What conclusions do you draw from the results of large-scale effectiveness studies such as the 1995 *Consumer Reports* study?
3. When graduate programs train their students in psychotherapy, to what extent should they emphasize common factors (e.g., forming and maintaining strong therapeutic relationships) as opposed to specific therapy techniques?
4. Consider the three-step sequential model of common factors. In your opinion, would men and women tend to move through the sequence identically? What steps might each group tend to emphasize or deemphasize?
5. What are the implications of the finding that the eclectic/integrative orientation has been the most commonly endorsed orientation among clinical psychologists in surveys since 1960?

LOOKING TOWARD GRADUATE PROGRAMS

The following table is a list of index entries in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018) relevant to the topics in this chapter (see page 25 in Chapter 1 for more information about the *Insider's Guide*). Keep in mind a few things as you consider these listings: They are current as of the 2018/2019 edition of the *Insider's Guide* (which typically revises every two years); they may have changed as faculty members retire, take new jobs, or develop new interests; and a graduate program's own website may have more extensive or up-to-date information.

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Counseling Process and Outcome Evidence-Based/Empirically Supported Treatments Intervention/Treatment Psychotherapy Process and Outcome Miscellaneous and Other—Therapeutic Relationship	Evidence-Based/Empirically Supported Treatments	Miscellaneous/Other—Evidence-Based Practice Miscellaneous/Other—Outcome Research Miscellaneous/Other—Psychotherapy Research

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

Clinician's Research Digest: Adult Populations
http://www.apa.org/pubs/journals/crd/index.aspx

Journal of Clinical Psychology
http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1097-4679

Journal of Consulting and Clinical Psychology
http://www.apa.org/pubs/journals/ccp/index.aspx

Journal of Psychotherapy Integration
http://www.apa.org/pubs/journals/int/index.aspx

Practice Innovations
http://www.apa.org/pubs/journals/pri/index.aspx

Psychology and Psychotherapy: Theory, Research and Practice
http://onlinelibrary.wiley.com/doi/10.1111/papt.2016.89.issue-2/issuetoc

Psychotherapy
http://www.apa.org/pubs/journals/pst/index.aspx

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12

PSYCHODYNAMIC PSYCHOTHERAPY

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This chapter and the three that follow each focus on a prominent psychotherapy orientation. We begin with the psychodynamic orientation for multiple reasons. For one, psychodynamic therapy came first historically. In fact, for much of the first half of the 1900s, the psychodynamic approach was so dominant that it was practically synonymous with psychotherapy itself. Also, many of the therapies that arose in later decades—including humanistic, behavioral, and cognitive therapies—were reactions against the psychodynamic approach (J. Gold & Stricker, 2017; Karpiak, Norcross, & Wedding, 2016). Many of the pioneers of the nonpsychodynamic therapies were actually initially trained in psychodynamic programs but later abandoned that approach to create something different.

But among many clinical psychologists, psychodynamic psychotherapy does not enjoy the reputation or popularity it once held. A leading scholar in psychodynamic therapy recently described its current status quite bluntly:

> There can be little doubt that psychoanalysis and the psychodynamic therapies that arise from it are currently in a beleaguered state. The average educated reader today is likely to be confronted with a series of statements in the popular press that are variations on the following themes: Freud is dead; psychoanalysis is more religion than science; psychodynamic psychotherapy has no research to support its efficacy; cognitive-behavioral therapy has surpassed psychoanalytic therapy as the most accepted form of psychotherapeutic

Learning Objectives

12.1 Identify the primary goal of psychodynamic psychotherapy.

12.2 Describe the techniques used by psychodynamic therapists to access the unconscious.

12.3 Explain the implications for psychotherapy of fixation at Freud's oral, anal, and phallic psychosexual stages.

12.4 Contrast contemporary forms of psychodynamic psychotherapy with Freud's original approach.

12.5 Discuss brief psychodynamic psychotherapy approaches.

12.6 Summarize the findings of psychodynamic psychotherapy outcome research.

> treatment; psychodynamic psychotherapy is too long and too expensive; and the fundamental psychoanalytic concepts are too vague to be rigorously studied. The common wisdom among psychoanalysts these days is that if a patient asks you if you are "a Freudian," the correct answer is "no." (Gabbard, 2009a, p. vii)

Despite the relative decline of psychodynamic therapy, it remains widely used and quite relevant, especially in certain pockets of the profession. Through adaptations of its traditional methods into novel approaches that better suit contemporary culture, it continues to exert significant influence on clinical psychology. In fact, it is worth emphasizing that the quote above appears in the preface of a full book devoted to evidence-based psychodynamic psychotherapy, the mere existence of which illustrates the continued vitality of psychodynamic psychotherapy in an era that emphasizes empirical efficacy.

DEFINING PSYCHODYNAMIC PSYCHOTHERAPY

For the sake of simplicity, in this chapter we will use the term **psychodynamic psychotherapy** to cover an extensive range of therapies. Our use of the term refers broadly to the pioneering work of **Sigmund Freud** and all subsequent efforts to revise and expand on it. Thus, it includes Freud's original approach to therapy, which in its classic form is known as psychoanalysis. Of course, Freud had many intellectual descendants, including some who were his contemporaries (such as Carl Jung, Alfred Adler, and Erik Erikson) and others whose ideas arrived in subsequent generations (such as his daughter Anna Freud, Harry Stack Sullivan, Frieda Fromm-Reichmann, Melanie Klein, Karen Horney, D. W. Winnicott, and Heinz Kohut). At various points in the evolution of his theory by others, Freud's original term *psychoanalysis* was replaced by terms such as *psychoanalytic psychotherapy*, *neo-Freudian therapy*, and *psychodynamic psychotherapy*, each of which has generated even more specific terms for its offshoots. Rather than trying to make demarcations and distinctions among so many overlapping terms, our purposes in this chapter are best served by choosing a single term—*psychodynamic psychotherapy*—to represent them all.

Web Link 12.1
Sigmund Freud

GOAL OF PSYCHODYNAMIC PSYCHOTHERAPY

The primary goal of psychodynamic psychotherapy is to make the unconscious conscious (Cabaniss, Cherry, Douglas, & Schwartz, 2011; Karon & Widener, 1995). Psychodynamic psychotherapists help their clients become aware of thoughts, feelings,

and other mental activities of which the clients are unaware at the start of therapy. The word **insight**, used often by psychodynamic therapists and clients alike, captures this phenomenon—looking inside oneself and noticing something that had previously gone unseen (J. P. Barber & Solomonov, 2016; Gibbons, Crits-Christoph, Barber, & Schamberger, 2007; Lacewing, 2014; McAleavey & Castonguay, 2014). Once we become aware of unconscious processes, we can make efforts to control them deliberately, rather than them controlling us.

Of course, the process of making the unconscious conscious presumes that an unconscious part of our mind exists in the first place. This fundamental idea—the existence of the **unconscious**—is one of Freud's most important and enduring contributions to clinical psychology (Kernberg, 2004; Kihlstrom, 2015; R. C. Lane & Harris, 2008). Before Freud, there was little acknowledgment among mental health professionals about any mental activity occurring outside our awareness. Freud changed the way we think about ourselves by proposing "mental processes that are outside the awareness of the individual and that have important, powerful influences on conscious experiences" (Karon & Widener, 1995, p. 26; see also Mendelsohn, 2015). According to Freud and his psychodynamic followers, not only does the unconscious exist, it also exerts a powerful influence on our day-to-day and minute-to-minute lives. In fact, they argue that unconscious processes underlie depression, anxiety, personality disorders, eating disorders, and all other forms of psychopathology that clinical psychologists treat. Thus, gaining access to it is vital (Kris, 2012).

Public Domain

Photo 12.1 Sigmund Freud (1856–1939), the leading figure in psychodynamic psychotherapy.

Accessing the Unconscious

Psychodynamic psychotherapists gain an appreciation of their clients' unconscious process in a variety of ways. Of course, these methods are quite **inferential**: Rather than understanding a client's unconscious in an empirical, factual way, psychodynamic psychotherapists understand it through inference, deduction, and conjecture. In more casual language, psychodynamic psychotherapists try to "read" their clients and hypothesize about their unconscious activity using the following processes.

Free Association

Free association is a technique in which psychodynamic psychotherapists simply ask clients to say whatever comes to mind without censoring themselves at all. The client's task is to verbalize any thought that occurs, no matter how nonsensical, inappropriate, illogical, or unimportant it may seem (Kernberg, 2004; Skelton, 2006). It may sound straightforward enough, but free association is no easy task. Consider how rare it is for

any of us to speak with complete spontaneity, without editing ourselves in one way or another. Perhaps the only times we hear such speech in daily life are when we listen to very young children or very intoxicated adults. Nonetheless, the words of people in such states of mind can be revealing about their innermost thoughts and feelings, according to the psychodynamic approach. If clients can feel relaxed enough to engage in free association (which may be rare even in psychotherapy), their unconscious processes can become more evident to the psychodynamic psychotherapist and, ultimately, to themselves.

It is important to distinguish free association from word association, a technique associated with Carl Jung. In word association, the therapist presents the client with a list of words. After hearing each word, the client is to respond with the first word that comes to mind. Jungian therapists have often developed hypotheses about clients based on the content of their associations or the amount of time it takes them to respond to certain words (C. S. Hall & Nordby, 1973). In contrast, free association involves no stimulus at all from the therapist.

Freudian "Slips"

According to psychodynamic psychotherapists, all our behavior is determined; there is no such thing as a random mistake, accident, or slip. So if a behavior can't be explained by motivations of which we are aware, unconscious motivations must be the cause. In this way, when we get something wrong or "forget" something, we reveal unconscious wishes. Psychodynamic psychotherapists who witness a client's slips of the tongue during a session or who hear clients' stories of such events may be able to glimpse the clients' underlying intentions. Although most examples of **Freudian slips** are verbal, they can be behavioral as well:

- Ray, a 52-year-old man who lived in New Jersey, was asked by his older sister Tanya to fly home to Ohio. The reason for the meeting was that their 80-year-old father had become quite ill in recent months, and although he currently lived alone, Tanya thought they should discuss assisted-living or nursing home options. Ray agreed to fly home, but a few hours before the flight, he lost his car keys. He found them at the last minute, but on the way to the airport he missed the airport exit off the highway (an exit he had taken dozens of times before). The combination of losing the keys and missing the exit resulted in Ray missing his flight. His "mistakes" uncover that, unconsciously, Ray dreaded the meeting and wanted to avoid it. He was fearful of the conflicts that would arise with Tanya as well as the sight of his father as unable to take care of himself.

- Liz and Amy, both 25 years old, have been very close friends since age 18, when they were roommates during their first year of college. At that time, Liz began dating Sean. They continued dating for the next 5 years and got married when Liz was 23. Liz and Sean just had their first baby, a boy named Benjamin. Amy is happy for them, and she is one of many to bring a gift for the new baby. In her card, Amy congratulated the couple on the birth of "Benjamine." Her unique misspelling of the baby's name combines the words "Benjamin" and "mine," revealing Amy's unconscious wish that she, rather than Liz, had dated, married, and had a child with Sean.

- Occasionally, network TV provides a good example of a Freudian slip. On a memorable episode of the sitcom *Friends*, Ross Geller and his fiancée Emily are getting

married. During the ceremony, when asked to repeat the words, "I take thee, Emily . . . ," Ross mistakenly replaces "Emily" with "Rachel." He tries in vain to explain it away as a meaningless slip of the tongue, but Emily (and, of course, all the viewers) recognize that although he may not realize it consciously, Ross truly wishes he were marrying his long-time friend Rachel rather than Emily.

Dreams

Although some emphasize **dreams** more than others, psychodynamic psychotherapists generally believe that our dreams communicate unconscious material (Cabaniss et al., 2011). Freud theorized that when we sleep, our minds convert **latent content** (the raw thoughts and feelings of the unconscious) to **manifest content** (the actual plot of the dream as we remember it). This process, called **dream work**, uses symbols to express wishes, which can result in unconscious wishes appearing in a very distorted or disguised form.

In psychodynamic psychotherapy, the therapist analyzes dreams by attempting to uncover the unconscious meaning behind them—essentially, undoing the dream work. Often, the therapist asks the client to help in the dream interpretation process by explaining the personal meaning of the symbols appearing in the dream. As an example, if a client has a dream about, say, a table, the therapist may be unsure about what the table represents. However, if the client explains that the particular table that appeared in the dream was reminiscent of the one she remembers from her childhood in her grandmother's house, the connotations of that symbol may become more apparent. Of course, even with the client's help, dream interpretations (and other attempts to understand the unconscious) remain inferential rather than factual.

Sigmund Freud (1900) famously called dreams the "royal road" to unconscious material. Although dreams are certainly still valued, the contemporary perspective is more inclusive of other paths as well: "Today the psychoanalytic view is that there are many 'royal roads' to the unconscious" (Kernberg, 2004, p. 18).

Resistance

Sometimes, when certain issues come up during the course of therapy, clients make it clear that they "don't want to go there." They communicate their reluctance in a variety of ways—some obvious, some subtle. They might change the subject, suddenly remembering something new they intended to discuss. They might fill sessions with talk of unessential topics. They might show up late to subsequent appointments or miss them altogether.

Psychodynamic psychotherapists have a name for this client behavior: **resistance** (Dewald, 1964; Gabbard, 2009c; Karon & Widener, 1995; LaFarge, 2012; R. C. Lane & Harris, 2008; Wolitzky, 2016). When clients sense that certain unconscious thoughts and feelings are being laid bare too extensively or too quickly, they feel anxious. That anxiety motivates them to create distractions or obstacles that impede the exploration of those thoughts and feelings. Psychodynamic psychotherapists could be frustrated by clients' resistance, but more often they are intrigued by it and use it to guide future efforts. When they notice resistance, they assume that the therapy has "struck a nerve," and although the client may not be willing to delve into the issue at the moment, the

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Photo 12.2 Psychodynamic therapists attempt to help clients access the unconscious in many ways, often including the analysis of dreams. In your opinion, how important is it for clients to uncover and interpret unconscious material during therapy?

resistance itself may be an important factor in the client's daily life and could become a productive topic of conversation later in therapy.

As an example, consider Talia, a 24-year-old client who has grown up with a physically abusive father. In therapy, she had no trouble voicing her resentment toward her father; in fact, it was a prominent theme of her sessions. Her psychodynamic psychotherapist, Dr. Harrison, noticed that Talia rarely spoke of her mother, who also lived in the home throughout Talia's childhood. At one point, during a discussion of her father's impact on her life, Dr. Harrison asked Talia whether her mother was aware of her father's abuse toward her. Talia replied, but in a way that sidestepped Dr. Harrison's question: "I just hate him so much for what he did to me." Later in the session, Dr. Harrison asked Talia if anyone knew about the abuse as it was happening to her. Talia abruptly changed the subject, suddenly remembering an unrelated event that had happened the day before. Talia missed her next appointment and was late to the one after that. Dr. Harrison developed the hypothesis that Talia may hold some strong unconscious feelings of resentment toward her mother, who may have known about the abuse but did nothing to stop it. Dr. Harrison also speculated that Talia was uncomfortable acknowledging this buried resentment of her mother, so she found ways to avoid the topic. In future sessions, Dr. Harrison pointed out to Talia her avoidance of this topic, and they discussed the possibility that Talia felt anger toward her mother, as well as Talia's reluctance to discuss that feeling.

Defense Mechanisms

Psychodynamic psychotherapists believe that by identifying clients' unconscious defense mechanisms and bringing them into the clients' awareness, they can improve the quality of their clients' lives (J. C. Perry, 2014). Before we examine the defense mechanisms in detail, however, let's review the personality components that, according to Freud, produce them.

Freud's structural model of the mind includes three forces, the interaction of which takes place largely outside our awareness: the id, the superego, and the ego. The **id** is the part of the mind that generates all the pleasure-seeking, selfish, indulgent, animalistic impulses. It seeks immediate satisfaction of its wishes, most of which are biological in nature, and is oblivious to any consequences. In contrast, the **superego** is the part of the mind that establishes rules, restrictions, and prohibitions. It tells us what we "should" do, and it often uses guilt to discourage us from overindulging in immediate pleasure. Whereas Freud believed the id was inborn, he theorized that the superego became a part of the mind through experiences with authority figures, especially parents. Essentially, the superego is an internalization of the rules and morals taught to each of us, and it stands in direct opposition to the id (Kernberg, 2004; B. E. Moore & Fine, 1990; Skelton, 2006).

So, according to Freud, our unconscious mental processes involve a constant battle between an id demanding instant gratification and a superego demanding constant restraint. How does the mind manage these battles? What does it do with the conflicted

impulses? These responsibilities fall to the **ego**, the third component in Freud's structural model of the mind. The ego is a mediator, a compromise maker between the id and the superego. It faces the challenge of partially satisfying both of these opposing forces while also meeting the demands of reality. The ego can be quite creative in the ways it handles id/superego conflict. Over time, it develops a collection of techniques on which it can rely. It is this set of techniques that Freud and his followers call **defense mechanisms** (J. P. Barber & Solomonov, 2016; Gabbard, 2005; R. C. Lane & Harris, 2008; Wolitzky, 2016).

Web Link 12.2
Glossary of Freudian terminology

So what are the common defense mechanisms? Although Sigmund Freud offered descriptions of many, his followers, including his daughter Anna Freud, added to the list (Dewald, 1964; A. Freud, 1936; Sandler & Freud, 1985). Let's consider some of the most commonly acknowledged defense mechanisms, keeping in mind that all of them occur unconsciously. Examples will follow the series of definitions.

- **Repression**. When the id has an impulse and the superego rejects it, the ego can *repress* conscious awareness of the impulse and id/superego conflict around it. In other words, the ego can take the impulse and the internal conflict it creates and "sweep them under the rug" so that we never even become aware we had them in the first place. *Denial* is a similar defense mechanism, but it usually refers to events that happen to us rather than impulses that come from within us.

- **Projection**. When the id has an impulse and the superego rejects it, the ego can *project* the id impulse onto other people around us. In this way, we try to convince ourselves that the unacceptable impulse belongs to someone else, not to ourselves. Essentially, we attribute our most objectionable qualities to others and, in the process, might cast ourselves as possible recipients of the others' unacceptable behavior rather than the ones with the impulse to carry it out ourselves.

- **Reaction formation**. When the id has an impulse and the superego rejects it, the ego can *form a reaction against* the id impulse—essentially, do the exact opposite. So when the id urges us to do something selfish, we don't simply resist the temptation; we do something selfless, as if overcompensating for the original id impulse.

- **Displacement**. When the id has an impulse and the superego rejects it, the ego can *displace* the id impulse toward a safer target. Rather than aiming the id's desired action at whom or what it wants, we redirect the impulse toward another person or object to minimize the repercussions—this way, the superego is somewhat satisfied as well. The phrase "kicking the dog" has been used to describe displacement, illustrating how the ego can reroute destructive urges.

- **Sublimation**. When the id has an impulse and the superego rejects it, the ego can *sublimate* it—essentially, redirect it in such a way that the resulting behavior actually benefits others. Unlike "kicking the dog," in which no one benefits from the behavior (especially the dog), sublimation allows the id to do what it wants, and in the process, others are helped rather than harmed.

To illustrate these five defense mechanisms, let's choose a specific id impulse—physical aggression—and imagine how the ego might manage the internal conflict caused when the superego rejects it. Your ego could repress the impulse, in which case

BOX 12.1

Metaphorically Speaking

If You've Been to a Movie Theater, You Understand Projection

When you're sitting in a theater, where's the movie? The knee-jerk answer is that the movie is on the screen in front of you, but that's actually incorrect. A *projection* of the movie is on the screen; the movie itself is in the projector behind you.

Projection as a defense mechanism works very similarly to projection in the movie theater, with the ego as the movie projector and the id impulse as the movie. The ego takes an id impulse and "projects" it out of the person and onto others. Thus, the person sees it in others but not in the self. As an example, consider Randy, who frequently has the id impulse, "I want to steal." By projecting this id impulse, Randy sees this impulse on others but not on himself. If Randy literally had a projector on his forehead (pardon the absurd image), his projection of the sentence "I want to steal" onto everyone around him would be obvious. When Randy sees others walking around with "I want to steal" written across their shirts (while remaining oblivious of the fact that the message actually originates inside himself), he may very well become convinced that he is surrounded by thieves. Randy's case illustrates that when an individual relies heavily on a single defense mechanism, that defense mechanism becomes a prominent feature of the individual's personality. To some extent, a defense mechanism can define a person or at least the kinds of problems the person brings to therapy. It would certainly be no surprise to find that Randy suffers from paranoia of one kind or another (perhaps paranoid personality disorder or a paranoid subtype of schizophrenia), because he views others as crooks and himself as a potential victim. Psychodynamic psychotherapists would aim to make Randy more aware of his own unconscious processes, including his tendency to project his own unacceptable unconscious wishes onto others.

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you would never be aware that you had it in the first place. It could project the impulse onto others, such that you became convinced that they—not you—had the impulse to attack. It could form a reaction against the impulse, in which case you would do the opposite of physical aggression—you might be exceedingly gentle and kind with others or dedicate yourself to nonviolent causes. It could displace the impulse to be aggressive

onto a safer target; if your id truly wanted to hit your boss, you might instead redirect the aggression toward a friend, family member, partner, pet, or even a punching bag or video-game enemy. Finally, it could sublimate the impulse, in which case you would behave in a physically aggressive way that actually benefited society—perhaps as a police officer subduing criminals or even as a surgeon whose incisions heal rather than harm.

Psychodynamic psychotherapists believe that some of these defense mechanisms are more mature or healthy than others (Dewald, 1964; S. Freud, 1905; Gabbard, 2005; Karon & Widener, 1995). For example, denial and repression are considered rather immature, largely because they don't effectively satisfy the id, so similar id demands resurface later. At the other end of the spectrum, sublimation is viewed as uniquely mature because it satisfies the individual's id impulses and a societal need simultaneously. Of course, the goal of psychodynamic psychotherapy is to help clients become aware of their unconscious processes, including their defense mechanisms. As clients become enlightened about defense mechanisms they use, they can exert some control over them and, in the process, move toward more mature ways of managing their internal conflicts.

Defense mechanisms may be easier to understand if we consider alternate definitions of some core Freudian terms: *id*, *superego*, and *ego*. Freud wrote in German, and some of his followers have argued that when his terms were translated into English, the translators made the curious choices of *id*, *superego*, and *ego*, which are actually Latin terms uncommon to everyday English. Some have argued that Freud's concept of the id would be better translated as "it": the part of you that is animalistic rather than human (not "he" or "she"). The superego might be better understood as the "over-me": an internalization of the rules and demands that came from authority figures, especially parents. And ego, therefore, refers to "me": the person negotiating between the demands for instant pleasure and the demands to follow rules (Karon & Widener, 1995; Sharpless & Barber, 2015; Truscott, 2010). This alternate terminology helps personify the ego and highlights the fact that defense mechanisms don't merely characterize some abstract component of the mind (the "ego"); they actually characterize the whole person and, in many cases, the problems he or she brings to a clinical psychologist.

Transference

Of all the ways to access a client's unconscious material, transference may be most essential to the psychodynamic approach. It is "generally regarded as the most important focus" of psychodynamic psychotherapy (Galatzer-Levy, Bachrach, Skolnikoff, & Waldron, 2000, p. 27) and is seen as "the most powerful tool" of those who conduct it (Karon & Widener, 1995, p. 27).

Transference refers to clients' tendency to form relationships with therapists in which they unconsciously and unrealistically expect the therapist to behave like important people from the clients' pasts. In other words, without realizing it, a client "transfers" the feelings, expectations, and assumptions from early relationships—usually parental relationships—onto the relationship with the therapist. Essentially, clients allow powerful previous relationships to distort their view of the therapist, and in subtle ways they "prejudge" the

therapist as a person whose responses will echo those of mom, dad, or some other important early figure (J. P. Barber & Solomonov, 2016; Gabbard, 2005, 2009c; S. H. Goldberg, 2012; A. Harris, 2012; R. C. Lane & Harris, 2008; Messer, 2013; Wolitzky, 2016).

Perhaps the concept of transference is best illustrated through an everyday example. (After all, psychodynamic psychotherapists believe that transference is not exclusive to therapy but present in all kinds of relationships.) Consider Asaan, a 7-year-old boy who has taken piano lessons from Ms. Terrell since age 4. Ms. Terrell is a harsh, demanding teacher. She expects excellence from her students, and when they fall short, she scolds them unsympathetically. She also makes insulting and discouraging comments such as, "That's terrible," "You're wasting your parents' money," and "How many times do I have to teach you this song before you get it right?" As you might expect, Asaan has developed strong feelings toward Ms. Terrell over the years. Specifically, he fears and resents her. When Asaan's family moves to another city, his parents arrange for him to take piano lessons from a new teacher, Ms. Wallace. At his first lesson with Ms. Wallace, Asaan makes a mistake, and—before Ms. Wallace responds at all—strong feelings well up inside Asaan. He anticipates Ms. Wallace's comments—he "knows" that she's thinking, "That's terrible," etc.—and starts to feel fear and resentment toward Ms. Wallace in exactly the same way he did toward Ms. Terrell. In short, Asaan unknowingly *transfers* the feelings and expectations from his early, formative relationship with a piano teacher (Ms. Terrell) onto his subsequent teacher (Ms. Wallace). What he doesn't yet realize is that, in reality, his new teacher is a world apart from his old teacher: Ms. Wallace is extremely supportive, kind, and complimentary toward her students. In time, Asaan may grow to appreciate Ms. Wallace more realistically, but at least at first, his unrealistic, unconscious transference toward her will distort his perception of her and the way they interact with each other.

What happened to Asaan and his piano teachers happens to all of us, according to psychodynamic theory, in a much broader sense. We all experience powerful early relationships in our formative years—especially with parents—and those relationships shape our expectations for future relationships. Thus, when we meet new people and begin to form friendships, romantic partnerships, or work relationships with them, our responses toward them might not be totally objective. Instead, to the extent that our new friends, partners, or coworkers evoke the important people with whom we had our primary, early relationships, we may unconsciously jump to conclusions about them and develop feelings toward them that aren't actually warranted. The role of the psychodynamic psychotherapist is to help clients become aware of their own transference tendencies and the ways these unrealistic perceptions of others affect their relationships and their lives. Once aware of these tendencies, clients can make conscious efforts to manage them.

Rather than learning about clients' transference indirectly through their descriptions of relationships with others, psychodynamic psychotherapists seek to experience that transference firsthand. In other words, psychodynamic psychotherapists presume that clients will bring the same transference issues to the client–therapist relationship that they do to many of the other relationships in their lives. This way, the therapist gets a direct, personal understanding of the expectations and emotions the client unknowingly assigns to new relationships. After the therapist identifies these transference tendencies in the client–therapist relationship, the therapist can call the client's attention to them—in other words, offer **interpretation** of the transference. These comments by the therapist to the client—observations, essentially, of the unconscious tendencies the client shows when he or she forms relationships—are crucial elements of this kind of therapy

(Ulberg, Amlo, Critchfield, Marble, & Høglend, 2014). In fact, some argue that interpretation "has always been the technique par excellence" of psychoanalysis and psychodynamic psychotherapies (P. Goldberg, 2012, p. 292). With any interpretation, it may take clients a long time to fully understand it, accept it, and see its impact on their day-to-day lives. After all, an interpretation can represent a drastically different explanation than the client had ever considered for his or her own behavior. For this reason, psychodynamic therapy often involves a lengthy **working through** process, in which interpretations are reconsidered and reevaluated again and again. Frequently, it takes many sessions for an important interpretation to "sink in" fully and take effect on the client's psychological functioning (Gabbard, 2009c; Wolitzky, 2016).

The **"blank screen" role** of the psychodynamic psychotherapist is essential to the transference process. Psychodynamic psychotherapists typically reveal very little about themselves to their clients through either verbal or nonverbal communication. (In fact, this was a primary reason why Freud had clients lie on a couch while he sat behind them in a chair, out of their line of sight.) Consider the example of Asaan and his new piano teacher, Ms. Wallace, once again. If Ms. Wallace decided that from the beginning, she was going to reveal as little as possible about her own personality—essentially, stay "blank" to Asaan—she could conclude that any feelings Asaan has toward her are based purely on transference from past relationships. If she gives him nothing to respond to—communicates very little verbally and nonverbally, keeps a neutral emotional tone—and Asaan nonetheless expresses resentment and fear toward her, Ms. Wallace can conclude that those feelings are left over from earlier relationships; after all, she couldn't have elicited them herself. For a piano teacher, this would be unusual behavior. But for a psychodynamic psychotherapist, it is quite common and purposeful.

One more note on the blank screen role—in the Internet age, it is increasingly difficult, and perhaps impossible, for therapists to maintain it. Research shows that many clients google their therapists, either before or after therapy has begun (Eichenberg & Sawyer, 2016). And when they do, they can find all kinds of information, both professional and personal. For example, one study found that over half of clients who searched online for information about their therapists found information about the therapist's family, age, and education. Over a quarter found information on the therapist's hobbies or interests, home address, and dating/relationship status. A smaller percentage found specific personal information related to sexual orientation, religion, politics, finances (Kolmes & Taube, 2016).

As a clinical example of transference, consider Marcus, a 30-year-old man seeing Dr. Wallace, a psychodynamic therapist, for anxious and depressive symptoms that stemmed mostly from ongoing relationship problems with his long-term girlfriend and his supervisor at work. At his first appointment, Marcus provided Dr. Wallace with some background information about himself and his relationship problems. Two things struck her about the way Marcus spoke to her. First, he dominated the conversation so much that she couldn't get a word in edgewise, and when she tried, he interrupted her and spoke over her, as if he didn't want to allow her to speak. Second, he used the phrase "Before you tell me I'm wrong . . ." many times. That is, he would make a statement to her, and rather than pausing or inviting her response, he would continue immediately with a defense of why his statement was right, as if responding to an imagined challenge. It's important to note that Dr. Wallace did not, in fact, challenge Marcus at all. She never interrupted him, disputed him, or even offered a quizzical look. Instead, she remained attentive and affirming to everything Marcus said, both verbally and nonverbally.

BOX 12.2

In My Practice . . .

In my practice, I once had a client who provided an example of transference that was hard to miss. Her name was Claudia, and she was 27 years old. As our clinical interview started, Claudia mentioned that she had been feeling mildly depressed, primarily about the death of a relative and a job loss that had occurred within the last couple of months. About 10 minutes into the interview, Claudia interrupted herself. She sheepishly said to me, "I'm so sorry. I know I'm boring you. You're probably thinking, 'I have so many other important things to do, and I have so many other clients who have more serious problems—this is a total waste of my time.' I feel bad that I'm making you listen to my stupid complaints. I'm so sorry—I'm just gonna leave now."

Claudia stood up and headed for the door. In that instant, I wondered if I had done anything to cause her to conclude that I thought she was a waste of my time. Was I yawning? Looking out the window? Staring at the clock? No, I hadn't done any of those things. I was listening, nodding, making good eye contact, asking questions—the kinds of things that a good interviewer should do. Before she reached the door, I urged her to stay. I assured her that I was in fact interested in what she was telling me. She didn't believe me at first, but after some hesitation, she returned to her seat and continued telling me about the relative who passed away and the job she lost. But after another 10 minutes, again, Claudia interrupted herself. Again, she explained that she just knew that I found her to be a waste of time. Again, she headed for the door. Again, I persuaded her to return to her seat. These self-interruptions happened twice more before the interview was over. And afterward, when I offered to make an appointment for another session, she seemed incredulous that I would want to spend more of my time listening to her.

As therapy continued with Claudia, I learned a lot about her childhood. For example, she grew up with a father who never had time for her. She recalled a scene that happened regularly in her house when she was a kid, in which Claudia tried to get her dad's attention, but her dad didn't even look away from the TV, newspaper, or computer screen. If she tried to speak to him anyway, he shunned her, saying something along the lines of "Can't you see I'm busy? Stop bothering me!" It doesn't take a great Freudian scholar to connect the dots between the way Claudia was treated by her father and the way she assumed I (especially as a male therapist) would respond to her in therapy. As therapy continued, I carefully and patiently offered Claudia this interpretation of her unconscious and unrealistic expectations toward me. We spent a lot of time working through this interpretation, and gradually she became more conscious of it.

The real benefit of making Claudia aware of the transference tendencies she showed with me involved her relationships with other people in her real life. In her past, she had often broken up preemptively with boyfriends ("I know they're going to lose interest in me. They're gonna think I'm a total waste of their time. So I might as well just end it now"). She had also frequently decided against applying for jobs she wanted and was actually qualified for ("My résumé can't offer them anything interesting. They're gonna think my application is just a waste of their time. Why even try?"). But after Claudia became more aware of how she transferred expectations from her dad onto me, she similarly became more aware of how she transferred expectations from her

dad onto other people in her life (often, but not always, men). She started seeing people in her life more realistically ("Maybe he doesn't think I'm a waste of his time . . ."), and responding to them accordingly. As a result, the quality of many areas of her life significantly improved.

To see this box come to life as a whiteboard video, please visit edge.sagepub/pomerantz5e.

Nonetheless, Marcus seemed convinced that she objected to almost everything he said. In effect, Dr. Wallace had been a blank screen, but when Marcus looked at it, he saw someone doubting and confronting him.

Why would he have seen her this way? The answer came in time, as Marcus explained to Dr. Wallace that throughout his childhood, his parents had consistently challenged, doubted, or confronted him whenever he tried to express himself. His parents repeatedly gave him the message that he couldn't simply say what he thought or felt. He had to back up such statements with effective arguments to support them; otherwise, they would prove him wrong. Without realizing it, Marcus had transferred what he learned from his relationships with his parents onto his new relationship with Dr. Wallace. But Dr. Wallace was not the first person onto whom Marcus had exhibited this transference—he had behaved very similarly with both his girlfriend and his supervisor (among others). In reality, neither his girlfriend nor his supervisor responded to Marcus like his parents had, but his experience with his parents had left him unconsciously biased toward assuming that others would be similarly challenging, and he responded accordingly. His responses caught his girlfriend and supervisor off guard. They didn't like the way he insisted on dominating conversations and winning arguments that seemed to exist only in his mind. By doing so, he caused unnecessary tension in these relationships, and that tension caused Marcus to feel anxious and sad.

Unlike his girlfriend or supervisor, Dr. Wallace was able to identify Marcus's transference—to "catch" him doing it and tactfully point it out to him—whenever Marcus unconsciously directed his expectations of challenge or confrontation toward her. (In other words, Dr. Wallace offered *interpretation* of Marcus's transference.) With continued discussion in therapy (*working through*), Marcus gradually became more aware of this previously unconscious transference tendency in himself. He was able to "catch" his own unrealistic transference feelings toward Dr. Wallace and, more importantly, toward his girlfriend, his supervisor, and other people in his life and replace them with a more realistic, objective appraisal. His relationships with his girlfriend and supervisor improved, and his own feelings of anxiety and depression decreased.

As a final note on transference, it's important to remember that therapists are people, too, and just as clients can transfer onto therapists, therapists can transfer onto clients. Psychodynamic psychotherapists call this transference by therapists toward clients **countertransference**, and, generally, they strive to minimize it because it involves a reaction to the client that is unconsciously distorted by the therapist's own personal experiences (L. J. Brown, 2012; Dewald, 1964; A. Harris, 2012; J. A. Hayes, Nelson, & Fauth, 2015; Maroda, 2010; Skelton, 2006; Wolitzky, 2016). One reason that many psychodynamic training programs require trainees to be clients in psychodynamic psychotherapy themselves is to become aware of their own unconscious issues so they won't arise as countertransference toward their own clients (Erwin, 2002; B. E. Moore & Fine, 1990).

BOX 12.3

Considering Culture

Culture-Specific Responses to the "Blank Screen" Therapist

Many psychodynamic psychotherapists make great efforts to remain "blank screens" to their clients. They don't self-disclose much at all through their words or their actions, so clients may learn very little about the therapist's personal background. Family, religion, values, hobbies, and the like remain unknown. This blank screen role facilitates transference, which is essential to the psychodynamic process. When a client reacts emotionally to a therapist whose own words and actions couldn't have provoked the reaction, the therapist can be sure that those feelings are the remnants of a previous relationship. The therapist can help the client become more aware of this transference and how it affects relationships in the client's real life.

But how do clients of various cultures respond to a "blank screen" therapist? Are all clients seeking therapists who play this role in their lives? Some have argued that clients outside mainstream U.S. culture might benefit most from therapists whose roles in their lives are at odds with the blank screen role of psychodynamic psychotherapists. For instance, Atkinson, Thompson, and Grant (1993) argued that among diverse clients, there is tremendous cultural variation in the services they want their therapists to perform or the functions they want them to serve. Specifically, these authors have identified eight distinct therapist roles that clients may seek:

- Adviser
- Advocate
- Facilitator of indigenous support systems
- Facilitator of indigenous healing systems
- Consultant
- Change agent
- Counselor
- Psychotherapist

Whereas some of these roles, especially those near the end of the list, may be compatible with the psychodynamic blank screen role, some of the roles near the top of the list may clash with it. For example, as an adviser or an advocate, it's likely that the therapist would be expected to reveal something personal, such as values and experiences. And certainly, as a facilitator of indigenous support or healing systems, the therapist would at the very least indicate a familiarity and endorsement of these systems and may also make known personal background information in the process.

Atkinson et al. (1993) argue that people less acculturated to mainstream U.S. culture tend to favor the first four roles on the list, whereas people with higher levels of acculturation tend to favor the last four. Indeed, clients with low levels of acculturation may be looking for something very different from what's traditional when they decide to seek help for a psychological or emotional problem. In your opinion, how rigid do psychodynamic psychotherapists have to be regarding the blank screen role? To what extent can it be adapted to accommodate diverse clients?

PSYCHOSEXUAL STAGES: CLINICAL IMPLICATIONS

Freud's psychosexual stages of development—oral, anal, phallic, latency, and genital—are among the most widely known aspects of his theory. Rather than restating the explanations found in introductory psychology textbooks, let's consider the implications most relevant to clinical psychologists and to the psychodynamic psychotherapy approach in particular.

Of the five stages, the first three have generally received the most attention from psychodynamic psychotherapists, especially regarding fixation (Karon & Widener, 1995). **Fixation** refers to the idea that as children move through the developmental stages, they may become emotionally "stuck" at any one of them to some extent and may continue to struggle with issues related to that stage for many years, often well into adulthood. Although fixation can happen for a variety of reasons, most often it occurs when parents do either "too much" or "too little" in response to the child's needs at a certain developmental point.

Oral Stage

Consider the **oral stage**, which takes place during roughly the first year and a half of a child's life. During this time, the child experiences all pleasurable sensations through the mouth, and feeding (breast or bottle) is the focal issue. Of course, kids whose parents mismanage this stage may display blatantly "oral" behaviors later in life: smoking, overeating, drinking, nail biting, and so on. Many of the consequences are not so obvious, however. According to psychodynamic theory, a primary issue at this stage is dependency. Babies are, after all, utterly dependent on others for survival and comfort. They cannot feed, clothe, bathe, protect, or otherwise take care of themselves, so they must depend on the adults in their lives. (Perhaps the adult equivalent of this is a hospital patient who is entirely debilitated by illness or injury and whose only option is to call the staff for help.) If parents overindulge children in the oral stage, children may learn that depending on others always works out wonderfully and, in fact, that other people exist solely to anticipate and meet their needs. Such children may develop overly trusting, naive, unrealistically optimistic personalities and, as adults, will form relationships accordingly. On the other hand, if parents are not responsive enough to children during the oral stage, children may learn that depending on others never works out and, in fact, other people have no interest in helping them at all. Such children may develop overly mistrusting, suspicious, and unrealistically pessimistic personalities and, as adults, will form relationships accordingly. These oral issues, especially in

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Photo 12.3 According to the psychodynamic perspective, the mouth is the focus of pleasurable sensations for children in the oral stage of development.

extreme form, are often at the root of clients' individual and interpersonal problems and the focus of psychodynamic psychotherapy.

Anal Stage

The **anal stage** follows the oral stage, occurring when the child is about 1.5 to 3 years old. Toilet training is a primary task of this stage, but it is not the only way children are learning to control themselves. Indeed, control is the central issue of this stage. At this age, adults (especially parents) begin to place demands on children regarding their speech and behavior. If parents are too demanding of children at this stage, children can become overly concerned about getting everything just right. In the bathroom, this may mean "no accidents," but, more generally, it means having everything in exactly the right place at the right time. These children often grow to become adults who think obsessively and behave compulsively in order to stay in control: They meticulously organize their desks, they program their daily schedules from start to finish, and they get their cars' oil changed every 3,000 miles exactly. By contrast, if parents are too lenient toward children at this stage, children can become lax about organization, and this trait can continue into adulthood: Their desks are covered in messy piles, their schedules are sloppy and haphazard, and their cars get oil changes "whenever." These "neat freak" or "slob" tendencies can have significant clinical implications, including anxiety disorders such as obsessive-compulsive disorder and relationship problems stemming from incompatible living styles.

Phallic Stage

The **phallic stage**, taking place from about age 3 to about age 6, is one of Freud's most controversial. In fact, many of the ideas originally contained in Freud's description of this stage, especially those closely tied to gender-specific biology, have fallen out of favor and are widely disputed by contemporary psychodynamic psychotherapists (Erwin, 2002; Karon & Widener, 1995). What remains is the fundamental idea implied by the Oedipus and Electra complexes: Children at this age wish to have a special, close relationship with parents. The parents' response to the child's wish is the crucial issue for clinical psychologists, because this parental response powerfully shapes the children's view of themselves. This view of the self—essentially, self-worth—is the key consequence of the phallic stage. Of course, the ideal situation is for parents to respond positively to kids' overtures. But when parents respond too positively, when they reciprocate the child's wishes too strongly, they overinflate the child's sense of self. Such children may grow into adults whose opinions of themselves are so unrealistically high that they strike others as arrogant or egotistical. Conversely, parents who reject their child's wishes for a special, close relationship can wound a child's sense of self-worth. These kids can grow up to become adults who devalue themselves and are overly insecure and self-doubting. (Felicia, the clinical example in the earlier section on transference, is an example.) As with the other psychosexual stages, the phallic stage often gives rise to issues discussed in psychodynamic psychotherapy, including disorders such as depression, dysthymia, anxiety, relationship problems, and any other issue that can involve questions of self-worth.

MORE CONTEMPORARY FORMS OF PSYCHODYNAMIC PSYCHOTHERAPY

Since its origins with Freud, psychodynamic psychotherapy has been reinvented in countless forms (Orlinsky & Howard, 1995). Most of these revisions have deemphasized the biological and sexual elements of the theory. For example, **ego psychology**, as exemplified by Erik Erikson and his eight-stage theory of development, revised Freud's psychosexual stages to highlight social relationships and emphasized the adaptive tendencies of the ego over the pleasure-based drive of the id. The **object relations** school, led by Melanie Klein, Otto Kernberg, Ronald Fairbairn, and others, deemphasized internal conflict (id vs. superego) and instead emphasized relationships between internalized "objects" (essentially, important people from the client's life; Shahar, 2015; P. Williams, 2012; Wolitzky, 2016). The **self-psychology** school of Hans Kohut and others emphasizes parental roles in the child's development of self, with special attention paid to the meaning of narcissism at various points, including in therapy (Karon & Widener, 1995; Skelton, 2006; Terman, 2012; Wolitzky, 2016).

Other revisions of the Freudian approach have cast doubt on Freud's theories regarding females and their development. Karen Horney was one such critic of Freud, publishing numerous articles and books in the early to mid-1900s that opposed many of Freud's ideas, including his assumptions that females felt inherently inferior and envious toward males. Her commentaries on Freud and her own gender-specific developmental theories represented "a courageous attempt to reform the accepted psychoanalytic ideas on women" (Westkott, 1986, p. 9). Moreover, Horney's writings influenced more recent feminist theorists, including Nancy Chodorow (1978) and Carol Gilligan (1982), who have further advanced theories of female development that account for the experience of girls and women more authentically than do Freud's original theories and do not represent variants of male development.

Throughout its history and in all its many variations, psychodynamic psychotherapy has always been among the longest and most expensive forms of psychotherapy—a mismatch for our current society, which is characterized by a desire for fast results and managed-care companies reluctant to pay for treatment they view as excessive (Galatzer-Levy et al., 2000; Sperling, Sack, & Field, 2000). As a result, most of the recent variations of psychodynamic psychotherapy have emphasized efficiency. Collectively, the many forms of **brief psychodynamic psychotherapy** have become far more common in recent years than the classic, orthodox version of Freudian psychoanalysis from which they derived (H. Levenson, 2010; H. Levenson, Butler, & Beitman, 1997; Steenbarger, 2008). The definition of *brief* in brief psychodynamic psychotherapy varies a bit, but it often refers to therapy lasting fewer than 24 sessions, which amounts to about 6 months of once-a-week sessions. With such a small window of time (by psychoanalytic standards), the therapist and client must quickly form an alliance, develop insights that facilitate new ways of understanding, and translate these insights into real-world changes. Compared with long-term psychodynamic psychotherapy, brief psychodynamic psychotherapy tends to be more successful when the client's problems are mild and narrowly defined, the therapist is active, and the focus includes the present rather than solely the past (Dewan, Weerasekera, & Stormon, 2009). Table 12.1 offers a more extensive comparison of brief and long-term approaches to psychodynamic psychotherapy.

Table 12.1 Comparison of Brief and Long-Term Psychodynamic Psychotherapy

Brief Psychodynamic Psychotherapy	Long-Term Psychodynamic Psychotherapy
Form therapeutic alliance rapidly	Form therapeutic alliance gradually
Focus on specific, narrowly defined problem	Focus on broad range of problems
Therapist's level of activity is relatively high	Therapist's level of activity is relatively low
Client's psychopathology is less severe	Client's psychopathology is more severe
Focus primarily on here and now	Focus on past and present
Client's ability to tolerate separation is high	Client's ability to tolerate separation is variable
Client has good object relationships	Client has poor to good object relationships

Source: Adapted from Dewan et al. (2009).

Some particular forms of brief psychodynamic psychotherapy have garnered significant attention in recent years. Let's consider two of them in detail as examples of contemporary variations of psychodynamic psychotherapy.

Interpersonal Therapy

Interpersonal therapy (IPT), which derives from the interpersonal school of psychodynamic thought of which Harry Stack Sullivan was a leader, was developed in the 1980s by Gerald Klerman, Myrna Weissman, and colleagues. It was originally created to treat depression, but it has since been used to treat numerous other disorders (Blanco & Weissman, 2005; Klerman, Weissman, Rounsaville, & Chevron, 1984; Lipsitz & Markowitz, 2016; Markowitz & Weissman, 2012; Stuart, 2017; Tong, Zwelling, & Doughan, 2017; Weissman, 1995). It is designed to last about 14 to 20 sessions, and, as such, its goals are more focused and limited than structural change of the entire personality. Its methods are outlined in a manual with specific therapeutic guidelines (Klerman et al., 1984).

The fundamental assumption of IPT is that depression happens in the context of interpersonal relationships, so improving the client's relationships with others will facilitate improvement in the client's depressive symptoms (Lipsitz & Markowitz, 2013). It focuses on current interpersonal relationships and role expectations and tends to deemphasize some of the aspects of more traditional psychodynamic psychotherapy related to intrapsychic structure and childhood fixations (E. Frank & Levenson, 2011; H. Levenson et al., 1997; Swartz & Markowitz, 2009; Weissman, 1995). Therapists practicing IPT have found that for most clients, especially those with depression, a few specific interpersonal problem areas tend to contribute to clients' problems: *role transitions*, such as becoming a parent or graduating from college; *role disputes*, such as entering a marriage; *interpersonal deficits*, such as a lack of social support; and *grief*, such as the reaction to loss of a loved one (Blanco & Weissman, 2005; Lipsitz, 2009).

IPT proceeds in three stages. The first stage (about 2 sessions in most cases) involves categorizing the client's problems into one of the four categories listed above (role transitions, role disputes, interpersonal deficits, and grief). The intermediate sessions (10–12 sessions) emphasize improving the client's problems as identified in the first stage. Common psychodynamic methods are used, including a focus on current emotions, explorations of transference, and resistance. Also, the intermediate stage often includes an educational component in which the therapist teaches the client about depression and its symptoms. The final stage (2–4 sessions) involves a review of the client's accomplishments, recognition of the client's capacity to succeed over depression without the therapist's continued help, and efforts to prevent relapse (Klerman et al., 1984; H. Levenson et al., 1997).

More recently, a variation of IPT has been developed. Known as *interpersonal and social rhythm therapy* (IPSRT), it is specifically designed for clients with bipolar disorder. IPSRT augments IPT with efforts to control and stabilize daily rhythms, sleep/wake cycles, and social interactions (Weinstock, 2015). Irregularities in these kinds of daily activities and interpersonal relationships can be especially disruptive to people prone to extreme variations in mood. By contrast, regularity in daily activities (e.g., going to bed and waking up at the same time every day) and interpersonal relationships (seeing friends predictably, rather than sporadically) can be protective. Clients in IPSRT are encouraged to make and follow detailed daily schedules and to track their behavior and mood in detail. Evidence supporting IPSRT as a beneficial treatment for bipolar disorder is sizeable and growing (E. Frank et al., 2005; E. Frank, Swartz, & Kupfer, 2000; T. R. Goldstein et al., 2014; Inder et al., 2015).

IPT is one of the few specific forms of psychodynamic psychotherapy for which researchers have gathered a sizable amount of empirical evidence (Didie, 2015). In other words, efficacy studies have found that IPT works for depressed individuals, and it appears on published lists of therapies that are known to be successful for particular disorders (e.g., Chambless et al., 1998; Nathan & Gorman, 2002). Most of the evidence of support for IPT comes from studies of depressed clients, but there is also some evidence that it works for clients with eating disorders, anxiety disorders, and some other problems as well (Blanco & Weissman, 2005; Bolton et al., 2003; E. Frank & Levenson, 2011; Markowitz & Weissman, 2009, 2012; Spinelli & Endicott, 2003). As such, it is a leader among psychodynamic therapies in terms of empirically supported efficacy.

Time-Limited Dynamic Psychotherapy

Time-limited dynamic psychotherapy (TLDP) is a modern application of the classic and often-referenced concept of the "corrective emotional experience" (F. Alexander & French, 1946): Clients will bring to therapy the same transference issues that they bring to many of their other relationships, and the therapist's task is to make sure that this time, the interaction will end differently. In other words, if the client's relationship with the therapist follows the same unconscious "script" as the client's other relationships, it may end badly, but if the therapist can make the client more aware of this script and offer a chance to enact a healthier, more realistic one, the "emotional experience" will be "corrective" or therapeutic (Betan & Binder, 2017; Binder, Strupp, & Henry, 1995; Steenbarger, 2008).

Web Link 12.3
Interview with Hanna Levenson, leader in the field of TLDP

TLDP is experiential in nature; the here-and-now relationship between therapist and client is the main tool for therapeutic change. Like most newer forms of psychodynamic psychotherapy, it is typically much briefer than classic psychoanalysis (about 20–25 sessions maximum; H. Levenson, 2010). The therapist's primary task is to identify the "script" that the client appears to be unknowingly following. This script is the by-product of previous relationships (often with parents), in which the client learned what to expect from others. The TLDP therapist assumes that the client's problems are at least partially due to an application of this script to inappropriate relationships or situations. In other words, a client may get into frequent arguments with a romantic partner because he "knows" what his partner may be thinking and feeling, but this "knowledge" is actually unrealistic and mistaken. When the client tries to enact the same script with the therapist, the therapist recognizes it, refuses to be provoked or prodded into it, and points out this process to the client. In this way, they do not perpetuate the outdated script, and the client is forced to develop a new, more realistic way of relating to others that is not bound by the assumptions of the script that he or she had been unconsciously following (Binder et al., 1995; Steenbarger, 2008).

When therapists conduct TLDP, they often use a visual diagram called the cyclical maladaptive pattern (H. Levenson, 1995). It is a working model of the client's primary issues organized into four categories: acts of self (how a person actually behaves in public; e.g., how a client interacts in a job interview); expectations about others' reactions ("I'm sure the interviewer didn't like me"); acts of others toward the self (the interviewer says, "Your application looks great. We'll call you in the next 2 weeks," and the client interprets this as rejection); and acts of the self toward the self (the client tells self, "You are such a failure," and spends next day alone and miserable). By identifying these four components of the cycle, TLDP therapists can help clients become more aware of specific thoughts and behaviors that contribute to the faulty script that they may enact, as well as healthier alternatives to these thoughts and behaviors.

HOW WELL DOES IT WORK?

The nature of psychodynamic psychotherapy makes it especially difficult to gauge its effects. From an empirical researcher's point of view, the challenge of defining and measuring the outcome of psychodynamic psychotherapy—and even the basic psychoanalytic concepts on which therapy rests, such as the unconscious, transference, insight, and defense mechanisms—can be extremely challenging (Gibbons et al., 2007; Luborsky & Barrett, 2006). How, exactly, do we know how well it has worked? Can we measure the extent to which the unconscious has been made conscious? Can we calculate the amount of insight a client has achieved or the extent to which his or her relationships have improved? These questions haunt psychodynamic psychotherapy and elicit criticism from those who prefer therapies of other kinds. Additionally, these questions help us understand why the methodology of many studies of the outcome of psychodynamic therapy are so widely disparaged and why such a small number of psychodynamic therapies have been manualized, subjected to empirical trials, or included on lists of treatments that work (Sandell, 2012).

BOX 12.4

Metaphorically Speaking

If You've Watched the Olympics, You Understand Allegiance Effects

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When we watch Olympic events such as gymnastics or ice skating on TV, we see not only the judges' scores but the judges' nationalities as well. In fact, in the on-screen graphic, each judge's individual score appears in a box just below the flag and abbreviation of the judge's country. Why is the judge's country relevant? Perhaps the assumption is that the judges' allegiance to their home countries limits their objectivity. Even if they try not to, the judges may root for a particular country, and that rooting may influence the way they judge the performances of the athletes. With the athletes' countries known to the judges, who themselves are citizens of certain countries, allegiance effects may be unavoidable in the Olympics.

Allegiance effects may be unavoidable in psychotherapy outcome research, too. The situation is very similar to the Olympics: The researchers who conduct the research and therefore judge the "performance" of various therapies are themselves either supporters or opponents of some of those therapies. Of course, these researchers should be completely objective, and they may believe that they are, but according to Luborsky et al. (1999), most are not. Their own preferences for certain kinds of therapy and against others appear to powerfully influence the results of the studies they conduct. In fact, Luborsky et al. report that allegiance effects can account for about two thirds of the variance in outcomes of treatment comparisons! Some more recent studies have corroborated the severity of allegiance effects in psychotherapy outcome studies. For example, a large-scale review of 30 separate meta-analyses on the allegiance effect concluded that the connection between the allegiance of the researcher and the outcome of the study is "substantial and robust" (Munder, Brütsch, Leonhart, Gerger, & Barth, 2013, p. 501). Along the same lines, others have suggested that the eye-opening 1999 study by Luborsky et al. may have actually underestimated allegiance effects (e.g., Stiles, 2009). But others have argued that Luborsky et al. overestimated the severity or importance of their conclusions about therapy outcome (e.g., Leykin & DeRubeis, 2009; G. T. Wilson et al., 2011). Just as Olympic judges might feel an underlying favoritism for the athletes from their own countries, so might psychotherapy outcome researchers for the therapies from their own orientations.

The Olympic judge's flag is revealed for all to see, but the psychotherapy outcome researcher's allegiance is often less obvious. When Luborsky et al. (1999) conducted their study on allegiance effects, they had to contact the researchers who had published psychotherapy outcome research, as well as those researchers' colleagues, to determine the researchers' orientations. In your opinion, should psychotherapy outcome researchers

(Continued)

(Continued)

be more forthcoming about their own orientations, perhaps including this information in the studies they publish, because these orientations may bring about biases? Should researchers make efforts to conduct studies in collaborative groups that contain members of various orientations so they counterbalance one another's biases? What other measures can psychotherapy outcome researchers take to limit the allegiance effect?

In spite of these methodological challenges, there have been many attempts to measure the outcome of psychodynamic psychotherapy. A large-scale review of psychodynamic and psychoanalytic outcome studies that, in total, included almost 2,000 clients treated by about 500 therapists across a wide range of settings suggested that the vast majority of clients improve substantially (Galatzer-Levy et al., 2000). Similarly, a review of many meta-analyses of psychodynamic treatment concluded that in general, across a wide range of clinical problems, psychodynamic treatment is beneficial (Shedler, 2010; Wolitzky, 2016). Other recent reviews of psychodynamic psychotherapy outcome studies have also touted empirical data supporting its use with a wide range of specific problems, including personality disorders, depression, bulimia, anorexia, and panic disorder and as well as other long-standing, complex clinical conditions that may encompass more than one diagnosis (e.g., Abbass, Hancock, Henderson J., & Kisely et al., 2006; J. P. Barber & Solomonov, 2016; Driessen et al., 2015; Gerber et al., 2011; Keefe, McCarthy, Dinger, Zilcha-Mano, & Barber, 2014; Leichsenring, 2009a; Leichsenring, Rabung, & Leibing, 2004; R. A. Levy, Ablon, & Kachele, 2012; Messer & Abbass, 2010; Sandell, 2012). However, other reviews of the status of psychodynamic therapies have pointed out that the number of empirically sound studies, particularly those that focus on a well-defined, single disorder, is quite small (Messer, 2015). In fact, for some major disorders, there isn't a single well-done empirical study—suggesting that the evidence that does exist should be interpreted cautiously (Driessen et al., 2010; Gibbons, Crits-Christoph, & Hearon, 2008; Leichsenring, 2009b). Certainly, psychodynamic psychotherapy lags behind some other forms of therapy, especially behavioral and cognitive therapy, in sheer volume of empirical data supporting its use with specific clinical problems. But these reviews indicate that such empirical data have been obtained for some disorders and continue to accumulate.

It is worth noting that some empirical evidence exists for particular components of psychodynamic psychotherapy—namely, interpretation of transference and countertransference reactions. Johansson et al. (2010) conducted a study in which 100 clients with depression, anxiety, personality disorders, or other interpersonal problems received weekly psychodynamic psychotherapy for 1 year. For half of them, therapists offered transference interpretations; for the other half, they did not. Outcome was measured immediately after therapy and also at 1-year and 3-year follow-ups. Results indicated that those whose therapists offered transference interpretations benefited more from the therapy and that this benefit correlated with a greater level of insight. In a more recent study with a similar design, clients in psychodynamic therapy who gained insight and awareness of their feelings via transference achieved the best outcomes (Høglend & Hagtvet, 2019). Additional research supporting the benefits of transference, especially for borderline and other personality disorders, has accumulated to such

an extent that a specifically designed *transference-focused psychotherapy* has now been developed for clients with these issues (Goodman, 2013; Kernberg, 2016; Normandin, Ensink, & Kernberg, 2015; Yeomans, Clarkin, Kernberg, & American Psychiatric Association, 2015; Yeomans, Levy, & Caligor, 2013). Regarding countertransference, J. A. Hayes, Gelso, and Hummel (2011) conducted multiple meta-analyses on countertransference and found that the presence of countertransference reactions by therapists correlated negatively with outcome; that is, the more countertransference, the worse the outcome for the client. They also found a positive correlation between therapist efforts to manage countertransference and therapy outcome; that is, the more therapists tried to keep their own biases toward clients "in check," the better the outcome was for clients.

Interestingly, in some meta-analyses, psychodynamic therapies are found to be effective but a bit less so than other forms of therapy. These slight discrepancies disappear, however, when allegiance effects are taken into account. **Allegiance effects** refer to the influence of researchers' own biases and preferences on the outcome of their empirical studies. Typically, the researchers who conduct empirical studies of psychotherapy outcome, including meta-analyses, are not psychodynamic in their own orientation. Instead, they tend to be behavioral or cognitive, which are the two broad categories of therapy to which psychoanalysis was found to be slightly inferior in some meta-analyses (Prochaska & Norcross, 2014).

Web Link 12.4
Allegiance effect

Allegiance effects can be quite powerful: Luborsky et al. (1999) evaluated many comparative reviews of the psychotherapy literature and comparative studies of psychotherapy and found a surprisingly strong relationship between the way therapies were rated and the orientation of the researcher doing the rating. (Some have since supported this finding, whereas others have refuted it; e.g., Leykin & DeRubeis, 2009; Stiles, 2009; G. T. Wilson, Wilfley, Agras, & Bryson, 2011. See Box 12.4 for more.) This is relevant for all psychotherapy orientations but especially for the psychodynamic approach, because a relatively small number of therapy outcome researchers have been psychodynamic.

BOX 12.5

Denise in Psychodynamic Psychotherapy

Denise attended 50 sessions over a 1-year period. At my request, she provided background information about her childhood, including the fact that she was the fifth of six children and that her parents both worked full-time. She explained that because of these factors, she received very little attention or praise from her parents throughout childhood. She described her depression stemming from recent changes at work and stated that the most upsetting part of the change was the new restaurant owner's insistence that she stop visiting with diners. She sorely missed their positive feedback about her dishes. In addition to sadness and other common depressive symptoms, Denise also listed a lack of energy that resulted in her being late to work and preparing dishes more slowly than usual. It was also noteworthy that Denise believed that the new owner may have made these changes with the intention of hurting Denise.

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(Continued)

(Continued)

Based on this information, my initial conceptualization of Denise included the following hypotheses:

- Denise was dealing with emotional issues from the phallic stage of development, and she may have been partially fixated at that stage. At the points in childhood when Denise sought a special, intimate relationship with a parent, the parent did not provide it. As a result, she lacked self-esteem and sought it from outside sources. The strongest of these sources—the feedback of the diners in her restaurant—had been cut off by the new owner, contributing greatly to her depression.
- Denise was struggling with unacceptable impulses of anger toward the new restaurant owner and was using defense mechanisms—projection, most obviously—to deal with them. Denise stated that she believed the new owner was trying to hurt her by instituting the new policies, but this seems an unlikely motivation for the new owner. Instead, it is probable that Denise is the one with the hurtful intentions toward the new owner, but rather than acknowledge them, she kept them unconscious by projecting them onto the owner, thus portraying the owner as the "bad" person and herself as the innocent victim.
- Denise's unconscious anger toward the new owner also revealed itself through her lateness and slow work. Although she attributed these behaviors to her lack of energy, they may have been unconsciously motivated acts of aggression toward the new owner and the new policies. Several of the specific stories she told—the time she "accidentally" dropped an entrée in the kitchen and the times she "forgot" to set her alarm—appear to fall in the category of Freudian slips.

My relationship with Denise was marked by numerous examples of resistance and transference. Regarding the resistance, Denise often changed the subject in subtle and clever ways when I asked questions regarding the new owner or her feelings toward her parents. Once, after a session ended in the middle of a particularly intense discussion of her work situation, she was 20 minutes late to her next appointment. Another time, she followed a session in which we examined her feelings toward her parents with a session in which she insisted on focusing on a seemingly minor argument with a previously unmentioned friend. Regarding the transference, Denise often seemed to assume that I didn't think she was worthy of my attention, and, in fact, she said so on many occasions. She also sought my praise on frequent occasions, sometimes directly asking me for feedback after describing something she had accomplished.

The main intention behind my interventions with Denise was to make her more conscious of her unconscious processes. On various occasions, I offered interpretations of her actions. I mentioned that perhaps she had hurtful feelings toward the new owner, rather than vice versa. I asked

questions that led her to consider whether her slow and sloppy performance at work may have been motivated by her own wishes. I discussed with her the role that the diners' feedback played in context of the attention and praise her parents did not provide. And, perhaps most important, I pointed out to her the unconscious tendencies (transference) that she brought to the relationship with me, even though our relationship did not realistically merit them: the assumption that I would dismiss her and her need for my praise and admiration. The discussion of transference was especially productive because it shed light on some of Denise's previous relationships as well. Slowly, with continued conversation in all these areas, Denise was able to achieve significant insight into her unconscious processes. By doing so, she was able to view others more realistically and control her motivations more deliberately, and her depressive symptoms eventually lifted.

CHAPTER SUMMARY

Psychodynamic psychotherapy is founded on the theories of Sigmund Freud. It presumes the presence of a powerful unconscious component of the mind, and its primary goal is insight or making unconscious processes conscious. It is a highly inferential approach to psychotherapy in which problems and therapy's impact on them are difficult to assess objectively or empirically. Psychodynamic psychotherapists gain access to clients' unconscious in numerous ways, including free association, Freudian slips, dreams, and resistance clients display during therapy. The ongoing unconscious conflict between the id and the superego requires the ego to mediate by employing defense mechanisms such as repression, projection, reaction formation, displacement, and sublimation. Reliance on particular defense mechanisms can characterize an individual's personality as well as the clinical issues she or he brings to therapy. Clients often experience transference toward their psychodynamic therapists, in which they unknowingly and unrealistically expect the therapist to relate to them as important people from the clients' pasts have related to them. Psychodynamic therapists often assume a "blank screen" role to facilitate this transference process and assume that the clients may have transferred similarly onto other people with whom they have formed relationships. Psychodynamic therapists also pay significant attention to clinical issues that may stem from fixation at an early psychosexual stage of development (e.g., oral, anal, or phallic). Psychodynamic psychotherapy has traditionally been relatively long-term, but shorter versions, such as ITP and TLDP, have emerged in recent decades, and evidence has begun to accumulate attesting to their efficacy, especially with depressed clients. Overall, the nature of psychodynamic psychotherapy has hindered the collection of empirical outcome data, but the data that have been collected suggest it is roughly as effective as other forms of psychotherapy.

KEY TERMS AND NAMES

allegiance effects 305
anal stage 298
"blank screen" role 293
brief psychodynamic psychotherapy 299
countertransference 295
defense mechanisms 289
displacement 289
dream work 289
dreams 287
ego 289
ego psychology 299
fixation 297
free association 285
Sigmund Freud 284
Freudian slips 286
id 288
inferential 285
insight 285
interpersonal therapy (IPT) 300
interpretation 292
latent content 287
manifest content 287
object relations 299
oral stage 297
phallic stage 298
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psychodynamic psychotherapy 284
reaction formation 289
repression 289
resistance 287
self-psychology 299
sublimation 289
superego 288
time-limited dynamic psychotherapy (TLDP) 301
transference 291
unconscious 285
working through 293

CRITICAL THINKING QUESTIONS

1. To what extent do you believe that insight, or making the unconscious conscious, is essential to overcoming psychological problems?
2. To what extent do you believe that Freudian slips accurately reveal unconscious wishes?
3. Box 12.3 lists eight distinct therapist roles that clients may seek. If you were the client, which would you seek? To what extent would a psychodynamic psychotherapist match that role?
4. Many psychodynamic graduate training programs require their trainees to undergo therapy themselves. If you were the client, how would you feel about the fact that your therapist had (or had never) been in therapy?
5. What efforts should psychotherapy outcome researchers make to minimize allegiance effects?

LOOKING TOWARD GRADUATE PROGRAMS

The following table is a list of index entries in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018) relevant to the topics in this chapter (see page 25 in Chapter 1 for more information about the *Insider's Guide*). Keep in mind a few things as you consider these listings: They are current as of the 2018/2019 edition of the *Insider's Guide* (which typically revises every two years); they may have changed as faculty members retire, take new jobs, or develop new interests; and a graduate program's own website may have more extensive or up-to-date information.

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Psychoanalytic/Psychodynamic Therapy Miscellaneous and Other—Countertransference Miscellaneous and Other—Object Relations	Interpersonal Therapy Psychoanalytic/Psychodynamic Therapy Miscellaneous—Brief Dynamic Therapy	Psychoanalytic/Psychodynamic Therapy

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

Psychoanalytic Psychology
http://www.apa.org/pubs/journals/pap/index.aspx

Psychodynamic Practice
http://www.tandfonline.com/toc/rpco20/current

Psychodynamic Psychiatry
http://guilfordjournals.com/loi/pdps

Psychotherapy
http://www.apa.org/pubs/journals/pst/index.aspx

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- eFlashcards
- Culture expert interviews
- In My Practice demonstration video
- Full-text SAGE journal articles
- Additional web resources
- Mock assessment data

13

HUMANISTIC PSYCHOTHERAPY

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Like most psychologists of his era, **Carl Rogers** was trained psychodynamically. But he didn't stay psychodynamic for long. He came to disagree with many of Freud's presumptions about therapy and, more fundamentally, about people (Cain, 2010; Demanchick & Kirschenbaum, 2015). Along with **Abraham Maslow**, Rogers pioneered the humanistic movement in psychology and its clinical application, **humanistic therapy**.

The humanistic approach to understanding people stood in opposition to the biologically based, id-dominated, cynical Freudian view that prevailed at the time. Maslow (1968) summarized its theoretical foundation:

Web Link 13.1
Carl Rogers

> Inner nature [of people] seems not to be intrinsically or primarily or necessarily evil. . . . Human nature is not nearly as bad as it has been thought to be. . . .
>
> Since this inner nature is good or neutral rather than bad, it is best to bring it out and encourage it rather than to suppress it. If it is permitted to guide our life, we grow healthy, fruitful, and happy. If this essential core of the person is denied or suppressed, [the person] gets sick sometimes in obvious ways, sometimes in subtle ways, sometimes immediately, sometimes later. (pp. 3–4)

Learning Objectives

13.1 Summarize the goal of humanistic psychotherapy.

13.2 Describe the three essential therapeutic conditions of humanistic psychotherapy.

13.3 Explain how humanistic therapists use the technique of reflection.

13.4 Discuss therapeutic applications of motivational interviewing.

13.5 Contrast the focus of positive psychology interventions with the focus of most other current forms of psychotherapy.

13.6 Review the findings of humanistic psychotherapy outcome research.

Rogers applied this theory to people with psychological problems and, in doing so, offered an approach to therapy that remains quite influential today. Despite the fact that he shared his most essential ideas about psychotherapy about half a century ago and died in 1987, a survey of more than 2,400 contemporary psychotherapists found that Carl Rogers was the single most prominent figure in terms of influence on the way they practiced psychotherapy (Cook, Biyanova, & Coyne, 2009).

To simplify our discussion, in this chapter we use the term *humanistic* to cover the family of therapies created by Rogers and his followers. At various times, the terms *nondirective*, *client-centered*, and *person-centered* have been used by Rogers and others to describe these approaches.

HUMANISTIC CONCEPTS: CLINICAL IMPLICATIONS

Picture a small plant, just sprouting from the soil. We can confidently make several assumptions about this plant. First, we assume that the plant has an innate tendency to grow. In other words, given the proper environment, healthy growth will naturally occur. Second, we assume that the plant's growth utterly depends on sunlight—without it, it simply won't thrive. Third, we assume that if sunlight is available only from certain directions, the plant will bend, twist, or even contort its growth to reach it. Certain branches may blossom while others wither, and the plant may take on an unexpected shape in the process, but the nourishment from the sun is so essential that the plant will alter itself quite drastically to attain it. Of course, if sunlight comes from all directions, the plant need not alter itself at all. It can simply bloom according to its own inherent potential.

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Photo 13.1 Carl Rogers (1902–1987), a leading figure in humanistic psychotherapy

There are compelling parallels between this plant and human beings, according to the humanistic approach. Humanists assume that people, like plants, arrive with an inborn tendency to grow. Humanists call this tendency **self-actualization** and presume that if the person's environment fosters it, self-actualization proceeds without interference (Cain, 2002, 2010). Humanists also recognize that people need certain things to live, and just as plants need sunlight, people need positive regard. **Positive regard**, from the humanistic point of view, is essentially the warmth, love, and acceptance of those around us. (Carl Rogers's frequently used term **prizing** may best capture this experience of receiving positive regard from others; e.g., C. R. Rogers, 1959.) As children, we bask in the glow of positive

regard from our parents; like plants with sunlight, we need it to grow. If we discover that our parents provide positive regard only when we behave in certain ways, we will emphasize certain aspects ("branches") of ourselves and suppress others in order to attain it. The end result may be a version of ourselves that is markedly different from the version that might have blossomed if our parents had provided positive regard no matter what.

GOAL OF HUMANISTIC PSYCHOTHERAPY

The primary goal of humanistic psychotherapy is to foster self-actualization. Humanists believe that psychological problems—depression, anxiety, personality disorders, eating disorders, and most other forms of psychopathology—are the by-products of a stifled growth process. People who seek professional help for psychological problems have within them the capacity and the will to grow toward health, but, somehow, their growth has been interrupted or distorted. The task of the humanistic therapist is, through the therapeutic relationship, to create a climate in which clients can resume their natural growth toward psychological wellness (Erekson & Lambert, 2015).

If we are all guided by the self-actualization tendency from the beginning of our lives, how could we find ourselves depressed, anxious, or otherwise struggling with psychological issues? The answer lies in the fact that the need for positive regard can, at times, override the natural tendency to self-actualize. That is, when we face an either/or choice between receiving positive regard from the important people in our lives and following our own natural inclinations, we may, out of necessity, choose the positive regard.

Problems arise when this positive regard is conditional rather than unconditional. Conditional positive regard communicates that we are prized "only if" we meet certain conditions. If you consider your own family or those of your best childhood friends, you can probably identify some of the **conditions of worth** that parents place on their children. These conditions aren't posted as a list on the refrigerator, but they are clearly communicated nonetheless: We'll love you only if you get good grades, dress how we like, adopt our values, excel in sports, don't gain weight, stay out of trouble, and so on. Usually, kids can keenly sense the conditions their parents place on their acceptance, and because they need their parents' acceptance, they do their best to meet these conditions. In the process, however, they often go astray of their own self-actualization tendency, which may have guided them in another direction. Thus, when they compare the selves they actually are—the **real self**—with the selves they could be if they fulfilled their own potential—the **ideal self**—they perceive a discrepancy. Humanists use the term **incongruence** to describe this discrepancy, and they view it as the root of psychopathology (J. C. Watson & Bohart, 2015). In contrast, **congruence**—a match between the real self and the ideal self—is achieved when self-actualization is allowed to guide a person's life without interference by any conditions of worth, and, as a result, mental health is optimized. That is, congruence happens when persons experience unconditional positive regard from others. No "only if" conditions are placed on them for acceptance, so they are free to develop and grow according to their own self-actualization tendency (Cain, 2010).

It is important to note that although conditions of worth originally come from others, they can eventually become incorporated into our own views of ourselves. That is, conditional positive regard from others brings forth conditional positive *self*-regard,

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Photo 13.2
Humanists believe that, like plants, people have an inborn tendency to grow and that certain conditions foster their growth.

whereas unconditional positive regard from others brings forth unconditional positive *self*-regard. The important people in our lives communicate to us what is lovable, acceptable, or "prizeworthy" about ourselves—the whole self or only certain aspects—and, eventually, we adopt those views in our evaluations of ourselves.

As an example of these humanistic principles, consider Mark, a first-year college student. Unlike many of his classmates, who are undecided about a major or career path, Mark long ago decided that he would become a lawyer. During his first month on campus, he declared himself a prelaw major, mapped out a 4-year plan of undergraduate courses designed to enhance his chances of getting into law school, and made a preliminary list of law schools to which he intended to apply. Mark's interest in law was strongly encouraged by his parents. Both of them, as well as Mark's only sibling, are accomplished attorneys themselves. During Mark's childhood, they frequently commented on Mark's potential as a lawyer—"With these good grades, you're on the path to a top-notch law school," "It's wonderful knowing that we'll be able to hand the firm over to you someday," and "Listen to the argument he's putting up about his curfew—he's gonna make a great courtroom attorney someday!" At various points in his childhood, Mark showed interest in other activities, such as acting, sports, and journalism, but his parents never paid much attention to those activities or accomplishments. They were much more interested in his debate club activities, which they viewed as a precursor to his legal career, than in his plays, soccer games, or school newspaper articles.

In his second semester, Mark took an art history course to fulfill an elective requirement. Although he was reluctant to admit it, he found himself fascinated by the subject—much more so than by law. During his free time, he found himself reading ahead in his art history textbook and searching related topics online. He even borrowed some painting materials from his roommate and tried his hand at painting. When his parents arrived for a visit, however, he hid his interest in art from them. He sensed that they would reject it and, more significantly, reject him if they sensed he was passionate about it. They had made it clear that their acceptance of him was contingent on choosing law as a career path, and Mark sensed that they might not support him—emotionally, financially, or in any other way—if he followed his own intrinsic interests. Mark eventually became a lawyer, and although he achieved some success, he was always unhappy that his artistic side was never allowed to grow. In fact, Mark came to realize that whereas the legal "branch" of himself received plenty of "sunlight," numerous other branches had been neglected. When he compared his real life—as a lawyer with no other developed interests, abilities, or skills—with the ideal self he could have been, he noticed an incongruence that left him feeling dissatisfied and unhappy. If Mark were to find himself in a meaningful relationship with someone who prized him "no matter what"—a humanistic

therapist, for example—he might be able to view himself more unconditionally, follow his own self-actualization tendencies, and achieve greater congruence between his real and ideal selves.

ELEMENTS OF HUMANISTIC PSYCHOTHERAPY

Because self-actualization is a primary, natural tendency in all people, the therapist need only make the conditions right for it to occur. The therapist does not directly heal the client, per se; instead, the therapist fosters the client's self-healing tendencies toward growth. The therapist accomplishes this by creating a therapist–client relationship characterized by **three essential therapeutic conditions** (Cain, 2002, 2010; C. R. Rogers, 1957, 1959; Tudor & Worrall, 2006).

Web Link 13.2
Three essential therapeutic conditions

Empathy

A therapist experiences **empathy** for a client when the therapist is able to sense the client's emotions, just as the client would, to perceive and understand the events of his or her life in a compassionate way. Empathy involves a deep, nonjudgmental understanding of the client's experiences in which the therapist's own values and point of view are temporarily suspended. The therapist sees life through the client's eyes and adopts the client's frame of reference (C. R. Rogers, 1980). Indeed, the term *client-centered therapy*, often used synonymously with humanistic therapy, reflects this emphasis on empathic understanding (Bozarth, 1997).

When a therapist empathizes accurately and communicates that empathy effectively, it can have a profound, positive impact on the client. Empathy can enable clients to clarify their own feelings for themselves and have more confidence in the emotions they experience. It can also make a client feel valued and supported as an individual (Cain, 2002, 2010; Gillon, 2007).

Unconditional Positive Regard

Unconditional positive regard (UPR) is, essentially, full acceptance of another person "no matter what." Carl Rogers (1959) stated that the therapist proving UPR to a client

> prizes the client in a total, rather than a conditional, way. [The therapist] does not accept certain feelings in the client and disapprove of others. . . . It means making no judgments. It involves as much feeling of acceptance for the client's expression of painful, hostile, defensive, or abnormal feelings as for [the client's] expression of good, positive, mature feelings. (pp. 13–14)

Recalling the plant metaphor, sunlight is to plants as positive regard is to people. So a therapist providing UPR is very much like unobstructed sunlight coming from

BOX 13.1

Considering Culture

Empathy Across Cultures

According to the humanistic approach, empathy is fundamental to successful psychotherapy. But does culture place limits on the extent to which a therapist can empathize? To what extent can therapists accurately, genuinely empathize with clients who differ markedly from themselves in terms of ethnicity, gender, age, or other variables?

Consider Stephanie, a 38-year-old woman seeking therapy for depressive symptoms arising after a recent miscarriage. Would a female therapist be more capable of empathizing with Stephanie than would a male therapist? If the female therapist had never had a miscarriage, would that limit her ability to empathize? What if the female therapist had never been pregnant?

Or consider Namrata and Amit, a married couple seeking therapy. Both Namrata and Amit moved from India to the United States as children, and they both speak Hindi and English fluently. They are expecting their first child in about a month, and they are arguing intensely about the language they will speak at home after the baby is born. Namrata believes that they should speak only English at home, in an effort to enhance their child's ability to compete in schools and job markets in which she expects English to be the dominant language. Amit believes that they should speak Hindi at home, in an effort to enhance their child's connection to their heritage and culture. If an Indian, bilingual (Hindi/English) therapist is available, would that therapist have a greater capacity for empathy than a non-Indian therapist or a therapist who spoke only English? If so, would it matter whether that therapist had children?

Finally, consider Faye, a 75-year-old woman struggling with issues of loss and health. Specifically, her spouse and several of her best friends have died in recent years, and her own health has declined significantly as well. Until 10 years ago, she was quite healthy and strong, but she now finds herself in a wheelchair and dependent on caretakers for basic tasks. Can a therapist half her age who has never experienced such loss or decline genuinely empathize with Faye?

In general, can the personal background or experience of the therapist influence the therapist's ability to empathize with clients? Are some clients' problems so specific to their gender, ethnicity, age, or other variables that only someone with the same background can truly appreciate how the clients may feel? Or are our emotional reactions universal even though the events that evoke them may differ? Little empirical research has been conducted on this topic (e.g., Eisenberg, 2000; Feldstein, 1987; T. Graham & Ickes, 1997; Lennon & Eisenberg, 1987), and that which has been conducted has been marked by inconsistent results.

As a clinical psychologist, would you feel confident in your capacity for empathy for all clients? As a client, would you seek a clinical psychologist whose cultural background matched yours, with the assumption that it would maximize the psychologist's empathic understanding?

every direction. UPR allows clients to grow in a purely self-directed way, with no need for concern about losing the respect or acceptance of the other person in the relationship. It contributes to a climate in which clients realize they are free to be wholly

true to themselves, without modifying, amending, or retooling themselves to meet the standards of another person. When therapists unconditionally prize clients, over time, the clients may come to prize themselves unconditionally, which facilitates higher levels of congruence and self-actualization. One empirical study measured changes in congruence in about 100 clients in a college counseling center seeking therapy for a variety of problems. Results indicated a strong correlation between increases in congruence and overall therapy benefits (N. Watson, Bryan, & Thrash, 2014).

Each of us has experienced relationships in which we were appreciated not for our whole selves but for some specific features of ourselves—personality traits, behaviors, even material things. Although it may not be stated explicitly, the other people in these relationships make it clear that they will continue to accept us as long as we show them the sides of ourselves that they like and hide the sides that they don't. According to humanists, such relationships impede growth and eventually cause us to drift away from our true selves. Therefore, as therapists, humanistic therapists make it a top priority to accept clients entirely and unconditionally. This provides an opportunity for clients to grow naturally into their own potential rather than being pressured by others to grow in various directions (Cain, 2010; Tudor & Worrall, 2006).

Genuineness

Empathy and UPR are worthless if they aren't honest. Humanistic therapists must, therefore, be genuine in their relationships with clients. They don't act empathic toward clients or act as though they unconditionally prize them. Instead, they truly are empathic toward clients and truly do unconditionally prize them. This **genuineness**—which Rogers and his followers have also called therapist congruence, because there is a match between the therapist's real and ideal selves—is the opposite of playing a role or putting up a front. When we sense others (friends, family, or therapists) doing that, we tend not to reveal much of ourselves. On the other hand, when we sense that others authentically care about us and accept us, we tend to open up and engage more fully in the relationship (Gillon, 2007; C. R. Rogers, 1959; Tudor & Worrall, 2006).

Being genuine with clients helps humanistic therapists establish therapeutic relationships that feel "real." Such relationships differ strikingly from therapist–client relationships in which the therapist hides behind a facade of professionalism; instead, the therapist's personality plays a more prominent role. As might be expected, Carl Rogers and other humanists encourage a relatively high degree of transparency by the therapist. Unlike the "blank screen" psychodynamic therapist, humanists tend to be more forthcoming and candid about their own thoughts and feelings during sessions. However, they understand that the sessions are for the benefit of the client, not the therapist, and their self-disclosures are guided by this goal (C. R. Rogers, 1957).

These three conditions—empathy, UPR, and genuineness—are the essential elements of the relationship between humanistic therapists and their clients, which, in turn, is the cornerstone of the humanistic approach to psychotherapy (Cain, 2010). As Carl Rogers (1961) explained it,

If I can create a relationship characterized on my part

- by a genuineness and transparency, in which I am my real feelings;
- by a warm acceptance of and prizing of the other person as a separate individual;
- by a sensitive ability to see his world and himself as he sees them;

Then the other individual in the relationship

- will experience and understand aspects of himself which previously he has repressed;
- will find himself becoming better integrated, more able to function effectively;
- will become more similar to the person he would like to be;
- will be more self-directing and self-confident;
- will become more of a person, more unique and more self-expressive;
- will be more understanding, more acceptant of others;
- will be able to cope with the problems of life more adequately and more comfortably. (pp. 37–38)

Necessary and Sufficient?

When Rogers described empathy, UPR, and genuineness as the three core conditions for successful psychotherapy, he wasn't merely suggesting that they might be effective for some clients. His claim was much bolder: Those three conditions were both necessary and sufficient for psychotherapy to be successful with any client (C. R. Rogers, 1957). In other words, Rogers argued that to facilitate growth and self-actualization in clients with any kinds of problems, the therapist must provide only empathy, UPR, and genuineness. No additional techniques or procedures are necessary.

Rogers's assertion that these three elements are both necessary and sufficient for successful psychotherapy has generated a significant amount of controversy and research. Through the mid-1970s, the research was generally supportive of Rogers's claim, but since that point, results have been more uncertain and inconsistent. More recent research suggests that Rogers's core therapy ingredients are probably necessary but not always sufficient for psychotherapy to succeed. Perhaps they are best understood as a prerequisite for good therapy, a set of conditions that may be enough to facilitate significant improvement in some clients or set the stage for additional therapeutic methods that will cause significant improvement in others. Another interesting way of understanding Rogers's three core conditions—a way supported by empirical research—is to appreciate them as essential parts of the therapeutic relationship, whether the therapist is explicitly humanistic or not. Stated differently, empathy, UPR, and genuineness appear to be *common factors*, which (as we discuss in more detail in Chapter 11) contribute heavily to the success of all kinds of psychotherapy (Bozarth, Zimring, & Tausch, 2002; Zuroff, Kelly, Leybman, Blatt, & Wampold, 2010).

BOX 13.2

In My Practice . . .

In my practice, I have often wrestled with Carl Rogers's claim that the three core elements of humanistic therapy—empathy, unconditional positive regard, and genuineness—are both necessary and sufficient to help clients. Actually, I've never wrestled with the "necessary" part. Rogers's three elements seem so central to a solid therapeutic relationship that it's difficult to imagine therapy going well without them. It's the "sufficient" part that I find more vexing.

I've had some clients for whom Rogers's three elements were clearly not going to be sufficient. One client, Anthony, a 21-year-old college student, arrived in my office struggling with a depression that he hadn't been able to shake for years. His tendency to think negatively about his life, especially events in his past, was remarkably strong. I could, and did, empathize with him as he recounted all of the reasons he felt so continuously sad. Although his gloominess had worn out quite a few friends and alienated him from his own family, I accepted and even "prized" Anthony no matter what (i.e., I provided unconditional positive regard). And all of my responses toward Anthony were genuine: When I empathized with and prized Anthony, I wasn't just playing the therapist role, but truly identifying with his emotional pain and valuing his true self.

But Anthony needed more than this combination of empathy, unconditional positive regard, and genuineness could provide. He needed more targeted, evidence-based techniques. I used cognitive therapy (covered in Chapter 15) to help Anthony challenge his illogical thoughts about some of his past decisions. For example, when he beat himself up about his decision to pass up a part-time job opportunity, I reminded him that the decision wasn't catastrophic and that similar opportunities were likely to arise in the near future. I also utilized behavioral activation (covered in Chapter 14) with Anthony. Specifically, I helped Anthony make a list of specific behaviors he does when he's not depressed (like exercising and socializing) and then helped him make and follow a schedule in which he did them regularly. In Anthony's case, the Rogerian elements built a great therapy relationship that served as a foundation for techniques specifically suited to his symptoms.

But it would be a mistake to think that every client needs those techniques. I have found that there are indeed some clients for whom empathy, unconditional positive regard, and genuineness are remarkably sufficient. I describe these clients as "self-propelled" (motivated to work on their own issues). They also tend to be insightful and only minimally impaired. For these clients, a therapist who facilitates their progress, rather than one who forges it, can be ideal. Consider Tiana, a 30-year-old woman. Tiana was a people-pleaser, the kind of person who always agreed with her friends and family, for fear of angering them or causing conflict. (This tendency wasn't strong or debilitating enough to meet criteria for dependent personality disorder, but it certainly leaned in that direction.) From my first meeting with Tiana, it was clear that she was well aware of this tendency and highly motivated to change it, especially since it had produced much unhappiness for her, such as staying in dating relationships far too long. Her insights seemed very much on-target as well. She didn't need much more than a good therapeutic relationship, founded on the Rogerian elements of therapy, to give her the venue to work out many of her own issues. Of course, I contributed, but not by imposing a structured technique on her. Instead, I brought empathy, unconditional positive regard, and genuineness to our discussions of her issues, and Tiana largely navigated her own improvement.

To see this box come to life as a whiteboard video, please visit ***edge.sagepub/pomerantz5e****.*

Therapist Attitudes, Not Behaviors

Whether empathy, UPR, and genuineness are necessary, sufficient, or both, it is important to remember that humanists view them as attitudes, not behaviors (Bozarth, 1997; Tudor & Worrall, 2006). Humanists balk at formulaic, mechanical approaches to therapy, and, as such, they tend not to offer many specific suggestions about *what* therapists should *do* with clients. Rather, they emphasize *how* therapists should *be* with clients:

> Contrary to the opinion of a great many psychotherapists, I have long held that it is not the technical skill or training of the therapist that determines his success—not, for example, his skillful dream interpretations, his sensitive reflections of feeling, his handling of the transference, his subtle use of positive reinforcement. Instead, I believe it is the presence of certain attitudes in the therapist, which are communicated to, and perceived by, the client, that effect success in psychotherapy. (C. R. Rogers, 1959, p. 10)

REFLECTION: AN IMPORTANT THERAPIST RESPONSE

Although they believe that the therapist's attitude is more vital than any particular therapist action, humanists generally agree that one therapist behavior—reflection—can contribute significantly to the success of psychotherapy. It serves as a mechanism by which empathy, UPR, and genuineness can be communicated and as an expression of the attitudes that humanists emphasize.

Reflection takes place when a therapist responds to a client by rephrasing or restating the client's statements in a way that highlights the client's feelings or emotions (R. J. Campbell, 2004). Reflection is not a mere parroting of the client's words to show that they have been heard but a comment by the therapist that shows the therapist's appreciation of the client's emotional experience. (In fact, humanists often use the phrase "reflection of feeling" instead of the shorthand "reflection" to illustrate the emphasis on emotion.) When they reflect, humanistic therapists mirror their clients' affect, even if that affect is not explicitly stated.

As an example, consider Rosa, a single mother of two children (aged 7 and 9 years) who works a full-time job. Rosa tells her humanistic therapist about a typical day: "I get up at 6:00, make the kids' lunches, take them to the bus stop, get myself ready for work, and get there by 8:30. I'm at work until 4:30, fight traffic to pick the kids up from after-school care by 5:00, and then make dinner. After dinner, I help the kids with their homework and get them to bed, at which point I have time to clean the house, pay the bills, and do the other stuff I need to do. The next day, it's the same thing all over again."

Rosa's therapist could assure Rosa that she was paying attention by merely repeating some of the facts: up at 6:00, at work by 8:30, pick up kids by 5:00, and so on. However, an effective reflection of feelings would pick up on the emotional connotations of Rosa's statements, perhaps communicated by her tone of voice or body language more than by the words she chooses: "It's a long, demanding day, and it sounds like you'd be exhausted

by the end of it. Seems like you might feel underappreciated and maybe frustrated about your situation. Is that how you feel?" By mirroring the emotions in Rosa's words rather than just the words themselves, her therapist expresses empathy. By doing so nonjudgmentally, she communicates UPR. And by doing so honestly, she communicates genuineness. In combination and over time, these three conditions contribute to a strong therapeutic relationship and facilitate Rosa's growth.

Late in his career, Rogers expressed some regret about the way "reflection of feeling" had been used by many inside and outside the humanistic movement. He was particularly unhappy with the fact that reflection had been mistakenly taught and misunderstood "as a technique, and sometimes a very wooden technique at that" (C. R. Rogers, 1986, p. 375). Above all, Rogers believed, reflection should be an attitude rather than a technical skill. And this attitude should include some humility, which can be lost when therapists reflect mechanically. When they reflect, therapists should not be telling clients how they feel but, instead, should be asking clients if their understanding of the clients' feelings is correct. In other words, therapists should not become overconfident in their ability to read clients' emotions and should always defer to the clients' expertise on their own feelings. Rogers went so far as to "suggest that these therapist responses be labeled not 'Reflections of Feeling,' but 'Testing Understandings,' or 'Checking Perceptions.' Such terms would, I believe, be more accurate [in communicating] a questioning desire rather than an intent to 'reflect'" (p. 375). Although the terms *reflection* and *reflection of feeling* have remained, Rogers's reminders about the way they should be understood and used are nonetheless important.

BOX 13.3

Metaphorically Speaking

If You've Looked in a Magnifying Mirror, You Understand Reflection

We look into mirrors all the time—when we try on clothes, brush our hair, put on makeup, shave, and so on. Why? In a pinch, we could probably complete most of these tasks adequately without the mirror. What does the mirror provide? Essentially, it gives us feedback on our appearance. We are able to better appreciate our physical selves when our faces and bodies are mirrored back to us. Without mirrors, we might have a less accurate sense of how we look.

When a humanistic therapist reflects a client's feelings, a similar process takes place. The therapist provides the client with feedback about the client's emotions. Clients are able to better appreciate their emotional selves when their feelings are mirrored back to them. Without such reflection, they might have a less accurate sense of how they feel.

Ordinary mirrors provide some feedback, but magnifying mirrors provide feedback more strongly. Magnifying mirrors amplify the features of our faces, showing us parts of ourselves of which we may not have been fully aware and about which we may not feel entirely comfortable. The emotional mirroring by humanistic therapists can have a comparable effect: It can highlight emotions about which clients may not have been fully aware

(Continued)

(Continued)

and about which they may not feel entirely comfortable. For this reason, wise humanistic therapists are always careful to reflect feelings in a cautious, caring way, making sure that clients aren't overly alarmed by what they might see in the emotional mirror.

Likewise, Carl Rogers (1986) always encouraged therapists to take a humble approach to reflection such that any attempt to reflect feelings comes across as more of a question than a statement. After all, a therapist wouldn't want to be like a circus mirror, reflecting a distorted version of the client's feelings back to the client. Just as people who are uncertain about their bodies might be mistakenly convinced by a circus mirror that they are fatter, thinner, shorter, or taller than they really are, clients who are uncertain about their emotions might be mistakenly convinced by an inaccurate therapist reflection that they are angrier, sadder, happier, or more jealous than they really are.

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ALTERNATIVES TO HUMANISM

Historical Alternatives

Throughout the history of clinical psychology, numerous forms of psychotherapy have been influenced significantly by humanism. Here, we will discuss two of the most notable of these historical approaches—existential therapy and Gestalt therapy. Although it is accurate to say that their heydays have passed, they nonetheless hold a significant place in the evolution of psychotherapy and continue to influence many therapists today.

Existential psychotherapy is an approach to therapy originally developed by Rollo May, Victor Frankl, and Irvin Yalom. It centers on the premise that each person is essentially alone in the world and that realization of this fact can overwhelm us with anxiety. This anxiety may take a number of forms and is the root of all psychopathology. In addition to the inescapable conclusion of aloneness, existential theory holds that other inevitabilities of human life, especially death, contribute to a powerful sense of meaninglessness in many people. Existential therapists place great emphasis on clients' abilities to overcome meaninglessness by creating their own meaning through the decisions they make. They especially encourage clients to make choices that are true to themselves in the present and future, rather than choices that are determined by restrictive relationships they have had in the past. They empathize with the clients' reactions to the unavoidable facts of existence, but through questioning and discussion, they aid clients

in assuming control and assigning significance to their lives (Frankl, 1963; May, 1983; K. J. Schneider & Krug, 2010; Yalom, 1980). Existential therapy does not lend itself to empirical study, and few of its proponents were inclined to pursue research of that type. However, a meta-analysis of existential therapies found (in spite of limits in the quantity and quality of studies) that it works well for some clients, especially those who are physically ill or concerned with the meaning of life (Vos, Craig, & Cooper, 2015).

Gestalt therapy was founded by Fritz Perls, and it emphasizes a holistic approach to enhancing the client's experience. This experience includes both mental and physical perceptions, and Gestalt therapists attend to both these aspects of client communication. In practice, Gestalt therapists encourage clients to reach their full potential, often through the use of role-play techniques. They deemphasize clients' past experiences and instead focus almost exclusively on the present moment (labeled as "the now"). Integration and awareness of all parts of the self is viewed as a sign of personal growth, and, as such, it is thought to correlate with psychological well-being (Fagan & Shepherd, 1970; E. K. Gold & Zahm, 2008; Mackewn, 1997; Perls, 1969).

Motivational Interviewing

In more recent years, new offshoots of humanistic therapy have emerged, many of which emphasize briefer approaches (Tudor, 2008). The prime example—a therapy that has amassed significant empirical evidence and widespread influence on the field—is **motivational interviewing (MI)**, developed by William Miller. Miller describes his MI approach to therapy as a revised application of basic humanistic principles (Hettema, Steele, & Miller, 2005; W. R. Miller & Rose, 2009). MI was originally developed to treat addictive behaviors such as substance abuse, but it has been used with a wide range of client problems. MI centers on addressing clients' ambivalence or uncertainty about making major changes to their way of life. Whereas many therapists might label such ambivalence as resistance, denial, or a lack of motivation, MI therapists acknowledge that it is a normal challenge for anyone facing the difficult decision of continuing with an unhealthy familiar lifestyle or committing to live in a more healthy but unfamiliar way. A key to the MI approach is that its practitioners don't pressure clients to change, because such tactics may backfire, resulting in clients arguing against their own improvement. Instead, they help clients see the discrepancy between their behavior and their own values (or as Rogers would call it, their incongruence). By doing so, they *elicit* motivation from within the clients, rather than *imposing* it from without. This enables the clients to activate their own intrinsic values as inspiration to change their behaviors (Rubak, Sandbæk, Lauritzen, & Christensen, 2005). In short, clinicians using MI help clients decide to change for themselves (W. R. Miller & Rollnick, 2002; W. R. Miller & Rose, 2009; Moyers, 1998).

Web Link 13.3
Motivational interviewing

In addition to being used as a full-fledged treatment, MI can be used to augment other kinds of therapy, especially as a "jump-start" at the outset. For example, a client beginning cognitive or behavioral therapy may be reluctant if the need or goals for therapy seem to have been imposed by others, such as the therapist or family members. If the therapist utilizes some MI principles at the outset to successfully elicit the client's own motivation, the cognitive or behavior therapy may progress more smoothly. In fact, one empirical study found that clients who got a single session of MI before starting

cognitive-behavioral therapy for an anxiety disorder showed greater engagement than comparable clients who underwent the same cognitive-behavioral therapy without the single session of MI (T. L. Barrera, Smith, & Norton, 2016).

The central principles of MI reveal its humanistic roots (Arkowitz & Westra, 2009; Dennhardt, Yurasek, & Murphy, 2015; W. R. Miller & Rollnick, 2002; Moyers, 1998; Westra & Aviram, 2013):

- *Expressing empathy*. Taking the clients' points of view and honoring their feelings about their experiences are vital to MI.
- *Developing the discrepancy.* MI therapists highlight how a client's behavior is inconsistent with his or her goals or values. This enhances the client's self-motivation to change and puts him or her (rather than the therapist) in the position to argue for a new way of living.
- *Avoiding argumentation.* MI therapists do not directly confront clients, even if clients are engaging in self-destructive behaviors. They recognize that clients must choose to change rather than being strong-armed by a therapist.
- *Rolling with resistance.* When clients express hesitancy to change, MI therapists accept and reflect it rather than battle against it. They respect that clients have mixed feelings about changing. For example, clients may simultaneously want to stop drinking when they recognize that it has become a major problem *and* believe that their drinking is actually under control or that the benefits outweigh the drawbacks.
- *Identifying "sustain talk" and "change talk."* Resistance to change is often communicated by clients as *sustain talk.* Sustain talk takes the form of client statements in favor of continuing the problem behavior: "I've always eaten unhealthy food, and so does my whole family. I wouldn't even know what else to buy at the grocery store or order in a restaurant." *Change talk*, on the other hand, is the statements clients make in favor of changing the problem behavior: "My weight and my cholesterol are getting out of control. I'm so out of shape and unhealthy—I really need to eat healthier." MI therapists acknowledge that both these voices are within the client. They compassionately understand the internal dispute and allow clients to work out a resolution for themselves. Research investigating the specific ingredients within MI that make it successful has started to point toward change talk, which has been found to predict positive behavioral change (W. R. Miller & Rose, 2009).
- *Supporting self-efficacy.* MI therapists make efforts to communicate to clients that they have the power to improve themselves. The role of the therapist is facilitative; it is the client who has the most power for change.

Many empirical studies have been conducted on the efficacy of MI, with very impressive results. Meta-analyses of MI outcome studies have indicated that it produces beneficial outcomes for a remarkably wide range of problems, including both psychological and physical symptoms. It has been found to improve substance-related disorders, gambling,

smoking, exercise, weight-loss efforts, anxiety disorders, depression, violence toward intimate partners, behaviors that increase the risk of HIV, cholesterol levels, diabetes management, and blood pressure—primarily by sparking change in the behaviors that can either worsen or enhance these issues (Arkowitz & Westra, 2009; Burke, Arkowitz, & Menchola, 2003; C. D. Jensen et al., 2011; Lundahl & Burke, 2009; W. R. Miller & Rose, 2009; Musser & Murphy, 2009; Parsons, Lelutiu-Weinberger, Botsko, & Golub, 2014; Rubak et al., 2005; J. C. Watson & Schneider, 2016). And as explained earlier, MI can be integrated into other forms of treatment as well, broadening its potential benefits (Hettema et al., 2005). On lists of evidence-based treatments that tend to be dominated by behavioral and cognitive therapies, MI is clearly a bright spot for humanistic therapies.

Positive Interventions and Strength-Based Counseling

The positive psychology movement, which emerged in the 1990s under the leadership of Martin Seligman and has expanded rapidly in the 2000s, overlaps with some of the fundamental underpinnings of humanism. **Positive psychology** is a broad-based approach that emphasizes human strengths rather than pathology, and cultivation of happiness in addition to reduction of symptoms in psychotherapy (A. L. Duckworth, Steen, & Seligman, 2005; Seligman, 2011). This approach acknowledges the inherent potential of individuals to develop and maintain positive attributes based on such assets as hope, wisdom, creativity, courage, autonomy, optimism, responsibility, and growth. Moreover, it suggests that bolstering these strengths is an often overlooked way of preventing psychological problems such as depression and anxiety or improving the lives of those who already experience them (Seligman, 2003; Seligman & Peterson, 2003). Although positive psychology does not claim to derive explicitly from humanism, positive psychology and humanism share a basic view of people as possessing inborn strengths and capabilities that can guide them throughout their lifetimes and buffer them from unhappiness, and a corresponding commitment to clinical work designed to enhance those strengths and capabilities.

Web Link 13.4
Positive Psychology Center

Unlike practitioners of most other current forms of psychotherapy who focus more exclusively on a disease-based model, therapists influenced by positive psychology assume a therapeutic role that "embraces both healing what is weak and nurturing what is strong" (Seligman & Peterson, 2003, p. 313; see also Rashid, 2009; Snyder, Lopez, & Pedrotti, 2011). The latter emphasis—nurturing what is strong in clients—is a contemporary echo of Rogers's original theories. In particular, it captures the essence of healthy growth inherent in the self-actualization tendency, a core of the humanistic approach (Joseph & Patterson, 2008).

Therapies that derive from positive psychology go by a variety of names, but they are most often labeled *positive interventions* or *strength-based counseling*. As summarized by Seligman (2011), these therapies look past mere diagnosis-based symptom reduction to the enhancement of a client's overall well-being, particularly such aspects as the client's positive emotion, engagement with life, relationships, meaning, and achievement. These approaches build on the findings that happiness is a protective factor against mental disorders (Layous, Chancellor, & Lyubomirsky, 2014; Lyubomirsky & Layous,

2013). Numerous other authors have made similar suggestions to incorporate a more holistic or comprehensive understanding of the client that includes strengths as well as weaknesses. For example, the "four-front approach," developed by Beatrice Wright and others, encourages therapists to consider four areas of each client's life (Wright & Lopez, 2002, as described in Snyder et al., 2011; see also Snyder & Elliott, 2005; Snyder, Ritschel, Rand, & Berg, 2006):

- Weaknesses and undermining characteristics within the person
- Strengths and assets within the person
- Destructive factors and resources that are lacking in the environment
- Resources and opportunities in the environment

Such an approach ensures that a therapist will not only appreciate the positive and the negative elements in a client's life but also delineate which of those elements stem from the client and which stem from the environment in which the client lives.

Parks and Layous (2016) describe seven basic categories of positive psychology techniques, each of which could take on a variety of forms:

- *Savoring*, in which clients intentionally focus on and extend, without distraction, moments of joy and happiness
- *Gratitude*, in which clients purposefully focus on reasons to be thankful by writing letters, journaling, visiting important people in their lives, or simply thinking about things that make them feel appreciative
- *Kindness*, in which clients deliberately do nice things for others, including the donation of money or time for charity, volunteering, tutoring, or similar activities
- *Empathy*, in which clients intentionally build a sense of understanding, forgiveness, and perspective-taking with others in their lives
- *Optimism*, in which clients purposefully cultivate positive expectations about the future and anticipate good things that may happen in their lives
- *Strength-based activities*, in which clients deliberately use (or write about) their personal strengths in meaningful or novel ways
- *Meaning*, in which clients intentionally remind themselves of their own values and set goals to live a life that falls in line with them

In addition to being full-fledged forms of therapy themselves, positive interventions and strength-based counseling can be blended into other forms of therapy as well (Kalata & Naugle, 2009; Ward & Reuter, 2011). Although they are relative newcomers to the field of psychotherapy, positive interventions and strength-based counseling strategies have begun to accumulate some empirical data supporting their benefits. For example, Seligman and Steen (2005) found that interventions designed to increase happiness levels in clients did

indeed achieve their goal and, at the same time, reduced depressive symptoms. Others have explored the use of positive interventions with individuals who have experienced traumatic events such as active military combat, emphasizing posttraumatic growth and resilience rather than posttraumatic stress and anxiety (e.g., Bonanno, Westphal, & Mancini, 2011; J. A. Lee, Luxton, Reger, & Gahm, 2010). Additionally, clients may find positive psychology interventions more acceptable than other types—for example, one study found that clients receiving therapy for depression found both positive interventions and cognitive-behavioral therapy beneficial but found the positive psychology intervention more satisfying, a finding that may translate into greater adherence and lower dropout rates (Lopez-Gomez, Chaves, Hervas, & Vazquez, 2017).

Emotionally Focused Therapy

Emotionally focused therapy (EFT) is a short-term humanistic therapy that has garnered significant empirical evidence and popularity in recent years. It emphasizes the expression, acknowledgment, and healing power of emotions in the present moment, as well as emotions that may have been "bottled up" for a long time. Therapists encourage clients to experience their feelings wholly and completely, and show them unconditional acceptance when they do so. They encourage clients to share their feelings toward others, sometimes using an "empty chair" technique in which clients speak directly to an imaginary person such as a family member or partner toward whom they have typically withheld their feelings (Fitzgerald, Johnson, & Thomas, 2015; L. S. Greenberg, 2004). A noteworthy feature of EFT is that it is frequently practiced with couples. EFT with couples involves each person experiencing his or her own feelings more authentically and accepting the authentic feelings of the partner more fully, which enhances closeness and the likelihood of sharing feelings in the future. The goal is to replace patterns of blaming and withdrawing with patterns of empathy and self-disclosure (L. S. Greenberg & Goldman, 2008; S. M. Johnson, Hunsley, Greenberg, & Schindler, 1999). Numerous studies have found EFT beneficial to couples and to individuals with depression, PTSD, and several other clinical issues (Dessaulles, Johnson, & Denton, 2003; L. S. Greenberg & Watson, 2006; Greenman & Johnson, 2012, 2013).

Other Contemporary Alternatives

Another modern adaptation of humanism has been developed by Arthur Bohart and Karen Tallman. The title of the book in which they describe their approach—*How Clients Make Therapy Work: The Process of Active Self-Healing*—illustrates its emphasis (Bohart & Tallman, 1999). Bohart and Tallman argue that therapy is most effective when the therapist recognizes that

> the client is a creative, active being, capable of generating his or her own solutions to personal problems if given the proper learning climate. For us, therapy is the process of trying to create a better problem-solving *climate* rather than one of trying to fix the *person*. (p. xi)

The therapist's role, then, is not of a technician but of a collaborator with clients whose views and opinions are respected. Bohart and Tallman are explicit in their intention of offering a modern therapy that goes against the current movement toward symptom-focused, manualized, technique-dominated approaches to therapy. Such approaches, they believe, place clients in a passive role and underestimate the clients' own abilities to improve their lives. Bohart and Tallman believe that therapists should mobilize clients to help themselves, rather than paternalistically presuming they cannot and applying prescribed techniques to them.

HOW WELL DOES IT WORK?

Despite the empirical challenges inherent to humanistic psychotherapy—how to define and measure self-actualization and how to translate the humanistic attitude into well-defined therapist behaviors—Carl Rogers was a pioneer of psychotherapy outcome research. His approach may not lend itself to empirical tests as much as some other approaches to therapy do (such as behaviorism); nonetheless, he often attempted to present his ideas as testable hypotheses and included with his theoretical writings many ideas for empirical studies (Bozarth et al., 2002; Cain, 2002; R. Elliott, 2002).

As a professor at Ohio State University in the early 1940s, Rogers was the first to audio-record psychotherapy sessions and play them back on 78 RPM phonographs, and along with training, research was a primary use for this technology (C. R. Rogers, 1942). Soon after, he and his colleagues published some of the earliest controlled studies of psychotherapy outcome (R. Elliott, 2002). By the 1960s and 1970s, however, few humanistic therapists were carrying on Rogers's tradition in empirical research. A resurgence of research interest took place in the 1990s, as represented in some recent meta-analyses of humanistic therapy outcome studies (R. Elliott, 1996, 2002; L. S. Greenberg, Elliott, & Lietaer, 1994). One meta-analysis (R. Elliott, 2002) was the largest in scope, incorporating 86 separate studies of humanistic therapy that collectively reported on the results of more than 5,000 clients' experiences. The results, after controlling for researcher allegiance effects, indicate that humanistic therapies are generally about as effective as the other major approaches to psychotherapy. Other studies have also found that humanistic therapies produce results similar to those of other therapy approaches, with the most impressive effects for depression, relationship problems, coping with long-term illness, substance abuse, eating disorders, and psychotic disorders (Angus, Watson, Elliott, Schneider, & Timulak, 2015; R. Elliott, Greenberg, Watson, Timulak, & Freire, 2013; J. C. Watson & Schneider, 2016).

Some recent empirical efforts toward determining how well humanistic therapy works have focused on its specific elements—namely, empathy, positive regard, and genuineness. Results of these studies have repeatedly found that each of these elements plays an important role in the success of therapy—not only for humanistic therapy but for any kind of therapy. In other words, the extent to which empathy, positive regard, and genuineness are present, particularly from the client's point of view, correlates significantly with the success of the therapeutic relationship and, ultimately, with the success of the therapy itself—whether or not the therapist makes a deliberate effort to incorporate or emphasize these three elements. So even if a therapist doesn't identify as a humanist per

se, he or she would be wise to consider empathy, positive regard, and genuineness—the elements Rogers always considered essential—as empirically supported components of therapy, along with any technique he or she may also choose to incorporate (R. Elliott, Bohart, Watson, & Greenberg, 2011; Farber & Doolin, 2011; Kolden, Klein, Wang, & Austin, 2011; Zuroff et al., 2010).

As a final note regarding outcome data of humanistic therapies, it is worth reiterating a point made in the section on motivational interviewing: MI has become, especially in recent years, a bona fide evidence-based psychotherapy for a wide range of psychological and physical problems (Arkowitz & Westra, 2009; Hettema et al., 2005; Jensen et al., 2011; Lundahl & Burke, 2009; W. R. Miller & Rose, 2009; Musser & Murphy, 2009; Rubak et al., 2005). In terms of "what works" empirically, MI undoubtedly leads the pack of the current generation of therapies derived from humanism.

BOX 13.4

Denise in Humanistic Psychotherapy

In our first sessions, Denise directed our conversations toward two topics—her current dissatisfaction with her job and some personal family history that she saw as relevant. She informed me that, as a kid, she felt loved by her mother and father but only as long as she didn't stray too far from their expectations for her. One of these expectations, Denise explained, involved cooking. During her elementary and middle-school years, she demonstrated a remarkable interest and talent in the kitchen, and with two working parents and five siblings, the family came to rely on Denise to prepare meals. She explained that she felt as though her family came to value her for what she could accomplish in the kitchen. Denise explained to me that, until recently, her job as a chef had given her similar opportunities for appreciation for her cooking, but the new owner's policy that she stay in the kitchen and not talk with the customers had put an end to that.

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Seeing this experience through Denise's eyes, I could certainly understand why she had been feeling depressed. She had learned early on in her life that the important people prized her only for certain parts of who she was, and, like any of us, she needed that prizing enough to meet their conditions.

Eventually, she came to see herself in very much the same way as her family saw her—in other words, their conditional positive regard had become conditional positive self-regard. I wondered if, in the process of doing what others wanted from her, she had neglected some other aspects of her identity. If so, that must have been a frustrating experience for her. Perhaps there was more to appreciate about Denise than her cooking abilities; in

(Continued)

(Continued)

fact, maybe her worth as a person wasn't tied to cooking at all.

In therapy, I did my best to communicate my empathy to Denise—about her experience growing up, as well as the recent changes at work. I remember some of our exchanges vividly. Once, after she had spent about 10 minutes listing the differences the new restaurant owner had imposed—"He makes the menu decisions, he won't let me talk with the customers," and so on—I tried to reflect the feelings I sensed through her words, tone of voice, and facial expressions: "You seem upset about these changes—disappointed and maybe a little angry, too—am I right about that? Do you have other feelings about it?"

Although many of the important people in Denise's life had valued her for certain things—cooking, to be specific—I had a different appreciation of her. I saw Denise (as I see all my clients) as a worthy person no matter what she chose to do, and I suspected that she had the potential to be more multifaceted than her family, her boss, and she herself had allowed her to be. After a number of sessions, as she began to feel more comfortable with me, Denise mentioned, with some hesitancy, that sometimes she didn't even like to cook and that occasionally she found it monotonous and boring. She seemed to expect disapproval from me about this, but I was just as accepting of this feeling as I was of any of Denise's feelings. A bit later, she mentioned that there were other activities she felt more passionate, or at least curious, about but had never allowed herself to think of them as anything more than "dreams" or "fantasies." I pursued this topic with interest, and Denise had the courage to explain that she had always been fascinated with repairing cars. Her brothers and her father had done a lot of this during her childhood, but Denise was told that it "wasn't for her." I felt that any of Denise's interests were worthwhile, as long as they were true to herself, and I did my best to make this known to her.

Another aspect of Denise's personality that emerged over time was her bitter anger toward her family (about her past) and her new boss (about her current situation). I didn't see this anger at the beginning of therapy, but as time went on, she revealed it first through occasional comments about being "bothered" or "perturbed" and eventually through rants and outbursts that involved shouting and tears. I believed that these moments of anger were important to Denise because they were honest expressions of emotion, and although she seemed to assume originally that such outward anger was unacceptable, I did my best to let her know that I accepted her no matter how she felt.

With time, Denise seemed to internalize my unconditional prizing of her. She seemed to accept herself more fully and completely, allowing herself to recognize a wider range of feelings and interests than she had before. She even explored professional opportunities outside cooking (including a car technician training program). And at her current job, she no longer missed the positive feedback from the diners as sorely as she had before, largely because she wasn't so dependent on it. Instead of seeking positive regard from others based on their conditions, Denise was able to give herself positive regard unconditionally.

CHAPTER SUMMARY

Carl Rogers and his colleagues founded the humanistic approach to psychotherapy on a view of people as inherently striving to grow in a positive, healthy way. This self-actualization tendency at times conflicts with the need for positive regard (or "prizing") from others, particularly when others provide positive regard conditionally. In these situations, individuals experience incongruence between their real and ideal selves, and psychological problems ensue. Humanistic therapists foster self-actualization in their clients by establishing a therapeutic relationship in which conditions of worth are absent and congruence is encouraged. More specifically, humanistic therapists provide the three conditions that Rogers identified as necessary and sufficient for therapeutic gain: empathy, unconditional positive regard, and genuineness. These three conditions are defined more as therapist attitudes than techniques, but one therapist response, reflection of feeling, is a key component of humanistic therapy. The fundamental relationship-based elements of humanistic therapy may constitute common factors across many forms of therapy practiced by therapists who don't identify themselves as humanistic per se. A variety of contemporary approaches, including motivational interviewing, positive interventions/strength-based counseling, and emotionally focused therapy are strongly influenced by humanistic principles. Motivational interviewing, in particular, has achieved high status among psychotherapy practitioners and researchers due in large part to an extensive, growing body of empirical research supporting its benefits for psychological and physical problems. Outside the work on motivational interviewing, empirical research on the benefits of humanistic therapy in general has not been as extensive in recent decades as for some other forms of therapy. Those outcome studies that have been completed suggest that, typically, humanistic psychotherapy is about as beneficial as most other approaches.

KEY TERMS AND NAMES

conditions of worth 313
congruence 313
emotionally focused therapy (EFT) 327
empathy 315
existential psychotherapy 322
genuineness 317
Gestalt therapy 323
humanistic therapy 311
ideal self 313
incongruence 313
Abraham Maslow 311
motivational interviewing (MI) 323
positive psychology 325
positive regard 312
prizing 312
real self 313
reflection 320
Carl Rogers 311
self-actualization 312
three essential therapeutic conditions 315
unconditional positive regard (UPR) 315

CRITICAL THINKING QUESTIONS

1. To what extent do you agree with the humanistic ideas, as stated by Abraham Maslow (1968), that our "inner nature is good or neutral rather than bad" and that "if it is permitted to guide our life, we grow healthy, fruitful, and happy" (pp. 3–4)?
2. What are some of the most common conditions of worth that you have seen parents place on children? What effects of these conditions of worth have you observed?

3. Carl Rogers argued that empathy, unconditional positive regard, and genuineness were not only necessary but also sufficient for psychotherapeutic benefit. In your opinion, for which clinical problems is this statement most and least valid?
4. In your opinion, which elements of motivational interviewing are most essential to the success it has demonstrated in the treatment of a wide range of psychological and physical problems?
5. Considering the discussion of cross-cultural empathy in Box 13.1, do you believe that a therapist who is culturally similar to a client has a greater capacity for empathy than an equally competent but culturally dissimilar therapist?

LOOKING TOWARD GRADUATE PROGRAMS

The following table is a list of index entries in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018) relevant to the topics in this chapter (see page 25 in Chapter 1 for more information about the *Insider's Guide*). Keep in mind a few things as you consider these listings: They are current as of the 2018/2019 edition of the *Insider's Guide* (which typically revises every two years); they may have changed as faculty members retire, take new jobs, or develop new interests; and a graduate program's own website may have more extensive or up-to-date information.

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Positive Psychology/Resilience	Motivational Interviewing	Miscellaneous/Other—Client-Centered/Experiential Psychotherapies
Miscellaneous and Other—Existential/Phenomenological	Person-Centered Therapy	Miscellaneous/Other—Existential/Humanistic
Miscellaneous and Other—Humanistic Psychology	Positive Psychology/Resilience	
Miscellaneous and Other—Motivational Interviewing	Miscellaneous—Existential/Humanistic	
Miscellaneous and Other—Person-Centered Interventions		
Miscellaneous and Other—Strengths, Optimal Functioning		

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

Journal of Consulting and Clinical Psychology
http://www.apa.org/pubs/journals/ccp/index.aspx

Journal of Humanistic Counseling
http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)2161-1939

Journal of Humanistic Psychology
http://jhp.sagepub.com/

Psychotherapy
http://www.apa.org/pubs/journals/pst/index.aspx

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- Self-quizzes
- eFlashcards
- Culture expert interviews
- In My Practice demonstration video
- Full-text SAGE journal articles
- Additional web resources
- Mock assessment data

14

BEHAVIOR THERAPY

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Behavioral Consultation 355
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Teacher Training 357
How Well Does It Work? 359

Perhaps the best way to transition from the previous chapters on psychodynamic and humanistic therapy to the current chapter on behavior therapy is to imagine that your clinical psychology class includes one very empirically oriented (and outspoken) student. When your professor discusses a psychodynamic therapy concept such as, say, defense mechanisms, your classmate raises his hand and asks, "How can you be sure that's what happens in the mind? Can you scientifically test that theory?" When your professor describes the three "necessary and sufficient" conditions for successful therapy according to the humanistic approach, your classmate interrupts to ask, "How, exactly, do you define empathy, unconditional positive regard, and genuineness—what specific therapist behaviors do they consist of? And if those conditions really do help a client self-actualize, how exactly would you be able to observe or measure that outcome?"

Learning Objectives

14.1 Discuss the origins of behavior therapy.

14.2 Explain the goal of behavior therapy.

14.3 Differentiate between classical and operant conditioning.

14.4 Describe behavior therapy techniques based on classical conditioning.

14.5 Review behavior therapy techniques based on operant conditioning.

14.6 Examine alternative approaches to behavior therapy.

14.7 Summarize the findings of behavior therapy outcome research.

Again and again, your professor would have little choice but to respond to your classmate's questions by conceding that the psychodynamic and humanistic approaches to therapy are not entirely empirical. In spite of their intuitive appeal and their clinical successes, they are characterized by speculations about mental processes that can't be precisely defined, directly observed, or scientifically tested. If your classmate insists on a therapy approach grounded in empiricism, he'll need to look elsewhere—to behavior therapy.

ORIGINS OF BEHAVIOR THERAPY

Behavior therapy is the clinical application of behavioral principles, which have theoretical and experimental roots extending back hundreds of years. A landmark in the history of behaviorism is the work of **Ivan Pavlov**, whose contributions took place in Russia in the late 1800s and early 1900s. Pavlov did not begin his career with aspirations related to psychology at all. In fact, he was a physiologist who studied the digestive systems of dogs. Routinely, Pavlov and his coworkers would present food to dogs and measure the amount of saliva the dogs produced—a natural, automatic response from any canine shown a potential meal. At one point, he noticed that the dogs were salivating before the food was presented. Pavlov first saw this untimely salivating as a problem, and he tried to minimize it so it wouldn't adversely affect the digestion data he was trying to collect. Later, he came to realize that he had inadvertently come across a remarkable phenomenon, which was ultimately labeled "classical conditioning." Through their experience, Pavlov's dogs had learned that food was often preceded by a particular stimulus: the sight of the researcher, the sound of the food being prepared, or similar events. Soon, the dogs were salivating as soon as they perceived a stimulus that, through repeated pairing with food, predicted food. Classical conditioning eventually replaced digestive research as Pavlov's primary interest, and, using a bell as the precursor to food, he conducted many pioneering studies on the topic (M. H. Hunt, 1993; Kazdin, 1978). A bit later in this chapter, we examine classical conditioning and its clinical applications in more detail.

Web Link 14.1
Website of journal *Behavior Therapy*

Web Link 14.2
Ivan Pavlov

The work of Pavlov and other Russian researchers made its way to the United States via **John Watson**. In the early 1900s, Watson fervently argued that the lessons learned from Pavlov's dogs applied to human behavior as well and that, as a result, psychology should refrain from focusing on the inner workings of the mind and should instead examine the ways conditioning shapes behavior. Watson was among the first prominent figures in American psychology to argue that psychology should study only overt, observable responses and the overt, observable stimuli that precede them rather than the inner workings of the mind that may occur in between. Feelings, thoughts, consciousness, and

other internal, mental processes were simply not suitable for scientific study, nor were they as powerful as conditioning in determining behavior (M. H. Hunt, 1993; Kazdin, 1978; J. B. Watson, 1924).

Web Link 14.3
B. F. Skinner Foundation

While Pavlov and Watson ushered in classical conditioning to the field of psychology, a second essential type of conditioning—operant conditioning—was heralded by Edward Lee Thorndike and **B. F. Skinner**. Actually, Thorndike's (1931) primary contribution, based on his research with cats, was a rather straightforward theory he called the **law of effect**. Essentially, Thorndike's law of effect stated that all organisms pay attention to the consequences (or effects) of their actions. Actions that are followed by pleasurable consequences are more likely to recur, whereas actions that are followed by unpleasant consequences are less likely to recur. Skinner effectively devoted much of his life's work to experimentation on the law of effect, and he made the case that operant conditioning, the mechanism by which the law of effect influenced behavior, was as great an influence on human behavior as classical conditioning. Skinner's research and writings became tremendously influential to the behavioral perspective on the origins of psychological problems and the techniques by which they could be treated (Bjork, 1993; M. H. Hunt, 1993; Kazdin, 1978; Vargas, 2015). Thus, through Pavlov, Watson, Thorndike, and Skinner, behaviorism evolved from basic research on animal behavior to an applied form of psychotherapy.

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Photo 14.1
B. F. Skinner's studies of operant conditioning play a crucial role in numerous forms of behavior therapy.

GOAL OF BEHAVIOR THERAPY

The primary goal of behavior therapy is observable behavior change. This goal stands in stark contrast to the goals of the psychodynamic and humanistic approaches, each of which emphasizes internal, mental processes—making the unconscious conscious and fostering self-actualization, respectively. In fact, the emergence and rise of the behavioral approach stemmed from strong dissatisfaction with numerous aspects of the psychodynamic approach that dominated through the early and mid-1900s (Craske, 2010). As a reaction against perceived weaknesses of the prominent approaches of the times, early behaviorists forged a new therapy differentiated from the outset by a distinct set of characteristics (Kazdin, 1978, 1980; M. G. Newman, LaFreniere, & Shin, 2017; Spiegler & Guevremont, 2010; Wenzel, 2017; Yates, 1970).

Emphasis on Empiricism

Behavioral therapists take the stance that the study of human behavior, whether normal or abnormal, should be scientific (Kazdin, 1978; Yates, 1970). As such, clinical psychologists

treating clients should employ methods that can be scientifically evaluated. Theories regarding the treatment of problem behaviors should be stated as **testable hypotheses**; in this way, they can be supported, refuted, modified, and retested. In contrast, if theories of change are not stated as testable hypotheses, they lack scientific rigor and might be best classified as conjecture, inference, or even guesswork. Of course, any scientific process involves data collection, and behavior therapy is no exception. Behavioral therapists regularly collect **empirical data** on their clients—as a baseline measure at the outset of therapy, at various points during the therapy to evaluate changes from session to session, and at the end of therapy as a final assessment of change (P. Grant, Young, & DeRubeis, 2005; Spiegler & Guevremont, 2010).

Expanding on the scientific nature of behavior therapy, Truax (2002) explains that "the endeavor of developing and testing hypotheses in the real clinical setting is much like that of a scientist" (p. 5). Indeed, behavior therapy is a clinical application of the five steps that are common to the scientific method across all disciplines. Table 14.1 summarizes this association between the scientific method and behavior therapy.

Table 14.1 Application of the Steps of the Scientific Method by Behavioral Therapists

Steps of the Scientific Method	How Applied by Behavioral Therapists
1. Observing a phenomenon	Assessing client behavior via observation, interview, or testing Defining a target behavior Establishing a baseline level of target behavior
2. Developing hypotheses to explain the phenomenon	Functionally analyzing target behavior to determine the factors that cause or influence it Establishing specific behavioral goals for treatment Planning interventions to alter behavior in preferred manner
3. Testing the hypotheses through experimentation	Implementing interventions as planned
4. Observing the outcome of the tests	Collecting data on changes in the target behavior Comparing data collected during or after treatment to baseline data Comparing data to goals
5. Revising the hypotheses	Modifying treatment plan as suggested by observed outcomes Restarting scientific process with revised hypotheses

Source: Adapted from Truax (2002).

Defining Problems Behaviorally

According to behavioral therapists, client behaviors are not symptoms of some underlying problem—those behaviors *are* the problem. For example, consider Ryan, who has a habit of excessively checking the front door of his apartment at bedtime to make sure it's locked. According to behavioral therapists, Ryan's door-checking habit is exactly that—a habit. It need not signify that a deeper, diagnosable problem resides within Ryan. Behavior therapists would see little benefit in defining Ryan's problem as obsessive-compulsive disorder, because to do so would suggest that the door checking is part of a cluster of symptoms that share a common source within Ryan's mind. They would prefer not to make unproven (actually, unprovable) inferences about the internal causes of Ryan's behavior and instead focus exclusively on Ryan's door-checking behavior itself as the problem to be addressed.

From the behavioral point of view, a benefit of defining problems in behavioral terms is that such definitions make it easy to identify target behaviors and measure changes in therapy (Spiegler & Guevremont, 2010). Such definitions can differ quite drastically from the definitions of their own problems offered by clients. As an example, consider Amber, a 30-year-old client who sought help from Dr. Tyler, a behavioral therapist, because Amber believed she "had depression." Amber entered therapy convinced that the depression was an entity, a "thing" within her mind that couldn't be seen directly but was nonetheless affecting her in numerous adverse ways. She said that the main "symptoms" of her depression were feeling sad and thinking pessimistically. Dr. Tyler redefined Amber's problems in two important ways. First, she asked Amber to describe the observable, measurable *behaviors* she most wanted to change. That is, rather than focusing on feelings and thoughts—none of which can be directly observed or measured—Amber was asked to focus on behaviors she performed too often or too rarely. With some help from Dr. Tyler, Amber was able to list many such behaviors: sleeping too much, missing work too frequently, crying too often, and exercising too little. Because these problems are actions that can be observed and measured, they are ideal targets for behavioral interventions. The second way Dr. Tyler redefined Amber's problem was to explain to Amber that these behaviors weren't symptoms of a deeper problem; instead, they were the problems themselves. After establishing **baselines**—measuring the pretreatment frequencies or durations of Amber's sleep, work attendance, crying, and exercising—Dr. Tyler proceeded by designing and implementing interventions that targeted each of Amber's problem behaviors. As therapy progressed, they measured Amber's behaviors on a regular basis and compared that data with her baseline data to assess her improvement.

Whereas most other types of therapy approaches endorse a medical model of psychopathology, the behavioral approach, as exemplified by Dr. Tyler, does not. That is, most other therapy approaches conceptualize psychological problems as medical doctors do physical diseases, especially with regard to the notion of symptoms as manifestations of underlying diseases. Shortness of breath, for example, might be a symptom of asthma; coughing, a symptom of bronchitis; abdominal pain, a symptom of an ulcer. Behavioral therapists don't adopt this medical-model view of psychological problems, largely because the connections between "symptoms" and the underlying "diseases" they indicate are largely speculative in psychology, whereas they are more concrete in

medicine. Because we cannot know empirically that a particular behavior is symptomatic of any particular underlying problem, behavioral therapists regard behavioral problems themselves as the most appropriate focus of treatment.

Measuring Change Observably

For behavioral therapists, measuring therapy outcome via **observable changes** goes hand in hand with defining the client's problems behaviorally from the outset. That is, whereas other kinds of therapists may measure change in clients in more inferential ways, behavioral therapists use more unambiguous indications of progress (P. Grant et al., 2005; Spiegler & Guevremont, 2010).

To illustrate, recall the psychoanalytic and humanistic therapists we studied in the previous chapters. They would have to look in the client's mind for some indication that their therapy has produced an effect. For empiricists such as behavioral therapists, this method of measuring change is unacceptable, primarily because it lacks objectivity. How do we know that the therapist (or the client, or anyone else for that matter) has a reliable, valid, unbiased take on how much insight the client has gained? How can we be certain that the growth a client has experienced is a result of an improved sense of congruence? Behavioral therapists reject **introspection** such as this—that is, they reject the notion that we can simply look inside the mind and attain an objective, accurate assessment of change. Instead, they focus on outward demonstrations of change in clients—overt behavior rather than covert mental processes—as indicators of client change.

Behavior therapists don't simply reject introspection as a means of measuring change in therapy; indeed, they believe that introspection should have no role at all in the clinical process. Therefore, when they consider the contributing factors to clients' problems, they emphasize external, environmental factors over internal personality traits. As an example, consider Jack, a 40-year-old man who is seeking therapy from Dr. Herrera, a clinical psychologist with a behavioral orientation. Jack is 100 pounds overweight and has gained 20 pounds in the past year. He explains to Dr. Herrera that he has tried to lose weight many times but always quits his exercise routine and returns to unhealthy eating habits—chips, cookies, fast food, and so on. At this early point, some therapists would already be speculating about Jack's internal personality characteristics as a cause of his overeating—for example, a psychodynamic therapist may consider an oral fixation. The problem with such a speculation, from a behavioral point of view, is that it cannot be supported or refuted, because it hinges on Jack's mental processes, which are not directly observable. Dr. Herrera makes no such speculations. Instead, the questions he poses to Jack involve observable aspects of the environment in which Jack lives. In other words, what events precede Jack eating a bag of chips or skipping a workout? What consequences follow these actions? In what settings and under what conditions do the unwanted behaviors take place? Questions such as these make it clear that Dr. Herrera, as a behavioral therapist, looks for the causes of Jack's problems not within Jack but in the world around him. In other words, Jack's too-frequent unhealthy eating and too-infrequent exercising are responses to stimuli and consequences from the world around him rather than expressions of some internal flaw in his personality.

TWO TYPES OF CONDITIONING

As discussed in the previous section, behavioral therapists contend that our behavior is the by-product of conditioning (also known as learning). If conditioning explains all behavior, then, by definition, it explains the acquisition of disordered behavior as well as the means by which such behavior can be modified by clinical psychologists (Craske, 2010). Behavioral therapists separate conditioning into two primary types: classical and operant.

Classical Conditioning

Classical conditioning is the type exemplified by Pavlov's dog studies. In those studies, dogs learned through experience that certain stimuli (a bell, e.g.) predicted that food would be delivered, and, as a result, they began to salivate in response to the stimuli. If we examine this process more closely, its four components can be identified. Let's start before any conditioning takes place, before the dog has had a chance to learn how things work in Pavlov's lab. In fact, let's imagine that the dog has just walked into Pavlov's lab for the first time, so everything it experiences or does is completely "unconditioned." In this situation, the dog will salivate when food is presented. So food is the **unconditioned stimulus**, which evokes salivation, the **unconditioned response**. No dog needs conditioning to learn that association—it's inborn. After spending some time in Pavlov's lab, the dog notices that when a certain bell rings, food comes a few seconds later. In other words, the dog is now becoming conditioned, or learning that the sound of the bell predicts the arrival of food. The dog soon begins to salivate in response to the bell. The sound of the bell, which had originally evoked no response at all in the dog, has become a **conditioned stimulus**, and salivating, when in response to the bell (not the food), is the **conditioned response**.

Several aspects of classical conditioning are worth emphasizing. First, it is important to note that classical conditioning is a rather passive type of learning. Pavlov's dogs didn't need to do much of anything to experience the conditioning, other than remain awake and alert. It's almost as if the classical conditioning happened to them—some combination of sights, sounds, smells, or tastes occurred in rapid succession, and they happened to be there to experience them. Second, a number of variables that can influence behavior surround the classical conditioning process. For example, the extent to which an individual demonstrates behavior that has been classically conditioned will depend on the extent to which generalization or discrimination takes place. **Generalization** occurs when the conditioned response is evoked by stimuli that are similar to, but not an exact match for, the conditioned stimulus. **Discrimination** occurs when the conditioned response is not evoked by such a stimulus (Kazdin, 1978, 1980). For example, if Pavlov conditioned a dog to salivate in response to the sound of a particular bell, would the dog also salivate to the sound of a slightly different bell? What about a very different bell or even a completely different sound, such as a drum? As a clinical version of the same questions, imagine that Wayne, a 16-year-old boy, developed a strong fear response to dogs after being viciously attacked by a golden retriever. To what extent would Wayne generalize his fear? Would

he be fearful of all golden retrievers, of dogs of any breed, of all animals of any kind? Conversely, might Wayne discriminate such that he responds fearfully to the one dog that attacked him but no others? The extent to which Wayne generalizes or discriminates his fear response would strongly influence the strategy of a behavior therapist helping Wayne overcome his fear.

Operant Conditioning

Operant conditioning occurs when the organism "operates" on the environment, notices the consequences of the behavior, and incorporates those consequences into decisions regarding future behavior (Sturmey, Ward-Horner, Marroquin, & Doran, 2007). Simply put, "the basic principle of the operant approach is that behavior is a function of its consequences" (Calhoun & Turner, 1981, p. 4). Operant conditioning is a more active style of learning than classical conditioning. For operant conditioning to take place, the organism must take an action of some kind. As a fictional example, imagine that one of Pavlov's dogs had, during a break in the experimenting, started to whine. If the whining brought forth treats, the dog would be more likely to whine again. If the whining brought forth a whack on the nose, the dog would be less likely to whine again. In either case, the point is that the dog experimented with a new behavior, paid attention to the outcome, and will use that outcome as a factor in future decisions about whining.

Skinner and other proponents of operant conditioning proposed that consequences shape all behavior, including behavior labeled as abnormal. In effect, all our actions are governed by **contingencies**, by "if . . . , then . . ." statements, including those labeled as abnormal. By changing the contingencies or revising the "if . . . , then . . ." statements that control a client's behavior, clinical psychologists can induce significant behavioral changes.

Thus, both classical and operant conditioning can be applied clinically to address problematic or unwanted behaviors in clients. Indeed, these two types of conditioning form the foundation for most behavior therapy techniques. Let's examine some of the most widely known and used techniques deriving from classical and operant conditioning.

TECHNIQUES BASED ON CLASSICAL CONDITIONING

Exposure Therapy

Simply put, **exposure therapy** is the clinical psychologist's version of "facing your fears." Phobias, according to the behavioral therapist, are best understood as the result of classical conditioning: A particular stimulus (spiders, heights, the dark, etc.) becomes paired with an aversive outcome (anxiety, pain, etc.). This pairing can be weakened and ultimately eliminated if the client experiences one without the other. That is, when the client is repeatedly "exposed" to the feared object and the expected aversive outcome does not take place, the client no longer experiences the fear response, which is a more

appropriate and rational way to react to such harmless stimuli (Hazlett-Stevens & Craske, 2008).

As a clinical example, let's reconsider Wayne, who was described earlier as having a fear of dogs as a result of a vicious attack. The essential task for a behavioral therapist conducting exposure therapy is to expose Wayne to dog-related stimuli. Based on his attack, Wayne has associated dogs with fear, but the truth is that for the vast majority of dogs, this is simply an inaccurate expectation. Unfortunately for Wayne, his avoidance of dogs since the attack has prevented him from having the opportunity to "unlearn" the association between dogs and fear. Through exposure therapy, his behavior therapist gives him exactly that opportunity, and the result of the exposure to dog-related stimuli is that Wayne's association between dogs and fear is unlearned.

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Photo 14.2 Facing feared stimuli—such as dogs, for someone previously attacked by a dog—is the essential element of exposure therapy.

The behavior therapist has several choices to make when conducting exposure therapy. One of the most important involves the imaginal versus in vivo nature of the stimuli to which the client will be exposed. In other words, the client can be asked to imagine anxiety-provoking objects (without ever being exposed to the real thing) or can be exposed to real-life (in vivo) items or situations that have produced fear (Hazlett-Stevens & Craske, 2008; Spiegler & Guevremont, 2010). In Wayne's case, **imaginal exposure** would involve visualizing dogs and dog-related items, whereas **in vivo exposure** would mean that Wayne would directly see, hear, and touch dogs. Imaginal exposure can also take the form of writing, such that clients are asked to write about things that cause them anxiety (rather than experiencing those things in vivo). Written exposures have been found beneficial in treatment of clients with severe anxiety stemming from a traumatic event, such as a car accident (Wisco, Baker, & Sloan, 2016; Wisco, Sloan, & Marx, 2013). Sometimes, in vivo exposures are interoceptive, or sensations within the client's own body. For example, a therapist may ask a client to climb a flight of steps to raise the client's heart rate if the client struggles with anxiety that occurs when he or she notices his or her heart beating fast (Abramowitz, Fabricant, & Jacoby, 2015).

Another important choice regarding exposure therapy involves the extent to which the client will be exposed to fear-inducing stimuli: gradually or all at once. The gradual approach is often called **graded exposure**, and it requires the client and therapist to collaboratively create an **anxiety hierarchy** in which they list about 10 stimuli that might induce fear. These stimuli are typically rated by the client on a scale from 0 to 100 in terms of the amount of subjective distress they produce and are then ranked in ascending order. Exposure begins at the lowest level and then proceeds through the hierarchy until the client reaches the highest level. Table 14.2 illustrates a graded in vivo hierarchy for Wayne's fear of dogs. Exposure that happens all at once rather than gradually is typically called **flooding** or **implosion** (Levis, 2008; Zoellner, Abramowitz, Moore, & Slagle, 2008). Although either of these all-at-once approaches can produce rapid change quickly, they can also be intolerable or even traumatizing for clients and are used cautiously and infrequently (S. Taylor, 2002; Yates, 1970).

Web Link 14.4 Exposure and response prevention

Table 14.2 Example of an Anxiety Hierarchy for Graded in Vivo Exposure for Fear of Dogs

Stimulus	Subjective Distress (0 = No Fear; 100 = Maximum Fear)
1. Hearing a dog bark in another room	5
2. Seeing a dog through a window	15
3. Standing within 20 feet of a dog on a leash	25
4. Standing within 10 feet of a dog on a leash	35
5. Standing within 5 feet of a dog on a leash	45
6. Standing within arm's length of a dog	55
7. Petting a dog for 1 second	65
8. Petting a dog continuously for 10 seconds	75
9. Petting a dog continuously for 1 minute	85
10. Petting a dog continuously for 3 minutes	95

Exposure therapies are most commonly used with clients who have phobias and other anxiety disorders, as well as PTSD (Rauch & Rothbaum, 2016). One particular application of exposure therapy, called **exposure and response prevention**, has received substantial empirical support for the treatment of obsessive-compulsive disorder, making it the clear treatment of choice for the disorder (Craske, 2010; M. E. Franklin & Foa, 2011; Gillihan & Foa, 2016). It involves graded exposure (as described above) to the obsessive thoughts (e.g., "My hands are filthy—I must wash them") or the situations that elicit such thoughts while simultaneously preventing the client's typical response (e.g., hand washing), which brings temporary relief but has come to interfere with the client's daily life (Rowa, Antony, & Swinson, 2007).

Systematic Desensitization

Systematic desensitization, a treatment also used primarily for phobias and other anxiety disorders, is quite similar to exposure therapy—in fact, exposure to anxiety-provoking stimuli is one of its key components—but rather than simply breaking the association between the feared object and the aversive feeling, systematic desensitization involves re-pairing (or **counterconditioning**) the feared object with a new response that is incompatible with anxiety. When exposure therapy works, the feared object is eventually paired with nothing (rather than the fear response), but when systematic desensitization works, the feared object is paired with a new response that replaces and blocks the fear response (Head & Gross, 2008; McGlynn, 2002; Wolpe, 1958, 1969).

Most often, the new response that replaces and blocks the fear response is relaxation. In fact, the first step of systematic desensitization is *relaxation training*, in which the behavior therapist teaches the client progressive relaxation techniques in which various muscles are systematically tensed and relaxed. Usually, the behavior therapist uses scripted instructions during relaxation training, and these instructions can be shared with the client in a variety of forms (written on paper, recorded on CD, etc.) to facilitate practice at home between sessions. Once clients learn and master the relaxation task (usually about half a dozen sessions), they move through an anxiety hierarchy as described in the section above on exposure therapy. The only difference is that prior to each exposure, the client achieves a relaxed state, such that exposure to the anxiety-producing stimulus becomes paired with relaxation, which inhibits anxiety. Thus, the feared object is no longer paired with anxiety but instead is paired with relaxation, and, gradually, the fear is overcome (Head & Gross, 2008; McGlynn, 2002; Wolpe, 1958, 1969).

In Wayne's case, a behavior therapist using systematic desensitization would construct the anxiety hierarchy and conduct the exposures just as in exposure therapy, but relaxation training would happen first and the relaxation response would be deliberately evoked during the exposure process to facilitate the replacement of anxiety with a new, anxiety-preventing response.

Assertiveness Training

Assertiveness training is a specific application of classical conditioning that targets clients' social anxieties. It is best suited for people whose timid, apprehensive, or ineffectual social behavior has a negative impact on their lives. In practical terms, it can help clients insist on appropriate service (in a restaurant, e.g.), ask someone out on a date, request a raise at work, communicate effectively with health care providers, or say no to an unreasonable demand from a friend or loved one.

Assertiveness training definitely includes elements of exposure therapy, and it may include elements of systematic desensitization as well (M. P. Duckworth, 2008; Gambrill, 2002). The exposure comes in the form of facing interpersonal fears. That is, people who have problems with assertiveness usually avoid situations in which assertiveness might be called for, so by simply exposing themselves to such situations and producing any kind of assertive response, they are taking a significant step forward. The counterconditioning component of systematic desensitization may also come into play, with assertiveness replacing relaxation as the new response that replaces and inhibits anxiety.

Assertiveness training usually begins with direct instructions from the behavior therapist with which the client is taught specifically what to say and do in a particular situation. Next, effective assertive behaviors are modeled for the client. The behavior therapist often does this modeling, but video-recorded models or even live actors can be used as well. Subsequently, the client is given an opportunity to rehearse the assertive behavior in a role-play situation, and the therapist provides specific, constructive feedback. Eventually, clients are given targeted homework assignments in which they are expected to practice their improving assertiveness skills in the "real world" (Gambrill, 2002).

As a clinical example, consider Deborah, a 26-year-old woman who got engaged a few months ago. She described to Dr. Paxton, her behaviorally oriented clinical

BOX 14.1

Considering Culture

Assertiveness Training and Collectivist Values

Assertiveness training, as a form of behavior therapy, is based on the assumption that assertiveness is a good thing: "A value stance . . . is associated with assertion training. It is assumed that people have a right to express their feelings in a way that subjugates neither others nor themselves, and that well-being includes this expression" (Gambrill, 2002, p. 121). Do all cultures hold this value equally?

Some would argue strongly that they do not. D. W. Sue and Sue (2008), for example, point out that "individualism, autonomy, and the ability to become your own person are perceived as healthy and desirable goals" in Western, individualistic cultures, but "not all cultures view individualism as a positive orientation. . . . In many non-Western cultures, identity is not seen apart from the group orientation (collectivism)" (p. 141). A lack of appreciation of this important distinction between individualistic and collectivistic cultures can lead therapists to overpathologize members of some cultures as overly dependent or unable to assert themselves.

In a discussion of cultural issues relevant to psychotherapy with Native American clients, Sutton and Broken Nose (2005) offer two case studies illustrating this concern. In the first, a Native American couple opens a restaurant, and all aspects of the new business are successful except for the fact that the couple chooses not to charge their family members for food. Moreover, they define family broadly enough to include distant relatives, in-laws, and the like. Because they serve free meals to so many people, the restaurant struggles financially. In the second case study, a Native American college student receives a fellowship to fund his studies but feels compelled to share his money, apartment, and time with siblings and other family members. Their presence causes significant financial and practical obstacles to the student's ability to succeed in school.

Although Native American culture is certainly not the only culture with some degree of collectivist values (many African and Asian cultures, among others, provide more examples), these scenarios highlight some of the sensitivities clinical psychologists should have when considering assertiveness training. In your opinion, in either of the two case studies described here, would assertiveness training be clinically effective? Would it be culturally appropriate? Should it be adjusted in some way to better match the client's cultural values? Should some alternative treatment be considered?

psychologist, that as she began to make arrangements for her wedding—the guest list, the menu, the flowers, the band, her dress, and so on—she became increasingly frustrated with others "making decisions for her." Specifically, she said that her mother and her sister had been telling her what to do, and she had had a difficult time telling them to "butt out." As an example, she described a recent trip to the wedding dress shop. She had been to the store alone previously and had tentatively chosen a dress for herself. When she went back to the store to try on the dress for her mother and sister, they pressured her to try on other dresses and eventually convinced her to buy a dress she had seen

but disliked when she was at the store the first time. Deborah wished she could have taken charge of the situation at the wedding dress shop but told Dr. Paxton that she just couldn't. She added that her day ended by crying about this experience to her fiancé, who, rather than responding with the empathy she sought, told her she needed to stand up to her mother and sister.

Dr. Paxton conducted assertiveness training with Deborah. First, she taught Deborah some specific, appropriately assertive phrases to say, such as, "Thanks for your input, but this is my decision to make" and "I really want you to be a part of this with me, but I need to have the final say." Dr. Paxton also coached Deborah on some specific behaviors involved in assertive responses, such as direct eye contact and appropriate volume and tone in her voice. Next, Dr. Paxton modeled these responses for her. Deborah then rehearsed the assertive responses, with Dr. Paxton playing the role of Deborah's mother or sister. Dr. Paxton offered some helpful feedback, including praise for saying the right things and reminders to make direct eye contact. Together, Deborah and Dr. Paxton came up with some homework assignments that could help Deborah "build up" to the confrontations she anticipated with her mother and sister. These homework assignments involved asserting herself to her mother and sister about less significant upcoming decisions in which they would probably pressure her, such as what she should order at lunch when they went out next Saturday. With time and practice, Deborah's skill in asserting herself increased while her anxiety about asserting herself decreased.

TECHNIQUES BASED ON OPERANT CONDITIONING

Contingency Management

Contingencies are the "if . . . , then . . ." statements that, according to behavioral therapists, govern our behavior. So if the goal is to change behavior, a powerful way to do so is to change the contingencies controlling it. Behavior therapists call this process **contingency management**. All behavior occurs because of its consequences, and if those consequences change, the behavior will change correspondingly (Drossel, Garrison-Diehn, & Fisher, 2008; Kearney & Vecchio, 2002; Villamar, Donohue, & Allen, 2008). Behaviorists emphasize that powerful but often overlooked contingencies—for example, "If I behave in a depressed way, I get attention from friends and family and I am excused from responsibilities"—can contribute to the development and maintenance of mental disorders (e.g., C. M. Anderson, 2007).

Reinforcement and Punishment

The consequences of a behavior—the words that complete the "then . . ." phrase in our contingencies—can be categorized as either reinforcements or punishments. **Reinforcement** is defined as any consequence that makes a behavior more likely to recur in the future. In contrast, **punishment** is defined as any consequence that makes a

behavior less likely to recur in the future. It's important to note the room for individual differences implied by these two definitions. Very few things are either reinforcing or punishing to all people in all situations. That which is punishing to one person may be reinforcing to another and vice versa. Behavior therapists who use contingency management remain aware of this phenomenon and therefore seek client input on all prospective contingencies (Drossel et al., 2008; Spiegler & Guevremont, 2010).

BOX 14.2

In My Practice . . .

In my practice, I have learned that reinforcement is in the eye of the beholder. So is punishment. Most often, this truth becomes evident when I work with parents to change the frequency of specific behaviors in their kids. For example, I worked with Jodi, a single mom of 11-year-old Jade, to get Jade to do her homework more often. Jade's teachers consistently told Jodi that Jade was one of the smartest kids in the class but that she didn't appear to take her homework seriously. In fact, she rarely completed her homework at all. Jodi acknowledged that she had never emphasized homework completion, but with middle school on the horizon, it was important for Jade to develop better homework habits. Jodi decided that she would offer a new puppy as reinforcement for homework completion. She settled on the idea of a puppy after seeing how much fun Jade always had with her cousins' puppy when they visited. With my help, Jodi developed a system in which Jade earned points for specific homework-related behaviors. When she reached specific point thresholds, Jade earned puppy-related stuff: a leash, a collar, dog toys, and ultimately the puppy itself. Jade made absolutely no progress. After considering a number of other explanations, we asked Jade directly what she thought about the reinforcement plan. "I don't want a puppy at all," she plainly said. Jodi was quite surprised: "Why? You get so excited every time we visit your cousins' house! You love playing with their puppy!" "That's because we're just *visiting* their puppy, mom. If I had my own puppy, I'd have to feed it, walk it, and clean up after it. I definitely don't want that much work." We quickly shifted the focus of that conversation to reinforcements that Jade might actually find reinforcing.

With another family, the focus was on punishment. In this case, Zachary and Julia couldn't figure out why sending their 13-year-old son Dominic to his room didn't change his behavior. Before they visited me, Zachary and Julia had used that punishment for a variety of behaviors they wanted Dominic to do less often, including teasing his sister, leaving food and dishes out, and talking back to his parents. Zachary and Julia were especially perplexed because that punishment had worked so well when their own parents used it on them during their own childhoods. Some important information came when I asked Dominic what he does when he gets sent to his room. "Oh, it's great," he told me. "In my room I have a TV, a computer, and of course I always have my phone in my pocket. So whenever my parents send me in there I either watch my favorite shows, play video games, or text my friends." Obviously, my next steps with Zachary and Julia focused on coming up with punishments that Dominic might actually find punitive.

To see this box come to life as a whiteboard video, please visit edge.sagepub/pomerantz5e.

Reinforcement and punishment can each be further divided into two types: positive and negative (Higgins, 1999). In this context, positive refers to adding a consequence, whereas negative refers to removing a consequence. So, simply put, **positive reinforcement** means "getting something good" (such as food), whereas **negative reinforcement** means "losing something bad" (such as pain). Notice that both of these would increase the likelihood of the behavior recurring. **Positive punishment** means "getting something bad," whereas **negative punishment** means "losing something good." Notice that both of these would decrease the likelihood of the behavior recurring (Sturmey, 2007).

Behavior therapists use both reinforcement and punishment during contingency management, but for most clinical situations, reinforcement is generally preferred (e.g., Higgins, 1999). When punishment is used, it must be used ethically, and it is most effective when it occurs immediately and consistently and is accompanied by the reinforcement of an alternate, more desirable response (Poling, Ehrhardt, & Ervin, 2002). **Aversion therapy** represents an example of the clinical use of punishment, in which an unwanted behavior (say, drinking alcohol) brings about an aversive stimulus (nausea or electric shock; Emmelkamp & Kamphuis, 2002; Spiegler & Guevremont, 2010).

As a clinical example of the use of the four varieties of reinforcement (reinforcement and punishment, each in positive and negative forms), consider the work of Dr. Howard, a clinical psychologist engaged in behavior therapy with Patty, a 15-year-old juvenile detention center inmate. Recently, Patty had been prone to bursts of anger and assault when escorted from her cell to daily classroom sessions. Some of Patty's assaults were so dangerous that they necessitated the use of physical restraints. The consequence for Patty's outbursts was dismissal from the day's schooling; she was returned to her cell, where she spent her time leisurely browsing through magazines approved for inmate possession. Dr. Howard analyzed Patty's outburst behavior and hypothesized that the contingency enacted by the staff was actually reinforcing, not punishing, the outburst response. Dr. Howard discussed with the staff four alternate contingencies, any of which could produce more desirable behavior from Patty:

- Positive reinforcement: If Patty attended her classes without any verbal or physical outbursts, then she received a new magazine of her choice.
- Negative reinforcement: If Patty attended her classes without any outbursts, then her ankle restraints—necessary with the onset of her assaults—would be removed for the next day.
- Positive punishment: If Patty engaged in any type of outburst, then she would receive a 2-hour detention in a cell without magazines.
- Negative punishment: If Patty engaged in any type of outburst, then all her magazines would be confiscated for the next day.

Extinction

When behavior therapists consider the contingencies that have maintained a behavior or new contingencies that may modify it, they often pay close attention to issues involving extinction. In the context of contingency management, **extinction** refers to the removal

BOX 14.3

Metaphorically Speaking

If You've Lost Money in a Soda Machine, You Understand Extinction and the Extinction Burst

Put the dollar in the slot, push the button, get the soda. Put the dollar in the slot, push the button, get the soda. There is no confusion about the contingency of the soda machine, and for many of us, soda-buying behavior is strengthened on a daily basis by the repeated delivery of the cold, bubbly, sweet reinforcement.

But what if the soda machine malfunctions? What if we put the dollar in the slot, push the button, and get nothing? Of course, in the long run, we'll stop putting dollars into that machine. In other words, our soda-buying behavior (at least at that particular machine) will be extinguished, because the expected reinforcement is no longer forthcoming. In the short run, however, our behavior is not extinguished; in fact, it intensifies. You've seen (or maybe you've been) the person who just lost money in the soda machine demonstrating the extinction burst—pushing every button, trying a different dollar, hitting the machine, kicking it, tipping it, and so on. It's as if we're responding to the soda machine's refusal to provide the soda with the behavioral statement, "Hey, give me what I'm used to getting!" When the unresponsive soda machine responds with the behavioral statement, "Too bad, you're not getting it," we walk away, but it may take a while for us to become entirely convinced.

Actually, the time it takes for us to become convinced that no soda is forthcoming, and for our soda-buying behavior to therefore be extinguished, depends on how consistently the soda machine stiffs us. If it is absolutely consistent, we'll quit pretty quickly. But if we know the machine to be fickle—sometimes the soda comes, sometimes it doesn't, sometimes it requires a good kick—it may take quite a while for our soda-buying behavior to become extinct. Perhaps a better example of such an inconstant, unpredictable machine is a slot machine, from which "you never know" what you might get (Kazdin, 1980). When behavior therapists use extinction methods with clients, which would you expect to be more successful—the consistent or inconsistent approach to denying expected reinforcement? More generally, how might the consistency of a contingency management technique influence its success in changing behavior?

of an expected reinforcement that results in a decrease in the frequency of a behavior (Culver, Vervliet, & Craske, 2015; Kearney & Vecchio, 2002; Miltenberger, Miller, & Zerger, 2016; Poling, Ehrhardt, & Ervin, 2002; Sturmey et al., 2007).

As an example, consider Wendy, an 8-year-old second grader whose parents brought her to Dr. Evans, a clinical psychologist with a behavioral orientation. Wendy's parents explained that in the past 2 weeks, Wendy had become extremely difficult at meal times: She cried and screamed about the food that had been prepared, saying she didn't like it and it made her stomach hurt. Wendy's parents expressed confusion at her behavior, specifically because it was the same food she had eaten many times before, and they had taken her to her pediatrician, who assured them that Wendy had no stomach ailments. When Dr. Evans asked Wendy's parents what happened after

Wendy cried and screamed at meal times, they explained that they typically allowed her to eat something else. When Dr. Evans inquired further, the parents added that they allowed Wendy to choose any food she wanted, and Wendy usually selected her favorite junk food. Dr. Evans developed a contingency management plan based on extinction. His conceptualization was that Wendy's crying and screaming behavior was being positively reinforced by the junk food she received after doing so. He explained this conceptualization to Wendy's parents and recommended that they remove those positive reinforcements—in other words, don't let her replace the family meal with junk food. They did, and although her behavior initially got worse, within a few days, Wendy stopped crying and screaming at meal time and resumed eating with her family as she had previously.

Wendy's case exemplifies an important aspect of extinction-based therapies: the **extinction burst** (Kazdin, 1980; Poling, Ehrhardt, & Jennings, 2002; Spiegler & Guevremont, 2010). Immediately after the reinforcement was removed, Wendy's crying and screaming increased—she did it more often and more intensely. Only after her parents "stood their ground" by continuing to withhold the reinforcement did Wendy's crying and screaming dwindle. It is important for behavior therapists and those working with them to anticipate the extinction burst that predictably occurs immediately after the removal of the reinforcement; otherwise, the person controlling the contingency might mistakenly think that the strategy is backfiring and resume the reinforcement again. If Wendy's parents had done this—if they had given in to her especially intense fits on the first days of the extinction process—they would have taught Wendy that if she ups the ante, she can still get what she wants. This would have strengthened, rather than extinguished, her crying and screaming behavior.

Token Economies

A **token economy** is a setting in which clients earn tokens for participating in predetermined target behaviors (Ghezzi, Wilson, Tarbox, & MacAlesse, 2008). These tokens can be exchanged for a number of reinforcements, including food, games, toys, privileges, time participating in a desired activity, or anything else deemed desirable by the client. In some token economies, clients can also lose tokens for engaging in undesired behaviors (Stuve & Salinas, 2002). Token economies are used most often in settings such as inpatient units, correctional facilities, and other sites where clients' behavior is under ongoing surveillance by supervisory staff. A strength of token economies is their versatility across clients. For example, on a psychiatric inpatient unit, different target behaviors may be identified for each client. One client may earn tokens for making the bed, another for taking a shower, and another for interacting with a group rather than staying alone. Of course, the success of the token economy depends on the perceived value of the reinforcements for which the tokens can be exchanged, so behavioral therapists are careful to select reinforcements that will motivate each client. If you have ever spent any time in Dave and Buster's or Chuck E. Cheese's, you're familiar with the fact that the same "prize" may be valued very differently by different people. When you were 6 years old, you might have spent a lot of time, energy, and money earning enough tokens (tickets) to exchange for stuffed animals, plastic toys, or candy. Today, you might not find those prizes quite as motivating, and your behavior would change accordingly. Likewise, a

poker chip may not have meant much to you when you were 6, but now, you may have a greater appreciation for its value as a token exchangeable for money (which itself is exchangeable for many reinforcements).

A potential limitation of token economies involves generalization (Spiegler & Guevremont, 2010). As discussed above, generalization refers to the application of a learned contingency to similar behaviors or situations. The goal of any token economy is not only to modify behavior in that environment but to modify it across all settings. For example, the psychiatric inpatients mentioned in the preceding paragraph would ideally apply the lessons learned about bed making, bathing, and socializing to the outside world in addition to the psychiatric unit in which they temporarily live. Behavior therapists can use a number of strategies to maximize generalization, including tapering clients off tokens gradually rather than all at once, using naturally occurring reinforcements (such as social praise) rather than artificial reinforcements, gradually increasing the delay between the behavior and the reinforcement, and providing reinforcement in as wide a variety of settings as possible (Stuve & Salinas, 2002).

Shaping

Contingency management is often based on reinforcing target behaviors in order to increase their frequency. Sometimes, however, the target behavior is so complex, challenging, or novel for a client that, at the outset of treatment, it simply can't be accomplished in its entirety. In these cases, behavioral therapists use **shaping**, which involves reinforcing successive approximations of the target behavior. Put another way, shaping is a technique in which the behavior therapist reinforces "baby steps" toward the desired behavior (Kazdin, 1980; Kearney & Vecchio, 2002; Miltenberger et al., 2016; Sturmey et al., 2007).

As an example, consider Dina, a 59-year-old client with serious depressive symptoms. Since she began feeling depressed about 3 months ago, Dina has become increasingly withdrawn. She has neither contacted any of her friends (of which she has many) nor returned their many calls. She has declined when her husband has asked her to go out to dinner, and she has refused invitations from her grown daughter to come to her house for a visit. Dr. Stein, Dina's clinical psychologist, conceptualizes Dina's problem from a behavioral point of view. That is, Dr. Stein has identified social behavior as the area for improvement and has specifically defined the goal as much more frequent social interactions for Dina: during each week, three phone calls with friends, one dinner out with her husband, and one get-together with her daughter. Dr. Stein realizes, however, that Dina is far from that level of social functioning at the moment. If she waits for Dina to complete all these tasks in a particular week, the wait may be excessive. So Dr. Stein uses a shaping strategy. First, she determines jointly with Dina a personally meaningful reinforcement: renting a DVD. Then she establishes the contingency for the first week: If Dina completes at least one social activity (a call to a friend, dinner out with her husband, or a get-together with her daughter), then she can rent a DVD of her choice. The next week, Dr. Stein raises the bar: at least two social activities, one of which must be in person. Dr. Stein continues to raise the bar each week until Dina is completing the full set of target behaviors for several consecutive weeks.

Web Link 14.5
Website of Christopher Martell

A key variable in any shaping program is the increment between each successive approximation. Behavior therapists must be careful not to make the steps between each new challenge too difficult for the client. By the same token, the steps should not be so small that therapy takes an unnecessarily long time. Thus, in Dina's case, Dr. Stein should adjust the amount by which she raises the bar if it is evident that Dina finds the tasks either too easy or too hard.

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Photo 14.3 Shaping is a behavioral treatment used to increase the frequency of complex behaviors. In this case, a single phone call to a friend is a step toward a greater level of social interaction.

Behavioral Activation

Behavioral activation is a form of behavior therapy, initially designed to treat depression, that has received significant attention and empirical support in recent years (Dimidjian, Barrera, Martell, Muñoz, & Lewinsohn, 2011; Mazzucchelli, Kane, & Rees, 2009; McCauley et al., 2016; Ros, Hernandez, Graziano, & Bagner, 2016; K. L. Schneider et al., 2016). It is based on the simple yet profound notion that in the day-to-day lives of depressed people, there is a shortage of positive reinforcement. So the goal of behavioral activation is to increase the frequency of behaviors that are positively reinforcing to the client. As a result, clients experience more positive emotions and become more fully engaged in their lives. Their tendencies to avoid unpleasant experiences diminish as their depression lifts (Craske, 2010; D. Dobson & Dobson, 2009; Hopko, Ryba, McIndoo, & File, 2016; Kanter, Busch, & Rusch, 2009; Martell, 2008; Puspitasari, Kanter, Murphy, Crowe, & Koerner, 2013; Ritschel, Ramirez, Cooley, & Edward Craighead, 2016; Shikatani, Fracalanza, & Antony, 2015).

At the outset of behavioral activation, one of the most important questions the clinical psychologist can ask is, "Are there things you are not doing now that you typically do when you are not depressed?" (Martell, 2009, p. 140). From the client's response to such questions, the client and psychologist can collaborate to form a list of rewarding behaviors and a plan to integrate them into the client's life. Often, clinicians help clients develop a structured routine to guide them through each day's events. Martell (2009) highlights that it is important to understand the function of a behavior from the client's point of view. In particular, it is crucial to know whether a client is engaging in a particular behavior because it brings positive reinforcement or enables the client to avoid something unpleasant. For example, a client may choose to go to the movie theater because the movie brings her pleasure or because doing so allows her to avoid an argument with her roommate about the overdue rent. From the point of view of the clinician conducting behavioral activation, the avoidance in the latter scenario is actually part of the problem, so if that is indeed the function of the behavior, an alternative behavior should be used instead.

It is clear that operant conditioning forms the basis of behavioral activation, but classical conditioning may play a role as well. In a manner that resembles the counterconditioning in systematic desensitization (described earlier in this chapter), behavioral activation takes advantage of the fact that depression and reinforcing activities are incompatible. In other words, behaviorists make the argument that it's impossible to feel the reinforcement of a favorite activity and feel depressed at the same time. In this way,

behavioral activation is a method of planning and encouraging activities that bring pleasure and preclude depression (Martell, 2008, 2009).

Although behavioral activation was created as a treatment for depression, research suggesting that it can also be quite successful in treating other problems is starting to accumulate. Those problems include anxiety disorders, psychotic disorders, personality disorders, schizophrenia, and chronic pain (Boswell, Iles, Gallagher, & Farchione, 2017; Hasson-Ohayon, Arnon-Ribenfeld, Hamm, & Lysaker, 2017; Mazzucchelli & Da Silva, 2016). In fact, some researchers have suggested that behavioral activation may be a common mechanism of change—in other words, the fundamental act of increasing the frequency of pleasant, rewarding activities that produce positive reactions like competence, relaxation, or agency may be a central part of therapeutic change across many disorders and therapy approaches (Dimaggio & Shahar, 2017).

Observational Learning (Modeling)

So far, all the clinical applications of operant conditioning that we have discussed involve clients learning directly from their own experiences. However, much of what we learn comes from contingencies we see applied to other people. This phenomenon is known as **observational learning** but has also been called *modeling* and *social learning*. As an example of observational learning, briefly reconsider the broken soda machine scenario described in Box 14.3. If the soda machine stole the dollar of the person in front of you while you were waiting your turn, would you step up and put your dollar in next?

In clinical practice, observational learning is a technique in which the client observes a demonstration of the desired behavior and is given chances to imitate it (K. A. Freeman, 2002; Spiegler & Guevremont, 2010). The client typically receives constructive feedback on these imitation efforts as well. The person acting as the model can be the therapist, another live model, or a model who has been video- or audio-recorded. The effects of modeling have been studied extensively by **Albert Bandura** and others (e.g., Bandura, 1977), producing a sizable body of knowledge regarding key variables in the modeling process. Among the findings is that models are most effective when they are similar to the client, an especially relevant point regarding client diversity in terms of cultural and demographic variables. For example, if Ana, a 19-year-old, homosexual, Hispanic, female college sophomore, is struggling with social skills, the success of an observational learning intervention may depend on the degree to which the model matches Ana in terms of age, sexual orientation, ethnicity, gender, and student status.

Observational learning strategies actually afford clients two different ways to learn. The first is **imitation**, in which the client simply mimics the modeled behavior. The second is **vicarious learning**, in which the client observes not only the modeled behavior but also the model receiving consequences for that modeled behavior. In other words, even without imitation, a client can learn to expect reinforcement or punishment for a target behavior by observing what the model receives (K. A. Freeman, 2002). In the case of Ana, vicarious learning would take place if Ana observed the model initiating a conversation with an unknown person and receiving obvious feedback—a kind greeting, a snub, or a neutral response—as a consequence. Of course, Ana's behavior therapist would try to ensure to the extent feasible that when the desired response is modeled, it is followed by a reinforcement.

ALTERNATIVES TO BEHAVIOR THERAPY

Behavioral Consultation

Behavioral consultation is an indirect way for a behavior therapist to modify a client's behavior. It differs from direct clinical services in that there are always three parties involved: the client, the consultee, and the consultant (therapist). The consultee is a person who spends significant time in the natural setting with the client and who has some control over the contingencies that govern the client's behavior. In many cases, the consultee is an adult who supervises a child in some capacity, such as a parent, teacher, or caretaker (Ehrich & Kratochwill, 2002). However, the consultee can also be seeking help with the behavior of an adult client, such as a corporate manager whose aim is to modify the behavior of an employee (Bailey & Burch, 2006). Some of the examples we consider in this chapter (such as Wendy, the girl who refused to eat the same dinner as her family) have involved a consultation component, but in behavioral consultation, the consultee serves as a true go-between, such that the consultant/therapist and the client may never meet each other.

Behavioral consultation is a flexible process, but it typically involves five stages (Ehrich & Kratochwill, 2002):

- *Initiation of the consulting relationship*, in which the roles and responsibilities of all parties are established.
- *Problem identification*, in which the target behavior is defined, usually through questions involving who, what, where, and when the behavior problem occurs. Baseline and goals are also determined.
- *Problem analysis*, in which the therapist identifies the reinforcement contingency that is maintaining the current behavior.
- *Plan implementation*, in which the consultee carries out the intervention as recommended by the consultant.
- *Plan evaluation*, in which the consultant and consultee measure the client's progress from baseline and toward goals.

As an example of behavioral consultation, consider Kathleen, a managing partner in a private law firm that employs about 20 lawyers. Kathleen seeks the assistance of Dr. Taguchi, a clinical psychologist and behavioral consultant, regarding productivity problems related to a recently hired administrative assistant, Pam. (To clarify, Dr. Taguchi is the consultant, Kathleen is the consultee, and Pam is the client.) Among other things, Kathleen describes one particularly problematic behavior that Pam performs on a regular basis: indulging in long lunches. Specifically, she states that Pam is often out of the office for 2 to 2.5 hours for lunch, always with friends from nearby offices, and that during this time Pam's work piles up. The attorneys in the office grow frustrated with Pam's absence, and they are unable to be as productive as they

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Photo 14.4
Behavioral therapists use parent training to help parents manage problem behaviors in their children, including problematic bedtime behavior.

could be otherwise. Kathleen explains that they had never had such a problem with previous administrative assistants, so she did not present Pam with a clear lunch policy when Pam was initially hired. Dr. Taguchi conceptualized the problem as a contingency in which Pam's long lunches were reinforced by social interaction and were not punished at all. Dr. Taguchi recommended that Kathleen determine a reasonable length to allow Pam for lunch and then either punish her for staying out too long or reinforce her for returning on time. Through discussion with Dr. Taguchi, Kathleen decided to use both the punishment and reinforcement strategies: If Pam stayed at lunch longer than 1 hour, her hourly pay was docked, and if Pam returned on time for an entire week, she was allowed to leave work an hour early on Friday. Kathleen implemented the plan, and Pam's behavior was quickly modified.

Parent Training

Parent training is a specific form of behavioral consultation in which parents seek help with problematic behaviors of their children. The range of problem behaviors for which parent training can be helpful is vast: Sleep-related problems, bedwetting, hyperactivity, oppositional and defiant behaviors, stuttering, fear of the dark, school phobia, and social skills are a subset (Forehand, Jones, & Parent, 2013; Forehand, Lafko, Parent, & Burt, 2014; Jones et al., 2013; Karpiak & Dishion, 2016; Leijten, Raaijmakers, de Castro, & Matthys, 2013; Schaefer & Briesmeister, 1989; Spiegler & Guevremont, 2010; van der Veen-Mulders, Hoekstra, Nauta, & van den Hoofdakker, 2018). Unlike child psychotherapy or family psychotherapy, parent training involves an arrangement in which the consultant (behavior therapist) may never meet the child directly. Parent training typically begins with a definition of the problem behavior, followed by instructions to parents to use reinforcement and punishment effectively. For example, psychologists may educate parents about shaping techniques to teach new complex behaviors or about the use of token economies; likewise, psychologists may educate parents about the use of mild punishment such as time-outs or loss of privileges. The intent is for the child's behavior to improve not only in the setting where the intervention occurs (e.g., home) but to generalize to other settings (e.g., school) as well (Moore & Patterson, 2008).

As an example of parent training, consider Heather, a single mother of a 7-year-old boy, Danny. Heather sought consultation from Dr. Ogden as a result of Danny's problematic bedtime behavior. Heather explained that each night, Danny would go

through the same bedtime ritual they had gone through for years—put on his pajamas, brush his teeth, read a story with mom, and say goodnight. However, in the past few months, Danny had started calling out, "Mommy, I can't sleep," after being in bed for about 10 to 15 minutes. When Dr. Ogden asked how Heather responded, she said that she often let him come out of his bedroom, thinking that perhaps he genuinely wasn't yet tired. Specifically, Heather explained that she often let Danny join her on the couch, where they watched TV and ate snacks for about an hour. Dr. Ogden saw Heather's behavior as strongly reinforcing to Danny and suggested that Heather no longer provide the reinforcement. Together, they discussed other viable options and decided that the best option would be no longer to allow Danny out of his room after bedtime. Of course, when Heather initially implemented this plan, Danny's behavior actually worsened (he demonstrated an extinction burst), but Dr. Ogden had warned Heather about this and encouraged her to remain consistent. When they evaluated the plan after several weeks, Heather said that Danny's behavior had improved, but she pointed out that a few times, Danny had defiantly come out of his room without permission, and Heather was unsure how to respond. Dr. Ogden recommended that Heather punish this behavior, and they agreed that losing desserts for the next day would be a suitable punishment. After this adjustment, Danny's coming-out-of-bed behavior was extinguished.

Teacher Training

Teacher training is quite similar to parent training, but the emphasis is on behaviors that take place at school. Many of these problem behaviors are interpersonal or disruptive in nature, but others are academic and involve refusal to complete assignments and similar task-related behaviors (Boulton, 2014; Fernandez, Gold, Hirsch, & Miller, 2015).

Often, when behavioral therapists serve as consultants to teachers, a primary task is to thoroughly analyze the consequences of the child's behavior. That is, the behavioral consultant can often help the teacher see all the reinforcement and punishment that a child receives for particular behaviors. As an example, consider Ms. Palmer, a third-grade teacher who sought behavioral consultation from Dr. Carr. Ms. Palmer explained that one of her students, Brian, was extremely disruptive in class. In particular, Brian frequently called Ms. Palmer "stupid" and "an idiot" during class. Ms. Palmer couldn't understand why this behavior persisted, because each time he did this, she punished him by sending him out of the room and later talking with him one-on-one about his inappropriate behavior. Through additional questioning, Dr. Carr uncovered two important facts: When Brian was sent out of the room, he was able to avoid the work the class was doing at that time, much of which Brian found overwhelming; additionally, the attention Ms. Palmer gave him during the one-on-one talks actually made Brian feel important, not punished. Thus, although Ms. Palmer intended to punish Brian's insulting behavior, she was actually reinforcing it. Ms. Palmer was able to find other means of responding to Brian's insults, including ignoring them, which resulted in a drastic reduction in the target behavior.

BOX 14.4

Denise in Behavior Therapy

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During my initial interview with Denise, she explained that she had been experiencing depression-related difficulties since a new owner introduced new policies at the restaurant where she worked as a chef. Denise spent a lot of time providing background information about her childhood. As a behavior therapist, I found this childhood information rather irrelevant to her current problems. Also, Denise spent a lot of time describing her thoughts and feelings regarding the situation at work, but, through behaviorally oriented questions, I tried to refocus Denise on overt, observable, measurable behaviors rather than inner mental processes such as thoughts and feelings. Eventually, Denise and I identified three specific behaviors that were suitable for intervention: getting to work on time, preparing dishes accurately, and exercising. We agreed that the aim of therapy would be to modify the frequency of each of these behaviors to more desired levels.

Once we had operationally defined these three behaviors, another important initial step was to establish a baseline for each. In other words, we assessed the frequency with which these behaviors were currently taking place to have a basis for comparison after treatment was implemented. Denise informed me that she was currently arriving to work on time only two times a week (out of 6 work days per week), preparing dishes inaccurately once per work day, and exercising zero times per week. Next, we set specific goals for each of these target behaviors: arriving to work on time 6 days per week, preparing dishes inaccurately only once per week, and exercising four times per week.

I combined behavioral activation and a contingency management approach incorporating shaping techniques to modify Denise's behaviors in each of these three areas. First, Denise and I explored a variety of options that might be effective reinforcement for her; we decided that time surfing the Internet, one of her favorite leisure activities, was the most reasonable choice. Then we established specific contingencies for each of the three target behaviors. The initial contingencies intentionally fell short of the final goal, because the plan was to shape Denise's behavior by reinforcing behaviors that were "steps in the right direction" and then "raise the bar" after each successful week. The initial contingencies were as follows: If she arrived at work on time on a given day, then she was allowed 20 minutes of Internet surfing; if she prepared all dishes accurately on any given work day, then she was allowed 20 minutes of Internet surfing; if she exercised for at least 30 minutes on any given day, then she was allowed 20 minutes of Internet surfing. Denise was quite successful in meeting these initial goals within 2 weeks, so we gradually began to increase the demands in order to earn the reinforcement. At the same time, Denise and I established and implemented a specific behavioral activation plan. She identified several activities that she finds positively reinforcing but had done infrequently as her depressive symptoms increased: shopping at the farmer's market, going to lunch with her sister, and talking with her best friend on the phone. We scheduled those events into Denise's weekly calendar, and she followed that schedule on a consistent basis. Within a relatively short time (about 8 weeks), Denise was meeting her final behavioral goals in all three areas.

HOW WELL DOES IT WORK?

In terms of empirical evidence supporting their benefits, behavior therapies are unquestionably highly supported. Lists and compilations of "treatments that work" regularly include disproportionately large numbers of behavioral treatments, especially in comparison with humanistic and psychodynamic treatments (e.g., Chambless et al., 1998; Fishman, 2016a; Nathan & Gorman, 2007). Behavior therapies have been found efficacious for an extremely wide range of disorders, but the disorders for which they have garnered the most support include anxiety disorders, depression, and children's behavior disorders (Prochaska & Norcross, 2014; Spiegler & Guevremont, 2010). To a remarkable extent, especially in comparison with many of the other therapy approaches covered in this book, behavior therapy has produced and provided efficacy data for specific interventions for specific problems (Neer & Beidel, 2015; Fishman, 2016a). The long list of specific behavior therapies with substantial evidence to support their use with specific disorders include exposure and response prevention for obsessive-compulsive disorder (Craske, 2010; M. E. Franklin & Foa, 2011; Olatunji et al., 2013); exposure for specific phobias, posttraumatic stress disorder, panic disorder, and social phobia (Deacon & Abramowitz, 2004; Foa & McLean, 2015; Wolitzky-Taylor, Horowitz, Powers, & Telch, 2008); behavioral activation for depression (Ekers et al., 2014); contingency management for drug and alcohol disorders (Petry, Martin, Cooney, & Kranzler, 2000; Petry, Alessi, & Rash, 2013); token economy for schizophrenia (Dickerson, Tenhula, & Green-Paden, 2005); parent training and teacher training for ADHD (S. W. Evans, Owens, & Bunford, 2014); and many more.

Some have pointed out that behavior therapies populate lists of empirically supported or evidence-based treatments because, more than any other form of treatment (e.g., psychodynamic, humanistic, cognitive), they lend themselves to the manualization and quantitative outcome measurement necessary for empirical testing. In other words, if the worth of a therapy is to be measured solely via quantifiable measures of objective, observable outcomes, it should be no surprise that behavior therapies are shown to be most successful (e.g., W. H. Silverman, 1996). Behavioral therapists have sought from the outset to create such a type of therapy, emphasizing an empirical, scientific approach to therapy as opposed to the more introspective, subjective approach of other methods, such as psychodynamic and humanistic therapies.

CHAPTER SUMMARY

Behavior therapy focuses on observable, measurable behavior rather than mental phenomena that can be only indirectly inferred. Problematic or undesirable behavior is viewed as the clinical problem, not as a symptom of some underlying, deeper disorder. The primary goal of behavior therapy is overt behavioral change, and its methods rely on empirical, testable hypotheses. Behavioral therapists believe that conditioning, either classical or operant, is the primary cause of behavior. As such, their clinical efforts

center on altering clients' learned contingencies (via such methods as contingency management, behavioral activation, or token economies) or breaking clients' learned associations (via such methods as exposure therapy, systematic desensitization, or assertiveness training). Principles of behavior therapy have also been used in an indirect, consulting capacity, whereby the behavior therapist helps a consultee (e.g., a parent or teacher) apply behavioral techniques with a client (e.g., a child or student). Empirical outcome studies have offered more empirical support for behavior therapy than for any other approach, and the disorders for which it has demonstrated empirical efficacy include anxiety disorders, depression, children's behavior disorders, and many others.

KEY TERMS AND NAMES

anxiety hierarchy 343
assertiveness training 345
aversion therapy 349
Albert Bandura 354
baselines 339
behavior therapy 336
behavioral activation 353
behavioral consultation 355
classical conditioning 341
conditioned response 341
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CRITICAL THINKING QUESTIONS

1. Do you believe that the law of effect is equally powerful in humans and animals?
2. To what extent do you agree that the primary goal of psychotherapy should be observable behavior change?
3. To what extent do you agree with the medical model of psychopathology?
4. In your opinion, what can behavioral therapists do to make imaginal exposure as similar to in vivo exposure as possible?
5. For what types of clinical problems does contingency management seem most and least likely to be beneficial?

LOOKING TOWARD GRADUATE PROGRAMS

The following table is a list of index entries in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018) relevant to the topics in this chapter (see page 25 in Chapter 1 for more information about the *Insider's Guide*). Keep in mind a few things as you consider these listings: They are current as of the 2018/2019 edition of the *Insider's Guide* (which typically revises every two years); they may have changed as faculty members retire, take new jobs, or develop new interests; and a graduate program's own website may have more extensive or up-to-date information.

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Applied Behavioral Analysis Behavioral Analysis/Therapy Miscellaneous and Other—Behavioral Treatments	Behavioral Therapy/Analysis Cognitive/Cognitive-Behavioral Therapy Miscellaneous—Behavioral Activation	Behavioral Analysis/Therapy Cognitive/Cognitive-Behavioral

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

Behavior Therapy
http://www.journals.elsevier.com/behavior-therapy/

Behaviour Research and Therapy
http://www.journals.elsevier.com/behaviour-research-and-therapy

Journal of Anxiety Disorders
http://www.journals.elsevier.com/journal-of-anxiety-disorders/

Psychotherapy
http://www.apa.org/pubs/journals/pst/index.aspx

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- eFlashcards
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- In My Practice demonstration video
- Full-text SAGE journal articles
- Additional web resources
- Mock assessment data

15

COGNITIVE PSYCHOTHERAPY AND MINDFULNESS-BASED THERAPIES

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Among today's clinical psychologists, **cognitive therapy** prevails. Surveys indicate a dramatic increase in the popularity of cognitive therapy, especially since the 1980s (e.g., Norcross & Karpiak, 2012). In fact, far more contemporary clinical psychologists endorse cognitive therapy as their primary orientation than any other single-school approach. In short, "cognitive therapy clearly has become the prominent approach to psychotherapy in the early part of the 21st century" (K. S. Dobson, 2012, p. 112).

The attraction to the cognitive approach may stem from a variety of factors. Cognitive therapy strikes a balance among some of the other psychotherapeutic options. Like behavioral therapy, it tends to be brief, structured, and targeted. However, like psychodynamic therapy, it focuses on important mental processes. In some ways, cognitive therapy actually represents a reaction against both the behavioral and psychodynamic approaches. Two important historical developments occurred around the same time, beginning in the 1950s and 1960s, contributing to the development of cognitive therapy:

- Strict applications of behavioral therapy—techniques based on operant and classical conditioning—didn't always work. Gradually, behavioral therapists and researchers began to recognize that cognition played a unique and important role in human behavior.

Learning Objectives

15.1 Describe the goal of cognitive psychotherapy.

15.2 Contrast the cognitive therapy approach with other major therapy approaches.

15.3 Explain the major components of Ellis's rational emotive behavior therapy.

15.4 Identify illogical automatic thoughts using Beck's list of common thought distortions.

15.5 Discuss the use of third-wave therapies in psychotherapy practice.

15.6 Summarize the findings of cognitive and third-wave therapy outcome research.

The extreme behavioral view that our actions are determined entirely by external stimuli gave way to the notion that internal mental processes can also exert a strong influence (Goldfried, 1995; O'Donohue, 2009).

• The eventual leaders of the cognitive therapy movement—**Aaron Beck** and **Albert Ellis**—grew disillusioned with the psychoanalytic method in which they and most of their cohorts were trained. They sought a new approach to therapy that addressed clients' symptoms more directly, focused less on the past and more on the present, and produced positive results more efficiently. Eventually, Beck and Ellis each broke from psychoanalytic tradition and forged separate but similar styles of therapy to achieve these goals (A. Ellis, 1962; Kuehlwein, 1993; Rosner, 2015a).

So cognitive therapy began as a revision of behavioral therapy within a context of increasing dissatisfaction with the psychodynamic therapy that dominated at the time (O'Donohue, 2009; M. G. Newman et al., 2017). Cognitive therapy still overlaps significantly with behavioral therapy: Many cognitive therapists use behavioral techniques also, and a large number of therapists straddle the line by identifying themselves as "cognitive-behavioral" (Wenzel, 2017). However, cognitive therapy is not just a minor variation of behavioral therapy in which cognitions complement conditioning. Undoubtedly, cognitive therapy has evolved into its own well-established and commonly practiced approach. And the evolution continues with the rise of third-wave cognitive therapies based on mindfulness and similar concepts.

Let's begin by considering cognitive therapy as it has traditionally been practiced. Then, we'll continue by considering the newer, mindfulness-based (or "third-wave") variants of cognitive therapy.

Web Link 15.1
Beck Institute for Cognitive Therapy and Research

GOAL OF COGNITIVE THERAPY

Simply put, the goal of cognitive therapy is logical thinking. The word *cognition*, after all, is basically synonymous with the word *thought*. Thus, cognitive therapists fundamentally presume that the way we think about events determines the way we respond (Lorenzo-Luaces, German, & DeRubeis, 2015; Onken, 2015). In other words, "individuals' interpretations and perceptions of current situations, events, and problems influence how they react" (J. S. Beck, 2002, p. 163). Psychological problems arise from illogical cognitions. For example, an illogical (or irrational or unrealistic) interpretation of a life event—a relationship breakup, an F on an exam, a comment from a friend—can cause crippling depression or anxiety. However, psychological wellness stems from logical cognitions. That is, when the cognitions appropriately match the event, they can lead

to more adaptive, healthy reactions. Therefore, the role of the cognitive therapist is to fix faulty thinking (A. T. Beck & Haigh, 2014; Bermudes, Wright, & Casey, 2009; D. A. Clark, Hollifield, Leahy, & Beck, 2009; D. Dobson & Dobson, 2009; K. S. Dobson, 2012).

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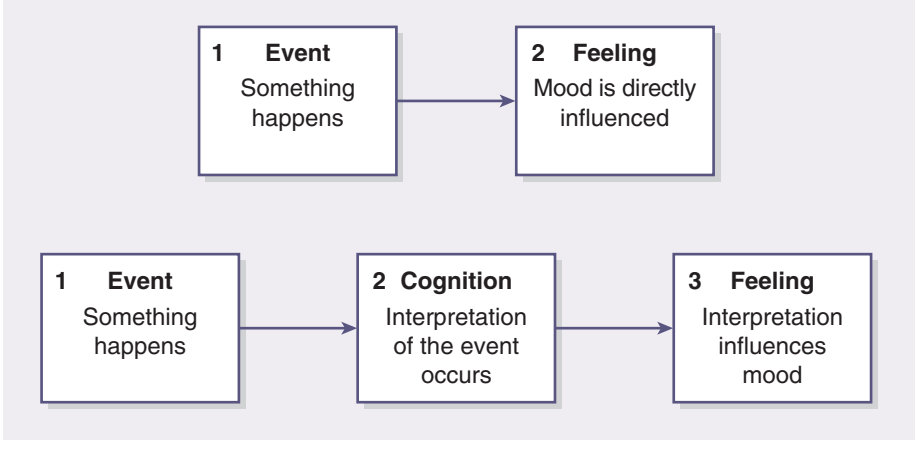
Photo 15.1 Aaron Beck (left) and Albert Ellis (right) are pioneers of the cognitive approach to psychotherapy, which is now more widely endorsed by clinical psychologists than any other single-school approach.

The Importance of Cognition

When they refer to **cognitions**, cognitive therapists use lots of terms interchangeably: *thoughts*, *beliefs*, *interpretations*, and *assumptions*, to name a few. Whatever we call them, we often overlook their importance in our day-to-day lives. When someone asks, "Why are you so happy?" or "Why are you so sad?" we typically point to a recent event that made us happy. We portray our experience as a two-step model in which things happen and those things directly influence our feelings. The truth, according to cognitive therapists, is that such a two-step model is flawed; specifically, it's missing an important step in the middle. The three-step model that cognitive therapists endorse goes like this: Things happen, we *interpret* those things, and *those interpretations* directly influence our feelings. Thus, "it is not a situation in and of itself that determines what people feel but rather the way in which they construe a situation" (J. S. Beck, 1995, p. 14). In other words, events don't make us happy or sad. Instead, the way we think about those events does. (See Figure 15.1.)

As an example of the power of our cognitions, consider an unexpected overnight snowfall. At the same time, three neighbors wake up, look out their windows, and see the ground covered in 6 inches of white, with more flakes continuing to fall. In the first

Figure 15.1 Two-Step Model (Ignoring Cognition) Versus Three-Step Model (Incorporating Cognition)



house lives a mail carrier who covers her route by foot. For her, the snow causes feelings of dread. It's important to recognize, however, that between seeing the snow and feeling the dread, she has thoughts: "This is going to be a miserable day. I'll be cold and wet, I might slip and fall, and my route will take much longer than usual." Her next-door neighbor owns a snowplow business. He wakes up, sees the same snow, and feels elated. Like his neighbor, somewhere between the sight of the snow and the resulting feeling, the snowplow driver thinks: "What a great day! I'm going to make a lot of money. Not to mention, I'll be able to help a lot of people." In the next house, a high school student who didn't study for today's biology exam sees the snow and feels tremendous relief. Between the sight and the feeling, he thinks: "Whew! School's going to be cancelled, and I'll have an extra day to study for that exam. What a lucky break." The same snowfall caused very different feelings in these three people, illustrating that it's not the events that happen to us but the meaning we assign to those events that shapes our feelings. Even if that process of assigning meaning happens automatically and within a split second, it nonetheless represents a crucial link to our feelings, including feelings characteristic of psychopathology such as depression or anxiety. Thus, these oft-ignored intermediary cognitions are a focal point in cognitive therapy.

Revising Cognitions

Once we accept the idea that cognitions determine feelings, revising them becomes the foremost task. Specifically, the goal is to ensure that the thoughts a person has about particular events rationally and logically correspond to the event itself. If they don't, they can lead to unnecessary and unpleasant feelings (D. Dobson & Dobson, 2009; A. Ellis, 2008; B. F. Grant, Young, & DeRubeis, 2005). For example, let's reconsider the mail carrier described above. It's reasonable for her to feel some degree of dread about the snow; after all, it will certainly make her day more difficult. But if the thoughts underlying her dread are illogical, they can make the dread excessive. That is, if she thinks, "I'll get fired if my route takes longer than usual today" or "I'll definitely fall on wet pavement and end up with a broken bone or a concussion" or "I may freeze to death on my route," she can cripple herself with anxiety or depression. The goals of the cognitive therapist would not be to make this mail carrier feel unrealistically positive—it would hardly be logical for her to jump for joy about this situation—but to help revise her thoughts so they make realistic sense. At the end of this process, she might be just a bit apprehensive or down, which, in comparison to being devastated by anxiety or depression, represents a significantly improved emotional state.

As we see later in this chapter, there are different methods of revising cognitions. (There are also different terms for it, such as *restructuring* or *modifying* cognitions.) In general, these methods follow a common three-stage sequence: Illogical cognitions are first identified, then challenged, and eventually replaced with more logical cognitions (J. S. Beck, 1995; A. Ellis, 2008; Leahy, 2003; C. F. Newman, 2016). The first step—the identification of illogical thoughts—should not be confused with the psychodynamic goal of making the unconscious conscious. Cognitive therapists do not delve into the unconscious depths of the psyche as do psychodynamic therapists. They do, however, acknowledge that some of our cognitions are **automatic thoughts**—that is, they take place in an instant and without any deliberation. (The student with the surprise snow day described earlier certainly didn't need to pause and ponder, "How do I feel about this snow? Hmm,

let me mull it over." The interpretation happened far more immediately.) As such, these cognitions can become so routine and habitual that they are hard to recognize. A primary responsibility of the cognitive therapist, especially early in therapy, is to assist the client in identifying automatic illogical or irrational thoughts.

The second step, in which the illogical cognitions are challenged, also takes a variety of forms. As we see later in this chapter, some therapists rely on the power of verbal persuasion to convince clients to abandon illogical beliefs, whereas others encourage clients to test the accuracy of their beliefs by performing assigned behaviors in the real world. The objective of either of these approaches is to cause clients to doubt the truth of their illogical beliefs and to reach the conclusion that these beliefs should be revised. This revising, the third step in this process, is often difficult for clients to do at first—it can feel foreign, since therapists are asking clients to think in ways opposite to the ways clients may have been thinking for many years (D. A. Roth, Eng, & Heimberg, 2002). The cognitive therapist may, therefore, take the lead in the initial attempts to revise the client's thoughts. Ultimately, however, the goal is for clients to be able to revise their own thoughts without therapist input.

The process of revising cognitions should always take place in a context of cultural sensitivity. There is no such thing as universally or absolutely logical thinking. A belief that is logical, rational, or adaptive for members of one culture may be illogical, irrational, or maladaptive for members of another culture. Culturally competent cognitive therapists are aware of the influence that their own cultural background has on their view of logical thinking and are careful not to impose their own cultural values on clients in the process of revising or restructuring clients' cognitions.

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Photo 15.2 According to the cognitive approach, events—such as an unexpected snowstorm—don't directly influence our feelings. Instead, the way we think about those events does. What are some possible thoughts about, for example, an unexpected snowstorm?

Teaching as a Therapy Tool

Cognitive therapists explicitly include in their duties the education of their clients about the cognitive approach. In other words, cognitive therapists often function as teachers with their clients. For example, they might use a combination of minilecture, handouts, and readings to explain to clients the difference between the two-step (events lead directly to feelings) and the preferred three-step (cognitions intervene between events and feelings) models of understanding the sources of our feelings. Moreover, they train clients to recognize illogical thoughts, to assign labels to them, and to track them in a particular written format. And, like any good teacher, cognitive therapists aspire for clients ultimately to be able to use the lessons learned to teach themselves rather than remaining dependent on the teacher (J. S. Beck, 1995; Olatunji & Feldman, 2008).

Homework

Another similarity between cognitive therapists and teachers is the assignment of **homework** (J. S. Beck, 1995; Kuehlwein, 1993; Olatunji & Feldman, 2008; P. Robinson, 2008).

Cognitive therapists strongly believe that much of the work of therapy is conducted between sessions. Much like the time between class meetings of a college course, the time between therapy sessions is used to explore and confirm the lessons learned during the meetings. In some cases, the homework is written: Clients are asked to keep a record of events, cognitions, feelings, and attempts to revise the cognitions to change the feelings they experience. (Later in this chapter, we examine written formats such as these in more detail.) In other cases, the homework is behavioral: Clients are asked to perform certain behaviors before the next meeting, typically for the purpose of examining the validity of an illogical thought. In either case, discussion of the homework will constitute a significant part of the subsequent session (Kazantzis & Dattilio, 2010).

BOX 15.1

Considering Culture

Cognitive Therapy With LGBTQ Clients

For many, life as a lesbian, gay, bisexual, transgender, or queer/questioning (LGBTQ) person today is quite different from what it was a generation or more ago, as society's attitudes toward members of the LGBTQ community appear to have become more accepting. However, Purcell, Swann, and Herbert (2003) argue that in spite of these social changes, negative attitudes toward LGBTQ individuals persist in our society and that these attitudes are reflected even among members of the LGBTQ community. In fact, these authors make the case that "internalized homophobia"—an aversion to homosexuality applied by gay individuals to themselves—pervades gay and lesbian culture to some extent. That is, gay or lesbian individuals may hold some beliefs that disapprove of their own sexual orientation or lifestyle, such as, "Homosexuality is wrong" or "My family/friends/religion/society will reject me if I come out." These beliefs could hinder self-respect and self-worth and contribute to depression or other clinically significant problems.

Are such beliefs consistent with contemporary societal values? Or expressed in terms

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of cognitive therapy, are such beliefs logical? To the extent that they are not, could cognitive therapy be helpful in identifying their logical flaws, challenging them, and replacing them with more logical thoughts? These questions illustrate the need for cultural competence and cultural self-awareness in clinical psychologists. It is essential for the clinical psychologist to understand these beliefs from the perspective of the clients—to see their world through their eyes—to appreciate whether such beliefs are sensible or misguided for them. Clinical psychologists should also be well aware of their own personal views on these issues and stop themselves from equating their own views with the "logical" way to think. What seems adaptive from the perspective of the

clinical psychologist may be maladaptive from the perspective of the client.

Of course, cognitive therapists' work with LGBTQ clients often focuses on cognitions that have nothing to do with their sexual orientation. Indeed, LGBTQ clients bring the same problems to therapy as heterosexual clients do (Martell, Safren, & Prince, 2004), but in addition, they may be struggling with some cognitions related to internalized homophobia, as described by Purcell et al. (2003).

What other diverse groups might experience similar "internalized" self-critical cognitions as a reflection of broader societal views? How might a culturally competent cognitive therapist address the logical or illogical nature of those cognitions?

A Brief, Structured, Focused Approach

Cognitive therapists strive to achieve a positive therapy outcome quite quickly—typically, in fewer than 15 sessions, but significantly longer in complex or severe cases (J. S. Beck, 1995, 2002; D. A. Roth et al., 2002). For outpatients, sessions typically take place once per week, eventually tapering off in frequency as the client improves. Several factors contribute to the efficiency of cognitive therapy, including its focus on the client's current problems (rather than extensive exploration of the past); a purposeful, goal-oriented focus on clearly identified symptoms; and structured therapy sessions (Grant et al., 2005; Olatunji & Feldman, 2008).

Web Link 15.2
Albert Ellis Institute

The structured nature of cognitive therapy sessions differs sharply from the free-flowing, spontaneous style of humanistic therapy (Pretzer & Beck, 2004). Whereas humanistic (or "client-centered") therapists allow clients to determine the topics to be discussed during a session, the amount of time spent on each, and the like, cognitive therapists set an agenda (J. S. Beck, 1995; A. Freeman, Pretzer, Fleming, & Simon, 1990). Typically, each session is sequentially organized into segments (see Table 15.1), and sometimes each segment is allotted a specific amount of time. Of course, the client has input on the content of the agenda for the session, but cognitive therapists usually shun therapy that lacks predetermined, explicit structure.

TWO APPROACHES TO COGNITIVE THERAPY

There are two widely recognized pioneers of cognitive therapy: Albert Ellis and Aaron Beck. As described earlier, each developed his own version of cognitive therapy at about the same time, and although each was influenced somewhat by the other, their approaches evolved independently for the most part. The two approaches unquestionably overlap in terms of their emphasis on improving clients' symptoms via correcting illogical thinking, but the terminology and, at times, the techniques they employ distinguish them from each other (DiGiuseppe, David, & Venezia, 2016). Let's consider each separately.

Table 15.1 Typical Sequential Structure of a Cognitive Therapy Session

1. Check on client's mood or emotional status and solicit brief updates on recent events.
2. Set and confirm the agenda for the current session.
3. Establish a link to the previous session, often by reviewing previous homework assignment.
4. Progress through the body of the current session, proceeding step-by-step through the agenda.
5. Develop and assign new homework assignment.
6. Summarize current session; solicit client feedback.

Sources: Beck (1995), Freeman et al. (1990), and Pretzer and Beck (2004).

Albert Ellis

For many years, Albert Ellis called his approach to therapy rational emotive therapy (RET), but later in his career, he altered the name to **rational emotive behavior therapy (REBT)**. We'll use the more recent name here, understanding that both refer to Ellis's version of cognitive therapy.

As the first two words of the REBT label indicate, Ellis's therapy approach emphasizes a connection between rationality and emotion (David, Cotet, Matu, Mogoase, & Stefan, 2018; Dryden, 2009, 2015; A. Ellis, 2008; A. Ellis & Ellis, 2011; O'Kelly & Collard, 2016). A. Ellis (1962) argues that if we can make our beliefs less irrational, we can live happier lives:

> The central theme of [REBT] is that man is a uniquely rational, as well as uniquely irrational animal; that his emotional or psychological disturbances are largely a result of his thinking illogically or irrationally; and that he can rid himself of most of his emotional or mental unhappiness, ineffectuality, and disturbance if he learns to maximize his rational and minimize his irrational thinking. (p. 36)

The ABCDE Model

One of Ellis's most enduring and clinically useful contributions is his **ABCDE model** for understanding and recording the impact of cognitions on emotions (also known as the ABC model) (e.g., David, 2015; Dryden, 1995, 2009; A. Ellis, 2008; A. Ellis & Ellis, 2011; A. Ellis & Grieger, 1977; A. Ellis & Harper, 1975). By creating this model, Ellis was able to frame the essential aspects of cognitive therapy into an accessible acronym that enabled its use by thousands of therapists and clients.

In the ABCDE model, A, B, and C represent the three-step model described near the beginning of this chapter: Events lead to thoughts, which in turn lead to feelings. Ellis's model simply replaces these three terms with more easily remembered terms:

***A*ctivating event** (A), ***B*elief** (B), and **emotional *C*onsequence** (C). According to Ellis, irrational beliefs are toxic because they function as rigid, dogmatic demands that we apply to ourselves—for example, "I must get an A in every class," "I need to be dating someone," or "I can't let my family down." Although these may be strong preferences, they are not, in fact, "musts" or absolute rules. Moreover, we tend to couple these demands with overestimations of the consequences of failure—"If I don't get an A, I'll flunk out of school and end up on the street"; "If I'm not dating anyone, I'm completely worthless"; or "If I let my family down, their disapproval will destroy me." Ellis sees flawed logic in all these self-statements and opportunity for therapeutic benefit in correcting them.

To accomplish this correction, Ellis's model adds two more steps, D and E. In his model, D stands for ***D*ispute**, and E stands for ***E*ffective new belief**. Specifically, the irrational belief (B) is the target of the dispute. The addition of this step is particularly important within Ellis's model of cognitive therapy. Ellis's model not only helps clients identify irrational beliefs (B) that may intervene with the events in their lives (A) and their subsequent feelings (C); it urges clients to dispute those beliefs as well. This can be an empowering experience for clients who have been stuck in an ABC sequence that leaves them feeling perpetually unhappy, anxious, and so on. When they realize that their experience need not stop at C (the unwanted feeling), that they have the right to challenge the belief that caused C and replace it with something more rational, therapeutic benefit is in the works. In Ellis's model, disputing often takes the form of pointed questions or statements that attack the irrational nature of beliefs or labels that can be assigned to irrational beliefs to discredit them. Regardless of the form of the dispute, if it is effective, it affords the client the opportunity to replace the original, irrational belief with an effective new belief (E) that is more rational and leads to less troubling feelings (Dryden, 2009; A. Ellis, 2008).

As a clinical example, consider Keyon, a 24-year-old man who recently earned a degree in accounting. Keyon sought therapy from Dr. Liu, a clinical psychologist with a cognitive orientation, because he was struggling with excessive anxiety. Specifically, Keyon was scheduled to take the Certified Public Accountant (CPA) exam in about 2 months, but his anxiety about the exam was interfering with his preparation. He intended to study for the exam, but when he tried, he was so anxious that he couldn't concentrate. In fact, just thinking about the CPA exam made Keyon feel panicky. After Dr. Liu educated Keyon about the cognitive model (specifically, Ellis's ABCDE model), they were able to identify steps A and C right away: The activating event was studying for (or thinking about) the CPA exam, and the emotional consequence was anxiety. With Dr. Liu's help, Keyon next identified two beliefs (B) that linked his thoughts of the CPA exam to his feelings of anxiety: "I absolutely have to pass the CPA exam on my first attempt" and "If I don't pass the CPA exam on my first attempt, my career is doomed, and that would destroy me." In the next step, disputing (D), Dr. Liu made efforts to question the logic of Keyon's beliefs:

> Who says you have to pass the CPA exam on your first attempt? I understand that's a preference, but is it a life-or-death necessity? Realistically, don't quite a few accountants fail the CPA exam on their first try? And don't many of them pass it later and go on to have successful careers? And even if you don't end up with the career in accounting that you envisioned, does that mean your life is ruined? There are plenty of ways for you to have a rewarding career that don't involve accounting at all.

In time, Keyon found himself persuaded by the strength of Dr. Liu's arguments and began to disbelieve his own irrational thoughts. Eventually, he was able to replace his original, irrational beliefs with effective new beliefs (E):

> I want to pass the CPA exam on my first attempt, but it's not an absolute necessity. If I pass it on a later attempt, that will probably work out fine also, and in the big picture, my happiness doesn't depend entirely on following the career path I've envisioned.

These new beliefs greatly reduced Keyon's anxiety.

The ABCDE model lends itself quite nicely to written format, and cognitive therapists often take advantage of this. It's likely that Dr. Liu, for example, would have taught Keyon how to view his experiences as ABCDE sequences and chronicle them accordingly. Typically, clients complete forms that are organized into A, B, C, D, and E columns. During sessions or as homework, in retrospect or as an event takes place, clients can sort their experiences into the five-column organizational structure provided by this type of journal form. By doing so, they train themselves to experience life in this sequence. In particular, they become more adept at identifying an irrational belief (B), constructing a dispute (D) in response to the belief, and generating an effective new belief (E). Of course, the goal is not for the clients to depend on this written format for the rest of their lives to feel happier; instead, a five-column ABCDE thought journal can serve as training wheels that stabilize clients while they learn to think more logically, and once they can stabilize themselves, the ABCDE process takes place within the mind, without any outside aids.

Table 15.2 summarizes the full ABCDE acronym, including an applied example from Keyon's therapy with Dr. Liu.

Table 15.2 Albert Ellis's ABCDE Model as Applied to a Clinical Example (Keyon)

A	Activating event	Studying for or thinking about the CPA exam
B	Belief	"I absolutely have to pass the CPA exam on my first attempt. If I don't pass the CPA exam on my first attempt, my career is doomed, and that would destroy me."
C	Consequence (emotional)	Anxiety
D	Dispute	"Who says you have to pass the CPA exam on your first attempt? I understand that's a preference, but is it a life-or-death necessity? Realistically, don't quite a few accountants fail the CPA exam on their first try? And don't many of them pass it later and go on to have successful careers? And even if you don't end up with the career in accounting that you envisioned, does that mean your life is ruined? There are plenty of ways for you to have a rewarding career that don't involve accounting at all."
E	Effective new belief	"I want to pass the CPA exam on my first attempt, but it's not an absolute necessity. If I pass it on a later attempt, that will probably work out fine also, and in the big picture, my happiness doesn't depend entirely on following the career path I've imagined."

BOX 15.2

Considering Culture

Are Some Beliefs Too Sacred to Dispute?

Cody, a 22-year-old college student, grew up in a family with strict religious beliefs. Among them: Premarital sex is sinful, and anyone who engages in it is an immoral person who will spend the afterlife in hell. Last week, Cody and his girlfriend of 2 years had sex for the first time. And his anxiety is through the roof. He seeks the help of a clinical psychologist, Dr. Talia Brown. Dr. Brown holds different religious beliefs than Cody does, and regarding premarital sex, she does not share Cody's beliefs at all. She believes that premarital sex is not necessarily sinful, and she disagrees that it will result in an eternity in hell. In fact, as she applies a cognitive perspective to Cody's case, Dr. Brown sees Cody's beliefs as irrational and distorted. Her plan to help Cody reduce his anxiety is to replace what's in his B column—"Premarital sex is sinful"—with something she believes to be more logical and less anxiety producing—"Premarital sex is OK in some circumstances."

A fundamental task in cognitive therapy is to dispute irrational beliefs, but who decides what's irrational? Moreover, are some client beliefs—religious, cultural, or otherwise—too sacred to dispute, even if the therapist questions their rationality? Is it appropriate for Dr. Brown to critically examine Cody's beliefs about premarital sex, or should she take a "hands-off" approach in the spirit of cultural sensitivity? In Dr. Brown's situation, many clinical psychologists would choose not to question any of Cody's religious beliefs, with the rationale that respect for Cody's religion is an important part of cultural competence. However, some psychologists disagree. Ridley, Ethington, and Heppner (2008) argue that when the working alliance between therapist and client is strong and the timing is right, *cultural confrontation* is preferable:

> Often [mental health professionals] assume that counselors should accept without question clients' cultural values. In their thinking, uncritical acceptance reflects an unbiased approach to counseling, especially as it pertains to clients whose cultural backgrounds are radically different from their own. . . . We maintain that in some cases, rigid and extreme adherence to cultural values not only is dysfunctional but also creates a great deal of psychological distress. We suggest that examination of clients' cultural impasses and at times cultural confrontation are essential competencies under the larger domains of multicultural counseling competence. (p. 378)

Indeed, cognitive therapists have long recognized the importance of cultural competence as they work with clients of diverse religious, ethnic, and other groups (e.g., Hays & Iwamasa, 2006; H. G. Koenig, 2005; Paradis, Cukor, & Friedman, 2006; Pargament, 2007). But the notion of cultural confrontation opposes much of the traditional wisdom about respecting clients' values by declining to challenge them. In your opinion, should Dr. Brown proceed with her plan to challenge Cody's religious beliefs about premarital sex? If she does, will Cody be receptive to her strategy? Will his anxiety decrease? If you were the client, how would you respond? Which beliefs of yours—religious, cultural, political, personal, or otherwise—would you see as too sacred for your psychologist to dispute?

Aaron Beck

Aaron Beck has always used the general term *cognitive therapy* to describe his technique. He originally developed his approach as a way to conceptualize and treat depression (e.g., A. Beck, 1976; A. Beck, Rush, Shaw, & Emery, 1979), but it has been very broadly applied since shortly after its inception. (In fact, his daughter, **Judith Beck**, has become a leader of the current generation of cognitive therapists and has spearheaded its application to many new problems.) An important part of Beck's theory of depression is his notion of the **cognitive triad**, in which he argues that three particular cognitions—thoughts about the self, the external world, and the future—all contribute to our mental health. Beck theorized that when all three of these beliefs are negative, they produce depression (Alford & Beck, 1997; J. S. Beck, 1995).

Web Link 15.3
Judith Beck

The essence of Aaron Beck's approach to cognitive therapy, like Ellis's, is to increase the extent to which the client thinks logically. And also like Ellis's, Beck's approach incorporates a way of organizing clients' experiences into columns on a written page. In Beck's brand of cognitive therapy, this form is known as a **dysfunctional thought record** (e.g., J. S. Beck, 1995, 2002; A. Freeman et al., 1990; Leahy, 2003), and although its headings differ a bit from Ellis's ABCDE acronym, they function similarly. Typically, the dysfunctional thought record includes columns for

- a brief description of the event/situation,
- automatic thoughts about the event/situation (and the extent to which the client believes these thoughts),
- emotions (and their intensity),
- an adaptive response (identifying the distortion in the automatic thought and challenging it), and
- outcome (emotions after the adaptive response has been identified and the extent to which the client still believes the automatic thoughts).

Conceptually, the columns in Beck's dysfunctional thought record correspond quite closely to the columns of Ellis's ABCDE forms. For example, in the fourth column ("adaptive response") of a dysfunctional thought record, clients perform essentially the same task they would in Column D ("Dispute") of Ellis's form. For this task, Beck created a vocabulary to identify common ways clients' thoughts can be distorted. This vocabulary has become a vital aspect of cognitive therapy. Let's consider it here in more detail.

Common Thought Distortions

An essential step in cognitive therapy is to discredit illogical automatic thoughts by labeling them. To facilitate this labeling, Beck and his followers have identified and defined a list of **common thought distortions** (e.g., A. T. Beck et al., 1979; J. S. Beck, 2002; Craighead, Craighead, Kazdin, & Mahoney, 1994; Leahy, 2003; C. F. Newman, 2016). Cognitive therapists teach these terms to clients, often using handouts or take-home

readings, and train them to use the terms when examining their own thoughts. Examples of these common thought distortions include the following:

- **All-or-nothing thinking:** Irrationally evaluating everything as either wonderful or terrible, with no middle ground or "gray area"
- **Catastrophizing:** Expecting the worst in the future, when, realistically, it's unlikely to occur
- **Magnification/minimization:** For negative events, "making a mountain out of a molehill"; for positive events, playing down their importance
- **Personalization:** Assuming excessive personal responsibility for negative events
- **Overgeneralization:** Applying lessons learned from negative experiences more broadly than is warranted
- **Mental filtering:** Ignoring positive events while focusing excessively on negative events
- **Mind reading:** Presuming to know that others are thinking critically or disapprovingly, when knowing what they think is, in fact, impossible

In Beck's cognitive therapy, when clients assign these thought distortion labels to illogical thoughts, the illogical thoughts grow weaker. Labeling thoughts as illogical enables the client to dismiss them and replace them with more adaptive and logical thoughts, which ultimately decreases the client's psychological distress. As a clinical example, consider Olivia, a 30-year-old woman who was recently divorced after a 3-year marriage and currently lives alone. In her first session with Dr. Zimmerman, a clinical psychologist with a cognitive orientation, Olivia explains that she feels depressed about being without a partner. The comments she made to Dr. Zimmerman could be summed up in these three beliefs: "I'm no good at relationships," "Living alone, even for a short time, is intolerable," and "A lot of my friends are married, but I'm not, so there must be something wrong with me." After educating Olivia about Beck's cognitive approach, including the list of common thought distortions, Dr. Zimmerman and Olivia got to work. Together, they attacked the flawed logic in each of Olivia's beliefs by labeling them as distortions. For example, when Olivia views herself as "no good at relationships," she's overgeneralizing from the recent divorce. Living alone may not be her preference, but to call it "intolerable" constitutes magnification. And to blame the divorce on herself—"there's something wrong with me"—is personalization that is unfounded and unfair. With repeated practice, Olivia developed the ability to identify and oppose her own illogical thoughts and replace them with more logical alternatives. She never became overjoyed about her divorce or the loneliness in her current life—such a reaction would also be illogical—but she was able to lift herself from a state of despair to a state of contentment and mild hopefulness, which made a tremendous difference in her day-to-day life.

Beliefs as Hypotheses

Beck argued that our beliefs are **hypotheses**, even though we may live our lives as if our beliefs are proven facts. Therefore, a potent way to expose a belief as illogical is

BOX 15.3

In My Practice . . .

In my practice, I have learned that when I use cognitive therapy, sometimes clients are reluctant to challenge an illogical belief. That seems counterintuitive—after all, shouldn't they welcome a dispute that helps them think more rationally and improve their emotional state? In some cases, I have found, there's a second illogical belief lurking behind the first one.

For example, consider Ayana, who was in therapy with me when she was a high school senior. Ayana had a remarkably negative self-image. That self-image was centered on a core belief of hers: "I'm stupid." Ayana's belief in her own stupidity was so powerful that it not only caused her great sadness, it also inhibited her from applying to college. As I explored this belief with Ayana, it became clear to me that it was not entirely based on truth. In fact, it seemed way off base. Ayana earned a solid B+ grade point average all the way through high school. She performed well in several honors courses. Her friends often turned to her for help with their homework. Clearly, there was plentiful evidence for Ayana's intelligence, but she focused only on the negative: She earned a D in a math class during the first semester of ninth grade, she had a couple of friends with better GPAs than hers, and she had recently bombed a physics test. I taught Ayana about the common thought distortions and helped her see that several of them could explain her irrational belief in her own stupidity: She was magnifying her few academic weaknesses and minimizing her many academic strengths; she was overgeneralizing from her worst academic moments to a broad image of herself as stupid; and she was using all-or-nothing thinking rather than giving herself the option of falling somewhere short of sheer genius but far, far above absolute stupidity.

Any of these arguments, I reasoned, should be enough to allow Ayana to escape from her false self-image as a stupid person. But she didn't escape. We discussed these disputes, and the effective new beliefs she could adopt, for several sessions. But she seemed more motivated to keep her self-image than to change it. Finally, I tactfully and kindly mentioned to Ayana what I was noticing. I asked her if there was a reason why she might be reluctant to see herself as a smart, capable young woman. Through tears, she revealed something important: "If I'm smart, I should apply to college. But if I go to college, my parents will resent me for it. I'll be the first one from our family to go to college. They will think I'm selfish. I can't do that to them." Now it made sense to me why Ayana might cling to the illogical belief that she's stupid. But was it possible that the belief about her parents' attitude toward Ayana going to college was itself an illogical belief? Much as Aaron Beck encouraged clients to test their beliefs as if they were hypotheses, I encouraged Ayana to bring up the topic of college with her parents, to see how they responded. To her surprise, they were completely in favor of it. Her parents mentioned nothing about selfishness or resentment and spoke extensively about how proud they would be. After a couple of conversations like this with her parents, Ayana was able to replace her inaccurate belief about her parents' view of college with the more accurate belief. That alteration, in turn, enabled her to replace her illogical belief that she was stupid with the logical belief that she was bright and capable. Soon, she was working on college applications.

To see this box come to life as a whiteboard video, please visit edge.sagepub/pomerantz5e.

to "put it to the test" in real life, just as scientists empirically test their hypotheses in the lab. Beck's approach to cognitive therapy often includes such personal "experiments," frequently in the form of homework, designed to bolster or undermine a client's beliefs (K. S. Dobson & Hamilton, 2008; Kuehlwein, 1993; D. A. Roth et al., 2002; C. F. Newman, 2016).

As a clinical example, consider Frank, a 45-year-old chain-restaurant manager who has held his job for 15 years but has become increasingly unhappy with it. This professional dissatisfaction is the main contributor to the depression for which he seeks therapy. He mentions to Dr. Morris, his cognitively oriented clinical psychologist, that he would like to look for another job, but adds, "I'm sure I wouldn't be able to get one," a belief that leaves Frank feeling dejected about the future. One effective strategy that Dr. Morris might employ would involve challenging the logic of Frank's belief that he is not employable elsewhere. Using Beck's dysfunctional thought record (or Ellis's ABCDE columns), Dr. Morris could try to use words to persuade Frank that his belief is illogical: Perhaps Frank is minimizing his skills and experiences, mind reading when in fact he doesn't know how prospective employers may evaluate him as an applicant, or engaging in some other cognitive distortion. To accompany this argument, Dr. Morris might also assign Frank some homework that will serve to test his belief. For example, Dr. Morris might ask Frank to create a résumé and highlight the parts of it that would be attractive to an employer (training, years of experience, etc.). Or Dr. Morris might ask Frank actually to test the market—respond to some want ads and see what kind of response he gets from employers. If these homework assignments result in some feedback inconsistent with Frank's hypothesis—that is, if his résumé actually looks good or if prospective employers show interest—the experience will force Frank to abandon the belief that he's unemployable. And by replacing that belief with the belief that he is indeed attractive to employers, Frank will be more hopeful and less vulnerable to depressive feelings.

Of course, when cognitive therapists encourage clients to test their hypotheses, they are careful to do so in a way that will effectively refute illogical thoughts (Kuehlwein, 1993). If they assign homework that confirms illogical beliefs, the efforts can backfire.

In practice, the approaches of Ellis and Beck overlap quite a bit. Cognitive therapists often incorporate elements of both styles of therapy into their own techniques. Although the terminology that Beck and Ellis use differs somewhat, their therapeutic goals are essentially the same: to identify and critically evaluate illogical thinking and replace it with more rational alternatives that ultimately alleviate psychological symptoms.

RECENT APPLICATIONS OF COGNITIVE THERAPY

Although cognitive therapy was originally targeted toward limited types of psychological symptoms, it is now applied almost universally across the range of psychological problems. In fact, it is increasingly used for problems outside the range of traditional mental disorders as well.

BOX 15.4

Metaphorically Speaking

If You've Seen Attorneys Argue in Court, You Understand How Cognitive Therapists Dispute Thought Distortions

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"Objection!" In the courtroom, this is how attorneys protest the unsound tactics of opposing counsel. Usually, the purpose of an objection is to interrupt an illogical argument. In other words, as soon as an attorney notices that the opposition is putting forth a logically flawed argument, the appropriate action is to insist that only logically sound statements be allowed in the argument and that any irrational statement be stricken from the record.

In a way, cognitive therapists teach clients to be their own defense attorneys in the "cognitive courtroom" of the mind. Specifically, cognitive therapists train clients to spot illogical thoughts, object to them, and insist that only logical thoughts be allowed. This process implies that there are two opposing voices in each of our minds, just as there are two opposing attorneys in a courtroom. These two voices are represented by the two columns in the cognitive model in which beliefs are articulated—Columns B (Belief) and E (Effective new belief) in Ellis's ABCDE model. Between them, Column D (Dispute) serves as an objection to the first, illogical voice (B) and an opportunity for the second, logical voice (E) to make a more logical statement.

To illustrate, let's consider an illogical belief held by Shannon, a 20-year-old college student: "If I fail an exam, I'm stupid." Or, stated as an accusation from another person, "If you fail an exam, you're stupid." If this accusation goes unopposed, as it would in a courtroom with only the prosecuting attorney in attendance, this case may end with the verdict that Shannon is, in fact, stupid. But if Shannon objects to the flawed logic in this belief—in Ellis's terms, if she disputes it effectively—Shannon will exonerate herself of the charge of stupidity. In truth, Shannon may be able to use several different disputes to support her objection—perhaps the test was unfair and many students failed it, or perhaps Shannon wasn't feeling well on the day of the test, or perhaps Shannon has earned As on most of her other tests in college. Any of these objections creates reasonable doubt about the accusation that Shannon is stupid and increases the likelihood that, in her "cognitive courtroom," she will be found innocent of this charge. As a result, she will avoid a "sentence" of depression. A key to this metaphor, and to cognitive therapy more generally, is that when illogical thoughts cross their minds, individuals feel entitled to defend themselves by objecting and correcting the illogical thoughts rather than simply allowing them to continue unchallenged.

The Third Wave: Mindfulness- and Acceptance-Based Therapies

In recent years, a new brand of therapies based on mindfulness and acceptance have become increasingly popular and empirically supported (S. C. Hayes, Villatte, Levin, & Hildebrandt, 2011; Masuda & Wilson, 2009; L. P. K. Morgan & Roemer, 2015). Collectively, they are often called "third-wave" therapies, referring to the evolution from strict behaviorism (first wave) to cognitive therapy (second wave) to these newer therapies (DiGiuseppe et al., 2016; W. C. Follette, Darrow, & Bonow, 2009; S. C. Hayes, 2004). As such, they should not necessarily be considered cognitive therapies in the strict sense; although they appear in this chapter on cognitive therapy, they also feature behavioral and other elements. We'll examine three particular forms of these therapies in detail: acceptance and commitment therapy, dialectical behavior therapy, and metacognitive therapy. But first, let's consider some commonalities among therapies of this kind.

Mindfulness lies at the core of the third-wave therapies (Dimidjian & Linehan, 2009; V. M. Follette & Hazlett-Stevens, 2016; S. C. Hayes et al., 2011; S. L. Shapiro, 2009). Mindfulness can be difficult to define, but its proponents have made attempts:

- Mindfulness "refers to being able to pay attention in the present moment to whatever arises internally or externally, without becoming entangled or 'hooked' by judging or wishing things were otherwise" (Roemer & Orsillo, 2009, p. 2).
- Mindfulness is "an innate human capacity to deliberately pay full attention to where we are, to our actual experience, and to learn from it. This can be contrasted with living on automatic pilot and going through our day without really being there" (Hick, 2008, p. 5).
- "The short definition of mindfulness . . . is (1) awareness, (2) of present experience, (3) with acceptance" (Germer, 2005, p. 7).
- Mindfulness is "the awareness that arises out of intentionally attending in an open and discerning way to whatever is arising in the present moment" (S. L. Shapiro, 2009, p. 555).

As the previous descriptions may suggest, mindfulness derives from Buddhist traditions, but it is typically used without any explicit religious ties. (Zen meditation can accompany mindfulness-based therapies, but they are often practiced without any meditation component. In fact, *mindfulness* is the term that often replaced *Zen* as Zen practices became more popular in the Western world and became less connected to religion practices or meditation (Lungu & Linehan, 2016).) Mindfulness promotes full engagement with one's own internal mental processes in a nonconfrontational way. This nonconfrontational approach is a key difference from the more traditional cognitive therapies of Albert Ellis or Aaron Beck. Whereas Ellis and Beck encouraged people to dispute and revise their thoughts, mindfulness-based therapists prefer to change people's *relationships to* their thoughts rather than the thoughts themselves (Olatunji & Feldman, 2008). So, rather than relating to thoughts as all-powerful determinants of reality or

truth, clients can learn to understand their thoughts as fleeting suggestions that may not require much of a reaction at all. Once the relationship with thoughts is changed in this way, individuals may find it easier to face unpleasant thoughts (or feelings or sensations) rather than avoiding them. That is, rather than engaging in **experiential avoidance**, as third-wave therapists call it, the individual can engage in **acceptance**: allowing these internal experiences to run their course without fighting against them. This can facilitate positive change for clients with a wide range of psychological problems (Dimidjian & Linehan, 2008; Farmer & Chapman, 2008; Niles et al., 2014; Orsillo, Danitz, & Roemer, 2016; Roemer & Orsillo, 2009; Spinhoven, Drost, de Rooij, van Hemert, & Penninx, 2014).

As a specific illustration of a clinical intervention that relies on mindfulness, consider *urge surfing*. Urge surfing is an approach to the treatment of addictive behaviors such as smoking or drinking, or any other behavior in which clients struggle with unwanted urges, that encourages clients to relate differently to their urges than they have before (S. Bowen & Marlatt, 2009; J. S. Harris, Stewart, & Stanton, 2017; Ostafin & Marlatt, 2008; Rogojanski, Vettese, & Antony, 2011; Shonin & Van Gordon, 2016; N. N. Singh et al., 2018). The goal is not to stop, suppress, or fight the urges, as other forms of therapy might encourage; instead, the goal is to experience them, to "ride" or "surf" them like a wave that will rise and then inevitably subside, and to realize that they are temporary and not all-powerful. As described by Lloyd (2009),

> Clients are taught to treat urges as though they were like waves in the ocean. Urges come on, grow in intensity, and eventually subside just like ocean waves. Moreover, like waves, urges tend to be brief. They do not grow and grow until the client has to do something before they will go away. Urges go away on their own. . . . [In this type of treatment,] clients are taught how to fully experience the urge in a different way—that is, to experience the urge for what it is: brief, nonlethal, of relatively predictable course, and most important, defeatable. (p. 669)

As this description illustrates, the critical element of this treatment is not the urge itself but the way the client responds to the urge. As with all third-wave or mindfulness-based therapies, the emphasis is on helping clients experience the urge (or feeling or thought) that arises in the moment and not overestimate its destructive power or underestimate their own ability to withstand it. It may be unpleasant, but it will pass, and the client can accept and survive it.

Mindfulness and acceptance are valuable new components of therapeutic practice, whether these concepts form the foundation of the therapy strategy or merely complement other approaches. That is, mindfulness and acceptance don't demand an either/or decision. Therapists can practice forms of therapy for which mindfulness and acceptance are core elements—as in the three therapies described in this section—or therapists can enhance other forms of therapy, such as psychodynamic, humanistic, or others, by blending in elements of mindfulness and acceptance (Hick & Bien, 2008). For example, one study found that for clients with OCD who were treated with exposure and response prevention, adding a mindfulness component to the treatment significantly increased their improvement (Key, Rowa, Bieling, McCabe, & Pawluk, 2017).

Acceptance and Commitment Therapy

Steven C. Hayes, a clinical psychologist and professor at the University of Nevada, Reno, is considered a leading figure in **acceptance and commitment therapy (ACT)**. According to Hayes and others, what is "accepted" in ACT is internal psychological experience, such as emotions, thoughts, and sensations (Bach & Moran, 2008; Forman, Juarascio, Martin, & Herbert, 2015; S. C. Hayes, 2004; S. C. Hayes & Strosahl, 2004). Too often, individuals who struggle with psychological problems have not accepted these private events but have dodged and ducked them via distraction. This kind of experiential avoidance can underlie all kinds of psychological problems (Eifert & Forsyth, 2005). It's a bit like a phobia, but the feared object is within rather than outside the individual. And just like phobias of dogs, snakes, or airplanes, avoidance is a common but ultimately unhelpful coping mechanism. So, in this context, acceptance means facing one's *internal* fears.

Web Link 15.4
Website of Steven Hayes

S. C. Hayes (2004) presents a couple of metaphors that clarify the basic tenets of acceptance within ACT. For one, he asks us to imagine our thoughts as a parade. In this parade, we are spectators, not participants. As our thoughts march by, we notice them, certainly, but how long can we let the parade flow without reacting to them—without joining the parade or trying to stop it or otherwise getting drawn in? The longer we can acknowledge our thoughts without compulsively reacting to them, according to Hayes, the better our chances of psychological well-being will be. In a second metaphor, Hayes compares the process of accepting internal experiences to jumping off a step rather than stepping down from it. Stepping down feels safer, he says, because we maintain control the whole time. Jumping down gives all the control to gravity. Clients, according to Hayes, need to "practice jumping" with regard to their thoughts, feelings, and sensations. That is, little by little, they need to stop battling for control with these experiences and simply trust that wherever the experiences drop them, they will be able to land safely and continue on their way.

The *C* in ACT refers to a commitment to one's own personal values. Of course, for many clients, clarification must come before commitment. That is, therapy must first help the client discover exactly what their personal values are. Once this happens, the client is poised to make a commitment to remaining true to these values in terms of their day-to-day decisions and behaviors. And the *T* in ACT refers to taking action consistent with one's own personal values; here, we see the behavioral emphasis inherent in ACT that connects the client's new way of thinking with new ways of living.

Hayes and others (e.g., Eifert & Forsyth, 2005; S. C. Hayes & Strosahl, 2004) describe a pair of acronyms to explain the problematic approach that underlies psychological illness, as well as their strategy for psychological wellness. Essentially, FEAR is replaced by ACT. To explain, FEAR stands for

- *Fusion* with inner experiences such as thoughts, feelings, and sensations that limits flexibility in responding;
- *Evaluation* of self, especially one's own inner experiences;
- *Avoidance* of unpleasant inner experiences via such means as distracting or numbing oneself; and
- *Reason-giving*, or leaning too heavily on rationalizations that sound legitimate but actually perpetuate unhealthy approaches to life.

ACT, on the other hand, stands for

- *Accepting* one's own inner experiences for what they are, and nothing more;
- *Choosing* directions in life based on one's core values, which will enhance life's meaning and purpose; and
- *Taking action* in matters large and small that are consonant with one's own values.

Dialectical Behavior Therapy

Dialectical behavior therapy (DBT) was developed by **Marsha Linehan** specifically for the treatment of borderline personality disorder (BPD; Koerner, 2012; Koerner & Dimeff, 2007; Linehan, 1993a, 1993b). The treatment has garnered a notable level of empirical support, such that it is now considered a treatment of choice for BPD, and in adapted form it is also used successfully for other disorders, especially those that involve emotion dysregulation (Cameron, 2015; Dimeff & Koerner, 2007; Kliem, Kröger, & Kosfelder, 2010; Lungu & Linehan, 2016; Lynch, Trost, Salsman, & Linehan, 2007; Paris, 2009).

DBT is based on a conceptualization of BPD as a problem of emotional regulation. In other words, across most aspects of their lives, individuals with BPD struggle to control the intensity of their feelings (Linehan, 1993a). Emotional dysregulation is thought to stem from two sources: biological predisposition and environment. Specifically, the environmental component is an invalidating interpersonal environment. That is, individuals who develop BPD often come from families who "communicate that the individual's characteristic responses to events (particularly their emotional responses) are incorrect, inappropriate, pathological, or not to be taken seriously" (Koerner & Dimeff, 2007, p. 3). Such an environment teaches the individual that only extreme emotional reactions will elicit a response from others, so they often communicate very intense displays of emotion when others might see such displays as excessive or unnecessary. This can explain why individuals with BPD are often very draining to friends and family (not to mention therapists) and why they threaten and attempt suicide at a high rate (Wheelis, 2009).

A few core practices are central to DBT: problem solving, validation, and dialectics. When DBT therapists work with clients on problem solving, they help clients "think through" stressful situations that might otherwise evoke an extreme emotional response. In a pragmatic way, they encourage clients to strategize for the best possible outcome, considering what will happen if they act on emotional impulses versus taking more deliberate actions. Additionally, the problem-solving component of DBT includes some discussion of the factors that might have inhibited effective problem solving in the past and how the client can overcome those factors in the present and the future. The validation component of DBT focuses on the client's feelings, which (as stated earlier) have typically not been validated in the past. Compared with empathy, which is commonplace among many therapies, validation as practiced in DBT is much stronger. It directly, persuasively communicates to BPD clients that their feelings are both important and a sensible reaction to their situation. The dialectics involved in DBT refer to efforts to resolve simultaneous, contradictory feelings or experiences of the client. For example, clients may express a wish to kill themselves yet not do so. In such a situation, the DBT

therapist, with a balance of respect and confrontation, can point out that the client has a wish to live as well as a wish to die and can discuss with the client ways in which the healthier wishes can be strengthened (Dimeff & Koerner, 2007; Koerner, 2012; Koerner & Dimeff, 2007; Wheelis, 2009). Another example of a dialectic involved in DBT is more broad-based: simultaneous acceptance and change of the client. The therapist should strive to communicate the dual message that they accept clients while at the same time pushing them to change, and to get clients to adopt the same attitude toward themselves. Striking a balance between the two is essential (and difficult), because endorsing acceptance too strongly can make clients feel invalidated regarding their emotional pain, while endorsing change too strongly can make clients feel invalidated regarding their overall acceptability (Lungu & Linehan, 2016).

Linehan (1993b) includes four specific modules of skills training within DBT. Collectively, they relate closely to the core components of DBT described above, but they are best described as problem-solving strategies that therapists teach to clients. The skills include

- emotion regulation, which involves identifying, describing, and accepting rather than avoiding negative emotions;
- distress tolerance, which emphasizes the development of self-soothing techniques and impulse control to help clients with BPD minimize such behaviors as suicide attempts, self-harm, and drug abuse;
- interpersonal effectiveness, which helps clients determine appropriately assertive social skills in order to preserve relationships that might otherwise be damaged by extreme emotional outbursts; and
- mindfulness skills, which encourage clients to engage fully in their present lives, including their internal experiences, such as feelings, thoughts, and sensations, without avoidance or evaluation.

Metacognitive Therapy

In traditional cognitive therapy, as practiced by Albert Ellis, the irrational belief is brought on by an "activating event" (A in the ABCDE model). The primary idea in the relatively new practice of **metacognitive therapy** is that the activating event can be a cognition itself rather than some external occurrence. Simply stated, people can become depressed, anxious, or otherwise psychologically unwell because of reactions to their own thoughts rather than their reactions to the things that happen to them (K. S. Dobson, 2013; P. Fisher & Wells, 2009; Wells, 2009). So the cause of our unhappiness is just as likely to be *thoughts about thoughts* as thoughts about external events.

Metacognitive therapists frequently refer to the cognitive attentional syndrome (CAS), a term that describes a brooding, ruminative, problematic thinking style that can underlie many psychological problems. The CAS can include two types of specific thoughts about worry, positive and negative—and both cause trouble. Positive beliefs about worry may sound like this: "Worrying helps me prepare for the future. If I don't worry, I could get blindsided by something. The last thing I want to do is stop worrying." Negative beliefs about worry may sound like this: "Oh no, I have started worrying. Once

I start, I'll never be able to stop. It's gonna be a terrible day now that the worrying is happening. This worry is totally out of control." Whatever the original, external event may have been—a failed exam, upsetting medical news, a relationship breakup, an unexpected financial expense—the client's thoughts about the event may pile up rather quickly, such that not just thoughts about the event but *thoughts about thoughts* about the event can be the most relevant trigger of the anxiety. As such, metacognitive therapists make thinking about thinking the primary focus of their interventions.

Web Link 15.5
Website of Marsha Linehan

Metacognitive therapy has been applied primarily to anxiety disorders, including obsessive-compulsive disorders, posttraumatic stress disorder, and generalized anxiety disorder. Although it is a relatively new treatment, evidence for its effectiveness with these disorders has started to appear (D. A. Clark & Beck, 2010; P. Fisher & Wells, 2008; Wells & King, 2006; Wells et al., 2008). A version of metacognitive therapy has also been used with clients who have schizophrenia, where the goal is to promote thinking about delusional thoughts (rather than immediate acceptance of those delusional thoughts; Moritz et al., 2014).

Cognitive Therapy for Medical Problems

The relationship between mind and body can strongly influence the way an individual deals with a medical problem. Of particular interest to cognitive therapists are the beliefs that medical patients hold regarding their illness, injury, or condition. How will it affect them? How will their family members respond or cope with it? How well will treatment work? What negative effects might treatment have? Irrational answers to these questions could unnecessarily hinder recovery, and, moreover, they could cause excessive worry or despair in the process.

In recent decades, numerous studies have indicated that cognitive therapy can have a significantly beneficial effect on the healing process and the ultimate prognosis of medical patients. For example, Jakes, Hallam, McKenna, and Hinchcliffe (1992) examined the effect of cognitive therapy on patients with tinnitus, an auditory problem involving excessive perception of noises. Some of these patients underwent a brief form of cognitive therapy in which their illogical beliefs about the disease were corrected. Compared with patients who did not receive this cognitive therapy, those who did demonstrated significant improvement regarding their level of distress about tinnitus. Cognitive therapy has been successfully applied to many other medical problems as well, including chronic headache, chronic pain, premenstrual syndrome, sexual disorders, spinal cord injuries, and brain injury (Carter, Forys, & Oswald, 2008; A. Freeman & Greenwood, 1987; Jay, Elliott, Fitzgibbons, Woody, & Siegel, 1995).

As an example of the potential of cognitive therapy to positively influence the lives of medical patients, consider Jackie, a 45-year-old woman recently diagnosed in the very early stages of breast cancer. Understandably, Jackie is distressed by the diagnosis, but some of her initial beliefs about the disease are in fact illogical, and these thoughts make her more distraught than she needs to be. For example, Jackie firmly believes, "I'm going to die," "I'll need chemotherapy, which will be so painful and miserable that I won't be able to live through it," and "My family and friends will distance themselves from me if they find out." Although these may be possibilities, Jackie may be overestimating their likelihood and, in the process, convincing herself of a worst-case scenario. In cognitive

therapy with Dr. Richards, her clinical psychologist, Jackie challenges the validity of these beliefs and learns to identify irrational thoughts and replace them with more rational thoughts. By the end of a brief course of therapy, Jackie remains concerned about her breast cancer but not excessively so. She is realistic rather than pessimistic: "I could die, but the odds are low because it was caught very early and I'm getting good care"; "I may not need chemotherapy, and if I do, it will be very unpleasant but tolerable"; and "I can't be sure how my family and friends will react, but their past behavior leads me to believe that most will be quite supportive." These new beliefs—free of catastrophizing, magnification, mind reading, or other distortions—produce in Jackie a better psychological state and a better medical prognosis as well.

Schema Therapy

Schema therapy is a relatively recent variation of cognitive and cognitive-behavioral therapy intended for clients who have borderline personality disorder or other long-standing, complex clinical issues. It is based on the idea that when young children experience poor parenting (such as parents who are rejecting, unloving, abusive, unavailable, etc.), they are likely to develop schemas, or deep-seated cognitions about themselves and their relationships to others, that match their experience but are maladaptive as they move forward in life. For example, a child may adopt schemas centering on abandonment ("People who are important to me will leave me"), mistrust ("People in my life will hurt me"), shame ("I must be a terrible, unlovable person"), or failure ("I'm untalented and will never succeed"). These schemas become the focus of therapy. The therapist not only helps the client see how these schemas may be false, but also uses the therapeutic relationship (in which the client is likely to act out these schemas) as a firsthand way to demonstrate their falsehood. The key difference between schema therapy and most other forms of cognitive therapy is the attention paid to the roots of the problematic cognitions—in other words, the focus on early childhood and how it shaped the cognitions, or schemas, that continue to have a negative effect on the client's current life (Dozois & Brinker, 2015; van Vreeswijk, Broersen, & Nadort, 2012; J. E. Young, Klosko, & Weishaar, 2003). Schema therapy has a notable and growing amount of evidence in the treatment of borderline and other personality disorders (Bamelis, Evers, Spinhoven, & Arntz, 2014; Farrell, Shaw, & Webber, 2009; Giesen-Bloo et al., 2006; Reiss, Lieb, Arntz, Shaw, & Farrell, 2014). It has also been used to treat clients with long-term depression and complex eating disorders (Pugh, 2015; Renner, Arntz, Leeuw, & Huibers, 2013).

HOW WELL DOES IT WORK?

The efficacy of cognitive therapy is strongly supported by a body of empirical evidence that is enormous and continues to grow. Lists of "treatment that work," including comprehensive books on the subject and websites (e.g., APA Division 12), mention cognitive therapy many times for many disorders (DiGiuseppe et al., 2016). The range of psychological disorders for which cognitive therapy works is expansive, including depression, anxiety disorders (especially generalized anxiety disorder), bulimia, posttraumatic stress

disorder, hypochondriasis, numerous personality disorders, and others (J. S. Beck, 2002; Cuijpers et al., 2014; Epp, Dobson, & Cottraux, 2009; Hanrahan et al., 2013; Higa-McMillan, Francis, Rith-Najarian, & Chorpita, 2016; Hofmann, Sawyer, Witt, & Oh, 2013; Querstret & Cropley, 2013; Rees & Pritchard, 2015; D. A. Roth et al., 2002). And the improvements that cognitive (and cognitive-behavioral) therapies provide may extend beyond mere symptom relief: One meta-analysis of 44 studies, covering over 3,300 clients, found that cognitive-behavioral therapy had a significantly positive effect not just on the disorder itself but on overall quality of life, including a broad range of physical and psychological variables (Hofmann, Wu, & Boettcher, 2014). And, as stated earlier, studies support the use of cognitive therapy for some medical problems and some personality disorders as well. Specific components of cognitive therapy have also received empirical support; for example, numerous studies have found that homework enhances therapy outcome in comparison with similar therapies that do not include homework (e.g., Dozois, 2010; Kazantzis, Whittington, & Dattilio, 2010).

Some of the evidence supporting the efficacy of cognitive therapy specifically focuses on REBT. One meta-analysis of dozens of REBT treatment outcome studies over a 50-year period found strong evidence for its ability to reduce irrational beliefs and produce positive outcomes (David et al., 2018). Another meta-analysis found that simply having rational beliefs—the primary aim of the REBT approach—significantly negatively correlates with psychological distress. In other words, as rational beliefs increase, the chances of psychological problems (broadly defined) drop. The most important rational beliefs appeared to be those related to self-acceptance (Oltean & David, 2018).

Evidence is accumulating for the third-wave, mindfulness-based therapies as well (Banks, Newman, & Saleem, 2015; Bos, Merea, Brink, Sanderman, & Bartels-Velthuis, 2014; Feliu-Soler et al., 2018; Norton, Abbott, Norberg, & Hunt, 2015). A large-scale meta-analysis of over 200 studies of mindfulness-based therapies, covering more than 12,000 clients, found robust effects across a wide range of clinical problems, especially anxiety disorders, depression, and stress (Khoury et al., 2013). Mindfulness-based therapies have also been found to have positive effects in meta-analyses that focus on specific problems like eating disorders, psychosis, and weight loss (Carrière, Khoury, Günak, & Knäuper, 2018; Linardon, Gleeson, Yap, Murphy, & Brennan, 2019; Louise, Fitzpatrick, Strauss, Rossell, & Thomas, 2018). DBT is particularly well established as a treatment that works for borderline personality disorder and an increasingly wide range of additional disorders (Dimeff & Koerner, 2007; Kliem et al., 2010; Koerner, 2012; Lynch et al., 2007; Paris, 2009). ACT has also accumulated an impressive amount of empirical evidence in the treatment of various anxiety and mood disorders and some other disorders as well (D. M. Davis & Hayes, 2011; Hofmann et al., 2010; Olatunji & Feldman, 2008; Roemer & Orsillo, 2009; Swain, Hancock, Hainsworth, & Bowman, 2013). Evidence is also growing for the benefits of schema therapy for borderline personality disorder and perhaps for other disorders such as chronic depression and complex eating disorders (Jacob & Arntz, 2015; Pugh, 2015; Renner et al., 2013).

As a result of this therapy's roots in the behavioral movement, cognitive therapists typically emphasize aspects of therapy that facilitate empirical evaluation, such as defining problems in terms that can be overtly measured and observed. If these terms are not blatantly behavioral, they may take the form of numerical ratings that clients assign to symptoms' severity (such as depressed mood or anxiety level) both before and after

cognitive interventions. So, although a client's anxiety may rate a 90 on a 0 to 100 scale before therapy, it may drop to 60 after a few sessions and to 20 by the time therapy is complete. Such objectivity facilitates empirical outcome research designed to determine how well cognitive therapy works.

BOX 15.5

Denise in Cognitive Psychotherapy

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In the first session with Denise, it was evident that her primary symptoms were depressive and that the event that precipitated these depressive symptoms was the change in ownership of the restaurant where she works as a chef. More specifically, the new owner's policies—instituting a menu over which Denise has no control and prohibiting her from visiting with customers in the dining room—seemed to have triggered Denise's depressive symptoms.

I proceeded to treat Denise using a style of cognitive therapy that included aspects of both Ellis's and Beck's techniques. First, I educated Denise about the cognitive model by explaining that although sometimes we experience life as if events directly cause feelings, there are, in fact, automatic thoughts or beliefs that intervene. I used Ellis's ABCDE model to further illustrate how Denise could understand her experience, and I provided her with a list of Beck's cognitive distortions to equip her with the tools she would need to dispute her irrational thoughts in Column D. Denise had no trouble filling in Columns A (activating event) and C (emotional consequence) of this model: The implementation of the new restaurant policies was the activating event, and her depressive symptoms represented the emotional consequence. Our next task was to identify the beliefs (B) that occurred between A and C. This was a bit of a struggle for Denise, but with some exploration and discussion, we identified this powerful thought: "I'm incompetent." Denise explained that she could think of no other reason why her new boss would bar her from determining the menu or from talking with diners. The boss's new policies, Denise believed, were motivated by the boss's desire to prevent Denise's incompetence from ruining the restaurant's business.

The next stage in therapy focused on Stage D (dispute), where I encouraged Denise to think critically about her belief (B) that she was incompetent. I asked a number of questions to facilitate this process, such as, "What evidence do you have that you are incompetent?" "What evidence do you have that you are not incompetent?" and "How else might we be able to interpret your new boss's policies?" At first, the notion of challenging her original (illogical) belief seemed foreign to Denise, and she struggled a bit with this task. But after a few sessions, it became more familiar and comfortable for her. Using homework assignments, I asked Denise to complete a five-column thought journal in which she identified not only activating events, beliefs, and emotional consequences but also disputes and effective new beliefs. These last two columns represented uncharted territory for Denise, but

(Continued)

(Continued)

with time she became quite adept at completing the entire table. Specifically, Denise applied the following thought distortion labels:

- Personalization (Maybe the new boss would have implemented these new policies regardless of the chef who held Denise's position; if so, Denise need not take them personally.)
- Mental filtering (Denise had been extremely competent as a chef and in other capacities throughout her life, yet she was focusing exclusively on this presumed slight by her new boss.)
- All-or-nothing thinking (Denise may not share her new boss's preferences, but there were still many things she loved about working as a chef. Just because her job wasn't perfect, she need not think of it as terrible. Instead, it may fall somewhere in between.)

Individually, any of these disputes was effective enough to negate Denise's illogical thoughts of incompetence, and in combination they were especially potent. By the end of therapy, Denise was completing Column E (effective new belief) with ease: "I'm not incompetent at all. My new boss's actions may have little or nothing to do with me personally; I have plenty of evidence that I am a competent person, and even if I don't like my job as much as I used to, it's still enjoyable and fulfilling in some ways." As a result of her disputes and effective new beliefs, Denise's depressive symptoms improved dramatically.

CHAPTER SUMMARY

The primary goal of cognitive therapy, which has become the most commonly practiced approach to psychotherapy among clinical psychologists, is to promote logical thinking. Cognitive therapists accomplish this goal by helping clients recognize and revise cognitions that are illogical or irrational during a course of therapy that is typically brief, structured, and problem focused. Their techniques involve teaching clients about the cognitive model, in which cognitions intervene between events and feelings, and assigning written or behavioral homework to be completed between sessions. Whether using Albert Ellis's ABCDE model (highlighted by disputing irrational beliefs and replacing them with effective new beliefs) or Aaron Beck's list of common thought distortions (e.g., all-or-nothing thinking, personalization, magnification/minimization), cognitive therapists enable clients to overcome psychological and behavioral problems by insisting on a logical response to the events of their lives. Although cognitive therapy originally targeted anxiety and mood disorders, it is now applied to most other psychological disorders, including personality disorders, and to other issues such as medical problems and minor psychological problems that fall short of diagnostic criteria. A large and increasing body of outcome studies suggests that cognitive therapy is highly efficacious for a wide range of psychological disorders, including mood disorders, anxiety disorders, eating disorders, and others.

KEY TERMS AND NAMES

ABCDE model 370
acceptance 380
acceptance and commitment therapy (ACT) 381
activating event 371
all-or-nothing thinking 375
automatic thoughts 366
Aaron Beck 364
Judith Beck 374
belief 371
catastrophizing 375
cognitions 365
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common thought distortions 374
dialectical behavior therapy (DBT) 382
dispute 371
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magnification/minimization 375
mental filtering 375
metacognitive therapy 383
mind reading 375
mindfulness 379
overgeneralization 375
personalization 375
rational emotive behavior therapy (REBT) 370

CRITICAL THINKING QUESTIONS

1. To what extent do you agree with the fundamental cognitive assumption that illogical or irrational thinking underlies psychological problems?
2. If you were the client, how would you respond to the assignment of homework by your therapist?
3. If you were a clinical psychologist practicing cognitive therapy, which of Albert Ellis's ABCDE columns would you expect clients to have the most trouble filling in?
4. In your opinion, do any of the common thought distortions that Aaron Beck and his followers defined seem to predispose individuals to particular types of psychological problems (e.g., anxiety, depression, others)?
5. What are the primary differences between traditional cognitive therapy and the more recently developed metacognitive therapy?

LOOKING TOWARD GRADUATE PROGRAMS

The following table is a list of index entries in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018) relevant to the topics in this chapter (see page 25 in Chapter 1 for more information about the *Insider's Guide*). Keep in mind a few things as you consider these listings: They are current as of the 2018/2019 edition of the *Insider's Guide* (which typically revises every two years); they may have changed as faculty members retire, take new jobs, or develop new interests; and a graduate program's own website may have more extensive or up-to-date information.

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Acceptance and Commitment Therapy Cognitive Therapy/Cognitive-Behavioral Therapy Dialectical Behavior Therapy Mindfulness	Acceptance and Commitment Therapy Cognitive/Cognitive-Behavioral Therapy Dialectical Behavior Therapy Mindfulness Miscellaneous—Rational Emotive Therapy	Cognitive/Cognitive-Behavioral Therapy Miscellaneous/Other—Acceptance and Commitment Therapy Miscellaneous/Other—Dialectical-Behavior Therapy

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

Cognitive Behaviour Therapy
http://www.tandfonline.com/loi/sbeh20#.V08XxvkrLX4

Cognitive Therapy and Research
http://link.springer.com/journal/10608

International Journal of Cognitive Therapy
https://www.springer.com/psychology/journal/41811

Journal of Cognitive Psychotherapy
http://www.springerpub.com/journals/journal-of-cognitive-psychotherapy.html

Psychotherapy
http://www.apa.org/pubs/journals/pst/index.aspx

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- eFlashcards
- Culture expert interviews
- In My Practice demonstration video
- Full-text SAGE journal articles
- Additional web resources
- Mock assessment data

16

GROUP AND FAMILY THERAPY

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Practical Issues in Group Therapy	396	*Assessment of Families*	402
Ethical Issues in Group Therapy	398	*Family Therapy: Essential Concepts*	406
How Well Does It Work?	399	*Ethical Issues in Family Therapy*	412
		How Well Does It Work?	413

Group therapy and family therapy represent a departure from the individual forms of therapy we've covered so far. Both feature multiple clients in the room together, and both emphasize interpersonal interaction. However, despite the fact that they share a chapter in this book, they differ markedly from each other. Group therapy and family therapy have quite separate histories, and they represent distinct specializations within the mental health field. It would not be unusual to meet a clinical psychologist who practices one regularly but the other rarely or never. Thus, this chapter is divided into separate sections for group and family therapy.

GROUP THERAPY

An Interpersonal Emphasis

Group therapy is practiced in a wide variety of forms, including adaptations of many well-known individual therapy approaches such as psychodynamic, behavioral, cognitive, humanistic-existential, and many others (Hopper, Kaklauskas, & Greene, 2008; MacKenzie, 2002; Shaffer & Galinsky, 1989). Most of these group therapies strongly emphasize **interpersonal interaction**. That is, most forms of group therapy take advantage of the fact that the group therapy experience itself is based on interacting with other

Learning Objectives

16.1 Describe Yalom's therapeutic factors for group psychotherapy.

16.2 Debate the ideal characteristics of a group therapy group (e.g., composition, cotherapists).

16.3 Explain limitations on confidentiality inherent in group therapy.

16.4 Summarize group psychotherapy outcome research.

16.5 Illustrate the systems approach of family therapy.

16.6 Discuss essential concepts of family therapy.

16.7 Analyze ethical issues that may arise for a psychologist practicing family therapy.

16.8 Review the outcome research for family therapy.

people (Marlin & Kasten, 2015). Group therapists' interventions often highlight the way group members feel, communicate, and form relationships with one another. More than any other component, it is this accent on interpersonal interaction that distinguishes group therapy from individual therapy (Burlingame & Baldwin, 2011). Whereas individual therapy is limited to a two-person interaction between client and therapist, group therapy allows for a far more complex network of relationships to develop. In group therapy, a client forms relationships not only with a therapist but also with a room full of other people. Thus, group therapy involves a greater range of interpersonal responses.

Irvin Yalom has emerged as a leading figure in the interpersonal approach to group therapy (Brabender, 2002; Burlingame & Baldwin, 2011; Friedman, 1989). In fact, the term *interpersonal* has been used as a label for his approach to group therapy, and his writings (e.g., Yalom, 1983, 2005) have influenced group therapists of all kinds. According to Yalom (2005), although it may be possible to conduct group therapy as a series of one-on-one interactions between therapist and clients, such an approach fails to "reap the full therapeutic harvest" (p. xv) that group therapy can offer. Instead, he encourages group therapists to recognize the unique opportunity that group therapy presents: the display of clients' problematic interpersonal tendencies in the group itself.

Yalom (2005) views these problematic interpersonal tendencies as central to clients' presenting problems, whether or not clients initially recognize it. In fact, he argues that all psychological problems stem from flawed interpersonal relationships, as exemplified by his comments about depression: "Therapists cannot, for example, treat depression per se. . . . *It is necessary first to translate the depression into interpersonal terms and then to treat the underlying interpersonal pathology*" (p. 24). This definition of depression differs considerably from traditional medical-model definitions that emphasize intrapersonal or biological roots of disorders. Instead, according to Yalom, an individual's disorder is a by-product of that individual's disturbed way of getting along with other people. If interpersonal relationship problems are the core of all psychopathology, it follows that a primary focus of group therapy would be the strengthening of interpersonal relationship skills.

Web Link 16.1
American Group Psychotherapy Association

Web Link 16.2
Irvin Yalom demonstrating group therapy

Therapeutic Factors in Group Therapy

Yalom (2005) describes 11 specific therapeutic factors by which group therapy benefits clients; Table 16.1 outlines the full list. Several of the factors that are especially vital to a fundamental understanding of group therapy are described here in more detail.

Table 16.1 Yalom's Therapeutic Factors for Group Psychotherapy

1. Instillation of hope
2. Universality
3. Imparting information
4. Altruism
5. Corrective recapitulation of the primary family group
6. Development of socializing techniques
7. Imitative behavior
8. Interpersonal learning
9. Group cohesiveness
10. Catharsis
11. Existential factors

Source: Adapted from Yalom (2005).

Universality

Clients with psychological problems often believe that no one shares their struggles. They may be unaware of the commonality of their problems, symptoms, or diagnoses. To find oneself in a room full of other people who have similar problems can be uplifting in and of itself. Yalom (2005) describes this phenomenon as **universality**.

If the group is organized around a single problem—eating disorders, panic disorder, or chemical dependency, for example—clients may experience this "we're all in the same boat" feeling right away. Universality is built into **homogeneous groups** such as these. In **heterogeneous groups**, however, universality may not be quite as obvious from the outset. With time, however, clients in heterogeneous groups often come to realize that although their symptoms may differ superficially, the fundamental issues that underlie them may in fact be quite similar. Whether homogeneous or heterogeneous, groups provide clients with the sense that they are not alone in a way that individual therapy cannot.

Group Cohesiveness

Group cohesiveness refers to feelings of interconnectedness among group members. This interconnectedness is characterized by feelings of warmth, trust, acceptance, belongingness, and value among group members. Participating in an ongoing group of this kind bolsters psychological wellness.

Yalom (2005) and others (e.g., Ormont, 1992) point out that group cohesiveness does not necessarily mean that every group interaction will be polite or courteous. In fact, an ongoing group interaction devoid of negative emotion may indicate that the group lacks cohesiveness. Truly cohesive therapy groups—such as cohesive families, work groups, or teams—are strong enough to tolerate anger, sadness, jealousy, disappointment, and the like. They understand that as people become meaningful to one another, they often elicit strong feelings, and they encourage the discussion of those feelings even when doing so is unpleasant or awkward.

Cohesiveness plays the same role in group therapy that the therapeutic relationship/alliance plays in individual therapy. It heals in its own right, and it also sets the stage for other healing factors to take effect (Burlingame, McClendon, & Alonso, 2011; Burlingame, McClendon, & Yang, 2018; Yalom, 2005). However, whereas a client's perception of the therapeutic alliance in individual therapy involves only one other person, the client's perception of group cohesiveness involves many other people. Thus, for group therapists, it is crucial not only to maintain a trusting, collaborative relationship directly with the client but to foster a group in which clients develop trusting, collaborative relationships with one another as well.

Interpersonal Learning

Learning from the in-group interpersonal experience—**interpersonal learning**—is at the heart of group therapy. Group therapists presume that interpersonal problems contribute to the client's reason for seeking therapy in the first place, that the same interpersonal tendencies will appear within the group, and that the lessons learned via interacting with fellow group members will generalize to clients' lives outside the group. In simple terms, group therapy provides an opportunity for "relationship practice." Much like a group of actors rehearses before a performance or an athletic team scrimmages before a game, members of a therapy group use group meetings as a place to identify problematic relationship tendencies and attempt to revise them. In other words, group members don't just talk about relationships; they form relationships with one another and work to improve them with the assumption that their gains will ultimately benefit their personal relationships outside the group as well.

Social Microcosm. As group therapy progresses, the group becomes a **social microcosm** for each member (Ormont, 1992; Yalom, 2005). In other words, the relationship tendencies that characterize clients' relationships with important people in their personal lives—romantic partners, friends, coworkers, children, siblings, and so on—will predictably characterize the relationships they form with their fellow group members. As Yalom (2005) explains,

> Clients will, over time, automatically and inevitably begin to display their maladaptive interpersonal behavior in the therapy group. There is no need for them to describe or give a detailed history of their pathology: they will sooner or later enact it before the other group members' eyes. (p. 32)

Thus, a client whose unwarranted suspiciousness of his partner is damaging his marriage will probably become overly suspicious of a fellow group member at some point,

a client whose passive-aggressive behavior has cost her many friendships may likely behave passive-aggressively toward a fellow group member, and a client whose pessimism has convinced him that searching for a rewarding job is pointless may pessimistically deem the group therapy process as similarly pointless.

One study examined the interpersonal tendencies that more than 200 group therapy clients demonstrated in relationships with both their fellow group members and their friends and family. They found that for most clients, there were striking similarities between the two sets of relationships, lending empirical support to the idea that group therapy clients tend to create the same kinds of relationships (and relationship difficulties) in therapy that they create in their outside lives (S. B. Goldberg & Hoyt, 2015). It makes sense, then, that improving those relationships in the social microcosm of group therapy can lead to improvements in relationships outside of therapy.

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Photo 16.1
Group therapists encourage clients to discuss the here-and-now relationships they develop with other members of the group.

The Here and Now. Because the group functions as a social microcosm, an essential task for the group therapist is continually to shine the spotlight on the way members relate to one another within the context of the group. So, rather than encouraging clients to talk about events that have happened in their lives outside the group ("there" and "then"), the group therapist steers clients toward examining the relationships among group members **here and now** (Kivlighan, 2014).

Specifically, Yalom (2005) promotes two activities among group members to maximize the use of the here and now: interaction (which he calls "experience") and reflection/discussion of that interaction (which he calls "illumination of process"). In other words, through the natural course of their continued interaction, group members communicate directly with one another—perhaps expressing happiness, boredom, dissatisfaction, envy, curiosity, confusion, or rage. The group therapist urges clients to reflect on this process, to talk about the way clients talk to one another, and to enlighten each member about their interpersonal tendencies. Once they learn about their tendencies, clients may decide to change them, which could improve the relationships they have within and, ultimately, outside the group.

Group therapy is one of the few social situations in which people are encouraged to talk with others about the way they relate to one another. More often, this is strongly discouraged in social relationships. For example, people who are interpersonally challenging, annoying, or frustrating might see numerous relationships end without receiving any feedback about their behavior. They might drive away friends, romantic partners, or employers without hearing directly from them exactly what they did to contribute to the demise of the relationship. For this reason, group therapy represents a unique opportunity for individuals to receive genuine, honest feedback on their behavior and its impact. This feedback often motivates clients to change their behavior, at which point the group becomes a place to try out and practice novel responses in anticipation of incorporating them into clients' lives outside therapy (Ormont, 1992; Yalom, 2005).

As an example, consider a group therapy scenario in which Ron, a very self-centered, narcissistic, and insensitive man, and Michelle, a timid and nervous woman, are among the members. In one session, when Michelle has been discussing her problems for a few minutes, Ron interrupts rudely with comments such as, "How long is this going to go on?" and "Just get over it already—I have some real problems to talk about." Michelle responds to Ron's interruptions by meekly apologizing for taking up so much time. Dr. Hollins, the group therapist, notices this interaction and asks the group, "What do you think about the way Ron and Michelle communicate with each other?" One by one, the group members other than Ron and Michelle offer very similar comments: Ron's behavior is rude and inappropriate, and Michelle's apology was unnecessary. They also add that they find Ron's rudeness unattractive and have a hard time respecting Michelle when she acts so timidly.

Dr. Hollins turns to Michelle and asks questions to explore her tendency to apologize rather than assert herself ("What stops you from confronting Ron about his behavior?" "What options do you have besides apologizing?") and later turns to Ron to explore his tendency to interrupt selfishly ("How does it make you feel to know that others see your behavior as rude and inappropriate?" "How else might you respond to Michelle in that situation?"). These lines of questioning allow Michelle and Ron to see their own behavior in a new light. Specifically, they see how their behavior impacts others and how others' reactions to their behavior subsequently affect their own happiness. Both Michelle and Ron find themselves motivated to change, and the group setting becomes a supportive place for them to practice doing so. The outcome is not only improved interpersonal behavior within the session but also real-life benefits for Michelle and Ron, whose relationships with their respective partners were on shaky ground largely because of Ron's self-centeredness and Michelle's submissiveness.

Practical Issues in Group Therapy

Group Membership

Groups typically include 5 to 10 clients, and many group therapists find that having 7 to 8 members in a group is ideal (Brabender, 2002; Yalom, 2005). Groups can be either open or closed in terms of enrollment. **Open-enrollment groups** allow individual members to enter or leave the group at any time. A strength of such groups is that at any given point in time, the group includes members at various stages of improvement. Thus, like college freshmen sharing classes with seniors, new group members have experienced models to follow. In **closed-enrollment groups**, all members start and finish therapy together, with no new members added during the process. In this type of group, cohesiveness can be easier to establish and maintain than in open-enrollment groups because of the stability of membership.

Unless a group centers on a particular diagnostic issue, group therapists are typically rather open about clients they might select for a particular group. In fact, greater variability in terms of presenting problems and demographic variables can create a more realistic social microcosm. Typically, characteristics that would cause a group therapist not to select a prospective member include those that would interfere with the client's ability to interact meaningfully with others and reflect on that interaction. Such characteristics often include psychosis, organic brain damage, acute crisis, and pragmatic issues

such as travel or transportation that would interfere with regular attendance (Friedman, 1989; Yalom, 2005).

Preparing Clients for Group Therapy

It is clinically wise to make concerted efforts to prepare new clients for the group therapy experience to correct any misconceptions and maximize therapeutic benefit (Brabender, Fallon, & Smolar, 2004). Some clients may mistakenly believe that group therapy is second-rate (in comparison to individual therapy), that they will immediately be forced to disclose intimate personal details to strangers, or that interacting with other people with psychological problems will somehow worsen their own symptoms (MacKenzie, 2002). In pregroup individual meetings, group therapists can assure clients that such notions are false and can provide realistic and encouraging data about how well their therapy works for most clients. In addition, group therapists can orient clients to the kinds of activities that will take place in group therapy and the behaviors that can maximize the benefit they receive, such as active participation and consistent, punctual attendance.

Developmental Stages of Therapy Groups

With time, therapy groups tend to move through a predictable set of stages. These stages have been assigned a variety of different labels (e.g., Brabender et al., 2004; MacKenzie, 2002; Tuckman, 1965; Yalom, 2005), but they have quite similar meanings. In general, clients in the initial stage of group therapy are cautious and concerned about whether they will be accepted into the group. After they overcome these initial concerns, they move onto a second stage in which there is some competition or jockeying for position within the social pecking order. Eventually, this competitive second stage gives way to a third stage in which cohesiveness forms, members feel closely connected and trusting of one another, and the group sessions become more consistently productive as clients learn about and improve on their interpersonal skills.

Cotherapists

In group therapy settings, two therapists often colead a single group. Such an arrangement presents numerous potential advantages and pitfalls. One advantage of a **cotherapist** is the mere presence of a second set of eyes and ears to notice the rich array of verbal and nonverbal communication inevitably produced by a room full of clients. A second advantage involves the cotherapists' ability to model collaborative relationships for group members. Among group members whose previous models of two-person teams (parents, etc.) used destructive or hurtful tactics, the power of two partners working together constructively can be compelling. A third advantage, at least for clients who grew up in two-parent homes, is a **recapitulation of the family group**. With one male and one female cotherapist, for example, a group can evoke the same dynamics as a traditional family, and the way clients from such families respond to each cotherapist—trusting one while fearing the other, for example—can shed light on their interpersonal tendencies with other important people in their lives (E. Shapiro, 1999; Yalom, 2005).

The potential pitfalls of cotherapy come into play when the cotherapists don't trust each other, compete with each other, or "step on each other's toes" by approaching group therapy with incompatible therapy orientations. Their styles need not be identical—in

fact, the group may benefit from somewhat distinct but complementary therapists—but cotherapists do need to be able to work well together both in and out of session. Of course, leading a group as a solo therapist is also a common practice.

Socializing Between Clients

Clients who socialize with one another outside therapy groups constitute a significant problem for group therapists. Often, group therapists educate clients about this potential problem and prohibit it during the pregroup preparation. Nonetheless, it happens with some frequency, as might be expected among clients who find themselves feeling increasingly connected to new people.

Web Link 16.3
Ethical issues in group psychotherapy

The negative consequences of **extra-group socializing** among clients, whether romantic or platonic, are many (Brabender et al., 2004; Yalom, 2005). For example, if Karen and Dena are members of the same therapy group and develop a friendship outside the group, they may become more loyal to their friendship than they are to the group as a whole. They may be reluctant to comment frankly on each other's behavior in front of the group, or they may save their most direct, meaningful exchanges for private moments rather than sharing them during group sessions. In this way, they can become spectators rather than active participants in the group process. Moreover, the exclusiveness of their relationship can adversely affect those members who feel "left out." Group therapists can respond punitively to socializing outside the group, but such behavior can also provide meaningful material for therapy sessions, especially if such defiant or secretive behavior is related to the reasons the client sought treatment in the first place.

Ethical Issues in Group Therapy

Confidentiality

Among the ethical issues especially relevant to group therapy, *confidentiality* is perhaps most concerning. The focus here is not on the group therapist maintaining clients' confidentiality, because that responsibility is assured by professional ethics and standards. Instead, the focus is on the possibility that fellow group members might violate a client's confidentiality (Brabender, 2002; Brabender et al., 2004; Lasky & Riva, 2006; Shaffer & Galinsky, 1989). At the outset, group therapists often require clients to agree in writing to hold material from group sessions confidential, but such contracts are "virtually impossible to enforce" (Shaffer & Galinsky, 1989, p. 266).

The consequences of one client violating the confidentiality of another can seriously affect the professional or personal life of the client whose confidentiality has been violated, especially considering how quickly information can spread on social media. Moreover, such breaches make the therapy group seem unsafe and untrustworthy, which decreases self-disclosure by other clients. Thus, it is important for the group therapist to do as much as possible to prevent such confidentiality violations. In addition to asking clients to commit to confidentiality during pregroup preparations (sometimes in the form of group therapy confidentiality contracts), therapists' preventive actions can include exemplifying impeccable behavior regarding confidentiality, frequent reminders about confidentiality, and group discussion of any confidentiality violations that do occur (Brabender et al., 2004; Knauss & Knauss, 2012; Riva & Cornish, 2018).

How Well Does It Work?

Although the outcome of group therapy has not been studied as extensively as that of individual therapy, the existing empirical research indicates that group therapy works. Specifically, meta-analyses and reviews have consistently found that group therapy is superior to no treatment and generally as effective as individual therapy, although a minority of comparative studies have found individual therapy to be slightly superior (e.g., Burlingame, Fuhriman, & Mosier, 2003; Fuhriman & Burlingame, 1994; Corey & Corey, 2016; McDermut, Miller, & Brown, 2001; McRoberts, Burlingame, & Hoag, 1998; L. Robinson, Berman, & Neimeyer, 1990; Rutan & Shay, 2017; Tillitski, 1990; Vandervoort & Fuhriman, 1991). Moreover, the range of disorders and presenting problems for which group therapy has been found effective is very broad. Also, as described earlier, empirical research indicates that cohesiveness within the group, much like the therapeutic relationship in individual therapy, is a major contributor to successful outcome in group therapy (Burlingame et al., 2011; Burlingame et al., 2018).

It is worth noting that, whereas the outcomes of group and individual therapies are often comparable, group therapy tends to be a less expensive mode of therapy. In this way, group therapy may be a particularly cost-effective approach to therapy (Brabender, 2002; Friedman, 1989; Markin & Kasten, 2015).

BOX 16.1

Denise in Group Therapy

Denise was remarkably quiet during the first five group therapy sessions. Clients often take time to develop trust and feel comfortable in the group, but Denise seemed to take longer than most. Eventually, during the sixth session, Denise spoke about the reasons why she had sought therapy in the first place. Our group—seven clients besides Denise, plus myself and my cotherapist—listened intently as Denise summarized the recent changes at the restaurant where she was the head chef, as well as the depressive symptoms she had experienced as a result.

After Denise had finished the summary and several of the group members had offered sympathy and understanding, my cotherapist posed a question to the group: "How do you feel about Denise's decision to share her story with the group today?" One by one, the group members each mentioned that they were very happy that Denise had become more engaged in the group. Many added that they hoped to hear more from her in upcoming sessions as well. I turned to Denise and asked, "What's your reaction to this consensus from your fellow group members?" Denise responded that she was surprised and relieved to hear it. Several group members seemed perplexed by this, so Denise explained:

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> I just haven't felt like it was OK for me to speak up. I never felt like I had the right to take group time away from someone else. No one was directly asking me what was going on with me, so I thought it would be rude just to demand everyone's attention. I thought that

(Continued)

(Continued)

would make everyone dislike me. So I just sat quietly, even though I really wanted to share this stuff as soon as I got here.

My cotherapist asked the group if they would have responded the way Denise assumed if she had spoken up earlier, and, of course, they all said no. By the end of that session, Denise seemed to have experienced a meaningful interaction with her fellow group members: She learned that although she assumed they'd dislike her for asserting herself, doing so made them like her more. She genuinely stated she would feel more comfortable speaking up in group in the future, and, in fact, she did so.

The next session, my cotherapist and I planned to pursue this topic with Denise. Our hypothesis was that the lesson Denise learned in the previous session might be applicable to her relationships outside therapy. In fact, we wondered if this issue might play a role in the very problems that caused her to seek therapy in the first place. As the session began, one of Denise's fellow group members got the conversation going by asking Denise a relevant follow-up question: "Have you ever told your new boss how you feel about the new policies at the restaurant?" Denise said no, she never had, and the thought of doing so made her nervous. She assumed that the new boss would dislike her and possibly even fire her for speaking up. Her fellow group members, my cotherapist, and I pointed out the parallels between her assumptions about her new boss and her assumptions about her fellow group members, which had proven completely false.

Over the next few sessions, Denise considered and gradually came to accept the idea that perhaps her tendency to believe that others want her to stay quiet rather than speak up led her to misperceive other people in numerous situations, including the situation with her new boss, and that the result was often sadness and dejection for Denise. Eventually, Denise had a meaningful discussion with her new boss about the new restaurant policies. The new boss was willing to compromise with Denise, allowing her to determine the menu on selected nights but standing firm on his new rule that Denise not visit with diners. Perhaps more important, the new boss had absolutely no hard feelings toward Denise for speaking up about this and, in fact, thanked her for doing so. Denise was quite happy and proud, not so much because she was able to determine the menu some evenings but because she had identified a flawed interpersonal tendency and corrected it both in and out of the group. As group therapy concluded for Denise, she felt especially optimistic about her ability to form healthy relationships and to prevent additional bouts of depression in the future.

FAMILY THERAPY

The System as the Problem

When the **family therapy** movement initially arose in the mid-1900s, it was considered revolutionary. Its leaders were seen as mavericks who had formulated an original

way of conceptualizing psychological problems and their causes (M. P. Nichols, 2010). Specifically, they opposed the widely accepted notion that psychological symptoms originated from solely *within* the mind of the individual. Instead, they believed that psychological symptoms were a by-product of the dysfunctional families in which the clients lived (F. W. Kaslow, 2011). One individual may exhibit the symptoms, but the problem actually belonged to the entire system (Rolland & Walsh, 2009).

Web Link 16.4
American Association of Marriage and Family Therapy

With this simple yet profound theory, the pioneers of family therapy opened the door to a fundamentally different mode of treatment in which family members worked together with a therapist to improve their interactions, which in turn strengthened the mental health of each member. Family therapists actually borrowed this **systems approach**, in which the whole is more than the sum of the parts, from philosophy and the sciences (Becvar & Becvar, 1982; S. D. Davis, Dattilio, & Gonzalez, 2015; Laszlo, 1972; Rolland & Walsh, 2009; Sexton & Stanton, 2016). Central to this systems approach is the idea that circular causality explains psychological problems better than does linear causality (J. M. Thompson, 2015). **Linear causality**, which tends to be endorsed by individual therapists, suggests that events from the past cause or determine events in the present in a unidirectional or "one-way street" manner. That is, the way a parent treats a child influences the child's subsequent behavior. **Circular causality**, on the other hand, suggests that events influence one another in a reciprocal way, such that a parent's and a child's behavior each affects the other continuously (more like a "two-way street"). Thus, family therapists tend to believe that regardless of the original cause of a problematic behavior, it is maintained over time by mutually influential, ongoing interaction among family members. See Figure 16.1 for a comparison of linear and circular causality.

Typically, family therapists have pointed to unhealthy **communication patterns** among family members as the type of interaction that most significantly contributes to psychological problems. Early in the history of family therapy, numerous theorists attempted to make connections between particular communication patterns and pathological outcomes in family members. Two well-known examples are the theory of the schizophrenogenic mother (Fromm-Reichmann, 1948)—cold, authoritarian mothers in combination with ineffectual fathers were thought to cause schizophrenic symptoms in their children—and the double-bind theory (Bateson, Jackson, Haley, & Weakland, 1956)—parents who consistently gave their children inescapable mixed-message commands also supposedly caused schizophrenic symptoms. Today, these two particular theories are neither widely held nor empirically supported, especially as findings regarding the biological factors underlying schizophrenia have been strongly established. However, they remain historically relevant as early attempts to link family communication patterns with symptoms of mental illness in individuals.

Another important feature of systems theory as applied to family therapy is its emphasis on

Figure 16.1 Unlike individual therapists, who tend to endorse linear causality, family therapists tend to endorse circular causality.

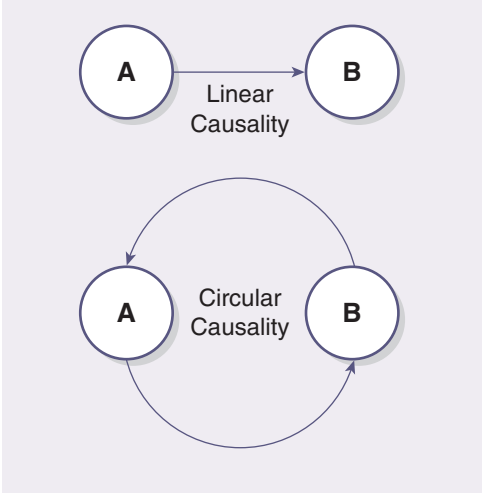

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Photo 16.2
Family therapists believe that improving relationships among family members will enhance the psychological wellness of individuals within that system. In your opinion, to what extent are psychological disorders due to problems in family relationships?

functionalism. Specifically, family therapists believe that although psychological symptoms may appear maladaptive, they are in fact functional within the individual's family environment. So even though the symptoms may qualify an individual as abnormal according to the *Diagnostic and Statistical Manual of Mental Disorders* (*DSM*), viewed in the larger context, those symptoms actually benefit the family's overall functioning in some way. For example, consider a family in which the parents' marriage is on shaky ground. Their fighting escalates to the point that their two children sense that a separation or divorce may be imminent. If one of the kids develops an eating disorder, attention-deficit/hyperactivity disorder, or another psychological problem, one effect of their symptoms may be that the parents stop arguing and unite in their focus on the child. In this way, the child's symptom actually serves the purpose of reducing the marital conflict and holding the family together. Family therapists do not believe that children consciously or deliberately plan these psychological problems, but they do suggest that family members can perceive tension within their own families and that on some level, this tension can contribute to the development of psychological problems in certain family members (Goldenberg & Goldenberg, 2007; M. P. Nichols, 2010).

The idea that family members can perceive tension within families relates closely to another core idea of systems theory: homeostasis. The notion of **homeostasis** suggests that systems have the ability to regulate themselves by returning themselves to a comfort zone or "set point." A thermostat is perhaps the most common example of a homeostatic system. If the thermostat is set at 72 degrees, the system is programmed to detect variations from this temperature and take action—cool or heat the room—to return the temperature back to 72. In much the same way, families have emotional or behavioral comfort zones, and when family members detect that the family is straying from theirs, they may take action to bring the family back. Family therapists often call these actions **feedback**, and these behaviors may, if viewed in isolation, qualify an individual for a mental disorder: conduct disorder, oppositional defiant disorder, depression, or substance abuse, to name a few. But in the view of family therapists, if the symptoms effectively return the family to a more familiar, comfortable, safe way of interacting, they have served a valuable and often unrecognized function within the context of the family (Hanna, 2007; Thompson, 2015).

Assessment of Families

Like clinical work with individuals, family therapy usually begins with a thorough assessment of the family's functioning. In fact, the assessment process continues throughout therapy. Assessment practices may vary according to the particular approach to family therapy that the therapist uses, but they typically focus on such issues as defining the presenting problem, understanding family members' beliefs about its causes, and appreciating the relationships within the family (Griffin, 2002).

An important initial step in assessing a family is to assess who, exactly, the family includes. The configuration of families can differ greatly across diverse groups. Some

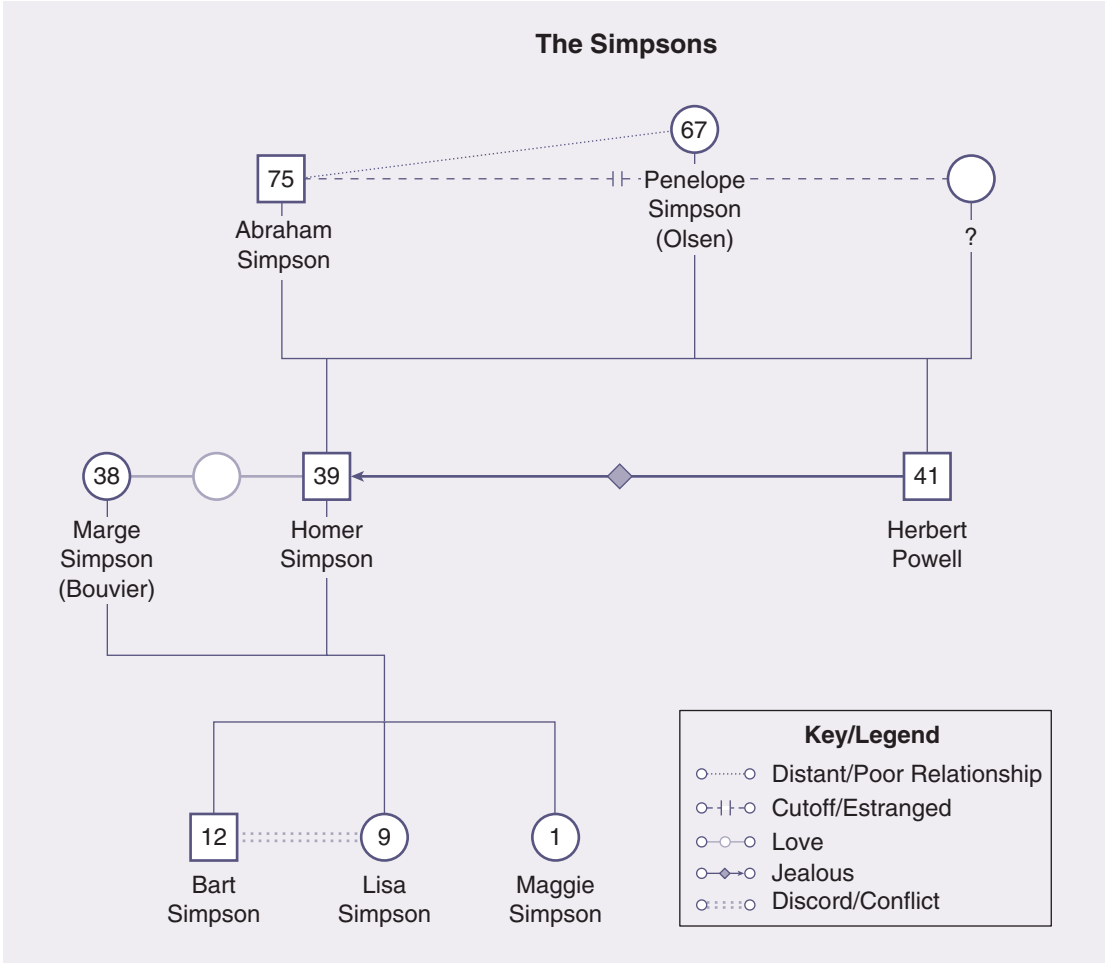
clients may include numerous "aunts," "uncles," "cousins," or other relatives who are not technically related but unquestionably function as family. Others may exclude close relatives who they see as extraneous or outcast from the family. Simple questions such as, "Who do you consider family?" or "Who lives with you?" can go a long way toward helping the clinical psychologist understand the client's perception of family (Fontes, 2008; Grieger, 2008). At times, such questions can yield different answers from members of the same family, and these differences in perception may actually be relevant to the problems for which the family is seeking help.

One helpful technique for understanding the family configuration and relationships within the family is the **genogram** (McGoldrick, Gerson, & Petry, 2008). A genogram is a pencil-and-paper method of creating a family tree that incorporates detailed information about the relationships among family members for at least three generations. It exemplifies the saying that "a picture is worth a thousand words" by graphically presenting not only the structure of the family relationships (marriages, offspring, etc.) but also the quality of those relationships (harmonious, very close, estranged, abusive, etc.). Generally, genograms use a consistent approach to notation: Males are represented by boxes and females by circles, relationships are represented by lines between two individuals, and the quality of these relationships is indicated by the types of lines selected. Not only is the result helpful in understanding the complexities of the family, but the process of creating a genogram can also be a constructive way for the therapist to engage with the family and begin to establish a healthy alliance. Figure 16.2 presents a genogram for the fictional television family from *The Simpsons*.

Another essential part of the family assessment process is an appreciation of the family's current developmental stage. Just as numerous theorists have put forth sequential, developmental stages for individuals (such as Freud's five psychosexual stages or Erikson's eight psychosocial stages), theorists have also offered sequential, developmental stages for families. Probably the most widely known of these theories is the **family life cycle** created by Betty Carter and Monica McGoldrick (1989, 2004). According to this theory, intact middle-class American families typically proceed through the following seven stages (McGoldrick, Carter, & Garcia-Preto, 2011):

- *Leaving home*. Single young adults become independent and self-sufficient.
- *Joining of families through marriage or union*. A new couple forms a new family system, and the partner is incorporated into existing family systems.
- *Families with young children*. Taking care of children; adjusting the marriage; and managing child-related, financial, and other responsibilities are among the primary tasks.
- *Families with adolescents*. Parents provide children with increasing amounts of independence and may adopt caretaking roles for their own parents as well.
- *Launching children and moving on in midlife*. Adjusting to the "empty nest," managing relationships with children's partners, and taking on the grandparent role are central.
- *Families in late middle age*. Focus shifts to managing declining health and adapting to new roles in the family and community.

Figure 16.2 Genogram of the Simpsons



- *Families nearing the end of life*. Accepting the realities and inevitability of death (one's own and those of family members) and, frequently, reversing the caretaker roles between older and younger generations exemplify this stage.

This set of stages may be a helpful way to conceptualize the evolution of a family through the years, but its authors and others emphasize that it must be modified to account for the wide variety of families that a family therapist may see, especially in countries as demographically or culturally diverse as the United States (Barnes, 2004; Carter & McGoldrick, 1989, 2004; McGoldrick et al., 2011). The list of variations to the traditional family on which the family life cycle is based is extensive: Divorced

families, step-families, single-parent families, families with gay/lesbian members, families of diverse or blended ethnicities or religions, families with parents in nontraditional gender roles, nonmarried cohabitating couples, couples without children, families that have experienced an unexpected or premature loss, and families with many years between offspring are only some of the families that might not match the prototype (Ahrons, 2011; McGoldrick & Carter, 2011; McGoldrick & Walsh, 2011; McGoldrick & Watson, 2011). As an example, Hines (2011) points out that for African American families living in poverty, the life cycle may be a bit accelerated, with family members becoming parents or grandparents sooner than in many other cultural groups. Also, families in these cultures have a relatively high frequency of chronic stress, sudden losses, and households headed by single females—all of which should be taken into account by the family therapist. With appropriate knowledge such as this, along with sensitive adaptation, the life cycle model can provide a useful starting point when working with families, especially during the assessment phase.

It is unfortunate but undeniable that clinical psychologists assessing a family must pay attention to the issue of abuse and violence (Birchler, Simpson, & Fals-Stewart, 2008; M. P. Johnson, 2008). Clinical interviews can reveal some information, especially if family members are interviewed separately (Holtzworth-Munroe, Meehan, Rehman, & Marshall, 2002). The **Conflict Tactics Scales (CTS)** is a more structured and formal way of assessing abuse and violence within families. The CTS is available in versions that assess treatment of partners by each other and treatment of children by parents or caregivers. It is an individually administered, objective, self-report questionnaire that includes about 40 items and takes less than 15 minutes to complete. It includes questions asking the individual about how they behave—speaking calmly, using insults, throwing objects, or hitting others, for example—when family conflicts arise. It has been widely used in clinical and research settings and has established reliability and validity (Straus, 2007; Straus, Hamby, Boney-McCoy, & Sugarman, 1996). Even when abuse or violence is not the presenting problem, it is important for the clinical psychologist to assess for these issues via interviews or other assessment tools such as the CTS.

As the assessment phase leads into the therapy phase, it can be critical for the family therapist to persuade the family that the problem is systemic rather than individual. Often, families enter the therapy process with an identified patient or a family member whose symptoms are most obvious or problematic. They may believe that the problem is contained within the **identified patient** and that the role of other family members is minimal or nonexistent. In fact, they may contact the therapist seeking individual therapy for a son, daughter, or other family member and have no expectation at all for family therapy. In these cases,

> an effort is made to dilute the idea that the presenting problem associated with an individual is encased within that person; instead an attempt is made to modify beliefs so that the problem is perceived to be a byproduct of the situation that characterizes the system. (Griffin, 2002, p. 794)

With family members successfully "on board," family therapy can commence with positive expectations.

BOX 16.2

In My Practice . . .

In my practice, family therapy often doesn't start as family therapy. Instead, it starts as child therapy, at least in the minds of the parents who seek treatment.

As an example, consider the Thompson family, which included mom Stephanie, dad Paul, and 8-year-old son Jacob. The initial call came from Stephanie. She explained that Jacob was experiencing a lot of anxiety and that she hoped I could help "him" (not "us") with it. (She was clearly casting Jacob as the "identified patient.") I met with all three family members, both together and separately. Stephanie and Paul described having "no idea" where Jacob's anxiety came from. They perceived the problem as something that belonged to Jacob and that they as parents preferred to distance themselves from it. My sense was that they were bringing Jacob to me much like they would take him to the dentist if he had a cavity: Beyond getting him to the office of an appropriate professional, they had no responsibility to participate in the treatment.

But when I spoke to Jacob individually, I began to see that his anxiety didn't come out of nowhere. Instead, it was the by-product of some dysfunction in their family system. Specifically, Jacob described how his parents bickered and argued in front of him all the time. When they argued about money, he feared they were going broke. When they argued about their relationship, he feared they would divorce. When they argued about him, he feared they didn't love him. Sometimes, his parents even pulled Jacob into their arguments, pressuring him to take sides. It became clear to me that any attempt to treat Jacob's anxiety in isolation would, at best, provide only temporary benefits. The whole system of family communication needed to be addressed.

Convincing Stephanie and Paul of this need was no easy task. After all, they assumed and favored a "hands-off" approach to their son's anxiety problems. The key was to assure Stephanie and Paul that if the family became the focus, they would not be blamed or scolded for their parenting. Instead, we would work together in a collaborative, nonjudgmental way to improve the way their family functioned. Additionally, I made every effort to help Stephanie and Paul understand the power they possessed to help their child. As Jacob's parents, they were uniquely capable of enacting changes at home that couldn't be accomplished through any other means. Eventually, and cautiously, Stephanie and Paul agreed that fixing the family was the best way to improve Jacob's anxiety (and the relationships between all of the family members), and they became active participants.

To see this box come to life as a whiteboard video, please visit edge.sagepub/pomerantz5e.

Family Therapy: Essential Concepts

Many varieties of family therapy have been developed and practiced. Most of the widely known approaches to individual therapy have been applied in the family context as well, including psychodynamic, humanistic-existential, behavioral, and cognitive styles (Goldenberg & Goldenberg, 2007; Hanna, 2007; M. P. Nichols, 2010; Rolland & Walsh, 2009). For example, when family therapists use a cognitive approach, they emphasize distorted or illogical beliefs that family members may have about each other or the relationships between family members. Replacing such beliefs with more realistic

BOX 16.3

Considering Culture

Gender, Abuse, and Family Therapy

Family therapists endorse the notion of circular causality, which contends that problematic behavior within families is part of a complex pattern in which numerous individuals play a role. In other words, although one person may exhibit the symptom, other family members may contribute to the problem in a reciprocal way.

How does this theory apply to abuse within families? Specifically, if one partner (e.g., a husband) acts violently toward the other (e.g., a wife), does the notion of circular causality suggest that her behavior somehow contributes to his violence? Is violent, abusive behavior indeed a systemic problem that multiple family members maintain, or does the problem belong solely to the abuser? If we conceptualize the problem systemically, are we blaming the victim?

Such critical questions have been raised repeatedly by feminist authors (e.g., Avis, 1992; Goldner, 1985; Hare-Mustin, 1978; Luepnitz, 1992; McGoldrick, Anderson, & Walsh, 1989; Silverstein & Goodrich, 2003). Family therapists are wise to give serious thought to these questions for a variety of reasons. For one, data on the incidence of violence of men toward women in couple relationships suggests that it is quite common: Each year, one in eight husbands engages in physical aggression toward his wife; at least 30% of married couples experience physical aggression at some point in the marriage; and when violence occurs in couple relationships, it tends to continue (Holtzworth-Munroe et al., 2002).

In addition to remaining sensitive to the possibility that violence may be occurring, family therapists should also remain sensitive to the messages they communicate to family members about violence. Those who pose the above questions about family therapy generally argue that attributing any responsibility to women being abused by men represents a particularly harmful form of blaming the victim and that the abuser is ultimately responsible for his own behavior. They further warn family therapists not to insensitively or mistakenly apply the notion of circular causality to family situations in which men abuse women.

In your opinion, how does the notion of circular causality apply to families or relationships in which men abuse women? How can family therapists most sensitively and therapeutically understand such problems?

or logical beliefs can help the family and its members function more effectively and happily (Dattilio, 2001, 2009; Dattilio, Piercy, & Davis, 2014; Friedberg & McClure, 2015). Griffin (2002) divides this wide range of styles into three broad categories:

- *Ahistorical styles*, emphasizing current functioning and deemphasizing family history (most common in contemporary clinical psychology)
- *Historical styles*, emphasizing family history and typically longer duration than ahistorical styles (including Bowenian/intergenerational family therapy and psychodynamic approaches)
- *Experiential styles*, emphasizing personal growth and emotional experiencing in and out of sessions (including the approaches of Carl Whitaker and Virginia Satir)

BOX 16.4

Metaphorically Speaking

If You've Been to a Car Repair Shop, You Understand Systems Theory

Imagine that there is something wrong with one of the tires on your car. It's showing uneven wear, and it loses air from time to time. Would you consider taking the tire off the car, taking the individual tire to a car repair shop, and asking the mechanic to repair or replace it? Or would you take the entire car in with the expectation that the mechanic would consider both the symptomatic tire and the other parts of the car to which it is connected?

These questions illustrate the role of systems theory in family therapy. In individual therapy, the client is seen in isolation, much like the tire detached from the car. The improved version will ultimately be placed back into the system from which it came. Perhaps the improvements will last, or perhaps the dysfunctional nature of the system will re-create the same problems. On a car, when the repaired or replaced tire is installed, we might see the same uneven tread or loss of air within a short time. In individual therapy, when the client returns to the family, we might see the same behavioral or emotional problem arise again. In fact, the early applications of systems theory to family therapy were motivated in part by the clinical observation that inpatients who showed great improvement in the psychiatric hospital often deteriorated when they moved back home (M. P. Nichols, 2010).

In family therapy, the therapist examines the entire family system in which the symptomatic individual lives, much like the mechanic checking all parts of the car that might influence the performance of the tire, including suspension, joints, axles, and the like. If the car is out of alignment, replacing the tire won't fix the problem but realigning the car will. Similarly, if an individual is displaying psychological symptoms because of a systemic problem within the family, realigning the family may be necessary to address the problem.

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In this section, we review some of the essential "classic" concepts of family therapy, most of which derived from historical approaches, that are still very relevant to contemporary practice. Then we consider three of the more commonly practiced contemporary ahistorical styles (narrative, solution-focused, and multisystemic family therapies) in more detail.

Classic Concepts

Family Structure. Every family has rules by which it operates. These rules are rarely explicit, but they nonetheless powerfully govern family members' behavior. Collectively, these rules are known as the **family structure**, and if the family structure is flawed, problems will ensue in the relationships among family members and may manifest as psychological symptoms in some family members (Minuchin, 1974; Sexton & Stanton, 2016).

The emphasis of structural family therapy, as practiced by **Salvador Minuchin** and others, is to improve the structure of families as a means of improving the functioning of their members. Structural family therapists emphasize **subsystems** within families (parental subsystems, sibling subsystems, etc.) and **boundaries** between those subsystems. These boundaries should be permeable enough to allow emotional closeness between family members but rigid enough to allow for independence as well. If boundaries are too permeable, family members can become **enmeshed**; if they are too rigid, family members can become **disengaged**. Thus, a primary task of structural therapy is uncovering and correcting and clarifying family rules, often with an emphasis on developing a clear hierarchy of power within the family. Like any organization—a company, the military, and so on—families simply run better when the structure is clear and sensible.

Differentiation of Self. According to **Murray Bowen** and his followers, a primary task for each family member is an appropriate degree of self-determination. In other words, healthy families allow each member to become his or her own person without sacrificing emotional closeness with other members of the family (Bowen, 1978; Sexton & Stanton, 2016). However, families sometimes experience problems in this regard, resulting in family members who remain overly connected with one another in an emotional sense. In other words, rather than achieving **differentiation of self**, the members of these families are **emotionally fused**. Such families have little tolerance for differences in feeling or belief; what one member thinks or feels, all members think or feel. When family therapists encounter a family such as this—often labeled an **undifferentiated ego mass**—the primary goal is to acknowledge the lack of differentiation, explore factors that may have caused or are currently maintaining it, and encourage relationships characterized by less fusion and more independence.

Triangles. When two people are in conflict, either one might decide to bring in a third party in an attempt to garner support. In a formal, legal sense, we see this phenomenon take place when attorneys bring in expert witnesses to bolster their cases in court. In families, a similar process occurs, but it can lead to significant problems, especially for the person being coerced into the triangle. A common scenario for **triangles** in families involves parents at odds and a child recruited to side with one parent. Unlike the expert witness in a court case, children are powerless to refuse parental attempts to triangulate and, as a result, can find themselves in a psychologically unhealthy predicament whereby they feel that they must "take sides" with one parent against the other. Although such situations can be obvious in families involving contentious divorce or child custody battles, adverse effects can also occur in intact families in which parents subtly pressure kids to side with them. With families whose triangulation tendencies have become problematic, family therapists often try to bring these tendencies to light and encourage the parties in the conflict to deal more directly with each other rather than involving third parties such as their children (e.g., Bowen, 1978).

More Contemporary Approaches

Solution-Focused Therapy. Families come to therapy because they have specific problems, and, according to solution-focused family therapists, therapy is successful when those problems are solved. **Solution-focused therapy** evolved from the **strategic family therapy** approach of Don Jackson, Jay Haley, and Cloé Madanes and shares its pragmatic emphasis (Goldenberg & Goldenberg, 2007; M. P. Nichols, 2010). But where strategic family therapists focused on problems, solution-focused family therapists focus almost exclusively on solutions (Berg, 1994; deShazer, Dolan, & Korman, 2007).

Web Link 16.5
Institute for Solution-Focused Therapy

The leaders of the solution-focused therapy approach, including **Steve deShazer** and **Insoo Kim Berg**, emphasize that family therapists should use **"solution-talk"** rather than "problem-talk" as much as possible. By discussing positive outcomes that the future may hold rather than unpleasant situations that characterize the present, clients begin to adopt a more positive point of view. For example, solution-focused therapists deliberately discuss "when" the problem will get better, rather than "if" it might (Corcoran, 2015). Similarly, therapists help clients focus on exceptions to their problems. That is, they remind clients that they already have the ability to help themselves, as evidenced by the fact that they have already experienced some moments or days that were better than others. Clients can sometimes assume their problems are ever present when, in fact, they are not. Likewise, improvements are not flukes but the result of efforts by the clients, even if they frequently overlook such successes (M. P. Nichols, 2010). So if a family says they argued slightly less in the past week than they had in the previous month, the solution-focused therapist seizes on this improvement and assumes that the family was instrumental in causing it: "What did you do to improve things last week? How can you continue to do more of the same in upcoming weeks?"

Solution-focused therapy uses a list of specific tasks and questions posed to clients to call attention to solutions rather than problems. For example, at the end of the first session, solution-focused therapists often present the **formula first-session task**, in which clients are instructed to take note of aspects of their lives in the upcoming week that they want to continue to happen. This forces clients to focus on the positive rather than the negative facets of their day-to-day lives. They also ask **exception questions** ("When was this not a problem for you? When was it not so bad?"), **miracle questions** ("If the problem disappeared, how would your life be different?"), and **scaling questions** ("On a scale of 1 to 10, how bad has the problem been in the past week? When has the problem been better? How did you make it improve at that time?"). Structured, positive-oriented questions such as these, along with an emphasis on quick resolution of presenting problems, have contributed to the increasing popularity of solution-focused therapy in recent years (M. P. Nichols, 2010).

Narrative Therapy. Also very popular among contemporary family therapists, **narrative therapy** highlights clients' tendencies to create meanings about themselves and the events in their lives in particular ways, some of which may cause psychological problems. As the name suggests, narrative therapists believe that the stories we construct about our own lives are powerful influences on the way we experience new events. In a sense, we "edit" our experiences in order to fit the story line we have developed. So if we cast ourselves as failures in our own life stories, we are likely to interpret events in ways that support such a tale. If we can revise our stories and recast ourselves as heroes, however, the same sequence of events might be interpreted in a more flattering way and our view of ourselves might be enhanced (e.g., M. White & Epston, 1990).

As an example, consider Phil, a 29-year-old man who has been married for 6 years. Phil and his wife, Jamie, have a 4-year-old daughter and are expecting another child in a few months. The primary reason why Phil and Jamie seek therapy is Phil's depression; specifically, both Phil and Jamie agree that Phil is extremely self-critical and dejected over career-related issues. Phil explains to Dr. Gupta, a narrative family therapist, that immediately after finishing college at age 22, he took an entry-level corporate job. Although he was quite capable and had advanced in the company, Phil found the work very unsatisfying and stifling. After discussing the matter at length with Jamie, at age 27 Phil decided to leave the corporate world and focus on his long-time passion, music. While Jamie continued to work full-time, Phil worked hard to establish himself as a guitarist/singer/songwriter, and he experienced quite a bit of success. He had regular gigs at numerous venues, had recorded several independent CDs that sold well locally, and had recently signed a contract with a record label.

In spite of these successes and the satisfaction he derived from his music, Phil saw his career change as a mistake and an embarrassment. He made less money as a musician than he would have in his corporate job, and some of his friends and relatives wondered why he would allow his career to go "off track." Phil also felt that his old job had more stability and prestige and that he wasn't adequately fulfilling his role as a husband and father with his current occupation. Phil's negative feelings about his career decision had a significant impact on his self-image and his relationships with his wife and daughter, both of whom wished to see Phil happier.

Dr. Gupta used the narrative approach to revise Phil's story from one of regret and failure to one of courage, authenticity, and success. If Phil's story had a title when therapy began, it might have been "I'm a Loser," and he found supporting evidence for this tale's theme in his decreased income and second-guessing friends and relatives. But Dr. Gupta offered another title—"A Self-Made Success"—and supported it by pointing out that Phil had made a brave decision early in his career that enabled him to pursue his passion and that he had, in fact, been quite successful in a short time in a difficult business. Dr. Gupta argued that keeping the corporate job might have been the safe or predictable thing to do and may have fit with society's expectations, but Phil's actions showed an independence, self-awareness, and determination that are remarkably admirable. With time, Phil began to see himself more heroically and was able to interpret life events accordingly.

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Photo 16.3 An appreciation of the cultural background of a family and each of its members is essential to culturally competent family therapy.

In some ways, narrative therapy is similar to cognitive therapy in its emphasis on altering clients' interpretations of events. In narrative therapy, those interpretations are seen as essential elements of the stories of clients' lives, and the process of changing them resembles editing. Although it is considered a form of family therapy, narrative therapy emphasizes the individual within the family, and it also tends to underscore clients' efforts to resist oppressive cultural or societal forces (M. P. Nichols, 2010).

Multisystemic Family Therapy. **Multisystemic family therapy** is designed for adolescents with long-term behavioral and emotional problems that involve legal offenses. These teens

encounter problems in multiple areas of their lives—they struggle to attend and perform well in school, their relationships with parents and other family members are conflictual, and they get into trouble with the law. As its name suggests, multisystemic family therapy extends beyond the teen's immediate family and incorporates many of these systems. It utilizes elements of cognitive, behavioral, parent-training, solution-focused, and other kinds of therapy. Multisystemic therapists meet with various members of the family in various combinations (and in various locations, including home, the school, and community settings). They also meet with school personnel such as teachers and principals. They steer teens toward peers that will be good influences and activities that will be beneficial to their futures. They train the adults around the teens to support and encourage such behaviors (S. D. Davis et al., 2015; Henggeler, Schoenwald, Borduin, Rowland, & Cunningham, 2009).

Evidence for multisystemic family therapy is often measured in very concrete ways. For example, some studies measure long-term incarceration rates of children whose families participated in multisystemic family therapy, and the results are often quite impressive, in comparison to comparable teens who had no therapy or individual therapy (Henggeler & Schaeffer, 2016; Letourneau et al., 2013; van der Stouwe, Asscher, Stams, Deković, & van der Laan, 2014; Wagner, Borduin, Sawyer, & Dopp, 2014). Other studies measure the outcomes in dollars and cents. For example, Dopp, Borduin, Wagner, and Sawyer (2014) found that every dollar spent on multisystemic family therapy for a juvenile offender's family saved more than $5 over the next 25 years in terms of taxpayer-funded services (mostly involving the legal system) that family members did not need. A similar study found that on average, treating a family with multisystemic family therapy produced a net savings of $26,000 over just a 16-month period (Dopp et al., 2014; Dopp, Schaeffer, Swenson, & Powell, 2018).

Ethical Issues in Family Therapy

Cultural Competence

Cultural competence is essential in any mode of therapy, but it is especially relevant in family therapy. Family therapists should have a full appreciation of and acceptance for the cultural background of the families with whom they work, including such characteristics as ethnicity and religion.

For therapists working with families, an appreciation of some cultural variables can be quite complex yet vital to the therapy process. The members of a family may actually come from or currently live in different cultural worlds. Partners, for example, might have been raised in different religions or different ethnicities. Or, a three-generation family living in the same house may include three distinct levels of acculturation. For example, the grandparents who immigrated may hold traditional values, the parents may be somewhat more assimilated, and the children may tend toward holding few or no traditional values. Or, a family with adopted children may include parents whose race or ethnicity is different from that of their children. In any of these cases, family therapists must be capable of considering culture not only as a context for the presenting problem but also as a potentially important aspect of the presenting problem itself.

Confidentiality

Family therapy can present some unique ethical quandaries with regard to confidentiality. Specifically, family therapists often find themselves in the difficult position of

having learned information from one family member in a private conversation—perhaps a phone call or a comment made in a brief one-on-one conversation before or after a session. Should the family therapist hold this information confidential from the other family members? Or should any information shared by any family member be shared with all family members? Does the answer depend on how serious or trivial this information may be to the rest of the family?

There are few easy answers to these questions (Knauss & Knauss, 2012; Wilcoxon et al., 2007). In general, it is best to set the ground rules for confidentiality during the initial informed consent process; that way, all family members understand up front that the family therapist will handle private conversations in a particular way. Some family therapists choose never to have private conversations with individual family members, whereas others allow them. To the extent that family therapists hold individual family members' comments confidential, they place a burden on themselves to recall during sessions not only what they know but also how they learned it and how much liberty they have to share that information with the whole family.

Diagnostic Accuracy

Family therapists face a dilemma regarding diagnosis: *DSM* diagnoses apply to disordered individuals, but family therapists work with disordered family systems. The *DSM* contains no diagnostic labels that apply to families, and despite a variety of efforts since the 1980s to create a "relational" diagnosis system of some kind, there is no widely accepted alternate to the *DSM* for definitively categorizing family or relationship problems (Beach et al., 2016; Lebow, 2015; Sexton & Stanton, 2016; Wamboldt, Kaslow, & Reiss, 2015). Therefore, especially when there is pressure to assign a diagnosis to qualify for payment from managed-care/insurance companies, family therapists may consider assigning a *DSM* diagnosis to one member of the family.

Family therapists often have justified reservations about diagnosing in this manner. For one, to diagnose only the identified patient may reinforce the family's assumption that the problem is individual, not systemic. Also, the identified patient can feel further stigmatized. And, of course, a diagnosis made when *DSM* criteria are not met represents both an ethical violation and insurance fraud (American Psychological Association, 2002). When a family's only means of paying for treatment hinges on a diagnosis, this can be a difficult issue for family therapists. Open and honest discussion with the family during the informed consent process can be helpful in resolving it (Wilcoxon et al., 2007).

How Well Does It Work?

Outcome studies of family therapy involve all the methodological difficulties of outcome studies for individual therapy, with one additional, complicating factor: Various members of the family may have quite different opinions about the result. For example, if family therapy results in previously uninvolved and lenient parents paying more attention to and placing appropriate demands on their children, the parents may view therapy as successful, but some of the children may not.

In spite of this and other methodological challenges, many studies of the efficacy of family therapy have been conducted. Although the amount of research is not as sizable as that for individual therapy, the result is generally the same: Family therapy works. More

specifically, family therapy tends to be about as effective as most other modes of therapy, and the differences among various family therapy approaches are typically insignificant (Bernal & Gomez-Arroyo, 2017; Griffin, 2002; Lebow & Stroud, 2016; Pinquart, Oslejsek, & Teubert, 2016; Riedinger, Pinquart, & Teubert, 2017; Sexton & Stanton, 2016; Shadish et al., 1993; Shadish, Ragsdale, Glaser, & Montgomery, 1995). Some specific problems for which family therapy has established success include schizophrenia, couples discord, eating disorders (especially anorexia) in adolescents, self-injury in adolescents, delinquency in adolescents, conduct disorder and other disruptive disorders in children, and numerous others (Glenn, Franklin, & Nock, 2015; Glick, Berman, Clarkin, & Rait, 2000; Goldenberg & Goldenberg, 2007; Jewell, Blessitt, Stewart, Simic, & Eisler, 2016; Lebow, 2006; Lock, 2015). As mentioned earlier in the chapter, some of this success has been measured in terms of financial benefits to society, particularly as multisystemic family therapy produces outcomes that reduce the need for taxpayer-funded services (Dopp et al., 2018). As in individual therapy, a strong therapeutic relationship is a key element to successful family therapy, as demonstrated in an increasing number of empirical studies (e.g., Escuerdo & Friedlander, 2017; Friedlander, Escudero, Heatherington, & Diamond, 2011; Friedlander, Escudero, Welmers-van de Poll, & Heatherington, 2018).

CHAPTER SUMMARY

Group therapy is characterized by an emphasis on interpersonal interaction. Irvin Yalom and other leaders in the group therapy field believe that flawed interpersonal styles contribute to clients' presenting problems and that focusing on interpersonal interactions within the group can have positive ramifications for clients outside the group. A number of unique therapeutic factors, including universality, group cohesiveness, and interpersonal learning, distinguish group therapy from individual therapy. Therapy groups can be heterogeneously or homogeneously composed, and they can adopt open- or closed-enrollment policies. Two cotherapists often conduct groups together, allowing both potential benefits and complications. Pitfalls of group therapy include extra-group socializing and breaches of confidentiality by fellow group members. Outcome studies suggest that group therapy is generally as effective as individual therapy.

Family therapy centers on a systems-based approach to psychological problems in which the symptoms of an identified patient are viewed as a by-product of a pathological pattern of interaction or communication within that person's family. Family therapists often use genograms, a sort of annotated family tree, as an assessment tool and conceptualize their clients developmentally on an adaptation of the family life cycle model of development. Many family therapists continue to be influenced by classic concepts of family therapy such as the subsystems and boundaries of structural family therapy or the differentiation and triangulation issues of Bowenian therapy, but newer approaches such as solution-focused, narrative, and multisystemic family therapy also exert a powerful influence on contemporary family therapists. As for all clinical psychologists, cultural competence is crucial for those practicing family therapy. Outcome studies of family therapy can be more methodologically complex than those for individual therapy, but those that have been conducted suggest the rate of success for its clients is generally comparable with that of individual therapy.

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Murray Bowen 409
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communication patterns 401
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CRITICAL THINKING QUESTIONS

1. In your opinion, which clinical problems seem best and least suited to group therapy?
2. In your opinion, which of Yalom's therapeutic factors for group psychotherapy seem most vital to its success? Why?
3. If you were a clinical psychologist, would you prefer to conduct group therapy alone or with a cotherapist? What factors might influence your decision?
4. To what extent do you agree with the fundamental assumption of family therapy that the system, rather than the individual, is pathological?
5. To what extent might the contemporary approaches to family therapy (especially solution-focused therapy and narrative therapy) also be applicable to individual clients?

LOOKING TOWARD GRADUATE PROGRAMS

The following table is a list of index entries in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018) relevant to the topics in this chapter (see page 25 in Chapter 1 for more information about the *Insider's Guide*). Keep in mind a few things as you consider these listings: They are current as of the 2018/2019 edition of the *Insider's Guide* (which typically revises every two years); they may have changed as faculty members retire, take new jobs, or develop new interests; and a graduate program's own website may have more extensive or up-to-date information.

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Adoption/Foster Care	Adoption	Family/Marriage & Family
Child and Family	Family/Family Therapy/ Family Systems	Miscellaneous/Other—Family Violence
Family/Family Therapy/Family Systems	Group Therapy	
Group Process and Therapy	Marriage/Couples/Intimate Partner Violence	
Marital/Intimate Partner Violence		
Marriage/Couples		
Miscellaneous and Other—Diverse Families		
Miscellaneous and Other—Family Violence		

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

American Journal of Family Therapy
http://www.tandfonline.com/loi/uaft20#.V08pn_krLX4

Couple and Family Psychology: Research and Practice
http://www.apa.org/pubs/journals/cfp/index.aspx

Families, Systems, and Health
http://www.apa.org/pubs/journals/fsh/index.aspx

Group Dynamics: Theory, Research, and Practice
http://www.apa.org/pubs/journals/gdn/

International Journal of Group Psychotherapy
http://guilfordjournals.com/loi/ijgp

Journal of Family Psychology
http://www.apa.org/pubs/journals/fam/index.aspx

Journal of Family Therapy
http://onlinelibrary.wiley.com/journal/10.1111/(ISSN)1467-6427

STUDENT STUDY SITE RESOURCES

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- eFlashcards
- Culture expert interviews
- In My Practice demonstration video
- Full-text SAGE journal articles
- Additional web resources
- Mock assessment data

PART IV

SPECIAL TOPICS

CHAPTER 17 • **Clinical Child and Adolescent Psychology**

CHAPTER 18 • **Health Psychology**

CHAPTER 19 • **Forensic Psychology**

17

CLINICAL CHILD AND ADOLESCENT PSYCHOLOGY

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How Well Does Psychotherapy for Children and Adolescents Work? 444

Clinical psychologists have worked with children and adolescents since the inception of the field. In fact, the focus of the very first psychology clinic, created by Lightner Witmer in the late 1800s, was the assessment and treatment of children with learning and behavioral problems (Reisman, 1991; Witmer, 1907). Along with the rest of the profession, **clinical child psychology** has grown tremendously (Carr, 2016; Lochman, Boxmeyer, & Powell, 2016). The membership of the **Society of Clinical Child and Adolescent Psychology** (Division 53 of the American Psychological Association) is currently in the thousands, and a sizeable number of graduate programs train psychologists specifically to work with children and adolescents (American Psychological Association Division 53, 2007).

Clinical child psychologists apply their professional skills in diverse ways. Some specialize in the assessment of problems related to behavioral, emotional, or intellectual functioning. Others are primarily therapists, working with children and their families to bring meaningful improvement to their lives. And, of course, some clinical child psychologists devote their professional lives to research or training. Many

Learning Objectives

17.1 Review major psychological issues of childhood.

17.2 Explain how a clinical psychologist could demonstrate a developmental perspective while assessing children or adolescents.

17.3 Describe the multisource, multimethod, multisetting approach to the psychological assessment of children.

17.4 Summarize the six broad categories of assessment methods psychologists may use with children.

17.5 Examine psychotherapy approaches that are designed for children and adolescents.

17.6 Explore the outcome research for psychotherapy with children and adolescents.

clinical child psychologists combine these applied and academic roles. In some cases, the applied work of clinical child psychologists overlaps with that of medical professionals. This cross-disciplinary field, known as **pediatric psychology**, promotes both mental and physical health of children with medical conditions.

In this chapter, we survey the field of clinical child psychology by exploring three major areas: psychological problems of children, assessment, and psychotherapy.

PSYCHOLOGICAL ISSUES OF CHILDHOOD

Disorders of Childhood

Some disorders are especially common among children, such as attention-deficit/hyperactivity disorder (ADHD), conduct disorder, oppositional defiant disorder, and separation anxiety disorder. Of course, children and adolescents can also be diagnosed with disorders that are more common in adults. Major depression, posttraumatic stress disorder, anorexia, bulimia, substance use, adjustment disorders, phobias, and generalized anxiety disorder are some of the many problems diagnosed in both adults and children. In some cases, the diagnostic criteria for these disorders are adjusted for children. For example, among the criteria for a major depressive episode, the *DSM-5* includes a note that in children and adolescents, irritable mood can replace depressed mood, and failure to gain weight according to growth expectations can replace weight loss. Among the criteria for specific phobia, *DSM-5* notes that children may express their fear and anxiety in unique ways, including crying, tantrums, or clinging to caregivers. For posttraumatic stress disorder, the *DSM-5* has a distinct set of criteria for children 6 years old and younger, including diminished interest in playing or playing that involves reenactment of the trauma (American Psychiatric Association, 2013).

Web Link 17.1
Society of Clinical Child and Adolescent Psychology

Some clinical psychologists who work with children have divided children's psychological problems into two broad classes: **externalizing disorders** and **internalizing disorders** (e.g., Kazdin, 2000). Externalizing disorders are those in which the child "acts out" and often becomes a disruption to parents, teachers, or other children. These disorders include ADHD, conduct disorder, and oppositional defiant disorder. Internalizing disorders, on the other hand, are often less noticeable because they involve maladaptive thoughts and feelings more than disruptive outward behavior. Depression and anxiety disorders exemplify internalizing disorders in children.

Resilience and Vulnerability

Why do some children develop psychological disorders, whereas other children do not? This is an essential question for clinical child psychologists, especially those who study the topics of **resilience** and **vulnerability**. It is indeed remarkable to notice that children from similar environments—in some cases, the same geographic area, ethnicity, socioeconomic level, school, and family—have very different kinds or degrees of psychological or behavioral problems (Bonanno et al., 2011). Research into factors that contribute to a child's vulnerability to psychological problems has yielded a list of risk factors that encompasses variables both within and around the child (e.g., M. C. Lambert, 2006; Laser & Nicotera, 2011; McKelvey, Conners-Burrow, Mesman, Pemberton, & Casey, 2015; Schroeder & Gordon, 2002; Stormont, 2007):

- Environmental factors, such as poverty, serious emotional conflict among parents, single parenthood, an excessive number of children in the home, neighborhood or community factors, and poor schooling
- Parental factors, such as poor parental physical health, poor parent mental health, low parent intelligence quotient (IQ), and hypercritical tendencies in the parent
- Child (internal) factors, such as medical problems, difficult temperament, low IQ, poor academic achievement, and social skills deficits

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Photo 17.1 Clinical child psychologists are particularly interested in factors that foster resilience rather than vulnerability to psychological or behavioral problems. In your opinion, which factors are most important?

The importance of family is apparent in the list above. Regarding parents in particular, the more psychological problems they have, the more psychological problems their children are likely to have (Schleider, Chorpita, & Weisz, 2014). One study assessed parents and kids when the kids were 3 years old, and then assessed the kids again at ages 4, 5, and 6. As the years went by, the kids whose parents had been diagnosed with a mental illness when the kids were 3 were significantly more likely to be diagnosed themselves. That relationship was especially true when the parents' diagnosis was ADHD, depression, an anxiety disorder, or a personality disorder (Breaux, Harvey, & Lugo-Candelas, 2014). In another study, children being treated for anxiety disorders were much more likely to be diagnosis-free, both immediately after treatment and at a 6-month follow-up, if their parents were nonanxious than if their parents were anxious (Hudson et al., 2014). Sibling relationships are also important factors in kids' vulnerability to psychological disorders (Dirks, Persram, Recchia, & Howe, 2015). A meta-analysis of 34 studies on the effect of sibling relationships on children's mental disorders found that those with warmer, more loving brothers and sisters had fewer internalizing and externalizing disorders. Those whose relationships with their siblings were full of conflict were more likely to have both categories of disorders (Buist, Deković, & Prinzie, 2013).

Fortunately, researchers have also examined the factors that contribute to resilience in children. After reviewing research on children who were somehow able to thrive in risky environments where many of their peers struggled, Grotberg (2003) sorted the differentiating factors into three categories: external supports, inner strengths, and interpersonal problem-solving skills. To these three categories, she assigned kid-friendly labels: "I Have" (external supports), "I Am" (inner strengths), and "I Can" (interpersonal and problem-solving skills). Box 17.1 presents a full description of Grotberg's resilience model. In many cases, interventions by clinical child psychologists can enhance these factors, effectively making the child more resilient to current and future stressors. By combining what researchers have learned about the factors underlying children's resilience and vulnerability, clinical child psychologists can work to prevent and treat children's psychological problems.

BOX 17.1

Factors Fostering Resilience in Children

I Have (external supports)

- People in my family I can trust and who love me
- People outside my family I can trust
- Limits to my behavior
- People who encourage me to be independent
- Good role models
- Access to health, education, and other services I need
- A stable family and community

I Am (inner strengths)

- A person most people like
- Generally a good boy/girl
- An achiever who plans for the future
- A person who respects myself and others
- Caring toward others
- Responsible for my own behavior
- A confident, optimistic, hopeful person

I Can (interpersonal and problem-solving skills)

- Generate new ideas or new ways to do things
- Work hard at something until it is finished
- See the humor in life
- Express my thoughts and feelings
- Solve problems
- Manage my behavior
- Ask for help when I need it

Source: Adapted from Grotberg (2003).

A final note on resilience and vulnerability in kids: It matters whether they believe they can change their personality traits. In other words, it makes a difference whether a child's "implicit theory" about her or his own characteristics is fixed ("I'm shy, and there's nothing I can do about it") or malleable ("I'm shy, but I can overcome it"). (The fixed theory is often called an "entity" theory, and the malleable theory is often called an "incremental" theory; Dweck & Leggett, 1988.) Specifically, the fixed approach is more likely to lead to mental health problems (Blackwell, Trzesniewski, & Dweck, 2007; Romero, Master, Paunesku, Dweck, & Gross, 2014; Yeager et al., 2014). A meta-analysis of 17 studies found that kids aged 4 to 19 had more frequent and more severe psychological diagnoses, both internalizing and externalizing, if they held a fixed/entity theory about themselves than if they held a malleable/incremental theory about themselves (Schleider, Abel, & Weisz, 2015).

ASSESSMENT OF CHILDREN AND ADOLESCENTS

The Developmental Perspective

In any competent assessment of a child or adolescent, clinical psychologists adopt a **developmental perspective**. In other words, it is essential to understand the child's behavior within the context of the child's developmental stage (e.g., Drabick & Kendall, 2010; LaGreca, Kuttler, & Stone, 2002). Problems of childhood may take on different meanings and call for different clinical interventions depending on their commonality for children at a given age. For example, imagine that Beth and her parents visit a clinical psychologist because of Beth's defiant, rule-breaking behavior. On several occasions, Beth has ignored her parents' curfews and prohibitions about smoking. A full, accurate understanding of this problem will depend entirely on Beth's age. If Beth is 7, this problem takes on very different connotations than it does if she is 17. Likewise, if Donyell's mother brings him to a clinical psychologist for a bed-wetting problem, whether Donyell is 3, 9, or 15 can make a significant difference in the conceptualization of this problem.

Of course, ethnicity or other diversity-related factors can influence the developmental maturation of children, as parents and other family members from a given cultural group may expect children to develop relatively quickly or slowly in certain areas (Park-Taylor, Ventura, & Ng, 2010). Clinical psychologists should appreciate these factors to grasp a child's problems more fully. For example, Hines and Boyd-Franklin (2005) describe a pattern common in African American families in which a "parentified" child takes on adult roles in a home with working parents or many younger children. This role "can enhance the parental child's sense of responsibility, competence, and autonomy" (p. 91) but can contribute to behavioral problems later if the child feels overwhelmed or if these parent-like duties conflict with the child's own personal needs.

BOX 17.2

Considering Culture

Parent–Child Relationships Across Cultures

Culturally competent clinical child psychologists appreciate that the manner in which parents raise their children can differ drastically according to cultural norms. Furthermore, culturally competent clinical child psychologists are knowledgeable about culture-specific norms in child rearing and amend their assessment and treatment accordingly to make them ideally suitable (Park-Taylor et al., 2010).

To illustrate the variety of parent–child relationships across cultural groups, consider the discrepancies among the following descriptions. Each excerpt comes from a separate chapter of the book *Ethnicity and Family Therapy* (McGoldrick, Giordano, & Garcia-Preto, 2005a):

Vietnamese families:

> "Even now, children are regarded as property by their parents, although this belief has loosened gradually since World War II as the culture has become more westernized. . . . Children are expected to follow their parents' advice in all aspects of life, including matters relating to marriage and career selection." (Leung & Boehnlein, 2005, p. 366)

Jewish families:

> "Jews tend to raise their children through reasoning and explanation of expectations, and Jewish parents have tended to be permissive, overprotective, and concerned about their children's happiness, at times at the expense of their own." (Rosen & Weltman, 2005, p. 675)

Haitian families:

> "In general, parents, who have near absolute power in Haiti, raise children with an understanding that they must be obedient and sometimes administer physical discipline with a belt or switch when they misbehave, which may on occasion be viewed as abuse in this country [the United States]. When child protective services intervene, parents can be stunned." (Menos, 2005, p. 133)

Asian Indian families:

> "In general, everyone, including older siblings and extended family, pampers children. A 4- or 5-year-old child is often less independent than children of European background." (Almeida, 2005, p. 387)

Iranian families:

> "There is no specific pattern to the punishment; depending on the father's mood, a child may be punished for a trivial act of misbehavior; at other times, a more

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Photos 17.2 and 17.3
The ways parents raise children can vary greatly across cultural groups. In your opinion, how should clinical child psychologists increase their awareness of these cultural norms?

> serious act may be laughed off or overlooked. A child may be slapped if he or she misbehaves in public or in front of a family guest, yet in private the same misbehavior might only merit a scolding. Western methods of disciplining, such as withholding favorite foods or sending children to their rooms, are rarely used." (Jalali, 2005, p. 457)

Arab families:

> "Another feature of the Arab family is its style of communication, which is described as . . . hierarchical, creating vertical as opposed to horizontal communication between those in authority and those subservient to that authority. This relationship . . . leads to styles of communication between parents and children in which parents use anger and punishment and the children respond by crying, self-censorship, covering up, or deception." (Abudabbeh, 2005, p. 427)

In addition to knowledge of such diverse cultural norms regarding parent–child interactions, culturally competent clinical child psychologists have abundant knowledge of their own cultural background, especially its influence on their expectations regarding parent–child relationships. With this knowledge, the clinical child psychologist can proficiently and sensitively treat a diverse range of clients.

A Comprehensive Assessment

An extensive amount of background information may be relevant to the psychological problems of a particular child; thus, clinical psychologists should strive to attain as much of it as possible. This background information can help the psychologist appreciate the full set of circumstances in which the child's presenting problems have emerged. Some of this background information relates directly to the presenting problem, whereas some is more contextual but nonetheless important. Among the questions for which a clinical psychologist might seek answers are the following (e.g., Schroeder & Gordon, 2002):

- *The presenting problem:* What, exactly, is the presenting problem? Do all parties (parents, child, teachers, etc.) agree on the definition of the problem? When did the problem arise? For whom is this problem most troubling?
- *Development:* What is the child's current state of physical, cognitive, linguistic, and social development? Have there been any developmental abnormalities during childhood or during the prenatal period? Has the child reached all developmental milestones at the expected points in time?
- *Parents/family:* What are the relevant characteristics of the child's parent or parents? What style of parenting is used? What parent factors (psychological, medical, other) might play a role in the child's problem? How might siblings, grandparents, or other family members influence the child?
- *Environment:* What is the child's larger environment outside the family? What relevant ethnic or cultural factors play a role in the child's behavior? Are there recent major events in the child's life that may factor into the current problems?

The clinical child psychologist may receive different answers to these questions from different people in the child's life. For this reason, it is important to rely on more than one source of information when assessing a child. More generally, a pluralistic approach to assessing children is sound clinical practice. In fact, Merrell (2008) advocates a **multisource**, **multimethod**, **multisetting approach** to assessment of children. Multisource assessment involves such parties as parents, relatives, teachers, other school personnel, and, of course, the child as sources of information regarding the child's problems. To rely on only one of these sources is to risk a one-sided or incomplete perspective of the problem (Tackett, Herzhoff, Reardon, Smack, & Kushner, 2013). Large-scale studies of assessments of over 25,000 children found that descriptions of a child's behavior frequently disagreed when those reports came from a parent and a teacher, or from a parent and the child. The disagreements were especially common when the clinical problem was an internalizing disorder (Rescorla et al., 2013, 2014). Multimethod assessment involves the use of different methods of data collection by the clinical psychologist, such as interviews, pencil-and-paper instruments completed by the child or those who know the child well, direct observation of the child's behavior, and other techniques. Multisetting assessment acknowledges that sometimes children's problems pervade all facets of their lives, but sometimes they are specific to certain situations. Thus, it is wise to solicit data from home, school, the clinician's office, and any other relevant setting.

As an example of the value of a multisource, multimethod, multisetting approach to assessing children, consider Ari, a 7-year-old boy whose parents brought him to Dr. Kim, a clinical child psychologist. Ari's parents described Ari as "impossible to manage": He simply won't listen to instructions and refuses to do what he is told. Ari's parents completed a rating scale that confirmed their comments, and they told numerous stories about episodes at home—bedtime, going to school, cleaning up his toys—during which Ari was unruly. To understand the problem more comprehensively, Dr. Kim requested Ari's parents' permission to contact Ari's first-grade teacher and a babysitter who watched Ari several times a week for a few hours at a time, which the parents granted by signing release forms. The information Dr. Kim received from Ari's teacher and babysitter was quite discrepant from the information provided by the parents. Via interview with Dr. Kim and on rating scale forms, the teacher and babysitter both described Ari as very compliant, obedient, and well-behaved. Dr. Kim also visited Ari's school to observe his behavior directly and found him not to be disruptive or disobedient at all. Thus, Dr. Kim was able to conclude that Ari's problem was not pervasive to all aspects of his life; instead, it was situational to his time at home when his parents were present. Dr. Kim was then able to proceed with an intervention specifically designed to improve Ari's parents' ability to communicate and manage contingencies in the home, which proved to be successful.

BOX 17.3

In My Practice . . .

In my practice, multisource assessment with children has involved quite a variety of sources. The lives of children today differ widely, particularly in terms of the people with whom a child spends significant time. Anyone who does spend significant time with a child is a potential source of helpful information. (Of course, an interview with any of these sources requires the permission of the parent or guardian of the child.)

In addition to the child clients themselves, certain sources of information—specifically parents and teachers—are extremely common in the psychological assessment of children. Typically, these are the adults who know the child best and can provide the most informed answers to questions about the child's presenting problems. But there are plenty of others. In various child assessments, I have spoken with extended family members, including grandmothers, grandfathers, aunts, uncles, cousins, and siblings. I have also spoken with various professionals at the child's school beyond classroom teachers. School counselors and teacher's aides have been most common within this category, but in one case, a child from a tiny school was so personally known to the principal and the school secretary that they each were able to provide detailed information about her. Another category of sources of information about child assessment clients are adults who work with the child outside of school. For example, one of my child assessment clients trained competitively in gymnastics. She spent many hours a week with her coach, and that coach was able to offer a perspective on her problem that neither her teachers nor her parents could match. Besides coaches, my assessments

(Continued)

(Continued)

of children have benefited from the input of tutors, neighbors, family friends, religious leaders, religious school teachers, and Big Brothers/Big Sisters. For teen clients, conversations with supervisors from clients' part-time jobs (with whom they often spend more time than their parents) have proven enlightening. And of course, for any child, speaking with a health professional (such as a pediatrician) or a previous mental health professional (such as a former psychologist) can be essential.

In some cases, the idea to speak to a particular source is mine. Based on something I hear in the initial interview with the child or parent, I bring up the possibility, and if I get parental permission, I seek out the conversation. In other cases, the child or parent directs me to the source. Over the years, the lesson I have learned is to respect the diversity of lifestyles across the wide range of children and families who seek psychological services. Along with that diversity comes a variety of sources, both traditional and nontraditional, who might be able to supply important convergent or divergent information about a child assessment client.

To see this box come to life as a whiteboard video, please visit edge.sagepub/pomerantz5e.

Assessment Methods

Clinical child psychologists use a vast array of assessment tools and techniques. In general, they can be categorized into six broad categories: clinical interviews, behavioral observation, behavior rating scales, self-report scales, projective/expressive techniques, and intellectual tests (Cashel, 2002; Krishnamurthy & Meyer, 2016; LaGreca et al., 2002; Merrell, 2008; Wodrich, 1997).

Interviews

When a clinical psychologist assesses an adult, very often the client is the only person with whom the psychologist speaks. The process is very different in child assessment, where the clinical psychologist typically interviews not only the child but also other people who, by virtue of their contact with the child, can shed light on the child's problem. Parents and teachers are perhaps most common, but depending on the problem and its circumstances, a number of other people can provide relevant data: siblings, grandparents, aunts, uncles, pediatricians, school administrators, coaches, family friends, child-care workers, nannies, or tutors, to name a few.

Interviewing those who know the child well—informants, as they are called in this role—is a vital skill for the clinical psychologist. When speaking with parents, it can be helpful to empathize with their experience of the child's problem, which may include frustration or sadness, and correct any misconceptions by reassuring them that the psychologist's role is not to assign blame but to work constructively and collaboratively with the parents to remedy the problem. Regarding interviews with individuals other than parents, it is always essential to obtain permission from parents/guardians prior

to contacting them. Once the contact occurs, it is best to be as professional, respectful, and concise as possible. This is especially true when contacting teachers, whose schedules often afford them little time for this important task. Recognizing the value of their time by contacting them at times of the day when they are most available to talk (i.e., are not teaching) and thanking them for their efforts afterward can facilitate the process (Kamphaus & Frick, 2005; Villa & Reitman, 2007).

Interviewing children themselves is an obviously crucial yet sometimes challenging component of the clinical child psychologist's job. Children rarely refer themselves to a psychologist; much more often, the referral comes from a parent, teacher, or pediatrician with concerns about the child's behavior (LaGreca et al., 2002). In fact, when children are brought to a clinical psychologist, typically "the focus is on *disturbing* rather than *disturbed* behavior" (Kazdin, 2000, p. 47). So, whereas other people may want the child to change, the child may be unmotivated or even resistant. In some cases, the child may be quite anxious about the interview and, depending on age and sophistication, may be under the misconception that the psychologist will do something scary or hurtful, along the lines of a shot or a physical examination at the pediatrician's office.

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Photo 17.4 When clinical child psychologists conduct a comprehensive assessment, the child's environment is one of many important areas to consider.

For this reason, establishing rapport is vital to interviewing children (Kendall, 2012). It can be helpful at the outset to engage in just enough small talk to make the child feel comfortable but not so much that the interview strays from its purpose. Also, the clinical psychologist's speech patterns should match those of the child, and the vocabulary should be comprehensible to the child as well (Kamphaus & Frick, 2005; Villa & Reitman, 2007). Above all, communicating genuine respect for children, concern for their happiness, and a wish to work together to improve their lives can help children feel comfortable in clinical interview situations. This holds true whether conducting structured or unstructured interviews, both of which are used commonly by clinical child psychologists (Nock et al., 2007; O'Brien & Tabaczynski, 2007).

Behavioral Observations

When the child's problem involves a behavior that can be directly observed, as in most externalizing disorders, clinical child psychologists often do so. Behavioral observation may require traveling to the setting where the behavior problem takes place, such as the child's school or home. Once there, the clinical psychologist typically uses a formal, systematic method of observing and coding the child's behavior. There are several published systems available for clinical psychologists to use for this purpose, including the Direct Observation Form, developed in conjunction with the Child Behavior Checklist; the Student Observation System, which is a component of the Behavior Assessment System

for Children; the Dyadic Parent–Child Interaction Coding System; and the Social Interaction Scoring System (Merrell, 2008; Merrell & Harlacher, 2008).

Behavioral observation systems can be either event or interval based. In an event-based system, the observer simply counts the number of occurrences of a target behavior within a relatively long time frame. For example, the observer may count the number of times an excessively chatty 8-year-old child talks to a classmate during a 30-minute academic period during a school day. In an interval-based system, the observer divides the full time period into smaller intervals and then notes whether or not the target behavior took place during each interval. So the observer may break the 30-minute time period into 30 intervals, each 1 minute in length, and determine the number of minutes out of 30 in which the child talks to a classmate. Either type of system allows for the clinical child psychologist to measure the target behavior empirically. This direct, systematic approach to assessing the problem can prove to be an important complement to indirect data solicited from children, parents, teachers, or others.

One concern regarding direct observation of behavior involves **reactivity**. That is, children's behavior may change simply because of their awareness of the presence of the observer. When reactivity is strong, the observation may be invalid. Thus, clinical child psychologists often wait to begin recording their observations until it seems that the children have become accustomed to the presence of the observer. The reactivity concern can be especially relevant in situations where the observer appears to be of a different cultural background than those in the room where the observation is taking place. Merrell (2008) offers the example of a clinical psychologist observing a child in a classroom in which all the students are of a different ethnicity than the psychologist.

When **naturalistic direct observation**—observation of a behavior in the place where it actually happens—is impractical, clinical child psychologists often conduct **analogue direct observation**. Analogue direct observation typically takes place in the clinic room, where the real-life situation is simulated. By definition, analogue direct observation may not perfectly match the real-world situation in which the problem behavior arises. Nonetheless, it can be an important addition to secondhand descriptions of the problem behavior.

Behavior Rating Scales

Behavior rating scales are standardized pencil-and-paper forms that parents, teachers, or other adults complete regarding a child's presenting problems. They typically consist of a list of behaviors, each of which is followed by a range of responses from which the respondent chooses the one most applicable to the child. For example, a behavior rating scale might include an item such as, "The child pushes other children," followed by a 5-point scale consisting of *very frequently*, *frequently*, *sometimes*, *infrequently*, and *very infrequently*. By summing scores across subscales and the full scale, clinical child psychologists can obtain objective data to be compared with published norms for children in the appropriate age range. Among the most commonly used behavior rating scales are parent and teacher versions of the Child Behavior Checklist (part of the Achenbach System of Empirically Based Assessment), for a wide range of problem behaviors; the Behavior Assessment System for Children, also for a wide range of problem behaviors; Conners' Rating Scale, for attention-related problems; and Children's Depression Inventory, for childhood depression (Achenbach, 2009a, 2015a, 2015b; Bérubé & Achenbach, 2007; Kamphaus, 2015; Kovacs, 2015; Merrell & Harlacher, 2008).

Advantages of behavior rating scales include their convenience, inexpensiveness, and objectivity. Disadvantages include the fact that they restrict respondents from elaborating on their responses and the possibility that the scale items do not adequately capture the child's problem behaviors.

Self-Report Scales

Just as assessment of adults often involves questionnaires completed directly by the client, assessment of children and adolescents often does as well. Of course, the use of these measures assumes that the client's reading level, attention span, and motivation to complete the test are appropriate. For these reasons, **self-report scales** are more commonly used with adolescents than with younger children (Cashel, 2002).

Self-report scales for children and adolescents are quite similar to self-report scales for adults. In fact, some of the most commonly used adult self-report scales, including the Minnesota Multiphasic Personality Inventory (MMPI), the Personality Assessment Inventory (PAI), and the Millon Clinical Multiaxial Inventory (MCMI), are available in adolescent versions. These are pencil-and-paper tests in which the adolescents read a series of statements and then mark the response (true/false or one of several choices on a continuum) that best describes them. Scores are derived by comparing the adolescent's patterns of scores to normative data based on age range. The adolescent versions of the MMPI, PAI, and MCMI are relatively long, broad-based measures of personality, but other self-report scales used with adolescents are more brief and targeted toward particular symptoms such as depression and anxiety.

As with adults, behavior rating scales and self-report scales to assess children or adolescents must be used in a culturally competent way. For example, the clinical child psychologist should ensure that the person completing the scale is fluent in its language and that the client's cultural background is represented in normative data with which the client's responses will be compared (Merrell, 2008).

Projective/Expressive Techniques

The projective and expressive techniques used in the assessment of children include some of the same tests used with adults, such as the Rorschach Inkblot Method, the Thematic Apperception Test (TAT), and sentence-completion techniques (covered in Chapter 10), as well as others that are more exclusive to children. Our discussion in Chapter 10 regarding the projective hypothesis and concerns about the reliability and validity of projective tests applies to their use with children as well.

The Children's Apperception Test is an adaptation of the TAT storytelling test that features animal rather than human characters, about which young clients are asked to tell a story. A more up-to-date and culturally sensitive alternative to this test is the Roberts Apperception Test (2nd edition), in which the characters are children of varied ethnic backgrounds engaged in various common interactions. Also current and culturally relevant is the Tell-Me-a-Story, or TEMAS, technique (Costantino, Dana, & Malgady, 2007; Costantino & Malgady, 1999; Costantino, Malgady, & Rogler, 1988; Teglasi, 2010). The TEMAS was specifically designed as a culturally sensitive alternative to the TAT. Clinicians can choose between full-color cards that depict nonminority, Latino/a, or African American individuals in a variety of situations, about which children (ages 5–18)

are asked to tell a story. The cards of the TEMAS incorporate a lower degree of ambiguity than the TAT, such that each card centers on a particular theme, setting, or issue. Normative data for the TEMAS is culture specific, with norms organized according to age, gender, and specific ethnicity (R. Flanagan, Costantino, Cardalda, & Costantino, 2008; Malgady & Colon-Malgady, 2008; Teglasi, 2010).

In addition to projective tests such as these in which clients respond with words, this category of children's tests includes some expressive tests in which clients respond with drawings. These drawings, when accurately interpreted, are believed to communicate important information about clients' personalities. In one such test, the Draw-a-Person test, the client is given blank paper and is simply instructed to draw a whole person. Similarly, in the Kinetic Family Drawing technique, the drawing consists of the client's family engaged in some activity, and the House-Tree-Person test requires a drawing of the three items listed in its title. Clinical child psychologists use the drawings produced by tests of this kind to develop hypotheses about the psychological functioning of the client. As in most projective or expressive techniques, drawing techniques tend to rely on inference (rather than empirically based processes) during the interpretation process, a tendency that brings their reliability and validity into question. Also, interpretation of drawing tests should take culture into account. Drawings that children create can reflect cultural as well as individual issues. For example, Esquivel, Oades-Sese, and Olitzky (2008) point out that Japanese children completing tests in which they draw human figures might be influenced by the popularity of anime and manga animation—which typically features unusually large eyes, among other common characteristics—among Japanese youth, and the relative infrequency of smiles within Japanese culture, which stems from a cultural norm of restraint rather than open expression of emotion.

Intellectual Tests

Intellectual tests for children generally consist of IQ tests and achievement tests, and they stand apart from the other methods described above. Whereas the methods described above assess behavioral and emotional functioning, IQ and achievement tests assess intellectual functioning. In a full psychological evaluation, a clinical child psychologist may use methods from all these categories, but in some common types of evaluations such as evaluations for specific learning disorder, the emphasis may be almost exclusively on intellectual tests. Both IQ and achievement tests for children as well as adults are described in detail in Chapter 9.

Among specific intellectual tests, the Spanish version of the Wechsler Intelligence Scale for Children—Fourth Edition (WISC-IV) has been noted as a significant improvement over previous editions of the WISC and a generally strong choice for Spanish-speaking youth (Park-Taylor et al., 2010). It was normed on children whose parents were from a variety of countries where Spanish is spoken predominantly, including Mexico, Cuba, the Dominican Republic, Puerto Rico, and others in Central America and South America. Of course, a drawback of this and many other translated tests is that although items were translated, content did not change. Tests that have culture-specific content or tests that aim to be culture fair by reducing or removing the impact of language—such as the Universal Nonverbal Intelligence Test (UNIT), described in detail in Chapter 9—can also be considered for clients with diverse cultural and linguistic backgrounds.

Web Link 17.2
Site for WISC-IV

The Frequency of Use of Specific Assessment Techniques

A 2002 survey of psychologists who specialize in working with children and adolescents measured the frequency with which specific tests were used during assessment procedures (Cashel, 2002). Respondents were American Psychological Association members who belonged to Division 53 (Clinical Child Psychology) or other divisions that similarly focus on children and assessment issues. Results indicated that interviews were used far more commonly than any specific assessment technique. Among noninterview techniques, clinicians reported using a somewhat different set of instruments with adolescents than with younger children, with greater reliance on self-report measures for adolescents and greater reliance on behavior rating scales (completed by parents or teachers) for younger children. Overall, it appears that the use of some tests (such as the WISC) is relatively common, but a significant amount of variation remains among clinicians. Tables 17.1 and 17.2 summarize the results of this survey. Although this survey is becoming dated and is in need of replication, more recent studies that have at least partially focused on the frequency of test usage with child clients suggest that many of its findings remain intact. For example, in a 2014 survey of specific assessment techniques taught in clinical psychology graduate programs, the WISC, Child Behavior Checklist

Table 17.1 Usage of Specific Assessment Techniques in Child Assessment

Assessment Technique	Percentage of Child Clinicians Who Use Technique	Type of Technique
Wechsler Intelligence Scale for Children	75	Intelligence test
Child Behavior Checklist	65	Behavior rating scale
Conners' Parent and Teacher Rating Scale	65	Behavior rating scale
Vineland Adaptive Behavior Scales	63	Semistructured interview and behavior rating scale
Sentence Completion	57	Projective/expressive
Draw-a-Person	56	Projective/expressive
Bender-Gestalt	56	Neuropsychological screen
Beery VMI	56	Neuropsychological screen
Peabody Picture Vocabulary Test	53	Intelligence test
Wechsler Individual Achievement Test	53	Achievement test

Source: Cashel (2002).

Table 17.2 Usage of Specific Assessment Techniques in Adolescent Assessment

Assessment Technique	Percentage of Child Clinicians Who Use Technique	Type of Technique
Wechsler Intelligence Scale for Children	70	Intelligence test
Sentence Completion	60	Projective/expressive
Child Behavior Checklist	53	Behavior rating scale
Rorschach Inkblot Method	48	Projective/expressive
Conners' Parent and Teacher Rating Scale	48	Behavior rating scale
Beck Depression Inventory	48	Self-report scale
Wechsler Individual Achievement Test	48	Achievement test
Minnesota Multiphasic Personality Inventory-Adolescent	46	Self-report scale
Child Behavior Checklist-Teacher Report Form	44	Behavior rating scale
Thematic Apperception Test	43	Projective/expressive
Youth Self-Report	43	Self-report scale

Source: Cashel (2002).

(ASEBA), and WIAT were among the child-focused tests highest on the list (Ready & Veague, 2014). Similarly, the WISC and Child Behavior Checklist (ASEBA) were among the child-focused tests highest on the list of a 2017 survey of tests used by practicing psychologists who conduct assessment (Wright et al., 2017).

PSYCHOTHERAPY WITH CHILDREN AND ADOLESCENTS

Although therapy with children and adolescents may look or sound quite different from therapy with adults, the techniques used by both types of therapists often originate from the same underlying theories. That is, the major approaches to therapy, including psychodynamic, humanistic, behavioral, and cognitive, have generated applications for children as well as adults.

That being said, children should never be mistaken for miniature adults, so therapy with children often necessitates some substantial adjustments from the adult model (André, Garcia, & Coyne, 2015). The assumptions in adult therapy include the client's willingness to be in therapy, motivation to change, and ability to sit relatively calmly and use words to describe experiences or express feelings for prolonged periods of time. When working with children and adolescents, none of these can be taken for granted. Moreover, when working with an adult, the client is typically the only person involved in the sessions. Children, however, don't come to therapy alone. Parents, relatives, teachers, or others may be involved, and it is preferable to keep them involved as allies in treatment. In fact, the therapeutic alliance is just as crucial in therapy with children and adolescents as it is in therapy with adults (Karver, De Nadai, Monahan, & Shirk, 2018; Shirk, Karver, & Brown, 2011), but in this case the alliance must include multiple parties—the child *and* the parent(s)—rather than just the client, as in adult therapy.

Web Link 17.3
Journal of Clinical Child & Adolescent Psychology

A full-fledged review of all psychotherapy methods for children and adolescents is beyond the scope of this chapter. In fact, Kazdin (2000) lists 551 distinct psychotherapy techniques for children and adolescents and considers this number a possible underestimate! Instead, we will sample various forms of child therapy being actively practiced today by clinical psychologists. Then we will review the empirical outcome literature on psychotherapy for children and adolescents to gain a current understanding of "what works."

Of course, with any form of therapy for kids (and clients of any age), cultural competence is essential. Elsewhere in this chapter and throughout this book, we have touched on many of the cultural variables that could be relevant when working with children or adolescents, including race, ethnicity, gender, sexual orientation, socioeconomic status, religion, and more. Some experts have argued that childhood and adolescence is itself a culture—with its own set of beliefs, customs, and ways of life—to which psychologists should be sensitive. A psychologist who is oblivious to this "youth culture," or the particular variation of youth culture in which their client lives, could jeopardize the therapeutic alliance and, ultimately, therapy outcome just as they would if they were oblivious to culture based on ethnicity, race, gender, or any other variable. The culture of children and adolescents may include many elements, but the use of technology, especially social media, may be among the most important. One study found that adolescent therapy clients' perceptions of their therapists' social media competency strongly correlated with the clients' rating of the therapeutic alliance (Pagnotta, Blumberg, Ponterotto, & Alvord, 2018). Specifically, 77 teens in therapy rated their therapists on such items as "My therapist values and respects my social media use," "My therapist is comfortable with different social media habits that I have," "My therapist demonstrates knowledge about the social media platforms I use," and "My therapist attempts to understand my problems within the context of my social media habits" (p. 341). The higher the ratings on those items, the higher the teens' ratings of the therapeutic alliance, which is a key predictor of therapy outcome.

Cognitive-Behavioral Therapies for Children

Cognitive-behavioral therapies for children are undoubtedly on the rise (James & Roberts, 2009; Ollendick & Shirk, 2011). Much more than any other kind of child

therapy, they represent the movement toward evidence-based treatment and reliance on empirical data; in other words, they have accumulated evidence that they work (Chorpita et al., 2011; Higa-McMillan et al., 2016; Warwick et al., 2017; Zhou et al., 2015). These therapies involve either behavioral or cognitive techniques, and, in many cases, techniques of both types are utilized together.

In most cases, this empirical evidence was obtained in psychotherapy outcome studies in which a particular, manual-based treatment was applied to a specific disorder. Although cognitive-behavioral treatments have been found beneficial for a variety of disorders in children, including depression and ADHD (J. F. Curry & Becker, 2008; Hoza, Kaiser, & Hurt, 2008; Sibley, Kuriyan, Evans, Waxmonsky, & Smith, 2014), they are most strongly supported for anxiety disorders. Indeed, there are specific, empirically supported treatments for children with obsessive-compulsive disorder (OCD), panic disorder, phobia (including school phobia/school refusal), and social phobia, among others (Higa-McMillan et al., 2016; Maynard et al., 2018; Ollendick & Pincus, 2008; W. K. Silverman & Pina, 2008; Storch et al., 2008; Warwick et al., 2017). In general, these child treatments rely on the same principles as their adult counterparts. For example, treatment of phobias in children centers on gradual exposure to the feared object or situation (often augmented by relaxation training); treatment of OCD in children centers on combining exposure with response prevention (prohibiting clients from the compulsive behavior, such as hand washing, that brings them temporary relief from anxiety-causing obsessive thoughts); and treatment of numerous childhood anxiety problems involves cognitive restructuring in which the child is taught to challenge illogical thoughts and replace them with more logical thoughts (Rapee, Schniering, & Hudson, 2009).

Of course, even if the principles are the same, the way these therapies are delivered to children often differs from the way they are delivered to adults. For example, cognitive-behavioral therapists working with children often adapt their interventions into games of various kinds (Knell & Dasari, 2011). D. B. Pincus, Chase, Chow, Weiner, and Pian (2011) describe several such game-based techniques, including "Bravery Bingo," in which a phobic child earns a token, to be placed on a Bingo board, for each successful exposure on the anxiety hierarchy. Pincus et al. also describe "Mr. OCD," in which kids practice cognitive restructuring by refuting a puppet (Mr. OCD) who exhibits flawed logic ("A monster's gonna get you tonight when you're sleeping") by offering more sound logical statements ("A monster has never gotten me before, and it's not gonna happen tonight either! There are no monsters in my room!"). Likewise, when cognitive-behavioral therapists assign homework to their child clients, they often use a much more deliberate reinforcement system to ensure that the homework gets done. Depending on the age and interests of the child (and the preferences of the parent[s]), they may use stickers, candy, privileges, or simply praise to encourage the child to complete assigned between-session tasks.

Two specific forms of treatment lean heavily toward the behavioral side of the cognitive-behavioral approach. One is *social skills training*, in which therapists teach kids behaviors that improve their interactions with others. Social skills training works well for kids with a wide range of problems, but it is used most often with kids who have autism spectrum disorder, social anxiety disorder, and ADHD. Many skills are targeted, including how to start a conversation, how to join other kids who are already interacting, how to express feelings appropriately, how to handle frustration, how to manage eye contact

and other nonverbals, how to manage volume and tone of voice, and more. The skills are modeled by the therapist and rehearsed during sessions, and successes are reinforced with rewards that are meaningful to the child. Social skills training tends to work best when children receive not only behavioral instruction but also an explanation for why that instruction will work and how it will benefit them in the long run (André et al., 2015; Block, Radley, Jenson, Clark, & O'Neill, 2015; Herbert et al., 2005; Laugeson, Gantman, Kapp, Orenski, Ellingsen, 2015; Spence, 2003).

A second behavioral treatment is *applied behavior analysis* (ABA), which relies heavily on operant conditioning principles like reinforcement, punishment, shaping, and extinction (W. W. Fisher & Piazza, 2015). ABA is an evidence-based therapy for children with autism spectrum disorder, and it is the most commonly utilized and recommended therapy for that disorder (Christon, Arnold, & Myers, 2015; T. Smith & Iadarola, 2015). An ABA therapist helps a child by first identifying and defining very specific behaviors to target. The goal can be an increase in frequency, such as a child making direct eye contact, getting dressed appropriately, or using the toilet alone. Or the goal can be a decrease in frequency, such as a child injuring others or himself/herself. A baseline of frequency is measured, and a frequency goal is set. Additionally, a functional analysis is performed so the therapist can better understand any contingencies that are influencing the behavior before ABA begins. Then, in the child's environment and at the child's pace, the behavior is subjected to consequences through contingency management. That is, the child receives rewards or punishments that are effective in changing the behavior in the desired direction. Progress is continuously and objectively measured. Emphasis is placed on changing behavior in a way that will generalize to new environments rather than being specific to the environment in which it was taught (Bailey & Burch, 2006, 2010; W. W. Fisher, Piazza, & Roane, 2011).

Self-Instructional Training

Donald Meichenbaum (1977, 1985, 2008) originally developed **self-instructional training** as a method for impulsive and disruptive children to gain greater control over their behavior (see also Meichenbaum & Goodman, 1971). It is still used with that population, but its use has expanded to many other problems of childhood (and adulthood) as well (e.g., Grace, Spirito, Finch, & Ott, 1993). Although it has also been called "guided self-dialogue," *self-instructional training* is the more common term.

Self-instructional training is essentially a form of cognitive therapy in which children are taught to "talk through" situations in which their behavior might be problematic to increase the likelihood that they will use a preferred behavior instead. The process involves a sequence of steps by which children hear instructions aloud and gradually incorporate those instructions into their own thinking. Once a target behavior or situation is identified, the therapist begins the process by modeling the appropriate behavior. The key to the therapist's modeling behavior is that as the child observes, the therapist guides himself or herself through the behavior by talking out loud. Next, the child tries the behavior, with the therapist again saying the instructions aloud. Once this is mastered, the child says the instructions aloud to himself or herself while completing the behavior. Then, the child whispers instructions to himself or herself, and, ultimately, the

child completes the behavior silently with only unspoken self-instructions. According to some developmental theorists (e.g., Luria, 1961), this sequence parallels the way children naturally and gradually take instruction from others, such as parents, and incorporate these instructions into their own thinking.

Self-instructional training is "designed to nurture a problem-solving attitude and to engender specific cognitive strategies that clients can use at various phases of their stress response" (Meichenbaum, 1985, p. 69). Rather than changing "cognitions" per se, therapists who use self-instructional training introduce new "self-statements" to their clients. That is, when a child faces a challenge and either lacks problem-solving skills or makes maladaptive self-statements such as, "I can't handle this" or "I have no idea what to do" or "I'm gonna get in trouble," the likelihood of an appropriate behavior is low. If those absent or negative self-statements could be replaced with more constructive, helpful self-statements that steer the child through the necessary steps, preferred behavior is much more likely to occur. Meichenbaum argues that constructive self-statements can be helpful at various points in time: in preparation for the stressor; in the midst of the stressful moment; and afterward, when reflecting on how the stressor was handled. During the training process, the child may need to say self-statements aloud, but with practice, the self-statements need be only thought rather than spoken.

As an example, consider Nicholas, a hyperactive 7-year-old second grader. Among the many behavioral problems he exhibits at school, his teachers describe transitions from one room to another as one of the most disruptive. When it is time for Nicholas's class to end a classroom activity, get up from their desks, and line up at the door to go to lunch, physical education, an assembly, or recess, Nicholas's tendency is to run excitedly around the room talking to his peers, which creates chaos and prolongs the transition for the whole class. The preferred behavior would be for Nicholas to put away his things and line up quietly at the classroom door, as he has been instructed many times. With Dr. Sanders, a clinical child psychologist, Nicholas works on improving this problem via self-instructional training. They reenact the classroom situation in Dr. Sanders's office, and at first, Dr. Sanders pretends to be the second grader. When it's time to go to the physical education class, Dr. Sanders says aloud to himself,

> Now, what am I supposed to do? Oh, I know, I'm supposed to put my papers and books and pencils in my desk, quietly walk over to the door, and line up with my friends. I can totally do that. No problem. Here I go—putting away all my stuff. Not running around or making noise. I'm doing good so far. Now I'm getting up, walking over to the door, still staying calm. Now I'm by the door, lined up, waiting quietly 'til the teacher says to go. All right! I knew I could do it!

Next, Nicholas tries the behavior himself, with Dr. Sanders talking him through it. Ultimately, Nicholas is talking himself through it, then whispering himself through it, and then silently instructing himself through it. After repeated practice in Dr. Sanders's office, Nicholas is performing this behavior with regularity in the classroom. Other problem behaviors were similarly addressed, and with each success, Nicholas grew more confident and capable of developing his own helpful self-instructional strategies and behaving in a nondisruptive way.

BOX 17.4

Metaphorically Speaking

If You've Had Dancing Lessons, You Understand Self-Instructional Training

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"Slow, slow, quick-quick; slow, slow, quick-quick." On the first day of dance class, these are the instructions that a dance teacher says aloud to students learning the fox trot. Although the rhythm might differ for the samba, the waltz, the mambo, or the swing, the basic teaching strategy is the same. At the outset of the lessons, the teacher talks the students through the dance by calling the steps out loud and performs the dance as a model. As the dancers begin to develop proficiency, the teacher becomes less vocal, and the dancers talk themselves through the steps. With more practice, their self-talk wanes to a whisper and then fades altogether, unnecessary because the steps have been incorporated into the dancers' thought processes.

Self-instructional training (Meichenbaum, 1977, 1985, 2008; Meichenbaum & Goodman, 1971) is very much like dance lessons. Rather than selecting a particular dance, the therapist and client (with input from parents and teachers) select a particular behavior on which to focus. The therapist then begins the process, much like a dance teacher on the first day of class, by modeling the behavior while saying aloud the steps that must be taken to complete the behavior effectively. Then the child attempts the behavior while the therapist calls out the steps. Next, the child attempts the behavior while making the self-statements aloud, followed by the self-statements being spoken under the client's breath, and, ultimately, only in the client's mind. All the while, the therapist offers feedback and encouragement, much like a dance teacher watching the class gradually learn to complete the dance steps less deliberately and more naturally.

Sometimes, a student enters a dance class with the firmly held belief, "I can't dance." Similarly, children often face difficult situations with firmly held self-statements such as, "I can't do this," "I'm gonna get in trouble," and so on. In both cases, the self-directed message is probably inaccurate and definitely maladaptive—in other words, it gets in the way of accomplishing the goal. Just as the dance teacher would work to revise the belief of the self-critical dance student, the therapist in self-instructional training works to replace the child's maladaptive self-statement with a series of constructive, encouraging self-statements that facilitate the successful completion of the target behavior.

Self-instructional training can be considered an example of the broader category of *problem-solving strategies* in cognitive-behavioral therapy (CBT) for kids. In other versions of problem solving, kids learn a sequence of steps that have already been created for the type of problem the child has (rather than entirely customizing one for each child), and often that sequence of steps follows an acronym that makes it easy for kids to remember. For example, a treatment program called Keeping Your Cool for kids with aggressive behavior problems uses the SPEAR acronym: S for stop (to identify the problem), P for plan (to consider solutions), E for evaluate (to choose one solution), A for act (to try the solution), and R for react (to determine if the solution worked) (Kendall et al., 2016; W. M. Nelson & Finch, 2008).

Besides problem solving, another emphasis in many CBT efforts for kids is *affective education*, in which the emphasis is the recognition, differentiation, and expression of emotions. For many kids, simply knowing what their feelings are and being able to share them can be major steps toward improvement. Even better is the experience of normalization—in other words, realizing (with help from the therapist) that everyone has feelings like theirs. One program that focuses on education is Coping Cat, in which kids with anxiety-related problems use a variety of techniques, including the creation of their own feelings dictionary and the recognition of emotions in the faces of other people in photos (Beidas, Benjamin, Puleo, Edmunds, & Kendall, 2010; Kendall, 1994; Kendall et al., 2016; Lenz, 2015; Podell, Mychailyszyn, Edmunds, Puleo, & Kendall, 2010).

Parent Training

Parent training is a form of behavioral therapy in which therapists teach parents to use techniques based on conditioning to modify problematic behavior in their children (R. H. Jackson & Leonetti, 2002). The parent training approach to children's behavior problems "utilizes parents as the primary agents of change for their children. It is, after all, the parents who construct and manage the child's environment" (Briesmeister & Schaefer, 1998, p. 1).

Like any form of behavior therapy, parent training begins with a clear, measurable definition of the child's problem behavior. Then, together with the parents, the therapist conducts a functional analysis to try to understand the contingencies that maintain the behavior. Next, again with the parents, the therapist explores different consequences for the behavior. Parents are instructed on contingencies to establish and how to communicate them to the child. Often, the therapist exemplifies the behavior for the parents, providing a model for a behavior that may not have been in their repertoire previously. Parents then apply the new contingency at home and report back to the therapist about the child's behavior using objective, quantifiable outcome measures. As the therapist educates the parents, many aspects of conditioning may be discussed, including reinforcement, punishment, extinction, generalization, discrimination, schedules of reinforcement (e.g., fixed or variable), and other concepts. Parent training programs have been developed for a growing number of childhood problems, including ADHD, conduct disorder, separation anxiety, school refusal, sleep/wake problems, and others. In some

cases, parent training programs have been employed on a large scale, such as the use of the Triple P–Positive Parenting Program and the Incredible Years training series in Head Start and other communitywide or government-funded agencies (Menting, de Castro, & Matthys, 2013; Sanders, Kirby, Tellegen, & Day, 2014). At the end of any such program of treatment, successful parent training yields not only improved behavior in the child but more competent and confident parents as well (Briesmeister & Schaefer, 1998).

As an example, consider Patty, a single mother of 3-year-old Miranda and 6-year-old Joy. Patty seeks the help of Dr. Fisk, a clinical psychologist, because Miranda has developed the habit of physically assaulting her big sister. She hasn't caused any real physical damage yet, but Miranda hits, kicks, and bites Joy on a daily basis. Usually, these actions are unprovoked, and sometimes the girls aren't even interacting before it happens. Dr. Fisk asks Patty what happens after Miranda acts aggressively toward her sister. Patty says that she takes Miranda on her lap and talks with her about the incident, usually for about 5 minutes. Patty says that during these talks, her tone of voice is always kind and concerned rather than punitive and that the 5-minute talks usually end with hugs and kisses and a promise from Miranda that she won't hurt her sister anymore. However, she inevitably does, at which point Patty repeats the same routine. In response to a question from Dr. Fisk, Patty mentioned that one-on-one talks with hugs and kisses happen very rarely other than in these postaggression moments. Dr. Fisk hypothesized that Patty's response to Miranda's behavior was actually reinforcing her aggressive behavior—Miranda's "payoff" was the hugs, kisses, and attention from Mommy. Dr. Fisk taught Patty to introduce new contingencies: A full day without any aggressive behavior earned Miranda a 5-minute snuggle at bedtime, and any aggressive behavior not only cost Miranda the 5-minute bedtime snuggle but also resulted in an immediate 3-minute timeout. If Miranda disobeyed the timeout, additional punishments, such as loss of dessert and TV privileges, were also systematically introduced. Besides offering these instructions to Patty, Dr. Fisk explained the behavioral concepts underlying them, including the likelihood of an extinction burst, in which Miranda's behavior might briefly intensify before it ultimately extinguished. With a full understanding of the reasoning behind it, Patty put the new contingency plan into effect and stuck with it, and within a short time Miranda's aggressive behavior was greatly reduced.

Play Therapy

Play therapy is a form of treatment unique to child clients. Typically used with younger kids (preschool or elementary-school age), it allows children to communicate via actions with objects such as dollhouses, action figures, and toy animals rather than words. Through play, children can reveal to clinical psychologists their emotional concerns and attempts to resolve them. Brems (2008) argues that play therapy has three basic functions: the formation of important relationships (i.e., the therapeutic relationship), disclosure of feelings and thoughts (e.g., expressing emotions, acting out anxieties), and healing (e.g., acquiring coping skills, experimenting with new behaviors). As it is practiced by clinical psychologists, play therapy can be completely unstructured, with the child given

free rein in the playroom to determine the course of the session, or structured to varying degrees, with the therapist instructing the child to play with certain objects or inviting the child to play a specific game (Cheung, 2006).

Of course, in any form of play therapy, cultural sensitivity is essential when furnishing the playroom with dolls, toys, and other objects for child clients. Children who find toys representative of and relevant to their own cultural backgrounds are more likely to engage in meaningful play and to find the play therapy experience rewarding. For example, Hinds (2005) encourages play therapists to offer African American children dolls with an array of authentic African American physical features, including variations in skin tone, hair length, and hair style. Similarly, Robles (2006) suggests that for Mexican American children, play therapists should include in the playroom traditional Mexican toys, including such games as *Juego de la Oca* (Game of the Goose) and *Serpientes y Escalaras* (Snakes and Ladders). Drewes (2005a) adds that any set of markers or crayons should include colors that represent multicultural skin tones and that any play kitchen should include multicultural items such as chopsticks or tortilla presses. Especially for children who may be struggling with issues of cultural acceptance, the presence of culture-specific objects in the playroom can itself help strengthen their ethnic and personal identities. In addition, Drewes (2005b) emphasizes that play therapists must be aware of various cultural norms that children may exhibit in the playroom in order to avoid misinterpretation; for example, across cultures, norms may differ in terms of personal space, eye contact, and the meaning of silence.

Meta-analyses of play therapy outcome studies suggest that they are generally effective (Lin & Bratton, 2015; S. A. Jensen, Biesen, & Graham, 2017). However, those meta-analyses also point out that the empirical quality of many play therapy outcome studies is questionable. Problems include the fact that play therapy outcome studies often lack therapy manuals, a focus on a specific diagnosis, or reliable and valid outcome measures. As a result, few if any play therapy outcome studies are truly randomized clinical trials, which makes it difficult for play therapies to attain the status of evidence-based treatments.

Play therapy has taken many forms, but two of the most commonly practiced stem from psychodynamic and humanistic theories.

Play Therapy—Psychodynamic

For psychodynamic clinical child psychologists, a child's play symbolically communicates important processes occurring within the child's mind, "revealing aspects of the child's internal life of which he or she may be unaware and unable to verbalize directly" (Marans, Dahl, & Schowalter, 2002, p. 388). So, as the child plays, the therapist is both a participant and an observer, interacting with the child but always monitoring the child's choices and actions. As in any form of psychodynamic therapy, the goal is to make the unconscious conscious, but with young children, it is assumed that they lack the verbal skills and attention span to converse directly with the therapist about their issues. Instead, the unconscious is conveyed through play, and the therapist's job is to infer these unconscious issues and, through interpretations, to make the child more consciously aware of them.

In **psychodynamic play therapy**, the playroom usually contains a variety of objects with which the child can choose to play. Psychodynamic play therapists may take note of

the objects the child chooses, especially if he or she appears to identify strongly with a particular toy. For example, a child who chooses a damaged doll and uses it as the primary character in make-believe adventures may have a damaged sense of self-esteem; a child who, every session, insists on using only the biggest, strongest truck in the box may be struggling with issues related to power. Of course, once certain objects are chosen, the therapist pays close attention to the way the child uses them, with the actions and themes that manifest in the child's play representing latent unconscious issues. The interpretations the therapist offers the child about their actions during play can help make the child aware of inner mental processes and increase the child's ability to make deliberate choices about behavior in the future (Marans et al., 2002; O'Connor, 2000).

As an example, consider 5-year-old Cassandra, an only child whose parents separated about 2 months ago. Cassandra currently lives with her mother, Tisha, and sees her father, Marquis, about once a week. Tisha brings Cassandra to Dr. Ball, a psychodynamic play therapist, because she is concerned about how the separation is affecting her. Tisha explains that she hasn't seen any "acting out" behavior, but Cassandra seems more withdrawn, more irritable, and less happy than she usually is. In an early play therapy session, Cassandra builds a small tower with blocks for a few minutes and, seemingly bored with the blocks, wanders over to the dolls. She chooses a doll and returns to the tower she had built. With her free hand, Cassandra slowly begins to knock the top blocks off the tower. At the same time, Cassandra puts the doll near the tower but not quite close enough to stop her free hand from knocking the tower over. She repeats this action in which her free hand destroys the tower while the doll is just unable to prevent it until the tower is completely leveled. All the while, she is narrating in a high-pitched, soft voice: "Oh no . . . save the tower . . . you have to do something . . . don't let the whole thing fall . . . do something." After the tower is destroyed, Cassandra's voice turns harsh and loud as she turns the doll to face her and says, "You are a bad girl! How could you let this happen? It's all your fault that the tower is wrecked!" Dr. Ball makes two inferences from this episode: Cassandra is very concerned about her parents' marriage (as represented by the tower) collapsing, and Cassandra (as represented by the doll) feels responsible yet powerless to safeguard it. The fact that Cassandra re-creates similar scenarios in subsequent sessions confirms Dr. Ball's thoughts. Over time, Dr. Ball talks about the episodes with Cassandra—what the doll is feeling, what her job is, whether it's fair to assign that job to her. Dr. Ball is eventually able to help Cassandra make the conscious connection between the doll's experience and her own, so Cassandra can consciously consider the questions as applied to herself—what Cassandra is feeling, what her job is, and whether that job is fair. Ultimately, Cassandra begins to accept that it's very normal to be sad, angry, or nervous about her parents' relationship but that blaming herself for its demise or expecting herself to protect it are unfair burdens. With this pressure lifted, Cassandra's mood improves despite her difficult situation.

Play Therapy—Humanistic

Humanistic play therapy involves many of the same activities, materials, and assumptions as psychodynamic play therapy does. Children play with objects in the playroom, therapists participate and observe, and the underlying assumption is that the activities and themes in the play express the inner workings of the child's mind. However, there are several key distinctions between humanistic play therapy (sometimes called

"child-centered" play therapy) and psychodynamic play therapy (O'Connor, 2000). Although interpretations are a key element of psychoanalytic play therapy, humanistic play therapists rarely offer interpretations, because they do not share the psychodynamic therapists' primary goal of making the unconscious conscious. Rather than interpreting clients' behavior, humanistic play therapists tend to reflect their clients' feelings, which may be expressed indirectly through their play activities. This reflection is an important part of the overall goal of humanistic play therapy, which is the same as humanistic therapy for adults—to facilitate self-actualization. This self-actualization happens in the context of an all-important therapeutic relationship based on true unconditional acceptance and genuine empathy. So as humanistic play therapists observe children and play along with them, they establish relationships in which the children are accepted wholly for who they are. This unconditional positive regard from the therapist is thought to bring about unconditional positive self-regard within the child. When therapists reflect children's feelings in a "real" and nonjudgmental way, children themselves become more accepting of their own emotional experience, which enhances congruence and, ultimately, self-growth (Landreth, 1991; O'Connor, 2000).

If Cassandra, the 5-year-old girl described in the psychodynamic play therapy section above, were seen instead by a humanistic play therapist, the course of therapy might have been somewhat different. Most notably, the act of interpretation would probably be absent; that is, the therapist would not make deliberate efforts to make Cassandra aware of her own unconscious processes. Instead, a humanistic play therapist would focus on Cassandra's emotional experience during the session (expressed directly or through the doll), emphasizing an unqualified acceptance and compassion for any of Cassandra's feelings. With time, a therapeutic relationship characterized by such honest care would have internalizing effects on Cassandra; that is, her own recognition and acceptance of her emotional experience would become enhanced. As a result, she would experience greater congruence between her real and ideal selves, which would in turn facilitate her natural tendency toward growth and self-actualization.

How Well Does Psychotherapy for Children and Adolescents Work?

A significant body of empirical research suggests that, in general, the efficacy of psychotherapy for children and adolescents is quite strong. In fact, the theme of such studies is quite similar to the theme of studies of therapy outcome for adults: Children and adolescents who undergo psychotherapy consistently demonstrate significant improvement in comparison with similar children who receive no treatment. Sometimes, in meta-analyses and reviews, various approaches to therapy are found to have roughly equal effects, but often, behavioral treatments are found to have a slight edge in efficacy over nonbehavioral approaches (Carr, 2008; D. L. Evans et al., 2005; Russ & Freedheim, 2002; Weisz, Weiss, Han, Granger, & Morton, 1995). A massive meta-analysis, spanning 50 years of psychotherapy that involved over 30,000 child and adolescent clients in more than 400 studies, concludes that psychotherapy for children and adolescents is generally effective. Specifically, it states that on average, a kid receiving treatment in a study has a 63% chance of a better outcome than a kid in the control condition; the best results are

generally for anxiety disorders; and, when considering outcome from the perspective of kids, parents, and teachers, CBT produced more consistently positive outcomes (Weisz et al., 2017).

With the general efficacy of child and adolescent psychotherapy firmly established, researchers have shifted their attention to more specific questions of particular forms of therapy for particular clinical problems (Carr, 2008; Chorpita et al., 2011; D. L. Evans et al., 2005). As mentioned earlier, the broad category of cognitive-behavioral therapies for children has demonstrated strong empirical efficacy with not only anxiety disorders but also a wide range of other presenting problems such as oppositional defiant disorder, depression, hyperactivity, enuresis (bedwetting), school refusal, and sleep disorders (Maynard et al., 2018; Ollendick & Shirk, 2011; Powers, 2002). Similarly, the broad category of play therapy has demonstrated benefits with a wide range of both externalizing and internalizing disorders (Bratton, Ray, Rhine, & Jones, 2005; Ray, 2006), but with some methodological shortcomings that are less common in studies of CBT. More specifically, a number of particular techniques, mostly behavioral in orientation, have proven successful in treating particular disorders, including these (Curry & Becker, 2008; Hoza et al., 2008; Kazdin, 2000; Nathan & Gorman, 2002, 2007; Rapee et al., 2009; Silverman & Pina, 2008; B. H. Smith, Barkley, & Shapiro, 2006; Storch et al., 2008):

- Systematic desensitization, modeling, and cognitive restructuring for phobia and other anxiety disorders
- Exposure and response prevention for OCD
- Parent-training techniques for oppositional defiant disorder and conduct disorder
- Parent training and classroom contingency management for ADHD
- Cognitive-behavioral techniques and interpersonal therapy for childhood depression

CHAPTER SUMMARY

Clinical child and adolescent psychologists work with a wide range of issues, including externalizing disorders (e.g., ADHD, conduct disorder, oppositional defiant disorder) and internalizing disorders (e.g., depression, anxiety). They also work in a preventive capacity to increase children's resilience and decrease their vulnerability to psychological problems. They incorporate a developmental perspective by remaining aware of clients' ages and developmental stages as a context for clinical issues. Assessments typically involve multiple sources of information, multiple settings, and multiple methods. Particular assessment techniques include interviews, behavioral observations, behavior rating scales, self-report scales, projective/expressive techniques, and intellectual tests. Therapy with children and adolescents takes on a wide variety of forms, many of which derive from the same basic orientations as adult forms of

psychotherapy. Cognitive-behavioral therapies for children have accumulated empirical support for numerous disorders, especially anxiety disorders and autism spectrum disorders. Self-instructional training and other problem-solving-based therapies teach children to "talk themselves through" difficult situations, parent training teaches parents to apply behavioral conditioning techniques to their children, and play therapy allows children to communicate and solve their problems via play with toys and other objects. In all aspects of their work, clinical child and adolescent psychologists maintain an appreciation of the cultural backgrounds of their clients and the families with which they live.

KEY TERMS AND NAMES

analogue direct observation 430
behavior rating scales 430
clinical child psychology 419
developmental perspective 423
externalizing disorders 420
humanistic play therapy 443
internalizing disorders 420
Donald Meichenbaum 437
multisource, multimethod, multisetting approach 426
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CRITICAL THINKING QUESTIONS

1. In your opinion, what are the most important factors in the development of resilience in children?
2. If you were a clinical child psychologist, how would you expect gender to influence the development of internalizing disorders and externalizing disorders?
3. In your opinion, how reliable and valid is the information children provide in clinical interviews? How reliable and valid is the information adults provide about children?
4. If you were a clinical psychologist, for what clinical problems would you expect self-instructional training to be most successful? Does the age of the child make a difference?
5. Under what circumstances would you expect parent training versus direct clinical intervention to be more successful in reducing a child's behavior problems?

LOOKING TOWARD GRADUATE PROGRAMS

The following table is a list of index entries in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018) relevant to the topics in this chapter (see page 25 in Chapter 1 for more information about the *Insider's Guide*). Keep in mind a few things as you consider these listings: They are current as of the 2018/2019 edition of the *Insider's Guide* (which typically revises every two years); they may have changed as faculty members retire, take new jobs, or develop new interests; and a graduate program's own website may have more extensive or up-to-date information.

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Adolescents/At-Risk Adolescents	Adolescents/At-Risk Adolescents/ Delinquency	Child & Family
Attention Deficit/Hyperactivity Disorder	Attention Deficit/Hyperactivity Disorder	Child Clinical/Pediatric
Autism/Asperger's Syndrome/ Developmental Disorders	Autism/Developmental Disabilities	Miscellaneous/Other—Parent and Infant Mental Health
Bullying	Child Abuse/Neglect	
Child Abuse/Neglect/Sexual Abuse	Child/Pediatric	
Child Clinical/Pediatric	Conduct/Disruptive Disorder	
Conduct/Disruptive Disorders	Developmental Disorders	
Learning Disabilities	Early Intervention	
Parent-Child Interaction/ Parenting	Learning Disabilities	
Psychopathology—Child/ Developmental	Parent-Child Interaction/Parent Training	
Miscellaneous and Other—Developmental Risk	Play Therapy	
Miscellaneous and Other—Play Therapy	Miscellaneous—Bullying	
	Miscellaneous—Child Bipolar Treatment	
	Miscellaneous—Early Childhood Services	
	Miscellaneous—Special Education	
	Miscellaneous—Youth and Family Agencies	

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

Behavior Analysis: Research and Practice
http://www.apa.org/pubs/journals/bar/index.aspx

Clinical Child and Family Psychology Review
http://link.springer.com/journal/10567

Clinician's Research Digest: Child and Adolescent Populations
http://www.apa.org/pubs/journals/crp/index.aspx

International Journal of Play Therapy
http://www.apa.org/pubs/journals/pla/index.aspx

Journal of Clinical Child & Adolescent Psychology
http://www.tandfonline.com/loi/hcap20#.V08VSvkrLX4

STUDENT STUDY SITE RESOURCES

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Visit the study site at **edge.sagepub/pomerantz5e** for these additional learning tools:

- Self-quizzes
- eFlashcards
- Culture expert interviews
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- Full-text SAGE journal articles
- Additional web resources
- Mock assessment data

18

HEALTH PSYCHOLOGY

Laura A. Pawlow and Andrew M. Pomerantz

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Health psychology is a relatively new subspecialty within clinical psychology. Its emergence corresponds with changes in the types of diseases that pose the greatest threat within our society. A century ago, those diseases were pneumonia, tuberculosis, and influenza, but today, they are diseases such as heart disease, cancer, and stroke. The newer killers have clear-cut behavioral components; they are worsened by smoking, poor eating habits, and lack of exercise. So it should be no surprise that the field of health psychology has surfaced in recent decades and has rapidly become popular among clinicians and researchers (L. T. Benjamin, 2005; T. W. Smith, Williams, & Ruiz, 2016).

The health psychology division of the American Psychological Association (APA Division 38) was founded in 1978, and today there are many journals dedicated to publishing research in this field (e.g., *Health Psychology*, *Journal of Behavioral Medicine*). Health psychology is primarily concerned with the ways our thoughts, feelings, and behaviors interact with our physical health. To become a competent health psychologist, a clinical psychologist needs training beyond the typical doctoral program requirements. This training can take the form of coursework or internships and often involves interaction with professionals from many health professions outside of psychology (S. L. Hanson & Kerkhoff, 2012).

Learning Objectives

18.1 Differentiate health psychology from behavioral medicine.

18.2 Summarize the impact of stress on physical well-being.

18.3 Describe interventions health psychologists use to treat patients experiencing stress.

18.4 Explain social support's impact on an individual's physical health.

18.5 Explore common areas of clinical focus in health psychology.

18.6 Debate the potential benefits and drawbacks of patient-centered medical homes.

18.7 Review the potential impact of cultural factors on the practice of health psychology.

Although the term *behavioral medicine* is often used interchangeably with the term *health psychology*, the two terms are not exactly synonymous. Before we explore the area of health psychology, let's start with some basic definitions.

DEFINITIONS: HEALTH PSYCHOLOGY VERSUS BEHAVIORAL MEDICINE

Health care professionals interested in the ways our minds influence our bodies are likely to be drawn to the broad, interdisciplinary field of **behavioral medicine**. Behavioral medicine involves the integration of knowledge from a wide variety of social sciences, including psychology, sociology, and anthropology, with knowledge from the medical disciplines. **Health psychology** is a subdiscipline of behavioral medicine that deals specifically with how *psychological* processes (e.g., cognitions, moods, social networks) interact with health and illness. A primary responsibility of health psychologists is the development, assessment, and application of programs designed to promote wellness; as such, they often work in hospitals or specialized clinics for such issues as weight management, smoking cessation, or pain control (Emery, Anderson, & Andersen, 2011; Lewis, Statt, & Marcus, 2011). They frequently work within primary-care teams, alongside physicians, nurses, dieticians, physical therapists, occupational therapists, and other health professionals (Goodheart & Rozensky, 2011; Kaholokula & Haynes, 2015). Health psychologists ask questions such as, "Why do students tend to get sick more often around exam week?" and "Will keeping a food diary help my clients lose weight?" Health psychologists might spend their days engaged in research or the application of such activities as promoting healthy behaviors (such as smoking reduction or cessation, abstinence from tanning beds, or the incorporation of more fruits and vegetables into the daily diet) and/or preventing illness through such means as teaching relaxation or coping skills.

Web Link 18.1
Website of APA Division 38

STRESS

Today's health psychologists have a wide selection of areas in which they can practice, but that was not always the case. The first applications of the field of health psychology dealt solely with the topic of **stress**: the psychological and/or physiological response to difficult or demanding internal or external circumstances (VandenBos, 2007). Although the terms

health psychology and *behavioral medicine* weren't coined until the 1970s, the idea that stress could negatively impact the body surfaced long ago. In 1932, Walter Cannon described the **fight-or-flight response**: When an organism perceives a threat, the body rapidly mobilizes energy reserves via the sympathetic nervous system and endocrine system to either fight or flee. For primitive humans, the body's behavioral responses were probably actually fighting or fleeing (perhaps from a snarling saber-toothed tiger), but today's "fighting" and "fleeing" look a bit different. Blaring your horn aggressively after being cut off in traffic and withdrawing from anxiety-provoking situations into self-medicating substances such as alcohol are modern variations of fighting and fleeing, respectively. More recently, some researchers have argued that the fight-or-flight strategy may not describe women's typical responses to stressful situations as well as it describes men's. These researchers argue that women may more often use a *tend-and-befriend* approach instead—one that emphasizes the benefits of strengthening (or tending to) family relationships and using social connections (befriending) to fight stress. This approach can produce a vital safety net for women and their families in times of stress (S. E. Taylor, 2006; S. E. Taylor et al., 2000; von Dawans, Ditzen, Trueg, Fischbacher, & Heinrichs, 2019).

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Photo 18.1 Stress, especially when chronic, can overwhelm our ability to adapt and can produce significant psychological and physical symptoms. Can you recall times when this has happened in your own life?

The duration of stressors can vary greatly. **Chronic stress** refers to the all-too-common syndrome defined by stress levels that are consistently high and unremitting, often due to a hectic, fast-paced lifestyle. Many of us have firsthand knowledge of chronic stress: Hamermesh and Lee (2007) found that 55% of adult American women and 43% of adult American men report feeling stressed "always" or "often." Unfortunately, our bodies are not set up to handle this long-term exposure to stress, and given a long enough period of time, we become exhausted and vulnerable to illness. This process was described by **Hans Selye** (1956) in his pioneering work on stress, which culminated in his **general adaptation syndrome** hypothesis: When confronted with a temporary stressor, our fight-or-flight system often works very effectively, but with repeated or prolonged exposure to stress, our bodies eventually wear out and break down.

STRESS AND PHYSICAL ILLNESS

Stress has been linked to myriad symptoms and diseases, including migraine headaches, osteoporosis, chronic back pain, cardiovascular disease, ulcers, diarrhea, acne, and fertility problems (Hafen, Karren, Frandsen, & Smith, 1996; Schneiderman & Siegel, 2007). The same bodily responses that originally provided us with the ability to save ourselves from an immediate danger have, in our society, become maladaptive conditions that lead to illness in many people. In fact, researchers at the American Institute of Stress (1997) estimate that 75% to 90% of all visits to health care providers result from stress-related disorders and that those who are stressed are six times more likely to be hospitalized for a physical ailment than those who are not (Kalia, 2002). Among other health problems, chronic stress can lead to

- increased levels of thyroid hormones, resulting in insomnia and weight loss;
- depleted endorphins, resulting in bodily pain;
- reduced sex hormones, leading to amenorrhea or infertility;
- the shutting down of the digestive system, resulting in nausea, bloating, and dry mouth; and
- an excess of cholesterol release that can contribute to blocked arteries, blood pressure increases of up to 400%, stroke, or aneurism. (Hafen et al., 1996)

In addition, one of the most debilitating effects of stress is the weakening of our immune system. Stress activates the **hypothalamic-pituitary-adrenal (HPA) axis**, which controls the release of our body's stress hormone, **cortisol**. Cortisol is beneficial in appropriate amounts; however, chronic stress leads to an increased level of cortisol, which leads to the deterioration of T-cells, essential cells in the immune system. T-cells both attack foreign pathogens and regulate other cells of the immune system. When dysregulated levels of cortisol lead to a decrease in T-cell functioning, the result is a weakened immune system. Or, more simply stated, chronic stress can make you sick. Sheldon Cohen's pioneering work on stress and the common cold found that those participants with the highest amounts of stress were more than five times as likely to become infected with the common cold virus (e.g., S. Cohen & Williamson, 1991). The study of this phenomenon of emotional stress setting the stage for physical illness is known as **psychoneuroimmunology (PNI)**. More specifically, PNI is the scientific field of study concerned with the interactions among behavior, the nervous system, and the immune system (Kropiunigg, 1993).

The deleterious effect of stress on the immune system has opened the door to research and clinical work on the psychology behind two common and deadly diseases: cancer and AIDS. Although it is important to note that there is absolutely no evidence that stress *causes* either of these diseases, there is much evidence to suggest that stress plays a role in the development and maintenance of both. For example, researchers found that being prone to stress was associated with the development of breast cancer (Faragher & Cooper, 1990). In a similar vein, one study of people with HIV reported that those under considerable stress had a disease progression with a significantly more rapid course (Lesserman et al., 1999). In partnership with other health professionals, clinical psychologists can play an important role in treatment or prevention of these illnesses.

STRESS AND COPING

We all know people who seem to be immune to even high levels of stress and others who respond to even minor stressors with psychological or physical ailments. A major difference between these types of people is the way they have learned to cope with their stress. **Coping** has been defined as the process of managing demands that are appraised as exceeding the resources of the person (Lazarus & Folkman, 1984).

What factors determine how well someone copes with stress? Like the answers to many questions in psychology, this answer involves both nature and nurture. On the

nature side, it appears that our genetic codes can be imprinted with susceptibility for struggling with stress. For example, one study involving 44 identical and 30 fraternal twin pairs suggested that genetic factors accounted for the majority of the variance in 57% of the coping variables studied (Mellins, Gatz, & Baker, 1996). It appears that this biological vulnerability to stress manifests itself in a tendency to overreact or become depressed in response to stress. In fact, researchers have even hypothesized that there is a genotypic inclination toward a **disease-prone personality** that predisposes people to stress-related illnesses such as arthritis, ulcers, and coronary heart disease (Scheier & Bridges, 1995; T. W. Smith & MacKenzie, 2006). The personality profile in question is one of marked anxiety, depression, pessimism, and defensive hostility (Guerrero & Palmero, 2010).

Yet studies such as the one described in the previous paragraph (Mellins et al., 1996) clearly show that genetics account for only about half the story. What nongenetic factors affect how we cope with stress? One important factor is the individual's *perception* of stress, which may or may not be a realistic appraisal. Indeed, the perception of stress (as opposed to the actual amount or degree of stress) can predict levels of stress-related illness (Hiramoto et al., 1999). In a study of women with breast cancer, the amount of distress—both psychological and physical—resulting from the disease was significantly related to the amount of control the women felt they had. Those who believed they had more control had significantly less distress than did those with perceptions of less control (Barez, Blasco, Fernandez-Castro, & Viladrich, 2009). So, whereas health psychologists can do little or nothing about a client's genetics, they commonly work with clients to challenge cognitive misperceptions and replace them with healthier points of view.

A common misperception involves the feeling of being completely devoid of control when faced with stress. However, when clients are challenged to look more carefully at the situation, they will often realize there are some aspects of the situation that are within their control. As an example, consider Max, a 64-year-old military veteran who for many years regularly attended a chronic pain clinic at his local Veterans Affairs hospital. Max benefited greatly from the services he received at the clinic, which had become an integral part of his pain management plan. On learning that the clinic would be closed due to an abrupt funding cut, Max was devastated. He quickly began to envision a future of miserable, constant back pain. Max's health psychologist, Dr. Phelps, sought to change Max's perception that without the clinic, he had no control over his own back health. She worked with him on compiling a list of the tips and techniques he'd learned from the clinic over the years and ways he could implement them on his own. With input from Max's physician, she also sought out relevant readings, pain management exercise videos, and other resources in the community, including a free weekly pain management support group in a neighboring town. In a relatively brief period of time, Dr. Phelps was able to work with Max to change his perception of being abandoned and helpless to one of having multiple sources of support. It is important to note that this process decreased not only Max's anxiety but his physical pain as well.

Another healthy coping skill that health psychologists can foster is **problem-focused coping**. Problem-focused coping emphasizes proactive, constructive attempts to take action about a stressful situation. This approach is exemplified in the people who, although realizing that the occurrence of a hurricane is completely out of their control, still work to be as prepared as possible by stockpiling emergency

BOX 18.1

Considering Culture

Physical and Psychological Expressions of Depression Across Cultures

According to the *Diagnostic and Statistical Manual of Mental Disorders—Fifth Edition* (*DSM*-5), a major depressive episode consists of a combination of at least five of nine possible specific symptoms. Most of these symptoms are psychological in nature (e.g., depressed mood, decreased interest in daily activities, impaired concentration, feeling worthless or guilty, thoughts of death or suicide). The other symptoms are more physical (e.g., appetite/weight changes, sleeping, fatigue, restlessness/slowed movements). But does depression express itself similarly across cultures? Specifically, in members of non-Western cultures, does depression tend to express itself through the body rather than the mind?

Some research suggests that the answer is yes. Specifically, Asian Americans have often been described as exhibiting depression (and some other mental illnesses) through somatic, bodily symptoms to a greater extent than do European Americans (e.g., D. W. Sue & Sue, 2008; S. Sue & Consolacion, 2003). (Of course, within Asian Americans and every cultural group, there is great heterogeneity.) In addition to the physical symptoms listed in the *DSM* definition, headaches, stomach aches, dizziness, aching muscles, and other physical ailments may be included in the experience of depression in many Asian Americans. These physical symptoms may replace the psychological symptoms common to depression in European Americans, such as sad mood and thoughts of death.

There are numerous possible reasons for this relative emphasis on physical symptoms among Asian Americans (D. W. Sue & Sue, 2008; S. Sue & Consolacion, 2003):

- Asian American culture tends to emphasize a holistic view of mind and body. With little or no boundary between these two aspects of life, Asian Americans may present emotional problems as somatic symptoms.
- The importance of collectivism within Asian American culture results in the discouragement of open displays of emotional distress. Such displays could threaten interpersonal harmony or expose personal weakness, neither of which tends to enhance group or family relationships.
- In Asian American cultures, physical illness tends to be less stigmatized than mental illness. Thus, bodily complaints allow clients to maintain a more socially acceptable appearance (i.e., "face") than do emotional symptoms.

For those who work with clients whose problems include both physical and psychological symptoms—health psychologists and many other health professionals as well—it is crucial to know that what manifests as physical may have psychological roots. Similarly, it is vital to respect the way clients experience depression and not to pressure clients to think about the problem in a way that is incompatible with their cultural values. So, for many Asian American clients, the psychologist would be wise to validate the physical manifestations of depression and to consider a reciprocal interaction between mind and body problems (D. W. Sue & Sue, 2008).

supplies and creating evacuation and communication plans. Research has strongly supported the idea that those who believe they can exert some personal control over stressful situations are better off both emotionally and physically than those who view themselves as having little to no control over stressors (Pittner, Houston, & Spiridigliozzi, 1983). When the stressor is less controllable—for example, a catastrophic injury to a friend or family member—health psychologists can promote *emotion-focused coping*, in which the emphasis is on changing the emotional reaction to the stressor (rather than the stressor itself). Flexibility between problem-focused coping and emotion-focused coping can be a beneficial overall strategy for coping with stress (Carver, 2013).

As an example of problem-focused coping, consider Sandra, a 19-year-old college student who developed ulcers and frequent headaches during her first two years in college. Although she had numerous sources of stress, she described her primary stressor as the constant worry that her parents would not be able to continue to support her financially and that without their help she would not be able to continue at her university. With the encouragement of Dr. Yu, a health psychologist who worked in conjunction with the physicians who treated Sandra's ulcers and headaches, Sandra began to search for aspects of this situation that she could control. For example, Dr. Yu helped Sandra identify other colleges that were more affordable, including some in the same city, where she might be able to transfer. Dr. Yu also helped Sandra make a list of other sources of income she could consider, including jobs, loans, and scholarships. Dr. Yu also suggested that Sandra meet with a financial aid specialist at her college, who offered even more resources to Sandra. The result was that by taking control and "doing something" about her problem rather than merely dwelling on it, Sandra's body and mind felt better. Her ulcers and headaches were less severe, and she felt more emotionally stable as well.

SOCIAL SUPPORT

Friends and family are important weapons against the damaging effects of stress. In fact, some argue that social support is the single most vital resource against stress (e.g., Hafen et al., 1996). **Social support** can be described as the perception that one has relationships with others who can provide support in a time of crisis and can share in good fortune as well. Social support is one of the most documented phenomena in health psychology; its benefits to physical health are ample and undeniable. In the words of Thomas Padus, "The breakdown in the social support structure precipitates a breakdown in the body's immune system" (as quoted in Hafen et al., 1996, p. 262).

House, Kahn, McLeod, and Williams (1985) purport that social resources are composed of three components: social network, social relationships, and social support. One's social network is the broad scope of all social contacts one has. In this large pool is a smaller pool of those with whom one has more meaningful social relationships. And social support refers to the quantity and quality of care and assistance those relationships bring. Thus, someone with a small social network who has a single

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Photo 18.2
Social support offers protection from the ill effects of stress.

meaningful social relationship accompanied by a large degree of social support might be better off than a "social butterfly" with a broad social network but no intimate friends. Interestingly, married men report that their main source of social support is their wives, whereas married women rely more heavily on other family members and friends. It is hypothesized that this may account for the fact that men are more likely than women to suffer physically from stress-related ailments after a divorce (Kiecolt-Glaser et al., 1996).

How, exactly, does social support protect our health? The answer is not yet entirely clear. Some hypothesize that the positive feelings associated with having social support allow us to perceive our stressors in a more manageable light. For example, the social support provided by church attendance was hypothesized to be a primary reason why those who attend religious services regularly are happier and healthier than those who do not (L. B. Koenig & Vaillant, 2009). Another hypothesis, strongly supported by the work of James Pennebaker, is that those who have strong social networks are more likely to confide in others. Pennebaker (1990) has studied the health benefits of confiding and has concluded, "Not disclosing our thoughts and feelings can be unhealthy. Divulging them can be healthy" (p. 82). His work has uncovered benefits to the immune system that are evident after the act of confiding in others. His immune-related findings are mirrored by those of health psychologist Janice Kiecolt-Glaser; one of her studies found clear evidence that those who are more lonely have the most compromised immunities (Kiecolt-Glaser et al., 1985).

A third hypothesis regarding how social support protects our health has to do with the sensation of touch. Research in this area was born out of the classic studies of Harry Harlow that compared the development of baby monkeys with and without access to other monkeys and found that monkeys who were touched and cuddled fared significantly better (Harlow & Zimmermann, 1959). Studies on humans have produced similar results. For example, premature babies who were stroked and massaged during their hospital stays were significantly healthier and heavier and had fewer physical problems at an 8-month follow-up period than did those who were not (S. Weiss, 2005). Another study reported that adults who enjoy regular, loving touch have stronger hearts and lower blood pressures, and they report feeling less stress (Engle & Graney, 2000).

Web Link 18.2
Journal of Health Psychology

CLINICAL APPLICATIONS

Today's health psychologists enjoy a wide menu of focus areas, expanding far beyond the original work on stress and health (see Table 18.1). The following sections will review some of the most popular current clinical applications of health psychology. In addition to the following examples of traditional client-based work, it is important to note that health psychologists can also work within a more broad, sociocultural context to craft behavioral health legislation (e.g., mandatory helmet laws, effective public service announcements), school and workplace policies (vending machine and smoking bans), and policymaking regarding health care reform.

Weight Management

Table 18.1 Common Areas of Clinical Focus in Health Psychology

Alcohol use	Irritable bowel syndrome
Smoking	Weight management
Illicit drug use	Eating disorders
Prescription drug abuse	Stress management
Pain management	Cancer
Chronic fatigue	Coping with medical regimens
HIV/AIDS	Compliance with medical regimens
Cardiovascular disease	Sleep disorders
Diabetes	Sexual disorders

In the United States, the prevalence rates for obesity and being overweight are at an all-time high. About 72% of adult Americans are overweight or obese, and the numbers continue to rise each year in both adult and child populations (Centers for Disease Control, 2016). The risks of carrying excess weight are numerous and well documented; obesity is estimated to cause more than 400,000 deaths annually (16.6% of total deaths) in the United States (Mokdad, Marks, Stroup, & Gerberding, 2004). And health care costs related to obesity are estimated at $117 billion (Y. Jackson et al., 2002), or 6% to 10% of our national health care expenditures (Andreyeva, Sturm, & Ringel, 2004). Health psychologists play a very important role in efforts to stem this national crisis (Butryn, Raggio, & Lowe, 2015; Fabricatore & Wadden, 2006; Martin, Stewart, Anton, Copeland, & Williamson, 2008; Rozensky, Pereira, & Whitehead, 2016; T. W. Smith et al., 2016). Proper nutrition and physical fitness are no doubt responsible for achieving and maintaining a healthy weight. However, recent research is suggesting that the problem is not that Americans are ignorant of what a healthy diet or exercise regimen is but, rather, that many have a hard time committing to these behaviors in the long term. As such, cognitive-behavioral therapy has become a very necessary and important component of successful weight management (Kaholokula & Haynes, 2015).

A common approach health psychologists use in treating people with excess weight is known as the **ABCDS of weight loss**: *A*ctivity increase, *B*ehavior change, *C*ognitive change, *D*ietary change, and *S*ocial support (Blair et al., 1996). Collectively, these components represent a multifaceted, long-term approach to maintaining a healthy weight. To illustrate the use of the ABCDS approach, consider Shannon, a 54-year-old woman who owned and operated a public relations office. Shannon was successful in many ways, but she had been significantly overweight since childhood, and her weight had increased through adulthood. To address her weight problem, Shannon sought the services of Dr. Gold, a health psychologist who worked with a multidisciplinary team of health professionals at a weight management center. Dr. Gold quickly learned that Shannon's many past attempts to lose weight had involved exercise (A) and diets (D) but nothing else (no B, C, or S). Moreover, her exercise and diet efforts had never proven successful, largely because she didn't stick with them.

Regarding her dieting (D) attempts, Shannon explained to Dr. Gold that her diets usually required a complete abstinence from her favorite foods, for which Shannon lacked the willpower over the long haul. Dr. Gold, with help from a dietician colleague, helped Shannon develop a more flexible meal plan (not a diet, per se) that would be more feasible as part of a long-term lifestyle change. Also, in collaboration with a personal trainer,

Dr. Gold helped develop an exercise plan (A) that involved five 30-minute workouts per week, some of which involved walking, running, or kayaking with her best friends (S). Also, to address the behavior change (B) aspect of the ABCDS plan, Dr. Gold and Shannon jointly created a list of smart decisions to make at restaurants (e.g., order or request smaller portions, substitute steamed vegetables for fried side dishes, avoid pre-meal items such as bread or chips) and grocery stores (reading nutrition labels, limiting purchases of certain high-calorie or high-fat foods).

To Dr. Gold's surprise, Shannon initially resisted Dr. Gold's suggestions. Shannon explained that they conflicted with some of her beliefs about eating and exercise. Specifically, Shannon believed that "successful weight loss means at least 5 pounds a week," "working out counts only if it lasts at least an hour," and "restaurants don't want to be bothered with special requests." An important aspect of Dr. Gold's job was to challenge these faulty cognitions (C). Dr. Gold persuasively educated Shannon about weight loss goals (1 to 2 pounds per week is more reasonable), exercise duration (any length "counts"), and the fact that most restaurants are very willing to accommodate customer requests. By addressing all the facets of the ABCDS approach, Dr. Gold enabled Shannon's success in maintaining a healthy weight—she lost 35 pounds in the first 6 months and kept that weight off for many years.

Besides the elements of the ABCDS model, health psychologists have identified other effective strategies to help with weight loss, including (Johnston, Moreno, & Foreyt, 2016):

- Self-monitoring—keeping track of eating and exercise behaviors
- Goal setting—creating reasonable goals that can produce frequent feelings of success
- Contracting—making promises to maintain exercise time, eat healthy food, etc.
- Problem solving—recognizing and getting around obstacles to weight loss instead of feeling helpless to do anything about them
- Stimulus control—keeping unhealthy foods out of the house, staying out of unhealthy restaurants, buying healthy foods at the grocery store, getting yourself to exercise locations
- Stress management—making lifestyle changes that lower the overall level of stress, using relaxation techniques, meditating
- Social support—discussing weight loss efforts, including setbacks, with people who care about you

An interesting line of research has focused on the role of **values affirmation** in promoting physical health, including weight management. By writing about the things they value most in life (e.g., relationships, service to others), patients focus their attention on their sense of personal identity and self-worth (D. K. Sherman & Cohen, 2006). These affirmations bolster self-control, because concentrating on one's most cherished values may lead to a decrease in impulsive behavior that is inconsistent with those values and only rewarding in the short term (Creswell et al., 2005). One notable recent study found that after completing

a single session of values affirmation, women weighed significantly less and had significantly smaller waist measurements almost 3 months later than did their peers in a control group that did not complete the values affirmation (Logel & Cohen, 2012).

Web Link 18.3
Values affirmation

Smoking

Like obesity, smoking is a massive public health problem in America (Martin et al., 2008). Although smoking rates have declined in recent decades, smoking remains a major cause of preventable death, accounting for more than 450,000 deaths in the United States each year (Niaura & Abrams, 2002; Scharf, Ferguson, Tindle, & Shiffman, 2016). Smoking has been proven to be a major risk factor in heart disease, cancer, vascular disease, stroke, and major depressive disorder (A. L. Haas, Munoz, Humfleet, Reus, & Hall, 2004). Also like obesity, the primary problem is not that people are ignorant of the dangers of smoking; instead, the problem lies in implementing a very difficult behavioral change: quitting.

Unfortunately, once it has become an addiction, smoking exerts very strong control over the physiological, psychological, and social aspects of the smoker's life. Some of the strongest weapons that health psychologists have in the battle against smoking are nicotine replacement, social support, stress management, and relapse prevention.

Nicotine replacement involves the use of over-the-counter or prescription alternatives to cigarettes, including gum or transdermal patches. The mechanism behind this form of therapy is twofold: to break the oral habit of smoking while at the same time slowly weaning the body off nicotine dependence through controlled dosing that decreases over time. Reviews of the efficacy of nicotine replacement therapy have supported it as an agent that often (but certainly not always) helps with smoking cessation (Cahill, Stevens, Perera, & Lancaster, 2013; Cepeda-Benito, 1993). Health psychologists can help educate clients on the research supporting this sort of therapy, how it works, where to obtain it (some can be purchased over the counter, and others need to be prescribed by a physician), and common side effects (e.g., headache, racing heart). In addition, they can help the client work through any potential barriers to compliance with the therapy before they begin (e.g., weighing the affordability and convenience of the various options).

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Photo 18.3
Weight-related problems are increasingly prevalent, so weight management has become a common focus of contemporary health psychologists.

An important aspect of smoking cessation is the availability of a nonsmoking social support network. Having smokers in one's social group has been shown to be both a significant detriment to smoking cessation and a significant predictor of relapse (Mermelstein, Cohen, Lichtenstein, Baer, & Karmarck, 1986; M. E. Roberts, Nargiso, Gaitonde, Stanton, & Colby, 2015). Helping the client create or strengthen a nonsmoking network is often an important first step of smoking cessation treatment. One way this might be accomplished is by first developing problem-solving strategies for limiting, ending, or dealing with relationships with other smokers (e.g., a spouse who smokes). This might include setting

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Photo 18.4
The well-established physical risks associated with smoking provide a strong motivation for health psychologists to help smokers quit.

such rules as limiting smoking to outside of the house and family vehicles. It is also important to create a list of the number and quality of nonsmoking social contacts across various situations (work, home, recreation). If this list is small, social network building may need to occur. The client would be encouraged to identify and start reaching out to other nonsmoking peers or to seek support through smoking cessation support groups (P. M. Smith, Reilly, Houston Miller, DeBusk, & Taylor, 2002). Finally, clients would also be encouraged to identify places where they typically smoke socially and plan alternate locations and activities that would be equally fulfilling. For example, clients who often dine out with friends might consider frequenting only smoke-free restaurants.

Some health psychologists use the acronym STAR to summarize the steps involved in successful smoking cessation: *S* for set a specific date for quitting (rather than leaving it unspecified); *T* for telling family and friends so they can provide support; *A* for anticipating specific challenges that might arise and coming up with constructive responses ahead of time; and *R* for removing items that might serve as cues or triggers to smoke, like lighters, ashtrays, and cigarettes (Scharf et al., 2016; Olenik & Mospan, 2017; World Health Organization, 2017).

Smokers often report that they smoke because they find it relaxing and stress reducing. As such, an important task for the health psychologist is to work with the client to develop and practice alternate replacement strategies for relaxation and stress reduction *before* the smoker begins the quitting process. Relapse prevention goes hand in hand with this concept, as its goal is to predict which factors are likely to trigger a relapse and to prevent those factors from occurring. Given that one of the most common triggers for a relapse is feeling stressed, the use of stress management techniques can play a strong role in relapse prevention. The **ABCs of relapse prevention** (Marlatt & Gordon, 1985) involve an examination by the client and the health psychologists of the events that occur before and after smoking. Specifically, *B* stands for Behavior (smoking), *A* stands for Antecedent, and *C* stands for Consequence. For example, if Julian recognizes that whenever he drinks alcohol at a bar, he's likely to smoke, he has identified drinking in the bar as an antecedent. If he also realizes that smoking a cigarette in the bar provides the consequence of reduced stress, he can identify alternate behaviors for reducing stress (e.g., exercising, watching TV, socializing with friends outside of bars). A health psychologist can be quite helpful in facilitating Julian's efforts by helping him recognize antecedents and consequences and by suggesting alternative behaviors.

Alcohol Use

Drinking alcohol occasionally or in moderation does not in itself constitute an alcohol problem. The use of alcoholic substances becomes pathological when certain criteria are met—for example, when the person can't cut down on alcohol, has strong cravings, or experiences repeated failures at work, school, or home related to alcohol (American Psychiatric Association, 2013). Two other hallmarks of problematic alcohol use are

tolerance and withdrawal. **Tolerance** occurs when the body requires increasing amounts of the substance to achieve the desired effect, and **withdrawal** occurs when cessation of the substance produces negative symptoms. Researchers estimate that about 4% of Americans use alcohol in a way that involves tolerance and withdrawal (e.g., B. F. Grant et al., 2004), which qualifies alcoholism as a large-scale social health issue, along with obesity and smoking.

Very similar to smoking cessation, the successful treatment of alcoholism includes the components of stress management, relapse prevention, and social support. As alcohol is also commonly used to reduce stress, health psychologists would work with clients to identify more healthy alternatives to manage stress, such as working out or calling a friend to talk. Identifying common precursors to drinking (going to a sports event) and developing strategies to avoid them (not carrying enough cash to purchase a drink) uses the same ABCs of relapse prevention that health psychologists use with smoking cessation clients. Also similar to working with smoking cessation clients, health psychologists would help clients who wanted to abstain from drinking identify and, if necessary, expand their social circle of nondrinkers and socializing options that are alcohol free.

The most commonly sought source of help for alcohol-related problems, the self-help support group Alcoholics Anonymous (AA), incorporates many of these strategies in its treatment plan. AA is a network of current and recovering alcoholics who share the goal of helping themselves and other members of AA stay sober. Although this group is not affiliated with health psychology, it uses many of the same principles, especially social support and relapse prevention. One important distinction between AA and health psychology is that AA incorporates a spiritual component to its recovery process.

An important distinction between the treatment for smoking and the treatment for alcohol problems is the fact that alcohol exhibits effects on the brain different from the effects of many other substances, including nicotine, and an addict might experience dangerous side effects as a result of weaning the body off the substance too quickly. As such, a period of medically supervised **detoxification** may be the most appropriate first course of action before proceeding to cognitive-behavioral techniques.

Pain Management and Biofeedback

Chronic pain—pain that lasts 6 months or longer—affects up to 35% of the population (Mantyselka, Turunen, Ahonen, & Kumpusalo, 2003). Although a large component of pain management is medically based (e.g., pain medications, surgery, or physical therapy), the fact that depression, anxiety, and many other factors can influence the experience of pain suggests that health psychology should also be an integral part of any comprehensive pain management plan (Vowels, Zvolensky, Gross, & Sperry, 2004; Rozensky, Tovian, & Goodheart, 2016).

One unique technique health psychologists employ in pain management is **biofeedback** (Austad & Gendron, 2018; Gevirtz, Alhassoon, & Miller, 2016; D. Shapiro & Schwartz, 1972). The purpose of biofeedback is to achieve control over the body via educating patients about bodily processes of which they are typically unaware. To bring these processes into the forefront of consciousness, a biofeedback machine converts some aspect of physiological functioning (e.g., heart rate, muscular tension) into a tone for the client to hear or a line on a graph for the client to see. So, for example, as heart rate increases, a

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Photo 18.5 Health psychologists use a variety of techniques to help clients who experience ongoing pain.

tone gets louder, or as shoulder tension declines, a line on a graph dips. The patient is guided through trial-and-error exercises to see what sorts of behaviors (e.g., deep breathing, changing postures, modifying thoughts, imagery) alter the bodily process in question. Once patients have learned the cause–effect relationships between their own behaviors and their bodies' responses, they can use them purposely to achieve the desired level.

As an example, consider Ronald, a 45-year-old client with chronic lumbar-region back pain. Before Ronald saw Dr. Booker, Ronald felt as though there was nothing he could do to control his back pain. As a health psychologist, Dr. Booker allowed Ronald to "see" his own back pain via a graphic biofeedback machine that displayed muscular tension as a visual graph. After trying a number of physical and mental exercises designed to influence the back pain, Ronald and Dr. Booker were pleased to find that two of Ronald's actions—elevating his legs and imagining himself on a quiet, calm beach—caused his back pain to decrease. This decrease in pain was evident both in Ronald's subjective experience and quite undeniably on the monitor he watched. After this discovery, Ronald was able to deliberately elevate his legs and imagine the beach scene in his daily life to reduce the spasms in his lower back.

Critics of biofeedback training have argued that its benefits are offset by its expense and time-intensiveness. Fortunately, an economical and relatively quick alternative has been found in an intervention originally developed long ago for anxiety disorders (Jacobson, 1938). **Relaxation training** involves teaching clients to consciously shift their bodies into a state of lowered tension and arousal. It can involve a variety of aspects, including guided imagery (e.g., imagining yourself in a gently swinging hammock), deep and controlled "belly breathing," and systematically tensing and then relaxing various large skeletal muscles such as those in the arms and legs. Relaxation training has been found to be successful in alleviating chronic pain, and it may work by increasing the release of endogenous opioids, our bodies' natural analgesics (McGrady et al., 1992). An additional benefit of relaxation training, regarding the stress associated with chronic pain, is that several studies have found beneficial effects of relaxation on immune system functioning (Pawlow & Jones, 2002, 2005).

Pennebaker's work on the therapeutic effects of confiding distressing thoughts to others also included the notion that writing about distressing events could be cathartic and even alter immune system functioning in a positive way (Pennebaker, Kiecolt-Glaser, & Glaser, 1988). A more recent study focusing on chronic pain found similar results: Chronic pain patients who wrote about their angry feelings surrounding their pain experienced improved levels of perceived control and decreased levels of depression (J. D. Graham, Lobel, Glass, & Lokshina, 2008). Thus, simply taking the time to journal one's thoughts surrounding one's pain may help alleviate some of the suffering due to that pain.

BOX 18.2

Metaphorically Speaking

If You've Used Exercise Equipment With a Heart Rate Monitor, You Understand Biofeedback

At the gym, you may have used a treadmill, stationary bike, or elliptical trainer that monitors your pulse via metal sensors in the handles. The machine then converts this pulse information to a heart rate (usually expressed as beats per minute, or BPM) and displays this information to you in real time throughout your workout. If your goal is to control your heart rate—raise it to a certain level or keep it below a certain level—you can take note of the cause–effect relationship between your actions and your heart rate. On the stationary bike, for example, you may learn that pedaling at a certain speed against a certain level of resistance brings about a certain heart rate.

Monitoring heart rate may be of secondary importance to many of the people exercising on these machines at the gym, but for clients using biofeedback, it is the primary goal. As the term suggests, a biofeedback machine provides the client with feedback about bodily activities such as heart rate, muscle activity, and the like. With the help of a health psychologist, clients can use biofeedback machines to learn which actions they can choose to take in order to bring about desired changes in the internal systems of their bodies.

This can be an empowering process. Without biofeedback, clients would be unable to know, for example, exactly how fast their hearts are beating. Moreover, it would be impossible for them to make connections between any particular behaviors they perform (e.g., intentionally slow breathing, mental imagery) and subsequent changes in heart rate. But with biofeedback, heart rate is instantly visible and continuously updated as various behaviors are attempted. Thus, a heart rate problem that a client may initially experience as uncontrollable becomes more directly manageable.

BOX 18.3

Sample of an Abbreviated Progressive Relaxation Training Script

The following is a small portion of a progressive muscle relaxation script (Bernstein & Borkovec, 1973):

> When I say the word "tense," that is your specific signal to tense the muscles in the group we are working as tightly and as immediately as possible. You will be holding the tension for 7 seconds. Do not begin tensing until you hear this cue. Likewise, when I say the word "release," that is your specific relaxation signal to immediately release the tension. Do not let the tension dissipate gradually, and do not release the tension until I give the cue. I will give you specific instructions on what muscle group to tense, as well as general guidelines on how to tense it. Please do not move

(Continued)

(Continued)

unnecessarily; however, feel free to move in any way that helps you to maintain a comfortable position at all times.

When I say "tense," I want you to tense the muscles in your dominant hand and forearm by making a tight fist. Ready? Tense. (Hold for 7 seconds.) Feel the muscles pull. Notice what it's like to feel tension in these muscles as they pull and remain hard and tight. Release. (Relax for 30 seconds.) Just let the muscles go, notice the difference between tension and relaxation, focus on the feeling in the muscle as it becomes more and more relaxed.

When I say "tense," I want you to tense the muscles in your dominant biceps by pushing your elbow down against the arm of the chair. Ready? Tense. (Hold for 7 seconds.) Feel the muscles pull. Notice what it's like to feel tension in these muscles as they pull and remain hard and tight. Release. (Relax for 30 seconds.) Just let the muscles go, notice the difference between tension and relaxation, focus on the feeling in the muscle as it becomes more and more relaxed.

When I say "tense," I want you to tense the muscles in the upper part of your face by lifting the eyebrows as high as you can and getting tension in the forehead and scalp region. Ready? Tense. (Hold for 7 seconds.) Feel the muscles pull. Notice what it's like to feel tension in these muscles as they pull and remain hard and tight. Release. (Relax for 30 seconds.) Just let the muscles go, notice the difference between tension and relaxation, focus on the feeling in the muscle as it becomes more and more relaxed.

When I say "tense," I want you to tense the muscles in the central part of your face by squinting your eyes very tightly and at the same time wrinkling up your nose to feel the tension in the upper cheeks and the eyes. Ready? Tense. (Hold for 7 seconds.) Feel the muscles pull. Notice what it's like to feel tension in these muscles as they pull and remain hard and tight. Release. (Relax for 30 seconds.) Just let the muscles go, notice the difference between tension and relaxation, focus on the feeling in the muscle as it becomes more and more relaxed.

Source: Bernstein & Borkovec (1973).

Compliance With Medical Regimens

In some cases, health psychologists work with clients whose problems do not stem from psychological contributors of medical illness. Instead, these clients need assistance complying with medical regimens (Goodheart & Rozensky, 2011; S. L. Williams,

Haskard-Zolnierek, & DiMatteo, 2016). Failure to comply with medical regimens can take many forms, including not taking medication, violating dietary restrictions, missing follow-up appointments, and not adhering to activity restrictions. There are many reasons why clients may not adhere to or comply with treatment regimens, including poor provider communication and fears about negative side effects. One study found that patients are less compliant with medication regimens when they are aware of the side effects, even if the side effects are mild and/or very unlikely to occur (Waters, Weinstein, Colditz, & Emmons, 2009). On average, patient noncompliance is seen in about 26% of all cases, with the highest rates in HIV, arthritis, gastrointestinal disorder, and cancer populations (DiMatteo, Giordani, Lepper, & Croghan, 2002).

Health psychologists can increase compliance with medical regimens in many ways. Sometimes they serve as liaisons between the provider and the patient to communicate the proposed treatment plan more carefully and thoroughly in a comprehensible and compassionate way. In fact, some research has suggested that simply hearing the message in a warm and caring tone leads to greater compliance (e.g., Sherbourne, Hays, Ordway, DiMatteo, & Kravitz, 1992). Another important aspect of reducing noncompliance involves thoroughly educating the patient on the proposed treatment, alternate options (including no treatment), the pros and cons of each, and the current research on the topic. Also, clinicians can aid the patient in securing or bolstering their personal support systems as another mechanism for boosting compliance. In a study of patients with chronic heart failure, those with higher levels of perceived social support demonstrated better medication, behavioral, and dietary adherence to their medical treatment plans (Sayers, Riegel, Pawlowski, Coyne, & Samaha, 2008).

Identifying patient barriers to compliance and working together to suggest time-limited options can also help. For example, a pain patient who doesn't "believe in touchy-feely stuff like relaxation" might be persuaded to give it an honest try for 2 weeks, and if she sees no effect, she can stop with the provider's blessing. And sometimes the health psychologist can adopt a direct problem-solving approach to help clients with problems that seem, to them, quite difficult. For example, Claire, a 78-year-old woman with numerous physical illnesses and conditions, was noncompliant with her medication regimen because she had an abundance of pills to take and was so terrified that she'd take the wrong pills at the wrong time that she chose to forgo all of them. Her health psychologist, Dr. Howard, enlisted Claire's daughter to purchase a cheap, monthly/daily pill case from the local pharmacy and to place her supply of pills in the appropriate boxes once a month. This simple step was sufficient to rectify the situation.

Coping With Medical Procedures

Health psychologists also play a significant role in helping medical patients cope with the stress of facing a medical procedure. These medical procedures may include surgery, therapies with predictable negative side effects (e.g., chemotherapy), painful physical therapy, or more experimental procedures such as deep brain stimulation. A primary duty of health psychologists in this role is to educate the client and demystify the medical procedure, thus increasing the client's sense of control and comfort. Much research has been conducted on the benefits of clear and detailed preintervention communication delivered in an empathic manner. For example, radiation therapy patients who receive this type of information show

better postoperative adjustment and healing than those who don't (J. Johnson, Lauver, & Nail, 1989). Also, the use of prerecorded videos to explain the steps of a complicated procedure such as heart surgery or a sensitive procedure such as a prostate screening has also been shown to be beneficial (e.g., Mahler & Kulik, 1998; K. Taylor, 2010).

Beyond optimizing communication, health psychologists also use cognitive-behavioral interventions such as relaxation training and learning to think differently about the unpleasantness of the situation (Ludwick-Rosenthal & Neufeld, 1988). As an example, consider Inez, a 50-year-old cancer patient who is about to begin chemotherapy. Inez is distraught about her cancer diagnosis, but she is almost equally distraught about the upcoming chemotherapy. When her health psychologist, Dr. Ricardo, asked Inez to explain her thoughts about chemotherapy, Inez tearfully stated, "It will go on for the rest of my life" and "The side effects will be so horrible—I won't be able to live through it." Dr. Ricardo showed compassion for Inez's concerns, but she also made sure to correct Inez's inaccurate understanding of chemotherapy: It was scheduled to last a few months, not a lifetime, and although most patients experience some unpleasant side effects, they do find that the side effects are ultimately tolerable. Moreover, Dr. Ricardo helped Inez focus on the improved health she could enjoy if the chemotherapy was successful (which, in Inez's case, was likely because her cancer was detected early and was considered very treatable). By correcting Inez's misinformed beliefs and shifting her focus to the possibility of a positive outcome, Dr. Ricardo was able to improve not only Inez's psychological well-being before the chemotherapy but also her chances of benefiting from it physically.

When health psychologists work with hospitalized children, a common goal is the minimization of separation anxiety (anxiety produced by separation from the primary caretaker, generally a parent). Techniques include the assignment of "substitute parents," in which a junior staff member such as a graduate student or student nurse is assigned to be the child's stand-in parent during any hours in which the actual parent is prohibited from visiting. Similarly, health psychologists can arrange for a young child's comfort items, such as a blanket or teddy bear, to be brought from home.

In some cases, a health psychologist may be asked to assess a patient's psychological readiness for a medical procedure before medical personnel decide to conduct it. In this capacity, health psychologists can screen out those patients whose state of mind would be likely to make a medical procedure ineffective or counterproductive. As an example, individuals for whom gastric bypass surgery is being considered routinely undergo a psychological evaluation before the surgery is scheduled. According to one study, almost 3% of gastric bypass applicants were deemed psychologically inappropriate to undergo the surgery. Specifically, these individuals, after an assessment including comprehensive clinical interviews, were found to be psychotic or at such a limited level of cognitive functioning that they were unable to fully understand and provide consent for the procedure under consideration (Pawlow, O'Neil, White, & Byrne, 2005).

Another medical procedure for which health psychologists commonly assess psychological appropriateness is organ donation. The medical community wants to be absolutely sure that donors understand the seriousness of their choice, including being able to comprehend all possible negative effects of the donation on both the donor and the recipient. It is also important in these cases to be sure that the donor is making the decision on their own and not being pressured by friends or family. Health psychologists are able to use their knowledge of both clinical assessment and the medical and psychological ramifications of organ donation to provide this important service.

A TREND IN HEALTH PSYCHOLOGY: PATIENT-CENTERED MEDICAL HOMES

Traditionally, patient care in the United States has followed a primary-care model. In recent years, however, our health care system has been facing a crisis brought on by a significant shortage of primary-care physicians as increasing numbers of physicians in training choose to focus their education on specialized, versus general, medical practice. Resulting problems include long delays for patients needing to see their primary-care physicians; shorter office visits, resulting in more confusion among patients; and a lack of coordination among primary-care physicians, specialists, and hospitals.

Patient-centered medical homes (PCMHs) have been proposed as a solution for delivering higher-quality and more cost-effective primary care (B. F. Miller et al., 2017). PCMHs aim to provide care that is patient centered, highly accessible and affordable, comprehensive, and coordinated. They are more holistic in nature, in that they do not view mental health as separate from physical health, and they are also more focused on prevention of disease than is the traditional primary-care model (Asarnow, Kolko, Miranda, & Kazak, 2017; N. J. Kaslow et al., 2007). In this model, each patient has a personal primary-care doctor responsible for arranging patient care with other professionals on the patient's care team (e.g., nurses, physician assistants, physical therapists, medical specialists), most of whom are housed in one central location. PCMHs utilize electronic health records for greater communication across providers and promote greater participation in the patient's own care by the patient and the patient's family. Given that this model views behavioral health as a key aspect of the patient's care, and because clinical health psychologists are uniquely educated to design, implement, and assess strategies that address the prevention and management of various health conditions such as obesity, AIDS, chronic pain, heart disease, diabetes, and cancer, health psychologists are particularly well trained to serve in leadership positions on PCMH care teams and are poised for "unprecedented workforce growth" opportunities even while other mental health employment opportunities are in decline (Beacham, Kinman, Harris, & Masters, 2012, p. 17).

Web Link 18.4
Patient-centered medical homes

Outcomes for patient-centered medical homes have been mixed. Some studies find significant benefits over other care strategies in terms of patient satisfaction, hospitalization rates, and emergency room usage (J. A. Alexander & Bae, 2012; Kern, Edwards, & Kaushal, 2016; K. M. Nelson et al., 2014), but others find either minimal or no benefits of this kind (G. L. Jackson et al., 2013; Werner, Duggan, Duey, Zhu, & Stuart, 2013).

CULTURAL FACTORS IN HEALTH PSYCHOLOGY

In all aspects of their work, it is crucial for health psychologists to fully appreciate the impact of cultural and diversity-related factors on the health- and illness-related experiences of their clients. In fact, research has indicated that health psychology interventions that have been adapted for particular cultural groups are typically more effective than similar interventions that have not been culturally adapted (M. Barrera, Castro, Strycker, & Toobert, 2012). As a starting point, it is vital for health psychologists to recognize the

disparities that exist in health care between various ethnic groups, as well as the divergent attitudes of members of these groups toward the medical establishment. There are significant differences in the quality of health care and the overall health status of the ethnic groups within the U.S. population, with whites and Asian Americans ranking near the top and African Americans and Native Americans ranking near the bottom in ratings of overall health status (LaVeist, 2005). The historical context of these rankings, including the centuries of oppression suffered by African Americans and Native Americans in the United States, cannot be ignored. A relatively recent example of this oppression is represented by the Tuskegee syphilis study, in which the U.S. Public Health Service intentionally and deceptively withheld adequate treatment from hundreds of African American men with syphilis from 1932 to 1972 in an attempt to learn more about the course of the disease.

These historical factors undoubtedly relate to the perceptions held by members of diverse groups regarding hospitals and physicians. For example, African American patients appear to demonstrate a greater mistrust of the health care system than do white patients (e.g., LaVeist, 2005). In a survey of medical patients (LaVeist, Nickerson, & Bowie, 2000), a significantly greater percentage of African American patients than white patients agreed with the statements, "Patients have sometimes been deceived or misled at hospitals," "Hospitals often want to know more about your personal affairs or business than they really need to know," "Hospitals have sometimes done harmful experiments on patients without their knowledge," and "Racial discrimination in a doctor's office is common." The same survey found that a significantly lower percentage of African American patients than white patients agreed with the statements, "Doctors treat African American and white people the same" and "In most hospitals, African Americans and whites receive the same kind of care." For health psychologists, appreciation of the discrepancies among diverse groups regarding health care experiences and attitudes is an essential component of cultural competence.

Cultural factors can also play a powerful role in determining the way medically ill people understand the source of their medical problems. According to Huff (1999), the perceived source of a sickness could fall into one of four categories:

- *Within the patient*—an infection, injury, or other biomedical irregularity
- *The natural world*—elements of the environment surrounding an individual, such as toxins or climate-related factors
- *The social world*—interpersonal conflict with others, especially those with whom the individual has close relationships
- *The supernatural world*—sorcery, witchcraft, ancestral spirits, or vengeful gods, for example

Huff (1999) explains that in Western cultures, it is typical for individuals to attribute illness to factors within themselves, and this attribution is quite consistent with mainstream Western medical care, which emphasizes biomedical causes of illness. However, in many non-Western cultures, it is quite common for individuals to attribute illness to one of the other three categories, each of which locates the cause of the illness outside the individual. So it is essential for health psychologists to remember that in various cultural groups, some individuals will maintain an external, rather than internal, locus of control regarding their health problems that may run counter to the assumptions of Western medicine. An appreciation of this difference, along with corresponding adjustments in practice, can ultimately enhance client health.

Culturally competent health psychologists also appreciate that, compared with white clients, members of ethnic minority groups may hold different expectations regarding the type of care they will receive and the types of interactions they will have with their caregivers (Huff, 1999; Kline & Huff, 1999). For example, in many of the largest ethnic minority groups in the United States, including Hispanic Americans, Asian Americans, Native Americans, and African Americans, reliance on culturally traditional healers and healing methods is common. These methods may include medicinal plants and herbs, massage, acupuncture, acupressure, prayer, chants, or other methods. Additionally, these methods often don't involve the type of quick, fact-based, yes/no questioning that characterizes many physician–patient interactions in Western culture. To overcome these discrepancies, health psychologists can play a facilitative role by educating both health care providers and diverse clients about the expectations of the other party, thereby increasing the effectiveness of the interventions.

Finally, it is important for health psychologists to suspend judgment about how clients from diverse ethnic backgrounds may perceive their illnesses or approach their treatment (Huff, 1999). The traditional Western way of understanding illness is just one of many, and culturally sensitive health psychologists openly recognize the validity in other ways of understanding as well.

CHAPTER SUMMARY

Health psychology is a burgeoning specialty area for clinical psychologists involving the interaction of psychological processes and physical health. Early efforts in health psychology focused primarily on issues of stress and its health-related consequences, but today health psychologists address a much wider range of issues. For example, they help clients manage their weight, quit smoking, reduce problematic alcohol use, and manage chronic pain. They also work to increase clients' behavioral adherence to medical regimens and their ability to cope with challenging medical procedures. They use a variety of specific therapeutic techniques, but their work often involves an educational component in which they enhance clients' understanding of the connection between their physical health and their psychological or behavioral functioning. In addition, health psychologists conduct psychological evaluations of individuals considering serious medical procedures such as gastric bypass surgery and organ donation. Recently, health psychologists have increasingly practiced via patient-centered medical homes in an attempt to better coordinate care among multiple professionals. Throughout their work, health psychologists maintain an appreciation of the cultural factors that may influence their clients' health-related experiences.

KEY TERMS AND NAMES

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behavioral medicine 450
biofeedback 461
chronic pain 461
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coping 452
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CRITICAL THINKING QUESTIONS

1. In general, to what extent do you agree with the notion that psychological factors contribute to physical health problems?
2. If clinical psychologists developed a successful, evidence-based treatment for the disease-prone personality (characterized by marked anxiety, depression, and hostility), what ramifications might this treatment have throughout the health care industry?
3. In your opinion, to what extent should physicians emphasize the creation of a social support network for their patients?
4. In your opinion, to what extent are psychological factors responsible for the large-scale problems with weight and obesity in the United States? To what extent do you think that people who struggle with losing excess body weight recognize these psychological factors?
5. What efforts can clinical psychologists make to increase their cultural competence regarding clients with physical health issues?

LOOKING TOWARD GRADUATE PROGRAMS

The following table is a list of index entries in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018) relevant to the topics in this chapter (see page 25 in Chapter 1 for more information about the *Insider's Guide*). Keep in mind a few things as you consider these listings: They are current as of the 2018/2019 edition of the *Insider's Guide* (which typically revises every two years); they may have changed as faculty members retire, take new jobs, or develop new interests; and a graduate program's own website may have more extensive or up-to-date information.

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
AIDS/HIV	AIDS/HIV	Health Psychology/Behavioral Medicine
Biofeedback/Relaxation	Biofeedback	Miscellaneous/Other—Biofeedback
Cardiovascular Health	Cardiovascular Psychology	Miscellaneous/Other—Oncology Psychology
Chronic Disease/Illness		

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Diabetes Health Psychology/ Behavioral Medicine Nicotine/Tobacco/ Smoking Oncology/Cancer Care Pain Management Psychoneuroimmunology Weight Management/ Obesity Miscellaneous and Other—Hypertension Miscellaneous and Other—Medical Issues	Health Psychology/ Behavioral Medicine/ Stress Oncology/Cancer Care Pain Management Weight Management/ Obesity Miscellaneous—Diabetes	

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

Health Psychology
http://www.apa.org/pubs/journals/hea/index.aspx

Health Psychology Review
http://www.tandfonline.com/loi/rhpr20#.V08Uc_krLX4

International Journal of Clinical and Health Psychology
http://www.journals.elsevier.com/international-journal-of-clinical-and-health-psychology/

Journal of Behavioral Medicine
http://link.springer.com/journal/10865

Journal of Health Psychology
http://hpq.sagepub.com/

Rehabilitation Psychology
http://www.apa.org/pubs/journals/rep/index.aspx

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- Full-text SAGE journal articles
- Additional web resources
- Mock assessment data

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FORENSIC PSYCHOLOGY

Bryce F. Sullivan and Andrew M. Pomerantz

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DEFINITION AND HISTORY

Forensic psychology—the application of psychological methods and principles within the legal system—dates back nearly to the origins of psychology as a recognized science. Psychologists with different academic backgrounds have been drawn to the field due to the interesting and diverse ways psychology and law intersect. Social psychologists are involved in researching and applying psychological science to issues such as jury selection and jury dynamics. Cognitive psychologists bring their expertise to bear on the issues of eyewitness testimony and its accuracy. Clinical psychologists, however, are especially qualified for and often involved in forensic psychology activities because of their extensive training in assessment, treatment, and psychopathology.

With different types of psychologists conducting research and practicing professionally in forensic psychology, it is not surprising that there are disagreements on what the field should be called (Ogloff, 2002). Should it be called law and psychology, psychology and law, criminal psychology, legal psychology, or forensic psychology? For clinical psychologists involved in professional practice, the term *forensic psychology* is most commonly used and most appropriate. Clinical psychologists working in forensic settings evaluate criminal defendants' sanity, assess legal competency to stand trial, assess families in child custody disputes, predict dangerousness, serve as expert witnesses, and engage in other interactions with the legal system that are outlined in this chapter (Roesch & Zapf, 2016). As this list indicates, the training and expertise of clinical psychologists fulfill many demands of the legal system, especially regarding clinical assessment. Otto and Heilbrun (2002) called forensic psychological assessment "the psychological assessment of persons for the purpose of assisting the legal fact finder" (p. 5), and clinical psychologists are

Web Link 19.1
American Academy of Forensic Psychology

Learning Objectives

19.1 Define forensic psychology.

19.2 Review the history of forensic psychology in the United States.

19.3 Summarize major issues that can arise when psychologists conduct forensic evaluations.

19.4 Describe the types of evaluations a forensic psychologist may be asked to perform.

19.5 Explore the types of treatment forensic psychologists may provide to forensic clients.

19.6 Investigate activities of forensic psychologists outside of assessment and treatment.

uniquely trained to conduct assessments that are comprehensive, valid, and reliable.

A variety of training opportunities involve the intersection of forensics and psychology, and there is little consensus about exactly what should be included in the training of a forensic psychologist. Within clinical psychology, some doctoral programs offer an elective course or two in forensic issues, others offer a specialized forensic track among other tracks, and still others are entirely devoted to forensic psychology and offer joint graduate degrees in law (JD) and psychology (PhD or PsyD). Additionally, various predoctoral and postdoctoral internships offer various types of forensic training (Burl, Shah, Filone, Foster, & DeMatteo, 2012; Marczyk, Krauss, & Burl, 2012).

Psychologists assisted the courts as early as the beginning of the 1900s, but it wasn't until the latter half of the 1900s that forensic psychology became a major subdiscipline within psychology (A. M. Goldstein, 2003). **Hugo Munsterberg** was among the first major promoters of the use of psychology in the legal arena. He wrote a best-selling book in 1908 titled ***On the Witness Stand*** with chapters on the memory of witnesses, untrue confessions, the detection of crimes, and other topics of interest to modern-day forensic psychologists. Munsterberg's aggressive promotion of psychology in the legal arena went beyond what were then its scientific limits, and his brash writings alienated prominent members of the legal field. Nevertheless, Munsterberg made significant contributions to forensic psychology by showing how psychological science could be applied to the field of law (Bartol & Bartol, 2006; Bazar, 2015a). Munsterberg also supported the growth of forensic psychology by mentoring students, such as the psychologist and attorney William Marston. In 1923, Marston presented testimony in a case that created the first standard for expert testimony (*Frye v. United States*, 1923) and paved the way for extensive use of psychological expertise in the courts.

In some ways, the development of forensic psychology mirrors the development of clinical psychology. Like clinical psychology, the field emerged around the end of the 19th century, but its most significant growth occurred after World War II. Although psychologists in the United States had been expert witnesses for the courts as early as the 1920s, it was not common for psychologists to serve in this role until several decades later. By the middle of the 20th century, psychologists were conducting evaluations to assist the court in matters of mental status, legal competence, and the sanity of defendants. Psychologists even presented expert evidence and were cited in the Supreme Court's brief in the watershed 1954 school desegregation case *Brown v. Board of Education* (1954). In this case, which led to the desegregation of public schools in the United States, psychologists presented research showing that segregation had a negative effect on the self-esteem of children, and their research was persuasive to the court (A. K. Hess, 2006a). Further signs of the growth of forensic psychology in recent decades

include the emergence of professional journals, standards of practice, training programs, professional certifications, and the **American Psychological Association Division 41 (American Psychology-Law Society)**. Forensic psychology has grown considerably in recent decades (DeMatteo, Marczyk, Krauss, & Burl, 2009; Heilbrun & Brooks, 2010; Roesch & Zapf, 2016).

Forensic work has evolved into a unique specialty area for clinical psychologists. Unlike most clinical work, it places the clinical psychologist in the midst of an adversarial relationship between parties in conflict with each other. As a result, "the good feelings and positive relationships common in the psychotherapeutic relationship do not exist in the forensic setting" (Knapp & VandeCreek, 2006, p. 161). Similarly, clinical psychologists working in forensic settings must acknowledge that when they assess an individual, their goal is not necessarily to help that individual per se but to help the legal system by offering information and recommendations about the individual. For example, when conducting forensic assessments, clinical psychologists must interact with clients in a "detached, neutral, and objective" manner rather than the "supportive, empathic" approach they might take in most other clinical settings (A. M. Goldstein, 2003, p. 6; see also E. Hunt & Kimonis, 2015). Moreover, the trust that characterizes, and to some extent facilitates, so many other clinical relationships is necessarily lacking in forensic work, especially assessments. This approach is necessary because the likelihood of clients falsely presenting themselves is simply too great (A. M. Goldstein, 2003).

Web Link 19.2
APA Division 41

Web Link 19.3
Journal of Forensic Psychology Practice

FORENSIC ACTIVITIES OF CLINICAL PSYCHOLOGISTS

Assessment Activities

Clinical assessment is a prevalent activity among clinical psychologists who work in forensic settings. The most common assessments are in the areas of child custody, competency, and presentencing evaluations, but assessments are also conducted in some parole decisions, personal injury lawsuits, workers' compensation hearings, preemployment screenings of legal personnel, guardianship cases, and Social Security disability evaluations (Bush & Heck, 2018; Dawson & Jacquin, 2015a; J. M. Wood, Garb, Lilienfeld, & Nezworski, 2002; Younggren, Boisvert, & Boness, 2016). A small number of forensic psychologists have even become involved in evaluations to determine whether people need emotional support animals (Boness, Younggren, & Frumkin, 2017). It is always important to consider ethical issues such as the limits of confidentiality when conducting a psychological examination, especially when conducting forensic evaluations, because the role of the psychologist is not as clear as it is in more common clinical situations. For example, when a psychologist conducts a police officer preemployment screening evaluation, the police department—not the applicant—is considered the client, and the information gained in the evaluation is shared with the police department senior staff. In all forensic evaluations, the question "Who is the client?"—the person being assessed, an attorney, the court system, or another party—should be clarified by the clinical psychologist from the outset (A. M. Goldstein, 2003; Hunt & Kimonis, 2015).

Additionally, in all forensic evaluations the limits of confidentiality should be made clear to the individual being evaluated by way of a written and oral informed consent process (American Psychological Association, 2002). Because the clinical psychologist cannot ensure complete confidentiality of the report once it is shared (typically with the party who requested it), care is taken to exclude from the report personal information that is not relevant to the reason the evaluation is being conducted. Although the report and test data are intended to remain confidential, an evaluation conducted for the courts or other legal entities may, unfortunately, become publicly available once it leaves the hands of clinical psychologists. For example, after professional boxer Mike Tyson was denied a license to box in Nevada in 2002 by the Nevada Athletic Commission, a comprehensive psychological evaluation of Tyson was conducted to evaluate anger management problems and other issues. Even though Tyson attempted to have the report kept private, it was made public and posted on various websites. Similarly, in the arena of preemployment testing, it seems that every spring the Wonderlic intelligence quotient (IQ) test results of some low-scoring college football players being evaluated by the National Football League (NFL) are leaked to the press. In 2012, the problem was so pronounced that the leaking of draft prospect test scores resulted in the NFL commissioner threatening "significant discipline" for anyone leaking these confidential scores (Krawcqynski, 2012).

Although forensic assessment represents a specialized area of clinical practice, many of the techniques used in forensic assessments are the same as those used more generally by clinical psychologists, including the clinical interviews, intelligence tests, and personality tests covered in earlier chapters of this book. A survey of highly credentialed and experienced forensic psychologists examined the use of specific tests for specific forensic purposes (Lally, 2003). The survey indicated that many of the tests receiving the strongest endorsement ("Recommended") were objective personality tests such as the Minnesota Multiphasic Personality Inventory-2 (MMPI-2) and intelligence tests with proven reliability and validity such as the Wechsler Adult Intelligence Scale (WAIS). On the other hand, many of the tests receiving the most negative rating ("Unacceptable") were projective personality tests such as the Rorschach Inkblot Method, Thematic Apperception Test (TAT), and projective drawing tests. It was also notable that these forensic experts strongly endorsed a multimethod approach to assessment in which numerous tests, rather than any single test, were used. Whatever assessment techniques are used in forensic evaluations, they should be solidly based on the science of psychology and not pseudoscience—the imposter of real science (Lilienfeld & Landfield, 2008b). Heilbrun and Brooks (2010) note that there has been progress in the development of special assessment measures for legal capacities. Table 19.1 describes the findings of the survey for several common forensic assessments. We will discuss these assessments and others in the remainder of this section.

While some of the assessment techniques used by forensic psychologists are the same as those used by clinical psychologists more generally, some specialized assessment skills are particularly important for forensic psychologists. Sageman (2003) describes three such skills: knowledge of the legal issues, addressing the demands of the legal system, and skill related to litigation (involvement in court cases). The first skill, knowledge of the legal issues, might involve an understanding of criminal responsibility, the definition of competency, and the definition of insanity. The second skill, addressing

Table 19.1 Opinions of Forensic Experts Regarding Psychological Tests Used for Various Forensic Purposes

Type of Assessment	Recommended Tests	Unacceptable Tests
Not guilty by reason of insanity (i.e., mental state at time of offense)	MMPI-2; WAIS	Projective drawings; TAT; sentence completion tests
Predicting dangerousness (i.e., risk for violence)	Psychopathy Checklist—Revised	Projective drawings; TAT; Rorschach; sentence completion tests
Competency to stand trial	WAIS; MacArthur Competence Assessment Tool—Criminal Adjudication	Projective drawings; Rorschach; TAT; sentence completion tests; Millon Clinical Multiaxial Inventory

Source: Adapted from Lally (2003), 491–498.

the demands of the legal system, often entails striving for neutrality regarding a client, predicting the future as well as possible, and gathering data about the case in a thorough and unbiased way. The third skill applies to litigation (court cases) and involves an understanding of each attorney's strategy, providing appropriate testimony, and deferring to others involved in the case when appropriate.

An issue that arises with greater frequency in forensic psychology than in clinical psychology more generally is *malingering*, in which the person being evaluated exaggerates or "fakes" symptoms in order to achieve some benefit (Chmielewski et al., 2017; Drogin & Biswas, 2016; R. Rogers, 2008). Many factors could motivate malingering. For example, a defendant in a criminal case might malinger in an attempt to avoid or reduce punishment for a crime. Or, a person might malinger in an attempt to seem more impacted (more depressed, more traumatized, more memory-impaired) in an attempt to maximize the compensation resulting from a lawsuit or to obtain worker's compensation or disability benefits (Aguerrevere et al., 2018; Pierson & Rosenfeld, 2015; S. Young, Jacobson, Einzig, Gray, & Gudjonsson, 2016). Assessment of malingering is a challenge, but psychologists often look for suspicious patterns in test responses. For example, in an intelligence test, the person might produce scores that fall far below observed levels of intellectual functioning, or fail items so easy that even a severely impaired person can typically perform them, or fail easy items yet succeed at difficult items. On personality testing, the person might endorse a high number of extreme or absurd symptoms, endorse only blatant (but not subtle) symptoms, or score high on the validity scales of such measures as the MMPI-2 (Knoll & Resnick, 2006; P. J. Resnick, 2007; P. J. Resnick & Knoll, 2008; R. Rogers, Sewell, Martin, & Vitacco, 2003).

An interesting question that has arisen among forensic psychologists who conduct assessments is whether data from social media sites—for example, Facebook, Instagram,

or Twitter—or elsewhere on the Internet can or should be accessed as part of the assessment. Indeed, this practice is on the rise and has become relatively common (Pirelli, Otto, & Estoup, 2016). Some forensic psychologists argue that information from social media can aid in difficult determinations, like a defendant's mental state at the time of the crime, which may have taken place weeks or months prior (Vitacco, Gottfried, & Batastini, 2018). One survey of mental health professionals found that 39% had in fact searched the Internet for information about a forensic examinee, and only 26% opposed this practice (Pirelli, Hartigan, & Zapf, 2018). The same survey also found that the majority of respondents agreed that the examiner should inform the examinee up front that the Internet search is taking place, give the examinee the opportunity to address information found in the search, and make clear in the final report the use of the Internet search. Another survey of forensic assessors similarly found that most had used social media data as part of an evaluation but that they often questioned the authenticity of that data and found it less helpful than other, more traditional and potentially more reliable sources of information such as medical records, police reports, criminal history, and examinee self-report (Coffey, Batastini, & Vitacco, 2018).

Predicting Dangerousness

When a legal case centers on the violent or dangerous behavior of an individual, the judge or jury often consider the likelihood that the individual will behave violently or dangerously again in the future. Clinical psychologists are often recruited to assist in **predicting dangerousness**, especially when the individual has a history of mental illness in addition to aggressive behavior (J. P. Singh, 2015). Decisions in these cases, which can include verdicts or sentences, are particularly challenging for the court system.

Clinical psychologists can assess the potential for future danger in a variety of ways, most of which can be placed into one of two categories: **clinical prediction methods** or **statistical prediction methods** (statistical prediction methods are also known as **actuarial prediction methods**; e.g., Dawes et al., 1989). In the clinical method, assessors use psychological tests, clinical interviews, clinical experience, and their personal judgments to make determinations of future dangerousness. In contrast, in the statistical method, assessors predict dangerousness according to a statistical or actuarial formula compiled from a comparison of an individual's characteristics with known correlations to future dangerousness (Grove, Zald, Lebow, Snitz, & Nelson, 2000). Such variables are typically objective rather than subjective and could include

- dispositional variables, such as age, race, sex, social class, and personality variables;
- historical variables, such as history of violence, work history, mental health history, and criminal history;
- contextual variables, such as current social supports, presence or availability of weapons, and current stress level; and
- clinical variables, such as current mental disorders, drug and alcohol abuse, and overall level of functioning. (Monahan, 2003)

By its individual and subjective nature, the clinical method differs from clinician to clinician, and these differences between clinicians tend to make the method less reliable. Additionally, the clinical method leaves room for biases and other forms of human errors common in subjective decision making. Statistical methods tend to be less flexible but more empirically sound due to their objectivity. In fact, Grove and his colleagues (2000) conducted a meta-analysis of studies in the area of clinical prediction and found that statistical predictions were about 10% more accurate than clinical predictions.

Experts in the field have debated the use of clinical versus statistical methods of prediction of dangerousness for decades, and some have suggested that the two approaches can be combined. In fact, a clinical prediction of dangerousness, although primarily subjective in nature, may incorporate some objective data about the client. Similarly, a statistical prediction of dangerousness, although primarily objective in nature, may incorporate subjective variables that the evaluator converts into objective ratings (e.g., personality traits, behavioral tendencies). Thus, predicting dangerousness may be seen as existing on a continuum encompassing both clinical and actuarial components (Litwack, Zapf, Groscup, & Hart, 2006; Skeem & Monahan, 2011).

Whereas clinical psychologists may emphasize clinical or statistical methods in particular cases, research has established some general guidelines for predicting dangerous behavior. Factors associated with dangerousness in the empirical literature include age (with younger people at higher risk), arrest record (with those with more arrests at higher risk), weapon availability (with those who have available weapons at higher risk), social support (with those low in social support at higher risk), and psychological symptoms (with those with active psychosis symptoms at higher risk). The research literature is clear on the fact that although there is a small relationship between the presence of mental illness and violence, the key psychological factors include the presence of substance abuse, psychotic symptoms, or psychopathy (Hemphill & Hart, 2003; Litwack et al., 2006; McNiel et al., 2002; Monahan, 2003).

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Photo 19.1 Clinical psychologists who work in forensic settings are often asked to predict the dangerousness of a client. In your opinion, are clinical prediction methods or statistical prediction methods preferable?

One of the most significant challenges associated with predicting dangerousness is the **base rate** problem (Grove et al., 2000). Specifically, in predicting dangerousness, base rates constitute a problem because they are so low. To explain, base rates report the incidence of something. For example, about 1% of the population meets the criteria for a diagnosis of schizophrenia; this means that schizophrenia has a base rate of 1%. Making predictions about phenomena with low base rates is very difficult when compared with making predictions about those with higher base rates, because the higher the base rate is, the more likely a prediction will be accurate based solely on chance occurrence. Trying to predict whether a person will show the active symptoms of schizophrenia based on genetic and environmental variables would be very difficult—even with full knowledge of all possible predictor variables. The same is true for predicting dangerousness.

When predicting dangerous or violent behavior, the clinical psychologist attempts to accurately identify those who actually will become dangerous (a "true-positive"

BOX 19.1

Metaphorically Speaking

If You've Bought Car Insurance, You Understand Clinical and Statistical Methods of Predicting Dangerousness

Why does car insurance cost more for some drivers than for others? The reason is that over the years, car insurance companies have compiled statistics indicating that drivers with some characteristics are more likely to get into accidents than are drivers with other characteristics. Using these statistics, the companies can predict with remarkable accuracy the likelihood of accidents for drivers with various combinations of characteristics. If you happen to have characteristics that make you, statistically speaking, a high risk to the insurance company, your premiums will rise to reflect this risk. The characteristics that car insurance companies consider include the driver's gender, age, driving record, and driving education/training, among others. As a specific example, statistics enable car insurance companies reliably to predict that male drivers under age 25 with a history of accidents and no driver's education training are especially likely to get into accidents in the future. If any of these driver characteristics were different—older than 25, female, no history of accidents, driver's education classes—the prediction would change accordingly.

When clinical psychologists attempt to predict the dangerousness of a forensic client, they can use a similarly statistical (also known as actuarial) method. In other words, they can base their predictions on objective characteristics of the individual that have been empirically tied to specific rates of dangerous behavior, including gender, age, history of dangerous or criminal behavior, history of substance abuse, and presence of psychological symptoms such as psychosis or psychopathy. Just as car insurance companies can create a formula to calculate the odds that drivers with certain characteristics will drive dangerously, clinical psychologists using the statistical method can create a formula to calculate the odds that forensic clients with certain characteristics will behave dangerously.

Imagine that a car insurance company decided to abandon the statistical approach to predicting accidents and replaced it with a method in which the insurance agents get to know each driver individually and then use their professional judgment to determine how likely the driver is to get into accidents in the future. In other words, each driver would meet with a car insurance agent and undergo an interview and a series of tests. The outcome of the interview and tests, rather than the driver's demographic characteristics or driving history, would determine the individual's insurance rate. This method is essentially similar to the clinical method that clinical psychologists might use when assessing a client for dangerousness. The clinical method involves the clinical psychologist assessing the forensic client as an individual and forming predictions about future dangerousness from the data obtained via the interviews and psychological tests included in this evaluation. Compared with the statistical method, the clinical method involves less reliance on aggregate group data (e.g., young adults, older adults, males, females) and more reliance on the professional judgment of the clinical psychologist. Thus, the clinical method not only allows for more consideration of individual characteristics but also allows for more subjectivity on the part of the assessor, who can be susceptible to bias and error.

If you owned a car insurance company, which method of assessment—a statistical formula based on the groups to which the driver belongs, or getting to know each driver individually—would you expect to be more accurate in the prediction of accidents? If you were a clinical psychologist working in forensic settings, which method of assessment—statistical or clinical—would you expect to be more accurate in the prediction of dangerousness?

prediction) and those who will not become dangerous (a "true-negative" prediction). It is impossible to make these identifications correctly in every case, and some mistakes can be far worse than others. In actual practice, to protect society, the clinical psychologists might lean toward identifying someone as more likely to be violent than he or she really is (a "false-positive" prediction). On the other hand, the cost of identifying someone as less violent than he or she really is (a "false-negative" prediction) can be devastating if that person goes on to kill or harm others, so clinical psychologists might be especially cautious about making such a prediction. Imagine, for example, the decision-making process involved in releasing from the court's supervision someone convicted of child abuse. The court and the clinical psychologist hired to assist in the process would try to predict this type of dangerous behavior accurately, but because it is so rare in the population (a low base rate), it is easy to see how false-positive decisions might be made because the risk of making false-negative decisions is so much higher for society. Even though clinical psychologists are far from perfect in predicting dangerousness, research has shown that mental health professionals can make better than chance predictions of violence in both short- and long-range situations (J. M. Wood et al., 2002).

Not Guilty by Reason of Insanity (NGRI)

According to the U.S. legal system, people who commit crimes should be punished only when they committed the crime of their own free will. If an individual was unable to control their actions due to a mental disorder—even if such actions were criminal in nature—the individual would not be held responsible for the crime but would rather be found **not guilty by reason of insanity (NGRI)**. Thus, a clinical psychologist's assessment of a defendant's mental state at the time of an offense can become a crucial component to legal proceedings. When defendants are found NGRI, they are not imprisoned in the usual sense, because such penalties were created as a means of deterrence and social control (Darley, Fulero, Haney, & Tyler, 2002). Imprisonment would not effectively deter someone who did not commit the crime of their own free will. Instead, the institutionalization of individuals found NGRI, which may take place in inpatient psychiatric units or similar facilities, is for their treatment and tailored for the individual's particular constellations of emotional and behavioral problems.

There are many misconceptions of the NGRI defense, some of which stem from highly publicized court cases in which the public perceives that the defendant may have sought an NGRI label solely to avoid punishment. An example is John Hinckley being found NGRI

after the 1981 attempted assassination of President Reagan. G. B. Melton et al. (2007) identified four common misconceptions regarding the NGRI defense. First, the general public may think this defense is used frequently, and, second, people may think it is often successful. In fact, less than 1% of felony jury trials involve an NGRI defense, and only about 25% of those trials result in an NGRI verdict (A. M. Goldstein, Morse, & Shapiro, 2003; Zapf, Golding, & Roesch, 2006). Third, people may think the NGRI defendant is often released without being institutionalized, but this is very uncommon. Typically, the person found NGRI is institutionalized in a mental hospital almost as long as he or she would have been incarcerated if convicted of the crime. Fourth, there is a misperception that individuals found NGRI are more dangerous than they really are. In the case of John Hinckley, more than three decades after the assassination attempt and over the objections of the U.S. Justice Department, in June 2009, a federal judge ordered that Hinckley be allowed extended visits to his mother's home and additional freedoms. He has been living at St. Elizabeth's Hospital in Washington, D.C., since the attempt on President Reagan's life in 1981.

The **M'Naghten test**, the first legal standard for the insanity defense in the history of the American legal system, emerged during the mid-19th century. In the M'Naghten case, the defendant (Daniel M'Naghten) was charged with attempting to kill the prime minister of England. This attempted murder was due to persecutory delusions M'Naghten held about Britain's Tory party and its leader, Prime Minister Sir Robert Peel. Instead of killing the intended target, however, he killed the prime minister's secretary, who was nearby. Although English common law had already recognized that "lunatic idiots" who acted as "wild beasts" due to their mental state were not responsible in the eyes of the law, it was the M'Naghten case that developed the formal test of insanity (R. G. Meyer & Weaver, 2006, p. 116). The court ruled,

> To establish a defense on the ground of insanity, it must be clearly proved that, at the time of the committing of the act, the party accused was laboring under such a defect of reason, from disease of mind, as not to know the nature and quality of the act he was doing; or, if he did know it, that he did not know he was doing what was wrong. (G. B. Melton et al., 2007, p. 206)

This test focused primarily on the cognitive ability of the defendant, and in the United States the "policeman at the elbow" or "irresistible impulse" test was added to include a volitional component. According to this test, it was argued that a person was legally insane if the person was unable to control his or her behavior even when a police officer was at his or her elbow. In 1954, the "product test" was instituted by the District of Columbia Court of Appeals in *Durham v. United States* (1954). The court wrote, "An accused is not criminally responsible if his unlawful act was the product of mental disease or defect" (pp. 874–875). This test was overturned in 1972 due to its definitional ambiguity, and legal experts turned their attention to clarifying criminal responsibility tests.

More recent tests of criminal responsibility were developed by the legal community. One of these legal tests was created by the American Law Institute (1962) and is used by many jurisdictions. The test reads,

> A person is not responsible for criminal conduct if at the time of such conduct as a result of mental disease or defect he lacks substantial capacity either to

> appreciate the criminality of his conduct or to conform his conduct to the requirements of the law. (Cited in R. G. Meyer & Weaver, 2006, p. 119)

By 1972, the American Law Institute rule was embraced by the majority of federal jurisdictions and many states (R. G. Meyer & Weaver, 2006). As a reaction to John Hinckley being found NGRI on all counts following his attempt to assassinate President Reagan in 1981, the conformity clause was removed from federal law, and the burden of proof about insanity became the responsibility of the defense, whereas burden of proof previously fell on the prosecution. Currently, legal standards for NGRI vary at the state and federal levels, but they all require that a mental disease or defect be present and operating at the time of the offense (A. M. Goldstein et al., 2003).

A recent legal development is the provision by which a defendant can be found **guilty but mentally ill**. About half of U.S. states have such a provision, and it is intended to provide a middle ground for jurors contemplating whether to assign criminal responsibility (and punishment) to defendants whose mental status is in question (Kimonis, 2015).

Another aspect of NGRI cases that is gaining increasing attention is the fact that different geographic areas, especially in different countries, may approach the issue of criminal responsibility (on which NGRI cases are based) quite differently. One review of the policies and practices of many countries found that some, including the United States and Canada, view the question as dichotomous (the person is criminally responsible or not), whereas others, such as Spain and the Netherlands, view the question on a continuum (the person is criminally responsible to a certain degree). Some psychological disorders, such as disorders that involve psychosis, were universally seen as signs of diminished criminal responsibility, but others, for example, personality disorders and psychopathy, were seen differently in different places (Grossi & Green, 2017). Evaluations involving the NGRI or guilty-but-mentally-ill strategies are especially challenging for clinical psychologists because they involve the assessment of mental state at the time of the crime, which may have been weeks or months prior to the time of the evaluation (Kimonis, 2015). Not surprisingly, there is often disagreement among experts' conclusions (Guarnera & Murrie, 2017). One study examined 483 psychological reports written by forensic evaluators about defendants in real court cases and found that when there were multiple reports about the same defendant, the conclusions of those reports agreed with each other only about half of the time. That rate was even lower when the evaluators were on opposite sides of the trial or when alcohol or drugs were involved in the crime. That rate was higher when diagnosis of a psychotic disorder (such as schizophrenia) or mental retardation (now known as intellectual disability) was present (Gowensmith, Murrie, & Boccaccini, 2013). Another study focused on cases with multiple evaluators in which the question was whether someone originally found not guilty by reason of insanity should be released. In such cases in which the person was released despite disagreement among the evaluators, 71.4% were rehospitalized. But in such cases in which the person was released after unanimous agreement among the evaluators, only about half as many (34.5%) were rehospitalized (Gowensmith, Murrie, Boccaccini, & McNichols, 2017).

Child Custody Evaluations

"**Child custody evaluations** may be the most complex, difficult, and challenging of all forensic evaluations" (Otto, Buffington-Vollum, & Edens, 2003, p. 179; see also Kuehnle,

Kirkpatrick, & Drozd, 2015). Indeed, child custody is an area fraught with challenges due to the importance of the decision and the competing interests of the parents. It is not, however, the parents' interests that are most important in child custody decisions. Instead, custody decisions are made based on a legal principle known as the "best interest of the child doctrine." This has not always been the case. In the 1880s, the father was typically given custody. This later gave way to the "tender years" doctrine (K. D. Hess, 2006) in which the mother was favored in custody disputes, particularly in the case of younger children. Because children, even adolescents, are not considered by the court to be able to make sound judgments (*Parham v. JR*, 1979), parents must make decisions for them. In a divorce situation, when the parents are in a custody dispute, even the parents are often not able to make well-reasoned decisions for the children. In such cases, the courts will sometimes appoint a **guardian ad litem** to make sure the rights of the child are protected. The guardian ad litem is a neutral party, often an attorney unaffiliated with either parent, appointed to avoid decisions being made by individuals who would have conflicts of interest.

After the death of Vickie Lynn Marshall (i.e., Anna Nicole Smith) in 2007, a Miami attorney was appointed by the court to serve as guardian ad litem to her 5-month-old daughter. A guardian ad litem was also appointed for Michael Jackson's children (Prince Michael, Paris, and Blanket) following his death in 2009. In both of these cases, following the death of the celebrity, legal claims were made by individuals claiming to be the parents of the various children. The duty of the guardian ad litem is to represent the interests of the children. Custody and legal cases such as these, in which many parties have interests that might conflict with the best interest of the child, highlight the important role a guardian ad litem plays in child custody disputes.

By the 1990s, child custody evaluations were likely to be conducted only in more complex cases involving psychopathology and parental conflict. Increasingly, these complex cases were handled by highly trained psychologists who based their findings on the scientific literature and solid research rather than on their clinical experiences or hunches (E. Ellis, 2008). Psychologists can participate in the child custody process in a variety of ways: as an evaluator of the parents or children, investigator, expert witness, mediator, research scientist, or treatment provider. To provide guidance to psychologists in these challenging situations, the American Psychological Association (1994) developed 16 standards for conducting child custody evaluations. These standards (see Table 19.2) outline both the purpose of and guidelines for conducting a child custody evaluation.

Even with such guidelines, child custody evaluations can be daunting tasks. Unlike other evaluations, in which only one person or one capacity is in question, child custody cases involve multiple people (and the relationships among them) and multiple capacities of each. Child custody evaluators not only meet with individuals directly involved in the case, such as the parents and children, but often talk with third parties or collateral sources as well, such as extended family members, teachers or day-care staff, mental health professionals, the family physician, and others who may know the family (Symons, 2010).

No two child custody evaluations are exactly alike, but some techniques are commonly used. More than 90% of evaluators conduct each of the following during a typical evaluation: clinical interviews with each parent, a clinical interview with the children, an observation of parent–child interaction, psychological testing with the parents, and a review of relevant documents. Additionally, more than 60% of evaluators conducted psychological testing with the children (Bow & Quinnell, 2001), although the parent

Table 19.2 American Psychological Association Guidelines for Conducting Child Custody Evaluations

I. Orienting Guidelines: Purpose of a Child Custody Evaluation
   1. The primary purpose of the evaluation is to assess the best psychological interests of the child.
   2. The child's interests and well-being are paramount.
   3. The focus of the evaluation is on parenting capacity, the psychological and developmental needs of the child, and the resulting fit.

II. General Guidelines: Preparing for a Child Custody Evaluation
   4. The role of the psychologist is that of a professional expert who strives to maintain an objective, impartial stance.
   5. The psychologist gains specialized competence.
   6. The psychologist is aware of personal and societal biases and engages in nondiscriminatory practice.
   7. The psychologist avoids multiple relationships.

III. Procedural Guidelines: Conducting a Child Custody Evaluation
   8. The scope of the evaluation is determined by the evaluator, based on the nature of the referral question.
   9. The psychologist obtains informed consent from all adult participants and, as appropriate, informs child participants.
   10. The psychologist informs participants about the limits of confidentiality and the disclosure of information.
   11. The psychologist uses multiple methods of data gathering.
   12. The psychologist neither overinterprets nor inappropriately interprets clinical or assessment data.
   13. The psychologist does not give any opinion regarding the psychological functioning of any individual who has not been personally evaluated.
   14. Recommendations, if any, are based on what is in the best psychological interests of the child.
   15. The psychologist clarifies financial arrangements.
   16. The psychologist maintains written records.

Source: © American Psychological Association (1994).

interviews, child interview, and parent–child observation are seen as more important than tests by most psychologists surveyed (Bow, 2006). The most common tests for children in custody evaluations include IQ tests and projective personality tests such as the TAT and the Children's Apperception Test, whereas the most common test for adults is the MMPI-2. Some specialized custody-specific tests are also often used (E. Ellis, 2008; Hagen & Castagna, 2001).

Competency to Stand Trial

The U.S. legal system is based on a principle in English common law dating back to the 17th century that holds that a person accused of a crime cannot be tried in court unless that person is mentally fit or **competent to stand trial**. Although the present focus is on the competency to stand trial, there are other important legal competencies such as those concerned with writing wills, entering into a contract, and consenting to medical treatment. All these types of legal competency can be affected by the mental status of the individual. In 1899, a federal appeals court gave competency a constitutional standing. The court ruled that "it is not 'due process of law' to subject an insane person to trial upon an indictment involving liberty or life" (*Youtsey v. United States*; cited in R. G. Meyer & Weaver, 2006, p. 94). The two primary reasons for the competency doctrine are that it provides for a dignified criminal process and it supports Sixth Amendment protections, including the right to effective counsel, the right to confront one's accusers, and the right to present evidence. A person with a mental impairment could be sufficiently disturbed so as to be unable to exercise these rights without a change in mental status.

The competency-to-stand-trial standard established in *Dusky v. United States* (1960) is the standard used throughout most of the United States. The Supreme Court stated that to stand trial, a defendant must have "sufficient present ability to consult with his attorney with a reasonable degree of rational understanding and a rational as well as factual understanding of proceedings against him" (cited in G. B. Melton et al., 2007, p. 127). This standard covers both the defendant's present or current capacity to understand the criminal process and the defendant's present or current ability to function within the process, otherwise known as *competency*. The definition of competency does not, however, include the defendant's prior mental status or mental status at the time of the offense; rather, such past mental states would be included in discussions of criminal responsibility. In other words, the time period of concern in assessments of competency is the present, and the time period of concern for criminal responsibility is the point in the past when the offense occurred. For competency, the defendant must have a rational understanding of the proceedings, and this is related to a person's current cognitive ability and mental state. Many psychologists use a clinical interview in which the defendants are assessed regarding their understanding of the legal system to determine whether they can participate in the legal process in their defense. They are evaluated regarding the particular situations and demands of the case. Various psychological instruments are often used in assessing competence, and there are some specialized tests available. The practice of using standard psychological tests instead of specialized competency instruments has been criticized because these tests have not been validated for assessing competency (Zapf, Viljoen, Whittemore, Poythress, & Roesch, 2002). Psychologists conducting competency-to-stand-trial evaluations also utilize information from third parties with detailed knowledge of the person being evaluated, including family members, friends, medical professionals, and employers (T. R. Phillips & Otto, 2015). In recent years, the use of videoconference to conduct these evaluations has been on the rise (Luxton & Lexcen, 2018).

Between 2% and 8% of all felony defendants undergo competency evaluations—a large percentage considering the large and increasing number of felony arrests (Zapf & Roesch, 2006). Some experts in the field recommend a **functional evaluation** of competency, based on the idea that competency to stand trial depends on the cognitive and

psychological demands of that particular case. Some cases are more complex than others, and some require different competencies than others, so competency should be judged relative to the demands of the case at hand. Thus, a defendant found to have a mental disorder is not necessarily incompetent for a given case, and a defendant found not to have a mental disorder is not necessarily competent for a given case (Zapf & Roesch, 2006).

Some jurisdictions have training programs to increase defendants' understanding of the legal system, because this is crucial to competency. Medications are also commonly used to treat defendants who are not competent to stand trial. While the incompetent defendant is hospitalized or incarcerated, the whole range of treatments available may be used to restore competency. Unless competency is established, the trial remains postponed. A person cannot, however, be held longer than "the reasonable period of time necessary to determine" whether competency will be attained in the near future (*Jackson v. Indiana*, 1972; cited in G. B. Melton et al., 2007, p. 137). Nor can a defendant be held longer than he or she would have been held if found guilty of the crime.

Culture can play an important role in competency evaluations. Some of the criteria used to make the determination of competency, including IQ tests and the *Diagnostic and Statistical Manual of Mental Disorders* diagnostic process, can be biased against certain ethnic and racial groups. Moreover, intelligence is defined differently by different cultural groups, so the intelligence reflected by scores on IQ tests derived from European American values does not necessarily capture such attributes as social competence, creativity, musical ability, physical fitness, and others that may be considered relevant to IQ in some cultures. Any competency evaluation should consider culture a factor in the interpretation of assessment data (Tsytsarev & Landes, 2008).

Web Link 19.4
Journal of Forensic Psychiatry & Psychology

Competency-to-stand-trial evaluations are not the only kind of competency evaluations that psychologists can perform in a forensic evaluation. For example, in cases involving the death penalty, the issue of competency to be executed, which centers on the person's mental capacity to appreciate the reason and consequences of execution, may arise (Phillips & Otto, 2015).

Commitment to Mental Institutions

Civil commitment to a mental institution is a process by which a person is involuntarily hospitalized by civil authorities for the welfare of the person and others. Civil commitments have a lower legal standard than criminal cases, but the standards are much higher now than in the pre-1970s era, when individuals were often institutionalized with little more than a family request (G. B. Melton et al., 2007). Critics such as Thomas Szasz (1963) have questioned the practice of hospitalizing individuals against their will and have argued that the practice can be abused to remove certain groups of undesirable people from society. In addition to losing their civil rights, the care of individuals who were institutionalized was often very poor. These poor standards of care and limited legal controls on civil commitments led to changes in the 1970s in the criteria, procedures, and outcome of civil commitments. The issue of civil commitment typically focuses on inpatient treatment but extends to court-ordered outpatient treatment as well, in which an individual is required by authorities to attend psychotherapy sessions on a regular basis but is not required to live in a psychiatric facility (Pinals & Mossman, 2015).

Decisions about involuntary commitment generally follow a two-step process in which jurisdictions initially grant emergency admission as determined by a clinician or

other authority, but judicial approval is required for continued confinement. All states have criteria for involuntary commitment that require the following conditions be met: The fundamental and first criterion for all state statutes is that the individual have a mental disorder. Second, some statutes require that the person be unable to appreciate the need for treatment because of the disorder. Third, the person must be dangerous to self or others due to the mental disorder. Fourth, most states also consider an individual's ability to care for basic needs as part of the commitment criteria. Fifth, some states require that there be a need for treatment. Finally, the commitment must occur in the least restrictive environment (G. B. Melton et al., 2007). Clinical psychologists, particularly because of their training in assessment, are well equipped to help determine the extent to which these criteria apply to individuals in question. In addition to the assessment measures they use, clinical psychologists often seek input from third parties, such as family members, friends, or employers of the person being evaluated, to help them with their decision (Pinals & Mossman, 2015).

Treatment and Other Forensic Activities

Treatment of Forensic Clients

Clinical psychologists are involved in the legal system in many ways, and providing treatment to address problems related to criminal activity or its consequences is vital among them. Treatment in a forensic context can take many forms. For example, as noted earlier, the primary purpose of hospitalizing individuals found NGRI is the treatment of their psychological and psychiatric symptoms. In the case of individuals found not competent to stand trial, the focus of treatment is typically increasing their knowledge of the judicial system relevant to their case. The treatment of both classes of individuals is primarily focused on short-term goals leading toward the person's reintroduction into society and community-based services in the first case and leading toward a trial in the second case. Individuals incarcerated in the prison system, on the other hand, typically have more long-range treatment needs.

The goals of sentencing criminals to incarceration include retribution, deterrence, incapacitation, and rehabilitation (G. B. Melton et al., 2007). Although the clinical psychologist may not be able to "cure" the offender in a global sense, treatment methods can be used to address the emotional and behavioral problems that contribute to unlawful behavior. Efforts in recent years have also been directed toward establishing "practice guidelines" that provide guidance to psychologists on the best treatment options (Heilbrun & Brooks, 2010). When working with criminal offenders, clinical psychologists and other therapists typically focus on four basic therapy tasks, either alone or in combination: crisis management, maintenance, outpatient psychotherapy, and targeted programs (Mobley, 2006). Management treatments are implemented when there is a crisis of some type. This would include someone engaging in self-harming or violent behaviors. Treatments of this type are brief, followed by a referral to long-range treatments. Maintenance treatments are used with incarcerated individuals who have more long-standing problems such as intellectual disability (intellectual developmental disorder, formerly known as mental retardation) and schizophrenia, and they often are held and treated in special housing units. Many types of outpatient psychotherapy are used

with prison populations, and successes or failures occur much as they do in more traditional settings. Programs targeting specific problems such as alcohol and drug treatment and sex offender treatment are also often available for incarcerated individuals. Such programs provide some advantages over outpatient treatment, including the ability to have more contact hours with the clients. For example, sex offender programs often last a year or longer and usually involve cognitive-behavioral strategies aimed at a number of treatment targets such as acceptance of responsibility, empathy, social functioning, deviant sexual interests, substance abuse, and relapse prevention (Marshall, 2006). From a rehabilitation perspective, reduced recidivism is a primary goal with all treatment interventions used in a prison population.

The cultural competence that clinical psychologists incorporate into their practices is especially relevant in prison settings. Incarcerated populations include a wide range of ethnic groups, including disproportionately high numbers of some minority groups (e.g., Layde, 2004; Tseng, Matthews, & Elwyn, 2004). Thus, whether serving as a therapist or an assessor, a clinical psychologist working with prisoners must be sensitive to the cultural background of each client, as well as to the prison culture in which the client currently lives.

Expert Witnesses

Clinical psychologists, along with psychiatrists and social workers, are the professionals who most commonly provide expert testimony to the court about mental health issues, and together they are involved in more than 1 million cases per year (G. B. Melton et al., 2007). Dating back to before the days of Hugo Munsterberg's grandiose claims about the benefits of using psychology in the legal system, attorneys and others have expressed cynicism about the role of psychology in the courts. This is partially due to the fundamental philosophical differences between science and law, but it is also rooted in a mistrust that arises from the idea that psychologists and other experts are "selling" their testimony. Although such a practice would be highly unethical, some claim that psychologists and other legal experts sell their opinions to the highest bidder and become "hired guns" (A. K. Hess, 2006b) or "whores of the court" (Hagen, 1997). Subtle or not-so-subtle pressures exist for an **expert witness** to testify in a manner that may be biased to support the case of the attorney who hired them. Given these pressures, it is important for psychologists to "*be effective advocates for their data*, whether or not that makes them effective advocates for the party that calls them to court" (G. B. Melton et al., 2007, p. 578).

As an example, consider a legal custody dispute in which two parents, Monica and Greg, are battling for custody of their two young children. Monica's attorney learns that 10 years ago, Greg received outpatient psychotherapy for depression. Monica's attorney contacts Dr. Redding, a clinical psychologist with expertise in depression, to serve as an expert witness in the custody trial. Because Monica and her attorney are hiring Dr. Redding to support their side of the argument, Dr. Redding may feel pressured to provide testimony that calls Greg's parenting potential into question and therefore strengthens Monica's case. That is, if Dr. Redding states that depression usually recurs throughout the life span, that it often gets more severe over time, or that children raised by parents who have been depressed have more significant problems than do those

raised by nondepressed parents, the court might be persuaded to place the children with Monica. However, Dr. Redding should not make these statements if they do not reflect an accurate, impartial appraisal of the current state of research on depression. Even if Dr. Redding's objectivity and honesty disappoint the attorney paying her to serve as an expert witness, Dr. Redding is obligated as an expert witness to state only that which is supported by valid and reliable findings relevant to the depression and parenting issues involved in this case.

An expert witness is approved for the court through a process called voir dire. In the **voir dire** process, the expert's education, training, and professional experiences are examined. Sometimes both sides stipulate—that is, accept without requiring proof—that a psychologist meets the criteria needed to be an expert witness, and other times the matter is hotly contested by the opposing attorneys. In the process of establishing the expert qualifications of psychologists, attorneys commonly ask many informational questions of the psychologist, such as, "Do you specialize in any particular areas of psychology?" "What has been your experience in these areas of professional practice?" "Are you licensed?" "Are you board certified?" and "Describe your education" (Ewing, 2003).

An expert witness is allowed to present significantly different kinds of testimony in court than a lay witness, who can testify only to what he or she saw, heard, or otherwise experienced. Experts approved by the court can provide both facts and opinions, can make inferences from facts, can provide information about disputed facts, and can help educate the jury or court about scientific information (Sales & Shuman, 2005). The ability to provide this range of testimony makes the expert witness a powerful contributor to the court's deliberations.

An important concern for clinical psychologists serving as expert witnesses is the admissibility of their testimony. In other words, what are the standards by which their testimony will be allowed or disallowed into the legal proceedings? Prior to 1993, expert scientific testimony was admissible in court if it was generally accepted in the expert's field. This standard was known as the Frye test, named after the 1923 *Frye v. United States* case. A 1993 Supreme Court ruling (*Daubert v. Merrell Dow Pharmaceuticals, Inc.*) supplanted the Frye test with a new set of standards. Under the current standards (known as **Daubert standards**, or Federal Rules of Evidence standards), the admissibility of evidence or testimony in court is based on its reliability and validity rather than its general acceptance in the field. So, according to the current standards, expert testimony can be allowed if it is deemed reliable and valid even if it is outside the boundaries of general acceptance in the field. For example, in psychological assessment, some question the admissibility of projective personality tests such as the Rorschach Inkblot Method as well as objective personality tests such as the Millon Clinical Multiaxial Inventory-III (J. M. Wood et al., 2002). Under the Daubert/Federal Rules of Evidence standards, the judge serves as the gatekeeper in deciding whether the expert evidence passes the threshold of reliability and validity. Since Daubert/Federal Rules of Evidence have been in effect, greater sophistication has been needed by judges who are required to evaluate research methods and experimental inferences (Faigman & Monahan, 2005).

Expert witnesses must also prepare themselves to deal with the possibility of cross-examination. Cross-examination is an especially stressful component of providing expert testimony, because it requires expert witnesses to defend their statements in response to challenging questions posed by opposing attorneys. Expert witnesses might

be asked to explain why they chose certain assessment measures, how reliable or valid those measures are, what scientific or professional sources support their conclusions, what alternate conclusions might be worth considering, and more. It is generally wise for expert witnesses to handle cross-examinations with honesty, admission of limitations, and a calm rather than combative or retaliatory approach (D. Meyer & Drogin, 2015).

BOX 19.2

Considering Culture

Cultural Competence in Forensic Clinical Psychology

Cultural competence is crucial in all areas of clinical psychology, and forensic work is no exception. Just as the general U.S. population represents a great variety of ethnicities and other cultural variables, the population of forensic clients is similarly diverse. Whether working in prisons or psychiatric institutions, or as independent consultants, whether conducting assessments, therapy, or other professional duties, clinical psychologists are ethically obligated to interact with clients in a culturally sensitive manner (American Psychological Association, 2002). In recent years, cultural issues within forensic psychology settings have received an increasing amount of attention from professionals in the field (e.g., Layde, 2004; Tseng et al., 2004). Among the primary concerns noted by these authors are the following:

- *Linguistic issues.* Many forensic clients may not be fluent in English. This lack of fluency may interfere with standard assessment procedures (including interviews and both written and oral tests) and therapy efforts. When available, multilingual clinicians or translators can address this problem effectively. One survey of psychologists who perform forensic evaluations found that over half had experience evaluating a Latino/a person with limited English proficiency. Some of these had used ad hoc interpreters (e.g., a client's family member, an employee or coworker of the evaluator who speaks Spanish), a practice that is inconsistent with guidelines and inferior to the use of a proper, professional interpreter with no connection to any party in the case (Canales, Kan, & Varela, 2017).
- *Culturally specific definitions of sanity and insanity.* Particular behaviors commonly viewed as symptoms of psychosis within mainstream U.S. culture may be viewed differently by members of various cultural groups. For example, hallucinations based on religious experiences may be understood as normal rather than abnormal in some cultural groups.
- *Cultural limitations of psychological tests.* As discussed in earlier chapters of this book, such issues as readability, language, and cultural representation in normative groups may limit the applicability of certain tests with members of certain forensic clients.

(Continued)

(Continued)

- *Lack of familiarity with the U.S. judicial system.* Some forensic clients, including many who have recently immigrated, may not fully understand the laws and judicial processes of the United States. This is especially relevant in competency-to-stand-trial evaluations but can also be relevant in numerous other aspects of forensic work.
- *Distrust of the legal system.* Forensic clients who are members of oppressed groups may be unwilling to cooperate with the efforts of the clinical psychologist working in a forensic setting. In such cases, the clinical psychologist should attempt to appreciate the historical factors that may play a role in the client's behavior.

Layde (2004) recommends that cultural competency receive heavy emphasis in the training of forensic mental health professionals. Specifically, Layde suggests that such training should include not only seminars, readings, and other didactic experiences but also supervised professional experiences involving direct exposure to forensic clients of diverse cultural backgrounds.

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Photo 19.2
Clinical psychologists working with forensic clients must be sensitive to varying abilities, knowledge, and beliefs stemming from diverse cultural backgrounds.

Patient's Rights

Throughout the United States, civil commitment laws exist that allow people to be committed to mental hospitals against their will. These laws are designed to provide mental health care for the individual and are not intended to be punitive. There have been, however, abuses of patients in mental hospitals in which many people were committed based on what today would be trivial reasons. In the first half of the 1900s, people could be committed by various legal mechanisms in different jurisdictions. Depending on the locale, the commitment process might be controlled by a judge, a justice of the peace, a lay jury, an asylum board, or, in some cases, only medical certification (Holstein, 1993). During this same period, medical treatments that were thought to hold great promise were attempted but were found to have significant negative side effects and could even cause patients significant harm. The new psychiatric medications of the 1950s caused long-term side effects such as tardive dyskinesia, with its lifelong symptoms of repetitive and involuntary movements, primarily of the face. Lobotomies were all too commonly conducted in mental hospitals in the mid-20th century. In a

lobotomy, the prefrontal cortex of the brain was purposefully damaged in the hope that the patient's maladaptive behaviors would decrease. Although normally reserved for the most disturbed patients, they were also used on patients with serious intellectual limitations, despite lack of scientific evidence that they were effective on any group of patients (Valenstein, 1986).

In the 1960s and 1970s, attitudes and legal opinions grew in support of improving the legal rights of patients. In 1975, the Supreme Court (*O'Connor v. Donaldson*) weighed in on the civil commitment of mental patients by ruling that having a mental illness was not a sufficient criterion for civil commitment. Instead, the Supreme Court made it the law of the land that the person must both have a mental disorder and be considered dangerous. As long as an individual is not dangerous and is capable of surviving with the help of family and friends, involuntary civil commitment is not an option. Thus, the methods described above regarding the ways clinical psychologists assess dangerousness, as well as the way they assess psychopathology more generally, are crucial to cases involving involuntary commitment to a mental institution.

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Photo 19.3 Clinical psychologists serve as expert witnesses in a wide variety of court cases. In your opinion, what are the greatest challenges involved in this responsibility?

In addition to examining liberties related to civil commitment, the courts have also identified other rights of patients in mental hospitals. In *Youngberg v. Romeo* (1982), the Supreme Court extended the rights of mental patients to include "reasonably safe conditions of confinement, freedom from unreasonable bodily restraints, and such minimally adequate training as reasonably may be required by these interests" (p. 307). The right to refuse medications was also provided by the federal courts in the 1970s. This right to refuse medication can be overridden if the patient has an emergency need for medications, is a danger to self or others, or is unable to make competent decisions (Stafford, 2003).

Consultations With Law Enforcement

Clinical psychologists are increasingly employed by law enforcement agencies to perform a variety of duties (Kitaeff, 2011; Scrivner, 2006). For example, clinical psychologists often conduct preemployment evaluations on candidates for law enforcement jobs or fitness-for-duty evaluations for current officers whose psychological status is under question. Clinical psychologists also work as consultants on a wide range of issues with police departments, and some even work full-time in large police departments. In addition, clinical psychologists often provide direct therapeutic interventions to officers, especially when the stress or trauma of the job is adversely affecting them. Clinical psychologists can be a great benefit to police departments by providing training in areas such as stress reduction, dealing with mentally disturbed people, multicultural training, and other areas of expertise.

Most psychologists who work with police departments do not engage in criminal profiling, despite the portrayal of psychologists in numerous movies and TV shows on this topic. When profiling is used, its purpose is to assist the investigators in providing

information on the perpetrator of the crime based on psychological factors. Although the results of criminal profiling shown in movies and television shows tend to be dramatic and unrealistic—often involving hunches and speculation rather than scientific or other forms of reliable and valid data—there have been some advances in this area of forensic science (Turvey, 2015; J. H. White, Lester, Gentile, & Rosenbleeth, 2011).

Of all the ways psychologists work with police departments, conducting **preemployment evaluations** is the most common, although such evaluations lack consistency and standardization (Dantzker, 2011). These assessments usually include objective personality tests such as the MMPI-2 to assess psychopathology, a measure of cognitive or problem-solving ability such as the Wonderlic Personnel Test, and a clinical interview. There are also tests specifically designed for preemployment screening of police officers, such as the Inwald Personality Inventory. When psychological tests are used in the selection process, there is always the possibility that the candidates will "fake good" on the test, thus making the results of the test difficult if not impossible to interpret (C. J. Rees & Metcalfe, 2003). The goal of the evaluations is to assess the extent to which the candidate has any psychological problems, cognitive limitations, or personality characteristics that might interfere with the duties of a police officer.

Fitness-for-duty evaluations are another type of assessment that clinical psychologists conduct with police personnel (Scrivner, 2006). These evaluations are usually requested after an extremely stressful experience by an officer, such as being shot or witnessing traumatic events in the line of duty. They are also sometimes used to assess whether an officer is able to continue working while he or she is receiving psychotherapy or medications for depression, anxiety, or other psychological problems. These evaluations are often more extensive than the preemployment exams, including more detailed clinical interviews, interviews with significant others, interviews with fellow officers and superiors, and the administration of a variety of psychological tests.

CHAPTER SUMMARY

Forensic psychology is a growing specialty area for clinical psychologists. Although mental health professionals have been active in the legal system for about a century, opportunities for clinical psychologists have developed rapidly in recent decades. In particular, the assessment skills of clinical psychologists have been put into practice in forensic psychology for various legal purposes, including the prediction of dangerousness of defendants, the determination of mental status when defendants plead not guilty by reason of insanity, the assessment of family members in child custody cases, the evaluation of defendants' competency to stand trial, and the evaluation of individuals for whom involuntary commitment to a mental institution is being considered. Clinical psychologists also provide treatment to incarcerated individuals and others in forensic settings. Additionally, clinical psychologists serve as expert witnesses in court cases and assist law enforcement agencies via assessment of officers and applicants, direct therapeutic services for officers and their families, and consultation to law enforcement organizations.

KEY TERMS AND NAMES

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CRITICAL THINKING QUESTIONS

1. In your opinion, what is the most fundamental difference between the forensic and therapeutic roles for a clinical psychologist?
2. If you were a clinical psychologist asked to predict the dangerousness of an individual, to what extent would you rely on clinical prediction methods versus statistical (or actuarial) prediction methods?
3. In what ways might an assessment of an individual's competency to stand trial differ from an assessment of an individual pleading NGRI?
4. If you were a clinical psychologist conducting a child custody evaluation, on which sources of data would you rely most heavily when making your recommendations?
5. If you were a clinical psychologist conducting preemployment evaluations for a police department, what assessment data would make you most likely to recommend that the department not hire an individual?

LOOKING TOWARD GRADUATE PROGRAMS

The following table is a list of index entries in the *Insider's Guide to Graduate Programs in Clinical and Counseling Psychology* (Sayette & Norcross, 2018) relevant to the topics in this chapter (see page 25 in Chapter 1 for more information about the *Insider's Guide*). Keep in mind a few things as you consider these listings: They are current as of the 2018/2019 edition of the *Insider's Guide* (which typically revises every two years); they may have changed as faculty members retire, take new jobs, or develop new interests; and a graduate program's own website may have more extensive or up-to-date information.

Research Areas	Specialty Clinical and Practica Sites	Program Concentrations and Tracks
Forensic/Psychology & Law	Divorce/Child Custody	Forensic/Psychology & Law—40
Malingering	Forensic	
Miscellaneous and Other—Juvenile Justice	Miscellaneous—Juvenile Justice	
	Miscellaneous—Juvenile Justice/Delinquent Youth	

KEY JOURNALS

Links available at **edge.sagepub/pomerantz5e**

Journal of Threat Assessment and Management
http://www.apa.org/pubs/journals/tam/index.aspx

Law and Human Behavior
http://www.apa.org/pubs/journals/lhb/index.aspx

Psychology, Public Policy, and Law
http://www.apa.org/pubs/journals/law/index.aspx

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- eFlashcards
- Culture expert interviews
- In My Practice demonstration video
- Full-text SAGE journal articles
- Additional web resources
- Mock assessment data

GLOSSARY

ABAB design: A particular type of research design for case studies in which a treatment is alternately applied and removed

ABCDE model: In cognitive therapy, a model for understanding and recording the impact of cognitions on emotions

ABCDS of weight loss: A common approach used by health psychologists in treating people with excess weight, including *A*ctivity increase, *B*ehavior change, *C*ognitive change, *D*ietary change, and *S*ocial support

ABCs of relapse prevention: A component of substance abuse relapse prevention involving an examination by the client and the health psychologist of the events that occur before (*A*ntecedents) and after (*C*onsequences) consuming the substance (*B*ehavior)

abnormality: Forms of behavior that are outside the normal range; often labeled mental disorders, psychiatric diagnoses, or psychopathology

Academy of Psychological Clinical Science: An organization of graduate programs subscribing to the clinical scientist model of graduate training in clinical psychology

acceptance: In acceptance and commitment therapy, allowing unpleasant thoughts, feelings, and other internal sensations to run their course without fighting against them

acceptance and commitment therapy (ACT): An increasingly popular form of psychotherapy emphasizing the acceptance rather than avoidance of an unpleasant internal psychological experience

acculturation: Response or adaptation to a new cultural environment, particularly with regard to adopting elements of the new culture or retaining elements of the original culture

achievement: In contrast to intelligence (what a person *can* accomplish intellectually), what a person *has* accomplished, especially in academic subjects

achievement tests: Type of standardized tests used to measure how much students have learned in specific, clearly defined content areas, including but not limited to reading, mathematics, science, and social studies

activating event: The *A* in the ABCDE model of cognitive therapy; the occurrence that initiates the sequence of mental events that may prompt illogical cognitions or beliefs

actuarial prediction methods: An approach to predicting dangerousness in which assessors predict dangerousness according to a statistical or actuarial formula compiled from a comparison of an individual's characteristics with known correlations to future dangerousness; also known as statistical prediction methods

all-or-nothing thinking: In cognitive psychotherapy, a common thought distortion in which the individual irrationally evaluates everything as either wonderful or terrible, with no middle ground or "gray area"

allegiance effects: In psychotherapy outcome research, the influence of researchers' own biases and preferences on the outcome of their empirical studies

American Psychological Association (APA): Prominent professional organization for psychologists of which many clinical psychologists are members

American Psychological Association Division 41 (American Psychology-Law Society): A division of the American Psychological Association devoted to issues related to forensic psychology

American Society for the Advancement of Pharmacotherapy (American Psychological Association, Division 55): A division of the American Psychological Association devoted to issues related to prescription privileges for psychologists

anal stage: In psychodynamic psychotherapy, the second of the psychosexual developmental stages, and the stage from which issues of control may emerge

analogue design: A research design that involves an approximation of the target client or situation as a substitute for the actual clients or situations

analogue direct observation: In contrast to naturalistic direct observation, a type of behavioral observation that typically takes place in the clinic room, where the real-life situation is simulated

anxiety hierarchy: In exposure therapy, a rank-ordered list of anxiety-provoking stimuli to which the client will be gradually exposed

aspirational: In contrast to enforceable, an approach to psychology ethics that emphasizes aspirational ideals describing how psychologists should strive to conduct themselves

assertiveness training: A form of behavior therapy based on classical conditioning in which clients improve on timid, apprehensive, or ineffectual social behaviors

assessment: A common professional activity of clinical psychologists in which clients are evaluated or appraised via clinical interviews, testing, observation, or other means

attending behaviors: Fundamental aspects of listening, including eye contact, body language, vocal qualities, and verbal tracking

attention: In the context of psychotherapy outcome research, the interest a therapist takes in the client that may constitute a common factor across therapies

automatic thoughts: In cognitive psychotherapy, cognitions that take place instantly and without any deliberation

aversion therapy: A form of contingency management therapy that emphasizes the use of punishment as a consequence for an unwanted behavior

Albert Bandura: A leading researcher in the area of observational learning, modeling, and social learning

base rate: In forensic psychology (and more broadly in all psychology), the rate at which a behavior occurs, whereby phenomena with a low base rate (low frequency) are more difficult to predict (e.g., dangerous behaviors)

baselines: In behavior therapy, pretreatment data on a problem behavior used as a basis for later comparisons

Aaron Beck: A leader in the field of cognitive psychotherapy and the developer of the concepts of the cognitive triad and common thought distortions, among others

Judith Beck: A leading figure in contemporary cognitive psychotherapy and the daughter of Aaron Beck

Beck Depression Inventory-II (BDI-II): The current edition of a brief objective test that assesses depression

behavior rating scales: In clinical child psychology, standardized forms that parents, teachers, or other adults complete regarding a child's presenting problems

behavior therapy: An approach to psychotherapy emphasizing empiricism, observable and quantifiable problems and progress, and a lack of speculation about internal mental processes

behavioral: An approach to clinical psychology emphasizing empiricism, observable and quantifiable problems and progress, and a lack of speculation about internal mental processes

behavioral activation: A form of behavior therapy for depression, the goal of which is to increase the frequency of behaviors that are positively reinforcing to the client

behavioral assessment: In contrast to traditional personality assessment, an approach to assessment that assumes that client behaviors are not signs of underlying issues; instead, those behaviors *are* the problems

behavioral consultation: An indirect alternative to behavior therapy whereby the therapist serves as a consultant to an individual such as a parent, teacher, or supervisor who ultimately implements the behavioral interventions with the client in the natural setting

behavioral medicine: An area of specialization for clinical psychologists and other health professionals focused on the impact of behavior on wellness and involving the integration of knowledge from a wide variety of social sciences, including psychology, sociology, and anthropology, with knowledge from the medical disciplines

behavioral observation: The direct, systematic observation of a client's behavior in the natural environment; also known as naturalistic observation

being self-aware: A general skill for clinical interviewers involving the ability to know how the interviewer might affect others interpersonally and how others tend to relate to him or her

belief: The *B* in the ABCDE model of cognitive therapy; the illogical cognition linking the activating event to the emotional consequence

Bender Visual-Motor Gestalt Test—Second Edition (Bender-Gestalt-II): The current edition of a brief neuropsychological screen in which clients are asked to copy simple geometric designs

Insoo Kim Berg: A leading figure in solution-focused family therapy

between-group design: A research design in which participants in different conditions, one of which is often a control group, receive different treatments

Alfred Binet: A pioneer in the assessment of intelligence who co-created the Binet-Simon scale in the early 1900s, which evolved into the Stanford-Binet Intelligence Scales

biofeedback: A type of treatment used by health psychologists for pain management clients designed to achieve control over the body via educating patients about bodily processes of which they are typically unaware

"blank screen" role: In psychodynamic psychotherapy, the therapist role in which little personal information is revealed to facilitate transference

body language: In the context of the clinical interview, a behavior that can communicate strong nonverbal messages

Boulder model: Alternate name for the scientist-practitioner model of graduate training stemming from the historic 1949 conference of directors of training in Boulder, Colorado

boundaries: An essential concept in structural family therapy; the divisions between family subsystems

boundaries of competence: The limits of competence that a clinical psychologist has an ethical obligation not to overstep

Murray Bowen: A leading figure in family therapy who is closely associated with the concept of differentiation of self

brief psychodynamic psychotherapy: A more efficient and increasingly common version of psychodynamic psychotherapy, typically lasting about 6 months or less

burnout: A state of exhaustion that relates to engaging continually in emotionally demanding work

John Carroll: A contemporary leader in the study of intelligence who has proposed the three-stratum theory of intelligence

case studies: Research methods that involve a thorough and detailed examination of one person or situation

catastrophizing: In cognitive psychotherapy, a common thought distortion in which the individual unrealistically expects catastrophic consequences

categorical approach: In contrast to the dimensional approach, an approach to diagnosis based on the notion that an individual either has or does not have a particular disorder

James Cattell: A leading figure in the study of intelligence who proposed separate fluid and crystallized intelligences

Dianne Chambless: A leading contemporary psychotherapy researcher who has argued strongly against the idea that all psychotherapy approaches are equally efficacious and in favor of a prescriptive approach to treatment

child abuse: In the context of clinical psychology, a condition that mandates the clinical psychologist to break confidentiality (per state laws) in an attempt to protect the welfare of the child at risk

child custody evaluations: Evaluations conducted by forensic psychologists for the purpose of providing recommendations for child custody

Children's Apperception Test (CAT): A projective test similar to the Thematic Apperception Test, created specifically for children

chronic pain: Pain that lasts 6 months or longer

chronic stress: Stress levels that are consistently high and unremitting

circular causality: In contrast to linear causality, a theory typically endorsed by family therapists whereby events influence one another in a reciprocal way

civil commitment: A process by which a person is involuntarily hospitalized by civil authorities for the welfare of the person and others

clarification: A clinical interview technique designed to make sure the interviewer has an accurate understanding of the client's comments

classical conditioning: Conditioning in which an unconditioned stimulus that produces an unconditioned response is paired with a conditioned stimulus such that the conditioned stimulus elicits a similar response (labeled as the conditioned response)

clinical child psychology: An area of specialization within clinical psychology focusing on issues of children

clinical interview: A common assessment technique used by clinical psychologists involving a purposeful conversation with clients characterized by questions and answers

clinical prediction methods: An approach to predicting dangerousness in which assessors utilize psychological tests, clinical interviews, clinical experience, and their personal judgments

clinical psychology: Rigorous study and applied practice directed toward understanding and improving the psychological facets of the human experience, including but not limited to issues or problems of behavior, emotions, or intellect

clinical scale: One of 10 scales on the MMPI and MMPI-2 indicating the extent to which an individual endorses symptoms of a particular category

clinical scientist model: A model of graduate training in clinical psychology that emphasizes empirical research over practice

clinical utility: In psychological assessment, the extent to which a technique benefits the clinician and ultimately the client in a meaningful way

closed-ended question: In contrast to an open-ended question, an interview question that allows for far less elaboration and self-expression by the client but yields a quick and precise answer

closed-enrollment groups: In contrast to open-enrollment groups, therapy groups in which all members start and finish therapy together, with no new members added during the process

code of ethics: Ethical guidelines for psychologists, originally published by the American Psychological Association in 1953 and most recently updated in 2002

cognitions: The focus of cognitive psychotherapy, the way individuals interpret the events that happen to them and determine their resulting emotions; also known as beliefs, interpretations, assumptions, or thoughts

cognitive: An approach to clinical psychology emphasizing illogical thought as the foundation of psychopathology and logical thought as the foundation of psychological wellness

cognitive therapy: An approach to psychotherapy emphasizing illogical thought as the foundation of psychopathology and logical thought as the foundation of psychological wellness

cognitive triad: A component of Aaron Beck's theory of depression whereby negative thoughts about the self, the external world, and the future contribute to depression

common factors: Factors common to all forms of psychotherapy that play an active role in client improvement and may explain the dodo bird verdict in psychotherapy outcome research

common thought distortions: In cognitive psychotherapy, particular ways in which a thought, cognition, or belief can be illogical

communication patterns: An area commonly targeted by family therapists as the source of psychological symptoms and an area for improvement in family therapy

competence: Sufficient capability, skill, experience, and expertise to complete particular professional tasks adequately

competent to stand trial: A required condition of the defendant in the U.S. legal system for which forensic psychologists often conduct assessments

Comprehensive System: The leading scoring system for the Rorschach Inkblot Method, created by John Exner

conclusion: A clinical interview technique that may involve summarization, an initial conceptualization, a diagnosis, or recommendations

conditioned response: In classical conditioning, the response elicited by the conditioned stimulus after the conditioned stimulus has been paired with the unconditioned stimulus

conditioned stimulus: In classical conditioning, the stimulus paired with the unconditioned stimulus that ultimately elicits the conditioned response

conditions of worth: In humanistic psychotherapy, the stipulations that individuals may place on their positive regard of others

confidentiality: As mandated by the code of ethics of the American Psychological Association, upholding the privacy of clinical information

Conflict Tactics Scales (CTS): A questionnaire designed to assess how individuals behave when family conflicts arise, often used to explore interpersonal abuse or violence

confrontation: A clinical interview technique used when the interviewer notices discrepancies or inconsistencies in the client's comments

congruence: In humanistic psychotherapy, consistency between the real self and the ideal self; the source of mental health

content scales: In addition to the clinical scales, scales of the MMPI and MMPI-2 that provide relevant clinical information

contingencies: The "if . . . , then . . ." statements connecting actions to outcomes that organisms learn through operant conditioning

contingency management: A form of behavior therapy based on operant conditioning in which the consequences following selected behaviors are changed to produce more desirable behavior

continuing education: Learning experiences by which clinical psychologists maintain their competence throughout their careers; often required by state licensing boards for license renewal

continuing education units (CEUs): Educational credits earned by licensed clinical psychologists by attending workshops, taking courses, reading selected material, or similar means; often required by states for license renewal

control group: The condition or group in a between-subject design that receives no treatment

coping: The process of managing the demands of stress

correlational methods: Research methods that examine the relationship that exists between two or more variables and in which causality is not implied

cortisol: The human body's stress hormone, the release of which is controlled by the hypothalamic-pituitary-adrenal axis

Paul Costa: With Robert McCrae, an author of the NEO Personality Inventory-Revised

cotherapist: A therapist who coleads group therapy sessions with another therapist

counseling psychologists: Professionals with some similarities to clinical psychologists but whose work tends to emphasize less seriously disturbed clients, vocational testing, and career counseling

counterconditioning: Re-pairing a conditioned stimulus with a response that is incompatible with the previously conditioned response; an essential component of systematic desensitization

countertransference: In psychodynamic psychotherapy, transference by the therapist toward the client

crisis interview: A type of clinical interview designed not only to assess a problem demanding urgent attention (most often, clients actively considering suicide or another act of harm toward self or others) but also to provide immediate and effective intervention for that problem

cross-sectional designs: In contrast to longitudinal designs, research designs that assess or compare a participant or group of participants at one particular point in time

crystallized intelligence: The body of knowledge one has accumulated as a result of life experiences

cultural competence: For clinical psychologists, the ability to work sensitively and expertly with culturally diverse members of a heterogeneous society

cultural concepts of distress: As listed in *DSM-5*, psychological problems observed in cultural groups from various parts of the world

cultural diversity: Heterogeneity in the cultural background of members of a society

cultural fairness: The extent to which an intelligence test is based on universal rather than culture-specific concepts or methods

cultural self-awareness: An important component of cultural competence by which the clinical psychologist recognizes the uniqueness of his or her own cultural perspective

Daubert standards: In forensic psychology, current standards for admissibility of the testimony of expert witnesses

defense mechanisms: In psychodynamic psychotherapy, techniques used by the ego to manage conflict between the id and superego

Patrick H. DeLeon: A prominent proponent of prescription privileges for clinical psychologists and a former president of the American Psychological Association

dementia praecox: Emil Kraepelin's term for a cluster of psychological symptoms similar to what is currently known as schizophrenia

dependent variables: Variables in an experiment that are expected to change as a result of changes in the independent variables

Steve deShazer: A leading figure in solution-focused family therapy

detoxification: A common and important component of the treatment of substance abuse problems designed to prevent any

dangerous side effects that may result from weaning the body off the substance too quickly

developing positive working relationships: A general skill for clinical interviewers often facilitated by attentive listening, appropriate empathy, genuine respect, and cultural sensitivity

developmental perspective: An essential aspect of clinical child psychology whereby clinicians understand the child's behavior within the context of the child's developmental stage

Diagnostic and Statistical Manual of Mental Disorders (DSM): The prevailing diagnostic guide for mental health professionals, containing the definitions of all mental disorders, published by the American Psychiatric Association

diagnostic criteria: Specific lists of symptoms used to define mental disorders

diagnostic interview: A type of clinical interview in which the primary purpose is to diagnose the client's problems

dialectical behavior therapy (DBT): A form of cognitive psychotherapy—and an example of an evidence-based treatment—developed by Marsha Linehan that has been found to be effective in the treatment of borderline personality disorder

differentiation of self: In family therapy, the healthy process whereby families allow each member to become his or her own person without sacrificing emotional closeness with other members of the family

dimensional approach: In contrast to the categorical approach, an approach to diagnosis based on the issue of severity of an individual's symptoms on a continuum or dimension, rather than the presence or absence of disorder

directive: In the context of the clinical interview, an approach by which interviewers obtain the information they need by asking clients specifically for it

discrimination: In classical conditioning, a process by which the conditioned response is not evoked by stimuli that are similar to, but not an exact match for, the conditioned stimulus

disease-prone personality: A personality characterized by marked anxiety, depression, and hostility that predisposes people to stress-related illnesses such as arthritis, ulcers, and coronary heart disease

disengaged: In structural family therapy, an unhealthy type of relationship among family members resulting from overly rigid boundaries between subsystems

displacement: In psychodynamic psychotherapy, a defense mechanism in which the ego displaces an id impulse toward a safer target

dispute: The *D* in the ABCDE model of cognitive therapy; challenging the illogical belief by labeling it as a particular type of thought distortion

Division of Clinical Psychology (Division 12): Division of the American Psychological Association devoted to clinical psychology

Dorothea Dix: A pioneer of reform in the treatment of the mentally ill in the United States and elsewhere in the 1800s

dodo bird verdict: A nickname for the common research finding that different forms of psychotherapy are roughly equally effective; derived from the line in *Alice in Wonderland*, "Everybody has won and all must have prizes"

dream work: In psychodynamic psychotherapy, the process of converting the latent content of a dream to its manifest content

dreams: In psychodynamic psychotherapy, an important means by which the client's unconscious material is communicated, and a common focus of interpretation

DSM-I: The first edition of the *Diagnostic and Statistical Manual of Mental Disorders (DSM)*, published in 1952

DSM-II: The second edition of the *Diagnostic and Statistical Manual of Mental Disorders* (*DSM*), published in 1968

DSM-III: The third edition of the *Diagnostic and Statistical Manual of Mental Disorders* (*DSM*), published in 1980

DSM-III-R: A revision of the third edition of the *Diagnostic and Statistical Manual of Mental Disorders* (*DSM*), published in 1987

DSM-IV: The fourth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (*DSM*), published in 1994

DSM-IV-TR: A text revision of the fourth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (*DSM*), published in 2000

DSM-5: The fifth and current edition of the *Diagnostic and Statistical Manual of Mental Disorders* (*DSM*), published in May 2013

duty to warn: Stemming from the *Tarasoff* case, the obligation of clinical psychologists to alert people toward whom their clients have made credible, serious threats

dysfunctional thought record: A form used in cognitive psychotherapy that organizes clients' experiences into columns on a written page

eclectic: A psychotherapy orientation that involves selecting the best treatment for a given client based on empirical data from studies of the treatment of similar clients

effective new belief: The *E* in the ABCDE model of cognitive therapy; the logical belief or cognition that replaces the original illogical belief

effectiveness: In contrast to efficacy, the success of a therapy in actual clinical settings in which client problems span a wider range and are not chosen as a result of meeting certain diagnostic criteria

efficacy: In contrast to effectiveness, the success of a particular therapy in a controlled study conducted with clients who were chosen according to particular study criteria

ego: In psychodynamic psychotherapy, the part of the mind that manages conflict between id and the superego and also meets the demands of reality; the part of the mind that generates defense mechanisms

ego psychology: A variation of psychodynamic psychotherapy emphasizing the adaptive tendencies of the ego over the pleasure-based drive of the id

Albert Ellis: A leader in the field of cognitive psychotherapy and the developer of rational emotive behavior therapy and the ABCDE model

emic: In contrast to *etic*, a perspective held by some psychologists emphasizing culture-specific norms and the appreciation of clients within the context of their own culture

emotional consequence: The *C* in the ABCDE model of cognitive therapy; the feeling that results from the illogical belief about the activating event

emotionally focused therapy (EFT): A short-term humanistic therapy emphasizing the expression and acknowledgment of emotions that has garnered significant empirical evidence and popularity in recent years among individual and couples therapists

emotionally fused: In family therapy, a type of family relationship resulting from incomplete differentiation of self whereby family members remain overly emotionally connected with one another

empathy: In humanistic psychotherapy, one of the three essential therapeutic conditions; the therapist's ability to sense the client's emotions just as the client would, to perceive and understand the events of the client's life in a compassionate way

empirical criterion keying: The method of test construction used in the creation of the MMPI, which involves identifying distinct groups of people, asking all of them to respond to the same test items, and selecting items that yield different patterns of responses between groups

empirical data: An essential feature of behavior therapy that can take the form of frequencies of problem behavior at various points in therapy

endogenous disorders: Emil Kraepelin's term for disorders caused by internal factors

enforceable: In contrast to aspirational, an approach to psychology ethics that emphasizes specific rules of conduct that can be violated

enmeshed: In structural family therapy, an unhealthy type of relationship among family members resulting from overly permeable boundaries between subsystems

ethical decision making: A process by which psychologists make decisions in ethically challenging situations

Ethical Standards: The section of the code of ethics of the American Psychological Association based on enforceable rules of conduct

etic: In contrast to *emic*, a perspective held by some psychologists emphasizing the similarities between all people and deemphasizing differences between cultural groups

evidence-based assessment: An approach to assessment emphasizing those methods that have strong psychometrics, clinical utility, and normative data and are sensitive to issues of diversity such as age, gender, race, and ethnicity

evidence-based practice: The practice of clinical psychology based on empirical research evidence in combination with clinical expertise and client characteristics, culture, and preferences

Examination for Professional Practice in Psychology (EPPP): A standardized exam used by states and provinces as a criterion for licensure

exception questions: In solution-focused family therapy, a technique whereby therapists ask families to recall situations when the problem was absent or less severe

existential psychotherapy: An approach to psychotherapy, related to the humanistic approach, that centers on the premise that each person is essentially alone in the world and that realization of this fact can cause overwhelming anxiety

John Exner: Creator of the Comprehensive System of scoring for the Rorschach Inkblot Method

exogenous disorders: Emil Kraepelin's term for disorders caused by external factors

experiential avoidance: A tendency to circumvent rather than experience unpleasant thoughts (or feelings or other internal sensations) that may contribute to a variety of forms of psychopathology

experimental condition: The condition or group in a between-subject design that receives the treatment being examined

experimental method: A research process that follows a number of discrete, sequential steps including observation of events, hypothesis development, definition of independent and dependent variables, empirical testing of the hypothesis, and altering the hypothesis according to results obtained

expert witness: In forensic psychology, a clinical psychologist or other mental health professional who provides expert testimony to the court about mental health issues

exposure and response prevention: A particular form of exposure therapy—and an example of an evidence-based practice—that has received substantial empirical support for the treatment of obsessive-compulsive disorder

exposure therapy: A form of behavior therapy based on classical conditioning in which clients gradually face a feared object or situation

external validity: The generalizability of research results, or the extent to which a particular finding is valid for different settings and populations

externalizing disorders: In clinical child psychology, disorders in which the child "acts out" and often becomes a disruption to parents, teachers, or other children

extinction: In behavior therapy, the removal of an expected reinforcement that results in a decrease in the frequency of a behavior

extinction burst: In behavior therapy, the initial increase in intensity of the unwanted behavior immediately after the expected reinforcement is removed

extra-group socializing: A problematic and discouraged behavior among group therapy members; interacting as friends, romantic partners, and so on, outside the group sessions

Hans Eysenck: An important figure in the history of psychotherapy outcome research whose negative review of psychotherapy outcome in 1952 inspired many subsequent empirical studies on psychotherapy outcome

family life cycle: A six-stage theory of family development that can be modified to account for the wide variety of families that a family therapist may see

family structure: In family therapy, the implicit rules that govern family members' behavior

family therapy: A form of psychotherapy in which family members attend sessions together and a primary goal is the improvement of dysfunctional characteristics of the family system

feedback: In family therapy, action taken by a family member to return the family to a state of homeostasis

fight-or-flight response: The response that takes place when an organism perceives a threat and the body rapidly mobilizes energy reserves via the sympathetic nervous system and endocrine system to either fight or flee

Celia Fisher: A leading figure in psychology ethics who served as chair of the committee responsible for creating the 2002 revision of the code of ethics of the American Psychological Association

fitness-for-duty evaluations: Psychological evaluations conducted by forensic psychologists, often for law enforcement agencies, to assess the extent to which current employees remain fit for duty after extremely stressful or traumatic experiences or in the midst of psychological problems

five-factor model of personality: A theory of personality thought by many to identify the fundamental, shared traits of human personality on which a dimensional approach to diagnosis may be based; also the basis of the NEO-PI-R personality test

fixation: In psychodynamic psychotherapy, unsuccessful resolution of the psychological tasks of a particular developmental stage

flooding: In exposure therapy, all-at-once in vivo exposure to feared objects or situations (in contrast to graded exposure)

fluid intelligence: The ability to reason when faced with novel problems

fluid reasoning: The ability to solve novel problems; also one of five factor scores of the Stanford-Binet Intelligence Scales—Fifth Edition

forensic psychology: The application of psychological methods and principles within the legal system

formula first-session task: In solution-focused family therapy, a technique whereby clients are instructed to take note in the upcoming week of aspects of their lives they want to continue

free association: A therapy technique in which psychodynamic psychotherapists simply ask clients to say whatever comes to mind without censoring themselves at all

Sigmund Freud: The pioneer of the psychodynamic approach to clinical psychology

Freudian slips: Verbal or behavioral mistakes determined, according to psychodynamic psychotherapists, by unconscious motivations

full-scale intelligence score: A feature of most contemporary tests of intelligence indicating overall or general level of intelligence

functional evaluation: An approach to competency-to-stand-trial evaluations based on the idea that competency to stand trial depends on the cognitive and psychological demands of that particular case

functionalism: In family therapy, the belief that although psychological symptoms may appear maladaptive, they are in fact functional within the family environment of the individual

"g": An abbreviation for general intelligence, as posited to exist by Charles Spearman and others (in contrast to "s," or specific intelligences)

general adaptation syndrome: The hypothesis developed by Hans Selye stating that when confronted with a temporary stressor, the fight-or-flight system works effectively to provide energy for immediate fight-or-flight needs, but when faced with repeated or prolonged exposure to stress, the body eventually fails to adapt and becomes vulnerable to illness

General Principles: The section of the code of ethics of the American Psychological Association based on aspirational ideals describing how psychologists should strive to conduct themselves

generalization: In classical conditioning, a process by which the conditioned response is evoked by stimuli that are similar to, but not an exact match for, the conditioned stimulus

genogram: A pencil-and-paper assessment technique in family therapy involving the creation of a family tree that incorporates detailed information about the relationships among family members

genuineness: In humanistic psychotherapy, the quality in the therapist of truthfulness, realness, or congruence, in contrast to playing the therapist role falsely

Gestalt therapy: An approach to psychotherapy, related to the humanistic approach, emphasizing a holistic approach to enhancing the client's current experience and often relying on the use of role-play techniques during therapy

graded exposure: In exposure therapy, a gradual approach to exposing clients to feared objects or situations

group cohesiveness: A therapeutic factor in group therapy; feelings of interconnectedness among group members and the equivalent of the therapist–client relationship in individual therapy

group level: One of three levels of the tripartite model of personal identity emphasizing the perspective that each individual is like some others

group therapy: A form of psychotherapy in which multiple clients participate in sessions together and interpersonal interaction is typically emphasized

guardian ad litem: In child custody evaluations, a neutral party (often an attorney unaffiliated with either parent) appointed to protect the rights of the child

guilty but mentally ill: In the U.S. legal system, a relatively new trial outcome in some states that constitutes a middle ground between full criminal responsibility and a finding of not guilty by reason of insanity

Halstead-Reitan Neuropsychological Battery (HRB): A relatively popular, comprehensive battery of eight standardized neuropsychological tests

harmful dysfunction theory: A theory of abnormality stating that the definition of disorder should include aspects of harmfulness (which is more socially determined) and dysfunction (which is more scientifically determined)

Starke Hathaway: With J. C. McKinley, one of the creators of the Minnesota Multiphasic Personality Inventory

health insurance/managed care: A method of payment for psychological services involving a third-party payer, in contrast to client self-pay

health psychology: A subdiscipline of behavioral medicine that deals specifically with how psychological processes (e.g., cognitions, moods, social networks) influence health and illness

here and now: A focus in group therapy on the present interpersonal interactions with fellow group members rather than events that have happened in clients' lives outside the group

heterogeneity: In the context of cultural diversity, the variety of cultural backgrounds among members of a society, both between and within particular cultural groups

heterogeneous groups: Therapy groups in which no single common characteristic (such as a diagnosis) is shared by all members

hierarchical model of intelligence: A model of intelligence in which specific abilities ("s") play an important role but are all at least somewhat related to one another and to a global, overall, general intelligence ("g")

homeostasis: In family therapy, the notion that systems have the ability to regulate themselves by returning themselves to a comfort zone or "set point"

homework: An important aspect of cognitive psychotherapy whereby therapists assign clients behavioral or written tasks

homogeneous groups: Therapy groups in which all members share a common characteristic, such as a diagnosis

hope: In the context of psychotherapy outcome research, the positive expectations that may constitute a common factor across therapies

humanistic: An approach to clinical psychology deriving primarily from the theories of Carl Rogers and Abraham Maslow, emphasizing the tendency toward healthy growth within each individual

humanistic play therapy: A form of psychotherapy with children emphasizing reflection of feelings in the context of a genuinely empathic and unconditionally accepting therapeutic relationship to facilitate self-actualization

humanistic therapy: An approach to psychotherapy deriving primarily from the theories of Carl Rogers and Abraham Maslow, emphasizing the tendency toward healthy growth within each individual

hypothalamic-pituitary-adrenal (HPA) axis: A system of structures (i.e., hypothalamus, pituitary gland, adrenal gland) that is activated by stress and controls the release of cortisol

hypotheses: In cognitive psychotherapy, according to Aaron Beck, the category of unproven theories in which thoughts, beliefs, and cognitions belong (in contrast to proven facts)

id: In psychodynamic psychotherapy, the part of the mind that generates and seeks immediate satisfaction for all the pleasure-seeking, selfish, indulgent, animalistic impulses

ideal self: In humanistic psychotherapy, the self that an individual could experience if he or she fulfilled his or her own potential, in contrast to the real self

identified patient: In family therapy, the family member whose symptoms are most obvious or problematic to other members

idiographic approach: In contrast to a nomothetic approach, an approach to research emphasizing or revealing the unique qualities of each person

imaginal exposure: In exposure therapy, exposure to anxiety-provoking objects via imagination, in contrast to in vivo exposure

imitation: In behavior therapy, a way in which clients can benefit from observational learning simply by mimicking the modeled behavior

implosion: In exposure therapy, all-at-once imaginal exposure to feared objects or situations (in contrast to graded exposure)

in vivo exposure: In exposure therapy, exposure to anxiety-provoking objects in real life, in contrast to imaginal exposure

incongruence: In humanistic psychotherapy, a discrepancy between the real self and the ideal self; the source of psychopathology

independent variables: Variables in an experiment that are manipulated by the experimenter and are typically hypothesized to influence dependent variables

index scores: A feature of most contemporary tests of intelligence indicating intellectual ability in one of a small number of broad areas

individual level: One of three levels of the tripartite model of personal identity emphasizing the perspective that each individual is like no other

inferential: A characteristic of psychodynamic psychotherapy referring to the clinician's reliance on deduction or conjecture rather than empirical or directly observable information

informed consent: The ethically mandated process of informing an individual about proposed activities (e.g., as a therapy client or research participant) and obtaining the individual's voluntary consent before proceeding with the activities

insight: A primary goal of psychodynamic psychotherapy; making the unconscious conscious

intake interview: A type of clinical interview in which the primary purpose is essentially to determine whether to "intake" the client to the setting where the interview is taking place

integrative: A psychotherapy orientation that involves blending techniques to create an entirely new, hybrid form of therapy

intelligence: Aptitude or intellect; the exact definition has long been a subject of debate among psychologists and may or may not include speed of mental processing, sensory capacity, abstract thinking, imagination, adaptability, capacity to learn through experience, memory, reasoning, or inhibition of instinct

intelligence tests: Psychological tests that measure a client's intellectual abilities

internal validity: The extent to which the change in the dependent variable is due solely to the change in the independent variable

internalizing disorders: In clinical child psychology, disorders that involve maladaptive thoughts and feelings more than disruptive outward behavior

interpersonal interaction: An emphasis of most forms of group therapy whereby therapists attempt to improve clients' abilities to relate to others in healthy ways

interpersonal learning: A therapeutic factor in group therapy; learning from the in-group interpersonal experience

interpersonal therapy (IPT): A specific, manualized, contemporary form of psychodynamic psychotherapy that emphasizes interpersonal relationships and has received empirical support for the treatment of depression

interpretation: In psychodynamic psychotherapy, the therapist's attempt to formulate and discuss with the client a hypothesized connection between unconscious material and client behavior

introspection: The process of looking inside the mind for evidence of mental processes or therapeutic change, rejected by behaviorists for its lack of objectivity

knowledge: General information accumulated over time via personal experiences, including education, home, and environment; also one of five factor scores of the Stanford-Binet Intelligence Scales—Fifth Edition

Emil Kraepelin: A pioneer of diagnostic categorization in mental health who was one of the first to assign formal labels to particular clusters of symptoms

latent content: In psychodynamic psychotherapy, the raw, unconscious thoughts and feelings represented by a dream

law of effect: The behavioral principle that actions followed by pleasurable consequences are more likely to recur, whereas actions followed by unpleasant consequences are less likely to recur

licensure: Status granted by states to clinical psychologists—once they have met educational, experiential, and exam-related requirements—that allows them to identify themselves as members of the profession and to practice independently

linear causality: In contrast to circular causality, a theory typically endorsed by individual therapists whereby events from the past cause or determine events in the present in a unidirectional manner

Marsha Linehan: The developer of dialectical behavior therapy, which has been found effective in the treatment of borderline personality disorder

listening: A simple yet vital task for the clinical psychologist, especially in the context of the clinical interview

longitudinal designs: In contrast to cross-sectional designs, research designs that emphasize changes across time, often making within-group comparisons from one point in time to another

Luria-Nebraska Neuropsychological Battery (LNNB): A relatively popular, comprehensive battery of 12 standardized neuropsychological scales

magnification/minimization: In cognitive psychotherapy, a common thought distortion in which the individual overemphasizes the importance of negative events and underemphasizes the importance of positive events

manifest content: In psychodynamic psychotherapy, the actual plot of the dream as remembered by the dreamer, which represents the latent content

manualized therapy: Forms of psychotherapy that follow a therapy manual, which provides detailed instructions for all phases of the treatment of a specific disorder; typically used in outcome studies supporting evidence-based treatments

marriage and family therapists (MFTs): Professionals with some similarities to clinical psychologists but who typically obtain a master's degree and focus on clinical work with couples and families

Abraham Maslow: A pioneer of the humanistic approach to clinical psychology

Robert McCrae: With Paul Costa, an author of the NEO Personality Inventory-Revised

Richard McFall: Author of the "manifesto" that served as the foundation for the clinical scientist model of graduate training in clinical psychology

Robert McGrath: A prominent proponent of prescription privileges for clinical psychologists and former president of the American Society for the Advancement of Pharmacology (Division 55 of the American Psychological Association)

J. C. McKinley: With Starke Hathaway, one of the creators of the Minnesota Multiphasic Personality Inventory

medical model of psychopathology: A method of defining mental disorders in which each disorder is an entity defined categorically and features a list of specific symptoms

Donald Meichenbaum: The creator of self-instructional training, a treatment method intended to help impulsive and disruptive children gain greater control over their behavior

mental filtering: In cognitive psychotherapy, a common thought distortion in which the individual ignores positive events while focusing excessively on negative events

mental status exam: A type of clinical interview often used in medical settings, the primary purpose of which is to assess quickly how the client is functioning at the time of the evaluation

Stanley Messer: A leading contemporary psychotherapy researcher who has argued strongly in favor of the dodo bird verdict and against a prescriptive approach to treatment

meta-analysis: A statistical method of combining results of separate studies (translated into effect sizes) to create a summation (or, statistically, an overall effect size) of the findings

metacognitive therapy: A relatively new form of cognitive therapy emphasizing thoughts about one's own thoughts, rather than thoughts about external events, as causal factors in psychopathology

microaggressions: Comments or actions made in a cross-cultural context that convey prejudicial, negative, or stereotypical beliefs and may suggest dominance or superiority of one group over another

Theodore Millon: A widely recognized scholar on personality disorders and the creator of the MCMI personality tests

Millon Clinical Multiaxial Inventory-IV (MCMI-IV): Current version of a comprehensive objective personality test that emphasizes personality disorders

mind reading: In cognitive psychotherapy, a common thought distortion in which the individual presumes to know that others are thinking critically or disapprovingly, when knowing what others think is, in fact, impossible

mindfulness: A key component of many recent forms of cognitive therapy promoting full engagement with one's own internal mental processes in a nonconfrontational way

Minnesota Multiphasic Personality Inventory (MMPI): A popular and empirically sound objective personality test for adults, originally published in 1943

Minnesota Multiphasic Personality Inventory-Adolescent (MMPI-A): Revision of the MMPI for adolescent clients, published in 1992

Minnesota Multiphasic Personality Inventory-2 (MMPI-2): Current version of the most widely used comprehensive objective personality test, published in 1989

Salvador Minuchin: A leading figure in structural family therapy

miracle questions: In solution-focused family therapy, a technique whereby therapists ask families to imagine life without the problem

mixed-group design: A research design that combines aspects of between-group and within-group designs

MMPI-2 Restructured Form (MMPI-2-RF): A relatively brief 2008 revision of the MMPI-2 in which the clinical scales include fewer items due to the omission of overlapping items

M'Naghten test: The first legal standard for the insanity defense in the history of the American legal system; based on a legal standard from England

Christiana Morgan: One of the creators of the Thematic Apperception Test, a popular projective personality test

motivational interviewing (MI): A contemporary variation of the humanistic approach to psychotherapy in which therapists empathically help clients see the discrepancy between their behavior and their own values when they experience ambivalence about making major changes

multiaxial assessment: An approach to assessment introduced in *DSM-III* by which mental health professionals can provide diagnostic information on each of five distinct axes or domains

multiaxial system: An assessment system used by recent editions of the *DSM* that allows mental health professionals to provide diagnostic information on each of five distinct axes or domains

multiculturalism: An approach in clinical psychology emphasizing the appreciation of cultural diversity and awareness of how techniques can be best applied to individuals of various cultural backgrounds

multimethod assessment: An approach to assessment incorporating multiple methods, including tests of different types, interview data, observations, or other sources

multiple relationships: A circumstance in which a clinical psychologist has a professional relationship with an individual and also has another type of relationship with the same person or has a relationship with someone closely associated with the person

multisource, multimethod, multisetting approach: An approach to the assessment involving multiple sources of information, multiple methods of obtaining information, and multiple settings in which information is solicited

multisystemic family therapy: A contemporary form of family therapy designed to address long-term adolescent behavioral, emotional, and legal problems through interaction with multiple systems, including immediate family, school, and community

Hugo Munsterberg: An important figure in the history of forensic psychology; an early promoter of the use of psychology in the legal arena and the author of *On the Witness Stand*

Henry Murray: One of the creators of the Thematic Apperception Test, a popular projective personality test

narrative therapy: A contemporary form of family therapy that emphasizes the revision of the stories that family members

use to explain and interpret the events in their lives such that the stories describe the members more positively

naturalistic direct observation: In contrast to analogue direct observation, a type of behavioral observation in which a behavior is observed in the place where it actually happens

naturalistic observation: The direct, systematic observation of a client's behavior in the natural environment; also known as behavioral observation

negative punishment: A form of punishment in which the individual "loses something good"

negative reinforcement: A form of reinforcement in which the individual "loses something bad"

NEO Five Factor Inventory (NEO-FFI): A short form of the NEO Personality Inventory-Revised that produces a less detailed personality profile

NEO Personality Inventory-3 (NEO-PI-3): The current edition of an objective personality test that emphasizes normal personality characteristics rather than forms of abnormality

NEPSY-II: A battery of neuropsychological tests designed specifically for children

neuropsychological tests: Psychological tests that focus on issues of cognitive or brain dysfunction, including the effects of brain injuries and illnesses

neurosis: Along with psychosis, one of the two broad categories of mental illness used in Europe in the 1800s; refers to disorders such as anxiety and depression in which the individual maintains an intact grasp on reality

nomothetic approach: In contrast to an idiographic approach, an approach to research emphasizing the determination of similarities or common qualities among people

nondirective: In the context of the clinical interview, an approach by which interviewers allow clients to determine the course of the interview

nonsexual multiple relationships: A type of multiple relationship in which a clinical psychologist has a professional relationship with an individual and also has another nonsexual relationship (e.g., friendship, business relationship) with the same person

John Norcross: A leading contemporary psychotherapy researcher and a champion of the integrative orientation to psychotherapy

normative data: A sample of test-result data gathered by creators of a psychological test and typically designed to accompany the test that constitutes a basis for comparison for individuals who take the test in clinical settings

not guilty by reason of insanity (NGRI): In the U.S. legal system, a possible finding whereby an individual was unable to control his or her criminal actions due to a mental disorder at the time of the offense and is therefore not held responsible for the crime

object relations: A variation of psychodynamic psychotherapy deemphasizing internal conflict (id vs. superego), and instead emphasizing relationships between internalized "objects" (essentially, important people from the client's life)

objective personality tests: Personality tests characterized by unambiguous test items, a limited range of client responses, and objective scoring

observable changes: An essential feature of behavior therapy; therapeutic changes that are directly observable rather than inferred

observational learning: In behavior therapy, conditioning that takes place when the individual observes contingencies applied to others rather than the self; also known as modeling and social learning

***On the Witness Stand*:** A popular early book on forensic psychology authored by Hugo Munsterberg

open-ended question: In contrast to a closed-ended question, an interview question that allows for individualized and spontaneous responses from clients

open-enrollment groups: In contrast to closed-enrollment groups, therapy groups in which individual members are allowed to enter or leave the group at any time

operant conditioning: Conditioning in which the organism "operates" on the environment, notices the consequences of the behavior, and incorporates those consequences into decisions regarding future behavior

oral stage: In psychodynamic psychotherapy, the first of the psychosexual developmental stages, and the stage from which issues of dependency may emerge

overgeneralization: In cognitive psychotherapy, a common thought distortion in which the individual applies lessons learned from negative experiences more broadly than is warranted

overpathologizing: Viewing as abnormal that which is actually normal; can be reduced by increasing cultural competence

paraphrasing: A clinical interview technique used simply to assure clients that they are being accurately heard

parent training: A form of behavioral consultation in which the therapist serves as a consultant to a parent, who ultimately implements the behavioral interventions with the child in the home

patient-centered medical homes (PCMHs): A relatively new model of care, often involving health psychologists, that emphasizes primary care that is high quality, cost-effective, comprehensive, holistic, and coordinated

Ivan Pavlov: An important figure in the history of behavior therapy; a researcher whose classical conditioning studies provided a foundation for many behavioral techniques

pediatric psychology: A specialty area within clinical child psychology focusing on the mental and physical health of children with medical conditions

Perceptual Reasoning Index: A measure of fluid reasoning, spatial processing, and visual-motor integration; also one of four index scores yielded by the Wechsler intelligence tests

Personality Assessment Inventory (PAI): An objective personality test for adults with a relatively low reading level and multiple clinical scales closely tied to specific diagnoses or problems

personalization: In cognitive psychotherapy, a common thought distortion in which the individual assumes excessive personal responsibility for negative events

phallic stage: In psychodynamic psychotherapy, the third of the psychosexual developmental stages, and the stage from which issues of self-worth may emerge

Philippe Pinel: A pioneer of reform in the treatment of the mentally ill in France in the late 1700s and early 1800s

positive psychology: A recent, growing movement within the mental health field that accentuates the strong and healthy rather than the pathological aspects of human behavior

positive punishment: A form of punishment in which the individual "gets something bad"

positive regard: In humanistic psychotherapy, the warmth, love, and acceptance of those closest to us; also known as prizing

positive reinforcement: A form of reinforcement in which the individual "gets something good"

postdoctoral internship: A 1- or 2-year internship occurring after the doctoral degree in clinical psychology is granted that consists of supervised clinical experience in an applied setting, often with a specialized focus

practitioner-scholar model: A model of graduate training in clinical psychology that emphasizes practice over empirical research

predicting dangerousness: The practice whereby a forensic psychologist assesses the likelihood that an individual will behave violently or dangerously in the future

predoctoral internship: A year-long internship occurring near the end of graduate training in clinical psychology that consists primarily of supervised clinical experience in an applied setting

preemployment evaluations: A psychological evaluation conducted by forensic psychologists, often for law enforcement agencies, to assess the extent to which the candidate has any psychological problems, cognitive limitations, or personality characteristics that might interfere with the duties of the job

premenstrual dysphoric disorder (PMDD): A provisional or proposed mental disorder characterized by severe premenstrual symptoms, possibly including depressed mood, anxiety, affective lability, and decreased interest in activities

prescription privileges: The ability to prescribe medication to clients, which constitutes a controversy among contemporary clinical psychologists

prescriptive approach: An approach to psychotherapy in which specific therapy techniques with the most empirical evidence are viewed as the treatment of choice for specific disorders

prizing: In humanistic psychotherapy, the warmth, love, and acceptance of those closest to us; also known as positive regard

problem-focused coping: An approach to coping with stress that emphasizes proactive, constructive attempts to take action about a stressful situation

Processing Speed Index: A measure of the ability to process simple or rote information rapidly and accurately; also one of four index scores yielded by the Wechsler intelligence tests

professional counselors: Professionals with some similarities to clinical psychologists but who attend master's programs in professional counseling that typically have relatively high admissions rates, and whose activities emphasize counseling and deemphasize assessment and research

projection: In psychodynamic psychotherapy, a defense mechanism in which the ego projects an id impulse onto other people

projective personality test: A personality test in which individuals are assumed to project their personality characteristics via responses to ambiguous or vague stimuli

proposed criteria set: A set of symptoms described in Section III of *DSM-5* and under consideration for inclusion as an official disorder in a future edition of the *DSM*

psychiatrists: Professionals with some similarities to clinical psychologists but who are medically trained, able to prescribe medication, and emphasize the biological aspects of clients' problems

psychodynamic: An approach to clinical psychology deriving from the theories of Sigmund Freud and emphasizing the goal of making the unconscious conscious

psychodynamic play therapy: A form of psychotherapy with children in which a child's play symbolically communicates important unconscious processes occurring within the child's mind

psychodynamic psychotherapy: An approach to psychotherapy deriving from the theories of Sigmund Freud, the primary goal of which is to make the unconscious conscious

***The Psychological Clinic*:** The first scholarly journal in the field of clinical psychology, founded by Lightner Witmer in 1907

psychoneuroimmunology (PNI): The field of study concerned with the interactions among behavior, the nervous system, and the immune system

psychosis: Along with neurosis, one of the two broad categories of mental illness used in Europe in the 1800s; refers to disorders in which the individual demonstrates a break from reality in the form of hallucinations, delusions, or grossly disorganized thinking

psychotherapy: Techniques and approaches used by clinical psychologists and other mental health professionals to alleviate psychological symptoms or improve some aspect of emotional, cognitive, or behavioral functioning

PsyD: A doctoral degree obtained by clinical psychologists who graduate from practitioner-scholar programs that has become an

increasingly common alternative to the more traditional PhD degree

punishment: In behavior therapy, any consequence that makes a behavior less likely to recur in the future

quantitative reasoning: The ability to solve numerical problems; also one of five factor scores of the Stanford-Binet Intelligence Scales—Fifth Edition

quasi-experimental design: A type of research design used when truly experimental design is not feasible

quieting yourself: A general skill for clinical interviewers involving the minimization of internal, self-directed thoughts that can preoccupy or distract the interviewer

randomized clinical trials (RCTs): A research method for assessing the efficacy of a manualized therapy in which participants who meet criteria for a particular diagnosis are randomly assigned to a group that receives the therapy or a group that does not receive the therapy

rapport: A positive, comfortable relationship between clinician and client, especially important in the context of the clinical interview

rational emotive behavior therapy (REBT): Formerly known as rational emotive therapy, the form of cognitive therapy developed by Albert Ellis and incorporating the ABCDE model

reaction formation: In psychodynamic psychotherapy, a defense mechanism in which the ego forms a reaction against the id impulse, resulting in a behavior opposite of the original id impulse

reactivity: In the context of behavioral observation, a problem whereby the client's behavior may change simply because of his or her awareness of the presence of the observer

real self: In humanistic psychotherapy, the self that an individual actually experiences, in contrast to the ideal self

recapitulation of the family group: A phenomenon in group therapy whereby a group evokes the same dynamics as a client's family of origin

reflection: In humanistic psychotherapy, a therapist response to a client involving a rephrase or restatement of the client's statements in a way that highlights the client's feelings or emotions

reflection of feeling: A clinical interview technique intended to make clients feel that their emotions are recognized, even if their comments did not explicitly include labels of their feelings

reinforcement: In behavior therapy, any consequence that makes a behavior more likely to recur in the future

relaxation training: The first step of systematic desensitization in which the behavior therapist teaches the client progressive relaxation techniques that induce a relaxation response incompatible with anxiety

reliability: In psychological assessment, the extent to which a technique yields consistent, repeatable results

Repeatable Battery for the Assessment of Neuropsychological Status (RBANS): A neuropsychological screen that focuses on a range of abilities, including visuomotor abilities, verbal skills, attention, and visual memory

repression: In psychodynamic psychotherapy, a defense mechanism in which the ego represses conscious awareness of conflict between id and superego

resilience: In contrast to vulnerability, the tendency to remain psychologically healthy in spite of the presence of risk factors that contribute to psychological problems in others

resistance: In psychodynamic psychotherapy, client behavior that impedes discussion or conscious awareness of selected topics or emotions

The Retreat: A residential treatment center in Connecticut founded by Eli Todd in which the mentally ill were treated in a humane and dignified way

Rey-Osterrieth Complex Figure Test: A neuropsychological screen involving the reproduction of a single, complex geometric figure

Carl Rogers: A pioneer of the humanistic approach to clinical psychology

Hermann Rorschach: Swiss psychiatrist who created the Rorschach Inkblot Method, a popular projective personality test

Rorschach Inkblot Method: The projective personality test created by Hermann Rorschach involving 10 ambiguous inkblots

Rotter Incomplete Sentences Blank (RISB): The most commonly used sentence completion test

"s": An abbreviation for specific intelligences that may not correlate with one another (in contrast to "g," or general intelligence)

Morgan T. Sammons: A prominent proponent of prescription privileges for clinical psychologists and a widely recognized expert on psychopharmacology

scaling questions: In solution-focused family therapy, a technique whereby therapists ask families to rate the severity of a problem over time and emphasize strategies the families have used to cause the reductions in severity

school psychologists: Professionals with some similarities to clinical psychologists but who work primarily in schools and often perform psychological and educational testing and conduct interventions with students

scientist-practitioner model: The traditional model of graduate training in clinical psychology with a dual emphasis on empirical research and practice resulting in the PhD degree

self-actualization: In humanistic psychotherapy, the inborn tendency to grow in a healthy way

self-instructional training: A treatment method intended to help impulsive and disruptive children gain greater control over their behavior

self-monitoring: In behavioral assessment, techniques in which clients observe and keep a record of their own target behaviors

self-psychology: A variation of psychodynamic psychotherapy emphasizing parental roles in the child's development of self, with special attention paid to the meaning of narcissism at various points, including in therapy

self-report scales: Assessment techniques that are completed directly by the client

Hans Selye: A pioneer in stress research who developed the general adaptation syndrome hypothesis

semistructured interview: Also known as a partially structured interview, a clinical interview that has some characteristics of structured interviews as well as some characteristics of unstructured interviews

Senior Apperception Test (SAT): A projective test similar to the Thematic Apperception Test created specifically for older adults

sentence completion tests: Projective personality tests in which the individual is asked to complete sentence stems

sexual multiple relationships: A type of multiple relationship in which a clinical psychologist has a professional relationship with an individual and also has a sexual relationship with the same person

shaping: In behavior therapy, reinforcing successive approximations of the target behavior

B. F. Skinner: A pioneer of behaviorism whose experimentation on the law of effect and operant conditioning formed the foundation for many forms of behavior therapy

social microcosm: A phenomenon in group therapy whereby the relationship tendencies that characterize clients' relationships with important people in their personal lives predictably characterize the relationships they form with their fellow group members

social support: Relationships with others who can provide support in a time of crisis and who can share in good fortune as well

social workers: Professionals with some similarities to clinical psychologists but who typically obtain a master's degree and whose work emphasizes the social and environmental factors that contribute to an individual's problems

Society of Clinical Child and Adolescent Psychology: Division 53 of the American Psychological Association, focusing on issues related to clinical psychology applied to children and adolescents

solution-focused therapy: A contemporary approach to family therapy that relies on solution-talk and similar techniques in attempting to solve families' problems

solution-talk: An essential aspect of solution-focused family therapy whereby therapists emphasize positive outcomes that the future may hold rather than unpleasant situations that characterize the present to help clients begin to adopt a more positive point of view

Charles Spearman: A pioneer in the study of intelligence who argued for the existence of "g," a general intelligence thought to overlap with many particular abilities

specific learning disorder: A diagnosis based primarily on the level of academic achievement falling significantly below expected levels of achievement for people of the same age

stages of change model: A theory of readiness to change asserting that psychotherapy clients may enter therapy in one of five stages: precontemplation, contemplation, preparation, action, or maintenance

Stanford-Binet Intelligence Scales: A widely accepted test of intelligence originally developed by Alfred Binet and Theodore Simon

Stanford-Binet Intelligence Scales—Fifth Edition (SB5): The current edition of a prominent test of intelligence for individuals across the life span

statistical prediction methods: An approach to predicting dangerousness in which dangerousness is predicted according to a statistical or actuarial formula compiled by comparing an individual's characteristics with known relationships to future dangerousness; also known as actuarial prediction methods

strategic family therapy: A pragmatic, problem-focused approach to family therapy from which solution-focused therapy evolved

stress: The psychological and/or physiological response to difficult or demanding internal or external circumstances

Structured Clinical Interview for *DSM-5* Disorders (SCID): A prominent structured interview technique appropriate for a wide range of *DSM-5* disorders

structured interview: In contrast to an unstructured interview, an interview with a predetermined, planned sequence of questions that an interviewer asks a client

Hans Strupp: A pioneering psychotherapy researcher who developed the tripartite model of assessing psychotherapy outcome

subcultures: Relatively small groups within a society that may not fully constitute cultural groups but whose members may nonetheless possess typical and culturally meaningful characteristics for the clinical psychologist

sublimation: In psychodynamic psychotherapy, a defense mechanism in which the ego redirects the id impulse in such a way that the resulting behavior actually benefits others

subsystems: An essential concept in structural family therapy; subgroups of family members separated by boundaries

subtest scores: A feature of most contemporary tests of intelligence indicating intellectual ability in one of many specifically defined areas

summarizing: A clinical interview technique that usually involves tying together various topics that may have been discussed, connecting statements that may have been made at different points, and identifying themes that have recurred during the interview

superego: In psychodynamic psychotherapy, the part of the mind that establishes rules, restrictions, and prohibitions

supplemental scales: In addition to the clinical scales, scales of the MMPI and MMPI-2 that provide relevant clinical information

systematic desensitization: A form of behavior therapy based on classical conditioning involving re-pairing (or counterconditioning) the feared object with a new response, such as relaxation, that is incompatible with anxiety

systems approach: A foundation of family therapy whereby clinical problems originate in the family system rather than in an individual family member

***Tarasoff* case:** A landmark legal case resulting in clinical psychologists' duty to warn potential victims of dangerous or harmful acts whose identity is revealed by clients during psychological services

teacher training: A form of behavioral consultation in which the therapist serves as a consultant to a teacher, who ultimately implements the behavioral interventions with the student in the school

technique: In the context of the clinical interview, the procedures and methods (e.g., types of questions) used by the interviewer

telepsychology: The use of technology, including the Internet, videoconferencing, smartphones, and text-based services, in the application of clinical psychology

test data: The raw data provided by clients during an assessment

test security: Protection of the integrity of psychological test materials by preventing them from entering the public domain

test selection: For a clinical psychologist conducting an assessment, the selection of tests that are appropriate in terms of the psychologist's competence; the client's culture, language, and age; and the test's reliability and validity, among other factors

testable hypotheses: In behavior therapy, an essential feature of theories underlying problem behaviors whereby theories can be empirically supported, refuted, modified, and retested

test-taking attitudes: The manner in which the test taker approaches a test, as measured by the validity scales of the MMPI and MMPI-2

Thematic Apperception Test (TAT): A projective personality test in which individuals create stories in response to ambiguous interpersonal scenes

therapeutic alliance: Also known as the therapeutic relationship or working alliance, and an established common factor in psychotherapy outcome research; a coalition or partnership between two allies (client and therapist) working in a trusting relationship toward a mutual goal

therapeutic relationship: Also known as the therapeutic alliance or working alliance, and an established common factor in psychotherapy outcome research; a coalition or partnership between two allies (client and therapist) working in a trusting relationship toward a mutual goal

therapy manual: A manual that provides detailed instructions for all phases of the treatment of a specific disorder; typically used in treatment outcome studies

third-party payer: Typically, health insurance or managed-care companies that pay some or all of a client's mental health expenses

Edward Lee Thorndike: A pioneer in the study of intelligence who promoted the idea that each person possesses separate, independent intelligences; also, a leading researcher in the area of operant conditioning and the law of effect

three essential therapeutic conditions: In humanistic psychotherapy, the three necessary and sufficient conditions that a therapist must provide for therapeutic benefit: empathy, unconditional positive regard, and genuineness

three-stage sequential model of common factors: A model explaining the order in which common factors contribute to psychotherapeutic change, beginning with support factors, moving on to learning factors, and concluding with action factors

three-stratum theory of intelligence: A contemporary theory that intelligence operates at three levels: a single "g" at the top, eight broad factors immediately beneath "g," and more than 60 highly specific abilities beneath these broad factors

Louis Thurstone: A pioneer in the study of intelligence who argued that intelligence should not be understood as a single, unified ability but as numerous distinct abilities that have little relationship to one another

time-limited dynamic psychotherapy (TLDP): A specific, contemporary form of psychodynamic psychotherapy in which the therapist makes efforts to form a "corrective" relationship with the client that does not follow the same unconscious "script" as the client's previous problematic relationships

Eli Todd: A pioneer of reform in the treatment of the mentally ill in the United States in the late 1700s and early 1800s

token economy: A form of behavior therapy based on operant conditioning in which clients earn tokens, exchangeable for reinforcements, for performing predetermined target behaviors

tolerance: A phenomenon whereby the body requires increasing amounts of a substance to achieve the desired effect

traditional personality assessment: In contrast to behavioral assessment, an approach to assessment that assumes that personality is a stable, internal construct; assessing personality requires a high degree of inference; and client behaviors are signs of underlying problems

transference: In psychodynamic psychotherapy, the tendency of clients to form relationships with therapists in which they unconsciously and unrealistically expect the therapist to behave like important people from the clients' past

treatment outcome: A common area of research for clinical psychologists focusing on the assessment of the benefits of psychological treatments

triangle: In family therapy, a phenomenon in which one or both of two family members in conflict attempt to bring a third member into the conflict to garner support

tripartite model: A model of assessing psychotherapy outcome developed by Hans Strupp and his colleagues that acknowledges the viewpoints of three parties (the client, the therapist, and third parties such as society, family, or managed-care companies)

tripartite model of personal identity: A three-level model of personal identity in which each individual is recognized as being entirely unique, similar to some other individuals, and similar to all other individuals

William Tuke: A pioneer of reform in the treatment of the mentally ill in England in the late 1700s and early 1800s

unconditional positive regard (UPR): In humanistic psychotherapy, one of the three essential therapeutic conditions; the full acceptance of another person without any conditions or stipulations

unconditioned response: In classical conditioning, the response elicited by the unconditioned stimulus before any conditioning has taken place

unconditioned stimulus: In classical conditioning, the stimulus that elicits the unconditioned response before any conditioning has taken place

unconscious: Mental activity occurring outside our awareness; a cornerstone of the psychodynamic approach

undifferentiated ego mass: In family therapy, a term used to describe an emotionally fused family

universal level: One of three levels of the tripartite model of personal identity emphasizing the perspective that each individual is like all others

Universal Nonverbal Intelligence Test-2 (UNIT-2): A recently published intelligence test that was designed as entirely language-free in an attempt to maximize cultural fairness

universality: A therapeutic factor in group therapy; the experience that one's problems are not unique but are common to others

unstructured interview: In contrast to a structured interview, an interview that involves no predetermined or planned questions and in which interviewers determine the course of the interview as it takes place

Vail model: Alternate name for the practitioner-scholar model of graduate training stemming from a 1973 professional conference in Vail, Colorado

validity: In psychological assessment, the extent to which a technique measures what it claims to measure

validity scales: Scales of the MMPI and MMPI-2 that inform the psychologist about the client's approach to the test and allow the psychologist to determine whether the test is valid and what kinds of adjustments might be appropriate to make during the process of interpreting the clinical scales

values affirmation: A relatively new clinical approach in health psychology in which clients are encouraged to clarify and affirm, often in writing, their own personal values, which often contrast with their unhealthy behaviors

Verbal Comprehension Index: A measure of verbal concept formation and verbal reasoning; also one of four index scores yielded by the Wechsler intelligence tests

verbal tracking: In the context of the clinical interview, the ability to repeat key words and phrases back to clients to assure the clients that they have been accurately heard

vicarious learning: In behavior therapy, a way in which clients can benefit from observational learning whereby the client observes not only the modeled behavior but also the model receiving consequences for that modeled behavior

visual-spatial processing: The ability to analyze visually presented information, including relationships between objects, spatial orientation, assembling pieces to make a whole, and detecting visual patterns; also one of five factor scores of the Stanford-Binet Intelligence Scales—Fifth Edition

vocal qualities: In the context of the clinical interview, pitch, tone, volume, and fluctuation of voice that are important to appreciate and use effectively

voir dire: In forensic psychology, a process by which an expert witness is approved for the court via examination of the expert's education, training, and professional experience

vulnerability: In contrast to resilience, the tendency to experience psychological problems in the presence of risk factors

Bruce Wampold: A leading contemporary psychotherapy researcher who has argued strongly in favor of the dodo bird verdict and against a prescriptive approach to treatment

John Watson: An important figure in the history of behavior therapy and early promoter of behaviorism in the United States

David Wechsler: A pioneer in the assessment of intelligence for various age groups, including adults, children, and preschoolers

Wechsler Adult Intelligence Scale (WAIS): A prominent scale of intelligence for adults

Wechsler Adult Intelligence Scale—Fourth Edition (WAIS-IV): The current edition of a prominent intelligence test for adults

Wechsler-Bellevue: The first intelligence scale created by David Wechsler in 1939, designed for adults

Wechsler Individual Achievement Test—Third Edition (WIAT-III): The current edition of a comprehensive achievement test for clients aged 4 to 50

Wechsler Intelligence Scale for Children (WISC): A prominent scale of intelligence for children

Wechsler Intelligence Scale for Children—Fifth Edition (WISC-V): The current edition of a prominent intelligence test for children

Wechsler Memory Scale—Fourth Edition (WMS-IV): A scale of visual and auditory memory involving both immediate and delayed recall; often used for neuropsychological purposes

Wechsler Preschool and Primary Scale of Intelligence (WPPSI): A prominent scale of intelligence for preschoolers

Wechsler Preschool and Primary Scale of Intelligence—Fourth Edition (WPPSI-IV): The current edition of a prominent intelligence test for preschoolers

withdrawal: A phenomenon whereby cessation of a substance produces negative symptoms

within-group design: A research design that involves comparisons of participants in a single condition to themselves at various points in time

Lightner Witmer: The founder of the field of clinical psychology, the first psychological clinic, and the first journal devoted to clinical psychology

working alliance: Also known as the therapeutic alliance or therapeutic relationship, and an established common factor in psychotherapy outcome research; a coalition or partnership between two allies (client and therapist) working in a trusting relationship toward a mutual goal

working memory: The ability to hold and transform information in short-term memory; also one of five factor scores of the Stanford-Binet Intelligence Scales—Fifth Edition

Working Memory Index: A measure of the capacity to store, transform, and recall incoming information and data in short-term memory; also one of four index scores yielded by the Wechsler intelligence tests

working through: An often lengthy component of psychodynamic therapy in which interpretations are reconsidered and reevaluated again and again

Irvin Yalom: A leading figure in contemporary group therapy, especially the interpersonal approach

York Retreat: A residential treatment center in England founded by William Tuke in which the mentally ill were treated with kindness, dignity, and decency

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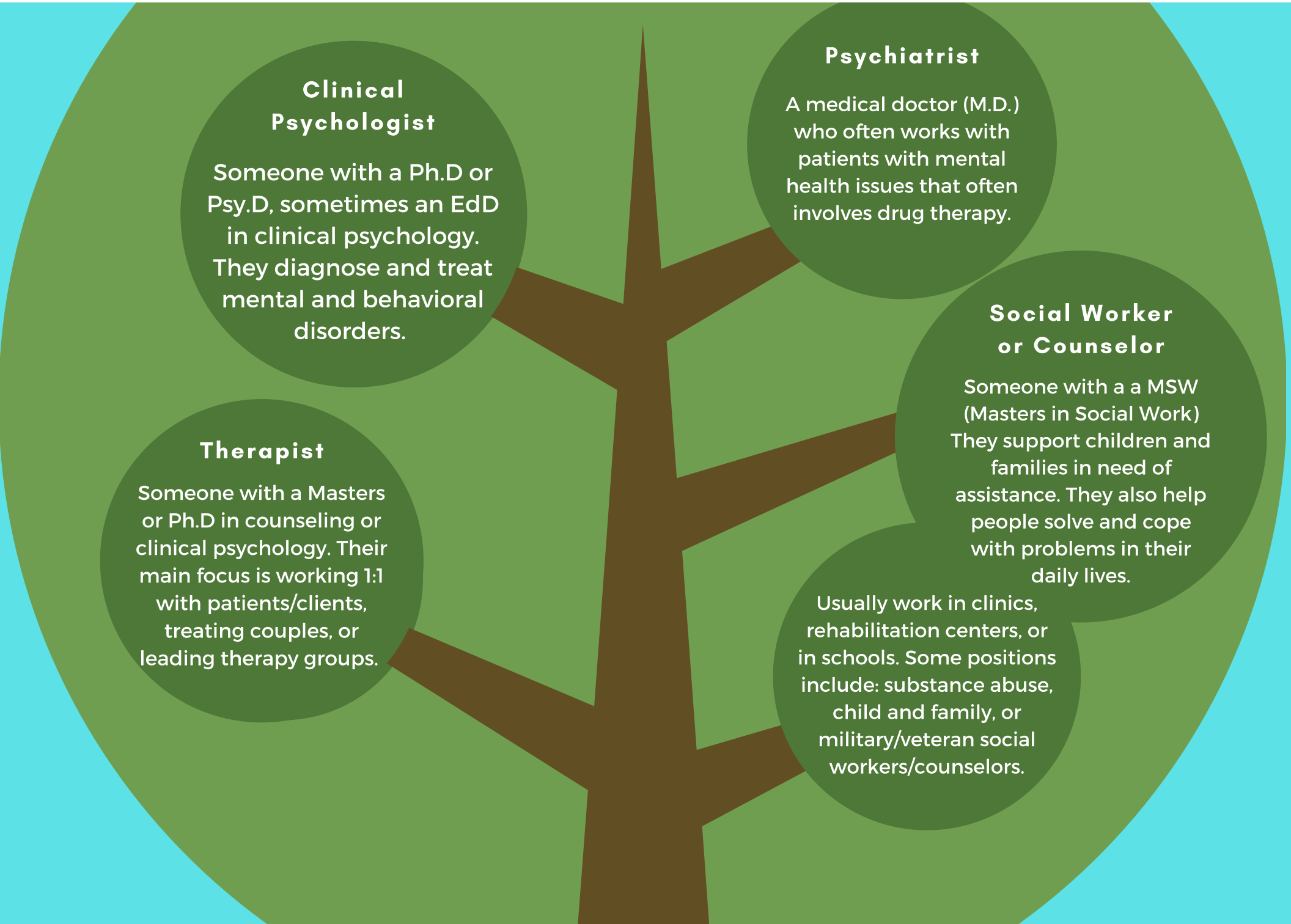

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DEGREE REQUIRED	MA/MS or PhD/PsyD/EdD	MSW	PhD, PsyD, EdD	MD
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CORRECTIONAL CASEWORKER
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JUVENILE OFFICER
PROBATION OFFICER
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CORE APPROACHES IN

COUNSELLING AND PSYCHOTHERAPY

FAY SHORT AND PHIL THOMAS

ROUTLEDGE

Core Approaches in Counselling and Psychotherapy

Core Approaches in Counselling and Psychotherapy is a comprehensive guide to the four main psychological approaches (humanistic, psychodynamic, behavioural and cognitive) and introduces several of the most common therapies used today. This textbook contains sufficient coverage to explain all of the most important elements of these core approaches and sufficient depth to provide a detailed analysis of the ten main therapies: person-centred therapy, psychoanalytic therapy, behaviour therapy, cognitive therapy, Gestalt therapy, transactional analysis, rational emotive behaviour therapy, cognitive-behaviour therapy, multimodal therapy and neurolinguistic programming.

The book focuses on the development of each approach and presents the associated therapy in its historical and psychological context, giving a deeper insight into the theories and clarifying the overlap between different therapies.

Presented in a unique style, with a clear layout, rigorous content and extensive online resources at www.routledge.com/cw/short, *Core Approaches in Counselling and Psychotherapy* is an invaluable asset for undergraduate and postgraduate students at all levels of study and is the ideal textbook for any degree or higher-level module in counselling.

Fay Short lectures on Social Psychology and Psychotherapy, alongside her role as Director for Teaching and Learning in Psychology, at Bangor University. She is a Chartered Psychologist with the British Psychological Society and a member of the British Association for Counselling and Psychotherapy, and has recently been awarded a highly prestigious National Teaching Fellowship award from the Higher Education Academy.

Phil Thomas is Lecturer at Coleg Llandrillo and Bangor University, where he is the Director of the counselling pathway within the Masters in Education. As a qualified teacher and therapist trained to NLP Master Practitioner level, he has a particular interest in the applications of therapy in education. He also provides counselling and supervision within his private practice.

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Fay Short and
Phil Thomas

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To my mum and dad for teaching me the meaning of life and my husband for giving my life meaning.
—Fay Short

To my family for their love, my tutors for their guidance, my students for their enthusiasm. Finally, to my clients who have given me the opportunity to put my learning into practice.
—Phil Thomas

To Elaine Ward, Tina Usherwood and Keith Mathews for being willing to demonstrate their excellence on video.
—Fay Short and Phil Thomas

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Preface

Counselling and psychotherapy have been described as a 'talking cure' and it is certainly true that these processes can offer a successful non-medicinal alternative in the treatment of depression, anxiety, psychosis, etc. But to suggest that therapy is simply a treatment for mental health issues or a crutch for the worried well is to miss the true value of this incredible process. An appreciation of counselling and psychotherapy can offer a unique insight into human nature and a deep understanding of how to connect with another person. Appreciation of the core approaches in counselling and psychotherapy can inform all of our interactions, within and beyond the therapeutic setting. Those who eventually pursue a career in healthcare will benefit immensely from this knowledge, but even those who choose a profession in the field of business or education or engineering – indeed, any field that involves some interaction with other human beings – will benefit from understanding those theories which explain human nature. Furthermore, successful application of these theories to our interpersonal interactions can significantly aid both our professional and personal relationships. There is, perhaps, no other academic area that can boast such a profound and wide-reaching effect on all aspects of our lives.

Permissions

Photograph of Aaron Beck appears courtesy of the Beck Institute for Cognitive Behavior Therapy, www.beckinstitute.org.

Photograph of Donald Meichenbaum appears courtesy of himself.

Photograph of Ulric Neisser appears courtesy of Cornell University.

Guide to the companion website

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Visit the companion website for *Core Approaches in Counselling and Psychotherapy* to discuss a comprehensive range of resources designed to enhance the teaching and learning experience for both students and instructors.

On the companion website you'll find the following range of resources with which you can engage with the content of *Core Approaches in Counselling and Psychotherapy*.

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- A glossary of key terms
- Therapy in action – four videos of therapy sessions, with an assessment form and transcript for each

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Introduction

This book aims to provide you with a comprehensive guide to the four main psychological approaches (humanistic, psychodynamic, behavioural and cognitive) and introduce you to several of the most common – and sometimes controversial – therapies used today (person-centred therapy, psychoanalytic therapy, behaviour therapy, cognitive therapy, gestalt therapy, transactional analysis, rational emotive behaviour therapy, cognitive-behaviour therapy, multimodal therapy and neurolinguistic programming). This textbook offers a guide to this subject for those studying at undergraduate or postgraduate level. You will find that the book contains sufficient coverage to explain all of the most important elements of the core approaches and sufficient depth to provide you with a detailed analysis of the main therapies. However, the content of the book is so clearly explained and the presentation of the book is so accessible that this textbook is also an ideal introduction for those who simply wish to learn more about this fascinating topic.

You will notice that this book is quite unique in style and format. It does not contain huge paragraphs of text. Instead, all information is presented in simple note style to improve digestion. Cognitive psychology teaches us that we cannot maintain attention for long periods of time, especially when the topic is difficult – long chapters filled with rolling paragraphs are almost impossible to understand when you are struggling with the core concepts. And it is just as impossible to resist copying these perfect paragraphs when you are trying to paraphrase to avoid plagiarism. It is far easier to read and understand short chunks of text highlighting the main points and then use this primary information as a foundation on which to build your own work. This book was written with the intention of providing lists of facts, brief explanations, clear relationships between concepts and an opportunity to quickly digest chunks of information for future use in essays, reports, exams or your interactions with others.

For further information on how to get the best from this textbook, please consult the following frequently asked questions.

HOW DO I READ THIS BOOK?

This book is designed to provide you with a comprehensive overview of the four main approaches to counselling and psychotherapy. This means that it contains a significant depth of analysis and this level of detail can be difficult to digest in one sitting. We recommend that you read this book like a magazine rather than a novel. Instead of starting at the

beginning and finishing at the end, we suggest that you start by flicking through the contents of the entire book and then move backwards and forwards through the various chapters according to your own needs as a student.

You should start exploring this book by reading through the contents list and getting an overview of the focus of each chapter. This is similar to the way that you might browse through the contents of a magazine. You should then skim swiftly through each chapter in order to gain a general understanding of the material covered in that section. Again, this is similar to the way that you might flick through a magazine in order to get a general overview of the different articles. Pay particular attention to headings and subheadings within the chapters as these will often give you an indication of the key topics covered in the text. It is often a good idea to ask yourself at the end of each chapter 'What does this chapter tell me?' If you can answer this question, move on to skim through the next chapter. If you cannot answer this question, skim back through the chapter.

Once you have a basic knowledge of the general contents, return to the start of the book and begin reading chapters of interest in more depth. You will notice that each chapter starts with a set of learning outcomes. These outcomes tell you what you should be able to do by the time you reach the end of the chapter. Once you have finished reading that chapter, return to the learning outcomes and ask yourself whether you can now meet these outcomes. If not, reread the chapter with the learning outcomes in mind.

Although it is not necessary to read this book from beginning to end, the chapters in this book are set out in a specific order to aid your learning. The first chapter is designed to provide you with a basic understanding of the therapeutic process, the following four chapters each focus on one of the core approaches with an associated therapy, and the final chapter describes several integrative/eclectic therapies. After skimming through the entire book, it is a good idea to start reading more thoroughly in the first chapter. This chapter will help you to establish a basic understanding of therapy in general. It is also a good idea to cover the chapters focusing on the four core approaches (Chapters 2–5) before exploring the integrative and eclectic therapies (Chapter 6). However, as noted above, there is no 'correct' order in which you should read this book – feel free to skim, flick forward, turn back, reread and jump sections according to your own desire for knowledge. Every chapter is self-contained and can be easily understood if read independently. However, the chapters are also designed to support each other and information can often be appreciated at a deeper level if you have first read the content of a previous chapter. This means that there may be occasions when you want to return to an earlier chapter to remind yourself about a key concept and this is highly recommended to support your learning process. For example, before starting the final chapter on integrative and eclectic therapies, you might want to reread the last section of the first chapter as this defines the concepts of integration and eclecticism.

Once you have read all of the chapters, you could reflect on the whole book by skimming through the summary tables. You will notice that every section in every chapter concludes with a summary table which contains all of the content for that section distilled into a set of key words. Consider these summary tables after you have finished reading the book. Look over the key words in order to check knowledge (can you explain and expand on every key term?), understand associations (do you appreciate the links between different topics?) and revise content (can you use these words as an aid to memory?).

IS THERE A DIFFERENCE BETWEEN AN APPROACH AND A THERAPY?

There is indeed a difference between an approach and a therapy, and this will be addressed in more detail in Chapter 1. However, it is good to get a basic understanding of this distinction before starting the book.

An approach could be described as the umbrella term for all theories and concepts converging on a similar set of principles. For example, the humanistic approach includes several theories relating to the positive nature of humans, the tendency towards self-actualisation and the importance of viewing the individual as the expert on the self. Theories and direction of focus may differ between different contributors to the approach, but the key principles are usually similar. For example, under the humanistic umbrella, Rogers focuses on the concept of congruence between the actual self and the self-concept whereas Maslow focuses on movement up the hierarchy of needs, but both theorists agree on the basic principle of self-actualisation.

Therapy could be described as the practical application of an approach. Several different therapies can operate under a single approach and each therapy may apply the theories within the approach in a slightly different manner, but all of the therapies under a single approach will share a common philosophy of human nature. For example, it is argued that both person-centred therapy and existential therapy operate under the humanistic approach because the core assumptions about human nature are similarly humanistic for each therapeutic application.

DOES THIS BOOK COVER ALL OF THE DIFFERENT APPROACHES AND THERAPIES?

There are far too many approaches and therapies in existence to be able to successfully describe them all in any one book. It is also important to note that practitioners are constantly developing new therapeutic methods so there could be several new methods created in the short space of time between writing this introduction and publishing this textbook.

This book focuses on four core approaches in psychology and ten different therapies. These four approaches were selected because they are generally agreed to be the most common perspectives in psychology. The therapies associated with each approach are regarded as the most typical examples of that approach. The remaining therapies presented in the final chapter of this book demonstrate interesting examples of integration and eclecticism.

WHAT ARE THE FOUR 'CORE' APPROACHES?

The four core approaches presented in this textbook are the three 'forces' and single 'revolution' in psychology. The three forces are numbered according to their impact on the field of psychology. Behaviourism and psychodynamic theory are regarded as the first two forces because they initially dominated the field until the humanistic perspective was introduced as a third perspective in psychology. In this textbook, however, we present the three forces in reverse order (humanistic, psychodynamic, behavioural). There are three reasons for covering these approaches in this order.

First, one of the most popular current forms of therapy combines the behavioural and cognitive approaches to produce cognitive-behaviour therapy. However, although they are commonly practised together, theoretically they are quite distinct with an independent background and development. It is important to achieve a firm grounding in the theories and concepts underpinning each before moving on to explore the combined application of cognitive-behaviour therapy (the combined approach is covered in full in Chapter 6). In order to fully understand how these two approaches are integrated, it is sensible to consider the two approaches consecutively. It was, therefore, logical to reverse the three forces so that the chapter on behaviourism precedes the chapter on the cognitive revolution.

Second, the humanistic approach often forms the foundation for all types of therapy. Almost all practitioners, irrespective of their own individual approach to psychotherapy, appreciate the importance of the therapeutic relationship. All modern therapies emphasise the value of the therapist adopting a non-judgemental attitude towards the client and highlight the benefits of ensuring that the client assumes responsibility for his or her own life. These are all key assumptions of the humanistic approach so it is logical for the reader to understand the basic principles of this approach before being introduced to alternative therapies.

Third, the humanistic approach provides an excellent therapeutic foundation for student readers. We are not suggesting that students should begin with this approach because it is the easiest. In fact, the humanistic approach is arguably the most difficult type of therapy given that the therapist is required to simply be with the client without being able to rely on activities and exercises to fill the time. However, although it is not the easiest, this approach is arguably the least destructive. As with many practical skills, one can only learn how to counsel effectively by engaging in the act of therapy. Unfortunately, there is a circular problem associated with this learning requirement: we should not risk the wellbeing of the client with an untrained therapist and yet all therapists need to practise therapy in order to train. The learner therapist intending to adopt a particular approach at the start of his or her training would be well advised to begin from a person-centred perspective. Unlike those therapies which may involve interpretation (psychoanalytic) or exposure to fears (behaviour therapy), the non-judgemental and non-advisory methods of person-centred therapy are least likely to do any harm to the client. For this reason, it is logical for the reader to be introduced to the humanistic approach at the start of his or her training.

WHY DOES THIS BOOK FOCUS ON BIOGRAPHIES OF KEY FIGURES?

We would argue that people who wish to pursue a career in therapy should have an interest in the lives of others. Therapists are frequently exposed to the life stories of others and invited to explore the significance of key events, so it is logical to assume that they should appreciate the importance of how beliefs and values are influenced by life history. This appreciation can significantly improve understanding of the core approaches in psychotherapy; the reader can link biographical information to the perspectives of the key figure in order to understand the development of the approach in a historical context. It can be very difficult to understand some of the nuances of an approach without

appreciating the origin of these concepts. For example, biographical details relating to the early childhood of Sigmund Freud give us an important understanding of his relationship with his mother and this context helps us to appreciate his theories relating to the Oedipus conflict. Instead of seeing a man who has plucked some apparently strange ideas out of thin air, we are able to see someone who has analytically reflected on his own feelings as a child in order to devise a theory to explain human nature. In this way, the biographies of these crucial figures help us to see how the key events in their life stories interact with the development of the theory.

We have carefully selected the biographies of those who have had a fundamental impact on the therapeutic approach. We do appreciate the significant contribution of modern theorists who may be shaping the current face of therapy, and we have incorporated many modern theories and concepts throughout this textbook. However, in terms of biographies, we have specifically focused on those who are generally regarded as the founding fathers of the approach. This focus will give you a unique insight into the original grounding of the theories in the appropriate historical and personal context.

WHY DOES THIS BOOK INCLUDE CASE STUDIES FOR EACH APPROACH?

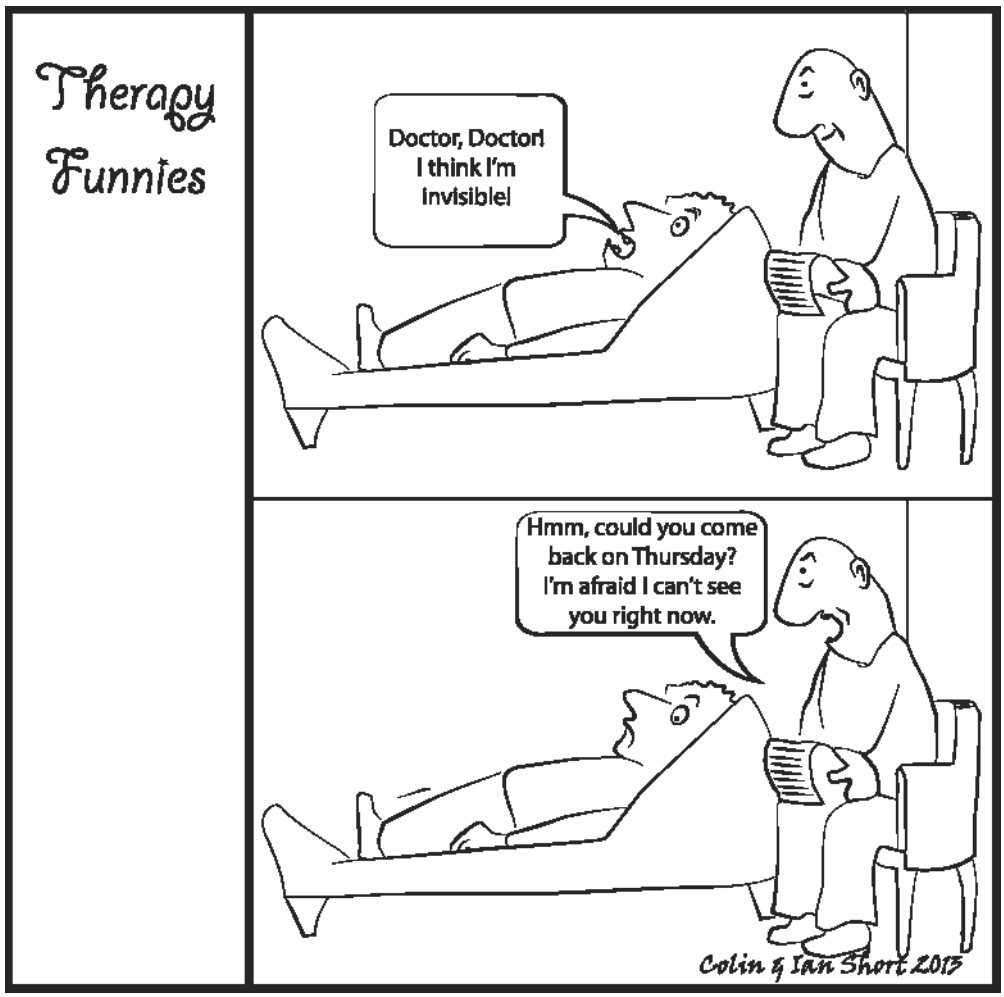
The reader will notice that each chapter ends with a novel case study involving a client presenting with a set of concerns and a therapist addressing these concerns using the therapy discussed in that particular chapter. Since each therapy is best suited for certain types of difficulties, each case study is unique in order to ensure that the examples provided are as clear as possible. To help the reader understand the work of the therapist, the case study will start with a brief description of the client presentation, including their self-reported history and symptoms. The reader will then be provided with a description of the session and a series of questions to direct the reader towards any crucial parts of the interaction. This session can be viewed by watching the associated video content of the therapy session in action, and a full written transcript of the session is available in the online resources. Please note that these videos depict REAL interactions; although the session has been arranged for the purposes of the video and the sessions will not continue after the recording, the interaction within the session is genuine. Neither individual is acting – the therapist is a qualified practitioner in that specific type of therapy and the client is reporting genuine experiences and emotions. The case study section of the chapter will conclude with a reflective report about the session from the perspective of the client. This will provide the reader with an insight into the client experience of each type of therapy.

The aim of these case studies is to provide the reader with a real-world example of therapy in action. Unlike many other fields of psychology, counselling and psychotherapy are not exclusively academic. In order to fully understand therapeutic approaches and methods, the reader must appreciate how these concepts can be applied in interactions with clients. The best way to present these interactions is in the form of case studies and we hope that readers are able to use these examples in order to further their own understanding and practice of counselling and psychotherapy.

We would now like to end this introduction by wishing you happy reading. We hope that this textbook serves as a knowledgeable guide on your journey through the fascinating subject of human nature. Perhaps more importantly, we hope that this book will inspire you to further your own learning experience by using your newfound understanding to support, encourage and assist your fellow man. To misquote the eminent Carl Rogers, education is a direction, not a destination.

Chapter 1

Counselling and psychotherapy



CHAPTER CONTENTS

- Introduction to counselling and psychotherapy
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INTRODUCTION TO COUNSELLING AND PSYCHOTHERAPY

This chapter aims to introduce the reader to the concepts of counselling and psychotherapy. Key concepts such as ethical issues and sociocultural diversity will be explored in this chapter. You will also be introduced to the idea of counselling and psychotherapy as career options and provided with a clear route to becoming a therapist in the UK. It is impossible to develop a good understanding of any specific therapy without first understanding the field in general, so this introductory chapter will provide you with a general overview of the fascinating field of counselling and psychotherapy.

LEARNING OUTCOMES

By the end of this chapter, you will be able to:

- appreciate the differences and similarities between the fields of counselling, psychotherapy and psychology
- discuss the different roles of the therapist and the client in the therapeutic relationship
- acknowledge the importance of maintaining boundaries and outline the ethical issues associated with therapy
- appreciate the differences between an approach, therapy and model, and discuss each of these with consideration of the differing methods adopted by practitioners

DEFINING COUNSELLING AND PSYCHOTHERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- outline the definitions of counselling
- appreciate the differences between counselling and psychotherapy
- recognise the career possibilities in the field of counselling and psychotherapy

Definitions of counselling and psychotherapy

UK versus US

American spelling = Counseling

British spelling = Counselling

Simplicity

For the sake of simplicity, this textbook will focus on 'counselling' (British spelling) whenever the term is used. However, in general, this book will use the terms 'therapy' and 'therapist' to refer to all aspects of the therapeutic encounter.

What are counselling and psychotherapy?

British Association for Counselling and Psychotherapy (BACP) definition of counselling and psychotherapy

> 'Counselling and psychotherapy are umbrella terms that cover a range of talking therapies. They are delivered by trained practitioners who work with people over a short or long term to help them bring about effective change or enhance their wellbeing.' (BACP, 2013)

British Psychological Society (BPS) definition of counselling

> 'Counselling is concerned with the interplay between psychological principles and the counselling process and is developed by substantial reflection on practice and research. Its understandings derive both from formal psychological enquiry and from the interpersonal relationships between practitioner and the client.' (BPS, 2013)

American Psychological Association (APA) definition of counselling

> 'Counseling psychology centers on typical or normal developmental issues as well as atypical or disordered development as it applies to human experience from individual, family, group, systems and organizational perspectives. Counseling psychologists help people with physical, emotional and mental disorders improve wellbeing, alleviate distress and maladjustment, and resolve crises. In addition, practitioners in this professional specialty provide assessment, diagnosis and treatment of psychopathology.' (APA, 2013)

World Health Organization (WHO) definition of counselling

> 'Counselling provides a supportive and non-judgemental atmosphere for people to talk over their problems and explore more satisfactory ways of living.' (WHO, 2013)

To summarise, counselling and psychotherapy include the following:

> Collaborative relationship between therapist and client
>
> Client seeks to enhance his or her emotional or psychological wellbeing
>
> Therapist seeks to provide a supportive environment in which the client can work towards seeking to enhance his or her emotional or psychological wellbeing

What does a therapist do?

Essentially, a therapist will provide support to those in need by offering a safe non-judgemental environment in which the client can explore his or her own issues

Specifically, a therapist may offer support for distinct clients and/or problems

Client groups may include:

> Individuals
>
> Groups
>
> Families
>
> Spouses/partners
>
> Children
>
> Adolescents
>
> Adults
>
> Older adults

- Employees
- Students

Problem types may include:

- Depression
- Stress
- Anxiety
- Low self-esteem
- Low confidence
- Poor anger management
- Insomnia
- Eating disorders
- Addictions
- Marital or relationship difficulties
- Sexual disfunctions
- Employment issues
- Financial concerns
- Prior trauma or abuse
- Bereavement

Modern public perception is much more positive towards therapy

British Association for Counselling and Psychotherapy (2010a) conducted a wide-scale review of public attitudes towards counselling and psychotherapy

Meta-analysis of current research in the field followed up by a telephone survey of 1440 adults in the UK

- 21% of respondents had consulted a counsellor in the past
- 88% of respondents agreed that consultation with a counsellor might make many people happier

What happens in counselling and psychotherapy sessions?

Therapy sessions can vary depending on the orientation of the therapist, issues raised by the client, and demands of the service in which the therapy is offered. However, in general, most therapy will follow this pattern

- Initial contact made by a referral agent or the client
- Client registers by completing an initial written assessment (usually in the form of a questionnaire)
- Client attends an initial meeting to assess his or her needs – risk assessments will also be completed at this time and further referrals may be made if the client is deemed unsuited for the specific type of therapy

- Client is allocated to an appropriate therapist (may be the therapist from the initial assessment) or added to a waiting list for the next available therapist (if the service is particularly busy)
- Client attends a first session with the therapist
  - First session typically involves information gathering and contracting – client is invited to tell his or her story, therapist will outline important policies and procedures (including confidentiality), and relationship boundaries will be established
- Client attends a series of sessions to work through his or her issues
  - Each session will typically last for one 'therapeutic hour' (50 minutes)
  - Sessions will often run once per week for an agreed time frame
  - Short-term therapy might run for as little as six weeks; long-term therapy could run for years
- If possible, client is given advance warning in the penultimate session so that the relationship can be drawn to a satisfactory close
- Client attends the final session with the therapist
  - Final session typically involves reflection on progress and ending the relationship
  - Client is encouraged to consider how s/he has changed during the therapy and to think about how life will continue beyond the therapeutic relationship
  - Both parties express their feelings about the end of the therapeutic relationship
- Client is asked to complete another written assessment (questionnaire) to determine the impact of the therapy and may also be asked to complete an evaluation of the service

Ending the therapeutic relationship can be difficult

- Both client and/or therapist are entitled to end the relationship at any time if it is felt that it is no longer providing any therapeutic benefit
- In the ideal scenario, client and therapist agree on a set number of sessions, complete the therapeutic process within those allotted sessions, and part on good terms with a positive ending
- However, it is often the case that therapy ends under less ideal circumstances
  - Clients may seek to end the therapy if they are not ready to address issues – ending therapy can be an effective way of avoiding difficult emotional investigations
  - Therapists may feel that the client is no longer benefiting from therapy but the client may be reluctant to be entirely independent
  - In both of these cases, the relationship end must be carefully managed by the therapist to encourage positive closure and ensure that the door remains open for the client to return to therapy in the future

How does a client access a therapist?

There are many ways in which an individual can obtain counselling support

- Referral by a GP or other health practitioner
  - Often the GP is the first point of contact for an individual who is struggling with his or her mental wellbeing
  - BACP (2010a) found that a GP or health practitioner had referred 65% of those receiving therapy
- Referral by a tutor, teacher, lecturer or employer
  - It is possible for an authority figure to submit a referral on behalf of someone in his or her care – for example, a tutor may refer a student to the student counselling service
  - This is less common as most therapeutic services insist on initial requests from the client rather than a third party
  - BACP (2010a) found that employers had referred only 7% of those receiving therapy
- Self-referral
  - Any individuals struggling with their own mental wellbeing can seek support from independent agencies advertised through websites, public directories, etc.
  - This approach carries the risk of self-referral to an inappropriate or illegitimate agency so it is important for individuals to check that the organisation is a member of a relevant governing body (like the BPS or BACP), employs only those who have been professionally trained and hold the relevant professional accreditations, and has full counselling insurance
  - BACP (2010a) found that 9% of those receiving therapy found a therapist independently through a public directory

Where do therapists work?

Sadly, there is a lack of employment opportunities available for those who qualify as therapists

- Many students have a specific job in mind and they are pursuing a career in therapy in order to do this job
- However, unless you are completely aware of exactly where you could apply for work and, more importantly, who will pay for this work, you may discover that your dream job does not actually exist
- For example, some students decide that they want to work with adolescent girls who are suffering with eating disorders – however, once they qualify as a therapist, they discover that there is no such organisation in their area so there are no paid positions available with this focus
- Assuming that you aim to work full time as a therapist, it is essential that you ask yourself 'who will pay me to do this job?'

Five main ways (other ways do exist but are less common) in which one can have gainful employment in counselling or psychotherapy in the UK

Care agencies

Most therapists work as paid or unpaid volunteers for organisations responsible for providing care to those experiencing specific difficulties, such as substance misuse, bereavement, etc.

Unpaid volunteering in care agencies is particularly popular with trainee therapists or recently qualified therapists working towards full accreditation with a professional body (you need a certain number of hours to gain accreditation)

Healthcare providers

Many therapists work with clients through healthcare organisations based in hospitals, general practices and community healthcare centres

88% of respondents in a telephone survey of 1440 adults in the UK felt that counselling should be available on the National Health Service (BACP, 2010a)

Education settings

Student therapists are widely available in university and college settings, and many schools now often fund a small service employing one or two therapists

Workplaces

Many large companies offer in-house confidential counselling services for employees

This recent move towards offering therapy through human resources is a positive step for therapy, especially since the BACP (2010a) survey suggests that 53% of employed respondents feel stressed by their jobs and 54% feel that the workplace should offer confidential counselling services

Private practice

Some therapists establish their own practices to support private paying clients

The BACP (2010a) survey found that 65% of respondents would be willing to pay for therapy

Legal loopholes in the UK today mean that any individual can open a practice, irrespective of qualifications or accreditation

It is absolutely essential that clients ensure that the selected therapist is qualified (at least a diploma, Bachelors or Master's degree in counselling or psychotherapy), accredited by a professional body (BPS, BACP, UKCP, etc.), and fully insured to practise as a therapist

Differences between counselling and psychotherapy

Psychology, counselling and psychotherapy

CLINICAL PSYCHOLOGIST

Clinical psychologists often work with individuals suffering severe physical or mental disorders and impairments

Clinical psychologists can be defined by focusing on the concept of the 'clinic' – it involves treatment of patients usually in a clinical setting

Clinical psychologists often work alongside medical doctors and neurologists to treat a patient population in health and social care settings (often the NHS)

COUNSELLING PSYCHOLOGIST

Counselling psychologists often work with individuals, couples or families experiencing mental health concerns

Counselling psychologists work in many different locations to support patients and non-patients in healthcare, prison, industry, education, etc.

Minimal difference in job role between clinical and counselling psychologists and many job adverts ask for either type of qualified individual

COUNSELLOR/PSYCHOTHERAPIST

At the present time, there is no recognised distinction between these two titles as both involve the same work role and accreditation process

Counsellors and psychotherapists often work with individuals, couples or families experiencing concerns and worries, or needing space to explore their own thoughts and feelings

Minimal difference in job role between counsellor/psychotherapist and counselling psychologist, although in practice it is often the case that the counselling psychologist will work with those who have significant mental health issues whereas the counsellor/psychotherapist will work with the 'worried well'

PSYCHIATRIST

Psychiatry is substantially different from psychology, counselling and psychotherapy

Psychiatrists are based on the medical model

- Psychiatrists are medical doctors with a specialism in mental health
- Psychiatrists work with patients in the health service
- Psychiatrists can diagnose, treat and medicate patients

Career paths in counselling and psychotherapy

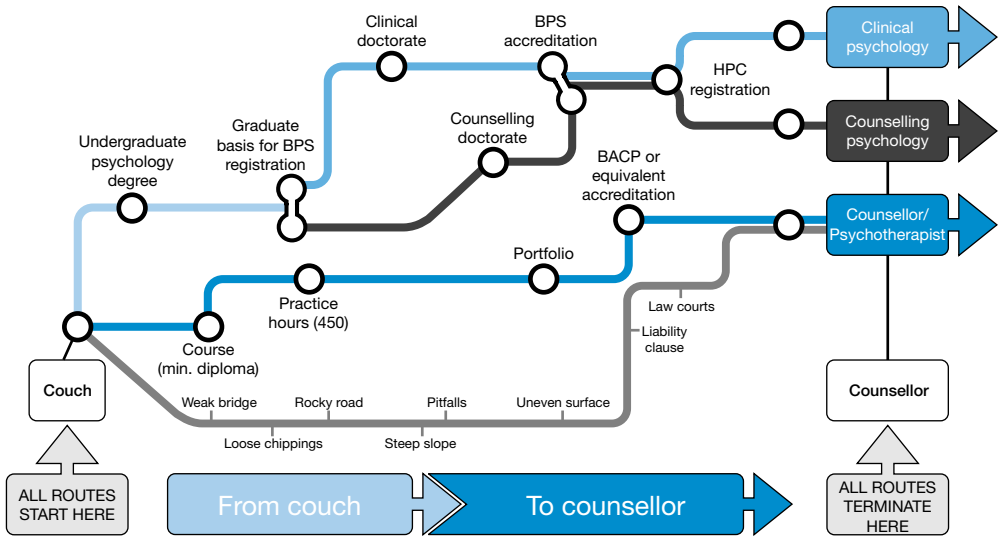

FIGURE 1.1 Couch to counsellor.

Important note

Please note that the information contained in this chapter is in accordance with current regulations in the UK in 2012 – please check on the websites of the relevant accrediting bodies for the most up-to-date information.

Clinical psychologist

Health Professions Council (HPC)

You must register with the HPC to be a chartered clinical psychologist

Health Professions Council is the governmental body responsible for governing all protected titles

You must register with the HPC to use the legally protected title 'clinical psychologist'

Registration with the HPC requires you to submit evidence of having completed an approved course (clinical psychology doctorate)

British Psychological Society (BPS)

You should register with the BPS to be an accredited clinical psychologist

British Psychological Society is the governing body for psychologists in the UK

Registration with the BPS requires you to complete a doctorate (PhD) in clinical psychology (accredited by BPS)

Highly competitive programme – usually need a BPS-accredited first-class degree and possibly a Master's in psychology

Counselling psychologist

Health Professions Council (HPC)

- You must register with the HPC to be a chartered counselling psychologist
  - Health Professions Council is the governmental body responsible for governing all protected titles
  - You must register with the HPC to use the legally protected title 'counselling psychologist'
  - Registration with the HPC requires you to submit evidence of having completed an approved course (counselling psychology doctorate)

British Psychological Society (BPS)

- You should register with the BPS to be an accredited counselling psychologist
  - British Psychological Society is the governing body for psychologists in the UK
  - Registration with the BPS requires you to complete a doctorate (PhD) in counselling psychology (accredited by BPS) or the BPS Qualification in Counselling Psychology (QCoP)
  - Highly competitive programme – usually need a BPS-accredited first-class degree and possibly a Master's in psychology

Counsellor/psychotherapist

Health Professions Council (HPC)

- Counsellor is not a legally protected title in the UK so you do not have to register with the HPC
  - Anyone can work as a counsellor, irrespective of qualifications or membership of professional bodies
  - In February 2011, the UK government recommended a system of voluntary registration and recommended that statutory regulation of titles only be considered when voluntary registers are not sufficient to manage risk
  - In March 2011, the UK government confirmed that it was not its intention to proceed with the statutory regulation of psychotherapists and counsellors

HOWEVER . . .

- Although it is not compulsory to hold the title with the HPC, it is recommended that counsellors and psychotherapists hold appropriate qualifications and 'volunteer' to register with appropriate professional organisations
  - In the absence of qualifications and professional accreditation, counsellors and psychotherapists are unable to get insurance, thus making them vulnerable to risk

British Association for Counselling and Psychotherapy (BACP)

You could register with the BACP to be an accredited counsellor

British Association of Counselling and Psychotherapy is the governing body for counsellors and psychotherapists (no distinction) in the UK

Registration with the BACP requires you to complete an accredited or equivalent course (minimum diploma level), complete 450 hours of practice in a supervised placement, and submit a detailed portfolio demonstrating reflection and continuous professional development

Arguably, the most common route to becoming an accredited counsellor

OR

Any other professional body for counselling and/or psychotherapy

Specific therapies may have their own professional bodies that can offer an appropriate level of accreditation

For example, the British Association for the Person-Centred Approach (BAPCA) or the British Association for Behavioural and Cognitive Psychotherapies (BABCP)

These routes usually follow specific training courses for that therapeutic orientation, rather than general counselling/psychotherapy courses

If searching for a professional body, students should look for those that are recognised at a national level and have their own set of ethical guidelines to which practitioners are expected to adhere

DECISIONS, DECISIONS . . .

It can be difficult to decide which career to pursue, especially since clinical psychology, counselling psychology and counselling/psychotherapy all seem very similar yet require very different training plans.

It is advisable to work backwards in order to identify the appropriate path. Try following these steps to form a personal career plan.

1. Search for your dream job as though you are ready to apply today. Consider websites, newspapers, etc.
2. Identify a job that meets your needs – look at the job description and/or contact the employer to find out exactly what you will be doing on a daily basis.
3. Once you have found an advert for your dream job, look for the job specification (may be titled 'essential/desirable requirements').
4. Identify the required qualifications for this job – do they want a BACP accredited counsellor? BPS accredited counselling psychologist? BPS accredited

clinical psychologist? Write out a plan to gain the relevant qualifications and accreditation to meet those job requirements.

5. Identify the required skills for this job – do they want you to have good teamwork skills or communications skills or information technology skills? Think about your experiences to date and try to find examples of your abilities to excel in these skills. If you cannot identify any experiences to demonstrate some of the skills, write out a plan to gain experiences that demonstrate your abilities to excel in those skills. People often ignore these skills until they are looking for a job and then they struggle to find past experiences to demonstrate the required skills – it is better to think about these in advance so that you can begin working towards bolstering your weaker skills.
6. Pull together your plans to set out actions for gaining your dream job!

Although this specific job is unlikely to still be available when you have gained all of the relevant qualifications and skills, you will be in a good position to find other similar roles.

SUMMARY

Counselling and psychotherapy definition: BACP, BPS, APA, WHO; collaborative, supportive, enhance emotional or psychological wellbeing; therapist offers safe non-judgemental environment to explore own issues; client access through health practitioner, teacher/employer or self-referral; often consists of 50-minute session once per week for a time-limited period; therapists work in care agencies, healthcare organisations, education establishments, workplaces, private practices

Differences: clinical psychologist (patients with mental and physical disorders), counselling psychologist (patients and clients with mental health disorders), counsellor/psychotherapist (worried well clients), psychiatrist (medical doctor with mental health specialism)

Careers: clinical and counselling psychologists must register with HPC (legally protected title) and gain accreditation with BPS (doctorate), counsellor/psychotherapists not required to register with HPC (not legally protected) but should register with professional body like BACP (diploma, 450 hours practice, portfolio)

THERAPIST AND CLIENT ROLES IN THE THERAPEUTIC RELATIONSHIP

LEARNING OUTCOMES

After reading this section, you will be able to:

- differentiate between helping and therapy
- outline the role of the client
- outline the role of the therapist
- discuss the nature of the relationship between the client and the therapist
- debate the nature and value of therapy in the modern world

Differences between helping and therapy

Therapy and helping may differ in many key areas

Contract

Boundaries

Direction

Advice

Criticism

Equality

Outcome

Contract

Therapy involves a short-term therapeutic agreement (formal or informal) between the client and the therapist

Other forms of helping may not involve a specific contract for the helper to provide personal support

For example, one friend may help another friend by listening to his worries, but they have not agreed to a specific therapeutic contract

Boundaries

Therapy involves a distinct type of relationship defined by strict boundaries and a code of ethics

Other forms of helping may not involve such strict boundaries or include a formal code of ethics

For example, a hairdresser may help a client by listening to her problems, but he is not abiding by a professional code of ethics

Direction

Therapy is often non-directive (particularly person-centred), and even those approaches that involve some direction (such as cognitive-behaviour therapy) will try to promote client-led activity whenever possible

Other forms of helping will usually involve the individual being directed towards a specific activity/goal/aim, etc.

For example, a debt advisor may help a client by telling them to set up a low-interest loan specifically to reach the goal of making the overall debt more manageable

Advice

Therapy does not usually involve the provision of advice – many people experience some confusion about therapy as they may expect the therapist to act as an expert offering specific advice and guidance, but this is very rarely the case in a therapeutic relationship

Other forms of helping will usually involve the individual being given advice or guidance on the most effective way to solve a problem or achieve a goal

For example, a teacher may help a student by advising them on how to revise in order to successfully pass an exam

Criticism

Therapy does not usually involve criticism, although it can be challenging for the client

Other forms of helping may use constructive criticism to highlight areas for improvement or motivate further action

For example, a personal trainer may help a client by criticising them in order to motivate them to work harder in the gym

Equality

Therapy involves equality between therapist and client – it is essential that the relationship between the therapist and client remain equal throughout the therapeutic process in order for the client to take ownership of their own growth and development

Other forms of helping may involve a power imbalance as the helper holds more expert power than the person being helped

For example, a lecturer may help a student in many ways but the relationship is always imbalanced because the lecturer has the power to award good or bad grades

Outcome

Therapy does not have a guaranteed outcome – it frequently focuses on the self-exploratory journey rather than the eventual destination, and goals established at the start may not be the desired outcome by the end of the process

Other forms of helping may set a specific outcome at the start and the relationship may be considered a failure if this outcome is not met by the end

For example, a dietitian may help a client by instructing them to eat a wheat-free diet to alleviate allergies and this will have a successful outcome provided that the client follows the advice

Distinctions between therapy, guidance and advice (Sutton & Stewart, 2002)

Advice

> Persuasive one-way exchange involving the advice-giver offering an opinion, making a judgement or making a recommendation

Guidance

> Encouraging a one-way exchange involving the guide showing the way, educating, influencing or instructing

Therapy

> Facilitative two-way collaborative and supportive relationship that allows clients to explore their problem, understand their problem and resolve or come to terms with their problem
>
> Therapists should not give advice as this would result in the therapist taking control of the life of the client rather than the client learning to control his/her own life (Sutton & Stewart, 2002)

HELPING VERSUS THERAPY

Consider the following literary and film figures – each of these characters adopted the role of 'helper' in some shape or form. Do you think that these characters could be classed as 'therapists'?

John Keating in *Dead Poets Society*

Emma Woodhouse in *Emma*

Yoda in *Star Wars*

Albert Pennyworth in *Batman*

Annie Sullivan in *The Miracle Worker*

Patch Adams in *Patch Adams*

Clarence in *It's A Wonderful Life*

Amelie in *Amelie*

The caterpillar in *Alice in Wonderland*

Role of the client in therapy

Reasons for attending therapy

Many people visit a therapist for many different reasons

- Clients may be suffering with a disorder such as depression, or have experienced a recent life event such as divorce, or be coping with a previous trauma such as bereavement, or simply be finding it difficult to make their own way through life without support

Most clients are not mentally ill and do not have a specific diagnosis

- Therapists use the term 'client' because this implies that they are accessing a service, rather than 'patient' because this implies that they have a disorder
- Client is often regarded as the 'worried well'

Clients may simply need somewhere safe to voice their feelings

- It is remarkably difficult to find a confidential and non-judgemental ear
- Many people know that friends and family want the best for them so they may find it difficult to avoid offering advice or opinions when you share your thoughts
- Many people do not want to burden others with their problems, especially since others have their own difficulties
- Many people feel that others may judge them if they are completely honest, and this is likely to be true if the disclosure presents them in a negative light
- Many people feel that others do not have time to listen to them talk (and they are often right)
- Therapy involves an hour of time dedicated to focusing on your problems without fear of judgement or future consequences of disclosure

'Counselling is a wonderful twentieth-century invention. We live in a complex, busy, changing world. In this world, there are many different types of experience that are difficult for people to cope with. Most of the time we get on with life, but sometimes we are stopped in our tracks by an event or situation that we do not, at that moment, have the resources to sort out. Most of the time, we find ways of dealing with such problems in living by talking to family, friends, neighbours, priests, or our family doctor. But occasionally their advice is not sufficient, or we are too embarrassed or ashamed to tell them what is bothering us, or we just don't have an appropriate person to turn to. Counselling is a really useful option at these moments.' (McLeod, 2003)

Carl Jung was once asked by his grandson how people coped before there were therapists and Jung replied, 'They had friends' (Spicer, n.d., cited in Sills, 2006)

High-risk clients

Most new clients will be asked to complete an initial assessment in the introductory therapy session or before attending the first therapy session

- Clinical Outcomes in Routine Evaluation Outcome Measure (CORE-OM) is a common measure for a therapeutic assessment

Assessments are designed to do the following

- Outline the current state of the client to provide the therapist with a preliminary understanding of the mental condition of the client
- Provide a measure by which the therapy can be evaluated – the client will often be asked to complete the same assessment again at the end of the therapeutic relationship to determine the impact of the therapy
- Identify any potential risk factors

Possible risk factors include the following

- Client is likely to engage in self-harm or attempt suicide
- Client is suffering harm or potential harm from another person
- Client is likely to attempt to hurt others (especially vulnerable others such as a child)
- Client is aware of harm or potential harm to others (especially vulnerable others such as a child)

Assessing the risk associated with a client can be a difficult task

- Often difficult to identify expressions or behaviours indicating genuine risk
- Client may say 'he makes me so angry that I could kill him'
  - This expression could be perceived as a flippant turn of phrase designed to exaggerate feelings of anger without any genuine intention to cause harm
  - This expression could be interpreted as a genuine threat to kill
- Client could self-harm by snapping an elastic band against the skin or making small nicks in the back of the arm with a clean knife
  - These actions could be perceived as the first steps towards more significant self-harm
  - These actions could be perceived as a way of dealing with difficulties in a non-life-threatening manner and this could prevent the client from reaching higher levels of emotional intensity and causing more significant self-damage
- In both of these cases, the therapist will need to take all factors into consideration to decide whether the client is high risk

If a client is identified as high risk, the therapist may need to implement certain policies and procedures

- Therapist may need to discuss the case with other professionals to decide on an appropriate course of action
- Therapist may need to establish some safe practice guidelines, such as not working alone with the client
- Therapist may need to breach confidentiality to report the client to the appropriate authorities

Client expectations

Managing expectations is essential to ensure that the client understands the possibilities afforded by therapy and does not expect something that cannot be delivered

- Most client dissatisfaction is caused by a lack of understanding of the true nature of therapy and this could be addressed by exploring expectations in the first session

Common inaccurate client expectations include the following

- 'Therapy will fix/cure me'
  - Therapy is not a magic wand
  - Therapy can be highly effective in supporting the client to address his or her own problems, but there are no guaranteed outcomes
  - Therapy is definitely not a quick and easy fix – it will take time and effort on the part of the client
  - Risks associated with this inaccurate expectation are that clients will disengage once they realise that they are not being 'fixed'
  - Other risks are that clients are referred to the service (by doctors, family, friends, court orders, etc.) with the expectation that this will be a cure
- 'My therapist is responsible for making me better'
  - Although the therapist is responsible for many things (such as the provision of a safe non-judgemental environment), the client is responsible for his or her own self-development
  - Client must be willing to engage with therapy and recognise that therapy involves work during and between sessions
  - This causes a huge problem during enforced therapy (such as court-ordered therapy) because the individual has not chosen to attend therapy so s/he is unlikely to engage with the process
- 'Therapy is an opportunity to have a little chat'
  - Therapy is hard work
  - Client must strive to address life issues and this can often be difficult and stressful
  - Therapist is likely to be challenging during the sessions and this can be difficult for the client to accept
  - Therapy should certainly not be a friendly little chat designed to make the client feel good about him/herself – instead, it should be hard work on the part of both client and therapist as they seek to explore and address difficult topics
- 'My therapist will be on call 24/7 in the event of an emergency'
  - Boundaries must be established at the start of the therapeutic relationship to ensure that the client does not assume that the therapist will be constantly available
  - This can be very difficult in the event of an emergency

It is incredibly heart wrenching to receive a midnight email from a distraught client, but the therapist who responds by inviting the client to an immediate meeting is likely to get the same email again and again (if the client is in immediate risk, the therapist should simply call the emergency services and then relinquish responsibility for that time period)

'Therapy will make me a better person and will ensure that bad things no longer happen to me'

Again, therapy is not a magic wand

Therapy cannot make someone a better person or guarantee that unhappy events will never strike again

However, therapy can help people understand and accept themselves and help them to explore solutions to their problems

'I believe that my therapist is my friend/loves me/desires me because s/he understands me so completely'

There is a risk of romantic affection in any therapeutic relationship, as clients may be surprised by the intense feelings aroused when the therapist genuinely demonstrates an interest in their lives

It is essential that the therapeutic relationship remains professional at all times – the therapist should never take advantage of the position of power and boundaries must be constantly held in place

It is important for therapy to explore the expectations of the client in the initial sessions

It is good practice for a therapist to ask the client what they expect to do in the therapy sessions and what they expect to gain from therapy

Therapist can then address any areas of confusion or inaccurate expectations at the start of the therapeutic relationship to minimise the risk of future problems

Role of the therapist in therapy

Traits of successful therapists

Following a review of hundreds of studies into therapy, Wampold informed an American Psychological Association symposium of the critical traits associated with effective therapists (Whitbourne, 2011)

Sophisticated interpersonal skills

Sensitive cultural awareness

Ability to inspire trust in the therapist and optimism about the therapy

Willingness to form an alliance with the client by explaining the basis of symptoms and developing an agreed treatment plan

Confidence about the therapy and attention to the progress of the therapy

Flexibility to adapt the therapy to the individual

- Self-insight to avoid boundary issues
- Evidence-based practice and continued professional development

According to the BACP (2010b), therapists should aspire to the following personal qualities

- Empathy: ability to communicate understanding of another person's experience from that person's perspective
- Sincerity: personal commitment to consistency between what is professed and what is done
- Integrity: commitment to being moral in dealings with others, personal straightforwardness, honesty and coherence
- Resilience: capacity to work with the client's concerns without being personally diminished
- Respect: demonstrating appropriate esteem to others and their understanding of themselves
- Humility: ability to assess accurately and acknowledge one's own strengths and weaknesses
- Competence: effective deployment of the skills and knowledge needed to do what is required
- Fairness: consistent application of appropriate criteria to inform decisions and actions
- Wisdom: possession of sound judgement that informs practice
- Courage: capacity to act in spite of known fears, risks and uncertainty

Aspiration, not perfection

- These lists of qualities and traits are good for aspiration purposes, but should not be seen as a yardstick for self-beating
- You should aspire to develop these traits while recognising that you will never actually be able to demonstrate all these qualities all the time

Values in therapy

Therapists must recognise their own values, beliefs, prejudices, biases, etc.

- Every single person holds values, beliefs, prejudices, biases, stereotypes, etc.
  - Existence in a social and cultural world will naturally lead to these tendencies
  - From a positive perspective, these tendencies can be helpful heuristics to simplify thinking
  - From a negative perspective, these tendencies can cause us to make assumptions, draw inaccurate conclusions or treat people unfairly
- Since we cannot ever counsel from a completely unbiased position, it is important to recognise our own values in order to acknowledge them if and when they affect our work

Tolerance and respect for others is crucial in effective therapy

By demonstrating these traits as a therapist, you are modelling these positive life habits for your client to develop

However, if you are unable to work with certain clients or problems, recognise this and acknowledge it as an unavoidable limitation in your client base – do not force yourself to work with those who make you uncomfortable (for example, those who have committed certain crimes)

Therapists must demonstrate good self-care

No need to be perfect, take time for yourself, do not assume responsibility for client, do not lose yourself in the problems of the client, do not feel obliged to self-disclose to your client, always remember that you are just as important as the client

Supervision

Every practising therapist must undergo regular supervision to support his or her therapeutic work

Both BPS and BACP guidelines suggest 1.5 hours of supervision for every month in contact with clients (although the BPS does suggest that this should increase in line with caseload)

Therapeutic supervision is a specific relationship designed to support therapists

The supervisor is NOT assuming the role of a line manager – s/he is not there to make sure that the therapist does a good job

The supervisor is NOT a therapist for the therapist – s/he is not there to give individual personal therapy to the therapist

The supervisor IS a professional point of support associated directly with the working life of the therapist

Supervision is essential for supporting the following

Sharing difficult client experiences

Exploring own emotional and psychological reactions to clients

Considering different options for working through client problems

Gaining a different perspective on client issues

Investigating possible directions for future continuous professional development

Therapeutic notes

Therapists are strongly advised to keep notes about their clients (client notes)

Notes can provide a clear record of the relationship for improving recall in later sessions and reflecting on development over time

'Practitioners are encouraged to keep appropriate records of their work with clients unless there are adequate reasons for not keeping any records. All records should be accurate, respectful of clients and colleagues and protected from unauthorised disclosure' (BACP, 2010b)

- However, therapists are not required by law to keep notes and the level of detail produced in the notes is a matter for debate

Some therapists like to keep detailed client notes . . .

- Notes can indicate the client life story, significant events reported by the client, significant events in the relationship with the therapist, thoughts and feelings reported by the client, and thoughts and feelings experienced by the therapist

But there are risks associated with producing detailed client notes . . .

- Client notes are not completely confidential because certain individuals and organisations can request or demand access
- Client notes should always be professional, appropriate and maintain as much confidentiality as possible to minimise risk
    - Notes should never include personal insults or slights, unsubstantiated suspicions or beliefs about a client, unsupported claims about the activities of the client
    - Notes should always be written in a factual manner – it is good practice to report the activities, thoughts and feelings of the client verbatim when possible and indicate that they were reported in such a way by the client him/herself

Certain individuals and organisations are legally entitled to access all client notes

- Client has a right of access to his or her own notes under the Data Protection Act 1998 and Freedom of Information Act 2000
    - But the therapist can insist that the client make a formal request for access to notes and follow a specific access procedure
- Courts have a right of access to any available evidence so they can request full disclosure of client notes to the defence and prosecution in civil and criminal proceedings
    - But therapists can request limited disclosure if they feel that some of the records are not relevant to the case
    - Courts can also request a report or statement from a therapist or even issue a '*subpoena duces tecum*' to bring the therapist to court to give evidence using the original notes
- Police may request access to client notes to conduct specific investigations
    - But police have no right to therapeutic records without a warrant issued by a judge – therapeutic records are exempt from police powers of search and seizure
    - In these cases, it is advisable to maintain confidentiality, ask the client to request access and disclose personally to the police, or request a court order for disclosure
- Solicitors may request access to client notes to support a legal claim
    - But solicitors have no right to therapeutic records without the explicit permission of the client

In these cases, it is advisable to maintain confidentiality, ask the client to request access and disclose personally to the solicitor, or request a court order for disclosure

For further details on appropriate disclosure of notes, please refer to Bond and Jenkins (2007)

Process notes and supervision notes

In addition to client notes, some therapists may keep notes specifically for personal or supervision purposes

Process notes focus on the therapeutic process with a client

Process notes are completely anonymous with no identifiable detail and little explicit content in relation to the client

Instead, process notes focus on the private thoughts and feelings of the therapist in relation to the sessions and the general process of the therapeutic exchanges

In this context, process notes are more about the therapist than the client so they are not subject to the same disclosure rules as client notes

Supervision notes reflect on the experience of supervision

Although these notes may refer to therapy sessions with clients, they will not contain any specific details about the client and will predominantly focus on the development of the therapist

Supervision notes are not subject to the same disclosure rules as client notes

Limits to confidentiality

Clients should have an opportunity to disclose in a safe, non-judgemental, confidential environment

Confidentiality is essential to ensure that the client feels able to discuss his or her life without restraint

In the absence of confidentiality, the client may lie or withhold information and this can make it very difficult to provide effective therapy

However, there are limits to the provision of confidentiality and all clients should be aware of these limits

Confidentiality may be limited to the organisation, rather than the individual

This means that other members of the organisation may have access to the client notes and may discuss the client case with the therapist

This is often done to allow smooth transfer of clients in the event that the therapist has to temporarily or permanently leave the practice

Confidentiality is unlikely to be maintained during supervision

Although most therapists will not provide identifiable details when discussing cases with a supervisor, it is likely that the case will be presented in sufficient detail to allow discussion during the supervision session

Supervisors are, however, also bound by confidentiality guidelines so this information should travel no further

Confidentiality can be breached in response to a court order

As noted in the section on client notes, the court can insist on the therapist revealing details about the client when this is relevant to the trial

Confidentiality can be breached if the client is identified as high risk

Please refer to the previous section explaining the identification of high-risk clients

If a client is identified as high risk through the initial assessment or during subsequent discussions with the therapist, then it is important to disclose appropriately in order to avoid negative ethical or legal consequences

Therapists have a legal obligation to report any client who is at risk of causing harm to self or others and any suspicion of child abuse or terrorist activity irrespective of whether the client is the victim, perpetrator or other

If the client is an immediate risk to self or others (perhaps even the therapist), in an excited state and/or has already hurt him/herself, then the therapist should immediately telephone the emergency services and explain the situation – the emergency services will probably send an ambulance in the event of harm and the police to escort the client to a place of safety (often a local mental health ward)

If the client indicates an immediate intention to harm self or others but is not in danger at that moment, then the therapist should report the disclosure to the head of the therapy service and the police as soon as possible

If the client indicates a future intention to harm self or others, then the therapist should contact the head of the therapy service to discuss the level of risk in order to decide whether to report the disclosure to the police (if in doubt, it is better to report to the police)

If the client indicates knowledge of or involvement in the current or future harm to a vulnerable other (such as a child), then the therapist must immediately report the disclosure to the head of the therapy service, the social services and the police

If the client is not classed as high risk but discloses something with legal or ethical consequences, then the decision to report the disclosure can be less clear

Walfish et al. (2010) surveyed therapists to reveal that many have experienced the client disclosure of an unprosecuted past crime

13% indicated client self-disclosure of murder

33% indicated client self-disclosure of sexual assault

69% indicated client self-disclosure of physical assault

Therapists are not legally bound to report all disclosures of past criminal activity if there does not appear to be a future risk of harm to self or others

Client confidentiality can be maintained in some of these cases in terms of reporting the disclosure, although the therapist may be forced to disclose in court

In these cases, the therapist must consider their own morals and the ethical guidelines of their professional body

If the client discloses a serious past crime (such as murder, assault, robbery, etc.) for which they or another identified individual have not been tried, then the therapist should carefully assess the risk of repeat offending posed by the client

If the client is deemed likely to offend again, then the therapist should report the disclosure

If the client is deemed unlikely to offend again, then the therapist must make his or her own decisions about whether the disclosure is warranted

In all cases of disclosure, it is a good practice for the therapist to outline the intention to disclose prior to disclosure

Therapists should have initially outlined the confidentiality policy to clients at the start of the therapeutic relationship and should reiterate this policy if it appears that the client is going to disclose something important

If the client continues with the disclosure then it can be assumed that they are aware that the information will be passed on, even if they do not appear to be happy about this at the time

Once the disclosure has been made, the therapist should gently explain that the information will need to be reported, outline exactly who will be contacted, and provide guidance on the likely outcome of the report (only to the extent that the therapist is able to advise)

It is important to consider the possible reaction of the client carefully before expressing an indication to disclose

Although this is a preferred method of disclosure, the therapist should not follow this procedure if the client is likely to respond with aggression or violence

In all cases of disclosure or uncertainty about disclosure, it is advisable for the therapist to take the issue to supervision for further consideration (see next section for details about supervision)

Please note that these general guidelines are not comprehensive and could change over time

If a therapist has a concern about the legal limits to confidentiality then s/he must contact his or her professional body to receive the current guidelines

Relationship between therapist and client

Secure

The therapeutic relationship should be secure

Interactions should be warm and friendly

Rapport between client and therapist is essential

Therapy is unlikely to be successful if the client or therapist feels uncomfortable or dislikes the other party

Client should feel safe in disclosing to the therapist

Client should trust that the therapist would maintain confidentiality and remain non-judgemental irrespective of the nature of disclosure (within the agreed limits)

Client must be able to be open and honest with therapist in order to explore issues, and therapist should be honest and congruent in return

Working

The therapeutic relationship should be a working relationship

Sessions should not be little chats, but rather a structured experience of self-exploration working towards an improved understanding of self and a healthier mental outlook

Structure and defined boundaries will help to ensure that the relationship stays at a working level rather than altering to a friend level

Client should become more independent over time, not more dependent on the therapist and therapy sessions

Therapy in the modern world

Therapeutic variety

Thousands of therapeutic methods in modern psychological interventions

Each method offers a new insight into human nature and a new set of techniques for improving mental wellbeing

Some of the less typical methods currently gaining popularity include bibliotherapy, cinema therapy and angel therapy

These are only three of a vast field of possible approaches and these three have been specifically selected as examples to demonstrate how modern therapy draws from self-help, media and spiritual concepts

BIBLIOTHERAPY

Clients are 'prescribed' books to support them through their problems

Recommended books are typically self-help therapy books containing information about the identified disorder and offering guidance or activities designed to encourage self-exploration and enhance coping skills

Book Prescription Wales is a scheme introduced by the National Health Service to support those with mild to moderate emotional problems by prescribing therapy books (NHS, n.d.)

Mental health professionals write out a prescription for their recommended book and the patient is able to fill the prescription at the local library

Systematic review of the effectiveness of bibliotherapy in mental health services suggests that it is beneficial for a range of conditions (Fanner & Urquhart, 2008)

CINEMA THERAPY

Wolz (2013) argues that cinema therapy can be a 'powerful catalyst for healing and growth'

Cinema therapy differs from book therapy because the films used during the treatment are not designed specifically to treat emotional problems (like self-help books), but are instead mainstream films produced for entertainment

- These films offer a creative experience to encourage freedom of thought, feeling and expression, rather than specific guidance on how to solve the emotional problem

Clients are 'prescribed' certain films that are associated with their own problem or emotional state

- For example, a client who is struggling to cope with a divorce might be encouraged to watch *Kramer versus Kramer*

Films offer an opportunity to explore an issue from an objective angle

- For example, the client who watches *Kramer versus Kramer* might be able to see the parallels between the plot and her own experiences and then use this parallel as an opportunity to discuss the situation from a more objective perspective (recognising both points of view to advise the characters)

Films also provide a cathartic release of emotion

- For example, the client who feels that she has to stay strong and not show her emotions about her own situation may have an acceptable outlet for her feelings by reacting to the events in the film

ANGEL THERAPY

Angel therapy combines traditional therapeutic methods with non-denominational spiritual healing

Virtue (2010) proposes that everyone has a guardian angel that seeks to guide him/her down his or her natural spiritual path

- Practitioners use the therapeutic relationship to work with the client to contact their own guardian angel to ask for healing, harmonisation and support

Therapy is generally regarded as pseudoscientific (or entirely non-scientific) and typically ignored in the scientific literature

- Norcross et al. (2006) conducted a Delphi poll (consensus of 101 experts) to identify psychological interventions that are unable to consistently generate treatment outcomes beyond base rates, placebos and time expectancies
- Angel therapy was identified alongside crystal healing, rebirthing therapy and prefrontal lobotomy as a 'certainly discredited' psychological treatment

Interested other

Importance of the therapeutic relationship is a concurrent theme across all therapies, irrespective of the unusual nature of the activities within the treatment

Stiles et al. (2008) suggest that all treatment methods are approximately equivalent in terms of outcome and thus it is the relationship between therapist and client that is the key to effective therapy

Common to all of the traditional (and less traditional) methods is the presence of an interested other

Whether the therapist is trying to identify films that parallel the situation of the client or consulting angels on behalf of the client, the other is showing an interest

Interested other is listening to the life story of the client, appreciating the magnitude of the problems experienced by the client, and seeking to help the client in some way

Technology in therapy (or therapy in technology)

New advances in computer technology and the introduction of the internet have had a massive impact on the field of psychotherapy

Modern therapy can be conducted in a variety of settings beyond the traditional and therapists have embraced these new opportunities to access a more diverse range of clients

ONLINE THERAPY

Cybertherapy can occur in real time through chat-rooms and Skype or be time delayed through email

Many therapists supplement their traditional relationships with clients by adding the extra dimension of online support

For example, clients who are unable to attend an appointment because they are out of the country could have a virtual session through a chat program

Some therapists complete all of the treatment through an online platform

For example, the Association for Counselling and Therapy Online (ACTO) lists over 42 registered therapists offering cybertherapy for depression in the UK

There are many benefits to cybertherapy

Access problems are reduced because the treatment is less costly (therapist does not have location overheads) and more convenient (therapists can see clients across great distances at any time of the day)

Distance between client and therapist may promote disinhibition, thus encouraging the reluctant client to be more open

However, there are also a number of significant disadvantages and limitations for this method

Technical faults during the session can be highly disruptive and may lead to misinterpretations by the client – consider the response if the contact accidentally disconnects immediately after a client has disclosed something extremely personal

Grohol (2011) noted that people may be reluctant to pay for online therapy since they are used to gaining access to online services for free – it is difficult to imagine

- why a client would pay £100 an hour to receive emails from a therapist when they could just access the free support information online instead
- Boundaries are more difficult to maintain when the client can email the therapist at any hour or request appointments during holidays, weekends and nights – it can be extremely difficult to retain the boundary if you happen to see an email threatening self-harm at 3am
- Herwitz (cited in Hoffman, 2011) argues that cybercounselling creates a 'perverse lower version of intimacy' – clients are able to hide behind the computer and avoid engaging in a genuine therapeutic relationship with the practitioner

It is interesting to note that the concept of distance therapy is not novel, although the application through the internet is new

- Many therapists throughout history (such as Freud and Jung) have communicated therapeutically with clients using non-face-to-face means, such as letters or telephone calls

THERAPIST-LESS THERAPY

Movement to an online platform provides a new potential for types of therapy that exclude a physical practitioner

Alongside traditional self-help information online, there are now interactive mobile phone and computer apps designed to provide therapy

- Apps guide users through exploratory activities to help raise self-awareness and achieve goals
- Cognitive-behaviour therapy is the most common therapy for delivery through a computerised application
- Many therapists integrate the use of apps into their practice and many have embraced the new technology as a more interactive form of self-help guide or homework worksheets
- However, apps also provide clients with an alternative therapeutic experience excluding the traditional therapist

On a more extreme level, modern advances in chatbots have produced the potential for a fully automated 'counsellorbot'

- Artificial semi-intelligence in the form of chatbots is able to mimic (to some extent) the art of human conversation and the most advanced chatbots are very similar to human respondents in the simulated environment
- These advances offer the potential for counselling to occur with a non-human therapist
- iTherapy (2005) is adopting chatbots to administer initial tests and establish the background story of the client before passing this information over to human counsellors
- The Machine called Abel (Genesis Interactive, 2009) offers teenage counselling services using a chatbot that can mimic teenage language in a counselling setting –

the creator argues that 'most teenagers hang out on the internet . . . through support and interest from appropriate social groups we can create an instance of ABEL to have the personality of a teenager, that speaks in their language (the local lingo) and comes across as someone who can relate with what they might be going through . . . at last, the teenagers will have a place to share their emotions – albeit with a machine, but an intelligent one' (Genesis Interactive, 2009)

Counsellorbots could potentially offer the ultimate in unconditional positive regard given the objective non-judgemental programming of the machine, yet perhaps fail to offer genuine empathy in a warm setting

These types of therapy raise new questions about the importance of the therapeutic relationship

Does therapeutic benefit depend on having a genuine interested other present in the relationship?

Or can there be therapeutic advantages to providing an outlet for self-exploration and emotion even in the absence of a receiver?

VIRTUAL REALITY AND AVATAR THERAPY

Modern use of virtual simulations through avatars has provided some novel options for implementing therapeutic interventions

Virtual reality has been used to provide safe graded exposure to fear-inducing stimuli during behaviour therapy (systematic desensitisation)

For example, there are large reductions in anxiety in phobia disorders following virtual reality exposure therapy (Parsons & Rizzo, 2008)

Simulated environments have also been used to provide wider and more diverse access to patient populations

For example, therapy through the online virtual world *Second Life* is feasible for a wide population and effective in reducing social anxiety disorders (Yuen et al., 2013)

Avatar therapy has been used to provide interactive constructs for exploring the various aspects of the self

For example, one novel approach to treating schizophrenic persecutory hallucinations is encouraging patients to create an avatar for the malevolent entity and then allowing the patients to slowly bring the entity under their control (Leff et al., 2013)

SUMMARY

Therapy versus helping: contract, boundaries, direction, advice, criticism, equality, outcome; distinction between advice, guidance and therapy

Client: many reasons for seeking therapy, not always mental illness, safe place to disclose; high-risk clients identified in initial assessment (CORE-OM), high risk = risk of harm to self or others, assessment is subjective, may need to breach

confidentiality with certain disclosures; client expectations must be managed at the start of the relationship

Therapist: Wampold lists traits of effective therapists (interpersonal skills, cultural awareness, inspires trust, forms an alliance, confident about therapy, flexible during therapy, self-insight, evidence-based practice and CPD); BACP lists traits to be an effective therapeutic person (empathy, sincerity, integrity, resilience, respect, humility, competence, fairness, wisdom, courage), aspiration not perfection; values, recognise own biases, demonstrate good self-care; attend regular supervision 1.5 hours per month to support work (not line manager or personal therapist); keep appropriate therapeutic notes (client, supervision, process), notes can be accessed by client and legal bodies; confidentiality is essential but limited, info shared with organisation and supervisor, confidentiality can be breached by court order or if client is high risk, legally obliged to disclose risk to children and terrorism – all other cases at the discretion of the therapist

Relationship: secure, honest, open, safe, warm, friendly; working, not 'little chats'; structured, defined, encouraging independence

Therapy in modern world: therapeutic variety, literally thousands of possible methods; bibliotherapy involves self-help book prescriptions (self-help), cinema therapy involves film prescriptions (media), angel therapy involves seeking guidance from guardian angels (spiritual) – regarded as pseudo-scientific; relationship (interested other) is critical in all methods; technology, online therapy through delayed email or real-time chat/Skype (enhances access, but less genuine relationship), therapist-less therapy through apps or counsellorbots (enhances access but eliminates relationship), virtual reality and avatar therapy

BOUNDARIES AND ETHICS IN THE THERAPEUTIC RELATIONSHIP

LEARNING OUTCOMES

After reading this section, you will be able to:

- discuss the importance of maintaining boundaries in the therapeutic relationship
- highlight the nature of the therapeutic contract
- discuss ethical issues in relation to therapy

Boundaries in therapy

Boundaries are defined as the rules that an individual holds about the appropriate and acceptable ways to interact with others

In therapy, boundaries refer to specific lines that should not be crossed by the therapist and/or client

- These lines are put in place to ensure that there is an appropriate amount of emotional, psychological and physical space between individuals
- Boundaries are a relatively new concept and research suggests that therapists did not consider boundaries in this context prior to 1990 (Totton, 2010)

Boundaries are essential in the therapeutic setting as the nature of the relationship between the therapist and the client could lead to misunderstandings

- Clear boundaries will ensure that the client remains aware that the therapist is operating as a professional and this will reduce the likelihood that the client may misinterpret empathy for affection or desire
- Clear boundaries will also reassure the client and the therapist as each individual in the relationship will be confident in their own role and the expectations associated with those roles

Both the BPS and the BACP state that all therapists should seek to establish clear boundaries

- Boundaries will provide a framework for working with clients
- Defining and maintaining boundaries is the responsibility of the practitioner

Boundaries should be established through collaboration between the therapist and client

- Initial sessions should focus on establishing the boundaries of the therapeutic relationship
  - Some boundaries will need to be established by the therapist irrespective of input from the client (such as rules regarding confidentiality), but even those boundaries should be discussed in the initial session
- Mearns and Thorne (2007) note that 'for the counsellor to impose boundaries without a consultative process with the client would be a denial of the essential equality of the relationship which it is hoped to establish'

Some boundaries can be simple to establish, whereas other boundaries may require careful consideration by both the therapist and the client

- Sometimes boundaries can easily be determined when considering extreme behaviour but can be more difficult to establish when considering milder forms of that behaviour
  - For example, present giving is generally regarded as inappropriate and most therapists would refuse an expensive gift like a diamond necklace
    - But it is difficult to adopt this rule for less extravagant gifts like a home-made cake without hurting the feelings of the client

For example, physical contact is generally regarded as inappropriate and most therapists would refuse to hug or kiss a client in greeting

But it is difficult to adopt this rule for less excessive contact like a touch to the back of the shoulder

Sometimes boundaries can be restricted by the nature of the therapeutic setting

For example, some therapists may work from home or offer home visits and this could result in future difficulties maintaining boundaries

Ryan (2010) noted: 'Working one-to-one from within a client's home generates a different dynamic. This is not a therapeutically neutral space; it is their home, a window on their personal world. Entering the client's home turns the conventional therapist/client relationship upon its head; there is a shift of power. In the therapy room the therapist controls the environment, the timing, and the ambience. In the client's environment the counsellor is in an alien environment, outside of their comfort zone . . . Even though the counsellor is sitting in their client's realm, there has to be separation between client and counsellor; the counsellor is there to counsel, not befriend, and boundaries are essential to keep that therapeutic distance'

Maintaining boundaries

Each therapist is different in their own expectations and requirements for clear boundaries. However, here are some possible suggestions for maintaining clear boundaries in the therapeutic relationship

Establish a therapeutic contract at the start of the relationship

Meet only at agreed times for sessions

Do not allow yourself to be 'on call', even for emergencies

Do not engage in a physical relationship with the client

Avoid self-disclosure if it is not relevant to the therapy

Try to focus the sessions on the therapy rather than simply 'a chat'

Refer to other agencies for problems beyond your own abilities

Therapeutic contract

Encouraging client autonomy

BACP ethical guidelines (2010b) relating to autonomy state that the therapist should 'engage in explicit contracting in advance of any commitment by the client' in order to ensure that both the counsellor and the client give fully informed consent

BPS (2009) noted that practitioners are responsible for making and reviewing 'clear and explicit contracts' that inform clients about confidentiality, record keeping and financial costs of therapy

Therapeutic contract helps to establish clear boundaries relating to the therapeutic relationship, thus encouraging client autonomy

Contracts provide an opportunity for the client to give fully informed consent to the therapeutic process

The contract launches the working alliance between the therapist and the client (Clarkson, 1992). The contract can 'make clients more informed about the process, more collaborative with their helpers, and more proactive in managing their problems' (Egan, 1994)

Advantages of a contract

Clients feel secure about exactly what is expected of them during the process

Clients are aware of all the practical aspects of the sessions such as length, duration, cost, etc.

Clients will feel in control of the therapeutic relationship as they have an active input in the agreement

Types of therapeutic contract (Dale, 2003)

Business contract

Duration of sessions, number of sessions, charges of sessions, cancellation expectations, etc.

Therapeutic contract

Reasons for attending therapy, process of therapy, responsibilities and expectations, limits of confidentiality, etc.

Contracts can vary in terms of style, content, depth, etc.

Some contracts can be very basic

Organisations such as the Samaritans provide drop-in centres with basic provision including little more than the offer of a safe space in which to spend as much or as little time as desired so these contracts are often unwritten verbal agreements only

Some contracts can be very complex

Private practices often use detailed written contracts to reduce the potential risk of litigation in the event of future disagreements with a client

Verbal or written contract

Advantages of verbal contract

Less official and formal so less intimidating to the client

Advantages of written contract

More permanent as a record of agreement to be used in the event of future problems

Recommended, not compulsory

Although contracts are strongly recommended in most practice guidelines (BPS, APA, BACP, etc.), they are not compulsory and some therapists do operate without contracts

Some therapists do not have an explicit contract, but they do engage in contracting activities in the initial session (discussing limits of confidentiality, agreeing session duration and cost, etc.)

Risks of operating without any type of contract (informal verbal agreement or formal written agreement) are that the therapist is deemed liable in the event of a disagreement relating to the promised provision

WHAT WOULD YOU DO?

Consider the following ethical dilemmas and try to imagine how you might react as a therapist.

- A man confesses to you that he murdered another man five years ago and has never stood trial for the crime. He says that it was a crime of passion committed under the influence of alcohol. Today, he is an upstanding citizen who strongly regrets his actions and wants to come to terms with what he has done to accept himself. There does not appear to be any risk that he will reoffend. What do you do?
- A female client is booked to see you from 4pm to 5pm on a Friday evening in January. When the session ends, it is dark outside and the weather is terrible. You know that your client will be walking home and that this walk will take her through a dangerous neighbourhood. You are also aware that you will probably be driving past her house on your way home. What do you do?
- Your client has been increasingly friendly over the last few sessions and constantly asks questions about your private life. In this session, the client confesses to having strong feelings towards you and invites you out for a coffee. The client explains that this would not be a breach of ethics because your client intends to terminate the therapy as soon as you agree to a date. What do you do?
- You run a private practice in your own home and you need to lead your clients through your hallway to reach your office. One of your clients is addressing issues of bereavement following the death of her baby. At the start of the third session, you notice that she stops to look at your family photographs in the hallway on her way through your house. During the session, she explains that the photographs make her feel uncomfortable and she asks you to remove them during her sessions. What do you do?

- You have recently been through a difficult divorce due to infidelity by your partner. Your client is currently wrestling with a decision about marrying her childhood sweetheart. She has described her fiancé as flirtatious and she suspects that he may have had an affair in the past. You believe that she would be making a mistake by marrying her fiancé and you are often tempted to give her advice on the basis of your experience. What do you do?
- You have recently suffered the loss of your adult daughter to cancer. You are finding it difficult to sleep and you have lost a lot of weight due to your tendency to forget to eat. You decided to return to work to try to take your mind off your sadness. Your client has recently given birth to a baby girl and she is attending therapy because she is feeling overwhelmed by the new arrival. You are finding it difficult to listen to her discuss her feelings about the new baby without reflecting on your own recent loss. In the last session, you had to leave the room for a few minutes to compose yourself because you began to cry. You are scheduled to see the client again tomorrow and you are dreading the session. What do you do?

Ethical guidelines

Ethical guidelines are designed to protect both client and therapist

Historically, some therapeutic practices have led to serious physical and psychological abuse of vulnerable clients

Recovered memory therapy (hypnosis, medication, dream analysis, etc. to recover repressed trauma) carries a high risk of false memory syndrome

The False Memory Syndrome Foundation reports a large number of cases of child abuse retractors who made accusations as a direct result of false memories (False Memory Syndrome Foundation, n.d.)

In 1992, Beth Rutherford accused her parents of sexually abusing her between the ages of seven and 14 (Loftus, 1997). Beth had 'recovered' memories during therapy of her father repeatedly raping her, her mother helping him by holding her down, being impregnated by her father on two occasions, and being forced to abort her fetus with a coat hanger. However, a medical examination revealed that she had never been pregnant and remained a virgin. Beth successfully sued her therapist for $1million in 1996

Rebirthing therapy carries a high risk of injury or even death

Rebirthing therapy is designed to address attachment disorders by simulating the birthing process to allow the formation of new attachments

Rebirthing therapy has been banned in some US states following the death of a 10-year-old child during the procedure (Josefson, 2001). Candace Newmaker

- was smothered in pillows and blankets then repeatedly crushed by therapists in an effort to simulate the birthing process in order to help her attach to her new adopted parents. Video evidence shown in court revealed that the child vomited, excreted and cried to be released, but the therapists responded by ignoring her or mocking her for not trying hard enough to be born. Eventually, after 70 minutes of torture, Candace died as a result of asphyxiation. The therapists responsible were convicted of reckless child abuse resulting in death and were sentenced to 16 years in prison

- Ethical guidelines issued by professional bodies are designed to reduce the risk of abuse and support the professional practice of therapist members

Formal ethical guidelines support the decision-making process for practising therapists

- Ethical issues can often be complex and controversial
  - Judgements about what is 'wrong' or 'right' can often reflect social conventions or personal opinions
  - For example, an individual may believe that it is 'wrong' to be unfaithful because it goes against our social convention of marriage
  - An individual may believe that it is 'wrong' to eat meat because it goes against his own personal beliefs
- Therapeutic ethical guidelines can offer a set of rules for all practitioners
  - These rules allow consistent behaviour across all ethical practitioners
  - However, many of these 'rules' are flexible to the extent that they are open to the interpretation of the practitioner and this can lead to the same complexities that the rules seek to reduce

Professional therapeutic bodies (BACP, UKCP, BPS, etc.) outline ethical guidelines by which member practitioners must abide

- Therapists must adhere to the ethical guidelines issued by their own professional bodies
  - Every working therapist should be a member (ideally, an accredited member) of a professional body
  - Each member must follow the ethical guidelines for that professional body
  - Therapists should avoid membership of professional bodies that do not provide ethical guidelines
- Some therapists choose to practise independently of any professional bodies
  - Membership and accreditation of professional bodies are voluntary (refer to information about protected titles with the HPC in the careers section of this chapter)
  - It is possible for some individuals to practise without being a member of any professional body; thus they operate without adhering to a formal set of ethical guidelines
  - While these practitioners may adhere to their own moral and ethical guidelines, therapists who are not members of any professional organisation will find it

difficult to obtain adequate insurance and may (or perhaps should) be viewed with mistrust

Some of the key values and principles for three major professional bodies (BACP, UKCP, BPS) are outlined in the next few pages. However, please note that this is just a summary of the main points and it is highly recommended that you consult the references link to read the full guidelines

BACP Ethical Framework for Good Practice in Counselling and Psychotherapy (2010b)

Values of counselling and psychotherapy

Respect human rights and dignity

Ensure the integrity of practitioner–client relationships

Enhance the quality of professional knowledge and its application

Alleviate personal distress and suffering

Foster a sense of self that is meaningful to the person(s) concerned

Increase personal effectiveness

Enhance the quality of relationships between people

Appreciate the variety of human experience and culture

Strive for the fair and adequate provision of counselling and psychotherapy services

Principles of counselling and psychotherapy

Fidelity: honouring the trust placed in the practitioner

Therapists should respect the trust of the client and avoid disclosure of confidential information

Autonomy: respect for the client's right to be self-governing

Therapists should ensure that advertisements and information about services are accurate, and they should protect the privacy of clients, engage in explicit contracting before gaining fully informed consent, report any conflicts of interest, and refuse to manipulate clients against their will (even for beneficial social ends)

Beneficence: a commitment to promoting the client's wellbeing

Therapists should act in the best interests of the client (especially those with diminished capacities), provide services on the basis of adequate training or experience, take advantage of supervision to support work with clients, and commit to continuous professional development

Non-maleficence: a commitment to avoiding harm to the client

Therapists should hold appropriate insurance, refuse to work when unfit due to illness or personal circumstances, seek to minimise harm to clients and assume responsibility to challenge poor practice in others, and avoid any incompetence, malpractice or client exploitation

Justice: the fair and impartial treatment of all clients and the provision of adequate services

Therapists should respect the human rights and dignity of all clients, be committed to equality of opportunity by avoiding discrimination on the basis of prejudice, provide fair access to services for clients with differing needs, remain aware of legal obligations, and remain alert to conflicts between legal and ethical obligations

Self-respect: fostering the practitioner's self-knowledge and care for self

Therapists should use supervision for professional support, seek continuous professional development, obtain appropriate insurance, engage in life-enhancing activities and relationships independent of relationships in counselling, and apply all of these principles to the self as well as clients

UKCP Ethical Principles and Code of Professional Conduct (2009)

General ethical principles

Best interest of clients

Therapist should treat all clients with respect, refuse to exploit or abuse clients, respect the autonomy of clients, not enter into a personal relationship with clients, not harm or collude in the harm of clients or others, and recognise how own behaviour outside of professional life can impact on clients

Diversity and equality

Therapist should actively consider issues of diversity and seek to prevent prejudice from impacting on treatment

Confidentiality

Therapist should respect and protect the confidentiality of clients, ensure consent where possible when information is to be disclosed, and explain to clients prior to the relationship those legal situations in which confidentiality cannot be guaranteed

Conduct

Therapist should recognise the impact of their actions on clients, seek to promote professional conduct at all times, and disclose any criminal convictions, investigations or disciplinary actions to the professional body

Professional knowledge, skills and experience

Therapist should disclose all qualifications to clients and the professional body to avoid any misleading assumptions

Communication

Therapist should explain all pertinent contractual points to the client and notify about ethical issues (including complaints procedure)

Obtaining consent

Therapist should explain the therapeutic approach to the client and gain fully informed consent to participate in therapy and/or any research associated with the therapy

Records

Therapist should keep appropriate records and store these records in accordance with data protection laws to safeguard client confidentiality

Physical or mental health

Therapist should not work when unfit to do so due to physical or mental illness or impaired by alcohol, drugs, etc.

Professional integrity

Therapist should report any breaches of ethical guidelines committed by themselves or colleagues

Advertising

Therapists should ensure that advertisements for services are fair, accurate and responsible

Indemnity insurance

Therapists should ensure that they are covered to practise by indemnity insurance

Complaints

Therapist should ensure that clients are aware of all complaints procedures and should raise ethical concerns to the professional body when observing breaches in the behaviour of colleagues

BPS Code of Ethics and Conduct (2009)

Ethical principles

Respect

Therapist should respect the dignity and worth of all people, maintain privacy and confidentiality, seek informed consent for any therapeutic intervention, and encourage and support client self-determination

Competence

Therapist should be aware of professional ethical guidelines, engage in ethical decision making, recognise the limits of their own professional competence, and recognise impairment due to health or personal issues and refrain from practice when unfit to do so

Responsibility

Therapist should avoid bringing the profession into disrepute through misconduct or allowing harm to clients or others, recognise those conditions under which the therapeutic relationship should be terminated, and seek to protect and debrief all research participants

Integrity

Therapist should be honest and accurate in dealing with clients and the general public, avoid exploitation and conflicts of interest, maintain personal boundaries, and address ethical misconduct in colleagues

WHAT WOULD YOU DO (AFTER FURTHER REFLECTION)?

Consider the following ethical dilemmas again and how you might react as a therapist based on the ethical guidelines available from professional organisations.

A man confesses to you that he murdered another man five years ago and has never stood trial for the crime. He says that it was a crime of passion committed under the influence of alcohol. Today, he is an upstanding citizen who strongly regrets his actions and wants to come to terms with what he has done to accept himself. There does not appear to be any risk that he will reoffend. What do you do?

> Remember that the therapist should maintain client confidentiality within legal limits and should seek to protect their own clients, clients of colleagues and members of the general public
>
> Consider the BACP principles of autonomy and justice, UKCP principles of confidentiality and professional integrity, and BPS principles of responsibility and integrity

A female client is booked to see you from 4pm to 5pm on a Friday evening in January. When the session ends, it is dark outside and the weather is terrible. You know that your client will be walking home and that this walk will take her through a dangerous neighbourhood. You are also aware that you will probably be driving past her house on your way home. What do you do?

> Remember that the therapist should not hold multiple relationships with the client and should seek to protect the client
>
> Consider the BACP principles of autonomy and beneficence, UKCP principles of best interest of clients and conduct, and BPS principles of responsibility and integrity

Your client has been increasingly friendly over the last few sessions and constantly asks questions about your private life. In this session, the client confesses to having strong feelings towards you and invites you out for a coffee. The client explains that this would not be a breach of ethics because your client intends to terminate the therapy as soon as you agree to a date. What do you do?

> Remember that the therapist should not hold multiple relationships with the client and the therapist and client are both entitled to terminate therapy at any time
>
> Consider the BACP principles of autonomy and non-maleficence, UKCP principles of best interest of clients and conduct, and BPS principles of responsibility and integrity

You run a private practice in your own home and you need to lead your clients through your hallway to reach your office. One of your clients is addressing issues

of bereavement following the death of her baby. At the start of the third session, you notice that she stops to look at your family photographs in the hallway on her way through your house. During the session, she explains that the photographs make her feel uncomfortable and she asks you to remove them during her sessions. What do you do?

- Remember that the therapist should seek to protect the client and should also protect their own wellbeing
- Consider the BACP principles of beneficence and self-respect, UKCP principles of best interest of clients and physical or mental health, and BPS principles of respect and integrity

You have recently been through a difficult divorce due to infidelity by your partner. Your client is currently wrestling with a decision about marrying her childhood sweetheart. She has described her fiancé as flirtatious and she suspects that he may have had an affair in the past. You believe that she would be making a mistake by marrying her fiancé and you are often tempted to give her advice on the basis of your experience. What do you do?

- Remember that the therapist should respect the free will of the client and seek to protect the client
- Consider the BACP principles of autonomy and beneficence, UKCP principles of best interest of clients and physical or mental health, and BPS principles of respect and competence

You have recently suffered the loss of your adult daughter to cancer. You are finding it difficult to sleep and you have lost a lot of weight due to your tendency to forget to eat. You decided to return to work to try to take your mind off your sadness. Your client has recently given birth to a baby girl and she is attending therapy because she is feeling overwhelmed by the new arrival. You are finding it difficult to listen to her discuss her feelings about the new baby without reflecting on your own recent loss. In the last session, you had to leave the room for a few minutes to compose yourself because you began to cry. You are scheduled to see the client again tomorrow and you are dreading the session. What do you do?

- Remember that the therapist should seek to protect the client and should protect their own wellbeing
- Consider the BACP principles of beneficence and self-respect, UKCP principles of best interest of clients and physical or mental health, and BPS principles of respect and competence

Can you imagine any other scenarios that could inspire an ethical dilemma? Devise three scenarios that could pose an ethical problem then work with a partner to identify the main principles associated with the scenarios and develop an action plan for working within the situations.

All professional bodies have similar ethical guidelines

Ethical considerations common to most professional bodies (BACP, UKCP, BPS, APA, BABCP, etc.) include the following

Therapist should provide all pertinent information at the start of the relationship (contracting) to obtain fully informed consent

Therapist should maintain client confidentiality (within legal limits)

Therapist should not hold multiple relationships with the client (client cannot be a sexual partner, family member, friend, etc.)

Therapist should not exploit the client in any way (sexual, financial, psychological)

Therapist should respect the free will of the client

Therapist should seek to protect the client, clients of colleagues and members of the general public

Therapist and client are entitled to terminate therapy at any time (given appropriate notice)

Therapist should protect their own wellbeing by withdrawing from work if unfit, using supervision to support practice, and seeking continuous professional development

ETHICAL CONTROVERSY

Ethical guidelines are designed to reduce the risks associated with therapy, but there are still many critics who argue that therapy carries ethical questions irrespective of these regulations. Reflect on the following questions and discuss the ethical issues with a colleague.

Does therapy lead to an unhealthy dependency on the therapist?

Despite all efforts to develop a collaborative relationship, it is inevitable that the client will view the therapist as an 'expert'

As an expert, the therapist wields a great deal of power and this power is very attractive to many clients

Clients who feel better as a result of therapy are reinforced for engaging with the therapeutic process; thus they may begin to believe that therapy is essential for them to be able to cope with everyday difficulties

Will therapy inevitably influence the client in the direction desired by the therapist?

Arguably, it is impossible to interact with another person without influencing the other to some extent

Therapy will inevitably influence the client – indeed, therapy presumably aims to influence the client in the direction of positive change

- However, it is difficult to state who should make the decision about which direction is 'positive' and which direction is 'negative'
- Therapists often seek to ensure that all decisions relating to change are made by clients, but there will be an inevitable influence of the therapist
- This means that the personal beliefs and biases of the therapist will impact on the life choices of any client

Does therapy encourage mental health disorders by providing reinforcing attention for reported symptoms?

- Many clients want to be the 'model patient' and seek ways to present exactly as they should
- Clients may learn about the diagnosed disorder and the intended treatment then report the correct symptoms to show that they are being a good client – sometimes they may be faking these symptoms and sometimes they may actually experience pseudo-symptoms as a result of the intense desire to please the therapist
    - For example, 'Lauri' reported her experiences of false memory syndrome as a patient in a multiple personality disorder therapy group (False Memory Syndrome Foundation, n.d.)
    - Lauri explained that she learnt the traits of multiple personality disorder (now known as dissociative identity disorder) in order to be the 'model patient'
    - Once she was exhibiting the behaviours associated with this disorder, her therapist explained to her husband, 'Because Lauri now has the behaviour, it follows that she has multiple personality disorder. Thus, some terrible abuse in her childhood must have caused it. So terrible that she's repressed those memories deep in her mind'
    - During subsequent regression hypnosis, the therapist invited Lauri to create an alter and then used leading questions to encourage this alter to describe childhood abuse
    - Eventually, Lauri reported a wide range of abuse, including taking part in Satanic rituals, being buried alive, drinking blood and being involved in the murder of a baby
    - Lauri was rewarded for recovering these memories through attention and sympathy – the therapist described her as his 'best' patient

Is the therapeutic field promoting the overdiagnosis of mental health disorders?

- Diagnosis of disorders such as depression and anxiety has increased substantially over recent years to the extent that one in four people will experience a mental health disorder at some point in the next year (Mental Health Foundation, n.d.)

Lutus (2013) is highly critical of psychology and psychotherapy in his report on the absence of science in the field of psychology

Lutus observes that the *Diagnostic and Statistical Manual of Mental Disorders* (DSM) classification of 112 disorders in 1952 rose to 374 disorders by 1994

Lutus argues that the DSM has begun to define many arguably normal emotional states as 'mental health disorders' – for example, he cites sibling rivalry disorder as one diagnosis that should not be classified as a mental health disorder

Lutus claims that this increase in the classification of arguably normal behaviours as 'disorders' will eventually lead to no behaviour being regarded as normal

SUMMARY

Boundaries: rules of appropriate interactions with others, lines between self and other, avoids misunderstandings, advised by professional bodies; established through collaboration between therapist and client, discuss in initial session, agree through consultation with client; maintaining boundaries, no hard rules as they may differ for each individual and situation, suggestions (establish contract at start, meet only at agreed times, do not be on-call, do not have physical relationship, avoid non-relevant self-disclosure), focus on therapy rather than a chat, refer if necessary

Contract: encourages client autonomy; type – business or therapeutic; style – basic or complex, verbal or written; recommended not compulsory; some make explicit contracts, some make implicit contracts, some have no contracts (not recommended)

Ethics: ethical guidelines aim to protect client and therapist; some forms of therapy can be extremely dangerous, recovered memory therapy carries risk of false memory syndrome, rebirthing therapy carries risk of injury or death; ethical issues can be complex and controversial with no clear right or wrong; professional bodies offer some rules, not recommended to operate independently of any professional body; BACP Ethical Framework for Good Practice in Counselling and Psychotherapy (fidelity, autonomy, beneficence, non-maleficence, justice, self-respect); UKCP Ethical Principles and Code of Professional Conduct (best interest of clients, diversity and equality, confidentiality, conduct, professional knowledge, skills and experience, communication, obtaining consent, records, physical or mental health, professional integrity, advertising, indemnity insurance, complaints); BPS Code of Ethics and Conduct (respect, competence, responsibility, integrity);

common guidelines irrespective of which professional body (informed consent, confidentiality, conflicts of interest, exploitation, free will, protect client and others, right to terminate, protect own wellbeing)

APPRECIATING DIVERSITY IN THE THERAPEUTIC RELATIONSHIP

LEARNING OUTCOMES

After reading this section, you will be able to:

- appreciate some of the cultural assumptions in mainstream therapy
- explain multiculturalism in the context of the therapeutic relationship
- self-reflect on your own cultural biases and understanding of cultural diversity
- critique the concept of multicultural therapy
- appreciate the importance of working holistically

Cultural assumptions in mainstream therapy

Modern western-centric therapy

Before we begin to explore our own attitudes to working with clients from different cultural backgrounds, it is worth noting that most therapies are driven by western culture and western values.

Looking at the main theoretical approaches to be explored in later chapters, we can begin to see the 'western-centric' lens through which therapy has been developed

Key figures responsible for developing the main approaches include

humanistic approaches: Rogers, Maslow

psychodynamic approaches: Freud, Jung

behavioural approaches: Watson, Skinner

cognitive approaches: Neisser, Beck

All of these key people are white, western, educated, middle-class, middle-aged men

Some women have made significant contributions to development of theory (for example, Melanie Klein, Karen Horney and Laura Perls), but their contributions have, in some cases, been played down due to the attitudes of men and the values of society in their time

However, these women are still white, western, educated, middle class and middle aged

As therapists, our own values and assumptions, stereotypes and biases are influenced by our society, our culture, our parents and our significant others

Similarly, all the psychological approaches and therapies discussed in this book will have been developed within the context of the time and place of the founder, who in turn will have been influenced by their upbringing (society, history, context, parents, etc.)

All approaches will have been influenced by the values and assumptions of the society in which the founder was living

This is one reason why this book includes historical and biographical details of the founding fathers of each approach

An understanding of these individuals gives an insight into the context for their approach and therapy

For example, the behavioural approach was developed at a time when behavioural psychology had established research into classical conditioning and assumed that behaviour was learned, could be unlearned and replaced by more effective learning (e.g. the installation and removal of a phobia)

The humanistic approach has basic assumptions (e.g. individuals striving for actualisation, the importance of the individual, etc.) that are influenced by the values of their time (the 'American Dream', the pursuit of happiness, etc.)

Individualistic versus collectivist cultures

All the therapies that developed from the first three forces of psychology had common values, including

Belief in the importance of the individual

Belief that individuals are the drivers of their own destiny

Belief that personal choice and responsibility should be valued

Belief that we all have the potential to change

All of these beliefs are common to the individualistic culture in western society

In comparison, collectivist cultures in eastern society may have very different values, including:

Belief in the importance of the group, family and society

Belief in the harmony of the individual in connection with the universe

Belief in the interconnectedness of opposites (dark/light, yin/yang, good/evil)

Belief in the importance of acting in a way that benefits society rather than the self

Therapies developed in the western world may make assumptions about the 'correct' goals for the client

For example, these therapies may focus on the importance of the individual client achieving independence and exercising free will

However, these goals might not fit with a culture in which it is expected that a person will contribute to the family and adhere to the family rules of honour

Multiculturalism in the therapeutic relationship

Monocultural to multicultural

In response to the monocultural perspective of traditional counselling theories, there have been developments in approaches to engaging clients from minority groups

The multicultural approach developed out of the civil rights movements and from equality legislation

Even in those therapies that emphasise the uniqueness of each individual and strive to understand the world from the client's perspective, the approach itself is grounded in implicit values and assumptions

A basic assumption in the development of a multicultural perspective is that traditional theories of therapy have little understanding of current cultural diversity

The culture in which a person is brought up influences the development of identity and is the basis for thoughts, feelings and behaviours

Therefore, culture should be at the centre of the therapeutic relationship rather than only noteworthy when there is an obvious cultural difference between therapist and client

Multiculturalism has been described as the fourth force in therapy, after the behavioural, psychodynamic and humanistic approaches (Pederson, 1991)

Sometimes the focus of multicultural therapy is described within narrow definitions and assumes that the goals are based around helping individuals assimilate into a new culture

However, the term 'multiculturalism' embraces a range of diversity and differences

Nelson-Jones (2005) identifies ten areas of difference

Culture

Race

Social class

Biological sex

Gender role identity

Marital status

Sexual orientation

Physical disability

Age

Religion or philosophy

Multicultural client

Clients from different cultural backgrounds will have different values and assumptions about the therapeutic process

For example, a client from a non-western society may value collectivism rather than individualism, and so will struggle with the basic goals of striving for self-actualisation, happiness, intimacy, authenticity, independence, etc. that run through many existing western approaches

Clients may present differently on the basis of cultural background

Some clients may view people of all ages as equal, whereas other clients may be deferential to older people

Some clients may believe that females should defer to males, whereas other clients may believe that there should be no difference in the treatment of each gender

Some clients may be comfortable discussing their own feelings, whereas other clients may feel that it is not appropriate to describe their own internal experiences

Some clients may be able to work openly with the therapist, whereas other clients may not feel comfortable making eye contact or may need to speak with a same-sex therapist

Some clients may feel that talking negatively about family is a form of betrayal, whereas other clients may enjoy an opportunity to vent feelings of anger and frustration about family members

Clients within the same culture may differ substantially due to subcultures

Even within any specific culture, there will be many subcultures

For example, one client may be a member of a 'drug culture' in which the values and behaviours differ from wider society

Those who are deeply religious will have very different values from those who are agnostic

Male clients may act, think and feel differently to female clients

These values are born out of our beliefs and are the principles that drive our behaviours

We are influenced by the culture and subcultures that we experience

In this way, everyone is a multicultural individual because it is the combination of our experiences that gives us a unique personal culture

This has clear implications for the multicultural therapist

Multicultural therapist

The Counselling Psychology Division of the American Psychological Association (APA) identified three areas for competencies (Sue & Sue, 1990)

Awareness of self (our assumptions, values and biases)

Awareness of the culturally different client's view of the world

Development of strategies and techniques

In order for therapists to be competent working with issues of diversity in a multicultural society, they need to

Understand their own cultural heritage

Be aware of their own beliefs, values and assumptions

Be sensitive to biases, stereotypes and prejudices

Understand the beliefs, values and cultural differences of others

Develop strategies and techniques for working with others

VALUES AND DIVERSITY

Using the ten areas of diversity noted by Nelson-Jones (2005), make a personal inventory of your understanding and awareness of each area.

Culture
Race
Social class
Biological sex
Gender role identity
Marital status
Sexual orientation
Physical disability
Age
Religion or philosophy

Once you have assessed your own understanding of each area, select an area in which you think that you have limited understanding and do some research into the beliefs, values, customs and practices of a group that differs from your own. For example, if you are an atheist who thinks that he is lacking in understanding of religion, then you could research the beliefs, values, customs and practices of those who actively follow the Catholic religion.

If you are part of a training group, spend some time assuming the role of a person from the chosen group, and role play a short therapy session. In the feedback explore what it felt like as client and therapist in the session.

Self-reflection to recognise bias and prejudice

Importance of self-awareness

There is value in therapists becoming more self-aware

Personal development is integral in training courses and ongoing continuous professional development

Every individual is born and raised in a specific time, society and culture

Cultures and subcultures in which we are raised will impact on our thoughts, feelings, values, beliefs, etc.

This can lead to cultural bias and this cultural bias can impact on how we behave towards others

Cultural bias is both inevitable and appropriate

It is impossible to be culturally neutral

We all make some assumptions about others

We are making an assumption if we decide to cross the street rather than walking past a gang of youths late at night

This could be regarded as stereotyping the youths and our behaviour could be interpreted as discrimination

However, most people would regard this as an acceptable level of assumption and the resulting behaviour would often be regarded as appropriate

We all hold some preferences in relation to others

We are demonstrating a preference if we accept a date with a man who is tall but refuse a date with a man who is short

This could be regarded as prejudice against shorter men and our behaviour could be interpreted as discrimination

However, most people would regard this as an acceptable preference and the resulting behaviour would be classed as appropriate

But cultural bias can sometimes lead to inaccurate assumptions and unfair preferences, and the resulting behaviour can be discriminatory and offensive

Best defence against damaging cultural bias is self-awareness

It is acceptable to hold some cultural biases, provided that you are aware of these biases and ensure that they do not impact negatively on others

EXPLORING ASSUMPTIONS AND VALUES

Read through the following therapeutic scenarios and consider how you would respond as the therapist. Are any of these particularly challenging to you? If so, why? Are there any in which you want further information? If so, what would you need to know?

1. An 18-year-old woman comes to you wanting to explore her thoughts and feelings around terminating her pregnancy.
2. A Chinese student is sent to you for a therapy session as he is struggling with living in a different culture. He appears to be reluctant to discuss how he feels,

agrees with all that you say, and does not make eye contact at all during the session.

3. A 35 year old with heroin addiction talks of the daily round of stealing and scoring.
4. A 72 year old mentions how he regularly drives home from the pub after a few pints and the odd whisky.
5. A 24-year-old woman speaks of how her behaviours are inconsistent with the faith she was brought up in. She is from a Jehovah's Witness family.
6. An elderly lady talks of her husband's terminal cancer, and how they have discussed her role in ending his life.
7. An Indian woman talks about her upcoming arranged marriage.
8. A young man is hesitant to talk, but wants to know your views on transgender issues.

Think of other scenarios which may challenge your beliefs and values, or raise awareness of your lack of understanding of cultural differences.

Finally, what assumptions do you think the author of this section has made in the scenarios?

Criticism of multicultural therapy

Cultural empathy

The Counselling Psychology Division of the American Psychological Association (APA) identifies the following areas for competencies

Awareness of self (our assumptions, values and biases)

Good practice to become more aware of ourselves

Our beliefs, values, biases and prejudices

Awareness of the culturally different client's view of the world

Good practice to increase our awareness of cultural differences

Others beliefs, values, biases and prejudices

Differences in the use of language and non-verbal communication

These competences will help us to build greater cultural empathy

Genuine understanding of the world of the client from the perspective of his or her own culture

Strategies and techniques

The Counselling Psychology Division of the American Psychological Association (APA) identifies the following area for competencies

Development of strategies and techniques

Need for this competency has been debated

Do we really need strategies and techniques to be multicultural?

Patterson (2004) has argued that the basic assumptions for multicultural therapy are flawed

If everyone is a multicultural individual, then all therapy is multicultural, and there is no need to develop multicultural therapy competencies

'Cultural groups are not pure and discrete, but overlapping. The process of globalization is blurring differences. The only workable product of a multicultural society is a society of individuals who must ultimately absorb different cultures into themselves. Thus, no discrete classifications are possible . . . Attempting to develop different theories, methods and techniques for each of these groups would be an insurmountable task . . . This approach is not only impossible, but also irrelevant and harmful in counseling the individual client' (Patterson, 2004)

Patterson (2004) advocates a 'universal system of counseling' and says that the essence of such a system has long been known

'It is what is known as client-centred therapy'

Working holistically

Ever-present cultural differences

Cultural (and subcultural) differences and the values that result from these differences are always present in the therapeutic room

Sometimes this is obvious and explicit

Sometimes it is subtle and implicit

But these differences are always present

Therapists need to be aware of these differences

Self-awareness to understand own cultural bias

Diversity awareness to understand the cultural biases of others

It may even benefit the therapist to gain some insight into other cultures

Some knowledge can be helpful and allow a deeper understanding of the world of the client

However, once the therapist is aware of these differences and holds some cultural knowledge, it is important to then put this knowledge on one side so as to be present for the client

Each client should be viewed as a unique individual rather than the simple output of a cultural context

The reality is that even the most culturally aware therapist cannot know the subtlety of the multicultural client

Indeed, having knowledge of the cultural background of an individual may lead the therapist to make assumptions about the client

For example, if the therapist 'knows' that a client from South East Asia will be deferential to authority figures, this may influence the initial behaviour of the therapist regardless of the individual

Would the therapist be being culturally empathic or stereotyping?

Key to success is to acknowledge cultures and subcultures, yet avoid assumptions on the basis of this knowledge

Important to recognise potential cultural differences and remain sensitive to potential problems in the therapeutic relationship as a result of these differences

For example, it is important to recognise that a female client from Saudi Arabia who follows a Muslim religion may have different values from a male therapist from Scotland who holds an atheist perspective

But the therapist must not make assumptions about an individual on the basis of knowledge about their culture

For example, one must not assume that the female client from Saudi Arabia who follows a Muslim religion is against abortion – it is important to give the client an opportunity to express her own opinions without forming these conclusions on the basis of cultural knowledge

Therapists offer core qualities irrespective of culture

Therapist must listen to the client's verbal and non-verbal communication

Therapist must strive to understand the client's world

If the therapist does not know something, she or he should be honest and ask – this will show interest, equalise any perceived power imbalance and develop the relationship

Therapist should be aware of difference, but should also look for similarities

Patterson (2004) noted that 'all clients are alike in one basic essential – they are all human beings'

SUMMARY

Cultural assumptions: modern western-centric therapy, devised by (mainly) white western educated middle-class middle-aged men; therapists influenced by own culture; individualistic (importance of individual, individuals drive own destiny, personal choice valued, all have potential to change) versus collectivist (importance of group, harmony between individual and universe, interconnectedness of opposites, important to act in ways that benefit group) cultures

Multiculturalism: monocultural to multicultural, ten areas of difference (culture, race, social class, biological sex, gender role identity, marital status, sexual orientation, physical disability, age, religion or philosophy); multicultural clients, cultures and subcultures; multicultural therapists, APA's three competencies (self-awareness, diversity awareness, strategies), needs (be aware of own values, understand values of others, understand own cultural heritage, develop strategies for working with others)

Self-reflection: value in being self-aware, cultural bias is inevitable and appropriate, but can lead to inappropriate discrimination if we are not self-aware of own biases

Critique: cultural empathy is important; APA recommends strategies and techniques for multiculturalism but strategies can be impossible and lead to bias; client-centred therapy may be a solution

Working holistically: ever-present cultural differences, need to be aware of differences (self-awareness and diversity awareness), but avoid assumptions on the basis of this knowledge, offer core qualities irrespective of difference, all clients are alike in being human

APPROACHES, THERAPIES AND MODELS

LEARNING OUTCOMES

After reading this section, you will be able to:

- distinguish between an approach, a therapy and a model
- distinguish between purist and non-purist perspectives
- distinguish between effective integration/eclecticism and ineffective syncretism
- define and discuss technical eclecticism, theoretical integration, assimilative integration and common factors
- describe and evaluate Egan's skilled helper model

Defining approaches, therapies and models

Approach

Approach could be described as the umbrella term for all theories and concepts converging on a similar set of principles

> For example, the humanistic approach includes several theories relating to the positive nature of humans, the tendency towards self-actualisation, and the importance of viewing the individual as the expert on the self

Theories and direction of focus may differ between different contributors to the approach, but the key principles are usually similar

For example, in the humanistic approach, Rogers focuses on the concept of congruence between the actual self and the self-concept whereas Maslow focuses on movement up the hierarchy of needs, but both theorists agree on the basic principle of self-actualisation

Four core approaches

Humanistic approach (third force)

Psychodynamic approach (second force)

Behavioural approach (first force)

Cognitive approach (revolution)

Therapy

Therapy could be described as the practical application of an approach to support human growth and development

Several different therapies can operate under a single approach and each therapy may apply the theories within the approach in a slightly different manner

All the therapies under a single approach will share a common philosophy of human nature

For example, person-centred therapy and existential therapy both operate under the humanistic approach

Some therapies can operate under two or more approaches by combining the theories within the approach in a novel manner

These therapies will combine the principles of human nature from two or more different approaches

For example, cognitive-behaviour therapy combines the cognitive approach with the behavioural approach

Some therapies do not operate under any approach, but instead select specific techniques from various different therapies that sit under different approaches dependent on the situation and client

These therapies argue that all the approaches hold some fundamental truths about human nature so they take the most relevant and appropriate parts from each without specifically adhering to all the principles of any

For example, multimodal therapy does not subscribe to the principles of any one approach

Model

Model could be described as a framework for applying a therapy

Models usually involve steps or processes which can be followed by the therapist irrespective of his or her chosen approach and therapy

For example, Egan's skilled helper model (Egan, 2007)

Approach > therapy > model

Approach provides a foundation for therapy

- Approach refers to the general theories about human nature and personality held by the practitioner
- Therapy is the practical application of these theories to improve psychological wellbeing

Therapy can be added to the framework of a model

- Therapy is the practical application of theories to improve psychological wellbeing
- Model is the framework for applying the various therapeutic techniques

Purist versus non-purist perspectives

Purist therapists

Purists exclusively use only one approach and usually remain within the remits of one specific therapy

- For example, a pure person-centred therapist would not agree with any theories or adopt any techniques which conflict with the core principles of the humanistic approach and person-centred therapy
  - Harley Therapy in London adopts a purist approach to both psychodynamic therapy and humanistic therapy with specific therapists rigidly adhering to their own field with no recognisable cross-over in terms of personality concept or therapeutic approach

However, few therapists maintain this strict approach

- Only 4.2% of therapists surveyed by Psychotherapy Networker (2007) identified themselves as purists
- Even those who do regard themselves as 'purists' often deviate towards other therapies to some extent by incorporating additional techniques
- For example, a behavioural therapist may acknowledge the importance of the therapeutic relationship by adhering to the core conditions of the person-centred approach

Non-purist therapists

Non-purists use more than one approach and will often combine elements of different therapies

- For example, a therapist might usually adopt the humanistic approach and use person-centred therapeutic methods, but use cognitive therapy techniques if this is deemed appropriate for the client

This open approach to therapy is regarded as integrative, eclectic or syncretistic

- Dependent on the theoretical motivation underpinning the act of incorporating various therapeutic approaches and techniques

Integration and eclecticism versus syncretism

Integrative therapy

Therapist aims to weld elements of two or more complementary approaches into a single whole

> Ambitious enterprise to bring together elements from different theories to create a new theory (McLeod, 2009)

Many modern practitioners adopt an integrative approach by combining elements of distinct theories to create an entirely new theoretical approach to therapy

> For example, cognitive-behaviour therapy melds the cognitive approach with the behavioural approach to create a new therapeutic method

Eclectic therapy

Therapist aims to bring together a range of therapeutic techniques

> Best or most appropriate ideas and techniques are brought together from a range of different approaches and therapies to suit the needs of the client (McLeod, 2009)

Many modern practitioners select those techniques deemed most appropriate for their individual client on the basis of need without specific subscription to any underlying approach

> For example, multimodal therapy uses a range of techniques from many different therapies without reference to a specific approach

Integration versus eclecticism

'Integration is the process of *bringing together* with the implication of making something whole and new. Eclecticism is a process of *selecting out*, with the implication of taking something apart' (Hollanders, 2000)

'An eclectic counsellor will select what is applicable to the client from a range of theories, methods and practices. Justification is based on the theory that there is no proof that any one theoretical approach works better than all others for a specific problem. Integrative counselling is when several distinct models of counselling and psychotherapy are used together in a converging way rather than in separate pieces' (BACP, 2010c)

Integration involves a combination of different approaches to produce an entirely new therapy on the basis of a new underlying approach

Eclecticism involves a combination of different therapies to produce an entirely new therapy without specific subscription to an underlying approach

'Eclecticism in practice and integration in aspiration' (Wachtel, 1991)

> Many therapists dislike the term 'eclecticism' as it is mistaken for syncretism
>
> Therapists tend to prefer the term 'integration' even when they are actually practising eclecticism

Both integration and eclecticism should involve careful consideration of client needs and individual circumstances

Provided that the therapist has evaluated the client carefully and has an understanding of the selected approaches or therapies, then integrative and eclectic therapies can be equally successful

Syncretism

Eclecticism involves carefully considered selection of appropriate therapeutic techniques based on a careful evaluation of the client's needs and a deep understanding of the different therapies available

Eclecticism should never be a hotchpotch of different things according to the whims and fancy of the therapist

Lazarus (1992) argued that the arbitrary selection of various random techniques for use in an unreasoned or spontaneous manner is a form of syncretism

Syncretism involves a mix of different approaches and therapies based on what is available to the therapist without an understanding of the underlying theories

Eclecticism versus syncretism

Even though the eclectic therapist might not subscribe to a single underlying approach, s/he is aware of the theoretical underpinnings of each of the approaches that s/he chooses to draw from and is aware of the empirical evidence supporting the therapies that s/he chooses to employ

Eclecticism can be highly effective

All good therapists adopt the eclectic stance (Corsini & Wedding, 2008)

In contrast, the syncretistic therapist would not understand the theoretical underpinnings of different approaches or be aware of the evidence for and against various methods, but would instead choose techniques based simply on a personal whim

Syncretism is at best useless and at worst dangerous

Please refer to Table 1.1 to see some examples of common approaches and therapies in counselling and psychotherapy today

Please note, however, that this list is in no way conclusive or exhaustive

There are many more therapies in the world and many therapists might argue against the categorisation in the table

This list is merely a simplified version of therapy and approach to help you to understand the nature of integration and eclecticism

Types of integration and eclecticism

Four types of eclecticism and integration (Dattilio & Norcross, 2006; Norcross & Beutler, 2008)

Theoretical integration

Assimilative integration

Common factors

Technical eclecticism

Theoretical integration

Theoretical integration will strive to integrate two (or more) core approaches into a novel single approach

This method has been used to create several new fields of therapy, and many of these new therapies have become even more popular than the preceding pure therapies

In short, theoretical integration involves the integration of two or more approaches

For example, cognitive-behaviour therapy combines the cognitive approach and the behavioural approach

Assimilative integration

Assimilative integration will focus on a single pure approach with the selective incorporation of techniques from some therapies under other approaches

In short, assimilative integration involves a willingness to selectively incorporate techniques from therapies under other approaches while remaining based within one specific approach

For example, one could argue that transactional analysis is an example of assimilative integration since it appears to have firm roots in the humanistic approach yet it has expanded to incorporate additional theories and techniques

Common factors

Common factors searches for common elements across multiple approaches and therapies in order to pull together the most effective theories into a single new approach and therapy

In short, common factors involves identifying similarities across different approaches to whittle down to a core approach based on commonalities

For example, family systems therapy is based on a number of common theories across different approaches

Technical eclecticism

Technical eclecticism utilises a variety of techniques from several different therapies without subscribing to any underlying theoretical approach

This method focuses on applied technique rather than theoretical underpinnings relating to a comprehensive concept of human nature and personality

In short, technical eclecticism involves the selection of techniques based on evidence for the success of the technique without subscription to any underlying theories

For example, multimodal therapy utilises a range of techniques without establishing direct links to any of the pure approaches

Egan's skilled helper model (Figure 1.2)

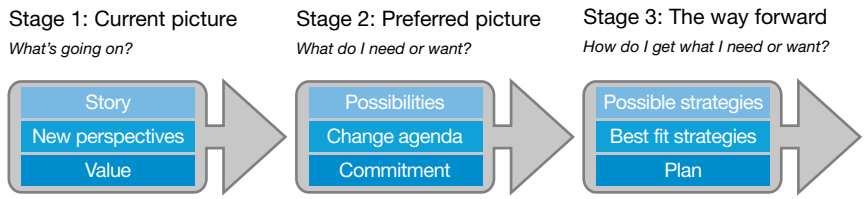

FIGURE 1.2 Skilled helper model. Adapted from Egan (2007).

Model

Egan's three-stage model (Egan, 2007) promotes a collaborative relationship between the therapist and the client designed to guide the helper through the process of empowering the client to solve their own problems and develop their own opportunities

Egan argued that previous theoretical models presented in psychology by scientists often failed to inform actual activity in the field by practitioners

- This situation led to a 'lose-lose-lose scenario' with scientists neglecting the real-life problems encountered in therapeutic relationships, practitioners using techniques lacking a scientific basis, and clients failing to reap the benefits of research-informed methods

Egan proposed a problem management and opportunity development perspective in the form of a model founded on the basis of extensive research yet applicable to real-life therapeutic relationships

- Egan stated that the aim of this model is to assist the therapist in helping the client to 'manage their problems in living more effectively and develop unused opportunities more fully'
- This skilled helper model provides a framework for helping by plotting the therapeutic process over three stages in order to highlight the appropriate use of skills during each step of the relationship from problem presentation to solution management

Egan's skilled helper model is arguably the single most enduring and influential helping model in history

- Significant impact on counselling, psychotherapy, psychology, business, nursing, coaching, etc. for over 30 years

Three stages

Therapeutic process is mapped through three stages

- First stage focuses on the current scenario by reviewing the problem situations and unused opportunities
- Second stage focuses on the preferred scenario by developing ideas about the possibilities for the future

Third stage focuses on strategy by determining a plan for realising the goals, solving the problem, taking advantage of the unused opportunities and striving for the preferred scenario

Stages do not follow a clear linear path with precise sequential steps, but rather act as a framework for the techniques and a guide to the process

First stage: current picture – what's going on?

The first stage of the skilled helper model aims to assist clients in exploring, identifying and clarifying their problem situations and unused opportunities

Clients need to be able to tell their story in their own words with the belief that they are being heard, acknowledged and accepted

However, although clients need to feel as though the therapist is listening to their story from their frame of reference, it is also important that the therapist encourages the client to explore their story in more detail with the aim of viewing the situation from a wider perspective and from the perspective of others

Key question: 'what is going on in your world?'

STORY

The first step of the first stage will focus on the client telling their story

This step forms the basis of the therapeutic relationship between the therapist and the client

Deffenbacher (1985) noted that the skilled helper should strive to build rapport with a client by listening in a non-judgemental manner, attending to both thoughts and feelings, and responding with honesty and care

It is essential that the therapist is able to establish this positive connection with the client during the early stages of the therapeutic process in order for the client to feel comfortable and confident enough to tell their story

The client has to trust the therapist in order to be completely honest about their story and this is the first step towards solving their problem

Key questions to encourage story telling will include 'tell me about . . .'

NEW PERSPECTIVES

The second step of the first stage will focus on identifying blind spots

People who are immersed in a problem will often fail to see the situation from a wider perspective and may neglect to account for the perspectives of others within the environment

Clients may experience blindness with regard to certain factors associated with their problem

Therapists should help clients to discover and challenge these blind spots by viewing the problem from a different angle, exploring alternative frames of reference, clarifying the problem details and investigating any unused opportunities

Key questions to address blind spots during this stage may include 'is there any other way of looking at this issue?' and 'how do you think a neutral observer might describe this situation?'

VALUE

The third step of the first stage will focus on the search for leverage

Therapists should help clients to focus on the issues that have a significant impact in their life

Egan (2007) suggests that the principles of leverage should be applied to determine which issue should be addressed: if applicable, priority must be given to crisis management, problems causing extreme pain for the client, problems viewed as particularly important by the client, problems for which the benefits outweigh the costs, problems that are manageable parts of a larger problem, and problems that impact on the general wellbeing of the client

Key questions to help the client prioritise their issues may include 'what is the most important part of all of this for you?' and 'what would you most like to focus on at this moment in time?'

Second stage: preferred picture – what do I need or want?

The second stage of the skilled helper model aims to assist clients in identifying their needs and desires in terms of goals based on an understanding of the current problem situations and unused opportunities

Some people move between problem and action without ever really considering exactly what they want from a solution

Therapist needs to focus on establishing the exact solution desired by the client

Key question: 'what do you want?'

POSSIBILITIES

The first step of the second stage will focus on exploring possibilities for the preferred scenario

Therapists should help clients conceptualise various new future situations as alternatives to the current problematic situation

Clients should be encouraged to be imaginative rather than realistic at this stage and therapists should try to keep these brainstorming activities fun and non-judgemental

Key questions to encourage the client to explore more ideas could include 'what do you really want to happen?' and 'what would your world be like if this problem did not exist and everything was perfect?'

CHANGE AGENDA

The second step of the second stage will focus on developing the range of possibilities into viable goals

Therapists should help clients to identify the realistic possibilities through a reality testing process

It is important that the potential goals are demanding enough to stretch the client, yet realistic enough to give the client an opportunity to succeed

This stage will often incorporate a number of challenging skills, as it requires the client to be honest about their own capabilities and options

Key questions to focus the client on goal setting could include 'which possible future scenario would be best for you?' or 'which future scenario do you think could be achieved?'

There are a number of important elements to consider when setting goals with the client and it may be helpful for the therapist to ask whether the goals are clear and specific, achievable and realistic, and measurable within a set time frame. These questions form the basis of the SMART model for goal setting: specific, measurable, achievable, realistic, time limited

COMMITMENT

The third step of the second stage will focus on establishing a commitment to change

Clients should attempt to conduct a cost/benefit analysis of the goal in order to assess the incentives for change

Key questions to encourage this analysis could include 'what will be better for you if you achieve this goal?' and 'what are the downsides of trying to achieve this goal?'

Third stage: the way forward – how do I get what I need or want?

The third stage of the skilled helper model aims to assist clients in developing action strategies for achieving their goals

Therapist should attempt to work with the client to establish a plan of action

Key question: 'how can you get what you want?'

POSSIBLE STRATEGIES

The first step of the third stage will focus on brainstorming action strategies

Therapists should encourage clients to include all possible strategies without limitations as even silly strategies might lead to realistic possibilities

Key questions to promote free thinking may include 'how could you achieve this goal?' and 'can you think of any wild and wacky ways of achieving this goal?'

BEST FIT STRATEGIES

The second step of the third stage will focus on selecting a set of strategies appropriate to the current situation of the client

Therapists should encourage clients to investigate both internal factors (client needs, values, preferences, etc.) and external factors (resources, support networks, etc.) to ascertain which strategies could be realistic and effective

Clients can be encouraged to focus on appropriate strategies through the use of key questions such as 'which of these ideas is most realistic?' or 'which of these ideas is right for you?'

PLAN

The third step of the third stage will focus on transforming the strategy into a plan of action

Therapists should help clients to formulate a plan of action designed to change the current scenario into the preferred scenario by asking key questions such as 'what will you do first?' and 'what exactly will you do next?'

These plans should be clearly outlined in a step-by-step guide with realistic time frames

SUMMARY

Approach/therapy/model: approach = umbrella term for all theories and concepts converging on a similar set of principles (humanistic, psychodynamic, behavioural, cognitive); therapy = practical application of an approach to improve psychological wellbeing (e.g. person-centred therapy under humanistic approach); model = framework for applying a therapy (e.g. skilled helper model)

Purist versus non-purist: purists exclusively use only one approach and usually remain within the remit of one specific therapy (only 4.2% regard themselves as purist); non-purists use more than one approach and will often combine elements of different therapies (regarded as integrative, eclectic or syncretistic)

Integration/eclecticism/syncretism: integration = combination of different approaches to produce an entirely new therapy on the basis of a new underlying approach; eclecticism = combination of different therapies to produce an entirely new therapy without specific subscription to an underlying approach; syncretism = random selection of techniques without understanding of underlying theory (ineffective)

Types: theoretical integration = integrate two or more approaches into a single novel approach; assimilative integration = focus on one core approach with selective integration of some techniques from therapies under other approaches; common factors = focus on common elements across multiple approaches to pull together the most effective theories into a novel approach; technical eclecticism = utilise techniques from therapies under different approaches without subscribing to any underlying approach

Egan's skilled helper model: problem management model, promotes collaborative relationship with helper guiding client towards solutions, arguably most enduring and influential helping model in history (30 years old); Stage 1 Present scenario, what's going on in your world? Story, new

perspectives, value; Stage 2 Preferred scenario, what do you want? Possibilities, change agenda, commitment; Stage 3 Strategy for A to B, how do I get what I want? Possible strategies, best fit strategies, plan

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TABLE 1.1 Examples of common approaches and therapies

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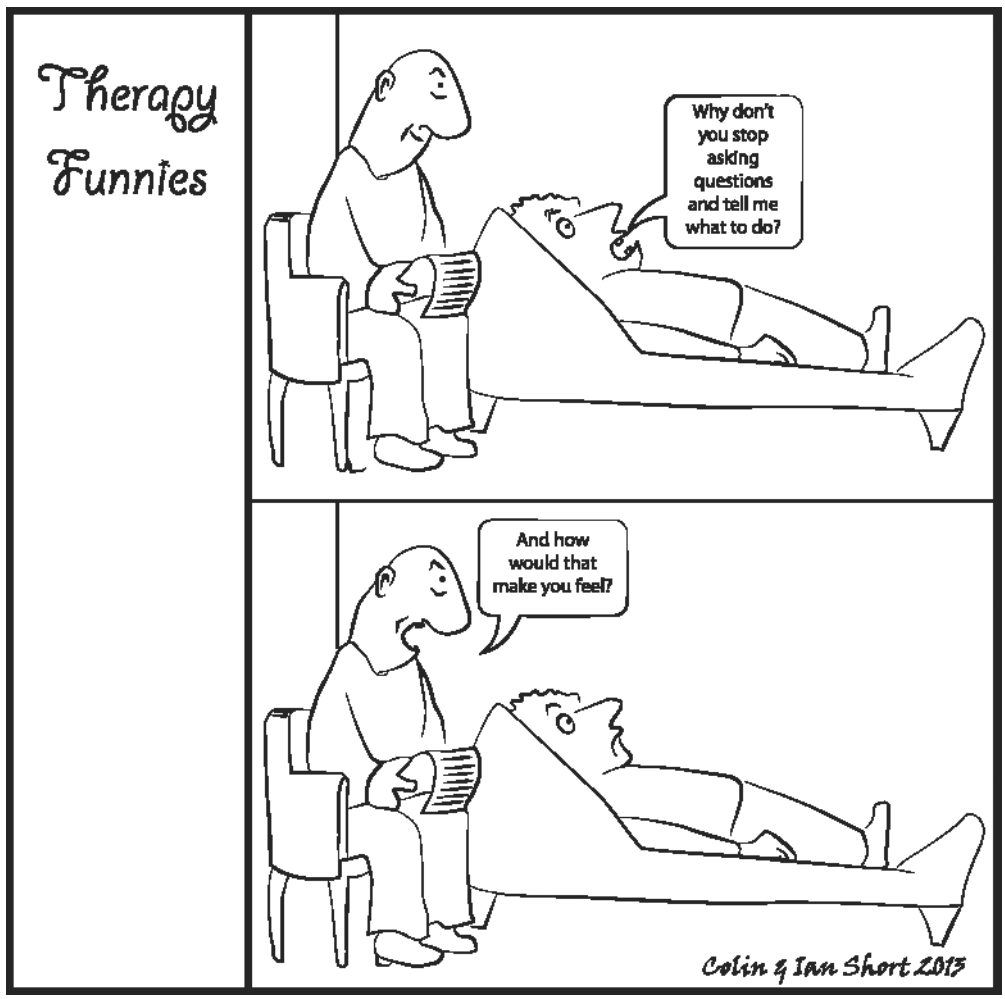
Chapter 2

Humanistic approach and person-centred therapy



CHAPTER CONTENTS

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INTRODUCTION TO HUMANISTIC APPROACH AND PERSON-CENTRED THERAPY

This chapter aims to introduce the reader to the humanistic approach to counselling and psychotherapy. Person-centred therapy will be explored as one example of a therapeutic method under the humanistic approach.

LEARNING OUTCOMES

By the end of this chapter, you will be able to:

- describe the development of the third force in psychology: the humanistic approach
- acknowledge the relative impacts of Carl Rogers and Abraham Maslow on the development of the humanistic approach
- discuss the core theories of human nature and personality from the humanistic perspective
- discuss the nature of the therapeutic relationship between therapist and client in person-centred therapy
- outline the main therapeutic techniques utilised in person-centred therapy
- appreciate the application of person-centred therapy in a real-world setting

DEVELOPMENT OF THE HUMANISTIC APPROACH

LEARNING OUTCOMES

After reading this section, you will be able to:

- list the three main forces in psychology
- discuss the development of the humanistic approach in a historical context
- acknowledge the main contributors to the development of the humanistic approach

Third force in psychology

Three forces in psychology

Behavioural approach

Psychodynamic approach

Humanistic approach

Humanistic psychology is the ‘third force’

- Reacted against psychodynamic and behavioural approaches dominant in early 20th century
  - First force
    - Behavioural psychology assumed that human nature was mechanistic and reduced humanity to the link between stimuli and response (see Chapter 4 for further information)
  - Second force
    - Psychodynamic psychology assumed human nature was deterministic and reduced humanity to basic biological drives (see Chapter 3 for further information)
  - Third force
    - Humanistic psychology is 'a response to the denigration of the human spirit that has so often been implied in the image of the person drawn by behavioural and social sciences' (Association for Humanistic Psychology, 2001)
- Reacted against the medical model
  - Medical model and psychiatry focus on illness
    - Aim of the medical model is to cure the patient
    - This makes the crucial assumption that the individual is sick and that the practitioner is an expert who needs to act on the individual in order to improve his or her mental health
  - Humanistic approach focuses on individuals
    - Aim of humanism is to offer a safe environment for an individual to re-establish control over his or her own world
- Reacted against the values of logical positivism
  - Logical positivism focuses on logic and empiricism to understand thoughts, feelings and behaviours
    - Logical positivism was developed by the Vienna Circle (Schlick, Godel, Carnap, Hempel, Neurath, etc.)
    - Logical positivism attempts to solve the demarcation problem (identify the line between science and pseudoscience) by suggesting that science should build a system of knowledge about the natural world using empiricisms (statements that can be verified as true)
    - Logical positivism emphasises rationalism (use of reason, intellect and logic to draw inferences about the world) and empiricism (observational evidence to draw conclusions about the world)
  - Humanistic approach focuses on the individual human experience, rather than objective scientific enquiry
    - Founding fathers of humanism were heavily influenced by existential phenomenology (Kierkegaard, Husserl, Heidegger, etc.)

- Existential phenomenology suggests that the human condition must be considered from the perspective of the individual as a unique whole seeking answers about reality
- Existential phenomenology emphasises the subjective nature of meaning and the importance of individual experience

Dedicated to a meaningful human-based perspective

- Psychology is predominantly the study of human nature and yet much of the focus appears to be on factors affecting all species
  - Drive for reward, drive for food, sex drive, etc. are present in almost all species across the world
- Humanists in the 1950s began to explore elements of the human that are overlooked in other fields of psychology
  - Essentially human phenomena such as love, hope, creativity, individuality, the nature of being, self-consciousness, self-awareness, philosophy, language, art and morality
- Friedman (1994) argued that the humanistic shift could be described as a move 'from determinism to self-determination, from causality to purpose, from manipulation to self-responsibility, from analysis to synthesis, from diagnosis to dialogue, from solution-oriented models to process, from degradation of human life to celebration of the human spirit'
  - This approach rejects the objective control of science in favour of an empathic understanding of humanity

Major advances in the humanistic approach

Initial interest in humanistic approach

Rogers suggested that psychology in general and therapy in particular should focus on the individual as a unique human being

- *Counseling and Psychotherapy* published in 1942
- *Client-Centred Therapy* published in 1951

Maslow suggested that psychology should focus on exploring human behaviour in terms of our desire to satisfy needs and argued that theories of motivation should be human centred rather than animal centred

- *A Theory of Human Motivation* published in *Psychological Review* in 1943
- *Motivation and Personality* published in 1954

Formal establishment of humanistic approach

Detroit meetings at the invitation of Maslow and Moustakas took place in 1957 and 1958

- Aim of establishing an association focusing on 'the subjective human being' (Rogers, 1962)

Cohen published the first book on humanistic psychology titled *Humanistic Psychology* in 1958

American Association of Humanistic Psychology formed in 1961

Establishment of this association marked the formal foundation of the humanistic approach

Journal of Humanistic Psychology launched in 1961

Initially known as The Phoenix

Humanistic Invitational Conference in Connecticut in 1964

Attendees included psychologists such as Rollo May, Gardner Murphy, Gordon Allport, J.F.T. Bugental, Henry Murray, Charlotte Buhler, Abraham Maslow and Carl Rogers

Attendees included other humanists such as Jacques Barzun, Rene Dubos and Floyd Matson

Conference focused on exploring why the first and second forces in psychology (behavioural and psychodynamic approaches) failed to explore the real issues, problems and complexities of the human condition

Attendees concluded that the 'third force' of humanistic psychology would 'offer a fuller concept and experience of what it means to be human' (Association for Humanistic Psychology, 2001)

American Psychological Association recognised status in 1972

Humanistic Psychology is Division 32 of the APA and produces its own academic journal, *The Humanistic Psychologist*

Modern humanistic approach

Breadth of the modern humanistic approach

One of the key principles of the humanistic approach is the idea of individuals being in a constant process of 'becoming'

Becoming means to be constantly striving to be more effective and moving towards self-actualisation

Therefore, it is not surprising that the approach itself is also not static

The approach too is in a process of continuous ongoing development

Rogers continued to develop the theories that underpin the humanistic approach, and welcomed others to research and develop their own ideas

Since producing his theory of personality, and later his ideas on the conditions for therapeutic change and the stages of the process of change, there have been other key theorists who have taken forward the theory and developed this approach further

Key figures in the humanistic approach

Carl Rogers (1902–1987)

Impact on the approach

- Introduced person-centred therapy as a therapeutic method of applying the principles of humanism
- Emphasised the importance of the therapeutic relationship in moving a client towards self-actualisation
- Argued that six conditions were necessary and sufficient for the therapist to encourage positive therapeutic change
  - It has become common in the person-centred approach to talk about the three 'core' conditions of unconditional positive regard, accurate empathic understanding and congruence (conditions 3, 4 and 5 of the original six), but it is important to recognise that Rogers argued that all six conditions were necessary and sufficient
- Focused on the client being 'fully received' (Rogers, 1961) and the idea of 'presence' (Rogers, 1980)

Selected works

- *Counseling and Psychotherapy* published in 1942
- *Client-centred Therapy: Its Current Practice, Implications and Theory* published in 1951
- *On Becoming a Person: A Therapist's View of Psychotherapy* published in 1961
- *Freedom to Learn* published in 1969
- *A Way of Being* published in 1980

Abraham Maslow (1908–1970)

Impact on the approach

- Developed a hierarchy of human needs to describe the nature of human motivation
- Highlighted the importance of addressing basic needs (such as physiological needs like food and safety needs like security of body) before being able to address higher needs (such as belonging needs like family and esteem needs like confidence) so that the individual can eventually strive towards self-actualisation

Selected works

- *A Theory of Human Motivation* published in *Psychological Review* in 1946
- *Motivation and Personality* published in 1954
- *Toward a Psychology of Being* published in 1968

Rollo May (1909–1994)

Impact on the approach

- Associated existential philosophy with the humanistic movement
- Provided insight into evil and suffering, and encouraged debate about the links between love and sex

Selected works

- *The Meaning of Anxiety* published in 1950
- *Existence* published in 1956
- *Love and Will* published in 1969

James Bugental (1915–2008)

Impact on the approach

- Published a manifesto for the third force in psychology
- Outlined five key postulates for the approach: human beings cannot be reduced to components, have in them a uniquely human context, have an awareness of oneself, have choices and responsibilities, and seek meaning, value and creativity

Selected works

- *Humanistic Psychology: A New Breakthrough* published in 1963
- *The Search for Authenticity* published in 1965

Eugene Gendlin (1926–current)

Impact on the approach

- Student and subsequent close colleague of Rogers
- Developed experiential focusing oriented therapy
- Interested in the process of how clients change, and how the therapist can help clients attend to their 'inner experiencing' through the use of empathic responding

Selected works

- *Focusing* published in 1978

Natalie Rogers (1928–current)

Impact on the approach

- Carl Rogers' daughter
- Developed person-centred expressive art therapy using empathy to help clients explore the personal meanings of their creativity

Selected works

- *The Creative Connection: Expressive Arts as Healing* published in 1993

Garry Prouty (1937–2009)

Impact on the approach

Significant contribution in engaging clients who need preparation to engage in the therapeutic process

Pre-therapy remains very close to the original humanistic concept proposed by Rogers but offers an opportunity to work with those who would ordinarily be excluded from the 'talking cure' (e.g. those unable to communicate)

Selected works

Pre-therapy published in 1976

SUMMARY

Three forces: behavioural, psychodynamic, humanistic; humanistic focused on essentially human traits, reacted against medical model and logical positivism, dedicated to a meaningful human-based perspective

Major advances: initial ideas proposed by Maslow and Rogers in 1940s; Rogers suggested psychology should focus on individuals as unique human beings, Maslow suggested psychology should be human centred and focus on satisfaction of needs; formally established in 1960s, Detroit meetings set up to establish approach in 1950s, American Association of Humanistic Psychology formed in 1961, *Journal of Humanistic Psychology* launched in 1961, Humanistic Invitational Conference held in 1964, APA recognised status in 1972; modern approach is constantly evolving and in a process of 'becoming'

Key figures: Rogers' person-centred therapy, Maslow's hierarchy of needs, May's existential philosophy, Bugental's humanistic manifesto, Gendlin's focusing, Rogers' expressive arts, Prouty's pre-therapy

PERSONAL AND PROFESSIONAL BIOGRAPHIES OF CARL ROGERS AND ABRAHAM MASLOW

LEARNING OUTCOMES

After reading this section, you will be able to:

- outline the personal and professional biography of Carl Rogers (1902–1987)
- outline the personal and professional biography of Abraham Maslow (1908–1970)
- appreciate the impact of these key figures on the development of the humanistic approach

Biography of Carl Rogers

'I speak as a person, from a context of personal experience and personal learnings'
—Rogers (1961)

Who was Carl Rogers?

Carl Rogers helped to found the humanistic movement

Possibly the most influential change in modern psychology

Carl Rogers introduced person-centred therapy

Possibly the most influential therapist in history

In answering the question 'who are you?'. . .

'I am a psychologist whose primary interest has been in psychotherapy' (Rogers, 1961)

Early years

Born January 8th 1902 in Chicago

Fourth of six children – father (Walter Rogers) was a civil engineer and mother (Julia) was a housewife

Successful affluent family

Strong Christian values: no alcohol or theatre, very little social life, lots of work

Solitary child who read incessantly and went on only two dates in high school

Educationally advanced from a young age

He entered school in second grade because he was reading before kindergarten

His family moved away from the temptations of suburban Chicago to a farm when he was 12 and the advanced methods of agriculture used on the farm instilled in him a love and understanding of science

Education

Agriculture major at Wisconsin College

He switched major after two years after a powerful religious student conference convinced him to seek a future in the ministry

History major at Wisconsin College

He visited China for an International World Student Christian Federation Conference

This experience brought him into contact with German and French enemies and he realised that likable individuals could hold extremely disparate attitudes about politics, religion, etc.

He emancipated himself from the religion of his parents and became an 'independent person'

Graduated with a BA in History in 1924

Family

Married Helen Elliot in 1924

He fell in love with his childhood friend and married her soon after college

This further increased the poor relations with his parents, although they did give reluctant consent to the wedding

He remained married to Helen Elliot until her death in 1979

'I cannot be very objective about this, but her steady and sustaining love and companionship during all the years since has been a most important and enriching factor in my life' (Rogers, 1961)

Raised two children

David was born in 1926 and Natalie was born in 1928

'. . . my son and daughter grew through infancy and childhood, teaching me far more about individuals, their development, and their relationships, than I could ever have learnt professionally' (Rogers, 1961)

Career

Attended the Union Theological Seminary in 1924

He petitioned the administration to be permitted to establish a credit-free seminar containing no instructor or syllabus in order for students to explore their own ideas

His petition was successful and subsequent discussion in these seminars began to form the philosophy of life evident in the humanistic approach

He decided that beliefs were so transient that it would be awful to work in a field in which one had to profess a set of beliefs in order to remain in the profession

He decided to seek a field of work in which he would have unlimited freedom of thought

Clinical psychology

Philosophy of Education at Teachers' College of Columbia University

Clinical work with children under Leta Hollingworth

MA in 1928

PhD in 1931

Granted a fellowship at the new Institute for Child Guidance

He was exposed to Freudian therapy through work with David Levy and Lawson Lowrey

Psychologist in the Child Study Department of the Society for the Prevention of Cruelty to Children in New York

He realised that mental disturbance might not have the sexual basis claimed by psychodynamic theory and that therapeutic work might not require direction and interpretation

'I was moving away from any approach which was coercive or pushing in clinical relationships, not for philosophical reasons, but because such approaches were never more than superficially effective . . . It is the client who knows what hurts, what directions to go, what problems are crucial, what experiences have been deeply buried. It began to occur to me that unless I had a need to demonstrate my own cleverness and learning, I would do better to rely upon the client for the direction of movement in the process' (Rogers, 1961)

Lecturer

Lecturer on Working with Troubled Children at University of Rochester in 1935

His work did not fit with the behaviourist approach of experimental psychology or the psychodynamic approach of psychoanalysis

He lectured under the Department of Sociology and Department of Education

He eventually fitted into the role of psychologist after the formation of the American Association for Applied Psychology

Professor of Clinical Psychology at Ohio State University in 1940

'Contrary to expectation, they offered me full professorship. I heartily recommend starting in the academic world at this level. I have often been grateful that I have never had to live through the frequently degrading competitive process of step-by-step promotion in university facilities, where individuals so frequently learn only one lesson – not to stick their necks out' (Rogers, 1961)

Professor of Psychology at University of Chicago in 1945

He established a therapy centre at the university where he served as Executive Secretary

Elected President of the American Psychological Association (APA) in 1946

Published *Client-centred Therapy* in 1951 to describe a new approach to therapy and outlined the core conditions for constructive personality change

Award for Distinguished Scientific Contributions by the American Psychological Association (APA) in 1956

Professor at University of Wisconsin in 1957

Joint post in Department of Psychology and Department of Psychiatry

Published *On Becoming a Person* in 1961

Researcher at Western Behavioral Sciences Institute in 1964

Center for the Studies of the Person in 1968

Published *Freedom to Learn* in 1969 outlining applications of the person-centred approach in education

Published *A Way of Being* in 1980

Founded a new centre focusing on therapy and psychoeducation

'We are a face-to-face community of people for whom the Person Centred Approach plays an important role in our personal and professional lives – as a way of Being and of doing. Our commitment is to maintain/build/create an environment where each of us can be heard, where we can be fully ourselves, where we accept and embrace each other as we are. Currently projects offer opportunities for Person Centred personal growth, community leadership development, creative use of conflict, publishing, library and bookstore, organizational development and psychotherapy training' (Center for the Studies of the Person, 2010)

Dedicated later years to applying theories to social conflict

He travelled the world delivering seminars and workshops with the aim of uniting conflicting forces

Protestants and Catholics in Belfast

Blacks and whites in South Africa

He was nominated for a Nobel Peace Prize in 1987

Death

Died February 4th 1987 in San Diego

Cause of death was recorded as heart failure

Significant learnings

'In my relationships with persons I have found that it does not help, in the long run, to act as though I were something that I am not' (Rogers, 1961)

'I have found it of enormous value when I can permit myself to understand another person' (Rogers, 1961)

'The more I am open to the realities in me and in the other person, the less do I find myself wishing to rush in to "fix things"' (Rogers, 1961)

'Evaluation by others is not a guide for me' (Rogers, 1961)

'It has been my experience that persons have a basically positive direction' (Rogers, 1961)

FOUNDING FATHER OF THE HUMANISTIC APPROACH: ROGERS

Consider the life story of Carl Rogers and try to answer the following questions.

1. What was the social and cultural context of the world in which Rogers developed his theories?
2. How might Rogers' life experiences have impacted on his perception of human nature?
3. How can this life story contribute to your understanding of the humanistic approach?

Biography of Abraham Maslow

'I was awfully curious to find out why I didn't go insane.'
—Maslow (as cited by Association for Humanistic Psychology, 2011)

Who was Abraham Maslow?

Abraham Maslow helped to found the humanistic movement

Possibly the most influential change in modern psychology

Abraham Maslow introduced the hierarchy of needs

His theories have been applied across many fields, including business, education, therapy, healthcare, social policy, etc.

Early years

Born Abraham Harold Maslow on April 1st 1908 in Brooklyn, New York

Father (Samuel Maslow) was a cooper and mother (Rose Scholofsky) was a housewife

Jewish family

Parents were uneducated first-generation Russian immigrants who were determined that the family should succeed in the US

He was expected to pursue a career in law, but preferred to read about psychology

Parents expected eldest son to assist with caring for the family

He was expected to study for many hours then assist father at work and care for six younger siblings

He had very few friends other than his cousins during his youth because most of his time was spent working

Extensive anti-Semitism within his neighbourhood

Non-Jewish neighbourhood with very few Jewish families

He was a shy and awkward child who suffered abuse from peers, teachers and neighbours

'I was a little Jewish boy in the non-Jewish neighborhood. It was a little like being the first Negro enrolled in an all-white school. I was isolated and unhappy. I grew up in libraries and among books, without friends' (Maslow cited in Hall, 1968)

Poor relationship with his parents

He had a poor relationship with his father

His father was determined that his son would succeed where he had failed, so he constantly pushed him in all academic pursuits

However, his relationship with his father did improve in later years and they were reconciled before his death

He had a particularly difficult relationship with his mother

'What I had reacted to and totally hated and rejected was not only her physical appearance, but also her values and world view, her stinginess, her total selfishness, her lack of love for anyone else in the world – even her own husband and children – her narcissism, her Negro prejudice, her exploitation of everyone, her assumption that anyone was wrong who disagreed with her, her lack of friends, her sloppiness and dirtiness . . .' (Maslow cited in Hoffman, 1988)

His negative feelings towards his mother were extremely strong and he never managed to overcome his hostility, even after extensive therapy as an adult – in fact, it is reported that he did not even attend her burial (Engler, 2008)

Educationally advanced from a young age

Attended Boys High School – top school in Brooklyn

However, he was unconvinced about his own capabilities and often had to be persuaded to exhibit his skills in any way

Perhaps a result of the negative opinions held (and often expressed) about him by his parents?

Education

City College of New York and Brooklyn Law School

Withdrew after three semesters against the wishes of his parents

Briefly attended Cornell, but had to withdraw for logistical reasons

Psychology major at University of Wisconsin

Graduated in Psychology with a BA in 1930, MA in 1931 and doctorate in 1934

Family

Married Bertha Goodman in 1928

Rebelled against parents by falling in love with his cousin and childhood sweetheart

Married when he was 20 years old and she was only 19 years old and still in high school

He describes this marriage as the true start to his life and they remained married until parted by death

Career

Research

Researched primate dominance behaviour at University of Wisconsin

Experimental behavioural research conducted alongside Harlow (attachment theory)

Researched human sexuality at Columbia in New York in 1935

Research conducted alongside Thorndike (theory of learning)

Adler introduced him to psychodynamic theory during his time at Columbia and this had a huge impact on his development as a psychologist

Lecturer

Taught psychology at Brooklyn College in 1937

He began to focus on the study of exceptional human beings when he met several intellectuals, including Benedict and Wertheimer

He developed many of his initial theories on motivation and the hierarchy of needs during his time in New York

Published *A Theory of Human Motivation* in *Psychological Review* in 1946 – paper outlined the hierarchy of needs

Chair of Psychology at Brandeis in Massachusetts in 1951

He worked alongside Goldstein who introduced him to the concept of self-actualisation (Goldstein, 1934)

He began promoting humanistic psychology in his role as Psychology Chair and this became the focus of his life's work

Published *Motivation and Personality* in 1954 to expand on the hierarchy of needs

Published *Toward a Psychology of Being* in 1968

Elected President of the American Psychological Association (APA) in 1968

Resident Fellow of the Laughlin Institute in California in 1969

Semi-retirement, but still updated his more influential books and wrote two more books

Dedicated later years to promoting humanistic psychology

He was responsible for controlling the new 'third force' in psychology and he focused exclusively on this psychological approach

Death

Died June 8th 1970 in California

Cause of death was recorded as a fatal heart attack while jogging in Menlo Park

FOUNDING FATHER OF THE HUMANISTIC APPROACH: MASLOW

Consider the life story of Abraham Maslow and try to answer the following questions.

1. What was the social and cultural context of the world in which Maslow developed his theories?
2. How might Maslow's life experiences have impacted on his perception of human nature?
3. How can this life story contribute to your understanding of the humanistic approach?

Significant learnings of Abraham Maslow

'The study of crippled, stunted, immature, and unhealthy specimens can yield only a cripple psychology and a cripple philosophy. The study of self-actualising people must be the basis for a more universal science of psychology' (Maslow, 1954)

'One of the goals of education should be to teach that life is precious. If there were no joy in life, it would not be worth living' (Maslow, 1954)

'If swindling pays, then it will not stop. The definition of the good society is one in which virtue pays' (Maslow, 1965)

'I suppose it is tempting, if the only tool you have is a hammer, to treat everything as if it were a nail' (Maslow, 1966)

'A first-rate soup is more creative than a second-rate painting' (Maslow, 1974)

SUMMARY

Rogers: founded humanistic movement and introduced person-centred therapy; *early years*, born 1902 in Chicago to an affluent Christian family, highly intellectual from a young age; *education*, initially studied agriculture but switched to history to enter ministry, graduated with BA in history; *family*, married childhood sweetheart and raised two children; *career*, moved away from religion and towards education and clinical therapy, took professorship in psychology in several institutes, introduced concept of client-centred therapy, outlined core conditions for constructive personality change, established the Center for the Studies of the Person, applied theories to education and social conflict, nominated for a Nobel Peace Prize; *death*, died of heart failure in 1987

Maslow: founded humanistic movement and developed hierarchy of needs; *early years*, born 1908 in New York to uneducated first-generation immigrants, Jewish family – anti-Semitism in neighbourhood, poor relationship with parents (especially mother), highly intellectual from a young age (possibly due to pressure to succeed from parents); *education*, studied law but withdrew against wishes of parents, studied psychology at University of Wisconsin – BA, MA, doctorate; *family*, married cousin and childhood sweetheart and raised two children; *career*, taught psychology at Brooklyn College, published *A Theory of Human Motivation* in 1946 (hierarchy of needs), Chair of Psychology at Brandeis, President of the APA, Resident Fellow of the Laughlin Institute, dedicated later years to promoting humanistic approach; *death*, died of a heart attack in 1970

HUMANISTIC THEORIES OF HUMAN NATURE AND PERSONALITY

LEARNING OUTCOMES

After reading this section, you will be able to:

- describe the organismic self and the self-concept
- discuss the impact of conditions of worth on the self-concept
- discuss the tendency towards self-actualisation
- describe the hierarchy of needs
- describe the characteristics of the fully functioning person
- [a] outline the 19 propositions of personality

Organismic self and self-concept

Organismic self

Actual self

Human organism generally provides the individual with trustworthy messages and will naturally strive towards enhancing the self (Mearns & Thorne, 1988)

Children are often completely congruent with their organismic selves

Child falls over = child feels pain = child begins to cry

Self-concept

Self-concept is defined as 'that organisation of qualities that the individual attributes to himself' (Kinch, 1963)

Configurations of self (Mearns et al., 2000)

Self-concept is not a unified entity

Individual may hold many different concepts of the self

Most of us hold contrasting views of ourselves

We might feel that we can be both spontaneous and cautious or loving and angry

For example, a man with an alcohol addiction may feel as though there are two parts to his self-concept: 'drunk me' and 'sober me'

Different configurations of the self may result in different and possibly conflicting thoughts, feelings and behaviours

For example, the man with an alcohol addiction may behave aggressively towards his partner when he is 'drunk me' but behaves in a loving manner when he is 'sober me'

It is important to note that these configurations are different parts of the SAME personality

This is NOT split personality disorder

Different 'parts' of the self are known as different 'configurations of self'

'Configurations of self' is used instead of 'parts of self' so that it is recognised that each 'part' is also made up of many different parts

For example, the man with an alcohol addiction may hold a 'drunk me' configuration that can be further divided into an 'angry me' and an 'impulsive me'

'Configuration is a hypothetical construct denoting a coherent pattern of feelings, thoughts, and preferred behavioural responses symbolised by the person as reflective of a dimension of existence within the Self' (Mearns et al., 2000)

Mearns et al. (2000) note that this notion of 'pluralism' in the self-concept (multiple parts of the self-concept) offers an 'adaptive versatility' because the different parts of the self can be used in different social environments

For example, a teenager might use her 'good daughter me' when at home and her 'rebel me' when with her friends

Perception of self

Social experiences will teach us about the person we 'should' be

Many of the different configurations of the self-concept have been created according to social influence

Children learn how they 'should' act, feel and think through interaction with others in their environment

Child falls over = child feels pain = child begins to cry = child is told that 'big boys don't cry' = child learns that pain should be buried and crying is a sign of weakness

Children eventually internalise these values and these values impact on their perception of the self

Child learns that crying is a sign of weakness = child believes that he is weak for wanting to cry = part of the self-concept of the child involves being a weak person = 'weak me'

OR

Child learns that crying is a sign of weakness = child believes that people who cry are weak = part of the self-concept of the child involves being a strong person who never cries = 'strong me'

Both of these self-concepts are equally damaging

'I am a weak person' can cause low self-esteem

'I never cry' can cause repression and emotional detachment

Perception of self is reinforced by interactions with others over time

Any interaction not fitting with the perception of the self (self-concept) will be ignored or distorted to fit with the self-concept

Self-concept involves being a weak person = individual is complimented for being a strong person = individual believes that the person giving the compliment does not really know him (information is ignored)

Self-concept involves being a strong person who never cries = individual is hurt and wants to cry = feelings of sadness are changed into feelings of anger (information is distorted)

Impact of conditions of worth on the self-concept

Positive regard (Standal, 1954)

Organismic self has a natural desire to receive positive regard from others

Human organisms naturally seek approval and recognition of success from family, friends, colleagues, superiors, etc.

Positive regard is rarely unconditional

Most of the significant people in our lives offer us conditional positive regard

Parents imply that they love us if we are good, spouses imply that they love us if we are faithful, friends imply that they love us if we are nice, etc.

Conditions of worth (Rogers, 1959) (Table 2.1)

Conditions of worth are the evaluations made by significant others about the correct way to act, think and feel in order to receive conditional positive regard

You must do these things in order to be loved, valued, etc.

You must meet these conditions in order to feel worthy

Conditions of worth are often internalised in childhood

Child falls over = child feels pain = child begins to cry = child is told that 'big boys don't cry' by mother = child stops crying = mother tells him that he is a good boy and gives him a hug = child learns that he is only lovable if he does not cry = child learns that crying is a sign of weakness and he will only be valued if he does not cry (condition of worth) = self-concept is influenced by this condition of worth = child believes that he is weak for wanting to cry or that he is a strong person who never cries

Tendency towards self-actualisation

Self-actualisation is the natural drive towards fulfilment

Goldstein (1934) coined the term self-actualisation to describe the driving force that works to move the organism through life towards achieving its true potential

In the absence of blocks and restrictions, all humans will naturally move towards self-actualisation

Each individual has a natural urge towards self-maintenance – we assimilate food, we defend ourselves when threatened, etc.

Child might be unable to find food alone yet all children will naturally know which behaviours to exhibit to seek out this goal (such as crying, suckling, etc.)

Each individual has a natural urge to mature – we start to walk, we start to have sex

Child might experience discomfort or pain when attempting to learn how to walk, and yet all children still strive towards this natural goal

Each individual also has a natural urge towards self-actualisation in a personal sense – we seek self-respect, autonomy, independence, etc.

Child will behave in specific ways designed to enhance their own self-respect and independence

From the perspective of an outside observer, some behaviour might not appear to be optimal for moving towards self-actualisation

However, if we view the world from the frame of reference of the individual and truly understand the private internal world of that individual, then we will see that even this behaviour is an attempt to move towards self-actualisation

Unacceptable behaviours such as stealing might initially appear to be actions moving away from enhancing the self, but we can understand these behaviours as attempts to self-actualise once we appreciate that the individual views financial gain as a means to achieve respect

Actualising tendency refers to the directional movement towards realisation, autonomy, perfection, acceptance, etc.

Note that the important factor here is not the achievement of self-actualisation but rather the constant growth, development and drive towards self-actualisation

All humans have this natural drive towards positivity but they are often confounded in their attempts to grow

Self-actualisation versus enlightenment

Enlightenment (or Nirvana or Knowing) often refers to a state of being

An individual may achieve this state of being through intense meditation or the movement of the self to a higher level of consciousness

Self-actualisation often refers to a process, drive or movement

An individual does not reach self-actualisation any more than one might reach 'running'

To be described as 'self-actualising', an individual must be moving towards his or her true potential

The actualising tendency is a state of movement, rather than a state of being

Congruency between organismic self and self-concept (Rogers, 1951)

In the presence of unconditional positive regard, you will become your 'real self' and experience self-respect, gain positive regard and value yourself as an individual

Your self-concept (perception of self) will be the same as your organismic self (true self)

This is known as congruence

In the presence of conditions of worth, you will develop an 'ideal self' and experience anxiety and gain only conditional positive regard

Your self-concept (perception of self) is not the same as your organismic self (true self)

This is known as incongruence

Self-actualisation is an attempt to reduce the space between the organismic self and the self-concept

Hierarchy of needs

Self-actualisation is a basic human need (Maslow, 1943)

Tendency towards self-actualisation is the highest need for the individual to achieve, but it can only be sought when all other needs are being met

Pyramid of human needs (Maslow, 1954)

Physiological needs

- These needs include food, water, sex, homeostasis, sleep, etc.
- These are the most basic needs for human life and must be met before the individual can seek to meet any higher needs
- If all the basic needs are unsatisfied, then the individual will first strive to meet the physiological needs before focusing on any other needs
- For example, a man who is starving will risk life and limb to seek food

Safety needs

- These needs include security of self, family and property
- Once physiological needs have been met, the individual will seek to meet safety needs
- These needs must be met before the individual can focus on attempting to meet the subsequent needs of love, esteem and self-actualisation
- For example, an insecure child who lives in fear of physical violence from her mother will seek protection rather than love

Love needs

- These needs include intimacy in terms of family, friends and partners
  - Important to note the distinction between love and sex
  - Love is a sense of belonging and affection
  - Sex is a basic physiological need for all animals
  - Although love can be demonstrated through sexual intimacy, these two needs are mutually exclusive – you can have love without sex and you can have sex without love
  - However, although this distinction does exist, many people fail to acknowledge the difference and still confuse the expression of sexual desire with the expression of love
- Once safety needs have been met, the individual will seek to meet love needs
- These needs must be met before the individual can focus on attempting to meet the subsequent needs of esteem and self-actualisation
- For example, a woman who feels unloved may show a lack of self-respect by acting in a clingy and needy manner towards her partner

Esteem needs

These needs include self-respect, respect of others, confidence and achievement

Once love needs have been met, the individual will seek to meet esteem needs

These needs must be met before the individual can focus on attempting to meet the highest need of self-actualisation

For example, an artist who believes that his father views him as weak and inferior may focus on succeeding in business, even though this is not a reflection of his true purpose in life

Self-actualisation needs

Self-actualisation is unique to each individual so it cannot be defined as a list of needs

Maslow (1943) explains that 'what a man can be, he must be' and states that self-actualisation involves the individual doing exactly what he is fitted to do: 'A musician must make music, an artist must paint, a poet must write, if he is to be ultimately happy'

Once all other needs have been met, the individual can focus on attempting to meet the highest need of self-actualisation

For example, a successful man who has sufficient resources to live in a secure environment with loving family can work on becoming his true self by doing what he is designed to do and accepting his true purpose in life

Peak experiences

Frankl (1946) suggested that self-actualisation is attainable (although perhaps only for short periods of time)

After achieving self-actualisation, the individual will feel happy with his or her life and enjoy peak experiences

Peak experiences are those events that bring absolute satisfaction and give a meaning or a purpose to a life

For example, a woman might have a peak experience after the birth of her child when she feels intense joy and believes that her life has a purpose

Frankl (1946) argues that this 'will to meaning' is the highest human need

Hierarchical structure (Figure 2.1)

Human needs are positioned in a hierarchy or pyramid

Lower needs must be met before the individual can focus on achieving higher needs

For example, a hungry man (physiological need) can hurt others because he is not seeking their love or respect (love and esteem needs)

A frightened child (safety need) will not be able to focus on learning in school (esteem need)

'Human needs arrange themselves in hierarchies of pre-potency. That is to say, the appearance of one need usually rests on the prior satisfaction of another, more pre-potent need. Man is a perpetually wanting animal. Also no need or drive can be treated as if it

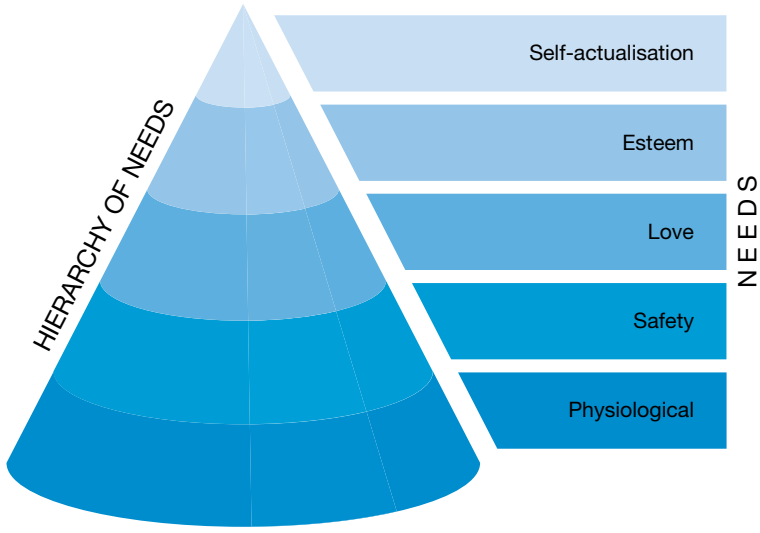

FIGURE 2.1 Maslow's hierarchy of needs. Adapted from Maslow (1943).

were isolated or discrete; every drive is related to the state of satisfaction or dissatisfaction of other drives' (Maslow, 1943)

But is this a simplistic perspective of human needs?

> How can we explain the man who risks his life (safety need) for the woman he loves (love need)?
>
> How can we explain the woman who does not eat for days (physiological need) because she is so engrossed in her work (esteem need)?
>
> Maslow (1943) argues that we often underestimate the importance of a need until it has not been met at a chronic and extreme level
>
> > Modern western society has adapted to ensure that physiological and safety needs are rarely neglected in the extreme
> >
> > 'The average American citizen is experiencing appetite rather than hunger when he says "I am hungry." He is apt to experience sheer life-and-death hunger only by accident and then only a few times through his entire life' (Maslow, 1943)
> >
> > Those living in this culture may underestimate the value of these lower needs to the extent that they can afford to risk losing them
>
> Those people who experience true deprivation of a basic need will be dominated by the desire to fulfil that need to the exclusion of all other factors
>
> > 'For our chronically and extremely hungry man, Utopia can be defined very simply as a place where there is plenty of food. Life itself tends to be defined in terms of eating. Anything else will be defined as unimportant. Freedom, love, community feeling, respect, philosophy, may all be waved aside as fripperies which are useless since they fail to fill the stomach. Such a man may fairly be said to live by bread alone' (Maslow, 1943)

Characteristics of the fully functioning person

The good life (Rogers, 1961)

Once basic needs (physiological, safety, love, esteem) are being met, the individual can strive to achieve congruency between the self-concept and the organismic self (self-actualisation)

An individual who is striving towards self-actualisation can be regarded as a fully functioning person

Fully functioning persons are experiencing 'the good life'

> 'The good life is a *process*, not a state of being' (Rogers, 1961)
>
> 'This process of the good life is not, I am convinced, a life for the faint-hearted. It involves the stretching and growing of becoming more and more of one's potentialities. It involves the courage to be. It means launching oneself fully into the stream of life. Yet the deeply exciting thing about human beings is that when the individual is inwardly free, he chooses as the good life this process of becoming' (Rogers, 1961)

Characteristics of a fully functioning person (Rogers, 1962)

An increasing openness to experience

> An individual who is fully open to experience will allow every single event in life to pass through his or her body and mind without being distorted by the use of defensive mechanisms
>
> > For example, a man who is open to experience is likely to accept his poor grades in an exam rather than distort them into evidence that he is a failure as a person
>
> Rogers (1961) argues that one aspect of 'the good life' is a 'movement away from the pole of defensiveness toward the pole of openness to experience'
>
> > He explains that an individual living the good life would be able to experience internal events so that he can fully accept negative feelings (such as fear, discouragement and pain) and positive feelings (such as courage, tenderness and awe)

Increasingly existential living

> An individual who experiences existential living is able to live fully in each moment
>
> > For example, many young children do not focus on what has happened previously or on what may happen in the future, but simply enjoy living in the present
>
> Rogers (1961) argues that each moment in life is new for an individual who is fully open to his new experience without defensiveness
>
> > He explains that this individual 'becomes a participant in and an observer of the ongoing process of organismic experience, rather than being in control of it' (Rogers, 1961)

An increasing trust in his organism

An individual who places trust in his own self could evaluate each situation and experience based on the available data, rather than relying on prejudice, bias, fear and defensiveness

For example, a woman who is able to interpret each situation based on the stimuli present and trusts her own instincts will be able to make better judgements concerning her career

Rogers (1961) suggests that this individual would have 'access to all of the available data in the situation, on which to base his behavior; the social demands, his own complex and possibly conflicting needs, his memories of similar situations, his perception of the uniqueness of this situation, etc.' (Rogers, 1961)

On the basis of all of this complex information, this individual could then 'permit his total organism, his consciousness participating, to consider each stimulus, need, and demand, its relative intensity and importance, and out of this complex weighing and balancing, discover that course of action which would come closest to satisfying all his needs in the situation' (Rogers, 1961)

Nineteen propositions of personality

Rogers' (1951) theory of personality outlined 19 propositions

1. 'Every individual (organism) exists in a continually changing world of experience of which he is the centre'

   The world around is constantly changing

   From our own perspective, we sit at the centre of this world because we are the main person in our own story

   For each of us, our experiences (what we consciously feel, see, hear, etc.) combine to create an internal private world

   Note that not all stimuli contribute to the private world – many senses are stimulated outside of conscious awareness

   This private world is our experiential field or phenomenal field

   Only you can truly understand your own phenomenal field because you are the only person who has had all of those unique experiences

2. 'The organism reacts to the field as it is experienced and perceived. This perceptual field is, for the individual, "reality"'

   We do not react to an objective actual reality

   Instead, we react to our own perception of reality based on our own phenomenal field

   This is why two people can react in completely different ways to exactly the same stimuli

On a philosophical note, actual reality is nothing more than the agreed acceptance of certain things as real based on overlap between many phenomenal fields

3. 'The organism reacts as an organized whole to this phenomenal field'

   It is impossible to adopt a reductionist approach to understanding the reactions of the individual

   You always react as an entire entity

      We could focus exclusively on the physiological reactions to stimuli only (for example, what happens to the body when you are sexually aroused), but this fails to account for the emotional, cognitive, social and behavioural reactions

   Strict physiological or behavioural approaches are insufficient

      We need to consider the organism as a whole

4. 'The organism has one basic tendency and striving – to actualize, maintain and enhance the experiencing organism'

   Each individual has a natural urge towards self-maintenance – we assimilate food, we defend ourselves when threatened, etc.

   Each individual has a natural urge to mature – we start to walk, we start to have sex

   These tendencies and strivings exist even in the face of adversity

   In the same way, each individual also strives towards self-actualisation in a personal sense – we seek self-respect, autonomy, independence, etc.

   Note that the important factor here is not the achievement of self-actualisation, but rather the constant growth, development and drive towards self-actualisation

5. 'Behaviour is basically the goal-directed attempt of the organism to satisfy its needs as experienced, in the field as perceived'

   Since our goal is to strive towards self-actualisation, then our behaviour is our way of attempting to move towards this goal

   Some behaviour might not appear to be optimal for moving towards self-actualisation

      However, if we view the world from the frame of reference and truly understand the private internal world of that individual, then we will see that even this behaviour is an attempt to move towards self-actualisation

   Unacceptable behaviours such as stealing might initially appear to be actions moving away from enhancing the self

      However, these behaviours can be understood as attempts to self-actualise once we appreciate that the individual views financial gain as a means to achieve respect

6. 'Emotion accompanies and in general facilitates such goal-directed behaviour, the kind of emotion being related to the seeking versus the consummatory aspects of the behaviour, and the intensity of the emotion being related to the perceived significance of the behaviour for the maintenance and enhancement of the organism'

    Significant behaviours are those that you believe will help you survive or move closer to self-actualisation

    Emotional intensity is directly related to the perceived importance of these behaviours

    Those behaviours that may have life or death consequences (for example, jumping out of the way of a racing car) will result in higher emotional intensity than those behaviours that have minimal consequences for the self (for example, cutting your toe nails)

7. 'The best vantage point for understanding behaviour is from the internal frame of reference of the individual himself'

    You are the only person who has had all of your unique experiences from your position in the world – this means that you are an expert about you

        If someone would like to truly understand you as an individual, s/he would need to view your behaviour while attempting to take into account your own private world

        They would need to stand in your shoes

    Our attempt at viewing behaviour from the private world or internal frame of reference of the individual is known as empathy

        Empathy is NOT simply feeling sorry for someone

8. 'A portion of the total perceptual field gradually becomes differentiated as the self'

    'Self' refers to the individual awareness of being

    Development of the conscious self occurs during childhood as we realise that some parts of the world belong to us (things under our control) and other parts of the world are external to us (things not under our control)

9. 'As a result of interaction with the environment, and particularly as a result of evaluational interaction with others, the structure of the self is formed – an organized, fluid but consistent conceptual pattern of perceptions of characteristics and relationships of the "I" or the "me", together with values attached to these concepts'

    Conscious awareness of self develops during childhood through interaction with the world

    We learn about ourselves as individuals by engaging with objects, environments, experiences, people, etc.

10. 'The values attached to experiences, and the values which are a part of the self structure, in some instances are values experienced directly by the organism, and in some

instances are values introjected or taken over from others, but perceived in distorted fashion, as if they had been experienced directly'

We will all learn the value of specific experiences – some things have a high value (enjoyable) and other things have a low value (not enjoyable)

We will learn some of these values through direct experience: for example, I enjoy eating sweet foods

We will learn some of these values by internalising the values of others: for example, I do not enjoy eating sweets and eating sweets is a bad thing to do

Internalisation of the values of others happens due to evaluation: for example, my parents tell me that I am a bad girl for eating sweets

We may then experience a conflict between our evaluation of the self and the evaluation of others: for example, I believe that I am a good girl but my parents tell me that I am a bad girl

In order to protect our threatened concept of self, we may then internalise the values of others: for example, I do not enjoy eating sweets and eating sweets is a bad thing to do

This act of internalising the values of others allows us to maintain our concept of self: for example, I can continue to believe that I am a good girl

11. 'As experiences occur in the life of the individual, they are either, a) symbolized, perceived and organized into some relationship to the self, b) ignored because there is no perceived relationship to the self structure, c) denied symbolization or given distorted symbolization because the experience is inconsistent with the structure of the self'

Some experiences in life will be entirely ignored because they are not related to you as an individual

You will experience many things every day that do not have a positive or negative impact on you so they are not noticed – you might pass a tree, postbox and shop on your way to work and never pay attention to them until the moment that you need to post a letter

Some experiences in life will be noticed because they meet a need (like the postbox) and they are consistent with your perception of yourself and the world

You tend to select from your sensory experiences those events that confirm your own view of the world

These experiences are then integrated into your concept of the world and yourself to further encourage you to believe that your perception is accurate

Some experiences in life will be noticed because they meet a need but they will cause conflict because they are inconsistent with your perception of yourself and the world

These experiences are often denied – for example, client with low self-esteem denies the experience of a compliment by saying that he 'doesn't really know me'

These experiences are often distorted – for example, client who has a self-concept of herself as a loving daughter distorts feelings of anger towards her father into a headache

12. 'Most of the ways of behaving which are adopted by the organism are those that are consistent with the concept of self'

Behaviours exhibited by us will be consistent with our perception of ourselves

All behaviours will be driven by desire to satisfy a need, but this behaviour must fit with what we believe about ourselves

For example, a person might have a strong need for food yet he will not steal because this behaviour does not fit with his concept of himself as an honest man

13. 'Behaviour may, in some instances, be brought about by organic experiences and needs which have not been symbolized. Such behaviour may be inconsistent with the structure of the self but in such instances the behaviour is not "owned" by the individual'

Contrary to the above proposition, some behaviour is almost automatic and exists irrespective of whether it matches the self-concept

In these cases, the individual may not admit to the behaviour

For example, a man defending himself against a sudden attack may respond irrespective of his perception of himself as a gentle person and he may then state that he did not know what he was doing and was not responsible for his actions

These behaviours (and the failure to accept ownership of them) can be particularly dangerous when they are the result of an ignored natural urge

For example, a man who constantly represses all sexual urges because they do not fit with his concept of himself as a 'pure man' might eventually commit sexual assault but deny ownership of his criminal behaviour

14. 'Psychological maladjustment exists when the organism denies awareness of significant sensory and visceral experiences, which consequently are not symbolized and organized into the gestalt of the self-structure. When this situation exists, there is a basic or potential psychological tension'

Any refusal to accept organic sensations (natural feelings and perceptions) because they do not match the self-concept will lead to anxiety

For example, a woman might hold a self-concept of herself as a loving mother resulting from the organic sensation of love for her child and internalised

conditions of worth from significant others informing her that a mother must be 'loving'

Any subsequent organic sensations of dislike towards her child will be ignored and the natural behaviour of aggression in response to these feelings will be refused – this will cause psychological tension

She will seek relief from this tension, but she cannot behave in a way that is not consistent with her self-concept

However, since her self-concept would justify her to use punishment on a bad child, she will begin to view all of his behaviour as bad so that she can gain relief by exhibiting her aggression

Her feelings and behaviour will have been distorted to fit with her self-concept

In the event that she loses control and expresses her dislike of the child, she would then deny ownership of this behaviour and claim 'I was not myself'

15. 'Psychological adjustment exists when the concept of the self is such that all the sensory and visceral experiences of the organism are, or may be, assimilated on a symbolic level into a consistent relationship with the concept of self'

Contrary to the previous proposition, acceptance of organic sensations and congruence (matching) between the organismic self (true self experiencing organic sensations) and the self-concept (perception of self) will result in relief from psychological tension

In relation to the previous example, the mother who dislikes her child will be able to cope with these negative emotions if her self-concept allows her to be a mother who experiences both love and dislike

16. 'Any experience which is inconsistent with the organization or the structure of the self may be perceived as a threat, and the more of these perceptions there are, the more rigidly the self structure is organized to maintain itself'

Any attempt by another person to draw the attention of the individual to things that are inconsistent with the self-concept will fail

We will naturally defend our own self-concept and we will provide evidence to support our own beliefs about ourselves

For example, a girl who holds a self-concept of herself as a poor student will defend herself vigorously against the 'attack' of those who claim that she is clever

17. 'Under certain conditions, involving primarily complete absence of threat to the self-structure, experiences which are inconsistent with it may be perceived and examined, and the structure of self revised to assimilate and include such experiences'

If an individual is placed in the unique position of experiencing a complete absence of threats to the self-concept, then they can begin to explore their own sense of self

If others do not judge but simply accept a person without insisting that they are something else other than what they present, that person may feel safe to explore their own organismic self

They might then start to assimilate their organic experiences into their sense of self and true personality change can occur

18. 'When the individual perceives and accepts into one consistent and integrated system all his sensory and visceral experiences, then he is necessarily more understanding of others and is more accepting of others as separate individuals'

Those who are congruent with the self can often be congruent with others

If a person denies some aspect of himself or herself then they are constantly defending against 'attack' from those who present conflicting evidence

This means that all behaviours exhibited by others can be viewed as threatening

Once the individual is able to accept all elements of the self, they are less inclined to feel threatened by others

This will lead to fewer miscommunications, better interpersonal relations and more acceptance of others

19. 'As the individual perceives and accepts into his self structure more of his organic experiences, he finds that he is replacing his present value system – based extensively on introjections which have been distortedly symbolized – with a continuing organismic valuing process'

An individual who is able to accept more organic experiences will begin to reject the values of others

This person will no longer feel the pressures of the conditions of worth because s/he will know that s/he is worthy regardless of behaviour

In the place of the old value system (shoulds, oughts and musts), a new value system will develop on the basis of genuine organic experience and the natural human desire to move towards growth, honesty and self-actualisation

SUMMARY

Organismic self and self-concept: organismic self (actual self) is based on experiences; self-concept (perception of self) is based on interaction with others and is sectioned into configurations of self

Conditions of worth: positive regard is rarely unconditional; values expressed by significant others are internalised to impact on the self-concept – I must do these things in order to receive positive regard (conditional positive regard) and be deemed worthy (conditions of worth)

Self-actualisation: natural drive towards personal fulfilment, can never actually achieve it but we are self-actualising when striving towards it; self-actualisation is different from enlightenment/Nirvana; those striving towards self-actualisation experience congruency between the organismic self and self-concept

Hierarchy of needs: pyramid of basic needs (physiological, safety, love, esteem, self-actualisation); lower needs must be met before seeking higher needs; peak experiences may be regarded above self-actualisation; hierarchy may be regarded as simplistic

Fully functioning person: lives 'the good life' by being congruent and meeting all needs; characteristics include increasing openness to experience, increasing existential living and increasing trust in organism

Nineteen propositions of personality: outlines human nature with links to therapeutic methods (explains organismic self, self-concept, self-actualisation, conditions of worth and core conditions of therapy)

THERAPEUTIC RELATIONSHIP IN PERSON-CENTRED THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- outline the six core conditions required for constructive personality change
- discuss the first condition of two persons in psychological contact
- discuss the second condition of the client being in a vulnerable or anxious state of incongruence
- discuss the third core condition of the therapist being congruent and integrated in the relationship
- discuss the fourth core condition of the therapist experiencing unconditional positive regard for the client
- discuss the fifth core condition of the therapist experiencing an empathic understanding of the client's frame of reference
- discuss the sixth condition of the therapist communicating empathic understanding and unconditional positive regard towards the client

- explain whether the core conditions are both necessary and sufficient
- recognise the seven stages of the client process
- debate the impact of the core conditions on relational depth

Six conditions for constructive personality change

Relationship is key in person-centred therapy

Relationship is the most important part of the therapy

Techniques, strategies, methods, activities, etc. are not encouraged in this type of therapy

Instead, the therapist focuses on developing a positive relationship with the client under the assumption that this will give the client the conditions to change in a positive way

Rogers (1957) outlined six therapeutic conditions for constructive personality change in the client. For constructive personality change to occur, it is necessary that these conditions exist and continue over a period of time

1. Two persons are in psychological contact
2. The first, whom we shall term the client, is in a state of incongruence, being vulnerable or anxious
3. The second person, whom we shall term the therapist, is congruent or integrated in the relationship
4. The therapist experiences unconditional positive regard for the client
5. The therapist experiences an empathic understanding of the client's internal frame of reference and endeavours to communicate this experience to the client
6. The communication to the client of the therapist's empathic understanding and unconditional positive regard is to a minimal degree achieved

No other conditions are necessary

If these six conditions exist, and continue over a period of time, this is sufficient

The process of constructive personality change will follow

Widely misreported to contain only three conditions for change (congruence, unconditional positive regard, empathy), all six conditions are actually described as both necessary and sufficient for constructive personality change

First condition of two persons in psychological contact

TWO PERSONS IN PSYCHOLOGICAL CONTACT

Therapeutic relationship is described as 'psychological contact' because it relates to the contact between two people engaged in the mental – as opposed to physical – relationship typical of therapy

This condition forms the basis for subsequent conditions as it is assumed that there is no potential for a relationship of any kind without initial contact forming between the two individuals

Most forms of therapy involve contact to some extent

Therapy at the student services at a university may involve one-to-one meetings and group meetings

Therapy for a sexual abuse support service may involve telephone discourse

IS CONTACT NECESSARY?

Advancements in modern technology, however, have led to some distortion of the boundaries of human contact

Many forms of therapy can now take place in an anonymous online setting and there is even the potential for a type of therapy to occur without the presence of an actual therapist

Some individuals use online blogs as an attempt to verbalise their own feelings and the return to these blogs at a future point in time allows the writer to reflect on their thoughts

This approach has been common for many years in the form of personal diaries; however, blogging offers one crucial addition to the typical diary in that the writer is actually exposing these thoughts to an anonymous reader

In this respect, the writer is able to share the depths of the self with another individual without forming an actual relationship with the other

CNN reported on the issue of blogging as therapy and cited one widowed blogger as stating 'Right after he died, people kept asking if I was in therapy, and I'd say, "No, but I have a blog"' (Grossman, 2008)

Second condition of incongruence in client

CLIENT IS IN A VULNERABLE OR ANXIOUS STATE OF INCONGRUENCE

Self-concept is incongruent with the organismic self

Individual experiencing a psychological event that is incompatible with their concept of the self may distort or ignore this event

Repeated exposure to experiences that do not fit with the self-concept could lead to vulnerability or anxiety

IS INCONGRUENCE NECESSARY?

Debate as to the extent of malleability and the number of facets of the self-concept

Self-concept is generally both organised and stable, thus giving the personality some degree of consistency (Purkey, 1988)

Self-concept is multifaceted to highlight the different parts or configurations of the individual (Mearns, 1999)

One could find an experience inconsistent with one part yet consistent with another part; for example, a man might feel that he is independent in one sense yet reliant on his partner in another sense. Thus it is difficult to state with certainty that an experience relating to dependency is consistent or inconsistent with the whole self-concept

Third condition of congruence in therapist

THERAPIST IS CONGRUENT AND INTEGRATED IN THE RELATIONSHIP

Congruency is otherwise known as genuineness as it refers to the degree to which one can relate to another individual in a sincere and honest manner

Genuineness is essential for both empathic understanding and unconditional positive regard

In the absence of congruence, it is very difficult for the therapist to demonstrate the remaining core conditions

MISINTERPRETATIONS OF THIS CONDITION

Success of the therapist depends on his or her achievement of congruence in all areas of life (akin to the attainment of self-actualisation)

It is not possible for the therapist to be entirely congruent at all times in life. Instead, Rogers (1957) suggests that the therapist need only be congruent and genuine in the moment in the therapeutic setting for the relationship to be successful

Therapy can only be genuine if the therapist discloses all thoughts to the client – Rogers himself disclosed his own feelings of boredom to a client (Landreth, 1984)

Aim of therapy is not for the therapist to express his or her own feelings

Instead, therapist should focus on acknowledging and accepting those feelings within him- or herself in order to determine whether the disclosure of such information might be beneficial to the client

IS CONGRUENCE NECESSARY?

Is it possible for a 'fake' therapist to be successful in a therapeutic relationship?

Patients with dysphoria could not detect a significant difference between congruent and incongruent behaviour in a videotaped therapy session (Hollander & Hokanson, 1988)

Conclusion was that there is a lack of interpersonal skills associated with depression, but these findings could also be taken as evidence to suggest that a client suffering from psychological disturbances cannot detect the level of congruency in the therapist

However, although these findings suggest that the patient may be unable to consciously detect the genuine therapist, it is not clear whether congruence and incongruence have a therapeutic impact on a subconscious level

Fourth condition of unconditional positive regard from therapist

THERAPIST EXPERIENCES UNCONDITIONAL POSITIVE REGARD FOR THE CLIENT

Rogers (1957) suggests that successful personality change requires the therapist to accept the client with genuine warmth (positive regard) irrespective of the personal traits or behavioural actions of the individual (unconditional)

> This condition highlights the importance of an understanding of oneself to be aware of any potential prejudice in order to maintain a non-judgemental attitude towards the client

Rogers regarded the human condition as inherently worthy, although he did acknowledge that the individual is capable of both positive and negative emotions (Kirschenbaum, 2004)

> For this reason, it is important for the therapist to appreciate that a person is doing the best that they are able to do at that time and any concerns about the behaviour exhibited by a client should be directed towards the actions rather than the character of the individual (Seligman & Reichenberg, 2009)

IS UNCONDITIONAL POSITIVE REGARD NECESSARY?

Is it possible for a therapist to feel genuine unconditional positive regard for a client?

> It is difficult to accept that any individual would be able to experience this level of regard for another, thus the importance of this condition is questioned because this capacity for regard does not appear to exist in reality
>
> Rogers (1957) acknowledges that the concept of unconditional positive regard operates on a continuum – he even suggests that the concept as an absolute and complete entity could never exist except in theory
>
> > This is an extremely comforting point as it is tempting to feel that one is a failure as a therapist if one is unable to feel genuine unconditional positive regard for the client at all times
> >
> > It is the attempt to maintain this non-judgemental attitude in the moment that will have the greatest impact on the success of the therapy

Fifth condition of empathic understanding from therapist

THERAPIST EXPERIENCES AN EMPATHIC UNDERSTANDING OF THE CLIENT'S FRAME OF REFERENCE

Rogers (1957) states that the fifth condition for successful therapeutic change is the therapist's ability to experience 'an accurate, empathic understanding of the client's awareness of his own experience'

> Empathic understanding requires the therapist to be able to perceive the world of the client from the client's frame of reference

Therapist should be close enough to the client to appreciate his/her own perspective, yet far enough away from the client to remain objective

In other words, to be able to step in and out of the shoes of the client or to be able to view the world through the eyes of the client (Kirschenbaum & Henderson, 1990)

May (1939) described empathy as 'walking with another person through the deepest chambers of his soul'

Essentially, the therapist should be walking at the side of the client through the journey towards self-actualisation

True empathy is difficult to obtain as it involves an immersion in the world of the client without a loss of the self

Many clients have never had the experience of being truly understood by another person and the expression of empathy by the therapist has resulted in many astonished responses, such as 'yes, that's it . . . I can't believe how well you understand me'

True empathy can enhance rapport, thus opening the lines of communication to a greater extent and allowing the client to engage in deeper self-reflection

Genuine warmth towards the client will be significantly influenced by the experience of empathy in the therapist

Often difficult to feel unconditional positive regard unless one can truly understand the world from the perspective of the client

For example, unconditional positive regard for a man who has abused a child might seem impossible, but a deep understanding of the world from his perspective might aid with this aim

Empathic state appears to be consistently important across all types of therapeutic relationships (both directive and non-directive)

Experts from ten different psychotherapeutic approaches were found to successfully build rapport with clients by being able to understand their feelings, following their line of thought, and never being in doubt about their meaning (Fiedler, 1950)

SYMPATHY VERSUS EMPATHY

Clients with a traumatic background (such as those recovering from rape or sexual abuse) may tempt the therapist to experience sympathy

Compassion for another person may stem from genuine warmth, but sympathy is unlikely to assist the client in exploring a deeper level of his or her own world

Expressions of sympathy ('I really feel for you') will disclose the emotions of the therapist without furthering the self-understanding of the client

True attempts to view the world from the client's frame of reference will help the therapist experience empathy

Expressions of empathy ('I am sensing that you find it really difficult to talk to your partner') can encourage the client to view his or her own feelings from a new perspective

PRIMARY EMPATHY

Primary-level empathy requires the therapist to respond to the feelings and behaviour expressed by the client (Egan, 2007)

- Therapist should actively attend to the client and indicate their understanding of exactly what was expressed by the client

ADVANCED EMPATHY

Advanced empathy requires the therapist to sense the expressions not communicated directly by the client (Egan, 2007)

- Self-disclosure, interpretations and challenges can all be used in advanced empathy
- Always need to be aware of the risks associated with advanced empathy – must ensure that the client always remains the expert

IS EMPATHY NECESSARY?

Is positive therapeutic change dependent on empathy?

- Reviews of 75 therapists and 15 research studies revealed no evidence of client change directly related to empathy or warmth in the therapist (Mitchell et al., 1973)

Sixth condition of communication between client and therapist

THERAPIST COMMUNICATES EMPATHIC UNDERSTANDING AND UNCONDITIONAL POSITIVE REGARD TO CLIENT

Effective empathic expressions depend on the level of communication between the client and therapist

- Unconditional positive regard, congruency and empathy are all redundant unless the client is able to detect these states in the therapist

MISINTERPRETATION OF COMMUNICATION

Concept of communication in a person-centred approach is often misunderstood to mean the excessive use of prompts such as 'uhuh' and the question 'how does that make you feel?'

- It is assumed that therapists merely reflect the words of the client and do little else to communicate

In reality, delicate communication of the inner experience of empathy and unconditional positive regard is essential in order to inspire change in the client

IS COMMUNICATION NECESSARY?

As noted for contact, advancements in modern technology have led to some distortion of the boundaries of human communication

- Many forms of therapy can now take place in an anonymous online setting and there is even the potential for a type of therapy to occur without the presence of an actual therapist (for example, iTherapy)

One-way online communication (such as blogs) does not require feedback and yet still appears to serve the purpose of encouraging self-exploration, self-reflection and self-expression

Necessary and sufficient?

Six conditions are reported to be both necessary and sufficient for constructive personality change

All six conditions are necessary

There is some debate regarding the necessity of these conditions

Please refer to the individual descriptions of each condition for discussion about whether each of these conditions is necessary for successful therapy

These six conditions are sufficient and no other conditions are necessary

Lots of research debating the sufficiency of these conditions alone and many other approaches utilise specific techniques – difficult to determine whether the conditions alone can inspire therapeutic success

Cases have been reported in which the client has reported that the therapist is congruent, accepting and empathic, yet the client did not improve (Lietaer, 2002)

Gelso and Carter (1985) argued that 'the conditions . . . are neither necessary nor sufficient, although . . . such conditions are facilitative'

Despite this debate, most therapists agree that the therapeutic relationship is extremely important and almost all therapists will strive to exhibit the core conditions

Kirschenbaum and Jourdan (2005) note that 'Rogers's core conditions may or may not be necessary or sufficient for effective psychotherapy (the debate is ongoing), but [as] a means to achieve a therapeutic alliance, the value of empathy, unconditional positive regard, and congruence is supported by the latest generation of psychotherapy process–outcome research'

Stages of constructive personality change

Rogers wrote about the six conditions in 1957, but it was not until 1961 that he developed his ideas for 'a process conception of psychotherapy', based upon a 1958 article of the same name

Rogers identified seven stages in the client's process

All stages can be viewed from an external frame of reference

Stages move from one to seven

In stage one, people are unlikely to come to therapy as they lack awareness of self, their views of the world are fixed and rigid, they are out of touch with their feelings, and they take no responsibility for their actions

If people become less rigid in thought processes and begin to get in touch with feelings (through being 'fully received' by another), then they may engage in therapy

Most therapy takes place with clients in stages two to six

If stage six is achieved, then stage seven will inevitably follow and is likely to happen outside the therapeutic relationship

THERAPEUTIC ALTERNATIVES

An individual will choose to seek therapy for many different reasons. Some people seek therapy because they want a safe place to vent frustrations, share feelings, let off steam and generally feel that someone is listening to their problems. Other people seek therapy because they want someone to offer an alternative viewpoint as they explore options with a view to making important decisions. In meeting these needs, it has been said that the modern therapist has taken the place of the religious elder (the more cynical amongst us might argue that the therapist has taken the place of family and friends!). However, modern life does offer many alternatives to therapy and these might be able to meet the needs listed above without involving any kind of therapeutic relationship. Consider trying some of the following activities and compare them with the traditional therapeutic experience to decide whether they are a weaker option, plausible alternative or appropriate addition to therapy.

- Write a diary or a blog describing your experiences and feelings every day for one week
- Express yourself about something that has upset you by talking to a pet
- Vent your frustration by screaming at an inanimate object (you might want to make sure that you do this in private to avoid attracting any unwanted attention)
- Visit a support website to read about issues that are causing you concern (for example, someone who is suffering after the loss of a beloved pet might find helpful information at www.livingwithpetbereavement.com/)
- Google search for a therapy chatbot and explain your current problems in virtual therapy (for example, ELIZA is a web-based person-centred therapist emulator found at www-ai.ijs.si/eliza/eliza.html and FreudBot is a web-based Freudian therapist emulator found at http://psych.athabascau.ca/html/Freudbot/test.html)

Disclaimer: please remember that these suggestions are designed to help you to think about the value of therapy and explore possible alternatives or supplementary options. They are not designed to address any serious personal or mental health issues and you should always contact a professional if you have any concerns in these areas.

Movement from stage one to stage seven involves the following

- Feelings unowned in the past to owned in the present
- Incongruence to congruence
- Experiences changing from remote and fixed to current and fluid
- Communication with self and others from closed to open
- Problems not recognised to accepted as part of the self
- Responsibility put onto others to self-responsibility

People can become less rigid in thought processes and get in touch with feelings through being 'fully received' by another

- Rogers (1961) argued that 'there is implied in this term ['received'] the concept of being understood, empathically, and the concept of acceptance'
- This explains how the six conditions can be used to support the process of therapy

Relational depth

Therapists should strive to achieve relational depth (Mearns & Cooper, 2005)

Relational depth is defined as a 'state of profound contact and engagement between two people, in which each person is fully real with the other, and able to understand and value the other's experiences at a high level' (Mearns & Cooper, 2005)

- Depth in this sense is related to 'honesty' or 'realism'
- A deeper relationship is one that is more honest, genuine, real and true
- This type of relationship is not superficial or artificial, but is instead an honest sharing of an encounter

Relational depth can be experienced within a moment ('a moment of intimacy' or 'a true connection during that conversation') or over a long period of time ('we are extremely connected' or 'we are always very intimate')

Relational depth versus core conditions

Core conditions form the foundation for standard person-centred therapy

- Core conditions indicate what conditions must be met in order for effective therapeutic change to take place (congruence, unconditional positive regard, empathy)

Relational depth goes one step further to explain the quality of the encounter

- Relational depth indicates the level of the intimacy between the therapist and the client

Concept of relational depth has been accused of simply being a new word for standard person-centred therapy

- Wilders (2007) argues that the description of relational depth is often nothing more than a description of person-centred therapy (except when the advice given to therapists conflicts with the core conditions – see point below)
- It could be argued that relational depth is simply a term to describe what happens when the therapist demonstrates the core conditions
    - Rogers used the word 'genuine' to describe the relationship, whereas Mearns and Cooper (2005) refer to the relationship in terms of 'depth'
    - 'Mearns and Cooper have failed to understand, or to experience for themselves, that when we, as real and genuine human beings, congruently offer empathy and unconditional positive regard to an other, then the more, not less, likely we are to have intimacy and closeness with our clients' (Wilders, 2007)
- Cooper (2007) responds to this criticism by suggesting that relational depth does not need to be viewed as novel or distinct from standard person-centred therapy
    - Instead, the intention is that the concept of relational depth will draw attention to the importance of in-depth relating to clients
    - Irrespective of whether this is already implied in traditional person-centred therapy, the concept of relational depth aims to bring it to the forefront and make it a critical component of the therapeutic aims

Concept of relational depth has been accused of being in conflict with the core conditions of person-centred therapy

- Purist person-centred therapy is entirely non-directional, yet advice for encouraging relational depth does occasionally indicate that the client may need to be directed or redirected
- Wilders (2007) argues that the concept of relational depth lacks an underlying coherent theory because it presents contradictory advice
    - Therapists are advised to 'be with' the client, thus suggesting that the therapist should follow rather than lead – this supports the principle of person-centred therapy
    - Therapists are invited to 'assume the right to confront or invade or challenge clients if they feel that this "encounter" will help to create a relationship of "depth"'. This conflicts with the principles of person-centred therapy because 'Relational depth at times gives free licence to therapists to offer suggestions, change the direction of the therapy, criticise, or . . . berate clients' (Wilders, 2007)
- Cooper (2007) responds to these criticisms by suggesting that it is illogical to assume that the same approach will work with all clients
    - Although non-directive methods may be highly effective for some clients, therapy should be flexible enough to incorporate a 'more directive way of engaging' if the therapist feels that this will improve the quality of the encounter (Cooper, 2007)

Value of controversy

Relational depth as a concept has sparked discussion and debate

> Perhaps it is not important whether it is a new theory or a contradictory theory or an incomplete theory . . .
>
> Simply reigniting the debate about the nature of the therapeutic relationship had a significant impact on research and development in person-centred therapy
>
> Rogers himself was both flexible and forward thinking, and he would no doubt have embraced the exploration with a view to further improving the therapeutic relationship

SUMMARY

Six conditions for constructive personality change: therapeutic relationship is the key component for successful therapy; for constructive personality change to occur, six conditions must exist and continue over a period of time (no other conditions, techniques or methods are necessary)

1. Two persons are in psychological contact
2. The first, whom we shall term the client, is in a state of incongruence, being vulnerable or anxious
3. The second person, whom we shall term the therapist, is congruent or integrated in the relationship
4. The therapist experiences unconditional positive regard for the client
5. The therapist experiences an empathic understanding of the client's internal frame of reference and endeavours to communicate this experience to the client
6. The communication to the client of the therapist's empathic understanding and unconditional positive regard is to a minimal degree achieved

Necessary and sufficient: small debate about whether all conditions are necessary in modern therapy; large debate about whether these conditions are sufficient for successful therapy

Stages: seven stages of the client process; client moves through the stages to become less rigid and more fully functioning

Relational depth: successful application of the core conditions will allow the therapist to work with the client at a deeper level; relational depth describes the intimacy present in a 'genuine' encounter; controversial theory which has sparked debate in the field, questions whether relational depth is just a new word for core conditions, questions whether relational depth conflicts with core conditions, but there is a value in controversy because it provokes discussion

THERAPEUTIC TECHNIQUES IN PERSON-CENTRED THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- appreciate the absence of specific techniques in this approach
- describe how the therapist uses genuineness, transparency and self-disclosure to demonstrate the core condition of congruence
- describe how the therapist uses a non-judgemental approach and sensitive cultural awareness to demonstrate the core condition of unconditional positive regard
- describe how the therapist enters the client's frame of reference by using primary and advanced empathy to demonstrate the core condition of empathic understanding

Absence of techniques

Successful therapy is not dependent on techniques

Successful therapy is dependent purely on the nature of the therapeutic relationship

> Constructive personality change requires only that the client and therapist engage in a positive therapeutic relationship

Person-centred therapy involves simply being with the client

> No use of games, exercises, activities, homework, etc.

Various methods can be taught and utilised to improve skills in communicating the core conditions of congruence, unconditional positive regard and empathy

> But these are simply interpersonal skills, rather than specific therapeutic techniques to be followed according to the instructions
>
> As noted by Thorne (1996), Rogers was horrified to find that the focus on the responses of the therapist became a list of techniques – the whole purpose of person-centred therapy is to be genuine in that moment, rather than to follow a set of protocols according to the book

Therapeutic relationship is more important than therapeutic techniques

Research exploring common factors across multiple types of therapy has often focused on the core conditions of person-centred therapy (Lambert & Barley, 2001)

> Analysis of over 100 empirical and meta-analytic studies of therapeutic interventions reveals that

40% of client improvement is dependent on external events unrelated to the therapy

15% of client improvement is dependent on the placebo effect

15% of client improvement is dependent on the therapeutic techniques used in the sessions

30% of client improvement is dependent on the nature of the therapeutic relationship

These findings are a little disturbing as they suggest that over half (55%) of client outcome is influenced by events that have absolutely nothing to do with what takes place in the therapeutic sessions

These findings also support the suggestion that the relationship itself (and the demonstration of the core conditions) is twice as important as the techniques employed in ensuring a positive outcome for the client

WHO I AM OR WHAT I DO?

Reflect on three people who have had a positive impact on your life; perhaps they inspired you or comforted you or assisted you in some way. Try to think about why you were influenced by these people . . .

Did they show you how to be a better person by demonstrating their own positive traits? Perhaps they were strong in the face of adversity and this taught you to have strength? This would be akin to a therapist demonstrating the core conditions in relation to the self (self-acceptance, etc.) in order to show the client how to move towards self-actualisation.

Did they offer you a warm and safe environment in which to share your feelings? Perhaps they were a shoulder to cry on or inspired you to express yourself? This would be akin to a therapist offering a positive therapeutic relationship in which the client can explore and develop on a personal level.

Did you benefit from something that they did? Perhaps they did something to help you or worked with you to complete a task? This would be akin to a therapist using specific activities and techniques to help a client.

All of the above describe situations in which you may have been significantly influenced by another person – perhaps even to a life-changing extent. However, the person-centred therapist would strive towards the first two situations because these are believed to offer the most lasting benefit to the client. In contrast, the third situation may only assist the client on one specific problem and may foster dependency rather than autonomy. Consider your own experiences to decide whether you agree with this perspective.

An APA task force (Norcross, 2011) explored the association between the therapeutic relationship and the outcomes resulting from therapy

- They concluded that the relationship makes 'substantial and consistent contributions to psychotherapy outcome independent of the specific type of treatment'
- These findings suggest that the relationship, rather than any specific techniques used in the treatment, has the greatest positive impact on the client

Lambert and Barley (2001) state that 'decades of research indicate that the provision of therapy is an interpersonal process in which a main curative component is the nature of the therapeutic relationship . . . clinicians must remember that this is the foundation of our efforts to help others'

Clients themselves often attribute the success of therapy to the core conditions demonstrated by the therapist, rather than any specific activity or task completed in the session

- Qualitative analysis of patients experiencing therapy revealed that those who described the treatment as 'successful' explained that the therapist was 'warm, attentive, interested, understanding, and respectful' (Strupp et al., 1969)

Demonstrating congruence

Genuineness, transparency and self-disclosure can be used to demonstrate congruence.

- Therapist must be congruent and integrated in the relationship (Rogers, 1957)
  - Congruency is the degree to which one can relate to another individual in a sincere and honest manner
  - Essential for both empathic understanding and unconditional positive regard

Genuineness to demonstrate congruence

Congruent therapist will demonstrate non-possessive warmth by conveying friendliness in all verbal and non-verbal communication

Congruent client will try to be genuine in all forms of communication

- Body language, gestures, expressions and tone must all match the sentiment being expressed to demonstrate true genuineness

Transparency to demonstrate congruence

Congruent therapist will not pretend to be an expert

- Therapist does not know more about the client than the client him- or herself
- Therapist may only be knowledgeable in one particular area of psychology or therapy, and this should be made clear to the client
- Therapist should not be afraid to ask for further information, admit that they do not know something or confess that they do not understand

Congruent therapist will be open about all qualifications and feel able to refer or signpost when meeting cases beyond the limits of their ability

It is perfectly acceptable for the therapist to admit to a client that s/he is unable to help with that issue and direct the individual to a more appropriate source of help

Congruent therapist will not attempt to hide things from the client, will seek informed consent and will always be transparent in motives and intentions

All goals should be agreed together

Client should not be secretly manipulated to achieve an aim

If confidentiality has to be broken, client should be honestly told about this in advance if at all possible

The traits discussed above enhance transparency and ensure that the therapist adheres to the British Association for Counselling and Psychotherapy ethical principle of autonomy (BACP, 1992)

Autonomy refers to the practitioner's respect for the client's right to be self-governing

BACP states that practitioners 'should ensure accuracy in any advertising or information given in advance of services offered, seek freely given and adequately informed consent, engage in explicit contracting in advance of any commitment by the client, protect privacy, inform the client in advance of foreseeable conflicts of interest or as soon as possible after such conflicts become apparent, and refuse to manipulate clients against their will, even for beneficial social ends' (BACP, 1992)

Self-disclosure to demonstrate congruence

Congruent therapist will try to be honest with the client about his/her feelings

Therapist should not lie or fake feelings to the client

Disclosure of feelings or personal information by the therapist can help both the client and the therapeutic relationship

Client can be helped if the disclosure normalises his or her own thoughts, feelings and behaviours

For example, client may feel that anxiety about motherhood makes her a 'bad mum', so a therapist who explains that she also found it difficult might help to normalise these feelings

However, it is important for the therapist to avoid trivialising the experience of the client or suggesting that the client is not unique

Client can be helped if the disclosure provides a new insight into a problem

For example, client may feel trapped in his current job so a therapist who explains what he did to find a new job might provide some new ideas for solving the problem

However, it is important for the therapist to avoid giving advice or instructing the client – information signposting should be nothing more than providing new information without any pressure to respond

- Therapeutic relationship can be helped if the disclosure enhances genuineness
  - 'Spontaneous sharing on the part of both parties is the essence of a genuine relationship' (Carkhuff, 1969)
  - However, it is important for the therapist to avoid directing attention away from the client and allowing the discussion to descend into a friendly chat
- Congruent therapist will not feel compelled to disclose in all situations
  - All disclosures should benefit the client (Patterson, 1985)
  - Some information can help the client to gain insight about his or her own actions
    - For example, a therapist who feels bored might wish to share this information because it might give the client insight about the way that s/he presents the self to others
  - Disclosures not intended to benefit the client should not be made
    - Therapist can explain that s/he does not wish to disclose this information or could ask the client how the information would help
    - 'Therapy is for the client, not the therapist' (Patterson, 1985)

Demonstrating unconditional positive regard

A non-judgemental approach and sensitive sociocultural awareness can be used to demonstrate unconditional positive regard

- Therapist must experience unconditional positive regard for the client (Rogers, 1957)
  - Successful personality change requires the therapist to accept the client with genuine warmth (positive regard) irrespective of the personal traits or behavioural actions of the individual (unconditional)

Non-judgemental approach for demonstrating unconditional positive regard

- Important to understand oneself to be aware of any potential prejudice in order to maintain a non-judgemental attitude towards the client
  - All people are entitled to hold their own views and values, but the therapist should strive to avoid imposing these on the client
  - Therapists should strive to avoid passing judgement on others
  - Ask yourself whether you would honestly behave differently in their circumstance
    - We often find that we judge others most harshly for those sins that we ourselves commit or wish to commit
  - Avoid judgemental expressions
    - Certain words and phrases tend to imply judgement: should, ought, must, got to, don't, etc.
    - If you have an urge to say 'you should . . .', try asking yourself why you think that they should

Sensitive sociocultural awareness for demonstrating unconditional positive regard

Important to recognise cultural differences and accept that others view the world in very different ways

- Different does not mean wrong
- Avoid stereotypes and assumptions
- Many stereotypes and assumptions about people are held so deeply that it is difficult to realise that they exist
  - Consider the concept of love – our understanding of love is often from the perspective of an individualistic culture so it is often difficult to appreciate the concept of arranged marriages and this can lead to assumptions of force
- Cultural awareness is dependent on understanding
- Acceptance can sometimes mean asking for more information so that you can truly understand the world of the client in order to avoid jumping to conclusions

However, cultural awareness is not dependent on knowledge

- Although it can be helpful if the therapist has some knowledge about the culture of the client, it is not always necessary for the therapist to be a member of that culture or to have a deep and comprehensive knowledge of cultural practices specific to the client
- Indeed, cultural knowledge can sometimes lead the therapist to disregard factors relating to the individual in favour of assumptions relating to the background culture
  - For example, a therapist who 'knows' that the culture of the client forbids premarital sex may assume that the client was forced to marry after falling pregnant – in reality, however, that client may have intentionally become pregnant in order to be permitted to marry the person that she loved
- Dyche and Zayas (2001) argued that we need therapists who are 'knowledgeable about the specifics of particular cultures but who are able to suspend this knowledge when with a client in order to listen openly and without assumptions'

Cultural understanding is sometimes classed as 'cross-cultural empathy' (Dyche & Zayas, 2001)

- Cross-cultural empathy helps the therapist understand the world of the client from a sociocultural perspective
- This deeper understanding and appreciation can help the therapist to accept the client unconditionally

Demonstrating empathic understanding

The client's frame of reference can be entered using primary and advanced empathy to demonstrate empathic understanding.

Therapist must experience an empathic understanding of the client's frame of reference (Rogers, 1957)

Successful therapeutic change requires the therapist to experience 'an accurate, empathic understanding of the client's awareness of his own experience' (Rogers, 1957)

Empathic understanding requires the therapist to be able to perceive the world of the client from the client's frame of reference

Frame of reference for demonstrating empathy

We all view the world from our own internal frame of reference

We do not see the world as it really exists, but instead we see the world based on our memories, beliefs, values, knowledge, experiences, feelings, etc. in addition to our perceptions and sensations

True understanding of another person can only occur if we attempt to view the world from their frame of reference

We must accept that this is never truly possible and we must ensure that we never become so immersed in the frame of reference that we cannot withdraw

But we must still strive to view the world in this way so that the client senses that we are trying to truly understand

Therapists can only enter the client's frame of reference if they listen carefully to the verbal and non-verbal communication from the client, attempt to identify the expressed feelings and underlying behaviours, and respond by showing understanding rather than judgement

Primary-level empathy (Egan, 2007)

Primary-level empathy requires the therapist to respond to the feelings and behaviour expressed by the client

Responses will usually involve active attending, paraphrasing content and reflecting feelings

Active listening and attending require the therapist to listen to the client, respond appropriately (use of minimal encouragers and clarification questions), and be aware of how they are sitting and acting towards the client

Paraphrasing content and reflecting feelings require the therapist to repeat the content and feelings of the client's expression

Paraphrasing will involve repetition of the content

Reflecting will involve repetition of the feelings

Mirroring will involve the direct repetition of specific words spoken by the client

Summarising will involve a brief outline of both the content and feelings indicated in the client's expression, linking thoughts, feelings and behaviours, and identifying themes that have emerged

Advanced-level empathy (Egan, 2007)

Advanced-level empathy requires the therapist to sense the expressions not communicated directly by the client

Self-disclosure, interpretations and challenges can all be used in advanced empathy

Self-disclosure involves sharing selected information about the therapist in order to enhance insight in the client

- Disclosure might be made to normalise – for example, 'Many people might feel like that in your situation, even I have had similar thoughts when I was grieving for my mum'
- But must beware of the risks of assuming that they are not unique or individual
- Disclosure might be made to indicate how the client presents – for example, 'When you raise your voice like that, I feel a little intimidated'

Interpretation involves detection of emotions on the basis of verbal and non-verbal language

- Interpretation might be made to help the client to see parts of themselves that they have not yet acknowledged or draw their attention to emotions not yet expressed
- For example, 'I notice that you clenched your fist when you said that and I wondered if perhaps you were feeling angry'

Challenges involve drawing the attention of the client towards any apparent contradictions

- Contradictions can exist between two sets of spoken words – for example, 'I notice that you said that you were happy for your sister but last week you mentioned feeling angry towards her, so I was just wondering what might be happening there?'
- Contradictions can exist between spoken words and body language – for example, 'I notice that you said that you were happy for your sister but you seemed to clench your jaw straight afterwards and I was wondering what might be happening there?'

Always need to be aware of the risks associated with advanced empathy

- High risk of focusing on therapist during self-disclosure so must ensure that the client remains the focus
- High risk of therapist appearing 'psychic' during interpretation so must ensure that the client remains the expert
- High risk of therapist appearing threatening during challenges so must ensure that the client remains comfortable sharing

SUMMARY

Absence of techniques: successful therapy is not dependent on techniques, success is dependent purely on the therapeutic relationship; research indicates that the main curative component in therapy is the nature of the therapeutic relationship

Congruence: therapist must be congruent and integrated in the relationship; genuineness; transparency; self-disclosure

Unconditional positive regard: therapist must experience unconditional positive regard for the client; non-judgemental approach; sensitive sociocultural awareness (cross-cultural empathy)

Empathy: therapist must experience an empathic understanding of the client's frame of reference; primary empathy, paraphrasing, reflecting, mirroring, summarising; advanced empathy, self-disclosure, interpretation, challenges, but be aware of the risks of advanced empathy

CASE STUDY DEMONSTRATING PERSON-CENTRED THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

Appreciate the application of person-centred therapy from the humanistic approach in a therapeutic setting

Recorded session: therapy in action

This chapter is accompanied by a recorded therapy session which is available for viewing.

This session lasts for one therapy hour (50 minutes) and it is presented as the initial session in a new therapeutic relationship. Prior to this session, the client will have completed an initial assessment questionnaire and the therapist will have read this paperwork to ensure familiarity with the case. This completed assessment and a full transcript of the recorded session are available.

No actors are used in this session. The client was one of the authors and the problem presented was genuine. The therapist is an experienced practitioner in the field. The only 'fake' aspect of this recorded session is that the client did not really seek therapy and this is not really the first session of a series of therapeutic contacts.

After the conclusion of the session, the therapist is invited to answer a few key questions about the session. This question and answer session lasts no longer than ten minutes. The transcript of this session is available.

Presentation of Phil Thomas: history and symptoms

Phil Thomas has been suffering with stress and anxiety over the last few months. He has recently experienced some minor life events and these have had an impact on his mental wellbeing. He feels particularly anxious about his tendency to procrastinate and he believes

that this is contributing towards his general feelings of stress. He has previously experienced person-centred therapy and he is generally positive about the possible outcomes for therapy. He would like to gain an understanding of his own behaviour in order to be able to reduce his own procrastination.

Therapy session: analysis

The therapy session can be sectioned as follows.

Introduction

- Gentle introduction incorporating humour
- Reflection on previous experience with therapy to establish possible expectations and beliefs about therapy
- Addressing any questions about the nature of person-centred therapy
- Outlining basic contractual details, especially the limits of confidentiality and the nature of sessions (length, number, etc.)

Story

- Client is invited and encouraged to share his story – this forms the largest part of the therapy session
- Focus on feelings and internal experiences, rather than the specific practicalities of the story

Goals

- Establishing goals for therapy is periodically explored throughout the session, rather than forming a conclusion to the session
- Goals are explored in terms of the current and ideal locations for the client

Ending

- Warning given prior to the end of the session
- Reflection on how the session was experienced by the client
- Invitation to return for future sessions

Key questions to consider in relation to this therapy session

How could the nature of this client be understood from the humanistic perspective?

- What are the various configurations of the self-concept of this client?
- What possible conditions of worth could be having an impact on this self-concept?
- What might be getting neglected in the hierarchy of needs to prevent self-actualisation?
- Which characteristics of the full functioning person is this client failing to experience?

What is the nature of the therapeutic relationship in this person-centred therapy session?

- Does the client seem incongruent?
- Does the therapist appear congruent in the interaction?

- Does the therapist seem to be experiencing unconditional regard for the client?
- Does the therapist seem to feel empathy for the client's frame of reference?
- How does the therapist communicate congruency, unconditional positive regard and empathy to the client?
- How does the client respond to congruency, unconditional positive regard and empathy from the therapist?
- How deep is the relationship between the client and therapist?

Which person-centred techniques are demonstrated in this therapy session?

- Is the therapist transparent in her actions?
- Is the therapist genuine and how is this displayed to the client?
- Does the therapist self-disclose and what is the effect on the client?
- Is the therapist non-judgemental and how is this understood by the client?
- Does the therapist recognise the sociocultural position of the client?
- Is the therapist entering the frame of reference of the client?
- How does the therapist show primary and advanced level empathy?

Personal experience of the client

I arrived at the session a little apprehensive, hoping that all would go well and that I would be a 'good' client. The opening few minutes of the session gave me time to settle, as the therapist clearly explained confidentiality, and briefly outlined her approach.

In terms of relationship, I felt that Elaine and I developed a good foundation within the session for future work. I felt listened to and understood. I sensed that Elaine had a good degree of empathy, that she made some real connections with how I experienced my issues in the session. This enabled me to explore deeper and come to some new awareness. I also was given the time and space to reflect upon my thoughts and feelings. I can see this more clearly with hindsight, but remember that there were times in the session when I felt discomfort with the nature of therapist responses; that they were reflections, rather than challenges. At the time I remember thinking that there was little happening, and I commented on more than one occasion that I felt stuck and that I didn't know what to do with it (the reflection, the new insight). I think at the time I was hoping that Elaine would step in and rescue me.

I think that says a lot about my process. I guess I need to feel that I am moving in some direction, any direction. From my experience, the person-centred approach doesn't give that kind of external direction (non-directive) and that any direction needed to come from me. I see how that is beneficial, but at the time it was a little uncomfortable.

During the session I became increasingly aware of internal incongruence; that there were 'parts' of me in conflict. Again, in reflection, I see these as 'configurations' and sense that there is an opportunity to have an internal dialogue between these configurations. This new awareness felt challenging, that here is something tangible to get into – an area to explore. I particularly liked Elaine's reflection that she felt sad for the part of me that I wanted to be rid of.

I left the session with a better sense of my issues, some areas were clarified, others areas were newly formed (e.g. internal conflicts). I had a sense that it would need to be a commitment to a process and that there were no short cuts to getting answers and resolutions. It felt like the beginning of a journey and that Elaine had the right qualities to be my guide.

—Phil Thomas

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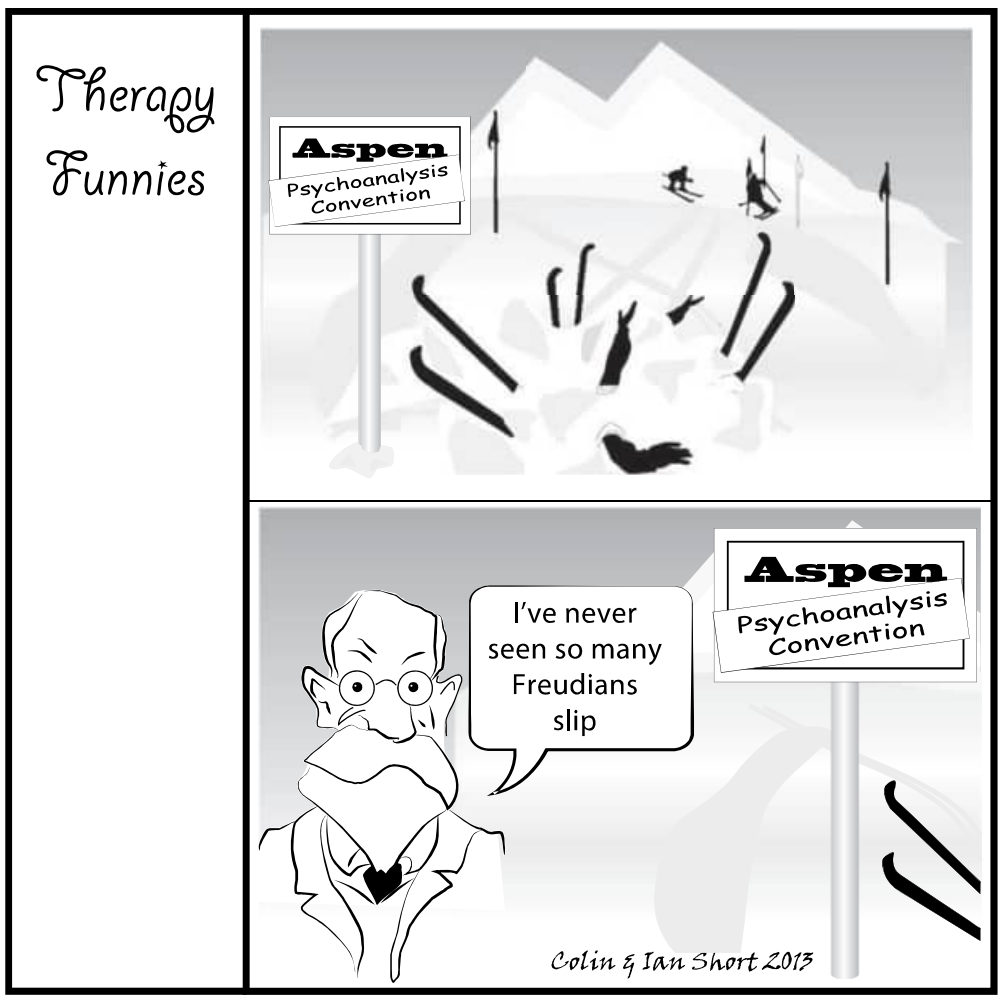
TABLE 2.1 My conditions of worth

Reflect on the following expressions and consider whether you have ever used them yourself or heard them being used by someone you love. Can you identify the unsaid implications of each statement?

Expression	Implication
Example: Real men don't cry	*Example*: If you allow yourself to experience and demonstrate the emotion of sadness then you are unworthy of the respect and love afforded to those who do not exhibit their emotions
Be a good girl/boy	
Don't act like such a big baby	
I wish you were more like your sister/cousin/friend/etc.	
You should know better than that	
You are so clumsy/lazy/stupid/useless	
Hurry up	
I wish you wouldn't do that	
You need to work harder	
I love you when you do that	

Chapter 3

Psychodynamic approach and psychoanalytic therapy



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INTRODUCTION TO PSYCHODYNAMIC APPROACH AND PSYCHOANALYTIC THERAPY

This chapter aims to introduce the reader to the psychodynamic approach to counselling and psychotherapy. Psychoanalytic therapy will be explored as one example of a therapeutic method under the psychodynamic approach.

LEARNING OUTCOMES

By the end of this chapter, you will be able to:

- describe the development of the second force in psychology: the psychodynamic approach
- acknowledge the relative impacts of Sigmund Freud and Melanie Klein on the development of the psychodynamic approach
- discuss the core theories of human nature and personality from the psychodynamic perspective
- discuss the nature of the therapeutic relationship between therapist and client in psychoanalytic therapy
- outline the main therapeutic techniques utilised in psychoanalytic therapy
- appreciate the application of psychoanalytic therapy in a real-world setting

DEVELOPMENT OF THE PSYCHODYNAMIC APPROACH

LEARNING OUTCOMES

After reading this section, you will be able to:

- list the three main forces in psychology
- discuss the development of the psychodynamic approach in a historical context
- acknowledge the main contributors to the development of the psychodynamic approach

Second force in psychology

Three forces in psychology

Behavioural theory

Psychodynamic theory

Humanistic theory

Psychodynamic psychology is the 'second force'

Developed alongside the behavioural approach as the two reacted against one another

First force

Behavioural psychology assumed that human nature was mechanistic and reduced humanity to the link between stimuli and response

Focused exclusively on observable external events (behaviour)

Second force

Psychodynamic psychology assumed human nature was deterministic and reduced humanity to basic biological drives

Focused exclusively on interpreted internal events (unconscious)

Rejection of the scientific method

Behavioural psychology was highly scientific as each principle could be tested in a controlled environment

In contrast, early psychodynamic theory was almost anti-scientific as each principle was based on internal unconscious drives that could only be detected through the subjective process of psychoanalysis

Although Freud considered himself a scientist, early psychodynamic theory did not apply the scientific principles of falsification to the study of the human psyche

This is often a criticism of the psychodynamic approach, but it is important to remember that science is limited by current available methods and it is not accurate to state that something cannot be true simply because it cannot be tested scientifically at that time in history

Modern neuroscience research into psychodynamic theories is now beginning to apply current techniques to exploration of the internal drives described by Freud and some of the findings are consistent with his theories

Neuropsychoanalysis is a new branch of psychology focusing on integrating neuropsychology and psychoanalysis (Solms & Turnbull, 2002)

Major advances in the psychodynamic approach

Initial interest in psychodynamic approach

Brucke presented the argument that every living creature contains dynamic energy governed by the first law of thermodynamics – energy can only be changed from one form to another, but it can never be destroyed

Lectures on Physiology published in 1874

Freud worked under Brucke at the University of Vienna in 1873

Freud was influenced by the lectures on physiology and applied these theories to the human psyche

Freud argued that human personality was a type of psychic energy subject to the same laws of thermodynamics

These early ideas formed the foundation of later psychodynamic theories

Freud and Breuer established the initial idea of repressed traumatic memories impacting on physical symptoms of hysteria during clinical work with hysteria patients

Studies on Hysteria published in 1895

Freud established the idea of dreams revealing repressed memories and unacceptable sexual and aggressive desires

The Interpretation of Dreams published in 1900

Freud began to establish the concept of the psychosexual stages of development as the psychodynamic theories began to form a cohesive approach to understanding human personality

Three Essays on the Theory of Sexuality published in 1905

Formal establishment of psychodynamic approach

Freud failed to get psychoanalysis accepted in the university system as an area of study and this led to the restriction of psychoanalysis training to private institutes (Jarvis, 2004)

Although his basic premises were accepted, the methods were regarded as too non-empirical for experimental psychology based heavily in the behavioural approach

In addition, many academics did not agree with his theories on infantile sexuality

Since his approach was not studied formally by academics, his theories remained free from the academic rigour normally applied to academic study and this led to the 'ossification' of theory (Malan, 1995)

Previous collaboration between Freud and Adler ended in 1911 and collaboration (and friendship) between Freud and Jung ended in 1914 (Bridle & Edelstein, 2000)

Both Adler and Jung struggled to accept the concepts relating to infantile sexuality, although they did agree on the general principles of the psychodynamic approach

Division of the approach at this stage helped to broaden the psychodynamic approach and establish psychoanalysis as the predominant therapeutic method across Europe and the USA

Freud eventually formulated his main theory of personality (structural theory) and this outlined the structure of the mind in terms of the id, ego and superego

The Ego and the Id published in 1923

Structural theory was widely accepted as demonstrating the structure of human personality, but debate continued to rage about the relative importance of each part of the personality on future behaviour

Jung established his own psychodynamic theories based on identifying personality types and archetypes

'Analytical psychology' is the Jungian school of psychodynamic theory

Psychological Types published in 1921

Adler establish his own psychodynamic theories based on a holistic approach to the person

'Individual psychology' is the Adlerian school of psychodynamic theory

Practice and Theory of Individual Psychology published in 1927

Alongside the development of these alternative branches of psychodynamic theory (Adlerian and Jungian), work continued on classic Freudian theory

Melanie Klein developed object relations theory in Britain after applying psychoanalysis to work with young children

The Psycho-Analysis of Children published in 1932

Anna Freud developed the concept of ego defence mechanisms and outlined how these are used in both beneficial and damaging ways

The Ego and the Mechanisms of Defence published in 1936

Both Klein and Freud worked in London using psychoanalysis with young children, and this overlap in activity but difference in perspective brought them into conflict

Eventually, each established her own branch of psychoanalysis, and training in the UK today is predominantly Kleinian whereas training in the US today is predominantly Freudian

Erikson devised an alternative set of developmental stages as a framework for psychoanalysis and this moved the theory away from infantile sexuality to focus on other aspects of human development (thus making it more acceptable to many academics)

Childhood and Society published in 1950 and revised in 1963

Horney presented a less male-oriented view of psychoanalysis, incorporating a new theory of self and a deep understanding of neurosis

Neurosis and Human Growth published in 1950

In the 1950s, the psychodynamic approach was eventually accepted in academia and introduced into the university system

Eysenck (1952) published a review of studies assessing the effectiveness of psychoanalysis

Although these studies have since been criticised, they do demonstrate the acceptance of the theory to the extent that it was regarded as appropriate for empirical evaluation

Bowlby's (1969) attachment theory also provided a framework within which to empirically test psychodynamic principles, particularly those relating to object relations

Modern psychodynamic approach

Today, the psychodynamic approach is accepted in academic circles to some extent

It remains a 'Cinderella paradigm' (Jarvis, 2004) to some extent, as it is still regarded as non-scientific

- Modern advances in neuroscience allow some empirical research of the core principles

Psychodynamic approach is split into the following schools of thought

- Modern psychoanalytic therapy focuses on the following
  - Freudian (classic psychoanalysis) – Sigmund Freud, Anna Freud
  - Kleinian (object relations psychoanalysis) – Melanie Klein
  - Independent (mix of Freudian and Kleinian) – Winnicott
- Alternative therapies derived from psychodynamic approach include the following
  - Analytical – Carl Jung
  - Individual – Alfred Adler

Key figures in the psychodynamic approach

Sigmund Freud (1856–1939)

Impact on the approach

- Founding father of psychodynamic theory and psychoanalysis
- Highlighted the effect of the unconscious on the conscious
- Established the structure of personality containing the id, ego and superego
- Presented the psychosexual stages of development, including the Oedipus and Electra conflicts

Selected works

- *The Interpretation of Dreams* published in 1900
- *Three Essays on the Theory of Sexuality* published in 1905
- *The Ego and the Id* published in 1923

Alfred Adler (1870–1937)

Impact on the approach

- Developed 'individual psychology'
- Established a 'holistic' perspective focusing on human behaviour in a social context
- Explained the concept of the 'inferiority complex'

Selected works

- *Practice and Theory of Individual Psychology* published in 1927

Carl Jung (1875–1961)

Impact on the approach

- Developed 'analytical psychology'
- Established the concept of the collective unconscious and 'psychic inheritance'
- Established the concepts of introversion and extraversion

Focused on explaining adult development and the midlife crisis

Discussed roles in terms of archetypes (such as the mother archetype)

Selected works

Psychology of the Unconscious published in 1912

Psychological Types published in 1921

Melanie Klein (1882–1960)

Impact on the approach

Developed 'object relations psychology'

Pioneered work in psychoanalysis for children, with focus on the analysis of play

Established the concept of 'internal objects' to explain how early interactions with parental figures through the paranoid-schizoid and depressive positions impact on later life

Selected works

The Psycho-Analysis of Children published in 1932

Envy and Gratitude published between 1946 and 1963

Narrative of a Child Analysis published in 1961

Karen Horney (1885–1952)

Impact on the approach

Founding mother of feminist psychology

Disagreed with Freudian psychoanalysis and argued that psychological traits (such as gender identity) were not stable factors based in biology, but were instead fluctuating states occurring in response to experience

Established a comprehensive theory of neurosis

Proposed a counterpart to penis envy – 'womb envy' occurs in men who feel envious of the ability to bear children

Selected works

Neurosis and Human Growth published in 1950

Feminine Psychology published in 1967

Anna Freud (1896–1982)

Impact on the approach

Followed in the footsteps of her father to practise psychoanalysis with patients and presented many of her own psychodynamic theories

Pioneered work in psychoanalysis for children and was heavily involved in charity work supporting families and children

Established the concept of defence mechanisms to protect the ego

Selected works

The Ego and the Mechanisms of Defence published in 1936

Donald Winnicott (1896–1971)

Impact on the approach

Introduced the concepts of the 'true self' and the 'false self'

Introduced the 'transitional object' as a means of transferring feelings of security from the mother to another object (such as a security or comfort blanket)

Emphasised the importance of play (in children and adults) for psychological wellbeing

Argued that mothers should trust their own intuition to be 'good enough' for the child

Selected works

Playing and Reality published in 1971

The Ordinary Devoted Mother and Her Baby published in 1949

Transitional Objects and Transitional Phenomena published in 1953

Erik Erikson (1902–1994)

Impact on the approach

Presented the psychosocial stages of development

Coined the phrase 'identity crisis' when working with troubled young adults

Focused on the social aspects of development with specific attention to the ego – different to the classic approach of focusing on the sexual aspects of development with specific attention to the drives of the id

Selected works

Childhood and Society published in 1950

John Bowlby (1907–1990)

Impact on the approach

Developed the concept of attachment theory based on study of maternal deprivation

Highlighted the effect of healthy childhood attachments on adult relationships

Selected works

Attachment and Loss: Attachment (Vol 1) published in 1969

Attachment and Loss: Separation Anxiety and Anger (Vol 2) published in 1973

Attachment and Loss: Sadness and Depression (Vol 3) published in 1980

Final thought . . .

'Anyone who wants to know the human psyche will learn next to nothing from experimental psychology. He would be better advised to abandon exact science, put away his scholar's gown, bid farewell to his study, and wander with human heart through the world. There in the horrors of prisons, lunatic asylums and hospitals, in drab suburban pubs, in brothels and gambling-hells, in the salons of the elegant, the Stock Exchanges, socialist meetings, churches, revivalist gatherings and ecstatic sects, through love and hate, through the experience of passion in every form in his own body, he would reap richer stores of knowledge than text-books a foot thick could give him, and he will know how to doctor the sick with a real knowledge of the human soul' (Jung, 1966)

SUMMARY

Three forces: behavioural, psychodynamic, humanistic; second force developed alongside behavioural but focused on the unconscious rather than the observable, non-scientific originally but recent research has focused on neuropsychoanalysis

Major advances: initial ideas proposed by Freud in late 1800s after exposure to thermodynamics by Brucke; Freud and Breuer introduced concept of repressed memory in 1895; Freud used dreams to reveal repressed memories and unacceptable urges in 1900, psychosexual stages presented by Freud in 1905; formal establishment began after 1920s, acceptance of behavioural approach in academia excluded psychodynamic approach, Adler and Jung diversified the approach and carried the theories across the US and Europe 1910 to 1920, Freud formulated structural theory in 1923, Adler established individual psychology in 1927, Klein introduced object relations in 1932, Freud outlined ego defence mechanisms in 1936, Erikson presented psychosexual stages in 1950, Horney presented a feminist view of psychoanalysis in 1950, accepted as an academic area in 1950s, Eysenck reviewed effectiveness of psychoanalysis in 1952, Bowlby's attachment theory gave a framework for testing psychodynamic theories in 1969; formally established and accepted as an academic area in 1950s; modern approach includes Freudian (classic), Kleinian (object relations), independent (derivatives include analytical (Jung) and individual (Adler))

Key figures: Freud's psychosexual stages and topographical and structural theories, Adler's individual psychology and inferiority complex, Jung's analytical psychology and collective consciousness, Klein's object relations and internal objects, Horney's feminist psychology and theory of neurosis, Freud's ego defence mechanisms, Winnicott's transitional objects, Erikson's psychosocial stages, Bowlby's attachment theory

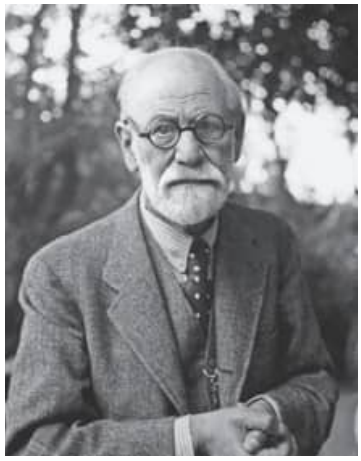
PERSONAL AND PROFESSIONAL BIOGRAPHIES OF SIGMUND FREUD AND MELANIE KLEIN

LEARNING OUTCOMES

After reading this section, you will be able to:

- outline the personal and professional biography of Sigmund Freud (1856–1939)
- outline the personal and professional biography of Melanie Klein (1882–1960)
- appreciate the impact of these key figures on the development of the psychodynamic approach

Biography of Sigmund Freud



'A certain degree of neurosis is of inestimable value as a drive, especially to a psychologist'
—Attributed to Freud (n.d.)

Who was Sigmund Freud?

Sigmund Freud founded the psychodynamic movement

Possibly the most influential perspective in psychology

Sigmund Freud introduced psychoanalytic therapy

Possibly the most influential therapist in history

Early years

Born May 6th 1856 in the Czech Republic

Named Sigismund Schlomo Freud

Shortened his name to Sigmund at the age of 22

Father (Jacob Freud) was a wool merchant and mother (Amalia Nathansohn) was a housewife

She adored her 'golden Sigi' and revolved the family home around his needs and desires

Her devoted nature established the role for women in his work, and his Oedipus complex was partly based on his own feelings for his parents

Jewish Galician family

In accordance with tradition, both parents favoured him because he was the firstborn son and carried the good omen of having been born with the caul (part of the amniotic sac remaining on the newborn) (Margolis, 1989)

'A man who has been the indisputable favorite of his mother keeps for life the feeling of a conqueror' (Freud, quoted in Jones, 1957)

Family lived in poverty with much of their income going towards the education of their eldest son

Educationally advanced from a young age

Reading Shakespeare at the age of eight

Entered secondary school a year early

Graduated with honours receiving a summa cum laude award (top 5%) in 1873

Education

Law to medicine

Initially intended to study law, but eventually decided on medical school after attending a lecture on the essay 'On Nature' by Goethe

'Anatomy is destiny' (Freud, 1912)

Medical student at Vienna University from 1873

Conducted research on the sexual glands of eels in Trieste then returned to Vienna to work in the laboratory of Ernst Brucke

First scientific paper was published at the age of 20

Attended Meynert's course in psychiatry to learn more about neurology

Awarded a delayed medical degree from the University of Vienna in 1881 after one year of compulsory military service in 1879

Wanted to work in research, but financial difficulties forced him to practise

Career

Initial research in neuroanatomy

Developed a new staining technique using gold chloride to view the microscopic world of the neuron and trace the connections between different parts of the nervous system (Freud, 1884)

First neurologist to trace the course of the spinocerebellar white matter tracts from the lateral portion of the spinal cord to the cerebellum (Amacher, 1965)

Close to being the first to describe the neuron theory of the central nervous system, and his work provided the foundation for later researchers to develop a unified theory of cerebral structure (Galbis-Reig, 2004)

Meynert's psychiatric clinic in 1883

Hermann Nothnagel's medical clinic in 1883

Worked as resident physician at the clinic and was exposed to the use of hypnosis as a treatment for psychiatric disorders

Published *Uber Coca* (an article on cocaine) in 1884

Discovery of cocaine-based anaesthetic

Freud discovered the analgesic properties of cocaine in 1884, but did not conduct formal experiments and instead charged a colleague (Konigstein) to complete the experiments while he visited his fiancée – his colleague never completed the studies

Before leaving, he also described his ideas to another colleague (Carl Koller) who used these theories to test cocaine as an anaesthetic during eye surgery – Koller is now recognised as the person who discovered the anaesthetic properties of cocaine

Modern anaesthetic would not be possible without these early findings about the analgesic properties of cocaine

Personal use of cocaine

Freud frequently used cocaine himself, and the drug may have influenced him when he reflected on childhood (possibly false) memories of sexual desire for his mother and sexual jealousy towards his father

Freud also prescribed cocaine to friend Fleischl von Marxow as a cure for morphine addiction – unsurprisingly, Fleischl became addicted to cocaine and eventually died at only 45 years of age

Freud was criticised significantly in medical circles for this tragedy – Erlenmeyer said that Freud has added a 'third scourge of humanity, cocaine' to morphine and alcohol (Borch-Jacobsen, 2012)

Fellowship to study with Jean-Martin Charcot in Paris in 1885

Charcot was a renowned neurologist who focused on the clinical applications of hypnosis

Observation of patient hysteria and the effects of hypnosis moved Freud from neurology towards psychopathology

Private medical practice in Vienna in 1886

Established a private practice focusing on nervous and brain disorders

Found that symptoms of hysteria could be eased by putting a person in a relaxed position (on the couch) and asking them to talk about whatever passed through their heads – he could then identify the true root of trauma and symptoms would be relieved by this revelation

Found that these procedures could be enhanced by the use of hypnosis and hypnotic suggestion – first use of his technique was in 1887

Published *On Aphasia* in 1891

Reviewed in detail the clinical presentation of aphasia

Argued against the theory that all aphasias are rooted in two or three structures and proposed a unified theory of the mind (Galbis-Reig, 2004)

Still regarded as a classic text containing crucial information about this disorder

Published *Clinical Study of the Unilateral Cerebral Paralyses of Children* in 1891 followed by his last manuscript in the field of neurology, *Infantile Cerebral Paralysis*, in 1897

Questioned the established view that the disorder was due to birth complications (lack of oxygen to the brain during birth) and suggested that the difficult birth was a symptom rather than a cause of the disorder

Argued that the symptoms indicated damage to the brain while the central nervous system was developing in the fetus

Regardless of his work, the birth complication explanation remained until only two decades ago when it was found that the disorder is in most cases due to damage to the brain during fetal development (Galbis-Reig, 2004)

Published *Studies in Hysteria* with Josef Breuer in 1895

Case study of Anna O (real name Bertha Pappenheim)

Anna O suffered a nervous cough, tactile anaesthesia and paralysis but no organic cause could be determined

Talking about traumatic experiences from her life (identifying the root cause of the problem) while under hypnosis appeared to calm her symptoms – this was known as the cathartic method

Formed the basis of the 'talking cure' (term coined by Anna O herself)

Coined the term 'psychoanalysis' in 1896 (Freud Museum)

Psychoanalysis eventually became the fundamental psychotherapeutic method and modern therapy would not have been possible without this initial framework

Began a process of self-analysis in 1897 (Freud Museum)

Abandoned the concept of hysteria and neurosis due to trauma

Began work on the theory of infantile sexuality and the Oedipus complex

Published *The Interpretation of Dreams* in 1900

He felt that this was his favourite book (Freud Museum) and it contained extensive dream analysis

Freud (1900) suggested that 'the interpretation of dreams is the royal road to a knowledge of the unconscious activities of the mind'

Commonly misquoted as 'dreams are the royal road to the unconscious'

Published *The Psychopathology of Everyday Life* in 1901

Foundation for the concept of the 'Freudian slip'

Slip of the tongue that can reveal the unconscious drives of the speaker

Published *Fragment of an Analysis of a Case of Hysteria* in 1905

Case study of 18-year-old Dora (real name Ida Bauer)

Dora suffered depression (suicidal thoughts) and a range of hysterical symptoms, such as a chronic cough and aphonia (loss of voice), with no recognised organic cause

Her father (an intimidating man) brought her to therapy and insisted that she held fantasies about the people in her life

She babysat for a married couple and reported that the female (Frau K) was having an affair with her own father and the male (Herr K) had propositioned her

She explained that her father wanted to exchange her for the married woman – Herr K would permit the affair between his wife and Dora's father to continue provided that he could have Dora in exchange

She was diagnosed as repressing her own desire for Frau K and, to a lesser extent, Herr K and her own father

Dora broke away from therapy after 11 weeks and Freud felt that the therapy ended too early to be successful – he later blamed himself for the lack of success as he felt that he had failed to notice key elements of transference during the therapy

Published *Three Essays of the Theory of Sexuality* in 1905

Outlined the psychosexual stages (oral, anal, phallic, latency, genital) and the Oedipus conflict, but these aspects did not appear in full in the first edition and were instead added at a later date (approximately 1915)

Attended the First International Psychoanalytical Congress in Vienna in 1908

Visited the USA with Jung to give the first lectures on psychoanalysis in America (Clark Lectures) in 1909

Freud was unimpressed with America – he described it as immature and repressed (Pruner, 1992)

In conversation with friends, Freud stated that 'America is the most grandiose experiment the world has seen, but, I am afraid, it is not going to be a success' (Pruner, 1992)

Published *Analysis of a Phobia in a Five-year-old Boy* in 1909

Case study of Little Hans – treated by his father under the guidance of Freud

Freud acknowledged that the material for this case study was gathered by the father and may have been biased (although he noted that few people other than a parent could encourage a child to express his feelings so openly)

- Little Hans had a phobia of horses, possibly relating to an incident in which he witnessed a horse falling in the street
- Fear of horses was equated to fear of his father and these feelings of anxiety were explained in terms of the Oedipus conflict
- Freud was less convinced by the Oedipus explanation of this case, but the parents were close followers of Freudian theory and the reports of the father did seem to fit with the theory

Published *Notes upon a Case of Obsessional Neurosis* in 1909

- Case study of Rat Man (real name possibly Paul Lorenz or Ernst Lanzer)
- Rat Man had heard about a military torture involving a bucket of rats tied to the bottom of a prisoner and then heated up so that the rats burrow into the anus to escape the heat
- Rat Man suffered obsessive fears that his father and fiancée would be a victim of this torture, and he felt compelled to carry out specific behaviours to cope with his thoughts
- Freud suggested that he was repressing homosexual fantasies and displacing aggressive impulses – guilt about his sexual desires and his aggressive feelings towards his father were transferred to fears about the fiancée and father suffering the rat torture
- Freud indicated that he made a complete recovery (although there is some controversy about this conclusion)

Delivered introductory lectures on psychoanalysis at University of Vienna in 1915

Published *From the History of an Infantile Neurosis* in 1918 (written in 1914)

- Case study of Wolf Man (real name Sergei Pankejeff)
- Wolf Man suffered with depression and apathy, and avoided engaging in life (work, family, etc.) to the extent that he was no longer self-sufficient
- He reported a terrifying dream from his childhood involving a dozen white wolves sitting in a tree outside his bedroom window – this dream was interpreted to be the result of the trauma of seeing his parents engaged in a primal scene ('coitus a tergo' or sexual intercourse from behind) at a young age
- Freud explained that Wolf Man was 'cured' after the initial therapy but returned to work on some transference issues after the war – after this second therapeutic process, Freud reported that Wolf Man was normal and behaved accordingly

Published *Beyond the Pleasure Principle* in 1920

- Introduced the concept of the death instinct and suggested that dreams could be forms of wish fulfilment

Published *The Ego and the Id* in 1923

- Outlined the structure of the mind (id, ego, superego)

Published *The Future of an Illusion* in 1927

- Established himself as an atheist and debunked religion on rational and scientific grounds

Exchanged letters with Einstein on the subject of war in 1932

'Why war?'

Moved to England to escape the Nazi invasion of Vienna in 1938

Continued seeing patients and working on his final books *An Outline of Psychoanalysis* and *Moses and Monotheism*

Family

Married Martha Bernays in 1886

Met in 1882 and were engaged within two months, but then spent four years apart before marrying in 1886

Freud famously wrote over 900 passionate letters to his fiancée and many of these provide an insight into his personality, attitudes and beliefs about masculine and feminine roles

'Woe to you, my Princess, when I come, I will kiss you quite red and feed you till you are plump. And if you are forward you shall see who is the stronger, a gentle little girl who doesn't eat enough or a big wild man who has cocaine in his body' (Freud, quoted in Farber, 1978)

After marriage, she was devoted to her husband and he used his feelings about her role in their relationship to form the basis of many of his theories about female nature

'I know, after all, how sweet you are, how you can turn a house into a paradise, how you will share in my interests, how gay yet painstaking you will be. I will let you rule the house as much as you wish, and you will reward me with your sweet love and by rising above all those weaknesses for which women are so often despised' (Freud, quoted in Farber, 1978)

Raised three sons and three daughters

Anna was born in 1895 and was to eventually work alongside her father and move the therapy forward after his death

Death

Freud was diagnosed with cancer of the jaw in 1923

He suffered more than 30 operations and was eventually fitted with a prosthetic jaw in place of his amputated jaw and palette – this prosthesis allowed him to eat, drink, talk and smoke his beloved cigars

Given his theories about oral fixation, it is interesting to note that Freud remained constantly addicted to the cigars that were responsible for such pain and eventual death

'A cigar is sometimes just a cigar' (attributed to Freud by Wheelis, 1950)

Freud was exiled from Vienna to London following the Nazi invasion in 1938

His Jewish status made his family potential targets

Nazis burnt his books and he was vilified by the regime

> 'What progress we are making. In the Middle Ages they would have burned me. Now they are content with burning my books' (Freud, quoted in Jones, 1957)

He sought to continue working with patients and writing his final book in the relative safety of London in wartime Britain

Freud eventually died as a result of a lethal injection of morphine at the age of 83

Freud had made an earlier agreement with his doctor and he reminded him of this pledge when he felt that he could no longer work

> 'My dear Schur, you certainly remember our first talk. You promised me then not to forsake me when my time comes. Now it is nothing but torture and makes no sense any more' (Freud, quoted in Schur, 1972)

Following agreement from his daughter Anna, fatal doses of morphine were administered and he died peacefully in his own study at home on 23rd September 1939

FOUNDING FATHER OF THE PSYCHODYNAMIC APPROACH: FREUD

Consider the life story of Sigmund Freud and try to answer the following questions.

1. What was the social and cultural context of the world in which Freud developed his theories?
2. How might Freud's life experiences have impacted on his perception of human nature?
3. How can this life story contribute to your understanding of the psychodynamic approach?

Significant learnings

'He that has eyes to see and ears to hear may convince himself that no mortal can keep a secret. If his lips are silent, he chatters with his fingertips; betrayal oozes out of him at every pore' (Freud, 1905)

'Psychoanalysis is in essence a cure through love' (letter to Jung, 1906, quoted in Bettelheim, 1984)

'It is always possible to bind together a considerable number of people in love, so long as there are other people left over to receive manifestations of their aggressiveness' (Freud, 1929)

'Homosexuality is assuredly no advantage, but it is nothing to be ashamed of, no vice, no degradation, it cannot be classified as an illness' (letter in response to an American mother's plea to cure her son's homosexuality, 1935, quoted in Grotjahn, 1951)

'The great question that has never been answered, and which I have not yet been able to answer, despite my thirty years of research into the feminine soul, is "What does a woman want?"' (Freud, quoted in Jones, 1957)

Biography of Melanie Klein

> *'One of the many interesting and surprising experiences of the beginner in child analysis is to find in even very young children a capacity for insight which is often far greater than that of adults'*
> —Klein (1955)

Who was Melanie Klein?

Melanie Klein revolutionised the psychodynamic approach and psychoanalytic therapy

- Established child psychoanalysis using the play technique
- Introduced the concept of object relations
- Introduced the psychological positions (paranoid-schizoid, depressive)

Early years

Born 30th March 1882 in Vienna

- Middle-class Jewish family (but religion did not play a role in family life)
- Father (Moriz Reizes) was a doctor – his family expected him to be a rabbi but he studied in secret to become a doctor and continued his education throughout his life

Happy childhood, despite the tragedy of loss

- Although Klein felt as though her father favoured her older sister, she always admired his intellectual drive – he died when she was only 18 (Segal, 1980)
- Klein had a closer relationship with her mother during childhood
  - Although this relationship became more difficult when her mother criticised her through a series of letters in her adult life
- Close to her siblings and devastated by the loss of her sister
  - Sister Sidonie died at the age of eight – she had taught Klein to read and write in the hope that she could pass on some knowledge before her own death (Segal, 1980)

Education

Passed her entrance exam to study at the gymnasium (responsible for preparing students for university) in 1898

- Initially wanted to become a medical doctor but instead studied humanities (art and history) at Vienna University after getting engaged

Family

Married Arthur Klein in 1903 at the age of 21

- Arthur was her second cousin and they met when she was just 17
- Married one day after her 21st birthday
  - Married while still in mourning for the loss of her beloved older brother – this marked the start of the depression that plagued her life
- Marriage ended her dreams of becoming a medical doctor, but she still retained her ambition and drive to help people throughout her life

Raised two sons and one daughter

- Final child Erich was born in 1914 – Klein did not wish to fall pregnant again and her depression increased during her pregnancy (Melanie Klein Trust, 1913)

Travelled frequently across Europe

- Moved to Budapest with her family in 1910
- Separated from her husband and moved with her children to stay with her in-laws in Rosenburg in 1919
- Moved to Berlin in 1920 and her marriage finally ended in 1924
- Moved to London in 1926 after the death of friend and analyst Karl Abraham
- Lived in London for the remainder of her life

Death of her son Hans in 1934

- Accidental death while hiking in Budapest
- Klein was too devastated to attend the funeral (Melanie Klein Trust, 2013)

Estranged from her daughter Melitta in 1942

- Melitta criticised her mother during the Controversial Discussions in 1932, and they remained estranged throughout her life (Donaldson, 2002)

Career

Initially introduced to psychoanalysis through her own depression

- Suffered depression and nervous anxiety reportedly associated with her difficult relationship with her mother followed by her death (Donaldson, 2002)
- As a result of her depression, she began psychoanalysis with Sandor Ferenczi in Budapest in 1914
- Ferenczi encouraged her to analyse her own children (Donaldson, 2002)
  - Klein used this experience to begin establishing a method of child psychoanalysis
  - She developed a 'play technique' for the psychoanalysis of children
  - Play is regarded as symbolic activity indicating underlying drives

Attended the Fifth Psychoanalytic Congress at the Hungarian Academy of Sciences in Budapest in 1918

- Attended a presentation by Freud – a self-confessed pivotal moment in her career (Melanie Klein Trust, 1913)

'I remember vividly how impressed I was and how the wish to devote myself to psychoanalysis was strengthened by this impression' (Klein, quoted in Grosskurth, 1986)

Became a member of the Hungarian Psychoanalytical Society in Budapest in 1919

Presented her first case study of her youngest son, Erich

Became a member of the Berlin Psychoanalytic Society in Berlin in 1923

Karl Abraham supervised her psychoanalysis work and she later received psychoanalysis from him

She incorporated his ideas about the death instinct into her work (Donaldson, 2002)

Published *The Development of a Child* in 1923

Case study of a child named Rita (two and a half years)

Therapy was successful and provided evidence for instinctual drives

Became a member of the British Psychoanalytical Society in London in 1926

Presented papers to the society resulting in critical acclaim

Ernest Jones organised a symposium at the British Psychoanalytical Society focusing on child analysis in 1927

Symposium was a response to a presentation delivered by Anna Freud to the Berlin Psychoanalytic Society earlier in 1927

Freud criticised the Klein approach (Melanie Klein Trust, 1913)

Sigmund Freud regarded the British symposium as a hostile reaction to his theories presented through his daughter (Melanie Klein Trust, 1913)

Sparked the initial divide between Kleinian and Freudian analysis

Published *The Psychoanalysis of Children* in 1932

Argued that children have a primary object relation to the mother

Published *A Contribution to the Psychogenesis of Manic Depressive States* in 1935

Introduced the psychological positions (particularly the depressive position)

Extraordinary Meetings of the British Psychoanalytical Society and Controversial Discussions were held in 1942

These meetings were designed to address the fighting between the members of the society

Anna Freud and Edward Glover argued that Klein was not a qualified psychoanalyst and Melitta Schmideberg (Klein's daughter) criticised the Kleinian method (Melanie Klein Trust, 1913)

Since both Melanie Klein and Anna Freud introduced the concepts of child analysis, discussions are often regarded as a direct battle between these two women

Klein did not, however, participate in the discussions in person until 1944 (Melanie Klein Trust, 1913)

'Kleinian' psychoanalysis established as an alternative to 'Freudian' psychoanalysis

Eventually, the society agreed to form two distinct subschools of psychoanalysis – Kleinian and Freudian

Also a 'middle' group containing those who remained independent and used a mix of techniques

Established the Melanie Klein Trust in 1955

She added her own money to start the running of the trust and invited her closest colleagues to be trustees (Melanie Klein Trust, 1913)

Published *Envy and Gratitude* in 1957

Most controversial of her works introducing the concept of primary envy

Published *Narrative of a Child Analysis* in 1961 (posthumous)

Case study of the analysis of a young boy

Synthesised and evidenced theories relating to object relations and the psychological positions

Death

Diagnosed with anaemia and visited Switzerland to recuperate in the summer of 1960

Unfortunately, her conditioned worsened and she returned to London to be admitted to hospital

Diagnosed with colon cancer after her return from Switzerland

Initially treated with an apparently successful operation but her health became poor again after a fall from her bed

Died on 22nd September 1960

FOUNDING MOTHER OF THE PSYCHODYNAMIC APPROACH: KLEIN

Consider the life story of Melanie Klein and try to answer the following questions.

1. What was the social and cultural context of the world in which Klein developed her theories?
2. How might Klein's life experiences have impacted on her perception of human nature?
3. How can this life story contribute to your understanding of the psychodynamic approach?

Significant learnings

'Feelings of love and gratitude arise directly and spontaneously in the baby in response to the love and care of his mother' (Klein, 1937)

'It was always part of my technique not to use educative or moral influence, but to keep to the psychoanalytic procedure only, which, to put it in a nutshell, consists in understanding the patient's mind and in conveying to him what goes on in it' (Klein, 1955)

'It is an essential part of the interpretive work that it should keep in step with fluctuations between love and hatred, between happiness and satisfaction on the one hand and persecutory anxiety and depression on the other' (Klein, 1955)

SUMMARY

Freud: founded psychodynamic approach and introduced psychoanalysis; *early years*, born 1856 in the Czech Republic, idolised by mother, moved to Vienna after father lost business, educationally advanced from a young age; *education*, originally studied law but moved to medicine after attending lecture On Nature, graduated as a medical doctor, scientific paper published at 20 years, wanted to work in research but forced to practise due to financial constraints; *family*, married after long engagement, wife was devoted and provided a template for female nature, raised six children; *career*, developed new staining technique to view neurons, first neurologist to trace course of spinocerebellar white matter tracts, almost first to describe neuron theory of central nervous system, studied hypnosis as treatment for hysteria, discovered analgesic properties of cocaine (controversy caused by personal use and prescribing as treatment for addiction), studied aphasia to propose unified theory of the mind, investigated cerebral paralysis to propose new explanation (recently found accurate), conducted case studies to establish a novel theory of personality and new approach to treatment of mental disorders (Anna O, Dora, Little Hans, Rat Man, Wolf Man), developed the topographical theory (conscious, preconscious, subconscious) and structural theory (id, ego, superego), outlined psychosexual stages and Oedipus conflict, coined the term 'psychoanalysis' and introduced concepts of Freudian slips and dream analysis; *death*, diagnosed with cancer of the jaw leading to eventual amputation, died following a lethal injection of morphine in 1939

Klein: revolutionised psychodynamic approach and introduced psychoanalysis for children; *early years*, born 1882 in Vienna, happy childhood marred by loss; *education*, passed exam to study at gymnasium, initially wanted to be a medical doctor but instead studied humanities at Vienna University after getting engaged; *family*, married second cousin at a young age, initially wanted to train as a doctor but focused on husband's career, raised two sons (one died at a young age) and one daughter (estranged in later life), onset of depression after death of brother and third unwanted pregnancy; *career*, depression led her to psychoanalysis, analysed her own children to develop a play technique for child psychoanalysis, accepted into various psychoanalysis societies and moved around Europe to eventually settle in London,

introduced the concept of object relations and psychological positions through case studies (Rita, young boy), established the Kleinian version of psychoanalysis as an alternative to Freudian after the Controversial Discussions in the British Society, eventually led to three forms of psychoanalysis (Freudian, Kleinian and Independent); *death*, died due to colon cancer in 1960

PSYCHODYNAMIC THEORIES OF HUMAN NATURE AND PERSONALITY

LEARNING OUTCOMES

After reading this section, you will be able to:

- acknowledge the role of unconscious drives in controlling conscious thoughts and behaviour (topographical theory)
- describe the structure of human personality and discuss the interactive roles of the id, ego and superego (structural theory)
- describe and evaluate the ego defence mechanisms
- discuss the psychosexual and psychosocial stages of personality development
- describe the patterns of attachment that recur in childhood and adult relationships
- explain the impact of object relations on psychological development

Topographical theory

Human behaviour is driven by the unconscious (Freud, 1900)

Conscious forms only a small part of the human psyche

Conscious awareness contains those things of which we are currently aware

Includes those things that we are thinking about at this moment

If we consider these psychodynamic theories from a slightly different angle, we could suggest that this is the equivalent of the short-term memory presented in the multistore memory model (Atkinson & Shiffrin, 1968); short-term memory is regarded as the mental desktop of the brain

Preconscious forms a slightly larger part of the human psyche

Preconscious is not usually within our awareness, but can be accessed if prompted

Includes all knowledge and information stored in the mind

If we consider these psychodynamic theories from a slightly different angle, we could suggest that this is the equivalent of the long-term memory presented

in the multistore memory model (Atkinson & Shiffrin, 1968); long-term memory is regarded as the mental storage tank of the brain

Unconscious forms the largest part of the human psyche

Unconscious is not within our awareness, but can be explored indirectly through psychoanalysis

Includes all drives, conflicts, desires, etc.

Two major instinctual drives

Life drive (Eros or the libido) to survive and thrive, sparks hunger, thirst, desire, etc.

Death drive (Thanatos or aggression) to self-destruct and return to the inanimate state, sparks aggression, risk taking, inertia, etc.

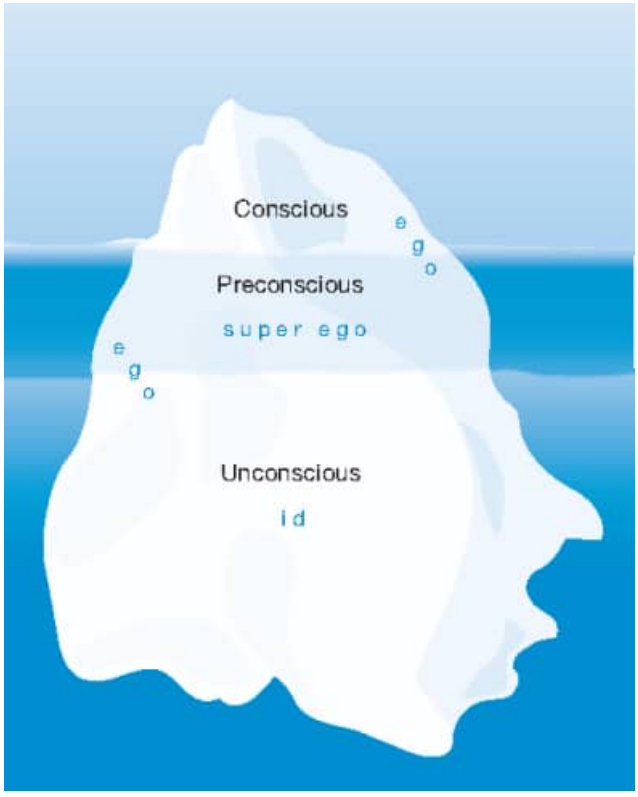


FIGURE 3.1
Topographical and structural theories: iceberg of the mind. Adapted from Freud (1900,1923).

Structural theory

Personality is composed of three interacting structures (Freud, 1923)

Id

Basic drive for gratification

Located in the unconscious

- Governed by instinctual drives
- Pleasure principle
- Present from birth
  - Young babies are entirely focused on their own satisfaction without awareness of the wider world

Ego

- Understanding of social constraints
  - Located across the conscious, preconscious and unconscious
  - Governed by an understanding of the realities of the social world
  - Reality principle
- Develops during the first three years of life
  - Child becomes aware of the needs and demands of others and responds to the urges of the id with consideration for the real world

Superego

- Moral and ethical restraints
  - Located in the preconscious
  - Governed by social (particularly parental) constraints
  - Similar to conscience
- Develops during the fifth or sixth year
  - Child accepts the moral values of caregivers to establish own set of rules for governing behaviour

Ego must be stronger than id and superego

- Ego needs to provide satisfaction for the id while avoiding upsetting the superego and remaining within the constraints of the real world
  - 'The poor ego has a still harder time of it; it has to serve three harsh masters, and it has to do its best to reconcile the claims and demands of all three . . . The three tyrants are the external world, the superego, and the id' (Freud, 1923)
  - 'The ego is not master in its own house' (Freud, 1917)

STRUCTURAL THEORY: SIMPSONS OF THE MIND

Imagine *The Simpsons* television cartoon . . .

- Homer is like the Id – childlike, demanding, spoilt, selfish, etc.
- Flanders is like the superego – self-righteous, ethical, judgemental, guilty, etc.
- Marge is like the ego – constantly struggling to appease Homer while impressing Flanders

Weak ego

Impulsiveness and selfishness would occur if the id became too strong

Judgemental attitudes and guilt would occur if the superego became too strong

Ego defence mechanisms

Intrapsychic conflict leads to anxiety (Freud, 1936)

Conflict between id and superego

Forbidden impulses are experienced by the id

Condemnation from the superego (and possibly society) would occur if the impulses were expressed openly

Ego will attempt to satisfy the urges of the id while avoiding condemnation from the superego by altering the presentation of the impulse

Ego defence mechanisms are normal and healthy methods to mediate this conflict by denying or distorting reality to protect the fragile balance of the psyche

Ego defence mechanisms operate in the unconscious so we are not aware of their impact on us unless it is interpreted during psychoanalysis

Freud (1936) described various defence mechanisms

Repression

Pushing traumatic memories into the unconscious

For example, forgetting an incident of sexual abuse in childhood

Regression

Reverting to a prior stage of development (usually in childhood)

For example, throwing a temper tantrum when your computer loses an important document

Denial

Claiming that an anxiety-provoking stimulus does not exist

For example, refusing to accept that a loved one has died

Projection

Ascribing own desires and emotions to other people

For example, accusing your partner of cheating on you instead of accepting own desire to have an affair

Displacement

'Kick the cat' syndrome

Redirecting emotions from the source to a less threatening target

For example, kicking the cat when you are angry with your computer

Sublimation

Expressing unacceptable impulses in an acceptable manner

For example, establishing a career in boxing to express aggressive urges

Reaction formation

Adopting a belief that is completely opposite to true feelings

For example, insisting that you would be very happy to get a third-class degree to hide the real desire to get a first-class degree

Compensation

Developing specific traits to make up for deficiencies in other areas

For example, buying a big car to make up for a small physique

IN MY DEFENCE . . .

For each of the following cases, explain how the ego defence mechanism might yield a positive and/or negative outcome.

A man in his late twenties has a criminal record for assault and burglary committed in his teenage years. He has not committed any crimes in recent years after starting a family with his childhood sweetheart and he has been employed as a labourer on a building site. However, he has just been told that he will be made redundant at the end of the week. He is angry with his boss and he is embarrassed about telling his partner the news.

A woman in her thirties has just lost her father. She has spent the last three years caring for him during the late stages of terminal cancer. Now that he has passed away, she is left with the task of distributing his property according to his last will and testament. She is devastated by the loss and overwhelmed by the responsibility of fulfilling his final wishes.

A teenage boy has had three dreams about being kissed by another boy in his class. He feels ashamed by these dreams when he wakes up and he thinks that his father would be angry if he knew about them.

A young girl was removed from her parents after a neighbour told the police that he had seen her father touching her in an inappropriate way. She is taken to live with foster parents, but the memory of the abuse prevents her from being able to form a close relationship with her new family.

An elderly man has been experiencing severe pains in his stomach and he has lost a significant amount of weight. His older brother died of colon cancer when he was 60 and he seemed to have similar symptoms before his death. However, the elderly man feels uncomfortable discussing his body with the doctor and does not want to worry his family.

Rationalisation

Formulating a logical and sensible, but false, reason to explain behaviour

For example, claiming that you failed your exam because you had a cold rather than accepting that it was because you did not revise

Intellectualisation

Focusing excessively on intellectual details rather than dealing with the associated emotions

For example, obsessing about the details of the funeral rather than focusing on feelings of grief after the death of a loved one

Psychosexual and psychosocial stages of development

Freud (1905) described the psychosexual stages from birth to adulthood

Oral stage – 0–1year

Focus of gratification is the mouth and oral exploration of the world

Child demonstrates a tendency to insert things into the mouth and a desire to suck on objects (typified by sucking at the breast of the mother)

Weaning is a key event during this stage

Child learns the importance of delayed gratification (does not always instantly get to suck on objects), independence of the self (not always able to control environment) and behavioural control (some behaviours lead to gratification, such as crying)

Too much or too little gratification at this stage of development can lead to an oral fixation

Orally aggressive individuals may chew gum and other objects

Orally passive individuals may engage in smoking, overeating, etc.

Both individuals will be passive, immature and gullible

Anal stage – 1–3 years

Focus of gratification is the bottom and elimination from the bowel and bladder

Child demonstrates a tendency to touch the bottom, eliminate bodily waste at inappropriate times, and play with faeces

Toilet training is a key event during this stage

Child learns how to resolve the basic conflict between the id and ego – id will want instant gratification in response to the urge to eliminate bodily waste but the ego has begun to encourage delayed gratification and parental demands will support this desire

- Too much or too little gratification at this stage of development can lead to an anal fixation
  - Child who seeks to retain bodily waste to spite the ego (and parents) and enjoy the sensation of pressure on the bowel and bladder will become anal-retentive (excessively organised and neat)
  - Child who seeks to expel bodily waste at inappropriate times to spite the ego (and parents) and enjoy the sensation of expulsion will become anal-expulsive (excessively disorganised and reckless)

Phallic stage – 3–6 years

- Focus of gratification is the genitals and sexual stimulation
- Child demonstrates a gradual understanding of the difference between males and females, a desire to explore their own and other bodies, and a tendency to play with their genitals
- Oedipus and Electra complex are key events during this stage
- Oedipus complex
  - Applicable to boys
  - As the boy begins to develop a sexual identity (I am a boy) and experience some sexual feelings (touching feels good), he will direct these feelings towards the main person in his life
  - Sexual desire is focused on the mother (I want to marry mummy)
  - Jealousy is directed towards the father who is viewed as more advanced (daddy is bigger than me) and already in possession of the mother (mummy loves daddy)
  - Id wants to kill father, but ego is aware that father would probably win a battle and could castrate the boy
  - Boy bonds with father to minimise this risk and possess the mother vicariously – male sexual role develops as boy identifies with father
- Electra complex
  - Applicable to girls, although Freud did argue against the use of this term
  - As the girl begins to develop a sexual identity (I am a girl) and experience some sexual feelings (touching feels good), she will direct these feelings towards the main person in her life
  - Sexual desire is initially focused on the mother but realisation of the absence of a penis means that this desire (can never possess mother) is redirected towards the father
  - Girl experiences both envy and desire for her father (penis envy) and resentment towards her mother (mother is assumed to have castrated her)
  - Girl bonds with mother to possess the father vicariously – female sexual role develops as girl identifies with mother

Too much or too little gratification at this stage of development can lead to a phallic fixation

Unresolved Oedipus or Electra complex will involve the development of inappropriate sexual roles due to a failure to identify with the appropriate person

Phallic character traits will involve recklessness, pride, narcissism and vanity

Latency stage – 6–12 years

Gratification is not sought during this stage because sexual feelings lie dormant

Child will use this time to pursue other, non-sexual interests prior to puberty

Genital stage – 12+ years

Focus of gratification is sexual interest in members of the opposite sex

Child will begin to develop sexual relationships with members of the opposite sex

Inability to focus on this stage of development due to fixation in one of the earlier stages can lead to frigidity, impotence and failure to develop satisfactory adult relationships

Erikson (1963) described psychosocial stages from birth to death

Infancy: trust versus mistrust – 0–1 year

Child begins to develop the ability to trust others based on relationships with caregivers, particularly during feeding

Successful completion of this stage leads to confidence and security

Unsuccessful completion of this stage leads to anxiety and insecurity

Early childhood: autonomy versus doubt – 1–3 years

Child begins to develop control over physical body through toilet training

Successful completion of this stage leads to independence and autonomy

Unsuccessful completion of this stage leads to shame and doubt

Preschool: initiative versus guilt – 3–6 years

Child begins to develop control over the surrounding environment through exploration

Successful completion of this stage leads to a sense of purpose

Unsuccessful completion of this stage leads to guilt

School: industry versus inferiority – 6–12 years

Child begins to develop coping skills for new social situations through school

Successful completion of this stage leads to competence

Unsuccessful completion of this stage leads to inferiority

Adolescence: identity versus role confusion – 12–18 years

Teen begins to develop a sense of self and personal identity through relationships beyond the immediate family

Successful completion of this stage leads to a strong sense of self

Unsuccessful completion of this stage leads to role confusion and a weak sense of self

Young adult: intimacy versus isolation – 18–35 years

Adult begins to develop intimate relationships with others

Successful completion of this stage leads to strong relationships

Unsuccessful completion of this stage leads to loneliness and isolation

Middle age: generativity versus stagnation – 35–60 years

Adult begins to develop a need to create things that will last after their death

Successful completion of this stage leads to accomplishment and usefulness

Unsuccessful completion of this stage leads to shallow involvement in life

Later life: integrity versus despair – 60+ years

Adult begins to develop self-reflection and the ability to reflect on life

Successful completion of this stage leads to wisdom and self-fulfilment

Unsuccessful completion of this stage leads to regret, despair and bitterness

Patterns of attachment

Attachment is a 'lasting psychological connectedness between human beings' (Bowlby, 1969).

Attachment is more than simple provision

Responses of rhesus monkey infants differ between carers who provide food versus carers who provide comfort (Harlow & Zimmerman, 1958)

Infants reared in isolation suffered physical and behavioural problems – unable to adjust to normal social life as adults

Infants reared with dummy mothers preferred to spend time with the dummy covered in soft towelling rather than the wire dummy holding the bottle – when frightened, infants would cling to the 'soft mother' rather than the 'providing mother'

Finding suggests that a connection with a warm figure to which the infant can cling is essential in the critical attachment period of the monkey

In humans, the absence of a warm attachment figure in the first two years leads to psychological problems in later life

In a monograph for the World Health Organization, Bowlby stated that 'the infant and young child should experience a warm, intimate, and continuous relationship with his mother in which both find satisfaction and enjoyment' (Bowlby, 1951)

- Widespread changes in institutional childcare have resulted from these theories – for example, introduction of parental visits for hospitalised children and foster parent placements for orphaned infants

Early relationships between primary caregivers (parent) and care receivers (child) establish attachments

- Attachment is an evolutionarily advantageous strategy for the vulnerable child to stay safely within close proximity of a figure who offers care and protection
  - Child develops a range of behaviours designed to maintain close proximity to the caregiver, such as crying, clinging, etc.
- In the first year of life the child forms an attachment with those who provide care (Bowlby, 1969)
  - 0–3 months = indiscriminate responses to any caregiver
  - 3–7 months = initial preference for most common caregivers
  - 7–9 months = special preference for main caregiver (attachment) with clear distress when the mother is absent (separation anxiety) and fear of unknown replacement caregivers (stranger fear)
  - 9–12 months = as the child matures, s/he will use the caregiver as a 'secure base' from which to explore the world; child negotiates with the caregiver to stretch the boundaries of proximity while retaining feelings of security
- Patterns of attachment emerge from these early experiences with caregivers
  - Ainsworth (1967) identified four distinct individual differences in infant attachment patterns through the 'strange situation' experimental paradigm
    - Infant and parent play in an unfamiliar space
    - Parent leaves the infant alone for a short while and infant responses are observed
    - Parent returns to the infant and infant responses are observed
  - Secure attachment
    - Child is able to cope with minor separation to explore the immediate environment comfortably
    - Child protests the departure of the caregiver and is comforted by the return of the caregiver
    - Child has a clear preference for the primary caregiver, but will accept comfort from strangers to some extent
    - Caregiver has formed a secure parental bond with the child
  - Avoidant attachment
    - Child is unaffected by separation
    - Child does not protest the departure of the caregiver and shows no response to the return of the caregiver

- Child treats strangers in the same way as the primary caregiver
- Caregiver encourages independence and offers minimal response to a distressed child

- Anxious attachment
  - Child is unable to cope with any type of separation and refuses to explore the immediate environment
  - Child is extremely distressed by the departure of the caregiver and refuses to be comforted by the return of the caregiver
  - Child has a clear preference for the primary caregiver and will not accept any comfort from strangers
  - Caregiver is inconsistent in responses to the child, and may display neglectful or abusive behaviours on occasions

- Research suggests that only 65% of children have a secure attachment pattern and the remaining 35% display avoidant or anxious attachment patterns (Prior & Glaser, 2006)

Patterns of attachment formed in childhood are repeated in adulthood relationships (Hazan & Shaver, 1987)

- Romantic relationships (in particular) tend to mimic the patterns established in childhood
  - Secure attachment
    - Adult is secure in the relationship
    - Adult feels confident about depending on others and being depended upon, without losing a sense of independence
  - Avoidant attachment
    - Adult will avoid close relationships
    - Adult is fiercely independent and does not like depending on a partner and being depended upon
    - Adult may be cold, distant or act out to push people away
  - Anxious attachment
    - Adult will feel very anxious about the relationship
    - Adult will lack independence and rely completely on a partner
    - Adult may feel angry or distressed in response to any suggestion that a partner is not similarly dependent
- Hazan and Shaver (1987) found similar distributions of attachment patterns in adults as those identified in infants
  - 60% reported a secure attachment in their relationships
    - 'I find it relatively easy to get close to others and am comfortable depending on them and having them depend on me. I don't worry about being abandoned or about someone getting too close to me'

20% reported an avoidant attachment in their relationships

'I am somewhat uncomfortable being close to others; I find it difficult to trust them completely, difficult to allow myself to depend on them. I am nervous when anyone gets too close, and often, others want me to be more intimate than I feel comfortable being'

20% reported an anxious attachment in their relationships

'I find that others are reluctant to get as close as I would like. I often worry that my partner doesn't really love me or won't want to stay with me. I want to get very close to my partner, and this sometimes scares people away'

Some research suggests that patterns of attachment recur through life, but the evidence to date is far from conclusive (Fraley, 2010)

Minor correlations between attachment styles to parents, partners and friends

One unpublished study identifies very little link between attachment pattern at the age of one and attachment pattern in later romantic relationships

Fraley (2010) noted that the 'existence of long-term stability of individual differences should be considered an empirical question rather than an *assumption* of the theory'

Object relations

Klein (1923) argued that adult relationships are based on interactions with significant others in infancy

Childhood relationships set a pattern of relating that is repeated in our connections with others as an adult

Self relates to objects – object relations

Self and objects

Self is the subject (me, myself and I)

Self exists at the centre of my universe and relates to objects around it

Objects are people or events that have a psychological impact on the self

Objects can be anything that is not categorised as the self

Objects can be whole or part

Whole objects are entire people

For example, the mother

Part-objects are only part of a person

For example, the breast of the mother

Infants tend to understand the world in terms of part-objects as they are focused on only those small parts of the world that satisfy needs

Whole objects develop when the child begins to understand the world from a wider perspective and appreciate that people are complete individuals

Objects can be external or internal

External objects are real things in the outside world

For example, my actual mother

Internal objects are internalised representations of things held in the subconscious

For example, my internal image of my mother

Internal objects are created through experience with external objects

Our subconscious creates internal objects following contact with external objects

For example, an internal representation of 'mother' is created in response to contact with the external actual mother in infancy

Once created, internal objects provide a template for interacting with future external objects

Future external objects can be associated with current internal objects so that our responses are predetermined

For example, an internal representation of 'mother' can be applied to people who exhibit mothering traits in the future

Objects can be 'good' or 'bad'

Good objects are those that fulfil the needs of the self

For example, the good breast is one that will feed the hungry child

Bad objects are those that fail to fulfil the needs of the self

For example, the bad breast is one that is absent or lacking milk when the child is hungry

Polarised views of 'good' and 'bad' minimise ambivalence in infancy

It is confusing for an infant to understand that every person has both positive and negative personality traits and that every person is capable of both kind and cruel actions

It is simpler for the infant to split each object into good and bad versions

For example, 'good' mother is kind and affectionate whereas 'bad' mother is unkind and withdrawn

Mother–infant dyad

Klein (1932) noted that the mother is a critical figure for the infant

Mother is the first external object for whom an internal object is created

Relationship with the external and internal mother-object will have a life-long impact on the individual

More specifically, the breast of the mother is the very first significant part-object in the world of the infant

Good breast will give milk and bad breast will deny milk when the child is hungry

This early relationship with the breast forms the basis of the psychological positions during childhood

Psychological positions

Klein (1935, 1937, 1957, 1961) argued that infants pass through two psychological positions during childhood

Paranoid-schizoid position – birth to six months

Depressive position – six months to one year

Paranoid-schizoid position

Characterised by anxiety, part-objects, splitting, projection and introjection

Vulnerable infant suffers a lot of anxiety

Infant experiences frustration whenever he does not have his needs immediately fulfilled

For example, hunger leads to anguish, frustration and fear of death (death instinct)

Anxiety leads to intense hatred towards the cause

For example, hatred of the breast because it is not satisfying his needs immediately

Infant deals with his anxiety by using defences

Negative (anxiety) and positive (satisfaction) feelings are projected into the world onto part-objects

Part-objects are split into either 'good' or 'bad' and this distinction is based on whether they satisfy or deny

Good and bad part-objects are then introjected and this internal part-object will impact on future thoughts, feelings and behaviours

For example, feelings of frustration and hatred towards the breast that fails to provide milk are projected onto the breast so that it is regarded as 'bad' and this 'bad breast' may impact on feelings about other providers in later life

Depressive position

Characterised by integration, reparation and loss

Infant begins to integrate the different part-objects into whole-objects

For example, the breast that feeds, hand that strokes and voice that soothes are all part of the same mother-object

Infant begins to integrate good and bad into holistic objects

For example, the 'good' mother who gives food and the 'bad' mother who denies food are the same person

This new understanding of the object (whole, good and bad) leads to regret about the previous hatred towards bad part-objects and fear that this hatred may have harmed the object

For example, the child who hated the bad breast now understands that it forms part of the whole mother, and this leads to regret for his previous hatred and fear that his aggression may have driven her away

Infant will experience guilt and a desire to repair the damage done by prior hatred and aggression

For example, this reparation deepens the bond of love between mother and child

Infant will also experience a sense of grief due to the loss of the completely 'good' object

For example, the child grieves the loss of the totally good mother now that she has been replaced by a mother who is a mix of good and bad elements

Positions in adulthood

Paranoid-schizoid and depressive positions can be retained to some extent in adulthood if these stages are not completed successfully in childhood

Those regressing to the paranoid-schizoid position may experience hatred leading to aggression

Those regressing to the depressive position may experience guilt and grief leading to depression

Transitional objects (Winnicott, 1953)

Transitional objects are the first 'not-me' possessions

Physical items that can be held, hugged, etc.

For example, blanket, stuffed toy, etc.

Transitional objects offer comfort to an infant because they are linked to the feelings attached to the mother-object

Infant experiences feelings of loss as the realisation of being an independent entity begins to highlight the mother as a separate entity – infant experiences the 'loss' of the mother during this transitional stage

At this point, feelings associated with the mother (warmth, security, etc.) are transferred over to an available object to provide a substitute in her absence

Transitional phenomena serve the same purpose as transitional objects, but these are thoughts or behaviours instead of objects

Transitional phenomena are often repetitive actions (such as rocking) or recurrent thought patterns (such as fantasies)

In adulthood, transitional objects can be used to provide comfort during times of difficulty

Many adults use objects that remind them of home when travelling as these transitional objects provide a feeling of security at insecure locations

Hotel chain Travelodge revealed that 35% of adults in Britain admit to sleeping with a teddy bear to alleviate stress and 25% of male respondents admit to taking the bear with them when travelling on business (Ahmed, 2010)

SUMMARY

Topographical theory: conscious, preconscious, unconscious (life/Eros and death/Thanatos instincts)

Structural theory: id, ego, superego; problems of a weak ego

Ego defence mechanisms: intrapsychic conflict between id and superego leads to anxiety, ego develops defence mechanisms to cope with anxiety; repression, regression, denial, projection, displacement, sublimation, reaction formation, compensation, rationalisation, intellectualisation

Psychosexual stages: oral, anal, phallic (Oedipus and Electra conflicts), latent, genital

Psychosocial stages: infancy, early childhood, preschool, school, adolescence, young adult, middle age, later life

Patterns of attachment: lasting psychological connectedness, more than provision, Harlow's monkeys; attachment forms between 3 and 9 months, three patterns shown in 'strange situation' (secure, avoidant, anxious); patterns repeated in adult relationships (though evidence is inconclusive), 60% secure, 20% avoidant, 20% anxious

Object relations: objects are people or events that relate to the self, whole or part-objects, internal objects created through experience with external objects, good or bad objects; mother is the first internal object; psychological positions include paranoid-schizoid and depressive; transitional objects and phenomena provide a security object in the absence of the mother

THERAPEUTIC RELATIONSHIP IN PSYCHOANALYTIC THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- describe and evaluate the 'blank screen' anonymity of the therapist
- discuss the theoretical assumptions underlying transference
- discuss the theoretical assumptions underlying counter-transference
- explain how transference and counter-transference are used in psychoanalytic therapy

Blank screen

Therapist remains an anonymous blank screen

Therapist does not present the inner self

No self-disclosures, no personal information, neutral responses

Client can project own needs, desires and values onto the therapist

Client is unable to form feelings about the therapist based on the actions or revelations of the therapist

Any apparent feelings towards the therapist must, therefore, give an indication of the inner world of the client

Feelings towards the therapist may be a projection of feelings towards other significant people in the life of the client

For example, reacting with anger to a therapist who has done nothing to provoke this response might indicate that the client has projected these feelings from someone in their own past to the therapist

This is known as transference

Transference

Twin concepts of transference and counter-transference

Freud noted that female patients would often indicate amorous intentions towards the therapist during psychoanalysis

For example, Freud was saved from one potentially embarrassing situation by the sudden intrusion of a housemaid just as a female patient had announced her affections by throwing her arms about his neck. Freud resisted the urge to attribute this behaviour to his own sexual desirability and focused instead on the nature and cause of the phenomenon

Observation revealed that the behaviour would often be exhibited at a point when the patient had begun to explore a previously repressed childhood trauma, typically relating to the Oedipus or Electra complex (Freud, 1905)

Freud argued that the patient would fail to recall the actual trauma, but instead would reproduce the feelings associated with those involved in the trauma and refer these feelings to the therapist through a 'mistaken mental connexion' (Racker, 1982)

These feelings were displaced from important figures in childhood, such as siblings and parents, to be repeated in reactance to the therapist as a strategy for resisting further disclosure of childhood events

In short, Freud proposed that feelings of love, hate, desire and fear were *transferred* from the original object to the therapist (Racker, 1982)

Definition of transference

Transference as a psychological concept refers to the redirection of feelings about one individual to another individual

> 'Transfer of feelings originally experienced in an early relationship to other important people in a person's present environment' (Luborsky et al., 2008)

Transference relationship can be classed as the cornerstone of psychoanalysis

> In a therapeutic setting, transference will occur when the client redirects feelings about a significant person in their life towards the therapist
>
> > Transference refers to the transfer of unconscious feelings about a significant figure from childhood (such as the father) to a current figure (such as the therapist)
>
> For example, a client is reminded of her father when she interacts with the therapist so her feelings of love and rebellion towards the father in the past may be transferred to the therapist

This may, however, be a simplified perspective of transference

> An alternative definition for transference would be 'an unconscious pattern of relating developed from early "object" relationships' made real in the current situation (Leiper & Maltby, 2004)
>
> Clients will transfer emotions from their own internal world created through experience in the past onto some aspect of the current external world
>
> This definition suggests that the client is trapped in an unconscious cycle of re-enacting old relationship patterns throughout their life and it is argued that the resulting experience of frustration can lead to significant psychological disturbance

Pattern of relationships

Transference reflects a constant pattern of old relationships repeatedly emerging in current life (Luborsky et al., 2008)

> Understanding the nature of transference in the therapeutic setting can provide the therapist with vital clues about the way that the client relates to others in the world beyond therapy
>
> For example, the client who transfers feelings of love and rebellion from her father to the therapist may have a tendency to demonstrate this behaviour in all areas of her life: trapped within this relationship, she may constantly seek to repeat her father/daughter interactions by transferring feelings about the father of her past into her current relationships. This could lead to a search for father figures in personal relationships followed by inevitable rebellion against this authority, resulting in the destruction of the relationship
>
> This hypothetical case provides evidence of how transference can be both beneficial to the client in the short term by providing her with a framework for relating to authority figures, yet damaging in the long term by trapping her in a cycle of rebellion against the desired authority figure

Positive and negative transference

Transference can result in both negative and positive feelings

- Anger and hostility may be examples of negative transference
- Love and desire may be examples of positive transference

Client could exhibit both negative and positive transference at the same time

- Similar to the manner in which one might experience both positive and negative feelings towards a parent
- For example, a client could experience feelings of love and seek to obtain the approval of the therapist whilst simultaneously experiencing feelings of anger and exhibit resentment towards the therapist

Transference or genuine emotion?

Important to be cautious about labelling all emotional responses to the therapist as evidence of transference

- Freud is reported to have said that 'a cigar is sometimes just a cigar' (Wheelis, 1950)
- It is likely that most behaviour may be based on genuine here-and-now emotions, rather than the displacement of emotions from the past
- Rogers (1951) noted that many attitudes of the client are 'of a reality, rather than a transference, nature'
  - For example, initial apprehension may give way to annoyance before eventually settling on a warm rapport (Rogers, 1951)
- Some clients will feel a natural liking or dislike on the basis of the behaviour exhibited by the therapist
- Some clients may feel sexually attracted to the therapist as a result of immediate desire rather than the misplaced attribution of past feelings
  - Indeed, it would be entirely normal for any individual to develop feelings for another person located in close proximity for a prolonged period of time
- Although these emotions may be inappropriate and should be dealt with accordingly by addressing boundaries or ending the therapeutic relationship, the existence of such emotions is not always evidence of transference
  - It is important to be able to distinguish between the transference of feelings and genuine here-and-now feelings in order to explore the revelations inherent in the former

Detecting transference

- Key features of transference (Leiper & Maltby, 2004)
  - Transference is typified by inappropriate and irrational emotions based in fantasy
  - Basis of these emotions will be protected by the distortion of reality to conform to the fixed pattern of response

Emotions will often be repeated indiscriminately across different settings

For example, a client who is transferring her early feelings of love for her father to her therapist might develop fantastical delusions of their supposed love affair, maintain these beliefs despite evidence to the contrary, and have a history of engaging in similar relationships and/or fantasies

This kind of transference can be dangerous for the therapist because he may find it extremely difficult to prove that they have not embarked on such a relationship

Counter-transference

Counter-transference = transference for therapists

Transference in the client during therapy is generally recognised as a natural response to human interaction

It is inevitable that s/he will base relations with the therapist on past relationships to some extent

But transference is not restricted to the client

Therapist may also experience similar emotional carryover effects

Therapists are human beings who cannot be entirely objective in their responses to clients as they will be subject to various biases, prejudices and desires

In a therapeutic setting, counter-transference may be exhibited on those occasions when the therapist reacts to the client in a manner that is not appropriate to that particular therapeutic relationship at that moment in time

Definitions of counter-transference

The term 'counter-transference' has been applied to a large number of psychological concepts in the therapeutic setting (Little, 2003)

Counter-transference refers to a specific unconscious attitude towards the client in response to transference from the client (objective transference – responding to transference from client)

Counter-transference refers to repressed elements of previous events leading to the displacement of past feelings from the source to the client (subjective transference – similar to transference in the client but acting in the opposite direction)

Counter-transference (objective)

Therapist responds to the transference exhibited by the client by acting in the role of the transferred figure

For example, a therapist treating a client who demonstrates transference by rebelling against the therapy in the same way that she rebelled against her own father might be experiencing counter-transference if he responds to the rebellious behaviour of the client by adopting the role of a critical father (responding to transference)

IDENTIFYING TRANSFERENCE

Read the following short transcript of a (rather inappropriate) therapy session and try to pick out examples of possible transference and counter-transference.

Therapist: Hello Suzy, what would you like to talk about today?

Client: I'm not sure. I just feel so lonely. I have just split up with my girlfriend and I feel so depressed.

Therapist: Could you tell me a little about this relationship?

Client: Well, at first, it was really good. We met at the gym when she showed me how to use the treadmill. She's a bit older than me and she was always teaching me stuff – she'd been to uni and had a good job. I've never really done anything like that – been working on the checkout at my supermarket since I was 15. She used to take me places. Like, we went down to London after we had been seeing each other for about a month. She wasn't really into clubbing like my other mates, but she took me shopping. She bought me a completely new outfit and a really expensive handbag. And I just felt really safe with her, you know? Even though London can be a bit rough, I knew that she'd take care of me.

Therapist: So when did things start to go wrong in your relationship?

Client: About six months ago. We were really close and then she just drifted away. Like, she used to wash my hair on the weekend and she used to cuddle me in bed, but she suddenly didn't want to do that any more. She said that she was too tired but it seemed like something more. I tried to talk to her about it but she said that I was just being demanding. I guess I was a bit demanding – it made me really angry that she wasn't, you know, with me any more, looking after me . . . I shouted at her a lot, sometimes I would smash up her stuff just so that she would pay me a bit more attention. But she just spent less and less time with me. Eventually, I stopped talking to her altogether. We were still living together but we didn't really communicate. When she tried to talk to me, I just didn't say anything back because I didn't want another argument. And then, last Sunday, I got home and she was gone. Packed her stuff and gone to her mum's. (Client begins to cry) I really miss her and it's all my fault. Do you think that I did something bad?

Therapist: No, not at all.

Client: But am I a bad person? I just don't know how to cope on my own.

Therapist: I don't think that you are a bad person. Relationships can be difficult and sometimes they do not work out. The important thing is to take care of yourself while you are going through this emotional distress.

Client: But I don't know how to do that!

Therapist: Well, my daughter has recently separated from her husband and she found it really helpful to join a local sports club. It got her out and about meeting new people.

Client: Do you think that I should do that?

Therapist: Yes, I do. . .

Counter-transference (subjective)

Therapist exhibits his own transference by redirecting feelings about a significant person in their life towards the client

For example, a therapist treating a client who demonstrates transference by rebelling against the therapy in the same way that she rebelled against her own father might be experiencing counter-transference if he behaves in a fatherly manner because he is reminded of his own past relationship with his daughter (transference acting in the opposite direction)

Use of transference and counter-transference in therapy

Person-centred therapy

Person-centred models of therapy highlight the importance of being able to recognise transference in the client and counter-transference in the therapist in order to prevent this phenomenon from impacting negatively on the therapeutic relationship

Rogers (1951) suggested that the client-centred therapist should strive to understand and accept client transference in the same way that one would strive to understand and accept any other client attitude

Similarly, counter-transference should be addressed through a process of self-reflection aimed at understanding the attitude within oneself as a therapist in order to ensure that this attitude does not have an impact on the client

Therapist should monitor and deal with any indication of counter-transference at the earliest stage through self-reflection, supervision, and continuous professional development

Minimising impact of transference and counter-transference

Can be accomplished by ensuring that the relationship between the therapist and the client remains professional at all times

Establishing fixed session times

Limiting personal interaction beyond the sessions

Refusing gifts or gratuities

Maintaining a degree of professional distance

It must be recognised, however, that it is practically impossible to completely eliminate transference in the therapeutic relationship

Even Rogers has been accused of fostering transference and counter-transference in his renowned exchange with Gloria: the client wishes that Rogers were her father and he responds by stating ‘you look like a pretty nice daughter to me’ (Thorne, 1992)

This aspect of the session has achieved some notoriety as it has been described as an indication that Rogers is unable to cope with transference; however, Thorne (1992) argues that this response from Rogers was essential in order to cement the relationship and further encourage the client to open up in future discourse

Transference and counter-transference are accepted within the relationship provided that the therapist remains aware of these forces and seeks to reduce the potential impact of these attitudes on the therapeutic process

Psychoanalytic therapy

In contrast to person-centred approaches to therapy, the psychodynamic approach actively uses transference and counter-transference to enhance understanding of the client

While transference is regarded as a dangerous phenomenon in most schools of psychotherapy, the interpretation of transference is essential in psychoanalytic therapy in order to explore the repressed experiences of childhood (Little, 2003)

Both Freud and Jung argued that transference was critical to the analytical process and that the exploration of transference was essential for the success of therapy. Indeed, Jung reportedly stated to Freud that transference was the 'alpha and omega of the analytical method' to which Freud remarked that Jung had 'grasped the main thing' (Jung, 1946)

Transference is allowed to develop in the psychodynamic relationship

Transference provides the therapist with crucial information about the childhood traumas of the client

For example, client who rebels against the therapist might be demonstrating transference of feelings from her father and this can provide crucial information about the nature of this past relationship

Transference provides the client with an opportunity to resolve past traumas by acting on the feelings within a safe environment, thereby potentially breaking the cycle of behaviour in the external world

Working through these transferences allows the client to become aware of these patterns of behaviour, address any unresolved conflicts, and seek to eliminate infantile regression by improving the relationship with the therapist

For example, a client may be able to resolve her anger issues in relation to her father by addressing her transferred feelings about her therapist – at the very least, the client will recognise this pattern of behaviour in her real-world relationships and possibly be able to break the cycle

SUMMARY

Blank slate: anonymous therapist does not reveal inner self so client can project own needs and values onto the therapist

Transference: redirection of feelings about one individual to another individual, demonstrates a pattern of relationships based on past experience; can involve positive and negative feelings; difficult to determine between transference

and genuine emotion, can be detected by the distortion of reality contrary to evidence

Counter-transference: transference experienced by the therapist; objective involves responding to transference from client, subjective involves transference in the opposite direction

Use of transference and counter-transference: person-centred therapy focuses on acknowledging transference so that it does not interfere with the therapeutic relationship; psychoanalytic therapy focuses on detecting transference to reveal information about past traumas and possibly resolve issues from the past

THERAPEUTIC TECHNIQUES IN PSYCHOANALYTIC THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- appreciate the differences between traditional psychoanalysis and modern psychodynamic therapy
- explain the value of the analytic framework
- explain the value and risks of interpretation in therapy
- discuss the use of transference analysis in therapy
- discuss the use of free association analysis in therapy
- discuss the use of dream analysis in therapy
- discuss the use of resistance analysis in therapy

Traditional versus modern psychoanalytic therapy

Traditional psychoanalytic therapy

Typically referred to as 'psychoanalysis'

Therapist is the expert analysing a patient to treat neurosis

Modern psychoanalytic therapy

Not often referred to as psychoanalysis, but instead described as 'psychodynamic therapy' or a more specific term (Kleinian, Freudian, brief psychodynamic, etc.)

Therapist is working collaboratively with the client to solve problems

TRADITIONAL VERSUS MODERN PSYCHOANALYTIC THERAPY

Traditional	Modern
Patient lies down on the couch and therapist is positioned on a chair out of sight	Client sits in a chair opposite the therapist (also in a chair)
Therapist strives to restructure personality of patient and focuses on fantasy material	Therapist focuses on practical concerns and strives to address specific objectives
Therapist will see patient several times every week and treatment lasts many years	Therapist will see the client once a week with treatment as short as 1–20 sessions
Therapist will remain silent as much as possible	Therapist will speak frequently and be active in the treatment process
Therapist offers no self-disclosure	Therapist might offer some self-disclosure
Therapist offers few expressions of reassurance, support or empathy	Therapist will try to uphold the core conditions of person-centred therapy (empathy, unconditional positive regard, congruence), provided it does not conflict with the goals of therapy (for example, transference analysis)

Analytic framework

Rules of the therapeutic relationship

Analytical framework refers to the basic rules of the therapeutic relationship

- Sessions of a particular length at a particular time each week
- Fees for sessions are non-negotiable
- Structure of each session follows a pattern of reflecting on last session, discussing current state, reflecting current session, confirming next session, and paying fees

Analytic framework should be fixed and structured

- Mirroring the preferred relationship between parent and child

Value of maintaining the analytic framework

- Therapist should strive to maintain the framework
  - Do not agree to meet outside of scheduled sessions
  - Do not offer discounts or free sessions
  - Do not interact on a personal basis
- Strict adherence to the framework has many benefits
  - Relationship remains professional (rather than a chat)
  - Relationship mimics preferred parent–child relationships, thus encouraging revealing examples of transference

- Relationship helps client to understand the value of structure

Importance of beginnings and endings

- Beginnings can tell us a lot
  - Explore the feelings that emerge in an initial meeting
- Endings can help the client more than anything else
  - Letting go of the important therapeutic relationship can be a positive step for those with attachment, bereavement and control issues

Breaches to the framework

Always work with breaches to the framework

- It is always easier to work with a genuine emotion in the here and now rather than exploring a past emotion
  - Whatever happens, work with it

Client failure to adhere to the rules of the therapeutic relationship can indicate something crucial

- Client might repeatedly invite the therapist out for a drink or bring small gifts to the session
- Client may be demonstrating a dependency that exhibits in all personal relationships and this can be explored in the here and now of the session

Client responses to necessary breaches in the framework made by the therapist can indicate something crucial

- The therapist who has to cancel a session (breaching the agreed rule of weekly meetings) may find that the client cancels the next session or seems withdrawn
- Client may be angry about the missed session so exploring this might give access to inner state in the here and now

Concept of 'holding'

Winnicott (1949) identified the concept of 'holding' as the actions of a 'good enough' mother

- An ordinary (good enough) mother will offer loving care for her child by creating a safe and secure environment (holding)
- Holding environment is created literally by holding the infant and metaphorically by protecting, feeding, bathing, touching and stroking the infant
- Infant feels secure in the holding environment and the feeling of being 'held' allows the child to explore the wider world without fear

Winnicott (1986) argued that psychotherapy should seek to parallel this early relationship between mother and child

- Therapist should strive to 'hold' (metaphorically, not literally!) the client by providing a safe and understanding environment in which the client can begin to explore

Within the safety of the therapeutic setting, the client is able to explore the true self while the therapist holds and constrains the emotional output within the session

Analytic framework provides this 'holding' environment for the client

Interpretation in therapy

Therapist explains meanings

Meaning of. . .

Transference

Free associations

Dreams

Resistance

See following sections for more details on interpretation in each of these areas

Client is able to assimilate the information revealed through the interpretations

Ego can accept this information because it is presented in a safe therapeutic environment

Winnicott suggested that 'a correct and well-timed interpretation in an analytic treatment gives a sense of being held physically that is more real . . . than if a real holding or nursing had taken place' (Casement, 1990)

Revelations must be carefully timed

Interpretation should be unknown by the client, but the client should be very close to discovering the information for themselves

Client will not accept interpretations if they are revealed too early (ego defence mechanisms)

Risks of interpretation

False memory syndrome

Memory for something that did not happen

Often something traumatic, such as abuse

People can mistakenly attribute an imagined event (mental imagery, thoughts, dreams, etc.) to a real event

Participants who are asked to imagine some actions and actually carry out some actions (such as combing hair or picking up an object) tend to recall actually carrying out actions that had been imagined (Goff & Roediger, 1998)

Misleading questions or descriptions by others can influence memories for real events (Loftus, 1979)

Recovered false memories

Suggestions, interpretations and questions can all implant false memories in the client (especially when also using hypnotherapy) (Loftus, 1979)

Client can assimilate inaccurate interpretations

If the therapist interprets client behaviour as evidence of repressed trauma and identifies the trauma as abuse, the client could begin to formulate false memories of this abuse

It may appear as though the client is recovering repressed memories, when in reality they are creating new memories

False memories can be exceptionally realistic

Many people have been prosecuted for crimes they did not commit, families have been torn apart, relationships have been destroyed . . .

Katrina Fairlie made accusations of child sexual abuse against her father in 2003 (Fairlie, 2010)

Ms Fairlie had been taken into hospital with stomach pains and was referred to a psychiatrist when a physical cause could not be obtained (later evidence suggests that the hospital failed to detect her inflated gall bladder due to chronic cholecystitis)

Extensive recovered memory therapy revealed a catalogue of abuse including protracted sexual abuse of her and other children at the hands of her father and 17 other men (even recalled observing her father beat a six-year-old girl to death with an iron bar)

Accusations destroyed the family and the career of her father (MP)

Subsequent investigation revealed absolutely no evidence to support these claims and Ms Fairlie later retracted all her accusations

Ms Fairlie claimed that her recovered memory was actually a false memory implanted by the therapist and she successfully sued the NHS for £20,000

Positive outcome for this case (Mr Fairlie did not face criminal proceedings as a result of the false accusations) but many other cases are less positive

Recommendations for therapists

Brandon Report (Brandon et al., 1997) is a set of training, practice and research recommendations produced by the UK Royal College of Psychiatrists

Advise psychiatrists to avoid any recovered memory therapy techniques, as there is minimal evidence to suggest that these memories are accurate and not false

'Evidence does not support the view that memory enhancement techniques actually enhance memory. There is evidence to support the view that these are powerful and dangerous methods of persuasion . . . there is sufficient evidence of distortion and/or elaboration of memories to assert that entirely new and false memories can be created, not only experimentally but also in clinical practice . . . The evidence suggests that this is true of . . . drug abreaction, hypnosis, age regression, dream interpretation, imagistic work, "feelings work", art therapy, survivors' groups. Therapist and/or patient expectations, reinforced by guided reading, particular techniques and survivors' group participation may

distort any existing memory or implant a wholly new one . . . Given the prevalence of childhood sexual abuse, even if only a small proportion are repressed and only some of them are subsequently recovered, there should be a significant number of corroborated cases. In fact there are none . . . There is no reliable means of distinguishing a true memory from an illusory one other than by external confirmation . . . There is no means of determining the factual truth or falsity of a recovered memory other than through external evidence . . . The damage to families if the accusations are untrue is immense. Patients who are mistakenly diagnosed as having been abused frequently end as mental health casualties. Where apparent improvement is based upon a false belief, there seems a serious possibility of further mental distress' (Brandon et al., 1997)

Transference analysis

Transference

Transference relationship can be classed as the cornerstone of psychoanalysis

In a therapeutic setting, transference will occur when the client redirects feelings about a significant person in their life towards the therapist

Transference refers to the transfer of unconscious feelings about a significant figure from childhood (such as the father) to a current figure (such as the therapist)

For example, client is reminded of her father when she interacts with the therapist so her feelings of love and rebellion towards the father of the past may be transferred to the therapist

Analysis of transference

Transference is encouraged and supported in psychoanalytic therapy

While transference is regarded as a dangerous phenomenon in most schools of psychotherapy, the interpretation of transference is essential in psychoanalytic therapy in order to explore the repressed experiences of childhood (Little, 2003)

Transference is allowed to develop in the psychodynamic relationship as it can . . .

Provide the therapist with crucial information about the childhood traumas of the client

For example, a client who rebels against the therapist might be demonstrating transference of feelings from her father and this can provide crucial information about the nature of this past relationship

Provide the client with an opportunity to resolve past traumas by acting on the feelings within a safe environment, thereby potentially breaking the cycle of behaviour in the external world

Working through these transferences allows the client to become aware of these patterns of behaviour, address any unresolved conflicts, and seek to eliminate infantile regression by improving the relationship with the therapist

For example, the client may be able to resolve her anger issues in relation to her father by addressing her transferred feelings about her therapist – at the very least, the client will recognise this pattern of behaviour in her real-world relationships and possibly be able to break the cycle

Free association analysis

Freedom of speech

Throughout therapy, clients are encouraged to say whatever enters their heads without any attempt at self-censorship

- No matter how silly, illogical, stupid, shocking or trivial it may appear
- Freedom of speech means that the client is able to share fully with the therapist
- Freedom of speech results in revealing Freudian slips and blocks

Freudian slip

- Free expression without self-censorship will sometimes lead to slips of the tongue
- These slips can reveal deeper emotions being repressed by the client
- Freudian slips provide a glimpse into the conflicts located in the unconscious

Blocks

- Things not said by the client can indicate areas of repressed trauma
- Since the client has free rein to say anything, refusal to discuss certain topics or problems expressing certain emotions could reveal a significant repression
- Blocks provide a glimpse into the conflicts located in the unconscious

Listening with the third ear

Interpretation of the apparently mundane thoughts verbalised by a client can lead to revelations

- Listening to the unconscious by interpreting the hidden meanings beneath speech is known as 'listening with the third ear' (Reik, 1948)
  - Therapist will search for surface and hidden meanings in the speech
  - Therapist will interpret Freudian slips and blocks

Free association activities

Therapist says a series of words and the client says the first thing that enters their head in response to the expressions

- For example, Tree, Horse, Table, Love, Work, Book, Father, Shelf, News
- Often a key word relating to an important part of the client history is integrated into the series of words (for example, 'father')

Client does not have time to self-censor so responses can be very revealing

For example, a client who is repressing his anger towards his father might respond to the word 'father' with the word 'hate' and this would reveal the repressed emotion

Free association activities are stereotypically associated with psychoanalysis

However, they are very rarely used in modern psychotherapy

Instead, therapists listen out for any natural free association during speech

Dream analysis

Royal road

Freud (1900) suggested that 'the interpretation of dreams is the royal road to a knowledge of the unconscious activities of the mind'

Defences are lowered during sleep so the ego is less likely to successfully hide repressed trauma

Dream analysis can reveal hidden conflicts, fears, feelings and needs

Latent and manifest

Although defences are lowered, the ego is still working to protect the sleeper

Unacceptable trauma and emotion will often be altered to a more acceptable form

Repressed conflicts, fears, feelings and needs are able to break through from the unconscious, but they are presented in a disguised, symbolised way

Manifest content of dreams

Content of the dream as it appears to the dreamer

Latent content of dreams

Hidden content in the form of symbols

Can be interpreted through psychoanalysis

Manifest content can reveal latent content

Therapist seeks to interpret dreams

Dreams are interpreted by considering the manifest content, suggesting possible meanings for different parts of the dream, observing reactions from client, exploring resistances, discussing the dream experience, etc.

Interpretation is unique to each client

Books on dream meanings cannot possibly be accurate because each meaning is unique to that individual and can only be accessed through extensive analysis

For example, a dream about searching for a lost ring could be interpreted in the following way

Feelings about father have been displaced to a neutral object (ring)

Abstract feelings of loss are represented in a visual form (searching)

Unacceptable emotions of anger and grief are symbolised by a dream event (searching for ring)

INTERPRETIVE DANGERS

Consider the following dreams and use your imagination to write two possible interpretations of the possible latent content of these dreams.

The dreamer is a 30-year-old woman who has suffered with depression since losing her job three months ago. She has previously described a happy childhood raised by a single mother.

> 'I was walking through a forest and it was beginning to get dark. I felt quite anxious that I might get lost and that I might never find my way out of the trees again. I was worried about how my mum might react if she never saw me again. It was not windy but all of the branches on the trees seemed to be moving and the leaves started to fall on me. I felt so helpless. I was weighted down with the leaves and I thought that I was never going to be free again. And then I heard a wolf howling in the distance and I knew that night had fallen. I was terrified and I struggled to escape the leaves before the wolf found me. I woke up tangled in my bedding and shaking with fear.'

The dreamer is a 30-year-old woman who has suffered with depression since losing her job three months ago. She has previously described an unhappy childhood raised by a single father with a drug addiction.

> 'I was walking through a forest and it was beginning to get dark. I felt quite anxious that I might get lost and that I might never find my way out of the trees again. I was worried about how my mum might react if she never saw me again. It was not windy but all of the branches on the trees seemed to be moving and the leaves started to fall on me. I felt so helpless. I was weighted down with the leaves and I thought that I was never going to be free again. And then I heard a wolf howling in the distance and I knew that night had fallen. I was terrified and I struggled to escape the leaves before the wolf found me. I woke up tangled in my bedding and shaking with fear.'

As you have probably noticed, these dreams are identical. But did you feel as though the manifest content might represent different latent content for each dreamer? Perhaps the first case seemed to indicate the woman's feelings of loss about her job and perhaps the second case seemed to indicate repressed trauma of abuse?

These examples highlight the individual nature of dream interpretation. They also demonstrate how easy it can be for the therapist to draw unsubstantiated conclusions based on the client history. It could be the case that a family friend abused the first client but the therapist is unaware of the existence of this person. It could be the case that the second client has moved on from the trauma of her childhood but feels overwhelmed by her recent unemployment. It is always difficult to draw any final interpretive conclusions so it is important to collaborate fully with the client throughout the process.

Resistance analysis

Defence mechanism

Resistance operates as an effective defence mechanism

- Some memories, thoughts and emotions are too traumatic to be experienced so resistance occurs to prevent these threatening experiences from entering consciousness
- Similar to the repression ego defence mechanism (Freud, 1936)

Resistance can hinder the therapeutic process

- Extreme pain or anxiety is avoided by resisting the process of therapy
- Since the aim of therapy is often to reveal these repressed traumas in order to work through them constructively, resistance is commonly employed unconsciously by the client to avoid facing these difficult revelations

Client may use a range of methods to resist the therapeutic process

- Refusing to discuss certain topics
- Using humour or sarcasm to hide true feelings
- Being flippant about important subjects
- Changing the subject away from the difficult area

Analysis of resistance

Analysis of the resistance can reveal repressed trauma

- For example, client who refuses to discuss school and cracks jokes about being a 'loner' when asked about friends may be demonstrating resistance and further analysis could reveal the trauma of childhood bullying

If resistance is overcome, repressed trauma can be addressed

- For example, client could begin to explore their feelings and experience a cathartic release of emotion in relation to this event – insight might also help them in understanding why they respond in certain ways and help them modify their behaviour to avoid getting caught in unhelpful cycles

BUT therapist should not suggest that all resistance is negative

- Remember that resistance is actually an effective defence mechanism to protect the fragility of our conscious emotional state – it is only negative when it hinders the therapeutic process

SUMMARY

Traditional psychoanalytic therapy: psychoanalysis; client on couch, therapist out of sight, focus on personality and fantasy, many sessions, silent therapist with no self-disclosure or empathy

Modern psychodynamic therapy: both on chairs, focus on practical concerns, possibly brief, interactive, core conditions

Analytic framework: rules of the relationship, should be fixed and structured to mirror parental relationship, beginnings and endings are crucial; breaches give an insight into the client; concept of 'holding' as the actions of a 'good enough' mother, therapy should parallel mother–child relationship by 'holding' the client

Interpretation: therapist explains meaning of expressions by client, revelations must be carefully timed; risk of false memory syndrome, Katrina Fairlie case, Brandon Report advises against memory recovery techniques

Transference analysis: transfer feelings from past significant others to therapist; encouraged in therapy, can reveal information about past relationships, can provide opportunity to resolve past trauma by acting on repressed emotions

Free association analysis: freedom of speech in therapy, no self-censorship; Freudian slips and blocks reveal deeper emotions; interpretation involves listening with the third ear; free association activity is rarely done in modern therapy, therapist listens to all speech content instead

Dream analysis: royal road to knowledge of unconscious; manifest (actual) and latent (hidden) content interpreted during psychoanalysis

Resistance analysis: defence mechanism, helpful for protecting self, can hinder therapy by hiding trauma; analysis of resistance can reveal repressed trauma and emotions; repressed trauma can then be addressed (but not all resistance is negative)

CASE STUDY DEMONSTRATING PSYCHOANALYTIC THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- appreciate the application of psychoanalytic therapy from the psychodynamic approach in a therapeutic setting

Recorded session: therapy in action

This chapter is accompanied by a recorded therapy session which is available for viewing. This session lasts for one therapy hour (50 minutes) and it is presented as the initial session in a new therapeutic relationship. Prior to this session, the client will have completed an

initial assessment questionnaire and the therapist will have read this paperwork to ensure familiarity with the case. This completed assessment and a full transcript of the recorded session are available.

No actors are used in this session. The client was one of the authors and the problem presented was genuine. The therapist is an experienced practitioner in the field. The only 'fake' aspect of this recorded session is that the client did not really seek therapy and this is not really the first session of a series of therapeutic contacts.

After the conclusion of the session, the therapist is invited to answer a few key questions about the session. This question and answer session lasts no longer than 10 minutes. The transcript of this session is available.

Presentation of Phil Thomas: history and symptoms

Phil Thomas has been suffering with stress and anxiety over the last few months. He has recently experienced some minor life events and these have had an impact on his mental wellbeing. He feels particularly anxious about his tendency to procrastinate and he believes that this is contributing towards his general feelings of stress. He has previously experienced therapy (though this was person-centred, not psychoanalytic) and he is generally positive about the possible outcomes for therapy. He would like to gain an understanding of his own behaviour in order to be able to reduce his own procrastination.

Therapy session: analysis

The therapy session can be sectioned as follows.

- Introduction
  - Gentle introduction
  - Reflection on previous experience with therapy to establish possible expectations and beliefs about therapy
  - Addressing any questions about the nature of psychoanalytic therapy
  - Outlining basic contractual details, especially the limits of confidentiality and the nature of sessions (length, number, etc.)
- Story
  - Client is invited and encouraged to share his story – this forms the largest part of the therapy sessions
  - Focus on feelings and internal experiences, rather than the specific practicalities of the story
- Goals
  - Establishing goals for therapy is periodically explored throughout the session, rather than forming a conclusion to the session
  - Goals are explored in terms of the current and ideal locations for the client

Ending

Warning given prior to the end of the session

Reflection on how the session was experienced by the client

Invitation to return for future sessions

Key questions to consider in relation to this therapy session

How could the nature of this client be understood from the psychodynamic perspective?

- Are there any unconscious drives likely to be controlling thoughts and behaviours?
- Is it evident that the id, ego and superego are in conflict?
- Are any ego defence mechanisms being exhibited?
- Does the client appear to be stuck at any psychosexual or psychosocial stage?

What is the nature of the therapeutic relationship in this psychoanalytic therapy session?

- Is the therapist a 'blank screen'?
- Is there any evidence of transference or counter-transference?

Which psychoanalytic techniques are demonstrated in this therapy session?

- Is the therapist using traditional psychoanalysis or modern psychodynamic therapy?
- Does the therapist direct or interpret the behaviour, thoughts and feelings of the client?
- Does the therapist analyse transference, resistance, defence mechanisms, free association or dreams?

Personal experience of the client

My apprehension at being a filmed client was very present for me at the start of the session. However, this subsided quickly as Tina gave an outline of what she would be hoping to focus on, and the 'permission' for me to free associate, to talk about whatever emerged as the session progressed. She was clear about being interested in relationships – in my early relationships, current relationships (including the therapist–client relationship), and my relationship with aspects of self.

As the session progressed, and I felt more comfortable and less anxious, some connections were made. Firstly how my not wanting to let people down might be something we needed to be aware of in the therapeutic relationship. I particularly liked the way she commented about things to 'notice', for example my feeling stuck early in the session, and losing my train of thought later. This felt very freeing, that I could talk about anything without it being pursued in this first session. I wasn't aware of being guarded or defensive in the session. The question asking whether I could connect thoughts and feelings to any earlier experiences was interesting, and again I felt comfortable enough to talk about my first day at secondary school, something that came to mind ('free association').

The session felt as though it had a direction, even though at times I seemed to be moving from one line of thought to another. I sensed that Tina was tracking content and making sense of my thoughts, acknowledging them, noticing them, and redirecting me as

appropriate. I always felt in control of the session despite this 'directing'. For example, about 20 minutes in, after my comments about my first day at school, she refocused me on what had happened about six weeks ago (the start of my current anxiety). This brought up a potentially important point about feeling valued.

I also liked the way Tina used the image that I offered (the cloak of invisibility) and gave me a new interpretation. From that I came to what I think was the most significant part of the session for me, the discrepancy between learning from mistakes, but wanting to do anything creative perfectly. This would be an area for future sessions.

I am aware of the possibility of transference in the session, but overall I felt a close connection to Tina. The relationship felt comfortable but challenging. I felt supported, listened to and understood. I really enjoyed Tina being so engaged in the process, in offering possible interpretations, in 'noticing', in guiding and gently directing. It gave me much to take away and reflect upon. I know that I would welcome the opportunity to work with Tina in future.

Phil Thomas

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Chapter 4

Behavioural approach and behaviour therapy

CHAPTER CONTENTS

- Introduction to behavioural approach and behaviour therapy
- Development of the behavioural approach
  - First force in psychology
  - Major advances in the behavioural approach
  - Key figures in the behavioural approach
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  - Biography of Ivan Pavlov
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- Behavioural theories of human nature and personality
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INTRODUCTION TO BEHAVIOURAL APPROACH AND BEHAVIOUR THERAPY

This chapter aims to introduce the reader to the behavioural approach to counselling and psychotherapy. Behaviour therapy will be explored as one example of a therapeutic method under the behavioural approach.

LEARNING OUTCOMES

By the end of this chapter, you will be able to:

- describe the development of the first force in psychology: the behavioural approach
- acknowledge the relative impact of Ivan Pavlov, John Broadus Watson and Burrhus Frederic Skinner on the development of the behavioural approach
- discuss the core theories of human nature and personality from the behavioural perspective
- discuss the nature of the therapeutic relationship between therapist and client in behaviour therapy
- outline the main therapeutic techniques utilised in behaviour therapy
- appreciate the application of behaviour therapy in a real-world setting

DEVELOPMENT OF THE BEHAVIOURAL APPROACH

LEARNING OUTCOMES

After reading this section, you will be able to:

- list the three main forces in psychology
- discuss the development of the behavioural approach in a historical context
- acknowledge the main contributors to the development of the behavioural approach

First force in psychology

Three forces in psychology

Behavioural theory

Psychodynamic theory

Humanistic theory

Behavioural psychology is the 'first force'

Psychology was initially defined as the study of the human mind and linked to work in medicine and philosophy

However, animal research in the early 1900s produced results that could be generalised to human behaviour (Thorndike, 1898; Pavlov, 1928)

This led to psychology being redefined as the study of human behaviour

Tabula rasa

- Behavioural approach focuses exclusively on observable and measurable behaviour
- Emphasises the importance of nurture
  - Humans are born as a *tabula rasa* or blank slate on which experience writes the patterns of their future behaviour

Major advances in the behavioural approach

Initial interest in the behavioural approach

Thorndike identified the Law of Effect in 1898

- Proposed that behaviour can be predicted and controlled according to consequences
- 'Animal intelligence: an experimental study of the associative processes in animals' published in 1898

Pavlov presented at the International Medical Congress in Madrid in 1903

- Reported physiological research revealing a system of conditioning to explain human behaviour
- Introduced the idea that psychological processes can be studied scientifically
- *Lectures on Conditioned Reflexes: Twenty-Five Years of Objective Study of the Higher Nervous Activity Behavior of Animals* published in 1928

Formal establishment of behavioural approach

Watson published the Behaviorist Manifesto in 1913

- Foundation of the behavioural approach
- 'Psychology as the behaviorist views it' published in 1913

Skinner introduced the concept of behaviour shaping through operant conditioning in 1938

- Established the principles for the experimental study of human behaviour
- *The Behavior of Organisms* published in 1938

Radical behaviourism was popular from 1940s to 1950s

- Objective study of behaviour with no focus on cognition, emotion, etc.
- All human processes are described as 'behaviour'
  - Internal states and mental existence are disregarded as hypothetical constructs that do not exist
  - Instead, internal processes are redefined as covert behaviours
  - Emotions are regarded as physiological reactions to external events
  - Thought is regarded as internal speech

Modern behavioural approach

Popularity of behaviourism decreased in the 1960s to 1980s

- Cognitive revolution redirected psychological research away from the study of behaviour to focus on theoretical models about the nature of the mind

Radical behaviourism softened into behaviour analysis

- Behaviourism that refused to accept the existence of cognitive states was rejected in favour of a less extreme version which focused on analysing behaviour as an external (measurable) evidence of cognitive states
- Staats (1996) argued that radical behaviourism failed as an approach in psychology because it did not incorporate other known phenomena, such as cognition
  - Psychological behaviourism should aim to unify all elements of psychology (biology, environment, cognition, emotion, etc.) into a single grand theory

Behaviour analysis adopts an interdisciplinary approach

- Theories include elements from other approaches such as cognitive, humanistic, etc.
- UK Society for Behaviour Analysis describes it as the science of learning
  - Experimental branch is the experimental analysis of behaviour and this aims to conduct research on how humans learn new behaviours
  - Applied branch is applied behaviour analysis (ABA) and this aims to promote behaviour change by analysing behaviours and developing interventions to modify these behaviours in a positive direction for individuals and society

Key figures in the behavioural approach

Ivan Pavlov (1849–1936)

Impact on the approach

- Classical conditioning
- Found that repeatedly pairing a neutral stimulus with an evocative stimulus eventually leads to a reflexive response to the neutral stimulus
  - For example, pairing saliva-inducing food with a neutral bell eventually causes a dog to salivate in response to the bell only

Selected works

- *Conditioned Reflexes: An Investigation of the Physiological Activity of the Cerebral Cortex* published in 1927
- *Lectures on Conditioned Reflexes: Twenty-Five Years of Objective Study of the Higher Nervous Activity Behavior of Animals* published in 1928

Edward Lee Thorndike (1874–1949)

Impact on the approach

- Established a theory of association through work on animal behaviour

Measured the time taken for cats to escape from a puzzle box by pressing a lever

Found that responses would initially be made accidentally (stepping on the lever) but were repeated when the consequence was desirable (escaping from the box)

Identified a learning curve in which behaviours are more readily completed if they have previously resulted in a satisfactory outcome

Law of Effect

Selected works

'Animal intelligence: an experimental study of the associative processes in animals' published in 1898

John Broadus Watson (1878–1958)

Impact on the approach

Behaviourism

Established the Behaviorist Manifesto in 1913

Produced the oft misquoted 'give me a dozen healthy infants'

Introduced the concept of adult behaviour resulting from childhood stimulus–response associations

Fear conditioning of Little Albert

Selected works

'Psychology as the behaviorist views it' published in 1913

Behavior: An Introduction to Comparative Psychology published in 1914

Conditioned Emotional Reactions published with Rayner in 1920

Behaviorism published initially in 1925 and revised in 1930

Burrhus Frederic Skinner (1904–1990)

Impact on the approach

Operant conditioning

Discovered that behaviour did not appear solely dependent on preceding or co-existing stimulus (classical conditioning), but was also dependent on the consequences of behaviour

Reinforcement and punishment occurring after the behaviour had an impact on the future occurrence of the behaviour

For example, rats rewarded with food when they press a lever are more likely to press the lever again in the future

Selected works

The Behavior of Organisms published in 1938

Arthur Staats (1924–current)

Impact on the approach

Transformed radical behaviourism into a human-oriented form of behaviourism known as psychological behaviourism – a unified approach aiming to integrate multiple psychological perspectives (biology, learning, cognition, emotion, etc.)

Introduced the concept of and coined the term 'time out' – temporarily removing a child from the situation as a method of punishing him/her for inappropriate behaviour, thus reducing the likelihood of the behaviour being repeated in the future

Selected works

Learning, Language, and Cognition published in 1968

Behavior and Personality: Psychological Behaviorism published in 1996

SUMMARY

Three forces: behavioural, psychodynamic, humanistic; first force in psychology, study of behaviour, *tabula rasa*, humans born as blank slates on which experience writes patterns for future behaviours

Major advances: initial ideas based on animal research by Thorndike who identified the Law of Effect in 1895 and Pavlov who presented classical conditioning at a conference in 1903; formally established through the Behaviorist Manifesto by Watson in 1913; Skinner introduced behaviour shaping through operant conditioning in 1938; radical behaviourism popular 1940s to 1960s (objective study of behaviour with no focus on cognition – all human processes are behaviours); modern approach is less radical, softened into behaviour analysis which adopts an interdisciplinary approach

Key figures: Pavlov's classical conditioning, Thorndike's Law of Effect, Watson's Behaviorist Manifesto, Skinner's operant conditioning, Staats' psychological behaviourism and time out

PERSONAL AND PROFESSIONAL BIOGRAPHIES OF IVAN PAVLOV, JOHN BROADUS WATSON AND BURRHUS FREDERIC SKINNER

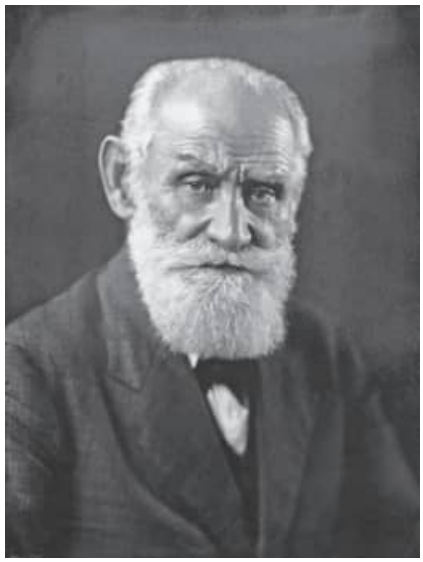
LEARNING OUTCOMES

After reading this section, you will be able to:

- outline the personal and professional biography of Ivan Pavlov (1849–1936)
- outline the personal and professional biography of John Broadus Watson (1878–1958)

- outline the personal and professional biography of Burrhus Frederic Skinner (1904–1990)
- appreciate the impact of these key figures on the development of behavioural therapy

Biography of Ivan Pavlov



'Perfect as the wing of a bird may be, it will never enable the bird to fly if unsupported by the air. Facts are the air of science. Without them a man of science can never rise.'
—Attributed to Pavlov (n.d.)

Who was Ivan Pavlov?

Ivan Pavlov established the concept of classical conditioning and this formed the foundation for the behavioural approach

Watson and Skinner also made critical contributions to this approach

Watson introduced the concept of behaviourism

Skinner established operant conditioning

Early years

Born on 14th September 1849 in Ryazan in Russia

Father (Peter Pavlov) was the village priest and mother (Varvara Uspenskaya) was a housewife

Eldest of 11 children – six siblings died before adulthood

Highly religious family

Active childhood with positive relationships with his parents and siblings (Asratyan, 1953)

Encouraged to pursue a religious career in the footsteps of his father

A serious injury following a fall placed him in the care of his grandfather and he did not attend formal schooling until the age of 11 (Asratyan, 1953)

Grandfather encouraged his educational development and he was educationally advanced from a young age

Able to read at the age of seven

Education

Originally intended on a religious career

Attended the Ryazan Theological Seminary to study for the priesthood

Transferred to the University of St Petersburg in 1870

Moved from theology to science after reading work by Charles Darwin

Graduated with a degree in natural sciences from the University of St Petersburg in 1875, then a medical qualification from the Military Medical Academy in 1879

Presented his doctoral thesis on the basic principles of the nervous system in 1883

Family

Married Seraphima (Sara) Karchevskaya in 1881

Early married life was marred by financial difficulties and a miscarriage

Sara became depressed after the death of her next child, Wirchik, during childhood

Remained married until the end of his life – Sara survived her husband by 11 years

Raised four sons and one daughter

Son Wirchik died in early childhood

Career

Academic at the Military Medical Academy from 1880 to 1925

Laboratory assistant and postgraduate student then lecturer to professor to chair in pharmacology and physiology

Organised and directed the Department of Physiology at the Institute of Experimental Medicine between 1890 and 1900 (Babkin, 1949)

Studied physiology of digestion in dogs

Demonstrated the role of the nervous system in controlling digestion – physiology maintains this position today (Babkin, 1949)

Explored the concept of 'psychic secretion' in 1895 (Nobel Lectures, 1967)

Psychic secretion refers to salivation at the distant sight of food, rather than the taste of food

Rejected the subjective interpretation of this phenomenon and developed a scientific approach to the study of the brain and nervous system

Discovered that psychic secretion could also occur in the absence of visual food stimuli if the food has previously been paired with an auditory stimulus

Reflexive (automatic) responses can be conditioned by associating the natural stimuli (e.g. food) with a novel stimulus (e.g. bell) – conditioned reflexes

Concluded that even psychic activity (thought processes) is reflexive (based on prior conditioning)

Presented research on the experimental psychology and psychopathology of animals at the 14th International Medical Congress in Madrid in 1903 (Nobel Lectures, 1967)

Defined conditioned reflexes

Emphasised the importance of these conditioned reflexes in terms of both physiological and psychological activity

Both physical reactions (behaviour) and psychological reactions (thoughts) occur in a reflexive way as a result of conditioning

Highlighted the importance of the cerebral cortex as the centre of organisation for all organism activity (Babkin, 1949)

Awarded a Nobel Prize for research on digestion in 1904 (Nobel Lectures, 1967) and the Order of the Legion of Honour in 1915

Published *Conditioned Reflexes: An Investigation of the Physiological Activity of the Cerebral Cortex* in 1927 and *Lectures on Conditioned Reflexes: Twenty-Five Years of Objective Study of the Higher Nervous Activity Behavior of Animals* in 1928

Two books collecting his most famous works to outline his findings relating to conditioned reflexes

Modern reference to conditioned reflexes regards it as Pavlovian conditioning (in honour of Pavlov) or classical conditioning (because it was the first form of conditioning to be identified)

Death

Died of pneumonia in Leningrad at the age of 86 on 27th February 1936

FOUNDING FATHER OF THE BEHAVIOURAL APPROACH: PAVLOV

Consider the life story of Ivan Pavlov and try to answer the following questions.

1. What was the social and cultural context of the world in which Pavlov developed his theories?
2. How might Pavlov's life experiences have impacted on his perception of human nature?
3. How can this life story contribute to your understanding of the behavioural approach?

Significant learnings

'We must painfully acknowledge that, precisely because of its great intellectual developments, the best of man's domesticated animals – the dog – most often becomes the victim of physiological experiments. Only dire necessity can lead one to experiment on cats – on such impatient, loud, malicious animals. During chronic experiments, when the animal, having recovered from its operation, is under lengthy observation, the dog is irreplaceable; moreover, it is extremely touching. It is almost a participant in the experiments conducted upon it, greatly facilitating the success of the research by its understanding and compliance' (Pavlov, 1893)

'One can truly say that the irresistible progress of natural science since the time of Galileo has made its first halt before the study of the higher parts of the brain, the organ of the most complicated relations of the animal to the external world. And it seems, and not without reason, that now is the really critical moment for natural science; for the brain, in its highest complexity – the human brain – which created and creates natural science, itself becomes the object of this science' (Pavlov, 1928)

Biography of John Broadus Watson

'Give me a dozen healthy infants, well-formed, and my own specified world to bring them up in and I'll guarantee to take any one at random and train him to become any type of specialist I might select – doctor, lawyer, artist, merchant-chief and, yes, even beggar-man and thief, regardless of his talents, penchants, tendencies, abilities, vocations, and race of his ancestors. I am going beyond my facts and I admit it, but so have the advocates of the contrary and they have been doing it for many thousands of years.'
—Watson (1930)

Who was John Broadus Watson?

John Broadus Watson established the concept of behaviourism and the Behaviorist Manifesto and this formed the foundation for the behavioural approach

Pavlov and Skinner also made critical contributions to this approach

Pavlov established classical conditioning

Skinner established operant conditioning

Early years

Born 9th January 1978 in Greenville, South Carolina

Father (Pickens Watson) was a farmer and mother (Emma) helped on the farm

Difficult childhood (Buckley, 1989)

Mother was a strict Baptist who demanded that her children abstain from sinful activities such as drinking

Father was a heavy drinker with a history of law breaking who had extramarital affairs and eventually left Watson's mother when he was only 13

Education

Absence of a father led him to rebellion against his mother and teachers (Buckley, 1989)

Occasionally in trouble with the law and had little respect for authority

Described himself as a poor student with no motivation for academia

However, his attitude towards education improved significantly when he was accepted to study at Furman University at the age of 16

Graduated with a Master's degree from Furman University in 1899 and earned a doctorate in psychology from the University of Chicago in 1903

Career

Taught at the University of Chicago from 1903 to 1908 and was associate professor at Johns Hopkins University from 1908 to 1920

Delivered a seminal presentation followed by a publication entitled 'Psychology as the behaviorist views it' in 1913

Regarded as the Behaviorist Manifesto

Outlined the founding principles of the behavioural approach

Argued that psychology was the science of human behaviour and thus it should be studied scientifically in laboratory conditions

'Psychology as the behaviorist views it is a purely objective experimental branch of natural science. Its theoretical goal is the prediction and control of behavior. Introspection forms no essential part of its methods, nor is the scientific value of its data dependent upon the readiness with which they lend themselves to interpretation in terms of consciousness. The behaviorist, in his efforts to get a unitary scheme of animal response, recognizes no dividing line between man and brute. The behavior of man, with all of its refinement and complexity, forms only a part of the behaviorist's total scheme of investigation' (Watson, 1913)

Published *Behavior: An Introduction to Comparative Psychology* in 1914

Established himself as the 'founder of behaviourism'

Proposed that human behaviour could be explained by generalising from animal research into conditioned responses

- Elected President of the American Psychological Association in 1915
- Published *Psychology from the Standpoint of a Behaviorist* in 1919
- Published *Conditioned Emotional Reactions* with Rayner in 1920
  - Famous Little Albert study
  - Applied the concept of Pavlovian classical conditioning to the development of phobias by conditioning a fear of rats in the child
  - Albert was raised in a hospital environment by his mother (wet nurse at the hospital)
    - He was a healthy child with no reported emotional disturbances
  - Nine-month-old Albert was tested for natural fear reactions
    - Initial tests revealed no fear of white rats, rabbits, dogs, monkeys, masks, cotton wool, etc.
    - Fear responses (starting, crying, falling over) were recorded for sudden loud noises (striking a steel bar out of sight)
  - Eleven-month-old Albert was then exposed to series of pairings between the sudden appearance of a white rat and the loud noise
    - Tests revealed fear responses (withdrawal from the stimuli, whimpering, crying) to the presentation of the white rat alone
    - Further tests found that this conditioned fear had generalised to some extent to visually similar objects (including rabbit, dog, fur coat, white hair) but not dissimilar objects (such as blocks, table, etc.)
  - Concluded that human fear can be conditioned
    - Criticised the Freudian perspective of sexual trauma in infancy to argue that adult fear is probably due to conditioning at a young age
  - Study was not criticised at the time of publication, but it has been criticised in more recent years and it was used to evidence the need for ethical guidelines (Buckley, 1989)
    - Criticisms focus predominantly on the ethics of inducing fear in a child and the validity of the conclusion that all phobias must have been conditioned at some time
    - Efforts to find the adult Albert concluded that he was likely to have been Douglas Meritte (Beck et al., 2009) – if Douglas was Albert, he passed away in 1925 from hydrocephalus at the age of six and there is no evidence to suggest that he suffered any ill effects from the experiments (although he was never deconditioned, no family stories indicate that he retained a fear of white furry objects)
- Forced to resign from Johns Hopkins University in 1920
  - Record of indiscretions with women had caused problems through his career
    - He was described as a 'ladies' man' with a history of sexual conquests and voted 'handsomest professor' in 1919 (Benjamin et al., 2007)

Affair with his graduate assistant Rosalie Rayner and subsequent divorce from his wife finally resulted in dismissal from his post (Buckley, 1989)

Although rumours have suggested that his dismissal was related to sexual experiments conducted within the department, this is unlikely to be the case (Benjamin et al., 2007)

Pursued a career in advertising after leaving academia

Unable to find work in academia despite his incredible career to date (Benjamin et al., 2007)

Vice President of J Walter Thompson Agency from 1924 to 1945

Applied behavioural principles to consumerism with impressive results and involved in some of the largest advertising campaigns of his time

Published *Behaviorism* initially in 1925 and revised in 1930

Maintained an interest in psychology despite no longer working in academia

Contained the oft quoted 'Give me a dozen healthy infants . . .' (see start of this biography)

Quote is often misrepresented with the final line omitted – this makes him appear controlling and obsessive

In reality, he recognised that his comments were extreme but made the point that it was no different to the extreme opinion taken by those who argue that we are born into our position in society

He actually recognised and valued the role of both nature and nurture

Retired in 1945

Awarded by the American Psychological Association for his contribution to psychology in 1957

Son collected his award in his place as he did not feel able to attend

Family

Married Mary Ikes in 1904

Mary was one of his students at the University of Chicago

Raised one son and one daughter with Mary

Watson reportedly had several extramarital affairs with students and the marriage ended in divorce in 1920 after Mary discovered his affair with Rosalie Rayner (Benjamin et al., 2007)

Married Rosalie Rayner

Rosalie was his graduate assistant at Johns Hopkins University and their affair caused controversy within the institute (Benjamin et al., 2007)

Raised two sons with Rosalie

Remained married to Rosalie until her premature death in 1935

Experienced difficulties in his later life (Buckley, 1989)

Watson struggled with his grief following the death of his second wife

Difficult relationship with his children which worsened after the death of his second wife

After retirement in 1945, he withdrew from society to live as a recluse on a farm in Connecticut

Death

Died in New York City on 25th September 1958

Reportedly died of liver cirrhosis

Burnt all of his unpublished work shortly before his death

'When you're dead, you're all dead' (Watson quoted in Buckley, 1989)

FOUNDING FATHER OF THE BEHAVIOURAL APPROACH: WATSON

Consider the life story of John Watson and try to answer the following questions.

1. What was the social and cultural context of the world in which Watson developed his theories?
2. How might Watson's life experiences have impacted on his perception of human nature?
3. How can this life story contribute to your understanding of the behavioural approach?

Significant learnings

'The Freudians twenty years from now, unless their hypotheses change, when they come to analyze [Little] Albert's fear of a seal skin coat – assuming that he comes to analysis at that age – will probably tease from him the recital of a dream which upon their analysis will show that Albert at three years of age attempted to play with the pubic hair of the mother and was scolded violently for it' (Watson & Rayner, 1920)

Biography of Burrhus Frederic Skinner

'Society attacks early, when the individual is helpless.'
—Skinner (1948)

Who was B.F. Skinner?

Skinner established the concept of operant conditioning and this formed the foundation for the behavioural approach

- Pavlov and Watson also made critical contributions to this approach
  - Pavlov established classical conditioning
  - Watson introduced the concept of behaviourism

Early years

Born on 20th March 1904 in Pennsylvania, USA

- 'Warm and stable' home environment (Skinner quoted in Vargas, 2005)
- Father (William Skinner) was a lawyer and mother (Grace) was a housewife
- Eldest of two children
  - Brother Edward died of a cerebral aneurysm at the age of 16

Intellectually advanced from a young age

- Active inventor (Vargas, 2005)
  - Designed a flotation system to identify ripe berries to aid his door-to-door elderberry selling business during his teens

Education

Graduated with a BA in English from Hamilton College in New York in 1926

- Did not enjoy the college lifestyle or the religious persuasion of the school

Dreamed of success as a writer between 1926 and 1928

- Returned home to live with parents while trying to write fiction

Spent a 'dark year' writing short newspaper articles before eventually deciding that his literary skills lacked life experience (Vargas, 2005)

Lived the life of a bohemian by travelling and working in New York City

Studied psychology at Harvard University between 1928 and 1931 (Smith & Morris, 2004)

Returned to school to study psychology after being motivated by books written by Pavlov and Watson

Completed his Master's in 1930 and PhD in 1931

Family

Married Yvonne Blue in 1936

Raised two daughters

Career

Researcher at Harvard University between 1931 and 1936 (Skinner, 1987)

During his PhD and post as a researcher, he worked jointly under the departments of psychology and physiology

Each department assumed that the other was engaged in active supervision, but in reality he was 'doing exactly as I pleased' (Vargas, 2005)

Developed the concept of operant conditioning while a graduate student between 1931 and 1936 (Smith & Morris, 2004)

Designed the operant conditioning chamber 'Skinner Box' for training animals using schedules of reinforcement

Chamber contains a food dispenser, response lever, loudspeaker and lights

Stimulus is presented through the loudspeaker and lights and animal must produce a specific response in order to receive reinforcement through the food dispenser

Alternative versions might include LCD screens to present specific forms of stimuli or electrified floors to deliver punishment

Experimented with rat behaviour using this apparatus to record responses to stimuli

Discovered that behaviour did not appear solely dependent on preceding or co-existing stimulus (classical conditioning), but was also dependent on the consequences of behaviour

Reinforcement and punishment occurring after the behaviour had an impact on the future occurrence of the behaviour

Labelled this new type of conditioning 'operant conditioning'

Lecturer in psychology at the University of Minnesota in Minneapolis between 1936 and 1945 (Smith & Morris, 2004)

- Published first book *The Behavior of Organisms* in 1938
  - Outlined the principles of operant conditioning and the impact of reinforcement on behaviour
- Created the baby tender or Aircrib for the birth of his second child in 1944 (Skinner, 1987)
  - Baby crib designed to reduce problems associated with a traditional crib
    - Risk of baby trapping legs between bars
    - Risk of suffocation from blankets
    - Problems with cradle cap and nappy rashes
  - Produced a fully enclosed, temperature-controlled, Plexiglass 'baby box'
    - Highly praised by his daughter Deborah who spent some time in the crib as an infant
  - Received a critical reception
    - Critics compared it to the Skinner Box and suggested that he intended to raise a baby in a controlled environment
    - Unfounded rumours circulated that his daughter was raised entirely inside the box and eventually committed suicide
    - In reality, the baby tender was a substitute for a cot only – the baby still spent most of life outside the crib
  - Despite experimental evidence highlighting the success of the baby tender, it never sold well as it seemed too much like a baby aquarium
- Ran Project Pigeon (Project Orcon) for the National Defence Research Committee during World War Two (Skinner, 1960)
  - Project Pigeon was pigeon-guided missiles
  - During World War Two, Skinner became aware that the use of bombs was limited by the lack of missile guidance systems
  - Received funding to train pigeons to guide bombs to their correct destination
    - Three pigeons in separate boxes in the nose cone of the missile
    - Each pigeon is presented with an image of what is located in front of the missile and trained to peck the target
    - Averaged direction of the pecks would guide the movement of the missile
  - Highly effective method
    - Pigeons are easily trained and demonstrate consistent responses
    - Pigeons continue pecking despite surrounding noise and falling sensation
    - Experimental trials found it to be a reliable technique for missile guidance
  - Controversial method was not used by the military
    - Funding was cancelled in 1944

No one was able to take seriously the idea of a bird guiding a missile, and most authorities found it difficult to place trust in such an unusual method (despite the reliable evidence)

Chair of psychology at Indiana University between 1945 and 1948 (Smith & Morris, 2004)

First meeting of the Society of the Experimental Analysis of Behavior was held here in 1946

Tenured professor at Harvard University between 1948 and 1990 (Smith & Morris, 2004)

Highly prolific in teaching, researching and writing

Devised a teaching machine in 1958 (Skinner, 1958)

Experience with his own daughters prompted him to explore applications for operant conditioning in education

Visit to his daughter's maths class made him realise that teaching violated every principle of conditioned learning

'Through no fault of her own, the teacher was violating almost everything we knew about the learning process' (Skinner quoted in Vargas, 2005)

Returned home to begin work on a teaching machine to improve established skills and a programme of instruction to develop new skills

Teaching machine randomly presented questions followed by immediate feedback

Essentially, this machine followed the same principles as the operant conditioning chamber – child was instantly rewarded for desirable behaviour

Programme of instruction presented material broken down into small steps and gradually reduced tutor support over time

Behaviour shaping technique helped the student to learn new skills

Retired in 1974

Death

Diagnosed with leukaemia in 1989

Fought valiantly against the progression of the disease and remained active in his career throughout the last year of his life

Delivered a talk for the American Psychological Association (APA) ten days before his death (Vargas, 2005)

Died on August 18th 1990

Completed his final article (based on the APA talk) on the day he died (Vargas, 2005)

Awarded the Lifetime Achievement Award by the APA in 1990

FOUNDING FATHER OF THE BEHAVIOURAL APPROACH: SKINNER

Consider the life story of B.F. Skinner and try to answer the following questions.

1. What was the social and cultural context of the world in which Skinner developed his theories?
2. How might Skinner's life experiences have impacted on his perception of human nature?
3. How can this life story contribute to your understanding of the behavioural approach?

Significant learnings

'Education is what survives when what has been learned has been forgotten' (Skinner, 1964)

'We shouldn't teach great books; we should teach a love of reading' (Skinner quoted in Evans, 1968)

'The real problem is not whether machines think but whether men do' (Skinner, 1969)

'A failure is not always a mistake, it may simply be the best one can do under the circumstances. The real mistake is to stop trying' (Skinner, 1972)

SUMMARY

Pavlov: established concept of classical conditioning; *early years*, born 1849 in Russia, religious family expected him to join priesthood but he pursued science, educationally advanced from a young age; *education*, doctorate in medicine in 1883 (thesis focused on principles of central nervous system); *family*, married and raised five children; organised and directed the Department of Physiology at the Institute of Experimental Medicine, explored psychic secretion to discover conditioned reflexes in dogs (classical conditioning), won Nobel Prize for research; *death*, died of pneumonia in 1936

Watson: established Behaviorist Manifesto; *early years*, born 1978 in South Carolina, difficult childhood characterised by rebellion after his father left the family; *education*, absence of father led to early rebellion against education, attitude improved after acceptance at university, doctorate in psychology from University of Chicago; *family*, married his student and raised two children, involved in a sex scandal when he divorced his wife following an affair with his graduate assistant Rayner, married Rayner and raised two children, recluse in later life after death of his wife; *career*, associate professor at Johns Hopkins University, presented the Behaviorist Manifest establishing

himself as the founder of behaviourism, argued that psychology was the science of human behaviour so it should be studied scientifically in lab conditions, president of the APA, conditioned fear in Little Albert, sex scandal meant he was forced to resign and he could not find a position in academia so worked in advertising, delivered the oft quoted 'Give me a dozen healthy infants . . .', awarded by APA for work in psychology (award collected by son); *death*, died of liver cirrhosis in 1958, burnt his unpublished work before his death

Skinner: established concept of operant conditioning; *early years*, born 1904 in Pennsylvania to a warm and stable family environment, intellectually advanced from young age; *education*, studied English and wanted to be a writer, PhD at Harvard working in psychology and physiology; *career*, discovered behaviour not dependent on co-existing stimuli and introduced concept of reinforcement and punishment to shape behaviour through experimentation with a Skinner Box while a graduate student at Harvard University, designed the baby tender 'Aircrib' after his daughter was born, ran Project Pigeon during WW2, tenured professor at Harvard, devised a teaching machine using operant conditioning principles, retired but continued writing until death; *death*, died due to leukaemia in 1989, lifetime achievement award by APA

BEHAVIOURAL THEORIES OF HUMAN NATURE AND PERSONALITY

LEARNING OUTCOMES

After reading this section, you will be able to:

- describe the stimulus–response model of classical conditioning, including explanations of how pure and second-order conditioning can shape behaviour
- describe the antecedent–behaviour–consequence model of operant conditioning, including explanations of how schedules of positive and negative reinforcement and punishment can shape behaviour

Classical conditioning: stimulus–response model (Figure 4.1)

Stimulus–response

Repeated pairings of an unconditioned and neutral stimulus will result in a conditioned response (Pavlov, 1928)

An unconditioned stimulus produces an automatic reflexive unconditioned response

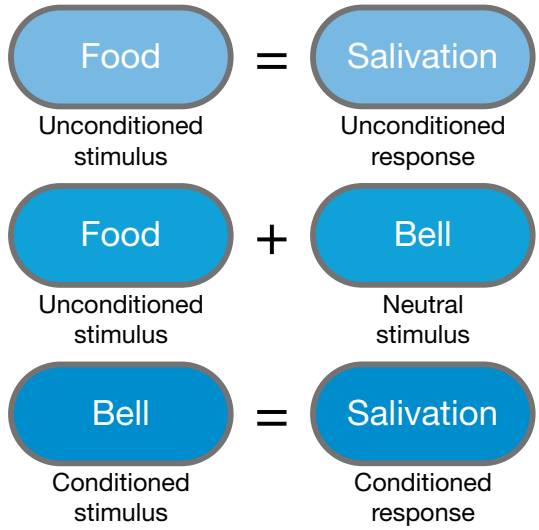


FIGURE 4.1 Classical conditioning in action. Adapted from Pavlov (1928).

For example, food (unconditioned stimulus) will automatically result in salivation (unconditioned response)

A neutral stimulus is repeatedly paired with an unconditioned stimulus to produce a conditioned stimulus

For example, pairing food (unconditioned stimulus) with a bell (neutral stimulus) will teach the individual to associate the two stimuli

A conditioned stimulus will produce the same automatic conditioned response after being paired with the unconditioned stimulus

For example, bell (conditioned stimulus) begins to automatically result in salivation (conditioned response)

S–R model

Stimulus–response

Model suggests that some stimuli are automatically paired with specific responses, and non-automatic responses can be associated with other 'neutral' stimuli through the process of conditioning

Conditioned fear

Fear can be classically conditioned (Watson & Rayner, 1920)

Little Albert was not naturally afraid of rats, but was naturally afraid of sudden loud noises

Little Albert began to demonstrate a fear of rats (cried in response to the sight of a rat) and other similar objects (white fur) after repeated pairings of the rat and a sudden loud bang

Pure and second-order conditioning

All negative human responses can be explained through pure and second-order classical conditioning

Pure classical conditioning

Direct pairing between unconditioned and neutral stimulus

We have a natural dislike of social disapproval so we will experience an unpleasant emotion (unconditioned response) in response to criticism (unconditioned stimulus)

Repeated or a single extreme pairing of criticism (unconditioned stimulus) and an exam (neutral stimulus) will teach a student to associate the two experiences

Student will experience an unpleasant emotion (conditioned response) in response to an exam (conditioned stimulus)

Second-order classical conditioning

PAIRING FOR LEARNING . . .

Try to identify the unconditioned stimulus and response and the conditioned stimulus and response in the following examples. Can you predict future behaviour on the basis of this conditioning?

A two-year-old girl is playing with her toys in the garden. Suddenly, her mother notices that the child has found a spider in the grass. She has picked it up and is about to put it into her mouth. Her mother is horrified and screams her name so that she does not eat the wriggling creepy-crawly. The little girl starts at the sound of the scream and drops the spider. She begins to cry so her mother hurries over to comfort her. He mother squashes the 'naughty' spider and cuddles the little girl until she stops crying.

A boy in his mid-teens steals a bottle of bourbon from the kitchen cabinet. He shares the bottle with his friends and, although he does not like the taste, he drinks several glasses. Shortly afterwards, he begins to feel dizzy and he is violently sick. He vomits repeatedly for the next two hours and has a painful headache for the next two days.

A young woman begins working at a local advertising company. Although she tries to talk to the other employees, she feels excluded and she often notices unpleasant comments directed towards her. One of the other women seems to be the ringleader of this nastiness – she constantly undermines her work and occasionally pushes her out of the way when they pass in the corridor. She tries to avoid this horrible woman by hiding in an empty office whenever she smells her strong flowery perfume. Eventually, she leaves the job for work in a different part of town.

Subsequent pairing between conditioned and neutral stimulus

One specific tutor (neutral stimulus) is always present during exams (conditioned stimulus) so the student learns to associate the two stimuli

Student will experience an unpleasant emotion (conditioned response) in response to the tutor (conditioned stimulus)

This could then be generalised to all tutors or all figures of authority

For example, child experiencing a school phobia could be explained in terms of classical conditioning

Operant conditioning: antecedent–behaviour–consequence model (Figure 4.2)

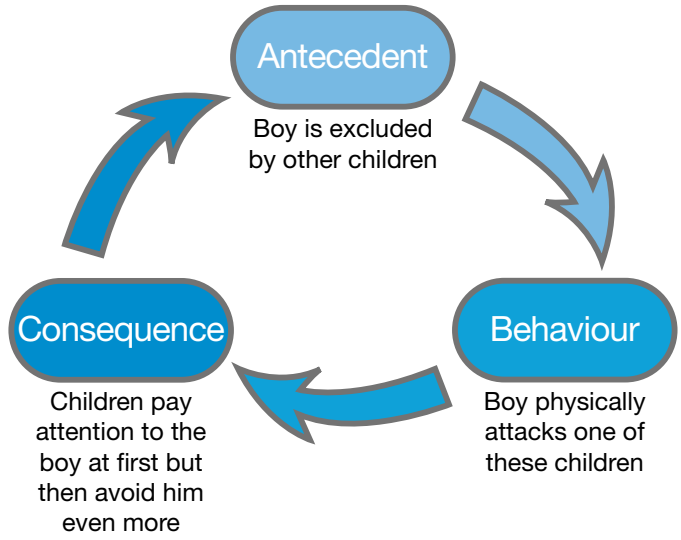


FIGURE 4.2 Operant conditioning in action. Adapted from Skinner (1938).

Antecedent–behaviour–consequence

Operant behaviour refers to active behaviours operating on the environment to generate consequences (Skinner, 1938)

All behaviour leads to consequences and these consequences will impact on the likelihood of that behaviour being repeated in the future

A-B-C model

Antecedent–behaviour–consequence

Model suggests that behaviour is prompted by stimuli in the environment (similar to S–R model) and results in specific consequences, then these consequences act as an antecedent for future behaviours

For example, a teenage boy who is excluded by his peers (antecedent) may eventually hit out at one of these other children (behaviour) – this will initially get their attention (rewarding consequence) but will then lead to further exclusion from his peers (consequence forms a new antecedent)

Positive and negative reinforcement and punishment

Hint: think of positive and negative in terms of maths (plus and minus/adding and subtracting) rather than value judgements

All negative human responses can be explained through reinforcement and punishment

Reinforcement

Anything that will strengthen the preceding behaviour

Student is disobedient in class (behaviour) and this results in more attention from the teacher and less ridicule from peers (reinforcement) so he is more likely to be disobedient again in the future

Punishment

Anything that will weaken the preceding behaviour

Student is obedient in class (behaviour) and this results in being ignored by the teacher and mocked by peers (punishment) so he is less likely to be obedient again in the future

For example, child misbehaving in school could be explained in terms of operant conditioning

Reinforcement is anything that will strengthen the preceding behaviour (Skinner, 1938)

Positive reinforcement

Addition of something pleasant results in the behaviour being more likely to be exhibited in the future

For example, giving a child a new toy when he is polite means that he is more likely to be polite again in the future

Negative reinforcement

Removal of something unpleasant results in the behaviour being more likely to be exhibited in the future

For example, reducing the number of sprouts on the plate after the child has been polite means that she is more likely to be polite again in the future

Punishment is anything that will weaken the preceding behaviour (Skinner, 1938)

Positive punishment

Addition of something unpleasant results in the behaviour being less likely to be exhibited in the future

For example, shouting at a child for being rude means that he is less likely to be rude again in the future

Negative punishment

Removal of something pleasant results in the behaviour being less likely to be exhibited in the future

For example, taking away a toy after a child has been rude means that she is less likely to rude again in the future

Combination consequences are the most effective

Punishment is less effective than reinforcement, but a combination of all elements is most effective for behaviour

For example, children who are disobedient in school may have learnt to exhibit this behaviour because being naughty leads to more attention from the teacher (positive reinforcement) and less ridicule from peers (negative reinforcement) whereas being good leads to bullying from peers (positive punishment) and less attention from the teacher (negative punishment)

Primary and secondary reinforcers and punishers

Reinforcers and punishers have an effect on the recipient

Effect can be natural or created artificially

Primary reinforcers or punishers

Naturally result in an automatic response

For example, food, water, sex, sleep are all natural reinforcers because they aid survival

Pain, hunger, thirst, tiredness are all natural punishers because they hinder survival

Secondary reinforcers or punishers

Need to be paired with a primary reinforcer to create a response

For example, money is a secondary reinforcer because society has associated money with the ability to gain food, water, sex, etc.

Poor exam grades are a secondary punisher because society has associated poor academic performance with difficulty gaining a job, thus leading to hunger, thirst, etc.

Training can utilise secondary reinforcers and punishers to avoid having to constantly provide primary reinforcement or punishment

Impractical to provide treats every time a child should be rewarded, so we can use stars on a chart as a secondary reinforcer by first associating with primary reinforcers

Difficult to distinguish between some primary and secondary reinforcers/punishers

Are respect, love and praise primary reinforcers? Or are they secondary reinforcers created as a result of pairing with the ability to gain resources such as food and sex?

Are humiliation, embarrassment and criticism primary reinforcers? Or are they secondary reinforcers created as a result of pairing with the loss of resources such as food and sex?

Answers depend on whether we regard emotional needs as instinctively important for survival

Schedules of reinforcement

Reinforcement (and punishment) can be delivered in two types of schedule

Continuous reinforcement

Partial reinforcement

Continuous reinforcement

Reinforcement is provided every single time the behaviour is exhibited

GETTING WHAT YOU DESERVE . . .

Try to identify the reinforcement and/or punishment in the following examples. Can you predict future behaviour on the basis of this conditioning?

A three-year-old boy throws a tantrum in the supermarket because he wants a chocolate bar. His mother is in a rush because she has to collect her eldest daughter from school and they are already running late. The little boy sits on the floor and refuses to move. He screams until his face turns red and other shoppers begin to stare at him. His mother eventually offers him a chocolate bar and he stops crying in order to eat it. She persuades him to stand up again and they complete the shopping without any further disruptions.

A young woman has a strong fear of travelling in cars. Unfortunately, she is employed in a shop that is a long distance from her home so her husband has to drive her to and from work every day. She spends the journey sitting tensely in the passenger seat with her eyes closed. He is aware of her fear and he tries to comfort her by constantly reassuring her in a gentle voice. Sometimes her fear will become so intense that she will begin to tremble and cry. On those occasions, her husband will pull over the car to hug and soothe her before continuing with the journey.

A father is trying to finish an important report at work, but his wife is sick so he has to take care of their two young daughters. The youngest daughter is drawing pictures in a book without making any noise. He is extremely relieved about this because it means that he can focus on his work. Unfortunately, his eldest daughter is singing noisily and constantly tries to climb on his knee. Eventually, he stops working to give his eldest daughter a lollipop to stop her singing and distract her while he finishes the report.

For example, child is given a sweet every time he scores well on a test in school

Highly effective method for learning behaviours, but can be extremely prone to extinction because the reinforcement has to be maintained constantly

Although effective, it is usually impractical to maintain continuous reinforcement or punishment so schedules will usually revert to partial reinforcement at some point

Partial reinforcement

Reinforcement is provided only at certain times when the behaviour is exhibited

For example, child is sometimes given a sweet when he scores well on a test in school

Highly effective method for maintaining behaviours and extremely resistant to extinction, but can be a slow process for learning behaviours

Four types of partial reinforcement schedule

Fixed ratio schedule: reinforcement is delivered after a set number of responses – for example, child is given a present after passing four tests at school

Variable ratio schedule: reinforcement is delivered after an unpredictable number of responses – for example, child is given a present after passing an unspecified number of tests at school

Fixed interval schedule: reinforcement is delivered after a set period of time has elapsed – for example, child is given a present every Monday provided that they are continuing to pass their tests at school

Variable interval schedule: reinforcement is delivered after an unpredictable period of time has elapsed – for example, child is given a present every so often provided that they are continuing to pass their tests at school

SUMMARY

Classical conditioning: stimulus–response model; repeated pairings of an unconditioned and neutral stimulus will result in a conditioned response; conditioned fear, Little Albert; pure and second-order conditioning

Operant conditioning: A-B-C model; active behaviours operating on the environment to generate consequences, positive and negative reinforcement and punishment, primary and secondary reinforcers and punishers, continuous and partial reinforcement (fixed ratio, variable ratio, fixed interval, variable interval)

THERAPEUTIC RELATIONSHIP IN BEHAVIOUR THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- debate the value of focusing on behaviour
- describe and evaluate psychoeducation in the therapeutic process

Behavioural focus

Advantages of behaviourism

No 'mind-reading'

It is impossible to know the thought processes or emotions experienced by another individual

Cognitive therapy claims to know these processes in order to be able to identify distortions or errors in thought and subsequently correct these inaccuracies

Psychoanalytic therapy claims to be able to see not just these thought processes, but beyond them into the unconscious processes of which even the client is unaware

Person-centred therapy avoids this problem by focusing only on what has already been revealed by the client

Behaviourism does not require the therapist to guess the inner state of the client – the therapist can simply focus on what the client actually does in relation to problems

Unclear relationship between behaviour, cognition and emotion

We often assume that we decide to do something (thought), do it (behaviour) and then feel good or bad due to the consequences (emotion)

Is this the case?

Evidence suggests that, at least in some cases, we act first and simply assume that we thought about it in advance

Consider an early morning alarm . . .

You hit 'snooze' and lie in bed thinking 'I have to get up, I have to get up, I have to get up, I have to get up . . .'

You know that you must get out of bed and you keep saying this to yourself and yet you do not move

But suddenly you are out of bed and walking to the bathroom!

What changed? Could it be that your last thought was slightly more forceful? Or could it be that some behaviours operate entirely independently of thought?

Perhaps we sometimes engage in a behaviour, justify the behaviour to ourselves afterwards, and then on reflection we assume that the thought process came first

This philosophical issue is currently being explored in science

Research evidence has found that neurological activity responsible for the intention to act can precede the conscious decision to act (Libet et al., 1983)

Participants were asked to make a voluntary response while brain activity was compared with the reported decision time

Analysis of the times revealed that the brain activated the intention to move before the participant made a conscious decision to move

These findings have been used to argue against the concept of free will

EDUCATION OR MANIPULATION?

Person-centred therapy would argue that the therapist should not seek to 'educate' the client as this implies that the therapist is an expert and may lead to the client being manipulated into behaving in a certain way. Consider the following cases and reflect on whether behavioural therapy is appropriate for the client. Try to think about the demands and risks of the therapy and those factors that might exclude someone from being able to engage with the treatment. You could also consider which alternative therapies might be more helpful for the client.

Susan has an intense fear of spiders. She is desperate to overcome this fear because she wants to go camping with her new boyfriend. She freezes every time she sees a spider and she had a panic attack last week when she found a spider in her bedroom.

Andrew has recently separated from his wife. He feels incredibly unhappy and frequently begins to cry during the day. He is convinced that he is a failure and he has stopped going to work. His brother is concerned about his wellbeing so he has forced him to see a therapist. Andrew, however, has made it clear that he would prefer to stay in bed – he does not believe that therapy will be helpful for him.

Christopher is a six-year-old boy with autism. He has lived in a residential home throughout his life and he has only limited contact with his birth mother. He does not communicate well with others and he engages in self-harm (banging his head against a wall) if he is introduced to any new people.

Claire is in a relationship with a man who has hit her on three separate occasions. She says that she is in love with him and she does not want to end the relationship. However, her mother is convinced that she needs help so she has persuaded Claire to see a therapist.

Alternative perspective

Cognition and emotion ARE behaviours

It could be argued that all human actions, including thinking and feeling, are types of behaviour and thus fall under the behaviourist category

If cognition and emotion are viewed as behaviours, then they can be modified in similar ways to all other behaviours using the core principles of classical and operant conditioning

No need to resort to deep analysis or complex interpretations

Therapeutic psychoeducation

Passive and active psychoeducation

Psychoeducation involves the practitioner teaching the client with the aim of improving mental health and wellbeing

Passive psychoeducation

Therapeutic intervention in which the therapist provides information and educational material (Donker et al., 2009)

Therapist might give out leaflets or handouts, direct the client towards websites or books, deliver lectures, etc.

Common form of online therapy

Evidence suggests that passive psychoeducation alone can have significant benefits on mental health and wellbeing (Donker et al., 2009)

Active psychoeducation

Therapeutic intervention in which the therapist provides specific advice and guidance in addition to general information and educational material (Donker et al., 2009)

Therapist might explicitly instruct the client to engage with certain activities or complete certain tasks (sometimes homework) in order to further educate the client on psychological processes

Collaboration between therapist and client

Behaviour therapy is often considered to be a psychoeducational approach (Weishaar, 1993, cited in Mkangi, 2010)

Therapist adopts a role similar to a teacher and client adopts a role similar to a student

Therapist collaborates with the client to develop new thinking patterns

Neither an expert nor a novice . . .

In contrast to the interpretative methods of the psychoanalytic approach and the non-directive methods of the person-centred approach, therapist will not establish him/herself as an expert or a novice with regard to the inner world of the client

Instead, the therapist will educate the client on basic behavioural principles and encourage the client to apply this new knowledge to his or her own life experiences

However, it can be very difficult to avoid being viewed as an expert when educating a client

By definition, teachers are viewed as more expert than students

Client is likely to assume that the therapist is an expert if s/he is using intervention techniques designed to teach the client

One way to avoid being viewed as an expert is to explore some issues in collaboration with the client

For example, therapist could suggest that both parties look for further information as homework between sessions

Education for the future . . .

This educational process will provide the client with the tools needed to address future problems and achieve independence by working towards personal goals without the constant support of a therapist

SUMMARY

Focus on behaviour: advantages of behaviourism, no mind-reading, unclear relationships between behaviour and cognition and emotion; arguably, cognition and emotion are behaviours

Therapeutic psychoeducation: passive and active psychoeducation can aid the collaboration between the client (student) and the therapist (neither expert nor novice); educational process provides client with tools to address future problems and achieve independence without the constant support of a therapist

THERAPEUTIC TECHNIQUES IN BEHAVIOUR THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- discuss exposure techniques, such as systematic desensitisation and flooding
- discuss contingency management techniques, such as time out and token economy
- describe and evaluate behavioural change programmes, such as personal and social skills programmes and the Nudge programme

Exposure therapy techniques

Reconditioning

Client is exposed to the fear-inducing stimulus in order to extinguish previous unhealthy associations and establish new healthy associations (Abramowitz et al., 2010)

- Unhealthy associations can be extinguished
  - Exposure to the associated stimulus without the negative consequences will lead to extinction of the conditioned response
  - Client who has been conditioned to fear dogs because of an early experience associating a dog (neutral stimulus) and a bite (unconditioned stimulus will automatically inspire fear)
  - Client might have this conditioned response of fear of dogs extinguished if he experiences the association between a dog (conditioned stimulus) and no bite (extinction of the association)
- Healthy associations can be established
  - Exposure to the associated stimulus with positive consequences will lead to establishment of a conditioned response
  - Client who has been conditioned to fear dogs because of an early experience associating a dog (neutral stimulus) and a bite (unconditioned stimulus will automatically inspire fear)
  - Client might have a new conditioned response of calm in the face of dogs if he experiences the association between a dog (conditioned stimulus) and relaxed music (unconditioned stimulus will automatically inspire relaxation)

Reconditioning can address unhealthy maintenance behaviours

- Some clients will adopt safety behaviours to protect themselves from anxiety, fear or other negative emotions
  - Avoidance is a common safety behaviour – for example, the client who fears blushing in front of strangers may avoid meeting people by staying at home all of the time
  - Objects and people can be used as safety behaviours – for example, the client who worries about getting lost might refuse to travel unless her husband accompanies her
  - Specific actions or attitudes can also be used as safety behaviours – for example, the client who fears being judged may adopt a brash outward appearance
- Safety behaviours are very reassuring in the short term because they offer comfort and reduce the undesirable emotions
- However, safety behaviours can work to maintain the problem in the long term
  - Individual learns that the problem is reduced only when the safety behaviour is employed

- Individual is reinforced for engaging in the behaviour so the safety behaviour is more likely in the future
- There is never an opportunity to learn that the safety behaviour is not necessary so the individual never addresses the actual problem

Many forms of 'helping' actually work to establish these counter-productive safety behaviours

- For example, the husband who tries to reassure his anxious wife by travelling with her on every trip

Exposure therapy works to expose the client to the fear-inducing stimulus without the presence of safety behaviours so that the client can learn to cope with the problem independently

- In some cases, safety behaviours may be permitted at the start of the process and then gradually removed during therapy

Two types of exposure therapy

- Systematic desensitisation
- Flooding

Systematic desensitisation

Client is systematically exposed to successively more anxiety-inducing stimuli whilst maintaining a stable level of relaxation (Wolpe, 1958)

Three steps to successful systematic desensitisation

- Relaxation training
  - Client is taught a meditative relaxation technique (such as progressive muscle relaxation or internal imagery relaxation) in order to be able to enter a relaxed state on command
- Anxiety hierarchy
  - Client is questioned on the topic of anxiety-inducing stimuli in order to produce a rank order of situations gradually increasing in anxiety severity
- Systematic desensitisation proper
  - Client is placed in a relaxed state then asked to imagine or experience the situation at the lowest rank of their anxiety-inducing scale while remaining relaxed
  - Once this can be accomplished successfully, client will begin to move up the scale with the aim of maintaining a relaxed state at each stage

Imagined versus *in vivo* exposure

- This technique can use imagined situations or actual stimuli
- Both options have costs and benefits
  - Imagined exposure is safer for the client
  - *In vivo* exposure is more realistic

Example of a client with a spider phobia

- Client is taught how to relax using the progressive muscle relaxation technique
- Client is asked to rank situations involving spiders according to anxiety – first stage might be seeing a spider at the other side of the room and last stage might be holding a spider in the hand
- Client establishes a relaxed state then imagines or experiences a spider at the other side of the room
- Client continues until able to maintain this relaxed state in the face of this situation and then moves on to imagine or experience the next stage on the ranked list
- Client will eventually get to the final stage and be able to hold a spider while maintaining a relaxed state

Highly successful technique which has been used for a wide range of problems (Spiegler & Guevremont, 2009)

- Phobia, anxiety, anger, asthma, insomnia, nightmares, motion sickness, etc.

Flooding

Client is exposed to the anxiety-inducing stimulus at a maximum level of intensity for a prolonged period of time (Wolpe, 1958)

Client is unable to use avoidance so must face the fear directly

- Clients will often use techniques to avoid their fears so they are never able to learn new associations or extinguish old associations
- This technique prevents clients from avoiding the fear or using any anxiety-reducing methods so that they realise that the anxiety-provoking stimulus does not actually lead to any real harm

Client is unable to maintain high levels of anxiety for a prolonged period of time

- Eventually the client will calm down and s/he will then have learnt that the stimulus is not dangerous or threatening

Imagined versus *in vivo* exposure

- This technique can use imagined situations or actual stimuli
- Both options have costs and benefits
  - Imagined exposure is safer for the client
  - *In vivo* exposure is more realistic

Example of a client with agoraphobia

- Client is exposed to the most anxiety-inducing situation in such a way that she cannot escape – perhaps she is deposited in the middle of a crowded supermarket
- Client will initially feel intense anxiety, but she is unable to reduce the intensity of the situation or maintain the anxiety for the prolonged period of time
- Eventually she will begin to feel calmer and she will learn that she is not in danger in the middle of the supermarket

Highly effective for fear-related disorders (Tryon, 2005), but often requires additional therapeutic support (Spiegler & Guevremont, 2009)

Limited range of disorders can be treated in this way – obviously, the client cannot be exposed to real harm so any excessive fears of genuine danger (such as fears of fire) cannot be treated through flooding

Contingency management techniques

Behaviour shaping

Client is provided with reinforcement and (to a lesser extent) punishment for specific behaviours in order to shape behaviour towards a desirable goal (Skinner, 1938)

Client will sometimes agree to a specific programme of behaviour or set of rules at the start of the treatment and then receive reinforcement or punishment for adhering to this treatment plan

For example, a client suffering substance abuse might devise a plan for reducing substance intake and the therapist will reward positive steps on this plan

Alternatively, clients who engage in inappropriate or self-damaging behaviours might not be specifically involved in devising the programme

For example, a client with severe intellectual disabilities might be encouraged to behave in a socially acceptable way by being rewarded for appropriate behaviour

Two types of contingency management

Time out

Token economy

Time out

Client is punished for failure to adhere to the programme or rules by being isolated from activities or people to some extent for a short time

Often used in cases of overexcitation or aggression in order to give the client time to calm down

Highly effective for children ('naughty step') or those with severe intellectual disabilities, but can also be used in other cases when the client can be taught to give him- or herself a time out to reflect on the situation

Contingent attention

Associated with time out, contingent attention programmes involve the client being reinforced or punished by being granted or denied attention

For example, child who is behaving badly is ignored whereas child who is behaving well is praised

Example of a client with anger management problems

Client is taught to recognise the first signs of anger and then remove himself from the situation before losing his temper

Client is encouraged to engage in a distracting time out followed by a reflective time out before returning to the anger-inducing stimulus

Distracting time out will reduce the anger by refocusing the client on a less provocative stimulus

Reflective time out will allow the client to think about the situation (ideally without returning to the angry state) in order to consider how best to handle the problem

Token economy

Client is reinforced with a token for engaging in a desirable behaviour or for adhering to the treatment programme

Tokens have no intrinsic value, but can be exchanged for reinforcing events at a later time

Material reinforcer, such as a chocolate bar

Service reinforcer, such as breakfast in bed

Privilege reinforcer, such as permission to go out for the evening

Eventually, client will be weaned off the tokens as s/he develops an internal guide for behaviour and can reinforce the self internally rather than relying on external rewards

PHYSICAL PUNISHMENT? NO, PSYCHOLOGICAL TORTURE IS MUCH MORE EFFECTIVE . . .

There is currently a wide debate relating to the effectiveness and morality associated with discipline for children. Consider the opposing arguments for corporal punishment versus psychological punishment and try to answer the following questions based on your knowledge of the area. It is inevitable that your personal opinions will influence your answers – this is absolutely appropriate provided that you remain aware of the difference between an answer based on evidence and an answer based on your own feelings.

1. Which type of discipline is most effective in instantly stopping a child from completing an action that is potentially dangerous (such as putting a hand in a fire)?
2. Which type of discipline is most effective in teaching a child right from wrong (such as not stealing)?
3. Is a smack across the bottom or withholding attention likely to have the greatest long-term effect on a child?
4. How does a parent decide which behaviours should be punished and how does the type of discipline relate to this decision?
5. Which type of discipline is most likely to lead to abuse and what are the potential long-term consequences of such abuse?

Highly effective for those with severe intellectual disabilities and has reported some success for children, but can also be used for other cases such as addiction

Example of a client with an addiction to cocaine

Client is asked to provide regular urine samples to indicate a period of abstinence from cocaine use

Client is rewarded with a voucher for a small dose of methadone

Client is eventually weaned off the methadone by reducing the dosage

Behavioural change programmes

Personal and social skills training

Personal and social skills include a wide range of behaviours

Anger management

Assertiveness

Appreciating the perspective of others

Problem solving

Conflict management

Peer negotiation and resistance skills

Active listening and effective communication

Increased acceptance and tolerance of others

These skills can be taught using psychoeducation, exposure therapy, contingency management, etc.

Highly effective in many applied settings

For example, Project ACHIEVE 'Stop and Think' social skills programme for schools in the US (Knoff, 2004)

Nudge

Behaviour can be positively influenced by gentle nudges (Thaler & Sunstein, 2008)

Nudges can be used at an institutional level by introducing policy changes that nudge people in a positive direction

UK government established the Nudge Unit (Behavioural Insights Team) in 2010

Aim of the unit is to find innovative methods of 'encouraging, enabling, and supporting people to make better choices for themselves' (Behavioural Insights Team, n.d.)

Nudges can be described as small actions to encourage positive behaviour change

For example, the Wales Centre for Behavioural Change has implemented a nudge at the local hospital by painting footprints across the floor to direct people towards the hand-washing stations

Nudges work by establishing a desired behaviour, assessing the current behaviour, seeking to reduce barriers to the behaviour, addressing logical errors and biases against the behaviour, then giving a small prompt to steer the individual towards the better behavioural option

For example, in 2010 the UK government offered subsidies for loft insulation but few people took advantage of the offer – analysis by the Nudge Unit revealed that the main barrier was laziness because people did not want the hassle of clearing the loft, so the Unit suggested that the insulation firm offer to clear the lofts themselves and this action increased uptake by 300% (Bell, 2013)

Nudges are used on a large scale to promote positive public actions

Nudges have been used to encourage the following behaviours

Public have been encouraged to pay their income tax on time by changing the wording on the letters

Drivers have been encouraged to park in their garages to reduce the risk of car theft by placing free skips on the street to allow people to empty their garages of rubbish

Walkers have been encouraged to recycle their rubbish by moving the recycling bin onto the actual footpaths

Patients have been discouraged from missing doctors' appointments by asking them to complete their own appointment cards

Drivers have been encouraged to donate to state parks by introducing an opt-out (rather than an opt-in) donation system

Jobseekers have been encouraged to seek work by establishing a commitment contract in the initial interview at the Jobcentre

Nudges are generally preferable to legislation because they still allow the individual to make a choice rather than enforcing a specific behaviour, thus impinging on individual freedom (Thaler & Sunstein, 2008)

However, if people are unaware that they are being nudged, it is debatable as to whether they really have a free choice

SUMMARY

Exposure therapy: reconditioning, unhealthy associations can be extinguished, new healthy associations can be formed, reconditioning addresses unhelpful maintenance behaviours; systematic desensitisation, relaxation training, anxiety hierarchy, systematic desensitisation proper, imagined versus *in vivo*, highly successful for range of problems (phobia, anxiety, anger, asthma, insomnia, nightmares, motion sickness); flooding, exposure to anxiety-inducing stimuli at maximum intensity for prolonged time, unable to avoid or maintain high anxiety so must eventually face the fear calmly, imagined

versus *in vivo*, highly successful for fear-related disorders but additional therapeutic support is needed

Contingency management: behaviour shaping, reinforcement and (to a lesser extent) punishment for specific behaviours to shape behaviour towards a desirable goal; time out, punished for inappropriate behaviour by exclusion from people or activities, contingent attention; token economy, reinforced for appropriate behaviour with a token to be exchanged for treats

Behavioural change programmes: personal and social skills progammes, Project ACHIEVE 'Stop and Think' social skills programme for schools in the US; nudge, behaviour can be positively influenced by gentle nudges, Nudge Unit (Behavioural Insights Team) use nudges on a large scale to promote positive public actions, preferable to legislation because they still allow choice (but is it choice if the individual is being subtly manipulated?)

CASE STUDY DEMONSTRATING BEHAVIOUR THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- appreciate the application of behaviour therapy from the behavioural approach in a therapeutic setting

Recorded session: therapy in action

This chapter is accompanied by a recorded therapy session which is available for viewing.

This session lasts for one therapy hour (50 minutes) and it is presented as the initial session in a new therapeutic relationship. Prior to this session, the client will have completed an initial assessment questionnaire and the therapist will have read this paperwork to ensure familiarity with the case. This completed assessment and a full transcript of the recorded session are available.

No actors are used in this session. The client was one of the authors and the problem presented was genuine. The therapist is an experienced practitioner in the field. The only 'fake' aspect of this recorded session is that the client did not really seek therapy and this is not really the first session of a series of therapeutic contacts.

After the conclusion of the session, the therapist is invited to answer a few key questions about the session. This question and answer session lasts no longer than 10 minutes. The transcript of this session is available.

Presentation of Fay Short: history and symptoms

Fay Short has an acute fear of heights. She has recently experienced high levels of stress at her workplace and some associated physical symptoms (including headaches and backache). However, her primary reason for seeking therapy is a desire to overcome an intense fear of heights. She has previously experienced person-centred therapy for a short space of time over ten years ago, but this was unrelated to her current fears. She would like to overcome these fears during therapy so that they no longer impact on her ability to take part in activities that involve heights.

Therapy session: analysis

The therapy session can be sectioned as follows.

- Introduction
  - Introduction outlining basic contractual details, especially the limits of confidentiality
  - Focus specifically on the problem presented by the client and an outline of the intended coverage in the session (map of the session)
  - Invitation to ask questions
- Story
  - Client is invited to explain how the problem began, how the problem has developed over time, and how the problem impacts on current life
  - Focus on actions carried out during experiences of the problem, including specific behaviours that are maintaining the problem
- Goals
  - Behaviour therapy is explained in terms of how reinforcement and punishment can shape and maintain behaviour
  - Explained how exposure to problem situations and structured reinforcement of appropriate behaviour can reduce problems
  - Identified a specific goal for therapy in the near future
- Ending
  - Summary of the session by the therapist
  - Reflection on how the session was experienced by the client
  - Explanation of how behavioural problems will be addressed in future sessions, including homework and treatment plan
  - Invitation to return for future sessions

Key questions to consider in relation to this therapy session

How could the nature of this client be understood from the behavioural perspective?

- What is the current behaviour that is causing a problem for the client?
- Has any previous classical conditioning established the current problem behaviour?
- Has the client experienced a pairing between a neutral experience and an emotive experience leading to the natural invocation of the problem behaviour?
- Are there any stimulus–response associations currently maintaining the problem behaviour?
- Has any previous operant conditioning established the current problem behaviour?
- Has the client experienced reinforcement or punishment to shape the current problem behaviour?
- Are there any consequences currently maintaining the problem behaviour?

What is the nature of the therapeutic relationship in this behaviour therapy session?

- Does the therapist focus on behaviour?
- Does the therapist explore thoughts and feelings and, if so, are these discussed in behavioural terms?
- Does the therapist adopt the role of an expert?
- Does the therapist use passive or active psychoeducational strategies?
- Does the therapist collaborate with the client to solve the current problem?

Which behaviour techniques are demonstrated in this therapy session?

- Does the therapist introduce any exposure techniques, such as systematic desensitisation or flooding?
- How does the therapist encourage reconditioning in the client?
- Does the therapist introduce any contingency management techniques, such as time out, token economy or contingent attention?
- How does the therapist encourage the reinforcement of positive change in the client?
- Does the therapist recommend or advise any specific programme of training to unlearn the old problem behaviours and learn new solution-focused behaviours?

Personal experience of the client

I was very anxious about this session because it was my first experience of behavioural therapy and my first experience of receiving therapy while being filmed. Keith reassured me in the first few minutes and I did manage to forget about the camera after we began talking.

I feel that the rapport with Keith was very positive. I felt comfortable talking to him about my experiences and I felt that he listened to me in an empathic manner. This was rather surprising because I had expected behaviour therapy to be less empathic than my previous experience of person-centred therapy. I did feel that I talked too much, especially at the start of the session. I felt that this did not fit my expectation of a behaviour session – he was less directive than expected and I was somewhat relieved to find that he was not going to demand that my fears were immediately flooded!

This session focused specifically on my behaviour during and after my scary experience in Cambodia. In particular, we explored those behaviours that I use to 'protect' myself and I realised that these behaviours have actually been maintaining my problem. For example, Keith explained that my tendency to avoid heights helps to reduce the discomfort and anxiety in the moment, but will actually teach me to avoid in the future because I am being reinforced for my avoidance. This was a surprise to me because I had never considered my behaviour in this way before – in fact, I always thought that it was very sensible for me to avoid heights given that I am so afraid. Keith also explained that my use of my husband as a support network could also be maintaining my problem – I had never considered that his comfort might be reinforcing my fears, but this made sense when it was explained in the session. On reflection, I feel that I learnt an awful lot about myself during this session.

I was very happy that Keith took the time to explain exactly what I could expect from behavioural therapy. This was very reassuring and his descriptions sounded considerably less scary than I had anticipated. He explained how the behaviours that had been maintaining my fear could be changed to reduce my fears in the future, and this gave me confidence to try out the exercises that he suggested. I left the session feeling reassured and optimistic about working with Keith on my phobia.

—*Fay Short*

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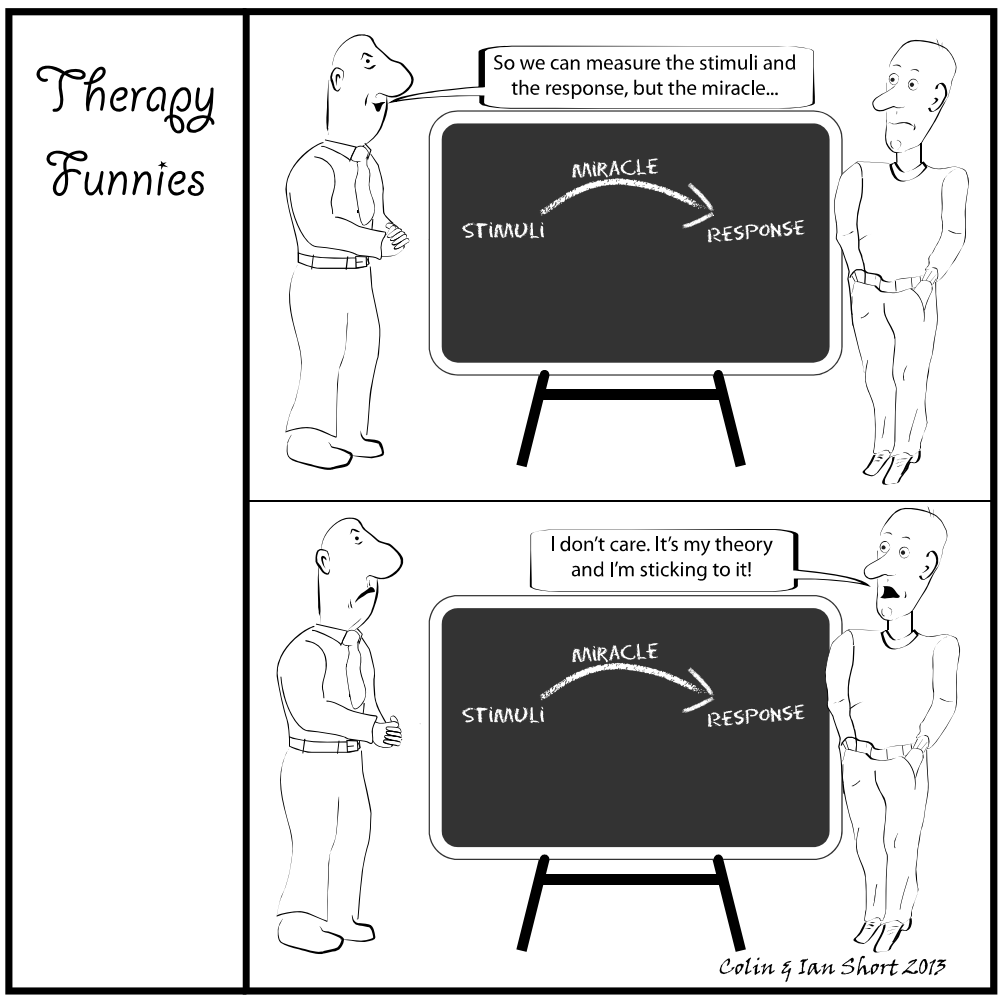
Chapter 5

Cognitive approach and cognitive therapy



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INTRODUCTION TO COGNITIVE APPROACH AND COGNITIVE THERAPY

This chapter aims to introduce the reader to the cognitive approach to counselling and psychotherapy. Cognitive therapy will be explored as one example of a therapeutic method under the cognitive approach.

LEARNING OUTCOMES

By the end of this chapter, you will be able to:

- describe the development of the cognitive revolution in psychology
- acknowledge the relative impact of Ulric Neisser and Aaron Beck on the development of the cognitive approach
- discuss the core theories of human nature and personality from the cognitive perspective
- discuss the nature of the therapeutic relationship between therapist and client in cognitive therapy
- outline the main therapeutic techniques utilised in cognitive therapy
- appreciate the application of cognitive therapy in a real-world setting

DEVELOPMENT OF THE COGNITIVE APPROACH

LEARNING OUTCOMES

After reading this section, you will be able to:

- list the three main forces in psychology and the subsequent revolution
- discuss the development of the cognitive approach in a historical context
- acknowledge the main contributors to the development of the cognitive approach

Cognitive revolution

Three forces in psychology

Behavioural theory

Psychodynamic theory

Humanistic theory

And then . . . a cognitive revolution

Revolution began in the 1950s as a response to the three main forces in psychology

Behaviourist psychology focused too heavily on behaviour, thus failing to appreciate the importance of cognitive mediation

Psychodynamic psychology focused too heavily on unconscious drives, thus failing to account for the possibility of testing cognitions empirically

Humanist psychology relied too heavily on introspection, thus failing to allow for expert guidance and understanding

Revolution was supported by advances in information technology

Cognitive approach adopted an information-processing approach

This approach assumed that we could explore the human mind using the analogy of the computer system because humans are essentially active information processors

Cyclical nature of research in psychology

Psychology was initially defined as the study of the human mind

However, animal research in the early 1900s produced results that could be generalised to human behaviour

This led to psychology being redefined as the study of human behaviour

Cognitive revolution brought psychology back to the mind

Psychology began to be defined as the scientific study of mind again

Major advances in the cognitive approach

Initial interest in the cognitive approach

Early psychology explored issues relating to the inner workings of the mind

Wundt explored immediate experiences of conscious processes (perception and attention)

Titchener explored the structures of the conscious mind (perception and attention)

Ebbinghaus explored verbal memory (memory)

James explored the functions of consciousness (memory)

Early psychological research sought to use scientific methods to study the human mind

However, it was not always scientific in the way that modern research would regard science

Wundt advocated introspection as a method for scientific observation, whereas today we would argue that an objective outsider should complete scientific observations

Titchener insisted on rigorous training for all subjects, whereas today we insist on naive participants

These methods were, however, scientific in the sense that they adopted empirical measures to explore human thought processes

Ebbinghaus developed methods to study memory that are still used in memory research today

Focus on the mind was lost when the behavioural approach dominated psychology

Behaviourism focused exclusively on observable and measurable behaviour

Concept of the mind was either ignored or regarded as another (less measureable) form of behaviour

- Behaviourism dominated psychology during the early 1900s
  - Work by Thorndike (1898), Pavlov (1928), Watson (1913) and Skinner (1938) emphasised the importance of controlled research measuring observable behaviours
- Behaviourism lost favour during the 1950s
  - Research into the limits of human performance was essential during the Second World War
    - Understanding how pilots react to large control panels or how snipers attend to potential targets under pressure was a crucial contribution to the war effort
  - Research focus returned to the investigation of mental processes and these could not be studied effectively within the behavioural paradigm
- Computer science provided a new understanding of information processing and proposed the possibility of the human mind as a type of information processor (computer analogy)
  - Information is encoded
    - Registering stimuli through your senses (eyes, ears, nose, etc.) and converting this information into a mental representation of reality
    - For example, hearing sounds emitting from the lips of a person and converting them into meaningful words and sentences
    - Similar to typing code into a computer
  - Information is stored
    - Holding information for a period of time in preparation for when it may need to be retrieved
    - Similar to storage in the hard drive of a computer
  - Information is retrieved
    - Recovering the stored information
    - Some information may be easy to retrieve
    - Some information may be difficult to retrieve
    - Similar to printing or bringing information up on the screen of a computer
- 'The magical number seven, plus or minus two' published by Miller in 1956
  - Miller provided empirical evidence for computational processes in the mind
  - Found that the human short-term memory is limited to seven plus or minus two chunks of information
    - This means that we can hold between five and nine chunks at any one time inside our heads
    - For example, you can remember seven digits (1, 4, 2, 6, 3, 9, 4) but the number of digits can be increased substantially if you chunk them (14, 28, 38, 24, 35, 41, 92)
  - Findings suggest that the human brain processes information and cannot be explained simply through the behaviourist paradigm

'A review of BF Skinner's *Verbal Behavior*' published by Chomsky in 1959

- Chomsky proposed language as an example of human activity that cannot be explained through a stimulus–response paradigm
  - Impossible to condition every word, phrase and expression so language provides evidence for active cognitive processes
- Turning point in the cognition-behaviour debate

A Study of Thinking published by Bruner in 1956

- Bruner proposed a paradigm for studying human thoughts, feelings and behaviours that differed from the accepted behavioural approach

Pinker (2002) identified five major assumptions of the cognitive approach that contributed to the move away from traditional behaviourist, psychodynamic and humanistic approaches

- Mental activity in the mind works through physical activity in the brain through a process of information, computation and feedback
- It is not possible for the mind to be a blank slate because blank slates do not do anything
- Finite combinatorial programs in the mind can generate an infinite range of behaviour
- Although there is some superficial variation in mental processes across cultures, basic mental mechanisms are universal
- The mind is a complex system composed of many interacting parts

Formal establishment of cognitive approach

Cognitive Psychology published by Neisser in 1967

- Neisser argued that perception (in particular, pattern recognition) was input, attention and problem solving were processing, and memory was output
  - 'The term "cognition" refers to all the processes by which the sensory input is transformed, reduced, elaborated, stored recovered, and used' (Neisser, 1967)
- Book had a huge impact on the field
  - Cognitive psychology became a common name and journals, courses and conferences were established in the field

Modern cognitive approach

Cognitive psychology is a thriving research field

- Rapid advancements in computer systems and artificial intelligence continue to influence the development of cognitive theories

Interdisciplinary work is common in modern cognitive psychology

- Cognitive neuropsychology is a common synthesis of research in cognition and research in neuroscience

Key figures in the cognitive approach

Albert Ellis (1913–2007)

Impact on the approach

- Founded rational therapy – original form of cognitive therapy
- Eventually developed rational therapy into rational emotive behaviour therapy (REBT) – earliest form of cognitive-behaviour therapy
- Emphasised the importance of rational thought processes and the dangers of thinking in terms of absolutes

Selected works

- 'Rational psychotherapy and individual psychology' published in 1957

Jerome Bruner (1915–current)

Impact on the approach

- Educational psychologist
- Introduced the concept of scaffolding for supporting the development of learners in education
- Found that poorer children overestimated the physical size of coins (but not similar-sized discs) due to their interpretation of the coins as high value ('New Look' experiments)
- Emphasised the importance of interpretation in reactions to stimuli and noted that this could not be explained with behavioural theories
- Formally established the study of cognitive psychology

Selected works

- 'Value and need as organizing factors in perception' published in 1947
- *A Study of Thinking* published in 1956

George Miller (1920–2012)

Impact on the approach

- Linguist and computer scientist
- Worked with Chomsky to critique the behavioural approach
- Argued that complex human processes (such as memory) could not be explained by measuring only the behavioural response
- Identified the magical number 7 (human short term memory holds +/- 7 chunks of information)
- Memory research supported the theory of cognitive processes

Selected works

- 'The magical number seven, plus or minus two' published in 1956

Aaron Beck (1921–current)

Impact on the approach

- Applied concepts from cognitive psychology to mental health
- Devised the Beck Depression Inventory
- Established the cognitive triad and cognitive distortions to explain depression and developed a cognitive approach to treatment

Selected works

- *Cognitive Therapy and the Emotional Disorders* published in 1975
- *Depression: Clinical Experimental and Theoretical Aspects* published in 1967

Noam Chomsky (1928–current)

Impact on the approach

- Linguist and political activist
- Worked with Miller to critique the behavioural approach
- Argued that the concept of learned verbal behaviour (Skinner) was an impossible application of behavioural theory and claimed that behaviourism was both simplistic and superficial as an explanation for human language
- Criticisms of the behavioural approach paved the way for the cognitive revolution

Selected works

- 'A review of BF Skinner's *Verbal Behavior*' published in 1959

Ulric Neisser (1928–2012)

Impact on the approach

- Often identified as the founding father of cognitive psychology
- Combined work in the fields of perception, attention, problem solving and memory to produce a comprehensive information-processing theory
- Highlighted the importance of ecological validity in cognitive research
- Challenged behaviourist assumptions

Selected works

- *Cognitive Psychology* published in 1967
- *Cognition and Reality* published in 1976
- *Memory Observed* published in 1982

Donald Meichenbaum (1940–current)

Impact on the approach

- Focused on changing cognitions by modifying the 'inner dialogue' through psychotherapy

Integrated cognitive and behavioural theories to form cognitive-behaviour modification – original form of cognitive-behaviour therapy (alternative to REBT proposed by Ellis)

Selected works

Cognitive-Behaviour Modification: An Integrative Approach published in 1977

SUMMARY

Cognitive revolution: behavioural, psychodynamic, humanistic; cognitive revolution supported by advances in information technology

Major advances: initial ideas focused on cognition (Wundt, Titchener, Ebbinghaus, James) using semi-scientific methods, then behaviourism dominated the field in the early 1900s, computer science and WW2 reinspired research interest in the mind, Miller and Chomsky and Bruner highlighted problems with radical behaviourism and proposed cognitive alternative in 1950s, Pinker gave five major assumptions leading to the move from behaviour to cognition (mental activity in mind worked through physical activity in brain, mind is not a blank slate because blank slates do not do anything, finite programs create an infinite range of behaviours, superficial variation across cultures but basic mental mechanisms are universal, mind is a complex system composed of many parts); formally established in 1960s with publication of *Cognitive Psychology* by Neisser; modern approach is an interdisciplinary thriving research field

Key figures: Ellis' rational therapy, Bruner's scaffolding for learning, Miller's magical number seven, Beck's cognitive triad, Chomsky's critique of behaviourism, Neisser as the founding father of cognitive psychology, Meichenbaum's cognitive-behaviour modification

PERSONAL AND PROFESSIONAL BIOGRAPHIES OF ULRIC NEISSER AND AARON BECK

LEARNING OUTCOMES

After reading this section, you will be able to:

- outline the personal and professional biography of Ulric Neisser (1928–2012)
- outline the personal and professional biography of Aaron Beck (1921–current)
- appreciate the impact of these key figures on the development of cognitive therapy

Biography of Ulric Neisser

'I was not "the father of cognitive psychology", just the godfather who named it.'
—Neisser (2002)

Who was Ulric Neisser?

Neisser is regarded as the father of the modern cognitive approach

- Challenged the behavioural approach to produce a comprehensive information-processing theory
- Highlighted the importance of ecological validity in cognitive research

Early years

Born on 8th December 1928 in Kiel, Germany

- Father (Hans Neisser) was an economist and mother (Charlotte) was active in the women's movement

Left Germany for the US when he was four years old

- Father came from a distinguished Jewish family and was an anti-Nazi Social Democrat so he moved his family out of Germany for their own safety shortly after Hitler established power
- Father took a position at the University of Pennsylvania

Happy childhood in Swarthmore, Pennsylvania

- Father earned a good salary as an academic

Minor difficulties during his teens due to German heritage

- Academically advanced (good grades, member of the Honor Society), but poor with sports, peers and girls
- Described himself as a 'weird outsider' (Neisser, 2002)
- Determined to be '100% American' to distance himself from the German enemy of the war (Neisser, 2002)

- Adopted the name Dick or Dickie because Ulric was too German
- Led to his rejection of the cultural and academic values of his German parents
- Failed to really appreciate his own heritage – did not even know that his father was Jewish or understand that he could have ended up in a concentration camp if they had remained in Germany (Neisser, 2002)

Expected to pursue a career in science

- Parents were highly academic and he felt in competition with his 'all-knowing' father (Neisser, 2002)
- Although he performed well academically, he did not really understand scientific theory and had never heard of psychology until he started university (Neisser, 2002)

Education

Studied psychology major at Harvard University from 1946 to 1950

- Changed his major to psychology after taking a course covering sensation, perception and learning
- Decided to become an experimental psychologist
  - Sought to 'fight the good fight against behaviourism' (Neisser, 2002)
- Also intrigued by parapsychology and the study of extrasensory perception
  - Established the Harvard Society for Parapsychology (Neisser, 2002)
  - Although he did not continue research in this area, he did attribute his love of controversial experiments to this early experience
- Graduated with a Bachelor's degree *summa cum laude* in 1950

Studied postgraduate psychology at Swarthmore College from 1950 to 1952

- Attracted by the recent appointment of Wolfgang Kohler which appealed to his appreciation of Gestalt psychology (perceived to be in opposition to behavioural)
- Little contact with Kohler during his degree, but did establish friendship with George Miller and moved away from Gestalt theories
- George Miller introduced him to information theory
- Graduated with a Master's degree in 1952

Studied postgraduate psychology at MIT from 1952 to 1954

- Followed Miller to MIT, but he was not interested in the research
- Instructor at Swarthmore from 1953 to 1954
- Returned to MIT to complete studies in 1954, but was still unhappy so decided to transfer to nearby Harvard

Studied postgraduate psychology at Harvard from 1954 to 1956

- Graduated with a doctorate in psychology in 1956

Family

Married Anna Pierce in 1952

- Raised three sons and one daughter
  - Divorced in 1964

Married Arden Seidler in 1964

- Arden already had three children who lived with the couple (Neisser's four children remained with their mother Anna)
- Raised one son alongside Arden's three children
- Remained married until her death

Career

Research and instructor fellowship at Harvard from 1956 to 1957

- Taught sensation and perception

Associate professor at Brandeis University from 1957 to 1965

Published *Searching for Ten Targets Simultaneously* in 1964

- Explored the fundamentals of human pattern recognition
- Provided evidence for parallel processing by demonstrating parallel visual search
  - Participants could search for ten letters as quickly as one letter

Sabbatical followed by position at the Unit for Experimental Psychiatry from 1965 to 1967

- Sabbatical was taken to complete his seminal book *Cognitive Psychology*

Published *Cognitive Psychology* in 1967

- Combined work in the fields of perception, attention, problem solving and memory to produce a comprehensive information-processing theory
  - Differed from the traditional model of information processing because it argued for parallel rather than linear processing
- Argued that perception (in particular, pattern recognition) was input, attention and problem solving were processing, and memory was output
  - 'The term "cognition" refers to all the processes by which the sensory input is transformed, reduced, elaborated, stored, recovered, and used' (Neisser, 1967)
- Challenged behaviourist assumptions without aggression
- Book had a huge impact on the field
  - Cognitive psychology became a common name and journals, courses and conferences were established in the field
  - Neisser became known as 'the father of cognitive psychology' (Neisser, 2002)

Professor of psychology at Cornell University from 1966 to 1983

- Introduced to Gibson who highlighted the importance of ecological validity and realism in research studies

- Published *Cognition and Reality* in 1976
  - Started while on sabbatical at the Center for Advanced Study in the Behavioral Sciences in Palo Alto in 1973
  - Highlighted the importance of ecological validity in cognitive research
    - Direct opposition to the controlled lab studies of the behavioural approach
  - Introduced the concept of the perceptual cycle
    - We hold schemata of information created by input of information which then influence the way that future information is input
    - For example, we hold a schema about how to behave in a restaurant based on our experiences in restaurants – this schema will then influence how we behave when we next visit a restaurant and this experience will reaffirm or conflict with our established schema
- Delivered a revolutionary keynote address at the first Practical Aspects of Memory conference in Wales in 1978
  - Emphasised the importance of addressing real-world problems using ecologically valid methods
  - Argued that psychology typically failed to study anything that is interesting or socially important (Neisser, 2002)
- Published *Memory Observed* in 1982
  - Emphasised the importance of the ecological study of everyday memory
  - Briefly covered the concept of false memories, leading to an invitation to serve on the Board of Scientific Advisors for the False Memory Syndrome Foundation
- Chaired professorship at Emory University from 1983 to 1996
- Published *Affect and Accuracy in Recall: Studies of 'Flashbulb' Memories* in 1992
  - Flashbulb memories are vivid recollections of specific moments in time
    - Almost as though a camera flashbulb has gone off in the head to capture the event on film
    - Usually associated with major events, such as your own wedding or a national disaster
    - For example, most people can vividly recall first hearing the news about the September 11th terrorist attack
  - Explored the types of errors common to flashbulb memories
  - Reported on a study which compared memories for a major event over time
    - Questioned students immediately after the explosion of the Challenger space shuttle in 1986
    - Questioned students again three years later to find huge discrepancies in memory
    - Students often held inaccurate memories and, in some cases, entirely false memories for the event

For example, one student claimed to recall a girl screaming about the disaster in the corridors at her dorm yet her original report indicated that she discovered the news from friends at lunch

Argued that the past is not stored like a video tape or photograph, but is instead distorted and reshaped based on subsequent experiences

For example, the student mentioned above may have subsequently experienced someone shouting about the Challenger disaster and modified her memory of her first exposure to the news to include this new experience

Chaired an American Psychological Association task force to address the concerns raised in response to the controversial book *The Bell Curve* in 1995

Produced a comprehensive overview of knowledge on IQ testing

Retired from Emory University to work as an emeritus faculty member at Cornell University in 1996

Death

Died on 17th February 2012 at the age of 83 following a long battle with Parkinson's disease

FOUNDING FATHER OF THE COGNITIVE APPROACH: NEISSER

Consider the life story of Ulric Neisser and try to answer the following questions.

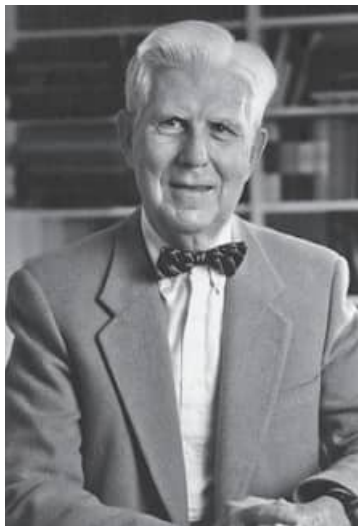
1. What was the social and cultural context of the world in which Neisser developed his theories?
2. How might Neisser's life experiences have impacted on his perception of human nature?
3. How can this life story contribute to your understanding of the cognitive approach?

Significant learnings

'Most of our oldest memories are the product of repeated rehearsal and reconstruction' (attributed to Neisser, n.d.)

'Cognitive processes surely exist, so it can hardly be unscientific to study them' (Neisser, 1967)

Biography of Aaron Beck



'Don't trust me, test me'
—Beck (cited by Blenkiron, 2005)

Who is Aaron Beck?

Beck founded cognitive therapy in the 1960s

- Cognitive therapy was built on the foundations of cognitive psychology laid down by Ulric Neisser
- Applied cognitive principles to the study of depression to formulate a new theory of depression and a new method of treatment

Early years

Born on 18th July 1921 in Providence, Rhode Island

- Father (Harry Beck) was a printer and mother (Elizabeth Temkin) was a housewife
- Youngest of three children (plus two children who passed away at a young age before his birth)

Depression marred his family

- Parents were Jewish immigrants from Russia
- Mother suffered with depression (Weishaar, 1993)
  - She lost two children prior to his birth and suffered with depressive mood swings
  - Her depression eased following his birth, but still existed to an extent throughout his life
  - Beck believed that he was a replacement for his deceased sister, but still felt proud that he had been able to cure his mother's depression by being born (Weishaar, 1993)

Life-threatening infection following an injury at the age of eight (Weishaar, 1993)

- Missed a great deal of school due to his illness

Experience left him with intense feelings of inferiority and fears of hospitals

Developed the ability to overcome these negative emotions by thinking rationally – first experience of the value of cognitive therapy

Education

Studied English at Brown University from 1938 to 1942 (Weishaar, 1993)

Elected member of the Phi Beta Kappa Society

Associate editor of the Brown Daily Herald

Awarded the Francis Wayland Scholarship, Bennett Essay Award and Gaston Prize for Excellence in Oratory

Graduated *magna cum laude* majoring in English and political science in 1942

Yale Medical School from 1942 to 1946

Initially interested in specialising in psychiatry, but was dissuaded by the irrational nature of theories in psychoanalysis

Graduated with a medical doctorate in 1946

Family

Married Phyllis Whitman in 1950

Met during his internship at Brown University (she was a student at the associated girl's school, the Hillel Foundation)

She matched Aaron Beck in her own career by completing a law degree and serving as a superior court judge in Pennsylvania

Raised two girls and two boys

Alice Beck – followed the career path of her mother and served as court judge

Judith Beck – followed the career path of her father and is now Director of the Beck Institute

Career

Resident in Pathology at Rhode Island Hospital from 1946 to 1948

Resident in Psychiatry at Cushings Veterans Administration Hospital from 1948 to 1949

Fellow in Psychiatry at the Austin Riggs Center from 1950 to 1952

Assistant Chief of Neuropsychology at Valley Forge Hospital in Pennsylvania from 1952 to 1954

Instructor then Associate in the Psychiatry Department of the University of Pennsylvania from 1954 to 1958

Assistant Professor then Associate Professor in the Psychiatry Department of the University of Pennsylvania from 1958 to 1971

Established the Depression Research Clinic focusing on empirical research into the causes and treatment of depression

Designed the Beck Depression Inventory in 1961

Currently the most widely used assessment scale for the clinical diagnosis of depression

Twenty-one items each presenting four options indicating current and recent emotional and cognitive states – each item is rated and the total score indicates the level of depression

For example:

0 I do not feel sad

1 I feel sad much of the time

2 I am sad all of the time

3 I am so sad or unhappy that I can't stand it

Diagnosis points vary, but are generally regarded as follows:

0–10 = Normal

11–16 = Mild mood disturbance

17–20 = Borderline clinical depression

21–30 = Moderate depression

31–40 = Severe depression

40+ = Extreme depression

Published *Depression: Clinical Experimental and Theoretical Aspects* in 1967

Introduced the cognitive triad and cognitive distortions

Professor of Psychiatry at the University of Pennsylvania from 1971

Cognitive revolution during the 1970s led to academic acceptance of cognitive therapy (Weishaar, 1993)

In psychology, shift from behavioural measures only towards acceptance of internal cognitive processes

In therapy, shift from psychoanalysis towards evidence-based methods

Beck was invited to give lectures in cognitive psychology at many institutions and is regarded as the founding father of cognitive therapy

Published *Cognitive Therapy and the Emotional Disorders* in 1975

Seminal book formalising the concept of cognitive therapy

Established the theories of automatic negative cognitions and explored their impact on a wide range of mental disorders: depression, eating disorders, drug addiction, anxiety disorders, etc.

Outlined clear methods for treating mental disorders

Founded the Beck Institute for Cognitive Therapy and Research in Philadelphia with his daughter in 1994

Aim of this institute is to provide training for therapists and treatment of clients using cognitive (today predominantly CBT) methods in order to encourage the growth of this field across the world

Currently retired, but remains Professor Emeritus of Psychiatry at the University of Pennsylvania and President Emeritus of the Beck Institute for Cognitive Therapy and Research

FOUNDING FATHER OF THE COGNITIVE APPROACH: BECK

Consider the life story of Aaron Beck and try to answer the following questions.

1. What was the social and cultural context of the world in which Beck developed his theories?
2. How might Beck's life experiences have impacted on his perception of human nature?
3. How can this life story contribute to your understanding of the cognitive approach?

Significant learnings

'Some authors have conceptualized depression as a "depletion syndrome" because of the prominence of fatigability; they postulate that the patient exhausts his available energy during the period prior to the onset of the depression and that the depressed state represents a kind of hibernation, during which the patient gradually builds up a new story of energy' (Beck, 1967)

SUMMARY

Neisser: Father of cognitive psychology; *early years*, born 1928 in Germany but raised in US, happy childhood but some rebellion during teens due to a desire to be 'all-American' rather than German; *education*, expected to pursue a career in science, introduced to psychology at Harvard University and graduated with a doctorate; *family*, married twice and raised five children; *career*, taught at Harvard then Brandeis University, published evidence for parallel processing, published *Cognitive Psychology* which outlined the foundation of the cognitive approach, professor at Cornell University, emphasised the importance of research with ecological validity, highlighted nature of schemata, professor at Emory, explored errors in flashbulb memories and worked to address problems of false memory syndrome, chaired a task force to address controversy about the bell curve and outlined key findings in intelligence, retired but remained emeritus; *death*, died from Parkinson's disease in 2012

Beck: founded cognitive therapy; *early years*, born 1921 in Rhode Island to Jewish Russian immigrants, mother suffered depression (eased somewhat after his

birth), sick as a child so missed a lot of school; *education*, graduated *magna cum laude* in English and political science, graduated with an MD from Yale Medical School; *family*, married once and raised four children; *career*, interested in psychiatry but not psychoanalysis, worked in several psychiatry departments before becoming professor at Pennsylvania, devised Beck Depression Inventory, introduced the concepts of the cognitive triad and cognitive distortions, cognitive revolution led to academic acceptance of his theories, outlined cognitive methods from treatment of mental disorders, founded the Beck Institute for Cognitive Therapy and Research with his daughter, currently retired but remains emeritus at Pennsylvania and the institute

COGNITIVE THEORIES OF HUMAN NATURE AND PERSONALITY

LEARNING OUTCOMES

After reading this section, you will be able to:

- discuss cognitive schemata
- describe the cognitive triad
- explain how cognitive distortions can influence thoughts, feelings and behaviours
- explain the link between cognition and behaviour

Cognitive schemata

Cognition

Cognition refers to the mental processes mediating between stimuli (external world) and response (our behaviour)

Cognition is a higher-level function of the brain

Cognition is similar to thought

Thought is usually a conscious process

Cognition is often unconscious and automatic

Cognition involves the following

Inputting – gaining information from the world

Example: noticing that people avoid eating with you and often make an unusual facial expression if they see you eating

Processing – interpreting this information for meaning

Example: interpreting the facial expression as disgust and realising that people do not like the way that you eat

Responding – behaving in an appropriate manner in response to social cues

Example: closing your mouth when you eat (or never eating in public again)

Cognitive schemata

Cognitive schemata are 'cognitive structures that represent knowledge about a concept or type of stimulus, including its attributes and the relations among those attributes' (Fiske & Taylor, 1991)

Thoughts, attitudes and beliefs about specific stimuli, such as other people, objects, events, situations, etc.

For example, our thoughts, beliefs and attitudes about school teachers

Schemata are created through experience

Our experiences will give us content to build up a schema for future use

For example, our experiences of teachers in school will combine to create a 'teacher schema' outlining our expectations for the teacher role

Four types of schemata

Person schema

Knowledge structures about specific people

Example: my dad is kind and generous, but he is not very good at telling jokes

Role schema

Knowledge about certain roles, including appropriate activities and boundaries for these roles

May be interpreted as stereotypes

Example: police officers can restrain and handcuff you, but they cannot kick or bite you

Event schema or script

Knowledge about events, including appropriate activities and behaviour

Example: you can shout and scream at a football match, but you cannot shout and scream in a lecture (well, not usually . . .)

Self-schema

Knowledge about the self

Example: I work as a lecturer and I love films

Self-schemata are complex because they are often multifaceted

Different theories argue that the self-schema is separated into different parts

Self-concept theory

- Two types of self-schema
  - Existential self – understanding that the self is an entity separate from others
  - Categorical self – understanding the various aspects of the self according to the schemata (knowledge) that we hold about the self

Self-discrepancy theory

- Three types of self-schema
  - Actual self – schema containing knowledge about how we are at the present time
  - Ideal self – schema containing knowledge about how we would like to be
  - Ought self – schema containing knowledge about how we should be

Social theory

- Three types of self-schema
  - Individual self – self as an individual with idiosyncratic traits (personal identity): schemata contain knowledge of things that make you different from every other person in the world
  - Relational self – self as human being who interacts with others: schemata contain knowledge of your personal relationships
  - Collective self – self as a member of the wider world (social identity): schemata contain knowledge of your wider associations

Schemata can operate as heuristics

Heuristics are cognitive short-cuts

- Information in the world is often limited, yet we are expected to make an appropriate response most of the time
  - Heuristics provide us with a simple judgement operation to avoid the complexities of dealing with each instance individually
- Schemata act as heuristics to provide us with a set of cognitions to allow us to quickly interpret a person or situation in order to determine a response
  - For example, our restaurant schema gives us a clear script outlining how to behave in a restaurant – we do not need to treat each restaurant as a novel experience because we can use the schema that has been already created through experience

Heuristics will not always lead to a correct inference, but it is fast and simple method for making an inference that is usually correct

- However, although heuristics can be effective in making accurate inferences in most situations, they are subject to certain errors and biases

Common errors

Overreliance on schemata

Assumption that every situation will match our schema can lead us to generalise too widely

Example: assuming that a dress code for one event will be similar in another event – this can lead to social anxiety

Overinfluence of extreme examples

Schemata based on extreme examples taken from individual cases or small samples can cause us to falsely generalise to the rest of the population

Example: media coverage of a few airplane accidents can lead us to infer that all airplanes are likely to crash – this could cause us to feel unnecessarily anxious about travelling in a plane

Ease-of-use errors

Recently digested information is likely to have a strong impact on our schemata, thus influencing our interpretations of future information

Example: your friend jumps out from behind a door shortly after you have finished watching a scary film so you are more likely to interpret the sudden event as a 'monster' because that thought is at the front of your mind (perhaps even unconsciously) – this could lead to high levels of anxiety

Ignoring base-rate information

Factual statistical information may be ignored if it is uninteresting or appears irrelevant resulting in inaccurate conclusions

Example: media images of ordinary people would be less likely to sell products, so the media present the 5% of people who are extraordinarily beautiful and we fail to appreciate the statistic, so conclude that everybody is beautiful – this could lead to low self-esteem by comparison

Schemata are both helpful and dangerous

Schemata are helpful in providing a clear and concise knowledge base for making judgements and decisions

We can respond quickly and efficiently to similar situations and people

For example, my general schema about firemen means that I can react quickly in response to the orders of a fireman even though I have never met him before

We can generalise to display appropriate behaviour in apparently novel situations

For example, my general schema about restaurants means that I will know how to behave in a new venue

We can guess the internal motivations of people around us based on prior information

For example, my schema about my 'funny dad' means that I know why he asked me why the chicken crossed the road

Schemata are also potentially dangerous

- We can commit social faux pas by making assumptions
  - For example, dressing in the wrong way for a nightclub because my schema is based on a different type of venue
- We can form stereotypes leading to discrimination
  - For example, believing that a young black male is going to try to steal my wallet because my schema is based on biased crime dramas
- We can make false judgements about ourselves and the world around us
  - For example, refusing to leave the house because my schema is based on exaggerated media reports

Cognitive triad (Figure 5.1)

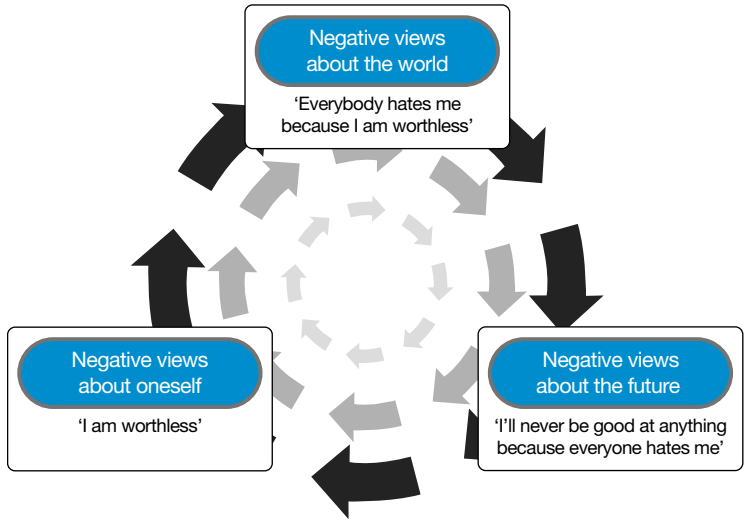


FIGURE 5.1 The cognitive triad. Adapted from Beck (1967).

The cognitive triad explains the nature of depression (Beck, 1967)

Thoughts relate to three domains

- Self
- World
- Future

Depression occurs when these three domains are viewed negatively

- The self is worthless
- The world is unfair
- The future is hopeless

Cognitive distortions

Three levels of cognition (Beck, 1967)

Level 1

- Negative automatic thoughts
  - For example, 'I've got an exam tomorrow and it will be awful'
  - As the term 'automatic thoughts' suggests, these thoughts happen without any conscious effort
  - Often when clients are asked what they were thinking that caused them to feel and respond in a certain way to an event, they answer 'I don't know'
  - However, once they think about the thoughts they had experienced, they can remember them
- Automatic thoughts are. . .
  - Negative, if they result in negative feelings
  - Closer to the surface than other levels of cognition
  - Usually specific to an event
  - Relatively easy to change

Level 2

- Dysfunctional assumptions
  - For example, 'I always do badly at exams'
- Dysfunctional assumptions are . . .
  - More general than automatic thoughts
  - Feed for automatic thoughts
  - More difficult to access and change than automatic thoughts

Level 3

- Core beliefs
  - For example: 'I am worthless'
- Core beliefs are . . .
  - Very general and affect the individual at identity level
  - Very difficult to access and change

Faulty negative automatic thoughts (Beck, 1975)

Distortions lead to faulty thinking

- Negative cognitive triad is usually due to cognitive distortions
- Many different types of cognitive distortion
  - Dichotomous thinking
  - Arbitrary inferences

Selective abstraction

Overgeneralisation

Magnification

Minimisation

Personalisation

Labelling and mislabelling

Dichotomous thinking

Viewing the self, others and the world in extreme categories

All-or-nothing thinking

Focusing on absolute terms, such as 'always', 'every', 'never', 'must', etc.

For example, assuming that you always fail exams and never succeed in your studies and everyone else is always getting better grades than you

Arbitrary inferences

Drawing conclusions on the basis of limited evidence

Mind reading and fortune telling

Assuming that others think badly of you and predicting that things will turn out badly in the future

For example, assuming that your lecturer thinks that you are stupid because you asked a question or predicting that you will get terrible questions in your final exam

Selective abstraction

Focusing on one specific detail to the exclusion of all other information

Mental filter

Paying attention to one negative aspect of a situation and ignoring all of the possible positive aspects of the same situation

For example, ignoring all of the positive comments on your marked assignment and focusing only on the one critical comment

Overgeneralisation

Generalising from a single stimulus to all stimuli

Sweeping generalised conclusions

Concluding that things will always be this way or that things will never change or that these things will always happen or . . .

For example, faulty thought that you will fail all of the exams on your degree because you failed one midterm

Magnification

Magnifying negative components

Making a mountain out of a molehill

Overestimating or exaggerating problems and difficulties

For example, overestimating how much work is required for your next assignment so that it seems like an impossible task

Minimisation

Minimising positive components

Playing down the positive

Underestimating or ignoring positives and benefits

For example, underestimating the value of the high grades obtained in your previous assignments

FIND THE DISTORTION

Identify which cognitive distortions are being demonstrated in each of the following examples.

Simon has recently separated from his girlfriend. He feels very angry – he describes his ex-girlfriend as a 'witch' and claims 'women are only after your money'. He insists that his relationships have always ended badly and says that he will never find a woman to be able to settle down.

Julie has not been out of her house for seven months. She believes that she is ugly and thinks that people will judge her. Before she withdrew into her home, she went to the local shop and felt that the person behind the counter was staring at her. She claims that this person was probably thinking that she looked deformed – she felt nauseous so had to leave the shop immediately. She wants to be able to go out again, but she does not feel that she will ever be able to face strangers without feeling dizzy and nauseous.

Karen was planning a trip to the seaside with her niece at the weekend. Unfortunately, it is raining heavily when she wakes up on Saturday morning. She is disappointed that she will not be able go on the trip and she feels as though she has let her niece down. She states that it is all her fault that they cannot go and explains that nothing ever seems to go right for her.

John has just returned from an annual review at work. Although his boss had stressed that his sales figures had been excellent and that his customer service was exemplary, John is particularly concerned because she also mentioned that there is room for improvement in his written reports. John feels that he is not capable of doing his work any more and thinks that he should begin to look for another post before he is dismissed.

Paul has been invited to a party on Saturday night. Although he would like to go out, he has decided not to attend the party because he thinks that it will be boring. He is certain that people will ignore him and he will feel uncomfortable. He decides to stay home instead, but then feels lonely and unhappy all evening.

Personalisation

Assuming personal attributions for negative events

It's all my fault . . .

Holding the self personally responsible for uncontrollable events

For example, blaming yourself for being stupid when you fail an exam (ignoring the fact that you were sick on the day)

It's all their fault . . .

Holding others personally responsible for uncontrollable events

For example, blaming your tutor for lecturing badly when you fail an exam (ignoring the fact that you did not revise)

Labelling and mislabelling

Assigning absolute and unchangeable labels

Hard to climb out of a box

Giving the self or others or situations labels which suggest that they will always exist in that particular way

For example, labelling yourself as 'stupid' so that you no longer even try to learn or labelling an assignment as 'impossible' so that you do not even attempt to complete it

Cognition and behaviour

Are thoughts, feelings and actions inevitably linked?

If we assume that cognition leads to behaviour . . .

We can predict emotions and behaviour based on cognition

We can change emotions and behaviour by changing cognition

However, this assumption is challenged in two ways

Cognition can follow (rather than precede) behaviour in some circumstances

Cognition can be entirely disassociated from behaviour in some circumstances

Cognition can follow (rather than precede) behaviour

Research evidence has found that neurological activity responsible for the intention to act (behaviour) can precede the conscious decision to act (cognition) (Libet et al., 1983)

Participants were asked to make a voluntary response while brain activity was compared with the reported decision time

Analysis of the times revealed that the brain activated the intention to move before the participant made a conscious decision to move

These findings have been used to argue against the concept of free will

Cognition can be disassociated from behaviour

There appears to be a big difference between what people think and how people behave

Early sociology study of establishments found that many places in the US claimed that they did not serve Chinese customers, but actually did when visited by the researcher accompanied by three friends from China (LaPiere, 1934)

Little link between attitudes about alcohol and alcohol consumption (Gregson & Stacey, 1981)

Only 9% of variability in behaviour can be explained by attitudes (Wicker, 1969)

Cognitive dissonance

Cognitive dissonance is a clear example of the divide between thoughts and behaviour

Although people may not always think and do the same thing, people do like to feel that their behaviours are consistent with their attitudes

Discrepancy between thoughts and behaviour can result in cognitive dissonance

Festinger (1957) claimed that cognitive dissonance is an unpleasant state of psychological tension generated by two or more inconsistent cognitions

Two or more bits of information do not fit together so the individual feels uncomfortable

People seek harmony in their attitudes and behaviours so they will often try to reduce cognitive dissonance by changing either the thought or the behaviour

Change behaviour to fit with thoughts

Change thought to fit with behaviour

Greater cognitive dissonance will result in greater attempts to change either behaviour or thought

For example, Simon does not like his job (thought) yet he continues to go to his work place every day (behaviour) – these two experiences are in conflict (cognitive dissonance) so he will change one cognition by either quitting his job (changing behaviour) or deciding that he actually likes his job (changing thought)

Susan is aware that smoking damages your health (thought) yet she continues to smoke (behaviour) – these two experiences are in conflict (cognitive dissonance) so she will either quit smoking (changing behaviour) or begin to argue that smoking is not really too damaging (changing thought)

Link between cognition, emotion and behaviour

Absence of a direct link between cognition and behaviour leads to questions about the appropriateness of seeking to change cognition in order to change negative emotions and behaviours

Often clients may recognise their own faulty automatic thoughts and acknowledge the alternative logical thoughts, but this awareness does not lead to an improvement in emotion and behaviour

For example, an individual might be aware that he is exaggerating when he says that he cannot cope with the loss of his job, but this awareness may not reduce the depression associated with this loss

SUMMARY

Schemata: cognition is the mental process mediating between stimuli (external world) and response (our behaviour); schemata are thoughts, beliefs and attitudes relating to specific stimuli, based on experience, four types (person, role, event/script, self); heuristics are cognitive short-cuts, can help process information efficiently but can be dangerous due to errors; common schema errors include overreliance on schemata, extreme examples, ease-of-use errors, ignoring base-rate information

Negative cognitive triad: self is worthless, world is unfair, future is hopeless

Cognitive distortions: three levels of cognition (negative automatic thoughts, dysfunctional assumptions, core beliefs); automatic faulty thoughts are caused by cognitive distortions; distortions include dichotomous thinking, arbitrary inferences, selective abstraction, overgeneralisation, magnification, minimisation, personalisation, labelling and mislabelling

Cognition and behaviour: if cognition leads to behaviour then we can predict and change behaviour by focusing on cognition, but evidence suggests that this link is not direct; cognition can follow (rather than precede) behaviour; cognition can be dissociated from behaviour, cognitive dissonance; absence of link between cognition and behaviour raises questions about trying to change cognitions; sometimes individual is aware that cognition is faulty but still cannot change behaviour

THERAPEUTIC RELATIONSHIP IN COGNITIVE THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- describe and evaluate client–therapist rapport in the therapeutic process
- debate the potential problems associated with cognitive therapy

Client–therapist rapport

Rapport is necessary but not sufficient

Rapport is necessary

- Core conditions of humanistic approach are important for establishing a positive working relationship
- Positive relationship is vital if the client is going to collaborate with the therapist to work towards therapeutic change
- Core conditions provide a firm foundation for cognitive therapies
  - Empathy, unconditional positive regard and congruence are crucial to the success of therapy
  - Young and Beck (1980) note that an effective cognitive therapist should communicate genuineness, sincerity and openness alongside warmth and concern – they also advise using and encouraging humour to establish positive rapport

Rapport is not sufficient

- Simply providing warmth, congruency, empathy and unconditional positive regard is not sufficient for effective change
- Cognitive techniques are essential for making lasting changes

Rapport is a desirable therapist strategy for developing a collaborative therapeutic relationship (Young & Beck, 1980)

- Rapport is described as a 'harmonious accord between people' (Young & Beck, 1980)
- In terms of cognitive therapy, rapport refers to the teamwork of the therapist and client working together in comfort and security
- Rapport requires the therapist to go beyond simply showing warmth and empathy – the therapist must also be flexible enough to adapt the therapy to suit the needs of each individual (Young & Beck, 1980)

Collaborative empiricism is crucial

Some clients may have the impression that the therapist can offer a 'fix' for their problems

- In reality, however, the process of cognitive therapy is exploratory with the aim of setting goals and moving towards those goals in collaboration

Effective therapy depends on the client and therapist collaborating to identify and correct faulty cognitive processes

- 'Collaborative empiricism means that the patient and the therapist become co-investigators both in ascertaining the goals for treatment and investigating the patient's thoughts. Methods of guided discovery are used to help patients to test their own thinking through personal observations and experiments rather than via cajoling or persuasion' (Beck & Dozois, 2011)

Two benefits of collaborative empiricism in the therapeutic relationship (Hutton & Morrison, 2012)

Clients are more engaged when treated as an equal partner by therapists

Clients who are involved in the decision-making processes during therapy will learn how to make better decisions in the future outside therapy

COLLABORATIVE EMPIRICISM OR MANIPULATION?

Imagine that you are a therapist working with a client who makes the following statement.

> *'I'm just feeling really confused about everything at the moment. I mean, I love my wife but we argue about starting a family all of the time. We only just got married but it feels like I never want to be around her. I find myself trying to get out of the house whenever I can. Me and John go to the game or the pub or just chill out on the sofa at his place. I can just be myself with him. I can talk to him and he really seems to get me. I don't know. I don't think that I am ever going to be happy.'*

Consider the questions that you might now ask in order to work with your client to identify faulty thinking (cognitive distortions), set goals and establish an action plan.

Imagine that you suspect that your client is attracted to his friend but refusing to accept his own feelings. What questions could you ask to test the theory that he might be gay?

Imagine that you suspect that your client loves his wife but is afraid about starting a family. What questions could you ask to test the theory that he might have a fear of commitment?

Consider the manipulative potential in the questions that you have devised. Your work with this client has the potential to steer him in specific directions and this is a huge risk for any type of therapy, but particularly those that seek to alter cognitive processes. It is essential that all therapists are aware of this risk so that they can structure their practice towards collaborative empiricism rather than manipulation.

Problems with cognitive therapy

Who decides which cognitions should be tested?

Many people hold cognitions which are arguably irrational and/or illogical without experiencing psychological problems

For example, many religious people hold beliefs that could be interpreted as irrational (indeed, Ellis concluded that religion was irrational)

It would not be appropriate for the therapist to decide that these cognitions must be tested and corrected if found to be illogical

Many people also hold cognitions that are undeniably irrational and/or illogical without experiencing psychological problems

For example, many people who play the lottery hold the belief that they will win one day but the statistical evidence suggests that this is illogical

It would not be appropriate for the therapist to decide that all cognitions must be tested and corrected if found to be illogical

Much cognition is based on subjective, rather than objective, factors

For example, cognitions about whether a person is attractive or unattractive are based on subjective opinions rather than unbiased logic

It would be very difficult for the therapist to address the logic of subjective opinion – instead, the therapist should focus on exploring counter-opinions

Therapists must ensure that they only address cognitions if these cognitions are causing problems for the client

What happens when a cognition contributing to the negative emotion is found to be logical?

Some cognition may be found to be accurate, but may still contribute to the problem

For example, cognitions relating to the loss of a loved one may be absolutely correct ('he is never coming back, I will always miss him, etc.') and these cognitions may be contributing to the depression of the client

Therapist may need to accept that these cognitions are logical, but explore any underlying implicit cognitions that lack logic

Therapist could instead address the implicit cognition 'I cannot cope with this' by focusing on how the client has managed to cope with this situation (or similar situations) up until this point

Is it ethical for a therapist to try to change minds?

Padesky (1993) acknowledges that changing beliefs can be therapeutic, but argues that the use of cognitive therapy for this sole purpose might undermine the empirical basis of the approach

Therapists who decide that a thought is illogical and guide the client towards changing their thoughts in that direction are arguably manipulating the mind of the individual in their own preferred direction

For example, the therapist who believes that abortion is morally wrong could guide the client towards an unwanted birth by questioning the logic of 'I will not be able to cope with a child right now'

Evidence for false memory insertion has often been associated with psychodynamic therapy, but there is a case to suggest that the process of changing cognitions could lead to similar problems

For example, the therapist who repeatedly questions the logic of the client who runs away from home could force the client to assume a defensive position by justifying the actions with a belief about abuse

Alongside individual ethical concerns, there are social concerns related to the potential for cognitive manipulation on a wider scale

Cognitive therapy (and CBT) are strongly promoted as the primary solution to many psychological problems

National Institute for Health and Clinical Excellence stated that computerised CBT should be offered to patients meeting the criteria for mild depression and individual CBT with antidepressant medication should be offered to patients meeting the criteria for moderate or severe depression (NICE, 2004, amended 2007)

UK government has highlighted CBT as the treatment of choice for those diagnosed with depression or anxiety and Health Secretary Alan Johnson promised a budget of £170 million in 2007 for the expansion of mental health therapies with the specific proviso that only evidence-based treatments such as CBT would be funded (Vaughan, 2007)

Given the widespread popularity of this therapeutic method, many (vulnerable) individuals are exposed to these cognitive techniques

Cognitive therapy has the potential to encourage individuals to think in ways that are more beneficial to society

Positivity, happiness and wellbeing have all been promoted as major targets for the government and cognitive therapy could be used as a method of encouraging these emotional states

Arguably, this could be interpreted as a means to control a dissatisfied general public in the face of increasing global and national difficulties

SUMMARY

Client–therapist rapport: core conditions are critical, rapport is a desirable strategy for developing a collaborative therapeutic relationship, but cognitive techniques are also essential because relationship rapport alone is necessary but not sufficient; collaborative empiricism is crucial to correct faulty cognitive processes experienced by the client

Problems with cognitive therapy: who decides which cognitions to test? What happens when problem cognitions are logical? Is it ethical to change minds?

THERAPEUTIC TECHNIQUES IN COGNITIVE THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- discuss methods used to validity test cognitions
- discuss the use of Socratic questioning in guided discovery

Validity testing cognitions

Cognitive validity refers to the accuracy of the thought

Some people hold cognitions that are demonstrably inaccurate

For example, a clinically underweight woman who believes that she is overweight holds an inaccurate cognition

In these cases, the validity of the cognition can be tested directly

Some people hold cognitions that are subjective and inconclusive

For example, a woman of average appearance who believes that she is ugly

In these cases, the validity of the cognition may not be tested directly, but weight can be added to the opposing argument

In therapy, the client is encouraged to test cognitions for validity to identify distortions (Beck, 1975)

Identifying distortions for validity testing

Distortions are not problematic *per se* but can cause problems in some cases

For example, an individual who believes that he has heard the voice of God offer words of comfort during difficult times might be classed as someone who is harbouring an irrational or illogical cognition; however, this belief is psychologically healthy because it does not lead to problems

In contrast, an individual who believes that he has heard the voice of God order him to kill someone might also be classed as someone who is harbouring an irrational or illogical cognition and this belief is likely to be dangerous and lead to problems

Cognitive therapy is used only for those cognitions that may be causing some problems

For example, a client suffering depression who holds the cognition that everybody hates him might benefit from testing this cognition for validity with the aim of reducing depressive symptoms

Defending the thought

Cognitions can be tested for validity by asking the client to defend the thought

- Client must provide evidence to support their belief in that cognition
  - For example, a client suffering depression who holds the cognition that everybody hates him might be asked to show evidence that everyone hates him
- The therapist will constantly question client to reveal the distortions and faulty logic until eventually the client must accept that the cognition is invalid
  - For example, a client suffering depression who holds the cognition that he is useless and everybody hates him might be asked 'what about your mum/girlfriend?', etc.
- Invalid cognitions can be discarded and new valid cognitions can take their place

TESTING COGNITIONS

Consider the following examples of potentially faulty cognitions and try to devise methods for testing these ideas against reality. It is important to consider the potential outcomes of these tests and how the client could be assisted in understanding these outcomes.

I need to drink to get through my day. I have a really stressful job and a couple of glasses of wine in the evening help me to wind down. Nothing else really does the trick. If I don't drink, I am going to be even more stressed and I won't be able to cope.

I can't go out because everyone will stare at me. And they will think that I look stupid and ugly. They will know that I am a freak and they will laugh at me. It is safer for me to stay at home.

I don't give presentations. I am just too nervous. If I try to talk in front of people, I will start to panic. I will go bright red and then my voice will go squeaky. I will start to feel hot and dizzy and I might pass out. That would be horribly embarrassing.

If I go up something high, I will start to panic and then I will definitely fall. My legs are not strong enough to support me and they will start to shake. I will not be able to stay safe and I could end up dying.

People think that I am stupid. I don't want to go to school because the teachers think that I'm thick. They talk down to me and treat me like an idiot. There's no point anyway because I'm not going to pass any of my exams.

I am rubbish at everything. I always mess it up. The world would be a better place if I was not here at all.

Experimentation

Cognitions can be tested for validity by asking the client to conduct an experiment

For example, a depressed client might be encouraged to test his belief that he always fails at everything in order to replace this faulty and irrational cognition with a more logical and rational cognition

Client who has recently lost his job might suffer depression because he believes that he is a failure

Therapist might encourage the client to test this belief that he is a failure by exploring past experiences and trying new experiences

He might be asked to consider previous tasks and list times when he has succeeded in order to determine whether he really does always fail at everything

He might be asked to try a new skill (such as cooking a particular meal) to determine whether he really does fail at everything

Socratic questioning

Socrates the teacher

Socrates was an educator in ancient Greece

Socrates taught Plato and Aristotle

Plato wrote much of his work because Socrates rarely committed his methods and theories to paper

Socrates appreciated the value of making the pupil find his or her own answers

Socratic questions are designed to encourage the learner to discover the answers by engaging in deeper thought about the issue in question

Socratic questions will help to establish critical thinking skills

Types of Socratic question

Six types of Socratic question

Questions aiming to clarify thinking

Ask the speaker to explain in more detail

For example, can you rephrase that?

Questions aiming to probe assumptions

Ask the speaker to question his or her own assumptions

For example, how could you verify or disprove that?

Questions aiming to establish evidence

Ask the speaker to provide real evidence or rationale

For example, can you give me an example of this?

Questions aiming to explore alternative perspectives

Ask the speaker to consider the argument from the viewpoint of another person

For example, what is the counter-argument for this?

Questions aiming to investigate possible implications

Ask the speaker to consider the possible outcomes

For example, what are the possible consequences of that?

Questions aiming to question the question

Ask the speaker to consider why that question was asked in the first place

For example, why was that question important?

SOCRATIC DINNER PARTY

Imagine that you are hosting a dinner party. Choose six guests from history, literature, film, etc. and list their names in the table below. Now try to think of some questions that you would like to pose to each of your guests. How could you use the Socratic method to explore these areas of interest?

Guest	Questions

Consider how your questions could guide the speaker towards disclosure. Is there a risk to this level of guidance?

Guided discovery in therapy

Socratic questioning can be used to guide discovery during the therapeutic process

> Effective guided discovery uses Socratic questions that the client has the knowledge to answer, draws the attention of the client to relevant information that exists outside current focus, and shifts the client from the concrete to the abstract so that the new conclusions can be used to re-evaluate previously held cognitions (Padesky, 1993)

Padesky (1993) identifies the following 'good' Socratic questions for therapy

> Have you ever been in similar circumstances before?
>
> What did you do?
>
> How did that turn out?
>
> What would you do now that you didn't know then?
>
> What would you advise a friend to do?

Padesky (1993) highlights the importance of Socratic questions in guided discovery

> Four stages to the guided discovery process
>
> 1. Gaining information
> 2. Listening
> 3. Summarising
> 4. Synthesising
>
> This cycle can be repeated throughout a session in order to guide the client towards discovering more about his own cognitions
>
> > *Client*: My kids hate me. I am just rubbish at everything.
> >
> > *Therapist*: You sound really angry about that. Have you managed to do anything at all this week? (*Gaining information to validity test the cognition*)
> >
> > *Client*: Not really. I was at work all week so I couldn't get to do anything with them at all.
> >
> > *Therapist*: So did you go to work every day this week? (*Listening has led the therapist to pick up on the fact that the client has been to work all week*)
> >
> > *Client*: Yeah. I had to finish a project for a customer so I was in every day until late. It meant that I couldn't get home before eight all week.
> >
> > *Therapist*: So you successfully went to work every day and managed to complete a project for a customer? (*Summarising to draw attention to the main points of the revelation*)
> >
> > *Client*: Yeah.
> >
> > *Therapist*: I am just feeling a little confused about how this all fits together. You mentioned earlier that you are rubbish at everything but how does this fit with your success in getting this project done? (*Analytical questions to encourage the client to apply the new information to the original cognition*)
> >
> > *Client*: Okay, I guess it doesn't. I am not rubbish at everything. I am just finding it hard to balance work and home.
> >
> > *The overgeneralisation distortion has now been identified and corrected to allow the client to now focus on the real problem*

SUMMARY

Validity testing cognitions: cognitive validity is the accuracy of thoughts, some thoughts are demonstrably inaccurate, some thoughts are subjective; therapy identifies distortions for validity testing (only those causing problems); validity testing is done by working to logically defend the thought (indefensible thoughts must be faulty) or experimentation (experiments to test the thought)

Socratic questioning: questions aiming to clarify thinking, probe assumptions, establish evidence, explore alternative perspectives, investigate possible implications, and question the question; guided discovery to lead the client to their own answers; Socratic questions used to aid guided discovery by gaining information, listening, summarising and synthesising

CASE STUDY DEMONSTRATING COGNITIVE THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- appreciate the application of cognitive therapy from the cognitive approach in a therapeutic setting

Recorded session: therapy in action

This chapter is accompanied by a recorded therapy session which is available for viewing.

This session lasts for one therapeutic hour (50 minutes) and it is presented as the initial session in a new therapeutic relationship. Prior to this session, the client will have completed an initial assessment questionnaire and the therapist will have read this paperwork to ensure familiarity with the case. This completed assessment and a full transcript of the recorded session are available.

No actors are used in this session. The client was one of the authors and the problem presented was genuine. The therapist is an experienced practitioner in the field. The only 'fake' aspect of this recorded session is that the client did not really seek therapy and this is not really the first session of a series of therapeutic contacts.

After the conclusion of the session, the therapist is invited to answer a few key questions about the session. This question and answer session lasts no longer than 10 minutes. The transcript of this session is available.

Presentation of Fay Short: history and symptoms

Fay Short has an acute fear of heights. She has recently experienced high levels of stress at her workplace and some associated physical symptoms (including headaches and backache). However, her primary reason for seeking therapy is a desire to overcome an intense fear of heights. She has previously experienced person-centred therapy for a short space of time over ten years ago, but this was unrelated to her current fears. She would like to overcome these fears during therapy so that they no longer impact on her ability to take part in activities that involve heights.

Therapy session: analysis

The therapy session can be sectioned as follows.

- Introduction
  - Introduction outlining basic contractual details, especially the limits of confidentiality
  - Focus specifically on the problem presented by the client and an outline of the intended coverage in the session (map of the session)
  - Invitation to ask questions
- Story
  - Client is invited to explain how the problem began, how the problem has developed over time, and how the problem impacts on current life
  - Focus on thoughts experienced during experiences of the problem, including rating the strength of the cognition
- Goals
  - Cognitive therapy is explained in terms of how thoughts can influence our feelings and behaviours
  - Explain how thoughts can be tested for validity to reduce problem cognitions
  - Identify a specific goal for therapy in the near future
- Ending
  - Outline the expected duration and content of future sessions (map of the sessions)
  - Explain that homework will be set each session, such as experiments to test validity of thought processes and diaries to record thoughts
  - Set readings as an initial homework
  - Reflection on how the session was experienced by the client
  - Invitation to return for future sessions

Key questions to consider in relation to this therapy session

How could the nature of this client be understood from the cognitive perspective?

- What does the client appear to think about herself?
- What does the client appear to think about the world?
- What does the client appear to think about the future?
- Does the client hold a positive or negative view of herself, the world and the future?
- Does the client hold thoughts based on available evidence?
- Is there any indication of any cognitive distortions?
- How do the thoughts held by the client impact on her feelings and behaviours?

What is the nature of the therapeutic relationship in this cognitive therapy session?

- Is the therapist honest and open?
- Does the therapist have a good rapport with the client?
- Does the therapist challenge any cognition held by the client?
- Are there any positive or negative effects of challenges to the cognition held by the client?
- Does the therapist suggest any experiments to test cognitions?

Which cognitive techniques are demonstrated in this therapy session?

- Which Socratic questions does the therapist use?
- How does the client respond to Socratic questions?
- How does the therapist encourage the client to identify cognitive distortions?
- How does the therapist encourage the client to defend the cognitive distortions?
- Does the therapist test any cognition for validity and what is the outcome of this test?

Personal experience of the client

I felt a little bit anxious about this session because it was my first experience of cognitive therapy as a client (and I was a little worried about being filmed disclosing potentially personal information!). However, my fears began to reduce as soon as I started talking to Keith. We established a good rapport and I felt comfortable sharing my experiences with him. He seemed to be genuinely interested in my story, and this was contrary to my expectations (I had previously experienced person-centred therapy and thought that cognitive therapy might be less relationship focused and more business-like).

I shared my story with Keith at the start of the session and I quickly forgot about the camera. I found his questions both engaging and frustrating at times – I often wanted to talk about how something 'felt' but he kept drawing me back to my thought processes. Although this was initially rather challenging, I did begin to see the value of this focus as he started to explain the way that my thoughts can impact on my feelings. His explanations in the later part of the session were particularly helpful and I think that I gained a good understanding of how my thoughts are helping to maintain my problem.

During this session, I realised several things for the first time. In particular, I realised that one of my primary concerns about heights relates directly to my own self-confidence. I have a firm belief about myself – I will look stupid if I panic – and this thought leads me to evaluate myself in a negative way. This means that I feel a strong need to constantly maintain control over myself and the wider world, and I recognised that this is both irrational and impossible. This increased understanding of my own fears has helped me to understand my anxiety a little better, and I now realise that it is only by facing this fear that I might be able to 'prove' to myself that it is not always 'stupid' to lose control. Alongside my low self-confidence and need for control, I also realised that I hold a more legitimate fear of falling. This seemed very sensible to me at the start of the session (and I guess that I believed that this was why my phobia was justified). However, during the session, it became clear to me that my fear is far more exaggerated than is reasonable in the circumstances.

At the end of the session, I had a clear understanding about cognitive therapy and a road plan for the future sessions. I was confident about my ability to tackle my concerns and I actually felt quite eager to get started on some of the readings suggested in the session.

—*Fay Short*

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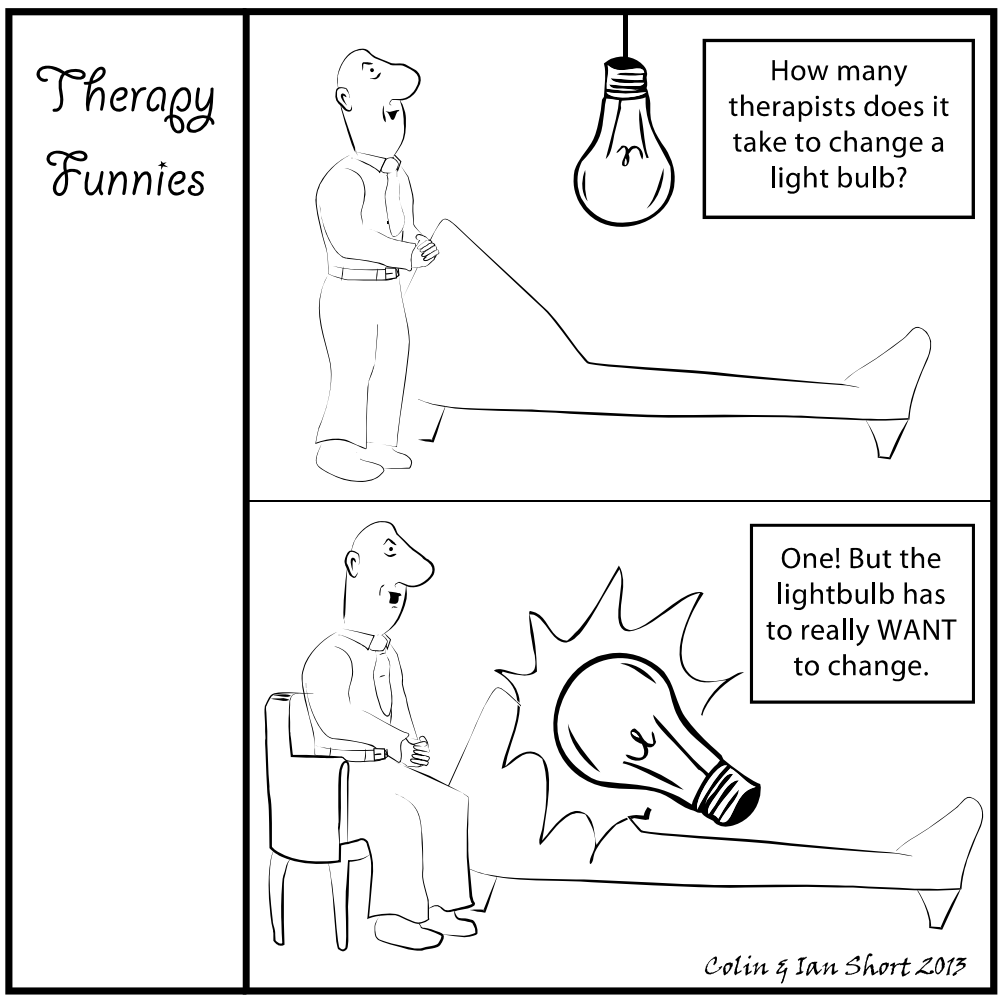
Chapter 6

Integrative and eclectic therapies



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INTRODUCTION TO INTEGRATIVE AND ECLECTIC THERAPIES

This chapter aims to introduce the reader to several integrative and eclectic therapies currently practised across the world. The terms 'integrative' and 'eclectic' are used here to describe any therapy that combines two or more approaches in order to produce a novel therapeutic method (please refer to Chapter 1 for more information on integration and eclecticism).

LEARNING OUTCOMES

By the end of this chapter, you will be able to:

- describe and evaluate a range of different counselling and psychotherapy methods and techniques
- explain the origin of each therapy and indicate how they relate to the core approaches in psychology

GESTALT THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- explain that Gestalt therapy is an integrative therapy combining psychodynamic and humanistic approaches
- discuss the history of Gestalt therapy with a brief biography of Friedreich 'Fritz' Perls
- explain the importance of holism and completion
- discuss phenomenological, field and existential perspectives
- explain how we have awareness of ourselves and the environment
- discuss the disturbances of the contact boundary
- explain the paradoxical theory of change
- discuss the dialogical relationship between therapist and client
- explain how language can be grounded in the here and now
- describe and evaluate experiential learning used in therapy
- appreciate the application of Gestalt therapy in a therapeutic setting (case study)

Integration of humanistic and psychodynamic approaches

Gestalt therapy is founded in the humanistic approach, but the founder arrived at the therapy through the psychodynamic approach

Integration of person-centred and psychoanalysis therapies

Often described as an existential-humanistic approach

Awareness, choice and responsibility are the underlying aims

While Gestalt therapy is not goal directed in the same way as some other approaches (for example, cognitive-behaviour or solution-focused therapy), it does have some general aims

Fritz Perls focused on raising awareness

Awareness of self, others and the environment

Awareness of self will lead to natural change

Awareness in the here and now of experiencing will ultimately lead to greater choice

Awareness of the present will also allow for previous unfinished business to emerge, and this can be brought into the present and worked with through experiments and exercises in the therapeutic exchange

Laura Perls focused on contact and support

Modern Gestalt therapists focus on the development of the relationship, as this is the vehicle through which change is achieved

Fritz did Gestalt therapy a disservice in his years of 'hot seat' confrontation

The infamous session documented with Gloria is interesting, but not representative of modern Gestalt therapy

Modern Gestalt therapy encourages clients to take responsibility for what they think, what they feel and how they behave

Client encouraged to be mature enough in relationships to be responsible for themselves and not to take on responsibility for others (thoughts, feelings and behaviours)

Client encouraged to be able to ask for what s/he wants and to be able to help others

Development of Gestalt therapy

History of Gestalt therapy

Established in the 1940s by Friedreich and Laura Perls

'Gestalt' is from the German for shape/pattern/configuration

Gestalt therapy originated from the psychodynamic approach (Perls was trained in psychoanalysis and the first book introducing Gestalt was described as a revision of Freud's theory)

Use of dream analysis and focus on the disturbances of the contact boundary draws from the psychodynamic approach

However, Gestalt is essentially an existential humanistic approach

Focus on the subjective experience of the individual in the here and now with recognition of the client draws from the humanistic approach

'Gestalt therapy is an existential-phenomenological approach and as such is experiential and experimental' (Perls, 1992)

Focus on the process (what is happening here and now in the therapeutic situation)

Focus on awareness of self, others and what lies between

First book to introduce Gestalt therapy was written by Perls in 1942 as *Ego, Hunger, and Aggression: A Revision of Freud's Theory and Method* – subtitle was changed in 1966 to *The Beginning of Gestalt Therapy*

The term 'Gestalt therapy' was coined by Perls, Hefferline and Goodman (1951) in their book *Gestalt Therapy: Excitement and Growth in the Human Personality*

New York Institute for Gestalt Therapy was established for workshops and seminars in the home of Fritz and Laura Perls in New York in 1951

Workshops and seminars were then established throughout the country and the APA convention held a special intensive Gestalt workshop in 1954

Laura was an often-overlooked major influence in the development of Gestalt therapy

While Fritz was the charismatic showman, Laura was the organiser and supporter

'Without the constant support from his friends, and from me, without the constant encouragement and collaboration, Fritz would never have written a line, or founded anything' (Perls, 1990)

Brief biography of Friedreich 'Fritz' Perls (1893–1970)

'I do my thing and you do your thing; I am not in this world to live up to your expectations; and you are not in this world to live up to mine; you are you, and I am I, and if by chance we find each other, it's beautiful; if not, it can't be helped'
—Perls (1969, Gestalt Prayer)

Who was Friedreich 'Fritz' Perls?

Perls founded Gestalt therapy with his wife in the 1940s

Emphasised the importance of the 'here and now' within the therapeutic relationship

New type of existential humanism

Early years

Born on 8th July 1893 in Berlin, Germany

Raised in a 'disturbed' family surrounded by fighting

He claims that his mother beat him with carpet-beating rods, but she failed to break his spirit (instead, he broke the rods)

Education

Expected to study law (following in footsteps of uncle) but actually studied medicine before and after serving time in the army in WW1

Qualified with an MD and worked with injured soldiers

Career

Trained in psychoanalysis in Vienna, and undertook analysis with William Reich

Reich went on to develop Reich body analysis

Focused on psychoanalysis and established a psychoanalytic training institute in South Africa after Hitler came to power

Travelled for some time as a psychiatrist but eventually settled in the US to write up his theories of therapy

Published *Gestalt Therapy* in 1951

Co-founded the New York Institute for Gestalt Therapy with Laura Perls in 1952

Moved from New York to Big Sur, California, and was involved at the Esalen Institute

Incorporated many Zen concepts into the therapy after staying at a monastery in Japan

Established a Gestalt community in Canada

In San Francisco in the 1960s, he became famous for his work and demonstrated his confrontational style in workshops

People were either very impressed or saw him as manipulative and meeting his own needs

He certainly seemed to enjoy the limelight!

Family

Married Laura Perls in 1930

Raised two children

As well as being his wife, Laura Perls also worked alongside her husband to develop and promote Gestalt therapy (Leibig, 1990)

Laura Perls was born on 15th August 1905 in Pforzhein, Germany

She became interested in psychology, aged 16, after reading about dream work in literature by Freud

She earned a doctorate in Gestalt psychology at Frankfurt University, then studied existential philosophy as a student of Buber and worked alongside Kurt Lewin

She was a member of the Berlin group of Gestalt psychologists

She was instrumental in keeping the New York Institute for Gestalt Therapy going while Fritz travelled the world demonstrating and promoting Gestalt

After the death of Fritz, she continued making contributions and promoting Gestalt therapy until her own death in 1990 at the age of 85, close to her birthplace in Germany

'I [Laura] was a Gestalt psychologist before I got into psychoanalysis. Fritz was an analyst before he got into Gestalt psychology. Sometimes it set us an insoluble conflict' (Perls, cited by Leibig, 1990)

Death

Died of heart failure on 14th March 1970 in Chicago, USA

Significant learnings

'Lose your mind and come to your senses' (Perls, 1970)

'Our dependency makes slaves out of us, especially if this dependency is a dependency of our self-esteem. If you need encouragement, praise, pats on the back from everybody, then you make everybody your judge' (attributed to Perls, n.d.)

'The only difference between the wise man and the fool is that the wise man knows he is playing' (attributed to Perls, n.d.)

'I am not in this world to live up to other people's expectations, nor do I feel that the world must live up to mine' (attributed to Perls, n.d.)

Holism and completion

'Gestalt' is a German word that has no direct equivalent in English. The closest words in English are *form*, *configuration* or *pattern*

Gestalt has a *wholeness*, a completion, about it

In order to understand something or someone, all aspects of the experience need to be taken into account and seen from the individual's perspective

Humans do not see things in isolation, but in context, and they have a predisposition to form patterns

We seek to complete 'Gestalts' and organise their experiences into meaningful 'wholes'

Anything not completed becomes 'unfinished business'

For example, we see a series of dots and we fill in the gaps to perceive it as a square

* * * *

* *

* *

* * * *

Or we see two familiar people shopping together, but we have never seen them spending time together before, so we fill in the gaps to make sense of the situation ('you'll never guess who I saw with . . .')

Figure and ground (Figure 6.1)

Rubin's vase (Rubin, 1915) is a well-known example of figure and ground

Depending on whether you see black or white as the 'figure', you will either see a vase or the profiles of two people

Figure comes to the forefront while the ground remains as the backdrop

Theory of figure and ground demonstrates how our attention works

This highlights how some things become important (figure) until they are completed when they can fade back into unimportance (ground)

For example, we see two familiar people shopping together, but we have never seen them spending time together before, so we fill in the gaps to make sense of the situation ('you'll never guess who I saw with . . .')

Seeing these two people together becomes the figure and everything else becomes the ground

Before seeing the two people, we might have been aware of being hungry (a previous 'figure')

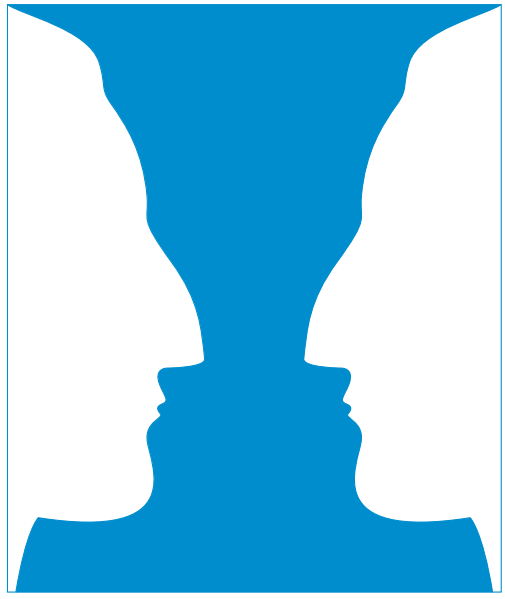

FIGURE 6.1 Figure and ground vase. Adapted from Rubin (1915).

But we are then distracted by them and forget about being hungry for a while

After we fill in the gaps (e.g. they must be in a relationship), we can go back to being aware of being hungry

Putting hunger on hold was temporary because, since we did not eat, it was an incomplete Gestalt (unfinished business)

Perspectives

Phenomenological perspective

Fritz and Laura Perls were greatly influenced by the works of phenomenological philosophers like Husserl, Heidegger and Sartre

Phenomenology looks at subjective experience from the first-person view (Woldt & Toman, 2005)

Each individual experiences the world through his or her senses

Experience is subject and creates individual realities

The best way to understand a client is to help explore his or her view of the world and to raise awareness in the present

The phenomenological perspective underpins humanistic therapy as it relies on understanding the client's world (empathy) rather than interpreting that world

This attitude towards the therapeutic encounter requires the therapist to be aware of her beliefs, attitudes, assumptions, etc. and to *bracket* off these personal experiences

Therapist must put personal experiences to one side to be available to meet the clients in their world, and to share the meanings they make of their moment-by-moment experience in the therapeutic encounter

The client is active in creating meaning, and the client and therapist together are active in co-creating the meaning of the 'what' and the 'how' of the relationship

Field theory perspective

Field theory is a concept developed by Lewin in the 1940s (Lewin, 1943)

Field theory explores interactions between an individual and the environment (total field)

The field is 'dynamic' in the sense that each part depends upon all other parts (Woldt & Toman, 2005)

This theory underpins the principle of holism – the therapist should view the client not as separate parts or separate from the environment

The client is the sum total of thoughts, feeling, behaviours, physical sensations, past and current experiences, social and cultural influences

We are not separate from our environment; we are part of it and influence it as it influences us

In Gestalt therapy, the immediate field is the therapeutic relationship in the moment-by-moment experience

Existential perspective

Existentialism states that the starting point of philosophical thinking must be the experiences of the individual (Woldt & Toman, 2005)

Existential choice lies at the heart of what it means to be human

We choose what we think, what we feel and how we behave

Therefore we have choice, maybe not for our situation but in how we respond to that situation

Individuals have to deal with four major themes or ultimate concerns in life (Yalom, 1980)

Isolation

Social beings have the need for interaction

Interactions give a sense of belonging

This interaction is important for self-esteem

Meaningless

The big 'meaning of life' question

We strive to make sense of, and give meaning to, our lives

Having a meaning provides the motivation to live fully

Mortality

We need to come to terms with the inevitability of death

- Some people turn to religion with the belief of an afterlife
- Family and friends may surround us, but we die alone

- Freedom
  - We give up some freedom in order to have relationships, thus we avoid isolation
  - But we still have freedom to choose the way we respond to situations (how we think, feel and behave)
  - Individuals have to act and take responsibility for their actions

Existential therapy places the subjective experience of the client central to the therapeutic process

Awareness of self and environment

Zones of awareness

There are three zones of awareness (Perls, 1969)

- The division of awareness into three zones is useful in helping the client focus on all aspects of self and identify the location of the focus of attention at any one time
- However, Gestalt therapy is holistic, thus it is noted that these zones are interconnected

Inner zone

- This is our inner world
  - It includes bodily sensations, tension, relaxation, emotions, etc.
  - It may not be within the awareness of the individual
- In therapy . . .
  - This information is not available to the therapist
  - But the therapist may help a client focus on this awareness by directing her to focus on what she can become aware of
  - For example, the therapist might say 'spend a minute becoming aware of what is going on for you inside – notice how you are sitting – notice where you have any points of tension – focus on your breathing – pay attention to your emotional state . . .'

Middle zone

- This is our thinking zone
  - It includes our images, fantasies and reactions
  - It makes sense of our experiences, giving meaning and understanding
- In therapy . . .
  - Therapist may help a client focus on this awareness by asking for specific client thoughts

For example, the therapist might say 'what are you thinking now?', 'what do you think of being asked to focus on what is happening inside?', 'notice any images you have, etc . . .'

Outer zone

This is about our contact with, or withdrawal from, the outside world

This is where an individual meets the environment

We need to be aware of what is going on in the world and how we relate to the world

In therapy . . .

Therapist may help a client focus on this awareness by directing focus towards the environment

For example, the therapist might say, 'be aware of what you can see around you – notice what sounds you can hear – how are you present in this relationship?'

Freedom of movement

In order to experience all aspects of life fully, we need to be able to move freely between these different zones

For example, we might be aware of how something in the environment (negative behaviour of a loved one) is affecting us internally (tightening in the stomach) and then make a decision (I need to challenge that behaviour)

However, it is possible to be predominantly focused in one area and this can lead to problems

For example, a teenager may be incapable of making decisions without knowing what other people would say about him or a woman may be obsessed with body image

Gestalt cycle

Also known as the cycle of awareness or cycle of experience

Gestalt cycle has seven stages (Clarkson, 1989) (Figure 6.2)

1. Sensation

Something emerges from the background and draws our attention

We sense and feel things

2. Awareness

These feelings and sensations come into awareness

Our awareness is raised

We become interested and notice the sensation

This then becomes a new figure

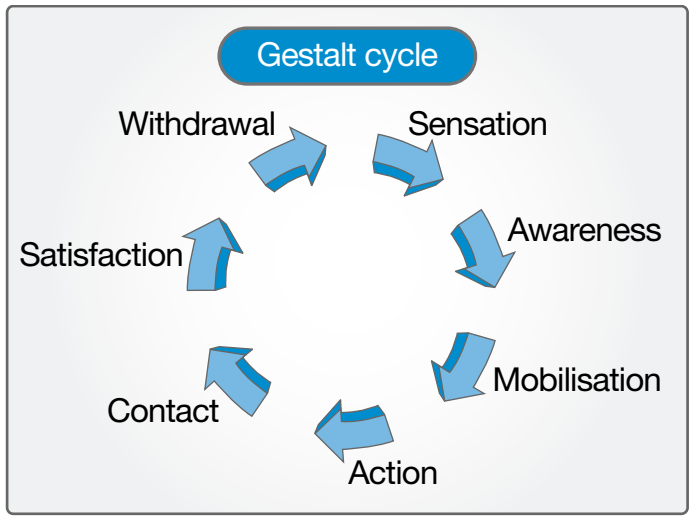

FIGURE 6.2 The Gestalt cycle. Adapted from Perls (1969).

3. Mobilisation
   - We prepare ourselves to take action in order to satisfy the need
   - Decisions are made and action is planned
4. Action
   - We engage in activity in order to satisfy the need
   - Action occurs at the boundary between self and the environment
   - It precedes full contact
5. Contact
   - We become fully involved in the action
   - We are fully engaged with others or the environment
   - At some point we become aware of a sense of completion
6. Satisfaction
   - We become more aware that the need for action has been satisfied. We can identify and enjoy the sense of satisfaction and completion
   - There is acceptance and integration and we are ready to withdraw
7. Withdrawal
   - We finish with the action and feel satisfied
   - We lose interest and let go of that particular figure
   - There is a move back to equilibrium
   - The next figure can now emerge for the cycle to begin again . . .

Cycles of Gestalt can be small or large and can run over a short term or long term

> For example, as you sit reading, you may become aware of a sensation in your body, maybe your back is stiff from sitting reading too long – you decide it is time for a break, maybe time to get up and walk around, stretch a little – you get up, engage in the activity of moving and after a few minutes feel ready to sit down again and resume reading – this is a complete simple Gestalt
>
> For example, a couple have been married for 22 years but the woman has slowly become aware of a growing dissatisfaction with the relationship. Having tried to ignore it, she now pays attention to the feeling, thinks and plans to do something about it – she may choose to leave the relationship, take action and fully engage in that process. This is a major life-changing event but is ultimately a good move for her and, after a time, she sees herself as a single woman again

If we move through all seven stages, we will complete the Gestalt and be available for the next figure to emerge

> This process of completion, or the desire to complete, is a key principle of Gestalt therapy, known as 'organismic self-regulation' (Yontef, 1993)
>
> However, at any stage, the cycle could become blocked by one of the boundary disturbances
>
> > If this happens, then the Gestalt is not completed and becomes unfinished business, retaining a hold over the individual

EXPERIENCING THE GESTALT CYCLE – PART 1

Stop reading for a minute and pay attention to what emerges from the temporary equilibrium.

Work through an example of a complete Gestalt – go through the seven stages and back to equilibrium, then try to describe your experiences below.

Sensation	
Awareness	
Mobilisation	
Action	
Contact	
Satisfaction	
Withdrawal	

Contact boundary

Distinction between self and environment

'A person exists by differentiating self from other and by connecting self and other'
—Yontef (1993)

Boundary between the self and the rest of the world allows us to . . .

- Establish a healthy sense of independence by acknowledging the boundary limits
- Make contact with the rest of the world by consciously reaching across the boundary

Firm yet permeable contact boundary

- Boundary must be firm enough to allow autonomy of thoughts, feelings and behaviours
- Boundary must be permeable enough to allow exchanges to occur between self and others

Disturbances of the contact boundary

Boundary can become too firm (impermeable) or too permeable (infirm)

- An impermeable boundary may lead to isolation
  - Isolation involves a loss of contact between the individual and the rest of the world
  - For example, the woman who insists that she does not need anyone in her life might be experiencing isolation
- An infirm boundary may lead to being unclear about the sense of self in relation to other
  - Blurred sense of self involves a tendency to blindly accept the opinions, beliefs and values of others
  - For example, the man who believes that he is stupid because his high school teacher once accused him of stupidity might have a blurred sense of self
- Seven patterns of boundary disturbance (Polster & Polster, 1988)
  - Retroflection
  - Introjection
  - Projection
  - Confluence
  - Deflection
  - Desensitisation
  - Egotism
- Conceptually similar to the ego defence mechanisms described in psychoanalysis
- Retroflection involves resisting the impulse to take action in the environment

Behaviour is internalised so that the individual behaves towards the self in the way that s/he would like to behave towards another OR in the way that s/he would like others to behave towards them

For example, self-harming in response to anger towards your boss

Introjection involves the internalisation of the environment without critical evaluation

Thoughts, attitudes and opinions of others are accepted and internalised without consideration of whether they are accurate and these introjected values are then imposed on the self

For example, student who believes that he is stupid because his high school history teacher once told him that he was stupid

Projection involves the attribution of unacceptable aspects of the self to the environment

Elements of the self which do not fit with the acceptable self-image are disowned and attributed to other people

For example, man who insists that everyone else is 'gay' because he is refusing to accept his own sexuality

Confluence involves a loss of the distinction between the individual and the rest of the world

Individual feels obligated or constrained to certain elements of the environment

For example, the woman who lives only for her children might be experiencing confluence

Deflection involves a failure to recognise or acknowledge aspects of the environment

Avoiding contact with or awareness of someone or some situation

For example, woman who wants to avoid confronting a colleague so instead does not look at her, talks about her rather than to her, remains vague when asked about the problem, and changes the subject when the disagreement is raised

Desensitisation involves a failure to attend to feelings and sensations

Feelings and sensations are neglected

This might be a bad thing; for example, we may ignore a chest pain because we are fearful of what it might mean

This might also be a positive thing; for example, when running a marathon we might become desensitised to the tiredness in our legs or a blister on our foot

Egotism involves dissociation with the environment to become a neutral observer

Egotism 'is characterised by the individual stepping outside of himself and becoming a spectator or a commentator on himself and his relationship with the environment' (Clarkson, 1989)

We are not truly in the experience if we are observing it from the outside

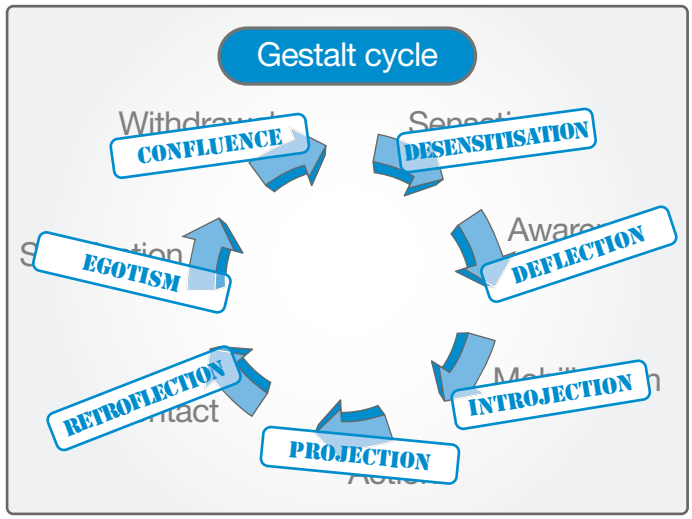

FIGURE 6.3 Boundary disturbances in the stages of the Gestalt cycle. Adapted from Perls (1969).

For example, people on holiday who only experience things through a video camera – they are so busy recording the holiday that they miss out on the first-hand experience of it

Disturbances at the contact boundary will interrupt a Gestalt from completing the Gestalt cycle

Seven boundary disturbances correspond to the seven stages of the Gestalt cycle (Figure 6.3)

Sensation may be interrupted by desensitisation

Awareness may be interrupted by deflection

Mobilisation may be interrupted by introjection

Action may be interrupted by projection

Contact may be interrupted by retroflection

Satisfaction may be interrupted by egotism

Withdrawal may be interrupted by confluence

Any incomplete Gestalts represent unfinished business

Many small Gestalts will be attended to in the short term

For example, we can delay going to the toilet when we are stuck in traffic by distracting ourselves for a short time, but the sensation will come back stronger to remind us that we need to take action

Other Gestalts are more complex

For example, we can delay the experience of bereavement by staying busy with funeral arrangements and this complex Gestalt takes a lot of energy to suppress while it remains incomplete

EXPERIENCING THE GESTALT CYCLE – PART 2

Think back to the previous activity – you stopped reading for a minute and paid attention to what emerges from the temporary equilibrium by working through an example of a complete Gestalt.

Consider the seven boundary disturbances – how could these have blocked your completion of this Gestalt?

How would it feel to be blocked in the completion of this Gestalt and how could you overcome the disturbances to ensure successful completion?

Go back through the seven stages with consideration of possible blocks at each stage, then write your experiences below.

Sensation blocked by desensitisation	
Awareness blocked by deflection	
Mobilisation blocked by introjection	
Action blocked by projection	
Contact blocked by retroflection	
Satisfaction blocked by egotism	
Withdrawal blocked by confluence	

Paradoxical theory of change

'Change occurs when someone becomes what he is, not when he tries to become what he is not'

—Beisser (1970)

This concept is about having respect for the self in the here and now

Acceptance of the self as it is right now, without any demand for change

Rather than trying to change to meet some preconceived idea of self, we need to become more aware and more accepting of ourselves in the present

Once we do this, we will have already have made a significant change and other changes will follow naturally

Change is not possible unless the individual is truly aware of the self as it currently exists

Awareness and acceptance of who you are now open the possibility of future change

We cannot change through coercion or pressure from others

Acceptance of the individual as we exist at that moment is essential in order for us to change

Consider attempting to move a stubborn donkey – the more you pull on his reins, the more he will insist on staying still

However, if you stop pulling and accept that he will stay as he chooses, then he will probably choose to move because he now has the freedom in which to make that choice

In therapy, the therapist must promote acceptance by focusing on the here and now

The therapist should demonstrate respect for who the client is at this moment in time, yet recognise his or her potential for growth and change

The therapist should provide a safe and non-judgemental environment focusing on how the client is at the present time in order to encourage the client to accept the self and open the possibility for future change

Topdog/underdog

Perls used the Gestalt terms 'topdog/underdog' to refer to intrapersonal conflicts similar to the Freudian concept of superego and ego

Clients come into therapy because they want to make changes in their lives, to overcome or manage problems, or to live life more fully

Gestalt therapists resist taking on the role of the instigator of the change process, as this sets them up as expert when they want to develop a relationship of equals

It would create a topdog/underdog relationship between therapist and client

'The Gestalt therapist believes that the topdog/underdog dichotomy already exists within the patient, with one part trying to change the other, and that the therapist

must avoid becoming locked into one of these roles. He tries to avoid this trap by encouraging the patient to accept both of them. One at a time, as his own' (Beisser, 1970)

Relationship between topdog and underdog shows the nature of internal conflict

The topdog is authoritarian, making demands ('You should/shouldn't' and 'you must/mustn't')

The internal response from the underdog is defensive, sometimes apologetic but always manipulative

The underdog has effective strategies for getting what it wants

The battle between topdog and underdog is ongoing

'The person is fragmented into controller and controlled. This inner conflict, the struggle between the topdog and the underdog, is never complete, because topdog as well as underdog fight for their lives' (Perls, 1969)

Dialogical relationship

Change process in Gestalt occurs through the relationship between client and therapist

Quality of the relationship is determined by the quality of the ongoing contact

Gestalt dialogue embodies authenticity and responsibility (Yontef, 1993)

Gestalt therapist does not manipulate the client towards a goal, but instead engages the client in dialogue

Directing a client towards a goal leads to a lack of responsibility, failure to accept ownership and dependency on others

Dialogue requires the therapist to show his or her true self and be honest (congruent) in the exchange

'Gestalt therapist says what s/he means and encourages the patient to do the same' (Yontef, 1993)

Four characteristics of dialogue (Yontef, 1993)

Inclusion

Inclusion is similar to the Rogerian concept of empathy

It is being aware of the experience of the client from their phenomenological viewpoint

It is without judgement, and so offers unconditional positive regard

This creates a safe place for the client to fully experience his or her own world

Presence

Presence in Gestalt is about the therapist being active in the process

The therapist will use appropriate self-disclosure statements to share thoughts, feelings and experiences, thus modelling to the client how to get in touch with

immediate phenomenological experiencing and helping the client to become more aware of self in the process

'In Gestalt therapy the therapist does not use presence to manipulate the patient to conform to pre-established goals, but rather encourages patients to regulate themselves autonomously' (Yontef, 1993)

Commitment to dialogue

Commitment to dialogue is about the contact between therapist and client

The dialogical relationship is central to the Gestalt therapeutic process, and again the therapist can model to the client how to 'do' the relationship

The therapist is committed to the process and to being an equal in that process – not intending to control or manipulate the client

Dialogue is lived

Put simply, the relationship between therapist and client is active and lived out in the here and now

There is immediacy about the ongoing relationship – it is done, not talked about

Along with verbal communication, non-verbal communication is also part of this lived dialogue

Immediate dialogue between client and therapist

'Gestalt therapy emphasizes that whatever exists is here and now and that experience is more reliable than interpretation. The patient is taught the difference between *talking about* what occurred five minutes ago (or last night or 20 years ago) and *experiencing* what is now' (Yontef, 1993)

The client may be talking about past events, but s/he should discuss them in terms of how s/he feels about them in that moment

The therapist does not assume the role of the expert to interpret the expressions of the client, but merely encourages the client to enhance self-awareness by remaining in the present and accepting responsibility for the self

Immediate dialogue within the client

The client may benefit from engaging in dialogue within the self

Two parts of the self may be split and the client can use various techniques to encourage a dialogue between these two halves

For example, the empty chair technique

Language in the here and now

Gestalt therapy aims to ground the client in the here and now

Language can often be used to distance the individual from the fullness of experience

Instead of owning the experience in the here and now, the individual can use language to make the experience seem distant or as though it is happening to someone else

For example, if the client talks of experiencing a feeling in the past, the therapist will invite her to bring it into the present ('Can you feel that anger here and now?') as it is only in the present that the feeling can be processed

Owning the expression

The client may suggest his or her own feelings, thoughts and behaviours, but use language to avoid ownership of these feelings, thoughts and behaviours

For example, in response to the question 'how did you feel when he shouted at you?', the client may say 'well, one often feels a little annoyed when someone is mean' or 'well, you tend to feel annoyed when someone is mean'

The therapist can draw attention to this use of language to encourage the client to change it to 'I' and own the expression

For example, the therapist might say 'who feels annoyed?' and encourage the client to say aloud 'I feel annoyed'

Accepting responsibility for the statement

The client may deny responsibility for an expression by asking the therapist to confirm that the expression is true

For example, the client might say 'but I shouldn't feel jealous, should I?'

The therapist can draw attention to the use of this language to encourage the client to change questions to statements in order to accept that s/he is responsible for the content of that statement

For example, the therapist might respond with 'turn that question into a statement' and the client would say 'I should not feel jealous' – now that they have accepted that statement as their own, they can consider whether they really believe it rather than relying on confirmation from someone else

Avoiding limiters (should and shouldn't/must and mustn't)

We all have our own internal rules and limitations, often introjected by parents and significant others in our childhood

For example, a client might say

You should work hard

You shouldn't get angry

I must try harder

I mustn't get upset

The therapist can challenge the client to see how this language affects him/her in the present

For example, the therapist might ask in response

In what ways does saying 'should' affect you?

What would happen if you did? (get angry or upset)

How does using the word 'must' limit you in the present?

Can you change 'you' to 'I'?

Establishing control over actions

The client may insist that external forces are responsible for his or her own behaviour by using words to imply that things are beyond their control

> For example, the client who feels angry that her husband never washes the dishes might say 'I end up washing them because I can't just leave them in the sink'

The therapist can draw attention to the use of this language to encourage the client to change 'can't' to 'won't' in order to accept that she is in control of her own actions

> For example, the therapist might question why the dishes 'can't' be left in the sink and encourage the client to rephrase the expression to 'I won't just leave them in the sink'

INCREASING AWARENESS OF LANGUAGE

Try to record yourself engaged in a natural conversation. This can be a bit tricky as you tend to behave differently when being recorded so it is often a good idea to leave the device recording for a long period of time and then focus on the middle section only.

Listen to the recording and note the number of times that you use language outside the here and now – are there any examples of you avoiding ownership by using second or third person, denying responsibility by asking questions, establishing limitations by asserting shoulds and musts, and claiming that external forces control actions by implying a lack of control?

Experiential learning in therapy

Staying in the moment

In Gestalt therapy, clients are encouraged to stay in the moment and not to escape into the past or the future

> For example, the client may be experiencing uncomfortable thoughts or feelings – rather than avoiding these uncomfortable sensations (maybe by changing the subject), the therapist would encourage the client to stay with the thought or feeling so as to fully experience it
>
> > Encouraging the client to be in the here and now eliminates anxiety. Perls (1969) argues that 'anxiety is the gap between the *now* and the *then*. If you are in the now you can't be anxious, because the excitement flows immediately into on-going spontaneous activity. If you are in the now, you are creative, you are inventive'

Clients can be encouraged to stay in the here and now by bringing the external issues into the real moment of the therapy

This is experiential learning

Client learns through real experiences in the moment

This can be done through experiments, exercises and dream work

Experiments

Experiments emerge from the dialogue between client and therapist

Therapist might notice something and invite the client to try an experiment on the basis of this observation

For example, the therapist might notice that the client is tense and invite him to try breathing deeply before repeating his previous statement in order to experiment with how that feels

Below is an example of an experiment

Dialogue begins after the therapist notices that the client has folded together his hands and is currently trying to pull them apart

Therapist: What are you doing with your hands?

Client: Well, my fingers are interlocked and I'm trying to pull them apart.

Therapist: Are they both equal or is one hand controlling the other?

Client: Well now you mention it, it feels like the right hand is in control.

Therapist: What would the left hand want to say to the right hand about this?

Exercises

While a Gestalt therapist may be creative and design experiments for the client in the here and now of the therapeutic relationship, there are also well-established techniques that are frequently used

These techniques are known as 'exercises' and trainee therapists are taught how to use these exercises in work with clients

Exercises are ready-made techniques preplanned to encourage therapeutic progress

The therapist can use these techniques with a client if it is deemed appropriate in that moment

Common exercises include internal dialogue exercises, reversal exercises, exaggeration exercises and repetition exercises

Internal dialogue exercises (empty chair)

Instead of running an internal dialogue, the client is encouraged to externalise the dialogue by using an empty chair to house the other person

This technique can be used to allow the client to converse between the different parts of the self, practise talking to someone who is not present, or explore the world from the viewpoint of another person

For example, a client who suddenly lost her mother might be encouraged to imagine that her mother is in the other chair and be given an opportunity to say all of the things she wanted to say to her (and perhaps answer for her mother too)

Reversal exercises

Clients are encouraged to try being the polar opposite of what they feel that they actually are

This technique can be used to encourage the client to accept different parts of the self by letting them try out these unfamiliar roles in a safe environment

For example, a client who insists on always being cheerful might pretend to be sad for a while to experience how it feels

The client may also be asked to try reversing roles

This technique can be used to encourage the client to view the world from different perspectives

For example, the therapist might ask 'if I had your problem, what would you be saying to me?' and the therapist and client could then role play this reversal of roles

Exaggeration exercises

The client is encouraged to exaggerate certain movements, postures or expressions

This technique can intensify the feelings associated with the behaviours and help the client to experience them fully in the moment

For example, a client who is experiencing repressed anger and revealing it subtly by gently clenching her fist might be encouraged to clench her fist even harder in order to accept the feelings

Repetition exercises

The client is encouraged to repeat certain words or phrases

The effect of repetition is similar to exaggeration – this technique can intensify the feelings associated with the behaviours and help the client to experience them fully in the moment

For example, a client might say 'he makes me really angry' and the therapist may then invite the client to repeat the statement over and over, or louder, until the client experiences the feeling in the here and now, can express it and feel some relief and release from it

Below is an example of a combination of exercises

Dialogue begins after the client states that she is angry with her father

Therapist: You say you are angry with him. What do you really want to say to him?

Client: What I *really* want say to him?

Therapist: Why not? It's safe to try it out here.

Client: Well, I think I'd want to say 'Screw you'.

Therapist: Go on then, say it. Imagine that he is sitting over here in this chair. Say whatever you need to say to him.

Client: (*quietly*) Screw you.

Therapist: Louder.

Client: (*louder*) Screw you.

Therapist: Again.

Client: Screw you. Screw you. (*shouting*) Screw you, you bastard.

Dream work

Perls was trained in psychoanalysis so he was influenced by dream work as a way of accessing the unconscious

In psychoanalysis, dreams are analysed and interpreted

In Gestalt, dreams are explored, brought into the here and now, and the client is encouraged to give them meaning

Through this meaning different aspects of the dream, which are believed to be different aspects of self, may be integrated

Dreams are the royal road to integration

For example, a client recalls a dream in which he is going to give a talk to a local charity group but cannot get to the venue because of road works – unsure of which way to go, he asks a policeman for directions but, rather than being helpful, the policeman is critical of him not knowing the way, and not being prepared enough by setting off too late. As the dream continues other obstacles prevent the client from making progress

Stage 1: therapist would first let the client recall the dream as above

Stage 2: therapist would ask the client to relive the dream as if it were happening now, so the client may start: 'I am going to give a talk for a local group. I am driving along but now see that the road ahead is closed and that there is a diversion. I am aware of feeling anxious that I might be late . . .'

Stage 3: the therapist may set up dialogues between different characters or different parts of the dream – she could ask the client to expand upon the dialogue with the policeman or ask him to speak as if he were the road works (what would the road works be saying to you? what would you say back to the road works?)

Stage 4: as the exercise progressed, the client would be encouraged to come to some understanding of the dream from his own (not the therapist's) point of view

It is important to note that all parts of the dream are parts of the dreamer, so the client is himself and also the policeman, the road works, the diversion, the

other obstacles, etc. Understanding this will give the client the opportunity to become aware of, and to integrate, these different aspects of the self

Below is an example of dream work with a client

Dialogue begins after the client mentions being affected by a dream ('it hasn't left me')

Therapist: Okay, tell me about what happened in your dream.

Client: Well, the dream starts off with me walking along a track, through some pine trees. It was a bright day and I could hear the wind in the trees. Around a bend I came across two oak trees, one fallen across the track. I scrambled over the tree, it was a little difficult but I managed and carried on, but then wondered why they were there, two oaks in a pine forest.

Therapist: Have you any thoughts on what it might mean?

Client: Well, I guess there are some obvious things, the tree blocking the path might mean the things I have to deal with in my job . . . or in my life.

Therapist: Tell me the dream again, this time as if you were reliving it in the present tense.

Client: Okay. I'm walking down a forest track, pine trees on either side. It's a nice day and I can hear the wind in the tree tops, although it is still here on the track. I'm going round a corner and I see a big tree across the path. It is one of two oaks, the other standing next to it. They look out of place in the middle of all the pine trees and I am sad that it's fallen because the other is one left on her own.

Therapist: Talk now as if you were the tree still standing. What would she say?

Client: She would say something like 'I am big and strong. I give shade on sunny days and shelter from the rain. Animals live in me.'

Therapist: And what would she say to you?

Client: She would say 'and you can rest here too'.

Therapist: And you would reply?

Client: 'I can't, I can't stop. I don't have time.'

Therapist: Say that again.

Client: I can't stop. I don't have time. I never have time *(she becomes quiet and reflective)*. I never have time *(she begins to cry)*.

There is no assumption that doing dream work will lead to any new insights, but this example confirmed for the client the more obvious messages (overcoming obstacles, always busy) and the less obvious (feeling sad about never having time)

Revisiting the dream, the client identified the oak still standing as her mother – her mother lived alone and, remembering the dialogue, the client decided to spend more time with her

FINDING THE MEANING IN DREAMS

As I was starting to write this section on Gestalt therapy, I had a dream that I had entered a marathon. I was not prepared for the start of the race and it began before I arrived. I needed to change, find a safe place for my valuables, and find the starting point. I became increasingly anxious because the race was well under way, and I knew I needed to phone someone to tell them that I had not yet started and that they should not expect me at the finish at the agreed time. I rarely remember my dreams but this was very vivid. The meaning was not lost on me. I spared my co-author the phone call, got my writing shoes on, and started running . . .

Think of a recent dream – try to relive the dream by thinking through all of the different parts and characters. Move around the dream taking on the role of significant parts and setting up dialogues. Stop when you are happy that you have made some sense of the meaning.

Case study demonstrating Gestalt therapy

Susan, 42, came into therapy through the occupational health department of a local company in which she is the marketing manager. She had been advised to visit the therapist due to her difficulties managing her workload and her recent absences due to depression. Her managerial role is 'pressured' and she feels trapped in her day-to-day life. The therapy was limited to six sessions.

In the following exchanges, therapist denotes therapist, client denotes Susan. Confidentiality statement and the contract have been discussed. The session is joined just after the therapist has invited the client to begin talking. However, she seems reluctant to speak and sits staring at her folded hands.

Therapist: If you do not feel comfortable talking about what brings you here, perhaps you can tell me what you are feeling right now?

Client: I just feel anxious (*client has folded her hands together and begins trying to pull them apart*).

Therapist: What are you doing with your hands?

Client: Well, my fingers are interlocked and I'm trying to pull them apart.

Therapist: Are they both equal or is one hand controlling the other?

Client: Well, now you mention it, it feels like the right hand is in control.

Therapist: What would the left hand want to say to the right hand about this?

Client: I'm not sure. Maybe it would say 'leave me alone'. Just leave me alone.

Therapist: So say it now. Be your left hand and say what you want to say to your right hand.

Client: Leave me alone. Just stop everything and leave me. I need some peace. I need to rest (*client is quiet for a few moments*). It makes me think about a dream that I had. It wasn't really a scary dream, but it hasn't left me since I woke up.

Therapist: Would you be happy to explore this dream a little? We could talk about your dream as though it is happening right now and you could think about what it all means to you.

Client: Yes. I think that I could do that.

Therapist: Okay, tell me about what happened in your dream.

Client: Well the dream starts off with me walking along a track, through some pine trees. It was a bright day and I could hear the wind in the trees. Around a bend I came across two oak trees, one fallen across the track. I scrambled over the tree, it was a little difficult but I managed and carried on, but then wondered why they were there, two oaks in a pine forest. Further on I suddenly felt lost, I could still hear the wind but when I looked up the trees are not blowing. I realised that the sound was not the wind, it was the sea . . . and then I was on a long beach. I could see some people in the distance so I ran, but the beach was so long that it seemed to take hours. When I got to the people I saw that they were statues, standing separately, facing the sea. A bit like those statues by that what's his name . . . you know, he did the Angel of the North. Anyway, the tide was coming in so I walked back up the beach and found a cafe but it wasn't a cafe, it was like an abattoir or maybe a place where they package up meat. It was full of carcasses. I'm a vegetarian. I was just about to leave when an old man came out of a room and said that he knew me. I knew that he didn't, that he was lying, and I got scared and ran away. He was chasing me, and I was running through deep wet sand. I saw a lighthouse in the distance and knew if I could get to it I would be safe. That's it. I woke up. It was so vivid, I don't usually remember my dreams.

Therapist: Have you any thoughts on what it might mean?

Client: Well, I guess there are some obvious things, the tree blocking the path might mean the things I have to deal with in my job . . . or in my life. Having to run for so long, I always feel like I'm running just to keep up at work. But I don't know what the statues or the lighthouse mean, or who the old man was.

Therapist: Tell me the dream again, this time as if you were reliving it in the present tense.

Client: Okay. I'm walking down a forest track, pine trees on either side. It's a nice day and I can hear the wind in the tree tops, although it is still here on the track. I'm going round a corner and I see a big tree across the path. It is one of two oaks, the other standing next to it. They look out of place in the middle of all the pine trees and I am sad that it's fallen because the other is one left on her own.

(*At this point the therapist has choices – he could set up a dialogue between the client and the fallen tree, but as the client has already attached a possible meaning to this, and has indicated an emotional response to the tree still standing, he chooses to explore this.*)

Therapist: Talk now as if you were the tree still standing. What would she (*picking up on the client's use of 'her'*) say?

Client: She would say something like 'I am big and strong. I give shade on sunny days and shelter from the rain. Animals live in me.'

Therapist: And what would she say to you?

Client: She would say 'and you can rest here too'.

Therapist: And you would reply?

Client: 'I can't, I can't stop. I don't have time.'

Therapist: Say that again.

Client: I can't stop. I don't have time. I never have time (*she becomes quiet and reflective*). I never have time (*she begins to cry*).

(*The dream work stops but the client is keen to continue later in the session after exploring issues around the pressures of her job – the dialogue picks up at the point of arriving at the statues.*)

Therapist: Now have a dialogue with one of the statues. What does the statue want to say?

Client: I am always here, always looking to the horizon. I feel nothing but am slowly being eroded by the wind and the sea. I'm rusting with salt water.

Therapist: And what does that mean to you?

Client: I think it's the part of me that is growing old. I've always been me, but I'm aware that some days I feel old, tired. I see younger people coming into the same types of job as mine, and I remember being their age, keen, full of life. I'm scared of rusting, eroding away.

Therapist: And the horizon?

Client: Ah yes, that fits. Always looking for the perfect life, just beyond my sight and out of reach.

Therapist: And the lighthouse?

Client: As I get closer I see it is no longer used, no light. It was once a beacon of safety, now it has outlived its usefulness. And as I say that I'm thinking of the fallen oak, and from nowhere I'm thinking of my dad. He died six years ago. I haven't thought about him for ages, but I know that the lighthouse and the oak are him (*she sobs*). He was wonderful, I still miss him.

Therapist: And the lighthouse would say to you?

Client: (*quietly*) It's okay. I'm still here in your dreams. I can still light your path, even if others can't see it.

Therapist: And you would say?

Client: I love you dad (*client begins to cry and she is unable to continue for a while*). But also I am just so angry with him.

Therapist: Are you feeling angry right now?

Client: Yes. He let me down. He left me on my own and I had to deal with everything without him.

Therapist: You say you are angry with him. What do you really want to say to him?

Client: What I *really* want say to him? It is hard to say. You shouldn't think that way about your father. Especially when he is gone.

Therapist: Who shouldn't think that way?

Client: I guess . . . I mean, I shouldn't.

Therapist: Can you say that?

Client: I shouldn't think bad things about my father (*she pauses*). But why shouldn't I? If that's how I feel. And that *is* how I feel.

Therapist: If you don't want to say it, then that's okay. But it's safe to try it out here if you want to explore how it feels to say it.

Client: Well, I think I'd want to say 'Screw you'.

Therapist: Go on then, say it. Imagine that he is sitting over here in this chair. Say whatever you need to say to him.

Client: (*quietly*) Screw you.

Therapist: Louder.

Client: (*louder*) Screw you.

Therapist: Again.

Client: Screw you. Screw you (*shouting*). Screw you, you bastard. You left me all alone with her.

The therapist noticed that Susan had referred to her mother for the first time here. Further exploration revealed that Susan had suffered verbal abuse at the hands of her mother throughout her childhood – she would frequently criticise her and this led to her feeling worthless. The therapist tried to keep this discussion in the room (rather than focusing on events in the past) by asking how the she felt today about her mother. Eventually, Susan was able to work through her issues towards her mother by writing her a letter.

ANALYSING THE CASE STUDY

Try answering the following questions using information about Gestalt therapy.

- What type of boundary disturbance might Susan be experiencing?
- Where is Susan being blocked in her cycle of awareness?
- Which parts of Susan are topdog and which parts of Susan are underdog?
- How does the therapist keep the discussion in the here and now?
- How is Susan encouraged to own the expression and assume responsibility for her thoughts?
- What experiments does the therapist use?
- What exercises does the therapist use?
- How is Susan helped through dream work?

SUMMARY

Humanistic and psychodynamic: Gestalt is founded in the humanistic approach but the founder arrived from the psychodynamic approach

Development: established in the 1940s by Perls (Fritz and Laura), first book to introduce Gestalt therapy by Perls in 1942, term 'Gestalt therapy' was coined in 1951, New York Institute for Gestalt Therapy was established for workshops and seminars in the home of Perls in New York in 1951

Holism and completion: see patterns as wholes, seek to complete things to avoid unfinished business; figure and ground highlight how we attend to things (figure) until they are completed then they fade from attention (ground)

Perspectives: phenomenological, field theory, existential

Awareness: zones of awareness (inner, middle, outer), freedom of movement between zones; Gestalt cycle (sensation, awareness, mobilisation, action, contact, satisfaction, withdrawal)

Contact boundary: distinction between self and environment, boundary should be firm yet permeable; disturbances lead to isolation or confluence, seven patterns of disturbance (retroflection, introjection, projection, confluence, deflection, desensitisation, egotism); disturbances correspond to Gestalt cycle

Paradoxical theory of change: must accept self in here and now for change to be possible; topdog/underdog

Dialogical relationship: authentic and responsible; inclusion, presence, commitment to dialogue, dialogue is lived; immediate dialogue within client and between client and therapist

Language in the here and now: owning the expression, change 'it' or 'you' to 'I'; accepting responsibility for statements, change questions to statements; avoiding limiters, change shoulds and musts to wants; establishing control over actions, change 'can't' to 'won't'

Experiential learning: staying in the moment; experiments, born of dialogue; exercises, ready-made preplanned techniques, internal dialogue empty chair exercise, reversal exercise, exaggeration exercise, repetition exercise; dream work, exploring meanings to allow integration of different parts of self, royal road to integration

TRANSACTIONAL ANALYSIS

LEARNING OUTCOMES

After reading this section, you will be able to:

- explain that transactional analysis is an integrative therapy combining psychodynamic and humanistic approaches
- describe the history of transactional analysis with a brief biography of Eric Berne
- discuss the Parent, Adult and Child of the ego-state model
- discuss the roles of positive strokes and negative strokes in reciprocal, crossed and covert transactions
- discuss the payoffs in the games people play
- describe therapeutic methods for creating the antithesis of games
- discuss the life positions and life scripts, highlighting the significance of early childhood messages
- map the process of therapy through the five key stages
- appreciate the application of transactional analysis in a therapeutic setting (case study)

Transactional analysis: humanistic and psychodynamic approaches

Transactional analysis (TA) is grounded in the humanistic and psychodynamic approaches

Integration of person-centred and psychoanalysis therapies

'Transactional analysis is a theory of personality and a systematic psychotherapy for personal growth and personal change' (International Transactional Analysis Association, 2013)

On the difference between TA and other theories, Berne said (with a smile)

'I think most other therapies . . . talk about thinking and feeling. Our question to the patient is not what do you think or how do you feel, but what are you going to do about it? . . . There's a story, that I think is a little unfair, but anyway it sort of illustrates what I mean, where a patient came in one day and said to the psychiatrist "by the way, I killed my wife last night and hid her body in the closet". And some people might say "ah ha, now we've got something to work with, what's your interest in closets?" Whereas we might say "why are you killing your wife?"' (Berne, 1966)

TA has applications in a number of areas

Theory is wide ranging

Theory of personality

- Theory of communications
- Theory of human development

TA has the main aim of autonomy (Berne, 1961)

- This means behaving, thinking and feeling in response to the reality of the here and now rather than responding to script beliefs
  - In order to move towards autonomy we need to consider three connected areas: awareness, spontaneity and intimacy
- Awareness
  - In order to behave, think and feel in response to the here and now, we need to be able to live in the present
  - We often spend time remembering the past and anticipating the future, not fully engaged in the present
  - We bring the past into the present though transactions, games and replaying familiar patterns
  - We are playing to gain strokes – these reinforce our life position and our early childhood script decisions and, when we have these scripts, we are not fully engaged in the present
- Spontaneity
  - Being spontaneous involves exercising choice in the present
  - It may feel risky because we are not playing out familiar games
  - Once we become aware of the present, we need to respond to it with awareness and through choice
- Intimacy
  - Intimacy involves being real in relationships
  - Being open and honest and not hiding behind the mask that we often present to the outside world
  - Speaking our real thoughts and showing our true feelings exposes us to others and makes us vulnerable, so we rarely do it
  - However, when we do show our true selves, we experience a richer, more meaningful relationship
    - For example, when we fall in love
  - This also happens in the intimacy of the therapeutic relationship
- Autonomy is a desirable aim but the journey towards it is challenging
  - It challenges us to become aware of the way our childhood decisions influence us in the present
  - It challenges us to break free from these old ways of behaving, thinking and feeling and to respond more appropriately to the here and now
  - It challenges us to take risks, to make ourselves vulnerable in order to achieve more meaningful, intimate relationships with self and others

Development of transactional analysis

History of transactional analysis

Introduced by Berne as an 'extra-Freudian' method of treating disturbance

- Often regarded as a logical step forward from psychoanalytic therapy, but the therapy arguably draws more on the humanistic approach
- Focus on interactions between people rather than unconscious drives

International Transactional Analysis Association established in 1964

- Approach began to be popular amongst practitioners, but still regarded as an offshoot by many researchers and purists

Controversial due to non-technical terminology

- Extremely popular with the general public
  - *Games People Play* (Berne, 1964) has sold more than five million copies
- Often regarded as pseudoscience by critics
  - Fails to present evidence for ego-states or games

Brief biography of Eric Berne (1910–1970)

'Awareness requires living in the here and now, and not in the elsewhere, the past or the future'
—Berne (1964)

Who was Eric Berne?

- Berne founded TA in the 1960s
  - Emphasised the importance of interactions between individuals and the games that people play within these interactions

Early years

- Born on May 10th 1910 and raised in Canada as Eric Bernstein
- Father died in 1921 and mother raised her two children alone

Education

- Qualified with an MD from McGill University in 1935

Career

- Studied psychoanalysis during his residency in psychiatry at Yale
- Initially worked at the San Francisco Psychoanalytic Institute under Erik Erikson, but began to diverge from traditional psychoanalysis
- Served in the Army Medical Corps during World War II, gaining the rank of major – discharged in 1945
- Focused specifically on the nature of intuition and social interactions
- Developed the term transactional analysis as a label for his style of therapy in the 1950s

Published *Transactional Analysis in Psychotherapy* in 1961

Published bestseller *Games People Play* in 1964

Founded the International Transactional Analysis Association in 1964

Published *What Do You Say After You Say Hello?* in 1972

Family

Married three times and raised four children

Death

Died on July 15th 1970 following a heart attack

Significant learnings

'The moment a little boy is concerned with which is a jay and which is a sparrow, he can no longer see the birds or hear them sing' (Berne, 1964)

'Each person designs his own life, freedom gives him the power to carry out his own designs, and power gives the freedom to interfere with the designs of others' (Berne, 1972)

Ego-state model of personality

Human personality is composed of three ego-states (Figure 6.4)

Structural model states that we are always in one of three ego-states at any point in time (Berne, 1964)

Parent

Child

Adult

Functional model states that we are always in one type of that ego-state (i.e. *how* we are in that ego state)

Nurturing Parent

Controlling Parent

Free Child

Adapted Child

Adult

Parent

Known as the exteropsyche state

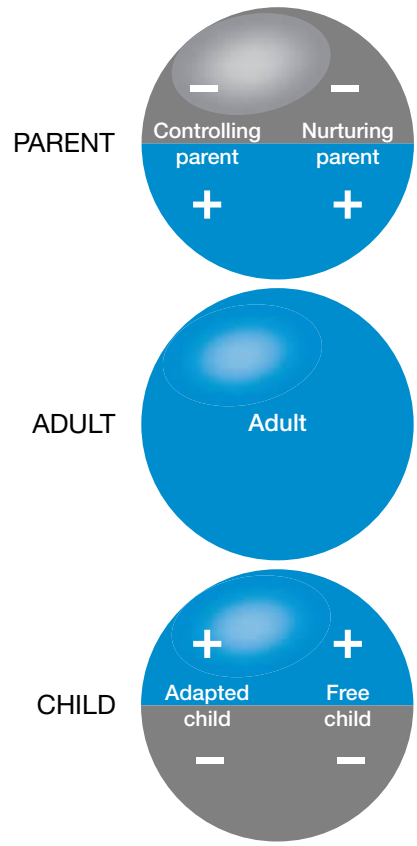

FIGURE 6.4 The three ego-states. Adapted from Berne (1964).

Perceiving reality from the perspective of a parent and responding to others in a parental manner

Influenced by the perceived actions of the real parents

Parent state can be caring and supportive (Nurturing Parent)

For example, I may adopt the Parent state by encouraging people to wrap up warm and worrying that they may catch a cold

Parent state can be critical and demanding (Controlling Parent)

For example, I may adopt the Parent state by criticising the way people dress and trying to make them wear something else

Child

Known as the archaeopsyche state

Perceiving reality from the perspective of a child and responding to others in a child-like manner

Influenced by own behaviours as a child

Child state can be spontaneous and fun loving (Free Child)

For example, I may adopt the Child state by spontaneously singing loudly in my car while waiting at traffic lights

Child state can adapt to rules or the demands of others (Adapted Child)

For example, I may adopt the Child state by asking permission from my friend before buying a new pair of shoes

Adult

Known as the neopsyche state

Perceiving reality from an objective viewpoint and responding to others in a logical manner unfettered by emotion

Influenced by logic, reason and a scientific understanding of the world

Adult state is usually located in the here and now and involves an appreciation of current feelings and logical interpretations of thoughts and beliefs

For example, I may adopt an Adult state when I recognise my desire to go to a party the night before a major exam but make a decision to revise at home instead because I know that the grades on the test will be important to me in the future

Adult ego-state is the rational, logical state and does not divide into different types of Adult

Ego-states do not determine actual roles

Ego-states are labels for the way that we behave, think and feel

Always indicated by a capital letter: Parent, Child, Adult (P, A, C)

Ego-state model is also known as the PAC model

Our roles indicate our actual relationships with others

I might be a parent to someone or a child to someone

Ego-states can conflict with our actual roles

For example, in the television sitcom *Absolutely Fabulous*, Saffy adopts a Parent state towards her actual parent Eddy and Eddy adopts a Child state towards her actual child Saffy

Ego-states are not intrinsically good or bad

Nurturing Parent may appear positive (supportive and caring), but this could become negative if the support was given to the extent that it encourages dependency

Controlling Parent may appear negative (critical and demanding), but this could become positive when we need someone to take charge of a situation

Functional ego-states can be beneficial

Parent state allows us to care for others

Child state allows us to relax and have fun

Adult state allows us to deal with problems logically

Dysfunctional ego-states can be damaging

Parent state can be controlling and overcritical of the self

Child state can be demanding and risk-taking

Adult state can make us overanalyse everything and not let us have any fun

GUESS THE EGO-STATE

Look at the scenarios below to identify the ego-states of the protagonists.

- Woman suffering with alcoholism who cannot resist spending all of her money on vodka as soon as she gets paid each week.
- Man scolding an employee in a condescending tone for arriving at work five minutes late.
- Teenager who refuses to go to a party the night before an exam because he wants to get good grades.
- Boy who helps his father into bed when he returns home drunk after a night on the town.
- Girl who looks up through her eyelashes and talks in a baby voice to persuade her boyfriend to buy her a new pair of shoes.
- Elderly person who carefully writes down her expenditure each week to ensure that it is covered by her pension.

We should move freely between different ego-states

Ego-states are not personality types used to categorise individuals

> Instead, they are designed to inform about specific interactions in that moment so our ego-states should change from moment to moment

It is possible for us to move between ego-states internally and externally

> For example, you are being playful by teasing your partner (Child) but you suddenly become concerned when she seems upset (Parent)
>
> It is cold and dark in the middle of winter and the alarm clock goes off – you lie in bed running an internal dialogue: 'just five more minutes (Child), but I really should get up now (Parent), but it's cold and I'm tired (Child), I've really got to go to work (Parent), okay if I don't get up now I'll be late (Adult)'

Exclusions occur when we exclude one or two ego-states from our interactions

It would be healthy to be able to move freely between ego-states and for us to have access to all ego-states as appropriate

However, we may have a tendency to use one or two particular ego-states most of the time

> For example, a reckless individual may frequently use the Child

We may also have a tendency to use one particular ego-state when we are with certain people or in certain situations

> For example, a normally mature woman might go into the Child whenever she is in the presence of her father
>
> For example, a confident adult might go into the Child when confronted by a police officer

If a person is mostly in Parent and Adult, with very little Child

> S/he will lack spontaneity and will find it difficult to have fun
>
> S/he will be serious and logical, and be very aware of the rules

If a person is mostly in Child and Parent, with very little Adult

> S/he will not be able to check reality and balance judgements
>
> S/he is likely to have wider mood and behaviour swings

If a person is mostly in Adult and Child, with little Parent

> S/he will not have a good awareness of the rules
>
> S/he may act on impulse but will be able to rationally assess the outcomes
>
> S/he may make the rules up as s/he goes along

ARE YOU EXCLUDING STATES?

Think about your interaction with someone that you know and try to answer the following questions.

- Do you rely on one or two ego-states a lot of the time?
- Do you use one ego-state when you are with this certain person?
- Do you use one ego-state when you are in a certain situation?
- Do you notice a particular ego-state in that person?
- How does your ego-state impact on your relationships with this other person?
- How does the ego-state of the other person impact on the relationship with you?
- What would happen if you changed your ego-state when relating to this other person?

Contaminations occur when we think that we are in one ego-state but are actually in another

We want to be rational and sensible so we assume that our responses are Adult

> However, we frequently respond in Parent or Child but justify our actions/thoughts/feelings by believing that we are Adult

If a person thinks s/he is in Adult but is really in Parent

> S/he is bringing beliefs that were introjected from parents in childhood into the present and thinks they are true
>
> For example, the belief that all Scotsmen are mean, all blondes are dumb, love will only hurt you, you cannot trust anyone, etc.

If a person thinks s/he is in Adult but is really in Child

> S/he is bringing beliefs that were formed in childhood into the present and thinks they are true
>
> For example, the belief that I am no good at maths, I will never get on in life, I must keep trying, they are all laughing at me, etc.

Therapy enhances understanding and control of our own ego-states

When we analyse our own ego-state in therapy, we can begin to understand some of our unhealthy behaviours, thoughts and feelings

> For example, a client who has a tendency to revert to Child state when dealing with an authority figure might benefit from recognising this in order to strengthen his or her own Adult state
>
> Using the ego-state model to understand different aspects of a person's personality is known as 'structural analysis'

Therapy also gives us choices about moving into and between ego-states, rather than going into them automatically

It would be healthy to be able to move freely between ego-states and for us to have access to all ego-states as appropriate

Transactions and strokes

Transactions refer to the interactions between people

Transactions involve the communication between two or more people

People communicate in one of the ego-states

Transactions are based on whether the state addressed is the state that eventually responds

Three types of transactions

Complementary (or reciprocal)

Crossed

Covert (or ulterior)

Complementary transactions

Each individual is addressing the ego-state of the other individual

Child to Child: Let's skip this lecture and have lunch instead

Child to Child: Okay, shall we get a burger?

Does NOT mean that the states have to match – simply means that the other person is in the state that you are addressing

Parent to Child: You have to get this work done at once

Child to Parent: But I want to go out tonight

Complementary transactions are often simple transactions

Adult to Adult: Hi, how are you?

Adult to Adult: Fine thanks, and you?

First person invited the second person to respond from Adult and the second did respond from Adult so the transaction is complementary

Ego-state addressed is the one that responded

Communication feels comfortable and may continue

Crossed transactions

Each individual is addressing a different ego-state to the one held by the other individual

Child to Parent: Shall I buy some new shoes?

Adult to Adult: It is your money so you can make your own decision

- Crossed transactions can have a negative impact on relationships
  - Child to Child: Let's skip this lecture and have lunch instead
  - Parent to Child: You're always doing that, do you want to fail?
  - Student invited her friend into Child, but the response was from Parent so the transaction is crossed
  - Communication is not comfortable, unless one person changes ego-states to get the communication back to complementary
  - For example, since the friend responded in Parent and invited the student into Child, the conversation could only become complementary again if the student replies from Child ('Aw but I don't want to go')
- When we have a communication that does not feel good, it is likely that there has been a crossed transaction

Covert transactions

- Each individual has a different implicit ego-state and explicit ego-state
  - Explicit Adult to Adult: We will need to stay late at the office tonight to finish this work
  - Explicit Adult to Adult: That is a very sensible idea
  - Implicit Child to Child: Let's stay behind and have sex
  - Implicit Child to Child: That sounds like fun
- Covert transactions occur when the message is on the explicit (or social) level but the real meaning is on an implicit (or psychological) level
  - Mother comes home from work to find a mess, points to a pair of shoes in the middle of the room and says 'What are these here?'
  - The answer, of course, is 'a pair of shoes' but the real meaning of the communication is 'pick these up and put them away'
  - It is likely that the person who receives the question ('what are these?') will respond by tidying the shoes away
  - Behavioural response holds the key to the real meaning of the transaction
- Covert communications are the basis for the playing of games

GUESS THE TRANSACTION

Look at the transactions below to identify which ego-states are being used and whether the transaction is complementary, crossed or covert (if covert, try to identify the underlying meaning).

Manager: You're late.
Worker: Sorry, it won't happen again.

Person 1: It's awful – the mess the bankers have got us into and they are getting bonuses!
Person 2: Yes, they should be ashamed of themselves.

Husband: I've had a heck of a day.
Wife: Huh, you should see what I had to put up with.

Wife: The bin's full.
Husband: Is it really?

Car salesman: This one is probably too sporty for you.
Customer: I'll take it!

Husband: Wrap up warm when you go out.
Wife: Aw but I wanted to wear my short skirt.

Wife: The weather is getting colder so we need to fix the boiler this weekend.
Husband: Aw but I wanted to go to the football match.

Strokes refer to the attention gained from other people

Every individual craves recognition, acknowledgement and attention

- Often we do not even care what type of attention we receive

Strokes can be verbal or non-verbal

- Verbal strokes
  - Other people give us attention by involving us in their conversation or referring to us when they speak
- Non-verbal strokes
  - Other people give us attention by doing things for us or to us

Strokes can be conditional or unconditional

- Conditional strokes
  - Other people give us attention only if we behave in a certain way
- Unconditional strokes
  - Other people give us attention irrespective of how we behave

Strokes can be positive or negative

- Positive strokes
  - Other people give us praise, compliments, kindness, love, etc. and we experience a positive feeling as a result
- Negative strokes
  - Other people give us criticism, punishment, cruelty, etc. and we experience a negative feeling as a result

We would always prefer to receive positive strokes, but we will settle for negative strokes if there are no other strokes available

For example, a child wants praise from his mother (positive stroke) but he learns that he is ignored when he is well behaved (no strokes) and receives attention in the form of punishment when he is badly behaved (negative strokes) – since some attention is better than no attention, he is constantly badly behaved

GUESS THE STROKE

Look at the scenarios below to identify whether the stroke is verbal/non-verbal, conditional/unconditional or positive/negative.

- Husband flashes his wife a withering look when she changes the channels on the television.
- Girlfriend receives a bunch of flowers through the mail with a card reading 'just because'.
- Mother spanks her child for misbehaving.
- Friend calls to say 'you are such a kind person' after receiving help moving to a new house.
- Boyfriend leans over to kiss his girlfriend when she brings him a cup of tea.
- Teacher says 'well done on this essay' to a student.

Therapy enhances our understanding of our own transactions and the strokes that we give and receive as a result of these transactions

When we analyse our own transactions in therapy, we can begin to understand some of the unhealthy patterns of relating

For example, a couple frequently engaged in crossed transactions might benefit from trying to create complementary transactions instead

When we analyse the strokes in our lives, we can begin to understand that some of our unhealthy tendencies are ineffective methods of gaining strokes and we can see how we impact on others by providing strokes

For example, client who has a tendency to give negative strokes and withhold positive strokes might benefit from realising how this impacts on her children

Games people play

Games involve a series of transactions leading to a predictable payoff

Games are sets of social behaviours that we habitually play when interacting with others without being consciously aware of our patterns

'A game is an ongoing series of complementary ulterior transactions progressing to a well-defined, predictable outcome. Descriptively, it is a recurring set of transactions . . . with a concealed motivation . . . or gimmick' (Berne, 1964)

Payoffs are the reason why we play games (Berne, 1964)

We are each seeking a payoff

Gaining attention

Gaining stimulation

Gaining confirmation that we are correct

Avoidance of anxiety

Avoidance of guilt

Avoidance of responsibility

Payoffs give us temporary satisfaction . . . but we ultimately feel dissatisfied and frustrated by the game

While games are being played, they are a source of strokes (Berne, 1964)

Payoff will give us a familiar, but negative, feeling

These 'racket feelings' originate in childhood and we set up a series of transactions (games) outside of awareness in order to experience the feelings

Racket feelings are familiar, predictable and safe . . . but ultimately negative

Different types of games (Berne, 1964)

Why don't you/yeah but

First player presents a problem and second player offers possible solutions to the problem, but the first player rejects every solution

Student: I'm worried that I'll fail the test.

Lecturer: Why don't you draw up a revision plan?

Student: Sounds good, but I never know when I'll be free

Lecturer: Well, why don't you try to revise for an hour a night?

Student: Yeah but I'm always tired after college and work.

Lecturer: Why don't you revise at the weekend then?

Student: Yeah but I have to work Saturdays and visit my mum on Sunday . . .

Payoff during the game for each player is the feeling that they are working towards a solution to the problem

If it weren't for you

Each player refuses to accept responsibility for their actions or the results of their actions

Wife: If it weren't for you, I could have a career by now

Husband: Well, if it weren't for you, I would have gone to university

Payoff during the game for each player is the feeling that their current problems are not their own fault

Now I've got you, you son of a bitch

First player sets up a trap for second player to allow the release of repressed anger

Mother: Where's your uniform?

Son: Upstairs in my room.

Mother: You said you were going to bring it down for the wash!

Son: I forgot.

Mother: You always forget. What about me? Running around after you, do you think I don't have a life too?

Payoff during the game for the first player is the release of emotions

Kick me

First player positions him/herself to receive negative strokes, often due to limited experience with positive strokes in the past

Daughter wears old clothes to go out

Mother: Why don't you make an effort, all your friends do?

Payoff for the first player is the receipt of negative attention

Why does this always happen to me?

First player positions him/herself to suffer multiple misfortunes

Employee takes on more work even though he is already under pressure to meet deadlines

He doesn't complete the task and gets disciplined

Payoff for the first player is inverse pride in the form of knowing that his/her misfortunes are worse than the misfortunes of other people

Karpman (1968) identified three positions in games (Figure 6.5)

Persecutor, Victim, Rescuer form a 'drama triangle'

Each player in a game will start in one position

Just before the end of the game there will be a switch in positions

Each player will end up with a negative feeling

Consider the 'Why don't you/yeah but' game

Player offering possible solutions to the problem is the Rescuer and the player seeking help is the Victim

Student: I'm worried that I'll fail the test.

Lecturer: Why don't you draw up a revision plan?

Student: Sounds good, but I never know when I'll be free

Lecturer: Well, why don't you try to revise for an hour a night?

Student: Yeah but I'm always tired after college and work.

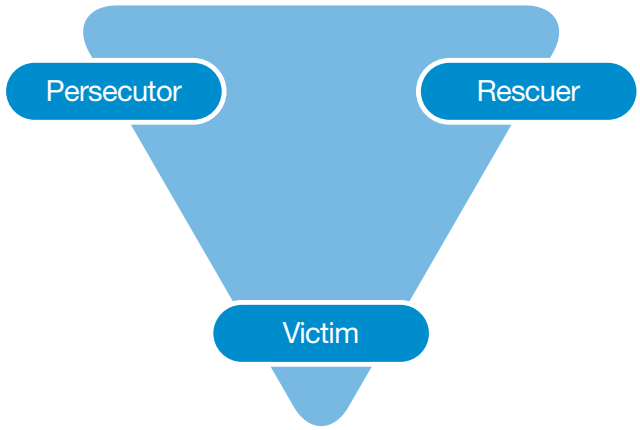

FIGURE 6.5 The drama triangle. Adapted from Karpman (1968).

Lecturer: Why don't you revise at the weekend then?

Student: Yeah but I have to work Saturdays and visit my mum on Sunday . . .

Student is the Victim and lecturer is the Rescuer

Both are getting some payoff from the game

Then the switch happens and the first player turns into the Victim and the second player turns into the Persecutor

Lecturer: Well, I don't know what else to suggest.

Student: So you can't help me at all then.

Lecturer is now the Victim and student is the Persecutor

Both feel dissatisfied with the outcome – familiar feelings may be helplessness (student) and uselessness (lecturer)

Antithesis of games

First rule of games

Once you know that you are playing a game, end the game

Refuse to take the opposing role in the game

For example, client adopts the Child ego-state by explaining how much she hates being married and you adopt the Parent ego-state by trying to give advice about improving her relationship – you realise that you are playing the 'why don't you/yeah but' game so you stop the game

How do you stop the game?

Move to the second rule . . .

Second rule of games

Break the game by depriving the actors of the payoff

Work out which game is being played, determine the payoffs for the actors (sympathy, vindication, etc.) and stop the payoffs

For example, client adopts the Parent ego-state by explaining that she has to do all of the work at home – you realise that she is playing the 'Kick me' game so you break the game by refusing to offer sympathetic responses

Break the game by responding from a different ego-state

Work out which ego-state is expected from the other player and adopt a different ego-state in response

For example, client adopts the Child ego-state by claiming that life is not fair and seems to want you to comfort him from the Parent ego-state – you realise that he is playing the 'Why does this always happen to me?' game so you break the game by adopting the Child ego-state instead

Life positions and scripts

Life positions are the basic views that we take about others and ourselves

Games reinforce our life position

One reason why we play games is to reinforce our position in life

Life positions are developed in childhood

We establish our views about others and ourselves based on early experiences

There are four life positions (Harris, 1967)

I'm OK, you're OK

I'm OK, you're not OK

I'm not OK, you're OK

I'm not OK, you're not OK

I'm OK, you're OK

A person in this position is confident, optimistic and engages well with life and others

This is the healthiest position from which to view self and others

No obvious games or racket feelings

This is the position that the therapist would assume in the therapy

I'm OK, you're not OK

A person in this position sees the self as superior to others

Critical of others (familiar ego-state Controlling Parent)

Often gets angry (racket feeling) with others

Starts all games as the Persecutor

I'm not OK, you're OK

A person in this position sees the self as inferior to others

Likely to appease and adapt (familiar ego-state Adapted Child)

Often strives to get away from others and feels depressed or afraid (racket feeling)

Starts all games as the Rescuer or Victim

I'm not OK, you're not OK

A person in this position sees the self and others as worthless and everything in the world as hopeless

Likely to avoid activities or maybe sabotage activities

Often feels pessimistic and hopeless (racket feeling)

Starts all games as the Victim

Life positions are important because they determine the racket feelings that we are likely to have, games we play, ego-states that we feel comfortable in, likely transactions that we have, and kind of strokes that we give and receive

All of these elements serve to consolidate our life script

Life scripts are the story of our lives (Steiner, 1966, 1974)

Life scripts refer to writing a life story with a beginning, middle and end

Much of the theory relating to transactions and strokes fits with a humanistic approach, but life scripts have a foundation in psychodynamic theory since they highlight the significance of childhood events, messages and decisions

Life scripts are created in childhood (Berne, 1964)

The basis of this script is written in infancy

Even before we can talk, we receive preverbal messages from our parents and significant others; for example, we will receive non-verbal strokes by being held close or neglected and we recognise facial expressions

When we have only a little language, we receive verbal messages that may reinforce these preverbal messages; for example, we will receive verbal strokes in the form of 'good girl' or 'well done'

More detail is added as we grow in childhood

Changes and additions will be made in adolescence

Scripts developed in childhood impact on adulthood

We are not aware of the influence that this script has on our lives

Early messages can be very powerful and hard to identify in adult life (Steiner, 1966, 1974)

Negative messages are known as injunctions

Injunctions will begin 'Don't . . .'

Positive messages are known as permissions

Permissions will begin 'You must/should/can . . .'

- Injunctions
  - Goulding and Goulding (1979) identified 12 injunctions that they saw repeatedly in their work
    - Don't exist
    - Don't be you
    - Don't be a child
    - Don't grow up
    - Don't make it
    - Don't do
    - Don't be important
    - Don't belong
    - Don't be close
    - Don't be well
    - Don't think
    - Don't feel
  - Injunctions often come from the Child ego-state of the real parents
    - Like all other people, your real parents will always be in the Child, Parent or Adult ego-state
    - In the Child state, your parents can give you injunctions
    - Injunctions start as preverbal messages and are likely to be delivered outside of awareness of the parent(s)
    - For example, one or both parents may neglect the baby, sending the message 'don't exist' (an infant will not have the ability to make rational decisions the way an adult can, so these decisions are likely to be more 'extreme')
    - Preverbal messages may then be reinforced by verbal messages later in infancy For example, the parent tells a child 'you were a surprise' or 'you weren't planned' or 'you were a mistake' or 'you were not wanted'
  - Injunctions influence our life script
    - We internalise (introject) these messages and they become part of our own Child ego-state
- Permissions
  - There are two types of permissions: rules and attributes
  - Rules and commands
    - For example, 'do what you're told', 'you should try harder', 'big boys don't cry', 'do your best'
  - Attributes
    - For example, directly to the child 'you're a good girl', 'you're not as clever as your sister'

- For example, indirectly to the child 'he's the naughty one', 'she's so good at looking after me', 'she's the quiet one'
- Permissions often come from the Parent ego-state of the real parents
  - Like all other people, your real parents will always be in the Child, Parent or Adult ego-state
  - In the Parent state, your parents can give you permissions
  - Permissions start as preverbal messages and are likely to be delivered outside of awareness of the parent(s)
  - For example, one or both parents may cuddle the baby whenever he is quiet, sending the message 'be quiet' (an infant will not have the ability to make rational decisions the way an adult can, so these decisions are likely to be more 'extreme')
  - Preverbal messages may then be reinforced by verbal messages later in infancy
  - For example, the parent tells a child 'be a quiet boy'
- Permissions influence our life script
  - We internalise (introject) these messages and they become part of our own Parent ego-state

Programs

- Alongside injunctions and permissions, we also receive information from our parents about how to do things
  - This information provides us with programs for how to interact with the world
  - For example, how to read, tie shoe laces, use a knife and fork, etc.
- Programs come from the Adult ego-state of our real parents and they are then introjected into our own Adult ego-state

In early childhood we receive thousands of verbal and non-verbal messages from parents and significant others

- Injunctions: restrictions and controls sent from the Child state of others and integrated into your own Child state
- Permissions: verbal rules, commands and attributes sent from the Parent state of others and integrated into your own Parent state
- Programs: useful 'how to . . .' messages sent from the Adult state of others and integrated into your own Adult state

These messages form the basis of the decisions that we make about self and others (life positions) and underpin the life script that we write for ourselves

- Scripts developed in childhood impact on adulthood
  - This life script is played out through our lives
  - We are not aware of the influence that this script has on our lives
  - Therapy would help a client become aware of things that are currently outside of awareness through script analysis

We are not always playing out our script – there are times when we are thinking, feeling and behaving appropriately in the here and now

Two factors that make it more likely that a person will enter a script (Stewart & Joines, 1997)

If the here-and-now situation feels stressful to the individual

If there is a similarity between a stressful event in childhood and the events experienced in the here-and-now situation

While the 'what' (the content) of the script is different for each individual, 'how' we live out the script (the process) falls into six main patterns (Berne, 1972; Kahler, 1978)

Always
Never
Until
After
Almost
Open ended

Always

A person with an Always script is likely to play out familiar patterns of behaviours (e.g. relationships) always with the same results ('why does this always happen to me?')

Alternatively s/he may always be doing the same thing (same job, same relationship)

S/he may have received strong consistent permissions in childhood to keep going, try hard to do well, make things work

In therapy, s/he could be challenged to consider how things could be different but, of course, she may also Always be doing therapy

Never

A person with a Never script never gets what s/he wants

S/he makes plans but never carries them out

S/he starts things but never finishes them

S/he may have received the injunctions 'Don't make it' and 'don't (do anything)' in childhood

In therapy, s/he should focus on a specific goal and be stroked for any success

Until

A person with an Until script will put things off

S/he will not enjoy herself until she has finished all her work

S/he will do the things she wants to do when the kids have left home

S/he may have received consistent permissions such as 'be perfect' and 'work hard' (do your homework, then you can play out) in childhood

In therapy she may be reluctant to commit to the process as she might not be a perfect client – she should be encouraged to work in the here and now, not to focus on the future

After

A person with an After script will not enjoy today for fear of tomorrow

His/her fun tonight will be spoiled by the thought that s/he will regret it in the morning

S/he may have good health but worry that s/he will become ill

S/he is likely to start a sentence with a positive then add a 'but' and a negative statement

S/he may have received strong injunctions such as 'Don't have fun' in childhood

In therapy, she should be encouraged to stop sentences (and thoughts) before the 'but' or change the sentence around so it ends with the positive element – therapy sessions should always end on a positive note

Almost

Kahler (1978) built on Berne's theory and divided the Almost script process into Almost Type 1 and Almost Type 2

A person with an Almost Type 1 script will get close to achieving their aim but will not quite do it

S/he will almost make it this time or almost finish this time

S/he will try really hard and will engage in activities to a point

S/he may have received 'try hard' but 'you'll never make it' messages in childhood

A person with an Almost Type 2 script does achieve their aim but does not take time to enjoy the achievement, as s/he has already got a new goal to focus on

S/he may well be very successful, getting promotion after promotion, but is rarely satisfied

In therapy, Type 1 should set small goals and achieve them whereas Type 2 should be encouraged to take time to experience the sense of achievement

Open ended

A person with an Open ended script will experience a lack of direction at a certain point in life

S/he might achieve something in the short term but then have a sense of 'what next?' before something else comes along

Bigger life events (e.g. children leaving home or retirement) may bring a bigger sense of 'what now?' and can result in anxiety or depression

In therapy, s/he would be encouraged to reframe this as a gift – s/he is free to write her own answer to 'what now?' free from the patterns present in the other script processes – s/he should work on small goals linked to larger long-term goals

Therapeutic process

Client–therapist relationship

Psychoeducational approach in which therapist and client take joint responsibility for the work

Therapist takes the life position 'I'm OK, You're OK'

- Provides the support needed for the client to move towards greater autonomy in their life
- Teaches the client about relevant theory
  - For example, the ego-state model, transactions, strokes and games

Clients will become more aware of their processes through this educational process

- With this awareness, they will be encouraged to explore being more spontaneous in relationships both inside and outside the therapy room
- Clients will experience intimacy in the therapeutic relationship

Lister-Ford (2002) identifies five stages in the process of therapy

- Stage 1
  - Client recounts his/her history
  - Therapist listens, reflects, questions to gather relevant information
  - Therapist teaches the relevant parts of theory when appropriate
  - Together they develop a safe, mutual therapeutic relationship
- Stage 2
  - Client becomes more aware of issues and the personal role in maintaining patterns and problems
  - Client notices the feelings that emerge during the process
  - Client begins to take more responsibility in the process
- Stage 3
  - Client begins to work through issues as feelings become stronger
  - Client begins to re-evaluate life
  - Challenging time in the process
- Stage 4
  - Client makes choices about how s/he wants to live with increased awareness
  - Client begins to challenge old script behaviours, thoughts and feelings
  - Client replaces old scripts with ones more appropriate to the here and now
- Stage 5
  - Therapy comes to an end
  - Client has more autonomy over her life
  - Therapist and client review the process and prepare for moving on

Active approach in which the therapist makes an initial assessment of the client's problem and contracts with the client to set effective goals

Treatment planning has three parts

Diagnosis

Assessing the problem and planning interventions

Contract

Agreeing the therapeutic goals with the client

Treatment direction

What techniques to use, for both short-term and long-term goals

Planning is ongoing and flexible throughout the process as new information emerges and as the client makes changes

Pragmatic in the interventions

Change in any part of the client's way of being changes the client

Interventions can be made at any point where they will benefit the client

For example, if a client has problems in communicating with figures in authority, she may be taught the ego-state model and how transactions work

This might bring about significant change for her and she might be happy with this change

The therapist will see the value in exploring her way of relating at deeper levels, to analyse script, parental messages, games, etc.

The short-term goal of improving relationships may be achieved

The long-term goal of becoming free from script is still in place

Further short-term interventions will be implemented to help the client become more aware and to spend more time making decisions from the Adult ego-state

Eventually the client may be able to remove old scripts and live in the here and now

Case study demonstrating transactional analysis

Roy, 33, came into therapy through the occupational health department of a local county council where he worked in health and safety. It was five months after the break-up of his third marriage. He was 'dissatisfied with life', wondering why 'it always happens to me'. In the aftermath of the ending of the relationship, he had begun to drink heavily again, and was now in danger of losing his job. The therapy was limited to six sessions, with the facility to extend, by agreement, in exceptional cases.

In the following exchanges, therapist denotes therapist, client denotes Roy. Confidentiality statement and the business contract have been discussed. The session is joined as Roy begins his story.

Client: I don't know where to start really. I think I'm here because my boss said that I needed help. I've missed a few days work recently and been late a few times, and she smelled drink on my breath and then made me go to occupational health, and here I am.

Therapist: Sounds like you don't really want to be here.

Client: *(hesitant, voice child-like, body language closed)* I don't really, talking about feelings to a stranger and all that . . . but if she's making me come I might as well give it a go.

Therapist: It's your choice. I don't want to work with someone who feels forced to come. You might find it useful to 'give it a go'. How about we decide at the end of the session?

Client: Okay *(pauses, then begins to talk, almost apologetically . . .)*. About six months ago my wife said she was leaving. Right out of the blue. I didn't see it coming, I thought we were doing okay. It's like . . . *(pause)* . . . it's just happened again.

Therapist: So it took you by surprise, but it's happened before.

Client: Yeah, twice. There goes my third marriage down the drain. Why does it always happen to me? I fall in love and then they leave me. I just never learn. I shouldn't trust them. All women are selfish bastards; they don't care about me, only about themselves. I should've listened to my dad. He hated them. 'Never trust a woman with your money or your life' he used to say. Anyway, after she left I just sat there, feeling sorry for myself I suppose. I managed to keep it together but felt crap, all sad and angry at the same time, so I phoned a few mates and we went for a drink. That was just over five months ago, and I've been drinking ever since.

Therapist: And the drinking, has that happened before as well?

Client: Yes, after both my other marriages. Only I don't want it to go the same way. First time, after my first marriage, I lost my job and then my house. I drank really heavily for nearly a year. Second time was shorter but worse. I got into trouble and was sent down. I'm not a violent man, but . . .

The therapist noted the way that Roy was describing his situation. His story was one of being badly treated by cruel and heartless women. The patterns emerging in the story suggested something else was going on in Roy's relationships. Later in the session, it emerged that Roy had actually begun to drink heavily during, not after, the marriages, and that when drinking, Roy would be verbally abusive to his partners. 'But I never hit them' he said. When asked about his childhood, he said his mother was a timid woman who often told Roy 'If it wasn't for you I'd have gone a long time ago.' He said he felt loved but not wanted by his mum. When asked to say more, he became quiet, reflective, almost tearful, and changed the subject. He spoke of being a frightened child. He remembers waiting for his dad to come home and would know his mood by the way he closed the front door. His father could be kind and loving, but could also be verbally abusive to Roy's mother and to Roy. Thursday nights were the worst. Roy's father was paid, in cash, on Thursdays and would come home drunk after the pubs closed. Most Thursdays Roy would lie in bed and listen to his father shouting, and then hitting his mother. 'But he never hit me', he said.

In subsequent sessions more of Roy's story emerged. The therapist encouraged Roy to take responsibility for his work in the sessions, and helped him identify patterns of behaviour, the roles he took in those patterns, the familiar feelings he experienced, the messages he received in childhood that influenced his thoughts, feeling and behaviours, and how all these linked to his script.

After the initial reluctance to engage, Roy found therapy useful, and was keen to gain insights. Translating this new awareness into behavioural change was harder, but Roy was well motivated to change.

Roy had the initial six sessions and then a further four, the maximum allowed under the employer scheme. At the end of the therapy, Roy felt that he had the knowledge and understanding he needed to make immediate changes (short-term management of problems), but knew that there was much to be done in order to free himself from his early childhood influences and decisions (longer term work to be free from script).

ANALYSING THE CASE STUDY

Try answering the following questions using information about transactional analysis.

- What ego-state was Roy in the early exchanges?
- What did the therapist do about this?
- What game is being played out in Roy's relationships?
- What position does he play and what position does he end up in?
- What position have his partners played and what position did they move to?
- What key script messages did he receive that may influence his relationships with women?
- What script process does Roy have?
- What do you think Roy's life position is?

SUMMARY

Humanistic and psychodynamic: main aim is autonomy (awareness, spontaneity and intimacy)

Development: introduced by Berne as an 'extra-Freudian' method of treating disturbance, International Transactional Analysis Association established in 1964, controversial due to non-technical terminology

Ego-state model of personality: Parent (critical and nurturing) perceives reality from the perspective of a parent and responds to others in a parental manner; Child (free and adapted) perceives reality from the perspective of a child and responds to others in a child-like manner; Adult perceives reality from an objective viewpoint and respond to others in a logical manner

unfettered by emotion; ego-states are not actual roles and are not good or bad; should have free movement between states; exclusions occur when we lack one or two states; contaminations occur when we are in a different state to the one we think that we are in; therapy enhances understanding and control of states

Transactions and strokes: transactions (complementary, crossed, covert) result in positive or negative strokes (verbal/non-verbal, conditional/unconditional)

Games: series of transactions leading to a predictable payoff; Why don't you/yeah but; If it weren't for you; Now I've got you, you son of a bitch; Kick me; Why does this always happen to me?; drama triangle, persecutor, victim, rescuer

Antithesis of games: first rule is stop the game; second rule is break the game by depriving player of payoffs and responding from a different ego-state

Life positions and scripts: positions include I'm OK, you're OK; I'm OK, you're not OK; I'm not OK, you're OK; I'm not OK, you're not OK; life scripts contain injunctions, permissions and programs; scripts are written in childhood, based on verbal and non-verbal messages received from parents and significant others; six patterns of playing out scripts (always, never, until, after, almost and open ended)

Therapeutic process: joint relationship responsibility, psychoeducational; five-stage process to increased awareness; treatment plan includes diagnosis, contract, treatment direction; pragmatic in interventions

RATIONAL EMOTIVE BEHAVIOUR THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- explain that rational emotive behaviour therapy is an integrative therapy combining the cognitive and behavioural approaches
- discuss the history of rational emotive behaviour therapy with a brief biography of Albert Ellis
- outline the ABCDE model
- discuss the dangers of musturbation: 'neurosis-inciting musts' of irrational beliefs
- describe and evaluate techniques for disputing irrational beliefs
- describe and evaluate techniques for cognitive restructuring
- describe and evaluate techniques for shame-attacking exercises
- appreciate the application of rational emotive behaviour therapy in a therapeutic setting (case study)

Rational emotive behaviour therapy (REBT): cognitive and behavioural approaches

Rational emotive behaviour therapy is grounded in the cognitive and behavioural approaches

- Integration of cognitive and behavioural therapies
  - Cognitive therapy and behavioural therapy are rarely practised independently any more
  - Instead, they are often integrated to produce a new form of therapy
  - Rational emotive behaviour therapy is one of these integrations

Development of rational emotive behaviour therapy

History of rational emotive behaviour therapy

Rational emotive behaviour therapy was founded in the 1950s

- Therapy initially introduced as rational therapy (RT)
- Rational therapy (RT) changed to rational emotive therapy (RET) in 1961
  - Irrational and rational thoughts were linked to emotion
- Rational emotive therapy (RET) changed to rational emotive behaviour therapy (REBT) in 1993
  - Irrational and rational thoughts and emotions were linked to behaviour
  - Earliest form of cognitive-behaviour therapy

History of REBT is the history of Albert Ellis

Brief biography of Albert Ellis (1913–2007)

'You largely constructed your depression. It wasn't given to you. Therefore, you can deconstruct it.'
—Attributed to Ellis (n.d.)

- Who was Albert Ellis?
  - Ellis founded rational emotive behaviour therapy in the 1950s
    - Introduced concept of emotion and cognition alongside behaviour therapy
    - First type of cognitive-behaviour therapy
    - Introduced the concept of irrational thoughts (decade before Beck) and explored their impact on a wide range of mental disorders – depression, sexual disorders, procrastination, anxiety, etc.
    - Established the non-profit Albert Ellis Institute in 1964
    - Author of over 600 papers and 50 books on REBT
    - REBT therapy was the earliest form of cognitive-behaviour therapy (CBT) – often described as the grandfather of CBT
- Early years
  - Born on 17th September 1913
    - Born in Pittsburgh and raised in New York City
  - Physically and emotionally distant family life
    - Father was a businessman – often absent from home on business
    - Mother was a housewife – emotionally absent from her children
    - Mother often failed in her duties as a mother and Ellis was usually left to care for his siblings
    - Ellis described his mother as a self-absorbed woman who frequently verbalised strong opinions on most things but failed to support her views with evidence (Ellis et al., 2005)
    - Ellis was the eldest of three children
    - Ellis recalls purchasing an alarm clock with his own money so that he could rise early enough to get his brother and sister ready for school (Ellis et al., 2005)
    - Parents divorced by the time Ellis was 12 years old
    - Ellis coped with the family tensions by immersing himself in books and became a studious teenage boy
  - Numerous childhood health problems
    - Eight hospitalisations (kidney disease, tonsillitis, streptococcus) between the ages of five and eight
    - Little support from his family during illness – few visits or efforts to console him (Ellis et al., 2005)
    - Instilled a strong sense of independence in the growing child
  - Experienced extreme fears of rejection in teenage years
    - Responded to his shyness and inability to talk to women by desensitising himself to rejection from the opposite sex

'I made verbal overtures to 100 different women sitting on park benches in the Bronx Botanical Gardens, got rejected for dating by all of them (one woman kissed me in the park, and made a date for later that evening, but never showed up!)' (Ellis, 1996)

Education

Graduated with a Bachelor of Arts degree in business from City University of New York in 1934

Initially intended to set up in accounting, make enough money to retire at 30, and then work on becoming a great American novelist

Unfortunately, the Great Depression resulted in poor business ventures and he had little success in fictional writing (Albert Ellis Institute, 2012)

Discovered a talent for non-fiction writing and focused on producing a treatise on sexual liberation – as many friends considered him an expert on sexual matters, he also began informal therapy at this time (Ellis et al., 2005)

Master's degree in clinical psychology from Teachers College, Columbia University in 1943

Doctorate in clinical psychology at Columbia University in 1947

Worked as a psychologist offering sex and family therapy in his own part-time private practice – psychology was unlicensed at that time in the US (Ellis et al., 2005)

Published several articles while working on his PhD

One article was a critique of paper-based personality tests (Ellis, 1946)

Trained as a psychoanalyst under Richard Hulbeck at the Karen Horney Institute

Extremely unusual for a non-MD to be accepted as a trainee

Although initially impressed with the effectiveness of psychoanalysis, he became quickly disillusioned with the lack of scientific support and empirical evidence for the methods (Albert Ellis Institute, 2012)

Career

Lecturer at New York University and Rutgers University in the late 1940s

Senior clinical psychologist at Northern New Jersey Mental Hygiene Clinic in the late 1940s

Chief psychologist of the State of New Jersey in 1950

Still maintained a private part-time practice in New York

Expanded his part-time practice to full time in 1952

His reputation as a sexologist helped increase the client base of his growing practice, and he was even involved as an expert witness in some legal cases (for example, cases against publishers of obscene materials) (Ellis et al., 2005)

- However, his reputation as an advocate of sexual liberation also hindered his ability to find a post as a lecturer in several major universities, including his own graduate school (Ellis et al., 2005)

Introduced rational therapy in 1955

- Abandoned psychoanalysis and focused on changing behaviour by confronting irrational thoughts and changing them to rational thoughts
- First paper presented to the APA convention in Chicago in 1956
- Published a seminal paper introducing the approach, 'Rational psychotherapy and individual psychology', in 1957

Established the Institute for Rational Living in 1959 (Ellis et al., 2005)

- Non-profit organisation aiming to educate the public
- Changed rational therapy to rational emotive therapy in 1961
- Irrational and rational thoughts were linked to emotion
- Published *Reason and Emotion in Psychotherapy* to integrate emotion into rational therapy in 1962

Purchased a mansion on 65th Street in Manhattan to house the Albert Ellis Institute in 1964

- Lived in the top floor of this building and continued working with patients
- Dedicated most of the profits from his work back to the institute, allegedly taking only $12,000 a year salary alongside the promise of lifetime accommodation and medical care (REBT Network, 2006)

Changed rational emotive therapy to rational emotive behaviour therapy in 1993

- Irrational and rational thoughts and emotions were linked to behaviour
- Earliest form of cognitive-behaviour therapy
- Published 'Changing rational emotive therapy (RET) to rational emotive behavior therapy (REBT)' to explain the change to include behavioural elements in 1995

Later life was marred with conflict and controversy (REBT Network, 2006)

- Unfortunately, following a serious illness, the board of trustees refused to continue paying for medical expenses in 2004 – he was able to continue living and working at home only by using his own savings to continue to pay for at-home nursing care
- Friday night workshops were a regular event at the institute for 40 years – workshops involved live demonstrations of REBT with audience members as clients
- Unfortunately, the board of trustees banned him from using the facilities at the institute for these workshops in 2005 and he was forced to relocate to a nearby rented building

Ellis sued the institute in 2005 after the board of trustees voted to remove him from the board and suspend him from all professional duties

Supreme Court concluded that Ellis should be returned to the board in 2006, but the trustees continued to limit his involvement and prevent his professional practice

Ellis eventually rejected the institute and claimed that the work of the institute no longer reflected his own mission statement

Despite threats from the institute to sue his advocates for trademark infringement, many fans of Ellis continued to teach and practise REBT according to his original guidelines

Family

Married three times (Ellis et al., 2005)

First marriage ended with an annulment

Second marriage ended with an amicable divorce

Third marriage to Debbie Joffe in 2004 lasted until his death

Debbie Joffe adopted the role of wife, caregiver and assistant, and she continued publicising REBT after his death (REBT Network, 2006)

No children

At least, no legitimate children . . .

Death

Died of natural causes at the age of 93 on 24th July 2007

Worked continually until his death

Until his final illness at the age of 92, he worked at least 16 hours every day (REBT Network, 2006)

One of his final messages to his supporters from his hospital bed that year was 'work harder'

Significant learnings

'Do, don't stew!' (Ellis, 1997)

'Nothing, yes nothing is awful, horrible. Or terrible, no matter how bad, inconvenient, and unfair it may actually be' (Ellis, 1999)

'Stop damning yourself and others by fully accepting the view that wrong, unethical, and foolish acts never can make you or them into bad or rotten people' (Ellis, 1999)

'There is no magic, no free lunch. Self-change, while almost always possible, requires persistent work and practice' (Ellis, 1999)

'Surrender your demand to be perfect' (attributed to Ellis, n.d.)

ABCDE model

ABC (Ellis, 1996)

REBT proposes that the ultimate goal for all humans is the attainment of happiness

Goal is intrinsic to our nature, yet it is often blocked by the myriad problems encountered on the pathway through life

Our responses to these blockages will depend on our beliefs about the world and ourselves

Illustrated in the ABC framework

A-B-C

Activating event (A)

An activating event (A) in the world may block the individual in their movement towards happiness

Belief about the event (B)

The event will be followed by a belief (B) about the event, and this belief can be logically based on the given evidence or illogically based on prejudices and superstitions

Emotional and behavioural consequences of belief (C)

This belief will influence the subsequent emotional and behavioural consequences (C), and these consequences can be psychologically healthy or unhealthy

For example, Susan is made redundant (A) and believes that this is evidence that she is a failure (B), thus she does not begin looking for another job and feels depressed (C)

Belief mediates between event and consequence

ABC model highlights the importance of belief as a mediating factor between an event and a consequence

Emotional disturbance is based on beliefs about events rather than the events themselves

Once you have gained this understanding, the locus of control is back in your hands

Although you may be unable to influence events, you are capable of changing your own beliefs about these events

For example, Susan is not looking for another job and feels depressed (C) due to her belief about being a failure (B) rather than the redundancy itself (A) – if she could change her belief then she would alter the consequences

Irrational beliefs can lead to negative consequences

Ellis did not propose that therapy should focus on how these irrational beliefs developed

Instead he argued that the therapist should focus on how the client is maintaining these beliefs and seek to dispute them

As a therapeutic response to the innate tendency towards irrationality, Ellis (1996) formulated two stages to be added to the original ABC framework

Disputing beliefs (D)

Effects of the new rational philosophy (E)

ABCDE (Ellis, 1996)

Model expanded to include D and E

Activating event (A)

An activating event (A) in the world may block the individual in their movement towards happiness

Belief about the event (B)

The event will be followed by a belief (B) about the event, and this belief can be logically based on the given evidence or illogically based on prejudices and superstitions

Emotional and behavioural consequences of belief (C)

LEARNING MY ABCS (AND MY DES)

Think of a negative experience in your own life and complete the following table to identify the ABCDE.

A. Activating event What was the negative event?	
B. Belief about the event What did you believe about the event?	
C. Consequences How did you feel and what did you do?	
D. Disputing the belief Could you believe something different about the event?	
E. Effects of the new rational philosophy How would this new belief make you feel and behave differently?	

This belief will influence the subsequent emotional and behavioural consequences (C), and these consequences can be psychologically healthy or unhealthy

Disputing the belief (D)

Therapy aims to identify, challenge and eventually dispute irrational beliefs through a process of questioning (D)

Effects of the new rational philosophy (E)

Disputed irrational beliefs will eventually give way to rational philosophies about the world and this will lead to fewer negative consequences in the future (E)

For example, Susan is made redundant (A) and believes that this is evidence that she is a failure (B), thus she does not begin looking for another job and feels depressed (C), but the therapeutic process helps her to question the belief that she is a failure by asking her to focus on other successes in her life (D) and she eventually accepts that she is not responsible for redundancies at her company, thus she is able to look towards the future with confidence and optimism (E)

Dangers of musturbation

'Neurosis-inciting musts' (Ellis, 1996)

Primary irrational beliefs usually involve irrational demands

Irrational demanding philosophy = must/ought/should/need

Rational preferential philosophy = like/want

Irrational demands are often absolutist

Individual cannot accept a world in which the demand is not met

Irrational beliefs can be defined as subconscious thought patterns about how the world must function based only on the consideration of the individual's own needs (Batte, 1996)

Derivatives of musts

Primary irrational beliefs (demands such as 'must') lead to an array of other secondary or derivative irrational beliefs (Ellis, 1995a)

Awfulising

Low frustration tolerance (I-can't-stand-it-itus)

Self/other/life downing

Three musts

Ellis (1996) identifies three 'neurosis-inciting musts' of irrational beliefs

Musts relating to the behaviour of the self

Musts relating to the behaviour of others

Musts relating to life

Musts relating to the behaviour of the self

'I must be thoroughly competent, adequate, achieving, and lovable at all times, or else I am an incompetent worthless person' (Ellis, 2003)

This belief will lead to the unhealthy emotional consequences of depression and guilt followed by unhealthy behavioural consequences of risk avoidance and procrastination

Musts relating to the behaviour of others

'Other significant people in my life must treat me kindly and fairly at all times, or else I can't stand it, and they are bad, rotten, and evil persons who should be severely blamed, damned, and vindictively punished for their horrible treatment of me' (Ellis, 2003)

This belief will lead to the unhealthy emotional consequences of rage and anger followed by the unhealthy behavioural consequences of violence and intolerance

Ellis (2003) suggested that this can lead to extreme actions such as feuds, wars, fights, genocide, and ultimately, an atomic holocaust

Musts relating to life

'Things and conditions absolutely must be the way I want them to be and must never be too difficult or frustrating. Otherwise, life is awful, terrible, horrible, catastrophic and unbearable' (Ellis, 2003)

This belief will lead to the unhealthy emotional consequences of self-pity and depression followed by the unhealthy behavioural consequences of procrastination and work avoidance

Musturbation is disputed in therapy (Table 6.1)

Disputing 'musts'

We recognise the irrational nature of thoughts once we realise that there is no good reason why the world 'absolutely must' function in a certain way

Disputed irrational beliefs will eventually give way to rational philosophies about the world and Ellis (1996) suggested that this will lead to a clearer insight into the main truths of the world

'We upset ourselves by holding inflexible beliefs, we continue to feel upset by clinging to these irrational beliefs, and we need to work hard to change our beliefs' (Ellis, 1996)

These insights allow us to fully accept all aspects of reality without conditions or assumptions

Disputing musts relating to the behaviour of the self

Self-acceptance encourages us to view ourselves as fallible human beings who are equally worthy as every other human being

Disputing musts relating to the behaviour of others

Other-acceptance encourages us to appreciate that other people are not obliged to treat people fairly, thus those who behave badly are equally worthy as every other human being

Disputing musts relating to life

> Life-acceptance encourages us to understand that life is rarely awful and almost always bearable irrespective of the fact that it might not always follow the path desired by the individual

TABLE 6.1 'I must' versus 'I would like'

Rephrase the following statements and consider the potential impact that the new expression might have on the speaker.

Demanding philosophy	Preferential philosophy	Impact on new philosophy
I must get an A in my exam		
My husband should be nice to me when I have a headache		
I need to win the lottery		
I ought to keep the house clean all of the time		
People should never cheat on their spouses		
I have to put my make-up on before I go out		
Everything has to go well in my interview tomorrow		
She should not treat me like that		
I have to lose two stone		
I must do better in the future		

Disputing irrational beliefs

Cognitive therapy focuses on cognitions

Cognitions are thoughts about the world

Client is encouraged to correct faulty thoughts, but this can fail to address the underlying beliefs

Sometimes the cognition might not actually be faulty

REBT focuses on beliefs

Beliefs are deeper held values about the world

Client is encouraged to detect potentially irrational beliefs, debate the rationality of these beliefs, discriminate between rational and irrational thought

Detect potentially irrational belief

Client is helped to recognise their tendency towards the derivatives of irrational beliefs (awfulising, i-can't-stand-it-itus, damning)

Client is then encouraged to reduce their problems by asking 'why is this so?' until s/he eventually arrives at one of the three neurosis-inciting musts

Debate the rationality of a belief

Client is encouraged to test beliefs using logic, personal experience and therapeutic experimentation

Logic can be used to determine whether the belief is essentially flawed or contradictive

Personal experience can be used to test whether the belief has had support in the past

Therapeutic experimentation can be used to test whether the belief is empirically true

Discriminate between rational and irrational thought

On the basis of the above debate, the client can determine whether the belief is rational or irrational

Irrational beliefs are discarded

Example of client with low self-esteem

Client claims that she is so ugly that no one could ever love her

Cognitive therapy would address only the cognition

Client has the potentially faulty cognition that 'people must be beautiful in order to be loved'

Client is invited to debate the rationality of this cognition by using logic (can we think of anyone who is ugly but still in a positive relationship?), personal experience (were you loved by your mother?), and experiments (go to the park and count the number of people holding hands – are all of these people attractive?)

Client is asked to determine whether it is rational to believe that she is so ugly that no one could ever love her

Client discriminates between this irrational thought and the more rational thought that she could find love by putting more effort into meeting people

BUT what if the client really is very ugly?

It is true that some people might fail to find love because they are so unattractive

But is it true that a person absolutely must be loved?

REBT would also address the deep underlying belief

Accept the cognition and begin to ask 'so what is bad about that?'

Do not need to argue with the cognition (people can be very stubborn about cognitions)

Instead, assume that it is true and focus on exploring the underlying belief

Client is encouraged to recognise the derivatives of the irrational belief, such as I can't stand to be unloved, the world is an awful place if I am unloved, I am a horrible person if I am unloved

Client is taught to detect the underlying 'must', such as I must be loved

Client holds a must relating to self: 'I must be loved'

Client is invited to debate the rationality of this 'must' by using logic (you are surviving as an unloved person so is it true that you absolutely must be loved?), personal experience (have you ever had any happiness in your life, even though you are unloved?), and experiments (do something to make yourself happy thus proving that you do not have to be loved)

Client is encouraged to change her irrational demand 'I must be loved' to a rational preference 'I would like to be loved'

Client is then released from the pressure of something that absolutely definitely must happen or else the world will end

We tend to feel blocked and trapped by fear when the must is too large, so removal of this pressure can free the client

Client can now focus on doing something to meet her desire for love

Cognitive restructuring

Irrational faulty beliefs replaced with rational constructive beliefs

Client is taught to adapt self-talk (Spiegler, 2008)

Self-talk is the inner monologue operating within each individual

Positive self-talk can be motivating, inspiring, problem solving, etc.

Negative self-talk can be demotivating, depressing, self-defeating, etc.

Client is taught the dangers of musturbation and encouraged to avoid the use of these absolute terms in their own self-talk

For example, client might be encouraged to recognise that the inner thought 'I must do well in this exam' is an example of musturbation and would be better changed to 'I would like to do well in this exam'

It is less damaging to fail to get something that you would like to have than it is to fail to get something that you absolutely must have

Adaptive self-talk is realistic and rational, thus it can help the client to work towards success yet accept potential failure without devastating effects

Shame-attacking exercises

Shame and embarrassment are uncomfortable but not lethal

Acceptance of the fact that things are rarely unbearable can be difficult without evidence to show that you can survive apparently impossible situations

Shame-attacking exercises are designed to give the client an opportunity to experience and survive risky and embarrassing situations (Ellis, 1995b)

Clients may be asked to engage in public activities that would normally result in shame, such as singing loudly, walking in a strange way, approaching strangers, etc.

Once the client has survived the shame then s/he has learnt that it is possible to cope with feelings of embarrassment

Repeated exposure to controlled shame will desensitise the client so that s/he is no longer constrained by the fear of shame

Example of a shy client

Client is asked to sing loudly in public

Experience is embarrassing, but not damaging

Client learns that s/he can cope with embarrassment and becomes less shy as a result

Risks of shame-attacking exercises

Shame can be a powerful emotion

Rather than providing a learning opportunity to manage and reduce fear of shame, poorly managed exercises can increase fear in the client

Repeated experiences of shame can lead the client to engage in more avoidance strategies or confirm negative self-beliefs held by the client

It is essential that the clients remain in control throughout the exercises so that they know that they can withdraw at any point if the experience becomes overwhelming

Exercises should be closely followed by extensive debriefing in the form of discussions about the experience through a positive framework of how the client successfully managed to cope with the discomfort

ATTACK THE SHAME!

Think carefully about the things that make you most anxious and consider the underlying cause of that anxiety. In many cases, people fear the experience of shame. For example, the person who loathes public speaking may argue that they are afraid that they will mumble or blush or faint but these things are not terrifying experiences in themselves – fear comes from the shame associated with being negatively judged. One way to reduce the fear of shame is to actively engage in shame-inducing activities until you eventually become desensitised to the feeling of embarrassment.

Find something that makes you squirm with embarrassment. If you are struggling for ideas, consider the following: singing in a public place, going to a party without wearing make-up, walking down the street with food on your shirt, making a speech in front of strangers, asking someone out on a date, or wearing a silly costume to shop at the supermarket. Once you have identified a shame-inducing activity, try to answer the following questions.

1. Imagine that you saw a stranger doing the shameful activity. What would you think about him/her?
2. Imagine that your best friend was doing the shameful activity. What would you think about him/her?
3. Imagine doing the shameful activity yourself – close your eyes and visualise the whole experience. How do you look and sound? How are people reacting? Do you feel shame or embarrassment? How can you learn to cope with these feelings?

Now that you have imagined the experience, it is time to try it out. Choose a time and location that is safe and appropriate then test out your shame-inducing experience. The aim is for you to feel embarrassed so do not be deterred once the discomfort arises; instead, focus on how well you are coping with the embarrassment and remind yourself that it is a temporary state that will not do you any lasting harm.

IMPORTANT
Please consider legal, moral and safety parameters when selecting your shame-inducing activity. Do not engage in any activity that is illegal (for example, being naked in a public place) or unsafe (for example, starting an argument with a stranger). Remember that you are in control of this learning experience and you are free to stop at any time if it becomes overwhelming.

Possible exercises are constrained by legal, moral and safety guidelines

Some of the most shame-inducing activities should not be encouraged due to potential risks

For example, a woman who is embarrassed about her body should never be encouraged to strip naked in public to face her shame (but she could try wearing a bikini on the beach)

Shame is to be expected in certain situations and the experience of shame can motivate people to change their behaviour for the better under some conditions

Experience of shame is sometimes used as an expression of regret and this can provide an insight into the world of the client

These types of shame are not appropriate for a shame-inducing activity

For example, a man who is ashamed of his poor treatment of his wife should never be encouraged to attack this shame by repeating the behaviour

Case study demonstrating REBT

Sarah, 19, is a student at a local college and she decided to seek therapy after the unexpected end of her relationship with her girlfriend. She discovered that her girlfriend was having an affair with another woman when she found a romantic text on her phone. Although she tried to save the relationship, her partner left her two weeks later and moved in with the other woman. She is devastated by this loss and she is convinced that the affair happened because she has recently gained weight. She insists that she 'will never find love again'. The therapy was limited to eight sessions, as per the policy for the college counselling service.

In the following exchanges, therapist denotes therapist, client denotes Sarah. Confidentiality statement and the business contract have been discussed. The session is joined just after Sarah has explained her story.

Client: So now I don't know what to do. I keep thinking about getting her back. I mean, if I lost some weight and then maybe just ran into her at the library . . . I just can't believe that this has happened. I can't stand it. I keep expecting her to be in the flat when I get home. How could she do this to me?

Therapist: It sounds like you are feeling very angry with her.

Client: I am. How dare she treat me like this? You just don't hurt people like this.

Therapist: Who doesn't?

Client: People. Her. She's just a horrible person for doing this.

Therapist: So you are saying that she should not treat you like this?

Client: Yes. She shouldn't.

Therapist: Why not?

Client: Because . . . because I don't like it.

Therapist: Okay. I understand that you don't like it when you are treated badly. But does that mean that she absolutely must not treat you like that? Is it true that people must always treat you in the way that you want to be treated?

Client: Well . . . no . . . but it would be nice if they did.

Therapist: I agree, it would be nice if people treated us kindly. Perhaps that could be a better way of expressing it.

Client: You mean, I would like to be treated nicely?

Therapist: Yes. People do not absolutely have to treat you nicely, but it would be nice.

Client: (*Sigh*) I bet she would have treated me better if I were thinner.

Therapist: You believe that she would have been nicer to you if you were thinner?

Client: I think so, yes. I'm never gonna find anyone to love me while I look like this.

Therapist: And is it very important to you that you find someone to love you?

Client: Well, yes. Obviously.

Therapist: Why is that obvious?

Client: Because everybody needs love.

Therapist: Do they?

Client: Yes! If you don't have love then you go through life alone.

Therapist: And what is so bad about that?

Client: Well, it means that you are alone. You spend all of your time on your own.

Therapist: And what is so bad about that?

Client: Obviously it is bad to be alone. It means that no-one is there for you.

Therapist: I know that I am repeating myself, but I have to ask one more time. What is so bad about being independent and not having anyone there for you?

Client: I . . . It just is. I don't know. It is just bad.

Therapist: I see. So you absolutely must be loved?

Client: Well, no. I don't absolutely have to be loved. I could live without being loved. It would be nice though.

Therapist: Okay. So it would be nice to be loved and it would be nice if other people treated you kindly.

Client: Yes.

Therapist: So what happens if you do not get those 'nice' things? Is it bearable?

Client: I guess so. I would like to be loved by someone who treats me well, but I guess that I can cope if that doesn't happen.

Over the subsequent sessions, the therapist worked with Sarah to identify her musturbatory beliefs about the end of the relationship. She identified her own tendency towards the beliefs that she must be loved (self-must) otherwise life is awful (awfulising) and unbearable (low frustration tolerance) and that other people must be kind to her (other-must) otherwise they are horrible people (other-downing).

Sarah was eventually able to understand that her current distress was caused by her interpretation of the break-up, rather than the break-up itself. She acknowledged that an alternative interpretation would probably lead to more positive feelings and behaviours. She also recognised that her own low self-esteem might be contributing to her difficulties so she decided to join a local theatre group in order to tackle her fear of being negatively judged.

Sarah decided to end the counselling relationship after five sessions as she felt more confident and content in her life.

ANALYSING THE CASE STUDY

Try answering the following questions using information about rational emotive behaviour therapy.

- What were the activating event, belief about the event and emotional/behavioural consequences of this belief for Sarah?
- How did the therapist dispute the belief and what was the effect of the new rational philosophy?
- What 'neurosis-inciting musts' did Sarah demonstrate?
- What derivatives of musts did Sarah experience?
- How were these musts disputed and what was the effect of this?
- How did the therapist encourage Sarah to change her self-talk?
- What shame-attacking exercises could be recommended for Sarah?

SUMMARY

Cognitive and behavioural: rarely practised independently; REBT was one of the first integrations

Development: founded by Ellis as RT in the 1950s and changed to RET in 1961 then REBT in 1993; first type of cognitive-behaviour therapy

ABCDE: activating event (A), belief about the event (B), emotional and behavioural consequences of belief (C), disputing the belief (D), effects of the new rational philosophy (E); belief mediates between event and consequence, irrational beliefs can lead to negative consequences

Dangers of musturbation: irrational demanding philosophy = must/ought/should/need; rational preferential philosophy = like/want; three absolutist 'neurosis-inciting musts' of irrational beliefs, musts relating to the behaviour of the self, musts relating to the behaviour of others, musts relating to life; musturbation is disputed in therapy, self-acceptance, other-acceptance, life-acceptance

Disputing irrational beliefs: cognition versus belief; detect and debate

Cognitive restructuring: adaptive self-talk is realistic and rational

Shame-attacking exercises: embarrassment is uncomfortable but not lethal; exercises involve doing embarrassing things to accept that you can survive unpleasant experiences such as shame; risks associated with some of these exercises

COGNITIVE-BEHAVIOUR THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- explain that cognitive-behaviour therapy is an integrative therapy combining the cognitive and behavioural approaches
- discuss the history of cognitive-behaviour therapy with acknowledgement of the wide number of contributors spanning many different psychological fields
- explain how cognition drives emotional reactions to external events
- explain how assessment and formulation are utilised through the therapy
- discuss the combined application of cognitive therapy and behaviour therapy techniques
- outline the structure of the therapeutic sessions throughout the therapy
- discuss the empirical evidence evaluating therapeutic efficacy
- describe and evaluate the use of computerised cognitive-behaviour therapy
- describe some 'third wave' therapies emerging from cognitive-behaviour therapy, appreciate the application of cognitive-behaviour therapy in a therapeutic setting (case study)

Cognitive-behaviour therapy (CBT): cognitive and behavioural approaches

Cognitive-behaviour therapy is grounded in the cognitive and behavioural approaches

Integration of cognitive and behavioural therapies

Cognitive therapy and behavioural therapy are rarely practised independently any more

Instead, they are often integrated to produce a new form of therapy

Cognitive-behaviour therapy is one of these integrations

Development of cognitive-behaviour therapy

History of cognitive-behaviour therapy

Behaviour therapy developed in the early 1900s out of the work of the behavioural psychologists (Pavlov, Watson, Skinner)

- It was a challenge to and developed alongside the psychodynamic approach, which was particularly dominant in the early 20th century
- In contrast to the psychodynamic approach, with its emphasis on instinctual drives and unconscious processes, the behavioural approach focused on observable and measurable behaviours and learning processes
- This allowed a more scientific approach that could be supported by empirical evidence (criticism of the Freudian approach)
- Behavioural approach found to be effective working on a variety of client issues, particularly anxiety issues and phobias
- However, not all clients responded to the approach and it was accepted that focusing purely on behaviours meant that thought processes were largely ignored – clearly, people were influenced by their beliefs, their thoughts and by the mental images they created

Cognitive therapy developed in the mid-1900s out of the work of cognitive psychologists (Neisser, Beck)

- Cognitive interpretations of emotional disorders (particularly depression) were instrumental in shifting attention from behaviour to cognition
- This laid the foundation for the integration of the cognitive and behavioural approaches – cognitive-behaviour therapy (CBT)

Prevalence of CBT

- In the UK, the National Institute for Health and Clinical Excellence (NICE) stated that CBT should be the treatment of choice for individuals suffering from anxiety or depression (NICE guidelines, 2004, amended 2007)
- Likewise in the US, the American Psychiatric Association stated that CBT should be regarded as the primary psychotherapeutic approach for patients with anxiety or depression (APA, 2000)
- As a result of these guidelines for the NHS, the UK government highlighted CBT as the treatment of choice for those diagnosed with depression or anxiety and, in order to cement this recommendation, a budget of £170 million was promised in 2007 for the expansion of mental health therapies with the specific proviso that only evidence-based treatments such as CBT would be funded
- Announcement of this policy met with a mixed response
  - Those in favour argued that the additional funding would improve the likelihood of recovery from mental illness
  - Those against argued that the effectiveness of CBT is no better than other forms of therapy and so this proposal will limit patient choice

Over a decade on from the first NICE guidelines, CBT is still generally considered the best approach for dealing with anxiety and depression, and it is the most widely used therapeutic method within the NHS

Multiple contributors

No individuals regarded as 'founding fathers/mothers' of CBT

Instead, multiple key figures have contributed to the development of this approach

Since CBT is an integrative approach combining behavioural and cognitive approaches, key figures identified for each of these approaches could be regarded as contributors to the development of CBT (please refer to previous chapters for brief biographies of these contributors)

Pavlov
Thorndike
Watson
Skinner
Staats
Ellis
Bruner
Miller
Beck
Chomsky
Neisser

Donald Meichenbaum is perhaps the main contributor to the modern concept of CBT

Brief biography of Donald Meichenbaum (1940–current)

'Any advocate that says "This is the cure" or proposes a "revolutionary" approach does not deserve your attention, nor your money.'
—Meichenbaum (quoted in Howes, 2008)

Who is Donald Meichenbaum?

Focused on changing cognitions by modifying the 'inner dialogue' through psychotherapy

Integrated cognitive and behavioural theories to form cognitive-behaviour modification – original form of cognitive-behaviour therapy (alternative to REBT proposed by Ellis)

Early years

Born 10th June 1940 in New York

Education

Graduated with a degree in psychology at City College, New York

Completed postgraduate studies at Illinois University

Career

Originally intended to pursue a career in industry, but focused on clinical psychology instead after working as a research assistant in a psychiatric hospital

Professor at University of Waterloo, Ontario, from 1966 to 1998

Published *Cognitive-Behavior Modification: An Integrative Approach* in 1977 to combine the classical behavioural techniques with the newly emerging cognitive techniques in order to produce a novel integrative therapy

Voted one of the ten most influential psychotherapists of the 20th century in a survey of 800 clinical and counselling psychologists in 1982 (Smith, 1982)

Published *Stress Inoculation Training* in 1985

Retired in 1998

Significant learnings

'I have trained and supervised psychotherapists all over the world and the major challenge is that they DO NOT LISTEN to their clients. Instead, they act as what I call 'Surrogate Frontal Lobes' for their clients by giving explicit advice' (Meichenbaum, quoted in Howes, 2008)

'The art of questioning is the psychotherapist's most valuable tool. Mainly, these are HOW and WHAT questions' (Meichenbaum, quoted in Howes, 2008)

'Psychotherapy is a noble profession and I am honored to be a member of this community' (Meichenbaum, quoted in Howes, 2008)

Cognition-driven emotions

'The mind is its own place and in itself, can make a Heaven of Hell, a Hell of Heaven.'
—Milton (Paradise Lost, Book 1)

Clients often assume that events in their lives cause them to feel emotions

'The presentation made me anxious'

'I'm angry because he ignored me'

However, if it were the event itself that caused the emotional response then everybody would feel the same and this is clearly not the case

'I'm excited about the presentation'

'I'm fed up because he ignored me'

It is the cognition (thoughts, beliefs, interpretations) that we have of the event that leads to the emotional response

CBT uses a basic ABC model (Ellis, 1996)

A is the event

B is cognition (our interpretation) of the event

C is the emotional (and then behavioural) consequence of B

CBT considers clients' problems, in terms of both creation and maintenance, as an interaction between . . .

Cognition

Feelings

Behaviours

Physical sensations

Cognition is considered to be the starting point in this process

If a client has a core belief that s/he is worthless and useless in all areas of life . . .

When they have an exam looming, this core belief will lead to a dysfunctional assumption about being bad at exams in general, which will lead to a negative automatic thought about the forthcoming exam ('it will be awful')

CBT therapists work with clients to help them become more aware of these negative automatic thoughts and to replace them with more realistic thoughts

Therapy may also aim to change dysfunctional assumptions and core beliefs

These are harder to change but can lead to change that may affect many areas in the client's life

REBT, an alternative type of CBT, considers changing beliefs to be the focus of therapy, leading to a deeper and more fundamental change

Assessment and formulation

Collecting information about the client enables the therapist to make an assessment

Information gathering would focus on the following areas

Symptoms
Predisposing factors
Precipitating factors
Triggers
Beliefs and assumptions
Maintenance processes

Symptoms

- Thoughts, feelings, behaviours and physical sensations of the problem and how these interact
- It is how thoughts, feelings, physical sensations and behaviours interact with each other that show the client's individual way of approaching the problem
- Client has a problem in relation to the environment (external factors which influence the client)
- The greater the detail collected on these interactions, the better informed are the client and therapist about the nature and extent of the problem
- For example, a claustrophobic client may describe feelings of doom, increased heart rate and thoughts of death in response to being in a small room

Predisposing factors

- Aspects of the client's history that may be influencing the current problem
- Events in the client's past, especially in childhood, are likely to have influenced beliefs and assumptions about self, others, the world and the future
- While the main focus of CBT is on the present and the future, understanding of significant past events may help in the understanding of how the problem may be maintained in the present
- For example, an adult client experiencing depression may be able to identify early parental messages of being unloved

Precipitating factors

- What was happening in the client's life at the time that the problem started
- Past events may predispose a client to a problem, but this will not be triggered until a life event (or series of events) occurs
- For example, a client may have been unaware of early parental messages of being unloved until the breakdown of his marriage

Triggers

- What is happening in the client's life just before the problem occurs
- These are the current factors that have triggered the problem
- For example, seeing a spider might result in a phobic reaction
- This will be worse depending upon other factors (for example, the size and proximity of the spider)

Beliefs and assumptions

- What beliefs and assumptions the client holds about self, others and the future
- If we have rigid, irrational beliefs about self, others and the world then these are likely to create and maintain problems
- Dysfunctional assumptions also underpin our negative automatic thoughts

Maintenance processes or cycles

- What role the client plays in maintaining the problem

CBT focuses on how the client keeps the problem going

Some common maintenance processes are

Avoidance: for example, someone with a fear of elevators will use the stairs

Withdrawal: for example, if a person thinks that they will appear stupid if they give an opinion in a group then they are likely to withdraw and not contribute to the discussion

Overvigilance: for example, someone who is scared of becoming ill is likely to be oversensitive to symptoms of illness and may misinterpret physical signs and symptoms

Much of this information will be collected in the first session with a client

Further details are likely to be added as therapy progresses, but there will be enough information gained early on with a client to develop a basic formulation

Assessment forms the basis of the initial formulation (Figure 6.6)

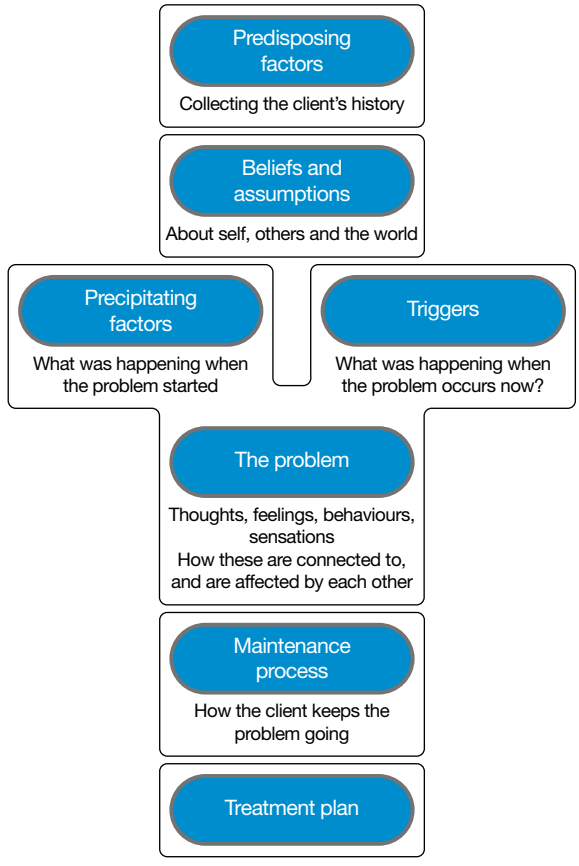

FIGURE 6.6 Formulation template.

'Formulation' is used to describe building up an understanding of the client's problem

This includes how the problem developed, how it is maintained and how it can be treated

Formulation will draw together all the information gathered during the assessment to provide a full picture of the situation

FORMULATE YOUR PROBLEM

Think of a situation that may be a problem for you. This could be something like giving a presentation at work or going inside an elevator or travelling by airplane.

Spend a few minutes thinking about how you 'do' the problem. Try to complete your own formulation by answering the questions in the boxes below.

Symptoms What do you think, feel and do during the problem?	
Predisposing factors Can you think of anything in your past that might be contributing to the problem?	
Precipitating factors What was going on in your life at the time the problem started?	
Triggers What happens to start the problem?	
Beliefs and assumptions What do you believe about the problem?	
Maintenance processes What might you be doing to maintain the problem?	

Once you have a clearer understanding of the problem, you can begin to think what you can do about it (treatment).

Formulation will then add a treatment plan in order to give both the therapist and the client a good understanding of how the therapeutic process will progress

This basic outline of therapy will develop as more information is explored in future sessions

A useful method of representing all the information gathered during the assessment and the subsequent treatment plan is by using a formulation template

This is a simple way of displaying all the key aspects of the client's problem

Application of cognitive therapy and behaviour therapy techniques

Cognitive-behaviour therapy is predominantly psychoeducational

The more the client knows about how the problem works and about the techniques to manage and overcome the problem, the more the client can become their own therapist

CBT for clients with mild depression or mild anxiety may encourage the client to use self-help guides or relevant books and manuals (bibliotherapy)

Another possibility is for clients to access computer-based CBT – this is one of the more recent developments of the approach

Wide variety of cognitive and behavioural techniques

CBT adopts a wide range of techniques from the fields of behaviourist psychology and cognitive psychology to provide an empirically based therapeutic approach to psychological disturbances such as depression and anxiety

Socratic questioning

Socratic questioning is an important technique that is used throughout the CBT therapeutic process

It has been noted as one of the essential competencies for a CBT therapist (Roth & Pilling, 2007)

Socratic questions can be asked at any stage in the process

Socratic questioning, named after the Greek philosopher Socrates, is a way of guiding a client to discover new perspectives by asking questions which she has the answer to but which may currently be outside her awareness

Such questions help the client focus on aspects of her thoughts, feelings, physical sensations and behaviours in order to gain new cognitive insights, which can then be the basis for behavioural change

Examples of Socratic questions

What were you thinking when . . . ?

What are you thinking when you feel . . . ?

How did you feel when . . . ?

What would it mean to you if . . . ?

What else did you think/feel when . . . ?

What did you do when . . . ?

What was happening before . . . ?

Were you aware of any physical sensations when . . . ?

What else could you have done . . . ?

How useful is it to believe . . . ?

Socratic questions can be used to . . .

Gather information in the assessment and formulation stage

Make connections between thoughts, feelings, sensations and behaviours

Challenge irrational beliefs and negative thoughts

Explore solutions and resolve problems

A good Socratic question will leave the client feeling empowered, that she has arrived at an answer, rather than being told by the therapist.

Consider the difference between the following examples

Client: My four year old just started screaming and so I hit him. I can't believe I did that.

Therapist: You could have counted to 10 to give yourself time to think

Client: Yeah, I suppose so

SOCRATIC QUESTIONS IN ACTION

Think of three different Socratic questions for each of the following client statements.

'I don't know what to do'

'Well, you would feel angry, wouldn't you?'

'I just keep crying for no reason'

'It's hopeless'

'It all started about a year ago'

'I started panicking'

'I felt sick when I woke up on the morning of the presentation'

'I don't care one way or the other'

'She shouldn't treat me like that'

'How do you think it made me feel?'

In this first example, the therapist has made the suggestion and so the client has no ownership

Client: My four year old just started screaming and so I hit him. I can't believe I did that.

Therapist: What would have been a better thing to do?

Client: I suppose I could have counted to 10 and . . .

In this second example, through the use of a Socratic question, the client comes up with her own answer

Identifying cognitive errors

Thoughts are connected to feelings

Depression is linked to thoughts such as 'Nothing will ever change. I'm worthless and unlovable'

Anger is linked to thoughts such as 'He shouldn't treat me like that'

Anxiety is linked to thoughts such as 'It's going to be awful'

Socratic questions can help the client to link feelings and thoughts

'What are you thinking when you feel that?'

'When you feel like that, how does that affect your thoughts?'

'When you think that about yourself, how does that make you feel?'

'What feelings do you have when you say that to yourself?

We all have faulty thinking from time to time

However, when people have problems these thinking errors are likely to be more frequent and more extreme

Some thinking biases go together with specific problems

For example, people suffering with depression are more likely to focus on negative events and filter out positive events

There are many common thinking errors (Beck, 1975)

All-or-nothing thinking

With this type of thinking there are no grey areas, only black or white

For example, 'If I don't get what I want now, I'll never get what I want'

Catastrophising

Worst-case scenario thinking

For example, 'If I fail my driving test, I won't be able to drive and I'll never get that job. I'll end up on the scrap heap'

Perfectionism

Setting standards that are unachievable

Often observed through the excessive use of 'should' and 'musts'

For example, 'I must do this perfectly or it will be/I am worthless'

Generalisations

Taking one event and making a global statement

For example, 'The car didn't start today. It always does that'

Focusing on negatives

Focusing only on the negative elements of life

Likely to become part of a process that reinforces depression

For example, 'Nothing good ever happens to me. Ever'

Discounting positives

Reducing the value or importance of the positive elements of life

A common error, magnified when we have problems

For example, 'He only complimented me because he wants something'

Mind reading

Dysfunctional assumptions based on limited information (e.g. reading body language)

For example, 'She's not speaking to me, that means she hates me'

Developing new perspectives

Once a client is aware of his thinking errors, there are techniques to help develop new perspectives and new ways of thinking

CBT therapists encourage clients to test the negative thoughts in order to identify errors

Whenever a client has a negative thought, s/he can be challenged to justify the thought

For example, 'Nothing good ever happens to me. Ever' could be tested by asking 'Has there ever been a time when something good did happen to you?'

'I must do this perfectly or it will be/I am worthless' could be tested by asking 'Who says you must do it perfectly?' or 'What would really happen if you didn't do it perfectly?'

Experimentation

Experiments can be used to empirically test potentially faulty negative cognitions

Experiments can be used to address actions that are maintaining negative behaviour

Some clients may use avoidance as a maintenance process – for example, staying in the house to avoid the possibility of a panic attack

What starts out as a logical strategy ('I had a panic attack yesterday, so I'll not risk it today') can turn into a major limitation (one day becomes two days, becomes a week, a month, a year . . .)

Behavioural experiments involve graded tasks set as homework between sessions

For example:

Day 1 walk to front gate

Day 2 step outside front gate

Day 3 walk 10 paces from front gate, etc.

These tasks would be agreed between the client and the therapist

This is a simple desensitisation hierarchy but could be combined with some measurement to record thoughts and intensity and duration of feeling as the task was carried out

Relaxation techniques such as focusing on breathing or positive visualisation may also be integrated into the task

Changing self-talk from negative to positive

Cognitive-behaviour modification focuses on client's self-talk to change unwanted behaviours

Helps clients become more aware of their present internal dialogue, highlights how this dialogue affects emotion and behaviour, and demonstrates how to change negative self-talk to more useful, coping talk

Similar to Beck and Ellis in that it highlights that unhelpful emotions and unwanted behaviours are a result of dysfunctional, negative thoughts, but it is often less confrontational

Meichenbaum (1977) identified a three-phase change process

1. Self-observation

   Becoming more aware of emotions and behaviours and the internal dialogue that influences them

2. Changing internal dialogue

   Once we become aware of unhelpful internal dialogue, we can begin to change it

   More positive dialogue will lead to more positive emotions and behaviours (which in turn reinforce the positive internal dialogue)

3. New skills

   Teaching clients more effective ways of coping in order for them to feel more in control

Homework tasks

CBT often involves setting the client homework tasks

Tasks will often involve collecting relevant data to 'measure' the problem, especially in the early stages of therapy

Collecting data is useful

Task engages the client in activity outside the session

Thoughts, feeling and behaviours noted as occurring at the time of the problem are likely to be more accurate than those reported in the session

Helpful to keep record of change because clients often forget how bad the problem was at the start of therapy by the time the changes have begun to take effect

Data serve as a therapeutic tool in itself as the client may benefit from focusing on what they are thinking when they become aware of the problem (mindfulness)

Measurements include . . .

Frequency (e.g. how often a client checks that the front door is locked)

Duration (e.g. how long does it take to get out of the house?)

Intensity (e.g. on a scale of 1 to 10 how bad did you feel?)

Methods of measurement include . . .

Rating scales

Thought diaries

Charts

COGNITIVE AND BEHAVIOUR TECHNIQUES IN THERAPY

Re-read the chapters on cognitive therapy (Chapter 5) and behaviour therapy (Chapter 4) then consider the following case studies. Try to identify the main issues for each client and outline a treatment plan incorporating both cognitive and behavioural techniques.

- Caroline is struggling with her dependency on alcohol after being made redundant. She is finding it difficult to get motivated to look for work and she feels that her only happy moments occur when she drinks wine in the evening.
- John is struggling to come to terms with the death of his dog. He has lived alone for most of his life and his dog has been his only companion over the last ten years. He feels like he does not want to continue living without his only friend.
- Janine has recently had a baby and she is terrified that the child is going to get sick or be injured. She tries not to go outside with the baby and she constantly checks on him at home. Although her husband tries to be supportive, her behaviour is having a major impact on their relationship.
- Drake is an excellent student but he has recently begun skipping classes and failing to submit assignments. He is terrified that he is going to fail his course and he regards any grade less than an A as a fail.
- Hazel suffers with acute social anxiety. She has a slight limp when she walks and she is extremely aware of this problem. She is convinced that people are laughing at her behind her back and she refuses to attend any public events in order to avoid being mocked.

Any measurements that the client is required to make should be . . .

- Easy to carry out
- Meaningful
- Specific
- Relevant
- Done as close as possible to the event

Structure of therapeutic sessions

Structure of a session will depend upon where the client is in the process

Beginnings and endings of the process will have different formats, with more emphasis on the client taking responsibility as the process develops

However, during the main phase of therapy, a 50-minute session is likely to run as follows

- Discussing the content of the session (5 minutes)
  - Discussion will include what both the therapist and client think would be useful to work on during that session
- Checking the previous week (10 minutes)
  - Update on what has been happening over the last week and how the client managed with the previous homework tasks
- Main content (25 minutes)
  - Specific work decided upon at the start of the session and including the application of CBT techniques
- Setting tasks for the next week (5 minutes)
  - Homework agreed by therapist and client
  - CBT requires clients to work on tasks between sessions – as the process of therapy progresses, the client should take more responsibility for setting tasks and managing the change process
- Review of the session (5minutes)
  - Checking out what the client thought was useful, and what was unhelpful, during the session

Empirical evidence for CBT

Due to the empirical nature of CBT, extensive research has been conducted to measure its impact on mental health

- Findings suggest that CBT is effective in reducing the symptoms of depression and anxiety
  - Controlled trials exploring treatment outcomes in severely depressed outpatients have found that CBT is as effective as antidepressant medication in reducing symptoms (DeRubeis et al., 1999)

While research to date has provided a significant body of evidence to support the hypothesis that CBT is effective in treating depression and anxiety, there are a number of criticisms directed at this empirical approach to treatment

- One initial criticism relates to the fact that effectiveness is measured in accordance with the scores obtained in questionnaires before and after the treatment process
  - These questionnaire measures are subject to participant bias since many clients may feel obliged to indicate an improvement due to a sense of obligation to their therapist
- Another criticism is that clients will have been selected on the basis of their desire to work towards improved mental health
  - CBT is inappropriate for clients who refuse to engage with the mini-experiments, activities and homework
  - Clients presenting with depressive symptoms associated with poor motivation or an unwillingness to engage with therapy would not be prescribed CBT
  - A recent meta-analysis of ten controlled studies comparing CBT with treatment as usual indicates only a modest therapeutic benefit for sufferers of major depression and no therapeutic benefit for sufferers of bipolar depression (Lynch et al., 2010)
- Another criticism argues that the suggestion that it is the most effective method of treatment is based on a logical error
  - Cooper et al. (2008) state that the conclusions drawn from recent research are an example of a logical fallacy: as more researchers subscribe to the CBT approach, more research grants will be awarded and more studies will be published about the benefits of CBT
  - The large number of supporting studies is assumed to indicate the efficiency of CBT, but it could instead indicate the popularity of the method

Research that has attempted to control for these criticisms finds little difference between CBT and other therapeutic methods

- Patients receiving cognitive-behaviour therapy, psychodynamic therapy or person-centred therapy through the NHS primary care services were compared on the Clinical Outcomes in Routine Evaluation – Outcome Measure (CORE-OM) to reveal no significant differences between the three approaches (Stiles et al., 2005; successfully replicated by Stiles et al., 2008)
- Patients receiving non-directive counselling or cognitive-behaviour therapy were compared on treatment satisfaction measures to reveal no significant differences between the two approaches (Ward et al., 2000)
- These findings illustrate the equivalence paradox or the 'Dodo verdict', as they suggest that all treatment methods are approximately equivalent in terms of outcome
  - 'Everybody has won and all must have prizes' (Caroll, 1865/1946, as cited in Stiles et al., 2008)

On the basis of the evidence, it could be concluded that CBT has a positive impact on the symptoms of depression and anxiety

- However, positive effect of CBT is possibly no greater than any other therapeutic method
- CBT not suitable for all sufferers of depression and anxiety
  - CBT requires a collaborative relationship with both the therapist and the client working towards a positive change in mental health
  - Techniques often involve homework assignments and active experimentation in the world beyond the counselling room
  - Therefore, CBT requires the client to be motivated to change and engaged with the therapeutic process
  - Even if the client is motivated to engage with therapy, some individuals may not feel comfortable with the directive nature of this approach
  - As CBT is not appropriate for all sufferers of anxiety and depression, it is essential that there are alternative psychological therapies available

Computerised CBT

Computerised CBT (CCBT) is offered to people meeting the criteria for mild depression

- Programs allow people to work through a CBT-based computer program at their own pace
- There are a growing number of such programs available, most offering telephone supports, along with a comprehensive self-help program
- Testimonials from users of these programs indicate that they are very effective, life enhancing and life changing
  - However, by their very nature, testimonials are given by those who are motivated to engage with the programs – those who drop out of the program are unlikely to give feedback

Advantages of CCBT

Widening access to treatment

- This is particularly valuable in rural areas where access to a CBT therapist may be restricted

Cost effective

- One of the key arguments in favour of developing and implementing computerised packages is the low running cost due to the absence of counsellor wages

Convenient to use

- Individuals can work through the program at a time and pace that is appropriate to them

'Cost-utility and cost-effectiveness analyses tend to be in favour of CCBT. On balance, CCBT constitutes the most efficient treatment strategy, although all treatments showed

low adherence rates and modest improvements in depression and quality of life' (Gerhards et al., 2010)

Disadvantages of CCBT

Requires self-motivation

- This can be a problem, especially for those suffering with depression
- Research highlights a high drop-out rate

Requires access to a computer and internet connection

- Issues around equality of access are clearly present
- While most people now have access to, and are confident in using, computers and the internet, this is not true for all

'Despite a short-term reduction in depression at post-treatment, the effect at long follow-up and the function improvement were not significant, with significantly high drop-out. Considering the risk of bias, our meta-analysis implied that the clinical usefulness of current CCBT for adult depression may need to be re-considered downwards in terms of practical implementation and methodological validity' (So et al., 2013)

Third wave cognitive-behaviour therapies

CBT continues to expand and develop

- Recent developments include . . .
  - Mindfulness-based cognitive therapy
  - Acceptance and commitment therapy
  - Dialectic behaviour therapy

Mindfulness-based cognitive therapy (MBCT)

MBCT was developed out of the work of Teasedale et al. (1995) when they challenged the assumption that CBT worked because it changed beliefs and replaced negative automatic thoughts with more positive thoughts

The alternative view offered was that it was the process of focusing on beliefs and thoughts

- Stepping back from beliefs and thoughts (seeing them from a new perspective) was therapeutic

MBCT encourages clients to . . .

- Focus on the here and now, be 'mindful' of the present, and not dwell upon past events or anticipate the future
- Be non-judgemental about events in order to see them as neither good nor bad
- Be aware that thoughts are not reality

By being in the here and now, clients cannot get into familiar cycles of avoidance, or of anticipatory anxiety

Mindfulness is about acceptance of our thoughts, rather than trying to enforce change

Acceptance and commitment therapy (ACT)

ACT is a mindfulness-based behavioural therapy developed out of relational frame theory (Barnes-Holmes et al., 2002)

Two parts to ACT

- Acceptance
  - Rather than trying to change thoughts, clients are encouraged to accept them and to see them for what they are
  - Thoughts influence the way we view the world (our representation of reality) but they are not in themselves reality
  - Once we become more accepting, we will be less rigid and more flexible
- Commitment
  - Client's commitment to make behavioural changes that are in keeping with the client's chosen values

To achieve a more fulfilling life, ACT identifies six core processes

- Contacting the present moment
  - Client is aware of and present in the here and now
- Defusion
  - Detaching from unhelpful thoughts and the worry or anxiety that they create
  - By taking a new position to view these thoughts, the client can be more objective in 'observing' them
- Acceptance
  - Once the client takes a different view of their thoughts, s/he can become more accepting of them
  - This differs from CBT in that ACT does not try to change the thoughts, just become more aware and accepting of them
- Observing self
  - With practice, the client can become more aware of their thoughts and feelings in the moment that they are having them
- Values
  - Focuses on what things are truly important
  - Big life questions: Who does the client want to be? What does the client want from life? What is significant and meaningful in the client's life?
- Committed action
  - Client takes action to move towards his/her chosen goals in keeping with his/her chosen values
  - Commitment to keep taking action even when that is difficult

Dialectical behaviour therapy (DBT)

Linehan (1993) developed DBT specifically for working with suicidal women who had been diagnosed with borderline personality disorder (BPD)

Four main areas of skills are taught (either individually or within a group setting)

- Mindfulness
  - Observing what is happening in the here and now, being fully present, and being non-judgemental
- Emotional regulation
  - Being able to identify and give a name to emotions
  - Being able to tolerate powerful, and at times overwhelming, emotions
- Distress tolerance
  - Being able to distract oneself from powerful emotions
  - Being able to step back from these emotions and accept them for what they are
- Interpersonal effectiveness
  - Having more effective communications to get what you need and want
  - Being assertive
  - Respecting and valuing self and others in relationships

These recent developments (or 'third wave') in CBT show that it is a flexible and changing approach; future developments will build upon this flexibility and show that this therapy is integrative in both practice and principle

Case study demonstrating cognitive-behaviour therapy

Margaret, 32, had come to counselling through a referral from her GP. The doctor simply stated 'generalised anxiety' as the reason for the referral. She described herself as 'sensitive' and 'emotional'. She had read up on CBT before coming and was hopeful that the course of therapy would 'sort her out'. The initial session was a counselling assessment in which the therapist sought to gather enough information to make a tentative formulation. It was agreed that there would be five further sessions and a review after the assessment session.

In the following exchanges, therapist denotes therapist, client denotes Margaret. Confidentiality statement and the business contract have been discussed. The session is joined as Margaret begins her story.

> *Client*: Well I suppose it started about a year ago, I was out shopping one day and I began to feel anxious, like nervous and panicky. It went away after a few minutes and I carried on doing the shopping. About two weeks later the same thing happened, only this time it was worse. I could really feel my heart beating fast and I thought I was going to pass out. It was awful. Ever since then it's been a real effort to go out. I have this kind of dread, and sometimes burst into tears for no reason. When it is bad like that I just have to get out of the shop – I leave my trolley full of food and escape . . . (*laughing*) . . . it's really embarrassing.

Therapist: Embarrassing when you were in the shop, or talking about it here?

Client: Both I guess.

Therapist: Okay, sometimes it is a little awkward to talk about things which we think are embarrassing, but it gives us some really useful information about how your anxiety happens. In CBT we like to understand how thoughts, feelings, behaviours and physical sensations interact with each other – how you *do* your problem, if you like.

Client: Yes, I remember reading that.

Therapist: The interesting thing is that two people could both get anxious about, say, shopping, but they might get anxious in very different ways. So let's concentrate on your way. Can you tell me in detail the most recent example of when this happened?

Client: Yes, just over a week ago, not long after I got the appointment with you for today, I was doing the weekly shop. I think I was already anxious even before I went out. Anyway, I was in the shop and I began to feel a bit breathless and nauseous. I held tight on to the trolley, and I remember thinking that it was so embarrassing, everybody looking at me and thinking I was so stupid. That's me, stupid.

Therapist: When you thought that, were you aware of any feelings?

Client: Well yes – I got this pain in my chest, I thought it was a heart attack but I remember reading about panic attacks and how people think they are going to die, so I told myself it was panic. That helped for about ten seconds, then I panicked again. I couldn't keep it all together, so I left the shop. After 10 minutes I went back in and finished the shopping.

Therapist: That sounds quite brave, going back in. How did you feel going back in?

Client: Yes, I was really proud of myself. I'd read a self-help guide on the internet, and I must have remembered it from there.

Therapist; And what did you tell yourself to help you go back in?

Client: I just said 'just pull yourself together – nobody cares, they're all too busy getting on with their own stuff'.

Therapist: Are you always this good at tackling the problem head on like that?

Client: Well no, that was the first time actually. Often if I'm feeling anxious before going out I just stay in. Why put yourself through all that?

Therapist: So sometimes you avoid the thing that gives you the anxiety. Makes sense in the moment, but actually can make it worse in the long run – if you avoid the problem you never challenge yourself to break the pattern. But last time you did, and felt really good because you managed.

In this exchange the client gives some detail to how she does the problem, and the therapist can begin to create the formulation. There is relevant information on:

Symptoms (thoughts, feelings, physical sensations and behaviours)

Triggers

Maintenance processes

There is also some useful information on management and possible treatment:

Client is keen to read up

Client broke the pattern of avoidance and felt 'proud' of her achievement

This suggests that the client will be willing to carry out homework tasks and is keen to be active in the change process

During the subsequent sessions, the therapist covered other areas of assessment to add detail to the formulation

What was happening in the client's life a year ago when the problem first started (precipitating factors)

Some information about the client's childhood experiences (predisposing factors)

It was agreed at the end of the session that the client would . . .

Keep a thought record for a week, noting when her thoughts were negative (frequency, duration and intensity)

Go to the supermarket where the last occurrence happened each day

Focus on breathing and positive self-talk whenever she felt anxious

Read the self-help guide recommended by the therapist

Margaret finished therapy after five sessions in a much less anxious frame of mind. She was a good example of a client who comes into therapy ready for change and ready to work to make that change

ANALYSING THE CASE STUDY

Try answering the following questions using information about cognitive-behaviour therapy.

- What symptoms, triggers,and maintenance processes did Margaret report?
- What cognitive techniques could be/were used during work with Margaret?
- What behavioural techniques could be/were used during work with Margaret?

SUMMARY

Cognitive and behavioural: rarely practised independently, usually combined to form CBT

Development: behaviour therapy developed in early 1900s, cognitive therapy developed in mid-1900s, integration began in later 1900s when Meichenbaum introduced cognitive-behaviour modification; highly prevalent today, recommended by UK NHS and NICE

Cognition-driven emotions: ABC model (activating event, belief, consequence); cognition instigates subsequent emotions, behaviours, etc.

Assessment and formulation: information gathering during assessment focuses on symptoms, predisposing factors, precipitating factors, triggers, beliefs and assumptions, maintenance processes; assessment forms the basis of the initial formulation, formulation draws all information together and adds proposed treatment to create a full plan

Cognitive and behaviour therapy techniques: CBT is psychoeducational, wide variety of cognitive and behavioural techniques; Socratic questioning; identifying cognitive errors; experimentation; changing self-talk from negative to positive; homework tasks

Structure of therapeutic sessions: discuss upcoming session (5 mins), check previous session (10 mins), main content (25 mins), set tasks for next session (5 mins), review session (5 mins)

Empirical evidence for efficacy: evidence suggests CBT effective in reducing symptoms of depression and anxiety; critics argue that measures are subject to participant bias and client selection bias, and conclusions contain a logical error; some research finds little difference between therapies, relationship is the critical factor; conclude that CBT does have a positive effect but no more than other methods and not suitable for all clients

Computerised CBT: offered for mild depression; advantages include widened treatment access, cost effective, convenient to use; disadvantages include need for self-motivation, requires access to internet, lacks relationship connection

Third wave CBT: mindfulness-based cognitive therapy (MBCT); acceptance and commitment therapy (ACT); dialectic behaviour therapy (DBT)

MULTIMODAL THERAPY

LEARNING OUTCOMES

After reading this section, you will be able to:

- explain that multimodal therapy is an eclectic therapy adopting techniques from across the approaches
- discuss the history of multimodal therapy with a brief biography of Arnold Lazarus
- explain the origin of psychological problems
- outline the initial assessment process
- describe the seven modalities and associated interventions for the BASIC ID
- explain modality and structural profiles

- discuss the tracking of the firing order of modalities and explain how bridging helps clients move from one modality to another
- appreciate the application of multimodal therapy in a therapeutic setting (case study)

Multimodal therapy: eclectic

Multimodal therapy is an eclectic mix of theories and techniques from across the approaches

> 'Multimodal therapy . . . provides Humanistic integration, systemization and a comprehensive "blueprint" for assessment and therapy. It deliberately avoids the pitfalls of *theoretical* eclecticism while underscoring the virtues of *technical* eclecticism' (Lazarus, 1981)
>
> Multimodal therapy (MMT) is a structured and systematic approach that can be described as 'technically eclectic'
>
> > Eclecticism generally means to select techniques from a variety of therapies to meet the needs of the client, without subscribing to the theory that underpins the approach from which the technique is taken
> >
> > MMT is pragmatic because it is prepared to use any technique that has been found to be effective
> >
> > > As long as there is a clear rationale and structure, the therapy can be designed specifically for each individual client
> >
> > Developed by Lazarus on the basis of many years of experience from many different approaches (Lazarus, 1981)
>
> 'Aim of Multimodal Therapy is to reduce psychological suffering and to promote personal growth as rapidly and as durably as possible' (Lazarus, 1981)

Multimodal therapy adds established techniques to a positive relationship

> Multimodal therapists accept the basic principles of developing an effective therapeutic relationship as identified by Carl Rogers
>
> > For example, therapist should provide the client with unconditional positive regards, congruence and empathy
>
> Lazarus (1981) noted 'parity as a way of life' in which he stated that all people are equal to one another
>
> > Clients often consider the therapist to be the expert, thus starting the relationship in a 'one down' position
> >
> > It is the job of the therapist to make the client an equal part of the therapeutic process
> >
> > The therapist might have knowledge and techniques to impart (like teacher/student) but the relationship itself is fundamentally one of equals

However, a range of techniques adopted from other therapeutic methods and tailored to specifically suit the individual client should supplement these core conditions

Therapist must be flexible in meeting the relationship needs of the client

If the client wants a supportive, listening relationship, then the therapist can tailor the treatment to suit this need

Alternatively, if a client wants to focus on goals in an active and challenging relationship, then the therapist can alter the treatment to suit this need

Lazarus (1981) described being flexible and adaptable to meet the needs of individual clients as being an 'authentic chameleon'

Development of multimodal therapy

History of multimodal therapy

As a systematic eclectic therapy, multimodal therapy draws upon many different theories without taking on board the underlying principles of any one approach

Lazarus, the founding father of the therapy, was trained in psychodynamic and humanistic approaches, attended seminars on behaviourism by Wolpe, experienced Adlerian therapy in London, and worked at Stanford University with social learning theorist Bandura

Initially focused purely on behavioural techniques, but later included cognitive techniques

This integration increased success with clients and led to the idea that an eclectic mix of established techniques could be the most effective therapeutic method

Began to incorporate other elements, such as social learning theory (people influence, and are influenced by, others and may learn through conditioning and modelling)

Eventually, expanded to include many therapeutic elements and developed the multimodal approach based upon seven different dimensions

Brief biography of Arnold Lazarus (1932–2013)

'Remember, think well, act well, feel well, be well!'
—Lazarus (2010)

Who was Arnold Lazarus?

Eclectic practitioner who sought to draw together successful elements of multiple therapies to create a new method of psychological treatment

Founder of multimodal therapy

Early years

Born 1932 in Johannesburg, South Africa

Education

- Studied an Honours degree and Master's degree in psychology at Witwatersrand University, Johannesburg
- Obtained a PhD in 1960
- Worked in private practice and lectured part time at Witwatersrand Medical School
- Training was primarily psychodynamic and humanistic
- Attended seminars by behaviourist Wolpe to learn about theories of conditioning
- Worked in the Marlborough Day Hospital in London for three months in 1957, encountering the Adlerian approach to psychotherapy

Career

- Worked with Albert Bandura (famous for his work on social learning theory) as a visiting lecturer in psychology at Stanford University in 1963
- Director of the Behavior Therapy Institute in Sausalito, California, in 1966
- Published *New Methods in Psychotherapy: A Case Study* in 1958
  - First use of the terms 'behaviour therapy' and 'behaviour therapist' in an academic paper
- In researching outcomes from behavioural therapy, he discovered that many clients relapsed over a period of time, but clients who had received cognitive as well as behavioural therapy had more successful outcomes over a period of time
- Many influences in his education, career and research led him to explore a broader approach to therapy
  - He argued that no one single approach could provide a comprehensive understanding of a person's development or problems
- Published *Broad Spectrum Behaviour Therapy and the Treatment of Agoraphobia* in 1966
  - Outlined a multidimensional approach to working with agoraphobics
- Developed a systematic therapeutic intervention based on cognitive and behaviour techniques in the 1970s
- Published *Behaviour Therapy and Beyond* in 1971 in which he advocated a broad approach to therapy
  - Influenced the development of cognitive-behaviour therapy
- Published *Multimodal Behaviour Therapy: Treating the 'Basic ID'* in 1973
  - Introduced the concept of the BASIC ID and outlined multimodal methods to address the various components of the BASIC ID
- Published *The Practice of Multimodal Therapy* in 1981
  - Outlined the process of multimodal therapy and presented the therapy without the 'Behaviour' title

Family

- Married in 1955
  - Raised two children

Death

Died on 1st October 2013, in Princeton, South Africa

Significant learnings

'We firmly believe that therapy is education rather than healing; that it is growth rather than treatment' (attributed to Lazarus, n.d.)

'My basic approach to therapy is very much that of problem solving. I see that people come into therapy with a whole myriad of different problems, and yet do not know how to solve some simple ones. And quite often simple solutions can have a tremendous, reverberating, positive effect' (Lazarus, n.d.)

'The therapist is on the client's payroll, and if one is not satisfied with what he or she is saying or doing, pull the plug' (Lazarus, 2010)

Basis of psychological problems

Psychological problems stem from numerous influences (Lazarus, 1981)

Misinformation

Incorrect beliefs and assumptions internalised from parents and significant others

For example, 'the world is a dangerous place', 'I must be perfect in all that I do', 'I must do for others', 'others should treat me well at all times'

Specific misinformation provided by others

For example, we may have been incorrectly informed that there is a long waiting list for referrals to a local agency to help with drink or drug addiction

Missing information

Lack of all of the pertinent information relating to the topic

For example, we may not know how to prepare for examinations

Defensive reactions

Avoiding situations that trigger negative feelings (such as anxiety)

This is understandable, but it does not help to challenge the anxiety

For example, a man with an anxiety about bridges will avoid bridges but this does not help him to deal with the underlying anxiety

Lack of self-acceptance

Drawing inappropriate conclusions from experiences based on beliefs about the self

For example, a woman goes for a job interview but is unsuccessful at getting the job, so she concludes that she is useless and will never get the job that she wants

Experience is assumed to be representative of the entire life, when it would often be better to look at the experience objectively in order to see what aspects could be improved

For example, the woman who failed to get a job could consider which aspects of the interview process could be improved so that she is better prepared for the next interview

Assessment process

Initial assessment is extremely important

Similar to most other approaches

- Developing relationship
- Establishing rapport
- Gathering information
- Assessing and determining a sense of direction

Lazarus (1981) identifies 12 areas that the therapist should be aware of during the first session

- Signs of psychosis
- Presenting problem and main antecedents
- Risk of harm to self or others, attitude towards self
- The way the client presented (appearance, body language, etc.)
- Key points in the client's history
- Maintenance factors (e.g. behaviours) that keep the problem going
- Goals that the client wishes to achieve
- Indication of particular therapeutic style appropriate for the client
- Suitability of therapy or the need to refer
- Client's strengths and resources
- Reason for engaging in therapy now
- Client's motivation and sense of hope

Multimodal life inventory

At the end of the first session (or, if there is a gap between initial contact and the first session, prior to the first session) the client will be asked to complete a multimodal life inventory

Detailed questionnaire that covers:

- Relevant aspects of the client's history, early relationships, etc.
- Details of the problem that the client is bringing to therapy
- Expectations of therapy
- Questions on each of the modalities

Inventory will provide the therapist with comprehensive information for the BASIC ID (Lazarus, 1973)

BASIC ID

BASIC ID is an acronym for seven modalities of human experiencing and functioning (Lazarus, 1973)

As individuals we think, feel, behave, have physical sensations, create images, interact with others and have biological functions – these are our modalities

The acronym covers these seven core modalities

Behaviour
Affect
Sensation
Images
Cognition
Interpersonal
Drugs

Behaviour

How we act and react in relation to self, others and events

Behaviour can be a reaction to change in another modality

For example, we become angry so we take action

Affect

Our feelings about self, others and events

Emotions are managed through one or more of the other modalities

For example, we reduce anxiety by using coping imagery

Sensation

Physical feelings and how our senses experience the world

Sensation and affect are closely connected

For example, we sense butterflies in the stomach so we feel excited

Sensation is influenced by other modalities

For example, we practise mindfulness so we feel relaxed

Images

Visual, auditory and kinaesthetic creations

Images influence other modalities

For example, flashbacks can create strong physical sensations and panic reactions

Cognition

Thoughts, values, attitudes and beliefs

These can be positive and life enhancing or negative and limiting

Cognition influences our feelings and behaviours

For example, if a person thinks 'I must always be in control' they will behave in ways to ensure they can be

Interpersonal

Relationships with significant others

These will influence the way we feel and behave

For example, we have an argument so we might feel angry and slam a door

Drugs/biology

This should be **B** for biology, but BASIC ID is more memorable than BASIC IB

Not only does it include drugs and medication, but also all that we take in, nutrition, exercise, fitness, hygiene, etc.

This modality influences and is influenced by other modalities

For example, we eat more healthily, we feel fitter and happier; we have a health scare, we stop smoking and drinking

As these seven modalities cover all aspects of the human experience, all need to be considered when assessing a problem

Failure to address one of the modalities may result in an important aspect of the problem being missed

Each modality influences, and is influenced by, other modalities so it is essential to understand each modality in the context of the other modalities

IDENTIFYING MODALITIES

Think of a problem that you have experienced in your life and identify all of the modalities associated with that event.

For example, you have a car accident (Behaviour), you get upset (Affect), replay the incident in your head (Interpersonal) and tell yourself 'I'm lucky to be alive' (Cognition) and feel sick (Sensation).

Look for examples in yourself and others throughout the next day – we can become very skilled at identifying modalities with practice.

Therapeutic interventions can be directed at specific parts of the BASIC ID

Therapy can apply any technique that has been found to improve client wellbeing in empirical studies to address specific problems in the BASIC ID

Interventions directed at Behaviour include . . .

Behaviour rehearsal

Therapist takes the role of the person to whom the client needs to speak

Client practises the communication until he feels confident enough to speak to the person directly

- Modelling
  - Therapist takes the role of client to show the client how to deal with a specific situation
  - Client can then practise the behaviour having had it modelled by the therapist
- Exposure
  - Client is encouraged to face a feared situation
  - Client could engage with a gradual, systematic exposure
  - For example, with a phobia of spiders, the client could be gradually introduced to spiders
  - Alternatively, client could be 'flooded' with the stimulus
  - For example, a client with a fear of bridges could be taken onto a bridge to stay there until anxiety reduces
- Recording and monitoring
  - Client will keep records and monitor thoughts and feelings
  - For example, a client who gets angry with his children may keep records of how often he gets angry, the triggers preceding the anger, the intensity of the anger, etc.
- Positive reinforcement
  - Client is encouraged to reward him/herself for achieving goals

Interventions directed at Affect include . . .

- Raising awareness of feelings
  - Client is helped to expand his feelings vocabulary and to be clearer about labelling feelings accurately
- Anxiety management
  - Teaching the client elements of relaxation training and coping strategies
  - Client can then either utilise these in actual situations that create anxiety or practise these by imagining anxiety-provoking situations within the therapeutic setting

Interventions directed at Sensation include . . .

- Relaxation training
  - Muscular relaxation training helps the client focus on the main muscle groups in the body, by tensing, holding and relaxing each muscle group in turn
  - Alternatively, the client may focus on specific muscle groups, for example the neck and shoulders, to manage a specific sensation

- Mindfulness
  - Helps the client focus and raises here-and-now awareness
  - There are connections between mindfulness and meditation
- Focusing on felt sense
  - Client is encouraged to relax and then to 'go inside' and pay attention to bodily sensations
  - Client can then be asked to stay with the sensation and label it appropriately

Interventions directed at Imagery include . . .

- Coping imagery
  - Clients imagine a situation in which they think they will not cope and go through the process step by step
  - At each point, they focus on how they are coping, despite any negative feelings that they see themselves experiencing
- Rational emotive imagery
  - Client imagines the worst-case scenario, experiences unhealthy negative emotions (e.g. fear) and, while keeping the negative image, changes the feelings to healthier emotions (e.g. concern)
- Changing the structure of the image
  - Technique to change the submodalities of a mental image (how the image is constructed)
  - For example, a mental image in colour and moving like a video can be changed to a black and white photograph
  - Emotional reactions to mental images will change when the structure of the mental image is changed

Interventions directed at Cognition include . . .

- Challenging negative automatic thoughts
  - Looking for evidence can challenge negative automatic thoughts
  - For example, a client may be anxious about going to the local shop because he believes that people will laugh at him so the challenge could be 'What is the evidence that people laugh at you in shops?'
  - If there is evidence that it has happened in the past, then the client can be challenged with the likelihood that it will happen again in the future
  - A more positive thought would be 'It has happened in the past but it is unlikely to happen again – however, if it did, it would not be the end of the world, and I could cope with it'

- Disputing irrational beliefs
  - Negative automatic thoughts may be underpinned by irrational beliefs
  - These beliefs are absolute statements of 'shoulds' and 'musts'
  - These can be disputed and replaced with more flexible rational beliefs
  - For example, 'They must treat me well at all times' can be disputed and replaced with 'I would prefer them to treat me well, but they don't absolutely *have* to and, if they don't, I can cope'
- Positive self-statements
  - Positive self-statements may be about coping in stressful situations
  - For example, 'I am a loving and lovable person'
- Correcting misinformation
  - People often maintain their problems because they do not update and correct misinformation
  - This may be incorrect factual information or statements about the self that were originally introjected in childhood but no longer apply in adulthood

Interventions directed at Interpersonal include . . .

- Assertiveness training
  - Clients who are passive in their communications with others benefit from assertiveness training
  - Assertiveness training may assist clients in learning to say 'no', asking for what they want, and demanding their rights
  - Role-play to rehearse situations would be used as part of assertiveness training
- Communication training
  - Clients can learn about effective communications (e.g. TA ego-state model to explore transactions)
  - Modelling and role-play would be used so that clients can practise starting and ending conversations

Interventions directed at Drugs/biology include . . .

- Smoking cessation programmes
- Alcohol/drug reduction programmes
- Nutrition awareness programmes
- Weight reduction programmes
- Exercise programmes
- Various other techniques designed to motivate and maintain progress towards achieving realistic goals

Profiling

Modality profiling

Modality profile is a chart drawn up to outline the presenting problem in the seven modalities

Profile remains open to change as more information is revealed and the therapy develops

Profile identifies interventions that may be used to deal with each aspect of the problem

MODALITY PROFILE

Below is an example of a modality profile for a student with examination anxiety. This modality profile is a work in progress and will be updated in response to the development of the therapy.

Modality	Problem	Intervention
Behaviour	Procrastination in revising Problems sleeping Avoiding lecturers	Realistic revision timetable Relaxation techniques Behaviour rehearsal
Affect	Anxiety Panic Anger	Anxiety management Breathing exercises Anger expression
Sensation	Nausea Headaches Palpitations Tension in shoulders and neck	Dietary information Relaxation techniques Breathing exercises Muscular relaxation training
Imagery	Negative image of self in exam Images of not coping Images of being unable to write	Positive self-talk Coping imagery Changing the structure of the image
Cognition	'I am going to fail' 'I will let my parents down' 'People will think I'm stupid' 'I am worthless'	Coping statements Challenging negative thoughts Disputing irrational beliefs Practising rational beliefs
Interpersonal	Avoiding friends Angry outbursts with family	Social skills training Strategies for coping
Drugs/biology	Missing meals Drinking lots of coffee	Nutrition education Information giving about effects of too much caffeine

Second-order profiling

Sometimes a client may be stuck in one area of the modality profile

In these cases, a second modality profile may be drawn up specifically for that area (Lazarus, 1973)

For example, the above client may become increasingly concerned about the palpitations he is having so a second-order profile about the palpitations might identify the following problems associated with this specific modality

Behaviour	Sits still for a while
Affect	Panic
Sensation	Shallow breathing
Images	Imagining having a heart attack
Cognition	I am going to die
Interpersonal	Withdrawal from friends
Drugs/biology	Not seeking medical advice

In this case, a new issue has emerged that needs to be explored – client is avoiding going to his doctor for fear of what might be discovered

This issue clearly impacts upon his presenting problem, and is likely to be resolved easily by correcting his misinformation and providing the relevant medical information with a visit to his GP

Therapy can then return to the presenting problem

PROFILING THE SOAPS

Take a character from a TV programme (soaps are a rich source of people with problems) and draw up a BASIC ID modality profile for their problem. Identify some possible interventions that would be appropriate for each of the seven dimensions

Structural profiling

Structural profile is how the client perceives him/herself in relation to each of the seven modalities (Lazarus, 2005)

Some people consider themselves to be 'thinkers', while others see themselves as 'doers'

Some people spend a lot of time creating images (daydreaming or visually planning), while others consider themselves as emotional, in touch with their feelings

Clients are asked to rate themselves for each of the dimensions (Lazarus, 1981) in response to questions such as:

B How active are you in getting things done?

Do you procrastinate?

A How much are you in touch with your feelings?

How emotional are you?

S How aware are you of bodily sensations?

How much do you focus on your senses in the here and now?

I Are you a daydreamer?

Do you consider yourself to be an imaginative person?

C How much of a thinker are you?

Do you like to think things through before acting?

I How important are other people to you?

How would you rate yourself as a social person?

D How healthy are you?

How much do you exercise and pay attention to nutrition?

Clients are then asked to rate themselves on a scale of 1 to 10 (Lazarus, 1981)

For example, the client who sees himself as very active would score 10 for D whereas the client who sees himself as overweight and taking no exercise would score 0 for D; this would then represent his current structural profile

Client could then be asked to draw up a preferred profile, and the differences between the two would indicate some areas to work on

For example, if a client wants to lose weight, have a better diet and become healthier, then his preferred D score may be close to 10; this would identify some work to be done on weight reduction and exercise planning

Any differences in the modality score for the current and preferred structural profile would indicate areas to focus on in the therapy

MY STRUCTURAL PROFILE

Spend a few minutes to think about yourself in relation to the seven dimensions of BASIC ID to draw up a current structural profile. Then think about how you would like to be to draw up a preferred structural profile. Note the differences between the two profiles and think of ways to start the change in the direction of the preferred profile.

Tracking and bridging

Tracking the firing order of the modalities (Lazarus, 2005)

Everyone has their own way of 'doing' their problem

This can be explored in the specific modality sequence (or order) in which the problem occurs

For example, two people who get anxious before going into a social situation may respond to the situation in very different ways

One person may become aware of feeling nauseous (S), label the feeling 'anxiety' (A), thinks 'It'll be an awful evening' (C), and decides to have a glass or two of wine to relax before going (B); this person has a firing order of S-A-C-B

Alternatively, another person may remember 'I've got that party tonight, I'm going to hate it' (C), visualises who will be there and sees herself being cornered by people she thinks are overbearing (I), feels 'anxious' (A), and senses the start of a headache (S); this person has a firing order of C-I-A-S

There are interventions for each of these modalities and in order to tailor the therapy to the individual, it is best to use the techniques in the same order as the modality firing order

For example, in the case given above, taking tablets for the headache may be appropriate but this does not stop the person from the negative thought, the negative image and the labelling of the feeling – it is dealing with the S of the C-I-A-S sequence

It would be better to deal with the modality that starts the sequence (Cognitive)

If the negative thought can be changed (e.g. 'I've got that party tonight and I'm not looking forward to it, but I can always leave early') then the other problem modalities may not follow (there may be a more positive image, no identifiable anxiety and no resulting headache)

Bridging between modalities (Lazarus, 2005)

As noted in the structural profile above, individuals have preferred modalities

We often hear expressions such as 'I'm a doer, not a thinker', 'She's very into her emotions', 'He's always daydreaming'

Individuals will also have least preferred modalities

Many people find it difficult to access and express their own feelings and they may answer feeling-questions with cognitions

However, in order to explore all areas of a problem, the therapist will need to help the client access some of the less preferred modalities

For example, a client may often answer Affect questions with Cognition response

'How did that make you feel?' (Affect question)

'I felt it was really unfair' ('unfair' is a thought, not a feeling)

In order to help the client access Affect, the therapist could stay with the same line of questions until the client gets to a feeling

'And how did you feel when you thought it was unfair?'

'Well, it's not right to be treated like that'

'And the feeling you were left with?'

'Well, angry I suppose'

These directed questions help the client move from Cognition to Affect

Bridging is an alternative way of engaging the client to help him/her access other modalities

For example, a client may often answer Affect questions with Cognition response

'How did that make you feel?' (Affect question)

'I felt it was really unfair' ('unfair' is a thought, not a feeling)

Therapist might note the Cognition response and ask the client about Sensations as a bridge to Affect

'What sensations were you aware of as this was happening?'

'Well, my breathing got shallower and faster and my heart started pounding'

'And how would you label that feeling?'

'Anger'

This uses the Sensation bridge to move the client from Cognition to Affect

Not only does bridging help the client access all modalities (and therefore become more fully connected to all aspects of self), it also helps build rapport between therapist and client

Case study demonstrating multimodal therapy

Carl, 17, was advised by his tutor to visit the therapist at his college. Although he is a good student, he struggled with his mock exams and he is becoming increasingly anxious about his final exams. His final maths exam will take place in two weeks time, but he has not yet begun to do any revision at all.

The first session was an initial assessment in which the therapist sought to find out the reason for attending therapy and establish a therapeutic relationship. It was agreed that there would be three further sessions after the assessment session. Carl was asked to complete the life inventory in order for the therapist to begin working with him to produce modality and structural profiles.

In the following exchanges, therapist denotes therapist, client denotes Carl. Confidentiality statement and the business contract have been discussed. The session is joined as the therapist begins to create the modality profile.

Therapist: Okay Carl, can you perhaps tell me a little bit about the revision problem?

Client: It's hard to explain really. I have always done so well at school, but I am just struggling with these exams. None of the information seems to stay in my head and

I think that I am going to fail everything and not get into university. I'm the only one in my family to go to university and my mum is so excited. She would be so angry . . . no, not angry . . . she would be so disappointed in me if I failed. But I just can't face doing the work at the moment.

Therapist: So can you tell me what happens when you come to do your work?

Client: Well, I decide that I am going to do it. But then I start to feel all hot and I keep imagining my mum's face when I get my failed results through the post and I feel sick so I have to just get away from my books for a while. So I play on the computer. And then another day passes and I haven't started revising yet.

Therapist: In multimodal therapy, we try to work out something called your BASIC ID. This is a profile of your problem so that we can really start to explore it fully. And you have already told me quite a lot that can fit in this profile. Let's write each bit down so that we can see it all together. So the B is behaviour and it sounds like your behaviour is procrastinating?

Client: Yeah. I play on my computer instead of revising.

Therapist: And the S is sensation and you said that you feel hot and sick?

Client: I do, yeah. All of my skin gets hot and I start to sweat then I get butterflies in my stomach and I start to feel sick. Sometimes my head starts to hurt too and I have to lie down for a while.

Therapist: You also mentioned an image and that is one of the I's.

Client: My mum's face. I keep picturing it when she reads the letter with my results.

Therapist: And the other I is interpersonal and that one focuses on your relationships with other people. I guess that the significant one here would be your mum?

Client: Mainly, yeah. But also my tutor. Mr Roberts has been so good to me and I don't want to let him down.

Therapist: So it sounds like your thoughts about the problem are that you are going to fail and your mum and your tutor will be disappointed in you? That would be the C for Cognition.

Client: They're gonna think that I'm stupid and that they shouldn't have wasted their time on me.

Therapist: Okay, so we have got most of the BASIC ID for your problem. But now that I am looking at it, I notice that there is something missing. The A is for affect or feelings. How do you feel about this problem?

Client: I just told you. I feel like my mum will be disappointed in me.

Therapist: And so you get butterflies and start to feel sick?

Client: Yes.

Therapist: So what name could you give to the feeling behind that sickness?

Client: Erm, I don't know. Worry, maybe?

Therapist: I guess worry sounds quite mild to me, but your feelings sound quite strong.

Client: Yeah, you're right. Not worry. Fear.

Therapist: Okay, so our A for affect would be fear?

Client: Yeah, that's it.

Therapist: Okay, so the last part of the BASIC ID is D for drugs.

Client: (*quickly*) I'm not on drugs!

Therapist: No, it doesn't really mean that. It means biology. Do you eat healthily and exercise?

Client: Oh. No, not really. I guess I eat a lot of rubbish at the moment. I haven't put any weight on, but I'm really sluggish and tired all of the time. I used to play football on a weekend, but I haven't done that for months. I keep saying that it's because I'm too busy revising, but I just sit on the computer and don't revise either.

Therapist: I think that gives us a lot to work with. Would you like to look over your BASIC ID profile? (*both look at the profile together*) One of the interesting things about problems is that everybody does them differently. Two people might seem to have the same problem, but they might feel very different or think in completely different ways. For example, two people might share a fear of dogs, but one might start the fear by shaking while another might start the fear by thinking that the dog is going to bite. So I am just wondering how you do your problem? Which bit of this BASIC ID comes first for you when you sit down to start working?

Client: Do you mean, which one do I feel first? Well, I guess that I feel hot first. Because I start thinking that I will never get through all of my work and I am going to fail and I picture my mum's face.

Therapist: So do you feel hot and then think that you are going to fail? Or do you think that you are going to fail and then feel hot?

Client: Erm, I guess that thinking comes first. I think that I'm gonna fail and then I picture my mum's face and then I feel hot.

Therapist: And when does the fear start?

Client: As soon as I think that I'm gonna fail.

Therapist: So you think that you are going to fail and then you feel fear and then you picture your mum's face and then you feel hot and sick?

Client: Yeah, that's exactly right.

In this exchange, the therapist successfully formulated an entire BASIC ID modality profile. In the next stage, the therapist works on designing interventions to address the modalities. Since Carl's firing order begins with Cognition, they decide to devise an intervention to tackle this area. The therapist plans to begin working on the structural profile in the next session, but Carl does not return to therapy.

Carl only ever attended this single session (he cancelled his next sessions by telephone), but he sent the therapist an email at the end of the academic year. The email thanked the therapist for helping him to understand his problem and said that he had been able to start revising straight after the session. His delay in starting work cost him in his final grades, but he did successfully pass all of his exams and he was accepted to study biology at university.

ANALYSING THE CASE STUDY

Try answering the following questions using information about multimodal therapy.

- What was the BASIC ID structural profile for Carl and which interventions could be used for each modality?
- How did the therapist use bridging in the session?
- How did the therapist track the firing order of the modalities?
- What would be expected in an actual and preferred structural profile for Carl?

SUMMARY

Eclecticism: technical eclecticism across multiple therapies, structured and systematic method, highlights importance of therapeutic relationship but also adds empirically supported techniques tailored to the individual

Development: Lazarus drew on many approaches to form this therapy, including humanistic, psychodynamic, behavioural, cognitive, etc.; therapy was introduced in a book by Lazarus published in 1973

Basis of psychological problems: all problems stem from misinformation, missing information, defensive reactions and lack of self-acceptance

Assessment process: initial assessment is crucial for developing the therapeutic relationship; multimodal life inventory provides information for the BASIC ID

BASIC ID: Behaviour, Affect, Sensation, Images, Cognition, Interpersonal, Drugs/biology; therapeutic interventions directed at specific parts of the BASIC ID: Behaviour (rehearsal, modelling, exposure, recording and monitoring, positive reinforcement), Affect (raising awareness of feelings, anxiety management), Sensation (relaxation training, mindfulness, focusing on felt sense), Imagery (coping imagery, rational emotive imagery, changing structure of image), Cognition (challenge negative automatic thoughts, dispute irrational beliefs, positive self-statements, correcting misinformation), Interpersonal (assertiveness training, communication training), Drugs/biology (health programmes)

Profiling: modality profile is a chart drawn up to outline the presenting problems and proposed intervention in the seven modalities; second-order profile involves profiling the seven modalities within one of the original modalities; structural profile is how the client perceives him/herself in relation to each of the seven modalities

Tracking: everyone has their own way of 'doing' their problem, can be explored in the specific modality sequence (or order) in which the problem occurs;

tracking the firing order of the modalities allows the treatment to focus on the original problem rather than later symptoms

Bridging: some clients will have most preferred (often Cognition) and least preferred (often Affect) modalities; bridging uses other modalities to move the client from the preferred modality to the least preferred modality

NEUROLINGUISTIC PROGRAMMING

LEARNING OUTCOMES

After reading this section, you will be able to:

- explain that neurolinguistic programming is an eclectic therapy adopting techniques from across the approaches
- discuss the history of neurolinguistic programming with brief biographies of Richard Bandler and John Grinder
- describe and evaluate representation systems in terms of sensory modalities and submodalities
- describe and evaluate the meta model with specific focus on faulty thought patterns involving distortions, deletions and generalisations
- discuss the techniques of anchoring and swishing
- discuss the controversies associated with neurolinguistic programming
- appreciate the application of neurolinguistic programming in a therapeutic setting (case study)

Neurolinguistic programming: eclectic

Neurolinguistic Programming (NLP) draws from a range of different therapies

Aims to unite all of the successful techniques and approaches in different therapies in order to establish a successful method of psychological change

NLP focuses on the use of language to make changes in the brain

Since therapy is known as the 'talking cure', it is logical that language should form the basis of the treatment and evidence suggests that language does influence thoughts and feelings

However, some practitioners have taken it a step too far by making excessive claims about the potential for NLP

Development of neurolinguistic programming

History of neurolinguistic programming

Bandler and Grinder (1975) introduced neurolinguistic programming as a method of psychotherapy designed to instil fast and effective change in those individuals experiencing psychological discomfort (such as phobias, depression, anxiety disorders, etc.)

This therapy was established following a review of current practices in psychotherapy to determine common methods yielding the greatest levels of success across many different approaches

Bandler and Grinder highlighted three highly successful therapeutic approaches: family therapy, Gestalt therapy and hypnotherapy

Family therapy

Bandler and Grinder were inspired to develop the meta model by the work of family therapist Virginia Satir

Satir argued that change is possible following the disruption of the status quo and subsequent conversion towards an improved state would only be possible if the foreign element creating the disruption could be effectively integrated into the life of the individual

Gestalt therapy

Bandler and Grinder were inspired to develop the meta model by the work of Gestalt therapist Fritz Perls

Perls focused on the concept of personal responsibility to devise the Gestalt Prayer:

'I do my thing and you do your thing; I am not in this world to live up to your expectations; and you are not in this world to live up to mine; you are you, and I am I, and if by chance we find each other, it's beautiful; if not, it can't be helped'

This prayer summarises the idea that the individual should focus on his or her own needs without projecting these needs onto others or allowing these needs to be influenced by the projections of others

Hypnotherapy

Bandler and Grinder were inspired to develop meta model by the work of hypnotherapist Milton Erikson

Erikson adopted an unconventional approach to psychotherapy using therapeutic metaphor and hypnotic suggestion to influence behavioural and emotional change in his clients

Brief biography of Richard Bandler (1950–current)

'We take the very best of what people do, synthesise it down, make it learnable and share it with each other – and that is what the real future of what NLP will be and it's gonna stay that way!'

—Attributed to Bandler (n.d.)

Who is Richard Bandler?

Co-founder of NLP

Early years

Born on 24th February 1950 in the US

Education

Graduated with a degree in philosophy and psychology from the University of California in 1973 and an MA in psychology from Lone Mountain College in San Francisco in 1975 (Clancy & Yorkshire, 1989)

Career

Ran a Gestalt group at the University of California in Santa Cruz

Worked with linguistic professor John Grinder in therapy sessions

Published *The Structure of Magic Volume 1* in 1975 and *Volume 2* in 1976 with John Grinder

Introduced the concept of NLP for effective therapeutic change

Published *Frogs into Princes* with John Grinder in 1979

Transcript of a seminar outlining the practice of NLP in therapy

Bandler and Grinder stopped working together in the 1980s

Rumour suggests that the separation was acrimonious (Renton, 2009)

Published *Using your Brain for a Change* with Steve and Connirae Andreas in 1985

Focused on the concept of manipulating submodalities to promote positive personality changes

Bandler filed a lawsuit for trademark infringement against Grinder in 2000 based on the claim that Bandler was the original creator and legal owner of the term 'neurolinguistic programming' (Wong, 2000)

Eventually both parties agreed to accept that they were co-creators of NLP and all claims were settled in 2000 (Grinder & Bostic St Clair, 2001)

Developed a new system of therapy called design human engineering

Ran a series of seminars on this new approach in 2000

Family

Reportedly abused alcohol and cocaine throughout his life (Clancy & Yorkshire, 1989)

Acquitted of the murder of Corine Christensen (Clancy & Yorkshire, 1989)

Corine Christensen was shot dead in her home in 1986

She lived with James Marino (convicted burglar and drug dealer) who was a friend of Bandler

Marino and Bandler were seen leaving the house after the shooting and Bandler's shirt was covered in her blood

In court, Bandler claimed that Marino shot his girlfriend because he believed that she was trying to have him murdered – in contrast, Marino claimed that

Bandler shot her because he believed that she was having a lesbian affair with his girlfriend

Bandler was acquitted in 1988

Significant learnings

'Most people plan by disaster. They think of what can go wrong and then they master it' (Bandler quoted in Bandler & Fitzpatrick, 2009)

'You can't just be happy, but you can learn to do things happily. Living happily entails paying attention to and enjoying the process of doing whatever it is you happen to be doing' (Bandler, 2010)

'Simply put, the meaning of life is the meaning you ascribe to it. What you believe is what you'll see, hear and feel; therefore it makes sense to pay special attention to the beliefs you hold' (Bandler quoted in Bandler & Thomson, 2011)

'In life you are given a bag of moments. You don't know how many of those moments you have. My courses are about learning how to maximize those moments' (Bandler, n.d.)

Brief biography of John Grinder (1940–current)

'Control is a fiction – a seductive illusion – choice is the point.'
—Grinder (2001)

Who is John Grinder?

Co-founder of neurolinguistic programming

Early years

Born on 10th January 1940 in the US

Education

Graduated with a degree in psychology from the University of San Francisco and a PhD in linguistics in 1961 (Preube, 2013)

Career

Served as a captain in the Special Forces (Preube, 2013)

Worked for the US Intelligence Agency (Preube, 2013)

Linguistics professor at University of California in Santa Cruz

Worked with psychology student and Gestalt practitioner Richard Bandler in therapy sessions and became interested in the effective use of language in therapy

Published *The Structure of Magic Volume 1* in 1975 and *Volume 2* in 1976 with Richard Bandler

Introduced the concept of NLP for effective therapeutic change

Published *Frogs into Princes* with Richard Bandler in 1979

Transcript of a seminar outlining the practice of NLP in therapy

Bandler and Grinder stopped working together in the 1980s

Rumour suggests that the separation was acrimonious (Renton, 2009)

Published *Turtles All The Way Down* with Judith DeLozier in 1987

Developed a New Code of NLP

Bandler filed a lawsuit for trademark infringement against Grinder in 2000 based on the claim that Bandler was the original creator and legal owner of the term 'neurolinguistic programming' (Wong, 2000)

Eventually both parties agreed to accept that they were co-creators of NLP and all claims were settled in 2000 (Grinder, 2001)

Published *Whispering in the Wind* with partner Carmen Bostic St Clair in 2001

Theoretical guide to the New Code of NLP with a specific set of recommendations, including focus on modelling as a critical component of NLP

Significant learnings

'Surely, if there is one thing all of us who have extended experiences inducing change with human beings (ourselves included, of course) through the application of NLP patterning can agree on, it is the futility of thinking in terms of controlling anything in the realm of human activity' (Grinder quoted in Grinder & Bostic St Clair, 2001)

Representation systems

Three sensory modalities

The human mind will represent information in accordance with particular senses

Visual

Auditory

Kinaesthetic

For example, an individual may process information by seeing internal images (visual), hearing an internal voice (auditory) or focusing on internal feelings (kinaesthetic)

These methods of internal representation of information are known as sensory modalities

Visual sensory modality

Visual individuals will focus on how something looks

They will think about things in picture form and may be more likely to recall information that has been presented in pictures

They are less distracted by noise but will have difficulties following verbal instructions

Visual people are often organised and orderly, and they will have a high regard for physical appearances

Auditory sensory modality

Auditory individuals will focus on how something sounds

They will think about things through an inner voice and may even talk to themselves or move their lips when thinking about a particular issue

They are more likely to recall verbal information, but they can be easily distracted by noise

Auditory people are often sequential, and they are more likely to respond to verbal instructions that have been worded in an appropriate way and delivered in a suitable tone

Kinaesthetic sensory modality

Kinaesthetic individuals will focus on how something feels

They will think about things in terms of feeling

They are more likely to learn by doing and will recall information if they have physically experienced an associated activity

Kinaesthetic people are usually tactile, and may respond well to physical rewards

Detecting preferred sensory modalities

Most people will have a preferred sensory modality indicating the way that their own mind structures and organises information (Bandler & Grinder, 1979)

Preferred sensory modalities will often be revealed through verbal cues

An individual with a particular preference for a distinct modality will often exhibit this preference by describing their world in those terms

Preference for visual sensory modality

An individual may display a visual preference by exclaiming that they can 'see' themselves trapped inside a cage or describing something confusing as 'misty' or 'cloudy'

Preference for auditory sensory modality

An individual may display an auditory preference by stating that they feel like the world is 'shouting' at them or that they cannot concentrate because they are 'surrounded by noise'

Preference for kinaesthetic sensory modality

An individual may display a kinaesthetic preference by explaining that they feel that there is a 'weight on their shoulders' or they are concerned that they cannot get a 'grasp' on their problem

Matching language

Effective therapists should strive to recognise the sensory modalities preferred by the client and match the language in order to build rapport

Bandler and Grinder (1979) illustrated the importance of matching language in accordance with the client's representational system

They present a comical imaginary dialogue between the client and the therapist to highlight the difficulties associated with these language conflicts

For example, the client states 'things feel really heavy in my life . . . I can't handle it' and the therapist responds with 'I can see that'

An observant therapist would have noticed that the client is demonstrating a kinaesthetic preference and would have responded accordingly

Therapists can enhance the effectiveness of their treatment by matching their language to the preferred sensory modality of the client

DETECTING SENSORY MODALITIES

Record yourself during conversation – try letting the recorder run for a long time to ensure that the discussion is natural. Now listen carefully to your own use of language. Do you use visual terms ('I see what you mean') or auditory terms ('I hear you') or kinaesthetic terms ('I get that feeling')? Try to identify your own preferred sensory modality.

Once you have experienced detecting sensory language, try to identify the preferred modalities of the people around you. Listen carefully to their use of language, particularly when they are using metaphors or analogies. Think about how you could match your own language to their preferred sensory modalities – see what happens when you intentionally mismatch and match language.

Submodalities

Human mind represents information in accordance with particular senses and these methods of internal representation of information are known as sensory modalities

Each of these sensory modalities can be further divided into submodalities in accordance with the specific representational system

Visual submodalities

Framed/panoramic
2D/3D
Colour/monochrome
Brightness
Sharpness
Focus
Location
Size
Shape
Style
Contrast

- Clarity
- Movement

Auditory submodalities

- Mono/stereo
- Volume
- Pitch
- Tone
- Variations
- Location
- Range
- Clarity
- Voice

Kinaesthetic submodalities

- Intensity
- Texture
- Pressure
- Heat
- Weight
- Location
- Movement

Submodalities are closely linked to emotional perception

- Our emotional responses to a memory will be associated with certain submodalities
  - For example, an individual may visualise a negative event in monochrome with a dark frame
- Submodalities can be used to explore feelings about a particular issue and, more importantly, alter feelings if required
  - For example, the above individual could improve their feelings about the negative event by mentally replacing the dark frame with a light frame and adding colour to the memory

Meta model

The meta model can encourage the client to address deeper issues

- Set of questions designed to challenge and expand the client's internal model of the world (Grinder & Bandler, 1981)
  - Meta model focuses on the translation of three faulty thought patterns
    - Distortions
    - Deletions
    - Generalisations

Distortions refer to inaccuracies or assumptions

Mind-reading distortions

Client claims to know the thoughts of another person

For example, 'he hates me'

We can challenge this distortion by asking the client how they know this information

Lost performance distortions

Client makes value judgements without identifying the source

For example, 'it is important to be liked'

We can challenge this distortion by asking the client who has made this judgement

Cause-and-effect distortions

Client assumes that one thing will cause another

For example, 'I will be miserable if he goes away'

We can challenge this distortion by highlighting the assumption and asking the client how or why one thing causes the other

Complex equivalence distortions

Client assumes that two things are synonymous

For example, 'I didn't get a good grade in this exam, I am rubbish at everything'

We can challenge this distortion by highlighting the assumption and asking the client if there are any cases when these two things are not synonymous

Presupposition distortions

Client assumes that something is true

For example, 'I am worried that my new boyfriend will be as unreasonable as my old boyfriend'

We can challenge this distortion by highlighting the assumptions and questioning their validity

Deletions refer to the exclusion of certain information

Simple deletions

Client omits certain information

For example, 'I am unhappy'

We can challenge this deletion by requesting further information

Lack of referential index deletions

Client fails to specify the relevant person or thing

For example, 'they are mean to me'

We can challenge this deletion by requesting further information

Comparative deletions

Client makes an unspecified comparison using general terms like 'best', 'worse', etc.

For example, 'he is a better person'

We can challenge this deletion by asking the client to define the terms

Unspecified verb deletions

Client uses a verb without specifying the details of the action

For example, 'she rejected me'

We can challenge this deletion by asking the client to define the verb in more detail

Nominalisation deletions

Client changes a verb into a noun

For example, 'he values his freedom'

We can challenge this deletion by asking the client to define the verb in more detail

Generalisations refer to the use of absolute sweeping statements

Universal quantifier generalisations

Client uses global terms such as 'all', 'every', 'never', etc.

For example, 'everyone thinks he is wonderful'

We can challenge this generalisation by highlighting the global term and then asking the client to consider exceptions and counter-examples

Modal operators of necessity

Client uses a verb implying obligations ('should', 'must', 'need to', etc.) to describe a verb

For example, 'I have to do it'

We can challenge this generalisation by asking the client to consider the effect of not meeting the obligation

Modal operators of possibility/impossibility

Client uses a verb implying prospects ('can/can't', 'possible/impossible', 'will/won't', etc.) to describe a verb

For example, 'I can't do it'

We can challenge this generalisation by asking the client to explain why these prospects exist

Dangers of challenging

Although this model could be helpful in highlighting faulty thinking, it is important to ensure that any challenges or confrontations with the client are handled delicately as they could make the client feel defensive

Therapists who have established a good rapport with a client could use these challenging questions to successfully help the client to understand their deeper feelings about problem issues, but care must be taken at all times

Anchoring and swishing

Anchoring

Anchors are based on the principles of classical conditioning

> Classical conditioning occurs naturally in many of our interactions: we might feel good when we hear a song that was in the charts during our happy childhood or we might feel bad when we smell a perfume that was worn by a teacher during our unhappy teenage years

Anchors use the same principles as conditioning to forge a link between a physical act (such as touching thumb to forefinger) and a desired emotional response (such as a feeling of calm) (Bandler & Grinder, 1979)

> Client visualises a positive scene in order to induce the desired emotion
>
> > For example, imagining relaxing by the sea in order to induce a feeling of calm
>
> Client performs the physical act as the feeling is ascending towards a peak
>
> > For example, touches certain part of the hand

Anchors can be established effectively by ensuring that certain key behaviours are completed during this process (Bandler & Grinder, 1979)

> First key to anchoring is that the desired emotional state should have a high intensity
>
> > Intense relaxation is more easily anchored than mild relaxation
>
> Second key to anchoring is that the physical act is unique to the anchor
>
> > Waving is not an appropriate act as it is too often fired unintentionally thus leading to the extinction of the association
>
> Third key to anchoring is that the physical act is easily replicable
>
> > It is unreasonable to expect a person to complete a complex difficult action in order to elicit the positive emotion
>
> Fourth key to anchoring is that the anchor is set a significant number of times
>
> > Process of setting the anchor should initially be repeated several times and this stacking of the anchor should be regularly refreshed
>
> Final, and perhaps most important, key to anchoring is that the physical act is precisely timed to coincide with the peak of the emotional experience
>
> > Anchor should always be set as the state is nearing the peak in the ascendancy to ensure that the physical act is not inadvertently associated with a diminishing emotion

Once anchor has been fixed, client can induce the desired emotion during negative situations by simply completing the physical act

> For example, the nervous student could induce a feeling of calm during an exam by simply touching thumb to forefinger

ANCHORING POSITIVITY

Try out the anchoring technique for yourself. Identify a positive emotion and then work to create that emotion within yourself. As the emotion reaches the peak, make a simple hand gesture. Repeat this process several times to ensure that the positive emotion is firmly anchored to the action. Once you have anchored the gesture, try it out when you are in a less positive frame of mind – does the gesture recreate the positivity?

Swishing

Swish pattern technique is used to redirect thoughts and behaviours from negative patterns to positive patterns (Masters et al., 1991)

Client is instructed to visualise the problem state from a fully associated position with the solution state in the left-hand corner of the problem image

- Swish is then completed by simultaneously switching the locations of the two images: the problem image shrinks and recedes to a distant point while the solution image explodes into view
- Swish between the two images is further supported by the use of submodalities: for example, the problem image may fade to monochrome while the solution image develops bright colours

Swish is repeated several times until it becomes difficult to visualise the problem image

- At the point when the problem image is difficult to visualise effectively, the swish pattern should be ceased in order to encourage the establishment of the new desired state
- Irrespective of alterations in the visualisation of the problem image, the swish should not be repeated more than ten times in any one session

SWISH AWAY NEGATIVITY

Try out the swishing technique for yourself. Identify a negative habit and then imagine yourself doing the behaviour on a big screen. Picture the alternative desired behaviour in the bottom left-hand corner. For example, if your negative habit is biting your nails, you might create a mental picture of your own hand with ragged fingernails coming up towards your mouth. Remember, you must be fully integrated into the image – you should not be observing as though it is happening to someone else. You could then imagine a pair of beautiful hands with long nails in the left-hand corner. Swish between these two images several times until it becomes more difficult to imagine the problem. What impact does this activity have on your subsequent behaviour?

Controversies in NLP

NLP is arguably the most controversial therapy in this textbook

> Some therapists regard it as pseudoscientific and consider it an unprofessional method of engaging with clients
>
> Some researchers have highlighted the exaggerated claims of proponents as evidence of weak theoretical foundations

Preferred representation systems

Little evidence for the use of preferred sensory modalities (Sharpley, 1984)

Eye movements were found to have no relation to the recall of visual, auditory and kinaesthetic stimuli (Wertheim et al., 1986)

No theoretical foundation or evidence

Roderique-Davies (2009) concluded that, despite three decades of use across thousands of organisations, there is little empirical evidence to support NLP techniques and no credible theoretical basis for NLP that is not strictly anecdotal

Tosey and Mathison (2007) argued 'the pragmatic and often anti-theoretical stance by the founders has left a legacy of little engagement between practitioner and academic communities'

Plagiarism?

Sharpley (1987) noted that several 'NLP' techniques are actually modifications of previous techniques from other therapies – they are marginally effective but cannot be classed as original

> But is therapy ever truly original? Or do all theories simply build on the previous theories? NLP is an eclectic therapy and has always admitted to borrowing successful techniques from other approaches

Despite the controversies surrounding this approach, there are some interesting and engaging techniques and these can be useful for the eclectic therapist

Case study demonstrating neurolinguistic programming

Laura, 26, has been struggling to reduce her weight for the last three years. Her doctor has advised her that she is experiencing many health difficulties relating to her diabetes because she is clinically obese yet continues to binge on sugary foods. She craves sweet things all of the time and regularly consumes six times the recommended daily allowance of sugar. She has visited an NLP therapist in the hope of finding a technique to control her eating habits. She is hoping to attend for only one session, but she is willing to attend a top-up session in the future.

In the following exchanges, therapist denotes therapist, client denotes Laura. Confidentiality statement and the business contract have been discussed. The session is joined just after Laura is completing her story.

Client: So I know that it is going to kill me, but I just can't stop. I picture food in my head all the time and I'm miserable on a diet. It feels like life isn't really worth living anyway if I have to be miserable. I just can't see myself ever being happy on salads.

Therapist: I can see your point, but could there be a compromise?

Client: Maybe. I guess. If I just ate a little bit less of everything, maybe.

Therapist: I get how that could work, but can you picture yourself stopping after only eating a little bit?

Client: (*sighs*) No. Once I start, I can't stop.

Therapist: Okay, so what if we come at this problem from another angle? Instead of stopping after starting, what if you just avoid some things altogether?

Client: I don't know. Like what?

Therapist: What is your favourite food? Absolute most favourite?

Client: Erm, probably chocolate biscuits. No, wait, chocolate cake.

Therapist: So let's leave that one alone. But what is your second favourite?

Client: Chocolate biscuits.

Therapist: So let's work with that one and our aim is for you to not eat chocolate biscuits at all. First of all, can you picture a plate of chocolate biscuits?

Client: Yes.

Therapist: Good. Try to make the picture in your head really vibrant. Make it bright and colourful. Now describe it to me – describe the whole picture including the plate.

Client: There are six biscuits and they are really rich and brown. They are on a bright red plate. One of the biscuits is broken and there is melting chocolate in the middle and it is oozing out onto the plate.

Therapist: And the colours are very rich?

Client: Yes. And it is really close to me. Just within reach. And they are really big.

Therapist: Gosh, they sound delicious! I feel hungry now!

Client: (*laughs*) They look delicious!

Therapist: Okay, now I would like you to work on changing that mental image. Let's start with the colour. Can you make the colours duller? Maybe less rich and more grey?

Client: (*frowning*) Yes, I guess so.

Therapist: You don't sound too happy about that?

Client: Well they look so delicious. I don't want to spoil them.

Therapist: That's good. Spoiling them is exactly what we are trying to do. If we spoil your mental image of them then you are less likely to want to eat them. Now try

making the colour even greyer. And move the image away from you. Move it back and shrink it a little. How do they look now?

Client: A bit pointless really. I don't know why I was so interested in them. It seems stupid that these little grey things are more important to me than my health.

Therapist: Let's talk a little bit more about your health. What does being healthy mean to you?

Client: I'd quite like to run a marathon. (*laughs*) But no, seriously, I know that is never going to happen. It would be good to be able to play with the kids though. I could maybe see myself doing that.

Therapist: Tell me about that image.

Client: Well, I guess that I can see myself at the park, pushing Karen on the swings, and maybe kicking a ball about with Jay. Not getting exhausted and needing to sit down all the time.

Therapist: Is that an easy image for you to create?

Client: No. It seems impossible at the moment.

Therapist: Now I want you to go back to that image of the biscuits – the first image – all big and bright. Imagine that this picture is on a huge television screen right in front of you. Have you got that?

Client: Yes.

Therapist: Okay, now imagine that there is a tiny little box in the bottom left corner of the screen. Inside that box is the image of you at the park, running around with Jay, pushing Karen on the swings. Try to make this image grey and fuzzy and small inside that little box.

Client: Okay, that's easy because it's hard to imagine me doing that anyway.

Therapist: Right, but now comes the tricky bit. On my word, you are going to do something called a swish. You are going to make the two images swop places. That big bright image of the biscuits will shrink to the little grey fuzzy image in the left corner. And the little grey image of you at the park will fill the screen in bright colours. Can you do that?

Client: I think so. I'll try.

Therapist: Okay, and SWISH.

In this exchange, the therapist has identified a preferred modality and begun working within that modality to introduce interventions. The rest of the session involves focus on the faulty thought patterns held by Laura. The session finishes with a repeat of the swish exercise. Laura found this session very helpful and she was able to reduce her craving for biscuits by repeating this technique at home. When she returned for a top-up session, she asked to focus on her desire for all sweet food and she has since been able to eliminate sugary treats from her diet.

ANALYSING THE CASE STUDY

Try answering the following questions using information about neurolinguistic programming.

- What was Laura's preferred sensory modality and how did the therapist detect this?
- What submodalities were used during this session and how might this have impacted on Laura's feelings about food?
- How was swishing used in this session?
- Could anchoring help Laura to overcome her problems with food?
- Can you anticipate any potential problems in the methods used by this therapist?

SUMMARY

Eclecticism: mix of techniques across multiple therapies; focuses on the use of language to make changes in the brain; very controversial and some proponents have taken the claims too far

Development: developed by Bandler and Grinder in 1975 as a therapeutic method of instilling fast effective change using the best techniques from a range of therapies; eclectic blend of family therapy, Gestalt therapy and hypnotherapy

Representation systems: sensory modalities (visual, auditory, kinaesthetic), detecting preferred modalities, matching language to representation system; submodalities, nature of the visual/auditory/kinaesthetic experience, closely linked to emotional perception, emotions can be manipulated by changing submodalities

Meta model: faulty thoughts; distortions (mind reading, lost performance, cause and effect, complex equivalence, presupposition); deletions (simple, lack of referential index, comparative, unspecified verb, nominalisation); generalisations (universal quantifier, modal operators of necessity, modal operators of possibility)

Anchoring: classical conditioning; anchor a physical act (such as touching thumb to forefinger) to a desired emotional response (such as a feeling of calm); keys to anchoring success, desired state should have high intensity, physical action should be unique and easily replicated, anchor is set a number of times, physical action is timed to coincide with peak emotion

Swishing: cognitive visualisation to redirect thoughts and behaviours from negative patterns to positive patterns; visualise vibrant problem state on a big screen with the dull solution state in the left-hand corner of the problem image then switch between the two

Controversies: questionable empirical evidence for preferred representation systems; limited theoretical foundation; accusations of plagiarism

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Conclusion

This book aimed to provide a comprehensive guide to the four main psychological approaches (three forces and a revolution). This guide to the psychological approaches and the main associated therapies was the primary focus of Chapters 2–5.

- Third force: humanistic approach and person-centred therapy
  - Development of the humanistic approach
  - Personal and professional biographies of Rogers and Maslow
  - Humanistic theories of human nature and personality
  - Therapeutic relationship in person-centred therapy
  - Therapeutic techniques in person-centred therapy
  - Case study demonstrating person-centred therapy
- Second force: psychodynamic approach and psychoanalytic therapy
  - Development of the psychodynamic approach
  - Personal and professional biographies of Freud and Klein
  - Psychodynamic theories of human nature and personality
  - Therapeutic relationship in psychoanalytic therapy
  - Therapeutic techniques in psychoanalytic therapy
  - Case study demonstrating psychoanalytic therapy
- First force: behavioural approach and behaviour therapy
  - Development of the behavioural approach
  - Personal and professional biographies of Pavlov, Watson and Skinner
  - Behavioural theories of human nature and personality
  - Therapeutic relationship in behaviour therapy
  - Therapeutic techniques in behaviour therapy
  - Case study demonstrating behaviour therapy
- Revolution: cognitive approach and cognitive therapy
  - Development of the cognitive approach
  - Personal and professional biographies of Neisser and Beck
  - Cognitive theories of human nature and personality
  - Therapeutic relationship in cognitive therapy
  - Therapeutic techniques in cognitive therapy
  - Case study demonstrating behaviour therapy

This book also aimed to introduce you to a range of modern integrative and eclectic therapies and this was the primary focus of Chapter 6.

- Gestalt therapy
- Transactional analysis
- Rational emotive behaviour therapy (REBT)
- Cognitive-behaviour therapy (CBT)
- Multimodal therapy (MMT)
- Neurolinguistic programming (NLP)

We said at the end of our introduction that we hoped that this textbook would serve as a knowledgeable guide on your journey through the fascinating subject of counselling and psychotherapy. Now that you have reached the end of this book, please take a moment to reflect on what you have learnt. If you have gained a little insight into some of the concepts discussed in this book to the extent that you are able to appreciate the approaches and evaluate the therapies, then we can be happy that we have achieved our goal. More importantly, if you now recognise that you have only just begun your therapeutic education and have been inspired to read further into these fascinating approaches and therapies, then we are thrilled to have exceeded our goal and delighted that we were able to offer some signposts at the start of your journey. We hope that you are able to apply your new understanding of counselling and psychotherapy in your professional and personal life, and we are confident that your deeper appreciation of human nature will benefit you in all aspects of your life.

If we have succeeded in motivating you to study this topic further, please log into our website to access further materials to support and enhance your learning experience. You will find all of the following available online:

- Glossary of key words as a reference guide
- Key word clouds to summarise content of textbook
- Mindmaps to support associative links between concepts
- Multiple choice questions to test recall of factual content
- Short answer questions to test understanding of material
- Essay questions to test critical appreciation of approaches and therapies
- List of resources (including films, books, documentaries, poems, songs and websites) to improve understanding and expand appreciation of the topics

We would like to take this opportunity to thank you for allowing us to join you on your educational journey and wish you lots of luck for your further travels in the future.

'I am not a teacher; only a fellow traveller of whom you asked the way.'
—George Bernard Shaw (1856–1950)

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COUNSELING APPROACHES

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CHAPTER - 6

COUNSELING APPROACHES

Topics Covered

- Psychoanalytic Approach
- Behavioral Approach
  - Cognitive Behavioral Therapy (CBT)
  - Rational Emotive Behavioral Therapy (REBT)
  - Transactional Analysis (TA)
- Humanistic Approach
  - Person Centered Therapy
  - Gestalt Therapy
  - Existential Therapy
- Other Contemporary Therapy
  - Adlerian Therapy/ Individual Psychology
  - Reality therapy
  - Family Systems Therapy
  - Feminist Therapy
  - Eye Movement Desensitization and Reprocessing (EMDR)
  - Eclectic Counseling
  - Integrative Counseling
  - Group Therapy

PSYCHOANALYTIC APPROACH

Sigmund Freud
(May 6, 1856 –September 23, 1939)

Psychoanalytic approach is a type of approach based upon the theories of Sigmund Freud, who is considered one of the forefathers of psychology and the founder of psychoanalysis. This therapy explores how the unconscious mind influences thoughts and behaviors, with the aim of offering insight and resolution to the person seeking therapy. Psychoanalytic therapy tends to look at experiences from early childhood to see if these events have affected the individual's life, or potentially contributed to current concerns.

This form of therapy is considered a long-term choice and can continue for weeks, months or even years depending on the depth of the concern being explored. Differing from several other therapy types, psychoanalytic therapy aims to make deep-seated changes in personality and emotional development.

Basic Philosophy: While each therapist will work in different ways according to the needs of the individual seeking therapy, many work on the following assumptions -

- Human beings are basically determined by psychic energy and by early experiences.
- Unconscious motives and conflicts are central in present behavior.
- Irrational forces are strong; the person is driven by sexual and aggressive impulses.
- Early development is of critical importance because later personality problems have their roots in repressed childhood conflicts.

Key Concepts: Normal personality development is based on successful resolution and integration of psychosexual stages of development. Faulty personality development is the result of inadequate resolution of some specific stage. Id, ego, and superego constitute the basis of personality structure. Anxiety is a result of repression of basic conflicts. Unconscious processes are centrally related to current behavior.

Developmental Stages

- *Oral stage* is centered on the mouth as a source of pleasure.
- *Anal stage* is centered on the anus and elimination as a source of pleasure.
- *Phallic stage* is centered on the genitals and sexual identification as a source of pleasure.
- ✓ Oedipus Complex is described as the process whereby a boy desires his mother and fears castration from the father, in order to create an ally of the father, the male learns traditional male roles.
- ✓ Electra Complex is described a similar but less clearly resolved in the female child with her desire for the father, competition with the mother; and thus, learns the traditional female roles.
- *Latency stage* is a time of little sexual interest in Freud's developmental view. This stage is characterized with peer activities, academic and social learning, and development of physical skills.
- *Genital stage* begins with the onset of puberty. If the other stages have been successfully negotiated, the young person will take an interest in and establish sexual relationships.

Structure of Personality

The personality has three parts - the id, the ego, and the superego.

- ➢ The id is present at birth and is part of the unconscious. The id is the site of the pleasure principle, the tendency of an individual to move toward pleasure and away from pain. The id does not have a sense of right or wrong, is impulsive, and is not rational. It contains the most basic of human instincts, drives, and genetic endowments.
- ➢ The ego is the second system to develop and it functions primarily in the conscious mind and in the preconscious mind. It serves as a moderator between the id and the superego, controlling wishes and desires. The ego is the site of the reality principle, the ability to interact with the outside world with appropriate goals and activities.
- ➢ The superego sets the ideal standards and morals for the individual. The superego operates on the moral principle which rewards the individual for following parental and societal dictates. Guilt is produced when a person violates the ideal ego denying or ignoring the rules of the superego.

Ego Defense Mechanisms

Ego defense mechanisms were believed by Freud to protect the individual from being overwhelmed by anxiety. He considered them normal and operating on the unconscious level.

Defense	Description	Example
Denial	arguing against an anxiety provoking stimuli by stating it doesn't exist	denying that your physician's diagnosis of cancer is correct and seeking a second opinion
Displacement	taking out impulses on a less threatening target	slamming a door instead of hitting as person, yelling at your spouse after an argument with your boss
Intellectualization	avoiding unacceptable emotions by focusing on the intellectual aspects	focusing on the details of a funeral as opposed to the sadness and grief

Projection	placing unacceptable impulses in yourself onto someone else	when losing an argument, you state "You're just Stupid"; homophobia
Rationalization	supplying a logical or rational reason as opposed to the real reason	stating that you were fired because you didn't kiss up the boss, when the real reason was your poor performance
Reaction Formation	taking the opposite belief because the true belief causes anxiety	having a bias against a particular race or culture and then embracing that race or culture to the extreme
Regression	returning to a previous stage of development	sitting in a corner and crying after hearing bad news; throwing a temper tantrum when you don't get your way
Repression	pulling into the unconscious	forgetting sexual abuse from your childhood due to the trauma and anxiety
Sublimation	acting out unacceptable impulses in a socially acceptable way	sublimating your aggressive impulses toward a career as a boxer; becoming a surgeon because of your desire to cut; lifting weights to release 'pent up' energy
Suppression	pushing into the unconscious	trying to forget something that causes you anxiety
Ego defenses are not necessarily unhealthy as you can see by the examples above. In fact, the lack of these defenses or the inability to use them effectively can often lead to problems in life. However, we sometimes employ the defenses at the wrong time or overuse them, which can be equally destructive.		

Role of the Counselor: To encourage the development of transference, giving the client a sense of safety and acceptance. The client freely explores difficult material and experiences from their past, gaining insight and working through unresolved issues. The counselor is an expert, who interprets for the client.

Goals of Therapy

- Helping the client bring into the conscious the unconscious.
- Helping the client work through a developmental stage that was not resolved or where the client became fixated.
- To reconstruct the basic personality.
- To assist clients in reliving earlier experiences and working through repressed conflicts.
- To achieve intellectual awareness.
- Help the client adjustment to the demands of work, intimacy, and society.

Therapeutic Relationship: The analyst remains anonymous, and clients develop projections toward him/her. Focus is on reducing the resistances that develop in working with transference and on establishing more rational control. Clients undergo long-term analysis, engage in free association, to uncover conflicts, and gain insight by talking. The analyst makes interpretations to teach them the meaning of current behavior as related to the past.

Techniques of Therapy: Interpretation, dream analysis, free association, analysis of resistance, analysis of transference. All are designed to help clients gain access to their unconscious conflicts, which leads to insight and eventual assimilation of new material by the ego. Diagnosis and testing are often used. Questions are used to develop a case history.

Free Association is a process where the client verbalizes any thoughts that may without censorship, no matter how trivial the thoughts or feeling may be to the client.

Dream Analysis is a process where the client relates their dreams to the counselor. The counselor interprets the obvious or manifest content and the hidden meanings or latent content.

Analysis of transference is a process where the client is encouraged to attribute to counselor those issues that have caused difficulties with significant authority figures in their lives. The counselor helps the client to gain insight by the conflicts and feelings expressed.

Analysis of resistance is a process where the counselor helps the client to gain insight into what causes form the basis for a hesitation or halting of therapy.

Interpretation is a process where the counselor helps the client to gain insight into past and present events.

Applications: Psychoanalytic therapy can be used by those with a specific emotional concern as well as those who simply want to explore themselves. Understanding why we are the way we are often brings with it a sense of well-being and a stronger sense of self. As psychoanalytic therapy is considered one of the more long-term therapy types, it is perhaps less useful for those seeking quick, solution-focused therapies. Psychoanalytic therapy is a gradual process that takes time; however the results can be life changing.

Some believe due to the nature of the therapy, psychoanalytic work is better suited to more general concerns such as anxiety, relationship difficulties, sexual issues or low self-esteem. Phobias, social shyness and difficulties sleeping are further examples of areas that could be addressed within psychoanalytic therapy. Critics have pointed out that the therapy may not be as useful to those with more specific or obsession-based concerns such as obsessive compulsive behavior, as you may be too concerned by your actions to participate fully.

How does psychoanalytic therapy work?

Psychoanalytic therapy is insight driven and therefore looks to foster change by helping you to understand your past and how events from your early life could be affecting you now. Sessions will vary according to where you are in the course of your therapy, but much of the time will be spent talking freely to your therapist in a safe, non-judgmental environment. The psychoanalyst will listen to your concerns and look out for patterns or certain events that may hold significance. In this type of therapy it is believed that unconscious feelings and childhood events play a key role in mental distress. As well as listening to you discuss your experiences; your therapist may use other techniques to help identify potential causes for your concerns.

Such techniques may include -

Free association: Free association involves you talking about whatever comes into your mind without censoring or editing the flow of memories/ideas. Your therapist will encourage you to speak freely to help you return to an earlier emotional state so they can better understand any recurrent patterns of conflict you may be experiencing.

Therapeutic transference: Transference relates to the way you may be transferring thoughts or feelings connected to influential figures in your life (for example your parents or siblings) onto your therapist. While this may not happen in every case, if it does your therapist should discuss transference with you to help you gain further insight into the way you deal with people in your daily life.

Interpretation: A key element of psychoanalytic therapy is interpreting and 'reading between the lines'. While your therapist is likely to stay relatively quiet and allow you to talk freely, they will occasionally interject with thoughts or interpretations of the topics you discuss. Your psychoanalyst may also ask you about your dreams; Freud wrote a lot on the subject of dream analysis and believed that dreams were important resources for understanding the unconscious.

Multiculturalism: Its focus on family dynamics is appropriate for working with many minority groups. The therapist's formality appeals to clients who expect professional distance. Notion of ego defense is helpful in understanding inner dynamics and dealing with environmental stresses. Its focus on insight, intra-psychic dynamics, and long-term treatment, though, is often not valued by clients who prefer to learn coping skills for dealing with pressing daily concerns. Internal focus is often in conflict with cultural values that stress an interpersonal and environmental focus.

Contributions: More than any other system, this approach has generated controversy as well as exploration and has stimulated further thinking and development of therapy. It has provided a detailed and comprehensive description of personality structure and functioning. It has brought into prominence factors such as the unconscious as a determinant of behavior and the role of trauma during the first 6 years of life. It has developed several techniques for tapping the unconscious. It has shed light on the dynamics of transference and counter-transference, resistance, anxiety, and the mechanisms of ego defense.

Limitations: Requires lengthy training for therapists and much time and expense for clients. The model stresses biological and instinctive factors to the neglect of social, cultural, and interpersonal ones. Its methods are not applicable for solving specific problems of clients in lower socioeconomic classes and are not appropriate for many ethnic and cultural groups. Many clients lack the degree of ego strength needed for regressive and reconstructive therapy. It is inappropriate for the typical counseling setting.

Psychodynamic Psychotherapy/Counseling: This approach stresses the importance of the unconscious and past experience in shaping current behavior. The client is encouraged to talk about childhood relationships with parents and other significant people and the therapist focuses on the client/therapist relationship (the dynamics) and in particular on the transference. Transference is when the client projects onto the therapist feelings experienced in previous significant relationships. The Psychodynamic approach is derived from Psychoanalysis but usually provides a quicker solution to emotional problems.

Psycho-synthesis: Sometimes described as 'psychology of the soul'. It is the name given to a series of actions that lead to a change or development which encourages personal growth by a bringing together of the whole person - the emotional, the mental, the physical and spiritual within a safe environment. Psycho-synthesis is useful for people seeking a new, more spiritually oriented vision of them.

BEHAVIORAL APPROACH: Cognitive Behavioral Therapy (CBT), Rational Emotive Behavioral Therapy (REBT), Transactional Analysis (TA)

BEHAVIORAL APPROACH

As the name implies, this approach focuses on behavior - changing unwanted behaviors through rewards, reinforcements, and desensitization. This therapy is based on the belief that behavior is learnt in response to past experience and can be unlearnt, or reconditioned, without analyzing the past to find the reason for the behavior. It works well for compulsive and obsessive behavior, fears, phobias and addictions. Someone whose fear of germs leads to excessive washing, for example, may be trained to relax and not wash his/her hands after touching a public doorknob.

Burrhus Frederic (B. F.) Skinner
(March 20, 1904 - August 18, 1990)

Behavioral therapy often involves the cooperation of others, especially family and close friends, to reinforce a desired behavior.

Basic Philosophy: Behavior is the product of learning. We are both the product and the producer of the environment. No set of unifying assumptions about behavior can incorporate all the existing procedures in the behavioral field.

Key Concepts: Focus is on overt behavior, precision in specifying goals of treatment, development of specific treatment plans, and objective evaluation of therapy outcomes. Therapy is based on the principles of learning theory. Normal behavior is learned through reinforcement and imitation. Abnormal behavior is the result of faulty learning. This approach stresses present behavior.

Role of the Counselor: Roles of the behavioral counselor are varies and include being a consultant, a reinforcer, and a facilitator. The counselor is active and may supervise other people in the client's environment to achieve the goals of therapy. Counselors using social learning may model the desired behavior, while respondent and operant conditioning counselors are more directive and prescriptive in their approach to the therapy goals. Use of tests and diagnosis varied greatly among behavioral counselors.

Goals of Therapy: Generally, to eliminate maladaptive behaviors and learn more effective behaviors. To focus on factors influencing behavior and find what can be done about problematic behavior. Clients have an active role in setting treatment goals and evaluating how well these goals are being met.

Therapeutic Relationship: The therapist is active and directive and functions as a teacher or trainer in helping clients learn more effective behavior. Clients must be active in the process and experiment with new behaviors. Although a quality client/therapist relationship is not viewed as sufficient to bring about change, a good working relationship is essential for implementing behavioral procedures.

Techniques of Therapy: The main techniques are systematic desensitization, relaxation methods, flooding, eye movement and desensitization reprocessing, reinforcement techniques, modeling, cognitive restructuring, assertion and social skills training, self-management programs, behavioral rehearsal, coaching, and various multimodal therapy techniques. Diagnosis or assessment is done at the outset to determine a treatment plan. Questions are used, such as what, how, and when (but not why). Contracts and homework assignments are also typically used.

Applications: A pragmatic approach based on empirical validation of results. Enjoys wide applicability to individual, group, marital and family counseling. Some problems to which the approach is well suited are phobic disorders, depression, sexual disorders, children's behavioral disorders, stuttering, and prevention of cardiovascular disease. Beyond clinical practice, its principles are applied in fields such as pediatrics, stress management, behavioral medicine, education, and geriatrics.

Multiculturalism: Its focus on behavior, rather than on feelings, is compatible with many cultures. Strengths include a collaborative relationship between counselor and client in working toward mutually agreed-on goals, continual assessment to determine if the techniques are suited to clients' unique situation, assisting clients in learning practical skills, an educational focus, and stress on self-management strategies. Counselors need to help clients assess the possible consequences of making behavioral changes. Family members may not value clients newly acquired assertive style, so clients must be taught how to cope with resistance by others.

Contributions: Emphasis is on assessment and evaluation techniques, thus providing a basis for accountable practice. Specific problems are identified, and clients are kept informed about progress toward their goals. The approach has demonstrated effectiveness in many areas of human functioning. The roles of the therapist as reinforcer, model, teacher, and consultant are explicit. The approach has undergone extensive expansion, and research literature abounds. No longer is it a mechanistic approach, for it now makes room for cognitive factors and encourages self-directed programs for behavioral change.

Limitations: Major criticisms are that it may change behavior but not feelings; that it ignores the relational factors in therapy; that it does not provide insight; that it ignores historical causes of present behavior; that it involves control and manipulation by the therapist; and that it is limited in its capacity to address certain aspects of the human condition. Many of these assertions are based on misconceptions, and behavior therapists have addressed these charges. A basic limitation is that behavior change cannot always be objectively assessed because of the difficulty in controlling environmental variables.

COGNITIVE BEHAVIOR THERAPY (CBT)

A combination of cognitive and behavioral therapies, this approach helps people change negative thought patterns, beliefs, and behaviors so they can manage symptoms and enjoy more productive, less stressful lives. Cognitive behavioral therapy (CBT) was developed by Aaron T. Beck, Albert Ellis, Maxie Maultsby, Michael Mahoney, Donald Meichenbaum, David Burns, Michael Mahoney, Marsha Linehan, Arthur Freeman, and others. Cognitive behavior therapy is a type of psychotherapeutic treatment that helps patients understand the thoughts and feelings that influence behaviors.

Dr. Aaron Temkin Beck
(July 18, 1921 to present)

CBT is commonly used to treat a wide range of disorders including phobias, addictions, depression, and anxiety. Cognitive behavior therapy is generally short-term and focused on helping clients deal with a very specific problem. During the course of treatment, people learn how to identify and change destructive or disturbing thought patterns that have a negative influence on behavior.

Most cognitive behavioral therapies have the following characteristics -

CBT is based on the Cognitive Model of Emotional Response. CBT is based on the idea that our thoughts cause our feelings and behaviors, not external things, like people, situations, and events. The benefit of this fact is that we can change the way we think to feel/act better even if the situation does not change.

CBT is Briefer and Time-Limited. It is considered among the most rapid in terms of results obtained. The average number of sessions clients receive (across all types of problems and approaches to CBT) is only 16. Other forms of therapy, like psychoanalysis, can take years. What enable CBT to be briefer are its highly instructive nature and the fact that it makes use of homework assignments. CBT is time-limited in that we help clients understand at the very beginning of the therapy process that there will be a point when the formal therapy will end. The ending of the formal therapy is a decision made by the therapist and client. Therefore, CBT is not an open-ended, never-ending process.

A sound therapeutic relationship is necessary for effective therapy, but not the focus. Some forms of therapy assume that the main reason people get better in therapy is because of the positive relationship between the therapist and client. Cognitive behavioral therapists believe it is important to have a good, trusting relationship, but that is not enough. CBT therapists believe that the clients change because they learn how to think differently and they act on that learning. Therefore, CBT therapists focus on teaching rational self-counseling skills.

CBT is a collaborative effort between the therapist and the client. Cognitive-behavioral therapists seek to learn what their clients want out of life (their goals) and then help their clients achieve those goals. The therapist's role is to listen, teach, and encourage, while the client's roles is to express concerns, learn, and implement that learning.

CBT is based on aspects of stoic philosophy. Not all approaches to CBT emphasize stoicism. Rational Emotive Behavior Therapy, Rational Behavior Therapy, and Rational Living Therapy emphasize aspects of stoicism. Beck's Cognitive Therapy is not based on stoicism.

Cognitive-behavioral therapy does not tell people how they should feel. However, most people seeking therapy do not want to feel they way they have been feeling. The approaches that emphasize stoicism teach the benefits of feeling, at worst, calm when confronted with undesirable situations. They also emphasize the fact that we have our undesirable situations whether we are upset about them or not. If we are upset about our problems, we have two problems - the problem, and our upset about it. Most people want to have the fewest number of problems possible. So when we learn how to more calmly accept a personal problem, not only do we feel better, but we usually put ourselves in a better position to make use of our intelligence, knowledge, energy, and resources to resolve the problem.

CBT uses the Socratic Method. Cognitive-behavioral therapists want to gain a very good understanding of their clients' concerns. That's why they often ask questions. They also encourage their clients to ask questions of themselves, like, "How do I really know that those people are laughing at me?" "Could they be laughing about something else?"

CBT is structured and directive. Cognitive-behavioral therapists have a specific agenda for each session. Specific techniques/concepts are taught during each session. CBT focuses on the client's goals. We do not tell our clients what their goals should be, or what they should tolerate. We are directive in the sense that we show our clients how to think and behave in ways to obtain what they want. Therefore, CBT therapists do not tell their clients what to do - rather, they teach their clients how to do.

CBT is based on an educational model. CBT is based on the scientifically supported assumption that most emotional and behavioral reactions are learned. The goal of therapy is to help clients unlearn their unwanted reactions and to learn a new way of reacting. CBT has nothing to do with 'just talking'. People can just talk with anyone. The educational emphasis of CBT has an additional benefit - it leads to long term results. When people understand how and why they are doing well, they know what to do to continue doing well.

CBT theory and techniques rely on the Inductive Method. A central aspect of rational thinking is that it is based on fact. Often, we upset ourselves about things when, in fact, the situation isn't like we think it is. If we knew that, we would not waste our time upsetting ourselves. Therefore, the inductive method encourages us to look at our thoughts as being hypotheses or guesses that can be questioned and tested. If we find that our hypotheses are incorrect (because we have new information), then we can change our thinking to be in line with how the situation really is.

Homework is a central feature of CBT. If when you attempted to learn your multiplication tables you spent only one hour per week studying them, you might still be wondering what 5 X 5 equals. You very likely spent a great deal of time at home studying your multiplication tables, maybe with flashcards.

The same is the case with psychotherapy. Goal achievement (if obtained) could take a very long time if all a person were only to think about the techniques and topics taught was for one hour per week. That's why CBT therapists assign reading assignments and encourage their clients to practice the techniques learned.

Basic Philosophy: Individuals tend to incorporate faulty thinking, which leads to emotional and behavioral disturbances. Cognitions are the major determinants of how we feel and act. Therapy is primarily oriented toward cognition and behavior, and it stresses the role of thinking, deciding, questioning, doing, and re-deciding. This is a psycho-educational model, which emphasizes therapy as a learning process, including acquiring and practicing new skills, learning new ways of thinking, and acquiring more effective ways of coping with problems.

Key Concepts: Although psychological problems may be rooted in childhood, they are perpetuated through reindoctrination in the now. A person's belief system is the primary cause of disorders. Internal dialogue plays a central role in one's behavior. Clients focus on examining faulty assumptions and misconceptions and on replacing these with effective beliefs.

Goals of Therapy: To challenge clients to confront faulty beliefs with contradictory evidence that they gather and evaluate. Helping clients seek out their dogmatic beliefs and vigorously minimizing them. To become aware of automatic thoughts and to change them.

Therapeutic Relationship: The therapist functions as a teacher and the client as a student. The therapist is highly directive and teaches clients an A-B-C model of changing their cognitions. The focus is on a collaborative relationship. Using a Socratic dialogue, the therapist assists clients in identifying dysfunctional beliefs and discovering alternative rules for living. The therapist promotes corrective experience that lead to learning new skills. Clients gain insight into their problems and then must actively practice changing self-defeating thinking and acting.

Techniques of Therapy: Therapists use a variety of cognitive, emotive, and behavioral techniques; diverse methods are tailored to suit individual clients. An active, directive, time-limited, present-centered, structured therapy. Some techniques include engaging in Socratic dialogue, debating irrational beliefs, carrying out

homework assignments, gathering data on assumptions one has made, keeping a record of activities, forming alternative interpretations, learning new coping skills, changing one's language and thinking patterns, role playing, imagery, and confronting faulty beliefs.

Applications: Has been widely applied to the treatment of depression, anxiety, marital problems, stress management, skill training, substance abuse, assertion training, eating disorders, panic attacks, performance anxiety, and social phobia. The approach is especially useful for assisting people in modifying their cognitions. Many self-help approaches utilize its principles. Can be applied to a wide range of client populations with a variety of specific problems.

Who could benefit from CBT?

This type of therapy is particularly helpful for those with specific issues as it is very practical (rather than insight-based) and looks at solutions. For this reason the therapy works well for those who -

- suffer from depression and/or anxiety
- suffer from post-traumatic stress disorder (PTSD)
- have an addiction
- want to change their behavior
- have anger issues
- eating disorders, such as anorexia and bulimia
- sleep problems, such as insomnia
- panic disorder
- have a phobia
- suffer from obsessive-compulsive behavior (OCD).

In some cases CBT is used for those with long-standing health problems such as chronic pain or irritable bowel syndrome (IBS) and chronic fatigue syndrome (CFS). While the therapy cannot cure such physical ailments, it can help people cope emotionally with their condition and lower stress levels. There is also up and coming interest in the use of CBT alongside medication to help those who suffer from hallucinations and delusions.

Multiculturalism: The collaborative approach offers clients opportunities to express their areas of concern. The psycho-educational dimensions are often useful in exploring cultural conflicts and teaching new behavior. The emphasis on thinking (as opposed to identifying and expressing feelings) is likely to be acceptable to many clients. The focus on teaching and learning tends to avoid the stigma of mental illness. Clients may value active and directive stance of therapist. Before too quickly attempting to change the beliefs and actions of clients, it is essential for the therapists to understand and respect their world. Some clients may have serious reservations about questioning their basic cultural values and beliefs. Clients could become dependent on the therapist for deciding what are appropriate ways to solve problems. There may be a fine line between being directive and promoting dependence.

Contributions: Major contributions include emphasis on a comprehensive and eclectic therapeutic practice; numerous cognitive, emotive, and behavioral techniques; openness to incorporating techniques from other approaches; and a methodology for challenging and changing faulty thinking. Most forms can be integrated into other mainstream therapies.

Limitations: Tends to play down emotions, does not focus on exploring the unconscious or underlying conflicts, and sometimes does not give enough weight to client's past.

When is CBT Used as a form of Therapy?

Scientific studies of CBT have demonstrated its usefulness for a wide variety of mental illnesses including mood disorders, anxiety disorders, personality disorders, eating disorders, substance abuse disorders, sleep disorders and psychotic disorders. Studies have shown that CBT actually changes brain activity in people with mental illnesses who receive this treatment, suggesting that the brain is actually improving its functioning as a result of engaging in this form of therapy.

CBT has been shown to be as useful as antidepressant medications for some individuals with depression and may be superior in preventing relapse of symptoms. Patients receiving CBT for depression are encouraged to schedule positive activities into their daily calendars in order to increase the amount of pleasure they experience. In addition, depressed patients learn how to change (restructure) negative thought patterns in order to interpret their environment in a less negatively-biased way. As regular sleep has been found to be very important in both depression and bipolar disorder, therapists will also target sleeping patterns to improve and regulate sleep schedules with their patients. Studies indicate that patients who receive CBT in addition to treatment with medication have better outcomes than patients who do not receive CBT as an additional treatment.

CBT is also a useful treatment for anxiety disorders. Patients who experience persistent panic attacks are encouraged to test out beliefs they have related to such attacks, which can include specific fears related to bodily sensations, and to develop more realistic responses to their experiences. This is beneficial in decreasing both the frequency and intensity of panic attacks. Patients who experience obsessions and compulsions are guided to expose themselves to what they fear in a safe and controlled therapeutic environment. Beliefs surrounding their fears (of contamination, illness, inflicting harm, etc.) are identified and changed to decrease the anxiety connected with such fears.

The same is true for people with phobias, including phobias of animals or phobias of evaluation by others (termed Social Anxiety Disorder). Those in treatment are exposed to what they fear and beliefs that have served to maintain such fears are targeted for modification. CBT is often referred to as a "first line treatment" in many anxiety disorders including generalized anxiety disorder, posttraumatic stress disorder, panic disorder, and obsessive-compulsive disorder and specific phobias.

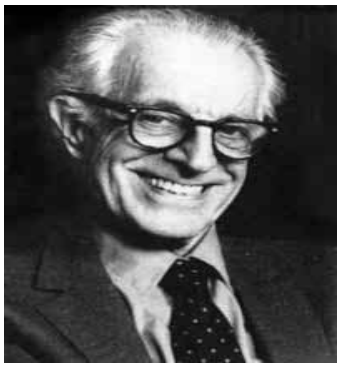
Over the past two decades, CBT for schizophrenia has received considerable attention in the United Kingdom and elsewhere abroad. While this treatment continues to develop in the United States, the results from studies in the United Kingdom and other countries have encouraged therapists in the US to incorporate this treatment into their own practices. In this treatment, often referred to as cognitive behavioral therapy for psychosis (CBT-P), patients are encouraged to identify their own delusional or paranoid beliefs and to explore how these beliefs negatively impact their lives.

Therapists will then help patients to engage in experiments to test these beliefs. Treatment focuses on thought patterns that cause distress and also on developing more realistic interpretations of events. Delusions are treated by developing an understanding of the kind of evidence that a person uses to support their beliefs and encouraging them to recognize evidence that may have been overlooked, evidence that does not support the belief. For example, a person who thinks that they are being videotaped by aliens may feel less worried when their therapist helps them to discover that there are no hidden cameras in the waiting room, or that a television remote does not contain any Alien technology within it.

CBT's focus on thoughts and beliefs is applicable to a wide variety of symptoms. While the above summary is certainly not comprehensive, it provides an overview of the principles of CBT and how they apply to the treatment of various psychological illnesses. Because CBT has excellent scientific data supporting its use in the clinical treatment of psychological illness, it has achieved wide popularity both for therapists and patients alike. A growing number of psychologists, psychiatrists, social workers, and psychiatric nurses have training in CBT.

RATIONAL EMOTIVE BEHAVIOR THERAPY (REBT)

Rational emotive behavior therapy, also known as REBT, is a type of cognitive-behavioral therapy. It was first called Rational Therapy, later Rational Emotive Therapy, then changed to Rational Emotive Behavior Therapy. It was first introduced in 1955 by Dr. Albert Ellis who had become increasing frustrated with the ineffectiveness of psychotherapy. Ellis drew from his knowledge of philosophy and psychology to devise a method which he believed was more directive, efficient, and effective.



Dr. Albert Ellis
(September 27, 1913 – July 24, 2007)

Basic Philosophy: REBT is focused on helping clients change irrational beliefs. The replacement of illogical and unrealistic ideas with more realistic and adaptive ones through direct intervention and confrontation by the therapist.

Key Concepts: Ellis suggested that people mistakenly blame external events for unhappiness. He argued, however, that it is our interpretation of these events that truly lies at the heart of our psychological distress. To explain this process, Ellis developed what he referred to as the ABC Model -

A – Activating Event: Something happens in the environment around you.

B – Beliefs: You hold a belief about the event or situation.

C – Consequence: You have an emotional response to your belief.

Role of the Counselor

- Counselors are direct and active in their teaching and correcting the client's cognitions.
- Ellis believes that a good REBT counselor must be bright, knowledgeable, empathetic, persistent, scientific, interested in helping others and use RET in their personal lives (Ellis, 1980).
- The counselor does not rely heavily on the DSM-IV categories.

Goals of the Therapy

- The primary goal is to help people live rational and productive lives.
- REBT helps people see that it is their thoughts and beliefs about events that creates difficulties, not the events or the situations.
- It helps the client to understand that wishes and wants are not entitlements to be demanded. Thinking that involves the words must, should, ought, have to, and need are demands, not an expression of wants or desires.
- RET helps clients stop catastrophizing when wants and desires are not met.
- It stresses the appropriateness of the emotional response to the situation or event. A situation or event need not elicit more of a response than is appropriate.
- It assists people in changing self-defeating behaviors or cognitions.
- RET espouses acceptance and tolerance of self and of others in order to achieve life goals.

Steps in Rational Emotive Behavior Therapy

The basic steps in Rational Emotive Behavior Therapy are -

Identifying the underlying irrational thought patterns and beliefs. The very first step in the process is to identify the irrational thoughts, feelings, and beliefs that lead to psychological distress. In many cases, these irrational beliefs are reflected as absolutes, as in I must, I should or I can't. According to Ellis, some of the most common irrational beliefs include -

- Feeling excessively upset over other people's mistakes or misconduct.
- Believing that you must be 100 percent competent and successful in everything to be valued and worthwhile.
- Believing that you will be happier if you avoid life's difficulties or challenges.
- Feeling that you have no control over your own happiness; that your contentment and joy are dependent upon external forces.
- By holding such unyielding beliefs, it becomes almost impossible to respond to situations in a psychologically healthy way. Possessing such rigid expectations of

ourselves and others only leads to disappointment, recrimination, regret, and anxiety.

Challenging the irrational beliefs. Once these underlying feelings have been identified, the next step is to challenge these mistaken beliefs. In order to do this, the therapist must dispute these beliefs using very direct and even confrontational methods. Ellis suggested that rather than simply being warm and supportive, the therapist needs to be blunt, honest, and logical in order to push people toward changing their thoughts and behaviors.

Gaining Insight and Recognizing Irrational Thought Patterns. Facing irrational thought patterns can be difficult, especially because accepting these beliefs as unhealthy is far from easy. Once the client has identified the problematic beliefs, the process of actually changing these thoughts can be even more difficult.

It is also important to recognize that while rational emotive behavior therapy utilizes cognitive strategies to help clients, it also focuses on emotions and behaviors as well. In addition to identifying and disputing irrational beliefs, therapists and clients also work together to target the emotional responses that accompany problematic thoughts. Clients are also encouraged to change unwanted behaviors using such things as meditation, journaling, and guided imagery.

Applications: REBT can be effective in the treatment of a range of psychological disorders including anxiety disorders and phobias as well as specific behaviors such as severe shyness and excessive approval seeking.

How does REBT work?

REBT's comprehensive approach works best for individuals desiring a scientific, present-focused, and active treatment for coping with life's difficulties, rather than one which is mystical, historical, and largely passive. REBT is based on a few simple principles having profound implications -

- You are responsible for your own emotions and actions.
- Your harmful emotions and dysfunctional behaviors are the product of your irrational thinking.
- You can learn more realistic views and, with practice, make them a part of you.
- You'll experience a deeper acceptance of yourself and greater satisfactions in life by developing a reality-based perspective.

REBT distinguishes clearly between two very different types of difficulties - practical problems and emotional problems. Your flawed behavior, unfair treatment by others, and undesirable situations, represent practical problems. Regrettably, your human

tendency is to upset yourself about these practical problems, thereby unnecessarily creating a second order of problems - emotional suffering.

REBT addresses the latter by helping you -

- Take responsibility for your distress. The first lesson in healthy emoting and relating was stated by the Roman philosopher Epictetus more than 2000 years ago - only you can upset yourself about events - the events themselves, no matter how undesirable, can never upset you.
- Recognize that neither another person, nor an adverse circumstance, can ever disturb you - only you can. No one else can get into your gut and churn it up. Others can cause you physical pain - by hitting you over the head with a baseball bat, for example - or can block your goals. But you create your own emotional suffering, or self-defeating behavioral patterns, about what others do or say.
- Identify your 'musts'. Once you admit that you distort your own emotions and actions, then determine precisely how. The culprit usually lies in one of the three core 'musts' -

  Must # 1 (a demand on yourself): I 'must' do well and get approval, or else I'm worthless. This demand causes anxiety, depression, and lack of assertiveness.

  Must # 2 (a demand on others): You 'must' treat me reasonably, considerately, and lovingly, or else you're no good. This must leads to resentment, hostility, and violence.

  Must # 3 (a demand on situations): Life 'must' be fair, easy, and hassle-free, or else it's awful. This thinking is associated with hopelessness, procrastination, and addictions.
- Ascertain what you're demanding of yourself, of your significant others, or of your circumstances. Not until you have discovered the 'must' can you then go on effectively to reduce your distress.
- Dispute your 'musts'. The only way you can ever remain disturbed about adversity is by vigorously and persistently agreeing with one of these three 'musts'. Thus, once you've bared them, then relentlessly confront and question your demands.
- Begin by asking yourself - What's the evidence for my 'must'? How is it true? Where's it etched in stone? And then by seeing - There's no evidence. My 'must' is entirely false. It's not carved indelibly anywhere. Make your view 'must'-free, and then your emotions will heal.
- Reinforce your preferences. Conclude, therefore -
  - Preference - 1: I strongly 'prefer' to do well and get approval, but even if I fail, I will accept myself fully.

- Preference - 2: I strongly 'prefer' that you treat me reasonably, kindly, and lovingly, but since I don't run the universe, and it's a part of your human nature to err, I, then, cannot control you.
- Preference - 3: I strongly 'prefer' that life be fair, easy, and hassle-free, and it's very frustrating that it isn't, but I can bear frustration and still considerably enjoy life.

Assuming that you take the above suggestions to heart and thereby greatly reduce your anxiety, hostility, depression, and addictions, what remains? Will you exist robot-like, devoid of human feeling and motivation? Hardly! Without your turmoil, you'll more easily experience love, involvement, and joy. And without your addictions, you'll be freer to engage in the gratifying experiences of spontaneity, commitment, and self-actualization. REBT will appeal to you if you relish quickly taking control of your own life, rather than remaining dependent upon a therapist for years. By giving you tools for identifying and overcoming the true source of your difficulties, it will prepare you to act in many ways as your own therapist. And by helping you to reinforce realistic, self-benefitting beliefs, it will enable you to eliminate present emotional and behavioral problems, and to avoid future ones.

TRANSACTIONAL ANALYSIS (TA)

The Transactional Analysis (TA) approach to counseling belongs to the Psychoanalytical School of therapy and was devised by Eric Berne a Canadian psychiatrist in the late 1940's. TA is a widely recognized form of modern psychology that involves a set of practical conceptual tools designed to promote personal growth and change. It is considered a fundamental therapy for well-being and for helping individuals to reach their full potential in all aspects of life. In counseling TA therapy is very versatile, for it can be used in a wide range of areas and incorporates key themes from humanistic, integrative, psychoanalytical, psychodynamic therapies.

Dr. Eric Berne
(May 10, 1910 – July 15, 1970)

Though it is commonly recognized as a brief and solution-focused approach, transactional analysis can also be applied as an effective long-term, in-depth therapy.

Basic Philosophy: TA has an optimistic view of human nature that believes that people can change despite life's events and that is never too late to change one's life. All decisions that are made in life can be re-decided.

Some basic assumptions are -

- All people are important and their concerns are valid.
- All people should be treated as equals and with respect.
- With a few exceptions, all people have the ability to think as adults.
- All people decide their own destiny and can make changes to it.

From these assumptions TA creates a form of therapy that is not only problem-oriented, but also goal-oriented. It aims to free the client from maladaptive patterns of behavior based on past thinking. In doing so, they will be able to choose a new direction in life. In fact, the basic goal of transactional analysis is to help a client make new decisions about their current behavior and change the direction their lives are taking. In other words, a person will learn to write their own life story instead of allowing it to be written for them.

Key Concepts: TA uses four major methods to analyze and predict behavior -

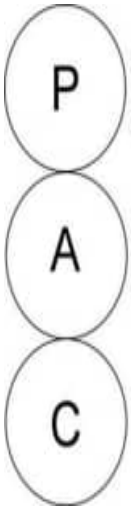
1. Structural analysis looks at what is happening within the individual.
2. Transactional analysis looks at what is happening between two or more people.
3. Game analysis looks at transactions between individuals leading to negative feelings.
4. Script analysis looks at the life plan the individual has chosen to follow.

Structural Analysis: TA is a dynamic theory at personality in the sense that uses the concepts of psychic energy and the distribution of energy. The structure of personality is based on the personality theory of parent, adult, child ego states. Eric Berne defines the ego states on the patterns of feeling, thinking and behaving. Any individual in a social grouping will exhibit one of these states. Structure analysis describes each person in terms of three ego states - *Parent ego state* contains the parental admonishments and values with dos, shoulds, and oughts. *Adult ego state* contains the objective, thinking, rational and logical ability to deal with reality. *Child ego state* is the source of childlike behaviors and feelings.

Parent Ego State: This is a set of feelings, thinking and behavior that we have copied from our parents and significant others. As we grow up we take in ideas, beliefs, feelings and behaviors from our parents and caretakers. If we live in an extended family then there are more people to learn and take in from. When we do this, it is called introjecting. For example, we may notice that we are saying things just as our father, mother, grandmother may have done, even though, consciously, we don't want

to. We do this as we have lived with this person so long that we automatically reproduce certain things that were said to us, or treat others as we might have been treated. Parental ego states are in two forms - (a) Controlling/ Critical Parent - which is negative, unsupportive, critical. (b) Nurturing Parent - which is supportive, helpful, nurturing, comforting.

Adult Ego State: It is about direct responses to the here and now. We deal with things that are going on today in ways that are not unhealthily influenced by our past. It is about being spontaneous and aware with the capacity for intimacy. When in our Adult we are able to see people as they are, rather than what we project onto them. We ask for information rather than stay scared and rather than make assumptions.



Parent Ego State
Behaviors, thoughts and feelings copied from parents or parent figures

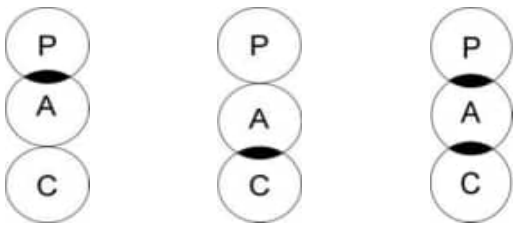
Adult Ego State
Behaviors, thoughts and feelings which are direct responses to the here and now

Child Ego State
Behaviors, thoughts and feelings replayed from childhood

Taking the best from the past and using it appropriately in the present is an integration of the positive aspects of both our 'Parent' and 'Child' ego states. So this can be called the 'Integrating Adult'. Integrating means that we are constantly updating ourselves through our every day experiences and using this to inform us.

Child Ego State: It is a set of behaviors, thoughts and feelings which are replayed from our own childhood. Berne says we all carry within ourselves a little boy or a little girl who feels, thinks, acts and responds just as we did when we were children. It also contains creative intuitive capacities such as rebellion, compliance etc. TA has identified three different forms of child ego state they are - Natural/ Free child; Adaptive child and Little professor.

Contamination Ego State: This occurs when we talk as if something is a fact or a reality when really this is a belief. Racism is an example of this. The 'Integrating Adult' ego state is contaminated in this case by the 'Parent' ego state.

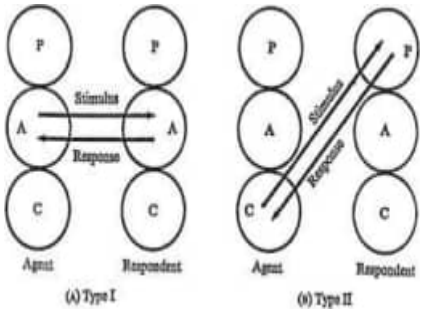


Parent Contamination | Child Contamination | Double Contamination (Parent & Child)

Both the Parent and Child ego states are constantly being updated. The 'Parent' and 'Child' ego states are echoes of the past. The 'Adult' ego state is a response to the here and now when a person is grown up and using grown up responses.

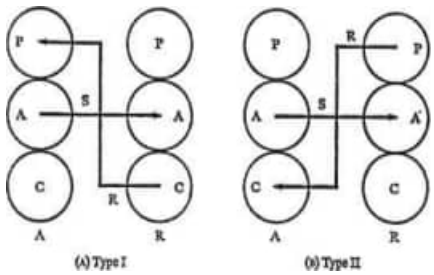
Transactional Analysis: The three ego states can be used as a way of analyzing transactions (communications) between people.

A transaction is a communication from A to B and the response from B to A. Transactional analysis helps the client to understand the three distinct types of transactions. They are complementary, crossed and ulterior.



Complementary Transaction

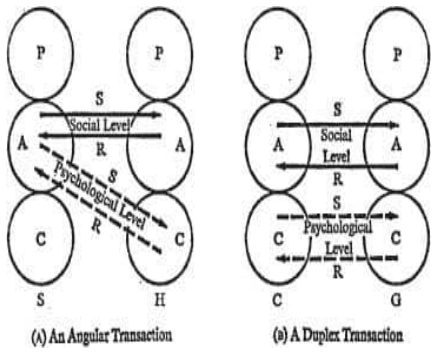
Complementary transactions are characterized by both people communicating from the same ego state. A complementary transaction occurs when a message, sent from a specific ego state, gets the predicted response from a specific ego state in the other person.



Crossed Transaction

Crossed transactions are characterized by both people coming from different ego states and resulting in an unexpected hurtful response. This happens between two ego states when a message sent from one person gets the unexpected response. The lines between the two ego states cross and communication breaks down.

Ulterior transactions are characterized by people coming from different ego states, but the responses appear to be from the same ego state. It involves more than two ego states at the same time. In this type of transaction two messages are exchanged one in the social level and one in psychological level. There are two types of ulterior transaction. They are Duplex transaction and Angular ulterior transaction.



Ulterior Transaction

Communication between people can be from one ego state to a different one or from one ego state to the same ego state. Normally communication will be from one ego state either to the same ego state or a different one. The person who first communicates will expect a reply to be from a certain ego state. If communication is from a different ego state to the expected one, then the communication may be

ineffective and the message may be lost, not received or disregarded by the person receiving it. If communication is from 'Adult to Adult' then it is likely to be the most effective communication for most of our communications.

There are three rules of communication' in TA -

1st Rule of Communication: So long as transactions remain complementary, communication can continue indefinitely.

2nd Rule of Communication: When a transaction is crossed, a break in communication results, and one or both individuals will need to shift ego states in order for the communication to be re-established.

3rd Rule of Communication: The behavioral outcome of an ulterior transaction (one where two messages are sent at the same time; one overt social and one covert psychological) is determined at the psychological level and not at the social level.

Game Analysis: Games are learned patterns of behavior. And most people play a small number of favorite games with a range of different people and in varying intensities. A game is an on-going series of transactions seemingly complementary, covertly ulterior which leads to a well defined pay off. Pay off are usually negative feelings like fear, anger, sadness for one or both the players. Game is defined as a series of complementary transactions, progressing to a well defined predictable outcome. Most of what happens in life is game. The client is encouraged to learn more satisfying ways of structuring time and acquiring strokes.

'First Degree' games are played in social circles generally lead to mild upsets not major traumas. 'Second Degree' games occur when the stakes may be higher. This usually occurs in more intimate circles, and ends up with an even greater negative payoff. 'Third Degree' games involve tissue damage and may end up in the jail, hospital or morgue. Chris Davidson (2002) has argued that world politics can involve 'Fourth Degree' games - where the outcomes involve whole communities, countries or even the world.

Games vary in the length of time that passes while they are being played. Some can take seconds or minutes while others take weeks months or even years. People play games for these reasons -

- to structure time;
- to acquire strokes;
- to maintain the substitute feeling and the system of thinking, beliefs and actions that go with it;
- to confirm parental injunctions and further the life script;
- to maintain the person's life position by 'proving' that self/others are not OK;

- to provide a high level of stroke exchange while blocking intimacy and maintaining distance;
- to make people predictable.

There are various ways to stop a game, including the use of different options than the one automatically used. We can -

- Cross the transaction by responding from a different ego state than the one the stimulus is designed to hook.
- Pick up the ulterior rather than the social message; e.g. when a person says "I can't do this, I'm useless". Rather than saying, "Let me do this for you", instead say, "It sounds like you have a problem. What do you want me to do about it?" (said from the Adult ego state)
- The opening message to the game always entails a discount. There are further discounts at each stage of the game. By detecting discounts we can identify game invitations and defuse them with options. A discount is when we minimize, maximize or ignore some aspect of a problem which would assist us in resolving it. Such as saying in a whiny voice "This is too difficult for me to do", so we automatically help them.
- Replace the game strokes. Loss of strokes to the 'Child' ego state means a threat to survival. We get a great many strokes from games, even if they are negative. However, if we don't obtain sufficient positive strokes, or give ourselves positive strokes, we will go for quantity rather than quality of strokes and play games to get them. This loss of strokes is also a loss of excitement that the game has generated.

Another way to think about this is to consider the game role we or the other person is likely to take. One way to discover this is to ask the following questions (John James, 1973) -

- What keeps happening over and over again?
- How does it start?
- What happens next?
- And then what happens?
- How does it end?
- How do feel after it ends?

We can then consider the reason we might have taken up a particular role, where we might switch to, and then consider how to do things differently. We need to consider what our own responsibility is in this - if the situation is too violent for us to get involved what options to we have? We could call for help; get others to come with us

to intervene and so on. We need to choose the appropriate assistance and take the action required.

Script Analysis: Script analysis involves the plans that a person makes for their life. Some of the common ones are -

- Never scripts result in a person who believes s/he is a never do well.
- Until scripts result in the person who continues to wait until they can deserve the reward.
- Always scripts result in a person continuing without change.
- After scripts result in anticipation of difficulty after a certain event.
- Open-ended scripts result in lack of direction after a given time or event.
- The ideal life script is one that is characterized by the "I am ok and you are ok".

"I'm OK - You're OK" is probably the best-known expression of the purpose of TA to establish and reinforce the position that recognizes the value and worth of every person. Transactional analysts regard people as basically "OK" and thus capable of change, growth, and healthy interactions. As Berne set his Psychology up, there are four life positions that a person can hold and holding a particular psychological position has profound implications for how an individual operationalizes his/her life. The positions are stated as -

The OK Corral (Franklin Ernst, 1971) Model

I am not Okay with me	You are Okay with me		I am Okay with me
	I am not OK You are Ok One Down Position Get away from Helpless	I am OK You are Ok Healthy Position Get on with Happy	
	I am not OK You are not Ok Hopeless Position Get nowhere with Hopeless	I am OK You are not Ok One-up Position Get rid of Angry	

You are not Okay with me

- I'm OK and you are OK. This is the healthiest position about life and it means that I feel good about myself and that I feel good about others and their competence.
- I'm OK and you are not OK. In this position I feel good about myself but I see others as damaged or less than and it is usually not healthy.
- I'm not OK and you are OK. In this position the person sees him/herself as the weak partner in relationships as the others in life are definitely better than the self. The person who holds this position will unconsciously accept abuse as OK.
- I'm not OK and you are not OK. This is the worst position to be in as it means that I believe that I am in a terrible state and the rest of the world is as bad.

Strokes: Strokes refer to compliments, acceptance and recognition, which are influential in how people lead their lives. TA therapy recognizes that we are greatly motivated by the reinforcement we get as children, and if this was dysfunctional, we are likely to adopt dysfunctional patterns of living as we get older.

Intimacy: Another motivation recognized in transactional analysis is intimacy. Similarly to strokes, if the intimacy a child experiences is dysfunctional, then they will learn that this type of intimacy is the best s/he can do to meet basic needs and communicate with others. This can lead to the development of repetitive patterns of behavior that can hinder a person's potential.

Re-decision: This refers to an individual's capacity to re-decide and make changes to certain decisions made as a child that stem from unconscious scripts. Re-decision reflects the assumption of TA therapy that individuals have the potential to lead their lives as they choose. This power is released after a re-decision is made while a client is in their child-ego state.

Contracts: Transactional analysis practice is based upon mutual contracting for change. Transactional analysts view people as capable of deciding what they want for their lives. Accordingly transactional analysis does its work on a contractual basis between the client and the therapist, educator, or consultant. This helps to set boundaries and realize expectations for both the client and the psychotherapist.

Ultimately, therapists will use these concepts in transactional analysis therapy to promote structural rearrangement and de-confusion of a client's child ego-state. They will encourage clients to challenge their current beliefs and the way s/he uses their life script. This will help them to better understand the direction and patterns of their life for themselves, and this awareness can help them to make the decision to change their behavior.

Role of the Counselor

- In TA the counselor and the client work together to establish the specific goals of therapy.
- The counselor contracts with the client for the changes that they desire.
- Counselors use their knowledge to address a concern introduced by the client rather than telling the client what issues need to focus on.

Goals of the TA

- The goal is to not only learn to adjust to life but to attain health and autonomy.
- Through the gaining of autonomy the client can be more aware, intimate and spontaneous living a life free of games and self-defeating life scripts.

Techniques of TA: In addition to structural, transactional, game and script analysis, TA counselors use other techniques. Some of these techniques are -
Treatment contract which is a agreement between the counselor and the client about what is to be accomplished and what responsibilities are agreed upon. *Interrogation*

involves forcing the client to answer from the adult ego state through a succession of confrontive questions. *Specification* is identifying the ego state that is the source of the transition. *Confrontation* is the process of pointing out inconsistencies. *Explanation* is a process where the counselor teaches the Adult ego state of the client a tenet of TA. *Illustration* is a story or example that is used to portray a point. *Confirmation* is a technique where the counselor directs the client's attention to a previously modified behavior that is reoccurring. *Interpretation* involves the counselor explaining to the child ego state the reasons behind the client's behavior. *Crystallization* occurs when the client realizes that game playing can be given up and the client can enjoy the freedom of choice in behavior.

How Does TA Work?

Transactional Analysis (TA) is a tool to know yourself, to know how you relate to others, and to discover how you have been living out a life plan based on unconscious relational patterns. These unconscious organizing patterns that begin at birth. They are reinforced and elaborated during a number of developmental ages. The results constitute a life 'script'. Change happens through the help of your therapist in becoming aware of the previously unconscious 'script' patterns. The therapist can then facilitate different outcomes so that you get what you want in life. TA offers many tools and useful approaches for individual, couple, and group therapy. TA is based on the assumption that all individuals can learn to trust themselves, think for themselves, make their own decisions and express their feelings.

TA therapy is based on the theory that each person has three ego states - parent, adult and child. These are used along with other key transactional analysis concepts, tools and models to analyze how individuals communicate and identify what interaction is needed for a better outcome. Throughout therapy, the TA therapist will work directly on here and now problem solving behaviors, whilst helping clients to develop day-to-day tools for finding constructive creative solutions. The ultimate goal is to ensure clients regain absolute autonomy over their lives. Eric Berne defines this autonomy as the recovery of three vital human capacities - spontaneity, awareness and intimacy.

Transactional analysis is a talking therapy and sessions are designed to explore an individual's personality and how this has been shaped by experience - particularly those stemming from childhood. This is achieved through skilful questioning and the utilization of various models, techniques and tools. Sessions can be carried out in the form of one-on-one counseling, or with families, couples or groups. The atmosphere that supports transactional analysis is non-judgmental, secure and respectful, ensuring that a positive relationship is forged between the therapist and client(s) in order to provide a model for subsequent relationships and communication that are developed outside of therapy.

In this setting the therapist works collaboratively with the individual to identify what has gone wrong in their communication and provide opportunities for them to change repetitive patterns that limit their potential. TA therapists recognize that we all have the potential to live the life we want, rather than the life we are programmed to live. Sometimes however this potential is hindered by repetitive patterns or 'unconscious scripts' that stem from childhood decisions and teachings.

Applications: TA can help -

- Develop awareness, options and skills in daily life, through the enhancement of your strengths and resources.
- Improve on your communication skills and help with managing conflict and difficult situations in your life.
- Foster intimacy, spontaneity, and autonomy within yourself and your relationships.
- Think clearly and raise your emotional intelligence.
- Develop new options and make better choices in your life.
- Experience and live the life you deserve.

Transactional Analysis can be said to be an effective method in the process of counseling. Recently its scope is increasing. Individuals especially the young adults are the needy clients of this transactional analysis counseling, because present generations carry different complex ego states. It enable the clients to identity the sorts of relationship.

The term 'cognitive-behavioral therapy (CBT)' is a very general term for a classification of therapies with similarities. There are several approaches to cognitive-behavioral therapy, including Rational Emotive Behavior Therapy, Rational Behavior Therapy, Rational Living Therapy, Cognitive Therapy, and Dialectic Behavior Therapy.

Cognitive Therapy: This approach aims to identify and correct distorted thinking patterns that can lead to feelings and behaviors that may be troublesome, self-defeating, or even self-destructive. The goal is to replace such thinking with a more balanced view that, in turn, leads to more fulfilling and productive behavior. Uses the power of the mind to influence behavior. It is based on the theory that previous experiences can damage self image and this can affect attitude, emotions and ability to deal with certain situations. It works by helping the client to identify, question and change poor mental images of them, thus altering negative responses and behavior. It can help pessimistic or depressed people to view things from a more optimistic perspective.

Cognitive Analytical Therapy: This combines Cognitive Therapy and Psychotherapy and encourages clients to draw on their own ability to develop the skills to change destructive patterns of behavior. Negative ways of thinking are explored in structured and directive ways, involving diary-keeping, progress charts, etc.

Dialectical Behavioral Therapy: DBT was developed from cognitive behavior therapy (CBT). The main aim of CBT is to change behavior, which is done by applying techniques with a focus on problem-solving, such as homework, diary cards and behavioral analysis. However, some people felt uncomfortable with the strong focus on change, and felt that their suffering and apparent loss of control over their lives were not understood. This caused them to become frustrated and even to drop out of treatment. Therapist sought to resolve this by the use of acceptance strategies. Acceptance strategies are added to the process of CBT which means that the therapist explores with their clients an acceptance that their behavior (e.g. self-harming, drinking, etc.), even though damaging in the long term, may be the only way they have learned to deal with intense emotions; and which might have led to positive short term benefits.

HUMANISTIC APPROACH: Person Centered Therapy, Gestalt Therapy, Existential Therapy

In the 1960s and 1970s, there was a growing interest among counselors in a 'third force' in therapy as an alternative to the psychoanalytic and behavioral approaches. Under this heading fall the person-centered approach, Gestalt therapy and existential therapy.

Humanistic psychology is concerned with human potential and the individual's unique personal experience. Humanistic psychologists generally do not deny the importance of many principles of behaviorism and psychoanalysis. They value the awareness of antecedents to behavior as well as the importance of childhood experiences and unconscious psychological processes. Humanistic psychologists would argue, however, that humans are more than the collection of behaviors or objects of unconscious forces. Therefore, humanistic psychology often is described as holistic in the sense that it tends to be inclusive and accepting of various theoretical traditions and therapeutic practices. The emphasis for many humanistic therapists is the primacy of establishing a therapeutic relationship that is collaborative, accepting, authentic, and honors the unique world in which the client lives. The humanistic approach is also holistic in that it assumes interrelatedness between the client's psychological, biological, social, and spiritual dimensions. Humanistic psychology assumes that people have an innate capacity toward self-understanding and psychological health.

Humanistic therapy helps individuals access and understand their feelings, gain a sense of meaning in life, and reach self-actualization. The aim of humanistic therapy is usually to help the client develop a stronger, healthier sense of self, also called self-actualization. It is also to help individuals access and understand their feelings to help to gain a sense of meaning in life. Humanistic theory sees each individual's personality as being composed of physical, intellectual, emotional, behavioral, creative, and spiritual elements. In order to help the client reach self-actualization, humanistic therapists help clients remove and replace the attitudes, behaviors, and beliefs that do not produce a positive state of being, and integrate the various components of their unique personalities so that each individual is more self-aware, mature, and authentic.

Key points of humanistic therapy are -

- Empathic understanding of the client's frame of reference and subjective experience.
- Respect for the client's cultural values and freedom to exercise choice.

- Exploration of problems through an authentic and collaborative approach to help the client develop insight, courage, and responsibility.
- Exploration of goals and expectations, including articulation of what the client wants to accomplish and hopes to gain from treatment.
- Clarification of the helping role by defining the therapist's role but respecting the self determination of the client.
- Assessment and enhancement of client motivation both collaboratively and authentically.
- Negotiation of a contract by formally or informally asking, "Where do we go from here"?
- Demonstration of authenticity by setting a tone of genuine, authentic encounter.

In humanistic therapy, there are two widely practiced techniques - gestalt therapy (which focuses on thoughts and feelings 'here and now' instead of root causes) and client-centered therapy (which provides a supportive environment in which clients can re-establish their true identity).

PERSON-CENTERED THERAPY

Dr. Carl Ransom Rogers
(January 8, 1902 – February 4, 1987)

Person-centered therapy (PCT) is also known as person-centered psychotherapy, person-centered counseling, client-centered therapy or Rogerian psychotherapy. PCT is a humanistic approach that deals with the ways in which individuals perceive themselves consciously rather than how a counselor can interpret their unconscious thoughts or ideas. It is a non-directive form of talk therapy that was developed by humanist psychologist Carl Rogers during the 1940s and 1950s. Today, it is one of the most widely used approaches in psychotherapy.

Client-centered therapy provides a supportive environment in which clients can re-establish their true identity. The world that we live in is judgmental. And many of us fear that if we shared with the world our true identity it would judge us relentlessly. Experience with these judgments results in people establishing a public identity to navigate the judgmental world. The ability to re-establish their true identity will help the individual understand themselves as they truly are; which is important as people suppress their feelings about issues because they are not supported, socially acceptable, or lead to unwanted judgment. The task of reestablishing one's true

identity is not an easy task and the therapist must rely on the techniques of unconditional positive regard and empathy. These two techniques are central to client-centered therapy because they build trust between the client and therapist by creating a nonjudgmental and supportive environment for the client.

Contemporary person-centered therapy is the result of an evolutionary process that continues to remain open to change and refinement. Rogers did not present the PCT as a fixed and completed approach to therapy. The four developmental periods of the approach are -

1. In the first period, during the 1940s, Rogers developed what was known as nondirective counseling as a reaction against the directive and traditional psychoanalytic approaches to individual therapy. Roger's theory emphasized the counselor's creation of a permissive and nondirective climate. Rogers also challenged the validity of commonly accepted therapeutic procedures such as advice, suggestion, direction, persuasion, teaching, diagnosis, and interpretation. This was based on his conviction that diagnostic concepts and procedures were inaccurate, prejudicial, and often misused. Nondirective counselors avoided sharing a great deal about themselves with clients and instead focused mainly on reflecting and clarifying the clients' verbal and nonverbal communications with the aim of gaining insight into the feelings expressed by clients.
2. In the second period, during the 1950s, Rogers renamed his approach client-centered therapy to reflect its emphasis on the client rather than on directive methods. This period focused on the phenomenological world of the client. Rogers assumed that the best vantage point for understanding how people behave was from their own internal frame of reference. He focused more explicitly on the actualizing tendency as the basic motivational force that leads to client change.
3. The third period, beginning in the late 1950s and extending into the 1970s, addressed the necessary and sufficient conditions of therapy. The process of "becoming one's experience" is characterized by openness to experience, a trust in one's experience, an internal locus of evaluation, and the willingness to be in process. He was interested in how people best progress in psychotherapy, and he studied the qualities of the client-therapist relationship as a catalyst leading to personality change. Client-centered philosophy was applied to education and was called student-centered teaching. It was also applied to encounter groups, led by laypersons in the 1960s.
4. The fourth phase, during the 1980s and 1990s, was marked by considerable expansion to education, industry, groups, conflict resolution, and the search for world peace. The theory became known as person-centered approach. This shift in

terms reflected the broadening application of the approach. Areas of further application include education, health care, cross-cultural and interracial activity, international relations, politics, and the achievement of world-peace.

Basic Philosophy

- Rogers viewed human nature as basically good.
- He believed that if given the appropriate environment of acceptance, warmth and empathy, the individual would move toward self-actualization.
- Self-actualization is the motivation that makes the individual move toward growth, meaning, and purpose.
- It is considered a phenomenological psychology whereby the individual's perception of reality is accepted as reality for the individual.
- Person-centered is often referred to as a self theory, because of Rogers's emphasis on the self being a result of the person's life experiences and the person's awareness of comparisons to others as the same or different.
- Rogers believed that most people were provided conditional acceptance as children, which lead them to behave in ways that would assure their acceptance. However, in their need for acceptance, the individual often behaved in ways that were incongruent with the real self. Thus, the greater this incongruence between the real self and the ideal self, the greater isolated and maladjusted the person became.

Key Concepts: Rogers identified six key factors that stimulate growth within an individual. He suggested that when these conditions are met, the person will gravitate toward a constructive fulfillment of potential. According to Rogerian theory, the six factors necessary for growth are -

- Therapist-Client Psychological Contact: This first condition simply states that a relationship between therapist and client must exist in order for the client to achieve positive personal change. The following five factors are characteristics of the therapist-client relationship, and they may vary by degree.
- Client Incongruence or Vulnerability: A discrepancy between the client's self-image and actual experience leaves him/her vulnerable to fears and anxieties. The client is often unaware of the incongruence.
- Therapist Congruence or Genuineness: The therapist should be self-aware, genuine, and congruent. This does not imply that the therapist be a picture of perfection, but that s/he be true to him/herself within the therapeutic relationship.
- Therapist Unconditional Positive Regard (UPR): The clients' experiences, positive or negative, should be accepted by the therapist without any conditions or

judgment. In this way, the client can share experiences without fear of being judged.

- Therapist Empathy: The therapist demonstrates empathic understanding of the clients' experiences and recognizes emotional experiences without getting emotionally involved.
- Client Perception: To some degree, the client perceives the therapist's unconditional positive regard and empathic understanding. This is communicated through the words and behaviors of the therapist.

Role of the Counselor

- The counselor sets up an environment where the client is safe to explore any aspect of the self. The counselor's job is to facilitate the exploration through a special "I - Thou" relationship of unconditional positive regard, empathy and warmth.
- The counselor uses psychological testing on a limited basis. The Q-sort is sometimes used in assessment by the person centered counselor. The Q-sort gives an indication of the incongruence between the perceived real self and ideal self.

Q-sort: A series of 100 statements are written on cards. The statements are self-descriptions, i.e. I am capable, I am dependent, I am worthless. The client is asked to read and sort each of these statements into nine piles from most like me to least like me. Then the stacks are recorded. The clientre-sorts the cards into what they would like to be like.

- The use of diagnostic categories is discouraged as incompatible with the philosophical view of the individual as unique. Diagnosis places the counselor in a position of authority and imposes a treatment plan.

Goals of Therapy: Person centered theory is directly concerned with the individual. The counselor facilitates the client toward -

- Realistic self-perception;
- Greater confidence and self-direction;
- Sense of positive worth;
- Greater maturity, social skill, and adaptive behavior;
- Better stress coping;
- More fully functioning in all aspects of their lives.

Therapeutic Relationship: The relationship is of primary importance. The qualities of the therapist, including genuineness, warmth, accurate empathy, respect, and non-judgmentalness and communication of these attitudes to clients are stressed. Clients use this real relationship with the therapist to help them transfer their learning to other relationships. Rogers based his hypothesis of the 'necessary and sufficient conditions for therapeutic personality change' on the quality of the relationship. Roger's

hypothesis is as follows -

- Two persons are in psychological contact.
- The first, or the client, is in a state of incongruence, being vulnerable or anxious.
- The second person, or the therapist, is congruent or integrated in the relationship.
- The therapist experiences unconditional positive regard for the client.
- The therapist experiences an empathic, understanding of the client's internal frame of reference and endeavors to communicate this experience to the client.
- The communication to the client of the therapist's empathic understanding and unconditional positive regard is to a minimal degree achieved.

Techniques of Therapy: This approach uses few techniques but stresses the attitude of the therapist. Basic techniques include active listening and hearing, reflection of feelings, clarification, and being there for the client. This model does not include diagnostic testing, interpretation, taking a case history, or questioning or probing for information. The techniques used in person centered therapy have changed over time. Three periods of time in which different techniques were stressed -

Nondirective Period (1940 - 1950): In this period of theory development, the counselor focused on listening and creating a permissive atmosphere. The counselor did not provide interventions, but communicated acceptance and clarification.

Reflective Period (1950 - 1957): During this period of time, counselors emphasized being non judgmental of the client, while responding to the client's feelings and reflecting the affect accurately.

Experiential Period (1957 - 1980): This is the period of the EWG - Empathy, Warmth and Genuineness. Empathy is the ability of the counselor to understand the emotions of the client and correctly communicate this understanding. Warmth is also referred to as acceptance and positive regard in person centered literature. Warmth is the ability of the counselor to convey an unconditional acceptance of the client's personhood. Genuineness or congruence is the ability to be who one really is without assuming roles or facades.

The counselor helps the client through accurate reflections of feelings, keeping the client focused on the concern, and clarification of feelings and information. The counselor uses open-ended questions or phrases to help the clients gain insight into experiences and necessary changes in their lives.

Applications: Has wide applicability to individual and group counseling. It is especially well suited for the initial phases of crisis intervention work. Its principles have been applied to marital and family therapy, community programs, administration and management, and human relations training. It is a useful approach for teaching,

parent/child relations, and working with groups composed of people from diverse cultural backgrounds. Today many people who are not practicing counseling use the approach to help guide them through day-to-day work and relationships.

Generally, person-centered counseling can help individuals of all ages with a range of personal issues. It has been found particularly useful in helping individuals to overcome specific problems such as depression, anxiety, personality disorders, eating disorders and alcohol addictions. These issues can have significant impact on self-esteem, self-reliance and self-awareness, but person-centered therapy can help individuals to reconnect with their inner self in order to transcend any limitations.

Multiculturalism: Focus is on breaking cultural barriers and facilitating open dialogue among diverse cultural populations. Main strengths are respect for clients' values, active listening, welcoming of differences, nonjudgmental attitude, understanding, willingness to allow clients to determine what will be explored in sessions, and prizing of cultural pluralism. Some of the core values of this approach may not be congruent with the client's culture. Lack of counselor direction and structure are unacceptable for clients who are seeking help and immediate answers from a knowledgeable professional.

Contributions: Rogers had a major impact on the field of counseling and psychotherapy. When he introduced his revolutionary ideas in the 1940s, he provided a powerful and radical alternative to psychoanalysis and to the directive approaches then practiced. Rogers was a pioneer in shifting the therapeutic focus from an emphasis on technique and reliance on therapist authority to that of relationship. One of Rogers's contributions to the field of psychotherapy was his willingness to state his concepts as testable hypotheses and to submit them to research. He literally opened the field to research. Person-centered therapy has demonstrated that therapist empathy plays a vital role in facilitating constructive changes in the client -

- Research consistently demonstrates that therapist empathy is the most potent predictor of client progress in therapy.
- Empathy is an essential component of successful therapy in every therapeutic modality.
- Empathy is a basic component of emotional intelligence.
- Client perception of therapist empathy, as opposed to that of external raters or therapists, is mot predictive of positive client outcome.

Limitations: Possible danger from the therapist who remains passive and inactive, limiting responses to reflection. Many clients feel a need for greater direction, more structure, and more techniques. Clients in crisis may need more directive measures. It

is difficult to translate the core conditions into actual practice in certain cultures. Clients accustomed to indirect communication may not be comfortable with direct expression of empathy or self-disclosure on the therapist's part. Another limitation is that this approach extols the value of an internal locus of evaluation. In some cases, clients are likely to be highly influenced by societal expectations and not simply motivated by their own personal preference.

Because the person-centered counselors place so much emphasis on genuineness and on being led by the client, they do not place the same emphasis on boundaries of time and technique as would psychodynamic therapists. They have very positive and optimistic view of human nature. The philosophy that people are essentially good, and that ultimately the individual knows what is right for them, is the essential ingredient of successful person centered therapy as 'all about loving'.

GESTALT THERAPY

Gestalt Therapy was developed in the late 1940s by Fritz Perls (a German-born psychiatrist) with his wife Laura Perls and further influenced by the likes of Kurt Lewin and Kurt Goldstein (Corsini & Wedding, 2000). It was developed as a revision to psychoanalysis and focuses on an experiential and humanistic approach rather than analysis of the unconscious; which was one of the main therapeutic tools at the time 'Gestalt Therapy' was employed.

Friedrich (Frederick) Salomon Perls
[Better known as Fritz Perls]
(July 8, 1893 – March 14, 1970)

Gestalt Therapy rejects the dualities of mind and body, body and soul, thinking and feeling, and feeling and action. According to Perls, people are not made up of separate components, i.e., mind, body and soul; rather human beings function as a whole. In doing so, one defines who one is (sense of self) by choice of responses to environmental interactions (boundaries). The word 'Gestalt' (German origin) refers to a "whole, configuration, integration, pattern or form" (Patterson, 1986). Gestalt Therapy focuses on the whole of the client's experience, including feelings, thoughts and actions. The client gains self-awareness in the 'here and now' (what is happening from one moment to the next) by analyzing behavior and body language and talking about bottled up feelings. In gestalt therapy, self-awareness is key to personal growth and developing full potential. The approach recognizes that sometimes this self-awareness can become blocked by negative thought patterns and behavior that can

leave people feeling dissatisfied and unhappy. It is the aim of a gestalt therapist to promote a non-judgmental self-awareness that enables clients to develop a unique perspective on life. By helping an individual to become more aware of how they think, feel and act in the present moment, gestalt therapy provides insight into ways in which s/he can alleviate their current issues and distress in order to aspire to their maximum potential.

Basic Philosophy

- The person strives for wholeness and integration of thinking, feeling, and behaving.
- Gestalt therapy is considered to be a here-and-now therapy focusing on awareness with the belief that when one focuses on what they are and not what they wish to become, they become self-actualized. The idea being that through self acceptance one becomes self-actualized.
- Individual naturally seeks to become an integrated whole, living productively.
- Gestaltists are anti-deterministic because they believe that people have the ability to change and become responsible.
- Gestalt focuses on the client's own inner world of interpretation and assessment of the present life situation.
- Gestaltists believe that individuals emphasize intellectual experience, diminishing the importance of emotions and senses, resulting in an inability to respond to the situations or events in their life.
- Gestaltists believe that thoughts, feelings, and reactions to past events or situations can impede personal functioning and prevent here-and-now awareness.
- The emphasis in on reality and not on embellished or imagined needs. The individual recognizes their internal need and meets that need through manipulation of the need and the environment.
- Difficulty may arise in several ways -
  - Loss of contact with the environment and its resources.
  - Loss of contact with self through over involvement with environment.
  - Fail to put aside unfinished business.
  - Loss the Gestalt resulting in fragmentation or scattering.
  - Experience conflict between what one should do and what one wants to do.
  - Experience difficulty with life's dichotomies, i.e., love/hate, pleasure/pain, masculinity/femininity.

Key Concepts of Gestalt Therapy: Several key concepts underlie Gestalt Therapy, many of which are similar to that of person-centered and existential therapy. The following are the key concepts of Gestalt Therapy -

Wholeness and Integration: Wholeness refers to the whole person or the individual's mind and body as a unit rather than as separate parts (Seligman, 2006). Integration refers to how these parts fit together and how the individual integrates into the environment. Often people who come to therapy do not have these parts fitting together in their environment, Gestalt Therapy is about facilitating clients to integrate themselves as whole persons and help restore balance in their environment.

Awareness: Awareness is one of the most important elements in Gestalt Therapy as it is seen as a "hallmark of the healthy person and a goal of treatment" (Seligman, 2006). When individuals are 'aware', they are able to self-regulate in their environment. There are two main causes lacking awareness - (a) Preoccupation with one's past, fantasies, flaws and strengths that the individual becomes unaware of the whole picture, and (b) Low self-esteem. There are three ways people may achieve awareness through therapy - (i) Contact with the environment - This is through looking, listening, touching, talking, moving, smelling, and tasting. This enables the individual to grow in his/her environment through reacting to the environment and changing. (ii) Here and now - This is the individual living in and being conscious at the present moment rather than worrying about the past or the future.
(iii) Responsibility - This refers to the individual taking responsibility for his/her own life rather than blaming others.

Energy and Blocks to Energy: Gestalt Therapists often focus on where energy is in the body, how it is used, and how it may be causing a blockage (Corey, 2005). Blocked energy is a form of resistance, for example, tension in a part of the body, not breathing deeply, or avoiding eye contact. Gestalt Therapy is about finding and releasing the blockages that may be inhibiting awareness.

Growth Disorders: Growth disorders refer to emotional problems that are caused by people who lack awareness and do not interact with their environment completely. In doing so, people are unable to cope with the changes in their lives successfully and, instead deal with the problems in a defensive manner (Seligman, 2006).

Unfinished Business: Unfinished business refers to people who do not finish things in their lives and is often related to people with a 'growth disorder' (Seligman, 2006). People with unfinished business often resent the past and because of this are unable to focus on the here and now. One of the major goals of Gestalt Therapy is to help people work through their unfinished business and bring about closure.

Role of the Counselor

- Counselor creates an environment for the client to explore their needs in order to grow.

- Counselor is fully with the client in the here-and-now with intense personal involvement and honesty.
- Counselor helps the client to focus on blocking energy and to positively and adaptively use that energy.
- Counselor also helps the clients to discern life patterns.
- Among the rules that counselor uses to help client -
  - The principle of the now requires the counselor to use present tense.
  - "I and Thou" means that the client must address the person directly rather than talk about them or to the counselor about them.
  - Making the client use the "I" instead of referring to own experiences in the second (you) or third (it) person.
  - The use of an awareness continuum that focuses on how and what rather than on why.
  - The counselor has the client convert questions into statements.
  - DSM-IV categories and standardized assessment is not considered necessary within this theory.

Goals of Therapy: To assist clients in gaining awareness of moment-to-moment experiencing and to expand the capacity to make choices. Aims not at analysis but at integration.

- Emphasis is on the here-and-now of the client's experience.
- The client is encouraged to make choices based on the now as opposed to past.
- Help the client resolve the past.
- Assist the client to become congruent.
- Help the client to reach maturity intellectually.
- Help the client shed neuroses.

Therapeutic Relationship: Central importance is given to the I/Thou relationship and the quality of the therapist's presence. The therapist's attitudes and behavior counts more than the techniques used. The therapist does not interpret for the clients but assists them in developing the means to make their own interpretations. Clients identify and work on unfinished business from the past that interferes with current functioning.

Techniques of the Therapy: Techniques in Gestalt therapy one of two forms, either an exercise or an experiment. (1) Exercises include activities such as - i. frustration actions; ii. fantasy role playing; iii. Fantasy; and iv. Psychodrama. (2) Experiments are unplanned creative interventions that grow out of the here-and-now interaction between the client and the counselor.

- Dream works in Gestalt therapy consist of the client telling the dream and then focusing their awareness on the dream from the perspective of each character or element in the dream.
- Empty chair is a process where the client addresses parts of the personality, as if it were an entity sitting an empty chair. The client may switch perspectives by switching chairs or may simply address the chair. The technique is contraindicated for severely disturbed clients.
- Confrontation is another of the Gestalt exercises that is very powerful. Basically, the counselor calls attention to the incongruence between the client's verbalizations and observed emotions or behaviors.
- Making the rounds is a group exercise in which the client is instructed to say the same sentence to each member of the group and then adding something personal to each person.
- "I take responsibility" is the phrase that follows each statement expressing statements or perceptions that the client states. The process is to help the client to integrate their internal perceptions and their behaviors.
- Exaggeration is over-dramatizing the client's gestures and movements to help gain insight into their meaning.
- May I feed you a sentence is the question that the counselor asks before giving the client a more specific expression of what the counselor believes is the underlying message of the client.

Applications: Ultimately, gestalt therapy is considered to help individuals gain a better understanding of how their emotional and physical needs are connected. They will learn that being aware of their internal self is key to understanding why they react and behave in certain ways. This journey of self-discovery makes the approach beneficial for individuals who can be guarded when it comes to their emotions, and find it difficult to process why they feel and act the way they do. It can also provide support and a safe space for individuals going through times of personal difficulty.

Gestalt therapy is considered particularly valuable for helping to treat a wide range of psychological issues - especially as it can be applied as a long-term therapy or as a brief and focused approach. It has been found effective for managing tension, anxiety, addiction, post-traumatic stress, depression and other psychological problems that can prevent people from living life to the full. Gestalt Therapy is also effective in counseling groups, couples, and families (Corsini & Wedding, 2000). Overall, people who participate in gestalt therapy tend to feel more self-confident, calm and at peace with themselves.

Multiculturalism: Its focus on expressing oneself nonverbally is congruent with those cultures that look beyond words for messages. Provides many experiments in working with clients who have cultural injunctions against freely expressing feelings. Can help to overcome language barrier with bilingual clients. Focus on bodily expressions is a subtle way to help clients recognize their conflicts. Clients who have been culturally conditioned to be emotionally reserved may not embrace Gestalt experiments.

Contributions: Main contribution is an emphasis on direct experiencing and doing rather than on merely talking about feelings. It provides a perspective on growth and enhancement, not merely a treatment of disorders. It uses clients' behavior as the basis for making them aware of inner creative potential. The approach to dreams is a unique, creative tool to help clients discover basic conflicts. Therapy is viewed as an existential encounter; it is process-oriented, not technique-oriented. It recognizes nonverbal behavior as a key to understanding.

Limitations: For Gestalt Therapy to be effective, the therapist must have a high level of personal development (Corey, 2005). Effectiveness of the confronting and theatrical techniques of Gestalt Therapy is limited and has not been well established. It has been considered to be a self-centered approach which is concerned with just individual development. Potential danger for therapists to abuse the power they have with clients (Corey, 2005). Lacks a strong theoretical base. Deals only with the here and now. Does not deal with diagnosis and testing.

How Does Gestalt Therapy Work?

Fundamentally, gestalt therapy works by teaching clients how to define what is truly being experienced rather than what is merely an interpretation of the events. Those undertaking gestalt therapy will explore all of their thoughts, feelings, behaviors, beliefs and values to develop awareness of how they present themselves and respond to events in their environment. This gives them the opportunity to identify choices, patterns of behavior and obstacles that are impacting their health and well-being, and preventing them from reaching their full potential. The unfolding of this therapeutic process will typically involve a range of expressive techniques and creative experiments developed collaboratively between therapist and client. These will be appropriate for the client and their specific problems. Below are some of the most common methods used -

Role-play: Role-play can help individuals to experience different feelings and emotions and better understand how they present and organize themselves.

Empty Chair/ Open Chair Technique: The empty chair technique is a "method of facilitating the role-taking dialogue between the patient and others or between parts of the patient's personality. It is generally used in a group situation" (Patterson, 1986). Two chairs are placed facing each other - one represents the patient or one aspect of the patient's personality, and the other represents another person or the opposing part of the personality. As the patient alternates the role, s/he sits in one or the other chair.

The therapist may simply observe as the dialogue progresses or may instruct the patient when to change chairs, suggest sentences to say, call the patient's attention to what has been said, or ask the patient to repeat or exaggerate words or actions. In the process, emotions and conflicts are evoked, impasses may be brought about and resolved, and awareness and integration of polarities may develop - polarities or splits within the patient, between the patient and other persons, or between the patient's wants and the social norms (Patterson, 1986).

Dialogue: A gestalt therapist will need to engage the client in meaningful and authentic dialogue in order to guide them into a particular way of behaving or thinking. This may move beyond simple discussion to more creative forms of expression such as dancing, singing or laughing.

Dreams: Dreams are used to bring about integration by the client. The focus of a client's dream is not on the unconscious, rather on projections or aspects of the dreamer (Seligman, 2006). The therapist would get clients to talk about their dream/s in terms of the significance of each role in the dream and this allows clients to take responsibility for the dreams and increase awareness of their thoughts and emotions.

Attention to Body Language: Throughout therapy, a gestalt therapist will concentrate on body language, which is considered a subtle indicator of intense emotions. When specific body language is noticed, the therapist may ask the client to exaggerate these movements or behaviors. This is thought to intensify the emotion attached to the behavior and highlight an inner meaning. For example, a client may be showing signs of clenched fists or frowning, to which the therapist may ask something along the lines of - "What are you saying with this movement?"

Gestalt Therapy deems that people cannot be considered as separate from their environment or from interpersonal relations. The individual is seen as being self-regulating and is able to motivate oneself to solve problems. Individuals are able to work towards growth and develop as their environments allow. A psychologically healthy person is someone who is self-regulating through the changes in life and has developed a sense of 'wholeness' between mind and body (Corsini & Wedding, 2000).

Use of Language: Gestalt Therapists choose language that will encourage change in the client. The following are ways that this can be accomplished (Seligman, 2006) -

- Emphasis on statements rather than questions to highlight a collaborative client-therapist relationship.
- 'What' and 'How' questions (when questions are used) to keep the client in the present and promote integration.
- 'I' statements are used to promote clients ownership of feelings rather than placing blame on others.
- The present tense is used so the focus is on the present rather than the past.
- Encouraging responsibility for clients of their words, emotions, thoughts, and behaviors so they recognize and accept what they are feeling.

Topdog - Underdog: A commonly utilized Gestalt technique is that of the topdog-underdog dialogue. This technique is used when the therapist notices two opposing opinions/attitudes within the client. The therapist encourages the client to distinguish between these two parts and play the role of each in a dialogue between them (Patterson, 1986). The tyrannical 'topdog' demands that things be a particular way whilst the 'underdog' plays the role of disobedient child. The individual becomes split between the two sides struggling for control.

Fantasy: Fantasy is used in Gestalt Therapy to increase clients' self-awareness of their thoughts and emotions and to bring about closure to unfinished business (Seligman, 2006). Therapists use guided imagery techniques (fantasy) to encourage clients to imagine situations such as what they would do in a certain situation or by projecting themselves into different roles.

Body as a Vehicle of Communication: Gestalt Therapy sees that not only are thoughts and emotions important to creating a feeling of 'wholeness' for the client, the physical sensations are also important. Seligman (2006) has identified three strategies to help with focusing attention on the physical sensations -

- Identification: Gestalt Therapists should be able to recognize physical signs of their clients. For example, a client might be tapping their feet on the ground. The therapist may say - "Become your leg and give it a voice?" This creates awareness of the client's physical sensations and emotions.
- Locating emotions in the body: Gestalt Therapists may ask clients where they are experiencing the emotion in their body. For example, a client may say they are feeling nervous about something. The therapist may ask where this is coming from in the body and the response from the client may be that the feeling is butterflies in the stomach. This helps the client to bring about more awareness into sensations and their emotions.
- Repetition and exaggeration: If there is repetition such as the example of the client tapping their feet on the ground, the therapist would get them to exaggerate the movement and talk about feelings that come up. This in turn focuses on the emotion and should help to release the blocked awareness.

Confusion: The technique of dealing with confusion of the client is about drawing attention to the client's hesitation in talking about something unpleasant. The hesitation can be shown through avoidance, blanking out, verbalism and fantasy (Patterson, 1986). By drawing attention to the hesitation, it creates self-awareness for the client and allows the client to work through the issue.

Confrontation: In Gestalt Therapy, confrontation means 'to challenge or frustrate the client'. The client is challenged with sensitivity and empathy on the part of the therapist to face the issues important to them. It is an invaluable tool for bringing clients into clear awareness of their realities, when used appropriately. However, confrontation is not a technique that can be used with all clients.

Gestalt therapy is a well-developed and well-grounded theory with a myriad of tenets, principles, concepts, and methods. Gestalt therapy is a sound science and a powerful means for facilitating and nurturing the full functioning of the human person with the potential of bringing about human healing, growth, and wholeness. It focuses on the integration between the 'whole' person and his/her environment. This therapy sees a healthy individual as being someone who has awareness in his/her life and lives in the here and now rather than focusing on the past or future. Gestalt Therapy has a number of successful techniques that are applicable in therapy today and may be utilized across a broad spectrum of emotional issues.

EXISTENTIAL THERAPY

Existential psychotherapy arose spontaneously in the minds and works of a number of psychologists and psychiatrists, namely Rollo May, Viktor Frankl, James Bugental,

Friedrich Nietzsche, Soren Kierkegaard, Jean-Paul Sartre, Irvin Yalom, Kirk Schneider, Stephen Diamond, and Myrtle Heery in Europe in the 1940s and 1950s.

Existential psychotherapy uses a positive approach that applauds human capacities while simultaneously maintaining a genuine perception of the limitations of the human being, human spirit, and human mind. Existential therapy shares many similarities with humanistic psychology, experiential psychotherapy, depth psychotherapy, and relational psychotherapy. Existential therapists believed drives in Freudian psychology, conditioning in behaviorism, and archetypes in Jungianism all had their own significance. Existential psychotherapy is not a specific technical approach that presents a new set of rules for therapy. It asks deep questions about the nature of the human being and the nature of anxiety, despair, grief, loneliness, isolation, and anomie. It also deals centrally with the questions of creativity and love. Out of the understanding of the meaning of these human experiences, existential psychotherapists have devised methods of therapy that do not fall into the common error of distorting human beings in the very effort of trying to help them.

Rollo Reece May
(April 21, 1909 – October 22, 1994)

Dr. Viktor Emil Frankl
(March 26, 1905 – September 2, 1997)

Basic Philosophy

- Existentialists believe that the individual writes their own life story by the choices that they make.
- Psychopathology is defined by existentialists as neglecting to make meaningful choices and accentuating one's potential.
- Anxiety is seen as the motivational force that helps the clients to reach their potential. Conversely, anxiety is also seen as the paralyzing force that prevents clients from reaching their full potential. Therefore, through awareness, this anxiety can be helpful in living more fully.
- Frankl shares that each person searches for meaning in life, and that while this meaning may change, the meaning never ceases to be.
- According to Frankl life's meaning can be discovered in three ways -
  i. by doing a deed (accomplishments or achievements),
  ii. by experiencing a value (beauty, love, nature, and arts),
  iii. by suffering (reconciling ourselves to fate).

Key Concepts: Humans are in a constant state of transition, emerging, evolving, and becoming. Being a person implies that we are discovering and making sense of our existence. Although the specific questions we raise vary in accordance with our developmental stage in life, the fundamental themes do not vary - who am I?; what can I know?; what ought I to do?; what can I hope for?; where am I going? The basic dimensions of the human condition include -

1. The capacity for self-awareness;
2. Freedom and responsibility;
3. Creating one's identity and establishing meaningful relationships with others;
4. The search for meaning, purpose, values, and goals;
5. Anxiety as a condition of living; and
6. Awareness of death and nonbeing.

Capacity for Self-Awareness: As human beings, we can reflect and make choices because we are capable of self-awareness. The greater our awareness, the greater our possibilities for freedom. We have the potential to take action or not to act; inaction is a decision. We choose our actions, and therefore we can partially create our own destiny. Existential anxiety, which is basically a consciousness of our own freedom, is an essential part of living; as we increase our awareness of the choices available to us, we also increase our sense of responsibility for the consequences of these choices. We are subject to loneliness, meaninglessness, emptiness, guilt, and isolation. We can choose either to expand or to restrict our consciousness. Self-awareness is at the root of most other human capacities, the decision to expand it is fundamental to human growth.

Freedom and Responsibility: Even though we have no choice about being thrust into the world, the manner in which we live and what we become are the results of our choices. Because of the reality of this essential freedom, we must accept responsibility for directing our lives. However, it is possible to avoid this responsibility for directing our lives, or exhibiting 'bad faith'. We are constantly confronted with the choice of what kind of person we are becoming, and to exist is never to be finished with this kind of choosing. Existential guilt is being aware of having evaded a commitment, or having chosen not to choose.

Striving for Identity and Relationship to Others: As human beings we strive for discover or create a self - that is our personal identity and we strive for connectedness with others and with nature. Rather than trusting ourselves to search within and find our own answers to the conflicts in our life, we sell out by becoming what others expect of us. Our being becomes rooted in the expectations, answers, values, and beliefs that come from the important people in our world.

Search for Meaning: Some of the underlying conflicts that bring people into counseling and therapy are centered in existential questions such as - "Why am I here? What do I want from life? What gives my life purpose? Where is the source of meaning for me in life?" Existential therapy can provide the conceptual framework for helping clients challenge the meaning in their lives. Questions that the therapist might ask are - "Do you like the direction of your life? Are you pleased about whom you are and what you want for yourself, what are you doing to get some clarity?"

Anxiety as a Condition of Living: Existential anxiety is conceptualized as the unavoidable result of being confronted with the 'givens of existence' - death, freedom, existential isolation, and meaninglessness. Existential therapists differentiate between normal and neurotic anxiety, and they see anxiety as a potential source of growth. Normal anxiety is an appropriate response to an event being faced. It can be used as a motivation to change. Neurotic anxiety, in contrast, is out of proportion to the situation. It is typically out of awareness, and it tends to immobilize the person. Being psychologically healthy entails living with as little neurotic anxiety as possible, while accepting and struggling with the unavoidable existential anxiety that is a part of living. When we make a decision that involves reconstruction of our life, the accompanying anxiety can be a signal that we are ready for personal change. Opening up to new life means opening up to anxiety. Existential therapy helps clients come to terms with the paradoxes of existence - life and death, success and failure, freedom and limitations, and certainty and doubt. Facing existential anxiety involves viewing life as an adventure rather than hiding behind securities that seem to offer protection. The therapist must guide clients in finding ways to deal with anxiety constructively. The therapist can help clients recognize that learning how to tolerate ambiguity and uncertainty and how to live without props can be a necessary phase in the journey from dependence to autonomy. When a client becomes more self-confident, the anxiety that results from an expectation of catastrophe will decrease.

Awareness of Death and Nonbeing: Awareness of death is a basic human condition that gives significance to living. It is necessary to think about death if we are to think significantly about life. Death provides the motivation for us to live our lives fully and take advantage of each opportunity to do something meaningful. If we realize we are mortal, we know that we do not have an eternity to complete our projects and that each present moment is crucial. Without being morbidly preoccupied by the ever-present threat of nonbeing, clients can develop a healthy awareness of death as a way to evaluate how well they are living and what changes they want to make in their lives.

Role of the Counselor

- Each client is considered a unique relationship with the counselor focusing on being authentic with the client and entering into a deep personal sharing relationship.
- The counselor models how to be authentic, to realize personal potential, and to make decisions with emphasis on mutuality, wholeness and growth.
- Existential counselors do not diagnosis, nor do they use assessment models like the DSM-IV.

Goals of Therapy: The goals of existential therapy are to enable people to -

- have the clients take responsibility for their life and life decisions.
- develop self-awareness to promote potential, freedom, and commitment to better life choices.
- help the client develop an internal frame of reference, as opposed to the outward one.
- identify factors that block freedom.
- successfully negotiate and come to terms with past, present and future crises.
- become more truthful with themselves.
- find clarity on what their purpose in life is and how they can learn from the past to create something valuable and meaningful to live for.
- understand themselves and others better and find ways of effectively communicating and being with others.

Therapeutic Relationship: The therapist's main tasks are to accurately grasp clients, being in the world and to establish a personal and authentic encounter with them. The relationship is seen as critically important. Clients discover their own uniqueness in the relationship with the therapist. The immediacy of the client/therapist relationship and the authenticity of the here-and-now encounter are stressed. Both client and therapist can be changed by the encounter.

Techniques of Therapy: Few techniques flow from this approach, because it stresses understanding first and technique second. The therapist can borrow techniques from other approaches and incorporate them in an existential framework. Diagnosis, testing, and external measurements are not deemed important. The most common technique used in existential counseling is the relationship with the client. Confrontation is also used by existential counselors, when they challenge the clients with their own responsibility for their lives.

Therapeutic Procedures: The therapy is a creative, evolving process of discovery that can be conceptualized in three general phases.

- The initial phase, counselors assist clients in identifying and clarifying their assumptions about the world. Clients are invited to define and question the ways in which they perceive and make sense of their existence. The counselor teaches them how to reflect on their own existence and to examine their role in creating their problems in living.
- The middle phase, clients are encouraged to more fully examine the source and authority of their present value system. This leads to new insights and some restructuring of their values and attitudes. Clients get a better idea of what kind of life they consider worthy to live and develop a clearer sense of their internal valuing process.
- The final phase focuses on helping clients take what they are learning about themselves and put it into action, to find ways of implementing their examined and internalized values in a concrete way. Clients typically discover their strengths and find ways to put them to the service of living a purposeful existence.

Applications: Existential therapy is especially appropriate for clients who are struggling with developmental crises, doing grief work, confronting death, or facing a significant decision. Some examples of these critical turning points that mark passages from one stage of life into another are the struggle for identity in adolescence, coping with possible disappointments in middle age, adjusting to children leaving home, coping with failures in marriage and work, and dealing with increased physical limitations as one ages. It can be useful for people who are on the edge of existence, such as those who are dying, who are working through developmental or situational crises, or who are starting a new phase of life. Can be applied to both - individual and group counseling, marital and family therapy, crisis intervention, and community mental health work.

Multiculturalism: Focus is on understanding client's phenomenological world, including cultural background. This approach leads to empowerment in an oppressive society. It can help clients examine their options for change within the context of their cultural realities, values of individuality, freedom, autonomy, and self-realization, often conflict with cultural values of collectivism, respect for tradition, deference to authority, and interdependence. Some may be deterred by the absence of specific techniques. Others will expect more focus on surviving in their world.

Contributions: Its major contribution is recognition of the need for a subjective approach based on a complete view of the human condition. It calls attention to the need for a philosophical statement on what it means to be a person. Stress on the 'I/Thou' relationship lessens the chances of dehumanizing therapy. It provides a

perspective for understanding anxiety, guilt, freedom, death, isolation, and commitment.

Limitations: The existentialists can be criticized on the grounds that they are excessively individualistic and they ignore the social factors that cause human problems. If a counselor consistently tells certain disenfranchised clients that they have a choice in making their lives better, they may feel patronized and misunderstood. It is highly focused on the philosophical assumption of self-determination, which does not take into account the complex factors that many people who have been oppressed must deal with. In many cultures it is not possible to talk about the self and self-determination apart from the context of the social network and environmental conditions. Another problem is the lack of direction that clients may get from the counselor. Many clients expect a structured and problem-oriented approach to counseling that is not found in the existential approach, which places the responsibility on the client for providing the direction of therapy.

Existential psychotherapy perceives the patient as an existing, immediate person, not as a composite of drives, archetypes, or conditioning. Instinctual drives and history are obviously present, but they come into existential therapy only as parts of the living, struggling, feeling, thinking human being in unique conflicts and with hopes, fears, and relationships. Existential therapy emphasizes that normal anxiety and guilt are present in all of life and that only the neurotic forms of these need to be changed in therapy. The person can be freed from neurotic anxiety and guilt only as s/he recognizes normal anxiety and guilt at the same time. The original criticism of existential therapy as 'too philosophical' has lessened as people recognize that all effective psychotherapy has philosophical implications. Existential therapy is concerned with the 'I-Am' (being) experience, the culture (world) in which a patient lives, the significance of time, and the aspect of consciousness called transcendence. Existential therapy always sees the patient in the center of his/her own culture. Most people's problems are now loneliness, isolation, and alienation. Our present age is one of disintegration of cultural and historical mores, of love and marriage, the family, the inherited religions, and so forth. This disintegration is the reason psychotherapy of all sorts has burgeoned in the twentieth century; people cried for help for their multitudinous problems. Thus, the existential emphasis on different aspects of the world (environment, social world, and subjective world) will, in all likelihood, become increasingly important. It is predicted that the existential approach in therapy will then become more widely used.

OTHER CONTEMPORARY THERAPY

- ADLERIAN THERAPY/ INDIVIDUAL PSYCHOLOGY
- REALITY THERAPY
- FAMILY SYSTEMS THERAPY
- FEMINIST THERAPY
- EMDR
- ECLECTIC COUNSELING
- INTEGRATIVE COUNSELING
- GROUP THERAPY

ADLERIAN THERAPY/ INDIVIDUAL PSYCHOLOGY

Adlerian Therapy, originated by Alfred Adler, is also called individual psychology and focuses on creating a therapeutic relationship that is co-operative, encouraging and practical. Alfred Adler was a major contributor to the development of the psychodynamic approach to therapy. After 8-10 years of collaboration, Freud and Adler parted company, with Freud taking the position that Adler was a heretic who had deserted him. Adler founded the Society for Individual Psychology in 1912.

Alfred W. Adler
(February 7, 1870 - May 28, 1937)

Other Freudian revisionists included Karen Horney, Erich Fromm, and Harry Stack Sullivan (neo-Freudians) agreed that social and cultural factors were of great significance in shaping personality. They are called neo-Freudian because they moved away from Freud's biological and deterministic point of view toward Adler's social-psychological and goal-oriented view of human nature. After Adler's death in 1937, Rudolf Dreikurs was the most significant figure in bringing Adlerian psychology. Dreikurs is credited with the idea of child guidance centers and training professionals to work with a wide range of clients. Adlerian counselors help clients look at their lifestyle and personal values to help them understand and question their usual patterns of behavior and hidden goals. It is a learning process that assists the client to move towards useful involvement and contribution to society.

Basic Philosophy

- Adlerian concept of social interest is the individual's feeling of being part of a whole, spanning the past, present, and the future. Adler believed that people were mainly motivated toward this feeling of belonging. He did not believe that social interest was innate but rather a result of social training.
- Adler expressed that people strove to become successful and overcome the areas that they perceived as inferior. He referred to this process of personal growth as

striving for perfection. Those who did not overcome feelings of inferiority developed an inferiority complex. Those who overcompensated for feelings of inferiority developed a superiority complex.

- Adler believed that a person's conscious behavior, not their unconscious, was the mainstay of personality development. Because of this concept, Adlerian theory emphasizes personal responsibility for how the individual chooses to interpret and adjust to life's events or situations.
- Maladjustment is defined in Adlerian theory as choosing behavior resulting in a lack of social interest or personal growth. Adler believed that misbehavior would take place when the person had become discouraged or when positive attempts at good behavior had failed to get the needed results. Encouragement to good behavior was often the recommended antidote to misbehavior.
- Another concept is that of teleology, which simply put means that a person is as influenced by future goals as by past experiences.
- Adlerian espoused the belief that the birth of each child changed the family substantially. He thought that the birth order of the children in the family influenced many aspects of their personality development. Briefly, characteristics of these birth positions are -

  Oldest children are usually high achievers, parent pleasers, conforming, and are well behaved. *Second born* children are more outgoing, less anxious, and less constrained by rules than first born. They usually excel as what the first born does not. *Middle children* have a feeling of being squeezed in and are concerned with perceived unfair treatment. These children learn to excel in family politics and negotiation. However, they can become very manipulative. This position also tends to develop areas of success that are not enjoyed by their siblings. *Youngest child* is the most apt at pleasing or entertaining the family. While they run the risk of being spoiled, they are also the most apt at getting what they want through their social skills and ability to please. They are often high achievers, because of the role models of their older siblings.

  Only children or children born seven or more years apart from siblings are more like first born children. Children with no siblings often take on the characteristics of their parents' birth order, as the parents are the only role models. While these children may mature early and be high achievers, they may lack socialization skills, expect pampering, and be selfish.
- Adler saw the family as the basic socialization unit for the child. He believed that children's interpretation of the events in their life was determined by the interaction with family members before the age of five. The family interactions

taught the children to perceive events and situations through certain subjective evaluations of themselves and the environment. These perceptions that guided the children's behavior were called fictions. Basic mistakes could be made based on these fictions. Adlerians believe that some of those mistakes are (Mozak, 1984) -

- Over-generalizing in which the individual believes that everything is the same or alike.
- False or impossible goals of security which leads the individual to try to please everyone in seeking security and avoiding danger.
- Misperception of life and life's demands which leads the individual to expect more accommodation than is reasonable and to interpret their failure to get accommodation as never getting any breaks.
- Minimization or denial of one's worth results in the individual believing that they cannot be successful in life.
- Faulty values results in a 'me first' mentality with little or no regard for others.

❖ Adler believed that life took courage or a willingness to take risks without knowing the outcome. He believed that a person with a healthy life style contributed to society, had meaningful work, and had intimate relationships. He espoused cooperation between the genders as opposed to competition. He believed that well adjusted people lived in an interdependent relationship with others in a cooperative spirit.

Key Concepts: It stresses the unity of personality, the need to view people from their subjective perspective, and the importance of life goals that give direction to behavior. People are motivated by social interest and by finding goals to give life meaning. Other key concepts are striving for significance and superiority, developing a unique lifestyle, and understanding the family constellation. Therapy is a matter of providing encouragement and assisting clients in changing their cognitive perspective.

Role of the Counselor

- The counselor is as a diagnostician, teacher and model.
- The counselor helps the client to explore conscious thoughts, beliefs and logic for behaviors that are not in the client's best interest or social interest.
- The counselor encourages the client to behave 'as if' the client were who they wished to be and often provides the client with 'homework' assignments outside the sessions.

Goals of Therapy

- To challenge client's basic premises and life goals.

- To offer encouragement so individuals can develop socially useful goals.
- To develop the client's sense of belonging.

Therapeutic Relationship: The emphasis is on - joint responsibility, mutually determining goals, mutual trust and respect, and equality. A cooperative relationship is manifested by a therapeutic contract. Focus is on identifying, exploring, and disclosing mistaken goals and faulty assumptions within the person's lifestyle.

Techniques of Therapy: Adlerians pay more attention to the subjective experiences of clients than to using techniques. Some techniques include gathering life-history data (family constellation, early recollections, personal priorities), sharing interpretations with clients, offering encouragement, and assisting clients in searching for new possibilities.

- Most commonly used are establishing *rapport*, *defining style of life* and *helping the client to gain insight*.
- *Confrontation* consists of challenging the client's private logic and behavior.
- Asking '*the Question*' consists of asking the client how their life would be different if they were well. The question often asked to parents is what would be the problem if this child were not the problem?
- *Encouragement* consists of the counselor supporting the client by stating the belief in the client's ability to take responsibility and change behavior.
- *Acting 'as if'* consists of instructing the client to behave 'as if' there were no problem or as the person that the client would like to be.
- *Spiting in the Client's soup* means that the counselor points out the purpose of the client's behavior. Afterward, the client may continue the behavior, but cannot do so without being aware of their motivation for engaging in the behavior.
- *Catching oneself* consist of helping the client learn to bring destructive behavior into awareness and stop it.
- *Task setting* consists of helping the client set short-term goals leading toward the attainment of long-term goals.

Therapeutic Procedures

Four phases not linear; and do not progress in rigid steps; can best be understood as a weaving that leads to a tapestry -

1. Establishing the proper therapeutic relationship.
2. Exploring the psychological dynamics operating in the client (an assessment).
3. Encouraging the development of self-understanding (insight into purpose).
4. Helping the client make new choices (reorientation and reeducation).

Establishing the Relationship: Working with clients is based on a sense of deep caring, involvement, and friendship. Adlerian therapists seek to make person-to-person contact with clients rather than

starting with 'the problem'. The initial focus should be on the person, not the problem. Therapists start by helping clients become aware of their assets and strengths, rather than dealing continually with their deficits and liabilities. A positive relationship is created by listening, responding, demonstrating respect for clients' capacity to understand purpose and seek change, and exhibiting faith, hope and caring. Adlerians pay more attention to the subjective experiences of the client than they do to using techniques. They fit their techniques to the need of each client. During this phase, the main techniques used are attending and listening with empathy, following the subjective experience of the client as closely as possible, identifying and clarifying goals, and suggesting initial hunches about purpose in client symptoms and actions. They attempt to access the core patterns in the client's life.

Exploring the Individual's Dynamics: The second phase consists of two interview forms - the subjective interview and the objective interview. In subjective interview the counselor helps the client to tell their story as completely as possible. What the clients says will spark an interest in the counselor and lead, to the next most significant question about the client. The Adlerian counselor is listening for clues to the purposive aspects of the client's coping and approaches to life. The subjective interview should extract patterns in the person's life, develop hypotheses about what works for the person, and determine what accounts for the various concerns in the client's life. Adlerians end a subjective interview with this question, "How would your life be different, and what would you do differently, if you did not have this symptom or problem?" The objective interview - (a) how problems in the client's life began; (b) any precipitating events; (c) a medical history; (d) a social history; (e) the reasons the client chose therapy at this time; (f) the person's coping with life tasks; (g) a lifestyle assessment.

The family constellation - Factors such as cultural and familial values, gender-role expectations, and the nature of interpersonal relationships are all influenced by a child's observation of the interactional patterns within the family. Assessment includes an exploration of the client's family constellation, including the client's evaluation of conditions that prevailed in the family when the person was a young child, birth order, parental relationship, family values, and extended family and culture. Early recollections - Asking the client to provide their earliest memories, including the age of the person at the time of the remembered events and the feelings or reactions associated with it. Early memories cast light on the 'story of our life', because they represent metaphors for our current views. The counselor might say "I would like to hear about your early memories. Think back to when you were very young, as early as you can remember, and tell me something that happened one time". The counselor will then ask - "What part stands out to you? If you played the whole memory like a movie and stopped it at one frame, what would be happening? Putting yourself in that moment, what are you feeling? What's your reaction? Three memories are usually considered a minimum to assess a pattern. Personality priorities - people rely on a number-one priority, a first line of defense that they use as an immediate response to perceived stress or difficulty (Kefir, 1981). Each priority involves a dominant behavior pattern with supporting convictions that an individual uses to cope. (a) Superiority - striving for significance through leadership, or accomplishment; (b) Control - looking for guarantees against ridicule - a need for complete mastery of situations; (c) Comfort - wanting to avoid stress or pain at all costs; (d) Please - avoiding rejection by seeking constant approval and acceptance. Integration - Integrated summaries of the data are developed and presented to the client for discussion. Mosak (2000) common basic mistakes - (1) Overgeneralizations; (2) False or impossible goals; (3) Misperceptions of life and life's demands; (4) Minimization or denial of one's basic worth; and (5) Faulty values.

Encouraging Self-understanding and Insight: Adlerians believe almost everything in human life has a purpose. When speaking of insight, Adlerians are referring to an understanding of the motivations that

operate in a client's life - a special form of awareness that facilitates a meaningful understanding within the therapeutic relationship and acts as a foundation for change. Disclosure and well-times interpretations are techniques that facilitate the process of gaining insight. They are hunches or guesses and they are often stated - "It seems to me that . . ."; "Could it be that . . ." or "This is how it appears to me . . ."

Helping with Reorientation: The action-oriented phase helping people discover new and more functional alternatives. People need to be reoriented to the useful side of life. The useful side of life involves a sense of belonging and being values, having an interest in others and their welfare, courage, the acceptance of imperfection, confidence, a sense of humor, a willingness to contribute, and an outgoing friendliness. Encouragement is the most distinctive Adlerian procedure and is central to all phases of counseling and therapy. As a part of the encouragement process, Adlerians use a variety of cognitive, behavioral, and experiential techniques to help clients identify and challenge self-defeating cognitions, generate perceptional alternatives, and make use of assets, strengths, and resources. Clients are encouraged to 'as if' they were the people they want to be. Clients are asked to catch themselves in the process of repeating old patterns. Commitment is an essential part of this process. Adlerian practitioners focus on motivation modification more than behavior change and encourage clients to make holistic changes on the useful side of living.

Applications: Because the approach is based on a growth model, it is applicable to such varied spheres of life as child guidance, parent/child counseling, marital and family therapy, individual counseling with all age groups, correctional and rehabilitation counseling, group counseling, substance abuse programs, and brief counseling. It is ideally suited to preventive care and alleviating a broad range of conditions that interfere with growth.

Multiculturalism: Its focus on social interest, collectivism, pursuing meaning in life, importance of family, goal orientation, and belonging is congruent with many cultures. Focus on person-in-environment allows for cultural factors to be explored. Some clients may view the counselor as an authority who will provide answers to problems, which conflicts with the egalitarian, person-to-person spirit as a way to reduce social distance.

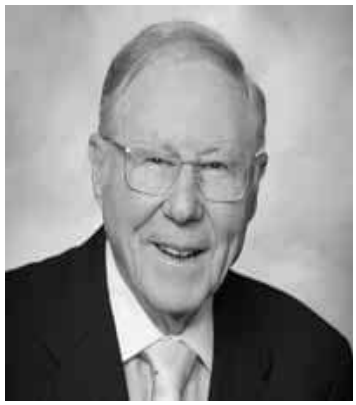
Contributions: One of the first approaches to therapy that was humanistic, unified, holistic, and goal oriented and that put an emphasis on social and psychological factors. A key contribution is the influence that Adlerian concepts have had on other systems and the integration of these concepts into various contemporary therapies.

Limitations: Weak in terms of precision, testability, and empirical validity. Few attempts have been made to validate the basic concepts by scientific methods. Tends to oversimplify some complex human problems and is based heavily on common sense.

REALITY THERAPY

Reality Therapy (RT) is based on a concept called 'Choice Theory' (originally called Control Theory). RT is a particular approach in psychotherapy and counseling. Reality therapy was developed by William Glasser, a psychiatrist in the 1960s.

Glasser believes that people who are behaving in inappropriate ways do not need help to find a defense for their behavior. Instead, they need help to acknowledge their behavior as being inappropriate and then to learn how to act in a more logical and productive manner. Reality therapy attempts to help people control the world around them more effectively so that they are better able to satisfy their needs. RT approach to counseling and problem-solving focuses on the here-and-now of the client and how to create a better future, instead of concentrating at length on the past.



Dr. William Glasser
(May 11, 1925 – August 23, 2013)

It emphasizes making decisions, and taking action and control of one's own life. Typically, clients seek to discover what they really want and whether what they are currently doing (how they are choosing to behave) is actually bringing them nearer to, or further away from, that goal. Reality Therapy is a considered a cognitive-behavioral approach to therapy; that is, it focuses on facilitating the client to become aware of, and if necessary, change, his/her thoughts and actions.

William Glasser maintains that people act on a conscious level and that they are not driven by instincts and the unconscious. Glasser believes that there is a health/growth forces in every person that seeks both physical and psychological health/growth. He separates these into the old brain or primitive physical needs and into the new brain or psychological needs. While the old brain contents itself with maintaining life, the new brain seeks belonging, power, freedom and fun. Glasser believes that identity or a healthy sense of self is necessary. A success identity comes from being loved and accepted. A failure identity comes from not having needs for acceptance, love and worth met. A person must experience identity before they can perform a task. Glasser espouses two critical periods of development in children -

- Children ages 2 to 5 first learn socialization and learn to deal with frustrations and disappointments. Children not getting support and love from their parents during this critical time begin to establish a failure identity.

- The second critical period is between 5 and 10 years or the early school years. Children who have socialization or academic problems may establish a failure identity.

Glasser suggests that human learning is a lifelong process; therefore, one can change one's identity at any time in one's personal history by learning what needs to be learned. Glasser believes that humans are self-determined. He believes that each person has within themselves a picture or perception of themselves. Each person then behaves in a way that is determined or controlled by this image of self so that the self image can be maintained. This control theory has three parts - 'B' is the behavior. 'C' is the control, where a comparison is made between the desired image and the image the behavior is producing. 'P' is the perception or the development of the image.

Basic Philosophy

- Humans are self-determining.
- Humans change when they determine their behavior is not getting them what they want & when they believe they have choices of behaviors.
- People choose total behavior, hence responsible for acting, thinking, feeling, and psychological states.
- Behavior is aimed to satisfy needs for survival, love and belonging, power, freedom, and fun.
- Acting and thinking are chosen behaviors (changing acting and thinking influences feelings and physiological states).
- Choice theory explains how people attempt to control their world and teaches ways to satisfy needs and wants in effective manners.

Key Concepts: The basic focus is on what clients are doing and how to get them to evaluate whether their present actions are working for them. People create their feelings by the choices they make and by what they do. The approach rejects the medical model, the notion of transference, the unconscious, and dwelling on one's past. The key terms are -

- Autonomy - state of accepting responsibility and taking control of self (life).
- Commitment - not skewing from the plan for change.
- Choice Theory - humans are internally motivated and have a purpose to control behaviors; humans are self-determined and create own destiny (all elements of behavior are interrelated).
- Cycle of Counseling - any means to create a positive environment in counseling, based on personal involvement and procedures for change to occur.
- Involvement - counselor's interest in and caring for clients.

- Paining Behaviors - people choose misery by developing symptoms (headache, depression, anxiety) because at the time they seem to be the best behaviors to execute for survival.
- Perceived World - the reality we experience and interpret.
- Psychological Needs - needs for belonging, power, freedom, and fun that drive people and explain behaviors.
- Quality World - perceptions and images we have to fulfill our basic psychological needs.
- Responsibility - satisfying personal needs while not interfering with people who fulfill their needs.
- Self-Evaluation - clients' assessment of current behaviors in order to determine if their behaviors are working and if behaviors are meeting their needs.
- Total Behavior - integrated components of doing, thinking, feeling, and physiology.
- WDEP System - procedures that are applied to the practice of reality therapy groups; strategies help clients identify their wants; determine direction behaviors take them, self-evaluations, and designs plans for change. Where, W = wants, explore wants, needs and perceptions; D = direction and doing, focus on what clients are doing and the direction they are headed; E = evaluation, challenge clients to evaluate their total behavior (continual basis); P = planning and commitment, assist in the formulation of realistic plans and making commitment to carry out plans.

Role of the Counselor

- Counselor is a teacher and a model to the client.
- Counselor creates an atmosphere of acceptance and warmth helping the client focus on the control of displayed thoughts and actions.
- Reality counselor used 'ing' verbs to help the clients describing their thoughts and actions, i.e., angering, bullying, intimidating, excusing.
- Focus of therapy is on the behavior that the client needs or wants to change and how to change that behavior in a positive manner.
- Reality therapy does not generally use formal assessment techniques or diagnostic categories.

Goals of Therapy

- Overall goal - assist clients to better meet their needs for love, belonging, survival, power, freedom, and fun.

- Help clients gain psychological strength, accept personal responsibility, regain control of lives.
- Challenge clients to examine what they are doing, thinking, and feeling.
- Help the client to develop a practical plan to accomplish their personal needs and desires.
- Counselor helps the client put the past behind and focus on the present and the outcome of present behaviors.
- To help people become more effective in meeting their needs.
- Teach client to self-evaluate behaviors and determine what they want to change.

Therapeutic Relationship

- Counselors are involved with clients from outset and create warm, supportive, and challenging relationship (clients must know that counselors are for them).
- Involvement with and concern for are conveyed throughout the counseling process.
- Once trust is established, counselors confront clients with reality and consequences of their actions.
- Counselors avoid - criticism, accepting excuses, and giving up on clients.
- Assist clients to continually assess effectiveness and appropriateness of their behaviors.

Techniques of Therapy: Various techniques may be used to get clients to evaluate what they are presently doing to see if they are willing to change. If they decide that their present behavior is not effective, they develop a specific plan for change and make a commitment to follow through. The main technique has three basic steps -

- Through involvement with the client, the counselor helps the client to see the reality and understand how a behavior is unrealistic.
- The counselor separates the client from the behavior and rejects the behavior without rejecting the client.
- The final step is to teach the client how to fulfill their needs realistically and positively.

- Glasser uses humor to point out absurdity without being sarcastic.
- Glasser uses confrontation to help the client accept responsibility for behavior.
- The eight steps that reality therapy uses to accomplish its goals and techniques -
  1. Establishing a relationship.
  2. Focusing on present behavior.
  3. Client evaluation of his/her behavior.
  4. Developing a contract or plan of action.

5. Getting a commitment from the client.
6. Not accepting excuses.
7. Allowing reasonable consequences but refusing to use punishment.
8. Refusing to give up on the client.

- Glasser's final technique is "Never Give Up".

Applications: It has been applied to -

- Youth offenders (detention facilities);
- Variety of behavioral problems;
- Individual and group counseling & marital and family counseling;
- Military;
- Drug and alcohol clinics;
- Teaching and administration;
- Social work, crisis intervention, and management & community development.

Multiculturalism

- Focus is on members, making own evaluation of behavior (including how they respond to their culture).
- Through personal assessment they can determine the degree to which their needs and wants are being satisfied.
- They can find a balance between retaining their own ethnic identity and integrating some of the values and practices of the dominant society.
- This approach stresses taking charge of one's own life, yet some clients hope to change their external environment.
- Counselor needs to appreciate the role of discrimination and racism and help clients deal with social and political realities.
- Respect difference in worldviews and cultural values.
- Skillful questioning will help assess how ethnic minorities have acculturated.

Contributions

- Short-term approach with wide range of clients.
- Due to the direct methods, it appeals to many clients who are often seen as resistant to therapy.
- Simple and clear concepts, can be used by parents, teachers, educators, nurses, ministers, managers, consultants, supervisors, counselors and social workers.
- Positive, action-oriented approach.
- Accepting personal responsibility and gaining effective control.
- People take charge of their lives.
- Focus for what clients are willing to do in the present

Limitations

- Discounts the therapeutic value of exploring the client's past, dreams, the unconscious, early childhood experiences, and transference.
- The approach is limited to less complex problems. It is a problem-solving therapy that tends to discourage exploration of deeper emotional issues.
- Direct questioning needs to be softened when working with ethnic minorities.
- At times choice is not an option, discrimination and racism are reality.
- Minority clients may be hesitant to state what they need.

FAMILY SYSTEMS THERAPY

A psychiatrist and theorician Murray Bowen produced a remarkable new theory of human behavior called 'Family Systems Theory', or 'Bowen Theory'. Family systems theory is also referred to as couple therapy, family therapy, and marriage therapy. This theory has the potential to replace most of Freudian theory and to radically change treatment approaches, not only in psychiatry, but in all of medicine. Potential applications of this therapy extend beyond the human family to non-family groups, including large organizations and society as a whole.

Dr. Murray Bowen
(January 31, 1913 – October 9, 1990)

Family therapy is a type of psychotherapy that works with families and couples. It helps families or individuals within a family, understand and improve the way family members interact with each other and resolve conflicts. It emphasizes family relationships as an important factor in psychological health. Family therapists tend to focus on what goes between people; that is, on how patterns of interaction within the family may foster or maintain the problem. A family therapist can be a member of any number of health professions. They may be psychologists, psychiatrists, social workers, nurses or simply people who have been trained in the use of one or more types of family therapy. Family therapy can take many different forms with the therapist using a variety of different approaches including supportive counseling, cognitive-behavioral techniques, psychodynamic techniques, or what is known as a systemic approach. Most practitioners are 'eclectic', using techniques from several areas, depending upon the needs of the client.

Basic Philosophy

- The family is viewed from an interactive and systemic perspective.

- Human functioning is governed by principles common to all life forms.
- Human life is guided primarily by emotional, instinctive forces that can be regulated to some extent by an individual's ability to think. Human emotional forces and behavior are influenced by the interplay among instinctual, biological, genetic, psychological, and sociological factors.
- The family is a single emotional unit made up of interlocking relationships existing over many generations.
- An individual's behavior across the life span is best understood as being closely related to the functioning of her/his family of origin.
- When the individual is seen in her/his family context, we can move beyond simplistic cause-and-effect thinking to a more comprehensive understanding of the multiple factors that interact across time to produce problems or symptoms.
- People are able to modify their automatic, emotional responses by studying patterns of automatic behavior in themselves and their family of origin, and developing greater objectivity in their appraisal of situations and their felt responses.
- Clients are connected to a living system; a change in one part of the system will result in a change in other parts.

Key Concepts

- Focus is on communication patterns within a family, both verbal and nonverbal.
- Problems in relationships are likely to be passed on from generation to generation.
- Symptoms are viewed as ways of communicating with the aim of controlling other family members.
- Key concepts vary depending on specific orientation but include differentiation, triangles, power coalitions, family-of-origin dynamics, functional versus dysfunctional interaction patterns, family rules governing communication, and dealing with here-and-now interactions.
- The present is more important than exploring past experiences.

Goals of Therapy: Most approaches are aimed at helping family members gain awareness of patterns of relationships that are not working well and create new ways of interacting to relieve their distress. Some approaches focus on resolving the specific problem that brings the family to therapy.

Therapeutic Relationship: The family therapist functions as a teacher, coach, model, and consultant. The family learns ways to detect and solve problems that are keeping members stuck, and it learns about patterns that have been transmitted from generation to generation. Some approaches focus on the role of therapist as expert;

others concentrate on intensifying what is going on in the here and now of the family session. All family therapists are concerned with the process of family interaction and teaching patterns of communication.

Techniques of Therapy: There is a diversity of techniques, depending on the particular theoretic orientation. Interventions may target behavior change, perceptual change, or both. Techniques include using genograms, teaching, asking questions, family sculpting, joining the family, tracking sequences, issuing directives, anchoring, use of counter-transference, family mapping, refraining, paradoxical interventions, restructuring, enactments, and setting boundaries. Techniques may be experiential, cognitive, or behavioral in nature. Most are designed to bring about change in a short time.

Applications: Applications vary depending on the particular approach to family therapy. Family therapy is frequently short-term and may be used in addition to other types of treatment, particularly for certain psychological disorders that require more in-depth treatment. Useful for dealing with -

- Marital distress;
- Problems of communicating among family members;
- Power struggles;
- Parenting issues;
- Crisis situations in the family;
- Child & adolescent behavior;
- Self-harm, drug, alcohol & other substance misuse;
- Supporting family members through separation, mediation and divorce;
- Emotional disorders including anxiety, depression, loss and grief;
- Anorexia, bulimia and other eating disorders;
- Supporting family members in step-family life;
- Emotional abuse or violence;
- Helping individuals attain their potential, and
- Enhancing the overall functioning of the family.

Multiculturalism: Many ethnic and cultural groups place value on the role of the extended family. Many family therapies deal with extended family members and with support systems. Networking is a part of the process, which is congruent with the values of many clients. There is a greater chance for individual change if other family members are supportive. This approach offers ways of working toward the health of the family unit and the welfare of each member. Some approaches are based on value assumptions that are not congruent with the values of

clients from other cultures. Concepts such as individuation, self-actualization, self-determination, independence, and self-expression may be foreign to some clients. In some cultures, admitting problems within the family is shameful. The value of keeping problems within the family may make it difficult to explore conflicts openly.

Contributions: In all of the systemic approaches, neither the individual nor the family is blames for a particular dysfunction. The family is empowered through the process of identifying and exploring interactional patterns. Working with an entire unit provides a new perspective on understanding and working through both individual problems and relationship concerns. By exploring one's family of origin, there are increased opportunities to resolve other relationship conflicts outside of the family.

Limitations

- Limitations include problems in being able to involve all the members of a family in the therapy.
- Members may be resistant to changing the structure of the system.
- Therapists' self-knowledge and willingness to work on their own family-of-origin issues is crucial, for the potential of counter-transference is high.
- It is essential that the therapist be well trained, receive quality supervision, and be competent in assessing and treating individuals in a family context.

FEMINIST THERAPY

Feminist Therapy derived from radical women uniting for change in the 1960s during the Feminist Movement. It was developed by a numbers of feminist therapists - Jean Baker Miller (1927 - 2006), Carol Gilligan (1936 -), Carolyn Enns (1946 -) and others. Feminist therapy grew out of concerns that established therapies were not helping women. Specific concerns of feminist therapists included gender bias and stereotyping in therapy; blaming victims of physical abuse and sexual abuse; and the assumption of a traditional nuclear family. In feminist therapy, gender is the primary focus; therapists must understand and be sensitive to how psychological oppression and socialization influences identity development. Feminist therapy also focuses on women's empowerment; communication, assertiveness, self-esteem and helping women discover how to break free from some of the traditional roles that they may feel are hindering their growth and development.

Feminist therapy also focuses on making gender issues transparent enough that women are both empowered and changed by engaging in actively understanding how certain issues affect their behavior (Mahaney, 2007). According to Corey (2008), feminist therapists understand that the client and the therapist are not equally yoked, but when the therapist gives the client a sense of control they are best able to understand their situation. The main goal of feminist therapy is not just to change the individual's situation or mindset, but also to form a revolution that changes the way society views gender issues.

Feminist therapy strives to prevent psychological distresses and inappropriate behavior in both genders. Also, feminist therapy attempts to remedy the stereotypes derived from the traditional views on gender which can affect one's identity.

Basic Philosophy

- Feminists criticize many traditional theories to the degree that they are based on gender-biased concepts and practices of being - androcentric, gendercentric, ethnocentric, heterosexist, and intrapsychic.
- This therapy is gender-free, flexible, interactionist, and life-span-oriented.
- Eight basis philosophies (separate views on social oppression to determine how to transform society. Each philosophy focuses on diversity, gender issues, and sexism) - Liberal; Cultural; Radical; Socialist; Postmodern; Women of color; Lesbian; and Global/International.
- Gender is central focus; therapists must understand and be sensitive to how psychological oppression and socialization influence identity development.

Key Concepts: Core principles that form the foundation for practice of feminist therapy are the personal is political, the counseling relationship is egalitarian, women's experiences are honored, definitions of distress and psychological illness are reformulated, emphasis on gender equality, and commitment to confronting oppression on any grounds.

Key Concepts
Androcentric theory - a theory that uses male-oriented constructs to draw conclusions about human nature.
Egalitarian relationship - relationship where power is balanced.

Ethnocentrism - believing that one's cultural group is superior to others, other groups are based on one's personal standards.
Gendercentrism - two developmental paths for men and women.
Gender-neutral theory - explains the differences between men's and women's behavior in terms of socialization.
Gender-role analysis - helps clients understand the impact of gender-role expectations.
Gender-role intervention - provides insight into ways how social issues affect problems.
Gender schema - psychological associations, interpretations are based on perceptions about gender.
Heterosexism - heterosexual orientation is normative and desirable; devalues same sex relationships.
Life-span perspective - human development is a lifelong process; personality patterns and behavior changes occur at any time during development.
Personal is political - client problems have social and political causes; therapists focus on helping clients change behaviors and become active participants to transform society.
Power analysis - emphasizes power differences between men and women in society; clients recognize the different types of power they possess and power is exercised.
Reframing - therapist changes the frame of reference for looking at behaviors; shift from intrapersonal (blame the victim) stance to consideration of societal factors that contribute to one's problems.
Relabeling - changing the label/evaluation that has been applied to clients' behaviors; shift is from negative to positive evaluation.
Relational-cultural theory - a perspective that suggests a woman's identity and self-concept develop in the context of relationships.
Self-in-relation - women's sense of self is largely dependent on they connect with others.
White privilege - invisible package of unearned assets White people enjoy are not offered to people of color.

- Societal gender-role messages influence how people view themselves and behave.
- Ethnicity, sexual orientation, and class may be important factors.
- Gender cannot be considered apart from other identity areas (race, ethnicity, sexual orientation, etc.).
- Problematic symptoms are viewed as coping/survival strategies.
- Six interrelated principles -
  - Personal is political;
  - Personal and social identities are interdependent;
  - Definitions of distress and mental illness are reformulated;
  - Use of integrated analysis of oppression;
  - Egalitarian relationship; and
  - Women's perspectives valued.

Goals of Therapy

- To bring about transformation both in the individual client and in society.
- Five specific goals are - Equality; Balancing independence and interdependence; Empowerment; Self-nurturance; and Valuing diversity.

- For individual clients the goal is to assist them in recognizing, claiming, and using their personal power to free themselves from the limitations of gender-role socialization.
- To confront all forms of institutional policies that discriminate on the basis of gender.

Therapeutic Relationship

- The therapeutic relationship is based on empowerment and equalizing power (egalitarianism).
- Therapists actively break down the hierarchy of power and reduce artificial barriers by engaging in appropriate self-disclosure and teaching clients about the therapy process.
- Structure of the relationship models how to responsibly identify and use power.
- There is mutuality, client-therapist authentic connection.
- Therapist demystifies counseling; client is an active partner in assessment and treatment.
- Teach clients to recognize how they define themselves and how they relate to others (gender-role expectations).

Techniques of Therapy: Although techniques from traditional approaches are used, feminist practitioners tend to employ consciousness-raising techniques aimed at helping clients recognize the impact of gender-role socialization on their lives. Other techniques frequently used include gender-role analysis and intervention, power analysis and intervention, bibliotherapy, journal writing, therapist self-disclosure, assertiveness training, reframing and relabeling, cognitive restructuring, identifying and challenging untested beliefs, role playing, psychodramatic methods, group work, and social action.

Applications: Principles and techniques can be applied to a range of therapeutic modalities such as individual therapy, relationship counseling, family therapy, group counseling, and community intervention. The approach can be applied to both women and men with the goal of bringing about empowerment.

Multiculturalism: Both individual change and social transformation are the ultimate goals of therapy. A key contribution is that both the women's movement and the multicultural movement have called attention to the negative impact of discrimination and oppression for both women and men. One criticism is that feminist therapy was developed by White, middle class, heterosexual women. Based on the feminist notions of collaborative relationships, self-determination, and empowerment, therapists need to assess with their clients the price of making significant personal

change. If this assessment is not made, clients in certain cultures may experience isolation as a result of making life changes or of assuming a new role.

Contributions

- Integrate concepts with other theories;
- Gender-free, flexible, interactional, life-span-oriented;
- Positive, egalitarian attitude toward women and men;
- Confront patriarchal systems;
- Emphasizes context of women's lives (not symptoms and behaviors);
- Sensitizes counselors to gendered uses of power in relationships;
- Battered women's shelters, rape-crisis centers, women's health and reproductive health centers;
- Establish policies to lessen discrimination of all types;
- Feminist principles applied to supervision, teaching, consultation, ethics, research, theory building, and the practice of psychotherapy;
- Community building, authentic mutual empathic relationships, sense of social awareness, and social change.

Limitations

- No neutral stance because advocate for change in social structure.
- Counselors may impose values on clients.
- No decision-making for clients.
- Feminist therapy is based on White, middle class, heterosexual women.

EMDR (Eye Movement Desensitization and Reprocessing)

Dr. Francine Shapiro
(February 18, 1948 to present)

Eye Movement Desensitization and Reprocessing (EMDR) is a comprehensive, integrative psychotherapy approach. It contains elements of much effective psychotherapy in structured protocols that are designed to maximize treatment effects. These include psychodynamic, cognitive behavioral, interpersonal, experiential, and body-centered therapies. Shapiro's (2001) Adaptive Information Processing (AIP) model posits that EMDR facilitates the accessing and processing of traumatic memories and other adverse life experience to bring these to an adaptive resolution. More than 20,000 practitioners have been trained to use EMDR since psychologist Francine Shapiro developed the technique in 1989. EMDR appears to be a safe therapy, with no negative side effects. After successful treatment with EMDR,

affective distress is relieved, negative beliefs are reformulated, and physiological arousal is reduced.

What Is EMDR?

Eye Movement Desensitization and Reprocessing is a powerful counseling technique which has been very successful in helping people who suffer from trauma, anxiety, panic, disturbing memories, post traumatic stress and many other emotional problems. Until recently, these conditions were difficult and time-consuming to treat. EMDR is considered a breakthrough technique because of its simplicity and the fact that it can bring quick and lasting relief for most types of emotional distress.

EMDR uses a technique called bilateral stimulation, (using right/left eye movement, or tactile stimulation), which repeatedly activates the opposite sides of the brain, releasing emotional experiences that are "trapped" in the nervous system. This assists the neuro-physiological system (the basis of the mind/body connection) to free itself of blockages and reconnect itself.

What Can You Expect From EMDR?

Although most research into EMDR has examined its use in people with PTSD (Post Traumatic Stress Disorder), EMDR is also used to treat many other psychological problems. They include -

- Injury or Loss of a Loved One
- Car Accident or Work Related Accidents & Injuries
- Witness of Violent Crimes/Post Traumatic Stress
- Test Anxiety/Relational Anxiety
- Trouble Sleeping/Worrying/Phobias/Fears
- Low Self-Esteem/Bad Temper/Panic Attacks
- Physical, Emotional, Verbal & Sexual Abuse/Rape
- Natural Disasters/Fire/Childhood Trauma & Abuse
- Depression/Anger/Sadness/Assault
- Eating Disorders/Performance Anxiety
- High Anxiety/Lack of Motivation
- Memories of a Traumatic Experience
- Fear of Being Alone
- Unrealistic Feelings of Guilt and Shame
- Difficulty in Trusting Others
- Relationship Problems.

The EMDR technique is most effective when used in conjunction with other traditional methods of therapy in treating these and many other emotional disorders. EMDR therapy can help clients replace their anxiety and fear with positive images, emotions and thoughts.

Eight Phases of Treatment

EMDR psychotherapy is an information processing therapy and uses an eight phase approach to address the experiential contributors of a wide range of pathologies. It attends to the past experiences that have set the groundwork for pathology, the current situations that trigger dysfunctional emotions, beliefs and sensations, and the positive experience needed to enhance future adaptive behaviors and psychological health.

Phase 1: History and Treatment Planning: Generally takes 1-2 sessions at the beginning of therapy, and can continue throughout the therapy, especially if new problems are revealed. In the first phase of EMDR treatment, the therapist takes a thorough history of the client and develops a treatment plan. Client and therapist identify possible targets for EMDR processing. These include recent distressing events, current situations that elicit emotional disturbance, related historical incidents, and the development of specific skills and behaviors that will be needed by the client in future situations.

Initial EMDR processing may be directed to childhood events rather than to adult onset stressors or the identified critical incident if the client had a problematic childhood. Clients generally gain insight on their situations, the emotional distress resolves and they start to change their behaviors. The length of treatment depends upon the number of traumas and the age of the problem onset. Generally, those with single event adult onset trauma can be successfully treated in less than 5 hours. Multiple trauma victims may require a longer treatment time.

Phase 2: Preparation: For most clients this will take only 1-4 sessions. For others, with a much traumatized background, or with certain diagnoses, a longer time may be necessary. One of the primary goals of the preparation phase is to establish a relationship of trust between the client and the therapist. During the second phase of treatment, the therapist ensures that the client has several different ways of handling emotional distress. The therapist may teach the client a variety of imagery and stress reduction techniques the client can use during and between sessions. A goal of EMDR is to produce rapid and effective change while the client maintains equilibrium during and between sessions.

During the second phase of treatment, the therapist ensures that the client has adequate methods of handling emotional distress and good coping skills, and that the client is in a relatively stable state. If further stabilization is required, or if additional skills are needed, therapy focuses on providing these. The client is then able to use stress reducing techniques whenever necessary, during or between sessions.

Phase 3: Assessment: Used to access each target in a controlled and standardized way so it can be effectively processed. Processing does not mean talking about it. For a single trauma reprocessing is generally accomplished within 3 sessions. If it takes longer, you should see some improvement within that amount of time. The clinician identifies the aspects of the target to be processed. The first step is for the person to select a specific picture or scene from the target event (which was identified during Phase One) that best represents the memory. Then s/he chooses a statement that expresses a negative self-belief associated with the event. Even if s/he intellectually knows that the statement is false, it is important that s/he focus on it. These negative beliefs are actually verbalizations of the disturbing emotions that still exist. During the Assessment Phase, the person identifies the negative emotions (fear, anger) and physical sensations (tightness in the stomach, cold hands) s/he associates with the target. The client also rates the disturbance using the 0 (no disturbance)-to-10 (the worst feeling you ever had) Subjective Units of Disturbance (SUD) scale. Then, the positive cognition should reflect what is actually appropriate in the present. At this point, the therapist will ask the person to estimate how true s/he feels his/her positive belief is using the 1-to-7 Validity of Cognition (VOC) scale.

Phases One through Three lay the groundwork for the comprehensive treatment and reprocessing of the specific targeted events. Although the eye movements (or taps, or tones) are used during the following three phases, they are only one component of a complex therapy. The use of the step-by-step eight-phase approach allows the experienced, trained EMDR clinician to maximize the treatment effects for the client in a logical and standardized fashion. It also allows both the client and the clinician to monitor the progress during every treatment session.

Phase 4: Desensitization: This phase focuses on the client's disturbing emotions and sensations as they are measured by the SUDs rating. This phase deals with all of the person's responses (including other memories, insights and associations that may arise) as the targeted event changes and its disturbing elements are resolved. This phase gives the opportunity to identify and resolve similar events that may have occurred and are associated with the target. That way, a client can actually surpass his/her initial goals and heal beyond his/her expectations.

During desensitization, the therapist leads the person in sets of eye movement (or other forms of stimulation) with appropriate shifts and changes of focus until his/her SUD-scale levels are reduced to zero (or 1 or 2 if this is more appropriate). Starting with the main target, the different associations to the memory are followed. For instance, a person may start with a horrific event and soon have other associations to it. The clinician will guide the client to a complete resolution of the target.

Phase 5: Installation: The goal is to concentrate on and increase the strength of the positive belief that the person has identified to replace his/her original negative belief. During this fifth phase of treatment, the goal is to accept the full truth of positive self-statement at a level of 7 (completely true). If the person is aware that s/he actually needs to learn some new skill, such as self-defense training, in order to be truly in control of the situation, the validity of his/her positive belief will rise only to the corresponding level, such as a 5 or 6 on the VOC scale.

Phase 6: Body Scan: After the positive cognition has been strengthened and installed, the therapist will ask the person to bring the original target event to mind and see if s/he notices any residual tension in his/her body. If so, these physical sensations are then targeted for reprocessing. Evaluations of thousands of EMDR sessions indicate that there is a physical response to unresolved thoughts. This finding has been supported by independent studies of memory indicating that when a person is negatively affected by trauma, information about the traumatic event is stored in motoric (or body systems) memory, rather than narrative memory, and retains the negative emotions and physical sensations of the original event. When that information is processed, however, it can then move to narrative (or verbalizable) memory and the body sensations and negative feelings associated with it disappear. Therefore, an EMDR session is not considered successful until the client can bring up the original target without feeling anybody tension. Positive self-beliefs are important, but they have to be believed on more than just an intellectual level.

Phase 7: Closure: Closure ensures that the person leaves at the end of each session feeling better than at the beginning. If the processing of the traumatic target event is not complete in a single session, the therapist will assist the person in using a variety of self-calming techniques in order to regain a sense of equilibrium. Throughout the EMDR session, the client has been in control and it is important that the client continue to feel in control outside the therapist's office. S/he is also briefed on what to expect between sessions (some processing may continue, some new material may arise), how to use a journal to record these experiences, and which techniques s/he might use on his/her own to help him feel calmer.

Phase 8: Re-evaluation: Opens every new session, the therapist checks to make sure that the positive results (low SUDs, high VOC, no body tension) have been maintained, identifies any new areas that need treatment, and continues reprocessing the additional targets. The next session begins with phase eight, re-evaluation of the previous work, and of progress since the previous session. EMDR treatment ensures processing of all related historical events, current incidents that elicit distress, and future scenarios that will require different responses. The overall goal is produce the most comprehensive and profound treatment effects in the shortest period of time, while simultaneously maintaining a stable client within a balanced system.

After EMDR processing, clients generally report that the emotional distress related to the memory has been eliminated, or greatly decreased, and that they have gained important cognitive insights. Importantly, these emotional and cognitive changes usually result in spontaneous behavioral and personal change, which are further enhanced with standard EMDR procedures.

How Does EMDR Work?

The counselor works gently with the client and asks him/her to revisit the traumatic moment or incident, recalling feelings surrounding the experience, as well as any negative thoughts, feelings and memories. They then hold their fingers about eighteen inches from the clients face and begins to move them back and forth like a windshield wiper (this can also be accomplished through hand taping or alternating auditory sounds). The client tracks the movements as if watching ping pong. The more intensely the client focuses on the memory, the easier it becomes for the memory to come to life. As quick and vibrant images arise during the therapy session, they are processed by the eye movements, resulting in painful feelings being exchanged for more peaceful, loving and resolved feelings.

Although a fairly new therapeutic technique, EMDR is meeting with much success all across the county. EMDR is a natural process. The client and the therapist become partners on a journey to help move traumatic and blocked energy. Together they work to transcend and free up the energy, so the client can return to their natural grounded state of being. The goal of this work is to help the client heal, so they can return to their life in peace.

ECLECTIC COUNSELING

As its name indicates, eclectic therapy is a therapeutic approach that incorporates a variety of therapeutic principles and philosophies in order to create the ideal treatment program to meet the specific needs of the patient or client. Instead of insisting upon strict adherence to one particular approach or school of thought, eclectic therapists employ elements from a range of therapeutic techniques, with the goal of establishing a course that is personally tailored to the patient or client. After all, the key here is to help the patient as quickly and as effectively as possible. Eclectic counseling is defined as the synthesis and combination of directive and non-directive counseling. It represents a middle status between the two extremes represented by the 'non-directive' technique on one hand and the 'directive' technique on the other. In eclectic counseling, the counselor is neither too active as in the directive counseling nor too passive as in the non-directive counseling. Counselor just follows the middle path between these two. Thorne (1973) finds that it is possible for a counselor to alternate between directive and non-directive methods even in the same interview without disrupting the non-directive permissive relationship with the client. He selects the techniques according to the requirements of the situation and the individual. The counselor must be competent and proficient in the use of all available methods. The validity of the results is determined by the skill with which any method is used. The critical factor is not what method is used but rather the skill with which it is used.

In eclectic counseling, the needs of a person and his/her personality are studied by the counselor. After this the counselor selects those techniques, which will be useful for the person. The main techniques used are reassurance giving information, case history, testing etc. In eclectic counseling the counselor first takes into consideration the personality and need of the counselee. S/he selects the directive or non-directive technique that seems to serve the purpose best. The counselor may start with the directive technique. When the situation demands, s/he may switch over to the non-directive counseling and vice-versa. An attempt is made to adjust the technique to the requirements of the situation and the individual.

Assumptions of Eclectic Counseling

- Passive methods must be used whenever possible.
- Active methods are used only with specific indications.
- In the early stages when the client is telling his/her stories passive methods are used. It permits emotional release.
- Complex methods are used only when simple methods fail.
- All counseling should be client centered.

- Client is given opportunity to resolve his/her problems indirectly.
- Directive methods are used where co-operations of other persons are needed for the solution.

Characteristics of Eclectic Counseling

- Objective and coordinating methods are used.
- In the beginning client active methods are used.
- Counselor remains passive.
- Importance is given to job efficiency and treatment.
- Principle of low expenditure.
- Professional efficiency and skill of counselors are must.
- Based on the need of client, directive and non-directive methods are used.
- Provide opportunity to client to find solutions to his/her problems himself/herself.

Hierarchy of Eclectic Practices: McBride and Martin (1990) advocate a hierarchy of eclectic practices and discuss the importance of having a sound theoretical base as a guide.

- The lowest or first level of eclecticism is really syncretism - a sloppy unsystematic process of putting unrelated clinical concepts together. It is encouraged when graduate students are urged to formulate their own theories of counseling without first having experienced how tested models work.
- The second level of eclecticism is traditional. It incorporates an orderly combination of compatible features from diverse sources harmonious whole. Theories are examined in greater depth.
- On a third level, eclecticism is described as professional or theoretical or as theoretical integration. This type requires that counselor master at least two theories before trying to make any combinations.
- A final level of eclecticism is called technical eclecticism. In this approach, procedures from different theories are selected and used in treatment.

Steps in Eclectic Counseling: The leading exponent of the Eclectic counseling Thorne (1973) suggested the following in the process -

1. Diagnosis of the cause.
2. Analysis of the problem.
3. Preparation of a tentative plan for modifying factors.
4. Securing effective conditions for counseling.
5. Interviewing and stimulating the client to develop his/her own resources and to assume its responsibility for trying new modes of adjustment.
6. Proper handling of any related problems which may contribute to adjustment.

Applications: Any condition that can be treated via any type of therapy is capable of being treated with eclectic therapy. Thus, individuals with addictions, substance abuse disorders, eating disorders, behavior compulsions, mood disorders, and other forms of emotional or psychological issues may be effectively treated by a therapist who embraces the philosophy of eclectic therapy.

Contributions of Eclectic Counseling

- This approach encourages counselors to use different types of strategies or approaches with clients. With such diverse approaches, counselor's awareness of techniques and treatments used would increase due to the past experiences.
- Diverse treatments and techniques used provide counselors with more flexibility as they have more options of treatments to choose from, which ultimately benefit the clients.

Limitations of Eclectic Counseling

- It is vague, opportunistic and superficial.
- Directive and non-directive cannot be mixed.
- In this, the question arises how much freedom should be given to the client? For this there is non-definite rule.
- Counselors often do more harm than good if they have little or no understanding about what is helping the client.

INTEGRATING COUNSELING

Integrative Therapy (IT) was developed in Paris at the end of the 1960s by a group of clinicians and researchers - psychologists and physicians - under the direction of Professor Dr. Hilarion G. Petzold. Integrative therapy or integrative counseling is a combined approach to psychotherapy that brings together different elements of specific therapies. Integrative therapists take the view that there is no single approach that can treat each client in all situations. Each person needs to be considered as a whole and counseling technique must be tailored to their individual needs and personal circumstances.

Integrative counseling maintains the idea that there are many ways in which human psychology can be explored and understood - no one theory holds the answer. All theories are considered to have value, even if their foundational principles contradict

each other - hence the need to integrate them. The integrative approach also refers to the infusion of a person's personality and needs - integrating the affective, behavioral, cognitive, and physiological systems within one person, as well as addressing social and spiritual aspects. Essentially, integrative counselors are not only concerned with what works, but why it works - tailoring therapy to their clients and not the client to the therapy.

Integrative Psychotherapy takes into account many views of human functioning. The psychodynamic, client-centered, behaviorist, cognitive, family therapy, Gestalt therapy, body-psychotherapies, object relations theories, psychoanalytic self psychology, and transactional analysis approaches are all considered within a dynamic systems perspective. Each provides a partial explanation of behavior and each is enhanced when selectively integrated with other aspects of the therapist's approach. The psychotherapy interventions used in Integrative Psychotherapy are based on developmental research and theories describing the self-protective defenses used when there are interruptions in normal development. Psychotherapy integration can be differentiated from an eclectic approach, in which a therapist chooses interventions because they work, without looking for a theoretical basis for using the technique. Eclectic practitioners are not bound by the theories or methodology of any one particular school. Instead, they may use what they believe or feel or experience tells them will work best, either in general, or to suit the immediate needs of individual clients. In contrast, integrative psychotherapy attends to the relationship between theory and technique. An eclectic therapist might experience a change in his/her client after administering a particular technique and be satisfied with a positive result. In contrast, an integrative therapist is curious about the 'why and how' as well.

Aim of Integrative Therapy

- Integrative counseling aims to promote healing and facilitate wholeness - ensuring that all levels of a person's being and functioning (psychological, physical and emotional health) are maximized to their full potential. Clients must be committed to self-exploration and open to identifying what factors in their life are perpetuating problems, and/or are causing current concerns.
- In particular, the integrative approach helps clients face each moment openly and authentically without having formed an opinion, expectation or attitude beforehand. This enables them to better focus on the fears and hurts that limit their psychological freedom, and recognize specific triggers that may be causing disruptive patterns of behavior.

- Through this awareness, integrative therapy helps to create a healthy alliance between mind and body - empowering clients to start setting goals and practicing new behaviors that will enable them to move beyond their limitations and discover greater life satisfaction. This will be worked towards alongside other goals that are drawn into therapy through the integration of other approaches. These will all be tailored to the client's personal limits and external constraints.

Integration of Approaches: The central premise of integrative counseling is that there are many ways in which human functioning can be explored and understood. This means integration can occur through a variety of modalities/systems of perspective. These may include - humanistic therapies; psychoanalytical and psychodynamic therapies; and cognitive and behavioral therapies. Each approach offers explanation and insight into human behavior, as well as a unique understanding of key factors that will result in changes to behavior and other areas of functioning such as cognition and emotions. These can be reinforced when selectively integrated with other elements of therapy. For example, if an integrative therapist is working with a client that has behavioral problems, they may want to start the therapy by working on adjusting behavioral functioning and reducing symptoms. This may involve applying cognitive behavioral techniques to help the client establish some control over their functioning before moving on to the next stage of therapy (i.e. working on improving, and gaining insight into the client's behaviors, emotions and thoughts). In this stage, the therapist may employ psychoanalytic techniques that recall childhood experiences and interpretation, dream analysis or analysis of transference.

There are many approaches to integrating psychotherapeutic techniques. Those approaches are as follows -

Common Factors Approach: The advantage of a common factors approach is the emphasis on therapeutic actions that have been demonstrated to be effective. The disadvantage is that common factors may overlook specific techniques that have been developed within particular theories. Common factors theory asserts it is precisely the factors common to the most psychotherapy that make any psychotherapy successful.

Technical Eclecticism: This approach is guided primarily by data that indicates what has worked best for others in the past. The advantage of technical eclecticism is that it encourages the use of diverse strategies without being hindered by theoretical differences. A disadvantage is that there may not be a clear conceptual framework describing how techniques drawn from divergent theories might fit together.

Theoretical Integration: This approach requires integrating theoretical concepts from different approaches. These approaches may differ in their fundamental philosophy about human behavior. While assimilative integration begins with a single theory and brings together techniques from different approaches, theoretical integration tries to bring together those theoretical approaches themselves and then develop a unified theory.

Assimilative Integration: This mode of integration favors a firm grounding in any one system of psychotherapy, but with a willingness to incorporate or assimilate, perspectives or practices from other schools. The therapist has a commitment to one theoretical approach, but also is willing to use techniques from other therapeutic approaches.

Therapeutic Relationship: The attitude and presence of an integrative counselor is another crucial element of integrative therapy. It is generally believed that the most effective model requires the therapist to be non-judgmental, interpersonal and intent on establishing a supportive and cooperative relationship with their client. They must also engage in deep, attentive listening without the pre-suppositions that can distort understanding. The meaningful contract empowers clients - helping them to explore and recognize patterns of behavior that need to be addressed through change and the setting of new goals. This aspect of integrative therapy is often referred to as the personal integration of therapists - they are committing themselves wholly to their client and their exploration of self.

Benefits of Integrated Therapy: A key advantage of integrative counseling is its flexibility and focus on the whole of an individual. The integration of different approaches means therapy can be tailored to meet a variety of needs and concerns. It can be particularly beneficial for those who want to overcome negative patterns of behavior caused by anxieties, fears, phobias or any other mental health issue that is greatly impacting life satisfaction (i.e. addictions, depression, past and current trauma, bereavement and low self-esteem). It has also been found useful in improving daily function in children with autism and learning difficulties. Often these problems can affect the four dimensions of human functioning - affective, behavioral, cognitive, and physiological systems.

Due to the in-depth exploration of issues and setting of goals, integrative counseling typically requires a substantial investment of time on the part of the client. Therefore it may not suit those who want a quick, solution-focused approach to personal development. The length of the therapy will depend on the client, the therapeutic goals set and the types of issues that are being addressed.

GROUP THERAPY

Joseph Hersey Pratt
(1872–1956)

Trigant Burrow
(1875-1950)

Paul Ferdinand Schilder
(1886-1940)

Irvin David Yalom
(1931 – to date)

Group therapy or group psychotherapy is a form of psychotherapy in which one or more therapists treat a small group of clients together as a group. The founders of group psychotherapy in the USA were Joseph H. Pratt, Trigant Burrow and Paul Schilder. All three of them were active and working at the East Coast in the first half of the 20th century. After World War II, group psychotherapy was further developed by Jacob L. Moreno, Samuel Slavson, Hyman Spotnitz, Irvin Yalom, and Lou Ormont. Yalom's approach to group therapy has been very influential not only in the USA but across the world. Group psychotherapy helps people who would like to improve their ability to cope with problems. Group therapy focuses on interpersonal relationships and helps individuals learn how to get along better with other people under the guidance of a professional. Group psychotherapy also provides a support network for specific problems or challenges. A group therapist appropriately selects people (usually 5 to 10) who would be helped by the group experience and who can be learning partners for one another. In meetings, people are encouraged to talk with each other in a spontaneous and honest fashion. A professionally trained therapist, who provides productive examination of the issues or concerns affecting the individuals and the group, guides the discussion. Groups can be as small as three or four people, but group therapy sessions often involve around seven to twelve individuals (although it is possible to have more participants). The group typically meets once or twice each week for an hour or two. During the session each member works to express their own problems, feelings, ideas and reactions as freely and honestly as possible. Accordingly, the group members have an opportunity to learn not only about themselves and their own issues, but also the value of helping other group members.

According to author Oded Manor in 'The Handbook of Psychotherapy', the minimum number of group therapy sessions is usually around six but a full year of sessions is more common. Manor also notes that these meetings may either be open or closed. In

open sessions, new participants are welcome to join at any time. In a closed group, only a core group of members are invited to participate. In many cases, the group will meet in a room where the chairs are arranged in a large circle so that each member can see every other person in the group. A session might begin with members of the group introducing themselves and sharing why they are in group therapy. Members might also share their experiences and progress since the last meeting.

Group therapy is a powerful tool for growth and change. In group therapy, 5-10 individuals meet face to face to share their struggles and concerns with the facilitation of 1-2 trained group therapists. The power in group therapy lies in the unique opportunity to receive multiple perspectives, support, encouragement and feedback from other individuals in a safe and confidential environment. These interpersonal interactions can provide group members an opportunity to deepen their level of self-awareness and to learn how they relate to others. Group therapy can be a safe and supportive place to experiment with new ideas and ways of being. For many emotional concerns, personal struggles, and relationship issues that college students face, group therapy is the most effective treatment modality. Because group therapy is so effective, the Counseling Center offers a wide array of groups to address the needs of our students. We feel confident that group therapy is one of the most effective and beneficial treatments to address the common issues faced by our students, but we understand that for many students, the idea of joining a therapy group can be daunting. Although many students are initially intimidated and hesitant about participating in group therapy, group members consistently describe group therapy as a very helpful and positive experience.

Principles of Group Therapy: In the 'Theory and Practice of Group Psychotherapy', Irvin D. Yalom (1995) outlines the key therapeutic principles that have been derived from self-reports from individuals who have been involved in the group therapy process -

- Instillation of hope: The group contains members at different stages of the treatment process. Seeing people who are coping or recovering gives hope to those at the beginning of the process.
- Universality: Sharing an experience with a group helps people see that they are not going through something alone. It also serves to remove a group member's sense of isolation, validate their experiences, and raise self-esteem.
- Imparting information: Group members can help each other by sharing information.

- Altruism: Group members can share their strengths and help others in the group, which can boost self-esteem and confidence.
- Corrective recapitulation of the primary family group: The therapy group is much like a family in some ways. Within the group, each member can explore how childhood experiences contributed to personality and behaviors. They can also learn to avoid behaviors that are destructive or unhelpful in real life.
- Development of socialization techniques: The group setting is a great place to practice new behaviors. The setting is safe and supportive, allowing group members to experiment without the fear of failure.
- Imitative behavior: Individuals can model the behavior of other members of the group or observe and imitate the behavior of the therapist.
- Interpersonal learning: By interacting with other people and receiving feedback from the group and the therapist, members of the group can gain a greater understanding of themselves.
- Group cohesiveness: Because the group is united in a common goal, members gain a sense of belonging and acceptance.
- Catharsis: Sharing feelings and experiences with a group of people can help relieve pain, guilt, or stress.
- Existential factors: While working within a group offers support and guidance, group therapy helps member realize that they are responsible for their own lives, actions, and choices.

Goals of Group Therapy

- Gain an understanding of personal problems and explore possible solutions.
- Give and receive feedback and support.
- Feel more connected to other students who are struggling with similar issues.
- Practice communication skills in a safe group setting.
- Learn more about how you come across to others.
- Improve your ability to identify your feelings and express them.
- Reduce social isolation.

Theories of Group Therapy: Following are the important group therapy models -

- Brief cognitive group therapy
- Cognitive-behavioral group therapy
- Strategic/interactional therapy
- Brief group humanistic and existential therapies
- Group psychodynamic therapy
- Modified dynamic group therapy (MDGT)
- Modified interactional group process (MIGP).

Brief Cognitive Group Therapy: Cognitive techniques work well in group therapy. The group is taught the basics of the cognitive approach, then individual members take turns presenting an event or situation that tempted them to abuse substances. Other members assist the therapist in asking for more information about the client's thoughts on the event and how it did or did not lead to substance abuse (or to negative feelings that might have led to use). Finally, the group members provide the client with alternative ways of viewing the situation.

Cognitive-Behavioral Group Therapy: The cognitive-behavioral approach focuses the group's attention on self-defeating beliefs, relying on group members to identify such beliefs in each other. The therapist encourages group members to apply behavioral techniques such as homework and visualization to help participants think, feel, and behave differently.

Strategic/ Interactional Therapies: The strategic therapist uses techniques similar to those used in family therapy to challenge each group member to examine ineffective attempted solutions. The therapist encourages group members to evaluate and process these attempted solutions and recognize when they are not working, then engages the group in generating alternative solutions. The therapist also works, where appropriate, to change group members' perceptions of problems and helps them understand what is happening to them. Typically, the therapist guides the process, while members offer suggestions and encouragement to each other as they identify and implement effective solutions.

The principles of solution-focused therapy are the same for group treatment as for individual therapy. These include client goal-setting through the use of the 'miracle' question, use of scaling questions to monitor progress, and identification of successful strategies that work for each client. The therapist works to create a group culture and dynamic that encourages and supports group members by affirming their successes. At the same time, the therapist works to restrain client digressions ('war stories') and personal attacks. The therapist tries to challenge group members - all of whom, unlike in family therapy, are seen as 'customers' - to take action to create positive change.

Brief Group Humanistic and Existential Therapies: Several approaches fall within this category. The transpersonal approach is useful in meditation, stress reduction, and relaxation therapy groups and can be adapted for clients who have substance abuse disorders. In dealing with issues of religion or spirituality, it is helpful to have other people talk about their perspectives. In this way, past degrading or punitive

experiences related to organized religion can be redefined in a more meaningful and useful context.

Gestalt therapy in groups allows for more comprehensive integration in that each group member can provide a piece of shared personal experience. Each group member plays a role in creating the group, and all of their perceptions must be taken into account in making a change. Role-playing and dream analysis in groups are practical and relevant exercises that can help clients come to terms with themselves.

Group Psychodynamic Therapy: Group psychodynamic therapy enables the group itself to become both the context and means of change through which its members stimulate each other to support, strengthen, or change attitudes, feelings, relationships, thinking, and behavior - with the assistance of the therapist.

Modified Dynamic Group Therapy (MDGT): On the basis of psychodynamic theory, a modified dynamic group therapy approach was defined for substance-abusing clients (Khantzian et al., 1990). Viewing substance abuse disorders as an expression of ego dysfunction, affect dysregulation, failure of self-care, and dysfunctional interpersonal relationships, MDGT falls in the intermediate length of time-limited group psychotherapy, with its basic structure defined by two meetings per week over a 26-week format. As a supportive, expressive group experience, MDGT provides substance-abusing clients the opportunity to evaluate and change their vulnerabilities in four primary areas: (1) accessing, tolerating, and regulating feelings; (2) problems with relationships; (3) self-care failures; and (4) self-esteem deficits. Congruent with this understanding of the origins of substance abuse, MDGP emphasizes safety, comfort, and control within the group context. Group facilitation is defined primarily by the therapist's ability to engage and retain substance abusers in treatment by providing structure, continuity, and activity in an empathic atmosphere.

Modified Interactional Group Process (MIGP): Time-limited MIGP is a synthesis of the work of several theorists (Yalom, 1995). The features that distinguish MIGP from a more traditional interactive process are the greater activity of the leader and the sensitivity to the development of a safe atmosphere that allows group members to examine relational issues without excessive emotional contagion. The atmosphere of safety is greatly enhanced by the therapist's adherence to group agreements or group norms and by the continued reinforcement of these agreements throughout the group process. The importance of confidentiality, the group's accepting responsibility for itself, and self-disclosure are all supported by the facilitator. Procedural agreements, including beginning and ending the group session on time and ensuring that each

member has a place within the circle, with any absences addressed, are part of the development of the safe environment. In this process, the therapist helps the clients recognize that they are the primary change agents. The group becomes a safe place both to give and to receive support. Although traditionally substance abuse groups tend to be confrontative, MIGP is far more supportive. This stems from the belief that denial and other defense mechanisms become more rigid when a person is attacked. Consequently, group members are encouraged to support one another and look for areas of commonality rather than use more shame-based interactive styles that attempt to 'break through denial'. MIGP focuses special attention in four areas of the client's life: gratification and support, vulnerability of self, regulation of affect, and self-care (Flores and Mahon, 1993). These four areas receive particular attention because they represent areas of vulnerability within the substance-abusing client that can easily lead to relapse and undermine recovery.

Preferred Group Counseling Techniques/Strategies

- Reflection - Allow clients to understand fully each person's comments and how they can relate to what they just stated.
- Active Listening - Allow clients to be very aware of their listening skills within a group and how important they are. This allows the group to be more open with each other when they know their group members are paying attention to their feedback.
- Clarification - Allow clients to be very concrete with others in the group of comments made.
- Summarizing - Allow clients to see the bigger picture of what was said in the group by doing a recap of what was said. Provides comfort within group.
- Linking - Allow clients to see how their problems connect with others' in the group and they have the same concerns.
- Encouraging - Allow clients to be more open with group and makes sharing personal feelings more inviting.
- Focus - Allow counselor and clients to give attention to groups concerns.
- Cutting Off - Allow counselors to keep the group on topic and give everyone a chance to share.
- Drawing Out - Allow counselor to bring forth quiet group members comments.
- Rounds - Allows each client to share within the group.
- Dyads - Allows clients to pair up with a partner and learn more about another client.
- Word or Phrase Round - Allow clients to broaden their vocabulary when describing how they felt about something within the group session.

- Comment Round - Allow clients to share their personal comments about that specific session.
- Icebreakers - Allow clients to warm-up to group members when beginning group.
- Modeling - Allow clients to see counselor as a role model in group work and the behaviors that can be applied to them personally.
- Use of Eyes - Allows counselor to have direct eye contact with group members that are sharing. It will allow the counselor to draw out quiet members to speak. Allow them to members that have lost interest in the topic.
- Tone Setting - Allow clients to establish a mood for their group. Rather it needs to be a tone that is serious, social, supportive, and formal.
- Use of Leader's Energy - Allow clients to see the counselor excitement about their new group, topic, and activities.
- Non-Judgmental - Allow clients to see that this is a positive experience and not to make anyone feel down.
- Empathy - Allow clients to convey to other group members that they understand and will not make fun of anyone.

Applications of Group Therapy: Group therapy has proven to be very effective for the treatment of depression and traumatic stress, such as post-traumatic stress disorder (PTSD) and sexual abuse survivors. Clinical cases have shown that the combination of both individual and group therapy is most beneficial for most clients. Group therapy has proven effective in treating feelings of isolation, difficulties with interpersonal relationships, aging, schizophrenia, bipolar disorder, anorexia, alcohol dependency, death & other losses, and lifestyle issues within a traditional culture.

Common Misconceptions about Group Therapy

I'll be forced to tell my deepest thoughts, feelings and secrets. You control what, how much and when you share. Most people find that when they feel safe enough to share what's troubling them, a group can be helpful and affirming. Even if you're not ready to talk about something, listening to others can help. What they're saying may apply to you.

Group therapy will take longer because I'll be sharing the time. Group therapy is often more efficient than individual therapy for two reasons - First, you can benefit from the group, even when you're quiet and listening carefully. Second, group members may bring up issues that strike a chord with you - things you may not have been aware of or brought up yourself.

Others in the group will verbally attack me. It's important for members to feel safe and leaders will help develop that environment within the group. While feedback is often hard to hear, groups can point out damaging behaviors in a respectful, gentle way, so you can hear and use the insights. As trust grows, group members generally experience feedback (and even confrontation) as a sign of caring.

I have trouble talking to people. I'll never be able to share in a group. Most people are anxious about talking in a group, but this almost always fades quickly. People also remember what it's like to be new to a group; you're likely to get a lot of support when you do begin talking.

Group therapy is second-best. Group therapy is the most effective treatment approach for many issues. When a group is recommended, it's because your therapist believes it's the best way for you to be helped. S/he can discuss the reasons for the recommendation with you.

Groups are confidential, meaning that what members disclose in sessions is not shared outside of the group. The importance of confidentiality and concerns about are reviewed with group members at the first meeting and every time a new member joins a group. For getting the most benefit from group therapy should do the followings -

- Attend the group regularly;
- Share your hopes and fears for participating in group;
- Share with the group what you're struggling with;
- Take some emotional risks in group;
- Be direct and honest with your feelings in group;
- Be willing to give and receive feedback;
- Share your own experiences and ask others about theirs; and
- Let the group know both positive and negative feelings that come up in group.

Contributions of Group Therapy

Group therapy allows people to receive the support and encouragement of the other members of the group. People participating in the group can see that others are going through the same thing, which can help them feel less alone.

Group members can serve as role models to other members of the group. By observing someone successfully coping with a problem, other members of the group can see that there is hope for recovery. As each person progresses, they can, in turn, serve as a role model and support figure for others. This can help foster feelings of success and accomplishment.

Group therapy is often very affordable. Instead of focusing on just one client at a time, the therapist can devote his/her time to a much larger group of people.

Group therapy offers a safe haven. The setting allows people to practice behaviors and actions within the safety and security of the group.

By working in a group, the therapist can see first-hand how each person responds to other people and behaves in social situations. Using this information, the therapist can provide valuable feedback to each client.

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EMERGING TRENDS IN PSYCHOLOGY

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ABSTRACT

Psychology is a developing control by utilizing other related orders. The wide scope of Psychology applications are distinguished and perceived by society. These angles with great number of chances are exceptionally reassuring, rousing and simultaneously entirely good for Psychology. It may not be full subtleties of momentum research in Psychology yet it gives a decent blueprint of latest things that are going on in Psychology. Psychology is the logical investigation of human conduct and intellectual (mental) measures. The Behavoiur has different measurements and qualities of conduct. Be that as it may, conduct is concentrated by other scholastic trains as well. Along these lines, which parts of conduct are the worries of Psychology, is the subject of examination and what is the connection of Psychology with other scholarly teaches. Everyone might have heard people using the terms ‘psychiatrist’, ‘psychologist’, ‘counselor’, etc. But, many of them perhaps, do not know the difference between them. Many of them also do not know, as to where do psychologists work. They think, Psychologists essentially work in mental clinics to treat anomalous/crazy individuals, or might be showing Psychology at schools and schools/colleges. While surely, these are where clinicians work; there are numerous different regions where information on Psychology is applied. We'll attempt to discover answer to the above inquiries. We'll examine about the different parts of Psychology and the wide scope of fields where clinicians apply the standards of Psychology, to take care of an assortment of human issues as well as improve the personal satisfaction.

INTRODUCTION:

Psychology accentuates on two angles, first is that Psychology as a science, A Science is characterized not by what it concentrates but rather by how it examines. Psychologists like some other researchers efficiently notice realities about individuals and coordinate these realities to show up at speculations. Psychology imparts to different sciences certain points, presumptions, methods of completing exploration and methods of building and changing speculations. Second viewpoint is Psychology is a collection of information that can be applied to help take care of an assortment of human issues. Notwithstanding the commitment in the region of exploration (formative, social, test, physiological Psychology), there are numerous different zones of expert applications. Clinical Psychologists work with specialists in a medical clinic setting or with other wellbeing experts in the network. Instructive analysts work with youngsters and teenagers in schools, universities, and nurseries and in the home. They work together with guardians and instructors to survey a youngster's advancement at school and give assistance and counsel. Guiding analysts work with people, families, couples or gatherings to improve individuals' prosperity, mitigate pain and assist them with tackling their own issues and take their own decisions.

RELATIONSHIP OF PSYCHOLOGY & OTHER BRANCHES OF KNOWLWDGE:

Psychology as a social science imparts normal attributes to other physical and natural sciences. All sciences have functional application to manage human issues and improve their personal satisfaction. Designing and innovation have created from the consequences of material science, science and arithmetic and have made life simpler and agreeable. Essentially, clinical sciences have contributed fundamentally to not just battle and control/kill numerous destructive illnesses, yet in addition fix/forestall numerous sicknesses. Analysts have likewise built up various methods to help individuals lead a fuller and more joyful life by advancing/upgrading their mental wellbeing. Analysts additionally fix and help in anticipation of different mental and enthusiastic issues (conduct problems). At times, the world's issues, for example, climatic change, contamination; repercussions of cataclysmic event, synthetic

debacles and so forth are so broad and genuine that nobody or two sciences can take care of such issues. A gathering of researchers from various fields need to, accordingly, meet up and mutually handle such issues. Unmistakably however Psychology has arisen as a free field of logical enquiry, by the by it has extremely cozy relationship with different sciences. Indeed, the very rise of Psychology as a science was conceivable due to advancements in different sciences like hereditary qualities, physiology, nervous system science, medication, physical science, and science. Its strategies for enquiry and examination just as application have all been affected, by advancements in different subjects, obviously, improvement in Psychology have thus, impacted the development and advancement of different orders like humanism, humanities, financial aspects, political theory, the board and so on Clearly despite the fact that we can consider various fields of logical information, the truth of the matter is that these fields depend for their development on the discoveries of different controls. Maybe, this will become clearer in the event that we can analyze in some detail the relationship of Psychology with certain different sciences. For this reason, we may amass different sciences in to the accompanying classes: organic sciences, actual sciences, clinical sciences and sociologies

APPLICATION OF PSYCHOLOGY IN SOCIETY:

A considerable lot of the essential current applications for Psychology spin around shielding individuals from passionate and actual mischief while furnishing them with the vital mental data transmission to deal with the mental dangers numerous individuals face day by day. Issues like connections, working environment stress and monetary challenges all can be influenced by mental side effects that require diagnosing and overseeing, which is the place where current Psychology becomes an integral factor and why it's so significant. Regardless of the well known generalization of the private practice advisor promoted by the media, Psychology experts really approach a wide assortment of enterprises and fields, from schooling and criminal equity to showcasing and legislative issues. Basically, Psychology helps individuals in enormous part since it can clarify why individuals act the manner in which they do. With this sort of expert knowledge, a clinician can assist individuals with improving their dynamic, stress the executives and conduct dependent on comprehension past conduct to more readily foresee future conduct. The entirety of this can assist individuals with having a more fruitful profession, better connections, more fearlessness and generally speaking better correspondence. Everybody utilizes Psychology consistently, regardless of whether they are chatting with companions, contending with an accomplice, or training their kids. The vast majority simply don't understand the science behind their choices. Psychology makes it simpler to live with others by understanding them more and working with their conduct. Underneath, University of Texas teacher James Penne dough puncher clarifies one way Psychology can be utilized to foresee sentimental similarity.

MAJOR EMERGING TRENDS IN PSYCHOLOGY:

The study of behaviors and thoughts of mind is ideally called as psychology. The behaviors we do, feel, think and say everything is determined by the functioning of the mind and this behaviors are used as a primary data for analyzing about how the mind works and this all theoretical testing’s of mind is called as psychology. Psychology is a multi disciplinary stream; it lies at the intersection of other different subjects like sociology, biology, medicine, physiology, linguistics, and anthropology.

Health Psychology: Even though psychology is employed in marketing, industries, education, fashion, social, and other varying fields, psychology in health and related sciences is exciting area and has very notable development. Health psychology focuses mainly on prevention and treatment of disease, promotion and maintenance of health, improvement of health care and policy information. Current research in health psychology is highly engaged on stress reduction, smoking cessation, weight management, daily nutrition improvement, and reduction of sexual behaviors. The realization of feelings, thoughts, and behaviors, psychological, environmental and other social factors depends on how good people understand the power/command of mind. Psychologically these factors play major roles in managing addictions, illness, health, along with disease genetic tendency.

Clinical Psychology: Clinical psychology answers the questions related to better understanding of relationship between physical wellness, health and illness, and connection among environment, body and mind. Psychologists find strategies to minimize/remove pain and known about pain abnormalities like analgesia, neuralgia and phantom limb pain. One of the difficult tasks for psychologists is, standard encouragement and/or motivating people towards positive thinking, to stick to medical direction and treatment. Psychology can help people towards physical fitness, support with chronic pain reduction, and finally improve the quality of patient everyday life. Clinical psychologies mental imagery domains have relevance to many social phobia, clinical disorders, depressions, and post-anxiety disorders.

Cognitive Science: Cognitive science is the study of how the mind works, functions, and behaves. As a scientific field of study, cognitive science requires applying multiple existing disciplines such as philosophy, neuroscience, or artificial intelligence in order to understand how the brain makes a decision or performs a task. Cognitive science seeks to understand intelligence and behavior which can help humans in many ways such as developing educational programs or building smarter devices.

Forensic Psychology: Forensic psychology is a specialty area that deals with issues related to psychology and the law. Those who work in this field of psychology apply psychological principles to legal issues. This may involve studying criminal behavior and treatments or working directly in the court system. This branch of psychology is defined as the intersection of psychology and the law, but forensic psychologists can perform many roles so this definition can vary. In many cases, people working in forensic psychology are not necessarily "forensic psychologists." These individuals might be clinical psychologists, school psychologists, neurologists, or counselors who lend their psychological expertise to provide testimony, analysis, or recommendations in legal or criminal cases.

Biopsychology: Biopsychology is a branch of psychology focused on how the brain, neurons, and nervous system influence thoughts, feelings, and behaviors. This field draws on many different disciplines including basic psychology, experimental psychology, biology, physiology, cognitive psychology, and neuroscience. People who work in this field often study how brain injuries and brain diseases impact human behavior. Biopsychology is also sometimes referred to as physiological psychology, behavioral neuroscience, or psychobiology.

Cross-Cultural Psychology: Cross-cultural psychology is a branch of psychology that looks at how cultural factors influence human behavior. The International Association of Cross-Cultural Psychology (IACCP) was established in 1972, and this branch of psychology has continued to grow and develop since that time. Today, increasing numbers of psychologists investigate how behavior differs among various cultures throughout the world.

Industrial-Organizational Psychology: Industrial-organizational psychology is a branch that applies psychological principles to research on workplace issues such as productivity and behavior. This field of psychology often referred to as I/O psychology works to improve productivity and efficiency in the workplace while also maximizing the well-being of employees. People who work in human factors are focused on improving how people interact with products and machines both in and out of the workplace. They might help design products intended to minimize injury or create workplaces that promote greater accuracy and improved safety.

Traffic Psychology: Traffic psychology is an emerging field that involves applying psychological principles to understanding driver behavior. Some areas in this field includes the designing vehicles that are safer and more ergonomic, Researching how people use transportation, Searching for ways to improve traffic safety and prevent auto accidents, Studying the relationship between driver behavior and traffic accidents Traffic psychology often involves a multidisciplinary approach, combining fields such as social psychology, behavioral psychology, and cognitive psychology. For example, traffic psychologists might assess how perception and cognition influence performance during a driving task. They might also look at how individual personality affects a driver's emotions, attitudes, and risk-taking behavior while driving.

Engineering Psychology: Engineering Psychology is an emerging specialty area of psychology that is still considered novel. This growing field is not widely researched and little data exists on salary rates and growth. As an increasing amount of focus is placed on the psychology behind consumerism, this field will likely explode over the coming years. Products designed under the guidance of an engineering psychologist are typically better designed from the start and can help eliminate upset customers and expensive redesigns, increasing the company's profitability. Engineering psychologists work in many different settings, including academia, governmental agencies and private businesses.

Gerontology Psychology: Gerontology is another rapidly growing area of psychology. The field of Geropsychology is a good option for psychologists who wish to pursue a psychology career that combines multidisciplinary work involving a wide range of problems. These may include depression, family relationships, anxiety, medical conditions, diminished mental capacity, retirement, changes in sexuality, facing possible poverty and many others. Professionals choosing a career in this area have the benefit of a variety of career options and places to work.

Psychology of Marriage & Family: Marriage and Family Psychology is yet another rising area within the field of psychology (the Bureau of Labor Statistics predicts a ten year rise of 29 percent). This field is closely related to mental health counseling in that many similar treatment methods are used in both types of therapy and intervention. Marriage and Family Psychologists or Therapists are mental health professionals with a minimum of a Master's Degree and typically two years of supervised clinical experience. Marriage and Family Therapists (MFTs or Family Therapists) are trained to assess, diagnose and treat mental health and substance abuse problems in an individual or small group. The field of family therapy is considered one of the cores mental health disciplines with roots in the notion that mental illness and family difficulties are best addressed as a family unit.

Military Psychology: The specialty area of Military Psychology is an area of psychology with a focus on military men and their families. This area is expected to see tremendous growth as Americans have remained engaged in active military combat for a number of years. Some of the responsibilities of a Military Psychologist include performing psychiatric evaluations, assessing and treating mental and emotional issues, and providing counseling services to service members and their families. In the United States, each branch of the military employs psychologists who work to treat active-duty and retired military men. Professionals who specialize in Military Psychology become experts in the responsibilities they conduct, depending on their area of specialization. Some Military Psychologists conduct research, perform tests, or treat mental and emotional disorders. It is the responsibility of Military Psychologists to conduct mental health evaluations before recruits are allowed to enlist in the service. These interviews help ensure that new recruits are mentally and emotionally stable enough to handle the stress of being in the military.

Neuropsychology: Neuropsychology involves the study of the relationship between the brain and behavior. Neuropsychologists often work with patients who have sustained a traumatic brain injury. This branch of psychology is concerned with how the brain and the nervous system affect an individual's thoughts and behaviors. Professionals within this area of psychology focus on how injuries or illnesses of the brain affect cognitive functions and behaviors. Neuropsychology involves the application of standardized tests and measures in the study of brain behavior relationships. Neuropsychological tests are used to evaluate cognitive deficits, while also being a part of the management, treatment and rehabilitation of cognitively impaired patients. Another part of Neuropsychology is the development of models and methodologies to be used in understanding normal and abnormal brain function.

Recreational Psychology: Another branch of psychology that is experiencing a great deal of job growth is the field of Recreational Psychology or Recreational Therapy. Recreational therapists are responsible for the planning, direction, and coordination of recreationally based treatment programs for individuals with mental illness, disability, injury, or other illness. These highly trained clinicians employ a range of different treatment modalities, including arts and crafts, games, drama, poetry, music, dance, sports and "community reintegration" field trips. The goal of these treatment modalities is to help preserve or enhance a client's physical, social, and emotional well-being.

Rehabilitation Psychology: The role of a Rehabilitation Psychologist or Rehabilitation Counselor is to work to help individuals with emotional and physical disabilities live a more independent lifestyle. They work directly with clients to minimize the personal, social, and professional effects that a disability can have on employment or independent living. Rehabilitation counselors help those with physical, mental, emotional, or social disabilities in a wide variety of ways. Some counselors work with students to implement strategies to most efficiently advance from school to work. Others help veterans cope with the mental or physical effects of their military service.

Sports Psychology: Another novel area of psychology is the field of Sports Psychology. The aim of this area is to fill the "hole" in competitive sports that cannot be filled by a coach (who focuses on the physical side of sport). It is the Sports Psychologist who focuses on an athlete's mindset and factors that influence their mindset. Sport Psychologists work with athletes of all levels to identify, refine and achieve their athletic goals. In some cases, athletes seek out help from a Sport Psychologist or other exercise and Sport Psychologists when they have a problem they cannot overcome by working alone or with a coach. Some common examples include anxiety or lost focus during competition. Another example might be difficulty communicating effectively with teammates, keeping their temper under control or having the motivation needed to complete exercise requirements. Sport Psychologists can also be helpful even when the athlete is not experiencing a issue. Sports Psychology is often present in non-athletic settings. Many of the same strategies that Sport Psychologists employ to train athletes, such as relaxation techniques, mental rehearsals and cognitive restructuring, are useful in the workplace and other applied settings.

CONCLUSION:

After the review of literature we can say that psychology have various emerging branches. It is very necessary to keep in mind that no single branch of psychology is more important or better than any other filed. All the specific areas contribute to our understanding of the many different psychological factors that influence the human behavior in different situations. By conducting research and developing new applications for psychological understanding, professionals working in every branch of psychology are able to help people better understand themselves, confront the problems they may face, and live better lives. We'll discuss about the various branches of psychology and the wide range of fields where psychologists apply the principles of psychology, to solve a variety of human issues and/or improve the quality of daily life.

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Guideline for the pharmacological treatment of hypertension in adults

World Health Organization

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Acronyms and abbreviations

ACE1	angiotensin-converting enzyme 1
ACE2	angiotensin-converting enzyme 2
ACEi	angiotensin-converting enzyme inhibitor
ARB	angiotensin-II-receptor blocker
BB	beta-blocker
BP	blood pressure
CAD	coronary artery disease
CCB	calcium channel blocker
CKD	chronic kidney disease
CRE	WHO Office of Compliance, Risk Management and Ethics
CV	cardiovascular
CVD	cardiovascular disease
DBP	diastolic blood pressure
DM	diabetes mellitus
DOI	declaration of interest
ECG	electrocardiogram
EML	Essential Medicines List
ERG	External Review Group
ESRD	end-stage renal disease
GDG	Guideline Development Group
GRADE	Grading of Recommendations Assessment, Development and Evaluation
HCW	health care worker (nonphysician)
HIC	high-income country
HTN	hypertension
LIC	low-income country
LMIC	low- and middle-income country
LVH	left ventricular hypertrophy
MACE	major adverse cardiovascular event
MI	myocardial infarction
MIC	middle-income country
NCD	noncommunicable disease
PEN	WHO package of essential NCD interventions
PICO	population intervention comparator outcome
QALY	quality-adjusted life year
RAAS	renin-angiotensin-aldosterone system
SBP	systolic blood pressure

Executive summary

More people die each year from cardiovascular diseases than from any other cause. Over three quarters of heart disease and stroke-related deaths occur in low-income and middle-income countries. Hypertension – or elevated blood pressure – is a serious medical condition that significantly increases the risk of heart, brain, kidney and other diseases. Hypertension can be defined using specific systolic and diastolic blood pressure levels or reported use of antihypertensive medications. An estimated 1.4 billion people worldwide have high blood pressure, but just 14% have it under control. However, cost-effective treatment options do exist.

In this guideline, the World Health Organization (WHO) provides the most current and relevant evidence-based global public health guidance on the initiation of treatment with pharmacological agents for hypertension in adults. The recommendations target adult, non-pregnant patients who were appropriately diagnosed with hypertension and counselled about life-style modifications.

The guideline provides new recommendations on the threshold for the initiation of pharmacological treatment for hypertension, as well as recommendations on intervals for follow up, target blood pressure to be achieved for control, and the cadre of health care workers who may initiate treatment. The guideline provides the basis for deciding whether to initiate treatment with monotherapy, dual therapy or single-pill combinations, as well as guidance for countries selecting medicines and algorithms for hypertension control for their national guidelines for hypertension management.

The guideline was developed in accordance with the WHO Handbook for Guideline Development. In brief, the WHO Steering Group, in collaboration with the Guideline Development Group, developed key questions and rated outcomes to identify those critical for the development of the guideline. Conflicts of interest were handled in line with the current Compliance, Risk Management and Ethics (CRE) policy and all members of the GDG were asked to fill in the standard WHO Declaration of Interest (DOI) forms, which were reviewed. An overview of systematic reviews of the evidence was used to build summary of findings tables according to the Grading of Recommendations, Assessment, Development and Evaluations (GRADE) approach. The Guideline Development Group developed recommendations, considering the certainty of the evidence; the balance between desirable and undesirable effects; resource requirements and cost-effectiveness; health equity; acceptability, patient values and preferences, and feasibility.

Recommendations

1. RECOMMENDATION ON BLOOD PRESSURE THRESHOLD FOR INITIATION OF PHARMACOLOGICAL TREATMENT
WHO recommends initiation of pharmacological antihypertensive treatment of individuals with a confirmed diagnosis of hypertension and systolic blood pressure of ≥140 mmHg or diastolic blood pressure of ≥90 mmHg.
Strong recommendation, moderate- to high-certainty evidence
WHO recommends pharmacological antihypertensive treatment of individuals with existing cardiovascular disease and systolic blood pressure of 130–139 mmHg.
Strong recommendation, moderate- to high-certainty evidence

WHO suggests pharmacological antihypertensive treatment of individuals without cardiovascular disease but with high cardiovascular risk, diabetes mellitus, or chronic kidney disease, and systolic blood pressure of 130–139 mmHg.

Conditional recommendation, moderate- to high-certainty evidence

2. RECOMMENDATION ON LABORATORY TESTING

When starting pharmacological therapy for hypertension, WHO suggests obtaining tests to screen for comorbidities and secondary hypertension, but only when testing does not delay or impede starting treatment.

Conditional recommendation, low-certainty evidence

3. RECOMMENDATION ON CARDIOVASCULAR DISEASE RISK ASSESSMENT

WHO suggests cardiovascular disease risk assessment at or after the initiation of pharmacological treatment for hypertension, but only where this is feasible and does not delay treatment.

Conditional recommendation, low-certainty evidence

4. RECOMMENDATION ON DRUG CLASSES TO BE USED AS FIRST-LINE AGENTS

For adults with hypertension requiring pharmacological treatment, WHO recommends the use of drugs from any of the following three classes of pharmacological antihypertensive medications as an initial treatment:

1. thiazide and thiazide-like agents
2. angiotensin-converting enzyme inhibitors (ACEis)/angiotensin-receptor blockers (ARBs)
3. long-acting dihydropyridine calcium channel blockers (CCBs).

Strong recommendation, high-certainty evidence

5. RECOMMENDATION ON COMBINATION THERAPY

For adults with hypertension requiring pharmacological treatment, WHO suggests combination therapy, preferably with a single-pill combination (to improve adherence and persistence), as an initial treatment. Antihypertensive medications used in combination therapy should be chosen from the following three drug classes: diuretics (thiazide or thiazide-like), angiotensin-converting enzyme inhibitors (ACEis)/angiotensin-receptor blockers (ARBs), and long-acting dihydropyridine calcium channel blockers (CCBs).

Conditional recommendation, moderate-certainty evidence

6. RECOMMENDATIONS ON TARGET BLOOD PRESSURE

WHO recommends a target blood pressure treatment goal of <140/90 mmHg in all patients with hypertension without comorbidities.

Strong recommendation, moderate-certainty evidence

WHO recommends a target systolic blood pressure treatment goal of <130 mmHg in patients with hypertension and known cardiovascular disease (CVD).
Strong recommendation, moderate-certainty evidence
WHO suggests a target systolic blood pressure treatment goal of <130 mmHg in high-risk patients with hypertension (those with high CVD risk, diabetes mellitus, chronic kidney disease).
Conditional recommendation, moderate-certainty evidence

7. RECOMMENDATIONS ON FREQUENCY OF ASSESSMENT
WHO suggests a monthly follow up after initiation or a change in antihypertensive medications until patients reach target.
Conditional recommendation, low-certainty evidence
WHO suggests a follow up every 3–6 months for patients whose blood pressure is under control.
Conditional recommendation, low-certainty evidence

8. RECOMMENDATION ON TREATMENT BY NONPHYSICIAN PROFESSIONALS
WHO suggests that pharmacological treatment of hypertension can be provided by nonphysician professionals such as pharmacists and nurses, as long as the following conditions are met: proper training, prescribing authority, specific management protocols and physician oversight.
Conditional recommendation, low-certainty evidence

1 Introduction

More people die each year from cardiovascular diseases than from any other cause. Over three quarters of heart disease and stroke-related deaths occur in low-income and middle-income countries *(1)*. Blood pressure is the force exerted by circulating blood against the walls of the body's arteries, the major blood vessels in the body. Blood pressure is written as two numbers. The first number (systolic) represents the pressure in blood vessels when the heart contracts or beats. The second number (diastolic) represents the pressure in the vessels when the heart rests between beats. Hypertension – or elevated blood pressure – is a serious medical condition that significantly increases the risk of diseases of the heart, brain, kidneys and other organs *(2)*. Hypertension can be defined using specific systolic and diastolic blood pressure levels or reported use of antihypertensive medications. An estimated 1.4 billion people worldwide have high blood pressure, but just 14% have it under control *(2)*. However, cost-effective treatment options do exist.

A World Health Organization (WHO) guideline dealing specifically with raised blood pressure was last published more than 20 years ago – in 1999 – and is now outdated. In 2007, a comprehensive guideline on cardiovascular risk included some recommendations on hypertension but this also needs revision and updating in the light of new evidence and practice *(3)*. Guidance is particularly needed on some controversial issues, such as when to start treatment and whether laboratory testing and cardiovascular risk assessment are needed prior to starting treatment. In the past decade, WHO has included diagnosis and management of hypertension in a total cardiovascular risk approach as part of the WHO package of essential noncommunicable disease (NCD) Interventions (WHO PEN) 2007, 2010 and 2013. However, this approach has not included the most recent advances in pharmacological treatment.

The WHO Essential Medicines List (EML) identifies all classes of antihypertensive drugs – angiotensin-converting enzyme inhibitors (ACEi), calcium channel blockers (CCB), angiotensin receptor blockers (ARB) and thiazide diuretics – as essential. In June 2019, the EML included single-pill combination medications for hypertension. This further supports the evaluation of all classes of antihypertensive drugs as well as single-pill combinations in this current guideline.

Scope and objectives of the hypertension guideline

The 2021 WHO hypertension guideline aims to provide the most current and relevant evidence-based global public health guidance on the initiation of treatment (with pharmacological agents) for hypertension in adults. The recommendations target the general adult, non-pregnant, hypertensive population.

Although several countries and professional societies have guidelines on the topic of hypertension, these are specific to the population of that particular country or the specific setting or constituency of the professional society. Recent shifts in hypertension management, such as moving away from using beta-blockers as a first-line agent or the increased research and adoption of combination therapies and single-pill combinations, are all additional reasons for new guidance. The *Guideline for the pharmacological treatment of hypertension in adults* will be the first global guideline in the past two decades on the topic and will have specific relevance to low- and middle-income countries (LMICs).

The guideline provides new recommendations on the threshold for the initiation of pharmacological treatment for hypertension, recommendations on intervals for follow up, target blood pressure to be achieved for control, and the cadre of health care workers who may initiate treatment. It provides the basis for deciding whether to initiate treatment with monotherapy, dual therapy, or single-pill combination, as well as guidance for countries on selecting medicines for hypertension control in their national guidelines for hypertension management.

This guideline will replace the guidance in the modules *Evidence-based protocols* and *Risk-based CVD management* of the HEARTS technical package, as well as the guidance in the WHO PEN package regarding the threshold for initiation of treatment, and for the preferred pharmacological treatment for hypertension.

The guideline does not address measurement of blood pressure or diagnosis of hypertension. It addresses pharmacotherapy in individuals with a "confirmed" diagnosis of hypertension, such as hypertension diagnosed when blood pressure is found to be elevated on two different days.

Although this guideline does not address modifiable risk factors for hypertension such as unhealthy diet, physical inactivity, consumption of tobacco and alcohol, and being overweight or obese, a comprehensive treatment plan for hypertension must include addressing these risk factors through life-style modifications and other interventions *(2)*.

Nonpharmacological approaches to treatment or prevention of hypertension include:

- reducing salt intake (to less than 5g daily)
- eating more fruit and vegetables
- being physically active on a regular basis
- avoiding use of tobacco
- reducing alcohol consumption
- limiting the intake of foods high in saturated fats
- eliminating/reducing trans fats in diet *(2)*.

The guideline does not address hypertensive crisis as it is focused on chronic blood pressure management in regular care settings.

The objectives of the hypertension guideline are to:

- provide a blood pressure threshold for the initiation of treatment for hypertension;
- determine if laboratory tests or cardiovascular risk assessment are required prior to initiation of treatment for hypertension;
- determine the pharmacological agents with which to initiate treatment;
- determine the need to initiate treatment with monotherapy, dual therapy, or single-pill combinations;
- provide targets for blood pressure control in hypertension;
- provide intervals for follow up for patients with hypertension; and
- determine how nonphysician health care workers can participate in the management of hypertension.

Target audience

Primary audience

Clinicians/health care providers at all levels of health care.

Secondary audience

National NCD/CVD programme managers, health care academics, policy-makers setting practice recommendations, students, and hypertension medicine manufacturers.

2 Method for developing the guideline

2.1 Guideline contributors

In order to develop the hypertension guideline, WHO established three groups:

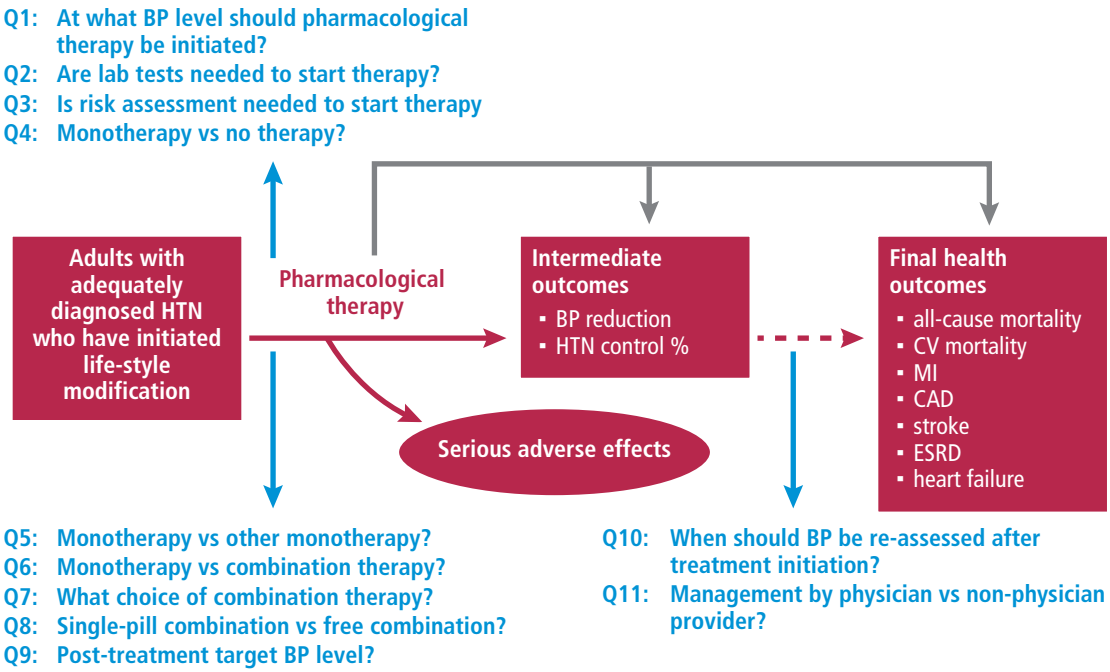
1. An internal WHO Steering Group to coordinate the guideline development process.
2. A Guideline Development Group (GDG), composed of hypertension (HTN) physicians, nephrologists, cardiologists, pharmacists, nurses, researchers, academics, policy-makers and representatives of patient groups, to review the evidence and develop recommendations. WHO selected the members of the GDG based on relevant expertise but it also considered appropriate representation by region and sex.
3. An External Review Group (ERG), composed of technical experts, representatives of HTN patient groups and ministries of health from low-resource countries, to provide peer review of the guideline and ensure recommendations are aligned with current global needs.

Annex 1 lists the contributors in each group. Annex 2 describes the process for declaring and managing conflicts of interest.

2.2 Analytical framework and PICOs

An initial GDG meeting was held in Geneva in July 2019 to determine the scope and PICO (population, intervention, comparison, outcome) questions of the guideline. The GDG first developed an analytical framework (Fig. 1) that demonstrates the impact of interventions on intermediate and final outcomes and displays the order of the key questions to better visualize and place them along the patient-flow pathway. Following this, and a preliminary scoping review and discussion between the steering group and methodologist, PICO questions were developed. These were considered, deliberated on, further refined and voted on during the meeting.

Fig. 1 Analytic framework for antihypertensive medication treatment



2.3 Outcome importance rating

Members of the WHO Steering Group, in consultation with the GDG and methodologist, developed a list of treatment outcomes most relevant to the care of individuals with HTN. The GDG then rated each outcome on a scale from 1 to 9 and indicated whether it considered each outcome critical (rated 7–9), important (rated 4–6) or not important (rated 1–3) for decision-making. The mean scores are provided in Annex 3.

2.4 Reviews of evidence

The WHO Steering Group, with the participation of the GDG, determined the scope of the guideline and identified eleven questions in the format of population, intervention, comparison, and outcomes (PICO) to guide the search for systematic reviews (Annex 4). Eleven overviews of reviews informed the guideline development process. A systematic search was carried out in PubMed, Embase, The Cochrane Library, and Epistemonikos to identify existing systematic reviews that answered the PICO questions published in 2015 or after. Suitable systematic reviews were then evaluated based on the following criteria:

- their methodology as appraised by the AMSTAR (Assessing the Methodological Quality of Systematic Reviews) tool;
- how directly they matched the PICO questions;
- whether they reported sufficient information to allow for an assessment of the certainty of the evidence (e.g. tables with characteristics of included studies, risk of bias assessments at the study level, results of meta analyses in forest plots);
- whether they reported evidence in subgroups of interest (e.g. patients with diabetes mellitus (DM), cardiovascular disease (CVD), chronic kidney disease (CKD) etc); and
- the date of the most recent review to ensure the most updated evidence was used.

The Systematic Review Team prioritized the most useful reviews for each question, comparison, outcome and subpopulation within questions, and included as many reviews as necessary to address each question comprehensively. (Web Annex A provides details of the search terms and strategy used.) No reviews were updated.

Two questions (PICO 2 and 10, see Annex 4) were not addressed in an existing systematic review, and evidence from primary studies was therefore used. In this case, the Systematic Review Team reviewed the list of studies used in existing guidelines, queried the GDG and content experts, and searched for primary studies.

A total of 159 systematic reviews and 17 additional primary studies were included. Most of these were traditional systematic reviews in which the authors conducted pairwise meta-analysis, whereas nine analysed individual patient data. The Systematic Review Team also identified several published network meta-analyses. (See Web Annex A for full details.)

Most of the included reviews were found to be of high certainty when assessed using the AMSTAR tool. Details of the selection process, and the reviews and studies included for each PICO question, are presented in Web Annex A.

Another overview of reviews was conducted to identify systematic reviews to inform other decision criteria in the evidence-to-decision framework, including people's values, resources, acceptability, feasibility and equity, presented in Web Annex A. This review was enriched by primary studies identified by GDG members. This review focused on evidence informing HTN management in low- and middle-income settings.

2.5 Certainty of evidence and strength of recommendations

The GDG rated the certainty of evidence and developed the recommendations using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach (4). When making recommendations, GRADE defines the certainty of a body of evidence as "the extent of our confidence that the estimates of an effect are adequate to support a particular decision or recommendation" *(5)*.

Members of the GDG, with the help of the methodologist, developed evidence profiles to summarize relative and absolute estimates of effects, and an assessment of the certainty of the evidence. One evidence profile for each comparison within a PICO question was constructed, using the online Guideline Development Tool GRADEpro (https://gradepro.org). When there were systematic reviews addressing the relative effects in specific subpopulations, separate evidence profiles were constructed for each subpopulation.

According to the GRADE approach, the certainty of the evidence can be high, moderate, low, or very low. Bodies of evidence from randomized controlled trials – which comprised almost the totality of those included in this guideline – start the assessment as high-certainty evidence but can be down-rated due to considerations of risk of bias, inconsistency, imprecision, indirectness, and publication bias. Table 1 presents an explanation of the different levels of certainty of the evidence *(5)*.

Table 1 Certainty of evidence and its implications

Certainty level	Definition
High	We are very confident that the true effect lies close to that of the estimate of effect.
Moderate	We are moderately confident in the effect estimate. (The true effect is likely to be close to the estimate of effect, but there is a possibility that it is substantially different.)
Low	Our confidence in the effect estimate is limited. (The true effect may be substantially different from the estimate of the effect.)
Very low	We have very little confidence in the effect estimate. (The true effect is likely to be substantially different from the estimate of effect.)

The strength of the recommendations reflects the degree of confidence of the GDG that the desirable effects (e.g. beneficial health outcomes) of the recommendations outweigh the undesirable effects (e.g. adverse effects). The strength of recommendations in this guideline was graded into two categories:

A strong recommendation is one for which the GDG was confident that the desirable effects of adhering to the recommendation outweigh the undesirable effects.

A weak or conditional recommendation is one for which the GDG concluded that the desirable effects of adhering to the recommendation probably outweigh the undesirable effects, but was not confident about these trade-offs.

2.6 Deciding upon recommendations

The GDG met virtually for four sessions in February 2021. Systematic reviews and GRADE tables were presented at the meeting. Formulation of recommendations and their strength were facilitated by the chair and supported by the methodologist, with the aim of achieving unanimous consensus. While the plan was to use a simple majority vote, full consensus was reached on all recommendations.

The GDG used evidence-to-decision tables to guide the process of developing recommendations. These tables addressed the certainty of evidence, the balance between desirable and undesirable effects, values, resource use and cost-effectiveness, equity, acceptability and feasibility. These tables are available in Web Annex B, and the full evidence profiles in Web Annex A. The WHO Steering Group noted remarks made by members of the External Review Group and considered incorporating these into the final guideline. Some evidence-to-decision frameworks were consolidated to provide recommendations that are more practical and implementable from an end-user perspective; hence, the 11 questions led to eight recommendations.

2.7 Funding

The development of this guideline was financially supported by the US Centers for Disease Control and Prevention and the World Health Organization.

3 Recommendations

3.1 Blood pressure threshold for initiation of pharmacological treatment

1. RECOMMENDATION ON BLOOD PRESSURE THRESHOLD FOR INITIATION OF PHARMACOLOGICAL TREATMENT
WHO recommends initiation of pharmacological antihypertensive treatment of individuals with a confirmed diagnosis of hypertension and systolic blood pressure of ≥140 mmHg or diastolic blood pressure of ≥90 mmHg.
Strong recommendation, moderate- to high-certainty evidence
WHO recommends pharmacological antihypertensive treatment of individuals with existing cardiovascular disease and systolic blood pressure of 130–139 mmHg.
Strong recommendation, moderate- to high-certainty evidence
WHO suggests pharmacological antihypertensive treatment of individuals without cardiovascular disease but with high cardiovascular risk, diabetes mellitus, or chronic kidney disease, and systolic blood pressure of 130–139 mmHg.
Conditional recommendation, moderate- to high-certainty evidence
Implementation remarks: ▪ Initiation of pharmacological hypertension (HTN) treatment should start no later than four weeks following diagnosis of HTN. If blood pressure level is high (e.g. systolic ≥160 mmHg or diastolic ≥100 mmHg) or there is accompanying evidence of end organ damage, initiation of treatment should be started without delay.

Evidence and rationale

The GDG reviewed evidence from 14 relevant systematic reviews that summarized data from a large number of randomized trials involving over 120 000 adult participants (Web Annex A). Evidence summaries are presented for the general population and for higher-risk populations (with diabetes (DM), coronary artery disease (CAD), prior stroke) and were presented for various systolic blood pressure (SBP) thresholds (Web Annex A).

The anticipated benefits of a lower blood pressure (BP) target (140 SBP in the general population and 130 SBP in a high-risk population) were reduction in mortality, cardiovascular mortality, stroke, myocardial infarction (MI) and heart failure events. The anticipated harms were mostly not serious side-effects, and some were a surrogate outcome, such as rise in creatinine that may not be clinically relevant. On average, treatment was associated with a reduction in deaths and cardiovascular events that ranged from 5 to 10/1000 and harms that ranged from 20 to 30/1000. The benefits were a reduction in severe events with significant morbidity and mortality whereas the harms were mostly not clinically significant.

In summary, the anticipated benefits were large and clearly outweighed the harms. The overall certainty varied from moderate to high, depending on the BP level and agent used.

Evidence-to-decision considerations

The value of antihypertensive therapy is well accepted by most patients, health care providers, health systems, professional societies and government agencies. From a patient's perspective, preventing cardiovascular events is highly valued. However, some individuals who are eligible for antihypertensive treatment may not present to care, may be lost to follow up, or are prescribed a treatment but fail to take/adhere to the treatment. Treatment may be perceived as low value from the perspective of an asymptomatic patient unless the person is convinced of a trade-off between immediate inconvenience/side-effects and potential long-term health gains *(6)*. The patient perception of an unfavourable cost-benefit may be further exacerbated by the requirement for out-of-pocket payment for appointments or medications. Therefore, the GDG considered that although there is important variability in stakeholder values, overall initiation of hypertension (HTN) medications is likely to be feasible and acceptable overall. Given that the barriers to accessing HTN care in low-income settings include low patient health literacy, lack of financial protections, and limited resources *(7)*, the GDG felt that health inequalities would probably be reduced by HTN treatment.

In terms of costs and resource requirements, the GDG acknowledged variability, based on the structure of a country's public health system and its economic status. Other costs, including human resources and medications, were considered moderate, given the benefits. Multiple sources of cost effectiveness are available from various countries, such as Argentina, Nigeria, the USA and UK *(8, 9, 10, 11, 12, 13)*, and for lower thresholds and higher-risk individuals *(14, 15, 16)*. Most cost-effectiveness estimates were clustered below USD 1000 per averted disability-adjusted life year (DALY) – well below the average 2017 GDP per capita for lower-middle income countries of USD 2188 (17), suggesting that pharmacological treatment could be very cost-effective for this group of countries. HTN treatment (treating all with BP ≥140/90 mmHg) has been shown to be cost-effective and a "best buy" intervention by the Kostova study *(8)*. Treating high-risk/CVD patients with baseline 130–139 mmHg has been shown to be cost effective, but not cost saving (SPRINT cost-effective analyses) *(15)*; value depends on maintaining the intervention effect for more than five years.

3.2 Laboratory testing before and during pharmacological treatment

2. RECOMMENDATION ON LABORATORY TESTING

When starting pharmacological therapy for hypertension, WHO suggests obtaining tests to screen for comorbidities and secondary hypertension, but only when testing does not delay or impede starting treatment.

Conditional recommendation, low-certainty evidence

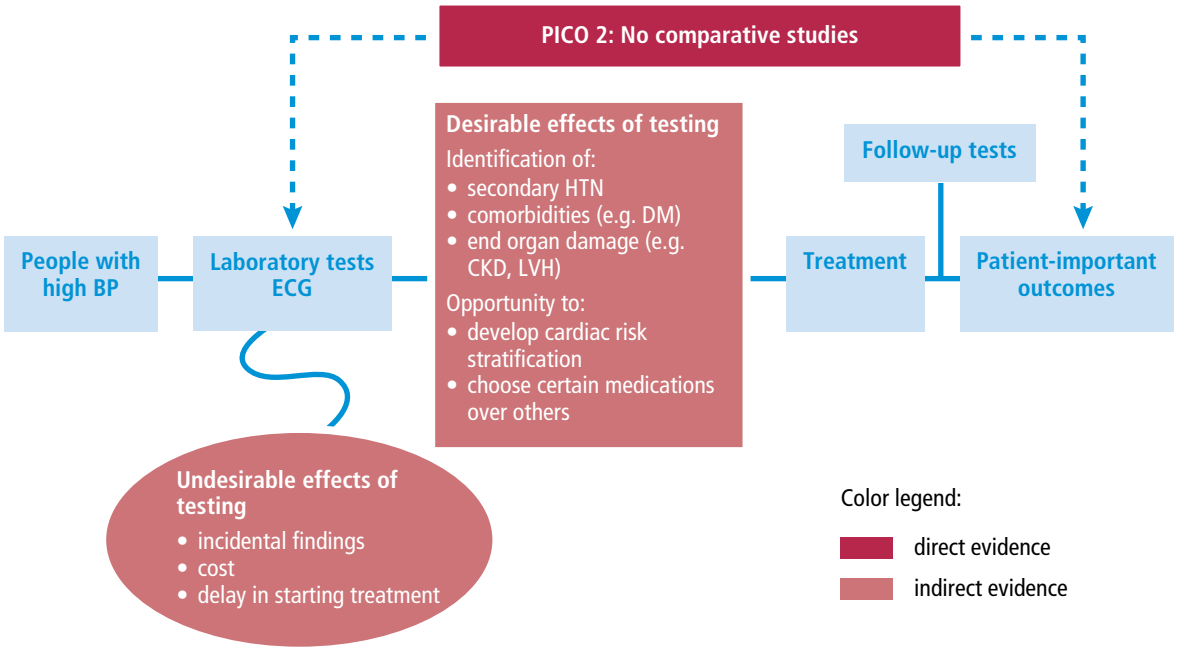
Implementation remarks:

- Suggested tests include serum electrolytes and creatinine, lipid panel, HbA1C or fasting glucose, urine dipstick, and electrocardiogram (ECG).
- In low-resourced areas or non-clinical settings, where testing may not be possible because of additional costs, and lack of access to laboratories and ECG, treatment should not be delayed, and testing can be done subsequently.
- Some medicines, such as long-acting dihydropyridine calcium-channel blockers (CCBs) are more suitable for initiation without testing, compared to diuretics or angiotensin-converting enzyme inhibitors (ACEi)/angiotensin-II receptor blockers (ARBs).

Evidence and rationale

Comparative studies that evaluate various testing strategies prior to the initiation of antihypertensive treatment were not identified, despite a search of the literature. Therefore, indirect evidence was sought to evaluate this question. An analytic framework was developed to conceptualize the rationale for obtaining diagnostic testing, such as laboratory tests and electrocardiogram (ECG), in this context (see Fig. 2). This framework identified the four most important reasons for obtaining testing, which are to diagnose secondary HTN, identify comorbidities (e.g. DM), identify end organ damage (e.g. chronic kidney disease (CKD) or left ventricular hypertrophy (LVH)), and for cardiac risk stratification.

Fig. 2 Framework for analysis



In terms of secondary HTN, various studies suggest a prevalence of 5–10% among patients with a diagnosis of HTN and a higher prevalence of 10–30% among patients with particularly high BP (e.g. over 175/115 mmHg) or BP that is resistant to treatment *(18, 19, 20)*. Morbidities and end organ damage that could be identified by testing patients with HTN are also common. One estimation indicates that 23%, 24% and 39% of patients with a diagnosis of HTN have one, two, three or more comorbidities respectively. Common comorbidities among patients with HTN that can be discovered via laboratory testing are hyperlipidaemia and diabetes, which have a prevalence of 56% and 27% respectively *(21)*. Testing at the point of starting HTN medications or for subsequent monitoring can also identify patients who develop certain adverse events after treatment (e.g. hyperkalaemia and acute kidney injury), thus providing a rationale for testing. Testing also had the additional advantage of identifying compelling indications for choosing certain medications over others. For example, identifying diabetes would favour the use of ACEis/ARBs, and identifying hyponatraemia would lead to not starting diuretics. Overall, the desirable effects of testing were judged to be at least moderate.

The framework also identified the most undesirable effects of obtaining testing, which were delay of starting treatment, cost and incidental findings. Delay of treatment was judged to be the most important concern since it can lead to losing the patient for follow up and the potential for adverse cardiovascular (CV) outcomes. Incidental findings on testing were thought to be less important. The undesirable effects of testing were judged to have smaller magnitude. On balance, desirable effects are likely to outweigh undesirable effects. The certainty of evidence across outcome was judged to be very low due to serious concerns about the indirectness of evidence.

Evidence-to-decision considerations

There is uncertainty about patients' values and preferences regarding the issue of testing before starting treatment for HTN. The cost of tests such as electrolytes, creatinine, lipid panel, glucose, HbA1C, urine dipstick, and ECG relative to overall costs of treatment and complications of HTN are small *(22)*. However, in less well-resourced settings, this cost can have a large impact. Furthermore, if additional tests like ECG or 24-hour ambulatory BP monitoring were added, the cost can become a barrier *(23)*. It is unknown whether testing would lead to cost saving or be cost-effective. Testing was judged to be acceptable to most stakeholders, particularly patients and health care providers, and to a lesser extent to those governing health systems. Requiring testing before starting HTN medications can exacerbate health inequities and may not be feasible in low-resource settings.

3.3 Cardiovascular disease risk assessment as guide to initiation of antihypertensive medications

3. RECOMMENDATION ON CARDIOVASCULAR DISEASE RISK ASSESSMENT

WHO suggests cardiovascular risk assessment at or after the initiation of pharmacological treatment for hypertension, but only where this is feasible and does not delay treatment.

Conditional recommendation, low-certainty evidence

Implementation remarks:

- Most patients with SBP ≥140 or DBP ≥90 mmHg are high risk and indicated for pharmacological treatment; they do not require cardiovascular (CVD) risk assessment prior to initiating treatment. CVD risk assessment is most important for guiding decisions about initiating pharmacological treatment for hypertension (HTN) in those with lower SBP (130–139 mmHg). It is critical in those with HTN that other risk factors must be identified and treated appropriately to lower total cardiovascular risk.
- Many CVD risk-assessment systems are available. In the absence of a calibrated equation for the local population, the choice should depend on resources available, acceptability and feasibility of application.
- Whenever risk assessment may threaten timely initiation of HTN treatment and/or patient follow up, it should be postponed and included in the follow-up strategy, rather than taken as a first step to indicate treatment.

Evidence and rationale

The most direct evidence is derived from an individual patient data meta-analysis by Karmali that compared the number of major adverse cardiovascular events (MACE) at five years when using a CVD risk assessment strategy (based on age, sex, body mass index, blood pressure, previous antihypertensive treatment, smoking, diabetes mellitus (DM), and history of CVD) vs BP levels alone for determining which patients receive treatment *(24)*. This analysis suggested that risk assessment can potentially prevent 310 MACE events in 1000 people over five years, which the GDG considered to be a moderate-to-large benefit. However, this evidence was indirect for many reasons, including the effect being dependent on the BP at presentation (graphs diverge at higher level of BP, compared with starting medications without risk assessment) and the fact that these trials did not actually randomize patients to the two strategies sought in PICO question 3 (see Annex 4). Furthermore this evidence should not suggest that people with intermediate risk would not have important treatment benefit.

There was no evidence of the undesirable anticipated effects of starting treatment based on cardiovascular risk assessment. However, delay in initiating care for HTN management and loss to follow up are important considerations, especially in low-resource settings.

The GDG deduced that benefits of risk assessment may not all be attributable to risk assessment per se, but rather to the various treatments provided for risk factors identified during risk assessment. The certainty of evidence across outcome was judged to be very low due to serious concerns about the indirectness of evidence. Overall, the desirable effects of risk assessment at or after initiating HTN medications outweighed the plausible undesirable effects, particularly when risk assessment was deemed not to delay the initiation of treatment.

Evidence-to-decision considerations

There is important uncertainty over the value stakeholders place on conducting a CVD risk assessment prior to starting pharmacological treatment, and it was noted that patient perspectives may vary, depending on the setting. In low-resource settings, patients may focus more on immediate treatment without having to bear additional costs for screening for other risk factors and treating them. Studies have also shown that in high-income countries (HICs) such as the United States, people of a lower socioeconomic status have lower control of blood pressure and higher CVD risk over the years *(25)*. Thus, in low-resource settings, adding one more step before initiating treatment may increase inequities, as those patients who have limited access to health care services may suffer delays in treatment or even end up not receiving HTN treatment at all.

In terms of costs, there is no direct evidence of whether treatment of HTN with or without risk stratification is more cost effective. The cost of implementation of CVD risk assessment should also account for capacity building of health care providers and the time taken to do so for each patient.

Cost of testing and delay in initiating care can be significant following a CVD risk stratification strategy in low-resource settings. Modelling by Gaziano et al. showed significant cost reduction using CVD risk-stratification before initiation of treatment in low-resource settings. However, screening costs, including the cost of obtaining risk-factor information, productivity costs due to work loss, care costs and travel time were not included in the analysis *(26)*.

A meta-analysis showed that proportional relative risk reduction in major CVD events from BP lowering did not differ substantially with the presence or absence of previous CVD events, coronary heart disease, or cerebrovascular disease. Hence, the absolute benefit of blood pressure lowering would be greatest in those with the highest absolute risk of CVD *(27)*.

3.4 Drug classes to be used as first-line agents

To develop a recommendation that is practical and implementable by end-users, the evidence-to-decision frameworks of PICO questions 4 and 5 (see Annex 4) were used to develop one recommendation.

4. RECOMMENDATION ON DRUG CLASSES TO BE USED AS FIRST-LINE AGENTS
For adults with hypertension requiring pharmacological treatment, WHO recommends the use of drugs from any of the following three classes of pharmacological antihypertensive medications as an initial treatment: **1. thiazide and thiazide-like agents** **2. angiotensin-converting enzyme inhibitors (ACEis)/angiotensin-receptor blockers (ARBs)** **3. long-acting dihydropyridine calcium channel blockers (CCBs).**
Strong recommendation, high-certainty evidence
Implementation remarks: ▪ Long-acting antihypertensives are preferred. ▪ Examples of indications to consider specific agents include diuretics or CCBs in patients over 65 years or those of African descent, beta-blockers in ischaemic heart disease, ACEis/ARBs in patients with severe proteinuria, diabetes mellitus, heart failure or kidney disease.

Evidence and rationale

Data from 32 systematic reviews were used to derive evidence on benefits and harms of various medication classes (19 for comparisons against placebo and 13 for head-to-head comparisons). These reviews summarized the results of a many large randomized trials (Web Annex A). The anticipated benefits were considered to be large. Mortality and major adverse cardiac events (MACE) reduction per 1000 treated people for the various classes were 3 and 14 (low-dose thiazide), 12 and 39 (high-dose thiazide), 23 and 48 (ACEi), 8 and 23 (CCB), 2 and 8 (beta-blockers) and 14 and no data for MACE (ARBs) respectively. The anticipated adverse events were judged to be moderate. Compared to placebo, 60 and 100 additional adverse events per 1000 treated people were observed for thiazides and beta-blockers respectively. Withdrawal from ACEi treatment and cough per 1000 treated people were 12 and 26 respectively. A systematic review of studies of pharmacotherapy for HTN in sub-Saharan Africa showed a rate of side-effects of CCB of 6% (headache), 2% (dizziness) 2% (ankle oedema) *(28)*.

In terms of the head-to-head comparisons among various classes, there was a smaller body of evidence, with less data available on hard endpoints and patient-important outcomes. Comparisons showed overall minimal differences in SBP or DBP. For example, ACEis/ARBs vs CCBs differed by less than 2 mmHg, and so did comparisons between ACEis/ARBs vs thiazide or ACEis vs ARB. There were more stroke events with beta-blockers than CCBs or ACEis/ARBs.

The anticipated benefits clearly outweighed the potential harms for three classes of medications: thiazide and thiazide-like agents, ACEis/ARBs, and long-acting dihydropyridine CCBs. The adverse events of these three classes were infrequent, usually mild, and can be managed or another agent can be substituted. The amount of BP reduction appeared to be a more major determinant of reduction in CV events than the choice between these three classes of antihypertensive medications, as was shown in several landmark trials (ALLHAT, VALUE, CAMELOT trials) *(29, 30, 31)*. This balance of benefits and harms was not as clear for beta-blockers as a first choice for HTN management.

In terms of potential subgroups of patients that may benefit more from specific medication classes, ALLHAT (Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial) suggested greater BP reduction in individuals of African descent with chlorthalidone than lisinopril, and that stroke was significantly less likely with the diuretic than with the lisinopril in this group of patients than in those of Caucasian ethnicity *(32)*. Other studies suggested benefit of diuretics or CCBs in patients over 65 years of age or of African descent, beta-blockers in patients with HTN who are post-myocardial infarction, ACEis/ARBs in diabetes mellitus, heart failure or kidney disease *(33, 34)*. Diuretics were likely to be the most efficacious medications, and CCBs the least efficacious medications for the prevention of heart failure.

The overall certainty of evidence varied from high to moderate for these three classes of medications when compared against placebo. It was noted that diuretic trials were older and practice patterns may have changed over time, and that the severity and stage spectrum of diabetes mellitus and chronic kidney disease varied in the available trials. In addition, evidence supporting the efficacy of antihypertensive drug therapy is derived from trials conducted in adults at high risk for CVD/atherosclerotic CVD. Since CVD risk increases with higher levels of BP, and given that risk factors for CVD tend to track together, the assumption of greater benefits using CVD risk could be attributed to this.

Evidence-to-decision considerations

The value of antihypertensive therapy is well accepted by most patients, health care providers, health systems, professional societies and government agencies. From a patient's perspective, preventing cardiovascular events is highly valued. However, some individuals who are eligible for antihypertensive treatment may evade efforts aimed at treatment or are prescribed a treatment but fail to take/adhere to the treatment. The asymptomatic nature of the disease and concern about adverse events are the likely driver for this perspective. Interviews of patients in England suggested greater acceptance of antihypertensive drug therapy with higher socioeconomic status.

In one study, as many as 35% of the Caucasians and 20% of the South Asians in the two lowest socioeconomic categories told their interviewer that they would not accept antihypertensive drug therapy *(35)*. Shahaj et al. *(6)* synthesized six qualitative and 29 quantitative reviews and identified a range of individual and social factors that affect treatment adherence, including familial (lack of support, need for separate meals), and environmental (sense of security, local amenities, healthy food availability). A review by Fragasso et al. *(36)* suggested that quality of life on antihypertensive therapy is an important issue because clinicians are asked to initiate drug therapy in mostly asymptomatic patients who are never happy to become, instead, symptomatic because of side-effects. Therefore, the GDG considered that there is important variability in stakeholder values, but overall initiation of HTN medications is likely to be feasible and acceptable overall. Considering the ample literature on disparities in adherence to BP medication regimes and cardiovascular outcomes based on race or socioeconomic status, treatment was judged to reduce health inequities.

In terms of costs and resource requirements, thiazide-like agents, ACEis/ARBs and long-acting dihydropyridine CCBs are available as generic drugs, are simple to manufacture, and should be available at low cost globally. Other costs related to workforce requirements, provision of infrastructure, laboratory testing, lost work time etc. are real but modest. Numerous modelling studies demonstrate cost effectiveness of antihypertensive therapy, which is especially beneficial in LMICs where large numbers of adults have untreated HTN, as long as medications are available at low cost. Models were available from many countries, including Bangladesh, Ghana and Nigeria *(37, 38, 39, 40)*.

3.5 Combination therapy

To develop a recommendation that is practical and implementable by end-users, the evidence-to-decision frameworks of PICO questions 6, 7 and 8 (see Annex 4) were used to develop one recommendation.

5. RECOMMENDATION ON COMBINATION THERAPY

For adults with hypertension requiring pharmacological treatment, WHO suggests combination therapy, preferably with a single-pill combination (to improve adherence and persistence), as an initial treatment. Antihypertensive medications used in combination therapy should be chosen from the following three drug classes: diuretics (thiazide or thiazide-like), angiotensin-converting enzyme inhibitors (ACEis)/angiotensin-receptor blockers (ARBs), and long-acting dihydropyridine calcium channel blockers (CCBs).

Conditional recommendation, moderate-certainty evidence

Implementation remarks:

- Combination medication therapy may be especially valuable when the baseline BP is ≥20/10 mmHg higher than the target blood pressure.
- Single-pill combination therapy improves medication-taking adherence and persistence and BP control.

Evidence and rationale

The GDG developed three PICO questions to address: monotherapy vs combination therapy as a first-line treatment for HTN, a comparison of the various combination therapies, and a comparison of single-pill combinations vs multiple-pill combinations. These three questions were addressed separately in the evidence profiles and evidence-to-decision framework, but eventually led to one recommendation. The evidence base consisted of six, seven and eight systematic reviews respectively (Web Annex A).

Evidence summaries demonstrate several comparisons of combination therapy to monotherapy. Data on mortality, MACE and other hard endpoints were imprecise. Combination therapy lowered SBP more than monotherapy did (e.g. standard dose CCB combined with ARB vs high dose CCB; or ACEi and ARB combination vs either drug class alone) and had fewer adverse events (standard dose CCB combined with ARB vs high dose CCB). Data on cardiovascular outcomes are limited from randomized trials. A large nonrandomized study from Italy (125 635 patients, age 40–85 years) evaluated those who started antihypertensive treatment with one drug vs a two-drug single-pill or free combination. Propensity score adjusted analysis suggests that an initial two-drug single-pill or free combination was associated with significant reductions in the risk of death (20%, 11–28%) and hospitalization for cardiovascular events (16%, 10–21%) compared with initial monotherapy *(41)*. Combination antihypertensive therapy may be associated with fewer side-effects due to use of lower doses of each drug.

A comparison of the various combination therapies suggested overall effectiveness of combination therapies that contained the three drug classes of diuretic, ACEi/ARB and CCB. Other desirable effects of a combination therapy are improved treatment adherence and persistence. However, many of these studies used a single-pill combination, thereby confounding the question of monotherapy vs combination therapy. A meta-analysis compared adherence and persistence between groups of patients taking antihypertensives as single-pill combinations vs free-equivalent components based on 12 retrospective database studies. Adherence, measured as the mean difference in medication possession ratio, was 8–14% higher with a single-pill combination. Persistence was also twice as likely *(42)*. A second systematic review demonstrated that simplifying dosing regimens results in significant improvements in medication adherence, ranging from 6% to 20% *(43)*.

The desirable effects of greater adherence/persistence, improved BP control, and potentially improved clinical outcomes of combinations of the three classes of antihypertensive therapy compared outweigh the undesirable effects such as side-effects, particularly when provided as a single-pill combination. The overall certainty in evidence was low across the outcomes of interest, noting that evidence was limited in terms of hard endpoints.

Evidence-to-decision considerations

In terms of stakeholder values and preferences about monotherapy vs combination therapy or the various combination therapies, data were minimal. No important variability in values was expected with regard to the critical outcomes. A systematic review demonstrated that simplifying dosing regimens results in significant improvements in medication adherence, ranging from 6% to 20% *(43)*. Considering the comparative ease of using a single-pill combination over multiple-pill combinations, and the anticipated impact on adherence and persistence, the GDG judged that from a patient perspective the single-pill option will be favoured by most.

Combination therapy is accompanied initially by a moderate increase in resource requirements, such as procurement, supply chain, and direct medication costs. Some combinations may be expensive, or not allow for exact dosing of both agents. However, the net benefit of improved BP control and reduction of major events associated with the hypertensive process compared to the increase in cost is large. BP control is also likely to be achieved sooner with combination therapy. Many modelling studies that evaluated combination vs monotherapy used a fixed dose (thereby not truly addressing the question). One model from Japan used data from randomized trials and compared low-dose combination therapy of controlled-release nifedipine (20 mg/day) plus candesartan (8 mg/day) vs titrated monotherapy of

candesartan. In the combination therapy group, higher efficacy and lower incremental treatment cost (dominance) were observed when compared to the monotherapy group *(44)*. A retrospective cohort study that used the 2008–2012 BlueCross BlueShield of Texas claims suggests that mean annual drug utilization costs were highest for a single-pill combination strategy. However, disease-related inpatient services utilization costs were lower compared with the up-titration strategy, which may offset initial costs *(45)*. In one model from China, olmesartan/amlodipine as a single pill was dominant, compared with olmesartan and amlodipine free combination and valsartan/amlodipine single-pill combination *(46)*. In a second study, there was a reduction in the cost of therapy of 33%, with a saving of USD 19 per patient/month after switching from free combination to the single-pill combination *(47)*.

Since single-pill combination therapy increases medication adherence and persistence, which could improve HTN control rates and decrease major clinical events, the impact on health equities is expected to be favourable. In terms of acceptability, combination therapy, including in a single-pill form, can initially be met with scepticism among stakeholders, including health care providers. However, this initial scepticism may improve once BP control improves. Despite effective, safe, affordable, and available pharmacological antihypertensive agents, the control rates of HTN are dismal worldwide, and over the last 5 to 10 years have been decreasing in some HICs, and in LMICS, in tandem with increasing major cardiovascular events. Over 30% of the world population has HTN and only 13.8% of cases are considered controlled (48). One major reason for this poor level of control (one in seven) is that most patients only receive monotherapy, whereas empirical evidence demonstrates that most patients require two drugs or more to achieve optimal and sustained control *(44, 46, 49, 50, 51, 52)*. The rationale for recommending a combination therapy, particularly in a single-pill approach, is based on the following considerations:

- most individuals with HTN will eventually require two or more antihypertensive agents to achieve BP control;
- the combination of two agents from complementary classes yields greater BP-reduction efficacy (at least equal to the sum of the efficacy of each chosen agent);
- lower doses of each agent are needed, which results in a reduction of side-effects and the fact that use of complementary classes of antihypertensive agents may mitigate the side-effects of each agent;
- adherence and persistence are increased; and
- simplified logistics can lead to fewer stock-outs and a reduced pharmacy inventory *(53, 54)*.

In terms of feasibility, a study from India compared prices of antihypertensive single-pill combinations and equivalent single-agent pills in the private health care sector. The results suggested that manufacturers have priced the combination higher than the price of its components. These data demonstrate that the price of combination pills could be lowered to match the combined price of the component, and that manufacturing costs and market forces do not present a barrier to the implementation of antihypertensive combination pills *(55)*. Thus, the intervention is likely feasible to implement. The GDG acknowledged some challenges to single-pill combinations, such as limited flexibility in modifying the doses of individual components, and difficulty in attributing side-effects to one of its components *(56)*.

Although randomized trials addressing this issue are not abundant, and those available are not sufficiently large or conducted for a long enough period to clearly address differences in major clinical events, the initial combination treatment approach has been in place for over 15 years in large health systems, such as the Kaiser Permanente system in the United States *(57)* and is a major component of the WHO Global HEARTS Programme and the PAHO HEARTS in the Americas Initiative *(53)*. Recently, combination antihypertensive medications in a single pill have been added to the WHO Essential Medicines List *(49)*. This approach has demonstrated general acceptance by government, public, and private stakeholders and is demonstrating success in increasing HTN control rates worldwide.

3.6 Target blood pressure

6. RECOMMENDATION ON TARGET BLOOD PRESSURES
WHO recommends a target blood pressure treatment goal of <140/90 mmHg in all patients with hypertension without comorbidities.
Strong recommendation, moderate-certainty evidence
WHO recommends a target systolic blood pressure treatment goal of <130 mmHg in patients with hypertension and known cardiovascular disease (CVD).
Strong recommendation, moderate-certainty evidence
WHO suggests a target systolic blood pressure treatment goal of <130 mmHg in high-risk patients with hypertension (those with high CVD risk, diabetes mellitus, chronic kidney disease).
Conditional recommendation, moderate-certainty evidence

Evidence and rationale

The evidence base consisted of five systematic reviews as well as a review of the SPRINT trial *(58)*. Evidence profiles were constructed for various BP treatment targets, based on age and comorbidities (Web Annex A).

The desirable effects of lower target BP (per 1000 treated patients) were: a reduction in mortality of 27 (for SBP <120 vs <130–139) and of 7 (for SBP 140/90 vs 150–160/95–105); a reduction in cardiovascular mortality of 40 (for SBP <130 vs <130–149) and 6 (for SBP 140/90 vs 150–160/95–105); and a reduction in stroke of 17 (for SBP <130 vs <140). The increase in serious adverse events per 1000 treated patients was 20 (for SBP <130 vs <140) and 5 (for SBP 140/90 vs 150–160/95–105).

Summary results from a systematic review focusing on adults 65 years and older by Murad et al *(59)*. suggests that treatment to a lower BP target in individuals 65 years or older leads to a significant reduction in all-cause and CVD mortality, chronic kidney disease, myocardial infarction, or stroke outcomes. Similar conclusions were provided by another systematic review by Reboussin et al *(60)*. Neither of these meta-analyses was able to account for the high risk of patients enrolled in the available trials – at least in SPRINT and ACCORD *(11, 61)*. Therefore, the GDG cautions against applying this evidence to lower-risk patients with raised BP or HTN – specifically, those not meeting trial eligibility criteria for SPRINT, ACCORD or SPS3 *(62)*. Network meta-analyses found a similar direction of effect but more optimistic beneficial effect sizes with intensive treatment *(63, 64)*.

In patients with comorbidity (CAD, DM, CKD) there is consistent benefit with lower targets (variable thresholds); however, data in these subgroups were imprecise and the evidence was less certain. Adverse events such as dizziness in intensive control group and ischaemia in patients with coronary artery disease can shift the balance of benefits and harms in those aged 65 years or older. Concern about lower adherence due to the need for extra patient and provider effort to reach lower targets should also be balanced against intensive control. The overall certainty of the evidence was judged to be moderate, with large benefits and moderate harms. The GDG made a judgement that the desirable effects outweigh the undesirable effects at a treatment goal of <140/90 mmHg in all patients with HTN without comorbidities and of <130 mmHg in high-risk patients with HTN – those with high CVD risk, diabetes, chronic kidney disease.

Evidence-to-decision considerations

From a patient perspective, HTN is often a silent disease and patients may not take antihypertensive medications as directed because the positive effects of these medications are not as obvious as potential side-effects *(61)*. Society and patients want to avoid premature mortality or disability. Serious adverse events are also feared, but their duration and severity are often not well characterized in trials. Lower targets are likely acceptable to other stakeholders, such as governments and providers, though there are usually several competing priorities and interests – especially the more acute demands of, and a higher priority placed on, acute conditions and health emergencies. Many well-known barriers to access to HTN care in low-income settings exist *(6)*. Investment in the primary health care platform required for effective HTN management is often a challenge. Countries with low rates of HTN control using more conservative BP thresholds may feel burdened by any request to set more ambitious BP treatment goals, even if only in selected high-risk patients.

Intensive treatment for selected patients adds complexity for health workers; emphasis on team-based care in low-resource settings means that simple, protocolized care is needed. Intensive treatment for some patients complicates treatment protocols and may lead to decisional overload, especially for health workers with more limited training and/or autonomy.

On the other hand, strict BP targets in the general population with HTN are likely to be less acceptable to stakeholders. Most available evidence is derived from high-risk patients receiving intensive treatment and not the general population living with HTN. Treating HTN will reduce health inequity because preventing CV events reduces mortality across the population. Uncontrolled HTN might be over-represented in vulnerable populations. Therefore, improvement of HTN treatment and control through better treatment and a lower BP target could reduce long-standing inequality.

Regarding costs, intensive treatment in the SPRINT trial meant one additional medication, one additional office visit, and one additional laboratory test evaluation on average, and additional titration visits per participant over 3.25 years, compared with standard treatment. In the United States, this translates to about USD 23 000 more per patient per year over their remaining lifetime *(14, 15)*. Health care costs are much less in countries other than the United States. Treating to lower BP targets will have diminishing returns in progressively lower-risk patients as the magnitude of benefit becomes smaller. A cost-effectiveness study of screening and optimal management of HTN, diabetes mellitus and chronic kidney disease in an Australian setting found that an intensive management of previously uncontrolled HTN compared with usual care resulted in an incremental cost-effectiveness ratio of AUD 2588. The study does not specify the target BP for the comparisons *(65)*. A SPRINT trial health economic analysis provided similar inferences *(48, 50)*.

3.7 Frequency of re-assessment

7. RECOMMENDATION ON FREQUENCY OF ASSESSMENT
WHO suggests a monthly follow up after initiation or a change in antihypertensive medications until patients reach target.
Conditional recommendation, low-certainty evidence
WHO suggests a follow up every 3–6 months for patients whose blood pressure is under control.
Conditional recommendation, low-certainty evidence

There was a minimal number of comparative studies that evaluated different follow-up lengths after initiation of HTN medications. One randomized controlled trial compared a follow-up interval of three months to an interval of six months in family practice clinics in Canada. Participants (age 30–74 years) had essential HTN that was controlled for at least three months before entry into the study. Mean BP, control of HTN, patient satisfaction and adherence to treatment were similar between the two groups *(66)*. A retrospective population-based cohort study of family-practice clinics in the UK *(67)* studied 88 756 adults with HTN (1986–2010). This study showed that in patients newly diagnosed with HTN, those with >1.4 months prior to initiation of treatment had a hazard ratio of 1.12 (1.05–1.20) for a major adverse cardiovascular event (MACE) compared to those who started treatment at <1.4 months. For patients who were initiated on treatment, those who waited >2.7 months before re-evaluation had a hazard ratio of 1.18 (1.11–1.25) for MACE compared to those reassessed at <2.7 months. In addition, when reviewing protocols of large HTN trials that demonstrated important improvement in cardiovascular events with BP control, such as ACCORD and SPRINT, the initial length of time to follow up was one month *(68, 69)*. This evidence indirectly suggests the appropriateness of this initial follow up in settings that conferred important benefit.

The anticipated desirable consequences of shorter follow up are better BP control and monitoring of side-effects, and perhaps improved adherence. Longer follow-up times are expected to lead to loss to follow up. A systematic review of the impact of interventions to improve medication adherence in adults prescribed antihypertensive medications suggested a decrease in adherence with an increase in time between intervention and follow up *(70)*. The undesirable consequences of shorter follow up are the burden on patients and the health system. Certainty relating to these effects is very low.

The GDG found no evidence related to the question of the optimal follow-up time after the point that the treated HTN patient achieves stable blood pressure control.

Evidence-to-decision considerations

Data on what patients consider significant in terms of the length of follow up after initiation of HTN medications are lacking. Many patients, particularly those aged 65 years or older, or who live alone, are likely to be reassured by more frequent monitoring of BP, which can identify early signs of clinical deterioration and provide a sense of security *(71)*. However, younger asymptomatic patients may have the opposite perspective and find that frequent monitoring interferes with work and family responsibilities. Telemonitoring may reduce the need for follow up, especially for patients living in areas remote from health care facilities. However, despite existing evidence on the effectiveness of telemonitoring for patients with HTN, there is no empirical evidence of its long-term outcomes or generalizability to patients with various backgrounds and educational levels *(72)*.

Data on costs, resources and cost-effectiveness were unavailable. Frequent follow up is anticipated to be associated with additional resource requirements, which may be offset by improved adherence, BP control and improved outcomes that are important to patients. Burden on the health system may be reduced by involving nonphysician providers in follow up.

The WHO GDG considered a one-month follow up after initiation of medications for HTN to be a reasonable approach, whereas intervals of 3 to 6 months can be applied when patients' BP is close to the target and stable. Due to the lack of comparative data, these recommended intervals should be viewed as suggestions and may be modified, based on feasibility and other contextual factors. Such intervals were judged to be feasible and acceptable to key stakeholders. The impact of such follow-up lengths on health equity is unclear.

3.8 Administration of treatment by nonphysician professionals

8. RECOMMENDATION ON TREATMENT BY NONPHYSICIAN PROFESSIONALS

WHO suggests that pharmacological treatment of hypertension can be provided by nonphysician professionals such as pharmacists and nurses, as long as the following conditions are met: proper training, prescribing authority, specific management protocols and physician oversight.

Conditional recommendation, low-certainty evidence

Implementation remarks:

- Community health care workers (HCWs) may assist in tasks such as education, delivery of medications, blood pressure (BP) measurement and monitoring through an established collaborative care model. The scope of hypertension care practised by community HCWs depends on local regulations and currently varies by country.
- Telemonitoring and community or home-based self-care are encouraged to enhance the control of BP as a part of an integrated management system, when deemed appropriate by the treating medical team and found feasible and affordable by patients.
- Physician oversight can be done through innovative methods such as telemonitoring or similar to ensure access to treatment is not delayed.

Evidence and rationale

PICO question 11 (see Annex 4) addressed BP management by nonphysician health care workers (HCWs) as well as self-management by patients. The evidence base for this question consisted of 11 systematic reviews (Web Annex A). The available evidence focused on evaluating care models in which BP control was managed by pharmacists, nurses, dietitians, and community HCWs. The outcome assessed in these studies were BP level and control. There was no data on cardiovascular events. Although the certainty of evidence was in general low, the magnitude of effect showed better control in 91–264 more patients per 1000 (pharmacist studies) and an SBP/DBP reduction of 1–8 mmHg (nurse/HCW/dietitian studies). No study showed that nonphysician management was inferior to physician management.

A systematic review by Greer et al., showed that pharmacist-managed care led to better BP control (relative risk 1.44 or 170 more controlled per 1000) with no obviously reported difference in adherence, clinical events or quality of life *(73)*. A systematic review by Anand has shown that in LICs and MICs, task sharing with pharmacists led to reductions of 8 mmHg SBP and 3.74 mmHg DBP. Similar results were yielded by task sharing with nurses (5.34 mmHg lower), dieticians (4.67 mmHg lower), and community HCWs (3.67 mmHg lower) *(74)*. Data on undesirable effects (harms) were unavailable, which may be due to publication bias, or reflect minimal harms.

In terms of self-management, a systematic review by Tucker *(75)* shows that self-monitoring by patients led to a 3.24 mmHg lower level SBP and 1.5 DBP, both statistically significant, and better BP control, as long as self-monitoring was remotely managed by a HCW. However, the study limitation was the inability to adequately blind participants to the intervention. There was minimal evidence about self-titration on BP medications.

The GDG also concluded that since the evidence was from HICs, it may be less applicable to other settings and that training of nonphysician HCWs varies considerably between countries. Overall the certainty of evidence was low, with large anticipated desirable effects and a small magnitude of undesirable effects.

Evidence-to-decision considerations

There is significant variability in patient and provider perspectives. Overall, society and patients want to reduce the risk of premature mortality or morbidity. Most of the available quantitative data were focused on remote monitoring and not specifically on whether patients preferred BP being managed by physicians vs other providers, which was the primary question. Limited information provided mixed results, with some patients appreciating some applications of self-care while others were concerned that being managed by others could harm the patient–doctor relationship, but these comments were related to use of home-monitoring devices. In some studies in which BP was managed by nonphysicians, there was good patient satisfaction and high retention, suggesting at least willingness, if not a preference, to have BP managed by nonphysicians *(76, 77)*.Conversely, in-depth interviews with a sample of patients in the UK explored nurse and pharmacist prescribing and demonstrated that patients had concerns about clinical governance, privacy and whether sufficient space was available to provide the service in community pharmacies. Participants had less concern about nursing management *(78)*. Another study from Scotland explored patients' perspectives on pharmacist prescribing and reported high patient satisfaction, but 65% stated that they would prefer to consult a doctor *(79)*. Presumably, health inequities are reduced, since task shifting in the public sector increases access to those using public health vs private health. Increasing access in underserved areas can improve inequities.

Regarding costs, Jacob et al. *(80)* synthesized data from 31 studies (24 from the US) and suggested that studies that use community team approaches project a cost around USD 200/person/year to implement, but with cost-savings for prevention of CVD outcomes such that net costs had a median cost of USD 65/person/year, with 10 studies indicating negative or cost-savings overall. Most cost/quality-adjusted life year (QALY) estimates were between USD 3888 and USD 24 000/QALY, with pharmacist-led programmes being more cost-effective than nurse-led ones. Only two were >USD 50 000/QALY out of 28 studies.

Most of the remaining cost data presented were related to self-monitoring and not to the question of physician-led vs nonphysician-led care. However, if it is assumed that nonphysician salaries are lower, then potentially costs will be lower, but that assumes that only limited effort by physicians is involved in any oversight of nonphysicians. Kulchaitanaroai et al. found similar results with a physician–pharmacist collaborative system *(81)*.

The two available analyses, by Jacob et al. and Kulchaitanaroai et al., focused on team-based interventions as opposed to specifically physician vs other provider, and it is not clear if incremental cost-effectiveness ratios fit countries in all economic categories, nor whether the countries' willingness- to-pay thresholds were analysed. All values appear to be below USD 50 000/QALY. For the US, the results were highly cost-effective, with most estimates well under USD 50 000/QALY but it remains unclear exactly how these may be translated to countries in lower economic categories. Even at USD 10 000/QALY, however, this would be acceptable for most MICs, though perhaps not for all LICs. However, if the costs were the same or lower in programmes led by nurses or pharmacists compared to those led by physicians then cost-saving was likely.

The GDG proposed four conditions that must be met for nonphysician-prescribing of antihypertensives. These focused on the prescribers having proper training, prescribing authority in their locale, working within specific management protocols and having physician oversight. Community HCWs were suggested as personnel who could assist with tasks such as education, delivery of medications, BP measurement and monitoring through an established collaborative care model.

Telemonitoring supervised by HCWs, and community- or home-based self-care, were considered as tools to enhance BP control as part of an integrated management system.

4 Special settings

4.1 Hypertension in disaster, humanitarian and emergency settings

Hypertension (HTN) is seen in a range of humanitarian crises and disaster settings (natural or humanmade). This includes, but is not limited to, the wars in Syria and Iraq, the impact of the Great East Japan Earthquake and Hurricane Katrina, and the living conditions of Palestinian refugees. The burden of HTN on those populations can be considerable *(82)*. There are very little data on HTN control, access to care and treatment, and patient understanding of HTN from Africa and Asia (except Japan), despite protracted refugee situations on these continents. Violent and protracted conflicts are disastrous to civilian populations and their health care systems, and result in interruptions to treatment and care *(83, 84)*. Armed conflicts are associated with increased short-term and long-term cardiac morbidity and mortality and increases in blood pressure (BP) *(85)*. Following exposure to conflict, research in military populations shows that post-traumatic stress disorder and severe injury are independent risk factors for the development of HTN *(86)*. The rates of treatment ranged from 53.4% to 98.1% of patients with HTN in this population *(87, 88)*.

There are currently no data regarding target BP or the best antihypertensive agent to treat disaster-related HTN. Opinion-based recommendation is that the target BP control level should be less than 140 mmHg for SBP and less than 90 mmHg for DBP. According to Kario et al, long-acting CCBs are preferred because they are metabolically neutral, and best at reducing BP variability, which is an independent predictor of clinical outcomes, especially stroke. In addition, the BP-lowering effect of long-acting CCBs is dose-dependent, and the degree of BP reduction that can be anticipated from these agents is known *(89)*. Despite the challenges of working in humanitarian settings, several agencies have produced guidelines for the identification and management of HTN. The WHO's Interagency Emergency Health Kit has included a supplementary module with antihypertensive medications since 2017, but it is unclear how widely these are being used *(90, 91)*. According to a personal communication from a physician who treated HTN in Syrian refugees, the treatment was variable, and dependent on whatever drug samples were available in the clinic. They had limited choices, including atenolol, lisinopril, and verapamil. Treatment was tailored to the patient's history. For example, patients with a history of coronary artery disease received atenolol and lisinopril, patients with diabetes received lisinopril, and patients with migraine received verapamil.

Assessment of HTN and appropriate resourcing to treat it should be a priority for agencies providing emergency and longer-term care for patients after or during humanitarian crises to prevent significant mortality and morbidity. Further studies are needed to accurately estimate prevalence of HTN in crisis-affected populations throughout the world and to evaluate the best treatment approach for this population.

Humanitarian crises and disaster settings (natural or humanmade) can affect health care and services in many different ways. A list of potential barriers that can affect the management of individuals with hypertension is as follows:

- significant decline of living standards
- loss/destruction of health care facilities
- flight of medical personnel causing shortage of medical care providers
- severe shortage of medicines
- lack or absence of essential supplies, equipment and materials
- compromise of the provision of primary and secondary health care
- interruption of water, food and electricity
- lack of morbidity and mortality data due to destruction of information systems and data collection
- high psychological stress burden on both general population and health care personnel.

4.2 COVID-19 and hypertension

Almost all available evidence suggests that hypertension increases the risk of severe COVID-19, defined as admission to intensive care, clinically defined severity or a combination of these; or mortality. It was sometimes unclear, however, whether this risk was independent of other risk factors *(92)*. Initial reports have identified higher rates of HTN among severely ill, hospitalized COVID-19 patients, with overall HTN rates of 50–56% *(93, 94)*. It had been unclear if this relationship was causal or confounded by age and other comorbidities associated with HTN, including obesity, diabetes and chronic kidney disease. Concerns regarding use of angiotensin-converting enzyme inhibitors (ACEis) in these patients were raised due to identification of angiotensin-converting enzyme 2 (ACE2), the monocarboxypeptidase that inactivates angiotensin II and thereby counters the activation of the classic renin–angiotensin–aldosterone system (RAAS), as the functional receptor for the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) *(95, 96)*. The WHO conducted a rapid review of evidence related to the use ACEis or ARBs in COVID patients which identified 11 observational studies. No studies were found that were designed to directly assess whether ACEis or ARBs increase the risk of acquiring COVID-19. After adjustment for confounders, history of ACEi or ARB use was not found to be associated with increased severity of COVID-19 illness. There were no studies that addressed the potential benefits and harms of initiating ACEis or ARBs as treatment for patients with COVID-19 *(97)*. Accordingly, discontinuation of ACEis or ARBs may yield worse outcomes than continuation of their use in patients with a diagnosis of COVID-19. In contrast to the uncertainty about the potential benefit of initiating RAAS blocker use in patients with COVID-19, there is a clear potential for harm in withdrawing these agents in high-risk COVID-19 patients with established myocardial injury, HTN or heart failure *(96)*. Most of the world's professional societies either recommend or strongly encourage continuing ACEis/ARBs in COVID-19-infected patients *(98)*. Further research that will address key unanswered questions about the role of the RAAS in the pathogenesis and possible treatment of COVID-19 and other coronavirus-based diseases is urgently needed. Prospective studies – in particular, ongoing randomized, placebo-controlled trials such as the Ramipril for the Treatment of COVID-19 (RAMIC) trial (ClinicalTrials.gov number, NCT04366050) may provide clearer insight regarding the effect of ACEis or ARBs in patients with COVID-19.

4.3 Pregnancy and hypertension

Hypertension, including chronic HTN, gestational HTN, pre-eclampsia, and eclampsia, is a very common medical condition in pregnancy *(99)*. Due to the adverse consequences of increased morbidity and mortality to both the women and fetus, HTN in pregnancy must be diagnosed, treated (when appropriate), and followed up diligently. It is important to note that up to 10% of pregnancy-related deaths are attributed to HTN, and its presence in pregnancy leads to long-term adverse cardiovascular consequences. Unfortunately, hypertensive disorders of pregnancy are markedly increasing *(100, 101)*. For instance, in the United States between 1998 and 2006, hypertensive disorders in pregnancy increased from 6.7% to 8.3%, chronic HTN in pregnancy increased from 1.1% to 1.7%, and pre-eclampsia/eclampsia from 0.9% to 1.2%. The potential serious consequences of HTN and pregnancy and the contraindication in pregnancy of some of the commonly prescribed pharmacological antihypertensive medications discussed below should be discussed with women who are or could become pregnant.

The normal haemodynamic state of pregnancy is one of systemic vasodilation accompanied by an increase in cardiac output and decrease in total peripheral resistance. This results in a normal decrease in BP in the second trimester. HTN in pregnancy is generally diagnosed when BP is ≥140 mmHg and/or ≥90 mmHg on at least two occasions, at least six hours apart. Chronic HTN is defined as a diagnosis of HTN before 20 weeks gestation, while gestational HTN is defined as a diagnosis of HTN at 20 weeks or later. Pre-eclampsia and eclampsia are pregnancy-specific medical conditions requiring immediate and specific medical management.

While BP treatment thresholds for HTN in pregnancy continue to change, it is generally recommended for both chronic and gestational HTN that pharmacologic treatment be initiated when the SBP is ≥160 mmHg and/or the DBP is ≥105 mmHg. In chronic HTN, frequently the woman has already been diagnosed with HTN prior to the pregnancy and thus may already be on chronic antihypertensive pharmacological therapy. In this case, the current regimen may be continued, with the caveat that the medication regimen may have to be changed to preferred medications, and certain antihypertensive medications that are contraindicated in pregnancy must be discontinued. The recommended treatment BP goal/target also has been subject to debate and is changing. For instance, achieving a lower BP target (DBP of 85 mmHg vs 100 mmHg) has recently been shown to decrease the maternal development of severe HTN while not increasing maternal or fetal risk. If target organ damage is present, initiating antihypertensive pharmacological treatment at a DBP of ≥90 mmHg should be considered.

As with most, if not all, other medical conditions requiring pharmacological treatment during pregnancy, the treatment considerations in HTN are no different from those of non-pregnant adults. Thus, since medications are not studied specifically for efficacy and safety in pregnancy, medication selection is usually based on long-term clinical use and experience. This usually means older medications that have had a substantial long-term track record of efficacy and safety are to be considered. For the pharmacological treatment of HTN in pregnancy, preferred medications include methyldopa, beta-blockers (particularly labetalol), CCBs (particularly nifedipine and, as an alternative, verapamil), and the direct-acting vasodilators (particularly hydralazine). There is evidence to suggest that among these agents, beta-blockers and CCBs appear to be more effective than methyldopa in decreasing the development of severe HTN later in the pregnancy. The use of thiazide diuretics has been debated, particularly if the individual is already chronically on a thiazide prior to the pregnancy. In this situation the thiazide diuretic may be continued during the pregnancy.

There are clear contraindications to the use of some antihypertensive medications during pregnancy. These include all the renin–angiotensin system inhibitors, such as the ACEis, the ARBs and, although not used any more, the direct-acting renin inhibitors, due to direct adverse effects on the fetus, and the mineralocorticoid receptor antagonist spironolactone due to fetal anti-androgen effects. The use of the beta-blocker atenolol is also contraindicated due to the observation of intrauterine fetal growth inhibition *(102)*.

In summary, HTN in pregnancy, manifested by the various hypertensive pregnancy disorders, is a very common medical condition. HTN pregnancy disorders have serious maternal and fetal consequences. There are currently several preferred oral antihypertensive pharmacological agents available to treat chronic HTN and gestational HTN during pregnancy. In addition, there are antihypertensive pharmacological agents that are contraindicated in pregnancy. There is evidence to support the pharmacological treatment of HTN in pregnancy at given BP thresholds without and with the presence of end organ damage to decrease the likelihood of the development of severe HTN later in the pregnancy. Even with the effective lowering of BP during the pregnancy and in the immediate post-partum period, the presence of hypertensive disorders of pregnancy significantly increases long-term CV risk, including future HTN, coronary disease, and stroke.

5 Publication, implementation, evaluation and research gaps

5.1 Publication

This guideline is available to download from the WHO website. Given that an overview of published systematic reviews was used for the development of the guideline, all reviews are already published and available online.

5.2 Implementation and dissemination

WHO regional and country offices, through their contacts with ministries of health, will encourage implementation at country level. WHO will provide technical assistance if substantial country adaptation is required. The HEARTS technical package, which is currently being implemented in 18 countries and has significant partner endorsement, membership and engagement, will be the platform used to implement and disseminate this guideline. The package will be revised to include the implementation tools from this guideline. Separate implementation aspects are being considered as implementation or derivative tools for nonphysician treatment and the treatment of hypertension in areas of humanitarian crises, following publication of the guideline. Implementation support will be extended to countries through all three levels of WHO.

5.3 Evaluation

WHO will monitor uptake and implementation of the guideline in national policies and programmes by reviewing the number of countries that have adapted or endorsed the guideline nationally.

5.4 Future updating of the guideline

The guideline is expected to be valid for a period of five years. This period reflects the fact that new research findings are likely to become available in the meantime but also represents a feasible time frame, considering the costs, time and other resources that are needed for the updating process. If the evidence base or user needs change before the five-year mark, consideration will be given to producing updates sooner.

5.5 Research gaps

Several research gaps were identified by the GDG according to the theme of the PICOs.

Thresholds to determine initiation of therapy and targets to achieve for control

- More evidence is required regarding treatment of those in the SBP 130–139 range who fall into one or more of the following subgroups: diabetes, chronic kidney disease, heart failure, 65 years or older.
- There is a need for better outcomes data from, for example, trials that include heart failure and cognitive impairment among outcomes.
- Clinical significance of adverse events registered in clinical trials needs greater clarity.
- There is a need to quantify the difference in estimates between blinded, placebo-controlled trials and unblinded, active control trials using a standard framework.

- There is a need for periodic analysis of trials in order to capture effects of changes over time in background epidemiology of CVD, non-BP treatments, competing risks, etc.
- More evidence is needed in LICs, MICs and other non-North American/European countries.
- An assessment of the feasibility, resource needs, and costs of intensive treatment in real clinical practice is needed. The resource commitment required for more intensive treatment in LMICs needs to be quantified.
- The opportunity cost of directing resources towards achieving SBP <130 in high-risk individuals needs to be established.
- Research is needed on the feasibility, acceptability, and efficacy of intensive treatment, especially in high-risk populations in LICs and MICs.

Laboratory tests to determine initiation of treatment

- A greater understanding of the essential tests to be performed in all patients to reduce costs and improve outcomes is required.

Role of cardiovascular risk in hypertension treatment

- An exploration is needed of key operational aspects of the implementation of a risk-based approach to CVD prevention and BP-lowering pharmacological treatment in primary health care settings.

Monotherapy versus combination therapy

- A comparison is required of long-term data about hard clinical endpoints between monotherapy and combination therapy.
- There is a need for research studies on real-world experiences, designed and statistically powered, to determine if there is a difference in clinical outcomes, such as reduction in MACE, mortality, and serious adverse events, between single-pill combinations vs multiple-pill combinations.
- Health economic analyses are needed to quantify cost-effectiveness and budget implications of implementing incremental initial combination therapy compared with initial monotherapy.

Frequency of re-assessment

- Criteria establishing the clinical definition of stable BP control will be needed to guide the selection of patients for less frequent follow-up visits.
- Research is needed for early and accurate identification of patients less likely to achieve BP control and less likely to follow up as requested by their health care provider.
- Better evidence is needed on the timing, frequency, and intensity of interventions that improve treatment adherence.

Team-based care for hypertension

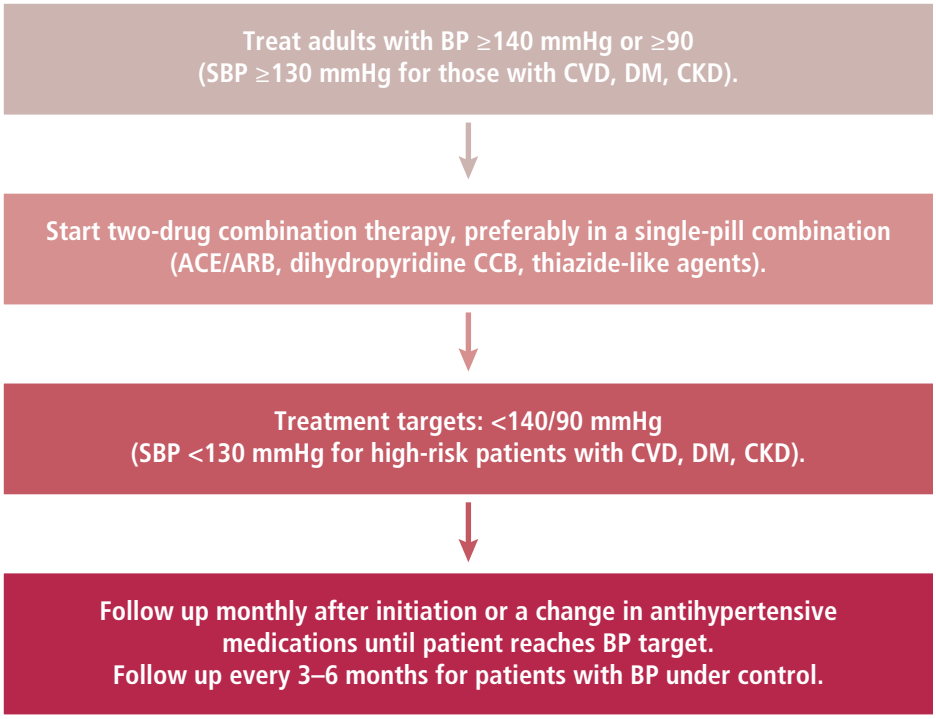
- Evidence is needed that remote monitoring and use of community HCWs/navigators can assist in the management of BP.
- Evidence of the feasibility, costs, and effectiveness of community/home-based monitoring of BP is needed.

6 Implementation tools

6.1 Guideline recommendations

Graphic summaries of the guideline recommendations are presented below in an algorithmic approach (Figs 3 and 4). This maps the recommendations to a patient-care pathway.

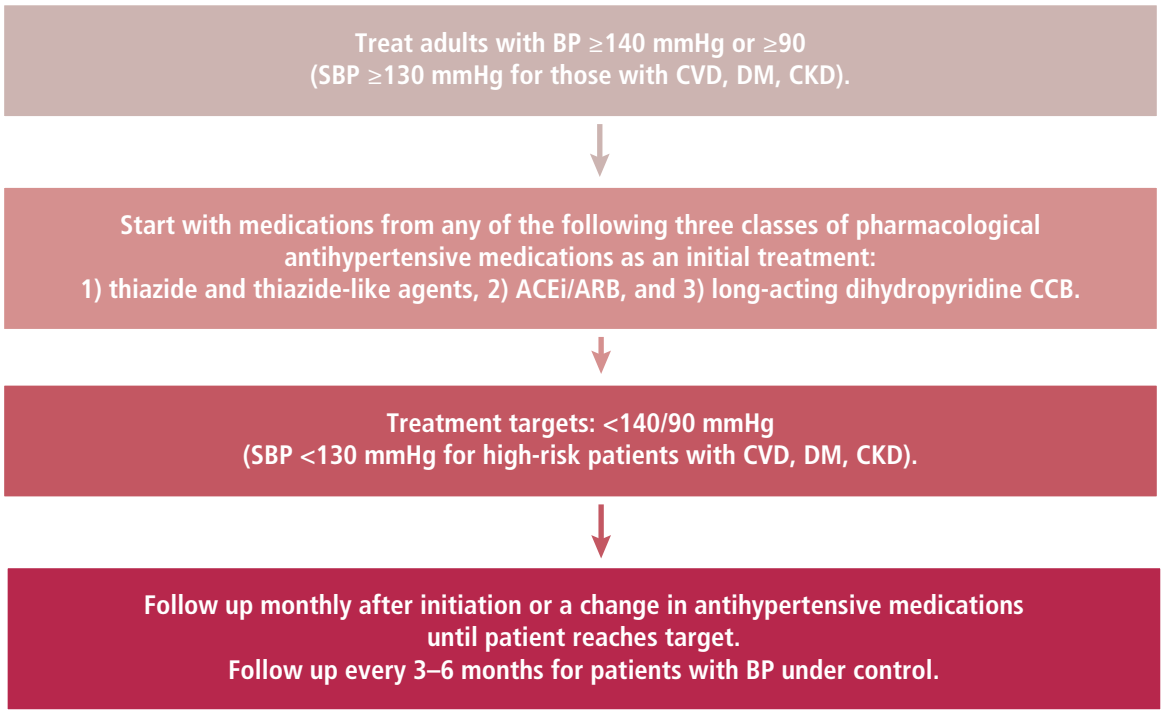
Fig. 3 An approach for starting treatment with a single-pill combination



Pharmacological treatment to be initiated under the following circumstances:

- A diagnosis of HTN has already been made.
- Initiation of pharmacological HTN treatment should start no later than four weeks after diagnosis of HTN.
- If BP level is high or there is accompanying evidence of end organ damage, initiation of treatment should be started without delay.
- Patient should be counselled about starting medication therapy.
- Basic laboratory testing (electrolytes, creatinine, lipogram, glucose, HbA1C, urine dipstick, and ECG) to occur as long as it does not delay treatment.
- A CV risk assessment can be conducted immediately (as long as it does not delay initiation of treatment) or at a later visit.
- Consider using diuretics or CCB in patients 65 years or older, or those of African or Afro-Caribbean descent, beta-blockers (BBs) post MI, ACEis/ARBs in those with DM, heart failure or CKD.

Fig. 4 An approach for starting treatment not using a single-pill combination (i.e. with monotherapy or free combination therapy)



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- A CV risk assessment can be conducted immediately (as long as it does not delay initiation of treatment) or at a later visit.
- Consider using diuretics or CCB in patients 65 years or older, or those of African or Afro-Caribbean descent, beta-blockers (BBs) post MI, ACEis/ARBs in those with DM, heart failure or CKD.

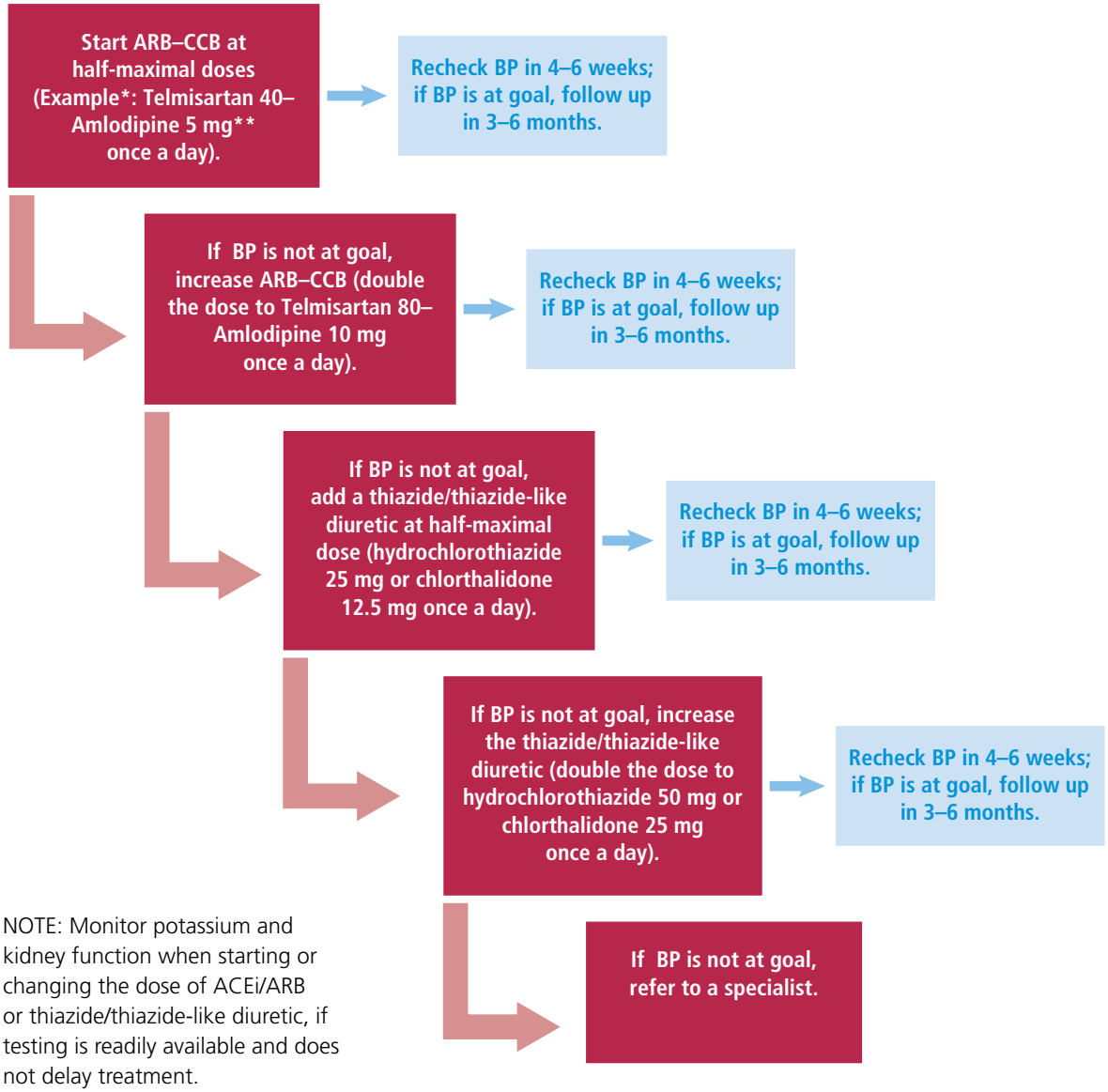
6.2 Drug- and dose-specific protocols

Two examples of suggested drug and dose-specific protocols are presented below (Figs 5 and 6). These should be viewed as examples and other approaches are possible.

Algorithm 1: Initiation of treatment with a single-pill combination

- Beginning treatment with two antihypertensive drugs from different classes is recommended when baseline BP is ≥20/10 mmHg above goal, and should be considered when baseline BP is ≥140/90 mmHg.
- Drugs affecting the renin–angiotensin system (ACEis, ARBs, and aliskiren) have been associated with serious fetal toxicity, including renal and cardiac abnormalities and death; they are contraindicated for use during pregnancy.

Fig. 5 Algorithm 1



NOTE: Monitor potassium and kidney function when starting or changing the dose of ACEi/ARB or thiazide/thiazide-like diuretic, if testing is readily available and does not delay treatment.

This protocol is contraindicated for women who are or could become pregnant. Neither an ACEI or ARB should be given to pregnant women.

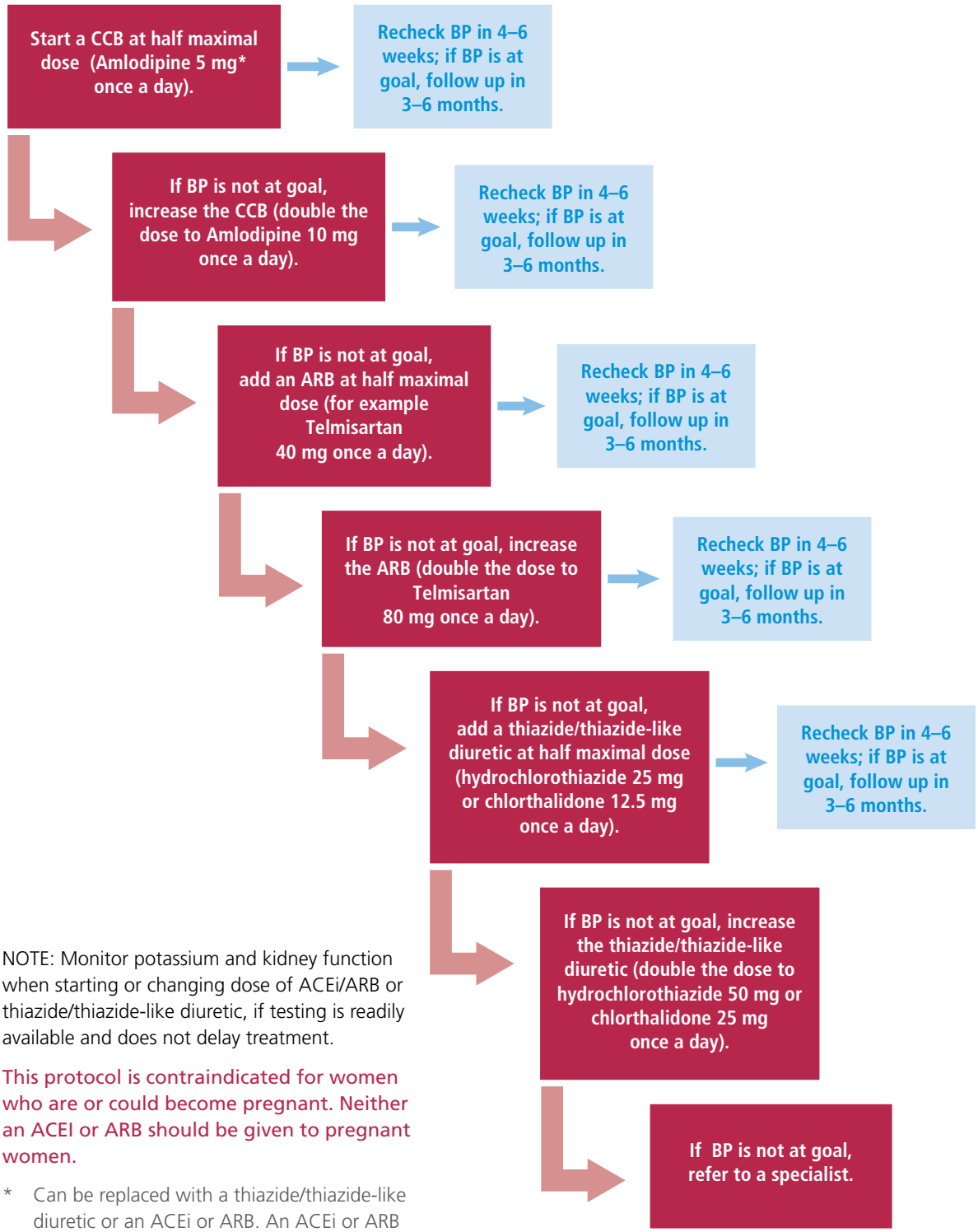
* The medications mentioned serve as examples and can be replaced with any two medications from any of the three drug classes (ACEis/ARBs, CCBs or thiazide/thiazide-like diuretics). Start two individual pills or, if available, both in a single-pill combination (fixed-dose combination).

** Can be replaced with other individual pills or, if available, other single-pill combinations (fixed-dose combinations).

Algorithm 2: Initiation of treatment not using a single-pill combination (i.e. with monotherapy or free combination therapy)

- A CCB, rather than a thiazide-type diuretic or ACEi/ARB, was selected as first-line medication if one agent is used, to avoid the need for electrolyte measurements or to alleviate concerns regarding potential change in GFR.
- Drugs affecting the renin-angiotensin-aldosterone system (ACEis, ARBs, and aliskiren) have been associated with serious fetal toxicity, including renal and cardiac abnormalities and death; they are contraindicated for use during pregnancy.

Fig. 6 Algorithm 2



NOTE: Monitor potassium and kidney function when starting or changing dose of ACEi/ARB or thiazide/thiazide-like diuretic, if testing is readily available and does not delay treatment.

This protocol is contraindicated for women who are or could become pregnant. Neither an ACEI or ARB should be given to pregnant women.

* Can be replaced with a thiazide/thiazide-like diuretic or an ACEi or ARB. An ACEi or ARB is preferred for patients with proteinuria.

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The World Health Organization (WHO) would like to thank the members of the Guideline Development Group, the scientists who provided systematic reviews and the external peer reviewers for their contributions to the development of these recommendations. Professor K Srinath chaired the meeting with the assistance of the deputy chair, Professor Nizal Sarafzadeggan.

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Overall coordination and writing of the guideline

The guideline process was coordinated by the WHO Department of Noncommunicable Diseases. The first draft was written by Taskeen Khan. Drafts were reviewed by the Guideline Development Group and External Review Group, and subsequently revised by Taskeen Khan.

Annex 2. Managing declarations of interest and conflicts of interest

The steering group followed the current Compliance, Risk Management and Ethics (CRE) policy.

All members of the Guideline Development Group (GDG) were asked to fill in the standard WHO Declaration of Interest (DOI) forms, which were reviewed. The WHO Secretariat reviewed the curriculum vitae of each potential participant and conducted internet searches (PubMed, Open Payments Data, Google Scholar) for information on potential financial and academic conflicts of interest related to the subject of the meeting. All DOIs are on file at the WHO Department of Noncommunicable Diseases.

The WHO Steering Group published the names and brief biographies of potential GDG members on the WHO website for more than two weeks, together with a description of the objective of the meeting, for public review and comment. There were no concerns raised about any members.

None of the declared interests was judged sufficient to affect any of the experts' objective judgement during the guidelines development process or on the recommendations, or therefore to preclude their full participation in the development of the guidelines.

During the course of the development of the guideline, one GDG member's status of conflict of interest changed as she accepted a temporary appointment as a staff member at WHO. She was removed from further GDG engagements immediately upon accepting appointment.

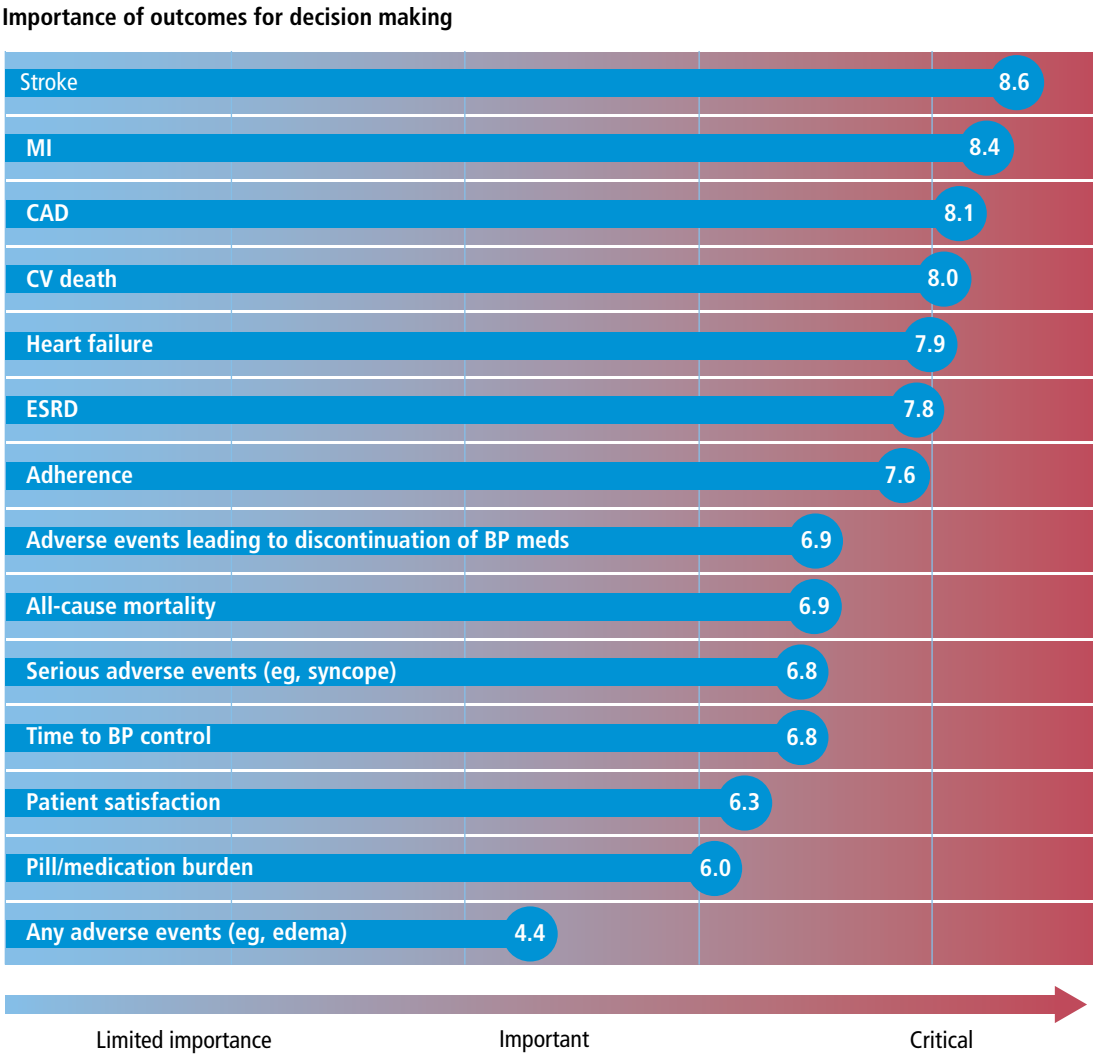
All members of the GDG and all meeting observers were required to sign a confidentiality agreement before participating in the meeting.

All members of the External Review Group (ERG) were asked to fill in the standard WHO DOI forms, which were reviewed. None of the declared interests was judged sufficient to affect any of the judgement during the review process, or therefore to preclude their participation as expert reviewers.

Annex 3: Treatment outcomes relevant to hypertension

Members of the WHO Steering Group, in consultation with the GDG and methodologist, developed a list of treatment outcomes most relevant to the care of individuals with hypertension. The GDG then rated each outcome on a scale from 1 to 9 and indicated whether it considered each outcome critical (rated 7–9), important (rated 4–6) or not important (rated 1–3) for decision-making (Fig. A3.1).

Fig. A3.1 Rating of outcomes



Annex 4: PICO questions

The eleven questions in population, intervention, comparison, and outcomes (PICO) format used to guide the systematic reviews. In addition, relevant subgroups (s) were identified.

1 At what level of blood pressure should pharmacological therapy be started to prevent cardiovascular events?

P	Adult men and women
I	Specific systolic and diastolic blood pressure thresholds:* systolic (mmHg): ≥120, ≥130, ≥140, ≥150 diastolic (mmHg): ≥80, ≥90
C	Placebo or systolic or diastolic blood pressure threshold that is higher than intervention thresholds
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, myocardial infarction, end-stage renal disease, cognitive impairment/dementia, heart failure events and adverse events
s	Based on different effect modifiers such as: estimated cardiovascular risk; pre-existing CAD, stroke, diabetes, age, sex, chronic kidney disease and race/ethnicity

* Each BP threshold in the intervention category (I) will be compared with a higher threshold. For example, I (<140) will be compared to C (≥140)

2 Is any laboratory testing necessary prior to initiation or during titration of pharmacological treatments

P	Adult men and women requiring antihypertensive treatment
I	Initiation or titration of antihypertensives without lab tests
C	Initiation or titration of antihypertensives with lab tests
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, myocardial infarction, end-stage renal disease and heart failure events, cognitive impairment/decline BP control Time to control BP Adherence Adverse effects Patient satisfaction
s	Individual drugs and doses Patients with no comorbidities Baseline blood pressure Type of lab test (ECG, blood, etc)

3 Should cardiovascular risk assessment be used to guide initiation of antihypertensive medications?

P	Adult men and women without pre-identified CVD
I	Initiating antihypertensives drug therapy based on a formal CVD risk estimation
C	Initiating antihypertensives drug therapy without formal CVD risk assessment (i.e. using only BP threshold)
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, myocardial infarction, end-stage renal disease heart failure events, cognitive impairment/dementia, and adverse events Proportion of people prescribed with antihypertensives BP levels
s	BP levels

4 In adults with hypertension requiring pharmacological treatment, which drugs should be used as first-line agents?

P	Adult men and women with hypertension requiring pharmacological treatment
I	BB, CCB, diuretics, ACEi, or ARB
C	Placebo
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, cognitive impairment/dementia, myocardial infarction, end-stage renal disease and heart failure events Adverse effects such as bradycardia, acute kidney injury, angioedema, asthma, electrolyte abnormalities or hypotension BP reduction and control (if data on CVD events are absent)
s	Based on different effect modifiers such as: estimated cardiovascular risk; pre-existing CAD, stroke, diabetes, age, sex, chronic kidney disease and race/ethnicity, level of baseline BP

5 In adults with hypertension requiring pharmacological treatment, which drugs (BB, CCB, diuretics, ACEi, or ARB vs BB, CCB, diuretics, ACEi, or ARB in head-to-head studies) should be used as first-line agents?

P	Adult men and women with HTN requiring pharmacological treatment
I	BB, CCB, diuretics, ACEi, or ARB
C	BB, CCB, diuretics, ACEi, or ARB (head-to-head studies)
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, cognitive impairment/dementia, myocardial infarction, end-stage renal disease and heart failure events Adverse effects such as bradycardia, acute kidney injury, angioedema, asthma, electrolyte abnormalities or hypotension Blood pressure reduction and control (if data on CVD events are absent)
s	Based on different effect modifiers such as: estimated cardiovascular risk; pre-existing CAD, stroke, diabetes, age, sex, chronic kidney disease and race/ethnicity, level of baseline blood pressure

6 In adults with hypertension requiring pharmacological treatment, which drugs (monotherapy using BB, CCB, diuretics, ACEi or ARB vs combination therapy using BB, CCB, diuretics, ACEi or ARB) should be used as first-line agents?

P	Adult men and women with HTN requiring pharmacological treatment
I	BB, CCB, diuretics, ACEi, or ARB
C	Combination therapy
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, cognitive impairment/dementia, myocardial infarction, end-stage renal disease and heart failure events Adverse effects such as bradycardia, acute kidney injury, angioedema, asthma, electrolyte abnormalities or hypotension BP reduction and control (if data on CVD events are absent)
s	Based on different effect modifiers such as: estimated cardiovascular risk; pre-existing CAD, stroke, diabetes, age, sex, chronic kidney disease and race/ethnicity, level of baseline BP

7 In adults with hypertension requiring pharmacological treatment, which combination therapy of two or more drugs (BB, CCB, diuretics, ACEi, or ARB) vs different combination therapy of two or more drugs (BB, CCB, diuretics, ACEi, or ARB) should be used as first-line agents?

P	Adult men and women with HTN requiring pharmacological treatment
I	Combination therapy of two or more drugs (BB, CCB, diuretics, ACEi, or ARB)
C	Different combination therapy of two or more drugs (BB, CCB, diuretics, ACEi, or ARB)
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, cognitive impairment/dementia, myocardial infarction, end-stage renal disease and heart failure events Adverse effects such as bradycardia, acute kidney injury, angioedema, asthma, electrolyte abnormalities or hypotension BP reduction and control (if data on CVD events are absent)
s	Based on different effect modifiers such as: estimated cardiovascular risk; pre-existing CAD, stroke, diabetes, age, sex, chronic kidney disease and race/ethnicity, level of baseline BP

8 In adults with hypertension requiring pharmacological intervention, is use of a single-pill combination of antihypertensive drugs associated with improved outcomes?

P	Adult men and women with HTN requiring pharmacological intervention
I	Single-pill combination (FDC) of antihypertensive drugs – five classes (any two or more from the five)
C	Pharmacological interventions that do not involve use of single-pill combinations
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, myocardial infarction, end-stage renal disease and heart failure events. Adverse effects Patient satisfaction Adherence BP level/change Number of antihypertensive medications
s	Based on different effect modifiers such as: estimated cardiovascular risk; pre-existing CAD, stroke, diabetes, age, sex, chronic kidney disease and race/ethnicity, level of baseline BP

9 What target blood pressure should pharmacological treatment aim to achieve?

P	Adult men and women
I	Specific systolic and diastolic blood pressure targets: systolic (mmHg): <120, <130, <140, <150 diastolic (mmHg): <70, <80, <90
C	Systolic or diastolic blood pressure targets that are higher than the intervention targets
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, myocardial infarction, end-stage renal disease heart failure events, cognitive impairment/dementia, and adverse events
s	Based on different effect modifiers such as: estimated cardiovascular risk; pre-existing CAD, stroke, diabetes, age, sex, chronic kidney disease and race/ethnicity

10 In adults with hypertension given pharmacological treatment, when should blood pressure be reassessed?

P	Adult men and women with HTN receiving a pharmacological intervention
I	Specific interval
C	Alternative interval
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, myocardial infarction, end stage renal disease and heart failure events Adverse effects BP control Adherence Patient satisfaction
s	Titration phase vs controlled HTN follow up, level of initial BP, other conditions, remote monitoring vs clinical visit

11 Can pharmacological management of hypertension be provided by nonphysician care providers?

P	Adult men and women
I	Pharmacological management by nonphysician care providers
C	Pharmacological management by medically qualified practitioners (doctors)
O	Death (all-cause mortality), cardiovascular death (death from MI, sudden cardiac death or stroke), stroke, myocardial infarction, end-stage renal disease and heart failure events BP control Adherence Serious adverse effects Patient satisfaction
s	Initiation vs follow up Self-care vs CHW vs nurse vs pharmacist vs physician assistant vs in or out of clinic Levels of care Rural vs urban settings Ethnicity

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Preface

HANDBOOK OF CLINICAL PSYCHOLOGY (VOLUMES 1 AND 2)

Over the past 3 decades the field of clinical psychology has periodically been reviewed in large tomes in multiauthor editions. Indeed, the senior editor of the present work has been involved in three such projects, published in 1983, 1991, and 1998. Since publication of the last of these works there have been major developments in clinical psychology, both with adults and with children and adolescents. Some of the most interesting innovations have appeared in the areas of ethics, legal issues, professional roles, cross-cultural psychology, psychoneuroimmunology, cognitive-behavioral treatment, psychopharmacology as practiced by clinical psychologists, political issues, geropsychology, parent training, pediatric psychology, assessment of child maltreatment and its remediation, and problems in infancy. The veritable explosion of new data has been fueled by the advent of new journals in highly specialized areas, in addition to the rapid dissemination of new ideas and data rapidly presented through electronic publication.

The plethora of data is distilled into 33 chapters for Volume 1 on adults and 32 chapters for Volume 2 on children. To enable the reader to easily traverse the two volumes we decided on a parallel structure. Therefore, each of the volumes is divided into six parts (Part I: General Issues, Part II: Theoretical Models, Part III: Research Contributions, Part IV: Diagnosis and Evaluation, Part V: Treatment, and Part VI: Special Issues). In both volumes a case illustration has been included in the section on Treatment. Authors here have been encouraged, to the extent possible, to use a relatively standard format: Identifying Features of the Client, Presenting Complaints, History, Assessment, Case Conceptualization, Course of Treatment and Assessment of Progress, Complicating Factors, Managed Care Considerations, Follow-up, Treatment Implications of the Case, and Recommendations to Clinicians and Students. In the volume on children and adolescents Developmental Factors and Parental Factors are added considerations that the authors have outlined.

These two volumes were conceived with graduate students and professionals in the field in mind. We believe that the material systematically presented herein is suitable both as a general reference work and separately as a major course text for clinical psychology with adults and clinical psychology applied to children. We, of course, eagerly anticipate the comments of both professors and students.

Many individuals have been involved in the development and fruition of this two-volume work. First and foremost we thank our contributors for taking time out

from their busy schedules to share their expertise with us. Second, we thank our friends at John Wiley and Sons for their appreciation of clinical psychology and the technical processing of this two-volume set. Third, once again we thank Carole Londeree for helping us to track all of the authors and manuscripts in her typical orderly and competent manner. Without her terrific help this could not have been accomplished. Finally, we thank Christopher Brown and Blake Kirschner for their wonderful technical work related to the indexes.

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PART I

General Issues

CHAPTER 1

Historical Perspectives

Thomas R. Kratochwill, Richard J. Morris, and Joseph Robinson

In this chapter, we trace the historical development of the assessment and treatment of children's and adolescents' mental health issues. Although our overview is brief (see several sources for more detailed historical accounts: Doyle, 1974; Dubois, 1970; Linden & Linden, 1968; P. McReynolds, 1975), we provide some perspectives on contemporary evidence-based assessment and treatment. An examination of the historical factors in assessment and treatment is important for several reasons. First, it is important to understand that many of the contemporary issues in evidence-based practice have their origin in past practices. Second, it is important to realize that many contemporary issues are related to social or even political concerns that have their origin in the past. Third, the past has sometimes provided or even imposed a structure on assessment and treatment practices. It is important to understand this structure to understand contemporary models and the scope of psychological practices. Finally, it is important to focus on historical factors to introduce a variety of scholarly perspectives into the discussion of the issues surrounding evidence-based practice. We first review historical features of diagnosis and assessment and then turn our attention to child and adolescent therapy. However, the conceptual, theoretical, and practice issues in these domains overlap.

ASSESSMENT AND DIAGNOSIS: ANCIENT INFLUENCES

Most historical treatments of the assessment literature typically begin with a discussion of the work of Galton in England and Cattell in the United States (i.e., many books on assessment begin with this period; e.g., Sunberg, 1977). However, assessment has a much richer history, attesting to the assumption that many features of contemporary assessment actually date back to the beginnings of recorded history. L. V. McReynolds (1974) traced the historical antecedents of the current practices in assessment beginning with antiquity and extending to the second half of the twentieth century. Four phases were reviewed: antiquity, the medieval period and the Renaissance, the Age of Reason, and the period from Thomesius to Galton. We adopt this framework in this section of the chapter.

Antiquity

An examination of early assessment practices shows that there was a close interplay between the methods used and the cultural views held during that particular time.

This perspective is not unlike the contemporary views in the United States that led to the development of special education services for children (Kratochwill, Clements, & Kalymon, 2007), with its emphasis on fair assessment practices for handicapped children. It is possible that the first personality assessment procedure was based on *astrology,* and that the first psychological "test" was the horoscope. Although astrology can be regarded as invalid on scientific grounds, it did contribute to (a) the view that individual personalities represent the focus of assessment, (b) the belief that the psychological makeup of the individual is predetermined, and (c) the development of taxonomical (diagnostic) categories.

Another early assessment strategy involved physiognomy, the interpretation of an individual's character from body physique. Physiognomics, also a very limited assessment procedure, assumed a relatively fixed conception of personality but shared some methodological features with naturalistic observation, not unlike the naturalistic observations conducted using behavior modification procedures (Kazdin, 1978). L. V. McReynolds (1974) noted that the longest continued assessment technique with some claim to rationality and one that remains with us today is physiognomy. Thus, work by Mahl (1956) and Gleser, Gottshalk, and Springer (1961) on speech patterns; by Hall (1959), Eibl-Eibesfeldt (1971), and Haas (1972) on methodology of movements; of Izard (1971) and Ekman and associates (Ekman, 1973; Ekman, Friesen, & Ellsworth, 1972) on emotions and facial expressions; and of Hess and associates (Hess & Polt, 1960; Hess, Seltzer, & Schlien, 1965) on the relation of pupil size to affect can be related to earlier physiognomic conceptions (cf. L. V. McReynolds, 1974).

Developments in assessment during early times were not always limited to the area of personality assessment. For example, Civil Service examinations were used in ancient China for selection purposes. Dubois (1966, pp. 30–31) notes:

> The earliest development seems to have been a rudimentary form of proficiency testing. About the year 2200 B.C. the emperor of China is said to have examined his officials every third year. . . . A thousand years later in 1115 B.C., at the beginning of the Chan dynasty, formal examining procedures were established. Here the record is clear. Job sample tests were used requiring proficiency in the five basic arts: music, archery, horsemanship, writing, and arithmetic. . . . Knowledge of a sixth act was also required—skill in the rites and ceremonies of public and social life.

Medieval Period and the Renaissance

L. V. McReynolds (1974) notes that during this period, the acceptance of humeral psychology and physiognomic strategies of evaluating people were widespread. Generally, this period supported the recognition of the individual, and so we again see an example of cultural influences on assessment practices. In some respects this period set the occasion for what would later be a debate on research methodology, especially surrounding the use of group versus single-case research design in therapy research (Kratochwill & Levin, 1992).

Age of Reason

The Age of Reason covers the period from approximately the middle of the sixteenth century to the second half of the eighteenth. A major theme of this period was the focus on individual differences, as reflected in some important works on assessment: Huarte's *Tryal of Wits,* Wright's *Passions of the Minde,* and Thomesius's *New Discovery*. During this period, the recognition of individual differences prompted measurement so that an individual's sense of well-being could be more fully realized.

From Thomesius to Galton

A significant contribution to assessment during this period, particularly in the nineteenth century, was phrenology. Phrenology bears a similarity to physiognomy, but whereas physiognomy emphasized assessment of external body features such as facial and other characteristics, phrenology emphasized the assessment of the external formations of the skull. Phrenology assumed that mental functions were based on specific processes localized in certain areas of the brain and that the intensity or magnitude of these functions was indicated in the contours and external topography of the skull (L. V. McReynolds, 1974).

Four positive contributions of phrenology that have a resemblance to contemporary assessment practices or activities were identified (L. V. McReynolds, 1974). First, there was an emphasis on individual differences. Second, the assessment paradigm emphasized the notions of assessor and subject, the systematic collections of data during a single session, and written reports that usually included qualitative profiles. Third, the phrenological movement helped advance objectivity through blind assessment and rating scales. Fourth, phrenology contributed to the development of a primitive taxonomical system, which included affective faculties (e.g., propensities, sentiments) and intellectual faculties (e.g., perceptive, reflective). This line of reasoning was likely influential in later conceptualizations of diagnostic and classification systems.

Implications

This brief historical overview of ancient influences points out that many contemporary assessment practices have their roots deep in our past. Noteworthy is the fact that the work of the phrenologists (and later, Quetelet's work on psychological statistics) set the stage for the emergence of Galton's contributions and the subsequent more modern era in assessment. It is interesting to speculate how some of the ancient procedures might have set the stage for child diagnoses specifically. L. V. McReynolds (1974, pp. 524–525) raises an interesting point:

> We know that such techniques as chiromancy, metaposcopy, and phrenology are in principle all totally invalid, yet I suggest that in the hands of insightful and discerning practitioners they may, at least on occasion, have been more valid than we suppose, even if for different reasons than their users, much less their clients imagined.

ASSESSMENT AND DIAGNOSIS: NINETEENTH-CENTURY INFLUENCES

During the nineteenth century significant developments were taking place in Western Europe and the United States that would shape the future of psychological and educational assessment (cf. Carroll, 1978; Dubois, 1970; Laosa, 1977). Specifically, events were occurring in France, Germany, England, and the United States that were to have a profound influence on assessment practices in child therapy and education.

France

Attention to two movements occurred in France that made a significant impact on the history of testing and assessment in general and child assessment in particular (Maloney & Ward, 1976). One movement, pioneered by Berhheim, Liebault, Charcot, and Freud, was focused on a new view of deviant behavior. The influence of this movement was to take abnormal behavior out of the legal or moral realm with which it had been previously associated and cast it as a psychological or psychosocial problem. This focus prompted psychological assessment rather than moral or legal sanction, as had been common prior to this period.

Also noteworthy was the movement called "the science of education." Jacques Itard, a French physician, taught Victor, the "Wild Boy of Aveyron," various skills. Many of the procedures used in Itard's work were similar to later behavior modification procedures that emphasized environmental stimulus and response changes during intervention. Itard's contributions also provided a background for Binet's work on measurement of intelligence.

Esquivol's (1722–1840) work, represented in his book *Des Maladies Mentales,* was influential in that he distinguished between "emotional disorders" and "subaverage intellect." According to his views, subaverage intelligence consists of levels of individual performance: (a) those making cries only, (b) those using monosyllables, and (c) those using short phrases but not elaborate speech. Thus, here we see the basis for an early classification scheme that could organize human behavior.

Germany

Although some of the work in France emphasized individual differences in pathology and cognitive ability, German scientists perceived individual differences as a source of measurement error. A significant contribution to the individual differences theme is found in the "Maskelyne-Kinnebrook affair." The difference between Maskelyne (the astronomer) and Kinnebrook (the assistant) in their measurement of the timing of stellar transits was later analyzed by Bessel. Bessel concluded that different persons had different transit tracking times, and that when all astronomers were checked against one standard, individual error could be calculated; thus, a sort of "personal equation" was developed (cf. Boring, 1950).

Another significant influence on assessment came from Wundt, who set up a psychological laboratory in Leipzig to study such processes as reaction time, sensation, psychophysics, and association. This work, as well as the general work

occurring on measurement, was helpful to popularize the notion of measurement of differences between individuals. Some Americans who studied with Wundt were G. Stanley Hall and James McKeen Cattell. Both of these individuals were to have a large impact on future child psychological assessment.

England

The work of Charles Darwin was most influential in psychological and educational assessment, particularly through his theory of evolution presented in 1859 in *Origin of the Species*. Darwin's work emphasized that there were measurable and meaningful differences among members of each species. Galton, Darwin's half-cousin, was influential in applying evolutionary theory to humans. In his 1869 book, *Hereditary Genius,* Galton argued that genius has a tendency to run in families. He was greatly influenced by the Belgian statistician Quetelet (1770–1864), who was the first to apply the normal probability curve of Laplace and Gauss to human data. This work translated into the notion that nature's mistakes were represented as deviations from the average.

Several implications of this work for child assessment and treatment are noteworthy. First, Galton's system of classification represented a fundamental step toward the concept of standardized scores (Wiseman, 1967). Second, in the application of Quetelet's statistics, Galton demonstrated that many human variables, both physical and psychological, were distributed normally. This concept is a direct precursor to the notion of a norm and application of standardization (Laosa, 1977). Third, a major influence of this work was to establish that certain variables should be subjected to quantitative measurement. Galton's work was significant in that it encouraged other efforts in the area of measurement of individual differences in mental abilities that were considerably more sophisticated than previous efforts (Cooley & Lohnes, 1976). Finally, through the application of the normal curve, individual performance or standing could be classified as deviant or even as a mistake of nature. We know that although Galton was influenced by the phrenologists, he rejected this form of assessment. He noted in 1906, "Why capable observers should have come to such strange conclusions [can] be accounted for . . . most easily on the supposition of unconscious bias in collecting data" (quoted in Pearson, 1930, p. 577).

United States

Early work in the United States contributed to what was called the "mental testing" movement, a major part of clinical and school psychology. Cattell (1860–1944) was the first to use the term "mental test," and he is generally referred to as the father of mental testing (DuBois, 1970; Hunt, 1961). Cattell also introduced experimental psychology into the United States. A significant contribution to assessment was that he advocated testing in schools; he was also generally responsible for instigating mental testing in the United States (Boring, 1950).

In 1895, Cattell chaired the first American Psychological Association (APA) Committee on Mental and Physical Tests. Although Cattell made major changes in the nature of testing, his work was not accepted unconditionally. For example, Sharp (1899) published an article questioning the reliability of mental tests. Wissler

(1901) compared the reliability of some of Cattell's psychological measures with various measurement approaches from the physical sciences and concluded that tests used in Cattell's lab showed little correlation among themselves, did not relate to academic grades, and were unreliable (Maloney & Ward, 1976). Even Wundt was not supportive of Cattell's focus on mental measurements (Boring, 1950). Nevertheless, Cattell's work, as well as work in France, promoted the development of a movement called "differential psychology."

INTELLIGENCE TESTING MOVEMENT

Around the beginning of the twentieth century, assessment was given a new impetus through the development of differential psychology (Binet & Henri, 1895; Stern, 1900; Stern & Whipple, 1914). Stern and Whipple (1914) suggested that mental age be divided by chronological age to produce a "mental quotient," a procedure, with refinements, that evolved into the IQ concept (Laosa, 1977).

The work of Binet and his associates was quite influential, although not necessarily in the direction that Binet had envisioned or desired (Sarason, 1976; Wolf, 1973). Binet initially focused his efforts on the diagnosis of "mentally retarded" children around the late 1880s. At this time he was assisted by Theodore Simon, with whom he later worked in the development of the first formal measure of intellectual assessment for children (Wolf, 1973). Based on a study conducted for the Ministry of Public Instruction, he focused efforts on predicting which children would be unable to succeed in school (Resnick, 1982). Binet noted that performance on his scale had implications for diagnostic classification and education. Resnick (1982, p. 176) notes:

> A scale of thirty questions was developed, each of increasing difficulty. Idiots were those who could not go beyond the sixth item, and imbeciles were stymied after the twelfth. Morons were found able to deal with the first twenty-three questions. They were able to do the memory tests and arrange lines and weights in a series, but no more . . . the test . . . was designed as an examination to remove from the mainstream of schooling, and place in newly developed special classes for the retarded, those who would be unable to follow the normal prescribed curriculum. As such, it was a test for selection, removing from normal instruction those with the lowest level of ability. Binet argued, however, that the treatment the children would receive in the special classes would be more suited to their learning needs. The testing, therefore, was to promote more effective and appropriate instruction.

Interest in testing the abilities of children was at a high level during this time. This interest was prompted, in part, by the growing population of children in schools due to natural population growth and immigration and the fact that students began to stay in school longer (Chapman, Terman, & Movement, 1979). With the growing number of children in schools, it became clear that not all children could profit from regular instruction. The policies and procedures for diagnosis and assessment of children during this time set a direction that has only recently been changed with new federal regulations that has, as one focus, to reduce the emphasis on an IQ-achievement discrepancy to make a determination of disability status for children with learning problems. The new focus in assessment is called "response

to intervention" and involves determining the child's response to instruction as part of the diagnostic process (Kratochwill et al., 2007).

Several American psychologists promoted Binet's work. For example, Goddard published the first revision of the Binet scale, and Terman developed the Stanford-Binet. Thereafter, the Binet scale was used to identify children who were regarded as "backwards" or "feebleminded." In 1911 the Binet scale was being used in 71 of 84 cities that administered tests to identify feebleminded children. However, the Binet scale was also being used experimentally to screen out and turn back "retarded" immigrants (Knox, 1914, as cited in Widgor & Garner, 1982).

The Stanford version of the Binet-Simon scale was originally published in 1916 by Terman; it was revised by Terman and Merrill in 1937 and 1960 and renormed in 1972 and subsequently. This translation and revision of Binet's earlier work firmly established intelligence testing in schools and clinics throughout the United States (DuBois, 1970).

Development of Group Testing

The assessment movement was given a major thrust through the development of group tests during World War I. Many assessment efforts during this time reflected a pattern of procedures similar to that used by Binet (T. E. Newland, 1977). Ebbinghaus demonstrated the feasibility of group tests, and some American psychologists (e.g., Otis, 1918; Whipple, 1910) recognized that the Binet-Simon scale could be adapted for group testing. However, there were important differences. Whereas the Binet-type items typically required a definite answer provided by the child, group tests usually called for recognition of a correct answer among several alternatives (Carroll, 1978).

A committee of the APA chaired by Robert M. Yerkes developed the Army Alpha and Army Beta group tests. The Army Beta (a nonverbal group test) was designed so as not to discriminate against illiterates and individuals speaking foreign languages. Although the impact of this development was to create a new interest in and role for testing, a review of the tests used revealed that the source of many tests was increasingly used for nonmilitary purposes (T. E. Newland, 1977).

Following the war, many clinical psychologists who were involved in wartime testing sought employment in the civilian ranks, and many became involved in the schools. Resnick (1982, p. 183) notes:

> Aiding this movement was Philander P. Clarxton, U.S. Commissioner of education, who communicated to school superintendents throughout the country about the reserve of trained people that could be tapped for the needs of the schools. He wrote enthusiastically about the "unusual opportunity for city schools to obtain the services of competent men." Among the services that they could render was "discovering defective children and children of superior intelligence."

This movement, in part, facilitated the use of group intelligence tests in the public schools for purposes of diagnosis and classification. Many of these tests were administered to identify children who could not profit from regular instruction. Although some schools had made provisions for special children, the intelligence tests

formalized the decision-making process for these special services. Also, between 1919 and 1923, Terman introduced the National Intelligence Test for grades three to eight and the Terman Group Test, for grades seven to 12 and found that the schools were most receptive (Resnick, 1982). Resnick reported that the most important use of the tests was for placement of children into homogeneous groups:

> Sixty-four percent of the reporting cities used group intelligence tests for this purpose in elementary schools, 56% in junior high schools, and 41% in high schools. Enthusiasm for the use of testing systemwide for this purpose was at a high level. In 1923, Terman's group test for grades seven to thirteen sold more than a half-million copies. (pp. 184–185)

The stage for the rapid development of ability tests was also set by such psychologists as Spearman, Thorndike, and Thurstone and their respective theories of intelligence. For example, Spearman developed an elaborate theory of the organization of human abilities in which he concluded that all intellectual abilities have a common factor, *g*, and a number of specific factors, *s*, which relate uniquely to each presumed ability. Spearman's two-factor theory was the basis on which tests examining specific abilities (Edwards, 1971) rather than global scores were developed (Laosa, 1977).

Thorndike viewed intelligence as comprising a multitude of separate elements, each of which represented a specific ability. Intelligence was also perceived as having both hereditary and environmental components. Thurstone concluded that there were seven primary mental abilities (in contrast to Spearman's *s* and *g* factors) and developed the Primary Mental Abilities Test to measure each specific ability.

Intelligence tests gradually evolved into major diagnostic instruments throughout the world. Such instruments became an important diagnostic tool for identifying children with cognitive disabilities. However, not all countries accepted the use of these tests. For example, in the Soviet Union such tests were banned in 1936 by the Communist Party because they were considered methods that discriminated against the peasants and the working class in favor of the culturally advantaged (Sunberg, 1977; Wortis, 1960). As an alternative, diagnosis in the USSR was based primarily on neurophysiological evidence. The neurologist and psychophysiologist, rather than the clinical psychologist, were primarily engaged in diagnosing children with mental retardation (Dunn & Kirk, 1963).

Work in these areas, as well as other contributions prior to and during this period, led to diverse views on the nature of intelligence and its assessment. A major contribution to the testing movement was the development of the Wechsler intelligence scales. Psychologist David Wechsler developed the Wechsler Adult Intelligence Scale (WAIS) by including a group of subtests from WWI vintage that he found valuable in his work with adults. His criterion of "general adaptability" (Wechsler, 1975) was extended downward in the development of the Wechsler Intelligence Scale for Children (WISC) and the Wechsler Preschool and Primary Scale of Intelligence (WPPSI). The work of Wechsler contrasted with that of Binet. Whereas Wechsler's scales emerged from work with adults and were later developed for use with children, Binet's emerged from work with young children and later were developed for use with older children (T. E. Newland, 1977).

PERSONALITY ASSESSMENT MOVEMENT

While tests of cognitive ability were rapidly evolving during the early part of the twentieth century, tests of personality were in their infancy. Although such devices as the Woodworth Personal Data Sheet were used in the military during WWI, the personality assessment movement received increased attention through the development of projective techniques such as the Rorschach Inkblot Test and the Thematic Apperception Test (TAT).

World War II, like the first war, did much to set the stage for rapid proliferation of testing practices. Indeed, psychological testing combined with the military need for assessment was one of the primary factors leading to the development of clinical psychology as an independent specialty (Maloney & Ward, 1976).

During the period following WWII, testing practices developed dramatically. Most tests developed during this period were tied to an intrapsychic disease model or state-trait conceptualization of behavior (cf. W. Mischel, 1968). Psychoanalytic theory generally accelerated assessment procedures that would reveal unconscious processes. Assessment practices emphasized an "indirect-sign" paradigm. Assessment was indirect in that measurement of certain facets of behavior were disguised or hidden from the client (e.g., such as in the TAT). Moreover, within the context of the intrapsychic model, testing practices were said to predict certain states or traits. The clinician's task was to administer a battery of tests to a child and look for certain signs of traits or states. An example of this approach was represented in the work of Rappaport, Gill, and Schafer (1945). In their classic book, the authors demonstrated how a battery of tests (e.g., TAT, Rorschach, WAIS) could be used to diagnose deviant behavior within the intrapsychic model (in this case, the psychoanalytic model).

Similar to the sign approach was the "cookbook" method of assessment, which reached a zenith during the mid-1950s (cf. Meehl, 1956). An example of this approach is the Minnesota Multiphasic Personality Inventory (MMPI; Hathaway & McKinley, 1943). As these authors noted, one of the presumed advantages of the cookbook approach was that "it would stress representativeness of behavioral sampling, accuracy in recording and cataloguing data from research studies, and optional weighting of relevant variables and it would permit professional time and talent to be used economically" (p. 243).

Emergence of Behavior Modification and Assessment

Behavior modification (also referred to as behavior therapy) and the related assessment procedures associated with this model have made a tremendous impact on psychology and education (Kazdin, 1978; Kratochwill & Bijou, 1987). As some historical reviews illustrate (Hersen, 1976; Kazdin, 1978), behavior therapy represented a departure from traditional models of assessment and treatment of abnormal behavior, both psychological and educational. Although the history of behavior therapy cannot be traced along a single line, practice was characterized by diversity of viewpoints, a broad range of heterogeneous procedures with vastly different rationales, open debates over conceptual bases, methodological requirements,

and evidence of efficacy (Kazdin & Wilson, 1978). Some reports of behavioral treatment followed Watson and Raynor's (1920) work in conditioning of fear in a child, but a significant impetus to behavioral assessment and treatment is commonly traced to the publication in 1958 of Joseph Wolpe's reciprocal inhibition therapy.

Independent of Watson's and Wolpe's work was research in the psychology of learning, in both Russia and the United States. Particularly important in learning research was operant conditioning, which Skinner brought into focus in the late 1930s (e.g., Skinner, 1938). The evolution of operant work into experimental and applied behavior analysis had an extremely important influence on the development of behavior therapy and assessment practices in general.

Although behavior therapy and assessment evolved considerably over the years, some general characteristics represented unities within the heterogeneity of evolving practice:

1. Focus upon current rather than historical determinants of behavior;
2. Emphasis on overt behavior change as the main criterion by which treatment should be evaluated;
3. Specification of treatment in objective terms so as to make replication possible;
4. Reliance upon basic research in psychology as a source of hypotheses about treatment and specific therapy techniques; and
5. Specificity in defining, treating, and measuring the target problem in therapy. (Kazdin, 1978, p. 375)

With the advent of behavior modification and its proliferation, a new assessment role also developed, particularly for clinical child and school psychologists. Behavioral assessment emphasized *repeated measurement* of some target problem prior (baseline), during, and after (follow-up) the intervention. Hersen and Bellack (1976a) noted that the psychologist's expertise in theory and application of behavioral therapy techniques (e.g., classical and operant conditioning) also enabled both an *assessment* and a *treatment* role to emerge in psychiatric settings. This focus was also to occur as a basis for the scientist-practitioner model of psychological services (Hayes, Barlow, & Nelson-Gray, 1999) and represented a strong foundation for the evidence-based treatment movement (Kratochwill & Stoiber, 2002) and what is now called response to intervention in psychological practice in schools (Brown-Chidsey & Steege, 2006; Kratochwill et al., 2007). Thus, the psychologist in various settings (e.g., clinics, hospitals, schools) became involved in direct service rather than engaged in only testing and diagnosis, although this was slower to evolve in schools. Behavior modification provided the impetus for these new roles and has continued to move practice forward in both prevention and treatment.

Developments in behavioral assessment also influenced the field of personality assessment in general. In many respects, assessment has acted as a barometer for the thinking of personality theorists. For example, a barometer of change in views about assessment has been the evolution of the title of the journal

specifically devoted to assessment in professional psychology (Goldfried, 1976). The journal, founded in 1936, was initially entitled *Rorschach Research Exchange*. Other projective techniques came into existence in the assessment process, and by 1947 the journal title was changed to *Rorschach Research Exchange and Journal of Projective Techniques*. Gradually, the more objective personality assessment techniques (e.g., the MMPI) were being used, and in 1963 the title was again changed to *Journal of Projective Techniques and Personality Assessment*. Projective techniques continued to show disappointing research results, and in 1971 this may have prompted the journal's change to the title *Journal of Personality Assessment*.

Nevertheless, there remained some doubt as to whether the future direction of assessment would take a distinct behavioral orientation. Even in 1963, when the journal *Behavior, Research, and Therapy* made its appearance, the issue was raised as to whether there would be a large enough readership to justify its existence (Brady, 1976). However, as Hersen and Bellack (1976a) documented, the future looked very positive, as reflected in major journals inaugurated in the United States between 1968 and 1970 (e.g., *Journal of Applied Behavior Analysis, Behavior Therapy, Journal of Behavior Therapy and Experimental Psychiatry*). Moreover, several other behaviorally oriented journals emerged (e.g., *Cognitive Therapy and Research, Biofeedback and Self-control*), and specific journals devoted primarily to behavioral assessment (e.g., *Behavioral Assessment, Journal of Behavioral Assessment*) were formed, although they were eventually changed.

CONCEPTUAL MODELS OF HUMAN BEHAVIOR: IMPLICATIONS FOR CHILD THERAPY

An extraordinary amount of theory and research has been generated that has a bearing on child therapy (Morris & Kratochwill, in press). As a result, tremendous amounts of data have been accumulated concerning the origins, development, influences, and variations in human behavior. Nevertheless, the wealth of information has clearly not resulted in any integrated view of human performance. Indeed, the current state of knowledge generated from the various conceptual models has not only resulted in the lack of an integrated view of human functioning, but has yielded various conceptual positions that are diametrically opposed and has spawned debate in the evidence-based practice movement (Kratochwill & Shernoff, 2004).

Because our understanding of human behavior is influenced by basic assumptions concerning the "why" of behavior, assessment and treatment practices often become inextricably interwoven with the particular conceptual model of human functioning held by the psychologist. Different models, with their different perspectives on behavior, yield vastly different assessment approaches and data that are used in making decisions relative to assessment and intervention. In this section, we review some models of human behavior that influence contemporary psychological practices. The models reviewed include the medical or biogenetic model, intrapsychic disease model, psychoeducational process test-based model, and behavioral model. These various models have been discussed by others in the professional literature, and due to space limitations we are not able to discuss them in detail or cover other

models (see Kratochwill & Morris, 1993, for coverage of other models). The models differ in their conceptualization of deviant behavior and assessment procedures and devices (sometimes), as well as the nature of the intervention implemented. Each model is discussed in the context of various components and considerations in its use.

Medical or Biological Model

Components

The medical model is one of the oldest approaches guiding assessment and treatment. The medical model can be applied in either a literal or a metaphorical context (Phillips, Draguns, & Bartlett, 1975). We view the model in its literal sense. That is, abnormal biological systems can be traced to some underlying biological pathology which is then treated. For example, defective hearing (symptom) may be traced to some type of infection (the cause), which may be treated with antibiotics. The prevalence of medical problems in children is actually quite high (see Bear & Minke, 2006, for an overview of various problems). A variety of health problems may be found in children in the school setting, including those children who are chronically ill and those with nutritional disorders (undernutrition, obesity), hearing and visual disorders, dental problems, disorders of bones and joints, infectious disorders, respiratory disorders, allergic disorders, urinary disorders, and drug-related problems. It seems clear that a medical model is appropriate to deal with the diversity of medical problems in the schools.

The medical model is a disease-based model. The pathology is assumed to be within the individual, although the causes may be environmental. Some theorists consider biological deviations to be the necessary and sufficient factors in the development of the pathology; others claim that chemical or neurological anomalies are the necessary but not sufficient condition for pathogenesis. Here, environmental conditions may or may not catalyze a constitutional predisposition to pathology.

Considerations

Medical model procedures are clearly justifiable when there is no basis for assuming physiological change in the organism as a result of the sociocultural environment. Controversial practices characterize medical model procedures when they are used to interpret measures of learned behavior (e.g., various forms of disruptive behavior in children, academic skill deficits). Although genetic, developmental, neurological, and biochemical factors all undoubtedly influence behavior, in reality these factors are not discrete entities. They are interwoven with one another as well as with environmental factors.

Applications of the medical model influence assessment and treatment in various ways. Organic factors may not always be the cause of an observed medical or physical problem. There is growing recognition that psychological factors may affect a physical condition and that physical symptoms may have no known organic or physiological basis. In the past, various concepts such as “psychosomatic” or

"psychophysiological" have been used to describe the psychological basis for physical or somatic disorders. However, such perspectives may be of limited usefulness because they imply a simplistic relation between psychological factors and a distinct group of physical disorders when, in fact, there may be a complex interaction of biological, environmental, psychological, and social factors contributing to various physical disorders (Siegel, 1983). Long ago, Lipowski (1977, p. 234) noted:

> The concept of psychogenesis of organic disease . . . is no longer tenable and has given way to the multiplicity of all disease . . . the relative contribution of these factors [social and psychological] varies from disease to disease, from person to person, and from one episode of the same disease in the same person to another episode. . . . If the foregoing arguments are accepted then it becomes clear that to distinguish a class of disorders as "psychosomatic disorders" and to propound generalizations about psychosomatic patients is misleading and redundant. Concepts of single causes and cirilinear causal sequences for example, from psyche to soma and vise versa, are simplistic and obsolete.

The point here is that even in the treatment of physical disease, psychological factors may be involved. Exclusive reliance on medical (drug) interventions may bias treatment in the sense that psychological (or other) aspects of functioning may be ignored. Problems most often arise when behavioral measures that can be influenced by a variety of environmental circumstances are used to assess the potential organic origins of a perceived symptom. The more the individual differences observed on a behavioral measure are influenced by environmental factors, the more the measure has the potential of being biased. Such a circumstance may arise when the environmental factors that influence the measure differ across cultural groups.

Psychodynamic Model

Components

The psychodynamic model proposes that maladaptive behaviors are symptoms resulting from underlying processes analogous to disease in the literal sense. This model is sometimes labeled the medical model in psychological and psychoeducational practice. Because conceptualization and treatment of abnormal behavior initially resided largely within the domain of medicine, the medical model was extended to treatment of abnormal behavior, both medical and psychological. The historical developments of the model are not reviewed in detail here; instead, the reader is referred to several historical sources that discuss this approach (e.g., Alexander & Selesnick, 1968; Kraepelin, 1962).

The psychodynamic approach can be characterized by the following:

> (a) uses a number of procedures, (b) intended to tap various areas of psychological functioning, (c) both at a conscious and unconscious level, (d) using projective techniques as well as more objective and standardized tests, (e) in both cases, interpretation may rest on symbolic signs as well as scorable responses, (f) with the goal of describing individuals in personological rather than normative terms. (Korchin & Schuldberg, 1981, p. 1147)

As is evident in this characterization, the psychodynamic approach is aimed at providing a multifaceted description and inferences. This process is said to promote a unique and individual approach to child assessment.

The psychoanalytic model represents one example of the psychodynamic disease paradigm, as do many other dynamic models of human functioning. The dynamic approach to assessment of deviant behavior is best elucidated in the context of assumptions held about the internal dynamics of personality (W. Mischel, 1968). Traditionally, dynamic approaches inferred some underlying constructs that account for consistency in behavior. Assessment is viewed as a means of identifying some sign of these hypothetical constructs, which are of central importance in predicting behavior. This *indirect sign paradigm* in assessment (cf. W. Mischel, 1972, p. 319) includes the use of a large variety of projective tests (e.g., Rorschach, TAT, Human Figure Drawings, Sentence Completion Tests) as well as objective personality inventories (e.g., MMPI-A for adolescents, MMPI-II, California Psychological Inventory) that are still used in contemporary practice.

A second feature of the traditional psychodynamic approach is that it assumes that behavior will remain stable regardless of the specific environmental or situational context. In this regard, test content is of less concern and may even be disguised by making items ambiguous, as is true in projective testing (Goldfried & Linehan, 1977). Indeed, a particular response to a projective test is rarely examined in view of the overt qualities of the situation in which the test occurred, but rather is interpreted in the context of a complex theoretical structure.

Considerations

The dynamic approach to assessment can be criticized on several grounds. One problem is the preoccupation with historical events, often in the absence of any verifying data. The second criticism relates to the emphasis during assessment on the individual's presumed unconscious beliefs, attitudes, motivations, and so forth, as interpreted through projections. Third, behavior is assumed to be a consequence of internalized pathological features. This assumption ignores evidence showing that many behaviors are situationally specific.

The use of various psychodynamic indirect measurement procedures has direct implications for child assessment and treatment. These measures continue to be used in clinical practice despite data indicating their low predictive validity (cf. Hersen & Barlow, 1976). When these issues became important, Goldfried and Kent (1972) noted that although the interpretation of certain signs on the Bender-Gestalt test (Hutt & Briskin, 1960) had no empirical support (cf. Goldfried & Ingling, 1964; Hutt, 1968), the revised version of the Bender-Gestalt manual presumably discounted these research findings and still recommended the use of questionable interpretations. A rather extensive literature on the comprehensive (predictive) validity of indirect measurement techniques (T. Mischel, 1971; W. Mischel, 1968) suggested that the predictions made on the basis of self-reports were equal to or superior to those made on the basis of indirect measurement techniques that are interpreted and scored by clinical experts. These findings held true for a wide variety of content areas (cf. W. Mischel, 1972).

Perhaps the most important issue that has been raised over traditional dynamic assessment is its relation to treatment. A number of authors over the years have noted that there was little relation between traditional assessment and treatment (Bandura, 1969; Goldfried & Pomeranz, 1968; Kanfer & Phillips, 1970; Peterson, 1968; Stuart, 1970). Thus, although traditional dynamic assessment may lead to a diagnosis that may in turn lead to the recommendation of a particular treatment, diagnoses resulting from traditional assessment methods cannot accurately predict what particular treatment mode should be implemented (Ciminero, Calhoun, & Adams, 1977; Stuart, 1970).

Psychometric Test-Based Model

The psychometric test-based model bears similarity to the psychodynamic disease model in that underlying processes, specifically process deficits, are said to account for learning and behavior problems. In many respects, this model can be considered a part of the dynamic model; however, a psychometric approach is characterized by the use of a variety of individual and group tests to compare individuals along various trait or construct dimensions. In trait theory approaches, various personality structures are said to account for an individual's behavior (W. Mischel, 1968, 1974). Trait theorists disagreed on what traits explained certain patterns of behavior, but generally agreed that certain behaviors were consistent across time and settings and that these patterns are expressions or signs of underlying traits.

In contrast to the psychodynamic position, trait assessors traditionally placed a high premium on objective administration and scoring of tests. Attempts usually were made to establish formal reliability and validity of the various measures used. On empirical grounds, historically this statistical approach proved generally superior to the more clinical method in predicting behavior (cf. Korchin & Schuldberg, 1981), but questions have been raised over the manner in which the research reflects the reality of decision making in actual clinical practice.

Closely related to the psychometric approach is the psychoeducational process model used by many practicing school psychologists. The model can be considered analogous to the psychometric trait model in that assessment focuses on internal deficits, except its context is psychoeducational rather than personality or emotionally oriented. Because a variety of cognitive, perceptual, psycholinguistic, psychomotor, and neuropsychological processes or abilities have been cited as causes of children's academic failure, norm-referenced cognitive (e.g., WISC, McCarthy, Stanford-Binet), perceptual (Bender Visual Motor Gestalt Test, Developmental Test of Visual Perception, Developmental Test of Visual-Motor Integration), psycholinguistic (e.g., Illinois Test of Psycholinguistic Abilities), and psychomotor (e.g., Purdue Perceptual-Motor Survey) tests are used to assess these abilities.

Most of these assessment procedures follow a diagnostic-predescriptive approach. Ysseldyke and Mirkin (1982, p. 398) noted:

> All of the diagnostic-perspective approaches based on a process dysfunction viewpoint of the nature of exceptionality operate similarly. When students experience academic difficulties it is presumed that the difficulties are caused by inner process dysfunctions or

disorders. Tests are administered in an effort to identify the specific nature of the within-child disorder that is creating or contributing to learning difficulties. Disorders or deficits are test named (e.g., figure-ground deficiencies, auditory sequential memory deficits, body image problems, eye-hand coordination difficulties, visual association dysfunctions, and manual expression disorders). Specific interventions are developed to "cure" the underlying causative problems.

Considerations

There are several important implications that can be raised with regard to the assessment tactics used in the psychometric model. First, because norm-referenced devices are commonly used in the model, the clinician must assume that clients tested have background and acculturation comparable to those on whom the test was standardized (cf. Oakland & Matuszek, 1977). Yet the point was frequently raised that standardized tests are biased and unfair to individuals from cultural and socioeconomic minorities because they reflect predominantly White, middle-class values and do not reflect the experiences and the linguistic, cognitive, and other cultural values and styles of minority individuals (Laosa, 1977). For example, although the norms for some tests (e.g., some group achievement and aptitude tests, the 1972 Stanford-Binet, WISC-IV) were generally good, norming on other instruments (e.g., Illinois Test of Psycholinguistic Abilities [ITPA], Leiter International Performance Scale, Slosson Intelligence Test) was quite inadequate.

A second issue is that research examining components of reliability and validity on various process measures has not been optimistic (Salvia & Ysseldyke, 1978; Ysseldyke & Salvia, 1974). For example, several early reviews of research on the ITPA (e.g., Bateman, 1965; Buros, 1972; Sedlak & Weener, 1973) drew attention to these limitations. The magnitude of the problems of inadequate norming, inadequate or incomplete reliability data, or questionable validity was nicely represented in data presented by Salvia and Ysseldyke (1978). Clearly, the potential for biased assessment practice is high given the poor psychometric properties of these instruments.

Behavioral Model

Components

Technically, there is no one model of behavior therapy, and contemporary behavior therapy, despite commonalities, is characterized by a great deal of diversity.* Historically, the different approaches in behavior therapy include applied behavior analysis (e.g., Baer, Wolf, & Risley, 1968; Bijou, 1970), mediational stimulus-response model (e.g., Rachman, 1963; Wolpe, 1958), social learning theory (e.g., Bandura, 1969, 1977b), and cognitive-behavior modification (e.g., Meichenbaum, 1974, 1977; Mahoney, 1974; Mahoney & Arnkoff, 1978.)**

*A detailed account of the history of behavior modification can be found in Kazdin (1978).

**These approaches are only briefly reviewed here. The reader is referred to Kazdin and Wilson (1978) as well as original sources within each approach for a more detailed presentation. The reader is also referred to Kratochwill (1982) and Kratochwill and Bijou (1987).

Applied Behavior Analysis

This form of behavior therapy developed from the experimental analysis of behavior (cf. Day, 1976; Ferster & Skinner, 1957; Sidman, 1960; Skinner, 1945, 1953, 1957, 1969, 1974). It emphasized the analysis of the effects of independent events (variables) on the probability of specific behaviors (responses). Contemporary applied behavior analysis focuses on behaviors that are clinically or socially relevant (e.g., various social behaviors, learning disorders, mental retardation, social skills) and adheres to certain methodological criteria (e.g., experimental analysis, observer agreement on response measures, generalization of therapeutic effects).

Advocates of applied behavior analysis use a more restrictive sense of the term "behavior" than other areas represented in the field of behavior therapy: behavior as the overt activity of an organism. Internal feelings and cognitions are typically not considered a major focus for the techniques of therapy, research, and practice. However, it must be stressed that applied behavior analysis focuses on the behavior of an individual as a total functioning organism, although there is not always an attempt to observe, measure, and relate an organism's entire response taking place at one time (Bijou, 1976; Bijou & Baer, 1978).

Many intervention procedures associated with applied behavior analysis are derived from basic laboratory operant research (e.g., positive and negative reinforcement, punishment, time-out, response cost, shaping, fading stimulus control; see Bijou, 1976; Gelfand & Hartmann, 1975; Kazdin, 1980; Sulzer-Azaroff & Mayer, 1977). Assessment emphasizes the individual application of these procedures and a functional evaluation of their effectiveness (Bijou & Grimm, 1975; Emery & Marholin, 1977). Behavior analysis refers to the study of organism-environment interactions in terms of empirical concepts and laws for understanding, predicting, and controlling organism behavior and repeated measurement of well-defined and clearly observable responses (Bijou, 1976; Bijou, Peterson, & Ault, 1968; Bijou, Peterson, Harris, Allen, & Johnson, 1969).

Neobehavioristic Mediational Stimulus-Response Model

The noebehavioristic mediational stimulus-response (S-R) model is derived from the work of such learning theorists as Pavlov, Guthrie, Hull, Mower, and Miller (e.g., Eysenck, 1960, 1964; Rachman, 1963; Wolpe, 1958). These approaches are characterized by "the application of the principles of conditioning, especially classical conditioning and counter-conditioning to the treatment of abnormal behavior" (Kazdin & Wilson, 1978, p. 3). Although intervening variables and hypothetical constructs play a role in assessment and intervention, covert activities are most commonly defined in terms of a chain of S-R reactions, with cognitive formulations de-emphasized.

A number of treatment procedures such as counterconditioning and systematic desensitization have been used to treat anxiety reactions, phobic patterns, and other strong emotional disorders in children (Morris & Kratochwill, 1983; Morris, Kratochwill, Schoenfield, & Auster, in press). Systematic desensitization, based originally on the principle of reciprocal inhibition (Wolpe, 1958), has been

successfully used to treat a wide range of child and adult problem behaviors (see Morris & Kratochwill, 1983; Morris et al., in press). Assessment in the mediational S-R model relies on survey schedules (e.g., fear survey schedules), self-support data, and direct measures of client behavior (as in the use of behavioral avoidance tests).

Cognitive-Behavior Therapy

A unifying characteristic of the cognitive-behavior therapy approach was an emphasis on cognitive processes and private events as mediators of behavior change (Kendall, 1981b). The source of a client's problems were said to be related to the client's own interpretations and attributions of his or her behavior, thoughts, images, self-statements, and related processes (Kazdin & Wilson, 1978). Contemporary cognitive-behavior therapy emanates from Ellis's (1962) rational-emotive therapy, Beck's cognitive therapy, and Meichenbaum's self-instructional training. Treatment strategies are quite diverse and include such techniques as problem solving, stress inoculation, self-instructional training, coping skills training, language behavior therapy, thought stopping, and attribution therapy. These techniques represent procedures not generally addressed by other behavior therapy approaches (e.g., applied behavior analysis).

Assessment in cognitive-behavior therapy tended to be quite broad-based, taking into account many different dimensions of behavior. Yet there was still an emphasis on defining the nature of the target problem, whether overt or covert. In some cases, a more traditional functional analysis of behavior, which emphasizes a careful examination of environmental antecedents and consequents as related to a certain response repertoire, was recommended (e.g., Meichenbaum, 1977).

Some specific purposes for cognitive assessment were outlined by Kendall (1981a, pp. 3–4):

1. To study the relationships among covert phenomena and their relationship to patterns of behavior and expressions of emotion.
2. To study the role of covert processes in the development of distinct psychopathologies and the behavioral patterns associated with coping.
3. To confirm the effects of treatment.
4. To check studies where cognitive factors have either been manipulated or implicated in the effects of the manipulation.

Social Learning Theory

Social learning theory is based on the original work of Bandura and his associates (e.g., Bandura, 1969, 1971, 1977b; Bandura & Walters, 1963) and has evolved considerably over time. Bandura noted that in addition to outcome expectation, a person's sense of his or her ability to perform a certain behavior mediates performance. He referred to these latter expectations as efficacy expectations or self-efficacy, and suggested that they have important implications for treatment. Psychological treatment and methods were hypothesized to produce changes in a child's expectations of self-efficacy, as in the treatment of phobic behavior. Self-efficacy was said to determine the activation and maintenance of behavior strategies for coping with

anxiety-eliciting situations. Self-efficacy expectations were also said to be modified by different sources of psychological influence, including performance-based feedback (e.g., participant modeling), vicarious information (e.g., symbolic modeling), and physiological changes (e.g., traditional verbal psychotherapy; cf. Kazdin & Wilson, 1978). Intervention procedures such as symbolic modeling (e.g., Bandura, 1971), direct modeling (Bandura, 1977b; Rosenthal, 1976), and self-modeling (Brody & Brody, 1977) have been associated with the social learning theory approach. For example, modeling historically has been used to treat a variety of children's fears (e.g., animal fears, inanimate fears, dental and medical fears; Morris & Kratochwill, 1983; Morris et al., in press), socially maladjusted children (e.g., social withdrawal, aggression), distractibility, and severe deficiencies (e.g., autism, mental retardation) as well as a wide range of academic behaviors (cf. Zimmerman, 1977). In all these approaches, social learning theory stressed that human psychological functioning involved a reciprocal interaction between the individual's behavior and the environment in that a client is considered both the agent as well as the target of environmental influence, with assessment focusing on both dimensions of behavior.

Unifying Characteristics

Despite apparent diversity among the different areas within behavior therapy, several dimensions set it apart from traditional forms of psychological assessment and treatment, particularly the test-based psychometric models and psychodynamic models (Hartmann, Roper, & Bradford, 1979). Contemporary behavior therapy consists of the following characteristics:

1. A strong commitment to empirical evaluation of treatment and intervention techniques;
2. A general belief that therapeutic experiences must provide opportunities to learn adaptive or prosocial behavior;
3. Specification of treatment in operational and, hence, replicable terms, and
4. Evaluation of treatment effects through multiple-response modalities, with particular emphasis on overt behavior. (Kazdin & Hersen, 1980, p. 287)

Behavior therapy has become very diverse over time and includes a number of therapeutic strategies that were excluded from the field during the early years (e.g., rational-emotive therapy). Although these characteristics are tied to the therapeutic aspects of the behavioral approach, each can also be conceptually representative of the behavioral approach to child treatment (Bergan & Kratochwill, 1990).

HISTORICAL CONTEXT OF CHILD THERAPY

The Early Years

As we noted previously, although the study of psychological problems among adults can be traced back as far as the ancient Greeks (if not further), the study

of child psychopathology represents a relatively recent phenomenon in the history of psychological treatment. The most famous early example of such study is Jean Itard's examination of and attempt to educate the Wild Boy of Aveyron beginning in 1799. However, an increased focus on problems of childhood would have to follow a greater focus on childhood as a distinct period in human development and an increased interest in the nature of psychological problems in children. The emergence of compulsory education marks an important milestone in the development of such interest, in part due to the large number of children who struggled in school.

The study of children in the United States was largely founded by G. Stanley Hall, whose work spanned 1883 through 1918 (Davidson & Benjamin, 1987). Hall focused on problems with education, and although he was not primarily concerned with psychopathology, his work helped promote a better understanding of the problems of childhood.

Clifford Beers played an important role in reversing the attitude toward the treatment of the mentally ill generally, and later this focus was applied to children. Beers became clinically depressed and suicidal as a Yale University law student. After his hospitalization, he wrote *A Mind That Found Itself* (1908), a book describing the poor treatment he received while institutionalized. The success of the book helped inform the public of the terrible conditions in state hospitals. Beers would form a number of organizations, including the Connecticut Society for Mental Hygiene (in 1908) and the National Committee for Mental Hygiene (in 1909), which promoted better conditions in state hospitals, better treatment methods, and research on the prevention and treatment of psychopathologies. This increased concern for the treatment of persons with mental illness brought about "mental hygiene" programs in schools (Morris & Kratochwill, in press-b).

Lightner Witmer is credited as the first clinical psychologist, although he had a strong interest in schools and has an APA Division 16 (School Psychology) award named after him. Witmer established the first psychological clinic in the United States, at the University of Pennsylvania, in 1896. His first client was a 14-year-old boy who had difficulty spelling (Benjamin, 2005). In the first year of the clinic's operation, Witmer and his students saw a total of 24 children between the ages of 3 and 16 years, who had such presenting problems as "learning difficulties, speech problems, and possibly chorea, hydrocephalus, and hyperactivity" (P. McReynolds, 1996, p. 238). Witmer published the first journal devoted to clinical psychology, titled *The Psychological Clinic*, in 1907 (Routh, 1996; Witmer, 1907/1996). The focus on children with problems in schools would eventually lead to the development of a specialty of psychology that involved practice in schools.

In 1909, the Cook County Juvenile Psychopathic Institute in Chicago was formed under the leadership of William Healy. The institute staff worked directly with juvenile offenders and stressed an interdisciplinary approach to studying juvenile issues (e.g., psychiatrists, psychologists, and social workers worked together on particular cases, taking into account multiple causes and perspectives). Aided by Beers's National Committee for Mental Hygiene, numerous child guidance clinics also developed across the country over the next several decades, and by 1930 there were about 500 child guidance clinics in the United States (Kanner, 1948).

Through the introduction of dynamic psychiatry and psychoanalysis during the early twentieth century, specifically the work of Sigmund Freud in Vienna and Adolph Meyer in the United States, the psychodynamic approach maintained that the origins of behavior problems lay in the past experiences (typically, childhood) of the person (Kanner, 1948), thereby turning attention to childhood. For example, with adult patients, these experiences were explored retrospectively by the psychiatrist or psychoanalyst, who attempted to draw causal relationships between these past experiences and the patients' present behavior. During the early years of the dynamic psychiatry movement, children were usually not seen in treatment by psychiatrists (Kanner, 1948). Nevertheless, the retrospective search for the relationship between early childhood events and present functioning aroused sufficient interest among professionals that some began to acquaint themselves specifically with the behavior problems of children as well as with the dynamics that contributed to their difficulties (Morris & Kratochwill, in press-a). This psychodynamic interest in children was not formally realized in the literature until the publication in 1909 of Freud's (1909/1963) detailed case of "Little Hans." Interestingly, although Freud formulated his etiological theory of phobias on the basis of Hans's symptoms and experiences, he did not treat Hans directly; Hans's father treated him under Freud's direction and supervision (Morris & Kratochwill, 1983). Although Hans's problem was treated successfully, it was not until at least 15 to 20 years later that Freudian psychoanalytic child therapy came into existence. This was largely due to the contributions and adaptations of Freud's work for children by Melanie Klein, Freud's student, and his daughter, Anna Freud, and the subsequent publication beginning in 1945 of Anna Freud's multivolume edited book series (with Hans Hartmann and Ernst Kris), *The Psychoanalytic Study of the Child*. The changes made by these women, as well as the earlier psychoanalytic therapy work with children first initiated by Hermine Hug-Hellmuth (the third woman to join Freud's Vienna Psychoanalytic Society during the 1900s), made Freud's therapeutic approach very relevant to children and contributed to its increasing popularity in the twentieth century and later influence in the development of many other forms of child and adolescent psychotherapy (Benveniste, 1998).

Some of the major methodological changes that were made for children in Freud's psychoanalysis were the substitution of play activities for the technique of free association and the use of drawings and dreams to understand a child's problems. For example, through the use of the medium of play, both Klein and Anna Freud discovered that children were able to represent to the child analyst their inner conflicts and perceptions of important relationships in their lives, as well as portray in their play their unique feelings about, perceptions of, and concerns about their various pleasurable and traumatic experiences (Morris, Li, Lizardi-Sanchez, & Morris, 2002; Warshaw, 1997).

Morris and Kratochwill (in press-b) identified some additional developments that influenced the focus of child therapy. First, as we noted previously, the intelligence testing movement had a tremendous influence on the study of children; in particular, it became possible to learn the extent to which a particular child differed from the norm in cognitive ability. It also demonstrated clearly the diversity of children in

terms of their comprehension of classroom instruction (Kanner, 1948) and spawned a decade of debate on bias in testing and eventually, in the twenty-first century, federal regulations that would expand the focus of special services for children in schools (Kratochwill et al., in press).

Second, the formation of professional associations contributed to the increasing emphasis on the treatment of children. The first professional association was the Association of Medical Officers of American Institutions for Idiots and Feeble-minded Persons, founded in 1876, which evolved into the present-day American Association on Mental Retardation. Its first president was Edward Seguin. In 1892, another professional association was formed, the American Psychological Association, in which Witmer was a charter member. G. Stanley Hall was the first president of APA, and James Cattell was one of the first members of the Council (Board of Directors of APA) and the fourth president of APA. In 1922, the Council for Exceptional Children was formed; it consisted primarily of educators and other professionals, although parents were members too. The fourth early association was the American Orthopsychiatric Association, founded in 1924 and consisting primarily of psychiatrists, applied psychologists, and social workers, although educators as well as other professionals and parents were also members. Each of these groups encouraged the formulation and conduct of research with children having behavior and learning disorders, as well as the sharing of information regarding effective psychological and/or educational interventions.

Third, a treatment approach emerged that was initially much less popular than psychoanalytic child therapy. This approach was behaviorism and later became known as the behavior modification or behavior therapy movement (Kazdin, 1978; Morris, 1985). As we noted earlier, behavior modification emerged largely from the experimental psychology laboratory rather than from direct interaction with patients and was based on theories concerning how people and animals learn to behave through S-R learning and conditioning rather than through the conscious or unconscious thinking found in psychoanalytic writings. The two most famous behaviorists associated with this movement are John B. Watson (1913, 1919), often referred to as the "father of behaviorism," and B. F. Skinner (1938, 1953), the behavioral researcher and theorist who extended Watson's behaviorist views and developed a learning paradigm that Skinner referred to as "operant conditioning." The behavior modification and behavior therapy procedures that were derived from various learning theories were largely confined for many years to research settings. In fact, it was not until the mid-1960s to the late 1970s that these procedures began being applied on a regular basis in children's residential treatment settings, regular and special education classrooms, and outpatient mental health settings (Morris, 1985; Morris et al., 2002).

Developments in behavior modification set the stage for the evidence-based practice movement, when many of the procedures were implemented in research that offered strong support for effective outcomes, first with adults and eventually with children. Behaviorism also set the stage for prevention models in educational settings with particular emphasis on such models as positive behavior support, a movement with its roots in applied behavior analysis (Crone & Horner, 2003; Crone, Horner, & Hawken, 2004).

DEVELOPMENT OF THE EVIDENCE-BASED INTERVENTION AND PRACTICE MOVEMENT

As has been evident in the previous sections of this chapter, a number of different events shaped the future of psychological assessment and treatment of children. As the need for services for children and families increased, attention turned to the training of professionals in psychology. By the 1940s, considerable attention had been given to the need for a training model in clinical psychology. Up to this point, a great variety of training curricula existed in training programs, and training was often rushed and inadequate. Some programs produced statistically sophisticated therapists, and others focused on more service-oriented practice (Hayes et al., 1999). The issue had been present at least since 1918, when Leta Hollingworth argued that a training model was critical in the development of clinical psychology as a profession (Benjamin, 2005). However, there was little resolution until 1949, when 73 professionals met in Boulder, Colorado, for 15 days to discuss the training of clinical psychologists. The result became known as the scientist-practitioner model (or the Boulder model) and would profoundly influence professional psychology in subsequent years and, eventually, the evidence-based practice movement.

Training Psychologists: Advent of the Evidence-Based Practice Focus

The events leading up to the Boulder Conference were perhaps as rooted in social trends as they were in professional trends. By the 1940s, there was growing concern for mental health in the United States. The year 1944 saw written instruments and assessments used on close to 20 million persons (Reisman, 1976). World War I produced an abundance of psychological casualties for whom too few professionals could provide adequate care. In the wake of World War II, as professionals took steps to improve the quality of their care, the Veterans Administration officially declared clinical psychology a health care profession. The next major step was to improve and homogenize the training of clinical psychologists. Pressure to do so began to come from a variety of sources.

In 1941, the American Association for Applied Psychology (AAAP) endorsed a day-long conference for the development of a committee for improved training of clinical psychologists. The subsequent report, largely focused on training for the treatment of psychopathology, supported a 4-year PhD program in psychology: a year for systematic foundation in psychology, a year for psychometric and therapeutic principles and practice, a year for internship, and a final year for dissertation (Shakow, 1942). Reactions were largely positive, and the AAAP formed the Committee on Training in Clinical Psychology (CTCP). The integration of the AAAP with the APA in 1944 prevented prompt action, but the seeds were planted for the organization of professional training.

With concern for the welfare of returning soldiers, an increasing shortage of psychologists in the public, and concern over the quality of professional training, the VA and the U.S. Public Health Service (USPHS) began to commit funds for the training of clinical psychologists. In 1947, the VA and USPHS requested that the

APA assist in determining ways to train greater numbers of professional psychologists (Baker & Benjamin, 2000). The APA turned to the CTCP with four requests: recommend a clinical psychology training program, develop standards for institutions training clinical psychologists, visit and study these institutions, and maintain contact with other organizations invested in these problems (e.g., the American Orthopsychiatric Association, the National Committee for Mental Hygiene; APA, 1947). The chair of the CTCP, David Shakow, sent committee members a copy of a report titled "Graduate Internship Training in Psychology" (Shakow et al., 1945) and asked that it be critiqued. The Shakow Report, as it would become known, garnered significant support as the most comprehensive set of clinical psychology training recommendations to date.

In 1948 and 1949, the CTCP visited a number of training programs and awarded accreditation to 43 of them (APA, 1949). Although the committee was largely impressed with the efforts put forth by these programs, significant concerns were raised. First, it was contended that the focus of training was too narrow. There was certainly a social need for emphasis on psychiatric hospitals and the severely disturbed patients, but the committee argued that clinical psychologists should be trained to deal with other social issues. Second, many programs taught a variety of clinical techniques, placing limited emphasis on research methodology and theory—an issue that would become central to the Boulder Conference later that year (Baker & Benjamin, 2000).

When the Boulder Conference began, the adaptation of the scientist-practitioner model was not entirely agreed on. There was great concern that it would be difficult to train all graduate students in both research and practice. The role of research in training likely was the most difficult topic covered in the conference. However, by the end of the conference, agreement was reached among the large majority. In the most comprehensive report of the conference, titled *Training in Clinical Psychology,* Victor Raimy (1950, p. 23) reported, "The original chasm seemed to have largely disappeared and recognition of the importance of including research training in the preparation of all clinical psychologists was generally accepted." In fact, the decision to recommend training in research and practice did not require a compromise, but instead was nearly a unanimous consensus.

Recommendation for preparation in both areas rested on five major considerations. First, although most students would focus their career on either research or practice (not both), all students should be trained in both to encourage "cross fertilization and breadth of approach" (Raimy, 1950, p. 81) within the profession. That is, dual training would benefit both foci through collaboration; in addition, the study of research and practice would deter programs' tendencies to produce narrow thought processes and subsequent actions. Second, the extreme lack of dependable knowledge in the field demanded (and still demands) a need for research in clinical psychology. As Raimy reports, "Participants at the conference displayed considerable humility with respect to present techniques" (p. 80). Third, it was believed that students of clinical psychology should be capable of carrying out both roles of clinician and researcher. The number of applicants greatly surpassed the number of available positions in training programs; therefore, participants in the conference concluded that applicants should be considered for acceptance only if they showed

promise in both areas of study. Fourth, it was noted that researchers would have a better knowledge of important research topics if they were directly involved with the clinical process. Fifth, services provided by clinical practice could provide financial support for research causes. This has become even truer in the current age of managed care. Increasingly, health care systems rely on evidence-based practices for success in the marketplace (Hayes et al., 1999).

The following summarizes the roles of clinical psychologists as concluded by participants at the Boulder Conference:

> The basic needs of society for the services of clinical psychology are of two major kinds:
>
> a. Professional services to:
> (1) Individuals through corrective and remedial work as well as diagnostic and therapeutic practices
> (2) Groups and social institutions needing positive mental hygiene programs in the interest of better community health
> (3) Students in training, members of other professions, and the public through systematic education and the general dissemination of information
> b. Research contributions designed to:
> (1) Develop better understanding of human behavior
> (2) Improve the accuracy and reliability of diagnostic procedures
> (3) Develop more efficient methods of treatment
> (4) Develop methods of promoting mental hygiene and preventing maladjustment. (Raimy, 1950, pp. 20–21)

Though the decision to endorse the scientist-practitioner model was clear, participants recognized that there would be great difficulties involved in combining both roles. In this respect, participants in the Boulder Conference clearly anticipated the scientist-practitioner gap. A major criticism of the scientist-practitioner model developed at the Boulder Conference is its demonstrated failure to involve clinicians in research. The model has been much more universally adopted in research than in practice. The average PhD psychologist has published very few, if any, research articles.

Recent Developments in the Evidence-Based Intervention and Practice Movement

Despite great momentum in research in professional psychology, the evidence base for treatment of a variety of childhood disorders and problems lagged behind outcome research with adults. Nevertheless, as attention turned to mental health issues in children, psychologists began to review the literature on prevention and intervention (therapy) outcome research. To perform this task they needed guidelines to review the growing literature on prevention and intervention. Over the past decade criteria have been produced for developing evidence-based interventions and guidelines for how such interventions should be evaluated (see Barlow, 2004; Kratochwill & Stoiber, 2002). With pressure mounting from managed health care, governmental agencies, and professional organizations, a major task force on

evidence-based interventions was formed by the APA Task Force on Promotion and Dissemination of Psychological Procedures (1995). The report stimulated considerable debate. The major controversial themes were identified: (a) The focus of research studies identified as evidence-based in the adult literature were identified as behaviorally oriented; (b) the criteria used were judged as potentially biased (e.g., methodological and statistical decision rules in the literature reviews); (c) the use of intervention manuals to standardize interventions was deemed problematic (e.g., manuals remove flexibility and adaptation to individual or group needs in psychotherapy); and (d) the generalizability of findings was deemed limited (i.e., whether and how research studies can be generalized to actual clinical practice).

Early in the evidence-based intervention movement school psychology explored evidence-based interventions and their implications for the field. Division 16 of the APA, the Society for the Study of School Psychology, and the National Association of School Psychologists supported the development of a Task Force on Evidence-Based Interventions in Schools (now co-chaired by Thomas R. Kratochwill and Kimberly Hoagwood).

The initial purpose of the School Psychology Task Force was to examine and disseminate the knowledge based on what prevention and intervention programs or approaches for children, youth, and families demonstrate empirical support for application in the school and community and to facilitate strong research methodologies, technologies, and innovations. The Task Force worked cooperatively with the APA's Division of Clinical Child and Adolescent Psychology (Division 53).

Another major direction taken in the evidence-based practice movement has involved the development of practice guidelines (White & Kratochwill, 2005). Various practice guidelines have been constructed by a variety of professional groups: the Agency for Health Care Policy and Research (AHCPR), the American Psychiatric Association, the National Institutes of Health, and the Division of Clinical Psychology (Division 12) of the APA. The AHCPR has developed guidelines for treating and diagnosing depression in primary care. The guidelines are primarily aimed at physicians and the general public. Practice guidelines of the American Academy of Child and Adolescent Psychiatry have been evolving for some time and provide a variety of foci for child intervention, including anxiety disorders, conduct disorders, and Attention-Deficit/Hyperactivity Disorder. The Division 12 Task Force (Chambless & Ollendick, 2001) provided a preliminary set of practice guidelines based on their review of empirically validated psychosocial interventions, resulting in a proposed three-category system of efficacy: well-established interventions, probably efficacious interventions, and experimental interventions (those not yet established as at least probably efficacious).

Over the past decade the Committee on Science and Practice of the Society for Clinical Psychology of APA Division 12 developed a procedural and coding manual for identification of beneficial interventions (Weisz & Hawley, 1998). The manual provided guidelines for identification, review, and coding of studies of psychological interventions for behavioral, emotional, and adjustment problems and disorders. The manual was designed to (a) identify intervention outcome studies related to the effects of psychological interventions; (b) code studies according to the committee's criteria and provide related information on characteristics of interventions

in the studies that evaluate efficacy; and (c) make a determination on whether the intervention is beneficial. The procedural and coding manual extended the work of the Division 12 Task Force and the efforts of the Division 53 Task Force by clarifying what was meant by "empirically supported interventions" and expanded on a variety of published literature and the work of various committee members.

As various professional groups and researchers began to review the literature, narrative reviews and the application of meta-analysis to child psychotherapy outcome contributed to an evolving positive landscape for effective interventions (e.g., Lonigan, Elbert, & Johnson, 1998; Casey & Berman, 1985; Kazdin, Bass, Ayres, & Rodgers, 1990; Kazdin & Weisz, 2003; Weisz, Weiss, Alicke, & Klotz, 1987). Specifically, between 1952 and 1993, more than 300 research investigations of psychotherapy involving children between the ages of 2 and 18 years indicated that the intervention groups scored higher on various outcome measures than 76% to 81% of children in control groups (Casey & Berman, 1985; Kazdin, 2002). Hoagwood and Erwin (1997) conducted a 10-year research review and found that three types of interventions have empirical support: cognitive-behavior therapy, social skills training, and teacher consultation. Modest support for interventions delivered within school-based consultation occurred through traditional literature reviews (Kratochwill, Sheridan, & Van Someren, 1988; Mannino & Shore, 1975) and meta-analysis (Medway & Updyke, 1985; Sibley, 1986). Summaries of evidence-based treatments with children and adolescents can be found on a Division 53 website (www.effectivechildtherapy.org).

As the cost of treatment of adult and childhood disorders became clear, psychologists became vocal about the importance and effectiveness of prevention programs (e.g., Biglan, Mrazek, Carnine, & Flay, 2003; Coie et al., 1993; Kratochwill, Albers, & Shernoff, 2004; Nation et al., 2003; Tolan & Dodge, 2005; Weissberg, Kumpfer, & Seligman, 2003; Weisz, Sandler, Durlak, & Anton, 2005). Meta-analysis of prevention programs has also demonstrated positive outcomes (e.g., Durlak & Wells, 1997; Weisz, Jensen-Doss, & Hawley, 2006). To assist psychologists in implementing effective prevention programs, a website jointly sponsored by APA and the Society for Prevention Research presents information on over 100 reviews of prevention research on a variety of topics (www.oslc.org/spr/apa/summaries.html).

Tremendous advances have been made in the evidence-based practice area of child prevention and treatment. In recent years, several textbooks that emphasize evidence-based prevention and intervention for child and adolescent problems and disorders have appeared (see Table 1.1). The future looks very positive with regard to improving the quality of life of children and families who experience a wide range of academic and mental health concerns. To advance practice in the field, a number of issues have surfaced that have an important bearing on the future of this movement (see Kratochwill et al., in press). Although it is not possible to review these issues in great detail here, a sampling will convey the complexity of the tasks. To begin with, major concerns surround the research methods of establishing the evidence base in this area, with calls for expanding the methodologies of quantitative research and even the framework for empiricism of the movement (see Slife, Bradford, Wiggins, & Graham, 2005). Concerns have also been expressed about the

Table 1.1 Coverage and Number of Chapters of Specific Topics Addressed in Each Book on Evidence-Based Intervention

Topic	1[a]	2[b]	3[c]	4[d]	5[e]	6[f]
Treatment research methods	E	E	C	E	E	E
Developmental issues	NC	NC	C	E	E	E
Ethical issues	NC	NC	NC	E	NC	E
Assessment	NC	E	E	NC	E	E
Diversity	NC	NC	E	E	E	C
Future directions	E	E	E	E	E	C
Clinical disorders or topics						E
Adherence to medical regimens			1			E
Anxiety disorders	2	1	1	2	4	1
Attention Deficit Hyperactivity	2	1		1	1	1
Eating disorders		1		1		
Conduct Disorder, Oppositional Defiant Disorder						1
Depression	6	1	1	6	2	1
Depression	2	1		4	1	
Developmental disorders (e.g., language)		1				1
Encopresis			1			
Enuresis			1	1		
Habit disorders			1			
Obesity						
Obsessive-Compulsive Disorder				1	1	1
Pain management			1			
Pervasive developmental disorders (Autism)		1		2		1
Medical or physical conditions		1				1
Posttraumatic Stress Disorder					1	1
Prevention					4	1
Psychotic disorders		1				
School refusal						
Sleep problems			1			
Substance abuse		1				
Suicide		1				
Tourette's syndrome		1				
Somatic disorders						1
Child sexual abuse or Child maltreatment						1
Ethical and legal issues						1
Mental retardation and Intellectual disability						1
Psychopharmacology						1

Note: C = Mentioned but minimally; E = Emphasized; NC = Not covered.

[a] *Psychotherapy for Children and Adolescents: Evidence-Based Treatments and Case Examples,* by J. R. Weisz, 2004, Cambridge: Cambridge University Press.

[b] *What Works for Whom? A Critical Review of Treatments for Children and Adolescents*, by P. Fonagy, M. Target, D. Cottrell, J. Phillips, and Z. Kurtz, 2002, New York: Guilford Press.

[c] *Treatments that Work with Children: Empirically Supported Strategies for Managing Childhood Problems*, by E. R. Christophersen and S. L. Mortweet, 2003, Washington, DC: American Psychological Association.

[d] *Evidence-Based Psychotherapies for Children and Adolescents*, by A. E. Kazdin, & J. R. Weisz, (Eds.), 2003, New York: Guilford Press.

[e] *Handbook of Interventions that Work with Children and Adolescents: Prevention and Treatment,* by P. M. Barrett and T. H. Ollendick (Eds.), 2004, New York: Wiley.

[f] *The Practice of Child Therapy*, fourth edition, by R. J. Morris and T. R. Kratochwill, in press-a, Mawah, NJ: Erlbaum.

limitations of the focus of research, which traditionally has not given attention to the transportability of the prevention and intervention programs (e.g., Schoenwald & Hoagwood, 2001). Limitations of the sample of participants in evidence-based research and the need for cultural adaptations of interventions have also been noted (e.g., Bernal & Scharron-Del-Rio, 2001). In addition, limitations of outcome assessment strategies and, in particular, the arbitrary nature of the instruments used in evidence-based intervention research have been noted (see Blanton & Jaccard, 2006; Kazdin, 2006). Despite these challenges, major commitments to advance the quality of practice with a firm foundation of scientific support are in place (Frick, 2007; Kratochwill et al., in press). As the evidence-based movement progresses it will be important to consider the historical evidence and its importance in teaching us that diversity of viewpoints can best advance science and practice.

SUMMARY

Assessment and intervention for children and adolescents experiencing academic and behavior problems have come far from their ancient roots. In this chapter, we summarized the major trends and events that led to the current state of child and adolescent therapy. We began with a synopsis of progress prior to the nineteenth century, which set the stage for increasingly systematic investigations of human characteristics and behavior. This focus was exemplified by the work of Itard, Wundt, Galton, Cattell, and others. During the first half of the twentieth century, intelligence testing and personality assessment emerged as significant roles for psychologists. Methodological differentiation rapidly became an interest during World War I, and intelligence testing became important for the assessment of schoolchildren. In the second half of the twentieth century, we witnessed the rise of behavioral modification and assessment methods as greatly important in child and adolescent psychology. In particular, these assessment and intervention technologies set the stage for a scientific basis for therapy that would shape the evidence-based practice movement in psychology.

In addition to a discussion of trends and events, we described major conceptual models that have been developed and persist today, including the medical or biological model, the psychodynamic model, the psychometric test-based model, and the behavioral model. Implications for their use in current practices were discussed. We addressed the history of child therapy, from its most famous early roots to modern trends, including the emergence of the scientist-practitioner training model and the evidence-based practice movement. A brief discussion of major concerns surrounding evidence-based practice was provided. The future looks bright for the evidence-based practice movement as the complexities of selection, implementation, and sustainability of treatment technologies are being addressed in research and practice.

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CHAPTER 2

Clinical Training

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Clinical child and pediatric psychology might be considered adolescents in the field of professional psychology (Prinstein & Roberts, 2006). They have matured to the point of cohesive self-definitions that are accepted by the professional community and that guide training within the specialties. But these developments are relatively recent. Although interest in psychological treatment of youth dates back to the end of the nineteenth century, the need for specific training in child-related service provision was not vocalized until more than 50 years later, and formal initiatives to develop training standards and implement them consistently lagged even further behind. Fortunately, for both clinical child and pediatric psychology, adolescence has come with emergence of clear and complex identities that are grounded in the broad foundations of clinical psychology, integrated with related fields of human development and health psychology, and sensitive to the importance of context and diversity in addressing the needs of youth and their families. Adolescence has also come with increased autonomy and respect from the professional community, thanks to the passionate efforts of leaders in these areas. Thus, clinical child and pediatric psychology have become legitimate specialties within clinical psychology.

The current chapter focuses on issues involved in preparing aspiring psychologists for careers in clinical child psychology and pediatric psychology. We begin by highlighting some important foundational issues in general clinical psychology. We then discuss the emergence and definitions of clinical child and pediatric psychology and examine their relationship to other fields in psychology and associated disciplines. Next, we discuss training models and important components of clinical child and pediatric training at the graduate, internship, and postdoctoral levels. Finally, we address some of the current challenges in clinical psychology training, such as balancing broad and general training with more focused depth, defining and assessing professional competence, and identifying and addressing workforce issues.

GENERAL CLINICAL PSYCHOLOGY FOUNDATIONS

The birth of clinical psychology as a profession was really the birth of clinical *child* psychology, when, in 1896, Witmer established the first psychological clinic to treat youth with behavior and learning problems (Hothersall, 2004). Thus began

an era in which clinical psychology began to define itself, through journals devoted to clinically relevant research and practice (e.g., *Psychological Clinic; Journal of Consulting Psychology*, now published as the *Journal of Consulting and Clinical Psychology*), establishment of psychological testing and treatment clinics (e.g., the Juvenile Psychopathic Institute in Chicago and Judge Baker Foundation in Boston), a focus on training (e.g., increased availability of textbooks, university courses, and internships; development of training standards), and establishment of professional organizations such as the American Psychological Association (APA) Section on Clinical Psychology (Edelstein & Brasted, 1991; Hothersall, 2004).

Although some of the earliest work in clinical psychology focused on youth, various social forces, notably the need to evaluate large numbers of military personnel during World Wars I and II, moved substantial focus to adults. Nonetheless, as a profession, clinical psychology did not differentiate between child- and adult-focused professionals. Indeed, the 1949 Boulder Conference on graduate training in clinical psychology, on which many of our standards of graduate education are still based, emphasized the need to avoid specialization and to focus on broad training across many age levels (Raimy, 1950).

Over the last 110-plus years, the profession of clinical psychology has grown, matured, and become increasingly complex (Prinstein & Roberts, 2006). Clinical psychologists engage in activities ranging from basic and applied research to assessment and diagnosis, prevention and intervention, consultation, teaching and training, and public policy development, in settings as diverse as universities and research institutions, medical settings, schools, businesses, and government. They address the behavioral and emotional disorders, personality disorders, physical and mental disabilities, medical issues, and environmental stresses affecting individuals from infancy to late adulthood. Thus, it is not surprising that many leaders in the field (e.g., Roberts, 2006; Tuma, 1983) have questioned the wisdom, and even the ability, of continuing to pursue the type of generalist training espoused at the Boulder Conference. In essence, it is no longer feasible (if it ever was) for clinical psychologists to master the universe of relevant literature and skills required for professional competence as generalists. This has created an interesting dilemma for the profession. On the one hand, current standards for training, as operationalized by the field's accrediting body (APA Committee on Accreditation; CoA), still call for broad and general training in clinical psychology. On the other hand, providing such generalist training by itself is increasingly likely to produce graduates who are ill prepared to make positive contributions to the profession.

So how can clinical training address the challenge of providing broad foundational education in clinical psychology without drowning students in an ever-expanding sea of knowledge? There are many possible answers to this question, and given the complexity of the issues, multiple solutions almost certainly need to be implemented. But one first step in addressing the question is to acknowledge that general training in clinical psychology does not have the same meaning that it did in 1949 and to create training environments that retain the core aspects of clinical psychology while allowing focused experience in subfields or specialty areas. Interestingly, several leaders in the field (e.g., Roberts, 2006; Roberts & Sobel, 1999; Tuma, 1983) argue that is exactly what has happened in most clinical

programs, albeit somewhat covertly. Roberts and Sobel suggest that "current use of the term *clinical psychology* is today not as generally clinical as it might have been in earlier times. Today, clinical psychology typically represents only adult clinical psychology and only superficially represents a generalist clinical psychology" (p. 483). From this perspective, clinical (adult) psychology and clinical child psychology represent parallel paths to balancing the demands of breadth and focus. This *Handbook*, with separate volumes for clinical adult and clinical child psychology, overtly acknowledges the legitimacy of *broad and focused* training in adult- and child-focused clinical psychology and is an important step in advancing the field.

EMERGING CHILD CLINICAL AND PEDIATRIC IDENTITIES

During the 30 years following the Boulder Conference, several leaders in psychology education noted the existence of large numbers of unserved youth with behavioral and emotional problems or health-related issues, but a shortage of professionals with sufficient training to address the needs of these youth (Ross, 1959; Tuma, 1983; Wohlford, 1978; Wright, 1967). The fields of clinical child and pediatric psychology began to emerge, with professional organizations (e.g., APA Section on Clinical Child Psychology, now Division 53; APA Society of Pediatric Psychology, now Division 54), journals, and conferences. Graduate programs offering specialty training also started to emerge across the country, but there was little consistency in what comprised clinical child or pediatric training experiences (Roberts, Borden, Christiansen, & Lopez, 2005; Tuma, 1983). Indeed, when Tuma surveyed members of APA child, pediatric, and youth and family sections and divisions, respondents reported receiving a wide array of educational experiences to prepare them for working with youth and families, ranging from on-the-job or postgraduate training to elective coursework to more formal degree programs. Thus, leaders in clinical child and pediatric graduate training mounted several efforts to define these specialties, including training conferences, task forces, and journal sections devoted to training issues.

Hilton Head Conference

In 1985, the National Conference on Training Clinical Child Psychologists was held in Hilton Head, South Carolina. Although the need for a conference was first identified in 1977, it was almost 10 years before community need and professional training and funding priorities coalesced to allow the conference to proceed. By the time of the conference, researchers had established that the training community was falling far short of the need for well-trained clinical child psychologists, APA had recognized the need for child services and established a position in their central office to monitor child-related issues, and several child-related organizations were moving forward to establish training guidelines (Roberts, Erickson, & Tuma, 1985; Tuma, 1986). Thus, conference participants were motivated and well informed and were able to make significant progress toward addressing issues of clinical child specialty training. By the end of the 3-day conference, participants had drafted

21 recommendations, including 10 recommendations specific to graduate training, seven targeting internship, and four targeting postdoctoral training in clinical child psychology (see Table 2.1).

Interestingly, several of the graduate program recommendations focused on child-relevant training for *all* clinical psychology programs, not just those with child tracks or emphases. Specifically, the Hilton Head participants proposed that

Table 2.1 Hilton Head Recommendations: National Conference on Training Clinical Child Psychologists

Recommendations for All Clinical Training Programs
All clinical psychology programs should:
Require a course on the developmental bases of normal behavior, emphasizing a life span developmental approach.
Provide experiences with "normal" children and youth, as well as with other age groups, in diverse, "normal" settings to all graduate students.
Provide didactic and practicum training sufficient for minimal competencies in assessment, psychopathology, and intervention (e.g., treatment, consultation, prevention) with children, youth, and families.
Recommendations for Clinical Child Training Programs
Clinical child psychology programs should:
Provide didactic and practicum training sufficient for minimal competencies in assessment, psychopathology, and intervention (e.g., treatment, consultation, prevention) with adults.
Provide knowledge of and experience with children from different minority backgrounds.
Recommendations for Graduate Program Structure, Implementation, and Evaluation
APA's Committee on Accreditation should:
Ensure that accreditation site visitor teams reviewing programs that train clinical child psychologists include a member with expertise in clinical child psychology.
Clinical child psychology programs should:
Foster an appreciation of the broad and diverse contexts within which children develop and clinical child psychologists work (e.g., family, school, community).
Have at least one clinical child faculty member.
Use the Division 37 Task Force Guidelines for Training Psychologists to Work with Children, Youth and Families (Roberts, Erickson, & Tuma, 1985) as guiding principles for implementing training programs.
Endorse the Boulder scientist-practitioner model of training.
Recommendations for Internship-Level Training
Internship and graduate programs should communicate regarding:
Matching internship site characteristics with intern talents and interest and evaluating intern performance.
Development and ongoing evaluation of an individualized training plan for the intern's experience.
Child-oriented internships should:
Provide a minimum of two thirds of child-related activity training.
Within the unique constraints of the internship site, provide a broad base of experience with regard to diverse populations served (across age, gender, socioeconomic status, ethnic origin, presenting problems, theoretical orientation, and experiences).
Incorporate research into the internship by allowing time and providing role models.
Include broad-based and varied didactic and supervised training in assessment, intervention, case management, and consultation, at a minimum, including prevention if possible. Experiences should emphasize ethical considerations and evaluation of interventions.
The Section on Clinical Child Psychology should:
Create a standing committee on internship training to expand on these recommendations.

Recommendations for Postdoctoral and Continuing Education

All clinical child psychologists should:

Participate in continuing education to stay abreast of burgeoning knowledge in assessment, consultation, and prevention relevant to clinical child psychology.

The Section on Clinical Child Psychology should:

Establish a task force to create and publish criteria for continuing education offerings relevant to clinical child psychology.

Establish a task force to conduct a survey of current postgraduate programming in clinical child psychology and establish guidelines for such programming, including specification of levels and types of supervision.

Attempt to gain representation on appropriate governance structures of APA (e.g., Education and Training Committee, subcommittees on continuing education).

From *Proceedings: Conference on Training Clinical Child Psychologists*, by J. M. Tuma (Ed.), 1985, Baton Rouge, LA: Section on Clinical Child Psychology. Adapted with permission.

all clinical programs require a course on normal life span development and provide experience with normal individuals across the life span (i.e., children, youth, adults). They also proposed that all clinical programs provide coursework and experiential training to ensure minimal competencies in psychopathology, assessment, and intervention with children, youth, and families. These recommendations speak to the importance that conference participants placed on general clinical psychology training, regardless of age-related specialization. However, with the exception of required coursework in developmental bases of behavior (a requirement for accreditation), these recommendations have not been fully embraced by many clinical programs.

Recommendations focusing on clinical child psychology graduate training emphasized similar breadth, encouraging minimal competencies in work with adults, as well as with children from diverse backgrounds and in multiple contexts. Because clinical child training was so unstandardized at that time, the Hilton Head Conference also addressed issues of program structure and evaluation. In particular, they specified that programs providing clinical child training should have at least one clinical child faculty member and that accreditation teams evaluating these programs should have at least one member with expertise in clinical child psychology. These recommendations seem fairly obvious today, but illustrate the challenge of many emerging training models: that they may be implemented or judged by people without suitable backgrounds or appreciation for the important aspects of such programs. Finally, conference participants endorsed the work of other conferences and task forces, supporting the Boulder model of scientist-practitioner training, as well as the recommendations of the Division 37 Task Force for Training Psychologists to Work with Children, Youth, and Families (Roberts et al., 1985). Although the Division 37 recommendations were endorsed as flexible guidelines for program development and clinical child programs have evolved in diverse ways, the scientist-practitioner identity remains a core aspect of clinical child psychologist training.

Recommendations regarding internship and postdoctoral training were much less developed than those for graduate training but continued to stress the importance of diverse training across ages, client demographics and presenting problems, contexts, and professional duties (e.g., assessment, intervention, consultation, research). The committee's suggestions also emphasized the role of both internship

and postdoctoral fellowships as opportunities for continuing education in empirical, practical, and ethical aspects of clinical child work. Probably the most specific and unique recommendation to come out of this conference was that clinical child psychology internships should devote two thirds of the internship year to child-related experiences. Because few other specific recommendations for internship or postdoctoral training emerged from the conference, the Hilton Head conferees called for new task forces to explore clinical child internship and postdoctoral training more fully. These task forces, which would be charged with developing detailed recommendations for training experiences, emerged in the years following Hilton Head.

Post–Hilton Head: Addressing Internships and Postdoctoral Training

Between 1986 and 1988, the Section on Clinical Child Psychology commissioned two task forces to establish guidelines for clinical child psychology internships (Elbert, Abidin, Finch, Sigman, & Walker, 1988) and postdoctoral training (Ollendick, Drotar, Friedman, & Hodges, 1998). In general, these task forces reiterated the core training goals articulated at the Hilton Head Conference and subsequent internship training conferences (Belar et al., 1989). For example, they recommended that both internship and postdoctoral fellowships include opportunities to work with children, adolescents, and families across a wide range of ages (including adults), with a range of problems, from diverse backgrounds, and in diverse settings. As training experiences, both task forces also called for a range of educational activities, with internships including a combination of didactic seminars, clinical and research experiences, and supervision, and postdoctoral fellowships including approximately 50% clinical service delivery, 25% clinical research, and 25% supervision. Recommended professional activities included assessment, case conceptualization, intervention, prevention, interdisciplinary consultation, and research. The internship task force was fairly specific in the types of assessment experiences they recommended (e.g., intellectual, ability, and personality assessment, interviews, behavioral observation) and the importance of developing competencies in communicating assessment results and incorporating them into intervention planning. Both task forces emphasized that specific experiences could vary widely across training sites, and they noted the desirability of exposure to diverse treatment approaches, supervisors (including nonpsychologists), and professional experiences. They also noted the importance of addressing the increasingly complex ethical and legal issues that arise as trainees prepare for independent practice in a variety of contexts. For instance, the internship task force recommended that all interns gain awareness of the juvenile justice system, child legal rights and custody issues, and general issues in forensic psychology. In sum, these tasks forces paved the way for a sequence of training in clinical child psychology that was focused yet diverse and flexible.

Developing Pediatric Psychology Identity

At the same time that the profession was defining goals and training recommendations for clinical child psychology, psychologists who worked with children in

medical settings were also defining the roles and preparation of pediatric psychologists. These early efforts did not take place in formal training conferences or task forces, but in surveys of the field and articles published largely within the pediatric psychology community. Perhaps the most common theme of training in pediatric psychology, aside from its obvious focus on integrated attention to psychological and medical health, was its variability. For example, in the late 1960s and early 1970s, Routh (1970, 1972) described pediatric psychology practicum experiences offered by some medical school departments of pediatrics. However, as Routh's surveys suggested, clinical psychology students were not necessarily the ones who took advantage of these opportunities; pediatric psychology practicum training was just as often sought by students in school, developmental, counseling, or other areas of psychology. By the 1980s, pediatric psychology training had become aligned with clinical psychology (La Greca, Stone, & Swales, 1989; Routh & La Greca, 1990). Variability in training experiences was still evident, though. La Greca and colleagues (1989; Routh & La Greca, 1990) surveyed university graduate programs that had produced at least four graduates identified as pediatric psychologists by virtue of their membership in the Society of Pediatric Psychology. Of 41 programs surveyed, over 66% provided pediatric psychology practica, but fewer than 25% offered a course on the topic. Interestingly, only 17% of programs provided both didactic and applied pediatric experience, and 27% provided neither! Routh and La Greca noted that it is conceivable that students might gain experience in pediatric psychology during graduate training through their research and then complete more formal training during internship or postdoctoral training. This is consistent with results of an earlier survey (La Greca, Stone, Drotar, & Maddux, 1988) suggesting that graduate training in clinical and developmental psychology provided the ideal foundation for later specialization in pediatric psychology. Nonetheless, pediatric psychology training at the graduate level has continued to evolve, despite the lack of formal training guidelines. It was not until the twenty-first century that such guidelines were published (Spirito et al., 2003); we discuss these guidelines in "Curriculum Domains."

CURRENT DEFINITIONS

Because the fields of clinical child and pediatric psychology are evolving, so are their definitions, varying depending on when they were written and by whom. Although definitions of clinical psychology, clinical child psychology, and pediatric psychology abound in articles and texts, perhaps the most widely accepted definitions are those adopted by the relevant divisions of the APA. These are also typically the definitions that divisions of other organizations forward to APA's Commission for the Recognition of Specialties and Proficiencies in Professional Psychology (CRSPPP) when they seek specialty status.

Clinical Child Psychology

The current definition of clinical child psychology supported by APA Division 53 Clinical Child Psychology and approved by CRSPPP (2005) states:

> Clinical child psychology brings together the basic tenets of clinical psychology with a thorough background in child and family development. Clinical child psychologists conduct research and provide services aimed at understanding, preventing, and treating psychological, cognitive, emotional, developmental, behavior, and family problems of children across the age range from infancy through adolescence. Of particular importance to the Clinical Child Psychologist is an understanding of the basic psychological needs of children and how the family and other social contexts influence children's socioemotional adjustment, cognitive development, behavioral adaptation, and health status. There is an essential emphasis on a strong empirical research base recognizing the need for the documentation and further development of evidence-based assessments and treatments in clinical child and adolescent psychology.

Central to this definition are several themes. First, clinical child psychology's foundations are those of clinical psychology generally, including a focus on assessment, diagnosis, prediction, treatment, and prevention of psychopathology and adjustment problems, and an emphasis on scientifically based knowledge and application. Because the populations of interest range from infancy through adolescence, clinical child psychologists must have a thorough understanding of child and family development. Thus, the clinical child psychologist's view of pathology is based on a developmental psychopathology framework, understanding child maladjustment in the context of normal behavior and development (Rutter & Sroufe, 2000).

A second theme is the role of context in children's adjustment. Clinical child psychology explicitly emphasizes the notion that child adjustment and development do not occur in a vacuum, but in context. Understanding the relevant contexts is important for conceptualizing youths' emotional and behavioral problems, as well as for treating them effectively. To put it colloquially, contexts are usually part of the problem, and almost certainly part of the solution. Thus, for clinical child psychologists, treating children in context is both a luxury (that many clinical adult psychologists, who see individual clients during an office visit, do not have) and a necessity. As our brief definition implies, the family context has central importance in children's development and adjustment. This makes sense given that the family has typically been with the children since birth, observing and participating in their development over time and across multiple contexts. Thus, clinical child psychologists must have an understanding of family structures and systems, as well as of the personalities and issues of individuals who make up the family. How family members relate to one another regarding such issues as rules, boundaries, and values and how they share experiences and emotions influence the lessons that youth learn and carry into their relationships and roles throughout life (Minuchin, 1985). In addition, understanding family members' health and adjustment is helpful for contextualizing youth adjustment problems that may have a genetic basis (e.g., Moffitt, 2005).

Beyond the family, several other contexts are important. Schools, neighborhoods, peer groups, and religious and extracurricular organizations can have a large influence on social, emotional, cognitive, and behavioral adjustment, especially as children reach school age and adolescence and spend more time in school and peer settings (Lerner, 2006). Broader contextual factors, such as the child's culture of origin and racial/ethnic group, should also be considered (La Greca & Hughes, 1999;

Roberts, 1998). Ecological models, such as Bronfenbrenner's (Bronfenbrenner & Morris, 2006) model that considers the reciprocal interplay between a developing individual and the many environments in which he or she interacts as the determining factors in youth adjustment, are mainstays of clinical child approaches to understanding and intervening with children. Thus, for clinical child psychologists, it is routine to go beyond individual or family-focused assessment and treatment and to incorporate school, peer, and community settings in assessment, consultation, and intervention.

Finally, clinical child psychology endorses the view that psychological theory and practice are necessarily based on science, and further emphasizes that well-trained clinical child psychologists will be scientist-practitioners, integrating both science and application into their training and professional identity. The scientist-practitioner training model was endorsed at the Hilton Head Conference and continues to be considered a central aspect of clinical child psychology training (La Greca & Hughes, 1999; Tuma, 1985).

Pediatric Psychology

Pediatric psychology is defined as a subspecialty of clinical child psychology, building on clinical psychology and clinical child psychology foundations but integrating knowledge of child health issues (e.g., disease process, health promotion) and application in medical settings (Roberts, 2003; Spirito et al., 2003). Defining this field, the Society of Pediatric Psychology (APA Division 54) states:

> Society of Pediatric Psychology is dedicated to research and practice addressing the relationship between children's physical, cognitive, social, and emotional functioning and their physical well-being, including maintenance of health, promotion of positive health behaviors, and treatment of chronic or serious medical conditions. In recognition of the close interplay between children's psychological and physical health, SPP has developed liaisons with several pediatric societies and many pediatric psychologists have become experts in the behavioral and developmental aspects of acute and chronic diseases. (American Psychological Association, 2006c)

Integrating across these definitions, a picture emerges of pediatric psychology as *clinical child psychology plus*. The Society of Pediatric Psychology is rooted in clinical child psychology as opposed to health psychology, and as Spirito and colleagues (2003) note, pediatric psychology is frequently integrated with or synonymous with clinical child psychology, particularly at the graduate training stage. Pediatric psychologists share clinical child psychology's scientist-practitioner approach to understanding, assessing, and treating youth adjustment and a training focus that integrates clinical psychology foundations with a developmentally and contextually sensitive approach to assessment and treatment. Pediatric psychologists should be competent to provide general clinical child services to youth and families (Roberts & Steele, 2003; Spirito et al., 2003).

To these clinical child foundations, pediatric psychologists add didactic training and experience in issues specific to child health. Pediatric psychologists must have an understanding of disease process, including how illnesses and their treatments

affect psychological and behavioral functioning, as well as of disease prevention and health promotion (Spirito et al., 2003). Research and applied activities of pediatric psychologists focus, for example, on how a healthy lifestyle can prevent conditions such as obesity and diabetes, or how chronic illness affects youths' self-concept and peer relations (Roberts, 2003). Pediatric psychologists also must master skills required for effective consultation and liaison with physicians and other professionals who address child health, as they are almost always part of an interdisciplinary team. Because of the already large demands of graduate training in clinical psychology, pediatric psychology experience often occurs primarily at the internship and postdoctoral levels (Drotar, Palermo, & Ievers-Landis, 2003; Spirito et al., 2003). Indeed, a recent Division 53 survey of clinical child and pediatric graduate programs indicated that only 3 of 45 programs that provide child-relevant training have a formal pediatric psychology track (Division of Clinical Child and Adolescent Psychology, 2006).

Relationship to Other Areas of Professional Psychology

General Clinical Psychology

As noted earlier, the scope of clinical psychology has evolved since its description at the Boulder Conference; as the knowledge base has expanded, the notion of a generalist, competent across all ages and problem areas, has become increasingly unrealistic. As Roberts (2006) suggests, few clinical training programs actually provide generalist training, but instead focus on adults. Clinical child and pediatric psychology also provide focused training on a subset of the life span. However, the core aspects of clinical psychology, including a focus on understanding, assessing, preventing, and treating maladjustment in the context of established scientific methods and broader foundations in psychology, are integral parts of training in clinical child psychology. In addition, although clinical child and pediatric training focus on children and families, they also include educational background in adult development and psychopathology and the ability to work with parents and other adults who are part of the child's life. And beyond a focus on individual clients, clinical child and pediatric psychology emphasize systems and contexts and conducting research and intervention in interdisciplinary teams.

Although clinical child and pediatric psychology are more focused than generalist clinical psychology, they could be considered broader than many clinical (adult) programs (Roberts, 2006). In their discussion of clinical training in the twenty-first century, Snyder and Elliott (2005) maintain that most clinical (adult) psychology has been characterized by a focus on individuals' weaknesses, or psychopathology. They call for clinical psychology to broaden its horizons to encompass a focus on people's strengths and to consider not just person factors, but also environmental context. Interestingly, as several authors note in their responses to Snyder and Elliott's thought-provoking article, this seems to be exactly what clinical child and pediatric psychology have been doing for years (Kendall, 2005; Roberts, 2005). Thus, clinical child and pediatric psychology training models may provide a guide for clinical adult programs that seek to broaden their perspectives to encompass client strengths and contexts.

School Psychology

School psychology shares many similarities with clinical child and pediatric psychology, including a focus on assessment, intervention, prevention, and health promotion with youth and families, a scientist-practitioner approach, and sensitivity to contextual and diversity issues in addressing youth adjustment (Reynolds & Gutkin, 1999). School psychology, however, emphasizes educational and learning processes and environments. Relevant practice may extend to learners of all ages, including adults, and often takes place in an educational setting. Assessment is frequently focused on evaluation of students' intellectual, achievement, psychological, and behavioral functioning, particularly with regard to how these areas influence learning effectiveness. Common activities of school psychologists also include consultation with parents and educators, program development and evaluation of educational programs, and advocacy for education-relevant legislation. Thus, the day-to-day activities of applied clinical child, pediatric, and school psychologists may look quite similar: involving work with children and adolescents in their relevant contexts, on issues that affect their health and adjustment, using approaches that are supported by empirical literature.

Similarities are also evident in research and academic training contexts. Like research conducted by clinical child and pediatric psychologists, school psychologists' research may examine issues such as child emotional, behavioral, and developmental disorders and health; child adjustment and health promotion; and contextual factors such as families, peer groups, schools, culture, ethnicity, and economic disadvantage. Again, the major difference is one of relative emphasis, with school psychologists more likely to research areas relevant to learning or adjustment within the learning environment. Finally, as one of the three types of accredited programs in professional psychology (along with clinical and counseling psychology), school psychology graduate program curricula include foundational training in diverse bodies of knowledge within psychology (e.g., social, cognitive, and biological bases of behavior), research methods, normal and abnormal development, and professional ethics, as well as area-specific knowledge in assessment, intervention, and prevention (APA, 2005). Like clinical child and pediatric psychologists, school psychologists complete a predoctoral internship and supervised postdoctoral experience for licensure.

TRAINING ISSUES

Training in clinical child and pediatric psychology addresses multiple purposes and, as such, serves many masters. Academically, clinical child and pediatric psychology are usually (although not always) part of psychology departments, along with other subdisciplines such as social, cognitive, developmental, experimental, and quantitative psychology. Housed within liberal arts and sciences colleges or divisions, department priorities typically involve the traditional combination of research, graduate and undergraduate teaching, and service to the campus and academic community. Professional psychology programs are somewhat unique within departments in their additional priority of training graduate students for careers in

professional health service. Thus, accreditation, applied practicum and internship training, and preparation for licensure must be considered. Because accreditation provides standards for evaluating professional training programs, we discuss accreditation issues first, and then examine the context, goals, and curricula of clinical child and pediatric training programs.

Accreditation Issues

As with all professional psychology training, clinical child and pediatric psychology programs participate in the accreditation process. Accreditation is a process of voluntary self-regulation by the field, with a primary goal of protecting students and the public by ensuring that programs meet at least minimal standards of content and performance (APA, 2005). Clinical programs have been accredited since the mid-twentieth century, when the need for federally funded psychology training prompted development of a system and criteria for accreditation (Sheridan, Matarazzo, & Nelson, 1995). Since then, accreditation guidelines and procedures have undergone several revisions that reflect the evolving nature of professional psychology, and even as this chapter is being written the process is under review and modification (Schilling & Packard, 2005). General accreditation guidelines are beyond the scope of this chapter (interested readers can refer to the CoA's *Guidelines and Principles for Accreditation of Programs in Professional Psychology*, APA, 2005, as well as recent revisions to the *Policy Statements and Implementing Regulations*, APA, 2006a). Here, we focus on select issues specific to clinical child and pediatric psychology training.

As currently defined, the purpose of accreditation is to evaluate the extent to which a training program has achieved the goals and objectives of its stated training model, defined within the more broadly accepted principles of professional psychology (APA, 2005). In other words, programs must meet field-defined criteria, but beyond that are encouraged (or at least allowed) to articulate and pursue innovative training models and practices. As Roberts et al. (1998) noted, this flexibility paved the way for clinical child psychology programs to begin to define themselves more specifically than as just clinical programs and to more fully embrace the guidelines for child-focused training articulated at the Hilton Head and later conferences. As with many guidelines, standards, and rules, these accreditation guidelines are aspirational, and many would argue that they have been applied inconsistently or incompletely. This has been evident in the challenges that some clinical child programs have faced when seeking accreditation or reaccreditation. For example, Roberts (2006) notes that his clinical child training program includes formal training in adult psychopathology in order to satisfy the CoA, although he questions whether adult-focused programs receive the same level of scrutiny regarding their coverage of child psychopathology. Similar challenges likely affect other clinical child and pediatric programs, as well as the accreditation teams faced with the task of evaluating specific and varied applications of clinical psychology training goals.

Given the importance of accreditation for training programs and the time and expense involved in the process, it is easy for *Accreditation* (with a capital A)

to become reified in a way that leads both training programs and accreditation committee members to forget that the process is essentially intended to serve the field rather than to define it. Whereas accreditation should promote high standards within the field and facilitate innovation and development, it is the field itself that must self-define, innovate, and evolve. Thus, continued articulation of how the fields of clinical child and pediatric psychology define optimal graduate training remains crucial for influencing future development and application of accreditation guidelines.

Academic Context

As noted, most clinical programs are housed within departments of psychology. Within the department, however, the structure of clinical child and pediatric psychology training can range from a stand-alone academic training program to a track or emphasis within a more general clinical program, to simply providing informal opportunities to gain child-specific experience. Several surveys conducted over the past 30 years have documented the diverse structures and academic settings that characterize clinical child psychology training programs (Mannarino & Fischer, 1982; Roberts, 1992; Zeman, Nangle, & Sim, 1999), and this diversity remains true today. Currently, Divisions 53 and 54 provide an online directory of programs that report providing clinical child or pediatric psychology training (Division of Clinical Child and Adolescent Psychology, 2006). Of the 45 programs listed as having child or pediatric training, only three (6.7%) report having separate clinical child training programs. Over half of the programs (24; 53%) provide formal child or pediatric tracks or specialties, but 40% (18) report providing no formal track.

Decisions about the context in which to provide clinical child and pediatric training are likely complex, encompassing philosophical, historical, and practical considerations, and an understanding of how these issues influenced training program development at each institution is obviously beyond the scope of this chapter. However, the Division 53 data do provide some suggestions of how context relates to program issues such as size of faculty and student body, course offerings, and practicum opportunities. In examining these data, it is important to remember that they represent self-reported program characteristics, and respondents may have used different definitions and counting conventions. Also, we examine descriptive data without evaluating the statistical significance of patterns. With these caveats in mind, we note several interesting findings.

Table 2.2 presents data on characteristics of programs listed in the Division 53 directory as providing clinical child and pediatric psychology training in separate programs, formal tracks, or informal opportunities. Although the separate programs tended to have a somewhat smaller overall clinical faculty than programs with or without formal tracks (approximately five versus nine clinical faculty), their child-focused faculty averaged about one more faculty member than other programs (approximately five versus four clinical child or pediatric faculty). For programs with tracks or informal child psychology training opportunities, child-focused faculty represented approximately 40% of total faculty. Similarly, child-focused students in programs with child or pediatric tracks represented just over 40% of their overall

Table 2.2 Program Characteristics for Clinical Child or Adolescent and Pediatric Psychology Training Programs (2005)

	Type of Clinical Training Provided			
Characteristics	No Formal Track or Emphasis	C or P Emphasis	Formal C or P Track	Distinct C or P Program
N	15	7	18	3
Faculty	*M (SD)*	*M (SD)*	*M (SD)*	*M (SD)*
All clinical faculty	8.93 (3.06)	8.71 (2.29)	9.83 (2.28)	5.00 (2.65)
Child clinical faculty	2.80 (1.65)	3.71 (1.55)	3.81 (1.10)	5.00 (2.65)
Pediatric faculty	0.57 (0.78)	0.57 (0.93)	0.75 (0.67)	0.67 (1.15)
Students				
All clinical students	32.33 (10.11)	43.14 (10.16)	38.18 (12.90)	31.33 (5.51)
Clinical child students	7.33 (5.08)	12.86 (5.40)	15.50 (8.17)	31.33 (5.51)
Pediatric students	1.47 (2.03)	0.57 (0.98)	2.24 (3.09)	6.00 (10.39)
Course and Opportunities				
Child clinical courses	2.60 (1.24)	3.00 (1.29)	4.28 (1.41)	9.67 (4.73)
Pediatric courses	0.27 (0.46)	0.14 (0.38)	0.56 (0.78)	1.33 (2.31)
Child practica	2.43 (1.57)	7.86 (7.86)	5.78 (5.40)	7.00 (3.61)
Pediatric practica	0.73 (1.03)	2.29 (2.14)	1.88 (2.64)	1.00 (1.73)

Note: C = Child; P = Pediatric.

student body. However, programs without formal tracks may attract fewer child-focused students, as less than 30% of their student body comprised child or pediatric psychology students. Programs overall averaged 30 to 40 clinical graduate students, but for all but the separate clinical child or pediatric programs, child-focused students were in the minority.

Course requirements and practicum opportunities varied widely, both across and within program type. Separate clinical child or pediatric programs required substantially more child-focused courses than their counterparts with tracks or informal child psychology opportunities. On average, separate programs required 11 child clinical or pediatric courses, likely because they are designing courses for their child-focused students rather than teaching some courses for students from multiple (i.e., child and adult) tracks. In contrast, child track programs required an average of approximately four child clinical or pediatric courses, and informal opportunity programs required only three such courses. There are probably benefits and costs to both approaches. For example, providing general clinical courses to adult and child track students should expose all students to broad education about issues that arise across the life span and across diverse clinical subspecialties. Unfortunately, many general courses focus primarily on adults, with perhaps 1 to 2 weeks of child-relevant material, shortchanging both child- and adult-focused students. Child-focused courses offer maximum opportunity for students to learn about the content, process, and current issues in clinical child and pediatric psychology from faculty who are expert in these areas. However, explicit attention to helping students gain exposure to adult development and psychopathology is also important. Program type seems to have important implications for practicum opportunities

available to graduate students. Separate child or pediatric programs and programs with formal tracks reported having an average of eight practicum placements that focused on child or pediatric training. In contrast, programs without a formal track offered only about three child-focused practica.

Finally, across program type, clinical child training seems more extensive than pediatric psychology training. Of 24 programs that reported explicitly on the emphasis areas offered by their program or track (versus, for example, responding yes to the question about a child or pediatric track), only three reported having a pediatric track, and two of these programs also had a clinical child track. Overall, programs reported more clinical child than pediatric faculty, graduate students, courses, and practicum placements. Indeed, on average, pediatric psychology opportunities accounted for no more than 10% to 20% of child-related training opportunities. This is consistent with Spirito et al.'s (2003) statement that specialized training in pediatric psychology often does not occur until the internship or postdoctoral stage of training.

Training Goals

In preparing graduates for careers in the science and practice of clinical child and pediatric psychology, the field has emphasized several goals and competencies. For example, in the Division on Child, Family, and Youth Services (APA Division 37) training guidelines that were the basis of some of the work at the Hilton Head Conference, Roberts and colleagues (1985, p. 72) outlined four primary goals of training:

1. Training should provide at least minimal entry level skills for providing professional psychological services . . . on behalf of children, adolescents, and their mentors (e.g., parents and teachers) in a range of settings (e.g., home, school, institutions, and medical clinics). . . .
2. Training should provide the scientific basis for professional psychological applications.
3. Training should include specialized research contributing to an understanding of children, families, and psychological development, as well as in the evaluation of therapeutic interventions.
4. Child-, youth-, and family-related issues should be integrated into psychology-based courses at all levels, regardless of specialty designation.

More recently, La Greca and Hughes (1999, p. 441, Table 5) identified five competencies important for clinical child psychologists: (1) multicultural competencies; (2) delivery and evaluation of comprehensive and coordinated systems of care; (3) collaborative and interprofessional skills; (4) empirically supported assessment and treatments for promoting behavioral change in children, families, and other systems; and (5) entrepreneurial and supervisory skills. La Greca and Hughes argue that these competencies, reflecting the thinking of many educators in child-focused professional psychology, are increasingly important for effective clinical child practice in our society.

Separated by almost 15 years of developments in clinical child and pediatric psychology, these two sets of goals represent both foundational goals that are central to the definitions of clinical child and pediatric psychology as well as more progressive goals that reflect the evolution of professional psychology. In particular, the theme of empirically based assessment and intervention targeting youth in their environmental and cultural contexts has echoed throughout even the earliest definitions of the field. Likewise, the importance of an interdisciplinary perspective is one of the hallmarks of clinical child and pediatric psychology and a central goal for training and professional competence. Development of a scientist-practitioner identity is also fundamental and is reflected in the clinical child psychologist's research training as well as professional competence in evaluation of service delivery. A clinical scientist identity, based on the newer clinical science training model that emphasizes preparing students for careers that include generation, application, and dissemination of clinical science research (Academy of Psychological Clinical Science, 2007; McFall, 1991), would also be consistent with this goal.

Roberts et al.'s (1985) goal of infusing child-related issues into all clinical psychology curricula, regardless of specialty designation, underscores the importance of having a broad developmental base in all clinical psychology training. Assuming that clinical child and pediatric psychology training would of course attend to child issues, this goal seems more relevant to general or adult-focused clinical training programs. The importance of this goal for clinical child and pediatric training seems to lie primarily in its call for all clinical psychology training to view child issues as a legitimate part of broad and general training.

Some of La Greca and Hughes's (1999) competencies reflect newer or elaborated goals that have emerged in concert with changing issues and needs in society. Multicultural competence, although implicit in early definitions and goals emphasizing contextual views of child adjustment, is an explicit part of current-day competence for child psychologists. Increased attention to multicultural issues is especially important given the growing diversity of the U.S. population as well as priorities noted by APA, the CoA, and the National Institutes of Health. Finally, entrepreneurial and supervisory skills extend beyond the traditional purview of clinical psychology, but are important for preparing clinical child and pediatric psychologists for both applied and academic careers. In applied settings, these skills are important for helping clinical child psychologists function effectively as part of treatment teams that may include junior or in-training members and for maintaining fiscally viable practices. In academic settings, having faculty who have received formal training in supervision is clearly an essential component of providing high-quality practica. Entrepreneurial skills are becoming increasingly important for academics. As funding for higher education becomes tighter, universities are encouraging their faculty and departments to develop diverse sources of funding for their research and training. On the national level, psychologists are becoming increasingly active participants in legislative activity relevant to health care and education (e.g., through the APA Practice and Education Directorates, the Society for Research in Child Development's policy fellowships). To effectively influence the business and legislative sectors, convincing nonpsychologists of the importance of supporting

our efforts to address child health, psychologists must speak the language of these professionals.

TRAINING CONTENT: CURRICULUM DOMAINS

As with all professional psychology programs, clinical child and pediatric psychology curricula are shaped in part by accreditation requirements for clinical programs. However, as noted earlier, accreditation requirements are designed to allow individual training programs to define their goals and methods for attaining them. So how can clinical child and pediatric programs best integrate their training goals and accreditation requirements into a curriculum that prepares competent psychologists? As several investigators have noted, this is apparently not an easy task to do well; even after definitions and goals of clinical child and pediatric psychology were developed, training programs differed widely in their curricula and did not always meet minimal training goals for these specialty areas (Roberts et al., 1998; Zeman et al., 1999).

Recent advances in professional psychology have paved the way for both greater specificity of clinical child and pediatric curricula and increased flexibility in implementation. First, starting in 1996, changes to accreditation guidelines and procedures facilitated innovation in how professional psychology programs can articulate, meet, and evaluate their training goals (Roberts et al., 1998). Second, the field has begun moving from activity-based to competency-based models of training. This is reflected in accreditation guidelines and in recent writings and conferences on professional competencies (Kaslow et al., 2004; Sumerall, Lopez, & Oehlert, 2000).

Both clinical child and pediatric specialties have articulated curriculum recommendations that fit with these developments. In 1998, Roberts and colleagues outlined a model for training clinical child psychologists based on work begun by the National Institute of Mental Health work group on improving training for child and youth service providers (Wohlford, Myers, & Callan, 1993). Roberts et al.'s training model included 11 domains of training considered important for clinical child psychologists. Several years after the clinical child training model was published, the Society of Pediatric Psychology charged a task force with developing similar guidelines for training pediatric psychologists (Spirito et al., 2003). The resulting guidelines, adapted from the clinical child model, include 12 domains of training, with 11 overlapping with the clinical child model and the final domain specific to pediatric psychology.

Although coursework is a common way to meet curricular requirements, competence in many of the domains of training requires practical experience. The clinical child and pediatric training models acknowledge this, describing student training as a three-level sequence: *exposure* to the content and processes of clinical psychology through didactic training and observation, *experience* that involves mentored practice of the relevant skill, and *expertise* that is obtained through extensive exposure and experience and enables independent practice. By the end of graduate training, students should have had at least some exposure to and experience with each

domain, although they may not have expertise in any. Internship is an opportunity to gain more experience and develop expertise in some areas, and by the time students have completed postdoctoral training and are ready for licensure, they should have established sufficient expertise to support independent practice with a variety of youth and family issues. Next we outline the 12 recommended domains of clinical child and pediatric psychology training, from graduate through postdoctoral training, and offer comments about potential ways of providing didactic, applied, and research training in these domains.

Life Span Developmental Psychology

From a developmental psychopathology perspective, complete understanding of youths' adjustment problems requires an understanding of individual and contextual factors in normal development (van Eys & Dodge, 1999). At a minimum, coursework should expose students to current theory and research on life span development across social, emotional, cognitive, behavioral, and physical domains, as well as to the role of individual (e.g., sex, race) and contextual (e.g., family, peer, culture) factors in development. Practical experiences in assessment and intervention should involve youth from a variety of developmental levels and contexts, with particular emphasis on issues such as how developmental factors impact youth psychopathology and response to treatment. For pediatric psychology students, didactic and practical experiences should focus on issues relevant to child and adolescent health, such as the impact of disease process on cognitive development or the role of adolescent autonomy seeking on adherence to medical regimens. Student research should also incorporate developmental issues, addressing age, gender, and context in the specific research questions and methods, or at least conceptualizing the research from a developmentally informed perspective.

Beyond the minimum, several training programs integrate clinical child and developmental training more fully. In their survey of child-focused clinical programs, Zeman et al. (1999) found that programs characterizing themselves as clinical-developmental required more developmental courses than other programs, averaging just over five required developmental courses versus none or one course required by clinical child programs or clinical programs with a child emphasis. These programs also encourage student research mentored by developmental faculty, attendance at developmental conferences (e.g., Society for Research in Child Development, Society for Research on Adolescence), and publication in developmental as well as clinical journals. An example of a clinical child developmental program is the joint effort of the clinical and developmental programs at the University of Missouri–Columbia. Students admitted to the clinical child developmental program essentially complete requirements of both clinical and developmental training areas. Practically, this means that in addition to courses required for the clinical child program (including courses in normal development, developmental psychopathology, child assessment, child interventions, and family systems and interventions), joint program students take two additional courses in cognitive, social, and emotional development; explicitly incorporate developmental content into

thesis and dissertation research and comprehensive examination; and participate in both clinical and developmental faculty labs.

Life Span Developmental Psychopathology

With foundational knowledge of normal development, students can conceptualize deviations from typical or adaptive functioning. During graduate training, courses and seminars should expose students to abnormal behavior across the life span, as well as current research on trajectories of risk and resilience. Supervised practica with various forms of psychopathology in children, adolescents, and adults allow students to gain experience conceptualizing and diagnosing psychopathology. Again, a variety of settings, including youth in clinic, family, community, school, and medical settings, give graduate students experience in incorporating context into their understanding of youth psychopathology.

Child, Adolescent, and Family Assessment

Becoming a competent clinical child or pediatric psychologist requires a thorough understanding of assessment issues, including how to evaluate the empirical foundations and utility of a variety of assessment methods, how and when to use them, and how to interpret their results for both applied and research purposes. Relevant assessments for youth go beyond tests or measures that youth complete (e.g., intelligence tests) and include evaluations of youth functioning by others (e.g., caregiver reports of adaptive functioning, school reports of academic achievement), as well as evaluations of the systems in which youth function (e.g., parent-child and parent marital relations). Assessment also goes beyond standardized tests and includes clinical conceptualizations, functional analysis, and a general hypothesis-testing framework. Both clinical work and research benefit from a scientific approach to generating questions and using appropriate assessment to address them.

Students should develop an understanding of how developmental factors such as age, sex, ethnic background, and cultural conventions can influence assessment validity, and how to conduct multimethod assessments that can maximize the strengths and contributions of the multiple methods and reporters. Finally, students should obtain experience in interpreting and disseminating assessment results and in assessment reports for clinical purposes as well as in research reports. Roberts et al. (1998) suggest a minimum of two semesters of coursework and associated practica devoted to exposing students to all relevant assessment issues and methods and allowing them to gain experience with several assessments and develop expertise with at least a few. Overall, understanding the conceptual and psychometric aspects of assessment seems most critical to competence as a clinical psychologist. If students leave graduate training with expertise in how to formulate assessment questions, design developmentally and contextually sensitive assessment plans, and integrate results from multimethod assessment, as well as exposure to the mechanics of assessment, they should be able to quickly master specific assessment tools as needed.

Intervention Strategies

Training in interventions should reflect the core values of clinical child and pediatric psychology: empirically and contextually based treatment that is developmentally and multiculturally sensitive and interdisciplinary. This is a tall order, calling for treatment that is both standardized and individualized. To be effective at delivering the standardized aspects of treatment, clinicians in training must be familiar with empirically supported treatment techniques and protocols, including how, when, and with whom to use them. For example, during graduate training, students should be exposed to many of the treatments considered well-established or probably efficacious, as defined by APA Division 12 (Division of Clinical Psychology) and Division 53 (Society of Clinical Child and Adolescent Psychology) task forces (Chambless et al., 1996, 1998; Lonigan & Elbert, 1998), including parent training and multisystemic therapy for oppositional and conduct disorders, behavioral parent training and problem-solving skills training for Attention-Deficit/Hyperactivity Disorder, interpersonal and cognitive therapy for depression, cognitive-behavioral therapy and family anxiety management for generalized and separation anxiety disorders, exposure and relaxation therapies for anxiety disorders, and behavior therapy for childhood obesity. Although exposure to some of these treatments may be limited to didactic coverage in courses or readings, experience with at least some of the treatment protocols should occur during graduate school, with additional opportunities to develop experience and expertise coming during internship and postdoctoral training.

Clearly, the list of evidence-based treatments is incomplete, and even when such treatments exist, they frequently do not have demonstrated equal applicability across multiple ages, socioeconomic groups, cultural groups, or family constellations. Thus, it is also essential for students to gain experience in how to handle situations in which there is no prescribed, empirically supported treatment. Two general approaches are valuable here. First, familiarity with the developmental and clinical literatures on the relevant treatment moderators can help guide treatment decisions. For example, clinicians might use the literature on differences in effective parenting styles across socioeconomic or racial/ethnic groups (McLoyd, 1998) to tailor family treatment for a low-income ethnic minority family. Second, using their scientist-practitioner framework, students should become expert at evaluating their treatments as they deliver them. This way, even treatments that do not have established empirical support can be appraised empirically.

Research Methods and Systems Evaluations

As noted, clinical application from a scientist-practitioner perspective includes using the empirical literature to guide treatment decisions and using empirical evaluation methods in the treatment context to assess treatment process and outcome. More broadly, advancing the literature that informs treatment, prevention, and health promotion requires sound research. Thus, students in clinical child and pediatric psychology should become proficient as both consumers and producers of applied scientific knowledge. Background in methods of conducting scientifically

sound research and analyzing resulting data is an integral part of all professional psychology training. However, clinical child and pediatric training programs should ensure that methods that are especially relevant to understanding youth and families, such as longitudinal and prospective designs and hierarchical analyses that can evaluate child data in the context of schools or families, are included in child-relevant training. For pediatric psychologists, familiarity with research issues and methodologies common to medical settings is also important, including issues of health care utilization, clinical trials, and epidemiological and multisite methods. In addition, because youth work so often involves systems consultation and program development, expertise in systems and program evaluation is important for competent and marketable clinical child and pediatric psychologists. Consistent with a scientist-practitioner approach, practicum training should include opportunities to empirically evaluate treatment progress, using single-case design and other relevant methodologies. Finally, to be truly effective, research must be disseminated. Thus students should develop skills in scholarly publication and presentation and grant writing.

Although internship is acknowledged as primarily a clinical training year, continued opportunity to develop expertise in clinical research is an important part of a scientist-practitioner identity. In their recommendations for internship, Elbert et al. (1998) proposed that, at a minimum, interns be encouraged to systematically review the literature as they conceptualize cases and plan treatment. In addition, many scientist-practitioner and clinical science internships provide protected time for interns to conduct research, either independently or as part of ongoing research at the internship site. Some programs (e.g., Case Western Reserve University, Medical University of South Carolina, University of Mississippi Medical Center, Western Psychiatric Institute and Clinic) provide seminars in grant writing and funding for research and conference attendance, and may even require journal article or grant submission as part of the internship experience. Rather than distracting interns from their clinical duties, these research activities enrich interns' integration of science and application. Similarly, although a primary goal of many postdoctoral training fellowships is preparation for licensure, clinical research and development of grant writing skills can be effectively integrated into the training experience (Drotar, 1998; Ollendick et al., 1988). Often, the more research-oriented internship sites will also have postdoctoral positions in which trainees can continue the clinical and research work begun on internship.

Professional, Ethical, and Legal Issues

Working with children and adolescents comes with many unique professional, ethical, and legal issues. At the most basic level, minor youth are, in most cases, not legally in charge of their own lives. Thus, issues of who decides whether and how to pursue treatment, who receives the treatment (i.e., who is the client), and who provides the treatment must all be decided for the child. Parents are responsible for seeing that their children receive appropriate educational, medical, and psychological services, and graduate students must be aware of issues of parent informed consent, as well as how parent rights and responsibilities are influenced by divorce,

abuse, and neglect. Graduate students must also know how to define the client in family treatment or school or medical consultations. In addition to identifying the client, child and pediatric work can be associated with challenges in identifying the treatment agent. In the interdisciplinary settings in which child and pediatric psychologists often work, psychologists must negotiate and define their roles on the treatment team, along with educators, health professionals, and other social service providers. In all of these tasks, the child's role as an active (albeit legally nonresponsible) participant must be appreciated. Training should address how to involve children and adolescents in assenting to, planning, conducting, and evaluating assessment and interventions, as well as how to balance legal issues with children's rights to privacy.

The types of challenges that children and adolescents face require sensitivity to specific legal and ethical issues. For example, parent divorce and custody issues, abuse and neglect, and delinquency all require familiarity with the legal system. Clinical child and pediatric psychologists should be aware of local, state, and federal laws that define abuse (e.g., the types of physical injuries or conditions that qualify) and neglect (e.g., the age at which children can be left unattended), reporting requirements, and the process by which such cases are handled in the county. Knowledge of how child truancy, other status offenses, and criminal behavior are handled is helpful in treatment of child and family treatment. Particularly by the internship and postdoctoral levels, experience with child forensic issues such as custody evaluation and expert testimony is warranted (Elbert et al., 1988). In pediatric psychology settings, graduate students should gain experience in dealing with issues such as life-threatening and terminal illness, developmental and neurological conditions, and painful medical procedures. In all of these instances, dealing with the affected youth is just one aspect of psychological care; clinicians must help youth address the challenge in the context of their ongoing development, and also help parents, siblings, and perhaps friends cope with the implications of the illness or condition.

Issues of Diversity

Given that considering youth in context is a hallmark of clinical child and pediatric psychology training, attention to the context of diversity is a natural extension of this training. During graduate training, students should gain exposure to and experience with multiple aspects of diversity, including definitions of what constitutes diversity, the influence of values and prejudice on societal and personal attitudes and behavior, and the role of diversity factors in psychological adjustment and treatment. For example, diversity should be defined broadly, to include gender and age, as well as racial/ethnic, cultural, socioeconomic, religious, sexual orientation, disability, and family constellation factors. Graduate coursework and practicum experience should expose students to information on youth with as many of these backgrounds or experiences as possible, with advanced practicum, internship, and postdoctoral training experiences selected to broaden these experiences. Issues that are important to address include the role of diversity factors in definitions of psychopathology, risk factors for certain psychological or medical conditions (e.g., conduct disorders,

depression, heart disease), health care utilization and acceptability, and culturally sensitive assessment and treatment.

Class and supervision time should also be used to allow students to explore their own values and attitudes to become more effective treatment providers for diverse youth and families. Students should gain an appreciation for the balance between considering diversity factors and understanding the individual child or family. Although research may provide valuable information about diverse groups, this information may not generalize to individual cases. Additionally, a child's sex, disability status, and racial/ethnic group do not define him or her, but simply represent one aspect of the child's identity. Thus, students should remember to consider youth as individuals within the context of diversity. Finally, in addition to gaining and using information on diversity, students should use their research as an opportunity to add to the knowledge base on issues of diversity. Although psychology has made great strides in understanding male and female youth across ages and from diverse backgrounds, much remains to be done, and the scientist-practitioner and clinical science backgrounds of clinical child and pediatric psychology students places them in an ideal position to conduct clinically relevant and contextually sensitive research.

Role of Multiple Disciplines in Service Delivery Systems

As we've noted throughout this chapter, interdisciplinary work is another hallmark of clinical child and pediatric psychology. Much of working effectively within an interdisciplinary team is part of a professional issues skills set, involving mastery of how to fit into existing systems, how to divide and share tasks, and how to understand the language and approaches of multiple professions. As such, students must gain familiarity with other disciplines' approaches and contributions to child health promotion and experience in participating in interdisciplinary teams. This should include coursework in areas such as consultation, family and community psychology, and public health. Practicum experiences should allow students to observe and participate in a broad range of systems, including families, schools, social service agencies, intensive (e.g., inpatient or day) treatment programs, and medical settings. Internships are often an ideal place for trainees to gain a variety of interdisciplinary experiences, and by completion of postdoctoral training, clinical child psychologists should be able to work effectively with the full range of professionals involved in child health care.

Prevention, Family Support, and Health Promotion

A comprehensive approach to dealing with child and adolescent adjustment issues includes attending not just to diagnosed disorders, but also to factors that place youth at risk for disorder or maladjustment and factors that promote resilience and healthy adjustment. Thus, clinical child and pediatric psychologists should have a knowledge base in youth and family healthy adjustment and prevention science, as well as research and applied competence in prevention and health promotion. Such experiences can come from coursework in typical development and resilience, applied experiences in schools or primary care facilities, and interventions that focus on health

promotion issues such as parent-child relationship enhancement, social skills training, healthy eating or obesity prevention, and reduction of risky sexual behavior.

Social Issues Affecting Children, Adolescents, and Families

Many of the challenges to youth adjustment are the social-environmental issues that influence children and families. Clinical child and pediatric psychologists should be aware of the sorts of social issues particularly relevant to youth, both generally and in their home communities. For example, as many as 50% of children have been exposed, either directly or indirectly (e.g., through media reports), to traumatic events such as natural disasters or violence in their homes, schools, communities, or the world (Bell & Allwood, 2007). Thus, trauma exposure is a major public health issue and one that all clinical child and pediatric psychologists will almost certainly encounter. Another important public health concern is youth access to mental health services; as many as 20% of youth are estimated to be in need of psychological services, yet only about one third of these youth receive needed services (Department of Health and Human Services, 1999). For both of these issues, clinical child and pediatric psychologists can make positive contributions in several ways, including identifying youth in need of treatment, participating in prevention programs to reduce risk for maladjustment, and participating in child advocacy efforts that can impact both individual children's lives as well as broader public policy and legislation. Involvement in Society for Research in Child Development's Office for Policy and Communications, APA's Public Interest Directorate, or state or local government can be effective ways for clinical child and pediatric psychologists to leverage their influence.

Specialty Applied Experiences

Roberts's (1998) final recommendation for clinical child training guidelines is that trainees gain applied experience in specialty practice relevant to assessment, intervention, and consultation with children, adolescents, and families. Although this is part of most of the preceding recommendations, its reiteration here emphasizes the importance of providing trainees with child-specific experience versus simply assuming that skills will generalize from adult-focused experiences. Roberts suggests that sufficient experience includes a minimum of closely supervised child practica in at least two different settings, internships that are at least half child-focused, and postdoctoral experience that provides further depth in autonomous child-related practice. Interestingly, Roberts's recommendation regarding internship is less than the two thirds child focus recommended at the Hilton Head Conference.

For the Society of Pediatric Psychology Task Force (Spirito et al., 2003), this curriculum domain centers on the importance of consultation and liaison skill development. Again, this domain underscores recommendations regarding interdisciplinary practice made throughout the Task Force's document. However, in addition to the need for pediatric psychologists to consult with parents and other professionals on child assessment and treatment, they will often serve in a training role. Specifically, pediatric psychologists should be able to educate health care professionals about principles of learning and development, psychological health (of youth and

of health care professionals), and psychological research (Drotar et al., 2003; Spirito et al., 2003). This is particularly important because many psychological and behavioral issues are first raised with children's primary care physicians.

Disease Process and Medical Management

The final curriculum domain, specific to pediatric psychologists, notes the importance of understanding medical issues that they will be likely to encounter. Trainees should participate in coursework, seminars, supervision teams, and applied experiences that familiarize them with the phenomenology of relevant medical issues, as well as psychological issues related to the medical issue and its treatment (e.g., adverse behavioral effects of medical treatments). Spirito and colleagues (2003) suggest that pediatric psychology students complete practica and internship rotations in primary and specialty medical clinics in order to work directly with medical professionals and medical issues.

TRAINING OUTCOMES

There are undoubtedly many ways that successful outcomes for graduate students in clinical child and pediatric psychology can be defined. During graduate training, success can be evaluated by examining student retention, performance on program tasks (e.g., course grades, percentage of students passing comprehensive examinations), participation in professional activities such as conference presentations and publications, and more subjective issues such as personal and financial stability (e.g., the toll that graduate training takes on students' health, personal relationships, and financial debt). Postprogram outcome variables might include student program completion statistics (e.g., the likelihood that students will complete their graduate training and obtain relevant internships, the total debt load incurred to obtain the graduate degree), professional competence (e.g., number of graduates who attain licensure, diplomate status, or other demonstration of competent practice), and career tracks and success (e.g., the number of graduates who secure relevant jobs, research and mentoring productivity of graduates in academic positions, career satisfaction and longevity). Obviously, many of these outcomes are fairly subjective, multiply determined, and may not even be accessible to graduate programs. For example, the field is currently struggling with the best ways to define and assess professional competence (Kaslow et al., 2004; Roberts et al., 2005; Rodolfa, Kaslow, Stewart, Keilin, & Baker, 2005). Similarly, graduate programs are often unaware of students' personal issues unless they are clearly affecting professional functioning, and even if aware, are often in a poor position to evaluate the extent to which graduate program stresses versus other factors contributed to student health problems, relationship breakups, or decisions to leave a program or the field. Despite these limitations, however, training programs should routinely collect and evaluate data on student outcomes. At a minimum, these data are useful for the self-evaluation that programs can use to identify areas for growth (and that CoA requires). Beyond that, sharing these data with potential students and professional colleagues has the potential to improve training in the field as a whole.

For many years, the Council of University Directors of Clinical Psychology (CUDCP; the training council for scientist-practitioner and clinical science–oriented clinical doctoral programs) has supported a resolution strongly encouraging all member programs to make trainee outcome data available to the public. Programs collect and share information on program attrition, time to degree completion for graduates, and the percentage of students who obtain predoctoral internships. In addition to helping prospective graduate students evaluate training programs, these data have been useful in evaluating how clinical programs fare on these basic outcomes. For example, recent years have seen an increasing imbalance between the number of internship slots available through the Association of Psychology Postdoctoral and Internship Centers (APPIC; 2006) match system and the number of students applying for these slots, with only 77% of applicants being matched to internships on National Match Day in 2006. However, a survey of CUDCP programs indicated that 86% of students from CUDCP programs secured internships on the 2006 Match Day, and 92% of students ultimately obtained an internship through the match, internship clearinghouse, or other methods (Collins & Callahan, 2006). Although these data are not specific to clinical child or pediatric training, they do suggest that students trained in scientist-practitioner programs are faring better than the national norms.

Recently, the CoA has instituted a requirement that all accredited training programs make similar data available to the public (APA, 2006b). Specifically, beginning January 1, 2007, the CoA requires that training programs provide information on time to completion (including mean, median, and percentages of students completing the program in under 5, 6, 7, and more than 7 years), program costs (e.g., tuition, fees, any certain or possible adjustments to costs such as waivers, fellowships, or assistantships), internship placement (including APPIC match rates and overall placement rates through other methods, rates of paid and half-time internships), and attrition (i.e., the number and percentage of students who fail to complete the program for any reason). Additionally, beginning in 2008, programs will be required to report the number and percentage of graduates who have become licensed psychologists in the preceding decade.

Beyond program completion, graduates' career trajectories provide an index of training success. Given the scientist-practitioner emphasis of most clinical child and pediatric psychology programs, careers that integrate research and applied activity seem the ideal outcome, although the nature of this integration might range from jobs in academic departments or medical centers that combine research, clinical teaching and supervision, and perhaps some clinical practice, to more applied jobs in which practitioners provide evidence-based treatments, participate actively in continuing education, and perhaps provide clinical supervision to doctoral students. For clinical science programs, the desired integration is likely to emphasize research more heavily, with graduates being effective contributors to clinical science research and dissemination. Although they do not examine child-focused programs specifically, a few studies have examined clinical programs' success in placing graduates in academic department faculty positions (Ilardi, Rodriguez-Hanley, Roberts, & Seigel, 2000), graduates' research productivity once they begin their own academic careers (Roy, Roberts, & Stewart, 2006), and graduates' scores on

the Examination for Professional Practice in Psychology (Yu et al., 1997). These studies provide useful information that individual programs can use to compare themselves to national norms. Interestingly, although these outcome variables all tended to relate significantly to reputation rankings such as the one published by *U.S. News and World Report*, they offer a more objective and specific way to assess training outcomes.

FUTURE CHALLENGES

Although exciting advances have shaped the growth of clinical child and pediatric psychology over the past 50 years, important challenges lie ahead as the field emerges from its adolescence into its professional adulthood. In addition, current issues that confront clinical psychology as a whole may provide distinct opportunities for leaders in child and pediatric psychology to creatively solve problems and improve training experiences for students. These challenges are the focus of several recent articles and conferences (e.g., the special issue of *Clinical Psychology: Science and Practice* on the present and future of clinical psychology; Kendall, 2006). Here, we discuss a few of these issues, include the challenges of designing curricula that balance breadth and depth, defining what it means to be a competent clinical psychologist, addressing impairment among students and faculty, and responding to dynamic market forces that shape the job outlook for clinical psychology graduates.

Broad versus Specific Training

As we discussed earlier, a central issue in the growth of clinical child and pediatric psychology has been how best to balance general (thus far primarily adult-focused) training in psychology with specialized coursework and clinical experiences that focus on child or pediatric psychology. As the field evolves in the twenty-first century, it is critical to evaluate whether broad and general training, as it is currently defined, keeps clinical psychology training stagnant. Further, we foresee attaining an appropriate balance as becoming more difficult as the technology of the field advances and expands. Clinical research is reaching deeper into existing problems and stretching into novel areas. For example, the development and validation of new assessments, preventions, and treatments translates into additional skills and educational content that students must master to be competent psychologists upon graduation.

The question remains whether specialization at the level of clinical child or pediatric psychology is appropriate and beneficial to students. Or put another way, are clinical child and pediatric psychology more of a specialty than clinical adult psychology? To date the field has taken steps to define itself as such, and many leaders in the field have embraced this definition. Yet, clinical child and pediatric psychology are not the only fields to deal with this issue. Leading clinical gerontologists, health psychologists, and clinical neuroscientists also struggle with these same training decisions (Roberts, 2006). What is potentially lost in specialization's overemphasis in one area is the scope of clinical psychology as a whole. As currently implemented by many programs, specialized child clinical or pediatric tracks often take a life span developmental and systemic approach to psychology that ensures broad

training (Prinstein & Roberts, 2006; Roberts, 2006). One potential resolution to these specialization issues is to define core curricula as a set of skills that are common to all psychologists, such as training in scientific methods, data analytic techniques, and general clinical skills. In addition, training in core areas of psychology would provide content that takes a life span developmental and systemic approach. Then the focus on clinical adult or clinical child psychology would be considered at the same level in terms of specialization. This would alleviate the "adult clinical plus" model that permeates many training programs, in which students who are interested in working primarily with youth populations end up taking similar numbers of clinical classes plus additional classes or seek additional training opportunities with children (La Greca & Hughes, 1999; Roberts, 2006). Clearly, however, the significant challenge of this approach is to provide students with sufficient exposure to the burgeoning literature in important areas without extending an already lengthy training process.

Graduate students in clinical psychology fall on both sides of the argument. In a recent survey of over 1,000 graduate students in programs affiliated with the CUPCP, almost three quarters of clinical psychology graduate students report specializing within their graduate training, and over half of those who reported not specializing plan on doing so before the end of their graduate training (Luebbe, Green, & Malcolm, 2006). Further, just under 50% of those who reported specializing indicated that their specialty was a recognized or formal track within their program, and over 60% of students reported that their specialty area was a large influence on their choice of graduate programs. Interestingly, students reported specializing in a variety of ways beyond the types of populations (e.g., child) with whom they work. Students also consider themselves specializing in certain disorders, research areas, theoretical models of therapy, and specific skills such as statistics. These results suggest that graduate students want to and already are specializing during their graduate training. This makes a strong argument for officially recognizing certain specialties. Yet, these results also suggest that the word "specialty" has varied meanings, and students are finding opportunities to specialize within their current programs regardless of whether such specialties are officially recognized. Overall, designing curricula that provide broad but focused training remains a critical issue as clinical child and pediatric psychology grow.

Professional Competence

As we described, the field of clinical psychology, including its regulatory and accrediting bodies, is currently in flux as it moves away from an activity-based model of training to one that is competency-based. An ongoing challenge for training programs not only in clinical child and pediatric psychology but also clinical psychology as a whole is to continue to produce professionals who are competent to practice psychology and who are committed to a process of lifelong learning that will enable them to stay up-to-date with the best practices available. The development of training models and lists of core competencies in clinical child and pediatric psychology (e.g., La Greca & Hughes, 1999; Roberts et al., 1998; Spirito et al., 2003) are good first steps toward establishing competency-based training.

Still, several questions remain during this transition. First, how will competency be assessed? Second, *who* will be responsible for such assessments? Finally, and especially pertinent to clinical child and pediatric psychology, how will competency among specialties differ from one another regarding content of competencies, unique contexts in which competency is needed, sequence of competency development, and assessment of competency? In the future, leaders in clinical child and pediatric psychology will likely be charged not only with the task of developing novel techniques to assess competency across the span of a career, but also for conducting the necessary research showing that such assessments are reliable and valid indicators of competency (Roberts et al., 2005). For example, although the Examination for Professional Practice in Psychology provides a reliable assessment of knowledge, it does not assess other foundational competencies such as skills and attitudes. Several authors have proposed use of additional assessment techniques such as standardized patients, self-assessments, or "360 degree evaluations" (Kaslow, 2004 Roberts et al., 2005).

Recently, the field has been quite active in attempting to identify specific competencies that professional psychologists should attain across their development and exploring useful ways of assessing and documenting competence. The 2002 Competencies Conference initiated by APPIC identified several areas in which competencies should be established (Kaslow et al., 2004) and paved the way for ongoing efforts by APPIC, APA's Board of Educational Affairs, the Council of Chairs of Training Councils (a meta-organization representing the training councils in professional psychology as well as other organizations interested in training), and other training-related organizations. Hatcher and Lassiter (in press) describe the results of a multiyear practicum competencies workgroup spearheaded by the Association for Directors of Psychology Training Clinics with input from the Council of Chairs of Training Councils. With additional input from clinic and training program directors from all areas of professional psychology, the workgroup developed a document that outlines practicum competencies considered necessary at the beginning of practicum training, that develop over the course of graduate practicum, and that should be evident at the end of practicum training in order for students to be considered ready for internship. Although the document does not specify how these competencies should be assessed, training programs are already exploring creative ways of incorporating these competencies into supervisor evaluations and student self-assessments. The APA's Board of Educational Affairs is also currently active in identifying competency benchmarks for professional practice in areas such as assessment and diagnosis and multicultural competency. These efforts promise to make a substantial impact on the way professional psychologists evaluate their training efforts.

Because of the unique populations served and the varied contexts in which clinical child and pediatric psychologists work, new and specific ways to assess competency may be needed. For instance, ratings by peers or other professionals in schools, hospitals, or juvenile justice systems may complement more traditional competency assessments. In addition, as competency-based models become the norm for clinical child and pediatric training programs, it is imperative that the field as a whole also conduct an ongoing self-assessment of sorts. As the definition

of a clinical child or pediatric psychologist evolves, so too must the knowledge, skills, and attitudes that compose the core competencies of these professionals. Leaders in the field must be willing to adjust and strengthen current models of training by adding new domains of competency, modifying current domains, or removing some domains altogether.

Professional Impairment

Professional competence among practicing psychologists, faculty supervisors, and students helps both to build the public's trust and to ensure its safety. Detecting and intervening with impaired professionals is therefore a vital responsibility of all clinical psychologists. Empirical studies have begun to show that graduate student impairment, whether it be lack of competence or experienced personal distress, is prevalent (e.g., Vacha-Haase, Davenport, & Kerewsky, 2004). Yet on the whole, far less has been written about this population than about interns or licensed professionals (Schwartz-Mette, in press). Further, given the vulnerability of the population with whom child and pediatric psychologists primarily work, professional competence is even more critical for students and supervisors in these fields. Also, as previously mentioned, clinical child and pediatric psychologists often work with multiple individuals in a variety of systems. This puts these students, not to mention practicing professionals, in a unique position as they potentially interact with a wider spectrum of the public than many psychologists, and as such may have to be multicompetent.

While APA ethical codes stipulate that psychologists should monitor themselves and peers for signs of impairment, graduate students may be particularly unlikely to do so for reasons associated with their professional inexperience, added stress during training, growing pains as they transition from undergraduate experiences to being a publicly recognized professional, and reservations regarding identifying impaired peers for fear of negative social consequences (Schwartz-Mette, in press). Thus, detection and appropriate intervention with impaired students often falls on the shoulders of academic training programs. Unfortunately, although both the Council on Accreditation's *Guidelines and Practices* (APA, 2005) and APA's *Ethical Principles of Psychologists and Code of Conduct* (APA, 2002) discuss impairment, neither document provides recommendations for resolving these issues. As such, most training programs are left with creating program-specific guidelines for intervening in situations in which impairment occurs, and these guidelines vary widely from program to program.

Several authors (e.g., Schwartz- Mette, in press; Vacha-Haase et al., 2004) have provided recommendations for addressing professional impairment. These include improvement of self- and peer monitoring through education about self-care and transition-related stress in graduate school, increased monitoring by clinical supervisors of students, and creation of more sensitive and open cultures in training programs. Additionally, more research is needed in the field that examines the unique issues that arise when child clinical and pediatric psychologists are impaired (e.g., prevalence of inappropriate conduct with children). Finally, training programs will need to continue to take steps toward managing students with

impairments through informal (e.g., varied clinical training cases to improve competency with specific populations) and formal (e.g., required counseling for emotionally distressed students) channels.

Marketplace Developments

The exponential growth of clinical psychology over the past 100 years has not come about without some growing pains, and changing market forces have altered the job outlook for many graduates of clinical psychology training programs. In the future, clinical child and pediatric psychologists have reason to be cautious, but also optimistic. One factor that directly influences the job market for new psychologists is the ratio of qualified applicants to jobs. On the bright side, job opportunities are predicted to grow faster than average for practicing psychologists in the next 8 years, especially for clinical and school psychologists (U.S. Bureau of Labor Statistics, 2006). Despite this growth, training programs are unfortunately producing even more psychologists than there are adequate training opportunities (i.e., internships) and perhaps market demand. For example, the ratio of internship applicants to available slots decreased from 1999 to 2002 to levels close to 1:1. Yet, this ratio has risen sharply from 2003 to 2006 (APPIC, 2006). If this trend continues in a similar fashion, the market may become oversaturated with psychologists. Although the APA Office of Research collects data on employment of new graduates, no up-to-date formal workforce analysis exists. Fortunately, the Office of Research is currently pursuing such an analysis.

Positions for academic clinical child and pediatric psychologists also appear to be growing. Many departments are receiving funding for new faculty lines, and additional positions formerly occupied by retiring faculty are opening. More universities and colleges are creating interdisciplinary positions, such as developmental psychopathologist positions with dual responsibilities in clinical and developmental training areas, or have growing departments or programs in human development and family studies. Joint doctoral programs in clinical and developmental psychology, like that at the University of Missouri–Columbia described earlier, may help to expand students' marketability in the academic job market. One challenge that remains for academic psychologists deals with research funding. Unfortunately, grant funding is reliant on political factors, and despite funding levels remaining relatively stable from agencies like the National Institutes of Mental Health in the past 5 years, the competition has grown more intense and success rates for awards have decreased sharply. It is uncertain whether this trend will continue in the future.

Dynamics in the marketplace are leading many graduates of clinical training programs, including those in child and pediatric psychology, to redefine their professional roles or to enter nontraditional employment opportunities. Generally speaking, fewer clinical psychologists are working in traditional office settings seeing clients for the 50-minute hour, and this is especially true for clinical child and pediatric psychologists. In turn, training programs are challenged to keep pace with these changes by adjusting their curricula to provide relevant coursework and experiences that prepare their graduates to be competitive job applicants. However, within this trend there is reason for child and pediatric psychologists to be

optimistic. Specifically, students who are trained in an ecological framework should excel in opportunities to work in a variety of settings where child clinical or pediatric psychologists traditionally have not. This will require, however, that graduates of clinical child and pediatric training programs take the lead in developing partnerships with community agencies and institutions to develop such niche positions.

SUMMARY

Although adolescents in the field of professional psychology, the future of clinical child and pediatric psychology looks promising. Defining these fields has not been an easy task, given the variety of roles, responsibilities, and endeavors undertaken by clinical child and pediatric psychologists. Further, as these definitions continue to evolve, training programs will have to modify their curricula to provide state-of-the-art instruction for their students. In this chapter, we have outlined several training models and important components of clinical child and pediatric training at the graduate, internship, and postdoctoral levels. Special attention was paid to outcomes of training, including what a competent clinical child or pediatric psychologist might look like in areas as varied as assessment and diagnosis, multiculturalism, research methods, and ethics. Such curriculum modifications are not without their challenges, and we ended this chapter with a discussion of issues that will need to be addressed both in the near future and several years down the road. Overall, training in clinical child and pediatric psychology has made considerable strides to date, and many training programs produce competent, skilled psychologists. Yet, we think it is important that leaders in clinical child and pediatric psychology not be content with the progress made thus far, but work diligently and proactively to address the challenges facing these fields.

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CHAPTER 3

Professional Practice

Daniel B. Chorney and Tracy L. Morris

The practice of professional psychology has been in existence for roughly one century. Most readers will be quite aware of the roots of clinical psychology in the independent psychiatric practice of Sigmund Freud in the late nineteenth century. Perhaps less well known, but no less pivotal, was the establishment of the world's first psychological clinic by Lightner Witmer at the University of Pennsylvania in 1896 (which specialized in the provision of services to children and their families).

Professional psychology has witnessed considerable growth over the intervening decades. In the period between World War I and World War II, the number of psychologists in the United States increased from 300 to 3,000. Prior to WWI the vast majority of psychologists were academics teaching and conducting research in colleges and universities, yet by 1930 over one third were working in applied settings (Finch & Odoroff, 1939). This proportion remains essentially the same today. Recent data indicate that of the 179,000 psychologists employed in the United States in 2004 (U.S. Department of Labor, 2006), nearly 60,000 were engaged in full-time clinical work (with an additional 18,000 engaged in some form of part-time practice; Kohout & Wicherski, 2003).

WHAT DEFINES PROFESSIONAL PRACTICE?

Professional practice is a relatively broad term used to describe all professionals within the field of psychology, not just those who choose to practice independently. Researchers, academics, clinicians—all share certain things in common that allow them to be called professionals, including, but not limited to, a shared professional code of ethics, years of graduate education in research and treatment of psychological disorders, and a license or state certification to utilize knowledge to help the public. The focus on using and evaluating empirical methods helps shape the way professional psychologists approach their work, and this rigorous training is just one aspect of what sets professional psychology apart from other helping professions (Todd & Bohart, 2003).

Professional psychologists provide a diverse range of services and skills. As Blau (1983, p. 57) points out, psychologists

> conduct a wide variety of psychotherapeutic interventions, assess intellect, aptitude, personality, neuropsychological functioning, and marital adjustment—they design, direct,

> and monitor various prevention and intervention programs . . . work in many diverse medical settings utilizing uniquely psychological methods and tools that are required in health service, rarely work autonomously, and ordinarily function in psychology departments or psychology services in university clinics, hospitals, prisons, community mental health centers, agencies, courts, and treatment centers.

While many skills are shared by all professional psychologists, distinctions in preparatory work, professional roles, and skills are important to note. These differences begin in graduate training, as some psychologists are trained under the scientist-practitioner model (commonly known as the Boulder model), first outlined in 1949. This approach stresses the importance of academic research in combination with supervised clinical experience, resulting in professionals who are highly competent in the assessment, evaluation, and treatment of psychological disorders using empirical methods. As an alternative, the practitioner-scholar model places a heavy emphasis on clinical training instead of research. Although still requiring a doctoral degree, students trained under the Vail model have reduced academic research requirements and are often associated with separate schools of professional psychology rather than research universities. The two models typically provide professional psychologists with different degrees: A PhD or doctorate in philosophy is awarded upon completion of graduate work, successful defense of a research dissertation, and a year of internship at a clinical setting approved by the American Psychological Association (APA). This compares to a PhD, or doctorate in psychology, which is awarded after 4 to 6 years in a professional program where the emphasis is on clinical skills and training, a year of internship training, and research that tends to focus on more qualitative methods directly related to clinical practice. Both qualify under APA's (1987a) definition of professional psychologist: "Psychologists have a doctoral degree in psychology from an organized, sequential program in a regionally accredited university or professional school." Both models, despite their differences, allow for the ability to produce psychologists for independent practice. The Boulder model's training emphasis on the development of an evaluative attitude toward one's own and others' work is not antithetical to independent practice, and the delivery of services in the private sector by independent practitioners does not necessarily mean these psychologists are no longer critical scientists (Cantor & Moldawsky, 1985).

Professional practice is also shaped by what psychologists actually do on a day-to-day basis. The next logical question, then, is What distinguishes professional psychology from other professions? The APA (1987b, p. 2), in its Model Act for State Licensure of Psychologists, has outlined and given a definition for the role of professional psychologists, stating:

> The practice of psychology is defined as the observation, description, evaluation, interpretation, and modification of human behavior by the application of psychological principles, methods, and procedures, for the purpose of preventing or eliminating symptomatic, maladaptive, or undesired behavior and of enhancing interpersonal relationships, work and life adjustment, person effectiveness, behavioral health, and mental health. The practice of psychology includes, but is not limited to, psychological testing and the evaluation or assessment of personal characteristics, such as intelligence, personality, abilities, interests,

> aptitudes, and neuropsychological functioning; counseling, psychoanalysis, psychotherapy, hypnosis, biofeedback, and behavior analysis and therapy; diagnosis and treatment of mental and emotional disorder or disability, alcoholism and substance abuse, disorders of habit or conduct, as well as of the psychological aspects of physical illness, accident, injury, or disability; and psycho-educational evaluation, therapy, remediation, and consultation. Psychological services may be rendered to individuals, families, groups, and the public.

Although a better description of how psychologists are trained, along with what skills they possess, the definition provided thus far only scrapes the surface of the many different faces, forms, and functions a professional psychologist may represent. Differences in theoretical orientation, specialization in a specific disorder, and the type of job psychologists choose all reflect what duties they will perform, how they will go about them, and with whom they will work and collaborate in daily functioning. The purpose of the present chapter is to examine in closer detail one career path a psychologist may take: private practice.

INDEPENDENT PRIVATE PRACTICE IN PSYCHOLOGY

The idea of private practice in psychology is as old as the profession itself. For many, simply saying the words aloud conjures up the image of couches and clients recollecting dreams and early childhood experiences while a therapist sits back and attempts to make sense of it all. This romantic and glorified image of the practice of psychology is easily brought to mind, yet it could not be further from the truth. Today's private practitioner lives in a world dictated by managed health care, empirically supported treatments, and constant financial uncertainty. The lure of private practice may be what motivates many people to enter the field of psychology, yet fewer and fewer psychologists are ending up in private practice. One does not have to look far for a book, article, or speech that outlines the demise of private practice in psychology. Starting up and running a business, building a referral and client base, handling an increasingly large and difficult managed care system, and dealing with legal liabilities are generally not considered reasons that people look forward to starting a private practice. Yet the profession survives. With all its limitations and challenges, the benefits of a career in private practice remain. Professional practice will inevitably remain a viable career path for many psychologists, despite the ever-changing face and demands of the profession itself.

If private practice is not meant for everyone, the question becomes For whom is it meant? In a recent survey of over 1,600 psychologists, 2% reported working full time in independent private practice. This number rises to 5% when group practice settings are included (Wicherski & Kohout, 2005). These numbers show a slight decline from the 1999 Doctorate Employment Survey (6% in total were in some form of practice). Although numbers may be declining and the environment is becoming more challenging overall, numerous advantages of establishing a private practice still exist.

RELATIVE ADVANTAGES AND DISADVANTAGES OF PRIVATE PRACTICE

As with any profession, certain qualities and individual characteristics draw people to the field of psychology. As stated earlier, a career in private practice is not for everyone, but it certainly can be rewarding for those who enjoy the benefits it has to offer. One of the most salient rewards is the notion of being your own boss. Although the same freedoms may not exist today as were available years before, establishing an independent private practice means more control over your hours, clientele, and area of specialization. For many people, the idea of going to work each day and not having to face one or multiple supervisors is a truly appealing notion.

Being your own boss brings with it a certain level of flexibility and independence, along with more responsibility. Numerous decisions must be made, such as deciding where to establish a practice (urban versus rural), what the operating hours will be (flexible and spread out or a regular 9-to-5 workday), and what types of clients you are willing to see (specialization or generalization). In addition to these important decisions, an independent private practitioner also has full control over business decisions and directs how the practice will grow (or shrink, depending on one's business skills).

With this control, the practitioner's decisions play a large role in deciding how much he or she will earn that year. Once a strong client base has been established and clients are being seen regularly, the potential for earnings increases dramatically. A key point here is that independent practice allows for the potential to earn a large salary; the median starting salary of a psychologist in independent private practice is $50,000 (this is reduced to $45,000 when in group practice), which compares to a median starting salary of $45,000 for an assistant professor on a 9-month contract (Wicherski & Kohout, 2005). Earning substantially more than this requires a large investment of time and effort (both on a daily and a long-term basis). How much is earned also will depend on what services are provided, and whether you decide to accept insurance or have a sliding scale.

Having more power over decisions and the daily running of your practice can certainly be an advantage, but it brings with it a high degree of responsibility and personal liability. As primary decision maker, the practitioner must deal with all the consequences of those decisions. Many of the advantages listed previously also have the potential to become disadvantages if not handled correctly. Independence and autonomy may be great for some, but others may require a certain level of structure and consultation in order to succeed; thus, what may be advantageous for some may be a burden for others. If one's practice is badly located, one's hours are too few or constrained, and one's area of specialization is too focused, one may soon realize how challenging private practice can be.

Financially there exists a high level of uncertainty and risk in establishing a practice, as it will take some time to develop a strong client base. Many may choose to begin in another setting and slowly build up a private practice on the side, before fully committing to such a potentially risky endeavor. As the practitioner becomes responsible for all billing, collections, insurance, and employee decisions that must

be made (Barnett & Henshaw, 2003), he or she quickly realizes how important it is to be familiar with basic business principles—something many psychologists are not aware of or prepared for when entering this area. Some graduate schools are leading the way in providing courses that address the financial and business aspects of working in the field of psychology, covering aspects such as information and financial management, contract negotiation, and planning and budgeting for the future. It may be of great benefit for those interested in pursuing a career in private practice to take such business classes while in graduate school to be better prepared for the multiple financial and business responsibilities associated with working independently or as part of a private practice.

Others may experience a sense of professional isolation when practicing independently. Although this may be solved by deciding to enter into or form a group practice instead, the number of colleagues one interacts with is very limited in comparison to working in a larger institution, such as a university or medical setting. This also becomes a challenge when consultation on a case is necessary, as an independent practitioner may not have anyone to immediately turn to for advice. Such concerns may be addressed, in part, by maintaining involvement in a larger network of individuals in the same profession (state professional organizations, local treatment providers, fellow private practitioners in the area). Contact with these individuals may help ease isolation and provide a social outlet and support when necessary or desired.

When working independently it is important to remember that in many cases, one truly is working independent of others. This may require the psychologist to wear many hats, including those of accountant, public relations person, cleaning staff, insurance policy expert, receptionist, real estate agent, financial analyst, bill collector, and office manager (Vineberg, 2005). Access to clerical or support staff who can help with paperwork, correspondence, scheduling issues, and numerous other day-to-day operational tasks will affect how business is conducted in the office, as much of this work is necessary to keep a business running smoothly. All of these duties may be challenging by themselves, yet the larger burden may be the time spent ensuring that they are done correctly. Personal freedom may come at the cost of sacrificed time, and one may have to work long hours and through the weekend just to ensure that all necessary business details are covered. It may be important to remember here that these administrative duties are all above and beyond actually conducting therapy and performing one's duty as a clinical psychologist.

GROUP PRACTICE IN PSYCHOLOGY

If the disadvantages of independent private practice seem to outweigh the benefits, a viable alternative is joining or forming a group practice. Group practices can be small, composed of a few psychologists or other mental health professionals, or much larger organizations consisting of an interdisciplinary team of providers working collaboratively and sharing resources. In this manner, many of the potential downfalls of independent practice can be avoided, or at the very least reduced.

Joining a group practice allows one to experience many aspects of independent practice before actually committing to it. The daily exposure to aspects of running

a business, handling client records and paperwork, and other daily activities associated with work in a practice may provide a good first step in making an informed decision as to whether a future career in independent private practice is actually desirable. By joining a group practice and getting your foot in the door, it may be easier to see if this type of work is suitable and agreeable without having to commit large amounts of time and money to opening an independent practice, only to find out it is not the best professional setting for your needs.

As mentioned earlier, a rather large financial risk accompanies a solo venture into private practice. One of the main advantages to joining a group practice is that the financial risk, investment, and capital associated with beginning a practice is distributed among group members. This may be especially appealing given recent data showing that 74% of new psychologists in practice subfields are starting their professional life with a median debt level of $67,500 (Wicherski & Kohout, 2005). As high as this seems, the level of debt grows substantially higher for those graduating with a Clinical PsyD, with a median debt level of $90,000 and 79% of new graduates reporting some level of debt. Given these numbers, and knowing that it may take a substantial amount of time before a regular client base is established and income becomes steady, it is readily apparent why the idea of sharing financial burden and risk is appealing. When joining an already established group practice, a referral base may already exist, which eliminates a large degree of work and reduces the amount of stress associated with building a clientele from scratch. Along with defraying the initial start-up costs, it may be financially beneficial overall to be in group practice. Despite the fact that generated income may need to be split among members in the group practice, this is counterbalanced by having all members of the group split the cost of expenses generated in running the practice. Everything from rental and operating fees to paying office staff can be shared, whereas an independent practitioner must pay for all of these costs without help from others. To pay for many of these expenses, a solo practitioner needs to keep overhead costs at a minimum, and often will have to perform multiple roles within the office (e.g., receptionist, therapist, and custodian). As one author puts it:

> When you consider solo versus group practice, you eventually have to ask yourself whether you want to hand over a sizable portion of your revenue to cover expenses that are paid solely by you, whether you want to minimize these expenses somewhat and add more work and headaches to your already busy schedule, or whether you want to hand over a sizable portion of your revenue to others and let them take care of the headaches so you can focus your energy on being a clinician. (Habben, 2005, p. 101)

Working alongside others in group practice eliminates the disadvantage of professional isolation and provides other benefits as well. Colleagues may help in providing knowledge and expertise on a variety of issues, including how to run a successful practice, dealing with especially difficult cases, and providing resources for referrals in the community. Colleagues also provide friendship and may help with expanding one's social and professional network. Despite working alongside others, much of the freedom of independent practice still exists, as often one can still determine to a large degree the type and amount of patients seen.

Although it may appear as though group practice solves any and all disadvantages of independent private practice, some disadvantages remain. First and foremost, the level and degree of control over decisions being made are no longer solely in one person's hands in group practice. Decisions regarding the hiring and firing of staff, the fees charged to patients, questions about referrals and providers, and all other business decisions must now be made as a group and no longer individually. To some this may seem appealing; to others this loss of independence may be enough to turn them away from group practice.

Similar to independent practice, income from group practice may be variable. Although the potential for large earnings still exists, a lot depends on the overall strength and productivity of the practice as a whole, as well as your own work. Fluctuating levels of productivity will provide varying levels of pay across time, and this may cause a high degree of frustration in anyone who prefers a steady and stable paycheck.

Another disadvantage arises primarily when one is seeking another job, as working in a group practice does not carry with it the instant credibility and respect by association as working for a university may provide (Habben, 2005). Considering the low name recognition a group practice has outside of its own community, it may be hard to communicate to a future potential employer the exact nature of work conducted, the environment and conditions, and the skills needed to work successfully in the practice being left.

PREPARING FOR LIFE IN PRIVATE PRACTICE

Perhaps one of the most difficult aspects of beginning a private practice is the transition one must make in one's sense of identity, a transition from therapist to businessperson. As an independent therapist, you are providing and selling a service: therapy. Graduate school and internships do little to prepare the training psychologist for life in private practice, and it may be difficult at first to view therapy as a marketable service provided for a consumer. One of the first questions that must be asked is What do I have to offer? The service a clinical psychologist provides is a complicated blend of the therapist's personality, character, and previous experience; the theoretical or philosophical approach and orientation he or she espouses; and the techniques and procedures used that reflect this approach (Earle & Barnes, 1999).

CHALLENGES IN STARTING AND MAINTAINING A PRACTICE

One of the first concerns in establishing a private practice is the start-up cost, which will vary immensely depending on what approach is taken and what sort of practice is wanted. Deciding to practice as part of a group may help to defer some of the initial costs, the first of which may be choosing where the practice will be and renting adequate office space. Things to consider here are the size and cost of your office space, transportation to and from your office and how accessible it is to clients, and what clientele you may draw depending on where your practice is located. The decision of where to practice also carries with it the weight of deciding

where you would like to live—at least for some considerable time. Even though your skills and services may be transportable, clients are not, as they are unlikely to follow you to where you end up moving, whether that be across town or across the country.

Other initial costs include licensing fees, which will vary depending on one's state, and malpractice insurance, which will also vary depending on what coverage is desired. These fees relate to where you decide to practice, as requirements for licensing vary across the country. Substantial costs may be incurred in furnishing the office (everything from chairs to filing cabinets to children's furniture and toys), computers, assessment materials, business cards and office supplies, utilities, professional membership dues, taxes, and possibly advertising.

Of all initial start-up concerns, perhaps the most important is establishing a steady flow of referrals. This is especially difficult if one decides to enter private practice immediately after completing internship and before having a strong base of referral sources from which to draw clients. The need to begin forming professional relationships and contacts during the internship year or even earlier may be crucial to help lay the foundation for a later referral base, especially if the goal is to stay within that community. Forming ties with others in the community (schools, local pediatricians, other clinics) can help build reciprocal referral relationships and help make one's availability, services, and presence known in the area. Other mental health professionals may be vying for the same clients, making this a difficult process. Again, good business sense and a strong marketing strategy can make a substantial difference here in attracting new clients.

Numerous legal issues will inevitably arise as part of private practice, and it is imperative that all steps have been taken to prepare before the time comes. Good malpractice insurance is required, along with having a solid understanding of HIPAA regulations, licensing laws, and state regulations. Consultation with an attorney may be of great benefit as you establish your practice.

MANAGED CARE

The topic of managed care in the United States almost immediately invokes strong feelings among health care professionals, including practicing psychologists. Whether liked or not, it must be accepted at present as the medium through which a large proportion of services are delivered. The current system of managed care services can be seen positively as being a significant source of referrals for one's practice, offering clientele ranging from private direct pay to third-party payers, patients charged on a sliding scale, and pro bono cases. Unfortunately, the demand it places on both patients and providers creates an uneasy tension, ultimately changing the face of mental health treatment in ways that many psychologists find detrimental.

In a 1998 survey of almost 16,000 licensed psychologists, 4 out of 5 reported that managed care had a negative effect on their professional work, with independent practitioners and psychologists in medical settings reporting the most negative effects (Phelps, Eisman, & Kohout, 1998). Their concerns included how managed care is changing clinical practice, excess precertification requirements of managed care panels, decreases in income, and ethical dilemmas created by managed

care. The authors conclude that the "marketplace-driven demands for integrated services, large and diversified group practices, and greater accountability have created obstacles and declining market opportunities that threaten the very existence of traditional psychological practice" (p. 35). Another 1998 survey (Murphy, DeBernardo, & Shoemaker, 1998) of 442 members of APA's Division 42 (Division of Independent Practice) found similar results, with only 2% of respondents reporting that managed care practices had a positive impact on quality of care. Moreover, 80% indicated that caps on the number of sessions interfered with treatment, and 84% of respondents stated that managed care companies have control over aspects of patient care and treatment that clinicians should control. Most respondents indicated that managed care had led to inappropriate treatment, insufficient treatment, or both. More specifically, they believed that overall, the number of sessions had been reduced, flexibility and room for clinical judgment had decreased, premature termination had become common, there were too many restrictions on patients served, referrals for medication had increased, use of protocols for treatment and the use of treatments outside of their primary orientation had increased, and the amount of assessment time had decreased. The negative effect of managed care on assessment is also seen in a recent survey of over 400 psychologists, in which over half reported that choice of tests, the number of tests, and the time invested in testing have been negatively impacted (Rupert & Baird, 2004). Other aspects of managed care identified as being stressful on psychologists are external constraints on services, managed care reimbursement rates, and excessive paperwork (Rupert & Baird, 2004).

Various difficulties may arise when trying to balance the needs of managed health care and the needs of patients. At times, it may be difficult to know exactly who one is working for, client or corporation. One issue that arises in independent practice is the balancing of a patient's confidentiality and privacy with the need to keep third-party providers informed about the patient's progress and need for continued treatment. In the previously mentioned study, 75% of psychologists strongly believed that contact with managed care and utilization review compromises patient confidentiality (Murphy et al., 1998). Here, as in other areas of practice affected by managed care, there appears to be a struggle between what is required by managed care and what psychologists believe is the ethical course of action. In the opinion of one practitioner, "There is nothing private about private practice, at least when it is practiced with a managed care patient. Private practice is an oxymoron in the age of managed care" (Wolf, 1999, p. 114).

Previous work has suggested that the pressures of managed care on clinicians, such as loss of autonomy, the stress of dealing with ethical dilemmas, and feelings of frustration after having treatment recommendations denied, may eventually reduce morale and increase the potential for burnout among psychologists (Kisch, 1992; Wooley, 1993). Some smaller surveys of the impact of managed care in specific states further reveal how the pressures involved with managed care may be compromising effective and ethical practice. A 1998 survey of psychologists in New Jersey (Rothbaum, Bernstein, Haller, Phelps, & Kohout, 1998) found that those with high managed care caseloads reported an increase in paperwork, a decrease in the overall number of sessions, and a drop in morale. Perhaps more important,

they reported pressure to compromise quality of care and ethical principles under these circumstances. Higher proportions of managed care caseloads also have been associated with working longer hours, having more client contact, receiving less supervision, reporting more negative client behaviors, experiencing more stress, being less satisfied with income levels, and showing higher scores on emotional exhaustion (Rupert & Baird, 2004).

The implications of these findings serve to underscore the importance of preparing graduates interested in pursuing a career in private practice for the demands and ever-changing nature of managed health care in the area of mental health. Some of the major concerns and stressors indicated in the numerous surveys conducted over the past decade (i.e., dealing with paperwork, resolving ethical dilemmas related to disclosure of confidential information to third-party providers) are issues that are rarely, if ever, taught or discussed with graduate students. The increasing importance of documentation and record keeping in the field, due in large part to the increase in lawsuits and malpractice claims, also needs to be stressed to those preparing for private practice. What may be more difficult to convey is how difficult and stressful it can become for practicing professionals to stay up-to-date with their documentation and administrative duties, along with the substantial amount of time this may take. As this is generally not the reason why most practitioners go into private practice, the commitment to these side duties may become a large source of strain and frustration, and practitioners should be aware of and ready for this when entering practice.

THERAPIST BURNOUT AND IMPAIRMENT

Once in private practice, it may be difficult to ascertain exactly how much work and stress one can handle independently versus when help may be required. Multiple stressors exist in the practice of psychology, many of which can potentially cause serious distress or impairment if allowed to continue unchecked. Sherman and Thelen (1998) examined potential causes of distress and impairment for psychologists, comparing stress associated with life events and stress caused from work-related factors. The authors found that malpractice claims, changed work situations (e.g., different work responsibilities, major changes in working conditions or hours), inadequate time for all obligations, office politics, and conflicts in relationships with colleagues were all among the most distressing work-related factors in clinical practice. The greatest impairment at work was caused by malpractice claims and restrictions imposed by managed care companies. Questions have been raised as to whether psychologists can adequately perform their duties while they are distressed, and although this debate continues, research findings (Nathan, 1986) indicate high correlations between life distress and life impairment and work distress and work impairment. Distress was defined as "the interference in ability to practice therapy, which may be sparked by a variety of factors and results in a decline in therapeutic effectiveness" (p. 79). These data appear to indicate that many psychologists are not only experiencing high levels of distress, but their work, and therefore their patients, are also suffering as a consequence. The study also found that reducing the average number of client contact hours per week was not related to

improvements in work distress or impairment, and that working in a private practice (versus group practice) did not appear to be related to levels of work distress, impairment, or professional satisfaction. This information may be especially salient for young professionals who are just beginning their practice, as their level of stress is likely to be high due to the inherent challenges associated with beginning a private practice in psychology.

Another potential cause of professional impairment that may be difficult for the independent practitioner to become aware of is burnout, which has three defining features: (1) emotional exhaustion, (2) depersonalization (defined as a negative and cynical attitude about one's clients), and (3) decreased sense of personal accomplishment and/or an increased likelihood of evaluating oneself and one's work negatively (Rupert & Morgan, 2005). The consequences of burnout impact both the therapist and clients, often resulting in impaired delivery and quality of services and significant stress to all individuals involved. Although research on burnout among psychologists has provided mixed results, a consistent finding is less burnout and lower levels of stress among independent practitioners when compared to agency-employed psychologists (e.g., Ackerly, Burnell, Holder, & Kurdek, 1988; Farber, 1985; Hellman and Morrison, 1987; Raquepaw and Miller, 1989; Vredenburgh, Carlozzi, & Stein, 1999). Different reasons have been hypothesized for why this finding exists, including that independent practitioners have greater control over their work activities and environment, they have fewer bureaucratic responsibilities, and they often have more say in choosing which clients they see and thus potentially see less disturbed clients. These reasons may have been true at the time these studies were conducted, but the more recent emergence of managed health care has significantly changed the environment of independent practice by reducing the amount of control and autonomy practitioners have, along with increasing financial uncertainty, creating more ethical challenges, and increasing therapist workloads. Recent studies examining burnout report similar overall rates among practitioners in comparison to studies conducted prior to the advent of managed care (Rupert & Morgan, 2005). This seems to suggest that although the health care environment may be changing in numerous ways, it does not appear to be increasing levels of burnout among practicing professionals. This does not suggest that burnout and emotional exhaustion are not serious problems faced by psychologists, as roughly 44% of respondents in the Rupert and Morgan investigation scored within the high burnout range.

Emotional exhaustion is a critical first sign of burnout. Factors related to emotional exhaustion include total hours worked, administrative and paperwork hours, managed care client percentage, negative client behaviors, and overinvolvement with clients. It is imperative that professional practitioners make efforts to safeguard their own mental health in order that they may have the energy and cognitive resources to be of maximum benefit to their clients. Potential strategies to help reduce or prevent burnout include increased self-monitoring and awareness of one's own levels of stress and symptoms of burnout, increasing one's social support network and working relationships with colleagues in the field, attending professional development workshops and seminars on prevention of burnout, and becoming involved in state or nationwide professional organizations that may provide help or support in areas causing the greatest amount of stress.

ETHICAL AND LEGAL ISSUES IN INDEPENDENT PRACTICE

Conflict of Interest

Certain pressures exist in private practice that are unique to this setting, one of the most notable being the pressure to keep one's clients happy. This is important for a variety of reasons—not only therapeutically, but also from a business perspective. In other settings, a client's deciding to leave therapy may have detrimental effects on the progress of therapy and that client's recovery. The same is true for a client terminating therapy when seeing a private practitioner, but now there is an added financial aspect from the point of view of the therapist. An unhappy client who terminates has an immediate impact on the income of the practice, as well as having the potential to affect new referrals if that client chooses to make his or her unhappiness known to friends, family, and colleagues. The issue may be even more complex in the practice of clinical child psychology, as the client typically encompasses the family, which may include two parents (who may disagree on the course of therapy) and siblings, as well as the child or adolescent for whom the referral was initially made. As many child referrals come from school systems, the satisfaction of teachers and school counselors also may be relevant factors. Within larger organizations where financial and legal responsibilities are largely shouldered by others, the practitioner may feel more comfortable addressing clients directly than might a professional in solitary practice (Applebaum, 1992). The importance of this matter is also reflected in the kind of clients an independent practitioner chooses to accept. One survey has shown that approximately 30% of practicing psychologists have avoided taking on certain clients due to fear of being sued (Pope, Tabachnick, & Keith-Spiegel, 1987). In this case, not only is there a fear of having a client terminate prematurely with a potential loss of income, but the therapist is acting defensively and preemptively to protect his or her own interests for fear of being drawn into a lawsuit. Although such fear is understandable in some ways, this still leaves the potential client in the community without active treatment—an ethically questionable situation that may not be such a great concern for psychologists who are not acting in constant fear of litigation.

Fees

Another issue concerns decisions regarding appropriate charges for services. Managed health care and insurance complicates this matter even further, yet the basic need to set appropriate fees still exists and the practitioner is left with a difficult decision to make. With only a certain amount of time and space for clients, decisions must be made in respect to who will be seen. Will the therapist choose to see clients on a first-come-first-served basis, give preference to cases he or she finds most interesting, or accept only those who have the greatest ability to pay? Apart from client selection considerations, the dilemma about specific fees is an obvious one: Charge too little and the practice may not survive despite having many clients; charge too much and risk having too few clients to pay the bills at all. Some practitioners may consider charging clients on a sliding-scale system, where fees are

based on the client's income level. However, the problem of making enough income to keep the business afloat will continue if the majority of clients are being charged the minimal fee allowed. Many clients may still not be able to pay the lowest fee allotted, leaving the therapist to decide how much time can be spent doing pro bono work. Further, regulations exist restricting differential fees for clients covered by insurance compared to those who are not covered.

The complication of reimbursement by insurance raises ethical issues for many private practitioners. In a 1987 survey of 456 practicing APA psychologists, Pope et al. found that over half the respondents reported altering an insurance diagnosis to meet insurance criteria—in other words, committing insurance fraud—either rarely (26.5%) or more frequently (35.1%). A more recent report indicates that 49% of psychologists believed other psychologists were altering patient diagnoses to help receive reimbursement, and 59% believed others were altering diagnoses to help patients get reimbursed (Murphy et al., 1998). This illegal and unethical behavior may be occurring more frequently in private practice settings, where the need to retain clients is high and oversight is low.

Termination

Similar to the dilemma of trying to keep the client happy and remain in therapy, the decision about when to terminate may cause a certain ethical conflict between client and therapist. Even if treatment goals are set out at the beginning of therapy, whether they are defined as reducing symptoms or achieving more broad life goals, the therapist is put in the difficult position of having to willingly terminate not only a client, but also a source of income for the practice. This dilemma can actually work both ways, with some therapists terminating treatment early if the client is particularly difficult or unrewarding to work with, and others prolonging therapy with clients who may have achieved all they initially set out to when entering into therapy. Again, the issue of personal needs versus client needs enters into the decision-making process while in private practice, with the potential consequences of these decisions affecting client and therapist, both monetarily and emotionally.

Leaving Practice

Psychologists choose to change their professional positions for many reasons. Whether due to financial difficulties, taking a new position elsewhere, disputes with colleagues, or simply a desire to discontinue practicing psychology, many psychologists will at one point in their career face the reality of leaving their current position or workplace. This may be an especially difficult situation for a private practitioner, who faces multiple legal and ethical dilemmas when leaving private practice. Professional ethics codes, including that of the APA (2002), discuss the importance and obligation for clinicians not to abandon their clients. As Barnett (2001, p. 183) succinctly states, "Simply leaving a practice without addressing each patient's treatment needs, arranging for a smooth transition if transfer of care is in order, or preparing patients for termination would appear contrary to the intent of relevant case law and the ethics codes of all mental health professions." Steps must be taken to ensure that the client's needs have been fully documented, met, and

followed up on when a transition takes place between therapy providers. This may be especially important, not only ethically but legally, for clients who present with any sort of imminent crisis or risk. In all circumstances, it is best to anticipate and avoid potential problems with terminating or transferring clients rather than deal with the consequences of inappropriate action after the fact.

When working with other professionals, it is important to clearly outline legal responsibility prior to entering into the business relationship, seeking consultation and guidance on noncompetition clauses, record ownership, and the disposal of client records. Noncompetition clauses prohibit a departing psychologist from setting up another practice for approximately 1 to 3 years within a certain geographical area. This essentially protects a practice from a psychologist who "use[s] the practice's reputation, resources, and referral sources and then, after establishing a local reputation, leaves the practice and opens an office in the local area, taking along the patients the practice provided" (Barnett, 2001, p. 184). Record ownership generally remains with the practice and must be maintained there in accordance with state laws and ethical standards. If the client chooses to seek treatment elsewhere, copies of records can be sent to the new therapist, but the original client record with treatment reports and notes remain the property of the practice. Prior to such events, detailed plans should be made concerning the retention, storage, and potential destruction of records in accordance with state laws and regulations. Practitioners should pay special attention to the required retention times of different types of patient records, both for liability reasons and in the event a client chooses to return at a later time. This may be especially important with underage clients, whose files are required to be stored until 3 years past their age of majority. Knowledge of applicable laws and regulations is of utmost importance, as these may differ across states. Specific requirements now exist in some states outlining the proper course of action should a client's file need to be destroyed, including the notification of the former patient in writing that his or her records will be destroyed, the date of the planned record disposal, and how the record or a summary of it may be obtained prior to that date.

Rural Practice

One of the first things that one must consider in establishing a private practice is where the practice will be located. A basic distinction can be made between deciding to open a practice in an urban or a rural setting. When choosing the latter, a number of ethical issues arise specific to practicing psychology in a rural community. A recent study (Helbok, Marinelli, & Walls, 2006) found that psychologists in small towns or rural areas are significantly more likely to encounter and engage in multiple relationship behaviors and situations, have a higher degree of visibility (feeling as though they are therapists 24 hours a day, running into clients in the community, participating in events in which clients are also participating), are more likely to prepare their clients for chance encounters, and are more likely to learn information about the client from sources other than the client. A private practitioner choosing to open practice in this setting must be willing to be known as an individual and a member of the community and not just as a therapist. The alternative is to restrict one's social activities by avoiding local establishments, which may not even be a

viable option in very isolated communities. Even if avoidance is a possibility, the personal toll this may take on the psychologist should be considered; therapists should not strive to isolate themselves from the community in which they live. In rural settings, clients are likely to know more about a therapist's private life and daily activities outside of practice, which may be difficult for some practitioners to accept and should be considered before choosing to operate in such settings.

Aside from operating within a small community defined by geographical and population standards, psychologists may also practice within a small community of persons who share something in common, whether that be religion, ethnicity, personal disability, sexual orientation, or a special interest. Maintaining boundaries may be difficult enough in an urban setting, but being a member of a small and visible minority group within a rural community may present even greater difficulties to the practicing psychologist. Here, where personal interests and beliefs may become known to the community, the psychologist may need to be sensitive to integrating these in the therapy setting, where a client's beliefs may be contrary to the therapist's and potentially interfere with the therapeutic relationship. Difficulties exist in striking a balance between being accepted in the community as a person and maintaining professional boundaries.

Boundaries of Competence

As mentioned previously, the financial burdens related to private practice require a certain amount of income to be maintained to keep the practice operating. This pressure, in some cases, may tempt therapists to practice outside of their boundaries of competence. This presents a complicated ethical dilemma, one where more shades of gray may exist than clear-cut answers. The APA Ethics Code (APA, 2002, p. 1063) states, "Psychologists provide services, teach, and conduct research with populations and in areas only within their boundaries of competence, based on their education, training, supervised experience, consultation, study, or professional experience." No clear definition of competence is given in the Code; however, Knapp and VandeCreek (2003, p. 49) assert that the "general rule is that psychologists can ascertain if they have become proficient in a certain area of practice after submitting their work to external feedback," although they acknowledge that this becomes difficult for practitioners to achieve once they have left their doctoral program and are practicing independently. Complicating matters further, the Ethics Code allows for provisions to this rule, whereby psychologists can provide services with different populations, techniques, areas, or technologies as long they "undertake relevant education, training, supervised experience, consultation, or study" (p. 52).

Faced with these dilemmas, it may be especially difficult for independent practitioners to know exactly when they are practicing outside of their area of competence. A balance must be struck between keeping a steady flow of new clients, being open to accepting people of varied diversity and backgrounds with different presenting problems, and ensuring that the skills necessary to treat these individuals are present and do not stretch the therapist's competency boundaries too far so that treatment itself becomes unethical, potentially causing more harm than benefit. With little or no external feedback, the independent practitioner must be able to evaluate his or

her own competencies, abilities, and weaknesses and make firm decisions as to who can be seen and who would be better suited by being referred to another mental health professional.

Cultural Competency

A recent sample of professional psychologists (Hansen et al., 2006) revealed that a large proportion are practicing culturally sensitive therapy with their clients. Respecting a client's worldview and individuality, being aware of personal and societal biases (both positive and negative), establishing rapport in racially/ethnically sensitive ways, and considering the impact of race/ethnicity in diagnosis were all endorsed by at least 80% of those responding. Other areas suggest need for improvement: 50% of the sample rarely or never made efforts to improve their multicultural competencies (by creating a plan and seeking consultation or feedback), negotiating therapist-client language differences, using various resources to augment treatment (literature, translators, and indigenous healers), referring a client to a more qualified provider, and using racially/ethnically sensitive data-gathering techniques. The authors of the survey provide some hypotheses as to why these behaviors are not occurring more frequently, most notably that it may not necessarily be lack of knowledge, but rather that clinicians may be constrained by institutional or managed care mandates. Many of the behaviors listed as not being practiced would require substantial time and effort on the part of the practitioner to remedy, time and effort that may simply not be available given the stressful nature and demands of private practice. Alternatively, the authors suggest that it may be an issue of choice, as 51% of respondents indicated that they believe they were very or extremely multiculturally competent. Considering themselves experts, these practitioners may not believe they need to be any more sensitive or competent than they already are. This study further found a significant difference in what the psychologists practiced and their beliefs, indicating that they did not consistently behave in a manner congruent with what they believed is important for competent practice with a multicultural population (Hansen et al., 2006).

Given this knowledge, it is important for independent practitioners to ensure that they are continually taking steps to improve their cultural sensitivity and awareness rather than simply believing they are competent in this area. As supervision and consultation are often difficult to come by when practicing independently, practitioners should take steps to create a professional development plan that addresses any weaknesses that may detract from their ability to provide psychological services to minority groups in their community. Other behaviors that may help include "increasing the amount of culture-specific case consultations, reading relevant literature, developing a referral network of competent clinicians and translators, and learning about indigenous resources" (Hansen et al., 2006, p. 72).

SPECIAL CONSIDERATIONS WITH CHILD CLIENTS

One area of independent practice that merits further attention is that of child and adolescent populations. The number of children and adolescents with emotional and mental disorders ranges between 14% and 22% (Brandenburg, Friedman, & Silver,

1990), yet the number of private practitioners whose clinical focus is on children and adolescents remains low, and some evidence points to a national shortage of clinical child psychology specialists (Hanley, 1994; Tuma, 1989). Culbertson (1993, p. 119) states, "The United States continues to have serious deficiencies in the delivery of its mental health services for children. . . . An obvious problem is personnel shortage—there simply are not enough mental health professionals training to work with children." Overall, a need exists for greater numbers of practitioners and improved quality of mental health care for children, as the majority of the 8 million children in need of some form of mental health treatment will go unseen (Roberts, 1994).

Given this state of affairs, there is a pressing need to increase the number of graduates from clinical child psychology specialty programs. The APA defines clinical child psychology as

> a specialty of professional psychology which brings together the basic tenets of clinical psychology with a thorough background in child, adolescent and family development and developmental psychopathology. Clinical child and adolescent psychologists conduct scientific research and provide psychological services to infants, toddlers, children, and adolescents. The research and practices of Clinical Child Psychology are focused on understanding, preventing, diagnosing, and treating psychological, cognitive, emotional, developmental, behavioral, and family problems of children. Of particular importance to clinical child and adolescent psychologists is a scientific understanding of the basic psychological needs of children and adolescents and how the family and other social contexts influence socioemotional adjustment, cognitive development, behavioral adaptation, and health status of children and adolescents. . . . A defining feature of Clinical Child Psychology is the emphasis on understanding children from the perspective of both normal development and of psychopathology. Particular attention is given to specialty-specific skills for working with such diverse populations and an appreciation for those cultural, ethnic, and other diversity-related factors that must be considered in clinical practice. (APA Commission for the Recognition of Specialties and Proficiencies in Professional Psychology, 2005)

While general practitioners continue to provide a valuable service, we contend that to best meet the needs of today's youth, professionals working with substantive child or adolescent caseloads should be trained specialists.

As alluded to previously, presenting problems of children and adolescents often warrant contact with many individuals beyond the specified client. Academic performance problems and disruptive behavior in school settings are common presenting complaints. Competent assessment and intervention necessarily warrant involvement of parents as well as school personnel. Further, observations in school or home settings may be particularly advantageous. Unfortunately, independent practitioners may find it difficult to take the time for visits off-site, and equally problematic is the difficulty in obtaining reimbursement for such services. We encourage professionals to continue to lobby for appropriate funding for empirically supported assessment and intervention strategies to properly serve child and adolescent populations. In the meantime, professionals may elect to make consistent use of school-home notes and phone calls to relevant parties (see Kelly & McCain, 1995).

SUMMARY

Professional psychology has undergone significant advances since its inception. Research efforts have allowed us to continually improve our ability to accurately identify and provide treatment for those in need of services. The practice of professional psychology may be as varied as the number of individuals engaged in the pursuit. Although no single theoretical platform has been adopted, the prevailing sentiment is toward increased accountability and primary implementation of empirically supported strategies. Future efforts must be directed toward increasing the number of clinical child specialists, as the demand currently exceeds availability and the shortfall is expected to increase as the number of children and adolescents suffering from mental health problems continues to rise worldwide. In addition, we would be remiss if we did not highlight the pressing need for training in business management practices for those planning to undertake careers in independent practice. The many challenges and hours of hard work invested in a career in private practice can ultimately result in an exciting and rewarding career path, with the long-term benefit of professional independence and flexibility in one's work as a practice grows over time. Though professional psychologists are no doubt confronted with challenges on a regular basis, the vast majority report that the perceived benefits far exceed the disadvantages. For many of us, seeing just one child or adult improve, either through the efforts of our research or through direct service provision, makes all the years of training worthwhile.

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CHAPTER 4

Ethical and Legal Issues

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This chapter is divided into two sections. The first addresses the Ethical Standards and Guidelines of the American Psychological Association (APA) and its committees as these pertain to children and adolescents. It is governed by Table 4.1, which lists the APA documents. The second section focuses on the investigative and practice guidelines published by national organizations such as the American Prosecutors Research Institute (APRI), the American Professional Society on the Abuse of Children (APSAC), and the American Academy of Child and Adolescent Psychiatry (AACAP). Comparative research reports relevant to the interface of clinical and forensic child and adolescent psychology are cited in this section. This discussion may aid the psychologist seeking to determine when a guideline assists or may compromise professional practice.

ETHICAL STANDARDS AND GUIDELINES OF THE AMERICAN PSYCHOLOGICAL ASSOCIATION

Fifteen publications of the APA that contain statements of ethical standards and policy and practice guidelines have been reviewed. Table 4.1 identifies these documents and which of these speak to children, adolescents, and their families. For a psychologist seeking guidance regarding an ethics or practice guideline issue, the document's name provides some orientation. No one guideline is an exclusive source for ethical standards and principles on a particular topic. Hence, familiarity with them all is important. Legal standards vary from state to state and are found in state statutes and case law. Both of these topics are beyond the scope of this chapter. State resources may be particularly helpful in assisting psychologists to find and apply ethical guidelines to particular cases and issues. Some state psychological associations publish a compendium of laws affecting the practice of psychology.* State licensing boards and associations may provide, on request, a listing and copy of laws defining the practice of psychology in that state. The ethics chair of the state psychological association may be of help with the same issues and also in identifying attorneys who specialize in defending psychologists in grievance and malpractice actions. Malpractice insurance carriers have an interest in assisting psychologists to practice

See A Compendium of Laws Affecting the Practice of Psychology (1987) Connecticut Psychological Association.

Table 4.1 Ethical Principles, Standards, and Policies Published by the American Psychological Association

Document Name	Applies
Ethical Principles of Psychologists and Code of Conduct (APA, 2002c)	✓
Determination and Documentation of the Need for Practice Guidelines (APA, 2005a)	
Guidelines for Psychological Practice with Older Adults (APA, 2004b)	
Record Keeping Guidelines (APA, 1993)	
Guidelines for Child Custody Evaluations in Divorce Proceedings (APA, 1994)	✓
Guidelines for Psychological Evaluations in Child Protection Matters (APA, 1999)	✓
Guidelines for Psychotherapy with Lesbian, Gay, and Bisexual Clients (APA, 2000)	✓
Guidelines on Multicultural Education, Training, Research, Practice, and Organization Change for Psychologists (APA, 2002d)	✓
Criteria for Evaluating Treatment Guidelines (APA, 2002a)	
Criteria for Practice Guideline Development and Evaluation (APA, 2002b)	
Guidelines for Providers of Psychological Services to Ethnic, Linguistic, and Culturally Diverse Populations (APA, 1990)	✓
APA Resolution on Outpatient Civil Commitment (APA, 2004a)	✓
Training Grid Outlining Best Practices for Recovery and Improved Outcomes for People with Serious Mental Illness (APA, 2005c)	
Policy Statement on Evidence-Based Practice in Psychology (APA, 2005b)	✓
Specialty Guidelines for Forensic Psychologists (1991)	✓

Note: Applies = Applies to children, adolescents, and their families.

defensively and to avoid litigation. For practicing psychologists who are insured, contacting your carrier for recommendations and for standards and case law that affect your practice can be helpful. Sometimes an attorney will offer a legal opinion.

Ethical standards and guidelines are often revised and updated. There is no one location at the APA to access all of its guidelines and statements regarding ethical conduct and professional guidelines. This also is true of the APA web page. It can be challenging to become informed and to stay abreast of national and state laws and professional standards and guidelines that effect clinical and forensic practice.

1. The Ethical Principles of Psychologists and Code of Conduct

The Ethical Principles of Psychologists and Code of Conduct (APA, 2002c) is the major ethics guideline for psychologists in the United States. It does not contain subsections that specifically refer to adolescents and children. The general principles and ethical standards apply in general to professional activity regardless of client's age. For professionals serving adolescents and children, the principles and standards are clear. It is expected that the professional will have appropriate training and experience; will be knowledgeable about assessment instruments and procedures and their strengths and limitations; will avoid multiple relationships, sexual involvement, and questionable treatments and practices; and will obtain informed consent from authorized guardians and others for professional contact with minors.

The Ethics Code applies to the activities of psychologists in their scientific, educational, and professional roles, and includes the clinical, counseling, and school practice of psychology, research, teaching, trainee supervision, and professional activities of psychologists in the public sector. The Ethics Code applies to activities in a number of contexts, including in person, postal, telephone, and Internet communications. Psychologists are advised to consider the Ethics Code of the APA as well as applicable laws and psychology board regulations and decisions. Psychologists are urged to be alert to potential conflicts between the Ethics Code, their own conscience, and state and federal law, and to seek to resolve any such conflicts in a responsible manner.

The APA Ethics Code lists five general principles that are "aspirational" in nature, that is, which have the intent to inspire psychologists to the highest ethical ideals. These aspirational principles are not meant to be a basis for imposing sanctions. The principles include striving to benefit those with whom one works; establishing relationships of trust; promoting accuracy, honesty, and truthfulness in professional conduct; recognizing that fairness and justice entitle all persons access to the benefits of psychological practice; and respecting the dignity and worth of all people as well as their rights to privacy and confidentiality. Psychologists are urged to be mindful and respectful of individual differences based on age, sex, gender identity, race, ethnicity, culture, national origin, religion, sexual orientation, disability, language, and socioeconomic status and, with these sensitivities and knowledge in mind, to strive to eliminate bias from professional practice.

Ten ethical standards are defined in the Ethics Code. These are:

1. Resolving ethical issues
2. Competence
3. Human relations
4. Privacy and confidentiality
5. Advertising and other public statements
6. Record keeping and fees
7. Education and training
8. Research and publication
9. Assessment
10. Therapy

Each of these ethical standards has a number of subsections.

Ethical Standard 1

Psychologists are required to take reasonable steps to correct or minimize the misuse or misrepresentation of their work. They are expected to abide by the law and take steps to resolve conflicts when conflicts between the law, professional practice, and other ethical standards and regulations are perceived. This includes recognizing conflicts between ethics and organizational demands and resolving them, attempting informal resolution of ethical violations with colleagues, and reporting

ethical violations when professional practice has been found to cause harm or there is concern that it may do so. Psychologists are expected to cooperate with ethical investigations and proceedings, not file or encourage the filing of frivolous ethics complaints or without good cause discriminate against someone who has been the subject of an ethics complaint.

Ethical Standard 2

The ethical standard of competence requires psychologists to guide their professional practice based on the limitations of their education, training, supervised experience, consultation, study, and professional experience. Psychologists are required to obtain the training, experience, consultation, and/or supervision to ensure the competency of their professional practice according to the characteristics of the population they serve. They are expected to take "reasonable steps to ensure the competence of their work" and to protect those who receive their services. When working in a forensic role, psychologists are expected to become "reasonably familiar with the judicial or administrative rules governing their roles."

Psychologists are expected to make ongoing efforts to maintain and increase their competence, ensure that their professional practice is based on scientific principles and professional knowledge, delegate work appropriately and only when there is reason to expect it can be performed competently, and avoid the violation of ethical principles and licensure laws in such delegation.

Ethical Standard 3

Psychologists are expected to avoid unfair discrimination, sexual harassment, other forms of harassment, doing harm through professional practice, and conflicts of interest and multiple relationships. Multiple relationships that would "not be reasonably expected to cause impairment or risk exploitation or harm are not unethical." Distinguishing when a multiple relationship, a conflict of interest, and the potential for harm may or may not exist is not explained. Psychologists may want to seek peer and other consultations to clarify ambiguous situations. Psychologists are expected to avoid exploitive relationships, cooperate with other professionals, seek informed consent for their work, and clarify the implications of third-party requests for services that are court ordered or mandated. Also, the psychologist informs the recipient about the nature of the services and about limits of confidentiality. Psychologists who deliver services in an organizational setting are subject to a number of ethical standards that extend the requirements of informed consent, definition of client, issues of confidentiality, and the definition of organizational versus individual rights and responsibilities. Psychologists are expected to make reasonable efforts to plan for continuation of services when service delivery is interrupted.

Ethical Standard 4

Psychologists have an obligation to take reasonable precautions to protect confidential information developed during the course of professional practice. This includes discussing the limits of confidentiality with clients and patients, obtaining

permission before recording the voice or image of a client, minimizing intrusions on privacy through professional activity, and obtaining appropriate consent for the disclosure of confidential information. Provisions are made for the disclosure of confidential information without consent. This arises in situations that are mandated by law as needed to perform professional services, to obtain appropriate professional consultation, to protect those involved from harm, and to obtain payment. Disclosure is limited "to the minimum that is necessary to achieve the purpose." Psychologists are expected to disguise the person or organization when using confidential information for public purposes (publications, lectures, etc.) unless there is written consent or legal authorization.

Ethical Standard 5

In advertising and other public statements, psychologists are required to avoid false and deceptive statements both directly and by others whom they have engaged to represent them. Psychologists adhere to the Ethics Code in media presentations and should not solicit testimonials from current therapy clients or exploit the vulnerability of persons who may be actual or potential clients for other business ventures.

Ethical Standard 6

Psychologists are expected to maintain satisfactory documentation of their professional activities and the confidentiality of records, facilitate the appropriate transfer of records, and may not withhold records that are required for emergency treatment because payment has not been received. Psychologists are expected to reach payment and billing arrangements as soon as it is feasible in a professional relationship, state fees clearly, discuss service limitation that may result from limitations in financing as soon as is feasible, inform clients of the intended use of collection agencies or other measures to collect fees, and offer an opportunity for prompt payment. Barter in return for professional services is allowed if the barter is not clinically contraindicated and the arrangement is not exploitive.

Ethical Standard 7

In education and training activities, psychologists are required to uphold APA ethical standards. In the design and delivery of education and training programs, APA standards for licensure, certification, and other goals must be met; program content must be accurately described, including specification of training goals and program completion requirements. Course content must meet ethical standards for professionalism and scientific merit. Unethical demands are not to be made on students. When therapy participation is a program or course requirement, students are allowed to select therapy from practitioners unaffiliated with the program. Here, too, dual relationships are avoided.

Ethical Standard 8

Ethical standards should be maintained in research and publication. This includes obtaining informed consent when this is required and protecting research participants from participation and nonparticipation. Conditions are specified for dispensing with informed consent requirements. These include research that would not be expected to create distress or harm, that involves the study of normal phenomena, and that offers anonymous report forms that do not place participant information or a person at risk. Excessive and inappropriate financial or other rewards for research participation should be avoided. Deception in research should also be avoided, though it can be justified under some circumstances. Psychologists must provide research participants with a debriefing opportunity and information. When they become aware that the research has harmed a participant, reasonable steps must be taken to minimize the harm. Animal research should be conducted in compliance with federal, state, and local laws and regulations and also with professional standards. Psychologists do not plagiarize or fabricate research results. Psychologists must accept publication credit only when it is deserved for work done and contributions made. The position of principal authorship is based on contribution and not status.

Ethical Standard 9

Psychologists provide opinions on the psychological characteristics of people only after proper evaluation and define factors that limit the nature and extent of their conclusions and recommendations. Sources of information are defined. Psychologists use assessment instruments with known validity and reliability for the population evaluated. Language preference and other factors are considered in assessment instrument selection. Assessment instruments are administered, scored, and adapted according to instrument and research guidelines. Informed consent is obtained for assessment and also for the use of interpreter services. This information is conveyed in the report. Issues of confidentiality are maintained. Impact assessment of interpreter involvement is included in a discussion of data limitations. Situational, personal, linguistic, and cultural characteristics of clients as well as test-taking ability are considered along with data characteristics when considering data interpretation and validity. The norms, validity, reliability, and any other special qualifications for the use of assessment and scoring services are acknowledged. Psychologists retain responsibility for test selection, scoring, and interpretation when these aspects are delegated to other persons or to an automated service. Psychologists take reasonable steps to ensure that explanation of assessment results with appropriate qualifications occurs. Test security is maintained.

Ethical Standard 10

Psychologists obtain informed consent for therapy; provide information about the course of therapy, fees, third-party involvement, and limits of confidentiality; and provide the opportunity for questioning and receiving answers. They explain the potential risks of therapy and the availability of alternative treatments and disclose

therapy techniques that do not possess established results. When the therapist is a psychologist in training and is supervised, this is disclosed and the name of the supervisor is provided. When psychologists become involved in treating couples and families, the psychologist defines the professional relationship of each person, identifies which person is the client, and avoids role conflict, such as being a couples or family therapist and then a witness for one party in a divorce proceeding. Issues of confusion and conflict are also resolved in a timely fashion when deciding whether to offer services to persons receiving mental health services elsewhere. Psychologists do not provide treatment to persons with whom they have had sexual intimacy and do not have sexual intimacy with current therapy clients. Psychologists do not engage in sexual intimacy with former clients for 2 years after the termination of therapy. Even after the passage of 2 years, a psychologist may be challenged to demonstrate that a sexual involvement with a former client is consistent with the APA Ethics Code.

Therapy is terminated appropriately, such as when the service is no longer needed or is producing no benefit or when the client might be harmed by its continuation. A psychologist may terminate therapy when threatened. Psychologists provide pretermination counseling when feasible and also suggest alternative service providers as appropriate.

2. Determination and Documentation of the Need for Practice Guidelines

Determination and Documentation of the Need for Practice Guidelines (APA, 2005a) is an aspirational document in intent and references statements that are recommended for professional behavior and conduct but that are not mandatory and do not require an enforcement mechanism. Their purpose is to assist the psychologist to provide high-quality services. The Committee on Professional Practice and Standards identified a need for legal and regulatory, public benefit, and professional guidance. This document reflects the recognition that laws, court decisions, professional interaction with the legal system, and changes in regulatory and administrative systems may affect the practice of psychology. For example, *Daubert v. Merrill Dow Pharmaceuticals*, 509 U.S. 579 (1993), defines how expert opinions are to be delivered in court and evaluated there. Guidelines for Child Custody Evaluations in Divorce Proceedings (APA, 1994), Guidelines for Psychological Evaluations in Child Protection Matters (APA, 1999), and Specialty Guidelines for Forensic Psychologists (APA, 1991) are intended to assist the psychological practitioner when interacting with the legal system in the defined legal contexts. This document found that practice guidelines were needed to improve service delivery; avoid harm; meet the needs of emerging, underserved, and vulnerable client populations; and assist psychologists in responding to public policy initiatives. Psychologists may also seek guidelines for new areas of technology, for work in new contexts and emerging areas of practice with new approaches based on scientific advances, and with risk management issues.

3. The Guidelines for Psychological Practice with Older Adults

The Guidelines for Psychological Practice with Older Adults (APA, 2004b) does not apply.

4. Record Keeping Guidelines

Record Keeping Guidelines (APA, 1993) speak to an important area of psychological practice with children and adolescents. Formulated in 1993, these guidelines provide an introduction to documentation issues that receive specialized attention in later sections of this chapter, particularly in reference to forensic practice and documentation of investigatory procedures. Psychologists have many reasons for maintaining records, and the nature and extent of the documentation may vary according to the service performed. Records provide a history and offer a basis for case review, for planning, and for implementation of services. The General Guidelines for Providers of Psychological Services determined that psychologists hold widely varying rules on the wisdom of recording the content of the psychotherapeutic relationship. With respect to record keeping, a 50-state review of laws governing psychologists found substantial differences among the states on this issue. There are differing requirements in areas of professional practice between situations calling for assessment and those calling for psychotherapy.

Well-documented records may help psychologists in professional liability cases where the professional action of a psychologist is at issue.

Records include information about the nature, delivery, progress, and results of psychological services. Because of their nature, they can be reviewed and duplicated. Records of psychological services should minimally include identifying data, dates of services, types of services, fees, assessments, intervention plans, consultation, summary reports, testing reports, release of information, consent forms, and other supporting data. Psychologists are expected to maintain a reasonable degree of accurate, current, and pertinent records with sufficient detail to permit an understanding of the service by another psychologist.

Psychologists maintain the confidentiality of records, make reasonable efforts to protect against their misuse, develop reliable record keeping procedures, and strive to assure that record entries are readable and completed in a time that is proximate to the service delivery, if not simultaneous with it. Records may be maintained in a variety of formats and media. In the absence of laws and regulations to the contrary, these guidelines offer the following time periods for record maintenance:

- Complete records are maintained for a minimum of 3 years.
- Thereafter, a summary is maintained for an additional 12 years.
- If the client is a minor, the record period is extended until 3 years after the age of majority.
- A simple 7-year requirement for the retention of the complete record is preferred. The practitioner is urged to consult specialty guidelines in this regard.

5. Guidelines for Child Custody Evaluations in Divorce Proceedings

Guidelines for Child Custody Evaluations in Divorce Proceedings (APA, 1994) is one of two APA guidelines that address professional practice issues with children and their guardians and parents in legal contexts. Child custody and parenting issues arise in several legal contexts, including parental divorce, guardianship, abuse and neglect proceedings, and termination of parental rights. These guidelines were written specifically for the context of parental divorce, are aspirational in intent, are not mandatory or exhaustive, and have the stated purpose of promoting the efficiency of psychological expertise when conducting child custody evaluations. Psychologists can provide an important service to the courts and to children and families by providing "competent, objective, impartial information in assessing the best interests of the child." Psychologists performing child custody evaluations are expected to possess specialized competencies in the evaluation of adults, children, relationships, and parenting ability. Specific guidelines are discussed under three headings:

- Purpose of a Child Custody Evaluation
- Preparing for a Child Custody Evaluation
- Conducting a Child Custody Evaluation

There are 16 guidelines.

Guideline 1

The primary purpose of the evaluation is to assess the best psychological interests of the child.

Guideline 2

The child's interests and well-being are paramount.

Guideline 3

The focus of the evaluation is on parenting capacity, the psychological and developmental needs of the child, and the resulting fit. This involves focused assessment of parenting capacity in connection with the psychological characteristics and developmental needs of each involved child. Consequently, both adult and child assessments occur, and these include the assessment of the functional ability of each parent to meet the child's needs. An interactional evaluation between each adult and child also occurs.

Guideline 4

The role of the psychologist is that of a professional expert who strives to maintain an objective, impartial stance. The psychologist does not act as a judge or as an advocating attorney and remains impartial, balanced, and objective.

Guideline 5

The psychologist gains specialized competence. This includes competency in performing psychological assessments of children, adults, and families, knowledge of child and family development and child and family psychopathology, and knowledge of laws that govern divorce and custody decisions. This includes adherence to the Standards for Educational and Psychological Testing (APA, 1985). When specialized issues arise that are outside the evaluator's expertise, additional consultation and supervision should be sought.

Guideline 6

The psychologist is aware of personal and societal biases and engages in nondiscriminatory practices.

Guideline 7

The psychologist avoids multiple relationships.

Guideline 8

The scope of the evaluation is determined by the evaluator based on the nature of the referral questions. These may broaden or limit the scope of the evaluation from an entire family to a single family member. Sometimes psychologists are asked to review and critique the report of another mental health professional or serve as an expert in a specialized area of psychological knowledge.

Guideline 9

The psychologist obtains informed consent from all adult participants and, as appropriate, informs child participants. This entails informing each adult participant about the purpose, nature, and method of the evaluation, the source of the evaluation request, to whom the report will be sent, and who pays the fees. Information is given to children as is developmentally and otherwise appropriate.

Guideline 10

The psychologist informs participants about the limits of confidentiality and the disclosure of information. Participants are to be informed that they are consenting to disclosure of the evaluations' findings. It is recommended that the psychologist obtain a waiver of confidentiality from all adult participants.

Guideline 11

The psychologist uses multiple methods of data gathering: clinical interviews, observation, psychological assessments, and documentation from at least two sources about important facts. This can include the review of school, health care, and other kinds of records, and interviewing persons who know the family and who may contribute useful information. When external information is deemed significant,

psychologists are urged to corroborate it by an additional source whenever possible.

Guideline 12

The psychologist neither overinterprets nor inappropriately interprets clinical or assessment data. The psychologist is urged to seek convergent validity from multiple sources of information, including interviews and test data, and to cautiously and conservatively interpret evaluation results with awareness of reliability and validity restrictions.

Guideline 13

The psychologist does not give any opinion regarding the psychological functioning of any individual he or she has not personally evaluated.

Guideline 14

Recommendations, if any, are based on what is in the psychological interests of the child. It is noted that the profession itself has not reached consensus about whether psychologists ought to make recommendations about final custody determination to the courts. When the psychologist does make a custody recommendation, this should be based on sound psychological data and reflect the child's best interests in the case at hand.

Guideline 15

The psychologist clarifies financial arrangements.

Guideline 16

The psychologist maintains written records. Reference here is made to the APA Record Keeping Guidelines (APA, 1993) and relevant statutory guidelines. Further discussion of record keeping in the particular area of investigatory interviews is found in a later section of this chapter.

6. Guidelines for Psychological Evaluations in Child Protection Matters

Guidelines for Psychological Evaluations in Child Protection Matters (APA, 1999) is the second APA guideline to address professional practice issues with children. The legal context is litigation in child abuse and neglect cases. This guideline contains useful subsections of 17 guidelines, a glossary of terms, and a bibliography of major professional publications in the field. There is also a useful listing of national information resources, including the American Academy of Pediatrics, the American Bar Association Center on Children and the Law, and the Clearinghouse on Child Abuse and Neglect Information. These guidelines were developed for psychologists conducting psychological evaluations in child protection proceedings and have the goal of promoting "proficiency in using psychological expertise."

They do not address the issues of psychologists acting in other roles, such as when serving as a therapist for an individual or family. These guidelines build on those already discussed, including Ethical Principles of Psychologists in the Code of Conduct (APA, 1992), Record Keeping Guidelines (APA, 1993), and Standards for Educational and Psychological Testing (APA, 1985).

Three stages are described for state intervention in the affairs of the parents and children in child abuse and neglect matters. In the first stage, an investigation occurs following a report of suspected child abuse and neglect. In the second stage, the state may obtain custody of a child and a rehabilitation plan may be developed. When the child cannot be returned to the family, there may be a hearing for the final disposition of the case that can include termination of parental rights. States vary in statutory and case law that applies to these matters. Psychologists may be called on to evaluate the parties and "may function as agents of the court, of a child protection agency, or of the parents." In one or in all of these roles, the psychologist may be called on to evaluate the child, the parents, the parent and child relationship, or parenting capacity; to assess specialized issues such as the impact on the child of abuse and neglect; to recommend therapeutic interventions for the parents and children; and to make predictions about future harm if the child is reconciled with the parents and if parental rights are terminated. Psychologists are urged to develop the specialized knowledge and competencies required to do this work and seek consultation as needed. There are 17 guidelines.

Guideline 1

The primary purpose of the evaluation is to provide relevant, professionally sound results or opinions in matters where a child's health and welfare may have been and/or may in the future be harmed.

Guideline 2

In child protection cases, the child's interest and well-being are paramount.

Guideline 3

The evaluation addresses the particular psychological and developmental needs of the child and/or parents that are relevant to child protection issues such as physical issues, sexual abuse, neglect, and/or serious emotional harm.

Guideline 4

The role of the psychologist conducting the evaluation is that of a professional expert who strives to maintain an unbiased, objective stance.

Guideline 5

The serious consequences of psychological assessment in child protection matters place a heavy burden on psychologists.

Guideline 6

Psychologists gain specialized competence.

Guideline 7

Psychologists are aware of personal and societal biases and engage in nondiscriminatory practice.

Guideline 8

Psychologists avoid multiple relationships.

Guideline 9

On the basis of the nature of the referral questions, the scope of the evaluation is determined by the evaluator.

Guideline 10

Psychologists performing psychological evaluations in child protection matters obtain appropriate informed consent from all adult participants and, as appropriate, inform the child participant. Psychologists need to be particularly sensitive to informed consent issues.

Guideline 11

Psychologists inform participants about the disclosure of information and limits of confidentiality.

Guideline 12

Psychologists use multiple methods of data gathering.

Guideline 13

Psychologists neither overinterpret nor inappropriately interpret clinical or assessment data.

Guideline 14

Psychologists conducting a psychological evaluation in child protection matters provide an opinion regarding the psychological functioning of an individual only after conducting an evaluation of the individual adequate to support their statements or conclusions.

Guideline 15

Recommendations, if offered, are based on whether the child's health and welfare have been and/or may be seriously harmed.

Guideline 16

Psychologists clarify financial arrangements.

Guideline 17

Psychologists maintain appropriate records.

The glossary of terms is useful for psychologists when legal issues define the professional practice context and the evidentiary standards that will be applied to psychological data and opinion. Many of the definitions contained in the glossary were taken from Working with Courts in Child Protection (National Center on Child Abuse and Neglect, 1995). Selected terms are defined (author's selection and sequence):

- *Abuse, emotional:* Also referred to as psychological maltreatment; generally defined as a repeated pattern of behavior that conveys to children that they are worthless, unwanted, or only of value in meeting another's needs; may include serious threats of physical or psychological violence.
- *Abuse, physical:* Generally defined as the suffering of a child or substantial risk that a child will imminently suffer physical harm, inflicted nonaccidentally on him or her by his or her parents or caretakers.
- *Abuse, sexual (child):* Generally defined as contacts between a child and an adult or other persons significantly older or in a position of power or control over the child, where the child is being used for sexual stimulation of the adult or other person.
- *Beyond a reasonable doubt:* Highest standard of proof used in cases where the loss of liberty interests are at stake (e.g., incarceration or loss of life). Generally defined as the highest degree of support or level of certainty (90% to 95% chance).
- *Clear and convincing:* Intermediate standard of proof used in cases where significant liberty interests are at stake (e.g., loss of parental rights, civil commitment). Generally defined as a degree of support or level of certainty (75% chance).
- *Expert witness:* An individual who by reason of education or specialized experience possesses superior knowledge of a subject about which persons having no particular training are incapable of forming an accurate opinion or deducing correct conclusions. A witness who has been qualified as an expert will be allowed (through his or her answers to questions posed) to assist the jury in understanding complicated and technical subjects not within the understanding of the average layperson. Experts are also allowed to provide testimony based on hypothetical scenarios or information and opinions that are not specifically related to the parties in a particular legal action.
- *Fact witness:* Generally defined as an individual who, by being present, personally sees or perceives a thing; a beholder, spectator, or eyewitness. One who testifies to what he or she has seen, heard, or otherwise observed regarding a circumstance, event, or occurrence as it actually took place or a physical object or appearance as it usually exists or existed. Fact witnesses are generally

not allowed to offer opinion, address issues that they do not have personal knowledge of, or respond to hypothetical situations.

- *Family preservation or reunification:* The philosophical belief of social service agencies, established in law and policy, that children and families should be maintained together if the safety of the children can be ensured.
- *Guardian ad litem:* Generally defined as an adult appointed by the court to represent and make decisions for someone (such as a minor) legally incapable of doing so on his or her own in a civil legal proceeding.
- *Maltreatment:* Generally defined as actions that are abusive, neglectful, or otherwise threatening to a child's welfare. Commonly used as a general term for child abuse and neglect.
- *Neglect:* Generally defined as an act of omission, specifically the failure of a parent or other person legally responsible for a child's welfare to provide for the child's basic needs and proper level of care with respect to food, shelter, hygiene, medical attention, or supervision. There are two types of neglect:

    1. Emotional: Generally defined as the passive or passive-aggressive inattention to a child's emotional needs, nurturing, or emotional well-being. Also referred to as psychological unavailability to a child.
    2. Physical: Generally defined as a child suffering, or at substantial risk of imminently suffering, physical harm causing disfigurement, impairment of bodily functioning, or other serious physical injury as a result of conditions created by a parent or other person legally responsible for the child's welfare or by the failure of a parent or person legally responsible for a child's welfare to adequately supervise or protect him or her.

- *Preponderance of evidence:* Lowest of the three standards of proof, and the standard applied in most civil actions; generally defined as a "probable" degree of certainty (e.g., "more likely than not," or a 51% chance).

7. Guidelines for Psychotherapy with Lesbian, Gay, and Bisexual Clients

In general, the Guidelines for Psychotherapy with Lesbian, Gay, and Bisexual Clients (APA, 2000) may be thought to apply to older adolescent and adult populations and therefore not directly applicable to the evaluation and treatment of children and adolescents in clinical and forensic contexts. Professional practice, however, indicates otherwise. Lesbian, gay, and bisexual persons are involved with children as parents, guardians, and foster parents and may individually or as couples be subject to child protection action. Children with gender identity, sexual orientation, or child abuse and neglect issues, or some combination of these, are seen in clinical and forensic evaluation cases and are often identified for specialized care. These guidelines, therefore, though not specifically addressing children and adolescents, involve populations with whom practicing psychologists are likely to have professional contact. The stated goals of these guidelines are to provide

practitioners with a frame of reference for the treatment of lesbian, gay, and bisexual clients and to provide basic information and further references in the areas of assessment, intervention, identity, relationships, and the education and training of psychologists. The guidelines are organized into four sections: (1) attitudes toward homosexuality and bisexuality, (2) relationships and families, (3) issues of diversity, and (4) education. Some of the 16 guidelines are articulated.

Guideline 1

Psychologists understand that homosexuality and bisexuality are not indicative of mental illness.

Guideline 2

Psychologists are encouraged to recognize how their attitudes and knowledge about lesbian, gay, and bisexual issues may be relevant to assessment and treatment and seek consultation or make appropriate referrals when indicated.

Guideline 3

Psychologists recognize that families of lesbian, gay, and bisexual people may include people who are not legally or biologically related.

Guideline 4

Psychologists are encouraged to recognize the particular life issues or challenges that are related to multiple and often conflicting cultural norms, values, and beliefs that lesbian, gay, and bisexual members of racial and ethnic minorities face.

Guideline 5

Psychologists are encouraged to recognize the particular challenges that bisexual individuals experience.

Guideline 6

Psychologists strive to understand the special problems and risks that exist for lesbian, gay, and bisexual youth. Here it is noted that lesbian, gay, and bisexual adolescents face specialized risks, including higher rates of estrangement from their parents, rejection by their parents, and increased risks of homelessness, HIV infection, and becoming victims of violence and of attempting suicide. Gay, lesbian, and bisexual youth may have complications of adolescent development and may require specialized therapeutic strategies and a secure therapy context to address sexual orientation issues.

Guideline 7

Psychologists make reasonable efforts to familiarize themselves with relevant mental health, educational, and community resources for lesbian, gay, and bisexual people.

8. Guidelines on Multicultural Education, Training, Research, Practice, and Organizational Change for Psychologists

The increasing diversity of the American population underscores the importance of these guidelines. The four stated goals in the Guidelines on Multicultural Education, Training, Research, Practice, and Organizational Change for Psychologists (2002d) are to provide psychologists with a means of addressing multiculturalism with basic information to enhance ongoing education and broaden the purview of psychology as a profession. The guidelines address U.S. ethnic and racial minority groups as well as persons from biracial, multiethnic, and multiracial groups. The guidelines indicate that approximately one third of the American population, according to the U.S. Census Bureau 2001 data, belong to a minority group. While racial/ethnic diversity varies greatly by state, region, and urban versus rural environment, practicing psychologists are likely to encounter minority clients in their practice. This increases the "urgency for culturally responsive practices and services." Psychologists are urged to become conversant with terms of culture, race, and ethnicity. Six guidelines are defined. These encourage psychologists to develop a commitment to cultural awareness, to recognize the importance of multicultural sensitivity and responsiveness, to employ constructs of diversity in psychological education, to conduct culturally centered and ethical psychological research, to apply culturally appropriate skills in psychological practice, and to support culturally informed organizational development. In psychological assessment, this may mean determining the applicability of particular tests and measures to minority populations. For example, Grisso (1998) in "Forensic Evaluation of Juveniles" reports that no relation was found between competent versus incompetent status and gender and race. Grisso found that three causal factors were related to a clinician's conclusion about incompetence: severe mental disorder, cognitive and intellectual disabilities, and age-related characteristics. By contrast, the Wechsler Intelligence Scale for Children—Fourth Edition (Wechsler, 2004) discusses the importance of age, gender, regional, ethnic, and racial characteristics of standardization populations. The guidelines explain that cultural sensitivity means taking account of the client's cultural context, using culturally appropriate assessment tools, and using a broad array of interventions. Cases may involve unusual combinations of client characteristics with variable nationality, racial, ethnic, religious, regional, and sexual characteristics and orientations. Some persons may be in transition from one group to another. Others may be confused about their own identity and group memberships. Professional practice can lead to work with illegal aliens and requires special assistance with language issues. Critical judgment is recommended when using standardized assessment tools and methods. Knowledge of helping practices used in non-Western cultures and culture-specific therapies may be indicated for particular ethnic communities.

9. Criteria for Evaluating Treatment Guidelines

Criteria for Evaluating Treatment Guidelines (APA, 2002a) is aspirational in intent and was developed for the purpose of evaluating the guidelines of health care

organizations, government agencies, and professional associations. These pertain to specific recommendations about treatments to be offered to patients. They are, therefore, patient-focused rather than practitioner-focused as are practice guidelines. The purpose of treatment guidelines is to establish the most effective treatments available, translating knowledge into clinical practice. Treatment efficacy is defined as ascertaining the effects of a given intervention as compared to an alternative intervention or with no treatment, thereby establishing the scientific basis for determining treatment benefit.

Twenty criteria are listed for the evaluation of treatment guidelines. These describe the scientific method to be used in evaluating treatment efficacy. This requires establishing the clinical utility and cost effectiveness of treatment and that guideline panels be composed by persons with a broad range of expertise.

10. Criteria for Practice Guideline Development and Evaluation

Criteria for Practice Guideline Development and Evaluation (APA, 2002b) is intended to promote quality and consistency in practice guidelines and to produce a high quality of psychological services. These guidelines are aspirational and not intended to take precedence over professional judgment. Practice guidelines consist of recommendations to professionals concerning their conduct and issues in particular areas of psychological practice.

11. Guidelines for Providers of Psychological Services to Ethnic, Linguistic, and Culturally Diverse Populations

Guidelines for Providers of Psychological Services to Ethnic, Linguistic, and Culturally Diverse Populations (APA, 1990) represents general and aspirational principles for psychologists who work with ethnic, linguistic, and culturally diverse populations. Nine guidelines are articulated. These include the recognition of research and practice issues related to the population serviced, respect for the cultural structure of the community, use of the client's language, consideration of culture in designing intervention, elimination of discriminatory practice in service delivery, and documentation of relevant cultural factors affecting the family.

12. The APA Resolution on Outpatient Civil Commitment

The APA Resolution on Outpatient Civil Commitment (APA, 2004a) addresses the increasing use of this power in courts and civil commitment venues across the country. Although this resolution does not address children and adolescents specifically, it is well known that children and juveniles are often subject to this procedure. It recognizes that persons with serious mental illness and psychological disabilities can be subject to the imposition of involuntary mental health services, including hospitalization and incarceration. There is recognition of the controversy about the legitimacy of outpatient civil commitment (OCC), how and when it should be applied, and the professional practice standards, clinical methods, and level of proof involved.

The resolution recognizes that people possess behavioral disorders severe enough to reach the point of dangerousness, that these persons and persons with serious mental illness have a right to treatment, and that psychologists often play a role in providing the functional assessments and opinions that lead to OCC. There is recognition that some populations may have a greater vulnerability to the imposition of OCC. This APA resolution states, "It is never the role of any mental health practitioner, of any discipline, acting in the role of a caregiver, to make decisions that infringe upon a person's right to consent to services." Psychologists are encouraged to promote the development of evidence-based clinical methods for determining risk and dangerousness and determining competence to make specific judgments and decisions, promote rehabilitation from serious mental illness, research alternatives to involuntary commitment, and provide professional expertise and consultation to legal and judicial authorities to ensure that legal processes and decisions are appropriately informed by scientific and clinical considerations.

13. Training Grid Outlining Best Practices for Recovery and Improved Outcomes for People with Serious Mental Illness

The Training Grid Outlining Best Practices for Recovery and Improved Outcomes for People with Serious Mental Illness (APA, 2005c) is intended to assist providers in identifying the most effective interventions and resources for advanced clinical training. This document applies to adults with serious mental illness.

14. American Psychological Association Policy Statement on Evidence-Based Practice in Psychology

The APA Policy Statement on Evidence-Based Practice in Psychology (2005b) addresses the integration of the best available research with clinical expertise and patient characteristics. The purpose is to promote effective psychological practice and enhance public health. Best research evidence relies on the use of scientific method, beginning with clinical observation and leading to randomized clinical trials. Patient groups include child, adolescent, adult, older adult, couple, family, group, organization, community, and other populations that receive psychological services.

Researchers and practitioners are encouraged to combine their efforts to improve psychological practice. Psychologists are encouraged to use their clinical expertise to conduct assessments and develop diagnostic judgments, case formulations, and treatment plans; to evaluate treatment progress; to seek additional resources when indicated; and to have a cogent rationale for clinical strategies. Psychological services are considered to be most effective when they are responsive to patient characteristics and other factors such as cultural, family, environmental, values, and treatment expectations. Some effective treatments are directed to the patient's environment, such as parents, teachers, and caregivers.

15. Specialty Guidelines for Forensic Psychologists

Specialty Guidelines for Forensic Psychologists (APA, 1991) is designed to guide forensic psychologists in monitoring their professional conduct. The goal is to

improve the quality of forensic psychological services. The guidelines are aspirational. They apply to such diverse positions in forensic practice as clinical forensic examiners, psychologists employed by correctional and forensic mental health systems, researchers who offer direct testimony about the relevance of scientific data to a psycholegal issue, trial behavior consultants, and psychologists who provide professional opinions to the courts through amicus briefs or who appear as forensic experts before judicial, legislative, and administrative agencies. The guidelines amplify the Ethical Principles of Psychologists in the context of the practice of forensic psychology. The guidelines are designed to conform to state and federal laws. They specify the nature of desirable professional practice. They are not meant to apply to a psychologist who is not informed at the time of service delivery that the service would be used for forensic purposes.

Forensic psychologists have an obligation to provide services that are consistent with the highest standards of their profession. Forensic psychologists provide services only in areas of psychology in which they have specialized knowledge, skill, experience, and education. It is obligatory for forensic psychologists to inform the court about the boundaries of their competence, have a reasonable understanding of the legal and professional standards that govern psychological expertise in legal proceedings, and recognize that personal and professional relationships may be inconsistent with forensic involvement. Forensic psychologists explain their fees and their personal and professional activities that may produce a conflict of interest. They acknowledge areas of competence and limitations of competence. They do not provide services on the basis of contingent fees. Forensic psychologists are encouraged to offer a portion of their professional services on a pro bono or reduced fee basis.

Forensic psychologists inform prospective clients of their legal rights with respect to the forensic service and obtain informed consent before proceeding with the evaluation. An exception may be when the evaluation is court ordered. When the client is unwilling to proceed, the evaluation should be postponed and the psychologist should take steps to assist the client to receive legal advice regarding participation. When the client does not have the capacity to provide informed consent, the client's legal representative is informed, and if the legal representative objects to the evaluation, the forensic psychologist notifies the court that ordered the evaluation. The forensic psychologist has an obligation to inform legal authorities about any source of conflict and take reasonable steps to resolve it.

Forensic psychologists inform their clients about the limits of confidentiality. Information is released pursuant to statutory requirements, court order, or the consent of the client. Forensic psychologists document and are prepared to make available to the court all evidence that formed the basis for forensic services. They anticipate the detail and quality of documentation required by judicial scrutiny and recognize that this standard is higher than the standard for general clinical practice. Forensic psychologists also recognize that they have a special responsibility to provide the best documentation possible under the circumstances. It is understood that psychological evidence is subject to rules of discovery, disclosure, and confidentiality. The duties and obligations of forensic psychologists regarding documentation apply "from the moment they know or have a reasonable basis for knowing that their data and evidence derived from it are likely to enter into legally relevant decisions."

As an expert, the forensic psychologist actively seeks information that will test plausible rival hypotheses. Unless an individual is handling his or her representation pro se, a forensic psychologist does not provide forensic services to any party contemplating a legal proceeding prior to that person's representation by counsel. An exception to this rule is the delivery of emergency mental health services, when a failure to provide such services would pose a substantial risk of harm to the defendant or to others.

Forensic psychologists seek third-party data only with prior approval of the legal party or pursuant to a court order.

Forensic psychologists seek to obtain independent and personal verification of data relied on. An effort is made to minimize sole reliance on hearsay evidence when offering expert testimony. It is recognized that many forms of data used by forensic psychologists are hearsay; therefore, an attempt to corroborate critical data is required. When the status of the data is uncorroborated, this is acknowledged by the forensic psychologist. The forensic psychologist acknowledges information gathered by others and avoids offering information that does not directly bear on the legal purpose of the professional activity. Forensic psychologists avoid giving evidence about the psychological characteristics of a person they have not examined. When it is not possible to conduct a direct examination, the forensic psychologist explains limitations on the reliability and validity of professional products, evidence, or testimony. The forensic psychologist is mindful of the Federal Rules of Procedure, which restrict testimony by the expert based on statements made by the defendant in the course of the forensic examination.

Forensic psychologists maintain a high level of public and professional communication. This includes taking reasonable steps to correct misuse or misrepresentation of their professional services. Access to information is restricted to individuals with a legitimate professional interest in the data and steps are taken to ensure that test results are released to a qualified professional and in compliance with Principle 16 of the Standards for Educational and Psychological Testing.

A special responsibility for fairness and accuracy is a feature of forensic psychological practice. This includes a fair and accurate evaluation of the work and opinions of other experts and parties. Forensic psychologists testify fairly and avoid partisan distortion and misrepresentation and errors of commission and omission. Forensic psychologists disclose all sources of their information and which information was relied on in formulating a particular opinion. Forensic psychologists "are aware that their own professional observations, inferences, and conclusions must be distinguished from legal facts, opinions, and conclusions."

PRACTICE GUIDELINES FOR INVESTIGATIVE INTERVIEWING

The first section of this chapter reviewed the ethical standards and practice guidelines of the APA with special reference to clinical and forensic practice with children, adolescents, and their families. Now we turn to three of the more comprehensive guidelines of professional organizations which address investigative interviewing with alleged child and adolescent victims of sexual abuse. The APRI, the APSAC,

and the AACAP have formally addressed the topic of investigatory interviewing guidelines and standards.

1. The American Prosecutor's Research Institute

Clearly, the most comprehensive contribution on this topic is from the APRI. Its publication, *Investigation and Prosecution of Child Abuse*, third edition (APRI, 2004), is a multidisciplinary guide for attorneys, medical professionals, and behavioral scientists involved in the investigation and prosecution of child abuse. Chapter 2, "Investigation," is of particular relevance for psychologists practicing in both clinical and forensic settings. It presents a thorough and concentrated discussion of interviewing guidelines and validation principles. Because the sponsoring organization in this case is federal in scope and extends its training and practice guidelines to all of the states, the principles contained therein can be seen as offering national guidelines in a legal context.

Interviewing the Child

The stated purpose of this section is to assist multidisciplinary interviewers and investigators, including psychologists and prosecutors, to interview children properly and to analyze the evidentiary record for completeness, accuracy, and error. Techniques for the interviewing of preschool, grade school, and adolescent children are distinguished from those used for adults. The interviewer is expected to understand the field of child abuse and the developmental capacities of children of different ages. This includes the ability to judge the quality of prior interviews conducted with all witnesses. Interviewers are expected to be thoroughly trained and to develop a protocol for conducting interviews that reflect these guidelines and that can be defended in court. It is emphasized that comfort in talking with children is required. Investigators are urged to start the interview with no assumption about whether abuse occurred or about the identity of a perpetrator or how the child might feel about the perpetrator. A stance of neutrality about the allegation is required as it has the best chance at generating the greatest amount of reliable information. Interviewers are admonished not to coerce or manipulate children or to educate them about what may have happened or how they should feel. The author refers the reader to Ceci and Bruck (1995) and Poole and Lamb (1998), who provide detailed discussions of research on investigatory error and its impact on child narration, including autobiographical event narration.

Preparing for the Interview

Preparation for the interview requires thorough review of all known information on the case. This includes police and child protection reports, medical findings, and other materials. When possible, the person who made the report, the physician who examined the child, and the parent, guardian, and therapist who may have been treating the child should all be contacted. Interested parties are cautioned not to rehearse the child prior to the forensic or clinical interview. Gathering this information in advance is seen as strengthening the interviewer's understanding

of the nature of the case, or the child, and of the special issues that could affect the interview process. This discussion acknowledges that there is expert opinion that recommends "allegation-blind" interviews in which the interviewer knows nothing about the background of the child or of the allegations. This approach is recommended to neutralize interviewer expectation and to minimize leading and confirmatory errors. The work of Cantlin, Payne, and Erbaugh (1996) is referenced in this connection; they compared allegation-blind and allegation-informed structured interviews. Planning is recommended for situations in which a child may be handicapped or speaks a different language. This will necessarily involve training of the interpreter and/or a support person for the handicapped child to prepare him or her to participate in an investigative interview. Such persons should be instructed not to speak for the child and not to show reactions to what the child says. The author suggests that interpreters also be trained to give verbatim accounts and not to independently question the interviewee.

Interview Location and Circumstances

It is recommended that a neutral location be found for the investigative interview and the number of interviews be limited. Police department facilities are to be avoided as well as the presence of uniforms and weapons. The interview room should be free of distractions and furnished to help the child to feel welcome. For example, some interview rooms have child-size furniture. Anatomical drawings and dolls are not to be in view. The interview location should provide the child with a feeling of privacy. Providing a child with food during the interview should be avoided, as should the promise of a snack when the interview is over. Interviews should be scheduled at a time when the child is likely to be alert and with the least disruption to the child's daily routine.

Who Should Be Present During the Interview

It is firmly recommended that children be interviewed alone by a single adult interviewer. Only in rare circumstances should a parent of a very young child be allowed to be present during the investigative interview. Caretakers should be informed in advance that they will not be in the interview room. In rare circumstances, the sex of the interviewer may be of significance and may require rescheduling to make a child more comfortable. It is recognized that more than one interview may be necessary to receive a complete account from the child. All investigative interviews should be conducted by the same interviewer. Children should not be reinterviewed about information previously disclosed.

How to Record Information from an Interview

The APRI expresses some flexibility on this issue, recognizing that there are jurisdictions that require audiotaping or videotaping of forensic interviews with children and others that do not. The APRI notes that a videotape can be a powerful tool for demonstrating the behavior and statements of children both in the hands of the prosecution and in the hands of the defense. When taping is performed,

high-quality equipment is recommended. When taping is not used, interviews are to be accurately documented. The APRI indicates that "skilled interviewers are able to fulfill both functions [interviewing and note taking]." Lamb, Orbach, Sternberg, Hershkowitz, and Horowitz (2000) demonstrate that this is not the case and that even trained forensic interviewers produce significantly incomplete and flawed "verbatim notes." Therefore, for the sake of complete documentation, Lamb et al. recommend video- and audiotaping.

Conducting the Interview

Recommended interview protocols share common features. These begin with rapport building, the interview then progressing through a series of stages. At the outset, the interviewer should help the child to feel physically comfortable. For video- or audiotaping, the child should be positioned to facilitate that recording. Although the child should be asked, "Do you know why you're here?," he or she should not be told that the parent or guardian has already informed the interviewer about the interview purpose. The child should be canvassed about his or her understanding of the purpose of the interview and the role of the interviewer. Then the child should be instructed about a few rules. The first rule is that the child should say what he or she knows but not to guess. The second rule is that the child should correct the interviewer if he or she makes a mistake. The third rule is to inform the child that if the interviewer asks a question a second time the reason is not to cause the child to give a different answer. The fourth rule is that the child should be explicitly told that he or she can answer any question with "I don't know" or "I don't remember." The fifth rule is to ask the child to report when he or she does not understand a question. The interviewer should be mindful of the child's age, developmental capacities, and attention span as well as the child's capacity to remember instructions. Open-ended questions are encouraged as the best questioning method.

Canvassing the child's capacity to differentiate between the concepts of truth and lie and telling the truth versus telling a lie is treated as optional. It is noted that younger children have difficulty with these concepts and that this procedure is essentially an attempt to assess the child's competence to take an oath. The use of this procedure is apparently widespread; it is treated as an ordinary interviewing issue by Sattler (1988) and others (APRI, 2004; APSAC, 2002; Sparta & Koocher, 2006). White and Quinn (1988, p. 274) indicate that such canvassing of children can be intimidating, confusing, and coercive. They refer to this as the "truth-lie paradigm," categorizing it as coercive in all of its forms. They note that the labeling of statements as true or untrue is the function of the judicial system. The use of this technique "signals to the child that the interviewer believes there exists a particular truth to be found." Condie (2003, p. 222) sees such questioning of a child as a "fruitless endeavor" because neither a yes nor a no answer by the child has any predictive value for the child's ability to report events accurately. Truth-lie canvassing and developmental competency assessment do not access the core testimonial issue of the child's eyewitness memory for personal experience. As such, they provide no indication of the child's capacity and willingness to provide accurate autobiographical reports.

When competency questioning is undertaken, it can begin with testing the child's capacity to give accurate and inaccurate responses to fact questions in the present. It assesses the child's developmental abilities to name colors, count, count instances of events, tell the time of day, distinguish relative positions, and so on. Questioning about autobiographical events can provide an indication of the capacity for narrative recall. The author suggests that such questioning be directed at events that lend themselves to external corroboration. Sample topics are how and with whom the child came to the interview location, what the child ate that day, and major recent and remote events in the child's life. These can be accessed with such questions as "Tell me how you got here today"; "Who brought you here today?"; "What did you eat for breakfast today?"; "Tell me some things that you did yesterday"; and "Tell me about some of the best and worst things that ever happened to you."

The third stage is the abuse inquiry. The APRI guidelines are flexible on this issue. The discussion includes the optional use of anatomical drawings to orient the child to the subject of the inquiry. The APRI notes that some interviewers, however, postpone the introduction of body drawings for a later point in an interview that has produced an abuse report and use body drawings for body identification purposes. The discussion notes that when body drawings are introduced prior to an abuse report, this procedure is open to the attack that such pictures are suggestive in and of themselves. Poole and Lamb (1998) provide a detailed discussion of the pros and cons of the use of anatomical drawings and dolls for investigative interviewing purposes. They conclude with the recommendation that the use of such explicit stimuli be postponed to the identification and validation phases of an interview and that they not be used in advance of an abuse report by the child. They point to research that shows that the use of such props increases the percentage of false reports.

The APRI states that it is important to avoid asking children to simply repeat their prior report with a question like "I want to know what you told your mom about the babysitter." A recommended alternative is "Did you tell your mom about something that happened with the babysitter?" These examples are offered to illustrate attempts to access abuse reports. The author advises that both alternatives are objectionable as they access prior reports rather than prior experience. It may be a better practice to probe the issue of prior reports after concluding the abuse interview. This allows for the chronological recapitulation of experience and the assessment of the instant eyewitness report of the child. This method attempts to minimize the compounding effect of potential contextual influences and distortions. It also allows for a distinction between the child's report to the investigator and the disclosure process itself, which is a subject of separate inquiry. The forensic interviewer should seek "investigatory independence" (White & Quinn, 1988) and avoid direct and indirect alignment with prior interviewers. When an abuse report is obtained, it can be compared to prior abuse reports. If no abuse is reported or if abuse is denied, a decision can be made to probe the causes for inconsistency of reporting.

The APRI offers a table on a continuum of question types. These range from questions that are open-ended to those that are closed-ended. Answers to open-ended questions generate more confidence than those obtained from closed-ended questions. Question types identified are general, focused, multiple choice, yes/no, and leading. The opinion is firmly stated that the most reliable information comes in

response to open-ended questions. Interviewers are encouraged to avoid questions that sound accusatory, have erroneous content, and introduce elements of threat, coercion, and bribes. Putting pressure on a child to speak or to continue an interview is discouraged.

Possible Questions for the Sexual Abuse Inquiry

The APRI lists four pages of possible questions that fall into subcategories of general, penetration, erection or masturbation, ejaculation, nudity, oral contact, and pornography. There are then lists of questions about physical and sexual abuse, fears and reasons for secrecy, determining who was involved, where the abuse took place, and alternative hypothesis testing. Most of these questions are phrased in the affirmative and are therefore suitable for situations in which the possibility of abuse was previously indicated in the interview. Examples are:

- How did the touching start?
- What did his penis look like?
- What happened to his penis after he made you touch it? It is considered more appropriate in advance of an abuse report to ask questions in an open-ended interrogatory form to explore the possibility of experience that may or may not have an abusive character. Some APRI examples are:
- Have you had any touches you didn't like or that made you feel uncomfortable?
- Do you remember what you were wearing when you and . . . were playing?
- Did you see pictures/books/magazines/movies/videos at . . . ?
- What happens if you do something you're not supposed to?
- Have the police ever come to your house? Why did they come?
- Has anyone asked you to keep a secret? What was the secret?

Open-ended examples of follow-up questions are:

- Tell me about that.
- Tell me more about that.
- Do you remember more about that?

It is important to remember that requests for additional information can constitute a demand in the mind of the respondent and can cause the production of information and detail to please the questioner. Children and adolescents are rarely self-referred for investigative interviews, and most have had prior interviews by parents and others. The author advises that even general and open-ended requests for additional information be preceded by inquiry concerning the presence or absence of additional experience and knowledge. Poole and Lamb (1998) report that with additional questioning children add credible detail to both true and false reports. The risk of added detail to false reports is heightened when respondents have been exposed to misinformation in prior interviews. The well-informed investigative interview

strives to offer respondents the opportunity to drop misinformation and false reports if these exist and to pursue interviewing in the least suggestive manner.

Direct questions are seen as sometimes necessary, and a distinction is made between more and less suggestive direct questioning. The example "Did something happen to your [child's word for vagina]?" is considered superior to "Did your dad do something to your [child's word for vagina]?"

Anatomical Dolls

The APRI allows for the use of anatomical dolls for demonstration purposes after an abuse report has been received. The interviewer is encouraged to use dolls that are chosen for the child and for the perpetrator, and whenever possible these dolls should be appropriate for age, sex, and race. A list of inappropriate uses of dolls includes:

- Presenting the dolls to the child unclothed.
- Encouraging play with the dolls.
- Naming the sexual parts of the doll for the child.
- Probing the doll's genitals or breasts.
- Using the dolls to demonstrate sexual behavior to the child.
- Failing to elicit a verbal description of the abuse before presenting the dolls to the child.
- Incorrectly clarifying what the child demonstrated.

The APRI advises interviewers not to ask adolescents to name body parts at the beginning of an interview and to use an anatomical doll only when this can be done in a way that is not insulting. This author advises interviewers not to introduce drawings or dolls until a verbal abuse disclosure has been made and to restrict the use of drawings and dolls to clarify important points. This admonition reflects the author's recommendation for best practice methods.

Ending the Interview

It is recommended that the interview be ended after determining that the child is no longer able to provide information. This can be preceded by such questions as "Is there anything else you want to tell me today?" and "Is there anything you want to ask me?" It is recommended that the child be thanked for participating. After the meeting with the child, it may be appropriate to meet with the parents or caregivers to provide them with information. This can include scheduling follow-up interviews, making treatment recommendations, and giving information about the investigatory and court process. Interviewers are encouraged to develop a protocol that defines the stages of the interview, including introduction, rapport building, instructions, methodology for beginning the abuse inquiry, methodology for the abuse inquiry itself, and closure.

Assessing Validity in Sexual Abuse Cases

There are 13 subsections to the validity discussion that lead to a summary of indicators of report validity in sexual abuse cases. The purpose is to determine "whether a report of sexual abuse is valid." The 13 points describe the process of evaluating the available evidence. Evidence includes eyewitness statements, confessions, medical evidence, the child's statements, emotional and behavioral characteristics of the child as seen by others, and assessment of the validity of the abuse report. No factors are thought to conclusively demonstrate the validity or invalidity of an abuse report.

1. Alternative Explanations for the Child's Statement The investigator is required to consider alternative explanations throughout the investigation. This includes consideration of the possibility of deliberate falsehood, misinterpretation of innocent contacts, and coaching. The author adds ruling out professional error to this list of requirements. These possibilities are to be translated into active steps in document review and in investigative interviewing before the inquiry is ended.

2. Spontaneity Spontaneous statements from children are considered more reliable as they are unlikely to be the result of prompting. The evidentiary record should be reviewed to determine whether there is reliable indication of a spontaneous disclosure by the child and which parts of the child's narration in the investigative interview possess this characteristic.

3. Consistency Consistency of child statements in an interview and across interviews and statements is an essential feature of a validation determination. Consistency is tracked from the point of initial disclosure through all forensic interviews as well as the child's testimony in court. Consistency is tracked for core elements rather than for exact wording. The validation review should be alert to the issue of rote recital, and when this is found the record should be "scrutinized carefully for the possibility of coaching."

4. Sexual Knowledge or Behavior That Is Developmentally Unusual This refers to the acting out of adult forms of sexuality, including sexual talk, rather than age-appropriate touching and talk about body parts, including genitalia, that may occur out of curiosity and play. Friedrich (1992) identifies explicit advanced sexual behaviors that occur with low frequency and are considered indicators for sexual abuse. These include:

- A child putting his or her mouth on another's sexual parts.
- A child asking to engage in sexual acts.
- A child masturbating with an object.
- A child inserting an object into his or her own anus or vagina.
- A child imitating intercourse.
- A child making sexual sounds.

- French kissing by a child.
- A child talking about explicit sexual acts.
- A child undressing others.
- A child asking to watch explicit sexual television or movies.
- A child using sexual words.
- A child imitating sexual acts with dolls.

Schoentjes, Deboutte, and Friedrich (1999) confirmed these data in a study of 917 2- to 12-year-old Dutch-speaking children in Belgium. More recently, however, Drach, Wientzen, and Ricci (2001) studied the utility of sexual behavior problems as diagnostic indicators of sexual abuse in a sample of 247 children evaluated for sexual abuse at a forensic child abuse evaluation clinic. It was found that non-sexually abused children were just as likely to have high scores on the Child Sexual Behavior Inventory as sexually abused children. These results suggest that the occurrence of seemingly advanced sexual behavior in relatively young children should be viewed with caution as an indicator of sexual abuse. Both the absence and the presence of advanced sexual behavior was evident in children who had been and who had not been sexually abused. This nullified the use of advanced sexual behavior as an abuse indicator.

5. Developmentally Appropriate Language Reports about a child's disclosures and the child's interview language should be analyzed for its developmental characteristics. The interviewer is urged to adapt questions to the child's developmental language and to use the child's language during all phases of the interview. It is thought that children who have been coached or who have borrowed language from other sources for use in abuse reports may use language that is developmentally unexpected or unusual. This analysis should allow for children who have precocious language ability, who have been exposed to multiple interviews, and who may have acquired language from the offender.

6. Play and Gestures Indicative of Abuse A suspicion of abuse may have arisen because of the observation of sexualized play. Or, after an abuse disclosure, reports may arise about previously observed sexualized play. Particularly in younger children, such play may be seen as a reenactment of an advanced sexual exposure or experience.

7. Idiosyncratic Detail Children are considered less likely to have fabricated an abuse report that contains personalized, experiential detail. Sensory detail referencing taste, smell, touch, size, pressure, or pain, and containing mechanical descriptions is considered validating of the abuse report. Ceci and Bruck (1995), however, demonstrate that such detail can be abundantly produced by children in response to improper and suggestive interviewing. Nonetheless, an idiosyncratic report in the absence of evidence of improper questioning can be compelling. The APRI advises that this can occur when the abuse report contains "such description and detail that there can be little doubt that the child witnessed the sexual act." The child may have knowledge about a feature of the perpetrator's body such as a tattoo or knowledge

of taste and texture or visual exposure to sexual activity. The child may recount highly personalized details that lend credibility to the abuse report. It is stated that "a child's total inability to furnish details that demonstrate familiarity with the sex act forced upon the child may indicate the absence of abuse."

8. Content of the Statement The investigator is called on to consider whether the actions described make sense. This includes considering the possibility that innocent physical touches were misinterpreted as sexual and whether the progression of events fits known patterns of abuse. This can include identification of some features that have been found in validated abuse reports, including a description of multiple incidents over time, a progression of sexual activity that includes grooming, a request for secrecy, the use of power and authority to gain compliance, and attempts to cause a child to retract a complaint.

9. The Child's Manner and Emotional Response The APRI notes that a child's emotional presentation may give a sense of whether or not an abuse account is genuine, and the investigator is called on to note all of the child's verbal and nonverbal emotions. The advantage of videotaping to capture this information is acknowledged. Hesitancy, tone of voice, and facial and bodily expressions can offer clues about fearfulness, guilt, aversion, and distaste. It is noted that the absence of emotionality does not automatically invalidate an abuse report.

10. The Existence of a Motive to Fabricate This section states, "The assessment should consider what motives the child may have to fabricate the abuse or the motives others may have to coach a child to allege abuse." But the discussion of this topic offers little guidance about how to investigate this issue and how to translate hypotheses into investigatory procedures and questions. No specific validation procedures or questions are offered. We are cautioned that the existence of a motive is not the equivalence of a fabrication and to be alert to offenders who may attempt to confuse motive with misdeed. The author recommends that validation questions be adapted to the facts of the case and should reflect a knowledge of the complainant's personality and relationship characteristics, including his or her relationship with the accused. When the complainant's biography includes a history of lying or this issue otherwise arises (Benedek & Schetky, 1987), this issue should be explored. For younger children, the author suggests the use of questions such as:

- Is this something that really happened or is it make-believe?
- Are you saying this just to get [the accused] in trouble?
- Are you saying this because you're angry/mad at [the accused]?
- Does somebody else think this happened even though you don't?
- Are you saying this because you want to live with. . . ?

For older children, the author recommends such questions as:

- Are you saying this to get out of the house?
- Are you saying this because [a valued person] was not being nice to you or not paying enough attention to you?
- Are you saying this to cover up because you wanted to do it (have sex)?

Other authors provide some help in identifying motives to fabricate. Kanin (1994) investigated 45 consecutive, disposed, false rape allegations over a 9-year period in a small metropolitan community that constituted 41% of the total forcible rape cases in that jurisdiction ($n = 109$). Kanin determined that the allegations appeared to serve three major functions for the complainants: providing an alibi, seeking revenge, and obtaining sympathy and attention. The sample consisted of young adults. Retrospective comparison reports with college populations produced similar results. These motives may apply differently to child and younger adolescent populations. For example, providing an alibi for consensual sexual behavior may be more applicable in an adolescent population. Seeking revenge and obtaining sympathy and attention may apply equally to child and adolescent populations that have the developmental capacity to be motivated in these ways.

Benedek and Schetky (1987) investigated child sexual abuse reports in child custody and visitation disputes. They noted that adolescents may allege sexual abuse for secondary gain because they are angry at a parent, jealous of a parent's boyfriend, or want someone excluded from the home. For younger children, they found that a history of lying, the use of rote phrases, reports from the perspective of an adult rather than from the perspective of a child, and child reports delivered only in the presence of an accusing adult were characteristics of false reports. Ash and Gyer (1991) discuss biased reporting in parents undergoing child custody evaluation. They studied 196 court-ordered child custody evaluations and found that parents give symptomatic child reports that suit litigation purposes. Everson and Boat (1989) reviewed false allegations of sexual abuse by children and adolescents in reports given to child protection workers; 88 child protection respondents who had worked on 1,249 sexual abuse cases were in the sample. The children ranged in age from preschoolers to adolescents. Child protection workers reported fewer than 2% of cases involving children under the age of 6 and 8% of those involving adolescents as cases of false report. Subsequent retraction by the child was found to be the major reason for determining an allegation to be false. Children were found to have insufficient credibility when the report itself was improbable, the details were insufficient, there were inconsistencies in the report, or there was conflicting evidence. Everson and Boat caution that obvious evidence of coercion on a child to recant should not be considered a circumstance for the invalidation of a complaint.

Jones and McGraw (1987) reviewed all reports of suspected sexual abuse made to the Denver Department of Social Services in 1983 ($n = 576$). Each report was investigated by that department and designated as either "founded" or "unfounded." Two of the evaluation categories include fictitious reports by adults and fictitious reports by children. Fictitious reports by adults included deliberate falsifications, misperceptions, and confused interpretations of nonsexual events. Fictitious reports by children included those that were falsely made and involved deliberate

falsifications and misperceptions. Fictitious accounts by adults composed 5% of the sample and fictitious accounts from a child were 1% to 2%. Among the features of fictitious accounts by adults were no child reports despite careful and repeated interviewing, an ongoing custody and visitation dispute, abuse and neglect histories of alleging adults, and emotional disturbance in alleging adults that included two professionals. Among the fictitious child accounts were those from children who ranged in age from 3 to 10, and the context of all was a bitter custody or visitation dispute. The seven mothers were found to be psychiatrically disturbed. Reference is made to other studies that found a lack of detail or of accompanying emotion in fictitious reports. Reference is also made to children who had been sexually abused in the past and made an erroneous subsequent allegation of reabuse. Also noted was the absence of threats or coercive elements in the fictitious allegations.

Ceci and Bruck (1995) discuss false allegations of sexual abuse after reviewing some of the same literature. Discussing the Jones and McGraw (1987) study, they note that of the 576 cases reported that year in Denver, 23% were judged unfounded and in 24% of the cases no clear judgment could be made because of insufficient information. Six percent of the unfounded cases were considered to be deliberate or malicious and another 17% to have been wrong but made in good faith. Ceci and Bruck consider 23% of the reports to be properly categorized as false, of which 6% were deliberate lies and the other 17% baseless. Ceci and Bruck considered the Everson and Boat (1989) numbers to be underestimates. They referenced a study by Faller in 1991 that examined the reports of 136 children of divorced families referred to a project on abuse and neglect in which 35% of the reports were found to be untrue. It was noted that Faller broke down the 35% rate of unsubstantiated cases into three subgroups consisting of false allegations made (8.8%), false allegations reflecting dynamics besides divorce (11%), and false allegations arising in an atmosphere of divorce acrimony (14%). High rates of unsubstantiated sexual abuse reports are the norm, according to Besharov (1994), who studied child abuse and neglect substantiation rates in 30 states in 1986, 1987, and 1988, with rates ranging upwards of 77% in Montana in 1988 and as low as 15% in Alaska in 1986. Kuehnle and Sparta (2006) wrote of a marked nationwide decline during the 1990s of child sexual abuse reports made and in the percentage of reports that are substantiated. These statistics are a reminder to remain open-minded regarding the outcome of any particular investigation. They underscore the importance of a validation effort in every case.

Investigatory error, primarily in the form of improper questioning of alleged child victims, is the central topic of the Ceci and Bruck (1995) analysis. This topic is addressed by the APRI for the purpose of avoiding mistakes but not addressed for the purpose of identifying mistakes already made and estimating their impact. For impact assessment of professional error in the fabrication of abuse reports, we turn to other sources. Mantell (1988) based his analysis on several hundred child protection, family, and criminal court cases he evaluated. He indicates that professional error is one of several kinds of error causing false sexual abuse allegations. The impact is treated as potentially total such that an allegation can emerge from the questioning of a child without prior existence in the child's utterances or experience. The Ceci and Bruck (1995) analysis provides a full discussion of this topic and identifies

categories of error sources in investigatory interviewing. Their analysis focuses on the effects of several forms of suggestive questioning, including interviewer bias, repeated questioning, negative stereotyping, the emotional tone of the interview and the effects of peer pressure, of interviews by adults of high status, of misleading information, of failure to test an alternative hypothesis, and of questioning with the request for visual imaging. This is an analysis of the confounding impact of external influence on autobiographical memory.

The *Report on Scott County Investigations* (State of Minnesota, 1985) shows the effects of repeated questioning, cross-germination (witness contagion), and private motives and pressures in generating graphically detailed false reports by teenagers and young adults of homicide and sexual abuse. This began with the interview of a 12-year-old boy who provided detailed accounts of homicides in 1984, describing seven children being stabbed, mutilated, and shot. Other children of varying ages provided reports of ritualistic torture and murder. After state authorities determined that no homicides had occurred, they turned their focus to allegations of sexual abuse and pornography. The continued interviewing of the children led to an increasingly large list of accused citizens and ultimately to the children reporting abuse by their parents. Children as old as 21 and 18 as well as 12 provided such reports. In some cases, children were interviewed together and prompted to discuss allegations in group settings.

In some respects, the 1989 edition of the APRI manual provides a more detailed account of error sources to be assessed in the validation phase of the investigation. It calls for the child to be interviewed alone, out of the presence of other adults, to make sure that the child is not "overwhelmed" or "bombed" by adult interviewers, and to assess the impact of family and support persons in providing approval and disapproval to the child witness. The interviewer is told to ask simple, direct, open-ended questions, avoid leading questions, and make sure that the information the child is giving is really coming from the child and not from other sources, such as the interviewer. Answers need to be clarified regarding their reliability. A child is never to be pressured to talk or continue an interview. If a child thinks that he or she will get into trouble for talking, the interviewer should find out with whom and for what reason. The interviewer should try to find out if any threats, promises, or rewards were offered or given to the child for talking. If the child says that no abuse occurred, the interviewer should ask the child if there is someone who thinks that something did happen. Other explanations for any behavioral or medical data that may indicate abuse should be considered. Other explanations for the child's sexual knowledge and behavior should be considered and explored. The use of nonleading questions is identified as an important assurance for the analysis of the investigatory interview that the child is using his or her own language for abuse reports. Contamination of the child witness by family members is listed as a key error source to be ruled out. Witnesses who are likely to be in contact with the child or with each other are to be told not to question or rehearse the child, not to investigate the case on their own, and not to compare notes. Members of the child's family are to be interviewed because they may be able to provide direct and indirect corroboration for the abuse report and can also provide observations of behavioral, emotional, and physical indicators of abuse as well as information about the relationship between the child

and the suspect. Family members and others also should be interviewed to rule out their influence on the child to make an abuse report. Other witnesses, such as teachers, nurses, and police officers, are to be interviewed for their information. The APRI recommends a high level of care in the documentation of statements by these witnesses, including videotaping the interview.

11. The Child Corrects the Interviewer The APRI advises that children who correct the interviewer are listening attentively and are not judged to be overly suggestible.

12. and 13. Medical and Forensic Evidence Medical and forensic evidence are additional indicators for analysis. They are listed in the Summary of Indicators. Neither is behavioral in character, and thus they are not included in this discussion beyond noting that although both are infrequent they can be decisive in establishing that abuse occurred and in the identification of the perpetrator.

This section concludes with Table 4.2, a summary of the 13 validation indicators for sexual abuse. Table 4.2 offers a quick overview of the investigatory process and evidentiary record. It includes, as the final indicator, the nature of the alleged offender statement, if any, placing it last as an important abuse indicator. Construction of a case-specific evidentiary chart after completion of a thorough validation review offers a useful perspective on the case and can provide a basis for the development of a more thoroughly informed opinion about the strengths and weaknesses of the total evidentiary record. This can assist the development of an opinion about the completeness and reliability of the procedures used. Oftentimes, investigators are fact witnesses. They report what they have done and the results they have obtained but do not provide professional opinions about its meaning. Interviewers should remain sensitive to the nature of their role and not overstep their boundaries. Whether the interviewer considers the evidence to be strong or weak, it remains the domain of the trier of fact to determine whether or not abuse occurred. The author reminds interviewers of research by Poole and Lamb (1998) and Horner, Guyer, and Kalter (1992) that demonstrates that psychologists are not able to reliably distinguish between true and false reports of sexual abuse.

Table 4.2 Summary of Indicators of Report Validity in Sexual Abuse Cases (APRI)

1. Medical evidence
2. Forensic evidence
3. The child's statement
4. Statement spontaneity
5. Consistency—both internally in the statement and across statements
6. Developmentally unusual sexual knowledge/behavior
7. Idiosyncratic detail
8. Developmentally appropriate language
9. Play and gestures indicative of abuse
10. The child's manner and emotional responses
11. Lack of a motive to fabricate
12. The child corrects the interviewer
13. The offender's statement

2. The American Professional Society on the Abuse of Children

The APSAC publishes seven documents that address investigatory interviewing of alleged child abuse victims:

- Practice Guidelines: Code of Ethics (APSAC, 1997a)
- Practice Guidelines: Psychosocial Evaluation of Suspected Sexual Abuse in Children, second edition (APSAC, 1997b)
- Practice Guidelines: Psychosocial Evaluation of Suspected Psychological Maltreatment in Children and Adolescents (APSAC, 1995c)
- Practice Guidelines: Photographic Documentation of Child Abuse (APSAC, 1995b)
- Practice Guidelines: Use of Anatomical Dolls in Child Sexual Abuse Assessments (APSAC, 1995d)
- Practice Guidelines: Descriptive Terminology in Child Sexual Abuse Medical Evaluations (APSAC, 1995a)
- Practice Guidelines: Investigative Interviewing in Cases of Alleged Child Abuse (APSAC, 2002)

This discussion focuses on the last of the practice guidelines, Investigative Interviewing in Cases of Alleged Child Abuse (APSAC, 2003). The content of this guideline overlaps with many aspects of the APA and APRI guidelines. This guideline opens by reminding us that "inadequate and improper interviewing can lead to errors in decision-making about child safety and criminal prosecution." The guidelines are recommended for use in conjunction with the APSAC Guidelines for the Use of Anatomical Dolls and for the Psychosocial Evaluation of Suspected Sexual Abuse in Children. Aspects of those guidelines will be discussed as well.

The stated purpose of the investigative interview is to obtain as complete and accurate a report from an alleged victim as possible. It is noted that most investigative interviews are conducted by child protective service workers and law enforcement officers and that other mental health and medical professionals may also participate in or conduct such interviews. Interviewers and investigators are expected to develop specialized knowledge. Interviews should be timed to occur as close to the event in question as is feasible and with appropriate notification of parents and guardians. A neutral environment for the interview should be selected whenever possible. Electronic recording in the forms of videotaping and audiotaping are identified as the most comprehensive and accurate method of documentation. The number of interviews should be governed by the need to gather a complete and accurate report from the child. This can often be accomplished in one interview, although some children require more than one interview. However, multiple interviews also carry risks. They have been associated with memory errors, acquiescence to interviewer expectation, and child distress. Interviewing by one interviewer is the preferable practice. The presence of additional interviewers is allowed with the recommendation that a lead interviewer be designated. Closed-circuit TV and two-way mirrors can enable other professionals to monitor the interview while remaining

out of sight. Parents are not to be present during the investigative interview unless there is an exceptional circumstance. Siblings are to be interviewed separately.

Investigators are urged to practice cultural competence and to take cultural features into account during the interviewing process and also in review. Children with special needs may require accommodation.

The APSAC is concerned about interviewer bias and recommends that interviewers approach the interview with an open mind about what may have happened. The customary practice is to know the specifics of an allegation before the interview. Interviewers are reminded that the purpose of the interview is "not merely to confirm prior suspicions" but to "elicit information from the child." It is noted that in some jurisdictions there is a standard practice that the investigative interview begin with an interviewer who is blind to the allegations. Bias is seen as impairing the capacity of the interviewer to receive and to objectively interpret information and as a source for interviewer error. Negative statements about suspected abusers are one source of error and suggestion. A number of interview aids, such as anatomical dolls, anatomical drawings, and doll furniture, may be used, but the timing and manner of their use is important. Investigators are expected to have specialized knowledge about these aids. The Practice Guideline: Use of Anatomical Dolls in Child Sexual Abuse Assessments (APSAC, 1995d) essentially dismisses the concerns that the use of anatomical dolls leads to erroneous reports of sexual abuse and leaves the timing of their introduction to the judgment of the interviewer.

This position contrasts with that of Ceci and Bruck (1995), who in their own studies and the review of the literature found that with the use of anatomically detailed dolls many children made both verbal and nonverbal reports of genital touching and digital penetration though neither had occurred. Younger children were found to be more susceptible to such error. Ceci and Bruck recommend that the dolls not be used for diagnostic purposes, stating, "We have reviewed enough studies that demonstrate that this tool has the potential for serious misuse, including misdiagnosis" (p. 186). Poole and Lamb (1998) also provide a detailed discussion of the anatomical doll issue and of the research literature. They indicate that there are no standardized procedures for the use of anatomically detailed dolls in investigative interviewing and that they are not a psychological test for the diagnosis of abuse. Although the APSAC allows for them, Poole and Lamb note that the APSAC-defined preferred practice is for their use to identify body parts, to clarify previous statements, and for a demonstration by non- or low-verbal children after there is an indication of abuse activity. Sparta and Koocher (2006) appear to support the use of anatomically detailed dolls in investigative interviews without discussing the timing and manner of their use, while also noting that some researchers are concerned about the error rates associated with their use because of a reduction in the accuracy of children's reports.

The APSAC states that "research does not confirm that anatomical dolls are inherently too suggestive or sexually stimulating." The APSAC specifically recommends the use of anatomical dolls for several purposes: identification of body parts, as a visual aid for direct inquiry about the child's personal experience with private parts, and as a demonstration aid. The APSAC specifically recommends the use of dolls with preschool children, the age group in which Ceci and Bruck (1995)

found the highest error rates for children's abuse reports of genital touches in pediatric examinations that did not contain such touches. Melton, Petrila, Poythress, and Slobogin (1997, p. 476) conclude that anatomical dolls are "undoubtedly the most controversial evaluation technique." They note that there is a consensus that play with anatomically detailed dolls cannot be used as a test of whether or not child maltreatment has occurred. They note that a question remains about how the dolls can be used "even as demonstration aids to clarify a child's statements" (p. 477). Ceci and Bruck indicate that jurisdictions vary dramatically in legal decisions about the use of anatomical dolls, ranging from those that ban their use entirely to others that permit children to use the dolls in court to demonstrate their own sexual abuse.

In its practice guidelines, the APSAC distinguishes anatomical drawings from anatomical dolls and provides little guidance for how anatomical drawings might be used or about their limitations. They are identified as an interview aid that can assist the child to identify names for body parts and functions and also the location of possible sexual touching or physical injuries. The author recommends the best practice approach, which is to avoid issues of suggestive questioning and reserve the use of anatomical drawings and dolls for the confirmation phase of an interview following a demonstration that their use was needed after other avenues of explanation had been exhausted (Poole & Lamb, 1998).

The APSAC recommends careful preparation for the forensic interview by developing a list of specific topics and the hypothesis to be addressed. The interviewer should appear relaxed and nonthreatening. The interview itself should be paced to mesh with the child's developmental and temperamental characteristics, and the interviewer should proceed slowly and never pressure a child to respond to questions. Especially with younger children, the interviewer should be aware of signals indicating fatigue and loss of concentration as well as the need for a break.

In regard to questioning strategies, the APSAC recommends minimizing the introduction of information to the child that might be adopted by the child in his or her report. (This is the very principle underlying objections to the use of anatomical dolls and drawings and good and bad touch instruction prior to a child abuse inquiry and an abuse report.) The use of open-ended questions is recommended. Yes/no and multiple choice questions are allowed. Yet the interviewer is encouraged to focus open-ended questions to specific topics that the child has introduced by saying "Tell me more about. . . ." In a later section, specific, closed questions such as those that have yes/no and multiple-choice formats are recommended after open-ended questioning has concluded and appears to have exhausted the information that a child can provide. When the multiple-choice format is used, the example given is, "Was he in the house, in the yard, or someplace else?" This author notes that the forced-choice format of this formulation can be reduced by including the options "I don't know" and "I don't remember" as approved alternatives and then making sure that the child has the ability and willingness to use them. It is also essential that such questioning first be clarified as relevant. This can be achieved by asking "Do you know where he was when . . . ?"

The APSAC expresses uncertainty about how to define leading and suggestive questions. White and Quinn (White and Quinn, 1988, p. 271) define a leading question according to Black's Law Dictionary, "as one which instructs the witness

how to answer or puts into his mouth words to be echoed back." The suggestive question is defined as "presentation of an idea especially indirectly as through association of ideas, bringing before the mind for consideration in the nature of a hint" (p. 271). White and Quinn state that the question "Your daddy did put his finger in your vagina, didn't he?" is leading, whereas the question "Did your daddy put his finger in your vagina?" is suggestive. Others, such as the APRI, might classify the second alternative as a direct question. There would probably be unanimity about the fact that the question contains information descriptive of bodily touching and in this case of the digital penetration of the genitalia and that it is not an example of a general, open-ended question to which a child might provide a wholly spontaneous and independent response. There also can be no controversy about the fact that the question is formulated in the vocabulary of the interviewer, not the vocabulary of the child, and that its use poses the concern of the respondent acquiescing to the question with confirmation. Whenever possible, it is recommended to avoid these risks by asking questions that call on the child to independently express the substance of an idea descriptive of action and experience.

The APSAC describes the following components of investigative interviewing. It begins with the interviewer introducing himself or herself and the interviewer's role, followed by rapport building in which there is a discussion of neutral topics. Developmental screening can then occur to assess the child's ability to understand and respond to questions. The competency check may occur in jurisdictions that require an assessment of the child's understanding of the difference between the truth and a lie. This is distinguished from the assessment of the child's ability to provide information about events that are known to have occurred, such as a recent birthday party. The ground rules of the interview are then explained. The child should be told that the interview is about things that really happened, that the child will be asked many questions, and that the child cannot guess about answers and can say "I don't know" and "I don't remember." The child is encouraged to correct the interviewer when the interviewer makes a mistake. If the interviewer asks a question that is too hard or difficult for the child to talk about, the child should let the interviewer know.

The APSAC recommends that the topic of abuse be introduced in one of a number of ways:

- "I understand something may have happened to you. Tell me about it from beginning to end."
- "I understand you had to go to the doctor. Is there a reason you had to go?"

The author finds these questions problematic and recommends avoiding them. They can convey bias, expectation, and alignment of the interviewer with authority figures who may be close to the child and instructed the child about the purpose of the interview and what to say. They may convey to the child that the interviewer wants the child to repeat prior statements or the opinions of others. Getting started does not require reliance on methods that contain error and that may mislead the respondent. Alternative questions are:

- Do you know why your . . . brought you here to speak to me?
- What were you told about what we are going to do and talk about?
- How do you feel about being here?
- What would you like to do and talk about?

Situation-focused questioning is also encouraged, such as asking a child about bath time or toileting, about secrets, or about things the child doesn't like. The APSAC says that anatomical drawings and dolls are allowed at this time and are useful for the purpose of assessing the child's body knowledge, focusing the discussion on bodies and body experience, and giving the child permission to discuss sensitive topics such as private parts.

Children should be asked to provide details about concerning events. Here the APSAC recommends the use of open-ended narrative prompts, such as "Tell me more about that" and "Then what?" This actually constitutes an inversion of questioning sequence from more specific (body parts and touch experiences) to the open-ended or general questions. When abuse reports occur, the child should be encouraged to provide detailed information, contextual information, and information about abuse witnessing by other persons. The APSAC states that anatomical drawings and dolls can be useful during this phase of the interview for clarification purposes about what the child is attempting to describe. It appears that the APSAC is recommending the use of anatomical drawings and dolls for two purposes: first, to focus the child's attention on the topics of interest (to the interviewer), and second, to clarify with the child the report that has been received.

The psychologist may wonder how to start an abuse inquiry if anatomical dolls and drawings and good-touch/bad-touch clarification are not used. Opening an abuse inquiry is not difficult. Many children know why they have been brought to the interview. Often, asking about this alone is sufficient to elicit the child's report and to set the stage for the forensic interview. When the child says that he or she does not know the reason for the interview, the interviewer can inquire about who brought the child, what was discussed, and whether the person who brought the child said what the reason is. If this does not lead to mention of the forensic topic, the child can be generally canvassed about major life experiences and about relationships with people, and then ultimately about general classes of positive and negative and also body experience. This can include questioning about general safety rules when crossing the street, with hot things, with water, with dressing and undressing, with toileting, and with body touching.

The APSAC does not discuss validation procedures. The reader is referred to the APRI guidelines, Ceci and Bruck (1995), Poole and Lamb (1998), and White and Quinn (1988) for a detailed discussion of this issue.

3. Policy Statement from the American Academy of Child and Adolescent Psychiatry: Guidelines for the Clinical Evaluation of Child and Adolescent Sexual Abuse

The AACAP (1990) states that the purpose of a clinical evaluation in a child sexual abuse case is to determine whether abuse has occurred, if the child needs protection,

and if the child needs treatment for medical or emotional problems. The guidelines were developed to assist clinicians performing those evaluations. The AACAP sets high standards for the qualifications of evaluators. They should possess sound knowledge of child development, family dynamics related to sexual abuse, and the effects of sexual abuse on the child. They should be trained in diagnostic evaluation of both children and adults and should be comfortable testifying in court. If not a child and adolescent psychiatrist or psychologist, clinicians performing these evaluations should be under the supervision of professionals with these qualifications. Role differentiation should occur so that the child or adolescent therapist is not the evaluator.

The child should be seen for the minimum number of times necessary to conduct the evaluation, with cross-agency sharing of information to avoid duplication and the unnecessary interviewing of the child. Optimally, evaluation teams should be formed that integrate local police and reporting agencies. Multiple interviews may be viewed by the child as a demand for more information, and the concern is specific that this alone may encourage confabulation.

The evaluation should be conducted in a relaxed environment. Comprehensive history taking is important and should include developmental history, cognitive assessment, prior history of abuse and other traumas, relative medical history, behavioral change, parental history of abuse as children, and family attitudes toward sexuality and modesty. Prior psychiatric disorders in the child or parents should be identified, and allegiances by a child to parents should be assessed.

In intrafamilial abuse, the family history and perspective of each parent should be obtained. Stressors other than sexual abuse that might account for the child's symptoms need to be identified. This can include a psychiatric assessment of each parent, particularly if there is a concern about a false allegation. This concern can arise when it is determined that the allegation is coming from a parent rather than from a child, when the parents are engaged in a dispute, and when the child is a preschooler. False allegations are seen as arising from other situations as well, such as the misinterpretation of a child's statement and the behavior of adults. Adolescents may make a false allegation because of vindictiveness or to cover up their own sexuality, and children who experienced prior sexual abuse may misinterpret or mischaracterize the behavior of an adult.

In conducting a clinical evaluation, it is required that the clinician maintain emotional neutrality, approach the case with an open mind, and seek out the unique characteristics of each case. Clinicians are urged to avoid leading and coercive questions and techniques and to allow the child to tell the story in his or her own words. The need for external corroboration distinguishes the forensic evaluation from the usual clinical evaluation; this can include the need to review medical and school reports and prior psychiatric evaluations and talking with other persons.

A number of characteristics enhance a child's credibility. These include event descriptions in the child's own language and from the child's own point of view, spontaneity, an appropriate degree of anxiety, idiosyncratic or sensorimotor detail, consistency of allegation over time, behavioral changes consistent with abuse, absence of motivation or undue influence for fabrication, and corroborating evidence.

In regard to anatomically correct dolls, AACAP states that they may be used for eliciting the child's terminology for anatomical parts and for allowing a child who cannot tell or draw what happened to demonstrate what happened. The AACAP strongly advises that dolls not be used to instruct, coach, or lead the child or as a shortcut for a more comprehensive evaluation, and that in these assessments it is not necessary to use anatomically correct dolls at all. California has barred the admissibility of evidence obtained through the use of anatomically correct dolls.

By contrast, children's drawings are seen as a helpful assessment tool. This can include spontaneous drawings or requesting a child to draw a male or female, to draw what happened and where, or to draw a picture of the alleged offender. Videotaping is encouraged. Psychological testing may be helpful but is not seen in and of itself as a method for determining whether or not sexual abuse occurred.

SUMMARY

Nine of the 15 APA publications addressing ethical standards and guidelines apply to children and adolescents. Three of the guidelines speak specifically to clinical forensic practice. Aspirational in nature, these three guidelines have much in common with one another, with APA ethical principles, and with central features of the remaining guidelines. They all call for a high level of science-based professional practice based on the principles of thoroughness and fairness. Psychologists are expected to practice and testify within the limits of their knowledge and expertise, to communicate their results and opinions clearly, to understand their role and the law when practicing in a legal context, and to avoid dual relationships, bias, and error in their work. High standards for record keeping and for the documentation of clinical and forensic work are emphasized. The increasing importance of cultural competence is the focus in several statements. Evidence-based practice is the standard. Psychologists performing child custody evaluations are expected to possess specialized competencies for the evaluation of adults, children, relationships, and parenting ability. These skills and a matching knowledge base are also expected of psychologists performing court-ordered studies in child protection matters; they are additionally expected to have a solid understanding of the field of child abuse and neglect. Psychologists do not offer opinions about persons they have not examined or about issues for which they lack clear and verifiable evidence.

The APRI has articulated the most complete statement regarding the investigative interviewing of suspected child victims of abuse concentrating on sexual abuse. The strengths and weakness of the APRI guidelines were reviewed, along with those of the APSAC and the AACAP. Whereas the APA Guidelines for Psychological Evaluations in Child Protection Matters deals with general practice issues, the APRI guidelines offer specific instruction for the forensic interviewer on how to conduct and validate an investigative interview within the larger context of a multidisciplinary forensic investigation. The strengths and weaknesses in the APRI and APSAC guidelines were identified and discussed with the help of behavioral science research. Research-based recommendations were made for alternative strategies and methods. The APRI model for a comprehensive review of the evidentiary

record was presented and discussed. Unresolved and controversial issues and best practices were identified.

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CHAPTER 5

Professional Roles

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Given that this volume of the *Handbook of Clinical Psychology* focuses on children and adolescents, it seems fitting that a chapter be devoted to a discussion of the various professional roles assumed by psychologists who work with children and youth. While these psychologists are often referred to by various labels, such as clinical child psychologists and pediatric psychologists, professionals with other backgrounds, such as school psychologists, also work with children and adolescents.

Child-oriented psychologists vary in the nature of their professional training (clinical psychology, school psychology, developmental psychology, specialty-specific training in clinical child or pediatric psychology); theoretical orientation (behavioral, cognitive-behavioral, interpersonal, psychodynamic, humanistic); the degree to which they focus on research, clinical activities, or both; the nature of the professional activities they engage in (teaching, research, assessment, therapy, consultation, advocacy); the types of child or adolescent problems they work with (mental health problems, physical health problems, school-related difficulties); and the settings in which they work (universities, medical centers, children's hospitals, pediatric clinics, private practice, mental health centers, schools). Given the diversity of training, theoretical orientation, professional activities, types of problems addressed, and settings, it should not be surprising that these professionals often function in multiple roles and engage in a range of professional activities in the context of these roles.

This topic of professional roles is of special relevance given the rapid growth of clinical child and pediatric psychology in the past 25 years and the degree to which this growth has resulted in expanded role opportunities and, in some cases, significant changes in the nature of the activities engaged in by psychologists serving children and adolescents.

ROLES AND ROLE-RELATED ACTIVITIES OF CHILD-ORIENTED PSYCHOLOGISTS

Before embarking on the task of highlighting the professional roles assumed by child-oriented psychologists, it is important to attempt to define what is meant by the term "role" and differentiate this term from the professional activities associated with these roles.

Considering several dictionary definitions, the term role, as it is used here, generally refers to a *consistent pattern of behavior* that is expected of someone based on his or her status or position in society (Merriam-Webster's New Collegiate Dictionary, 11th ed.). From this perspective, functioning in the role of a clinical child or pediatric psychologist involves engaging in a pattern of professional behavior that is consistent with one's prior training as a specific type of child health service provider. At the simplest level, the term "professional role" as applied to a clinical child psychologist can be best thought of as a general *construct* used to summarize the range of interrelated specialty-specific behaviors that are engaged in as one trained in clinical child psychology carries out his or her professional activities.

Closely associated with the term role is the pattern of specialty-specific professional activities, engaged in by the psychologist by virtue of his or her professional training, that define him or her as a specific type of health care professional. As regards the various roles of child-oriented psychologists, some function in the role of clinical child psychologist, and others function in the role of pediatric psychologist. Still others function in the role of school or applied developmental psychologist. Both clinical child and pediatric psychologists may engage in somewhat similar activities to a greater or lesser degree. For example, both may engage in assessment, intervention, consultation, research, or various other professional activities with children and families. But, while there may be some similarity in terms of the general types of activities engaged in, the roles may differ significantly in the way these activities are carried out, the problems and clinical populations dealt with, and the settings in which these professional activities take place.

As but one example, the clinical child psychologist's activities might involve an interview and administering a well-chosen selection of psychological tests to assess for the presence of Attention-Deficit/Hyperactivity Disorder (ADHD) and possible comorbid conditions (e.g., learning disability, anxiety or depressive features), collaborating with a child psychiatrist in the treatment of the child's core ADHD symptoms and using cognitive-behavioral interventions to treat the child's diagnosed anxiety or depressive disorder, in a private practice or mental health clinic setting. In contrast, a pediatric psychologist's assessment activities might entail interviews with the child and parent, consultation with the child' physician and nursing staff, and behavioral observations at meal time to determine factors contributing to a 5-year-old's frequent vomiting and failure to eat. Treatment might involve behavioral interventions designed to teach the parent how to reward successive approximations to normal eating and avoid attending to statements of stomach upset and instances of vomiting. Other interventions might involve additional forms of reinforcement for increased caloric intake, as well as other contingencies following vomiting episodes, with such treatment taking place on a pediatric floor of a university-affiliated medical center. Thus, although both may engage in similar activities (e.g., assessment, consultation, intervention), the nature of the specific activities involved may differ depending on whether the clinician is functioning in the role of clinical child or pediatric psychologist.

It should also be emphasized that child-oriented psychologists may function in roles that represent a greater or lesser degree of specificity. For example, one clinical child psychologist may engage in a wide range of child-related professional

activities such as assessment, therapy, teaching, and research, and another may function more specifically in the role of child psychotherapist. Likewise, a pediatric psychologist may functional broadly in this professional role (e.g., working with children displaying both acute and chronic illnesses and engaging in a variety of child or adolescent consult-liaison activities) or may restrict his or her professional activities, as in the case of the pediatric psychologist whose work primarily involves conducting pretransplant evaluations with children and adolescents and working with children who have recently been transplanted (who may think of himself or herself as a pediatric transplant psychologist). Finally, whereas some clinical child or pediatric psychologists assume roles that involve varying levels of specialization, as described earlier, others simultaneously function in multiple professional roles (e.g., administrator *and* clinical child psychologist), each of which may vary in terms of breadth and complexity and relationship to one another.

Despite general agreement that there is an insufficient number of adequately trained child-oriented psychologists to meet the mental health needs of children, there is an increasingly large number of psychologists who, by virtue of their training and/or professional experience, currently define themselves as clinical child or pediatric psychologists and engage in a wide range of professional activities with children, youth, and families. In the sections to follow we provide an overview of the diverse range of professional activities and functions that are associated with these professional roles. This is followed by a brief commentary on the role of school psychologists, as they also often provide clinical services to children and adolescents.

CLINICAL CHILD AND ADOLESCENT PSYCHOLOGY

The area of clinical child and adolescent psychology represents a rapidly growing area of professional psychology. A capsule summary of the multiple roles and functions played by professionals working in this area can be seen in the ways this specialty has been self-defined.

The following definition of the specialty of clinical child psychology has been provided in the archival description of the specialty submitted to the Commission for the Recognition of Specialties and Proficiencies in Professional Psychology (1998), upon the formal recognition of the specialty by the American Psychological Association in 1998:

> Clinical Child Psychology is a specialty of professional psychology, which brings together the basic tenets of clinical psychology with a thorough background in child, adolescent and family development and developmental psychopathology. Clinical child and adolescent psychologists conduct scientific research and provide psychological services to infants, toddlers, children, and adolescents. The research and practices of Clinical Child Psychologists are focused on understanding, preventing, diagnosing, and treating psychological, cognitive, emotional, developmental, behavioral, and family problems of children. Of particular importance to clinical child and adolescent psychologists is a scientific understanding of the basic psychological needs of children and adolescents and how the family and other social contexts influence socioemotional adjustment, cognitive development, behavioral adaptation, and health status of children and adolescents. There is an essential

emphasis on a strong empirical research base recognizing the need for the documentation and further development of evidence-based assessments and treatments in clinical child and adolescent psychology.

Of special note in this definition is that the training of clinical child psychologists represents (or at least should represent) a blending of developmental psychology and clinical psychology in a framework that places emphasis on the developmental aspects of psychopathology, which has as its focus understanding the mechanisms of development and behavior change. Central to the focus on developmental psychopathology is the belief that the study of atypical development can add to the understanding of normal development and, conversely, that the methods and approaches used in normative developmental science can shed light on the etiology, course, and outcome of psychological disorders. It is this focus on training in child, adolescent, and family development *and* developmental psychopathology, along with the tenets of clinical psychology, that distinguishes the well-trained clinical child psychologist from the general clinical psychologists who happen to also work with children. It needs to be emphasized that clinical child psychology is more than just the knowledge of clinical psychology methods applied to children.

This definition also highlights the fact that clinical child psychologists work with a wide range of individuals, displaying a wide range of difficulties, through the application of a wide range of assessment and treatment methods. In the sections to follow we elaborate on these topics.

Populations Served

Clinical child psychologists work with children of all ages, ranging in age from infancy and childhood through the adolescent years, and their families. Nevertheless, children of some ages tend to be seen more frequently in clinical practice. For example, Tuma and Pratt (1982) found that the most common age groups seen were children between the ages of 5 and 10 (29%). The next most common age groups seen were adolescents 14 to 18 (23%), preadolescents 11 to 13 (22%), and preschoolers below 4 (13%). These child clinicians also reported seeing a significant number of adults, with 38% reporting working with patients over age 18.

Clinical child psychologists work with children who display a wide range of problems. Unlike the pediatric psychologist, who is likely to focus more on child and adolescent health-related problems, the clinical child psychologist is likely to focus more on mental health difficulties, although the distinction between the activities of clinical child and pediatric psychologists can be blurred on occasion. Problems that clinical child psychologists are often called on to assess and treat can range from easily managed problems such as bedwetting and soiling and helping parents manage a child with a difficult temperament to internalizing problems such as anxiety and depression and externalizing conditions such as Oppositional Defiant Disorder, Conduct Disorder, and ADHD. Clinical child psychologists also see children with learning disabilities, as well as those with severe psychological disorders such as mental retardation, pervasive developmental disorders, Schizophrenia, and pediatric Bipolar Disorder. Child clinicians are often called on to deal with problems

that may or may not be defined in terms of a specific diagnosis. Here, examples include working with children of divorce, those who have been vicariously exposed to marital abuse, those who have experienced physical or sexual abuse, or those who have been impacted by hurricanes or other natural disasters.

Practice Settings

For those clinical child psychologists engaging in applied clinical work, practice settings include private practice, children's residential treatment centers, mental health clinics, and public health agencies, schools, medical school clinics. Those whose primary role is university teaching (often in departments of psychology) may engage in part-time clinical activities in the university setting. As will be seen later, child clinicians who are more health-oriented often practice in children's hospitals or sometimes in private practice pediatric settings.

Some figures regarding clinical practice settings have been provided by Tuma and Pratt (1982) in their survey of 358 clinical child psychologists who were members of what is now the Society of Clinical Child and Adolescent Psychology. The data provided by this survey suggest employment in a wide variety of work settings, with most respondents working in more than one setting. Most reported working in medical school settings, primarily in departments of pediatrics and psychiatry (51%). The second largest work setting was private practice (44%). Other work settings included graduate departments of psychology (21%), schools (15%), mental health centers (11%), child guidance centers (9%), children's hospitals (9%), and general hospitals (5%). A much smaller number were employed in residential and state hospitals.

More recent survey findings regarding practice settings of 162 child-oriented psychologists (including clinical child and pediatric psychologists and school psychologists) have been provided by Cashel (2002). Of these clinicians, 29% reported working in a medical center or other hospital settings. Other settings included college or university (22%), school system (17%), outpatient clinic (11%), and community mental health center (1.9%). As in the Tuma and Pratt (1982) survey, many of the respondents in this study worked in multiple settings, with approximately 31% working exclusively in private practice.

Taken together, these findings suggest that, across time, the primary practice settings of clinical child psychologists appear to be medical schools and hospitals, private practice, and university settings where the child clinician likely also plays a major role in clinical training and supervision. In other settings the numbers are considerably smaller, with surprisingly few clinical child psychologists working in community mental health centers.

Theoretical Orientations

Of interest is the theoretical orientation of clinical child psychologists, especially as this may show changes over time. This issue is of relevance to clinical practice as approaches to both assessment and treatment are usually, to some degree, tied to the theoretical orientation of the clinician. And, to the extent that the primary theoretical orientation of clinicians changes over time, this is likely to have implications both

for the role functions of the clinician and for the type of training those aspiring to be clinical child psychologists will need to meet changing role functions.

Data relevant to the issue of theoretical orientation have been provided by the two surveys cited earlier. In the Tuma and Pratt (1982) survey, 28% of the 358 clinical child psychologists surveyed classified themselves as having a psychodynamic orientation, 25% described their orientation as behavioral, and 4% described their orientation as humanistic-existential; the remainder described their orientation as eclectic.

These findings regarding theoretical orientation can be contrasted with the findings 20 years later (Cashel, 2002). Here, the majority of child clinicians described themselves as cognitive-behavioral (52%), and an additional 7% described themselves as having a behavioral orientation. Thus, almost 60% of the respondents endorsed either a behavioral or cognitive-behavioral orientation. Respondents endorsing other orientations included psychodynamic (9%) and interpersonal (5%), with 25% labeling themselves eclectic.

These findings suggest that during the 20-year period between 1982 and 2002, the theoretical orientations of child clinicians have changed significantly, with a vastly increased number of clinical child psychologists now endorsing a behavioral or cognitive-behavioral theoretical framework (and presumably assessment and treatment methods based on this framework), with a concurrent reduction in clinicians claiming a psychodynamic orientation. These findings would seem to be consistent with the increasing focus on both evidence-based assessment (Kazdin, 2005; Mash & Hunsley, 2005) and evidence-based child treatment methods (Ollendick & King, 2004) that currently characterize clinical child psychology.

Role-Related Professional Activities

As suggested earlier, functioning in the role of clinical child psychologist can involve participating in an extensive array of professional activities that serve to define the specialty. Among the most prominent of these activities are assessment, treatment, and conducting research on a wide range of topics relevant to children and families. A sizable number of clinical child psychologists, especially those employed in academic settings, function as teacher and clinical supervisor, in addition to their involvement in research and perhaps part-time clinical work. A somewhat smaller but still significant number of clinical child psychologists also work in the area of prevention and engage in consultation activities. In the sections to follow we highlight the work of clinical child psychologists in a number of these areas and, where relevant, the ways these activities have changed as a function of the evolution of the specialty.

Assessment Activities

Assessment has long been a major professional activity of clinical child psychologists of various theoretical orientations, and for some practitioners, providing psychological assessments for children and adolescents is their primary professional activity. Indeed, assessment is one of the most frequently engaged in professional activities of psychologists working with children. Assessment can involve

open-ended interviews, the use of intelligence tests, and the assessment of academic achievement, adaptive behavior, visual-motor functioning, as well as the use of projective tests and various questionnaires to assess personality dynamics and what are thought to be enduring personality characteristics. Assessment can also involve the use of more structured interviews to assess specific symptoms of psychopathology, the use of objective behavior problem checklists designed to assess behavioral characteristics of the child, and approaches to assess overt behaviors and the factors that elicit (antecedents) and maintain (consequences) these behaviors in the natural environment. Also commonly used are measures designed to assess aspects of parenting stress and family environment as well as other variables thought to be relevant to the functioning of the child and his or her family. Given that assessment is essential to making an accurate diagnosis of childhood problems and is usually assumed to be an essential prerequisite to the development of an appropriate treatment plan, it is not surprising that clinical child psychologists have typically devoted a significant amount of time to assessment-related activities.

It is noteworthy that the nature and, to some extent, the *type* of involvement by clinical child psychologists in assessment activities has changed over the years. For example, in the Tuma and Pratt (1982) survey, the average amount of professional time spent in assessment activities with children, adolescents, parents, and families was 28%, as compared to 41% of their time being devoted to treatment-related activities. According to the authors, assessment and treatment were the two primary professional activities engaged in by these clinical child psychologists.

These authors also provided information regarding the nature of the assessment measures used by this sample of clinical child psychologists. Here, 83% reported using intelligence tests, 54% reported using the Rorschach, 53% the Thematic Apperception Test (TAT), and 60% figure drawings. A total of 53% indicated that they used the Bender Gestalt test, 51% indicated that they used achievement tests, 18% said they used the Minnesota Multiphasic Personality Inventory (MMPI), 17% reported use of sentence completion tests, and only 13% reported the use of behavioral assessment measures. As can be seen, apart from the use of intelligence and achievement tests, the child clinicians in this survey appeared to rely heavily on the use of projective tests of various types, to the exclusion of more behaviorally oriented assessment measures.

These assessment preferences can be compared to those found by Cashel (2002), where respondents reported spending 27.3% of their time conducting psychological assessments, supporting the view that assessment continues to be a major professional activity engaged in by child-oriented clinicians. When these respondents were asked to indicate their use of various assessment measures, the Wechsler Intelligence Scale for Children was, by far, the most frequently used measure. When asked to rank other assessment approaches by order of importance, the clinical interview was ranked as most important, followed by behavioral observations, behavior ratings, self-report inventories, and, finally, projective tests (Cashel, 2002).

These findings suggest that although some clinical child psychologists still use the Rorschach, the TAT, figure drawings, and other projective approaches, the perceived value of these measures has declined, while behaviorally focused assessment measures have become increasingly more popular and appear to be more highly

valued by today's clinical child psychologist. This trend parallels the increasing focus on evidence-based assessment, where the preference is not only for useful assessment methods that are supported by research findings but also for measures that are useful in monitoring treatment effectiveness (Kazdin, 2005). Not only has the focus on evidence-based practice likely had some impact on child assessment practice, but the nature of child assessment has also been impacted by managed care. Today there seems to be less of a focus on the use of standardized test batteries and more on assessment methods relevant to specific problem areas. In the Cashel (2002) survey, it was found that the utilization of many previously popular test measures such as the Rorschach and MMPI seems to have declined due to decreased reimbursement for such measures. Although assessment continues to be a major professional activity of clinical child psychologists, the nature of the assessment measures used by child clinicians today is increasingly moving away from subjective measures and projective techniques and toward more objective, behaviorally focused, and more empirically supported assessment measures.

Intervention Activities

Along with assessment, providing treatment for children, adolescents, and their families is one of the defining professional activities of clinical child psychologists. Indeed, many clinical child psychologists in private practice function primarily in the role of child and family therapist.

It can be recalled that Tuma and Pratt (1982) found that, on average, clinical child psychologists spent over 40% of their professional time engaging in treatment-related activities with children, adolescents, and families. The types of therapy engaged in included counseling with parents, play therapy, individual psychotherapy, behavior therapy, and family therapy. Not surprisingly, the type of therapy varied as a function of the child's age, with parent counseling, play therapy, and behavior therapy being used by 63%, 44%, and 36% of the therapists, respectively. With school-age children, play therapy (53%), family therapy (37%), and individual psychotherapy (35%) were the most widely used approaches, with fewer therapists endorsing parent counseling and behavior therapy (29% each). Individual psychotherapy (74%), family therapy (46%), and counseling with parents (32%) were the most widely used therapeutic approaches with adolescents.

Today, clinical child psychologists engage in a wide range of interventions with children and families. These continue to include many of the traditional approaches to child treatment described by Tuma and Pratt (1982; e.g., parent counseling, play therapy, individual psychotherapy, family therapy, behavior therapy) but with perhaps a much greater focus on behaviorally oriented parent training approaches and behavior management approaches applied in the school, community, and health care settings. Today, we also see the widespread use of more recently developed cognitive-behavioral approaches to treatment that have been found useful in dealing with children and adolescents displaying psychological difficulties such as anxiety, depression, anger management, and Obsessive-Compulsive Disorder (O'Donohue, Fisher, & Hayes, 2003), as well as multimodal interventions for children and adolescents exhibiting complex problems.

This movement toward the use of more behavioral and cognitive-behavioral child therapies appears due to two interrelated factors: the changing theoretical orientation of child-oriented clinicians and the coming of age of evidence-based treatments. As noted earlier, 25 years ago the two predominant theoretical orientations of clinical child psychologists were psychodynamic and behavioral, with somewhat more child clinicians endorsing the psychodynamic perspective and likely engaging in treatments that were in line with their orientation (Tuma & Pratt, 1982). More recent surveys suggest that almost 60% of child-oriented clinicians now define themselves as having cognitive-behavioral or behavioral orientations, with fewer than 10% identifying themselves with the psychodynamic perspective (Cashel, 2002). This represents a dramatic shift in theoretical perspective that, along with the increased focus on the development of evidence-based treatments, likely relates to the increased use of behaviorally related approaches to intervention that we see today.

Teaching and Training Activities

As was briefly noted earlier, surveys conducted 20 years apart by Tuma and Pratt (1982) and Cashel (2002) have both found that a sizable number of clinical child psychologists work in some type of academic setting. Both found that approximately 20% worked in college and university settings, often in graduate departments of psychology. Likewise, both found that many worked in medical school settings (51% in the Tuma and Pratt survey and 29% in the Cashel survey). Employment was usually in departments of pediatrics or psychiatry.

Clinical child psychologists employed in these settings are frequently involved in training of some type. In university departments of psychology, where the academic training of clinical child psychologists often takes place, teaching can take several forms. It can involve teaching basic psychology courses (e.g., introductory psychology, personality, abnormal psychology) or teaching academic courses relevant to specialty training (e.g., developmental psychology, developmental psychopathology, child assessment, child treatment, evidence-based practice). Teaching in this setting can also involve a supervisory role in the context of various clinical practica, where the primary focus is on helping the psychology trainees develop important conceptual and clinical skills. Importantly, teaching and mentoring may also take place in the context of the student's research training, which, ideally, should at some point involve research with clinical populations so the trainee can learn how clinical knowledge can inform research and how good clinical research can inform clinical practice.

For those clinical child (or pediatric) psychologists working in medical schools, teaching formal didactic courses may be less common than teaching seminars relevant to aspects of clinical practice and the clinical supervision of trainees (e.g., interns, postdoctoral fellows, psych techs). Not infrequently, psychologists working in this setting are asked to contribute to the behavioral science curriculum of medical students or to coteach courses such as "Introduction to Patient Evaluation," which are often designed to help medical students learn clinical interviewing and relationship-enhancement skills that will serve them well as professionals working with patients.

The teaching possibilities for clinical child psychologists are many. They are also important, as the quality of training that future clinical child psychology (or other) trainees receive is likely related not only to the trainees' ultimate level of professional competence but also to the quality of care that the trainees provide children and families upon completion of training.

Research Activities

Over the years, the predominant model of graduate training for clinical psychologists has been the Boulder or scientist-practitioner model (Raimy, 1950; Shakow, 1948). This model has also been repeatedly endorsed as the preferred model for training both clinical child and pediatric psychologists. As the basic premise of the scientist-practitioner model is that the clinical child psychologist should be trained as both a scientist and practitioner, it is not surprising that clinical child psychologists often adopt the roles of both researcher and practicing clinician. Although relatively few clinical child psychologists in private practice devote a significant amount of time to research, adoption of these dual roles is not at all uncommon for clinical child psychologists who work in academic or medical school settings, where they are likely to also function in the roles of teacher and supervisor.

The range of research topics addressed by clinical child psychologists is too broad to cover here but includes research related to the development and validation of assessment measures (e.g., structured and semi-structured interview measures, intelligence and achievement measures, measures of development and adaptive behavior, behavior problem checklists, personality questionnaires, computer-based assessment measures); research related to the development and documentation of the efficacy and effectiveness of various approaches to child treatment; research related to the etiology, correlates and developmental course of various types of child psychopathology; and research on factors that may place the child at risk for later psychopathology (e.g., genetic factors, difficult temperament, divorce, abuse), just to name a few areas.

One area that deserves special attention relates to efforts to identify empirically supported psychosocial treatments for children and adolescents. This work, which has a similar focus to the evidence-based practice movement in medicine (Sackett, Richardson, Rosenberg, & Haynes, 2000), resulted from the efforts of the Society of Clinical Psychology's (APA, Division 12) Task Force on Promotion and Dissemination of Psychological Procedures (Chambless, 1996; Chambless & Hollon, 1998).

This task force, which began its work in 1995, initially attempted to develop criteria for determining psychological and behavioral approaches to treatment that were well supported by research findings. Research support was to be determined by studies of treatment efficacy, with significant treatment-related change being documented in well-controlled research investigations, emphasizing internal validity. Using the criteria adopted by the task force, it was possible to categorize psychosocial treatments of various types into two efficacy-related categories: well-established treatments and probably efficacious treatments.

To qualify as a "well-established treatment," two adequately designed group studies (from different researchers), demonstrating the treatment to be significantly better than a pill placebo, psychological placebo, or other treatment or equivalent to an already established treatment were required. Alternatively, designation as a "well-established treatment" could result from a series of > 9 well-controlled, single-subject design studies that compared the experimental treatment to some other treatment. In both instances, the nature of the samples had to be adequately described and the treatment based on a treatment manual.

Treatments could qualify as "probably efficacious" based on (1) two well-controlled studies that demonstrated the treatment to be superior to a wait-list control group or (2) one or more studies that met criteria for "well-established treatments" (see previous) but conducted by the same researcher. Probably efficacious treatments could also be documented via a series (here > 3) of single-subject design studies by a single investigator that met the well-established treatment criteria. Again, treatment was to be based on a treatment manual and sample characteristics had to be clearly specified (Chambless & Hollon, 1998).

As Ollendick and King (2004) have noted, using criteria such as these, the task force was able to identify a total of 25 largely adult-focused treatments that were judged to be empirically supported to a greater or lesser degree (18 well-established treatments; 7 probably efficacious treatments). Of these, only three treatments for children were identified as well-established treatments, and one met criteria for a probably efficacious treatment. These included behaviorally oriented interventions for children with developmental delays, enuresis and encopresis, oppositional-defiant behavior (well-established treatments), and tics (probably efficacious).

Given that the initial focus of this task force was largely on adult-related treatments, the Society of Clinical Psychology and the Section on Clinical Child Psychology (Division 12; Section 1, presently Division 53 of the American Psychological Association), both subsequently developed task forces to determine empirical support for various approaches to child and adolescent treatments. The combined efforts of these two groups provided varying levels of evidence-based support for a number of additional child treatments (Lonigan, Elbert, & Johnson, 1998). Ollendick and King (2004) have also cited other attempts to delineate the degree of support available for child/adolescent treatments, using somewhat similar research criteria. Here, they cite the work of Nathan & Gorman (1998) and Roth, Fonagy, Parry, & Target (1996). Taken together, these efforts highlight a range of child treatments that have been judged as well-established or probably efficacious (Chambless & Ollendick, 2001; Ollendick & King, 2004). These treatments are listed in Tables 5.1 and 5.2.

While the information presented in Tables 5.1 and 5.2 highlights the significant advances that have been made in documenting the evidence base for a variety of approaches to child treatment, it is important to note that the treatments listed here are not likely to represent a final tally of effective psychosocial treatments for use with children and adolescents. For example, ongoing research related to the treatment of adolescent depression is providing increasing support for the role of interpersonal therapy as an evidence-based treatment for this disorder (Mufson,

Table 5.1 Examples of Well-Established Child Treatments

Problem or Disorder	Type of Treatment
Attention-Deficit/Hyperactivity Disorder	Behavioral parent training
	Classroom behavior modification
Anxiety disorders	None
Autism	None
Depression	None
Enuresis	Behavior modification
Encopresis	Behavior modification
Obsessive-Compulsive Disorder	None
Oppositional Defiant Disorder/ Conduct Disorder	Behavior parent training
	Functional family therapy
	Videotaped modeling
	Multisystemic therapy
Phobias	Graduated exposure
	Participant modeling
	Reinforced practice

From "Empirically Supported Treatments for Children and Adolescents: Advances toward Evidence-Based Practice" (p. 8), by T. H. Ollendick and N. J. King in *Handbook of Interventions that Work with Children and Adolescents: Prevention and Treatment,* P. M. Barrett and T. H. Ollendick (Eds.), 2004, Hoboken, NJ: Wiley. Adapted with permission.

Dorta, Moreau, & Weissman, 2004). Research related to the treatment of other types of child psychopathology is currently underway. As these research efforts continue, hopefully evidence will be found for the effectiveness of an even wider range of approaches that are useful in the treatment of child and adolescent problems.

Finally, although lagging behind research related to child treatment, the developing focus on evidence-based assessment (Mash & Hunsley, 2005; Kazdin, 2005) is also rapidly becoming an active area of research focus for clinical child psychologists.

Prevention Activities

Many psychologists focus on ways of treating problems already displayed by the child, adolescent, or family; the focus of other clinical child psychologists is on ways of preventing the initial development of childhood disorders (primary prevention) and/or on the early detection of problems so as to decrease the duration of the problem and minimize its effects (secondary prevention). A third form of prevention (tertiary prevention) is designed to minimize the impact of an already existing problem or disorder; in many ways, this can be equated with therapy.

Examples of primary prevention include programs designed to reduce teenage pregnancy and/or risky sexual behavior, programs designed to enhance seatbelt or car seat usage, and child-proofing homes to prevent unintentional injuries. Also included are programs designed to prevent substance abuse and family interventions to prevent child abuse in high-risk families.

Examples of secondary prevention include programs such as those designed to detect autism in very young children so that effective behavioral treatments can be used to teach social and language communication skills. Another example is the

Table 5.2 Examples of Probably Efficacious Child Treatments

Problem or Disorder	Type of Treatment
Attention-Deficit/Hyperactivity Disorder	Cognitive-behavior therapy (CBT)
Anxiety disorders	CBT
	CBT + family anxiety management
Autism	Contingency management
Depression	Behavioral self-control therapy
	Cognitive-behavioral coping skills
Enuresis	None
Encopresis	None
Obsessive-Compulsive Disorder	Exposure/response prevention
Oppositional Defiant Disorder/ Conduct Disorder	Anger control training with stress inoculation
	Anger coping therapy
	CBT
	Assertiveness training
	Delinquency prevention program
	Parent-child interaction therapy
	Problem-solving skills training
	Rational emotive therapy
	Time out + signal seat treatment
Phobias	Imaginal desensitization
	In vivo desensitization
	Live modeling
	Filmed modeling
	CBT

From "Empirically Supported Treatments for Children and Adolescents: Advances toward Evidence-Based Practice" (p. 8), by T. H. Ollendick and N. J. King, in *Handbook of Interventions that Work with Children and Adolescents: Prevention and Treatment*, P. M. Barrett and T. H. Ollendick (Eds.), 2004, Hoboken, NJ: Wiley. Adapted with permission.

Early Head Start program, which is designed to provide early education, health, and family support services for at-risk children up to age 3 from low-income families (Raikes & Love, 2002). Research has found that children benefit from this program in terms of increased cognitive development, more intellectually stimulating home environments, and the development of better paternal parenting skills (McNeil & Bernard, 2003).

Justifiably most often thought of as treatment, an excellent example of tertiary prevention is the use of parent-child interaction therapy (Brinkmeyer & Eyberg, 2003) to reduce disruptive behavior in preschool children with oppositional-defiant behavior, with the goal of preventing the long-term negative outcomes of adult antisocial behavior.

Work in the area of secondary prevention (early intervention) has been demonstrated to be of clear importance and to be of documented value in minimizing some of the negative impacts of childhood problems that, if left untreated, would likely result in negative long-term outcomes. Unfortunately, inadequate funding of primary prevention programs has left the potential value of many such programs unrealized (Boles, Mashunkashey, & Roberts, 2003). With increased levels of federal funding for the development and dissemination of such programs, work in the

area of primary prevention could become a more attractive professional activity for clinical child psychologists.

The Training of Clinical Child Psychologists

Over the years there has been a dramatic increase in the number of clinical child psychology training programs, which has been prompted, in part, by the shortage of a sufficient number of adequately trained psychologists to meet the mental health needs of children (Knitzer, 1982). In the 1976–1977 edition of the American Psychological Association's *Graduate Study in Psychology,* only eight graduate programs self-identified as offering specialty training in clinical child psychology (Johnson, 2003). A survey of graduate training programs in 1982 was able to identify only 15 doctoral programs that provided clinical child specialty training (Roberts, 1982). By 1995, the Directory of Graduate Programs in Clinical Child/Pediatric Psychology (Tarnowski & Simonian, 1995) listed more than 100 programs in the United States and Canada that *self-reported* offering training in clinical child or pediatric psychology, although many of these would likely not meet strict criteria for a formal training program. Indeed, it seems likely that the number of graduate training programs offering quality specialty training in clinical child psychology is well below this number.

Given the need for *adequately trained* clinicians to meet the mental health needs of children *and* the increasing number of programs offering training in clinical child and pediatric psychology, considerable attention has been given to developing training guidelines to ensure adequate preparation of those entering the field. Over the years, three efforts in this regard are especially noteworthy.

First was the work of a task force of the APA Division of Child Youth and Family Services (Division 37), which proposed initial general guidelines (academic coursework, research experiences, and applied training in assessment and therapy) for training psychologists for working with children, youth, and families (see Roberts, Erickson, & Tuma, 1985). This work was followed by the national conference on "Training Clinical Child Psychologists," held at Hilton Head, South Carolina, in May 1985. Here, participants, who were leaders in the fields of clinical child and pediatric psychology, endorsed the general recommendations of the Division 37 Task Force and a scientist-practitioner model of training that emphasized training clinical child psychologists to function both as scientist and clinician and agreed on a range of other recommendations for clinical child training at the graduate, internship, and postdoctoral levels (Johnson & Tuma, 1986; Tuma, 1985).

A more recent attempt to integrate and elaborate on the Hilton Head conference guidelines resulted from the efforts of a National Institute of Mental Health Center for Mental Health Issues Task Force, which held a subsequent training conference at the University of Kansas in 1993. The results of these efforts, detailed by Roberts et al. (1998), represent an elaborate and well-considered framework for the professional training of individuals desiring to work with children and families. It was recommended that trainees be exposed to a wide range of didactic and applied training experiences in (a) life span developmental psychology; (b) life span developmental psychopathology; (c) child, adolescent, and family assessment;

(d) intervention strategies; (e) professional, ethical, and legal issues pertaining to children, youth, and families; (f) research methods and approaches to system evaluation; (g) issues of diversity; (h) prevention, family support, and health promotion; (i) the role of multiple disciplines and service delivery systems; (j) social issues affecting children, youth, and families; and (k) specialized applied experiences in assessment, intervention, and consultation.

These recommendations indicated that clinical training in these areas should be designed to progress sequentially from simple exposure to the development of expertise in various areas, that they should involve structured research experiences relevant to the specialty, and that internship training should build on predoctoral training and provide a foundation for postdoctoral work. This framework appears to provide an excellent foundation for clinical child specialty training. Building on these guidelines, additional training recommendations related specifically to pediatric psychology training have been outlined by Spirito et al. (2003).

A significant issue in clinical child psychology training is that, at present, there are few quality assurance mechanisms to ensure that graduate programs purporting to offer specialty training do, in fact, offer training consistent with recommendations repeatedly endorsed by professional organizations representing the specialty. This is, in part, due to the fact that historically there has been no mechanism for the accreditation of specialty programs apart from clinical, counseling, and school psychology. Thus, although graduate programs in these areas are required to conduct self-studies and meet APA Committee on Accreditation standards to initially get or maintain accreditation, there are currently no provisions to ensure that clinical training in specialty tracks (where most of clinical child training takes place) is consistent with existing specialty training guidelines. It is clear that many existing clinical child psychology training programs provide quality graduate education in this area, as evidenced by a strong clinical child training faculty, a formal sequence of required clinical child training experiences (that are consistent with specialty training guidelines), and their having a strong track record of turning out successful clinical child psychologists. Quality assurance mechanisms are necessary to ensure that all programs purporting to offer clinical child training meet these standards.

Clinical Child Psychology: Present and Future

Clinical child psychology, along with pediatric psychology, has grown rapidly over the past several decades. Since the 1960s, when the first interest groups related to clinical child psychology were organized, the field has grown in the number of professionals identified with the area and in the number of graduate programs designed to train clinical child psychologists. Likewise, as noted, the area has developed clinical training guidelines in an initial attempt to ensure that those trained in the area are trained well.

After many years of being viewed as a subspecialty of clinical psychology, clinical child psychology has now been formally recognized as a separate specialty in professional psychology, and the primary organization representing the area, the Society of Clinical Child and Adolescent Psychology, has evolved from being a section of the APA Division of Clinical Psychology to being an APA division in

its own right, as has the Society of Pediatric Psychology. In the short time since its inception, research productivity of the members of the specialty has grown at an ever-increasing rate. Much of this research has been published by two journals maintained by these divisions, the *Journal of Clinical Child and Adolescent Psychology* and the *Journal of Pediatric Psychology*, which have grown from humble beginnings to currently enjoying the status of first-rate psychology journals. Cutting-edge research in the area of clinical child psychology is also regularly presented at the biannual Kansas Conference on Clinical Child and Adolescent Psychology, and current research in the area of pediatric psychology is presented at the biannual National Conference on Child Health Psychology.* The growth of the area is also highlighted by the fact that it is now possible for evidence of excellence in clinical child practice to be recognized through specialty board certification in clinical child psychology through the American Board of Clinical Child and Adolescent Psychology.

As suggested in the formal petition to the APA for specialty recognition in 1998, the area of clinical child psychology has indeed become a "well-developed, legitimate, and formally recognized area of clinical and research specialization, characterized by the development of an ever increasing body of specialized knowledge and a vibrant, diverse, and specialized area of practice" (Commission for the Recognition of Specialties and Proficiencies in Professional Psychology, 1998, p. 2). Given the rapid growth of the specialty during the past 20 to 25 years, questions arise as to where the specialty of clinical child psychology will go in the future. Here, change and advancement are likely on several fronts.

First, it seems likely that the evolving nature of the field will bring about significant changes in the practice of clinical child psychology in the years to come and result in new and different role functions for those identified with this area. As noted earlier, it appears that clinical child psychology has begun to experience a significant shift in the theoretical orientation of new clinical child psychologists coming into the field. There is clear movement toward an increasing number of practitioners with behavioral and cognitive-behavioral theoretical orientations and fewer clinical child psychologists with a predominantly psychodynamic orientation. Given this trend, in the future we will likely continue to see a de-emphasis on training in projective techniques and other more subjective assessment measures and an increased emphasis on more objective and evidence-based assessment measures. This is likely to be accompanied by a corresponding training emphasis on more evidence-based treatments for childhood disorders, which at this point seem to be interventions that are primarily behavioral or cognitive-behavioral in nature. These already evident trends in clinical training will likely also be reflected in the clinical practice of those clinical child psychologists entering the workforce.

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*The National Conference on Clinical Child and Adolescent Psychology was founded in October 1994 by Michael Roberts and colleagues in the Clinical Child Psychology program at the University of Kansas. The National Conference on Child Health Psychology (formerly the Florida Conference on Child Health Psychology) was founded in 1988 by James H. Johnson and Suzanne Bennett Johnson and their colleagues in the Department of Clinical and Health Psychology at the University of Florida.

It is also likely that the impact of managed care will continue to influence clinical practice in terms of assessment and treatment approaches as evidence-based assessment and treatment methods are likely to become increasingly favored for reimbursement.

Consistent with these projections, it is safe to predict that the next several years will see an increased focus on clinical child psychology research related to various aspects of evidence-based practice. This will likely include expanded efforts in researching evidence-based treatments for children and adolescents, including treatments other than those based on behavioral and/or cognitive-behavioral principles, and increased research focus on evidence-based assessment as well as research on the *integration* of evidence-based assessment and evidence-based treatment methods.

As regards child and adolescent treatment we can also expect to see a significant increase in research on the effectiveness of those child treatments that have been determined to be efficacious. It is clearly important to conduct research to document significant treatment effects in the research setting; it is also essential to document the effectiveness of these treatments in applied clinical settings. This research on effectiveness is essential and will likely become a major focus of child treatment research in the near future. Related to the issue of effectiveness, one can safely predict an increasing focus on finding ways of disseminating empirically supported treatments so they are likely to be used in clinical practice as well as researching ways of individualizing manualized treatments to ensure treatment fidelity while also making them more flexible and hence more palatable to practitioners (Kendall & Beidas, 2007). Continued advances in this area of research and the increased dissemination of evidence-based approaches to treatment are likely, over time, to have a significant impact on the role functions of clinical child practitioners in terms of their increasing reliance on empirically based methods of treatment. This would be a very positive outcome for children.

The changing roles of clinical child psychologists have also been highlighted by Prinstein and Roberts (2006) who have noted that, upon completing their graduate training, new clinical child psychologists are increasingly obtaining positions in new settings such as schools, primary care, and corporations that involve new role-related activities such as consulting, public policy, program evaluation, and the supervision of nonpsychology health care professionals. Regarding this trend, Prinstein and Roberts suggest that significant modifications of professional training will likely be needed to prepare clinical child psychologists for assuming these new employment opportunities and diverse professional roles in the future.

PEDIATRIC PSYCHOLOGY

Primary to the science and practice of pediatric psychology is an emphasis on health- and illness-related issues. More specifically, "pediatric psychology is a field of science and practice that addresses a range of physical, psychological, developmental, health, and illness issues affecting children and their families" (Roberts, Mitchell, & McNeal, 2003, p. 3). This often involves working with children, and their families, who are struggling with chronic (cystic fibrosis, asthma, diabetes) or acute

(enuresis, abdominal pain, headaches) health conditions. Compared to clinical child psychologists, pediatric psychologists are also more frequently involved in prevention and health promotion activities, such as promoting more healthful lifestyle behaviors and practices. Some psychologists who work with pediatric populations on health-related issues prefer to be referred to as child health psychologists. Those who differentiate between the two terms note that the term pediatric psychologist is more often associated with psychologists who focus on treatment of children with chronic health conditions, whereas child health psychologist often denotes a wider scope of practice, including work with acute and chronic health conditions, as well as general prevention and health promotion. However, for the purposes of this chapter, we use these terms interchangeably.

Practice Settings

There are a wide variety of settings in which pediatric psychologists practice their craft. The practice of pediatric psychology originated in inpatient medical centers dedicated to the treatment of disease and illness (Roberts et al., 2003). A recent survey of the members of the Society for Pediatric Psychology, Division 54 of the APA, found that pediatric psychology is still primarily a hospital-based field, with a majority (63%) of pediatric psychologists working in hospital settings (Opipari-Arrigan, Stark, & Drotar, 2006). Practice in inpatient settings often involves consultation services and treatment of children exhibiting less than optimal adjustment to illness or illness management behavior. Carter and colleagues (2003) completed a case-controlled study examining the nature of referrals to pediatric consultation and liaison service in 104 children. Reasons for referrals included problems in coping and adjustment (34.6%), medication and treatment noncompliance (30.7%), depression (29.8%), pain management (15.4%), anxiety (15.4%), illness exacerbation (14.4%), acute evaluation (13.4%), new medical diagnosis (13.4%), difficulties in parent coping (14.4%), ongoing treatment (12.5%), decision making regarding treatment (9.6%), differential diagnosis (9.6%), acting out (7.7%), general support (6.7%), and family conflict (6.5%), with many patients having multiple reasons for referral.

Hospital-based services may also involve consultation and clinical practice activities in emergency departments conducting assessments for depression or suicide risk, assessment and triage for victims of acute trauma or acute illness exacerbations, or assessment of children in acute psychosocial crisis. Medically based practice by pediatric psychologists also involves practice in medical outpatient clinics either independently or as part of multidisciplinary teams. The latter may include teams headed by pediatric subspecialties from endocrinology, pulmonology, gastroenterology, hematology, or rheumatology addressing health and coping issues related to diabetes, cystic fibrosis, asthma, inflammatory bowel disease, pediatric feeding aversion, sickle cell disease, juvenile rheumatoid arthritis, or other conditions. Pediatric psychologists also practice alongside health care professionals in primary care settings addressing acute health conditions (i.e., recurrent abdominal pain, headaches, toileting and soiling issues, sleep problems), as well as general emotional and behavior problems in children (Schroeder, 1979).

The science and practice of pediatric psychology is not limited to medical settings. Whether primarily involved in practice, training, research, or a combination of the three, pediatric psychologists also engage in role-related activities essential to the field in a variety of other settings. Whereas Opipari-Arrigan et al. (2006) found that most respondents worked in hospital settings (with about half in academic medical centers), a number of other practice settings were also noted. These included private practice (22%), academic psychology departments (5%), academic departments other than psychology (15%), mental health agencies (3%), and school systems (2%). Pediatric psychologists can also be found working or volunteering in camp settings designed for children suffering from chronic health conditions (Roberts et al., 2003). The following sections turn to the broad role and role-related activities engaged in by pediatric psychologists working in various settings.

Role-Related Professional Activities

As with clinical child psychologists, pediatric psychologists engage in a wide range of specialty-specific role-related activities. These include consultation, assessment, treatment, health promotion, promoting public health policy, training, and research. The ways pediatric psychologists carry out these activities in the context of their professional roles are detailed in the following sections.

Consultation Activities

As noted earlier, consultation is a critical role for pediatric psychologists, as our medical colleagues often represent the primary sources of patient referrals. The level of involvement between pediatric psychologists and medical colleagues during consultation may vary as a function of the case involved, the proximity of colleagues, or the presence of a formalized structure to support consultation. Consultation may be limited to brief discussions regarding general principles and strategies for dealing with children and families, provision of advice relevant to a specific case, brief involvement in assessment or development of a treatment plan, or active long-term involvement in patient treatment and case management. There are various models for the level of collaboration (Drotar, 1995; Roberts & Wright, 1982). For example, pediatric psychologists can function independently from medical colleagues (e.g., in their own outpatient clinic), providing assessment, diagnosis, and, if necessary, treatment of referred patients. Alternatively, they can function as part of a collaborative team in the assessment and treatment of children and families. This may involve working side by side with medical colleagues in multidisciplinary clinics, marked by active involvement and shared responsibility by team members leading to collaborative case conceptualization and treatment recommendations. It is within the rubric of consultations that pediatric psychologists often perform a number of functions, listed next.

Assessment Activities

As in all domains of applied clinical psychology, assessment is a central function of pediatric psychologists. In the ideal situation, most assessment leads to the

development of recommendations or potential treatment options to help address areas of concern. Assessment by pediatric psychologists commonly focuses on child and family behaviors, interaction patterns, coping strategies that impact adjustment to the child's medical condition, including the challenges associated with managing the child's illness, the implications for the long-term health of the child, and social and emotional sequelae of the health condition. Common strategies to gather information include (a) structured or nonstructured interviews with children and their parents; (b) completion of self-report or parent-report questions; (c) completion of questionnaires by schoolteachers or other professionals working with the child; (d) direct observations of family interactions patterns, child social skills, or completion of illness management tasks; and (e) self-monitoring of emotions, cognitions, or behaviors.

A couple of examples may help to illustrate the assessment process. Children with cystic fibrosis are often required to engage in a variety of illness management tasks, including increased caloric intake, airway clearance (chest physiotherapy and medications to loosen mucus in the lungs), pancreatic enzyme replacement therapy, and antibiotic medications to fight bronchial infections. Adherence to these treatment tasks, whether in children with cystic fibrosis or other illness conditions, is one of the most common and challenging referrals to pediatric psychologists. Accurate assessment of the actual adherence to various treatment tasks and barriers hindering adherence (including antecedents and consequences of engaging in or not engaging in certain behaviors) are critical to developing appropriate and effective intervention plans that ultimately can impact the child's long-term physical health and quality of life. Interviews (focusing on the child, parents, the medical staff, and possible school personnel), behavioral checklists and self-report questionnaires, and written logs can be critical to documenting adherence and critical barriers. In addition, over the past 20 years electronic monitoring devices have become an invaluable tool in providing more accurate assessment of adherence to various illness management tasks. These include metered dose inhalers and medication event monitoring systems that record the date and time of vial openings and closings, as well as chest physiotherapy assistance devices, such as the Flutter or the high-frequency chest compression vest, which vibrates to loosen mucus in the lungs and can track adherence to chest physiotherapy.

Assessment may also involve evaluation of child and family functioning in anticipation of challenging future medical procedures or health issues. A common example is pretransplant evaluation of children presenting for organ transplantation. Pretransplant evaluations assess a wide variety of critical strengths, potential barriers to graft maintenance, and risk factors that may impact a child's or family's ability to follow the pre- and posttransplant treatment regimen, as well as posttransplant adjustment and coping. These include, but are not limited to, an assessment of (a) adherence to treatment regimen (taking prescribed medications, adherence to dietary recommendations), (b) understanding of the transplant process; (c) emotional coping and adjustment of child and parents or legal guardians; (d) presence of oppositional or disruptive behavior; (e) academic or social functioning; (f) social, emotional, and logistical support available to the family; (g) family communication; (h) family's ability to continue to complete daily life management tasks (grocery

shopping, laundry, helping siblings with schoolwork, etc.); (i) family finances; and (j) other stressors such as additional family illness, potential job loss, and marital conflict. Accurate and reliable assessment from a multidisciplinary team is critical to establish a coherent picture of how these different areas ultimately can impact the child's and family's long-term health, quality of life, and maintenance of the graft.

Finally, assessment may involve psychoeducational assessment, including formalized measures such as the Wechsler Intelligence Scale for Children, the Stanford-Binet Intelligence Scale, or the Wechsler Individual Achievement Test to assess both cognitive functioning and academic achievement. Such testing may be advised to provide a baseline and track potential changes in cognitive functioning for children undergoing radiation or chemotherapy treatment for cancer or those who have experienced head trauma or other medical conditions that can lead to deterioration in cognitive functioning over time.

Treatment Activities

Treatment of children with acute or chronic health conditions is one of the most commonly recognized functions of pediatric psychologists. However, it is clear that treatment by pediatric psychologists (and clinical child psychologists), as well as others, does not occur in isolation. Prior to engaging in intervention efforts practitioners must travel a long road of didactic and applied clinical training to develop skills and competencies in evidence-based approaches to treatment. Moreover, prior to and throughout the course of treatment, assessment is essential to elucidate targets and pathways for change and to monitor ongoing progress. Research, either by the practitioner delivering the intervention or, more likely, by other psychologists and health care professionals, is necessary to provide data on best practices and evidence-based treatments to guide and inform intervention efforts. In fact, treatment-related research is becoming critically important, given the growing emphasis on evidence-based treatment in pediatric psychology. These treatments are highlighted by the ongoing series on empirically supported treatments in the *Journal of Pediatric Psychology*.

Potential child health-related difficulties that require intervention are numerous, but commonly involve coping and adjustment issues, adherence to treatment recommendations, or pain management. Problems of coping and adjustment may relate to death and dying, limits of daily functioning, family communication and conflict, school reentry, social interaction difficulties, or improving self-esteem. Adherence to treatment recommendations may involve increasing adherence to medication regimens, dietary recommendations (for cystic fibrosis, diabetes, inflammatory bowel disease, constipation), airway clearance (for cystic fibrosis), or increased physical activity (for obesity, Type 2 diabetes, juvenile rheumatoid arthritis). Pain management and coping with medical procedures may involve the use of diaphragmatic breathing, muscle relaxation, cognitive imagery or other active coping strategies to assist with injections, lumbar punctures, nausea associated with chemotherapy, or pain associated with sickle cell crisis or tumors. Treatment efforts often focus not only on the ill child, but also the parents and other members of the immediate and extended family, given the

bidirectional influences between the child and his or her immediate family members. As with other functions performed by pediatric psychologists, collaboration with other health care and school professionals in the child's environment is essential as effective treatment often requires intervention at multiple levels in the child's environment.

Teaching and Mentoring Activities

Many pediatric psychologists spend valuable time training and mentoring the next generation of pediatric psychologists in clinical service delivery, research methodology, grant writing, and professional practice. Mentoring may involve formal and/or informal interactions of students, interns, and postdoctoral fellows. Senior faculty can also play a valuable role in the mentorship of junior colleagues and trainees. As pediatric psychologists strive to promote the value of their services to their medical colleagues and hospital administrators, education of professionals in other health care disciplines can be invaluable. This can take the form of active involvement in teaching trainees from other health care disciplines or engaging in informal teaching interactions with more senior medical colleagues (Drotar, Spirito, & Stancin, 2003; Roberts et al., 2003). This topic is reviewed in more depth later in this chapter.

Health Promotion Activities

Pediatric psychologists are becoming increasingly involved in health promotion activities such as developing and implementing lifestyle interventions to improve dietary and physical activity habits, reducing risk for childhood injuries, reducing drug and alcohol abuse, reducing tobacco use, parent training for adolescent mothers to reduce the risk for child neglect or abuse, and osteoporosis prevention. For example, Stark and colleagues (Stark, Janicke, McGrath, Mackner, & Hommel, 2005) demonstrated the effectiveness of a behavior intervention designed to improve calcium intake and bone mineral content in children with juvenile rheumatoid arthritis and inflammatory bowel disease. More recently, pediatric psychologists have ventured into the public health domain and are working to help promote public health agendas through more broad-based community initiatives (Drotar et al., 2003). There are a variety of examples of quality research in this area. One excellent example is Planet Health (Gortmaker et al., 1999), a 2-year school-based intervention that has been demonstrated to have positive effects on the weight status of middle school youth (Gortmaker et al., 1999). Another obesity prevention program is the GEMS project (Girls' Health Enrichment Multisite Studies), which is an ongoing, multisite community-based program designed to identify and implement effective obesity prevention strategies for African American girls (Rochon et al., 2003). Finally, Levy and colleagues (Levy, Brugge, Peters, Clougherty, & Saddler, 2006) have recently reported on a community-based participatory research intervention that has demonstrated reductions in respiratory symptoms of pediatric asthmatics living in public housing. These types of programs demonstrate the positive impact that innovative prevention and health promotion efforts can have on children's long-term health and quality of life.

Research Activities

As greater demands are placed on the profession to provide evidence supporting the effectiveness of psychological interventions for health-related problems and as competition for the provision of services is growing from social workers and other medical professionals, research activities are taking on greater importance for pediatric psychologists. Many of our medical colleagues have recognized what pediatric psychologists have to offer as active members of a multidisciplinary team, and subsequently are very willing to collaborate on research endeavors related to pediatric and child health psychology. One of the most important areas for future collaboration is in the area of treatment outcome research, where there has been a growing emphasis on developing evidence-based assessment and treatment methods. There is also a greater push to promote translational research and practice, to improve access to effective mental health care in diverse populations and community settings (Glasgow, Lichtenstein, & Marcus, 2003; Strauman & Merrill, 2004).

A major focus of treatment outcome research is to enhance adherence to treatment recommendations in children struggling with chronic and acute health conditions such as diabetes, cystic fibrosis, and asthma. Another common theme of current research in pediatric psychology is the identification of mediators and moderators of positive coping and adjustment in children with chronic health conditions. Finally, pediatric psychologists are at the forefront of integrating modern technology to assist in data collection and distribution of treatments for children and families, including telehealth, accelerometers to track physical activity, metered dose inhalers to track medication use, and palm pilots to assist children with accurate self-monitoring of dietary intake or physical activity.

Pediatric Psychology Compared to Clinical Child Psychology

In many ways, the field of pediatric psychology can be considered a subspecialty of clinical child psychology. Strong training in clinical child psychology is essential, but not sufficient, to function as a pediatric psychologist. Skills in assessment and intervention and knowledge of the role of developmental processes and child psychopathology are critical for pediatric and clinical child psychologists. However, there are significant differences between these disciplines. Pediatric psychologists place a much greater emphasis on medically related issues and regularly work with children and adolescents experiencing acute and chronic health conditions. Pediatric psychologists spend more time working in medical settings and collaborating with medical professionals from a variety of disciplines. As a result, they must have a greater understanding of disease processes, medical management of illness, broader health systems issues, and the referral and consultation process than clinical child psychologists.

Frequently the time allotted to pediatric psychologists to complete assessments or intervention is less than that typically allocated to clinical child psychologists (Tuma & Grabert, 1983). Children often are admitted to inpatient units for only a few days, or weeks, at a time. Many families may drive great distances for expert pediatric care. Primary care settings, another common venue for pediatric

psychologists, are often fast-paced environments with a large number of patients, allowing for a limited amount of time with each family. In multidisciplinary clinics, often there is only a brief amount of time allotted to each member of the treatment team to work with a family. Moreover, whether in primary care settings, outpatient clinics, or hospital settings, pediatric psychologists may be forced to work with a child and/or family members in inpatient or exam rooms, with numerous interruptions. Given these limitations, brief, targeted, and flexible therapies that can address critical problematic behaviors or coping skills in a short period of time are often emphasized.

Training for Pediatric Psychology

As noted in the preceding sections, there are a variety of role-related professional activities and opportunities for practice open to pediatric psychologists. Not surprisingly, students pursuing a career in pediatric psychology require training to help them develop a blend of general or core skills similar to those developed through training in clinical child psychology, as well as specialized training related to the unique roles, settings, and contexts in which pediatric psychologists function. For many students, core training in child therapy and research skills comes first through coursework and in-house practicum training in graduate school. Fundamental therapy process skills and behavioral, cognitive-behavioral, interpersonal, and family systems intervention strategies form a foundation on which more specific training in pediatric psychology can build.

Regarding those elements seen as necessary for adequate training in this area, the Society of Pediatric Psychology's Task Force on Training of Pediatric Psychologists (Spirito et al., 2003) has outlined 11 domains of training that are believed to be necessary to develop the requisite knowledge and build the general and specific skills necessary for pursuing a career in pediatric psychology. These include training in the following: (a) life span developmental psychology; (b) life span developmental psychopathology; (c) child, adolescent, and family assessment; (d) child and family intervention strategies; (e) research methods and systems evaluation; (f) issues of diversity; (g) the role of multiple disciplines in service delivery systems; (h) prevention, family support, and health promotion; (i) social issues affecting children, adolescents, and families; (j) consultation and liaison roles; and (k) disease process and medical management.

One of the most salient aspects of training, unique to pediatric psychologists, is the need to have opportunities to work in interdisciplinary settings with professionals and trainees from other health professions (Spirito et al., 2003). Primary care and other outpatient clinical settings offer a fast-paced environment, where pediatric psychologists must think on their feet, remain flexible, build skills in effective but targeted brief therapy interventions, and learn to effectively communicate findings and impressions while also coordinating care with other health professionals. Given that most university academic settings have limited access to hospital-based training experiences, most in-depth hospital-based experiences usually occur during the internship and postdoctoral training experiences, although there are certainly exceptions to this rule.

Once again, the changing health care environment highlights the continuing need for additional treatment outcome research by pediatric psychologists to demonstrate the worth of our services to both patients and health care professionals. Conducting successful treatment outcome research in medical or community settings requires flexibility and teamwork, and knowledge of disease processes and medical systems, a deep respect for the issues and challenges presented to families of children struggling with acute and chronic health conditions, and research methodology considerations are critical. Almost all doctoral students in clinical psychology receive training in research methods and the evaluation of treatment outcomes. However, training students to conduct treatment outcome research studies on the scale required to support judgments of long-term efficacy and effectiveness necessitates more specialized training and experience. This may include, but are not limited to (a) training in participant recruitment and retention, (b) assessment methodology unique to specific health conditions, (c) longitudinal research design and corresponding data analytic approaches appropriate to such designs, (d) data management programs and protocol, (e) supervision and training of research team members and interventionists, (f) facilitating and maintaining collaboration with medical colleagues, (g) dissemination and translational research approaches, and (h) ethical considerations in the conduct of treatment outcome research.

Given the nature of the current health care environment with its increasing limitations on reimbursement for services provided by pediatric psychologists and the constraints on fiscal survival and success, "pediatric psychologists must be advocates not only for our services but also for the resources that support them" (Kronenberger, 2006, p. 648.). Thus, training in basic business management practices such as marketing of services, accounting, and financial management seem increasingly relevant to the ability to survive and, eventually, flourish as a profession. This may start with coursework during graduate training, but should be supplemented with seminars and hands-on practice during internship and postdoctoral training. In addition, as funding sources become more limited, training in grant writing will be beneficial for research and clinical pediatric psychologists. Again, these skills can be built through both didactic coursework and seminars, as well as hands-on experience assisting mentors with their applications or by submitting applications for their own National Research Service Awards during graduate and postdoctoral training.

Future Directions and Key Issues for Pediatric Psychologists

There are a number of key issues facing the field of pediatric psychology that will ultimately impact the direction of future practice as well as the viability of the field. Many of these issues were outlined by an expert panel of pediatric psychologists (Brown & Roberts, 2000), and some have been discussed previously in this chapter. The most important issue identified by respondents was related to potential threats to the viability of pediatric psychology. In response to this threat, there is a growing need to demonstrate the effectiveness of intervention protocols through controlled treatment outcome studies, a greater emphasis on interventions that improve health status as opposed to just coping with illness, and a greater need for

research examining the cost offset associated with psychological interventions. It will also be important for pediatric psychologists to assess alternative sources of funding and reimbursement (Drotar, 2004), as many of the services provided by pediatric psychologists, such as improving adherence, adjustment to chronic illness, consultation with medical colleagues, and pain management, are not technically in the categories of the *Diagnostic and Statistical Manual of Mental Disorders*. Increasing awareness of what pediatric psychologists can do in medical settings by marketing our services to medical professionals as well as patients may help ensure the viability of pediatric psychology as a field of practice (Bradford, 2004).

Many of the populations and issues researched by pediatric psychologists involve small numbers of patients in hospital settings. As such, future research, especially treatment outcome research, will require multisite investigations to enroll adequate numbers to provide sufficient power to test study hypotheses.

Most pediatric psychologists are congregated in large academic medical centers. As a result, many people have to travel long distances for pediatric care as they do not have access to pediatric psychology services in their home town (Kronenberger, 2006). Thus, a significant opportunity for future growth of pediatric psychology lies in outpatient, primary care pediatric settings. Development and behavioral problems often first present to primary care providers, who have been referred to as the gatekeepers to the health care system. Such a setting would seem ideal for pediatric psychologists to make significant contributions to the health status and quality of life of children and families and to demonstrate their usefulness and the benefit of their services to medical colleagues. The call for pediatric psychologists to build practices in primary care settings has been echoed over the past few decades, yet few psychologists have successfully accomplished this transition. Renewed efforts in this direction could provide big dividends to pediatric psychologists and may be vital for the future of our field.

New and emerging technologies provide pediatric psychologists with a variety of opportunities to expand their service net. Genetic testing, for example, has the potential to provide families with information on the risk of children developing life-threatening diseases, chronic conditions, and development irregularities (Patenaude, 2003). There are a variety of opportunities for research to ascertain the impact of knowledge of risk on adjustment and decision making, as well as the best ways to communicate information to parents and children (Tercyak, 2003). Another emerging technology is telehealth, which provides opportunities to bring treatment to children and families in underserved areas, but expertise and research will be required to determine effective strategies to intervene using this new medium (Harper, 2003).

Prevention and health promotion are areas that hold the potential for long-term health benefits and medical cost offsets. As noted earlier, there are numerous opportunities for pediatric psychologists to make valuable contributions in these areas. Included here is work in nutrition and exercise to impact weight status and physical fitness, new and more comprehensive efforts to reduce unintentional injuries, multimodal parent training programs to reduce abuse and neglect, and approaches to osteoporosis prevention, just to name a few. Historically, reimbursement for prevention activities has been limited; however, the growing emphasis on translation

and dissemination research and practice may provide more opportunities for pediatric psychologists to provide mental health services in underserved communities (Glasgow, Klesges, Dzewaltowski, Bull, & Estabrooks, 2004; Glasgow et al., 2003).

SCHOOL PSYCHOLOGY

Fagan and Wise (2000, p. 4) define a school psychologist as "a professional psychological practitioner whose general purpose is to bring a psychological perspective to bear on the problems of educators and the clients educators serve" and emphasize the "educational and psychological foundations" of the discipline. Unlike clinical child and pediatric psychology, school psychology is unique in terms of its focus on the functioning of individuals, groups, and systems in the school setting. As an area, school psychology also differs from clinical child and pediatric psychology in that the majority of practicing school psychologists have a master's or educational specialist degree rather than the doctorate, with the specialist degree being the most common. A National Association of School Psychologists membership renewal survey in 1999 found that 54% of school psychologists held a specialist degree, whereas only 26% held a doctoral degree (Fagan & Wise, 2000).

Not surprisingly, the most common setting for school psychologists is in schools, with approximately 80% of practitioners working in elementary or secondary schools. Private practice (3% to 5%) appears to be the second most common setting for school psychologists, and government agencies, community mental health centers, developmental disability centers, and universities are among some of the other work settings (Fagan & Wise, 2000). It should be noted that doctoral-level school psychologists frequently work with clinical populations outside of the school setting and engage in clinical activities similar to those performed by clinical child psychologists.

Role-Related Activities of School Psychologists

School psychologists engage in a variety of role-related professional activities, such as psychoeducational assessment, intervention, consultation, and, to a lesser degree, research and program evaluation, training, and supervision. These activities are not mutually exclusive as school psychologists often perform these activities concurrently.

Assessment Activities

School psychologists have historically spent much of their time conducting psychoeducational assessments. Indeed, Reschly (2000) has suggested that school psychologists spend approximately 50% to 55% of their time engaged in activities related to psychoeducational assessments. These assessments typically serve to gather information regarding the child's current difficulties to help plan interventions and/or provide a basis for special class placement or classroom accommodations. Presenting problems dealt with by school psychologists can relate to a range of possible areas, including academic, social, emotional, and behavioral functioning. Assessments can include multiple components, including review of school

records, classroom observation, and interviews with parents, teachers, and other persons who may be able to provide information related to the child's presenting difficulties. They can also include testing of cognitive ability, academic achievement, and perceptual and motor skills, as well as the administration of standardized rating scales measuring behavior problems, personality, and adaptive functioning (Fagan & Wise, 2000). It is noteworthy that the assessment activities of school psychologists are to some degree shifting away from an emphasis on psychoeducational testing toward more functional assessments designed to generate and evaluate interventions (Reschly, 2000).

Intervention Activities

School psychologists report spending approximately 20% of their time engaged in intervention-related activities, including planning, implementation, and evaluation of interventions (Reschly, 2000). These interventions can take various forms, including both individual and group counseling and planning and implementing classroom-based intervention. Individual counseling can focus on helping children learn skills and coping strategies to deal with a range of difficulties in the areas of academic, social, emotional, and behavioral functioning. Group counseling is often used to help children learn and use adaptive skills related to peer interactions, dealing with anger, and stressful situations such as divorce or death of a parent. Classroom-based interventions can involve collaborating with the child's teacher to arrange a particular seating arrangement, selecting an appropriate classroom placement, finding additional academic supports such as tutoring, and implementing behavior management programs in the classroom.

Consultation Activities

School psychologists are reported to devote approximately 23% of their time to consultation activities (Reschly, 2000). Consultation has been viewed as a collaborative, problem-solving endeavor between school psychologists and other personnel, often teachers, to help ensure that assessment-based recommendations are implemented (Fagan & Wise, 2000). Consultation can involve multiple components, including problem identification, assessment, and intervention planning, implementation, and evaluation. It has also been conceptualized as a way to help address the potential shortage of school psychologists in the future, given the potential to impact beyond the individual. For instance, consultation with a teacher can lead to interventions that have an impact on a whole classroom, and consultation on a systemic level may have the potential to effect positive change in an entire school district. Consultation has also been viewed as a way of promoting preventive strategies by providing children with supports, increasing their subjective sense of well-being, and building competencies (Meyers, Meyers, & Grogg, 2004).

Other Role-Related Activities

Unlike clinical child and pediatric psychologists, only a small percentage of school psychologists appear to be involved in conducting applied research and program

evaluation. Survey findings suggest that only about 2% of school psychologists engage in these activities (Reschly, 2000). This low number may be due in part to the fact that the large majority of school psychologists work in school settings, where research may not be supported (Fagan & Wise, 2000). Research that is conducted by school psychologists is likely most often conducted by PhD-level school psychologists who work in university settings.

Some school psychologists engage in activities related to the training of students, including teaching and supervision. School psychologists based in university settings likely engage in such roles to a significant degree, whereas school psychologists in other settings may engage in these activities on a part-time or adjunctive basis (Fagan & Wise, 2000). Other school psychologists function as administrators of psychological services or as supervisors of other school psychologists.

Factors Influencing Role-Related Activities

The amount of time school psychologists engage in various activities may be influenced by several factors. As mentioned earlier, laws and funding mechanisms can exert a significant influence. Historically, state and local education agencies have been most invested in having school psychologists conduct assessments to determine children's eligibility for special education services (Reschly, 2000).

The ratio of school psychologists to students can also influence the opportunity for school psychologists to engage in different roles. Generally, the more favorable the ratio, the more school psychologists are able to engage in activities beyond psychoeducational assessment, such as intervention and consultation (Curtis, Hunley, & Grier, 2002).

The level of training of the school psychologist is likely to be important, with doctoral-level practitioners generally being less involved in psychoeducational assessment activities and more involved in intervention, consultation, and applied research and program evaluation (Curtis et al., 2002). The availability of other providers in the setting, such as social workers and clinical psychologists, can influence the number of opportunities school psychologists have for intervention activities (Fagan & Wise, 2000).

Fagan (2002) noted that a consistent theme in the school psychology literature over the past several decades has been the call for school psychologists to expand their roles beyond assessment and become more involved in activities related to intervention and consultation. He notes that despite this consistent theme, research suggests that school psychologists continue to spend approximately half of their time involved in activities related to psychoeducational assessment, and report generally positive job satisfaction ratings.

The Expanding Role of the School Psychologist

Although the large majority of school psychologists work predominantly in school settings and spend a significant amount of time engaging in assessment-related activities, a significant number of doctoral-level school psychologists have expanded their range of clinical activities well beyond assessment and function in roles that

do not differ measurably from clinical child psychologists. School psychology programs are also now providing opportunities for students pursuing doctoral training in school psychology to obtain subspecialty training in the area of pediatric school psychology. Such training is designed to prepare school psychologists to engage in health-related activities such as serving as liaisons between schools and health care agencies to address the medical, educational, psychological, and community needs of children with or at risk for emotional and behavioral disabilities. As such, doctoral-level school psychologists are becoming increasingly similar to professionals in the area of pediatric psychology in terms of role functions.

SUMMARY

As can be seen from this overview, the specialties of clinical child and pediatric psychology have grown rapidly during the past 25 to 30 years. Indeed, in a recent paper, Prinstein and Roberts (2006) characterized these specialties as currently in their "professional adolescence" and as "grown up and striving for autonomy." As these two specialties have evolved, the roles of those working in these areas have become more and more complex, with child and pediatric psychologists assuming more varied role-related activities as health care providers and taking a prominent role in research focusing on the development of evidence-based practice. While primarily working in school-related settings, doctoral-level school psychologists also often engage in clinical activities that overlap significantly with the work of both clinical child and pediatric psychologists.

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PART II

Theoretical Models

CHAPTER 6

Psychodynamic

Sandra W. Russ

What is the future of psychodynamic psychotherapy with children? Although psychodynamic treatment has a rich theoretical and clinical underpinning, there is a relatively small empirical base for treatment effectiveness. Because of the nature of psychodynamic treatment, it has been difficult to carry out empirical studies. These studies are beginning to occur, but there is a long way to go to catch up with cognitive-behavioral approaches. The psychodynamic approach encompasses two categories of information important to the field of child psychotherapy. One is a number of psychological constructs that provide a way of understanding child development. Many of these constructs have empirical support. The second category is a wealth of psychotherapy techniques with children that have emerged from a psychodynamic understanding of the child. Of particular importance is the use of pretend play in therapy. This chapter reviews the psychological constructs and the treatment techniques. Empirical evidence is presented as much as possible, with guidelines for future research. Implications for treatment and prevention programs are discussed.

PSYCHODYNAMIC THERAPY

The psychodynamic approach focuses on the internal world of the child (Russ, 2006). What is the child thinking and feeling, and how does he or she experience other people and their environment? A developmental perspective is used to understand underlying cognitive, affective, and interpersonal processes (Shirk & Russell, 1996). The level of development of these processes and the interaction among them largely determines the child's behavior, relationships, and internal state. The psychodynamic framework is applied to understand these internal processes and childhood disorders. This psychodynamic understanding determines the specific intervention approach and techniques to be used with a particular child.

Historically, psychodynamic approaches evolved from psychoanalytic theory and therapy. As Fonagy and Moran (1990) pointed out, many forms of psychodynamic therapies are based on the psychoanalytic conceptualization of child development. Children's play was especially important in psychoanalytic approaches in that it was used as a vehicle for communication with the therapist. Also, the therapist actively worked to establish a positive relationship with the child (Freud,

1927). Anna Freud (1966) made significant contributions to psychoanalytic theory with her work on the ego and mechanisms of defense. Melanie Klein, in a different approach, developed early concepts in object relations theory and interpersonal relations (Tyson & Tyson, 1990). Klein also utilized active interpretation of children's play. A major goal of psychoanalytic and psychodynamic therapy is to return the child to normal developmental pathways (Freud, 1965; Shirk & Russell, 1996). Psychodynamic therapy differs from psychoanalytic therapy in that it has more focused treatment goals, is less frequent and intensive, and is more flexible in terms of types of interventions used and the integration of other theoretical perspectives and techniques (Fonagy & Moran, 1990; Russ, 1998; Tuma & Russ, 1993).

In most forms of psychodynamic therapy, the therapist and child meet individually once a week for a session of 45 to 50 minutes. The basic understanding between the therapist and the child is that the therapist is there to help the child express feelings and thoughts, understand causes of behavior, and form a relationship with the therapist (Freedheim & Russ, 1992). Goals of treatment are also discussed. Play is a major tool in the therapy. Traditionally, the child has usually structured the hour, chosen the topic to discuss, chosen the toys to play with, and has determined the pace of the therapy. But as therapy has become more short term, therapists are becoming more active and directive within the psychodynamic framework in focusing on specific topics and directing the play (Russ, 2004). Also, for most child therapists, parent guidance, parent therapy, and parent education and consultation with the school are essential components to the therapy. For a review of the practical issues that arise in psychodynamic therapy and for case presentations, see Chethik (2000), Kessler (1966, 1988), and Russ (2004).

How does change occur in psychodynamic approaches? Russ and Freedheim (2001) and Russ (2004, 2006) reviewed mechanisms of change in therapy most relevant to psychodynamic approaches. Some of the mechanisms of change are similar to those of other therapy approaches. Catharsis or the release of emotion and expression of feelings is thought to be therapeutic by a number of different schools of therapy (Axline, 1947; Freud, 1965). Expression of negative feelings in particular is important for many children (Moustakas, 1953). Helping children to know that they can safely express feelings through talk or play is a major task of the therapist. The therapist labels the feeling expressed by the child or puppet, accepts the feeling, and tries to understand the feeling. Words help put the feeling into a context for the child, thus making the feeling less overwhelming.

Another mechanism of change is a corrective emotional experience. The therapist accepts the child's feelings and thoughts. Often, the child's learned expectations are not met. The relationship between the therapist and child is especially important for a corrective emotional experience to occur. For example, a child expresses angry thoughts and feelings about his or her mother. The therapist, contrary to the child's expectations, is not angry or punishing; rather, the therapist is accepting of the feelings and works to understand the reasons for the anger. After a number of these therapeutic events, a corrective emotional experience occurs (Kessler, 1966). The automatic connection between angry thoughts or feelings and anxiety or guilt gradually decreases (or extinguishes). The child then becomes more comfortable with these feelings and thoughts. In children, expression of thoughts and feelings often

occurs through play. The therapist works with those expressions and, if possible, connects them to the child's daily life.

With some children, the therapist is directive and helps the child think about alternative ways of viewing a situation and generates problem-solving strategies. Role-playing and modeling of coping strategies are used by the therapist. Singer (1993) offers examples of modeling techniques during therapy. Although this mechanism of change is not often associated with psychodynamic therapy, it is frequently part of an intervention within a psychodynamic conceptualization of a case. Shapiro, Friedberg, and Bardenstein (2006) have proposed that life education also occurs in psychodynamic psychotherapy. Life education involves giving information about general human functioning. They pointed out that Anna Freud (1968) included educative components that taught adaptive skills.

The major mechanism of change associated with psychodynamic therapy is insight and the working-through process. It is this mechanism that differentiates psychodynamic therapy from other forms of interpersonal therapy. One goal of the therapist is to help the child reexperience major developmental conflicts or situational traumas in therapy. When underlying conflicts are a major issue, therapy has the goal of cognitive insight into origins of feelings and conflicts, causes of symptoms, and links between thoughts, feelings, and actions (Sandler, Kennedy, & Tyson, 1980; Shirk & Russell, 1996). Verbal labeling of unconscious impulses, conflicts, and causes of behavior helps lend higher order reasoning skills to understanding problems. Well-timed interpretations by the therapist attempt to make these links and to help the child understand the causes of his or her feelings and behavior. However, in many cases, especially with young children, cognitive insight does not occur. Rather, emotional reexperiencing, emotional working-through, and mastery of material occur and result in behavior change and changes in internal feeling states. This is an important mechanism of change and is often overlooked in the literature.

Messer and Warren (1995) stated that the goal of making the unconscious conscious needs to be modified in therapy with many children. In Erikson's (1963) concept of mastery, the child uses play to gain mastery over traumatic events and everyday conflicts. Resolving conflicts through play is part of normal child development. For normal children who are using play, cognitive insight into underlying issues does not usually occur. But the emotional processing that is occurring helps the child resolve the issues. This concept of emotional processing of negative events and traumas will be discussed further at a later point. Therapists who are using play in therapy may be helping the emotional processing occur without cognitive insight in many cases. Freedheim and Russ (1992) described the process of playing out conflict-laden content until the conflict and negative affect are resolved. The psychodynamic therapist helps guide the play, labels the feelings, describes the action, and makes interpretations to facilitate conflict resolution and the working-through process.

Another major mechanism of change with seriously emotionally disturbed children is the development of better object relations, which enables attachment and more stable day-to-day functioning. Many children have deficits in internal structures (such as internal representations) and processes that result in serious emotional

disorders. Deficits in self/other differentiation and object representations and object relations can result in serious problems with self-esteem regulation, reality testing, attachment, and impulse control. In these children, there are major deficits in underlying cognitive, affective, and interpersonal processes. Psychodynamically informed structure-building approaches are based on conceptualizations by Mahler (1968) and Kohut (1977). The therapist attempts to be seen as a caring, empathic, stable, and predictable figure. Development of good object relations is a major goal of therapy with these children. Gilpin (1976) stressed that the role of the therapist is to become an internalized object for the child. The therapist serves as a stable figure who helps the child develop these ego functions as much as possible. The relationship between the therapist and the child is probably the most important aspect of the therapy in helping this process to occur. The expression of empathy by the therapist for the child's experience is an essential technique that enables the child to develop.

Different forms of psychodynamic therapy emphasize different mechanisms of change. The two major forms are insight-oriented therapy and structure-building approaches. Shapiro et al. (2006, p. 117) list four objectives for psychodynamic therapy:

1. Increased self-understanding
2. Increased acceptance of feelings and wishes
3. Replacement of (unconscious) defense mechanisms with conscious coping strategies
4. Development of realistically complex and positive schemas for relationships between self and others

Which objectives and change processes are emphasized depends on the child's particular set of problems.

Insight-Oriented Therapy

The form of therapy most closely associated with the psychodynamic approach is insight-oriented therapy; it is most appropriate for the child with anxiety and internalized conflicts (Tuma & Russ, 1993). This approach is appropriate for children who have age-appropriate ego development, show evidence of internal conflicts, can trust adults, and, for young children, can use play effectively. Insight-oriented therapy is most appropriate for children with internalizing disorders involving anxiety or depression. Children with internalizing disorders often experience internal conflicts, have good development in major areas, and have good attachment and object relations. Goals of therapy are to resolve conflicts. The major mechanism of change is insight and working through. The therapist actively interprets thoughts and feelings in play and in talk about the child's life. As mentioned earlier, often cognitive insight does not occur. Rather, the child reexperiences difficult emotions and processes these feelings and develops a better understanding of them. Through the use of play and the therapist's reflection and interpretation of thoughts and

feelings, the child expresses "forbidden fantasy and feelings, works through and masters developmental problems, and resolves conflicts" (Freedheim & Russ, 1983, p. 982).

The technique of interpretation is important in facilitating insight and/or emotional reexperiencing. Shapiro et al. (2006) offer a good description of the various types of statements by the therapist that help bring new information to consciousness. They stated that interpretations are "causal explanations; they attempt to explain why the client feels, thinks, or behaves in some particular way" (p. 124).

Often, pretend play is a vehicle for the occurrence of insight and working through. The emotional resolution of conflict or trauma and negative feelings occurs in play. The play process has been thought of as a form of conflict resolution. For example, Waelder (1933) described the play process as one in which the child repeats an unpleasant experience over and over until it becomes manageable. Erikson (1963) presented the concept of mastery, in which the child uses play to gain mastery over traumatic events and everyday conflicts. During this process the therapist labels and interprets the play. The therapist will also tie the observations and interpretations to events in the child's life.

Insight and working through can be helpful for children with good internal resources who have experienced a major trauma. For example, Altschul (1988) described the use of psychoanalytic approaches in helping children to mourn the loss of a parent. Chethik (1989) discussed "focal therapy" as therapy that deals with "focal stress events" (p. 194) in the child's life. Chethik listed events such as death in the family, divorce, hospitalization, and illness as examples of specific stressors. Focal therapy focuses on the specific problem and is usually of short duration. It is a psychodynamic approach that uses insight and working through. Chethik viewed this approach as working best with children who have accomplished normal developmental tasks before the occurrence of stressful events.

Structure-Building Approaches

A second major form of psychodynamic therapy is the structure-building approach, which is used with children with deficits in object relations and interpersonal processes (Russ, 1998). For children with impaired object relations, self/other boundary disturbances, and problems distinguishing reality from fantasy, the therapist uses techniques that foster the development of object permanence, self/other differentiation, modulation of affect, and impulse control. The major mechanism of change is the development of these internal processes. The development of object relations is especially important. Mahler (1968) articulated the separation-individuation process and described the development of object constancy and object representations. Blank and Blank (1986) have pointed out that object relations plays a major role in the organization of other intrapsychic processes. Children with severely impaired object relations, such as borderline children and children with psychotic and characterological disorders, have early developmental problems in a variety of areas. Also, they tend to have severe dysfunction in the family and often a genetic predisposition.

In a structure-building approach, empathy on the part of the therapist is a much more important intervention than interpretation (Kohut & Wolfe, 1978). Kohut and

Wolfe discussed the failure of empathy from the parent that is the major issue in faulty parent-child interaction. Because of the frequency of this occurrence in the interaction between the child and parent, empathy from the therapist around the history of empathic failure becomes an important part of therapy. It is the empathy from the therapist that results in the therapist's being internalized and becoming a stable internal figure for the child. Chethik (1989) and Russ (2004) present cases of psychotherapy with borderline children. Often, help with problem solving and coping is used with these children as well. Therapy with these children is usually long term (1 to 2 years).

PSYCHODYNAMIC CONSTRUCTS

There are a number of psychodynamic constructs and hypotheses that are associated with psychodynamic theory. Research that investigates these constructs and hypotheses helps to build empirical support for psychodynamic theory. Although not directly investigating psychodynamic therapy, empirically supported constructs should guide therapy research in terns of what to focus on and what therapy techniques to emphasize. Westen (1998) reviewed the research literature and identified a number of psychodynamic constructs and propositions that have been supported. Much of the research has been carried out in cognitive, social, developmental, and personality psychology. Often, psychodynamic terminology is not used or referred to. Concepts are placed in a more contemporary cognitive-affective framework. Westen identified the following constructs and principles of psychodynamic theory that have received significant empirical support: unconscious processes, ambivalence and conflict, importance of childhood origins of personality and social dispositions, mental representations of self and others, and developmental dynamics.

Primary process thinking is another construct that has a large body of research support (Holt, 1977; Suler, 1980). Primary process thinking refers to drive-laden oral, aggressive, and libidinal material and illogical thinking related to that material (Holt, 1977). Holt described primary thought as an early, primitive system of thought that was drive-laden and not subject to rules of logic or oriented to reality. Affect and affect-laden cognition are a large component of primary process.

A large number of studies have found a relationship between access to primary process thinking and creativity, as psychoanalytic theory predicts, in adults and children (Russ, 1996; Suler, 1980). Russ has discussed these relationships within current cognitive-affective conceptualizations. The finding in a large number of studies that access to primary process thinking relates to measures of creativity supports the psychoanalytic theory that access to drive-laden thinking is important in the creative process. Repression of threatening drive-laden thoughts, images, and memories leads to a general intellectual restriction, which in turn constricts associative thinking. Access to a broad network of associations is important in most measures of creativity. Empirical support of this psychoanalytic theory is important in psychotherapy, where one goal is to help the child gain access to unconscious, repressed content so that the child can bring higher order problem-solving skills to daily problems and stressors.

Pretend play is a key resource for children. The important role of play in child development is not a concept that is unique to psychodynamic thinking. However, play has been a major technique in psychodynamic therapy and is viewed as having a central function in child development.

Pretend Play

Play has been a part of therapy with children since Melanie Klein and Anna Freud first began using play techniques in child psychotherapy in the 1930s. Currently, play is used in therapy from a variety of theoretical traditions. As of 1992, play in some form was used in child therapy by a majority of clinicians, as reported by Koocher and D'Angelo (1992).

In the child therapy literature, four broad functions of play emerge as important in therapy (Russ, 2004). First, play is a natural form of expression in children. Chethik (1989) refers to the language of play. Children use play to express feelings and thoughts. Chethik stated that play emerges from the child's internal life and reflects the child's internal world. Therefore, children use play to express affect and fantasy and, in therapy, to express troubling and conflict-laden feelings.

Second, the child uses the language of play to communicate with the therapist. By understanding these communications, the therapist develops a therapeutic relationship with the child (Chethik, 1989). The therapist actively labels, accepts, empathizes with, and interprets the play expressions, which, in turn, helps the child feel understood (Russ, 1995). These play expressions are important in enabling mechanisms of change to occur in therapy. For example, a corrective emotional experience can occur by the therapist accepting and understanding angry feelings expressed in play. Also, for many children, the feeling of empathy from the therapist around the play facilitates change in their internal representations and interpersonal functioning.

A third major function of play is as a vehicle for the occurrence of insight and working through. The emotional resolution of conflict or trauma and negative feelings is a major mechanism of change in psychodynamic child therapy. Children reexperience major developmental or situational trauma in therapy, and many of these conflicts are expressed in play. The play process itself has been thought of as a form of conflict resolution. During this process the therapist labels and interprets the play. The therapist will also tie the observations and interpretations to events in the child's life. Although there is controversy in the psychodynamic literature about how much interpretation to use (Freud, 1966; Klein, 1955), there is general agreement that working through and mastery are important mechanisms of change.

The whole area of emotion and how play helps process and regulate emotion is a key area for investigation. Shields and Cicchetti (1998) defined emotion regulation as the ability to modulate emotions and engage in an adaptive way with the environment. Mennin, Heimberg, Turk, and Fresco (2002) concluded that an emotion regulation perspective would have as goals of treatment to help individuals (a) become more comfortable with arousing emotional experience, (b) be more able to access and utilize emotional information in adaptive problem solving, and (c) be better able to modulate emotional experience and expression. Play in therapy

can aid with these three processes. The insight, emotional reexperiencing, and working-through process involve some additional components. For the child who is repressing or avoiding negative affect or thoughts and images that arouse conflict, play can help the expression of that content. Although some of the negative affect (e.g., anxiety) might be extinguished after repeated play, the emotional reexperiencing and working-through process involves more than extinction of emotion. It involves building a meaningful narrative around the emotion or trauma (Gaensbauer & Siegel, 1995). In addition, insight is applied to experiences and behaviors to which the insight pertains (Shapiro et al., 2006). Operationalizing the components of insight and working through and identifying how play facilitates this process is an important next step for research. It will also be important to identify when this type of process in therapy is important for a child, and when it is not necessary.

A fourth major function of play in therapy is to provide the child opportunities to practice with a variety of ideas, behaviors, interpersonal behaviors, and verbal expressions. Because play is occurring in a safe environment in a pretend world, with a permissive, nonjudgmental adult, the child can try out and rehearse a variety of expressions and behaviors without concern about real-life consequences. In some forms of play therapy, the therapist is quite directive in guiding the child to try new behaviors. Knell (1993) developed a cognitive-behavioral play therapy approach that actively uses modeling techniques and a variety of cognitive-behavioral techniques. Many of these can be used within a psychodynamic framework.

Although these functions of play occur in normal play situations, the therapist builds on these normal functions by enhancing the play experience. The therapist creates a safe environment, gives permission for play to occur, actively facilitates play, and labels the thoughts and feelings expressed. For the psychodynamic therapist, interpretation specifically aids conflict resolution.

For most children younger than 10 years in psychodynamic therapy, therapy is a mixture of play and talk. How play is used in psychotherapy depends on the child's ability to use play, developmental level, age, ability to verbalize, and the overall treatment approach and goals. A number of practical issues arise in using play in therapy. For suggestions about how to get the child started in using play, kinds of play materials, how much the therapist should engage in play, how much to interpret, and how much to set limits, see Chethik (1989), Singer, (1993), and Russ (2004).

RESEARCH WITH PSYCHODYNAMIC THERAPY

There has been a strong movement in the field of child psychotherapy to identify empirically supported treatments (Lonigan, Elbert, & Johnson, 1998). Most of the therapy outcome studies have been efficacy studies that are conducted under controlled conditions that involve random assignment, control groups, and single disorders. Effectiveness studies, on the other hand, are clinical utility studies that focus on treatment outcome in real-world environments. Empirically supported treatment reviews have focused on efficacy studies (Kazdin, 2000). In the child psychotherapy area, Kazdin concluded that the empirically supported treatments for children are few in number and consist mainly of cognitive-behavioral treatments.

Weisz and Weiss (1993) pointed out the low number of psychodynamic outcome studies in the literature. Psychodynamic treatments have not been investigated in the kind of controlled conditions necessary for efficacy studies. Criteria for efficacy studies include random assignment, specific child populations, use of treatment manuals, and use of multiple outcomes with blind raters.

Russ (2006) reviewed the research literature relevant to psychodynamic psychotherapy. Fonagy and Moran (1990) proposed that one reason for the paucity of research is that the rather global approach and broad goals of psychodynamic approaches do not lend themselves easily to carefully controlled outcome studies. Russ (1998) has proposed that the mechanisms of change are also broad and multiple: Therapy is individualized and the therapist is making decisions about intervention on a moment-to-moment basis. However, specificity in research is essential for the field to progress. An important conclusion from Weisz and Weiss (1993) was that studies showing positive results in psychotherapy research tend to be ones that focus on a specific problem, with careful planning of the intervention. In 1983, Freedheim and Russ stated, based on the literature to date, that therapy research needed to become very specific and investigate which specific interventions affect which cognitive, affective, and personality processes. Shirk and Russell (1996) developed a framework for conceptualizing intervention research that ties specific treatment processes to underlying cognitive, affective, and interpersonal processes within a developmental framework. They stress the importance of investigation of specific processes and change mechanisms. Psychodynamic conceptualizations such as the effect of conflict or of deficits in object relations should be testable within their framework.

There has been progress in the area of psychodynamic psychotherapy with children. Fonagy and Moran (1990) applied many of the current guidelines for psychotherapy outcome research to evaluating the effectiveness of child psychoanalysis. Their studies are models for how to investigate the efficacy of psychodynamic approaches. They have carried out different types of studies that are well suited to the psychoanalytic or psychodynamic approach. In one study (Moran & Fonagy, 1987), they used a time series analysis to study the 184 weeks of treatment of a diabetic teenager. Time series analysis investigates whether or not there is a time-bound relationship between events. A relationship was found between major themes in analysis and diabetic control. Moran and Fonagy concluded that the interpretation of conflicts in the treatment brought about an improvement in diabetic control. The improved control led to temporary increases in anxiety and guilt. The improved diabetic control appeared to increase the likelihood of manifest psychological symptomatology. This pattern fits the psychodynamic understanding of brittle diabetes.

In a second study, an inpatient program for diabetes was evaluated (Fonagy & Moran, 1990). Eleven patients received psychotherapy and medical supervision, while the comparison group received medical treatment with no psychotherapy. The analytic treatment was well defined and based on the psychoanalytic understanding of brittle diabetes as being caused by unconscious emotional factors influencing the significance of the disease or its treatment regimen. This leads to disregard for normal diabetic care. The goal of therapy was to make conscious the conflicts and

anxieties that were interwoven with the diabetes treatment regimen. The treated group showed significant improvement in diabetic control. None of the untreated group showed improvement. Improvement was maintained at 1-year follow-up. Fonagy and Moran stressed the importance and feasibility of systematic and specific intervention research that investigates changes in psychic structure, such as affect regulation and empathy, as a result of psychoanalysis.

Shirk and Russell (1996) reviewed a number of research programs and measures relevant to assessing change in psychodynamic psychotherapy. Wallerstein (1988) and Kernberg (1995) have coding systems for dimensions relevant to psychodynamic treatment. The Affect in Play Scale (Russ, 1993) can be used to measure change in play processes in psychodynamic therapy.

Muratori, Picchi, Bruni, Patarnello, and Romagnoli (2003) evaluated short- and long-term effects of time-limited psychodynamic psychotherapy for children with internalizing disorders. Fifty-eight outpatient children (6 to 10 years old) who met *DSM-IV* criteria for depressive or anxiety disorder were assigned to therapy or community services. Children were assessed at baseline, 6 months, and 2-year follow-up using the Children's Global Assessment Scale (C-GAS) and Child Behavior Checklist (CBCL). Group assignment was not random. Assignment to the treatment group occurred if there was an opening; if no opening was available, the child was referred to community services. There was no significant difference at baseline between the groups on major variables. Therapy consisted of 11 sessions that followed a protocol. There were five parent-child sessions, five sessions with the child alone, and a final parent-child session. There appear to be clear guidelines about therapist intervention in each session. The treatment approach was to point out the nature of the core conflictual theme and connect it to the child's symptoms and the representational world of the parent. In the child sessions, the therapist labeled feelings and connected mental content and symptoms. Play was seen as an important part of the therapy. Therapists were trained in this approach. Their interventions were videotaped and discussed weekly to monitor adherence to the protocol.

Results revealed major improvements in the treatment group on the C-GAS and CBCL. Improvements on the C-GAS were at 6 months, and CBCL findings were at 2-year follow-up. The changes in the internalizing syndrome on the CBCL at the follow-up only is evidence for the "sleeper effect," a delayed response to the treatment. Muratori et al. (2003) concluded that changes in the shared representational world brought about by the therapy resulted in changes in parent-child interaction only after a period of time. Also, the treatment group sought mental health services less than the comparison group during the 2-year period. This study followed many methodological guidelines for adequate research design described by Weisz and Weiss (1993). The sample was relatively homogeneous (anxious and depressed); a treatment protocol was followed; and therapists were trained and monitored to make sure they adhered to the intervention techniques. In addition, there was a theoretical rationale for why this approach should affect this syndrome. This is the type of rigorous research that needs to occur in the psychodynamic area.

Trowell et al. (2002) compared psychodynamic therapy with psychoeducational topic-focused discussions for 71 girls, 6 to 14 years old, who had experienced

sexual abuse. Children were randomly assigned to treatment groups and manualized treatment was used. Sessions were limited to a maximum of 30. Both treatment groups showed improvement, with the psychodynamic therapy group showing significantly more improvement on the "persistent avoidance of stimuli" measures on a PTSD scale. In a review, Ritvo (2006) pointed out that this study and the Fonagy and Moran (1990) study of children with diabetes both focused on children dealing with specific stress rather than with a clinical diagnosis. She stated that this focus on life stress is consistent with the emphasis that the psychodynamic approach places on adaptation and function rather than on clinical diagnosis per se.

We also need more studies that investigate specific mechanisms of change in therapy. A good example of the kind of study that needs to occur is that by Wiser and Goldfried (1998). Although this study was conducted with adults, it is a good model for the child area because it investigates specific types of interventions on specific processes. They investigated the effectiveness of therapeutic interventions on emotional experiencing. The expression of emotion in session has been thought to be important for different mechanisms of change to occur. An important question with adults and children is how to facilitate emotional experiencing in the session. Wiser and Goldfried coded interventions by therapists with anxious and/or depressed outpatient adults in either psychodynamic or cognitive-behavioral therapy. They found that interventions of reflections and acknowledgments, affiliative and noncontrolling interventions, and interventions highlighting nonspecific client content were associated with maintaining high emotional experiencing in both approaches (reflections were nonsignificant in the psychodynamic approach but may be significant with a larger sample size). Interestingly, no type of intervention was associated with a shift to higher emotional experiencing. Some interventions were related to shifts to low experiencing: those that were lengthy and those that were affiliative but moderately controlling. It is important to know which techniques to use to increase emotional experiencing and which to decrease emotional experiencing. Both types of interventions are therapeutic at different times with different populations. The Wiser and Goldfried study is an excellent example of the type of therapy process research that needs to be carried out with children.

Aspects of the relationship with the therapist are also important to investigate in psychodynamic therapy. Using meta-analytic procedures, Shirk and Karver (2003) reviewed associations between therapeutic relationship variables and treatment outcomes in child and adolescent therapy. Results were a modest association (.20), which was similar to that seen in adult studies. This relationship was found across diverse types and modes of child treatment.

Target (2002) reported on results from a retrospective chart study of treated children at the Anna Freud Center who were in psychodynamic or more intensive psychoanalytic treatment. Although this study was retrospective and there was no randomized design or comparison groups, the results are important and do provide guidance. Looking first at anxiety or depressive disorders, 72% of those treated for at least 6 months showed clinically significant improvement (Target & Fonagy, 1994). Phobic disorders were most likely to remit and depressive disorders least likely. Variables that identified children most likely to improve were higher IQ, younger age, longer treatment, good peer relations, poor overall adjustment of the

mother, the presence of anxiety in the mother, concurrent treatment of the mother, and absence of maternal antisocial behavior.

In the chart review of disruptive disorders, improvement rates were lower than for the internalizing disorders, with one third terminating therapy prematurely. Of those who stayed in treatment, 69% were no longer diagnosable upon termination. Predictors of improvement were the presence of an anxiety disorder, absence of other disorders, younger age, intensive treatment, longer treatment, maternal anxiety disorder, treatment of the mother, and foster care for the child. Target and Fonagy (1996) concluded that psychodynamic therapy was effective with disorders involving anxiety, even when the anxiety was coupled with disruptive disorders. A different picture emerged with the disruptive (externalizing) disorders. When therapy was effective, it was when intensive treatment was used. They reported that it was the oppositional disorders that were most likely to improve and conduct disorders the least likely.

It appears that before recommending psychodynamic therapy for externalizing disorders in which anxiety is not a major factor, several issues should be considered. Cognitive-behavioral approaches have been found to be very effective for this problem. For example, parent-child interaction therapy is very effective with Oppositional Defiant Disorder (Bodiford-McNeil, Hembree-Kigin, & Eyberg, 1996). If psychodynamic therapy is recommended, two things should be kept in mind and shared with the family: (1) There is no solid empirical support for psychodynamic therapy with externalizing disorders, and (2) the intensive therapy that would be attempted is costly.

The research studies that do exist in the area support the use of psychodynamic therapy for internalizing disorders where anxiety and conflict are important factors. Shapiro et al. (2006) concluded that psychodynamic therapy's focus on play, emotion-dominated thinking, and magical fantasies makes it appropriate for young children. They concluded that no treatment approach has been proven effective for anxious children under 7 (except for simple fears and phobias) or for depression below age 9. The play intervention research that exists also supports the use of a psychodynamic conceptualization for reducing anxiety and negative affect.

Play Intervention Studies

Russ (1995, 2004) reviewed studies investigating the effect of play interventions on specific types of child problems in specific populations. I call these studies play intervention because the focus is highly specific, usually involving only a few sessions with the child. These studies have focused on reducing anxiety in children in medical settings or dealing with separation anxiety. In two different studies, puppet play reduced anxiety in children facing medical procedures when compared with a control group (Cassell, 1965; Johnson & Stockdale, 1975). Rae, Worchel, Upchurch, Sanner, and Daniel (1989) found that children in a therapeutic play group showed significantly more reduction in self-reported hospital-related fears than children in three other treatment groups. This was a well-done study that controlled for time spent with an adult, verbal support, and play activity. They concluded that fantasy activity in the play resulted in fear reduction. Milos and

Reiss (1982) found that thematic play reduced separation anxiety in preschoolers when compared with a control group. Also, the quality of play ratings were significantly negatively related to posttest anxiety. High-quality play was defined as play that showed more separation themes and attempts to resolve conflicts. They concluded that their results supported the notion that play can reduce anxiety associated with psychological problems. The results are consistent with psychodynamic theory.

A well-designed study by Barnett (1984) investigated separation anxiety. Expanding on the work of Barnett and Storm (1981), Barnett found that free play reduced distress in children following a conflict situation the first day of school. Seventy-four preschool children were observed separating from their mothers and were rated anxious or nonanxious. These two groups were further divided into a play or no-play story-listening condition. For half of the play condition, play was solitary. For the other half, peers were present. The story condition was also split into solitary and peers-present segments. Play was rated by observers and categorized into types of play. Play significantly reduced anxiety in the high-anxious group. Anxiety was measured by the Palmer Sweat Index. There was no effect for low-anxious children. For the high-anxious children, solitary play was best in reducing anxiety. High-anxious children spent more time in fantasy play than did low-anxious children, who showed more functional and manipulative play. High-anxious children also engaged more in fantasy play when no other children were present. Barnett suggested that play was used to cope with a distressing situation. The findings supported her concept that it is not social play that is essential to conflict resolution, but rather imaginative play qualities that the child introduces into playful behavior.

The results of these studies suggest that play helps children deal with fears and reduce anxiety and that something about play itself is important and serves as a vehicle for change. Both the Milos and Reiss (1982) study and the Barnett (1984) study suggest that fantasy and make-believe are involved in the reduction of anxiety. The studies effectively controlled for the variable of an attentive adult. Results also suggest that children who are already good players are more able to use play opportunities to solve problems.

These research findings are consistent with psychodynamic theoretical and clinical literature suggesting that play therapy assists children with internal conflict resolution and mastery of internal issues, as well as with external traumas and stressful life events. As a result of this conflict resolution and problem solving, anxiety is reduced. Psychodynamic approaches also suggest the use of insight, conflict-resolution approaches for children whose fantasy skills are normally developed and who can use play in therapy.

Future research should focus on the mechanisms that account for the finding that play reduces anxiety. Conceptualizations and research from other theoretical frameworks could apply to play therapy. For example, Harris (2000) views play as helping the child construct a situation model that is revisable. Children go back and forth between an imagined world and reality and develop a new cognitive appraisal of the situation. Pennebaker's work is also relevant: Adults who develop a coherent and meaningful narrative of an event through an emotional

writing exercise have improved mental and physical health (Pennebaker & Graybeal, 2001). Children may be developing meaningful and coherent narratives in play therapy.

Future research should investigate techniques in therapy that facilitate play in therapy. Russ, Moore, and Pearson (2006) developed play intervention protocols that resulted in increased fantasy and imagination and affect in play in first- and second-grade children. In five 30-minute sessions with a play trainer, the child was instructed to play out various story scripts. In the affect play group, the child was asked to play out stories with different emotions and to express emotion. In the imagination play group, the child was asked to play out stories with a high fantasy content and high organization of the story. The instructions and prompts by the play trainer were standardized. The affect play group had higher scores on all play abilities when compared to a control group on the outcome play measure (Affect in Play Scale). The imagination play group had higher scores on the affect play variable. The results suggest that a brief standardized play intervention can improve play skills and increase emotional expression in play in psychotherapy. This technique could be helpful for children with constricted affect such as those with PTSD or depression. By having children make up various stories and directing them to express affect, with modeling and reinforcement by the therapist, children could increase affect expression quickly. This technique could be especially useful in time-limited therapy.

SHORT-TERM PSYCHODYNAMIC PSYCHOTHERAPY

Short-term psychotherapy (6 to 12 sessions) is a form of psychodynamic psychotherapy frequently used with children (Messer & Warren, 1995). The practical realities of HMOs and of clinical practice in general have led to briefer forms of treatment. Often, the time-limited nature of the therapy is by default, not by plan (Messer & Warren, 1995). The average number of sessions for children in outpatient therapy is 6 or fewer in private and clinical settings (Dulcan & Piercy, 1985).

There is little research or clinical theory about short-term therapy with children (Messer & Warren, 1995). A few research studies have shown that explicit time limits reduced the likelihood of premature termination (Parad & Parad, 1968), and that children in time-limited therapy showed as much improvement as those in long-term therapy (Smyrnios & Kirby, 1993). The time is right for the development of theoretically based short-term interventions for children. Messer and Warren suggest that the developmental approach utilized by psychodynamic theory provides a useful framework for short-term therapy. One can identify the developmental problems and obstacles involved in a particular case. They also stressed the use of play as a vehicle of change and, as Winnicott (1971) has said, of development. They suggest that the active interpretation of the meaning of play can help the child feel understood, which, in turn, can result in lifelong changes in self-perception and experience. In other words, the understanding of the metaphors in a child's play could give the child insight or an experience of empathy, or both. This lasting change could be accomplished in a short time.

As previously discussed, Chethik (1989) developed focal therapy to deal with specific stressful events in the child's life. Basic principles of psychodynamic therapy are applied in this short-term approach, with the basic mechanism of change being insight and working through. Chethik views this approach as working best with children who have accomplished normal developmental tasks before the stressful event occurred.

In general, brief forms of psychodynamic intervention are seen as more appropriate for the child who has accomplished the major developmental milestones. Proskauer (1969) stressed the child's ability to quickly develop a relationship with the therapist, good trusting ability, the existence of a focal dynamic issue, and flexible and adaptive defenses as criteria for short-term intervention. Messer and Warren (1995) concluded that children with less severe pathology are more responsive to brief intervention than children with chronic developmental problems. The Muratori et al. (2003) study that found that short-term intervention with anxious and depressed children was effective is consistent with this conclusion. The research and clinical literature suggest that internalizing disorders are most appropriate for brief psychodynamic intervention (Russ, 2004). The therapist is active, at times directive, and uses all mechanisms of change in the therapy. Insight and working through are essential, but modeling, rehearsal, and problem-solving strategies are also part of the therapy. Children with major deficits in object relations and those with early developmental problems need longer term structure-building approaches.

Shelby (2000) described the importance of using developmentally appropriate interventions in brief therapy with traumatized children. Working with traumatized children in Sarajevo, she used play and drawing. She described an experiential mastery technique in which children drew pictures of the thing that frightened them. Children were encouraged to verbalize their feelings about the drawing. They were also instructed to do anything they wanted to the drawing. Shelby described how this and other developmentally appropriate brief play interventions, although not empirically tested, helped a number of these traumatized children to integrate traumatic events and return to "normal developmental functioning" (p. 72).

Structured play techniques would be especially useful in short-term therapy. The MacArthur Story Stem Battery (MSSB), although designed as an assessment tool, can be used to structure the play situation. In an innovative approach, Kelsay (2002) used the MSSB to structure play therapy. The MSSB is a set of story beginnings (e.g., parents arguing over lost keys), and the child is asked to complete the story. The therapist can choose appropriate story stems tailored to the issues that the child is dealing with. This structured approach could move the therapy to central issues more quickly.

Structured play techniques have also been used with very young children. Gaensbauer and Siegel (1995) described structured play techniques with toddlers who have experienced traumatic events. They conceptualized the mechanisms of change when play is used as being similar to those in older children with PTSD. With these very young children, the therapist actively structures the play to recreate the traumatic event. Gaensbauer and Siegel outlined three purposes of structured play reenactment. First, play enables the child to organize the fragmented experiences into meaningful narratives. Second, the interpretive work by the therapist helps the

child understand the personal meanings of the trauma. Third, there is desensitization of the anxiety and fear and other negative emotions associated with the trauma. They stressed that the key element that enables the child to use play adaptively, rather than in a repetitive fashion, is the "degree to which affects can be brought to the surface so the child can identify them and integrate them in more adaptive ways" (p. 297).

RESEARCH GUIDELINES

Given the results of the research in the area, research guidelines suggested by Russ (2006) that would be the most fruitful and have the most immediate impact on the field would focus on the following:

- In young children, well-controlled efficacy and effectiveness studies with anxiety disorders, PTSD, depression, and focused problems involving anxiety and conflict should be carried out with psychodynamic therapy.
- Short-term interventions should be investigated and refined.
- Mechanisms of change in therapy, especially insight and emotional working through, are needed to identify mechanisms leading to anxiety reduction and reduction of negative affect.
- Play processes and techniques that facilitate play in therapy should be researched.

Demonstration of the effectiveness of structure-building approaches with seriously emotionally disturbed children is also necessary. Because the therapy is so long term and complex with these children, single-case designs would be especially appropriate.

SUMMARY

The psychodynamic approach focuses on the internal world of the child. A developmental perspective is used to understand underlying cognitive, affective, and interpersonal processes. A psychodynamic understanding determines the specific intervention approach and techniques to be used with a particular child. The mechanisms of change most associated with psychodynamic therapy are insight, emotional reexperiencing, and working through. A key question for future research is to identify the components of these mechanisms of change. In therapy, pretend play is frequently used as a vehicle for change. There are a number of constructs in psychodynamic theory that inform therapy. Empirical support for these constructs lends support for psychodynamic therapy. Outcome research on the effectiveness of psychodynamic therapy is sparse. A few studies do offer empirical support. Research in play intervention also suggests that the use of play in therapy can reduce anxiety. Future research should focus on the use of psychodynamic therapy with internalizing disorders in young children, play facilitation techniques, short-term therapy, and the components and role of the insight and working-through process.

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CHAPTER 7

Family Systems

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This chapter provides an overview of major contributions of family systems thinking to the field of clinical psychology. Consistent with the volume's other chapters, it attends to theory, research, assessment, and treatment issues. To provide additional substance for some of the central conceptual issues advanced in this chapter, a section on treatment considerations develops a clinical case example in an outline paralleling those of other chapters covering major treatment approaches. In so doing, we strive to clarify how clinicians working from a family systems approach might work with children and families in a manner quite distinct from, though often complementary to, interventionists working from psychodynamic, behavioral, or cognitive approaches.

What our field refers to as "family systems thinking" is actually a conceptual and philosophical framework that reflects a wide array of quite different approaches and influences—indeed, there *is* no single "systems" theory. What ties together this collection of rather disparate influences and approaches is the central, guiding notion that problem behavior exhibited by a child or an adult can never be understood devoid of its relational context. Further, from a systems view, the behavioral problems evident in any individual family member are perhaps best understood as manifestations of dysfunction within the broader family unit. From this basic tenet follow distinctive lines of inquiry in research studies, and approaches in clinical formulation, assessment, and treatment, that often differ substantially from paths followed in other theoretical traditions and frameworks.

We begin this chapter with a review of some of the major assumptions of systems approaches and a brief recounting of some of the historically important forms of influence in shaping this approach to psychopathology and its treatment. We summarize roles that empirical research has played in the family systems field. From this base, we outline how family systems principles might organize clinical work with a case referred for the treatment of ostensible problems in an individual family member. We close the chapter with some reflections on the value of systems-informed approaches in clinical work.

BASIC CONCEPTS AND PRINCIPLES

Historically, the systems theory developments that inspired and were infused by early family therapies presented exciting and revolutionary challenges to the

practices and theories of the times. Systems theory was held to differ from psychoanalytic approaches by emphasizing relational as opposed to intrapsychic processes, and by taking a holistic and multidirectional view of pathological behavior and its treatment (Bateson, Jackson, Haley, & Weakland, 1956; Lidz, Cornelison, Fleck, & Terry, 1957a, 1957b; Wynne, Ryckoff, Day, & Hirsch, 1958). But the sheer number of different thrusts within this emerging field and the complexity of some of the early theoretical formulations and writings meant that it would be many years before coherence in the field began to emerge. Accounts of "systems theory" and its pertinence to families and individual development often seek to trace the history of how principles governing physical systems came to be applied to social systems. In such accountings, special homage is often accorded the creative writings of von Bertanffly (1968). As Patricia Minuchin (1985) noted, however, the writings of Bateson (1972, 1979) and other family theoreticians more directly concerned with the functioning of human systems (e.g., Jackson, 1957; S. Minuchin, 1974; Watzlawick, Beavin, & Jackson, 1967) had a more formative impact on most practicing clinicians of the time. Among the most essential of systems tenets, outlined by P. Minuchin (1985) and others (e.g., Bornstein & Sawyer, 2006; Cox & Paley, 2003) are the following:

- Systems are organized wholes, and their constituting elements or subsystems are interdependent.
- Interconnected subsystems have their own integrity, are organized hierarchically, and are separated by boundaries.
- Patterns in a system are circular and not linear.
- Stable patterns are maintained over time through homeostatic processes.
- Open systems do adapt, change, reorganize, and develop.

Several important talking points follow from these central tenets. First, when clinicians consider any family system, they should find it possible to identify both a family group reality wherein patterns of organization governing the functioning of the overall unit operate as organized patterns in their own right, and a variety of subsystem realities that may or may not themselves mirror the same patterning and rules of the broader system. Most families are composed of multiple subsystems—among them, marital subsystems, parent-child subsystems, sibling subsystems, parent-grandparent and child-grandparent subsystems in multigenerational family systems, and so on. When clinicians and researchers assess family dynamics, they regularly detect logical interconnections among different subsystem levels; for example, high levels of distress in the marital subsystem typically coincide with evidence of impaired parenting of one or more children in the family (Erel & Burman, 1995). They also find that subsystem functioning both affects and is affected by broader systemic functioning, as when marital strife between a husband and wife within their dyadic marital subsystem comes to disrupt collaboration in effectively coparenting children at the family group level (Belsky, Crnic, & Gable, 1995; McHale, 1995). In other cases, such linkages are not as overt or easily detected, as when family members as a unit collude to obscure problematic

alignments or abuse within a particular dyadic subsystem in the family (Wynne, 1961).

Besides these interconnections between subsystems and the functioning of the family group, the functioning of the overall unit is also properly seen as constraining and organizing interactions at all other levels. For example, in Bowen's (1978; see also Friedman, 1992) conceptualization of the family's emotional system, family members are described as having developed emotional interdependencies to the point where the emotional system through which they are connected has evolved its own principles of organization. The resulting organizational structure then comes to influence the functioning of various individuals and dyads far more than any of them alter the functioning of the full system (S. Minuchin, 1974). For example, a variety of different interaction patterns and sequences (e.g., children diffusing conflict between parents; parents exhibiting preoccupation and overprotection of children) may be seen in families whose structure is guided by the covert rule that "family members must protect one another" (see Nichols & Schwartz, 1998); although changing any particular sequence would not alter this governing structure, altering the structure would reshape interaction patterns throughout the system.

Of course, if an interventionist was successful in meaningfully changing any given aspect of a family's entrenched pattern, this would have the effect of perturbing other patterns. Such temporary ripple effects and adjustments have the potential to affect more enduring change in the problematic family pattern—but only to the extent that the short-term adjustments could be sustained. Unfortunately, supporting enduring pattern changes is very difficult, in part because systems show marked propensities for preserving equilibrium. New patterns of behavior by a child or adult that challenge established family patterns and rules are often quickly countered by adjustments elsewhere in the system that are knowingly, or unconsciously, intended to reestablish the previously enduring status quo. Said differently, because everyone's behavior has become interdependent, the family members collectively collude to maintain their existing patterns. Moreover, repetitive patterns are especially unlikely to change if they have evolved to serve some important protective function for the system as a whole (Madanes, 1981; Selvini Palazzoli, Boscolo, Cecchin, & Prata, 1978), as when a child's misbehavior deflects attention away from a more serious rift in the parents' marital relationship.

The notion of circular causality in family systems (Bateson, 1979) is also central in family systems conceptualizations. Seen as a hallmark of systems thinking that stands in contrast to the linear thinking implicit in other approaches, the principle of circular causality is that the patterns families develop are actively maintained by all of the family's participant members. Such group and subsystem reciprocity is seen as largely responsible for the enduring nature of family patterns through time. As an example, extreme emotional overinvolvement by a child's mother, to the point where she interferes with that child's own capacities for autonomous emotional regulation, is often accompanied by pronounced disengagement by the child's father (in some families, the pattern is reversed, with intense father-child involvement co-occurring with maternal disengagement). Whether fathers disengage in response to intensive maternal involvement or whether maternal overinvolvement evolves as a response to inadequate father involvement is an immaterial question; all parties

(including the family's children) collude to help maintain system functioning by steadfastly keeping to the pattern that has evolved through a process of checks, balances, and recurring feedback loops.

Given these powerful corrective feedback loops self-regulating families and maintaining their enduring patterns, how do families ever change and evolve? Different theorists provide different perspectives, but most find consensus on certain points. First, all living systems do change in form (Speer, 1970). They do so in response to major and sustained perturbations either from within or from the outside that disrupt established patterns (Hoffman, 1981; Prigogine, 1973). The initial reactions of an intensively challenged system are to mobilize familiar modes of response. But as such responses fail to accommodate the new challenge, the system begins advancing previously untested, untried methods of adaptation until ultimately happening upon solutions relevant to addressing the challenge. During this period of duress, the system is in relative disarray. Indeed, it operates temporarily at a lower level of integration, as its previous modes of adaptation are abandoned in favor of newer and less refined adaptations. However, following this period of destructure and disintegration, newer structures and solutions effectively replace the old, becoming integrated into the system's patterning and modes of action. This process of disequilibration and reorganization (Block, 1982) typically eventuates in the emergence of a more complex and differentiated system, though devolution, as in the case of incestuous families, is also possible (Alexander, Sexton, & Robbins, 2002). The operating principle is that change is expectable and inevitable. In human systems this is largely because change is regularly initiated by individual development within the broader system.

Where, then, is the connection with von Bertanffly's general systems model? Extrapolating from the biological systems model, family theorists have often drawn connections to notions of systems as open, cybernetic, or closed (Bateson, 1972 1980; White, 1986, 1989). Open systems, von Bertanffly (1968) explains, have the intrinsic capacity for change, whereas change in closed (including cybernetic) systems occurs only in response to outside sources. Don Jackson (1957) was among the first to draw a parallel between closed information systems in states of homeostasis and the consistency of family disturbances and symptomatic behavior that resisted change. Appropriating the metaphor of homeostasis from general systems theory had appeal, as families stuck in recurrent maladaptive patterns could be viewed as being in a state of equilibrium, behavioral redundancy, and relationship balance. The homeostasis metaphor has remained a very influential one, though it is flawed in many ways. As Moyer (1994) has pointed out, whereas human systems are open rather than closed, theoretical concepts of cybernetics and homeostasis properly describe closed systems and hence are not truly applicable. For example, open (as opposed to cybernetic) systems are intrinsically capable of change, do possess the potential for free interchange of information, energy, or matter, and are composed of components that are simultaneously actors and acted upon (in a circular, rather than linear, dynamic).

Nonetheless, characterizations of disturbed family systems as closed, insular, and resistant (if not impermeable) to adaptive informational exchanges with the outside world have remained a recurring and robust metaphor in many clinical

writings. Certainly, though, not all theorists find it necessary to invoke open-closed and cybernetic metaphors when describing closure and insularity; Bowen (1978), for example, observed simply that family members withdraw from outside relationships and insulate themselves when emotional forces within the family are in opposition and internal anxiety and tension mount. Such closed or relatively closed-off family systems are usually seen as dysfunctional when contrasted with healthier open system functioning (Olsen, Sprenkle, & Russell, 1979; Satir, 1972), though open systems too can be plagued by serious problems, such as excessive and chaotic interaction, prolonged conflict, and ambiguity (Constantine, 1983).

We now turn to several of the most influential theoretical streams and concepts emanating from the related fields of family therapy and family psychology.

Historically Important Ideas and Lines of Influence: Family Therapy and Family Psychology

The view that children's behavioral problems are inextricable from their family group context and that they are most effectively addressed through family collective approaches has been active for nearly a century. Richmond (1917), echoing practices of the social work movement of her time, posited that child adjustment problems had to be understood in their family context, advocated work with family units, and emphasized the palliative role of family cohesion. Yet even as she wrote, common practice then as now was for children's mothers (and seldom fathers) to attend such family sessions. In clinical psychology, mainstream practice did not follow suit in assuming social work's collectivist stance through most of the first half of the twentieth century, even though positions developed by Sullivan, Horney, and Fromm emphasized the interpersonal nature of psychiatric disorders. The work of these pioneering figures, however, was not truly systemic in that treatment approaches seldom moved to involve family members; such inclusion was usually construed as a violation of client confidentiality.

During the 1950s, John E. Bell, Nathan Ackerman, and a handful of other clinicians broke from these long-standing traditions and began conducting conjoint therapy when seeing problem-referred children. This significant reconceptualization of work with child cases helped catalyze the gradual spread of family-level approaches and came to spark many generative debates and discussions among clinical psychologists. Herein were the gentle beginnings of a family therapy movement, guided by the clarion call that individual psychopathology signaled dysfunctional family systems. Distressed children were said to have accepted an assignment from the family to be its "identified patient" (IP), while the child's symptomatology came to be understood as serving a "messenger" function communicating dire problems at work in the broader family system. Yet even as the IP broadcast these messages, his or her problems were also playing some functional role in protecting the family and maintaining its balance (as when redirecting parents' attention away from their own strife and onto the child's problems). Moreover, this function often eluded the conscious awareness of some or all of the family's members. Based on this premise that child behavior problems were telegraphing broader system dysfunction, family practitioners argued that it was senseless to excise a child from his or her family

system to treat the child's behavior problems, and then return him or her to the very same context that had catalyzed the development and supported the maintenance of those problems to begin with. Rather, logic dictated attempting to effect change at the root source, alleviating the child's suffering by fostering meaningful change in the functioning of the child's family system.

There were many different pathways toward such ends charted by pioneers of the family therapy movement. Given space limitations, we highlight only some of the principal and most enduring contributions of various perspectives and underscore a few commonalities among approaches. However, we caution readers that by doing so we necessarily obfuscate much of the important detail and richness of theory differentiating the disparate approaches, and we recommend consulting original sources for fuller readings of theory exposition. As an orienting comment, it is probably fair to say that all of the major early schools of thought—including psychoanalytic, Bowenian, contextual, experiential, strategic, structural, communication, and behavioral therapies—shared a few things in common. All acknowledged mutual contributions by all family members to the problems they encountered, shared the therapeutic goals of helping family members assume greater responsibility and delimit blame, and aspired to clarify and enhance intrafamily communication patterns (Kaslow, 1982; Nichols & Schwartz, 1998). How the therapies sought to do so, what each took as their point of entry, how much each privileged the intrapsychic world or privileged external interpersonal relations, and how much emphasis each gave to the role of past as opposed to contemporary, here-and-now relationships—even insofar as differing in defining who actually constituted the family (and in multigenerational and extended kin frameworks, how far this reach needed to go)—varied greatly. Yet all approaches sought essentially similar ends: greater role flexibility and adaptability, greater clarity and specificity of communications, a more equitable balance of power among the family's coparenting figures, and promotion of greater individuality and differentiation of members within a cohesive family collective. Hence family cohesion, adaptability, and communication are some of the common themes tying together different approaches (Olson, Russell, & Sprenkle, 1980).

In most accounts Ackerman, who was a psychoanalyst and child psychiatrist by training, is hailed as the pioneering figure in the family therapy movement; his early publications, *The Unity of the Family* and *Family Diagnosis: An Approach to the Preschool Child,* are counted among the field's seminal works. Perhaps the most provocative conceptual advances early on in the family therapy movement, however, emanated from work with families of young men and women diagnosed with Schizophrenia. Among the innovative clinicians bucking a therapeutic tradition of segregating schizophrenic individuals from their families when providing treatment (and involving the family members only on a need-to-know basis) was Murray Bowen, who strove to treat the schizophrenic person's entire family unit. Bowen believed that the introduction of an outside person into disturbed relational systems had the potential to modify relationships within that system, but he emphasized the importance of the therapist's keeping the intensity of therapeutic focus within the family unit rather than siphoning it into an intense transference relationship. This stance, which broke with psychoanalytic convention, was one defining feature of the Bowenian approach. Bowen's theory described how individual

family members in troubled families experienced difficulty differentiating within a relatively closed-off emotional family system (in 1974, Bowen portrayed families with schizophrenic members as characterized by an "undifferentiated family ego mass"), and he was among the first family theorists to explicitly outline the role that multigenerational transmission played in family pathology. But perhaps the most central of all Bowen's many major contributions was his exposition of emotional triangles as the basic building blocks of families, and of how a third person helps to stabilize inherently unstable dyads. While similar ideas were developed by several other theorists, as outlined later, Bowen helped specify why it is that triangulation breeds stress; to the extent that one family member becomes responsible for or tries to change the relationship of two others, that person shoulders the stress for the others' relationship. Stress within emotional triangles thus came to be seen as a positional phenomenon; assuming responsibility overburdened younger, less differentiated family members, in particular, by trapping them in problematic relationships that they were ill equipped to handle (Friedman, 1992).

Elsewhere, Gregory Bateson and his colleagues in Palo Alto, California (among them Jay Haley, John Weakland, and later Don Jackson), had been developing a communication theory of Schizophrenia, from which surfaced the notions about family homeostasis discussed earlier. According to the group's double-bind theory, an individual's psychosis was understandable in the context of pathological family communication. When individuals receive contradictory messages or injunctions communicated on different (manifest content and metacommunication) levels, the theory went, they are inextricably trapped in contradictory, double-binding conundrums offering no clear channel for resolution or escape (Bateson et al., 1956). The Palo Alto group portrayed the cognitive distortions endemic to Schizophrenia as one adaptation to such insoluble double-binding communications. Despite its elegance and allure, double-bind theory as a theory for Schizophrenia did not stand up to clinical and empirical scrutiny and ultimately fell into disrepute. However, several innovative conceptualizations emanating from the group's work, including the coexistence and significance of multiple communication levels in family interactions and the notion that destructive relationship patterns endure as a result of self-regulating interactions, have had lasting impact.

As a family theory, many of the seminal ideas of the Palo Alto group, such as the double-bind notion, and Jackson's (1965) "quid pro quo," were limited in how much they could explain because they actually described dyads rather than full family systems. However, Haley developed an enduring interest in triads (which he later developed further in his writings about cross-generational coalitions and perverse triangles). This concept was influential in helping to bridge two of the more influential family systems approaches: strategic and structural family therapy (for greater historical perspective on this connection, see Nichols & Schwartz, 1998).

A third set of investigations of family dynamics in Schizophrenia that also had lasting impact in family thinking was that of the psychodynamic theorist Theodore Lidz and his colleagues (1957a, 1957b). Among Lidz's key contributions was his challenging of the blame burden that had historically been foisted exclusively upon mothers. Though he did not absolve mothers, Lidz did consider the potentially damaging role of fathering relationships in families, seeding a field now flourishing as

coparenting theory (McHale et al., 2002; McHale, 2007a). Focusing on what he termed role reciprocity, Lidz outlined ways in which parents of disturbed youth had failed to achieve mutually supportive, collaborative, and coordinated parenting stances within the family. In some cases, the problem was one of marital schism, whereby parents chronically undermined one another and competed for their children's loyalties. In other families, which he characterized as showing marital skew, one of the parents held inordinate sway in the family while the other passively gave way to the domineering parent. In Lidz's published cases it was typically the father who was the one who acceded, though the dynamic could also be flipped. In both kinds of family systems, children were caught in the middle, triangulated into the problematic interadult relations.

In many of these early initiatives, there was an implicit focus on nuclear family systems and problems that eventuated when power dynamics or family roles deviated from what were seen as optimal pathways. As will be detailed shortly, some of these presumptions have been critically evaluated by contemporary family scholars, including Hare-Mustin and Leupnitz. Yet even in advance of these critiques, not all theorists and clinicians were limiting the focus of their family conceptualizations and interventions to nuclear family units; in several approaches, present-day family dynamics were framed as temporal extensions of family lineages. These multigenerational approaches, in turn, laid important foundational groundwork for contemporary approaches that view family dynamics as embedded not just within extended and historical kin relations, but also within a broader network of systems that themselves impact the family's adaptations (Henggeler & Bourdin, 1990). One example of multigenerational thinking already mentioned is Bowen's work. In contrast with seemingly related psychodynamic approaches that had long maintained that people inherit, introject, and bring forward patterns from their historical past, Bowen's perspective on multigenerational transmission was that the past never left, and that families and family emotional systems press up against and interlock daily with prior generations. Friedman (1991) captured this position's essence well in observing that for Bowen, the intergenerational evolutionary flow had more power to format the structure of relationships than did the logic of their current connections. As should be apparent, true adoption of such a framework demands a thorough rethinking of the scope of work required to dislodge a family's symptomatic patterns and sustain newer forms of functioning.

Another seminal perspective on the historical embeddedness of contemporary child and family adjustment problems is Boszormeny-Nagy's contextual family therapy. As did Bowen, Nagy emphasized the continuing powerful influence of relations with families of origin, including parents' indebtedness to their origin families (innovatively cast as a ledger of owed balances and obligations). Many of these ideas were developed in Boszormeny-Nagy and Sparks's (1973) *Invisible Loyalties: Reciprocity in Intergenerational Family Therapy,* an influential work that illustrated the ways in which origin family dynamics reach into contemporary family processes even when parents are consciously unaware of or insist on having "disowned" them. Like Bowen, Nagy emphasized the unearthing of hidden ties and loyalties and the repair of unsolved problems. Unlike Bowen (who sent families home to do such work), Nagy gathered together various members of

the extended family network to address ruptured and strained relationships. The multigenerational perspective also helps to account for how it is that children in a contemporary family system come to be assigned (or to inherit) particular roles that reincarnate kindred roles that have existed in the family for generations. Attending only to the realities of the child's current-day nuclear family unit would not allow for detection of the source behind the historically entrenched family role the child has unwittingly stepped into and been expected to play.

Extended kin networks were also well attended to by the school of structural family therapy, though unlike the intergenerational approaches Salvador Minuchin's (1974) structural approach focused primarily on here-and-now family interaction and functioning. In the structural school of thought, the key core relational patterns in families are those revealed during family interactions. It would be inaccurate to imply that S. Minuchin did not give credence to the past. Rather, the structuralist perspective maintained that what is important about the past is reenacted in present transactions and is evident in current behavior. S. Minuchin, collaborating with colleagues Jay Haley (with whom he worked from 1967 to 1976), Braulio Montalvo, and Bernice Rossman, cultivated many of his seminal theses through work with low-income, inner-city families at the Child Guidance Clinic of Philadelphia; Haley later left to develop a distinctively different problem-solving therapy (see later discussion). Among the most influential core concepts contributed by structural family theory are those of family structure, boundaries, and subsystems, and in his therapeutic practice S. Minuchin worked to reconfigure nonadaptive family coalitions. He and his colleagues mapped family structure (the recurring and enduring patterns of interaction that had come to organize and structure daily family life) and then worked to foster change through techniques relying on the creation or amplification of challenges within here-and-now, ongoing family interactions. To be successful, they found it necessary to join and then work from within the family system, calling on action-oriented approaches as they worked directly with in-session enactments to help introduce, promote, and support change. By encouraging and supporting new patterns of behavior that introduced structural shifts in the family's hierarchical structure, alliance patterns, and boundary adherence, structural therapists sought to move families toward more adaptive functioning—with the identified child or adolescent client following suit. S. Minuchin and Haley each understood and respected family developmental issues as they worked to reconfigure nonadaptive family coalitions. The structural approach went further, however, to take stock of the family's entire relational field, including extended kin, other supportive figures, and helping professionals and institutions.

Perhaps most central to understanding S. Minuchin's theory are his views of hierarchy and boundaries. In structural theory, adaptive and healthy family systems are hierarchically organized, such that parents or parenting adults are clearly in charge as the system's executives. Families, as indicated earlier, can be composed of any number of different subsystems, including marital, coparenting, parent-child, and sibling subsystems. In well-functioning families, the adaptive emotional growth and development of children and other family members is contingent on the existence of appropriate boundaries between these subsystems. Conversely, when hierarchies break down, boundaries are violated, or children are triangulated

into adult-adult subsystems, symptomatic behavior is far more likely. S. Minuchin outlined a number of different ways in which children can be triangulated in family systems. In some cases, parents may compete for a child's loyalty and one may succeed in establishing a coalition with the child that excludes the other parent. In other cases, parents respond to distress in their relationship by deflecting their distress and anxiety to the needs or problems of a child. This solution provides short-term relief for the adults' relational distress and can artificially bond them together, as such detouring often results in psychosomatic or other emotional problems for the pathologized child. As did many other family theorists, S. Minuchin concurred that the identified patient's symptomatology played an important role in maintaining family homeostasis, keeping the family system together, and preventing it from fragmenting (S. Minuchin, Rosman, & Baker, 1978).

There has been a fair degree of clinical and research-based support for many of these basic suppositions (Andolfi, 1978; Buchanan, Maccoby, & Dornbusch, 1991; Camara & Resnick, 1989; Johnston, Campbell, & Mayes, 1985; S. Minuchin et al., 1978; S. Minuchin, Chamberlain, & Graubard, 1967; Rosenberg, 1978), and several integrative models have been derived in recent years using classic structural family therapy as their base. These include an attachment-based family therapy promulgated by Diamond, Siqueland, and Diamond (2003); a biobehavioral family model outlined by Wood, Klebba, and Miller (2000); a multidimensional family therapy propounded by Liddle (2000); and a multisystemic family therapy model, which explicitly takes into consideration other systems in which families are embedded, advanced by Henggeler, Schoenwald, Borduin, Rowland, and Cunningham (1998).

There have been other recent and major elaborations of practice promulgated by S. Minuchin's protégés at the Philadelphia Child and Family Therapy Training Center, expanding the basics of S. Minuchin's approach rather significantly. Earlier forms of structural family therapy emphasized a relatively rapid joining of the family by a strong and directive therapist who worked first to understand and accept the family's version of reality, and then intervened actively to alter here-and-now processes. By contrast, present-day practitioners often assess far beyond here-and-now family processes in an effort to more fully understand the family in its historical and larger community context and to take stock of potentially relevant individual biological, affective, and psychological processes of family members (Jones & Lindblad-Goldberg, 2002). They rely less on appealing to the therapist's expertise in establishing a hierarchical relationship with the family and fully engage families in planning and evaluation of assessment and treatment. They attend to the creation of relationships that will promote and nurture socioemotional competencies of family members, including emotion regulation and promotion of attachment.

Jones and Lindblad-Goldberg (2002) maintain that these shifts were natural ones as greater information than was available to S. Minuchin 30 years earlier came to the fore about the particular family and parenting processes that promote or constrain socioemotional development. They also point out that with the recent shifts in practice comes a more demanding accountability, from a focus on organizational functionality as the central treatment outcome to a focus on establishment of growth-promoting practices and strength of emotional connections. For the latter to succeed,

change will almost always be necessary at multiple system levels, including those beyond child and family at the child-family community interface. As Nichols and Schwartz (1998, p. 316) mused in commenting on the challenges of work with inner-city multiproblem families, "When between five to ten public agencies are involved in a family's life, and when the family is up against the crushing weight of poverty and racism, family therapy can seem quite puny. Recognizing limits has made therapists rethink their roles." We develop this systems-within-systems principle in the case example provided later in this chapter.

One other major effort from the early golden years of family therapy not yet discussed was Jay Haley's development of strategic family therapy. Following his work with the Palo Alto group and S. Minuchin, and influenced heavily by Milton Erickson's creative use of unorthodox therapeutic techniques and hypnosis to bring about rapid therapeutic change, Haley founded the Family Therapy Institute in 1976 and began offering brief problem-solving therapies. Recognizing that families in crisis sought rapid relief and seldom committed to participation in therapy for extended periods, Haley's time-compressed approach moved to initiate change in families not via insight and understanding but by the family carrying out directives prescribed by the therapist. Haley's goals were structural, and like S. Minuchin, he sought to address hierarchy and boundary problems that created and maintained problem sequences in families. Though his ideas owed in part to the cybernetic concept of positive feedback loops, Haley emphasized longer (and sometimes, much—month-long or more—longer) chains of sequence as well as chains that involved three, rather than two, people. Where he diverged from his earlier colleagues was in his calculated approach to effecting sequence and thereby system change, and in his view that family interactions could be understood as human struggles for control and power (Haley, 1976).

In strategic therapy, the therapist takes an authoritative and often manipulative stance designed to counter family resistance. Families are actively directed and maneuvered to undertake activities that amplify and exaggerate the troublesome symptoms that were actually serving as communication metaphors (see also Madanes, 1981, 1984). Strategic interventions are often subtle, usually covert, and frequently paradoxical (such as instructing families not to change and to exaggerate their disturbing symptoms). These kinds of interventions have been critiqued by many writers as both controversial and risky, not only because families inevitably deteriorated, at least temporarily, when following paradoxical directives, but also because the ultimate risk-benefit ratio and conditions for effective and safe use of paradoxical interventions were never empirically determined (Fruzzetti & Jacobson, 1991). Nonetheless, the strategic movement was one of the most active approaches on the scene during the second half of the twentieth century and gave rise to a number of creative variants (e.g., Selvini Palazzoli et al., 1978).

Prior to family therapy's embrace of multisystems thinking, most approaches had stayed squarely focused on family interiors, working with family systems as bounded entities unto themselves. As we've outlined, for some theorists this included the family's embeddedness in their multigenerational histories, and for others a lateral expansion to extended kin networks—but seldom did the views take stock of the broader organizational role of cultural attitudes and historical cohort.

This situation changed dramatically following searching critiques by Rachel Hare-Mustin (1978), Deborah Leupnitz (1988), and others who decried the gender bias inherent in the existing major schools of thought. For example, cybernetic theory, in repudiating unilateral control and power in systems (the components of which, instead, continually and circularly influenced one another in recurring feedback loops), implicitly absolved any particular system member of blame. When problems existed in a system, all of the system's components were involved and hence all could be said to bear equal responsibility. Such a posture, however, ignored the realities of the subjugation of women in a patriarchal society and effectively blamed victims in crimes against them. The implication that husbands and wives contributed equally to battering, for example, and bore equal responsibility for changing such problems reified cultural patriarchy.

Similarly, feminist writers drew attention to seldom examined historical realities driving one of the more common clinical presentations in two-parent families, in which mothers were cast as overinvolved and ineffectual and fathers as problematically underinvolved. Sadly, family therapy's seeming advance over psychodynamic theory in absolving mothers of primary blame for the child's ills (by factoring fathers into the family equation) had actually changed precious little. Mothers were still viewed negatively and pathologically, without recognition that they had been thrust by society into economically dependent, emotionally isolated, and hyperresponsible positions. Feminist therapists held that although a change of family circumstances was in order, it was not because incompetent mothers required help but because men's assumption of their responsibilities to their children would enable mothers to move out of the crazy-making situation that had been foisted upon them (McGoldrick, Anderson, & Walsh, 1989; Walters, Carter, Papp, & Silverstein, 1988).

In that systems approaches had introduced a revolutionary paradigm shift in the field of clinical psychology, it seems fitting that in closing this section we underscore how feminist theory in turn challenged existing systems beliefs. Feminist theory also had an impact on a final development highlighted here, which itself also deconstructed many of the unique expositions of earlier family theories. One critique of the early family systems work has been that its focus on cybernetics, systems, and patterns reflected a misguided search for grand theories that could explain human behavior, universal laws that could foster the understanding, diagnosis, repair, and resolution of most problems. This quest and its emphasis on the knowable and objectifiable were roundly criticized in postmodern critiques maintaining that there is no tangible reality, only the mental constructions of observers. Among the first theorists to broach this issue in family therapy were Paul Watzlawick (1984) and Lynn Hoffman (1985, 1988), who questioned the confidence and assuredness that had become such a central force in so many family therapies of the time. Collaborative approaches, such as the one introduced by Harlene Anderson (1993), followed soon thereafter, and took a position of *not* knowing in order to enable genuine conversations with families, and sticking firm to this stance rather than deftly shifting later into a position of expert. This approach, as did others, had its flaws, but it laid important groundwork for other critiques and new approaches, including narrative therapy, that were to follow.

In some ways, the essential tenets of narrative therapy threw into disarray all that family theories before it had promulgated. For decades, family therapy had stood in sharp contradistinction to psychodynamic approaches that viewed psychopathology as borne and bred within individuals; family therapy's basic tenet, as emphasized throughout this chapter, was that such problems were in interactions and relationships. But narrative theory eschewed this perspective as well, positing that problems were embedded in the stories that families lived by and that had come to organize their behavior. The focus shifted from the ways people behaved to the ways they constructed meaning. To move toward health, narrative approaches held, families needed to become aware of and discard these disempowering and self-defeating narratives they had been induced into, replacing them with alternate, health-promoting stories to govern their lives and future travels (Freedman & Combs, 1996; White, 1995). A major metaphor in such work was the externalization of problems: construing problems as forces threatening the family from the outside, rather than circulating within.

In many ways, narrative approaches incorporate very little of the systems thinking outlined throughout this chapter. Indeed, their social constructivist base deemphasizes dysfunctional family interactions and instead targets the malevolent influences of cultural biases and institutions. In so doing, narrative therapies provide a fresh, nonpathologizing view of families as capable and powerful. There is no question that their emphasis on collaboration and on expanding beyond families to frame families' problems infused creative new wisdom into the family field. It is less clear whether the accompanying wholesale rejection of systems theory by many narrative theorists was necessary. There remains relatively little evidence that any one system is inherently more or less helpful for aiding any particular problem, and there seem no inherent logical inconsistencies in retaining the valuable aspects of systems thinking while simultaneously attending to the humanizing, ethical, and empowering dimensions so celebrated in narrative frameworks (Eron & Lund, 1996). An ongoing challenge for those working in the field will be to evaluate when eclecticism in approach enhances work with families, and when it dilutes the power of particular approaches. Greater eclecticism and openness, greater collaboration with families, and refound respect for family competencies, worldviews, and inner wisdom have all enhanced modern practice, but in the end the approaches assumed by therapists are valuable only to the extent that they help effect real and meaningful change for families and family problems.

Family systems approaches and family therapy owe their place in the contemporary landscape of clinical psychology to the creativity and charismatic leadership of a relatively small group of brilliant individuals. This history has been well chronicled in a variety of sources, though this initial period of growth of the family therapy movement has been portrayed as a competitive and dogmatic one, guided by various leaders supported by groups of avid disciples emphasizing points of difference between the various approaches. As late as the 1980s, there had been relatively little integration across different approaches despite the similarities in many of the field's basic concepts. However, this circumstance has since changed. There has been a healthy and beneficial cross-pollination both across different schools of

thought within family systems approaches and in integrating important conceptualizations from other fields and disciplines, including developmental psychology and individual approaches to psychotherapy.

We have attempted to highlight some of the major areas of consensus among the various approaches. The majority of contemporary approaches view and treat families as systems, attend to forces that promote stability of behavior patterns and that delimit change, emphasize the triadic nature of human relationships while also recognizing the multiple systems in which families are embedded, and appreciate in one form or another the protective function of healthy boundaries not just for individuals and subsystems, but also for the family as a cohesive entity traveling through time together. Most schools concur that family groups function best when they are cohesive; freely, openly, and directly exchange information; and attend to the different developmental needs of different family members at changing points in the family life cycle. In addition, most perspectives view families as adaptive to the extent that they can show flexibility to adapt to shifting life circumstances, solve problems effectively, include adequate hierarchical structure, and support the individual autonomy and growth of all family members.

Beyond these similarities, there remain a variety of important differences in emphasis and scope. These include whether the proper focus of conceptualization and change efforts needs to be primarily on behavior and interactions, or whether it should take account of private inner experience of family members; the extent to which change efforts are enhanced by taking stock of individual character and personality dysfunction; and the extent to which family dynamics are seen as emanating from inner turmoil as opposed to being poor adaptations to the presses of an oppressive external world and culture. There are still disputes about whether the path to health will be charted through changes in behavior, action, and interaction or whether insight is required, and whether all family members must be a part of effective family solutions or whether it is possible to affect and sustain meaningful systemic change through work with individuals and subsystems. These things said, contemporary therapies rarely adhere tenaciously to theoretical dogma, and most now benefit from the convergence of ideas and rapprochement that began during the last decade of the twentieth century.

In an era of intense scrutiny on all forms of mental health intervention, cast in the name of accountability, family systems–based approaches to mental health and treatment have been called on to prove their merit to a widening external audience. In 1998, summarizing the current state of the field, Nichols and Schwartz anticipated this emerging challenge for family scholars and practitioners as they wrote "If family problems cause psychological problems, we should be able to demonstrate it; and perhaps more important, if we say we can resolve problems with family therapy, we'll probably have to prove it" (p. 498). The burgeoning field of family psychology, outlined later in the chapter, has largely shouldered the first part of this dual charge, while assumption of the second charge has also been formally taken on, though by a relatively smaller group of active researchers whose work has largely been informing recent national policy. These developments are summarized in the next section.

RESEARCH

If not already clear, from the days of the Palo Alto group's scholarly pursuit of communication theory, research has always been central to the understanding of family systems. Over the past quarter century, however, an interesting split of sorts has occurred. The growth of a family research field has been flourishing as empirical studies of families have benefited from advances in observational and recording capacities, digitalization, statistical modeling, and other scientific technologies and breakthroughs. Concurrently, a field bearing the name of family psychology has taken firm root. It is broad in scope and has been formally recognized as a new specialty within contemporary psychology. It is represented by a flagship journal, the *Journal of Family Psychology*, which debuted in 1986 and joined *Family Process* (an outlet that has been around since 1962, when Jay Haley served as its first editor) as the major source of breaking knowledge for professionals about family systems. Yet much of the scholarly work conducted under the flag of family psychology remains unfamiliar to active family practitioners.

The lack of dialogue is unfortunate, if understandable. James Alexander, who is among the small group of family psychologists whose work is well known to national policymakers, invoked Albert Einstein in a recent publication: "Not everything that can be counted counts, and not everything that counts can be counted" (Alexander et al., 2002, p. 17). Einstein's observation captures a rather interesting paradox in the family field: On the one hand, early family therapists took pains to insist that problem sequences in families are observable, identifiable, and deeply meaningful, and yet family therapy thinking has been replete with notions that would seem to defy objectification and standard measurement. Concepts such as triangulation, boundaries, hierarchy, differentiation, legacies, family projection processes, undifferentiated ego masses, rubber fences, and the like would seem to defy objective measurement. There have been both prevailing sentiments that any reductionist objectification of such inherently fluid family processes strips them of any true meaning, and contentions that empirical research is of little creative or additive value as it typically does little more than confirm established knowledge. Perhaps of greater concern, there is the fear that in objectifying complex structures and processes, researchers run the risk of obscuring and overlooking alternative interpretive frames. Indeed, from the postmodern perspective, everything is a construction and there is no objectifiable truth.

The discussion of operationalization and standardization has sometimes become most heated and strained when the topic shifts from the valid assessment of family processes to the assessment and efficacy of family therapies. Putting aside for the moment whether it is possible to evaluate important family group processes in meaningful ways (family psychologists have made a number of important strides in this arena, summarized later), a variety of writers have advanced the position that manualized treatments, insistence on adherence to treatment protocols, and evaluation of success in terms of statistical effect sizes undermines the creativity and fluidity that have been the hallmark of family therapy since its inception (Henry, 1998). This is certainly an extreme position, and not all practitioners share this concern that the empirical movement and move to more standardized treatment

regimens has a greater potential for harm than good (see, e.g., Jones & Lindblad-Goldberg, 2002).

Addressing this tension, Nichols and Schwartz (1998, p. 337) pointedly note that the research efforts of some of the major centers studying multiproblem families, including Henggeler's and Alexander's groups, Gerald Patterson's and Marion Forgatch's studies at the Oregon Social Learning Center, and major initiatives administered by Tolan and Gorman-Smith in Chicago and by Liddle and Szapocznik in Miami "provide credibility for family therapy within the larger mental health field, credibility that has been strained by the eccentric practices and unsubstantiated outcome claims" of earlier family models. Further, questioning the necessity and wisdom of casting blueprints as straitjackets, certain writers have decried the characteristically minimal dialectic between practicing family therapists and family therapy research groups. Nichols and Schwartz, for example, highlight both clinicians' ethical mandate to evaluate the efficacy of their interventions and the modern-day reality that family therapists must demonstrate the efficacy of their approaches to third-party payers. Echoing this latter point, Alexander and his colleagues (2002) noted that family-based approaches have been oddly absent from best practice lists generated by psychology's empirical treatments movement. They point out that this absence is out of keeping with accumulating empirical evidence for the effectiveness of family psychology interventions, evidence that has been gathered in research studies that have for the most part managed to sidestep the major criticisms typically levied against most other randomized clinical trial studies.

A comprehensive review of the efficacy literature generated by family psychology intervention scientists is beyond the scope of this chapter. Interested readers are referred to a comprehensive and readable volume published by the American Psychological Association, *Family Psychology: Science-Based Interventions*, edited by Liddle, Santisteban, Levant, and Bray (2002). It provides scholarly reviews and summaries of advances in family intervention research, including reviews of programs designed to enhance positive parenting practices through parent management training procedures, and other programs targeting parents' capacities for assuming responsibility (providing greater structure and closer monitoring), expressing warmth, and decreasing conflict and use of damaging disciplinary practices (e.g., Forgatch & Knutson, 2002; Liddle, 1995; Schoenwald & Henggeler, 2002; Szapocznik & Kurtines, 1989). Although initial work for some of these models was carried out in demonstration projects conducted in university settings, where conditions could be tightly controlled and titrated (e.g., Henggeler et al., 1986), there have since been concentrated efforts to render standardized family-based approaches within existing naturalistic contexts (homes, schools, community agencies, residential treatment facilities). An example of the more ambitious and intense work in this regard is multisystemic therapy provided within a family preservation model, introduced as a desirable alternative to outside-of-home placements of severely disturbed youth (Fraser, Nelson, & Rivard, 1997). In such work, outlined by Schoenwald and Henggeler (2002), anywhere from 2 to 15 hours of service per week is devoted to families over a concentrated 4- to 6-month period, with clinicians carrying low caseloads (4 to 6 families per clinician) but on call 24 hours 7 days a week.

Overall, empirically validated treatment approaches are informed by a systems perspective and by the contextual tradition of family therapy, but also focus broadly on behavioral, cognitive, and emotional transactions within the family and between the family and the influential social systems in which it resides. Such approaches generally see the focus of classic, pragmatic family therapies on intrafamilial interactions as insufficient to achieve and sustain meaningful gains with multiproblem youth. Most attend in one way or another to how children's intra-individual qualities (genetics, temperament) can challenge parenting systems and how social disadvantage, neighborhood, schools, and other contextual factors press on and interact with family processes. Many of these initiatives emphasize prevention and early identification of risk, and most are attentive to both cultural and ethic differences. Approaches vary a good deal with respect to how unabashedly they cast therapists into a role of expert educator, but all aim to affect some form of hierarchical change wherein parents assume greater responsibility for oversight and discipline of children while working to help promote greater family cohesion. Children themselves are often explicit targets of skill building as well, beyond the parenting and family work.

The other major focus of the family psychology field has been in providing evidence for ways families contribute in formative ways to the individual problems of their members. Such work often takes on a more decidedly linear focus than that endemic to cybernetic and circular perspectives, though researchers are often aware of this bias and episodically devote specific and concentrated energies to pursuing bidirectional, if not transactional, hypotheses. Again, a comprehensive review of this body of work is far beyond the scope of this chapter, but interested readers are referred to relevant overviews and contributions by P. Minuchin (1985), Cox and Paley (2003), McHale and Grolnick (2002), and Bornstein and Sawyer (2006). A special edition of the journal *Development and Psychopathology* (Davies & Cicchetti, 2004) includes a broad array of empirical papers presenting work from leading researchers involved in this area.

Much research in this tradition has been concerned with establishing and clarifying the nature of dynamic linkages between different family subsystems. Investigations of couple relationships, for example, reliably tie relationship functioning to both parenting behavior and child adjustment (e.g., Hetherington, 2006; Kaczynski, Lindahl, & Malik, 2006), as would be predicted by most theoretical frameworks in the family therapy field. Studies typically document child adjustment problems in the face of destructive marital conflict (Cummings & Davies, 1994), parenting disturbances in the face of marital distress (Erel & Burman, 1995), and child disturbances as a consequence of insecure parent-child attachments (Arend, Gove, & Sroufe, 1979; Speltz, DeKlyen, & Greenberg, 1999; Vondra, Shaw, & Swearingen, 2001; Weiss & Seed, 2002). Studies are often quite explicit in outlining the specific nature of dynamics in families, as when mothers in marriages devoid of intimacy are at greater risk for boundary violations and seductive behavior with children (Jacobvitz, Hazen, & Curran, 2004). They substantiate the cross-generational transmission of relationship patterns, as when parents who deny the significance of attachment-related experiences in their own origin families parent their infants in such a way that the infants likewise end up developing avoidant patterns of

attachment toward them (Steele, Steele, & Fonagy, 1996; Ward & Carlson, 1995; Zeanah, Benoit, & Barton, 1993). Such research studies are also attentive to context, as when outlining how changes in circumstances within or outside the family can alter risk trajectories (Egeland, Jacobvitz, & Sroufe, 1988).

Beginning in the mid-1990s, a new line of empirical investigations began appearing in the child and family development literature, explicitly assessing family group-level dynamics and the relationship of these whole family processes to children's adaptation (e.g., McHale & Cowan, 1996; McHale & Fivaz-Depeursinge, 1999). In particular, this work began paying explicit attention to specific coparenting dynamics in nuclear family systems, including hostility and competitiveness, alliance cohesion and support, and disconnection (Feinberg, 2003; McHale et al., 2002). A parallel line of inquiry also published during this time, fully systemic in nature, examined the development of family alliances from the prenatal period on through the toddler and preschool years (Favez et al., in press; Fivaz-Depeursinge & Corboz-Warnery, 1999). Among the most significant findings from these complementary lines of work are indications that early coparenting dynamics and family alliances can be forecast on the basis of both intrapsychic and interpersonal factors assessed during the pregnancy (McHale et al., 2004; von Klitzing, Simoni, & Burgin, 1999); coparenting processes are related to but also exert influences in the family that are independent of marital relationships and dynamics (Bearss & Eyberg, 1998; Katz & Low, 2004; McHale, 1995; Schoppe-Sullivan, Mangelsdorf, Frosch, & McHale, 2004); family alliance types consolidate early, are markedly stable through time, and affect children's social and emotional development in a host of important ways, even as early as the infant, toddler, and preschool years (Belsky, Putnam, & Crnic, 1996; Caldera & Lindsey, 2006; Fivaz-Depeursinge, Frascarolo, & Corboz-Warnery, 1996; Frosch, Mangelsdorf, & McHale, 2000; McHale, 2007b; McHale & Rasmussen, 1998); and family dynamics are sensitive to infant contributions from the very earliest months (Fivaz-Depeursinge, Favez, Lavanchy, de Noni, & Frascarolo, 2005; Fivaz-Depeursinge & Favez, 2006; McHale & Rotman, 2007). Related work has substantiated the protective effect of functional family hierarchies (Shaw, Criss, & Schonberg, 2004), and conversely, the destructive effects of family triangulation (Kerig, 1995).

These various lines of inquiry each, in their own way, have provided impetus for one form of intervention or another targeting family factors shown to increase risk. For example, if couple relationship problems before a baby's arrival are shown to foreshadow parenting and relationship problems afterward, then strengthening the couple relationship might be expected to help promote early parenting and couple adjustment (C. P. Cowan & Cowan, 1992; Shapiro & Gottman, 2005; Silliman, Stanley, Coffin, Markman, & Jordan, 2002). If maternal insensitivity places infants at risk for cultivating insecure attachments, interactive guidance and other forms of mother-infant psychotherapy targeting mothers' ability to read and respond warmly, contingently, and accurately to infants' signals may help mitigate such risk (McDonough, 1993; Robert-Tissot, Cramer, & Stern, 1996). If such reading is impeded or overwhelmed by maternal preoccupation with unresolved origin family issues, maternal representations may serve as an additional target for such interventions (Cramer, 1995; Fraiberg, 1980).

An interesting question from a systems perspective, of course, is whether long-term gains can be sustained if a single dyad in the family, be it husband-wife, mother-infant, or (as in several recent initiatives targeting father involvement; e.g., McBride & Mills, 1993) father-infant, is prioritized without parallel consideration of the ripple effects such interventions may cause throughout the family system (McHale, 2007a). Herein lies one crucial and ongoing challenge for systemically informed interventionists and clinical researchers of all sorts: establishing the extent to which an intervention with an individual or a dyad is sufficient to change a family's course, even if not all family members are the focus of intervention. Though the possibility of affecting meaningful systemic change without engaging all key players within the family would seem antithetical to most family systems theories, dyadic interventions are actually standard practice in the infant mental health field, as well as being an unfortunate reality of most community-based practice (P. A. Cowan, Cowan, Cohen, Pruett, & Pruett, in press). Hence, both basic and research efforts will continue to be crucial in helping to document the extent to which interventions seeking to effect enduring change show clinically meaningful results even without attending to all important relationships and networks in case formulations and intervention.

The vigorous dialogue about such issues that is active in family research circles exemplifies a level of inquiry and discourse healthy for all systems. For researchers, there is a never-ending revisiting and reworking of existing frameworks contingent on new data, even as certain bulwarks of the field (organizing effects of security of attachment, multirisk likelihood attendant to destructive marital conflict, and protective facets of supportive coparenting relations) provide necessary coherence. When there are episodic moves toward premature closure—some of the most vocal critiques of the movement toward empirically validated treatments have been along this line—there are corrective responses from within the research community. Episodically, someone can be counted on to stir the pot vigorously and threaten deconstruction of all held dear, as did Belsky, Campbell, Conn, and Moore, in one 1996 report failing to uncover significant stability in attachment security, either at the level of avoidance, security, and ambivalence (Mary Ainsworth's standard ABC classification rubric) or secure-insecure classifications. A more recent example involved Cook and Kenny's (2006) revisiting of data from a 1989 study by Cole and Jordan. Cole and Jordan had adapted Olson and colleagues' Cohesion and Adaptability scales on the Family Adaptability and Cohesion Evaluation Scale (FACES) instrument so that the scales' individual items reflected dyad-specific (mother-child, mother-father, father-child) rather than family-level processes. They had then asked each of the three family members to rate each of the three subsystems. Reanalyzing Cole and Jordan's data using advanced statistical analyses (EQS; Bentler, 1993), Cook and Kenny found that systematic variance at dyad subsystem levels was *rarely* significant after they took into account variance that was attributable to individuals within the family (mothers, fathers, or children). Taking the argument to its conceptual level, they speculated that in some circumstances what appears to be a systemic pattern can be as or more parsimoniously attributed to a single individual's influence, as when a father who does not like to be involved with people in general has difficulty being engaged with both his spouse and his children. Such a

problem, which when framed systemically would underscore a dialectic between dyadic (marital and parent-child subsystems), could actually be traced most fundamentally to the father's personality and habits. Any empirical finding like this is clearly delimited by the fragilities of instrumentation, yet such cyclical rekindling of old debates helps keep the field and its pursuits fresh, self-evaluative, and noncomplacent.

Perhaps somewhat surprisingly, despite the many, many volumes published on measurement issues in family research (e.g., Hofferth & Casper, 2006), assessments and evaluations of families in clinical practice seldom rely on instruments developed in laboratory research. In fact, most family assessments and evaluations do not rely on standardized instruments at all. In the next section, we summarize common practices and issues in family assessment and evaluation.

ASSESSMENT AND EVALUATION

As may not be surprising given the vast array of different approaches and areas of emphasis represented by different schools of thought, no standard or universally agreed upon set of assessment practices or techniques for evaluating families has ever emerged. Many students trained in family therapy approaches are introduced to some of the innovative ways of gathering information about families, such as completing detailed genograms (McGoldrick & Gerson, 1999)—sometimes indispensable in multigenerational formulations and interventions—and some are introduced to structured checklists or inventories (which help to organize and structure behavioral therapies). But most family therapists do not define formal assessment phases during which they systematically partake in standard assessment protocols or procedures. This is certainly not to imply that assessments are not conducted or are thought of as unimportant, but only that a cataloguing of different assessment tools in this section would badly misrepresent this facet of clinical work with families. Rather, we summarize here some of the major issues attended to by most systems therapists in the early stages of working with families.

Besides explicitly seeking information and clarity during family interviews about certain critical life events (e.g., domestic violence, sexual abuse, extramarital affairs, drug and alcohol abuse), most family systems approaches give special credence to the behavioral sequences and interactions that are revealed during early contacts with families. Such interactions are thought to reveal much of what is necessary to know about family communication patterns, boundaries, rules, and hierarchies. What family members do during assessments is as important as and sometimes more important than what they say, and hence very few family assessments take place without some direct observation of the family's interaction patterns. How early assessments are conducted, how directive therapists are during initial evaluations, and what the goals of early contacts are do differ from school to school, though there are some commonalities. First, most family therapists take pains during the early stages of work with families to listen to and take stock of the family's account of the presenting problem. During this stage, each family member is usually allowed a voice, and therapists acknowledge each person's perspective on the problem. This process is usually navigated in as open-ended a way as is

possible before additional detail is sought. The additional detail, when solicited, is geared to identify important details and to contextualize the problem with respect to time, place, and key players involved (including important individuals and systems outside of the immediate family). Information is also obtained about how the family has already tried to deal with the problem.

On the basis of these early assessments, interventionists are able to draw some preliminary hypotheses about structural and communication problems in families. From both session content and from observation of interaction sequences, the family hierarchy and the roles of different family members in the family's dynamics begin to become apparent. From observed family process patterns and sequences, the therapist is able to speculate about triangles, coalitions, and problematic boundaries within the system. This is particularly so when all of those principally involved in the maintenance of the problematic patterns are in attendance; understanding of family dynamics and communication patterns in families often shifts significantly when a sibling or live-in grandparent who had been absent during earlier family sessions subsequently attends a session and a different sequence pattern or alliance structure is revealed. Ultimately, however, an important aim for most therapies, not just structural approaches, is to develop an understanding of the invisible structures maintaining the referral problem.

Most therapies proceed from a general appreciation of the family life cycle and of normal family process. However, such understandings are often very global in scope and not always fully informed by contemporary developmental and family data. Most clinicians appreciate that the transition to new parenthood brings expectable strains that test the husband-wife dyad and increase risk for disengagement and/or depression by new mothers or fathers; that the early coparenting dynamic established in families often must reorganize as children move from infancy through the toddler and preschool years, since the behavioral challenges presented by mobile, verbal children introduce fresh new strain into the family system; and that adolescence requires parents to negotiate new relationships with their children to allow for greater autonomy and choice, without pulling the rug out from under them or affording too much freedom and not enough guidance. Less well understood are how normative developmental transitions are shaded by cultural expectations or by the origin family expectancies of the child's parents. For example, unlike most European and African American family systems, cosleeping and hand-feeding well through the preschool years is normative in many Asian heritage families. This cultural understanding is necessary to learn about during clinical assessments of Asian families, lest the family's parenting practices be cast as emotional overinvolvement. On a different front, cultural views of sexuality during early adolescence are also important to assess, as such views can be dramatically different in families where both parents became sexually active during their own early teens than they are in families where parents' sexual activity did not bloom until early adulthood. In research labs, developmental studies of family systems have uncovered revolutionary new data suggesting that even infants as young as 3 to 4 months are recruited into and become capable of actively shaping coparenting and family process (Fivaz-Depeursinge & Favez, 2006). Although very few families of infants this young come to the attention of family therapists because of ostensible problems in the

infant, assessing a family system without including infants would be as incomplete as omitting key members from assessments of families of older children.

Beyond the assessment of families' presenting problems and the patterns sustaining them, and the cultivation of an understanding of family adaptation apropos their position in the family life cycle (informed by relevant cultural and subcultural shading), a centrally important assessment task in most modern family therapy approaches is identifying contributory roles played by systems outside the family. Although this is most acutely so in assessments completed with multiproblem families (Henggeler et al., 1998), external systems do come into play for the majority of families with children and adolescents. However, though most assessments take stock of the family's interface with other systems, the extent to which such information is formally solicited during the early phase of family evaluations differs greatly from therapy to therapy. Typically, information about significant others in the family would be explicitly pursued during questioning in the earliest contacts with families, whereas information about outside systems (schools, the mental health system, the legal system, other social agencies) would be gathered as indicated as case material was shared by the family. More than occasionally, especially in cases where there is significant involvement by systems outside the family, some contact with these systems will ultimately be indicated, though in most approaches such contacts are more likely to take place later in the course of treatment than they are during the initial family assessment phase.

Finally, it is important to emphasize that although most family therapies do not systematically evaluate the psychological functioning of each individual family member, neither do they ignore developmental problems or psychopathology in individuals when it is present. Over the long history of the family field, there have been times when such individual problems were downplayed or cast as secondary and relatively inconsequential to the more significant relationship problems that entrapped the family, though this is seldom the case any more. As outlined earlier, therapy approaches now do consider the extent to which individual biological, affective, and psychological processes of family members are relevant (Jones & Lindblad-Goldberg, 2002), as such an understanding can help, rather than obfuscate, reasons why certain families get stuck.

To illustrate some of the major concepts described in the chapter, we now turn to a case conceptualized and treated from a family systems approach. In so doing, we underscore how systemic approaches help guide both the formulation of the problem and the path toward intervention chosen in a manner that might diverge substantially from courses charted by other major theoretical approaches in this volume.

Case Illustration

The case we have chosen to illustrate family systems intervention is one involving work with the Jackson family, a married heterosexual couple and their 14-year-old son, John, the family's identified patient. John's parents describe him as having shown precipitous deterioration in many areas of functioning, citing school failure (tardies, cuts, failing grades, warnings from administration), marijuana and

(*Continued*)

alcohol use, and symptoms of depression (sleep disturbance, lethargy, lack of motivation, constriction of activities). For his part, John expresses anger and rejection of his mother and an alignment with his father in the marital breakdown presently occurring. John recently engaged in individual psychotherapy, though he also "forgot" several appointments. His parents were alarmed at the downward spiral in John's behavior and requested a psychiatric evaluation for medication.

John's father works as a business executive, and his mother has been a homemaker throughout the marriage. The father travels frequently and states that work has always been his primary involvement. Both parents agree that he is a good provider and concur that he has had only peripheral involvement in the domestic life of the family. Indeed, the coparenting partnership could be characterized as disconnected and skewed since John's birth. John's father took on very little of the hands-on, day-to-day work of parenting and almost always deferred to John's mother when child-related decisions had to be made. As individuals, the parents' personalities are also distinctively different. John's father impresses as low-key, intellectual, introverted, and conflict avoidant; when conflict escalates between him and his wife and/or between his wife and John, his response is typically to withdraw to his home office. By contrast, John's mother wears her emotions on her sleeve. She is extraverted and socially skilled, yet often impresses as dependent in her interpersonal style. The couple's long-term marriage (the first for both parents) became progressively less intimate over the years as the mother invested most of her attention in her parenting involvement with John, while the father invested in his successful career. Three years ago, the mother began an affair with a man in the community. She became increasingly preoccupied with this involvement, and as she did her parenting deteriorated. She became indulgent and less attuned and attentive to John's needs, and not surprisingly conflict between mother and son also escalated.

Presenting Complaints

The initial telephone contact was made by the father, who wanted to come in with his son to work on his son's acting-out at home. After some inquiry as to the family situation, the family therapist convinced the father that John's mother also needed to be included in the intake session. In the initial session, where all three family members were seen, the father and son sat close together on the therapist's couch. They exchanged frequent glances throughout the session, and on a few occasions finished one another's sentences. They both also engaged in verbal attacks of the mother, who remained quiet, self-recriminatory, and tearful during the session. The father reported that he had learned of his wife's affair only recently from a friend and had immediately confronted his wife, who confirmed its 3-year duration. The father described being furious, and announced his plans to separate and divorce. He had told his wife that she could not continue to live in the family's home, and she obliged by moving to a nearby apartment. The father said he has been torn because he cannot himself assume parental responsibility for John, given his work demands, whereas John's mother, whom John won't talk to, remains completely available. The father is unequivocal in asserting that he intends to separate from his wife, but he also understands that the son and his mother need to have a relationship. He acknowledges that he has discussed the affair with John and was appalled to have discovered that John had been colluding with his mother in keeping the affair a secret. This complicity has added to John's and John's father's shared rage and contempt for the mother.

During the session, the son reported, "My life used to be great, but now it sucks. . . . I hate my mother. She had an affair with this scumbag and she made me keep it a secret from my dad for 3 years. She used to bad-mouth my dad, but now I know she was lying. I can't trust anything she says. I wish my dad would divorce her so I wouldn't have to deal with her ever again." John is vehement in his insistence that he has no desire for contact with his mother.

John's mother expresses only remorse and seems accommodating of anything that might enable John's father to reconsider staying married. She states that she now recognizes that the main reason she pursued an affair was because of the increasing loneliness and frustration she felt in her attempts to rekindle intimacy with her husband. She says that she agreed to move out of the family home voluntarily (though she hopes it is temporary) and notes that since she moved out, John's father has made some changes so as to become more available to parent their son. She adds, however, that her husband is also calling on his extended family and friends to provide coverage when he is unavailable and laments that these individuals have also been quite disparaging of her to their son. She expresses concern that John has been unsupervised for the most part after school and reports that he has regularly been having friends over to the house to party. Currently, the mother comes to the home once or twice a week, at the request of the father (and over the objection of their son) to drive the son to school and to cook meals. She seems desperate to persuade her husband to consider reconciliation, fearing that separating will only add further stress to the son.

Both parents are extremely concerned about the son's functioning and agree that addressing his deterioration is the priority over sorting out their marital issues. Because the father seems resolute in his intent to separate and divorce, the stance the family therapist elects to take in closing the intake session is that the structure of their family needs to transition to more independent parenting of the son, consistent with a postseparation shared custody arrangement. The mother was clearly devastated by the therapist's recommendation and ongoing work was necessary to help her acknowledge that this was the only viable structure that would permit the couple to quickly begin addressing their son's significant dysfunction.

Family System Assessment

In keeping with contemporary structural family frameworks, John's current symptoms were presumed to derive from the mutual influences of both individual and family system factors, as well as factors external to the family system. On an individual level, John's adjustment was hampered by several individual vulnerabilities (including the attentional deficit, the current substance abuse, and the growing depression). Each of these factors would require ongoing assessment and monitoring because John's ability to respond positively to structural systemic interventions could potentially be affected adversely by these individual-level factors. For this reason, coordination with an effective individual psychotherapy and perhaps, at some phase of the treatment, a referral for psychiatric assessment may be indicated and pursued.

With respect to the context in which John's symptoms are developing, the marital subsystem had clearly deteriorated, perhaps irrevocably. The affair, triangulating a third party into the marital relationship, compromised the marital

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subsystem further, with the breach of trust, humiliation, and subsequent anger and contempt prompting the husband's unilateral decision to sever the marital partnership. Equally, the parental subsystem had been markedly compromised; the previous functionality of the mother's primary parental role, which had gradually deteriorated as she began neglecting domestic matters, took a further downward turn once the affair came to light because John began actively resisting her efforts to parent. While John looked to be intensely aligned with his father, the father remained substantially unavailable and was lacking in necessary parenting skills. Moreover, given the destructive conflict in the marital subsystem, the already disconnected coparenting subsystem became compromised even further.

The uncertain status of the Jackson's nuclear family given the current separation placed their family system in a family life cycle limbo. That is, it was far from clear whether the therapeutic aim should be working to restore greater functionality in the nuclear family system or working to restructure the Jacksons as a binuclear family (which would be the structural shift that would have to occur with a separation and divorce). The coparenting team's executive authority in structuring the family vis-à-vis the son had been undermined by multiple factors, including the mother's reliance on John as a confidant and the alignment John presently had with the father against John's mother. Both of these developments had disrupted the functional hierarchy in which the coparental subsystem trumped the parent-child dyadic subsystems.

Problems in the current coparenting alliance were numerous, dominated by the lack of parenting support between partners, leaving each to fly solo. For this family, the absence of a functional coparenting alliance was especially disastrous, given the father's historical absenteeism and lack of competence when he was in charge and John's newfound unwillingness to allow his mother to parent him. Successful coparenting was further compromised by ineffective child-focused communication and decision making, stemming from the adults' self-focus, hostility, and open conflict in the marriage. John's escalating symptomatology only placed added stress and pressure on this already compromised coparental subsystem. The lack of boundaries and structure (agreements and coordination about when each parent is to be parenting, if and when John's mother should and could be in the family home, who was responsible for coordinating with John's school, therapist, tutor) resulted in chaos, conflict, and further neglect of John's now considerable needs. From a family systems perspective, John's increasing symptomatology is readily cast as a response to the untenable role demanded of him in keeping his mother's affair a secret, and as a cry for help for a chaotic and disorganized family system.

This understanding of the family interior is insufficient, however. Effective family systems approaches also attend to systems outside the family currently (or anticipated to be) affecting the family system. For the Jackson family, the family therapist identified several such systems. First, there was the family's mental health care, specifically, an individual therapist whom John had been seeing. While consultation with outside therapists is often indicated in family work when such individuals are involved, it turned out to be especially pivotal in this case. There were some indications that the individual therapist may have unwittingly introduced some additional, complicating boundary problems that affected the family. By seeing John and his father conjointly but failing to involve John's mother at all in the treatment, she appeared to have unintentionally gotten entangled in the couple's marital conflict and reinforced John's alignment with his father. In their

conjoint meetings, it turned out, father and son had frequently raged to the therapist about John's mother, even as the mother's phone calls to the therapist had sometimes gone unanswered for as much as a week or more, by mother's report.

To assess the role being played by the mental health system, the family therapist obtained a waiver of confidentiality to connect with John's individual therapist. Such contact was in order both to facilitate assessment of the individual therapist's perspectives and goals and to permit the family therapist to deepen her understanding of the family system so that the interventionists could try to coordinate a treatment plan together that would be working toward parallel, appropriate goals. Similarly, given John's school problems it was important to initiate contact with a relevant liaison at the school. This was essential both to better understand the nature of John's current problems at school and to permit creation of a functional coordination of home-school support for him. Finally, given the strong inkling that John's father was preparing to initiate the process of divorce, potential legal implications had to be considered so that any future involvement with the legal system could circumvent the parents' involvement in an adversarial process certain to exacerbate the family's already high level of conflict.

Case Conceptualization

For over a decade, the Jackson family had sustained a reasonably stable and functional family adaptation, although it was one characterized by a compartmentalized and skewed parenting partnership in which John's mother carried out the active parenting (cementing a strong mother-son dyadic subsystem) as the father assumed an economic provider role (functioning only marginally in both marital and father-son dyadic subsystems). This once stable structure had gradually deteriorated over time, as avoidance and alienation in the marriage ultimately precipitated the mother's undertaking of an extramarital affair. As the mother's long-standing availability and quality of parenting the son deteriorated to the point of neglect, conflict increased between the son and his mother, while the father continued to avoid involvement. Keeping the affair secret from the father (seen by many family therapists as exemplifying a pathological coalition between mother and son) had infused additional toxicity into this triadic family system. The Jackson family's dynamics are a vivid example of family subsystems mutually influencing one another, both in contributing to the escalating dysfunction of the system and in calling out for an intervention that attended to each subsystem.

John's symptoms were both a functional cry for help for his family system and a reflection of marital and parental subsystem breakdowns. The family's hierarchy and many of its boundaries (mother-son, marital-parental) had disintegrated and now needed considerable support to restructure. The limbo state of the separation and divorce decision complicated matters by impairing the Jacksons' ability to functionally respond to the family life cycle crisis. The possibility of a high-conflict custody dispute necessitated structuring any intervention to protect the family and treatment process from being cast into the legal adversarial system. The professional systems outside the family were not adequately supporting the family. Coordination was poor (as there was no functional connection with the son's school), and the son's therapist was aligned with the father.

Every individual member and every dyadic relationship subsystem in this family was showing a significant need for intervention, and the deleterious impact of each subsystem on the other (affected through the family's mutual feedback

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loops) likewise needed to be interrupted. Given the multiple problems, a clear prioritization of need was required, and the therapist in this case chose to work from a directive, structurally based approach. The approach involved joining with the family quickly and building a collaborative working alliance by utilizing the parents' mutual concern for their son's welfare.

Complications

A number of complications became apparent only as the case progressed, introducing difficult decisions about how best to intervene. First, as already indicated, consultation with John's individual child therapist confirmed that her role in helping John process the emotional trauma of the family breakdown had been severely compromised by her unwitting but problematic entanglement in the marital conflict. Given the complications this triangulation introduced, the family therapist and John's therapist needed to process whether it would be in the Jacksons' best interests for the individual therapist to withdraw from the case and support John's engagement with a new individual therapist who specialized in high-conflict coparenting cases.

Second, the seeming inevitability of the family's move into a divorce transition led the therapist to incorporate proactive structural and psychoeducational interventions into the work. One focus of this work was on preempting new adversarial processes, inherent in the legal system, so as to keep them from infiltrating and further escalating the marital conflict and endangering the family and its members. From the outset, the family therapist also had to decide whether to advocate reconstitution of the nuclear family structure (which was the mother's clear desire) or to assist the family in restructuring to a binuclear family. Because the creation of a clear and functional structure would be the key to restoring adaptive functioning in this family, the family therapist decided to strongly advocate for disengaging the marital and parenting dyads. This stance, which prompted initial distress and resistance from John's mother, ultimately helped to enable the work of differentiating marital issues (which in a binuclear structure were not relevant targets for the therapy, except as they interfered with functional coparenting) from coparenting issues (which became the agreed-upon focus in helping the family to move toward greater functionality).

Finally, given the severity of the son's current dysfunction and the family's seeming inability to cooperate adaptively, the family therapist posed the possibility that the son may need to be placed outside the home. Individual factors such as the boy's substance use and the possibility of his school failure or dropout, complicated by the inadequate functional parental involvement and supervision, questioned whether continued in-home placement for the son would be a viable option. The out-of-home placement recommendation had not been considered by the parents, who were both resistant to the idea for multiple reasons. Their unified resistance was used to motivate all the family members to work on issues with focus and urgency.

Course of Treatment

The initial intervention occurred when the father contacted the family therapist asking to come in with the son, and the therapist insisted that the whole family come in together for the intake. The presenting problem was then reframed by the suggestion that resolution of John's symptoms needed to be addressed in the broader family system. The therapist highlighted the seriousness of the son's symptoms and used the family's immediate resistance to an outside-the-home

placement to motivate the parents to prioritize the provision of better structure and support to the son and stabilize the marital and parenting subsystems. This was handled by the therapist in a directive manner, and an explicit plan was worked out with the couple to implement a binuclear family structure. The specific plan called for a nesting residential arrangement, wherein the parents shared a single apartment where first one, and then the other, resided when not parenting John. John, however, stayed in the family home with whichever parent was actively parenting. This arrangement allowed for (a) a stabilization of the basic family structure—there was no time line imposed or immediate decision called for about separation and divorce—at the same time as (b) the marital and parenting subsystems were functionally separated. Treatment was then able to progress to structuring (by creating a schedule for) the parenting time, thereby assuring that appropriate responsibility and supervision would be occurring. The boundaries between spousal issues and parenting issues were functionally reinstituted, and John's parents and the therapist mutually agreed to move spousal issues to the back burner and to focus on parenting issues as primary.

Work with the parental subsystem became the primary thrust of therapy. Disengaging the couple's focus on the marriage and keeping their energies child-focused, with exchanges taking place principally over e-mail (augmented by occasional phone contacts and in family sessions), contained their conflict and made the coparenting relationship more manageable. Sessions often clarified when spousal relationship issues were compromising coparenting, and the parents were gradually able to attain better boundaries with regard to these issues. Conjoint work with the mother and son addressed the trauma induced by the affair and the secret, working on healing the relationship breach that had occurred so that the son could gradually accept more nurturance and parenting from his mother. In the beginning, parenting time was titrated between mother and son based on the boy's tolerance of being with her. There was more variability later, depending on advances and setbacks in the mother-son reunification work, and so titration continued to be monitored in an ongoing way by the family therapist. Work with John's father addressed his need to step up and assume greater parenting responsibility, to make himself more available, and to work on his deficient parenting skills. Coparenting during these early phases was coordinated by the family therapist, with explicit focus on ongoing information exchange and active decision making about the son.

As this work began, the conversations with the boy's individual therapist commenced. It became clear rather quickly that she was aligned with the father-son coalition and that her identification with the son's rejection of the mother was especially strong. She reluctantly acknowledged that she had probably alienated the mother and possibly also compromised her primary therapeutic role by conducting conjoint parent-child work with only the father and son. However, because she was steadfast in her conviction that she would almost certainly be unable to form a working alliance with the mother, she gradually acknowledged (and helped persuade the reluctant father) that it seemed best to have the son referred to another therapist. The family therapist assisted the family with a referral to a child therapist who specialized in high-conflict cases and established a collaborative team approach with this therapist from the beginning of their involvement. The family therapist also contacted the school counselor (who had not heard from either parent) and coordinated a meeting between relevant school staff and both parents. The school counselor and parents set up a plan

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to support the son's functioning at school, which included structures for regular feedback from the school.

Progress was soon evident in multiple parameters of this family system's functioning. John's performance at school improved, and he gradually began working through his anger toward his mother, allowing her to reconnect as his primary parent. John's mother, who had ended the affair, redoubled her attentive parenting efforts and worked diligently to prove her commitment to preserve the marriage. John's father felt content with the moratorium on pursuing separation and divorce. He willingly accepted the parent coaching that the therapist offered and soon began looking forward to and enjoying his scheduled parenting time without John's mother present. Considerable work had to be done to help John's father support the healing of the mother-son relationship, despite his own continuing anger with her. The work of coparenting continued to be effectively coordinated by the family therapist; structures for communication were put into place, parenting issues that came up were discussed and decisions made in conjoint sessions with the parents, and written documentation of specific decisions made at these meetings were provided to John's parents to reinforce these structural shifts. Responsibility and accountability were supported throughout this process, from which a functional parental hierarchy came to be established with a transformation to a more equitable and shared parenting arrangement. This arrangement was established and documented in a formal parenting plan that evolved from the treatment, so that when the father made a decision to file for divorce 6 months after the work began, the family (despite some setbacks) was ultimately able to move collaboratively with only minimal legal-adversarial involvement to a shared custody arrangement. The family therapist–child therapist team continued to support the family during this transition.

Implications of the Case

John's individual symptoms were understood as occurring in the context of this family's triadic family process. Therefore, the goals of family systems treatment were broader than immediate resolution of the identified patient's symptoms. They were focused on addressing relational processes in the marital, coparental, and parent-child subsystems and in the extrafamilial system that was impacting the family. Helping the family to functionally restructure as they negotiated a transition from nuclear family to binuclear family was a critical focus of the treatment. Work on clarifying and reinforcing functional subsystem boundaries and reestablishing the hierarchy between parents and son remained a focus throughout the treatment. Given the destructive potential of the marital breakdown, structural disengagement of the marital couple (through the nesting residential arrangement and coordination of their coparenting) was an essential intervention in this case. Similarly, assessment of and direct intervention addressing extrafamilial involvement (with the son's individual therapist and relevant school personnel) was needed to create a collaborative professional context around the family system. Psychoeducation and recommendations about more collaborative divorce processes helped the parents resist the pulls from the legal adversarial family court system to escalate their spousal conflict. As the Jackson family moved through the divorce transition, the son benefited from a more balanced, competent, shared parenting arrangement, a marked change from the almost exclusive maternal involvement that had characterized his family experience prior to the beginning of treatment.

As illustrated in earlier sections of this chapter, there are myriad ways that family therapists of different schools might have chosen to approach and formulate this case, but there would also be a number of likely commonalities. For example, most systems-oriented clinicians would have sought to observe and assess the triadic or whole family process and to evaluate problems in the family's hierarchy and boundaries during initial sessions. Most also would have looked beyond the family's interior dynamics to understand the roles being played by interrelated systems—in this case, the school, legal, and mental health systems.

The therapeutic stance and the kinds of information gathered and privileged during this process would have varied from approach to approach, of course. The approach the family therapist took in working with the family was more old school than new age, in that a clear hierarchical arrangement installing the therapist as expert was set up from the outset to help stabilize the family quickly as a number of crucial decisions were made. The therapist was quite active in proposing an appropriate course of action, rather than coaxing the family to make these decisions themselves.

Different schools may very well have attended to other aspects of the case unaddressed in the work summarized here. For example, from certain frameworks, a spotlight may have been trained on John's father's financial and executive planning power and control in the family situation and on his mother's long-standing one-down position. Or focus may have been given to the cutoff of marital intimacy John's mother felt prior to embarking on the affair and/or to her empowerment as an individual (both absent in the case formulation) in the service of supporting and sustaining postdivorce parenting adaptation. The casting of out-of-home placement as an exterior threat to the family (which thereby mobilized their cohesiveness in combating the threat) resonated well with narrative approaches. Multigenerational theorists would have sought different insights to this case, perhaps unwrapping family members' collusion with the father in denigrating the mother to her son in case formulation.

Though there would almost certainly be differences in emphasis depending on the theoretical framework, the path toward recovery for the adolescent charted by helping to restructure the family, clarify boundaries, and strengthen the coparenting alliance was guided by a distinctively systems-driven approach. While remaining sensitive to the potential role played by intra-individual factors, the case exemplifies major ways in which therapists working from systems frameworks formulate and intervene in a manner differing in emphasis from those of other major traditions represented in this volume.

SUMMARY

Family therapy's revolutionary insight was that the problems of any given individual are a function of the whole family, and this systems viewpoint continues to guide virtually all contemporary family therapy practice. The field has undergone a dizzying number of changes in its 50-year history, and in at least certain schools of practice there have been some major breaks with long-standing traditions of focusing on interaction over cognition and on privileging relationships over individuals. Yet the hallmarks of systems approaches—thinking triadically about behavior problems, focusing on family process over content, understanding that

families are open systems embedded in extrafamilial systems—remain as fresh as ever. They will undoubtedly continue to have enduring impact and, when embraced, promise to infuse critically important insights into effective clinical practice.

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CHAPTER 8

Applied Behavior Analysis

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GENERAL OVERVIEW

The purpose of this chapter is to provide readers with basic information on applied behavior analysis (ABA). Important terms used throughout the chapter are defined first. Then, as context is highly relevant to ABA, a context for the work is provided. This context setting is accomplished through brief discussions of the philosophy and history associated with applied behavior analysis. Historical applications funnel to a discussion of the current uses of ABA. Finally, ethical considerations related to the use of ABA are covered.

Important Terms

Antecedent: A stimulus that acts as a signal or cue for future behavior either to repeat historically reinforced behavior or to avoid historically punished behavior (Baldwin & Baldwin, 2001).

Behavior: The organism's interaction with the environment in which there is a change in the environment due to an overt or covert action of the organism (Johnston & Pennypacker, 1993; Newman, Reeve, Reeve, & Ryan, 2003).

Conditioned stimulus: A stimulus that was formerly neutral that gains an ability to elicit respondent behavior through association with an unconditioned stimulus; was originally labeled a conditional stimulus (Newman et al., 2003).

Consequence: A stimulus that follows a behavior and results in a change in frequency of a behavior and/or in the probability of a behavior occurring and influences the significance of an antecedent stimulus as a cue for future behavior (Baldwin & Baldwin, 2001).

Differential reinforcement: A procedure in which one response is reinforced and another is extinguished; this process is involved in highly skilled and complex behaviors such as throwing a ball (Skinner, 1953).

Discrete trial training: A procedure used to teach new skills, in which there is a distinct block of time during which there is an instruction or prompt (an antecedent), a measure of emitted (or nonemitted) behavior, and consequential reinforcement (Newman et al., 2003).

Neutral stimulus: A stimulus that does not elicit a response without conditioning but can come to elicit a response after it is paired with an unconditioned stimulus and thus becomes a conditioned stimulus (Newman et al., 2003).

Operant conditioning: A procedure in which there is a behavior-environment interaction that changes the probability of a behavior (or the frequency of a behavior) occurring due to a consequence that contingently follows such behavior. This will also change the function of an antecedent consequence (Baldwin & Baldwin, 2001).

Punishment: A contingency between a behavior and a consequence that decreases the probability of that behavior occurring in the future under similar circumstances (Newman et al., 2003).

Reinforcement: A contingency between a behavior and a consequence that increases the probability of that behavior occurring in the future under similar circumstances (Baldwin & Baldwin, 2001; Leslie, 1996).

Respondent conditioning: Also known as classical conditioning or Pavlovian conditioning, a procedure in which a neutral stimulus is paired with an unconditioned stimulus and results in the neutral stimulus becoming a conditioned stimulus with the ability to elicit a response similar to that of the unconditioned stimulus with which it was paired (Newman et al., 2003).

Shaping: A process in which successively closer approximations, or variations, of a desired behavior are reinforced; learning occurs in a stepwise fashion and involves changing mastery criteria necessary to receive reinforcement so that more and more sophisticated behavior is required (Baldwin & Baldwin, 2001).

Target behavior: A behavior of interest in applied behavior analyses that can be increased or decreased depending on what is desired (Newman et al., 2003).

Unconditioned stimulus: A stimulus that elicits an innate or unlearned response, or a stimulus that leads to a response without teaching (reflex); originally called unconditional stimulus (Newman et al., 2003).

THE PHILOSOPHY OF APPLIED BEHAVIOR ANALYSIS

The Structure of Scientific Revolutions was published in 1962. In that seminal text, Thomas Kuhn wrote that it was impossible for a scientific community to conduct science without a set of foundational beliefs. Applied behavior analysis was still an emerging field in the early 1960s; however, the idea of communally shared beliefs has long been present in the philosophy underlying the work. As will become evident throughout this chapter, applied behavior analysis emphasizes not only recognition, but also explicit discussion of such a perspective.

The perspective on which applied behavior analysis is based is behaviorism. The term behaviorism can be used to refer to a philosophy of science or to a general approach to the scientific study of psychology (Lattal & Chase, 2003). At the most basic level, behaviorism refers to the idea that psychology can be explained through objective study, without making causal attributions to internal processes. There are

several varieties of behaviorism and definitions of behavior, but those who subscribe to behavior analytic work, often known as radical behaviorists, define behavior as anything that an integrated organism does in relation to context.

Defining Behavior

Defining behavior has been a key charge of many philosophers and psychologists throughout history. Aristotle attempted to understand behavior. Descartes did as well; he made a link between bodily movement and mechanical, as opposed to supernatural, causes. Descartes wrote that the soul and the body were two separate entities (i.e., he described an inherent dualism). He also proposed that some behaviors were caused by the soul, whereas other (less complex) behaviors were more mechanical, or determined by biological factors. Although his dualistic and mechanistic approach is quite different from much of the thought throughout behavior analysis, Descartes's work did set the stage for others, including Pavlov, to study reflexes (Leslie & O'Reilly, 1999). Classical conditioning principles as discovered and documented by Pavlov in the early 1900s play a crucial role in the historical development and the current conduct of applied behavior analysis. Therefore, Pavlov's work is elaborated further later in this chapter. For purposes of this philosophic discussion, it is sufficient to note that Pavlov's studies on salivation in dogs led to a link between biology and observable behavior.

Materialistic Approach

The emphasis on behavior as biological opened the door to an empirical analysis of materialistic philosophy. Materialism refers to the assumption that everything is composed of physical matter, and interactions between such material result in observable behavior. All supernatural explanations are rejected from a materialistic perspective (Cziko, 2000). Although early Greek philosophers, including Leucippus and Democritus, and early Western thinkers, such as Epicurus, proposed that behavior could be explained through the combination and separation of atoms, societal pressures about religion prevented materialism from taking hold until much later. There was some advancement of materialism in the seventeenth century when Hobbes wrote of the human body as a machine and LaMettrie and d'Holbach explained learning as resulting from changes in the brain. However, it was Newton's discovery of gravity and related physical principles that allowed for matter to be examined in a systematic manner (Cziko, 2000). Thorndike's studies on instrumental learning in animals, which will be elaborated on later, followed a materialistic approach and greatly advanced behaviorism.

Selectionism

Another, related advancement in philosophical thought and the study of behavior came when Darwin's work on evolution was published. Darwin's theory allowed for behavioral observations in biological contexts (Leslie & O'Reilly, 1999). The behavior analytic approach to studying psychological events parallels selectionism, the idea that characteristics of species evolve in response to success or effectiveness

(of survival and reproduction) in the environment. Behavior, like species development, is based on environmental contingencies; it is determined. Individual behavior and cultural behavior evolve in response to their effectiveness on and in context (Skinner, 1981). This emphasis on context, and behavior in interaction with context, is the reason it is so important to analyze and describe the philosophy underlying applied behavior analysis.

Contextualistic Approach

Contextualism refers to a philosophy of science that states that behavior is inseparable from the context in which it occurs. From a behavior analytic perspective, context can include the individual's learning history and internal states as well as the current physical environment. Contextualism emphasizes empiricism and experience. It is the opposite of mechanism, and involves the investigation of behavior at many levels that influence each other (Capaldi & Proctor, 1999). Contextualists believe that mechanistic attempts to study behavior lessen the quality of interpretation. Although early behaviorism was more mechanistic, and there are current behavioral works conducted from a mechanistic approach, there is wide consensus that ABA is contextualistic. One cannot perform ABA without careful consideration of context (Dougher & Hayes, 2000; Rosnow & Georgoudi, 1986). It is necessary to look at external factors that play a role in initiating and maintaining behavior.

Applied behavior analysis is built on a specific type of contextualism: functional contextualism (Biglan & Hayes, 1996). This term refers to a natural science that uses empirically based rules to predict and influence behavior with precision, scope, and depth (Biglan, 1995). From a behavior analytic perspective, prediction and influence are related to the function that is served by behavior. Related to that, causal variables are limited to those that are observable and manipulable. Given that internal variables do not meet those criteria, applied behavior analysis rejects the idea that a mind or soul causes behavior. Applied behavior analysis thus is monistic.

Pepper (1942) wrote that philosophical systems fall into one of four basic worldviews. Worldviews are characterized by root metaphors, which are how scientists attempt to understand the world, and by truth criteria, which are the measure by which analyses are evaluated. The root metaphor in functional contextualism is the act in and inseparable from its historical and current context, and the truth criterion is related to the function of the analysis. Truth depends on whether the goal is met (Biglan & Hayes, 1996). This is referred to as successful working, which was originally discussed in works on pragmatism by Pierce, Dewey, James, and Darwin. In fact, several behavior analysts have noted that contextualism and pragmatism occur symbiotically (e.g., Baer, Wolf, & Risley, 1968).

Pragmatism

Pragmatism emerged in the late nineteenth century as a mix of empiricism and psychology. More specifically, British empiricists, German Romantic idealists, Kant, Hegel, and others who worked in areas of philosophy, science, and education influenced pragmatism. Charles W. Pierce was the first in the United States to talk and

write about pragmatism. He talked about making ideas clear, which was related to using operational definitions and focusing on observable units of analyses as well as using successful working as a truth criterion. This is linked to behavior analysis in that the idea is to have correspondence between the goal and outcome after the analysis is complete. John Dewey and William James both expanded on Pierce's work. James took Pierce's work, which emphasized experimental (basic) work, and applied the philosophy to concerns about human behavior (Lattal & Laipple, 2003). James's work was part of an important shift from philosophy to the real world (Baum, 1994). Thus, our discussion shifts to a focus on the development and use of behavior analytic principles in applied settings.

A BRIEF HISTORY OF APPLIED BEHAVIOR ANALYSIS

Applied behavior analysis refers to the process of utilizing the principles of behavior in an effort to improve socially important behaviors (Baer et al., 1968). Principles of behavior were discovered through the use of carefully designed experiments, which have provided general information about respondent and operant conditioning as well as more specific knowledge about reinforcement, punishment, extinction, stimulus control, and their application. These principles are utilized in techniques such as shaping, chaining, token economies, and self-management—all of which were historically designed as methods to decrease maladaptive behavior with social relevance. Applied behavior analysis as we know it today began in the 1950s. At that time, researchers first applied the behavioral principles outlined by B. F. Skinner (1938) in his seminal work, *The Behavior of Organisms,* to clinical work with humans. However, it is important to also consider those figures and events that made early contributions to the development of behavioral principles.

Sechenov

Ivan Michailovich Sechenov (1829–1905) is considered the founder of Russian physiology (Koshtoyants, 1863/1965). At a time when the philosophical and psychological establishment emphasized mind-body duality, Sechenov took a monistic and materialistic approach to the study of mental processes, which he believed could be explained physiologically (Koshtoyants, 1863/1965). In his *Reflexes of the Brain,* Sechenov (1863/1965) proposed that all forms of behavior, including thoughts, emotions, and the holding of virtuous beliefs, were reflexes of the brain. Furthermore, he emphasized the importance of using strictly objective methods in the study of mental processes, and he stressed the significance of the role of the environment in an organism's behavior. Sechenov's ideas influenced subsequent Russian physiologists, most notably Ivan Pavlov (Koshtoyants, 1863/1965).

Pavlov

Ivan Petrovich Pavlov (1849–1936) shared Sechenov's materialistic approach, viewing psychological phenomena as physiological in origin. He also adhered to stringently objective methods in his study of higher nervous system activity (Pavlov, 1927/1928). Pavlov conducted his revolutionary research over the course

of 25 years, studying the gastric and salivary reflexes of dogs. His work on gastric reflexes earned him acclaim in physiology, not to mention the Nobel Prize, but it was his subsequent discovery and delineation of the principles of respondent, or classical, conditioning that made him a significant figure in psychology as well.

In the course of his research on gastric reflexes, Pavlov (1927/1928, p. 6) began to notice "psychic secretion" of saliva in his dogs. Specifically, the dogs began to salivate in the presence of external stimuli other than food (e.g., the experimenter who brought the food, the sound of the footsteps of said experimenter). Pavlov set about investigating this phenomenon in a strictly objective manner. So important to Pavlov was a precise experimental method in his studies of nervous system processes that a specially designed laboratory with a trench around the outside was constructed to ensure experimental control (Pavlov, 1927/1928). In his earliest experiments, Pavlov surgically transplanted the dog's salivary ducts to the outside of the cheek or under the chin for the purposes of observation and precise measurement. This surgical procedure made it possible to count the number of drops of saliva or measure the amount of colored fluid in a tube that was displaced by the dog's saliva. When food was presented to the dogs, they began to salivate. Pavlov considered the salivation response to food an unconditioned, or inborn, reflex that did not need to be trained, and food was considered an unconditioned stimulus. He then began preceding the presentation of food to the dogs with the sound of a beating metronome. After several trials in which the dogs heard the metronome and then received food, the sound of the metronome alone resulted in salivation. The metronome had become a conditioned stimulus, and the salivation in response to it was a conditioned response. Although at the time it was called a "conditioned reflex," today the term "reflex" is usually thought of in terms of the stimulus-response relationship. The conditioned response is a response that was previously an unlearned, or unconditioned, response to an unconditioned stimulus and is now occurring in response to a conditioned stimulus.

Pavlov (1927/1928) and his fellow researchers replicated these early experiments many times, conditioning a variety of stimuli to elicit salivation, including the buzzing of electric bells, the application of electric current, thermal stimuli, and the cessation of continuous metronome beating. In the process, Pavlov outlined several key principles:

- *Timing:* For a conditioned response to be established, the neutral stimulus must be presented such that it precedes or occurs at the same time as the unconditioned stimulus.
- *Extinction:* Repeated presentation of the conditioned stimulus without the unconditioned stimulus will result in a weakening of the conditioned response.
- *Differentiation, or discrimination:* Presentation of stimuli similar to the conditioned stimulus in the absence of the unconditioned stimulus will result in elicitation of the conditioned response only by the specific conditioned stimulus.
- *Generalization:* If discrimination training is not applied, the conditioned response may be elicited by stimuli that are similar to the conditioned stimulus even without direct pairing with the unconditioned stimulus.

Thorndike

At about the same time that Pavlov began investigating conditioned reflexes, Edward Lee Thorndike (1874–1949) was investigating other basic principles of learning in animals (Jonçich, 1968). Thorndike's work in animal, or comparative, psychology is considered part of the functionalist movement, a parent of behaviorism (O'Donnell, 1985). In his dissertation research at Columbia University, Thorndike conducted objective, methodical investigations of stimulus-response connections, or associations (Jonçich, 1968). The basic experiment consisted of placing a hungry kitten (and eventually hungry dogs and lonely chicks) in one of several puzzle boxes he designed, from which the animal could escape by actions such as pulling on the loop of a cord or pressing a lever (Thorndike, 1898). Food was placed visibly outside of the box (in the case of lonely chicks, other chicks were outside of the box), and Thorndike observed and recorded the animal's behavior and the time it took for the animal to escape the box. He plotted the time to escape over several trials on a graph and was one of the first researchers to capture evidence of learning in this way (Chance, 1999). Thorndike noted that, once the animal was successful, the time to escape generally consistently decreased with successive trials. He concluded that an association is formed between a certain situation and a certain response and that the resulting pleasure (obtaining food or company) served to strengthen this connection (Thorndike, 1898). In later works, Thorndike came to refer to this finding as the law of effect (Hearst, 1999).

Watson

If animal psychology can be thought of as a bridge between functionalism and behaviorism, John Broadus Watson (1878–1958) defiantly carried the flag of psychology across this bridge and planted it firmly on the other side. After 12 years of behavioral research with animals, Watson (1913) published "Psychology as the Behaviorist Views It," an article that called for a radically different approach to the science of psychology. He argued that psychology should abandon consciousness as its subject of study and forsake introspection as its method. Watson advocated psychology as an objective science, the goal of which would be the prediction and control of behavior through the study of observable responses to stimuli, using a precise and reproducible experimental method. The approach would be the same in studying animals and humans. Although Watson viewed the evaluation of the behavior of any species as important in its own right, he also criticized psychologists for failing to produce results that could be practically applied. Watson thus began the movement called behaviorism, which profoundly shaped psychological thought for much of the twentieth century (Bergmann, 1956).

Following his 1913 paper, Watson and others conducted behaviorist research that reflected a commitment to studying practical problems. His subsequent research on emotional responses as conditioned during childhood is a good example. In their famous experiment with 11-month-old Albert B., Watson and Rosalie Rayner (1920) endeavored to condition fear of an animal in the infant, to demonstrate that this conditioned emotional response could be transferred to

similar objects, to show that the response would persist over time, and to develop a method to remove the conditioned fear reaction. In the first three endeavors, they were successful. Prior to training, Albert showed no fear response to animals and objects presented to him, including a white rat, a rabbit, a dog, masks with hair, and cotton wool. He did exhibit a fear reaction in response to the noise made by a hammer being struck against a steel bar by an examiner standing behind him. To condition the fear response to a previously neutral animal, Watson and Rayner began trials in which the presentation of a white rat to Albert was paired with the striking of the hammer on the bar behind him. Within seven pairings of the two stimuli, the fear response had been completely conditioned, and presentation of the white rat alone was sufficient to elicit crying and escape behaviors in Albert. Upon further testing, the researchers determined that the conditioned fear response had transferred (or generalized) to a rabbit, a dog, a seal fur coat, cotton wool, Watson's hair, and a Santa Claus mask. Furthermore, the fear response to such objects was still present 1 month after the trials. Watson and Rayner planned to attempt the removal of the conditioned fear response in Albert, but the infant left the hospital where the experiments were conducted before any such attempts could be made.

Four years later, however, with assistance from Watson, Mary Cover Jones successfully unconditioned an existing fear response to white rabbits and similar objects in a 2-year, 10-month-old boy named Peter (Jones, 1924). To uncondition (or recondition) Peter's fear response, Jones first presented the rabbit to Peter on several occasions with other children in the room who eagerly interacted with the rabbit. Later she began pairing the presentation of the rabbit (successively closer, at first in a cage and then out of its cage) to Peter along with food and candy. By the end of the sessions, Peter showed no fear of the rabbit or similar objects. This was a significant early application of behavioral principles to a human problem. Through the work of B. F. Skinner, such applications would become an influential force in addressing countless human difficulties.

Skinner

Burrhus Frederick Skinner (1904–1990) graduated college with a major in English. After a year and a half as a struggling writer, he took a job in a bookstore in Greenwich Village in New York (Skinner, 1976). There he read Bertrand Russell's *Philosophy* (1925) and Watson's *Behaviorism* (1924/1930). Not long after, he applied and was accepted to graduate school in psychology at Harvard. In addition to psychology courses, Skinner also took classes in physiology under the department chair William Crozier. He studied Pavlov's work, and he began conducting research in the spirit of Pavlov and Watson's stimulus-response paradigm. Specifically, Skinner began investigating eating behavior in rats, which he postulated was a series of reflexes (Vargas, 2001). However, he soon observed that not all of the rats' behaviors were elicited by stimuli. Some behaviors were, instead, controlled by the stimuli that followed them. After 8 years of methodically investigating this phenomenon, Skinner (1938) published *The Behavior of Organisms,* in which he delineated the basic principles of behavior analysis.

In the studies summarized in *The Behavior of Organisms,* Skinner (1938) examined how the likelihood of a rat's emitting a target behavior could be changed by altering the consequences (in this case, the delivery of a food pellet) of that behavior. For these experiments, he constructed an "experimental box," which included a release chamber, a food tray, a waterspout, and a small brass lever that was connected to the food-dispensing magazine and to a recording device. Skinner used white rats because they were "cheap" (p. 47) and "amazingly stable" (p. 48). He chose lever pressing as the operant of study because it was an unconditioned response that occurred spontaneously in the rats from time to time before conditioning occurred. The food pellets he made contingent on responding were Purina Dog Chow.

In a typical experiment, a hungry rat would be placed behind the release door of the experimental box for 1 to 2 minutes before the experiment began (Skinner, 1938). This was done to control for any effects of the experimenter having just handled the rat. Once the door was opened, food would periodically be dispensed into the food tray from the magazine. The lever would not be connected to the magazine during this time, because it was first important for the rat to learn to respond to the tray only when it heard the magazine and not in the absence of the sound. Once this discriminative responding was established, the lever was connected and conditioning began. Any spontaneous lever pressing would now result in a food pellet being dispensed from the magazine into the tray. Lever pressing was recorded electrically in the form of a graph, with the total number of responses plotted over time. Because Skinner was interested in rate of responding, this was calculated from the slope of the lines created on the graphs. During initial conditioning, in general only four lever presses resulting in a food pellet being released into the tray were required for the rate of responding to increase dramatically. Skinner continued these experiments, delineating principles and demonstrating that the manipulation of reinforcement contingencies could powerfully shape behavior. He labeled these principles operant conditioning (Skinner, 1938).

Classical versus Operant Conditioning

Before continuing with the history of applied behavior analysis, it is important to note similarities and differences between classical, or respondent, conditioning as first described by Pavlov (1927/1928) and operant conditioning as described by Skinner, and to outline additional principles important in operant conditioning. Whereas respondent conditioning relies on antecedent stimuli predicting the presentation of another stimulus, operant conditioning relies on the consequent stimuli that follow a behavior. In other words, a consequent stimulus contingently following a behavior changes the future probability of the behavior recurring. Such a consequent stimulus is called a reinforcer.

In a situation in which a reinforcer is presented contingent on the occurrence of a particular behavior and subsequently increases the future probability of that behavior occurring, that reinforcer is referred to as a "positive reinforcer" (Newman et al., 2003). In a situation in which a reinforcer is removed contingent on the occurrence of a particular behavior and subsequently increases the future probability

of that behavior occurring, that reinforcer is referred to as a "negative reinforcer" (Newman et al., 2003). Note that in each case the reinforcers increase the future probability of a target behavior occurring. One factor that will increase the likelihood of a stimulus acting as a reinforcer is an organism's deprivation status (e.g., the longer an animal has been deprived of food, the greater the chance of food acting as a reinforcer, whereas a satiated animal's behavior may not be reinforced by food). Deprivation is an example of an establishing operation, which is a change in or characteristic of the environment that affects the power of stimuli to serve as antecedents and consequences that affect the initiation and maintenance of behavior (Newman et al., 2003).

In respondent conditioning, extinction refers to weakening of the conditioned response following presentation of the conditioned stimulus in the absence of the unconditioned stimulus (Pavlov, 1927/1928). In operant conditioning, extinction refers to a procedure or process in which the rate of a behavior of interest ultimately decreases. It involves the removal of the reinforcement contingency that produced operant conditioning, either by no longer presenting the reinforcing stimulus or by breaking the contingency that maintained the behavior of interest (Leslie, 1996, p. 57; Newman et al., 2003). In terms of behavior or process, extinction is an observed decline in the frequency of a response (Leslie, 1996).

Similar to the change in future probability of the occurrence of a target behavior (mentioned earlier in terms of reinforcement), there are also operant processes that decrease the likelihood of future behavior. As is the case with reinforcement, there are two types of punishment: positive and negative. Positive punishment is the contingent presentation of a stimulus following a particular behavior that subsequently decreases the future probability of that behavior occurring (Newman et al., 2003). Negative punishment is the contingent removal of a stimulus following a particular behavior that subsequently decreases the future probability of that behavior occurring (Newman et al., 2003). In instances of both reinforcement and punishment, positive and negative do not follow the popular connotations of "good" and "bad," but rather the presentation or removal of a stimulus, respectively. Reinforcement and punishment occur not in response to intentions, but in response to environmental contingencies that either increase or decrease the future probability of behavior, respectively (Newman et al., 2003).

The concepts of reinforcement and extinction are similar to Thorndike's instrumental behavior, but are spoken about in slightly different ways. Thorndike spoke in terms of the behavior as being influenced by pleasure or satisfaction (Leslie, 1996). Operant behavior, or more specifically, reinforcement, simply states that a reinforcer is a stimulus event that increases the probability of the behavior occurring in the future regardless of the organism's subjective evaluation (Skinner, 1953).

Operant behavior is generally thought of as voluntary behavior, and respondent behavior is thought of as involuntary behavior. Respondent behavior refers to a class of responses that are elicited by certain conditioned or unconditioned antecedent stimuli (Johnston & Pennypacker, 1993). Operants are a class of behaviors that may or may not be topographically similar and produce functionally similar consequences (Baldwin & Baldwin, 2001; Johnston & Pennypacker, 1993). The two classes of behavior are not mutually exclusive.

One way that these types of behavior overlap is that consequences can be classically conditioned. A primary, or unconditioned, reinforcer is one that is able to work effectively as a reinforcer without any prior learning or conditioning (Newman et al., 2003; Skinner, 1953). When a neutral consequence is paired with an established reinforcer, it becomes a reinforcer as well, that is then referred to as a conditioned, or secondary, reinforcer (Newman et al., 2003). One type of conditioned reinforcer is called a generalized reinforcer. A generalized reinforcer is a reinforcer that has been paired with more than one primary reinforcer and can often be exchanged for another reinforcer (Newman et al., 2003; Skinner, 1953). The benefit of a generalized reinforcer is that it has an effect on behavior despite the deprivation status of the organism. Examples of generalized reinforcers are tokens, successful control of our environment, attention, approval, affection, and the submissiveness of others (Skinner, 1953).

An area of additional overlap is that operant and respondent conditioning both affect antecedent control, albeit in different ways. An antecedent stimulus is said to have stimulus control when it reliably increases or decreases the probability of a response based on the organism's history, or experience (Baldwin & Baldwin, 2001; Leslie, 1996). Respondent conditioning involves an antecedent's ability to predict the presentation of an unconditioned stimulus. In operant conditioning, a behavior can be emitted in response to a particular antecedent stimulus but not another; these stimuli are called discriminative stimulus and S delta, respectively (Newman et al., 2003). Also in operant conditioning, one function of an antecedent is to set the occasion for an organism to repeat or avoid a past behavior that was met with reinforcement or punishment, respectively (Baldwin & Baldwin, 2001). In other words, in one situation the antecedent stimulus predicts the presence of an unconditioned stimulus (conditioned stimulus, or conditioned elicitor), and in the other case, it indicates the availability of a reinforcing or punishing consequence if a certain behavior is emitted.

Related to this issue, Skinner (1938) used the term "differentiation" to refer to operant processes, unlike Pavlov's definition, which referred to respondent responses. In *The Behavior of Organisms,* Skinner distinguished between differentiation and discrimination. Differentiation refers to distinction of the form of the response that will elicit the reinforcer from other responses in the same situation. Discrimination refers to the ability to distinguish which antecedent stimulus signals the availability of a reinforcer.

Skinner and many others continued basic laboratory research on behavioral principles (the experimental analysis of behavior) following the publication of *The Behavior of Organisms*. It became evident through such studies that reinforcement leads to maintenance of behavior, even when reinforcement is not continuous. In other words, a reinforcer does not have to immediately follow every response for behavior to be repeated (Leslie, 1996). Ferster and Skinner (1957) wrote about this phenomenon in *Schedules of Reinforcement*. The authors reported the results of comprehensive studies with pigeons, and they outlined the effect on behavior of various schedules of reinforcement. Specifically, they outlined fixed ratio (FR), fixed interval (FI), variable ratio (VR), and variable interval (VI) schedules:

- *FR:* A schedule of reinforcement in which a behavior is reinforced after a set number of responses.
- *FI:* A schedule of reinforcement in which a behavior is reinforced at the first response that occurs after a set interval of time elapses.
- *VR:* A schedule of reinforcement in which a behavior is reinforced after a variable number of responses occur that average around a value required for delivery of reinforcement.
- *VI:* A schedule of reinforcement in which a behavior is reinforced after a variable time interval elapses that averages around a value required for delivery of reinforcement.

Early Applications to Human Behavior

Ferster and Skinner's (1957) book as well as Keller and Schoenfeld's (1950) *Principles of Psychology,* which made Skinner's ideas more widely available, spurred continued *basic* research and necessitated the publishing of the *Journal of the Experimental Analysis of Behavior* (*JEAB*) in 1958 (Michael, 2004). However, it was Skinner's (1953) *Science and Human Behavior* that encouraged work in *applied* behavior analysis, or behavior modification. In this book, Skinner advocated the application of the science of behavior to human issues. He contended that human behavior could, in fact, be predicted and controlled by identifying and manipulating relevant environmental contingencies. With *Science and Human Behavior,* Skinner suggested how the basic principles outlined in *The Behavior of Organisms* apply to human individual behavior, to the behavior of groups, and to the behavior of organizations. Subsequent researchers in the 1950s and 1960s, including Skinner's students, were among the first to demonstrate that behavioral principles could indeed be applied to improving the human condition. Such effects were widely studied in work with children, adults, individuals with developmental disabilities, and individuals in mental health institutions.

During the early experiments in which Skinner outlined the importance of reinforcement, he discussed human analogues, but these plans did not come to fruition (Vargas, 2001). One very early demonstration of the principles of operant conditioning in a human was conducted by Paul R. Fuller (1949). Fuller precisely applied Skinner's methodology to operantly condition vertical right arm movements in an 18-year-old man with a severe developmental disability and limited mobility. Specifically, prior to reinforcement, he recorded the baseline rate of the participant's right-arm movements, then he reinforced the target movements with a sugar and milk solution until the rate of responding increased to desired levels, and finally he reestablished the baseline rate of responding using extinction. This experiment demonstrated that behavioral principles could indeed be applied to behavior change in humans. However, technically it was not applied behavior analysis because it did not improve a socially important behavior for the participant.

In one of the first studies that could be considered applied behavior analysis, graduate students of Skinner, Nathan Azrin and Ogden Lindsley (1956), increased cooperative behavior among children playing a game using operant conditioning

techniques, without giving any instructions to cooperate. The cooperative behaviors were reinforced using jellybeans. These behaviors increased when reinforced and decreased when reinforcement was no longer given. Free operant techniques were subsequently used to decrease a variety of problematic behaviors and to increase effective behaviors in children. These techniques involve altering contingencies involving free operants, which are behaviors that can occur at any time, without any specific cue, and often occur naturally (Newman et al., 2003). Early studies showed that the application of behavioral principles was effective in decreasing, for example, thumb sucking (Baer, 1962), tantrum behavior (Williams, 1959), and enuresis (Wickes, 1958). During this time, free operant techniques were also used with children to increase imitation (Baer & Sherman, 1964), walking (Harris, Johnston, Kelley, & Wolf, 1964), and appropriate verbal responses (Hart, Allen, Buell, Harris, & Wolf, 1964). Behavioral techniques were soon applied to the classroom setting, and researchers demonstrated that disruptive behaviors could be eliminated through extinction (ignoring such behaviors maintained by attention) and that appropriate classroom behaviors could be increased using positive reinforcement (Zimmerman & Zimmerman, 1962).

With adults, early studies in applied behavior analysis included increasing personal statements of opinion through positive reinforcement (Verplanck, 1955), reducing stuttering by making the presentation of an aversive stimulus (a loud tone) contingent upon it (Flanagan, Goldiamond, & Azrin, 1958), reducing the rate of tics in a musician by removing continuous music whenever a tic occurred (Barrett, 1962), and eliminating hysterical blindness by positively reinforcing use of visual cues (Brady & Lind, 1961). Techniques utilizing the principles of operant conditioning were also shown to be successful in treating obesity (Ferster, Nurnberger, & Levitt, 1962) and marital difficulties (Goldiamond, 1965).

Much of the early work in applied behavior analysis was conducted with individuals with developmental disabilities and with individuals in residential psychiatric facilities. Early studies successfully utilized the behavioral procedures of shaping, extinction, time-out (a period of removal from opportunities for reinforcement), and differential reinforcement to reduce problematic behaviors in children with autism (Wolf, Risley, & Mees, 1964) as well as positive reinforcement to increase the behavioral repertoires of these children (Ferster & DeMyer, 1962).

Of note, the use of differential reinforcement to reduce problematic behaviors often makes use of different intermittent reinforcement schedules. In procedures such as these, the problematic behaviors are often extinguished and other (desired) behaviors are reinforced (Leslie, 1996). There are five types of differential reinforcement schedules: differential reinforcement of other behavior (DRO), differential reinforcement of incompatible behavior (DRI), differential reinforcement of alternative behavior (DRA), differential reinforcement of low rates of responding (DRL), and differential reinforcement of high rates of responding (DRH; Leslie, 1996). In a DRO, any behavior other than the problematic behavior is reinforced. In a DRI, a particular behavior that is incompatible, or mutually exclusive, with the problem behavior that is reinforced. In a DRA, reinforcement is provided for alternative behavior to the problem behavior. In the DRL procedure, reinforcement is

delivered after a specific interval has passed in order to reduce the problem behavior by progressively lengthening the required interval. In opposition to this procedure, the DRH schedule delivers reinforcement only when a behavior is emitted rapidly (Baldwin & Baldwin, 2001; Leslie, 1996).

Additional early studies illustrated that parents and teachers of children with developmental disabilities such as autism and mental retardation could be trained to successfully utilize behavioral procedures to manage child behavior in the home and school settings (Bijou, Birnbrauer, Kidder, & Tague, 1966; Hawkins, Peterson, Schweid, & Bijou, 1966). In the school setting, the application of behavioral principles (including the use of secondary reinforcers such as stars or check marks that could be accumulated upon assignment completion and exchanged for other desired items) was shown to be effective, not only in eliminating inappropriate behaviors in these children, but also in facilitating learning of school subjects, including reading, writing, and arithmetic (Birnbrauer, Bijou, Wolf, & Kidder, 1965).

Behavioral principles as applied to work with residents in psychiatric institutions were effective in reducing a wide range of problematic behaviors in these settings, including vocalizations related to the experience of hallucinations (Lindsley, 1959), refusal to eat (Ayllon & Haughton, 1962), and hoarding behaviors (Ayllon, 1963). Once trained, nursing staff could effectively utilize techniques such as extinction and differential reinforcement to reduce problematic behaviors and increase adaptive behaviors in their patients, with researchers assuming a supervisory role (Ayllon & Michael, 1959).

Many of these institutional studies used various procedures to decrease or increase specific behaviors in one or two participants. Ayllon and Azrin (1965) endeavored to standardize and extend applications of behavior analysis to facilitate improvements in a wide array of behaviors in large numbers of adult patients in a psychiatric facility. Working with an entire ward population (up to 44 patients) at Anna State Hospital in Anna, Illinois, Ayllon and Azrin used a reinforcement program to increase participation and performance in ward jobs. These jobs included dietary assistant, cleaning assistant, laundry assistant, and recreational assistant. Because salient reinforcers for each patient tended to vary and many such reinforcers were impractical to deliver consistently and immediately following job participation (e.g., attention from the psychologist), the researchers used conditioned, or secondary, reinforcers in the form of specially designed tokens. Performance of each task was rewarded with a given number of tokens, which could then be exchanged in various amounts at designated times for a variety of desired reinforcers, for example, attention, privacy, and consumables. The program dramatically increased job participation and performance among patients, and it was quickly adapted for use by other professionals in facilities for children with mental retardation and juvenile offenders (Ayllon & Azrin, 1965). Programs such as this would become known as token economies (see Ayllon & Azrin, 1968).

Characteristics of Applied Behavior Analysis as Science

Several of the early endeavors in applied behavior analysis were published in *JEAB* and were collected in edited works such as *Case Studies in Behavior Modification*

(Ullmann & Krasner, 1965) and *Control of Human Behavior* (Ulrich, Stachnik, & Mabry, 1966). However, the body of applied research continued to grow and soon necessitated the publication of a counterpart to *JEAB*, the *Journal of Applied Behavior Analysis* (Risley, 2001). In the first issue in 1968, Baer et al. summarized dimensions relevant to studies in applied behavior analysis. In addition to the obvious characteristics of being "applied, behavioral, and analytic," the authors suggested that such studies should also be "technological, conceptually systematic, and effective" and "should display some generality" (p. 92). To be considered applied, studies should consider problems that are important to society and to the individual. As behavioral, the studies should precisely measure what an individual does and should include measures of reliability among those observing such behavior. Baer and colleagues suggested that in an analytic study, the experimenter must reliably demonstrate control over the behavior under study, such that the effect on behavior is replicable.

Two experimental designs that can illustrate such control are the reversal design and the multiple baseline design (Baer et al., 1968). In a reversal design, baseline levels of the target behavior are measured, then the variable of interest is applied and the target behavior is measured for change. If change occurs, the variable of interest is removed, and the target behavior is again measured. If the variable of interest was responsible for the behavior change, when it is removed the behavior change will no longer be observed. Finally, the variable of interest is reapplied, and the target behavior is measured to determine if the change recurs. If it does, there is evidence of behavioral control. When reversing a target behavior is not desired or impossible, the multiple baseline design can be used to demonstrate control (Baer et al., 1968). In this design, multiple target behaviors are identified and baseline levels are measured. A variable of interest is applied to one behavior and changes in that behavior are recorded, as well as the lack of change in the other behaviors. The variable of interest is then applied to another target behavior, and if it also changes while other behaviors go unchanged, there is evidence that this variable is successful in controlling behavior.

To be technological, Baer and colleagues (1968) suggested, the procedures used in studies in applied behavior analysis should be sufficiently described and should include specific details so that the study can be replicated. By conceptually systematic, the authors meant that such studies should include a discussion of the behavioral principles that are utilized in the procedure. With regard to effectiveness, the authors contended that studies in applied behavior analysis should strive for effects that are of sufficient size to be useful to the individual and those influenced by the individual's behavior. Finally, the authors indicated that such studies should aim for behavior change that has generality over time, across settings, and/or across similar behaviors.

Research including many of these dimensions of applied behavior analysis continued into the 1970s and 1980s, and, although work continued in educational and institutional settings, applied behavioral applications expanded into new arenas. Behavioral research provided benefits to community programs; for example, researchers used the principles of reinforcement (Kohlenberg & Phillips, 1973) and stimulus control (O'Neill, Blanck, & Joyner, 1980) to reduce littering and increase

proper waste disposal. In addition, behavioral techniques were successfully used to increase use of seat belts (Geller, Paterson, & Talbott, 1982). Behavioral principles were also applied successfully to improving health-related behaviors, including increasing medical appointment keeping (Friman, Finney, Rapoff, & Christophersen, 1985) and pursuit of dental care (Reiss & Bailey, 1982), as well as facilitating healthier dietary choices (Stark, Collins, Osnes, & Stokes, 1986). Early studies in the area of sports psychology indicated that application of behavioral principles could facilitate, for example, practice attendance among athletes (McKenzie & Rushall, 1974) as well as skills acquisition in sports such as football, tennis, and gymnastics (Allison & Ayllon, 1980). Studies of applied behavior analysis were also conducted in industrial settings. Applications in the business world were effective in changing a host of work-related behaviors, including reducing tardiness (Hermann, DeMontes, Domínguez, Montes, & Hopkins, 1973) and enhancing job performance (Pierce & Risley, 1974).

In the 20 years following the first publication of the *Journal of Applied Behavior Analysis,* research in behavior analysis and modification continued to proliferate, and several additional journals were developed to publish related work, including, *Behavior Therapy* in 1970, the *Journal of Organizational Behavior Management* in 1977, and the *Behavior Analyst* in 1978 (Miltenberger, 1997). In 1987, Baer, Wolf, and Risley again examined the dimensions of applied behavior analysis. They concluded that the components applied, behavioral, analytic, conceptual, technological, effective, and capable of generalization continued to be essential in studies of applied behavior analysis. They also acknowledged that the complexity of the definitions of these terms had increased. The authors suggested that studies after 20 years and into the future may include consideration of how socially important problems are identified, more complex experimental design to take into account contextual factors, consideration of the potential for methods to be adapted to additional settings, measures of social validity, and increased focus on interventions at both the individual and system levels.

CURRENT APPLICATIONS

To implement any psychological intervention effectively, it is imperative that the clinician and treatment team first conduct a comprehensive, detailed assessment. This assessment is done to best ascertain the nature of the problem and assist in subsequent treatment planning. Although many assessment procedures are based on a conceptualization of behavior as a manifestation of an underlying disorder, behavioral assessment assumes that an individual's behavior is context-dependent. Therefore, there is recognition that behavior may not be stable across various situations; thus, behavior should be assessed in a situation-dependent manner (Galassi & Perot, 1992). Behavioral assessment utilizes an ideographic approach, focusing on addressing contingencies that directly influence an *individual's* behavior. Given the need for detailed assessment that considers individual-specific context and contingencies, applied behavior analysis includes a procedure known as functional analysis to determine the form and function of relevant behavior.

Functional Analysis of Behavior

A crucial step in applied behavior analysis is conducting the functional assessment. A functional assessment allows the practitioner to identify and describe the problem behavior and the antecedents and consequences of that behavior in the context within which it occurs. The individual performing the behavior cannot be separated from current and historical contexts.

The objective of the functional assessment, or functional case conceptualization, is to obtain data that will be used to develop an effective treatment strategy that produces the desired outcome (S. C. Hayes, Nelson, & Jarrett, 1987). This is accomplished by identifying the purpose that behavior serves and hypothesizing, based on that information, which antecedents and consequences might be manipulated to decrease maladaptive behavior and optimize positive change.

Behavior analysts differentiate between functional assessment and functional analysis. Functional assessment refers to the procedures involved in clarifying antecedents and consequences; functional analysis refers to the direct manipulation of environmental contingencies in order to observe their impact on the target behavior (Gresham, Watson, & Skinner, 2001). In more complicated cases, functional analysis may include psychotherapy that focuses on reducing aversive (or appetitive) conditioning in addition to the direct altering of environmental factors. Proper functional analysis continues throughout treatment, and plans are modified if evaluation indicates that the treatment is not effective. Thorough analysis includes consideration of respondent and operant contingencies as well as their interactions (Wilson & Murrell, 2002).

Applied Behavior Analysis for Childhood Disorders

Although there are some concerns about the traditional use of behavior analysis, myriad complex behavioral issues have been dramatically improved through the application of applied behavior analytic procedures in home, school, and primary care settings. In addition, whereas pharmacological interventions have also been shown to be helpful with childhood disorders, research suggests that a combination of pharmacological and psychosocial interventions, particularly behavioral ones, can often maximize treatment gains (Hinshaw, Klein, & Abikoff, 2002).

Concerns, Common Obstacles, and Considerations

In addition to concerns about the use of traditional behavior analysis, there are common obstacles that occur during implementation. Therefore, several factors need to be considered in design and evaluation of applied behavior analyses. Of particular importance are generalizability, ease of implementation, and experimental design.

Pelham, Wheeler, and Chronis (1998) criticized behavioral interventions by pointing out that many treatment gains that are initially shown within the therapeutic milieu fail to generalize after the intervention is completed. Therefore, it is important to use interventions outside of a laboratory setting or therapeutic environment to increase the probability of the behavior generalizing to more natural settings. To do this, it is often necessary to train others to implement an intervention.

In child cases, it is important for involved parents and teachers to be motivated and able to carry out the ABA (Kazdin, 2000; Pelham et al., 1998). To enhance treatment success outside of the therapy setting, it is therefore important to identify all factors involved in enhancing and maintaining parents' and teachers' adherence to the treatment protocol. In addition, for functional relationships between independent and dependent variables to be established, it is imperative that interventions be implemented accurately and with integrity (Gresham, Gansle, & Noell, 1993). In other words, one wants to ensure that the individuals responsible for implementing the intervention are doing so correctly and consistently.

Many assessment techniques and interventions focus on the environmental contingencies regulating the child's behavior. Allen and Warzak (2000) emphasize the importance of also examining environmental contingencies that control parent, teacher, and clinician behavior. For example, many treatments are most successful when the parents are directly involved in implementing interventions for the child. Therefore, it is also important to examine the antecedents and consequences of parental behavior. The use of idiographic assessment increases the likelihood that appropriate contingencies will be identified. Often, environmental contingencies are directly related to the parent's adherence to treatment guidelines when implementing interventions (Allen & Warzak, 2000). It should be noted that contingencies regulating teachers' and clinicians' behavior should also be considered because they, too, can influence the effectiveness of the intervention.

Another factor related to effectiveness is strength of experimental design. Many behavioral studies have failed to operationally define the independent variable (such as the intervention) and to measure the accuracy with which it is implemented (Gresham et al., 1993). These flaws in experimental design can lead to failed attempts at using and evaluating behavior analytic interventions. To make scientific advances, the independent and dependent variables must be operationally defined. This is especially important in facilitating accurate replication (Gresham et al., 1993).

When generalizability, ease of implementation, and experimental design are considered, ABA evidences great success. Behavioral principles and applied behavior analytic techniques can be successfully used with children with Autistic Disorder and other pervasive developmental disabilities, Attention-Deficit/Hyperactivity Disorder (ADHD), and pediatric medical conditions such as elimination disorders (e.g., enuresis and encopresis), sleep problems, and a variety of habit disorders (trichotillomania, thumb sucking, and hair pulling). In addition, ABA is effective in treating children with Conduct Disorder and dealing with parenting problems that result from such symptomatology. It is also successfully used to address academic issues in classroom settings, with both children and teachers. The following discussion is meant to provide a sampling of some of the ways behavioral principles and the behavior analytic process, including functional assessment of problem behaviors, are currently applied to the treatment of various clinical problems across settings.

Pervasive Developmental Disorders

Applied behavior analysis facilitates the discovery and evaluation of interventions with individuals with developmental delays, such as autism spectrum disorders

and mental retardation (Frea & Vittimberga, 2000). In fact, a perusal of the current behavior analytic literature indicates that the majority of studies are being conducted with individuals with mental retardation or developmental delays (Carr, Coriaty, & Dozier, 2000; Reid, 1991).

Lovaas and colleagues (e.g., Lovaas, 1987) conducted early studies examining the impact of utilizing operant procedures to address behavioral problems in children with autism. This early work used discrete trial training to evoke and reinforce appropriate behaviors. Punishers were used to decrease inappropriate behaviors such as self-stimulatory behavior, self-injury, and incorrect vocalizations. A discrete trial is an intense, brief cycle of instruction between the instructor and the child that is used to facilitate the learning of new information and communication. One benefit of discrete trial training is that professional and trained paraprofessionals and family members can implement the procedures with the child (Smith, 2001). A typical discrete trial consists of five representative components: the cue (or discriminative stimulus), a prompt in which the instructor aids the child in responding, the actual response, a consequence, and an intertrial interval that is a brief pause before the next trial begins (Smith, 2001). In recent years, these discrete trial interventions have been modified to facilitate their use in classroom settings.

Lovaas's Young Autism Project at UCLA serves as the prototype of discrete trial training and is aimed at children under age 4. It is a time-intensive program that involves 40 hours per week of one-to-one training sessions for 3 years. It has shown success in helping children with developmental disabilities to acquire new information and new discrimination among cues.

Attention-Deficit/Hyperactivity Disorder

Direct contingency management, clinical behavior therapy, and self-management skills are three commonly used behavioral interventions for treatment of ADHD (Hinshaw et al., 2002). Contingency management procedures involve direct attempts to manipulate antecedent and reinforcement procedures to decrease problem behaviors and increase prosocial ones. Examples of contingency management procedures are time-out, in which the child is prevented from receiving reinforcement; token economy procedures; and regulating feedback from parents and/or teachers (Kronenberger & Meyer, 2001). Contingency management procedures are typically used in more professional settings, and treatment gains are often more pronounced than they are in the child's home environment (Pelham et al., 1998). Outpatient clinical behavior therapy is successful in helping parents learn to incorporate contingency management procedures in the home to facilitate treatment consistency and enhance treatment outcome (for a review, see Pelham et al., 1998).

Both direct contingency management and clinical behavior therapy place considerable demands on those implementing the programs. Self-management is a child-focused behavioral intervention for ADHD that enables the child to identify, monitor, and reinforce his or her own behaviors, thereby reducing demands on parents and teachers (Braswell & Bloomquist, 1991). Because self-management requires the child to focus on and monitor his or her own behaviors accurately in order to earn reinforcement, it not only reduces symptoms, but it also fosters

independence and personal control over behavior (Barry & Messer, 2003). Self-management can be implemented in a regular classroom or at home with some initial help from the teacher or parents, respectively. This technique is very effective at increasing on-task behaviors and improving academic performance while decreasing disruptive behaviors (Barry & Messer, 2003). Due to the self-monitoring and self-evaluation components, it is likely that self-management is more appropriate for older middle school, junior high, and high school students.

Behavioral Pediatrics

Behavior problems are often seen in primary care settings, and behavior pediatrics represents the merger between applied behavior analytic principles and pediatric medicine (Blum & Friman, 2000). Behavioral pediatrics is a broad area of study that focuses on pediatric behavior problems (such as sleeping dysfunction and elimination disorders) and health (Blum & Friman, 2000). Behavioral principles are used successfully to treat elimination disorders such as enuresis and encopresis. A number of behavioral interventions have been identified and studied to treat nocturnal enuresis in children, and the most well-known and accepted practice is the urine alarm ("bell and pad") and cleanliness training intervention (see Mowrer & Mowrer, 1938, as cited in Kronenberger & Meyer, 2001, for a description). Operant theorists suggest that the success of the bell and pad intervention among children can be attributed to negative reinforcement: specifically, the absence of the alarm and having to perform cleanliness training, which includes cleaning themselves and the toilet, changing their clothes, and changing the bed linens.

Applied behavior analysis is used with children who have a variety of other pediatric conditions, including asthma (e.g., Renne & Creer, 1976), diabetes (e.g., Epstein et al., 1981), feeding disorders (e.g., Coe et al., 1997), and other medical problems. Application of behavioral principles is also effective in increasing child and parent compliance with medical procedures and treatment (see Matthews, Spieth, & Christophersen, 1995). For example, in one study, contingency management was used to increase cooperative behavior in 3- to 6-year-old children during dental work (Allen & Stokes, 1987). Escape and attention were provided when children displayed nondisruptive behaviors during successively more invasive presentations of dentistry tools. Children were also rewarded with tangible reinforcers (i.e., stickers and toys) for remaining cooperative during increasing periods of time. Physiological arousal as well as disruptive behavior decreased from baseline (Allen & Stokes, 1987).

Conduct Disorder and Parenting Problems

Behavioral therapies based on operant learning have been used to address childhood conduct problems and to enhance the child's ability to formulate appropriate solutions to problems. Brestan and Eyberg (1998) reviewed current literature in hopes of identifying effective psychosocial treatments for Conduct Disorder, Oppositional Defiant Disorder, and associated behavior problems. Despite variability in methodological rigor, the more recent and well-designed studies were shown to be effective in decreasing negative behavior. One such treatment is

problem-solving skills training (PSST). This treatment, which is "probably efficacious," is often used to assist children in formulating prosocial responses. Games and actual social situations are used to decrease the likelihood of aggressive behaviors (Brestan & Eyberg, 1998). Recent modifications of PSST rely heavily on applied behavior analytic principles such as shaping appropriate behavior and extinguishing maladaptive responses (Kazdin, 2000).

Given the environmental contingencies that promote misconduct and possibly later antisocial behavior, PSST is often used in conjunction with parent management training (PMT; Weisz, 2004), a home-based intervention that is based on the premise that many maladaptive behaviors are due to dysfunctional parent-child interactions. Parents are taught underlying behavioral principles and skills to enhance the quality of interactions with their child, decrease problem behaviors, and facilitate adaptive behaviors. More specifically, parents are taught to implement reinforcement and punishment procedures and set appropriate rules and boundaries. In addition, the work focuses on enhancing problem-solving skills (Kazdin, 2000).

Parent-child interaction therapy (PCIT) is another evidence-based treatment designed to enhance interactions between parents and children and decrease parental stress (Eisenstadt, Eyberg, McNeil, Newcomb, & Funderburk, 1993). It is commonly used with preschool children and emphasizes teaching communication skills and behavioral principles to parents to implement in play situations with their children. The parents are given feedback on their use of the skills during the ongoing session. Initially, PCIT involves a child-directed play interaction in which the child leads the activities; this is followed by the parent-led interaction in which the parent utilizes appropriate communication and feedback to direct the play session. Finally, appropriate discipline is taught. Sessions use positive and negative reinforcement strategies to increase prosocial behaviors and decrease maladaptive ones (Eisenstadt et al., 1993).

Parent training programs often focus on enhanced consistency in efforts to maintain treatment gains. Most often, this work with youth and parents involves consistent application of reinforcement strategies and limited use of punishment procedures to facilitate behavior change (Thorpe & Olson, 1997). However, childhood behavior problems are characteristic of a number of childhood disorders and are often embedded within a problematic environmental context that maintains the child's problematic behaviors and makes lasting changes difficult. Kazdin and Whitley (2003) found that an added treatment component that specifically addressed parental stress improved overall therapeutic change when used with other models (such as PSST and PMT). Similarly, Weinberg (1999) found that training programs for parents of children with ADHD enhanced parental comprehension of behavior management principles and reduced overall stress related to parenting a child with ADHD.

Classroom Settings

In addition to substantial contributions to special education, applied behavior analysis is effectively used in general classroom settings. Historically, applied behavior analysis has focused on consequential procedures such as contingency management procedures using reinforcement and punishment. Due to ethical concerns, such as

excessive reliance on punitive or punishment procedures, applied behavior analysis has begun to shift focus and now also includes emphasis on antecedent control (Dunlap, Kern, & Worcester, 2001).

Token economies used in the classroom setting are well-established treatments for ADHD (Pelham et al., 1998). Whole-classroom token economies to manage disruptive classroom behavior may be more practical to implement than individualized token economies (Filcheck, McNeil, Greco, & Bernard, 2004). Token economies reward children for prosocial behaviors by awarding them with secondary reinforcers, such as tokens, that can be exchanged for a more powerful reinforcer. Occasionally, tokens can be taken away from the child contingent upon maladaptive behaviors. Removing tokens from the child is called response cost (Kronenberger & Meyer, 2001). Appropriate implementation is crucial to the success of such token economies.

Implementation of an intervention is an important matter in the classroom. A study by Greenwood, Terry, Arreaga-Mayer, and Finney (1992) looked at the possible moderating role of implementation factors in students' achievement in a classwide peer tutoring program. They instructed teachers to observe and record the students' successes and failures during periods when students were responsible for tutoring peers in spelling tasks. Students who were challenged by and mastered the material received the best implementation of the program. Interaction between the students was also seen to have an effect on student performance.

To enhance student performance, Lindsley (1992) expressed a hope for classroom settings to employ a behavioral precision teaching (PT) approach instead of the traditional lecturing approach. This method intends to provide more individualized, rapid progress, using a format in which the teacher acts more as a facilitator, and the students are responsible for their progress in terms of timing performance, counting, and daily charting (Lindsley, 1992). This approach involves the utilization of repeated practice with an error-correction procedure in place, along with aims expressed in terms of fluency and monitored with the use of standard celeration charts (McDowell & Keenan, 2001). The fluency factor is an important component in PT because it takes into account both accuracy and rate of responding, with the latter shown to be more sensitive to environmental changes than percentage correct (Lindsley, 1992). The application of rate of response and cumulative recording to the classroom setting comes from a free operant conditioning background (Lindsley, 1992). The use of PT in classrooms has allowed for exponential learning gains, such as seen in Morningside Academy with gains of two to three grade-level advances per year from 1980 to 1987 (Lindsley, 1992). (See Lindsley, 1992, for a brief review of important features involved in PT.)

The use of PT procedures provides gains in fluency and endurance, or attention span, in learned material (McDowell & Keenan, 2001). McDowell and Keenan reported a case study in which the use of PT procedures with a child diagnosed with ADHD resulted in an increase in correct responding, a decrease in incorrect responding, and an increase in on-task behavior. A return to baseline procedure prior to the acquisition of goals shows a decrease in performance, whereas a return to baseline after goals are acquired shows maintenance in rate of correct responding and on-task performance (Lindsley, 1992; McDowell & Keenan, 2001).

Concern with frequency of response can be seen in other classroom procedures as well. Programmed instruction is a technique based on reinforcement contingencies in which a correct answer is contingent upon a specified preceding behavior (Kritch & Bostow, 1998). In other words, instructional material is presented in a sequence that requires successively more correct responding. Typically, students are required to fill in the blanks in lessons as they work through these programs at their own pace. They are provided feedback and the correct responses immediately after providing an answer. Kritch and Bostow conducted a study that underscores the importance of frequency in responding in programmed instruction. The authors found that students in the high-density response requirement condition (i.e., the condition that demanded frequent response) performed better on tests than students who spent equivalent time on the task without any requirement that they respond. The participants in the high-density responding group were also better able to apply the knowledge than other groups.

Despite the widespread success of behavioral interventions in the classroom, Axelrod, Moyer, and Berry (1990) indicated that many teachers fail to use behavioral interventions. Teachers are rarely taught how to implement behavioral strategies prior to entering the classroom. Axelrod and colleagues suggest that behavior analysts should focus on facilitating the accurate use of behavioral principles by individuals other than behavior analysts. One way to address this problem is to teach behavior analysis to teachers; another solution is to train clients to monitor and adapt their own behavior.

Outpatient Therapy

As mentioned, applied behavior analysis is successful in treating myriad childhood behavioral disorders, pediatric medical disorders, and accompanying parent problems. There is a recent trend, often known as clinical behavior analysis, to rely on behavior analytic principles to address more complex psychological disorders. To best understand the emergence of clinical behavior analysis, a brief history of behavior therapy is required.

The Waves of Behavior Therapy

Traditionally, behavior therapy has shown a commitment to empiricism by examining overt behaviors and modifying environmental contingencies. The goal of early behavioral treatment was to reduce problematic behaviors and/or symptoms. Less overt psychological phenomena, such as thoughts and emotions, were of little interest to early behavior therapists. This lack of interest likely resulted from difficulty in manipulation, as behavior analysis (a radical behavioral perspective) never underestimated such factors. The focus on elimination of problematic behaviors and alleviation of symptoms represents *first-order* change processes: the direct targeting of symptoms and behaviors in clinical treatment with the goal of their alleviation and/or elimination. This first wave of behavior therapy was criticized, however, for not providing an adequate account of human language and cognition (Eifert & Forsyth, 2005).

The second wave of behavior therapies sought the integration of cognitive and behavioral models to facilitate treatment development and address criticisms in the literature. The application of direct change strategies to behavior and cognition has evidenced success in the reduction of psychological symptoms (S. C. Hayes, 2004). However, although research suggests that cognitive-behavior therapy (CBT) works with a variety of disorders, dismantling studies and meta-analyses suggest that researchers do not have a comprehensive understanding of *why* it works. It appears as if CBT might not work through its proposed mechanisms (e.g., Jacobson et al., 1996). Without an understanding of the mechanisms through which a treatment works, it is less likely that researchers and practitioners will be able to effectively reach the goals of behavior analysis: to predict and control (or influence) behavior.

Given the need to address language and cognition *and* have a treatment that is closely linked to examinable and manipulable explanatory mechanisms, many clinicians and applied researchers alike have made attempts to design and utilize behavior analytic treatments that address complex phenomena. These treatments, known as the *third-wave* behavior therapies, often focus on acceptance of private psychological events (e.g., thoughts and feelings) to facilitate behavior change. Further, recent therapies have incorporated spirituality, dialectics, Eastern wisdom traditions, and interpersonal functioning and values to broaden the scope of existing treatments. Consistent with behavior analysis, third-wave therapies rely on a philosophy of contextualism and pragmatism that emphasizes the function of the behavior (what purpose it serves and whether it results in positive or negative reinforcement) rather than its form (S. C. Hayes & Bissett, 2000).

These newer approaches encourage deeper analysis of private events in both assessment and treatment phases (S. C. Hayes & Brownstein, 1986). The third wave includes such treatments as dialectical behavior therapy (Linehan, 1993), mindfulness-based cognitive therapy (Segal, Williams, & Teasdale, 2002), integrative behavioral couples therapy (Jacobson, Christenson, Prince, Cordova, & Eldridge, 2000), functional analytic psychotherapy (Kohlenberg & Tsai, 2003), and acceptance and commitment therapy (S. C. Hayes, Strosahl, & Wilson, 1999). Third-wave behavior therapists may utilize first-order change techniques in their work. However, the ultimate and overarching goal of third-wave treatments is to facilitate *second-order* change. Second-order change refers to a shift in an individual's reactions and responses to his or her problems, regardless of whether the problems are decreased or eliminated. In other words, second-order change strategies target the processes of thinking and feeling rather than the content. This shift is achieved through the use of techniques that serve two primary purposes: (1) to alter the function (or meaning) of specific behaviors through the consideration of contextual factors and (2) to expand one's behavior repertoire so that, in the presence of distressing private events, one may flexibly choose to behave in a multitude of ways (S. C. Hayes, Strosahl, Bunting, Twohig, & Wilson, 2004.)

Overview of Acceptance and Commitment Therapy and Relational Frame Theory

Acceptance and commitment therapy (ACT; S. C. Hayes et al., 1999) has been influential in the emergence of third-wave behavior therapies. The developers describe

ACT as "a therapy approach that uses acceptance and mindfulness processes, and commitment and behavior change processes, to produce greater psychological flexibility" (S. C. Hayes et al., 2004, p. 13). The two halves of this description map onto the previously mentioned purposes of second-order change techniques. Work that increases a nonjudgmental awareness of private events and their relation to an individual in the present moment (i.e., acceptance and mindfulness processes) serves to facilitate a change in function of private events. Values assessment, behavioral contracting, behavior activation, and related techniques (i.e., commitment and behavior change processes) are used to expand behavior. Depending on their pragmatic utility in the ever-changing treatment context, didactics, paradox, metaphors, and experiential exercises are used to achieve the goal of psychological and behavioral flexibility.

Acceptance and commitment therapy is widely applied behavior analysis. Like other applied behavior analytic treatments, the work draws on the philosophy of functional contextualism and pragmatism as well as the principles of conditioning. Therapy begins with a functional assessment and includes ongoing functional analysis that identifies overt problem behaviors along with antecedents and consequences. In addition, the analyses include assessment of covert events, especially values and barriers to value-consistent behavior. These barriers may include variables in the external environment and/or conditioned aversives.

Proponents of the ACT model distinguish between direct and indirect conditioning. Direct conditioning (respondent and operant learning) results from personal experience or direct sensory perception of another's experience. In contrast, indirect conditioning emerges from verbal behavior. For example, a mother may become worried that her child will die of an obscure disease that she heard about on television, though neither she nor anyone she knows has the disease. When a problematic private event has been directly conditioned, such as when a child who has been bitten by a dog is scared in the presence of dogs, an ACT therapist will do exposure as it is traditionally done (e.g., in this case, a systematic desensitization, changing environmental contingencies, to the physical presence of dogs). When psychological distress is the result of indirect learning, exposure is labeled defusion, and acceptance and mindfulness processes are targeted in treatment. The intent of traditional exposure is to reduce elicitation, arousal, or avoidance in the presence of noxious stimuli. Defusion is intended to serve a different purpose: the development of wide and flexible behavior in response to noxious stimuli.

Most often, clients will present for treatment with a desire to eliminate problems. This is consistent with the traditional exposure paradigm. However, when given a choice, many will work for the possibility of a broader life. In those cases, defusion techniques are included. From an ACT perspective, clients are repeatedly asked to choose the therapeutic direction and goals. If the client chooses to work for the possibility of a broader life, successful therapy shifts the focus from a negative reinforcement paradigm to a positive reinforcement paradigm in which the emphasis is on utilizing values to create a rich, vital, and meaningful life. This emphasis on a rich life and on values can make the difficult work of treatment feel more dignified. Values, then, can be seen as powerful conditioned reinforcers. How such positive reinforcement develops with only the possibility of broader experience can be explained by the processes associated with human language and cognition.

Acceptance and commitment therapy is based on an underlying theory of human language and cognition called relational frame theory (RFT; S. C. Hayes, Barnes-Holmes, & Roche, 2001), which is an operant account of indirect learning processes. Using traditional behavioral principles, or the principles of direct conditioning, such as generalized operants (Baer et al., 1968) and positive and negative reinforcement, RFT explains how humans derive relationships between stimulus events that have never been directly conditioned and suggests how these processes may underlie psychopathology.

Consider the following interactions as an example of how direct conditioning is related to RFT. A mother smiles and asks her child, "What am I doing?" The child responds by saying "Smiling," and the mother says, "Yes, good job! I am smiling!" The mother then frowns, asks the same question, and responds similarly when her child says, "Frowning." Thus, the child has been directly trained and reinforced for correctly answering the question "What am I doing?" Although the content changes, a generalized response develops, one in which the child's behavior of providing the corresponding label has been positively reinforced through the use of praise. The mother then says, "Show me your smile," and the child responds with a smile. The child is reinforced for discriminating smiling from other facial expressions as well as performing the act. A few years later, through similar reinforcement strategies, the child is taught that the sound *smile* goes with the letters s-m-i-l-e. Without any additional training or reinforcement, a child with typical verbal capacities will know (or derive) that the facial expression is the same as the written word.

Words then become functionally equivalent to the stimuli they represent. This indirect conditioning ability, or relational conditioning, helps us balance short- and long-term contingencies, plan for the future, and categorize and sort information. However, when these relational processes become the primary source of stimulus control, these indirect conditioning principles can reduce our contact with actual environmental contingencies. Thus, our ability to react flexibly in situations is diminished, and this contributes to psychopathology. The ACT model suggests that psychological rigidity occurs as a by-product of language and verbal behavior. More specifically, humans tend to react to verbally constructed rules as though they are true rather than as what they are: verbally constructed rules, essentially thoughts. This phenomenon is referred to as *cognitive fusion* (S. C. Hayes et al., 1999). Further, humans also engage in *experiential avoidance,* which refers to the tendency to decrease, fix, or control the form or frequency of internal events, even when doing so contributes to psychological distress (S. C. Hayes, Wilson, Gifford, Follette, & Strosahl, 1996). Experiential avoidance may be adaptive in situations that are transient and time-limited. However, when it is primarily a means of addressing problems in daily living, it can contribute to decreased quality of life (Wilson & Murrell, 2004).

Returning briefly to how ACT attempts to increase quality of life, the model includes six primary and overlapping components: acceptance, defusion, contact with the present moment, self-as-context, valuing, and committed action. *Acceptance* refers to a nonjudgmental awareness of private events, along with the recognition that struggling with the content of those events is ineffective. *Defusion* work, as previously discussed, involves assisting clients in recognizing that words, thoughts,

feelings, and other such private events are not literal, real, or requiring of certain response. *Contact with the present moment* simply means that there is an ongoing effort by both the clinician and the client to remain focused on the here-and-now, as opposed to reliving the past or ruminating about the future. Although this is a process worked on in session, the skills ideally generalize to daily life. The aim of *self-as-context* is to differentiate between the content of psychological experiences and the context in which those experiences occur. In *valuing* work, clients are encouraged to identify what is important to them; therapists, along with clients, are expected to voice their values and make *committed action* to carrying out behaviors in accordance with those values.

Acceptance and Commitment Therapy with Children, Adolescents, and Families

There is growing empirical evidence to support the use of ACT with adults dealing with a variety of clinical disorders and behavioral problems (for a meta-analytic review, see S. C. Hayes, Luoma, Bond, Masuda, & Lillis, 2006). Researchers and clinicians are becoming interested in utilizing ACT for the treatment of childhood disorders as well as for the difficulties associated with parenting. Traditional ACT protocols are being modified to facilitate their applicability to children and adolescents. Such modifications in implementation include the use of more concrete language and multiple exemplars. Similarly, ACT techniques that are creative, interactive, and experiential rather than didactic or instructive are emphasized (Greco, Blackledge, Coyne, & Ehrenreich, 2005). Although treatment topics are chosen by the client, it is wise for the clinician to be aware of topics that are of typical relevance at various stages of development. For example, a clinician working with a 6-year-old should be prepared to discuss school entry difficulties. In addition, familiarity with games and television shows targeted toward young children is a plus. A clinician working with an adolescent may need to know more about current fashion trends and, more important, how the trends are related to burgeoning independence.

There are a few ACT treatment outcome studies that have been conducted with children, adolescents, and parents (for a review of current literature, see Murrell & Scherbarth, 2006). Case studies have shown promising results for youth with anorexia, anxiety, and chronic pain. One randomized clinical trial showed that ACT was more effective than medical treatment as usual for idiographic pain, and there are two ongoing randomized clinical trials that investigate the effectiveness of ACT in reducing delinquent behaviors. As with all child treatments, collateral information is often necessary while doing ACT, especially for functional analysis and other assessment. Information from parents is very important.

High levels of stress are often found among parents of children with psychological problems such as autism spectrum disorders and childhood aggression. From an ACT perspective, parenting deficits are thought to be associated with the parents' attempts to avoid painful emotions, thoughts, and experiences. Parents can make poor parenting choices to decrease their own negative psychological experiences. Consequently, this powerful motivation to avoid negative emotions

may actually prohibit the development of new, more adaptive parenting skills (Coyne & Wilson, 2004). Blackledge and Hayes (2006) utilized an ACT group therapy format to address depression and stress among parents of children with autism. Significant decreases among depression levels, psychiatric symptoms, and parental stress were found at a 3-month follow-up after the group ended. Treatment consistent with ACT principles has also been effective in working with parents of aggressive children. For example, in one case study, ACT treatment components were used to facilitate more value-consistent behavior on the part of the parent (Coyne & Wilson, 2004). As a result, child misbehavior and parental stress decreased after the 3-month treatment ended.

Murrell and Scherbarth (2006) encourage use of sophisticated research methodology to determine the effectiveness of ACT with children, adolescents, and families. Specifically, they point to the need for studies with larger sample sizes and controlled designs. To date, most of the ACT studies with children and adolescents involve case studies and uncontrolled group designs. To establish ACT as a credible and empirically based approach for use with children and adolescents, more rigorous research designs are needed.

ETHICAL CONSIDERATIONS

Utilizing a new, not yet empirically supported approach like ACT comes with ethical concerns. As with many topics in psychology, there are ethical concerns about the utilization of more traditional applied behavioral analysis as well. Common misperceptions of behavior analytic procedures, particularly negative reinforcement programs, may result in ABA having a negative connotation among the general public. Misunderstandings of terminology can be easily corrected. However, the majority of concerns about the use of aversive procedures are not easily allayed, because there is not a clear-cut line of right and wrong.

Punishment: To Use or Not to Use

There is a debate over the use of aversive procedures, such as punishment, and what reasonably justifies the use of such procedures (see Axelrod, 1990, for a description). Iwata (1988) advised that behavior analysts should make attempts to refrain from the use of aversive procedures to modify behavior, as there are several negative consequences that might come from that process. One such consequence is that those who implement aversive procedures also acquire aversive properties. This could decrease demand of their services. Another possible consequence is that the addressed problems will not benefit from the use of aversive procedures. This could leave behavior analysts in the negative position of developing punishment procedures for populations that exhibit the most extreme negative behaviors. Iwata notes that aversive procedures should be utilized only as a last resort; in other words, in addressing maladaptive behavior, preference should be given first to using positive reinforcement programs.

Iwata (1987) states that the attempt to extinguish behavior can be detrimental when there is a potential of an extinction burst, a phenomenon in which behavior

that is no longer reinforced increases in frequency, magnitude, and variability before it decreases (Newman et al., 2003). When the behavior in question is a life-threatening or harmful action, a temporary increase may be detrimental to the client. If unsuccessful attempts to extinguish behavior result in premature termination of the procedure, the behavior might inadvertently be reinforced at high levels (Iwata, 1987; Newman et al., 2003).

However, Iwata (1987) has also recognized the need for procedures such as punishment. Leslie (1996) suggests taking three steps when considering whether an aversive procedure should be used: First, define all involved terms; second, specify procedures to be considered; third, assess the possible consequences of the intervention, of the alternatives, and of taking no action. Leslie also stresses the need for an individualized assessment of the behavior and situation to examine important contextual aspects and the personal history of a client. General ethical principles may not be applicable because one solution will not be appropriate to all situations. There are multiple factors that should be considered in the decision of whether to use an aversive procedure. As mentioned, there is harm in failing to successfully implement an intervention. In addition, it is possible that behavioral effects such as aggression may emerge as a result of aversive procedure utilization (Leslie, 1996).

People often worry about the use of aversive procedures, yet they are already present in our natural environment (Axelrod, 1990; Leslie, 1996). The use of punishment allows for rapid response suppression (Leslie, 1996). When considering no intervention or alternatives to aversive procedures, Axelrod asks why one would want to waste a client's time with a procedure that is known to be less effective when there is a procedure available that is known to work. This is especially important when a problematic behavior could be life-threatening if allowed to continue, such as head banging or running into a busy street.

Sometimes Applied Behavior Analysis Is Painful

Sometimes punishment is necessary. Sometimes, even when punishment is not utilized, applied behavior analytic procedures can be painful. Treatment can be difficult. Another issue to consider is the balance between quality of life and the means of an intervention (L. J. Hayes & McCurry, 1990). One example is that of a cancer patient. If left untreated, the cancer may eventually be fatal. Yet while undergoing treatment, the side effects of a cure can cause an individual to be sick. The patient will need to come to a decision: to be sick now in order to feel better later, or to refuse treatment. An issue as complex as this would benefit from an empirical analysis of the phenomenon, which in turn carries its own ethical issues.

What to Do

Punishment, because it is easy and often effective, is reinforcing. Yet, there are clearly negative consequences related to its use. Some have suggested that an empirical understanding of topics such as punishment may allow us to reduce the use of such procedures. L. J. Hayes and McCurry (1990) remind us that aversive procedures serve a function for the greater good, but we cannot forget to contemplate

the ends versus the means of an intervention. Finally, the complexity of these issues highlights the need for an idiographic approach.

SUMMARY

Applied behavior analysis is an area of psychological research and treatment endeavoring to use behavioral principles in an effort to improve socially important human behavior (Baer et al., 1968). Philosophically, applied behavior analysis can be seen as a materialistic, deterministic, contextual, and functional approach to science. Historically influenced by Russian physiologists, American animal psychologists, and early behaviorism's call for psychology as an objective science, applied behavior analysis was realized in the 1950s and 1960s following the work of B. F. Skinner (1938, 1953) as researchers began using the principles of operant conditioning to predict and influence human problem behavior. These early researchers applied behavioral principles and techniques, manipulating the antecedents and consequences of behavior, in the successful elimination of undesirable behaviors and the development of adaptive behaviors in schoolchildren, adults, individuals with developmental disabilities, and individuals in mental health institutions. In the 1970s and 1980s applications of behavior analysis broadened to examine and improve community programs, health-related behaviors, athletic performance, and work-related behaviors. Today, applied behavior analysis continues to be a burgeoning area of research and to offer effective, empirically supported methods of psychological treatment useful in addressing a host of human difficulties, including problems associated with ADHD, Oppositional Defiant Disorder, parenting, elimination disorders, developmental delay, and serious mental illness in home, school, and institutional settings. With the emergence of third-wave behavior therapies, techniques rooted in applied behavior analysis are being successfully extended to work in outpatient settings with individuals and couples grappling with a host of psychological difficulties.

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CHAPTER 9

Cognitive-Behavioral Theory

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Cognitive-behavioral therapy (CBT) continues to evolve from its early roots in behavior therapy, and it is often represented in summaries of empirically supported interventions. Since the emergence of behavior therapy in the late 1950s and early 1960s, cognitive and behavioral theories have merged to produce a theoretically complex combination of therapeutic approaches known today as CBT, though some have continued to debate the merits of adding cognitive components to the traditional behavioral model (Beidel & Turner, 1986; Reitman & Drabman, 1997). After a period of development focusing primarily on adult psychopathology, recent years have witnessed the emergence of a wide range of CBT interventions and related research for children and adolescents. Dobson and Dozois (2001, p. 4) argue that CBT interventions for children are characterized by the following assumptions: " (a) Cognitive activity affects behavior, (b) cognitive activity may be monitored and altered, and (c) behavior change may be achieved through cognitive change."

One important facet of cognitive-behavioral theory is the assumption that "cognitive activity" and "behavior" are fundamentally different. Indeed, several authors have regarded cognitive activity as a subcategory of behavior (see Watson, 1924). In this chapter, we use the term "cognitive" to represent covert behavior, including thoughts and images. We use the term "overt behavior" when it is important to distinguish it from covert behavior. In addition to the three assumptions noted by Dobson and Dozois (2001), there are other key propositions within the cognitive-behavioral framework. Namely, the environment, overt behavior, and covert behavior all influence each other. Finally, Dobson and Dozois also distinguish cognitive-behavioral therapy from "pure" forms of behavior therapy and cognitive therapy. For example, in their view, applied behavior analysis for individuals with profound mental retardation and cognitive therapy focusing completely on memories of traumatic events would be viewed as theoretically distinct.

Kendall (2006, p. 7) regards the cognitive-behavioral framework as placing "greatest emphasis on the learning process and the influence of the models in the social environment, while underscoring the centrality of the individual's mediating/information processing style and emotional experiencing." This definition includes several key components, including (a) learning from direct experience, (b) social learning, and (c) cognitive and emotional mediation. Both Kendall and

Dobson and Dozois (2001) emphasize the mediational perspective. That is, in addition to environmental influences, a person's thoughts and feelings are believed to make a unique contribution to behavioral health.

The theoretical foundation of any form of therapy is regarded as critical by some authors who argue the merits of therapeutic techniques derived from a unifying theory. Hayes, Strosahl, and Wilson (1999, pp. 13–14), for example, distinguish between *theory* and *technique* by describing theory as "statements that have broad applicability" and technique as the "practical details" of therapy. The value of a well-developed theory is the ability to adapt techniques to new problems, develop new techniques, and add coherence to applied psychology (Hayes et al., 1999). Interestingly, most of the techniques used in CBT were derived from either behavioral theories or cognitive theories but not a unified "cognitive-behavioral" theory, and many have argued for a more flexible approach to case conceptualization that incorporates elements of many theoretical perspectives (A. Lazarus, 1996).

This chapter first provides brief descriptions of behavioral and cognitive theories, followed by a discussion of the continued integration of the two theories as they relate to children and adolescents. We close with a few comments about the future of the cognitive-behavioral framework.

BEHAVIORAL THEORY

Although there were many early contributors to the development of behavioral theory, John B. Watson's (1924) *Behaviorism* represented one of the first comprehensive efforts to describe behavioral theory. Elaborating on themes first developed in his "behaviorist manifesto" (Watson, 1913), Watson (1924) argues that behaviorism ought to focus on observable phenomena rather than consciousness. Observable responses were said to include what the organism "does," including overt actions such as speaking aloud and even covert actions such as speaking to oneself (i.e., thinking). Thus, cognitions are included in Watson's account of human behavior, and in his view, thinking is subvocal speech, observable, at least in principle, to the thinker. By contrast, psychodynamic theory held that many important motivations and explanations of human behavior were subconscious, neither observable nor available for further analysis (except indirectly through dream analysis or via unconscious responses to ambiguous or neutral stimuli, etc.). Watson also emphasizes the importance of stimuli (both environmental and physiological) that elicit observable responses.

In his early writings, Watson (1924) makes several important observations. First, he describes several kinds of responses that do not appear to require conditioning. For example, following Pavlov, he observed that responses to stimuli that are essential to the survival of the species (e.g., stimuli signaling danger or the availability of food) do not require elaborate conditioning histories. That being said, Watson also gathered data suggesting that these basic behavioral repertoires could be modified by experience. Ultimately, he concludes that much human behavior is learned via classical conditioning processes. Throughout his text, Watson consistently contrasts his view of learning with the prevailing view (in the 1920s and 1930s) that heredity alone accounts for most human behavior. Watson's work with an 11-month-old

infant, Albert, yielded one of the most potent examples of conditioning (Watson & Watson, 1921). Using classical conditioning procedures developed by Pavlov, Watson and Watson demonstrated that Albert showed no fear response (e.g., avoidance) to a white rat until experimenters repeatedly paired loud sounds (produced by striking a metal bar) with the introduction of the rat. After just a few pairings of the rat and sound, Albert began to show a fear response (i.e., crying and crawling away), even when the rat was presented in the absence of the sound. Albert's fear also transferred (or generalized) to similar stimuli, such as a rabbit. In other work, Watson and colleagues describe how one might use classical conditioning to alleviate fear, including (a) having other children handle the stimulus without fear and (b) deconditioning the fear by gradually presenting the feared stimulus without the accompanying aversive stimulus. These two techniques are today called participant modeling and in vivo desensitization, respectively, and both are counted among the handful of empirically supported interventions for anxiety problems (Ollendick & King, 1998). Watson and other early behaviorists went on to apply their conditioning model broadly to a wide range of human behaviors, including what they termed "emotional reactions" such as rage and love.

Over the course of the next few decades, B. F. Skinner continued to shape behavioral theory through extensive basic research and provocative theorizing about the implications of laboratory demonstrations of operant conditioning. Influenced by Thorndike's (1911) law of effect, Skinner gradually came to distinguish his work from the work of other behaviorists, including Watson. Whereas Watson tended to emphasize the stimuli that elicit responses, Skinner's work devotes greater attention to consequences of the behavioral response (see Skinner, 1953). One of the most important concepts to emerge from Thorndike's, and later, Skinner's work is the explicit recognition that under certain circumstances, humans and other organisms tend to repeat behaviors that are followed by particular events or consequences. Behavior can be strengthened through *positive reinforcement* by adding something (e.g., social, tangible, physical consequences) and can also be strengthened through *negative reinforcement,* typically by removing something aversive.

Skinner's theory also accounts for reductions in the strength of behavior. Specifically, *operant extinction* involves withholding the presentation of consequences maintaining a given behavior, whereas *negative punishment* involves reducing the frequency of behavior by the removal of a consequence following the performance of a behavior. *Positive punishment* involves weakening behavior by adding aversive stimuli contingent on the performance of a behavior. Skinner includes antecedents in his analyses of reinforcement contingencies, specifically focusing on antecedent stimuli (e.g., discriminative stimuli) that control behavioral responses because of a history of reinforcement in the presence of those stimuli. The three-term contingency describes the important relation that exists between *antecedent* stimuli, the *behavior,* and its *consequence* (i.e., the so-called A-B-Cs). Under certain circumstances, setting events that alter the discriminative functions of the antecedent stimuli are also included in the analysis (Iwata, Smith, & Michael, 2000). Ultimately, Skinner made many other important contributions to behavioral theory and applied behavior analysis, including the concepts of reinforcement schedules, shaping, fading, and functional analysis, just to name a few.

After an initial period of laboratory work with nonhumans, Skinner's students and colleagues rapidly expanded the application of learning principles utilizing laboratory methods to study, and influence, the behavior of children in child care, institutional, and educational settings (see Bijou & Baer, 1961, for an early example). Even today, the manipulation of antecedents and consequences is the basis for many empirically supported interventions for youth (Ollendick & King, 2000). Over the past few decades, applied behavior analysis has flourished, especially among individuals with developmental disabilities (see this volume's Chapter 8 for a more detailed account).

Like Watson, Skinner (1953, 1957) makes an explicit effort to address the role of cognition in his theory of human behavior. Skinner distinguishes between *private events* (e.g., thoughts, images, sensations, feelings) and *public events* (i.e., overt behavior), but does not afford private events a causal role independent of events (either contemporary or historical) occurring outside of the individual. For example, Skinner (1963, p. 953) once commented that "an adequate science of behavior must consider events taking place within the skin of the organism, not as physiological mediators of behavior, but as part of behavior itself." Put another way, Skinner considered both private events and public events to be forms of behavior that could serve as links in a "causal chain," ultimately determined by environmental factors.

In the late 1960s and early 1970s, Bandura initiated a series of studies examining the impact of social (or vicarious) learning, or what later became known as "modeling." Bandura (1974) agreed with Skinner's view that much human behavior was shaped by external consequences, but he suggests that learning would be exceptionally slow if humans were limited to learning via exposure to direct contingencies alone. Bandura (1973) argues that learning through observation is not only more efficient than direct contingency shaping; it also permits the individual to avoid aversive consequences. For example, in one of the landmark studies investigating observational learning, Bandura found that children who watched an aggressive model (i.e., live or video) demonstrated more aggressive acts with an inflatable doll than those who observed a nonaggressive model. Children observing the aggressive model also demonstrated more aggressive behavior than those who did not observe any model (see Bandura, 1973).

Although both Skinner and Bandura view verbal behavior as important to understanding behavior, Skinner (1957) emphasizes the role of the speaker and developed a fairly complex taxonomy of verbal stimuli that functioned as antecedents or consequences. Taking a different approach, Bandura places greater importance on cognitive variables (e.g., self-efficacy) and affords these variables a causal role in a wide range of human activity. Whether or not cognitions play a causal role in overt behavior continues to be debated (see Reitman & Drabman, 1997, on the relative merits of such debate).

COGNITIVE THEORY

Rational-emotive behavior therapy (REBT) was developed by Albert Ellis in the 1950s. First termed rational therapy, Ellis (1962) later adopted the name rational-emotive therapy (RET) to differentiate his theoretical approach from others sharing

the same name and to distance himself from the philosophical position known as rationalism. When fellow theorists and critics pointed out that many of his therapeutic techniques were essentially compatible with behavioral theory, Ellis (1999) incorporated behavior more explicitly, changing RET to rational-emotive behavior therapy. Long regarded as an inclusive therapeutic approach (A. Lazarus, 1989), REBT liberally incorporates elements of behavioral theory and practice, thus making it difficult to distinguish many of the various techniques used in REBT from behavior therapy.

The essence of Ellis's REBT is summarized by the acronym ABCDE (Dryden & Ellis, 2001; Ellis & Bernard, 1983). In this model, "A" represents an event that *activates* specific irrational thinking (e.g., a child gets a poor grade on an assignment). "B" represents *beliefs* that are irrational and illogical (e.g., "I can't stand to get a bad grade"). "C" represents the emotional and behavioral *consequences* of irrational and illogical beliefs (e.g., sadness and withdrawal). The focus of therapy, therefore, is to intervene at the level of beliefs (B), as opposed to the particular activating event or its consequences per se. In describing persons with psychological problems, Ellis (1989, p. 217) remarked, "Their emotional problems and their behavioral dysfunctions mainly stem from their grandiose demands on themselves, on others, and on environmental conditions." The role of the therapist, therefore, is to focus on the client's irrational thinking in order to produce a widespread change in emotional and behavioral functioning. The major efforts at producing cognitive and behavioral change occur in the final two steps of the model, where "D" represents the therapist *disputing* irrational thinking and beliefs (in conjunction with other REBT techniques) and "E" represents the healthier, more *effective,* state of emotional, behavioral, and cognitive functioning.

Although much of his theory places prime importance on a universally held irrational belief system, the REBT model reflects Ellis's (1962) view that all human existence consists of the primary domains of thought, emotion, and overt behavior. Ellis and Bernard (1983, p. 9) stated that people "almost never experience *pure* thoughts, feelings, or actions." Therefore, although Ellis suggests that cognitions can elicit behavioral and emotional consequences, he emphasizes that thoughts, emotions, and overt behavior are part of a holistic process in which no single domain determines activity in other domains.

Ellis contrasts REBT with humanistic models that emphasize the importance of unconditional support of the client as the primary goal in therapy. He argues that although the client should be valued, the dysfunctional thoughts and overt behavior of the client should be vigorously challenged (Ellis, 1962). In fact, Ellis is known for his often confrontational approach to therapy. Early critics of his theory questioned his forceful challenges to the client's irrational thinking, but Ellis argued that such efforts were necessary to maximize the effectiveness of therapy.

Aaron T. Beck is another highly influential figure in the history of CBT. Developing his theory independently of Ellis, A. Beck (1963) was one of the first researchers to offer a detailed account of the role of cognition in psychopathology. In one of his earliest works, he describes a pattern of cognitive distortions and themes in his adult patients with depression, and concludes that the actual source of his patients' depressed mood was distorted and irrational thinking (A. Beck, 1963). For example,

he observed that many patients perceived themselves "as incapable, incompetent, and helpless," regardless of their actual life circumstances (p. 327). In addition to having negative views of themselves, other common negative beliefs encompassed aspects of the future and the world. Collectively, these negative beliefs about the self, future, and world became known as the "cognitive triad" (DeRubeis, Tang, & Beck, 2001).

In his original formulation of the theory, A. Beck (1963) refutes the idea that mental health problems such as depression are driven by the affect experienced by the patient. Rather, he suggests that the affective, behavioral, and social facets of depression result from distorted thinking. Through interviews and extensive clinical experience with depressed patients, he began to notice certain patterns of cognitive distortions. Some of the most common forms of depressive cognitions were arbitrary inference, selective abstraction, overgeneralization, and exaggeration (A. Beck, 1963). For example, selective abstraction refers to the tendency for some persons to attend to only negative facets of a particular situation and neglect other details (A. Beck, Rush, Shaw, & Emery, 1979). In the case of depression, a person may recall a past conversation, focusing only on perceived subtle criticisms, while ignoring complimentary information.

A. Beck (A. Beck et al., 1979) has also described people with depression as shifting cognitively from "mature" thinking to more "primitive" thinking, which in turn leads to negative affective consequences. Mature thinking is characterized by reflection and insight that allows the individual to form a multidimensional, changeable, and relatively nonjudgmental self-concept. On the other hand, primitive thinking is absolute and global, leading the person to view themselves as unchangeable and fundamentally flawed (A. Beck et al., 1979). Early work by A. Beck (1963) focuses on depression, but he later suggests that cognitive theory could be applied to other disorders such as externalizing problems and anxiety (A. Beck, 1999; A. Beck, Emery, & Greenberg, 2005). Additionally, although cognitive theory focuses primarily on cognition, A. Beck and others have argued that the relation of cognition and affect may be complex (see Kohlenberg & Tsai, 1991). In any event, both Ellis and A. Beck were united in their hope that the explicit focus on cognitions would promote more efficient alleviation of negative affect (A. Beck et al., 1979).

A. Beck's (A. Beck et al., 1979) early work implies that there are different levels of belief, and his theory has evolved to incorporate these observations. For example, Judith Beck (1995) describes three levels of belief: automatic thoughts, intermediate beliefs, and core beliefs. *Automatic thoughts* are situation-specific and idiosyncratic (e.g., "I'm going to fail this test"). *Intermediate beliefs* are essentially rules or assumptions that a person applies to his or her life (e.g., "Things never work out for me"). Finally, *core beliefs* are both global and fixed (e.g., "I am a failure"). In this theory, automatic thoughts and intermediate beliefs are considered to be a reflection of a related core belief. Thus, the ultimate goal of treatment for a particular disorder is to modify irrational thinking at the level of the core belief, although treatment often begins with related automatic thoughts and intermediate beliefs.

Both Ellis and A. Beck have acknowledged the importance of reciprocal relations between cognition, the social environment, and overt behavior. For example,

if the host of a holiday party excuses herself to greet other guests, a person prone to depression and/or social anxiety may react with (a) the automatic thought "She thinks I'm boring," (b) the intermediate belief "Nobody wants to talk to a person like me," and (c) withdrawal from further social interaction. The withdrawal response may then prompt others to avoid the individual in that social situation. Consequently, the individual's intermediate belief (i.e., "Nobody wants to talk to me") may be supported by his or her subsequent experience at the party. Interestingly, the depressed or anxious individual is also likely to discount evidence contrary to the distorted hypotheses, reasoning "People only spoke to me because I was standing near the bar" and so forth. This sequence of events illustrates the reciprocal nature of individual cognition, overt behavior, and social conditioning that can maintain distorted thinking (A. Beck et al., 1979).

Although both Ellis and A. Beck postulate that irrational thinking maintains maladaptive behaviors, neither regards irrational thinking as the ultimate cause of psychopathology. Rather, both point to biological and environmental influences as important factors in the development of irrational thinking (A. Beck et al., 1979; Bernard & DiGiuseppe, 1989). According to A. Beck et al., a person's genetics and childhood environment may initiate a pattern of irrational thinking, which then leads to, and maintains, maladaptive behaviors and psychopathology. A. Beck suggests that most people lead life governed by rational thinking. However, irrational thoughts formed early in life could be triggered at some point by environmental stress, allowing for a shift from a pattern of rational thinking to irrational thinking.

The term cognitive-behavioral therapy is often used to refer specifically to the combination of A. Beck's cognitive techniques with other behavioral techniques. At other times, CBT is used as a generic term to refer to all manner of cognitive and behavior therapy blends. Although admitting many similarities, Ellis (1980; Ellis & Bernard, 1983) distinguishes between REBT and CBT. For example, Ellis states that CBT tends to focus more on cognitive change as it affects behavior, whereas REBT is primarily concerned with cognitive change as it leads to both emotional change and a broader change in one's philosophy of life, thus ostensibly affecting one's ability to cope with future problematic situations. Additionally, Ellis suggests that CBT focuses more on situation-specific cognitions and behaviors, and REBT addresses more global, irrational thinking and beliefs.

Most cognitive therapists began their work focused on problems confronted by adults in outpatient psychotherapy, so it is only recently that CBT has begun to formulate a coherent theoretical model to guide work with children. Ellis and Bernard (1983) were among the first to discuss the application of cognitive therapies to children and adolescents. Specifically, they noted that "although preferential RET is highly suitable for many bright adolescents, it may require too much philosophical analysis and more of an application of rigorous scientific method than many average youngsters, not to mention most younger children, are capable of fulfilling" (p. 8). Thus, children and many adolescents may benefit more from behavioral techniques and cognitive techniques that address situation-specific cognitions rather than belief systems.

A. Beck has written little on the subject of applying cognitive theory to children and adolescents. For example, a description of manualized cognitive therapy

treatment for adolescents who have attempted suicide seems more or less consistent with adult CBT as it progresses from the first session to termination, with no discussion of changes necessary to the therapy due to the developmental stage of adolescence (Henriques, Beck, & Brown, 2003). Recently, a few researchers have begun to examine the applicability of cognitive therapy to young children and adolescents. Specifically, Doherr, Reynolds, Wetherly, and Evans (2005) describe two studies that experimentally examined the ability of young children (5 to 7 years old) to complete cognitive tasks that would be required for cognitive therapy. Their findings indicate that, for the most part, the 5- to 7-year-olds possessed the cognitive abilities necessary to participate in cognitive therapy, although materials used for this therapy required alteration. For example, cognitive-behavioral therapy with adults often involves the use of a dysfunctional thought record in which clients record situations, automatic thoughts, and rational responses (J. Beck, 1995); some children may be able to talk about these concepts but would have difficulty writing them down independently.

At this time, the existing literature on adapting cognitive techniques for children is limited, and researchers and clinicians attempting to apply cognitive techniques to children rely on many theoretical and practical assumptions that were developed through research with adults (Derisley, 2004). On the other hand, many empirically supported interventions for youth (Ollendick & King, 2000) have been influenced by Ellis and A. Beck. Perhaps most important, changes in children's negative self-talk seem to mediate treatment outcome (Treadwell & Kendall, 1996). In the next section we present a brief review of CBT models developed more explicitly for children or that emerged based on work with child and adolescent populations.

FURTHER INTEGRATION OF BEHAVIORAL AND COGNITIVE THEORIES

While many theorists in the 1970s debated the strengths and weaknesses of the behavioral and cognitive models, some argued for an integrationist perspective including both theories. Cognitive therapists included many behavioral techniques, and some behaviorists focused more on covert behavior. These integrationist approaches ultimately had an important influence on the development of CBT interventions for children and adolescents.

Many of the integrated theories have been included under the broad label of "self-management" and include a variety of behavioral and cognitive approaches. Though sometimes referring to specific techniques, when the term self-management is used broadly, it implies that the person's behavior is partly controlled by external and internal stimuli and that the person is capable of manipulating those external and internal stimuli to achieve long-term goals (Rokke & Rehm, 2001). Thus, the derived self-management (or self-control) therapies focus on teaching learning and cognitive principles to clients so that they can apply them in their own lives. Self-management techniques therefore include monitoring one's own overt and covert behavior (i.e., self-monitoring), efforts to alter or change one's own external and internal antecedents to behavior (i.e., stimulus control), and the application of both external and internal consequences to one's own behavior (i.e., self-reward and

self-punishment; Mahoney & Arnkoff, 1978). Some self-management theorists place greater emphasis on adults (e.g., Goldfried, Decenteceo, & Weinberg, 1974); others focus more on children (Bandura, 1973; Meichenbaum, 1977). Theories of self-efficacy, problem solving, and self-instruction all fit within the self-management model.

Bandura (1997) first significantly influenced behavioral theory by adding social learning; he later continued to shape cognitive-behavioral theory with his self-efficacy model, which places primary importance on the beliefs of the individual. Although Bandura's writings are generally supportive of Skinner's behavioral theory, he suggests that earlier behavioral theories are incomplete, rather than inaccurate. For example, Bandura's research demonstrates that the consequences that follow a model's behavior play an important role in the behavior of the observer. That is, modeling is often more powerful when the model receives a reward, and Bandura (1977a) suggests that this increases the observers' expectations that reinforcers would likely follow their behavior as well. Indeed, basic research has shown that an individual's beliefs about a given reinforcement schedule could prove a more powerful predictor of behavior than the actual reinforcement schedule itself (Baron, Kaufman, & Stauber, 1969). As a result, social learning theory has increasingly emphasized the importance of cognition in predicting behavior. In particular, Bandura's recent work focuses on outcome expectations and self-efficacy. Outcome expectations are a person's predictions about future consequences, and self-efficacy describes the "beliefs in one's capabilities to organize and execute the courses of action required to produce given attainments" (Bandura, 1977b, p. 3). Bandura (1977b) describes four primary ways to develop self-efficacy; that is, self-efficacy develops when (1) mastery experiences result in the individual successfully obtaining a reinforcer, (2) social modeling involves observing others having success, (3) social persuasion is used to enhance the individuals' belief that they too can be successful, and (4) people are taught to read their own physical and emotional states and use these as a guide to success. Reflecting his greater emphasis on the role of cognitions in influencing behavior, Bandura (2004) now uses the term "social cognitive theory" to describe his theoretical model.

Cognitive problem solving is another model that emerged from the integrationist perspective. D'Zurilla and Goldfried (1971) present problem solving as the "self-directed cognitive-behavioral process by which a person attempts to identify or discover effective or adaptive solutions for specific problems encountered in everyday living" (D'Zurilla & Nezu, 2001, p. 212). D'Zurilla and Nezu draw largely from the work of R. Lazarus (1966; R. Lazarus & Folkman, 1984) in developing their "coping" or problem-solving model. In this model, stress is defined as a relationship between a person and the environment in which the person's coping skills are strained. Specific stressors include life's major problems as well as daily hassles. D'Zurilla and Nezu distinguish between two types of coping responses. *Problem-focused* coping responses are used to change something about a situation (e.g., communication style) and are best employed when the person has some control over the problem situation. On the other hand, *emotion-focused* coping responses are an attempt by the person to change his or her own emotional response

(e.g., relaxation techniques to decrease anxiety or anger) and are best used when the person does not have control over the problem situation.

Problem solving involves first identifying a situation requiring a coping response and then generating an adaptive cognitive coping response, a behavioral coping response, or both (D'Zurilla & Nezu, 2001). D'Zurilla and Nezu also distinguish between being *oriented* to a problem and then *solving* the problem. That is, before one can start generating effective solutions, one must (a) recognize that there is a problem, (b) attribute the problem to normal life rather than an internal stable deficiency, (c) see the problem as significant to one's well-being, (d) be willing to put time and effort into solving the problem, and (e) have high self-efficacy and an expectancy for change. Evidently, their problem-solving model was influenced by Bandura's work. Actually solving problems includes several components as well: (a) formulating the problem, (b) generating several possible solutions, (c) choosing the best solution, (d) implementing the solution, and (e) evaluating the solution's effectiveness. Interestingly, formulation of the problem might include cognitive and/or behavioral influences, and the solutions might be cognitive and/or behavioral in nature.

Meichenbaum (1977) offers yet another integrationist perspective, and he explicitly positioned his book *Cognitive-Behavior Modification* as an attempt to "bridge the gap" between behavior therapy and the cognitively focused therapies of Ellis and A. Beck. Meichenbaum also cites Luria (1959, 1961) and Vygotsky (1962) as important influences on his approach. Based on Luria and Vygotsky, Meichenbaum argues that the behavior of young children is first directed by their parents' verbal behavior. Children then begin to direct their own behavior through overt speech, and finally, inner speech is developed to direct their own behavior. Meichenbaum adds that once a child becomes proficient at a task, the inner speech for that task is no longer emitted. Thus, he hypothesized that maladaptive child behavior is partly the result of deficits in inner speech. For example, Meichenbaum theorizes that children with high rates of impulsive behavior have deficits in producing and effectively using inner speech. His theoretical focus on the progression of inner speech development was the impetus for the development of self-instruction training, which he applied to children with high rates of impulsive and hyperactive behavior. Self-instruction training assumes that children may have a deficit in that they do not produce enough effective inner speech to direct their behavior.

Meichenbaum was influenced by both social learning theory and the problem-solving model. Following from research suggesting that children imitate their parents' verbal reasoning (Maccoby, 1968), Meichenbaum extends Bandura's social learning theory (which focuses on modeling overt behaviors) to modeling inner speech, a form of covert behavior. Thus, self-instruction training first uses "cognitive modeling," in which adults model the use of problem-solving skills by talking aloud while performing a task. Using this technique, the adult verbally models defining the problem, focusing attention on task requirements, and providing self-praise and corrective feedback. The second step is for the child to perform the task while being instructed by the adult. The next steps involve the child imitating the adult with overt speech, then later with whispered speech, and finally with inner

speech. Meichenbaum (1977) also discusses integrating behavioral components, such as employing external reinforcers in addition to the use of self-praise.

Meichenbaum (1977) offers some important distinctions between his theory and those offered by Ellis and A. Beck. Specifically, he suggests that Ellis and A. Beck place a greater emphasis on the *presence* of maladaptive thoughts, and Meichenbaum places a greater emphasis on the *absence* of adaptive thoughts. He uses impulsive behavior as a prime example, arguing that it is not the presence of maladaptive thoughts that is problematic per se, but the absence of adaptive thoughts prior to impulsive behavior. Meichenbaum also suggests that some maladaptive thinking is normal, but that it is the degree of adaptive thinking that distinguishes normal functioning from abnormal functioning. Kendall and Hollon (1979) suggest that this distinction is the cognitive equivalent of behavioral excesses and behavioral deficits, and more recently, Kendall (2000) has used the terms "cognitive distortions" and "cognitive deficiencies" to explore this issue. Practically, this distinction suggests that for some problems maladaptive thoughts need to be directly challenged, whereas for other problems the focus of therapy should be on teaching more adaptive thoughts.

Meichenbaum's (1977, 1985) integrative theory also led to the development of stress inoculation training, another cognitive-behavioral approach to therapy. Stress inoculation training includes self-instruction training but incorporates additional components. The label "stress inoculation" is an analogy to medical inoculation, which builds antibodies to prevent diseases. In stress inoculation therapy, the concept is to build coping skills (i.e., "psychological antibodies") to prevent psychological problems (Meichenbaum, 1985). Meichenbaum gives credit to other investigators for independently developing similar ideas. For example, he cites Goldfried et al. (1974) as integrating Ellis's RET with behavioral theory. Overall, coping skill-based therapies involve teaching clients both more adaptive social behavior and cognitive coping skills. In the case of anxiety, a client is taught physiological relaxation (e.g., deep breathing and muscle relaxation) and adaptive self-statements. Clients first practice using these skills in a nonstressful situation (i.e., a therapy session) before deploying the skills in a more stressful situation.

Commenting on the diverse integrationist approaches to CBT that were emerging in the late 1970s, Mahoney and Arnkoff (1978) categorized cognitive-behavioral interventions as falling into three categories: cognitive restructuring, coping skills, and problem solving. The *cognitive restructuring* category includes the work of Ellis and A. Beck. In these therapies, maladaptive thinking is considered to be the primary cause of distress for the person. Cognitive restructuring therapies often use some behavioral techniques and carefully monitor the effects of treatment; however, the ultimate goal is to replace the person's maladaptive cognitions (internal problems) with more adaptive cognitions. The *coping skills* category includes Meichenbaum's stress inoculation therapy and other therapies that focus more on adapting to external challenges by increasing coping skills. Thus, though these therapies also address cognitions, successful therapy is determined by changes in behavioral responses rather than cognitive ones. Finally, the *problem-solving* category is perhaps the broadest in scope. These therapies focus on teaching the person a general strategy for identifying problems and generating solutions. Thus, the

focus for any given problem could be cognitive, behavioral, or both, and the range of possible solutions could include making changes to cognitions, overt behaviors, and/or the environment. Clearly, many empirically supported interventions with youth have been directly influenced by the work of cognitive-behavioral therapists with an integrationist perspective.

APPLYING THE COGNITIVE-BEHAVIORAL FRAMEWORK TO YOUTH

Cognitive-behavioral theory was largely developed by clinical researchers working with adult populations, and in recent years discussion has revolved around the extent to which the CBT framework applies to work with children and adolescents. Braswell and Kendall (2001) highlighted the importance of cognitive development in child CBT. Specifically, children's memory, attention, verbal expression, comprehension, and conceptual reasoning all might affect their response to treatment. Adaptation of CBT for children and adolescents would seem to minimally require that they be capable of sharing private events with the therapist and able to distinguish rational and irrational thoughts. To the extent that the child or adolescent is unable or unwilling to perform these tasks, the role that cognitive interventions such as cognitive restructuring can play in the therapy will be limited.

Another point of departure for adult and child CBT is that collaterals (e.g., parents and teachers) are unlikely to be directly involved in therapy for adults. Also, children rarely seek out therapy themselves, making the process of gaining their trust and engaging them in the therapeutic process especially challenging. Recognizing the importance of client motivation for treatment, Braswell and Kendall (2001) note that the manner in which CBT is presented must be altered to maximize the likelihood of positive therapy outcomes. Specifically, "a therapist must be able to teach in a playful manner and play in a way that teaches" (p. 250). To compensate for both limited cognitive capacity and motivation, Braswell and Kendall recommend greater emphasis on the incorporation of play (e.g., games, puppets), more role-plays, and the use of visual aids to supplement language-based aspects of therapy. For example, whereas adults often rate their fear on a 0 to 100 scale, in child CBT for anxiety children report on fears by choosing from an array of pictures (e.g., a row of faces depicting progressively greater fear). Some therapists use cartoon characters with thought bubbles as a tool in sharing thoughts.

Cognitive-behavioral therapy with children typically incorporates the four major components of behavioral skills training—modeling, instructions, rehearsal, and feedback (Miltenberger, 2004)—as well as manipulation of behavioral contingencies. The focus on behavioral skills training is to replace an undesirable behavior with a desirable behavior. That is, the goal is to teach the learner what to do in addition to what not to do. *Modeling* includes showing the learner how to perform a behavior, and examples include modeling behaviors for children (e.g., how to share) as well as modeling behaviors for their parents (e.g., how to use time-out). *Instructions* describe the desired behavior to the learner; examples include verbal instructions as well as written instructions. *Rehearsal* provides the learner with several opportunities to practice the desired behavior in front of the therapist.

Finally, the therapist provides *feedback* based on rehearsal, which includes praise and positively valenced critiques of the rehearsal. Thus, in the feedback component, behavioral contingencies are used to shape the learner's behavior, and the learner's behavior is ideally reinforced within the context of the role-play. Behavioral contingencies are employed outside of the therapy setting to increase generalization. Consistent with Bandura's (1974a) social learning theory, behavioral skills training increases self-efficacy through mastery experiences, social modeling, social persuasion, and sometimes through changing internal physical and emotional states.

Since their inception, behavioral and cognitive theories have drawn on other theoretical traditions. For example, the cognitive-behavioral framework is particularly compatible with *ecological systems theory* (Bronfenbrenner, 2005). The ecological systems perspective considers both characteristics of the individual and those of the family, school, and community; CBT is being increasingly used in schools (Mennuti, Freeman, & Christner, 2006). Importantly, both ecological systems theory and the cognitive-behavioral framework acknowledge that genetics and other biological influences play a role in the development of psychopathology. However, it is often difficult to determine the degree of bidirectional influence on behavior stemming from environmental and biological factors and, further, to distinguish normal behavioral variation from deviant developmental trajectories (Jensen, Knapp, & Mrazek, 2006).

In the following subsections, we discuss the application of cognitive-behavioral theory to specific problems for youth defined by the *Diagnostic and Statistical Manual of Mental Disorders,* fourth edition, text revision (*DSM-IV-TR;* American Psychiatric Association, 2000). Although our focus is on depression, anxiety, conduct problems, attention problems, and parenting stress, the cognitive-behavioral framework has also been successfully applied to many other problems affecting youth, including eating disorders and obesity (Wilfley, Passi, Cooperberg, & Stein, 2006), abuse (Deblinger, Behl, & Glickman, 2006), academic problems (Harris & Graham, 1996), and chronic pain (Masek, Russo, & Varni, 1984).

Depression

Cognitive-behavioral theorists have offered several accounts of potential etiological factors involved in youth depression (i.e., Major Depressive Disorder and Dysthymic Disorder). Lewinsohn (1974) hypothesizes that a lack of positive reinforcement could lead to depression. Some research has supported the notion that children with depression lack the social skills needed to obtain social reinforcement (Rudolph & Clark, 2001). Stressful life events (e.g., hospitalization, moving) have also been implicated in the development of depression among adolescents (Lewinsohn, Allen, Seeley, & Gotlib, 1999; Seligman, 1975). For example, in a study of over 1,500 adolescents, Lewinsohn et al. found that three or more stressful life events in the previous year appear to be the threshold for making predictions about the onset of a first depressive episode. Interestingly, recurrent depressive episodes could not be predicted using this threshold.

Consistent with a more cognitive (mediational) account of adolescent depression, experiencing several stressful life events could influence depression through

the development of a depressive attributional style (Seligman, 1975). Interestingly, negative attributional style does not predict depression when a high number of life stressors are present, but it does predict depression when there are lower levels of stress (Lewinsohn, Joiner, & Rohde, 2001). In a meta-analysis of 28 studies involving youth, depressive symptoms were correlated with internal (i.e., blame oneself), stable (i.e., not changeable), and global (i.e., applies to everything) attributions for *negative* life events and external, unstable, and specific attributions for *positive* life events (Gladstone & Kaslow, 1995). The influence of maladaptive thinking as a contributor to child and adolescent depression has also received support in the literature. Negative self-appraisals have been found to be predictive of depressive symptoms in children in sixth and seventh grade (Hoffman, Cole, Martin, Tram, & Seroczynski, 2000), but the study did not distinguish between cognitive distortions and potentially accurate negative self-statements. In another study, children with depression rated their own social status lower than peers rated them (Rudolph & Clark, 2001). Furthermore, McGrath and Repetti (2002) studied fourth- through sixth-grade students and found that depressive symptoms were predictive of later cognitive distortions; however, cognitive distortions were not predictive of an increase in depressive symptoms over 3 years. This study suggests that cognitive distortions may play a greater role in the maintenance of depression in children than in the development of depression.

The cognitive-behavioral framework allows for parental influence on the development of childhood depression. For example, recent research suggests that childhood depression is associated with family conflict (Pilowsky, Wickramaratne, Nomura, & Weissman, 2006). Also, though the study was correlational, parents of youth with depression appear to provide less praise and are more critical than parents of nondepressed youth (Puig-Antich et al., 1993). Meichenbaum's theory of cognitive modeling suggests a possible mechanism for the transmittal of negative self-appraisal. For example, a parent's critical statement "You are lazy" may later be covertly modeled "I am lazy."

Anxiety

As noted previously, Watson (1924) was among the first to examine how children's fears could be acquired through classical conditioning. According to Mowrer's (1939) two-factor theory, once acquired, fear-related behaviors can be maintained through negative reinforcement (e.g., through escape or avoidance behaviors). Following Bandura's social learning theory, Rachman (1991) notes that fears can develop both by watching other people engage in fearful behavior and by receiving other fear-provoking information through language. A. Beck's (A. Beck et al., 2005) theory of cognitive distortions was originally developed for depression but has been extended to anxiety disorders as well.

DSM-IV-TR anxiety-related problems that have been diagnosed among children and adolescents include Separation Anxiety Disorder, Panic, Agoraphobia, Specific Phobia, Social Phobia, Obsessive-Compulsive Disorder, Posttraumatic Stress Disorder, and Generalized Anxiety Disorder. With the exception of Separation Anxiety Disorder, the diagnostic criteria applied to adult anxiety problems and those applied

to child anxiety problems are nearly identical. Lang's (1968) tripartite model of anxiety is among the most influential of the cognitive-behavioral theories of anxiety and is composed of physiological (i.e., activation of the sympathetic nervous system), behavioral (e.g., fight-or-flight responses), and cognitive (i.e., fear-related speech) dimensions. More recently, a fourth dimension, emotional responding, has been explored (Southam-Gerow & Kendall, 2002). These four response dimensions are considered to be common to all of the anxiety disorders.

Research indicates that anxious children view ambiguous situations as more threatening than nonanxious children (Barrett, Rapee, Dadds, & Ryan, 1996). Barrett et al. also reported that parents play a significant role in children's anxiety. Specifically, like their children, parents of anxious children also interpret ambiguous situations as more threatening than do parents of nonanxious children. Furthermore, after anxious children had a chance to converse with their parents following an ambiguously threatening situation, the children's choice of avoidant behavior increased. The authors suggest that parents modeled anxious coping and reinforced anxious behavior by offering reassurance following avoidant responding.

Cognitive-behavioral approaches to child and adolescent anxiety have been designed to address concerns at each level of the model (Kendall & Suveg, 2006). For example, at the physiological level, children may learn relaxation techniques (e.g., diaphragmatic breathing, muscle relaxation, visual imagery). Cognitive rumination may be addressed by teaching children how to generate rational coping responses to cognitive distortions. At the behavioral level, the majority of treatment alternatives for anxiety continue to emphasize the role of exposure-based techniques such as systematic desensitization. Most recently, CBT researchers have begun to emphasize the role of the social environment in maintaining social anxiety. Thus, parents and teachers and other important influences (e.g., siblings, peers) on the child have been more aggressively recruited to participate in treatment. For example, in some cases, it has been suggested that to maximize the power of modeling and social reinforcement, parents must first be taught to manage their own anxiety before treatment for their child is initiated (Silverman & Kurtines, 1996).

Conduct Problems

As with internalizing problems, both environmental and cognitive variables are hypothesized to play a key role in the development of externalizing problems such as Conduct Disorder and Oppositional Defiant Disorder. Drawing both on Bandura's social learning theory and Skinner's operant model, Gerald Patterson's (Granic & Patterson, 2006) coercion model has been shown to account for significant variance in the development and maintenance of externalizing problems. According to the coercion model, parents may accidentally teach their children to exhibit externalizing problems. The cycle begins with parent demands for compliance. Children typically respond to requests for compliance with relatively mild coercive behavior (e.g., whining, aggression). For the coercive process to advance, parents then withdraw demands contingent on the child's aversive behaviors. Interestingly, both the child's behavior (via withdrawal of the demand) and the parent's permissive behavior (via the child's withdrawal of aversive behavior) are negatively reinforced. As

a result, the child becomes more likely to use coercive behavior in the future, and the parent is more likely to give in. The development of this pattern of behavior is deemed the "coercive cycle" or "coercion trap" and, if unabated, is hypothesized to lead to antisocial behavior.

Cognitions are believed by many to play an important role in externalizing problems. For example, in addition to studying anxiety in children, Barrett et al. (1996) also investigated oppositional behavior, with similar results. That is, consistent with Dodge (1986), children categorized in the oppositional group interpreted ambiguous situations as more threatening than did children in a control group. In fact, the differences observed in the oppositional group were even greater than the differences observed in the anxious group. Although the anxious and oppositional groups each viewed the situation as threatening, the groups differed in their choice of responses to the situation. As might be expected, the anxious group chose generally avoidant responses, and the oppositional group chose aggressive responses. As in the anxious group, parents of oppositional children viewed the ambiguous situation as more threatening than parents in the control group. Remarkably, parent-child discussions following the first exposure to the ambiguous stimulus also increased the likelihood of an aggressive child response.

Attention Problems

Kendall's (2000) distinction between cognitive distortions and cognitive deficiencies is relevant to both internalizing and externalizing problems; however, the distinction is perhaps most relevant to Attention-Deficit/Hyperactivity Disorder (ADHD). That is, researchers tend to characterize ADHD as a problem of cognitive deficiency rather than a problem of cognitive distortion, thus potentially rendering the disorder more difficult to treat (Barkley, 1997; Hinshaw, 2006). Indeed, cognitive-behavioral interventions for ADHD have generally been regarded as less effective than interventions for anxiety, depression, and conduct problems. Current state-of-the-art treatment for many children diagnosed with ADHD is a combination of stimulant medication and behavioral interventions in the classroom and at home (MTA Cooperative Group, 2004).

Despite their apparent utility for other problems affecting youth, CBT interventions for ADHD have received only inconsistent support (Pelham, Wheeler, & Chronis, 1998). On the other hand, some authors have claimed some success in shifting the control of behavior from external consequences to internal ones. As noted previously, Meichenbaum (1977) presented self-instructional training as one promising alternative. Some research has even supported using CBT procedures with preschoolers labeled impulsive, and treatment gains have sometimes generalized to the classroom (e.g., Bryant & Budd, 1982). However, little research has been conducted on self-instructional training as a treatment option for ADHD.

Because approximately 50% of children with ADHD also have significant social problems (Pelham & Bender, 1982), social skills training has become increasingly popular with this population (Hupp, Reitman, Northup, O'Callaghan, & LeBlanc, 2002; Pfiffner & McBurnett, 1997). Indeed, it is well known that effective social

skills training should incorporate elements of behavioral therapy such as teaching problem-solving skills in social contexts (Barkley, Edwards, Laneri, Fletcher, & Metevia, 2001; Fenstermacher, Olympia, & Sheridan, 2006). By and large, the effectiveness of such efforts in improving short-term outcomes for children diagnosed with ADHD is promising (see Pfiffner & McBurnett, 1997) but remains speculative over the long term.

Parenting and Parenting Stress

Cognitive-behavioral therapy has the potential to affect parents in at least four ways. First, CBT teaches parents how to more effectively manipulate antecedents and consequences associated with their child's problem behavior. Second, CBT often employs parents as models for desirable, adaptive behaviors. Third, CBT may address the parent's own psychopathology (e.g., depression, anger). Fourth, beyond simply addressing the need for some parents to obtain treatment of their own psychopathology, CBT has begun to focus more attention on the role of parenting stress. Specifically, researchers have become interested in how stress may influence parental perceptions of the child. The first three issues have been discussed in previous sections; we now examine how the cognitive-behavioral framework may be applied to parent perceptions of child behavior.

Cognitive-behavioral theorists have recently begun to examine the extent to which parenting behavior may be influenced by parental attributions about child behavior (Anastopolous & Farley, 2003; Sanders, Cann, & Markie-Dadds, 2003). In an experimental manipulation of parents' attributions, Slep and O'Leary (1998) randomly assigned 40 mothers to one of two conditions. In one condition, the mothers were given feedback that their child was primarily responsible for his or her own disruptive behavior (i.e., child-responsible group). In the other condition, parents were told that their child's disruptive behavior was developmentally appropriate (i.e., child-*not*-responsible group). This feedback was actually not based on anything other than random assignment, though parents were told it was based on a previous interview. During observations of a parent-child interaction that followed the bogus feedback, mothers in the child-responsible group demonstrated significantly more harsh parenting behaviors than parents in the child-not-responsible group. Thus, experimentally manipulated parenting attributions played a significant role in the parent's behavior. This line of research has now led to the development of child abuse prevention programs that incorporate components targeting parental attributions for change (Sanders et al., 2004).

SUMMARY

Cognitive-behavioral theory has been described as a big tent with room for many complementary (and some competing) theories. Cognitive-behavioral therapists can choose from an increasingly large number of intervention options, but there are at least three major unifying factors underlying CBT for children and adolescents. Although these factors are not uniformly characteristic of all research and theory in CBT, these general principles continue to offer guidance concerning the

future path that CBT research and practice may take. First, CBT for children should emphasize the development of a collaborative relationship between the therapist and the client (or family) as well as a focus on the response of the individual to treatment. Second, consistent with the view offered by Wolpe (1993) and other CBT founders, empirical data should serve as the basis for future modifications of CBT theory and practice (Reitman & Drabman, 1997). The development of manualized and, more recently, modularized treatment approaches are recent changes in practice that were driven by empirical data suggesting that the treatment outcomes could be enhanced with improvements in treatment integrity. Third, CBT therapists emphasize the importance of client activity outside of the therapy room, including homework exercises between sessions, self-monitoring, and a variety of other techniques that ensure that improvements are noted in session ultimately result in meaningful change (i.e., change that generalizes and can be maintained over time; see Stokes & Baer, 1977).

Despite the development of manualized treatments for children and adolescents, no two children, parents, or families that participate in CBT will have the same therapeutic experience. Moreover, as noted previously, treatment targets and developmental considerations may dictate the use of procedures that emphasize or de-emphasize the inclusion of cognitive variables. For some youth, overt behavioral problems may be the only focus of intervention, whereas other youth might be more likely to benefit from a therapeutic approach emphasizing cognitions. Also, as noted by Kendall (2000), cognitive-behavioral therapists may focus on cognitive distortions or cognitive deficiencies.

Nor is it presently necessary to make a hard choice between CBT approaches based on their overt and covert characteristics. Overt and covert approaches can certainly be used in a complementary fashion. For example, Mennuti, Christner, and Freeman (2006) recently enumerated common CBT procedures based on this overt-covert dimension. Drawing on their work, we have drafted a similar table (see Table 9.1) that more directly contrasts the overt behavioral technique with its covert counterpart. Though not comprehensive, this table provides a useful index of common cognitive techniques and nicely summarizes the therapeutic tools available to contemporary cognitive-behavioral therapists.

Table 9.1 Examples of Techniques Used in Cognitive-Behavioral Therapy

Overt Behavior Techniques	Covert Behavior Techniques
Self-monitoring	Thought record
Participant modeling	Cognitive modeling
Social skills training	Self-instructions
Competing response training	Rational responses
Positive activity scheduling	Increasing positive thoughts
Tokens and reinforcers	Self-praise
Role-playing	Mental rehearsal
Behavioral experiments	Socratic questioning
Progressive muscle relaxation	Visual imagery
Exposure to external stimuli	Exposure to internal stimuli

As can be seen in Table 9.1, monitoring can be employed to track either overt or covert behavior. Similarly, the covert counterpart to participant modeling is Meichenbaum's cognitive modeling. Social skills training is commonly used to overcome social deficits, and self-instruction training can be used to deal with cognitive deficits. Competing behavioral responses can be generated for overt or covert behavior (e.g., irrational thoughts). Increases in positive reinforcement theoretically can be achieved through positive activity scheduling (see Lewinsohn et al., 1999) or techniques aimed at increasing positive thoughts (see Henriques et al., 2003, for an example called the Hope Kit). Additionally, as seen at the bottom of Table 9.1, anxiety treatments may involve exposure to overt or covert (e.g., imaginal) stimuli. For example, a "worry chair" (e.g., making oneself worry for a set amount of time each day) is designed to expose a client with Generalized Anxiety Disorder to covert stimuli (e.g., worry), and interoceptive exposure (e.g., intentionally making oneself dizzy) is designed to produce covert exposure to panic symptoms (see Barlow, 2002, for a detailed account).

The last pairing in Table 9.1 hints at the newest theoretical wrinkle in the cognitive-behavioral framework. Specifically, Hayes (2004) describes "three waves" of behavior therapy. The first wave was heavily influenced by the work of Watson and Skinner. As shortcomings of the first wave became apparent, a second wave emerged. Influenced largely by Ellis and A. Beck, this second wave is sometimes referred to as the "cognitive revolution." Mahoney and Arnkoff (1978) summarized such innovations during this time as cognitive restructuring, coping skills, and problem solving as part of the behavior therapy tradition. The third wave of behavior therapy is thus characterized as a response to the shortcomings of the second wave. For example, Hayes (2004, p. 645) argues that it is "possible to alter the function of thoughts without first altering their form." Hayes goes on to present alternatives to directly attempting to modify cognitive distortions. These most recent additions to the CBT model are presently referred to as acceptance- or mindfulness-based therapies, and they incorporate principles borrowed from theology, Middle Eastern philosophy, and meditation.

One of the best known of the acceptance-based approaches is acceptance and commitment therapy (ACT; Hayes et al., 1999). In ACT, clients are instructed to learn to accept their cognitive distortions and emotional reactions without having to avoid or change them. Clients are taught to simply observe cognitions and, further, that the only cognitive change that is needed is to accept the fact that it is okay to have cognitive distortions and intense emotions. Similar therapies are mindfulness-based cognitive therapy (Teasdale et al., 2002) and dialectical behavior therapy (Linehan, 1993). Although acceptance- and mindfulness-based therapies are relatively new, early research appears encouraging (Hayes, Masuda, Bissett, Luoma, & Guerrero, 2004). Very little research has been devoted to ACT for youth, but a few case examples have yielded promising results (Heffner, Sperry, Eifert, & Detweiler, 2002; Wicksell, Dahl, Magnusson, & Olsson, 2005; see also Greco, Blackledge, Coyne, & Ehrenreich, 2005; Murrell, Coyne, & Wilson, 2004). As with other innovations within the cognitive-behavioral framework, ACT does not diminish the value of the other cognitive-behavioral techniques. Rather, it appears to offer yet another treatment alternative.

Throughout this chapter, CBT has been discussed primarily in the context of individual or single-family psychotherapy; however, cognitive and behavioral techniques hold significant promise for group therapy. In fact, CBT for adolescent depression may be enhanced by group social support (see Clarke, DeBar, & Lewinsohn, 2003). Because CBT is suitable for group administration it may also prove useful when incorporated into prevention programs. There are several forms of prevention: indicated, selective, universal, and health promotion models (Weisz, Sandler, Durlak, & Anton, 2005). Indicated prevention is intended to address the needs of individuals with significant symptoms that fall short of diagnosis. Selective prevention programs are for groups of individuals who share risk factors for developing problems (e.g., exposure to a traumatic event). Universal prevention programs are designed to limit the effects of potential risk factors for all youth (e.g., everyone in a school district), without specifically identifying individuals with certain risk factors. Finally, recent trends in the prevention literature point toward a health promotion focus that emphasizes building competencies within an entire population rather than focusing on risk reduction. An evidence base is quickly emerging for cognitive-behavioral prevention and health promotion programs at all four levels (Weisz et al., 2005).

In a recent symposium of former presidents of the Association for Behavioral and Cognitive Therapies (ABCT;* Albano, 2006), a number of topics relevant to the development of CBT were discussed, and two issues were central to this chapter and the future of CBT for children and adolescents. The first issue concerns core principles. Specifically, other than empiricism, are there any remaining core principles associated with CBT? If the answer is no, then theoretical considerations must always take a backseat to what works. Indeed, many early behavior therapists seemed to suggest that data should trump theory in behavior therapy, and that no aspect of the therapeutic model was to be regarded as sacred (see Reitman & Drabman, 1997). Nevertheless, in the symposium, Ollendick (2006) noted that he was initially drawn to the behavioral approach by something akin to core principles, specifically, its emphasis on functional assessment and idiographic case analysis. Wilson (2006) suggested that the validity of core theoretical principles should be subjected to rigorous analysis in dismantling studies, and he noted the substantive contributions made by Jacobson and Rachman in challenging the status quo via such an approach.

A second major issue discussed in the symposium concerned how best to go about exporting or disseminating behavioral work. Heimberg (2006) pointed out that many clients still do not benefit from cognitive-behavioral therapy and warned against complacency. Wilson (2006) suggested that efforts to disseminate existing treatment packages were unlikely to be successful without the elucidation of simpler, universal change principles underlying CBT. Interestingly, there were different ideas presented regarding how to best refine CBT. On the one hand, some viewed the future as requiring more sophisticated studies (with greater numbers of

*Reflecting a greater emphasis on cognitions, the association changed its name from Association for the Advancement of Behavior Therapy to Association for Behavioral and Cognitive Therapies in 2005.

participants) examining the mediators and moderators of treatment outcome; others thought more progress could be made by returning to idiographic approaches (to better understand change mechanisms at the level of the individual). Ultimately, this collection of ABCT past presidents suggested that much progress had been made on both theoretical and practical levels, yet there was broad consensus that much remained to be done.

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PART III

Research Contributions

CHAPTER 10

Statistical Considerations: Moderators and Mediators

Barbara Jandasek, Grayson N. Holmbeck, and Brigid M. Rose

In recent years, researchers who study pediatric populations have begun to posit complex theoretical models to explain developmental and psychological phenomena of interest. These models include longitudinal developmental pathways, risk and protective factors, and cognitive appraisal and coping processes (Thompson & Gustafson, 1996). This increase in theoretical complexity has necessitated an increase in terminological sophistication and the use of multivariate statistical strategies. Research based on mediational and moderational models has emerged as a critical method for testing competing theories about developmental pathways and other concepts central to clinical child and pediatric psychology. This chapter represents a compilation of the authors' previous works regarding mediation and moderation (Holmbeck, 1997, 2002; Rose, Holmbeck, Coakley, & Franks, 2004).

Specifically, the goal of this chapter is to provide an overview of mediators and moderators and their application to both research and practice in clinical child and pediatric psychology. First, definitions of each term and applied examples are provided, including more complex uses of these terms (e.g., moderated mediation, mediated moderation) and common definitional inconsistencies found in the literature. Next, statistical strategies used to evaluate mediator and moderator effects are reviewed (i.e., multiple regression and structural equation modeling techniques), including a presentation of strengths and limitations, common statistical inconsistencies, and applied statistical examples. Third, mediator and moderator effects in longitudinal research, intervention and outcome studies, and the study of risk and protective factors are discussed. Finally, we provide recommendations for applying mediational and moderational research to clinical practice.

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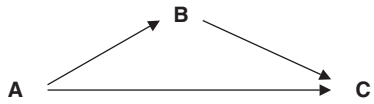

Figure 10.1 Mediated relationship among variables (A = Predictor; B = Mediator; C = Criterion/outcome).

DEFINITIONAL ISSUES

What Is a Mediator?

A mediator is an explanatory link in the relationship between two other variables (Figure 10.1; see Table 10.1 for definitions of all important terms). Often a mediator variable is conceptualized as the mechanism through which one variable (i.e., the predictor) influences another variable (i.e., the criterion; Baron & Kenny, 1986; Holmbeck, 1997, 2002). In other words, a mediator specifies how (or the mechanism by which) a given effect occurs (Baron & Kenny, 1986; James & Brett, 1984). Other terminology, such as *indirect effect* and *surrogate* or *intermediate endpoint effect,* is sometimes used interchangeably with mediation (MacKinnon, Lockwood, Hoffman, West, & Sheets, 2002), although these terms may not be conceptually identical (Holmbeck, 2002).

Suppose, hypothetically, that a researcher finds that parental intrusiveness is negatively associated with child adherence to a medical regimen. Although interesting, this finding does not tell us very much about processes that underlie the relationship between intrusiveness and adherence. By testing for mediational effects, a researcher can explore whether a third variable (e.g., child independence) might account for or explain the relationship between these variables. Continuing with the example, suppose that child independence mediates the

Table 10.1 Definition of Terms

Term	Definition
Mediator	The variable or mechanism by which a predictor influences an outcome variable (Figure 10.1).
Moderator	A variable that influences the strength or direction of a relationship between a predictor and an outcome (Figure 10.2).
Moderated mediation	A mediational model is significant only at certain levels of a moderator variable (Figure 10.3).
Mediated moderation	A moderational (or interaction) effect is mediated by another variable (Figure 10.4).
Protective factor	A variable that decreases the likelihood of a negative outcome under adverse conditions (Figure 10.7).
Resource factor	A variable that positively influences outcome, regardless of the presence of adversity (Figure 10.7).
Vulnerability factor	A variable that increases the likelihood of a poor outcome under adverse conditions (Figure 10.8).
Risk factor	A variable that negatively influences outcome regardless of the presence of adversity (Figure 10.8).

relationship between intrusiveness and adherence (more intrusive parenting → less child independence → less medical adherence). That is, parental intrusiveness impacts negatively on level of child independence, which in turn contributes to poor medical adherence (see Holmbeck, Shapera, & Hommeyer, 2002, for a similar example). In intervention studies, mediator effects may describe critical components of an intervention through which therapeutic change occurs (MacKinnon, Krull, & Lockwood, 2000, see later in chapter for more detail).

It is critical to note that a mediational model is, by its nature, a causal model. In other words, a mediational hypothesis is appropriate when one seeks to explain the causal process, or mechanism, by which an independent variable impacts a dependent variable (MacKinnon et al., 2000). Even if one's data do not permit causal conclusions (e.g., the data are cross-sectional and nonexperimental), the theory that underlies such a model is still inherently causal because it suggests a direction of influence.

One additional clarification about mediator variables is warranted. Typically, mediator mechanisms are proposed only after a significant association between the independent variable and the dependent variable has been fairly well established in the literature (i.e., indicating that there is, in fact, an effect to mediate). In other words, one does not typically begin study in a new research area by proposing mediational models. Such models are usually a natural extension of a well-established body of literature. Continuing with the earlier example, if one was interested in examining variables that mediate the link between parental intrusiveness and child medical adherence, one would want to be very confident that this link would be significant in one's own data set based on past research conducted on the variables.

What Is a Moderator?

Unlike a mediator, a moderator is a variable that influences the strength or the direction of a relationship between a predictor variable and a criterion variable (Figure 10.2). In other words, moderation describes an interaction effect whereby the impact of the independent variable on the dependent variable varies as a function of the level, or value, of the moderator variable. Moderator variables are frequently described as individual (e.g., sex, coping) or contextual (e.g., social support, family environment) characteristics (Grant et al., 2006). Suppose a researcher finds that familial stress is negatively associated with child psychological adjustment. Although this finding may be of interest to the researcher, it may be that the effect becomes more or less robust in the presence of other contextual variables. In fact, the researcher may develop specific theories about conditions that determine

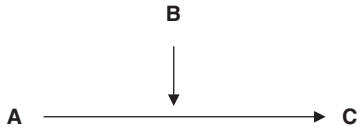

Figure 10.2 Moderated relationship among variables (A = Predictor; B = Moderator; C = Criterion/outcome).

the strength of the relationship between stress and adjustment. For example, the strength or the direction of the relationship between stress and adjustment may depend on the type of coping used by the family. That is, a significant association may emerge *only* when a child copes in a maladaptive manner. By testing coping style as a moderator of the relationship between stress and outcome, the researcher can specify certain conditions under which family stress predicts child adjustment. Such findings would likely have implications for future interventions.

Diagrammatic Distinction between Mediators and Moderators

Mediators and moderators can also be differentiated diagrammatically (see Figures 10.1 and 10.2; Baron & Kenny, 1986; J. Cohen & Cohen, 1983). A mediator (B; top of Figure 10.1) falls in the causal pathway between two variables (A and C in Figure 10.1; James & Brett, 1984). That is, if A is significantly associated with C, and A influences B, and B influences C, then B is a mediating variable between A and C. On the other hand, if A is expected to be related to C, but only under certain conditions of B, then B is a moderator variable. The moderator (B) can be drawn to indicate that it has an impact on the relationship between A and C (see Figure 10.2). Although some variables are more likely to be moderators than mediators (e.g., gender), some variables could serve either function—depending on the conceptual model under investigation (for examples where "coping strategies" were tested as both mediators and moderators in competing models, see Lewis & Kliewer, 1996; Sandler, Tein, & West, 1994). Moreover, both moderators and mediators can be specified within the same model (e.g., moderated mediation; James & Brett, 1984; for examples of this strategy, see Holmbeck, 1996; Simons, Lorenz, Wu, & Conger, 1993).

Applied Examples of Mediation and Moderation

An example will illustrate the distinction between moderated and mediated effects. This example is based on Kaczynski, Lindahl, Malik, and Laurenceau's (2006) study of marital conflict, parenting, and child adjustment. The authors hypothesized that maternal and paternal parenting would function as a mediator between marital functioning and adjustment difficulties (i.e., internalizing and externalizing symptoms) in children. In other words, marital conflict was expected to negatively impact the quality of parenting, which, in turn, would increase child maladjustment. Additionally, the authors tested a moderational effect: that parent gender would impact the nature of the relationship between marital conflict and disrupted parenting behavior. Specifically, this relationship was expected to be stronger for fathers than for mothers. Their findings fully supported the hypothesized mediational model. The moderational model was found to be valid as well, but only for families of boys.

As noted, the same variable can serve as either a mediator or a moderator (or both), depending on the research question. For example, Quittner (1992) provided a useful example of how a single construct (i.e., social support) could potentially serve as either a mediator *or* a moderator. She proposed two competing theoretical models in an attempt to study the role of social support in the relationship between parenting-related stress and psychological distress in parents of children

with hearing impairment and seizure disorder. She reasoned that parenting stress is more likely to have adverse effects on adjustment when parents have low levels of social support (i.e., social support is a moderator). On the other hand, it could also be that parenting stress undermines one's ability to garner social support, which in turn impacts on parental adjustment (i.e., social support is a mediator).

In her analyses, Quittner (1992) found that the relationship between parental stress and psychological distress did not vary as a function of level of social support. That is, social support did not *moderate* or alter the strength or direction of the relationship between parental stress and psychological distress. Having ruled out social support as a possible moderator, she tested social support as a mediator. Initially, she found that parenting stress (the predictor) was significantly associated with psychological distress (the outcome). Parenting stress was also a significant predictor of social support (mediator). Specifically, parents who experienced more stress tended to have fewer social contacts and to perceive themselves as less supported. Moreover, Quittner found a significant relationship between social support (the mediator) and psychological distress (the outcome), such that lower levels of social support were associated with higher distress ratings. When she tested the relationship between parental stress and psychological distress in the presence of this mediator, the strength of this previously significant relationship dropped significantly. She was able to conclude that the relationship between stress and psychological functioning was mediated by social support (parenting stress → social support → psychological distress) and that reduced social support may be one mechanism by which parental stress is linked with parental psychological distress.

Complex Models: Moderated Mediation and Mediated Moderation

Other types of theoretical models can be proposed that include both mediational and moderational effects. For example, with a moderated mediational model (Figure 10.3), a researcher can test whether a mediational model is significant only at certain levels of a moderator variable. Building on the earlier mediational example (i.e., parental intrusiveness → child independence → medical adherence), we might hypothesize that the entire mediational relationship will hold for only one gender and not the other. If this were found to be the case, this mediational model would be moderated by gender (i.e., moderated mediation). Similarly, mediated moderation is possible (Figure 10.4). For example, suppose that the association between intrusiveness and adherence is moderated by child gender (i.e., intrusiveness ×

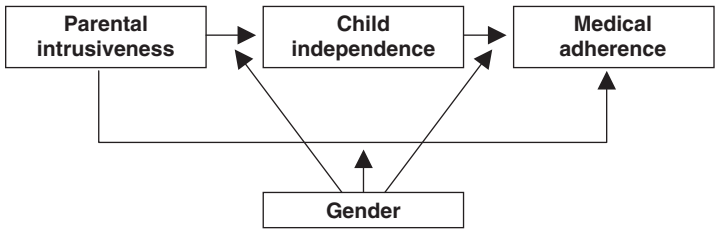

Figure 10.3 Moderated mediation.

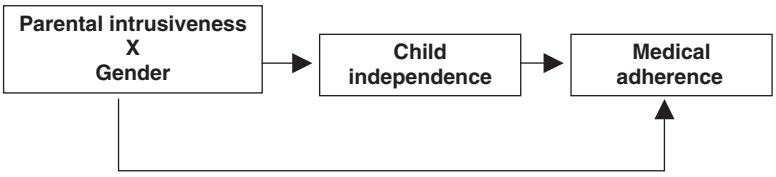

Figure 10.4 Mediated moderation.

gender → medical adherence; see later discussion regarding statistical methods for testing moderator effects as interaction terms). The investigator may then be interested in determining the mediational processes (e.g., child independence) that account for this significant moderational effect. One then tests whether the significant interaction (moderational) effect is mediated by child independence. Although statistically similar, it is important to note that moderated mediation and mediated moderation are not equivalent hypotheses when viewed conceptually. The former is based on the notion that an entire mediational model is significant only at certain levels of a moderator. The latter is based on the notion that a significant moderational effect is mediated by some variable of interest. The reader is referred to Muller and colleagues' (Mulleret, Judd, & Yzerbyt, 2005) article for a more detailed discussion regarding definitional issues and analytical procedures for mediated moderation and moderated mediation.

Common Definitional Inconsistencies

A lack of conceptual and statistical clarity in the study of mediated and moderated effects has become particularly prevalent in mental health literatures where investigators seek to examine factors that mediate or moderate associations between selected predictors and adjustment outcomes. In the child clinical and pediatric psychology literatures, for example, models of predictor-adjustment relationships have become quite complex (e.g., Grych & Fincham, 1990; Thompson, Gil, Burbach, Keith, & Kinney, 1993). Investigators working in these areas have found it necessary to invoke conceptual models that include mediated and moderated effects. Despite the appearance of several useful discussions of differences between mediated and moderated effects (e.g., Aldwin, 1994; Baron & Kenny, 1986; James & Brett, 1984), there continue to be inconsistencies in the use of these terms. More specifically, several types of definitional problems occur with some regularity: (a) vague or interchangeable use of the terms, (b) inconsistencies between terminology and the underlying conceptualization of the variables employed, and (c) a mismatch between written text and diagrammatic figures (see Holmbeck, 1997, for examples).

One concerning issue is that definitions used in the literature are not always consistent, in that they do not adhere to established statistical definitions (see Baron & Kenny, 1986) or the necessary A → B → C causal relationship. Although it is not the case that all investigators must adhere to the same definitions of all terms, it is likely that progress in the field will be hampered if the same term is used in different ways by different scholars. Research studies also often confuse or inappropriately

interchange the terms "mediation" and "moderation" (Holmbeck, 1997). Baron and Kenny provide some examples of this problem from the social psychology literature. For example, some research has referred to variables such as age of onset and socioeconomic status (SES) as mediators. However, it is likely that the effect described is moderational in nature. For example, variables such as age of onset and SES presumably dictate conditions under which a stressor is (or is not) associated with problematic outcomes (i.e., these variables are more likely to serve a moderational than a mediational role). Studies sometimes refer erroneously to an interaction effect as a mediation effect (Holmbeck, 1997). The opposite problem is also possible, where authors describe a moderational effect as mediation without making a clear case for how their variables could serve a mediational function. To be considered a mediational effect, the conceptual relationships between the variables of interest must adhere to a mediational model (A → B → C).

One potential cause for confusion regarding mediational models lies in the distinction between *temporal* and *causal* models (Holmbeck, 1997). For example, variables such as coping and appraisal are often misrepresented as mediators. This error is likely attributable to the use of models illustrating *temporal* pathways (see Thompson & Gustafson, 1996, for several examples of temporally oriented models). For example, one might provide a flowchart whereby a stressor is followed *in time* by various coping processes that are, in turn, followed by changes in psychosocial adjustment (i.e., stressor → coping → adjustment). Flowcharts like these are temporal (rather than mediational) in nature because they specify a time-based sequence of events rather than a causal sequence of events. For example, this type of model could illustrate how a stressor, a coping process, and changes in psychosocial adjustment are all placed *in time* relative to each other. The model becomes causal (and mediational) only if the investigator posits causal associations among the variables. The model just discussed would become mediational if the investigator hypothesized that the stressor actually impacted on or changed the way a person copes (e.g., the stressor causes a person to use more avoidance coping), which, in turn, impacted on subsequent levels of adjustment. Because researchers do not usually expect a stressor to influence or cause changes in the style with which a person copes, coping is not usually viewed as a mediational variable (although it *is* often viewed as a moderational variable).

Frequently, terminological, conceptual, and statistical errors are all present in the same study, such as when investigators conceptualize a variable as a moderator (e.g., coping strategies are hypothesized to serve a protective or buffering function), use the term mediator (rather than moderator) to describe the impact of the variable, provide a figure where the variable is presented as a mediator (rather than as a moderator), and conduct statistical analyses that test neither mediation nor moderation. When such a mismatch between terminology, theory, figures, and statistical analyses exists, findings become particularly difficult to interpret.

STATISTICAL STRATEGIES

Now that the conceptual differences between mediators and moderators have been clarified and their potential application within research has been addressed, the next

portion of the chapter is devoted to a more complete discussion of how mediational and moderational effects are tested statistically. For a more detailed overview of statistical methods and post-hoc probing (introduced later), as well as Statistical Package for the Social Sciences (SPSS) syntax, the reader is referred to relevant papers by Holmbeck (1997, 2002). This chapter focuses primarily on two types of statistical strategies: multiple regression (as suggested by Baron & Kenny, 1986, and as used by several investigators) and structural equation modeling (SEM; see Tabachnick & Fidell, 2007, for a relatively straightforward discussion; also see Bollen, 1989; Byrne, 1994; Hoyle, 1995; Jaccard & Wan, 1996; Joreskog and Yang, 1996; Mueller, 1996). Although SEM is often considered the preferred method because of the information it provides on the degree of fit for the entire model after controlling for measurement error (Peyrot, 1996), proper use of regression techniques can also provide meaningful tests of hypotheses. Moreover, for investigators working in areas of study where sample *N*s are often relatively small, such as pediatric psychology, use of regression techniques (as opposed to SEM) may be necessary because of power considerations (see Tabachnick & Fidell, 2007, for a discussion of sample size and SEM).

Statistical Approaches for Testing Mediation

This section addresses mediational analyses. Techniques utilizing both regression and SEM are presented, with a focus on the importance of evaluating the potency of a mediator. Criticisms of and alternatives to common approaches used in testing mediation are discussed, followed by an applied statistical example.

Regression Approach to Testing Mediated Effects

According to Baron and Kenny (1986), four conditions must be met for a variable to be considered a mediator: (1) The predictor, A, must be significantly associated with the hypothesized mediator, B (letters refer to variables in Figure 10.1); (2) the predictor, A, must be significantly associated with the dependent measure, C; (3) the mediator, B, must be significantly associated with the dependent variable, C, after controlling for the effect of A; and (4) the impact of the predictor, A, on the dependent measure, C, is less after controlling for the mediator, B. A corollary of the second condition is that there first has to be a significant relationship between the predictor and the dependent variable for a mediator to serve its mediating role. In other words, if A and C are not significantly associated, there is no significant effect to mediate. Such a bivariate association between A and C is not required in the case of moderated effects (nor is it required in the case on an indirect effect, as discussed later).

These four conditions can be tested with three multiple regression analyses (see Eckenrode, Rowe, Laird, & Brathwaite, 1995, for an example that includes figures as well as a complete explanation of this data analytic strategy). This strategy is similar to that employed when conducting a path analysis (J. Cohen & Cohen, 1983; Nie, Hull, Jenkins, Steinbrenner, & Bent, 1975). The significance of the A $\rightarrow$ B path (condition 1 above) is examined in the first regression, after controlling for any covariates. The significance of the A $\rightarrow$ C path (condition 2) is examined in

the second regression. Finally, A and C are employed as simultaneous predictors in the third equation, where C is the dependent variable. Baron and Kenny (1986) recommend using simultaneous entry (rather than hierarchical entry) in this third equation so that the effect of B on C is examined after controlling for A, and the effect of A on C is examined after controlling for B (borrowing from path analytic methodology; Nie et al., 1975). The significance of the B → C path in this third equation is a test of condition 3. The relative effect of A on C in this equation (when B is controlled), in comparison to the effect of A on C in the second equation (when B is not controlled), is the test of condition 4. Specifically, A should be less highly associated with C in the third equation than was the case in the second equation (see later discussion for a more detailed example). In other words, one is interested in determining whether the predictor → outcome effect is less after controlling for the mediator. The question that arises in this type of analysis is How much reduction in the total effect is necessary to claim the presence of mediation? The degree to which the effect is reduced is an indicator of the potency of the mediator. As Baron and Kenny discuss, it would be unusual in psychology for this A → C effect to be reduced from significance to zero.

In the past, some researchers have reported whether the predictor → outcome effect drops from significance (e.g., $p < .05$) to nonsignificance (e.g., $p > .05$) after the mediator is introduced into the model. This strategy is flawed, however, because a drop from significance to nonsignificance may occur, for example, when a regression coefficient drops from .28 to .27 but may not occur when the coefficient drops from .75 to .35 (Holmbeck, 2002). In other words, it is possible that significant mediation *has not* occurred when the test of the predictor → outcome effect drops from significance to nonsignificance after taking the mediator into account. On the other hand, it is also possible that significant mediation *has* occurred even when the statistical test of the predictor → outcome effect continues to be significant after taking the mediator into account. Clearly, a test is needed for the significance of this drop. This significance in the drop of the total effect on inclusion of the mediator in the model is computed using the Sobel test (Sobel, 1988).

Failure to test the significance of a mediated effect is likely to lead to false conclusions. One risks both false-positive and false-negative conclusions when one fails to test the significance of a mediated effect (Holmbeck, 2002). Suppose one tests the utility of a mediational model and seeks to determine whether the significance of the total effect drops to nonsignificance after the mediator is taken into account. Also suppose that this "drop to nonsignificance" criterion is used as the basis for whether significant mediation has taken place. If an initial total effect was just below the $p < .05$ threshold of significance (e.g., $p = .049$) and then dropped so that the significance level was now just above the $p < .05$ threshold (e.g., $p = .061$), one might conclude that significant mediation has occurred. Upon further analysis (using the strategy employed here), however, one may find that this represents a false-positive conclusion. On the other hand, if the original total effect was well under the significance threshold (e.g., $p = .001$) and remained under the threshold (e.g., $p = .040$) after accounting for the mediator, one might conclude that no mediation occurred (which would likely be a false-negative conclusion). Indeed, analyses using the Sobel equation may reveal that significant mediation

had occurred in this latter case. As noted earlier, the "drop to nonsignificance" criterion is flawed. It is also worth mentioning that there may be other reasons for false-negative conclusions (i.e., Type II errors). For example, there may be limited power to detect an effect, or the measures may be weak (e.g., they may have low reliability).

The reader is referred to an article by Sobel (1988, p. 56) that includes a complete explanation of how to apply the significance test and compute confidence intervals for the indirect effect (for empirical examples, see also Colder, Chassin, Stice, & Curran, 1997; Ireys, Werthamer-Larsson, Kolodner, & Gross, 1994; Lustig, Ireys, Sills, & Walsh, 1996). Holmbeck (2002) provides a detailed explanation, in addition to a computational example. Finally, an interactive website is available that conducts the Sobel test (with significance tests) if path coefficients and standard errors are entered (www.psych.ku.edu/preacher/sobel/sobel.htm).

Structural Equation Modeling Approach to Testing Mediated Effects

The SEM strategy is often preferred when the investigator has more than one measured variable for each of the constructs (or latent variables) assessed. Assuming there is a latent predictor variable (A), a hypothesized latent mediator variable (B), and a latent outcome variable (C), one would first assess the fit of the direct effect (A → C) model (Hoyle & Smith, 1994). Assuming an adequate fit, the investigator then tests the fit of the overall A → B → C model. Assuming the overall model provides an adequate fit, the A → B and B → C path coefficients are examined. At this point, the A → C, A → B, and B → C paths (as well as the A → B → C model) should all be significant (which is analogous to the regression strategy discussed earlier).

The final step in assessing whether there is a mediational effect is to assess the fit of the A → B → C model under two conditions (where the first condition is nested within the second): (1) when the A → C path is constrained to 0, and (2) when the A → C path is not constrained. One then examines whether the second model provides a significant improvement in fit over the first model. As noted earlier, improvement in fit is assessed with a significance test based on the difference between the two model chi-squares. If there is a mediational effect, the addition of the A → C path to the constrained model should not improve the fit. In other words, the previously significant A → C path is reduced to nonsignificance (i.e., it does not improve the fit of the model) when the mediator is taken into account (which is again analogous to the regression approach).

An additional consideration in using SEM to test for mediational effects is the important distinction between *indirect* and *mediated* effects. In this case, we define an indirect effect as a mediational effect where the A → C path is not significant (i.e., there is no effect to mediate, and only the A → B and B → C paths are significant). An example is used to highlight this distinction. Capaldi, Crosby, and Clark (1996) recently conducted a longitudinal study using structural equation modeling, where they concluded that the effect of aggression in the family of origin on aggression in young adult intimate relationships was mediated by the level of boys' antisocial behaviors during adolescence. On the other hand, Capaldi and her colleagues appear

to have found that the direct path between the predictor and the criterion was not significant even though the predictor-mediator and mediator-criterion paths were significant. Although there appears to be evidence for an indirect effect between predictor and criterion, the findings suggest that the mediator does not (and cannot) significantly account for the predictor-criterion relationship (because there was not a significant relationship between predictor and criterion in the first place; Hoyle & Smith, 1994). Thus, Capaldi et al.'s findings fit the criteria for an indirect effect but do not fit the criteria for a mediated effect (as they have been defined here). In the case of such an indirect effect, one must be conservative when discussing interpretations of links between predictor and criterion because one cannot claim that the predictor and the criterion are significantly associated.

Criticisms and Alternative Approaches for Testing Mediation

MacKinnon (MacKinnon et al., 2002; see David MacKinnon's website: www.public.asu.edu/~davidpm/ripl/mediate.htm) has argued recently that the regression approach to testing mediation may be overly conservative, with low power and inaccurate Type I error rates. In addition, MacKinnon and colleagues (2002, p. 87) criticize the Baron and Kenny (1986) approach, commenting that it does "not provide the full set of conditions for the strong inference of a causal effect of the independent variable on the dependent variable through the intervening variable." He and others are currently considering alternative methods in testing the significance of indirect effects (MacKinnon et al., 2002; Shrout & Bolger, 2002; also see David Kenny's website: http://davidakenny.net/cm/mediate.htm).

Some statisticians have argued that the Baron and Kenny (1986) approach, although appropriate in some cases, is overused (Spencer, Zanna, & Fong, 2005). These authors articulate several disadvantages of this approach, including that it (a) requires that the constructs of interest be easily measurable and that their measurement not interfere with the underlying processes (e.g., avoidance of priming effects); (b) relies on correlational, as opposed to causal, evidence; and (c) must adhere to statistical assumptions of multiple regression, which can, at times, be difficult to fulfill. Consideration of experimental designs to support mediational effects is suggested as an alternative to more traditional approaches, especially in cases where the constructs, or processes of interest, are easily manipulated.

Applied Statistical Example of Mediation

In a study of parental overprotection and adjustment in children with spina bifida, Holmbeck and colleagues (Holmbeck, Johnson, et al., 2002) found that parents of children with spina bifida tended to be more overprotective than parents of typically developing comparison children. The authors reasoned that group-related differences in the children's cognitive ability might explain or mediate the observed group differences in parenting style because developmental and intellectual delays might elicit more protective behavior from their parents. Specifically, it was hypothesized that level of cognitive functioning would mediate the relationship between group status and parental overprotective behavior (Figure 10.5).

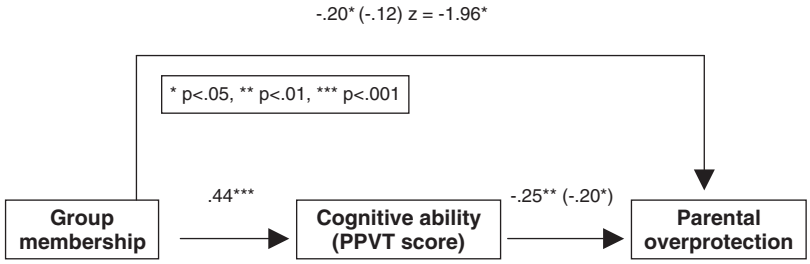

Figure 10.5 Mediational model: Cognitive ability as a mediator of the relationship between group status (spina bifida versus comparison sample) and parental overprotection.

As discussed previously, a mediational model is supported when four statistical criteria are met: (1) The predictor variable is significantly associated with the criterion outcome variable; (2) the predictor variable is significantly associated with the mediator; (3) the mediator is significantly associated with the outcome variable, after controlling for the predictor; and (4) the previously significant predictor → outcome relationship is significantly diminished when effects of the mediator are controlled (Barron & Kenny, 1986; Holmbeck, 1997). These four conditions can be tested with three regression equations (with steps 3 and 4 being tested with a single regression analysis; Holmbeck, 1997, 2002).

In the current example, the first step involved demonstrating that the predictor variable, group status (i.e., spina bifida versus comparison sample), was significantly associated with the outcome variable, overprotection. If the predictor and outcome variables were not found to be significantly associated, there would be no effect to mediate and further testing would not be necessary. In support of their hypothesis, Holmbeck and colleagues (Holmbeck et al., Holmbeck, Johnson, et al., 2002) found that group status significantly predicted parental overprotection, as measured by self-report questionnaires and observer ratings, so they proceeded to test condition 2 (the relationship between the predictor and the mediator). As expected, the second regression showed that the predictor variable (group status) was significantly associated with cognitive ability (as measured by the Peabody Picture Vocabulary Test—Revised; PPVT-R; Dunn & Dunn, 1981), thus supporting the second criterion for mediation.

Regarding the third and fourth mediational criteria, it was necessary to show that PPVT scores significantly predicted parental overprotection when group status was held constant (condition 3) and that the strength of the group status → overprotection effect dropped significantly when the mediator was controlled (condition 4). This was achieved by using simultaneous entry (rather than hierarchical entry) to enter group status and PPVT scores as predictors with parental overprotection as the dependent variable. The results of the analyses provided the information necessary to confirm the mediational effect. First, it was observed that the mediator (PPVT) was a significant predictor of overprotection, with group status controlled (condition 3). Second, in support of condition 4, it was shown that the relationship between group status and overprotection dropped significantly in this last analysis (when PPVT scores were controlled), as compared with the strength of the

relationship observed in the first regression model (when PPVT scores were not controlled). The significance of the drop was tested using Sobel's equation (Figure 10.5; Holmbeck, 2002; Sobel, 1988). This suggests that the mediator (PPVT score) accounted for a significant portion of the variance in the relationship between the predictor (group status) and the outcome (overprotection).

Statistical Approaches for Testing Moderation

Although statistical approaches used to test moderational effects primarily rely on both regression and SEM, this section focuses on regression. The use of various SEM techniques to evaluate moderation has demonstrated significant progress in the past several years, and a discussion of these approaches is, therefore, beyond the scope of this chapter. The reader should be aware that analytic strategies utilizing SEM may be advantageous when one has multiple indicators for the latent variables under investigation. Additionally, SEM analyses are generally less vulnerable to measurement error compared to those utilizing regression (Jaccard & Wan, 1995; Peyrot, 1996; Ping, 1996). For more information regarding SEM approaches used to test moderation, the reader is referred to articles by Moulder and Algina (2002) and Schumaker (2002).

Regression Approach to Testing Moderated Effects

Although the manner in which moderators are tested statistically varies somewhat depending on whether the predictor and/or moderator are continuous or dichotomous (Baron & Kenny, 1986; Mason Tu, & Cauce, 1996), the general strategy is the same regardless of the nature of the variables involved. As noted earlier, a moderator effect is an interaction effect. The preferred strategy is to use the variables in their continuous form (if they are not dichotomies) and employ multiple regression techniques (J. Cohen & Cohen, 1983; S. Cohen & Wills, 1985; Jaccard, Turrisi, & Wan, 1990; James & Brett, 1984; Mason et al., 1996).

The predictor and moderator main effects (and any covariates, if applicable) are entered into the regression equation first, followed by the interaction of the predictor and the moderator (e.g., Fuhrman & Holmbeck, 1995). Depending on the investigator's conceptual framework, the main effects can be entered in a hierarchical, stepwise, or simultaneous fashion (J. Cohen & Cohen, 1983). For example, in analyses involving marital conflict as a predictor and family structure as a moderator, marital conflict and family structure could be entered in any order or simultaneously. The interaction term is represented by the product of the two main effects (e.g., marital conflict $\times$ family structure) and "only becomes the interaction when its constituent elements are partialled" (J. Cohen & Cohen, 1983, p. 305; also see Aiken & West, 1991; Evans, 1991; Friedrich, 1982; Holmbeck, 1989). Thus, although the main effects may be entered in any order, they must be entered prior to the interaction term for the product of these two terms to represent the interaction when it enters the equation.

Given the manner in which the interaction is computed, the main effects (i.e., the predictor and the moderator) will be highly correlated with the interaction term, which can produce "ill-conditioning" error messages when using some statistical

software packages. To eliminate problematic multicollinearity effects between first-order terms (i.e., the independent variable and the moderator) and the higher order terms (i.e., the interaction terms), Aiken and West (1991) recommend that the independent variable and the moderator be "centered" prior to testing the significance of the interaction term. To center a variable, scores are put into deviation score form by simply subtracting the sample mean from all individuals' scores on the variable, thus producing a revised sample mean of zero. Such transformations have no impact on the level of significance of the interaction terms or the simple slopes of any plotted regression lines (see Holmbeck, 2002).

Post-Hoc Probing of Moderational Effects

Failure to conduct post-hoc probing with moderational effects may lead to false conclusions. When one tests for the presence of a moderational effect with multiple regression, one examines whether an interaction between two variables (one independent variable and a moderator) is a significant predictor of an outcome variable, after controlling for the effect of the two predictors. The presence of a significant interaction tells us that there is significant moderation (i.e., that the association between the predictor and the outcome is significantly different across levels of the moderator or that the association is conditional on values of the moderator), but tells us little about the specific conditions that dictate whether or not the predictor is significantly related to the outcome.

For example, if one was interested in whether the association between a parenting variable (e.g., father psychological control; Holmbeck, Shapera, et al., 2002) and an outcome (e.g., school grades) is moderated by group status (e.g., spina bifida versus an able-bodied comparison sample), one would test the interaction of psychological control and group as a predictor of school grades after controlling for the parenting and group main effects. If the interaction is significant, this tells us that the slope of the regression line (i.e., simple slope) that represents the association between parenting and grades for the spina bifida sample is significantly different from the slope for the comparison sample. Unfortunately, the significance of the interaction effect does *not* tell us whether either of the simple slopes is significantly different from zero. In other words, we do not know, based on the initial significant interaction effect, whether the relationship between parenting and grades is significant for the spina bifida sample, the comparison sample, or both samples. Post-hoc probing of the interaction effect via computation of the simple slopes with statistical tests is necessary to gain this information (see Holmbeck, 2002, for computational examples). Additionally, such information also facilitates the plotting of regression lines in figure form.

If an investigator is examining moderational effects by testing the significance of interaction terms, it is likely that the investigator has hypothesized previously that the impact of a predictor on an outcome is conditional on the level of a moderator variable. Suppose one has predicted that "A" will be related to "B" for males, but not for females. A significant "A $\times$ gender" interaction effect only tells you that A is related to B differentially as a function of gender; unfortunately, this statistical test does not answer the research question of interest. Only the post-hoc

probing procedure will tell you if A is significantly associated with B for males, but not for females. In the past, some investigators have merely plotted significant interaction effects and interpreted the significance of regression line slopes based on visual inspection, without conducting post-hoc probes. This strategy is likely to lead to false-positive results; one may be more likely to conclude that a slope is significantly different from zero based on eyeballing than via statistical tests. Thus, post-hoc probing is a critical step in the evaluation of a moderator effect.

One might also be tempted to employ post-hoc probing strategies that differ from those suggested here. For example, if one has isolated a significant interaction effect between a dichotomous variable and a continuous variable, one might choose to examine the bivariate correlation between the continuous predictor and the outcome at each level of the dichotomous moderator. Similarly, if one had found an interaction between two continuous variables, one might be tempted to examine the bivariate correlation between one of the continuous predictors and the outcome at high and low levels (usually based on a median split) of the other continuous variable.

Although this bivariate correlation approach is superior to doing no post-hoc probing, this strategy is less desirable for several reasons. First, the bivariate correlation strategy does not provide the investigator with a regression equation. Without such an equation, the plotting of findings is not a straightforward task. Second, when one generates regression line equations with slopes and intercepts, the slope is in the same metric as the outcome. Given the slope, one is able to determine the increase (or decrease) that will occur in the value of the outcome as a function of a 1 unit increase (or decrease) in the predictor (at a particular level of the moderator). Third, by computing the regression equations one can determine mathematically where the regression lines cross (see Aiken & West, 1991, pp. 23–24), which may be of practical or theoretical interest. Finally, the post-hoc strategy discussed here allows for greater flexibility in computing and plotting regression lines. In the case of an interaction between two continuous variables, one can use the + 1 *SD* convention *or* a variety of other values. When using the bivariate correlation strategy, one typically uses only the median split approach. An additional drawback of the median split strategy is that it yields a correlation for a fairly diverse subsample of participants (i.e., the association between the predictor and the outcome for all individuals above or below the median on the continuous moderator). The post-hoc strategy discussed in this chapter allows one to examine associations between predictor and outcome at any possible value of the moderator.

Applied Statistical Example of Moderation

The following example demonstrates how a moderational effect can be tested using multiple regression analyses. In a conference presentation drawn from the data set described earlier, the authors investigated the moderational influence of cognitive ability on the longitudinal relationship between early parental psychological control and adjustment in children with spina bifida 2 years later (Rose, Buck, Hagstrom, Skelnik, & Stafiej, 2002). Psychological control, or the use of guilt induction and coercive behavior as methods of influencing a child's behavior, has been shown to have detrimental effects on child adjustment in many developmental contexts.

However, it was hypothesized that psychological control might be less harmful in the context of spina bifida because children with greater physical and/or cognitive impairment tend to spend more time at home with their family and tend to be more dependent on their parents for assistance with basic needs. As a result, it is possible that they are less distressed by more intense parental involvement. Moreover, the authors predicted that intellectual ability (as approximated by PPVT score) would moderate the relationship between psychological control and adjustment, such that psychological control would be less strongly predictive of psychological maladjustment (as measured by the Children's Depression Inventory, CDI; Kovacs, 1992) in children who were more cognitively impaired. At the most complex level, it was hypothesized that associations between psychological control and adjustment would be the least strong among those with spina bifida *and* those with low IQ and most strong among those in an able-bodied comparison sample *and* those with high IQ. Thus, 3-way interactions (i.e., moderated moderation) among psychological control, group status, and PPVT scores were tested.

To demonstrate a moderational effect using multiple regression, the main effects and interaction effects of the predictor variables (i.e., group, psychological control, PPVT score) on the dependent variable (CDI score) were analyzed. Before the regression analyses were run, all continuous predictor variables (e.g., IQ) were centered around zero to prevent multicollinearity among the predictors and the interaction terms in the equation and to allow for proper testing of simple slopes. To do this, each predictor variable was transformed by subtracting the sample mean for that variable from all individual scores. Dichotomous variables (e.g., group) were recoded as 0 or 1. To allow for the testing of interaction terms, new variables were created consisting of all possible 2-way and 3-way products of the centered and/or dichotomous predictor variables. These new variables were then tested as interaction terms in the regression analyses.

Once these steps were complete, all predictor variables (i.e., group, psychological control, and PPVT) were entered in the first step of the regression equation to test for main effects. All possible 2-way products of the predictors (i.e., group $\times$ psychological control, group $\times$ PPVT, psychological control $\times$ PPVT) were entered on the second step. Finally, to test the 3-way interaction (in this case, group $\times$ psychological control $\times$ PPVT), all possible 3-way products of the predictors were entered in the third step.

In the current example, contrary to expectations, there were no significant 2-way interactions between PPVT and psychological control. In other words, PPVT did not appear to moderate the relationship between parental psychological control and child depression across the entire sample. The other 2-way interactions were also nonsignificant. However, there was a significant 3-way interaction (group $\times$ PPVT $\times$ psychological control), indicating that there was a significant difference in the interactions among the predictor variables (PPVT $\times$ psychological control) for the two groups. Once significant interactions were identified, simple regression slopes representing the predictor $\rightarrow$ outcome relationship at high and low levels of the two moderator variables were plotted and tested for significance.

Probing of the 3-way interaction revealed that the interaction between psychological control and PPVT in the prediction of depression varied by subsample.

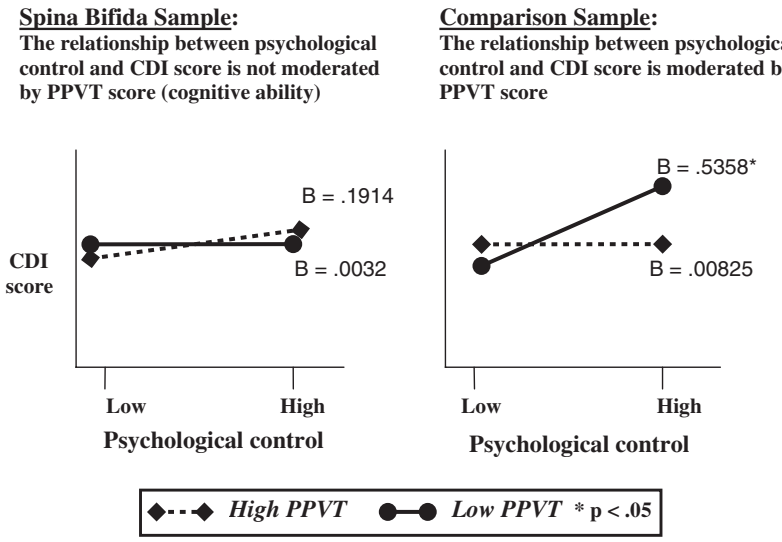

Figure 10.6 Three-way moderational model: In children with spina bifida there was no relationship between parental psychological control and child depressive symptoms. However, in the comparison sample, the relationship between psychological control and depressive symptoms was moderated by cognitive ability of the child (i.e., the relationship was significantly positive among children with relatively low PPVT-R scores).

Among children with spina bifida, the association between psychological control and depression was not significantly moderated by PPVT (Figure 10.6). In fact, psychological control and depression were not significantly related in this condition. However, among children in the comparison sample, the interaction between psychological control and PPVT was significant (Figure 10.6). Consistent with the hypotheses, PPVT scores appeared to influence or moderate the relationship between psychological control and depression in this group. On the other hand, and contrary to the hypotheses, tests of the simple slopes at two levels of the moderator (high PPVT and low PPVT) showed that there was a significant positive relationship between psychological control and depression among children with *low* PPVT scores. Among children with *higher* PPVT scores, the relationship between psychological control and depression was nonsignificant. Thus, it was concluded that the effects of psychological control on depression were moderated by condition status (i.e., psychological control was not associated with depression in children with spina bifida) and by level of intellectual functioning (among comparison children, higher intellectual ability appeared to protect against the effects of psychological control on depression).

Common Statistical Inconsistencies

In addition to the challenges already mentioned regarding inconsistent use of definitional terms in the literature, problems involving statistical testing of mediation and moderation also appear. These concerns fall into three general categories:

(1) use of data analytic procedures that fail to test for mediated and moderated effects when one of these effects is hypothesized, (2) a mismatch between terminology and statistical analyses (mediation is hypothesized, but moderation is tested), and (3) a lack of clarity in discussing implications of statistical results. These challenges often result in varying and discrepant interpretations of findings (see Holmbeck, 1997, for a detailed discussion and examples).

Inappropriate application of statistical strategies often occurs when a variable is included in a model that represents a response to another variable in the model. This issue is likely related to the use of temporal flowcharts that are subsequently interpreted as mediational models (see earlier discussion regarding diagrammatic inconsistencies). Variables such as coping strategies, cognitive appraisals, and causal attributions cannot exist in isolation; they exist only in relation to variables that have preceded them (e.g., marital conflict, a chronic illness). One cannot exhibit a coping strategy in response to marital conflict, for example, if there is no marital conflict in the first place. Some have also argued that such response variables are the mechanism through which the independent variable influences the dependent variable and are therefore best thought of as mediators. In many investigations, such variables are included in a box which is placed, connected by arrows, between antecedent (e.g., stress) and outcome (e.g., adjustment) variables.

Investigators who employ variables such as coping strategies and cognitive appraisals as mediators rarely provide a complete rationale for how these variables could serve a mediational function (as has been defined in this chapter). To do so, they would need to select a specific coping strategy (e.g., denial) and propose how such a coping strategy is expected to be employed with greater (or lesser) frequency when there are higher (or lower) levels of some illness parameter (e.g., severity of illness; Frese, 1986; A $\rightarrow$ B in Figure 10.1). They would also need to propose that higher (or lower) rates of certain maternal adjustment outcomes are expected when this particular coping strategy is employed with greater (or lesser) frequency (B $\rightarrow$ C in Figure 10.1). Finally, they would need to propose that the illness parameters are expected to be associated with the maternal adjustment outcomes (A $\rightarrow$ C in Figure 10.1).

RESEARCH APPLICATIONS IN CLINICAL CHILD PSYCHOLOGY

Now that mediators and moderators have been defined and differentiated and statistical approaches have been discussed, the next portion of this chapter focuses on the relevance and importance of mediator and moderator analyses in clinical child and pediatric psychology research. Interestingly, the study of mediators and moderators can be applied to a wide range of research questions. Specifically, mediational and moderational processes can be proposed in correlational and regression-oriented predictive utility studies, group differences research, longitudinal investigations, complex model testing (with structural equation modeling), studies of interventions including randomized clinical trials, and temporal relationship models in which "What precedes what?" questions are of interest. In this section, we discuss the application of mediation and moderation across three types of research

studies: longitudinal studies, risk and protective factor research, and intervention trials.

Longitudinal Research

Longitudinal mediational analyses allow researchers to test specific causal theories about time-ordered relationships among variables and the particular mechanism or pathway by which a relationship occurs. Such analyses differ from standard mediational analyses only insofar as they include variables gathered at different points in time. For example, information gathered using longitudinal data can advance what is known about the development of health behaviors, psychopathology, and problem behavior by allowing researchers to test the degree to which more immediate or proximal etiological factors can explain links between more remote or distal risk factors and adjustment, thereby contributing to intervention and prevention efforts (Weisz & Hawley, 2002). Furthermore, longitudinal analyses allow for a life span developmental perspective, which is helpful in understanding variability in adjustment trajectories (Cicchetti & Rogosch, 2002). In recent years, longitudinal mediational analyses have been used creatively to explain developmental pathways to resilience or maladjustment, beginning in early childhood and continuing through late adolescence (Masten et al., 1999). Knowledge of high-risk time points or critical transitional periods in child development may also facilitate the development of more effective treatments and interventions (a point discussed in more detail later; Weisz & Hawley, 2002).

Unlike research that is based solely on cross-sectional data, the use of mediational models with multiple data points enables the researcher to clarify the temporal sequencing of causal processes by ruling out bidirectional pathways (Holmbeck, 1997). Continuing with the mediational example discussed earlier, an investigator can propose a longitudinal hypothesis by arguing that high levels of parental intrusiveness are likely to be associated with *subsequent* decreases in child independence. The researcher might also hypothesize that low levels of child independence are likely to be associated with *subsequent* decreases in medical adherence. Longitudinal data would not only permit an examination of these hypotheses but would also allow the investigator to rule out the possibility that lower levels of child independence or lower levels of adherence are associated with *subsequent* increases in parental intrusiveness (an equally plausible scenario with a causal direction opposite from that proposed in the original model). Such hypotheses could not be tested with cross-sectional data.

At a minimum, two data points are needed to begin testing causal time-ordered mediational hypotheses (although three or more are preferred; Cole & Maxwell, 2003). To test the predictor → mediator link, initial Time 1 data are used for the predictor and Time 2 data are used for the mediator *after* controlling for Time 1 mediator data. More generally, partialing out the Time 1 dependent variable data from the Time 2 dependent variable data essentially converts the Time 2 dependent measure into a residual change variable. That is, a given participant's residual score represents the change in rank for this participant (relative to the other participants in the sample) on the dependent measure between Time 1 and Time 2. Similarly,

when examining the mediator → outcome link, initial Time 1 data are used for the mediator and Time 2 data are used for the criterion outcome, after controlling for the Time 1 data on the criterion outcome (Cole & Maxwell, 2003). One continues in this way when testing all pathways in the mediational model. It is important to note that if one has 3 waves of data, one could use Time 1 data for the predictor, Time 2 data for the mediator (controlling for Time 1 mediator data), and Time 3 data for the criterion outcome (controlling for Time 2 criterion outcome data).

Latent growth curve modeling is another statistical method by which to evaluate mediational processes using longitudinal data. Cheong, MacKinnon, and Khoo (2003) describe this type of mediational analysis as measuring two distinct parallel processes: (1) *growth* in the mediator and (2) concurrent *growth* in the outcome variable. This approach builds on the traditional A → B → C mediational model, in that the independent variable affects the growth (or trajectory) of the mediator, which in turn impacts the growth (or trajectory) of the outcome variable. In their study, Cheong and colleagues evaluated the effects of a prevention program geared toward reducing drug use and improving health behaviors among high school football players. Their results supported a longitudinal mediational model; exposure to the prevention program was found to increase perceived importance of team leaders over time, which, in turn, resulted in improvement in nutrition behaviors over time. Although this approach requires large sample sizes and multiple data points, it holds several advantages: It (a) allows for individual differences in development, (b) measures the impact of change in the mediator on *change* in the outcome, and (c) is able to model various trajectory shapes, including curvilinear trajectories (Cheong et al., 2003).

Longitudinal data can also be useful for testing moderator effects within a longitudinal or developmental framework. Using Cole and Maxwell's (2003) approach described earlier, it is necessary to control for earlier levels of the outcome variable when testing the relationship between a predictor (or a predictor-moderator interaction) and a later outcome. MacKinnon and Lockwood (2003) describe longitudinal moderational analyses that parallel Cheong and colleagues' (2003) approach to longitudinal mediation. These authors utilize growth modeling techniques to capture moderational effects on trajectories of change, or growth, over time. For example, one could test whether the impact of peer relationships on growth in deviant behavior during adolescence varies based on ethnicity.

Risk and Protective Factor Research

The study of risk and protective factors often implies mediational and moderational processes. Generally, research on risk and protective factors is focused on understanding the adjustment of youth who are exposed to varying levels of adversity. There is evidence that both contextual factors (e.g., SES, family-level functioning, peer relationships) and developmental variables (e.g., cognitive skills, autonomy development) can significantly influence outcomes for individuals living under adverse conditions, and thus serve a moderational role (Coaklay, Holmbeck, Friedman, Greenley, & Thill, 2002; Holmbeck et al., 2003; Holmes, Yu, & Frentz, 1999; Masten et al., 1999; Rutter, 1990). Risk and protective processes have been

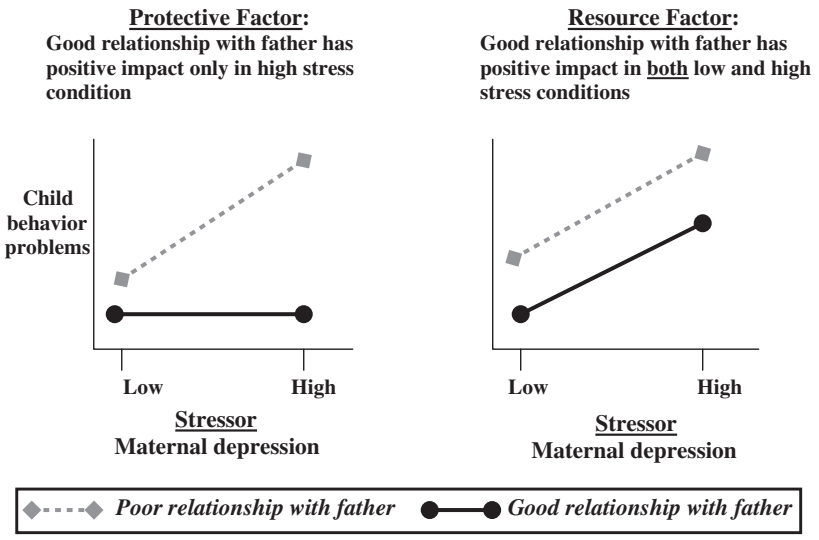

Figure 10.7 Factors associated with favorable outcomes: Protective and resource factors.

explored in the study of resilience, a term used with increasing frequency. Resilience refers to the process by which youth successfully navigate stressful situations or adversity and attain developmentally relevant competencies (Masten, 2001). Appropriate application of the terms commonly used to identify processes that influence the adjustment of youth exposed to adversity is necessary to promote terminological consistency. Thus, a brief description of several terms, and their relevance to moderational analyses, is presented using hypothetical examples (presented in Figures 10.7 and 10.8).

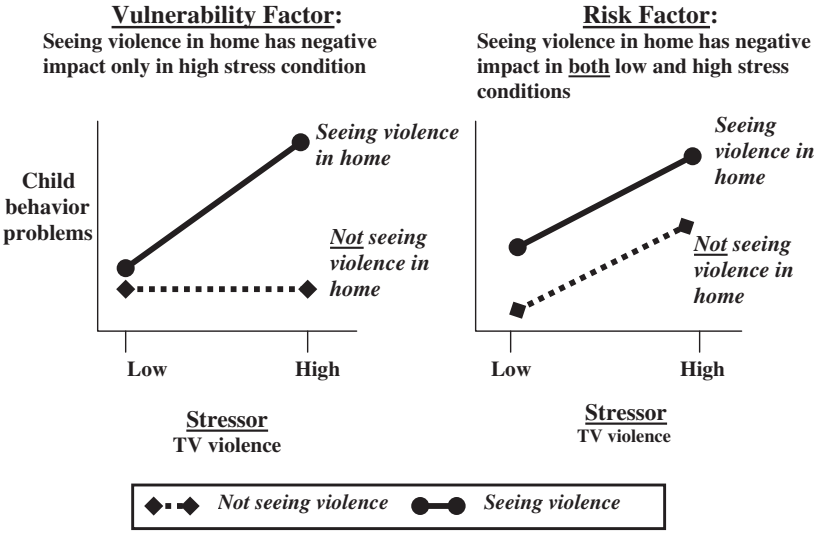

Figure 10.8 Factors associated with unfavorable outcomes: Vulnerability and risk factors.

Protective versus Resource Factors

A protective factor either ameliorates negative outcomes or promotes adaptive functioning. To isolate a true protective factor, however, there must be a particular stressor that influences the sample under investigation. The protective factor serves its protective role only in the context of adversity; a protective factor does not operate in low adversity conditions.

Protective factors are contrasted with resource factors. Specifically, a factor that has a positive impact on the sample *regardless* of the presence or absence of a stressor is a resource factor (Rutter, 1990). For example, if a positive father-child relationship reduces behavior problems only in children of depressed mothers, but has no impact for children of nondepressed mothers, then the father-child relationship would be conceptualized as a protective factor (Figure 10.7). However, if the positive father-child relationship reduces behavior problems in all children, regardless of mother's level of depression, then it would be conceptualized as a resource factor (Figure 10.7; Rutter, 1990). A model may also identify a positive father-child relationship as both a protective and a resource factor if it reduces behavior problems in children who have depressed mothers *more* than in children who have nondepressed mothers, but if it also produces a significant reduction in behavior problems for all children, regardless of level of maternal depression. It is important to note that a protective factor represents a moderational effect (see the statistically significant interaction effect in Figure 10.7), whereas a resource factor represents an additive effect. Statistically, a resource factor emerges as two main effects.

Risk versus Vulnerability Factors

Risk and vulnerability factors operate in much the same way as resource and protective factors, but in the opposite direction (Figure 10.8). A vulnerability factor is a moderator that increases the chances for maladaptive outcomes in the presence of adversity (Rutter, 1990). Similar to a protective factor, a vulnerability factor operates only in the context of adversity. By contrast, a variable that negatively influences outcome regardless of the presence or absence of adversity is a risk factor (Rutter, 1990). For example, witnessing violence in the home environment is conceptualized as a vulnerability factor if it only increases behavior problems in children who are also exposed to a stressor, such as viewing extensive violence on television (Figure 10.8). A vulnerability factor is a moderator and is demonstrated statistically with a significant interaction effect. Witnessing violence in the home can be conceptualized as a risk factor if it results in an increase in child behavior problems for all children, regardless of the amount of TV violence witnessed. As with resource factors, a risk factor represents an additive effect (i.e., two main effects). A model may also identify a factor as being both a risk and a vulnerability factor if it increases the chance of a maladaptive outcome in samples with and without exposure to a stressor, but increases the chances for maladaptive functioning significantly more in the sample with the stressor.

To summarize, if a factor significantly promotes or impairs the chances of attaining adaptive outcomes in the face of a stressor, then it operates via protective or

vulnerability mechanisms, respectively. In these cases, the factor serves a moderational role. However, if a factor significantly promotes or impairs the chance of attaining adaptive outcomes without differentiating between the presence or absence of a stressor, then it is conceptualized as operating via resource or risk mechanisms, respectively. Given that new research in the field has found that predictor → outcome effects may be moderated by developmental and contextual factors, there is an impetus for investigating these mechanisms at a more process-oriented level.

Intervention and Prevention Research

Research that involves a randomly assigned intervention as the predictor variable provides a particularly powerful design for drawing conclusions about causal mediational relationships (Clingempeel & Henggeler, 2002; Howe, Reiss, & Yuh, 2002). These types of models have three important strengths. First, significant mediational models that involve interventions inform us about the mechanisms through which treatments have their impact (Kraemer, Wilson, Fiarburn, & Agras, 2002; Rutter, Pickles, Murray, & Eaves, 2001; Weersing & Weisz, 2002). Simply put, with such models one is able to ask *how and why* an intervention or prevention program works (Kraemer et al., 2002; MacKinnon & Lockwood, 2003).

Second, as noted by Collins, Maccoby, Steinberg, Hetherington, and Bornstein (2000), if a manipulated variable (i.e., the randomly assigned intervention) is associated with change in the mediator, which is in turn associated with change in the outcome, there is significant support for the hypothesis that the mediator is a *causal* mechanism. One is more justified in invoking causal language when examining mediational models in which the predictor is manipulated than in mediational models in which no variables are directly manipulated.

Third, when one isolates a significant mediational process, one has learned that the mediator may play a role in the maintenance of the outcome (e.g., problem behavior). In this way, knowledge about mediational processes in the context of randomized clinical trials (RCTs) informs us about etiological theories of disorders (Kraemer et al., 2002). It is also important to note that when examining mediational processes in RCTs, one must demonstrate (via the research design and the timing of the data collections) that changes in the mediator *precede* changes in the treatment outcome (Kraemer, Stice, Kazdin, Offord, & Kupfer, 2001).

As an example of this strategy, Forgatch and DeGarmo (1999) examined the effectiveness of a parenting-training program for a large sample of divorcing mothers with sons. They also examined several parenting practices as mediators of the relationship between the intervention and child adjustment (see Figure 10.9 for an example of this type of model). Compared to mothers in the control sample, mothers in the intervention sample showed improvement in parenting practices. Improvements in parenting practices were linked with improvements in child adjustment (although, interestingly, the intervention did not have a direct effect on child outcome). As indicated in Figure 10.9, such models not only allow one to test potential mediators within an experimental design, but they also allow one to examine the differential utility of different mediators. In other words, one can determine which mediator best accounts for the effectiveness of a given treatment.

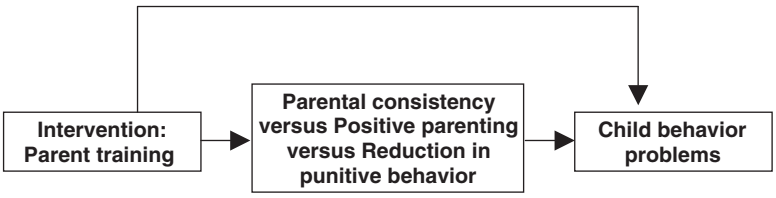

Figure 10.9 Mediators in intervention research: Parenting behaviors as mediators of the relationship between parent training (intervention) and child behavior (outcome).

At a more complex level of analysis, one could posit moderated mediational models. Such a model would not only suggest processes by which a treatment has an effect (i.e., a mediational process) but also which subsamples demonstrate the mediated treatment effects. Findings based on such complex models would provide important information that could be applied to modify the intervention so that it targets relevant mediational processes for specific subgroups of treatment participants.

Kraemer and colleagues (2002) have distinguished between moderators and mediators in the context of RCTs. According to these authors, a moderator is present prior to treatment and a mediator occurs during the treatment process. For example, a moderator might be a pretreatment characteristic of the individual (e.g., gender, IQ, initial severity of symptoms) or a contextual treatment variable (e.g., the setting of the treatment, of the level of exposure to the intervention). A mediator, on the other hand, is a variable that is expected to change as a result of treatment (e.g., parenting behaviors). In other words, while mediators inform *how* and *why* change occurs, moderators provide information regarding *for whom* the intervention or prevention is effective (MacKinnon & Lockwood, 2003). Given these definitions in the context of RCTs, Kraemer and colleagues maintain that the same variable cannot be both a moderator and a mediator (because of their differential temporal relationship to the treatment).

The ability of mediators and moderators to elucidate the impact of an intervention emphasizes the need to hypothesize and measure potential mediational and moderational effects. Unfortunately, although program effects are frequently reported, mediators and moderators of treatment often remain unspecified (MacKinnon & Lockwood, 2003; Weersing & Weisz, 2002). Measurement of potential mediators not only helps us to understand how and why an intervention program is successful, but also illuminates why certain programs may fail. In addition, evaluation of treatment mediators and moderators may facilitate informed modification of intervention efforts and improved cost-effectiveness of services (MacKinnon & Lockwood, 2003).

Research Findings and Future Directions

Grant and colleagues (2006) recently conducted a review of literature investigating mediational or moderational effects in the context of the relationship between stressors and child or adolescent adjustment. Their paper highlights several important

overall findings and challenges, as well as future directions for research incorporating mediation and moderation.

Examination of past moderational research revealed that demographic variables, such as age, sex, and ethnicity, were most frequently analyzed, with relatively few studies focusing on the role of cognition, competencies, and coping. Although age, sex, and ethnicity were not consistently significant across studies, significant findings did reveal some interesting patterns. For instance, it appears that for younger children, symptoms, as reported by parents, may be more strongly associated with stressors, whereas self-report may be more strongly related for adolescents. In terms of gender, girls may be more likely to develop internalizing symptoms in response to a stressor, whereas boys may be more prone to externalizing symptoms. Finally, associations between stressors and maladjustment may be stronger for White youth than for youth of other ethnicities. The moderational role of cognition was consistently supported, especially in the development of depressive symptoms. In terms of individual protective factors, academic competence appears to function as a buffer more consistently than social competence and coping. Finally, social support may protect against negative effects associated with exposure to violence and abuse.

The authors point to several challenges that permeate moderational research. First, inclusion of moderators in analyses frequently lacked theory. That is, analysis of potential moderators largely consisted of non-hypothesis-driven inclusion of demographic variables, with sparse research devoted to theory-based moderator variables, such as cognition. Second, measures and definitions of potential moderators were inconsistent, precluding generalization across findings. Finally, assessment of environmental moderators overrelied on use of child report, which can be problematic given possible common source variance and confounding with other child characteristics (e.g., competence).

In contrast to moderational studies, research studies investigating mediational effects were, for the most part, theory-driven. Although some evidence of statistical inconsistencies and use of inappropriate statistical analyses was found, sophistication of analyses appeared to increase over time. Across studies, family process (e.g., parenting behavior) emerged as a consistent mediator in the relationship between stressors and psychosocial adjustment. The authors noted that the majority of mediational research utilized multiple regression or SEM to analyze mediational effects. Weaknesses of mediational research included an overreliance on cross-sectional data and a failure to test competing mediational models.

This review indicates several future directions for mediation and moderation research in the area of clinical child and adolescent psychology. First, routine incorporation of conceptual models and theory grounded in developmental psychopathology would strengthen research pertaining to moderation. Second, utilization of consistent measures and definitions of moderator variables, such as coping, would promote cross-generalization across studies. Third, research would benefit from inclusion of other sources and methodologies in assessing environmental context beyond child report, such as observational measures and parent report. Fourth, longitudinal data are needed to explore mediational and moderational variables that impact on developmental phenomena. Fifth, recent criticisms of the multiple regression approach for analyzing mediation encourage exploration of newer

statistical methodology based on recent recommendations (e.g., Cheong et al., 2003; Cole & Maxwell, 2003; MacKinnon et al., 2002; Shrout & Bolger, 2002). Finally, mediation research should incorporate tests of competing mediational and integrative models investigating how family, child, and environmental variables operate in conjunction with each other (i.e., mediated moderation and moderated mediation).

IMPLICATIONS OF RESEARCH FOR CLINICAL PRACTICE

Now that the applications of mediational and moderational analyses in the field of research have been reviewed, it may be helpful to address briefly the relevance of this type of research to clinical practice. This chapter has reviewed the utility of mediational and moderational research for a variety of purposes: confirming or refuting theories about developmental pathways, identifying circumstances that promote or inhibit successful adaptation to chronic healthy or life stressors, and identifying specific mechanisms associated with positive outcomes following a therapeutic or clinical intervention. These accomplishments are important not only for the researcher, but also for the clinician seeking empirically validated prevention or intervention techniques.

For example, suppose that a clinician is treating a young child with a chronic illness (such as diabetes). The child lives with a single parent who is suffering from Major Depressive Disorder and is unable to provide consistent care. Mediational and moderational research can be of use to a clinician treating a child living under these circumstances. First, mediational research involving random assignment to intervention conditions can provide information about the relative efficacy of various treatments or combinations of treatments (e.g., medication, dietary changes, exercise, and all possible combinations of these). Mediational research can also clarify which specific aspects of traditionally accepted treatments are most responsible for observed changes, so that unnecessary components of the treatment may be eliminated. Mediational research such as this allows the clinician to make more informed decisions about treatment (e.g., which treatment is best, based on the means by which the treatment is shown to operate) and also provides information about critical points for intervention in a child's development.

Moderational research is also readily applicable to the clinical setting. First, it can clarify whether a treatment is equally effective when applied in the presence of various demographic, genetic, and environmental factors (e.g., child gender, family medical history, level of parental involvement). For example, perhaps a treatment found to be most effective in a sample of children in a rural setting is not very effective when applied to children living in an urban area. Or perhaps risk and protective factor research suggests that an intervention is beneficial only under particular conditions or circumstances. By considering these issues, moderational research fine-tunes the conclusions drawn from typical intervention studies. Through awareness of the variables that moderate the appropriateness of a given treatment, a clinician can tailor interventions to the individual rather than providing a "one size fits all" treatment.

SUMMARY

In this chapter, we provided definitions of mediated and moderated effects, reviewed statistical approaches, and discussed how such effects can be examined in longitudinal, intervention, and risk and protective factor research. The need for consistency in the use and analysis of mediation and moderation in the child clinical and pediatric psychology literatures was highlighted. It is recommended that care be taken in discussing these processes and that investigators be clear about what statistical approaches are appropriate for a given hypothesis. Because research in pediatric and child clinical psychology has important treatment, prevention, and public policy implications, appropriate modeling and statistical techniques are needed to move the field toward greater understanding. Additionally, this research could benefit from (a) greater inclusion of theory and complex models, (b) consistency in definitions and measures of mediational and moderational constructs, (c) integration of multiple methods and sources to assess environmental variables, (d) use of longitudinal data, (e) exploration of newer statistical methodologies, and (f) generation and testing of complex models.

To conclude, we provide a list of important concepts pertaining to mediational and moderational research. This list, which incorporates the major points of this chapter, is intended to serve as a guide and inspiration for future mediational and moderational research:

- If a researcher is interested in examining a relationship between two variables, the use of mediational and moderational models can lead to deeper and more comprehensive knowledge about the relationship by providing information about the conditions under which the two variables will be associated (moderation) and also about the intervening processes that help to explain the association (mediation).
- The same variable can either mediate *or* moderate a relationship between a predictor and an outcome, depending on the nature of the research question and the role the variable is thought to play in the relationship.
- Mediational models are usually proposed after an association between predictor and criterion variables is well established in the literature. If there is no established relationship, one may be better served by first examining associations among the predictor and criterion variables.
- Mediational models are causal models that illustrate a pathway of influence among variables. In this chapter, we distinguished between causal and temporal models.
- Risk, protective, vulnerability, and resource factors can all be identified and illustrated with the use of a moderational model.
- Mediational models are particularly informative and compelling when they are tested with longitudinal data and/or when they are tested in a randomized clinical treatment research design.
- The statistical strategies used to test mediational and moderational effects are distinct. The models differ both conceptually and statistically.

- In choosing a statistical approach for testing mediation and moderation, one should take into account the nature of one's data (e.g., sample size), as well as the strengths and weaknesses of each approach.
- Post-hoc probing is necessary to understand the nature of mediational and moderational effects.
- It is possible to propose complex theoretical models that include mediated moderation and moderated mediation.
- Mediational and moderational models are useful not only to researchers. Because they can provide information about developmental pathways and conditions affecting treatment efficacy, these models can be useful to clinicians as well.
- With the use of mediational and moderational models, researchers can test competing theories about relationships among variables of interest. By directly testing two or more alternative models, a researcher can determine statistically which theoretical model best captures or explains observed relations among the variables of interest.

As has been emphasized throughout this chapter, attention to mediation and moderation has great potential to enrich the field of clinical child and pediatric psychology. It is hoped that this chapter will encourage more widespread and accurate use of these concepts and techniques to enhance the scientific contribution of research.

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CHAPTER 11

Single-Case Research Designs

Kurt A. Freeman and Eric J. Mash

Clinical child and adolescent psychology concerns itself with the assessment and treatment of abnormal behavior or psychopathology. Most branches of contemporary clinical psychology are firmly grounded in the scientific method or empirical approach to studying psychopathology and methods of its amelioration. This is affirmed by the adoption of the Boulder model of training by most clinical psychology doctoral training programs (though the development and growth of professional schools of psychology arguably have detracted from this emphasis), which espouses integration of scientific methods with clinical practice. For years, there has been concern about the disconnect between clinical psychology training in the scientist-practitioner model and real-world clinical practice (Hayes, Barlow, & Nelson-Gray, 1999). However, many argue that instead of abandoning training in research methods, the mandate remains to identify methods of bridging the gap between scientific endeavors and clinical practice through the identification of research methods and evidence-based practices applicable to clinical settings (Gelso, 2006; Hayes et al., 1999; Weisz, 2004; Weisz, Jensen-Doss, & Hawley, 2006).

Scientific inquiry assists in understanding the cause-effect relationships between various life events and situations and the clinical phenomena with which clients present. Such an understanding can be developed either via group or single-subject design methodologies, depending on the particular questions of interest and constraints present. In fact, though there are misunderstandings about each approach (e.g., Aeschleman, 1991) and some suggest or imply that one approach may be better than the other (Bailey & Burch, 2002; Gliner, Morgan, & Harmon, 2000), no one type of design is inherently better and each addresses important questions about clinical phenomena of interest.

Since the early days of the behavioral sciences there has been some debate regarding the most appropriate approach to scientific inquiry (Leary, 2001). Today most researchers in the behavioral sciences, including those in clinical child and adolescent psychology, take a nomothetic approach. This involves studying larger groups of individuals to identify general principles and broadly applicable generalizations (Blampied, 2001). Based on this perspective, using group designs (both between- and within-group designs) is most appropriate. They can assist with answering questions such as the percentage of individuals who respond a particular way given certain conditions (e.g., the percentage of youth with depressed mood

who respond to cognitive-behavior therapy) and the association between variables (e.g., child age or gender and treatment outcome). Others, however, have argued that the idiographic approach must be included to ensure that behavioral processes are understood at the level of the individual (for discussion, see Blampied, 2001; Leary, 2001). Idiographic research, therefore, focuses on describing, examining, and comparing the performance of an individual against his or her own performance at different times or in different situations, and thus typically involves single-case research designs. The current chapter focuses on the latter designs.

The primary goal of this chapter is to provide a discussion of single-case designs (SCDs). Specifically, an introduction to the various SCDs currently in use is described in terms of the processes involved in their use, as well as the benefits and weaknesses of each. Although some have argued that SCDs are most appropriate for testing the relation between particular variables of interest and targeted behavioral patterns because one can experimentally demonstrate a cause-effect relationship (e.g., Bailey & Burch, 2002), we intend to provide no judgment as to whether group or single-subject designs should be given greater value. In fact, the American Psychological Association's task force charged with determining guidelines for classifying empirical support for psychological treatments indicated that either can be an appropriate method for determining whether an intervention is efficacious (Chambless et al., 1996; Task Force on Promotion and Dissemination of Psychological Procedures, 1995), an issue that is discussed more thoroughly in a later section. Instead, our focus is on developing a better understanding of SCDs. We emphasize how SCDs are useful in clinically oriented research, but are also applicable to clinical practice.

CHARACTERISTICS OF SINGLE-CASE RESEARCH DESIGNS

When evaluated from a group design perspective, SCDs may be considered limited in scope and application, or even meaningless. For example, some have argued that SCDs are limited in terms of internal and external validity (Aeschleman, 1991). However, such concerns are grounded in the group design perspective of scientific inquiry, and taking such an approach may be misleading (Hillard, 1993). To appreciate the complexity and utility of the various single-subject designs, one needs to understand their fundamental premises and characteristics.

Repeated Observation of Phenomena of Interest

Single-case designs share in common repeated observation of the phenomena of interest across time under standard conditions (Holcombe, Wolery, & Gast, 1994). Specifically, measurement of the dependent variable occurs multiple times, across at least two separate conditions, which are usually referred to as *baseline* and *intervention*. Such a method of data collection is based on the perspective that determining whether the dependent variable differs from one condition to the next in a meaningful way requires multiple data points within each condition. Doing so allows the researcher to investigate whether desired trends in the data are evident (see "Visual Inspection" section) and protects against making assumptions about

differences that may be spurious or the result of natural variation in the dependent variable of interest.

Often in research, the dependent measure utilized is some form of observable behavioral phenomenon (e.g., aggressive behavior of a young child, out-of-seat behavior of a child with Attention-Deficit/Hyperactivity Disorder, duration of crying by a depressed adolescent). However, there is nothing inherent in SCDs that requires the dependent variable be of such form. End products (e.g., weekly weight loss for overweight children), scores on questionnaires (e.g., obtained scores on the Children's Depression Inventory), or subjective ratings (e.g., measurements of anxiety using the Subjective Units of Distress Scale) can be obtained repeatedly in SCD research, allowing for inferences regarding the impact of the independent variable on measures of outcome. Multiple examples abound regarding the use of various types of dependent measures in a single-subject research paradigm (e.g., Glicksohn, Gvirtsman, & Offer, 1997; Nugent, 1992).

In clinical practice, repeated observation can serve an important role in the therapeutic process. Specifically, if the clinician and/or client gathers data in a systematic and repeated manner, decisions can be made regarding whether to alter interventions when monitoring suggests a lack of progress. How one goes about making changes in clinical interventions will determine whether the practitioner adheres to methods that retain sufficient experimental rigor so as to empirically demonstrate causal relations between changes in the dependent and independent variables. However, because the goal of most clinical endeavors is not to demonstrate experimental control but to document change in the clinical phenomena of interest, such violations may be less important than when engaged in research practices.

Participants as Their Own Experimental Controls

Similar to what occurs in within-subjects group designs, participants serve as their own controls in SCDs. However, there is an avoidance of aggregating obtained data across cases with SCDs. Instead, they are based primarily on demonstrating experimental control through replication of the impact of an independent variable on a case-by-case basis. The main analytic process used to determine experimental control involves comparing an individual's behavior across experimental conditions, rather than comparing across individuals. This is viewed as the relevant comparison, because the real question is whether an individual's behavior has changed as a result of treatment relative to pretreatment behavioral presentation (Morgan & Morgan, 2001). For instance, a child psychologist is more concerned with whether depressive symptoms of an adolescent client have changed as a result of therapy relative to preintervention level of symptoms rather than whether the client's symptoms changed relative to a statistically derived group mean. Note that intrasubject comparison across experimental conditions as an empirical test of the effect of the independent variable on the symptoms of interest is different from completing a normative assessment (i.e., comparing a particular individual's performance or behavioral level to normative data) to determine if a person's current level of functioning is within normal limits. This latter practice is often an important issue clinically and has some relevance to determining whether changes observed are

clinically significant or meaningful (Atkins, Bedics, McGlinchey, & Beauchaine, 2005).

Given the focus on intrasubject variability as the primary issue of importance, SCDs are particularly relevant for psychotherapy research (Hayes et al., 1999; Hilliard, 1993; Lundervold & Bellwood, 2000; Morgan & Morgan, 2001). Typically, psychotherapy involves repeated interactions between the therapist and the client (e.g., child, adolescent, family) as a means of producing change. Further, there is usually a period of assessment in which the therapist learns about the client's presenting problems (analogous to baseline in research scenarios) and then treatment, during which the therapist creates conditions in an effort to produce change in the client's presenting problem. Like the SCD, psychotherapy as a process allows for comparison across conditions.

Replication of the Experimental Effect

Experimental replication of observed effects is one of the primary principles of scientific inquiry (Morgan & Morgan, 2001). Through replication, scientists are able to provide information about the reliability of observed effects, as well as to self-correct when initial findings are not consistently obtained. Unfortunately, replication studies are lacking in the behavioral sciences, including clinical child and adolescent psychology. This is likely due to a host of reasons, an important one being the energy and expense of replicating studies that rely on large groups of participants.

A hallmark of experimentally sound SCDs (i.e., those that allow for a determination of the functional relation between the independent variable and changes in dependent variables) is replication. That is, SCDs set as the standard repeated demonstration of the impact of the independent variable as the measure of experimental control. Support for the interpretation of the independent variable (e.g., training parents in methods of giving clear instructions to their children) as the cause of the change in the dependent variable (e.g., rate of child compliance to adult instructions) increases each time one demonstrates that the latter changes as a function of a change in the former. Because of this, each participant in single-subject research is repeatedly exposed to baseline and treatment conditions. This is in stark contrast to the most common group design used in psychological research, the between-groups design. In the between-groups design, the experimenter compares the aggregate level of the dependent variables across at least two groups: the comparison group, which is not exposed to the independent variable, and the treatment group, which is. In such a design, the experimenter rarely introduces, withdraws, and reintroduces the independent variable while obtaining measurement of the dependent variable across conditions. Instead, there is reliance on statistical analysis to determine whether the obtained difference would occur by chance versus as a result of the treatment.

Replication occurs across participants as well. The nomenclature describing SCDs implies that only one participant is involved (e.g., single-subject design, *n* of 1 designs). However, most research using these designs involves multiple participants (Horner et al., 2005; Leary, 2001), though there is no gold standard

for the number of individuals who should be included in a single study to allow for adequate assessment of intervention effects across individuals. Each time an intervention effect is observed with a different participant, confidence that the independent variable is responsible for the change increases. Thus, when there is a small effect, researchers strive to have more participants to increase confidence in interpretation of the source of the effect. The number of participants needed may be fewer if the effect is large.

There are instances in which repeatedly introducing and removing an intervention is impractical or unethical. For instance, one may implement an intervention focused on skills acquisition (e.g., teaching a child with an intellectual disability how to tie his shoes). In this instance, maintenance of the behavior is probable even if intervention is withdrawn because of the naturally rewarding properties of engaging in the behavior. In other words, once one has mastered the skill of tying one's shoes, the skill may not return to baseline levels when the intervention used to establish that skill is removed. When the intervention is designed to address a serious or threatening behavior (e.g., suicidal self-harm behavior), removing an effective intervention may be considered unethical. In these instances, researchers will rely on SCDs that more heavily focus on intersubject replication and, in some cases, cross-situational replication as a means of investigating experimental control of the independent variable on changes observed in the dependent variable (see "Multiple Baseline Designs" section).

Changing One Variable at a Time

Another essential tenet of SCDs is that only one variable should be altered at a time when moving from one phase of the experiment to the next (Barlow & Hersen, 1984; Hersen, 1982). For example, if a researcher is determining effective interventions for treating nocturnal enuresis, he or she would want to introduce only one intervention (e.g., the urine alarm) at the start of the treatment phase. If more than one variable is changed, it is impossible to determine which variable is producing change in the dependent variable. Using two strategies, such as decreasing evening fluid intake *and* introducing a particular behavioral intervention (e.g., the urine alarm) at the same time, would result in a situation in which it would be impossible to determine which produced any noted improvements that occur. Thus, unless the investigator is actually evaluating an intervention with multiple components (see later discussion), standard practice is to hold all variables constant except for one independent variable as the investigation moves from the baseline to the treatment phase.

Although the general guideline is to change only one variable at a time as one moves from baseline to treatment, as with any rule there are exceptions. Specifically, if a researcher is interested in the combined effect of several independent variables, as is often the case with many empirically supported interventions (Kazdin & Weisz, 2003; Mash & Barkley, 2006), it is acceptable to change more than one variable. For example, Freeman and Piazza (1998) were interested in the effects of a combined intervention for treating food refusal (i.e., stimulus fading, escape extinction, and reinforcement) and therefore introduced several strategies as a package intervention.

In this way, the authors were able to determine the impact of the combined treatment on the targeted behavior. It should be noted, however, that if investigators are using such an approach to experimentally demonstrate control over the dependent variable, all independent variables should be introduced simultaneously, rather than progressively. In this way, the package intervention essentially becomes one independent variable. Although by using a package treatment it is impossible to determine which component of the intervention is producing the change, or whether it is the components in concert, such an approach allows one to determine if the combined intervention impacts the dependent variable.

Utilizing a package intervention is becoming more common as psychotherapies become increasingly manualized (Drozd & Goldfried, 1996; Scaturo, 2005). Thus, at first glance, using SCDs to empirically test the effect of manualized interventions with different client populations or clinical settings seems reasonable. Theoretically this is true, but there are some challenges to testing manualized psychotherapies using SCDs. Specifically, manualized treatments typically involve multiple components that are applied in a sequential, versus simultaneous, fashion. For example, popular manuals to guide parent management training for young children with disruptive behavior disorders typically start by training parents in various positive parenting practices (e.g., use of labeled praise, token systems), followed by discipline strategies (e.g., time-out, response cost). Because of this, using relevant components of SCDs (e.g., repeated measurement) may be useful for documenting clinical outcome but may not be appropriate for empirically demonstrating cause-effect relations. Of course, if the package intervention is implemented such that all components are introduced simultaneously, one can systematically withdraw components of treatment over time to evaluate which components are essential (Kazdin, 2003; see also Jones, 1993).

Graphical Representation and Visual Analysis of Data

Data presentation and analysis differ markedly between SCDs and group designs. The most obvious difference is that data are presented for individual participants rather than as a summarized group aggregate. Additionally, information is typically presented in real-time, graphical representation, allowing the researcher to easily examine the data. Data are represented in the manner that most accurately captures the variable of interest (e.g., frequency per observation, duration of the behavior), typically with each data point representing one observation period (e.g., 1 day, one class period, one session).

Although data gathered using SCDs can be subjected to statistical analysis (for attention to this issue, see Parker et al., 2005; Todman & Dugard, 2001; see also Perone, 1999, for argument for experimental control versus statistical control), traditionally analysis involves visual inspection of data attending to level, trend, and variability within and across phases (Horner et al., 2005; Kazdin, 2003). *Level* refers to mean performance within a condition or phase; *trend* refers to the rate of increase or decrease in the hypothetical best-fit line of the variable being measured within a phase, or slope; *variability* refers to the amount that data vary around the mean or slope within a phase. Hayes and colleagues (1999) suggest that *course* is

also an important consideration when analyzing data gathered via SCDs. They define course as a more flexible method of investigating trend, allowing for a determination of whether data follow a linear or curvilinear pattern. Finally, latency to change in the dependent variable after changes in the independent variable is also important for understanding the impact of the latter on the former (Kazdin, 2003). Specifically, the closer in time between the introduction of the treatment and the change in the dependent variable, the more confident one is in the source of the change. By attending to each of these variables while visually inspecting data obtained, one can make determinations about the reliability and consistency of the intervention effect.

Compared against statistical analysis of data, visual inspection focuses much more heavily on clinical significance than statistical significance (Kazdin, 2003; Zhan & Ottenbacher, 2001). In clinical psychology, there has been increasing focus on clinical significance of intervention effectiveness versus simply statistical significance. Clinical significance refers to value or meaningful difference that an intervention has on a person's functioning or everyday life (Kazdin, 1999). Criteria have been suggested regarding how to determine clinical significance when using group designs (e.g., Jacobson, Roberts, Berns, & McClinchey, 1999; Jacobson & Truax, 1991), and some researchers argue that to determine clinical significance one must analyze the data at the level of the individual (Jacobson & Truax, 1991).

Single-case designs are well suited for attention to clinical significance of intervention effects. Because analysis relies on visual inspection, emphasis is placed on ensuring that there is a large effect present to say with confidence that the independent variable produced a meaningful change in the dependent variables. That is, to trip the belief operant that there is an effect present, single-subject researchers demand notable changes in dependent variables from baseline to intervention phases. Of course, for clinical purposes, there still may be valid reasons to complete norm-based assessment using standardized instruments to determine whether the change altered behaviors of interest so that they are now within typical or nonclinical limits (Hartmann, Roper, & Bradford, 1979).

VALIDITY AND SINGLE-CASE DESIGNS

Paramount to any research methodology is the ability to adequately address issues of internal and external validity. Without careful attention to these variables, obtained findings have little scientific meaning and thus are not helpful to the scientific or practitioner community. *Internal validity* refers to the extent to which one can say with confidence that a change in the dependent variable is the result of the independent variable. Multiple threats to internal validity exist and must be accounted for in order for research findings to be interpretable. A complete discussion of the factors that may threaten the internal validity of research findings is beyond the scope of this chapter. For a more in-depth discussion of these issues, readers are referred to Campbell and Stanley (1966), Scotti, Morris, and Cohen (2003), and Shadish, Cook, and Campbell (2001).

When using group designs, one of the primary methods for controlling for threats to internal validity is through random assignment of participants to experimental

groups (e.g., treatment, no treatment). Though some have argued for the use of randomization procedures in SCDs (e.g., Edgington, 2007), such a practice is not typical. Instead, SCDs adequately address threats to internal validity in several critical ways through other processes. For example, through use of repeated measurement of behavior across time, repeated demonstrations of the impact of independent variables, and intra- and interparticipant comparison, they control for threats such as maturation and history effects. Extraneous variables that threaten internal validity are assumed to be constant across baseline and treatment conditions, and thus any notable differences across the conditions can be attributed to the independent variable. How this is controlled for depends on the type of SCD used. Additionally, because data analysis involves frequent plotting and visual inspection of data, researchers using SCDs have the ability to observe whether aberrant or unusual patterns of behavior emerge during the course of an experiment and can go about searching for (and potentially controlling) the source of variability accounting for those patterns (Morgan & Morgan, 2001; Perone, 1999).

External validity relates to generality of findings and thus refers to the ability to extrapolate findings observed with participants from a particular study to a larger group of individuals who are similar on characteristics considered important. External validity of findings obtained using SCDs is enhanced with each replication of intervention effect across participants, settings, or conditions and/or different measures of the same independent variable (Horner et al., 2005). Thus, SCDs address external validity through two types of replication: direct and systematic replication (Sidman, 1960). Direct replication is accomplished by demonstrating the impact of the independent variable across participants with similar characteristics. As the number of replications of the impact increases across participants, the extent of generality is identified (McReynolds & Thompson, 1986). Systematic replication (i.e., varying some aspect of the original experimental conditions) is also used to address external validity. Thus, using SCDs to demonstrate replication of treatment effects across clinical disorders, settings, clinicians, and so forth allows for the assessment of the extent to which the findings are externally valid.

CHOOSING A BASELINE

With few exceptions, the experimenter utilizing an SCD initiates the process by completing a period of observations of the dependent variable as it occurs in the absence of the independent variable. Conventionally, this first phase is referred to as the *baseline* and is labeled with an "A" when representing the data on a graph. The data pattern that emerges during this phase of the research serves as the point of comparison for patterns of data during other (often intervention) conditions. Thus, careful attention to the patterns that develop during baseline data collection is important, with particular attention to level and trend (Bailey & Burch, 2002; Kazdin, 2003).

In basic research, investigators are typically concerned with establishing baseline responding under tightly controlled conditions that produce patterns of data with little variability (e.g., Perone, 1991; Sidman, 1960). In contrast, applied researchers usually collect baseline data during naturally occurring conditions

(Barlow & Hersen, 1984; Hersen, 1982). Rather than creating the baseline conditions, the applied researcher often simply documents the occurrence of the dependent variable in the typical environment. Thus, obtaining the level of control over the environment necessary to produce data with minimal fluctuation can be difficult. Further, applied researchers, because they are often working with human participants who may be experiencing significant distress, may not have the luxury of continuing baseline observations until a narrowly defined pattern of data emerges. As such, the definition of "stability" or "pattern" is typically viewed differently when in the applied context. This is not to suggest that applied researchers should not be vigilant in minimizing or eliminating variability in data patterns due to measurement errors or other extraneous variables that are controllable (and for which control is warranted). Rather, the point is that the acceptable amount of control, and thus the amount of variability, differs considerably across basic and applied research environments. The remaining discussion focuses on issues primarily relevant to applied research situations.

When conducting baseline observations, applied researchers and clinicians are interested in gathering a sample of the dependent variable sufficient to serve as a standard against which to compare patterns of data produced during other conditions. To date, there remains no definitive criterion for the right length of baseline. The most important consideration when determining whether one's baseline observation period is adequate is that of consistency in data patterns, which then allows one to predict future data patterns in the absence of any change to experimental conditions (Kazdin, 2003). Once a pattern of consistency in the desired direction has developed (e.g., the baseline pattern is increasing if one expects the independent variable to produce decreases), one can introduce the independent variable and obtain interpretable findings. It is conventional to consider a minimum of three data points as sufficient for determining whether consistency or pattern exists (Barlow & Hersen, 1984). However, one can easily find published examples of single-subject research baseline periods that include fewer or greater numbers of baseline observations. Thus, although most single-subject researchers strive for a minimum of three baseline data points, the ultimate criterion of being able to adequately assess the impact of the independent variable prevails.

Although various combinations and permutations could occur, several basic baseline patterns have been described (Barlow & Hersen, 1984; Hersen, 1982). Each has particular implications for the interpretation of the impact of the independent variable on the dependent variable. Figure 11.1 (top graph) depicts a stable baseline using a 6-point baseline observation period with hypothetical data. Here there is minimal variation with no significant increasing or decreasing trends across observations. Introducing the independent variable following this baseline period would result in unambiguous interpretation of its effect, easily determining whether there is no change (i.e., pattern remains the same during treatment), improvements (i.e., data patterns either increased or decreased, depending on the goal of the intervention), or worsening (i.e., again, data patterns either increased or decreased, depending on the goal of intervention).

The second general pattern of baseline is a worsening of the problem (i.e., increasing or decreasing trend). Take, for example, a situation in which a researcher

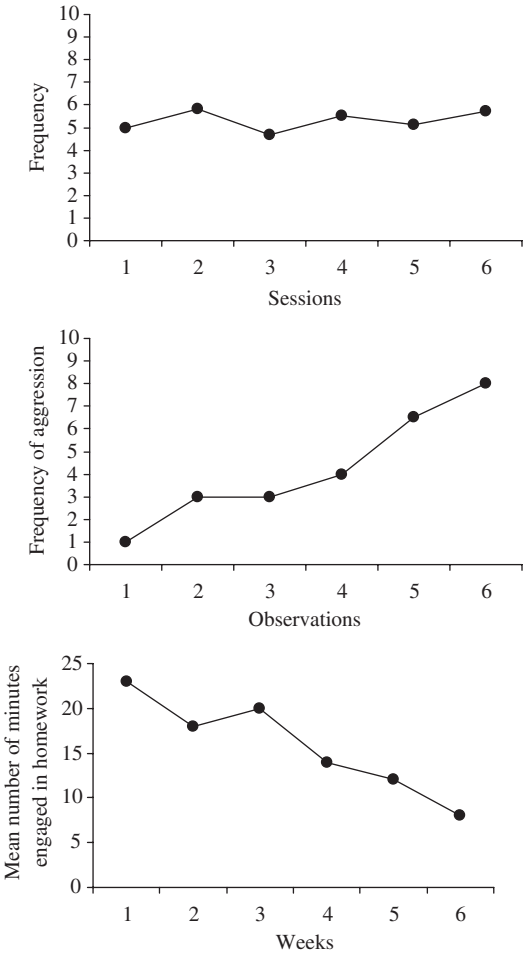

Figure 11.1 The stable baseline: hypothetical data demonstrating slight but negligible variation in data patterns (top graph). The increasing or decreasing baseline: hypothetical data on aggressive acts demonstrating an increasing baseline (middle graph) and hypothetical data on the number of minutes per evening spent engaging in homework (bottom graph), both representing worsening baseline trends.

is measuring the rate of aggressive acts displayed by a youth at recess (see Figure 11.1, middle graph). An increasing baseline pattern would document more frequent occurrence of the aggressive acts as baseline continued, suggesting that the behavior is worsening over time. Alternately, Figure 11.1 (bottom graph) depicts a decreasing trend in the mean number of minutes spent completing homework per evening across successive weeks, which is also indicative of a worsening of the problem during baseline. Both baselines are acceptable patterns for treatment comparison because meaningful interpretations of the impact of the independent variable can be made. Specifically, if the independent variable produces a reversed pattern of data (i.e., decrease in the number of aggressive acts or increase in minutes spent on homework), then the original pattern serves as an adequate comparison. An increasing or decreasing baseline trend becomes problematic, however, when

the independent variable does not produce a reversal in the direction of the behavior pattern. Specifically, such trends may mask whether an intervention has a detrimental effect, such as a worsening of the problem (Hersen, 1982). However, if there were a marked shift in the slope of the data (significantly steeper in either direction), one would be able to determine that the independent variable was having an adverse impact on the dependent measure.

In contrast to increasing or decreasing trends that represent a worsening in the dependent measure, the same data patterns that indicate improvements are troublesome as baseline patterns. Returning to the previous examples, one would not want a baseline that indicates a decreasing trend in aggressive acts or an increasing trend in number of minutes spent engaged in homework. These patterns are problematic as points of comparison because one could not determine whether continued improvements following the start of treatment are the result of naturally occurring factors (i.e., those that resulted in the initial improvement) or of the treatment.

The fourth pattern is the variable baseline (see Figure 11.2), which can be relatively common in applied research given difficulties in establishing tight experimental control over environmental variables. Although one could aggregate the data in a manner that minimizes the variability (e.g., by averaging the data collected over two sessions), this simply masks the variability and does not alter the basic pattern. Sidman (1960) recommended that, when a researcher is faced with such a pattern in baseline data, he or she should seek out and eliminate the extraneous factors that produce such variability. However, as has already been mentioned, this may be difficult or excessively time consuming for the applied researcher.

The utility of the variable baseline as a point of comparison depends on the impact of the independent variable. If the treatment produces a decrease in the variability of the data pattern *and* a change in the desired direction, then it is appropriate to conclude that the treatment had the desired impact. However, if variability is reduced, but the problem remains at unacceptable levels, then interpretation of the independent variable as useful would not be warranted. In this latter situation, although the dependent measure is impacted by the independent variable, the impact is not in the desired direction. Thus, one would not want to retain the treatment.

Another pattern of baseline is the variable-stable baseline. Here, the pattern is initially variable, but then becomes stable. Such a pattern may be achieved by

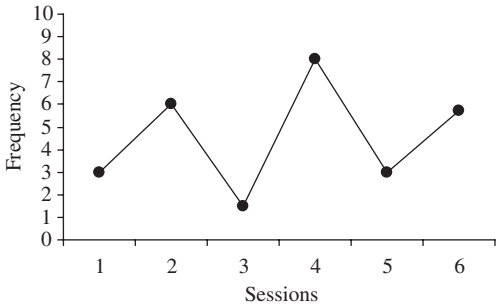

Figure 11.2 The variable baseline.

extending the baseline observation period for sufficient length so as to have the dependent measure come under the control of the baseline conditions. Such an approach might be warranted, for instance, if baseline conditions are novel to the participant. Because the independent variable is introduced following a period of stability, interpretation of its impact is the same as with the stable baseline: unambiguous. However, although perhaps the ideal scenario when initially obtaining variable data, practical and/or ethical constraints may limit the applied researcher's ability to extend the baseline to a sufficient length to establish stability (e.g., as in the case of a brief inpatient hospitalization, or when the dependent measure is severe self-injurious behavior).

The stable-variable baseline presents particular challenges as a point of comparison. Although unstable patterns of baseline data are not in and of themselves problematic, the stable-variable pattern suggests the influence of extraneous variables introduced after the initiation of baseline measurement. This may make it difficult to evaluate the impact of treatment as the extraneous variable may interfere with the independent variable. In these instances, attempts to seek out and control (or at least identify and describe) these outside influences are important.

SPECIFIC SINGLE-CASE DESIGNS

Now that defining characteristics of SCDs and issues related to evaluating baseline data patterns have been presented, the discussion is extended to encompass methods of introducing the independent variable to allow for appropriate comparisons. Specifically, defining features of commonly used SCDs are described.

Exploratory or Quasi-experimental Designs

There are times when it is not necessary or is impractical to utilize rigorous experimental SCDs when attempting to gather useful information about clinical phenomena of interest. Clinical psychology, as well as many other fields, is replete with many examples of interesting anecdotal descriptions of unique cases or systematic yet nonscientific observations of unusual phenomena. Gathering information using *exploratory* methods contributes to our understanding of phenomena of interest and often serves as the impetus for more in-depth and thorough analysis. In this section, we review two types of SCDs that allow for exploration of phenomena of interest without applying rigorous experimental criteria.

The Case Study

A case study is the "detailed study of a single individual, group, or event" (Leary, 2001, p. 321). The field of clinical psychology abounds with case studies of interesting, unique, and challenging clinical cases. In fact, early endeavors in this field to more rigorously analyze variables contributing to clinical phenomena relied heavily on the in-depth analysis of individual clients (Bailey & Burch, 2002). Even today, there are peer-reviewed journals devoted to publishing case studies based on psychotherapeutic work (e.g., *Clinical Case Studies*), a testament to the fact that there is much to be learned about the clinical phenomena experienced by our

clients and the experience of attempting to address those problems by presenting descriptive, typically qualitative summaries of observations and client reports.

Although case studies lack attention to systematic or replicable collection of information and are without controls to threats to validity, they have several beneficial characteristics (Kazdin, 2003). They serve as a source for information and ideas about clinical problems, as well as a source for developing therapy techniques. Further, they can be a beneficial way of studying phenomena of interest that are rare. A well-documented case study can also be used to challenge principles or practices that are considered well established by describing how they did not appear to apply to a given individual. Thus, though case studies are not experimental in nature, they play an important role in clinical psychology.

The Simple A-B Design

As was previously mentioned, the hallmarks of SCDs include repeated observation of the behavior of interest across time and systematic changes in independent variables. The least complex SCD utilized to accomplish these goals is the A-B design. With this design, the experimenter collects measures of the dependent variable during baseline and then again during the treatment condition. Thus, the effect of the independent variable is demonstrated once. The A-B design moves beyond the clinical case study in regard to experimental rigor because of its reliance on repeated and systematic collection of data on the dependent variable of interest.

However, in consideration with other SCDs, the simple A-B design lacks experimental rigor. Specifically, inclusion of only one phase of each of the components (i.e., baseline and treatment) decreases the ability to adequately assess experimental control or internal validity. Theoretically, it is possible that a change in an extraneous variable affecting the individual occurred concurrently with introduction of treatment. If or when this occurs, making inferences about the impact of the independent variable separate from the impact of the extraneous variable is impossible. Take a situation in which a clinician is working with an adolescent experiencing depressive symptoms. Suppose, after gathering baseline data on daily subjective assessment of mood, the clinician initiates behavioral activation strategies. Coinciding with this, the adolescent's family also initiates services with a psychiatrist, who prescribes a medication. If subjective ratings of mood change, the clinician will be unable to determine whether this change is due to the behavioral activation strategies, the medication, or a combination of both. Several other threats to internal validity (e.g., maturation, instrument decay, statistical regression to the mean) are also not well accounted for by the simple A-B design.

Despite limitations in the ability to make definitive conclusions about the causal role of independent variables using the A-B design, it can be useful given certain circumstances, and it offers advantages over the uncontrolled case study methodology. Specifically, for problems that have been resistant to change for significant periods of time, noticeable improvements observed during the B phase offer some support for the intervention as the causal mechanism. Further, the A-B design is appropriate when practical or ethical limitations decrease one's ability to remove the intervention as a means of demonstrating experimental control (e.g., treating

severe, life-threatening behavior). In these situations, the scientist-practitioner is at least gathering data to document whether change is occurring, regardless of whether effects can be fully ascribed to the specified intervention. Finally, for practitioners, reliance on A-B designs can be an important component of documenting that change occurred in a client's presenting problems, even if methodological weaknesses prevent making strong statements about the source of the change. Use of this design may be particularly valid for practitioners when they are implementing empirically supported interventions that have been previously evaluated via more rigorous methods and when one is primarily interested in documenting change rather than determining the source of change, a process referred to as Level II research (Hawkins & Matthews, 1999).

Classic Single-Case Withdrawal Designs

One of the most common categories of SCDs used in research involves systematically and repeatedly introducing and withdrawing an intervention and documenting the impact of those changes on the dependent variable of interest. This group of designs has been referred to as equivalent time-samples designs (Campbell & Stanley, 1966), withdrawal designs, and reversal designs (Kazdin, 2001). In this section, we describe common designs that are the result of permutations of the basic procedure of introducing and withdrawing intervention.

The A-B-A Design

The A-B-A design corrects the primary limitation of the A-B design by including a return to the original baseline conditions. This serves as a means of confirming experimental control over the dependent measure by allowing for repeated demonstration of experimental effect. If the dependent measure returns to a pattern that is similar to the original baseline, then stronger assumptions about the causal role of the independent variable can be made. In contrast to the A-B design, one is less concerned that some extraneous variables coincided with the introduction of the treatment phase. Although this may happen once, the possibility of another change in extraneous variables occurring at precisely the same time that the intervention is both introduced and withdrawn is so remote that it does not constitute a realistic threat to internal validity.

Although the A-B-A design corrects for problems of the simple A-B design, it has limitations. Most important for the clinical researcher, completing an investigation while the participant is exposed to baseline conditions is problematic. Let us return to the example of the adolescent client with depressive symptoms. Assume that an A-B-A design was used in which the B phase consisted of effective use of behavioral activation strategies. Ending the investigation while the participant is exposed to baseline conditions would be problematic if those conditions produced a return in data patterns similar to the original baseline. In this example, the client would be left to deal with his or her problems without the aid of intervention. Although perhaps not required to demonstrate experimental control, clinically and ethically it would seem prudent to reintroduce the intervention so as to address the presenting complaint.

The A-B-A-B Design

By introducing an additional treatment phase, the investigator creates the A-B-A-B design. This design controls for the limitation of the A-B-A design and is more methodologically rigorous due to the fact that three opportunities occur to compare the behavior patterns during baseline and treatment conditions (i.e., A to B, B to A, and A to B).

To illustrate the A-B-A-B design, consider Friman et al. (1999), who investigated the impact of an intervention for bedtime behavior problems exhibited by two children, ages 3 and 10 years. Both children exhibited bedtime problems in the form of calling out from their bedroom and leaving their room after their parents had put them to bed. Baseline data were collected across 15 days, during which time the parents were instructed to respond to the bedtime problems as they would typically. Treatment began on the 16th day and involved parents providing each child with a "bedtime pass" good for one free trip outside of the room past bedtime (e.g., get a drink of water, get another hug). Once the pass was used, it was relinquished to the parents, and no further trips were allowed. The pass was returned to the child on the subsequent night. During intervention, the parents were instructed to ignore all crying and to simply physically guide the child back into his room if he were to leave without the use of the pass.

Results of the experimental analysis are presented in Figure 11.3. Somewhat variable but increasing trends are noted in the dependent measure for both participants during the initial baseline. Following introduction of the intervention, a marked decrease in the targeted behavior occurred, to zero for the 10-year-old participant. Thus, it was demonstrated quite clearly with both participants that the behavior changed in the desired direction during the initial intervention phase. This is tempered slightly, however, with the 3-year-old participant. For him, as intervention continued the behavior problems began to increase again after the initial decrease. Reintroduction of the baseline conditions (i.e., withdrawal of the bedtime pass) resulted in the increased occurrence of the bedtime behavior problems, which decreased to zero for both participants upon the readministration of the intervention. Obtaining zero rates of behavior with the 3-year-old participant during the second intervention phase adds strength to the determination of experimental control given the initial variability in the first intervention phase.

The B-A-B Design

Although single-subject research typically starts with an initial baseline period, there may be instances in which this is not feasible. For example, perhaps the investigator is able to gather data only after some form of intervention is put into place. Or perhaps a participant's behavior problems are so severe that intervention is needed immediately. In either case, it is possible to use the B-A-B design. Although not as complete as the A-B-A-B design (because it does not allow for as many repeated comparisons of the experimental effect), it is more advantageous than the A-B-A design because it ends on a treatment phase (Hersen, 1982).

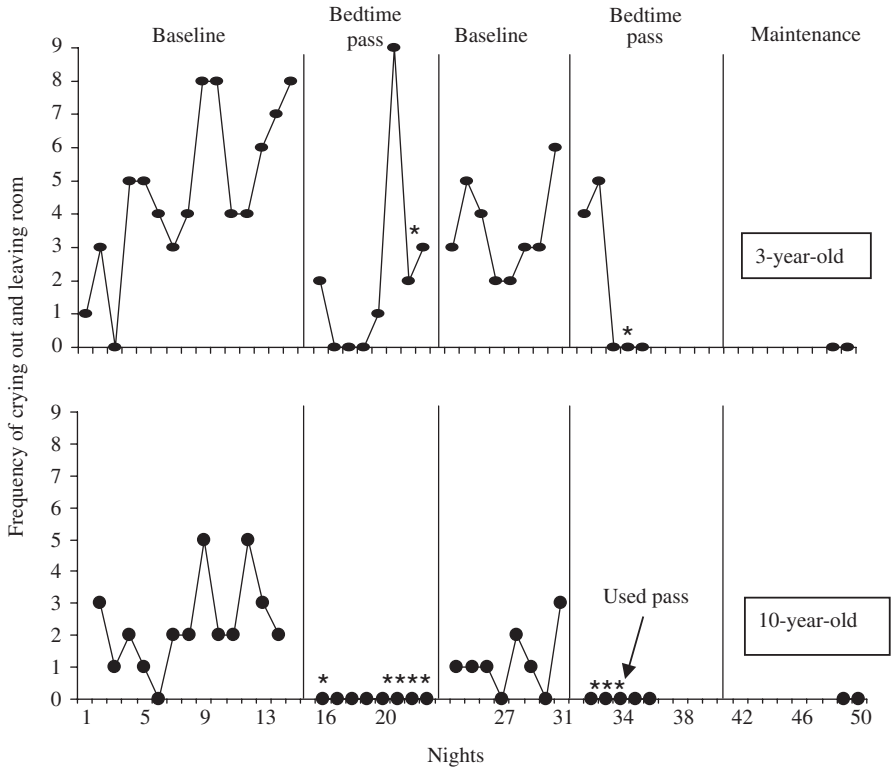

Figure 11.3 Nightly frequency of crying and leaving the room for the 3-year-old child (top) and the 10-year-old child (bottom). Asterisks indicate use of the pass. *Source:* "The Bedtime Pass: An Approach to Bedtime Crying and Leaving the Room," by P. C. Friman et al., 1999, *Archives of Pediatric and Adolescent Medicine, 153,* pp. 1027–1029, figure 1. Copyright 1999 by *Archives of Pediatric and Adolescent Medicine*. Reprinted with permission.

Variations of the Withdrawal Designs

The underlying strategy employed to demonstrate functional control using the various withdrawal designs (e.g., A-B-A, B-A-B)—the repeated introduction and withdrawal of the intervention—allows for various extensions of the basic designs. Thus, perhaps endless permutations or formations of the basic components of the designs can be created to address particular questions of interest. Several examples of such designs are considered next, although the list is not meant to be exhaustive.

A-B-A-B-A-B Design

The basic components of the A-B-A design can be extended, resulting in multiple withdrawals and administrations of the independent variable (e.g., the A-B-A-B-A-B design). Using such a design adds further support for the demonstration of functional control of the independent variable over the dependent variable. By demonstrating multiple times that the data patterns change in the expected direction when the treatment is added and withdrawn, the researcher can be more confident

that the effect is due to the particular treatment variables. Although this might not be necessary when the treatment effect is large (i.e., when there is a significant difference in the data pattern between baseline and treatment conditions), repeatedly demonstrating the desired effect may be necessary when the effect is small (i.e., when the difference between the two conditions is small). Observing minimal or moderate changes multiple times may provide more convincing evidence that the independent variable is producing the impact, above and beyond natural variability.

A-B-A-C-A-C Design

Introducing a condition in which the effects of a second independent variable are assessed, as with the A-B-A-C-A-C design, also can extend the basic A-B-A design. In such a design, the investigator compares the impact of treatment conditions B and C on the dependent variable. However, with such a design, one is not able to compare the *relative* effects of the different independent variables because they are confounded by the extraneous variable of time (Hersen, 1982). Further, order effects may confound the findings. Perhaps the conditions present in C have an impact only after the participant is exposed to the conditions present in B. With such a design, one is unable to determine whether this is the case, and thus there is a threat to the internal validity of the study. If completing intersubject comparisons, one can control for this by using a counterbalancing technique (i.e., participants 1 and 3 are exposed to the conditions in the following order: A-B-A-C-A-C, whereas participants 2 and 4 are exposed to them in the following order: A-C-A-B-A-B). By using counterbalancing, the researcher is able to determine whether the independent variables produce similar effects regardless of order of presentation. Thus, if all participants respond to condition C in similar ways regardless of when they were exposed to this condition, then one has greater confidence in the effect of C alone.

Interaction Designs

As stated earlier, single-subject research typically involves a change in only one variable at a time across conditions. This rule can be applied in a way that allows the investigator to evaluate the combined effects of multiple independent variables by using a withdrawal design. Specifically, by introducing a condition in which two independent variables are present, it becomes possible to assess the impact of the combined intervention relative to a single intervention. In some situations, a particular treatment may produce minimal impact or may produce an impact that is less than that which is desired. By extending the basic components of the A-B-A design, one can assess the impact of adding additional treatment components.

Assume, for example, that a researcher is interested in treating teen smoking. Figure 11.4 shows hypothetical data in this scenario. After establishing a baseline of the number of cigarettes smoked per day, the impact of a reinforcement-based treatment is evaluated. As is shown, although the number of cigarettes smoked decreases somewhat, the desired goal of zero cigarettes smoked is not achieved. Thus, the investigator adds an additional component to the intervention, the use of a nicotine patch. This results in a condition labeled “BC” because it involves

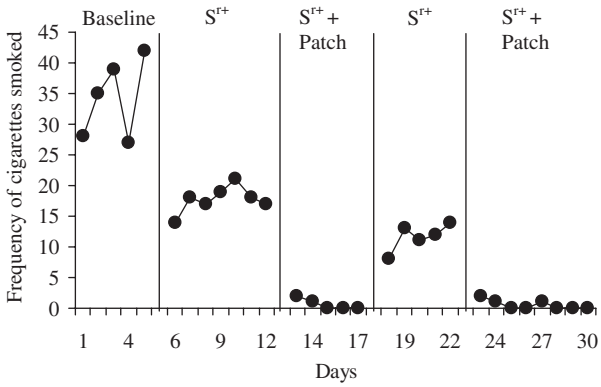

Figure 11.4 Number of cigarettes smoked per day. Hypothetical data of the assessment of interventions for teen smoking.

both components (i.e., reinforcement + nicotine patch). As shown in Figure 11.4, the combined intervention results in decreases in the dependent variable to zero. When the investigator withdraws the nicotine patch, the data patterns return to being similar to the original treatment phase. Finally, the combined intervention is reintroduced, resulting in the elimination of smoking. As illustrated, it is possible to demonstrate the impact of the combined intervention with such a design. Put another way, one is able to investigate the relative impact of condition C over condition B. The number of combinations is theoretically endless, potentially resulting in conditions such as BCD, BCDE, and so forth.

STAGE-PROCESS DESIGNS

Although the withdrawal designs have proven remarkably useful in scientific inquiry, partly because of the flexibility in applying the basic concepts of repeated observations and replication of experimental effect in multiple ways, there are situations in which their use is not appropriate. Withdrawal designs may be considered inappropriate due to a variety of reasons (e.g., ethical, practical). As mentioned earlier, withdrawal of a treatment may be unethical if the behavior problem is severe or life threatening. Also, it may not be possible to withdraw or remove certain therapeutic strategies. For example, once instructions have been given to a participant, it is impossible to remove them. Although the researcher could stop giving them in a particular condition, the fact that the instructions were given earlier could result in a change that is not removable. As another example, if the intervention results in behavior that is supported beyond the treatment conditions, then withdrawal of the intervention may have no impact on the behavior. For instance, if a clinician teaches someone to be appropriately assertive, the reactions of others may maintain use of assertive behavior regardless of whether the clinician removes the intervention. In this scenario, the lack of a return to baseline data patterns might suggest a lack of functional control of the independent variable. However, such is likely not the case; instead, other controlling variables continue to maintain the behavior patterns in the absence of the intervention. Because of these limitations, various other SCDs

are used. These include the multi-elements design, multiple baseline designs, and changing criterion designs.

The Multi-Elements Design

The multi-elements design (also referred to as the simultaneous treatment design or the alternating treatments design) differs from other SCDs in that multiple conditions (i.e., baseline and treatment conditions, two or more treatment conditions) are conducted in rapid succession, with the order of presentation typically determined through random selection, and compared against each other (Miltenberger, 2001). For example, perhaps baseline conditions are in effect on one day, treatment conditions the next, and so forth. Thus, unlike withdrawal designs, the effects of the different experimental conditions are evaluated across the same time frame. This helps eliminate the possibility of extraneous influences on the dependent variable during only one experimental condition. In other words, any extraneous variable is going to impact the dependent variable during all conditions because they are occurring essentially in conjunction. One is able to rule out the extraneous influence as the cause of any differences noted across conditions.

With the multi-elements design, often there are three phases: baseline, comparison (rapid alternation between two or more conditions), and the use of the effective intervention (Holcombe et al., 1994). In some situations, however, the baseline might not be necessary. This may be particularly true if one of the comparison conditions is a baseline condition. One treatment condition is judged to be superior if it produces data patterns in the expected direction at a level that is greater than other conditions. Another component of the multi-elements design is that an equal number of sessions of each condition should be conducted. To ensure discriminated responding across conditions, researchers often pair separate but salient stimuli with each condition. This design may be particularly useful if the investigator is comparing interventions that have an immediate effect, and when the dependent measure is particularly sensitive to changes in stimulus conditions (i.e., reversible; Holcombe et al., 1994).

The multi-elements design can be used to compare a variety of different variables of interest. For example, the design can be used to compare different assessment conditions, such as in functional analysis research (e.g., Anderson, Freeman, & Scotti, 1999). In addition, this design can be used to compare a treatment condition to baseline (e.g., Freeman & Piazza, 1998) or to compare multiple treatments (e.g., Kahng, Iwata, DeLeon, & Wallace, 2000). Such an approach allows the investigator to assess which intervention may prove to be the most effective in changing the targeted dependent variable. The benefit of such a design over other designs is the ability to make such treatment comparisons in a relatively short amount of time.

Multiple Baseline Designs

There are three types of multiple baseline designs: (1) multiple baseline across behaviors, (2) multiple baseline across persons, and (3) multiple baseline across settings (or situations; Hersen, 1982; Miltenberger, 2001). With the multiple baseline design across behaviors, the impact of an intervention across different behaviors

emitted by the same person is evaluated. As such, this is a within-subjects design. The intervention is applied sequentially to the different (presumably) independent behaviors. An example is evaluating the use of a particular reward strategy in affecting a child completing chores, cleaning his room, and completing his homework. The second design—multiple baseline across persons—involves the evaluation of the impact of a particular intervention across at least two individuals *matched* according to relevant variables who are presumed to be exposed to identical (or at least markedly similar) environments. For example, the investigator may compare an intervention across two students who attend math class with a particular teacher, one who attends in the morning and one who attends in the afternoon. Finally, with the multiple baseline across settings design, a particular intervention is applied sequentially to a single participant or group of participants across independent environments (e.g., home and school).

Technically, there must be at least two separate dimensions (i.e., behaviors, settings, or persons) present to utilize a multiple baseline design, although convention suggests a minimum of three or more. Multiple baseline designs are characterized by the presence of only two conditions: baseline and treatment. However, unlike the simple A-B design, treatment is introduced in such a way that one is able to evaluate experimental control of the independent variable. Specifically, the baseline condition is extended for increasing lengths of time as the intervention is introduced with the other dependent variables. Thus, these designs are particularly useful for studying irreversible effects because replication is achieved without withdrawal and reintroduction of the independent variable (Perone, 1991).

An assumption underlying the multiple baseline designs is that the dependent measures (i.e., behaviors, persons, settings) are functionally independent. If this is the case, a change in one dependent variable as a result of the independent variable should not produce changes in the other dependent variables. Treatment effects, therefore, are inferred when the dependent measure changes only when the intervention is applied to it. Variables still exposed to baseline conditions should show little to no change when the treatment is introduced with the other dependent variables. In this way, the multiple baseline designs are weaker than other designs (Barlow & Hersen, 1984; Hersen, 1982). Specifically, experimental control is inferred based on the comparison of nontreated dependent variables as compared to the treated variables, as well as the replication of the effect across behaviors, persons, or settings.

If the dependent variables are not functionally independent (i.e., treatment with one variable produces changes in the other variables), then the ability to establish experimental control is compromised. Determining a priori whether dependent variables are independent can be difficult, however. Take, for example, a situation in which the dependent measures are three problematic behaviors of a student in a classroom (i.e., talking out of turn, throwing spit wads, getting out of seat without permission). As an intervention, the investigator instructs the teacher to respond in a particular way to the behaviors (e.g., differential reinforcement of competing behavior) and evaluates the impact of the intervention using a multiple baseline design. The student may well learn to change each of the behaviors simply by exposure to the treatment for one behavior (e.g., differential reinforcement when the student

talks out of turn affects other behaviors). As this example illustrates, as an artifact of the lack of independence, intervention with one dependent variable may produce changes in the other variables, resulting in the *inability to demonstrate* experimental control. This is problematic because the intervention may be the controlling variable in this situation, but this cannot be demonstrated in a convincing manner.

Kazdin and Kopel (1975) provided three recommendations for addressing problems of dependence across dependent variables. First, they recommend selecting dependent measures that are as topographically distinct as possible as a means of increasing the likelihood that they are independent. However, such an approach still relies on successful guessing, as topographically distinct behaviors may still be functionally related. Second, they suggest that investigators utilize four or more baselines as compared to two or three. They argue that, by increasing the number of baselines, one is increasing the likelihood of selecting measures that are independent. However, as pointed out by Hersen (1982; Barlow & Hersen, 1984), the probability of interdependence may be enhanced with a larger number of dependent variables. Third, Kazdin and Kopel recommend that, when faced with the occurrence of dependence across variables, the investigator withdraw and then reintroduce the independent variable. By doing so, it may be possible to demonstrate adequate experimental control. Although the investigator originally may have selected the multiple baseline design to avoid using a withdrawal design, this may be necessary to adequately demonstrate control.

To a certain extent, there are competing clinical and research goals when considering multiple baseline designs. From a research standpoint, the goal is to identify target dependent variables that are less likely to be influenced by changes in other targeted dependent variables. However, clinically, one is often interested in designing interventions that not only address targeted behavior patterns, but are also likely to produce change that generalizes to other relevant behaviors or contexts. For instance, a therapist working with an adolescent with social anxiety is likely interested in implementing interventions that address anxious responding not only in the classroom setting, but also in the lunchroom, during extracurricular activities, and elsewhere. The clinical goal is to teach skills that will generalize to relevant contexts. Of course, whether generalization happens naturally (e.g., teaching use of relaxation skills in the classroom results in the person also using those skills in the lunchroom) or must be programmed explicitly (e.g., prompting the adolescent to use skills across settings) will be determined only as one progresses.

Nonconcurrent Multiple Baseline Design

The presumption of the multiple baseline designs is that the measurement of the different dependent variables occurs simultaneously. In this way, the designs control for threats to internal validity such as history effects. However, there may be situations in which it is particularly difficult to simultaneously observe multiple individuals who meet the specified criteria, thus limiting one's ability to utilize the multiple baseline across persons design. This may be particularly true in applied or clinical research (Hayes, 1985). In such situations, it may be possible to utilize the nonconcurrent multiple baseline design, originally described by Watson and Workman (1981).

With the nonconcurrent multiple baseline across persons design, each baseline length is predetermined (e.g., 3, 6, 9 days). Then, when a participant with the requisite features is available, he or she is exposed to a randomly selected baseline period. From this point, the methodology is conducted in the same manner as the simple A-B design: baseline observations are conducted for the predetermined length, followed by the application of treatment. In this way, baseline and treatment data are obtained from multiple participants, allowing for adequate between-persons comparisons. If a participant fails to display an acceptable baseline pattern, that individual is dropped from the investigation, although his or her eventual reaction to treatment may still be useful as a replication (Watson & Workman, 1981).

Some (e.g., Harris & Jenson, 1985a, 1985b) have argued that the nonconcurrent multiple baseline design across persons is equivalent to a series of A-B designs with replication. This nomenclature has been recommended because multiple baseline designs rely on simultaneous data collection as a means of demonstrating adequate control over threats to internal validity. As described earlier, by showing that the change in the dependent variable occurs only when the treatment is introduced, the investigator has confidence in the causal relationship. This is strengthened when data with several individuals are collected simultaneously because the design controls for history effects. Harris and Jenson argued that the nonconcurrent multiple baseline design does not control for such effects because the data are not collected across individuals simultaneously; thus, the individual is not exposed to the same environment.

By varying the length of baseline, however, the nonconcurrent multiple baseline design does control for threats to internal validity (Hayes, 1985; Mansell, 1982). Specifically, each time the investigator demonstrates that the dependent measure changes when the treatment is introduced, regardless of the length of baseline, the likelihood that an extraneous variable produced the change for each participant is greatly reduced. Further, by using increasing lengths of baseline, the researcher controls for the possibility that the exposure to baseline conditions naturally produces changes in the dependent measure. Thus, although generally considered one of the weaker SCDs, the nonconcurrent multiple baseline across persons design can serve as a useful methodology if other factors limit the ability to utilize a more stringent design.

Changing Criterion Design

The changing criterion design shares features with both the simple A-B design and the alternating treatments design (Hersen, 1982; Miltenberger, 2001). Specifically, this design is characterized by the presence of only one baseline and one treatment phase. However, what differentiates this design from the A-B design is that the treatment condition is defined by the sequential introduction of different performance goals. In other words, the treatment phase is applied until the targeted dependent variable achieves a specified level of performance. At that time, the goal (i.e., criterion) of performance is altered, and the intervention continues until the behavior again achieves the desired level. Changes in the criterion occur until the dependent measure is occurring at the desired terminal level. As such, the changing

criterion design is particularly well suited for situations in which the investigator is interested in evaluating shaping programs that are expected to result in increases or decreases in the dependent measure (e.g., decreased cigarette smoking, increased level of exercise; Hersen, 1982). Evaluation of the intervention as the causal agent occurs through two comparisons: between the occurrence of the dependent measure during baseline and during treatment, and between the occurrence of the dependent measure across the different levels of the intervention. If the dependent variable changes in the desired direction only when the criterion changes, then the investigator can have confidence in the controlling nature of the independent variable.

SINGLE-CASE DESIGNS AND EMPIRICALLY SUPPORTED TREATMENT

In 1993, the Division of Clinical Psychology of the American Psychological Association (APA) initiated efforts to define the criteria for determining whether treatments for clinical problems are adequately supported by empirical evidence. The task force charged with this assignment, now referred to as the Committee on Science and Practice (CSP), issued their first report in 1995 (Task Force, 1995), with updates offered in the following years by the committee (Chambless et al., 1996) and independently by members of the committee (Chambless & Hollon, 1998). The CSP provided guidelines or decision rules for defining empirically supported treatments (EST), as well as a list of psychotherapies meeting those criteria at the time. Establishing a process for determining whether an intervention approach is scientifically grounded, as well as a list of ESTs, has been viewed by many as important for promoting evidence-based practice within the fields of psychotherapy and clinical psychology (Herbert & Gaudiano, 2005), though others have questioned both the exercise and the criteria offered (for a summary, see Beutler, 1998).

The importance of evidence-based practice (a broader term used in other disciplines such as medicine and special education) has increased over the years and is now influenced by multiple sources. For instance, increasingly insurance companies are looking to providers to utilize therapies with known efficacy (Hayes et al., 1999; Morgan & Morgan, 2001). Additionally, certain federal initiatives and funding agencies are increasingly giving funding priority to proposed studies that utilize research methods to determine effectiveness of interventions (Horner et al., 2005; McDonald & O'Neill, 2003). Currently, many fields, including clinical psychology, tout between-group designs as the most scientifically rigorous method of determining effectiveness, with randomized clinical trials seen as the gold standard. However, SCDs can play an important role in determining empirical evidence for effectiveness.

The CSP (Task Force, 1995; Chambless et al., 1996) as well as modifications made since (Chambless & Hollon, 1998) provide guidelines for determining whether an intervention is "well established" or "probably efficacious." Initial CSP statements (Chambless et al., 1996; Task Force, 1995) included guidelines for determining whether an intervention was empirically supported using either group or single-subject designs. Regarding the latter, they recommended considering an intervention's efficacy to be well established if a large series of studies using SCDs

demonstrate intervention effects, and the intervention to be considered probably efficacious if a small series of studies supported the intervention. Chambless et al. specified that the series of studies needed to include nine or more participants total to be considered a large series.

Horner and colleagues (2005) provided greater specificity regarding how SCDs may be used to determine empirical evidence for interventions or practices. Though their discussion occurs in the context of educational practices, it has relevance for clinical child and adolescent psychology. Specifically, they provide five standards that must be met for SCDs to demonstrate evidence of efficacy for a practice (e.g., intervention, strategy, approach):

> (a) The practice is operationally defined; (b) the context in which the practice is to be tested is defined; (c) the practice is implemented with fidelity; (d) results from single subject research document the practice to be functionally related to change in dependent measures; and (e) the experimental effects are replicated across a sufficient number of studies, researchers, and participants to allow confidence in the findings. (pp. 175–176)

They elaborate on these standards, providing important information regarding suggested guidelines for ensuring that each is met. In particular, they suggest more rigorous criteria for the number of studies and participants using SCDs than the CSP guidelines. Specifically, they recommend that, for an intervention or practice to be considered efficacious, there be a minimum number of five studies using SCDs, that the studies be conducted by at least three different researchers or research teams, and that the number of participants included in those five studies total at least 20. Adherence to these criteria ensures that the effect of the independent variable on the dependent variable has been repeated multiple times by several researchers, increasing the internal and external validity of findings (Horner et al., 2005).

SUMMARY

An overview of the basic characteristics of various single-subject research designs has been provided, based in the context of clinical child and adolescent psychology. Single-case designs are distinguished from group design approaches to scientific inquiry by several features, including repeated observation of the dependent variable, replication of treatment effects, intrasubject and intersubject comparisons, visual analysis of individual participant data, and systematic manipulation of independent variables. Further, direct and systematic replications are used to establish the generality of the findings. Together, these features allow the investigator to guard against, or to detect when present, threats to internal and external validity. Thus, their use in psychological and psychotherapy research can allow investigators to answer questions of interest in a manner that meets the requirements of the scientific method (Hayes et al., 1999; Hilliard, 1993; Morgan & Morgan, 2001).

A particular strength of SCDs is their flexibility, allowing an investigator to change the purpose of the study when warranted as the investigation progresses (Barlow & Hersen, 1984; Holcombe et al., 1994). In fact, some have argued that the defining features of SCDs allow researchers to be more intimately involved with

their subject matter as compared to group designs (e.g., Perone, 1999). Assume, for example, that baseline data are variable. Rather than simply moving to the treatment phase in which the impact of a particular intervention is assessed, the investigator may wish to determine the source of the variability (Sidman, 1960). In this way, the investigation becomes an extended assessment that involves the identification of extraneous variables affecting the dependent variable. Once this is done, the researcher may wish to introduce a treatment (which, consequently, may be more effective now that sources of variability have been identified and can be accounted for).

Or perhaps the researcher was initially interested in the impact of a particular intervention on the dependent variable. For instance, perhaps a researcher is investigating the impact of time-out as an intervention for a noncompliant child using an A-B-A-B withdrawal design. Baseline data are collected on the rate of compliance per day. Then use of time-out as a consequence for noncompliance is introduced, which produces no change in the occurrence of the noncompliance. If this were a group design comparing the effects of time-out on noncompliance against either no treatment or an alternative treatment, the investigator would need to continue to utilize the exact procedures across all participants to assure experimental integrity, or consider the research unacceptable and stop the experiment altogether. With the SCDs, however, the investigator can extend the original lack of findings by evaluating another treatment (e.g., praise for compliance). Specifically, if there is no difference in the occurrence of the dependent variable between the initial baseline and treatment phases (A and B phases), it would be appropriate to introduce the second intervention (C phase) without returning to the original baseline conditions. If this produced an effect, withdrawal of the intervention would involve a return to the B phase. In other words, because data patterns do not differ between A and B, it is possible to use B as the comparison against patterns during the C phase. Thus, SCDs may be more economical than group designs in that they are malleable and changeable based on the data obtained.

In this way, single-subject designs are useful not only in the context of psychological research, but also in the context of demonstrating the effectiveness of clinical interventions (Hayes et al., 1999; Hilliard, 1993; Kazdin, 2003). Psychotherapy as a process involves, first, hypothesis building (i.e., developing reasonable explanations for the client's current conditions) and, second, hypothesis testing (i.e., assessing the effectiveness of an intervention expected to be effective based on the original hypothesis). Initial interventions may not be effective, and thus clinicians may experiment with others until improvements are noted. The flexibility of SCDs can allow clinicians to collect data and manipulate variables in a way that allows for adequate demonstration of therapeutic outcome as a result of specific interventions. Therefore, their use in psychotherapy can facilitate a clinician truly operating as a scientist-practitioner.

Multiple investigations have demonstrated that most graduates of clinical psychology programs, including those who come from scientist-practitioner programs, are involved in clinical endeavors rather than research pursuits (for review, see Gelso, 2006). As such, increased emphasis on training in empirical designs that have relevance for clinical settings seems prudent; in fact, cases for such an emphasis

have been made (Blampied, 1999; Hayes et al., 1999). As Hayes and colleagues noted, one of the major challenges in fostering greater connection between practice and scientific inquiry has been the lack of research designs appropriate for clinical settings. Single-case designs seem to have several distinct advantages over group designs in relation to conducting research in applied settings (e.g., applied at the individual level, more malleable so as to be responsive to information learned through the process); thus, they seem to have a greater probability of being incorporated into clinical practice.

Rigorously applying SCDs to clinical practice presents challenges, however. Specifically, as articulated at various places throughout this chapter, the requirements needed to experimentally demonstrate that changes in the dependent variable are the result of an intervention using SCDs can be at odds with the realities of clinical practice. A fundamental issue is identifying the variables (e.g., context, type of behavior, question being asked, population being served, resources available) for deciding which SCD is most appropriate for use in a particular clinical context for a specific purpose. For instance, if a treatment has already been established as evidence-based, then a simple A-B design may be appropriate as a mechanism of monitoring change. Further, reporting to the scientific community multiple applications of an intervention's effects using simple A-B designs will help support clinical application of the intervention that was originally tested in highly controlled conditions. On the other hand, because many interventions (including those identified as empirically supported according to APA criteria) have not been subjected to effectiveness trials or used with special populations (e.g., rare disorders, with individuals with comorbid disorders), use of SCDs provides an opportunity to establish effectiveness of the treatment in a clinical context and with variations of client characteristics that were not included in randomized clinical trials. One can envision a multistage process in which SCDs are initially used in clinical contexts to establish treatments as evidence-based; following this, treatments can be used more clinically in a single-case decision-making framework that meets certain purposes (e.g., case formulation, treatment design, monitoring progress) but that would not require withdrawal or multiple baseline designs, in part because the intervention had been established as effective in previous research.

The issue of how to successfully incorporate SCDs into clinical practice is complex, and full attention to it is beyond the scope of this chapter. We raise the issue here to encourage people to consider how to integrate SCDs into clinical practice, and how the pursuit of doing so is beneficial for not only clinicians and their clients (e.g., through documentation of change, better informed decision making) but also for the research community (e.g., by verifying previous outcomes with unique populations, by raising additional questions to be addressed through more comprehensive or precise research).

Single-case designs have critical underlying assumptions and basic characteristics that separate them from group designs, while still ensuring that data are collected and analyzed in an experimentally rigorous manner. Single-case designs are typically aligned with operant traditions of behavioral psychology, in particular Skinnerian radical behaviorism and the applied branch of this science, given ardent endorsements by critical leaders in this field (Sidman, 1960; Skinner, 1974).

However, as has been noted (e.g., Hayes, 1981), these designs are largely theory-free and can be useful for any investigator (whether a researcher or a practitioner) who gathers time-series data in a systematic fashion. Thus, regardless of theoretical premise of a given researcher or practitioner, SCDs may be appropriate for determining causal or functional relations between changes in the dependent variable and introduction of the independent variable. This is not to suggest that solid training in the philosophical and theoretical context in which these designs were developed is not important (Blampied, 1999), but instead to support their broad utility in the diverse field of clinical child psychology. Further, their utility is exemplified by the fact that there are frequently calls for more single-subject research in various fields of psychology and beyond (e.g., Cook, 1996; Hrycaiko & Martin, 1996; Lundervold & Bellwood, 2000). Not only are SCDs useful for explicit research purposes, but they can also be used to assist practitioners in determining the impact of their efforts. Because of this, understanding SCDs is important for both researchers and clinicians alike.

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CHAPTER 12

Personality in Childhood and Adolescence

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There has been a great deal of research on the personality characteristics of youth over the past 3 decades. In part, this blossoming of research has stemmed from the availability of models of general personality theories and the broadening of research within the adult literature on general personality. Much more research is being conducted on children and adolescents, and there has been a greater emphasis in better understanding the etiology, general structure, and stability of personality of youth and its connection to psychopathology. This area of study is somewhat controversial, as noted by the current nomenclature's lack of formal acknowledgment of personality and personality disorders in children and adolescence. However, our current nomenclature does suggest that personality disorders can be recognized in childhood and adolescence, and the research has been advancing to show that there are likely broad identifiable personality traits in children and adolescents.

The purpose of this chapter is to review aspects of research on early emerging individual differences during the years from infancy through late adolescence. First, we provide a general description of temperament and personality and discuss how the two have been conceptualized over time. In this section of the chapter we also discuss the potential links between temperament and personality and examine whether the two might be one and the same. Second, we describe what is known about the structure of personality in children and adolescents. We discuss four primary models of general personality developed over the past several decades and attempt to integrate the competing models. Third, we review the stability of childhood personality over time. We examine the data on personality models, showing the existence of both stability and change across major age groups. Fourth, and importantly, we explain some of the potential processes involved in continuity and change. Research examples of several processes are provided to shed light on this important issue. In this section of the chapter we also discuss Thomas and Chess's (e.g., Thomas & Chess, 1977; Thomas, Chess, & Birch, 1968; Thomas, Chess, Birch, Hertzig, & Korn, 1963) notion of the "goodness of fit" between the child's temperament and personality and his or her predominant environment. Fifth, we discuss childhood personality and its connection to both positive (adaptive) and negative (psychopathology) outcomes. This section of the chapter is important if

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we are to better understand how normal personality traits can lead to meaningful and fulfilling lives, and it also sheds light on how certain personality traits might be linked to psychopathology. Finally, we look at the *Diagnostic and Statistical Manual of Mental Disorders*, fourth edition, text revision (*DSM-IV;* American Psychiatric Association, 2000) model of pathology and its application to personality problems in youth. There is not an easy alliance between general models of personality and the *DSM-IV* categories of pathology, although researchers are beginning to bridge these gaps. The chapter concludes with suggestions for future research in this area, highlighting the exciting research on the near horizon for examining personality in children and adolescents as it relates to both positive and negative outcomes.

DESCRIPTION OF PERSONALITY IN CHILDREN AND ADOLESCENTS

Few would argue that children differ in their personalities. Some children are outgoing and exuberant, whereas others are more quiet and reserved. Some are irritable and emotionally labile; others show equanimity from an early age. Some are aggressive and others are gentle. These individual differences, although observed by parents, peers, teachers, and psychologists in everyday life, are not always easily described in research and clinical practice (Caspi & Shiner, in press; Shiner, 2006).

Traditionally, personality has been viewed as a stable and enduring set of characteristics that affects one's perceptions of oneself, others, and the world. The *DSM-IV* defines personality traits as "enduring patterns of perceiving, relating to, and thinking about the environment and oneself that are exhibited in a wide range of social and personal contexts" (American Psychiatric Association, 2000, p. 630). A personality disorder, in turn, is defined as an enduring pattern of inner experience and behavior that has deviated markedly from the expectations of the individual's culture, is pervasive and inflexible, has an onset in adolescence or adulthood, is stable over time, is evident across a broad range of personal and social situations, and leads to distress and impairment (p. 630).

Although there have been numerous studies of personality in adult samples, there has been limited research in the direct emergence or development of child personality and the concomitant problems that can stem from personality (Bernstein, Cohen, Velez, Schwab-Stone, Siever, & Shinsato, 1993). There is also little information on change or stability in childhood and adolescent personality. Part of the concern, when thinking about personality in children and adolescents is a general uneasiness that the identification of traits (or disorder) suggest little opportunity for change. Some scientists and clinicians believe that personality has not yet crystallized in children or adolescents; in this case, the existence of personality and/or personality disorder makes little conceptual sense. Others (e.g., Shapiro, 1990) question only whether general personality can be identified and personality disorders diagnosed before adolescence, when adultlike identity is thought to coalesce (Blos, 1967; Erikson, 1968). This is a nontrait and nondevelopmental approach because it does not consider the process by which, at each phase of development, an age-appropriate identity and personality exist and continue to form. However, researchers have noted

that ignoring the existence of a general blueprint for personality and the developmental line of the structuring of identity and personality means ignoring the ways personality development can occur across time but also how it might be positively or adversely affected at any age (Roberts & Wood, 2006; Shiner, 2006; Tackett, 2006).

BRIEF HISTORY OF THE STUDY OF PERSONALITY IN CHILDREN AND ADOLESCENTS

Individual differences in infants and children have focused on temperament and personality. The scientific study of early temperament can be traced to the seminal work of Thomas and Chess, who initiated the New York Longitudinal Study to study the significance of biologically based temperament in infancy and childhood (Thomas, Chess, Birch, Hertzig, & Korn, 1963). Thomas and Chess viewed temperament as a style of behavior and abilities. Their work challenged the way social development was studied at the time because they emphasized the two-way interaction of biology and environment in shaping children's outcomes. Prior to this view, many researchers and clinicians viewed child personality and outcome as being influenced almost exclusively by rearing environments. Thus, the work of Thomas and Chess was an important step in research because it suggested that there are preexisting personality styles from which individuals then operate.

Temperament has been defined differently by different researchers, and the link between temperament and personality is not fully understood (Frick, 2004). There are a variety of points of view on this topic: (a) that temperament is different from personality, (b) that temperament is linked to certain personality traits, and (c) that temperament is personality. Developmental and child psychologists traditionally describe stable and observable differences in young children by relying on temperamental constructs such as negative emotionality (Thomas & Chess, 1977) or emotionality (Buss & Plomin, 1984), sociability (Buss & Plomin, 1984) or surgency (Rothbart & Derryberry, 1981), task persistence (Thomas & Chess, 1977) or effortful control (Rothbart & Derryberry, 1981), and activity level (Buss & Plomin, 1984; Goldsmith & Campos, 1982; Putnam, Ellis, & Rothbart, 2001; Thomas & Chess, 1977). Researchers assume that temperament differences are expressions of neurobiological mechanisms that have a strong genetic basis (Mervielde, De Clercq, De Fruyt, & Van Leeuwen, 2005). In contrast, personality traits are mainly used to chart stable latent differences in adults, presumed to be partly influenced by temperament and interaction with the environment. According to this model, temperament can be considered a direct precursor to personality and, thus, is the earliest expression of individual difference.

Temperament is thought to be most directly observable during infancy and toddlerhood (Goldsmith et al., 1987) and to make up the entirety of personality during these early years (Caspi & Shiner, in press; Shiner & Caspi, 2003). It is thought that as children develop, early temperament traits develop into broader, more inclusive higher order personality traits as well as increasingly differentiated lower order traits (Buss & Finn, 1987; Caspi, Roberts, & Shiner, 2005; Caspi & Shiner, in press; Lahey, 2004; Rothbart & Ahadi, 1994; Shiner, 1998). Specifically, personality

develops as children progress through various cognitive and emotional stages that increasingly allow them to interact with, experience, and respond to the world in more complex ways (Caspi, 2000; Roberts & Wood, 2006; Rothbart & Ahadi, 1994). Thus, the structure of personality may change as children gain new skills (e.g., motor or language), gain the capacity to regulate emotions, and develop a sense of self (Roberst & Wood, 2006; Rothbart & Ahadi, 1994; Shiner, 1998, 2006; Shiner, Masten, & Tellegen, 2002). However, despite growing consensus that temperament traits make up the core of later personality, a clear understanding of the developmental relationship between temperament and personality is lacking (Halverson et al., 2003; Shiner & Caspi, 2003). This model is not embraced by all researchers, and the idea of temperament as "style" has been critiqued as vague and unnecessarily narrow. In fact, there are a variety of definitions of the term temperament (see Table 12.1), and some researchers claim that behavioral styles even early on are both biologically and environmentally influenced (see Frick, 2004; Plomin & Daniels, 1984), raising the question of whether the distinction between temperament and personality is necessary.

The resurgence of trait psychology, and especially the preponderance of the five-factor model (FFM; Digman, 1990), challenged both the viewpoint that personality could not be assessed in youth and that it could not also be considered a biologically based constellation of differences in youth. To elaborate on this point, personality psychologists have made several key and compelling arguments about the applicability of personality to children and adolescents. First, personality psychologists tend to agree that five broad dimensions—neuroticism, extraversion, openness, agreeableness, and conscientiousness—can be considered the basic dimensions underlying adult personality. Personality psychologists interested in the developmental antecedents of this FFM later showed that these five dimensions are also applicable to individual differences in children and adolescents (Digman, 1963; Digman & Inouye, 1986; Goldberg, 1993; John, Caspi, Robins, Moffitt, & Stouthamer-Loeber, 1994; Kohnstamm, Halverson, Mervielde, & Havill, 1998;

Table 12.1 Definitions of Temperament

Source	Definition
Buss (1995)	"Inherited personality traits that appear during the first two years of life and endure as basic components of personality." (p. 49)
Goldsmith (1996)	"Individual differences aspect of emotionality apparent early in development." (p. 221)
Kagan & Snidman (1991)	"Initial, inherited profiles that develop into different envelopes of psychological outcomes." (p. 856)
Rothbart & Derryberry (1981)	"Individual differences in reactivity and regulation assumed to have a constitutional basis." (p. 40)
Thomas & Chess (1977)	"The constellation of behaviors exhibited at any one age-period [and] are the result of all the influences, past and present, which shape and modify behaviors in a constantly evolving interactive process." (p. 171)

Note: Frick (2004) identified these definitions with the exception of Thomas and Chess earlier and also outlined specific dimensions of temperament.

Lamb, Chuang, Wessels, Broberg, & Hwang, 2002). In a key study, John et al. introduced the basic personality dimensions to developmental psychologists and demonstrated that the "little five" predict externalizing problem behavior in children.

McCrae and his colleagues (McCrae et al., 2000) cogently argued against the artificial distinction between temperamental constructs and personality traits, noting there are strong empirical and conceptual links between the domains of temperament and personality. The defining characteristics of temperament variables also apply to traits, including genetic basis, early observability, and pervasive impact on a range of behaviors (McCrae & Costa, 1997). In the past decade, behavior genetic studies have consistently documented the strong genetic basis of FFM traits (Jang, McCrae, Angleitner, Riemann, & Livesley, 1998), with heritability estimates ranging from .40 to .60 depending on the trait and method used for indexing the trait. McCrae and Costa (1996) conceptualized the FFM dimensions as basic tendencies shaping interactions with the environment, resulting in characteristic (mal)adaptations such as interests, values, and attitudes but also problem behavior and psychopathology. In another key study, Mervielde et al. (2005) compared the major models of temperament with the FFM and showed similarities and conceptual overlap between them. Frick (2004) commented that much of the work on temperament, personality, and psychopathology has been separated largely by discipline boundaries, which may account for differences in the terminology but not necessarily differences in the underlying concepts. Caspi et al. (2005) recently concluded that temperament and personality increasingly appear to be more alike than different.

Despite differences in terminology and perhaps conceptual differences, it has been suggested that the field of temperament and childhood personality needs a unifying framework to allow organization and integration of empirical findings and to facilitate communication among researchers within the field, as well as with researchers in clinical, developmental, and personality psychology subdisciplines (Frick, 2004; Halverson et al., 2003; Lonigan, Vasey, Phillips, & Hazen, 2004; Shiner & Caspi, 2003).

The popularity of the FFM in adult personality research has contributed to the growing body of literature documenting investigations of the five-factor structure in children and adolescents (e.g., Caspi & Shiner, in press; Digman, 1990; Graziano & Ward, 1992; John et al., 1994; Lamb et al., 2002; Martin, Wisenbaker, & Huttunen, 1994; Widiger & Trull, 2007). These studies tend to find evidence for the five-factor structure, although there is some evidence to suggest that openness to experience may not emerge until adolescence (Lamb et al., 2002). Halverson et al. (2003) in a novel study used a bottom-up approach to the structural debate by attempting to construct a cross-cultural, cross-age measure of childhood personality. These researchers collected parental descriptors of children (ages 3 to 12) across eight countries and, using a combination of rational (sorting done by focus groups) and factor analytic techniques, derived a somewhat new model of childhood personality. The resulting model consisted of a five-factor structure, although openness to experience was linked to only a single intellect scale, which reflects characteristics such as precociousness, intelligence, and speediness in comprehension. Overall, the

authors concluded that results of their effort produced a preliminary comprehensive taxonomy of childhood personality.

Shiner and Caspi (2003) also proposed a preliminary comprehensive taxonomy for childhood personality, which the authors recently extended (Caspi et al., 2005; Caspi & Shiner, 2006). They based their proposed taxonomy on a review of the literature on childhood and adolescent personality and also integrated existing and emerging work on the structure of personality. They initially defined a classification system consisting of four superordinate traits and 11 lower level traits, recently extending the taxonomy to include a fifth higher order trait (Caspi et al., 2005; Caspi & Shiner, 2006). The higher order traits mapped onto the five factors in the Big 5. They noted that questions remain over whether the fifth factor in the FFM, openness to experience, has a direct analogue in childhood (Shiner, 2006) or whether it emerges later in development (Caspi & Shiner, 2006). These two recent attempts to provide a preliminary classification system for childhood and adolescent personality virtually agreed on the relevant higher order traits. This convergence on the superordinate structure is particularly compelling given the different research approaches used to construct the two taxonomies.

Taken together, these studies provide a good starting point for discussing childhood personality within a comprehensive framework. Moreover, the framework put forth will allow for greater communication in the area of childhood personality, as well as linking areas of childhood and adult personality (Caspi & Shiner, 2006). These findings suggest that many researchers believe that the FFM should be chief in both child and adult research investigations, although we believe that it is also critical to reevaluate and revise theories as new evidence arises. Although the childhood measures have been developed from both bottom-up and top-down perspectives, we briefly review the adult models of personality and attempt to integrate the research on these models to better inform the models that have resulted for child and adolescent samples.

GENERAL MODELS OF PERSONALITY

Researchers (Watson, Kotov, & Gamez, 2006; Widiger & Trull, 2007) have noted that there are superordinate factors to the structure of human personality in adulthood, and the field is somewhat divided between two or three prominent models of personality. Watson et al. asked the question, If there is a stable factor structure for adult and child personality, why are there so many different models? These authors noted that despite a variety of models and some disagreement, these models are closely related or converge, and it has been shown that they can be easily integrated with each other (see Clark & Watson, 1999; Markon, Krueger, & Watson, 2005; Watson, Clark, & Harkness, 1994). We discuss each of these basic models of personality in turn. Then we show how the three major models can be integrated, drawing on the work of Clark and Watson (1999; see also Watson, Kotov, & Gamez, 2006).

The Big 5 Factors of Personality

The Big 5 factor model of personality developed out of a number of attempts to understand the lexicon or natural language of trait descriptors (John & Srivastava,

1999; McCrae et al., 2000). Factor analytic studies of these descriptors consistently revealed five broad factors: extraversion, agreeableness, conscientiousness, neuroticism, and openness to experience (or imagination, intellect, or culture). This structure has been shown to be robust, with the same five factors emerging in both self- and peer ratings (McCrae & Costa, 1987), in analyses of both children and adults (Digman, 1997), and across a wide variety of languages and cultures (McCrae & Costa, 1997).

The Big 3 Factors of Personality

The Big 3 factor structure of personality is based on a smaller set of superordinate factors: negative emotionality (paralleling neuroticism), positive emotionality (paralleling extraversion), and disinhibition versus constraint (paralleling conscientiousness; Clark & Watson, 1999; Markon et al., 2005). This structural format emerged from the seminal and pioneering work of Eysenck (Eysenck, 1997) and his colleagues. Eysenck originally created an influential two-factor model consisting of neuroticism and extraversion. Subsequent analyses led to the identification of a third broad dimension, labeled psychoticism. Although thus titled, this third dimension is better viewed as assessing individual differences in disinhibition versus constraint (Watson & Clark, 1993).

Other theorists subsequently posited similar three-factor models. For instance, Tellegen (1985) proposed a structure consisting of negative emotionality (paralleling neuroticism), positive emotionality (cf. extraversion), and constraint (which is strongly negatively correlated with psychoticism). Watson and Clark (1993) delineated a similar model, with factors named negative temperament, positive temperament, and disinhibition (versus constraint), respectively. In his reformulation of the California Psychological Inventory, Gough (1987) introduced the higher order "vectors" of self-realization, internality, and norm-favoring, which represent the low ends of neuroticism, extraversion, and psychoticism, respectively.

The Interpersonal Circumplex Model of Personality

Based on the work of Sullivan and Leary (Leary, 1957; Sullivan, 1953; Wiggins & Broughton, 1985; Wiggins & Pincus, 1989, 1994) proposed that personality was made up of octants: assured-dominant (PA), arrogant-calculating (BC), cold-hearted (DE), aloof-introverted (FG), unassured-submissive (HI), unassuming ingenuous (JK), warm-agreeable (LM), and gregarious-extraverted (NO). There exists considerable empirical literature on the correlates of the interpersonal circumplex model (e.g., Kiesler, 1996; Plutchik & Conte, 1997; Wiggins, Trapnell, & Phillips, 1988). Accordingly, the interpersonal circumplex model provides elements from a model of general personality and attempts to map the interpersonal circumplex onto Big 5 factor models that are generally well understood and are delineated in further detail next.

Integrating the Big 3, Big 5, and Interpersonal Circumplex Models of Personality

The accumulating data establish that the Big 5 model essentially represents an expanded and more differential version of the Big 3 (Markon et al., 2005; Rothbart,

Ahadi, Hershey, & Fisher, 2001; Watson, 2005). Most notably, the neuroticism and extraversion dimensions of the Big 5 essentially are equivalent to the negative emotionality and positive emotionality factors, respectively, of the Big 3. Thus, negative emotionality and positive emotionality are common to both models and constitute what Watson et al. (2006) refer to as the basic Big Two of personality. Moreover, the disinhibition versus constraint dimension of the Big 3 is a mixture of (low) conscientiousness and agreeableness. More specifically, according to Watson et al. (2006), disinhibited individuals tend to be impulsive, reckless, and irresponsible (i.e., low conscientiousness), as well as uncooperative, unsympathetic, and manipulative (i.e., low agreeableness). Finally, openness is strongly associated to the Big 3, although it does show a moderate positive association with positive emotionality (Digman, 1997; Markon et al., 2005). The interpersonal circumplex also maps onto the Big 5; many believe this personality framework taps two higher order dimensions, dominance and love. Dominance is most closely linked to extraversion and love is most closely linked to agreeableness and possibly openness (McCrae & Costa, 1989). Putting these data together, Clark and Watson (1999) noted that one can transform the Big 3 into the Big 5 by (a) decomposing disinhibition versus constraint into agreeableness and conscientiousness and (b) incorporating the additional dimension of openness. In addition the interpersonal circumplex can be subsumed under this broader umbrella of personality components and traits. Not surprisingly, of the somewhat varying models of personality in research, the FFM or Big 5 model of personality appears to subsume the others and thus is the most commonly used measure of personality traits.

It is important to note that some researchers have found flaws with this model. Specifically, although the Big 5 is the most common model, Block (1996, 2001) argued that the word "approach" might be more applicable than "model," since there is such an array of models that have not converged as of yet. He claimed that there are deficiencies in the FFM becoming the general perspective on personality, and pointed to the absence of an overall explanation for the five-factor approach. Others (Briggs, 1989) have voiced similar concerns, stating that there is no theoretical explanation for why these five dimensions were more coherent than other models' dimensions. On the other hand, no one has offered an explanation as to why this model is an invalid system of measure.

It is probably safe to say that the general consensus among researchers is that this model can be useful in describing the personality structure from which variant patterns can develop, thus forming the uniqueness of personality of an individual. Although it has shown reliable results when viewing adult differences, there have been limited findings with regard to children and the FFM. This could be for a number of reasons. For instance, there could be concerns about the reliability of self-reports and developmental and maturity levels, as well as diverging ideas about the age at which personality begins to develop from baseline attachments. In addition to these factors, reliability of parent reports of young children, or semantics, could have limited the degree to which research is conducted with children and the Big 5.

However limited, the FFM is increasingly credited with being one of the best currently available models for the study of personality, and research has been aiming to find a universality of personality as well as using it to track the development of

personality. Yamagata et al. (2006) conducted a study with 1,200 twins from Canada, Germany, and Japan to determine if the FFM identifies general personality structure that can be considered universal across cultures. These researchers found a significant congruence across genetic and environmental factors in relation to the FFM across all three countries, therefore providing a basic, universal genetic component in addition to the environmental factors involved in personality development.

A study by McCrae et al. (2002) examined changes in personality during adolescence, using the FFM to directly compare the adolescent results to the adult scores. They argue that adult personality measures can be meaningful when used with adolescent samples, as demonstrated by De Fruyt, Mervielde, Hoekstra, and Rolland (2000) and Parker and Stumpf (1998). Parker and Stumpf measured a sample of intellectually gifted adolescents and found that self- and parent reports were valid and reliable and yielded the same factor structure seen in adults. It is possible that psychologists underestimate the capacity of adolescents to relate to the adult personality items. Not only do adolescents understand the factors in the adult measure, but the factors seem to easily describe personality among adolescents when personality is evaluated by a variety of sources, including parents, teachers, and other adult observers. McCrae et al. (1999; see also McCrae et al., 2002) focused on the personality changes that occur during adolescence and used the FFM to examine each specific factor of neuroticism, extraversion, openness, agreeableness, and conscientiousness, predicting that adolescents would exhibit high neuroticism and extraversion and low agreeableness and conscientiousness. Although only moderate correlations were found with regard to high neuroticism, openness was also moderately elevated in adolescence. However, De Fruyt (De Fruyt et al., 2000) has noted that there is a need for continuity in measures. More specifically, it is possible that if there were a child version of the FFM the correlations may be stronger (or weaker). Either way, more research about the transition between childhood temperament and developmental maturity and later personality development should be examined not only with the FFM, but with other, equally strong measures of personality, such as the three, four, six or seven-factor models. We elaborate on this point next.

ASSESSING PERSONALITY AT A YOUNG AGE

A comprehensive assessment of age-specific indicators of traits is crucial to studying personality at a young age, especially in development. Different approaches have been adopted to assess FFM dimensions in children and adolescents, and as mentioned, work in this area is still in its infancy. Often, FFM measures—initially developed for adults—are downwardly extended and used to describe differences in younger age groups (see, e.g., studies using the Neuroticism Extraversion Openness Personality Inventory—Revised [NEO-PI-R] to assess adolescents' personality: De Clercq & De Fruyt, 2003; De Fruyt, Mervielde, Hoekstra, & Rolland, 2000; Goldberg, 2001; McCrae et al., 2002). Other studies adapted the phrasing of personality items for younger age groups (e.g., the junior version of Eysenck's Personality Questionnaire measure; Eysenck, 1963; Eysenck, Makaremi, & Barrett, 1994) to be more developmentally appropriate. However, it can be argued that these downwardly extended measures are probably not ideal for fine-grained

assessment of childhood and adolescent personality differences and especially not to index developmental change (De Clercq, De Fruyt, & Van Leeuwen, 2004). Therefore, according to De Clercq and De Fruyt an alternative approach that is more sensitive to subtle personality differences and developmental status at a variety of ages should be developed on the basis of the full range of personality differences observable prior to adulthood.

De Fruyt and his colleagues (De Fruyt et al., 2000; Slotboom, Havill, Pavlopoulos, & De Fruyt, 1998) have offered several suggestions for research and tool development in this area. Specifically, the lexical approach to personality description provides a convincing justification for the development of a comprehensive child and adolescent personality taxonomy. Mervielde and De Fruyt (1999) adopted this approach to construct such a taxonomy for classification of a large pool of parental personality descriptions in Flemish children between 6 and 13 years old (Kohnstamm et al., 1998). These authors subsequently developed the Hierarchical Personality Inventory for Children (HiPIC), representing the content of parental descriptions in short sentence items referring to concrete and observable behavior. The HiPIC can be considered a lexically based measure of the active parental vocabulary, in contrast to the NEO-PI-R, in which the facets are not derived empirically but are selected after a careful search of the adult personality literature. The HiPIC items span five broad domains: extraversion, benevolence, conscientiousness, emotional instability or neuroticism, and imagination. Some domain labels differ from the lexical adult Big 5 (Goldberg, 1993), although there is quite a bit of overlap in the content. It is beyond the scope of this chapter to cover comprehensively the measures used to assess the Big 5, but it should be noted that age-specific measures ought to undergo development and be tested. Researchers and clinicians alike are referred to Caspi and Shiner (2006) for a review of the existing measures, which might serve as a starting point for the development of newer measures with a developmentally sensitive theoretical framework. We comment on some important developmental factors that are likely to have an impact on personality development when considering the interaction between biology and environment and the two-way interplay that can lead to various outcomes for children and adolescents.

PERSONALITY DEVELOPMENT IN CHILDREN AND ADOLESCENCE

Developmental theories conceive a number of child and adolescent cognitive developments as well as puberty as important stages for social and personality development (Roberts & Wood, 2006), involving changing social interactions with parents and peers and increasing societal influences (e.g., Crick & Dodge, 1994; Roberts & Wood, 2006). Of course, we are increasingly aware that development occurs across the life span and that we are in the midst of development at any given age and developmental period. Nonetheless, the importance of adolescence as a key transitional phase has been acknowledged by a number of prominent theorists and researchers. Adolescence was acknowledged by Erikson (1950, 1968) as a critical stage in our development. Erikson considered adolescence a second individuation stage in which change is likely. Similarly, behavioral theories (Robin & Foster,

1989) emphasize that during adolescence, learning processes and contingencies are embedded in novel social networks and environments, including changing peer groups. In addition, organismic theories such as Piaget's (1983) theory of cognitive development and subsequent brain architecture suggest that newly acquired cognitive structures influence the way children and adolescents interact with their environment.

Personality theories such as Cloninger's (Cloninger, Svrakic, & Przybeck, 1993) theory of character development discern qualitatively different life strategies that an individual has to master before a more advanced developmental phase or level can be met. These theories emphasize developmental discontinuities, but it remains to be established whether these discontinuities reflect changes in basic tendencies or whether they are restricted to changing characteristic adaptations (McCrae & Costa, 1996; Shiner & Caspi, 2003). The demonstration of different forms of stability across the FFM trait hierarchy in childhood and adolescence would underscore McCrae and Costa's five-factor theory and therefore requires that the previously reviewed theories of personality development account not only for trait change but also for trait stability. We agree that there are some reasons to believe that personality would be stable across the life span; however, there are also compelling reasons with respect to maturation to expect both stability and change and that there would be a variety of outcomes for youth (both positive and negative) even if they manifested similar constellations of traits (e.g., extraversion and openness).

During the past several decades, research has uncovered considerable knowledge about the child's developing personality, including the emergence of a sense of identity, affect modulation, thinking style, and relationship with the external word (Mroczek & Little, 2006). With respect to the development of a sense of self, for example, it has been found that recognition of oneself by name in a mirror occurs by 3 years of age; a sense of shame, implying self-consciousness, emerges before 2 years of age (Lewis, 1993). Another example is empathy; a basic component of interpersonal functioning because of its role in the relationship between self and others, empathy develops in early childhood (Hoffman, 1977), evidencing clear signs by 2 years of age (see Salekin, 2006). These factors could also impact the development of personality in that youth might evaluate their basic personality characteristics and start to shape the various outcomes that result from their general personality makeup. Therefore, another component of developmental status and maturity—thinking style and the presence of a concrete operational reasoning system (Piaget, 1950)—is apparent by middle childhood and persists into adulthood. Differences in style of reasoning could impact the eventual outcome for youth. All these developmental considerations need a point of reference. That is, we need to know how stable personality is across childhood and adolescence and, more generally, across the life span.

Stability of Personality across Time: Continuity and Change

In the past decade, there has been a wealth of studies on personality continuity that were recently summarized in two meta-analyses of longitudinal data on differential (Roberts & DelVecchio, 2000) and mean-level (Roberts, Walton, & Viechtbauer,

2006) continuity. Differential continuity describes the degree to which the relative differences among individuals remain stable across time. Mean-level stability refers to the extent to which personality scores change over time. Longitudinal designs are required to investigate differential stability, looking at trait correlations across time. Mean-level stability can be studied with the use of longitudinal data (Roberts, Walton, Viechtbauer, 2006). Moreover, mean trait scores from cross-sectional age cohorts are useful for mean-level stability comparisons (McCrae et al., 1999).

Roberts and DelVecchio (2000) recently conducted a meta-analysis of differential stability and examined whether and when stability peaks during the life course, challenging Costa and McCrae's (1994; McCrae & Costa, 1984, 1990) claim that personality is "set like plaster" after the age of 30. They analyzed 3,217 test-retest correlation coefficients from 152 longitudinal studies and demonstrated an increase in stability from .31 during childhood to .54 in young adulthood, rising to .64 by age 30, with the highest stability of .74 observed between the ages of 50 and 70 years. These data suggest that personality is less stable during the preadult years; however, there was still a considerable amount of stability.

McCrae and colleagues (2002) examined both mean-level and differential changes during adolescence in self-descriptions using the NEO-PI-R, including an analysis of continuity at the level of the individual. These researchers found that mean-level personality scores for extraversion, agreeableness, and conscientiousness were stable between the ages of 12 and 18; however, neuroticism appeared to increase in girls and openness increased in boys and girls. A 4-year longitudinal study of intellectually gifted students showed a considerable degree of rank-order instability across the two assessment points. Stability coefficients for boys across a 4-year interval ranged from .31 (agreeableness) to .49 (conscientiousness) and for girls from .30 (neuroticism) to .63 (conscientiousness). Individual-level continuity analyses relying on the reliable change index (Jacobson & Truax, 1991) indicated that about 60% of the sample did not change over the 4-year interval for each of the FFM dimensions. A similar analysis for college-aged individuals (Robins, Fraley, Roberts, & Trzesniewski, 2001) showed that almost 80% were stable on each of the FFM dimensions, again suggesting that differential stability is somewhat lower during adolescence.

The meta-analysis of Roberts et al. (2006) on mean-level change during adolescence (10 to 18 years) showed a small increase for openness ($d = .23$) and a significant increase for social dominance ($d = .20$). No significant mean-level changes were reported for social vitality, agreeableness, and conscientiousness during adolescence. Studies examining stability of personality prototype classification using three personality profiles (resilient, undercontrolled, and overcontrolled children), provided partial evidence of ipsative stability (De Fruyt, Mervielde, & Van Leeuwen, 2002). However, according to De Fruyt et al., it was unclear whether the instability to prototype classification for some individuals should be attributed to real developmental change measurement error or the procedure to derive prototypes.

Taken together, these important studies and others (Funder, Parke, Tomlinson-Keasey, & Widaman, 1993; Graziano, 2003; Halverson, Kohnstamm, & Martin, 1994; Mroczek & Little, 2006) lead to two general conclusions. First, the level of

personality continuity in childhood and adolescence is higher than often expected (e.g., compared with young adulthood; Roberts et al., 2006). Second, personality change occurs further into adulthood than one might expect. Caspi and colleagues (2005) argued in this respect that personality trait development is not a continuity-versus-change proposition, but that continuity and change coexist. These authors concluded that the major challenge for developmental theories will be to account not only for trait continuity but also for trait changes. We return to this point later in the chapter when we cover contextual factors and what Thomas and Chess (1977) and recently Shiner (2006) have referred to as "goodness of fit."

PERSONALITY AND LIFE OUTCOMES

The importance of studying personality is that certain traits will be able to forecast certain outcomes for children and adolescents. Quite often in psychology and psychiatry, the fields have focused on the negative aspects of personality; this is particularly apparent in the *DSM-IV* and its successive versions, which have focused on personality disorder. Fortunately, there is also a movement in the field to focus on positive and healthy outcomes for children and adolescents. This section of the chapter highlights some previous work in the area of child and adolescent personality that has emphasized both positive and negative outcomes (Caspi & Bem, 1990; Caspi & Shiner, 2006; Rothbart & Bates, 1998; Shiner, 2006; Shiner & Caspi, 2003). Specifically, we focus on positive outcomes such as academic achievement, social competence with peers, rule-abiding conduct, the development of romantic relationships, and work competence in order to build on Shiner's and others work in this area (Funder et al., 1993; Masten et al., 1995; Shiner, 2006). With respect to negative outcomes, this section covers children's internalizing and externalizing behaviors. Taken together, the studies reviewed in this section point to two important conclusions: Children's traits are differentially linked with specific outcome, and these patterns are often replicated across studies, and there is convincing evidence for the consequential nature of children's individual differences or personality (Shiner, 2006). Typically, the prediction afforded by children's temperament and personality traits is modest in size. We take each of the Big 5 dimensions and tie each to positive and negative outcomes to build on Shiner's work in this area (2006). Although we do not examine personality disorders as conceptualized in the *DSM-IV,* we provide a road map for how this approach might allow for further study of *DSM-IV* disorders (see Table 12.2).

Extraversion: Negative and Positive Outcomes

It has been noted that children's differences in extraversion predict a complex set of outcomes (Shiner, 2006). There are some positive outcomes associated with extraversion. Shiner has noted that for some children extraversion is associated with positive and meaningful peer relations concurrently and across time into late adolescence and early adulthood. Moreover, extraversion predicts positive changes in social competence across time (Shiner, 2000; Shiner, Masten, & Roberts, 2003). Similarly, extraversion in early adolescence predicts growth in perceived social

Table 12.2 Outcomes Associated with Each Big 5 Factor Trait

Trait	Positive Outcome	Negative Outcome
Extraversion	Social competence Promotes good health	Antisocial behavior Callousness
Neuroticism	Conscience development Guilt when expected	Poor relationships Relationship conflict Relationship abuse Relationship dissolution Less competent parenting Risk for unemployment
Conscientiousness	School adjustment Educational achievement Occupational attainment Job performance	Obsessive
Agreeableness	Social competence Positive parenting Responsible parenting	
Openness	Exploring Friendliness Academic achievement	Exposure to risks

Note: This table is intended to be illustrative and not an exhaustive list of positive and negative outcomes.

support from early adolescence to late adolescence (Asendorpf & van Aken, 2003a, 2003b). Extraverted children are more likely than introverted children to have a life history of stable, high-quality romantic relationships by age 30 (Shiner, 2006; Shiner et al., 2003). Childhood extraversion is associated with lower levels of later internalizing symptoms (Chen et al., 2002).

Although being extraverted has some positive aspects to it, research has shown that some of the outcomes for those scoring high on extraversion measures are negative and may even point to certain types of pathology. For instance, a wide variety of studies have linked early signs of extraversion to heightened risk of externalizing symptoms and aggression (Chen et al., 2002; Frick & Scheffield-Morris, 2004; Huey & Weisz, 1997; Markey, Markey, & Tinsley, 2004). As Shiner (2006) indicated, the mechanisms for why extraverted children are more likely than introverted children to develop antisocial behavior are not yet clear; they may have stronger impulses that they must learn to regulate, or they may be more likely to engage in antisocial behavior with peers. Outcomes of shyness and inhibition essentially parallel those found with low extraversion, including lower social support (Newman, Caspi, Moffitt, & Silva, 1997) and higher rates of anxiety symptoms (Kagan, Snidman, Zentner, & Peterson, 1999). Socially inhibited and/or shy boys may experience more negative life outcomes than shy girls. According to Caspi and his colleagues (Caspi, Bem, & Elder, 1989; see also Gest, 1997), this might be because shyness is more socially sanctioned for girls than for boys.

Neuroticism: Negative and Positive Outcomes

Neuroticism and the broader negative emotionality trait are associated with negative outcomes in a variety of areas. Children high on negative emotionality are

at risk for a wide variety of social difficulties (see Eisenberg, Fabes, Guthrie, & Reiser, 2000; see also Shiner, 1998 and Muris, Schmidt, Merckelbach, & Schouten, 2001). Negative emotionality in children predicts both internalizing and externalizing psychopathology, suggesting a broad bandwidth of psychopathology (Lengua, 2002; Rothbart & Bates, 1998). Caspi and Shiner (2006; Shiner, 2006) have noted that it is important to consider the outcomes for anxiety and fear separately from the outcomes of irritability given that there is some heterogeneity in the neuroticism concept. According to these authors, this is important because in some cases, the lower order traits predict similar outcomes but may do so for quite different reasons. For example, they note that higher anxiety or fear and higher irritability both predict greater social difficulties (Asendorpf & van Aken, 2003a, 2003b; Eisenberg, Pidada, & Liew, 2001) and lower occupational attainment in adulthood (Caspi et al., 1989; Judge, Higgins, Thoresen, & Barrick, 1999). Yet, anxiety and fear may cause such difficulties as preventing children from fully participating in social and work spheres, whereas children's irritability may exert its effects through alienating other youths. Importantly, in other cases, these two components of negative emotionality may predict different outcomes altogether. For example, anxiety and fear are associated with greater risk for internalizing symptoms, and irritability with greater risk for externalizing symptoms (Rothbart & Bates, 1998; Shiner, 2006).

Although high negative emotionality spans a broad range of negative outcomes, it is equally important to recognize that higher fearfulness may promote positive outcomes for some children. For example, more fearful children experience greater guilt when they believe they have done something wrong, and this guilt proneness in turn appears to mediate the association between children's level of fear and their later compliance with rules (Kochanska, Gross, Linn, & Nichols, 2002). These findings are consistent with research documenting that childhood fearfulness protects against the development of externalizing behavior problems, including aggression (Raine, Reynolds, Venables, Mednick, & Farrington, 1998). These findings have also been linked to conscience development and the possibility of protecting against psychopathy-like traits in youth (Salekin, 2006). Like high extraversion, high fear predicts a complex mixture of both positive and negative outcomes.

Conscientiousness: Negative and Positive Outcomes

There is little to chart the negative outcomes for being conscientious. It might be that extreme forms of conscientiousness lead to obsessiveness and perhaps other forms of anxiety, but to our knowledge there are no data to support this supposition. Rather, scientific work on children's conscientiousness documents what might be considered somewhat obvious findings. That is, greater self-control appears to promote many good outcomes (Shiner, 2006). First, children's attention, self-control, and carefulness predict later academic achievement, even when controlling for earlier measurement of academic ability and IQ (Martin, Olejnik, & Gaddis, 1994; Shiner, 2000; Shiner et al., 2003). Childhood conscientiousness predicts positive changes in academic achievement from childhood to adulthood (Shiner, 2000; Shiner et al., 2003). These positive changes continue into adulthood, resulting in happiness and

well-being in other areas. Specifically, conscientiousness in adolescence forecasts adult career success in terms of both income and occupational status. Conscientiousness also predicts job satisfaction (Judge et al., 1999). Children's attention and self-control, regardless of the method of indexing these traits, also appear to promote the development of rule-abiding behavior rather than externalizing, antisocial behavior (Shiner, 2006). Put another way, children who are impulsive, inattentive, or careless are at risk for developing externalizing behaviors (Ackerman, Brown, & Izard, 2003; Olson, Schilling, & Bates, 1999; Shiner, 2000, 2006). Again, it is important to note that these links between early self-control and lower risk of externalizing problems may be mediated in part by more self-controlled children's tendencies to develop a stronger conscience (Kochanska & Knaack, 2003; Salekin, 2006). Finally, components of conscientiousness in childhood, including effortful attention, self-control, and carefulness, have predicted children's concurrent and later social competence with peers in several studies (Lamb et al., 2002; Hawley, 2006; Shiner, 2000).

Agreeableness: Negative and Positive Outcomes

Similar to conscientiousness, childhood agreeableness has few known negative outcomes, but instead points to many different positive life outcomes (Graziano & Eisenberg, 1997). With respect to negative outcomes, high compliance, an aspect of agreeableness, could be problematic for children and may be a particular risk factor for negative outcomes for girls (Pulkkinen, 2002). Although there are few data on the negative outcomes for agreeableness, it will be important for future work to address whether extremely high agreeableness poses adaptive risks for children.

Greater childhood agreeableness is associated with better peer relationships concurrently and across time (Graziano, Jensen-Campbell, & Finch, 1997; Shiner, 2000, 2006). However, some authors have noted that the name of this trait is somewhat misleading if it suggests that its importance is confined to interpersonal relationships. Childhood agreeableness is also linked to greater academic attainment and a reduced risk of externalizing behaviors and delinquency (Laursen, Pulkkinen, & Adams, 2002; Shiner, 2000, 2006; Shiner et al., 2003). In one longitudinal study, childhood agreeableness predicted positive work competence 20 years later, when childhood academic achievement and IQ were controlled (Shiner et al., 2003).

As noted previously, agreeableness has some heterogeneity to its lower order traits, and the components of this higher order trait include at least two major lower order traits: prosocial tendencies and antagonism (which includes aggressiveness). Findings for children's prosocial tendencies parallel those for agreeableness. For example, a stronger prosocial orientation in childhood predicts positive social adjustment and academic achievement. Moreover, and relatedly, a stronger prosocial orientation predicts fewer externalizing behaviors over time (Chen, Li, Li, Li, & Liu, 2000). Alternately, findings for antagonism and aggression typically are like those found for low agreeableness. More antagonistic, aggressive children are at risk for a number of negative outcomes, including academic failure, peer rejection, job difficulties and unemployment, and later conflictual romantic relationships (Kokko

& Pulkkinen, 2000; Moffitt & Caspi, 1998; Rubin, Bukowski, & Parker, 1998; Shiner, 2006). Boys who are chronically aggressive are at a heightened risk of serious delinquency and even violence (Nagin & Tremblay, 1999).

Thus, the traits underlying both conscientiousness and agreeableness promote positive adaptation in many different domains, and there are not a lot of data to suggest negative outcomes. As noted, these two higher order traits appear to share a common core of behavioral constraint or self-control, with conscientiousness focused on task-related behavior and agreeableness focused on interpersonal behavior, which may account for their concomitant positive outcomes. It is interesting to note, however, that the positive outcomes of conscientiousness are not restricted to the domains of achieving and striving. Nor should it be interpreted that the positive outcomes of agreeableness are limited to interpersonal domains. Rather, there is more likely a greater array of positive benefits that stem from each higher order construct. More detailed research on how highly conscientious and agreeable children approach a variety of situations will help shed light on why the two traits share such similar positive outcome profiles.

Openness to Experience: Negative and Positive Outcomes

Few studies have shown negative outcomes for the openness to experience construct. However, it is possible that being too open could leave one slightly more vulnerable. Research is needed in this area to determine if there are some negative aspects associated with extreme levels of openness. For the most part, research on the links between openness to experience and intellect shows positive outcomes. However, research in this area is more limited than for the other four higher order traits; yet, the findings generally are consistent. Openness to experience has been shown to have concurrent relations with academic achievement in samples of school-age children and young adolescents (Graziano et al., 1997; John et al., 1994; Measelle, Ablow, Cowan, & Cowan, 1998). Moreover, the size of the correlation varies from modest to strong, and this difference in magnitude could stem from the differences in trait descriptors for this dimension across indexes, with some emphasizing intellectual characteristics and others less broadly covering this aspect of the higher order trait (Shiner, 2006). Because openness and intellect in childhood are moderately associated with IQ or other measures of intellectual ability (Lamb et al., 2002), it is not clear to what extent these traits predict academic achievement; thus, these traits differ from the other Big 5 traits in that they may show more narrowly focused importance for adaptation (Caspi & Shiner, 2006; Shiner, 2006).

MODELS OF PERSONALITY AND THEIR CONNECTION TO PSYCHOPATHOLOGY

One of the questions that has fueled scholarly and popular interest in temperament is whether children with particular temperaments are at a greater risk for negative outcomes, including psychopathology. Much of the early interest in temperament traits was initiated by Thomas and Chess (1977), who suggested that early individual differences could set off a chain of transactions between the child and the

environment that could ultimately end with clinical disorders. The previous section highlighted general negative and positive outcomes associated with each of the Big 5 personality factors; this section focuses more specifically on the connection between the five factors and psychopathology.

Shiner and Caspi (2003) noted that although individual differences in personality shape individuals' adaptation over time, childhood personality also plays a key role in the development of psychopathology. Much of the current research on personality and psychopathology simply documents correlations between temperament or personality traits and aspects of psychopathology, without articulating how the two domains may be connected (Caspi & Shiner, 2006; Hoyle, 2000; Maziade et al., 1990; Rothbart & Bates, 1998). Shiner and Caspi (2003) and more recently Tackett (2006) present conceptual models of the possible associations between personality and psychopathology in childhood and adolescence. These childhood models have parallels to those developed by Clark, Watson, and Mineka (1994) and Widiger, Verheul, and van den Brink (1999). The models posit several different links to pathology. First, these authors all suggest that psychopathology may represent the extreme end of a continuously distributed personality trait or cluster of traits (this is also referred to as the spectrum model). Second, personality may set in motion processes that cause the development of psychopathology (referred to as the vulnerability model). Third, personality may protect against the development of psychopathology in the face of stress and adversity (referred to as the resilience model). Fourth, personality may influence the form and prognosis of a disorder, even if the personality trait is not a component or cause of the disorder (referred to as the pathoplastic model). Fifth, psychopathology may influence the course of personality development itself (referred to as the scar model; Shiner & Caspi, 2003; Tackett, 2006; see Table 12.3).

Over the past 2 decades, a growing body of research has attempted to explain the nature of these relationships (for reviews, see Krueger & Tackett, 2003; Widiger & Clark, 2000; Widiger et al., 1999), building on current research investigating the etiology and structure of both personality and psychopathology. However, this work has primarily focused on adult populations, despite the growing evidence of robust associations between personality traits and mental disorders in children and adolescents. Tackett (2006) has argued that to promote understanding and further investigation of the relationship between personality and psychopathology, existing work with children and adolescents must be integrated into research on adults to create a broader developmental picture.

Table 12.3 Models Depicting the Connection between Personality and Pathology

Model Name	Path (or Relation) to Pathology
Spectrum	Pathology is an extreme form of personality (viewed along a continuum).
Vulnerability	Personality sets in motion and is the cause of pathology.
Resilience	Personality may protect against or buffer against pathology.
Pathoplastic	Personality may influence the form of the disorder even if it is not the cause.
Scar	Pathology affects personality change.

Evidence for each of these models is limited in part because the study designs needed to detect the hypothesized relations are complex and time consuming. Nonetheless, there is some emerging research in this area. For instance, the one longitudinal study found that childhood antisocial behavior problems predicted an increase in neuroticism, or negative emotionality, in adulthood even after controlling for the level of neuroticism in childhood (Shiner et al., 2002); this model provides some support for the scar model, which suggests that childhood psychopathology may cause changes in personality later in life. However, without a measurement of personality before the onset of Axis I pathology, one cannot infer whether these changes in personality traits were a result of the Axis I disorder (Tackett, 2006).

There is also some evidence for the pathoplasty model. One area of study has concentrated on comorbidity between anxiety disorders and Conduct Disorder (CD) in children. This literature has largely found that disruptive children with high levels of inhibition or shyness, which would be related to the higher order factor of neuroticism, have less severe manifestations of CD and better prognosis compared to those with low levels of inhibition (Kerr, Tremblay, Pagani, & Vitaro, 1997; Walker et al., 1991). The bulk of research evidence to date appears to back the vulnerability and dimensional models.

The vulnerability model splits childhood pathology into externalizing and internalizing disorders. Several types of childhood and adolescent problem behaviors are typically categorized as externalizing pathology, such as Oppositional Defiant Disorder (ODD), Conduct Disorder, Attention-Deficit/Hyperactivity Disorder, and substance use. Disorders characterized by antisocial behaviors, such as ODD and CD, have received the greatest attention from the vulnerability perspective. In general, this literature has primarily identified conscientiousness and neuroticism as potentially relevant personality risk factors for the development of later psychopathology.

A number of studies have found that characteristics related to impulsivity (which has been related to low conscientiousness) were significantly related to antisocial behaviors throughout childhood (Hirshfeld et al., 1992; Raine et al., 1998; Tremblay, Pihl, Vitaro, & Dobkin, 1994) and adolescence (Lynam et al., 2000; Salekin, Lesitico, Trobst, Schrum, & Lochman, 2005). Some longitudinal studies that followed individuals into adulthood have also shown strong relationships between early characteristics of impulsivity and later antisocial behaviors. This finding has been substantiated by studies with measurements of impulsivity in adolescence (Sigvardsson, Bohman, & Cloninger, 1987; White, Bates, & Buyske, 2001), middle childhood (Farrington & West, 1993), and as early as 3 years of age (Caspi, 2000; Caspi, Moffitt, Newman, & Silva, 1996; Henry, Caspi, Moffitt, & Silva, 1996). In addition to the aforementioned findings related to low conscientiousness, early observations of negative emotions have been connected with antisocial behaviors in middle childhood (Renken, Egeland, Marvinney, Mangelsdorf, & Sroufe, 1989). Augmenting this work on general antisocial behavior, other lines of research have focused on relationships across time between personality traits and specific subtypes of antisocial behavior (Lynam, 1996; Moffitt, 1993; Moffitt, Caspi, Dickson, Silva, & Stanton, 1996; Salekin, 2006; White et al., 2001). Another strand of research

has measured personality traits related to the construct of psychopathy, such as callousness and impulsiveness, and found them to predict antisocial behavior across childhood (Lynam, 2006; Lynam et al., 2005; Salekin, 2006; Salekin & Frick, 2005; Salekin et al., 2005; Salekin, Neumann, Leistico, DiCicco, & Duros, 2004; Tremblay et al., 1994).

A larger body of work has demonstrated that measures of disinhibition predict externalizing pathology in early and middle childhood (Caspi, Henry, McGee, Moffitt, & Silva, 1995; Eisenberg et al., 2000 2004; Mun, Fitzgerald, Von Eye, Puttler, & Zucker, 2001; Rende, 1993; Rubin, Burgess, Dwyer, & Hastings, 2003; Silverman & Ragusa, 1992) and later into adolescence (Olson et al. 1999). Related to these results, other studies have reported that high levels of inhibition predicted lower levels of externalizing psychopathology in childhood and adolescence (Schwartz, Snidman, & Kagan, 1996; Sigvardsson et al., 1987; Tremblay et al., 1994).

Another major domain of disorders, typically referred to as internalizing pathology, consists primarily of problems with anxiety and depression. One personality construct that has received a great deal of attention regarding its connection with anxiety disorders is often called behavioral disinhibition (BI; Tackett notes that this is typically described as a temperament). In the context of personality, BI would represent a combination of high neuroticism and extraversion. Having said this, a substantial literature has reported a relation between BI and later internalizing problems (see Hirschfeld-Becker et al., 2003). Indices of BI in toddlerhood and early childhood predicted anxiety disorder in early middle childhood (Biederman et al., 1993, 1990; Hirschfeld et al., 1992; Rosenbaum et al., 1993) and adolescence (Hayward, Killen, Kraemer, & Taylor, 1998; Schwartz, Snidman, & Kagan, 1999).

A smaller set of studies has examined the links between personality characteristics and depression. These studies suggest that traits reflecting extreme inhibition in early childhood predict depression in adulthood and may show a significant relation with suicide attempts when combined with traits marking low conscientiousness (Caspi, 2000; Caspi et al., 1996). Another study reported that high levels of neuroticism and high levels of novelty seeking in adolescence predicted later suicidal behavior (Fergusson, Beautrais, & Horwood, 2003). Although the dimensional models are also very compelling, we do not cover them in as great of detail in this chapter. The essence of this model is that personality is pathology when the personality is extreme. One example of this is the work of Lynam (2006) and Salekin (2006), who have examined the extreme form of some personality traits that could in fact be psychopathy. They have found a potential connection between personality and externalizing disorders. The dominant model linking internalizing disorders to personality traits is the tripartite model, which has recently been applied to children (Clark & Watson, 1991; Clark et al., 1994).

In summary, the research on the connection between personality and pathology provides preliminary support for both vulnerability and spectrum approaches to conceptualizing the personality-psychopathology relationship in children and adolescents, with the vulnerability, or risk factor, approach receiving the greatest research attention. Specifically, neuroticism and conscientiousness have been implicated as particularly relevant in the development of later psychopathology. The

resilience or protective factor model also has support (Chiccetti & Luthar, 1990). There is less support for alternative conceptualizations, such as the scar and pathoplasty models, due to the lack of research. Although the spectrum model has not been a driving hypothesis to the extent that the vulnerability model has, it is difficult to disentangle evidence for these two approaches. Indeed, research supporting a vulnerability explanation for the personality-psychopathology relationship may also support a dimensional conceptualization, and these two approaches may ultimately work in concert to provide the best explanation. More direct tests of the scar, pathoplasty, and spectrum models will help elucidate a comprehensive approach to conceptualizing the personality-psychopathology relationship. Ultimately, we may find that all models are necessary to create such a comprehensive picture, possibly differing for various types of psychopathology or various individuals (Tackett, 2006).

CONNECTIONS BETWEEN BIG 5 AND THE *DIAGNOSTIC AND STATISTICAL MANUAL*

Some researchers have argued that the personality superfactors are too broad to capture all the interesting variation in children's personality, and distinctions at the level of more specific, lower order traits are necessary (McAdams, 1992; Tackett, 2006). Such is the case, for example, if using structural models of personality traits to distinguish between different personality disorders (Lynam & Widiger, 2001; Widiger, Trull, Clarkin, Sanderson, & Costa, 2002); recent work suggests that categorical personality disorders are distinguished by particular patterns of dimensional lower order personality traits, as shown earlier. The advantage of broad, higher order superfactors is their substantial bandwidth; the disadvantage is their low fidelity (Robins, John, & Caspi, 1994). There is much to learn from studying variation at different levels of the personality trait hierarchy, but this does not obviate the need to understand how different levels are associated hierarchically to each other.

The connection between the Big 5 and *DSM-IV* disorders is not all that well known. Research appears to be progressing along independent lines with little integration of the two (e.g., Brent, Zelenak, Buckstein, & Brown, 1990), similar to the problems with the research on temperament and personality (Bernstein, Cohen, Skodol, Bezirganian, & Brook, 1996). Although there is some connection between Axis I childhood pathology and the Big 5 (e.g., the connection between CD and neuroticism), much less has been conducted on the *DSM-IV* personality disorders themselves. Some have argued that the impact of *DSM* personality disorders on functioning can be as profound for a young person as for an adult (Kernberg, Weiner, & Bardenstein, 2000). For example, some preliminary research indicates that suicide is more likely for adolescents diagnosed with impulsive, dramatic, Avoidant or Dependent Personality Disorder (Brent et al., 1994), and suicidal behavior is more severe in older adolescents and young adults who have both Borderline Personality Disorder and depression (Friedman, 1982). Researchers should attempt to make connections between *DSM* disorders and the Big 5 and eventually integrate the models and then possibly tie the personality concepts to the *DSM-IV* Axis I disorders (see Table 12.4). Some significant strides have been made in this regard (see Krueger, 1999; Krueger, McGue, & Iacono, 2001; Tackett & Krueger, 2006).

Table 12.4 Connecting the Big 5 to Specific Childhood Disorders

Externalizing (high extraversion, low conscientiousness, and low agreeableness)	Internalizing (high neuroticism, low extraversion, high conscientiousness, low agreeableness)
Oppositional Defiant Disorder	Separation anxiety
Conduct Disorder	Depression
Attention-Deficit/Hyperactivity Disorder	Generalized anxiety
Substance abuse	Obsessive-Compulsive Disorder
	Eating disorders

Processes Underlying Continuity and Change in Personality Development

As noted in the preceding sections, children's early temperament and personality traits show both substantial continuity and change (Shiner, 2006). By the preschool years, individual differences are moderately stable. Children's individual differences moderately predict many aspects of development, including both positive and negative outcomes. Thus, early temperament and personality traits meaningfully predict later personality and life outcomes for some children (including both positive and negative outcomes, including psychopathology), but other children appear to undergo significant change over time. As Shiner points out, this finding regarding stability and change requires explanation: What is the process that underlies both the stability and the change? In part, the answer lies in contextual factors. This issue is hardly new, as Thomas and Chess (Thomas et al., 1963) raised the question of explaining continuity and change in their early theoretical work on temperament. Specifically, they argued that each early temperamental trait could yield a variety of subsequent outcomes, depending on the goodness of fit between the child's temperament and the context in which the child flourished, languished, or simply did okay. For example, some infants with a difficult temperament may continue to exhibit irritable, hard-to-manage behavior as children, whereas other infants with similarly challenging early temperaments may learn to manage their intense negative emotions if their parents can successfully adapt their parental style, and subsequently lead successful and happy lives. This means that a variety of intervening processes can play important roles in determining the eventual outcomes of children's early individual differences. These differences may promote personality continuity by shaping children's experience and engagement with their environment.

Caspi and Shiner (Caspi & Shiner, in press; Shiner & Caspi, 2003) compellingly cover the processes that facilitate stability. Briefly, these authors talk about elaboration, which could lead to further solidifying of a given personality style. According to these authors, the process of elaboration may involve at least six processes, thought to develop at different stages of development. The first of these processes starts in the first few months of life and includes learning processes and environmental elicitation. In early to middle childhood, after the emergence of necessary cognitive functions, the second set of processes—environmental construal and social comparison processes—may influence personality development.

Third, in late childhood and adolescence, environmental selection and manipulation emerge. These processes come online once the self-regulatory functions have developed. Shiner and Caspi describe these six processes through which endogenous biological and experiential learning sources of influence may serve to elaborate an initial disposition over time so that it increasingly organizes thought, emotion, and action. The theoretical models are in place; research is now needed on each of these processes in relation to different temperament and personality traits.

Other processes promoting continuity and change range from genetic or biological processes to more broad, contextual processes involving classrooms and neighborhoods (Kagan, 1998; Shiner, 2006). Genetic influences may play a role in shaping the continuity versus discontinuity of early individual differences. The MacArthur Longitudinal Twin Study (Saudino & Cherny, 2001) showed that for all of the traits except shyness, twins' stability in temperament—in terms of activity, task orientation, affect-extraversion, shyness at home, and behavioral inhibition in the lab—was derived from genetic influences; for shyness, both genetic and shared (familywide) environmental factors contributed to stability. On the other hand, sources of change varied for the traits; changes in shyness across time derived from both shared and nonshared (child-specific) environmental influences, whereas change in affect-extraversion was due to nonshared environmental influences only. These results fit well with other research on children and adults suggesting that, whereas personality stability appears to be due to genetic factors, change is due to environmental influences (Saudino & Plomin, 1996). Research in this area is preliminary. The Louisville Twin Study has found evidence for genetic influences on change in temperament in the first 4 years of life (Loehlin, 1992), but clearly, more work is needed to establish more definitively the genetic influences on change in children's individual differences, particularly focusing on both positive and negative outcomes.

Another factor that could be related to stability and change may have to do with the permutations of traits and other characteristics that accompany personality. Shiner (2006) has referred to these factors as intrapersonal factors. The point here is that the mix or makeup of children's characteristics is another source of stability and change. Allport's (1937, p. 48) classic definition of personality assumes the importance of intra-individual psychological organization: "Personality is the dynamic organization within the individual of those psychophysical systems that determine his unique adjustment to his environment." Researchers have recently attempted to understand how this "dynamic organization within the individual" may shape the stability and change in youths' personalities. These types of intra-individual characteristics have been investigated as potential moderators of personality stability and change and include other personality traits, IQ, and adaptation.

To elaborate on this point, Eisenberg and colleagues (2000) demonstrated that children's negative emotionality often moderates the links between aspects of conscientiousness and children's social behavior, such that children with poor attentional and behavioral control have poorer quality social functioning when they are also high on negative emotionality. In addition, poor self-control is particularly associated with later externalizing, antisocial behavior for children with higher levels of negative emotionality (Henry et al., 1996). Second, and importantly, children's

intelligence may moderate the stability and outcomes of their personalities. Brighter children may find it easier to handle challenging aspects of their personalities. For example, a longitudinal study of children from age 4 to age 10 demonstrated that IQ predicted the stability and change in behavioral inhibition or shyness. This study found that more intelligent children became increasingly less inhibited over the 6-year span (Asendorpf, 1994). Third, children's relative success or failure in important developmental domains (e.g., academic achievement, social competence with peers) may influence personality development. Research previously reviewed makes clear that childhood personality shapes later adaptation in work, school, and relationships (Shiner, 2006). It is true that the converse can occur, whereby life adaptation may influence children's typical ways of feeling, thinking, and behaving (Shiner & Masten, 2002; Shiner et al., 2002). Clearly, in the future it will be important to examine the patterning of children's individual differences in personality, IQ, and adaptation in order to understand the circumstances leading to stability versus change (Shiner, 2006).

Other factors impacting personality development are parenting, neighborhood context, schools, and the like. As noted previously, Thomas and Chess's (Thomas et al., 1963) early work on temperament highlighted the potential role of family environment in moderating the outcomes of children's early individual differences. This idea was at the core of their goodness-of-fit model of personality development. Recent research has shown several replicable patterns of interactions between childhood temperament and parenting in the prediction of a variety of outcomes (Bates & McFadyen-Ketchum, 2000; Gallagher, 2002; Hawley, 2003; Putnam, Sanson, & Rothbart, 2002). In a handful of studies, it has been found that parenting practices moderate the links between temperament and later externalizing problems. Specifically, when children are high on irritability or score low on agreeableness, several maternal parenting variables predict higher externalizing behaviors. These predictor variables include unskilled disciplining tactics, negativity toward the child, and a lack of monitoring and control over the child (Bates, Pettit, Dodge, & Ridge, 1998; Belsky, Hsieh, & Crnic, 1998; Hagekull & Bohlin, 2003; Rubin et al., 2003; Shiner, 2006; Stoolmiller, 2001). Some research has shown that parenting affects attachment, which impacts emotion regulation and developing extraversion (Sroufe, Carlson, Levy, & Egeland, 1999). Attachment was also found to predict neuroticism and openness (Hagekull & Bohlin, 2003). It appears that skillful, warm parenting may protect highly negative and hard-to-manage children from developing externalizing problems. These findings are similar to those noted by Dishion and Patterson (2006). Because it is possible that genetic factors can account for the covariance of children's temperaments and family variables, future work on goodness of fit and contextual factors should utilize genetically informative research designs (Caspi & Shiner, 2005; Shiner, 2006).

Another source of children's personality continuity and change involves the broader context in which children develop, including the neighborhoods in which they grow and the schools they attend. Children's school environments, peer relationships, and neighborhoods could all have important impacts on whether children's personalities remain stable and whether their personalities lead to either poor or good outcomes (Shiner, 2006). Although these broad contextual factors have

received less research attention than more proximal family processes, a number of recent studies have found evidence of moderating effects of children's contexts on their personality development (Chang, 2003; Dodge et al., 2003; Gazelle & Ladd, 2003). These studies suggest that peer relations, teacher warmth, and neighborhood warmth, safety and stability moderate the links between poor self-control and criminal involvement. Impulsive youth in better-off neighborhoods or who remain in school show less criminal behavior than similarly impulsive youth who live in poor neighborhoods or who drop out of school prematurely (Henry, Caspi, Moffitt, Harrington, & Silva, 1999; Lynam et al., 2000). Taken together, these studies provide clues that broader contexts are, in fact, important moderators and help chart the course of personality development.

CONCLUDING COMMENTS

Despite substantial progress in the decades since Thomas and Chess's (Thomas et al., 1963) classic work on temperament, some aspects of childhood personality development remain quite limited. Great strides have been made with respect to better understanding the connection between temperament and personality. In addition, the higher order structure of personality appears to be gaining further resolution. Yet, although these aspects of personality development have made remarkable progress, some of the same old debates continue to arise in personality research, particularly as it applies to children and adolescents. These include whether it is developmentally appropriate to look for and examine personality in children, and whether it should be assessed in child samples. Much of the concern has to do with whether personality is stable across the life span. Shiner and Caspi (2005) have argued that it is not an either/or argument with regard to stability, but rather both stability and change occur, and the process that underlies that change requires much greater understanding. This does not negate the importance of studying and using the concepts of personality with children and adolescents. The research cited in this chapter shows that not assessing personality means not understanding it. Along with these concerns, the connection between personality and psychopathology is not well understood. Importantly, Caspi and Shiner and Tackett (2006) have pointed the way to some important methods and research designs to advance our understanding of personality as it relates to positive and pathological outcomes in children and adolescents.

Although significant strides have been made in the area of personality in children and adolescents, this chapter signals the need for study in several other areas. The following are suggestions for potentially productive avenues of work on early personality structure that stem from the already important work conducted on children and adolescents. The research reviewed in this chapter leads to some natural conclusions and may provide a road map for where we need to go with personality investigations. These suggestions have in mind the goal of examining the long-term outcomes of childhood personality and the process shaping personality continuity and change from childhood to adulthood. One area of research that is needed is to continue to work on measures that are developmentally appropriate for children and adolescents. Although some work has been done in this area (De Fruyt et al.,

2006), efforts that focus on bottom-up approaches might be informative. As noted by Caspi et al. (2005), the field could benefit from continued creative measures of individual differences and an openness to the need to refine measures. Relatedly, another area of importance for research studies is to continue to examine the higher order structure of personality in children, particularly the structure of individual differences from the very earliest ages (infancy through to age 7 or 8). Shiner (2006) has noted that this is a particularly important area of study because the developmental changes during this period are rapid and sweeping. This chapter also signals the need for future research to specify lower order traits that can be identified in children. Many lower order traits examined in adults may prove to exist in children (e.g., integrity, talent, unconventionality).

Aside from structural issues, much more work is needed on the developmental mechanisms (or processes) at work. Although many of the findings on the long-term outcomes of children's personalities are robust and consistent, future work should examine the pathways linking personality and positive outcomes (Shiner & Masten, 2002). Scientists need to understand more about the mediating processes (Funder, 2001). As Shiner (2006) noted, many studies linking early personality with later outcomes span long periods of time. It could be very informative to also use more short-term longitudinal studies to provide an up-close glimpse of the processes at work. An avenue of research that needs work is the study of the genetic, intra-individual, family, neighborhood, and other contextual factors promoting personality continuity and change in children. Again, it will be important to explore the processes that actually mediate stability and change. Finally, much more research is needed to model the relation between personality and psychopathology. Caspi and Shiner (2005) and Tackett (2006) have provided explicit models for how these relations could be tested, and their guidelines offer a very good starting point.

Discoveries in all these areas may lead to interventions to help children with challenging temperaments or personality styles. Research is also likely to help us understand the features in children and adolescents that promote healthy development and well-being. Future work may serve to enrich understanding of how resilience emerges across development. Each of these avenues of research, and each of these research aims, will be very informative for intervention programs designed to help youth grow and flourish.

SUMMARY

This chapter reviewed research on the development of personality in children and adolescents. It covered the differences between the terms temperament and personality and examined the structure of personality in children and adolescents. An important aspect of studying personality is examining how stable it is across time. This chapter shows that personality is stable but that it also changes across time, even well into adulthood. This chapter also examined both the positive and negative outcomes associated with each of the Big Five dimensions. In addition, this chapter examined the connection between personality and psychopathology. Finally, the current chapter points to directions for future research that could shed light on the development of personality across the life span.

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CHAPTER 13

Treatment Research

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This chapter provides a broad overview of principles, practices, and issues related to planning, conducting, and evaluating treatment research with children. We begin by discussing relevant issues, considerations, and obstacles in conducting child research. These include developmental considerations, measurement issues, informed consent, and confidentiality. The ensuing sections recapitulate the evolving debate concerning clinical efficacy and clinical effectiveness, and the evolution of empirically supported treatments (ESTs) for children. Methodological approaches that address the selection and evaluation of outcome measures appropriate for children—including the clinical significance of findings—are highlighted afterward. The last two sections provide broad coverage of research designs frequently used in child research and discuss factors relevant to identifying a research sample, including sample size and power.

MEASUREMENT ISSUES AND DEVELOPMENTAL CONSIDERATIONS

Measurement Issues

Conducting research with children requires broad knowledge of measurement issues related to maturation, selection of dependent measures, and the degree to which informants (parents, teachers, observers) agree with one another when providing information about children's behavior. This is due to several factors. Children's behavior is qualitatively different from that of adults and thus requires special consideration with respect to measurement and observation. Extrapolating adult measures for use in child research—by changing wording, modifying administration procedures, and making other adjustments—frequently fails to capture the intended underlying construct. Childhood behavior problems typically entail a broad range of symptoms, and these difficulties can affect functioning in multiple situations and settings. Observations of core and secondary symptoms may consequently vary significantly depending on setting and task demands. For example, children with Attention-Deficit/Hyperactivity Disorder (ADHD) typically experience chronic behavioral and academic difficulties at home and at school, and there is generally good agreement among observers. In contrast, children with internalizing disorders are frequently underrecognized until daily functioning is sufficiently

impaired to render their suffering more transparent. Multiproblem and multisituation assessment and analysis are thus the rule rather than the exception in child research.

Finally, children's rapidly emerging and changing verbal and cognitive abilities require careful consideration for research studies in which children are asked questions about themselves, others, and internal (e.g., mood/affect, anxiety, concentration) or external (e.g., activity level, impulsiveness) states. This consideration is complicated by extant research demonstrating differences among raters (e.g., parents, teachers, and trained observers), across settings (e.g., home and school), and for different types of behavior problems, as discussed later in the chapter.

Developmental Considerations

Treatment research with children is an intriguing but complicated domain of clinical psychology due to developmental factors that play a key role in our understanding of and assessing maladaptive behavior. For example, many behaviors associated with or mistaken as maladjustment and emotional disturbance are relatively common during childhood. Estimated prevalence rates based on parent reports indicate that fears and worries (43%), temper tantrums (80%), bedwetting (17%), and restlessness (30%) occur frequently in 6- to 12-year-old children (Lapouse & Monk, 1959) but do not necessarily portend later psychological dysfunction. In a similar vein, most children exhibit stranger anxiety around 8 months of age that mirrors their ability to discriminate a familiar face from an unfamiliar face. Separation anxiety becomes evident shortly after stranger anxiety begins, characterized by distress and an inability to be readily comforted by others in the absence of the child's parents or primary care providers. This is a normal response pattern in very young children, whereas excessive anxiety concerning separation from major attachment figures in later years accompanied by other behaviors—such as reluctance to attend school, unrealistic worry that an untoward calamitous event will separate the child from his or her parents, and somatic complaints—may signal the onset of a clinical disorder. Oppositional behavior is a normal developmental phenomenon in 18- to 36-month-old children; however, its persistence and accompaniment by other behavior problems is considered maladaptive in later years.

Behavioral and emotional problems evident in less severe forms during different developmental stages frequently represent normal developmental progressions in children, whereas their persistence may predict developing psychopathology. Consider the heterotypic continuity of ADHD symptoms as an example. Symptoms of ADHD arise early in childhood; however, their presence does not necessarily portend a persistent pattern of ADHD beyond 3 years of age in an estimated 50% to 90% of children so characterized (Palfrey, Levine, Walker, & Sullivan, 1985). Continuation of early ADHD-like symptoms to 4 years of age, however, is highly predictive of clinical hyperactivity at 9 years of age (Campbell, 1990). Thus, the early onset, degree, and persistence of symptoms past 4 years of age is highly predictive of a clinical diagnosis (indicating continuing and worsening impairment) and continuing difficulties throughout adolescence and early adulthood.

Other problem behaviors show clear developmental trends and are not considered maladaptive until late childhood. Lying and destructiveness are common exemplars. An estimated 50% of boys and girls engage in lying by age 6, based on parent report. Frequency of lying decreases to approximately 25% and 13% by age 7 in boys and girls, respectively, and continues to follow a downward trend as a function of increasing age in most children. Destructive behavior shows a similar developmental trend, peaking at 3 and 5 years of age in girls and boys, respectively, with a clear downward trend for both sexes through age 13 (MacFarlane, Allen, & Honzik, 1954).

Some maladaptive behaviors remain stable over time in terms of their frequency, but change with respect to topography or form. The manifestation of childhood aggression is a good example. Threatening, pushing, and shoving in young children frequently evolve into verbal and physical assault in older children, who continue to manifest aggressive behavior (Dishion & Patterson, 2006).

The foregoing examples inform us that many problematic behaviors in children will diminish or remit over the course of normal development and may not be appropriate or high priorities for intervention unless the behavior is sufficiently impairing and more rapidly alleviated by clinical treatment. They also highlight the importance of understanding the typical frequency and topography with which specific behavioral and emotional problems occur in children, and how base rates change over the course of development. Failure to appreciate and control for these variables may inadvertently result in attributing change to treatment rather than maturation or other historical events. Study design must also plan for possible topographical changes in maladaptive behavior if outcome assessment spans several years (e.g., in some cross-sectional and longitudinal studies). This may necessitate using different age norms or even different measurement instruments, which introduces unwanted error into the protocol.

A different set of developmental considerations comes into play when research studies require investigators to assign children to groups based on the presence or absence of a clinical disorder. Structured or semi-structured clinical interviews, coupled with other instruments and measures, are characteristically used to render a clinical diagnosis for purposes of group assignment (for reviews, see Orvaschel, 2006; Rapport, Timko, & Wolfe, 2006). Skilled interviewers have little difficulty interviewing adult informants, whereas interviewing children about their emotional state can be extremely difficult, and at times impractical, depending on developmental factors. Children's emotional development significantly affects their ability to respond to clinical interview questions about themselves and how they feel. The ability to label emotions and talk to others about feelings develops by about 2 years of age, and children have a reasonably well-developed range of emotional displays and labels by school age. Children can report changes in positive emotions before negative ones, but most experience difficulty describing variability in negative emotions until late childhood or early adolescence. Conscious awareness of and the ability to report on emotions as a state or system is not fully developed until middle or late adolescence (Haviland-Jones, Gebelt, & Stapley, 1997).

MEASURING AND ASSESSING CHILDREN'S BEHAVIOR

Informant Ratings of Children's Behavior

Behavior checklists and rating scales play a prominent role in child research. They serve as an important source of information concerning a child's behavior in different settings, how others judge behavior, and the extent to which behavior deviates from age- and gender-related norms. Information gleaned from rating scales contributes to the diagnostic process, and several scales serve as treatment efficacy measures.

Standardized scales are available that provide broad (e.g., internalizing, externalizing behavior problems) and narrow (e.g., measures of particular clinical disorders or states such as depression) indices of behavior, as well as for particular constructs (e.g., self-esteem, perceived competence) and other types of functioning (e.g., classroom performance, adaptive behavior, peer perceptions). Incorporating rating scales in research necessitates examination of the scale's psychometric properties (i.e., whether the instrument provides valid and reliable information with respect to what it purports to measure), general knowledge concerning the degree to which different informants can be counted on to provide valid information about a child's behavior, and factors that influence informant ratings. A judicious review of an instrument's psychometric properties is a necessary first step in constructing a research protocol. This is accomplished by checking with the publisher or author of the instrument, conducting a literature search of the instrument's psychometric properties, and researching specialty texts that cover tests and rating scales (e.g., *The Fifteenth Mental Measurements Yearbook,* edited by Plake, Impara, & Spies, 2003).

Extant research examining the degree to which raters agree concerning the presence of behavior problems reveals several, relatively consistent trends. In general, informants who interact with the child in the same environment (e.g., parents) tend to show better agreement in their reports of behavior than those who interact with the child in different environments (e.g., parents versus teachers versus mental health workers; Achenbach, McConaughy, & Howell, 1987). Agreement between parents, however, is less than ideal, as highlighted in a recent meta-analytic review (Duhig, Renk, Epstein, & Phares, 2000). Correspondence between mother and father ratings of children's behavior problems varied depending on the category of problem behaviors examined. For example, mean correspondence between parents for internalizing behavior problems was .45 as opposed to .63 and .70 for externalizing and total behavior problem scores, respectively—perhaps because externalizing behavior problems are easier to judge and/or more consistent across situations or settings (Achenbach et al., 1987; Walker & Bracken, 1996). Although a correlation of .45 is considered a moderate level of agreement between raters, it nevertheless indicates that only 20% ($.45^2$) of the variability in one parent's ratings can be explained (predicted) by the variability in the other parent's ratings. Higher correspondence between parents was reported for adolescents than for younger children when examining both internalizing and externalizing behavior problems, and the family's socioeconomic status appears to exert small but significant effects

on parent ratings. Agreement among raters also varies for clinical and nonclinical samples, is more variable for particular types of behavior and emotional problems, and is unacceptably low when comparing children's self-ratings to adult ratings (Achenbach et al., 1987).

The foregoing summary indicates that parents and teachers can be relied on to provide reasonably reliable ratings of children's behavior in the context in which they observe children for broad indices of behavior problems such as internalizing and externalizing behavior problems. Diminished correspondence between informants is evident, however, for ratings of more discrete types of behavior problems, when informant ratings are based on different settings, and when comparing children's own ratings to those of adults. Investigators need to consider these issues during the research design stage and minimize error variance by utilizing a multitrait, multimethod approach. This approach recognizes the undesirable shared overlap (variance) among similar measures and raters of specific constructs and behaviors and informs the investigator how to maximize the unique contribution of measures relevant to the variables studied (cf. Burns & Haynes, 2006, for a review).

Measurement Limitations

Limitations common to most rating scales include their reliance on subjective judgments and multiple threats to internal validity. These include halo effects, response bias, intensity and immediacy effects, and rater expectation bias (Harris & Lahey, 1982; McClellen & Werry, 2000). The underlying assumptions that Likert rating formats reflect interval-level measurement (i.e., that the unit of measure between 2 and 4 is identical to the difference between 1 and 3, and that these behavioral units are consistent across scales) and that all behavioral and emotional problems should be equally weighted (count the same when endorsed at the same level) represent additional psychometric challenges to clinical rating scales. For example, two children could receive identical Conduct Disorder rating scale scores, yet be dramatically different with respect to behavioral disturbance. One child might receive four 2-point ratings (indicating the highest level of severity or frequency) for items such as lying, stealing, cursing, and truancy, whereas the other child receives four 2-point ratings for breaking and entering, rape, firearm violation, and physical aggressiveness. Their total scores would be identical, yet the second child's symptoms are clearly more serious than the former's. Other factors, such as the presence of psychopathology among informants, the informant's experience with children, and other demand characteristics, represent additional challenges to clinical child rating scales.

Several desirable psychometric properties have not been thoroughly established for many child rating scales. For example, scale validity is usually accomplished by demonstrating that scores derived from one instrument correlate with those derived from an already established scale of the same latent variable, or by demonstrating that children known to be highly active (e.g., ADHD) score significantly higher than normal peers on the scale. Most scales meet at least one of these two criteria. Extant research, however, reveals that even when scale scores are correlated, they may be unrelated to objective measures of the same trait (e.g., Rapoport, Abramson,

Alexander, & Lott, 1971; Stevens, Kupst, Suran, & Schulman, 1978). For example, when measured by step counter, nearly 64% of children rated as clinically hyperactive were less active than the most active child rated as being normal by the teacher (Tryon & Pinto, 1994). Collectively, these shortcomings limit the interpretability and usefulness of activity rating scales.

The diagnostic utility of most rating scales is unknown; however, this situation has improved considerably over the past several years. Four metrics address this concern: sensitivity, specificity, positive predictive power (PPP), and negative predictive power (NPP). Sensitivity and specificity indicate the proportion of the group with a target diagnosis who test positive and negative on a measure, respectively. These two indices are useful for examining the overall classification accuracy of rating scales and other instruments, but are not particularly valuable to clinicians who are unaware of a child's diagnostic standing prior to referral. The statistics most relevant for this purpose are PPP and NPP. As it applies to rating scale utility, PPP indicates the conditional probability that a child exceeding a rating scale cutoff score meets criteria for a particular diagnosis such as ADHD (i.e., the ratio of true positive cases to all test positives). In contrast, NPP indicates the conditional probability that a child who doesn't exceed an established cutoff score will not meet criteria for a particular clinical diagnosis (i.e., the ratio of true negative cases to all test negatives). High values (e.g., $>.80$) for all four indices are desirable.

Finally, it is becoming increasingly common for investigators to create specific measures for assessing outcome by borrowing from measures with established psychometric properties and reconfiguring them for the study. For example, items contained in adult rating scales may be reworded and rekeyed (i.e., changed from a 1–5 to a 1–3 metric scale) before administering them to children. Others borrow from different scales that reflect a similar latent construct (e.g., anxiety) and combine or select specific items thought to better reflect the construct. These well-intentioned efforts ignore the backbone of psychological measurement: psychometric theory. Altering a measure, no matter how strong its existing psychometric properties, requires the investigator to reestablish its psychometric properties with the intended population (e.g., children) before using it in research. The researcher must also consider relevant developmental phenomena when reconfiguring rating scales and other questionnaire instruments. For instance, young children are unable to describe variability in negative emotions until late childhood or early adolescence, and usually cannot reliably discriminate between behavioral descriptions on a 5-point frequency or severity scale.

Interview Measures of Children's Behavior

Conducting detailed clinical interviews with children and their caregivers is a common method for screening participants, determining group assignment, and obtaining study data for child research investigations. Structured and semi-structured clinical interviews provide unique information beyond the data gained from rating scales and currently represent the only assessment option that allows clinicians to probe for the onset, course, and duration of endorsed symptoms—a necessity for differential diagnosis. As a result, they are considered the gold standard for

assessment and diagnosis. The distinction between structured and semi-structured interviews lies in the degree of freedom granted to the clinician to stray from a given script and ask open-ended, probing questions in response to symptom endorsements made by the interviewee. As a general rule of thumb, semi-structured interviews evoke more detailed information concerning presenting symptomatology owing to the extensive probing and clarification permitted (e.g., by asking for examples) but require greater clinical acumen and training.

The development of structured and semi-structured interviews represents the recognition of both the questionable reliability and validity of unaided clinical interviews and the frequent disagreement between parent and child reports of symptom endorsement and severity. Structured and semi-structured clinical interviews result in decreased error compared to unstructured clinical interviews arising from both internal (e.g., differential training level of clinicians, clinician biases) and external (e.g., discrepancies between informants) sources. The interviews typically require between 1 and 2 hours to complete, depending on a range of factors. These include the clinician's experience, the informant's ability to remain focused and recall historical information, and the severity, range, and duration of presenting problems (Table 13.1). The time investment limits their practicality for repeated use (e.g., for assessing treatment effects). Financial investment varies significantly across interview schedules. Some are free and available online (e.g., Schedule for Affective Disorders and Schizophrenia for School-Age Children [K-SADS]), whereas others have an initial cost of $600 in addition to $2,000 fixed training costs (e.g., Child and Adolescent Psychiatric Assessment [CAPA]). All but one of the semi-structured interviews covers all major *DSM-IV* diagnoses for school-age children through age 18. Some of the semi-structured interviews provide separate versions for parents, children, and adolescents. None of the available semi-structured interviews includes teacher versions. Clinicians must use other instruments to obtain school-related information for purposes of establishing impairment across multiple settings.

Strong convergent and discriminant validity is reported for most of the available semi-structured interviews, and test-retest reliability suggests stability of diagnosis over 1 to 3 years in clinical samples (Pelham, Fabiano, & Massetti, 2005). High sensitivity and specificity are typically reported for the semi-structured interviews, with only limited information available pertaining to PPP and NPP. The lack of PPP and NPP metrics is due in part to the use of semi-structured clinical interviews as the gold standard from which the predictive power of other measures (e.g., rating scales) is established.

The CAPA appears to be the most extensively developed of the clinical interviews, but it requires up to 2 weeks of classroom instruction and an additional 1 to 2 weeks of practice to acquire the necessary certification (Angold & Costello, 2000). This training is estimated to cost $600, and there is an additional fixed cost of $2,000, which may limit widespread usage among clinicians, especially when schedules such as the K-SADS demonstrate adequate reliability and validity and are available online at no cost. The CAPA, however, may be superior to other available schedules due to several excellent features. The instrument provides for an intensity rating that varies by three symptom groupings: intrapsychic phenomena

Table 13.1 Clinical Child Diagnostic Interviews

Measures	Age Range	Time (min)	Test-Retest (Kappa)	Symptom History	Disorders Considered	Scoring Format	Instrument Cost ($)	Training Required
Structured Interviews								
Diagnostic Interview Schedule for Children IV (DISC-IV)	6–17 (P) 9–17 (C)	90–120 (P) 45–90 (C)	0.79 (P) 0.42 (C)	4 weeks/ 12 months	All major Dx	Y/N	150– 2,000	2–3 day training module
Diagnostic Interview for Children and Adolescents IV (DICA-IV)	6–17	60–120	NR (P) 0.32 (C) 0.59 (A)	4 weeks/ 12 months	All major Dx	Y/N	1,000	2–4 weeks
Children's Interview for Psychiatric Syndromes (CHiPS)	6–18	40	0.4	NR	All major Dx	Y/N	115	NDR
Semi-structured Interviews								
Schedule for Affective Disorders and Schizophrenia for School-Age Children (K-SADS)	6–17	30–90	0.63	6 months/ lifetime	All major Dx	0–3	Free online	CTR
Semi-structured Clinical Interview for Children and Adolescents (SCICA)	13–18	60–90	0.57 (AP Scale)	NR	Does not correspond with *DSM-IV*		110–295 25 for 50	NDR
Child and Adolescent Psychiatric Assessment (CAPA)	9–17	20–210 $M = 66$ (P) 22–150 $M = 59$ (C)	NR for ADHD 0.55 for CD	3 months	All major Dx	0–3	600 + 2,000 Fixed Costs	BA
Interview Schedule for Children and Adolescents (ISCA)	8–17	120–150 (P) 45–90 (C)	Between 0.64–1.0	NR	All major Dx	0–3		CTR

Notes: Properties are equivalent for parent and child/adolescent version unless otherwise indicated.
A = Adolescent; AP = Attention Problems; BA = Bachelor's level training; C = Child; CTR = Clinical training required; Dx = Diagnosis; NDR = No degree requirements; NR = Not reported; P = Parent.

such as worrying, qualitatively different symptoms such as psychosis, and conduct disturbances. Training and coding are based on a detailed glossary, and thoroughly investigated symptoms are matched to appropriate glossary definitions and levels of severity. Formal rules are provided for the use of screening, mandatory, and discretionary questions.

The Diagnostic Interview for Children and Adolescents, *DSM-IV* edition (DISC-IV), provides separate versions for children (ages 6 to 12) and adolescents (13 to 18) based on field testing of interview questions with different age, gender, and racial groups. Two to 4 weeks of training at a cost of approximately $1,000 are required to reach the desired level of competence. The duration of training is based on clinician experience and includes topics such as age-appropriate probe questions and maintaining a child's interest through techniques such as tone of voice and appropriate nonverbal gestures. Reliability estimates corroborate research findings indicating that children are less reliable reporters of externalizing symptoms but more reliable reporters on internalizing symptoms relative to their parents. Computer versions are available, but initial research suggests poor reliability compared to the standard interview format.

The K-SADS is currently the most widely used semi-structured clinical interview. The Present-Lifetime (PL) version collects information from the parent regarding both current symptomatology as well as symptomatology at its most frequent and severe levels in the past. A separate interview is conducted with the child, and a third pair of ratings is generated based on integration of the parent and child reports with historical information and other data (e.g., rating scales). The K-SADS-PL focuses on chronology, treatment, impairment, and severity of symptoms. The initial screening interview consists of 82 items covering all major *DSM-IV* diagnostic categories. Cutoff scores are used to determine the need to administer the in-depth supplementary sections available for each diagnostic category, thus shortening administration time by allowing the clinician to skip supplementary sections based on negative endorsement of key screening questions. Interrater reliability estimates for the K-SADS are among the highest of any of the semi-structured clinical interviews. Extensive knowledge of diagnostic and symptom subtleties is required owing to lack of formal training requirements, and clinicians must use their judgment to interject noncued verbal probes to clarify informant responses and elicit examples of problematic behavioral and emotional symptoms.

Observational Measures of Children's Behavior

Behavioral observation is a time-consuming but valuable approach for obtaining detailed information about children and the environments in which they learn, play, and interact with others. It refers to a process in which observers record motor, verbal, and interactive behaviors of children using carefully defined operational criteria as opposed to evaluative judgments (e.g., rating scales) or mechanically activated devices (e.g., movement monitors, physiological monitoring). As such, many consider it the sine qua non of child assessment. An advantage of using behavioral observations is that the technique is virtually unrestricted with respect

to context. Observations can be conducted in naturalistic settings (e.g., to record classroom, playground, or bus riding behavior), analogue settings (e.g., a clinic room furnished to resemble a classroom), or specialty clinics (e.g., to assess parent-child interactions as part of a treatment program for children with Conduct Disorder, to assist a diagnostic evaluation for autism).

Numerous observational coding schemas are available for research purposes. Some permit coding of a wide range of behaviors (e.g., parent-child interactions, general classroom deportment) and are best suited for particular settings (e.g., home or classroom observations). Others permit highly refined and detailed observation, recording, and measurement of more discrete types of behavior (e.g., gross motor activity). Assessing interobserver agreement is nearly always required for research studies and involves pretraining on the selected coding schema and arranging for multiple observers to code the targeted behaviors simultaneously while in the setting or based on taped recordings. Detailed information concerning how to select particular stimuli, responses, and specific recording techniques; how to operationalize particular types, forms, and classes of behavior; and how to calculate interobserver reliability is readily available in classic texts on child behavior assessment (Kazdin, 1982; Ollendick & Hersen, 1998).

Direct observations represent the gold standard for experimental and outcome research for many investigators, but their use in clinical assessment is limited by several factors. No two research teams or commercially available observation systems define behaviors in exactly the same way, and research indicates that differences in observational schema can produce widely discrepant results in collected data (see Kofler, Rapport, & Alderson, in press, for a meta-analytic review).

Commercially available observation systems shown in Table 13.2 are available for school-age children and typically require between 10 and 30 minutes of observation. Multiple observation days are required to produce representative and reliable data; however, some systems offer software versions of their product, allowing researchers to record behavior directly onto a personal digital assistant (PDA) or laptop computer.

Direct observations are commonly used in school and institutional settings that require a formal functional analysis to determine whether antecedent and consequential stimuli or events contribute to a child's maladaptive behavior. These specialized assessments are typically conducted by highly trained school psychologists and individuals specializing in applied behavior analysis.

Although direct observations by independent observers can provide more objective and valid data than any of the other assessment methods discussed here, their relatively high temporal cost, coupled with the general lack of norms, suggests that their usefulness for assessment and group assignment may be limited to situations where large discrepancies exist among informant reports.

An alternative and less costly procedure for obtaining information relevant to children's classroom functioning is to ask classroom teachers to answer specific questions about a child's day or to save work samples. For example, desk checks ("Is the child prepared for class?"), teacher records of verbally intrusive behavior, and the percentage of daily academic assignments completed correctly appear to be useful for discriminating between children with and without ADHD (Pelham

Table 13.2 Mechanical and Observational Assessment Tools

Instrument/ Distributor	Age Range	Recording Length	Norms	Software Available	Cost
Mechanical					
Actigraphs Ambulatory Monitoring MiniMitter MTI, Inc.	Any	22 days per 32 kilobytes of memory	N	Y	Starter: $1,000+ (with necessary software and reader interface); $500–$2,000 for each additional actigraph
Actometers[a] Model 108 Engineering Department Times Industries Waterbury, CT 06720	Any	Variable	N	Y[b]	*NR*
Pedometers (available at sporting goods stores) Stand-alone with data downloadable to PC	Any	Range: 99,999 steps (~5.25 miles) to 1,000 miles	N	N Y	$10–$40 $125–$400+
Direct Observations					
ADHD BCS Barkley, 1990	*NR*	15 min.	N	N	*NR*
AET-SSBD Sopris West	School age	15 min.	Y	N	Kit: $108 (includes all three parts of SSBD)
ADHD-SOC Checkmate Plus, Ltd.	School age	16 min.	N	N	Kit: $25
BASC-2 SOS AGS, Inc.	School age	15 min.	N	Y	25 forms @ $33
BOSS Harcourt Assessment	School age	15 min.	N	Y	Kit: $120
COC Abikoff, 1977/1980	School age	32 min.	N	N	NR
DOF ASEBA	5–14	10 min.	Y	Y	50 forms @ $25
SECOS Saudargas, 1997	Grades 1–5	20 min.	Y	N	
Noldus Observer Noldus Information Technology	Any	Variable	N	Y	Observer Basic 5.0 $1,795 Observer Video Pro 5.0 $5,850

Notes: AET-SSBD = Academic Engaged Time Code of the SSBD; BCS = Behavior Coding System; BOSS = Behavioral Observation of Students in Schools; COC = Classroom Observation Code; DOF = Direct Observation Form; NR = not reported; SECOS = State-Event Classroom Observation System; SOC = School Observation Code; SOS = Student Observation System.

[a]Many studies report either using the Kaulins & Willis actometers (no longer manufactured) or enlisting a jeweler to modify a self-winding wristwatch as described by Schulman & Reisman (1959).

[b]Eaton, McKeen, & Saudino (1996) provide SAS syntax for performing group-level data analysis based on actometer readings.

et al., 2005). These methods have the potential for objectivity that characterizes the independent observation methodologies but await critical evaluation to determine their utility for rendering diagnostic judgments at the individual level.

Potential Obstacles to Conducting Research with Children

Knowledge of regulatory statutes, the role and function of institutional review boards (IRBs), and general procedures for ensuring and obtaining informed consent is essential for conducting research with children. Detailed information concerning procedures and regulatory statutes related to studies involving children can be obtained from the Department of Health and Human Services website (www.nihtraining.com/ohsrsite/guidelines/guidelines.html). These regulations provide the minimum standards for protecting human subjects, particularly under the subpart D section, which includes detailed information concerning the involvement of children in research.

Regulations require the *assent* of the child or minor and the permission or *consent* of the parents or legally authorized guardian when children are involved in research. Assent is a child's affirmative agreement to participate in research. The standard for assent is the ability to understand, to some degree, the purpose of the research and what will happen if one participates.

Consent refers to the permission or agreement of parents or guardian to the participation of their child or ward in research. There are two forms of parental consent. *Passive* parental consent is a procedure that requires parents to respond only if they *do not* want their child to participate in a research project. *Active* parental consent requires the parent to return a signed consent form indicating whether or not they are willing to allow their child to participate. The latter method is recommended, although many researchers prefer the passive consent procedure because of the larger participant pool typically achieved. As a general rule for research involving minimal risk, it is sufficient to obtain informed consent from one parent.

Informed consent consists of three primary elements: knowledge, volition, and competency. Parents or guardians of children must be knowledgeable about all aspects of a study, which includes a thorough description of the facts, plausible risks, and potential sources of discomfort associated with an experimental study that may affect their decision to permit their child to participate. This information must be presented to the child's parents or legal guardian in an understandable fashion, followed by an opportunity for them to ask questions and clarify any issues that might not be thoroughly understood.

Volition refers to the process in which participants (or, in the case of minor children, their parents or guardian) agree to participate in a study free of coercion or threat. Subjects (or, in the case of minor children, their parents or guardian) must be able to decline or withdraw from participating at any time preceding or during an experiment without penalty, and should be informed of this provision (i.e., participation must be entirely voluntary).

Ensuring competency in child research typically entails the parents' or guardian's ability to render an educated decision and give consent for their child to participate based on thorough knowledge and understanding of the study.

A consent form signed by the parents or guardian of children indicates their willingness to have their child participate in the study based on an informed decision. These forms traditionally undergo formal review by an IRB or agency committee that evaluates the research proposal, consent procedures, and the form itself. Consent forms vary by design, but must contain a thorough description of the study, inherent and potential risks and benefits, procedures, and issues concerning confidentiality and how the study data may be used. Components of an informed consent form are described in Table 13.3.

Institutional review boards serve to assess the risks, possible benefits, and discomforts associated with a research project. Before beginning a project, an appropriate IRB committee must consider and formally approve all research proposals. University-based projects will typically be reviewed by the institution's IRB panel, and additional approvals may need to be obtained from other agency boards (e.g., hospitals, school systems, National Institute of Mental Health), depending on the selection of participants and whether the project is funded or sponsored by private or federal agencies. Many universities and research centers currently mandate

Table 13.3 Components of Informed Consent Forms

Section of the Form	Purpose and Contents
Overview	Presentation of the goals of the study, why this is conducted, who is responsible for the study and its execution.
Description of procedures	Clarification of the experimental conditions, assessment procedures, requirements of the subjects.
Risks and inconveniences	Statement of any physical and psychological risks and an estimate of their likelihood. Inconveniences and demands to be placed on the subjects (e.g., how many sessions, requests to do anything, contact at home).
Benefits	A statement of what the subjects can reasonably hope to gain from participation, including psychological, physical, and monetary benefits.
Costs and economic considerations	Charges to the subjects (e.g., in treatment) and payment (e.g., for participation or completing various forms).
Confidentiality	Assurances that the information is confidential and will only be seen by people who need to do so for the purposes of research (e.g., scoring and data analysis), procedures to assure confidentiality (e.g., removal of names from forms, storage of data). Also, caveats are included here if it is possible that sensitive information (e.g., psychiatric information, criminal activity) can be subpoenaed.
Alternative treatments	In an intervention study, alternatives available to the client before or during participation are outlined.
Voluntary participation	A statement that the subject is willing to participate and can say no now or later without penalty of any kind.
Questions and further information	A statement that the subject is encouraged to ask questions at any time and can contact an individual (or individuals) (listed by name and phone number) who is available for such contacts.
Signature lines	A place for the subject as well as the experimenter to sign.

Source: Clinical Psychotherapy: Developing and Identifying Effective Treatments, by A. E. Kazdin, 1988, New York: Pergamon Press. Reprinted with permission from Allyn & Bacon and the author.

that all persons involved in the conduct of research—from the investigators to the research assistants—take a specific training course and pass an exam concerning the ethical and appropriate conduct of research. Several of these are offered online (e.g., www6.miami.edu/citireg/), and results are forwarded to the host university's IRB as a precondition for conducting research.

CHILD RESEARCH SETTINGS

General Dimensions

A research study may be conducted in a laboratory (e.g., clinic, university laboratory) or an applied setting (e.g., school, home, hospital, community mental health center). The advantage of most laboratory settings is that they typically allow for maximal control over the experiment. Their most apparent disadvantage is that the nature of the setting is usually quite different from most real-life situations, which calls into question the generality of results. The advantage of applied settings is that they frequently represent real-life situations or environments, which allows for greater generalization of results. Their most obvious disadvantage is that they usually afford less control than a laboratory setting. Typically, one thinks of true experimental and quasi-experimental designs as taking place in a laboratory setting and observational designs as taking place in an applied setting. This is because the researcher attempts to exert a high degree of control in true and quasi-experimental designs, whereas few or no controls over extraneous variables are implemented in observational designs. True or quasi-experiments, however, can be conducted in applied settings, and observational studies can occur in a laboratory setting. Single-case research designs are used in both settings.

Gaining Access to Educational Settings

Conducting research in applied settings such as a school involves striking a balance between maintaining the requisite degree of scientific rigor and respecting the mission and structure of the educational setting. Research protocols must be planned such that disruption of the school or classroom routine is minimized. The investigator must be mindful that in the educational setting, classes (as opposed to individuals) are frequently the units of analysis. For protocols involving individual administration, pulling a child from some class periods (e.g., recreational reading) may be less disruptive than others (e.g., small group science activity) for the class as a whole and the child in particular. Moreover, schools may be especially protective of certain groups of students, such as children with developmental disabilities or behavioral/emotional problems. Conducting research in educational settings consequently requires careful planning and organization to maintain scientific rigor and accommodate the structure of the school. The ensuing discussion is particularly relevant for group studies in which a large sample of participants is required.

Subject recruitment is a necessary first step for most research projects, and schools are a logical source for studies involving children. The choice of school depends on a number of factors, which vary in degree of relevance depending on the focus of the investigation. Among the factors to consider are the socioeconomic

status of the community served by the school and the ethnic distribution of the school's population of students. For investigations involving multiple groups, the number of students enrolled in the school becomes an important consideration, in light of the power needs of the study's design to detect group differences.

Gaining access to research participants involves a stepwise progression to enhance the likelihood of a proposal's acceptance by educational institutions. Prior to contacting schools, it behooves the investigator to obtain endorsements from related agencies and the district superintendent of education—this smoothes the way to the engagement phase of decision making for the school principal. Subsequent contact with the principal should first be handled formally through an individually addressed letter that briefly explains the importance and nature of the study and mentions the endorsements received, followed by a telephone call approximately 1 week later to secure an appointment for a meeting.

Prior to contacting schools, the investigator must prepare to address any questions or concerns that may be forthcoming. Petosa and Goodman (1991) outline five phases of decision making that school officials undergo and discuss them in relation to decisions surrounding research proposals. In the *legitimacy phase,* the credibility of the investigator and the relevance of the study are of central concern. Representatives of the research project who contact the school should be prepared to address questions concerning the expertise and affiliation of the chief investigator, the source of funding for the project, and the project's relevance for the education and welfare of children or the community at large. In the second, *information-seeking phase,* the concerns of school officials involve the impact of the project on the day-to-day operation of the school. The investigator should consider the timing of the data collection with the school's schedule in mind. For example, the weeks preceding the winter holiday break and the end of the school year are often hectic for school personnel. Other times of the year, depending on the school system, are largely devoted to mandated standardized achievement testing. School officials will also wish to know the specific requirements of the project to gauge its impact on the school's mission and function. The investigator should be prepared to address questions pertaining to the measures to be administered, the class time required for participation in the project, the costs of participation accruing to the school (e.g., the extent of involvement of school staff), and the resources that will be available to the school (e.g., teacher training and curricular materials). During this phase, school officials will also be vigilant of any potential controversy that may arise due to the study's methodology, including parent objections to particular rating scales or survey questions and the potential for wasted classroom learning time.

In the ensuing *expression of limitations phase,* education officials are likely to express concerns over idiosyncratic situational factors that may pose obstacles to the school's participation in the project. The resourceful investigator should prepare to address concerns over issues of *fit* between the study's requirements and the school's structure, without undermining the integrity of the research project. If the investigator is successful, all parties enter the *engagement phase* of mutual problem solving, in which various ways of increasing the feasibility of the study are considered, such as finding the optimal dates and times and allocating physical space for data collection.

In the final, *commitment phase,* all parties make specific plans for the school's participation in the research project. Olds and Symons (1990) recommend presenting a *data collection participation form* for the principal's signature to secure the school's commitment. The initial start date for the study should be made far enough in advance to allow the school time to block the dates set aside for the data collection in its master calendar and to disseminate the information to parents and school staff. At least 1 week prior to the agreed-upon date for data collection, a follow-up reminder to the school should be made by telephone. At that time, information concerning any changes in the school's normal schedule or routine—such as assemblies or field trips—made subsequent to the meeting with the principal should be solicited.

The foregoing information and sequence of steps pertains primarily to schoolwide and classroom studies but is also applicable to conducting single-case research. Because single-case research impacts a classroom rather than a school, the key contact becomes the teacher and, as a courtesy, the principal, rather than the district superintendent. The decision-making process on the part of school personnel remains the same, however, and the investigator should follow the same suggestions for proactive planning and organization.

Additional considerations are warranted once project personnel are in the school to facilitate school support and cooperation. Gaining and maintaining the support of school personnel depend on the behavior and professionalism of research associates. These individuals need to be visible but unobtrusive, professional in appearance and demeanor, friendly and polite without exception, and respectful of school policies, procedures, and personnel. Research associates must be mindful of the school's mission to educate children in a safe environment and, from the outset, solicit information from school staff about such routine procedures as signing in and out upon entering and leaving the campus and whether identification badges are required. Most important, researchers should adhere to the original contract as agreed to by the school principal and refrain from such deviations as adding measures, collecting data on other dates, and attempting to deliver any intervention not previously agreed upon.

The research design and methodology are of prime concern to the investigator, but respect for the mission and structure of the educational setting is equally important. Demonstrating respect by minimizing disruption of daily operations increases the likelihood that the research proposal will be approved and ensures that the data collection process will run as smoothly as possible. Moreover, following through with promised benefits after the data have been collected represents a sound investment, as the school may be amenable to participation in future research projects. One means of accomplishing this is to volunteer to provide a teacher workshop that includes summarizing the study's results.

Clinic and Laboratory Investigations

Clinic- and laboratory-based investigations provide researchers with unique opportunities to manipulate carefully designed independent variables and observe their effects on dependent variables. These types of studies tend to have unparalleled

heuristic value and frequently test predictions stemming from extant models of child psychopathology. For example, a recent model of ADHD hypothesized a causal relationship between working memory and activity level in children (Rapport, Chung, Shore, & Isaacs, 2001; Rapport, Kofler, Alderson, & Raiker, in press) and challenged the notion that hyperactivity represents a ubiquitous deficit unrelated to task and setting demands (Porrino et al., 1983). Testing predictions stemming from the working memory model would be difficult to accomplish in a typical classroom for a variety of reasons. Measuring classroom activity level by utilizing actigraphs or behavioral observations does not represent a serious obstacle; however, introducing working memory tasks at varying degrees of difficulty in a counterbalanced manner to specific children and appropriate controls might prove impractical, if not unfeasible. Moreover, obtaining appropriate control over ambient noise levels (including ongoing student conversations or comments), lighting, chair type and placement, distractions, movement, and other setting characteristics is necessary to ensure that changes in the dependent variable (e.g., activity level) are due to differences in working memory demands (e.g., manipulating stimulus set size) rather than extraneous factors. The research clinic provides an ideal setting for these types of investigations.

Research clinics also enable investigators to scrutinize a vast array of personal and interpersonal processes by means of analogue investigations. These studies establish pseudo-situations that mimic important aspects of a child's life, either at home (e.g., asking parents to interact with their children following specific guidelines) or at school (e.g., having children attend a simulated classroom). Results of these studies often have important diagnostic and treatment implications. For example, observations of children with autism interacting with their parents in an analogue setting reveals a high degree of diagnostic sensitivity based on the child's reciprocal social interaction, play, stereotypic behaviors, gesturing, and communication (Filipek, Accardo, & Baranek, 1999). Teaching parents more adaptive ways to interact with their children in analogue settings has also shown exceptional promise for reducing Oppositional Defiant Disorder (Reid, Webster-Stratton, & Hammond, 2003).

CLINICAL EFFICACY, CLINICAL EFFECTIVENESS, AND EMPIRICALLY SUPPORTED THERAPIES

Overview

The terms *clinical efficacy* and *clinical effectiveness* emerged in response to calls for increased use of empirically validated treatments for individuals suffering from mental health problems. Debate evolved between research centers demonstrating that particular treatments resulted in significant client improvement following prescribed protocols, and the failure to find similar levels of improvement in applied settings such as community mental health centers. Hypothesized variables contributing to the lack of generalization of treatment effects across settings included differences in therapist training and orientation, use of manualized therapies to minimize therapist influences and maximize treatment integrity, and the extensive

oversight and supervision typically found in research centers. The essential concern was that treatments developed and tested in highly controlled settings were not enjoying the same level of success in applied settings.

Clinical Efficacy

The term clinical efficacy was coined to reflect the situation in which a particular treatment has proven useful and beneficial in treating psychological problems in children through controlled research. For example, Stein and colleagues (Stein, Jayxoc, & Kataoka, 2003) found that children treated with cognitive-behavior therapy for 10 weeks experienced significantly fewer symptoms associated with Posttraumatic Stress Disorder and depression, as well as improved psychosocial functioning and classroom behavior, relative to a 3-month wait-list control group in a randomized clinical trial.

Additional conditions are required to satisfy the recommended criteria for establishing clinical efficacy. The therapy must be significantly superior to no treatment, placebo, or an alternative treatment, or equivalent to a treatment already established as efficacious. These comparisons must be accomplished in the context of a randomized controlled trial, equivalent time-samples design, or controlled single-subject experimental design, and replicated by an independent research team.

Empirically Supported Treatments

Psychological therapies meeting clinical efficacy criteria are considered empirically supported treatments. Empirically supported treatments are psychological treatments proven effective for specific populations through controlled research (Chambless & Hollon, 1998). Meta-analytic reviews attest to the clinical efficacy of many therapies for children by computing and analyzing effect size estimates across studies. Thus, studies rather than subjects become the focus of analysis, and an overall effect size based on reviewed studies provides compelling evidence of the magnitude of the difference (in standard deviation units) between treatment and nontreatment groups. For those unfamiliar with meta-analysis, effect size estimates are calculated based on recommended formulas (Hedges, 1982) and involve subtracting treatment means from control group means and dividing by the pooled standard deviation of the two groups. The result is a number that reflects the difference between the two groups—for example, before and after therapy—in standard deviation units. Thus, an effect size of 1.2 indicates that the treated children are 1.2 standard deviations higher on a particular measure (e.g., in social competence) relative to the untreated group. Larger effect sizes are usually desirable and indicate greater mean differences in outcome measures between the treated and untreated or comparison groups.

Effect size metrics reported in meta-analyses can also be related back to a central clinical question. For example, Kofler et al. (in press) examined how often children with ADHD were on-task relative to control children in regular classroom environments. They used best case estimation to determine the magnitude of on-task deficits after accounting for significant moderators and obtained an effect size of 1.40. This effect size was translated onto the control group distribution to estimate

actual percentage rates, and revealed that children with ADHD were on-task an average of 74.7% compared to 87.9% for typically developing children.

Some of the published criteria for establishing a therapeutic intervention as an EST are dubious. For example, the superiority of the intervention must be demonstrated in at least two independent research settings *or* with a sample size of three or more subjects for single-subject design experiments also in at least two independent research settings. This means that a treatment can be considered clinically efficacious based on results from only six children—a rather weak case for suggesting that treatment effects are likely to generalize to other children with similar difficulties and in similar settings. The moniker *possibly efficacious* reflects the situation in which a single, well-controlled study supports the treatment's efficacy, highlighting the critical need for replication.

Clinical Effectiveness

In contrast to clinical efficacy, clinical effectiveness addresses whether the treatment is useful in applied clinical settings, that is, its clinical utility. The distinction between the two terms reflects the frequent failure to replicate treatment effects produced in highly controlled research environments—such as university- or hospital-based research clinics—in regular psychological service delivery systems such as community mental health centers and private practice settings. Variables likely to contribute to a treatment's lack of utility in applied settings, despite proving successful in controlled settings, are reviewed elsewhere (Chambless & Hollon, 1998; Kazdin, 1988). In response to these criticisms, several researchers have attempted to establish clinical effectiveness in applied settings; however, traditional design controls (e.g., random assignment to groups, nontherapeutic waiting-list groups) are certain obstacles to these efforts.

Relevant Issues

Determining whether a treatment or intervention is effective is not as straightforward as it may seem. Effectiveness varies significantly based on the type of measure (self-report, standardized rating scale, direct observation, clinical interview, objective measure such as an actigraph), the source of measurement (child self-ratings, teachers, parents, therapists), and the area of functioning assessed (e.g., social deportment versus academic functioning).

Many measures—and particularly rating scales—currently used in research studies have inadequate measurement units, insufficient psychometric properties, and lack age and sex norms. As an example, none of the available child activity level rating scales has a standard unit of measurement—that is, a meaningful unit of activity or movement that is equivalent within and across measures and raters. Accurately defining activity level is severely compromised as a result and leaves us in the uncomfortable position of describing differences and changes in children's activity level by referring back to the scales used to initially quantify the behavior. The moderate correlations reported between activity rating scales only partially mitigates this concern when contrasted with correlations between rating scales and objective measures of children's activity level. For example, correlations between activity

rating scales and actigraphs are generally between .32 and .58. These values indicate that 66% to 91% of the variability in activity rating scale scores is not linearly related to variability in actigraph scores in the same children measured at the same time. This finding probably reflects the fact that children's activity rating scales tend to reflect other aspects of behavior and not just activity level. The wording of most scale items and reliance on factor analytic scale construction methodology contribute to this phenomenon because descriptions of activity level may correlate highly, but usually reflect a broader range of behavior than just movement.

STATISTICALLY AND CLINICALLY SIGNIFICANT CHANGES

Overview

The importance of changes that occur following an intervention can be quantified in three ways. The first and most common approach is statistical significance testing, which examines group differences and estimates the reliability of change. Statistical significance testing calculates how certain we are that changes from pre- to posttest are robust—for example, not due to other factors such as measurement error. It does not tell us, however, about the magnitude of this change or its impact on children's lives. Effect size and correlational approaches can provide meaningful estimates of the magnitude of change, but these estimates do not necessarily have clinical or practical significance. Clinical significance refers to the practicality of an intervention—whether or not it makes a meaningful difference in the daily lives of children or those who interact with the child.

Intuitively, larger effects are more likely to be clinically relevant. The magnitude of effect, however, is not necessarily related to clinical significance. Kazdin (2004) has argued that an intervention can be clinically significant—or insignificant—with large, small, or no changes from pre- to posttest. As examples, he offers the possibility that a large but clinically insignificant effect may be obtained on a cognitive or laboratory task that has little to do with daily living. Likewise, a clinically significant finding of no change may occur when symptom deterioration was expected. The impetus of these examples is to demonstrate that a measure's ecological validity—its relevance to real life—must be considered when selecting measures used to test clinical significance.

Evaluating Clinical Significance

Several methods currently are used to evaluate clinical significance (Kazdin, 2004). Subjective evaluation methods rely on the judgments of the child or relevant others and ask whether they can discern meaningful improvement in their daily lives. Measures of social impact may also inform clinical significance, for example, whether there was a change in arrest record frequency, days worked, or missed school days. Absolute change methods are interested in determining whether a problem or symptom has been eliminated, or whether the child continues to meet diagnostic criteria for a psychiatric disorder.

The most common type of clinical significance testing is the *comparison method,* which asks whether members of the treatment group are distinguishable from

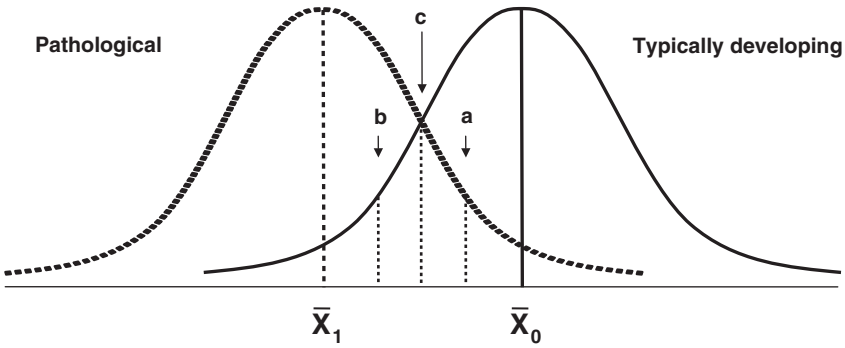

Figure 13.1 Clinical significance cutoff scores based on different criteria.

members of a typically developing (normal control) comparison group after treatment. Comparisons can be made to normative data or to a normal control group. Several commercially and freely available rating scales and direct observation systems provide standard scores based on normative data from nationwide community samples. Clinical significance may be demonstrated when a child changes from the clinical (usually 1.5 or 2 *SD* above the mean) to the normal (usually less than 1.5 *SD* above the mean) range following treatment. Comparison with rating scale norms is potentially problematic, however, due to different assessment conditions (Kazdin, 2004). For example, norms are typically based on a single test administration, whereas treatment outcomes are based on multiple administrations. Comparison with results from a concurrently collected normal control group is preferred.

Absolute change and two normal comparison, cutoff criterion methods are shown in Figure 13.1. X_0 represents the mean of the control group; X_1 represents the pathological group pretreatment mean. Three distinct cutoff scores have been proposed for determining clinical significance. The first (a) represents an absolute change method; the other two (b and c) estimate *normalization:* clinical significance in relation to a normal control group. Some argue that clinically significant improvement has occurred if a child's posttest functioning is (a) outside the range of the pathological group, defined as 2 standard deviations above the pathological control group pretreatment mean. Others argue that the child must fall (b) within the range of the normal population as defined by a score within 2 standard deviations of the normal group mean. A third proposal represents a compromise between these two options, defining normalization as occurring when a child is functioning (c) closer to the mean of the normal group than the pathological group.

In practice, children are classified as deteriorated, unchanged, improved, or normalized. The first three are determined by creating a confidence interval around the children's pretreatment score.* Children falling within the confidence interval are considered unchanged; those falling outside of it are labeled deteriorated

*Estimated true score is used if regression to the mean is considered a threat. See Speer (1992) for a discussion and simple method for determining whether regression to the mean is likely affecting posttreatment scores.

or improved. For example, children with a score of X_0 at pretreatment would be considered improved if they scored at or above cutoff score *a* in Figure 13.1. Children who improved and exceeded the predetermined cutoff score (*b* or *c*; see Figure 13.1) are considered normalized.

The difference between statistical and clinical significance is illustrated in the following example. Rapport, Denney, DuPaul, and Gardner (1994) evaluated the effectiveness of methylphenidate (Ritalin) on the classroom performance of children with ADHD. They used three outcome measures: teacher ratings of children's classroom deportment, academic efficiency scores (percentage of assigned class work completed correctly), and observed rates of on-task behavior in the classroom. Traditional significance tests revealed that all three outcome measures were significantly improved as a function of medication. Were these changes meaningful? Rapport and colleagues used the Jacobson and Truax (1991) model, correcting for regression to the mean as recommended by Speer (1992). As depicted in Figure 13.2, they found that between 50% and 78% of children were normalized, meaning that they were functioning within the normal control group range. Interestingly, although statistical significance was demonstrated for all three outcome variables, the rates of improvement or normalization were quite different. Most noticeable is the difference between teacher ratings of behavior, with 94% of ADHD children rated as improved or normalized, and actual classroom academic assignments completed correctly, with only 53% of ADHD children improved or normalized. Their findings also illustrate the importance of selecting relevant outcome

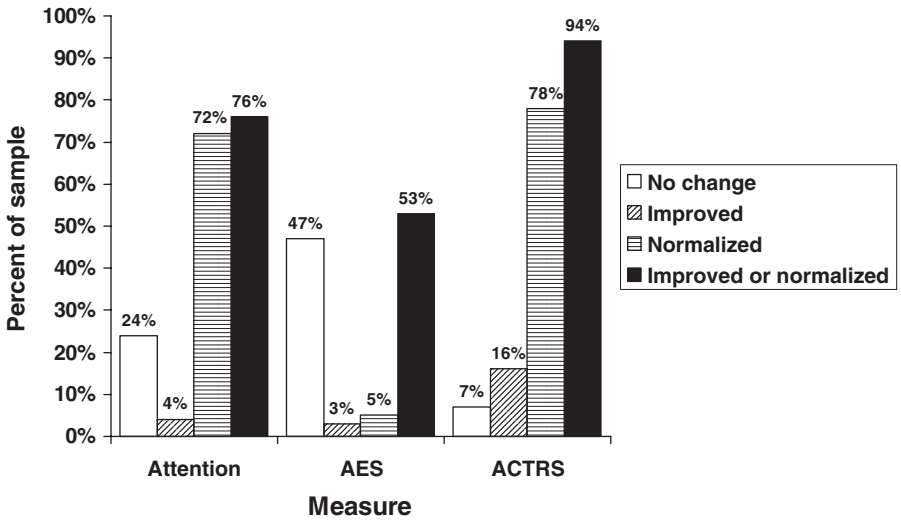

Figure 13.2 Clinical status of the group ($n = 76$) collapsed across methylphenidate dose conditions for three classroom measures. AES = Academic Efficiency Score; ACTRS = Abbreviated Conners Teacher Rating Scale. *Source:* "Attention Deficit Disorder and Methylphenidate: Normalization Rates, Clinical Effectiveness, and Response Prediction in 76 Children," by M. D. Rapport, C. Denney, G. J. DuPaul, and M. J. Gardner, 1994, *Journal of the American Academy of Child and Adolescent Psychiatry, 33*(6), pp. 882–893. Reprinted with permission.

measures. Teachers rated 94% of ADHD children as improving significantly when taking medication. When looking at their schoolwork, however, only 53% made actual improvements compared to their pretreatment performance. In most situations, when relevant outcome measures are used, statistical significance will be a necessary but not sufficient condition for clinical significance. Likewise, large effects may be more likely to be clinically significant effects. The presence of one or both of these conditions, however, does not demonstrate clinical significance.

In summary, clinical significance testing operates at the individual rather than group level. Beyond statistically reliable group differences, it is interested in the practical effects of an intervention—whether treatment makes a noticeable difference in the day-to-day lives of the clients we serve. Clinical significance can be defined subjectively by asking children about changes in their behavior, socially by comparing impact measures (e.g., school absences or test scores), or statistically using absolute and normative comparisons.

RESEARCH METHODS

Research *methodology* refers to the principles, practices, and procedures that direct research, whereas research *design* refers to the organized plan used to examine questions of interest. Both aspects represent important knowledge that guide the planning, implementation, and analysis of a research project.

Selection of Measures

Relationship between and among Variables

Selection of measures will depend on your specific research question as well as the level of understanding you wish to accomplish with respect to the phenomenon studied. For example, research may focus on examining *correlates,* or relationships between or among variables without consideration of time sequence or causality. Other studies may focus on identifying *risk factors,* or characteristics that precede and increase the likelihood of some event, outcome, or behavior pattern that may be modifiable. *Marker variables*—characteristics that precede and increase the likelihood of some event or behavior, but are not modifiable or have no effect on the outcome if modified—may be the focus of other investigations. Other measures will be more appropriate for questions concerning *treatment outcome* or *prevention* in children (for a detailed discussion, see Kazdin, 1999). Establishing causal factors, wherein one variable precedes and influences some other variable, represents the purest form of research if one embraces prediction of human behavior as the sine qua non of research.

Studying variables (*moderators*) that influence the direction, magnitude, and nature of a relationship between or among variables represents important avenues for clinical research, particularly when *protective factors* (a special case of moderator variables) can be identified that reduce the likelihood of an undesirable outcome. Researchers examine moderator variables when they suspect that the relationship among variables may differ because of the influence of some other variable. For example, if the relationship between type of aggressive behavior and a diagnosis of

Conduct Disorder is different for boys and girls, sex may be considered a moderator variable. Several moderating variables may be important to examine in research with children, including age, sex, ethnicity, height, weight, and motivation.

Once a causal relationship is established in the literature, researchers may wish to focus on explaining the process or mechanism by which the process evolves (*mediator variables* such as psychological or biological mechanisms). In this situation, a previously unconsidered variable is hypothesized to intercede between two related variables and reflects an indirect effect of one variable on another. For example, Rapport, Scanlan, and Denney (1999) found that the relationship between children's early attention problems and later scholastic achievement was mediated by differences in working memory.

In summary, a correlational relationship represents the most basic level of understanding between or among variables. A deeper level of understanding is achieved if it can be determined that one or more variables preceded the occurrence of other variables in time. To establish a causal linkage, investigators must provide evidence of a correlation and ordered temporal relationship between or among variables, complemented by three additional considerations: Other influences must be ruled out as possible explanations for the findings; the findings must be replicable across different samples from the same population; and a valid explanation concerning the mechanisms and processes through which the causal variables operate and are related to outcome variables must be explicated.

Types of Dependent Measures

The foregoing summary suggests that researchers need to consider the level of understanding they wish to achieve with respect to their research topic before selecting particular measures. Selection of measures will depend on a host of factors (e.g., availability of instruments or instrumentation, time constraints, costs) and can include a wide range of instruments and techniques. Popular exemplars include self-report inventories, behavior rating scales completed by others, and physiological instrumentation. Consultation of specialty texts on child assessment is advised for those interested in reviewing the broad range of instruments and techniques available (e.g., see Mash & Terdal, 1997; Hersen, 2006; and current volume).

Innovative measures can be created to assess discrete incidences (e.g., stereotypic movement, drooling) or categories of behavior (e.g., bizarre speech) but must meet expected standards of psychometric rigor (e.g., interrater reliability, validity). Computerized assessment is commonly used to assess important constructs such as vigilance (e.g., continuous performance tasks) and behavioral inhibition (e.g., stop-signal paradigm) in children. Incorporating current or archival indices of target behaviors or latent constructs reflects an additional method of data collection available to child researchers. For example, Fergusson and Horwood (1995) used arrest records as an index of later delinquency in children with early conduct problems and ADHD. Duration of hospital stay, school attendance, earned grades, completed academic assignments, and retention rates are other common examples of measures with high face validity used by researchers.

As a general rule of thumb, try to incorporate multiple indices of behavior that have strong psychometric properties. That is, select instruments, indices, or observations that provide valid and reliable measures of both the specific behaviors of interest as well as other potentially related areas of functioning. Doing so demonstrates that identified behavior problems or changes in behavior are not confined to a particular type of instrument or recording procedure and permits measurement of corresponding improvement in other domains related to the child's functioning. For example, designing a treatment outcome study to improve a child's academic performance in the classroom may result in corresponding improvement in the child's attention and classroom behavior. In such cases, teacher rating scales coupled with direct observational recording procedures that measure domains other than just academic performance are needed to document the broader spectrum of treatment gains related to the independent variable. Including collateral measures that reflect areas not expected to change (unless the researcher is expecting a generalized treatment effect) or undesirable change is necessary in some research protocols. The former provides evidence for discriminate validity (e.g., only the targeted area changes with the introduction of the independent variable), whereas the latter addresses the possibility of emergent symptoms or undesirable change due to the intervention.

Studies of specific clinical groups (e.g., ADHD, childhood depression), learning problems (e.g., academic deficits), or maladaptive behavior (e.g., peer aggression) may require measures different from those used to evaluate change due to intervention. For example, structured or semi-structured clinical interviews complemented by specific rating scales are traditionally used to define, classify, or describe children with a particular clinical disorder (i.e., serve as grouping variables), whereas other instruments, observations, or ratings may be used to measure change associated with intervention (i.e., serve as dependent variables). Therapy effects are strongest for outcome measures that match the problems targeted in treatment rather than through nonspecific or artifactual effects (Weisz, Weiss, Han, Granger, & Morton, 1995).

RESEARCH DESIGNS

Single-Subject Research Designs

Single-case research designs are valuable methodological tools used to evaluate different types of research questions involving individuals and groups. Consider some of the important scientific discoveries in psychology over the years. Single-case research was the principal paradigm in Wundt's (1832 to 1920) investigations of sensation and perception, Ebbinghaus's (1850 to 1909) studies of human memory, Pavlov's (1849 to 1936) classic experiments in respondent conditioning, and B. F. Skinner's (1904 to 1990) research in operant conditioning. These designs are particularly relevant to understanding children and the environments in which they live, where changes in everyday life may be of greater importance than obtaining a statistically significant change between groups (i.e., greater clinical relevance). Experimentation at the level of the individual case study may also provide greater insights with respect to understanding therapeutic change.

In contrast to the heavy reliance on established psychometric techniques and instruments required by between-group research, single-case methodology begins with identifying the focus of investigation (i.e., designating target behaviors) and proceeds to selecting potential strategies of assessment. When deciding on target behaviors, try to select observable behaviors and environment events as opposed to covert behaviors such as thoughts and ideas or hypothetical constructs such as anxiety and self-concept. Doing so will facilitate objective measurement and agreement between observers. Definitions should be written with absolute clarity to avoid ambiguity, and the boundaries of the defined target behavior must be clearly specified to minimize inference concerning whether a particular behavior qualifies as an occurrence or a nonoccurrence. On the surface, this may appear to be an easy task, but consider selecting classroom attentiveness ("on-task" in educational parlance) as a target behavior. Should off-task or inattentive behavior be recorded if a child is gazing up at the ceiling while working on a math assignment? Perhaps the child was thinking about the problem at hand or performing mental arithmetic. Are children permitted to ask peers for assistance when working on a problem, or allowed to leave their seat to sharpen a pencil during an academic work period? Does a momentary glance away from one's work count as off-task behavior? These and other definitional parameters must be decided and agreed upon, and preferably subjected to extensive pilot testing, prior to beginning formal observation and data collection. Once established, clear definitions of target behaviors will permit the recording of reliable baseline data and, in turn, serve as the traditional yardstick by which change is measured.

Assessing Behavior

Assessment of behavior can be accomplished in many different ways and will depend on what is being assessed and which method of recording best suits the needs of the researcher. The most commonly used methods of assessment include using frequency measures, classifying responses into discrete categories, counting the number of children or events, and measuring behavior based on discrete units of time.

Frequency counts are used when dealing with discrete behaviors that require a relatively constant time interval to perform. The first criterion enables observers to know when a designated behavior begins and ends, whereas the second permits recorded behavior to be treated as similar units for purposes of comparison. Consider the situation in which a researcher records the frequency with which a child speaks out of turn in a classroom. The child may blurt out an answer on one occasion and on another may turn and talk with a peer for 10 minutes or longer. Clearly, the two incidents are not comparable, and an alternative assessment method such as time interval recording should be considered. For situations in which frequency measures are taken for different periods of time (e.g., 20 minutes on day 1 and 40 minutes on day 2), calculate the frequency per minute or response rate by dividing the frequency of responses by the number of minutes observed each day. This metric, frequency per minute or response rate, will yield data that are comparable for different durations of observation.

Discrete categorization is used in situations in which behavior is best defined by categorical assignment. Commonly used categories include appropriate–not appropriate (e.g., social interactions between children), complete–not complete (e.g., classroom assignments), and discrete behaviors that form a functional (e.g., getting dressed in the morning, wherein each article of clothing counts as a discrete behavior) or correlated but unrelated (e.g., performing household chores) response chain.

Counting children is often used to assess the effectiveness of an intervention program. Examples include counting the number of children who perform a designated target behavior while on a school field trip, or calculating the percentage of children who complete their daily academic assignments in a classroom each morning. Contingencies can be introduced subsequently to reinforce daily academic assignment completion rates, for example, by scheduling in-class free time if at least 80% of the class meets the established criterion or for children who independently complete the assigned work.

Interval recording is used in situations in which the researcher wishes to obtain a representative sample of a target behavior during particular times of the day. Common examples of appropriate child behaviors include staying in one's seat during an academic work period, appropriate talk with peers, and paying attention. A block of time (e.g., a 30-minute daily or every other day observation period) is divided into a series of shorter intervals (e.g., 15-second observation blocks followed by 5-second recording blocks), and the target behavior is recorded as occurring or not occurring during each 15-second observation interval. This example would yield three observation blocks per minute and 90 intervals of recorded behavior per day. Variations of the interval recording method are common (e.g., time sampling) and might involve brief observations of a target behavior throughout the day rather than being confined to a single block of time.

Recording *duration* is another time-based method for observing behavior and is more appropriate for recording behaviors that are continuous rather than discrete acts. Common examples include observing ongoing social interactions between or among children or the total time required to complete an academic assignment. In these cases, the total duration or time interval that the behavior is performed serves as the dependent variable. An interesting but infrequently used variation of the duration method involves recording elapsed time before a particular behavior is performed (i.e., response latency). An example is to record how long a child takes to perform a particular behavior or chain of behaviors following adult instruction. Contingencies might be established subsequently based on the child's compliance within an increasingly shorter time interval over several days or weeks.

Design Types

The essence of single-subject research lies in its ability to demonstrate experimental control of an independent variable (IV) over one or more dependent variables (DV) by means of shrewd design and graphical illustration. Although statistical procedures exist that can be used to assess outcome effects associated with single-subject case studies, the more traditional means of demonstrating experimental control is to

provide a compelling visual (graphical) illustration that even a Doubting Thomas would acknowledge as evidence. To convince the scientific audience, an IV is introduced using a variety of design options such that its effects on the DV are systematically produced, reproduced, and/or eliminated, as in a carefully choreographed dance. Detailed descriptions of specific types of single-subject designs are provided in this volume (see Chapter 11).

Group Experimental Designs

Overview

Despite the numerous benefits associated with single-subject design methodology, the fact remains that an overwhelming majority of studies in psychology involve the comparison of groups, not individuals. Group designs are conventionally classified as true experimental designs, quasi-experimental designs, and observational designs, and may be used in a variety of contexts.

A true experimental design is a design in which the researcher manipulates one or more independent variables and measures one or more dependent variables. The researcher chooses what independent variables to manipulate, how they are manipulated (e.g., which levels to include), and what dependent variables to measure, based on the nature of the research question. For example, a researcher might be interested in determining what type of therapy works best for children with school phobia. Therefore, the research question has determined that the independent variable is *type of therapy* and the dependent variable is *school phobia*. Now, the researcher must decide whether any other independent or dependent variables should be included in the study and operationally define the independent variable(s) and the dependent variable(s).

Operationally defining independent variables refers to deciding what the levels should consist of, whereas operationally defining dependent variables refers to deciding exactly how to measure them. For example, the researcher needs to decide what levels of therapy to include (e.g., behavior therapy, cognitive therapy, cognitive-behavior therapy), whether or not to include a control group (condition), and how to measure school phobia. In the context of a between-subject design, a control group is a randomly assigned group that receives either no experimental treatment or a substitute for the experimental treatment. In the context of a within-subject design, a control condition is a condition (administered to all participants) in which either no experimental treatment or a substitute for the experimental treatment is administered. Control conditions (groups) are sometimes needed to control for nonspecific treatment effects (e.g., some children might believe that simply taking a pill helps because it's "medicine").

A quasi-experimental design is a design that is set up to emulate a true experimental design but includes one or more independent variables that cannot be manipulated by the researcher (often referred to as quasi-independent variables). These include variables such as sex, ethnicity, height, and weight, or clinical diagnoses such as ADHD or Reading Disorder. Another reason for the inability to manipulate certain variables may be due to ethical reasons such as substance abuse, smoking, exposure to harmful toxins, and cancer. The limitation to quasi-experimental

designs is the inability to express a causal relationship for quasi-independent variables.

Nonspecific Treatment Effects

Nonspecific treatment effects are any effects brought about by the experiment besides the treatment, such as being aware of what the experiment is about, contact with the experimenter, and discussing the experiment with other people. Nonspecific treatment effects are basically extraneous (nuisance) variables related to participants' perceptions concerning the experiment. Related to nonspecific treatment effects are placebo effects, in which participants improve simply because they believe that they are receiving treatment. If a control condition (group) is included, it is important that there are no other differences between the control condition and the experimental conditions except for an absence or substitute for the experimental treatment. Any other differences would introduce extraneous (nuisance) variables. Control conditions are necessary only when there is concern that nonspecific treatment effects may have an effect on the results. Nonspecific treatment effects are possible only if the participants' perceptions about the experiment can have an effect on the results. There are many situations in which participants' perceptions cannot have an effect on the results. For example, if a researcher is interested in studying what type of teaching method works best for teaching mathematics to third-grade children, a control condition consisting of teaching no mathematics would be benighted. Conversely, a child with social phobia may believe that taking a pill provides him with the necessary confidence to complete an in vivo exposure protocol.

There are three main classes of experimental designs: the between-subject design, the within-subject design, and the mixed-subject design, each with multiple variations possible.

Between-Subject Designs

The between-subject design is a design in which participants are randomly assigned to different treatment groups (levels of the independent variable) and each treatment group receives a different experimental condition. For example, in a single-factor experiment (an experiment with only one IV) examining two methods for teaching science to third-grade children (Technique 1 and Technique 2), half of the students would be randomly assigned to receive Technique 1 and half to receive Technique 2.

In a factorial experiment—an experiment with more than one independent variable—examining different methods for teaching science to third-grade children (Technique 1 and Technique 2) and the mode of presentation (teacher versus computer), one fourth of the students would be randomly assigned to receive Technique 1 via computer, one fourth to receive Technique 1 via teacher, one fourth to receive Technique 2 via computer, and one fourth to receive Technique 2 via teacher.

Advantages of the between-subject design as compared to the within-subject design include no carryover effects and no order or sequence effects. Disadvantages

of the between-subject design include the fact that many more participants are required than for the within-subject design, and the between-subject design yields less power than the within-subject design.

Ideally, there are an equal number of participants in each of the different treatment groups and assignment to the different treatment groups is random. Random assignment to treatment conditions simply means that each participant has an equal probability of being assigned to any given treatment group. Both ideals, equal number of participants per treatment group and random assignment to treatment groups, may not be possible for a given study. In the case of an equal number of participants per treatment group, the total number of participants may not be equally divisible among the various treatment groups or some participants may drop out of the study or miss the day that data are collected. In the case of random assignment of participants to the different treatment groups, this may not be possible. As in the example concerning methods of teaching science, it might be difficult to randomly assign half of each class to receive a different teaching method. The teacher cannot very well teach half of her class at a time, and even if he or she could, this would introduce the confound of which method is taught first. Both problems (unequal sample size and nonrandom assignment) can frequently be handled statistically. It is important, however, to take into account both unequal sample size and nonrandom assignment when analyzing data (i.e., one should not ignore them and analyze the data as if there were equal sample size and random assignment).

Another important issue is the loss of participants from the experiment, referred to as *attrition*. If the loss of participants is random (e.g., roughly an equal number of participants dropped out of each of the experimental groups), there is no problem and certain statistical techniques may be used to correct the situation. If, however, the loss of participants is not random but due to some aspect of particular treatment conditions (e.g., almost every participant who dropped out was in one particular treatment condition), then little can be done to salvage the experiment other than conducting a post-hoc test to determine whether attrition significantly biased one group relative to the other. For example, one may need to demonstrate that children with the most deviant scores did not drop out of one of the two groups receiving treatment, or that noncompleters were not significantly different on particular measures relative to completers.

As an additional note, the problems associated with randomly assigning some children within a given classroom to receive one level of the independent variable (treatment) and other children from the same classroom to receive other levels of the independent variable is not uncommon to research conducted in school settings (or even clinic or hospital settings). For example, it may not be possible for the teacher to teach some students using one teaching method and the other students using different teaching methods. When it is not possible to randomly assign children from the same class to different levels of the independent variable, an alternative is to randomly assign different classrooms to the different levels of the independent variable. This type of design is called a *hierarchical design* and requires special analysis. In a hierarchical design, the levels of at least one independent variable are nested under the levels of another independent variable, and the remaining independent variables are fully crossed. For example, if each level of IV 2 (e.g.,

Table 13.4 Hierarchical Design

	Technique 1		Technique 2
Classroom 1 Classroom 2		Classroom 3 Classroom 4	

classrooms) appears with only one level of IV 1 (e.g., teaching method), then IV 2 (classrooms) is said to be nested under IV 1 (teaching method). Examining two different methods for teaching science to third-grade children, two third-grade classes could be randomly assigned to receive Technique 1 and two third-grade classes could be randomly assigned to receive Technique 2. The design for this model is diagramed in Table 13.4.

The separate columns for Technique 1 and Technique 2 indicate that the model is not fully crossed, as classrooms (1, 2, 3, 4) are nested under techniques (1, 2). In this experiment, the independent variable *classroom* (classrooms 1, 2, 3, 4) is an extraneous (nuisance) variable. It is included in the design and analysis because it might have an effect on the dependent variable, and including it allows its effects to be isolated. If a hierarchical design is used, it is incorrect to analyze the data as if students were randomly assigned to each level of the independent variable (e.g., teaching method). There are additional complications that may arise when employing a hierarchical design, for example, when the levels of the nested variable (classroom) cannot be randomly assigned to the levels of the variable it is nested under (teaching method). Hierarchical designs are considered balanced if they satisfy two criteria. First, there must be an equal number of participants in each treatment combination (e.g., students in each class for each teaching method). Second, there must be an equal number of levels of the nested variable under each level of the other independent variable (e.g., two classes under each level of treatment method). If these criteria are not satisfied, the model is considered unbalanced and more complicated to analyze than balanced designs.

Within-Subject Designs

Every participant receives every treatment condition (levels of the independent variables) in a within-subject design. In the single-factor experiment on teaching techniques, every participant would receive Technique A and Technique B. In the factorial experiment involving teaching technique and mode of presentation, every participant would receive Technique A via computer, Technique A via teacher, Technique B via computer, and Technique B via teacher. A within-subject design may have children participating in the different levels of the independent variable simultaneously, or children may complete one level of the independent variable before participating in the next level. When participants must complete one level of the independent variable before participating in the next level, this is commonly referred to as a *crossover design*.

Within-subject designs present some special problems, including carryover effects, order effects, and sequence effects. Because participants receive every treatment condition, the effects of one treatment condition may carry over to the next

treatment condition. For example, once children are exposed to a particular reading method, they may be irreversibly changed. The investigator cannot simply cross them over into an alternative method and assume that nothing has been gained during the earlier experience. The order in which the treatment conditions are presented to the participants may also have an effect on the results. Finally, the sequence in which the treatment conditions are presented may have an effect. For example, in a study of perceived heaviness of objects, whether someone lifted a 10 pound object and then a 20 pound object or lifted a 20 pound object and then a 10 pound object would have an effect on the perceived heaviness of each object. Order and sequence effects may sound similar; however, order effects refer to the ordinal position in which the condition is presented in (e.g., first, second, third). Sequence effects, in contrast, have to do with which treatment follows which other treatment. As an example, consider the two sequences A—B—C and C—B—A. In both sequences, B has the same ordinal position (second). However, in the first case B follows A and in the second case B follows C; therefore, the sequence is different. Counterbalancing procedures are frequently used to control for problems related to carryover effects, order effects, and sequence effects.

Counterbalancing means that participants are exposed to the different treatment conditions in different orders. Ideally, one or more participants could be exposed to every possible order of treatment conditions. This is not possible with more than a few treatment conditions, as the number of possible orders increases rapidly (number of possible orders equals $n!$, such that three treatment conditions produces six possible orders: $3 \times 2 \times 1$). When the number of possible orders is too great, only a subset may be used. One method of obtaining a subset of the possible orders of treatment conditions is to simply randomly select a subset of the possible orders. This may pose a problem, however, because if the order or the sequence of treatment conditions has an effect on the dependent variable, random selection of treatment orders will not control for these effects.

Another method that not only controls for the order effect of treatment conditions but also allows for the order effect to be analyzed separately is the Latin square design. The Latin square design has as many orders represented as there are treatment conditions. Therefore, if there are four treatment conditions (Rx = treatment), four orders (indicated by uppercase letters A, B, C, D) would be represented as depicted in Table 13.5. Next, a random selection procedure is invoked to determine which of the four treatments corresponds to which letter (i.e., the order treatments will be administered). For example, if it was randomly determined that A = Treatment 3, B = Treatment 1, C = Treatment 4, and D = Treatment 2, then the matrix is designed as in Table 13.5.

Table 13.5 Simple Latin Square Design

A = Rx 3	B = Rx 1	C = Rx 4	D = Rx 2
B = Rx 1	C = Rx 4	D = Rx 2	A = Rx 3
C = Rx 4	D = Rx 2	A = Rx 3	B = Rx 1
D = Rx 2	A = Rx 3	B = Rx 1	C = Rx 4

Table 13.6 Balanced Latin Square Design

A = Rx 3	B = Rx 1	C = Rx 4	D = Rx 2
B = Rx 1	D = Rx 2	A = Rx 3	C = Rx 4
C = Rx 4	A = Rx 3	D = Rx 2	B = Rx 1
D = Rx 2	C = Rx 4	B = Rx 1	A = Rx 3

The Latin square design ensures that every treatment condition appears in every possible order (first, second, third, etc.). Therefore, if the order in which participants experience the different treatment conditions has an effect on the dependent variable, the Latin square design will cancel out that effect. The Latin square design also allows for the comparison of the different treatment orders (by including order as another independent variable) to determine whether there is a significant difference between the different orders. As mentioned earlier, in some situations both *order* and *sequence* may have an effect on the dependent variables. The balanced Latin square design controls for the effects of order and sequence effects and allows these differences to be compared. It ensures that treatment condition appears in every possible order (first, second, third, etc.) and that each treatment condition is preceded and followed by every other treatment condition exactly once (e.g., Rx 1 is preceded once by Rx 2, Rx 3, and Rx 4), as depicted in Table 13.6.

As a result, if order or sequence of treatment conditions affects the dependent variable, the balanced Latin square design will cancel out these effects. This design also allows for the comparison of the different treatment orders and sequences (taken together, not separately) to determine whether there is a significant difference between them. In Table 13.6, the rows indicate the order in which treatments will be assigned, and the columns indicate the sequence in which treatments will be received. As in the simple Latin square design, a random selection procedure is invoked to determine which treatment is assigned to which letter.

Mixed-Subject Designs

The mixed-subject design is a design in which one or more independent variables are between-subject variables (e.g., different participants randomly assigned to different levels of the independent variables) and one or more independent variables are within-subject variables (e.g., all participants receive all levels of the independent variables). For example, in the experiment involving teaching technique and mode of presentation, every participant would receive Technique A and Technique B (teaching technique is a within-subject variable), and half of the subjects would be taught by a computer and half would be taught by a teacher (mode of presentation is a between-subject variable). All of the issues discussed concerning between-subject designs and within-subject designs apply to mixed-subject designs.

Extraneous Variables

A major advantage of true experimental designs is that they are the only method that allow causal relationships among variables to be proven, which includes ruling out outside or extraneous influences. Thus, it is important to use a design in which

nothing else differs between the experimental groups except for the experimental conditions (levels of the independent variable). Any other differences besides the treatment conditions (called extraneous or nuisance variables) can call into question the causal inference.

Extraneous variables can sometimes affect all treatment conditions equally, either weakening or strengthening their effects. For example, in the study comparing teaching technique and mode of presentation for schoolchildren, if the treatment conditions were administered immediately following recess, it is possible that participants would be too wound up to pay attention, thus weakening the effects of all of the treatment conditions. When an extraneous variable varies systematically with the different treatment conditions, it is referred to as a *confounding variable* and poses an even greater danger to the interpretation of the results. When a confounding variable is present, any changes in the dependent variable cannot be attributed to the treatment condition with absolute certainty. Using our teacher study example, if everyone receiving Technique A via teacher had one teacher and everyone receiving Technique B via teacher had a different teacher, it could not be determined whether the different technique or the different teacher was responsible for any differences found in the dependent variable. Thus, it is important to keep everything constrained equally across the different treatment levels except for the treatment itself. This is not always possible, especially when conducting research in school and clinic settings. It does not necessarily invalidate the study to have extraneous or confounding variables present; however, it does limit the degree of causality that you can attribute to your treatment conditions.

Correlational Designs

In observational designs, the researcher simply measures variables as they occur naturally in the environment, which limits the degree to which causality can be inferred. Many observational studies, however, are conducted to examine whether a relationship exists between two or more variables or to predict certain variables (criterion variables) from other variables (predictor variables). Observing behavior and recording predictor and criterion variables is another method used in observational studies. For example, a researcher could observe children in the classroom and record the number of times they raise their hands to answer questions and the frequency of praise they receive from the teacher. Alternatively, observational studies may rely on self-report methods such as teacher or parent questionnaires that measure the variables of interest. Observational studies may also be conducted by obtaining ratings about children by people who know them, such as parents, teachers, or peers, or use combinations of these techniques.

Longitudinal and Cross-Sectional Designs

Overview

Longitudinal designs involve measuring participants repeatedly over an extended period of time, perhaps years or even decades. Cross-sectional designs and longitudinal designs are ways of obtaining different types of information. Cross-sectional

designs are best used for answering questions concerning how a treatment works at one point in time. Longitudinal designs are best used for answering questions concerning developmental change. For this reason, they can be especially informative when studying children because of their rapid developmental changes. Cross-sectional designs may be set up to assess different age groups simultaneously (e.g., children at ages 2, 6, 10, 14, 18), which would answer questions concerning possible age differences. However, it is possible that differences may exist between children of different age groups at a single point in time because they have different histories (cohort effects). Therefore, the children who are 2 years of age at the same point in time as other children who are 14 years of age might not display the same characteristics when they become 14 years of age. Longitudinal designs involve examining the long-term effects of some event or intervention. An additional advantage to longitudinal designs is that because information is collected with measures repeated at multiple time periods, error variance is reduced, often allowing for the detection of small behavior changes. Additionally, participants are compared to themselves at different points in time.

Potential Limitations

Disadvantages of longitudinal designs are mostly tied to the length of time required and the inherent cost associated with the study. Because the data are collected over long time periods, procedures and measures may become outdated and new procedures and measures may be developed. This leads to a major quandary: Should the outdated procedures and/or measures be continued so that differences may be compared across time, or should the new procedures and/or measures be adopted because they are better? One possible solution is to continue using the old procedures and/or measures to ensure accurate comparison and to adopt new procedures and/or measures as they are developed. Participant attrition is also a major problem with longitudinal designs because it is quite possible that the group that remains at the end of the study is not representative of those who dropped out along the way. Additionally, participant attrition increases the probability of a Type II error (failing to find a treatment effect when one exists) and decreases the generality of the results. Another problem is the potential confound between the effects of personal age and the effects of historical period (e.g., children growing up during a particular decade may be exposed to environmental or other events such as war that younger cohorts are not exposed to).

PARTICIPANT DEMOGRAPHICS AND SAMPLING

Identifying a Population of Interest

After formulating a research question and establishing a basic design, the next step is to consider how best to define and describe the population of interest. Review recent articles in credible journals to gain an understanding of how others have accomplished this task. For example, most research studies dealing with children contain basic sociodemographic information, such as children's age and grade (mean and *SD*), estimated level of intelligence, family socioeconomic status, sex, and ethnicity.

This information is included for other participants (e.g., parents) when relevant to the study (e.g., when studying parental attitudes toward medication compliance in children).

Detailed information is also included relevant to identifying, categorizing, or describing the population of interest. This may involve (a) a clear description of how a diagnosis is ascertained (e.g., using structured or semi-structured clinical interviews combined with using rating scale cutoff scores), (b) how a group is identified (e.g., a learning disability discrepancy formula), or (c) describing the characteristics of a select group of children (e.g., children attending a special education classroom). The use of diagnostic monikers (e.g., ADHD, Social Phobia) for identifying research samples is conventionally based on instruments with acceptable psychometric properties. This approach usually entails the administration of a structured or semi-structured clinical interview coupled with parent and teacher rating scales. Inclusion criteria for a particular diagnostic group is typically based on meeting *DSM-IV* criteria for the disorder in addition to exceeding an identified cutoff score, such as 2 standard deviations above the mean or the scale's identified range for clinical diagnosis. Exclusion criteria vary according to the research question posed. For example, a researcher studying children with learning disabilities may wish to exclude children with ADHD to determine how well an innovative intervention works with reading disabled children independent of ADHD. Conversely, a different research team investigating whether working memory deficits are unique to language-impaired children as opposed to a more generalized deficiency associated with psychiatric disability (e.g., common to many childhood disorders) may wish to include children with and without comorbid ADHD.

Alternative methods for identifying research samples are available, depending on the nature of the investigation. For example, children with poorly developed social skills who do not meet diagnostic criteria for a clinical disorder may nevertheless benefit from a social skills training program. Specific measures of social skill deficits, such as parent-teacher ratings and even direct observations, can be used if they reliably identify the sample.

Sampling

As a general guideline, researchers must sufficiently describe characteristics of their sample that might affect the generalizability of findings as discussed earlier and determine how best to obtain the research sample from the targeted population. This is particularly important for group designs but also applies to single-subject designs.

There are two classes of samples, *probability* and *nonprobability* samples. Probability samples are samples in which every member of the population has a known probability of being selected for inclusion, whereas nonprobability samples are samples in which the probability of being selected for inclusion is unknown.

Four well-known types of probability samples are *simple random samples, stratified random samples, systematic samples,* and *cluster samples*. A simple random sample is one in which every member of the population has an equal probability of being selected, whereas a stratified random sample is a random sample in which the proportion of certain characteristics in the population (e.g., race, ethnicity, culture,

sex, age, education, income, SES) are matched in the sample. For example, if sex is considered important to a study and the population under study consists of 58% boys and 42% girls, the sample reflects these same proportions. Samples may also be stratified on several different characteristics simultaneously. For the sample to be a stratified random sample, sampling would be random within each characteristic considered important to the study.

A systematic sample is a sample selected in a nonrandom fashion. For example, if 10% of the child population is to be included in the sample, then every 10th child would be chosen for inclusion.

A cluster sample is a sample in which clusters (groups) are randomly selected rather than individuals. For example, if researchers are interested in sampling grade school children, they would start with a list of grade schools and randomly select which schools to include. Within each school, you could include all of the students, randomly select students for inclusion, or randomly select additional clusters (e.g., grade levels).

The most commonly used types of nonprobability samples include the *convenience sample,* the *stratified convenience sample,* and the *snowball sample*. A convenience sample is a sample of participants that is convenient for the researcher to obtain (e.g., college undergraduates, hospital patients, child referrals to a university-based specialty clinic). A stratified convenience sample is the same as a stratified random sample, except that the participants are selected for convenience rather than randomly. A snowball sample is a sample that is created by having the initial participants (e.g., children's parents) suggest additional possible participants; these additional participants suggest additional possible participants, and so on. Because researchers do not typically have access to the entire population they wish to study, most studies use nonprobability samples.

Sample Size and Power

An important issue concerning samples is determining how many participants should be included. The main issue concerning sample size is that you want to recruit a sufficient number of participants to have a powerful test, but not more than you need, as this can be costly and time consuming. *Power* refers to the probability of finding a significant treatment effect when one truly exists (probability of rejecting a false null hypothesis). Power can be increased by setting alpha equal to 0.05 rather than 0.01, increasing the size of the treatment effect (increasing the between-condition or group variation), or reducing the error (reducing the within-condition or group variation). One method of reducing the within-condition variation (error) is to increase the number of participants. As sample size increases, within-group variation (error) decreases.

Convention generally holds that .80 is the minimum acceptable level of power. This means that if there truly is a treatment effect, your statistical test has an 80% probability of finding that treatment effect (rejecting the null hypothesis). Before you conduct an experiment, you should do a power analysis to determine how many subjects are needed to achieve (at least) 80% power. Conducting a power analysis requires the researcher to make several educated guesses concerning

the data (e.g., size of treatment effect and the population standard deviation). Specialty texts (see Cohen, 1988) and software (GPower: www.psycho.uni-duesseldorf.de/abteilungen/aap/gpower3/) for computing a power analysis are readily available. Researchers recognize that a certain degree of controversy exists concerning hypothesis testing, because with a large enough sample, even trivial treatment effects may be statistically significant. For this reason, researchers should specify the minimum interesting treatment effect (i.e., the minimum effect that would be of interest) to be found with 80% power before conducting a power analysis.

SUMMARY

Conducting research with children is a multifaceted enterprise that requires broad knowledge of research methods, research design, psychometric theory, statistics, and child development, coupled with a healthy dose of curiosity and tenacity. Textbooks and coursework are useful resources for novice researchers, but there is no substitution for working in an active research laboratory under the guidance of an accomplished mentor.

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PART IV

Diagnosis and Evaluation

CHAPTER 14

Assessment, Diagnosis, Nosology, and Taxonomy of Child and Adolescent Psychopathology

Thomas M. Achenbach

This chapter addresses a host of interrelated issues that are especially salient for those who work with children. (For brevity, I use "children" to include "adolescents.") These issues are also becoming increasingly salient for those who work with adults, so much so that they are shaping drafts of the forthcoming fifth edition of the American Psychiatric Association's *Diagnostic and Statistical Manual of Mental Disorders* (*DSM-V*). The issues are organized in terms of four interrelated aspects of what can be understood as continuous processes. These four aspects are assessment, diagnosis, nosology, and taxonomy.

In clinical services, assessment is often viewed as an initial step that then leads to diagnosis. Diagnosis, in turn, is shaped by nosologies (classifications of diseases) that specify the criteria for the disorders to be diagnosed. And taxonomies systematically represent similarities and differences among the phenomena to be distinguished from one another in terms of taxonomic constructs. Assessment data are essential for diagnosis, nosology, and taxonomy. However, the data sought and the procedures for seeking the data are shaped by diagnostic, nosological, and taxonomic concepts and criteria. Figure 14.1 illustrates the continuous process that includes assessment, diagnosis, nosology, and taxonomy. Although everyday clinical services focus primarily on assessment and diagnosis, these enterprises are ultimately structured according to nosological and taxonomic systems. Multiinformant, multicultural, and multitaxonomic perspectives are needed to take account of variations in the phenomena to be distinguished and in the models for representing the phenomena.

The clinical phenomena on which this chapter focuses include behavioral, emotional, social, and thought problems occurring between the ages of 18 months and 18 years with sufficient severity to warrant the term "psychopathology" and a likely need for professional help. The chapter does not focus on problems that result primarily from severe mental retardation or known physical abnormalities. The prospects for significant changes in the *DSM-V* argue for viewing existing *DSM* diagnostic criteria and categories as provisional and subject to change, despite their current influence on practice and reimbursement. Consequently, the *DSM* nosology

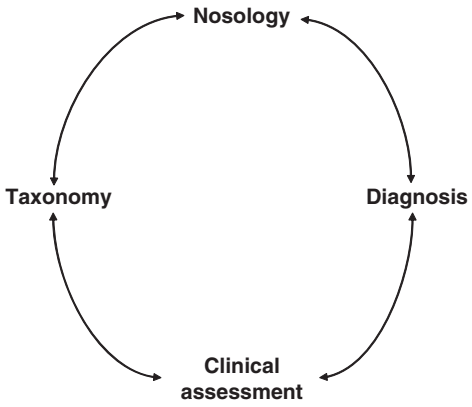

Figure 14.1 Schematic relations among assessment, diagnosis, nosology, and taxonomy.

is viewed as but one way to represent psychopathology rather than as embodying immutable truths concerning disorders and how to identify them. The *International Classification of Disease,* tenth edition (*ICD-10;* World Health Organization, 1992) provides a different nosology for representing psychopathology in terms of diagnostic categories, but it, too, is being revised. Continuing research on these and other ways of representing psychopathology is needed to optimize relations between clinical practice and advances in scientific findings.

CLINICAL ASSESSMENT

Clinical assessment refers to procedures for identifying the distinguishing features of problems, disorders, or cases, as well as relevant strengths. Many tests, questionnaires, inventories, interviews, and other procedures have been developed for the clinical assessment of children. Some assessment procedures, such as projective tests and unstructured clinical interviews, tend to be highly inferential. In using these assessment procedures, the practitioner makes inferences from responses to stimuli and questions in order to construct idiographic representations of the individual being assessed. Less inferential assessment procedures obtain responses whose meanings are more nomothetic. In other words, they are used to determine individuals' standing on scales or to place them in diagnostic categories, rather than to yield idiographic representations on the basis of the practitioner's interpretations. Assessment procedures also differ with respect to whether data obtained for individuals are judged in relation to norms versus nonnormed decision rules, such as those provided in the *DSM*.

Assessment procedures have long been used in clinical evaluations of children. However, the particular procedures, objectives, and types of conclusions have varied widely. In response to massive efforts to advance evidence-based treatment (EBT), Mash and Hunsley (2005) have argued for greater attention to what they call "evidence-based assessment" (EBA) as a necessary component of scientific support for clinical practice. Considering the flood of publications on EBT, it is

surprising how little attention has been paid to obtaining assessment data with which to choose the most appropriate treatment. Much of the EBT literature appears to assume that practitioners automatically know what should be treated and need only apply the appropriate EBT. The lack of explicit attention to systematically using assessment to determine what should be treated may reflect the fact that participants in EBT trials are often recruited on the basis of a priori definitions of the problems to be treated. Consequently, children are admitted to EBT trials if they are deemed to have the target disorder. However, criteria for recruiting participants for EBT trials may not be readily translatable into practical assessment procedures for helping clinicians decide which EBT to use with each client.

Mash and Hunsley (2005) have done a great service by calling attention to the failure of EBT to include generalizable assessment procedures. Nevertheless, their use of the term "evidence" may be somewhat ambiguous in "evidence-based assessment." In EBT, "evidence" appropriately refers to the informational basis for concluding that a particular kind of treatment is worthwhile. However, when applied to assessment, "evidence" is the *product* of the assessment process as well as the informational basis for deeming the assessment process to be worthwhile. In other words, assessment should *obtain evidence* on which to base choices of treatment, to judge the progress of treatment, and to evaluate the outcomes of treatment. Because assessment should serve these functions by empirically obtaining appropriate information about the case being assessed, I use EBA to stand for "empirically based assessment." This in no way weakens Mash and Hunsley's argument that EBT cannot stand alone without EBA. My own way of emphasizing their point is to observe that, "without EBA, EBT may be like a magnificent house with no foundation" (Achenbach, 2005, p. 547).

Obtaining Assessment Data on Children

Every individual differs from every other individual in countless ways. Consequently, no clinical assessment procedure can identify everything that is unique about each child and about the family members who are also important parts of each child's case. To focus on the subset of characteristics deemed most relevant to clinical decision making, assessment instruments are typically designed to obtain data according to particular diagnostic, nosological, and/or taxonomic models, as outlined in the following sections.

Diagnostically Based Assessment Instruments

Assessment instruments designed to make diagnoses include interviews consisting of questions about whether criteria for particular diagnoses are met. Several standardized interviews for making *DSM* diagnoses of children have been published. *ICD-10* diagnoses can also be made from some of the interviews, but most focus mainly on *DSM* diagnoses. Some diagnostic interviews, such as the Diagnostic Interview Schedule for Children (DISC; Shaffer, Fisher, Lucas, Dulcan, & Schwab-Stone, 2000), are highly structured in that they use precisely scripted questions to obtain yes-or-no answers as to whether each criterion is met for particular *DSM*

diagnostic categories. Other diagnostic interviews, such as the Schedule for Affective Disorders and Schizophrenia for School-Age Children (K-SADS; Ambrosini, 2000), are semi-structured in that interviewers are free to adapt the questions according to their clinical judgment.

In addition to ranging from highly structured to semi-structured, diagnostic interviews also vary in the degree to which conclusions are based only on the *respondent's* endorsement or denial of each *DSM* criterion versus being based on the *interviewer's judgment* of whether each criterion is met. The DISC is an example of a respondent-based interview, whereas the Child and Adolescent Psychiatric Assessment (Angold & Costello, 2000) is an example of an interviewer-based interview that requires interviewers to decide whether each *DSM* criterion is met.

When the first standardized *DSM* diagnostic interviews for children were developed in the 1980s, it was discovered that children's replies to interview questions often differed greatly from what parents and others reported about the children (Costello, Edelbrock, Kalas, Kessler, & Klaric, 1982; Herjanic & Reich, 1982). Consequently, parent versions of the interviews have been developed, and child versions have come to be used primarily with preadolescents and adolescents, rather than with younger children. However, even when restricted to preadolescents and adolescents, *DSM* diagnoses made from child interviews do not agree well with *DSM* diagnoses made from interviews with the children's parents. For example, in the massive Methods for Epidemiology of Child and Adolescent Disorders (MECA) Study of 1,283 9- to 18-year-olds, the mean kappa was .12 for agreement between diagnoses made from DISC interviews with children and their parents (P. Jensen et al., 1999). To provide some perspective on the meaning of the kappa statistic, Jacob Cohen (1960), the inventor of kappa, showed that the magnitude of kappa approximates the phi coefficient, which is the Pearson *r* scored from 2 $\times$ 2 tables (e.g., tables displaying relations between yes-versus-no categorizations of diagnoses made from child and parent interviews). Low agreement between parent and child reports of psychopathology is by no means limited to DISC interviews, as meta-analyses have yielded a mean *r* of only .25 between reports of child psychopathology by children and their parents, averaged across many instruments (Achenbach, McConaughy, & Howell, 1987).

In addition to interviews, another approach to assessing children in terms of *DSM* diagnoses is via questionnaires for rating *DSM* symptom criteria. An example is the ADHD Rating Scale IV, which consists of the *Diagnostic and Statistical Manual of Mental Disorders,* fourth edition (*DSM-IV;* American Psychiatric Association, 1994) symptom criteria for the inattentive and hyperactive impulsive types of Attention-Deficit/Hyperactivity Disorder (ADHD; DuPaul, Power, Anastopoulos, & Reid, 1998). Parent and teacher versions are available, on which each item is rated 0 to 3.

An example of a questionnaire for rating a broader range of *DSM* diagnostic categories is the Adolescent Symptom Inventory 4 (ASI-4; Gadow & Sprafkin, 1998). Parents and teachers rate symptoms for several *DSM-IV* diagnoses on 0 to 3 scales. A parallel self-rating questionnaire, the Youth Inventory-4 (YI-4; Gadow & Sprafkin, 1999), includes many of the same symptoms as the ASI-4.

Empirically Based Assessment Instruments

A key difference between the *DSM*-based instruments and what are here called empirically based instruments can be described in terms of their top-down versus bottom-up approaches. The *DSM*-based instruments are top-down in the sense that they start from the diagnostic categories and criteria that are listed in the *DSM*. The *DSM*-based assessment instruments are then constructed by wording the diagnostic criteria as questions that are either answered in interviews or are rated on questionnaires. The data sought with the *DSM*-based instruments are thus entirely shaped by the *DSM* nosology. Equally important, the conclusions drawn from assessment with these instruments are yes-versus-no judgments of whether criteria for particular *DSM* diagnoses are met.

By contrast, the empirically based instruments are bottom-up in the sense that their development starts with a large pool of assessment items that are chosen for their potential ability to identify children who are likely to need help for psychopathology. The items are pilot-tested by having them rated by appropriate respondents, such as parents, teachers, and children. Items that survive pilot testing are then used to assess large general population and clinical samples to determine how well they discriminate between children who have been referred for mental health services and demographically similar children who have not been referred for such services. Items that are found to be easily understood by the respondents, to be appropriately distributed, and to significantly discriminate between clinically referred and nonreferred children then become candidates for the scales of the assessment instruments. Items scored for large samples of children are subjected to statistical analyses, such as exploratory and confirmatory factor analyses (EFA and CFA), to identify sets (syndromes) of problems that tend to co-occur. The generalizability of the statistically derived syndromes is tested by performing similar analyses of items rated for children of each gender and different ages, by different informants, and in different cultures. These steps are outlined in Table 14.1.

Table 14.1 Developing Empirically Based Instruments for Assessing Child Psychopathology

1. Assemble pool of items that describe children's problems.
2. Formulate instructions and rating scales appropriate for the informants (e.g., parents, teachers, children).
3. Pilot-test and revise instruments based on informants' responses.
4. Use instruments to assess large clinical and epidemiological samples.
5. Do statistical analyses to identify syndromes of co-occurring problems.
6. Construct scales and profiles for scoring syndromes from the assessment forms.
7. Construct norms based on scores obtained by samples that are representative of relevant populations.
8. Test reliability and other psychometric properties.
9. Test validity in relation to external criteria.
10. Test replicability of syndrome structure in multiple cultures.

From *Multicultural Understanding of Child and Adolescent Psychopathology: Implications for Mental Health Assessment,* by T. M. Achenbach and L. A. Rescorla, 2007b, New York: Guilford Press. Adapted with permission.

Some of the steps required to develop empirically based instruments and scales for scoring them have been followed in the development of many instruments. For example, factor analyses were used in the development of some scales scored from the Behavior Assessment System for Children (Reynolds & Kamphaus, 2004). However, the results of the factor analyses were overridden when they contradicted the authors' judgments of what the scales and their constituent items should look like.

For another instrument, the Strengths and Difficulties Questionnaire (SDQ; Goodman, 1997), a priori scales were constructed and factor analyses were subsequently performed to see whether the obtained factors would match the scales (e.g., Dickey & Blumberg, 2004; Goodman, 2001; Koskelainen, Sourander, & Vauras, 2001; Muris, Meesters, Eijkelenboom, & Vincken, 2004; Rønning, Handegaard, Sourander, & Mørch, 2004). The factor analytic results varied considerably, perhaps owing at least partly to variations in methodology, samples, and criteria for determining whether the factors matched the SDQ scales.

Two families of widely used instruments have been developed according to steps like those outlined in Table 14.1. C. Keith Conners (1997, 2001) has developed a family of instruments primarily for assessing ADHD and related problems. The items comprising the instruments are rated on 0-1-2-3 scales by parents (Conners Parent Rating Scale—Revised; CPRS-R), teachers (Conners Teacher Rating Scale—Revised; CTRS-R), and youths (Conners-Wells Adolescent Self-Report Scale). The item pools and scales differ across the three instruments.

The CPRS-R and CTRS-R are scored for ages 3 to 17 and have been used much more widely than the youth form. The CPRS-R and CTRS-R both have factor-analytically derived scales designated as follows: Oppositional, Cognitive Problems/Inattention, Hyperactivity, Anxious-Shy, Perfectionism, and Social Problems. The CPRS-R is also scored on a scale designated as Psychosomatic, which cannot be scored from the CTRS-R because it lacks somatic items. Although the main scales were derived and then replicated by using factor analysis, items were eliminated if they lacked "conceptual coherence" with the factor on which they loaded (Conners, Sitarenios, Parker, & Epstein, 1998, p. 282). The CPRS-R and CTRS-R also have scales that paraphrase the *DSM-IV* symptom criteria for ADHD.

A family of instruments developed according to all the steps outlined in Table 14.1 includes the Child Behavior Checklist for Ages 6 to 18 (CBCL/6–18), Teacher's Report Form (TRF), and Youth Self-Report (YSR; Achenbach & Rescorla, 2001). Together with interview and observational instruments covering these ages, plus other instruments for ages 1.5 to 5, 18 to 59, and 60 to 90+, the CBCL/6–18, TRF, and YSR are components of the Achenbach System of Empirically Based Assessment (ASEBA; Achenbach & Rescorla, 2004; Rescorla & Achenbach, 2004). In addition to the factor-analytically derived scales, ASEBA instruments also have *DSM*-oriented scales comprising items identified by international experts as being very consistent with *DSM* diagnostic categories, as well as scales for assessing strengths. The problem items of most ASEBA instruments are rated 0 = not true, 1 = somewhat or sometimes true, and 2 = very true or often true.

As noted earlier, correlations between parents' and children's reports of child psychopathology are quite modest, as indicated by an *r* of only .25 in meta-analyses

(Achenbach et al., 1987). This low level of agreement means that parent reports and self-reports cannot be substituted for one another. Furthermore, agreement between adults who play different roles with respect to the children they rate is also modest, with meta-analyses yielding a mean *r* of only .28 between reports by parents versus teachers versus mental health workers versus observers. The *r*s averaged .60 between pairs of adults who play similar roles with respect to the children they rate (pairs of parents, teachers, mental health workers, observers). However, even these higher correlations do not indicate enough agreement to enable reports by any one informant to routinely serve as proxies for reports by other informants who play similar roles with respect to the children.

Although the meta-analytic findings were published in 1987, cross-informant agreement continues to be quite modest. As noted by De Los Reyes and Kazdin (2005, p. 483), the modest cross-informant agreement reported in the 1987 meta-analyses "has come to be one of the most robust findings in clinical child research: Different informants' (e.g., parents, children, teachers) ratings of social, emotional, or behavior problems in children are discrepant." Moreover, cross-informant discrepancies are not restricted to ratings of childhood problems. Meta-analyses of correlations between self-reports and others' reports of adult psychopathology have yielded mean *r*s of .43 for internalizing problems (depression, anxiety, somatic complaints) and .44 for externalizing problems (aggressive and rule-breaking behavior) when the probands and informants completed parallel instruments (Achenbach, Krukowski, Dumenci, & Ivanova, 2005). When probands and informants completed different instruments, the mean *r* was only .30. The fundamental fact is that agreement between different people's reports of psychopathology is simply too low to allow data from any single person to serve as a gold standard. Consequently, assessment, diagnosis, nosology, and taxonomy must all take account of the differences between the pictures obtained from different people.

The ASEBA instruments are designed to explicitly compare and contrast data from different informants. For preschoolers, the Child Behavior Checklist for Ages 1.5 to 5 (CBCL/1.5–5) is designed to be filled out by parent figures, and the Caregiver-Teacher Report Form (C-TRF) is designed to be filled out by day care providers and preschool teachers (Achenbach & Rescorla, 2000). These forms are scored on parallel syndrome scales that were derived from EFA and CFA of large general population and clinical samples. The scoring software provides profiles of scores and systematic comparisons of ratings by up to eight informants, as illustrated in Figures 14.2 and 14.3.

For school-age children, the scoring software compares scores from up to eight CBCL/6–18, TRF, and YSR forms. Furthermore, to take account of multicultural variations in norms, the scoring software enables users to display scale scores in relation to percentiles and *T* scores based on normative data obtained in many different societies (Achenbach & Rescorla, 2007a). For example, if a child lives in a society where problem scores tend to be relatively low, the user can elect to display the child's scale scores in relation to norms based on data from low-scoring societies. On the other hand, if a child comes from a low-scoring society but currently resides in a society where scores tend to be relatively high, the user can elect to display the child's scale scores in relation to norms for high-scoring

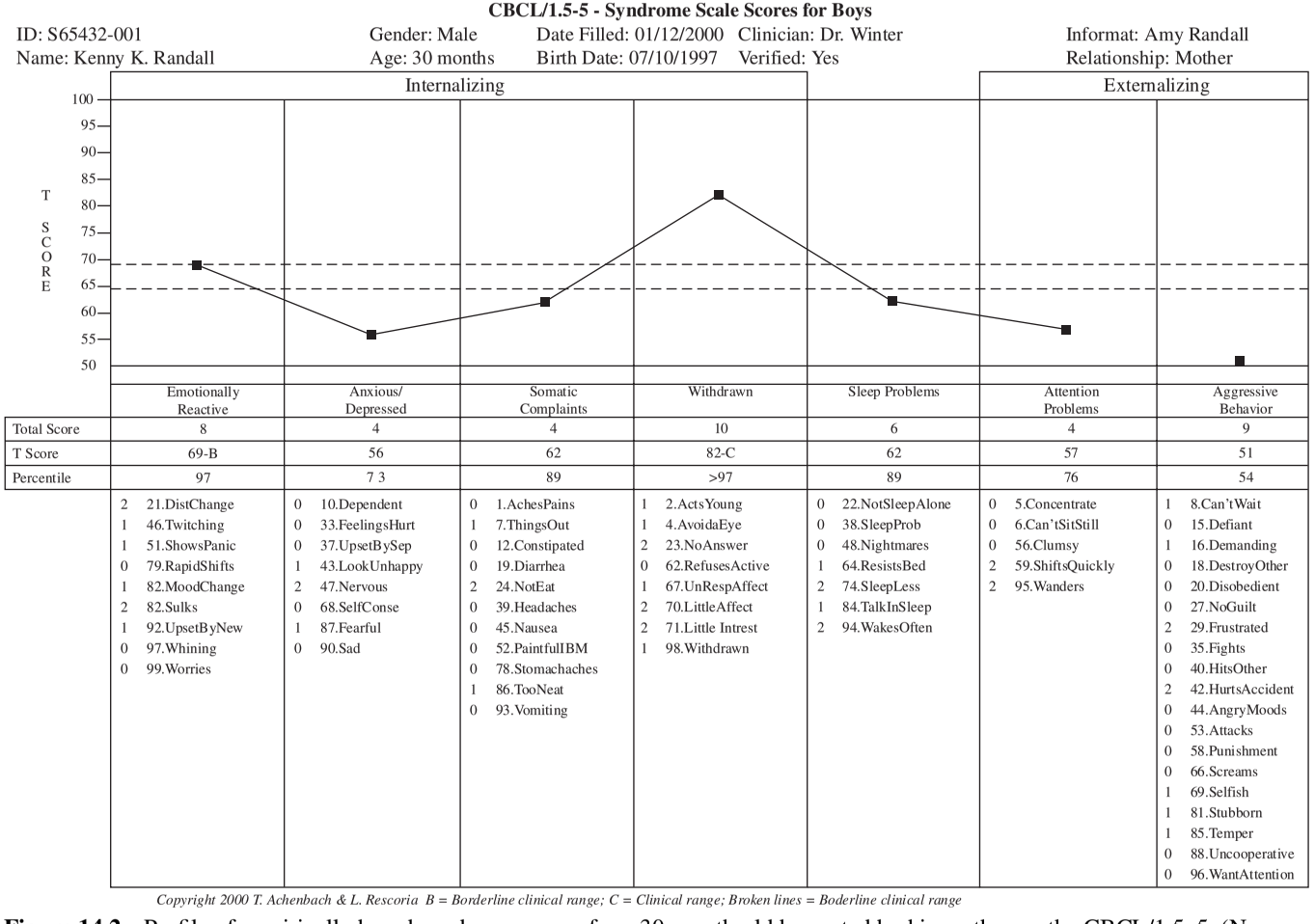

Figure 14.2 Profile of empirically based syndrome scores for a 30-month-old boy rated by his mother on the CBCL/1.5–5. (Names are fictitious.) *Source: Manual for the ASEBA Preschool Forms and Profiles,* by T. M. Achenbach and L. A. Rescorla, 2000, Burlington: University of Vermont, Research Center for Children, Youth, and Families. Reprinted with permission.

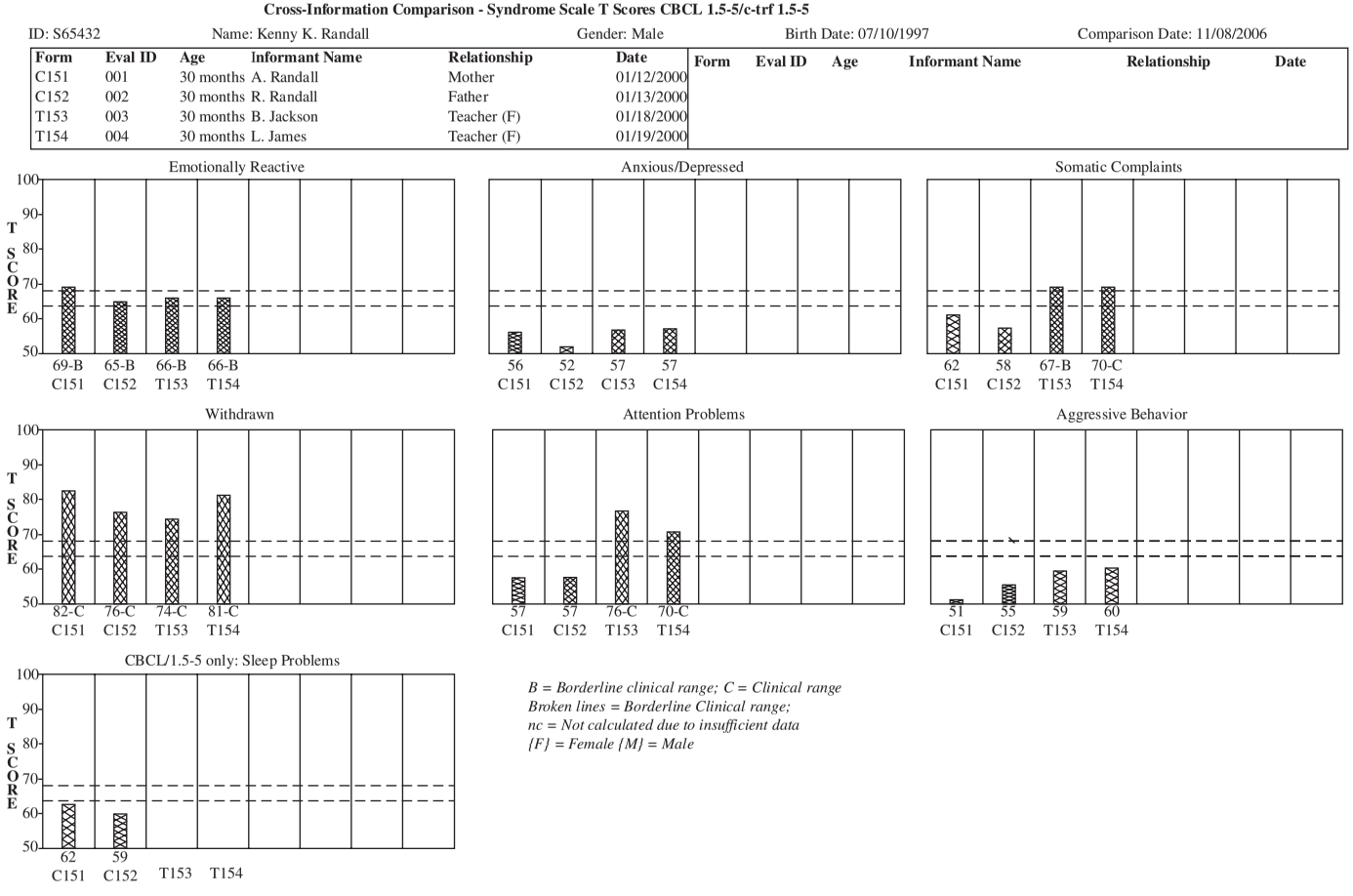

Figure 14.3 Cross-informant comparisons of syndrome scores for a 30-month-old boy rated on the CBCL/1.5–5 by his mother and father and on the C-TRF by two preschool teachers. (Names are fictitious.) *Source: Manual for the ASEBA Preschool Forms and Profiles,* by T. M. Achenbach and L. A. Rescorla, 2000, Burlington: University of Vermont, Research Center for Children, Youth, and Families. Reprinted with permission.

societies but can also display the child's scale scores in relation to norms for low-scoring societies like the child's home society. In this way, the user can determine whether the child's scores are clinically deviant according to both sets of norms, one set of norms, or neither set of norms. And, because the child's scale scores can be displayed for parent, teacher, and self-ratings, the user can compare scores obtained from each informant to whichever set of norms the user deems appropriate. Translations of ASEBA forms in more than 80 languages facilitate multicultural applications (Bérubé & Achenbach, 2008).

Obtaining Assessment Data on Parent Figures

Clinical assessment of children typically involves parent figures, including biological, adoptive, and stepparents. Parent figures may also include relatives, such as grandparents, aunts, uncles, and older siblings, and unrelated caregivers, such as foster parents, group home parents, the boyfriend or girlfriend of a child's parent, and child care workers. Parent figures (hereafter collectively designated as "parents") often instigate clinical assessment or seek it in response to urging by school, medical, welfare, or law enforcement personnel. Parents are key sources of information about children's functioning, are apt to affect children's problems and strengths, and are often pivotal in determining what will be done to help children.

Because child clinical cases typically need to be conceptualized in relation to parents, assessment of parents is a key component of the process of deciding how a child's problems should be viewed. When evaluating a child, mental health practitioners usually interview parents to learn about the background of the referral, to form impressions of the parents, and to discuss options. Parents may also be asked to complete instruments for assessing their child, such as *DSM*-based interviews, empirically based rating forms, and questionnaires focused on particular referral problems, such as ADHD, anxiety, or depression.

Because parents' reports and discrepancies between parents' reports are apt to be shaped by characteristics of the parents, it is helpful to illuminate these characteristics by assessing parents with instruments that parallel those for assessing their children. Although *DSM*-based interviews are typically too costly and intrusive to administer to parents, self-administered empirically based adult assessment instruments are available that parallel the ASEBA instruments for children (Achenbach & Rescorla, 2003).

The ASEBA adult instruments include parallel self- and other-report forms that tap adult versions of many of the strengths, problems, and scales that are scored from the child instruments, plus characteristics that are more specific to adults. To take account of differences between reports of adults' functioning by themselves and by others, the scoring software for the adult instruments can compare scores obtained from self-reports by each parent about himself or herself and reports by the parent's partner or someone else who knows the parent. Because the self- and other-report forms can each be completed in about 15 minutes at home, in a waiting room, or on the Internet, no professional time is needed to obtain the assessment data.

The profiles and comparisons of self- and other-reports can help the practitioner see relations between these different pictures of the parents' functioning and what the parents report about the child who is being evaluated. This approach enables the practitioner to efficiently document parental characteristics relevant to understanding children who are being assessed. The practitioner may also elect to show the parents the profiles obtained from the self- and other-reports to highlight similarities and differences in how their functioning is perceived. Viewing the profiles and cross-informant comparisons can teach parents about differences between perceptions of functioning and may motivate them to change their own behavior.

CLINICAL DIAGNOSIS

Since the publication of *DSM-III* (American Psychiatric Association, 1980), diagnosis has become an increasingly central focus of mental health services, training, research, and reimbursement. The introduction of explicit diagnostic criteria and fixed decision rules in *DSM-III* probably contributed to the ascendance of diagnosis. It was hoped that the change from inferential narratives in *DSM-II* (American Psychiatric Association, 1968) would produce more reliable diagnoses, etiological discoveries, and prescriptive treatments. Although the more explicit diagnostic criteria have led to much more detailed documentation of problems, the hopes for greater reliability of *DSM-III* diagnoses than achieved for *DSM-II* diagnoses were not realized for childhood disorders (Mattison, Cantwell, Russell, & Will, 1979; Mezzich, Mezzich, & Coffman, 1985). Furthermore, the hopes for etiological discoveries and for treatment specificity based on *DSM* diagnoses have not been realized either (Kupfer, First, & Regier, 2002; Rounsaville et al., 2002).

Although the term "diagnosis" may imply authoritative identification of a particular disorder, the Greek roots of *dia* = "apart" and *gignoskein* = "to know" pertain to distinguishing a particular condition from other conditions. In other words, diagnoses involve distinguishing among a variety of possibilities rather than simply identifying a condition whose identity can be understood without reference to similarities and differences between it and other conditions. Accordingly, the meaning of each possible diagnosis depends on how it is differentiated from other diagnoses within a system of diagnoses. The assessment data on which diagnoses are based must therefore ultimately be linked to a system that distinguishes among the phenomena that different diagnoses are supposed to represent. However, to clarify the relations between assessment and diagnosis and the nosological and taxonomic systems to which they should ultimately be linked, it is important to distinguish among the different meanings of diagnosis that sometimes confuse efforts to advance knowledge of child psychopathology. At least three meanings of diagnosis need to be distinguished, as outlined in the following sections.

Diagnostic Processes

Diagnostic processes consist of gathering data with which to make formal diagnoses and diagnostic formulations. Diagnostic processes can be thought of as constituting that portion of assessment that is directed toward making formal diagnoses and

diagnostic formulations. Although assessment thus includes diagnostic processes, it is not necessarily limited to diagnostic processes because it may include obtaining data on strengths, problems, and other variables that are not included in diagnostic criteria. However, assessment that is focused only on determining whether diagnostic criteria are met, such as *DSM*-based interviews, can be viewed as limited to diagnostic processes.

Formal Diagnoses

Formal diagnoses are made by determining the category of a nosology to which a case belongs. Diagnosis in this sense is what a leading psychiatric diagnostician called "the medical term for classification" (Guze, 1978, p. 53). When it is determined that a child meets criteria for a category of a nosology, such as *DSM* or *ICD,* this determination constitutes a formal diagnosis of the child, that is, a matching of the child's problems to a category of a nosology.

Diagnostic Formulations

Diagnostic formulations involve putting together various kinds of information about a case into a comprehensive picture of the case. For children, the information may include developmental history, family dynamics, physical liabilities, stressors, formal diagnoses, and prospects for change. In clinical practice, the diagnostic formulation may say more about the child's needs and the indications for particular treatments than formal diagnoses do.

To prevent undue reverence for diagnosis from clouding our thinking about relations among assessment, diagnosis, nosology, and taxonomy, it is important to be clear that the *DSM* and *ICD* consist of formal diagnoses but do not specify either diagnostic processes or diagnostic formulations. Although the *DSM* has been said to provide "operational definitions" for diagnoses (Rapoport & Ismond, 1996, p. 13), it does not in fact specify assessment operations (i.e., diagnostic processes) for most childhood disorders. About the only operationalized diagnostic criteria specified by *DSM-IV* are "an IQ of approximately 70 or below on an individually administered IQ test" (American Psychiatric Association, 2000, p. 49) as a criterion for mental retardation, plus low performance on relevant tests for learning disorders (Widiger & Clark, 2000). Thus, although diagnostic processes and formulations are certainly essential clinical activities, they are not within the purview of the *DSM* or *ICD* nosologies. Instead, these nosologies are classifications of formal diagnoses, as addressed in the following sections.

NOSOLOGICAL CATEGORIES

The *DSM* and *ICD* are nosologies, that is, systems for classifying diseases (called "disorders" in the *DSM*). They provide criteria for defining categories of disorders and for distinguishing particular disorders from other disorders. Since the publication of *DSM-III* (American Psychiatric Association, 1980), the *DSM* has included explicit criteria for each disorder and has specified fixed decision rules for determining whether a child has a particular disorder. Each criterion must be judged in a

yes-or-no fashion as either being met or not met. The *DSM* provides guidelines for describing some disorders as mild, moderate, or severe, but if the required number of criteria is not met, the disorder is deemed absent. Conversely, if the required number of criteria is met, the disorder is deemed present. The standard version of *ICD-10* (World Health Organization, 1992) does not provide such explicit diagnostic criteria or fixed rules for formal diagnoses, although a little-used research version of *ICD-10* does provide explicit criteria (World Health Organization, 1993).

Challenges of Comorbidity

Since the advent of explicit diagnostic criteria in *DSM-III,* numerous studies have found that many children qualify for multiple diagnoses. Termed "comorbidity," the tendency for disorders to co-occur could be very informative. For example, if many children diagnosed as having Disorder A are also diagnosed as having Disorder B, this could suggest that one of the disorders causes the other or that both disorders have a common cause. Comorbidity between Disorder A and Disorder B could also suggest that having one disorder makes a child more vulnerable to the other disorder, perhaps as a consequence of weakened resistance or of adverse effects of treatment for the initial disorder. However, because the *DSM* does not have well-validated markers for distinguishing each childhood disorder from each other disorder, it is also possible that apparent comorbidity between disorders reflects a lack of clear boundaries between them (Achenbach, 1991; Caron & Rutter, 1991). In other words, the diagnostic criteria for different nosological categories may not accurately represent truly different disorders. Nosologies may thus imply distinctions between disorders without evidence for true differences between the disorders.

As an example of a possible "distinction without a difference," two studies found that 96% of boys who met *DSM-III* criteria for Conduct Disorder (CD) also met criteria for Oppositional Defiant Disorder (ODD; Faraone, Biederman, Keenan, & Tsuang, 1991; Walker et al., 1991). Furthermore, field trials for the *Diagnostic and Statistical Manual of Mental Disorders,* third edition, revised (*DSM-III-R;* American Psychiatric Association, 1987) found that 84% of clinically referred children who met criteria for CD also met criteria for ODD (Spitzer, Davies, & Barkley, 1990). These findings certainly suggest that the nosological distinction between CD and ODD did not reflect two truly different disorders.

Unidirectional versus Bidirectional Comorbidity

Computing comorbidity in the other direction, that is, as the percentage of children diagnosed with ODD who were also diagnosed with CD, might yield a different picture. For example, if the proportion of children diagnosed with ODD were much greater than the proportion diagnosed with CD, the percentage of those diagnosed with ODD who had comorbid CD would be lower than the percentage of children diagnosed with CD who had comorbid ODD. Thus, if 200 children in a sample are diagnosed with ODD but only 100 are diagnosed with CD, the percentage of children who have ODD and also have CD could not exceed 50%. However, if all 100 of the children diagnosed with CD were among the 200 diagnosed with ODD, 100% of the children with CD would have comorbid ODD.

To avoid the misleading results of unidirectional computations of comorbidity rates, McConaughy and Achenbach (1994) recommended that comorbidity be computed as the mean of the comorbidities obtained by computing comorbidity in each of the two possible directions. Computing comorbidity bidirectionally in our hypothetical example would yield 100 CD/200 ODD = 50% + 100 ODD/100CD = 100%, for a bidirectional comorbidity of 50% + 100%/2 = 75%. Unfortunately, many studies of comorbidity fail to do this.

Although unidirectional computation of comorbidity makes it difficult to draw generalizable conclusions about precise rates of comorbidity, it is certainly clear that comorbidity rates have remained high even after *DSM* criteria were modified to reduce the amazingly high rates found for diagnoses such as CD and ODD (e.g., Angold, Costello, & Erkanli, 1999). In addition to high comorbidity between particular pairs of formal diagnoses, high comorbidity is also evident in findings that large proportions of children qualify for multiple diagnoses in community samples (e.g., Bird, Gould, & Staghezza, 1993) and clinical samples (e.g., A. Jensen & Weisz, 2002). In other words, children who have enough problems of a particular type to qualify for a diagnosis often have enough other kinds of problems to qualify for additional diagnoses as well.

Implications of Comorbidity

What are the implications of the high comorbidity rates found for formal diagnoses? As indicated by the 96% comorbidity found between CD and ODD in two studies, some comorbidity rates are so high that they raise questions about whether nosological distinctions between formal diagnostic categories accurately reflect true differences between disorders. If the different nosological categories do not in fact discriminate between different disorders, then comorbidity may be an artifact of having multiple categories for the same disorders.

The criteria for ODD and CD were changed from *DSM-III* to *DSM-III-R* to *DSM-IV* to reduce the overlap between these two diagnostic categories. However, it may still be an open question as to whether there is a meaningful difference between ODD, designated in *DSM-IV* as "[a] pattern of negativistic, hostile, and defiant behavior," and CD, designated as "[a] repetitive and persistent pattern of behavior in which the basic rights of others or major age-appropriate societal norms or rules are violated" (American Psychiatric Association, 2000, pp. 98, 102). Furthermore, the 15 criterial symptoms for CD are grouped into subcategories designated "aggression to people and animals," "destruction of property," "deceitfulness or theft," and "serious violations of rules." Because only 3 of the 15 criterial symptoms need to be judged present to warrant a diagnosis of CD, some children who qualify may have three aggressive symptoms, whereas others may have three rule-violating symptoms, and still others may have three deceitful symptoms or various combinations of aggressive, destructive, deceitful, and rule-violating symptoms. Consequently, diagnoses of CD may include very different patterns of behavior problems, some of which overlap considerably with ODD criteria, whereas others do not. Although changes in criteria since *DSM-III* may have reduced comorbidity between formal diagnoses of ODD and CD, overlaps between ODD kinds of problems and

particular subsets of CD problems may still artifactually contribute to apparent comorbidity.

Another implication of high comorbidity rates is that multiple diagnoses can greatly complicate use of treatments that are designed for particular disorders. For example, suppose that Treatment X has been found effective with children who were selected for having Disorder A. Does this mean that Treatment X should also be used for children who have Disorder B and Disorder C, as well as Disorder A? Even if the answer is yes, should additional Treatments Y and Z be used because they have found been effective with Disorders B and C?

If formal diagnoses of behavioral, emotional, social, and thought problems validly discriminate between disorders that are as distinct as physical diseases such as cancer, measles, and diphtheria, it would make sense to use multiple specific treatments for children who qualify for multiple diagnoses. However, because neither physical etiologies nor other physical abnormalities have been identified as underlying nosological categories for children's behavioral, emotional, social, or thought problems, treatments cannot be aimed at different physical abnormalities marked by different formal diagnoses. Instead, treatments are aimed at altering behaviors, feelings, and thoughts that may overlap among nosological categories. Consequently, research is needed not only on the effectiveness of specific treatments for specific diagnoses but also on the effectiveness of treatment for the many children who qualify for multiple diagnoses.

A related implication of comorbidity is that children who qualify for multiple diagnoses supposedly have multiple disorders. If such children in fact have multiple disorders, it could be hypothesized that they would have worse outcomes than children who have only one diagnosis. Doss and Weisz (2006) tested this hypothesis by analyzing the effects of initial severity on particular empirically based syndromes versus the effects of multiple syndrome deviance on treatment outcomes. They found that initial severity (high scores) on particular syndromes had much larger effects on outcomes than did deviance on multiple syndromes. Although these analyses did not directly test the effects of comorbidity among categorical diagnoses, such tests may be quite impractical, as discussed in the section on taxonomy.

New Directions in Nosology

A major thrust in planning for *DSM-V* involves possible quantification of at least some aspects of the diagnostic criteria (Helzer, Kraemer, Krueger, Wittchen & Regier, in press). Although quantified diagnostic criteria are often referred to as "dimensions," dimensions are not the only relevant forms of quantification. Consequently, I use the more comprehensive term "quantitative" to include dimensional as well as other quantitative approaches. Quantitative approaches have long been advanced for nosologies of personality disorders (Trull & Durrett, 2005), which primarily concern adult psychopathology. Quantitative approaches to nosologies of other adult disorders have also been proposed (Krueger & Markon, 2006). As will be evident in the section on taxonomic systems, quantitative approaches to child psychopathology have spawned extensive taxonomic research on psychopathology, including groupings compatible with existing categorical nosologies.

Although categorical and quantitative approaches are often assumed to be incompatible with one another, Kraemer (in press) has pointed out that each categorical diagnosis in the *DSM* nosology can be viewed as representing two (binary) values on a dimension. The two values are positive (the proband is deemed to have the disorder) versus negative (the proband is deemed free of the disorder). Most *DSM* diagnoses are specified in terms of a list of criterial symptoms and a cut point on the symptom list for distinguishing between positive and negative values (i.e., diagnosis present versus absent). The number of symptoms deemed present thus constitutes a score that ranges from 0 to *N* with *N* being the total number of symptoms on the list.

Kraemer (in press) argues that dichotomization is especially detrimental when, as is usually the case, there is no strong empirical basis for choosing cut points to discriminate between those who truly do versus do not have a particular disorder. To illustrate the detrimental effects of dichotomization, Kraemer cites studies of genetic influences on psychopathology, where unrealistically large samples are needed to detect genetic effects when the phenotypic characteristics are dichotomized rather than quantified. Kraemer also cites examples of research on treatment in which quantification of outcome measures revealed significant differences between the effects of different treatment conditions that were not detected when the same measures were analyzed dichotomously.

Although it has been argued that clinicians need categorical diagnoses (e.g., First, 2005), Kraemer (in press) provides examples of clinical diagnoses of hypertension, diabetes, and cancer that all involve quantitative specifications. She argues that the increasing importance of genetic, imaging, and biochemical data for research, training, and treatment related to psychopathology makes it essential to quantify aspects of the *DSM-V,* rather than deferring quantification to some unspecified future date, as was done when quantification of previous editions of the *DSM* was proposed. Preparations for the next edition of the *ICD* also include possibilities for quantification.

TAXONOMIC CONSTRUCTS

As discussed in the preceding section, nosologies are classifications of disorders. Like many classifications, they are constructed for the convenience of users. Accordingly, nosological categories may be constructed according to a variety of procedures and principles that are not necessarily consistent among all the categories. In a general medical nosology, for example, categories of bone fractures may be organized quite differently from categories of infectious diseases, cancers, and cardiac disorders.

Within the *DSM* and *ICD* nosologies, categories for disorders defined mainly in terms of behavioral and emotional problems are organized very differently from categories for mental retardation, learning disabilities, and dementias. Rather than being based directly on scientific principles or research findings, the *DSM* and *ICD* nosological categories were constructed mainly on the basis of negotiations among various stakeholders. A key consideration was whether the nosologies would be useful to clinicians, third-party payers, and those responsible for health care

records. Maintenance of continuity with previous editions of the nosologies was also an important objective.

The Nature of Taxonomies

Taxonomies are classification systems that are intended to capture intrinsic distinguishing features of the phenomena that are classified (Gove, 1971). Consequently, all taxonomies are classification systems, but classifications that do not apply scientific principles to distinctions among the classified phenomena are not taxonomies.

One view of taxonomies is that they should arrange entities into "natural categories on the basis of key features they share" (Millon, 1991, p. 246). Consistent with this view, it has been hoped that a taxonomy of psychopathology would "carve nature at its joints" (Kendell, 2002, p. 7). The goal of basing taxonomies on natural categories that carve nature at its joints is inspired largely by taxonomies of plant and animal species that are thought to distinguish among "natural kinds," each of which is inherently different from other natural kinds. However, no particular criteria have been found to consistently distinguish among species or even to provide an overarching definition of what "species" means. Consequently, biological species are now viewed as taxonomic constructs, rather than as embodying categories that are inherent in nature (Levin, 1979; Mayr, 1987). Considering that even taxonomies of biological species thus consist of constructs, current concepts of psychopathology are certainly unlikely to embody natural categories. Because it seems unrealistic to expect current concepts of psychopathology to yield taxonomies of natural kinds, Zachar (2000, p. 167) has argued that taxonomic constructs should be based on "patterns (of psychopathology) that can be identified with varying levels of reliability and validity."

If we acknowledge that natural taxonomic categories of psychopathology are unlikely to be discovered in the near future (if ever), it may be more realistic to consider multiple ways of taxonomically organizing data on psychopathology. This means that we should compare and contrast different ways of taxonomically organizing data on psychopathology. To avoid reifying any single set of taxonomic constructs, it may be helpful to view data on child psychopathology from multiple taxonomic perspectives. Considering the differences typically found between reports by different informants, it is also essential that taxonomic constructs take account of multi-informant variations. And because the patterning and prevalence of problems may differ among cultural groups, taxonomic constructs should take account of multicultural variations.

Nosologically Based Taxonomic Constructs

Although nosologies are classifications constructed for the convenience of users, it is assumed that each of their categories represents a particular type of disorder. In effect, each category represents a taxonomic construct that is defined in terms of the diagnostic criteria for the category. For example, the diagnostic criteria for CD define a taxonomic construct that is assumed to be different from the taxonomic constructs of ODD and ADHD. As illustrated in Figure 14.4, the approach to defining these taxonomic constructs can be described as top-down because it

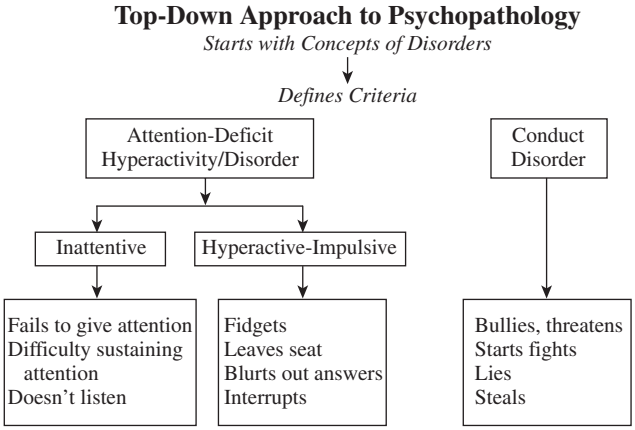

Figure 14.4 The top-down approach to defining taxonomic constructs. *Source: Multicultural Understanding of Child and Adolescent Psychopathology: Implications for Mental Health Assessment,* by T. M. Achenbach and L. A. Rescorla, 2007b, New York: Guilford Press. Reprinted with permission.

starts with a concept of a nosological category that is negotiated by the formulators of the nosology. The formulators of the nosology then work downward from the concept of the category to diagnostic criteria for determining which children fit the category.

Operationalizing Nosologically Based Criteria

Although diagnostic criteria have been made explicit in *DSM-III, DSM-III-R,* and *DSM-IV,* they are not specified in terms of operations for assessing individual children to determine which criteria are met. Instead, separate assessment procedures must be used to operationalize the diagnostic criteria to determine which children meet criteria for which diagnoses. Standardized diagnostic interviews, such as the DISC (Shaffer et al., 2000) and K-SADS (Ambrosini, 2000), and *DSM* rating forms, such as the ADHD Rating Scale-IV (DuPaul et al., 1998) and the YI-4 (Gadow & Sprafkin, 1999), operationalize criteria for those nosological categories that are based primarily on behavioral, emotional, social, and thought problems. However, each operationalization may yield different conclusions about which children meet criteria for which categories. For example, a mean kappa of only .03 was found for agreement between "definite" *DSM* diagnoses made from the DISC administered to mothers and their children versus the K-SADS administered to the same mothers and children (P. Cohen, O'Connor, Lewis, Velez, & Malachowski, 1987). When the diagnostic criteria were relaxed to include agreement between "possible" diagnoses, the mean kappa rose to only .14. Thus, even when criteria were operationalized by means of two of the most thoroughly researched and widely used interviews, the interviews failed to agree on their assignments of individual children to *DSM* categories. Furthermore, meta-analyses have yielded a mean kappa of only .15 between *DSM* diagnoses of children made from standardized interviews,

including the DISC and K-SADS, and diagnoses made from clinical evaluations of the children (Rettew, Doyle, Achenbach, Dumenci, & Ivanova, 2006).

DuPaul et al. (1998) reported point-biserial correlations averaging .31 between various combinations of *DSM* ADHD diagnoses and scores from the ADHD Rating Scale-IV completed by parents and teachers. When Gadow et al. (2002) selected the highest point-biserial correlations between 11- to 18-year-olds' self-ratings on YI-4 scales and *DSM* diagnoses in the youths' clinic charts, they found a mean correlation of .25 for diagnoses of ADHD, ODD, CD, depressive disorders, and anxiety disorders. Although not quite as low as the kappas, these modest correlations further indicate that major challenges remain for operationalizing the nosological taxonomic constructs.

Empirically Based Taxonomic Constructs

Empirically based assessment instruments were presented earlier in the chapter. Some of these instruments were explicitly designed to obtain data from which taxonomic constructs could be derived. For example, development of the CBCL/6–18, TRF, and YSR started with diverse assessment items that were then culled, revised, and augmented on the basis of extensive pilot testing. Subsequently, EFA and CFA were used to derive syndromes of co-occurring problems (Achenbach & Rescorla, 2001). The statistically derived syndromes serve as taxonomic constructs, where "constructs" are defined as "objects of thought constituted by the ordering or systematic uniting of experiential elements" (Gove, 1971, p. 489).

Although parent, teacher, and self-ratings differed with respect to some specific items, as well as with respect to the raters' perspectives, EFA and CFA identified eight similar syndromes in ratings by all three kinds of informants. The items common to the syndromes derived from the CBCL/6–18, TRF, and YSR define "cross-informant" syndrome constructs. Individuals' ratings of a particular child on the specific items composing the CBCL/6–18, TRF, and YSR versions of the syndromes operationalize the cross-informant syndrome constructs in ways that reflect variations in the raters' perceptions of the child. The cross-informant correlations between ratings on virtually all instruments, including the empirically based instruments, are inevitably limited by factors such as differences in the informants' roles (e.g., mother versus father versus math teacher versus history teacher versus self), knowledge and memory of the child's behavior, interpretations of assessment items, and candor. Because the empirically based approach derives syndromes statistically from data on large samples of real children, this approach to taxonomy can be described as working from the bottom up, as illustrated in Figure 14.5.

Relations between Nosologically Based and Empirically Based Taxonomic Constructs

Despite differences in their origins, some of the nosological categories include criterial symptoms analogous to the problem items comprising some of the empirically based syndromes. As an example, many of the criterial symptoms for ADHD have counterparts among the problem items that comprise the empirically based

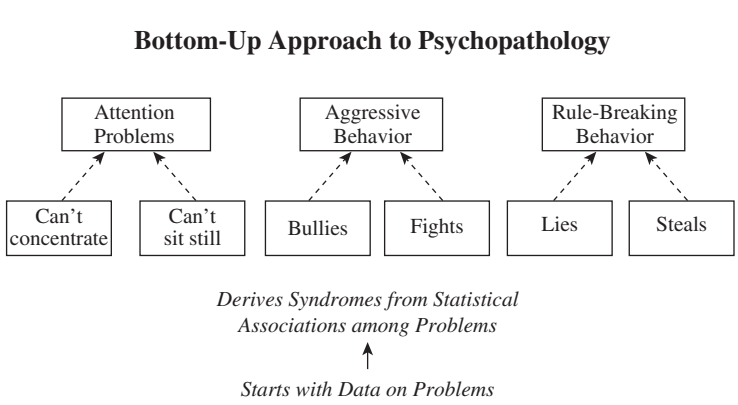

Figure 14.5 The bottom-up approach to defining taxonomic constructs. *Source: Multicultural Understanding of Child and Adolescent Psychopathology: Implications for Mental Health Assessment,* by T. M. Achenbach and L. A. Rescorla, 2007b, New York: Guilford Press. Reprinted with permission.

attention problems syndrome. On the other hand, some nosological categories group criterial symptoms differently than the empirically based syndromes group the corresponding problem items. For example, the nosological category of CD includes both overtly aggressive symptoms and nonaggressive rule-breaking symptoms. By contrast, factor analyses of the CBCL/6–18, TRF, and YSR, as well as of other instruments (Quay, 1986), have yielded separate syndromes of overtly aggressive versus rule-breaking behavior. Conversely, criterial symptoms for depression and anxiety are grouped in separate nosological categories, whereas factor analyses of the CBCL/6–18, TRF, and YSR, as well as other instruments (Achenbach & Rescorla, 2003), have yielded a syndrome that includes both depressive and anxiety problems.

Despite differences between the ways particular kinds of problems are partitioned, as well as differences between the categorical nosological criteria versus the quantification of items and syndromes, multiple studies have reported significant associations between *DSM* diagnoses and scores on empirically based syndromes (e.g., Bird, Gould, Rubio-Stipec, Staghezza, & Canino, 1991; Graetz, Sawyer, Hazell, Arney, & Baghurst, 2001; Steinhausen, Metzke, Meier, & Kannenberg, 1997; Verhulst, van der Ende, Ferdinand, & Kasius, 1997).

To facilitate cross-walks between nosological and empirically based approaches to taxonomy of child psychopathology, a top-down "expert judgment" approach has been used to construct *DSM*-oriented scales from the same pools of items that were factor analyzed to identify syndromes. This has been done for preschoolers, school-age children, adults, and the elderly (Achenbach, Dumenci, & Rescorla, 2003; Achenbach, Newhouse, & Rescorla, 2004; Achenbach & Rescorla, 2003). The *DSM*-oriented scales were constructed by having expert psychiatrists and psychologists from many cultural backgrounds identify problem items that they judged to be very consistent with particular *DSM-IV* categories. For nosological categories for which enough items were identified by a substantial majority of experts, the items were used to construct *DSM*-oriented scales. Like the empirically based

syndromes, the *DSM*-oriented scales are displayed on profiles in relation to age-, gender-, and informant-specific norms.

Multi-informant Perspectives

Previous sections have summarized findings of low to moderate levels of agreement between reports of psychopathology by different informants. Each informant may contribute clinically valuable information about a child's functioning and how it is perceived by the informant. The differences among the informants' reports are as valuable as the similarities for indicating the possible situational specificity versus cross-situational consistency of a child's problems. Differences among informants' reports may also reveal outlier informants whose perceptions of the child are idiosyncratic enough to warrant exploration of the informant's interactions with the child. Such findings may warrant interventions that focus on a particular informant or on the informant's interactions with the child.

To help clinicians efficiently document and compare reports by multiple informants, software is available that displays *DSM*-oriented scale scores, as well as syndrome scores obtained from any combination of up to eight forms (Achenbach & Rescorla, 2007a). As illustrated in Figure 14.6, bar graphs indicate standard scores (*T* scores) on each *DSM*-oriented scale in relation to norms for peers of the child's age and gender, as rated by different kinds of informants, such as parents, teachers, and children themselves. In each bar graph, the broken lines demarcate a borderline clinical range. Scores beneath the bottom broken line are in the normal range, whereas scores above the top broken line are in the clinical range. By looking at the bar graphs, the clinician can quickly identify clinically elevated scores and can also identify similarities and differences among informants' reports. If nosologically based taxonomic constructs are quantified, as exemplified by the *DSM*-oriented scales and as proposed for *DSM-V* (Helzer et al., in press), scores on such constructs can also be used for systematic cross-informant comparisons.

Multicultural Perspectives

Today's clinical caseloads include people from many different backgrounds. Native-born minority groups and immigrants from other regions and countries may need to be evaluated in relation to norms different from those of the majority groups in the societies where they are being clinically evaluated (Achenbach & Rescorla, 2007b). Before considering norms for different groups, it is important to determine whether particular taxonomic constructs are appropriate for representing psychopathology among different cultural groups. The multicultural generalizability of the statistically derived syndromes of the ASEBA instruments has been tested by performing CFA on large samples of parents' ratings of their children in 30 societies, teachers' ratings of their students in 20 societies, and youths' self-ratings in 23 societies (Ivanova, Achenbach, Dumenci, et al., 2007; Ivanova, Achenbach, Rescorla, Dumenci, Almqvist, Bathiche, et al., 2007; Ivanova, Achenbach, Rescorla, Dumenci, Almqvist, Bilenberg, et al., 2007). The societies differed in many ways from the United States, where the syndromes were originally derived. Examples of societies that differed in geographical region, language, ethnicity, religion, and

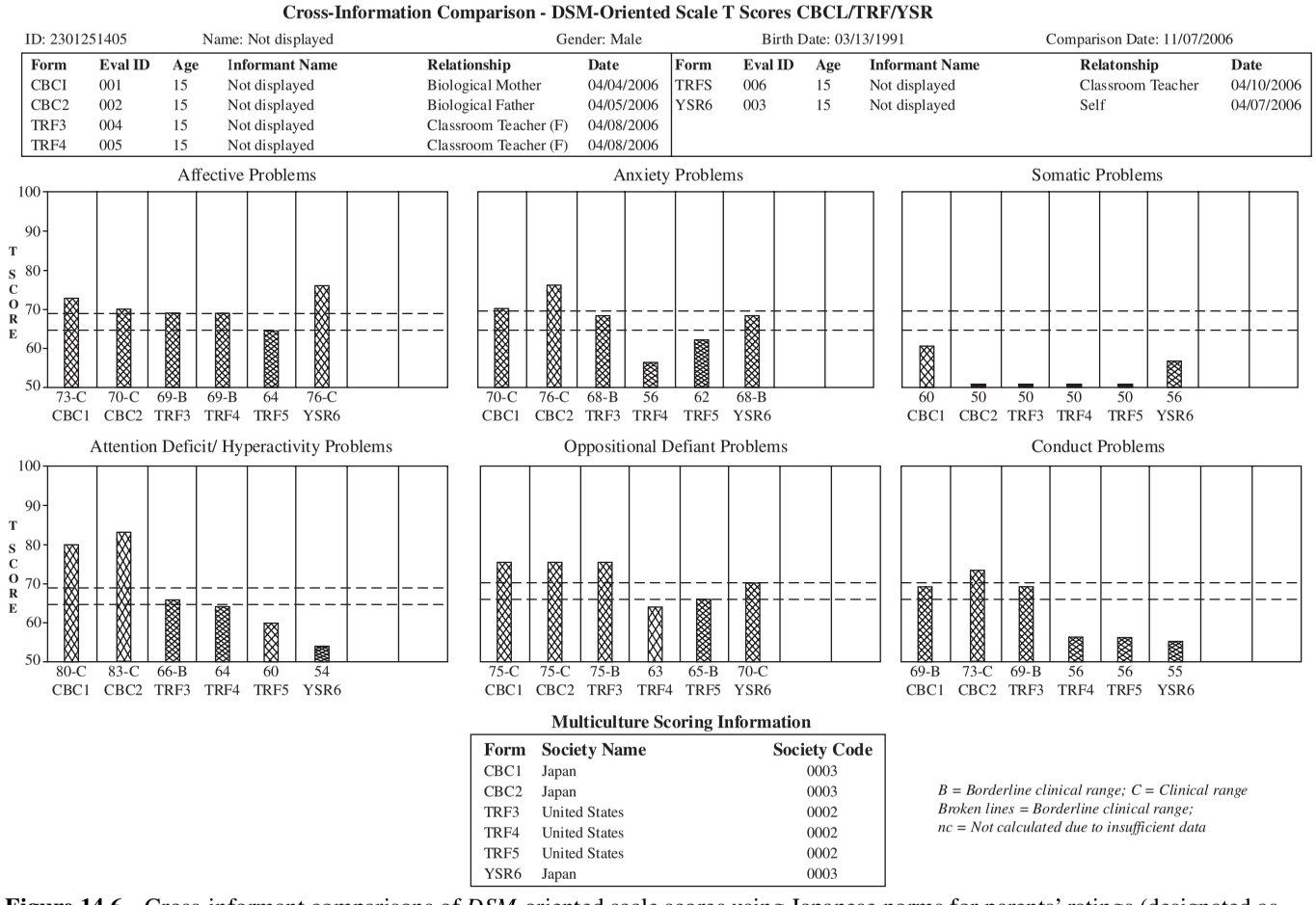

Figure 14.6 Cross-informant comparisons of *DSM*-oriented scale scores using Japanese norms for parents' ratings (designated as CBC) and youth's self-ratings (designated as YSR) and U.S. norms for teachers' ratings (designated as TRF). *Source: Multicultural Supplement to the Manual for the ASEBA School-Age Forms and Profiles,* by T. M. Achenbach and L. A. Rescorla, 2007a, Burlington: University of Vermont, Research Center for Children, Youth, and Families. Reprinted with permission.

political systems are Iran, China, Japan, Thailand, Korea, Ethiopia, Lebanon, Russia, and Turkey. Societies from the Caribbean and from western, eastern, northern, and southern Europe were also included.

To the surprise of the researchers, the syndromes derived primarily from U.S. samples were found to fit the data for all the other societies. The taxonomic constructs embodied by the statistically derived syndromes thus represent patterns of parent, teacher, and self-ratings of behavioral, emotional, social, and thought problems for at least this large array of very diverse societies. At this writing, no multicultural studies have tested taxonomic constructs embodied in the *DSM* or *ICD* nosologies for children's disorders.

The multicultural CFA support for the statistically derived syndromes justifies use of the syndromes for assessing children from many societies. However, this does not necessarily mean that the distributions of scale scores are similar enough to justify use of the same norms for different societies. The distributions of scale scores have been statistically compared for the United States and all the societies that were included in the CFA tests of the statistically derived syndromes (Rescorla et al., 2007, 2007a, 2007b). A great deal of overlap was found between the distributions of problem scores from all societies and the mean total problems scores from most societies were clustered within a narrow range. However, some societies had mean total problems scores that were low or high enough to warrant separate norms.

To take account of the relatively low scores, one set of norms was constructed by averaging the scores for societies whose mean total problems scores were more than 1 standard deviation below the mean of all the societies (called the "omnicultural mean"; Ellis & Kimmel, 1992). Another set of norms was constructed by averaging the scores for societies whose mean total problems scores were more than 1 standard deviation above the omnicultural mean. A third set of norms, for the middle scoring societies, consists of norms for U.S. samples, which are about in the middle of the middle group and which have been used in many clinical and research applications.

When evaluating individual children, clinicians can elect to have the children's syndrome and *DSM*-oriented scale scores displayed on profiles and cross-informant bar graphs in relation to norms for the low-scoring, middle-scoring, or high-scoring societies. The clinician can also choose to have percentiles and *T* scores from the low-, middle-, or high-scoring societies displayed. Equally important, the clinician can see how the child's scale scores compare with different sets of norms. This is especially useful for immigrant children and for children whose parents and/or teachers have very different cultural backgrounds.

As an example, for the cross-informant comparisons shown in Figure 14.6, the clinician elected to have the parents' CBCL/6–18 scale scores displayed in relation to norms for the high-scoring societies, because the parents had emigrated from a high-scoring society, maintained its customs, still spoke its language, and completed a translation of the CBCL/6–18 in that society's language. Because their child attended a mainstream school in the host society where TRF scores were in the middle-norm group, the clinician elected to have the TRF scale scores displayed in relation to the middle norms. Because YSR scores in the youth's home culture were in the high-norm group, the clinician elected to have the YSR scale scores displayed in relation to the high norms. However, because the host society's YSR scores

were in the middle-norm group, the clinician subsequently had the cross-informant comparisons displayed again with the youth's YSR scores displayed in relation to the middle norms. This enabled the clinician to see whether any scales would be clinically elevated that had not been clinically elevated in relation to the high norms. Note that all scales in the cross-informant comparisons are displayed in terms of *T* scores, but the user decides whether to display scores for each informant in relation to *T* scores based on the low-scoring, middle-scoring, or high-scoring societies.

Implications for Clinical Practice

Taxonomic constructs may seem irrelevant to clinical practice. However, they provide foundations for assessment, diagnosis, nosology, and treatment. Nosological categories such as ADHD, ODD, CD, Major Depressive Disorder, and Generalized Anxiety Disorder shape many aspects of clinical practice. Although such categories may seem to have an intrinsic reality, changes in the categories from one edition to another edition of a nosology, the lack of operational definitions, poor agreement among "gold standard" diagnostic procedures, and high comorbidity indicate that current nosological categories cannot be equated with categorically distinct disorders. Instead, the categories can be viewed as representing implicit taxonomic constructs. As such, they should be subjected to comparisons with alternative taxonomic constructs. For child psychopathology, statistically derived syndromes offer one alternative set of taxonomic constructs for which there is considerable empirical support. Options for scoring both statistically derived syndromes and *DSM*-oriented scales from the same sets of parent, teacher, and self-ratings can help clinicians view children's problems from both taxonomic perspectives. Multi-informant and multicultural comparisons of both kinds of scales further enhance opportunities for flexibly adapting clinical assessment to the needs of children from diverse backgrounds, as seen from different informants' perspectives.

SUMMARY

Assessment, diagnosis, nosology, and taxonomy can be understood as interdependent aspects of continuous processes. This chapter focused on these processes as they pertain to behavioral, emotional, social, and thought problems occurring between the ages of 18 months and 18 years with sufficient severity to warrant the term "psychopathology" and a likely need for professional help.

Clinical assessment refers to procedures for identifying the distinguishing features of problems, disorders, or cases, as well as relevant strengths. Some assessment instruments are designed to determine whether children meet criteria for *DSM* diagnoses, whereas other instruments assess broader spectra of functioning. Evidence- (or empirically) based assessment is necessary for appropriate applications of evidence-based treatment and other clinical services. Comprehensive clinical assessment of children should include assessment of parent figures.

Diagnosis has become a primary focus of clinical services since the *DSM-III* introduced explicit diagnostic criteria. Confusion often arises from failure to distinguish clearly among diagnostic processes, formal diagnoses, and diagnostic

formulations. Most relevant to nosology and taxonomy, formal diagnoses are made by determining how a case should be categorized.

Nosologies such as the *DSM* and the *ICD* are systems for classifying diseases (or disorders). Since the introduction of explicit diagnostic criteria by *DSM-III,* many children have been found to meet criteria for multiple diagnoses. Although comorbid diagnoses may provide useful information, the implications of comorbidity have been clouded by failure to compute it bidirectionally and by rates of comorbidity between some diagnoses that are so high as to cast doubt on whether they really represent different disorders. Quantitative approaches that are being advocated for *DSM-V* could help to alleviate problems arising from excess comorbidity among diagnoses as well as to bring diagnoses more in tune with advances in genetics, imaging, and biochemistry.

Taxonomic constructs may seem irrelevant to clinical practice, but they provide foundations for assessment, diagnosis, nosology, and treatment. The lack of known natural categories for psychopathology means that we should think in terms of multiple ways of taxonomically organizing data on psychopathology to avoid reifying any single set of taxonomic constructs. Nosological categories can be viewed as taxonomic constructs, although criteria for the constructs must be operationalized in a top-down process of developing assessment procedures such as diagnostic interviews. However, even the most thoroughly researched and widely used diagnostic interviews have yielded negligible agreement for child diagnoses made from other diagnostic interviews and from clinical evaluations.

Empirically based taxonomic constructs have been derived from statistical analyses of data obtained with empirically based assessment instruments. The generalizability of the syndromes derived from this bottom-up process has been supported by CFA of large samples from many societies.

There are important points of contact between nosologically based and empirically based taxonomic constructs. Both kinds of constructs have been operationalized via a common pool of assessment items. For both kinds of constructs, assessment data from multiple informants and normative data from multiple cultures are needed for clinical research and practice in the twenty-first century.

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CHAPTER 15

Diagnostic Interviewing

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It is well recognized by those involved in children's mental health that proper assessment and diagnosis of the diverse childhood psychopathologies are critical undertakings. The importance of these issues has been reflected in the numerous changes and developmental adaptations made to the *Diagnostic and Statistical Manual of Mental Disorders* (*DSM*). The criteria provided in the *DSM* are considered the basis for all major psychiatric diagnoses and constitute the necessary requirements for reimbursement or coverage by major health management organizations. Over the past 25 years, the *DSM* has undergone several changes specific to childhood disorders. In so doing, the *DSM* has begun to recognize that children are not miniature adults and that the identification of various disorders in childhood requires developmental adaptations (Ollendick & Hersen, 1983, 1989, 1998). For example, with the third revision of the *DSM,* a section specific to disorders that are usually first diagnosed in infancy, childhood, or adolescence was introduced. In addition, with each revision to the *DSM,* increasingly more diagnostic criteria that incorporate developmental considerations have been put forth. To illustrate, in the current *DSM-IV* (American Psychiatric Association, 1994, pp. 416–417), the diagnostic criteria for Social Phobia provide the following distinctions for children:

> (A) In children, there may be evidence of the capacity for age-appropriate social relationships with familiar people and the anxiety must occur in peer settings, not just in interactions with adults; (B) In children, the anxiety may be expressed by crying, tantrums, freezing, or shrinking from social situations with unfamiliar people; (C) In children, this feature (i.e., recognition of the fear as excessive or unreasonable) may be absent; and (F) In individuals under age 18 years, the duration is at least 6 months.

Similar modifications have been made to other psychiatric disorders to make them more applicable to children and adolescents for whom separate categories do not exist.

Increased attention to the issue of proper evaluation of child psychopathology has also been reflected in the actual role of children in the assessment and diagnostic process. In the past, parents were thought to be the best informants regarding

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a child's behavior and were, therefore, the only ones included in the assessment process (Loeber, Green, & Lahey, 1990; Rutter & Graham, 1968; Rutter, Tizard, & Whitmore, 1970; Williams, McGee, Anderson, & Silva, 1989). Any information collected from or reported by the child was considered secondary to the parents' reports. Reliance on parental report was primarily based on the assumption that children lacked the cognitive sophistication to respond appropriately in an interview format. For example, researchers reported concern that children would be inattentive to the assessment process, lack insight regarding their feelings, and be unable to accurately describe their own behaviors and symptoms (Edelbrock & Costello, 1990; Herjanic, Herjanic, Brown, & Wheatt, 1975; Schwab-Stone, Fallon, Briggs, & Crowther, 1994). Thus, the amount of credence placed on the child's report was minimal, and parents were considered the experts regarding the child's outward behaviors as well as their internal thoughts and feelings.

This view changed dramatically following seminal studies by Lapouse and Monk (1958), who included mothers and their children in the interview process, and Rutter, Graham, and colleagues (Rutter & Graham, 1968; Rutter et al., 1970; Rutter, Tizard, Yule, Graham, & Kingsley, 1977; Rutter, Tizard, Yule, Graham, & Whitmore, 1976), who designed and psychometrically validated the first structured interview procedures for children. The consideration of the child as a separate and valuable respondent by these researchers as well as their careful evaluation of the interview process paved the way for today's inclusion of multiple informants in clinical interview procedures. Currently, most researchers and clinicians consider a child assessment incomplete if information from numerous sources (e.g., teachers, parents, doctors), including the child, is not obtained (Ollendick & Hersen, 1984, 1993). Thus, children are now viewed as reliable, valid, and valuable informants regarding their subjective thoughts, behaviors, and emotions (Chambers et al., 1985; Edelbrock, Costello, Dulcan, Kalas, & Conover, 1985; Herjanic et al., 1975; Hodges, McKnew, Cytryn, Stern, & Kline, 1982; Kazdin, French, & Unis, 1983; Moretti, Fine, Haley, & Marriage, 1985; Verhulst, Althaus, & Berden, 1987). Consequently, companion diagnostic interviews for children and their parents have been developed (and revised) in line with the changes in the *DSM*. To demonstrate their psychometric soundness (e.g., reliability, validity, clinical utility), careful evaluations have been conducted with both the child and parent diagnostic interviews.

In general, "reliability refers to the consistency of measurements," whereas "the validity of a test refers to the extent to which a test measures what it is supposed to measure, and therefore the appropriateness with which inferences can be made on the basis of the test results" (Sattler, 1988, pp. 25, 30). Although a measure can be reliable while lacking validity, both of these psychometric features are necessary to proclaim an interview psychometrically sound.*

*The kappa statistic (i.e., κ) is commonly used in studies that assess reliability with dichotomous data. The kappa is a correlational statistic that corrects for chance and ranges from -1.00 (total disagreement) to 1.00 (pure agreement). Although the kappa statistic evaluates agreement beyond the level of chance, it is also sensitive to base rates, resulting in lower κs for less prevalent symptoms or disorders. Interpretations vary, but typically $\kappa s < .40$ are considered poor, $\kappa s = .40$ to .70 are considered acceptable, and $\kappa s > .70$ are considered good or excellent.

Important distinctions in the literature regarding reliability are those among different indices of interrater, test-retest, and multiple-informant agreement. *Interrater agreement* refers to the amount of consensus between two observers of the same child or of the same child's behaviors at the same point in time and in the same setting. With the diagnostic interviews, this is usually accomplished in one of two ways: live observation with simultaneous coding or video or audiotape recordings that are coded following the diagnostic interview. In general, interrater reliability tends to be high and is frequently higher than the other two types of reliability. *Test-retest reliability*, also called within-informant reliability, refers to the amount of consistency, or agreement, found regarding the same measure of behavior over time. In diagnostic interviews, this type of reliability customarily examines the level of symptom or diagnostic agreement *in the same informant* from one assessment time to another. *Multiple-informant agreement*, on the other hand, concerns the consensus of ratings made regarding a particular child *by different individuals* (e.g., parent-child, parent-teacher). Although discrepancies can affect all types of reliability, research findings suggest that more pronounced differences are found with multiple informants. That is, in contrast to generally acceptable levels of interrater and test-retest agreement, multiple-informant reliability studies have consistently yielded poor results. This has been so for child behavior rating scales as well as diagnostic interviews (cf. Achenbach, Dumenci, & Rescorla, 2002; Achenbach, McConaughy, & Howell, 1987; Grills & Ollendick, 2002).

In general, *validity* refers to how well a test measures what it is supposed to measure. Common types of validity examined include content, criterion-related, and construct validity. *Content validity* refers to the degree to which questions in the interview represent all aspects of the disorder and cannot be examined with traditional statistical procedures. Basically, it refers to the extent to which the diagnostic interview taps the various symptoms of a given disorder. *Criterion-related validity*, on the other hand, concerns the degree to which a measure like a diagnostic interview predicts an outcome on another measure such as adjustment, and is typically separated into concurrent (i.e., both measures obtained at the same time) and predictive (i.e., one measure obtained prior to the other) validity categories. Finally, *construct validity* measures how effectively the information obtained from the interview agrees with the theoretical construct being investigated (i.e., depression). Construct validity is typically determined via positive correlations with established measures of the construct under investigations (convergent validity) and nonsignificant associations with measures previously established to measure theoretically diverse constructs (divergent validity).

Establishing the validity of psychiatric diagnoses based on diagnostic interviews is frequently problematic because no gold standard exists with which to compare the findings. Consider the validity of diagnostic interviews and less structured and more traditional clinical interviews. Without a definitive comparison or standard, it cannot be determined whether it is the diagnostic interview or the clinical interview that is more accurate. Indeed, there are myriad factors that could influence either indicator, for example, the need for clinicians to obtain a diagnosis in order to receive insurance reimbursement or the inability of structured diagnostic interviews

to allow for additional inquiry of symptoms (see A. Jensen & Weisz, 2002, for further discussion of factors that could affect each). Moreover, very few studies have been conducted with regard to the validity of diagnostic interviews, and oftentimes estimates of validity are put together based on various methodological designs. For example, examination of content validity cannot be statistically evaluated, and yet the face validity of structured interviews is typically considered certain as items are derived from the *DSM* diagnostic criteria. A bit of circular reasoning is involved with such conclusions.

Criterion-related validity of the concurrent type is also difficult to determine. These findings are troubling as agreement between diagnoses generated by diagnostic interviews and real-world clinicians is often poor (cf. A. Jensen & Weisz, 2002), despite the use of varied methodologies and interviews. For example, A. Jensen and Weisz examined diagnostic interview diagnoses and diagnoses determined at community health clinics and found agreement not significantly better than chance for most individual disorders (e.g., Dysthymic Disorder) and broad categories of those disorders (e.g., mood disorders). Similarly, Lewczyk, Garland, Hurlburt, Gearity, and Hough (2003) found poor agreement with diagnostic interviews and diagnoses generated by a multidisciplinary team that included diverse mental health practitioners, including a child psychiatrist, a child psychologist, and a social worker. Overall, studies have shown poor validity when diagnoses obtained from diagnostic interviews are compared to clinician diagnoses at outpatient clinics (e.g., Ezpeleta, de la Osa, Doménech, Navarro, & Losilla, 1997), inpatient clinics (Pellegrino, Singh, & Carmanico, 1999; Vitello, Malone, Buschle, & Delaney, 1990), and on admission (Weinstein, Stone, Noam, & Grimes, 1989) and discharge (Aronen, Noam, & Weinstein, 1993; Welner, Reich, Herjanic, Jung, & Amado, 1987) from both of these settings.

Predictive validity has been demonstrated in only a few studies. For example, ratings of disruptive behavior disorders on diagnostic interviews and later problematic behaviors (e.g., delinquency, oppositional behavior) have been demonstrated (Jewell, Handwerk, Almquist, & Lucas, 2004). Furthermore, significantly better predictive validity (over a 6-month period) was found for child reports of Conduct Disorder on diagnostic interviews than for children who received diagnoses only from clinicians (and not reported by themselves) and a control group of children for whom no reports of externalizing problems were reported.

With regard to construct validity, diagnoses derived from diagnostic interviews are used to validate information obtained from rating scales, self-report instruments, and behavioral observations. In addition, some research has examined this process in reverse; that is, questionnaires and rating scales have been used as the criterion, and diagnoses obtained from diagnostic interviews have been examined in relation to these measures. For example, empirical research has confirmed the validity of diagnoses formulated using the child and parent versions of the Anxiety Disorders Interview Schedule (see later discussion) by showing that scores on child and parent rating scales converge in expected ways with diagnoses (e.g., Weems, Silverman, Saavedra, Pina, & Lumpkin, 1999; Wood, Piacentini, Bergman, McCracken, & Barrios, 2002). Wood et al. evaluated the validity of Anxiety Disorders Interview Schedule: Child and Parent Versions (ADIS-C/P) diagnoses of Social

Phobia, Separation Anxiety Disorder, Generalized Anxiety Disorder, and Panic Disorder in children and adolescents referred to an outpatient anxiety disorders clinic. In addition to youth and parents being administered their respective versions of the ADIS interviews, they were administered the Multidimensional Anxiety Scale for Children (MASC; March, Parker, Sullivan, Stallings, & Conners, 1997). Strong correspondence was found between ADIS-C/P diagnoses and empirically derived MASC factor scores corresponding to each of these diagnoses and disorders, with the exception of Generalized Anxiety Disorder. For example, MASC social anxiety factor scores, but no other factor scores, were significantly elevated for children meeting *DSM-IV* criteria for Social Phobia on the ADIS-C/P. Despite these positive findings, there is a need for research that evaluates the validity of diagnoses formulated with interview schedules beyond looking at diagnoses' convergence and divergence with questionnaire data (Jarrett, Wolff, & Ollendick, 2007). Ideally, other markers, including what have come to be known as endophenotypes, could be used for this purpose.

In addition to establishing the correct diagnosis, diagnostic interviews are also frequently used for case conceptualization and treatment planning purposes (Silverman & Ollendick, in press). Such purposes refer to the clinical utility of the instrument being used. Diagnoses themselves yield powerful information about treatment targets and potential treatment plans. For example, the anxiety and worry of Generalized Anxiety Disorder (GAD) is associated with one or more of the following six *DSM-IV* symptoms: being restless, being easily fatigued, having difficulty concentrating, being irritable, having muscle tension, and having difficulty falling or staying asleep. Any one or more of these problems (i.e., target symptoms) could lead to highly specific treatment plans designed to address them. This approach to prescriptive interventions is illustrated in the early work of Eisen and Silverman (1993, 1998) and is captured well in a recent book on modularized treatment by Chorpita (2007). In an early study, Eisen and Silverman (1993) showed that cognitive-behavior therapy was most effective for children with overanxious disorder (the *DSM-III-R* precursor to GAD) when they were matched with specific symptoms. For example, children with primary symptoms of worry and obsessions responded more favorably to cognitive therapy, whereas children with primary symptoms of somatic complaints responded more favorably to relaxation training aimed at dealing with physiological and somatic complaints. This early study was replicated in a second study by Eisen and Silverman (1998), in which these authors again showed that although both treatments were generally effective, only the prescriptive treatments produced sufficient improvements for the treated children to meet exacting positive end-state functioning criteria. Similar effects of matching were shown by Ollendick, Hagopian, and Huntzinger (1991) with separation anxious children and by Ollendick (1995) with adolescents with Panic Disorder and Agoraphobia. Thus, in these studies, treatments were shown to be maximally effective when they were closely matched with the symptom presentation of the individual child or adolescent. That is, specific diagnoses when supplemented with symptom profiles led to superior outcomes.

Likewise, diagnostic interviews have been used for treatment evaluation purposes. For example, in treatment outcome studies, 100% of youth necessarily meet

diagnostic criteria at pretreatment. In evaluating treatment outcome, of interest is the diagnostic recovery rate at posttreatment and follow-up; most studies report 60% to 80% of participants as recovered and no longer meeting diagnostic criteria at posttreatment and follow-up. As such, diagnostic interviews are extremely useful and provide one of the most rigorous ways of determining treatment outcomes and providing an evidence base for various treatment interventions (Ollendick, King, & Chorpita, 2006).

Finally, before proceeding to a review of specific diagnostic interviews, brief mention should be made of what has come to be called the treatment utility (a form of clinical utility) of measures such as diagnostic interviews. As we have noted, if one wishes to use the treatment that possesses the most research evidence for a given disorder, it is important to first have *confidence* (i.e., have reliable and valid information) that the youth or groups of youths with whom one is working are in fact accurately diagnosed with the disorder under study, rather than some other clinical disorder. Second, it is important to have confidence about the *specific* type(s) of disorder that the youth is diagnosed with. For example, with the anxiety disorders, it is crucial to know the exact disorder so that appropriate exposure tasks can be assigned in and out of the treatment sessions (e.g., expose the child to social evaluative situations if the child has a Social Phobia; to separation situations if the child has Separation Anxiety Disorder). However, as Nelson-Gray (2003) has pointed out, empirical verification of treatment utility is yet to be demonstrated. That is, it has not been empirically tested whether different outcomes result for children who have received diagnoses assigned with a diagnostic interview schedule and are given an intervention versus a group of children who have *not* been given a diagnostic interview schedule and receive the same treatment. Although such may be so, it seems to us that starting an intervention without this information would be premature.

As is evident, a host of issues surrounding diagnostic interviews exist, including their reliability, validity, and clinical utility. Although many questions remain about the psychometric properties of these interviews, much progress has been made. A number of reliable, valid, and clinically useful interviews have been developed, a topic to which we next turn our attention.

DIAGNOSTIC INTERVIEWS

Generally speaking, diagnostic interviews are considered to be a formal and full examination of a particular problem area or areas. There are several expectations imposed on these interviews: (a) They are psychometrically sound measurements, (b) they have specific rules regarding the content and order of the questions and topics being addressed (e.g., asking whether depressed mood is present prior to asking the possible effects of the depressed mood) as well as the manner of recording responses, and (c) some degree of structure is imposed with regard to final diagnostic decisions (Weiss, 1993). The degree of correspondence with these specifications appears to be directly related to the amount of structure imposed by the interview. Various descriptions have been applied to distinguish among diagnostic interviews, with the most common distinction pertaining to the

structure (semi- or highly structured) of the interview. Specifically, distinctions have been made among unstructured, semi-structured, and highly structured interviews.

Unstructured interviews give complete discretion to the interviewer as to what, when, and how questions will be asked and recorded. Although these interviews typically contain questions pertaining to the child's history (e.g., medical, psychological, developmental, academic, social), presenting problems (e.g., antecedents, behaviors, consequences), environmental contexts (e.g., home and family, school), and treatments (i.e., past and present), there are no set procedures used across clinicians. Semi-structured interviews also tend to allow leeway for the interviewer regarding the order in which questions are asked, the manner in which the questions are phrased, and the way responses are recorded. Emphasis is placed on obtaining consistent and reliable information; thus extensive training is generally required for administration of these interviews to ensure that clinical discretion will be applied judiciously. Highly structured interviews, on the other hand, are more restrictive in the amount of freedom allotted to the interviewer. With these interviews, it is generally expected that examiners ask all questions in the same manner and order, as well as record all responses in a prespecified manner. Clinical judgment is reduced with highly structured interviews, such that specific and/or extensive training is not required for their administration. In fact, highly structured interviews are commonly administered by laypersons (e.g., individuals without a formal degree in psychology, psychiatry, or social work), and several have been converted to computer-based formats (e.g., Diagnostic Interview Schedule for Children [DISC], Diagnostic Interview for Children and Adolescents [DICA]; see later discussion). Although highly structured formats allow for more confidence in the exactness of the interview's administration and perhaps more reliable findings, the rigidity of the interview may also make it seem impersonal, hinder the establishment of rapport, and interfere with reliability and validity by not providing the interviewee the opportunity to report all difficulties or to explore them in full depth. As a result, the use of highly structured interviews may result in unanswered questions for the clinician that might have been addressed in a less structured format (e.g., knowing a child feels sad most days does not answer questions of potential precipitants, etiological factors, responses by others in the child's environment).

In addition to differentiations based on interview structure, a variety of descriptors have been used in the literature (i.e., respondent-based versus interviewer-based, glossary-driven versus clinician-driven). Nevertheless, each of these distinctions concerns the degree of latitude allotted the interviewer, as well as the type of research with which the interview was designed to be used. Initially, highly structured interviews were used for epidemiological research, and semi-structured interviews were used in clinical research and randomized controlled trials. However, currently these distinctions seem less meaningful, with all research teams but that of the DISC considering their interviews semi-structured and some creating separate epidemiological and clinical interview versions (e.g., Schedule for Affective Disorders and Schizophrenia for School-Age Children [K-SADS]; see later discussion).

For the most part, the format of companion parent and child interviews is similar; each contains (a) an introductory section designed to help build rapport with the informant (e.g., demographics, school, free time activities) and elicit initial information regarding presenting problems; (b) symptom-specific modules (e.g., mood disorders) that usually begin with screener questions and determine whether to skip or continue with follow-up questions (i.e., regarding frequency, intensity, duration, and interference); and (c) the determination of diagnostic presence or absence based on algorithms or clinical judgments. Several of the most commonly utilized diagnostic interviews are discussed next, with general psychometric information provided. Overall, acceptable test-retest and interrater reliability coefficients have been reported for each of these interviews; however, findings for multiple-informant reliability have been more varied (cf. Grills & Ollendick, 2002), with little data published on the validity of these instruments.

The Anxiety Disorders Interview Schedule for Children for DSM-IV: Child and Parent Versions

The ADIS-CP for *DSM-IV* (Silverman & Albano, 1996; Silverman, Saavedra, & Pina, 2001) is a downward extension of the adult Anxiety Disorders Interview Schedule for *DSM-IV* (Brown, Di Nardo, & Barlow, 1994) and is most detailed in its coverage of the anxiety disorders, relative to the other interview schedules. The ADIS for *DSM-IV:* C/P and its previous *DSM-III* and *DSM-III-R* versions (Silverman, 1991) have been used most frequently in the youth anxiety disorders research literature, including randomized clinical trials. The ADIS for *DSM-IV:* C/P contains a series of modules that cover all the anxiety disorders described in *DSM-IV,* as well as sections that cover the most prevalent disorders of childhood and adolescence (e.g., Attention-Deficit/Hyperactivity Disorder [ADHD], depressive disorders) and screening questions for most other disorders (e.g., eating disorders, enuresis). Additional questions are included that allow interviewers to obtain information about the history of the problem as well as situational and youth cognitive factors influencing anxiety. In addition, the ADIS for *DSM-IV:* C/P contains clinician severity rating scales that assess for degree of impairment or interference in youth functioning associated with the specific anxiety disorder endorsed by the youth and parent, respectively. Specifically, based on the information obtained from the child and parent versions of the interview, interviewers assign the degree of distress and interference associated with each disorder (0 = None to 8 = Very severely disturbing/impairing) with respect to the youth's peer relationships, school work, family life, and personal distress. Each module in the interview also contains questions that allow interviewers to assign 0 to 8 ratings on the youth's fear and avoidance of diverse situations relevant to a particular disorder (e.g., Social Phobia). Similar to the adult ADIS, clinician severity ratings of 4 (definitely disturbing/impairing) or higher are viewed as clinical diagnoses, and those less than 4 are viewed as subclinical or subthreshold.

Acceptable test-retest reliability (7 to 14 days) estimates have been reported for the ADIS for *DSM-IV* Child (aged 7 to 16; $\kappa = .61$ to .80), Parent ($\kappa = .65$ to 1.00), and combined ($\kappa = .62$ to 1.00) diagnoses (Silverman et al., 2001). Likewise,

interrater agreement for earlier versions of the ADIS have generally fallen within the acceptable to excellent range for videotape ($\kappa = .45$ to .82; Rapee, Barrett, Dadds, & Evans, 1994) and live observer (.35 to 1.00; Silverman & Nelles, 1988) paradigms. In contrast, findings for multiple-informant reliability have been poor for both the most recent and earlier versions of the ADIS (Grills & Ollendick, 2003; Rapee et al., 1994).

Schedule for Affective Disorders and Schizophrenia for School-Age Children

There are currently several different forms of the K-SADS, all of which include corresponding parent and child (ages 6 to 18) versions. The three most current (*DSM-IV* applicable) and researched are Present state (P-IVR; Ambrosini & Dixon, 1996); Epidemiological (E-Version 5; Orvaschel, 1995); and Present/Lifetime (P/L; Kaufman et al., 1997). Each version of the K-SADS has undergone extensive revisions, which have been detailed elsewhere (see Ambrosini, 2000; Kaufman et al., 1997). Although the primary focus is affective disorders, these three K-SADS versions also assess for several additional disorders (see Table 15.1). The primary difference among these versions is the diagnostic time frame; that is, the K-SADS-P-IVR assesses disorders from the current episode (within the week preceding the interview) as well as from the past 12 months, whereas the K-SADS-E and P/L diagnose current and lifetime disorders (with a focus on the most severe past episode). In all cases, the interviews are conducted with the parent or primary guardian first, and the same interviewer then administers the corresponding version to the child. Information is gathered regarding the child's current symptoms, as well as from when the informant considered the symptoms worst during the specified time frame (i.e., past 12 months or lifetime). Discrepancies in reports of observable behaviors are addressed in a joint parent/child interview prior to the clinician's final ratings. Finally, based on all of the information obtained (e.g., parent, child, schools, charts), the clinician determines summary severity scores for each symptom and makes a diagnosis using accompanying criteria checklists. Research diagnostic criteria (RDC) are used whenever possible, with *DSM* diagnostic criteria used for those disorders not covered by RDC.

Using varied approaches (i.e., videotape, audiotape, live or joint observation), acceptable interrater reliability has been reported for the most recent (K-SADS-P IV, $\kappa = .80$ to 1.00) and earlier (K-SADS-P IIIR, $\kappa = .46$ to .91; K-SADS-E, $\kappa = .51$ to .77; Ambrosini, 2000; Ambrosini, Metz, Prabucki, & Lee, 1989) K-SADS versions. Likewise, test-retest studies have generally revealed adequate reliability coefficients for the K-SADS (P-IIIR, $\kappa = .46$ to .91; E, $\kappa = 1.00$ major depression past episode; P/L, .55 to 1.00). Despite these, agreement between parent and child reports on earlier versions of the KSADS, administered by the same interviewer, has varied greatly ($\kappa = –08$ to .96, Chambers et al., 1985: $\kappa = -.07$ to .52, Angold et al., 1987; $\kappa = .05$ to .66, Hodges, McKnew, Burbach, & Roebuck, 1987), and information on the most recent K-SADS is not currently available. In addition to these reliability estimates, predictive and construct validity have been demonstrated for the K-SADS interviews (see Ambrosini, 2000; Ambrosini et al., 1989).

Diagnostic Interview for Children and Adolescents

The DICA (Reich, 1998) is a semi-structured instrument, with parent and companion child (ages 6 to 12) and adolescent (ages 13 to 18) versions. The most recent revision of the DICA corresponds with the latest revisions to the *DSM* (*III-R, IV*) and is relatively unique in that it examines lifetime diagnoses (cf. Reich, 2000; Rourke & Reich, 2004). In addition to diagnostic sections (see Table 15.1), the DICA includes questions about psychosocial stressors, risk and protective factors, and perinatal, delivery, and early child development (parent interview only). Following completion of the interview, the interviewer pursues problematic areas which are then resolved by consultation with the DICA manual and/or discussion with more experienced clinicians or primary investigators. There are also more structured versions of the DICA, specifically a computerized version, which can be administered by a trained interviewer or completed by the informant alone.

Table 15.1 Summary of Diagnoses Possible with the Structured Interviews Reviewed

Diagnoses	K-SADS-P IV, -E, -P/L	ADIS-IV	DISC-IV	DICA
Major Depressive Disorder	Y	Y	Y	Y
Dysthymic Disorder	Y	Y	Y	Y
Bipolar Disorder	Y	N	N	N
Mania	Y	N	Y	Y
Hypomania	Y	N	Y	Y
Cyclothymia	Y	N	N	N
Anorexia Nervosa	Y	Y (screens for)	Y	Y
Bulimia Nervosa	Y	Y (screens for)	Y	Y
Enuresis	Y (E, P/L, only)	Y (parent version)	Y	N
Encopresis	Y (E, P/L, only)	N	Y	N
Tic disorders	Y (E, P/L, only)	N	Y	Y
Schizophrenia	Y	N	Y	N
Schizoaffective Disorder	Y	N	N	N
Generalized Anxiety Disorder	Y	Y	Y	Y
Separation Anxiety Disorder	Y	Y	Y	Y
Obsessive-Compulsive Disorder	Y	Y	Y	Y
Simple Phobia	Y	Y	Y	Y
Social Phobia	Y	Y	Y	Y
Agoraphobia	Y	Y	Y	N
Panic Disorder	Y	Y	Y	Y
Posttraumatic Stress Disorder	Y	Y	Y	Y
Oppositional Defiant Disorder	Y	Y(parent version)	Y	Y
Attention-Deficit/ Hyperactivity Disorder	Y	Y	Y	Y
Conduct Disorder	Y	Y(parent version)	Y	Y
Substance disorders	Y	Y	Y	Y
Personality disorders	Y	N	N	N

Generally acceptable 1-week test-retest reliability for the current DICA version has been reported for the adolescent ($\kappa = .59$ to .92) and child ($\kappa = .32$ to .65) versions. Examinations of earlier DICA versions revealed poor parent-child agreement for most diagnoses ($\kappa = .00$ to .58, Reich, Herjanic, Welner, & Gandhy, 1982; $\kappa =$ .00 to .35, Brunshaw & Szatmari, 1988; $\kappa = .11$ to .42, Sylvester, Hyde, & Reichler, 1987; $\kappa = -.05$ to .68, Boyle et al., 1993). In addition, at the symptom level, Herjanic and Reich (1982) found kappa coefficients ranging from .00 to .87; however, coefficients for only 16 of the 185 questions examined obtained acceptable status. A notable exception was found by Welner and colleagues (1987; $\kappa = .49$ to.80) through inclusion of probable cases of parent-child agreement (i.e., one symptom lacking or duration uncertain). Test-retest and multiple-informant agreement for the current DICA were not found.

The Diagnostic Interview Schedule for Children

The only interview considered highly structured, the DISC was originally designed for epidemiological surveys of children that were conducted by laypersons. The most recent revision (DISC-IV; Shaffer, Fisher, Lucas, Dulcan, & Schwab-Stone, 2000) contains criteria for disorders that can be diagnosed in children and adolescents using either the *DSM-IV* or *ICD-10* (see Table 15.1). Interviewers are trained to read questions exactly as written, with the majority of responses requested in a yes/no format. The DISC-IV inquires first about symptoms from the past 12 months. If the informant responds positively to this initial inquiry, information regarding symptoms from the past 4 weeks is obtained. The DISC is scored using a computer program (SAS) and algorithms that match *DSM-IV* (or *ICD-10*) criteria. In addition to the paper version, the DISC has computerized (C-DISC-4.0), Spanish, Present State, Teacher, Voice, Young Adult, and Quick versions (cf. Shaffer et al., 2000) and is considered the most highly researched and utilized of the parent/child companion interviews (A. Jensen & Weisz, 2002).

The DISC-IV diagnostic scales have evidenced generally acceptable test-retest reliability for parent ($\kappa = .43$ to .96) and combined parent/child ($\kappa = .48$ to .86) versions, with slightly less reliable findings for the child version ($\kappa = .25$ to .92, Fisher et al., 1997). Similar test-retest reliability rates were found for prior DISC versions in community and clinical samples (P. Jensen et al., 1995; Schwab-Stone et al., 1996). In addition, excellent interrater reliability has been reported for earlier versions of the DISC (Shaffer, Schwab-Stone, Fisher, & Cohen, 1993). Recent examinations of parent-child agreement have found poor reliability with the DISC-IV ($\kappa = -.08$ to .25, Crowley, Mikulich, Ehlers, Whitmore, & MacDonald, 2001; P. Jensen et al., 1999).

To summarize, several well-researched semi- or highly structured interviews exist for researchers and clinicians. As shown in Table 15.1, the majority of these interviews cover basic *DSM-IV* diagnoses for childhood disorders, with some focused more on specific areas than others (e.g., the ADIS and anxiety disorders versus the K-SADS and depressive disorders). In addition to procedural differences (e.g., level of structure, computer administration capability, same versus different interviewer

for parent and child components) among these interviews, several have also been adapted for use in multiple languages. For example, in addition to the standard English versions, there are currently Chinese (Ho et al., 2005), French (Breton, Bergeron, Valla, Berthiaume, & St. Georges, 1998), and Zhosa versions of the DISC, as well as a Greek version of the K-SADS (Kolaitis, Korpa, Kolvin, & Tsiantis, 2003). Therefore, although specific cases where the primary focus of study (e.g., epidemiological studies) or presenting problem (e.g., a disorder not assessed by all interviews, such as Bipolar Disorder) guides the choice of diagnostic interview, given the similarities among interviews, most often the clinician's or researcher's discretion or preference guides this decision. In fact, Hodges (1994) suggested that there is not one best interview, but that researchers and clinicians should determine which interview to use based on the sample and focus of their endeavor. Finally, it is important to note that the previously described interviews do not represent an exhaustive list of those available. Additional commonly utilized interviews are the Child Assessment Schedule (Hodges, Kline, Stern, Cytryn, & McKnew, 1982) and the Child and Adolescent Psychiatric Assessment (Angold, Prendergast, Cox, & Harrington, 1995), among others.

LINGERING ISSUES INVOLVED WITH DIAGNOSTIC INTERVIEWS

One of the most consistently raised difficulties involved in using parent-child diagnostic interviews concerns the discordant reports received. That is, symptom and diagnostic agreement within and across informants is generally poor—a finding that has led several researchers to examine potential factors that may influence multiple informants' reports and thus contribute understanding to the discrepancies found (cf. Grills & Ollendick, 2003). In general, these variables can be described in terms of whether they pertain to the interviewee, the interviewer, or the interview itself.

Beginning at the *interviewer* level, establishment of rapport is an important aspect of any clinical relationship. Although diagnostic interviews attempt to incorporate rapport-building strategies (e.g., beginning with general information), the lengthy interview format likely impedes rapport, in the traditional sense, from being fully developed (LaGreca & Stone, 1992; Verhulst et al., 1987). Individuals who have not established rapport with the interviewer may not feel comfortable disclosing diagnostic information, requesting clarification, or stopping the interview (Breton et al., 1995), thus potentially affecting the reliability and validity of the reports obtained. Unfortunately, this is likely often the case, as clinicians and researchers tend to use diagnostic interviews early in the assessment process to probe for areas of concern. In addition, despite requiring specific training in their administration, individual and across-site differences and errors likely remain. Furthermore, differences in clinician experience levels may affect interviews, as one with more experience may complete the interview in a more timely, fluid, and comprehensive manner. This latter issue would be particularly relevant for semi-structured diagnostic interviews (e.g., ADIS), as there is more clinical probing and leeway permitted with these instruments. An additional source of confounding can occur when the same interviewer administers both interviews. For instance, information

from the first interview may alter the interviewer's opinions and subsequently the style (nonverbal cues) and emphasis given to probing certain domains as well as the recording of responses in the second interview. With multiple informants, joint meetings may be held to resolve discrepant reports; however, the clinician cannot be sure that the final consensus accurately represents both informants' opinions or the concession of one of them (Angold et al., 1987; Sylvester et al., 1987).

Interview methodological problems are another area of concern for researchers. For example, it has been suggested that the sequence of disorder presentation in the interviews may affect informants' reports. Sylvester et al. (1987) noted that questions regarding externalizing disorders are asked first with the DICA and that this may influence parents' reports by focusing them on these outward behaviors. Thus, later questions about internalizing symptoms may be overshadowed by this earlier focus (Sylvester et al., 1987). This notion was recently supported by findings of order effects in the total number of symptoms endorsed, impairment ratings, and diagnostic criteria met for both parent and child reports on the DISC (P. Jensen et al., 1999). In contrast, the ADIS requests information on all anxiety-related symptoms early in the interview, with externalizing disorders probed later. Comparison of these measures and order effects may prove another area important for future studies. Similarly, parent-child concordance may be affected by differences in the informants' emotional state at the time of the interview. To illustrate, parents may overemphasize behavioral problems if the child was behaving especially poorly earlier that day or in the waiting room. They may also overemphasize behavioral problems if they themselves are having a difficult day or are under stress of one type or another. For children, poor reliability may be a reflection of their desire to quickly complete an interview (e.g., by saying no to all the probes) or impatience or boredom with the interview process (Edelbrock et al., 1985; P. Jensen et al., 1995; Reich, 2000). Potential differences concerning the degree of structure (semi- versus highly) imposed by interviews has also been reported. Whereas semi-structured interviews may lead to improved reliability over highly structured interviews (i.e., DISC) by allowing for more clinical probing (Verhulst et al., 107), highly structured interviews may limit confounding characteristics (e.g., interviewer experience, differences in probing across administrations). Thus, it is not surprising that studies conducted for the various interviews have revealed similar findings regardless of the degree of structure associated with the interview (Grills & Ollendick, 2003). It is important to note that these clinical interviews must be considered in the context of the diagnostic system on which they are based. If diagnostic criteria are not presented in a manner that allows for their adequate assessment, this will be reflected in the interviews as well. Furthermore, if the boundaries between normal and pathological child disorders are unclear, assessment strategies will also likely produce blurred and discrepant results. Thus, reliability and validity problems on multiple interviews may speak to the need for further alterations or amendments to the diagnostic system itself.

The most attention has clearly been devoted to *interviewee* (parent and child) characteristics proposed to affect agreement, including those particular to children and/or parents. For example, it is often presumed that younger children will evidence less agreement with their parents due to factors (e.g., verbal,

cognitive, memory, and attentional) related to their ability to understand and describe their feelings and behaviors (Breton et al., 1995; Edelbrock et al., 1985; Rapee et al., 1994; Schwab-Stone et al., 1994). Several researchers have indicated that structured interviews may require information that is too complex or beyond the cognitive capabilities of young children (Breton et al., 1995; Brunshaw & Szatmari, 1988; Edelbrock et al., 1985; Ezpeleta et al., 1997; Herjanic & Reich, 1982; Piacentini et al., 1993; Schwab-Stone et al., 1994; Valla, Bergeron, Berube, Gaudet, & St Georges, 1994; Welner et al., 1987; Young, O'Brien, Gutterman, & Cohen, 1987). Indeed, the majority of research studies have found improved ratings within (i.e., test-retest) and across (e.g., parent/child, clinician/child) informants for older children regardless of the interview used (Edelbrock, Costello, Dulcan, Conover, & Kalas, 1986; Edelbrock et al., 1985; Ezpeleta et al., 1997; Grills & Ollendick, 2002; P. Jensen et al., 1999; Rapee et al., 1994; Reich et al., 1982; Silverman & Eisen, 1992; Verhulst et al., 107). It appears that both the wording (e.g., length, vague concepts such as "most" and "fair," number of concepts included in the question) and types (e.g., yes/no versus open-ended, duration, age of onset, clustering, frequency, concrete information versus comparative judgments) of questions included in these interviews can be problematic. Concerns like these have been supported by studies revealing low overall understanding of concepts (16% to 56%) and time dimensions (24% to 30%) of DISC questions when children (ages 9 to 11) were asked to explain their answers (Breton et al., 1995). Likewise, Perez, Ezpeleta, Domenech, and de la Osa (1998) reported that the most reliable questions on the DICA-R were those containing no time concepts, clustering of symptoms, or impairment ratings and no peer comparisons or subjective judgments, externalizing (versus internalizing) content, and shorter duration (for younger children, but not adolescents). Although not all researchers have found support for age differences on the reliability of interviews (Angold et al., 1987; Boyle et al., 1993; Briggs-Gowan, Carter, & Schwab-Stone, 1996; Earls, Smith, Reich, & Jung, 1988; Perez et al., 1998; Thompson, Merritt, Keith, Murphy, & Johndrow, 1993), a higher minimal age for structured interviewing has been proposed by some (e.g., age 10+ for the DISC; Edelbrock et al., 1985; Schwab-Stone et al., 1994). Importantly, studies in this area also demonstrate aspects of potential revision for child structured interviews. For example, it may be helpful for interviewers to use multiple examples, visual aids, and explanations to ensure children's comprehension.

Interestingly, the child's age may also influence parent reports, as older children typically spend more time outside the home (e.g., with friends, in extracurricular activities), potentially decreasing parents' knowledge of their child's behaviors (Edelbrock et al., 1985). Although this notion has not been well-studied, it certainly remains an important consideration and further supports the need for both adolescent and parent reports.

Motives may also influence reliability. For example, symptoms may be overreported to gain services for a child that the parent feels are necessary to obtain. In addition, parent-child concordance may be disrupted if children intentionally hide information from their parents (e.g., drug use, truancy; Edelbrock et al., 1985), perhaps to avoid negative consequences (e.g., punishment, forced treatment)

resulting from disclosure. Alternatively, children may attempt to conceal emotional problems while in the presence of others, such as their peers or the clinician, to portray a positive image of themselves (DiBartolo, Albano, Barlow, & Heimberg, 1998; Grills & Ollendick, 2003; P. Jensen, Traylor, Xenakis, & Davis, 1988; Silverman & Rabian, 1995; Stavrakaki, Vargo, Roberts, & Boodoosingh, 1987). Similarly, parents may underreport symptoms in an attempt to conceal the child's behavioral problems (Crowley et al., 2001; Victor, Halverson, & Wampler, 1988), perhaps to prevent stigma or conceal family problems (e.g., marital conflicts). These social desirability concerns could then lead to the appearance of overreporting on the part of another informant. Social desirability may also influence concordance within or across informants if disagreeing with others or one's own earlier report is perceived as being viewed unfavorably.

Another commonly discussed area concerns the type of disorder on which the informants are reporting. That is, externalizing (e.g., hyperactivity, aggression), as compared to internalizing (e.g., anxiety, depression), symptoms are generally considered easier to observe and to report on by others, resulting in increased diagnostic agreement. Likewise, the severity of the disorder may influence reliability, perhaps through relation with more open dialogue about problems, more pronounced or observable symptoms, or less reliable answers (Cohen, O'Connor, Lewis, Velez, & Malachowski, 1987; Perez et al., 1998). Whereas findings with the various diagnostic interviews support the former proposal, such that increased reliability is often found for externalizing behaviors (Bidault et al., 1995; Chambers et al., 1985; Costello & Edelbrock, 1985; Edelbrock et al., 1986; Ezpeleta et al., 1997; Herjanic & Reich, 1982; Hodges, Gordon, & Lennon, 1990; P. Jensen et al., 1988; Kolko & Kazdin, 1993; Orvaschel, Weissman, Padian, & Lowe, 1981; Silverman & Eisen, 1992), the influence of disorder severity does not appear to be supported by multiple-informant reliability studies (e.g., Angold et al., 1987; Grills & Ollendick, 2003). Likewise, there remains debate as to whether parental psychopathology contributes to decreased interview reliability, perhaps due to greater sensitivity to their children's symptoms, misinterpreting of their child's behaviors, or projection of their own symptoms onto the child (Angold et al., 1987; Breslau, Davis, & Prabucki, 1988; Briggs-Gowan et al., 1996; Fergusson, Lynskey, & Horwood, 1993; P. Jensen et al., 1988; Kashani, Orvaschel, Burk, & Reid, 1985; Kolko & Kazdin, 1993; Miles et al., 1998; Moretti et al., 1985; Rapee et al., 1994). Poor family communication (Bidault-Russell et al., 1995) and family conflict or stress may also influence reliability. For example, children and parents from families characterized by high conflict have been found to evidence poorer agreement on the ADIS (Grills & Ollendick, 2003).

Although gender is also commonly examined, Klein (1991) has argued that there are no theoretical grounds on which to expect gender differences to occur. Indeed, findings for gender differences have been exceedingly mixed, and review of studies reporting them reveals generally few and small differences (cf. Grills & Ollendick, 2003). Furthermore, there remain myriad potential factors that have not yet been well examined for their influence on the reliability and validity of reports on clinical interviews (e.g., ethnicity, socioeconomic status). Although determining the role of these variables awaits future research, it seems likely that similar findings to

those already described will result. That is, despite some trends regarding particular interviewee characteristics, findings have not been particularly robust for any one area.

In sum, various interviewer, interviewee, and interview characteristics may influence the reliability and validity of clinical interviews. Future research should attempt to clarify the role of these characteristics to limit additional variance that may be introduced through them. The most robust findings appear to be related to children's comprehension of the interview itself.

Several procedures have been proposed to manage discordant findings with interviews: (a) basing diagnostic decisions on just one of the reports (Silverman et al., 2001) or only when both reports agree, (b) following specific guidelines for the consideration of each report (e.g., rules of evidence, P. Jensen et al., 1999; Shaffer et al., 2000), and (c) applying rules for how reports should be combined (e.g., ADIS Composite Diagnoses; Silverman & Albano, 1996; "or" rules; Bird, Gould, & Staghezza, 1992). Applying rules for combining reports appears most advantageous as this allows discrepancies to be interpreted as informative, not problematic (Boyle et al., 1996; Schwab-Stone et al., 1996). That is, parent and child reports should not be viewed as interchangeable, but complementary.

In light of these findings, the general consensus in the youth assessment area (Silverman & Ollendick, 2005, in press) is that parents and children should both be involved in the assessment of youth symptoms and diagnoses. One is not necessarily more right or wrong than the other. By carefully considering each source's information, there is increased likelihood that no child or adolescent is denied services (Comer & Kendall, 2004; Foley et al., 2005). De Los Reyes and Kazdin (2005), in a comprehensive review of the informant discrepancy literature in clinical child and adolescent psychology, further proposed that in addition to obtaining information of youths' problems from multiple informants, efforts should be made to collect information from these informants about their perceptions of why the youth is exhibiting these problems, as well as their perceptions of the desired treatment of the youth for these problems. De Los Reyes and Kazdin further discussed how gathering this type of information can serve as the basis for an assessment model that can guide research and clinical practice to help explain informant discrepancies in clinical child and adolescent research.

SUMMARY

A host of issues surrounding diagnostic interviews exist, including their reliability, validity, and clinical utility. Although many questions remain about the psychometric properties of these interviews, much progress has been made and a number of reliable, valid, and clinically useful interviews have been developed and are in wide use today. These interviews vary along a number of lines, including the amount and extent of structure embedded in them (unstructured, semi-structured, and highly structured). The semi-structured and the highly structured interviews enjoy the best psychometric properties. It is also evident that a number of complications can arise when using diagnostic interviews during the assessment process. For example, it is a common practice in the assessment of children to include multiple informants

since it is believed that adults alone cannot provide all of the information that is needed concerning the child's behavior. However, inclusion of multiple informants also creates a dilemma, as disagreements commonly occur. A number of potential variables related to the interviewer, the interviewee, and the interview itself have been considered in terms of their influence on the reliability, validity, and clinical utility of these instruments. However, few, if any, clear or consistent patterns have emerged. It seems likely that there is a complex interaction among the various factors that cannot be ignored and that will require additional research. Nonetheless, diagnostic interviews represent one of the most reliable, valid, and clinically useful instruments for diagnosis, treatment planning, and treatment evaluation. They are an indispensable part of our clinical armamentarium.

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CHAPTER 16

Intellectual Assessment

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The practice of estimating intellectual abilities dates back to the beginnings of scientific thought. Greek philosophers and modern-day psychologists alike have attempted to understand the nature of human intelligence and to devise methods by which to measure intellectual capabilities. These intellectual assessments often play important roles in diagnoses of learning disabilities, academic placement, occupational selection, and medical treatment plans. Additionally, intellectual assessments routinely have a significant role in the conceptualization of neurocognitive sequelae following traumatic injuries. Thus, the methods by which we estimate intellectual functioning, and the scientific integrity of these assessment tools, are critical to our practice as psychologists.

The goals of this chapter are to provide an overview of the history of intellectual assessment and common definitions of intelligence. We additionally review traditional intellectual assessment tools, as well as more recent additions to the intellectual assessment arsenal. We discuss the impact of intellectual assessment in the context of psychoeducational assessments and review legislation related to intellectual assessment in the context of educational settings. Finally, we provide two clinical case examples to reiterate the role of cognitive assessment in clinical practice.

HISTORY OF INTELLECTUAL ASSESSMENT

We begin our discussion of intellectual assessment with a review of historical perspectives. This vantage point is critical because little can be understood devoid of knowledge of its history.

Speculations into the nature of human intelligence can be traced back to the Greek philosopher Plato (ca. 428/427–348/347 B.C.E.) who referred to *reason* as the highest functioning part of the soul. Since that time, philosophers and scientists have engaged in dialogue pertaining to the nature of cognitive processing and intelligent behavior. Beginning in the sixteenth century, attempts were made to differentiate mental deficiencies from mental illness. Prominent philosophers and psychologists involved in this work include Jean Esquirol, Sir Francis Galton, Alfred Binet, and James Cattell.

Jean Etienne Esquirol

Jean Esquirol (1772–1840) was among the first scientists to differentiate mental deficiency, or "idiocy," from mental illness (Sattler, 2001). He regarded mental deficiency as a condition in which the intellectual facilities are not fully developed whereas mental illness involved the loss of functioning once possessed by the afflicted individual (Zusne, 1957). To differentiate these groups, Esquirol developed the first rudimentary test of cognitive functioning, in which he used a language criterion to determine mental retardation.

Sir Francis Galton

Sir Francis Galton (1822–1911) is commonly referred to as the founder of the testing movement (Kail & Pellegrino, 1985; Sattler, 2001). A diversified scholar, Galton's published work significantly contributed to the fields of geography, biology, psychology, and criminology. However, he is most highly recognized for his contributions to the field of behavioral genetics (Clayes, 2001). Greatly influenced by his cousin, Charles Darwin, Galton focused much of his work on the heritability of mental functioning. In his published work, Galton theorized that superior mental functioning was passed genetically from parent to offspring and used the bell-shape curve to describe differences in intellectual functioning (Sattler, 2001; Seligman, 2002). Additionally, Galton developed the statistical concepts of *regression to the mean* and *correlation,* which allowed for the empirical study of heritability of traits over time and among family members. His significant contributions to the fields of behavioral genetics and psychology paved the way for future statisticians and psychometricians, and his statistical concepts are still used as the gold standard in test construction and validation (Bynum, 2002).

Alfred Binet

While Esquirol developed the first crude measure of cognitive functioning, Alfred Binet (1857–1911) is credited with developing the first practical test of intelligence (Kail & Pellegrino, 1985; Sattler, 2001). In 1904, Binet was commissioned by the French government to develop an instrument that identified mentally retarded children in need of alternative education. In collaboration with his student, Theodore Simon, Binet created an intelligence test that consisted of 30 subtests measuring a broad range of cognitive abilities, including language skills, abstract reasoning, and memory functioning (Siegler, 1992). This test, commonly referred to as the 1905 Scale, compared the testing subject's performance to a chronological cohort deemed by their teachers to be of average intellectual functioning. The scale was revised in 1908 and adapted to English by Lewis Terman of Stanford University in 1916. Terman's adaptation, referred to as the Stanford-Binet, included culturally appropriate items and reported the child's level of performance in terms of an intelligence quotient (IQ) consisting of the child's mental age divided by chronological age multiplied by 100.

Although the Binet-Simon scale was created as a diagnostic instrument, Binet was equally interested in developing a theory of intelligence (Kail & Pellegrino,

1985). In fact, his definition of intelligence as consisting of complex cognitive processes including abstract reasoning, memory and comprehension greatly influenced the design of the Binet-Simon testing instrument. However, the practical demands of creating and maintaining the Binet-Simon scales overshadowed his pursuits in intelligence theory, thus marking a division in psychometrics between those scientists most interested in theory and those most interested in diagnostic measurement. The theory versus measurement discourse continues among psychologists.

James McKeen Cattell

James McKeen Cattell (1860–1944) was the first American psychologist to emphasize the study of individual differences and cognitive functioning (Kail & Pellegrino, 1985; Sattler, 2001). Cattell is often remembered for his work on theories of personality and bringing an empirical and scientific perspective to theories of personality. Rather than following in the footsteps of his predecessors and focusing on the specific *outcome* of his research, Cattell was more concerned with the *process* of research and shifting psychology to an empirical science based on observable behavior and subjected to statistical analysis. He believed that sensory, perceptual, and motor processes were the fundamental elements of cognitive functioning and, consequently, sought to measure these phenomena in the tests he designed. In his work, he attempted to use *mental tests,* a term he coined, to predict college success. Although the measures he used did not correlate well with educational achievement, Cattell was among the first scientists to empirically study cognitive functioning.

DEFINITIONS OF INTELLIGENCE

We now turn our attention to commonly employed definitions of intelligence. There are basically two main approaches utilized to understand and define intelligence. *Factor analytic theories* involve understanding intelligence by examining patterns of individual differences on assessments and then attempting to describe these patterns based on statistical principles. Theorists embracing this paradigm include Charles Spearman, Philip Vernon, Lewis Thurston, and Raymond Cattell.

Alternatively, *information-processing theories* describe the *components* of intelligent thought that translates external stimuli into an internal concept leading to an appropriate behavioral response. Theorists in this camp include psychologists such as Joseph Campione, Ann Brown and John Borkowski, and Robert Sternberg.

Factor Analytic Theories

Throughout the twentieth century, the factor analytic approach has dominated intelligence theory, research, and practice. Scientists adopting this approach attempt to understand intelligence by examining patterns of individual differences on tests measuring various cognitive skills such as vocabulary, general knowledge, visual-spatial processing, abstract reasoning, and arithmetic. These theories, also known as psychometric theories, statistically analyze test performance to identify the basic units, or *factors,* of intelligence. Although providing insight into the structure of intelligence, there is much debate regarding whether intelligence represents a

global, general ability or multiple abilities. The following section presents theories supporting a general factor of intelligence as well as those supporting a multifactor approach.

Spearman's Two-Factor Theory

Charles E. Spearman (1863–1945) was a prominent British psychologist best known for his statistical contributions to the study of intelligence (Kail & Pellegrino, 1985; Sattler, 2001). In his landmark 1904 publication, Spearman (1904a) introduced the statistical concept of factor analysis in an effort to measure factors associated with general intelligence. More specifically, he examined patterns of correlations among IQ subtest scores to look for consistency of performance with the reasoning that if intelligence is global, there would be a high degree of association among subtests. Once correcting for subtest attenuation (for more information, see Williams, Zimmerman, Zumbo, & Ross, 2003), Spearman found evidence in support of a two-factor theory of intelligence in which a general factor, or *g,* superseded more specific factors, or *s*. According to this theory, the *g-factor* accounts for complex cognitive processes such as deductive and analytic reasoning. It is available to the same individual across different intellectual acts and explains consistency in performance across these acts. In contrast, the *s-factors* account for more rudimentary mental processes, including processing speed, recognition, and recall. They are unique to the intelligent act and individuals can exhibit strengths and weaknesses across factors.

Vernon's Hierarchical Group Factor Theory

Heavily influenced by Spearman's work, Philip E. Vernon (1905–1987) adopted a factor analytic approach to intelligence. At the top of his hierarchical model was Spearman's *g,* which accounted for the largest source of variance in intelligence (Kail & Pellegrino, 1985; Sattler, 2001). Underlying *g* were two major group factors: verbal-educational ability (v:ed) and practical-spatial-mechanical abilities (k:m). These major group factors were subdivided into minor group factors, which were further decomposed into several more basic factors. For example, *k:m* subsumed more specific abilities such as spatial, mechanical, and physical abilities. Spatial ability, which was a minor group factor, could then be broken down into three specific factors: spatial orientation (i.e., ability to orient oneself in space relative to environmental markers), spatial relations (i.e., mental rotation of objects), and spatial visualization (i.e., complex mental manipulation of objects). Similar to Spearman's theory, Vernon believed that factors at the top of the hierarchy were more general abilities that affected a wider range of intelligent behaviors, and those at the bottom of the hierarchy involved skills unique to a specific act.

Thurstone's Theory of Primary Mental Abilities

Lewis L. Thurstone (1887–1955) was an established statistician, psychometrician, and intelligence theorist. Among his many contributions to the field of psychology was the development of a new method of factor analysis that allowed for the

exploration of multiple latent factors within a set of variables (Sattler, 2001). The use of this new analytic method challenged Spearman's theory of a unified intelligence in support of seven primary mental abilities: word fluency, verbal comprehension, spatial visualization, number facility, associative memory, reasoning, and perceptual speed (Thurstone, 1938). Moreover, Thurstone found that people with similar IQ scores often displayed vastly different profiles on the primary mental abilities, which he felt further substantiated his claim of multiple intelligence factors. Later studies conducted on a more heterogeneous group of people, however, found fewer discrepancies in primary mental abilities and, instead, supported the notion of a unitary intelligence. Thurstone conceded a statistical compromise that accounted for the presence of both a general factor and the seven primary mental abilities.

The Cattell-Horn Theory of Fluid and Crystallized Intelligence

Among many accomplishments, Raymond Bernard Cattell (1905–1998) is widely recognized for his contribution to intelligence theory. Along with his student, John Horn, Cattell proposed a model of intelligence describing general intelligence as a conglomeration of different abilities falling into two categories: fluid abilities (Gf) and crystallized abilities (Gc) (Horn & Cattell, 1966). Fluid abilities include novel problem-solving skills and the ability to think and act quickly. They are innate abilities that are free from education and acculturation. Examples of fluid abilities include spatial reasoning and symbolic classification. Fluid abilities have been shown to peak during adolescence and then gradually decline as the individual progresses into adulthood. They are more susceptible to brain injury and physiological degeneration than crystallized abilities. In contrast, crystallized abilities include acquired knowledge and skills that are the direct result of education, personal motivation, and culture. They develop throughout adulthood and are less susceptible to physiological degeneration. Examples of crystallized intelligence include general information and language skills.

Information Processing

Whereas the psychometric approach to intelligence focused on factor *structure,* the information-processing approach to intelligence is concerned with the *processes* involved in intelligent acts. Information processing theories elaborate on the structure of intelligence by describing the internal workings of intelligent thought and how it results in behavioral output. Rather than studying *factors,* information-processing theorists describe the *components* of intelligent thought that translates external stimuli into an internal concept and designates appropriate behavioral responses.

Campione, Brown, and Borkowski's Theory

In 1978, Joseph Campione and Ann Brown proposed an information-processing theory of intelligence that was later expanded on by John Borkowski (1985). The theory describes intelligence as consisting of both an architectural system, which describes the structure of cognition, and an executive system, which describes the processes involved in intellectual thought. Of particular importance to the model

were the executive skills of metacognition, which are skills used to monitor and regulate cognitive processes. The metacognitive skills utilized in intelligent thinking include (a) *planning* steps in problem solving, (b) *monitoring* effectiveness of the steps, (c) *revising* steps as needed, and (d) *evaluating* the efficacy of the problem-solving steps. An example of a metacognition is the selection of a rehearsal strategy to learn a word list or telephone number.

Sternberg's Triarchic Theory

Robert J. Sternberg (1985) proposed a triarchic theory of intelligence in which he divided intelligence into three different information-processing components. *Metacomponents* are higher order components that, much like Campione's metacognitive skills, are used to monitor and regulate cognitive processes. They involve the ability to recognize a problem as it arises, select a lower order strategy to correct the problem, monitor the efficacy of the strategy, and evaluate the outcome of the solution. *Performance components* are lower order components used to execute a solution to an identified problem, such as inferring the relationship between two objects. *Knowledge-acquisition components* are involved in learning new information and storing and retaining the information in long-term memory. Three knowledge-acquisition components involved in cognitive processing are (1) selective encoding, which involves distinguishing relevant from irrelevant information; (2) selective combinations, which is the process by which previously encoded information is combined in a new way that maximizes interconnectedness; and (3) selective comparison, in which the previously encoded and connected information is related to information stored in memory.

COMMONLY EMPLOYED MEASURES OF INTELLIGENCE

The impact of theory on the development of intellectual assessment tools cannot be overemphasized. Theories of intelligence have played an important role in the development of many of the traditional and more recent tools designed to measure intellectual abilities. Although many theorists focused mainly on the development and study of their unique theoretical assumptions regarding intelligence, their work has been instrumental to assessment tools that have been generated to measure intellectual capabilities. This section reviews traditional and more recent intellectual assessment tools, highlighting the theoretical basis of each measure where applicable.

Traditional Measures of Intelligence

Traditional measures of intellectual assessment include the Stanford-Binet Intelligence Scale, Wechsler Scales of Intelligence, and the Woodcock-Johnson Test of Cognitive Functioning.

Stanford-Binet

The Stanford-Binet Scale was developed to provide a global estimate of intellectual functioning in children. The most recent version, the Stanford-Binet Intelligence

Scales, 5th edition, was released in 1986 and incorporates into its arsenal of subtests a measure of working memory, which has been shown to relate to reading and math achievement (Johnson & D'Amato, 2004). The Stanford-Binet typically takes 45 to 75 minutes to administer and covers a wide age range (2 to 85 years).

The Full-Scale IQ score is based on the administration of 10 subtests and provides a global measure of cognitive ability (Johnson & D'Amato, 2004). The test battery includes two index scores: Verbal and Nonverbal IQ. The Verbal index requires the examinee to speak, read, and comprehend age-appropriate language. The Nonverbal index requires minimal language and, instead, emphasizes the use of fine motor coordination to maneuver toys and puzzle pieces and point to correct answers. The subtests on the Stanford-Binet measure concepts of fluid reasoning, knowledge, quantitative reasoning, visual-spatial processing, and working memory.

The Stanford-Binet has numerous strengths. For example, it has an expansive age range (2 to 85) and can be used in assessing a variety of disorders where verbal ability is limited (e.g., autism, hearing impairments, limited English-speaking skills). Internal consistency reliability is considered to be good, ranging from.95 to.98 for IQ scores; criterion validity using the individuals tested on the SB4 and SB5 was high ($r = .90$; Kush, 2004).

In spite of these strengths, the Stanford-Binet has been criticized for being racially and culturally biased, although efforts have been made to standardize recent editions on more heterogeneous populations. The Stanford-Binet was further criticized for its emphasis on speed, causing older adult test takers to be penalized for protracted test completion. As such, Binet's mental age norms were thought to be inadequate when applied to adult populations. Furthermore, historically, the Binet scales did not consider the deterioration of intellectual ability that coincides with aging.

Wechsler Intelligence Scales

David Wechsler is best known for his development of a widely used series of intelligence tests, including the Wechsler Preschool and Primary Scale of Intelligence, the Wechsler Intelligence Scales for Children, the Wechsler Adult Intelligence Scales, and the Wechsler Abbreviated Scale of Intelligence. Although a student of Spearman, Wechsler believed that the concept of general *g* was too narrow to adequately describe intellectual functioning. Rather, it was best described in terms of both verbal and nonverbal abilities. As a result, the Wechsler scales consist of several subsets designed to assess abilities within the domains of verbal and nonverbal intelligence, although an estimate of global cognitive functioning is additionally provided. The Wechsler scales have undergone numerous revisions and updates (approximately every 10 years) and are among the most widely used assessment instruments for intellectual functioning across the life span.

Strengths of the Wechsler scales include an adequate standardization sample and good satisfactory reliability and validity. The ability to estimate scores on other Wechsler scales is another advantage for both screening and reevaluation purposes (Keith, 2004).

In spite of efforts to improve on the original Binet scales, the first Wechsler scales were not without flaws. One criticism was the inaccuracy in population representation. The first Wechsler scale's normative sample was insufficient, comprising 1,081 Whites from the eastern United States (Kaplan & Saccuzzo, 2005). Not only was this sample underrepresentative of the country's various regions, but it was racially and ethnically restricted as well.

Woodcock Johnson Scales

The Woodcock Johnson scales are based on the Cattell-Horn-Carroll (CHC) theory of cognitive abilities, a derivative of Spearman's *g* (Kaufman & Lichtenberger, 2006). As previously mentioned, Spearman's theory states that a single *g* sufficiently describes cognitive abilities. This is in contrast to the Wechsler scales, which describe intelligence in terms of abilities across multiple domains, including verbal and performance capabilities.

The Woodcock Johnson scales consist of two separate batteries measuring cognitive ability and achievement. The WJ III cognitive battery consists of 10 subtests (e.g., Verbal Comprehension, Auditory Working Memory) and the extended form consisting of 10 additional subtests (e.g., Picture Recognition, Decision Speed). The cognitive battery provides two indices of general intelligence: a GIA score (general intellectual ability) and a BIA score (brief intellectual ability). The GIA is considered to be a highly reliable score, with the highest median reliability coefficient across ages of.97 on the standard battery (Sandoval, 2004). The BIA score is obtained by administering three subtests only and has a high reliability coefficient of.95 (Cizek, 2004).

The Woodcock Johnson scales have numerous strengths. Construct validity using factor analysis shows the 7-factor CHC cognitive model is more closely linked than any of the other six models frequently used in deriving cognitive batteries; however, more data on validity would be useful (Sandoval, 2004). The split-half method was used in computing reliability for a coefficient of.80 for 38 tests and greater than.90 for 11 tests (Sandoval, 2004). The WJ-III software, manuals, and test materials are impressively modern, but hand scoring is cumbersome. Another advantage of the WJ-III is that the user can administer only selected subtests that can be individually tailored to meet specific testing requirements of individual subjects without sacrificing validity. Extended subtests are available to acquire additional information (Sandoval, 2004).

Recent Measures of Intellectual Functioning

Recently, a number of psychologists have developed assessment tools in attempts to address shortcomings of more traditional measures. Specifically, these assessments have sought to demonstrate applicability across diverse populations, incorporate developmental and educational perspectives into the assessments, and provide a method to assess intellectual functioning among nonverbal and non-English-speaking clients. These assessments include the Kaufman Adolescent and Adult Intelligence Test, the Differential Abilities Scales, and the Test of Nonverbal Intelligence.

Kaufman Adolescent and Adult Intelligence Test

The Kaufman Adolescent and Adult Intelligence Test (KAIT), one of the newest measures of cognitive functioning, was derived from Horn and Cattell's theory of fluid (Gf) and crystallized (Gc) intelligence, Luria's definition of planning ability, and the formal operational stage of development in Piaget's theory (Kaufman & Lichtenberger, 2006). It is a standardized measure of old (crystallized) and new (fluid) learning which can be consolidated into the construct of general intelligence. The core battery consists of Definitions, Auditory Comprehension, Double Meanings, Rebus Learning, Logical Steps, and Mystery Codes (Kaufman & Lichtenberger, 2006).

The KAIT contains a Gf scale and a Gc scale. The Gf scale measures one's adaptability and flexibility when solving both verbal and nonverbal new problems. The Gc scale measures acquired facts, problem-solving ability, and concepts using stimuli dependent on schooling, acculturation, and verbal conceptual development for success, and a composite IQ score.

The strengths of the KAIT include its ease in administration, scoring, and manipulation (Flanagan, 2004). The KAIT yields high test-retest reliability coefficients, ranging from.87 to.97 (Flanagan, 2004). This test also exhibits high construct validity for Caucasian, African American, and Hispanic populations (Flanagan, 2004). It seems to be an adequate assessment of fluid ability and is stimulating to examinees (Flanagan, 2004).

A weakness of the KAIT is that the guiding theory of fluid and crystallized intelligence was included after the test had been standardized and, therefore, raises the question regarding the theoretical basis for the development of the test (Keith, 2004).

Differential Abilities Scale

The Differential Abilities Scale (DAS) was originally developed from the British Ability Scales (Aylward, 2004). The DAS consists of 17 cognitive and three achievement subtests and assesses individuals from 2 years, 6 months to 17 years, 11 months. The cognitive battery generates a General Conceptual Ability score, and the three achievement subtests yield measures of math, spelling, and reading. Internal consistency reliability coefficients of the DAS (based on item-response theory) ranged from.70 to.92 on individual subtests (Aylward, 2004). Correlations with the DAS and other cognitive measures are high (Reinehr, 2004).

The strengths of the DAS include the test's incorporation of a developmental and educational perspective and practicality in evaluating children suspected of having developmental delays, hearing or language difficulties, mild mental retardation, and learning disabilities (Aylward, 2004). The DAS includes special education categories in its normative sample. The chief limitation is the complexity in administration, scoring, and interpretation (Reinehr, 2004). The record form is a lengthy 22 pages and the summary page requires a 12-step process to complete (Reinehr, 2004). Additionally, the manual suggests that administration can be completed in less than $1\frac{1}{2}$ hours, yet this is difficult at best (Reinehr, 2004).

Test of Nonverbal Intelligence

The Test of Nonverbal Intelligence (TONI) was developed to evaluate aptitude intelligence, abstract reasoning, and problem solving in a nonverbal format. Nonverbal tests are useful to the mental health and education fields in that they allow for differential diagnoses, treatment, and placement recommendations. The TONI is brief, taking only 15 to 20 minutes to administer, and is a substitute intelligence measure for those who do not speak English or suffer from sensory deficits (Atlas, 2004). The TONI's Intelligence score was created from the Ravens Progressive Matrices Test (Atlas, 2004). It is used primarily in testing non-English-speaking, learning-disabled, emotionally disturbed, gifted, dyslexic, and ADHD populations (Atlas, 2004). Although the TONI is a popular instrument for nonverbal assessments, it has low concurrent validity, ranging from.53 to.63 (DeMauro, 2004).

PSYCHOEDUCATIONAL ISSUES RELATED TO INTELLIGENCE TESTING

There are numerous issues related to the use of intellectual assessments and testing in formal educational settings. Children are routinely placed in classrooms (either gifted classrooms or classrooms designed to provide children school-based interventions for learning difficulties) based on the outcomes of routine psychoeducational evaluations. Currently, approximately 5% of children attending public schools are diagnosed as having learning disabilities (LD), diagnoses that are routinely based on a discrepancy between intellectual abilities and academic achievement. Typically, these evaluations are conducted by clinical psychologists or school psychologists and frequently include assessment of intellectual functioning, along with achievement-based assessment measures. Intellectual assessments routinely serve as the benchmark for competence and capability.

All children suspected of having handicaps or disabilities impacting school-based performance must be identified and classified accordingly. A multidisciplinary team often is involved with this identification and typically includes a clinical and/or school psychologist, special education teacher with expertise in the area of the suspected disability, the regular education teacher, the parents, and sometimes the student. The team is formed to evaluate whether or not a student meets criteria for a disability using behavioral observations, a review of educational records, additional testing, and any other information necessary to determine eligibility for accommodations (McIntosh & Decker, 2005). Educational testing assesses the domains of intelligence and achievement, which have been established as substantially related (McIntosh & Decker, 2005). To evaluate learning disabilities, initial screening measures are administered to determine if the discrepancy between IQ and achievement meet state guidelines for qualification. To meet inclusion criteria for a learning disability, there must be a substantial difference between expected ability (estimated by intelligence tests) and actual academic performance (Gordan, Lewandowski, & Keiser, 1999). Thus, the role of accurate intellectual assessment is paramount in the determination of learning difficulties.

Outcomes from psychoeducational assessments often lead to specific accommodations for school-based interventions. The intent of these school-based interventions is to increase the likelihood that the student will be able to achieve academically at a level commensurate with his or her intellectual abilities. Obviously, the importance of the validity of the assessments utilized and the interventions subsequently employed is paramount. The significance of these assessments has led to the establishment of specific legislation aimed at increasing the likelihood that accommodations are employed and routinely monitored.

This section reviews the major legal issues related to utilization of intellectual assessments as a component of psychoeducational testing. Specifically, we provide an overview of 504 plans and individualized educational planning. In the general context of the discussion of intellectual assessment, this overview is important because intellectual abilities routinely serve as the starting point for assessment of learning disabilities (and, thus, psychoeducational accommodations).

504 Plan

Section 504 of the Rehabilitation Act of 1973 (34 CFR §104.3j; Rehabilitation Act, 1973) and the Americans with Disabilities Act (ADA) of 1990 (ADA; PL 101-336, 1990) were designed to protect the civil rights of students with disabilities and provide protection against discrimination for those with disabilities. Section 504 defines a disability as substantially limiting one or more major life activities, and this limitation is determined by comparing the individual's performance to the average performance of the general population (Smith, 2001). This definition includes the prevention of any form of discrimination against individuals who are otherwise qualified or capable with the exception of their disability limitations (Smith, 2001).

Section 504 applies to all entities that receive federal funding, but there is no federal funding attached to compliance with Section 504. The ADA applies to almost every entity in the United States, with the exception of churches and private clubs (Smith, 2001). If an individual is protected under the IDEA, that individual is also protected under Section 504. However, if an individual qualifies for services under Section 504, that individual may not necessarily qualify for services under IDEA (McIntosh & Decker, 2005). The basic requirements of Section 504 are nondiscrimination, equal access to free appropriate public education, and participation opportunity in the same academic and extracurricular activities as all other students (Smith, 2001). Section 504 plans may be implemented at any level of education (i.e., kindergarten through 12th grade, as well as at the collegiate level).

Under Section 504, "A qualified [college] student with a disability is one who is able to meet a program's admission, academic, and technical standards either with or without accommodation" (Thomas, 2000, p. 250). If a postsecondary institution receives federal aid of any sort, it is mandated to comply with Section 504. This compliance is verified by filing a document stating that the college does not discriminate based on disability, providing such a notice to recipients of the program, identifying an employee to coordinate efforts to comply, conducting an evaluation of programs, and adopting grievance procedures (Thomas, 2000).

During the preadmission process, colleges are prohibited from requesting information regarding disabilities; yet postadmission inquiries can be made confidentially to determine adequate accommodations (Thomas, 2000).

In determining whether a student is qualified under Section 504, any criteria that may adversely impact the admission of a student with a disability, such as first-year grades, must be excluded from admission requirements. Any admission test used to evaluate a student's performance must be designed to accommodate those with disabilities (Thomas, 2000). Accommodations are necessary only for students with appropriate documentation and should not result in the fundamental alteration of academic standards (Thomas, 2000). Once a student with a disability has been admitted and a review of his or her qualifications is conducted, assessment of that student's ability given the requested accommodations is necessary (Thomas, 2000). The caveat with accommodations is that they should not allow a student with a disability an unfair advantage over those students without disabilities (Thomas, 2000).

Individualized Education Plan

The IDEA mandates that all children, regardless of exceptionality, be entitled to free and appropriate public education in the least restrictive environment. Per the least restrictive environment clause, it is initially assumed that children with specific learning disabilities and/or behavior difficulties may be educated in the regular classroom with the help of supplementary aids and services. The IDEA further requires that each state develop and implement a plan for special education services, as outlined in an individualized education Program (Jacob & Hartshorne, 2003).

Typically, following a psychoeducational evaluation that includes an intellectual assessment, a multidisciplinary team is formed to develop and implement the individualized education program (IEP). A growing trend is including the student in his or her IEP meeting (Arndt, Konrad, & Test, 2006).

The IEP includes the student's present level of performance, defined as both academic achievement and functional performance. The annual goals are outlined to monitor progress in the general education curriculum, and the individualized education program must be reviewed, and if necessary revised, annually (Gartin & Murdick, 2005). Quarterly reports or other periodic reports are required to monitor a student's progress toward annual goals. To meet these goals, the IEP contains a description of the services to be offered and the extent to which the individual participates in general education with his or her peers. In compliance with state and district wide assessments, the IEP must include a statement regarding the appropriate accommodations to measure the student's academic achievement and functional performance. Finally, the IEP must specify a target start date for services to be provided.

CULTURAL ISSUES RELATED TO INTELLIGENCE TESTING

The number of children in the United States whose primary language is not English continues to grow; this trend has been apparent for the past two decades (Lam, 1993). This enormous growth of children in the United States whose primary language is not English and who vary in their rates of acculturation has occurred at a

time when public schools are placing increasing emphasis on the use of standardized tests and intellectual assessments to make placement decisions (Lam, 1993). In 1991, Torres reported that approximately 5 million students were inappropriately (because of their unique cultural background) administered standardized assessments in U.S. school systems. Obviously, that number has likely risen since 1991.

There are inherent problems related to the validity and reliability of intellectual assessments and standardized tests administered to individuals whose background differs from that of the normative population. As Lam (1993) discusses, legal cases aimed at challenging the validity of placement decisions based on standardized tests administered to non-English-speaking or diverse populations have led to mixed decisions.

Lam (1993) provides an excellent review of examiner and test-taker concerns related to administration of standardized tests to students whose primary language is not English. He describes five basic assumptions that clinicians should be mindful of when administering standardized assessments (e.g., the aforementioned intellectual assessments) to these children. First, the test-takers need to be able to follow the direction of the tests and have no linguistic barriers that limit their performance. Second, the test selected should have appropriate content and be consistent with the unique training the child has received prior to testing. In other words, the test administered (especially in the case of standardized achievement tests) should accurately reflect the child's exposure and past curricula. Third, the test-takers should have a previous history of taking standardized tests and some familiarity with this process in order for the results to be considered accurate reflections of their true abilities. Fourth, motivation plays an important factor in the validity of the results. A child who has a long-standing history of academic failure and/or difficulties attributable to cultural and language barriers may not be properly motivated to perform on standardized tests. The examiner should be aware of the impact of this lack of motivation on the interpretability of the results. Five, the examiner needs to be aware of any adverse reactions the test-taker may have to being assessed. Cultural barriers, language barriers, and past history of academic failures and struggles may lead to anxiety that impacts overall performance. While this type of anxiety can be found among the normative population, it has the potential to lead to greater difficulty among a non-English-speaking population. It is critically important that examiners take into account the test-taker's cultural background, specifically language barriers and previous educational curricula, when interpreting the results of standardized tests routinely used in the school setting. At a minimum, results should always be interpreted cautiously and judiciously.

CLINICAL APPLICATION

In an attempt to demonstrate the importance of intellectual assessment in the context of academic settings and medical treatment planning, two clinical case studies are reviewed next. The first case involves a 16-year-old, Caucasian American male referred for psychoeducational assessment following multiple academic failures at the college level. The second case involves a 10-year-old, African American male referred for academic difficulties following treatment for a pediatric brain tumor.

Case Illustration

C.R. was referred for a psychoeducational evaluation to assess for a specific learning disorder. According to his parents' report, he experienced difficulty learning foreign languages and failed several foreign-language classes in high school. C.R. reported difficulty remembering and pronouncing words and that spelling [was] always difficult for him. At the time of the referral, C.R. was required to take 2 years of foreign-language classes as a requirement for high school graduation. There was no history of psychiatric disorders or attention problems.

Cognitive testing revealed that C.R.'s overall ability fell within the average range, with little discrepancy between his verbal and nonverbal abilities. This pattern suggested that he performed equally well on verbal and performance-based tasks. On achievement testing, C.R. performed at a level commensurate with his academic history and estimated cognitive ability on tasks related to reading comprehension, pseudo-word decoding, numerical operations, math reasoning, and listening comprehension. However, his achievement scores for spelling and written expression were drastically lower than expected based on his educational background. Thus, he was generally achieving academically at a level consistent with his estimated cognitive functioning except for tasks involving written expression and word reading. This pattern suggested the presence of a learning disorder. More specifically, the discrepancy between his IQ and achievement test scores met criteria for the diagnosis of Disorder of Written Expression. Because academic curricula, including foreign languages and literature classes, rely heavily on written expression, C.R.'s difficulty in this domain was interpreted to negatively impact his academic performance in this content area and special classroom and curriculum accommodations were recommended to provide him with the optimal learning environment.

Case Illustration

C.T. was referred for a psychological evaluation by his mother for academic difficulties. At the time of testing, he was 5 years posttreatment (tumor resection, radiation, chemotherapy) for a pediatric brain tumor, with no signs of progressive disease. Although historically a good student, C.T. began to exhibit academic problems 2 years prior to the referral. More specifically, his mother reported that C.T. would "study for a test and know the material, but forget it" when he took the exam. This pattern resulted in a significant decline in his academic performance, with grades plummeting from primarily B's to mostly D's and F's. His mother reported that it appeared to take him longer to read than other children his age and that he had a hard time focusing while studying. C.T. indicated that he, also, felt it took him longer to read, but if given the time he could process the information. C.T. stated that he knew "the information, but when it [was] time for the test [he forgot] it all." To study, C.T. reported that he read the material several times and then tested himself on the material. Furthermore, he indicated that his difficulty seemed to be global and was not specific to one particular subject. He reported no sleep difficulties prior to a test

(*Continued*)

and no sudden change in appetite. He indicated that although he got nervous before a test (i.e., worried about not performing well, sweaty hands), he "never panicked" and did not feel that his nervousness interfered with his performance. Thus, his academic difficulties were not likely the result of test or performance anxiety. There was no reported family history of substance abuse, learning disorders, psychiatric disorders, or attention problems. He exhibited no significant behavior problems and denied the use of alcohol or drugs. Baseline testing (prior to diagnosis) indicated superior intellectual functioning.

C.T. was administered a comprehensive psychoeducational test battery to determine the nature of his recent academic declines. This battery included, among other instruments, the Wechsler Intelligence Scale for Children, Fourth Edition (WISC-IV) and the Wechsler Individual Achievement Test. Results of the WISC-IV indicated that C.T.'s Full-Scale IQ score was in the average range. Although consistent with scores of peers his age, his score was significantly below baseline testing and suggested a decline in intellectual functioning. Moreover, his verbal working memory index and processing speed fell within the low average range. Although these scores were not significantly lower than those of peers his age, the pattern showed a trend toward deficient memory functioning and processing speed in that the scores were two thirds of a standard deviation below average performance. The results from this achievement test suggested that C.T. performed at a level consistent with or better than his estimated cognitive ability on all academic tasks; thus, there was no evidence of a specific learning disorder.

Taken collectively, the pattern of results indicated a significant decline in cognitive functioning that was likely attributed to problems with memory and processing speed. This pattern is consistent with the cognitive late effects observed among pediatric cancer survivors. Thus, his academic difficulties were interpreted to be the result of cognitive late effects secondary to the disease process itself and/or aggressive treatment protocols (tumor resection, radiation). C.T. was diagnosed with Cognitive Disorder NOS and appropriate treatment recommendations were made.

The significance of intellectual assessment in both of these clinical cases is apparent. Accurate intellectual assessment assisted with the case conceptualization and, ultimately, the treatment planning for both clients. In the case of C.R., estimated intellectual abilities provided the basis to demonstrate a discrepancy between aptitude and achievement necessary for a diagnosis of Disorder of Written Expression and appropriate accommodations at school. In the case of C.T., declining intellectual functioning along with deficient memory and processing speed led to an efficacious treatment plan for treatment-related cognitive deficits and baseline to measure longitudinal changes postinjury.

SUMMARY

Intellectual assessment has a long-standing history in the disciplines of philosophy and psychology. Numerous theorists have developed theories of intelligence that have sought to best describe overall intellectual abilities of clients and patients.

Factor analytic theories attempt to understand intellectual functioning by examining patterns of individual differences on tests of cognitive skills. Information-processing theories of intelligence focus on the processes believed to be involved in intellectual activities by attempting to describe the internal workings of intelligent thought. Theories of intelligence have impacted both the definitions of intellectual functioning and the assessment tools utilized to measure intellectual abilities.

Intellectual assessments are clinically important for a variety of reasons. These measures routinely serve as the starting point of psychoeducational evaluations for learning disabilities. Baseline estimates of intellectual functioning are commonly utilized to demonstrate discrepancies between achievement (academic) and ability (intellectual prowess) necessary for the diagnosis of learning disabilities.

Intellectual assessments are also important in the classification of employees and in the context of military service. Assessment utilizing intelligence tests, specifically crystallized versus fluid intelligence, provides important diagnostic information for professionals working with patients who have suffered brain injuries and/or insults.

As a result of these multifaceted applications of intellectual assessments, it is important for the practitioner to remain abreast of novel and innovative approaches to assess intellectual abilities. It is equally important to remain mindful of the shortcomings of these assessment tools in light of their numerous ramifications, especially with respect to cultural biases that may be present. Finally, it is imperative that practitioners utilizing intellectual assessment tools accurately convey findings to clients and patients in such a manner that the information is easily understood.

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CHAPTER 17

Pediatric Neuropsychological Evaluation

Leslie D. Berkelhammer

"Left brained, right brained—I call that bird brained." Nearly 20 years ago, this was the opening sentence in the first lecture of my first neuropsychology professor, Ursula Kirk, EdD, at Teachers College of Columbia University. With characteristic dry wit, she encouraged us to think critically at a time when hemispheric specialization was all the rage. Neuropsychology is no longer the black box it once was and appreciation of brain-behavior relationships is more commonplace. Advances in our understanding of cognitive neuroscience, genetics, neuroimaging, and medical technology have enhanced neuropsychological research and clinical practice. Perhaps the challenge before today's researchers, clinicians, and students is best phrased by the late Ray Mulhern, PhD, of St. Jude Children's Research Hospital, who exhorted, "Don't think—know." Integration and consolidation of interdisciplinary findings in design research and clinical protocols is essential to advance the field and to gain recognition of the unique contributions of neuropsychologists working collaboratively with multidisciplinary colleagues in an academic medical setting.

DEFINITION OF A CLINICAL NEUROPSYCHOLOGIST

Clinical neuropsychology is a specialty formally recognized by the American Psychological Association, Canadian Psychological Association, and many other international bodies. In 1989, Division 40 of the APA first published the definition of a clinical neuropsychologist. The Houston Conference on Specialty Education and Training in Clinical Neuropsychology, held in 1997, formalized the programmatic and competency-based components of an integrated scientist-practitioner model for specialty training in clinical neuropsychology. The Houston Conference defined a clinical neuropsychologist as

> a professional psychologist trained in the science of brain-behavior relationships who specializes in the application of assessment and intervention principles based on the scientific study of human behavior across the life span as it relates to normal and abnormal functioning of the central nervous system.

In 2001, the National Academy of Neuropsychology updated the 1989 APA Division 40 definition with wording consistent with that of the Houston Conference. The

NAN definition further delineated the methods of a clinical neuropsychologist as one who

> uses psychological, neurological, cognitive, behavioral, and physiological principles, techniques and tests to evaluate patients' neurocognitive, behavioral, and emotional strengths and weaknesses and their relationship to normal and abnormal central nervous system functioning. The clinical neuropsychologist uses this information and information provided by other medical/health care providers to identify and diagnose neurobehavioral disorders, and plan and implement intervention strategies.

SCOPE OF PRACTICE

Some neuropsychologists are primarily clinicians or researchers; others combine these roles. Both those in academic medical centers and those in practice settings provide opportunities for training and mentorship. Professional organizations serve the interests of their constituents. The four primary professional organizations are the International Neuropsychological Society, the National Academy of Neuropsychology, Division 40 of the APA, and the Society for Neuroscience. A web-based, interactive database of programs offering training in clinical neuropsychology at the graduate, internship, and postdoctoral levels was recently added to the Division 40 website (www.apa.div40.org).

The Web has made access to information easier while also raising the bar for expectations of immediate responses and professional exposure. Many consumers of neuropsychological services now seek information on providers via the Internet that was previously based on a trusted referral. For professionals and future practitioners, opportunities for self-education are easily accessed on the Web, particularly for neuroanatomy tutorials and interactive brain atlases (e.g., www.digitalanatomist.com). Listservs such as the Pediatric Neuropsychology and Women in Neuropsychology listservs are useful resources for a wide audience, with discussions about particular cases, diseases and disorders, practice issues, and career development.

HYPOTHESIS-DRIVEN ASSESSMENT

Neuropsychological assessment is a hypothesis-driven assessment of higher brain functions resulting in an integrative analysis of findings in the context of a neurodevelopmental-systems approach and detailing recommendations for addressing the presenting problems as well as those that may be revealed in the course of the evaluation. Whether a neuropsychological test is conducted to profile the strengths and weaknesses of an individual or a population, the potential for changing how others view the presenting problem or research question is considerable. At an individual level, neuropsychological testing samples multiple cognitive domains and contextualizes findings within a neurodevelopmental approach. Indeed, skilled clinicians are able to translate test results into meaningful relevance for parents and educators such that the feedback session serves as a therapeutic intervention. Families and educators often change their perspective of the child and

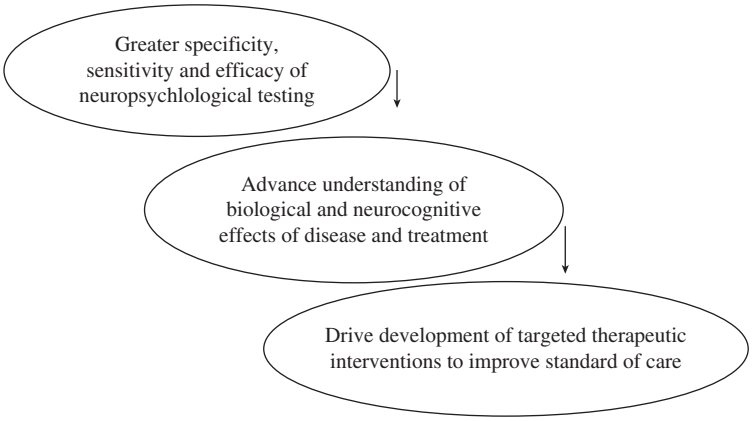

Figure 17.1 An integrative model of neuropsychological evaluation, treatment, and research.

reset expectations for parenting and teaching demands based on this more informed view. Further, the feedback session allows the child to gain a better understanding of his or her ability profile (Roman, 2004). Within and across groups, analysis of performance via empirical research affords opportunity to advance understanding of biological and neurocognitive effects of disease and treatment. Thus, neuropsychological data of individual and group performance drive development of targeted therapeutic interventions and have potential to improve standards of care (see Figure 17.1).

Pediatric or child neuropsychology is the evaluation, treatment, investigation, and teaching of brain-behavior relationships in children with a neurological, medical, or psychiatric disorder that negatively affects cognitive, psychological, and/or behavioral functioning. At its core, pediatric neuropsychology examines neurodevelopmental factors and the effects of disability, disease, and trauma in comparison to typical development and premorbid baseline abilities. In this chapter, I share with you some of my thoughts about child neuropsychological assessment. I describe the development of the discipline, distinguish pediatric neuropsychology from other child psychology specialties, and offer clinical tools for practice.

A NEURODEVELOPMENTAL APPROACH

The neuropsychologist approaches the testing session with an informed plan, based on a thorough review of the child's medical and school records, to test hypotheses regarding underlying neurological and developmental factors contributing to the child's presenting problem. It's not enough to simply document that the child's reading comprehension is at the 76th percentile compared to same-age peers. Rather, analysis of the subcomponents and reasons for the deficits as well as recommendations for accommodations and strategies for compensatory methods must be delineated within a brain-behavior and developmental-systems approach.

Bernstein (2000, p. 434) describes her neurodevelopmental model in terms of an interaction effect between the variables of brain, context, and development: "The

clinical assessment is not just a tally of skills made in a vacuum; it is a principled undertaking in the context of a theory of the organism." Neuropsychological testing is a dynamic (nonstatic) environment wherein the examiner, examinee, and examination are influenced by those same three factors (Bernstein, Kammerer, Prather, & Rey-Casserly, 1998).

Analogous to brain maturation and child developmental theory, the specialty of pediatric neuropsychology has evolved from infancy through childhood. The work of Bernstein and colleagues suggests the coming of age of child neuropsychological assessment. In essence, they have extended Kaplan's pioneering principle of the process approach inherent in clinical assessment of an individual to a more global view of the systems in which the particular child has lived, lives, and will live. Importantly, findings must take into account the four overlapping systems in which the child is affected and affects: medical, school, family, and community.

Levine (1991) has facilitated cross-talk among educators and more neuropsychologically based professionals interested in the interface of developmental pediatrics, neuropsychology, and pedagogy. He describes eight neurodevelopmental constructs (attention, temporal sequential ordering, spatial ordering, memory, language, neuromotor functions, social cognition, and higher order cognition) and their applications to specific curriculum objectives, instructional strategies, and accommodations for learning.

INDICATIONS FOR NEUROPSYCHOLOGICAL EVALUATION

A neuropsychological evaluation is indicated in cases in which brain-based impairment in thinking or behavior is suspected. Examples of relevant conditions include cognitive changes associated with medical conditions (e.g., childhood cancers, sickle cell disease, hematological disorders, HIV/AIDS, epilepsy) or medical treatments (e.g., brain surgery, radiation, chemotherapy, shunt placement, bone marrow transplantation, certain medicines) and brain injury, including stroke. Specific areas of difficulty may include language problems, memory difficulties, organizational problems, emotional difficulties, or behavior problems. Evaluation may be needed to characterize initial functioning and monitor change in children with potentially unstable conditions (e.g., children undergoing treatments with known cognitive risks, children at risk for stroke, children with a history of seizure), make a diagnosis, update academic or vocational plans, or assess for psychosocial problems related to a medical condition. A neuropsychological evaluation is particularly useful for tracking progress in rehabilitation after brain injury or neurological disease. Neuropsychological evaluations may confirm or clarify a diagnosis, identify changes in functioning since prior examinations, provide a profile of strengths and weaknesses to guide rehabilitation or educational or vocational services, identify beneficial compensatory strategies, or result in referrals to other specialists (e.g., educators, neurologists, psychologists, psychiatrists, social workers, vocational counselors).

SHARED VARIANCE AMONG CHILD PSYCHOLOGY SPECIALTIES

Neuropsychology is "the study, evaluation, and treatment of known and suspected brain disorders using the methods of psychology," according to the definition posted on the National Academy of Neuropsychology (NAN) website. By comparison, the American Psychological Association (APA) website posts definitions for clinical and pediatric psychology:

> The field of clinical psychology integrates science, theory, and practice to understand, predict, and alleviate maladjustment, disability, and discomfort as well as to promote human adaptation, adjustment, and personal development. Clinical Psychology focuses on the intellectual, emotional, biological, psychological, social, and behavioral aspects of human functioning across the life span, in varying cultures, and at all socioeconomic levels.

Finally, "Pediatric psychology is an interdisciplinary field addressing physical, cognitive, social, and emotional functioning and development as it relates to health and illness issues in children, adolescents, and their families."

Neuropsychological evaluations may include some of the same tests as those used to conduct a psychoeducational evaluation. However, a noticeable difference will be evident in the breadth and depth of skills measured and interpretation in relationship to brain processes. Neuropsychological assessment is not about the tests. It's about the dynamic testing environment and the theoretical approach of the professional who interprets the findings and ensures that a qualitative analysis of the child's performance is every bit as important as the quantitative results.

Commonalities and differences exist across specialties in child psychology. Pediatric neuropsychology incorporates elements of clinical, school, counseling, and organizational psychology. Pediatric neuropsychologists must be well versed in developmental psychopathology, knowledgeable about pedagogy and school systems, familiar with counseling principles, and informed about organizational behavior when rendering feedback and conducting cognitive rehabilitation to both the child and the family.

QUANTITATIVE AND QUALITATIVE DESCRIPTION OF GLOBAL AND SPECIFIC FUNCTIONS

A neuropsychological assessment provides quantitative and qualitative data on the domains of intelligence, attention, executive functions, language, learning and memory (verbal and nonverbal), visual-spatial, motor, behavioral, psychological, adaptive, and academic skills. However, select areas will be emphasized over others, based on the reasons for referral and data that emerge during the evaluation. The clinical interview is the foundation for the evaluation as it sets the tone for the patient and allows the examiner a period of focused data gathering on which to refine the preliminary testing. The Pediatric Neuropsychology Questionnaire (Baron, Fennell, & Voeller, 1995) is a particularly useful structured interview.

Components of the evaluation include the following: presenting problems and reasons for referral; relevant developmental, family, medical, educational, and psychosocial history; an integrative assessment and analysis of cognitive domains and skills within and across domains; comparison of the student's performance to same-age peers for normative comparison; description of the individual's relative strengths and weaknesses; specific recommendation for intervention; and prognosis, given all these considerations taken together.

Assessment of global cognitive abilities, or general *g*, should allow for comparison of verbal and nonverbal skills as well as analysis of specific subtests and the underlying skills presumed to be measured. Evaluation of attention and concentration abilities should compare auditory versus visual attentional skills, attention span, sustained and divided attention, and processing of simple versus complex information. Executive functions influence the expression within multiple domains and are particularly reliant on the connectivity of neural systems. Throughout the evaluation, task analysis of ability to prioritize, plan, sequence, analyze, reason, and make decisions should be prominent.

Language assessment should explore components such as expressive and receptive abilities, comprehension, verbal fluency, repetition, naming, speech production, conversational speech, organization of verbal output, and pragmatics (e.g., prosody, rambling speech, and nonverbal aspects of speech: eye contact, turn-taking, gestures, interpersonal space). Similarly, information processing should examine auditory versus visual abilities, response speed, and the relative pacing of simple versus complex information. Learning and memory assessment should also be multimodal and quantify functioning visual, verbal, auditory, tactile, and motor skills. There are several well-normed measures that allow the neuropsychologist to analyze which component of memory is problematic for the individual being examined: encoding versus retrieval. Recommendations should account for intra-individual performance in short- versus long-term memory and for simple versus complex information for both verbal and visual-spatial material.

Evaluation of current levels of academic achievement compared to peers is an integral part of the neuropsychological evaluation. Again, the goal should not be limited to mere reporting of ability levels in the core areas of mathematics, reading, spelling, and written expression. Instead, data-driven conclusions about the neuropsychological underpinnings of these expressed deficits should be offered. Recommendations to maximize strengths and minimize weaknesses should also be considered along the same lines. In sum, neuropsychological findings should inform opinions about the *goodness of fit* between the child and his or her learning environment.

Psychological, social-emotional, behavioral, and adaptive functions should be assessed. Recommendations for interventions and skill building should be grounded in an integrated understanding of the child's neuropsychological abilities. For example, the child with Attention-Deficit/Hyperactivity Disorder may require social skills training and cognitive rehabilitation to compensate for deficient frontal feedback loops that limit awareness of his or her effect on others. Temperament, or the biological bases of personality, seems to be underexamined by neuropsychologists, although there appears to be some common ground with child neuropsychology.

NEUROPSYCHOLOGICAL REPORT

The neuropsychological report is an important document. Often, there are multiple consumers of the report, each interpreting the information from the viewpoint of his or her respective disciplines (e.g., physician, educator, coach, and parent). Yet, each of these consumers relies on the information in the report to get a sense of the child's abilities under ideal conditions and to extrapolate to the situation in which the consumer of the report interacts with the child. Accordingly, the report should contain an integrated analysis of the child's cognitive abilities, academic achievement, and psychological functioning. The neuropsychologist should provide quantitative and qualitative data to elucidate multiple factors contributing to the presenting problems, including relevant neurological, medical, and developmental issues. Thus, the report is often a critical tool in decisions regarding school planning, educational program placement, medical care, and treatment monitoring.

By any measure, evaluation is an expensive procedure. Cognitive screening may be appropriate to prioritize those in need of a more comprehensive evaluation, estimate restricted functions, and generate epidemiological data. Regardless of the comprehensiveness of the testing, documentation of the child's strengths and weaknesses, comparison to age-level expectations, recommendations for instruction and curriculum, accommodations to optimize learning and development, and identification of children in need of additional services is expected.

COGNITIVE REHABILITATION

The emergence and development of the subspecialty of cognitive rehabilitation is a logical progression from the discipline's early emphasis on testing and test development (Sohlberg & Mateer, 2001). Cognitive rehabilitation is a subspecialty that employs functional activities to a systematic application of remediation strategies designed to ameliorate impaired brain functions after injury based on a thorough evaluation of the patient's acquired deficits and residual strengths. Treatment goals include strengthening premorbid adaptive skills and behaviors, establishing new patterns of cognition via compensatory strategies for impaired neurological systems, establishing new patterns via external compensatory supports, and adapting to reduced levels of ability. Learning therapy to improve acquired cognitive deficits by instruction and rehearsal of compensatory strategies with environmental modifications can have a profound difference on quality of life following injury or disease. Butler and colleagues (Butler & Copeland, 2002; Butler & Mulhern, 2005) have demonstrated the efficacy of cognitive rehabilitation in children treated for cancer. Moreover, evidence-based outcomes have been demonstrated on functional magnetic resonance imaging studies (Ogg, Zou, White, et al., 2002; Laatsch, Thulborn, Krisky, Shobat, & Sweeney, 2004).

FUTURE DIRECTIONS

As understanding of the biological basis of behavior and cognition become more important, and there are medical or biological treatments to be applied, the only

method of functionally monitoring such treatment is neuropsychological assessment. We need to be at the forefront of working with those who are devising treatments for diseases caused by both innate and acquired injuries to monitor functional outcomes of both biological and behavioral treatments.

—Elsa Shapiro

Historically, neuropsychologists have relied heavily on their expertise in assessment and test development to thoroughly document the cognitive and psychological strengths and weaknesses of the patient. Over time, demand for shorter test batteries, multidisciplinary treatment team involvement, and application of test findings to cognitive rehabilitation, forensics, and policy development has emerged. I have long posed a challenge to myself and colleagues with a two-word question: "So what?" That is, although quantification of an individual's neuropsychological profile in comparison to normally developing peers on standardized measures is important, relevance to the varied environments in which children operate is critical.

These are, indeed, exciting times for those who conduct neuropsychological assessment, treatment, and research and train the next generation of clinicians and researchers. Opportunities to incorporate neuropsychological theory and novel applications in forensic settings, day treatment programs, and automated environments push the envelope. Virtual reality applications to neuropsychological assessment and cognitive rehabilitation of acquired brain injury and neurological disorders are poised to further the field as we move beyond assessment and toward evidence-based outcomes (Schultheis, Himelstein, & Rizzo, 2002). Building on the well-received text, *Neuropsychological Evaluation of the Child*, Baron (2004) describes the growth and development of pediatric neuropsychology in the context of maturation of the field, development of cognitive rehabilitation models applied to a range of disorders, and increased cross-fertilization from allied health, psychology, neuroscience, and neuroradiology (Baron, in press).

I have included several clinical forms and a sample neuropsychological report in six appendixes. Appendix 1 is a disease-specific clinical intake form with content customized to known areas of cognitive vulnerability and contact information for the multidisciplinary team serving this group of patients. Appendix 2 is a test results summary grid provided by my colleague Jordi Bernabeu, PhD, of the La Fe Hospital in Valencia, Spain. He has found it helpful to explain test results to persons unfamiliar with specific tests using a common metric of z-scores with domains color-coded across multiple measures. I believe that we sometimes overestimate the readability of our reports. Even with tables embedded in the report to render scores into more meaningful functional categories, the consumer—multidisciplinary treatment team providers, parents, and educators—may not easily appreciate the psychometrics of reported data. Appendix 3 is a 24-hour preliminary feedback form on which test abbreviations and relevant scores can be entered by domain and categorical outcome. I have found this form helpful in rapidly communicating with the referral source. Appendix 4 is a preferred tests form, with measures grouped by domain and presented in a check box format to aid the examiner before and during the testing session. Given the proliferation of commercially available tests, the list is not exhaustive; it reflects my own selection biases. Appendix 5 is a recommendations form letter,

summarizing medical background and specific accommodations to be implemented in the school setting. Appendix 6 is a neuropsychological sample report.

SUMMARY

In this chapter, child neuropsychological evaluation was defined as a hypothesis-driven assessment of higher brain functions. Indications for and components of neuropsychological testing were defined. The importance of the neuropsychological report as a descriptive analysis of cognitive, psychological, and behavioral strengths and weaknesses, contextualized within a neurodevelopmental-systems framework, was explored. Professional qualifications, scope of practice, and differences between select child psychology specialties were described. Progress and future directions for the field of pediatric neuropsychology were discussed.

An integrated model of neuropsychological testing, treatment, and research was presented. Greater specificity, sensitivity, and efficacy of neuropsychological assessment was proposed as a means to advance understanding of the biological and neurocognitive effects of disease and treatment. In turn, these refinements were proposed as a catalyst to drive development of targeted therapeutic interventions and yield improvement in standards of care. As a specialty area of practice, research, and teaching, the field of neuropsychology has a bright future. Creative and well-trained neuropsychologists will, most certainly, further established lines of research and explore new applications with an increased reliance on emerging biotechnology.

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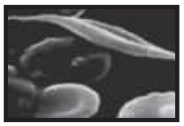
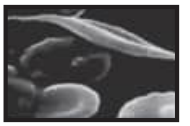
APPENDIX 1: CLINICAL REFERRAL FORM



CONFIDENTIAL
CLINICAL REFERRAL FORM
Sickle Cell Disease



Name: _____ **MRN #:** _____
Date of Testing: _____ **Date of Birth:** _____ **Age:** _____
Grade: _____ **School:** _____
Referring Physician: _____
SCD Genotype: ☐ HbSS ☐ HbSC ☐ HbSβ° thalassemia
☐ HbSβ± thalassemia ☐ HbSβ+ thalassemia

Therapy: ☐ None ☐ Chronic Transfusion ☐ Hydroxyurea ☐ Transplantation
Education: ☐ Retained _____ ☐ Sp. Ed. _____
Psychiatric: ☐ Diagnosis _____ ☐ Medication _____
☐ Psychotherapy _____ ☐ Pica ☐ Sleep Disorder
TCD: ☐ Normal ☐ Conditional ☐ Abnormal
MRI: ☐ Infarct _____ ☐ Other _____

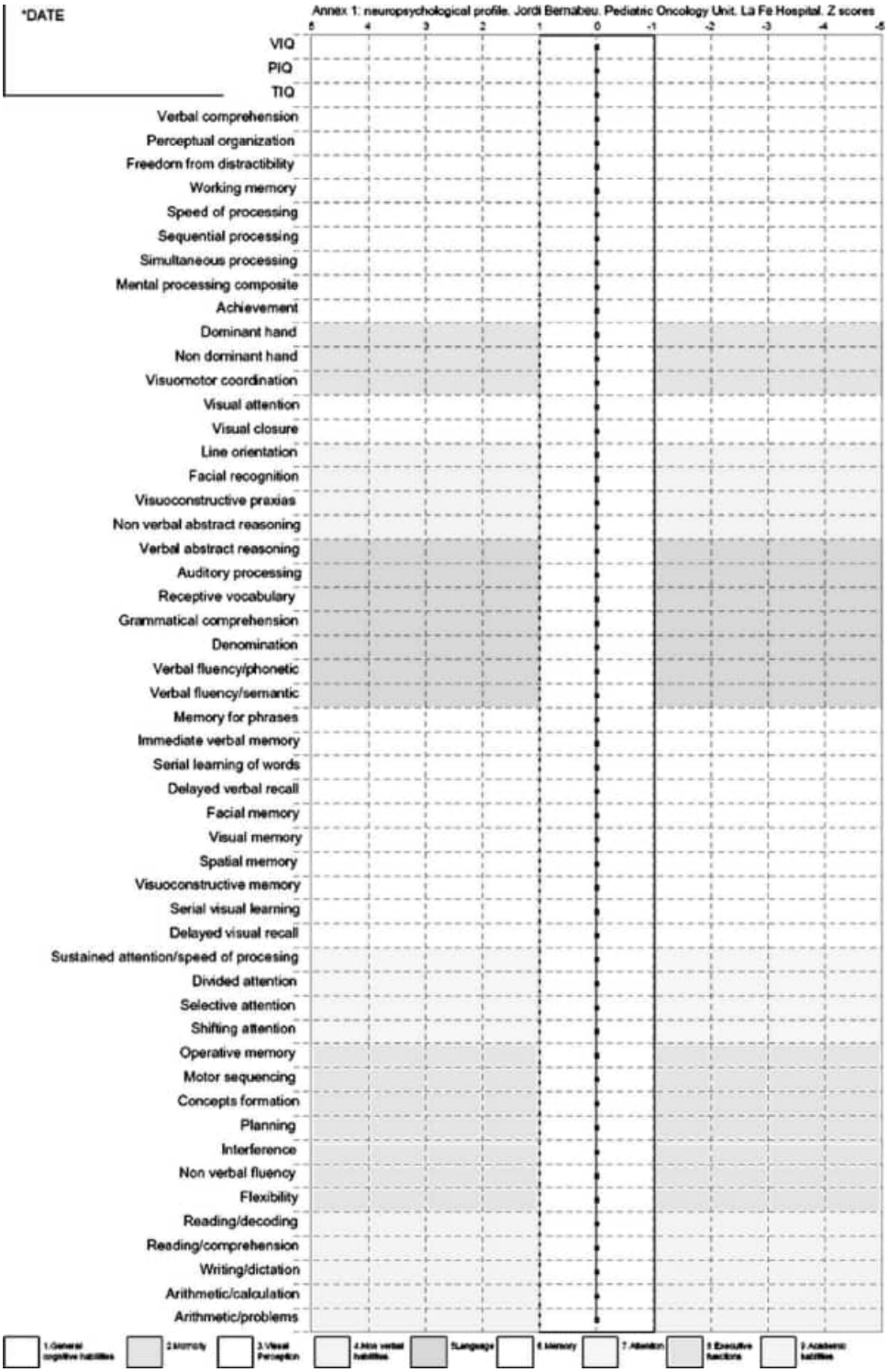
Presenting Problem(s) & Reason for Referral:

Evaluate:
☐ Intelligence
☐ Attention ☐ Processing Speed
☐ Executive Functions
☐ Working Memory ☐ Cognitive Flexibility ☐ Planning and Organization
☐ Fluency ☐ Task Initiation ☐ Task Persistence
☐ Learning and Memory
☐ Language
☐ Visual-Spatial
☐ Motor
☐ Psychosocial ☐ Behavioral / Adaptive ☐ Pain / Coping
☐ Academic Achievement
☐ Reading ☐ Math ☐ Writing ☐ Spelling

CONTACT INFORMATION FOR SCD TEAM

		Phone	Page
Neuropsychologist			
Nurse Case Managers	*Infant & Toddler (0–4 years)*		
	School-Age (5–13 years)		
	Teen (13–18 years)		
	Hydroxyurea (all ages)		
Educators			
Social Worker			
Physicians			

APPENDIX 2: TEST RESULTS SUMMARY GRID



APPENDIX 3: 24-HOUR PRELIMINARY FEEDBACK FORM

CONFIDENTIAL
PRELIMINARY RESULTS OF NEUROPSYCHOLOGICAL TESTING
Feedback provided within 24 hours as professional courtesy. Report to follow within 30 days.

Name: _____ **Medical Record #:** _____
Date of Birth: _____ **Date of Testing:** _____
Age: _____ **Grade:** _____
Examiner: _____ **Supervising Neuropsychologist:** _____
Diagnosis: _____ **Referring Physician:** _____
Reason for Referral:
_____
_____

	Impaired	Borderline	Below Average	Average	Above Average	Superior	Very Superior	Not Tested
Intelligence								
Attention & Executive								
Learning & Memory								
Language								
Visual-Spatial & Motor								
Psychosocial, Behavioral & Adaptive								
Academics								

IMPRESSIONS:

_____ _____
Neuropsychology Fellow Licensed Neuropsychologist

APPENDIX 4: PREFERRED TESTS FORM

CONFIDENTIAL
PREFERRED TESTS FORM

Please evaluate this patient's skills within the following domains and, if indicated, using the tests indicated. It is understood that clinical judgment may require changes to this plan.

NAME: _____ **MEDICAL RECORD #:** _____
DATE OF TESTING: _____ **DATE OF BIRTH:** _____
AGE: _____ **GRADE:** _____ **SCHOOL:** _____
DIAGNOSIS: _____ **MEDICATION:** _____
REFERRAL QUESTION: _____
BEHAVIORAL OBSERVATIONS: _____
_____
_____
_____
_____

☐ **Intellect and Global Cognitive**
- ☐ Bayley Scales of Infant and Toddler Development, Third Edition (Bayley-III)
- ☐ Bayley-III Screening Test
- ☐ Bracken Basic Concept Scale, Third Edition: Receptive (BBCS-3:R)
- ☐ Differential Abilities Scale (DAS)
- ☐ Kaufman Assessment Battery for Children, Second Edition (KABC-II)
- ☐ Kaufman Brief Intelligence Test, Second Edition (KBIT-2)
- ☐ McCarthy Scales of Children's Abilities
- ☐ McCarthy Screening Test (MST)
- ☐ Miller Assessment for Preschoolers (MAP)
- ☐ Mullen Scales of Early Learning
- ☐ NEPSY-II: A Developmental Neuropsychological Assessment, Second Edition (NEPSY-II)
- ☐ Raven's Progressive Matrices
- ☐ Stanford-Binet Intelligence Scales, Fifth Edition (SB-V)
- ☐ Universal Nonverbal Intelligence Test (UNIT)
- ☐ Wechsler Preschool and Primary Scale of Intelligence, Third Edition (WPPSI-III)
- ☐ Wechsler Intelligence Scale for Children, Fourth Edition (WISC-IV)
- ☐ Wechsler Intelligence Scale for Children, Fourth Edition, Integrated (WISC-IV-Integrated)
- ☐ Wechsler Abbreviated Scale of Intelligence (WASI)
- ☐ Wechsler Adult Intelligence Scale, Third Edition (WAIS-III)
- ☐ Woodcock-Johnson Tests of Cognitive Abilities, Third Edition (WJ-III COG)

☐ **Academic**
- ☐ Gray Oral Reading Test, Fourth Edition (GORT-4)
- ☐ Kaufman Test of Educational Achievement, Second Edition (KTEA-II)
- ☐ School Motivation and Learning Strategies Inventory (SMALSI)
- ☐ Wechsler Individual Achievement Test, Second Edition (WIAT-II)
- ☐ Woodcock-Johnson Tests of Achievement, Third Edition (WJ-III ACH)

☐ **Attention and Executive Function**
- ☐ Behavior Rating Inventory of Executive Function (BRIEF)
- ☐ Conners' Continuous Performance Test II, Version 5 (CPT II)
- ☐ Delis-Kaplan Executive Function System (D-KEFS)
- ☐ Clock Drawing
- ☐ Test of Variables of Attention (TOVA)
- ☐ Test of Everyday Attention in Children (TEA-Ch)
- ☐ Wisconsin Card Sort Test: Computer Version 4 (WCST: CV4)

☐ **Learning and Memory**
- ☐ California Verbal Learning Test, Children's Version (CVLT-C)
- ☐ California Verbal Learning Test, Second Edition (CVLT-II)
- ☐ Children's Memory Scale (CMS)
- ☐ Rey-Osterreith Complex Figure Test
- ☐ Wechsler Memory Scale, Third Edition (WMS-III)
- ☐ Wide Range Assessment of Memory and Learning, Second Edition (WRAML-2)

☐ **Language**
- ☐ Clinical Evaluation of Language Fundamentals, Fourth Edition (CELF-IV)
- ☐ Controlled Oral Word Association (COWA) / Verbal Fluency
- ☐ Expressive One-Word Picture Vocabulary Test, 2000 Edition (EOWPVT-2000)
- ☐ Comprehensive Test of Phonological Processing (CTOPP)
- ☐ Peabody Picture Vocabulary Test, Third Edition (PPVT-III)
- ☐ Receptive One-Word Picture Vocabulary Test (ROWVT-2000)
- ☐ Test of Written Language, Third Edition (TOWL-3)
- ☐ Writing Sample: ☐ Cookie Theft Test ☐ TOWL-3 ☐ WIAT-II
- ☐ Test of Word Reading Efficiency (TOWRE)

☐ **Visual-Spatial and Motor**
- ☐ Beery-Buktenica Developmental Test of Visual Motor Integration, Fifth Edition (Beery VMI)
- ☐ Finger Tapping Test
- ☐ Grooved Pegboard Test
- ☐ Hooper Visual Organization Test
- ☐ Judgment of Line Orientation, Short Form (JLO-S)
- ☐ Luria Sequential Hand Movements
- ☐ Wide Range Assessment of Visual-Motor Abilities (WRAVMA)

☐ **Psychosocial, Behavioral, and Adaptive**
- ☐ Adaptive Behavior Assessment System, Second Edition (ABAS-Second Edition)
- ☐ Behavior Assessment Scale for Children, Second Edition (BASC-2)
- ☐ Parenting Relationship Questionnaire (PRQ)
- ☐ Vineland Adaptive Behavior Scales, Interview Edition, Survey Form (VABS)
- ☐ Visual Analog Mood Scale (VAMS)
- ☐ PedsQL
- ☐ Pain Rating Scales (Faces)
- ☐ Sickle Cell Self-Efficacy Scale: Adolescent Version (SCSES-A)

APPENDIX 5: RECOMMENDATIONS FORM LETTER

[Letterhead]

[Director of Special Education]
[School or District]
[City, State Zip Code]

RE: **Patient Name & Medical Record Number**

Dear [Director of Special Education]:
[Full Name (Nickname)] completed a neuropsychological evaluation on [date(s)]. Recommendations and accommodations are summarized below. If you have any questions, please do not hesitate to contact me at [phone]. Respectfully,

[Signature & Title]

MEDICAL BACKGROUND
Diagnosis:
Treatment:
Vision Impairment ☐ Yes ☐ No
Hearing Impairment ☐ Yes ☐ No
Gross/Fine Motor Impairment ☐ Yes ☐ No
Speech Impairment ☐ Yes ☐ No
Assistive Device(s):

RECOMMENDATIONS
Due to the diagnosis, results of neuropsychological testing, and the educational impact of diagnosis and treatment, [Name] meets criteria to receive special education services and accommodations under the category [classification] as per [federal law #]. Specific recommendations are as follows:
☐ Regular classroom with accommodations
☐ Resource room for the following subjects: _____
☐ Self-contained classroom ☐ Full-time aide ☐ Part-time aide
☐ Abbreviated school day ☐ Part-time school attendance/part-time homebound

Instructional Strategies
☐ Review, preview, and summarize new material
☐ Specific, immediate, and frequent feedback
☐ Notebook color coded to textbook color
☐ Neuropage or other electronic prompting device
☐ Copy of material to be copied from book or board
☐ Copy of best notes in an anonymous manner
☐ Study guide
☐ Review assignment 1:1 to ensure understanding
☐ Modify assignment format (photo essay or skit)
☐ Other:

Classroom Modifications
☐ Preferential seating
☐ Second set of books
☐ Assignment book or Electronic organizer
☐ Abbreviated assignments, reduced homework
☐ Extended time to complete assignment
☐ Elevator or extra time to navigate hallway
☐ Color coded map of school
☐ Adaptive P.E.
☐ Other:

Testing and Grading Accommodations
☐ Extended time for testing
☐ Extended time to complete assignments
☐ Modify test format (word bank, multiple choice)
☐ Extra grade opportunities (redo items)
☐ Oral testing
☐ Portfolio-based assessment
☐ Quiet environment with few distractions to take tests
☐ Modified grading scale
☐ Read aloud/sign for internal instructions
☐ Read aloud/sign for test items
☐ Use of calculator
☐ Talking or electronic device with Braille display
☐ Word processor w/ or w/o talk-text technology
☐ Reads into auditory recorder and plays back
☐ Other:

APPENDIX 6: SAMPLE NEUROPSYCHOLOGICAL REPORT

CONFIDENTIAL
NEUROPSYCHOLOGICAL REPORT

KEISHA ST. JUDE
Dates of Testing: 7/14 and 27/2005
Dates of Feedback: 9/12/05 (mother, legal guardian)
9/21/05 (grandmother, durable power of attorney for health care decisions)
Date of Birth: 5/20/1988

REASON FOR EVALUATION: Keisha is a 17 year, 2 month old African American young woman with HbSS sickle cell anemia (SCA), status post cerebral vascular accident (CVA) at age 6 years old, who was referred by _____, MD for a comprehensive neuropsychological evaluation. A neuropsychological evaluation has not previously been conducted. The purpose of this neuropsychological evaluation was to evaluate her intelligence, attention, executive function, memory, language, behavioral, and psychosocial abilities for educational, vocational, and treatment planning.

BACKGROUND INFORMATION: Keisha had a documented CVA (stroke) in 1994 at age 6 with reported symptoms of slurred speech and weakness in her right hand and leg. Of note, Keisha and her mother reported a 2nd possible neurological event approximately 3 years ago which was previously not known to the treatment team prior to the clinical interview conducted as part of this neuropsychological evaluation. Symptoms of slurred speech, "eyes looking funny", weakness, and "not feeling good" were reported. Allegedly, the emergency room physicians in her home state determined that Keisha had an "allergic reaction" despite her mother reportedly stating that Keisha has a prior history of stroke secondary to SCA and denial of any known source for the "allergic reaction". It has been years since Keisha's last brain MRI and an updated MRI is pending.

Keisha's primary caretaker has been her maternal grandmother for most of Keisha's formative years. Keisha's biological mother, with whom she currently resides, remains her legal guardian but has awarded durable power of attorney for health care decisions to the grandmother. Keisha returned to Memphis to live with her mother and four younger half-siblings, reportedly, to remove her from school problems and the influence of negative peers as she had acquired a considerable history of multiple fights and association with delinquent peers.

Keisha and her mother denied basic knowledge of the effect of stroke on her cognitive and academic abilities. They also denied understanding of the importance of hydration and Desferal for iron overload due to chronic red blood cell transfusions. These medical and SCA management issues have been discussed at length on multiple occasions and by multiple providers with Keisha, her mother, and grandmother.

Keisha disclosed a prior episode of suicidal ideation with plan approximately one year ago (cutting wrists), which was previously unknown to the treatment team and grandmother. It was during a low point in her life at the time when she relocated to her current residence, thus being "made" to leave her boyfriend and grandmother. At the time, Keisha reportedly discussed her feelings with a trusted aunt who provided comfort and agreed not to tell anyone about Keisha's suicidal ideation with plan. Keisha denies current suicidal ideation or plan. She contracted to tell her grandmother or other responsible adult if she has suicidal ideation in the future. Grandmother was advised to ensure Keisha's immediate safety if suicidal ideation or plan were to become known.

There is a history of receiving special education services while in the 5th and 6th grades. However, Keisha and her mother have reportedly refused special education since the 7th grade. Keisha was retained in the 9th grade and is not currently receiving special education services in her current 9th grade placement. She enjoys gym and science. Keisha's career aspiration is to become a nurse.

BEHAVIORAL OBSERVATIONS: Keisha was accompanied by her mother to the first testing session and her grandmother to the next session. Initially, Keisha showed a defensive, passive response style and seemed somewhat despondent. However, she responded well to support with open communication and warmed to the examiner and the testing experience. On the second and third visits, Keisha smiled broadly upon greeting this examiner and seemed attentive and cooperative. Overall, Keisha presented as a very likeable young lady who demonstrated good effort on testing, although she lacked confidence in her performance and seemed to have limited ability to appreciate how her past choices have contributed to her current difficulties. Keisha tolerated participation well. Results are considered valid.

PROCEDURES:
Behavioral Observations
Clinical Interview with Keisha, mother, and grandmother
School consultation with two of Keisha's teachers
Wechsler Abbreviated Scale of Intelligence (WASI)
Tests of Variables of Attention (TOVA)
Wide Range Assessment of Memory and Learning-Second Edition (WRAML-2):
    Story and Design Memory subtests
Rey-Osterreith Complex Figure Test
Woodcock-Johnson Tests of Achievement-Third Edition (WJ-III):
    Word Attack, Reading Fluency, Calculation, and Spelling subtests
Wechsler Individual Achievement Test: Written Expression subtest
Behavior Assessment System for Children (BASC): Self-Report, Parent, and Teacher Forms
Childhood Depression Inventory: Short Form (CDI-S)
Feedback session with Keisha & her grandmother (9/12/05)
Feedback session with Keisha & her mother (9/14/05)

CPT CODE 96117, Neuropsychological Testing
Billable hours = 12 (7 hours testing; 5 hours scoring, interpretation, and documentation)

RESULTS:
Intelligence: Intellectual functioning fell within the lowest end of the *Borderline* range for verbal and *Impaired* range for nonverbal abilities, as measured on the Wechsler Abbreviated Scale of Intelligence (WASI) (VIQ=70, 2nd percentile; PIQ=54, ‹1st percentile). The WASI is an abbreviated test of general cognitive abilities, comprised of 4 subtests measuring verbal (i.e., word knowledge and verbal reasoning) and nonverbal (i.e., pattern analysis and nonverbal problem solving) skills compared to same-age peers. There was a significant difference between her obtained verbal and performance score. Therefore, the full scale IQ is not reported. The discrepancy is attributable to her T-score of 35 on the Similarities subtest as compared to T-scores of 20, 21, and 20 on the Vocabulary, Block Design, and Matrix Reasoning subtests, respectively. That is, the Similarities subtest requires associative reasoning to compare how two objects or concepts are alike. In contrast, on subtests requiring higher level language and conceptual reasoning skills, Keisha consistently performed three standard deviations below the mean. Qualitatively, Keisha's performance on a subtest of word knowledge was notable for her concrete responses and simple word associations. On a task of nonverbal problem solving, she demonstrated good task persistence, although an efficient trial and error approach lacking a systematic plan.

Attention and Executive Function: Keisha performed within the *Impaired* range on measures of attention and executive function. On a computerized test of visual attention, the Test of Variables of Attention (TOVA), Keisha demonstrated significant difficulty maintaining her attention after 5 to 6 minutes of a low-response (boring) task. Similarly, verbal and nonverbal subtests of the WASI requiring higher level conceptual reasoning were *Impaired* (i.e., Block Design=21 T-score, Similarities=35 T-score, and Matrix Reasoning=20 T-score). Further, Keisha's ability to analyze and copy a complex visual-spatial design indicated *Impaired* ability to integrate and synthesize the component parts into an integrated whole figure (Rey-Osterreith Complex Figure: Copy=9 of 36 raw score).

Learning and Memory: Learning and memory for language presented in context (i.e. 2 paragraph-length oral stories) was *Average* on the Wide Range Assessment of Memory and Learning, Second Edition (WRAML-2: Story Memory: Immediate recall=10 scaled score, Delayed Free Recall=9 scaled score). Cueing, using a multiple choice recognition format, was helpful (Delayed Recognition Recall=11 scaled score). Visual spatial learning and memory of simple to moderately complex designs was *Below Average* (WRAML-2: Design Memory= 7 scaled score). In contrast to her recall for verbal material, cueing was not helpful on the delayed visual-spatial recognition recall on which her performance declined to the *Borderline* range (WRAML-2: Delayed Recognition Recall=5 scaled score). It may be that Keisha's impaired sustained attention and decision-making abilities may have negatively influenced her demonstrated visual-spatial recall abilities. On a more complex task, copy and recall of the Rey-Osterreith Complex Figure was *Impaired* (Copy=9 raw score, Immediate recall=8.5 raw score, Delayed recall=7 raw score of 36 points, respectively). Despite several attempts at the copy task, she was unable to integrate the component parts. However, Keisha was able to retain the information initially acquired. This suggests that Keisha's deficiency in this area may be due to difficulty with visual-spatial analysis and integration rather than a memory problem.

Language: Possible expressive and receptive language problems were suggested. Qualitative analysis of Keisha's conversational speech was remarkable for sparse output, misarticulation, and word substitutions. For example, Keisha substituted the word "summer" for "day" in recalling the target phrase of the "hottest day in July". On several occasions, Keisha requested repetition of the target words. Her responses seemed to indicate that she did not hear the ending of select words presented in isolation. The possibility of an auditory processing deficit was discussed with Keisha who agreed that she often has difficulty understanding the speech and language of others. She agreed that her misunderstanding of language frequently results in miscommunication impacting socialization and school. Keisha reported that she sometimes studies material presented orally, only to get a poor test grade due to having learned erroneous information.

Academic: Tests of academic achievement were administered. Keisha's performance on both the Wechsler Individual Achievement Test, Second Edition (WIAT-II) and Woodcock-Johnson Tests of Achievement, Third Edition (WJ-III) fell within the *Impaired* range relative to same-age peers in reading, math, and spelling. Reading comprehension and speed on short passages was *Impaired* on the WIAT-II (Reading Comprehension=46 standard score, Reading Speed=4th quartile). The following subtest age- and grade-equivalent scores are reported for the WJ-III:

WJ-III Subtests	Age-Equivalent	Grade-Equivalent
Word Attack	7-7	2.1
Reading Fluency	10-1	4.7
Calculation	9-8	4.1
Spelling	13-3	7.1

Qualitatively, Keisha demonstrated some understanding of elementary rules of phonics, although she was inconsistent in her application of select rules. She was able to demonstrate accurate ability to read for understanding on simple sentences requiring basic reflective thought and

judgment of everyday events. It may be that residual effect of Keisha's stroke sometimes causes her to misinterpret the visual code essential for reading and application of rules of sound to symbol associations. Further, it may be that increased complexity of the task contributes to performance declines.

In terms of mathematical skills, Keisha was able to add, subtract, multiply, and divide single-digit problems. She was able to subtract two-digit numbers that did not require "borrowing", add a three-column straight addition problem, and sometimes multiply a two-digit number by a one-digit number. Keisha was observed to count on her fingers for single-digit arithmetic problems. She lacked automaticity for basic multiplication problems. For example, although Keisha was able to correctly mentally multiply 8×5, she required considerable time and effort to produce the correct response. Keisha remarked that she "goes through the times table" in her head to arrive at most answers on multiplication problems.

Written expression, although well below age- and grade-level expectations, was notable for the suggestion of functional writing skills. Keisha responded well to structure and support. Spontaneously, Keisha wrote the following to the scenario of the WIAT-II: "I am writing to the editor of the newspaper informing them why uniform is a bad opinion. First, everybody have on the same color everyday and the staff have a problem finding out who we are when we be in the cafeteria." Examiner prompting resulted in Keisha adding: "I think that uniforms are hot because most of the time there is no air on in the building and we can't be in the uniform all the time." Examiner encouragement yielded additional information: "A reason uniforms are good is because some kids can't afford cloths like everybody else. And they need to wear uniform because their parents can't afford those expenise cloths or shoes."

In clinical interview, Keisha remarked that it is often difficult for her to demonstrate what she knows and particularly difficult to elaborate on her thoughts in writing. Keisha agreed that she tends to exert less effort on academics because of repeated failure experiences and frustration with her lack of abilities. Moreover, attempts to work independently without adequate academic support for her efforts have not been positive and are, therefore, avoided. Keisha added that she enjoys reading, if the material is at her level.

Psychosocial and Behavioral: In clinical interview, Keisha acknowledged a prior history of suicidal ideation with plan (see Background Information section) but denies current suicidal ideation or plan and contracted to tell grandmother or another responsible adult if suicidal ideation or plan should return. Other significant psychosocial and behavioral concerns were reported, including current gang association and ongoing issues with anger control and poor interpersonal relations. Keisha acknowledged that she now has a reputation as a "fighter" and that it is difficult for her to change this expected pattern of behavior, even though she verbalized desire to change. Keisha reports that there have been times when she has been "provoked" into a fight and has not seen an alternative acceptable behavior. For example, a boy reportedly poured chocolate milk on her jacket. In another circumstance, Keisha fought to defend the honor of her stepfather subsequent to peer teasing. Of note, Keisha endorsed desire to "drop out" of school, which was new information to her mother and grandmother. Risk for school failure is assessed as extremely high.

School consultation with Keisha's English teacher was remarkable for her commitment to Keisha's success and her concern about Keisha's psychosocial problems. The teacher described that Keisha's behavior and attitude has improved following a recent intervention. That is, the teacher approached Keisha after she noted that Keisha seemed to be having difficulty with a female classmate. The teacher spoke privately with Keisha and presented her with a supportive challenge: "You're not going to let somebody else bring you down are you?" Since that time, the girls have not bothered each other in the classroom and Keisha's efforts in class seem considerably more appropriate.

The Behavior Assessment System for Children (BASC) was administered to Keisha, her grandmother, and her teacher. Responses of Keisha's grandmother, who has been her primary caretaker for many years, indicated clinically significant problems in several areas. Of note, Keisha's grandmother endorsed critical items of concern that Keisha sometimes says that she wants to kill herself, die, or wishes she were dead and often threatens to hurt others. The following T-score, percentile, and descriptive ratings are provided:

	T-Score	%ile	Rating
Conduct Problems	70	95	Mild
Anxiety	77	98	Mild
Depression	88	99	Moderate
Hyperactivity	66	93	At-Risk
Attention Problems	68	95	At-Risk
Withdrawal	64	91	At-Risk
Leadership	26	1	Mild
Social Skills	30	3	Mild
Externalizing Problems Composite	74	97	Mild
Internalizing Problems Composite	81	99	Moderate
Behavioral Symptoms Index	80	99	Moderate

Similarly, Keisha's English teacher indicated problems of clinical significance in the following areas:

	T-Score	%ile	Rating
Anxiety	77	99	Mild
Learning Problems	75	98	Mild
Withdrawal	81	99	Mild

In contrast, Keisha's responses on the BASC: Self-Report Form failed to indicate any problems of clinical significance and her response pattern fell within the *Caution* range, suggesting lack of forthrightness in her responses. However, Keisha did endorse critical items of "I can't seem to control what happens to me", "Nobody ever listens to me", and "I get into fights at school." Similarly, Keisha's responses to the Childhood Depression Inventory: Short Form (CDI-S) failed to indicate problems of significance. It may be that Keisha's responses underestimated her true feelings. Indeed, she cried during the initial feedback session and acknowledged that her mood is frequently sad, which she tends to mask with anger/socially-acting out behaviors and that she is frequently upset about her school failure. Keisha, her mother, and grandmother are in agreement that counseling, which has been strongly and repeatedly recommended by the SCA treatment team at the hospital, seems well indicated and may be helpful. However, the family has had difficulty following through on prior recommendations for counseling.

DSM-IV DIAGNOSES:

294.9 Cognitive Disorder NOS; status post cerebral vascular accident (CVA)
296.3x Major Depressive Disorder, Recurrent
312.82 Conduct Disorder, Adolescent-Onset Type

SUMMARY: Keisha is a 17 year, 2 month old African-American female with HbSS sickle cell anemia who is status post cerebral vascular accident (CVA) approximately eleven years ago. She was retained in the 9th grade and is currently receiving special education services in high school. Clinical interview revealed a possible second CVA, previously unknown, that was allegedly diagnosed as an "allergic reaction" by emergency room physicians in Arkansas approximately three years ago. An updated MRI is scheduled for 9/21/2006.

Results of this comprehensive neuropsychological evaluation revealed *Impaired* cognitive abilities (WASI: VIQ=70, 2nd percentile, *Borderline*, PIQ=54, 0.1 percentile, *Impaired*). Significant deficits in attention and executive function were indicated. Keisha was unable to sustain her attention for more than 6 minutes on a low-response (boring) task. On verbal and nonverbal tasks requiring high-level conceptual reasoning, Keisha's performance was deficient compared to same-age peers. Learning and memory for language presented in context, such as stories, was *Average*. However, visual-spatial learning and memory for simple to moderately complex designs was *Below Average* and dropped to *Impaired* with increased complexity of the material. Cueing was mildly helpful for verbal but not for nonverbal information. Possible deficits in expressive and receptive language functions were suggested. Academic achievement was approximately 8 years below age-level expectations in reading and math compared to approximately 4 years below age-level expectations in spelling.

Significant problems in psychosocial and behavioral functioning were reported by multiple informants at home, school, and with peers. Specifically, Keisha in addition to her mother, grandmother, and teachers described problems of clinical significance in the areas of anxiety, depression, hyperactivity, inattention, withdrawal, leadership, social skills, learning, and behavior. Ongoing problems of clinical significance were reported for physical aggression to peers and gang involvement. There is a reported history of suicidal ideation with active plan, approximately one year ago, which Keisha confided in a trusted aunt. Current suicidal ideation and/or plan were denied. Keisha contracted to immediately seek out a responsible adult if suicidal ideation and/or plan should occur in the future. Despite these difficulties, Keisha presents as a likeable young woman who seems distressed about her school failure yet lacks understanding as to how to alter her current maladaptive academic and psychosocial behaviors. Risk for school failure and drop out are assessed as extremely high. Keisha's knowledge and age-appropriate responsibility for medical management of her sickle cell anemia is well below expectations. Without significant intervention, the prognosis for this young lady is not positive.

RECOMMENDATIONS:

1. Continued consultation with the Director of the STAR Program at SJCRH, to further consider educational advocacy and vocational planning is strongly recommended. A highly structured academic program, with appropriate support and objectively rendered consequences, is strongly recommended. Teachers will need to become aware of Keisha's true cognitive abilities so as to structure expectations appropriately. It should be noted that Keisha seems to respond well to praise, reinforcement, and reward. Keisha should be encouraged to pursue her vocational aspirations at a level commensurate with her abilities and interests. Educators with the STAR Program were informed of the results of this evaluation.

2. Immediate need for counseling continues to be strongly recommended, particularly given newly disclosed report of a prior history of suicidal ideation with a plan. Current suicidal ideation and/or plan was denied. However, Keisha presented as tearful and fragile when discussing this history and her current psychological state. Continued assistance to the family in following through with this prior recommendation for counseling is of high priority. The LCSW was informed about these results. The family was informed and appeared to understand that if Keisha becomes suicidal in the future, they should take her to the emergency room to ensure safety. Keisha contracted for safety and agreed to inform her mother, grandmother, or other responsible adult if she were to become suicidal in the future. Keisha's current gang association and extremely high risk for school failure and "drop out" were discussed at length with Keisha and her mother and grandmother. This was allegedly new information to the mother and grandmother.

3. Recommendation for a speech/language evaluation is made. Of note, there were times when Keisha did not seem to hear the ending of select words, despite repetition. She reports and demonstrates considerable difficulty elaborating on her thoughts in both oral and written language. This relative weakness may be a reflection of her global cognitive abilities. However, additional evaluation seems warranted and may help elucidate findings and contribute to educational and vocational planning.

4. Educational recommendations for additional time on tests and assignments, reduced homework load, and ability to complete written assignments in alternative formats are suggested. For example, portfolio-based assessment and independent learning projects that allow a creative and hands-on demonstration of her knowledge should be encouraged.

5. Opportunities for appropriate and supervised peer socialization are strongly recommended. Participation in the Adolescent Support Group for teens with sickle cell anemia continues to be recommended. In the past, Keisha has declined participation in this group. Of note, in the interview, she expressed interest in attending the group. This information was communicated to the Teen Clinic Nurse Coordinator and Social Worker who co-lead the monthly Teen Group. Efforts will again be initiated to encourage Keisha to participate in the Teen Group. Also, Keisha's English teacher, Ms. Smith, described two positive peer socialization clubs available at school. She agreed to, again, encourage Keisha to consider participation.

6. Keisha, her mother, and grandmother were informed of the results of this neuropsychological evaluation and appeared to agree with and understand the findings. It was a pleasure to evaluate Keisha and to work collaboratively with her mother, grandmother, and school-based professionals. If there are additional questions, please do not hesitate to contact me at 901-495-5040.

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Leslie Berkelhammer, Ph.D.
Chief Neuropsychologist, Sickle Cell Program
St. Jude Children's Research Hospital

cc: Mother (legal guardian)
Grandmother (durable power of attorney for health care decisions)
Relevant providers on the sickle cell disease multidisciplinary treatment team

CHAPTER 18

Assessment of Psychopathology

Stefan E. Schulenberg, Jessica T. Kaster, Carrie Nassif, and
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INTRODUCTION

Psychological assessment has long been a core aspect in the training and practice of applied psychologists, with trends over time characterized by remarkable consistency (Archer, Maruish, Imhof, & Piotrowski, 1991; Archer & Newsom, 2000; Camara, Nathan, & Puente, 2000; Cashel, 2002; Watkins, Campbell, Nieberding, & Hallmark, 1995). Applied psychology includes such fields as clinical psychology, counseling psychology, school psychology, and neuropsychology. Assessment comprises many areas (e.g., personality, behavior, intelligence) and is conducted in many professional contexts (e.g., inpatient, outpatient, forensic, medical, educational and occupational settings). For clinicians who work directly with consumers in applied settings, some spend a significant portion of professional time conducting assessments, with the remainder of time divided among other activities, such as therapy, research, teaching, supervision, consultation, and administration (Archer & Newsom, 2000; Cashel, 2002; Watkins et al., 1995). Assessment of personality and psychopathology is common, particularly among clinical psychologists (Camara et al., 2000). As for youth assessment, testing is a major aspect of psychological services that is experiencing swift expansion in measure development, with a concomitant challenge for clinicians to keep up with this growth (Geffken, Keeley, Kellison, Storch, & Rodrigue, 2006; Kamphaus, Petoskey, & Rowe, 2000).

The purpose of this chapter is to discuss issues in the assessment of personality in older children and adolescents, with particular regard for the evaluation of psychopathology. A comprehensive approach to youth clinical assessment is presented, including a discussion of the referral questions, informed consent and assent, rapport, clinical and collateral interviews and collateral records, test and measure selection, cognitive and achievement testing, self-reports via objective personality inventories, and behavioral rating scales (completed by youths, their parents, and their teachers) and observations. The controversy surrounding projective techniques is addressed, followed by a discussion of report writing and feedback. Potential complicating factors to conducting comprehensive assessments are outlined, as well as suggestions for developing assessment competency. The perspective of the authors is that of clinical psychology; however, the chapter has relevance to other

applied fields of psychology. Although a number of areas are discussed, the major focus is on how each area contributes to the understanding of psychopathology, and therefore greatest attention is paid to personality and behavioral assessment, where behavior is viewed as an external manifestation of personality.

THE IMPORTANCE OF IDENTIFYING PSYCHOPATHOLOGY IN OLDER CHILDREN AND ADOLESCENTS

Clinicians frequently are called on to assess disruptions in personality and behavior for a variety of referral questions, such as accurate diagnostic classification and assisting in psychological care and treatment monitoring. Given the variety of problems with which youths present, be they internalizing (depression, anxiety) or externalizing (Attention-Deficit/Hyperactivity Disorder or ADHD, Conduct Disorder), early intervention, accurate diagnosis, and congruence between consumer abilities and resources and treatment recommendations are critical. Kasen, Cohen, Skodol, Johnson, and Brook (1999) hypothesized that experiencing any disorder is disruptive to the general developmental process, but especially so with regard to social and interpersonal skills. A delay or lack of resolution of these developmental tasks would logically manifest in a higher risk of subsequent disturbed personality functioning. Additionally, Ramklint, von Knorring, von Knorring, and Ekselius (2003) found that the presence of a Major Depressive Disorder, disruptive disorder, and substance use disorder in childhood or adolescence each independently increases the odds of personality disorders in adulthood. Finally, youths with Conduct Disorder appear to be at increased risk for developing Antisocial Personality Disorder in adulthood, the connection of which is made in the *Diagnostic and Statistical Manual of Mental Disorders,* fourth edition, text revision (*DSM-IV-TR;* American Psychiatric Association, 2000; Kimonis & Frick, 2006).

Comprehensive assessment of psychopathology can aid the clinician in demarcating the line between adaptive and maladaptive functioning, accurately identifying problematic symptoms and behaviors, and developing a thorough and appropriate treatment strategy that is tailored to the unique attributes of the youth. Are the behaviors or symptoms extreme variations of otherwise normal development (and likely to be transient)? Or are the symptoms or behaviors no longer within the acceptable range and indicative of a potentially severe or chronic pathological syndrome that is causing impairment in functioning? Assessment is well suited to providing illumination to such important questions.

ASSESSING PSYCHOPATHOLOGY IN OLDER CHILDREN AND ADOLESCENTS

Clarifying the Referral Questions

Prior to beginning the assessment, it is necessary to clarify the referral questions and establish a collaborative working relationship with the consumer (Brenner, 2003; Harvey, 2006; Schroeder & Gordon, 2005; Snyder, Ritschel, Rand, & Berg, 2006). What is it that the referral source (e.g., parents or legal guardians) wishes to learn

about the youth's functioning? Working with the referral source throughout the evaluation is integral to conducting quality psychological assessments, in selecting the tests that are most appropriate to answering the referral questions effectively, and in tailoring the report to the referral questions (Brenner, 2003; Michaels, 2006; Snyder et al., 2006). With well-defined referral questions that take into account areas of difficulty and strength, clinicians can shape the evaluation and the report to the unique needs of the individual. Vague referral questions result in less precise evaluations, despite the efforts of clinicians to determine what problems may be evident. In many cases, the referral source will be a parent requesting assistance with better understanding a child's functioning or current problems. However, referral sources may also involve physicians, other mental health professionals, schools, or the courts.

Informed Consent and Assent

A variety of issues must be covered to obtain informed consent (American Psychological Association, 2002; Urbina, 2004). What purpose does the evaluation serve? How will the results be used? Who will receive a copy of the results? Will verbal feedback be available given the context of the assessment? Are the consumers apprised of the nature of confidentiality and the limits thereof? It is also important to determine the extent of the information requested to guard the youth's privacy. For instance, a clinical or forensic evaluation will contain more personal data than an evaluation conducted during a school-based assessment. When conducting assessments with older children and adolescents, informed consent must be obtained from the parent or legal guardian, and obtaining assent from the youth being tested is highly important to facilitate cooperation and build rapport.

Assent entails the explanation of the nature and purpose of the evaluation in language appropriate for the youth's developmental level. What is the youth's understanding as to why he or she is being assessed? Obtaining assent is an intricate and evolving issue, as it is unclear at what age a child may make an informed decision. Although offering a youth the opportunity to assent can be an affirming, rapport-building action on the part of the clinician, the question arises as to whether a youth with serious emotional problems referred by his or her parents should be able to refuse (Merrell, 2003). Depending on the maturity of the child, it is helpful to discuss the pros and cons of compliance, how the information will be used, the limits of confidentiality, and the like, offering youths the opportunity to have their questions answered.

Establishing Rapport

Through the clarification of the referral questions and by working with youths (assent) and their parents or legal guardians (informed consent), the chances that rapport will be healthy are enhanced significantly. The importance of finding ways to encourage youths and their parents to devote themselves throughout the assessment process cannot be overstated. Absence of rapport can negatively influence the evaluation to the extent that the results may be invalid, and it is necessary to prepare individuals and encourage them to be personally invested to do their best

on measures of ability and to respond frankly on personality instruments (Urbina, 2004). To build rapport with children, the examiner should be flexible and patient and possess an understanding of youth development; in addition, discussing topics familiar to youths, such as movies, sports, or computers, may be of assistance (Morrison & Anders, 1999).

As for adolescents, these youths often present difficulties for clinicians due to response styles characterized by resistance or denial (McCann, 1998). For instance, when adolescents are asked to complete the Minnesota Multiphasic Personality Inventory—Adolescent (MMPI-A) as a part of an assessment battery, they are essentially being asked to complete a long, exam-like questionnaire, which results in the possibility that they will not give their best effort (Archer & Krishnamurthy, 2002). Here we return to the issue of assent. It is important that the purpose of testing and the use of the results be explained (adolescents should be made aware that total confidentiality cannot be provided given their status as a minor), with opportunities for questions to be asked and answered, and with a clear indication that feedback of the results will be provided; such efforts demonstrate respect and create opportunities for youths to become involved (Archer, 2005; Archer & Krishnamurthy, 2002). It has been suggested that a firm, task-oriented approach may be helpful to solicit participation that will lead to meaningful results (Archer, 2005; Archer & Krishnamurthy, 2002). Other ways of increasing rapport include noting where restrooms and drinking fountains are, praising effort during testing, and offering adequate breaks (Archer & Krishnamurthy, 2002).

Conducting Interviews: Considering the Cultural Context

Adolescence is a time of transition, of expanding cognitive, physical, social, and personality development. The perceived and actual number of stressors experienced varies from youth to youth and is moderated by a diverse array of variables (e.g., coping skills, acculturative stress), and thus there may be a high level of variation when considering individual development.

For psychopathology to be accurately identified and understood, it must be assessed not only in the developmental context, but in the context of culture, gender, language, and socioeconomic status as well (Achenbach & Rescorla, 2007; Jones et al., 2001; Kuperminc & Brookmeyer, 2006; Leong, Levy, Gee, & Johnson, 2007). From culture to culture there may be different emphases on the importance of some personality traits being expressed over others, how youths should be socialized and values transferred, and how normal and unusual behaviors are defined. Thus, a behavior that is problematic in one cultural context may not be problematic in another. Clinicians also need to be aware that variations within cultures (from individual to individual) are just as significant as variations across cultures (Achenbach & Rescorla, 2007; Jones et al., 2001; Leong et al., 2007).

Cultural variables that clinicians should consider include acculturation, racial/ethnic identity, and minority status stress, all of which can significantly impact the results obtained and interpretations made in an assessment (Jones et al., 2001; Leong et al., 2007; Strack, Dunaway, & Schulenberg, in press). A variety of measures are available to assess these constructs; however, they are not used routinely

in clinical practice. Further research and developmental efforts must be undertaken with individuals of varied ages and backgrounds, and especially with youths, so that at some future point clinicians will be able to routinely and systematically measure these variables as part of a larger assessment battery when clinically indicated.

Another way that these variables may be examined is through clinical interviews. What values are important to the family? How strongly does the youth identify with his or her cultural background, and how does this affect his or her worldview? What communication style is most comfortable to the youth and his or her family? What is the family's perspective on mental illness, assessment, and mental health treatment? If there is a previous history of mental health assessment or intervention, this should be determined to assess the potential impact on the youth and his or her family. In the case of multicultural assessments, this may be of particular relevance given that some individuals may have had negative experiences that could influence the current assessment, and those with limited experience could be influenced by their expectations about what will happen (Allen, 2002; Strack et al., in press). These circumstances reflect on the importance of rapport and informed consent with parents or legal guardians and assent with youths.

Clinical Interviews: Selecting a Format

There are a variety of kinds of interviews that can be conducted, generally falling under the categories of unstructured, structured, and semi-structured. Unstructured interviews consist of open-ended questions that allow the interviewee to talk with little guidance from the examiner. Strengths of unstructured interviews are flexibility and facilitation of rapport (Schroeder & Gordon, 2005; Wiger & Huntley, 2002). Although follow-up information can be gathered based on what is learned during the interview, the content and quality of the information gathered, as well as the diagnoses assigned, may be substantially influenced by the training, experience, and perspective of the interviewer (Wiger & Huntley, 2002).

Structured interviews (increased exactitude, less flexibility) increase the psychometric properties of the data gathering process (Faul & Gross, 2006; Karg & Wiens, 2005; Wiger & Huntley, 2002). Questions are asked to determine the presence or absence of clinical symptoms and minimize the role of interviewer bias. Often there are versions for children and parents available. Structured interviews use the same questions across interviewees to maximize the chances that diagnostic areas are not overlooked, and they may help uncover problematic areas not identified in other sources (such as collateral records); however, although structured interviews tend to result in increased precision, the rigid format may interfere with rapport and the collecting of data not included on the interview form (Wiger & Huntley, 2002).

Semi-structured interviewing, a combination of the unstructured and structured approaches, is often used by clinicians to account for diagnostic precision and the uniqueness of the interviewee (Wiger & Huntley, 2002). It consists of a more guided approach, but also allows for following leads and inquiring further about information gathered. The interviewer decides which areas need to be addressed, such as school, home, friends, behavior, and history of the problem. While it is

generally the case that a semi-structured approach is useful in terms of collecting data and meeting the needs of the interviewee, clinicians themselves must be flexible in terms of their own style to account for important moderating factors (e.g., culture). There are numerous sources available for information on formal structured and semi-structured interviews (e.g., Loney & Frick, 2003; Orvaschel, 2006; Rourke & Reich, 2004; Shaffer, Fisher, & Lucas, 2004). The remainder of this discussion focuses on the standard clinical interview, in which a clinician probes a variety of symptom constellations and areas, using flexibility as necessary to evaluate for diagnoses and also to collect information relevant to understanding the youth's unique attributes. We begin with the parent or legal guardian interview.

Interviews with Parents or Legal Guardians

When conducting interviews, the clinician should be collecting data in such a way as to understand what events are occurring in the youth's life, when symptoms manifested themselves, whether there is an external cause evident, and whether symptoms wax and wane in accordance with certain events. Depth of understanding of contextual factors and how they intertwine with symptoms, coupled with an understanding of the youth's unique patterns of strength, will afford the clinician greater opportunities to discern whether a diagnosis is warranted, as well as to plan appropriate intervention, if needed.

When interviewing parents or legal guardians it is important that they feel comfortable and that the examiner is in tune with their concerns. It is ideal to interview both parents; however, this may not be possible given family dynamics (e.g., work schedule conflict, divorce). Interviews with parents or legal guardians offer a great deal of information about the youth's history and current functioning. In addition to providing information, parents can be brought on board early in the process, which can be beneficial when it is time to present and work on treatment interventions (Schroeder & Gordon, 2005).

Data to be gathered during an interview generally includes duration, frequency, and severity of youth behaviors and symptoms. Interviewing the child or adolescent independently, as well as together with parents or legal guardians, can yield information as well as family dynamics. Various symptom constellations should be addressed, whether or not they appear as part of the presenting complaint, to clarify the diagnostic picture. Other information to be gathered includes behavioral observations, developmental history (e.g., meeting of developmental milestones), family physical and mental health history, social history (e.g., friendships, relationships with siblings), losses in the family, and medical (e.g., injuries, medication allergies) and mental health (e.g., previous diagnoses, treatment) histories. Data should also be collected on the child's or adolescent's educational and cultural background, as well as occupational history in the case of some older adolescents. The interviewer can also inquire about parenting style, approach to discipline, the youth's behavior at home, and youth strengths. Focusing on these and related areas has been described to varying degrees in a number of sources (e.g., Merrell, 2003; Schroeder & Gordon, 2002 2005; Snyder et al., 2006). Finally, it is important to convey to parents the idea that positive action may be taken, and if additional data

are needed it is paramount to explain the rationale and methods consistent with the collaborative approach (Schroeder & Gordon, 2005).

Interviews with Older Children and Adolescents

With regard to conducting interviews with youths, behavioral observation and life history data are quite important to the overall quality of the assessment and should not be disregarded in favor of interviews with parents or legal guardians. Rather, a clinician may gather relevant information from parents or legal guardians and the youth being evaluated. Clinicians interviewing youths should expect to see variations based on such variables as development, education, and culture.

School-age children (ages 6 to 12) typically think in factual, literal, and rigid terms (they are able to consider some, but not all, eventualities) and are likely to participate little in discussions characterized as abstract or involving feelings and attitudes (Gudas & Sattler, 2006). Children may be concerned with how well they are doing on psychological tests and with pleasing the interviewer and/or deferring to the wishes of their parents, legal guardians, or other authority figures. Rapport building with youths is needed to account for such potential complicating factors.

Adolescents, in comparison to school-age children, think less concretely and are able to consider a substantially wider range of alternatives and to test hypotheses (Gudas & Sattler, 2006). They are able to report on symptoms and to consider elements of time and place; however, although they can think with increasing abstraction, problems with risk taking and invincibility may be evident (Gudas & Sattler, 2006). Their peer group tends to take on more influence in the determination of behavior, a role that typically was previously held by parents or legal guardians. Although this differentiation is often regarded as a part of developing into independent adulthood, it has specific implications during the interview. Adolescents tend to be more concerned about how they fit in and how they will be judged by others than with reporting how they actually experience things. They may be actively searching for an identity, including how they relate to ascribed gender and cultural roles. The adolescent interviewee may present at once as being genuinely both naive and blasé. Having an approachable yet frank presence can help foster a vitally important sense of acceptance and build rapport necessary to facilitate the interview.

As with older children, adolescents are capable of providing information useful to skilled clinicians. It is important for youths to know that their views are important to the clinician, and for the clinician such data are imperative in the process of developing and refining clinical hypotheses. In interviews of youths, especially adolescents, it is important to establish confidentiality and the limits thereof. With regard to interview content, it is important to assess such areas as drug and alcohol use, behaviors harmful to self or others, and sexual activity (Wiger & Huntley, 2002), in addition to the other areas outlined earlier.

Collateral Interviews

With appropriate consent, discussion with other relevant individuals, such as extended family members, teachers, and case managers, may provide important

contextual data about the youth's functioning across multiple settings. Information from as many sources as are reasonable will improve the ability of the interviewer to gain a comprehensive picture of the youth (Boggs, Griffin, & Gross, 2003). Such interview data may also be combined with behavior rating scales (discussed later) to enhance the clinical picture. For additional information on interviewing techniques and procedures as they relate to both clinical and collateral interviews, an array of sources is available (Groth-Marnat, 2003; Merrell, 2003; Morrison & Anders, 1999; Sattler & Hoge, 2006; Sharp, Reeves, & Gross, 2006; Wiger & Huntley, 2002).

Review of Collateral Records

The review of collateral records is of importance in improving the quality of psychological assessments. Are there certain medical or physical conditions that have a bearing on the current presentation? Are school records available that reveal academic performance or behavioral problems? In the case of adolescents in particular, if they are demonstrating behaviors that bring them into contact with the law, are police records available? Having access to collateral data is critical to understanding context, as a basis to facilitate discussion with the youth and his or her parents or legal guardians, and as a comparison with test data. Collateral records aid the clinician in establishing the youth's history, facilitating an individualized case conceptualization for the presenting complaints.

Selecting Tests and Measures for the Evaluation

There is a distinction between testing and assessment. A test is standardized, reliable, valid, and clinically useful (Hunsley, Lee, & Wood, 2003), whereas assessment involves the integration of data from multiple sources, such as tests (e.g., Wechsler Intelligence Scale for Children, fourth edition [WISC-IV]) and nontests (e.g., collateral records). Psychometric properties of a test may vary depending on the individual being tested (based on cultural or socioeconomic factors), the context (a test may be psychometrically sound for one clinical purpose but not another), and the composition of the test (tests with multiple subscales will have different psychometric properties for each scale). In a survey of psychologists who work with adolescents, Archer and Newsom (2000) learned of an array of factors relevant to instrument selection, including soundness of psychometric properties, relationship between the instrument and the referral question, the relevance of the generated data for treatment planning, provision of comprehensive information pertinent to psychopathology and diagnosis, and simplicity of scoring and administration.

As for a practical example of how tests may be selected, consider a relatively common referral question: "My teenage son is getting into fights in school, he's struggling academically, and he is having trouble paying attention to his teachers. What's wrong? Does he have ADHD?" An assessment battery may be composed of interviews with the youth and his parents; collateral interviews (e.g., with a teacher); a review of collateral records (e.g., report cards and other school records); intelligence and achievement testing (e.g., WISC-IV, Wechsler Individual Achievement Test, second edition [WIAT-II]); a continuous performance test to assess for difficulties with attention; self, parent, and teacher ratings (e.g., Behavior Assessment

System for Children, second edition [BASC-2]); and additional measures of functioning as needed (e.g., MMPI-A, Children's Depression Inventory [CDI]). Why such a comprehensive battery? Because ADHD is comorbid with learning disorders, externalizing behavior disorders such as Oppositional Defiant Disorder and Conduct Disorder, and internalizing disorders such as depressive and anxiety disorders (Sattler, Weyandt, & Willis, 2006). A thorough assessment facilitates a detailed diagnostic formulation. As a case in point, inattentiveness may be taken for ADHD, when in some cases it may be related to depression or anxiety. Aggression could be part of a larger constellation of symptoms meeting criteria for a diagnosis such as Conduct Disorder. A partial assessment may miss the mark.

Another consideration for clinicians when selecting a test is the degree of fit between the test and the language and cultural background of the respondent (Butcher & Pope, 2006; Flores & Obasi, 2003). This means determining that the norms, standardization, reliability, and validity are acceptable for the individual being evaluated and that the instrument is in the person's preferred language, all of which are necessary to minimize erroneous clinical interpretations (Flores & Obasi, 2003; Jones et al., 2001; Leong et al., 2007; Strack et al., in press). It is heartening that in recent years additional attention has been paid to multicultural issues in assessment (both within the United States and internationally), and this emphasis will likely continue; however, it is disheartening that empirical research continues to lag behind the need (Leong et al., 2007; Strack et al., in press; Wood, Garb, Lilienfeld, & Nezworski, 2002).

Cognitive and Achievement Testing: Clues to Personality Style?

In addition to clinical and collateral interviews and collateral records reviews, cognitive and achievement testing may reveal clues about personality functioning that may be evaluated using more systematic, standardized, and comprehensive methods. Of the intelligence tests used, the Wechsler scales are among the most commonly employed by clinicians and are quite common in adolescent and child batteries specifically (Archer & Newsom, 2000; Camara et al., 2000; Cashel, 2002; Kamphaus et al., 2000). The current edition of the Wechsler scales appropriate for older children and most adolescents is the WISC-IV (Wechsler, 2003).

Why mention a cognitive test in a chapter on assessment of psychopathology in older children and adolescents? As many clinicians know, behavioral observations are a heavily emphasized part of clinical assessments, and observing a youth's performance on the WISC-IV, and on achievement tests such as the WIAT-II (The Psychological Corporation, 2002), may reveal hypotheses about how the youth interacts with his or her world that may be examined subsequently through other methods. Looking at the WISC-IV specifically, a variety of tasks are asked of the youth, including those that tap performance-oriented abilities (e.g., Block Design) and verbal-oriented abilities (e.g., Comprehension). The administration of each subtest yields behavioral observation information for the clinician. For example, as noted by Sattler and Dumont (2004), during Block Design, where the task involves the examinee putting together blocks to match a series of designs, a number of hypotheses about performance may be generated. Is the youth deliberate or

impulsive? Does the youth persist in the task or give up easily? Fumbling too much may be indicative of anxiety, youths very slow in their work may be experiencing boredom, and youths who constantly check their work may be insecure (Sattler & Dumont, 2004).

Observations similar to these may be collected throughout a test such as the WISC-IV and also may be made in the area of achievement testing. In the case of the WIAT-II, what is the youth's attitude toward perceived success and perceived failure? Does the youth show resilience or quickly give up? How does the youth respond to the praising of effort? Such clinical observations and interpretive hypotheses are valued by many clinicians as a means to better understand the functioning of youths, but clinicians must exercise caution in relying on behavioral observations in isolation or otherwise placing too much emphasis on them. Intelligence and achievement tests are not standardized for these purposes, and interpreting clinical observations appropriately can be difficult, as there are many reasons why a youth may respond in a certain way in a certain context. Also, there is no guarantee that informal observations or interpretations relating to personality style will translate to the youth's typical environments. There is a danger for clinicians to form erroneous impressions early and stay with them. Multiple and appropriate standardized sources of data should be considered, and systematic evaluation of disconfirming data should be undertaken, prior to arriving at a diagnostic formulation.

Objective Comprehensive Measures of Psychopathology

Objective instruments of personality and psychopathology are typified by standardization and norms, as well as straightforward items, response format (e.g., true-false), and scoring (Segal & Coolidge, 2004; Urbina, 2004). As compared to specific domain measures (discussed later), comprehensive measures tend to be characterized by several hundred items and require added time to administer; however, they tend to have better norms and a wider range of applications and generate many scores (Hogan, 2007). Comprehensive measures are often desired in clinical assessments given the thorough data they provide across a range of domains. These data are generally useful for hypothesis testing and treatment planning. Commonly used comprehensive measures in youth assessments are the MMPI-A and the Millon Adolescent Clinical Inventory (MACI; Archer & Newsom, 2000; Cashel, 2002), both of which are considered to be effective tools (Groth-Marnat, 2003).

The Minnesota Multiphasic Personality Inventory—Adolescent

The MMPI-A (Butcher et al., 1992) is composed of 478 true-false items, is based on a large and diverse normative sample, and is used in a variety of settings to assess emotional functioning and/or refine clinical hypotheses in adolescents between the ages of 14 and 18 (Archer & Newsom, 2000; Baer & Rinaldo, 2004; Butcher & Pope, 2006; Graham, 2006). The normative sample consisted of 805 males and 815 females, and data on a clinical sample were also gathered (420 boys, 293 girls; Baer & Rinaldo, 2004). The MMPI-A contains a number of validity scales to detect random or unusual responding, and its many clinical (e.g., Depression, Psychasthenia, Psychopathic Deviate), content (e.g., School Problems, Conduct Problems), and

supplementary scales (e.g., Alcohol-Drug Problem Acknowledgment, Alcohol-Drug Problem Proneness) provide the clinician with a wealth of potentially useful data.

Raw scores are translated into *T* scores (uniform or linear translation, depending on the scale), and though there is no clear cutoff for scale elevations, scores as low as 60 to 64 possibly have interpretive significance (Baer & Rinaldo, 2004; Butcher & Pope, 2006; Graham, 2006). Interpretation is undertaken knowing the larger context of the adolescent's life history and presenting problems, proceeding with an examination of the MMPI-A's validity scales; if the profile is valid, the clinical scales are interpreted, with content and supplementary scales aiding in the refinement of clinical hypotheses (Baer & Rinaldo, 2004; Butcher & Pope, 2006). Thus, although the MMPI-A yields many data regarding an adolescent's personality functioning, clinicians must be aware of the psychometric properties of the many validity, clinical, content, and supplementary scales. For further information regarding developmental, psychometric, and interpretive considerations, numerous resources are available (e.g., Archer, 2005; Archer & Krishnamurthy, 2002; Archer, Krishnamurthy, & Stredny, 2007; Archer, Zoby, & Stredny, 2006; Butcher & Williams, 2000; Frick & Kamphaus, 2001).

There are a variety of positive aspects to using the MMPI-A, including provision of comprehensive information, useful adolescent-related material and norms, simplicity of administration, and research support with a wide variety of clinical populations (Archer & Newsom, 2000; Butcher & Pope, 2006). The measure is available in a standardized recorded version, it has been translated into Spanish with adolescent norms available for Spanish-speaking youths in the United States, and a variety of translations are available in languages such as Arabic, Chinese, Korean, and Hebrew, to name a few (Butcher & Pope, 2006).

Negative aspects of using the MMPI-A include the length of the measure and the test-like demands placed on the examinee, a demanding reading level for many adolescents (about seventh grade), time demands (administration, scoring, interpretation), and cost when considering the context of a managed care environment (Archer & Newsom, 2000). Additionally, Frick and Kamphaus (2001) criticized the MMPI-A's normative sample for undersampling Hispanic youths and 18-year-olds. Finally, although some multicultural studies have been conducted (e.g., Cashel, Rogers, Sewell, & Holliman, 1998; Gumbiner, 1998; Negy, Leal-Puente, Trainor, & Carlson, 1997), Baer and Rinaldo's (2004) review noted that in some cases there were no differences between racial/ethnic groups studied, but that when differences were found it was difficult to determine what the findings meant, or whether the differences were clinically meaningful. More multicultural research is needed (Gómez, Johnson, Davis, & Velásquez, 2000; Leong et al., 2007).

The Millon Adolescent Clinical Inventory

The MACI (Millon, Millon, & Davis, 1993) is a 160-item, true-false, self-report instrument based on the theory of personality and psychopathology put forth by Millon (Millon & Grossman, 2005; Tringone, 2002; Tringone, Millon, & Kamp, 2007). It is composed of 31 scales across four domains: Personality Patterns, Expressed Concerns, Clinical Syndromes, and Reliability and Modifying Indices (Millon &

Grossman, 2005; Tringone, 2002; Tringone et al., 2007). The MACI requires approximately a sixth-grade reading level and is useful for evaluating areas of concern, difficulties with personality styles, and psychopathology with adolescents ranging in age from 13 to 19 (Millon & Grossman, 2005; Tringone, 2002; Tringone et al., 2007). The measure has four separate norm tables (13- to 15-year-old males and females and 16- to 19-year-old males and females) seen in clinical settings, with the overall norm group consisting of over 1,000 adolescents from the United States and Canada (McCann, 2006; Tringone, 2002; Tringone et al., 2007).

Item development comprised a three-stage process: theoretical-substantive, internal-structural, and external-criterion (Millon & Grossman, 2005; Tringone, 2002; Tringone et al., 2007). The MACI utilizes base rate scores (BR), as opposed to *T* scores, where BR cutoff scores of 85 and 75 indicate trait prominence and presence of a trait, respectively (McCann, 2006; Millon & Grossman, 2005). Interpretation occurs in the context of the youth's history and involves systematic review and subsequent integration of the Modifying Indices (validity indicators), Personality Patterns (e.g., Inhibited, Unruly, Oppositional), Expressed Concerns (e.g., Sexual Discomfort, Social Insensitivity, Family Discord), and Clinical Syndromes (e.g., Substance-Abuse Proneness, Delinquent Predisposition, Anxious Feelings, Depressive Affect; McCann, 2006; Millon & Grossman, 2005; Tringone, 2002). The Expressed Concerns scales target areas that may be problematic to troubled youths, while elevations on the Personality Patterns and Clinical Syndromes scales signify psychopathology necessitating professional intervention (Millon & Grossman, 2005; Tringone, 2002). For more information on developmental, psychometric, and interpretive considerations, a variety of sources are available (e.g., McCann, 1999, 2006; Meagher, Grossman, & Millon, 2004; Pinto & Grilo, 2004; Tringone, 2002; Tringone et al., 2007).

The MACI may be useful in some clinical situations, such as when distortions in personality development appear to be evident, so long as such use is supported by the empirical literature and data are not used in isolation. Proponents of the MACI note that its advantages include its multistage construction, brevity, design for use with adolescent clinical populations (tapping both Axis I and Axis II symptoms), inclusion of Modifying Indices, and theoretical basis (Millon & Grossman, 2005; Tringone, 2002; Tringone et al., 2007). Noted disadvantages include a sixth-grade reading level being necessary (audiocassette administration is available for clients who speak English and Spanish), and more severe types of psychopathology such as thought and psychotic disorders, bipolar disorders, and paranoid ideation are not directly measured (McCann, 2006; Millon & Grossman, 2005; Tringone, 2002; Tringone et al., 2007). Moreover, few multicultural studies involving the MACI have appeared in the literature (for examples with Mexican American youths, see Blumentritt, Angle, & Brown, 2004; Blumentritt & Wilson VanVoorhis, 2004).

Objective Specific Domain Measures of Psychopathology

The MMPI-A and the MACI cast a broad net, assessing many different kinds of behavioral and emotional difficulties. Other measures assess specific constructs and therefore tend to have fewer items (faster administration times), yield fewer scores (perhaps a single score), and have norms and applications that are more limited

(Hogan, 2007). Specific domain instruments are often desirable to clinicians given their focus on a specific problem and the ease with which they may be combined with other data, such as comprehensive objective measures and interview data. Specific domain measures straddle many clinical areas, commonly depression and anxiety. Although a comprehensive review is not possible in a single chapter, we mention a few measures prior to discussing a recent constellation of specific domain measures that appears promising pending future research.

One of the most popular specific domain measures is the CDI (Kovacs, 1992). The CDI may be used with youths between the ages of 7 and 17. It includes 27 items of depressive symptoms with a choice of three response options (0, 1, or 2). Scores are summed, with higher scores considered to be more symptomatic. Raw scores are converted to *T* scores to yield a global score and five subscales: Negative Mood, Interpersonal Problems, Ineffectiveness, Anhedonia, and Negative Self-Esteem. The CDI is generally considered to be reliable (Campbell, 2006b; Merrell, 2003). However, given mixed discriminant validity support, and because it is more an indicator of distress as opposed to a diagnostic indicator, it should not be used in isolation (Fristad, Emery, & Beck, 1997; Schroeder & Gordon, 2002). This is excellent advice for any instrument, especially those of the specific domain variety. Sitarenios and Stein (2004) review the CDI's psychometric characteristics.

Additional self-report measures of depression include the Reynolds Child Depression Scale (RCDS; Reynolds, 1989) and the Reynolds Adolescent Depression Scale, second edition (RADS-2; Reynolds, 2002). The 30-item RCDS covers ages 8 to 13 years and is considered to thoroughly assess symptoms of depression, with good psychometric properties and adequate norms (Merrill, 2003; Reynolds, 2006; Schroeder & Gordon, 2002). The RADS-2 contains 30 items utilizing a 4-point response format and is used with youths between the ages of 11 and 20 (Campbell, 2006b; Reynolds, 2004, 2006). It yields *T* scores with a mean of 50 and a standard deviation of 10 and includes four factorially derived subscales in addition to the total score: Dysphoric Mood, Anhedonia/Negative Affect, Negative Self-Evaluation, and Somatic Complaints (Campbell, 2006b; Reynolds, 2004, 2006).

As for the measurement of anxiety, one self-report measure that has been receiving increasing attention is the Multidimensional Anxiety Scale for Children (MASC; March, 1997), which assesses anxiety in youths ages 8 to 19 years. The MASC's 39-item, 4-point response format yields *T* scores and is composed of a Total Anxiety score, Inconsistency Index, and Anxiety Disorder Index, as well as four factor scores: Physical Symptoms, Harm Avoidance, Social Anxiety, and Separation/Panic (Campbell, 2006a; Schroeder & Gordon, 2002). In reviews of the MASC, authors have noted the measure's reliability (particularly the internal consistency of the Total Anxiety score), convergent and discriminant validity support, and apparent utility in distinguishing youths with and without anxiety disorders (Campbell, 2006a; Schroeder & Gordon, 2002; Woodruff-Borden & Leyfer, 2006).

Beck Youth Inventories for Children and Adolescents, Second Edition

The Beck Youth Inventories for Children and Adolescents (BYI-II; Beck, Beck, Jolly, & Steer, 2005) are a recently available group of specific domain measures.

The original Beck Youth Inventories for Emotional and Social Impairment (Beck, Beck, & Jolly, 2001) were composed of five separate 20-item scales (Anxiety, Depression, Disruptive Behavior, Anger, and Self-Concept) that could be used separately or in combination with youths ages 7 to 14 (Beck et al., 2001; Bose-Deakins & Floyd, 2004; Steer, Kumar, Beck, & Beck, 2001). The scales appear to be internally consistent for clinical screening purposes, with growing evidence for convergent validity (Beck et al., 2001; Bose-Deakins & Floyd, 2004; Steer et al., 2001). However, Bose-Deakins and Floyd noted the need for additional discriminant validity data.

Given that the BYI-II was recently published, limited research is available beyond the manual at this time. The BYI-II continues the format of the original (second-grade reading level, five inventories of 20 questions each that may be used separately or in combination) with an expanded age range (7 to 18). Although initial psychometric properties appear promising, additional psychometric data are needed. Although the Anger, Anxiety, Depression, and Disruptive Behavior scales assess potentially problematic areas, it is noteworthy that the Self-Concept scale assesses positive attributes such as self-worth and may be one means of evaluating a youth's strengths. Considering that there are many specific domain measures available to assess personality and psychopathology in youths, it will be intriguing to see where these measures will eventually fit.

Behavior Rating Scales

Behavior rating scales are increasing in availability and popularity among clinicians (Archer & Newsom, 2000; Cashel, 2002; Kamphaus et al., 2000). They are advantageous in terms of their ease of use with a variety of respondents (youths, parents, teachers), cost effectiveness, and coverage of a wide variety of clinical domains such as ADHD and other disruptive behavior disorders (Elliot, Busse, & Gresham, 1993; Kamphaus et al., 2000; Piacentini, 1993). They are useful in providing data about low-frequency behaviors (e.g., assaults) that direct observation sessions may miss, assessing youths who are unwilling or unable to provide information about themselves, and accessing data in youths' typical environments, such as school and home settings, by those who know them well (Merrell, 2003). Potential disadvantages include their being subject to the perceptions of raters (e.g., response biases) and that they tend to focus on current functioning with little guidance as to etiology or intervention strategies (Elliot et al., 1993; Frick & Kamphaus, 2001; Merrell, 2003; Piacentini, 1993).

Rating scale systems frequently used by clinicians are the BASC, currently revised as the BASC-2, the Child Behavior Checklist (part of the Achenbach System of Empirically Based Assessment, or ASEBA), and the Conners Rating Scales (Archer & Newsom, 2000; Cashel, 2002; Kamphaus et al., 2000). For recent reviews of the ASEBA, see Achenbach (2007) and Achenbach and Rescorla (2006); for the Conners Rating Scales—Revised, see Kollins, Epstein, and Conners (2004). The rest of this discussion focuses on the most recently revised of these major rating systems, the BASC-2.

The BASC-2 (Reynolds & Kamphaus, 2004; see also Kamphaus, VanDeventer, Brueggemann, & Barry, 2007) contains separate rating forms for teachers, parents, and youths. The various 100- to 139-item Teacher Rating Scales (TRS) assess adaptive and problematic behaviors evidenced by the youth in the school setting, with the specific form used depending on the child's age (ages 2 to 5, 6 to 11, and 12 to 21). Clinical scales include Aggression, Anxiety, Atypicality, Depression, and Attention Problems. Adaptive scales include Adaptability and Functional Communication. The 134- to 160-item Parent Rating Scales (PRS; ages 2 to 5, 6 to 11, and 12 to 21) prompt parents to provide data on adaptive and problematic behaviors in the home and community. The 139- to 185-item Self-Report of Personality (SRP) forms provide information on the youth's perception of his or her functioning (child, ages 8 to 11; adolescent, ages 12 to 21; college, ages 18 to 25). Clinical scales include Attention Problems, Anxiety, Depression, Hyperactivity, Sense of Inadequacy, and Social Stress, and adaptive scales include Self-Reliance, Self-Esteem, Relations with Parents, and Interpersonal Problems. In general, there are many similarities across forms in terms of clinical and adaptive scales, item content, and composite scores, although there are some variations based on the age of the respondent and the rater (Kamphaus et al., 2007). Raw scores are converted to norm-referenced *T* scores to facilitate interpretation, and each of the sets of forms contains scales to evaluate the quality of obtained data (validity and response set indicators; Kamphaus et al., 2007). The BASC-2 also contains a Structured Developmental History (a data gathering tool) and a Student Observation System (a classroom observation method based on momentary time sampling).

With regard to how a rating system such as the BASC-2 may be used in a psychological evaluation, BASC-2 forms may be used to gather self-reports of youths (SRP), their parents or legal guardians (PRS), and their teachers (TRS). Such ratings generally provide psychometrically sound data to accompany clinical and collateral interviews. When feasible and depending on the clinical context, separate ratings may be sought from each parent, and if the youth has multiple teachers, more than one rater may be requested. Thus, rating scale data could be collected from three to five people (youth, one to two parents or legal guardians, one to two teachers) to get a better idea as to behaviors and symptoms that are considered problematic, to what degree, and in what context. Problems may be occurring at home and not in school, or vice versa; problems may be seen with one parent and not another, or with one teacher but not another. Such differences could be related to the clinical presentation or the environment, which has both diagnostic and treatment implications.

Behavioral Observations in Different Settings

As part of a multifaceted assessment, behavioral observations in the school, home, and other germane settings may be helpful in facilitating the development of a comprehensive picture of the context of the youth's behavior and how he or she interacts with others. Observations may take on a variety of forms, ranging from time-limited and informal to time-involved and intensely structured (Gudas & Sattler, 2006; Winsor, 2003). Although observations may be a rich source of data

that enhance the assessment, similar to interviews the quality of the data is largely dependent on the skill and objectivity of the observer and on the recording system that is employed (Urbina, 2004; Winsor, 2003).

The school is one environment where observations are typically conducted. The observation of older children (as opposed to adolescents, where observation tends to be more intrusive) in their school environment may be undertaken with the permission of the parents or legal guardians, the school, and teachers. With regard to aid in hypothesis generation, is the child inadvertently being reinforced for negative behaviors by the attention of a teacher? Are there alternative behavioral strategies that may be implemented by teachers to reinforce the child for more adaptive behaviors? Collecting behavioral observations may be helpful in certain cases, but they are somewhat limited by factors such as reactivity (behavior change in response to being evaluated) and the lack of ability to monitor internal affective states (Franz & Gross, 1998; Merrell, 2003; Tryon, 1998; Winsor, 2003). Direct observations in the schools are also substantially limited by the amount of time involved. Clinicians working under time constraints at community mental health centers, for example, may be hard-pressed to conduct observations of behaviors for this reason. For further information on direct observations, such as defining target behaviors, and the different forms of coding procedures (e.g., event recording, duration recording, time sampling) and observation systems, many references are available (e.g., Merrell, 2003; Miltenberger, 2001; Tryon, 1998; Winsor, 2003).

PROJECTIVE TECHNIQUES: A PSYCHOMETRIC CONTROVERSY

Projective techniques have been, and continue to be, an enduring aspect of the training and practice of the clinician (Camara et al., 2000; Hogan, 2007; Watkins et al., 1995), and they are commonly employed with children and adolescents (Archer & Newsom, 2000; Cashel, 2002). Projective techniques rely on vague stimuli and a free-response format. The idea is that with freedom to answer, the person's deep-seated personality characteristics will shape the response, revealing important information about underlying desires, motivations, perceptions, and so on (Hogan, 2007; Segal & Coolidge, 2004; Urbina, 2004). Examples of popular projective techniques are the Rorschach, the Thematic Apperception Test (TAT), the Children's Apperception Test, sentence completion forms, and projective drawings such as Human Figure Drawings, House-Tree-Person, and the Kinetic Family Drawing (Hogan, 2007; Urbina, 2004; Watkins et al., 1995).

On the Controversial Nature of Projective Techniques

Some have called into question the psychometric relevance of projective techniques to assessment and diagnosis, noting that they are not science but pseudoscience, and that in the rare cases where they are useful they do not appear to add much beyond other methods, such as interviews and objective inventories (Hunsley et al., 2003; Lilienfeld, Wood, & Garb, 2001). Another criticism centers on the lack of standardization among many of the techniques and the question of how well

clinicians adhere to procedures with the techniques that are standardized (Hunsley et al., 2003). Standardization is necessary to counter problems with subjectivity in administration, scoring, and interpretation. Despite such criticisms, projective techniques continue to be used with great frequency. Proponents argue that they are a source of insight beyond what objective inventories can provide, that they assess personality at a deeper level. Another justification is that they may be useful as a means to settle in to the assessment, perhaps as an "ice-breaker" (Hogan, 2007, p. 522).

Projective Drawings

Projective drawings may be popular for their rapport-building capabilities or as a means of facilitating discussion, because youths often enjoy such activities (Cashel, 2002; Hunsley et al., 2003). Some clinicians use projective drawings to get a sense of the youth. Others use specifics of the drawing, or signs, to make diagnostic inferences about personality and psychopathology (Lilienfeld et al., 2001). The validity of using signs is quite limited (Lally, 2001; Lilienfeld et al., 2001), and the diagnostic relevance of projective drawings to assessments has often been questioned (e.g., Motta, Little, & Tobin, 1993; Palmer et al., 2000; Stawar & Stawar, 1989). Some have argued that they should not be used in personality assessments given their psychometric and diagnostic shortcomings and considering the lack of scientific data on the variability of drawings by age, gender, and interest in the task (Hunsley et al., 2003; Motta et al., 1993). Such data are lacking with regard to multicultural variations in drawings as well.

With regard to formal scoring systems, some clinicians use them, but it does not seem to be a common practice. Structured scoring systems may fare better given the integration of multiple pieces of data (Lilienfeld et al., 2001). However, even the utility of one of the most recent attempts to establish a psychometrically sound, structured scoring system, the Draw-a-Person: Screening Procedure for Emotional Disturbance (Naglieri, McNeish, & Bardos, 1991), has been criticized as flawed, with poor validity and limited peer-reviewed, published data available (Lally, 2001; Motta et al., 1993). Others indicate that the psychometric success of such systems is "modest" (Hogan, 2007, p. 542), or holds "very limited promise" (Hunsley et al., 2003, p. 65). Ultimately, although some use projective drawings as a means of communicating with young clients and in establishing rapport, clinicians must exercise due caution given that the nature of the task pulls for impressionistic interpretation.

Sentence Completion Techniques

Sentence completion techniques also enjoy widespread use, as is evident from surveys of clinical assessment practices. They are not one technique, but rather a group of forms with similar formats, where the respondent completes a series of sentence fragments, or stems, the process of which is thought to reveal something about certain aspects of personality style (Hogan, 2007; Holaday, Smith, & Sherry, 2000; Rogers, Bishop, & Lane, 2003). Given the variety of sentence completion techniques available, clinicians may not be aware which form is best for which client,

the psychometric properties behind the form, or even the name of the form they are using (Holaday et al., 2000; Rogers et al., 2003). There is also some evidence that clinicians tend to use these measures informally, without recognized scoring procedures (Holaday et al., 2000). Interpreting sentence completion techniques through impressionistic methods, that is, based on clinical intuition, without a standardized and psychometrically sound scoring system, will leave clinicians susceptible to errors in decision making.

Storytelling Techniques

The Thematic Apperception Test (Murray, 1943) consists of 31 cards, where the goal is to solicit a narration from the respondent as to the content of the card. Through the narration the respondent is thought to reveal underlying attitudes, drives, and the like, and cards may be selected based on whether the respondent is male or female, a youth or an adult (Hunsley et al., 2003). A major problem with the TAT is standardization, as examiners select different sets of cards (different cards, different numbers of cards, and/or different orders of cards) for use with different examinees (Groth-Marnat, 2003; Hunsley et al., 2003; Lilienfeld et al., 2001). Each set of cards results in a different technique, and psychometric properties need to be established for each set. This property of the TAT also makes the evaluation of research difficult, as studies do not always report the combination of cards that were used. Although some scoring systems are available, there is currently no widely used, well-validated administration and scoring system for the TAT. Thus, those who use it in formal assessments risk misdiagnosis due to reliance on impressionistic interpretation (Hunsley et al., 2003; Lilienfeld et al., 2001). Finally, although a number of derivatives of the TAT have been developed for a variety of populations (e.g., the Children's Apperception Test; Bellak, 1954; see also Bellak & Abrams, 1997), the same caveats apply to the use of any narrative technique with regard to the importance of norms and standardized administration and scoring procedures in generating interpretations that are empirically supported as accurate and clinically useful.

The Rorschach

The Rorschach consists of 10 inkblots (one inkblot per card), and respondents reveal the way they perceive their world through their verbal responses to each card (Ritzler, 2005; Urbina, 2004). There are a variety of coding and scoring systems for the Rorschach; Exner's Comprehensive System (CS; Exner, 2001, 2003) is the most popular (Ritzler, 2005; Weiner, 2005). There are Rorschach proponents (e.g., Ritzler, 2005; Weiner, 2003, 2005), but there are those who raise serious psychometric criticisms. For example, it has been argued that the Rorschach's CS norms may result in erroneous clinical impressions (i.e., respondents may look like they have more pathology than they do or have psychopathology when they do not) and that there is a lack of data and inconsistent psychometric support for many of the CS scores (Garb, Wood, & Lilienfeld, 2005; Hunsley et al., 2003; Lilienfeld et al., 2001; Wood et al., 2002; Wood, Nezworski, Lilienfeld, & Garb, 2003). Psychometric data must be established for each Rorschach score, which is

quite a daunting task given the number of variables that make up the CS system. Psychometric properties will differ from variable to variable.

Some uses have been acknowledged even by critics of the method, as in the case of the Rorschach's apparent ability to assist in the evaluation of thought disorders (Garb et al., 2005; Hunsley et al., 2003; Wood et al., 2003). Beyond these rare points of consensus, however, the discussion is polarized, with impassioned debates characterized by point/counterpoint arguments appearing in various professional journals in recent years. The debate will likely continue for some time and is quite relevant to the assessment of older children and adolescents. It is not uncommon to find those who describe or otherwise support the use of the Rorschach with youths (e.g., Erdberg, 2007; Weiner, 2003, 2005). However, Wood et al. warn that the technique may overpathologize children, which may result in harm due to the possibility of inaccurate conclusions being drawn and a resulting incongruence of fit between the youth and recommended treatment options (Garb et al., 2005; Lilienfeld et al., 2001; Wood et al., 2003).

Projective Techniques: The Bottom Line

Clinicians are advised to carefully consider the arguments and data when selecting measures for formal clinical evaluations. Instruments incorporated into a psychological assessment battery should be selected with the referral questions in mind; be standardized and psychometrically sound with regard to norms, administration, scoring, and interpretive procedures; possess utility for the intended clinical purpose; and be appropriate for the context and the person being evaluated. Clinicians must make their own decisions whether to include projective techniques and be prepared to justify them based on these considerations. Lilienfeld et al. (2001) warn clinicians against the use of projective techniques, except in the rare cases where the merit of scoring and interpretation has been supported. For research-based guidelines on incorporating projective techniques into clinical practice, the reader is referred to Garb, Wood, Lilienfeld, and Nezworski (2002). As for the future of projective techniques, it is important for those who advocate their use to decide on standardized procedures (norms, administration, scoring, and interpretation) so that evaluation of their psychometric properties may take place through well-designed research studies (Hunsley et al., 2003).

THE PSYCHOLOGICAL REPORT: ENHANCING WRITTEN COMMUNICATION

In general, a good psychological report is one that explains the nature of the referral, documents observations of the person's behavior, lists the instruments and techniques used in the assessment and the dates employed, notes reliability and validity issues pertaining to the assessment, clearly and adequately explains results, offers diagnostic impressions, contains a summary and recommendations, and is signed with adequate credentials evident (Braaten, 2007; Brenner, 2003; Groth-Marnat & Horvath, 2006; Harvey, 1997, 2006; Koocher, 2005). The content of the report should be tailored to the individual (Groth-Marnat & Horvath, 2006), and the pros

and cons of diagnosis should be considered. When an integrated assessment supports the presence of a diagnosis, clinicians should make diagnostic distinctions that are appropriate, while also considering the diagnosis that is the least stigmatizing. A diagnosis may be helpful in fostering the implementation of services, as well as respecting the consumer's right to knowledge; however, less severe diagnoses may be associated with less stigmatization and emotional distress, and with youths there is the added danger of others reviewing the report, with certain elements being integrated into other documents and becoming part of the youth's records (Michaels, 2006). Although there are numerous variables to consider when writing a psychological report, a few important points are delineated here relating to inclusion of client strengths, improving readability, and using computerized interpretative narratives. For further general information on report writing, the reader is referred to Groth-Marnat (2003); Lichtenberger, Mather, Kaufman, and Kaufman (2004); and Sattler and Hoge (2006).

Inclusion of Strengths

Personal (e.g., good social skills) and environmental (e.g., solid social network) strengths should be highlighted, as often evaluations remain deficit-oriented and general (Brenner, 2003; Groth-Marnat & Horvath, 2006; Snyder et al., 2006). A separate section on strengths could be included in the psychological report (Groth-Marnat & Horvath, 2006; Snyder et al., 2006), or the material could be integrated into the narrative of the report and become a focal point of the feedback session. Focusing on an individual's assets is perhaps one means of alleviating stigmatization and emotional distress (Michaels, 2006), as well as empowering a youth to make adaptive and effective choices. Incorporating client strengths is a challenge given that it is more labor-intensive during the assessment and report writing phases; however, it results in a more thorough and appropriate case conceptualization (Snyder et al., 2006).

Improving Readability

Psychological reports need to be geared toward the consumer to be of utility (Brenner, 2003; Harvey, 2006). Reports are read by a variety of individuals with varied academic backgrounds, such as psychologists, educators, court employees, and, increasingly, clients and parents of youths (Groth-Marnat & Horvath, 2006; Harvey, 1997, 2006; Snyder et al., 2006). A major problem is that reports continue to be written at a level that is difficult for consumers to understand (Groth-Marnat & Horvath, 2006; Harvey, 1997, 2006), and use of technical terms may result in the misinterpretation of information, which is potentially harmful to the consumer (Michaels, 2006). To increase the readability of psychological reports, difficult and technical words, wordiness, passive verbs, and acronyms should be minimized (avoided if possible), sentences should be shortened, and subheadings employed (Brenner, 2003; Groth-Marnat & Horvath, 2006; Harvey, 1997, 2006). These efforts are critical to writing clear, consumer-oriented reports, although additional report writing time is required, which may be an added stressor for clinicians working in environments characterized by time constraints (Harvey, 2006).

Computerized Interpretive Narratives

Computerized interpretive narratives are often available with the computer scoring packages of personality instruments such as the MMPI-A and the MACI. These computerized narratives must be carefully tailored to the individual being assessed to combat interpretations and language that are hard to understand, lengthiness, and incorrect hypotheses and recommendations (Harvey, 2006). They should also be viewed as one source of data, confirmed by other sources (Michaels, 2006). Clinicians should be thorough consumers of the quality (e.g., validity) of the computerized interpretive packages that they use (Butcher, 2003; Garb, 2000; Snyder, 2000), as clinicians retain responsibility for the applicability of computerized interpretations and the content eventually included in the written psychological report (American Educational Research Association, American Psychological Association, & National Council on Measurement in Education, 1999; American Psychological Association, 2002; Association of State and Provincial Psychology Boards, 2001; Groth-Marnat & Horvath, 2006; Schulenberg & Yutrzenka, 2004).

THE FEEDBACK SESSION: ENHANCING ORAL COMMUNICATION

Conducting feedback sessions is important for cultivating relationships with consumers of assessment services (Brenner, 2003). Feedback sessions afford additional opportunities for clinicians to clarify hypotheses, refine the precision of their reports, work to ensure that the content of the report is understood, and emphasize the importance of following recommendations (Brenner, 2003; Harvey, 2006; Snyder et al., 2006). When communicating assessment results, clinicians are advised to tailor feedback to the educational level of the intended recipient and to be sensitive to cultural issues related to diagnosis and treatment. Communicating in clearly understandable language may be difficult, however, as many terms commonly used by professionals have different meanings from professional to professional, and therefore feedback provided to consumers may be substantially different based on the clinician providing the interpretations (and also by the consumers receiving feedback, who may be relying on their own definitions; Harvey, 2006).

Consumers should feel that they have been heard, that they are a part of the process, and that they have opportunities to pose questions, have concerns addressed, and comment on recommendations (Braaten, 2007; Brenner, 2003; Harvey, 2006; Snyder et al., 2006). Comments from parents or legal guardians and the youth being assessed can encourage the clinician to clarify the findings, particularly if there is a discrepancy between the test results and other data (e.g., collateral information). As for recommendations, they should be presented in a clear, straightforward, and convincing manner, relate to the referral questions, and take into account cultural variables, community resources, client strengths, and the ability of individuals (e.g., parents, teachers) to carry them out (Brenner, 2003; Gudas & Sattler, 2006; Harvey, 2006; Jones et al., 2001; Snyder et al., 2006). Explanation of recommendations is particularly important, as there may be a variety of barriers to implementing them (e.g., transportation, cultural differences; Geffken et al., 2006; Jones et al., 2001).

Potential barriers to following through with recommendations may be examined, and the clinician may problem-solve with consumers to minimize treatment nonadherence (see Geffken et al., 2006, for a discussion of recommendation follow-through).

Parents or legal guardians whose youths are undergoing psychological evaluations typically want to know if there is something "wrong," and they may experience anxiety at the prospect of the evaluation results and have questions about the testing and the implications of the findings. Parents or legal guardians may feel defensive regarding their youth's presenting problems (Lichtenberger et al., 2004), and some may need to be referred for treatment for their own psychological problems. The emotional responses of the parents or legal guardians should be addressed; however, the focus of the feedback session should remain on the youth (Braaten, 2007). Thus, clinicians must work to strike a delicate balance between providing information that is nonjudgmental and supportive, yet is accurate to the situation (Braaten, 2007; Lichtenberger et al., 2004). Starting the feedback session with positive comments about the youth, reiterating these positives throughout the evaluation, focusing on how parents or legal guardians can help their child, and conveying a sense of hope are generally useful strategies (Braaten, 2007).

A more complicated question is when and how to provide personality assessment feedback to the youths themselves. What age is appropriate for youths to hear this type of information? Could this information be detrimental? Young children who are able to read should not review their personality descriptions; however, it is appropriate to provide them with verbal feedback consistent with their developmental level (Lichtenberger et al., 2004). Older children and adolescents may be involved in feedback to the degree possible given a variety of variables, such as their developmental level, parental wishes, and the nature of the feedback. Focusing on a youth's strengths provides a critical starting point on which to build. Involving them in the assessment early on by building good rapport can be helpful later in involving them in the feedback session. Adolescents in particular may be interested in obtaining answers to their questions, and in some cases it is useful to provide feedback to the youth on an individual basis, as well as to the youth and his or her family together (Lichtenberger et al., 2004). Assessment and feedback should provide supportive and tactful illumination on areas of concern, identifying ways that youths may navigate their environments with greater effectiveness.

PSYCHOLOGICAL ASSESSMENT WITH YOUTHS: COMPLICATING FACTORS

Thus far we have presented a thorough model for the assessment of psychopathology in youths. Although the model presented is comprehensive, it should be noted that many clinicians work in different environments or under circumstances that complicate the thorough assessment of psychopathology. We now turn to a discussion of complicating factors: the forensic environment, mental retardation and developmental disabilities, and managed care.

The Forensic Environment

Increasingly, clinicians are conducting psychological assessments in juvenile forensic settings. To be effective working in the forensic environment, clinicians must be able to work with youths, families, attorneys, and legal officials, as well as understand the differences between clinical and forensic roles. Clinical roles tend to be supportive, whereas forensic roles tend to be more neutral. Additional intricacies relate to the referral questions, informed consent and assent, limitations of confidentiality, identifying who exactly is the client, establishing the role of the evaluator, and determining the content and length of the psychological report, as well as who will receive feedback and how feedback will be conducted (e.g., Ackerman, 2006; Butcher & Pope, 2006; Gudas & Sattler, 2006; Koocher, 2006). Is the client an attorney, a youth's parents, or a court? Does the referral relate to a specific legal issue, and if so, how is the assessment of personality relevant? The answers to such questions are important, as they have a bearing on how forensic interviews are conducted to a degree that requires specialized training (Gudas & Sattler, 2006).

With regard to the selection of measures appropriate to the clinical or forensic task at hand, specific guidelines are available (Butcher & Pope, 2006; Heilbrun, 1992), and clinicians should be prepared to justify their decisions (Gudas & Sattler, 2006; Koocher, 2006). In addition, given the forensic nature of the evaluation, the concern that adolescents may malinger is prominent (Butcher & Pope, 2006; McCann, 1998; Steffen & Ackerman, 1999). One means of countering this problem is by using objective personality inventories with validity indicators to aid in the identification of unusual response styles. As part of a comprehensive battery of assessment tools, the MMPI-A has some effectiveness in this regard (Butcher & Pope, 2006; McCann, 1998), and the MACI's utility is also viewed as promising (McCann, 1998, 2006). Thus, both the MMPI-A and the MACI are potentially useful in juvenile forensic settings (e.g., Archer et al., 2006; Butcher & Pope, 2006; McCann, 2006).

Working with Youths with Mental Retardation or Developmental Disabilities

Another issue that may arise is the assessment of psychopathology in youths with mental retardation or developmental disabilities (MR/DD). In an article focusing on working in a high-management group home setting with youths with MR/DD and who also demonstrated sexual offending behaviors, Schulenberg (2003) noted the impossibilities inherent in assessing these youths with self-report measures such as the MMPI-A and the MACI given the reading difficulties with this population. However, the potential assessment utility of having others complete standardized behavior rating scales in this context was noted.

Managed Care

In their review, Wood et al. (2002) noted the apparent impact of managed care on assessment practices, with psychologists facing increasing accountability to justify the utility, cost-effectiveness, and incremental validity of their services. In Cashel's (2002) survey, although the overall impact of managed care on test and technique

utilization was minimal, the largest and most noticeable declines in reported test usage were with measures such as the WISC, Rorschach, TAT, and MMPI-A. Similar findings were noted in Archer and Newsom's (2000) survey, with clinicians reporting less use of several measures as a result of managed care limitations, including the Rorschach, the TAT, the Wechsler scales, and the MMPI-A. For some clinicians, managed care may result in reduced activity in the area of psychological assessment, with fewer measures included in the battery (making batteries less comprehensive) and with diminished reimbursement (Archer & Newsom, 2000). With specific regard to the continued use of projective measures, although they continue to be popular among clinicians, it is possible that they may experience an eventual decline in the area of youth assessment. If such a decline were to occur, it could be due to the psychometric criticisms that have been raised, the increased use of behavior rating scales, and/or managed care (Archer & Newsom, 2000; Cashel, 2002; Groth-Marnat, 2003; Kamphaus et al., 2000).

SUMMARY

Delineating a specific standard for youth assessments is difficult given that they are conducted in a variety of environments (Palmiter, 2004). Our approach in this chapter was to paint a broad stroke with regard to major issues in the assessment of psychopathology in youths. In doing so a variety of issues were covered, beginning with the referral questions, informed consent and assent, rapport building, and measure selection, and continuing with clinical and collateral interviews, review of collateral records, and intelligence and achievement testing. The objective assessment of personality, as well as the psychometric controversy surrounding the use of projective techniques, was also discussed, and the potential utility of behavior ratings scales (administered to youths, their parents, and their teachers) and behavioral observations were underscored. Finally, issues in report writing and feedback were outlined, followed by complicating factors to the assessment process. Given space limitations, it was beyond our scope to provide an in-depth view of each area. For example, despite the emphasis on the importance of psychometric properties, a thorough treatment of each instrument discussed simply was not possible, nor was it possible to discuss all of the numerous measures available in a given area despite clinical popularity or psychometric support in certain circumstances.

To make sense of the wide array of instruments available, clinicians must be trained in test construction and psychometrics. They should be aware of the psychometric pros and cons of their tools and their applicability to the diagnostic question in the specific context of the assessment.

In addition to psychometrics, to competently conduct personality assessments with youths, clinicians must understand relevant developmental, environmental, personality, psychopathological, diagnostic, ethical, and multicultural variables. Assessments should be thorough, taking into account multiple sources of data through a variety of means, to generate hypotheses about a youth's functioning and to arrive at a precise and comprehensive case formulation. Thorough training in the administration, scoring, and interpretation of psychological tests, as well as strength identification, integration of findings, development of individualized

recommendations, and constructive methods of report writing and feedback provision is also necessary. These recommendations are consistent with other sources denoting standards for assessment competency (e.g., Krishnamurthy et al., 2004). For additional information on ethical and professional standards in assessment, see the publications by the American Educational Research Association, American Psychological Association, and National Council on Measurement in Education (1999); the American Psychological Association (1991, 1993, 2002); and the Association of State and Provincial Psychology Boards (2001). For those looking for additional information on psychometric properties of instruments useful in the personality and psychopathological assessment of older children and adolescents, beyond consulting the respective manuals, the *Mental Measurements Yearbook* (www.unl.edu/buros/), a comprehensive database for psychometric reviews of a variety of measures, is a useful place to begin. In addition, readers should consult the respective chapters, texts, and empirical articles that continue to increase in availability.

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CHAPTER 19

Behavioral Assessment

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One of the most important approaches to evaluating children and adolescents is behavioral assessment. No single definition of behavioral assessment exists, but the approach is often thought of as a problem-solving-oriented means of gathering information about current, observable behavior and what maintains behavior over time. Behavioral assessment is an idiographic approach involving detailed evaluations of specific behavior to develop a prescriptive treatment plan. A prescriptive treatment plan is one tailored to the individual needs of a given child. Although behavioral assessors derive from different traditions within psychology and other professions, the overarching goal of designing a customized treatment regimen for a particular child has remained largely unchanged over time (Mash & Terdal, 1997).

Behavioral assessment methods focus on *forms* of behavior that can be clearly defined and monitored. Examples include aggression in the form of slapping a classmate and fear in the form of avoiding a large dog. Frequency, duration, intensity, and other topographies of behavior may be assessed as well. Behavioral assessment methods also focus on *functions* of behavior, or antecedents and rewarding consequences that maintain behavior. Examples include teacher attention after slapping a classmate and reduction of unpleasant physical sensations after avoiding a large dog.

Behavioral assessment has been differentiated from global or traditional forms of assessment in several ways (Reitman, 2006). First, behavioral assessment emphasizes current, contextual factors in a child's life that are well defined and observable. Global forms of assessment such as intelligence and personality tests are designed to obtain information about general characteristics of a child such as problem-solving ability or psychological discomfort. In contrast, methods of behavioral assessment are designed to garner information about *molecular* characteristics of a child, such as length of attention when listening to a teacher command and amount of time spent in school.

Second, behavioral assessment emphasizes instability of behavior across settings. Global forms of assessment assume a more trait-like approach to behavior, wherein a child's cardinal characteristics, such as extraversion, will largely determine actions across settings. In contrast, behavioral assessors assume a child's behavior to be more situation-specific. A child may, for example, be relatively

gregarious and talkative with family members at home but somewhat inhibited and soft-spoken in peer-based social situations.

Third, behavioral assessment emphasizes multiple samplings of behavior to understand a child's actions. Global forms of assessment such as achievement tests are often given once to obtain information about general characteristics of a child, such as academic status. In contrast, behavioral assessors employ *ongoing* techniques, such as daily logbooks, to examine patterns of actions across settings and over time. In related fashion, behavioral assessment relies less on historical factors and more on present aspects of behavior and commonly involves multiple sources of information.

Fourth, behavioral assessment emphasizes direct, on-site methods such as behavioral observation and self-monitoring to gather information about specific aspects of a child's behavior. Global forms of assessment include more circuitous analyses of a child's behavior, such as a neuropsychological test in an office setting to indirectly assess brain dysfunction. In contrast, behavioral assessors actively evaluate a child's behavior in direct, natural environments such as home, school, playground, and social events.

Finally, behavioral assessment may be more appropriate for children with diverse ethnic backgrounds compared to traditional assessment. Because behavioral assessment emphasizes concrete and proximal influences on behavior, such as learning processes, the process may be freer of cultural bias compared to standardized tests. In addition, behavioral assessment techniques can be quite useful for identifying different behaviors across different cultural contexts (Castillo, Quintana, & Zamarripa, 2000; Suzuki, Alexander, Lin, & Duffy, 2006).

This chapter covers a central model of behavioral assessment, functional analysis, in addition to specific techniques endemic to the model. These techniques include interview, analogue assessment, behavioral observation, rating scales, self-monitoring, sociometric assessment, physiological assessment, and activity measures.

FUNCTIONAL ANALYSIS

A central model of behavioral assessment is functional analysis, defined as examining and manipulating antecedents and consequences of a behavior of interest to provide information about the relationship of these events to the behavior (McComas, Hoch, & Mace, 2000). Functional analysis is often linked to terms such as functional assessment and descriptive and experimental functional analysis. *Functional assessment* is a broader term that refers to developing general hypotheses about the function of problem behavior (Gresham & Lambros, 1998). A clinician addressing a child with temper tantrums in public, for example, might develop a hypothesis that the behavior is driven by attention and tangible rewards from parents. In this case, the child's parents may be inadvertently rewarding the tantrums by quickly attending and providing candy to quash the behavior.

Functional assessment typically consists of descriptive and experimental functional analysis. *Descriptive functional analysis* refers to indirect methods of gaining information about the relationship of a behavior to antecedent and consequent

events. Typical methods of assessment in descriptive functional analysis are interviews, rating scales, review of records, self-monitoring, and naturalistic observations. *Experimental functional analysis* refers to direct methods of gaining information about the relationship of a behavior to antecedent and consequent events. Experimental functional analysis involves standardized procedures to identify key maintaining variables of behavior and rule out competing explanations for why behavior continues to occur (Mace, 1994).

Functional analysis has been a key staple for assessing and treating developmental and disruptive behavior disorders in youth. Functional analysis is typically utilized for externalizing problems such as self-injurious behavior, aggression, tantrums, property destruction, and pica (Hanley, Iwata, & McCord, 2003). Functional analysts typically assess maintaining variables of behavior, such as sensory reinforcement, escape from aversive situations, attention, and tangible rewards. However, functional assessment may be broadened to include other contextual variables that trigger and reinforce behavior. Such assessment usually leads to effective treatments; such as training persons with developmental disorders to communicate desires more effectively so maladaptive behavior is not necessary (Durand & Merges, 2001).

Two examples of functional assessment using descriptive and experimental procedures are presented here. Field and colleagues (Field, Nash, Handwerk, & Friman, 2004) evaluated a 12-year-old male with disruptive behavior that included noncompliance and other misbehaviors in a residential group home. Noncompliance was defined as opposition to commands, rules, or stated expectations within the home. Descriptive functional analysis to garner hypotheses about the boy's behavior was conducted via teacher interviews, case file review, behavioral observations, and an interview with the boy. These data led the authors to hypothesize that attention was largely responsible for noncompliance. Specifically, attention typically followed noncompliance, and the boy often escalated noncompliance to force attention from others.

Experimental functional analysis then consisted of a reversal design that alternated two treatments. Standard treatment within the group home, which consisted of token economy and teacher interactions, was compared to modified treatment involving increased positive teacher interactions with the boy. Modified treatment led to a substantial decrease in noncompliance, thus supporting the original hypothesis that the misbehavior was reinforced by attention. In this case, the experimental functional analysis assessment strategy also consisted of effective intervention.

Kearney designed a functional assessment model for school refusal behavior, a behavior with many internalizing components. In this model, youths are thought to refuse school for one or more of several reasons: to avoid school-based stimuli that provoke negative affectivity, to escape aversive social and/or evaluative situations, to pursue attention from significant others, and to pursue tangible rewards outside of school. Descriptive functional analysis consists of a structured diagnostic interview, in-session behavioral observations, and a rating scale (School Refusal Assessment Scale—Revised) that measures relative influence of each of the four hypothesized functions of school refusal behavior (Kearney, 2002, 2006).

Procedures for experimental functional analysis include establishing parameters for a child's school day that best predict attendance and nonattendance (Kearney & Albano, 2007). If a child were suspected of refusing school to escape an aversive evaluative situation, such as tests, then the child's attendance could be evaluated on typical school days and days when the child is exempted from tests. Similarly, if a child were suspected of refusing school for tangible rewards outside of school, then attendance could be compared for typical school days and days when parents provide substantial tangible reinforcement for being in class. A functional assessment approach has been shown to be effective for predicting which prescriptive treatment strategy is best for youths who refuse school for different functions (Chorpita, Albano, Heimberg, & Barlow, 1996; Kearney, Pursell, & Alvarez, 2001; Kearney & Silverman, 1999).

The remaining portion of the chapter focuses on methods of behavioral assessment with children that may be used within a functional analytic model. These methods include interviews, analogue assessment, behavioral observation, rating scales, self-monitoring, sociometric assessment, physiological assessment, and activity measures.

BEHAVIORAL ASSESSMENT MEASURES

Interviews

Interviews with children and parents are the most common form of assessment used by mental health professionals. Although many professionals use unstructured interviews, clinicians and others who emphasize behavioral assessment tend to utilize structured interviews that are psychometrically stronger. In the area of behavioral assessment, structured interviews may be diagnostic or functional in nature. Structured diagnostic interviews are typically linked to *DSM* criteria for mental disorders (American Psychiatric Association, 2000). The interviews are designed to provide detailed information about the form of a child's mental disorder and help an assessor derive a formal diagnosis.

Diagnostic interviewing is discussed in more length in another chapter of this volume, but a brief list of measures is presented here:

- Anxiety Disorders Interview Schedule for DSM-IV: Child and Parent Versions (Silverman & Albano, 1996)
- Child and Adolescent Psychiatric Assessment (Angold & Costello, 2000)
- Child Assessment Schedule (Hodges, McKnew, Cytryn, Stern, & Kline, 1982)
- Children's Interview for Psychiatric Syndromes (Weller, Weller, Fristad, Rooney, & Schecter, 2000)
- Diagnostic Interview for Children and Adolescents (Welner, Reich, Herjanic, Jung, & Amado, 1987)
- Interview Schedule for Children (Kovacs, 1985)
- National Institute of Mental Health Diagnostic Interview for Children (Shaffer et al., 1996)

- Schedule for Affective Disorders and Schizophrenia for School-Age Children: Present and Lifetime Version (Kaufman et al., 1997)

Each of these interviews has demonstrated good reliability and validity. Though most cover externalizing disorders in youth in great detail, less attention has been paid historically to internalizing problems. However, the Anxiety Disorders Interview Schedule for DSM-IV: Child and Parent Versions (ADIS: C/P) has been recognized as the gold standard for assessing anxiety-related disorders in youth. In addition, the interview contains sections for other internalizing disorders, such as depression, as well as sections for externalizing, eating, developmental, and psychotic disorders. A section for school refusal behavior is also available and concentrates on yes/no questions regarding whether a child has problems attending school, whether a child is nervous when attending school, whether a child is upset about performing poorly in school, and whether a child has been medicated for nonattendance.

Structured diagnostic interviews have strengthened the child assessment process and are commonly used to steer treatment and evaluate outcome. Problems with these interviews have been noted, however. First, the interviews provide an overall view of a child's difficulty but may not provide specific information necessary for developing a detailed treatment plan. Second, the interviews are not particularly sensitive to developmental or cultural differences. An older adolescent with Attention-Deficit/Hyperactivity Disorder (ADHD), for example, is likely to present with different symptoms than a younger child with the disorder. In related fashion, a Hispanic child and a European American child may differ with respect to symptoms of Separation Anxiety Disorder. Third, the interviews are not generally designed for ongoing use or for multiple sources of information. A snapshot of a child's behavior is obtained, but behavioral changes that occur over time may not be examined (Kearney, 2005).

Interviews within a behavioral assessment paradigm may also be functional in nature and help address problems of structured diagnostic interviews. These interviews may be unstructured or structured, but more specifically concentrate on operationally defined forms of behavior and what maintains behavior over time. These interviews can be shaped in various ways to accommodate different developmental levels and can provide detailed information to help a clinician form a unique treatment plan.

Behavioral and functional interviews may be given to gather multiple sources of information; a sample excerpt with a parent is presented here. Contrast these questions with the general yes/no questions for the ADIS: C/P:

Therapist: What specifically does Brian do on the playground before school?

Parent: He just won't go into school.

Therapist: Okay, what does he say and do during that time?

Parent: He cries, clings to my leg, and sometimes tries to run away. When the teacher comes out to line everybody up for school, he runs behind me and holds tight.

Therapist: Does he do this every day?

Parent: No, not every day, but I notice it's worse on Mondays and Tuesdays or after a long break from school.

Therapist: How do you and the teacher react when Brian does this? What specifically do you and she do?

Parent: I try to comfort and reassure him as much as possible, giving him a lot of hugs even as the other kids are going into the school. The teacher tells Brian to come into the school but has to deal with the other kids, so she often leaves me and Brian behind.

Therapist: Does the extra attention and reassurance you give Brian help him go to class?

Parent: No, he just clings more and cries and I feel so bad that I just take him home.

Note that these questions are very specific about one episode of school refusal behavior. Questions surround the form of Brian's particular behaviors, when the behaviors occur, and what reinforcer (attention, in this case) might be maintaining the behavior. These kinds of questions can be asked of different people and at various times during the assessment and treatment process. The questions can also be tailored to the developmental level of a child. If Brian were an older child, for example, questions about his cognitions prior to entry into school might be particularly salient.

Behavioral and functional interviews are flexible enough to cover a wide range of material relevant to a particular case, especially the following topics:

- Specific behavioral, cognitive, and physiological aspects of a child's difficulty
- Effects of misbehavior on a child's and family's daily functioning
- Current life stressors
- Contextual or distal variables or risk factors that might influence a child's behavior, such as family dynamics, marital conflict or divorce, poor financial and social support, maltreatment, school-based threats, temperament, and physical and medical status
- Antecedents and consequences pertaining to specific episodes of misbehavior, such as sensory reinforcement, negative reinforcement, attention seeking, and pursuit of tangible rewards
- Child's skill levels in various areas, such as social, verbal, and academic
- Pertinent cultural variables such as language differences, normative behaviors for a particular culture, and perspectives on child misbehavior

Behavioral and functional interviews can be more structured as well. A good example is the Functional Diagnostic Profile modified for youths with selective mutism by Schill and colleagues (Schill, Kratochwill, & Gardner, 1996). The assessment protocol initially consisted of soliciting general information about a particular child with selective mutism to derive hypotheses about the function of the child's behavior. In this case, several hypotheses were formed. The child's mutism

was thought to be potentially related to several conditions: (a) escaping aversive requests, directives, or task demands; (b) avoiding attention from unfamiliar others; and (c) deriving attention from familiar others. Following this assessment process, the researchers engaged in analogue assessment to experimentally support these hypotheses and design individualized treatment that eventually led to increased verbalizations. A greater explication of analogue assessment is described next.

Analogue Assessment

Analogue assessment refers to behavioral observations typically conducted in a controlled setting, such as a therapist's office or research laboratory (Barkley, 1991). The observations are often done within a behavioral assessment paradigm to confirm hypotheses about the function of behavior. The premise behind analogue assessment is that a child's behavior in the contrived setting will mirror behavior in the child's natural environment. In the case presented by Schill and colleagues (1996), the child's mutism was explored under several conditions: allowing the child to escape a demand, asking the child to read before others, and providing positive reinforcers for speech such as play and parent praise. The child's responses under this set conditions helped the researchers confirm expectations about what maintained selective mutism.

A popular form of analogue assessment is a *behavioral assessment test* or *behavioral avoidance test,* in which a child role-plays a given situation or performs in some way before others. Behavioral assessment tests are commonly used to evaluate anxiety- and fear-based behavior in children. Among children with Social Phobia, for example, behavioral assessment tests may include conversing with others, writing or solving problems before others, speaking before others, reading a story aloud before others, and taking tests, such as subsections of an intelligence or achievement test (Beidel, 1991; Beidel, Turner, & Morris, 1999).

Behavioral assessment or avoidance tests may be used as well for youths who fear specific stimuli. A test in this situation might involve asking a child to approach a feared animal as close as possible, or spend as much time as possible in school. As a child approaches a feared stimulus, ratings of discomfort may be solicited and distance to the stimulus can be measured. Behavioral assessment or avoidance tests have been used to evaluate youth phobic of dogs, snakes, water, blood, heights, medical procedures, strangers, school-related events, and darkness, among other stimuli (Barrios & Hartmann, 1997).

Behavioral assessment tests have also been used to evaluate externalizing behavior problems in children. Children with ADHD, for example, have been subjected to controlled observations of their behavior in playrooms divided into grids to measure excess movement across the grids. Analogue classrooms with desks have also been used to measure attention problems, out-of-seat behavior, and inappropriate vocalizations. Analogue assessment procedures can help evaluate social skills, aggression, noncompliance, and academic behaviors such as completing worksheets (Barkley, 1991, 1997; Elliot & Gresham, 1991; Gable, Hendrickson, & Sasso, 1995; M. A. Roberts, 1990). Parent-adolescent conflict and interpersonal problem solving may be evaluated in this manner as well (Goldstein, 1999; Robin & Foster, 1989).

Traditional analogue assessments are common but inconsistent methods for evaluating children's behavior. The assessments have several advantages, including control of stimulus conditions, practicality, cost-effectiveness, and evaluation of multiple motor and other behaviors at one time. However, the assessments have variable test-retest reliability and construct validity because they have not been well standardized (Mori & Armendariz, 2001).

An increasingly popular form of analogue assessment among adults is *virtual reality assessment,* which involves placing a person in a digitally created environment that mimics a particular setting and allows for interaction within that setting. A person afraid of heights, for example, may interact in virtual scenes involving progressively higher places. The assessment can thus be conducted quite confidentially, can be manipulated to include many different contexts, and is amenable to examining different response systems at one time. Virtual reality assessment for children remains largely in development, with particular focuses on eating disorders, Social Phobia, ADHD, learning difficulties, and pain (Lannen, Brown, & Powell, 2002; Riva, Wiederhold, & Molinari, 1998; Rizzo et al., 2006).

Behavioral Observation

Analogue assessments are contrived observations often preferred because of their practicality. Direct naturalistic observations, however, represent the purest form of behavioral assessment and functional analysis. Analogue assessments often target a specific area of interest, such as approach toward a feared stimulus, whereas direct naturalistic observations usually involve a much wider range of target behaviors and contextual variables.

Behavioral observations typically concentrate on specific topographies of overt behaviors. The process typically begins by defining a behavior of interest in operational terms. Aggression, for example, may be operationally defined as a slap to the head of a classmate or verbal insult given to a sibling. In related fashion, aspects of a certain behavior may be observed and recorded. Behavior may be recorded with respect to intensity, duration, frequency, and latency. A rater may record the length of time between a teacher's command and a child's compliance. Behavior accuracy, such as number of mistakes on a spelling test, can be assessed as well.

Behavioral observations also include contextual variables or events that surround and influence behavior. Many researchers observe and code valence, which may involve classifying a behavior into positive, negative, neutral, and other categories. In addition, key antecedents and consequences to behavior are commonly observed for purposes of functional analysis. A child who slaps a classmate may do so with an angry facial expression (negative valence) immediately after a frustrating test (antecedent) and prior to a harsh teacher reprimand and attention from classmates (consequences; Watson & Steege, 2003).

Direct observation can involve simple systems of narrative recording as well as more complex research-based observational systems. Narrative recording involves simple written accounts of behavior as it transpires, though more sophisticated accounts may include recording topographies such as frequency and intensity. Raters

who are observing and interacting with a child simultaneously may find it difficult to maintain an ongoing log of transpired behavior, so interval recording is common. In this system, a rater may simply record the presence or absence of a behavior in blocks of time, such as every 20 seconds or 5 minutes (Skinner, Rhymer, & McDaniel, 2000).

Behavioral observations in research settings typically involve greater complexity. Many have been designed to evaluate youths with disruptive behavior disorders because of the overt nature of problems such as noncompliance. Some of these observational systems are highly contrived and qualify more as analogue assessments. These often involve analyses of free play, parent-directed play, and parent-directed chores (M. W. Roberts, 2001). Research-based observation systems for more naturally occurring behavioral phenomena have also been designed, however. These most prominently include systems for evaluating children's social and coping skills, play, and parent-adolescent communications (Anderson, Moore, Godfrey, & Fletcher-Flinn, 2004; Margolin et al., 1998; Pellegrini, 2001; Pretzlik & Sylva, 1999).

Whether simple or more complex behavioral observations are used, a key goal is to derive interrater reliability or consistency of ratings across two or more observers. At a basic level, interrater reliability can be calculated as the number of agreements divided by the number of agreements plus the number of disagreements times 100. Thus, 12 agreements and 3 disagreements would result in interrater reliability of 80%. At more complex levels, interrater reliability can be calculated in several ways, though a predominant way is Cohen's kappa.

Direct observations of behavior carry several strong advantages. Observations can supply a plethora of data regarding forms and functions of behavior and are often essential components of a detailed assessment plan for complex behaviors. In addition, observations in natural settings have high external validity. Observations can supply data about the molecular characteristics of a given behavior, such as a child's problem-solving method when completing a math worksheet or subtle changes in tone during parent-adolescent communications. Knowledge about these molecular characteristics is often invaluable for designing an individualized treatment plan.

A primary disadvantage to behavioral observation is that covert or low-frequency behaviors are not amenable to the process. Behavioral observation is not particularly useful for assessing internal mood states, visceral responses, or secretive behaviors. In addition, behavioral observations are subject to reactivity and other demand characteristics that could bias the assessment process. Observer drift, where raters deviate from the original operational definition of a targeted behavior, can reduce validity of the process as well. Finally, behavioral observations require substantial time, effort, training, and, ideally, multiple raters. As such, observations are not as practical or cost-effective as rating scales, which are described next.

Rating Scales

A particularly well-developed area of behavioral assessment for children is rating scales. *Rating scales* or questionnaires are instruments wherein children and

significant others endorse the presence or absence of various physiological, cognitive, and behavioral symptoms. Rating scales have several strong advantages, including ease of use, good reliability and validity, excellent coverage of many topics, sensitivity to treatment effects, standardization, and extension to multiple sources of information, such as parents and teachers. In addition, rating scales are good for assessing low-frequency behavior and behavior not amenable to direct observation, such as internalizing constructs of worry or sadness.

Rating scales also have several disadvantages, however. A key problem is rater bias, especially among parents and teachers, who may rate a child more negatively or positively than is actually the case. Measurement error is endemic to rating scales as well, and can come from variance in behavior across time as well as variable rater, setting, and instrument characteristics. In addition, rating scales have poor applicability to very young children and correlate inconsistently with real-life behavior. Developmental differences between children and adolescents are often not fully considered. Finally, rating scales provide summaries of perceptions of behavior rather than direct measures of actual behavior (Merrell, 2000; Myers & Winters, 2002a).

A full explication of all rating scales in all areas of child functioning is beyond the scope of this chapter, but a summary is presented here of the major rating scales available for children, parents, and teachers. Measures were chosen based on their psychometric strength, widespread use in the field, and applicability to designing an individualized treatment plan for a child. In addition, only the most prevalent areas of mental disorder among children and adolescents are covered here: anxiety, depression, ADHD and other disruptive behavior disorders, and developmental disorders. Primary measures in other areas are listed as well.

Anxiety Disorders

Anxiety disorders are the most common mental disorders in children, and research into the assessment and treatment of these problems has burgeoned in the past 2 decades. Because anxiety is largely an internalizing problem, the development and use of rating scales has become widespread in this area. Developers of anxiety rating scales must naturally contend with a phenomenon that changes quickly over time, but several scales appear to have excellent psychometric strength (Kearney & Bensaheb, 2007; Myers & Winters, 2002b).

Rating scales for anxiety disorders may be divided into traditional measures that assess broad swaths of anxiety and more recent measures that specifically target more narrow types of anxiety. Traditional child self-report measures include the Revised Children's Manifest Anxiety Scale (RCMAS), State-Trait Anxiety Inventory for Children (STAIC), and Fear Survey Schedule for Children—Revised (FSSC-R). The RCMAS measures physiological anxiety symptoms, worry and oversensitivity, and fear and concentration. The STAI-C assesses situation-specific and global anxiety. The FSSC-R contains subscales for fear of failure and criticism, injury and small animals, danger and death, medical topics, and a category of unknown fears (Ollendick, 1983; C. Reynolds & Richmond, 1985; Spielberger, 1973). Recent measures of general anxiety constructs include the Multidimensional Anxiety Scale

for Children, which measures social anxiety, separation anxiety and panic, physical anxiety, and harm avoidance, and the Screen for Child Anxiety Related Emotional Disorders, which measures somatic and panic symptoms, general and separation anxiety, and school and social phobias (Birmaher et al., 1999; March, 1997).

Though useful, traditional measures of anxiety are gradually becoming replaced by more specific measures that cover particular aspects of anxiety. With respect to social anxiety, for example, the Social Anxiety Scale for Children—Revised (SASC-R) and Social Phobia Anxiety Inventory for Children (SPAI-C) are most common (Beidel, Turner, & Morris, 1995; La Greca & Stone, 1993). These measures cover fear of negative evaluation and social avoidance and distress (SASC-R) as well as assertiveness and concerns about social encounters and public performance (SPAI-C).

With respect to Obsessive-Compulsive Disorder, the Leyton Obsessional Inventory—Child Version is a brief rating scale for compulsions, obsessions and incompleteness, and cleanliness (Bamber, Tamplin, Park, Kyte, & Goodyer, 2002). Finally, the Trauma Symptom Checklist for Children is useful for measuring children's reactions to trauma and the Child PTSD Symptom Scale is useful for measuring criteria necessary for a diagnosis of Posttraumatic Stress Disorder (Briere, 1996; Foa, Johnson, Feeny, & Treadwell, 2001).

Depression

Depression is a highly prevalent disorder among youth in general and adolescents in particular. The most robust and common child self-report measure of depression is the Children's Depression Inventory (CDI), which purportedly measures symptoms of negative mood, interpersonal problems, ineffectiveness, anhedonia, and negative self-esteem (Kovacs, 1992). Factor analysis of the measure, however, has yielded constructs for dysphoria-sadness, social self-esteem, and oppositional-misbehavior (D. Cole, Hoffman, Tram, & Maxwell, 2000). The CDI has been used extensively to assess youths with depression, to distinguish depressed youths from other populations, and to examine developmental, psychological, and ethnic differences in children and adolescents. Another common child self-report measure of depression is the Reynolds Child (and Adolescent) Depression Scales, which are based on *DSM* criteria (W. Reynolds & Mazza, 1998).

Internalizing problems of childhood, which include anxiety and depression, can also be measured via parent and teacher ratings. The most common measures for doing so are the Child Behavior Checklist (CBCL), Teacher Report Form (TRF), Conners Rating Scales—Revised (CRS-R), and Child Symptom Inventory 4 (CSI-4; Achenbach & Rescorla, 2001; Conners, 1997; Sprafkin, Gadow, Salisbury, Schneider, & Loney, 2002). The CBCL and TRF have an Anxious/Depressed subscale. The CRS-R has subscales for Anxious-Shy and Perfectionism, and the CSI-4 has subscales related to generalized, separation, and social anxiety.

Attention-Deficit/Hyperactivity Disorder

In contrast to anxiety and depressive problems, externalizing problems of childhood are more commonly rated by parents and teachers. A popular rating scale for

ADHD is the ADHD Rating Scale IV, which measures symptoms of hyperactivity-impulsivity and inattention in home and school settings. The scale is linked to *DSM-IV* criteria and can successfully distinguish youths with and without ADHD (DuPaul, Power, Anastopoulos, & Reid, 1998). Related measures include the ADHD Symptoms Rating Scale and Attention Deficit Disorders Evaluation Scale, second edition (Holland, Gimpel, & Merrell, 2001; McCarney, 1995). In addition, the Strengths and Weaknesses of ADHD Symptoms and Normal Behavior has been designed as a dimensional measure of ADHD symptoms. Items are presented in Likert-scale format so raters can judge the strength of certain symptoms such as inattention (Collett, Ohan, & Myers, 2003).

General rating scales such as the CBCL, TRF, and CRS-R are also commonly used to assess symptoms of ADHD. The CBCL and TRF have a subscale for Attention Problems, and the CRS-R has several ADHD subscales for Cognitive Problems/Inattention, Hyperactivity, and an ADHD index. The CSI-4 is useful for examining symptoms of ADHD across settings. The Conners ADHD/DSM-IV Scales may also be used to assess more specific symptoms of ADHD, such as trouble concentrating and difficulty playing quietly (Conners, 1999). Finally, the Behavior Assessment System for Children, second edition (BASC-2) allows for parent and teacher ratings of many different behaviors from the preschool period to adolescence. Relevant subscales regarding ADHD include Attention Problems and Hyperactivity (C. Reynolds & Kamphaus, 1998).

Other Disruptive Behavior Disorders

General scales for ADHD also contain subscales relevant to other disruptive behavior disorders such as Oppositional Defiant and Conduct Disorder. The CBCL and TRF contain subscales for Rule-breaking and Aggressive Behavior, the CRS-R contains an Oppositional subscale, the CSI-4 contains subscales for Oppositional Defiant and Conduct Disorder, and the BASC-2 contains subscales for Aggression and Conduct Problems.

Measures more specifically designed for oppositional and conduct problems have also been developed. The Eyberg Child Behavior Inventory (parent) and Sutter-Eyberg Student Behavior Inventory—Revised (teacher) evaluate the number of difficult behavior problems and the frequency with which they occur (Eyberg & Pincus, 1999). The ACQ Behavior Checklist is an amalgam of items from the CBCL, Conners parent scale, and Quay-Peterson Revised Behavior Problem Checklist that assesses aggressive and delinquent behavior (Achenbach, Howell, Quay, & Conners, 1991; Quay & Peterson, 1996). Finally, the Conduct Disorder Scale contains ratings and yes/no questions regarding key aspects of oppositional and conduct problems such as argumentativeness and cruelty to others (Gilliam, 2002).

Developmental Disorders

Rating scales for developmental disorders such as autism and mental retardation have also been widely developed in recent years. These scales are typically completed by clinicians or teachers. A popular one for autism is the Childhood Autism Rating Scale, which solicits ratings for difficulty relating to people, odd body

movements, intense reactions to change, and verbal and nonverbal communication (Schopler, Reichler, & Renner, 1988). In addition, the Autism Screening Instrument for Educational Planning 2 (ASIEP-2) assesses for sensory, relating, body concept, language, and social self-help behaviors. The Autism Behavior Checklist is part of the ASIEP-2 and can be completed by parents and teachers (Krug, Arick, & Almond, 1995). Other prominent measures include the Gilliam Asperger's Disorder Scale, Gilliam Autism Rating Scale, second edition, and the Social Communication Questionnaire (Gilliam, 2001, 2006; Rutter, Bailey, & Lord, 2003).

Clinician-administered rating scales are commonly used to assess adaptive behavior in youths with mental retardation. A predominant measure is the Vineland Adaptive Behavior Scales, second edition, which assesses communication, daily living skills, socialization, motor skills, and maladaptive behavior (Sparrow, Cicchetti, & Balla, 2005). In addition, the American Association on Mental Retardation's AAMR Adaptive Behavior Scale—School (second edition) includes domains of independent functioning, physical development, economic activity, language development, prevocational and vocational activity, numbers and time, self-direction, responsibility, socialization, and maladaptive behavior (Nihira, Leland, & Lambert, 1993). Other prominent measures in this area include the Adaptive Behavior Assessment System (second edition) and Scales of Independent Behavior—Revised (Bruininks, Woodcock, Weatherman, & Hill, 1996; Harrison & Oakland, 2003).

Other Disorders

A full explication of all rating scales for all other disorders and problems that apply to children and adolescents is beyond the scope of this chapter, but some prominent ones are listed here:

- Adolescent Substance Abuse Screening Inventory (Miller, 2001)
- Dyslexia Screening Instrument (Coon, Waguespack, & Polk, 1994)
- Eating Attitudes Test (Foreyt & Mikhail, 1997)
- Pediatric Pain Questionnaire (Varni, Thompson, & Hanson, 1987)
- Scales of Social Competence and School Adjustment (Walker & McConnell, 1995)

Self-Monitoring

Self-monitoring refers to observing and recording one's own behavior as well as antecedents and consequences to the behavior. At a basic level, self-monitoring involves asking a child to record whether a specific behavior occurred or did not occur during a certain time period. A child may, for example, record whether he or she fully completed homework during a prescribed evening time. Or a child may record the number of hours spent in school per day. In addition, a child may record events just prior to and following a behavior, such as parent prompts to do homework and playing with friends after school. Recordings are typically done in paper-and-pencil fashion but can be done via self-analysis of audiotaping or videotaping.

Self-monitoring is utilized within a behavioral assessment paradigm for several reasons. First, self-monitoring is intricately linked to self-regulation, as people more aware of their own behavior often become better at controlling the behavior. This is especially true of excess maladaptive behavior; the more one self-monitors and becomes aware of excess maladaptive behavior, the less one tends to engage in the behavior. Self-monitoring is often linked to self-regulation processes of goal setting, self-evaluation, and self-reinforcement to reduce maladaptive behavior (Agran & Wehmeyer, 2006).

A salient example is habit reversal training to treat Trichotillomania (compulsive hair pulling) in children. One component of habit reversal training for this population is awareness training in which a child is instructed to increase attention toward subtle movements that eventually lead to hair pulling. Awareness training may be supplemented with self-monitoring and other behavioral assessment measures to modify behavior associated with less conscious thought (Azrin, Nunn, & Frantz, 1980; Romaniuk, Miltenberger, & Deaver, 2003).

Second, self-monitoring is important for educating youths about a targeted behavior. For example, psychoeducation for anxious children includes a detailed explication of the three response systems of anxiety: physiological, cognitive, and behavioral. A clinician treating an anxious child may ask him or her to provide daily ratings of physiological distress as well as lists of problematic thoughts and avoided situations. These personal examples are then used to explain how one anxiety response system leads to another. In one case, for example, a child's aversive physiological symptoms may lead to worry about harm from the symptoms and subsequent avoidance of a certain situation (Kearney & Albano, 2007).

Third, self-monitoring is important for assessing treatment outcome. Global measures of behavior are not particularly sensitive to treatment changes over time, but measures such as daily ratings of distress, hours spent in school, frequency of aggressive acts, and words spoken audibly can provide much more information about patterns of behavior change across time (Kearney & Silverman, 1999). Finally, self-monitoring may be useful for assessing secretive behaviors and helps alleviate pressure on others, such as teachers, to supply data for behavioral analysis.

Basic self-monitoring, such as recording the presence or absence of a behavior, may be more appropriate for younger children, those with a singular, primary behavior problem, and those who need initial training in how to self-monitor behavior. More sophisticated self-monitoring strategies can involve recording multiple behaviors, providing ratings of severity and intensity of the behaviors, and rating internal sensations or mood states. Children and adolescents are generally considered to be the best reporters of their own internal mood states, so self-monitoring is particularly relevant to youths with anxiety and depressive disorders. However, self-monitoring has also been used to assess aggression, socially appropriate behavior, and academic tasks such as paying attention, staying seated, and completing work problems accurately (C. Cole, Marder, & McCann, 2000).

Formal methods of self-monitoring have been developed for anxious children (Beidel, Neal, & Lederer, 1991). These methods involve daily diaries in which children list anxiety-provoking situations, time and setting of the situations, and responses to the situations. Children may also be asked to endorse an illustration

that best represents how they felt in a given situation and to rate their level of avoidance, physical symptoms, and coping skills. Daily diaries such as these have adequate reliability, but a key problem is that children are not typically compliant with the task, nor do they tend to engage in the task for more than 2 weeks (Beidel et al., 1999). In addition, the quality and accuracy of self-monitoring data may be questionable given children's limited cognitive development, distractibility, and lack of cooperation.

To address these concerns, clinicians should carefully tailor self-monitoring to the individual needs of a given child, simplify the procedure as much as possible, provide detailed training on how to engage in self-monitoring, link incentives and negative consequences to compliance and noncompliance, and ask parents and others to closely supervise self-monitoring without unduly influencing a child's responses. In addition, child self-monitoring may be supplemented with reports from informed others such as parents and teachers to assess accuracy (C. Cole et al., 2000).

Sociometric Assessment

Sociometric assessment refers to measures of children's social relationships and standing among peers. Sociometric assessment is important for identifying children with poor social competence who may be at risk for emotional, behavioral, and academic problems. This assessment approach is also used to identify children with exceptional social skills and who have particularly good social standing within a peer group. In many cases, sociometric assessment is conducted in school settings such as classrooms or play areas. Procedures for sociometric assessment include peer nominations, rating scales, and rankings.

Peer nominations typically involve asking children to privately identify or sort peers or classmates along certain dimensions. In many cases, a simple listing is provided of names of children a child most and least likes to play with. Although younger children are often asked to nominate three classmates for a given question, older children and adolescents may provide longer lists. Peer nominations are usually scored by dividing the total number of nominations by the total number of raters.

From Chan and Mpofu (2001), the following questions are often asked as part of a peer nomination sociometric procedure:

- What three classmates do you like the best?
- What three classmates do you like the least?
- Which three people do you admire most in your class?
- Which three people do you respect most among your classmates?
- What three people would you wish to be like in your class?
- Which of your classmates try hard and get good grades?
- Which of your classmates do not try hard and get poor grades?
- Which of your classmates get angry much of the time?
- Which of your classmates is most cooperative?

- Which of your classmates is a good leader?
- Which of your classmates start fights or become disruptive in class?

Peer nominations have been used to classify children into social groups. Children who receive many positive peer nominations tend to be popular or socially accepted, and those who receive many negative peer nominations tend to be rejected. Neglected or isolated students generally receive no peer nominations of any sort, though controversial students are linked to many positive and negative peer nominations (Elksnin & Elksnin, 1997; Terry & Coie, 1991). In addition, peer nominations may be helpful for identifying children with Social Phobia who engage in avoidant self-isolation (Kearney, 2005).

Peer nominations have several advantages, including ease of use, portability, and linkage to social groupings that have good psychometric stability. Peer nominations also have disadvantages, however. First, only those children actually nominated by peers are available for analysis. As such, some students may be mistakenly classified as isolated and a full examination of a classroom's social dynamics may be impaired. Second, peer nominations may work less well in racially diverse classrooms where student groups may not readily interact (Chan & Mpofu, 2001).

An alternative to peer nominations is peer ratings, in which all children in a classroom or social setting rate all other peers in that setting. Ratings may be linked to the questions listed earlier for peer nominations, such as rating a particular child on a 1 to 5 Likert-type scale where 1 = not at all liked and 5 = very much liked. Ratings can also be obtained for traits such as cheerfulness, humorousness, and disruptiveness. In many cases, children are presented with a grid that lists names of relevant peers as well as spaces for ratings. Peer ratings are then averaged for each child. Peer sociometric ratings can also be supplemented by teacher ratings (Hintze, Stoner, & Bull, 2000).

A key advantage of peer ratings over peer nominations is that data for all children in a particular setting may be obtained. In addition, ratings are more amenable to psychometric analysis and have been shown to be reliable and valid measures. The dimensional aspect of peer ratings allows for a better way of assessing social acceptance of a particular child. A downside of peer ratings is they may be less useful than nominations for categorizing a child's social status. Also, younger children may have difficulty understanding a rating system; in this case, peer nominations and ratings may be integrated, or a simpler rating scale may be used (Chan & Mpofu, 2001).

Sociometric assessment also includes rankings whereby all children are ranked by all other children in a particular social setting. In this way, all children are included in the assessment process and ratings are unnecessary. Rankings may come from teachers and other knowledgeable adults as well. Rankings may involve comparison to a standard for a particular characteristic, such as likeability, or comparison against all other peers. Rankings could also be made for each possible peer pair, such as student A, B, C, and so on versus all other students. Unfortunately, rankings require substantial effort on the part of a clinician and raters, and so are rarely used except in conjunction with other sociometric techniques (Mpofu, Carney, & Lambert, 2006).

Physiological Assessment

Physiological assessment, sometimes referred to as *psychophysiological assessment,* includes measures designed to directly evaluate the relationship between physiological processes and thoughts, emotions, and behavior. Physiological assessment concentrates on motor and visceral responses that underscore higher order cognitive processes and behavior to increase the psychometric strength of other forms of behavioral assessment. Physiological assessment with children remains less well-developed than other forms of behavioral assessment, especially with respect to traditional measures such as electrocardiogram for changes in heart rate and electroencephalogram for changes in brain wave activity. With newer techniques such as functional magnetic resonance imaging, however, physiological assessment is rapidly becoming a dominant force.

Traditional physiological assessment primarily includes heart rate, blood pressure, and measures of skin conductance and resistance, perspiration, and muscle tension. Heart rate, blood pressure, and vasomotor activity (constriction and dilation of blood vessels) have been commonly measured via electrocardiogram, a device that records electrical activity of the heart. However, simple pulse rate and systolic/diastolic values are often obtained without formal use of an electrocardiogram (Chakrabarti & Stuart, 2005; King, 1994).

Skin conductance and resistance are measured via electrodermal devices; electrodes are placed on a person's palm, middle and index fingers, or wrist and forearm. An electrical signal is introduced and the device measures the length of time necessary for the signal to travel from one electrode to another. Greater conductance or less resistance to the signal generally indicates greater autonomic arousal. In related fashion, perspiration may be measured by the number of active sweat glands in a certain skin area, often the palm (palmar sweat index), by means of a plastic impression method. Muscle tension is typically assessed via electromyogram, a device that measures muscle contraction and atonia (Kohler & Schuschel, 1994; Latzman, Knutson, & Fowles, 2006; Tedroff, Knutson, & Soderberg, 2006).

These traditional physiological measures have been prominently used to study children with anxiety disorders, with variable results. Children with anxiety disorders generally show increased heart rate, perspiration, and muscle tension. However, these increases are not consistently found, nor are they always maintained over time (Beidel, 1991; Schniering, Hudson, & Rapee, 2000; Turner, Beidel, & Epstein, 1991). With respect to externalizing behavior problems, low resting and task-related electrodermal activity has been associated with psychopathy and conduct problems. Low resting heart rate and high heart rate reactivity have been associated with aggression and conduct problems (Lorber, 2004).

These traditional forms of physiological assessment are gradually giving way to more advanced techniques. These include electroencephalograph, computerized axial tomography, magnetic resonance imaging, functional magnetic resonance imaging, and positron emission tomography. The electroencephalograph traditionally involved multiple electrodes placed on a person's head to measure brain wave activity and evoked response potentials, but more contemporary devices involve elaborate and less intrusive head covers that contain high-density electrodes. Brain

wave measurement has been particularly salient with respect to child sleep problems, learning disorders, ADHD, and seizure disorder (Collins & Rourke, 2003; Kuntsi, McLoughlin, & Asherson, 2006; Tay et al., 2006).

Neuroimaging methods now produce computer-generated pictures of specific brain sections. Computerized axial tomography is used to assess structural abnormalities of the brain such as tumors or enlarged ventricles potentially related to mental disorder. Iodine is injected and brain tissue density is assessed by detecting the amount of radioactivity from X-rays that penetrate the tissue. Magnetic resonance imaging produces higher quality brain images without radiation by having a person lie in a large cylindrical magnet as radio waves are beamed through the magnetic field. Functional magnetic resonance imaging is a more advanced technique, whereby brain images are taken rapidly to assess metabolic changes in the brain.

Positron emission tomography is an invasive procedure whereby radioactive molecules are injected into the bloodstream and emit positrons that collide with electrons. The resulting light particles emitted from the skull are utilized by a computer to construct a full image of the brain. Neuroimaging techniques have received greater attention in the child literature, particularly with respect to major problems such as traumatic brain injury, Schizophrenia, autism, mood disorders, ADHD, and learning disorders (Ernst & Rumsey, 2000).

Physiological assessment with children may also include biochemical assessment, such as analysis of metabolites for neurotransmitters or toxicology examinations. Although much basic information about child behavior may be obtained from physiological assessment, key disadvantages include its prohibitive cost and need for technical expertise. In addition, physiological assessment has long been criticized for providing information not particularly useful for developing an individualized treatment plan. An exception may be biofeedback, where children are provided feedback regarding internal responses and trained to regulate them to help reduce problems such as hyperventilation and accelerated heart rate (Hirshberg, Chiu, & Frazier, 2005).

Other Areas

Behavioral assessment for children can consist of other measures as well. Children's activity levels, for example, can be measured via behavioral observation or rating scales, but objective *activity measures* include the following:

- Pedometer, a device that records steps taken while walking
- Accelerometer, an electronic device that measures accelerations produced by body movement
- Actometer, a modified watch that records frequency of movement

These measures are commonly used to assess youths with ADHD but can also be useful for evaluating inactivity from a developmental disorder or sedentary lifestyle (Rapport, Kofler, & Himmerich, 2006; Sirard & Pate, 2001).

Behavioral assessment can also include review of archival or record data. Case history files, academic and school attendance records, medical and psychiatric

reports, completed written or oral schoolwork, report cards, and audiotapes and videotapes can be useful sources of information. A particularly innovative use of archival data has involved analysis of home movies of youngsters very early in life to identify emerging patterns of developmental disorders such as autism (Werner & Dawson, 2005).

Behavioral assessment also includes methods of evaluating cognitions more specifically than the rating scales described earlier. *Thought listing,* for example, requires a child to maintain a written log of thoughts before, during, and after a critical event such as an oral presentation. Thought listing may be done in naturalistic settings such as a classroom, but many children have difficulty identifying or listing their thoughts in these situations. In research settings, therefore, thought listing is sometimes utilized as part of a behavioral assessment test where a child engages in a task while videotaped. The clinician and child can then review the videotape and the child can recall various cognitions during certain segments (Kendall & Chansky, 1991; Spence, Donovan, & Brechman-Toussaint, 1999).

An alternative method is a *think-aloud procedure,* in which a child is asked to verbalize thoughts and emotions before, during, and after a behavioral assessment test. Solicited thoughts and emotions can then be categorized with respect to content and valence, such as positive, negative, or neutral. Unfortunately, this procedure is highly subject to reactivity and the information received may not be useful for developing a treatment plan. As such, researchers have modified the think-aloud procedure to focus on simple ratings of distress or perceived threat as well as open-ended questions about a child's cognitions, the responses from which are then coded (Bogels & Zigterman, 2000; Chansky & Kendall, 1997).

SUMMARY

Behavioral assessment is a problem-solving-oriented means of gathering information about current, observable behavior and what maintains that behavior over time. Methods of behavioral assessment include interviews, analogue assessments, behavioral observation, rating scales, self-monitoring, sociometric assessment, physiological assessment, and activity and other measures. A central model of behavioral assessment is functional analysis, or examining and manipulating antecedents and consequences of a behavior to provide information about the relationship of these events to the behavior. Methods of behavioral assessment emphasize molecular aspects of behavior, situation-specific behavior, and ongoing sampling of overt and current behavior in multiple contexts and from multiple sources of information. Behavioral assessment is thus a key aspect of evaluating children with myriad behavior problems and mental disorders.

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PART V

Treatment

CHAPTER 20

Play Therapy

Sue C. Bratton, Dee Ray, and Garry Landreth

Identifying early mental health interventions tailored to the maturational needs of children is a growing concern in the United States (New Freedom Commission on Mental Health, 2003). Increases in societal problems that directly involve and impact young children has focused greater attention on this issue. The most recent U.S. surgeon general's report on mental health described the shortage of appropriate services for children as a major health crisis and estimated that, although at least 1 in 10 of all children suffer from emotional and behavioral problems severe enough to impair normal functioning, fewer than half receive treatment (U.S. Public Health Service, 2000). Thus, providing families with greater access to interventions that are responsive to the distinct needs of young children is critical, not only to diminish unnecessary suffering, but to prevent the development of more serious impairment across the life span.

Play therapy is founded on the principle that children must be approached and understood from a developmental perspective. Children are not miniature adults, and just as their world is a world of concrete realities, so too their means of communicating their world is on a child's level through play (Landreth & Bratton, 1998). Play is to the child what verbalization is to the adult. Play provides a developmentally responsive means for expressing thoughts and feelings, exploring relationships, making sense of experiences, disclosing wishes, and developing coping strategies (Landreth, 2002).

Although play therapy has been a treatment of choice for children since the early 1900s, the formation of the Association for Play Therapy (APT) in 1982 established play therapy as a specialized treatment modality within the field of mental health. The Association's influence, along with an increase in university-based play therapy training programs, provided the momentum for the substantial growth of the field over the past 25 years. Today, play therapy is widely used among clinicians to successfully treat a range of emotional and behavioral problems (Bratton & Ray, 2000, 2002). According to APT's web site (2006), approximately 5,000 mental health professionals identify themselves as play therapists. The growing interest in the field is also evident in the more than 3,500 play therapy publications describing its use, rationale, and efficacy, the majority of which were produced after 1980 (Center for Play Therapy, 2006).

RATIONALE FOR PLAY THERAPY

Play has long been considered significant in children's lives. As early as the eighteenth century Rousseau (1762, 1930) acknowledged the importance of observing children's play as a means of understanding them; however, it was not until the early twentieth century that the symbolic function of play was recognized. Froebel (1903) proposed that children's play has a conscious and unconscious purpose and thus is meaningful and important. More recently, the value of play has been recognized by early childhood experts as the way children learn about themselves, others, and their world as they try to organize and make sense of their experiences. Frank (1982) suggested that play is the way children learn what no one can teach them.

Play has been used by child therapists since the early twentieth century to provide children with a medium for self-expression. Most children below the age of 11 lack a fully developed capacity for abstract thought, which is a prerequisite to meaningful verbal expression and understanding of complex issues, motives, and feelings (Piaget, 1962). The child's world is one of concrete realities; thus, children must be afforded a concrete means of expressing their perception of self, others, and the world. Children can more meaningfully express their inner world through the symbolic representation provided by toys, with toys and play materials serving as their language. Although children often use play to directly express their experiences, in play therapy it is the symbolic function of play that is most significant. Sometimes children's symbolization of their experiences is quite apparent; at other times the connection may seem rather remote. Regardless, children use symbolic play to develop mastery and a sense of control over their world as they reenact their experiences directly in the safety of the therapeutic milieu.

In play therapy, then, play is viewed as the vehicle for communication between child and therapist on the assumption that the child will use play materials to directly or symbolically act out feelings, thoughts, and experiences that the child is not able to express meaningfully through words alone (Axline, 1947; Kottman, 2001; Landreth, 2002; O'Connor, 2001; Schaefer, 2001). Play allows children to bridge the chasm between their experiences and understanding, thereby providing the means for insight, learning, problem solving, coping, and mastery. In play therapy, regardless of the reason for referral, the therapist has the opportunity to enter into and experience the child's world and actively deal with the issues that brought the child to therapy. Axline described this process as one in which the child has the opportunity to play out feelings as they emerge, getting them out in the open, and either learning to appropriately control them or abandon them.

DEVELOPMENT OF PLAY THERAPY

As early as the eighteenth century play was recognized as essential to children's healthy development, but it was not until the early twentieth century that play was introduced into a therapeutic setting. The use of play in therapy originated with the classical child analysis case of Little Hans by Sigmund Freud. Freud saw Hans only once for a brief visit and conducted treatment by advising Hans father of ways to respond, with suggestions based on the father's notes about Hans play.

Generally acknowledged as the originators of play therapy, Anna Freud (1928) and Melanie Klein (1932) used play as a substitute for verbalized free association in their efforts to apply analytic techniques to their work with children. David Levy's (1939) development of release play therapy, along with the structured approach of Gove Hambidge (1955), marked the next advance in the field. In these two approaches play materials were structured by the therapist to induce catharsis; this is in contrast to psychoanalytic methods, in which the analyst made no attempt to direct the child's play.

Virginia Axline (1947) was the first to apply nondirective therapeutic principles in play therapy based on her trust in children's capacity to resolve their own problems through their play. Her work and writings in the late 1940s and 1950s, including her accounting of play therapy with Dibs (Axline, 1964), heralded perhaps the most significant development in the field of play therapy and popularized play therapy as a psychotherapeutic treatment modality for children. Building on Axline's work, Haim Ginott (1961), Clark Moustakas (1953), Louise Guerney (1983), and Garry Landreth (1991, 2002) contributed significantly to the widespread acceptance and practice of what is now more commonly referred to as child-centered play therapy (CCPT). Landreth (1991, 2002) provides an in-depth description of the principles and procedures of CCPT.

The development of filial therapy by Bernard and Louise Guerney in the early 1960s marked a significant and innovative development in the field of play therapy. The Guerneys were the first to develop a model for training and supervising parents in client-centered play therapy methods to use with their own children (Guerney, 2000). Since the late 1980s the use of filial therapy by practitioners has increased tremendously. Guerney (2000) credited this growth, in part, to the efforts of Landreth (1991, 2002) and his protégés. Building on the work of the Guerneys, Landreth developed a more condensed, 10-session parent training format based on his experience that time and financial constraints often hindered parents' participation. Landreth and Bratton (2005) formalized the 10-session training format in a text, *Child Parent Relationship Therapy (CPRT): A 10-Session Filial Therapy Model.* Grounded in the principles and procedures of child-centered play therapy, the CPRT parent training model emphasizes a balance of didactic and supervision experiences in a 2-hour weekly support group format and requires parents to conduct weekly, videotaped play sessions at home. The CPRT protocol was recently manualized by Bratton, Landreth, Kellam, and Blackard (2006), allowing for replication of the model. Although originally conceived as a group model for training parents, filial therapy has also been successfully adapted for individual parents and couples (VanFleet, 1994) and for training teachers, mentors and other paraprofessionals who play a significant role in children's lives (L. Jones, Rhine, & Bratton, 2002; Morrison, 2006; White, Flynt, & Draper, 1997).

The field of play therapy grew dramatically during the 1980s and 1990s as various theorists, academicians, and practitioners developed specific play therapy approaches based on their theoretical views and personal experiences with children, including gestalt play therapy (Oaklander, 1994), Jungian play therapy (Allan, 1997), Adlerian play therapy (Kottman, 1995), developmental play therapy (Brody, 1997), cognitive-behavioral play therapy (Knell, 1997), ecosystemic play therapy

(O'Connor, 2000), and prescriptive play therapy (Schaefer, 2001), to name a few. The sheer number of contributors to the development of this discipline prevents mentioning each.

Play therapy has grown over its 100-year history to include a variety of theoretical schools of thought. Although they may differ in their approach, they all embrace the therapeutic and developmental properties of play: "to help [children] prevent or resolve psychosocial difficulties and achieve optimal growth and development" (APT, 2001, p. 20). It would be impossible to describe all the various theoretical approaches to play therapy within the scope of this chapter; thus, in the practice and case study sections, we focus on the principles and procedures of the play therapy approach with the longest history of use and the strongest base of research support, CCPT.

RESEARCH IN PLAY THERAPY

Identifying mental health interventions that are effective for young children is a critical need for the twenty-first century (U.S. Public Health Service, 2000). In spite of play therapy's growing popularity among clinicians, the scientific community has been slower to embrace this modality as an empirically sound intervention. Scientifically proving the effectiveness of any therapeutic intervention is essential to its widespread acceptance and use. Although the field of play therapy has a history of over 60 years of continuous research, historically play therapy studies have been hindered by inadequate research design, most notably the lack of a control group and small sample sizes. More current research has utilized well-designed methodologies and supports the use of play therapy as an effective treatment that is responsive to children's unique developmental needs.

Play therapy research dates back to the 1940s, demonstrating one of the longest histories of research of any psychological intervention. In 1942, Dulsky attempted to study the relationship between intellect and emotional problems. He inadvertently established the effect of nondirective play therapy, which was to improve social and emotional adjustments, yet no improvement was shown on intellect. Although the research demonstrated a positive effect, several flaws in the research are noted, including a lack of control or comparison group, and insufficient description of treatment and participants. Since Dulsky's 1942 study, an approximate count of play therapy research, excluding filial therapy research, includes 103 studies, of which 71 were published in professional journals and 32 remained in dissertation form. The majority of play therapy studies demonstrate some positive effect of play therapy on the participants.

An examination of the play therapy research focused on filial therapy modality reveals 37 studies using a pre-post control group design, 27 of which were published in professional journals. Of the 37 filial therapy outcome studies, 27 utilized the 10-session CPRT model (Landreth & Bratton, 2005), originally introduced by Landreth (1991). Of the 27 studies examining the effects of the 10-session model, 24 were conducted under the supervision of the CPRT authors to ensure treatment integrity. Videotaped and live supervision was used to assure adherence to the procedures originally developed by Landreth and recently formalized in a published treatment

manual (Bratton et al., 2006). A detailed summary of filial therapy research from 1967 to 2005 can be found in Landreth and Bratton. For the purpose of this chapter, we focus our discussion on research examining the effects of play therapy conducted directly by professionals.

Current Play Therapy Research

Since 1990, 36 (27 published) research studies on the impact of play therapy have been conducted. These recent studies have demonstrated the positive impact of play therapy on general behavioral problems (Raman & Kapur, 1999; Shashi, Kapur, & Subbakrishna, 1999), externalizing behavioral problems (Flahive & Ray, in press; Garza & Bratton, 2005; Karcher & Lewis, 2002; Kot, Landreth, & Giordano, 1998; Schumann, 2005), internalizing problems (Packman & Bratton, 2003), self-efficacy (Fall, Balvanz, Johnson, & Nelson, 1999), self-concept (Kot et al., 1998; Post, 1999), anxiety (Baggerly, 2004; Shen, 2002), depression (Baggerly, 2004), speech problems (Danger & Landreth, 2005), and diabetes treatment compliance (E. Jones & Landreth, 2002).

Meta-Analytic Research on Play Therapy

LeBlanc and Ritchie (1999) published the initial results of their meta-analysis of play therapy outcomes summarizing the results of 42 controlled studies, with an effect size of .66 standard deviation. The researchers further detailed their study in a later publication citing that benefits of play therapy appear to increase with the inclusion of parents and optimal treatment duration (LeBlanc & Ritchie, 2001). Using Cohen's (1988) guidelines for interpretation of effect size, .66 denotes a moderate treatment effect, similar to effect sizes (ES) found in other child psychotherapy meta-analyses (Casey & Berman, 1985, ES = .71; Weisz, Weiss, Han, Granger, & Morton, 1995, ES = .71).

Bratton, Ray, Rhine, and Jones (2005) conducted the largest meta-analysis on play therapy outcome research. This meta-analysis reviewed 180 documents that appeared to measure the effectiveness of play therapy from 1942 to 2000. Based on stringent criteria for inclusion—use of a controlled research design, sufficient data for computing effect size, and the identification by the author of a labeled "play therapy" intervention—93 studies were included in the final calculation of effect size. The overall effect size was calculated at .80 standard deviation, interpreted as a large effect, indicating that children receiving play therapy interventions performed .80 standard deviation above children who did not receive play therapy. LeBlanc and Ritchie (2001) and Bratton et al. (2005) included filial therapy research in their definitions of play therapy. Filial therapy is a parental intervention based on CCPT.

Bratton et al. (2005) coded specific characteristics of play therapy that impacted or had no impact on play therapy outcome. Effect sizes for humanistic (ES = .92) and nonhumanistic (ES = .71) play therapy interventions were considered to be effective regardless of theoretical approach. However, the effect size reported for the humanistic approach was in the large effect category, whereas the ES for the nonhumanistic approach was in the moderate category. This difference in effect may be attributed to a larger number of calculated humanistic studies (n = 73)

compared to nonhumanistic studies ($n = 12$). When play therapy was delivered by a parent (ES = 1.15), the effect size was much larger than when delivered by a mental health professional (ES = .72), indicating the importance of involving parents in treatment to increase the chances of a successful outcome. This finding was similar to the conclusions of LeBlanc and Ritchie (2001), who reported parent involvement as a predictor of play therapy outcome. Treatment duration was also a factor in the success of play therapy. Optimal treatment effects were obtained in 35 to 40 sessions. However, many studies with fewer than 14 sessions also produced medium and large effect sizes. Age and gender were not found to be significant factors from which to predict the effects of play therapy; play therapy appeared to be equally effective across age and gender. An effect size was not calculated for ethnicity due to the lack of specificity in the reporting of ethnicity in individual studies. In addressing presenting problems, the researchers encountered difficulty distinguishing specific diagnoses and symptoms due to the variation among the studies. However, 24 studies were coded as investigating internalizing problems; analysis revealed an effect size of .81. An effect size of .78 was calculated for the 17 studies coded as examining the effects of play therapy on externalizing problems. For the 16 studies that addressed a combination of internalizing and externalizing problems, the effect size was .93. These results indicated that play therapy had a moderate to large beneficial effect for internalizing, externalizing, and combined problem types.

Individual Study Research on Play Therapy

Ray (2006) highlighted specific empirical studies that support the use of CCPT with children with behavioral and emotional challenges. The following studies are the most recent supportive research on CCPT that were identified as adhering to current experimental rigor. Kot et al. (1998) found that among child witnesses of domestic violence between the ages of 4 and 10 years, children who participated in 12 CCPT sessions scored significantly higher on self-concept than children who did not receive play therapy. Their mothers further reported significantly fewer externalizing behavior problems and total behavioral problems than mothers of children who did not receive play therapy. Among Hispanic children identified by parents or teachers as demonstrating behavioral problems at school, parents reported that those children who participated in 15 CCPT sessions demonstrated significantly fewer externalizing problems than children who participated in a small group curriculum-based guidance program, with the greatest improvement in conduct problems (Garza & Bratton, 2005). The children who participated in CCPT also demonstrated moderate improvements in internalizing problems. Flahive and Ray (in press) found that, compared to the control group, after 10 sessions of a child-centered approach to sand tray therapy for preadolescents in upper elementary school, teachers reported that the students significantly improved in their internalizing and externalizing problem behaviors, and parents reported improvement in children's externalizing problem behaviors. In this study, the wait-list control group continued to experience increased behavioral problems, both internalizing and externalizing.

Fall et al. (1999) and Post (1999) found that children who participated in CCPT demonstrated statistically significantly greater self-efficacy and self-esteem over their peers who did not participate in play therapy. Shen (2002) explored the effect of CCPT on Chinese children who experienced a devastating earthquake in Taiwan. Children who participated in 10 group sessions of CCPT demonstrated a significant decrease in anxiety and suicide risk compared to children who did not participate in play therapy. E. Jones and Landreth (2002) examined the efficacy of 12 sessions of intensive individual play therapy sessions on anxiety levels and medical compliance of 30 insulin-dependent children, ages 7 to 11 years. Compared with the control group children who attended the same camp, the 15 experimental group children demonstrated significant improvement in medical compliance (giving shots, following diet, etc.), when measured immediately following camp and at a 3-month follow-up. These are just a few examples of recent research conducted on CCPT that support the validity of the intervention in helping to reduce childhood problems and improve emotional conditions.

Historical Play Therapy Studies

The focus on these exemplary play therapy research studies by no means devalues the significant contribution that many other historical play therapy studies have made to the field. A rich history of play therapy efficacy has been established in the areas of hospitalized children with symptoms of anxiety (Cassell, 1965; Clatworthy, 1981; Johnson & Stockdale, 1975; Rae, Worchel, Upchurch, Sanner, & Daniel, 1989), self-concept (Crow, 1990; Gould, 1980; House, 1970; Perez, 1987), social adjustment (Cox, 1953; Oualline, 1976; Pelham, 1972; Thombs & Muro, 1973), and behavioral difficulties (Brandt, 2001; Gaulden, 1975; Hannah, 1986; Quayle, 1991). Detailed accounts of these studies can be found in the compilation of play therapy research summarized in Bratton and Ray (2000).

PLAY THERAPY IN PRACTICE

As stated earlier, the scope of this chapter prevents a description of the practice of play therapy from all theoretical viewpoints. Although the various play therapy approaches may differ philosophically and technically, they are alike in their use of the therapeutic and developmental properties of play to help children benefit from treatment. In this section, we focus on the principles and procedures of CCPT. Of the many well-established play therapy schools of thought, CCPT has the longest history of use, the strongest research support, and, according to a survey of APT members (Lambert et al., 2005), the largest number of followers.

Role of Therapist and the Therapeutic Relationship

The goal of child-centered play therapy is to establish conditions so that the child can experience growth and integration. Raskin and Rogers (2005) postulated that if the therapist is successful in conveying genuineness, unconditional positive regard, and empathy, then the child will respond with a changed personality organization.

Child-centered play therapists believe that a child client's experience in the counseling relationship is the factor that is most meaningful and helpful in creating lasting, positive change.

Given the opportunity, children will play out their feelings and needs in a manner or process of expression that is similar for adults. Although the dynamics of expression and the vehicle for communication are different for children, the expressions (fear, satisfaction, anger, happiness, frustration, contentment) are similar to those of adults. Children may have considerable difficulty describing what they feel or how their experiences have affected them. If permitted, however, in the presence of a caring, sensitive, and empathetic adult, they will reveal inner feelings through the toys and materials they choose, what they do with and to the materials, and the stories they act out. Children's play is meaningful and significant to them, for through their play they extend themselves into areas they have difficulty entering verbally.

Because the child's world is a world of action and activity, play therapy provides the therapist with an opportunity to enter the child's world. The child is not restricted to discussing what happened; rather, the child lives out at the moment of play the past experience and associated feelings. If the reason the child was referred to the therapist is aggressive behavior, the medium of play gives the therapist an opportunity to experience the aggressive behavior firsthand as the child attempts to break toys or to shoot the therapist with a dart gun, while also allowing the child opportunities to learn self-control by responding with appropriate therapeutic limit-setting procedures.

Axline (1947) identified eight basic principles that guide the therapist in child-centered play therapy. These basic principles are consistent with a person-centered philosophy of working with children by emphasizing the primacy of the counseling relationship. These principles insist that the therapist (1) must develop a warm, friendly relationship with the child; (2) accepts the child unconditionally, without wishing the child were different in some way; (3) establishes a feeling of permissiveness in the relationship so that the child feels free to express himself or herself; (4) recognizes and reflects the feelings of the child to create understanding for the child; (5) respects the child's innate ability to solve his or her own problems and offers the child the opportunity to take responsibility for his or her own behavior; (6) does not attempt to direct the child's actions or conversation, but allows the child to lead the way; (7) recognizes the gradual nature of the child's process and does not try to rush counseling; and (8) establishes only those limitations that are necessary to anchor the child's experience in counseling to the world of reality.

Without the presence of play materials, the therapist could only talk with the child about the aggressive behavior the child exhibited yesterday or in the past weeks. In play therapy, whatever the reason for referral, the therapist has the opportunity to experience and actively deal with that problem in the immediacy of the child's experiencing.

How Play Therapy Facilitates Change in Children

Play therapy assists children's development by helping them learn to know and accept themselves. Most of what is learned in the play therapy relationship is not

cognitive learning, but experiential, intuitive learning about the self that occurs over the course of therapy. Landreth (2002) described the following learning experiences:

- *Children learn to respect themselves.* The play therapist maintains and communicates constant regard and respect for children regardless of their behavior, whether they are playing passively, acting out aggression, or being whiny and insisting on help with even the simplest tasks. Children sense the therapist's respect, feel respected, and, because of the absence of evaluation and an ever present acceptance, they internalize the respect. Once children have respect for themselves, they learn to respect others.
- *Children learn that their feelings are acceptable.* By playing out their feelings in the presence of an adult who understands and accepts even the intensity of their feelings, children learn that all of their feelings are acceptable. As children begin to experience that their feelings are acceptable, they begin to be more open in expressing them.
- *Children learn to express their feelings responsibly.* Once children have expressed their feelings openly and been accepted, those feelings lose their intensity and can more easily be controlled appropriately. This is a freeing process for children to experience in that they are free to go beyond those feelings.
- *Children learn to assume responsibility for themselves.* In the natural process of development, children strive toward independence and self-reliance but often are thwarted in their efforts by adults who, although well intentioned, take charge by doing things for them and thus deprive children of opportunities to experience how being responsible for themselves feels. In the play therapy relationship, the therapist believes in children's ability to be resourceful and so resists doing anything that would deprive children of the opportunity to discover their own strength. As the therapist allows children to struggle to do things for themselves, they discover what that responsibility feels like.
- *Children learn to be creative and resourceful in confronting problems.* When children are allowed to figure out things for themselves, to derive their own solutions to problems, and to complete their own tasks, their own creative resources are released. With increasing frequency, children will tackle their own problems and experience the satisfaction of doing things all by themselves. Although children may at first resist the opportunity to solve their own problems, the creative tendency of the self will come forth in response to the therapist's patience.
- *Children learn self-control and self-direction.* If no opportunities are available to experience being in control, learning self-control and self-direction is not possible. Although this principle seems simplistically obvious, the absence of opportunities like this in children's lives is conspicuous when one takes the time to carefully observe children's interactions with significant adults. Unlike most other adults in children's lives, the play therapist does not make decisions for children or try to control them either directly or subtly. Limits on children's

behavior in the playroom are verbalized in such a way that children are allowed to control their own behavior. Because control is not applied externally, children are able to make their own decisions.

- *Children gradually learn, at a feeling level, to accept themselves.* As children experience being accepted just as they are with no conditional expectations from the therapist, they gradually, and in sometimes imperceptible ways, begin to accept themselves as worthwhile. This is both a direct and an indirect process of communication and learning about self. Acceptance is an attitudinal message communicated verbally and nonverbally by all that the therapist is and does. The children first feel acceptance and then know it by being accepted nonjudgmentally for who they are, just as they are, with no desire that they be different. This increased self-acceptance is a major contributing factor to the development of a positive self-concept.
- *Children learn to make choices and to be responsible for their choices.* Life entails a never-ending series of choices. How can children learn how to make a choice if they are not afforded the opportunity of making choices, of being indecisive, of struggling, of wanting to avoid, of feeling incapable, of being anxious, and of feeling apprehensive that their choice will be unacceptable to others? Therefore, the therapist avoids making even simple choices for children.

Structuring the Relationship and Play Sessions

Beginning the Relationship

For most children, the therapeutic relationship is like no other they have experienced, one in which they are fully accepted and valued for who they are. The play therapist maintains and communicates a constant regard and respect for children. The structuring of the relationship begins with the initial interactions between therapist and child in the waiting room. The therapist's task is to give full attention to the child, communicating the child's importance. (*Note: Prior to the initial meeting with the child, the therapist has met with the parents without the child present to fully focus on the parents' needs and concerns, gather background history, and inform parents of what to expect.*)

In the initial meeting between therapist and child, the therapist should crouch down, make eye contact, and greet the child. Following a short introduction, the therapist can say, "We can go to the playroom now. Your mom (or the person who brought child) will wait here, so she will be here when we come back from the playroom." The play therapy relationship is further structured by how the therapist introduces the child to the playroom. Verbal communication should be kept to a minimum at this point. No amount of words will convey to the child that this is a safe place. The child learns this through experience. The therapist should choose words carefully to communicate to the child freedom, self-direction, and the parameters of the relationship. The therapist might say something such as "Sarah, this is our playroom, and this is a place where you can play with the toys in a lot of the ways you would like to." This statement conveys responsibility for direction, freeing the child. Boundaries on this freedom are conveyed by the words "in a lot of the ways,"

which in effect communicates limits on behavior. The playroom is not a place of complete freedom where children can do anything they want.

Playroom and Materials

Landreth (2002) suggested that the dimensions of a playroom be 12 by 15 feet. However, an open space of almost any size will suffice. In school buildings, we have used janitorial closets, old book rooms, and portable buildings for a playroom. If a therapist is very limited on space, he or she can develop a traveling playroom by maintaining a box or bag of materials that can be used in an office or conference room.

Because toys and materials are part of the communicative process for children, careful attention must be given to the selection of appropriate items. The rule is selection rather than accumulation. Play areas and playrooms containing an assortment of randomly acquired toys and materials often resemble a junk room and doom the play therapy process to failure. Toys and materials should be carefully selected for (a) the contribution they make to the accomplishment of the objective of play therapy and (b) the extent to which they are consistent with the rationale for play therapy. Not all play materials automatically encourage the expression of children's needs, feelings, and experiences. Because toys and materials are used by the child in the act of play to communicate a personal world to the counselor, toys should reflect the cultural experiences of children. Landreth (2002) suggested that toys and materials should be selected with the following goals in mind:

- Establishment of a positive relationship with the child.
- Expression of a wide range of feelings.
- Exploration of real life experiences.
- Testing of limits.
- Development of a positive image.
- Development of self-understanding.
- Opportunity to redirect behaviors unacceptable to others.

The following toys and materials are recommended because they can encourage a wide range of expressions and can be easily transported in a tote bag or stored out of the way in a corner or in a closet. They can be grouped into three broad categories:

1. Real-life toys, such as a doll (small size with soft body); a bendable doll family; a small cardboard box with rooms indicated by strips of tape or felt pen marker (could double as container for materials); doll house furniture (cost will probably necessitate purchase of plastic furniture and thus set limits on how it is used); a nursing bottle (plastic); two play dishes and cups (plastic or tin); spoons (avoid forks because of sharp points); a small car; a small airplane; and a telephone.

2. Acting-out or aggressive release toys, such as handcuffs (spring-release type without key because of possible embarrassment if child cannot get them off), a dart gun, toy soldiers (30-count size is sufficient because of the difficulty of picking up more than that), a pounding bench, and a rubber knife.
3. Toys for creative expression and emotional release, such as crayons (8-count size to avoid too many choices; break some in half and peel paper off to convey that it is not necessary to be neat and precise); newsprint; blunt scissors; pipe cleaners; popsicle sticks; Play-doh (place it in small plastic container to keep it pliable; the top of the cardboard Play-doh can may fray and allow air to enter can); hand puppets; a small plain mask (Lone Ranger type); a Nerf ball (a rubber ball bounces too much); Gumby (bendable, nondescript figure); scotch tape; nontoxic glue or paste; and an empty vegetable can (doubles as container for toy soldiers).

An inflatable punching toy (Bobo) is desirable if the room allows. Additional toys and materials can be added by counselors who have a permanent place for play therapy. Landreth (2002) provided a complete list of play materials in a fully equipped play therapy room. Highly structured materials and electronic toys should generally be avoided because they may interfere with rather than facilitate children's expressions. Because toys are the child's words, they should be in good condition.

Facilitative Responses

The following therapist skills are considered to be essential to the child-centered play therapy process. These are specific skills expected to be demonstrated in most play therapy sessions. The extent to which they will be used depends on the therapist's intuitive understanding of the child's needs in the immediacy of the session.

Nonverbal Skills

Play therapy is heavily reliant on nonverbal skills. Because play therapists believe that play is the language of children, the verbal world becomes less important in a play therapy session. Nonverbal skills are critical to any person-centered approach, but especially to play therapy. Nonverbal skills are demonstrated when the therapist leans toward the child and is physically directed toward the child at all times, showing keen interest in the child throughout the session. As with counseling adults, the therapist should strive to be congruent with how the child expresses himself or herself. The therapist should not only match the level of affect displayed by the child but should also convey a sense of genuineness. The skill of matching verbal response with nonverbal response is representative of the therapist's level of genuineness with the child. Mental health professionals who are new to play therapy often find this skill challenging, as they are not yet comfortable with responding to children's nonverbal play. Just as the therapist would not flatly respond, "You're happy that you got that to stick" (after the child looks over with a big smile after

finally getting the dart to stick on the wall after 10 unsuccessful attempts), the therapist should avoid being so animated that he or she overwhelms the child. Again, congruence and genuineness are the key.

Verbal Skills

How the therapist verbally responds to the child is almost as impactful as the words chosen. Effective delivery of responses includes a focus on succinct, interactive responses and rate of responses. Because play therapy is offered to young children and because play therapy recognizes the limited language ability of these children, the importance of short therapeutic responses is key. Lengthy responses quickly lose the child's interest, confuse the child, and often convey a lack of understanding on the part of the therapist. Therapists match their level of interaction to that of the child. If the child is quiet and reserved, the play therapist will slow his or her responses; if the child is highly interactive and talkative, the play therapist will match this level of energy with an increased number of responses.

Child-centered play therapy verbal responses can be structured into categories that help facilitate growth in the child. The following are several relevant categories of verbal responses. Therapist responses should reflect the developmental level of the child; responses to a 3-year-old and an 8-year-old should be different. The examples that follow are appropriate for most young children.

Reflecting Nonverbal Play Behavior: Tracking The therapist tracks or reflects the nonverbal content of the child's play behavior simply by stating what he or she observes. Reflecting the child's nonverbal behavior allows the child to know that the therapist is interested and accepting of the child. It also helps the therapist immerse himself or herself into the child's world. Examples of tracking behavior are "Hmm, you're not sure what you want to play with" (as a child walks around, briefly touching several different toys) or "You decided to play with that" (as the child picks up the plastic horse and puts in the sandbox).

Reflecting Content Reflecting verbal content in play therapy is identical to reflecting content in talk therapy. To reflect content, the play therapist paraphrases the verbal interaction of the child. Reflecting content validates the children's perceptions of their experience and helps to clarify their understanding of themselves (Landreth, 2002). An example of reflecting content is "You know a lot about dinosaurs" (after the child shares detailed information on dinosaurs).

Reflecting Feeling Reflecting feeling is the verbal response to emotions expressed by children in play therapy. The reflection of feeling can sometimes be threatening to a child and should be presented carefully. Reflecting feeling helps a child become aware of emotions, thereby leading to the appropriate acceptance and expression of such emotions. Examples of reflecting feeling are "You are really angry with that guy" (after the child throws the big elephant across the room, saying, "He stepped on everyone. I hate him") and "You're frustrated that it's not working out the way you wanted" (when the child tries unsuccessfully to cut a heart design out of paper

and then crumples it up). As stated previously, therapist nonverbals and intonation of voice are as important, if not more so, as the words used.

Facilitating Decision Making and Returning Responsibility One of the play therapist's goals is to help children experience a sense of their own capability and to take responsibility for their expression of capability. Responses that facilitate decision making or return responsibility help children experience themselves as able and empowered. Examples of responses that facilitate decision making or return responsibility are "In here, you can decide what color you want it to be" or "In here, that's up to you" (when the child wants to draw a picture and asks, "What color of paper should I use?") and "Show me what you want me to do" (when, without making an attempt, the child points to her wet painting on the easel and asks, "Can you take this off for me?"). The latter response allows the therapist to see if the child already knows how to use the clips holding the paper to the easel or is quite capable of doing so once she tries. Many children come to play therapy with little experience of figuring things out for themselves. In their experience, adults have all the answers, so there is no point in their trying. The rule of thumb in play therapy is to never do for a child what the child can do for himself or herself (Landreth, 2002).

Facilitating Creativity and Spontaneity Helping children experience their own sense of creativity and freedom is another benefit of play therapy. Acceptance and encouragement of creativity sends a message to children that they are unique and special in their own way. Experiencing the freedom of expression allows them to develop flexibility in thought and action. Examples of responses that facilitate creativity or spontaneity are "You can make whatever you want" (when the child picks up craft material sticks and asks, "What do I make with these?") and "You've got a lot of ideas about want you want to make today" (when the child moves from making one craft project to another in play session).

Esteem Building and Encouraging Encouraging children to feel better about themselves and their abilities is a constant objective for the play therapist. The use of esteem-building statements works to help children experience themselves as capable. Examples of esteem-building and encouraging responses are "You're trying lots of different ideas. You're not giving up" (when the child has trouble getting the glue bottle open and tries several different solutions) and "You did it" or "You kept at it until you found a way" (when the child tries and tries to reach the top shelf, and after a few attempts, succeeds).

Facilitating Relationship Because the therapeutic relationship serves as a model for all intimate relationships, the play therapist seeks to respond to any attempt by the child to address the relationship. Relational responses help the child learn effective communication patterns and express the therapist's care for the child. Examples of responses that facilitate the relationship are "You're wondering what I think about that" (when the child is drawing a picture and stops to look up at the therapist but says nothing) and "You want us to look just alike" (when the child puts a hat on

the therapist and says, "Now yours is like mine") and "You're mad at me, because I wouldn't do what you wanted" (after the therapist declines to eat the sand given by the child, and the child responds, "You're no fun, I don't want to come here anymore"). Relationship responses should always include a reference to the child and a reference to oneself as therapist.

Limit Setting In an attempt to provide an environment for a child that allows self-direction and self-responsibility, minimal limits are established in child-centered play therapy. The goal is to help the child move toward the ability to self-limit. Typically, limits are set when children attempt to damage themselves, another person, and certain expensive or irreplaceable toys or if their behavior impedes counselor acceptance.

Landreth (2002) proposed a specific method for setting limits in play therapy. This method has been widely adopted by play therapists, regardless of theoretical orientation, as the initial response to setting a limit in the playroom. The A-C-T model of limit-setting is Acknowledge the feeling, Communicate the limit, and Target an alternative. In this model, the play therapist first recognizes and addresses the child's feelings in the moment: "You're really excited about your project." Then the therapist sets a short, concrete, definitive limit: "But it's not for stapling to the walls." Finally, the therapist provides an alternative to the action: "You can tape it to the walls." When children have directed energy in the moment, it is important to provide them an alternative for that energy so that they do not feel the need to act on impulse. Although there are other methods for setting limits, the A-C-T model is short, direct, and works effectively.

Involving Parents as Partners in the Therapeutic Process

Prior to the first play session, the play therapist should meet with the parents without the child present. The content and length of the initial parent meeting will vary with the therapist's theoretical and personal approach to involving parents in their child's therapy and the importance placed on intake assessment. The therapist's primary task during this session is to form a connection with the parents, focusing entirely on the parents' needs and concerns and conveying to them their importance in their child's therapy. The initial parent consultation provides the play therapist with the opportunity to inform parents what they need to know about play therapy and what to expect during the process. During this initial session, the therapist will explain the play therapy process to the parents, including the description of the playroom, how the therapist responds to the child, and how the process is helpful to the child. The therapist will also want to inquire about special considerations for the child that would impact the therapeutic process. For example, some young children have a history of difficulty separating from their parents. It is helpful to know that information prior to the child's first visit for play therapy so the therapist can discuss possible options with the parents and coach them on how to respond if the child clings to them.

The therapist will also take this opportunity to convey the importance of confidentiality. Although, in most states, children do not have the legal right to

confidentiality, the play therapist emphasizes the need for children to experience confidentiality of specific actions and words in the playroom. In play therapy, it is expected that parents will not observe their children's play so that the child can feel free to fully express thoughts, feelings, and actions. In the first consultation, the therapist will also review the importance of attendance, indicating that sporadic participation in play therapy interferes with progress.

Finally, the play therapist highlights the need for a final session if the parents decide to prematurely terminate therapy. Often, when parents decide to terminate therapy without the support of the therapist, the process is abrupt and disconcerting to the child. A final session allows children to experience a positive end to a relationship that has been important to them. Play therapists approach the initial consultation with an attitude of relational alignment. For parents to invest in the therapeutic process, they need to feel that the play therapist is understanding and accepting of their plight. An initial consultation is not a time for education and instruction, which can quickly become overwhelming and feel judgmental to the parents.

The play therapy process encourages consistent contact with parents on an individual basis for subsequent consultations. Meeting with parents on a regular basis (at least monthly) from 30 minutes to an hour allows the play therapist to develop a relationship with the parents that will, consequently, support the child's progress. The length and frequency of parent meetings is determined by the unique needs of each family. In continuing consultations, therapists typically check with the parents about new events at home and school. To keep parents involved in their child's play therapy and help them develop a better understanding of their child, the play therapist will discuss play themes exhibited by the child during play and discuss the relevance to the child's home and school life. For example, a therapist may share that the child in play therapy seems to exhibit a significant need for power and control and responds well to choice giving by the therapist. After helping the parents better understand the child, a therapist may then teach a short concept. Using the example just given, the play therapist may follow the explanation of the theme with teaching how choice giving works and then practicing with the parents. Common skills for play therapists to teach parents are choice giving, encouragement, reflection of feeling, problem-solving techniques, returning responsibility to the child, and limit setting (Bratton et al., 2006).

Facilitating consistent consultations with parents encourages continuation of the child's therapy, improves in parenting skills, and fosters better systemic relationships within the therapeutic context. For some children who are experiencing significant difficulties at school, similar consultations with an invested teacher can be helpful and provide greater consistency in how significant adults respond to the child.

In addition, the use of CCPT principles, including involving parents more fully in their child's therapy through filial therapy training, can impact the family system by encouraging the adults in the system to make use of CCPT's relational principles, thereby creating a safer, more enhancing environment for children. The scope of this chapter prevents a detailed description of filial therapy procedures. Landreth and Bratton (2005) provide a comprehensive description of their 10-session filial therapy model based on CCPT principles and skills.

Group Play Therapy Applications

The skills used in child-centered individual play therapy are the same skills used in child-centered group play therapy. As in individual play therapy, a child-centered play therapist recognizes the group members' ability to develop their own potential to move in a positive direction (Landreth & Sweeney, 1999). Ginott (1961) recognized a child's "social hunger" as a need to interact and develop relationships with other children. Group play therapy feeds this social hunger by placing children together in an environment that provides for freedom of expression and understanding of multiple worldviews. A child's appropriateness for a referral to group play therapy is dependent on developmental readiness and presenting concerns (Landreth & Sweeney, 1999). White and Flynt (1999) advocated the use of child-centered group play therapy due to its focus on developmental appropriateness, variation of function (applicable for preventive, remedial, or crisis intervention counseling), vicarious learning and cathartic effect, positive socialization, and growth through insight.

The practicality of using child-centered group play therapy is limited by space and scheduling. In response to this limitation, the play therapist may be able to serve only two to three children in a group setting. This number allows for comfortable movement in the space, while still providing the benefits of a social environment.

Multicultural Applications

Play therapy, including filial therapy, has been used as an effective treatment modality for children across cultures (Bratton et al., 2005; Landreth & Bratton, 2005). Because of the acceptance of the child as a holistic organism, negating any preconceived labels or designations, child-centered play therapists have the opportunity to serve all children while embracing their cultural and ethnic backgrounds. As the play therapist seeks to fully know and understand each child, it is imperative that the therapist recognizes the significance of the child's culture and is open to learning about the child's and family's unique social and cultural environment. Person-centered philosophy is sometimes criticized for its individual focus that potentially ignores the contextual nature of some cultures (Kirschenbaum, 2004). However, a well-trained child-centered play therapist will recognize the impact of culture and embrace the culture as well as embracing the individual. Because toys and materials are used by the child to communicate a personal world to the counselor, toys should reflect the cultural experiences of the child.

Therapist Training

Mental health professionals who practice play therapy should have specific training in child development, as well as didactic training and supervised experience in play therapy theories and procedures. Children are not miniature adults. Because talk therapy may not be developmentally appropriate, they require an alternative form of communication. Thus, those who provide therapy for children require specialized training. All too often, graduate students receive no training to counsel children and then are assigned to an internship setting that requires them to work exclusively with

children. Many universities do not have specific play therapy training; however, play therapy training and supervision is widely available through conferences and workshops. To find training near you, check with the Association for Play Therapy (www.a4pt.org) and the Center for Play Therapy (www.centerforplaytherapy.com).

Case Illustration

The following case provides a glimpse into the process of play therapy with a young child with a history of multiple interpersonal traumas. The child participated in play therapy and filial therapy with the first author for a total of 55 sessions over a 20-month period using CCPT procedures presented in the practice section. My initial objective was to establish a therapeutic relationship in which the child would feel secure and safe enough to explore all his experiences, including those that in the past had been too threatening to admit to awareness. Axline (1947) described the optimal play therapy climate as "good growing ground," where the child can fully experience all aspects of self and realize potential for growth. Specifically, this case illustration shows the child's movement toward trusting primary adults in his life, integration of his early experiences into his self-concept, and increased levels of self-acceptance and self-regulation. Regular parent consultation and filial therapy training were important components of treatment. As part of the filial therapy training, parents participated in 16 supervised play sessions with their child during the treatment period. Identifying information has been changed to protect the child and his family.

Identifying Features of the Client

Dom, an approximately 6-year-old male of Chinese descent, was referred to play therapy by his adoptive parents, both of whom are of Anglo descent. He was adopted at the age of approximately 4 years of age. Prior to then, he lived at an orphanage in China. According to sketchy records, he was abandoned at the orphanage when he was approximately 1 year old by a neighbor and given the name Dom.

At the time of intake, Dom had recently decided he wanted to be called Andy. He was living in the southwestern region of the United States with his adoptive parents, Carol and Tom; a 14-year-old stepsister, Ann (biological child of parents); and a 7-year-old stepbrother, Chris, also adopted from China when Andy was $5^1/_2$ years of age. His mother was 45 years old and employed as a teacher in the private school that Andy and Chris attended. Andy's father, also 45 years old, was employed as a sales professional and traveled several days a week. The parents shared similar values and traditional Protestant beliefs. Both were actively involved in their children's extracurricular activities.

Presenting Complaints

During a phone consultation with Andy's mother, she reported concerns about his mood swings, violent outbursts, running away, stealing (at home), and refusing to let her leave home without him. Secondary concerns included eating and sleeping difficulties, gender identity issues, and possible sexual abuse from when he lived in the orphanage. She further disclosed that Andy had been diagnosed by the school psychologist with Reactive Attachment Disorder and that he was receiving special education services for a learning disability and speech

and language delays. Carol's stress was obvious in her voice. I immediately scheduled her and her husband for an intake and initial parent consultation, emphasizing the importance of both of them attending. I allotted 2 hours to permit sufficient time to focus on Carol's and Tom's needs as parents of this child and to help them feel supported, while also allowing them time to tell Andy's story, a story that appeared to be rather complicated for such a short life.

Although both Tom and Carol attended the initial intake session, the following information, unless otherwise specified, was reported by Andy's mother. Intake was conducted 1 week after Andy's sixth birthday, during the summer after he had completed kindergarten. Carol began by relaying her concerns almost verbatim to her phone report, with Tom nodding but not adding anything. When asked what they enjoyed most about Andy, Carol described Andy as charming, adding, "He is very affectionate and funny and loves to cuddle." However, she quickly refocused on her concern about his episodes of sudden violent outbursts that she described as "coming out of nowhere." She reported that Andy purposely damaged things during his violent outbursts, adding that he had put holes in the wall and broken windows in their home. She reported that when he was in a "rage," she physically restrained him until he calmed down, adding that Andy generally wanted her to continue to hold him and "cuddle" after an episode.

Carol expressed a high level of frustration about not being able to leave the house without Andy. When she tried, he would chase the car until she came back. At the time of intake, she had resigned to not leaving him except at school and to occasionally try to run an errand after he was asleep. Because Carol was employed as a reading specialist at Andy's school, administrators allowed him to come to her classroom when his teacher determined that he needed his mother. Both boys came to her classroom after school while she completed her workday. When I inquired about when she had any time to herself, she joked, "Only in the bathroom . . . and even then, not always!" Although she did not complain during the intake, my impression of Carol was that she was highly stressed in her role as parent and feeling resentful.

Carol and Tom both expressed their alarm that Andy had lately begun stealing from them. When confronted, he would first deny that he took anything, and then immediately tell Carol where she "might find it." Andy's speech delay, possible learning problems, and eating and sleeping problems were briefly discussed, but those issues were not at the top of their list of current concerns.

When asked about Andy's adjustment to his adoption experience and his new family, Carol stated that he had adjusted well to their family, and Tom agreed. She reported that he refused to discuss that he was adopted or that he had lived anywhere other than with them, adding, "Andy tells people that he was born in my [Carol's] tummy." Andy acknowledged the difference in his skin color by telling Carol that she needed to "get a tan like mine." Carol found this humorous and endearing. She believed that he had blocked out all his early memories from the orphanage. Tom said little during the intake, other than that he agreed with Carol's report. Tom acknowledged that Andy is more of a problem for Carol than for him and that Andy was much closer to Carol than to him. Tom acknowledged that because he traveled several days a week, he had much less responsibility for caretaking of Andy and Chris. Tom reported that although Andy seemed to obey him better than Carol, Andy would not stay with him without Carol present.

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Tom explained that Carol was generally the one who restrained Andy when he was out of control, because he was afraid that he would hurt Andy.

Socially, Andy was described by his parents and teacher as "acting younger than his age." He enjoyed playing with a few younger children; however, his preference was to play at home with his mother. The decision had been made that Andy would repeat kindergarten in the fall with the same teacher.

Carol's and Tom's responses on the Child Behavior Checklist (CBCL) administered at intake supported their worries about Andy's behavior at home and his performance at school. Carol's scores on the CBCL (reported in Table 20.1) fell in the clinical range in several areas. Tom's responses on the CBCL revealed a pattern of concern similar to Carol's, but showed a lesser degree of concern. A psychiatric referral for Andy was briefly discussed, but Tom and Carol were adamantly opposed to medication as a treatment option.

Table 20.1 Carol's Scores on the Child Behavior Checklist

Elements	*5 months Prior	July Pretreatment	December 5 month	May 9 month	August 12 month	December 16 month	April 20 month
Total problems	64C	65C	61B	59	59	60	58
Externalizing problems	69C	72C	67C	63B	62B	62B	60B
Syndrome scales							
Aggressive behavior	71C	72C	69B	65B	63	64	62
Rule-breaking behavior	70C	73C	65B	60	60	65B	62
Thought problems	66B	65B	58	57	58	61	58
Social problems	70C	68B	60	63	60	61	59
DSM-oriented scales							
Conduct problems	72C	73C	66B	64	63	64	63
Oppositional defiant	68B	71C	64	62	61	60	58
Affective problems	70C	70C	65B	64	61	61	60
Competency scale scores							
School	27C	24C	27C	29C	35B	30C	31B

Note: Only CBCL scales that were elevated at pretreatment intake are included.
B = Borderline clinical range; C = Clinical range.
*Administration by school psychologist 5 months prior to intake.

Carol and Tom seemed to recognize that Andy's behavior stemmed from his early experiences and genuinely wanted to understand how to help him. During this initial consultation, I believed that I had forged a strong connection with Andy's parents and that they would be active partners in the therapeutic process. We discussed what they could expect during the process of play therapy and the important role that they would play in Andy's treatment. I reviewed *What Parents and Children Need to Know about Play Therapy* (Center for Play Therapy, 1998) and gave them a copy to take home. Because Andy was genuinely remorseful

after he calmed down after his violent outbursts, and Carol had already been responding by saying, "I know you don't like it when you feel this way," we mutually decided that the next time he had an episode, she would take that opportunity to explain about coming to play therapy and how it could help Andy. Because of Carol's worries regarding Andy's inability to separate from her, we decided that the three of us would meet together for Andy's first introduction to the playroom and me.

Developmental History

Very little is known about Andy's developmental history prior to adoption at age 4. His birth date and exact age were unknown, and his adoptive parents had no pictures of him until he was about 3 years old. Sketchy orphanage records revealed that when Andy was a small baby, his birth mother abruptly abandoned him with a neighbor under the pretense of running an errand, and that the neighbor kept him for an unspecified length of time prior to leaving him at the orphanage at what officials guessed to be around the age of 1 year. Andy was initially abandoned at the orphanage with no information, and these brief details were later reported by the neighbor. The orphanage staff guessed at his age based on this limited information and gave him a birth date of July 1.

Orphanage records stated that Andy never talked but that he was a very affectionate and loving baby and child. His developmental milestones were not recorded, but he was mostly toilet-trained at the time of adoption. His gross motor skills and coordination were reported by his adoptive parents to be above average. When Carol and Tom first met Andy in the orphanage they noticed that he appeared cross-eyed but that he was able to focus when they came close to him. Andy had just turned 4 years of age in July when Carol and Tom brought him from China to join their family in the United States. A checkup by the family pediatrician determined that Andy's eye condition was so severe that he could see very little and only from very close up. After being in the United States a little over 1 week, he underwent major eye surgery. Carol recalled the week directly after surgery as very stressful for Andy and for her. As part of his eye treatment, Andy wore glasses with a very strong correction.

Within a month of his arrival, 4-year-old Andy was enrolled in the prekindergarten program at Carol's school. He was described as "extremely affectionate, clingy, and shy" and as "not wanting to be away from Carol. . . ever!" At the age of 5, Andy was enrolled in kindergarten at the same school. During his second year with Carol and Tom, Andy began exhibiting some anger outbursts and uncontrollable crying, but not to the current extent. Because of his immaturity, Andy was scheduled to repeat kindergarten. As noted earlier, he had recently requested that everyone call him Andy, saying that Dom was a stupid name (his official American name was Andy Dom Smith).

Parental Factors and Family Dynamics

Tom and Carol had been married for 18 years at the time of intake and reported satisfaction with their marriage and lifestyle. Neither reported any remarkable history. Neither set of grandparents lived close, so the family had little contact with them other than an annual summer vacation at the beach with Carol's parents. Tom and Carol reported no more than the usual problems in rearing their

(*Continued*)

14-year-old daughter, Ann, whom they described as loving but stubborn and strong-willed as a child. Both parents stated that they felt closer to Andy than to Chris (who had been recently diagnosed with Fetal Alcohol Syndrome), adding that Ann and Andy also enjoyed a close relationship. Ann appeared to assume a significant amount of responsibility in helping with Andy and his brother, including discipline. Family church activities were important, although more so to Carol. Carol was the primary disciplinarian; however, she reported, "When Tom speaks the children tend to listen to him more than they do to me." She also noted that when Tom got involved in discipline it was when "he has had enough," typically when Andy and Chris are fighting and their behavior and emotions are out of control. Andy tended to be the catalyst for most fights and enjoyed being "the boss." The brothers reportedly played together fairly well because Chris would generally go along with what Andy wanted to do.

In later consultations with Carol, she disclosed feeling overwhelmed with the role of primary caretaking, and that she did not feel supported by Tom. Carol's self-report was supported by her scores on the Parenting Stress Index (PSI; Abidin, 1983), reported in Table 20.2.

Table 20.2 Parent Stress Index: Percentile Rankings for Carol's Scores

Elements	July Pretreatment (%)	December 5 month (%)	May 9 month (%)	August 12 month (%)	December 16 month (%)	May 20 month (%)
Total stress	85	83	75	50	65	55
Child domain	96	93	80	55	75	60
Adaptability	99+	98	85	55	80	70
Reinforces parent	90	80	80	55	80	55
Demandingness	85	90	65	65	75	55
Mood	85	85	60	50	60	60
Acceptability	85	90	50	50	60	40
Parent domain						
Isolation	90	85	80	70	80	70
Attachment	85	80	75	65	75	50
Spouse	85	85	80	70	80	70

Note: Scores at or above 85% are considered high. Only those scales that were elevated at time of intake are included.

Assessment

Information was gathered through interviews, informal observation, and formal assessment. Andy presented as a much smaller than average (by American norms) 6-year-old male who had just completed public school kindergarten. Formal assessment by the school district 5 months prior to his referral for play therapy confirmed significant language delays and revealed cognitive delays in several areas. However, the school psychologist determined that intelligence test results could not be considered reliable due to his significant receptive and expressive language delay. The CBCL (Achenbach, 2001) was also administered as part of the school's assessment battery. Carol's scores indicated clinical levels of concern regarding Andy's externalizing behaviors. In addition, Andy had been diagnosed by the school psychologist with Reactive Attachment Disorder (American Psychiatric Association, 1994). Carol and Tom's description of Andy's current behavior and early history were consistent with diagnostic criteria for this disorder, particularly Andy's ambivalent and contradictory approach/avoidance response to Carol and, to a lesser degree, his excessive familiarity with relative

strangers. It made sense that Andy's early history placed him at risk for difficulties in developing appropriate and stable attachments.

I administered the CBCL (Achenbach, 2001) and the PSI (Abidin, 1983) to Tom and Carol. Carol's elevated scores on the CBCL were consistent with her self-report at intake and similar to, although slightly higher than, her report at the time of the school administration 5 months prior. Carol reported clinical levels of concern in Externalizing and Total Problems, School Competency, and several subscales. Her scores are shown in Table 20.1. Tom's scores on the CBCL reflected a similar pattern of behavioral concern, but at consistently lower levels than Carol. Carol's scores on the PSI were higher than what I expected based on her initial verbal report and, as shown in Table 20.2, indicate high levels of Total Stress and high levels of stress related to the Child Domain. Tom did not complete the PSI during intake. The CBCL and PSI were administered approximately every 4 months for the duration of play therapy to provide a measure of treatment outcome. Although both parents were asked to complete both instruments at each point of measurement, only Carol complied. Tables 20.1 and 20.2 show CBCL and PSI results at time of assessment, along with results from repeated measurements of the PSI and CBCL over the treatment period.

Informal assessment supported the school's findings regarding Andy's significant expressive and receptive language deficits. Immature social-emotional behavior was noted by parents and confirmed through informal observation over the first several weeks of therapy. Consistent with parent report, Andy's motor skills and coordination were observed to be very well developed for his age. Because he was receiving special education services for a learning disability and speech and language delays, and because the school recommended retesting after 1 year, no further testing was done at time of intake.

Case Conceptualization

My understanding of Andy and his treatment needs was based more on what I believe all children need rather than on the facts presented during intake. Andy needed to experience a secure and consistent relationship, both in play therapy and within his family, in which he could fully experience all aspects of self and become the person he was meant to be. That does not mean that I view all children the same, nor does that mean that the background history and information presented by Carol and Tom were not useful. It provided me with a context for understanding Andy's experiences played out in therapy as well as an understanding of the child's system, including sources of support to tap into and elements within the system that were likely hindering Andy's optimal growth.

Andy was displaying many of the symptoms one would expect from a child with his history: hypervigilance to noise (exacerbated by his inability to see much during the first 4 years of his life), difficulty going to sleep and staying asleep, eating difficulties (refusing to eat much, but wanting to save the food for later), inappropriate affection with relative strangers, and incessant fear that his adoptive mother would leave him. Andy's early experiences were characterized by a series of abandonment and loss of primary caregivers—first his mother, then the neighbor who cared for him for several months, then caregivers that came and left in the orphanage—and the loss of his cultural identity and the only home Andy could remember in his young life. That these experiences likely occurred

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during the time in his development when separation anxiety was at its peak confirmed for Andy that his fears were real: his significant caregivers (mother, neighbor) left and did not come back. These early traumatic events likely compounded the impact of his hospitalization shortly after he arrived in the United States and his brief but traumatic separation from his new mother.

Based on Andy's behavior as reported by his parents, and consistent with my experiences of Andy as we began our relationship, I guessed that his early attachment experiences had been inconsistent; sometimes his needs had been responded to and his cries had been soothed, and other times they had not. He seemed to view the world as sometimes safe and secure and sometimes unpredictable and scary. These inconsistent early experiences with primary caregivers seemed to have resulted in two internalized and contradictory messages: (1) "I cannot depend on you. I must not be worthy of being loved and cared for. I have to take care of myself. I don't need you," and (2) "I need you, but I don't trust you. I cannot let you out of my sight, because you might not come back when I need you. I have to make you like me so you will not want to leave me." Thus, Andy approached relationships with ambiguity, which was very puzzling to Tom, and even more so to Carol, who felt rejected and confused when Andy pushed her away. I knew that helping Carol to understand that Andy's contradictory approach/avoidance response to her was "normal" given his history would be a top priority as we began therapy.

I believed that a significant factor in Andy's progress would revolve around helping Carol and Tom better understand how Andy's previous experiences impacted his current functioning. As they began to see the relationship between his early experiences, his unmet needs, and the resulting behavior that he chose to get his needs met, I believed that they would be able to accept him more fully. Several events from Andy's early experiences seemed particularly salient to his current functioning, as evidenced by his repeatedly playing out related themes. Because Andy was able to symbolically, and at times quite literally, express his world to me through play therapy, I was able to help Carol and Tom better understand Andy's needs and how they could meet them. This type of collaborative feedback proved particularly helpful to Carol.

A wise mentor once told me that if I focus too much on the problem I will lose sight of the child. With a case as complicated as this one proved to be, I had to be vigilant to not let that happen. My primary objective would be to first establish a consistent relationship in which Andy would feel secure and safe enough to explore all his experiences, including those in the past that were too threatening to admit to awareness. I would need to be patient in developing this kind of relationship with Andy, whose experience had taught him that relationships were not consistent, safe, or secure. It was my job to provide him with a corrective experience, an experience in which he would come to perceive me as an adult who was trustworthy, consistent, and predictable, one who would not leave him prematurely. An additional objective was for Andy to experience our relationship as unconditional, one in which he did not have to earn my approval or acceptance by impressing me or being overtly affectionate (his mode of connecting when we first began our relationship). My hope was that as Andy experienced unconditional acceptance in our relationship, he would begin to alter his self-perception of "damaged . . . no good," not worthy of love, to one of "lovable" and worthy of positive regard. Through these experiences, I hoped that Andy would

form new perceptions of his world as safe and predictable. I believed that by providing Andy with this kind of experience he would be ready for the most critical therapeutic task of all, to experience this kind of relationship with his adoptive parents. I believed that as Andy grew to allow himself to fully experience his parents' love for him, he would no longer feel the need to push them away for fear that they would eventually leave him.

Andy's current family situation was a significant strength, a strength that I would build on during his therapy. Carol and Tom seemed amenable to participating in additional parent or family therapy services that might be recommended as an adjunct to play therapy. Although there were parenting strategies that I believed would help Carol and Tom to respond more effectively to Andy's needs, their interactions did not appear to be contributing to Andy's difficulties in any significant way. As is true for most parents and caregivers of children who have experienced trauma, I believed that Tom and Carol would benefit from direction and support regarding how to talk to Andy about his early experiences, particularly his complicated adoption story. Andy's refusal to stay with Tom was an issue that we would focus on to provide Carol with some relief and hopefully reduce her level of stress related to parenting Andy.

Helping Andy's parents learn how to establish a healthy attachment with their son would be critical to his healing; therefore, I planned to incorporate CPRT into our overall treatment plan. Additional goals that would be addressed through CPRT were (a) fostering greater understanding of Andy and his experiences, (b) normalizing Carol and Tom's understanding and response to Andy's behavior problems, (c) enhancing parent-child attunement through creating mutually enjoyable and developmentally responsive interactions, and (d) strengthening parenting skills and confidence in responding to Andy's unique and challenging needs. However, my first priority in our parent consults was to address several safety issues around Andy's violent outbursts and to help Carol become more attuned to what precipitated the events as well as develop strategies that she could implement to prevent or de-escalate the situation. One strategy that I expected would be useful was helping Carol identify times throughout the day when she could initiate holding Andy close in a loving and playful manner, rather than waiting for him to get his "holding" needs met during the aftermath of his "rages."

Course of Treatment and Assessment of Progress

Andy's mother brought him to therapy and participated in regular parent consultation. Andy attended 55 play therapy sessions, approximately 45 minutes in length, over a 20-month period. I saw Andy weekly for the first 15 weeks of treatment and after that an average of 3 times per month with the exception of summer. I met with Carol every 2 to 3 weeks to provide support and help her develop a better understanding of Andy by describing themes I observed in his play, the connection between his play and his early experiences, and how these themes were relevant to his behavior at home and school. I also used this opportunity to introduce brief CPRT parenting skills that I believed Carol could implement successfully, including reflecting feelings (particularly anger), limit setting, and choice giving (Bratton et al., 2006). Carol competed a weekly parent report form during each session while Andy was in play therapy

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to keep me informed of any changes or significant events at home or school over the past week, as well as to rate Andy's behavior and mood compared to the week before. This was a very effective communication tool for Carol and me, providing an efficient means for us to stay connected on the weeks that we didn't formally meet.

As therapy progressed, I suggested many strategies that Carol and Tom could implement at home to promote the work being done in play therapy. They also participated in intensive filial therapy training using the CPRT protocol (Bratton et al., 2006), meeting 2 times per week over 6 weeks in the summer, when they both had breaks from teaching. During that time Andy participated in 10 supervised play sessions with his father, and Carol conducted play sessions with Chris. In the final phase of counseling, Carol participated in 6 additional weeks of supervised filial play sessions with Andy.

The playroom where Andy and I met was a fully equipped playroom similar to Landreth's (2002) description. I made sure there were baby dolls and doll figures with varying skin tones, including dolls representative of Andy's ethnicity. Andy was immediately drawn to a Native American baby doll with thick, short black hair that he initially decided was a girl. As Carol was not aware of any particular toys that Andy had played with in the orphanage, the only other culturally responsive toys included were musical instruments, food items, and eating utensils.

During a brief meeting with Andy and his mother prior to his first play therapy session, I introduced Andy to the playroom, saying, "Andy, this is the playroom where we will come each week. In here you can play with the toys in lots of the ways that you'd like," and then I allowed him time to explore. I bent down to be on his level and inquired if there was anything he wanted to ask or know more about. When he shook his head, I asked him if his mother had told him why they were here today. He informed me (with very limited English), as he looked at his mother, "Get mad and break things." I responded to let him know that lots of kids come here for the same reason, and added that this was a special place where he could play and talk about things that bothered him. Andy looked up at me and smiled, and although he didn't let go of his mother's hand, I believed that in those brief moments we had made a tentative connection. I ended the initial meeting by loaning Andy and Carol a child's storybook on play therapy to read together to help Andy feel more secure about coming to play therapy. I followed up with a phone call to Carol to discuss her concern that Andy would not separate from her and go to the playroom with me in the upcoming week and to formulate a plan for how we would respond if that occurred.

Andy was sitting in his mother's lap when I greeted him in the waiting room to take him to the playroom for our first session. I noted that he seemed a little anxious. I leaned down and made eye contact and said, "Hi, Andy. Hi, Carol. Andy, remember I showed you and your mom the playroom last week? It's time for us to go back to the playroom and play" (I slowly stood up). As if on cue (and following the instructions I had given her during intake), Carol stated, "Andy, I'll be right here waiting for you when you get through playing!" He smiled up at me, but then hesitated and looked at his mother. I quickly added, "You can choose: Should we run to the playroom or walk?" His face lit up, and he jumped up and said "Run!" as he took off down the hall with me running behind him trying to

catch up. That started a weekly ritual of Andy running to beat me to the playroom. Although I gave Andy the choice of running or walking to the playroom to take his focus off separating from his mother, I was surprised he went so easily. Based on Carol's report of his extreme difficulty leaving her, I had prepared her for the eventuality of coming to the playroom with us for at least the first week. (As Andy and I ran to the playroom, I was thinking to myself that this child has more resiliency than his mother gives him credit for!) As we entered the playroom, I reminded Andy, "You can play with the toys in lots of the ways you'd like to. We will have 45 minutes to play, and then it will be time to go back to the waiting room where your mom is."

Andy's first four sessions were characterized by his need to win my approval and affection by impressing me, playing close to me, and by overt demonstrations of affection, which included touching me, patting my head, and putting his face right up next to mine. Based on his early history, it was not a surprise that he acted overly familiar with me. I knew that no amount of words would convince Andy that I already accepted him without condition—that he did not need to win my acceptance and approval. He would have to experience my acceptance firsthand before he could come to see himself as acceptable. During the fourth session, he used the flashlight to connect with me in a playful way by turning off the light in the room, then coming close to me and shining the flashlight so we could see each other's faces, saying, "You see me; I see you." I responded in kind, "I see you; you see me." He enjoyed repeating this over and over, often moving his face closer and farther away and giggling, but without trying to touch me. He seemed to have discovered a safe and fun way to connect with me.

Andy's expressive language was very underdeveloped and his speech was often difficult to understand. Many of his interactions with me consisted of one- to two-word utterances. Among his favorites were "Hey!"; "No good"; "Bad"; "What?"; "What that?"; and later, an exclaimed "Yes!" to show pride in his accomplishments. Nevertheless, he managed to communicate his world to me quite eloquently through his play. As might be expected from a traumatized child, Andy was initially hypervigilant to every noise he heard in and outside of the playroom. These disruptions made it difficult for him to focus. It was no wonder that learning was so difficult for him!

In the beginning phase of developing our relationship, Andy worked hard to gain my approval by showing me he was strong and could do difficult things, including hammering nails, lifting the sandbox and other heavy objects in the playroom, and climbing to the top of the shelves. After he successfully completed these tasks, he would often make a "muscle" with his arm to show me how strong he was. One of the first ways that he chose to impress me with his strength and ability was hammering nails into the log stump in the playroom. Because the log and hammer were next to where I usually sat, this also allowed him to remain very close to me. During our third session, as he stood by me and looked around the room, he kept glancing over at the log and hammer. I reflected, "Looks like you're wondering about that." Andy looked at me and said, "Dangerous" (obviously a response he had heard from adults). I reflected his concern, "You think maybe that would be dangerous to play with," and he shook his head no, as if to say he did not think it was dangerous. I responded, "You're not sure.

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Well, in here it's up to you to decide if you want to play with that," letting him know that he was free to decide what to play with, just as adult clients are free to talk about what they want to talk about. Andy looked at me and looked at the log wistfully, letting me know he really would like to play with it. I reflected, "Looks like you're thinking you might like to try it out. There are some safety glasses on the hook." Andy immediately picked up the hammer and a nail and exclaimed a word I did not understand, but his actions and affect clearly communicated what he was trying to say. I replied, "You decided you want to hammer!" This brief interaction revealed Andy's struggle between his internal motivation and desire, wanting to hammer, and the external message he had received from adults that hammers are dangerous. Because I was able to respond to his internal world, he became more focused in his play and his anxiety was reduced. After he carefully hammered the first nail in, he looked at me and exclaimed, "Look!" I leaned over to look closely at his work and reflected his obvious excitement: "You wanted me to see. You got it all the way in!" He proceeded to hammer in several more nails with strength and precision and looked up at me and showed me the muscle in his arm, to which I replied, "You're really strong." He nodded his head in agreement. Then I understood better what the hammering was all about and added, "It took a lot of strength to get those nails all the way in." Although hammering initially served as an activity that Andy chose to gain my admiration, later this activity served other important functions, including demonstrating competency and expression of anger. *(A note of caution: because children can accidentally injure themselves with a hammer, I always sit close by when a child chooses this activity.)*

The fifth session marked a shift in Andy's play. He was now feeling safe enough to leave my side and explore the playroom. He delighted in his discovery of a western-type hat that had a whistle attached by a string and proceeded to spend a considerable amount of time exploring the toys on the shelves and happily blowing the whistle. He noticed the baby bottle for the first time and filled it with water from the sink, sucking contentedly on it for approximately the last third of our session while he continued to wear the hat. When I told Andy that we had 5 minutes left to play, he went to the sandbox, climbed in, lay back against the edge, and sucked on the bottle with a glazed look on his face. When I announced that our time was up, he shook his head with the bottle still in his mouth, letting me know that he wasn't ready to leave. I was patient and restated the limit as I leaned down to his level and made eye contact and reflected his desire to stay longer: "Andy, I know that you'd like to stay longer, but our time is up for today. You can drink some more from the bottle next week." I repeated similar statements three more times, letting him know I understood it was hard to leave when he was enjoying himself. Still he shook his head and remained in the sandbox. Ordinarily, I would tell my students that this is a time when you need to be more firm in your tone of voice, but I sensed that there was something different about this experience. I realized in that moment that I wished that Andy didn't have to leave either. I understood that the sucking was serving a very important function for him, but that it was also important that he experience the consistency and predictability of our time together. I bent down even closer to him and said, "Andy, I wish we didn't have to leave either, but we do. It's time for us to go back to the waiting room where Mom is." Focusing my responses on our relationship, rather than on Andy's behavior, seemed to be what he needed—a

lesson that I shared with Carol in our next parent consultation. This brief but powerful interaction with Andy reminded me how important it is for me to trust my experience with children—that when I do, they let me know what they need from me.

Our next several sessions together were characterized by Andy's fascination with the whistle and the bottle (although his sucking was less intense), while continuing to show me how strong and competent he was at hammering. Blowing the whistle seemed to be serving an important expressive function for Andy. During session 7, as he happily blew to his own beat while he played, he suddenly stopped and looked at me, saying "Take whistle home?" Then he added, "Please . . . I bring back next time." For a brief moment as I looked into his pleading eyes, I almost gave in; however, I knew that this, too, was part of the consistency and predictability of the play therapy experience that Andy needed so desperately to experience. Instead, I let him know that I understood how much he wished that he could take it home and explained the playroom rule: "All the toys are for staying here in the playroom, but it will be here for you to play with next week." He seemed to be considering the rule, but stopped playing with the whistle. (Because of Andy's delay in language development, I sometimes wondered how much he understood.) A few minutes later I told Andy that we had 5 minutes left (a time when he often introduced significant play themes or play sequences). He began to paint for the first time, painting all the colors on top of each other to make a blob of brownish-black. The following brief interaction seemed to reveal Andy's inner conflict between his own wishes and needs and trying to please others, especially "Mommy."

Andy: Mommy be mad (as he began to use his hands to smear the paint).

S: You're worried that Mommy will be mad.

Andy: Messy (as he went to the sink and washed the paint off hurriedly).

S: You're worried that Mom wouldn't like it if you got messy. But before I could get the words out, he began to take his shoes off and looked over at me to see what I thought (as if to ask my approval).

S: Looks like you're not sure if that's okay. In here you can decide if you want to take your shoes off or leave them on. (Andy takes both shoes off.) You decided you wanted them off. (I responded to his internal desire.)

Andy went back to the paints and picked up the paint brush and looked at me and said something I did not understand as he proceeded to paint one hand and then the other, then sat down and began to paint the bottom of his feet, enjoying the feel of the brush (much like the sensorimotor play of a younger child).

Andy: Tickles (a big smile on his face).

S: You like how that feels.

S: Andy, we have 1 minute left to play, then it's time to wash up to go home. (I was careful to make sure that we washed all the paint off his feet and hands before we went back to the waiting room to prevent potential guilt feelings that Andy might experience if Carol reacted adversely to a "messy Andy.")

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In our next parent consultation, I reminded Carol of the importance of Andy's wearing old clothes to play therapy so that he could get messy if he needed to, and we discussed what making "messes" might symbolize for him.

For the next 2 sessions, Andy repeated this ambivalent play theme of "Mommy be mad" and "Not supposed to get messy" and the obvious joy of getting messy, smearing paint on the paper and painting his hands and feet. My responses followed a similar pattern, reflecting that he was concerned about what his mom would think, but focusing on his enjoyment of the experience and always making sure he was returned to his mother mostly clean. Andy's play seemed to reflect his continued internal struggle between perceived external demands of pleasing others (hypervigilance), including an awareness that his "out-of-control" behavior was unacceptable ("not okay to be messy"), and his internal need to express all aspects of self, even the "messy and chaotic."

Andy reveled in his newfound freedom in the playroom. As he experienced the safety and freedom to fully express himself, his play became more focused and meaningful. It was during this phase of play therapy that he began to display more of his conflicted and difficult feelings, most notably his sense of being damaged or "no good." He would inspect all the toys, and any that had the slightest flaw or did not seem to work properly were deemed "no good" and thrown away. During this period, he would sometimes mumble "No good" up to two dozen times in a session.

During our parent consult in week 9, Carol announced that Andy's "stealing" was now happening at school as well as at home. It was obvious from Andy's play ("Mommy be mad") that he perceived that Carol was angry and upset with him, although she denied it initially. On the surface Carol was trying to appear understanding, but underneath her façade she was embarrassed and upset with Andy. Her response seemed exacerbated by the fact that she taught at the same school and was concerned about what her colleagues would think. I helped her understand that Andy was picking up on her underlying feelings, even though she was not directly expressing them to him. We again discussed Andy's hypersensitivity to nonverbals. This problem was significantly impacting Carol's acceptance of Andy; thus we needed to actively address possible solutions.

My first priority was to help Carol and others to reframe how they viewed Andy's behavior. It seemed obvious that Andy's stealing, as described by Carol, was not malicious. Thus I asked Carol to use the term "taking things that belong to others" instead of "stealing" when talking to Andy (and others) about his behavior. Although I could have made guesses about several possible underlying reasons why Andy was taking things that did not belong to him, the explanation that seemed to help Carol and others normalize Andy's actions was that this behavior could best be understood as a function of a significant developmental delay (one of many regressed behaviors he demonstrated). I explained to Carol that her description of the events and Andy's response when confronted were consistent with what one would expect from much younger children who simply see something they want or need and take it. I reminded her of a toddler's worldview: "If I want it, it's mine!" She agreed that if Andy were 2 years old she would not see his behavior as stealing. Carol and I devised a plan for how to respond to Andy at home and at school if and when he initiated this behavior again. Carol and I talked with Andy's teacher at school to discuss the plan, and Carol also

elicited school administrators' help. We all approached this as if Andy's behavior was a function of development and that it was our job as adults to help further his development by helping him learn the concept of ownership and people's related feelings about their property. Andy's parents and the school were instrumental in helping Andy move through this phase over the next several weeks. This is just one example of many instances in which they proved to be incredible allies throughout the therapeutic process!

Andy's play during the next several weeks could best be described as disorganized and chaotic. This relatively short phase was marked by the most intensely negative affect during our time together. Themes of "bad boy" and "no good" persisted. He began to actively deal with the issue of stealing. For the next 4 weeks he was focused on trying to figure out a way to take the whistle from the playroom. Each week he would retrieve the hat with whistle from the shelf, take it behind the puppet theater, cut the string to get the whistle off, and then try to figure out how to take it out without my knowing. He was generally agitated and often unfocused in his play once he had the whistle cut off. The first time he hid the whistle under the beanbag chair behind the puppet theater. He kept asking me how much time we had left and for the first time wanted to leave early. After I stated that there were 5 minutes left, he went behind the theater and put the whistle in his pocket and then came over to me and pulled it out and showed me "his whistle" that he had "brought from home." I responded that it looked just like the one we had in the playroom. At first he tried to convince me that it was his and then said, "Just kidding," and put it back on the shelf. I let him know that I understood how much he liked the whistle and wished that he had one like it.

During the next 2 sessions his level of agitation was at an all-time high. In each session, Andy immediately took the hat and whistle behind the theater, cut the whistle off, and stuffed it in his pocket. His play was highly disorganized and he was clearly distressed. (I was having difficulty being with him in this experience and found myself wanting to rescue him. I had to work very hard to not let my need to protect Andy from these difficult feelings interfere with his need to work through them in his own way. I found myself needing to seek peer supervision over the next few weeks to process my own feelings and avoid imposing my needs onto Andy.) During session 13, with the whistle already pocketed, Andy spent 15 minutes throwing toys, repeating, "No good, no good." I reflected, "Nothing seems to be going right today." He seemed to calm down for a moment, and with less intensity than before, he hit himself in the head (with his hand) and said, "Stupid." I went over to him and reflected, "Andy, I know you're upset about something, but you're not for hitting." He paused and looked up at me and quietly said, "I want to go." I knew it was important to maintain the structure and consistency of our time together, so I held firm and replied with all the empathy I was feeling for this little boy, "Andy, I know that you're having a hard time in the playroom today, but we have 20 minutes left and then it will be time to go." He insisted, "I don't want to play anymore." I replied softly, "That's up to you. . . . You can choose to play or not to play in the 20 minutes that we have left." He did not try to leave, but every few minutes he looked at the clock and asked me how much time we had left. When I told him we had 1 minute left, he went straight to the door with obvious relief. At this point, he still had the whistle in his pocket. As he got to the door, I gently responded, "Andy, I think maybe you put the whistle in your pocket

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when you were playing with it earlier. Remember, the toys are for staying in the playroom." First he shook his head no. "Maybe you should check to make sure." He put his hand in his pocket and said, "I forgot," and pulled out the whistle and handed to me. As I put it back on the shelf I reminded him that "it will be right here for you next week." (I tried to remember each week to check the playroom in advance to make sure I kept my promise.) The next session followed a similar pattern; however, Andy was noticeably less agitated, and when it was time to leave he pulled the whistle out of his pocket without any prompting and said, "Oh yeah," and handed it to me matter-of-factly.

The change in Andy's affect in our next session was remarkable. He was very excited to see me and ran straight to find the hat and whistle and put it on his head. It was the first time in 5 weeks that he had not cut the string and hidden the whistle. He put the hat on and marched around the room happily blowing the whistle (very free), all the while sucking on the baby bottle for the first time since he became preoccupied with the whistle. He spent the majority of the session with the whistle in one hand and the bottle in the other, much of the time marching to a beat of suck-suck—whistle-whistle. Whereas his play over the past 4 weeks was constricted, agitated, and chaotic, his play this week was spontaneous and joyful. It was during the last 5 minutes of this session that Andy first introduced the "spinning game," his name for a game he devised that involved his directing me to spin him around while he sat in the swivel chair that I normally sat in. There were several variations of this game, but they all had in common a theme of mother-baby play. Little did I know just how important this game would become!

This tumultuous phase in Andy's play seemed to represent his struggle for self-regulation as he internalized the concept that the whistle was not his and that although he wanted it, he could now choose to play with it and give it back (just as outside of the session, he was struggling with understanding this concept on a larger scale). The combined impact of his being able to play out his struggles in the safety of the playroom with me as a supportive witness to his expression of difficult feelings, and his parents' and teacher's changed attitude and response to his behavior at home and school resulted in an almost complete halt in the stealing behavior. Carol reported that Andy was no longer taking things at school and that he had mostly stopped this behavior at home. Her scores on the CBCL and PSI completed after session 15 supported her verbal report of Andy's improvement in this area as well as in the frequency and intensity of his anger outbursts.

Andy's play over the next 10 sessions was characterized by greater spontaneity in self-expression and increased positive affect. For a child that had begun therapy being so hypersensitive to what he needed to be in order to win others' approval, this movement was just what I had hoped for. During this time, he found the baby dolphin and chose it to become a central figure in his play. He often carried "Baby" around while he played or gave it to me to hold. He determined that the dolphin was male and was especially gentle with it. Andy continued the spinning game, which became increasingly more intimate. He would often lie in the chair and suck from the bottle, gazing at me and sometimes giggling while he directed me to spin him around and around.

It was also during this time that he introduced his favorite game, "hide-and-find." With the exception of a few breaks, this was a game that Andy delighted in

up until our termination, sometimes spending over half of our session in this activity. From the first time he introduced this activity, with "Close eyes, I hide," I sensed that this game was very important to him. I wanted to play my part just the way he needed me to; thus I inquired, "How will I know when you're ready?" Andy said, as if it were obvious, "Count to 10," and after I did, "Find me!" he yelled. I slowly began to look for him. (I was wondering to myself, Does he need me to find him easily or take a long time? Because of his abandonment issues, I believed that how I responded to my task of finding him was vital.) In a soft voice, I asked, "Are you going to be easy to find or hard to find?" Andy paused a minute and said, "Hard!" So I began a long search around the room, amplifying my movements and efforts to find him, often exclaiming variations of "I bet you're over here. . . . Hmm, not here . . . You hid really well. . . . I wonder where you are?" When he would giggle, I would reflect, "I hear something. You must be over there." Each time we played this out, no matter whether he chose to be easy to find or hard to find, he always gave me clues to let me know when he was ready to be found. We were in synch with each other; I knew when and how he needed me to respond. Again, Andy had chosen a playful and safe way to deepen our relationship and work through his early experiences of what I guessed to be an inconsistent response to his nurturing needs. He was learning that I could be trusted to consistently respond to what he needed. Through this experience and his experience of being with a loving family, I hoped that Andy would begin to see the world as secure and predictable—a place where he could depend on adults to keep him safe and respond appropriately to his needs.

Andy was clearly working on attachment and trust issues that I believed could be more effectively addressed by involving Carol and Tom in his play therapy. We discussed starting CPRT training in the near future so that Andy could play out his attachment needs with his parents. We agreed that Andy was making good progress, including a reduction in his violent outbursts, a marked decrease in stealing, and fewer sleep problems. We tentatively discussed ending therapy by the end of the school year.

During this time, Carol reported that Andy was showing more interest in what he was like as a baby. When he saw babies on television, he would ask, "Did I look like that?" When he saw pictures of babies, he would ask, "Was that me?" Once again, I recommended that Carol help Andy make an adoption scrapbook to help him understand his adoption story, and we practiced how she could explain events to him in appropriate words (suggestions that Carol continued to resist; it was, after all, a difficult story to explain). However, as a result of our discussions, Carol and Tom came up with a remarkable plan. They had arranged for a 17-year-old exchange student from China to live with them for the next academic school year. Because of Andy's increased interest in adoption issues and Tom's work schedule which prevented him from starting the CPRT and filial training, we decided to continue play therapy until the end of the school year and begin CPRT the first week in June, when his schedule allowed him to travel less.

Sessions 25 through 28 marked the introduction of another significant play theme, although I had no idea what it meant at the time. Andy's play was highly

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organized and purposeful. He clearly had a plan in mind. He began to use the dress-up clothes trunk as an important part of his play, first by taking everything out and trying on most of the clothes to see how he looked. He would often dress up both of us so that we looked similar and exclaim, "I girl like you!" I was a bit puzzled by what this meant for Andy, but he took great delight in this activity. However, what puzzled me most was that he ended this play sequence each week with a ritual of putting the lit flashlight in the trunk, along with the baby dolphin, and closing the lid. Before he left the playroom, he would open the lid to make sure the light was still on. He seemed to gain a great deal of satisfaction from his play during these weeks. Although at this point I did not comprehend the meaning his play held, I conveyed my understanding that it was important to him.

By this time, school was out and Andy had successfully completed his second year of kindergarten. Carol's scores on the CBCL and PSI (Table 20.1 and 20.2) reflected her weekly reports and comments regarding Andy's improved behavior, most notably in the areas of sleep, stealing, and anger outbursts.

In keeping with our plan, Carol and Tom began intensive CPRT training twice a week, following the treatment protocol outlined by Bratton et al. (2006). We had 6 weeks before they and their family would be going on vacation. During that time I did not see Andy in play therapy, because he was coming to the clinic twice a week to have play sessions with his dad as part of the CPRT and filial therapy training. Carol conducted her play sessions with Andy's stepbrother, Chris. Carol and Tom and I decided that this was a good opportunity to focus on helping Andy develop a closer bond with Tom, in hopes of relieving Carol of her full-time parenting role. Andy played out some of the same themes with his dad that he played with me, particularly doctoring the baby and sucking from the baby bottle. I had prepared Tom for the possibility that Andy might want to suck from the bottle, but Tom struggled to accept that behavior from his now almost 7-year-old son. He understood that it was important to Andy and that he should be accepting, but as he said, "That's easier said than done." We role-played how Tom could respond in a way that would convey his acceptance and encourage Andy to involve him in his play, but what seemed to help most was asking Tom to remember when Anne (his 14-year-old biological daughter) was a baby and how he had enjoyed holding her close and giving her a bottle. Andy had not had the opportunity to have that experience with him or Carol (and maybe not with anyone).

Both Tom and Carol were motivated students and became fairly proficient in demonstrating the CCPT play skills taught in CPRT. Each session began with a review of skills, role-play, and questions, immediately followed by Tom and Carol taking turns conducting play sessions with their respective child of focus, while I supervised through the two-way mirror. After both parents had conducted a play session, I met briefly with them to give feedback and to process their play sessions. I focused primarily on helping them better understand what Andy and Chris were communicating through their play. The children were supervised by clinic staff when they were not participating in play sessions. During this time, both Tom and Carol reported that this was the least stressful time since they had adopted both boys. They agreed to continue their weekly play sessions while they were on their monthlong vacation at the beach. Because Andy seemed to be actively addressing his attachment needs, we decided that I would see

Andy for at least one more month after they returned to allow for appropriate termination of our relationship.

Andy and I resumed play therapy 2 weeks before he would start first grade and the week before Dong, the exchange student, arrived from China. I had not seen Andy in 5 weeks and had not had a play session with him in 11 weeks. Carol completed the fourth measurement of the CBCL and PSI to assess progress over the summer months, when the only treatment provided had been CPRT (and going to the beach for a month!).

Andy was very excited to see me and pulled me down the hall running. He ran ahead and when I entered the room, he was inside the trunk covered with clothes. He delighted in jumping up and yelling, "Boo!" Andy's play seemed to pick up where he had left off, with a fascination for the dress-up clothes and the trunk. With 5 minutes left, I guessed that he had something important in mind as he blocked the door and taped it so "we can't get out." He quickly threw the rest of the clothes from the trunk and climbed inside, trying several different positions to see how he could make himself fit. Then he climbed in again and curled up in a distinctly fetal position and told me he wanted me to close the lid. I clarified that he would let me know when I should open it. He told me he would knock, and he demonstrated that he would knock three times. He replayed this scenario three more times, staying inside only a few seconds each time. He obviously had something else in mind, as he very purposefully retrieved the flashlight from its spot on the shelf to take it with him inside the trunk. He turned on the light and curled into a little ball and told me to shut the lid "all the way." This time he stayed for almost a minute, although it seemed like several minutes to me. I glanced at my watch and noted that we had 1 minute left and bent down and whispered into the trunk, "Andy, we have 1 minute left." He immediately knocked on the lid to let me know he was ready to come out and ran over to get the baby dolphin and the baby bottle of juice. He carefully laid them inside the trunk on a soft piece of fabric, along with the lit flashlight and shut the lid. Again, I had no idea what all of this meant for Andy, only that he was working hard and had a definite plan in mind, one that he first conceived of 12 weeks ago!

Following this session, Carol and I met for our regular consultation. She looked more relaxed than I had ever seen her. She discussed Dong's arrival from China in the coming week and how excited Andy was that he and Dong would be sharing a room. Then Carol said, "I just remembered what I wanted to tell you. Andy said the strangest thing to me this week. He asked me if he had a flashlight when he was in his mommy's tummy." (By now, Andy was able to acknowledge that he came from someone else's tummy—not Carol's.) Carol disclosed that Andy had continued to be preoccupied with babies over the summer. He loved to watch shows on cable television that showed babies being born and persistently asked, "Was that me?" They had recently purchased *The Miracle of Life,* which was now Andy's favorite movie. Now, Andy's flashlight-in-the-trunk play made perfect sense: He was trying to make sense of where he came from, quite literally! Carol's scores on the fourth measure of the CBCL and PSI in August reflected her self-report of "how relaxing and uneventful" the summer had been. The CPRT filial training, 1 month at the beach, and continuing the home play sessions seemed to be just what the doctor ordered for this family.

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"I got new big brother!" Andy exclaimed when I greeted him the next week. He was obviously excited, and much of his play revolved around his "new big brother." He played inside of the trunk again with the baby dolphin and the flashlight, but with less intensity. He did not ask me to close the lid. This was the end of Andy's important in-the-trunk play until our closing session. In the next 5 sessions, his play was primarily focused on issues related to attachment and mastery. His play showed increased levels of reciprocity and sharing. He continued to suck the bottle, but with much less intensity (his sucking need seemed to have decreased in response to the play sessions with his mom and dad over the summer). We spent extended periods of time playing the spinning game, which had become increasingly more intimate, sometimes with the light off and sometimes with Andy directing me to shine the flashlight under my chin or his so we could see each other. He delighted in inventing several variations of baby-like games in which he directed me to touch and count his fingers and toes while I spun him around. During this time, he also began to express negative feelings toward me directly. One day I was 15 minutes late for a session and he wouldn't look at me as we walked to the playroom (no running today!). When we entered the playroom, he looked at me and then kicked the inflatable bop bag. I reflected, "Seems like you might be mad that I was late. It's okay to tell me that you're mad." Andy shook his head slightly as if to say he was not mad. It seemed important to help him understand that it is okay for people to get mad at each other and that our relationship could recover. I bent down to Andy and gently said, "You were expecting me to be here today when I was supposed to, and I wasn't. I'm sorry" (something I should have said earlier).

Dong's arrival seemed an impetus for Andy to more directly address issues around the orphanage, his adoption, and cultural identity. Themes of attachment, trust, and mothers and babies (often looking for each other) continued over the next 15 weeks. His play became more complex and imaginative. During this time he often had difficulty leaving the playroom. Carol and I discussed the importance of Andy's play themes and his need for the safety of the play therapy relationship to play out and express these confusing experiences. We decided he should continue in play therapy.

Session 36 marked Andy's first attempts to play out what seemed to be experiences that were directly related to his abandonment by his birth mother and his orphanage experience. The baby dolphin had long been a favorite and a meaningful toy for him. I already sensed that he used the dolphin to represent himself. (Andy had previously identified it as a baby and as a male during in-the-trunk play.) He took the baby dolphin and for the first time buried it deep in the sand. He abruptly left it there and began to hammer nails in the log for the first time in several weeks. I saw this behavior as a "play disruption," a sudden shift from play that seems to hold specific meaning for the child in response to anxiety or discomfort (Findling, Bratton, & Hensen, 2006). Dealing with traumatic material can be retraumatizing; thus it was important for Andy to approach this at his own pace. Later he went back to the sandbox and looked at me and said, shaking his head, "Don't know where he is." I simply reflected, "Not sure where he is," not knowing who it was that didn't know where the baby was. I waited patiently to see if a story would unfold. Andy found the large dolphin (which I interpreted as a mother or caregiver figure) and a few other caregiver-figure animals (a seal, a walrus, and a polar bear) off the shelf and placed them carefully in the

sandbox. He again abruptly went back to hammering (an activity that he had long ago mastered). After I announced that we had 5 minutes left, Andy went to the sandbox and hurriedly moved the "mother" animals all around on top of the sand as if they were all looking for the baby, then had the large dolphin dig down and find the baby and exclaim, "There he is!" He then took the little dolphin and made it flip up in the air as if it were jumping out of water. I reflected, "It looks like he's happy that they found him." (Inside, I was struggling with staying with Andy's expression of wanting his mother to find him and knowing that that will not happen. My need to try to protect him from being hurt had surfaced again.) I wondered if Andy's play represented wishful thinking that he could go back and find his mom (or remembered "mother figure"), like Dong would go back to his family in China at the end of the year.

From the beginning of Dong's stay Carol and I had discussed the importance of helping Andy understand that Dong would be going back to China at the end of the school year. Because of Andy's thematic play, I made a note to further discuss these issues with Carol, along with adding more facts to his adoption story. I also found a child's book on adoption that more closely mirrored Andy's own experience than the books that Carol had been able to find. *We Adopted You, Benjamen Koo* (Girard, 1989) became a favorite of Andy's; he could relate to the boy, whose skin color was different from his parents', who did not know his real birthday or name, and who would probably "never know his birth mother."

Using animals rather than people figures to make sense of his adoption experience seemed to provide Andy with the safety and distance he needed. On two occasions he initiated brief play with mother dolls with two different skin colors, but both times he abandoned the play. Over the next several weeks Andy was preoccupied with babies and mothers. In our clinic playrooms, we have several sets of animal families grouped together. He readily identified the "babies" and separated them from the mother-figures, devising various scenarios of what happened next. Often he would place the babies across the room from the mothers, sometimes out of sight behind the puppet theater. My responses mirrored his actions; sometimes the babies could not find the mothers and sometimes the mothers could not find the babies. In Andy's play there were always more babies than mothers, which seemed to reflect his orphanage experience. Another theme that emerged was that the baby dolphin began to receive better care than the other babies; the others often were thrown down from the shelf, but the baby dolphin was always gently handed to me to hold. (I took that as a positive sign of self-nurturing, and that Andy was internalizing messages from his adoption story that Carol and he had been working on. I knew that the story conveyed the message, "We chose you out of all the children," and other messages that conveyed that he was "special.") Andy returned to burying the baby dolphin deeply in the sand, but with a different story line this time. For the first time he identified the large dolphin as a mother and then handed it to me as if I knew what I should do.

S: What am I supposed to do? (soft voice).

Andy: Look for baby.

S: Is he going to be hard to find or easy to find?

Andy: Hard (I started to look in the sandbox).

(*Continued*)

Andy: No—everywhere, not here. (Now, I understood, I was supposed to search for the baby dolphin the way I searched for Andy when we played hide-and-find.)

S: I may need some help if you want me to find him. (After I had spent a long time looking for the baby and not getting a sense of what Andy needed from me. I was not sure who I represented—birth mother, orphanage caregiver, or maybe even Carol; neither did I understand if he wanted me to find the baby or not, so I was careful to let him fully direct this play sequence. Andy never did direct me to find the baby, but he seemed to enjoy that I was methodically looking all over the playroom.)

S: I'm looking everywhere I can think to look, but I can't find him. . . . I wonder where he is? (Andy looked at me and shrugged his shoulders.)

S: I don't know where he is either, but I'm going to keep looking. . . . I'm not giving up.

Andy seemed content that I (as the mother dolphin) looked everywhere and never gave up. He also seemed content or resigned to the fact that in this session the mother never found the baby. Andy seemed to be integrating his previous experiences into a positive memory of being wanted and special, even if his mother had left him. Just before we left the playroom each week, he was always careful to return the dolphin to its place on the shelf. It seemed important to him that the baby was left in its safe and predictable spot. In subsequent sessions Andy's mother–baby animal play continued, but with less focus on the mother dolphin. He no longer buried the baby dolphin and began to include the polar bear, seal, and walrus as more central figures in his play. Although he occasionally continued to place the mother dolphin in the sandbox, the other figures played a more active part in his play. These figures seemed to represent his current family, with the polar bear taking on an increasingly nurturing role. Often, he would make a "home" just for the polar bear and baby dolphin, and the other animals were left out. Consistent with Carol's report, although Andy would now stay with Tom while she went out for a brief period of time, he still preferred to spend time just with her. Although we were making progress, Andy was not at the point with Carol that he had fully internalized the important attachment messages "When you leave me, I know you will come back" and "I can count on you to meet my needs." Andy was clearly working on attachment issues and making sense of his adoption experience; thus Carol and I decided to continue play therapy and reevaluate monthly.

During this phase of therapy, hide-and-find and the spinning game were central to Andy's play activity. He continued to fill the bottle with juice each week, but often simply drank from it more than he sucked. He would often fill it and leave it on the shelf (in case he needed it). He explored the limits of our relationship by purposefully, yet playfully, choosing to break limits that neither hurt the room nor me, such as shooting me in the back with the dart gun and spraying me with water. The only new play that he introduced during this time was related to making sense of his cultural identity. He introduced two puppets, "Shake Your Bootie Guy," a ferocious-looking dragon who spoke English, and "Donga," a cute little girl puppet with a big smile that made baby-like noises that sounded like the Chinese equivalent of "ga-ga-goo-goo." From Carol's report, Andy did not know any Chinese-language words other than a few words he had learned from Dong. Andy incorporated Shake Your Bootie Guy and Donga primarily into our

hide-and-find play, varying how he used them. I took it as a positive sign that both puppets were important to him. Toward the end of his therapy, Andy began to choose a similar, but friendlier-looking version of Shake Your Bootie Guy to sometimes represent Donga. I thought this might be a sign that he was beginning to integrate his two cultures.

The fifth measure of treatment outcome was administered after session 42, at the end of the fall semester. Carol reported overall improvement in Andy's behavior, although school continued to place additional stress on Andy, which placed additional stress on Carol. Her scores on the CBCL and the PSI reflected Andy's school stress, as well as her stress related to parenting a more stressed Andy. Although he continued to be much improved from when he started play therapy, his scores were slightly more elevated than they had been prior to the beginning of this school year. Carol and I decided that Andy would continue in play therapy. I encouraged Carol to resume her home play sessions with Andy, at least during the holidays.

Sessions 43 to 45 marked a brief but tumultuous shift in Andy's play, with a marked increase in expression of negative affect. He was clearly feeling conflicted about something. His anger was expressed more directly toward me, although I suspected this was transference of his feelings toward his mother. On two occasions he stated, "I'm gonna make a big mess" and proceeded to follow through with his threat, purposefully knocking toys off the shelves that he knew would not break. (I thought to myself that this was progress: Andy seldom expressed his anger directly and in such a controlled manner.) I reflected that he seemed angry and that he was showing me how mad he was. The second week that this happened, I wondered aloud, "Seems like maybe you're still mad about something." Andy exclaimed, "Yeah—you're mean!" He then proceeded to hammer the nails really hard for several minutes until he seemed spent, and I reflected, "It feels good to hit it really hard when you're mad." Andy spent the rest of the session focusing on our relationship by playing hide-and-find in the dark with the flashlight. During our last 5 minutes, he initiated the spinning game with a new twist; he filled the bottle with juice, took his glasses off, and pushed the chair so it was touching my legs. Then he showed me he how he wanted me to just barely move the chair back and forth (it felt like rocking). He lay across the chair and contentedly sucked while he gazed into my eyes with a glazed look. It was a very intimate moment, the kind he needed to be having with Carol and Tom. I made a mental note to talk with Carol about taking a break from play therapy and beginning filial therapy in the next couple of weeks.

When Andy and I returned to the reception area, Carol was insistent that she needed to talk to me before they left. I knew by her tone of voice that it must be important, and so did Andy. I bent down to let Andy know that his mom and I would be back in 15 minutes (that was all the time I had before another appointment) and that he would stay with Jane (the receptionist) while we were gone. Andy grabbed his mom around the waist and stated adamantly that he didn't want her to go and was not going to let her go. Although on the surface his words and actions conveyed one message, I had the strongest sense that the real issue was that he did not want me to hear what his mother was planning to say, that he believed that what she had to say would negatively impact my view of him. I did something I have never done before; I knelt down beside Andy and whispered so that only he could hear, "Andy,

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there's nothing your mom could say to me that would ever change how I feel about you." Although, in general, I believe that my actions with children speak much louder than my words, I sensed that Andy needed to hear this. I knew that he experienced my acceptance of him as unconditional, but his past experiences told him that things can (and will) change. He looked at me for a few moments as if he was deciding if I was telling the truth, then smiled at me as if to say "Okay" and went over to watch the cartoon playing in the waiting room. Our relationship had passed a test. I was reminded once again of the importance of trusting my intuitive sense of what children need from me.

Andy's academic performance and behavior at school were bothering Carol, and her frustration was carrying over to home, where she felt unsupported. Although Andy seemed to be experiencing more problems at school than he had in a while, Carol's extreme reaction was my main concern. Her escalating level of stress was an indication that I needed to shift my focus to strengthening the relationship between Carol and Andy and at the same time provide Carol with some much-needed parental support. Carol was initially unsure about taking a break from play therapy to focus on filial therapy, but I emphasized to her that right now Andy needed her more than he needed me. Coincidentally, I also sensed at this moment that Carol needed me more than Andy did. We decided that she would begin filial play sessions with Andy in 2 weeks during our regularly scheduled appointment time.

At the beginning of our next session, I explained to Andy that he and I would be taking a break from playing together: "You and I will come to our playroom two more times, and after that, you and Mom will get to play together in the family playroom." (I always use a different playroom for parent-child play sessions if I am concurrently seeing the child in play therapy.) In Andy's next two sessions with me, he showed me that he was secure and confident in our relationship. The spinning game and sucking on the baby bottle were his primary play activities. Instead of just counting fingers and toes, Andy taught me how to sign "I love you" as I spun him around.

For the next 4 weeks, I coached and supervised Carol's play sessions with Andy. I spent time supporting her worries and concerns about Andy and about her being a good enough mother for him. We actively worked on increasing her enjoyment and acceptance of Andy for who he is, school difficulties and all. Andy enjoyed seeing me at the clinic, but was eager to play with his mom. Carol reported feeling much less stressed, and we both agreed that Andy's behavior had improved markedly by the end of the 4 weeks. A return to filial play sessions proved to be just what Carol and Andy needed. With the end of the school year 5 weeks away, Carol and I decided that it would be a good time to end treatment, at least for now.

Planning for the ending of therapy is important for all children, but even more so for children who have had multiple experiences of abandonment and loss. I knew that how I approached the ending of our relationship would be critical so that Andy would not perceive me as yet another important adult (and female) who was leaving him. I explained to Carol that I would meet with her and Andy briefly before Andy's play session in the following week to talk with both of them about termination. I explained to Carol that I would be letting Andy know why I believed that he no longer needed to come, and that I would ask Andy and Carol what they thought about ending our sessions. I encouraged Carol to say to Andy

what she had reported to me about the improvements she was seeing. Because Andy and I had such a long-standing relationship, I decided that 4 weeks would be an appropriate termination phase. In our meeting with Andy, Carol did a remarkable job of conveying to Andy all the positive changes that she had seen over the past several months. Although Andy did not say much, he seemed to understand that we would be ending soon. Marking the weeks on a calendar I gave him, I told Andy he would come to the clinic four more times, and that each week he could decide if he wanted to play with his mom or with me. Because he had been playing most recently with his mother, it seemed important to give him the option of play therapy with me or filial play sessions with his mom.

For the first of our termination sessions, Andy decided to have a filial play session with his mother, but afterward he announced, "All the next times, just you and me play." His play over the last three sessions was primarily constructive and showed his mastery over the playroom. He didn't introduce any new play and seemed to be replaying our time together, almost like fast-forwarding a favorite home movie. He briefly played out some of his more destructive, confused, and chaotic play, but without the previous intensity. Mostly, he enjoyed repeatedly playing his favored hide-and-find and spinning game, while making sure he had some time for hammering the log in our last moments together.

When I went to meet Andy in the waiting room for our last time together he showed me the cupcake he had brought. He whispered in my ear that he needed a lighter and showed me the candle he had in his pocket. (I later found out that he wanted Carol to let him bring a lighter and, understandably, she refused.) It seemed very important to Andy, so I retrieved a lighter from my office. He directed me to light the candle on the cupcake and sing "Happy Birthday," after which he blew out the candle and said, "Do it again!" Andy's birthday was 2 months away, but he seemed to have found a perfect way to celebrate our ending and at the same time, his beginning. We repeated the scenario 12 times, with Andy sometimes joining in the singing. He looked into my face with a huge smile the entire time I sang to him. After the 12th round of "Happy Birthday," he said that we could now eat the cupcake and carefully cut it in half with a play knife from the kitchen.

"Spinning game!" Andy announced, which I had long ago learned meant he would sit in the swivel chair and that I would sit in the smaller chair. After a few rounds of playing our traditional spinning game, touching fingers, counting, and signing, he went to get the flashlight and announced with a huge grin, "Hide and find!" At that point in our relationship, we needed few words to communicate meaningfully with each other. Andy spent most of the rest of the session hiding in all the places he had hid over the months we had been together, starting with the oven in the play stove that he was no longer able to fit in and ending with the dress-up clothes trunk that had been such an important part of his play. With each new hiding place he would giggle as he told me without my prompting whether he was going to be "easy to find" or "hard to find."

After I announced that there were 5 minutes left in the session, he came back and sat down by me and began to tell me about going to his grandmother's beach house for the summer (it was very unusual for Andy to sit and talk!). He let me know that he is going to have two birthday parties, one here with his friends and his "real" birthday party at his grandmother's house. Then Andy looked up

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at me and said, “I won’t see you.” I assured him that although we wouldn’t see each other for a while, he could always call me. I wrote down the clinic phone number and gave it to him. Then he looked up at me for a few moments and said quietly, “You might need to call me.” I leaned over and quietly replied, “Sounds like you’re worried about us not seeing each other. Anytime you need me or want to come back here, you can call or tell your mom.” Then Andy responded with a big smile, “I know! I give you my phone number,” as he wrote the number and gave it to me. I smiled in response and exclaimed, “Now we both know how to call each other if we need to.” Andy seemed very satisfied that he had figured this out; it was important for him to know that although I would not physically be with him, I was not abandoning him, that I would be available if he needed me. Andy then jumped up and ran behind the puppet theater and yelled, “Guess where I am?” (without turning out the lights). I stood up to begin our usual ritual of hide-and-find, prepared to search all around the room for him, when after just a few moments (not waiting for me to find him, as he had always done in the past), he jumped from behind the puppet theater, threw his arms wide in the air, and with a huge smile on his beautiful face, exclaimed, “HERE I AM!” I could not have summed it up any better! I responded by throwing my arms in the air and exclaimed, “Yes, THERE YOU ARE!”

I am always amazed by how children know what they need to bring closure to the therapeutic relationship if they have proper notice. Just thinking about our last moment together brings a huge smile to my face. Andy was free; he no longer needed me. He was well on his way to fully becoming all that he was meant to be.

In the following week, Carol and I met for a final consultation without Andy to bring closure to our relationship, to review Andy’s progress, and to devise a plan that would support Andy’s continued progress without therapy. We briefly reviewed Carol’s scores on the CBCL and PSI and discussed Andy’s progress since the beginning of treatment. Andy’s posttreatment CBCL scores indicated normal functioning in 8 of the 10 areas that were initially elevated, with an average mean decrease of 8.5 across all areas. Carol seemed almost surprised at her initial scores on the CBCL and how much differently she now viewed Andy’s behavior. Tears came to her eyes (for the first time in our 20 months together) when we reviewed the PSI outcomes and she realized how much her stress related to parenting Andy had been affecting their relationship. She finally seemed to understand that her view of Andy had a considerable influence on Andy’s view of himself. We discussed, among other things, his school placement for next year, strategies to minimize Andy’s stress related to school, involvement in extracurricular sports that he excelled in, and continuing special family activities at home, including reading Andy’s adoption story, “Chinese Night,” and weekly filial play sessions. Carol’s recent experience with Andy in their play sessions had been particularly rewarding for her, and she seemed highly motivated to continue. But I suspected that without support, she might have difficulty following through once the stress of the new school year began. We discussed the importance of follow-up and agreed to talk after Andy had started second grade in approximately 6 months (or sooner, if Carol saw a need).

Complicating Factors

Children’s ability to benefit fully from treatment is dependent on their caregivers and other systemic factors. When parents are unwilling or unable to commit to

participating in the therapeutic process or when children remain in an environment that is unsafe or is contributing significantly to the presenting issue, therapy is compromised. In this particular case, neither of these scenarios was an issue. Quite the contrary, the systemic factors related to this case were highly facilitative of Andy's growth in therapy. I noted no significant complications to the successful outcome of Andy's treatment.

Managed Care Considerations

In this particular case, managed care considerations were not an issue. Andy was seen at a community clinic with a sliding-scale fee. For those families that rely on managed care systems, educating third-party payors about the length of time of treatment needed for children with a history of interpersonal trauma is critical to ensure that these children receive the help they need. It is also imperative that the value and necessity of working with parents as part of a child's therapy be acknowledged and reimbursed by third-party payors. Although cases such as the one presented are complicated and require longer term therapy, play therapy research shows this method is effective in only a few sessions (Bratton et al., 2005). More research is needed to examine the efficacy of play therapy with specific populations and to determine optimal length of treatment relative to presenting concern.

Follow-Up Treatment

As a matter of practice, with the termination of all child clients I encourage parents to report back in around 4 to 6 months on how their child is progressing. At the time of this writing, 6 months had passed since termination of this case. During a follow-up phone consultation, Carol reported that school continued to be a challenge for Andy, but he was now better able to cope with school-related stress. His extracurricular involvement in tae kwon do and basketball seemed to be an effective outlet and a source of feeling competent. Andy excelled in both sports and particularly enjoyed his family's cheering him on in basketball. He was receiving educational support services in speech and language, but was on grade level in math and science. Although Andy continued to have occasional anger outbursts, the intensity and duration were greatly reduced. He had been sleeping in his own room for the past 9 months, and it had been well over a year since he had exhibited any stealing at home or school. Thus, according to Carol, there was no need for additional counseling at the time of follow-up. I followed up on the suggestions I had given her in our final consultation. She confessed that since school had started she had not been consistent with any of the family activities we had discussed, but acknowledged their importance and committed to setting aside 30 minutes each week for an activity of Andy's choosing. I encouraged Carol to call if any concerns arose and discussed the possibility that Andy might need additional counseling at certain points in his development.

Implications and Recommendations for Practice

Andy was an amazing child. In the face of overwhelming circumstances, he used play to fully experience all aspects of his life, both positive and negative, and assimilate those experiences into a more integrated self. What a privilege it was to be with him on this journey! Andy illustrates children's remarkable resiliency

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and ability to master difficult situations if allowed to do so at their own pace and in their own way. Play provided the way for Andy to express, make sense of, and develop mastery over his confusing experiences. I could never have devised a treatment plan that was as brilliant as Andy's internal plan for what he needed in order to heal and overcome his early experiences.

Just as children have an innate drive for mastery and self-actualization, they also possess a protective tendency to suppress that which is painful until they are strong enough to allow difficult and confusing memories into their experience. Thus, the nonthreatening nature of CCPT was particularly appropriate to allow Andy to choose how, when, and to what to degree he would address his worries. Creating a safe, accepting, and respectful relationship provided Andy with the "good growing ground" (Axline, 1947) he needed to support his movement toward fully becoming the person he was meant to be. With no specific direction from me, he actively used his play to confront difficult issues in his life, including abandonment, loss, self-identity (including cultural identity), adoption, attachment, and belonging. Children who have experienced trauma are often triggered by what would be normal events for most children. Through my understanding of what Andy communicated to me in his play, I was able to help his parents and teachers identify potentially retraumatizing events (bedtime, separation, doctor visits) and devise strategies for altering or minimizing their impact on him.

Through the process of play therapy, Andy discovered new dimensions of himself that resulted in a revised self-image and more congruent and satisfying behavior. Outwardly, he demonstrated marked improvement in targeted behaviors, most notably self-regulation and a secure attachment to his parents, particularly to Carol. Andy was blessed to have a home with such remarkable parents as Carol and Tom. In my view, an invested parent is one of the most important tools in play therapy, or any child therapy. Their active participation in almost anything I suggested and willingness to implement changes were critical factors in Andy's growth. Their involvement in CPRT served as both curative and protective factors. It strengthened their confidence in their parenting and gave them a sense of empowerment as they successfully learned to utilize therapeutic play skills with Andy and saw what a difference they were making. Tom and Carol ended therapy armed with the confidence and skills they needed to continue to help Andy reach his potential. Providing parents with the skills and experiences to build a closer parent-child relationship is an essential element of therapy for a child like Andy, whose early experiences disrupted the formation of a strong bond with a caregiver.

For myself, my experiences with Andy on his complicated journey of growth and healing forced me to confront my own issues around wanting to protect children from difficult and overwhelming experiences. Andy did not need me to protect him from his feelings. He needed me to facilitate and witness the full expression of all of his experiences, both positive and negative. Peer consultations during some of the more difficult periods in Andy's therapy helped ensure that my needs were not interfering with Andy's need to fully convey all aspects of himself. Because children so readily express painful events and related emotions through the symbolism of play, play therapists may be more vulnerable to their own unresolved issues surfacing in the context of the child's playing out difficult experiences.

This case has shown how using a child-centered play therapy approach allows child therapists to join children at their level. The toys and materials provide children with a developmentally sensitive and personal means of expressing feelings, experiences, thoughts, and wishes in a way that child therapists can understand and respond to. The implication of this case is that play therapy is an effective and responsive treatment for young children. Furthermore, this case illustrates the value of involving parents as partners in the play therapy process for optimal treatment effects.

SUMMARY

Play therapy has grown over its over 100-year history to include a variety of treatment methodologies and theoretical schools of thought that embrace the therapeutic powers of play. Play therapy is based on developmental principles, and thus provides, through play, developmentally appropriate means of expression and communication for children. Enjoying a rich history of literature and research dating back to 1942, play therapy is the longest standing modality of psychotherapy developed specifically for children. Through early studies and recent meta-analyses, play therapy demonstrates a base of evidence to support its effectiveness.

This chapter has focused specifically on the principles and procedures of child-centered play therapy. In practice, CCPT offers a clear method of treatment, in which the relationship between child and therapist is the catalyst for all therapeutic change. The child's experience within the counseling relationship is the factor that is most meaningful and helpful in creating lasting, positive change. Hence, the therapist works toward creating an environment of acceptance and understanding that helps the child feel safe to fully express thoughts and feelings, moving toward more positive functioning. Following guiding CCPT principles (Axline, 1947; Landreth, 2002), the therapist develops a warm relationship with the child, accepts the child unconditionally, allows permissiveness within the relationship, reflects the child's feelings, respects the child's ability to solve problems, does not attempt to direct the child's actions, recognizes the gradual nature of the child's process, and establishes minimal limits.

For further effectiveness in play therapy, the play therapist is trained to work within the system that supports the child. Working with parents on understanding and responding to their children's needs is critical to the progress of play therapy. Parent consultation is one method of alignment with parents so that the therapist makes consistent intimate contact with the parents to sustain their commitment to their child's play therapy and to offer brief opportunities for education. Filial therapy is a more in-depth approach and provides extensive education in play therapy procedures and group support to parents so that they are empowered to use therapeutic play skills with their child.

This chapter has served to provide an introduction to the modality of play therapy for child therapists. The rationale and history of play therapy, as well as the research in the field, presented a base for its therapeutic use. The practice of CCPT was enumerated through descriptions of the therapist role, objectives, structure, therapist responses and attitudes, and applications of play therapy concepts. The case study

offered a concrete example of how play therapy is applied in a real-life setting, working with a child with a complicated presentation. Play therapy serves as a developmentally responsive intervention for children, allowing therapists to speak the language of the child. Skill in using play therapy is an essential tool for mental health professionals who work with children. Therapeutic play allows children to express themselves fully and at their own pace with the assurance that they will be understood and accepted. Through this experience, children learn to express, accept, respect, and take responsibility for themselves and their feelings, which leads to creative insight, learning, problem solving, coping, and mastery.

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CHAPTER 21

Behavior Modification

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Behavior modification is the field of psychology devoted to understanding and changing human behavior. In behavior modification, assessment procedures are used to record the problem behavior and identify the factors that contribute to its occurrence, and intervention procedures are used to help people change their behavior. Behavior modification procedures are based on basic principles of behavior discovered in experimental research with laboratory animals and humans over the past 70 years (e.g., Skinner, 1938; Ullmann & Krasner, 1965; Ulrich, Stachnik, & Mabry, 1966). As behavior modification is currently practiced, it is synonymous with the field of applied behavior analysis (Baer, Wolf, & Risley, 1968; Miltenberger, 2004).

In behavior modification there is an emphasis on observable behaviors, both behavioral excesses and behavioral deficits. Objective definition and measurement of behavior is a hallmark of behavior modification. Behavior change is accomplished by changing environmental events occurring at the time of the behavior, antecedents that evoke the behavior, and consequences that maintain the behavior. Behavior modification is largely based on operant conditioning principles and, to a lesser extent, on respondent conditioning principles (Miltenberger, 2004). Behavior modification has been successful in helping people of all ages and ability levels change a wide variety of socially significant behaviors in a wide variety of natural contexts (J. Carr & Austin, 2001; Gross & Drabman, 2005; Hersen & Rosqvist, 2005; Sugai & Horner, 2005; Vollmer et al., 2000).

This chapter provides an overview of the field of behavior modification, starting with issues in assessment and research design, followed by a review of basic behavioral principles that form the foundation of behavior modification procedures. Next, procedures to establish new behaviors are reviewed, the functional approach to assessing and changing behavior is presented, and various behavioral interventions are described.

USING DATA IN BEHAVIOR MODIFICATION

Observing, recording, and displaying data to evaluate treatment and make treatment decisions are important activities that help define behavior modification.

Observing and Recording Behavior

In behavior modification emphasis is placed on objective measurement of behavior through direct observation procedures. A number of steps are involved in the process of recording behavior (Miltenberger, 2004).

Defining the Behavior

The first step in the process of recording behavior is to define the target behaviors in objective terms. Target behaviors are chosen based on their social importance (Baer et al., 1968; Wolf, 1978) and may include problematic behaviors to be decreased (behavioral excesses) and desirable behaviors to be increased (behavioral deficits). Target behaviors are defined using action verbs so that two independent observers can record their occurrence reliably.

Logistics of Recording

Once the target behaviors are defined, the next steps are to choose an observer and the time and place of observation. The observer may be the person engaging in the behavior (self-monitoring) or a care provider (e.g., parent, teacher, staff member), therapist, or researcher. The observation period will be the time and place in which the behavior is most problematic or in which the behavior has the highest likelihood of occurrence.

Recording Method

The next step in the process of recording behavior is to decide on the use of event recording or a sampling procedure. With event recording, every instance of the behavior is recorded during the observation period. The four dimensions of behavior that can be measured are frequency (number of times the behavior occurred), duration (the time from onset to offset of the behavior), intensity (a measure of the force of the behavior), and latency (the time from some stimulus to the onset of the behavior). One or more dimensions are chosen for recording based on their relevance to the target behavior.

Sampling procedures, including interval and time sample recording, are not designed to record every instance of the behavior but rather to provide an estimate of the occurrence of the behavior. In *interval recording,* the observation period is divided into smaller consecutive intervals of time and the behavior is recorded as occurring or not in each interval. *Time sample recording* involves recording the occurrence of the behavior in brief intervals of time separated by longer intervals of time without observation. The outcome of interval or time sample recording is reported as the percentage of observation intervals in which the behavior occurred.

Recording Instrument

The final step is to choose a recording instrument. The most common recording instrument is a data sheet prepared for the particular client and target behaviors in question. The data sheet, designed to be convenient and easy to use, provides a

space for recording each instance of the behavior as it is observed. Other recording instruments are counters or timers, stopwatches, PDAs or other hand-held devices, and laptop computers, to name a few. The recording instrument is designed to be practical, to minimize disruption of ongoing activities during behavior recording, and to facilitate immediate recording of the behavior as it occurs.

Interobserver Agreement

Behavior recording procedures are designed to produce information that is as reliable as possible. One index of reliability commonly used in behavior modification is interobserver agreement. Interobserver agreement is the degree to which two independent observers agree on the occurrence of the behavior during the same observation period. Interobserver agreement is a necessary component of research in behavior modification, although it is used less often in clinical applications of behavior modification. The reader is referred to Miltenberger (2004), Bailey and Burch (2002), and J. Cooper, Heron, and Heward (2007) for details on calculating interobserver agreement.

Indirect Measures

In behavior modification, emphasis is placed on direct observation of behavior. However, in some instances, indirect measures of behavior, including product measures and retrospective reports, are used. Product measures may be used when the target behavior produces permanent changes in the environment (products) that can be measured. Examples of product recording are the number of assignments completed in an educational setting, the number of units assembled in a factory setting, and the number of hairs pulled by a person diagnosed with Trichotillomania. Retrospective reports about the occurrence of behavior can be gathered through interviews or questionnaires. Retrospective reports of behavior are not used as the sole assessment method but are typically used during the initial phase of assessment to plan direct observation procedures.

Data Display and Research Design

In behavior modification, the target behavior is recorded before, during, and after intervention is implemented. By recording before the intervention, a baseline level of the behavior is established and the effects of the intervention can be assessed by looking at changes in the behavior. A graph is the tool used in behavior modification to identify changes in the behavior from baseline to intervention. In a graph, the level of the behavior is plotted on the vertical axis (y axis) and time is plotted on the horizontal axis (x axis) so that changes in behavior over time can be easily discerned.

In most clinical applications of behavior modification the target behavior is recorded and displayed on a graph in two phases: baseline and intervention. However, when research is conducted in behavior modification, more sophisticated graphs are used that demonstrate a functional relationship between environmental

events (the intervention) and the target behavior. To demonstrate a functional relationship between intervention and the target behavior the researcher must:

- Reliably record the target behavior.
- Show that the target behavior changes when (and only when) the intervention is implemented.
- Provide a replication of the effect. In essence, the intervention is implemented two or more times and the behavior changes each time the intervention is implemented, but only when the intervention is implemented.

A number of research designs are used in behavior modification to demonstrate a functional relationship. These include the ABAB reversal design; multiple baseline designs across behaviors, participants, or settings; alternative treatment design; and changing criterion design. See Miltenberger (2004), Bailey and Burch (2002), or J. Cooper et al. (2007) for descriptions of these behavior modification research designs.

BASIC PRINCIPLES

Behavior modification procedures are based on a core set of basic behavioral principles that describe functional relationships between specific types of environmental events and behavior. These basic behavioral principles are reinforcement, extinction, punishment, stimulus control, and respondent conditioning.

Reinforcement

Reinforcement is a fundamental principle that underlies many behavior modification procedures. First studied extensively by Skinner (1938) and Ferster and Skinner (1957), reinforcement is defined as the process in which the consequence of a behavior strengthens the future probability of the behavior. Understanding the principle of reinforcement is important for understanding the causes of behavior and for strengthening appropriate behavior. There are two forms of reinforcement: positive reinforcement, in which behavior is strengthened when it results in the delivery of a positive reinforcer, and negative reinforcement, in which behavior is strengthened when it results in escape or avoidance of aversive stimuli. The consequences of the behavior involved in both positive and negative reinforcement may consist of changes in the physical environment or the behavior of others.

Reinforcement is influenced by a number of factors, including immediacy, contingency, establishing operations, individual differences, and magnitude (see Miltenberger, 2004, for discussion of these factors). One important factor that is increasingly utilized in behavior modification procedures is the establishing operation (e.g., Vollmer & Iwata, 1991). An establishing operation is an environmental event that makes a reinforcer more potent and evokes the behavior that produces that reinforcer (Michael, 1982).

Extinction

Extinction occurs when the reinforcing consequence maintaining a behavior is no longer present following the behavior, and as a result, the behavior decreases and eventually stops occurring. For behavior maintained by positive reinforcement, extinction occurs when the positive reinforcer no longer follows the behavior. For behavior maintained by negative reinforcement, extinction occurs when the individual is no longer able to escape or avoid the aversive stimulus following the behavior. In each case, when the behavior is no longer reinforced, it will stop occurring.

One phenomenon frequently seen during extinction is an extinction burst (Lerman & Iwata, 1995). In an extinction burst, there is a temporary increase in the behavior or the occurrence of novel or emotional behaviors when the target behavior no longer produces the reinforcer. The extinction burst is a transient phenomenon, and the target behavior will eventually stop occurring if the reinforcer continues to be withheld following the behavior. Understanding the extinction burst is important for the successful use of extinction in clinical practice.

Punishment

Punishment occurs when the consequence that follows a behavior decreases the future probability of the behavior. There are two variations of punishment. In positive punishment the behavior is followed by the addition of an aversive stimulus (also called a punisher); in negative punishment, the behavior is followed by the removal of a positive reinforcer. In both cases, the behavior is weakened. As with reinforcement, punishment occurs naturally in our contact with the physical environment and in our interactions with others. Some behavior modification procedures are based on the principle of punishment (e.g., time-out and response cost). The same factors that influence reinforcement influence punishment: immediacy, contingency, establishing operations, magnitude, and individual differences (Miltenberger, 2004). See Lerman and Vorndran (2002) for an in-depth discussion of punishment.

Stimulus Control

Although reinforcement, extinction, and punishment are the basic principles that determine whether a behavior will continue to occur, the effects of these operations are relatively situation-specific. Reinforcement strengthens behavior in situations that are similar to the ones in which the behavior was reinforced in the past. Likewise, extinction and punishment weaken behavior in situations that are similar to the ones in which the behavior was subjected to extinction or punishment procedures in the past. The increased probability that behavior will occur in specific situations (in the presence of specific stimuli) is referred to as stimulus control. A behavior is under the stimulus control of a specific stimulus when it is more likely to occur when that stimulus is present.

Stimulus control is developed through stimulus discrimination training. In stimulus discrimination training (also called discrimination training), the behavior is reinforced in the presence of one stimulus (a discriminative stimulus or S^D) and is not reinforced in the presence of other stimuli. As a result, when the S^D is present

in the future, the behavior is more likely to occur. The S^D has stimulus control over the behavior. The S^D will continue to have stimulus control over the behavior as long as the behavior continues to be reinforced in its presence.

Although discrimination training makes the behavior more likely to occur in the presence of the S^D, the behavior is also likely to occur in the presence of stimuli that are similar to the S^D. The more similar a stimulus is to the S^D, the more likely the behavior is to occur in its presence (Guttman & Kalish, 1956; Lalli, Mace, Livezey, & Kates, 1998). The occurrence of behavior in the presence of stimuli that are similar to the S^D is called generalization.

Respondent Conditioning

Reinforcement, extinction, punishment, and stimulus control are all operant conditioning principles. In operant conditioning, behavior is controlled by its consequences (reinforcement strengthens behavior, and extinction or punishment weaken behavior). Although operant behavior comes under the stimulus control of antecedent stimuli (the S^D), stimulus control is developed through the differential consequences that are present when the S^D is present. Whereas operant behavior is controlled by its consequences, respondent behavior is controlled by its antecedents. Respondent behavior is elicited by a prior stimulus, either an unconditioned stimulus or a conditioned stimulus.

Respondent conditioning occurs when a neutral stimulus is paired with an unconditioned stimulus (US). The US elicits an unconditioned response (UR) and, as a result of its pairing with the US, the neutral stimulus comes to elicit a response similar to the UR. The neutral stimulus is then called a conditioned stimulus, and the response it elicits is called a conditioned response. Anxiety, anger, and inappropriate sexual arousal are examples of conditioned responses that may be targets of intervention in behavior modification procedures.

Behavior modification procedures, the actual techniques used to help people change their behavior, are based on the basic principles just described. Behavior modification procedures are used to establish new behaviors, to identify the variables contributing to the occurrence of behaviors, and to help people increase or decrease existing behaviors.

PROCEDURES TO ESTABLISH NEW BEHAVIORS

When a desirable behavior is not currently in a person's repertoire, a number of procedures can be used to help the person acquire the behavior. These procedures include shaping, prompting and fading, chaining, and behavioral skills training procedures.

Shaping

Shaping is the differential reinforcement of successive approximations of a target behavior. Shaping is used to help an individual engage in a new behavior (e.g., a child saying a word for the first time), a new dimension of an existing behavior (e.g., a patient speaking with increasing voice volume), or a previously exhibited

behavior that is no longer occurring (e.g., a stroke patient walking again; Jackson & Wallace, 1974; Miltenberger, 2004; G. W. O'Neill & Gardner, 1983). To use shaping, the target behavior is defined and then a starting behavior is chosen. The starting behavior (or first approximation), a behavior that already occurs at least occasionally, is reinforced a number of times until it occurs consistently. Then it is no longer reinforced (extinction), and the next approximation to the target behavior is reinforced instead. Once this behavior occurs a number of times, it is no longer reinforced, and a closer approximation to the target behavior is then reinforced. This process continues until the individual engages in the target behavior. Once the target behavior is exhibited it is reinforced consistently to strengthen it and maintain its occurrence over time.

Shaping, a procedure that came from experimental research with laboratory animals (Skinner, 1938, 1951), has been used extensively in animal training (Pryor, 1985) and with humans to help individuals acquire a number of socially significant target behaviors in areas such as education, developmental disabilities, rehabilitation, and sports (Hagopian & Thompson, 1999; Horner, 1971; G. W. O'Neill & Gardner, 1983; Scott, Scott, & Goldwater, 1997). Shaping is used when more efficient behavioral acquisition strategies (e.g., prompting, instructions, modeling) are not possible.

Prompting and Fading

Prompting is used when a desired behavior is not currently occurring or not occurring in the correct circumstances (in the presence of the S^D). Prompting is a procedure in which an antecedent stimulus is delivered to evoke the behavior in the presence of the S^D so that the behavior can get reinforced. Fading is the gradual elimination of the prompt once the behavior is occurring in the presence of the S^D. The prompt gets the correct behavior to occur at the right time, and fading eliminates the prompt to get the behavior under the stimulus control of the S^D.

Prompts can include the behavior of another person (response prompt) or a change in an antecedent stimulus (stimulus prompt). There are four types of response prompts: verbal, gestural, modeling, and physical. A *verbal prompt* is when the verbal behavior of another person is used to get the correct behavior to occur in the presence of the S^D. A *gestural prompt* is when a physical movement or gesture of another person gets the correct behavior to occur. A *modeling prompt* is when a demonstration of the behavior gets the correct response to occur. A *physical prompt* is when another person physically assists or guides a person to engage in the correct behavior. Physical prompting typically consists of hand-over-hand guidance.

When prompting is used to evoke the correct behavior, the trainer first presents the S^D and, if a response does not occur immediately, the trainer provides the prompt. When the prompted response occurs, the trainer provides a reinforcer.

Consider the case of a training program to teach a child with autism to hit a baseball off a tee as part of a recreational program. The S^D is the ball on the tee on home plate on a baseball field in front of the child while she holds the bat. In the presence of the S^D, hitting the ball is the correct response and praise from the

coach is the reinforcer. A verbal prompt might be to say "Hit the ball" while the child is standing in position holding the bat. If the child hits the ball after receiving the prompt, the behavior can be reinforced. If the child does not hit the ball after receiving the verbal prompt, a gestural prompt such as pointing to the ball on the tee can be delivered. If the child does not engage in the correct behavior following the gestural prompt, a modeling prompt can be delivered. A modeling prompt might consist of hitting the ball off the tee yourself as the child watches. If the modeling prompt does not evoke the correct behavior, then a physical prompt can be used. A physical prompt might involve holding the child's hands on the bat and swinging the bat with the child to hit the ball. The use of hand-over-hand guidance in this way would result in the correct response and praise from the coach.

In the baseball example, once the child is hitting the ball correctly with the coach's physical guidance, the coach will fade the physical prompt. Over consecutive trials, the coach will provide less and less assistance (e.g., he will help the child start to swing and let her finish with no assistance) until the child swings the bat to hit the ball without any further assistance. Once prompts are no longer needed, the behavior is under the control of the S^D (the ball on the tee). See J. Cooper et al. (2007), Miltenberger (2004), and Sulzer-Azaroff and Mayer (1991) for more information on the use of prompting and fading.

Chaining

A chaining procedure is used to teach the learner to engage in a chain of behaviors. A behavioral chain (also called a stimulus-response chain) consists of a number of behaviors, each consisting of an S^D and a response, that occur together in a sequence. Each response in the chain creates the S^D for the next response in the chain. For example, consider the behavior of getting a drink from a water fountain. There are at least three behaviors in the chain: pushing the handle for water, bending over the stream of water, and drinking the water. In this example, the water fountain is the S^D. The first response is turning on the water. This behavior creates the second S^D (the water stream). The second response is bending over the water stream. This behavior creates the third S^D (the water stream in proximity to or touching your mouth). The third response is drinking the water.

Before using a chaining procedure to teach a chain of behaviors, you must conduct a task analysis. A task analysis identifies every S^D and response in the chain of behaviors (e.g., Bellamy, Horner, & Inman, 1979). Careful observation of someone engaging in the chain of behaviors is used to develop the task analysis.

Three procedures are used to teach a chain of behaviors: forward chaining, backward chaining, and total task presentation (Miltenberger, 2004).

Forward Chaining

In forward chaining you use prompting and fading to teach the first component in the chain. After the learner engages in the first response when the first S^D is presented without any prompts, prompting and fading are used to teach the first two components. At this time, the learner will engage in the first response when the first S^D is presented. When the first response creates the second S^D, the learner is

prompted to engage in the second response. Once prompts are faded, the learner can now engage in the first two responses in the chain without prompts. Then prompting and fading are used to teach the first three components, and so on until all of the component behaviors occur together.

Backward Chaining

In backward chaining, the last S^D is presented and prompting and fading are used to get the last response to occur. Once the last response is under the stimulus control of the last S^D, the next to last S^D is presented and the next to last response is prompted. Once the next to last response occurs, it creates the last S^D and the learner will then engage in the last response. After the last two behaviors in the chain occur without prompts, the third to last S^D is presented and prompting and fading are used until the last three responses occur without prompts. This process continues until the learner engages in the whole chain of behaviors without prompts.

Total Task Presentation

In total task presentation, you present the first S^D and physically guide the learner through the entire chain of behaviors from start to finish. Over trials, you fade the physical guidance to shadowing until the physical prompt is eliminated. This fading method, in which physical prompting is faded to shadowing and eventually eliminated, is called graduated guidance. Total task presentation is typically used with more capable learners, and forward or backward chaining is used with learners with significant disabilities.

Behavioral Skills Training

Behavioral skills training (BST) is one final strategy for helping a learner acquire new skills. Behavioral skills training consists of four training components: instructions, modeling, rehearsal, and feedback (e.g., Himle & Miltenberger, 2004). With instructions and modeling, the skills are described and demonstrated for the learner. Rehearsal is an opportunity for the learner to practice the skills in a simulation or role-play situation. Finally, feedback consists of praise for correct performance of the skills and correction or further instruction when the skill is not performed correctly. Typically, BST is used for behaviors that can be demonstrated and practiced in a simulation or role-play format. Various skills have been taught with BST procedures, including social skills, parenting skills, job-related skills, child safety skills, and clinical interviewing skills (Bakken, Miltenberger, & Schauss, 1993; Himle, Miltenberger, Flessner, & Gatheridge, 2004; Johnson et al., 2005; Miltenberger & Fuqua, 1985).

To conduct BST, you first define the target behaviors and all of the situations in which the behaviors are to be performed. In teaching abduction prevention skills, for example, the target behavior consists of saying no, running away, and telling an adult when faced with an abduction lure. All of the different ways an adult might try to lure a child would then be identified and incorporated into training (e.g., offering the child candy or a toy to leave, asking for assistance, invoking

authority). Once the target behaviors and S^Ds (situations) are identified, training begins with instructions and modeling. The trainer describes the behavior in detail and the situations in which the behavior is to be performed. The trainer then models the behavior in the context of a role-play of one of the relevant situations.

After receiving instructions and modeling, the learner then has the opportunity to rehearse the behavior. The trainer role-plays the same situation that was just modeled and the child practices the skill in response to the situation. For example, the trainer might tell a child, "Pretend I am a stranger that walks up to you in your front yard and asks you to help me look for my puppy down the street. Show me what you would do if this happened." The trainer then role-plays the situation, delivers the lure, and observes the child's response. Immediately after the child rehearses the behavior, the trainer provides feedback. The trainer provides descriptive praise for any aspect of the behavior the child performed correctly and then tells the child how the behavior could be improved (if needed). The trainer then immediately conducts the same role-play and provides the child the opportunity to rehearse the behavior again after receiving feedback. Once the child demonstrates the correct behavior in the role-play, other situations are incorporated into role-plays so that the full range of situations and responses is trained.

PROCEDURES TO INCREASE OR DECREASE EXISTING BEHAVIORS

A number of behavior modification procedures are used to help individuals decrease undesirable behaviors and increase desirable behaviors. The functional approach to assessment and intervention starts with a functional assessment to identify the antecedents and consequences influencing the behavior (R. E. O'Neill et al., 1997). Functional interventions, designed to address the antecedents and consequences of the behavior, include extinction, differential reinforcement, and antecedent control procedures (Miltenberger, 2004). Changing the antecedents and consequences of the behavior and promoting functionally equivalent alternative behavior alter the conditions that influence the behavior, resulting in the greatest likelihood of change in the behavior.

Functional Assessment

Functional assessment is a process of gathering information on the target behavior and the antecedents and consequences that are functionally related to the occurrence of the behavior (Lennox & Miltenberger, 1989). The outcome of a functional assessment is a hypothesis about the reinforcing consequences maintaining the behavior and the antecedents that evoke the behavior (R. E. O'Neill et al., 1997).

There are four categories of reinforcing consequences that could be maintaining a target behavior: social positive reinforcement (e.g., attention, tangible, or activity reinforcers provided by another person), automatic positive reinforcement (some form of sensory stimulation arising directly from the behavior), social negative

reinforcement (escape or avoidance of aversive activities or interactions mediated by another person), and automatic negative reinforcement (escape from aversive stimulation such as pain, discomfort, negative emotional responding; Iwata, Vollmer, Zarcone, & Rodgers, 1993). A functional assessment produces information that enables you to identify the type (or types) of reinforcement maintaining the problem behavior.

Antecedents that are identified in a functional assessment are the events, people, and circumstances that are reliably associated with the occurrence of the behavior. Antecedents can include S^Ds that occur immediately prior to the behavior and establishing operations (EO) whose onset may be proximal or more distal to the occurrence of the behavior. Identifying the S^D and EO associated with the behavior allows you to alter these antecedents to influence the problem behavior and alternative behaviors (Smith & Iwata, 1997).

There are three methods for conducting a functional assessment: indirect assessment, direct assessment, and functional analysis.

Indirect Assessment

An indirect assessment involves interviews, questionnaires, or rating scales in which individuals who have frequent contact with the client (informants) provide information from their recall of antecedents and consequences associated with the behavior (Lennox & Miltenberger, 1989; R. E. O'Neill et al., 1997). R. E. O'Neill et al., for example, developed the Functional Analysis Interview Format to gather detailed functional assessment information. The advantage of an indirect functional assessment is that it is easy and convenient to administer and does not take much time. The limitation is that it does not involve direct observation of the behavior as it occurs but instead relies on the informant's recall of events.

Direct Observation Assessment

A direct observation functional assessment (also called ABC recording) involves direct observation and recording of the behavior and its antecedents and consequences as the behavior occurs (Lennox & Miltenberger, 1989; R. E. O'Neill et al., 1997). Observations occur across a number of instances of the behavior until patterns in the antecedents and consequences are evident in the data. ABC recording can involve descriptive recording or the use of a checklist. In descriptive ABC recording, the observer writes down a description of antecedents, behavior, and consequences as they occur. To be of most value, the descriptions must be detailed and objective. In the checklist method, the observer has a checklist of possible target behaviors, antecedents, and consequences and places a checkmark in the appropriate column to designate the particular antecedents and consequences that were observed at the time of the behavior. The advantage of ABC recording is that the antecedents and consequences are observed as they occur, which results in objective (and likely more accurate) recording of the events. The limitation is that it takes more time to conduct multiple observations of the individual in the natural context (group home, school, etc.).

Functional Analysis

A functional analysis involves direct observation of the target behavior as antecedents and consequences are systematically manipulated to identify their influence on the behavior (e.g., E. G. Carr & Durand, 1985; Iwata, Dorsey, Slifer, Bauman, & Richman, 1982). For example, you could provide attention following each instance of the target behavior to see if the behavior increased, thus demonstrating that attention functioned as a reinforcer. Likewise, you could provide escape from demands contingent on the behavior or tangible reinforcers contingent on the behavior to see whether these events functioned as reinforcers for the behavior.

A functional analysis might be used to test a hypothesis, in which case one reinforcer might be manipulated, or a functional analysis might test a number of possible reinforcers. For example, to test the hypothesis that attention is reinforcing the target behavior, a functional analysis would alternate conditions in which attention is provided contingent on the behavior and conditions in which attention is withheld contingent on the behavior (e.g., Arndorfer, Miltenberger, Woster, Rortvedt, & Gaffaney, 1994). If attention is a reinforcer, the behavior will increase in the first condition relative to the second. To test a number of reinforcers, you might alternate conditions in which attention, escape, or tangible reinforcers are made contingent on the behavior. An alone condition would also be used to see if the behavior persisted in the absence of social consequences (e.g., Iwata et al., 1982). The advantage of a functional analysis is that, because antecedents and consequences are manipulated and resulting changes in the target behavior are recorded, a functional relationship between the antecedents and consequences and the behavior is demonstrated. The limitation of a functional analysis is that it takes substantial time and expertise to conduct and requires temporarily reinforcing the undesirable behavior.

Once a functional assessment is conducted and hypotheses about the antecedents and consequences of the behavior have been developed, functional interventions (described later) are chosen based on the hypotheses.

Extinction

Extinction, described earlier, is a basic principle of behavior. Extinction is also a behavioral procedure when it is used to help an individual eliminate an undesirable behavior (Lerman & Iwata, 1996). To use extinction, you must first conduct a functional assessment to identify the reinforcer maintaining the problem behavior. Once the reinforcer is identified, extinction involves eliminating the reinforcer after each instance of the behavior. If the behavior no longer gets reinforced, it will cease to occur. However, the successful use of extinction in clinical practice requires careful consideration of a number of questions (Miltenberger, 2004):

- Have you identified the reinforcer for the behavior? You cannot assume a particular reinforcer is maintaining a problem behavior; you must identify the reinforcer from a careful functional assessment. Sometimes, multiple reinforcers are maintaining a problem behavior in different contexts.

- Can you eliminate the reinforcer? Eliminating the reinforcer can be a problem when the behavior is reinforced by other people and you do not have influence over those individuals or when the behavior is automatically reinforced.
- Is extinction safe to use? In some cases of aggressive or self-injurious behavior, if the parent or teacher no longer responds to the behavior, it might put individuals at risk for harm. Even though the adult's response to the behavior may be reinforcing the behavior (i.e., attention), during extinction it may be necessary for the adult to respond to interrupt the behavior and assure the individual's safety. In such cases, the reinforcing consequence may be attenuated if it cannot be totally eliminated.
- Will individuals using extinction be able to deal effectively with an extinction burst? In most cases when extinction is used, the problem behavior temporarily worsens (increases in frequency, duration, or intensity) before it decreases. The parents, teachers, or others using extinction must be alerted to the probability of an extinction burst and be prepared to deal with it effectively. If they are not prepared, they might accidentally reinforce a more extreme problem behavior when it emerges during an extinction burst.
- Can consistency be maintained over time with all people? For extinction to be effective, it must be implemented each time the problem behavior occurs. If some instances of the behavior are reinforced only at some times by some individuals, then the behavior will likely persist and possibly worsen due to intermittent reinforcement.

When extinction is used clinically, it must be used in the context of other functional interventions as well. Not only should the problem behavior result in extinction, but differential reinforcement should be used to strengthen more desirable behavior. In addition, antecedent conditions should be altered to prevent the occurrence of the problem and evoke desired behaviors while using extinction.

Differential Reinforcement

Differential reinforcement involves the use of reinforcement to increase desirable behavior and extinction to decrease the occurrence of the problem behavior. There are three variations of differential reinforcement: differential reinforcement of alternative behavior (DRA), differential reinforcement of other behavior (DRO), and differential reinforcement of low rates of behavior (DRL; Miltenberger, 2004).

Differential Reinforcement of Alternative Behavior

In DRA, reinforcement is provided for a desirable alternative behavior, and extinction is used for the problem behavior. The goal of DRA is for the desirable behavior to increase in frequency and replace the problem behavior as it is decreased through extinction. For DRA to be effective it is important to reinforce the desirable behavior each time it occurs while withholding reinforcement for every instance of the problem behavior. In addition, the reinforcer chosen for the desirable behavior should be a known reinforcer. When possible, it is best to reinforce the desirable behavior with the same reinforcer that was maintaining the problem behavior prior

to the use of extinction. For example, if the problem behavior was maintained by attention, attention should be provided as a reinforcer for the alternative behavior. In this way, the desirable alternative behavior is functionally equivalent to the problem behavior because it produces the same outcome that used to produce the problem behavior (e.g., E. G. Carr, 1988).

Differential reinforcement of alternative behavior may involve positive reinforcement or negative reinforcement. In the case of positive reinforcement involving attention, attention no longer follows the problem behavior and instead is delivered following each occurrence of a desirable alternative behavior (Durand & Carr, 1991, 1992). In the case of negative reinforcement involving escape from aversive activities, escape is no longer contingent on the problem behavior and is delivered instead contingent on the occurrence of a desirable behavior (Allen & Stokes, 1987; Marcus & Vollmer, 1995). Steege et al. (1990) used DRA to decrease self-injurious behavior that was maintained by escape from grooming activities. In this case, a child with intellectual disabilities bit his wrist while an adult was brushing the child's teeth and the biting was reinforced by the termination of tooth brushing. When DRA was used, wrist biting no longer resulted in termination of tooth brushing (extinction). Instead, the child was prompted to press a switch to activate a tape-recorded message asking for a break. When this alternative behavior occurred, it resulted in a break (termination of tooth brushing for a brief period of time). When pushing the button to activate the taped message was reinforced with escape but wrist biting was not, the child quit biting his wrist and engaged in the alternative behavior instead.

Differential Reinforcement of Other Behavior

Differential reinforcement of other behavior is a procedure that utilizes extinction for the problem behavior and reinforcement for the absence of the problem behavior. If the problem behavior no longer produces a reinforcer and the reinforcer can be acquired only for the absence of the behavior, the behavior will decrease. Before using DRO you must first identify the reinforcer maintaining the problem behavior. Then you identify the reinforcer to be delivered contingent on the absence of the problem behavior. In most cases, the reinforcer delivered for the absence of the problem behavior will be the reinforcer that was maintaining the problem. Next, you identify the interval for the delivery of the reinforcer. The interval will be based on the average time between occurrences of the problem behavior in baseline. For example, if a problem behavior occurs on average 3 times per minute, the average time between responses is 20 seconds and the DRO interval should be set at 20 seconds or less.

To use DRO you provide the reinforcer after each interval of time in which the problem behavior does not occur. If the problem behavior does occur at any time, the reinforcer is withheld and the interval is reset. Again, the problem behavior must be absent for the entire interval for the reinforcer to be delivered. Once the problem behavior begins to decrease in frequency and the reinforcer is being delivered for its absence in the majority of intervals, the interval is gradually lengthened. As the interval is lengthened, the problem behavior must be absent for longer periods of

time before the reinforcer is delivered. The goal when lengthening the DRO interval is to end up with an interval length that is practical for the parent or teacher to implement. Although it will require the complete attention of the caregiver to deliver the reinforcer after every 20-second interval with the absence of the problem behavior, once the interval is lengthened to a more manageable time (such as 30 minutes) the caregiver will be able to resume normal activities while implementing DRO (e.g., Cowdery, Iwata, & Pace, 1990; Iwata, Pace, Cowdery, & Miltenberger, 1994; Vollmer & Iwata, 1992).

Differential Reinforcement of Low Rates of Behavior

Differential reinforcement of low rates of behavior is a procedure in which a reinforcer is delivered if a problem behavior happens fewer than a specified number of times in an observation period. The goal is to provide a reinforcer for a lower rate of the behavior. Consider the case of a child with Attention-Deficit/Hyperactivity Disorder who gets up from the dinner table on average more than 10 times per meal. In a DRL procedure, the parent might arrange a contingency in which the child gets dessert (a reinforcer) only if she gets up from the table fewer than 5 times during the meal. Once the child is successful, the number of acceptable responses can be lowered to a more desired level. When using DRL, it is important to provide the child with a way to keep track of the number of responses that have occurred. In this case, the parent might bring to the table a chart with five boxes and place an X in a box each time the behavior of getting up from the table occurs. The child would have to have at least one blank box at the end of the meal to receive the reinforcer.

In another form of DRL, a reinforcer is delivered only if x amount of time has elapsed since the last response. In this form of DRL, the timing of responses is important. As the interresponse time increases, the response rate decreases. An example of this form of DRL is from a study by Lennox, Miltenberger, and Donnelly (1987). The authors worked with adults with mental retardation who engaged in rapid eating at meals. To decrease the rate of eating to make the individuals more socially acceptable, the authors used a DRL procedure in which a bite of food was allowed in the mouth only if 15 seconds had elapsed since the previous bite of food. If 15 seconds had not elapsed since the previous bite, the authors blocked the participants from putting the bite in their mouth. In this way, the time between bites increased to more than 15 seconds (thus, eating rate decreased).

Antecedent Control

Antecedent control procedures involve the manipulation of S^Ds, EOs, or response effort to make the occurrence of a problem behavior less likely or evoke the occurrence of desirable behaviors (Miltenberger, 2004).

To use antecedent control to increase desirable behavior you can present the S^D or an EO for the behavior or decrease response effort for the behavior. Presenting an S^D for a behavior makes the behavior more likely to occur because it has been reinforced in the presence of the S^D in the past. An example might be to send a student to a quiet room with textbooks on a desk to increase the likelihood of studying. Presenting an EO makes a behavior more probable because it increases

the reinforcing value of the outcome of the behavior (Vollmer & Iwata, 1991). In the case of a training program with a child with autism in which small edible reinforcers are used (bits of favorite cereal), the trainer can run the training program just before meals, when the child has not eaten in a few hours, as a way to make the edible reinforcers more potent. The trainer can also make sure the child does not have access to the favorite cereal until the training session as a way to make the cereal more reinforcing. Decreasing response effort for a behavior makes the behavior more probable than a functionally equivalent alternative behavior that requires more effort (e.g., Horner & Day, 1991). When a child with tantrum behavior maintained by attention is taught how to ask for attention, the child is more likely to ask for attention if asking takes less effort than engaging in the tantrum behavior and produces an equal amount of attention.

To use antecedent control to decrease a problem behavior, you can remove an S^D or EO for the behavior or increase response effort for the behavior. A behavior is less likely to occur in the absence of the S^D, so removing the S^D is one way to make the behavior less likely. If watching television and playing computer games interfere with doing homework, then turning off the television and computer removes the S^Ds for these competing behaviors to make them less likely (and make doing homework more likely). If the EO for a particular reinforcer is removed, the reinforcer is no longer potent and the behavior producing that reinforcer will be less likely. In the case of the child whose tantrums are reinforced by parent attention, frequent noncontingent attention from the parent will make the tantrums less likely because the reinforcing value of parental attention is decreased through satiation (e.g., Vollmer, Iwata, Zarcone, Smith, & Mazaleski, 1993). Noncontingent reinforcement can make behavior maintained by various forms of reinforcement (attention escape, tangibles) less likely by altering the EO through satiation (Wilder & Carr, 1998). Increasing response effort for a problem behavior makes the behavior less likely because the individual will be more likely to engage in a less effortful, functionally equivalent alternative behavior. For example, when a student with multiple disabilities in a special education classroom slapped students near him, the behavior was reinforced by teacher attention (the teacher came over to him, expressed concern, and explained why the behavior was inappropriate). In this case, the student was moved to a location in the classroom where other children were out of reach. Because it now required more effort to slap another student, the student was more likely to engage in alternative behavior (e.g., raising his hand) for teacher attention.

To use antecedent control procedures successfully, you must identify relevant antecedents through a functional assessment and alter the antecedents that are functionally related to the target behaviors. Numerous studies have demonstrated the effectiveness of antecedent control procedures for problem behaviors maintained by positive reinforcement and negative reinforcement (Luiselli, 2006).

Punishment

In behavior modification problem behaviors are addressed first through a functional approach to assessment and intervention. After conducting a functional assessment,

extinction, differential reinforcement, and antecedent control procedures are utilized, often with success. In cases where problem behaviors are not successfully treated with these functional interventions, punishment may be considered (Iwata, 1988; Miltenberger, 2004). The most commonly used punishment procedures are based on the principle of negative punishment and include time-out and response cost. Although positive punishment procedures involving the delivery of aversive stimulation (such as the use of contingent shock) have been used in the past, such procedures are rarely if ever used in contemporary behavior modification.

Time-out

Time-out from positive reinforcement ("time-out" for short) is a procedure in which, contingent on a problem behavior, a child is removed from a reinforcing environment for a brief period of time. Time-out is a negative punishment procedure because the child is removed from all sources of positive reinforcement contingent on the problem behavior and the behavior decreases. The child is typically escorted to a nearby room or an area devoid of any access to reinforcers. Time-out has been shown to be a successful intervention for a range of childhood problem behaviors (e.g., Mathews, Friman, Barone, Ross, & Christophersen, 1987; Porterfield, Herbert-Jackson, & Risley, 1976; Rortvedt & Miltenberger, 1994).

For time-out to be effective, a number of factors must be considered (Miltenberger, 2004):

- First, the environment from which the child is removed (the time-in environment) must be reinforcing (must contain positively reinforcing activities, interactions, and/or stimuli). If the time-in environment is not reinforcing, time-out will not function as negative punishment.
- Second, time-out must be initiated immediately contingent on the problem behavior.
- Third, time-out must be brief (a few minutes in duration) to provide minimal disruption of ongoing educational or other important activities.
- Fourth, attention or other reinforcers must not be available in the time-out area. If the time-out room or area is reinforcing, time-out will not be effective.
- Fifth, the child should not get any attention while being taken to time-out. Contingent on the problem behavior the parent should simply state the problem behavior and that the child has to go to time-out and proceed to escort the child to time-out without any further discussion.
- Sixth, the child should not escape from time-out until released by the parent when time-out is over.
- Seventh, the time-out room or area must be safe.

Response Cost

Response cost is a procedure in which, contingent on the occurrence of the problem behavior, a specified amount of a reinforcer is removed to decrease the behavior.

Losing the opportunity to have dessert contingent on problem behavior at supper, or loss of a portion of an allowance contingent on the occurrence of a problem behavior are examples of response cost that might be used by parents. In each case, the loss of the reinforcer makes the problem behavior less likely in the future. Response cost procedures are used frequently by governments and other organizations (e.g., speeding tickets, fines for parking in handicapped parking spaces, fees for bounced checks, late fees from credit card companies). Response cost procedures are used effectively for a variety of child behavior problems as well. For example, Little and Kelley (1989) use response cost to decrease noncompliance to parental requests; Long, Miltenberger, and Rapp (1999) use response cost to decrease thumb sucking; and Rapport, Murphy, and Bailey (1982) used response cost to decrease hyperactive behavior in children.

For response cost to be effective, the loss of the reinforcer (or a statement that the reinforcer will be lost) should occur immediately contingent on each instance of the problem behavior. The magnitude of the reinforcer loss should be sufficient to function as a punisher. Finally, ethical issues must be considered such that the loss of the reinforcer does not harm the child in any way or violate the child's rights (e.g., loss of meals, for example, is not an acceptable form of response cost).

Promoting Generalization

In behavior modification, generalization is defined as the occurrence of target behavior outside of the treatment or training setting in the natural environment. For example, if a child learns social skills in treatment sessions with the school psychologist, generalization occurs when the social skills are used with peers at school or in other natural settings. A number of strategies can be used to promote generalization (e.g., Stokes & Baer, 1977; Stokes & Osnes, 1989):

- Provide reinforcement for the target behavior when it occurs in the natural environment. In the social skills example, the school psychologist could provide praise whenever she observes the child engaging in good social skills with peers.
- Teach skills that will contact natural contingencies of reinforcement. The child should be taught the kind of social skills likely to generate a reinforcing response from peers.
- Modify natural contingencies of reinforcement. The psychologist might talk to the student's teachers and have them provide praise for the student's appropriate use of social skills.
- Incorporate a variety of relevant stimulus situations in training so that the child has practiced and received reinforcement for engaging in the correct behavior in a wide range of situations he might encounter in the natural environment.
- Incorporate stimuli from the natural environment into the training situation (common stimuli). An example would be to have some same-age peers in the social skills training sessions so the student practices with the kinds of peers he is likely to encounter in school.

- Teach self-generated mediators of generalization. This strategy might include teaching the student some simple self-instructions that he could use in social situations to cue himself to use the social skills that he had learned in training sessions.

Another form of generalization occurs when a target behavior that decreased in treatment sessions also decreases in the natural environment. To promote a generalized reduction in problem behavior, it is important to reinforce a functionally equivalent alternative behavior so that the desirable behavior takes the place of the problem and produces the same type and amount of reinforcement that the problem behavior did before treatment (E. G. Carr, 1988). It is also important to maintain extinction or punishment contingencies across situations and over time in case the problem behavior occurs again for some reason (e.g., spontaneous recovery).

Habit Reversal

Habit reversal is a procedure developed by Azrin and Nunn (1973) as a treatment for nervous habits (hair pulling, thumb sucking, nail biting, skin picking, and other repetitive self-directed behaviors) and motor and vocal tics. Habit reversal procedures were also applied successfully to stuttering by Azrin and Nunn (1974). Since the original studies describing these procedures, numerous studies have demonstrated the effectiveness of the procedures or streamlined versions of the procedures (Miltenberger, Fuqua, & Woods, 1998).

Habit reversal is a multicomponent procedure with the following components: (a) motivational procedures involving review of all the inconveniences, embarrassment, or social and personal problems caused by the habit behavior; (b) awareness training in which the client describes the habit behaviors and learns to detect each instance of the habit as it occurs or as it is about to occur; (c) competing response training in which the client learns to engage in a physically incompatible motor response contingent on the occurrence of the habit behavior or in anticipation of the habit behavior; and (d) social support procedures in which a parent or other adult helps the child control the habit behavior by prompting the child to use the competing response when necessary, praising the child for using the competing response at the appropriate time (DRA), and praising the child when the habit behavior is not occurring (DRO).

Habit reversal procedures are taught to the child with the parent present in a few treatment sessions. Booster sessions are then used as needed to help the child continue to use the procedures successfully over time (Rapp, Miltenberger, Long, Elliott, & Lumley, 1998; Romaniuk, Miltenberger, & Deaver, 2003). Although habit reversal has been used successfully for child habit behaviors, research has shown that it may not be effective for younger children (Long, Miltenberger, & Rapp, 1999; Rapp, Miltenberger, Galensky, Roberts, & Ellingson, 1999) or individuals with disabilities (Long, Miltenberger, Ellingson, & Ott, 1999).

Token Economy

A token economy involves the systematic application of conditioned reinforcers called tokens to promote and maintain desirable behaviors (Ayllon & Azrin, 1968). In a token economy, tokens are delivered for desirable behavior and later exchanged for backup reinforcers. Tokens maintain their effectiveness as conditioned reinforcers because they are paired with already established backup reinforcers. The following steps are involved in developing and implementing a token economy (Miltenberger, 2004):

1. Identify and define the target behaviors you want to strengthen through the use of the token economy.
2. Identify the objects that will used as tokens. Tokens must be able to be delivered immediately to the client following the desirable behavior. They can consist of small objects such as poker chips, pennies, or marbles that can be accumulated by the client. They can also include points, checkmarks, hole punches in a card, or anything else that can be delivered and accumulated easily.
3. Identify the backup reinforcers. Because tokens maintain their power as conditioned reinforcers by being paired with backup reinforcers, it is important to choose items or activities that are known to be powerful reinforcers for use in the token economy. Furthermore, it is best to have a variety of backup reinforcers available to help avoid satiation.
4. Identify the number of tokens that will be delivered for each of the target behaviors. Typically, more tokens are delivered for more difficult or more important target behaviors.
5. Identify the exchange rate for backup reinforcers. You need to determine how many tokens are required to exchange for each backup reinforcer. Larger or more valued reinforcers will require more tokens.
6. Identify a time and place for exchanging tokens for backup reinforcers. Clients will be receiving and accumulating tokens each day for engaging in desirable behavior. Unless a time and place is set for exchanging tokens, the clients may seek to purchase backup reinforcers numerous times daily, thus disrupting ongoing activities.

To use a token economy effectively, it must be planned in advance, taking into consideration each of these steps. In addition, staff must be trained and supervised to carry out the program successfully. Finally, it is important to develop the token economy in such a way that the tokens cannot be acquired in any way other than by engaging in the target behaviors and that the backup reinforcers cannot be acquired in any way other than with tokens. If backup reinforcers are freely available, individuals will not work to earn tokens to exchange for them.

Behavioral Contracts

A behavioral contract is a written agreement specifying a desired behavior and a consequence for the behavior (O'Banion & Whaley, 1981). By writing a behavioral contract, a person is making a commitment to engage in a specified behavior in a specified time frame and arranging a consequence as a reinforcer for the behavior. Typically, a contract manager implements the contract with the person seeking behavior change. It is the contract manager's job to verify the occurrence of the behavior and carry out the consequence specified in the contract. For example, a student might agree to a behavioral contract with a school counselor in which the student commits to being on time for school each day for a week. When agreeing to the contract, the student gives the counselor five personal checks for $10 each. Each day the student gets to school on time, she gets one of the checks back. If she is late, the check for that day is sent to charity. Knowing that she will avoid the loss of $10 by arriving on time helps motivate the student to engage in the behavior.

The essential components of a behavioral contract include (a) a clear description of the target behavior to be engaged in or avoided, (b) the time frame in which the target behavior (or its absence) should occur, (c) how the target behavior will be measured, (d) the consequence for engaging in (or not engaging in) the target behavior, and (e) signatures of the individual seeking behavior change and the contract manager (Miltenberger, 2004).

Case Illustration

Christy was a 25-month-old girl referred by her mother for feeding difficulties. She was seen by a psychology graduate student under the supervision of a faculty member in clinical psychology.

Identifying Features of the Client
Christy was 25 months of age at the time of referral. She lived at home with her parents and 5-year-old brother. Due to her feeding problems, she had a gastric tube and received supplemental feedings each night while she was asleep.

Presenting Complaints
The problem for which her mother sought help was a feeding problem involving food refusal. Christy refused to accept bites of food presented to her on a spoon by closing her mouth, turning her head, placing her hands in front of her mouth, and trying to get away from the table. Another problem was gagging and food expulsion, which occurred to a lesser extent than food refusal. Gagging was most likely to occur when coarser foods were eaten or larger bites were attempted. Christy refused most foods except baby foods, pureed foods, soft foods such as ice cream and pudding, and shakes.

History and Developmental Factors
Christy's mother had an uncomplicated pregnancy and delivery. Within 10 days after birth, during which time Christy cried, grimaced, and vomited frequently during bottle feeding, she was diagnosed with gastro-esophageal reflux. She received a medication but the problem continued for the next 4 months. Because Christy did not gain weight at the appropriate rate in the first 6 months of life,

a gastric tube was implanted and she received supplemental feedings while asleep in bed. Her medication was also changed and the reflux problem abated. However, Christy exhibited food refusal problems when her mother started feeding her baby food and then table food.

At the time of referral, Christy was developmentally normal in all areas except oral-motor development, where she was slightly delayed. However, she did have the ability to chew and swallow food cut up into small pieces. She was at the 30th percentile for height and the 2nd percentile for weight.

Parental Factors

Both parents were present in the home and committed to solving Christy's feeding problem. Her mother was at home with her while her father worked for the first 2 years of her life, so her mother was the one who usually fed Christy. Christy started day care 2 months before the referral.

Assessment

Assessment was conducted through interviews with the mother, direct observation of meal times by the graduate student therapist, and behavioral recording by her mother. The behaviors recorded during direct observation were (a) number of bites presented, (b) number of bites accepted, (c) type of food, and (d) amount of food consumed. Functional assessment information was gathered through the interview and through direct observation to determine the types of food refused and parental responses to food refusal.

Case Conceptualization

It appeared from history that food refusal was initially negatively reinforced by escape from aversive stimulation arising from reflux. It also appeared that food refusal was reinforced through avoidance of gagging. At the time of referral, both positive and negative reinforcement appeared to play a role in maintaining food refusal. Christy escaped from eating foods that were more coarse or required more chewing by engaging in food refusal behaviors. In addition, when she refused to accept bites of food, her mother provided substantial coaxing and other attention in an attempt to get her to eat. Furthermore, when she refused to eat food being served at the meal, her mother gave her preferred foods instead so that she was consuming some calories. Therefore, the food refusal appeared to be maintained in part by attention and the receipt of preferred foods. Gagging was probably reinforced in part by attention, preferred food, and escape as her mother expressed great concern each time she gagged, let her stop eating, and let her have access to the pureed foods or baby foods instead.

Course of Treatment and Assessment of Progress

Treatment consisted of (a) a shaping program, (b) differential reinforcement involving the use of attention and bites of preferred food for accepting bites of nonpreferred food, and (c) extinction in which attention and preferred foods were no longer given as a consequence for food refusal.

In the shaping procedure, Christy was asked to eat very small portions of nonpreferred foods at the beginning of treatment, and the portion size was gradually increased across days and weeks. In the differential reinforcement procedure, each time she ate the nonpreferred food that was presented to her,

(*Continued*)

she received her preferred food. She also received enthusiastic praise for each bite of nonpreferred food she accepted. In the extinction component of treatment her mom and dad were taught to withhold their attention for any refusal behaviors or gagging. They were also taught not to provide preferred food following food refusal behaviors.

The graduate student therapist went to the family home and modeled the intervention procedures at a few mealtimes and coached the mom and dad as they used the procedures in subsequent meals. Once the parents were successfully using the procedures and stated that they were comfortable with the procedures, home visits were terminated and office visits were continued to discuss progress and troubleshoot any problems. Initially, mealtimes were videotaped to collect data on Christy's refusals and gagging. When it was evident that refusals were declining greatly and the amount eaten was increasing, videotaping was no longer conducted. Christy's mother continued to record type and amount of food consumed and number of refusals.

Complicating Factors

There were no complicating factors making assessment or intervention more difficult in this case.

Managed Care Considerations

The intensity of the intervention, conducted at supper time in the home a few times per week, would be difficult to carry out in a typical managed care environment. Because a graduate student receiving practicum training was involved in the case, the home visits and increased frequency of sessions could be accomplished.

Follow-up Treatment

Christy continued to make improvements over the course of 3 months, engaging in far fewer food refusal behaviors and eating larger quantities of food. Her weight increased into the 10th percentile after 3 months of treatment. Although she continued to be fed through the gastric tube each evening, the amount of supplementation was gradually decreased.

Implications and Recommendations to Clinicians and Students

This case illustrates how a functional approach to assessment and intervention can effectively address a serious child behavior problem. The intervention procedures were chosen based on the results of the functional assessment, which identified the antecedents and consequences maintaining the problem behavior. The antecedents and consequences were then addressed in treatment to change the variables contributing to the problem. Clinicians and students are encouraged to utilize a functional case conceptualization and a functional approach to treatment as best practice when using behavior modification.

SUMMARY

Behavior modification is a field of clinical psychology that emphasizes objective measurement of behavior before and after intervention to document the effects of the

intervention. Intervention involves the application of behavioral principles to help people change socially significant behaviors. Behavior modification procedures are aimed at changing aspects of the physical or social environment to bring about changes in behavior. Behavior modification procedures are used to help people develop new behaviors (to overcome behavior deficits) and to help people stop engaging in undesirable behavior (to decrease behavioral excesses).

When using behavior modification procedures to help people address behavior problems, a functional approach to assessment and intervention is utilized. A functional assessment is conducted first to identify the variables contributing to the problem behavior. Interventions are then developed based on the results of the functional assessment information. Interventions seek to alter the antecedents and consequences that contribute to the problem behavior while promoting functionally equivalent alternative behavior to replace the problem behavior. Behavior modification procedures are successful when they produce desired behavior changes that generalize to the client's everyday environment.

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CHAPTER 22

Parent Training

Mark W. Roberts

Surely parents were given advice about how to raise their difficult children long before the introduction of parent training to the child clinical literature in the 1960s. McLoughlin (1982) cites explicit American efforts dating back to 1815. What is surprising is that professional therapists attempted to treat emotionally disturbed children during the first 60 years of the twentieth century without involving parents! This chapter traces the therapeutic use of parents to socialize defiant, aggressive children from its humble beginnings at midcentury to the relatively robust protocols of today. The entire parent training literature is too vast to be integrated into a single book, let alone a chapter, so a specific focus must be adopted. First, only articles regarding overt problems of conduct are considered. Specifically excluded are literatures on teaching parents to promote skills in children with disabilities (e.g., S. L. Harris, Wolchick, & Milch, 1982), to facilitate normal adaptive skills such as toileting (e.g., Azrin & Besalel, 1979), or to treat internalizing disorders (e.g., Barmish & Kendall, 2005) or covert problems (e.g., stealing, vandalism). Second, the chapter is limited to children in the preschool and middle childhood periods (roughly 2 to 12 years of age), eliminating publications addressing infants and adolescents. Third, only parent training based on social learning theory is considered (Miller & Prinz, 1990), excluding parent training based on humanistic, dynamic, or family systems theory (Rinn & Markle, 1977). Fourth, the use of parent training as a primary prevention tool (Sanders, 1999) or a secondary prevention strategy (Conduct Problems Prevention Research Group, 2004) is beyond the scope of this chapter. The chapter does presume to accurately represent parent training studies that treat problems of overt social aggression in 2- to 12-year-old children from a social learning perspective.

The chapter is organized into six sections in which the current status and limitations of knowledge are reviewed. A brief history of parent training, its purpose and origins, is presented, which includes references to major review articles and identification and discussion of six ongoing research domains. The theoretical basis for parent training is examined from a historical perspective with an eye toward case conceptualization and treatment selection. The various methods of teaching skills to parents are compared in terms of effectiveness and efficiency. Current parent

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training curricula are reviewed, with a focus on treatments for noncompliance and aggression plus brief discussions of negative vocalizations and situational issues. A case illustration and articulation of future challenges conclude the chapter.

A BRIEF HISTORY OF PARENT TRAINING

The Early Years: 1959 to 1973

Children disobey parents and teachers; they fight with other children; they scream and yell at adults who attempt to discipline them or deny their requests; they argue with "uncooperative" parents, siblings, and peers; and they talk rudely to those who attempt to thwart their interests. Essentially, the presenting problems of oppositional and disruptive children are problems of social interaction, including parent-child, teacher-child, sibling-child, and peer-child interaction. It is to the credit of the small set of clinical child researchers who launched parent training research in the 1960s that they recognized the inextricable relationship between child social aggression and the social contexts in which it flourishes.

A case study by Williams (1959) is often cited as the first published account of parent training, in which the parents of a 21-month-old were instructed to ignore his bedtime screaming. The recommended extinction procedure did successfully covary with decreased tantrum duration at bedtime. Boardman's (1962) florid account of a conduct disordered 6-year-old is often cited as the second documented parent training case. "Rusty" defied his mother, left home and school without permission, set fires, and destroyed property. Boardman met with the family just once, but followed up with several phone calls. Room time-outs and logical consequences were prescribed. Rusty reacted with violent resistance, running away, and continued defiance until an intense spanking episode appeared to suppress his coercive efforts. Positive reinforcement for cooperation had also been recommended by Boardman, but there was no evidence of its use. Boardman's treatment plan elicited concerned and scholarly comments from Bandura (1962), who noted the limitations of punishment-only strategies and recommended the systematic use of positive reinforcement. Intriguingly, the 1962 Boardman and Bandura exchange foreshadowed some of the basic questions faced every day by clinicians, only some of which have been resolved by research. Can we treat social aggression by reinforcing socially appropriate alternative responses or periods of inhibition? Must we actively teach alternative responses to replace coercion, or might such skills already reside within the youth's repertoire? How might we alter natural contingencies among siblings and peers to support alternatives to social aggression, given the effort and nonreinforcement that often accompanies "social skills"? If discipline is necessary to block reinforcers for aggression, how do we advise parents to manage the child's aversive reaction to the recommended discipline routine?

Neither Williams nor Boardman articulated the case for parent training. The case studies, single-case designs, and uncontrolled group studies that followed in the next decade, however, did offer explicit rationales for using "mothers as behavior therapists" (Wahler, Winkel, Peterson, & Morrison, 1965, p. 113) to "reprogram the social environment" (Patterson, McNeal, Hawkins, & Phelps, 1967, p. 181).

Review articles of published studies from the 1960s and early 1970s summarized the many reasons advanced in these vanguard publications for abandoning individual psychotherapy with socially aggressive youth and replacing it with the new therapeutic modality, parent training. Three good reviews from this period are available: Berkowitz and Graziano (1972), Johnson and Katz (1973), and O'Dell (1974). Each reviewer looked at the same set of studies a bit differently, but induced three rather similar themes.

First, all reviewers noted that traditional therapy (i.e., relationship-based, insight-oriented) occupies a minor amount of time in the life of a child (e.g., 1 hour a week) and is performed in an unusual situation (i.e., an office playroom with an unfamiliar adult). Even if successful, therapeutic gains might not generalize to home, school, or neighborhood settings where the social aggression was manifest, let alone persist in those natural settings. In contrast, parents interact almost continuously with the young child referred for treatment. Generalization of treatment effects to the home setting and across time could be directly programmed by providing parents with guidance on how to perform the therapy themselves at home. The parent would assume the primary therapy role, while the licensed clinician would assume a consulting role. Essentially, the first theme culled by reviewers of the period was one of quantitative advantage and setting relevance for parents over traditional therapists.

Second, Berkowitz and Graziano (1972) and O'Dell (1974), and to a lesser extent, Johnson and Katz (1973), inferred a theoretical rationale for parent training from the initial spate of publications. Because parents are intrinsically involved in the socialization of their own children, and because referred children present as socialization failures (i.e., significant aggression and defiance beyond the age of 4 years), parents with socially aggressive children are likely to be unwitting contributors. Typical parenting errors might include providing attention for misbehavior; succumbing to the relentless drone of crying, arguing, or sultry mood displays by providing previously denied objects or activities; allowing natural reinforcers for aggression to operate by ignoring sibling fighting; or modeling angry, defiant responses with one's adult partner. Whatever the parenting mistakes might be, the therapeutic implications of the new theoretical position were clear. A psychopathogenic home environment could overwhelm any therapeutic change a clinician might achieve in a clinic playroom, since the pattern of child management in the home had not been addressed and would continue to elicit and reinforce social aggression. The extreme case of the new theoretical perspective was that targeting social aggression without changing the behavior of parents would simply fail.

Third, all three reviews called for improved research methodology. Published studies at that time often failed to embrace basic research procedures needed for replication and to establish internal validity. These basics include pretreatment measures of child deviance, objective definitions of parenting skills and the techniques of teaching those skills, measurements of parent acquisition and use of therapeutic skills, measurements of child reactions to changed parent behavior, and adequate experimental controls. The early work clearly set the stage for the very productive science of the next 3 decades, which addressed methodological shortcomings, proposed theoretical models, and expanded the research questions.

Patterson, Wahler, and Hanf

Three early entries into this literature warrant special recognition: Gerald Patterson at the University of Oregon (and later the Oregon Social Learning Center, hereafter OSLC), Robert Wahler at the University of Tennessee, and Constance Hanf at the University of Oregon Medical Center. Each contributed substantially to measurement, treatment, and theory during a sensitive period in the development of parent training based on social learning. A new measurement strategy often leads the way in science, and Patterson, Hanf, and Wahler did just that. Each developed microbehavioral observational systems to record parent-child event sequences. Although Patterson and Wahler focused on home settings under general conditions (Reid, 1978; Wahler, House, & Stambaugh, 1976) and Hanf on clinic settings under specific simulated conditions (Hanf, 1968), each coding system could detect frequencies of child and parent behavior, as well as conditional probabilities that revealed the stimulus control and contingent relationships among the behavioral events. Verifiable, observational data were used to confirm the presence of child deviance, identify likely parenting mistakes, provide treatment outcome data, and contribute to theory development.

Patterson and his colleagues defined 29 behavioral codes that sampled an alternating sequence of child and family member interaction in a general home context. Wahler's observational system is similar. He defined 24 child and parent codes, but used an interval-sampling approach, rather than the event-sequence approach pioneered by Patterson. The Patterson and Wahler observational systems provided data that served as the foundation of theoretical advancements, including the measurement of coercive processes (Patterson, 1982; Patterson, Reid, & Dishion, 1992; Wahler & Dumas, 1987). In contrast, Hanf defined nine standard laboratory situations derived from parent interviews about the conditions associated with preschool children's misbehavior (e.g., a parent denies a child's request). Unlike Patterson and Wahler, Hanf's codes did not apply to all measurement situations, and only some codes operationalized an event sequence (e.g., child compliance given a parent's command). Nevertheless, Hanf's observational system directly influenced the coding systems constructed by Forehand (Forehand & McMahon, 1981, pp. 183–218) at the University of Georgia, Eyberg (Eyberg, Bessmer, Newcomb, Edwards, & Robinson, 1994) at the University of Florida, and Webster-Stratton at the University of Washington, who adopted Eyberg's Dyadic Parent-Child Interaction Coding System. By moving the field of child clinical psychology toward observational methods and away from a reliance on interviews, symptom checklists, and personality measures, Patterson, Wahler, and Hanf created the primary tool needed for the development of parent training as a legitimate alternative to traditional psychotherapy.

Hanf and Patterson also developed influential treatment protocols (Eyberg, 1992), based in part on the earlier work of Wahler. In 1965, Wahler and his colleagues trained two parents of oppositional children to use differential attention, and one parent to use differential attention and a room time-out, based on the time-out procedure first defined by Wolf, Risely, and Mees (1964). Wahler (1969a) subsequently published one of the most widely cited empirical demonstrations of

the combined use of social reinforcement and time-out by parents to reduce oppositional behavior. Concurrently, Patterson and his colleagues (Patterson, Reid, Jones, & Conger, 1975) were developing a general set of learning-based strategies to use with parents of socially aggressive 3- to 14-year-old youth. The OSLC introduces parents to social learning principles; specific misbehaviors and prosocial activities are then defined and tracked by the parent in the home; some combination of token reinforcement, token fines, and room time-outs are tailored to the individual child to modify behavior and replace the parenting errors presumed to maintain child misbehavior; the token economy is then extended to address classroom misbehavior as needed; finally, the token economy is gradually replaced by more typical contingencies (e.g., an allowance system). Hanf's (Hanf, 1969; Hanf & Kling, 1973) treatment protocol targeted a narrower developmental range, 2 to 7 years, and focused on noncompliance. It was not Hanf's (Hanf, 1968) intent to focus exclusively on noncompliance. Indeed, she attempted to construct clinic simulations to measure and treat the full range of parent-reported child misbehaviors for this age group. Unfortunately, her outcome measures were limited to noncompliance and parent-child play quality, to the exclusion of other social aggression targets like fighting, tantrums, and negative talk. Hanf's treatment protocol for noncompliance recommended that parents first improve the quality of play with their child in order to enhance the mother's "social desirability" (Hanf & Kling, 1973, p. 7). Parents are taught to discriminate and positively attend to the child's age-appropriate play, to inhibit instructions and questions, and to ignore minor misbehavior. Hanf's first treatment stage is commonly referred to as "the child's game" by the Forehand group (McMahon & Forehand, 2003, *Helping the Noncompliant Child* [HNC]) and "child-directed interaction" by the Eyberg group (Eyberg, 1988, parent-child interaction therapy [PCIT]). Hanf's second stage of therapy taught parents to issue direct commands, praise compliance efforts, warn the child given noncompliance, and use a chair time-out system for continued noncompliance. Hanf's second treatment stage is referred to as "the parent's game" in HNC projects and "parent-directed interaction" in the PCIT tradition. It would be very difficult to find a parent training publication with preadolescent externalizing disordered children that did not include some or all of the Wahler, Patterson, or Hanf treatment protocols. Arguably, Patterson, Wahler, and Hanf, in conjunction with their prominent students (e.g., Chamberlain, Dumas, Eyberg, Forehand, and Webster-Stratton), and the students of their students, have produced much of the knowledge on social learning–based parent training reported in this chapter. Joining these direct descendants is Matthew Sanders (1999) of the University of Queensland in Australia and his many students reporting on the Triple P-Positive Parenting Program.

Discoveries and Reactions

Parent training has been reviewed many times since the early 1970s in peer-reviewed journals, book chapters, and monographs. Journal reviews focusing exclusively on social learning–based parent training are instructive of the maturation of the field across the past 3 decades. The scholarly reader is directed to the work

chronologically: Moreland, Schwebel, Beck, and Wells (1982), Griest and Wells (1983), Miller and Prinz (1990), Graziano and Diament (1992), Taylor and Biglan (1998), Brestan and Eyberg (1998), and Nixon (2002). In some cases, parent training reviews are embedded within the larger topic of treatments for antisocial children (Dumas, 1989; Estrada & Pinsof, 1995; Kazdin, 1987, 1997; Kazdin & Weisz, 1998). Journal reviews have also included more focused aspects of the parent training literature, such as noncompliance (Houlihan, Sloane, Jones, & Patten, 1992), father involvement in parent training (Coplin & Houts, 1991), and applications with Attention-Deficit/Hyperactivity Disorder youth (Chronis, Chacko, Fabiano, Wymbs, & Pelham, 2004). Noteworthy book chapters include Forehand's (1977) initial review of the noncompliance literature; O'Dell's (1985) efforts to clarify the different methods of teaching skills to parents; theoretical contributions by Eyberg (Foote, Eyberg, & Schuhmann, 1988), Wahler and Dumas (1987), and Patterson, Dishion, and Chamberlain (1993); and a relatively recent comprehensive parent training chapter by McMahon (1999). Monographs have also contributed to organizing the field, from those presenting PCIT and HNC traditions to clinicians (Hembree-Kigin & McNeil, 1995; McMahon & Forehand, 2003) to more general parenting applications (Briesmeister & Schaefer, 1998; Dangel & Polster, 1984; Mash, Handy, & Hamerlynck, 1976). The OSLC contributions are presented in the five-volume series, *A Social Learning Approach to Family Intervention* (Patterson, 1982; Patterson, Reid, & Dishion, 1992; Patterson et al., 1975; Reid, 1978), which spans 1975 to 1994, plus a more recent collection of invited chapters in Reid, Patterson, and Snyder (2002).

Consolidating the parent training literature to capture its essential themes and historical transitions is a daunting task. McMahon (1999) conceptualized three stages. The first stage witnessed the development of parent training as a viable model of intervention for socially aggressive children, which corresponds nicely to this chapter's previous discussion of the "Early Years: 1959 to 1973." McMahon's second stage was stimulated by Forehand and Atkeson's (1977) clarion call for research to evaluate and promote generalization of parent training effects from clinic to other relevant settings (e.g., home, community), to untreated siblings, to untreated child misbehaviors, and across time. Part of the promise of parent training, relative to individual psychotherapy, was exactly these generalizations. It was hoped that once the parent understood social learning principles and had gained some degree of reinforcement (and associated self-confidence) for the successful application of new skills, the parent would continue to use effective child management skills in the home (promoting generalization from clinic to home and across time) and to apply principles and specific skills to new problems (generalization across responses) and with other family members (sibling generalization). McMahon's third stage emphasized the development and empirical evaluation of methods to enhance and expand basic prescriptive parent training to address newly identified child problems (e.g., cognitive distortions), relevant contexts (e.g., unsupervised postschool periods), and barriers to successful treatment (e.g., maternal depression, marital discord). See the review on enhancements to parent training by Miller and Prinz (1990) and the earlier call to reformulate the field to "behavioral family therapy" (Griest & Wells, 1983).

A complementary model to McMahon's (1999) three stages is offered next. Six distinct and unending waves of research can be conceptualized, each a reaction to discoveries made at different points in time, and each wave a contributor to progress in the others.

Wave I

The initial spate of reviews (e.g., Berkowitz & Graziano, 1972) challenged the field to empirically test the effectiveness of parent training approaches to externalizing disorders. Clear progress has been made. Social learning theory–based parent training now meets APA's Division 12 criteria as an "empirically supported treatment" (Brestan & Eyberg, 1998; Chambless & Ollendick, 2001; Kazdin & Weisz, 1998). Meta-analytic reviews of parent training effects, contrasted primarily with wait-list control conditions, have reported average effect sizes of .86 (Serketich & Dumas, 1996) and .56 (Weisz, Weiss, Han, Granger, & Morton, 1995). Barlow and Stewart-Brown (2000) reported effect sizes from .20 to .68 for observational data following group-based parenting programs (e.g., Webster-Stratton, 1981a). Far less is known about how well social learning–based parent training compares to alternative family-based treatments, such as Parent Effectiveness Training (Gordon, 1970), family systems theory (Minuchin, 1974), or Systematic Training for Effective Parenting (Dinkmeyer & McKay, 1973, 1976). These counseling approaches to parent training were positively reviewed during the 1970s (Reisinger, Ora, & Frangia, 1976; Rinn & Markle, 1977; Tavormina, 1974, 1975), although more recent social learning-oriented reviewers have found little empirical support for such models (McMahon, 1999; Miller & Prinz, 1990). A meta-analytic review of Parent Effectiveness Training (Cedar & Levant, 1990), however, arrived at a different conclusion. Treatment comparisons appear to be hampered by disagreements over participant inclusion criteria and outcome measures. When overt misbehaviors are measured (Patterson, Chamberlain, & Reid, 1982; Wells & Egan, 1988), social learning programs have usually outperformed the alternative models, but not always (Bernal, Klinnert, & Schultz, 1980). A second ongoing task of Wave I research is dismantling the major treatment programs to determine the active components to successful treatment (e.g., Roberts, 1985).

Wave II

Generality of treatment effects constitutes the second ongoing research domain, corresponding to McMahon's second stage. Much knowledge has been gained (e.g., Sanders & James, 1983) since Forehand and Atkeson (1977) articulated the issues. Generalization of parenting skills from clinic to home settings has been documented multiple times by all major research programs (HNC: e.g., Peed, Roberts, & Forehand, 1977; PCIT: e.g., McNeil, Eyberg, Eisenstadt, Newcomb, & Funderburk, 1991; the group discussion videotape modeling [GDVM] protocol developed by Webster-Stratton: e.g., Webster-Stratton, Kolpacoff, & Hollinsworth, 1988; Triple P: e.g., Sanders & Christensen, 1985; and OSLC: e.g., Patterson, 1974). It is reasonable that home data would confirm setting generalization effects, since clinic work is designed explicitly to reprogram the home environment, giving parents new

methods to manage child misbehavior. Generalization to school settings, however, has quite understandably yielded mixed results, since the teacher is not involved in the parent training, nor are relevant classroom conditions targeted by parent trainers (e.g., independent seat work, peer conflict on the playground). As early as 1969 Wahler (1969b) demonstrated independence of the home and school settings, which necessitated specific interventions designed for the school. Similarly, HNC researchers have not found generality to school settings, despite positive home setting outcomes (Breiner & Forehand, 1981). In contrast, PCIT scientists, using a very similar Hanf-based protocol, have found positive generalizations to untreated school settings (McNeil et al., 1991). Durability of treatment effects has also been repeatedly demonstrated with young noncompliant children (McMahon, Wells, & Kotler, 2006, pp. 162–172) for periods up to 14 years (Long, Forehand, Wierson, & Morgan, 1994). Wahler's (1980) data, however, documents the clear failure to maintain treatment gains over even brief periods when treatment samples include impoverished or isolated mothers. Some evidence of generality to untreated siblings has been detected by various groups (e.g., Eyberg, Boggs, & Algina, 1995; Humphreys, Forehand, McMahon, & Roberts, 1978), which is reasonable, given shared conditions and treatment procedures when both children are in the same developmental period. In contrast, response generalization seems less likely, since untargeted child misbehaviors are likely to involve different psychosocial conditions (e.g., a sibling conflict versus a parental instruction), different functions (e.g., attention seeking versus task avoidance), and different topographies (e.g., hit versus argue). Nevertheless, it is possible that response classes exist to the degree that conditions and functions overlap, despite differences in response topography (Wahler, 1975). Unfortunately, such overlaps are rare (Sanders & James, 1983), although a few have been reported (Wells, Forehand, & Griest, 1980). It appears that parent trainers are usually required to program change across a child's set of presenting problems, rather than focusing on a narrow band (e.g., noncompliance).

Wave III

Techniques for training parents effectively and efficiently have drawn the attention of many researchers, especially O'Dell (1985) at the University of Mississippi and Webster-Stratton (e.g., 1994) at the University of Washington. Given the centrality of parent training techniques to this chapter, research representative of Wave III is reviewed later in some depth.

Wave IV

Treatment failures and dropout data led Wahler (1980) and his colleagues to study the adverse impact of social disadvantage and social isolation on parenting practices. Specifically, inattention, inadequate monitoring, and indiscriminate responding characterize the child management patterns of poor, uneducated, socially isolated women. Wahler and Dumas (1987, 1989) hypothesized that distal setting events (e.g., an aversive interchange with a relative) disrupt parent attention, monitoring, and recognition of the stimulus events actually associated with coercive child behavior, especially for disadvantaged parents. Subsequently, Wahler, Cartor,

Fleischman, and Lambert (1993) developed and evaluated synthesis training, a supplement to basic parent training that has the look and feel of traditional insight-based psychotherapy, centered on each mother's child care experiences and associated extrafamilial events. Synthesis training combined with basic parent training was found to improve child behavior and reduce indiscriminate parenting, whereas basic parent training alone was ineffective with these socially disadvantaged and isolated parents. Teaching complex parenting skills to multistressed, uneducated parents remains a critical and unsolved problem in the literature.

Wave V

Parent pathology has been found to increase risk for inadequate parenting and poor responses to parent training interventions (Griest & Wells, 1983, pp. 42–46; McMahon, 1999, pp. 168–170; McMahon et al., 2006, pp. 149–152; Miller & Prinz, 1990, pp. 294–295). Specifically, parental depression has been associated with attenuated parental monitoring and inaccurate judgments of child behavior, poorer treatment outcomes, and dropout. Similarly, marital distress may disrupt the parent training process. Meta-analyses performed by Erel and Burman (1995) and by Krishnakumar and Buehler (2000) reported significant average correlations between marital distress and child misbehavior. Although distressed couples may profit from parent training to treat externalizing disorders (Forehand, Griest, Wells, & McMahon, 1982), follow-up data reported by Dadds, Schwartz, and Sanders (1987) indicated that marital distress adversely impacts treatment durability. This fifth research wave implies that interventions addressing parental depression and marital discord should be screened for, and treated if necessary, to render parent training effective. Fortunately, empirically supported treatments exist for both adult depression and marital distress (Chambless & Ollendick, 2001). In addition to depression and marital discord, the OSLC group (e.g., Patterson et al., 1992, pp. 102–112) and developmental psychopathologists (e.g., Frick & Loney, 2002) have repeatedly noted the empirical association between parents' antisocial behavior and child externalizing disorders. It is not difficult to understand the adverse effects of an antisocial parent who models angry, aggressive reactions to normal frustrators and interacts coercively with both society and the developing child. Moreover, antisocial adults may be court-ordered to treatment, bringing little motivation to accommodate to therapist recommendations. Parental substance abuse can also disrupt parenting and appears to be exacerbated by disruptive children, creating an event chain rendering both parent substance abuse and child misbehavior more likely (Pelham & Lang, 1993). Unlike parental depression and marital distress, there are no robust outpatient treatment protocols for Antisocial Personality Disorder or substance abuse, leaving the parent trainer with substantive challenges. For the reader interested in the extensive work on parent pathology and child developmental problems, see the recent volume edited by McMahon and Peters (2002).

Wave VI

Variables that promote or degrade parent engagement in the parent training process are now recognized as quite important. See discussions by Miller and Prinz

1990, pp. 299–300), McMahon (1999, pp. 163–164), and Patterson et al. (1993, pp. 65–67) regarding the relatively small and emerging literature on the therapeutic alliance. Although the child's misbehavior is the precipitating factor for the clinic referral, the parent is the mechanism of change, and hence the individual with whom a therapeutic alliance is central. Chamberlain, Patterson, Reid, Kavanagh, and Forgatch (1984) found that high rates of parent refusal statements ("I can't" and "I won't") during the first two treatment sessions are linked to dropout and are associated with maternal depression, paternal social disadvantage, and high base rates of antisocial child behavior. Erstwhile parent trainers inherently place many demands on parents in sessions (e.g., to role-play a new skill) and between sessions (e.g., to use time-out for noncompliance), some of which are very difficult as a result of complexity, child resistance, and/or associated parent emotionality. Parents may have even attempted some variation of a therapist-recommended routine at some point in the past and met with blatant failure. Consequently, many parents argue with the therapist and/or passively avoid therapeutic activities between sessions, to the great consternation of therapists. The OSLC group has consistently called for therapist support and supervision (e.g., Patterson, 1985) to mitigate the potentially adverse effects of nonresponsive parents on therapists.

Balancing traditional supportive therapeutic processes (e.g., active listening skills, reframing, empathic communications, and genuine regard for family members) with prescriptive parent training is an important goal for all therapists. Balancing, however, does not imply abrogating one's professional responsibility to confront difficult problems for the sake of parent comfort, even if that confrontation requires complex and demanding interventions. Flexibility is recommended. Tailoring interventions to accommodate the idiosyncratic issues of a specific family is always appropriate, given that the recommended procedures are theoretically reasonable and the outcome can be monitored. For example, a therapist might opt for a relatively slow shaping process with an unusual but salient positive reinforcement menu, despite an available, efficient, and socially valid standard treatment strategy. Clearly, much is unknown about the dialectical process between prescriptive therapies like parent training and needed therapeutic alliance building.

THEORETICAL FOUNDATIONS OF PARENT TRAINING

Operant Conditioning

The continuing strength of parent training protocols rests on operant principles. Early clinical interventions were attempts to apply learning principles to the treatment of child misbehavior. Patterson (1982, p. 2) views the seminal publications by Skinner (1953), Bandura and Walters (1963), and Ullmann and Krasner's (1965) edited collection of *Case Studies in Behavior Modification* as setting the stage for applied behavior analysis with parents in a therapist role. Early efforts to positively reinforce prosocial child behavior with parent attention or token reinforcers, to extinguish minor misbehaviors via parent ignoring, and to punish aggression and noncompliance with time-outs, fines, or privilege losses are all operant techniques. After a decade of data collection and clinical trials, the OSLC group articulated a

series of conceptual breakthroughs. First, they defined socially aggressive behavior as aversive and contingent (Patterson, 1982, p. 12). An aggressive act is aversive if interacting agents (parents, teachers, siblings, or peers [PTSP]) report it as unpleasant and work to actively escape from it or avoid it. An aggressive act is contingent if its conditional probability is greater that its base rate probability (i.e., demonstrating it is not displayed at random). The OSLC data convinced most scientists that socially aggressive child behavior could coerce interacting agents (PTSP) into changing their own behavior in exchange for temporary cessation of the child's aggression. A "reinforcement trap" (Patterson, 1982, p. 145) was hypothesized to exist, in which the child coerces positive or negative reinforcement from PTSP with aggressive behavior, while PTSP is negatively reinforced by aggression termination. Patterson's widely accepted coercion theory is, therefore, based on rather simple operant processes, but its implications for the psychopathology of childhood social aggression are profound:

1. Aggression is likely to be stable across time, since the child and PTSP are both reinforced, and resistant to extinction, since reinforcement schedules are likely to be intermittent.
2. The child and/or PTSP will escalate to even greater intensities of aggression if either party withholds expected reinforcers (i.e., an extinction burst).

No mechanisms beyond positive reinforcement, negative reinforcement, schedules of reinforcement, and extinction are necessary to appreciate implications 1 and 2. Treatment implications of coercion theory are equally important:

3. Parents and teachers must block reinforcers coerced by social aggression; treatments that do not do so will likely fail.
4. Because most siblings and peers are not sufficiently mature to assume any responsibility to change on behalf of the target child, therapists must teach parents and teachers to guide and protect siblings and peers if the child's aggression is to be successfully treated; otherwise, siblings and peers will continue to unwittingly reinforce child aggression.
5. Parental ignoring will be an ineffectual treatment for child coercion, which is negatively reinforced by parent withdrawal of events aversive to the child (e.g., instructions, denials, discipline efforts), since ignoring allows the negative reinforcement process to operate.
6. Implications 1 through 5 render the successful treatment of socially aggressive children very difficult, especially for middle childhood youth who have experienced thousands of reinforced coercive trials over the course of many years and who can violently oppose parent and teacher efforts to adjust.

Important assessment and case conceptualization strategies are also implicated by coercion theory. Assessments should identify the idiosyncratic set of stimulus events that discriminate the occasion for coercion, as well as guide hypotheses about the parenting errors that are likely to reinforce coercion. The specific stimulus

events (e.g., parent "Stop X" instructions) and settings in which they occur (e.g., bedtime, unsupervised sibling play) must be known to construct a comprehensive yet family-specific treatment plan. Because aggressive behavior is partly defined by its contingent quality, coercion theory implies there will be a set of 1 though K events, presented by PTSP, each of which are discriminative for coerced reinforcement. Figure 22.1 is offered as a heuristic device for the practitioner. Common interpersonal events that elicit social aggression are included in the first two columns of the figure. Any given referred youth will display some subset of the many possible event sequences displayed in the figure. Knowing the relevant operant sequences, however, is not an adequate assessment. See the recent chapter by McMahon and Frick (2005) for a full discussion of evidence-based assessment procedures for externalizing disordered youth.

Figure 22.1 requires elaboration. It is based entirely on coercion theory as presented by Patterson and his many colleagues (Patterson, 1982; Patterson et al., 1992, 1993) and Hanf's (1968) identification of common situations provoking child misbehavior. The figure attempts to characterize typical event sequences consisting of

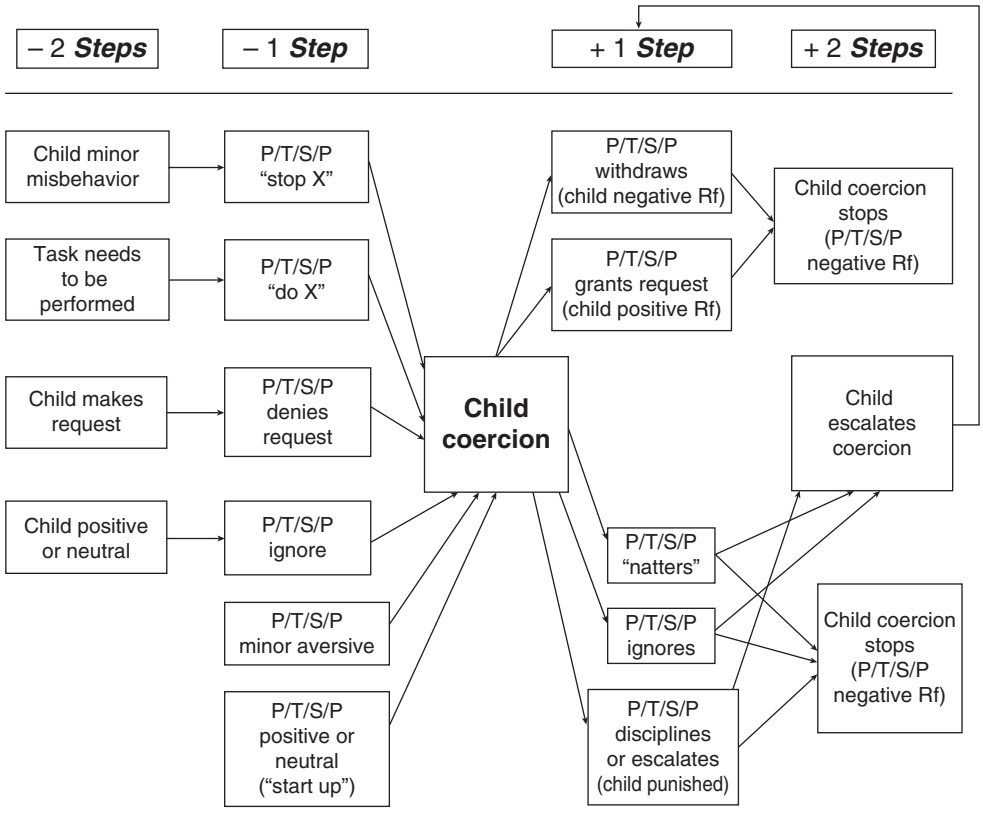

Figure 22.1 Coercive event sequences based on the work of Patterson, other OSLC scientists, and Hanf, designed as a heuristic device for clinicians to conceptualize the specific clinical presentations of referred families. P/T/S/P = Parent/Teacher/Sibling/Peer; Rf = Reinforcement.

six antecedent events ("–1 Step") and five consequent events ("+1 Step"), anchored in the middle by child coercion. Child coercion is defined as overt, contingent, and aversive behavior that is most commonly expressed by the subjects of this chapter as fight, noncomply, negative verbalization, and/or tantrum. Up to two behavioral events that precede coercion ("−2 Steps" & "−1 Step") and two that follow coercion ("+1 Step" & "+2 Steps") are needed to conceptualize the operant mechanisms. At "−2 Steps" an identifiable behavior or condition is present (e.g., child teases his sibling) that increases the likelihood of the "−1 Step" event (e.g., parent instructs child to inhibit teasing). Note that a "−2 Steps" condition has not been identified for all "−1 Step" events. For example, a host of different events probably precede "P/T/S/P minor aversive" (e.g., sibling teases the target child) that are difficult to characterize. Similarly, what precedes the positive or neutral behavior of family members that sometimes evokes child coercion (known as "Start Up"; Patterson et al., 1992, pp. 47–54) is currently unknown. Child coercion is likely to follow "−1 Step" events, especially in clinic-referred youth. At "+1 Step" the child may be reinforced, ignored, or punished for coercion in an indiscriminant manner, resulting in the hypothesized intermittent reinforcement schedule.

The concept of "nattering," introduced by John Reid (Patterson, 1982, p. 69), may be unfamiliar to some and requires discussion. Nattering is interpreted as a hostile, verbal outburst by PTSP (e.g., threatening gestures, angry tone of voice, criticisms, stop instructions, empty threats) that fails to discriminate subsequent discipline but may provide positive reinforcement in the form of contingent attention and certainly provides a hostile verbal model. Figure 22.1 supposes that on some occasions, nattering is followed by child cessation ("+2 Steps"), negatively reinforcing the natterer, but more often than not, nattering simply prolongs the aversive interchange and sets the occasion for escalated intensities of social aggression (Patterson, 1982, pp. 67–84; Patterson et al., 1992, pp. 39–54). The model hypothesized in Figure 22.1 also captures the paradoxical concept of punishment acceleration (Patterson, 1976), in that referred children are much more likely than normal controls to react to discipline efforts ("+1 Step") with escalated intensities of coercion ("+2 Steps"), throwing PTSP back to "+1 Step" options, where the PTSP may finally relent and provide reinforcement, attempt to ignore, or continue the struggle by escalating the intensity of punishment or by nattering.

Patterson's (1976) concept of the aggressive child as both "victim and architect" of his or her own environment is germane within the operant framework. The model provided in Figure 22.1 implicates both PTSP and the child. Certainly the operant model implies that PTSP lack skills. These important socialization agents are reinforcing social aggression on an indiscriminant, intermittent basis. To that extent, the child is a "victim" of dysfunctional child management. On the other hand, the operant model exposes the child's unintended role as "architect" of the very environment that sustains his or her psychopathology. First, aggressive child actions coerce their own reinforcement from the many unwitting interactive agents who seek short-term relief from the child's aversiveness via misguided child management strategies. Second, to avoid the aversive child coercion altogether, PTSP are likely to inhibit presenting "–1 Step" events, thus abrogating their normal socialization roles.

Social Learning Theory

Most parent trainers ascribe to the theoretical framework provided by social learning theory (Bandura, 1977), rather than relying on the more narrow principles of operant learning. Indeed, Patterson's published series is entitled *A Social Learning Approach to Family Intervention*. Social learning theory embraces the full range of biopsychosocial variables that interact continuously during the child's development. Consequently, social learning theory provides a better theoretical home for evidence-based parent trainers than the more limited operant model. For example, it is presumed that biological-based temperaments impact parenting (e.g., Wahler & Dumas, 1987), that models of antisocial behavior promote coercion (Frick & Loney, 2002), that modeling and discussing child management skills can result in parent knowledge and performance acquisitions (e.g., Webster-Stratton, 1994), that social-cognitive processes influence overt social behavior in middle childhood (e.g., Dodge, 1980), and that a family's cultural background is an important consideration in parent training (Forehand & Kotchick, 1996). Each of these biopsychosocial processes are well beyond the confines of operant mechanisms.

Developmental psychopathology is consistent with social learning theory. The relevant examples of this are the child-onset (i.e., "early starter") Conduct Disorder model, nicely summarized in McMahon et al. (2006, pp. 147–152), the Patterson stage model (e.g., Patterson et al., 1992, pp. 115–141), and Frick's (2004) recent review of the developmental psychopathology of severe antisocial behavior. Child-onset Conduct Disorder models are based on the continuous interaction of biopsychosocial variables from a developmental perspective across the changing social contexts of the child's cultural group from infancy to adolescence. Of particular relevance to clinicians are two classes of findings from the child-onset developmental model. First, inept parental discipline and inadequate parental monitoring have been repeatedly found to increase the risk of social aggression (Patterson et al., 1993). Consequently, parent trainers must invariably evaluate and improve parent discipline and monitoring. Second, social (e.g., Dodge, 1980) and academic (e.g., Hinshaw, 1992) skill deficits are very likely to emerge in externalizing disordered youth by middle childhood (Patterson et al., 1992, pp. 120–131). Consequently, parent trainers should assess sibling and peer skills and classroom function and, if needed, intervene or refer to address the identified deficits.

The empirical and theoretical work of Wahler and Dumas with impoverished, socially isolated mothers deserves special comment here. It is consistent with the cognitive elements of social learning theory (Wahler & Dumas, 1987, p. 616) and adds to our theoretical understanding of seemingly intractable dysfunctional parenting. Specifically, Wahler and Dumas (1989) argue that multistressed parents (e.g., impoverished, socially isolated, depressed, abusive) operate with trait-like response classes, such that setting events irrelevant to parent-child interaction adversely impact temporally distal parent monitoring, judgments, and responses to children. Coercion theory cannot explain the apparent overgeneralization of parent responding from a child-irrelevant context (e.g., friend, employer, spouse) to child encounters. Moreover, this parental failure to attend to the differences in social context and respond adaptively may be a significant cognitive barrier to parent

training with multistressed parents. All social learning theory–based parent training programs attempt to teach parents to discriminate specific child actions, track those actions in the home, and respond in the prescribed manner. In opposition to these well-intended recommendations, multistressed parents appear to operate with macro-, trait-like cognitive strategies that link seemingly disparate social encounters. It is currently unknown if synthesis training, a very cognitive, insight-based therapeutic process, will reduce this barrier to treatment success.

Attachment Theory

Eyberg and her colleagues (Foote et al., 1998) have oriented the child clinical field to the special importance of parental sensitivity to child signals. Parental sensitivity and the establishment of synchronized turn taking have long been found to facilitate secure attachments and promote general socialization (Maccoby & Martin, 1983, pp. 26–37). Developmental psychologists have repeatedly found specific positive associations between mother-child reciprocity and successful socialization (e.g., Kochanska, 1997; Stayton, Hogan, & Ainsworth, 1971). In contrast, parents of aggressive children are likely to be indiscriminate and aversive (e.g., Dumas & Wahler, 1985) in comparison to normal control parents. Eyberg's child-directed interaction (in Hembree-Kigin & McNeil, 1995) and McMahon and Forehand's (2003) child's game are treatment components in noncompliance programs that actively teach the parent to be sensitive to age-appropriate child signals during free-play conditions. Eyberg argues that by enhancing the responsiveness of parents to appropriate child signals, therapists will strengthen the parent-child relationship in a manner analogous to the development of secure attachments in normal infants. Whaler and Dumas would support this position, since child-directed interaction skills teach the parent to discriminate and attend to age-appropriate child behavior. Even operant devotees would applaud the parental attention to "good" behavior found in child-directed interaction. Because teaching parent responsiveness is a fairly widespread clinical practice, the theoretical linkage to normal child development is important. Indeed, child-directed interaction looks like the nurturance component of Baumrind's (1967) classic conceptualization of the successful authoritative parent, who combines firm control with nurturance and communication skills. A second intriguing linkage is to traditional humanistic-based play therapy (Axline, 1969), with the major procedural exception, of course, that parents are trained to perform the therapist role.

Summary of Theoretical Perspectives

Parent trainers earn their daily living with operant strategies. Coercive processes must be minimized if the child is to gain a normal socialization path. The much broader shoulders of social learning theory, however, are embraced to consider the wider range of biopsychosocial variables needed to assess, conceptualize, and treat externalizing disorders. Finally, attachment theory, stripped of its psychodynamic overtones, provides a basis for the common goal of strengthening the parent-child relationship.

TECHNIQUES FOR TRAINING PARENTS

Training techniques can be dimensionalized in terms of therapist time and degree of contact. At the least intrusive, least expensive level, therapists can impart skills didactically via discussion, handouts, and programmed reading materials, all of which can be used in group formats to add even more efficiency. Modeling is usually conceptualized as more intrusive than didactic approaches, especially in vivo modeling, since it requires an actor in addition to the therapist. Videotaped modeling, however, does not require an actor and is very compatible with group training (e.g., Webster-Stratton, 1994). Because discussion, written materials, and modeling are passively received by parents, the clinician remains uncertain if acquisition has occurred. In contrast, role-playing and guided practice are more intrusive, active, and informative strategies, since the parent must perform overtly. In role-play formats the parent responds to an actor (or therapist) simulating child behavior. During guided practice the therapist provides immediate feedback to the parent while the parent interacts with his or her own child. Guided practice provides direct evidence of both parent acquisition and child reaction to the recommended protocol. Finally, techniques to prompt and motivate home use of newly acquired skills include specific homework assignments (accompanied by written materials for reference), visually posted behavioral tracking systems, and phone contacts.

Psychologists tailor the set of training techniques to the child's presenting problems, developmental level, social-cultural circumstance, and treatment protocol. For example, introducing a home token system for use with aggressive grade-school siblings would likely require discussion, written materials, a home tracking system, modeling, and role-playing. In addition to the parent, both siblings would be actively involved in all facets of the training to gain their input regarding the reinforcement menu, as well as their awareness of the new rules, the reasons behind the rules, and all the changed contingencies. In contrast, introducing differential attention skills to the parents of a defiant 3-year-old would emphasize guided practice using a bug-in-the-ear during free-play conditions. Moreover, the 3-year-old would not be involved in discussion with the therapist about goals, rationales, and choices.

Psychologists also adjust their training techniques, based on parent performance. For example, role-playing might reveal a parent tendency to repeat instructions quickly, despite verbal feedback. Consequently, the therapist might revert to several modeling trials before attempting a second try at the more demanding role-play routine. Conversely, efficiencies can be obtained by sampling a parent's performance following minimal training, for example, brief discussion. If the advice is specific and meaningful, a parent might adjust his or her behavior immediately and with little difficulty (e.g., "Ignore whichever child is crying and play with the other sibling"). The data will inform the clinician if this 3-second intervention worked!

A brief and representative review of empirical findings is offered.

Didactic Strategies

Simple advice giving in the absence of other techniques has been described (Boardman, 1962; Holland, 1969), but not studied with adequate experimental control.

Obviously, verbal instructions are an inherent component of all other training strategies (i.e., "Do X under Y conditions for Z reasons"). The therapist's interpersonal skill in communicating these instructions is likely an important component of engaging (or failing to engage) the parent in the therapeutic process, represented by Wave VI of ongoing research in parent training discussed earlier.

Written materials can be a sufficient intervention under limited circumstances. Clark et al. (1977) used a multiple-baseline design with observational data to test the effects of a written manual of advice for parents on how to shop successfully with preadolescent children. Procedures included providing information about shopping, differential attention, token reinforcement for rule following, and token loss for misbehavior. Child misbehavior and ineffective parenting declined as a function of reading the manual (Green, Clark, & Risley, 1977). Similarly, McMahon and Forehand (1978) used a brochure to instruct parents on a mealtime routine, consisting of differential attention, instruction giving, and brief room time-outs. Again, a multiple-baseline design indicated that children reduced their overt misbehavior and parents displayed new skills. Sloane, Endo, Hawkes, and Jenson (1991) found that written self-instructional materials alone helped most parents provoke some improvement in their child's compliance.

Together, these three multiple-baseline projects reveal the potential contribution of a well-written manual or handout for changing overt parent behavior and subsequent child adjustment. Each project was limited, however, by the use of recruited subjects. Connell, Sanders, and Markie-Dadds (1997) evaluated the effects of written materials on parents of oppositional children relative to a wait-list control group. The positive results, however, were limited to improved questionnaire data. When parents have been randomly assigned to reading-only conditions and compared to those experiencing more active training techniques, reading-only is usually less effective on performance measures (Eyberg & Matarazzo, 1980; Flanagan, Adams, & Forehand, 1979; Nay, 1975; O'Dell et al., 1982), but can be equivalent (O'Dell, Krug, Patterson, & Faustman, 1980; O'Dell, Mahoney, Horton, & Turner, 1979). See McMahon and Forehand (1980) for an early review of self-instructional studies and McMahon and Forehand (1981) for a discussion of the many parameters (e.g., parent reading level) that should be considered when using or researching written instruction techniques.

Modeling

Modeling specific skills has a long history (Mash & Terdal, 1973) and a sound theoretical basis in social learning theory (Bandura, 1977). Several years ago Webster-Stratton (1981b) developed a training protocol in which videotape modeling is the primary technique. Parents in small groups (8 or 9) view vignettes of parents displaying specific skills with children in a variety of normal social contexts. After viewing each vignette (about 2 minutes), a therapist prompts discussion of the critical events, helps the group discuss how to apply the skill at home, and assigns homework. A programmatic series of publications has largely supported the efficacy of Webster-Stratton's GDVM protocol. Performance measures of parental responsiveness during play contexts improved relative to wait-list controls

posttreatment (Webster-Stratton, 1981a) and were maintained at 1-year follow-up (Webster-Stratton, 1982). Early conclusions were attenuated, however, by the use of recruited subjects and nonspecific play observation conditions in a clinic (i.e., a condition not likely to provoke coercion). More recently, the GDVM protocol was found to be as effective as an individually administered videotape (IVM) modeling procedure and a group discussion protocol, relative to a wait-list condition (Webster-Stratton et al., 1988). Importantly, participants in these later projects were referred children who met a questionnaire criterion for deviance, and observational data were collected in the home. Consumer satisfaction with GDVM and group discussion protocols has been rated higher than for IVM, suggesting the importance of the therapist contribution (Webster-Stratton, 1989); moreover, parents found that GDVM led to skills that were easier to implement at 1-year follow-up than either IVM or group discussion alone. Therapist consultation appears to improve the effectiveness of IVM (Webster-Stratton, 1990). Because highly stressed families (cf. Wahler & Dumas, 1987) were less likely to profit from IVM (Webster-Stratton, 1992), GDVM can be augmented by parent communication, problem solving, and self-control components (Webster-Stratton, 1994), a nice example of McMahon's Stage 3 in parent training research. The enhanced program yielded increased benefits in consumer satisfaction, child pro-social knowledge, and problem-solving and communication skills not targeted by GDVM.

Recently, Foster and Roberts (2007) replicated the effectiveness of videotape modeling combined with didactics to improve parent skills in both play and clinic task conditions, but also replicated the need for therapist monitoring of initial parent use of time-out (Roberts & Powers, 1990). In addition, technique comparison studies have generally favored more active techniques, which include modeling, over didactic strategies and wait-list on performance measures (Eyberg & Matarazzo, 1980; Flanagan et al., 1979; Nay, 1975), although O'Dell et al. (1982) found that written instructions compared well with videotape modeling and a more intrusive procedure, live modeling plus role-playing.

In summary, modeling, like written instructions, has been demonstrated to be an important technique for parent trainers. Modeling is invariably more effective than no training and competes well with alternative techniques. The effectiveness of Webster-Stratton's GDVM protocol is truly impressive. The opportunity to discuss the modeled event sequence with other parents under the watchful eye of a therapist appears to be an important component of GDVM. If videotapes are self-administered, a therapist consultant is recommended, especially to assist the parent in initial use of potentially emotion-provoking routines such as time-out.

Role-Playing and Guided Practice

Role-playing has three distinct advantages over all other techniques: flexibility, control, and the opportunity to present retrials immediately. A little imagination and a few props allow psychologists to tailor role-play analogues to family-specific problem conditions (e.g., bedtime, shopping, car trips, sibling aggression, toilet training). For example, the trainer might say, "Let's pretend this doll represents your child and these four chairs are the family car. When the doll starts fighting

with his pretend brother (another doll), you practice stopping the 'car' just like you saw me do it a moment ago." There is no limit to creative simulations. Second, the therapist has complete control over the doll, enabling task difficulty to be modulated to accommodate the parent's acquisition rate. Third, immediate retrials of exactly the same stimulus array can be presented to practice a skill or to correct an error. For example, given an error, the parent trainer can discuss the error, the targeted response, the reason for the targeted response, and then immediately present the task again. No other training technique allows this intensive, individualized practice. Written materials and videotape models are inflexible; guided practice puts the trainer at the mercy of the child, who often exhibits nongraduated task difficulties, will not pause for discussion, and may not give a second chance at the same misbehavior. Only in vivo modeling with an actor matches the flexibility and control of role-playing, but lacks the retrial component, since performance is not required. Consequently, parent trainers often provide live models followed by a role-play to teach specific event sequences.

Several experiments (Flanagan et al., 1979; Nay, 1975; O'Dell et al., 1982; Webster-Stratton, 1984) have found that live modeling plus role-playing outperformed didactics and no training on performance measures, but sometimes, equivalent acquisition rates are found (O'Dell et al., 1979, 1980). Powers and Roberts (1995) contrasted a standard training protocol, toy cleanup, with an augmented simulation procedure designed to enhance generalization. Parents who rehearsed skills to mastery via modeling and role-playing for wake-up/dressing, mealtime, and cleanup/bedtime conditions performed better during unstructured clinic and home conditions than parents trained solely under toy cleanup conditions, suggesting that multiple, content-valid exemplars should be incorporated into modeling and role-play regimens.

If the parent is provided prompts and feedback during interaction with his or her child, the training technique is considered "guided practice." A bug-in-the-ear device is often used to allow the therapist in an observation room to talk to a parent in an adjacent clinic room without the child's awareness. The "bug" is a small radio receiver worn by the parent like a hearing aid; the radio signal is transmitted locally to the clinic room via an antenna. Hanf (1969) introduced the procedure into the literature, which is now a staple of HNC and PCIT parent training programs that target 2- to 7-year-old disruptive and oppositional children. Note that guided practice loses its applicability in middle childhood, as youth become savvy to one-way windows and clinic simulations lack credibility. The training advantages to guided practice are obvious: immediate social reinforcement, corrections for errors of commission, and prompts for errors of omission. Therapist support via the bug can be gradually faded as the parent displays mastery, allowing the parent to practice with the child under more home-like conditions. Because the target child is providing the behavior display, programming for setting generality is maximized. Unlike all other training techniques, a psychologist using guided practice has the opportunity to observe the child's reaction to the recommended changes in parent behavior, which allows the psychologist to make procedural adjustments as needed. Because young defiant children are especially prone to react to discipline with coercion (Roberts & Powers, 1988), a phenomenon consistent with

theory and data (e.g., Patterson, 1982; Roberts, 1982b), it is particularly important to monitor initial time-outs with overtly defiant children (Roberts & Powers, 1990). Guided practice provides that mechanism. Only one thorough test of this position, however, is available. Webster-Stratton (1984) randomly assigned recruited parents to individual therapy, which made use of live modeling, role-playing, and guided practice with a "bug," to GDVM and a no-treatment condition. Both treatment groups performed equally well posttreatment and better than wait-list controls; no behavioral differences were present at 1-year follow-up. Neither treatment group, however, manifested a significant decrease in child noncompliance ratios, which is inconsistent with data from HNC and PCIT outcome studies. It is hypothesized that guided practice fills an important niche in the tools of the parent trainer, but no group comparison data currently support that contention.

Homework, Posted Tracking Systems, and Phone Contacts

To facilitate transfer of training from clinic to home settings, specific homework is routinely assigned (e.g., "Use the compliance program every time you give an instruction at home this week"). In addition, some form of record keeping system is provided to parents to monitor progress at home. These participant-observation devices are usually posted somewhere central (e.g., the refrigerator), thereby serving a prompting role in addition to their use as treatment outcome data. If a therapist anticipates parent difficulty in implementing a recommended child management routine (e.g., initial time-outs, a bedtime program), phone calls can be scheduled to provide support and guidance. Parents are invariably invited to contact the therapist during working hours if problems arise at home and can be provided after-hours emergency contact systems if needed.

Although homework, participant observation, and phone support are well integrated into parent training programs, only participant observation has been studied systematically. The parent daily report (PDR; Chamberlain & Reid, 1987) is the best example. Parents are contacted by phone and report on the occurrence or nonoccurrence of specific misbehaviors during the prior 24 hours. Test-retest reliability of PDR data has been adequate in clinic samples ($r = .60$) and excellent in recruited samples ($r = .82$); interparental agreement has been modest ($r = .56$), but given differential daily interaction with the child, would not be expected to be a sensitive index of accuracy. Importantly, PDR data have converged well with total aversive behavior rates collected by professional observers in three samples ($.46 < r < .69$). Moreover, normative data are available, indicating an average of 5.3 misbehaviors per day ($SD = 3.1$) in 4- to 10-year-olds. Unfortunately, the PDR system is more of a research tool than a practical clinical instrument at this point, and does not serve the prompting role of posted tracking systems. Most clinicians cannot make daily phone calls, even for brief periods, and most clinicians want daily frequency counts rather than binary data. Parent-collected daily frequency counts of targeted sibling aggression (Olson & Roberts, 1987) have yielded good evidence of internal consistency (i.e., the odd/even day correlation for sibling aggression was $r = .88$), but nothing is known about the accuracy or concurrent validity of such parent-collected frequency data.

CHILD MANAGEMENT SKILLS

The core curricula of all modern social learning–based parenting programs were established by the early work of Patterson, Wahler, and Hanf. Six generic parenting skills can be identified. First, parents are introduced to the social learning principles that underlie the acquisition and maintenance of externalizing disorders (see Figure 22.1) and the use of operant, cognitive, and developmental principles to treat the child's coercive patterns. Patterson and Gullion's (1968) *Living with Children,* a programmed textbook, served as the prototype for all future efforts (e.g., McMahon, Forehand, & Griest, 1981; Patterson, 1975) to help parents understand why their child misbehaves and how they could be part of the solution. Second, parents are taught to define, observe, and record the presence of targeted misbehaviors and prosocial accomplishments in the home setting. Participant observation provides more than assessment and treatment outcome data; it enhances the parent's awareness of child acts that will become discriminative for implementing therapist-recommended interventions and could help multistressed parents with the cognitive appraisal problems identified by Wahler and Dumas (discussed earlier). Third, parents are trained to attend to age-appropriate child behavior while ignoring coercive behavior maintained by parent attention. There are two distinct and different purposes behind differential attention skills. First, clinic-referred parents commonly provide attention to misbehavior in the form of reprimands, "Don't" instructions, and/or nattering in general (e.g., Patterson, 1976), which at worst positively reinforces coercion and at least models arguing. Teaching parents to withhold attention for specific misbehaviors can effectively extinguish some coercive acts, such as negative verbalizations (Herbert & Baer, 1972), tantrums during time-out administrations (Hobbs & Forehand, 1975), and defiant talk during compliance training (Roberts & Hatzenbuehler, 1981). Note, however, that differential attention can make some misbehaviors worse (Herbert et al., 1973). Moreover, differential attention does not play a substantive role in treating noncompliance (Roberts, 1985), since noncompliance appears to be reinforced by task avoidance rather than misguided parental attention. For an early discussion of differential attention, see Sajwaj and Dillon (1977), and for a modern procedural description of the effective use of differential attention, see McMahon and Forehand (2003, pp. 106–129). The second purpose of differential attention skills is to improve the parent-child relationship via responsiveness during play. Eyberg and her many students (e.g., Foote et al., (1998) emphasize parental responsiveness during play contexts as a mechanism to enhance the parent-child bond, as discussed earlier under attachment theory. Fourth, because noncompliance is a foundational problem in externalizing disorders (Loeber & Schmaling, 1985; Patterson et al., 1992), parenting programs invariably teach parents to use verbal instructions that provoke child attention and provide precise information. Verbal instructions that specify a specific motor task, followed by a 5-second pause, elevate compliance ratios of 3- to 7-year-olds relative to experimental control subjects (Roberts, McMahon, Forehand, & Humphreys, 1978). In addition, eye contact has been experimentally shown to increase compliance ratios in middle childhood (Hamlet, Axelrod, & Kuerschner, 1984), a finding supported by a group comparison project with 3- to 7-year-olds (Hudson & Blane, 1985). Specifically,

nonclinic parents used more eye contact than parents of referred children. In addition, Hudson and Blane found that instructions issued in close proximity, at the child's eye level, while orienting toward the relevant objects and using a pleasant tone of voice were more likely in parents of normal children than clinic-referred children. Wruble, Sheeber, Sorensen, Boggs, and Eyberg (1991) demonstrated that direct commands (e.g., "Come here") were more effective than indirect commands (e.g., "Would you come here?") in a recruited sample of 3- to 5-year-olds. These data are consistent with the common recommendation to use an imperative grammatical form of verbal instruction rather than the interrogative form in which choice is implied. Fifth, most parenting programs introduce some form of discipline to block the reinforcers previously coerced by the child. Time-outs, token fines, privilege losses, or work chores are routinely recommended for noncompliance and aggression, if not for other targets. Sixth, token reinforcement systems are introduced to provide a mechanism to deliver activity and material reinforcers for prosocial behavior on the assumption that social reinforcers (e.g., praise) are relatively weak compared to coerced reinforcers (e.g., task avoidance), especially for referred middle childhood youth (e.g., Patterson et al., 1975; Wahler, 1969a). Successful applications of token systems are not limited to middle childhood, however, and can be used successfully with children as young as 3 years (Christophersen & Mortweet, 2001, pp. 223–239).

Noncompliance Programs

The classic Hanf protocol for teaching noncompliant 2- to 7-year-olds to obey parent instructions has been repeatedly evaluated and come through 35 years of empirical scrutiny relatively unscathed. Figure 22.2 presents the often displayed familiar event sequence (e.g., McMahon & Forehand, 2003, p. 131). Parents are first taught to present effective instructions as discussed earlier. If the child makes any motoric

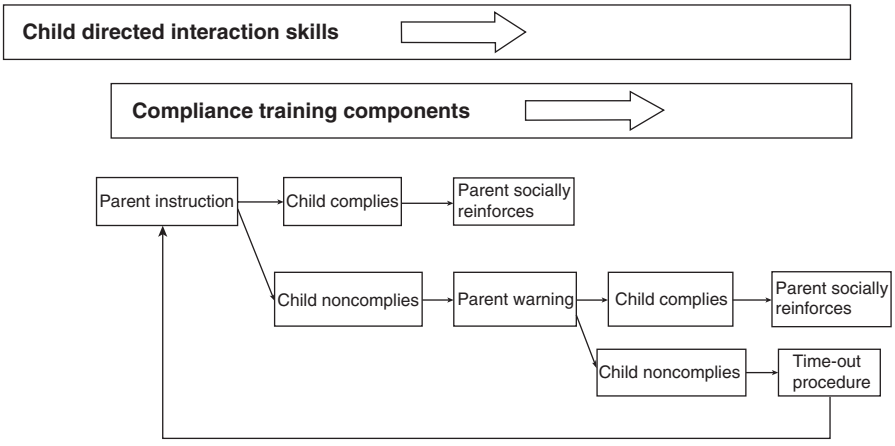

Figure 22.2 Recommended compliance training protocol for 2- to 7-year-olds based on the sequential strategy of current HNC and PCIT programs.

effort to comply within 5 seconds, social reinforcement is provided. If the child makes no effort within 5 seconds of the termination of the instruction, a warning is issued. Child effort within 5 seconds of the warning is socially reinforced, whereas a lack of effort results in a time-out administration. Upon completion of time-out, the original instruction is re-presented and the process is recycled as needed until the child complies. It sounds simple, but there are endless complexities. Fortunately, some empirical guidance is available.

Because the parent's game and cleanup conditions frequently provide the context for teaching compliance-eliciting skills in both HCN and PCIT traditions, the many successful demonstrations of compliance acquisition focus on child initiations to perform discrete acts (e.g., "Put the toy on the shelf"). Less is known about motivating children to display sustained effort to complete a substantial task (e.g., "Clean your room") or to cease an ongoing response (e.g., "Stop yelling"). The ability to sustain effort in the absence of an adult requires complex self-regulatory skills that emerge gradually (e.g., Maccoby & Martin, 1983, pp. 35–37), highlighting the need for parents to discriminate noncompliance from normal developmental errors. For example, parents might instruct an 8-year-old to clean his or her room, but not a 3-year-old for whom parent proximity, modeling, and repeated instructions would be needed. A different set of issues involves "Don't" instructions. We know that even 2-year-olds can inhibit actions given an instruction (Minton, Kagan, & Levine, 1971), but the immediate cessation is less relevant than the resultant frequency of the targeted misbehavior. For example, does the child's base rate of teasing his sibling decline, given compliance to the adult instruction "Stop teasing"? That is currently an empirical question, addressed at the level of the individual child. For physical aggression, however, some general knowledge exists. Jones, Sloane, and Roberts (1992) found clear evidence that parent "Stop fighting" instructions can function like nattering, socially reinforcing misbehavior and allowing coerced reinforcers from siblings to operate. Because the child stops fighting briefly, the parent is negatively reinforced for issuing "Stop fighting" instructions, as predicted by coercion theory (Figure 22.1), despite an unchanged or worsened fight frequency. Similarly, Kendall, Nay, and Jeffers (1975) observed that adolescent peer aggression in a residential facility increased when an instruction or warning procedure preceded time-out. Therefore, parent trainers are advised to consider the type of instruction used by the parent and adjust the treatment plan according to developmental variables and empirical evidence. Using a noncompliance protocol to reduce physical aggression is contraindicated.

Hanf recommended a 5-second interval as the window for child initiation of effort to instructions and warnings, which intriguingly has some observational support (Wruble et al., 1991), despite its clearly arbitrary roots. If the child does display effort during the 5-second window, social reinforcement is recommended. Interestingly, social reinforcement is not necessary for compliance acquisition or maintenance (Roberts, 1985; Roberts, Hatzenbuehler, & Bean, 1981) in overtly noncompliant referred preschoolers, failing to augment instruction and time-out effects. Noncompliance appears to be negatively reinforced by task avoidance rather than misguided parental attention, and compliance acquisition, at least in

previously noncompliant 2- to 7-year-olds, appears to be negatively reinforced by active avoidance of time-out. Compliance acquisition in normally developing toddlers may be quite a different process, however, and involve such variables as infant temperament and parental responsivity, as discussed earlier. Hopefully, all children, including previously noncompliant ones, eventually internalize a standard of conduct to obey legitimate authority and, subsequently, regulate their own behavior rather than obey simply to avoid discipline. Nothing in these empirical findings and theoretical speculations indicates that parents should not be taught to praise child compliance. Theoretically, every opportunity to respond positively to prosocial child behavior is an opportunity to promote the parent-child relationship. The high consumer satisfaction (McMahon, Tiedemann, Forehand, & Griest, 1984) and social validity (Forehand, Wells, & Griest, 1980) of HNC protocols suggest that the Hanf two-stage protocol, which includes social reinforcement for compliance and mastery skills in child-directed interaction (PCIT) or the child's game (HNC), is thoroughly justified. Consequently, child-directed interaction is included in Figure 22.2 as the first step in compliance training. Nevertheless, the scholarly skeptic is not yet convinced that parent social reinforcement of child compliance is important, given the current limited set of dependent variables. See Forehand (1986) for a more thorough look at this issue, and the ensuing rejoinder by Lutzker, Touchette, and Campbell (1988) and the Forehand reply (1988).

Warnings to obey instructions (or be sent to time-out) have proven useful when the instruction is to initiate a discrete task (Roberts, 1982a). The warning component has reduced the number of time-outs needed to elicit compliance criterion responding and has not attenuated compliance acquisition rates relative to no-warning control conditions. The concern has always been that warnings will function like nattering, which appears not to be the case for instructions to initiate behavior. As noted earlier, however, fighting is not well served by "Don't" instructions and warnings, suggesting that the type of instruction interacts with the utility of warnings. If parent trainers are trying to manage a misbehavior other than noncompliance (e.g., tease, yell, dangerous climbing) with "Don't" instructions, the base rate of the misbehavior should be used to evaluate the protocol, rather than compliance to the instruction or warning.

The time-out component of Hanf-based programs has been repeatedly shown to be a critical treatment component (Roberts et al., 1981), yet one plagued with difficulty (Roberts, 1982b, 1984). Time-out procedures are complex. Early reviews (K. R. Harris, 1985; MacDonough & Forehand, 1973) explicated the many parameters of time-out (e.g., duration, location, release rules), some of which are now understood. Decades ago Leitenberg (1965) queried whether time-out is an aversive condition. It is. Overtly noncompliant children will fiercely resist initial time-outs (Roberts, 1982b), and successfully treated referred children will persistently comply to avoid them (Roberts, 1985). The more noncompliant the child at baseline, the more intense the time-out resistance (Roberts & Powers, 1988, table 2). If parent trainers allow children to successfully escape from time-out, compliance acquisition is significantly degraded (Bean & Roberts, 1981; Roberts & Powers, 1990). Moreover, contingency awareness does not mitigate the defiant child's behavioral insistence on testing the limits of the newly imposed contingencies (Roberts, 1984).

Fortunately, the spanking component of Hanf's original time-out maintenance procedure proved unnecessary (Day & Roberts, 1982; Roberts, 1988). Brief room time-outs maintained by a barrier suppress chair time-out escape efforts just as well as spanking, while avoiding corporal punishment and modeling aggression. That finding does not imply that putting a barrier or door between the parent and an angry child is a pleasant process. For just that reason, Roberts and Powers (1990) and Foster and Roberts (2007) have called for clinicians to monitor initial time-outs, support parents and children through the process, and adjust the procedures as needed. A legitimate and unanswered question focuses on time-out location. Why not use the prototype room time-out (Wolf et al., 1964) and avoid the escape problems of chair time-outs? The early work of Wahler and the OSLC group did just that. The argument voiced by HNC and PCIT programs is that 2- to 7-year-olds are better served by chair time-out conditions that can be visually monitored for safety and readily adapted to community settings. This argument, however, is rational rather than empirical. For a recent discussion of time-out parameters, see Roberts (2005).

Aggression Programs

The OSLC database has yielded some important empirical insights into the interrelationships of aggressive boys, their siblings, and their mothers. All parties appear to elicit, model, and reinforce physical aggression, yielding about one fight per hour, with the most likely victim a male sibling (Patterson, 1984, 1986). Years ago O'Leary, O'Leary, and Becker (1967) established the basic intervention for sibling aggression: immediate chair time-out combined with reinforcement for cooperative sibling interaction. An important distinction between treatments for noncompliance and aggression was anticipated by the O'Learys. The noncompliance protocol promotes its prosocial replacement skill (i.e., compliance) via an instruction and two reinforcement mechanisms: time-out avoidance and social reinforcement. In contrast, suppressing aggression with time-out or other strategies (e.g., token fines) does not directly provoke an acceptable replacement skill. Indeed, siblings might actually avoid each other, as was observed by O'Leary et al. The immense social skill training literature is designed specifically to teach a repertoire of nonaggressive strategies to solve myriad interpersonal sibling and peer conflicts (see McMahon et al., 2006, pp. 178–186), and, although well beyond the scope of this chapter, needs attention from parent trainers working with sibling aggression. The social skill assessment and training issues are complex and unresolved. For example, does the child know how to "share," or "take turns," or "ask nicely," or "take 'no' for an answer," or "be verbally assertive," given a specific and complex social condition (Roberts, Arnold, & Mangum, 1992)? What is the best way to teach those skills (Thomas & Roberts, 2006), let alone motivate performance of those skills in relatively unmonitored home settings?

Immediate time-out is effective at suppressing sibling aggression in 2- to 7-year-olds (Allison & Allison, 1971; Jones et al., 1992; O'Leary et al., 1967; Olson & Roberts, 1987). Referred to as a "standing rule" by HNC theorists (McMahon & Forehand, 2003, pp. 150–155), both children are immediately sent to

time-out for fighting, effectively blocking all reinforcers that might be coerced by the aggressive act and simultaneously calming down both children via the quiet release contingency inherent in time-out routines. Following time-out, parents are taught to discuss the event-sequence leading up to the fight to enhance awareness of the conflict situation and alternative solutions. Does the parent trainer also need to program reinforcement for aggression inhibition or prosocial alternatives? Leitenberg, Burchard, Burchard, Fuller, and Lysaght (1977) have done so successfully with recruited families. Moreover, there is a huge literature on peer aggression that indicates success with various combinations of discipline, reinforcement, and skill building (e.g., Fehrenbach & Thelen, 1982). Therefore, a comprehensive and sequential approach to sibling aggression is recommended and is displayed in Figure 22.3: reinforcement for inhibition, immediate time-out, and skill building, in that order. Each component can be added as needed, based on parent records of sibling aggression at home. Developmental limitations, however, preclude most 2- and 3-year-olds from the skill-building and reinforcement components, given cognitive and attentional limitations. Aggressive siblings under age 4 might profit from immediate material reinforcement for cooperative play if a therapist could be inserted into the home (O'Leary et al., 1967), but such a routine seems impractical for parents and would lack social validity. Reducing sibling aggression without an immediate time-out seems especially unlikely in children under age 4. It is true that one can find theoretical arguments and some empirical support from developmental psychology (e.g., Brody & Stoneman, 1987) and family systems therapy (e.g., Dinkmeyer & McKay, 1973) to ignore sibling aggression. Others vehemently disagree (Bennett, 1990; Ihinger, 1975), noting that a quiet tyranny of the mighty could emerge, consistent with coercion theory (Figure 22.1). It is difficult to imagine how a delayed reinforcer for inhibition and/or skill building role-played in a clinic could override the immediately coerced reinforcers for sibling aggression in children under 4. In contrast, middle childhood youth (7 to 12 years old) might be effectively discouraged from sibling aggression by reinforced inhibition and/or social skill building alone (see the peer aggression literature reviewed by

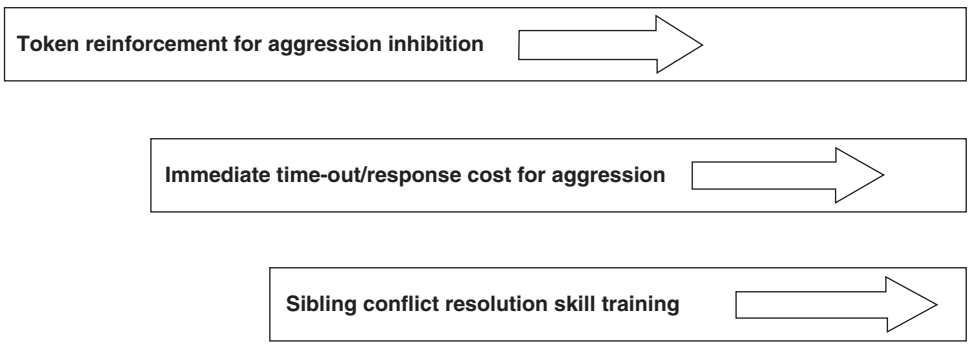

Figure 22.3 Recommended sibling aggression protocol for 2- to 12-year-olds based on a sequential strategy. Immediate time-out or response cost is needed if reinforcing inhibition is insufficient; skill training is needed if the combined reinforcement and time-out/response cost programs are insufficient.

McMahon et al., 2006, pp. 178–186). If sibling aggression still persists, then immediate time-outs, token fines, privilege losses, or work chores are indicated.

Other Programs and Situational Variables

Unlike noncompliance and aggression, which are motoric actions, tantrums and negative verbalizations are both vocal, suggesting extinction processes might be successful. If the function of coercive vocalizations is to provoke adult attention (e.g., nattering), differential attention procedures can effectively extinguish verbal coercion (e.g., Herbert & Baer, 1972). If the function is to terminate time-out conditions, a quiet release rule is effective (Hobbs & Forehand, 1975). If the coercive verbalizations covary with instruction presentations, simply persisting with the instruction routine can extinguish the negative verbal output despite socially reinforcing compliance in the presence of negative talk (Roberts & Hatzenbuehler, 1981). Therefore, teaching parents to persist with the stimulus conditions (Figure 22.1, "–1 Step") that provoke the negative vocal outburst is just as important as ignoring negative vocalizations (Figure 22.1, "+1 Step") in other contexts (e.g., sibling tease, disagree, bossy talk). Extinction, however, can be an aversive routine for many adults, given an extinction burst, and is sometimes impossible to implement (e.g., the child may climb on parent and scream). Teaching the parent to assume a physical posture and activity (e.g., turn one's back, leave the area, stand up) that makes ignoring viable is one option. Some unignorable emotionally aroused preschoolers, however, may need a time-out to successfully structure the extinction procedure for the parent. Only the quiet contingency of the time-out needs to be met to terminate time-out conditions for unignorable tantrums. In middle childhood, specific negative verbalizations (e.g., cursing) might be targeted within a larger token system by reinforcing inhibition and fining specified negative talk with tokens, backed up by activity or material reinforcers. Contingent chores (Fischer & Nehs, 1978) have also been used to suppress negative verbalizations in middle childhood.

Three social contexts require accommodations and are briefly discussed: bedtime, community, and car. Defiance, aggression, and tantrums at bedtime can befuddle the parent trainer if standard daytime treatments are applied. Time-out on a chair in the living room may function like "time-in"; differential attention may lead to reinforcing the child for refusing to go to bed, since staying up is ignored and attention for remaining in bed cannot compete with unrestricted access to the common rooms; token reinforcers for successful bedtime behavior are often delayed until morning. Traditional routines targeting bedtime crying, such as ignoring and graduated extinction (i.e., a gradual and systematic increase in parent response latency to crying), are effective (Kuhn & Elliot, 2003) but aversive to parents and presume that the child will remain in bed. Repeated "walk-backs" to bed or ultimately a shut door may prove necessary. Edwards and Christophersen (1994) recommend a sequential strategy, in which daytime self-quieting is targeted prior to attempting a bedtime extinction procedure (which requires children to calm themselves). Daytime ignoring and/or time-outs with quiet contingencies for tantrum and negative verbalizations provide practice opportunities for successful self-quieting. Recently,

Burke, Kuhn, and Peterson (2004) tested a clever positive reinforcement system (a prize under the pillow, left by the Sleep Fairy) that warrants replication with an oppositional sample. See Christophersen and Mortweet (2001, pp. 99–122) for a recent review of the psychopathology and empirically supported interventions for managing sleep problems.

Car and community situations often fail to provide the rich array of play options found at home (e.g., a long car trip, watching mother shop, sitting through big sister's piano recital), increasing the chance of coercion to provoke reinforcers. Further, community settings may be discriminative for misbehavior, since some parents actively mollify the child in public to avoid the embarrassment of child tantrums. For those two reasons, planned activity training combined with modified time-out routines are recommended. Sanders and Dadds (1982) helped parents identify and promote age-appropriate activities in specific community settings (e.g., quiet toys and games for the car; reinforced shopping participation; outdoor toys for a teenage sister's soccer match) that significantly enhanced generalization of parent training effects to community settings. Time-out locations can be defined in any specific, low-traffic area, backed up by the two private locations found in the community: bathrooms and the parent's car (Christophersen & Rapoff, 1979, p. 371; McMahon & Forehand, 2003, pp. 155–157). Sibling aggression in the car or seat belt refusal may necessitate stopping the car in a safe location and declaring time-out until all children are sitting quietly.

Some children may not adjust to community or car settings despite good planned positive activities and appropriately modified time-out routines. Consequently, parent trainers should consider inserting more powerful token systems tailored to the community setting (e.g., Barnard, Christophersen, & Wolf, 1977) to reinforce rule following and participation in planned positive activities. In addition, parent trainers can shape community skills by initially assigning brief, repetitive exposures under rich reinforcement conditions combined with simulation practice in the clinic. Data collected by the parent in the target setting can be used to adjust the treatment protocol.

Case Illustration

Ms. Smith and her three children, ages 7, 5, and 2, were referred to the clinic by their pediatrician for evaluation of conduct problems and possible therapy. During the standard interview, Ms. Smith reported that the 5-year-old, Zachary, was her main concern. Zach had been asked to leave a day care facility that summer for repeatedly fighting with peers and was now "in trouble" in kindergarten. Ms. Smith reported that Zach had always been "difficult" to raise. He argued, disobeyed, tantrummed, and was even aggressive with her when he didn't get his way, his requests were denied, or when she attempted to spank him or send him to his room. Zach was very bossy with his 2-year-old brother, which led to angry outbursts and aggression. In contrast, the 7-year-old sister had been "very easy" to raise and seemed to disregard the two boys. Ms. Smith voiced worry that her 2-year-old was learning "bad habits"

from Zach. Questionnaire data from the Child Behavior Checklist (Achenbach & Rescorla, 2000) and the Eyberg Child Behavior Inventory (Eyberg & Pincus, 1999) both yielded elevated scores on Social Aggression scales for Zach, but not for either sibling. The Slosson Intelligence Test Revised 3 (Slosson, 2002) was administered to screen for delays in Zach's cognitive/verbal development. He was normal (SS = 92). Maternal depression was also screened and fortunately absent (Beck Depression Inventory II = 9, "minimal"; Beck, Steer, & Brown, 1996). Since Ms. Smith was a single parent, a marital adjustment scale was not administered, but an instrument like the Marital Satisfaction Inventory R (Snyder, 1997) is recommended to screen for marital conflict. Because neither depression nor marital conflict was present, referrals to address these issues were not needed. A phone call to Zach's kindergarten teacher revealed disruptive behavior and occasional aggression, but no teacher interest in assistance from an outside professional at that time. The school had its own system for managing misconduct (an escalating response-cost system). His teacher did request (and was eventually mailed) a copy of his psychological evaluation, findings, and recommendations.

Home record data collected by the mother during the 2-week evaluation period yielded 3.6 fights per day (usually with the 2-year-old brother) and over 5 tantrums per day. In addition, Zach ran away from the house twice when confronted by his mother over conflict with his brother, provoking necessary but distressing "chase" episodes. Clinic observations under child's game conditions showed a responsive mother (64% positive or neutral reactivity, given a signal from Zach or the 2-year-old), but a very high rate of questions plus instructions (8.3 per minute) and a virtual absence of positive maternal signals (praise + descriptions + imitations = 0.2 per minute). The 7-year-old was not available for the observational measures since she was in school when the family attended clinic sessions. Under cleanup task analogue conditions, Zach's compliance ratio was 22%. Ms. Smith, however, repeated her instructions so quickly that her beta instruction ratio was almost 75% (i.e., repeated + vague instructions divided by the total number of instructions issued); her praise ratio was 0%. A compliance test was administered to ensure that Zach's defiance was not simply an artifact of maternal instruction style. It was not; his compliance ratio, given well-presented instructions to pick up toys, was 3%. He also became quite angry during the compliance test, yielding cry and back-talk ratios over 50%.

Given the interview, questionnaire data, home record data, clinic observations, and school information, Zach met criteria for Oppositional Defiant Disorder. Using Figure 22.1, it could be seen that several paths were relevant to help conceptualize the case. Coercion was very likely, given maternal or sibling instructions to "Do X" or "Stop Y," as well as denials of Zach's requests. Fortunately, there was no evidence that minor aversive events (e.g., disagree with siblings), ignoring, or positive or neutral events set the occasion for Zach's misbehavior. Zach played alone quite well. Discipline efforts, however, were ineffectual and provoked escalating intensities of coercion (e.g., he had physically hurt his mother on more than one occasion when she attempted to guide him to his room). Common parenting errors (i.e., Figure 22.1, "+1 Step"), although not observed, presumably reinforced Zach's coercion on a partial reinforcement schedule. Social learning–based parent training was recommended.

(*Continued*)

In addition, Zach's school progress was to be monitored to ensure that programs implemented in that setting were successful.

Child-directed interaction skills were targeted first. Within three sessions Ms. Smith became rather skillful at giving attention to age-appropriate play when interacting with Zach and his 2-year-old brother in the clinic playroom. Her responsiveness index remained high (over 70%), while the quality of her responses became more positive (praise + descriptions + imitations exceeded 4 per minute). It was difficult for her to learn to inhibit question asking, since she seemed to want to "teach" skills via this route, but she eventually adopted a more child-centered strategy, limiting questions plus instructions to under one per minute. She also had difficulty learning to ignore the boys when they became bossy, angry, or disruptive with each other, but finally learned to turn her back and play with the other sibling or simply model quiet toy play until one or both siblings began to play more appropriately. Ms. Smith was asked to set aside some morning time (e.g., 15 to 20 minutes) to use child-centered play with her 2-year-old and a similar time after school with all three children. In addition, she was instructed to look for opportunities throughout the day to socially reinforce child initiations with her new listening skills. Handouts on responsive play skills and the principles of shaping and positive reinforcement were provided for review and future discussion.

Compliance training, which began during the third treatment session, was more difficult. During modeling and role-playing of instruction, social reinforcement, warning, and time-out skills, Zach displayed interest in his mother's reactions to the doll, but when queried about why the doll went to time-out, Zach refused to talk to the therapist. During the initial guided practice with Zach, his mother, wearing a bug-in-the-ear, met with immediate defiance and time-out resistance. Zach refused to go to the time-out chair and physically fought his mother when she guided him to the backup room. He banged on the door and yelled angrily for the entire 60-second period; when escorted back to time-out for his second opportunity at chair time-out conditions, he repeated the entire sequence of escape, a furious 1-minute room backup, and subsequent replacement on the time-out chair. During his third experience with chair time-out conditions, he inhibited escape for about 5 seconds before arising and throwing the chair against the wall. He was immediately escorted to the backup room for a third and final time; there to remain until a 5-second quiet rule was met. Ten minutes later he finally extinguished his crying and yelling sufficiently for the therapist to instruct the mother to open the door and say, "Since you're quiet, you can come out now." Zach immediately reinitiated screaming and refused to leave the time-out room. Pouting after initial time-outs is not unusual, but Zach was extreme. The therapist had to enter the playroom, prop the time-out backup room door open (because Zach kept slamming it), and prompt the mother to play positively with the 2-year-old out of Zach's visual field. This process also allowed the therapist to provide social support and compassion for the mother beyond what could be done over the radio device. Zach was ignored. Four minutes later he emerged from the time-out room. Rather than re-present the disobeyed instruction immediately (as in Figure 22.2), the therapist and mother continued in a responsive play mode with the younger sibling and Zach until Zach began to play and talk normally. The therapist then left the room and resumed compliance training via the bug. Zach immediately disobeyed the same cleanup instruction and was sent to

time-out again. The entire coercive cycle repeated itself, although greatly toned down. Zach remained on the time-out chair after just one 60-second trip to the backup room and became quiet within 3 minutes. He again refused to leave time-out when told he could, but when the mother began playing with his little brother, Zach left the chair and began playing (the therapist did not have to enter the playroom this time). Once Zach began playing, the therapist instructed the mother to say (via the bug), "Okay Zach, we still need to put the stuffed animals back in the toy box." Zach started yelling, but when given the warning, he picked up one toy despite continued verbal protests. His mother praised his efforts and continued to use the basic Figure 22.2 system under guidance via the bug. Zach met the criterion of 10 compliant responses in a row (despite the necessity of three warnings, which he obeyed). Ms. Smith was instructed to give him a big hug and inform Zach that he could play the way he wanted to now. Responsive play was resumed until both boys were playing and talking appropriately. The 2-year-old was then issued similar, but very simple motor tasks (e.g., "Come here"; "Take this (toy)"; or "Put it (location)," always accompanied by close proximity, eye contact, and a gesture. The 2-year-old met compliance criterion with only one time-out and virtually no time-out resistance. A record card was sent home to monitor chair time-outs for disobeyed warnings and needed room backups (using the child's bedroom). Because Zach had been extremely resistant to time-out, phone contacts were scheduled during the next few days and until the mother was confident she could administer the Figure 22.2 program. Handouts on the compliance program and the principles of extinction and punishment were sent home and reviewed at a later session.

During the fifth clinic visit, the home record data indicated that Zach had required from one to four chair time-outs per day for disobeyed warnings and room backup procedures about half the time. His mother did report that Zach often obeyed warnings, which pleased her, and the younger brother obeyed almost all the time if she "made it simple" as instructed. The mother had clearly gained some confidence in the compliance routine, but felt helpless about the boys' fighting. A no-fighting rule (Figure 22.3) was instituted in addition to reviewing responsive play and compliance protocols. Zach sat beside his mother while the therapist sent two dolls to time-out for fighting over a toy. An effort was made to discuss the conditions, mistake, and optional responses, but Zach would have none of it. Ms. Smith, however, role-played the post-time-out effort at incidental training regarding the conflict conditions (e.g., brother 1 had taken brother 2's toy), the mistake (i.e., fighting), and the optional responses (e.g., using an assertive instruction or seeking maternal help). During the responsive play sample, Zach pushed his brother intentionally; the mother was prompted to implement the no-fight rule, marching Zach immediately to the time-out chair while labeling the contingency in an authoritative manner (i.e., "Since you hit your brother, you must go to time-out"). Zach actually stayed on the time-out chair, but refused to talk to the mother during the incidental learning procedure after time-out. There was no more fighting during that clinic session. Time-out for fighting was added to that week's home record card and monitored thereafter. A token system was also initiated. Zach was shown spaces on the home card where stickers would be entered for remembering his no-fight rule; he could earn up to two on school

(*Continued*)

days and three on the weekends. Both Zach and his brother were shown the clinic store where stickers could buy prizes the next week. His mother was instructed to insert stickers for specified periods of aggression inhibition and lavish praise for rule following.

During the sixth through eighth clinic sessions home cards were reviewed, stickers were rewarded with clinic prizes, responsive play and clinic task conditions were observed, and rule protocols were reviewed. Zach's little brother always received a prize, despite not participating in the token system. A set of planned positive activities and adaptations for time-out, if needed, were modeled and role-played for shopping and car situations. Both boys watched with great interest as Ms. Smith role-played with the dolls. Shopping and car opportunities were monitored on the home record card and discussed with Ms. Smith during follow-up sessions. During the eighth session an allowance system was established for Zach and his older sister to substitute for the token system. Both older siblings earned a set amount of money each week (10 cents per day for Zach) for completing at least one household chore each day. In addition, Zach was fined 5 cents for every time-out, which had dropped to about four per week at that point. The token reinforcement system for aggression inhibition was then discontinued. Because fighting rates had declined dramatically, sibling conflict resolution skill training (Figure 22.3) was not considered necessary. Zach was, however, still having trouble in school. A school psychologist had been called in by the principal to address continued classroom disruption and occasional fighting. Following phone consultation and one lengthy meeting, a program was jointly evolved that included a token reinforcement system for rule following and classroom productivity, token fines and privilege losses for fighting, and a weekly social skill session with two other boys and two "peer helpers" led by the school psychologist.

After eight treatment sessions, two follow-up sessions, and 3 months, Zach's home data finally looked normal for a 5-year-old (i.e., 1 to 2 time-outs per week, no room backups), but school problems persisted. Long-term follow-up was planned to provide continued support and educational services to the mother, as well as consultation services for the school district as needed.

FUTURE CHALLENGES

Current parent training treatment modalities have limitations. In particular, effective discipline procedures for middle childhood have not been identified. Theoretically (Figure 22.1), the coerced reinforcers of violence must be blocked to make progress with oppositional and conduct disordered grade-schoolers. Unfortunately, parent trainers lack an effective procedure to enforce time-outs, token fines, privilege losses, or work chores with those 8- to 12-year-olds who fiercely resist discipline efforts. Their 2- to 7-year-old counterparts can be safely escorted to time-out as needed. In contrast, older children can intensify the level of violence toward the parent, thwart the implementation of the discipline, coerce denied reinforcers, and/or run out of the house! Increasing the level of care to psychiatric facilities, therapeutic foster care, community residential programs, or juvenile detention centers is the current solution, but it is not satisfactory.

Similarly, some middle childhood referrals continue to display unacceptably high rates of aggression and defiance, despite the establishment of a seemingly good

set of treatment procedures. Frick (2004) has identified a subset of externalizing disordered children he calls "callous-unemotional" who may not respond to family-based interventions. Specifically, family dysfunction may be less important than temperamental and physiological differences for this minority of violent clinic referrals, suggesting a need for supplemental treatment modalities. Unfortunately, there are no specific and effective interventions for instilling guilt for transgressions, acceptance of responsibility for mistakes, empathy for victims, or identification with mainstream social activities. These important self-regulatory processes emerge gradually in normal children during socialization. Parent trainers often observe phenomena that suggest improvement in these important covert processes by many of our child clients, but not all. The discussion of "child effects" is not new in our field (Lytton, 1990), nor resolved (Dodge, 1990; Wahler, 1990). There is, however, no doubt that some young males seem untreatable. Relentless and remorseless appear apt descriptors to this author. Psychiatric medication and higher levels of care appear to be their current destination.

SUMMARY

Parent training is based on the recognition that externalizing disorders of childhood are rooted in coercive patterns of social interaction. Intentional, aversive child behavior is very likely to coerce parents, teachers, siblings, and peers into unwittingly reinforcing common referral symptoms, such as disobedience, fighting, negative talk, and tantrums. Parent training evolved as a new treatment modality to address the reinforcement traps perpetuating the child's social aggression. The pioneering work of Patterson, Wahler, and Hanf led the way toward the sophisticated theoretical models, assessment strategies, and treatment protocols of today. Social learning–based parent training is now widely recognized as an empirically supported treatment. Parents can learn child management skills, use those skills at home, and reduce socially aggressive child behavior. Improvements observed in clinic settings generalize to home settings and persist over time. Basic child management skills include knowledge and use of social learning principles, observation, differential attention, responsiveness, instruction giving, discipline, and token systems. Basic methods of teaching skills to parents include didactic strategies, modeling, role-playing, guided practice, and homework. Both parent group and individual family approaches have been effective.

Problems not initially considered became evident as clinical scientists pursued the dropouts and treatment failures from parent training programs: poverty, social isolation, parent pathology, and therapeutic disengagement head the list. Moreover, there are some child presentations we still cannot manage on an outpatient family therapy basis (e.g., violence in middle childhood), let alone successfully treat. Parent training is a work in progress, as witnessed by the ongoing waves of clinical research discussed in the chapter. Nevertheless, routine defiance, aggression, negative talk, and tantrums, especially for oppositional and disruptive children under 8 years of age, can be successfully treated at home and in other relevant situations via parent training. Under those conditions, parent training is the empirically supported treatment of choice.

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CHAPTER 23

Cognitive-Behavioral Treatment

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If you have ever observed or listened closely to children in clinical situations, at school, or in a playground, you have most likely heard at least a few of the following:

- "She said no when I asked to sit at her lunch table. I'll never have any friends."
- "I can't do this math problem—I'm a total idiot!"
- "If I don't get this homework done perfectly, it just shows how stupid I am."
- "I'll never hit the ball, not even if I try a million times."

For a century or more, we have known that children aren't little adults (Kingery et al., 2006). However, like adults, they too experience cognitions that reflect irrationality or overgeneralization or catastrophic thinking. They may predict that small upsets in peer relationships will continue "forever"; they may interpret spelling errors as total failure; and they may describe normative fluctuations in mood as making life utterly impossible and overwhelming. Such patterns of thinking, in concert with individual personality characteristics and contextual factors, interact to determine both adaptive and maladaptive psychological functioning. Although prevalence rates vary with methodologies used, recent epidemiological estimates have found that roughly 15% to 22% of youngsters will suffer from psychological issues and their related functional impairment during childhood and adolescence (Costello, 1989; McCracken, 1992).

Cognitive-behavioral models posit that, like adults, children and adolescents have faulty beliefs and automatic thoughts that negatively impact their behaviors and contribute to the development and maintenance of psychopathology. Cognitions refer to beliefs, schemas, attributions, perceptions, or mental representations of events. Cognitive-behavioral theory is phenomenological in that it assumes that behavioral responses are a function of an individual's perception of events, rather than the events themselves. In other words, we imbue our worlds with meaning and respond to them in terms of these meanings. Attributions we make about the world may be more or less accurate and, similarly, more or less helpful in directing our behavior.

Recent approaches argue that cognitions may function as "rules." They suggest that because the behavioral responses associated with such thoughts are thus

rule-governed, they may be especially intractable (Hayes, Strosahl, & Wilson, 1999). Very often, we don't have control over environmental contextual factors that maintain maladaptive behaviors. However, we do have access to thoughts and emotions: We can monitor internal psychological events, notice how they appear to be connected to behavioral responses, evaluate them, and, if we choose, engage in attempts to change them. Changes in thoughts are linked with changes in behavior, and thus, cognitive-behavioral theorists ascribe them causal status. As such, cognitive-behavioral approaches contend, our thoughts may be ideal targets for treatment.

Cognitive-behavioral therapy (CBT) has enjoyed a wealth of empirical attention over the past few decades. In the past 10 years or so, these approaches have been adapted for use with children and adolescents. Although originally developed for use with adults, child-focused CBTs are now numerous and have been investigated for youth of varying ages and across a variety of disorders. Whereas early efforts in treatment development involved downward extensions of adult models of CBT (e.g., based on the work of Ellis and Beck), newer approaches reflect more sophisticated thinking about the developmental nature of child psychopathology, representing more bottom-up innovations (e.g., Kendall & Hedtke, 2006; Kendall, Kane, Howard, & Sigueland, 1990). Interventions have been designed in more contextually and developmentally sensitive ways, thus expanding the utility of CBTs in broader domains and with more diverse populations. Recent advances include applications with very young children (Freeman, Garcia, Coyne, Leonard, & Compton, 2007), family-based CBT models (Barrett, Dadds, & Rapee, 1996; Freeman, Garcia, Fucci, Karitani, Miller, & Leonard, 2003), and those incorporating acceptance- and mindfulness-based approaches (Semple, Reid, & Miller, 2005), to mention just a few. Moreover, CBT has been adapted for use across multiple settings, including primary care, behavioral health, and with severely emotionally disturbed youth; for multiple purposes, including prevention and intervention efforts; and with diverse racial-ethnic populations. This chapter provides an overview of contemporary cognitive-behavioral treatments with children, discusses current research in CBT efficacy, with particular focus on childhood depression and anxiety, and illustrates an innovative use of family-based CBT for young children through a case study depicting intervention for a young child with Obsessive-Compulsive Disorder (OCD).

OVERVIEW

Cognitive-behavioral therapy is based on principles of operant and respondent conditioning. Consistent with a social learning perspective, CBT models assume that an individual's behaviors are a function of antecedent and consequent control. Contingencies may be directly experienced or observed. However, perceived environmental contingencies, in concert with individual characteristics such as biological predispositions, temperament, or personality, as well as particular contexts, interact to influence behavioral responses and dynamically developing repertoires. Cognitive-behavioral therapy holds that although distorted or irrational thoughts alone are not sufficient to precipitate psychological difficulties, they are a major

contributor to maladaptive behavioral repertoires. As such, unrealistic or unhelpful thoughts and emotions are primary treatment targets in cognitive-behavioral approaches. Clinicians strive to help clients modify or reframe irrational thinking to effect desired changes in behavior. Thus, CBT dually targets client thoughts and behaviors to ameliorate symptoms and improve functioning. This is accomplished through a collaborative relationship between clinician and client that is at times didactic and at times experiential. Often, clinicians use Socratic questioning to call client attention to maladaptive thinking patterns and to challenge the logic of such cognitions. Treatment tends to be structured, goal-oriented, and time-limited and may be delivered in an individual or group format.

Although there is variation across different treatment manuals and clinical issues, CBT consists of three fundamental components: psychoeducation, cognitive restructuring, and behavioral experiments. *Psychoeducation* is perhaps the most didactic element of CBT and serves as an important first step in treatment. Children are educated about the role that cognitions play in their emotions and behavior. Clinicians help children recognize and monitor their response patterns and appraisals of problematic situations. By educating children about how their response patterns work, they may exhibit greater facility in applying their knowledge and new skills after treatment concludes and across different situations. Though psychoeducation is the first step in treatment, the education process continues for the duration of a given intervention.

Cognitive restructuring is the central cognitive feature of CBT and was originally defined by Albert Ellis (1962) and Aaron Beck (1976). Cognitive restructuring requires that children strive to do two things: first, identify their erroneous or distorted thoughts or beliefs, and second, dispute them. This "rational disputation" is the hallmark of this therapeutic approach. Contemporary conceptualizations of cognitive restructuring include a variety of techniques meant to help children and adolescents reframe or challenge their own beliefs, automatic thoughts, and schemas. Techniques include, but are not limited to, perspective taking, problem solving, replacing negative self-evaluations with positive self-statements, revising distorted views of social interactions and relationships, as well as more experiential exercises such as guided imagery and relaxation. Although these strategies may differ somewhat among CBTs, their overarching goal is to assist individuals in modifying the content of their thoughts, reducing their believability, and thus supporting the eventual development of more functionally useful behavioral responses.

Another core component of CBT is a set of skill-building techniques that strive to help children develop behavioral competencies for coping with difficult or feared situations. In this component of treatment, children are asked to try out a variety of responses, including testing their faulty appraisals, automatic thoughts, and beliefs through behavioral experiments. Two examples of this set of techniques are exposure with response prevention (EX/RP) in the treatment of OCD and behavioral activation in the treatment of depression. Exposure with response prevention consists of exposing children to feared stimuli, then preventing their ritualized response. In so doing, children experience their discomfort rise and fall and learn that the situations they have avoided are not catastrophic after all. Behavioral experiments can either be *in vivo*, whereby children practice engaging in the

behavior, or *imaginal,* whereby they visualize the behavior and its consequences. In practicing this technique, children learn that they need not engage in rituals to reduce their anxiety and that they may use more adaptive coping skills to address their fears. This technique is most commonly found in treatment for OCD, although exposure is used more broadly across numerous child and adolescent anxiety disorders.

Behavioral activation involves pleasant event scheduling, which assists depressed children and adolescents in more frequently engaging in behaviors that will allow them to experience rewarding consequences. Engaging in behaviors that are either inherently reinforcing or that bring children in contact with naturally occurring rewards helps children to refute their beliefs of hopelessness and the intractability of their negative moods. It may also create behavioral momentum by increasing the density of reinforcers experienced.

Although each of these components is present in some degree in cognitive-behavioral treatments, some approaches rely more heavily on more traditional behavioral techniques (e.g., treatment for child and adolescent OCD), and others use more explicitly cognitive tools (e.g., treatment for depression or Generalized Anxiety Disorder) that emphasize the modification of thoughts, beliefs, and appraisal processes. The degree to which each of these strategies is used is in part a function of the childhood disorder targeted and in part a function of the age of the child or adolescent in treatment. Cognitive-behavioral therapy also incorporates a number of other techniques, such as problem solving and skills training, in the treatment of adolescent depression.

DEVELOPMENTAL CONSIDERATIONS

When using cognitive-behavioral therapy with children and adolescents, it is critical to evaluate the developmental stage of the client. Careful evaluation of the cognitive and emotional functioning of a given child will help guide the selection of therapeutic techniques and strategies: Children 8 and younger may have great difficulty with tactics such as thought monitoring and rational disputation, whereas older children might benefit greatly from these exercises.

Similarly, parents may be helpful as architects of their young children's attempts at behavior change, given that young children are far more dependent on parents for structure and guidance. The purpose of enlisting parents in treatment is twofold: First, they can ensure that behavioral changes and homework are occurring at home; second, the clinician can intervene in any parent behaviors that may inadvertently exacerbate the child's problems. In contrast, adolescents may benefit more if given greater autonomy to effect behavioral changes at their own pace, rather than as dictated by parents.

It is also crucial to consider the contextual factors that may impact a child's or adolescent's behavioral presentation. These include home, school, and peer environments. Depending on a child's age and interests, the relative importance of these factors may vary over time. For adolescents, peer relationships may stand out as a critical treatment issue. Young children, on the other hand, may be more influenced by their parents. Additionally, youngsters' functioning may differ across each of

these environments; thus, using multiple informants (e.g., child, teacher, parent) to describe behavior in each of these domains is recommended.

Finally, a developmental perspective is important in helping to determine whether behaviors are timely and normative or reflective of psychopathology. One example of this is the topography of normal fears across childhood and adolescence. Fear of monsters under the bed is quite normative (if distressing) in preschool-age children; however, such a fear in a teenager might represent a hallucinatory event. Conversely, teens are often concerned with violence and unfairness in the world at large, but a 6-year-old overly concerned with terrorism might merit assessment for anxiety or trauma.

EMPIRICAL SUPPORT FOR COGNITIVE-BEHAVIORAL THERAPY WITH YOUTH

Cognitive-behavioral treatments have been used with a variety of child and adolescent disorders, including anxiety disorders (Albano & Kendall, 2002; Berman, Weems, Silverman, & Kurtines, 2000; Flannery-Schroeder, Choudhury, & Kendall, 2005), OCD (Franklin et al., 1998; J. M. March, Mulle, & Herbel, 1994; Piacentini, 1999; Pediatric OCD Treatment Study [POTS], 2004), depression (Compton, Burns, Egger, & Robertson, 2002; Crisp, Gudmundsen, & Shirk, 2006), eating disorders (Lock, 2005; G. T. Wilson & Sysko, 2006), Attention-Deficit/Hyperactivity Disorder (ADHD; Hinshaw, 2005), conduct problems (Southam-Gerow, 2003), substance abuse (Kaminer & Slesnick, 2006; Kaminer & Waldron, 2006), and trauma (Cohen, 2005; Cohen, Mannarino, & Knudsen, 2005). Gains following CBT appear robust over time (Albano & Kendall, 2002; Dadds et al., 1999; Flannery-Schroeder et al., 2005). A thorough review of the treatment outcome literature is beyond the scope of this chapter; however, in the following we describe the relative efficacy of CBT across childhood and adolescent depression and anxiety and briefly describe applications with each, as well as recent innovations in their use.

Cognitive-Behavioral Therapy for Child Depression

Data suggest that prevalence rates of Major Depressive Disorder (MDD) among children and adolescents are increasing (Cross-National Collaborative Group, 1993; Kessler et al., 1994; Kessler & Walters, 1998). Moreover, suicide is the third leading cause of death in children ages 10 to 19 (Centers for Disease Control and Prevention, 2004). Yet, in an overwhelming majority of youngsters, MDD goes undiagnosed for a year or more, and few receive treatment even by the end of adolescence (Jensen, 2006; Kessler, Avenevoli, & Merikangas, 2001; Leaf et al., 1996). Thus, the need for accurate detection, diagnosis, and effective treatment of depression during childhood is critical. Applications of cognitive-behavioral treatments constitute a strong attempt to meet that need.

Cognitive models of MDD highlight the role of maladaptive beliefs in development and maintenance of depression, although there is some variability across conceptualizations. Recent theoretical work has stressed the importance of evaluating cognitive vulnerability in context and has relied on diathesis-stress

conceptualizations. As originally described by Beck, Rush, Shaw, and Emory (1979), depressed individuals are thought to have systematic biases in thinking that lead to idiosyncratic interpretations of events. Additionally, individuals who are vulnerable to depression appear to have schemas that guide information processing and may potentiate negative self-attributions that are at the heart of depression. Finally, cognitive models characterize vulnerability to depression in terms of the negative cognitive triad: negative perceptions of the self, the world, and one's future. All these increase an individual's risk for depression. Depression has also been explained in terms of Seligman's (1975) model of learned helplessness and as a function of negative attributional style (Abramson, Seligman, & Teasdale, 1978). This latter term refers to the tendency to attribute poor outcomes to internal, global, and stable factors and to ascribe positive outcomes to external, specific, fleeting, or transient factors. Some data suggest that links between attributional style and depression get stronger as children age (Nolen-Hoeksema, Girgus, & Seligman, 1992) and with the increasing sophistication of abstract thinking (Turner & Cole, 1994).

Research on CBT for adolescent depression has found it to be a promising treatment, although not without shortcomings (Birmaher et al., 1996; Brent, Birmaher, Holder, Johnson, & Kolko, 1995; Kolko, Brent, Baugher, Bridge, & Birmaher, 2000). Very recently, the most comprehensive study ever conducted on the efficacy of CBT for adolescent depression was completed (Jensen, 2006). The Treatment of Adolescent Depression Study (TADS) was an incredibly sophisticated, 6-year-long, multisite, randomized controlled trial comparing cognitive-behavioral therapy alone, fluoxetine + CBT, fluoxetine alone, and a placebo control in a sample of 439 adolescents with moderate to severe levels of depression (March et al., 2004). Results indicated that the combined CBT + fluoxetine treatment led to statistically significant reductions in depression compared to placebo. Data also revealed that neither fluoxetine alone nor CBT alone fared better than placebo, although fluoxetine alone had a slight advantage over CBT (March et al., 2004). However, other studies have shown that CBT and CBT + antidepressant medication were similarly effective in symptom reduction (Melvin et al., 2006). Severity of depression, comorbid anxiety or dysthymia, and family factors, such as parent psychopathology, parent conflict, low socioeconomic status, and exposure to stressors, predict attenuated treatment gains (Birmaher et al., 1996; March et al., 2004). In the TADS trial, age, expectations about treatment, severity, complexity, and chronicity also contributed to poorer outcomes (Brent, 2006; Curry et al., 2006). Further work is needed to better elucidate the efficacy of CBT and to develop more effective interventions. However, initial data are positive.

A meta-analysis of published studies on internalizing disorders indicated the following components as key features of CBT for depression in children: cognitive restructuring, problem-solving skills, decision-making skills, relaxation training, coping strategies, and skill generalization strategies (Compton et al., 2002). These combined strategies get at the main goals of CBT, which are to challenge and change thoughts, and then change behaviors. Recent data examining mediators of treatment response suggest that CBT effects treatment outcome though reducing cognitive distortions (Kolko et al., 2000). However, such data are scarce, and more research is needed (Chorpita, Brown, & Barlow, 1998; Kazdin, 2004).

The treatment of depression via CBT is a common practice for older children and adolescents, although randomized clinical trials are far more rare in younger children. However, one innovative cognitive approach to childhood depression is contextual emotion regulation therapy (CERT; Kovacs et al., 2006). The model on which CERT is based focuses on the emotion regulatory aspects of depression. In particular, youngsters who are unable to regulate their distress or dysphoria are at increased risk of developing symptoms of depression. Thus, treatment focuses on teaching children coping skills to reframe their experiences of compromised mood in terms of coping with accumulating stressors in their daily lives. A small, open trial in 15 children ages 7 to 12 years found that 53% of completers had full remission of dysthymia, and at 1-year follow-up this number grew to 92% (Kovacs et al., 2006).

Cognitive-Behavioral Therapy with Anxiety Disorders

Recent epidemiological data suggest that between 12% and 20% of children and adolescents experience clinically significant anxiety disorders (Achenbach, Howell, McConaughy, & Stanger, 1995; Velting, Setzer, & Albano, 2004). Although anxiety may change in presentation, it tends to be stable over time and if left untreated results in significant functional impairment in social, academic, and vocational domains (Gosch, Flannery-Schroeder, Mauro, & Compton, 2006; Velting et al., 2004). As was the case for depression in childhood, only a small proportion of children who suffer anxiety receive adequate assessment and treatment services. Thus, making effective treatments available is essential.

Anxiety has been conceptualized in terms of respondent, operant, and social learning principles. With regard to respondent conditioning models, the association of a neutral stimulus repeatedly paired with an unconditioned stimulus may lead to an unconditioned anxiety response. Unpleasant internal states, in addition to observable environmental stimuli, can also come to have elicitative properties (Gosch et al., 2006; Watson & Raynor, 1920). Mowrer (1960) argued that a respondent conditioning paradigm was not sufficient in and of itself to explain anxiety. He proposed a two-factor theory, which relies on both respondent and operant models. Operant models suggest that avoidance behaviors arise and are maintained by the reduction of fear and/or positive reinforcement by significant others, such as parents (Gosch et al., 2006; Skinner, 1969). Social learning models of anxiety hold that individuals may begin to doubt their self-efficacy in anxiety-provoking situations and thus learn avoidant repertoires vicariously, through observing others (Bandura, 1977). Cognitive models of anxiety, such as Rachman's (1991) neoconditioning theory, stress the importance of verbal constructions of events.

The use of cognitive-behavioral approaches for the treatment of childhood anxiety disorders has been well-studied. Empirical data suggest the efficacy of CBT across a broad range of anxiety and anxiety-related disorders, including simple phobias, Social Phobia, Generalized Anxiety Disorder, Agoraphobia, and Separation Anxiety Disorder (Albano & Kendall, 2002; Berman et al., 2000; Flannery-Schroeder et al., 2005). Findings are positive, and largely consistent, although some

data suggest that children and adolescents who struggle with anxiety and comorbid depression may be less likely to complete or benefit from treatment. For example, in one study of exposure-based treatment for anxiety in children ages 6 to 17, results indicated that children in the treatment failure group were significantly more likely to have comorbid depression (Berman et al., 2000).

With regard to maintenance of treatment gains in anxious children and adolescents, CBT has been found to have significant effects at long-term follow-up (Albano & Kendall, 2002; Flannery-Schroeder et al., 2005). For example, one investigation of how children fared several years out from termination of treatment indicated maintenance of gains from CBT for anxiety at an average of 7.4-year follow-up (Kendall, Safford, Flannery-Schroeder, & Webb, 2004). This may indicate that the skills and patterns learned during CBT for childhood anxiety remain with the child throughout subsequent years and even into different developmental stages. Cognitive-behavioral therapy has been found to be equally efficacious at 1-year follow-up, regardless of whether it was delivered in an individual or group format (Flannery-Schroeder et al., 2005). Delivery of CBT in a group format provides an efficacious yet cost-effective treatment.

Cognitive-behavioral therapy has also been found to be an effective tool for preventing the development of anxiety in at-risk children. A study by Dadds and colleagues (Dadds, Spence, Holland, Barrett, & Laurens, 1997) examined the effects of a school-based cognitive-behavioral intervention program for anxiety disorders in at-risk children ages 7 to 14. Treatment focused on the use of coping strategies in group-based exposure exercises and included child management skills sessions for parents. Results indicate that children in the intervention group had significantly lower rates of anxiety disorders at 6-month follow-up. Intervention also reduced the rates of anxiety in those children who had mild to moderate levels of anxiety at onset. At 2-year follow-up, participants in the intervention group reported significantly lower rates of anxiety when compared to the monitoring group (Dadds et al., 1997).

Cognitive-behavioral approaches to anxiety disorders in children and adolescents contain the same core components common to CBT for other issues. Generally speaking, these include psychoeducation, coping skills training, exposure tasks, and contingency management procedures (Gosch et al., 2006). As in CBT for depression, psychoeducation aims to educate the children and their parents about the cognitive, affective, and physiological components of anxiety. Child behavioral responses to feared situations are typically conceptualized in two ways. Attempts to avoid feared situations or stimuli are described to children as "avoidant coping" and are discouraged. More appropriate replacement behaviors, classed as "coping, or bravery-based responses," are encouraged (Kendall, 1990; Pincus, Eyberg, & Choate, 2005). One well-researched treatment manual targeting younger children uses the analogy of a "scaredy cat" versus a "coping cat" to assist children in understanding and engaging in treatment (Coping Cat; Kendall, 1990).

Cognitive restructuring for child anxiety focuses on teaching children to test their hypotheses about feared situations by engaging in approach rather than avoidance behaviors. For example, socially anxious children often come into treatment

believing they "cannot" tolerate their fear in social situations, or with certainty that they will do "something stupid," peers will laugh at them, and they will feel greatly embarrassed. Cognitive restructuring then revolves around approaching these thoughts as hypotheses and testing them via actually placing children in feared social situations and asking children to self-monitor their fear-based cognitions and check the degree of truth of these cognitions. Children may also be coached to use self-instructions, or positive self-statements, as an additional strategy to support the development and generalization of approach-based responses.

Exposure-based treatment components involve approaching feared situations in a hierarchical manner. Children practice "behavioral experiments," which consist of approaching feared stimuli so that they may watch their level of anxiety rise and then fall until habituation is reached. Children then progress through treatment until mastery is reached at each step; in other words, they have learned and effectively used their approach-based, rather than avoidance-based, behaviors.

Many manualized treatments for childhood anxiety have been developed in recent years and reflect innovative uses of CBT for a spectrum of anxiety disorders. For example, the Coping Cat Program (Kendall, 1990) is widely used and has versions for use with adolescents and school-age children. Other novel uses of cognitive principles are Pincus and colleagues (2005) adaptation of parent-child interaction therapy (PCIT), a well-established behavioral training program typically used with parents of young oppositional children, for use with young separation-anxious children. Two treatment phases in PCIT target the development of a warm and responsive relationship between children and caregivers and then focus on imparting appropriate behavior management techniques. To make it more relevant to separation-anxious children, Pincus and colleagues added a "bravery directed interaction" module in which parents are taught how to encourage approach behaviors and set limits around avoidance. This component specifically targets parent cognitions that may impede their encouragement of their children's approach attempts (Pincus et al., 2005). Other innovative uses include recent adaptations of CBT for groups of young separation-anxious children in the context of a summer camp Ehrenreich, Goldstein, Wright, & Barlow, under review) and for young children with OCD (Freeman et al., 2003).

Cognitive-Behavioral Therapy for Obsessive-Compulsive Disorder

Recent data suggest that CBT is a treatment of choice for childhood OCD (Pediatric OCD Treatment Study, 2004). When treating adults and older children, CBT with EX/RP is considered the first-line intervention (Franklin et al., 1998; J. M. March et al., 1994; Piacentini, 1999). Only recently has the efficacy of CBT, and in particular, EX/RP, been carefully reviewed and studied in children and adolescents (J. March, 1995; Piacentini, 1999). The publication of the Pediatric OCD Treatment Study, a multisite, randomized controlled trial, reported that CBT and CBT plus augmentative pharmacotherapy with serotonin reuptake inhibitors were efficacious in treating pediatric OCD. Although CBTs have been developed and effectively used for older children and adolescents, there has been growing interest in adapting this approach to younger children.

Until very recently, there were no cognitive-behavioral treatments available for children 8 and younger with OCD. However, Freeman et al. (2007) formulated a manualized CBT with EX/RP for early childhood OCD in children ages 4 to 8. Initial results from a small randomized clinical trial indicated that this treatment resulted in more positive gains for children in a family-based cognitive-behavioral treatment condition compared to those in a family-based relaxation treatment (Freeman et al., 2007).

Cognitive-Behavioral Therapy for Young Children with Obsessive-Compulsive Disorder

Onset of OCD has been documented in children as young as 3 years and may be more common in younger children than previously believed (Hollingsworth, Tanguay, Grossman, & Pabst, 1980; Swedo, Rapoport, Leonard, Lenane, & Cheslow, 1989). Although the presentation of OCD is similar across child and adult populations, the developmental context of young children complicates how cognitive-behavioral treatment may best be adapted. When using CBT with young children, it is important to consider (a) their developmental level and (b) their dependence on and involvement in their family system. Failure to address these issues would constitute significant treatment obstacles for the use of CBT with younger populations.

Developmental Issues

It is important to consider developmental factors in treating early childhood-onset OCD for numerous reasons. First, cognitive-behavioral treatments for anxiety and related disorders in children often involve the use of client-generated exposure hierarchies both in and between sessions. Young children may be unable to accurately collaborate in creating an exposure hierarchy due to their more limited skills in recognizing and articulating the level of distress or discomfort elicited by particular stimuli. This may lead clinicians or parents who assist with between-session homework assignments to implement exposure hierarchies that are too easy or too difficult. It is also important to note that early-onset OCD cases are more likely to have comorbid tic disorders and "just-right" obsessions (Geller, Biederman, Griffin, Jones, & Lefkowitz, 1996; Geller et al., 1998; Pauls, Alsobrook, Goodman, Rasmussen, & Leckman, 1995). It is difficult for children of any age to describe "just right" urges, but it is considerably harder for very young children to describe these urges. In a related vein, compulsive behavior in young children is not always perceived as ego-dystonic, but rather, as "just what kids do." Thus, it is important to include the observations of parents in developing an appropriate hierarchy. Additionally, it is useful to revisit the hierarchy in a systematic way during each session, to continually assess its level of appropriateness, and to provide opportunities to incorporate new information as young children progress through the treatment.

When treating young children, it is essential to use simple, concrete, child-friendly descriptions of symptoms or exposure tasks. For example, in J. S. March and Mulle's (1998) cognitive-behavioral treatment for older children and teens with

OCD, the disorder is described in psychobiological terms. With young children, it may be more useful to conceptualize OCD symptoms as a "hiccup of the brain." Younger children may be asked to label OCD symptoms with a silly name or to draw a picture of OCD as a "worry monster." Active participation in exposure may be described as "bossing back the worry monster" (e.g., J. S. March & Mulle, 1998, p. 152). Where possible, relying more on experiential tools in treatment is helpful. For example, active modeling of approach behaviors by clinicians and parents is particularly helpful for younger children.

When considering a plan for assessment during cognitive-behavioral treatment, it is helpful to evaluate symptoms and functional impairment not just at the outset of treatment, but throughout treatment, at termination, and, if possible, at follow-up check-in appointments. Systematic assessment assists in accurately and comprehensively describing treatment gains, as well as possible plateaus and exacerbations of symptoms. Before engaging in treatment, it is helpful to perform a structured clinical interview to assess the breadth of possible symptom areas given the high rate of comorbidity in childhood anxiety disorders, but especially due to higher rates and different patterns in young children relative to older children (Geller et al., 1996, 1998; Pauls et al., 1995). In particular, OCD is often comorbid with other anxiety disorders, mood disorders, tic disorders, and ADHD. With older children, the Anxiety Disorders Interview Schedule (both parent and child versions) is useful.

Family Contextual Issues

Consideration of family contextual factors that may impact how young children will respond to cognitive-behavior therapy is also critical. Young children are far more dependent on their parents for assistance and support than are older children. Parents of younger children are more likely to inadvertently reinforce or even actively accommodate their child's rituals (Lenane, 1989, 1991; Pollack & Carter, 1999; Steketee, 1997). Young children are particularly likely to involve family members in rituals, especially in the form of reassurance seeking (Lenane, 1989; Leonard et al., 1989; Rettew, Swedo, Leonard, Lenane, & Rapoport, 1992). Moreover, parents of younger children may have greater difficulty tolerating the distress that OCD symptoms precipitate in their children (Freeman et al., 2003). Thus, parents are often trained as coaches or cotherapists in CBT approaches with younger children. However, children who experience early childhood-onset OCD are far more likely to have parents who may struggle with their own OCD symptoms, other anxiety disorders, or depression (Nestadt et al., 2000; Pauls et al., 1995; Rosario-Campos et al., 2005). This is complicated by the fact that very young children rely on parents for guidance and direction in remembering and practicing exposure skills at home between sessions. This constellation of factors may lead to significant disruption of family routines and emotional distress as families progress through exposure-based treatment.

Parents struggling with their own compromised mental health may be less willing or able to actively support their children in treatment. In addition, they may have more difficulty tolerating their own or their child's negative affect during

challenging exposure exercises (Freeman et al., 2003). They may also have difficulty modeling adaptive, approach-oriented coping strategies. Studies of child anxiety have shown an effect of parental anxiety on child outcome (Cobham, Dadds, & Spence, 1998). Whether parents use avoidant- or approach-oriented coping strategies with their own anxiety issues, they may be at risk of undermining treatment, whether intentionally or not. This may occur through overt resistance during sessions, noncompliance with between-session homework tasks, or potentially, more subtle indicators such as providing distraction for children during EX/RP or "bailing out" of EX/RP tasks before habituation occurs.

Family accommodation has been associated with a host of additional negative factors, such as less developed problem solving, higher rates of depression in family members, and more expressed anger (Amir, Freshman, & Foa, 2000; Przeworski et al., 1999). Because EX/RP requires children to resist engaging in rituals, such a treatment approach often causes a great deal of distress and disruption of the family. This, unfortunately, is a typical pattern in families of individuals with OCD and may be especially problematic in families of younger children. Thus, effective use of behavior management strategies is critical to support the effective use of exposure and ritual prevention, especially with regard to between-session homework.

It is not uncommon for parents to disagree about the nature or intensity of symptoms or about their engagement in treatment. Thus, clinicians should carefully consider viewpoints of both parents, in addition to children's own report of their symptoms. It is fairly common to receive discrepant reports about an individual child's symptoms across parents and the anxious child. In particular, children may report more distress than parents, although given young children's less developed ability to appreciate and express complex or intense emotions, the reverse may be true here.

Because of these complex issues, the direct involvement of families in treatment is essential. Direct involvement in planning, exposure, coaching, and psychoeducation may improve the child's treatment outcome, adherence, and motivation. Research has indicated that the addition of a developmentally appropriate family component to CBT leads to more positive outcomes (Waters, Barrett, & March, 2001; Wood, Piacentini, Southam-Gerow, Chu, & Sigman, 2006). This may be especially true for younger children with OCD.

The cognitive-behavioral treatment illustrated in this chapter reflects current clinical treatment outcome research investigating family-based CBT for young children (ages 4 to 8) with OCD. This treatment explicitly addresses young children's cognitive and emotional sophistication, the role of parents or caregivers, and common comorbid conditions (externalizing and internalizing disorders). To date, a preliminary clinical trial has concluded, and a larger multisite replication and extension is under way (e.g., Freeman et al., 2003). Initial results are promising. The following case study illustrates the adaptation of cognitive-behavioral treatment for a young child struggling with OCD and his family. It also describes the use of new advances in cognitive-behavioral treatments, including acceptance and mindfulness techniques.

Case Illustration

Identifying Features of Client

Max was a 5-year-old Latino boy presenting to an outpatient child anxiety treatment center that is part of a Department of Child and Family Psychiatry at a large public hospital in an urban setting. Max was brought in for an initial evaluation by his mother and was accompanied by his older sister, Nina, who was 9, and his infant brother, Carlos, who was 7 months old. Max appeared slightly tall and underweight for his age. He made limited eye contact with his interviewers. He was neatly groomed, although his clothing fit poorly and seemed to be several sizes too large. He was extremely polite during the interview, often replying "Yes, sir" and "No, sir" to the interviewer. He fidgeted and shifted in his chair throughout the evaluation and appeared to make some subtle facial grimaces, lifted his eyebrows, and moved his eyes upward and from side to side. Max spoke quickly and quietly during the interview and at times seemed distracted from what he was saying. This tendency was so pronounced that the interviewer prompted him several times to finish what he had started to say.

Max spoke English quite clearly, without accent, although the family spoke primarily Spanish in the home. Max's mother spoke some English, but often relied on Max or his older sister to translate the interviewer's questions and comments. The family emigrated from the Dominican Republic 2 years ago, when Max was 3. They moved to the United States to join other members of their extended family who live and work here. Max's father worked for an upholstery cleaning company. His mother stayed at home with the children and also cared for Max's maternal grandmother, who was infirm. The family rented a small second-floor apartment in a predominantly Latino neighborhood.

Presenting Complaints

Max's mother and Max himself served as informants. Max was referred for evaluation by his family physician after a visit during which his mother complained that he was often noncompliant and was very anxious about getting sick. Max's mother described him as "stubborn," always wanting things his own way, even if it bothered other members of the family. Her primary complaint was that Max spent a great deal of time in the bathroom, often using up all the soap, shampoo, and toilet paper so that there was none left for the rest of the family. Often, there was no hot water left for showers or baths. This was a particular problem due to the family's limited budget. Max also insisted that his mother wash his toys when he felt they were dirty, and he grew quite distressed if they were not "clean enough" or if she refused to wash them.

In addition to this, Max also was a "picky eater." Max's mother reported that she often cajoled and scolded him to eat his meals, but typically, he would take only a few bites. Max described feeling that there was something "wrong" with his food. According to his mother, he tended to get particularly distressed if different types of food "touched" or got mixed together on the plate. Max's mother also described incidents during which Max would come down to breakfast and refuse to eat anything, and she reported that he would not say why. When asked directly about this by the interviewer, Max simply sat silently. Finally, Max reportedly tended to hoard a number of items that did not appear to have any sentimental value, notably, used tissues, rocks, and twigs. Max threw tantrums if he was prevented

from "doing things his way," according to his mother. He also sometimes hit or screamed at his older sister if she tried to "boss him," according to Max.

The amount of time Max spent washing, eating meals, refusing to speak, and hoarding objects had grown over time, and at the time of evaluation, constituted a significant proportion of the day. Max often missed opportunities to play with his older sister and with neighborhood children due to his involvement in ritualized behavior. In addition, his distress had also grown over time. At first, he did not appear to notice how disruptive his symptoms were and simply thought of them as "what I do." However, over the past several weeks, he had become more aware of them, concurrent with their increasing intensity. According to his mother, he described having more "bad days" than good days, and he sometimes cried in frustration at feeling that he "had to" continue engaging in rituals.

Max's mother reported that he was having difficulties at school, in addition to his struggles at home. According to her, his teacher was concerned that he had difficulty sitting still in class, completing tasks and art projects, and practicing holding and using pencils correctly. He also had some difficulty making friends and tended to play by himself on the playground. Occasionally, he got into physical fights with the other children, although he often denied this and would not tell his mother any details about the altercations.

Max's father reportedly did not think there was anything wrong with Max. According to Max's mother, he viewed Max as very similar to how he himself was as a child. Max's father felt strongly that Max's behavior was a "family issue" and that the family should not be seeking special help from another doctor. He believed that Max would "grow out of it" and was not supportive of Max's initial evaluation visit.

Developmental Factors

A number of factors were considered essential to adapting a cognitive-behavioral treatment for Max and his family. These included, first, assessment of Max's rituals as developmentally normative versus clinically significant. Second, it was important to determine Max's ability to recognize and express his anxiety in such a way as to contribute to building an accurate, workable fear hierarchy.

Parental Factors

Given Max's young age and relative dependence on his mother for guidance and limit setting, family accommodation of his rituals constituted a significant obstacle to treatment. In Max's case, there was already some evidence of his mother's acquiescence to his requests that she assist in his rituals. Moreover, other family members, especially his older sister, were considered unwitting participants in helping maintain his OCD-related behaviors.

In addition, several other familial factors stood out as potentially critical factors in using CBT effectively. First, his parents disagreed about the nature of Max's symptoms. His mother appeared concerned, at least enough to follow the advice given her by her family physician. However, Max's father saw his symptoms as more of an adjustment issue that Max would outgrow. Thus, the viewpoints of both parents, in concert with Max's own, were considered. In light of the parents' disagreement, the clinician decided to carefully observe Max's level of anxiety during sessions and to foster his ability to communicate how he felt throughout the treatment. Additionally, psychoeducation provided during CBT

(*Continued*)

helped to create more consistency in how Max and his parents understood his symptoms, his level of distress, and the functional impairment the symptoms created.

Given that Max often grew oppositional when limits were set on his compulsive behavior, his family may have been involved in subtle or overt accommodation of these symptoms. Again, educating the parents with regard to the nature of Max's noncompliance or distress around thwarted rituals was central to treatment. Highlighting the parents' distress when observing their son's discomfort when he was prevented from engaging in rituals enhanced parent treatment compliance. It was also useful to enlist the help of his older sister to prevent her inadvertent accommodation of his compulsions.

Finally, because Max and his family were Dominican, and perhaps more important, were recent immigrants dependent, at least in part, on an extended family network, the therapist took great care in assessing cultural differences in how young children were treated in both Max's immediate and extended family. It was also important to accurately conceptualize the degree to which the family had acculturated. In less acculturated Latino families, children may be given less autonomy than might be expected in the majority culture in the United States. However, there is such diversity in Latino groups that it is important that clinicians do not make assumptions about cultural differences (Szalacha et al., 2003). Both of these related factors impacted Max and his family's ability to understand, implement, and benefit from cognitive-behavioral treatment. Max's clinicians made every attempt to elicit and welcome discussion of cultural differences and to structure CBT in a culture-specific, culturally acceptable way. Failure to do so might have resulted in treatment resistance or even dropout.

History

Max reached all developmental milestones on time. As an infant, he had difficulties with transitions and novelty, often becoming overwhelmed when experiencing new situations, toys, or people. He stayed at home with his mother until the year of his initial assessment, when he began kindergarten at a local public school. At the beginning of the school year, Max had separation difficulties and often cried inconsolably when his mother dropped him off at school. He often sought reassurance about when his mother would pick him up from school and where he should wait for her when she did so.

Max had engaged in ritualistic behavior since he was 4, but this became more pronounced upon his entry into preschool. At age 5, one of Max's more involved rituals involved complicated sequences of events, which he requested his mother to perform, at bedtime. For example, Max required his mother to lay out his pajamas in a particular way on the bed and help him take off his play clothes in a particular order. After his pajamas were put on in a circumscribed sequence, Max brushed his teeth until they "felt" clean, which often went on for more than 5 minutes. Following that, he required his mother to line up his animals, tuck him in "just so," read three bedtime stories, and kiss him twice on each cheek. She then had to say "I love you" and "Don't let the bedbugs bite." Failure to perform any of these behaviors or doing them out of sequence caused Max great distress. If that happened, Max often cried until they could repeat the sequence correctly. Repeating morning routines resulted in lateness to school as well as general irritability toward his teacher and peers.

Most recently, Max had become concerned with "germs" and often complained to his mother that things were dirty and need to be cleaned. He made frequent attempts to have his mother wash his toys. She acquiesced because this placated him. Max was also very worried about toileting and spent an excessive amount of time cleaning himself after using the bathroom. This often resulted in dry, irritated skin and a great deal of discomfort. Other areas of concern for Max were that his food may be contaminated and that he felt compelled to keep nonsentimental items. His collection of used tissues, for example, constituted two large black garbage bags in his room. He became very distressed when any family member touched or moved the bags.

In general, the frequency and intensity of Max's rituals were exacerbated by stressful events. In addition, Max appeared to tic more during symptom exacerbations. Onset of his symptoms was gradual, rather than acute. As a toddler, he hated sticky or greasy substances on his hands and typically asked his mother or sister to wipe or wash them. He would tantrum until they did so.

Max did not have a history of significant medical problems. His mother had discussed Max's "picky eating" with the family physician, who reported that Max was "on the thin side" but that his weight and growth were within normal limits. Max's mother reported that she suffered from "nervous spells" and that Max's father was "moody." No other history of mental health issues was reported.

Assessment

Because Max was 5 and the Anxiety Disorders Interview Schedule for *DSM-IV* has not been normed in young children, a broader structured clinical interview, the Schedule for Affective Disorders and Schizophrenia for School-Age Children—Present and Lifetime Version (K-SADS-P/L; Chambers et al., 1985; Kaufman et al., 1997) was used to establish his diagnostic status. The K-SADS-P/L is a semi-structured, clinician-rated interview that yields *DSM-IV* diagnoses across the Axis I domains. The interview is administered to the parents (or primary caretakers) regarding the child, and to children 6 years and older. The K-SADS is routinely used to assess psychiatric diagnoses in children as young as 5 years (Hirshfeld-Becker et al., 2004; Youngstrom, Gracious, Danielson, Findling, & Calabrese, 2003).

The Children's Yale-Brown Obsessive Compulsive Scale (CY-BOCS), a 10-item semi-structured clinician-rated interview, was administered to assess OCD symptoms and severity at baseline, at midtreatment, at termination, and at all follow-up check-ins. The CY-BOCS rates obsessions and compulsions on 5-point scales on five dimensions (time, interference, distress, resistance, control) and yields a total obsessions score (0 to 20), a total compulsion score (0 to 20), and a combined total score (0 to 40). This scale has adequate psychometric properties and, more important, has been shown to be sensitive to change in both behavioral therapy and pharmacotherapy treatment studies. The CY-BOCS has been successfully used with children as young as 6 years (J. S. March & Leonard, 1998).

Because Max demonstrated some facial tics, the presence of a tic disorder was assessed with the Yale Global Tic Severity Scale (Leckman et al., 1989). This is a clinician-rated scale used to assess severity of motor or phonic tics, as well as related impairment. Motor and phonic tics are each rated on a 6-point scale

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regarding number of tics present, frequency, intensity, complexity, and interference. Motor and phonic tic scores may range from 0 to 25, and the combined total score ranges from 0 to 50. There is also an impairment score that provides a score from 0 to 50 that reflects overall burden due to tics. This instrument has demonstrated excellent psychometric properties (Leckman et al., 1989).

In addition to these measures, more general indexes of functioning in both home and school environments were sought. The Child Behavior Checklist Parent and Teacher Report forms (CBCL; Achenbach & Rescorla, 2001) is a widely used scale with well-established psychometric properties used to assess an array of behavioral problems in children ages 1.5 to 18 years.

Max's clinician was interested in both the impact of treatment and the rate at which Max's behaviors changed. The clinician also wanted to provide feedback to Max's mother regarding her son's progress, in the hopes that when treatment hit a rough patch, the feedback would be a motivating factor for the family. Thus, symptoms were assessed at several points throughout the treatment. To assess symptoms at midpoint, termination, and follow-up appointments, the CY-BOCS and CBCL were used, as well as unstructured interviews with Max's mother and teachers. General indexes of functional impairment were used, including Max's self-reported distress and level of satisfaction with school, his peer relationships, and his interactions with his family.

It is important to note that the assessment measures just described have not been used extensively with minority groups. However, they were considered the most appropriate measures available for use in this case. Thus, the scores ascribed to Max from these instruments were interpreted with some caution and were reported to Max's family and teachers with the appropriate caveats.

Case Conceptualization

Max is a 5-year-old Dominican male who, after participating in an extensive assessment procedure, was diagnosed with clinically significant OCD and secondary Tic Disorder. Although the externalizing disorders ADHD and Oppositional Defiant Disorder were considered, Max's presentation did not warrant assignment of these diagnoses.

Obsessive-Compulsive Disorder in children, as well as in adults, is characterized by recurrent, upsetting obsessions and repetitive, purposeful compulsive behaviors. Max's obsessions involved contamination fears: excessive and irrational worry about getting severely ill through exposure to germs and microbes, as well as contaminating others. He exhibited a number of compulsions related to these fears, including excessive hand washing, showering, and wiping after using the toilet. Max also had concerns about germs or poison in his food, although these worries included a general sense that "something was wrong" with his food when different food items touched on his plate. His compulsion regarding this was to refuse to eat or request a new plate of food items that did not touch. Max also felt compelled to hoard items that had no sentimental value and did not constitute items that would be considered developmentally normative to save. At school, Max reported that when he made pictures or held his pencil, it did not feel "right." Thus, he avoided coloring and writing altogether. If that was impossible, he would continue to make new pictures until he created one that was "perfect."

Max exhibited a number of motor tics, particularly facial grimaces, eyebrow movements, and shifting his eyes from side to side. Occasionally he did exhibit mild vocal tics, including clearing his throat and sniffing. However, the tics did not tend to co-occur and waxed and waned over time and across situations. They grew more pronounced in stressful situations or when Max felt vulnerable. The frequency and intensity of these tics was mild to moderate, and often Max did not realize that he was engaging in them. His teacher and the other children had begun to notice them at school; however, they did not constitute a significant source of distress for him.

Although Max was described by his mother as oppositional and noncompliant, those behaviors typically occurred in the context of ritual prevention. Rather than reflecting clinically significant oppositionality, they were more a manifestation of Max's frustration and anxiety when he felt compelled to engage in ritualized behavior. For example, Max's refusal to talk, although suggestive of oppositionality, actually represented his compulsive attempt to "protect" his family from contamination by germs Max breathed out. Max was also assessed for ADHD, in part because of a high level of observed activity as well as some symptoms typically attributed to inattention. However, interviews suggested that these were more reflective of his being distracted by obsessive fears. Given the high rate of comorbidity of OCD, ADHD, and Tourette's Disorder, Max would be watched over the course of treatment to remain vigilant for any changes in his symptom presentation.

Max's family had certain strengths on which to draw as they progressed through treatment. Namely, they were part of an extended family network that may be helpful in providing social support. Additionally, the fact that Max's mother sought the help of her family physician, and followed that advice to seek the services of a clinical psychologist suggested that she recognized a problem in her child's behavior and was amenable, at least initially, to participating in treatment.

Max's family was also characterized by what might constitute significant obstacles to treatment. These included many of the developmental factors outlined earlier, including Max's young age, his dependence on his parents, the family's potential difficulty in understanding the rationale of treatment, and issues with emotion regulation around his symptoms. Parental issues, including frank accommodation of Max's symptoms, his mother's level of anxiety and potential difficulty tolerating her own and Max's negative affect, and her lack of support from her husband, were complicating factors.

Course of Treatment and Assessment of Progress

Given Max's symptom presentation, in concert with developmental and familial issues outlined earlier, it was decided that Max would benefit from a family-based cognitive-behavioral approach, specifically adapted to a child of his age. This treatment was developed by researchers at Rhode Island Hospital/Brown Medical School in the Pediatric Anxiety Research Clinic (Freeman et al., 2003). The treatment was designed specifically to treat OCD in young children and, as such, does not directly target Tic Disorder. Treatment of choice for clinically significant tics would be habit reversal. However, given that Max's tics were not as pronounced or distressing as his symptoms of OCD, treatment was designed in a sequential way to address his OCD first.

(*Continued*)

The family-based cognitive-behavioral treatment was designed to address the unique needs of young children with OCD. In particular, it attends to young children's cognitive and socioemotional issues, the critical role of the family and their beliefs and behaviors around OCD symptoms, and the potential need for training in appropriate behavior management skills. Therapeutic material is tailored to younger children by including simple, concrete, child-friendly metaphors and experiential exercises. A great deal of attention is paid to young children's relatively less sophisticated emotion recognition and expression skills. Thus, a "child tools" section of each therapy session includes strategies to help children understand the treatment rationale and develop skill in acknowledging and expressing their anxiety along a continuum. The treatment also incorporates basic behavioral management strategies that parents may utilize in many situations, including, but not limited to, the child's OCD symptoms. These include overt modeling of approach behaviors, as well as psychoeducation regarding accommodation of symptoms. As an additional measure, the treatment deals with parents' and children's thoughts and feelings about and within their relationship and considers the role these may play in parent responses to child symptoms. Thus, both child and parent behaviors are targets of treatment.

Treatment consists of five main components, which are described next. This particular treatment was designed and its feasibility and efficacy compared to a family-based relaxation treatment (Freeman et al., 2007). It comprises 14 sessions. The first two sessions are with the parent(s) alone and generally last 90 minutes. The remaining 10 sessions include both parents and child and last approximately 60 minutes. Follow-up sessions are added as needed to address any residual OCD symptoms or symptom exacerbations that may occur over time. Because this is a manualized treatment, there is a suggested structure in the sequence of presentation of each element. However, as with most CBT manualized treatments, flexible use tailored to each individual client's needs is encouraged, with the constraint that all treatment components are eventually addressed.

Psychoeducation

During this portion of the treatment, the family is given education about the nature of OCD. Psychoeducation occurs during the first two sessions for the parents and the third for the child, according to the manual. However, in practice, it is also woven throughout sessions as needed. The main goals of this treatment component are to describe OCD as a neurobiological disorder such that it is understandable to the family. Oftentimes, parents who do not understand the nature of a psychological illness such as OCD are likely to blame their child or interpret symptoms as oppositionality or stubbornness. Similarly, they may be likely to require their child to "just cut it out." Providing detailed information conceptualizing OCD as a brain-based behavioral disorder helps to redirect parent change strategies in ways that may be more useful for their child. Thus, a second goal of this therapeutic component is to correct misattributions about OCD.

Children with OCD may blame themselves for their symptoms. They may also see their symptoms as part of themselves and view them as impossible to change. A core goal of psychoeducation is to help children view OCD symptoms as separate from themselves. This is accomplished through a number of strategies, including asking children to draw a picture of their OCD, asking

them to think in terms of "bossing back" (see J. S. March & Mulle, 1998) their OCD instead of allowing the OCD to "boss" them, and drawing a map of what parts of their life are under their own control and what parts are under the control of the OCD. The overarching goal of therapy is to increase the domains of the children's life in which they, and not OCD, are in charge.

During this portion of the treatment, it is of utmost importance to describe to parents what they can expect. Treatment is designed to facilitate the child to participate in exposure and response prevention in a graduated way both within and outside of session. Parents are asked to become part of the treatment team and, as such, act as coaches for their child throughout the course of therapy. Because exposure and response prevention are inherently uncomfortable and stressful for children and parents, therapists must take care to address this issue in an honest and realistic way. Parents should be told that the treatment is often stressful and may disrupt family routines. If this is not carefully discussed, parents may have difficulty accepting their child's or their own affect during exposure exercises. This can impede the progress of therapy, lead to setbacks, and, ultimately, to dropout.

The following is a sample clinical transcript of how the therapist approached Max's parents:

Therapist: First and foremost, it is important for you to know how exposure works and what it may be like for your family to go through it. Helping Max participate in exposure will be hard, although we will start with easier exercises, and then move on to harder ones. It will likely be stressful for you as parents to watch Max go through the treatment, and watch him feel anxious. He may get scared, sad, or angry. Those are normal responses, and come from being prevented from engaging in rituals, which he now uses to make himself feel better. Even though he will feel uncomfortable, and it may be hard for you to watch and participate in that, it is important that he continue the treatment, but also that he does his exposure exercises until his level of distress goes down. Through doing this, he will learn that whatever anxiety and distress he feels is simply his OCD bossing him around. And if we as a team help him put his foot down and stand up to his OCD, he will learn that he doesn't have to do his rituals. Eventually, our goal is to help him get more of his life back, so that he can do more of the fun things he wants to do, without being disrupted by his OCD symptoms. So if it gets difficult for you, try to keep that in mind if you feel your resolve weakening.

To increase both parent and child participation, it is important to make the process of psychoeducation as simple and interesting as possible. This not only assists therapists in building rapport with the family, but also sets the tone for therapy. Presenting the information in several modalities and using visual imagery, metaphors, and concrete examples make it more memorable.

The following is an example of how a therapist described the nature of OCD to Max:

Therapist: Sometimes we all get thoughts that pop into our heads over and over again, like when a song gets stuck in your head. The problem is that sometimes those thoughts are *worries* and they scare or upset us. When someone

(*Continued*)

gets lots of worries that keep popping into their head over and over again, and they feel scared or upset a lot, we might call these thoughts (for example) "The Worry Monster." To try to make those worries stop, a person might try doing some different things. Like, if someone felt dirty, they might wash their hands, or if they felt like they said something wrong, they might need to repeat it or apologize for it. These behaviors are called (for example) "re-do's." The problem is that you feel better for a second, but then you worry again and you have to do these things again and again. (Coyne, McVey-Noble, Freeman, & Garcia, 2005, p. 5)

To describe to Max how OCD might impair his functioning, the therapist relied on a comical, child-friendly metaphor:

Therapist: Imagine you bought a hermit crab and it was really cute and cool to play with. Every day you fed it and changed its water and hung out with it. That would be fun, right? Then imagine that it started to grow really big, and it started to follow you everywhere. Sometimes it even ate your lunch and your favorite dessert, and it annoyed you when you were playing with your friends and wouldn't let you watch your favorite TV shows, and it made a lot of noise when you were trying to go to sleep at night, and maybe even tried to sleep in your bed, which would squish you. Pretty silly, but not much fun, right? Then, imagine that it started being a really bratty crab and giving you little pinches unless you paid attention to it and gave it snacks all the time. Not nice. That's what happens when those worries take over and start being the boss of you. So, let's make you the boss of you and your worries! (Coyne et al., 2005, p. 2)

Parent Education

In addition to educating parents and children about the nature of OCD symptoms and distinguishing those symptoms from normative developmental rituals and any comorbid symptoms of other anxiety disorders, therapists also teach parents skills for behavior management. Although parents may enter treatment with their children at various levels of knowledge and performance of appropriate behavior management strategies, it is important to help parents strengthen this skill set and adapt it to their child's OCD symptoms. Tools addressed in this program include differential attention, modeling, and scaffolding. These skills are taught in session and practiced in the home. Therapists are trained to troubleshoot incorrect or inconsistent use of these skills and to help foster better usage, especially with regard to the child's OCD symptoms.

Differential attention refers to removing attention to a child's attempts to resist or refuse to engage in exposure or homework exercises, while attending to the child's approach behaviors. Use of this tool reflects the hypothesis that parent attention is an important motivational factor in CBT for early-onset OCD, as previous research has shown that family members often accommodate children's symptoms.

As previously discussed, parents learn to model approach behaviors for their children. Reasons for teaching this particular skill come from a social learning perspective that assumes that the parents' modeling of approach behaviors will help motivate and support their children in engaging in exposure and exposure-based homework during the course of treatment.

Finally, scaffolding (or coaching) involves parents' assessment (guided by the therapist) of their child's progress through treatment and encouragement of their

child to engage in increasingly difficult exposures. It involves empathic encouragement and includes eliciting the child's feelings, use of a brainstorming approach to collaboratively come up with a way to approach rather than avoid the stimulus, choosing a method of approach, and implementation of that method. Clinicians may teach parents by example or in a more didactic way. Approaching the potentially hesitant child in this way helps to engender and maintain the child's positive momentum and may prevent treatment resistance or fallbacks. It also assists the child in learning emotional and behavioral regulation and the internalization of these responses. This is thought to promote generalization of gains made across settings and time.

The following is a sample of how the therapist addressed the brainstorming component of scaffolding with Max's parents:

Therapist: It is often helpful to brainstorm with your child reasons *why* he should do EX/RP. You might remind him about the hermit crab metaphor [see previous example], and say that OCD doesn't go away simply if you ignore it—it just gets bigger! Another way is to use the idea of "bossing back" OCD. You might ask Max who he wants to be the boss—himself, or mean old Mr. Worry.

Externalizing Obsessive-Compulsive Disorder and Exposure/Response Prevention

Exposure and response prevention is a core component of cognitive-behavioral treatment for OCD. This involves graduated exposure to feared stimuli and prevention of discomfort-reducing responses until habituation is reached. Generalization of exposure practice and response prevention is facilitated by the use of a token reward system implemented by parents. To that end, both parents and children learn to externalize, or separate, OCD symptoms from themselves. For young children, this is best accomplished through having them "boss back" OCD. Children also learn how to use a "fear thermometer" to help them report gradations in their experience of discomfort, whether to structure an anxiety hierarchy or to communicate rising and falling levels of anxiety during exposure. Parents are encouraged to use rewards to encourage children to practice EX/RP at home between sessions.

The following is an example of how a clinician educated Max about the use of a fear thermometer:

Therapist: Do you know how a thermometer works? (Child nods.) Well, my feelings thermometer works exactly the same way. When you don't feel scared at all, the red stuff stays down at the bottom. If you are sort of scared, the red is somewhere near the middle. If you are really, really scared, the red is all the way at the top. Can you think of how scared you might be if you were sitting on your couch, watching your favorite show, eating popcorn? Can you show me where the red might be on my thermometer? (Give child a chance to point out fear level, typically at the bottom.) What about if you were really really scared of heights, and you had to parachute out of an airplane? Where would the red be then? (Give child a chance to point out fear level, usually near the top.) Well, we are going to use this thermometer to figure out how much your OCD bugs you when you don't do your re-dos (compulsions).

(*Continued*)

Family Process Components

Family accommodation of symptoms is addressed very specifically via this component of treatment. This is often a process that continues throughout the course of therapy, as parent insight and ability to discriminate between accommodating and nonaccommodating responses grows. Sometimes parents are angry with or hostile toward their children, given that child symptoms have the potential to significantly disrupt family routines. In a related vein, parents may have great difficulty tolerating their own distress when watching their children struggle with symptoms. Thus, parents are encouraged to discuss these emotions, and in turn, to engage in more adaptive problem solving with regard to their child's symptoms. Finally, in this treatment component, therapists may highlight for parents their own anxious interpretations and behaviors about their child's symptoms.

The following reflects a conversation between the clinician and Max's parents regarding their feelings about his OCD:

Therapist: What is it like for you when you see Max struggling to wash his hands over and over?

Parent: Well, it's really hard. It breaks my heart to see him so anxious. I'd do anything to make him feel better.

Therapist: What are some things you do when you see him like that?

Parent: I try to tell him it's okay, but that doesn't help at all. I plan on extra time getting anywhere, because I know that Max will have to spend some time washing and checking his hands. I buy really gentle soap so that his hands don't get so chapped. I even write notes to his teachers when he is having a bad day, so that they will be more understanding and let him spend extra time in the bathroom, and he won't get in trouble.

Therapist: That sounds like it's pretty distressing for you as well, and that you try to do a lot of things to give Max the space he needs to do his rituals.

Parent: It is hard for me, but it's definitely harder for Max. I do what I can to help.

Therapist: get how important it is for you to help Max. I want you to consider this: What if Max had a medical disorder, like diabetes, and you had to teach him to manage his sugar intake?

Parent: Okay.

Therapist: What if he threw a really big fit one day, because he really wanted a piece of chocolate cake? Say his sugar was fairly high already that day.

Parent: Well, he couldn't have it. I wouldn't let him have it.

Therapist: But what if he got really upset?

Parent: Well, it just wouldn't be an option—it would make him sick.

Therapist: But it would be hard to say no, wouldn't it?

Parent: Sure, but I would know I was doing what was best for him.

Therapist: OCD is exactly like that. If Max is allowed to do his rituals, then he will likely keep having his OCD symptoms. If you help him learn to resist them, even though it will be hard, you will be doing what is best for him.

Contingency Management

In addition to the parenting tools described earlier, therapists also address potential (or actual) resistance to engaging in EX/RP by teaching parents appropriate limit setting. Sometimes young children develop compensatory strategies to cope with their OCD, such as oppositional behavior when required to approach a feared stimulus. If this is the case, clinicians may impart to parents how to effectively use response-cost procedures to address child resistance or token systems to encourage EX/RP practice. Once parents learn this component, their skills in this area may generalize to other domains of their children's behavior that are not specific to their OCD.

Mindfulness, Acceptance, and Values

What have been called "third wave" behavior therapies have developed technologies for addressing thoughts and emotions in a slightly different way than traditionally defined cognitive-behavior therapy. Acceptance and commitment therapy (ACT) is one such treatment paradigm. It may be classed as a cognitive-behavioral therapy in that it addresses the role of cognition in behavior. Thus, individuals progressing through the therapy learn to attend to their internal psychological experiences and the meanings with which they imbue external events. However, whereas traditional cognitive therapies conceptualize maladaptive behavior as the product of faulty cognition and therefore strive to change the content of such mentation, ACT stresses mindful awareness and acceptance of unpleasant or unwanted thoughts. As such, individuals are taught in an experiential way that they may simply notice and detach from their thoughts without evaluation or response. Doing so permits them to engage in valued, personally meaningful activities. Accruing evidence suggests that ACT may work through different mechanisms than CBT (Hayes, 2005).

Core components of ACT include mindfulness or awareness and valuing. Incorporating these components of ACT into CBT approaches may help to address motivational issues in treatment. Mindfulness has been defined in similar ways across the literature and generally means moment-to-moment awareness of one's experience—including thoughts, feelings, and physiological responses—without judgment. With regard to parents, mindful acceptance may help parents identify without response their thoughts and feelings about their child. This may minimize parent reactivity to their child's expression of strong emotion when rituals are prevented, and thus may help parents resist accommodation. For children, learning to be mindfully aware of their own thoughts and feelings in a detached way may augment their approach attempts during exposure exercises. Moreover, it may facilitate children's attending more fully to their actual experience of exposure, rather than their "stories" or expectations about how it will go. In this way, they may be more sensitive to actual environmental contingencies, and thus may benefit from exposure more fully. Finally, teaching mindfulness skills to children may help inoculate against mental rituals or "white knuckling" and engender more willingness to engage in exposure. This may also reduce the possibility of parents' coercion of the child into the exposure.

Here is an example of how mindfulness was used to increase the efficacy of exposure with Max:

(*Continued*)

Therapist: I am going to show you a way to be what we call "mindful" when you are doing your exposure work. We're going to practice being scientists. Do you know what scientists do?

Max: They find out stuff, I guess.

Therapist: How do you think they do that?

Max: I don't know. Maybe they watch stuff to see what happens?

Therapist: That's exactly right. Scientists watch stuff and see what happens. But here's the trick—they watch stuff without trying to change it.

Max: I don't get it.

Therapist: Think of it this way. When you eat an ice cream cone, what do you think about?

Max: Nothing, I guess. I just eat it.

Therapist: Yup. And you probably notice that it's cold, and that it's sweet, and maybe a little melty if it's hot out, and that the cone is crunchy, like a cracker, but sweet too. Well, that's like what a scientist does—noticing stuff, without trying to change it. That's what we call being mindful. Can we try to practice that right now?

Max: I guess so.

Therapist: Okay. Close your eyes. . . get comfy in your chair. . . notice how your body feels. . . where it touches the chair, where your feet touch the floor. . . notice all the sounds you hear. . . see how many different ones you can notice. . . see if you can notice how warm or cool the room is. . . see if you can tell what color the light is on your eyelids. . . see if you can notice your breathing, gently, in and out. . . notice that the air is cool when you breathe it in, and warmer as you breathe it out. Now notice what you are thinking. . . if you are seeing pictures in your head, see if you can print them on billboards on a highway. . . imagine that each thought or picture in your head is on a billboard. . . and just look at it until it passes and a new image pops in.

The therapist practiced this skill with Max in the context of his neutral thoughts, feelings, and bodily sensations, and then progressed onto distressing thoughts and images in the context of exposure exercises. Because mindful awareness constitutes exposure to feared psychological events (called *defusion* in the ACT literature), it is helpful to encourage children to practice it in a graduated way, just as they would with more traditional modes of exposure.

Mindfulness and acceptance are components that have been used in a number of different cognitive-behavioral therapies, including those targeting reduction of stress, depression, and anxiety (Hayes, 2005; Hoppes, 2006; Levitt & Karekla, 2005; Roemer, Salters-Pedneault, & Orsillo, 2006). *Valuing* is a component unique to ACT. It refers to activities and ideals that are meaningful and vital to an individual (K. G. Wilson & Murrell, 2004). Values are not goals; rather, they are directions in which to move. As such, engaging in a valued activity is inherently reinforcing, and self-renewing, since values cannot be reached, but rather, serve to direct behavior. Clinicians may harness valued action in treatment to help motivate parents and their children to engage and actively participate in treatment. For example, if parents feel that helping children engage in EX/RP is consistent with the value of being a good parent, this may inspire better treatment adherence, as well as dignify their often difficult work in and outside of session.

The following is a sample of how the clinician worked with Max's mother to identify her values with regard to Max:

Therapist: (After a brief, experiential centering exercise, eyes closed) I want you to imagine Max 13 years from now, at his high school graduation. There are many people at the ceremony, graduates, their families and friends. You see Max from where you are sitting. . . . As part of the ceremony, there is an announcement that one of the graduates will give a speech. There is silence, and you see Max slowly rise and walk toward the podium. When he reaches it, he clears his throat, and looks for you in the crowd. Notice his face, his eyes, the way he looks at you. He smiles. Imagine that he begins to speak about his struggle with OCD, and your role in that struggle. I want you to take a few minutes and imagine where you would like for him to be at that moment in his life, and how you would like him to describe your role in his struggle. (Pause.)

Mother: I want him to be happy, to be successful, to have a good life. I want to fight for him. I want him to say I fought for him, and that I helped him. That I didn't give up on him.

Therapist: (Pause) Is that what is most important to you?

Mother: Yes. More than anything. I want to protect him.

Therapist: What if our work here, even when it is at its hardest, is the best way to protect Max? What if it is a step toward Max having a successful, good life? What if this is one way you can fight for him, right now, by helping him to do his exposure work?

Mother: I would do it.

Therapist: Even when it is hard for him, and for you. . . and it will be, if it is to work. . . I want you to keep that picture of him in your mind, and in your heart. That will be our goal—to do the work here, now, that will help Max get to that place in the future, where he is living the life he wants, and doing the things he most cares about. You will help him get there.

Complicating Factors

Many factors that may complicate a family's progress through treatment have been detailed in previous sections. Two broad areas—developmental issues and parenting issues—were identified. Given Max's young age, his developmental stage and skills in emotion recognition and expression, care should be taken by his clinician to ensure that encouragement of these skills is woven into treatment. Similarly, given his developmental level, Max is more embedded within his family and dependent on his parents than an older child might be. This highlights the importance of motivating parents to become involved and to maintain their involvement in Max's treatment. One obvious difficulty is Max's parents' contrasting views of his OCD symptoms. His mother views them as impairing and emblematic of a pathological process; his father views them as a phase that he will outgrow. A clinician working with this family would be well advised to help build consensus between the parents, as well as to be vigilant for any parent behaviors that might undermine Max's treatment.

Parents may have difficulty tolerating their child's distress as the family progresses through treatment. This may lead to parent accommodation of child symptoms in the name of reducing child distress and minimizing disruption to

(*Continued*)

the family. Clinicians should offer support for parents in this area and take opportunities to notice, empathize with, and normalize perceived parent stress. Additional treatment components, such as motivational interviewing and acceptance and mindfulness-based techniques may be helpful with regard to facilitating treatment adherence in both parents and children, although data on these approaches are limited (Barmish & Kendall, 2005; Gance-Cleveland, 2005; Semple et al., 2005).

Max's Dominican culture may also prove to be a complicating factor in treatment, especially if the clinician is unaware of (or worse, insensitive to) important cultural differences that may impact treatment. Unfortunately, there is very little research involving anxiety in Hispanic and Latino cultures, and less in its particular subgroups (Varela, Vernberg, & Sanchez-Sosa, 2004). There is some evidence, however, that Hispanic Americans of Dominican descent may frame their troubling anxiety symptoms using idioms such as *nervios,* which may reflect a more benign, culturally sanctioned conceptualization (Salman, Diamond, Jusino, Sanchez-LaCay, & Liebowitz, 1997). This could suggest that Dominican families may view mental health issues, and anxiety in particular, as stigmatizing or shameful. This may result in a family's attempts to minimize or conceal anxiety symptoms and their related interference in family function (Salman et al., 1997). This view may be more or less pronounced, depending on differences in the level of acculturation within a particular family and its members. In Max's case, it may in part explain his father's minimization of his son's OCD symptoms.

When faced with families with racial and ethnic origins different from their own, clinicians should remain open and maintain the stance that clients are the experts on their own culture, degree of acculturation, and culturally accepted practices with regard to mental health issues. If issues of shame or stigma are identified, it may be useful to address these with the family in a nonevaluative way. It is important for clinicians not to make assumptions, but rather, to approach those from diverse cultural or ethnic groups with openness and respect throughout the course of treatment.

Treatment resistance constitutes another broad obstacle to treatment progress. For example, one hindrance to cognitive-behavioral treatment for young children with OCD is failure to complete homework tasks between sessions. To address this issue, clinicians should plan and troubleshoot homework assignments with great care. Some parents are more likely than others to comply with homework assignments. Reasons for variability in compliance include level of family disorganization, time constraints, parent accommodation and unwillingness to tolerate distress or disruption of family routines, poor planning, overzealousness, and failure to make treatment a priority. Clinicians should spend time assessing for these issues and making as detailed a homework plan as possible. It is also helpful if the development of such a plan is undertaken collaboratively, with compromises on the part of both family and clinician. Clinical judgment must be wisely used in terms of weighing issues of family skill, commitment to treatment, and logistical problems against the necessity of the child's engaging in sustainable, slightly difficult EX/RP exercises throughout the week. It is also useful to elicit information from the family about what worked and what did not, in the service of better planning future homework assignments.

Within-session resistance, whether from parents or children, may also be a problem. Parent tools, including differential attention, positive verbal praise, scaffolding, use of a token system, and limit setting techniques, have been

incorporated into the treatment to address child resistance. Clinician attention to family process issues, such as parents' emotions and attributions about their child's symptoms, parents' role on the treatment team, and how parents cope with their own anxiety around treatment, is essential to treatment success.

Children, young children in particular, may have difficulty generalizing the skills that they learn in session to their lives more broadly, especially as time passes after treatment termination. Between-session homework addresses this issue, to some degree. As Max's family gained mastery over particular rituals, the therapist encouraged them to practice in a wide variety of settings, from home to school, restaurants, movie theaters, and elsewhere. The family was also encouraged to try impromptu practice; in other words, whenever Max's symptoms crept up, his parents were asked to remind him to use his skills to "boss back" his OCD. The therapist also educated the family with regard to the tendency of OCD symptoms to wax and wane, or shift in topography, so that they might be watchful and ready to use their skills as needed. The package of skills taught in therapy is conceptualized as a life change—one that the family must continue to practice as needed should symptoms return in times of stress or transition. Finally, the therapist highlighted major developmental events in Max's life—transition to first grade, increasing demands in school, puberty—that might herald periods when Max might be vulnerable to symptom recurrence or difficulty implementing his EX/RP skills.

Managed Care Considerations

Cognitive-behavioral treatments for children and adolescents are oftentimes constrained by managed care. Psychological practice undertaken in a managed care setting must address careful measurement of client progress and outcomes to justify continued reimbursement by insurance providers. There are many downsides to this situation, including reliance on "quick and dirty" assessment methods that may not capture the full complexity of a family's struggle, the fact that families that do not progress quickly may exhaust their allotted mental health services, and that some families have less access to needed mental health services because they cannot afford insurance. Some may view the increased accountability of health care providers, emphasis on developing more effective treatments, and greater emphasis on careful tracking and measurement of meaningful outcomes as positive consequences of managed care (Fonagy, Target, Cottrell, Phillips, & Kurtz, 2002). For Max's treatment, the clinician included multiple methods of measuring treatment gains, as detailed earlier.

Follow-Up Treatment

Follow-up visits were useful to continue to assess Max's symptom picture. In addition, booster sessions may be offered as his symptoms wax and wane or change in topography over time. Just as it is important to meet Max at his developmental level, it is also important to consider the demands of developmental transitions: progressing from kindergarten to first grade, the beginning of each school year, the onset of puberty, the increasing autonomy that comes with adolescence, and so on. The option of follow-up treatment is especially critical at these junctures, as families may need additional help tailoring their use of EX/RP skills in more developmentally sensitive ways.

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Implications of the Case

The case described in this chapter illustrates that cognitive-behavioral therapy can indeed be adapted to younger populations. However, such adaptations must take into account issues such as the phenomenology of different mental health issues in young children, their increased reliance on and embeddedness in the family, and levels of cognitive and socioemotional sophistication.

Despite the increasingly recognized need, there still exists a gap in the treatment literature on the application of CBT to young children with OCD. The majority of the published treatment studies do not include children under the age of 7 or 8, nor do they address early developmental considerations. Though current clinical trials are working to adapt CBT with EX/RP for children ages 4 to 8, there still exists the need to address the unique developmental factors for this younger age group and its possible moderators of treatment. It is for this reason that treatment formulation for young children must be approached by considering all of these factors as well as any other developmental factors unique to a particular child. Comprehensive assessment of a child's developmental attainment and stage will allow for the augmentation of current CBT approaches that have been found to be highly successful in treating OCD.

Recommendations to Clinicians and Students

Increasingly, CBT approaches for children also address parent issues, such as stress, deficits in skills acquisition or performance of appropriate behavior management techniques, and parents' own struggles with mental health issues (Barmish & Kendall, 2005; Barrett et al., 1996; Chronis, Chacko, Fabiano, Wymbs, & Pelham, 2004). Given the increasing diversity of the U.S. population, coupled with disparities in access to and utilization of mental health services to minority populations, it is essential for clinicians to keep apprised of empirical research regarding the use of CBT with these groups. Additionally, the increasing use of acceptance and mindfulness-based techniques and motivational interviewing may open up new ways to improve and broaden outcomes of more traditional CBT approaches. Finally, although we as a field have accrued a good deal of evidence detailing the efficacy of CBT, we need to turn our focus to the mechanisms of treatment, as well as mediators and moderators of treatment outcome (Kazdin & Nock, 2003). Without investigation in this area, we will not be able to make continued advances in the broad goals of helping to ameliorate mental health concerns so that individuals may live more effective, meaningful lives.

SUMMARY

Cognitive-behavioral treatment approaches have been successfully developed and used with a variety of psychological issues in children, adolescents, and their parents. Although empirical data suggest that CBTs are efficacious in many domains and across many disorders, there is still room for growth and refinement in how they are conceived, operationalized, applied, and evaluated. Recent developments include applications with diverse groups, in diverse settings, using different media (e.g., virtual reality, computer-based programs, use of palm devices), and with younger children. In addition, clinical treatment researchers have been increasing their emphasis on mechanisms of action in treatment and mediators and moderators

of treatment outcome. Taken together, these new directions promise continued sophistication, efficacy, and, hopefully, effectiveness in CBTs as the empirical literature continues to advance.

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CHAPTER 24

Primary Care Behavioral Pediatrics

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Children are not born civilized and socialized. They do not emerge from the womb predisposed to use the toilet appropriately, sleep through the night on their own, wait their turn, sit still or be quiet when stillness or quiet are called for, or to exhibit any of the myriad behavioral requirements of civilized society. Unfortunately for parents, teachers and other caretakers, some children will resist most, and most children will resist some, critical aspects of the socialization and education processes in this culture, and a vast number and array of child behavior problems are the result (e.g., American Academy of Pediatrics, 2001; Christophersen, 1982; Christophersen & Rapoff, 1979; Costello & Shugart, 1992; Earls, 1980). For example, nutritional and maturational health is predicated on food preferences that include the major food groups, and yet many children, apparently satisfied with milk and pureed food, resist parental attempts to introduce new tastes and textures into the daily diet. Adaptive child performance during the day is dependent on receipt of adequate sleep at night, yet many children resist parental efforts to impose a reasonable bedtime. Most parents, preschools, and many day care programs require full toilet training during the third year of life, yet many children resist parental training efforts. Success in most life situations requires a reasonable amount of instructional control, yet many children resist following important adult instructions. There are many other examples, and they generally emerge in situations that require child adherence to or conformance with the family, school, or societal standards or requirements. There are also a large number of child problems that do not involve child resistance so much as child inability to perform, maintain, reduce, or cease important behavior. Although all of these problems pose risks for the child and family, they mostly represent skill deficits rather than pathologies, especially in young children (e.g., Christophersen, 1982, 1988; Friman, 2002; Earls, 1980; also see Horwitz, 2002). An obstacle for mental health practitioners interested in addressing these problems is that they are much more likely to surface initially in primary medical care (PMC) than in mental health service (MHS) settings (e.g., American Academy of Pediatrics, 2001; Costello, 1988; Costello, Burns, et al., 1988; Costello, Edelbrock, et al., 1988; Costello & Shugart, 1992; Dulcan et al., 1990; Goldberg, Roghmann, McInerny, & Burke, 1984; Hickson, Altemeier, & O'Connor, 1983; Horwitz, Leaf, Leventhal, Forsyth, & Speechley, 1992; Lavigne et al., 1993; Starfield et al., 1980).

Although the skills children need to overcome these problems can emerge through vicarious experiences, most skills have to be trained directly by parents or caregivers. Unfortunately, there is an implicit assumption in this culture that those who provide the training should be able to do so without benefit of MHS. Although there is substantially more cultural permission to seek assistance with child behavior problems in PMC settings, as well as a much greater likelihood of parents doing so, the structure of typical PMC visits imposes substantial limits on the assistance actually made available (e.g., Finney et al., 1990; Wissow et al., 2002). Mental health service is vouchsafed by the culture only when child behavior problems reflect significant deviations from cultural norms (e.g., it meets criteria for a psychiatric diagnosis; American Psychiatric Association, 1994); correspondingly, costs accrued from obtaining MHS for such problems are usually reimbursable through third-party payers. Some assistance for problems that are not considered clinically significant is available outside of MHS and PMC (e.g., websites, books, television shows, lay counseling), which is fortunate for parents because they appear to be mostly on their own when training children whose behavior problems are not considered clinically significant. Unfortunately, the quality of that assistance is hard to determine and impossible to guarantee to any meaningful degree, and reimbursement for it is not available from third-party payers. In addition to that assistance, there are multiple child training methods that are an integral part of the parenting process in this country, and virtually all parents have direct access to them. Prominent examples include praising, scolding, yelling, warning, restrictions, and rewards (e.g., Blum, Williams, Friman, & Christophersen, 1995; Peters, 1998; Regalado, Sereen, Inkelas, Wissow, & Halfon, 2004) and a range of methods collectively referred to as time-out (Friman, 2005c; Friman & Finney, 2003). How well the nonprofessional assistance and freely available training methods work can be estimated through inspection of epidemiological surveys of children whose behavioral concerns meet criteria for diagnostic classification; presumably these are children whose behavior has been exposed to some combination of the assistance and the methods. Major surveys indicate that between 11% and 25% of children in this country have behavioral problems that meet criteria for a psychiatric diagnosis (cf. American Academy of Pediatrics, 2001; Costello, 1988; Costello, Burns, et al., 1988; Costello, Edelbrock, et al., 1988; Costello & Shugart, 1992; Dulcan et al., 1990; Goldberg et al., 1984; Hickson et al., 1983; Lavigne et al., 1993; Starfield et al., 1980). These data suggest that a substantial portion of child behavior problems, although not initially deviant from cultural norms, devolve to something much more serious and that a very large number of parents need more help addressing the problems than is made available to them outside of professional contexts.

The two most logical sources of professional help are PMC (e.g., pediatricians) and MHS (e.g., psychologists). The former source is utilized by virtually all parents, although mostly for medical care. Nonetheless, child behavior problems are much more likely to be initially presented in PMC than MHS. Unfortunately, service seeking for children with behavior problems presenting in PMC is restricted by three challenges: limited time, training, and reimbursement. Regarding limited time, early time-and-motion studies showed that the length of the average visit in

PMC for children was between 8 and 15 minutes, with only a small portion of that time devoted to behavior (Christophersen, 1982; Reisinger & Bires, 1980); more current research shows that these early findings have not changed appreciably in recent times (Cooper, Valleley, Polaha, Begeny, & Evans, 2005). Regarding limited training, the curriculum for physicians planning to work in pediatric settings has historically included very little training for assessing and treating behavior problems (e.g., Friedman, Phillips, & Parrish, 1983; Glasscock, O'Brien, Friman, Christophersen, & MacLean, 1989), although the American Academy of Pediatrics is attempting to remedy this problem by requiring a rotation in developmental-behavioral pediatrics (2001; Hagan, 2001). The average rotation, however, is only a month long, which is certainly better than no training at all but is hardly sufficient to develop the expertise necessary to identify and manage the range and number of behavior problems that present in PMC. Regarding reimbursement, physician provision of behavior assessment and treatment involves services that managed care companies are increasingly reluctant to cover (Kelleher, Scholle, Feldman, & Nace, 1999), resulting in non-revenue-generation time by PMC providers and/or limits on services for child behavior problems supplied by PMC providers.

In MHS, providers typically have both the time and the training to competently assess and treat child behavior problems, and they have a service code and nosological system that can lead more effectively to reimbursement for related services than the system used in PMC. Unfortunately, whereas very few parents have reservations about seeking advice about child behavior problems from providers in PMC, many parents have substantial reservations about seeking advice from providers in MHS. Among the reasons for these reservations are the routine use of psychopathology in the conceptualization of child behavior problems, the related stigma associated with the failure of tactics available outside professional contexts to address the problems, the common reliance on strong medication in treatment, high costs, and fear of permanent labeling by third-party payers. To phrase this point in blunt colloquial terms, most parents are loath to see a shrink about their children's behavior problems unless those problems have become severe (e.g., Friman, 2002; Leinwand, 2006).

As a result of the limited services in PMC and limited service seeking in MHS, a large gap in behavioral health care for children has opened up in this country. In that gap is a large number and array of child behavior problems. By definition, these are addressed only with tactics made available outside of professional contexts, and, as indicated by data from epidemiological research, a large portion worsen over time (cf. American Academy of Pediatrics, 2001; Costello, 1988; Costello, Burns, et al., 1988; Costello, Edelbrock, et al., 1988; Costello & Shugart, 1992; Dulcan et al., 1990; Goldberg et al., 1984; Hickson et al., 1983; Lavigne et al., 1993; Starfield et al., 1980). A professional movement focused on addressing this mounting problem involves primary care behavioral pediatrics (PCBP). Primary care behavioral pediatrics does not currently have a widely accepted technical definition, and yet a seemingly noncontroversial one can be abstracted from various descriptive papers. Specifically, PCBP involves behavioral health care that employs applications drawn from the behavioral and child developmental sciences to

address child behavior problems that populate the intersection between child PMC and MHS. Although the problems that present in PCBP can involve secondary (curative) or tertiary (rehabilitative) features, PCBP upholds the long-standing tradition in pediatric medicine of emphasizing prevention over cure or rehabilitation. In the words of Stanford Friedman (1975, p. 515), an early architect of the general field of behavioral pediatrics, "Curative and rehabilitative orientation [is] always second best to preventing the disease or defect in the first place."

DESCRIPTION OF PRIMARY CARE BEHAVIORAL PEDIATRICS TREATMENT

Essentially pragmatic, PCBP treatment is much more concerned with problem resolution than with explication of hypothetical underlying psychogenic causes or constructing elaborate theories about such causes. Additionally, PCBP is concerned mostly with problems that have not devolved sufficiently to fit major diagnostic categories, such as suicidal behavior, Major Depressive Disorder, Conduct Disorder, Bipolar Disorder, Posttraumatic Stress Disorder, or psychosis. These major disorders and conditions are the ones most likely to generate complex theorizing about mental illness, and they are generally outside the domain of PCBP. Nonetheless, it is important that PCBP practitioners be able to recognize major disorders so that they can refer clients to professionals who do specialize in the appropriate evaluation and treatment. The therapy provided for those problems that remain in PCBP typically involves two intersected kinds of therapy: supportive health education and prescriptive behavioral treatment.

Supportive Health Education

Supportive health education involves provision of information about the appropriate classification and developmental, learning, and prognostic context of presenting problems. As an example, there is a widespread cultural ignorance and misunderstanding of the etiology of elimination disorders, and belief in characterological and/or psychopathological defects as major contributing factors has historically been widespread (Friman, 1986, 2002). Relatedly, incontinence is a major contributor to child abuse (Helfer & Kempe, 1976; Kempe, Helfer, & Krugman, 1997; Schmitt, 1987). To address this problem, PCBP supplies health education about elimination processes, ordered and disordered, summarized nicely by the term "demystification" first offered by Levine (1982) and now used by multiple others (e.g., Christophersen & Friman, 2004; Friman, 1986; Friman & Jones, 1998, 2005). As another example, supportive health education often involves the reassurance that many child behaviors reported in primary care are actually developmentally normal and will resolve with the tincture of time. For example, common problems such as thumb sucking, periodic separation anxiety, and crying at bedtime in 3-year-olds, although often of concern to parents and disruptive of the family (Earls, 1980; Costello & Shugart, 1992; Christophersen, 1982, 1988; Christophersen & Rapoff, 1979), are usually benign and not deviant from a developmental perspective. If such

problems perpetuate beyond certain age limits, substantially worsen, or begin to cause health concerns, supportive health education is insufficient, and prescriptive behavioral treatment recommendations are provided.

Prescriptive Behavioral Treatment

If a behavior problem does not resolve following the provision of supportive health education and does not involve a major psychiatric condition (e.g., major depression) at the boundary of PCBP, practitioners recommend a series of therapeutic steps for the parents and child to follow at home (or school). The therapeutic recommendations typically emphasize procedure over process, and most procedures used are derived from the more pragmatic parts of the behavioral sciences, particularly those focused on learning and development (Blum & Friman, 2000; Christophersen, 1982; Friman, 2005a; Friman & Blum, 2003; Friman & Piazza, in press). Practitioners of PCBP tend to interpret behavior problems not as representative of pathologies per se, but as representative of skill deficits whose signature behaviors are currently aggravating health and development and that could devolve to actual pathologies if not remedied. Treatments that work are valued for their own sake, and their importance is not diminished because they are at odds with this or that theory. Efficiency, effectiveness, and acceptance are valued over and above theoretical consistency, precision, and scope.

Treatment, although predominantly verbal and thus consistent topographically with traditional child psychotherapy, differs from it in at least two important ways. First and perhaps most fundamental, parents (or primary caregivers) rather than children are often the direct recipients of treatment (i.e., supportive health education and the recommendations that make up the prescriptive behavioral treatment regimens). Children are, of course, the ultimate recipients of PCBP treatment regardless of its form, and they are often present during its delivery and even participate in its preparation. But the fundamental vehicle for PCBP therapy is educational and/or prescriptive advice that pertains to the parent portion of parent-child interactions. Although the comparison is not perfect, it may be helpful to view PCBP therapy as a specialized form of parent training. The critical point, however, is that although child problems are the reason for PCBP therapy, parents are the proximal recipients of the therapeutic advice pertaining to those problems.

Second, because of limitations on time and an emphasis on procedure in pediatric settings, PCBP treatments are often brief and protocol-driven. In this respect they differ substantially from the process-based, time-intensive interventions that characterize traditional child psychotherapy. However, PCBP therapy is consonant with the increasing emphasis on empirically supported treatment and manualized practice in the contemporary practice of psychotherapy (e.g., Christophersen & Mortweet, 2001). The therapeutic armamentarium of the PCBP therapist includes a variety of procedures, each with abundant empirical support, including (but not limited to) time-out, contingency management, home-school notes, simple token economies, and various empirically supported procedures for simple habits, chronic incontinence, bedtime struggles, feeding problems, mild depression, and most of the anxiety disorders.

WHO ARE THE PRACTITIONERS?

Because of its preventive emphasis, clinical expertise in the major psychiatric problems of childhood is not a prerequisite for practitioners in the field. In some respects, such expertise could even handicap practitioners by predisposing them to interpret routine child problems as evidence of a latent or occult pathology (Friman, 2002). For example, hair play, twirling, and pulling in toddlers, although potentially problematic and certainly important enough to address in a behavioral pediatric visit, is not necessarily reflective of psychopathology or indicative of true Trichotillomania (Drews & Friman, 2003; Friman & Hove, 1987). Thumb sucking in preschoolers is more likely a benign source of self-soothing than a malignant sign of oral fixation or regressive personality disorder (Friman, Byrd, & Oksol, 2001). Soiling in young children is much more likely to result from constipation than from psychic mechanisms such as resentment, regression, or anal fixation (Field & Friman, 2006; Friman, 2003). Daytime wetting in first graders is more likely due to incomplete training than insufficient personality (Christophersen & Friman, 2004). Nocturnal wetting is almost always due to developmental variables (e.g., reduced functional bladder capacity) and only rarely to psychological variables suggestive of pathology (Friman, 1986, 1995, in press-a; Friman & Jones, 1998, 2005). There are many other examples.

Because highly specialized psychiatric expertise is unnecessary for effective practice, and because the child problems that occupy the intersection between MHS and PMC are diverse, the field of PCBP is professionally eclectic. Thus primary care physicians (e.g., pediatricians, family practitioners) can specialize in PCBP just as readily as pediatric, school, and clinical child psychologists, and there are specialized training programs for various types of professionals. One essential qualification for all PCBP practitioners is some knowledge of the biological variables that are functionally related to child behavior problems. The primary reason this specialized knowledge is needed is that several behavioral problems that respond well to PCBP intervention involve significant medical (i.e., biological) dimensions. The overall field includes the evaluation and treatment of at least three categories of problems: (1) routine behavior problems, (2) behavior problems with significant medical dimensions, and (3) medical problems with significant behavioral dimensions. Although a fourth category can plausibly be described, one that emphasizes the interaction between medical and behavioral issues more than is typical of categories 2 and 3 (e.g., Blum & Friman, 2000; Friman, 2005a; Friman & Piazza, in press), this chapter focuses only on these three categories, presenting a descriptive PCBP analysis of and therapeutic approach to a representative problem from each area.

COMMON BEHAVIOR PROBLEMS IN PRIMARY CARE SETTINGS

As indicated, a substantial portion of children seen in PMC have symptoms that meet criteria for a behavioral or emotional disorder (e.g., American Academy of Pediatrics, 2001; Costello, 1988; Costello, Burns, et al., 1988; Costello, Edelbrock, et al., 1988; Costello & Shugart, 1992; Dulcan et al., 1990; Goldberg et al., 1984;

Hickson et al., 1983; Lavigne et al., 1993; Starfield et al., 1980), and a much larger percentage exhibit behaviors or emotions that cause their parents concern and/or cause some functional impairment for the child but do not meet criteria for a disorder (American Academy of Pediatrics, 2001; Costello & Shugart, 1992; Earls, 1980; Christophersen, 1982, 1988; Friman, 2002). The problems parents report to PMC providers vary with the age and developmental level of their children. During infancy, parents are most likely to report concerns about their children's excessive crying and problems initiating and maintaining sleep. During the preschool years, parents are most likely to report concerns about their children's oppositional behavior, toileting practices (especially resistance), limited attention span, overly selective eating, and fears or worries (Earls, 1980; Lavigne et al., 1993). During the school years, parents continue to report concerns about problems first encountered during the preschool years that have persisted, as well as new concerns about academics, school behavior, and peer problems (Kanoy & Schroeder, 1985).

Many of these behavior problems will ultimately resolve without the direct intervention of a professional. For this reason, PMC providers often initially describe presenting problems as fitting within the range of normal expectations and recommend that parents let their children grow out of it. The "tincture of time" approach, however, does little to address the substantial stress that behavior problems can place on a family while they are occurring, a fact that is highlighted by the increased risk of child abuse that occurs in association with many problem behaviors (Helfer & Kempe, 1976; Kempe et al., 1997; Schmitt, 1987). Furthermore, children do not always grow out of their behavior problems, and, in most cases, it is very difficult to distinguish those who will from those who will not. Without treatment of problems that persist, children can have serious consequences later in life. For example, untreated, persistent oppositional behavior can devolve into major conduct problems that require extraordinary therapeutic interventions for remission (Caspi & Moffitt, 1995). Unresolved toileting problems can lead to serious medical problems, such as megacolon, urinary tract infection, and the unstable bladder of childhood (Christophersen & Friman, 2004; Friman, 1986, 1995; Friman & Christophersen, 1986; Friman & Jones, 1998). Untreated bedtime problems can lead to habitually disrupted sleep patterns, family discord, and child maltreatment (Blampied & France, 1993; Blum & Carey, 1996; Edwards & Christophersen, 1994; Ferber, 1985). Unresolved school problems can lead to incomplete education, truancy, and school failure, which, in turn, are instrumental in the development of delinquency, drug use, and ultimately criminal behaviors (Friman, 2000). There are many other examples; collectively, they underscore the benefit of brief, problem-specific, effective treatment for children whose behavior problems have persuaded parents to discuss those problems in PMC, even if they involve low-intensity, high-frequency problems such as those just mentioned. The next section discusses one example in detail.

Bedtime Problems

Teaching children to go to bed, go to sleep, and stay asleep throughout the night is difficult for many families in this culture (Blampied & France, 1993; Blum

& Carey, 1996; Edwards & Christophersen, 1994; Ferber, 1985; Friman, 2005c). A representative survey indicated that at least 30% of families contend with this problem three or more nights a week (Lozoff, Wolf, & Davis, 1985). The difficulties reported by parents include bedtime struggles such as resistance to going to bed, fussing and crying while in bed, and night waking with fussing, crying, and unauthorized departures from the bedroom. These problems are important because they disrupt parents' and their children's sleep, which, in turn, decreases the subsequent capacity of both to execute emotional self-control (Teitelbaum, 1977). Although PCBP advice about sleep is needed from the age of 3 months on, its importance is greatly amplified if bedtime struggles continue after 6 months of age. One intervention for child bedtime problems employed by many pediatric providers (and often parents acting on their own) involves medication. Although sometimes effective in the short term, all soporific medication results in unwanted side effects, and sleep-related benefits are usually lost when the medication is withdrawn. Additional information on medication for bedtime problems is available in published reviews of the research on them (e.g., Edwards & Christophersen, 1994).

Treatment for Bedtime Problems

The cardinal component of the most effective behavioral interventions for bedtime problems involves one of the first documented and most frequently used in behavioral psychology: extinction. As children develop sleep habits, they often learn to associate specific environmental factors with self-quieting and the induction of sleep. Misinformed parental efforts to help children sleep often result in problematic sleep associations that mitigate the process of falling asleep. For example, for most children sleep induction is enhanced by the presence of a parent to sooth and cuddle them and thereby ease the transition from wakefulness to sleep. These parental activities can contribute functional status to a constellation of stimuli that influence children's behavior in bed. With the parent present, rapid sleep induction typically occurs. Unfortunately, when the parent is absent at bedtime, the child is left without the stimulus that is most powerfully associated with sleep. The children's response to parental absence typically involves prolonged and intensive crying that resembles an extinction burst (Blampied & France, 1993; Ferber, 1985; Edwards & Christophersen, 1994; Friman, 2005b). This response motivates most parents to intervene either by soothing or disciplining their child, which, unfortunately, usually worsens the problem. Soothing responses to crying can reinforce the crying; disciplinary responses to crying often provokes more crying; and both responses limit the child's learning of self-quieting skills (Blampied & France, 1994; Edwards & Christophersen, 1994; Ferber, 1985; Friman, 2005b; Lozoff et al., 1985). Not surprisingly, after failed attempts to solve the problem themselves, parents whose children exhibit bedtime problems often seek professional advice. The four procedures most likely to be prescribed by PCBP practitioners are described next.

Complete Extinction This approach to bedtime problems involves no visits by the parent to the child's bedroom after the child has gone to bed. In effect, the child

is left to cry it out. Generally, complete extinction works more rapidly than other approaches, but it presents problems that mitigate its overall effectiveness (Adams & Rickert, 1989; Edwards & Christophersen, 1994; Rickert & Johnson, 1988). For example, the child's crying can be highly aversive during the first nights of implementation. If the family lives in an apartment complex, or the treatment is implemented during a season when windows are open, the crying and screaming can draw attention from the neighbors, with predictably problematic consequences. Additionally, extended crying and screaming differentially affects parents. Discord is possible (probable) when crying is substantially more aversive for one parent than the other (Adams & Rickert, 1989; Ferber, 1985). Thus, complete extinction is a straightforward behavioral approach to bedtime problems in children but has limited social validity. In an attempt to improve social validity, sleep researchers have developed other, multicomponent methods that employ extinction but decrease its intensity and aversiveness for parents through the inclusion of other procedures.

Graduated Extinction This procedure involves advising parents to ignore bedtime problem behavior for specific time intervals that gradually increase. The optimal length of the intervals has not been established empirically, but expert advice (Ferber, 1985) and related research studies (e.g., Adams & Rickert, 1989) recommend beginning with a 5-minute interval on the first episode, 10 at the second, and 15 minutes for subsequent episodes on night 1. These intervals increase over the course of a week, ending with 35 minutes for the first episode on night 7, 40 for the second, and 45 minutes for all subsequent episodes and nights. Although children can tantrum for longer than 45 minutes at night, published data (Adam & Rickert, 1989; see also Edwards & Christophersen, 1994) and a large amount of clinical experience described by Ferber (1985) suggests that very few do.

An important question about graduated extinction is why it works. At first glance, it would seem to be the perfect procedure for teaching children to cry gradually longer and longer, culminating in a reliable 45-minute bout reproducible throughout the night. This, however, is not what happens. No research has yet addressed why, and the following explanation is speculation. The procedure is perhaps more appropriately labeled a differential reinforcement schedule than scheduled extinction. The parents visit the room only if their child is crying; thus, visits are contingent. However, increasing the response requirement to 45 minutes of crying may stretch the schedule so much that the reinforcing effects of sleep supersede the reinforcing effects of parental visitation.

Positive Routines This procedure involves a hybrid of extinction and a rewarding bedtime ritual. In the procedure, parents determine when they would like their child to go to sleep and when the child typically falls asleep. Beginning shortly before the time the child typically falls asleep, parents engage the child in several quiet activities lasting no longer than 20 minutes total. During the activities, the parents issue easily followed instructions and richly supply rewarding responses for compliance. In this respect, the procedure resembles a widely studied procedure for increasing instructional control called behavioral momentum (Mace et al., 1988). In positive

routines, as in behavioral momentum, the child is given a series of instructions with a high probability of compliance, followed by a terminal instruction whose probability of compliance is low. In this case, the terminal instruction is "Now stay in bed and go to sleep," or something equivalent. If at any time after the completion of the routines and the terminal instruction the child leaves the bed, the parent places the child back in bed, telling him or her that the routine is over and it is time for bed. Crying and verbalizations are ignored. At specified intervals (e.g., 1 week) the parents move the positive routine back in time 5 to 10 minutes. They continue this backward movement until they arrive at the bedtime they prefer for their child. For example, the child may typically fall asleep at 9:15, and the parent's preferred bedtime might be 8:15. Thus, it could take between 6 and 8 weeks for parents to arrive at the preferred bedtime. Experimental comparison of the positive routines procedure with scheduled extinction showed that both produced substantially improved bedtime behavior for children, but that the parents using positive routines reported significantly improved marital relations, suggesting a more socially valid procedure (Adams & Rickert, 1989).

The Bedtime Pass The bedtime pass program involves (a) requiring that children get into bed; (b) providing them with a small, laminated note card exchangeable for one free trip out of the bedroom or one visit to them by the parent after being put to bed in order to have a small request satisfied (e.g., for a drink, hug, or visit to the bathroom); (c) surrender of the pass after it is used; and (d) extinction thereafter. In the initial study, the program eliminated the high rates of crying out, calling out, and coming out of the bedroom after bedtime exhibited by two children, ages 3 and 10 years. Additionally, these successful results were achieved without an accompanying extinction burst during initial intervention periods, and a large group of sample parents rated the intervention as more acceptable than extinction alone (Friman et al., 1999). The pass program was subsequently replicated in a single-subject analysis of four 3-year-old children (Freeman, 2006) and a randomized trial involving nineteen 3- to 6-year-old children (Moore, Friman, Fruzetti, & MacAleese, 2007).

Two plausible possibilities for how the pass achieves its effects have been proposed. First, the pass program is analogous to differential reinforcement of alternative behavior (with use of the pass as an alternative to crying, calling, or coming out from the bedroom and the parents' response as reinforcement), which is a well-established method for reducing disruptive behavior (Vollmer & Iwata, 1992). Second, the use of the pass involves the reinforcement of a request made by the child. A small line of recent research shows that reinforcing appropriate behavior in disruptive individuals and subsequently placing disruption on a course of extinction can reduce the disruption without producing an extinction burst (e.g., Bowman, Fisher, Thompson, & Piazza, 1997).

This brief discussion of child bedtime problems and their treatment is by no means complete. It merely describes a PCBP analysis and four representative PCBP interventions that are the most frequently used and have the most empirical support. There are other interventions used by parents (and prescribed by some professionals) but usually avoided in PCBP (e.g., medication, the family bed; see Friman,

2005b, for description) as well as others that almost certainly will become part of the PCBP armamentarium if they generate sufficient empirical support. Examples include the sleep fairy intervention, a variation on the tooth fairy (Burke, Kuhn, & Peterson, 2004), and the teaching of self-quieting skills during the day to promote a better transition to bed at night (Edwards, 1993; Edwards & Christophersen, 1994).

The general premise of this section is that child bedtime concerns represent a very important and common presenting complaint in PMC, one that is not appropriate for conventional medical evaluation and not optimally responsive to conventional medical treatment (e.g., medication). The vast majority of bedtime concerns fit within the category of common behavioral problems: They are high-frequency, low-intensity presenting complaints. As with most common behavior problems, bedtime problems have a fundamental behavioral and learning basis; thus, not surprisingly, they respond well to behavioral- and learning-based evaluation and treatment. Yet conventional pediatric training does not emphasize this type of evaluation and treatment, and even if it did, most pediatricians would not have sufficient time during the average visit to accomplish it (cf. Christophersen, 1982).

As indicated, PCBP research has contributed several behavioral- and learning-based treatments for bedtime problems; however, the research agenda is far from complete, especially in light of the extent of the problems in this culture. For example, little research on the actual function of bedtime problems has been reported and, although the assumption that the primary function is attention seeking has served the field well, there may be other functions. Additionally, although the treatment procedures described here extend the clinical effectiveness of PCBP practitioners, new procedures, especially those with high social validity, are needed. More generally, new effective and socially valid treatments are needed for all of the common behavior problems presenting in pediatric settings, and PCBP is uniquely well positioned to address that need.

BEHAVIOR PROBLEMS WITH SIGNIFICANT MEDICAL DIMENSIONS

A fundamental assumption of PCBP is that behavior occurs mostly as a function of environmental circumstances, but this assumption does not disavow the influence of medical (i.e., physiological) variables. Rather, many physiological variables are indeed seen as determinative, but are themselves the result of environmental behavior contingencies that occurred in a phylogenetic context (Skinner, 1966). Physiological variables often play an initiating role in behavioral problems that present in PMC. For example, an immune response can lead to a rash, a rash to scratching, scratching to tissue abrasions, and tissue abrasions to scratching unrelated to the rash. Child stomach pain brought on by physiological variables (e.g., flu virus) can lead to missed school. While the child is home from school, the child's behavior is influenced by reinforcers involving avoidance of schoolwork as well as contact with sympathetic responses from caregivers. These influences can, in turn, result in complaints of stomach pain that do not involve physiological variables, a condition sometimes referred to as recurrent abdominal pain (Finney, Lemanek, Cataldo, Katz, & Fuqua, 1989). Hair loss caused by medical problems

(e.g., alopecia areata) can lead to social avoidance, the extended duration of which can lead to a deterioration of social skills that can, in turn, lead to social avoidance even after hair has grown back. There are many other examples; the one elaborated upon here involves constipation, a physiological variable that can cause common toileting problems, the most serious of which is retentive encopresis.

Retentive Encopresis

Descriptive Features

Definition Encopresis, a common presenting complaint in pediatrics (3% to 5% of all referrals), is a disorder in which children either voluntarily or involuntarily pass feces into or onto an inappropriate location, usually their clothing (Friman & Jones, 1998; Wright, 1973; Wright, Schaefer, & Solomons, 1979). Encopresis is not diagnosed if the problem is exclusively due to an anatomic or neurologic abnormality that prevents continence. The current criteria from *DSM-IV* (American Psychiatric Association, 1994) are (a) inappropriate passage of feces at least once a month for at least 3 months, (b) chronological or developmentally equivalent age of 4 years, and (c) not due exclusively to the direct physiological effects of a substance (e.g., laxatives) or a general medical condition, except through a mechanism involving constipation. The *DSM-IV* distinguishes two subtypes of encopresis, one with constipation and overflow incontinence (i.e., retentive encopresis) and one without these symptoms (i.e., nonretentive encopresis). For reasons that will be discussed later, this section focuses only on encopresis with constipation.

Relevant Physiology The large intestine, or colon, is the distal end of the alimentary tract, which is sequentially composed of the esophagus, stomach, small intestine, and colon. A thorough review of the colonic system is beyond the scope of this chapter (for more thorough reviews, see Weinstock & Clouse, 1987, or Whitehead & Schuster, 1985). Some rudimentary description of the system, however, is necessary to understand the physiology that supplies the logic of effective treatment. The colon is a tubular organ with a muscular wall. It connects the small intestine to the rectum and anus. It has three primary functions: fluid absorption, storage, and evacuation. Extended storage and planned evacuation are the defining features of fecal continence. Movement of waste through the colon is achieved through muscular contractions called peristalsis, which produce a wave-like motion of the colon walls. As the waste moves through the colon moisture is absorbed from it, creating semisolid feces. Movement through the colon is potentiated by a variety of external events that instigate muscular contractions in the colonic wall. For example, eating a meal increases colonic contractions (referred to as the gastrocolonic reflex), and moving about will have a similar effect (referred to as the orthocolonic reflex).

Usually the rectum, a hollow receptacle at the distal end of the large colon, contains little or no feces, but prior to defecation muscular contractions in the colonic wall propel feces into the rectum. This results in distension, which stimulates sensory receptors in the rectal mucosa and in the muscles of the pelvic floor, resulting in relaxation of the internal sphincter, which facilitates defecation. This process is involuntary, but the child can constrict the anal canal and inhibit

defecation by contracting the external anal sphincter and the functionally related puborectalis muscle. When the urge to defecate is suppressed this way, the rectum accommodates the retained stool through the adaptive pliance of its structure and terminates the reflex relaxation of the internal sphincter. The urge gradually decays, and some of the fecal matter in the rectum is returned to the descending colon by retroperistalsis.

Etiology Between 80% and 95% of encopresis cases can be traced to a primary causal variable: constipation (Hatch, 1988; Levine, 1975, 1982; Wright et al., 1979). Although definitions for constipation vary, children who frequently go two or more days without a bowel movement are probably prone to constipation. A common complaint by the parents of encopretic children is that they deliberately soil their clothing (Wright et al., 1979), but this accusation is usually false (Levine, 1982). The primary cause of soiling is fecal retention (constipation). In most cases, retention is not caused by characterological or psychopathological problems (Friman, Mathews, Finney, & Christophersen, 1988; Gabel, Hegedus, Wald, Chandra, & Chaponis, 1986). Retention is usually the result of a constellation of factors, many of which are beyond a child's immediate control (Levine, 1982). These factors include a constitutional predisposition (i.e., slow gastrointestinal transit time), diet, insufficient leverage for passage of hard stools, and occasional or frequent painful passage of hard stools, resulting in negative reinforcement for holding stools (Christophersen & Rapoff, 1983). Although retention is frequently attributed to sexual abuse, an association between fecal retention and accidents has not been established (e.g., Mellon, Whiteside, & Friedrich, 2006). For some children, especially those with extreme constipation and/or treatment failure, there is an increased threshold of awareness of rectal distension, a possibly weak internal sphincter, and/or a tendency to contract the external sphincter during the act of defecation (Meunier, Marechal, & De Beaujeu, 1979; Wald, Chandra, Chiponis, & Gabel, 1986). The combined effect of all these factors is a lowered probability of voluntary stool passage and a heightened probability of fecal retention.

Chronic fecal retention results in fecal impaction, which results in enlargement of the colon. Colon enlargement results in decreased motility of the bowel system and, occasionally, in involuntary passage of large stools and frequent soiling due to seepage of soft fecal matter. The seepage is often referred to as paradoxical diarrhea because the children retain large masses of stool and thus are functionally constipated, but their colon allows passage of soft stool around the mass, which results in diarrhea (Christophersen & Rapoff, 1983; Levine, 1982).

That fecal impaction is related to retentive encopresis has been established by several investigators, primary among whom are Davidson (1958), Levine (1975), and Wright (1975). All independently reported that 80% of their patients had fecal impaction accompanying fecal incontinence at the first clinic visit. Subsequent to his 1975 report, Levine and his colleagues developed a simple clinical procedure to identify fecal impaction from an X-ray of the lower abdomen (Barr, Levine, Wilkinson, & Mulvihill, 1979). As a result of the improved diagnostic method, Levine revised his initial 80% estimate of fecal impaction's coexistence with fecal incontinence to 90% (Christophersen & Rapoff, 1979).

Differential Diagnosis There are rare anatomic and neurologic problems that can lead to fecal retention and soiling. Anatomic problems include a variety of malformations and locations of the anus that are detectable on physical exam and require medical management (Hatch, 1988). Hirschsprung's disease or congenital aganglionosis is a disorder in which the nerves that control the muscles in the wall of the colon or in an entire section of the colon are absent, causing severe constipation. Its incidence is approximately 1 in 25,000, and it usually causes severe symptoms in infancy (Levine, 1975). Thus the clinical presentation itself should prevent the astute clinician from mistaking one for the other. The possible exception is ultrashort segment Hirschsprung's disease, which has a more subtle clinical picture. However, the existence of this condition is controversial, and, even if it does exist, proper collaboration between pediatrician and behavior analyst should ensure timely diagnosis.

Evaluation

The initial PCBP evaluation should begin with a physical examination that usually includes a routine check of history, abdominal palpation, rectal examination, and sometimes an X-ray of the abdomen to determine the extent of fecal impaction. A barium enema is rarely necessary unless features of the exam suggest Hirschsprung's disease. The evaluation should also include a behavior assessment that begins with a behavioral interview that involves critical questions related to constipation. These include asking whether there is ever a long period between bowel movements, whether bowel movements are atypically large (stop up the toilet), whether fecal matter ever has an unusually foul odor, whether fecal matter is ever hard, difficult, or painful to pass, and whether the child ever complains of not being able to feel the movement or make it to the toilet on time. An additional question that pertains more to treatment history than to pathogenesis is whether the child ever hides soiled underwear. Affirmative answers to any or all of these questions are highly suggestive of retentive encopresis, and hiding underwear indicates a history that includes some form of punishment.

The encopresis evaluation is usually the first step in treatment because it always includes a substantial amount of supportive health education. Encopresis is not well understood outside of the medical community, and the child's parents are likely to be under the influence of the characterological and psychopathological interpretations that are prevalent in Western culture (e.g., Friman, 2002; Horwitz, 2002). The parents' interpretation of the condition is also likely to influence how the children view their problem. Thus, the encopresis evaluation can actually begin treatment by providing accurate information that demystifies the problem. Last, the evaluation should include questions about diet and timing of meals. Low-fiber diets and irregular meals can be contributing factors in encopresis.

Treatment

During the past 15 years, several descriptive and controlled experimental studies have supported a multicomponent approach to treatment of chronic retentive encopresis partly derived from the pioneering work of Davidson (1958), Christophersen

and Rainey (1976), Levine (1975), and Wright (1975). As indicated earlier, the first component can be addressed within the evaluation. Specifically, the entire elimination process, including its disordered manifestations, should be demystified (Christophersen & Rapoff, 1983; Levine, 1982). Generally this means providing information about bowel dynamics and the relationship of the problem to constipation (Levine, 1982). Second, if there is a fecal impaction it should be removed with enemas and/or laxatives (Christophersen & Rapoff, 1983; Levine, 1982; O'Brien, Ross, & Christophersen, 1986). Third, the child should sit on the toilet for about 5 minutes one or two times a day (O'Brien et al., 1986; Wright, 1975). Fourth, the parents should promote proper toileting with encouragement and not with coercion. Additionally, they should not reserve all of their praise and affection for proper elimination; a child should be praised just for sitting on the toilet (Christophersen & Rapoff, 1983; Levine, 1982; Wright, 1975). Fifth, a stool softener such as mineral oil (Davidson, 1958) or glycerin suppositories (O'Brien et al., 1986; Wright & Walker, 1977) should be used to ease the passage of hard stools. Sixth, dietary fiber should be increased in the child's diet (Houts, Mellon, & Whelan, 1988; O'Brien et al., 1986). Seventh, to increase and maintain motility in the child's colon, the child's activity levels and fluid intake should be increased (Levine, 1982). Eighth, during toileting episodes the child's feet should be on a flat surface. Foot placement is crucial to the Valsalva maneuver (grunting push necessary to produce a bowel movement; Levine, 1982; O'Brien et al., 1986). And ninth, the child should be rewarded for all bowel movements in the toilet (Christophersen & Rainey, 1976; Levine, 1982; O'Brien et al., 1986; Wright & Walker, 1977).

The literature on this approach (or variations thereof) has progressed sufficiently to lead to group trials. For example, in a study of 58 children with encopresis, 60% were completely continent after 5 months, and those who did not achieve full continence averaged a 90% decrease in accidents (Lowery, Srour, Whitehead, & Schuster, 1985). There are multiple other examples that document similar levels of success (e.g., Stark et al., 1997). However, not all children succeed with the conventional approach; for these children, augmentative methods have been developed. These methods developed through study of behaviors associated with treatment failure (Stark, Spirito, Lewis, & Hart, 1990). Incorporating behavior management methods relevant to the behaviors, teaching parents to use them, and delivering treatment in a group format resulted in an 83% decrease in accidents in 18 treatment-resistant children with encopresis, and the results were maintained or even improved at 6-month follow-up (Stark, Owens-Stively, Spirito, Lewis, & Guevremont, 1990).

The general premise of this section—that medical variables can influence or cause behavior problems—is not controversial, even in sciences such as behavior analysis that are devoted to the study of environmental cause (e.g., Friman & Piazza, in press). Constipation is one such variable, and it can cause behavior problems ranging from minor toileting resistance to encopresis, and there are many others (e.g., anorexia due to gastroesophogeal reflux, restricted activities due to pain). Because of the medical component of these problems, there are frequently serious health consequences for unsuccessful treatment that compound any

psychological or behavioral components. For example, extreme fecal retention can be life-threatening, and even routine cases seriously decrease social standing and increase social distancing in affected children. Because of the behavioral component of these problems, a solely medical intervention is insufficient for effective treatment. Needed are treatments that deliver or aid the delivery of the medical components of treatment while addressing the behavioral components. Thus, PCBP is an appropriate context. Additionally, although there are many types of behavior problems that stem from physiological influences, for the medical dimensions the most frequently occurring problem is some form of noncompliance with a treatment regimen. As an example, cooperation with prescribed treatment for encopresis is so necessary for success that instructional control training is frequently either a component of treatment or a prerequisite for it (Friman, 2003). Furthermore, although reasonable progress has been made in improving pediatric compliance, encopresis remains one of the most chronic problems in all of pediatric medicine.

MEDICAL PROBLEMS WITH SIGNIFICANT BEHAVIORAL DIMENSIONS

The immediately preceding section discussed the influence of medical variables on child behavior problems, with emphasis on constipation and its influence on toileting resistance and retentive encopresis, problems that are frequently seen in PMC. This section discusses the influence of behavior on medical variables with emphasis on the behavioral treatment of physiologically based behavior problems that are also frequently seen in PCBP. For decades health-based sciences have demonstrated relationships between child behavior and health. For example, eating nutritious food and engaging in modest exercise can improve child cardiovascular health. Obtaining sufficient sleep can help sustain immunological responses, emotional resiliency, and adaptability. Maintaining adequate personal hygiene can decrease child susceptibility to infectious disease. There are many other examples. More recently, PCBP has demonstrated a variety of healthful outcomes from changes in behavior (e.g., Finney, Miller, & Adler, 1993; Friman & Christophersen, 1986; Irwin, Cataldo, Matheny, & Peterson, 1992; Stark et al., 1993). A review of this literature is beyond the scope of this chapter, and thus the focus here is on biofeedback, a treatment involving the manipulation of behavioral variables to improve health; treatment of nocturnal enuresis is offered as the primary example.

Nocturnal Enuresis

Biofeedback Perspective

Biofeedback involves the use of electrical or electromechanical equipment to measure and increase the salience of stimuli associated with pertinent physiologic processes and training patients to discriminate and control them in order to improve their own health. The penultimate goal of biofeedback is to train individuals to alter the physiologic processes in healthful directions, and the ultimate goal is to train them to do so without biofeedback (Culbert, Kajander, & Reaney, 1996; Friman, in press-b). Most biofeedback-type treatments require sophisticated instrumentation

and specialized training and thus may not be incorporated readily into primary care practices (e.g., anorectal manometry combined with electromyography for treatment of fecal incontinence). However, nocturnal enuresis, a physiologically based behavioral problem that is one of the most frequent presenting behavioral complaints in PMC, is highly responsive to urine alarm treatment, a minimally technical, uncomplicated form of biofeedback that is readily used in PMC (Christophersen & Friman, 2004; Friman, 1986, 1995, in press-a; Friman & Jones, 1998, 2005). In fact, urine alarm treatment is the most empirically supported treatment for nocturnal enuresis, by a wide margin, despite more than 50 years of research devoted to discovery of effective physiologically based treatments (e.g., medication). There are at least two medications (DDAVP and imipramine) that can improve enuresis, but with significantly smaller magnitude and duration of effects than the urine alarm. This section discusses descriptive features of nocturnal enuresis as well as its evaluation and treatment.

Descriptive Features

Definition The current criteria for enuresis (nocturnal and diurnal) from *DSM-IV* (American Psychiatric Association, 1994) are (a) repeated urination into bed or clothing; (b) at least two occurrences per week for at least 3 months or a sufficient number of occurrences to cause clinically significant distress; (c) chronological age of 5 or, for children with developmental delays, a mental age of at least 5; and (d) not due exclusively to the direct effects of a substance (e.g., diuretics) or a general medical condition (e.g., diabetes). There are three subtypes of enuresis: nocturnal only, diurnal only, and mixed nocturnal and diurnal. There are two courses: The primary course includes children who have never established continence, and the secondary course involves children who, after establishing continence, resume having accidents. This section primarily discusses nocturnal enuresis (although some discussion of diurnal enuresis is supplied), which is estimated to occur in as many as 20% of first-grade children (Christophersen & Friman, 2004; Friman, 1986, 1995, in press-a; Friman & Jones, 1998, 2005).

Relevant Physiology The bladder (detrusor) is an elastic hollow organ with a muscular wall. Its shape resembles an upside-down balloon with a long narrow neck; it has two, primarily mechanical functions: storage and release of urine (Vincent, 1974). Extended storage and volitional release are the defining properties of urinary continence. In infancy, distension of the bladder leads to contraction of the bladder and automatic (nonvolitional) evacuation of urine. As children mature, the capacity of the central nervous system to inhibit bladder contraction increases, typically coinciding with the development of continence in early childhood (Berk & Friman, 1990; Koff, 1995).

The components of the urogenital system that are under volitional control and that are necessary for continence skills are the muscles of the pelvic floor. Except during imminent or actual urination, these muscles remain in a state of tonus or involuntary partial contraction, which maintains the bladder neck in an elevated and closed position (Vincent, 1974). Even after initiation of urination has begun, contraction of the pelvic floor muscles can abruptly raise the bladder neck and terminate urination.

But for children with nocturnal enuresis, these urinary inhibitory responses are either not present or are sporadic (Friman, 1995; Friman & Jones, 1998; Houts, 1991; Mellon, Scott, Haynes, Schmidt, & Houts, 1997).

Etiology Although nocturnal enuresis has a strong genetic basis, its exact cause is unknown. For decades there has been a sustained effort to link it to causal psychopathology, but contemporary research (Friman, Handwerk, Swearer, McGinnis, & Warzak, 1998), as well as several reviews of older research (Friman, 1986, 1995, 2002; Friman & Jones, 1998), suggest that most enuretic children do not exhibit clinically significant psychopathology, and when they do, it is more likely to be an outcome than a cause of nocturnal enuresis. Physiologically oriented studies of nocturnal enuresis suggest that some affected children may have difficulty concentrating their urine during the night and, thus, produce more urine nocturnally than their nonenuretic peers (Lackgren, Neveus, & Stenberg, 1997; Rittig, Knudsen, Norgaard, Pedersen, & Djurhuus, 1989). The overall importance of this factor, however, is controversial because the proportion of enuretic children with urine concentration problems may be small (Eggert & Kuhn, 1995). Finally, nocturnal enuresis may be viewed as merely a deficit in the skills necessary to prevent urination while asleep (Houts, 1991).

Evaluation

As with encopresis, the initial stage of an enuresis evaluation should include a physical examination. Once a thorough history has been obtained and preliminary information about enuresis has been shared with the parents and child, PCBP goes no further until all possible pathophysiological causes of enuresis have been ruled out. There are several such causes, some of which are common (e.g., urinary tract infection) and some rare (e.g., diabetes), and they must be ruled out before treatment begins. There are several other elements necessary for a full evaluation of enuresis, and these are well documented in multiple other papers (e.g., Christophersen & Friman, 2004; Friman, 1986, 1995; Friman & Jones, 1998; in press-a).

Treatment

The two most common treatments for nocturnal enuresis are the urine alarm and the medications DDAVP and imipramine. These medications can provide symptomatic relief; approximately 25% to 40% of children will be dry most nights when taking them. However, when the medications are stopped, the enuresis usually returns (Moffatt, 1997). The urine alarm is a moisture-sensitive switching system, the closing of which (when the child wets the bed) rings the alarm. Repeated pairing of awakening by the alarm with episodes of wetting is consistently described in the literature as the single most effective treatment for enuresis (Christophersen & Friman, 2004; Friman, 1986, 1995; Friman & Jones, 1998, in press-a). Its success rate is higher (approximately 75%) and its relapse rate lower (approximately 41%) than any other drug- or skill-based treatment.

The urine alarm is a simple form of biofeedback treatment because its primary function is to provide feedback for a physiological event, urination, which in

enuretic episodes occurs beneath awareness. The feedback (ringing of the alarm) increases the salience of the urination and, thus, aids in the ultimate establishment of urinary self-control. The mechanism by which the alarm improves enuresis, however, is still unknown. Changes in secretion of hormones that affect the ability to concentrate urine (Friman, 1995; Friman & Jones, 1998; Houts, 1991) or alterations in the brain's inhibition of bladder contraction are at least theoretically possible, but have not been investigated. Early conceptualizations described the mechanism as classical conditioning, with the alarm as the unconditioned stimulus, bladder distention the conditioned stimulus, and waking the conditioned response, but most children successfully treated with the alarm learn to sleep through the night without wetting (or waking). Subsequent conceptualizations emphasized a negative reinforcement paradigm in which the children repeatedly awakened by the alarm avoided it by urinating in the toilet (initially) and (ultimately) holding their urine until a more convenient time. The current prevailing account involves a combination of the two previous accounts with classical conditioning of pelvic floor muscles and operant conditioning of volitional behaviors related to continence (Houts, 1991). The key distinguishing feature of the current account is that children are not necessarily trained to awaken to the alarm; most merely learn to engage their urinary inhibition system even if they are asleep, a skill that would be difficult to teach without the sensory feedback provided by the alarm.

Alarm-based treatment produces cure slowly, and during the first few weeks of alarm use, waking, if it occurs at all, occurs only after a complete voiding. Recent research using the size of the urine stain on the soiled sheets as the dependent measure showed that prior to accident-free nights, the stain grew increasingly smaller on successive nights, suggesting a graduated process of continence attainment (Ruckstuhl & Friman, 2003). In other words, the feedback properties of the alarm gradually but inexorably strengthened the skills necessary to avoid it. The core skill involves contraction of the pelvic floor muscles, resulting in sustained elevation of the bladder neck, resulting in the cessation or, preferably, prevention of urination. There is also the possibility of increased sensory awareness of urinary need and waking to urinate, but this is a less likely outcome and actually inferior to sustained, accident-free sleep throughout the night.

However, increasing sensory awareness of urinary need prior to daytime accidents is also a key component in the most empirically supported treatment for diurnal enuresis. Only two studies are available (very little research has been conducted on diurnal enuresis), and the first used a much simpler conceptualization (Halliday, Meadow, & Berg, 1987). Specifically, this early study merely suggested that the alarm served as a reminder to urinate. A subsequent study utilizing the biofeedback conceptualization was conducted with a young girl who was initially unresponsive to urinary urge and onset, but who rapidly became responsive with use of the alarm (Friman & Vollmer, 1995). The decreasing latency between alarm onset and appropriate response was characteristic of learning curves resulting from alarm-based treatment for nocturnal enuresis and biofeedback treatments in general.

As indicated earlier, most biofeedback treatments are much more technically complex than the urine alarm, and they are used for a broad range of physiologically

based behavioral concerns that often initially present in PMC. Among the physiological processes that can be monitored are muscle tension, skin temperature, respiratory rate, blood pressure, and skin moisture (perspiration). Biofeedback devices sensitive to these processes have been used in treatment of a wide variety of disorders, including headaches, other varieties of chronic pain, asthma, bruxism, anxiety disorders, sleep disorders, and dysfunction of the autonomic nervous system (Culbert et al., 1996). Additionally, evidence is mounting that biofeedback can generate operant responses that, in turn, lead to control over physiologic processes long thought to be outside of volition, such as skin temperature and blood pressure. For example, verbally based awareness-enhancement methods have been shown to alter the level of mediators of the immune system in saliva (Olness, Culbert, & Uden, 1989) and to decrease the recurrence of chronic mouth ulcers (Andrews & Hall, 1990). Collectively, the large body of research documenting the effectiveness of the urine alarm, along with the even larger literature on the effectiveness of biofeedback treatment for a broad range of medical conditions, underscores the research and clinical potential represented by the influence of behavioral variables on physiology.

Case Illustration: A Routine Case of Bedtime Problems

Identifying Features of the Client

Tom was an 8-year-old Caucasian boy who lived at home with his natural parents, his 12-year-old brother, Bill, and his 3-year-old sister, Suzie. His medical, developmental, psychiatric, and educational histories were unremarkable. He attended a local parochial school and was in the third grade. According to his teachers and his parents his grades were excellent and his behavior was very good. For example, the school used a demerit system and he had received only one demerit at the midyear point and that was for misplacing an assignment. At school, he had several good friends, and during his initial clinic session he identified three of them as best friends. When asked what he liked about his best friends, he mentioned that they were nice to him, funny, and shared similar interests.

Presenting Complaint

Tom's presenting complaint included disruptive emotional behavior after bedtime. Approximately 4 weeks prior to the initial visit, Tom began waking up 2 to 3 hours after bedtime and crying out for his parents or coming into their room to summon them if they did not heed his call. When asked about the nature of his distress, he typically used a range of complaints, including nightmares, upset stomach, noises in his bedroom, and worry over school the following day. In response, the parents either allowed him to sleep in their room until the following morning or took him back to his room, at which point one parent would lie with him until he fell asleep. These responses to his distress actually appeared to worsen rather than improve his nighttime concerns, and by the time of the initial appointment, he had begun exhibiting distress at or even prior to bedtime, often refusing to go to bed without being accompanied by a parent.

(*Continued*)

Parental Factors

The parents described their marriage as good and their relationship with all three of their children as loving and supportive. They did exhibit a difference of opinion about how to manage Tom's bedtime concerns. The father thought that too much attention was being paid to these concerns, whereas the mother thought Tom needed the extra attention. Due to Tom's distress, his mother's concern, and the conflict between the parents, the mother presented the problems to Tom's pediatrician, who initially recommended that the parents ignore Tom's nocturnal distress. The parents found this advice impossible to follow because of Tom's extraordinary emotional response, and subsequently the pediatrician recommended they seek services from my office.

During the interview, the parents revealed several potentiating factors, the most significant of which appeared to be a recent move from a small home in a well-established neighborhood to a large home in a new suburb. Tom had shared a bedroom with his sister in his previous home that was very close to their parents' room. He had his own room in the new home, and it was on the opposite end of the second floor from the parents' room. In addition, Tom's mother disclosed that she was an anxious person who had had trouble with bedtime, sleep onset, and nightmares as a child. She also reported that several of her relatives, including her own mother, had histories of anxiety problems. Thus she empathized with her son and said it was very hard for her to ignore him when he was distressed at night. According to Dad, Mom was overly solicitous with Tom not only at bedtime but in response to any form of distress, physical or emotional. Mom reported feeling that Dad was possibly too emotionally distant when it came to addressing Tom's distress and felt that she needed to do extra to supplement Dad's presumed insufficient response. Thus she was the parent targeted by Tom at night.

History

A typical day interview revealed that bedtime was at 8:30 and included a well-established bedtime routine involving hygiene (e.g., bath), dressing (e.g., pajamas), stories, hugs, and lights out. Immediately prior to the initial episode, the bedtime routine had been nonproblematic. However, the parents reported an early history of resistance to bedtime, resulting in periodic stays by Tom in their room. Although their memory of these events was somewhat hazy, they reported that the peak incidence occurred around the age of 3. Additionally, Mom reported that Tom had been mildly colicky. Neither of Tom's siblings had exhibited bedtime problems, and both had their own room and their own routine, which they followed regularly. However, the intensity of Tom's distress at night had begun distressing his siblings at the time of the referral, especially his younger sister. The parents were fearful that she would begin to mimic his responses in order to obtain their attention.

The remainder of the typical day interview revealed little of clinical significance. For example, Tom's morning routine was regularly completed without disruption or instances of significant countercontrol. There was no resistance to the seatbelt on the car ride to school. Upon returning home from school he routinely had a snack, did about 15 minutes of homework, and then played with friends in the neighborhood or his siblings. The family evening meal was a sit-down version, and there were no reported instances of significant misbehavior during it. He

watched approximately 1 hour of television a day and was allowed to play video games for 1/2 hour.

However, he had no household responsibilities that were regularly enforced. For example, the parents had requested that he make his bed on a daily basis and also bring his dishes to the sink after the evening meal, but, on average, he made his bed once a week and it usually had to be redone by his mother. He also rarely brought his dishes to the sink. The parents reported that bedroom time-out was their primary disciplinary method but that they employed it rarely and could not remember a recent instance. As the parents described expectations and discipline, their tone took on a confessional note and they seemed somewhat embarrassed by how easy they had made life for Tom. Nonetheless, behavior screening via the parent version of the Child Behavior Checklist (Achenbach, 1991) did not reveal any significant externalizing problems; the only elevations were in the Anxiety subscale.

Case Conceptualization

My clinical impression of this case was that it involved a form of separation anxiety brought on by problematic adjustment to a recent move, a related dramatically altered bedroom configuration, and nocturnal separation between child and parent. Tom was probably predisposed to anxiety and worry as a function of family history stemming from mother's own history and symptoms that, in turn, had their own family history leading back through a grandparent. The difference of opinion between parents about how best to approach Tom's distress and the limited extent to which household expectations of him had been enforced and discipline had been imposed may have increased his susceptibility to clinically significant experiential avoidance problems such as separation anxiety (Friman, in press-c; Hayes, Wilson, Gifford, Follette, & Strosahl, 1996). In other words, he may not have been prepared to manage the sudden onset of bedtime distress because he had had little experience addressing daytime distress. The progression of his overt distress that was occurring an hour or two after bedtime prior to the referral to distress occurring at, and increasingly before, bedtime at the time of the referral suggests that Tom's condition was worsening and that he was at risk for more generalized problems.

Course of Treatment

The treatment prescribed involved the bedtime pass procedure. In addition, Tom was taught relaxation exercises, to visualize pleasant settings and events when distressed, and controlled breathing. In the bedtime pass portion of treatment, Tom was given a bedtime pass after he had completed his bedtime routine and was in bed. He was informed that he could use the pass to summon his parents to address a simple request or to seek them out for the same purpose. Once the request had been satisfied, he would surrender the pass and spend the remainder of the evening in bed alone. An incentive was added to the program, allowing Tom to turn in unused passes for use in a simple reward program. In that program, Tom identified something that he wanted his parents to purchase for him and that they agreed was appropriate (a video game). Tom and his parents drew a picture of the item using a dot-to-dot format. For each unused pass, Tom was allowed to connect one dot, and when the dots were completely connected, the parents purchased the item for him.

(*Continued*)

Estimate of Progress

Immediately upon implementation of the program, Tom completed three nights without calling out or coming out of his bedroom, thus earning three dots on his dot-to-dot incentive program. On the fourth night, a strong storm moved through the area and Tom called out for his parents and subsequently joined the entire family in the basement because of a tornado warning. Because of the extraordinary nature of this night, he was allowed to keep his pass and turn it in for a dot. However, the next night he again called out for his parents and used his pass even though the weather was calm. Subsequently, he stopped calling out or coming out and earned 12 continuous days of nonpass use. On the 13th night he had a bad dream and called out for his parents and used his pass. In the 3 weeks since that night, he has not used the pass, has earned his reward, and the parents asked if they could discontinue the pass program and transfer the dot-to-dot reward program to other accomplishments.

Complicating Factors

The only complicating factor in this case involved the disagreement between the parents over how to manage Tom's nocturnal distress, and it was resolved easily with the pass program. Tom's enthusiastic response to the pass program and to the exercises he was taught seemed to reassure the parents that they had pursued the right course of action.

Managed Care Considerations

Primary managed care considerations in cases such as this one involve third-party reimbursement. There were no concerns here, however, because Tom met criteria for a covered condition, specifically Adjustment Disorder with Anxiety Features. Only three visits were needed to resolve the problem, and although a fourth was scheduled, it was canceled by the parents by telephone because there was no apparent need. Therefore, the diagnosable condition along with the short number of visits rendered any managed care concerns moot. After the third and final treatment session, follow-up was accomplished by telephone just prior to the scheduled fourth visit, and during the telephone call, the parents expressed no concern. Tom also participated in the call and expressed his excitement about mastering bedtime and also earning his video game.

Implications of the Case

This case represents a classic example of PCBP treatment for routine behavior problems. The problem initially surfaced in PMC and failed a course of treatment provided there. Subsequently, the PMC provider referred the case to a PCBP clinic, where it was quickly classified and effectively treated with just three clinic visits. Additionally, a generalized anxiety condition did not occur, and thus the brief course of treatment could be characterized as preventive. This last point is important because I suspect Tom was at risk for more generalized conditions given his history and the spread of symptoms that had occurred prior to the first visit. Thus, this case represents the fundamental goal of treatment in primary care behavioral pediatrics: prevention.

RECOMMENDATIONS TO CLINICIANS AND STUDENTS

My primary recommendation is to consider following a similar approach to problems of a routine sort that are referred to practitioners by PMC providers. Specifically, routine problems initially presenting in PMC, as was the case with this one, do not necessarily require a comprehensive psychiatric evaluation before treatment is pursued. A simple behavioral screening in conjunction with a good, verbally based diagnostic interview can direct the clinician to prescribe treatment for the vast majority of referral problems or more evaluation for those problems that appear to involve clinically significant dimensions that extend beyond the initial description of the presenting complaint. In the majority of cases where treatment is pursued, the treatment itself becomes a form of assessment because response to it directs the next course of action. In situations where the problem resolves, such as this one, no further clinical care is needed. In situations where it does not, additional assessment and treatment revision are needed.

A second recommendation is to consider modifying treatment protocols in accord with the presentation of the case, even if an empirically supported treatment is used. The safest way to do this is to include all of the components of the empirically supported treatment and add other components in accord with presumed clinical benefit. In this case, the bedtime pass program was used in a fashion consistent with its description in the two experimental small *n* studies (Freeman, 2006; Friman et al., 1999) and randomized clinical trial (Moore et al., 1997) mentioned in the section on common behavior problems, and it was augmented by visualization, relaxation, and a breathing exercise, along with an incentive system. Whether these augmentations added to the active ingredients in treatment or merely enhanced its social validity isn't possible to determine from a case report. Nonetheless, the added components did appear to please Tom as well as his parents and may possibly have made the entire treatment protocol more acceptable. Whether this can be demonstrated empirically is a question for future research.

SUMMARY

The three domains of care—common behavior problems, behavior problems with significant medical dimensions, and medical problems with significant behavioral dimensions—the representative problems used to illustrate them, and the case illustration were supplied to provide an overview of PCBP. The main purpose of this chapter was to present PCBP as a partial solution to problems occupying the gap in behavioral health care between PMC and MHS for children with behavior problems. Because many more providers are needed, another purpose of the chapter was to increase interest in PCBP. Generally, the field favors those whose orientation to practice is guided by science more than art, whose claim to expertise is predicated on empiricism more than clinical or ex cathedra authority, and whose methods are typified more by procedure than process. Thus, there are similarities between PCBP providers and those in some branches of psychology (e.g., behavior therapy, applied behavior analysis, pediatric psychology) but not in others (e.g., psychoanalysis, existentialist psychology, human potential psychology).

Although many of the behavior problems addressed in PCBP are clinically unremarkable, some actually are, resemble, or if unresolved can lead to clinically significant conditions, and thus they require some level of professional assessment prior to intervention. For example, most bedtime problems are functional, but they can chronically resist lay attempts at resolution, resulting in multiple other problems that put not only children at increased risk, but also the parents and family. Furthermore, a small percentage of problems actually involve biologically disordered sleep processes, and these need to be identified and referred to appropriate specialists. As another example, most cases involving child incontinence are also functional, but they can easily be under- or overinterpreted and correspondingly under- or overtreated, resulting in serious health consequences for affected children. Furthermore, a small percentage of cases actually involve pathological processes, and these need to be identified early and referred to appropriate specialists. There are many other examples, generally involving behavior problems resulting from a mismatch between situational requirements for child exhibition of adaptive behavior and/or inhibition of maladaptive behavior and child behavioral skill levels (Blum & Friman, 2000; Christophersen, 1982; Friman, 2005a; Friman & Blum, 2003; Friman & Piazza, in press).

The PCBP perspective on these problems is that most do not involve pathology, many will resolve with time and health education, some will require prescriptive treatment, and all can benefit from some level of professional assessment to determine which is which. As indicated, the problems usually emerge as a function of the friction between child preference and/or skill level and the requirements inherent in socialization and education processes. If unresolved, the problems can devolve into bona fide pathological conditions, but the PCBP perspective is that little scientific or clinical benefit is gained from the assumption that the presence of a behavior problem necessarily occurs as a function of an underlying psychopathology (e.g., Friman, 2002; also see Horwitz, 2002). This perspective, unfortunately, appears to be at odds with the vast majority of psychology and psychiatric literature on child behavior problems, which is focused almost entirely on detection of psychopathology with minimal regard for detection of child health. In fact, very few clinical assessment instruments are even designed to detect healthy child behavior. Behavior assessment instruments used in the research on and practice of child behavior problems ask questions about symptoms or behavior problems, and the typical intent is to determine whether a given child has significantly more symptoms or problems than the children of a similar age in the group used to norm the instruments. The de facto definition of child behavioral health within clinical child psychology and psychiatry appears to involve a composite of symptoms and problems that are below a threshold established for clinical significance, not a composite of healthful behaviors. In other words, appropriate functioning is conventionally viewed as the absence of illness, not necessarily the presence of health.

Because it is much more concerned with prevention than cure or rehabilitation, PCBP provides a health-based alternative to the psychopathological context that pervades mainstream clinical child psychology and psychiatry (Friman, 2002; Horwitz, 2002). This is not to say that PCBP disputes or disregards the possibility of child psychopathology. Standard practice in PCBP involves referral to appropriate

specialists when presenting problems are deemed to be beyond the bounds of primary care or when they prove resistant to interventions attempted with PCBP. This referral practice is as much a part of PCBP as it is any other branch of pediatrics. The essential point here, however, is that the conventional assumption in PCBP is one of child behavioral health until proven otherwise.

Perhaps the best way to view the PCBP approaches to behavior problems is as a first line of defense. To use a medical metaphor, the typical intent is to strengthen children's immunity against threats to behavioral health. The problems that pose such threats typically involve mild to moderate child behavior problems, most of which are responsive to changes in practices by parents, other caregivers, and teachers. Although some of the problems may meet diagnostic criteria for clinical conditions (e.g., enuresis, encopresis, simple phobias, Oppositional Defiant Disorder), the problems are usually in their early stages and are much more responsive to changes in teaching or training practices than problems that have been chronic for years. Thus, even though PCBP provides treatment, the context of care is still characterized as preventive rather than curative or rehabilitative. And, as the provision of PCBP services widens, the gap in behavioral health care for children in this country could correspondingly narrow.

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CHAPTER 25

Peer Intervention

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Peers are being used as intervention agents with increasing frequency. No doubt, this growth is largely attributable to the many advantages of peer interventions, perhaps the most prominent of which are convenience and cost-effectiveness (Covert & Wangberg, 1992; Fowler, 1988; Topping & Ehly, 1998). Though most often used with elementary-school-age children and adolescents, these interventions have also targeted preschoolers (e.g., Carey, 1997; Kohler, Strain, Hoyson, & Jamieson, 1997) and adults (e.g., Moore, 2005). The wide scope of peer interventions is evidenced not only in their application across developmental levels, but also in the diverse range of targeted clinical problem areas and disorders. By no means an exhaustive list, peers have been used in interventions aimed at improving the social and communication skills of children with autism (e.g., Kohler et al., 1997), reducing medical fears (e.g., Elkins & Roberts, 1986), decreasing delinquency and aggression (e.g., Becker, Hall, Ursic, Jain, & Calhoun, 2004), resolving conflicts (e.g., Johnson & Johnson, 1996), promoting health behaviors (e.g., Agha & Rossem, 2004), and preventing suicide (e.g., Bagley, Tse, & Hoi-Wah, 1994).

This chapter provides an overview of peer interventions as they have been used with children and adolescents. Given the variety of populations, targets, and intervention formats, we begin the chapter with a review of recent categorization efforts and the delineation of specific intervention types. This is followed by coverage of more general issues in intervention implementation. Among these are important considerations regarding the peer interveners themselves, which include selection, training, and often overlooked concerns over the relative benefits and risks of participation. Continued discussion of general implementation issues, including advantages, disadvantages, ethical considerations, and evaluation, is followed by a review of applications targeting specific clinical problem areas. The chapter closes with an illustrative case example.

PEER INTERVENTION FORMS

The frequent and widespread use of peer interventions has given rise to a number of attempts to define and categorize the various approaches. One area of agreement is that such interventions are a method of delivery rather than a specific treatment technique (e.g., Shiner, 1999). Their common element is that a child, rather than a

therapist or teacher, implements the intervention (Shiner, 1999). Of course, adult involvement is still needed, but should not include continual prompting of the child's intervention efforts (Fowler, 1988). Common roles for adults are training the interveners, supervising the intervention, maintaining systems to monitor behavior, and arranging for rewards (Fowler, 1988). Some intervention formats may require greater adult supervision to ensure appropriate implementation, and younger children may require more adult support than older children or adolescents. Even with such adult involvement, however, peer interventions are "designed to be by and for young people; they themselves largely determine what is relevant in terms of information and how it is to be delivered" (Backett-Milburn & Wilson, 2000, p. 94).

Researchers have categorized past efforts in a number of different ways. Distinctions have been made between methods emphasizing skill development and those focusing on knowledge acquisition (Topping & Ehly, 1998). Shiner (1999) distinguishes between the constructs of peer development (i.e., a focus on the development of the interveners) and delivery (i.e., the provision of services). Perhaps the most widely discussed and useful distinctions are based on the roles of the children involved. That is, the child can either intervene directly or indirectly and either be the intervener or the target.

Peer interventions can be direct or indirect (Kalfus, 1984; Odom & Strain, 1984). In indirect approaches, an adult implements the intervention and the child is used in a supplementary role (e.g., as a model, employing group reinforcement contingencies). For instance, to reduce fears of medical procedures, children observed the responses of another child going through the same procedures (Elkins & Roberts, 1986; Stokes & Kennedy, 1980). In direct approaches, the peer implements the intervention and works to directly influence the target child's behavior. For example, Krantz, Ramsland, and McClannahan (1989) had one child with autism prompt others with the disorder to engage in conversations about sports. Most peer interventions appear to fall into the latter category. In the review of specific approaches that follows, we therefore include several direct approaches (e.g., education, mentoring, and counseling) and only one indirect approach (i.e., modeling).

The second distinction focuses on the rigidity of roles and whether a child can be both the intervener and the target or can serve only one of those functions (e.g., Fowler, 1988; Johnson & Johnson, 1995). Fowler describes three basic models characterizing the relationship between the target and the intervening child. In the "subject as client" model, the target child receives an intervention implemented by another child who generally exhibits desired behaviors (e.g., good social skills and teacher compliance) and is well-liked by peers. As such, one child or a small group of children are selected, by adults, to perform the role of intervener. In the "subject alternating as client and therapist" model, all group members have the opportunity to be both therapist and client and to switch between these roles. Members of the group may differ in their intervention needs, and some children may not be in need of intervention. Specific criteria may be used to choose the child who will serve as intervener on a particular day. For instance, the child may be required to exhibit a minimum level of a particular behavior. In the "subject as therapist" model, the target child serves as the intervener rather than the "client." This encourages the

child to focus on the positive behaviors that he or she is rewarding in another child and may result in new positive interaction patterns between the child and his or her peers. Similarly, Johnson and Johnson described two models of the roles in peer counseling interventions. In the "cadre approach," a small number of children are trained to serve as counselors to aid other children in their problem-solving efforts. In the "student body approach," the entire student population is trained in intervention strategies (e.g., problem solving, conflict resolution) and has the opportunity to serve as counselors. These authors suggest that an advantage to this approach is that all children learn new strategies and can apply them outside of a formal intervention.

Having discussed common elements and distinguishing features, we now turn to a review of specific peer intervention types, including modeling, prompting and reinforcement, initiation, tutoring, education, mentoring, and counseling.

Peer Modeling

In peer modeling approaches, one or more competent children are used to exemplify the desired behaviors (Schunk, 1998). Three features are required: one or more models, one or more observers (learners), and a task to be performed (Schunk, 1998). Generally, a target child is paired with a more competent peer model whose behavior can be imitated. A common distinction is between mastery and coping models. The mastery model engages competently and skillfully in the desired behavior. In contrast, the coping model still needs to improve and learn to overcome problems. Thus, through observing a peer who is a mastery model, the target child can see what the desired behavior should look like. By observing a coping model, the target child can learn how to improve performance and handle difficulties or failures, in addition to the possibility of being comforted by the realization that other children face such challenges. Research has shown that both coping and mastery models can be effective in changing children's behavior (e.g., Schunk and Hansen, 1985). Regardless of whether a mastery or a coping model is used, the model and behavior should be selected carefully to reduce the chance of incorrect modeling or inability to learn the behavior (Schunk, 1998).

Another variant commonly used in social skills interventions is the "peer pairing" approach (e.g., Bierman & Furman, 1984; Prinz, Blechman, & Dumas, 1994). Target children are paired with a more competent peer, who receives no training and very little guidance. For instance, Prinz and colleagues used teacher ratings to select aggressive and nonaggressive socially competent children. These two types of children were then paired and given instructions (e.g., "Find out something that made your partner laugh this week") for performing role-plays. The socially competent peer always enacted the role-play first in an attempt to model appropriate behaviors; however, this peer was not given any prior instructions.

The effectiveness of modeling interventions depends on characteristics of the target child, the model, and the behavior to be learned. The modeled behavior is more likely to be followed if the model is the same sex as the target child and is perceived to have high power and the target child has high self-efficacy for and places importance on performing the action (Bandura, Ross, & Ross, 1963;

Grusec, 1992). When behaviors are rare or difficult to elicit or the model may have an unpredictable response (e.g., coping with medical procedures), video modeling of the desired behavior may be preferable (e.g., Gilbert et al., 1982). Observation of the model may be sufficient for simple behaviors, but further intervention is often required for more complicated actions (Schunk, 1998). When modeling alone is insufficient, the target child will need to practice the observed behavior and receive feedback regarding performance.

Modeling can be used to teach a variety of behaviors, including cognitive strategies, social skills, and coping methods. In modeling cognitive strategies, other children can be asked to verbalize their thought processes in solving problems. For example, Schunk and Hanson (1989) used two types of coping models to exhibit cognitive processes involved in learning math skills. The coping-emotive model initially made negative statements (e.g., exhibiting low self-esteem or the perceived difficulty of the task) and then positive coping statements (e.g., regarding the use of concentration or working hard), and the coping-alone model verbalized only positive coping statements. Peers have also been used to model social skills, such as conversational skills (e.g., Kohler et al., 1997) and play behaviors (e.g., Pierce & Schreibman, 1995). As part of a larger intervention, Pierce and Schreibman taught children to model social skills, including complex play behaviors and appropriate verbalizations during play in an attempt to improve the social behavior of children with autism. Others have used children to model emotional coping with stressful or anxiety-provoking events. For instance, Elkins and Roberts (1986) had children observe the emotional coping responses (e.g., talking about and successfully coping with fear) of a peer model undergoing an operation in an attempt to reduce fear of medical procedures.

With its versatility and potential for teaching novel responses, modeling has been included in interventions targeting quite a variety of clinical problems, such as coping with medical fears (e.g., Elkins & Roberts, 1986; Gilbert et al., 1982; Stokes & Kennedy, 1980), ameliorating academic difficulties (e.g., Schunk, Hanson, & Cox, 1987), and improving language skills (e.g., Kohler et al., 1997) and affective behavior in children with autism (e.g., Gena, Couloura, & Kymissis, 2005). In a particularly interesting application, Gena et al. used video modeling to help preschool-age children with autism better regulate their emotional expressions in reaction to situations eliciting the emotions of sympathy, appreciation, and disapproval. Videotapes exhibited age- and sex-matched peers displaying appropriate verbal and facial responses for each situation. Targeted children evidenced 100% accuracy in emotional responses within 10 sessions of video modeling.

Peer Prompting and Reinforcement

Peers are often used to prompt target children to engage in particular behaviors or to reinforce the appropriate use of behaviors. Prompting includes instruction and reinforcement for appropriately following the instruction (Odom & Strain, 1984). This strategy has been used to increase autistic children's language skills (e.g., Krantz et al., 1989), improve autistic children's social skills (e.g., Odom & Strain, 1986), and improve healthy eating behaviors in obese children (Foster, Wadden,

& Brownell, 1985). For example, Foster and colleagues conducted an intervention to improve health-related behaviors of obese children using behavioral change strategies that included a peer intervention component. In this component, eighth graders were trained to reinforce the eating behaviors of selected second- through fifth-grade students. A few times a week, children met with their peer intervener, who examined their lunchbox and provided verbal praise and stickers based on food choices. When compared to control children, those receiving the intervention lost significantly more weight and evidenced greater improvement in self-concept.

Peer Initiation

In peer initiation interventions, other children are used to initiate or maintain social interactions, such as play or conversation, with the target child (Carey, 1997; Guevremont, MacMillan, Shawchuck, & Hansen, 1989; Odom & Strain, 1984). It is sometimes difficult to ensure consistent implementation by the peer initiators, and adult prompts or rewards may be beneficial (Carey, 1997; Fowler, 1988). All in all, peer initiation interventions are relatively straightforward, and even preschoolers are able to successfully implement them (Carey, 1997).

Peer initiation has been included as a component of social skills training studies targeting children with autism (e.g., Goldstein, Kaczmarek, Pennington, & Shafer, 1992; Sasso, Mundschenck, Melloy, & Casey, 1998) and behaviorally disordered or developmentally delayed preschoolers (Goldstein & Wickstrom, 1986). Goldstein and Wickstrom (1992) taught two nonhandicapped peers to intervene with three target children with behavioral problems and developmental delays. The peer interveners prompted social interaction through actions such as establishing eye contact or a joint focus, responding to the target child's verbalizations, and describing play behaviors. The teacher used rewards (tokens that could be traded in for stickers or toys) and prompts (e.g., suggested new strategies) to ensure the cooperation of the interveners. All three target children exhibited and maintained improvements in the frequency of social interactions.

Peer Tutoring

Kalfus (1984) describes peer tutoring as needing to include any two or more of the following: instructions to the student, prompts for correct responding, praise, corrective feedback, and ignoring of specific student behaviors. Peer tutors are generally thought of as individuals who focus on ameliorating academic problems, and tutoring programs can take a structured or unstructured format (Kalfus, 1984). Structured programs provide formal training and clearly delineate who is the student and who is the tutor, whereas unstructured programs do not. The independence conferred on the tutor can vary across programs (Topping & Ehly, 1998). Tutors may be provided with specific materials or a detailed curriculum. In contrast, other tutors may be expected to design their own materials and curriculum content.

These programs have been applied effectively at the classroom (see Greenwood, Maheady, & Delquadri, 2002, for a review) and individual (Calhoon, 2005; DuPaul, Ervin, Hook, & McGoey, 1998) levels. Though specific academic skills

are generally targeted, broader improvements in classroom behavior and on-task performance are often seen. In one example, DuPaul and colleagues used a classroomwide peer tutoring program to help improve the mathematics, spelling, and reading skills of target children diagnosed with Attention-Deficit/Hyperactivity Disorder (ADHD). Children were paired with each individual taking on the role of tutor. The tutor was given academic materials, including correct answers so that he or she could provide praise and reinforcement to the target child. Targeted ADHD children exhibited improvements in their on-task behavior as well as specific academic skills. In an example of a more structured program, Wright and Cleary (2006) implemented a cross-age reading skills tutoring intervention in which one child served exclusively as the tutor and the other as student. The tutor received four sessions of training that included appropriate behavior for tutors, the use of praise and other forms of encouragement, how to manage problem behavior, and instruction in the specific tutoring activity. Overall, children exhibited an increase in reading fluency of one word per week.

Peer Education

Peer education programs serve as a method of dispersing important or sensitive information throughout the peer group (Topping & Ehly, 1998). Children can, and already do, provide information to each other in a shared language (Mathie & Ford, 1998; Topping, 1996). Unlike teachers and other adults, peers can impart knowledge without being perceived as someone with authority. Children and adolescents may be unlikely to speak freely with individuals whom they know in other roles (e.g., teacher, administrator) that carry the weight of authority. Formalizing previously informal methods of knowledge transmission serves to increase the accuracy and credibility of the information communicated (Mathie & Ford, 1998; Topping, 1996; Topping & Ehly, 1998).

Peer education programs have been used to address substance abuse (Botvin, Baker, Filazzola, & Botvin, 1990), bullying (Salmivalli, 2001), and the prevention of HIV/AIDS (Agha & Rossem, 2004) and suicide (Sandoval, Davis, & Wilson, 1987). These educational programs can take a variety of forms. Some are comprehensive and ongoing, whereas others are focused and brief. An example of the former is a bullying prevention program described by Salmivalli. Students gave a schoolwide presentation (including information about bullying, participants' roles in bullying, and dramatic skits demonstrating bullying) and led small group discussions in classes (focusing on classroom climate, bullying problems, and how students could reduce bullying). During a weeklong period, the topic was also integrated in several school activities. Examples included coverage of conflict negotiation topics in the school news and related posters displayed in art classes. Following the intervention, reductions were found in self-reported victimization across all grades and peer-reported bullying for seventh-grade girls. In contrast, an HIV/AIDS prevention program with Zambian adolescents was conducted in single sessions lasting less than 2 hours (Agha & Rossem, 2004). Adolescents attended one discussion-oriented session focused primarily on condom use and abstinence.

The educators provided factual information, performed skits, and provided the adolescents with informational pamphlets. Adolescents exhibited improvements in their beliefs regarding abstinence and condom use.

Peer Mentoring

Unlike other peer interventions where similarity in age is generally preferred, peer mentoring is often characterized by an age discrepancy (Topping & Ehly, 1998). In this relationship, the mentor exhibits more experience, skill, or knowledge in a particular area and fills the role of passing this information on to the target child. Sometimes, peer mentors and their mentees are chosen from different schools in an effort to smooth future school transitions. For instance, in one mentoring program, youth in a secondary school were chosen as mentors for primary school students (Dearden, 1998). The mentors served as resources during the transition to the secondary school.

Mentoring interventions have also been used in efforts to scaffold appropriate behaviors (e.g., Fair, Vandermaas-Peeler, Beaudry, & Dew, 2005), improve school connectedness (e.g., Karcher, 2005), and decrease violent or delinquent behavior (e.g., Becker et al., 2004). In a cross-age mentoring program, high school students served as mentors to fourth- and fifth-grade students (Karcher, 2005). Mentors and mentees were given the opportunity to select their pairs, and the mentees included several children whom teachers had identified as being at high risk for academic failure. Meetings between the dyads occurred in a group setting twice a week for 2 hours. The mentoring program positively improved school and parent connectedness in mentees. In the Caught in the Crossfire intervention, Becker and colleagues employed young adults from backgrounds similar to that of target adolescents as peer mentors and provided them with training in counseling skills, anger management, cultural competency, and resource identification. Target adolescents were individuals who had been hospitalized as the result of a violent injury. For many target adolescents, these young adults served as role models for up to a year following the initial hospitalization. Mentees were less likely to be arrested or have any criminal involvement when compared to youth receiving no intervention, but there was no difference between the groups in injury and death rates.

Peer Counseling

Despite their nonprofessional status, peer counselors serve functions similar to those of professional counselors (Topping & Ehly, 1998). They offer help to other children and adolescents by listening, providing empathy, and using problem-solving skills (D'Andrea & Salovey, 1983; Topping & Ehly, 1998). These services can be offered in groups or individually, in person or over the phone, and can involve the provision of information and resources rather than advice giving.

Most often, peer counselors address typically occurring stressors and life issues, such as conflict resolution, relationship problems, and school transitions (Bell, Coleman, Anderson, Whelan, & Wilder, 2000; Johnson & Johnson, 1995; Morey, Miller, Rosen, & Fulton, 1989). For instance, in one high school peer counseling

program, the issues most frequently discussed were relationship problems (Morey et al., 1989). Another peer counseling program addressed conflict resolution (Bell et al., 2000). Teachers and students selected a small group of sixth through eighth graders to serve as peer mediators for the school, which included students in the first through eighth grades. When conflicts arose, teachers had the option of referring children to these counselors or to the principal. Over a 6-week period, 32 of the 34 conflicts mediated by peers were deemed successful; that is, the mediator and disputants came to a written agreement and the conflict was not referred to the principal for follow-up.

Individuals presenting with more serious concerns (e.g., depression and suicide) have also been served by peer counselors (e.g., Bagley et al., 1994; Boehm & Campbell, 1996). When difficult or complicated problems arise, peer counselors should have a method for referring the counselee for professional services or other appropriate resources (Leader, 1996; Topping & Ehly, 1998). This is often the case with hotlines developed for adolescents. Callers to such hotlines present with concerns ranging from family and peer relationship problems to sexuality, academic difficulties, suicide, and drug and alcohol use (Boehm, Schondel, Marlowe, & Manke-Mitchell, 1999). An example is Teen Line, a counseling hotline staffed by adolescents trained to handle a variety of calls, including those from suicidal individuals, and provide appropriate resources as needed (Boehm & Campbell, 1996). Additional supports for the peer counselors included an available adult and the ability to trace the calls of suicidal individuals.

GENERAL ISSUES AND CONSIDERATIONS

Identifying and Selecting Peer Interveners

To start, it is important to delineate who constitutes a "peer." In a very broad definition, peers can include everyone from close friends to virtual strangers involved in a shared activity (Coleman & Hendry, 1990). In peer interventions, peers are often described as nonprofessionals who are similar to the target child in age, status, and knowledge (Ehly & Vasquez, 1998; Leader, 1996). Though using age as the sole factor in establishing similarity is a rather common practice, Shiner (1999) points to other identity characteristics, such as ethnicity, sexual orientation, social class, and sex that may be more salient features.

In addition to ensuring similarity, there are a variety of additional criteria and methods for selecting peer interveners. Many times, children are chosen because they exhibit exemplary behaviors (Fowler, 1988). Similarly, there is evidence that children who are highly respected and socially accepted can have a more significant influence on behavior change (Hamburg, 1989). Exhibiting appropriate behaviors and being socially accepted, however, are not absolute requirements. Kalfus (1984) points out that a variety of children (e.g., mentally retarded, learning disabled, juvenile delinquents) have served as effective peer agents in past interventions. Additional selection criteria might include an interest in performing the role and a consistent ability to follow adult instructions (Fowler, 1988; Kalfus, 1984; Strain, 1981). Others include a specific grade-point average, a certain time commitment, or

enrollment in a course designed to provide appropriate skills and training (Covert & Wangberg, 1992).

Although some criteria can be broadly applied, others are specific to the characteristics of the intervention (e.g., fixed versus flexible roles), the amount of adult supervision provided, or the clinical problem being addressed (Fowler, 1988). For instance, good verbal and other communication skills may be desirable in verbally based interventions, but may be unnecessary for a modeling intervention (Hamburg, 1989; Kalfus, 1984). Counseling interventions may require adequate problem-solving abilities and social competence (Leader, 1996), and regular school attendance is an important element for school-based interventions (Fowler, 1988; Odom & Strain, 1984; Strain, 1981).

Selection methods vary considerably as well. The first component of the selection process is choosing potential interveners from a broader population; this has been done in a number of ways. Bell and colleagues (2000) had students choose a pool of potential interveners from which teachers or school administrators then selected appropriate individuals. In another intervention, teachers or school staff provided recommendations, and the potential interveners were given the ultimate decision regarding participation (Kim, McLeod, Rader, & Johnston, 1992). Chapman (1998) suggests using sociometric nominations of social acceptance as a method of identification to ensure that interveners are drawn from various subgroups of the population. Greater diversity of interveners in terms of race, cultural background, interests, personality, or other characteristics improves the chances that a target child will have someone with whom he or she can identify (Chapman, 1998; Covert & Wangberg, 1992). Sociometric measures of acceptance have been the major selection method for social skills interventions (e.g., Fantuzzo et al., 1988).

Selection often includes an effort to determine whether or not the peers meet recommended criteria. Generally, the potential peer interveners and others (e.g., teachers) are included in this process in order to assess the ability to intervene, intellectual and emotional capabilities, commitment, tolerance, and empathy (Chapman, 1998; Leader, 1996). Chapman recommends including the intervention recipients, the potential interveners, school or community personnel, and the program coordinators in this step. Leader described use of a multiple-stage process for selecting adolescents to serve as counselors for a teen hotline. Potential peer counselors were asked to complete a questionnaire and were individually interviewed by a clinician. This was followed by a group interview and an orientation meeting with their parents.

Training Peer Interveners

Once selected, interveners need sufficient instruction to adequately implement the intervention. As mentioned, the selection process often elicits interveners with specific personality characteristics or skills. Although this expertise should be respected, these individuals will benefit from further training (Leader, 1996). The format, duration, and components of training vary across interventions. Training may include lectures on important topics, modeling of techniques, role-playing, pre- and postassessments of skills, and continued monitoring and supervision (e.g., Covert & Wangberg, 1992; Fowler, 1988; Leader, 1996; Martin, Martin, & Barrett, 1987).

A number of peer interventions include instruction on similar topics and skills, such as communication and listening skills; delineation of responsibilities, requirements, and rules; confidentiality; and enumeration of additional community resources or referral sources (Bishop, 2003; Covert & Wangberg, 1992; Fowler, 1988). Bishop recommended that certain core skills (e.g., active listening, empathy) be taught to train individuals in supporting others, regardless of the specific intervention to be used. Of course, most interventions will also necessitate the addition of more tailored instruction topics. For example, training for an adolescent suicide prevention counseling program included topics such as depression and characteristics of individuals who are suicidal (Martin et al., 1987).

The depth and breadth of training needed will depend on the type of intervention. Little or no training might be needed to teach a child to model appropriate behaviors, whereas in-depth training may be needed for peer education or peer counseling formats. In one school intervention, for example, training for peer mediators required sessions held once every 2 weeks over a 2-month period (Bell et al., 2000). To ensure the maintenance of training benefits, two booster sessions were held. For another intervention, adolescent hotline counselors participated in a 13-week training program that included two weekly sessions focusing on listening skills and adolescent psychology (Leader, 1996). Training also involved the observation of others fielding phone calls and participation in monthly supervision sessions.

Studies examining training intensity needs for peer counseling interventions have been inconclusive. For example, Guttman (1987) randomly assigned peer counselors to either a high-training or minimal-training condition. Counselors in the high-training condition participated in a yearlong training and practicum, whereas those in the minimal-training condition participated in three workshops, each 4 hours long. Both groups exhibited improvements; the only group difference was that the minimal-training group used a greater number of statements identifying the group or group process. In another study, Cooker and Cherchia (1976) found that adolescents were able to competently facilitate adolescent groups on drug use after only 8 hours of training. Although there is not agreement regarding the optimal duration or intensity of treatment, there is consensus that supervision or adult monitoring should continue beyond the constraints of formal training (e.g., Covert & Wangberg, 1992; Fowler, 1988; Leader, 1996).

Considering the Peer Interveners Themselves

In addition to the benefits afforded to target children, peer interventions have been shown to have positive effects on the interveners such as improvements in classroom behavior, grade point average (Garner, Martin, & Martin, 1989), mental health (Silver, Coupey, Bauman, Doctors, & Boeck, 1992), and self-esteem (Kim et al., 1992). Indeed, Hamburg (1989, p. 9) suggests that "the long-term effects of peer counseling may be more important for the peer leaders than for the youngsters counseled. By being useful to others, adolescents can build a more durable basis for self-esteem."

Some gains may come from participating in training and gaining the skills necessary to intervene. In an intervention described by Silver and colleagues (1992),

inner-city adolescents who participated in a 3-month counselor training and accepted positions as peer counselors exhibited improvements in their mental health. The training, however, did not promote such improvements in all of the adolescents. Those who declined counseling positions or who were not judged to be adequately skilled following training did not exhibit similar gains. In another peer counseling intervention, Kim et al. (1992) reported that trainees in their intervention evidenced improvements in empathic responding and personal growth (i.e., communication skills, emotional control, helping others, and interpersonal relations).

Unfortunately, there is an absence of research examining the potential risks for peer interveners. In the absence of empirical findings, we are left to speculation. For instance, Fowler (1988) suggests the possibility that the target child could retaliate against the peer implementing treatment. Perhaps worthy of some exploration is the potential for risks associated with increased deviant peer affiliation. In well-publicized research, Dishion and his colleagues have demonstrated iatrogenic effects for group interventions that bring together at-risk youth (e.g., Dishion, McCord, & Poulin, 1999). They reviewed evidence from two experiments suggesting that youth participating in group interventions had increased negative behaviors (e.g., smoking, delinquent behavior) in short- and long-term follow-ups. Dishion and colleagues suggest that "deviancy training" may be occurring within adolescent groups and interfering with intervention effects. That is, deviant peers may socialize each other to engage in antisocial behavior. Though it seems plausible that such processes could operate in peer interventions, there are unique elements that are likely to afford some protection. For instance, peer interveners typically pass through selection procedures and are often chosen based on their social competence. As such, they may not be as prone to deviancy training.

Advantages and Disadvantages of Peer Interventions

Using youth as interveners has several advantages relative to more traditional adult-mediated approaches (e.g., teacher-implemented educational strategies, psychotherapy). Their convenience and cost-effectiveness enhances their applicability to school settings (Covert & Wangberg, 1992; Fowler, 1988; Topping & Ehly, 1998). In this way, they have the potential of expanding service provision to larger numbers of children and establishing links between at-risk children and needed professional services (Henriksen, 1991). They also allow adults to focus efforts on other activities and interventions (Kalfus, 1984; Topping & Ehly, 1998). Even if not always meeting the same quality standards as adult-mediated approaches, peers have the advantage of being able to implement without delay (Greenwood, Carta, & Kamps, 1990; Topping & Ehly, 1998). They are in a unique social position that allows them to observe behaviors that may go unnoticed by adults and to effect change on other children's behavior (Carey, 1997; Fowler, 1988). More immediate feedback and consequences are also possible. Another advantage is the facilitation of maintenance and generalization. Typically, the intervention is carried out in the target child's natural environment; thus, the peer intervener, as well as other relevant stimuli, remain in place after treatment and serve as cues for appropriate responding in the future (Stokes & Osnes, 1989).

Given the social milieu in which children operate, peers may be especially important for influencing and changing behavior. This may be particularly true during adolescence, when greater amounts of time are spent with peers, who become a primary source of social support. Also important to consider is that such peer influence can be positive or negative. Well before adolescence, for example, children can unintentionally contribute to unwanted behaviors. Fowler (1988) suggests four main ways in which children reinforce negative behaviors. First, children positively reinforce the negative behavior. For instance, a child who uses aggression to resolve a conflict may be subsequently accepted by the peer group. Second, peers negatively reinforce the behavior. In this case, a child who is being teased might hit the bully, resulting in the cessation of taunting. Third, alternative positive behaviors are ignored or inappropriately reinforced, resulting in their eventual diminishment. Thus, a child may attempt to enter group play appropriately (e.g., asking nicely to play, attempting to take turns with a toy) but be denied access to the group. Over time, the child may cease to use these appropriate behaviors if they do not result in the desired outcome. Fourth, positive behaviors or social attempts are met with negative peer responses. In this case, children may respond to an appropriate social request, such as asking to share a toy, with aggression or by making negative comments (e.g., "I don't want to play with you"; "I don't like you").

Including peers in intervention has other advantages. For instance, having another child model appropriate behavior can be more beneficial than using an adult model (Schunk, 1998). The peer model is likely to be perceived as more similar to the target child and is less likely to demonstrate a perfect performance. As such, the target child may better identify with the child model and perceive himself or herself as able to mimic the model's behavior. In addition, children might be more likely to use services provided by their peers than more traditional services (Covert & Wangberg, 1992). For example, the use of traditional counseling services may carry a greater stigma than meeting with a school's peer counselor. For some topics, youth may be perceived as more expert than adults and better able to understand the child's or adolescent's experience (Mathie & Ford, 1998). When attempting to educate adolescents about sexual behavior or social decision making, for instance, it may be difficult to separate an adult from his or her other roles (e.g., a teacher), increasing discomfort and decreasing willingness to openly engage in discussions. Another adolescent may be perceived as having a better understanding of the youth's real-life struggles with these topics.

Despite these advantages, there are some limitations. Increased adult involvement can undermine efficiency and cost-effectiveness and might suggest that the intervention is no longer truly peer-mediated (Fowler, 1988; Kalfus, 1984). Indeed, the many advantages of using peers notwithstanding, Kalfus suggests that adults may be more efficient and effective in producing behavior change. For example, an adult may be better able to teach new skills or effectively implement rewards. Another disadvantage of using peers relates to the need for informed consent and confidentiality (Ehly & Vasquez, 1998). The peer intervener might need to be privy to sensitive information regarding the target child's behavior and need for treatment. As discussed earlier, there may also be risks for the peer intervener. Finally, as illustrated in our case example, these interventions are often implemented in

schools and other nonclinic settings, and their feasibility for clinicians not already working within such environments may be an important concern.

Ethical Considerations

The use of peers as intervention agents raises some unique ethical considerations. One of the major issues is maintaining client confidentiality. The peer intervener may need to be made aware that treatment is occurring and learn private information. The ability of the intervener to understand and protect confidentiality should be carefully considered in the selection process. Such concerns may be greater in some forms of intervention than others. For instance, peer education interventions rely primarily on the dispensation of information and may not include disclosure of personal information. In contrast, peer counseling interventions often involve discussion of sensitive, private matters that would require counselors to have an understanding of and respect for confidentiality. Furthermore, peer counselors should be trained on the limits of confidentiality and when it may be appropriate to seek adult support or intervention (Ehly & Vasquez, 1998).

A second major concern pertains to informed consent. In some interventions, target children may not need to know that they are participating in an intervention. In fact, the intervention may be more successful if the target child does not realize another child has been instructed to help them. For example, in a social skills intervention, the target child may be naive to the fact that other children are being instructed to initiate social interactions (e.g., Middleton, Zollinger, & Keene, 1986). The degree to which the intervener participates voluntarily is also a significant ethical concern. Individuals selected as peer interveners should have the option of whether or not to participate (Greenwood et al., 1990). Some children may feel more comfortable in this role than others. Given the responsibilities placed on the peer intervener, he or she should be informed of all aspects of the role in the intervention and be allowed the choice of whether or not to be involved.

Evaluating Peer Interventions

A variety of approaches have been used to evaluate peer interventions, ranging from assessments of the interveners' abilities to reports of treatment satisfaction and objective measures of behavioral changes (Covert & Wangberg, 1992). Fowler (1988) recommends the use of a number of different methods to assess intervention effects, including consumer satisfaction surveys, daily assessments of effectiveness, assessment of generalization and maintenance of behavior changes, and assessment of behavioral changes in the intervener. Compliance with treatment protocols and the accuracy with which youth implement the intervention or reward systems should also be examined. Although the evaluation of peer interventions is important for many reasons (e.g., determining behavior change, funding, and gaining continued commitment from schools), the quality of program evaluations varies dramatically, and a number of studies do not use any form of objective evaluation (Covert & Wangberg, 1992).

In spite of such variation in outcome assessment quality, there is a fair degree of empirical support for the effectiveness of peer interventions. As indicated in

our review thus far, peer interventions have been effective in addressing a wide range of presenting concerns. Published reviews of two of the major forms of peer intervention have also been favorable. In a review of peer education programs, Mellanby, Rees, and Tripp (2000) reported that, in general, peer-mediated programs were at least as effective as adult-led programs in improving knowledge and inducing behavioral changes. Johnson and Johnson (1996) completed a review of school counseling programs evaluating the knowledge gained from training as well as outcomes at the individual and school level. After conflict resolution training, students were found to have made improvements in their management of conflicts and use of resolution skills. In peer mediation programs, high levels (i.e., over 80%) of successful resolutions were generally obtained, although some programs had lower rates (e.g., 58% to 74%) and not all programs resulted in changes to the overall school climate.

CLINICAL APPLICATIONS

Not surprisingly, with their flexibility and many other advantages, peer interventions have been applied to many different clinical problem areas. Our review of peer intervention forms included many example applications. In this section, we provide a selective review highlighting the targeted problem areas.

Communication and Social Skills Deficits

Related interventions for communication and social skills deficits generally take the form of peer modeling, peer social initiations, or peer prompting and reinforcement. Children targeted have included those with autism, developmental delays, and cognitive deficits, as well as those evidencing social withdrawal. In an intervention aimed at improving social and communication skills in children with autism, normally developing preschoolers were used as models of appropriate play behavior (Kohler et al., 1997). Target children's social interactions with other children increased during the peer intervention phase and were maintained at follow-up. Peer modeling, however, is not always successful with autistic children. For instance, Charlop and Walsh (1986) used time delay and modeling procedures to facilitate affection responses in autistic children. The children were quick to learn the behaviors through the time delay technique, but peer modeling was ineffective. The authors suggest that the loosely structured environment of the modeling intervention may have contributed to its failure.

In a unique application, Krantz and colleagues (1989) trained an adolescent with autism to prompt three other autistic children in an effort to increase conversational abilities. Through a series of role-play sessions, the adolescent prompter was trained to talk about sports and ask five questions that could be applied to a variety of topics (e.g., "Do you like basketball/baseball?"). The target children evidenced improved conversational skills in dyadic and group conversations. Moreover, the skills generalized to new situations and were maintained at a 1-month follow-up. Other evidence suggests that peer initiation interventions may be most successful when the target child is paired with one child instead of multiple children. In a study

by Sasso and colleagues (1998), children with developmental delays and autism showed greater improvement in social behaviors when paired than in a similar intervention conducted in triads.

Socially withdrawn children have also been the focus of such interventions. Guevremont and colleagues (1989) trained classmates to initiate and maintain interactions with two withdrawn children. Significant improvements in each child's social behavior were observed during and following the intervention. Similar success was observed in an intervention for four blind, socially withdrawn elementary school children (Sisson, Van Hasselt, Hersen, & Strain, 1985). Peers were used to initiate social behaviors in the target children, and increases in social behaviors were maintained at a 4-month follow-up assessment. In a treatment for socially neglected children, Middleton and colleagues (1986) selected accepted or popular peers to act as interveners and provided them with weekly instruction on behaviors to use with the target children (e.g., greeting or sitting next to at lunch). Also included were instruction and modeling components in which the peer modeled proper behavioral responses and offered suggestions for coping with unwanted behaviors. Socially neglected children who received treatment showed increases in peer nominations during posttest assessments.

Academic Problems

Tutoring is the most common form of peer intervention used to address problems of an academic nature, although other approaches (e.g., mentoring, modeling) have been used as well. Intervention targets have included specific academic skills, such as writing, spelling, and reading (e.g., Braaksma, Rijlaarsdam, & van den Bergh, 2002; James, Charlton, Leo, & Indoe, 1991), as well as school behaviors, such as work skills and attention to task (e.g., DuPaul et al., 1998). Formats of these interventions have also varied in terms of who is selected to intervene and the rigidity of roles. Competent peers may be used in the fixed role of intervener (e.g., Topping, Peter, Stephen, & Whale, 2004) or children may alternate between being the target child and the intervener (e.g., Kohler, Schwartz, Cross, & Fowler, 1989).

Several studies have suggested that children learn academic skills from peer models better when the model's competency level is similar to their own (e.g., Schunk et al., 1987). For instance, young teens were assessed on their ability to learn argumentative writing in three different conditions (Braaksma et al., 2002). Two conditions involved observing another child who was either a good or a weak model. The final condition was a control condition in which the children had to write short arguments. Individuals learned better from the model that was more similar to them in competence. Similarly, children had greater self-efficacy for learning and actually learned math skills better when observing a peer of similar competence gradually learn the skill than when observing a child who had already mastered the skill (Schunk et al., 1987). Notably, however, repeated observations of a mastery model resulted in similar effects.

Other interventions have been aimed at improving general academic skills. Kohler and colleagues (1989) selected three low-achieving students to receive a peer intervention aimed at improving independent work skills. Rather than assume

solely the client role, these children and their classmates alternated roles daily. Regardless of the role assumed, the target children exhibited improvements in work skills (i.e., on-task behavior); however, task completion and accuracy did not improve.

Bullying, Aggression, and Disruptive Behavior

Peer programs aimed at bullying, aggression, and disruptive behavior have taken on a number of formats, including counseling, mentoring, mediation, and education. In his suggested strategies for preventing school violence, Stephens (1997) encourages the implementation of peer counseling and mediation programs. In his words:

> Students represent some of the best agents for promoting and maintaining a safe and secure campus. An effective peer counseling program can head off many problems before they reach explosive and violent levels. In addition, the students who are trained as counselors become wonderfully influential resources for nonviolent problem solving. (p. 74)

Bullying has been spotlighted recently, and a number of programs focus on providing support services to victims. For example, Boulton (2005) developed one-on-one peer counseling programs for victims in two secondary schools in the United Kingdom. When students were asked about their previous and future use of these programs, 22% stated that they had previously used the program and the majority said they would definitely (37%) or maybe (60%) use the program if needed in the future. In choosing the people they were most likely to go to for help with bullying, students chose close friends over the peer counselor, but the peer counselor was rated higher than a teacher or sibling. There was no difference in the likelihood of seeking help for bullying between the peer counselors and school counselor or parents.

Other programs focus on the aggressors or individuals who exhibit disruptive behaviors. For example, the Caught in the Crossfire program involved peer mentoring in a community setting for adolescents who sustained violent injuries (Becker et al., 2004). This program was developed to reduce the likelihood of reinjury or future involvement in delinquent activities. Recipients were less likely to be arrested or have any criminal involvement when compared to individuals who did not receive the intervention. Rather than target individuals at high risk for violence, other programs use a broader approach by intervening schoolwide. For instance, sixth-grade students received a violence prevention program taught by a teacher with or without a peer leader (Orpinas, Parcel, McAlister, & Frankowski, 1995). Individuals who received the teacher plus peer leader intervention exhibited greater attitude changes toward violent responses.

Peer Mediation and Conflict Resolution

Although in some ways similar to peer counseling, peer mediation is a more focused intervention strategy aimed at resolving conflicts between two individuals. In a schoolwide implementation of a peer mediation program, Bell et al. (2000) observed important changes in student conduct. Thirty students in the sixth through eighth

grades were trained to be peer mediators for the entire school population of first through eighth graders. Teachers were given the option of referring students to the program or to the principal. During the 6-week study period, 34 mediations were conducted, and the percentage of suspensions and fighting decreased. Johnson and Johnson (1995) evaluated the effectiveness of conflict resolution training using both student body and cadre approaches. Overall, the results indicated that conflict resolution strategies and problem solving can be learned by the students and used in real-world settings.

Suicide and Depression

Suicide and depression interventions are generally in the form of prevention and detection programs or counseling programs. Some interventions encompass a two-tiered system of primary (curriculum-based prevention program) and secondary (peer counseling) prevention components (Sandoval et al., 1987). Further, a number of peer counseling programs address depression and suicide prevention but do not focus exclusively on these issues (e.g., Boehm et al., 1999). For example, one teen counseling hotline received 441 calls pertaining to depression or suicide over a 5-year period (Boehm & Campbell, 1996). Researchers in Hong Kong developed a schoolwide suicide prevention program that was designed to address concerns with available alternatives (Bagley et al., 1994). The Health Intervention Training—Mutual Aid Network promotes positive social support among students using a "student-body approach" in which all students serve as both intervener and client.

Health Behaviors

Health behaviors have been addressed mainly through peer education programs focusing on topics such as sexual behaviors (Agha & Rossem, 2004), AIDS and HIV prevention (Backett-Milburn & Wilson, 2000), substance abuse (Botvin et al., 1990), and asthma education (Shah et al., 2006). A review of studies comparing youth-led and adult-led health education programs indicated that participants in youth-led programs evinced knowledge gains, attitude changes, and behavioral outcomes that were as positive, or more so, as in adult-led programs (Mellanby et al., 2000).

Backett-Milburn and Wilson (2000) described an HIV/AIDS peer education program in which educators worked together in groups. With this enhanced flexibility, the peer educators could capitalize on individual strengths. In addition, while some members led the groups, others were able to work behind the scenes to plan sessions or search for information and resources. In another intervention focused on sexual health and HIV prevention, peer educators conducted single sessions with groups of Zambian adolescents (Agha & Rossem, 2004). The trained educators used dramatic skits to depict sexual behavior models. At follow-up, the intervention group showed significant improvements in their approval of condom use and intent to use condoms. At the 6-month follow-up, intervention adolescents were more likely to think that it was normal for both women and men to suggest using a condom.

Coping with Illness or Medical Procedures

Acknowledging the negative impact of chronic illness on children's social relations and development, coping interventions have also targeted peer interactions (Harbeck-Weber, Fisher, & Dittner, 2003). For instance, Greco, Shroff Pendley, McDonnell, and Reeves (2001) included the best friends of adolescents with diabetes in a group intervention. In this informal peer intervention, the friends learned diabetes knowledge and ways to better support the adolescent with diabetes. A novel intervention approach, STARBRIGHT, capitalizes on computer technology in a peer intervention for hospitalized children (Holden, Bearison, Rode, Rosenberg, & Fishman, 1999). The STARBRIGHT program connects hospitalized children (ages 9 through 19) through an online community. Less formal than many other peer interventions, it allows children to gain support from others experiencing similar conditions. Children using STARBRIGHT reported less intense pain and reduced anxiety compared to those receiving standard pediatric care alone (Holden et al., 1999).

More conventional modeling approaches have been frequently used to help children with medical fears (e.g., Elkins & Roberts, 1986; Melamed & Siegel, 1975). In a normative population of children, modeling was examined by Elkins and Roberts to see whether medical fears could be reduced. Third- and fifth-grade children were randomly assigned to view one of four videotapes. Peer models were used in two videotapes (either one child alone or two children talking together), an adult was used in another videotape, and the last film was a control video in which a child models coping but outside the hospital setting. Children high in medical anxiety reported reduction in their anxiety after viewing all three medically related films; a similar effect was not seen in children with low medical anxiety. Similar methods have been used with children who have significant medical fears or who are uncooperative with medical treatments. For instance, as part of a larger intervention, highly uncooperative children observed peers undergoing a dental procedure and being reinforced for appropriate behavior just prior to their own treatment (Stokes & Kennedy, 1980). Children's uncooperative behaviors (e.g., body movements, complaints, crying) were reduced following the intervention.

Case Illustration

Identifying Features of the Client

Bobby D. is a 7-year-old boy who was diagnosed with Autistic Disorder at age 3. Bobby attends first grade at a public elementary school and engages in an intensive applied behavioral analysis program for children with autism and other developmental delays approximately 20 hours per week. Currently, Bobby lives with both parents and two siblings (4-year-old sister and 9-year-old brother).

Presenting Complaints

Bobby currently divides his weekdays between a structured applied behavioral analysis program and a regular classroom with the support of a behavioral specialist. He also receives individual speech and language therapy 3 days a week. All of these services are coordinated by his school as part of his individual education plan. At home, Bobby's parents receive

additional support from a behavior specialist contracted through a community agency who provides extra instruction on activities such as communication skills (e.g., asking for objects) and daily living skills (e.g., mealtime behaviors).

More recently, Mr. and Mrs. D. sought some adjunctive intervention to help Bobby improve his social behavior. On the advice of school personnel, they contacted a clinical psychologist who was in private practice and had frequently consulted with Bobby's school on cases involving autism spectrum disorders. Although Bobby's social behavior (e.g., eye contact and use of gestures) had improved, it was still of major concern. He is relatively high-functioning and has adequate communication skills, as he is able to speak in simple sentences and occasionally uses nonverbal gestures or sign language to communicate. According to his most recent speech and language evaluation, Bobby has mildly delayed expressive language and moderately delayed receptive language. At the present time, Mr. and Mrs. D. would like to see improvements in Bobby's ability to appropriately respond to and engage with his peers.

Developmental Factors

Like many children with this disorder, Bobby exhibited developmental delays in the communication and social domains. Though he started to walk and crawl on time, his first spoken words came at about 27 months. His parents also noted his frequent resistance to eye contact, lack of response to his name, and failure to reciprocate social initiations. Mrs. D. reported a normal pregnancy and on-time delivery without medical complications. Bobby was a relatively healthy child and had no major illnesses, injuries, or hospitalizations.

Parental Factors

Bobby lives in an intact, two-parent household in which both parents work full time. As such, Bobby spends several hours a day with a behavioral specialist while his parents are at work. Mr. D. has some college education and Mrs. D. completed high school. Both Mr. and Mrs. D. are involved in Bobby's treatment and work with his behavior specialist to provide Bobby with discrete trial training to address daily living skills such as appropriate mealtime and bedtime behaviors.

Bobby's behavior and special needs contribute to Mr. and Mrs. D.'s daily stress. At times their marriage has become strained due to differing opinions regarding Bobby's care. Another concern is that the additional attention given to Bobby will have detrimental effects on their other children. Mr. and Mrs. D. appear to be coping well with this stress and are involved in a support group for parents with children with autism.

History

Prior to intervention efforts starting at age 3, Bobby had very limited language abilities. He exhibited severe delays in both expressive and receptive language. As previously mentioned, Bobby's language skills have improved and he now evidences only mild to moderate delays. At times, Bobby uses jargon and echolalic speech, but he can communicate using simple sentences.

Bobby has also exhibited a number of delays in the acquisition of social skills. When initially diagnosed, Bobby rarely made eye contact and made very few

(*Continued*)

attempts at social referencing. For instance, he would occasionally grab one of his parent's hands and pull him or her toward a requested item. However, he never made attempts to show his parents completed activities or gain their assistance when injured. When around his siblings or peers, Bobby rarely made any attempts to engage them and often ignored other children's social bids. For instance, in preschool, he would sometimes push other children away when they approached him to engage in a joint activity. At this time, Bobby has made improvements in his eye contact such that he consistently makes eye contact with familiar individuals (e.g., parents, teachers, siblings, familiar peers). He still has difficulty in his peer interactions and does not spontaneously engage in reciprocal play behaviors.

Assessment

The consulting psychologist conducted a clinical interview with Bobby's parents. In addition, Mr. and Mrs. D. each completed the Achenbach Child Behavior Checklist for ages 6 to 18 (CBCL) and his teacher, Ms. Baker, completed the Teacher Report Form (TRF) and Social Skills Rating System (SSRS). Bobby was also observed in his classroom and on the school playground to assess for the frequency and quality of interactions with peers. Based on information obtained from the parental interview and observations of Bobby's behavior, the psychologist completed the Childhood Autism Rating Scale (CARS), a clinician-completed scale assessing abilities or domains that are frequently impaired in children with autism.

On the CBCL, Mr. and Mrs. D.'s responses both resulted in clinical elevations on the Social Problems, Attention Problems, Externalizing, and Total Problems subscales. Mrs. D.'s responses also resulted in an elevation on the Withdrawn/Depressed subscale. On the TRF, Ms. Baker's responses resulted in clinical elevations on the Social Problems, Attention Problems, Aggressive Behavior, Externalizing, and Total Problems subscales. Overall, Ms. Baker endorsed a greater severity of symptoms than either Mr. or Mrs. D. Her SSRS endorsements indicated that Bobby's social skills were well below that of more normative comparison children, and his overall conduct was somewhat more problematic. Frequently occurring behaviors included temper tantrums, whereas behaviors such as making friends easily and giving compliments to peers were endorsed as never occurring.

Behavioral observations in both the classroom and on the playground revealed that Bobby rarely engaged in social interactions with his peers. During a 20-minute play period, Bobby was observed to interact with his peers for less than 5 minutes. All interactions were initiated by peers, but Bobby often ignored such initiations. On two occasions, Bobby was observed to push other children away. In the classroom, he ignored other children even during a group task. While three other children in his group interacted and worked together to complete a puzzle, Bobby was observed to stare off into space and rock in his chair.

On the CARS, Bobby received an overall score that placed him in the mild autism range. Domains in which he exhibited the most significant delays included his relations to others, interactions with objects, nonverbal communication, and emotional responding.

Case Conceptualization

A major characteristic of autism is impairment in social functioning, including difficulties initiating or maintaining social interactions, impaired social relationships, and a lack of reciprocity or sharing of social or emotional experiences. Similar to other children with autism, Bobby has difficulty engaging in social interactions with peers and often refuses or ignores other children's social bids. Bobby exhibits other behaviors characteristic of autism, such as rocking, spinning, and repetitive play behaviors. Unlike other children his age, Bobby's play activities are generally limited to playing with trains or railroad-related toys and pictures. When looking at pictures of trains or playing with trains, he often becomes immersed and has difficulty being redirected to other activities. At this time, the major treatment concern is to increase the frequency of Bobby's social interactions with his peers. Improvements in social and other domains are difficult to achieve, and progress is incremental.

Course of Treatment and Assessment of Progress

The major goals of treatment were to improve Bobby's responses to other children's social initiations. The treatment approach was based on that used by Goldstein and colleagues (1992). Based on Ms. Baker's report, three socially competent peers were selected from Bobby's classroom to facilitate the intervention. The psychologist obtained Mr. and Mrs. D.'s permission to involve these children in intervention efforts. Prior to the start of the intervention, the psychologist discussed the responsibilities and behaviors required with each peer intervener and their parents. Parental consent and child assent were obtained. The parents were informed of the nature of the intervention and the need for confidentiality. The children were told that they had been selected to play with Bobby, but were not explicitly informed of the treatment aspect of this play to minimize responsibility for maintaining confidentiality. Bobby was not informed that he was the target of an intervention. The children were trained in separate steps to use several strategies to promote social interaction. Consistent with Goldstein and colleagues, the steps included (a) establishing joint attention by moving into Bobby's visual space, (b) saying "Bobby," (c) repeating "Bobby" and physically tapping him if he remained unresponsive, (d) starting a conversation and listening to see if Bobby responded, and (e) continuing to speak to Bobby. Each peer had individual play sessions with Bobby in which these steps were practiced. To gain peer intervener compliance, rewards were given following each play session for following the steps and engaging in conversations with Bobby.

Bobby's progress during these sessions was monitored by a teacher or teacher's aide who recorded the number of steps that were required to gain Bobby's attention and whether or not he engaged in conversation with the peer. To assess generalization, the psychologist observed Bobby on a biweekly basis in his regular classroom setting or on the playground. As with the baseline observations, the frequency of Bobby's interactions with peers was tallied. Over a 4-week period, Bobby exhibited substantial improvement in his responses to the three peers during the one-on-one play sessions. However, these improvements failed to generalize to other, less structured settings.

(*Continued*)

Complicating Factors

Although Bobby's behavior improved in the one-on-one play sessions, it failed to generalize to other situations, such as unstructured play with other peers. During her observations, the psychologist noticed that Bobby's responses to peer initiations had not improved and that peers rarely attempted to initiate such interactions. Even those peers who engaged in the one-on-one play sessions were rarely observed to attempt social interactions outside of these sessions. To address these concerns, the peer prompters were instructed to also initiate social interactions outside of the individual sessions. Teachers took note of these initiations and rewarded interveners for attempts outside of the individual sessions. Over the next 6-week period, Bobby's peers engaged in more frequent initiation attempts. Further, Bobby's responses to these attempts improved such that the frequency of ignoring initiations decreased dramatically and aggressive behavior (e.g., pushing children away) was eliminated completely.

Managed Care Considerations

Receiving compensation for services provided in schools is often a challenge. In this case, only the time spent with the parents in the clinic may be billable according to most managed care guidelines. The time spent training and supervising the peer interveners, as well as coordinating and monitoring the ongoing intervention, would probably not be covered by most insurance companies. Travel to and from the school would likely be an additional unrecoverable expense. Being directly contracted by the school as a consultant is a preferable arrangement in cases such as the one presented here.

Follow-Up Treatment

Once the frequency of Bobby's social interactions and the quality of his responding had improved to reasonable levels, the active peer intervention was terminated. Continued observation and teacher report indicated that Bobby was engaging in more frequent social interactions with his classmates and rarely ignored other children's social bids. Subsequently, a follow-up treatment was started that consisted of attempts to use peers to model more complicated and extended social interactions.

Implications of the Case

Bobby's case highlights several concerns regarding the implementation of peer interventions. In this case, some difficulty in achieving generalization was encountered. Although the peer interveners initiated interactions with Bobby during structured play sessions, they failed to initiate in other situations in which they were not directly reinforced. This emphasizes the need for continued monitoring of progress and supervision of peer interveners. Bobby's situation also illustrates some of the challenges in obtaining consent and maintaining confidentiality. Given the young age of the children involved in this intervention, they were not fully apprised of the therapeutic nature of their involvement. Further, Bobby was not made aware that he was being targeted in these social initiations.

Recommendations to Clinicians and Students

Despite their many advantages and seeming simplicity, peer interventions pose some significant challenges. Given issues relating to reimbursement and

availability of peers to intervene, a majority of these interventions take place in a school setting. Even those interventions in community settings occur outside of a typical clinic setting. Added demands include the time required for training the peer interveners, consulting with teachers, and providing ongoing supervision and monitoring. In combination with the unique concerns regarding informed consent and confidentiality, the feasibility of this intervention form is an issue for most practitioners. Such interventions may be best suited to those already working within the school setting.

SUMMARY

With their many inherent advantages, peer interventions have grown in popularity and have been used in efforts to address a wide variety of clinical problem areas and disorders. For many reasons, peer interventions may be preferable to adult-led treatments. Due to their similar social status, children may be perceived as more accessible and can be convenient for implementing treatment. Information may be more readily received when coming from one's peer, and the stigmas associated with treatment seeking may be lessened. The convenience and cost-effectiveness of peer interventions enhances their applicability to school settings, thus potentially expanding service provision to large numbers of children. Using peers as treatment agents and implementation in the natural environment also facilitates maintenance and generalization.

With these advantages come some considerations and potential concerns. Though often touted for their convenience and cost-effectiveness, peer interventions are not as straightforward as they may seem. Clinicians must carefully weigh the pros and cons relative to more traditional adult-mediated approaches. Setting is one consideration, as these interventions are most frequently implemented in the child's natural environment and seem most amenable to school environments. A variety of particular intervention forms are available, and one most suitable to the target child and presenting problem must be chosen. Implementation requires a carefully considered approach to the selection, training, and supervision of the peer interveners, as well as sensitivity to the unique ethical challenges posed by peer interventions. Securing the cooperation of others in the child's environment, such as teachers and other school personnel, and obtaining quality outcome data are additional considerations.

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PART VI

Special Issues

CHAPTER 26

Cultural Issues

Laura Johnson and Christina Tucker

Culture is the lens through which we view children and adolescents. Culture provides the frames we use to label, categorize, and make sense of childhood development and behaviors. It defines our relationship to children, what is considered normal and desired child behavior, and what is considered pathological. Culture indisputably impacts childhood psychopathology. How disorders are expressed and communicated, whether help is sought for children, treatment processes and desired outcomes are all influenced by culture. Psychologists and other mental health professionals must have a thorough understanding of culture and the ways it impacts child and adolescent mental health, psychopathology, service utilization, assessment, and treatment (American Psychological Association [APA], 2003; Constantine & Sue, 2005; Cuellar & Paniagua, 2000).

In this chapter, we first provide an overview of major sociocultural changes occurring globally and in the United States that have implications for children. With this contextual frame we highlight the need for cultural competence in the assessment and treatment of children. Next we define culture and describe its complexity, outlining the ways that it influences the status of children, child development, socialization processes, family and social relationships, and beliefs about child and adolescent psychopathology. We introduce key cultural concepts, such as acculturation, ethnic identity, and worldview, and provide suggestions for working with culture at each phase of assessment and treatment. Problems related to culture, cultural adaptations to treatment, and unique populations are also discussed.

A CHANGING MULTICULTURAL CONTEXT

Global Perspectives

Worldwide increases in population mobility and economic globalization are bringing together people from all cultures at unprecedented rates (Global Commission on International Migration [GCIM], 2005; Population Resource Center [PRC], 2005; N. Kaufman & Rizzini, 2002). Children and adolescents, accounting for 39% of the world's population, are increasingly growing up in a multicultural context (GCIM, 2005). Economic globalization and infrastructure development have resulted in increased opportunities for living, working, and studying in other countries. There is an overall increase in voluntary migration, usually from rural to urban areas

and from developing countries to economically developed countries (GCIM, 2005; PRC, 2005). There is also an increase in forced migration, resulting in 25 million children uprooted from their home due to natural disasters, ethnic conflict, and war (United Nations High Commission for Refugees, 2001). While the importance of the extended family network remains high on a global level, population mobility breaks up traditional kinship networks and strains family and community supports.

Currently, the global population exceeds 6 billion, and over 1 billion children will be born in the next decade (PRC, 2005). Most of these children will be children of color, many will be poor, and most will live in developing countries (GCIM, 2005; PRC, 2005). Critical challenges facing these children include poverty, violations of human rights (e.g., child trafficking, involvement in armed conflict), and social justice concerns, such as the widening gap between rich and poor and between developed and developing countries (GCIM, 2005; PRC, 2005). A lack of sociopolitical power, racism, environmental threats, health risk behaviors, and global population pressures are other critical concerns facing today's children (GCIM, 2005; N. Kaufman & Rizinni, 2002; Lerner, Fisher, & Weinberg, 2000; PRC, 2005). While population rates are on the rise, natural resources are dwindling, and cultural ideology (based on ethnicity, religion, or social class) often frames the argument over resources and supplies the justification for violent conflict. Trends in population growth and mobility are expected to continue, resulting in more opportunities and demands for intercultural contact and communication (Martin & Nakayama, 2005).

Cultural Diversification of the United States

The United States is becoming an increasingly diverse, multicultural society. Children today are more likely to belong to an ethnic minority group (Hernandez, 2004, 2006). As illustrated in Figure 26.1, ethnic minority children are quickly becoming the new statistical majority. Immigrant families and children account for the greatest growth in the ethnic minority population, and about 84% come from countries in Latin America and Asia (Hernandez, 2004). Current estimates indicate that approximately 35 million foreign-born persons live in the United States, with about 1 in 5 children having at least one foreign-born parent (Hernandez, 2004; PRC, 2005). The top countries of origin are Mexico (30.7%), the Philippines (4.5%), India (4%), Mainland China (3.4%), and Vietnam (3%), with all other countries combined accounting for 45.6% (PRC, 2005).

Implications for Children

The changing multicultural context in the United States and worldwide has implications for how children and adolescents will interface with educational, legal, occupational, and health and mental health care systems (Hernandez, 2004; GCIM, 2005). These institutions were designed from a particular cultural perspective (European American) and, at least historically, were for the benefit of a population with that same cultural perspective (Marsella & Yamada, 2000). Educational, economic, health, and mental health disparities are well documented (U.S. Department of Health and Human Services, 2001). Compared to European Americans, ethnic minorities in the United States face lower educational status, higher unemployment,

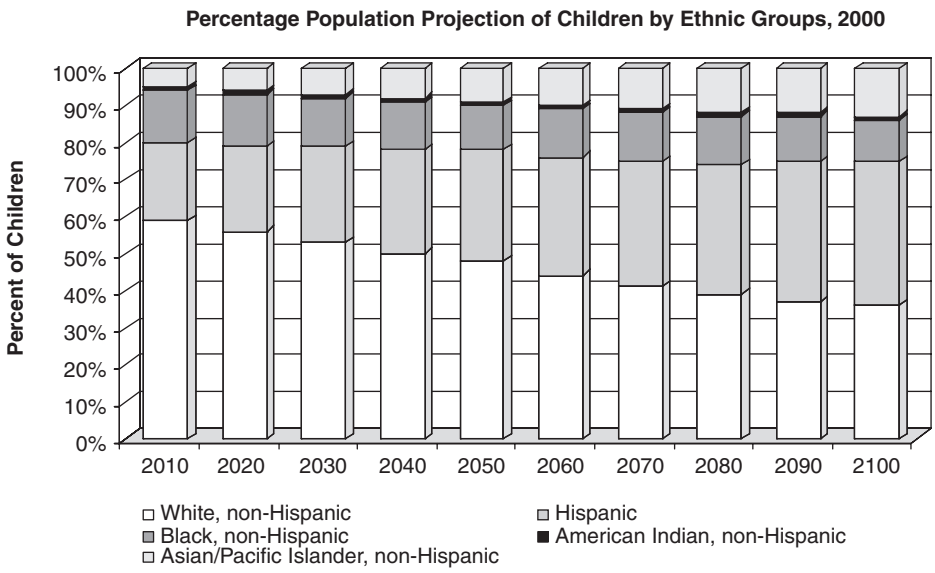

Figure 26.1 Changing ethnic profile of children in the United States. *Source:* "Demographic Change and Life Circumstances of Immigrant Families," by D. J. Hernandez, April 2004, *Future of Children*, pp. 17–47. Reprinted with permission.

higher rates of poverty, a higher risk of health and mental health problems, and increased rates of victimization and incarceration (Cuellar & Paniagua, 2000). Children living in immigrant families have higher poverty rates, are more likely to have poorly educated parents, and are more likely to live in linguistically isolated households than native-born families (Hernandez, 2004).

Despite the cultural and linguistic diversification of the United States, ethnic minority and immigrant populations continue to underutilize outpatient professional psychological services. Those who do seek help are more likely to be misdiagnosed, receive inappropriate treatment, drop out of treatment early, and receive fewer positive benefits overall compared to European Americans (Kurasaki, Sue, Chun, & Gee, 2000; Sue, Chun, & Gee, 1995; Sue, Zane, & Young, 1994). Psychologists have an ethical imperative to understand these disparities and to work toward clinical, scientific, and educational excellence in the applicability and effectiveness of their approaches (APA, 2003; Whol & Aponte, 2000).

Cultural Competence

Cultural competence is a product of general, cross-cultural, and culture-specific competencies that allow for effective functioning in the context of cultural differences (APA, 2003). It applies to individual and organizational practices and spans the areas of education, research, clinical training, and practice in psychology (APA, 2003; Constantine & Sue, 2005; D. Sue, Arredondo, & McDavis, 1992). Operationally, cultural competence is often described as three interrelated components of awareness, knowledge, and skills (Arredondo et al., 1996; D. Sue et al., 1992).

Awareness

Clinician *awareness* is commonly viewed as a prerequisite for the development of cultural competence. This requires clinicians to become aware of themselves as cultural beings. Clinicians are encouraged to intentionally bring their attention to the role of culture in their own life and to consider how it impacts their own values, beliefs, and behaviors, including biases and stereotypes about diverse children and families (APA, 2003). Skills in self-reflection can be developed and awareness enhanced through written reflection assignments, such as journal writing or reaction papers, case studies, small experiential exercises, group discussions, and coaching (Hong, Garcia, & Soriano, 2000; Martin & Nakayama, 2005).

Clinical self-awareness must extend to the profession. It is crucial to understand the sociocultural and historical context in which mainstream (i.e., European and American) theories of psychological assessment and treatment emerged (Pedersen, 2004). Although individuals of European or North American ancestry account for only one sixth of the world's population, a Western worldview and cultural perspective dominates modern psychological therapies and practices (Marsella & Yamada, 2000). Psychologists are gaining increased awareness of the cultural embeddedness of their approaches to defining, assessing, and treating childhood psychological problems.

Knowledge

Beyond awareness, clinicians must enhance their *knowledge* and understanding of culture and how it impacts children, adolescents, and the assessment and treatment process. This includes general cultural concepts applicable across many cultural groups and culture-specific information related to a particular cultural group (APA, 2003; Constantine & Sue, 2005). General knowledge includes a clear understanding of the nature of culture and how it impacts the expression of pathology, beliefs, and attitudes about childhood problems, expectations for the assessment and treatment process, and the clinician's relationship with the family.

Specific knowledge about one's *own culture* and *other cultures* is also necessary. Clinicians can enhance their cultural knowledge by reading literature, attending continuing education conferences or workshops, watching films, and reading professional books and journals as well as popular books or magazines. Experiential-based activities, such as visits to ethnic neighborhoods, international travel, second-language learning, interviews with diverse others, volunteer work, and practicum experiences, are additional ways of enhancing one's knowledge of other cultural groups.

Clinicians should undertake particular efforts to educate themselves about the major cultural groups with whom they are likely to come in contact. Additionally, when children or their family members come from cultural backgrounds with which the clinician is unfamiliar, clinicians should learn about relevant culture-specific group norms, including, but not limited to, sociopolitical history or history of oppression, cultural beliefs and values, communication styles, and attitudes toward mental health and illness (Pedersen, 2004; D. Sue & Sue, 2003). Although

it is important to be knowledgeable about group norms, there is vast within-group variability.

Skills

Clinician *skills* for cultural competence include reflective thinking, establishing rapport, bridging cultural differences, intercultural communication, and accurate and culturally appropriate assessment, diagnosis, conceptualization, and treatment (APA, 2003; Arredondo et al., 1996; Constantine & Sue, 2005; Hays, 2001). In the following sections, we provide information to further clinician awareness and knowledge about the nature of culture and how it impacts the lives of children and adolescents.

THE NATURE AND COMPLEXITY OF CULTURE

A better understanding of culture and its role in shaping beliefs and behaviors is essential for clinicians working in a multicultural environment. Although there are well over 150 definitions of culture, most describe culture as being learned and shared among a group of people and passed on through the generations (Kluckholm & Kroeber, 1952). Culture includes external factors, such as style of dress, language, food, traditions, kinship structures, and social patterns, and behavioral norms, including norms for expression of psychological problems (Draguns, 2000; Paniagua, 2000). Culture can be considered a macro-template for predicting and explaining child and adolescent behavior. However, this view of culture is deceivingly simple. Culture is quite complex and has a number of features that present challenges in the clinical setting. A failure to understand the complexity of culture can result in an over- or underemphasis on culture, stereotyping, misdiagnosis, misuse or misinterpretation of tests, or culturally incompatible treatment (Canino, Canino, & Arroyo, 1998; Canino & Spurlock, 2000).

Culture Is Pervasive and Subjective

Culture has a pervasive influence in the lives of children. It encompasses a wide range of human behaviors and activities and ways of seeing, understanding, and interacting with the world. It influences our sense of time and time consciousness, relational styles, patterns of communication, sense of personal space, family structures, worldview, and ways of thinking, categorizing, and problem solving. Culture includes overt aspects, which are driven by internal features, such as beliefs, attitudes, norms, and values, that are largely unobservable and subjective. These are the unwritten rules of culture. An iceberg analogy is often used to illustrate that the observable aspects of culture are just the tip of the iceberg. Not only are most cultural factors unobservable, but they are also out of our awareness. Like fish in water we go about our daily lives, behaving and interpreting from a particular cultural frame without the awareness that we are doing so. Intentionally bringing culture into one's awareness (and keeping it there) can help clinicians recognize cultural influences and biases (Pedersen, 2000).

Culture Is Multidimensional and Dynamic

Culture is multidimensional, and children and families may describe their culture in relation to one or more dimensions or sociocultural reference groups, such as sociorace, ethnicity, tribal or clan affiliation, language group, religion, socioeconomic status, age or generational status, gender, sexual orientation, disability, geographic or regional location, nationality, and political affiliation (D. Sue & Sue, 2003; Hays, 2001). Although these groups have common features that are shared with others, the importance of different dimensions of culture (and therefore their relative importance in assessment and treatment) is in a state of flux as children develop and move in and out of various cultural milieus.

What cultural dimensions are salient will vary from child to child and family to family across time and contexts. The importance of various cultural factors and identity dimensions may change as a function of developmental stage. For example, ethnic identity and sexual orientation may be more relevant for adolescents than for young children. Relevance may also be spurred by an event, such as experiencing an act of racism. Many ethnic minority children and adolescents report that they were unaware of their race until they were called a racist name. Other events that require a child or adolescent to recognize and adapt to another culture, such as moving to a new country, a new school, or even a different state or region, can highlight the importance of different aspects of culture and identity.

Culture Is Uniquely Expressed

Cultural influences interact with individual and environmental factors to manifest idiographically in each child. The relative importance of different dimensions will vary based on a child's learning history, life experiences, and context. For example, in the United States, where there is a history of slavery, segregation, and oppression and no shortage of racial discrimination today, "being Black" is often a salient feature in determining the worldview and identity of African American children, adolescents, and their parents (Helms, 1990). In other countries, religion, social class, or tribal affiliation may be more contextually salient. Status as a member of the majority or minority may also influence a child or family's worldview or identity. Sexual identity may be more salient for gay and lesbian youth than heterosexual youth, and religious identity may be more salient for Muslim youth growing up in a majority Christian environment. It is important to note that cultural contexts change as well, bringing to the forefront different dimensions. For example, prior to the Belgian colonization of Rwanda, one's ethnicity (e.g., Hutu or Tutsi) was largely irrelevant, but from 1993 to 1994 it became the basis for who would kill or be killed.

Culture Is Interactive

Children and their families are involved in multiple cultural systems and interactions. Children and youth move in and out of different cultural contexts and social environments in their day-to-day lives (Akiba & Garcia-Coll, 2004; Bronfenbrenner, 1979, 1989; Cuellar, 2000). As illustrated in Figure 26.2, culture

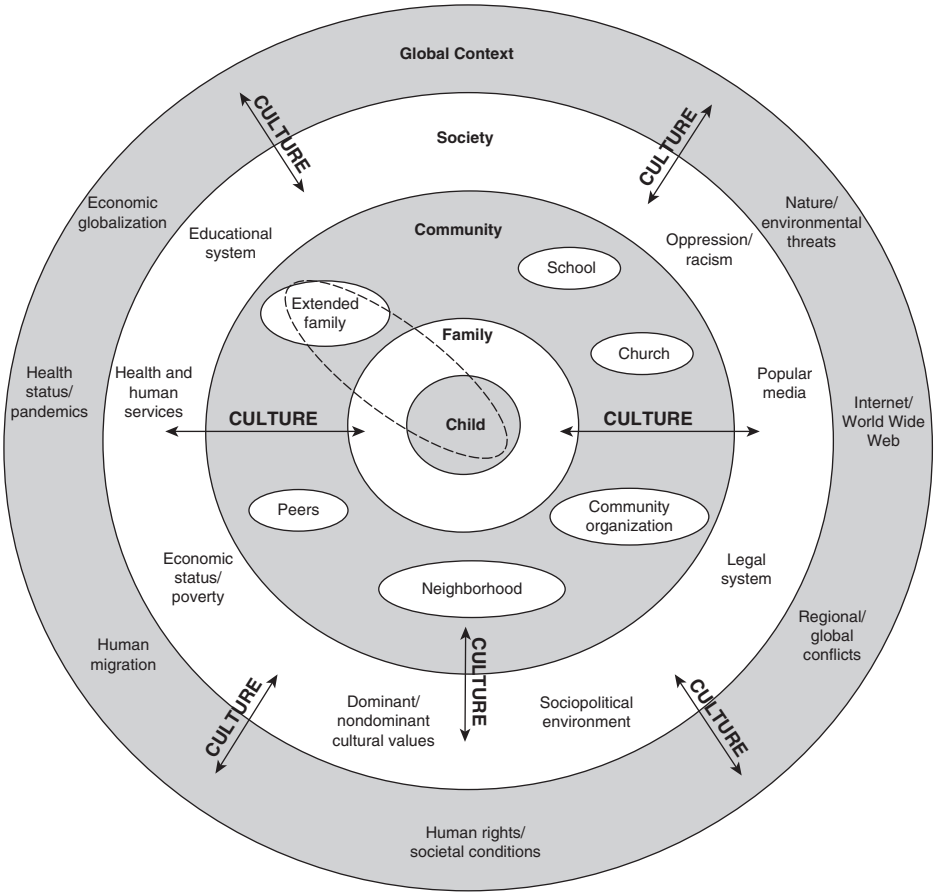

Figure 26.2 Interactive and reciprocal influence of culture across ecological contexts.

has a multilevel and reciprocal influence across children's contexts, at the proximal, microsystem level as well as at the more distal, macrosocial level. Family and community cultures interact with larger societal cultural systems and global influences. There may be drastic differences in communication, accepted norms, rules, and behaviors in the various microsystems of home, community, and school, or between the child's community and the larger society. Clinicians should consider how culture impacts the child's functioning and fit within his or her range of daily experiences. This includes the cultures of the home, school, neighborhood, and community and how they interact with each other (Bronfenbrenner, 1979, 1989).

Macrosystem factors include globalization, sociopolitical atmosphere (e.g., human rights, degree of democracy and freedom), socioeconomic conditions (e.g., poverty, educational opportunities), environmental threats related to industrialization and urbanization (pollution, safety concerns, overcrowding, noise, connection to natural environment), and overall health status (epidemics, pandemics, health risks). Of key importance is that children and adolescents are engaged in a reciprocal relationship with culture. The ecological model of child development shows

that children play an active role in defining, interpreting, and shaping their cultural environment (Brislin, 2000; Bronfenbrenner, 1979, 1989).

THE IMPACT OF CULTURE ON CHILDREN

Cross-cultural research in psychology, anthropology, and sociology has revealed much about cultural variation in the role and status of children, approaches to education and socialization, cultural and social norms for development, family structure, and social relationships (Brislin, 2000; Schmidt, 2006; Tseng, 2003). Clinicians should be aware of these differences, as they have important implications for assessment and treatment.

Role and Status of Children

Around the world there is considerable cultural variation in children's roles and status, influencing how they are viewed and treated (Brislin, 2000; Petersen, Steinmetz, & Wilson, 2005a; Tseng, 2003). For example, children and adolescents living in more individualistic cultures, such as in the United States and Western Europe, may be viewed as separate, equal individuals deserving of respect. They may have considerable choice in their day-to-day lives and are frequently given a voice in family and community matters. In the context of the one-child policy in China, it has been suggested that Chinese children are likely to be pampered and doted on by parents and grandparents. However, their status as only children in a collectivist society also brings an expectation for filial piety, increased responsibilities for future caretaking, and intense pressure to succeed (Feng, 2002; Tseng, 2003). Other cultures view children as property or as essentially providing a service. In some parts of the world, for example in New Guinea, children as young as 8 may be responsible for growing, harvesting, and cooking food for themselves and their younger siblings.

The responsibilities, role, and status of children can influence risk and protective factors that may enhance or reduce their physical and psychological health. This has implications for a child's overall development; the supports, services, and resources afforded them; and their potential to be neglected or abused (Tseng, 2003). Gender, class, and cultural issues tend to interact in this regard, often conferring male and class privilege though reduced social status and fewer economic and educational opportunities for young girls worldwide (United Nations Children's Fund, 2004).

Education and Socialization

Cultural factors shape the ways that children and adolescents are educated, socialized, enculturated, and disciplined. Major differences in child-rearing practices are found cross-culturally in terms of who is involved in education and socialization of children and also in the methods used to teach children (Brislin, 2000; Petersen et al., 2005a). In a Western setting, teaching styles favor a direct approach emphasizing written history, direct verbal communication, and linear and dialectical thinking. Critical thinking and Socratic questioning are often encouraged, and the uniqueness of each individual is valued. Teaching in the United States, in particular,

is considered child-centered, with the responsibility for the child's learning placed with the parent or teacher. In other cultures, approaches to teaching, and often parenting, tend to place more responsibility on the child and have high expectations for obedience and respect for authority (Petersen, Steinmetz, & Wilson, 2005b). Blending in and working for the good of the whole is valued, and personal distinction is frowned upon. Asking questions of persons in authority (teachers or parents) is desirable behavior in one context, whereas in another it brings ridicule and shame.

Cultural differences exist in tolerance for misbehavior and in approaches to punishment (e.g., short or long term, mild or severe) and types of punishment (e.g., physical; suspension of social, material, or activity rewards; emotional response such as shame or embarrassment, or withdrawal of love; Brislin, 2000; Peterson, Steinmetz, & Wilson, 2005c; Tseng, 2003). Families may have different approaches for boys and girls, or for children of different ages. Discipline may be used consistently or haphazardly (Tseng, 2003). In the United States, African American parents often have a more authoritarian approach to child rearing than European American parents. American Indians, on the other hand, may value a natural unfolding of development (Brislin, 2000). Misunderstanding of the cultural norms for parenting may lead one to incorrectly jump to assumptions about abuse or neglect with these and other populations.

Sense of Self, Family, and Social Relationships

How one views the self and an emphasis on different aspects of the self is a fundamental concern to multicultural assessment. The idea of the unique, autonomous, and bounded self and the American ideal of rugged individualism are in contrast with the view of the majority of the world's population, which is more interdependent or sociocentric in nature. In these cultures, the boundaries of the child's sense of self extend to include siblings, parents, grandparents, aunts, uncles, and other extended family members; members of the community, ethnic group or tribe, and nation; ancestors; even the solar system and the universe (Brislin, 2000). The nature of interpersonal relationships that children and adolescents share with others varies from culture to culture, and the differences in closeness of relationships and the kinship structures can be drastic. American ethnic minority groups and immigrants are more likely to have important extended family structures and obligations than Euro-American children and families. Immigrant children are more likely to live with extended family members (Hernandez, 2004). Clinicians should pay careful attention to flexible family roles and assumptions related to the closeness and the importance of significant relationships of children. Ideas about dependence, enmeshment, and differentiation of the self should be examined critically (Hays, 2001; Tseng, 2003).

Normative (and Valued) Development

Definitions about what is healthy and desired child and adolescent development and behavior vary culturally. Different modes of behaving (passive versus active), communication (oral versus written, verbal versus nonverbal), relating to others

(independent versus interdependent), and acquisition of different skills (e.g., motor versus language skills) vary culturally (Brislin, 2000). So, too, do the ways that development is shaped by a child's family and society and how poor (or abnormal) development is tolerated and handled. Cultural differences in expectations and valued processes and outcomes are commonly seen between individualist cultures and collectivist cultures (Triandis, 1995). It is useful for clinicians to keep in mind that worldwide, the majority of cultures (including all major ethnic minority groups in the United States) are rooted in collectivism. Individualistic cultures tend to value independence in behavior and relationships, competitiveness, and autonomy. Egalitarianism and fairness are also valued, and individuality and uniqueness are encouraged. On the other hand, traditional collectivist cultures tend to stress an interdependent sense of self, cooperative behaviors, and dependence, and value social responsibility over personal interests.

In collectivist cultures, consideration of children's attachments may be expanded to include attachment and bonding relationships with extended family and other important caregivers. Developmentally, 2- to 4-year-old children in a Eurocentric culture may be encouraged toward autonomy, whereas other cultures may view too much independence as problematic and thus place a greater value on shame. In some sociocultural contexts, child initiative may be expressed as aggression, whereas other cultures may value a "being" over "doing" orientation toward activity (Tseng, 2003). The age at entering adulthood also varies. Instead of moving toward independence, some cultures may emphasize adolescents' understanding and fulfilling their family or societal roles. Gender expectations and roles will interact with both age and cultural differences, and these roles may be more or less defined and rigid.

Variation in Psychopathology

How problems of a psychological nature are defined is determined by culture, and behaviors can be understood only within their cultural context. Culture impacts the incidence of various problems of childhood, symptom expression, help seeking, and treatment process (Aponte & Johnson, 2000; Draguns, 2000). Culture may directly or indirectly contribute to pathology, shaping the clinical picture as a whole or at level of symptom manifestation (Jenkins, 1996).

Some challenges to clinicians include cultural issues related to mind-body dualism, somatic presentation, the role of spirituality, ideas of selfhood, and categories of emotion and related semantics (Angel & Williams, 2000; Draguns, 2000; Good & Good, 1986). This separation of mind and body and the associated symptoms stems from Western dualism and runs contrary to holistic views of the mind and body that dominate non-Western thinking. How one views the self and emphasis on different aspects of the self will impact the cognitive and affective aspects of disorders and their associated meanings. This has implications for the prominence or absence of symptoms and how they are expressed. Statements assessing depression, for example, emphasize the self ("I feel sad"), whereas individuals from non-Western cultures may be more apt to express dysphoria in terms of their relations to others, for example, concerns about disappointing or embarrassing other

individuals (Manson, 1995). Symptoms associated with anxiety and the content of phobias also vary greatly across cultures.

Variations in the language of emotion and differential emphasis on particular emotions also present difficulties. *Indeterminacy of meaning* refers to problems that arise in the semantic translation of concepts and symptoms and the assumptions made about emotional categories (Good & Good, 1986). For example, Western-based diagnostic systems (e.g., the *Diagnostic and Statistical Manual of Mental Disorders* [*DSM*]) may construe guilt, shame, and sinfulness as generally having the same meaning, whereas other cultures may see important differences. In other instances, symptoms or disorders that are conceptually distinct (e.g., anxiety and depression) according to Western nosology may be presented as fused (e.g., feeling "sad and bothered"; Good & Good, 1986).

Another issue of concern for clinicians is how to respond to somatic and spiritual or religious presentations of problems. Somatic symptoms play a more central role in both the experience and expression of mental disorders worldwide. Somatic distress is common in non-Western populations and in American ethnic minority groups (Kleinman, 1987). Similarly, spiritual and religious factors may play a greater role in understanding culturally diverse populations. Spiritual concerns may include concerns related to religious identity, loss of faith, religious beliefs that a child's disorder is a sign from God (blessing or punishment), problems of the soul, angered ancestors or God(s), concerns related to being cursed though witchcraft or sorcery, or problems interacting within a religious community. These symptoms may be puzzling to Western clinicians and are not well integrated into Western conceptions of mental illness. Spiritual or religious beliefs may also come into play as cultural strengths, coping resources, paths to treatment, or the means of treatment.

Cultural Strengths and Resources

Thinking about the complexity of culture and the many ways that it can impact children and their families, it is easy to see how clinicians may become overwhelmed with all that cultural competence requires of them. Culture can also be a great help to clinicians, allowing them to tap natural strengths and coping resources of children and families. Identifying family cultural values and resources can help the clinicians develop treatment plans that are acceptable to the family and also work with their strengths and supports (Hill, 1993). Some of the cultural strengths and values of the major ethnocultural groups in the United States are shown in Table 26.1.

CULTURAL CONCEPTS IN ASSESSMENT AND TREATMENT

Responding to culture effectively in the clinical setting requires that we unpackage culture into its many parts (or cultural factors). Defining and assessing specific cultural concepts can allow for a deeper understanding of culture as it impacts children within and across cultural groups. Included among these, and essential for accurate assessment and treatment, are etic and emic approaches, enculturation, acculturation, cultural identity, worldview, and explanatory models.

Table 26.1 Cultural Strengths and Resources of U.S. Ethnic Minority Groups

African Americans	American Indians	Asian Americans	Latinos/as
Strong desire to achieve in education	Success in learning important cultural values of tribe	Family expects high academic performance from children; Peer support for academics	Academic success supported by family and community
Strong work ethic	Strong work ethic	Intense loyalty to employer	Vigorous work ethic
Flexible family roles; Extended kinship and supports; Connection to other African Americans	Diverse family structures; Communal values of sharing within group and extended family	Family roles are hierarchical and interdependent	Value marriage, importance of immediate, extended family, and friends
Strong spiritual and religious orientation	Sense of connection between nature and self	Strong sense of connection among family members	Importance of interpersonal relationships and sensitivity

Etic and Emic Approaches

Western European and American psychological research and practice traditionally relied on a nomothetic or etic approach aimed at discovering universals (Lewis-Fernandez & Kleinman, 1995). An idiographic, or emic, approach emphasizes the importance of the local cultural context in shaping phenomena (Munroe & Munroe, 1980). Kleinman (1977, 1980) offers a conceptual distinction between the etic construct of "disease/disorder" and the emic construct of "illness." Disorder refers to the malfunctioning of biological or psychological processes. Illness is a cultural construct that refers to the expression and experience of the disorder. It includes the personal and social meanings of the disorder and how it impacts the lives of children and their families (Kleinman, 1977, 1980). Use of the two perspectives is considered essential to adequate assessment and treatment planning (Lewis-Fernandez & Kleinman, 1995). The right balance between etic and emic approaches largely depends on the child's and family's level of enculturation and acculturation.

Enculturation

Enculturation refers to the process of socialization into a particular cultural, ethnic, or social group (Aponte & Johnson, 2000; Brislin, 2000). Enculturation occurs directly and indirectly through teaching, coaching, reinforcing various behaviors, social role modeling, media, and other environmental and contextual influences. Enculturation of children is an ongoing process and is impacted by the parents' enculturation history, ethnic identity, social class, and current living context. Cultural knowledge, awareness, pride, and feelings of connectedness in one's ethnic or cultural group (i.e., a strong ethnic or cultural identity) are possible outcomes of enculturation. Successful enculturation also results in skills for interacting effectively

within one's cultural group (Aponte & Johnson, 2000). In some cases, children and adolescents will not be well enculturated. Children of refugees and immigrants may reject parental attempts at enculturation, whereas other children are not exposed to their cultural heritage due to parental and/or societal pressure to assimilate into the majority cultural group. Enculturation processes and outcomes interact with acculturation processes that occur as a result of contact with other cultures.

Acculturation

Acculturation Process

Acculturation refers to changes in values, beliefs, and behaviors that occur as a result of sustained contact with a second culture (Berry, 1997; Berry & Kim, 1988). Due to the increase in cultural contact worldwide, the concept of acculturation has garnered increased attention and application over the past few decades. It is considered crucial in the experiences of refugees, immigrants, indigenous groups, ethnic minorities, and sojourners, such as international students and business travelers (Aponte & Johnson, 2000; Berry, 1997; Roysircar, 2004). Additionally, attention has recently been given to "the other side of acculturation," referring to acculturation processes occurring in dominant majority cultures as a result of immigration and other population trends that cause increased intercultural interactions.

Whereas acculturation generally refers to changes resulting from group-level encounters between two or more cultures, *psychological acculturation* refers to psychological changes occurring at the individual level (Berry, 1998). Psychological acculturation is of key importance in assessing culturally diverse children because it involves changes across a number of domains, including language, cognitive style, attitudes, style of relating, and identity (Berry, 1986).

Acculturative Stress

In the context of acculturation, children and families may experience *acculturative stress.* This refers to mildly pathological and disruptive behaviors and experiences that are commonly generated during acculturation. Symptoms of acculturative stress may include family conflict, academic or behavioral problems, depression, anxiety, physical complaints, anger, identity confusion, and substance abuse (Berry, Kim, Minde, & Mok, 1987). Several group and individual factors interact to impact the amount of acculturative stress experienced by children and families. These include (a) macrosocial influences of the child's environment, such as legal constraints, discrimination, degree of tolerance for cultural diversity; (b) factors associated with the background of the acculturating family, such as worldview, social and familial network structure, cultural distance from American society; and (c) individual factors such as age, sex, language proficiency, personality, coping skills, and cultural knowledge (Aponte & Johnson, 2000; Berry, 1986, 1997). Two other factors affecting acculturative stress are (d) phase of acculturation and (e) acculturation strategy (Berry, 1986, 1997, 1998). Phase of acculturation refers to the pattern of adjustment over time (Berry, 1986). Different patterns of responding to the demands of acculturation have been referred to as acculturation strategies or modes (Berry, 1997).

Modes of Acculturation

Of the many factors impacting acculturation problems in children and families, the mode of acculturation may be the most important in predicting overall adjustment outcomes. Four different modes are assimilation, separation, marginalization, and integration. The modes of acculturation account for the relative degree of cultural contact and participation by families and the amount of value placed on such actions (Berry, 1997). As seen in Table 26.2, those with an *assimilation* strategy try to disengage completely from their culture of origin in hopes of being completely absorbed and accepted into the dominant host culture (LaFromboise, Coleman, & Gerton, 1993). Assimilated children may operate effectively within the new social, cultural, and academic environment, but they may sacrifice their sense of identity, their cultural or familial supports, and groundedness in their culture of origin. Families and children using the *separation* strategy will segregate themselves from mainstream American society and establish relationships primarily with others from their own ethnic or cultural group. Often living in ethnic pockets or neighborhoods, these families may garner much needed social support. However, as these children and their parents face demands to speak English, perform academically, access

Table 26.2 Acculturation Modes and Approaches to Working with Children and Families

Mode of Acculturation	*Positive* Contact, Affiliation, and Effectiveness with *Dominant/Majority* Culture	*Lack of* Contact, Affiliation, and Effectiveness with *Dominant/Majority* Culture
Positive Contact, Affiliation, and Effectiveness with *Own Culture*	**Bicultural** Children and families have an integrated identity; skills to interact across multiple, cultural settings; ability to access services and benefits from heritage culture and majority culture. Consider bicultural approaches; adapt mainstream approaches; integrate culturally related or traditional approaches; use culture as a resource.	**Separated** Children and families are supported in their community but separated from the dominant culture. Conflicts arise in negotiating school, legal, health, or other systems. Mistrust related to racism, oppression, or cultural differences may exist. Consider culturally related and traditional approaches; be aware of biases and miscommunication.
Lack of Contact, Affiliation, and Effectiveness with *Own Culture*	**Assimilated** Children and families function reasonably well in dominant society and systems. May lack cultural identity and traditional supports; internalized racism or inability to fully assimilate may cause conflict. Consider mainstream approaches and culturally adapted approaches.	**Marginalized** Children and families lack effectiveness in both cultures. Without resources, natural coping strengths, and the ability to access help, these families are likely to suffer from multiple hardships, including psychological problems. Consider bicultural approaches.

resources, and/or negotiate school, legal, or health care systems, they are likely to experience high levels of acculturative stress (Berry, 1997; LaFromboise et al., 1993). *Marginalized* families will experience the highest levels of acculturative stress, as they lack the behavioral repertoire needed to interact with members of their own culture and with the dominant society. They have the greatest risk of psychological maladjustment (Berry, 1997; Berry & Kim, 1988).

Children and families with a certain degree of integration into the majority culture will experience the fewest problems (Berry, 1997). The *bicultural* strategy suggests that children can acquire the skills and psychological flexibility to function effectively in both their culture of origin and the majority culture. Achievement of bicultural effectiveness serves as both a coping strategy for and a protective factor against acculturative stress (Berry, 1997, 1998; LaFromboise et al., 1993). In practice, bicultural effectiveness may involve the techniques of *code switching* (completely shifting behavior depending on the context) and *culture blending* (merging aspects of two or more cultures; Garcia Coll & Magnuson, 1997; Szapocznik, Santisteban, Kurtines, Perez-Vidal, & Hervis, 1984).

Acculturation modes are not dichotomous, and families can be at the extremes or in the middle on either dimension. Moreover, children and their families usually do not undergo acculturation at the same rate or approach it in the same way. For example, among immigrant and refugee families, children and youth may tend toward assimilation, while their parents, fearing loss of their traditional heritage, may tend toward separation.

Cultural, Ethnic, and Racial Identity

Cultural identity is a multidimensional construct that results from a dynamic combination of different aspects of social identity, such as ethnic, racial, and gender identity (Aponte & Johnson, 2000; Schmidt, 2006). Theory and research related to cultural, ethnic, and racial identity are prominent contributions to the field of multicultural psychology (Atkinson, Morten, & Sue, 1998). One's cultural identity is usually evolving, but at any point, it represents an outcome of one's age and developmental stage, processes of enculturation and acculturation, and immediate and historical contextual situations. A strong or well-developed sense of one's ethnic or cultural identity is considered to be a protective factor, or developmental asset, among youth (Phinney, 1990).

Numerous models of cultural, ethnic, and racial identity have been developed for various groups, for example, Black racial identity development (Cross, 1971, 1995; Helms, 1990); minority identity development (Atkinson et al., 1998); person of color identity development (Helms, 1995); homosexual identity development (Cass, 1979); biracial identity development (Root, 1990); and White racial identity development (Helms, 1990, 1995). Most of the models describe a process of development from a lack of awareness of oneself as a cultural being and the implications of culture (or race) to an integrated and balanced identity. Along the way, individuals may experience dissonance and confusion about cultural differences (often spurred by an event), and at other times may develop a strong sense of cultural identity and pride coupled with complete immersion into their own cultural heritage

and background. In the more advanced stages of identity development, individuals incorporate many identity dimensions and reference groups into their overall self-concept. This includes the ability to respond objectively to members of the dominant cultural group. Integrated identity awareness often extends to advocacy or social action to benefit culturally marginalized or oppressed groups. Although these models often use a stage process to describe identity development, development does not always occur in a linear fashion. Children may have characteristics of more than one stage at any time (Helms, 1995).

Worldview

Worldview refers to psychological and value orientations that influence how people think, behave, and interpret and experience situations, including the assessment and treatment process. Several models draw attention to different aspects or conceptualizations of worldview. Kluckhohn and Strodtbeck's (1952) orientation model (as cited in Mio, Barker-Hackett, & Tumambing, 2006) is based on four dimensions: (1) orientation to time (e.g., past, present, future), time flexibility, and time consciousness; (2) relationship to nature (e.g., harmony or control over nature); (3) view of the self and social relations (e.g., independent or interdependent); and (4) activity orientation (e.g., focus on being or doing). Other concepts of worldview highlight the two dimensions of locus of control and locus of responsibility, suggesting that unique worldviews emerge based on their combination (D. Sue, 1978). A strong belief in fate among parents, or feeling incapable of controlling or changing the behavior of their children or their circumstances, would have major implications for the assessment and treatment process. Other commonly discussed worldviews are collectivism and individualism (Triandis, 1995). Clients from individualistic cultures tend to place the individual's rights and needs before that of the group, whereas clients from collectivistic backgrounds emphasize the rights and gains of the group over the individual. Another concept of worldview, the minority worldview, highlights one's status as a minority (e.g., ethnic, gender, sexual orientation) and related experiences of oppression as influencing worldview (Mio et al., 2006). These aspects of worldview may be extremely important in guiding the behaviors of the client and the clinician.

Explanatory Models

Explanatory models are beliefs about an episode of illness that give meaning to the illness and convey how the illness is understood or conceptualized by the child and the child's family. These beliefs play a key role in understanding the illness experience of the child and family, their treatment choices, patterns of mental health service utilization, and treatment outcomes (Kleinman, 1980; Pelto & Pelto, 1996; Weiss & Kleinman, 1988). Explanatory models contain information about the cause of the illness, its onset, the probable course of the illness, its severity, and level of impairment. Eliciting an explanatory model will provide useful information about fears associated with the illness, the main problems it has caused, and, importantly, what kind of treatment the family is expecting and what they hope the outcomes will be (Chrisman & Johnson, 1996; Kleinman, 1980).

CULTURE AND ASSESSMENT

The influential nature of assessments and their results can dramatically affect the lives of children and their families. All clinical interactions are to some extent intercultural. Therefore, it is safe to assume that culture is always operating in *some* capacity in *every* case, and the need to address culture when conducting assessments is always applicable. When interpreting test results, making diagnoses, and conceptualizing cases clinicians should undertake a cultural formulation of the client. The *DSM-IV* (Appendix I) offers clinicians an outline for things to keep in mind while composing the cultural formulation. This formulation, in addition to other assessment procedures, will assist the examiner in understanding the relative importance of culture and the extent to which procedures, instruments, and interpretations need to be adjusted. In this section, we discuss the cultural formulation along with other cultural aspects of assessment, such as assessing the context, identifying biases of the examiner and the tests, interviewing the family, selecting and administering tests, adapting tests, assessing the child and family's culture, and interpreting the results.

Identify Cultural Bias in Assessment and Testing

The *DSM-IV* Outline for Cultural Formulation stresses that clinicians must identify cultural elements of the relationship between the client and the clinician (APA, 1994). Bias in testing and assessment may be related to the examiner, the approach to assessment, and the testing tools and procedures. Determining which questions to ask and to explore, deciding which instruments to administer, and interpreting the results are all influenced by unique, culture-bound beliefs. Clinician self-awareness is key to accurately observing and interpreting the verbal and nonverbal behaviors of clients. For instance, the Western idea that a high level of eye contact is acceptable and preferred may be contrary to that in other cultures, and a lack of eye contact may be misinterpreted as a negative characteristic. Sullivan (1954, as cited in Takushi & Uomoto, 2001) states that clinicians should continually ask themselves about alternative explanations for behaviors (e.g., "Could this mean something different from what readily comes to mind?" or "What might the client mean by that?").

The cultural formulation also states that clinicians must identify cultural differences between themselves and their clients that may influence the therapeutic relationship. This suggests that clinicians need to identify their personal biases, stereotypes, and assumptions about diverse children and families. Clinicians should be aware of within-group variability. Each family has its own cultural traditions and belief systems and each child should be thought of as coming from his or her own unique amalgam of cultural influences. Failure to account for within-group variability and a reliance on stereotypes or group norms can lead to misdiagnosis or inappropriate treatment (Canino & Spurlock, 2000). It is impossible to learn all of the details of every cultural system, but it is the clinician's responsibility to understand the limits of his or her own cultural knowledge and seek out additional knowledge as needed.

In addition to clinician biases, many standardized tests are biased. In general, assessments favor European American and middle-class children, while disadvantaging ethnic minority children (Kaufman & Lichtenberger, 2002). If interpreted incorrectly, results can be particularly harmful to children in school settings. To provide a culturally responsive assessment to diverse children, the following steps are suggested:

- Research the cultural context of the referral question.
- Conduct a culturally informative and sensitive clinical interview.
- Select and administer tests in a culturally responsive way.
- Use the cultural formulation to guide assessment and interpretation.

Research the Cultural Context of the Referral Question

It is essential that the child's behaviors be considered in multiple contexts (i.e., within the microsystems of home, peer network, school). This gives the examiner information regarding the development of the child's behaviors as well as the responses of parents, siblings, teachers, and extended family. Working with the culturally diverse child, clinicians should try to understand any discrepancies regarding how the behavior is viewed across contexts. It is possible for a behavior to be interpreted as extremely disruptive, offensive, and maladaptive within one context; however, the same behavior may be considered adaptive and culturally appropriate within the child's family system or community. During this process, it is important to consider not only maladaptive behaviors, but also the child's strengths.

The *DSM-IV* Outline for Cultural Formulation states that the clinician must gather information regarding the cultural factors related to the psychosocial environment and levels of functioning for each client. Important cultural and historical information may be gathered from records prior to testing. For instance, knowing the regions a child has lived in prior to the current context, the origin of the child's family and ancestors, and the child's first language could be helpful for better understanding the nature of the question, the family's cultural perspective, and whether an interpreter will be needed. If possible, the examiner should obtain information regarding the client's socioeconomic status, family history, language, and acculturation experiences before the interview. While cultural information gathered prior to the assessment can be helpful as a starting point, a culturally sensitive and informative interview is essential for understanding the unique perspective of each child and family.

Conduct a Culturally Informative Clinical Interview

This portion of the assessment is critical for obtaining culturally valid information. During a parental interview there are a number of goals to be considered. The examiner should first offer information regarding the process of the assessment and answer any questions to help ease possible anxiety or cultural discrepancies. The examiner should be aware that it is considered offensive in some cultures to ask

personal questions and should therefore address this possibility in the beginning. Perhaps a cultural broker could be used to facilitate this process in a more culturally responsive manner. The examiner should explain not only confidentiality and its limitations but also the purpose of the questioning. It should be clear to the parents that the intent of the examiner is not to offend but to gather information to gain a full picture of the child's environment. However, the examiner should also explain that if anything is asked that is deemed too personal by the parents, they may choose not to disclose, and they should understand that the examiner means no disrespect.

It is essential to gain the parents' trust to obtain all the information needed regarding the child's history and current functioning. The clinician may also want to ask about any previous experiences with mental health and let the client's parents explain their frustrations or satisfactory experiences. This gives the examiner a better picture of the level of trust the parents have in mental health providers in general and any cultural misunderstandings that may have occurred in the past. In the case of school or court referrals, parents may feel that an evaluation insinuates that there is a deficit or something abnormal with their child; if the "maladaptive" behaviors being assessed are viewed as adaptive within their own culture, the parents may believe an assessment is unnecessary. This may result in defensiveness, and parents may withhold information they consider evidence of such a deficit. It is the examiner's responsibility to inform the parents of the objective and neutral nature of the assessment.

Next, the examiner should obtain a familial and developmental history of the child. It should also be noted what languages are spoken within the home as this may be a factor in language acquisition and development. The examiner should inquire about any incidences reported in the records, such as previous illnesses or developmental milestones, and what the parents' explanatory models are regarding these occurrences. Information to be considered by the clinician includes the following:

- The child's cultural context within all settings.
- Year or time period each developmental stage occurred within the child's life and the correlating events or cultural beliefs.
- Client's language usage and comfort speaking the clinician's language (may require an interpreter or referral).
- Cultural strengths the client may gain from membership in a particular cultural group.
- Cultural information apparent in how the parents and family interact with the child.

Assess Child's Enculturation, Acculturation, and Ethnic Identity

In addition to the issues presented by the clients, psychologists must also keep in mind generational history (how many family members in the country, circumstances of the move, etc.); citizenship or residency status (e.g., number of years in the country); fluency in standard English or other language; extent of family

support or disintegration of family; availability of community resources; level of education; change in social status; and work history and level of stress related to acculturation (APA, 2003). According to the *DSM-IV* Outline for Cultural Formation, psychologists must explore these issues and their influence in the progression of treatment. Hays (2001) suggests the following questions for inquiring about the client's enculturation and acculturation:

- On what terms did you or your family arrive here (refugees, students, etc.)?
- How long have you lived here?
- What language did you first learn to speak? What is spoken in your home? What language do you prefer?
- What is your cultural (or ethnic) heritage or background?
- What is your involvement with _____ (Americans and American culture, own ethnic culture)?
- Would you mind telling me about your religious upbringing? Do you have a religious (or spiritual) practice now?
- What is it like to grow up as a boy/girl in your culture and family?
- Would it be okay if I ask you about your family?
- How is your family's economic situation now? Are there ways it impacts your family life or child's problem?
- Are there ways in which discrimination (prejudice, racism) impacts your family or your child's problem?
- Are there ways in which your (or your child's) disability contributes to the problem?

Assess the Family's Explanatory Model

Assessing the explanatory model of the child and family will help the clinician understand the child's and the family's perspectives on the problem. The model will reveal important cultural and/or personal beliefs about the child's problem that have implications for assessing and treating the child. Research suggests that the family's explanatory model may be just as important as, if not more important than the child's in determining help seeking. Kleinman (1980, p. 106) suggests several questions for exploring explanatory models:

- What do you call your child's problem? What name does it have?
- What do you think caused your child's problem?
- Why do you think it started when it did?
- What does your child's illness do to him or her? How does it work?
- How severe is it? Will it have a short or long course?
- What do you fear most about your child's problem?

- What are the problems your child's illness has caused? What is the impact on the family?
- What kind of help (assessment/treatment) do you think your child needs? What are the most important results you hope to get from treatment?

These questions may be asked during the clinical interview, but keep in mind communication norms and sensitivities. Rather than barrage the family with direct questions, questions can be incorporated more naturally into a conversation about the family's concerns. Young children in particular may not have a complete explanatory model; some parts may be missing, vague, or fused together. The beliefs will change over time, and the child's and parents' views may be different from each other. Depending on the age, developmental stage, and cultural background of the child, consideration of the child's perspective may be more or less important in the family's eyes. The clinician needs to consider the family structure when making efforts to elicit or communicate the child's perspective. For example, having the child vocalize a difference or disagreement with a parent, either alone or in front of the parent, could cause a loss of face, shame, or embarrassment for the parent. Moreover, giving equal weight to a child's or adolescent's explanatory model may threaten the parent's authority and jeopardize the examiner's credibility.

Select Appropriate Test Instruments

Clinicians must understand and be knowledgeable about the reliability, validity, standardization, and appropriate use of any procedures or instruments used in an assessment (APA, 2002). The APA Ethics Code urges psychologists to "use assessment instruments whose validity and reliability have been established for use with members of the population tested" (p. 13). The Ethics Code states that "when such validity or reliability has not been established, psychologists must describe the strengths and limitations of test results and interpretation" (p. 13). It is the responsibility of the examiner to understand the instrument's cultural limitations and adapt procedures and interpretations of results as needed.

The number of school-age children coming from minority backgrounds is increasing rapidly; however, assessment instruments normed on nonmajority populations are sparse. Padilla (2001) states that using a test with individuals of cultural or linguistic groups who were not included in the standardization group calls into question the reliability and validity of the results. Instruments appropriate for use with children from diverse cultural backgrounds and varying linguistic abilities are not commonly available in educational and psychological settings. According to Torres (1991, as cited in Padilla, 2001), studies have shown that approximately 5 million students are inappropriately tested each year by standardized assessment instruments. This is of great concern, particularly in special education, where standardized testing is mandatory to give a child a diagnosis and secure appropriate support. Students' results may be deflated, resulting in inappropriate academic labeling and placement, such as being retained in a grade or placed on a low-expectation education track (Padilla, 2001).

Suggestions for Making Standardized Tests More Culturally Responsive

Although no test can be considered culture-free, there are some tests considered to be culturally responsive in testing with minorities. Some of the instruments were developed with a particular ethnicity in mind; there are also many available for use with larger diverse populations. For instance, the Wechsler Intelligence Scale for Children III, Woodcock-Johnson Psychoeducational Battery—Revised, Child Behavior Checklist, and the Kaufman Assessment Battery for Children are all considered appropriate for use with children from diverse populations (Canino & Spurlock, 2000).

In an attempt to lessen the cultural bias toward diverse populations, many clinicians choose to alter the administration of the instrument. For instance, when giving timed tests, it is important to understand that the emphasis in Western culture on quick responding as an indicator of intelligence is culture-based. Some cultures (e.g., Asian) may emphasize careful consideration of a question before responding and may further consider a careful, unrushed, and thoughtful response to be a sign of intelligence (Gopaul-McNicol & Thomas-Presswood, 1998). As a result, some clinicians eliminate (or extend) the time limits for some subtests or items (Canino & Spurlock, 2000).

Gopaul-McNicol and Thomas-Presswood (1998) advocate altering vocabulary portions of instruments to allow the child to use the vocabulary item within a sentence (Canino & Spurlock, 2000). This allows the child to contextualize the word and may give the examiner a better picture of the true knowledge the child has regarding the vocabulary word. Another method used to alter testing administration and minimize cultural bias is to advocate the use of paper and pencil to work out math problems. This not only allows the examiner to see the process the examinee uses but may also help the examinee in the task. It may be beneficial for the examiner to model the required behavior prior to giving actual test items. Doing so may clarify directions that were misunderstood due to a language deficiency or cultural misunderstanding. It also gives the child the opportunity to ask questions regarding the task and minimizes the possibility that the child's behaviors during testing are misinterpreted as low levels of ability rather than a lack of clarity (Paniagua, 1994). Adaptations should be made with awareness and knowledge of the child's acculturation level and language usage and fluency. Some strategies for culturally responsive testing include:

- When using an interpreter or cultural broker, set up a meeting prior to the assessment to discuss the process and possible cultural concerns.
- Explore all possible venues to explain the client's responses, scores, and behaviors during testing.
- If possible, current scores should be compared relative to the client's prior performance instead of against others' scores or the norm.
- Avoid the use of standardized personality tests for diagnostic purposes with clients of minority identities.

Interpretation and Case Conceptualization

The clinician's overall interpretation of the client's situation stems from his or her own way of perceiving the client's presentation of problems. This evaluation is guided by the clinician's professional orientation, sociocultural background, cultural awareness, and experiences. It is the clinician who tailors the interviewing process, test selection and administration, interpretation and perception, and ultimately the understanding and reporting of the problem. Therefore, it is essential that clinicians compose a thorough case conceptualization that takes culture into account and is alert to possible biases. It is through this culturally relativistic prism that test scores and behaviors should be interpreted.

CULTURE AND TREATMENT

Culturally responsive treatment can be developed through collaborative negotiation between the clinician and the family (Hays, 2001). How and what cultural material should be integrated vary with each child and family. According to the *DSM-IV* Outline for Cultural Formation, information about the client and the client's cultural background should be organized into an overall approach for treatment that links the conceptualization to the treatment. Culture frames beliefs of the clinician and of the family about the best route to recovery and the desired result. Therapeutic theories, orientations, and approaches have their own cultural roots and reflect a particular worldview and philosophy (Pedersen, 2004). Whether the goal of therapy is to work for a reduction of psychological symptoms or for an acceptance of them is a value-laden question (Tseng, 2003).

Ideally the treatment plan will flow from two sources: the clinician's conceptualization (based on theoretical orientation and comprehensive assessment) and the perspective and goals of the child and family. Cultural factors should be considered at every phase in the treatment process.

Next we consider culture across the phases of treatment, from before treatment begins to building rapport, establishing roles and boundaries, and planning and adapting treatments.

Establish a Relationship

First, clinicians should take the necessary time to establish rapport, respect, and credibility with the family (Hays, 2001; Hong et al., 2000; Okazaki, 2000). This asks the clinician to develop an appropriate level of intimacy with the child and other family members. What level is appropriate and the expected degree of formality will vary based on cultural, gender, and age-related factors, as well as the nature of the problem. Clinicians should consider the use of titles, differences in communication (e.g., language usage and fluency, paralanguage), and social interaction (formal versus informal, gender roles). With clients from patriarchal and authoritarian cultures, it will be important to pay special attention to establishing respect and credibility with the father; in other cultures it could be a maternal grandmother who deserves special consideration. Necessary family support for treatment may hinge, to some extent, on the support, approval, and cooperation of these individuals. With

other groups, such as African Americans, efforts to connect equally with all family members, including the child and other siblings, will be valued.

Therapists should also appreciate that some families will have a need to contextualize the clinician, or otherwise understand something about the clinician or the clinician's background (Hays, 2001). To build trust and engender respect and rapport, a clinician may engage in small talk, use appropriate self-disclosure, and adapt language style by avoiding jargon and using the language of the family members (e.g., idioms, ways of describing) when possible. Providing concrete help to the family or otherwise quickly addressing a family concern can also build trust. Clinicians should be mindful of personal reactions or biases related to race or culture that may result in mistrust, suspicion, hostility, ambivalence, overreliance or overfriendliness, excessive guilt, anger, or denial of ethnicity on the part of the family or the clinician. All of these interfere with the therapeutic relationship (Vaughn, 2004; Wohl, 2000). In all cases, a clinician's skills in conveying warmth, openness, and respect can go a long way in building a working alliance with the child and family members. A certain degree of humility and appropriate use of humor are also considered to be essential to building intercultural relationships (Hays, 2001).

Understand and Address Treatment Barriers

Understanding and addressing obstacles to treatment barriers is essential to equitable service provision. Ethnic and cultural minorities and low-income and other disenfranchised groups face a number of barriers in accessing and benefiting from professional psychological treatment. These barriers include a lack of affordability, accessibility, acceptability, and appropriateness in current services. Efforts to address the psychological needs of children and adolescents should be particularly attuned to culture-based beliefs about psychological problems, such as stigmas and beliefs regarding from whom and what type of help is needed. A lack of familiarity with mental health services or counseling, cultural mistrust due to racism, or previous negative experiences with mental health services are additional factors that prevent psychological help seeking. Instead, families may seek informal help from family and friends; from religious, spiritual, or other community supports; or from medical providers. Psychological services may be sought only as a last resort, when problems have escalated or the family is in a state of crisis. Moreover, service may be sought due to outside pressures (e.g., legal or school referral) rather than family choice. Once help is sought, clients may experience communication barriers or institutional or individual racism and treatment that may be culturally incongruent with their values and beliefs.

The nature and extent of these treatment barriers will need to be assessed and addressed for each child and family. In general, clinicians should be aware of common barriers within their treatment context (location, service population, type of service). Culturally competent responses to such barriers might include sliding fee scales; transportation vouchers; flexible operating hours; community-based or school-based service provision; flexible treatment modalities, such as group, family, and culturally adapted approaches; and culturally sensitive and knowledgeable clinicians.

Primary and secondary prevention approaches to child and adolescent problems should be considered for their nonstigmatizing nature. Likewise, strengths-based approaches that view culture as a resource and aim to build protective factors and resilience in children and adolescents are encouraged (Hays, 2001; Lerner, Brentano, Dowling, & Anderson, 2002). Coordination of services and collaboration among stakeholders, such as community participation in planning and evaluation, can also aid in successful programming.

Determine the Family's Familiarity with the Treatment Setting

It is important to determine, early on, the child's and family's familiarity with psychotherapy and expectations for treatment. This may include discussing their previous experiences with mental health services or other interactions with the health care system. Personal and social meanings and beliefs about seeking psychological help should also be explored. There may be a lack of familiarity with the concept and practice of Western counseling and psychotherapy. Even when psychological treatment is sought, families may continue to have a certain level of cultural mistrust and apprehension entering the therapy setting. Certain routine intake procedures, such as signing consent and confidentiality forms, may increase distrust, especially if the need for such forms is not well explained (Hays, 2001). Additionally, the typical therapy format relies on a direct style of communication to identify and address personal problems in the presence of a professional (the psychologist or clinician), who is usually a stranger. This format may be very foreign to families coming from cultures that rely on familiar social and religious supports, such as community leaders, village elders, family members, priests, religious leaders, or shamans for addressing such problems. Counselors and therapists who lack cultural knowledge can exacerbate the problems and feelings of unease.

Conduct Pretherapy

Families that are unfamiliar with the therapy setting (e.g., ethnic minorities, immigrants, rural populations, and lower income populations) may benefit from pretherapy, which refers to time spent orienting the child and family to the therapist and the treatment setting. The family should be invited into the therapist's culture (Hays, 2001). Pretherapy may include added time for rapport building, education about what to expect in therapy, knowledge about the therapist's credentials and approach, and demystification of the therapy process. At this time, the role of the therapist will be clarified and so will the general expectations of the child and family members. Some cultures do not have a history of keeping regular, scheduled appointments that begin and end at a certain time. Additionally, children and families with no transportation or a lack of child care for siblings may have difficulty keeping appointments.

In many traditional approaches to helping, payment or reciprocity for service is not required on a regular basis, but is offered as a donation after treatment. Expectations regarding payment should be handled in a delicate and sensitive manner. Procedures related to confidentiality should be given careful attention, as parents with strong authoritarian family roles may expect that the clinician will automatically

share all information with them (Tseng, 2003). The child's parents will need to be carefully informed of limits to confidentiality, such as child abuse reporting laws. The time spent in pretherapy can also be used to further establish credibility and rapport and to gather cultural information.

Gather Cultural Information

Next, attempts are made to learn about and understand the cultural perspectives of the child and family by crossing into the client's culture (Hays, 2001). That is, the clinician should learn about the child's cultural context and background. The client and the client's family are good sources for relevant cultural information. However, it is not appropriate to rely solely on clients, and the clinician should seek information about a child's cultural background and group-related norms outside of the therapy session. There are a number of excellent books and book chapters, scientific and practice-related journals, and Web-based resources that provide information specific to different cultural and minority groups. Important background information includes sociopolitical and historical events, history of oppression, worldview, communication and language issues, cultural variation in symptom expression, stigmas and related help-seeking patterns, cultural strengths, values, and supports.

In learning about cultural norms and values, clinicians should attempt to suspend judgment about a client's cultural beliefs and remain mindful of their own, culture-based reactions to areas of difference (Martin & Nakayama, 2005). In addition to learning about cultural norms and values, identifying the strengths of children and families and mobilizing them as a resource in treatment is one of the most important skills for effective practice with diverse clients (Hays, 2001).

Clarify Clinician Roles and Boundaries

Working in a multicultural context, careful attention should be given to clarifying the function and roles of the therapist, determining the nature of the therapeutic relationship, and defining the boundaries of the relationship (Constantine & Sue, 2005; Tseng, 2003). The role of clinicians and the nature of the therapist-client relationship should be viewed from a cultural perspective. Practitioners from multicultural, feminist, and humanistic approaches, for example, may encourage enhanced authenticity and genuineness in the client-therapist relationship. Boundaries are not as rigidly defined, but continuous in nature, with shifts readily occurring in areas such as self-disclosure, giving and receiving gifts, multiple roles, bartering, and nonsexual touch and other expressions of care. Indeed, when grounded in one's theoretical approach and well thought out in each case, relaxed boundaries in some of these areas are considered culturally competent service provision. Although research findings are inconclusive, practitioners suggest that appropriately timed self-disclosure by the clinician can engender feelings of trust and respect and facilitate the development of rapport with youth and with some cultural groups. Acceptable practices vary greatly across different treatment settings and populations.

Proponents of multicultural approaches also call on psychologists to take on multiple helping roles beyond traditional individual therapy (Atkinson et al., 1998).

With a host of concerns related to diverse clients' environment or external context, a case management approach is often called for to assess and treat children with awareness of their familial, cultural, and ecological contexts. A task for the clinician may be building parents' skills for successfully advocating for their children and accessing needed resources and services. Other clinician roles may include conducting groups, intervening in systems, serving as a consultant or cultural broker, and implementing community-based preventive interventions (Atkinson et al., 1998).

Involve the Family

Working with the family members of children and adolescents is a common practice generally, and in the context of cultural difference can be a great asset (Wilson, Kohn, & Lee, 2000). Family members, including parents and siblings, grandparents, aunts and uncles, and close others in the child's family network can be valuable resources of collateral and/or corroborating information about the occurrence of various behaviors and how they are interpreted. Family members can provide a base for assessing cultural perspectives and for determining the extent to which various behaviors or other psychological manifestations are culturally acceptable. Family members will also be knowledgeable about culturally appropriate coping choices and ways of responding. It is important that clinicians be aware of culturally diverse family hierarchies, rules for relating among family members, and taboos. For example, it may be inappropriate for a family member to mention the cause of a parent's death (Tseng, 2003).

Address Communication Differences

Even when clinicians and their clients are able to communicate in the same language, cultures vary greatly in their communication styles (whether they are direct or indirect, high or low context), communication patterns, and norms and rules for verbal and nonverbal expressions (Martin & Nakayama, 2005). Clinicians should pay attention to the language used by themselves and the children and families with whom they work. Efforts should be made to use simple and clear language and to minimize the use of psychological jargon, idioms, and colloquialisms. The process of communication and rules for formality and nonverbal communication, such as eye contact, gestures, and body movements, vary widely. Clinicians should clarify the intended meaning of any expression with which they are unfamiliar. An incorrect interpretation can result in minor misunderstandings or serious mistakes, such as misdiagnosis. It is important for clinicians to know that a client who says "Yes, yes" does not necessarily mean "I agree with you." How symptoms are expressed is also a matter of culture, and expressions may be culturally stylized (Tseng, 2003).

Direct communication about emotions or discussions of certain other matters may be taboo, or considered very private. If an indirect style is used, the clinician may need to adjust his or her communication style and seek consultation on specific indirect methods, such as making use of proverbs or stories that reflect cultural wisdom. Clinicians should proceed with caution and determine cultural

norms for discussing family matters, sexual or religious practices, money, suicide, and ancestors (alive or dead). Individuals with a collectivist background may consider it improper to discuss family members or family matters, because to do so would bring great shame to the family. Clinicians should explain their need for the information and ask permission first (e.g., "May I ask you about your family?"). Invite clients to share and then pay attention for signs (nonverbal and verbal) that you might have overstepped cultural boundaries (Pedersen, 1997).

Work with Interpreters

Children and their families should have access to treatment in a language that they completely understand and in which they can comfortably express themselves (APA, 2003; Constantine & Sue, 2005). With 19% of families in the United States speaking a language other than English in the home, mental health practitioners will increasingly be providing services to individuals with limited English proficiency (LEP). In all cases in which children, youth, or their families are considered LEP or prefer speaking in their native language, clinicians should arrange for the services of a professional interpreter. Under Title VI of the 1964 Civil Rights Act, institutions and service providers that receive federal funds are required to provide such interpretation services. Professional interpreters are trained to provide direct, simultaneous, or sequential interpretation in a professional and ethical manner. They often have had training in confidentiality, dual roles, and medical or psychological terminology.

Interpretation is provided verbatim and in the first person. Typically, interpreters will sit beside the client and family. During the session, an interpreter often takes a rather technical role, serving as the conduit for communication between the clinician and the patient. Eye contact, to the extent that it occurs, will largely remain between the clinician and the child and family. If the interpreter needs to interject or further explain something, he or she may enter the conversation by saying "The interpreter would like to say. . . . " This may be done to clarify a communication, explain a concept, or provide important cultural information. If trained professionals are not available, for example in rural locations, telephone interpreters can be arranged. Some things to consider when selecting an interpreter are sex and ethnic background, as these may influence client disclosure and trust. Despite their level of training, brief pre- and posttherapy sessions with interpreters can provide a chance to discuss therapeutic concepts that may be difficult to translate, such as relaxation training. Presessions may help prepare the interpreter for a potentially difficult (e.g., crisis) situation.

Use a Cultural Consultant

Cultural brokers may be useful in accessing cultural knowledge or advice. Who is a cultural broker? This role is rather loosely defined, but may include a professional (e.g., anthropologist, sociologist, psychologist, or other mental health professional) with experience or training in a cultural population. A cultural broker could be a person with knowledge and experience in the culture or community in question, such as a community leader, spiritual person, pastor, or other traditional helper.

Cultural brokers can provide background, group-level cultural information, or case-specific consultation. Interpreters can also serve as cultural brokers and have the benefit of providing this service as treatment sessions unfold and directly afterward. Interpreters serving as cultural brokers may be able to further elaborate or explain culturally embedded concepts and recognize and head off cultural clashes. For example, the interpreter may inform the clinician that, if a concept is literally translated, it will offend the client. Cultural brokers may even assist the therapist in coming up with a culturally acceptable way to address a sensitive issue, such as suicidality.

Negotiate a Common Explanatory Model

In selecting and developing a treatment plan, clinicians should begin with what the family presents (their explanatory model) and make appropriate adjustments.

After assessing the family's explanatory model, clinicians can compare it to their own (Chrisman & Johnson, 1996; Lewis-Fernandez & Kleinman, 1995). Studies suggest that, more often than not, the models of clinicians differ from their patients' (Cohen, Tripp-Reimer, Smith, Sorofman, & Lively, 1994; Kleinman, 1980). Disparate beliefs, left unexplored, may result in poor patient-practitioner communication and lead to misdiagnosis, inappropriate treatment, treatment nonadherence, early attrition, and reduced treatment outcomes (Chrisman & Johnson, 1996; Kleinman, 1980; Wohl, 2000). The following steps are suggested to negotiate a common model with children and their families.

- Start with what the family presents.
- Share your own model, using client language when possible.
- Bridge the models by emphasizing similarities and discussing differences.
- Make modifications that are culturally salient for the family.
- Agree on a common model and a treatment plan that emphasizes strengths.
- Continue to monitor and modify as needed.

First, the clinician should summarize the family's view of the problem and expectations and goals for treatment. This will convey an accurate understanding and highlight misunderstandings that the clinician has. Next, the clinician can share his or her conceptualization of the problem and suggested treatment. Next, clinicians should make efforts to engage the family in a collaborative approach to goal setting and treatment planning. Even when there are drastic differences in practitioner and family explanatory models, clinicians can emphasize similarities and work toward shared goals. To the extent that the clinician has built credibility and trust, families may be more receptive to new ideas or methods proposed by the therapist. Reframing or psychoeducational approaches can be used to present treatments in culturally palatable ways. Strengths-based approaches are also suggested, and because they are congruent with cultural and family values may be more likely to engender family support (Hays, 2001).

Adjust the Therapeutic Frame

Efforts to make treatments culturally relevant and meaningful often involve making adjustments to the therapeutic frame. Although some families may not be familiar with formal, Western-based psychological treatments, families all over the world are familiar with the idea of talking to or otherwise seeking help from a caring and concerned person. Keeping this basic similarity in mind, clinicians can make a variety of technical and theoretical adjustments, ranging from minor adjustments to mainstream approaches to drastic changes in the foci, goals, and process of treatment so that treatments are developed largely from cultural (emic) approaches (Koss-Chiono, 2000).

Adaptations should be made with consideration of the family's cultural perspective and available resources and services. Simple adjustments in mainstream approaches can lay the groundwork for effective services. For example, although many therapists are taught not to give advice, families from diverse backgrounds may seek out the therapist's guidance in various dilemmas (Tseng, 2003). Clinicians who can provide concrete help to families in solving problems are adjusting the relational frame to meet the family's expectations and needs.

Other therapeutic adjustments may come into play as treatment foci, goals, and processes are selected. Goals should be culturally congruent and contextually relevant. Which behaviors are targeted for change should be considered in light of the child's multiple contexts. For example, working to increase assertiveness with a Chinese adolescent may be helpful in the school environment, but problematic at home. These issues should be explored thoroughly before treatments are pursued.

Adapt Treatments

Clinicians consider a number of factors when selecting the treatment mode and format, and culture should be among these. Culturally sensitive and adapted treatment programs may garner increased patient support and satisfaction.

Level and Mode of Intervention

Family approaches to treatment are often suggested for ethnic minority and culturally diverse youth, as are small groups and peer-based and systems-level interventions (Aponte & Bracco, 2000; Wilson et al., 2000). Community approaches are considered preferable to individual approaches in meeting the needs of ethnic minorities. Typically, they emphasize ecological validity, a multiproblem focus, commitment to interagency collaboration, employment of bicultural and bilingual staff, development of cultural competency, strengthening social support networks, and use of prevention and consultation strategies (Aponte & Bracco, 2000; Cuellar, 2000).

Culturally Related and Culture-Specific Approaches

Treatments range widely in terms of intensity, focus, modality, and the degree to which cultural material is incorporated. Many clinicians choose to adapt standard European-based strategies such as behavioral, cognitive, interpersonal, or family

systems, by incorporating cultural values, strengths, and resources into therapeutic goals and techniques (Hays, 2001). Examples of adjustments in a cognitive-behavioral treatment might include using proverbs, religious verses, or other cultural metaphors to help counter negative thoughts, and using family to help support home-based exposures. Culture-based coping, such as meditation or prayer, could be integrated into a stress reduction program, and other cultural rituals related to healing may be included in an overall treatment plan. For example "having a sweat" (i.e., in a sweat lodge) may be recommended alongside other treatment recommendations for an American Indian adolescent living on an Indian reservation. Alternative approaches to treatment include narrative, testimonial, and bibliotherapy, as well as the use of creative, expressive, and somatically based therapies (Hays, 2001).

Culture-specific approaches are based on specific cultural frameworks. Most of these interventions provide cultural education based on traditional values, and then use these values as the underpinnings of the treatment. For example, clinicians may use concepts of *familismo, simpatia,* and *respeto* as themes when working with Latino/a children and adolescents, and themes related to collective responsibility, self-determination, or spirituality when working with African American youth. Culture-adapted and culture-specific treatments may make use of ethnically similar role models, discussion of cultural attitudes regarding risk behaviors, ethnic identity development, and issues related to racism and discrimination (Belgrave et al., 1994). Traditional cultural customs, stories, songs, and other cultural materials may be used, and treatment should be in accessible locations where families can draw support from community leaders or organizations. Some culture-specific approaches make use of a "rites of passage" paradigm to facilitate, formally recognize, and celebrate the physical and emotional development of youth. Although there has been little controlled research on the therapeutic efficacy of these interventions, cross-sectional research has indicated that a grounding in traditional cultural values is negatively associated with substance abuse, anxiety, hostility, juvenile crime, and depression (Belgrave et al., 1994).

Two culturally related treatments are cuento therapy for Latinos/as (Constantino, Malgady, & Rogler, 1986) and Afrocentric "rights of passage" programs for African American youth (Brookins, 1996; P. Hill, 1992). Cuento therapy draws on the themes in Hispanic cultural stories of heroes, heroines, and moral messages. Afrocentric programs have youth undertake a series of behavioral tasks and related rituals that signify growth, maturity, and a commitment to self and community. These programs aim to educate and empower adolescents and enhance their life skills, cultural identity, self-esteem, and academic performance. One obvious concern is the practicality and costs of designing culture-specific treatments. Moreover, to make treatment decisions based on the ethnic background of the child and family may be considered an ethnic slur and a violation of ethical and clinical rules of practice (Wohl, 2000). Instead, multiple cultural factors will overlap and manifest idiographically, accounting for vast diversity within ethnic groups.

Work with Traditional Healers

Children and families may benefit from traditional approaches, such as in the case of culture-bound syndromes or when families are highly *separated* from mainstream

society. Clinicians should develop awareness and understanding of indigenous healing and helping methods so that they can better understand their client's cultural perspective and provide referrals (Koss-Chioino, 2000). Clinicians should recognize that there are several parallels between the functions of traditional healers and those of psychologists. Traditional healers vary greatly in type, level of training, and approaches to treatment, and include shamans, priests or pastors, imams, herbalists, curanderos, and spiritualists. Traditional and indigenous healing approaches are gaining increased attention, and it is suggested that clinicians learn about practicing local healers and develop respectful relationships with them (Koss-Chioino, 2000).

Select a Focus for Treatment

The focus of treatment may or may not be culturally related. Clinicians will assess and treat children referred for common child and adolescent problems involving conduct, social relations, mood, mental retardation or other developmental delays, learning problems, and adjustment difficulties. Although there may be phenomenological variation in the presentation of these disorders, if the presenting problem is directly related to or influenced by cultural factors, then a more direct focus on cultural issues is needed. Some common treatment foci that are culturally related are school adjustment, acculturation-related problems, and identity concerns.

School Adjustment

Children from low socioeconomic backgrounds, ethnic minorities, immigrants, and refugees all face formidable challenges in the school system. The school system in the United Sates, for example, has a pedagogical philosophy valuing written expression, direct verbal communication, student-centered approaches, competitiveness, and individual achievement. To the extent that children live in households or communities with different cultural values, or have a history of schooling in another system, adjusting to school can be quite a challenge. Language problems, cultural differences in learning style or behavioral norms, discrimination, and the inability of parents to help with schoolwork are challenges faced by immigrant and ethnic minority youth.

Among refugees and immigrants, language issues, difficulty making American friends, and differences in educational systems are some of the acculturative problems reported (Vinokurov, Trickett, & Birman, 2002). Other problems are discrimination by peers, such as name-calling and teasing (Bemak & Greenberg, 1994). A lack of previous access to formal education can make the adjustment even more difficult for some immigrant and refugee youth. Self-efficacy will be important for adjustment in the school setting, with some studies suggesting an impact on academic skills and performance, computer skills, and interpersonal skills. A thorough assessment will be needed to determine the appropriate type, target, and level of intervention (e.g., a system- or organizational-level change or child behavior change).

Acculturation Problems

The seriousness of symptoms and the degree of psychological difficulty experienced by children and families in cultural conflict will vary considerably. Berry (1997) breaks symptom severity into three levels of distress: (1) mild to moderate cultural conflict, (2) acculturative stress, and (3) psychopathology. Mild to moderate cultural conflict is explained by the need to learn new ways of behaving (Ward, 1997). These conflicts arise as certain behaviors are unlearned or shed, while being replaced with a new behavioral repertoire. If behavioral shifts are not easily made, the second level of difficulty, acculturative stress, will occur (Berry, 1997). When major conflicts are experienced, a psychopathology perspective is helpful. Children and their family members may be in crisis or otherwise suffering from incapacitating levels of depression, anxiety, and/or other mental disorders needing serious attention (Berry, 1997).

Parent-Child Acculturation Gap

Children of immigrants and refugees often acculturate more rapidly than their parents and older family members. They may adopt new behaviors and values that are incongruent with their parents' expectations. Refugee and immigrant youth may become ashamed of their parents, reject their culture of origin, and rapidly assimilate to America's youth-oriented society. Parents are often fearful that their children will forget their cultural values, and they may become upset as their adolescents adopt new styles of dress, wear makeup, or engage in unacceptable social behaviors, such as dating and attending parties (Sourander, 2003).

Children may learn English before their parents do, and power shifts may occur as children take on new roles interpreting for the family. At the same time, parents may lose their status and effectiveness in the new culture (Tousignant et al., 1999). These shifts in power dynamics can be quite upsetting for children and adults, especially those coming from more traditional, patriarchal families or collectivist cultures. Family conflict may result, which can easily escalate to child or domestic abuse situations that did not exist prior to migration (Bemak & Chung, 2000; Colic-Peisker & Walker, 2003).

Bicultural Effectiveness

In working with mild to moderate acculturation concerns, clinicians may employ strategies for culture learning and social skills acquisition. A bicultural (or multicultural) approach will reinforce the value and importance of maintaining effectiveness in the culture of origin and the new culture. Some suggestions for addressing the acculturation gap between parents and children include the following:

- Reframe family conflict as acculturation differences that normally occur.
- Identify cultural similarities and differences.
- Teach families about the process of cultural adjustment and how it can play out in families.

- Examine the cultural values and demands faced by the child outside the home (e.g., in school, with peers).
- Help youth understand the value in their heritage culture.
- Learn skills needed to be effective in different cultural environments (i.e., bicultural competence).

Bicultural effectiveness programs have been developed to reduce intercultural conflict among Cuban American youth and their parents (Szapocznik et al., 1984) and to reduce substance abuse among American Indian adolescents (Schinke et al., 1988).

Identity Concerns

Questions related to one's personal and social identity, such as "What kind of person am I? Do others like me? Do I belong to a group? What is my ethnic/racial or other cultural group? How do others see me as a part of this group?" and "How do I see myself as a part of this group?" are common beginning with middle-age children and may be especially so for ethnic and cultural minority youth (Schmidt, 2006). Efforts to nurture a positive social and cultural identity can be integrated into standard treatment approaches. Effort to enhance a child's awareness of and sense of belonging to his or her cultural group may include bibliotherapy (including cultural stories, myths, proverbs), attendance at cultural events, cultural and personal values exploration, identification of cultural strengths and resources, learning about the cultural or sociopolitical history of one's cultural group, and appreciation of or participation in creative or expressive arts and artistic traditions of one's group. Approaches that work toward an integrated (e.g., bicultural or multicultural) identity offer maximal benefits across settings.

Mistakes and Modifications

Once the treatment focus and techniques are agreed upon, clinicians need to consider cultural factors and influences throughout treatment. During the entire process, clinicians should make efforts to maintain therapeutic rapport, monitor the child's and family's progress and commitment, be mindful of mistakes that are made, and make modifications as necessary. Maintaining a good working alliance with children and credibility with families requires ongoing attention and effort. Clinician roles, boundaries, and style of interaction may need adjusting as treatment and the level of familiarity increases. Latino families, for example, may expect the clinician to shift from an interactional style favoring *formaliso* to one of *personalismo* (Hays, 2001). Cultural impasses may occur as new material comes up and/or new techniques or interventions are discussed or implemented.

Multicultural psychologists point out that mistakes will be made in treating the culturally diverse. Successfully recognizing and responding to cultural missteps is an important clinical skill (Pedersen, 1997, 2000). Awareness of changes in relating or treatment nonadherence may indicate a potential miscommunication. Mistakes may be addressed directly or indirectly (as culturally appropriate) and

modifications made in the technique or approach. Respect, humility, and humor are considered among the most important cross-cultural skills and can be quite effective in maintaining relationships in spite of mistakes (Hays, 2001; Pederson 1997, 2000).

Termination

Cultural expectations of the relationship may impact termination. Families may not understand such an artificial termination and may feel a need to maintain the relationship. Clinicians may help terminating children and families by linking them with community supports and also by scheduling spaced-out sessions and booster sessions or otherwise inviting families to return for additional treatment (Tseng, 2003).

SPECIAL POPULATIONS AND EMERGING TOPICS

Children Living in Poverty

Children and adolescents growing up in poverty are affected on multiple levels by compounding factors. High poverty often goes hand in hand with lack of education or poor education; lack of good nutrition and health care; unsafe, high-crime neighborhoods; lack of supervision; few social supports; poor environmental conditions; and uprooted cultural values (Vargas & Koss-Chiono, 1992). The background of children with low socioeconomic status emphasizes survival. These children's behaviors could be the direct result of living without proper care or sustenance, lack of formal education, and an insecure environment. Clinicians need to keep in mind the influence their own worldview may have when interpreting the behaviors of children from such groups. Flexible clinician roles, case management approaches to treatment, and community-based approaches may be helpful.

Immigrant and Refugee Children and Youth

Challenges faced by immigrant and refugee children and youth include changes in daily living, language, behavior, education system, and socialization practices. Clinicians should facilitate new culture acquisition and also encourage retaining cultural ties (see skills under "Bicultural Effectiveness"). Immigration status, legal or illegal, will have implications for the degree of stress experienced and service available. Immigrant youth who do not have legal status or who have undocumented family members face challenges accessing health and educational resources and are vulnerable to legal retribution, unfair and unsafe labor practices, family separation, and added stress related to their uncertain status and need for secrecy.

Refugee children's adjustment problems are underscored by multiple losses, family separation, refugee camp experiences, and trauma (Bemak & Greenberg, 1994). Resettling refugee children and adolescents report depression, irritability, withdrawal, low self-image, isolation, estrangement, loneliness, a loss of power, feeling out of place, sleep problems, feeling that their old values are useless, and a sense of being devalued (Sourander, 2003). Refugee and immigrant services

organizations provide some services for refugees and immigrants, such as language classes, assistance in finding housing and enrolling children in school, accessing health care services, and limited economic support. Clinicians should consider a case management approach to service provision and bicultural effectiveness.

Ethnic Minority Children and Adolescents

Ethnic minority children and adolescents face discrimination, racism, and prejudice related to their minority status (Porter, 2000). Stress associated with racism is one factor accounting for higher rates of stress-related health problems in ethnic minority groups. Overall, ethnic minority youth are at increased risk for developing emotional and behavioral problems. Issues facing ethnic minority youth include physically threatening environments, conflicting messages about appropriate identity development, economic hardship, and limited access to supportive services that may place many ethnic youth at a significant disadvantage when compared to their White counterparts (Hoberman, 1992). Experiences of oppression and racism further impede their healthy psychosocial adjustment and development. Signs of maladjustment associated with these many contextual factors include substance abuse, mood disturbances, academic underachievement, premature sexuality and childbearing, delinquency, underemployment, and instability of family life (Barbarin, 1993).

Biracial Children

Children of biracial or multiethnic backgrounds may have unique concerns related to their personal and social identity. A child's physical appearance may impact his or her self-identity and also influence parental identification. Models of biracial or multiracial identity development can help clinicians get a sense of where biracial children are in their identity development (Root, 1990). For biracial children, cultural conflicts regarding values, family roles, and parenting may arise in multiple contexts. If parents and families successfully negotiate these cultural differences, problems may not arise, but conflict in these areas could result in inconsistency, confusion, and family stress. Depending on the parents' origins, the child could be attempting to understand and negotiate multiple cultural influences. The clinician must assess the child's identification with each culture and the impact this may have across contexts.

Internationally Adopted Children

Rates of international adoption in the United States have increased dramatically in the past decade (Friedlander et al., 2000). More and more children are growing up with parents and siblings from a different language and cultural background from themselves and, importantly, who look different from themselves. These children may face teasing, name-calling, and discrimination, just as ethnic minority children do. White parents may be ill equipped to understand and/or address their adopted child's experiences with racism or struggles with racial or ethnic identity.

Families may also experience conflict with other family members or the community regarding their decision to adopt an ethnically different child.

Children with a history of institutionalization (e.g., living in an orphanage) face health problems, delays in cognitive or socioemotional development, sensorimotor integration difficulties, and problems with attachment and bonding. Additionally, adoptive parents of these children may be dealing with unresolved fertility issues, disappointments related to the child, or problems related to family integration. As young children mature, issues related to ethnic and cultural identity may emerge to an even greater extent. Family activities and routines that promote bonding and relationships among all family members can be encouraged. Families may need assistance resolving cultural conflicts and skills training on how to become a bicultural (or multicultural) family (Friedlander et al., 2000).

Gay, Lesbian, Bisexual, and Transgender Youth

When dealing with children and adolescents who identify themselves as gay, lesbian, bisexual, or transgender (GLBT), the clinician must be aware of not only the client's current functioning within that identity but also the past experiences associated with it. Worldwide, GLBT youth face discrimination and social marginalization related to their sexual or gender orientation. Models of identity development indicate that GLBT individuals may undergo a series of stages of confusion, identification, and acceptance of their sexual orientation (Cass, 1979). When this occurs, how it occurs, and the outcomes will vary from person to person. Individual and contextual factors certainly come into play when assessing each client's experience of this process. In an atmosphere of family support and openness, the client is psychologically equipped to handle outside pressures or discrimination. However, in an environment with strong family, religious, or cultural sanctions against homosexuals, children and adolescents may have to face these societal pressures alone.

Given these factors, it is important to give GLBT youth a safe and open environment to express their experiences, frustrations, and ideas. The worldview of GLBT individuals is one that has developed alongside experiences of discrimination, fear, and possibly danger (Herek, 1995, 2000). It is important for the clinician to recognize the multidimensional nature of the client and address GLBT issues as they relate to the presenting problem.

NEW DIRECTIONS IN CULTURALLY RESPONSIVE TREATMENT

Consistent with an increased global interest in positive psychology and cultural competence, new directions in child and adolescent approaches to treatment view children and culture as resources (Lerner et al., 2002). Positive youth development (PYD) supports the notion that cultivating social, emotional, and cognitive competencies in children and adolescents can head off, or at least ameliorate some emotional and behavioral problems (Larson, 2000). Programs employing PYD promote

resilience by helping children and youth to better understand their environment and act in ways that promote healthy development (Dworkin, Larson, & Hansen, 2003; Lerner et al., 2002).

These programs may be particularly appropriate for addressing the mental health needs of culturally diverse youth (Johnson & Johnson-Pynn, 2007). They offer a broad-based approach that is widely culturally acceptable and accessible (i.e., often school- or community-based). Nurturing cultural strengths and community connections is integral to most PYD programs. Program outcomes, such as social competencies, positive bonds with people and institutions, positive self-regard, self-efficacy, positive ethnic identity, and social justice values, have obvious benefits for culturally diverse youth (Catalano, Berglund, Ryan, Lonczak, & Hawkins, 2002; Lerner, 2002).

SUMMARY

In this chapter, we have described the nature and functioning of culture as it impacts children and families and the process of assessment and treatment. First we discussed important contextual factors, such as the changing demographic landscape in the United States and the increasing multicultural nature of society, that highlight the need for cultural competence in child assessment and treatment. We discussed the complex, multifaceted, subjective, and dynamic aspects of culture that must be understood in order to fully appreciate culture's influence and intricacies. Next we outlined some of the ways that culture interacts with other factors to affect child development, socialization, education, values, and norms. We also described how culture results in variations in mental heath functioning and psychopathology in children, shaping symptom expression and communication, help seeking, illness beliefs and expectations, and desired outcomes for assessment and treatment.

Next, we provided an overview of key cultural constructs to consider in assessment and treatment. Acculturation, ethnic identity, worldview, and explanatory models were defined, as was their potential to influence the assessment and treatment process. We then discussed how culture may come into play at each phase in the assessment and treatment process. We integrated suggestions for attending to culture into each section and provided practical ideas, such as how to build a satisfactory relationship with clients, how to ask clients about their culture, how to adapt one's communication style, how to incorporate cultural strengths into treatment, and how to work with interpreters and cultural brokers. Culture-related problems, such as acculturative stress and identity concerns, were discussed, as were culturally responsive interventions.

Finally, we called attention to the unique concerns of certain groups of children, including refugee and immigrant children, gay and lesbian youth, bicultural children, ethnic minorities, children living in poverty, and internationally adopted children. In closing, we point to the promise of positive youth development programs as culturally responsive approaches to meeting the mental health needs of significant numbers of culturally diverse children and adolescents. Although this chapter covered many aspects important for understanding and working with clients from a variety of cultures, it is impossible to sufficiently address cultural issues in

assessment and treatment in a single chapter. Clinicians working in the context of cultural differences are encouraged to make ongoing efforts to enhance their cultural competence in assessing and treating children and adolescents.

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CHAPTER 27

Divorce and Custody

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Divorce is a common transition for growing numbers of parents and children in the United States. Each year over a million American children experience their parents' divorce (U.S. Census Bureau, 2000). As part of the divorce, families encounter the legal system as they attempt to make plans for how children will be raised in the new family structure. Clinical psychologists may become involved in several ways, such as by providing mediation and custody evaluations that influence the way family life will proceed after divorce. Thus, clinical psychologists may benefit from information on divorce and custody issues for these litigating families.

This chapter provides an overview of divorce and custody issues relevant for clinical psychologists. It introduces a general model of divorce adjustment, followed by a review of critical factors affecting child outcome for divorcing families. Then it presents three areas that affect postdivorce legal decisions: guiding legal principles, custody and visitation arrangements, and child support. Finally, it focuses on mediation and custody evaluations as mechanisms for assisting families through the divorce transition.

A GENERAL MODEL OF DIVORCE ADJUSTMENT

Perhaps the most widely accepted theoretical model of divorce involves a process perspective. Specifically, divorce is viewed not as a single isolated event, but as a complex chain of marital transitions and family reorganizations that alter roles and relationships and affect individual and family adjustment (Hetherington, Bridges, & Insabella, 1998). Prior to divorce, conflict within the family may increase, accompanied by periods of physical separation between spouses and attempts at reconciliation. Divorce itself may lead to additional disruptions in daily routines, roles, activities, and social relationships, including continued shifts in living arrangements and visitation, such as physical separation, accompanied by changes in contact between the child and the nonresidential parent. Even after the divorce occurs, changes may continue, brought about by parental efforts to establish new partnerships, including dating and nonmarital cohabitation. These changes present children and

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families with adaptive challenges. The response to these challenges is influenced by previous functioning and experiences and subsequent stresses and resources (Amato, 2000; Hetherington et al., 1998).

Converging research indicates that parenting practices mediate many of the contextual changes that occur during and after divorce (e.g., Simons & Associates, 1996). Thus, divorce precipitates changes in family life that have the potential to disrupt effective parenting practices. These practices include monitoring children's whereabouts, providing consistent discipline, using constructive problem-solving strategies, and engaging in positive involvement. Disruption in parenting subsequently increases the likelihood of children's exhibiting behavioral and emotional difficulties.

Empirical sources suggest there are four key factors that have implications for how children and families adapt: economic hardship, interparental conflict, parental distress, and cumulative transitions and related life stress. These factors are related (e.g., parents experience distress when conflict is high and economic resources are low), but each has potential challenges and implications for child adjustment.

Economic Hardship

With divorce, families often encounter dramatic reductions in income, with per capita declines averaging between 13% and 35% in national samples (R. Peterson, 1996). Economic pressure may mean that families are unable to meet material needs, fall behind on debt payments, and have to cut back on everyday expenses to live within available means (Conger, Ge, Elder, Lorenz, & Simons, 1994). Economic hardship, when combined with high levels of life stress, is associated with increased psychological distress and decreased parenting skills. Not surprisingly, some parents become distracted and emotionally or physically unavailable to their children (Simons et al., 1996). Such disruptions in parenting lead to problematic behavior in children (Forgatch, Patterson, & Ray, 1996). With sufficient time, stress usually abates, although families with lower income may continue to experience disruptive events. The increase in economic hardship following divorce is a potent factor in the emergence of problematic outcomes for children and thus a key reason why courts attempt to determine and enforce child custody payments.

Interparental Conflict

Research indicates that the frequency and intensity of conflict, as well as more overt forms of conflict (e.g., verbal and physical aggression), are related to problematic adjustment in children, regardless of marital status of the parents (Amato, 1999). In fact, interparental conflict has been shown to be a better predictor of child maladjustment than changes in the parents' marital status (Davies, Harold, Goeke-Morey, & Cummings, 2002). Interparental conflict appears to increase children's emotional distress, disrupt peer relations and academic performance, and lead to physical health difficulties. Children who have a family history of exposure to interparental conflict (particularly if conflict is unresolved) show increasing sensitivity or reactivity over time, with reactions ranging from fear and sadness to anger and aggressiveness. Children may be particularly sensitive when conflict concerns

child-related issues such as child rearing and custody arrangements. Conflict over these topics may cause children to feel shameful, to blame themselves for the divorce, or to fear becoming involved in the conflict (Grych, Harold, & Miles, 2003). Some theorize that exposure to highly conflictual situations undermines children's sense of emotional security, as evidenced by emotional and behavioral problems (Davies et al., 2002). Others show that high levels of conflict disrupt parenting practices, so that mothers show more negativity and inconsistent discipline, and fathers become less involved with their children, in contrast to counterparts in low-conflict marriages (Gonzales, Pitts, Hill, & Roosa, 2000).

Parental Distress

Divorce is one of the most stressful experiences that an individual may experience. Many individuals experience a variety of adjustment problems, such as increased risk of psychopathology, higher incidence of motor vehicle accidents, elevated drinking and drug use, alcoholism, suicide, and even death (see review by Amato, 2000). More recently separated and more emotionally attached divorcing individuals have poorer immune functioning, greater levels of depression, and other health problems (Kiecolt-Glaser, McGuire, Robles, & Glaser, 2002). For some, divorce may bring about positive effects, such as increased personal autonomy, overall happiness, social involvement, and career development (Hetherington & Kelly, 2002).

Adults who experience difficulties in adapting to stressful postdivorce changes may have problematic relationships with their children (Tein, Sandler, & Zautra, 2000). These problematic relationships interfere with adjustment in children (e.g., Martinez & Forgatch, 2002). In the context of the divorced family, parental adjustment assumes a critical role, given that child adjustment typically follows from how well adults in the household fare. Children's own adjustment difficulties also contribute to the custodial parent's level of stress and quality of parenting (Forgatch et al., 1996). So processes that operate within the parent-child relationship appear to contribute to the long-term adjustment of each family member.

Cumulative Transitions and Related Life Stress

Adjustment problems also arise from transitions that surround relationship dissolutions and consequent repartnering in both formal and informal unions (e.g., Martinez & Forgatch, 2002). After divorce, most adults enter into new relationships; one study found that at the time of the divorce filing, half of the parents already had begun to date, and over one quarter were in a serious romantic relationship (Anderson et al., 2004). Within 10 years of separation, about two thirds of families with two or more children have experienced a nonmarital cohabitation (Bramlett & Mosher, 2002). Parents who remarried had dated an average of three to five people prior to meeting the person they would eventually remarry (Montgomery, Anderson, Hetherington, & Clingempeel, 1992). These relationships introduce additional transitions and reorganizations within the family (Raley & Wildsmith, 2004). Echoing findings discussed earlier, the effect of repartnering transitions on child adjustment is mediated by parenting practices (DeGarmo & Forgatch, 1997).

Short-Term and Long-Term Adjustment to Divorce

Children's adaptation to divorce depends on the particular combination of stress and resources in their lives (Amato, 1993). As stresses begin to decrease and resources become more plentiful, many children who had exhibited problematic adjustment will improve. However, problems with adjustment may persist when stressors such as financial hardship, interparental conflict, and disrupted parenting continue.

POSTDIVORCE DECISIONS

Divorce calls for a restructuring of parental rights and responsibilities along two broad areas relating to child rearing: custody and visitation, and child support. Before discussing each area, we present information on the overarching or guiding legal principles in decisions related to child custody.

Guiding Legal Principles in Child Custody Decision Making

Divorce law and the determination of child custody are established through state statutes. The most prevalent criterion for determining placement of children after divorce is the Best Interest of the Child standard, which emerged in the 1960s and 1970s as an alternative to the Tender Years Doctrine, an earlier standard favoring placement with mothers. The Tender Years Doctrine itself supplanted a custody preference for fathers, when children were considered legal property. Most jurisdictions now explicitly state a gender-neutral preference for custody decisions (see O'Donohue & Bradley, 1999, for more details).

The Best Interest of the Child standard set forth a list of factors to be used in determining child placement after divorce, although states vary in their relative consideration and weighting of these factors. Judges too have great latitude in applying the standard for making determinations in custody cases. Buehler and Gerard (1995) provide a compilation of all Best Interest factors used throughout the United States (see Table 27.1).

Before considering Table 27.1 in detail, note that a critical distinction should be made with respect to legal custody versus physical custody. Legal custody refers to the relative level of continuing responsibility and involvement each parent will have in decisions about the child, ranging from joint, where custody is shared, to sole, where custody is assigned to one parent. Physical custody refers to the daily residential and supervisory arrangements for the child. In practice, these two decisions are independent; for example, parents may be awarded joint legal custody, but one parent may receive physical custody (i.e., the child resides primarily within that parent's household and sees the other parent at established times). Or there may be some form of joint physical custody awarded. Although less clearly defined than sole physical custody (Buchanan & Williams, 2006), a minimum criterion in use for joint physical custody is that children spend at least 25% of the time with each parent (Bauserman, 2002).

With respect to Table 27.1, factors 1 to 5 correspond to the criteria outlined in the Uniform Marriage and Divorce Act (1970); the remaining factors consist of

Table 27.1 "Best Interest of the Child" Factors

Criteria Described in the Uniform Marriage and Divorce Act (1970)

1. Wishes of the child's parent or parents as to his/her custody
2. Wishes of the child as to his/her custodian
3. Interaction and interrelationship of the child with his/her parents, his/her siblings, and any other persons who may significantly affect the child's best interest
4. Child's adjustment to his/her home, school, community
5. Mental and physical health of all individuals involved

Criteria Related to the Primary Caretaker Standard

6. If one parent, both parents, or neither parent has provided primary care of the child
7. Capacity and interest of each parent to provide for the emotional, social, moral, material, and educational needs of the child
8. Capacity and disposition of each party to provide the child with food, clothing, medical care, and other material needs

Criteria Related to Safety and Well-being of Children

9. Evidence of domestic violence, abuse, or neglect
10. Evidence that substance use affects the child's well-being
11. Moral fitness of the parents

Criteria Related to the Cooperative Parent Factor

12. Which parent is more likely to allow the child frequent and continuing contact with the noncustodial parent
13. Nature and extent of coercion or duress used by a parent in obtaining an agreement regarding custody
14. Ability of both parties to cooperate and make decisions jointly
15. Child support noncompliance

Other Criteria

16. Out-of-state residence
17. Length of time the child has lived in a stable, satisfactory environment and the desirability of maintaining continuity
18. Permanence of custodial home
19. Nature and amount of contact with both parents
20. Love and affection existing between the child and each parent
21. Age and sex of the child
22. Parents' respective home environments
23. Physical, emotional, mental, religious, and social needs of the child
24. Health, safety, and welfare of the child
25. Child's cultural background

Source: "Divorce Law in the United States: A Focus on Child Custody," by C. Buehler and J. M. Gerard, 1995, *Family Relations, 44,* pp. 439–458.

outgrowths or restatements of these basic criteria. It is noteworthy that the first two factors involve the direct wishes of the parties involved, although the preference for the child's wishes typically is reserved for older children. Standards are emerging for representing these and other child interests in custody cases (e.g., Ackerman, 2006; American Bar Association [ABA], 2003; American Law Institute, 2002; American Psychological Association [APA], 2002; Elrod, 2003).

Factors 6 to 8 in Table 27.1 address the preference for placing the child with the primary caretaker, the individual who has provided the daily pattern of care.

In principle, this preference appears determinative and relatively easy to establish (Buehler & Gerard, 1995). In actual practice, however, it may lead to inequitable physical custody arrangements, such as granting sole physical custody to a parent who has performed only marginally more than the other with respect to caretaking. An alternative standard has evolved, called the approximation rule, with custody arrangements attempting to resemble what had been in place prior to the divorce. Emery (1999) has advocated for this preference because it combines the best aspects of the primary caretaker and joint custody standards. The determinate standard of time spent caring for the child is used to establish the custody arrangement, thereby accommodating a wide variety of potential predivorce parenting situations.

Factors 9 to 10 in Table 27.1, domestic violence and substance abuse, arguably may override other factors as they directly impact the safety of the child (Buehler & Gerard, 1995). Most states prohibit custody for a parent with verified domestic violence. Factor 11 mentions "moral fitness," a vague term often equated with sexual promiscuity. Seven states, for example, outlaw cohabitation between unmarried partners, which may indirectly affect awards of child custody.

Factors 12 to 15 in Table 27.1 relate to the "cooperative or friendly parent," such that parents who act to restrict access to the child in the absence of compelling reasons of safety may face greater challenges in obtaining custody and visitation. Relatedly, efforts to restrict access (along with other behaviors such as efforts to "brainwash" the child from wanting to have a relationship with the other parent and unfounded accusations of abuse) have been the focus of a controversial proposed psychiatric disorder for understanding children who become resistant to contact with the noncustodial parent, called Parental Alienation Syndrome (Gardner, 1998). More recent views, however, have recast the situation of the child who is resistant to visitation more broadly, now called the *alienated child;* sources of resistance now include the behaviors of both parents, along with developmental preferences and reactions to the divorce. In support of this reformulation, results of two cross-sectional studies show child resistance to visitation arising from multiple determinants, such as prolonged litigation, parental lack of warmth, and the child's own emotional and social problems. Johnston (2005) has integrated this work into Figure 27.1.

Finally, as listed in Table 27.1, Buehler and Gerard (1995) provide factors relating to contact and continuity. Out-of-state residence, for example, is likely to affect practical access to the child. Clearly, the information in Table 27.1 shows that no single approach will fit all potential situations. In the case of incarcerated fathers, for example, judges must weigh the desire of the parties to maintain contact, along with a consideration of whether the visit itself may be difficult for the child, or whether written or phone contact may be more appropriate (Menno, 2003). Moreover, issues such as the nature of the offense, length of prison stay, and past history may affect the form that custody will take.

Thus, in practice, there is wide latitude given to the courts in determining the nature of custody and visitation arrangements. Buehler and Gerard (1995) propose a presumption of favoring parental agreements, such that parents forward their preference for custody and visitation to the court, which the court will honor. In the absence of an established parental agreement, these authors propose a presumption

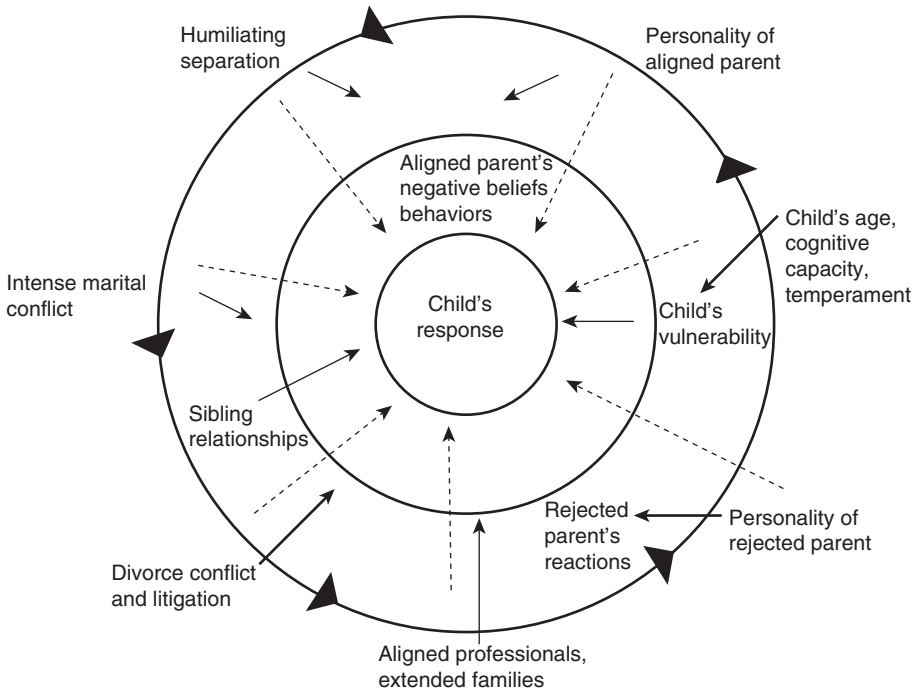

Figure 27.1 A model of factors predicting child alienation. *Source:* "Children of Divorce Who Reject a Parent and Refuse Visitation: Recent Research and Social Policy Implications for the Alienated Child," by J. R. Johnston, 2005, *Family Life Quarterly, 38*(4), pp. 757–775. Reprinted with permission.

of shared legal custody, sole physical custody to the primary caretaker, and substantial access by the nonresidential parent.

Custody and Visitation Arrangements

In this section, we consider differences across custody arrangements identified in previous research, issues related to maintaining contact with the nonresidential parent, and special considerations that impact custody and visitation arrangements.

Common Practices and Comparisons across Custody Arrangements

Following divorce, up to 80% of children are in the sole physical custody of their mother (Emery, 1999). The prevalence of sole father custody, however, is increasing rapidly, with fathers more likely to be awarded custody when children are adolescent (Maccoby & Mnookin, 1992). Buchanan, Maccoby, and Dornbusch (1996) note that mother and father sole custody arrangements were similar on a wide variety of dimensions. Differences were noted in emotional closeness and monitoring, which favored mother custody households; this can be explained in part by the lower rate of behavior problems for children in mother custody households. It also should be noted that fathers essentially may be taking on more challenging situations to begin with, given that they gain custody under more difficult family circumstances than

mothers (Emery, 1999). Sole custodial father families are less likely to be granted child support awards, and nonresidential mothers are less likely to comply with the child support awards given (Meyer & Garasky, 2003).

In response to social and economic changes beginning in the 1970s, there has been an increased preference for parents to share custody, with the hope of maximizing involvement of both parents in their children's lives. There has been a call for current legal practices to recognize the child's need to have both parents actively involved in his or her life (Menno, 2003). The movement toward recognizing the importance of both parents has coincided with the tendency for states to deregulate custody arrangements, allowing parents to establish individualized arrangements (Buehler & Gerard, 1995). Additionally, developmental research has stressed the role that both parents play in children's lives as well as the need for custody arrangements to be flexible (J. B. Kelly & Lamb, 2000). When awarded, joint custody is more common for children in middle childhood and for families with higher income (Cancian & Meyer, 1998; Maccoby & Mnookin, 1992).

As opposed to comparisons across different kinds of sole custody homes, meta-analytic evidence suggests that children in joint custody arrangements fare better across a variety of indexes; in fact, they showed comparable levels of adjustment with children from never-divorced households (Bauserman, 2002). Better adjustment appears to be accounted for by greater closeness between the child and both parents. In addition, interparental conflict is lower in joint custody homes. Finally, children from joint custody families were more likely to see their father weekly, have more overnight stays, and benefit from more child support (Seltzer, 1998). However, the families that enter into joint custody arrangements may themselves be different from those that enter into sole custody arrangements; as described earlier, states likely award joint custody to families that demonstrate that they can get along well enough to manage the close contact and coordination between parents that such arrangements may require. So better outcomes may be due, in part, to selection effects.

What factors are associated with acquiring sole custody? Several studies have shown that parents are more likely to receive custody when they are the plaintiffs and have higher income and education levels compared to their spouse, but not when they have prior marriages, children from prior marriages, or marital infidelity (Cancian & Meyer, 1998; Fox & Kelly, 1995; R. F. Kelly, Redenbach, & Rinaman, 2005; Seltzer, 1998). Fathers are more likely to receive sole custody if ending a long-term marriage; they are less likely to receive custody when they are unemployed or owe child support arrears or when the family is eligible for welfare support (Cancian & Meyer, 1998; Fox & Kelly, 1995). Mothers are more likely to receive custody when they are older, when the oldest child in the family is female, or when there are high levels of conflict (Cancian & Meyers, 1998; Fox & Kelly, 1995; R. F. Kelly, et al., 2005). Joint custody is more likely when all the children are male, both parents work and own homes (Cancian & Meyer, 1998), and when fathers are more educated (R. F. Kelly et al., 2005).

In addition to sole and joint custody, split custody is available to families with more than one child. In this individualized custody arrangement, some of the children may live with the father while other children may live with the mother (Ellis, 2000).

Continuing Contact with the Nonresidential Parent

A critical issue with sole custody arrangements concerns continued contact with the nonresidential parent, typically fathers, given prevailing custody arrangements. Adults who experienced parental divorce in childhood report a sense of loss due to the lack of contact with their father (Laumann-Billings & Emery, 2000) and a wish to have been more involved with him while growing up (Fabricius & Hall, 2000). Some level of contact is necessary for closeness (Fabricius, 2003), especially for younger children (Leventhal, Kelman, Galatzer-Levy, & Hraus, 1999; Pruett & Johnston, 2004), though contact alone is not always a sufficient determinant for relationship quality (Berscheid & Peplau, 1983; Buchanan & Williams, 2006).

When fathers were able to maintain strong emotional ties with their nonresidential children, their children had greater psychological and behavioral adjustment during adolescence (Buchanan et al., 1996; McLanahan & Sandefur, 1994; Seltzer, 1994; Whiteside & Becker, 2000). Sheer contact with the nonresidential father may not be directly tied to children's adjustment (Amato & Gilbreth, 1999; Hetherington & Kelly, 2002; Marsiglio, Amato, Day, & Lamb, 2000; McLanahan & Sandefur, 1994; Seltzer, 1994; Whiteside & Becker, 2000), although its absence is certainly reflected in other ways, such as adult children's sense of loss mentioned earlier. Rather than a direct effect on children's adjustment, the impact of contact between the noncustodial parent and the child on the child's well-being may depend on other factors, most notably the amount of conflict between the divorced parents (e.g., Buchanan & Williams, 2006; Lamb et al., 1997). High interparental conflict may undermine the closeness between the nonresidential father and child, or involve children directly in marital disputes. Relatedly, conflictual parents are less likely to carry out visitation plans, even when court-imposed (Pruett, Williams, Insabella, & Little, 2003; Whiteside & Becker, 2000). Conflict over visitation is one of the primary reasons for the courts to determine that supervised visits should occur. In a multisite study of supervised visitation, over three-quarters of the families reported that anger toward the other parent was the primary reason for the court determination of supervised visitation (Pearson & Thoennes, 2000). Those most satisfied with supervised visits were the families who had reported fighting at drop-off and pickup times.

Special Considerations

There is a considerable controversy over whether to encourage or discourage overnight stays when children are infants and toddlers. Much of this controversy concerns a study by Solomon and George (1999a, 1999b) in which overnight stays were linked to problems in infant-mother attachment. However, infant adjustment to the overnight stays could be promoted with parental cooperation and maternal support. Results highlight the need to consider more than simply the child's age when making decisions about overnight contact.

There also is legal recognition that high-conflict custody cases represent a substantial risk factor for the involved children and families (High-Conflict Custody Cases, 2001). The Family Law section of the ABA, in conjunction with the Johnson

Foundation (2001), developed standards of practice to recognize and address the challenges that these children and families face.

Given high rates of geographic mobility in the United States, parental relocation following separation and divorce is increasingly problematic as joint custody has become the preferred custody arrangement (J. B. Kelly & Lamb, 2003). Joint physical and/or legal custody assumes that both parents maintain geographically close residences to enable contact between children and parents. Relocation by one or both parents makes this difficult to maintain (Austin, 2000). Relocation of one parent also may reduce financial support and lead to a more distant relationship between that parent and the child (Braver, Ellman, & Fabricius, 2003). Relocation may depend on the children's own developmental status (J. B. Kelly & Lamb, 2003) or, as Warshak (2000) suggests, on unique considerations that require a case-by-case analysis.

Child Support

Though the emotional upheaval that typifies divorce is most commonly at the forefront of issues for families experiencing this transition, financial problems also present challenges to child well-being. Single-parent families are more likely to be poor (Bartfeld, 2000). Thus, children have a greater chance of living in poverty after divorce.

Public policy in the United States has increasingly sought to address this issue by establishing the financial responsibility of both the custodial and the noncustodial parent for their children. As early as the nineteenth century, public policy began to enforce parental financial responsibility for dependent children. More sweeping changes to child support policy appeared during the twentieth century, coinciding with the dramatic increase in separation and divorce in American families. Beginning with the Social Security Act of 1935, government support was granted to women and their dependent children based on need, specifically in cases where one parent was not providing any financial support. Additional policies since then have addressed the need for parents to assume greater parental responsibility for their children, such as the Uniform Interstate Family Support Act in 1992 and the Personal Responsibility and Work Opportunity Reconciliation Act in 1996. Currently, child support is regulated by the Child Support Enforcement Program, a partnership of federal, state, and local authorities to collect child support (U.S. Department of Health and Human Services, 2005). Current policy efforts to enforce child support have yielded benefits: Children are less likely to live in poverty (Edin, 1995), their families are less reliant on government support, and their parents take on greater financial responsibility for their children (Altman, 2003).

More generally, child support represents an important resource on which children rely (Amato & Gilbreth, 1999). Increases in their standard of living likely improve children's health and educational attainment (Furstenberg, Morgan, & Allison, 1987; King, 1994; McLanahan, Seltzer, Hanson, & Thompson, 1994). There is evidence, for example, that child support shows positive effects on children's school achievement (Argys, Peters, Brooks-Gunn, & Smith, 1998; Knox, 1996). A meta-analysis of 14 studies shows child support related to greater academic success and fewer externalizing problems in children (Amato & Gilbreth, 1999).

Developmental gains from child support may be more important for younger children (King & Soboleski, 2006) and for involuntary types of payments (Argys et al., 1998; Hernandez, Beller, & Graham, 1995).

Though the general consensus favors child support enforcement, establishing guidelines for child support awards is a more complex issue. Before legislative changes in the 1980s, child support was typically awarded on a case-by-case basis (Rogers, 1999). However, policy changes with the Family Support Act sought to ensure a more equitable system that made use of economic data to estimate the financial costs associated with raising children (Beld & Biernant, 2003). Within these federal guidelines, individual states were free to establish their own formulas; the more common formulas are described next. More detailed reviews of these and other formulas can be found elsewhere (Beld & Biernant, 2003; Blumberg, 1999; Morgan, 2004).

> *Income Shares.* Developed by the U.S. Department of Health and Human Services, the basis for this formula is to provide children with the same proportion of financial resources and support that would have been available to them had their parents remained married.
>
> *Percentage of Obligor Income.* Developed by the state of Wisconsin, this formula takes into account the parent's income, the amount of time the child spends with each parent, and whether the parent is supporting other children. Formulas may differ based on the complexity of the family arrangement, although the basic support guidelines assume that the noncustodial parent is responsible for providing 17% of gross income for 1 child, 25% for 2, 29% for 3, 31% for 4, and 34% for 5 or more children.
>
> *Cost Shares.* The cost shares formula takes into account the costs of raising children, as well as tax benefits associated with dependent children.
>
> *Hybrid Models.* Evolving out of some of the limitations with previous formulas, various hybrid models have emerged, such as one proposed by the ALI [American Law Institute] that allows for greater discretion in deciding costs associated with living arrangements. (Bartlett, 2001; Czapanskiy, 2001; Harris, 2001)

As with any public policy, the various formulas for determining child support have come under fire. Some charge that states do not raise award standards often enough to fairly reflect the actual cost of raising children, given federal legislation that requires updates only every 4 years (Venohr & Griffith, 2005; Venohr & Williams, 1999). Others criticize specifics of the formulas, charging that the income shares and the percentage of obligor income models allow custodial parents to achieve a higher standard of living than nonresidential parents (Rogers, 1999). In fact, formulas such as income shares were designed to protect the primary custodial parent and residential children from a precipitous drop in standard of living after the divorce, although Braver and colleagues (2003) counter that the drop may not be as drastic as once thought. Criticism also is levied because many of the formulas assume that one parent will have sole custody, rather than acknowledging the growing numbers of shared custody arrangements and situations in which visitation and residence vary over time (Melli & Brown, 1994). In fact, child support awards typically do not factor in costs of rearing children for the nonresidential parent until time spent in that household approaches the level of joint custody. Fabricius and Braver (2003) provide evidence for a linear relationship between household

expenses and amount of contact, suggesting that nonresidential expenses are being underestimated. Findings by Garfinkel, McLanahan, and Wallerstein (2004) differ, perhaps because of sample characteristics. The use of a nationally representative, multi-informant sample may be required to address this issue further.

There also is disagreement with whether child support should end when children achieve majority status or after receiving additional post–high school education and training. Current formulas and legislation typically do not consider expenses incurred after children turn 18 years old. New state laws are beginning to address this issue, although there is evidence that some divorced parents are voluntarily providing equal proportions of support to their college-age children (Fabricius, Braver, & Deneau, 2003).

Finally, difficulties surround international child support enforcement. There is currently no global legislation regarding child support, although the United States has reached agreement with several nations on a case-by-case basis (Duncan, 2004). Efforts have been made to develop international policy, but there is no obligation to adhere to the legislation.

A recent report published by the U.S. Census Bureau provides a report card of sorts on child support awards, actual receipts, and enforcement, based on the Current Population Survey reports from 1994 to 2004 (Grall, 2006). It showed that the proportion of children living in poverty after divorce has dropped, from one third in 1993 to less than one fourth in 2003. However, only half of custodial parents had received all their child support award due; the average amount of child support in arrears was $5,100, for a combined total of $37 billion past due. Many custodial parents acknowledge receiving noncash support, such as gifts and coverage of expenses, and over half of child support awards included health insurance provisions from the nonresidential parent.

What would help the parents without primary custody maintain their child support payments? With respect to policy, Freeman and Waldfogel (2001) found that compliance with child support was greatest when paired with stronger enforcement laws. Additionally, greater compliance occurs when child support awards explicitly state the amount to be paid (J. Peterson & Nord, 1990). With respect to individual considerations, greater compliance is linked to access to children, as measured by actual contact (Argys, Peters, & Waldman, 2001; Seltzer, Schaeffer, & Charng, 1989) or geographic proximity (Teachman, 1991). Greater compliance occurs when the award is perceived as fair (Lin, 2000) and when mediation has been used (Emery, Matthews, & Kitzmann, 1994). Greater compliance also occurs when noncustodial parents believe they have some say in the child's life and how the money will be used (Braver et al., 1993; Weiss & Willis, 1985). Perceptions of control influence the likelihood of child support payments indirectly through visitation. Payment of child support is more likely among nonresidential parents who see their child frequently (Seltzer et al., 1989).

Other factors also may decrease the likelihood of child support receipt. Noncompliance is associated with more marginalized groups, such as the less educated and the poor (J. Peterson & Nord, 1990). It also is associated with families with greater interparental conflict (Hetherington & Stanley-Hagen, 1999) and domestic violence, although victims of domestic violence are less likely to seek and receive

child support awards (Menard & Turetsky, 1999). Though recent policy efforts have been made to ensure the safety of those seeking child support, greater advances need to be made in securing the safety of victims of domestic abuse in seeking child support.

Finally, there are few established policies or laws that dictate how child support should be handled among divorces involving stepparents. Few states have laws requiring continuing support from the stepparent following divorce (Hans, 2002; Fine & Fine, 1992).

ASSISTING FAMILIES WITH THE DIVORCE TRANSITION

When assisting parents with the divorce transition, clinical psychologists may make use of various strategies, some of which place them in an advisory role to the family or legal community. This section discusses two such strategies: mediation and custody evaluation.

Mediation

Reflective of dissatisfaction with litigation, mediation has evolved as an alternative. Mediation is essentially cooperative in nature, having originated from the field of negotiation. Thus, mediation is based on the fundamental premise of self-determination: Each parent makes a voluntary effort to reach mutually agreed upon decisions about a parenting plan.

Mediators are typically attorneys, psychologists, or social workers who assist with identifying needs and interests and facilitating an agreement between the parents. The format used is goal-oriented and short term, usually lasting from 1 to 5 sessions. The mediator acts as an advocate for both parents in attempting to create a mutually agreeable solution. Emery and Wyer (1987) outline four aims for mediation, otherwise known as the four "Cs": to reduce *conflict,* to increase *cooperation,* to allow individuals *control* over key decisions affecting their lives, and to reduce *costs* associated with divorce and litigation.

There has been rapid growth in the use of mediation in divorce over the past 2 decades, with some states mandating an attempt at mediation in custody disputes. When mandated, parents must attend one or more meetings with a mediator in an attempt to reach an agreement, but they are not required to reach an agreement. Additionally, some states have implemented programs to educate parents about the impact of divorce and related transitions on children's well-being.

Typically, mediators seek to generate a mutually agreed upon, workable coparenting plan that defines the respective boundaries of each parent and, in doing so, minimizes the need for future contact. The process of mediation helps to model how the coparenting relationship should operate in future interactions, with communication that is less emotional and more businesslike in tone. A distinct advantage with mediation is the evolution of a detailed, uniquely tailored parenting plan. Such plans may generate greater satisfaction, thereby yielding better adherence.

Reviewers of related research have concluded that mediation appears to have moderate success (J. B. Kelly, 1996; Saposnek, 1998). Benefits are cost savings incurred when litigation can be avoided, improved parental satisfaction, and increased compliance for mediated agreements compared to those reached through adjudication. Perhaps because of the increases in parental satisfaction, parents tend to report lower levels of conflict at 1 year after the divorce, although such results are less consistently seen over the long term. Mediation does not appear to garner benefits to parent and child psychological adjustment. Also, court-mandated mediation appears to be less successful at producing an agreed upon parenting plan. Finally, mediation may be less appropriate in situations involving domestic violence and child abuse, where there are substantial discrepancies in bargaining power, or when parents are severely disturbed.

Custody Evaluation

When collaborative efforts such as mediation fail, families may turn to the courts for adjudication. Maccoby and Mnookin (1992) note that about 50% of families settle out of court, with custody uncontested; about 30% contest custody but settle through attorney negotiations out of court; another 11% resolve custody issues with mediation; the remaining are given mandatory custody evaluations consistent with the local courts, with about half of those settling out of court after the evaluation is conducted. Thus, approximately 4% of the sample proceeded to trial, and about half of those cases settled during the course of the trial. Only 1.5% of the sample completed a custody trial and received a final determination by the judge. The authors conclude that about 25% of the sample had entrenched legal battles over their children. These families were not distinguished from the larger sample with respect to race, education, and income; discriminating factors were children younger than 3 and mothers not engaged in paid employment.

Custody evaluations have become a routine part of litigation, either sponsored by the parties themselves or initiated by the courts. The custody evaluation comprises an assessment of the child, the parents, and the family by a psychologist. Because the stakes are high—an evaluation carries weight in determining parental rights relative to the child—custody evaluations are often difficult and demanding work. Adding to the difficulty are evaluations that involve accusations of child abuse, negligence, and family violence. Either parent may retain the psychologist, but generally most psychologists prefer to be retained by the court or jointly by the attorneys of both sides (Ackerman & Ackerman, 1997).

Although state statutes differ, generally the evaluation assesses "parenting capacity" and its fit with the psychological and developmental needs of the child: (a) parental capacity of the prospective custodians: their knowledge, skills, and abilities; (b) psychological functioning and developmental needs of the child (can also include the wishes of the child in some cases); and (c) functional ability of each parent to meet the child's needs. Thus the assessment includes the mental and physical health of all family members involved; the child's adjustment at home, school, and community settings; and the interaction of the child with parents and siblings. The APA (1994) directs evaluations to focus on three issues: (1) the needs of the

child, (2) the abilities and capacities of the parents, and (3) the best fit between the two.

Custody evaluations can be extensive, as the psychologist conducts clinical interviews with each parent and child (and any cohabiting romantic partner), reviews all relevant documents, conducts parent-child observations, conducts psychological testing on parents and children, and talks with teachers, attorneys, and therapists. The total time to conduct a custody evaluation (excluding court testimony time) ranges from 18 to 28 hours, and fees average $3,000 (Bow & Quinnell, 2001). As such, custody evaluations are often a critical part of the divorce proceedings, and may have a pivotal effect on the determination of the so-called ultimate issues of a case: custody and visitation rights.

In this legal setting, the custody evaluator is a professional expert who is called on to inform and advise the court in a balanced and impartial way, in contrast to a judge, who renders a final decision, or an advocating attorney, who represents the interest of one party. More recently, this role of imputed impartiality has come under increased scrutiny. Tippins and Wittmann (2005) state that custody evaluations are most valuable (and potentially least biased) when focused on facts and observations rather than clinical inference or judgments. The authors propose a four-level model for directing clinical involvement in custody cases, with the levels reflecting the degree of clinical inference involved (see Figure 27.2). At Level I, the lowest level of clinical inference, custody evaluators make observations about actual behaviors and test outcomes and processes occurring as part of the evaluation. Examples the authors provide include "She hung her head low and was often tearful" and "There is an elevation on scale two of the MMPI-2 [Minnesota Multiphasic Personality Inventory-2]."

At Level II, the custody evaluator begins to make simple inferences about Level I data. Examples the authors provide include "The father's style of parenting is very authoritarian" and "The mother's blue mood, the child's report of her chronic sadness, and her MMPI-2 elevations suggest that she is moderately depressed." In contrast, at Level III, the evaluator makes reference to custody-specific constructs, such as "parenting capacity" and "potential psychological risks of primary custody with this parent." The authors provide this example: "Multiple days away from the mother, this child's primary psychological parent, will be emotionally stressful." Finally, Level IV comprises the evaluator's custody-related recommendations about the ultimate issues, such as a proposed access schedule and physical custody. The authors direct evaluators to be circumspect in their evaluation, charging that specific empirical support may be lacking in making recommendations at Level IV. They and others (Karras & Berry, 1985; Melton, Petriala, Poythress, & Slobogin, 1997) recommend that custody evaluators avoid making specific recommendations about what a court "should" do. Ironically, specific recommendations are often viewed as what is most valuable about custody evaluations (Bow & Quinnell, 2004). Some argue instead that custody evaluators should provide recommendations, but that judges should view them as advisory in nature (Stahl, 2005).

If custody evaluators cannot be impartial, they should attempt to withdraw from a case, or at a minimum provide information to the court about factors that affect their impartiality. Custody evaluators also should avoid multiple relationships. In

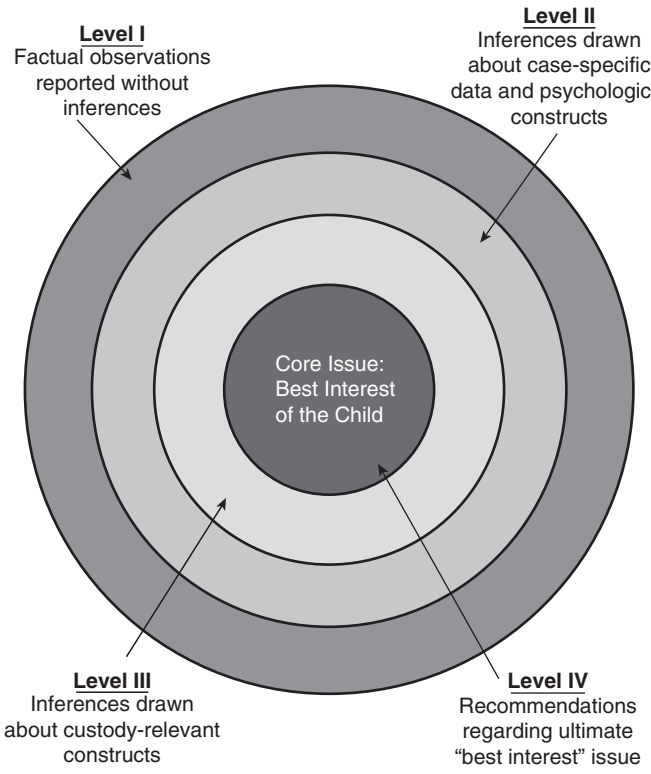

Figure 27.2 A model for directing clinical involvement in custody cases. *Source:* "Empirical and Ethical Problems with Custody Recommendations: A Call for Clinical Humility and Judicial Vigilance," by T. M. Tippins and J. P. Wittmann, 2005, *Family Court Review, 43*(2), pp. 193–222. Reprinted with permission.

cases where they have served in a therapeutic role, they should avoid providing custody evaluations. They may testify as a witness, although recommendations are generally to decline such a role unless ordered by the court to do so. During the custody evaluation, they should refrain from accepting any of the involved parties as therapy clients, and even after the evaluation is complete, they are urged to be cautious in agreeing to treat former participants of the custody evaluation.

Custody evaluators should have a thorough understanding of ethical guidelines for conducting custody evaluations. For example, APA guidelines specify that the custody evaluator obtain informed consent from each adult participant, including identifying the assessment tools and techniques that will be used. Informed consent also specifies fees, who will pay, and the limits of confidentiality, including what information will be disclosed. In conducting the evaluation, the psychologist should refrain from inappropriately interpreting the data and should use multiple methods of data collection.

The APA standards do not address many of the issues involved in child custody evaluations, and thus are considered aspirational in nature. Additionally, custody evaluations have a high risk for malpractice suits and ethical complaints (Kirkland &

Kirkland, 2001). Thus, the psychologist should follow closely the ethical guidelines and other risk management procedures. See Craig (2005) for other guidelines that may apply for custody evaluations.

Because of the nature of the work, custody evaluators should have competency in psychological assessment of adults, children, and families. They should be knowledgeable in child development, family development, adult and child psychopathology, and research on children and divorce. Further, they should be familiar with applicable legal standards, especially for their state or jurisdiction. Their work should reflect acceptable clinical and scientific methods. Their evaluation should be based on multiple methods, with important facts or opinions based on at least two sources of information.

Additionally, the psychologist should use standard and commonly used assessment techniques in custody evaluations. Measures that have been frequently used can be found in Bow and Quinnell (2001) and Hagen and Castagna (2001). The measure most frequently used by psychologists in evaluating child custody issues is the MMPI. This test is often used to assess parenting capacity and emotional adjustment of each party. It also allows for detecting impression management, or the degree to which individuals are consciously inflating or concealing various aspects of themselves. This test has been normed on child custody litigants (e.g., Strong, Greene, Hoppe, Johnston, & Olesin, 1999) and has been used to identify parents who attempt to alienate the child from the other parent (Siegel & Langford, 1998). Craig (2005) recommends that evidence from the MMPI should be explored further as part of the workup. However, Bradley (2004) warns of potential cultural biases in this instrument and others. The Millon Clinical Multiaxial Inventory (MCMI) is being used increasingly in child custody evaluations. There is some controversy over the use of the MCMI because it lacks established norms for differentiating clinical from nonclinical respondents (Butcher & Miller, 1999) and there are issues with scale interpretation. Craig recommends caution when using the MCMI for custody evaluations. Hysjulien, Wood, and Benjamin (1994) report the use of various instruments, including the Achenbach Child Behavior Checklist (McConaughy & Achenbach, 1996), FACES III (Edmon, Cole, & Howard, 1990), and the Parent Attachment Structured Interview (Roll, Lockwood, & Roll, 1981). Bradley recommends the Child Perception of Parent and Parent Awareness Skills Survey (Bricklin, 1995) and the Parental Discipline Techniques Self Report Instrument (Gardner, 1997).

SUMMARY

Parental divorce has the potential to create considerable risk for children, but there is large variation in how children respond to the experience. Key risk factors for maladjustment after divorce are economic hardship, interparental conflict, parental distress, and cumulative transitions and related life stress. Clinical psychologists may provide mediation or custody evaluation, both of which require a sophisticated knowledge of legal, ethical, and clinical issues. The work is painstaking and difficult, but offers help to families attempting to make plans for how children will be raised in the new family structure.

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CHAPTER 28

Child Maltreatment

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> It was a sunny day in a popular vacation spot, crowded with children and adults in the pool area, some swimming, some lying about on chairs, and some in conversation. A mom, lying face down, was oblivious to the world. A dad, sitting in a nearby chair, seemed quite uncomfortable. Perhaps he did not favor the heat. His daughter, about 12 years old, was restlessly standing in front of him. She approached and sat on his knee, moving from side to side. Finally, she got up, turned around and straddled him. The frankness of that straddle, his delay in responding to this movement, and my audible response of shock put this man in motion and he roughly pushed her off him. The child's face registered no emotional reaction, just a watchful blankness. The child left her dad and came in the pool beside me. I tried to make friendly conversation—"Hi." "The pool water is really warm!" "Do you like to swim?" The total lack of response from the child in any dimension indicating registering the prior event or registering my communication efforts led me to entertain several hypotheses: non-English speaker, hearing-impaired, cognitively delayed, or fearful of connecting to another (the first three could later be ruled out). After a little while, the father said words to the mother. He called his daughter. Child and father walked away together, leaving the mother behind who had remained frozen in her ideal tanning position.

Child maltreatment disturbs your peace. If it is clear, it sets you in motion. If it is unclear, it may nag, percolate, and unsettle you in the background workings of your mind. You struggle because the pieces of information do not align well and, likely, the child is not talking. For the safety of the child, you move carefully through ruling out more hypotheses than is typical. In child clinics, the assessment question may come as broadly as "What is going on?" and in person, your colleague seems at a loss. The clinician may observe or have reported such behaviors as public self-stimulation, "daydreaming" during a conversation, back arching from a light touch, aggression or emotional overreactivity at the slightest provocation, or selective mutism. As hyperactivity is a nonspecific symptom, relevant to Attention-Deficit/Hyperactivity Disorder, genetic syndrome disorders, children of newly clinically depressed mothers, and so on, maltreatment leaves scant symptom

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profiles. This is due to the possibility of maltreatment affecting every developmental domain—difficulties sustaining attention and learning, behavioral hyperactivity and aggression, depression, anxious clinging to or overreaching with friendly adults, nonnormative fears, and bizarre or nonnormative behaviors—potentially layered on top of preexisting problems.

Childhood maltreatment is no disorder. It is the result of adult action serving adult purposes or stemming from adult deficits. People who interact regularly with maltreated children may note that things seem "off." Clinical examples include a kindergarten child repeatedly referring to himself as "having the devil inside him," and teachers describing a child sibling pair as "weird" given their unusual behaviors (e.g., licking a classmate's hand) and another child as "Dr. Jekyll/Mr. Hyde" given his extreme behavioral and affective changes. For other children, their adaptation may not include individual signs, but may take on compulsive caregiving of younger siblings, sublimating their own distress. Other children may find avenues for more positive development by becoming essentially members of another family, be it via other family members, school and work relations, or in their neighborhood. Finally, for some children, symptomatology is not evident or may be delayed or very restricted in its expression, even in the context of high-impact trauma (e.g., witnessing one parent's murder by the other parent) or chronic maltreatment. Regardless of symptom expression, we have child protection laws acknowledging that it is not the child's responsibility to fend off violence from adults.

Professionals wrestle with the legal requirement to report suspected maltreatment and the ethical requirement to intervene and document concerns. Documentation is one way to establish a pattern of concerns, even if the initial report is not investigated or substantiated by child protective services (CPS). With clear injury patterns, indications of patterned behavior, and verbal disclosures, the clarity of reasonable *suspicion* of abuse, neglect, or risk increases. Professional hesitation regarding reporting and quantifying confidently "reasonable suspicion" in less than clear situations has reasonable grounds. These include concern about civil litigation challenging "good faith" reporting; the fact that the short-term and long-term child effects of CPS involvement have received little research in its 30 years of existence; the concern that the only CPS service offered will be investigation; and the puzzlement, given current clinic wait-lists, that most children are returned to the home within 6 months. Further, there is a community threshold for reportable abuse or neglect within the practical reality of limited resources.

Erring on the side of caution is a prudent approach when considering that U.S. figures of mortality from child abuse are likely underestimates (Herman-Giddens et al., 1999). Using North Carolina medical examiner data over a 10-year period, nearly 60% of fatal cases were recorded as due to battery, not child abuse specifically. The probable assailant was male (66%) and a biological parent (63%). It is generally accepted that much serious abuse and neglect go undetected by formal systems. The World Health Organization (WHO) estimates that 57,000 children die yearly from maltreatment fatalities (Krug, Mercy, Dahlberg, & Zwi, 2002). In the United States, falling is the most common reason given initially, with the child being found unresponsive as the next most prevalent (Collins & Nichols, 1999).

Most deaths occur when the mother and her unrelated boyfriend live in the home (Stiffman et al., 2002). Another large category is neglect deaths from starvation, exposure to heat or cold, and failure to safeguard. As a risk factor for childhood death, any protection system involvement for neglect or physical and/or sexual abuse dramatically increases mortality, especially for children under 1 year of age (Sabotta & Davis, 1992).

When maltreatment is repeated, we can only acknowledge all of our system failures, from public monitoring to front-line professionals (community recreation, faith-based institutions, education, health, substance abuse, mental health, hospital emergency, police, domestic violence shelters, etc.) to child protective services. Part of our citizenry social contract is to provide a safety net for expectant and at-risk parents, and government is the ultimate partner in providing leadership in this.

Most United Nations member states have committed to an action plan to uphold the child's human rights to be protected from abuse and to have the basic necessities of life provided. The U.S. surgeon general's "Healthy People 2010" targets specific reductions in rates of child abuse and related fatalities and research goals of understanding the magnitude of new cases and evaluating the impact of existing interventions. Child maltreatment is properly regarded as an issue of human rights, public health, mental and physical illness prevention, and economics. The remainder of this chapter addresses the definition and scope of the problem of maltreatment, using brief case descriptions to illustrate these. An overview of theoretical models and current knowledge on outcomes is presented. Finally, current gold standards in intervention are highlighted. We end with thoughts on future directions for research and practice and provide a summary for this chapter.

DEFINITION AND SCOPE OF THE PROBLEM OF CHILD MALTREATMENT

Barber and Sibert (2000, p. 743) note, "The initial recognition of the possibility of abuse is in some ways the most important step . . . and one should never walk alone" in the child protection process. Thus, the first step is to have maltreatment in mind as an active possible hypothesis to rule out, as well as to know local applicable laws and key consultation experts. U.S. figures (Federal Fiscal Year, October 2004–September 2005) place the annual substantiated rate of abuse or neglect at about 12.10/1,000 children. Most (75%) are first-time victims. The recurrence rate for substantiated abuse over 6 months is about 6% (U.S. Department of Health and Human Services [USDHHS], 2007). Canadian rates of entries into child protection, which may reflect substantiated high risk as well as substantiated discrete events, are about 22/1,000 children (Canadian Incidence Study; Trocmé et al., 2005). Consistent over recent years, neglect is the most common form presenting to CPS (62.8%), with psychological abuse the least frequent in the United States (7.1%) and sexual abuse the least frequent in Canada (3%). With the exception of sexual abuse, parents are the vast majority of perpetrators, with mothers somewhat higher than fathers (83.4% abuse by a parent acting alone or with another person; USDHHS, 2007). Dramatic gender differences exist only with sexual abuse, the majority

being female victims and the majority of perpetrators being male. Younger children show the highest victimization rate (16.1/1,000 children, birth to age 3), with decreasing rates across ages to teens. However, substantial levels of maltreatment do occur in adolescence (12–15 year-olds: rate of 10.2/1,000; 16–17 year-olds: rate of 6.2/1,000; USDHHS, 2007). It should be noted that victims in CPS have an estimated 6-month maltreatment recurrent rate of 94.6%, as determined by the 2006 Child and Family Services Reviews (USDHHS, 2007). Further, community survey data suggest that physical abuse is more than 40 times more prevalent and sexual abuse is more than 15 times more prevalent than official rates (Wekerle, Miller, Wolfe, & Spindel, 2006).

Neglect

Neglect is challenging to define as it covers a wide range of failure-to-care events. Neglect can be willful disregard of child needs or more passive acts of omission. Neglect examples include abandonment or desertion; failure to provide a home or a safe, sanitary shelter; supervisory neglect; neglect of personal hygiene; refusal or delay in providing normative health care (e.g., specific concern visits, such as ear infection, maintaining recommended immunization schedule, routine dental care); nutritional neglect; failure to meet standards for education (e.g., high absenteeism); and failure to meet special needs (e.g., failure to treat asthma from secondary smoke, failure to treat learning disorders with educational support). Exposure to physical hazards (e.g., exposed wiring, broken glass, nonsecured and loaded guns, illegal drugs, dangerous or unhygienic pets, asking child to perform dangerous activities, driving with child while intoxicated) is another area of concern. Neglect is a chronic condition with high recidivism, experiencing the greatest rate of placement in out-of-home care. In the United States, parents are perpetrators in 86.6% of cases. Neglect is more prevalent among younger caregivers, households with a primary male caregiver, and families with low socioeconomic status (Trocmé et al., 2005).

The U.S. case-level statistics indicate that neglect accounts for 60% of children receiving clinical services (USDHHS, 2007). The rate of cases of neglect reported to CPS is also around 60%, or 7.4/1,000 children (USDHHS, 2007). The most common form of investigated neglect is "failure to supervise leading to physical harm," where the child suffered or was at substantial risk of suffering physical harm because of the caregiver's failure to monitor and protect the child adequately. This form of neglect constitutes 37% of the substantiated forms of maltreatment (3.06/1,000 children) among new reports to CPS (Trocmé et al., 2005).

Research indicates that caregiver substance abuse is strongly linked to neglect (Takayam, Wolfe, & Coulter, 1998; Wekerle & Wall, 2002; Yampolskaya & Banks, 2006). Among a range of caregiver vulnerability factors, caregiver substance abuse was the single best predictor of neglect, although the composite of vulnerabilities was the best overall predictor (Wekerle, Wall, Leung, & Trocmé, 2006) after controlling for other child and parent characteristics. Other significant single predictors were socioeconomic disadvantage, caregiver history of childhood maltreatment, and caregiver lack of social support. Recent caregiver violent partnership interacted with total caregiver vulnerability, indicating an exacerbating impact, more

than doubling the likelihood of a substantiated versus nonsubstantiated neglect finding. Addressing caregiver vulnerabilities early on, when the family first comes to the attention of CPS, seems a critical move.

Some definitions of neglect also include emotional neglect (a marked indifference to children's need for affection, attention, and emotional support), as well as exposure to chronic or extreme spousal abuse or intimate partner violence (IPV). These subtypes can be summarized as involving three major forms of neglect: (1) lack of emotional support, affection, or both; (2) lack of protection from family conflict, violence, or both; and (3) lack of protection from community violence (Wekerle et al., 2006). A recent epidemiological survey of randomly selected teens on the active caseload of an urban center found that the teens reported higher prevalence for emotional neglect items than physical neglect items, as measured by the Childhood Trauma Questionnaire (Bernstein & Fink, 1998). Item examples include often or sometimes "did not have enough to eat" (37%) and often or sometimes "parent too drunk/high to care for them" (27%; Wekerle et al., 2006).

Case Illustration

Robert, age 4, was observed playing on the roof of his house by his neighbors, who called the child protective agency. When the social worker investigated the home situation, she found Robert and two other siblings, Nancy, age 5, and Alice, age 2, living in unsafe and filthy conditions in the care of their mother, age 30, and her boyfriend, who was the children's stepfather. The mother, Carol, reported that her partner, Frank, had a cocaine habit and used much of what she received from social assistance to buy drugs. He was physically and verbally abusive to her and threatened to hurt the children if she told anyone about the abuse that she was experiencing. They had been together for 1 year; previously, Carol had been with another man, the father of two of the three children, who had also been abusive to her. Carol had grown up in a home where she had been sexually abused by a maternal uncle; when she disclosed this to her mother, she was not believed, and so left home at the age of 15 years. She also witnessed violence between her parents. Carol dropped out of school in grade 10 and had hopes of eventually finishing her education, but was struggling to first look after the children. She acknowledged that she occasionally left the children locked in a room together while she went out to buy groceries; it was during one of these occasions that Robert had climbed out the window onto the roof.

Physical Abuse

According to the World Health Organization (WHO, 2006), physical abuse is defined as the intentional use of physical force against a child that results in, or has a high likelihood of resulting in, harm to the child's health, survival, development, or dignity. This includes exposure to such behaviors as hitting, beating, kicking, shaking, biting, strangling, scalding, burning, poisoning, and suffocating. Some physical violence against children in the home occurs in the context of parental discipline attempts in response to normative child behaviors such as constant crying,

whining, toilet-training issues, defiance, and sibling fighting. Closed-fisted strikes, striking with a hard object (belt, board, bat), and throwing the child into objects (furniture, wall) are extremes on a continuum of corporal punishment. Public debate continues about whether spanking constitutes a form of maltreatment; however, there is growing consensus that certain types of spanking—for example, of a teenager, or spanking resulting in injury, such as a bruise—is abusive. Multiple bruising is more consistent with abuse, with slap marks being common; evidence of severe physical injury describes a minority of substantiated physical abuse cases. In the United States, 16.6% of cases were physical abuse (about 2/1,000 children; USDHHS, 2007). In Canada, the rate of substantiated physical abuse (or deemed at high risk of) was 5/1,000 children: 41% were hit with a hand; 21% shaken, pushed, or thrown; 19% were hit with objects; and 8% were punched, kicked, or bitten (Trocmé et al., 2005).

The U.S. Developmental Victimization Survey (Finkelhor, Ormrod, Turner, & Hamby, 2005) found that over half of community children (age 2–17) surveyed experienced a physical assault in the past year. Poly-victimization (four or more different types of events) best described 18% of surveyed children (Finkelhor, Ormrod, & Turner, 2007). One state population survey found physical abuse in the form of kicking, slapping, or punching was most common among teenagers (Theodore et al., 2005). Also, shaking of children less than 2 years of age was reported by caregivers more often than previously recognized. It was estimated that for every child who sustains a serious medical injury as a result of shaking, an estimated 150 children may have been shaken yet were undetected. These authors noted that comparing their survey rate to those from the National Child Abuse and Neglect Data System places physical abuse at more than 40 times higher than the official state-level statistic. Bruising is suspicious for abuse when the marks occur are recent, when the bruising is over nonbony areas (ears, abdomen, anterior chest), and when bruise patterns capture the image of an object (e.g., a hand). Considerable force is needed to cause a bone fracture; this type of abuse is seen mostly in children 6 months and under (Barber & Sibert, 2000). Although most bruising is the result of physically abusive discipline, some actions are more sadistic and planful.

One type of physical abuse more likely to present in hospital emergency rooms and to hospital-based abuse and neglect specialty teams is Munchausen by proxy syndrome (MBPS), in which the caregiver (typically the mother) purposefully makes the child ill (e.g., induces vomiting, administers poison), fabricates the medical complaint and/or psychological symptoms, or subjects the child to medical testing repeatedly, including invasive procedures. Moving from expert to expert increases the amount of medical testing. Fabricated and induced illness is also a diagnostic term (Craft & Hall, 2004), as is fictitious disorder by proxy (*DSM-IV-TR;* American Psychological Association, 2004). These conditions are considered uncommon, although their incidence is unknown.

Although there is no clear understanding of parental motivation in these cases, it has been suggested that such behaviors result from a desire to be recognized as a caring parent, to gain positive adult attention from medical professionals, to prevent a partnership breakup, to avoid criminal culpability, and for secondary economic gain. Others may witness the illness crisis but not the initiation of the

medical event, yet the MBPS caregiver is present consistently. Girls and boys seem equally targeted. Younger children tend to be the focus, although MBPS can span childhood and include participation of a teenager in self-harm. The symptoms cease or decrease when the child is separated from the MBPS caregiver. In a study on male MPBS caregivers, perpetration of physical abuse (e.g., bone fractures) or sexual abuse emerged in the background data collection (Meadow, 1998). Fathers often themselves had Munchausen syndrome or a somatoform or factitious disorder; no father was in regular employment or had been in long-term employment, and about a third of fathers were receiving a disability allowance. Unlike MBPS mothers, fathers were noted by hospital staff as overbearing, aggressive, and quick-tempered. Other situations more likely to occur in MBPS families included unusual pet deaths, home fires, and formal complaints filed by a parent about the hospital.

In a British study that used covert video recordings of suspected MBPS in a hospital ($N = 39$ children; median age 9 months), evidence of abuse (suffocation, poisoning, arm breaking) was captured in the majority of cases. Risk of abuse extended to other children in these families. About 30% of siblings had died suddenly and unexpectedly (most were previously categorized as sudden infant deaths); abuse was documented in a further 36% (Southall, Plunkett, Banks, Falkov, & Samuels, 1997; see article for case examples of sequential observation notes from recordings). Compared with controls, the abused children were less likely to have been born prematurely and more likely to present with bleeding from the nose or mouth, and more of the abusive parents were diagnosed as having personality disorders. These parents were not deemed psychotic; they were observed to attack their children in the absence of challenging child behavioral conditions (e.g., when child was asleep). Maternal allegations of their own maltreatment or the presence of domestic violence were common. The likelihood of a problematic childhood is strengthened by the finding that over 80% of the mothers in one study showed insecure adult attachment of a dismissing or disorganized type, reflecting an undervaluing of close relationships and a frightened and frightening caregiver (Adshead & Bluglass, 2005). Although only a minority of these mothers scored in the personality disorder range, about half showed clinically elevated scores, mostly on Cluster B personality problems (*DSM-IV-TR;* American Psychiatric Association, 2000). Given the planful and persistent nature of this behavior and the harm to the children, the potential role of antisocial and dissociative parental tendencies and subdisorder level of clinical issues need to be studied.

In one follow-up study of maternal MBPS ($n = 17$) with coordinated intervention for the individual (offending) mother and family, the majority of children were functioning reasonably well in their biological family homes (Berg & Jones, 1999). Parents who persistently denied abuse and culpability or had a severe parental personality disorder were not considered suitable for treatment; in these cases, alternative care (fostering, adoption) was maintained. Despite the severity of the initial abuse, in other cases family reunification was achieved and maintained in the context of treatment with attachment-based psychotherapy, cognitive-behavior therapy, and systemic family therapy, along with in-depth assessments of child outcomes, intensive liaison with key professionals local to the family, and involvement of other family members and friends in the treatment process. One of the

13 treatment cases re-presented with MSBP. As these children ranged from 2.5 to 12 years at the 2-year follow-up mark, what remains unclear is the impact of developmental level on outcome and recidivism assessment. Full assessment of sibling outcomes and longer term follow-up are important research goals. For instance, when all siblings are considered, reabuse rates of 20% have been reported (Davis et al., 1998).

It should be noted that criminal and clinical approaches to MPBS are distinct, with intentionality difficult to confirm and more central to legal objectives (Pankratz, 2006). It is important to consider in MSBP, and indeed in all forms of maltreatment, that positive, appropriate, and affectionate parent behavior is likely to be observed or reported. Observational studies of parenting have clearly shown that aversive parenting and positive affect are independent (Solomon & George, 1999). In the vast majority of substantiated abuse cases, parents do declare that they love their child. Affective switching and reactivity are prominent in problematic personality styles, as well as in personality, mood, and anxiety disorders. Careful attention to context is critical in understanding the range of parental behaviors historically and currently demonstrated. Erring on the side of caution for child safety is a reasonable professional stance.

Case Illustration

Susan, age 9 years, was noted by her teacher to wear long sweaters even in summer. She presented as subdued and rarely interacted with the other children; in class, she seemed distracted at times. During an outdoor track and field day, when the children were in gym clothes with short sleeves, the teacher noted that Susan had two or three bruises brown to red in color on each forearm. When the teacher met with Susan and asked her about the bruises, Susan said that she had fallen off her bicycle. In fact, the injuries were not consistent with such an explanation; they fit the pattern of grab marks. The teacher was alert to the fact that these physical marks did not fit with a fall from a bicycle and contacted the local child protection agency.

Psychological and Emotional Abuse

According to the WHO (2006), emotional abuse involves both isolated incidents and a pattern of failure over time on the part of the caregiver to provide a developmentally appropriate and supportive environment. Acts in this category may have a high probability of damaging the child's physical or mental health, or physical, mental, spiritual, moral, or social development. Acts include the restriction of movement; a pattern of belittling, blaming, threatening, frightening, discriminating against, or ridiculing; and other nonphysical forms of rejection or hostile treatment. In the United States, as a single type, the psychological abuse rate was .9/1,000 children in 2004 (USDHHS, 2005). In Canada, the rate of substantiated emotional abuse was 3/1,000 children in 2003 (Trocmé et al., 2005). Emotional abuse is considered to be an undercurrent to other types of maltreatment.

Case Illustration

Tiffany was a 15-year-old waiting to go to court for her sexual abuse victimization by her biological father. As part of a victim impact statement, she spoke of the severe impact of his emotional abuse and how it was connected to the sexual abuse. She recalled that only two hugs in her life from her dad were nonsexualized. There was an unwritten rule that you were not allowed to close or lock the bathroom door when you were using it. She was given many household jobs, living with a single dad. When she failed to comply according to his view, she was required to eat her dinner from the dog bowl. When younger, she was required to kneel in a corner for long periods of time as a parental sanction. Her father played her compliance off her brother's, referring to who was his "best child" that day. Tiffany viewed her father as part of her biology, but never as a dad.

Exposure to Intimate Partner Violence

One type of child maltreatment that has been recognized only recently is exposure to IPV. Empirical studies examining this type of abuse first appeared in the 1980s (Kitzmann, Gaylord, Holt, & Kenny, 2003). The WHO defines IPV as

> any behaviour within an intimate relationship that causes physical, sexual or psychological harm to those in that relationship. It includes acts of physical aggression (slapping, hitting, kicking or beating), psychological abuse (intimidation, constant belittling or humiliation), forced sexual intercourse or any other controlling behaviour (isolating a person from family and friends, monitoring their movements and restricting access to information or assistance. (WHO Intimate Partner Violence and Alcohol Fact Sheet, page 1)

Most *reported* IPV is committed by men against women (Krug et al., 2002). Some surveys suggest that similar proportions of men and women experience IPV; however, abused women experience more physical and emotional impairment in relation to IPV than men (Archer, 2000; Tjaden & Thoennes, 2000). Violence against men by women and between same-sex partners is important for future research to consider at greater depth.

Two recent reviews underscore the broad range of health problems experienced by children exposed to IPV (Bair-Merritt, Blackstone, & Feudtner, 2006; Kitzmann et al., 2003). In a meta-analysis of 118 original studies, Kitzmann and colleagues examined the psychosocial effects of IPV; child outcomes included psychological, social, emotional and behavioral, and academic measures. About 63% of children who witnessed interparental violence were "doing more poorly" than the average child without such exposure. The authors emphasize that effect size did not depend on outcome type. Children who witnessed violence had levels of adjustment problems that were similar to those exposed to physical abuse. A second review that included 22 research articles examined the physical health outcomes of IPV exposure (Bair-Merritt et al., 2006). Exposure to IPV among children was associated with increased risk-taking behavior, such as alcohol abuse and risky sexual activities, during adolescence and adulthood.

There is substantial overlap between exposure to IPV and other types of maltreatment. In a review focusing on the co-occurrence of these exposures, Osofsky (2003) referred to several studies concluding that in homes where a woman is physically abused, 60% to 75% of children have also experienced physical abuse. Children residing with a mother exposed to physical abuse are at increased risk of sexual abuse, either by the mother's partner or a perpetrator outside the home (McCloskey, Figueredo, & Koss, 1995). Osofsky suggests that the co-occurrence of child maltreatment and IPV is important "from a risk and protective factors framework" since it may be more difficult to ensure adequate safety and protection in these families. As a general rule for all maltreatment concerns, it is essential when one subtype of child maltreatment is identified to consider the co-occurrence of other types of abuse and neglect. Research is clear on the predominance of severe male-to-female battering as opposed to female violence. However, child abuse and neglect is concerned with the negative impact on the child, irrespective of the sex of the primary perpetrator.

Case Illustration

Tyrone was working full time while his wife, Jan, was at home with their five young children. When Tyrone came home from work, he completely took over all child care activities. Jan was always finding fault with Tyrone and put him down in front of the children, calling him "fat," "useless," and "a loser," often within a string of expletives. The children, ranging in age from preschooler to preteen, were typically present during fights that often escalated to Jan's threatening to kill Tyrone and slapping, hitting, scratching, and pulling his hair. On several occasions she gave him a black eye, which was noted at his place of work. Tyrone always gave excuses for his marks, such as that he fell over something or the kids got carried away in rough-housing. The children would reenact these scenarios at school, causing concern among the teachers, and the youngest child would use inappropriate language, including swear words and "sex." Generally, Jan presented well outside the home for short durations, confining most of her explosiveness and inappropriate behaviors to the household, although she had a history of suicidal gestures. She alternated rage with a hopeless, helpless depression and would lock herself in her room for days. She had a history of suicidal gestures and had threatened self-harm in front of her children. She had no friends or social life outside of her family and had cut off all contact with her own and Tyrone's family. Eventually, Tyrone's boss confronted Tyrone on the repeated patterns of his injuries and harassing phone calls from Jan to his place of work. Tyrone's boss initiated intervention for Tyrone, which led to other system involvement.

Sexual Abuse

Sexual abuse is defined by the WHO as the involvement of a child in sexual activity that he or she does not fully comprehend, is unable to give informed consent to, or for which the child is not developmentally prepared, or else that violates the laws or social taboos of the society. Children can be sexually abused by both

adults and other children who, by virtue of their age or stage of development, are in a position of responsibility, trust, or power over the victim. Authors of a U.S. community survey study noted that the rate of childhood sexual abuse generated from their self-report data is more than 15 times higher than the official state-level statistic (Theodore et al., 2005). For 2004, 1.2/1,000 children was the official sexual abuse rate (USDHHS, 2005), keeping the same rate over 5 years. Three-quarters of perpetrators were friends or neighbors. Victims are predominantly in the age 12 and up category, although 9% are in the 4- to 7-year category and 2% are in the 3 and under category. In Canada, for 2003, the rate was less than 1/1,000 children (Trocmé et al., 2005).

Case Illustration

Scott, age 9, lived with his mother, Susan, who worked part time as a teacher; his father, Brant, a physician; and his younger brother, Stephen, age 7. The family got along well together with no major stressors, although Brant worked long hours. Scott was an excellent student attending grade 3 and loved all sports. The couple had been married 5 years before Scott was born; both children were planned pregnancies. The family did many activities together; one night a week, Susan attended an art class and Brant worked in the local walk-in clinic. The parents arranged to have a 16-year-old son of close friends, Bruce, as a babysitter. This arrangement appeared to work well for about 6 weeks, until Scott became increasingly resistant to his mother's attending art class. On two occasions, he said that he felt ill and expressed tearfully that he wanted her to stay home. When Susan questioned Scott about his reluctance at her going out, Scott disclosed that on two occasions, Bruce had made him "suck his penis." Bruce had told Scott that this was the latest thing at school, and all teenagers were doing it. Scott became tearful while telling his mother what had happened; he expressed confusion and not wanting to do this, but Bruce was someone that he and all the other kids looked up to, so he thought it must be okay.

MALTREATMENT-ASSOCIATED MALADJUSTMENT

Childhood maltreatment is not a psychiatric disorder but an exposure. It is associated with a range of impairment in areas of basic functioning (eating, sleep), cognition (attention, memory, learning, academic achievement), affective style (modulation, regulation, mood disorders), motivation, and relationships. The challenges in managing tendencies toward both avoidance and overreactivity can create secondary problems in somatization, substance abuse, behavioral risk taking, and aggression toward oneself and others. These problems can persist, yielding implications for partnering and parenting.

The cycle of violence is a clinical reality, particularly when IPV is included with abuse and neglect, but, empirically, is of modest predictive significance in many studies. Across types of maladjustment, a parent's history of maltreatment no more than doubled his or her risk of abusive parenting. This suggests that parental psychiatric status and socioeconomic disadvantage are more weighty, proximal predictors,

compared to a parent's own childhood history (Hurley, Chiodo, Leschied, & Whitehead, 2003; Renner & Slack, 2006; Sidebotham & Heron, 2006). In reviewing generational physical abuse (parent and offspring), most studies fell short methodologically, with the strongest study estimating a relative risk of 12.6 (Ertem, Leventhal, & Dobbs, 2000). In a longitudinal study that involved 224 former male victims of sexual abuse, who were identified based on a nationwide search of clinical and social service case records, 26 had subsequently committed sexual offenses, in most cases involving children outside their family (Salter et al., 2003). Thus, it would seem appropriate to consider a cycle of dysfunction and disadvantage, in addition to a cycle of violence.

Maltreatment is linked with lower school achievement (Wekerle & Wolfe, 2003), which has been accounted for by such factors as school engagement and social competence (e.g., Shonk & Cicchetti, 2001). One prospective, longitudinal study of substantiated abused and neglected children found that only 22% were deemed resilient in adulthood (e.g., no period of homelessness, consistent employment, and no juvenile or adult arrests), which significantly differed from matched controls (McGloin & Widom, 2001). As such, maltreatment may be one of the most modifiable barriers to living a "normal" life, including freedom from mental illness (DeBellis, 2001).

Psychiatric Disorders Associated with Childhood Maltreatment

Mood, anxiety, substance use, eating, and personality disorders in youth will be considered. Research in childhood maltreatment is uneven; most studies have focused on the areas of sexual and physical abuse, and on adults, mostly females. To provide an overview of the associated risk of the considered disorders, odds ratios have been compiled into a table on adult female functioning (Table 28.1).

Mood Disorders

Depressive symptoms have commonly been associated with childhood maltreatment (Danielson, de Arellano, Kilpatrick, Saunders, & Resnick, 2005; Gover, 2004; Meyerson, Long, Miranda, & Marx, 2002). As can be seen from Table 28.1, depression and anxiety problems far outstrip other maladjustment for female survivors of childhood maltreatment. In a community-based longitudinal study, individuals with a history of child abuse and neglect reported a two- to threefold increased likelihood of Dysthymic Disorder and Major Depressive Disorder during adolescence and young adulthood, as compared to nonabused individuals, even after statistically controlling for certain confounding variables (Brown, Cohen, Johnson, & Smailes, 1999). In a meta-analysis of child sexual abuse, victims exhibited significantly greater social withdrawal and internalizing symptoms, such as depressed mood, anhedonia (i.e., lack of pleasure in previously reinforcing activities, such as leisure, sex), and feelings of worthlessness and guilt (Kendall-Tackett et al., 1993). Other common emotional effects include guilt (i.e., feeling responsible for the abuse), helplessness and hopelessness, sleep disturbance, appetite disturbance, and low self-esteem. According to the *DSM-IV-TR* (APA, 2004), the symptoms of depression for children and adolescents are the same as those for adults. However, it is suggested that different symptoms may be more prominent at different ages.

Table 28.1 Adult (Female) Maladjustment Based on Childhood History of Maltreatment

Diagnoses/problems	Physical Abuse	Sexual Abuse	Emotional Abuse	Child Neglect
Anxiety	39.4 (33.8–45.1)	42.9 (35.3–50.4)		
Major Depressive Disorder	23.3 (18.2–28.3)	28.5 (20.8–36.2)	2.7 (2.3–3.2)	4.1 (1.9–8.8)
Alcohol abuse	9.9 (7.0–12.8)	10.3 (5.9–14.8)		1.65
Other substance use disorders	4.1 (2.7–5.5)	6.6 (2.5–10.7)		1.7 (1.5–2.0)
Posttraumatic Stress Disorder	1.03 (1.00–1.06)	10.2 (7.1–14.5)	5.67 (2.74–11.74)	3.5 (1.5–7.9)
Antisocial personality	4.1 (2.5–5.7)	4.7 (2.2–7.2)		1.32 (0.6–2.88)
Criminality	1.09 (1.01–1.19)	1.14 (1.02–1.27)	b = 2.06	1.7 (1.2– 2.3)
Suicidal ideation	2.77 (1.26–6.12)			
Suicide attempt	1.64 (0.63–4.28)		b = 0.57	3.03 (1.11–8.33)

Note: Odds ratios and confidence intervals are presented unless otherwise specified. *Source:* H. A. Bergen, G. Martin, A. S. Richardson, S. Allison, & L. Roeger (2003); A. C. Boudewyn & J. H. Liem (1995); K. Brewer-Smyth, A. W. Burgess, & J. Shults (2004); J. Brown, P. Cohen, J. G. Johnson, & E. M. Smailes (1999); D. P. Chapman, C. L. Whitfield, V. J. Felitti, S. R. Dube, V. J. Edwards, & R. F. Anda (2003); E. Evans, K. Hawton, & K. Rodham (2005); C. Evren, S. Kural, & D. Cakmak (2005); J. G. Johnson, P. Cohen, J. Brown, E. M. Smailes, & D. P. Bernstein (1999); P. R. Joyce, J. M. McKenzie, S. E. Luty, R. T. Mulder, J. D. Carter, P. F. Sullivan, et al. (2003); S. J. Kaplan, D. P. Covitz, S. Salzinger, F. Mandel, M. Weiner, & V. Labruna (1999); D. G. Kilpatrick, K. J. Ruggiero, R. Acierno, B. E. Saunders, H. S. Resnick, C. L. Best, et al. (2000); D. G. Kilpatrick, K. J. Ruggiero, R. Acierno, B. E. Saunders, H. S. Resnick, & C. L. Best (2003); J. B. Kingree, D. Phan, & M. Thomson (2003); D. E. Nelson, G. K. Higginson, & J. A. Grant-Worley (1995); C. S. Widom (1999).

For example, somatic complaints and irritable mood are especially common during childhood and adolescence. Sleep and appetite disturbances in children and adolescents, as in adults, may vary, with some patients oversleeping and/or overeating (hypersomnia and hyperphagia, respectively), while others present with difficulty falling or staying asleep and/or having no appetite.

Anxiety Disorders

General symptoms of anxiety, nightmares, inappropriate fears of certain places, and clinging to parents are commonly found among maltreated children (Giardino & Giardino, 2002; Kendall-Tackett et al., 1993). Child inpatients with histories of exposure to abuse were described by their parents as being significantly more afraid of being left alone with others, exhibiting more suspicion, and getting more upset when touched, as compared to nonabused inpatients (Kolko, Moser, & Weldy, 1988). Based on self-report, sexually abused children were significantly more anxious than both clinical and normal controls (Mannarino, Cohen, & Gregor, 1989). Physical abuse has been shown to increase the likelihood for any anxiety disorder, as well as for Generalized Anxiety Disorder (GAD), indicating excessive worries (Flisher et al., 1997).

The initial response to single-event trauma may be hyperarousal, notably, hyperactivity, disorganization, and disruption in maintaining routines. However, as trauma is repeated, the adaptation response can become more complicated by dissociation, affective disturbance, and prolonged gaps in historical memory. With

a single-event trauma, there is an emphasis on pathological fear and erroneous associations (people, object) with danger (Tolin & Foa, 2002). Most of the focus has been on the Posttraumatic Stress Disorder (PTSD) diagnosis, although subclinical symptom levels are important as they have been linked to significant functional impairment in youth (Carrion, Weems, Ray, & Reiss, 2002; Putnam, 1998). From age 8 to 10 years, PTSD reactions are more in line with adult diagnostic criteria, with females more likely to show PTSD-like responses and males to show aggression responses (Dyregrov & Yule, 2006).

Beyond the presence of a traumatic stressor, there are three main symptom clusters, with the *DSM* diagnostic requirement that criteria be met in each of the clusters. The three-cluster symptom model consists of (1) reexperiencing the trauma (e.g., flashbacks, nightmares); (2) avoidance of trauma-related cues and emotional numbing; and (3) problems with arousal (e.g., sleep onset, maintenance; exaggerated or diminished startle response). Some of these symptoms may manifest in other ways, including somatic symptoms (e.g., stomachache, headache to support escape or avoidance), self-destructiveness (e.g., cutting to support release from numbing), and hostility (e.g., lashing out under hyperarousal; sexual play as trauma-based reenactment).

In a clinical sample (most had multiple traumas and prior child protective services involvement), PTSD was commonly comorbid with other anxiety and mood disorders (Carrion et al., 2002). The most intense PTSD symptoms were irritability and anger, distressing dreams, and feelings of detachment from others. The most frequent symptoms were avoidance of trauma-related thoughts, feelings, and cues; distressing recollections; inability to recall important aspects of the trauma; distressing dreams; and difficulty concentrating. Full diagnosis for PTSD was predicted by severe and frequent hypervigilance, exaggerated startle, and detachment from others. The most common symptom cluster was reexperiencing or intrusions about the trauma. Clearly this symptom pattern of sleep and concentration difficulties must challenge the PTSD child and youth in school.

Similarly, in a study of randomly selected teens involved with the child protection system, reexperiencing was among the most common self-reported PTSD symptoms ("bad dreams or nightmares"; "remembering things that happened that I didn't like"). As is consistent with developmental level and abstract thinking about a shortened future, teens reported feeling a vague sense of impending doom ("feeling afraid something bad might happen"; Wekerle et al., 2006). In a study of teen inpatients, emotional abuse was the only consistent predictor of PTSD symptoms across all three clusters when considering all types of abuse and neglect simultaneously. Sexual abuse history also predicted the reexperiencing symptom cluster (Sullivan, Fehon, Andrew-Hyman, Lipschitz, & Grilo, 2006). Taken together, PTSD symptomatology may be as important to consider as the issue of meeting full criteria for the disorder. Given that the likelihood of meeting all three cluster criteria is greater in older children, current *DSM* criteria for PTSD may not be sensitive to the developmental manifestation in younger children. Symptomatology of PTSD has been shown empirically to mediate maltreatment and negative outcome (e.g., Risser, Hetzel-Riggin, Thomsen, & McCanne, 2006; Wekerle et al., 2001); therefore, targeting PTSD symptomatology is a fruitful intervention avenue.

Research has suggested that those with PTSD tend to have lengthier histories of maltreatment and/or broader exposure to violence (Deblinger, Steer, & Lippman, 1999; Linning & Kearney, 2004; Wood, Foy, Layne, Pynoos, & James, 2002). This may be why sexual abuse tends to predict PTSD diagnosis better than physical abuse, if both are present. Symptoms of PTSD may be associated with behavior inhibition problems (Ruchkin, Schwab-Stone, Koposov, Vermeiren, & Steiner, 2002). One study of youth from shelter facilities found that PTSD was comorbid with other mood or anxiety disorders, most notably dysthymia, major depression, GAD, panic with and without agoraphobia, and specific phobias (Linning & Kearney, 2004). In studies of adult PTSD, the majority of PTSD diagnoses occurred among females and in the context of other diagnoses, notably depression, where PTSD contributes to functional impairment across social, physical, emotional, and general health domains (Cloitre, Koenen, Gratz, & Jakupcak, 2002). The adult literature is investigating the tenet that chronic maltreatment is more likely to manifest as PTSD "personality" or complex PTSD, with characterological symptoms (impulsivity, aggression, self-destructiveness), pervasive despair and hopelessness, somatization symptoms, transient dissociative episodes, and difficulties in maintaining close relationships. In this way, PTSD has become a central organizing construct for personal and interpersonal style for the maltreated individual.

Overall, PTSD may be a relatively underassessed condition, whereas depression or anxiety may be noted rather than the more complex symptom picture. However, given that studies suggest reexperiencing is a frequent outcome, sleep patterns and distressing or intrusive dreams and thoughts should be considered in the context of coping with maltreatment. Further, with adolescence and young adulthood, the issue of personality involvement is raised.

Behavioral Problems and Disorders

Children exposed to maltreatment, especially those who are physically abused, show problems with aggression (Joshi, Daniolos, & Salpekar, 2004a, 2004b; Kendall-Tackett et al., 1993), and it is a problem that continues to emerge in adulthood (see Table 28.2); Also, see Botash and Ricci (2006). Social competence is further challenged by the tendency for aggression to co-occur with withdrawing behaviors in maltreated samples (e.g., Rogosch & Cicchetti, 1994). Physically abused youth show greater sensitivity to anger displays and threat cues (Pollak & Kistler, 2002, Pollak & Tolley-Schell, 2003) and more hostile attributions toward others (Dodge, Pettit, Bates, & Valente, 1995). During adolescence, physical abuse victims are more likely to engage in delinquency and risk-taking behaviors (Giardino & Giardino, 2002). Prevalence studies of disruptive behavior disorders among physically abused youth, as compared to those who are nonabused, reveal high rates of Conduct Disorder (CD) and Oppositional Defiant Disorder (Flisher et al., 1997; Pelcovitz et al., 2000; Pelcovitz, Kaplan, Goldenberg, Mandel, Lehane, & Guarrera, 1994). Both physically abused males and females are 9 times as likely to meet criteria for CD, even after controlling for potential mediating factors (e.g., parental psychopathology, lack of family cohesion, and lack of parental support; Kaplan et al., 1998). However, other research suggests that *females,* rather than

Table 28.2 Signs of Maltreatment

Type of Abuse	Signs
Physical Abuse	*Bruising of soft tissue (e.g., head, neck, trunk, arms):* → Pattern bruising in the shape of objects or hand marks. → Imprint of large, multiple, clustered bruises; bruises at different stages of healing (multiple color bruising). *Burns:* → Normally occur on hands, feet, and genitalia. → Burn patterns: • Inflicted burns are symmetrical when forced. • Accidental burns are irregularly shaped. → Cigarette burns: • Inflicted cigarette burns appear 7 to 10 mm round and have deep central crater. • Accidental cigarette burns appear oval in shape. → Scald burn (with hot liquid) is a common cause of childhood burns, both accidental and inflicted: • Accidental burns with hot liquids are irregularly shaped, with patches and splash marks. • Immersion burn injury occurs when a child is dunked in hot liquid; it can leave a stocking- or glove-like burn pattern found on the hands or feet. *Bites:* → Human bites cover a larger surface area than animal bites and tend to tear or crush (3 cm or greater space between canines indicates the bite was most likely by an adult). *Other:* → Inadequately explained bone fractures (X-ray evidence of history of multiple fractures). → Brain injuries from blunt head trauma: • Shaken baby syndrome: cerebral contusions, subdural hemorrhages, injury to cervical spinal cord, rib fractures; may present with seizures, decreased consciousness, respiratory difficulty, irritability, lethargy, vomiting, and sleep apnea.
Emotional Abuse	Unexplained physical symptoms/complaints with no medical basis. Sleep disturbances. Extreme dependence. Inability to trust. Feelings of shame and guilt. Child's own sense of his or her potential is limited. Child fails to thrive. Withdrawal. Poor self-esteem.
Neglect	Child's inability to adequately grow and develop to or above the 3rd percentile in height and weight, with no organic reason. Malnutrition, poor growth, consistent hunger, distended stomach. Untidy appearance, inappropriate dress, poor hygiene. Pale or listless appearance. Repeated or unattended lice. Recurring and unexplained patterns of minor injuries. Aches, pains, migraine, gastrointestinal problems.

(*continued*)

Table 28.2 (*Continued*)

Type of Abuse	Signs
Sexual abuse	Trauma to genital or anal area: →Abrasions or bruising of the genitalia. →Abdominal, genital, or rectal pain. →Genital or rectal discharge; unexplained sexually transmitted infection. Findings sometimes seen following sexual abuse but also other causes: History and other investigation is important in diagnosing abuse. → Frequent urinary tract or yeast infections.
Exposure to Intimate Partner Violence	Increased risk of physical harm or injury due to proximity to an act of family violence. Headaches. Stomach pains (e.g., ulcers). Difficulty sleeping/insomnia. Difficulty with memory. Hearing and articulation problems. Anxiety-related increases of physical issues (asthma, tics, eczema, epileptic fits, enuresis/encopresis).

Note: This table is an expanded version of Cheng, Munn, Jack, & MacMillan, 2006 based on American Academy of Pediatrics (n.d.); American Humane Association (2006); E. D. Bariciak, A. C. Plint, I. Gaboury, & S. Bennett (2003); J. Bays & D. Chadwick (1993); W. Bernet, W. Ayres, J. E. Dunne, E. Benedek, G. A. Bernstein, E. Bryant, et al. (1997); S. E. Chaney (2000); C. Doyle (1997); K. M. Drach, J. Wientzen, & L. R. Ricci (2001); J. Emans, E. Woods, N. Flagg, & A. Freeman (May 1987); E. Evans, K. Hawton, & K. Rodham (2005); J. Fantuzzo, R. Boruch, A. Beriama, M. Atkins, & S. Marcus (1997); J. W. Fantuzzo & C. U. Lindquist (1989); M. E. Herman-Giddens, G. Brown, S. Verbiest, P. J. Carlson, E. G. Hooten, E. Howell, et al. (1999); V. M. Herrera & L. McCloskey (2003); G. Hornor (2005); N. Kellogg & The Committee on Child Abuse & Neglect (2005, August); W. J. King, M. MacKay, & A. Sirnick (2003); J. R. Kolbo, E. H. Blakely, & D. Engleman (1996); L. Kos & T. Shwayder (2006); G. W. Lambie (2005); S. Maguire, M. K. Mann, J. Sibert, & A. Kemp (2005); B. W. Mayer & P. Burns (2000); S. S. Mudd & J. S. Findlay (2004); K. Nimkink & P. K. Kleinman (2001); *Ontario Association of Children's Aid Societies* (n.d.); V. J. Palusci, E. O. Cox, T. A. Cyrus, S. W. Heartwell, F. E. Vandervort, & E. S. Pott (1999); M. D. Peck & D. Priolo-Kapel (2002); C. Powell (2003); A. Slep & R. Heyman (2006, August); R. Wright, H. Mitchell, C. M. Visness, S. Cohen, J. Stout, R. Evans, et al. (2004, April).

males, are at high risk for CD when physically abused (MacMillan et al., 1999) or when experiencing multiple forms of maltreatment (Romano, Zoccolillo, & Paquette, 2006). Similarly, E. Nelson and colleagues (2002) found an increased risk for the development of CD and Antisocial Personality Disorder among women who report histories of childhood sexual abuse. When compared to physical abuse and exposure to marital violence, childhood sexual abuse appeared to be the strongest predictor of self-reported violent and nonviolent criminal behavior among females (Herrera & McCloskey, 2003). Epidemiological work suggests that type of maltreatment exposure is not specific in explaining the increased risk of conduct problems (MacMillan et al., 2001).

Substance Use Disorders

There has been greater recognition of the overlap between substance abuse and maltreatment, both in terms of childhood history and in perpetrators of maltreatment (Wekerle & Wall, 2002). Both physical and sexual abuse are linked with a greater

likelihood of early-onset smoking, drinking, illicit drug use, and injection drug use and nonmedical prescription drug abuse compared to nonmaltreated children and adolescents (De Marco & Phelan, 2004; Kessler et al., 1996; National Clearinghouse on Child Abuse and Neglect, 2004; Raghavan & Kingston, 2006; Vermeiren, Schwab-Stone, Deboutte, Leckman, & Ruchkin, 2003). In a study using a nationally representative probability sample, investigators found that having a history of physical or sexual abuse doubled the risk of alcohol, marijuana, or other hard drug abuse or dependence in the past year. The victimizations reported by this sample occurred prior to nonexperimental drug use, suggesting that victimization is a risk factor for later abuse or dependence (Kilpatrick et al., 2000). Further, sexual abuse history directly predicts both earlier age of substance abuse and higher reported rates of lifetime traumatic events among adults, raising the issue of an interplay between substance use and revictimization (Raghavan & Kingston, 2006).

Drug and alcohol use may function as a coping mechanism for increased stress brought on by childhood abuse. Alcohol and other illegal drug use may reduce symptoms of hyperarousal and unpleasant emotions and produce emotional numbing or euphoria (Cohen, Mannarino, Zhitova, & Capone, 2003; Kilpatrick et al., 2000). The frequent comorbidity between PTSD and substance use is consistent with a self-medication hypothesis (Cohen et al., 2003; Stewart & Israeli, 2002).

Youth with a history of maltreatment may enter service systems through a number of doors. Aarons et al. (2001) found high rates of lifetime prevalence of substance use disorders (SUDs) among youth from child welfare (19%), mental health (24%), and juvenile justice (41%) systems, whereas comparable community rates are less than 10%. Although child welfare youth seemed to be less vulnerable than those from other entry points, this study did not investigate the association between child maltreatment across service site and SUDs. This study is important, however, as child welfare youth are not often included as a population for epidemiological study.

Eating Disorders

Elevated risk for eating disorders or weight problems among teens reporting sexual abuse or physical neglect has been noted, above and beyond the risk from parental psychopathology, childhood eating disorders, and difficult temperaments (Johnson, Cohen, Kasen, & Brook, 2002). Sexually abused teens also appear to report significantly higher levels of distress about their weight more often than nonabused adolescents (Wonderlich et al., 2000). Women diagnosed with Bulimia Nervosa and those who have recovered for at least 1 year reported significantly higher levels of physical and psychological abuse (Rorty, Yeager, & Rossotto, 1994).

Personality Disorders

An interesting prelude to personality research is the finding based on story retelling tasks that maltreated children display from an early age (e.g., 3.5 years) affective splitting, and that greater negative emotionality is evidenced thereafter. Most nonmaltreated individuals would develop a positivity bias, being able to see the positives in themselves, others, and situations (Ayoub et al., 2006). Thus, it is not a surprise that childhood abuse has been linked with an increased risk of personality

disorders (Battle et al., 2004). Specifically, documented and self-reported physical abuse is significantly associated with Antisocial Personality Disorder, even after controlling for previous sexual abuse, neglect, and characteristics of other personality disorders. When controlling for these mediating factors, childhood sexual abuse was significantly associated with Borderline Personality Disorder (Johnson, Cohen, Brown, Smailes, & Bernstein, 1999). Borderline Personality Disorder has been frequently cited as the personality disorder most commonly associated with sexual abuse (Battle et al., 2004; Zanarini et al., 1997).

Suicidal and Self-Harm Behaviors

A history of childhood maltreatment is a significant risk factor for suicide (Giardino & Giardino, 2002; Wozencraft, Wagner, & Pellegrin, 1991). For instance, a Turkish study of teenagers revealed that the rate of suicide attempt for those with a history of childhood maltreatment was 7.6 times higher than for those without such a history (Zoroglu et al., 2003). Autopsy studies of youth have found an association between physical abuse and suicide (Kaplan, Pelcovitz, Salzinger, Mendel, & Wiener, 1997; Riggs, Alario, & McHorney, 1990). These findings are consistent with prospective longitudinal community studies (Brown et al., 1999; Johnson et al., 2002). Johnson and colleagues (2002) found an increased risk of suicide attempts in early adulthood with a history of maltreatment, even after adjusting for demographic characteristics and psychiatric symptoms and disorders in childhood and early adolescence. It is suggested that interpersonal skills may be the linking factor, with the maltreated individual experiencing greater social isolation and/or antagonistic interactions with others, which in turn adds risk for suicidal behavior. A study reviewing the risk factors for youth suicide reported that youth with a history of childhood sexual abuse had an increased risk for suicide up to about 12 times that of youth with no reported history of abuse (Beautrais, 2000). Individuals with histories of sexual abuse who report suicidal ideation tended to be older at the time of abuse and to have remained in their home after abuse was detected (Wozencraft et al., 1991). The relationship between childhood sexual abuse and suicidality was greatly reduced, but not eliminated, after controlling for a wide range of potentially confounding factors (Fergusson & Mullen, 1999). Thus, the increased risk for suicide from maltreatment may be partly accounted for by other factors.

A significant portion of the empirical literature demonstrating the association between maltreatment and *nonsuicidal* self-injurious behavior has focused on adults (e.g., van der Kolk, Perry, & Herman, 1991; Wiederman, Sansone, & Sansone, 1999). For example, 62% of subjects who self-mutilate reported being sexually or physically abused at a young age (Favazza & Conterio, 1989). However, a number of studies specifically focusing on childhood maltreatment and adolescent self-injurious behavior exist as well. In one inpatient adolescent sample, those in the self-mutilation group reported histories of physical and sexual abuse more often than those in the control group. However, when all types of maltreatment were controlled simultaneously, only sexual abuse and emotional neglect appeared to be significant predictors of self-mutilation (Lipschitz et al., 1999). Similarly, more sexually abused adolescents reported deliberate self-harm, as compared to the

non-sexually abused adolescents. In this study, 30% of the abused teenagers reported deliberately harming themselves five or more times (Martin, Bergen, Richardson, Roeger, & Allison, 2004). Zoroglu and colleagues (2003) found a 27-fold increase in self-mutilation in adolescents reporting a history of maltreatment compared to those who did not. It is advised that clinicians working with maltreated populations be mindful of, and regularly assess, potential risk for self-injurious behavior.

Dissociation

Dissociation is considered a psychological avoidance strategy, or an unconscious attempt to become emotionally numb via altering one's level of conscious self-awareness. To accomplish an emotionally disconnected state, one's emotions, thoughts, and behavioral experiences are separated from one another (Giardino & Giardino, 2002). In children and adolescents, dissociation can take the form of a trance-like state (e.g., abruptly distracted, sudden change in head orientation in midconversation) or manifest as a sudden change in behavior (e.g., one moment cooperative and the next moment oppositional or aggressive). Typically, activities during the dissociative episode are not remembered during nondissociative states. Dissociation may manifest as functional amnesias for sets of behaviors, a sense of depersonalization or derealization (e.g., problems with tracking of own activities in real time; sense of detachment from others; sense of being outside of oneself; repressing pain sensations in parts of the body), and experiences in identity alterations (e.g., true self versus projected, public self; chameleon-like personality with different interactants, pretending to be someone else; Putnam, 1996). Dissociation is thought to be a key contributor to the maltreated child's fragmented sense of self. As early as the preschool years, physically and sexually abused children exhibit signs of dissociation, although this is not as apparent for neglected children (e.g., Macfie, Cicchetti, & Toth, 2001). There is evidence for differential relationships between dissociation and physical and sexual abuse (Joshi et al., 2004a, 2004b; Giardino & Giardino, 2002). Using a longitudinal, prospective design, Kisiel and Lyons (2001) evaluated the prevalence of dissociation among physically and sexually abused youth using self-report and parent report questionnaires. Whereas sexually abused children reported significantly higher levels of dissociation compared to nonabused subjects, physically abused children did not. However, parental report of child dissociative symptoms indicated a significant association with both physical and sexual abuse histories. Dissociation may be more likely when maltreatment involves physical violence and multiple perpetrators (Trickett, Noll, Reiffman, & Putnam, 2001).

Asymptomatic Victims

Maltreatment covers a wide range of behaviors, from single events to chronic exposure. Within types there is also substantial variability; for example, fondling is the most frequently substantiated form of sexual abuse, as compared to penetration acts (Trocmé et al., 2005). The impact of maltreatment can manifest in myriad ways, suggesting that there is no single symptom profile associated with maltreatment (Kendall-Tackett et al., 1993).

A meta-analysis of sexual abuse studies found that between 21% and 49% of the subjects appeared asymptomatic at the time of assessment (Kendall-Tackett et al., 1993). McLeer and colleagues (McLeer, Deblinger, Atkins, Foa, & Rolphe, 1988) suggest that psychiatric sequelae can develop well after the time of the actual abuse (McLeer et al., 1988). Such a "sleeper effect" (Fergusson & Mullen, 1999) may emerge at developmentally sensitive times; for example, in teen dating, maltreatment-relevant threat cues may be heightened (e.g., physical proximity, physical contact, push-pull affect, sexual response; Wekerle et al., 2001). Maltreatment characteristics predict symptomatology, such as child age at onset, the type of abuse, the frequency and severity of abuse, the relationship between the victim and the perpetrator, the family environment, and family psychiatric history (Boney-McCoy & Finkelhor, 1995; Flisher et al., 1997; National Clearinghouse on Child Abuse and Neglect, 2004). The likelihood of having adjustment problems after experiencing abuse increases if the abuse is severe, violent, and lengthy in duration, and if limited family support is available (Boney-McCoy & Finkelhor, 1995; Fergusson & Mullen, 1999). Elements such as coping style, attachment status, and perceptions of self-identification as abused may influence distress and adjustment levels (Wekerle & Wolfe, 2003; Wekerle et al., 2001). For assessment of adjustment and trauma-related issues, see Cohen, Mannarino, Murray, and Igelman (2006), Kolko (1996), Wekerle et al. (2006), and V. Wolfe (in press).

Continued research on child and adolescent adjustment postmaltreatment is needed; the majority of work to date has been conducted on adults, mostly with retrospective reporting on maltreatment, with some important twin and prospective longitudinal studies. As Table 28.1 demonstrates, the importance of childhood maltreatment is underscored when considering adult disorders and impairment. As with adults, a similar pattern in impairment domains is shown for youth: substance abuse, antisociality and aggression, anxiety, depression, PTSD, and suicidality. We are forced to acknowledge that it is hard for many abused individuals to cope and to stay alive.

THEORETICAL MODELS OF MALTREATMENT-ASSOCIATED MALADJUSTMENT

Social Learning and Attachment Theories

Caregiver vulnerabilities of various kinds have been linked to compromised parenting, for example, anxiety and depression, substance abuse, aggression and antisociality, social support, and poor physical health (e.g., Egeland, Yates, Appleyard, & van Dulmen, 2002; Eiden & Leonard, 2000; Lyons-Ruth, Lyubchik, Wolfe, & Bronfman, 2002; MacMillan & Munn, 2001; Pine et al., 2005; Rogosch, Cicchetti, & Toth, 2004; Shea, Walsh, MacMillan, & Steiner, 2005). Developmental research shows that inadequately responsive parenting increases children's risk for problems with future relationships, managing emotions, self-efficacy, and violence (Dube et al., 2001; Kim & Cicchetti, 2003; National Institute of Child Health and Human Development Early Child Care Research Network, 2004).

According to social learning theory, children form a schema for how stress is regulated in a close relationship through their experience of interacting with their caregivers. Longitudinal research has shown, for example, that maternal depression typically precedes child aggression and hyperactivity, as well as affective disturbances (Elgar, Curtis, McGrath, Waschbusch, & Stewart, 2003). Early abusive relationships inform relationship schema, wherein maladaptive stress–maladaptive coping is learned via modeling and experience with reinforcement schedules. For example, attack can end the stress from the interactant by causing harm and shutting down conflict or interaction, promoting the interactant to acquiesce, appease, and remove himself or herself from the immediate environment. Maltreated individuals may maintain developed dysfunctional coping partly as the result of their dysfunctional relationship strategy.

According to attachment theory, the end goal of the early caregiving relationship is to protect us from harm and enhance our exploratory behavior. The confidence in the protection that subsequently develops serves as both the source for stress regulation and a model of stress regulation to be internalized. This internal model is carried into future situations as a central means to regulate stress (Goldberg, 2001). An abusive relationship turns the source of predictable comfort and first choice for support into a source of fear, confusion, and hesitancy, thereby compromising the development and practice of stress-regulating strategy (Lyons-Ruth & Jacobvitz, 1999). Thus, from both social learning and attachment perspectives, relationship functioning represents a key ongoing area of adaptational challenge for maltreated children (Rutter & Sroufe, 2000). This heightened vulnerability has been tied, theoretically, to holding a conceptualization of relationships that are organized along power abusive, victim-victimizer dimensions (e.g., Crittenden & Ainsworth, 1989; Crittenden & Claussen, 2002), as attachment theory (Bowlby, 1970, 1972, 1980, 1982) advances that both sides of a relationship are learned.

Maltreated victims may, by implication, have a more developed (and more generally applied) conceptualization of their victimizer. Defensive processes (e.g., including perceptual information that is consistent with one's relationship schema and excluding perceptual information that is inconsistent with the relationship schema) are thought to interfere with being open to new input, thereby increasing the likelihood of schema maintenance. Maltreated teens would be expected to select partners who would contribute to the maintenance of distressed relationships (Crittenden & Ainsworth, 1989).

Much recent evidence has accumulated to show that maltreated youth, relative to their nonmaltreated counterparts, exhibit indicators of self-system dysfunction, including poorer self-representations, self-esteem, and ego control (e.g., Cicchetti & Rogosch, 1997; Toth, Cicchetti, Macfie, & Emde, 1997). Such disorganization has been linked with higher trauma-related symptoms, specifically dissociation, in adolescents who had childhood histories of maltreatment and other trauma (Carlson, 1998). The PTSD trauma symptomatology is also functionally related to relationship disturbance, including adolescent dating violence (Wolfe, 1999; Wekerle et al., 2001). Dissociation, for example, may contribute to asynchrony in and abrupt withdrawal from interacting. Hypervigilance may lead to an agitated, highly reactive response in interactions.

Developmental Traumatology Theory

According to the theory of developmental traumatology, early exposure to the intense traumatic stress of maltreatment, compounded by the underdevelopment of an optimally functioning stress-regulating system, will result in an overreactive stress system (Cicchetti & Walker, 2001; DeBellis, 2001; DeBellis & Putnam, 1994). Such a long-term effect of maltreatment experience on stress and adverse health outcomes has been linked to the impairment of the hypothalamic-pituitary-adrenal axis, the psychobiological stress system that is sensitive to the effect of early environment. In turn, this compromises other stress-related systems, including the immune, neurotransmitter, and sympathetic nervous systems, and may be associated with structural changes to the brain (DeBellis, 2001). Nowhere is this convergence of malfunctions across difference stress systems more apparent than in PTSD.

In the developmental traumatology model, PTSD symptomatology is seen as the key mediator of negative outcomes. Such symptoms may persist and overwhelm youth who are disadvantaged by their stress response mechanisms, thereby interfering with their sense of self-continuity and ability to cope, leading to maladjustment (Bradley, 2000; Chandler & Lalonde, 1998). Due to maltreatment-related cues directly leading to aversive experience, escape-avoidance mechanisms may be reinforced. Substance abuse and self-harming may be efforts to self-medicate in response to PTSD symptoms via reducing negative affect, numbness, dissociation, tension, and hypervigilance or enhancing positive affect. However, progression to more harmful behaviors (e.g., abuse of cocaine, opiates) may exacerbate affect problems and increase victimization risk (e.g., Ginzler, Cochran, Comenech-Rodriguez, Cauce, & Whitbeck, 2003).

Personality Trait Model

Personality research posits that people differ on a set of underlying hypothetical dimensions, and these represent stable dispositions that are inherent in the individual, manifest early in life, and generalize across a variety of situations. Research on maltreating caregivers has highlighted the role of the caregiver's personality in limiting his or her ability to cope with stress associated with parenting, thus predisposing the individual to abusive behavior (Marziali, Damianakis, & Trocmé, 2003). To further understand the role of the caregiver's personality in the perpetration of child maltreatment and spousal violence, the model suggests that certain personality traits are a risk factor for child and partner abuse, operating singly or in conjunction with other risk factors such as substance abuse (Flett & Hewitt, 2002). In addition, it has been shown that personality may serve as a mediator in the relation between the individual's own history of childhood maltreatment and the perpetration of partner violence (Ornduff, Kelsey, & O'Leary, 2001). This indicates that the development of personality is shaped by the abusive experience during early childhood and consolidated into a stable behavioral and cognitive tendency that individuals bring to close relationships, perpetuating the cycle of violence and abuse (see also Pickering, Farmer, & McGuffin, 2004).

It has also been suggested that childhood maltreatment and individual personality traits contribute synergistically to adverse behavioral and health outcomes. For

example, Blackson, Tarter, and Mezzich (1996) showed that the adverse effect of parental discipline by substance-abusing fathers on their child's behavioral adjustment is exacerbated by the child's difficult temperament, an indicator of poor affect regulation. Similarly, Trocki and Caetano (2003) have shown that children who were exposed to family violence were more likely to develop depressive symptoms and abuse substances during adulthood if they also have a difficult temperament. Child response to trauma may be moderated by preexisting psychopathology, attributional or coping style, and temperament (Feiring, Taska, & Lewis, 2002). The mechanism underlying such interaction between personality and maltreatment can be conceptualized along gene-environment interaction effects. Because of its effect on the monoamine (e.g., serotonin, norepinephrine, and dopamine) level at the prefrontal cortex, different polymorphisms of genes associated with the monoamine system have been linked to individual differences in temperamental affect regulation, measured in terms of maternal report as well as behavioral observation (Benjamin, Ebstein, & Belmaker, 2002). In a now classic study, Caspi and colleagues (2002) found that male children who were maltreated were more likely to exhibit externalizing behavior if they had the low monoamine oxidase A activity genotype. Similarly, Kaufman et al. (2004) demonstrated that maltreated children who receive low social support and carry the short allele of the serotonin transporter gene promoter polymorphism were twice as likely to develop depression, even compared to their nonmaltreated counterparts who carry the same polymorphism. Maltreated children who carry the short allele but receive high social support had only minimal increase in the depression score, indicating the buffering effect of a positive environment. Recently, in validating the gene-environment interaction effect observed among maltreated children, Kim-Cohen and colleagues (2006) examined two different types of gene-environment correlation that may have potentially confounded the interaction effect: (1) The abusive parent may have transmitted to the child both an adverse environment and a genetic susceptibility for behavioral maladjustment, and (2) children who carry the risk genotype may behave in ways that are more challenging to an underskilled, stress-vulnerable parent. To address these issues, the researchers tested for the former by controlling for maternal antisocial personality and for the latter by assessing whether exposure to child maltreatment is predicted by the child's genetic polymorphism. Results indicate that even after controlling for the two types of gene-environment correlation, the interaction effect remains significant, indicating that children who carry the polymorphism that calls for low monoamine oxidase A activity were more vulnerable to the exposure to physical abuse.

One point to consider in reviewing these studies of gene-environment interaction in maltreated samples is the potential difference in maltreatment experience even between genetically identical twins. The shared environment may not be shared in a chronically chaotic and dysfunctional household. Such family dynamics as role reversals (e.g., parentification), scapegoating (e.g., targeting one child), and changing household rules with changing caregivers, adults, and siblings may lead to very different daily realities, as well as different maltreatment events, when considered in detail. Further, rather than the actual event, it may be the perception of the experience of maltreatment that differs, even when children are similarly maltreated

by the same perpetrator. For example, one child may view himself or herself as a victim, whereas the other embraces spirituality, such as forgiveness of the perpetrator, seeing a greater value in individual suffering, believing he or she has a special relationship with God, and so on. In our clinical experience, one mother identified that God had allowed such substantial abuse when she was a child since he knew she was "strong enough" to "take it" and survive. In research, youth differed in their acknowledgment of exposure to physical abuse even when endorsing physical abuse items with clear injury outcomes (Wekerle et al., 2001). Capturing the maltreatment event, the perception of the maltreatment, and the "theory" or story (or lack thereof) created to understand the maltreatment may be important aspects of a shared environment.

SCREENING FOR CHILD MALTREATMENT

During the 1980s and 1990s, attempts were made to screen individuals to determine risk for committing child maltreatment (Brayden, MacLean, Bonfiglio, & Altemeier, 1993; Caldwell, Bogat, & Davidson, 1988; Center on Child Abuse Prevention Research, 1996; Kotelchuck, 1982; McCurdy, 1995). One study examined the association between staff assessment of a participant's risk for committing abuse based on personal and familial characteristics such as economic difficulties, and scores on the Child Abuse Potential Inventory, a self-report measure (McCurdy, 1995). The predictive validity of this strategy for subsequent abuse, however, was not determined. Two investigations followed families prospectively and measured risk of future maltreatment (Brayden et al., 1993; Center on Child Abuse Prevention Research, 1996). Such risk approaches have shown only limited ability to predict future maltreatment. For example, the interview used by Brayden and colleagues had a positive predictive value of 6.6% and a sensitivity of 55.6% for physical abuse. The main factor that precludes use of child maltreatment screening approaches continues to be the unacceptably high false-positive rate (Caldwell et al., 1988; Kotelchuck, 1982); individuals are identified as potentially at risk for abusing a child and yet do not go on to do so. Certain child, parental, and social factors are important to consider in any assessment of child maltreatment, including evaluations of parents or caregivers to determine their ability to care for a child (often referred to as a parenting capacity assessment).

Table 28.2 outlines physical and other signs linked with an increased risk for occurrence of child maltreatment. Although the presence of these variables increases the risk of maltreatment, it is important to recognize that abuse or neglect can occur without any risk indicators present. At the same time, many individuals and families with one or more of these indicators present are not experiencing child maltreatment. Adverse events for children in the home tend to cluster together (multiple forms of maltreatment, parental alcohol and drug abuse, IPV, parental marital discord, divorce or separation, parental criminality; Dong et al., 2004) and may also be reflected in the community context (e.g., safe, affordable, low crime, high in parenting resource areas).

There is a strong tendency among both researchers and child protection agencies to focus exclusively on mothers when considering issues of child neglect. Yet

some research suggests that the involvement of a father (biological or the mother's partner) protects against abuse and neglect (Biller & Solomon, 1986; Dubowitz et al., 2000; Egeland, Jacobvitz, & Sroufe, 1988; Quinton, Rutter, & Liddle, 1984; Turcotte, Dubeau, Bolté, & Paquette, 2001). Other research, however, found that male partners were rarely a significant source of support to neglecting mothers (Polansky, Chamlers, Buttenwieser, & Williams, 1981). The research suggests that an analysis of the family type leads to a better understanding of the problem of neglect (Mayer, Dufour, Lavergne, Girard, & Trocmé, 2003). This underscores both the challenge and the importance of conducting a careful history with all family members. The assessment of the child in multiple locales (at home during family routine times, such as meals; at school; in service centers) and in multiple contexts (separating and returning to parent, response to strange adults, bonding with teacher and school staff) may be helpful. It is essential that all forms of maltreatment be considered as simultaneous working hypotheses for the clinician in order to capture possible polyvictimization (Finkelhor et al., 2007).

Another issue that has arisen more recently is screening women for exposure to IPV. Citing the prevalence, consequences, and costs associated with IPV (Campbell et al., 2002; Ulrich et al., 2003), some organizations advocate screening as a key aspect of identification and response; the American Medical Association (1992), the Family Violence Prevention Fund, and the American Academy of Pediatrics, among others, have recommended universal screening to their members (Bair-Merritt, Mollen, Yau, & Fein, 2006). Taket and colleagues (2003, p. 674) have asserted that "routine inquiry is the only way to increase the proportion of women who disclose abuse and who may benefit from intervention." These recommendations disregard the fact that no evidence exists regarding the possible harm associated with universal screening.

Those relying on evidence-based approaches favor caution. Systematic reviews by the Canadian Task Force on Preventive Health Care, the U.S. Preventive Services Task Force, and Ramsay and colleagues (MacMillan & Wathen, 2003; Nelson, Nygren, McInerney, & Klein, 2004; Ramsay, Richardson, Carter, Davidson, & Feder, 2002; U.S. Preventive Services Task Force, 2004; Wathen & MacMillan, 2003) have concluded that there is insufficient evidence to recommend universal screening for IPV. Efforts have been made to assist health care providers to use the best available evidence to address the issue of IPV with their clients (Ferris, 2004; Liebschutz, Frayne, & Saxe, 2003; MacMillan & Wathen, 2003a, 2003b; Rhodes & Levinson, 2003). The key question, however, is not whether to screen universally; the issue is the lack of evidence for the effectiveness of interventions for IPV. Identification is only the first step.

ASSESSMENT OF CHILD MALTREATMENT

In most regions in North America, it is the responsibility of child protection agencies to carry out investigative interviews with children and family members when child maltreatment is suspected. Although there is no evidence to support general screening of children for child maltreatment, it is important to include questions about exposure to the four main types of child abuse and neglect (physical, sexual,

and emotional abuse and neglect) in any diagnostic assessment of a child that involves emotional or behavioral problems (some clinicians separate exposure to IPV as a type of maltreatment, while others include it as a form of emotional abuse).

Clinicians must ask clients directly whether they have been abused. A history of victimization form that focuses on the behavioral aspects of maltreatment may be an appropriate prelude to the clinical interview for age-appropriate children. Examples of appropriate forms are the Juvenile Victimization Questionnaire (Hamby & Finkelhor, 2004), Childhood Trauma Questionnaire (Bernstein & Fink, 1998), and Childhood Experience of Violence Questionnaire (Walsh, MacMillan, Trocmé, Dudziuk, & Boyle, 2000). The use of behavioral referents and the avoidance of labels are critical and need to be contextualized (e.g., discipline, relationships). Such paper-and-pencil or computer-administered questionnaires may be helpful because they do not require individuals to self-label and conceptualize their experiences as abuse or neglect. An initial family interview is a good time to explain the goals, perceptions of concerns, expected procedures, and ethical issues (i.e., harm to self, harm to others, child abuse and neglect) of assessment. In maltreating families, it is important to understand who in the family is the person in power so that the clinician can support child empowerment. Going directly to the child may lead to more conflict for the child if the perception is that a child broke an adult's rule. Research shows that most children do not tell.

Children will most typically indicate their maltreatment behaviorally rather than in terms of verbal disclosures. Children will not respond with disclosures readily to the caregiver unless there is a "talking" and warm relationship with a reliable caregiver. For example, in one case, a child did not disclose sexual abuse when directly queried by an older sibling witnessing the child's distress after an incident had occurred with a neighbor. In another example, three male siblings did not disclose bizarre punishment (i.e., hanging a child upside down over an apartment balcony to elicit compliance) by their mother's boyfriend. In this case, the mother herself reported in a family session with the boyfriend present that she was physically abusive to her children (e.g., punching child in the face, knocking out child's teeth). Also present in the family assessment session was the maternal grandmother, who declared a "no tears" rule for this extended family. In all the discussions of violence, no distress was expressed by any family member. Thus, it is important for the clinician to maintain openness to disclosures of violence from a variety of family members and in a variety of presentational forms.

In the event of a disclosure, the maltreatment utterances and clinician questions and responses should be recorded verbatim in writing, in real time, and dated and signed. Clinicians need to be prepared to relay their concerns in terms of observed and reported behavior or signs, the sequential actions leading to the discovery, incongruence and improbabilities in explaining physical signs (see Table 28.2), the presence of risk indicators, as well as the basic facts (alleged perpetrator name, relationship, address, child level of fear of current danger or risk of victimization, knowledge of other victims and/or contact with other children, whether others were told about the abuse or neglect). The person who receives the information directly should act on the report or suspicion forthwith and should not engage the alleged

perpetrator or other caregiver in the report. Such limits to confidentiality should be clearly conveyed prior to clinical intervention.

All clinicians involved in the care of children should be alert to entertaining a hypothesis of maltreatment. A general principle to assist in identifying abuse injuries is when the history is not consistent with the physical findings. Even when the client is an adult or a couple, the best interests of the child remain the first priority. As a helping professional, the clinician has entered into the social contract that makes protection of children a priority over family privacy and unification or adult treatment goals. Discipline-specific codes of ethics and legal statutes exist to ensure child health and safety. The duty to report overrides most privileged information, including with police and religious authorities. Beyond familiarity with applicable laws, clinicians may consult local CPS to discuss the situation and obtain guidance anonymously. Abuse or neglect does not have to be proven for a report to be made; "reasonable" or "probable" grounds for suspicion are required. With the best interests of the child in mind, reports are made in "good faith," and malicious reporting is rare. Failure to report carries with it penalties such as fines, disciplinary action from the professional licensing body, civil suits, and prosecution by authorities. For further details, see texts by Wekerle et al. (2006), Reese & Ludwig (2001), and an article by Sirotnak, Grigsby, and Krugman (2004).

A helpful approach to understanding the context of symptomatology is to query caregivers and youths about a typical day. This allows the clinician to understand who does what generally and how the day proceeds, including levels of child self-care and sibling responsibilities, eating times, hygiene, affection displays, and discipline routines. It may be helpful to compare the child's perspective with that of siblings and caregivers. Examples of broad questions that provide opportunities to ask about child maltreatment are: What things do you worry about? What happens in your family when someone gets in trouble? Has anyone ever touched you in a way that made you feel uncomfortable or upset? Has anyone ever made you do something sexually you did not want to do? It is generally not appropriate to ask detailed questions about a disclosure of maltreatment because in many jurisdictions, a child protection agency worker needs to obtain the disclosure. Children should not be subjected to investigative interviews by clinicians without specialized training in this area. Maltreatment victims hope for an empathic response, reassuring them that telling was the right thing to do, allowing them to express their feelings and thoughts about the disclosure; the clinician should convey positive messages but not promises that cannot be kept. Helpful messages include conveying that the children did not do anything to provoke or deserve their maltreatment, that abuse and neglect are against the law, that you are sorry that they have had to go through this or suffer so, that they are not alone, and that other children have experienced it. The child should be monitored and supported postdisclosure with a known professional (i.e., CPS staff, family doctor). In cases where abuse is suspected but not substantiated by CPS investigation, the risk of subsequent reports and substantiation is high. As with many clinical phenomena, a good predictor of future behavior is past behavior.

TREATMENT AND PREVENTION OF CHILD MALTREATMENT

There are several promising and effective interventions on different aspects of maltreatment effects and violence prevention (for a review of the evidence base and available interventions, see J. Cohen et al., 2006). Recent infant-parent psychotherapy reports indicate success in altering the insecure attachment status of infants (about 66% were abused and/or neglected in the first year of life), as compared to low-income normative controls. There were substantial increases in secure attachment, and disorganized attachment was prevalent among the treatment-as-usual community maltreated infants (Cicchetti, Rogosch, & Toth, 2006). Certain interventions appear to hold promise for neglected children and families: Group play training and resilient peer-mediated treatment for withdrawn preschoolers may improve social behavior; therapeutic day treatment may improve self-esteem; and multisystemic therapy and parent training may improve parent-child relations, reduce parenting problems, and decrease family social problems (National Clearinghouse on Family Violence, 2006).

Currently, the two most researched and supported interventions are home visitation models (HVM) for younger children and trauma-focused cognitive-behavioral therapy (TF-CBT) for preschool and school-age children and adolescents. Research studies are under way to assess effectiveness in community settings, including child protection populations. Further research is needed in terms of entry into these interventions. Both HVM and TF-CBT are very promising approaches to support child protection services work in the area of referring families whom CPS does not serve but who are reported to the system, if families are interested and deemed appropriate. For example, a dating violence prevention program, the Youth Relationships Project, was found to significantly reduce the self-reported trauma symptomatology and dating violence among CPS and at-risk youth over a 2-year postintervention, as compared to youth receiving typical agency service (Wolfe et al., 1996).

Home Visitation Models

The prevalence and consequences of maltreatment have prompted many efforts in primary prevention in the past 2 decades. The U.S. Advisory Board on Child Abuse and Neglect identified child maltreatment as a national emergency in 1991 and promoted an expansion of Hawaii Healthy Start, a program of paraprofessional home visitation that had shown some evidence of effectiveness in nonrandomized trials (Duggan et al., 1999). Healthy Families America (Daro & Harding, 1999) was modeled after Hawaii Healthy Start and disseminated nationally. Currently, Health Families America programs are running in 430 locations in the United States and Canada.

In the past 10 years, randomized controlled trials have been conducted on Hawaii Healthy Start and Healthy Families America in New York State, San Diego, and Alaska. The trials have found significant effects on some domains of self-reported parenting behaviors (including abusive behaviors) and beliefs. The Alaskan study showed positive effects on some aspects of the child's environment, child

development, and parent-reported child behavior at age 2, but the San Diego trial showed that the effect on child development was not sustained. The programs showed no effect on official reports of child abuse or neglect, childhood injuries, or other objective measures that are proxies for maltreatment (Olds, 2006). In 2004, the Washington State Institute for Public Policy estimated that Healthy Families America has incurred a $4,500 (U.S.) per family loss in investment.

The 1991 home visitation recommendations of the Advisory Board were based on evidence from research on the Nurse-Family Partnership Program (NFP; Olds, Henderson, Chamberlin, & Tatelbaum, 1986), a substantively different program from that offered by Hawaii Healthy Start and Healthy Families America. The NFP is a program of prenatal and infancy home visiting by nurses (not paraprofessionals) for low-income mothers having their first child. Nurses follow detailed visit-by-visit guidelines in their efforts to (a) help women improve the outcomes of pregnancy by promoting healthy prenatal behaviors, (b) improve child health and development by promoting parents' competent care of their children, and (c) enhance parents' life-course development by encouraging pregnancy planning and parents' education and work. The nurses help families make use of needed health and human services and attempt to involve other family members and friends in the pregnancy, birth, and early care of the child. The NFP has a schedule of visits beginning before the end of the second trimester in pregnancy, as follows: weekly for the first four visits, every 2 weeks for the remainder of the pregnancy, every week for 6 weeks in the postpartum period, every 2 weeks until the infant is 21 months old, then once a month for the next 3 months until the child is 2 years old.

The NFP has been tested in three randomized controlled trials over 3 decades. The first trial was conducted in Elmira, New York ($N = 400$, 89% White). The second, conducted in Memphis, Tennessee ($N = 1139$, 92% Black), was designed to determine if the results from Elmira could be replicated when a health department delivered the NFP and when it served low-income Blacks in an urban area (Kitzman, Cole, Yoos, & Olds, 1997). The Denver trial ($N = 735$, 45% Hispanic) was designed to determine whether the NFP might be delivered effectively by paraprofessionals who shared many of the characteristics of the families they visited (Olds et al., 2002). The Denver trial also included a randomly assigned group visited by nurses and a control group. At randomization and in subsequent assessments, treatment and control groups were essentially equivalent in terms of demographics, and high rates of sample retention were maintained in each trial.

In each of the trials, women were randomized to receive either home visiting during pregnancy and in the first 2 years of their children's lives or comparison services that consisted of some combination of free transportation for prenatal and well child care and sensory and developmental screening and referral of the child for potential problems. All three studies employed a variety of data sources, including maternal interviews, developmental tests of the child, and medical and administrative data. The randomized controlled trials fell somewhere between efficacy and effectiveness trials, in that the samples had few restrictions, nearly entire populations of low-income first-time mothers were registered in the communities where the trials were conducted, and the interventions were conducted in collaboration with local agencies (Olds, 2002).

In Elmira, the intervention reduced rates of state-verified reports of child abuse and neglect. A 15-year follow-up showed that this effect was increased: the nurse-visited group had 48% fewer verified reports of maltreatment than did the control group, and among mothers who were at higher risk, 77% fewer verified reports compared to control group counterparts (Olds et al., 1997). However, rates of maltreatment found in official records in Memphis and Denver were too low in the first 2 years of children's lives to be reliable outcomes. Rates in Elmira were about 3 times higher than in Memphis and Denver, and 6 to 7 times greater for high-risk families. Child protection service records typically pick up only a fraction of actual maltreatment; therefore, the results in Elmira may be an underestimate of the effect. As well, a surveillance bias could have been operating in the study; maltreatment could be detected at a greater rate in nurse-visited families because the nurses are mandated reporters.

Because of the low rates of substantiated reports of maltreatment in Memphis, the research used childhood injuries reported in the medical record as a proxy (Kitzman et al., 1997). Program effects on injuries were similar to the pattern in Elmira; program effects were enhanced among children born to mothers with low psychological resources. In Denver, the health care delivery system made it impossible to trace children's utilization patterns reliably, so infants' early emotional expressions (which are associated with child abuse and neglect) were measured. Nurse-visited 6-month-olds born to mothers with low psychological resources had higher rates of observed emotional vitality and lower rates of emotional vulnerability than their control group counterparts. As the children matured, the program produced effects on language and executive functioning consistent with better prenatal and caregivng environments.

The Elmira follow-up showed that the program effect on child maltreatment decreased as the level of IPV increased, although the program had no impact on the incidence of IPV in this trial. The moderating effect of IPV was specific to child maltreatment and did not attenuate program effects on any other outcomes (Eckenrode et al., 2000).

It is noteworthy that nurse-visited women in the Denver trial reported significantly less IPV from partners during the 6-month interval prior to the 4-year interview than those in the control group (6.9% versus 13.6%, $p = .05$; Olds et al., 2004). Future research should be focused on this interplay between IPV and child maltreatment and efforts made to enhance IPV-effective components for these families.

The best evidence for prevention of physical abuse and neglect in high-risk groups comes from the nurse home visitation program developed by Olds and colleagues in 2007. MacMillan and colleagues examined the effectiveness of an intensive program of nurse home visitation in preventing recidivism of physical abuse and neglect among CPS-involved families referred because of at least one child in the family experiencing either physical abuse or neglect. Families were randomly assigned to a control group, which received CPS standard service such as risk assessment, parenting education, or referral, or the intervention group, which was provided standard CPS service plus a 2-year program of nurse home visitation. Both the control and the intervention group were followed up annually

over 3 years. Results showed no statistically significant reduction in recurrence of either child physical abuse or neglect; the recidivism rate for both intervention and control groups was high. The only promising finding was that among those families whose involvement with child protection lasted less than 3 months, there was a reduction in recurrence of physical abuse but not neglect. This suggests the possibility that early intervention with families involved with CPS might lead to less recurrence of child physical abuse. While the NFP has been shown to be effective in preventing maltreatment and associated outcomes, the trials by Olds and colleagues and MacMillan et al. (2005) have substantial differences. The NFP (Olds et al., 2007) targeted high-risk first-time mothers and began prenatally, whereas the study by MacMillan and colleagues (2005) involved a much broader range of families. In the NFP, the nurses assisted mothers prenatally, and then after the birth of the infant worked closely with mothers in helping them understand the needs of their infants. Although the original rationale for assessing a program of nurse home visitation with a sample of families involved with the CPS was based on the positive findings of the NFP, it was structured very differently from the NFP. The trial was designed to evaluate an intervention that paralleled the in-home services being offered by CPS in general, but at a more intensive level than was available in the community. The negative findings, though disappointing, emphasize that randomized, controlled clinical trials in the CPS population should be a priority for research. Disseminating findings from strong methodological studies should be a priority for CPS practice and policy, regardless of whether the outcome was positive or negative. Further, there remains an urgent need for rigorous evaluation of the effectiveness of parenting programs that are specifically designed to treat physical abuse and neglect, either independently or as part of broader packages of care. Studies need to measure outcome broadly to assess effects on child abuse and neglect per the legal definitions (CPS reports) as well as objective measures of parenting (e.g., home behavioral observations). A recent systematic review indicates treatment effectiveness of some programs on associated aspects of parenting; currently, the evidence on reduced maltreatment rates is limited for treatment compared to prevention with at-risk samples (Barlow, Johnston, Kendrick, Polnay, & Stewart-Brown, 2006).

Clearly, the effects seen in the NFP were not observed in the paraprofessional visiting model of Hawaii Healthy Start and Healthy Families America. The Denver trial of the NFP showed that nurses produce a larger and broader range of beneficial effects in, among others, infant caregivng and language development versus paraprofessionals. However, a one-size-fits-all model is not effective. The NFP model showed substantial limitations when interparental violence was present; the positive findings in reduction of child maltreatment were not maintained under such conditions (Eckenrode et al., 2000). This may be the best option for high-risk, young mothers prenatally or beginning their parenting career. Program implementation and fidelity are critical. In 1996, the U.S. Justice Department recommended replication of the NFP in community settings. This led to the creation of the NFP National Service, which develops and maintains procedures for ensuring faithful reproduction of the program's essential elements as it is replicated in new communities. The NFP is currently operating in over 170 sites, serving 20,000 families per

year in 250 counties in the United States. The Washington State Institute and the RAND Corporation estimate that the NFP saves $17,000 (U.S.) for every family served.

Trauma-Focused Cognitive Behavior Therapy

When the diagnosis of PTSD was first formulated in 1980, it was not believed to be relevant to children and adolescents. Child stress reactions were considered short-lived (Masten, Best, & Garmezy, 1990). The empirical basis for such views, however, consisted mostly of reports from parents and teachers, who may underreport child distress (Handford, Mayes, Mattison, & Humphrey, 1986). As the assessment moved to asking children and adolescents themselves about their thoughts, feelings, memories, and emotions following trauma, debilitating and long-lasting PTSD symptoms were noted (Dalgleish, Meiser-Stedman, & Smith, 2005). Pediatric PTSD symptomatology is consistent with children behaviorally enacting their emotional reaction to trauma (Dyregrov & Yule, 2006). For example, it was observed that young children show less emotional numbing and have more problems in reporting avoidance; they require rather complex cognitive introspection.

Consequently, the need for a thorough assessment following exposure to trauma is crucial to the implementation of age-appropriate and developmentally sensitive interventions to capture the intense feelings of shock, threat, terror, horror, or helplessness. In a recent review (Strand, Sarmiento, & Pasquale, 2005), assessment and screening tools currently available to researchers and practitioners were divided into four categories in terms of how trauma is constructed: those that screen for (1) both a history of exposure to traumatic events and the presence of symptoms of trauma, (2) only a history of exposure, (3) symptoms of PTSD or dissociation, and (4) symptoms other than or in addition to PTSD. Alternately, Perrin, Smith, and Yule (2000) categorize these instruments into four different modes: (1) *DSM*-criteria-based semi-structured interview, (2) parental interview, (3) unstructured child interview, and (4) child self-report. In evaluating these instruments, Strand et al. argued that most instruments are not designed to be age-specific, raising questions about the influence of age on outcome. Specifically, only 8 of the 33 measures available are designed for children 10 years or younger, and fewer of these are geared for the very young child. In addition, Perrin et al. point out that since many of the *DSM* PTSD criteria are not developmentally sensitive, existing semi-structured interviews may have only limited usefulness for young children. Further, because children are better reporters of internalizing symptoms and parents are more accurate reporters of externalizing or behavioral symptoms (Greenwald & Rubin, 1999), only instruments that collect information from both sources can provide a balanced account. Very few self-report measures were developed specifically to assess child PTSD. Instead, the majority of these instruments are adapted from adult questionnaires with little normative child data available (see McNally, 1991; Nader, 1997). Clearly, there is a need to develop instruments that measure more developmentally sensitive child PTSD.

The treatment, however, has centered on Trauma-Focused Cognitive Behavior Therapy (TF-CBT), which can be used to target trauma deriving from all types

of maltreatment, although it was initially developed for sexually abused children (Berliner, 2005; Cohen & Mannarino, 1996, 1998; Cohen, Mannarino, & Deblinger, 2006; Cohen, Mannarino, & Knudsen, 2005; Deblinger, Stauffer, & Steer, 2001; Stovall-McClough & Dozier, 2004). A systematic review of outcome studies on sexually abused children concludes that some outcomes have been more successfully demonstrated than others (e.g., decreases in child depression, improvements in parenting practices), and some outcomes have yet to be assessed. Overall, more controlled outcomes studies on various populations are encouraged (Macdonald, Higgins, & Ramchandani, 2006).

Treatment proceeds along 12 to 16 weekly sessions. In addition to the recently published manual (Cohen et al., 2006), free Web-based training is available (www.musc.edu/tfcbt/). TF-CBT has demonstrated efficacy for multiply traumatized children, maintaining treatment gains at 1-year follow-up (Cohen, Deblinger, Mannarino, & Steer, 2004; Deblinger, Mannarino, Cohen, & Steer, 2006). While not direct targets of treatment, parental PTSD and depression symptoms have been reduced with TF-CBT, further enhancing the likelihood of adaptive parental support. As is the case with most protocols, where additional intervention is deemed necessary, referral is provided. This is seen as particularly important when parental psychiatric issues are interfering with parenting capacity.

A key issue with TF-CBT is to engage a nonoffending caregiver as an active agent of change, providing the daily continuity needed to support the child's new learning (relaxation, leisure skills) and ability to talk openly about the maltreatment (emotions vocabulary, assertive communication) and to give the child confidence in the caregiver as a protector (caregiver affect management, parent training for child behavior management). Further, relaxation skills can be widely applied by the child and supported by the parent to address sleep, hyperarousal, and somatic issues. For the child, graduated exposure to the traumatizing maltreatment, addressing the phobic response to trauma memories and cues, is addressed in the creation of a narrative. This achieves the goal of decoupling maltreatment cues from debilitating anxiety. The child's story of the maltreatment can be in pictorial, audio, written, or typed form and is developed over time and at the child's pace. The practice of the cognitive components, addressing cognitive errors (e.g., dwelling on the worst possibilities, self-blame for maltreatment) via thought-stopping, positive self-talk, and practice applying the cognitive triangle (linking feeling, thoughts, behaviors), is achieved in therapist-child discussions during rereading and reworking of the narrative in a manner consistent with the CBT gains. Opportunities for corrective relationship experiences are available in individual sessions with the therapist and ongoing supportive work with respect to parenting. In this way, TF-CBT addresses the common relationship issues in maltreatment: impaired trust, social incompetence, poor relationship problem-solving skills, problems in sustaining relationships, and hypersensitivity in interactions. Another key maltreatment issue, affect regulation, is addressed via skill development in self-soothing, distress tolerance, and managing negative affect (sadness, fear, anger, anxiety). Risk behaviors (sexualized, aggressive, unsafe) are addressed in skill-building components.

The flow of TF-CBT sessions moves from assessment to assessment feedback, parallel child and adult individual sessions, and conjoint child-parent sessions.

During assessment, the central aspect of TF-CBT (i.e., the trauma narrative) as a skill is evaluated and practiced in a nonthreatening context. Children share their feelings, thoughts, and bodily sensations about a positive experience or favorite activity. The TF-CBT components form the acronym PRACTICE, reflecting the specific session goals across time: Psychoeducation and parenting skills, Relaxation, Affective modulation, Cognitive coping and processing, Trauma narrative, In vivo mastery of trauma reminders, Conjoint parent-child sessions, Enhancing future safety and development. Although caregiver involvement is optimal, significant PTSD symptom improvement has been documented with child-only work. Clinicians are advised to utilize their clinical judgment in determining the initial focus of treatment, which may need to be crisis management, reinstitution of the child's routine, and child behavior stabilization. Clinicians should assess for fragile states, as creating the trauma narrative could worsen any suicidal ideation or substance abuse. An alternative treatment strategy based on dialectical behavior therapy (Linehan, 1993) that targets suicidal risk and other multiproblem situations, particularly for adolescents, developed by Miller, Rathus, and Linehan (2007), is described in Wekerle et al. (2006).

SUMMARY

Child maltreatment is a serious challenge to adaptation and development across domains and across the life span. Children who experience multiple and/or chronic exposure to maltreatment events adapt less well to normative challenges in life. When maltreatment effects are evident, development is compromised as misses in early skills weaken the foundation for more complex skill attainment. In particular, maltreatment appears to carry an additional cognitive load for the individual such that academic performance is compromised, information integration is slower, affective splitting is more likely, emotion-laden and self-referential communication is more difficult, and negative biases in emotionality, attributions, and behavior (harm to self, harm to others) are more likely. Particular disorder vulnerabilities occur in the domains of mood, anxiety, substance abuse, aggression, eating, and personality. Further, survival is challenged with suicidal ideation and attempts, and suicide is a real possibility. Relationship functioning and self-preservation are consistent themes, with greater insecurities and disorganization in attachment, more problematic peer relationships, and greater likelihood of dating and intimate partner violence. Consistent with theoretical formulations, PTSD symptomatology appears to be central, at least for female victims. TF-CBT is a first line of treatment for those with impairing levels of symptomatology. The bulk of the data on maltreatment, though, continues to urge us toward violence prevention and early intervention. The NFP for first-time socially disadvantaged mothers shows the best evidence for prevention of maltreatment and associated outcomes such as injuries. However, once the line of abuse and official system involvement has been crossed, developing and testing approaches for parents continue to be pressing research questions.

Abuse and neglect of children are entirely unnecessary adult actions that can leave much damage to address in their wake. Even so, most maltreated individuals do not

go on to maltreat their partners and children. Many maltreated individuals achieve high functionality in some domains, but may continue to struggle in others (e.g., romantic relationships). Given the social contract that children's human rights be ensured and protected, research must continue to be productive in understanding the full experience of maltreatment for the victim, support the broad uptake of effective programs while continuing to rigorously test them in their field settings, and thank those maltreated individuals and their families for participating in research to intervene, treat, and, most of all, prevent abuse and neglect.

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CHAPTER 29

Autism Spectrum Disorders and Related Developmental Disabilities

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This chapter addresses the pervasive developmental disorders (PDD) included in the *Diagnostic and Statistical Manual of Mental Disorders*, fourth edition, text revision (*DSM-IV-TR;* American Psychiatric Association, 2000). These disorders include Autistic Disorder (autism), Asperger's Disorder, Pervasive Developmental Disorder—Not Otherwise Specified (PDD-NOS), Rett's Disorder, and Childhood Disintegrative Disorder. Given the sheer amount of information currently available on all of the PDDs, we highlight aspects of the disorders that are particularly pertinent to clinical psychologists.

This chapter first addresses terminology related to the PDDs. In particular, the increased attention to specific PDDs by diverse groups, such as consumers, educators, legislators, and basic researchers, has produced an imprecise umbrella term, autism spectrum disorder (ASD). The lack of clarity in terminology has produced, and continues to produce, significant confusion and miscommunication. Second, we address the three factors common across Autistic Disorder, Asperger's Disorder, and PDD-NOS: social impairment, language and communication impairments, and restricted, repetitive, and/or stereotyped behaviors. This discussion is important because these three areas vary greatly within and across each of these disorders. Issues related to assessment and diagnosis are discussed next. This section introduces the reader to commonly used assessments as well as the issues surrounding diagnostic evaluations for individuals with an ASD. A discussion of intellectual functioning and mental retardation follows. Comorbidity, trends in prevalence, early intervention, and etiology of ASD are covered next. Rett's Disorder and Childhood Disintegrative Disorder are discussed afterward, as these disorders are rare in occurrence and have distinct differences from the other three disorders. Next, treatment for the core symptoms of the PDDs is discussed. A brief section on family issues is last.

TERMINOLOGY

A review of the terminology over the past decade shows that the term pervasive developmental disorder as defined by the *DSM-IV-TR* (American Psychiatric Association, 2000) is often used synonymously with autism spectrum disorder.

Unfortunately, many studies applying either term to describe participants, samples, or population fail to specifically denote which disorders, if any, are included in these categories. Moreover, there is no standard operational definition of ASD, as the term is not in the *DSM-IV-TR,* nor have other attempts to adequately and consistently define this term been made. For consistency purposes, we suggest conceptualizing ASD as a spectrum including Autistic Disorder, Asperger's Disorder, and PDD-NOS. When referring to ASD, Rett's Disorder and Childhood Disintegrative Disorder are not included, mostly due to the low occurrence of each disorder and their specific course.

An additional problem is the use of descriptive labels for presumed subcategories of Autistic Disorder that also have yet to be adequately defined (Volkmar & Klin, 2005). For example, the term "high-functioning autism" is widely used yet has no reference in the *DSM-IV-TR* (American Psychiatric Association, 2000). The term typically refers to individuals with a diagnosis of Autistic Disorder who have adequate verbal language abilities and an average or above average level of cognitive ability (Baron-Cohen, O'Riordan, Stone, Jones, & Plaisted, 1999; Escalante-Mead, Minshew, & Sweeney, 2003; Klin & Jones, 2006). Individuals described as having high-functioning autism present clinically similar to individuals with Asperger's Disorder. However, researchers debate whether there are meaningful differences between individuals diagnosed with Asperger's Disorder and those described as having high-functioning autism (Lincoln, Courchesne, Kilman, Elmasian, & Allen, 1998; Miller & Ozonoff, 2000). This is an important area that has begun to receive more attention, as it might have implications for the study of etiology, response to treatment, type of treatment recommended, and prognosis. Using the term high-functioning autism as a specifier that is not always consistently defined creates potential diagnostic overlap. This can lead to problems when conducting research and when attempting to diagnose an individual with a PDD using the current *DSM* classification system.

Clearly, more precise definitions for Autistic Disorder, Asperger's Disorder, and PDD-NOS are needed. An examination of the *DSM-IV-TR* (American Psychiatric Association, 2000) diagnostic criteria for the PDDs might be helpful in understanding how similar these disorders are to each other. Table 29.1 lists the major diagnostic criteria in the areas of social skills, communication, and behavior for each PDD. Notice that the social skills and behavior criteria are the same for both Autistic Disorder and Asperger's Disorder. Under the PDD-NOS category, the criteria define a less severe impairment than in Autistic Disorder and Asperger's Disorder for social and behavior areas. Given the overlap of symptoms and characteristics of these disorders, the next sections describe the main impairments (social, communication, and behavioral) in turn.

As indicated in Table 29.1, describing the characteristics in the social, communication, and behavioral domains of Autistic Disorder, Asperger's Disorder, and PDD-NOS involves discussion of many similar and overlapping characteristics. We describe these characteristics under the term ASD, given the current trend in the literature. However, characteristics and symptoms specific to a particular disorder are noted.

Table 29.1 Major *DSM-IV TR* Criteria for Pervasive Developmental Disorders

Impairments	Autistic Disorder	Asperger's Disorder	PDD-NOS	Rett's Disorder	Childhood Disintegrative Disorder
Social	Qualitative impairment in social interaction (at least two specific impairments needed)	Same as Autistic Disorder	"A severe and pervasive impairment in development of reciprocal social interaction" (p. 84)	Loss of social engagement early in the course (although often social interaction develops later)	Clinically significant loss in social skills; abnormalities of functioning in qualitative impairments in social interaction
Communication	Qualitative impairment in communication (at least one specific impairment needed)	No clinically significant general delay in language	Impairment in verbal or nonverbal communication skills	Severely impaired expressive and receptive language development (after a period of normal development)	Clinically significant loss in expressive/receptive language; abnormalities of functioning in qualitative impairments in communication
Behavior	Restricted, repetitive, and stereotyped patterns of behavior, interests, and activities (at least one specific impairment needed)	Same as Autistic Disorder	Presence of stereotyped behavior, interests, and activities	Loss of previously acquired purposeful hand skills between 5 to 30 months of age with subsequent development of stereotyped hand movements	Clinically significant loss in adaptive behavior/motor skills; abnormalities of functioning in restricted, repetitive, and stereotyped patterns of behavior, interests, and activities, including motor stereotypes and mannerisms

Adapted from *Diagnostic and Statistical Manual of Mental Disorders*, Fourth Edition, Revised, by the American Psychiatric Association, 2000, Washington DC: Author.

SOCIAL IMPAIRMENT

The impairment in social skills, social interactions, social reciprocity, and social competence in individuals with Autistic Disorder, Asperger's Disorder, or PDD-NOS is central to the diagnosis. In this section we describe the specific impairments associated with each of the disorders in turn.

Autistic Disorder

Leo Kanner (1971) identified social impairment as the primary hallmark of autism (his term was "early infantile autism"). That is, the children's "inability to relate themselves in the ordinary way to people and situations from the beginning of life" (p. 140). Indeed, developmental research supports this claim. For example, infants show preferential attention toward social stimuli (i.e., faces) over other types of stimuli within the first few months of life (Sigman, Dijamco, Gratier, & Rozga, 2004), with initial social development occurring between the infant and caregiver. As children grow, social behavior begins to become increasingly differentiated; children develop peer relationships, pro-social skills, and an increasing capacity for self-regulation (Singer, 1996). In children with Autistic Disorder, however, retrospective studies suggest that aberrant social development starts within the first few months after birth (Goin & Myers, 2004; Osterling, Dawson, & Munson, 2002).

One profound social impairment in individuals with Autistic Disorder is a lack of eye contact. Unlike children with mental retardation or other developmental disabilities (but not a PDD), children with Autistic Disorder may not engage in consistent, predictable eye contact with parents, caretakers, and others (Carter, Davis, Klin, & Volkmar, 2005). Similarly, individuals with Autistic Disorder do not readily initiate other social behavior involving eye contact, such as joint attention, which in typically developing children develops in infancy (Mundy & Burnette, 2005).

Social impairment is not only considered to be the core deficit, but is also complex, as it varies with chronological age and developmental level. Considered a primary deficit, social impairment has been implicated in the aberrant development of other areas, such as language abilities, emotion regulation, and behavioral patterns of rigidity, stereotypies, or self-stimulatory behavior (Bachevalier & Merjanian, 1994; Happe & Frith, 1995). For example, some children have difficulty using nonverbal cues to modulate their social interaction (e.g., continuing a conversation when the listener is looking at his or her watch repeatedly and yawning; Lord, 1990).

Typically, social competence is markedly below the cognitive ability of an individual with Autistic Disorder (and more broadly ASD). For example, a 7-year-old child with Autistic Disorder with cognitive ability in the mental retardation range may neither display eye contact nor respond to the greeting of another person, whereas a same-age child with Autistic Disorder but of average intelligence may display partial eye contact and may have a limited repertoire in responding to a greeting. Both children show deficits in social interaction and social skills when compared to their typically developing peers.

Social skills vary by age as well. By adulthood individuals may develop some social skills that are typical in childhood, while other social skills important in

adulthood remain deficient (e.g., relating to others on an intimate level). In addition, some individuals will have more diverse and advanced social abilities than others.

Play skills are also impaired (Kasari, Freeman, & Paparella, 2006). By the end of 2 years of age, most typically developing children demonstrate functional and symbolic play skills. The play skills of children with Autistic Disorder are severely diminished compared to typically developing children. Children with Autistic Disorder may seek out play items but use them in a self-stimulatory manner. Most children with Autistic Disorder show difficulty using a toy or object in a more social, playful manner (Thomasa & Smith, 2004). A preference to play alone and with only a few specific toys is usually observed, whereas the typically developing child would more likely prefer to play with others, explore a variety of toys, and engage in other social interactions with his or her environment.

Deficits in expressive social communication (e.g., gestures) and engaging in eye contact, joint attention, and social reciprocity (Lord, 1990) are observed. It is also the case that abilities to initiate social behavior and understand the rules governing social interaction are impaired (Wing, 1981). Further, individuals with Autistic Disorder are typically not responsive when others are in distress (Travis, Sigman, & Ruskin, 2001). There may also be difficulty with perception of affective content (Hobson & Lee, 1989). One study showed that adolescents with Asperger's Disorder are significantly impaired in their ability to detect the emotional content of videotaped scenes that lack speech content cues as compared to typically developing adolescents (Koning & Magill-Evans, 2001). A series of studies using the Vineland Adaptive Behavior Scales, which is a norm-referenced assessment, found that socialization deficits are greater than expected based on overall developmental level (Sparrow, Balla, & Cichetti, 1984; Volkmar et al., 1987). Additionally, these studies demonstrated that children with Autistic Disorder exhibit significantly lower socialization scores than children with mental retardation.

Decoding of emotions from facial expressions is also impaired relative to typically developing children. It appears that individuals with ASD do not rely on facial expressions for emotional information. In addition, they may have difficulty relying on other situational cues to determine a person's emotional state (Egan, Brown, Goonan, Goonan, & Celano, 1998).

Asperger's Disorder

Individuals with Asperger's Disorder may exhibit all or many of the social impairments just described. The socioemotional and social communicative deficits in Asperger's Disorder are primary deficits. Individuals with Asperger's Disorder usually have significant difficulties identifying ("reading") and understanding social cues. It appears that individuals with Asperger's can learn cues and facial expressions to correctly identify emotions, but do not show generalization of these skills to spontaneous use (Grossman, Klin, Carter, & Volkmar, 2000; Serra, Minderaa, van Geert, & Jackson, 1999). Egan et al. (1998) found that individuals with ASD were unable to use prosody (tone of voice) and facial expression to identify emotions. A more recent study by Lidner and Rosén (2006) supported these results. These authors found that children with Asperger's Disorder are able to utilize verbal content

to assist them with identifying emotions. Grossman et al. found that this is not the case for individuals with Autistic Disorder; individuals with Autistic Disorder are unable to compensate using verbal ability to identify emotions.

Individuals with Asperger's Disorder typically desire friendships and want to make connections with others. However, they lack social skills to do so in a meaningful manner. Because individuals with Asperger's Disorder are intellectually typical with regard to their cognitive and verbal language abilities, they are aware that they are "socially different" from others. These individuals, therefore, are at a higher risk for developing depression or anxiety as a result, especially in adolescence and adulthood. Because of the compromised ability of individuals with Asperger's Disorder to understand and interpret social behavior appropriately, forming sustainable relationships is a challenge

Pervasive Developmental Disorder—Not Otherwise Specified

Individuals with PDD-NOS share many of the characteristics of both Autistic Disorder and Asperger's Disorder. However, the severity of this impairment may not be to the same degree. Thus, a separate section that covers PDD-NOS is redundant, even though this is not a reflection of a less important diagnosis.

COMMUNICATION

Children with Autistic Disorder either develop verbal language later than typically developing children or do not develop functional language at all. Fifty percent of children with Autistic Disorder do not develop spoken communication (Sigman & Capps, 1997). For individuals who do develop spoken language, abnormal features are typically present (Sheinkopf, Mundy, Oller, & Steffens, 2000; Sigman & Capps, 1997). For example, Sheinkopf et al. noted that preschool children with Autistic Disorder showed a rate of verbalization similar to normally developing children for their developmental level. However, the types of verbalizations and vocal quality were noticeably abnormal (e.g., growling, squeals).

Echolalia is a characteristic of speech patterns in approximately 75% of individuals with Autistic Disorder (Prizant, 1983). Echolalia is defined as the repetition of the speech of another person or oneself and is often present in typically developing infants. Echolalia can be immediate (i.e., repeating a word, phrase, or sentence immediately after it is spoken by someone else), delayed (i.e., repeating speech heard in the past), or mitigated (e.g., imperfectly repeating speech heard in the past). Echolalia is not unique to Autistic Disorder, as it is an abnormal speech quality observed in other psychiatric disorders and sometimes as a result of brain damage (Sigman & Capps, 1997). There is some question in the field as to whether there is a communicative function of echolalia for individuals with ASD (Tager-Flusberg, Paul, & Lord, 2005). Some researchers suggest that children with ASD use echolalia to engage in a conversation because they do not understand how to use speech to communicate (Loveland & Tunali-Kotoski, 2005).

Prosody is impaired in individuals with ASD. Prosody is the rhythm of spoken language; it helps the speaker add specific meaning to a phrase or sentence. For

example, when asking a question, most individuals raise their voice slightly at the end of the sentence. This helps the conversational partner understand that the intent of the sentence is to ask a question. Individuals with ASD typically have abnormal intonation and use of rhythm and stress in speech production. Other characteristic speech abnormalities are pronoun reversals, articulation difficulties, and idiosyncratic use of speech, including the development and use of neologisms (Paul, Augustyn, Klin, & Volkmar, 2005). By age 5 or 6, most children with Autistic Disorder still show a delay in spoken language. If speech is present by this age, it is typically an indicator of better prognosis for the child (Tager-Flusberg et al., 2005).

In typically developing children who have not acquired speech, for example, due to a specific speech delay, the use of nonverbal communication is usually intact. A child who is deaf, for instance, is able to acquire nonverbal forms of communication (e.g., including sign language or other communication methods) and will compensate through the use of physical gestures, eye contact, pointing, and body language to communicate with others. For individuals with Autistic Disorder, these areas of nonverbal communication are either absent or noticeably impaired (Buffington, Krantz, McClannahan, & Poulson, 1998; Kasari, Sigman, Mundy, & Yirmiya, 1990; Tager-Flusberg et al., 2005).

Communicative intent is the planned purpose of a specific communication interaction. There are many purposes or goals for communicating with others; for instance, communicating that one is feeling sad, hungry, or that one would like to read a book located in an unreachable location, are all different yet important reasons to communicate to another person. Communicating with another person is categorized as either declarative (e.g., asking a friend what he received for his birthday) or regulatory (e.g., asking a friend to pass a piece of birthday cake; Landa, 2005; also known in behavior analysis terms as tact and mand, respectively). One of the significant characteristics of communicative intent in individuals with Autistic Disorder is that the intent is typically toward gaining access to an object or activity, not an interaction or conversation with a person. Thus children will use communicative gestures to gain access to an object, but not use such gestures in a social way.

There has recently been much attention devoted to joint attention deficits in ASD. Joint attention is a form of communicative intent that typically develops in infancy. Joint attention involves gestures such as pointing and use of eye contact that serve to focus another person's attention on an object or event for the purpose of shared enjoyment (Landa, 2005). Joint attention begins to emerge between 9 and 18 months (Bakeman & Adamson, 1984). Deficits in joint attention in children with ASD are noticed at an early age and have been well documented in the literature. These deficits include failure to orient to social stimuli, speak sounds, follow eye gaze, and follow pointing in conjunction with eye gaze (Leekam, Hunnisett, & Moore, 1998; Mundy, Sigman, & Kasari, 1994). Children with ASD also exhibit deficits in the initiation of joint attention, including a lack of showing objects of interest and failure to use declarative pointing and referential looking (MacDonald et al., 2006). Deficits in responding to joint attention appear to be less severe than the initiation to joint attention (MacDonald et al., 2006).

Interestingly, it does not appear that joint attention skills are delayed per se, as they sometimes fail to develop after more advanced social skills, such as nonverbal

pointing to request an item (Mundy et al., 1994). MacDonald et al. (2006) reported relatively no differences in the use of joint attention behaviors between 2-, 3-, and 4-year-olds with autism. Carpenter, Pennington, and Rogers (2002) reported that the development of joint attention skills in children with autism does not follow the same pattern of development of these skills in typically developing children. These authors suggested that there might be a direct relationship between language ability and joint attention skills. Joint attention has also been shown to be related to social development (Schertz & Odom, 2004) with regard to emotion regulation and the ability to develop connections with other individuals, given that joint attention is related to the ability to share enjoyment and other forms of affect with others (Schertz & Odom, 2004; Wetherby, Prizant, & Schuler, 2000).

Most communication is used in the context of a social interaction. Given that individuals with ASD have pronounced difficulties in the area of social interaction, it follows that these individuals also have difficulties in aspects of communication that require interaction with others. Presuppositional skills represent the ability to produce the necessary quality and quantity of communication given different contexts. This skill emerges in preschool-age children and improves as the child's working memory, social cognitive skills, and metacognitive and linguistic skills develop. Discourse management, which typically develops by age 4, includes conversational skills, such as turn taking, starting and ending conversations, and staying on the topic of conversation (Landa, 2005).

Pragmatic language skills include the ability to adapt communication to listeners and situations. Although typical children become adept at pragmatic language skills as early as the preschool years, individuals with ASD show a wide variety of skill deficits in this area. Specific pragmatic language deficits include asking inappropriate questions (that may result in embarrassment or discomfort to the conversational partner), inappropriately interrupting others, inability to start or end conversations, and perseverating on a topic of interest. When neologisms are used in conversation, confusion may occur, especially when the individual with ASD does not acknowledge that the conversational partner does not understand the neologism. Individuals with ASD have difficulty understanding sarcasm, jokes, and puns. They tend to interpret the meaning of a joke, for example, at face value, or literally (Emerich, Creaghead, Grether, Murray, & Grasha, 2003; Van Bourgondien & Mesibov, 1987).

To converse with another person requires numerous skills in addition to the ability to produce and understand spoken language. A high percentage of children with Autistic Disorder do not acquire verbal skills but do use augmentative or alternative ways to communicate. Thus, these children should be encouraged to engage in conversations with others (Loveland & Tunali-Kotoski, 2005).

Asperger's Disorder

Characteristics similar to those just described are present in Asperger's Disorder. Although there is no delay in the acquisition of language in Asperger's Disorder, the way individuals with this disorder communicate to others, particularly in the

social context, is noticeably impaired. For instance, the topic of conversation with someone with Asperger's Disorder may be narrow in scope and idiosyncratic. The individual will typically attempt to keep the conversation focused on a specific topic of interest to him or her, be indifferent to requests to change topics, have considerable difficulty taking turns in the conversation (i.e., social reciprocity), and have difficulty ending the conversation.

Receptive language difficulties include making inferences from others' nonverbal behavior and with certain elements of speech. For example, perspective taking can be difficult, as well as recognizing indirect social cues. Other elements of receptive language that are difficult for individuals with Asperger's Disorder are nonverbal (e.g., eye contact, facial expressions) and paralinguistic (e.g., tone of voice, prosody) cues and discriminating the meaning of a phrase, joke, or pun (i.e., they might have difficulty distinguishing something that has a literal versus a sarcastic meaning). Individuals with Asperger's Disorder tend to have a pedantic style of speech and prefer factual information.

BEHAVIOR

The broad category of restricted and repetitive behaviors in individuals with Autistic Disorder and Asperger's Disorder includes self-stimulatory behavior (e.g., repetitive finger movements, gazing at lights, rocking, toe-walking), stereotypical motor behavior (e.g., posturing), preoccupation with objects and activities, insistence on sameness or inflexibility in routines, and restrictive interest in objects or activities (South, Ozonoff, & McMahon, 2005; M. Turner, 1999). Children with ASD tend to have a narrow range of interests and activities, whereas adults have a somewhat wider range of interests but continue to be more resistant to change. Depending on routine and familiarity, the rigid or stereotyped thinking processes characteristic of ASD make abstraction and comprehension difficult. These clusters of behaviors are not unique to ASD. In fact, they may be seen in individuals with psychiatric conditions (e.g., Obsessive-Compulsive Disorder) and neurological conditions (e.g., Tourette's Disorder).

Recent studies are beginning to identify types or categories of atypical behaviors in ASD (Bishop, Richler, & Lord, 2006; Carcani-Rathwell, Rabe-Hasketh, & Santosh, 2006). In the current literature, the atypical behaviors within the category of restricted and repetitive behaviors are classified as either "lower order" or "higher order" behaviors. Lower order behaviors include stereotypical motor movements (e.g., hand and finger mannerisms) and preoccupations with parts of objects. Higher order behaviors include circumscribed interests, adherence to routines, and resistance to change (Bishop et al., 2006). In addition, Cuccaro et al. (2003) and Bishop et al. reported that younger children more readily engage in lower order behaviors (i.e., sensory-motor behaviors), whereas older children with ASD engage in higher order behaviors (i.e., insistence on sameness).

Bishop et al. (2006) point out that identifying specific atypical behaviors with different subtypes of ASD may be helpful in locating underlying genetic components and pathology among the ASDs. For instance, M. Turner (1999) and Lord, Rutter, and Le Couteur (1994) found that certain atypical behaviors (e.g., repetitive

sensory-motor behaviors) were associated with lower levels of adaptive functioning as measured by the Vineland Adaptive Behavior Scales (Sparrow et al., 1984). However, both studies indicated that other atypical behaviors, such as difficulty with or resistance to change, were not associated with adaptive functioning. In addition, there have been advances in research linking repetitive behaviors to specific neurobiological regions (Lewis & Bodfish, 1998; Pierce & Courchesne, 2001) as well as chromosomal regions (Alarcón, Cantor, Liu, Gilliam, & Geschwind, 2001; Cuccaro et al., 2003). Certainly, though more research needs to be conducted, the field is coming to understand the behavioral aspects of ASD much better now than in the past.

Disruptive behaviors are common in children and adolescents with ASD. Common aggressive behaviors observed and reported in individuals with ASD and other developmental disabilities include physical aggression (e.g., hitting, kicking, biting, shoving, pinching), verbal aggression (e.g., yelling, calling names), and destructive behavior (e.g., throwing objects, breaking objects of others). Assessment and interventions for aggression are discussed in the treatment section of this chapter. Other types of disruptive behaviors commonly observed in children with ASD are sleeping and eating problems.

It has been noted that individuals with ASD have significant feeding and sleeping issues. These two problems are serious given the impact they have on parental, family, and child functioning. However, these problems are not part of the *DSM-IV-TR* diagnostic criteria. In the context of this chapter, feeding problems are maladaptive feeding behaviors not due to a medical condition and/or without a medical explanation. They include aversion to food, refusal to eat, gagging or choking responses to food, and restrictive eating patterns (i.e., eating only certain very specific types of food).

Ahearn, Castine, Nault, and Green (2001) suggest that feeding problems in children with ASD may be categorized under the restrictive patterns of interests and activities characteristic of ASD. The *DSM-IV-TR* (American Psychiatric Association, 2000) includes a diagnostic category of feeding problems, which includes failure to thrive. Many children with ASD eat enough to not meet this criterion; however, their restrictive eating patterns might lead to significant nutritional deficiencies that will cause developmental problems later in life. Ledford and Gast (2006) reported that even though children with ASD may not be considered at risk for an eating disorder, the long-term health-related consequences of having a restricted diet should be taken into consideration and treatment of feeding problems in young children with ASD should be a priority.

Ledford and Gast (2006) conducted a comprehensive review of the literature on descriptive studies of feeding problems in ASD and also on interventions for feeding problems. Based on their review, these authors concluded that 46% to 89% of children with ASD have feeding problems. The authors reviewed nine intervention studies, all of which demonstrated positive outcomes of the interventions. It should be noted that the interventions were behavioral and included differential reinforcement, stimulus fading, escape extinction, and appetite manipulation (simultaneous and sequential presentation of a nonpreferred food with a preferred food).

Children with ASD have significantly more sleep problems than typically developing children (Hoffman, Sweeney, Gilliam, & Lopez-Wagner, 2006). Parents reported that these sleep problems include bedtime resistance, parasomnias, night wakings, sleep anxiety, sleep disordered breathing, and sleep-onset delay. However, children with autism are not observed to be sleepier during the daytime than children who do not have sleep problems. Sleep problems tend to decrease as children age, according to the study by Hoffman et al.

Empirically supported treatment of sleep difficulties in ASD has not been documented. However, traditional behavioral methods for sleep intervention do show some promise for children with ASD. It is necessary to understand the etiology of sleep disorders in ASD to provide more efficacious treatments for these individuals. Many parents of children with ASD report that their children have sleep difficulties that exceed those of typically developing children.

Hoffman et al. (2006) surveyed parents of children with ASD and parents of typically developing children and found that children with ASD have significantly more sleep problems in the following areas: (a) bedtime resistance, (b) sleep-onset delay, (c) sleep duration, (d) sleep anxiety, (e) night wakings, (f) parasomnia, and (g) sleep disordered breathing. These authors also found that as children with ASD age, the severity of some sleep problems decreases. A previous paper by Hoffman et al. (2005) demonstrated a positive correlation between sleep difficulties and autism symptomatology. Research is needed to evaluate sleep interventions for children with ASD and the impact that these interventions might have on learning, family and parental stress, and other symptoms of ASD.

ASSESSMENT AND DIAGNOSIS

Autistic Disorder is typically diagnosed before the age of 3. It is quite common for a diagnosis to be given around the age of 18 to 24 months (Robins, Fein, Barton, & Green, 2001). However, the diagnosis of young children with ASD can be quite difficult. There are no reliable biological markers for ASD; thus diagnosis relies on the use of instruments designed specifically for the diagnosis of ASD. These include the Autism Diagnostic Observation Schedule—Generic (ADOS-G; Lord, Rutter, DiLavore, & Risi, 2000), the Autism Diagnostic Interview—Revised (ADI-R; Rutter, LeCouteur, & Lord, 1995), and the Childhood Autism Rating Scale (CARS; Schopler, Reichler, & Renner, 1980). A recent study by Ventola et al. (2006) indicated that the use of the ADOS-G, CARS, and clinical judgment based on the *DSM-IV* (American Psychiatric Association, 1994) criteria demonstrated strong agreement with each other. Even though these results are encouraging, as they suggest that clinicians might be able to use different assessment instruments, more research needs to be conducted to determine which method for diagnosing ASD has the highest sensitivity and specificity.

Autism spectrum disorders are heterogeneous, and each individual with an ASD will present differently in certain areas. In fact, a diagnosis of Autistic Disorder is sometimes more straightforward than a diagnosis of high-functioning autism, Asperger's Disorder, or PDD-NOS. This is because some of the characteristics of these other disorders are more subtle and may not appear to be severe enough

according to a measurement instrument. Distinguishing between the boundaries of PDD-NOS, Asperger's Disorder, and high-functioning Autistic Disorder remains a challenge to both researchers and clinicians. As more research is conducted on diagnostic instruments, the symptom clusters and developmental trajectories specific to different ASDs should be illuminated.

The psychological assessment of an individual with ASD should include, but not be limited to, an understanding of the individual's current developmental level, his or her verbal and nonverbal strengths and weaknesses, and adaptive functioning (i.e., daily living skills). Because children with ASD have deficits in other areas than those listed in the *DSM-IV-TR,* the use of instruments that assess cognitive, language, motor, and adaptive skills are important to include in a diagnostic assessment battery. Standardized assessment instruments that can be utilized to assess standard intellectual functioning include the Wechsler Scales (the specific scale depends on the age of the individual); to assess mental processing and acquired language, the Kaufman Autism Behavior Checklist II; to assess developmental abilities, the Differential Ability Scales; and for children with severely limited language abilities, the Leiter-R. For developmental assessment, the Mullen Scales measure nonverbal, language, and motor skills. The Vineland Adaptive Behavior Scales measure adaptive functioning. Speech, language, and communication assessments should evaluate basic speech and language skills (including receptive and expressive language, prosody, paralinguistics, metalinguistics, and narrative skills) as well as social communication skills (such as joint attention, reciprocal social and communicative interactions, and self-regulation).

Importantly, when providing assessment, consultation, or intervention services to an individual with ASD, there will be numerous other professionals with whom a clinical psychologist is likely to interact for a variety of reasons (i.e., assessment, treatment, referral). Professionals range from early intervention service coordinators, special education teachers, and certified behavior analysts to a variety of family members, vocational skills training professionals, and psychiatrists. Of course, this is a short list, as the number and type of professionals depend on a host of factors, such as the client's age and level of functioning and the services available.

Above all, it is important to understand the child's functional skills, that is, skills the child can use on a day-to-day basis to communicate, interact socially, and accomplish tasks. Clinically it may not be important for a person with ASD to be able to do complex math problems consistent with his or her chronological age, but it may be far more important for that person to tell time on a digital clock in order to participate in an environment where schedules must be kept.

INTELLECTUAL FUNCTIONING AND MENTAL RETARDATION

Mental retardation (MR) is diagnosed when an individual's cognitive ability is assessed as an IQ of 70 or less, with adaptive skills deficits, and the age of onset is before 18 years (American Psychiatric Association, 2000). Given the criteria required for a diagnosis of Autistic Disorder, the criteria for impaired adaptive skills and age of onset overlaps. Thus, the critical aspect of diagnostic importance

for MR is strictly the cognitive skills deficit (Edelson, 2006). For years, reports of a rate of 70% to 80% of children with Autistic Disorder with comorbid MR have been suggested (Bohman, Bohman, Björck, & Sjöholm, 1983; Chung, Luk, & Lee, 1990; Gillberg & Coleman, 2000). A recent report suggests that between 40% and 55% of children with Autistic Disorder have comorbid MR (Chakrabarti & Fombonne, 2001).

Edelson (2006) examined retroactively the statistics in previous reports and studies to ascertain the nature of support for the claim of the incidence of MR in Autistic Disorder. Edelson provided an excellent and critical review of over 215 studies. For articles in which the methodology for determining MR was described (only 15.7%), the incidence of MR was 71.89%. Interestingly, Edelson found that 157 of the 215 articles reported the diagnosis of MR with autism, but did not provide a methodology for diagnosing MR.

Instruments used to assess cognitive ability are limited because when assessing the cognitive ability of children with ASD, poor motivation in particular may significantly attenuate performance. Given the variability of behavior in some children with ASD, it is difficult to say whether additional factors (e.g., attention, language comprehension, unfamiliarity with structured interactions) negatively influence performance on cognitive tests. Ritvo and Ritvo (2006) insist that whether a child with autism has MR does not really matter; what matters is whether the child can carry out daily activities, including academic ones. These authors suggest that the focus of treatment should be on social and communication deficits, not on cognitive and academic deficits. Future prevalence surveys should include the whole spectrum of Autistic Disorder and other ASDs to ascertain a more accurate estimate of the prevalence of MR (Freeman & Van Dyke, 2006).

COMORBIDITY OF PSYCHIATRIC DISORDERS

Comorbidity refers to the co-occurrence of two disorders where the occurrence of one disorder does not exclude the occurrence of the other (Krishnan et al., 2002). Over the past 10 years, comorbidity issues in ASD have appeared increasingly in the literature, especially with regard to anxiety disorders in ASD.

A recent study by Leyfer et al. (2006) piloted an instrument called the Autism Comorbidity Interview—Present and Lifetime Version (ACI-PL), which was modified from the Schedule for Affective Disorders and Schizophrenia for School-Age Children (K-SADS; Chambers et al., 1985). The ACI-PL is a semi-structured interview and is similar in format and scoring to the K-SADS. In the two pilot samples, the ACI-PL demonstrated adequate specificity, sensitivity, and reliability for three *DSM* diagnoses: (1) Major Depressive Disorder, (2) Obsessive-Compulsive Disorder, and (3) Attention-Deficit Hyperactivity Disorder. This study found that the most frequent disorder comorbid with Autistic Disorder was Specific Phobia, with a comorbidity rate of 44%. Obsessive-Compulsive Disorder was the second most common (37%), and the most common type of compulsion was performing a ritual in a certain way. The full diagnosis of Attention-Deficit/Hyperactivity Disorder (ADHD) is the third most common comorbid disorder (31%). Interestingly, 55% of individuals with Autistic Disorder also had subsyndromal ADHD. The authors

note that the inattentive type was more common in individuals with Autistic Disorder (whereas the hyperactive type is most common in the typically developing population). Ten percent of individuals with Autistic Disorder had experienced at least one episode of depression.

Overall, this study found that approximately 30% of the pilot sample had on average two comorbid diagnoses along with Autistic Disorder. The pilot sample tended to represent children with Autistic Disorder who would be described as high functioning, as 67.71% of children tested had Full-Scale IQ scores greater than 70, with a mean age of 9 years.

Comorbidity is an important area in ASD, given that the diagnostic criteria for ASD appear to be broadening (despite no change in the *DSM-IV-TR;* American Psychiatric Association, 2000) and the field is beginning to learn more about these comorbid disorders. The data Leyfer et al. (2006) provided should be accepted cautiously, given that this was a pilot study. However, the results do confirm an increased comorbidity of ASD with other *DSM* diagnoses reported by other investigators (Gillot, Furniss, & Walter, 2001; Kim, Szatmari, Bryson, Streiner, & Wilson, 2000; Towbin, Pradella, Gorrindo, Pine, & Leibenluft, 2005).

CURRENT TRENDS IN THE PREVALENCE OF AUTISTIC SPECTRUM DISORDERS

Historically, autism has been considered a very rare disorder. The *DSM-IV-TR* (American Psychiatric Association, 2000) reports the prevalence of Autistic Disorder to be 5 per 10,000, ranging from 2 to 20 per 10, 000. At the time of publication of the *DSM-IV-TR,* prevalence data were lacking for Asperger's Disorder, PDD-NOS, Rett's Disorder, and Childhood Disintegrative Disorder. Over the past 6 years, since the latest revision to the *DSM,* epidemiological research has been published providing new estimates of the prevalence, not only of Autistic Disorder, but also for Asperger's Disorder, PDD-NOS, and ASD (Fombonne, 2005). Thus, information about the prevalence of ASD in the *DSM-IV-TR* is no longer accurate.

In a recent review of prevalence studies from 1966 through 2004, Fombonne (2005) suggested that the current prevalence of Autistic Disorder is 13 per 10,000, of PDD-NOS is 21 per 10,000, of Asperger's Disorder is 2.6 per 10,000, with a prevalence of 60 per 10,000 for all PDDs. Croen, Grether, Hoogstrate, and Selvin (2002) conducted a study that investigated the change in prevalence of ASD in California from 1987 to 1994. The prevalence of ASD rose from 5.8 per 10,000 in 1987 to 14.9 per 10,000 in 1994. Interestingly, the prevalence of mental retardation showed the opposite trend; in 1987, the prevalence was 28.8 per 10,000 compared to 19.5 per 10,000 in 1994.

The male-to-female ratio of Asperger's Disorder, Autistic Disorder, and PDD-NOS is 4:1. Notably, when a female is diagnosed with Autistic Disorder, she typically has a more severe form (i.e., lower cognitive abilities, increased behavioral problems, and more severe social and communicative impairments).

Questions loom as to why there has been an increase in the prevalence of ASD. Croen et al. (2002) propose that factors accounting for the increase in prevalence rate may include improvements in identification of ASD and changes in the criteria

for a diagnosis of ASD in the *DSM-IV-TR* (American Psychiatric Association, 2000) compared to the *DSM-III* (American Psychiatric Association, 1980). Children diagnosed with Autistic Disorder in the *DSM-IV-TR* may represent a broader phenotype than if they were diagnosed using the *DSM-III*. Even though the precise answer to this question is not available, it is the case that we now have assessment tools to identify ASD at a younger age and more accurately, increased awareness and knowledge in communities and among practitioners, and increased services available for ASD (Croen et al., 2002; Fombonne, 2005). Regardless of the reasons for the increase in prevalence of ASD, early identification of the disorder is crucial for early intervention services to begin (Romanczyk et al., 2005).

EARLY IDENTIFICATION

Early identification and early intervention are necessary to provide positive outcomes for children with autism. Of the 3.75 million infants born each year in the United States, 20% are at risk for developmental disabilities due to pre- and postnatal factors (Haber, 1991).

Two federal laws that mandate service provision to children in the early intervention years (birth to 3 years) are Public Law 99-457 and Public Law 105-17. Public Law 99-457 was enacted in 1986 to provide early intervention services for at-risk infants and children (Haber, 1991). In addition, Public Law 105-17 allows eligible infants, toddlers, and children with disabilities to receive both appropriate and free early intervention services (birth to 35 months) and public education (starting at 36 months). Public awareness of the importance of early intervention services for children with autism has also increased. Several studies have shown both the short-term and long-term benefits for children receiving appropriate early intervention services (Anderson & Romanczyk, 1999; Guralnick, 1996; Lovaas, 1987; Smith, Eikeseth, Klevstrand, & Lovaas, 1997). These services are required to be free and it is mandatory for all states to follow the statutes in both public laws. Early identification, which leads to early intervention, is extremely important for individuals with ASD. Numerous studies have demonstrated impressive gains in language, cognition, and social development when individuals received early intervention services (Bondy & Frost, 1995; Harris & Handleman, 2000; National Research Council, 2001; L. Turner, Stone, Pozdol, & Coonrod, 2006).

The importance of early identification and intervention has spurred the development and validation of many screening measures. Given research on early signs of ASD, a diagnosis of ASD can be made before 24 months of age. Screening instruments are not substitutes for a diagnostic assessment, but do allow for the higher probability of identification of children who may be at risk for a developmental disability, such as ASD, in order to receive a more comprehensive diagnostic evaluation. Multiple screening measures are available, including the Checklist for Autism in Toddlers (Baird et al., 2000), the Modified Checklist for Autism in Toddlers (Robins et al., 2001), the Social Communication Questionnaire (Rutter, Bailey, Berument, Lord, & Pickles, 2003), and the Screening for Autism in 2-Year-Olds (Stone & Ousley, 1997).

Table 29.2 Clinical Clues for Autism

Delay or absence of spoken language
Looks through people; not aware of others
Not responsive to other people's facial expressions or feelings
Lack of pretend play; little or no imagination
Does not show typical interest in or play near peers purposefully
Lack of turn taking
Unable to share pleasure
Qualitative impairment in nonverbal communication
Not pointing at an object to direct another person to look at it
Lack of gaze monitoring
Lack of initiation of activity or social play
Unusual or repetitive hand and finger mannerisms
Unusual reactions, or lack of reaction, to sensory stimuli

Excerpted from *Clinical Practice Guideline for Autism/Pervasive Developmental Disorders: Assessment and Intervention for Young Children Ages 0–3 Years* (Guideline Technical Report. no. 4217), Table III-4, by the New York State Department of Health, Early Intervention Program, 1999, Albany, New York: Author.

The New York State Department of Health published *The Clinical Practice Guideline for Autism/Pervasive Developmental Disorders* in 1999. There are three versions of the guideline: *The Guideline Technical Report, Quick Reference Guide,* and *Report of the Recommendations.* Common to each version are basic recommendations for assessment and intervention methods for children with autism or other PDD. Within the assessment content area is a table listing 13 clinical clues recommended for the purpose of screening children for autism or other PDD. These clinical clues were taken from empirical studies that listed diagnostic criteria for autism according to the *DSM-III-R* and *DSM-IV-TR.* Each clue has been validated by empirical studies; has been shown to have a sensitivity or specificity of greater than 50%; is relevant to the *DSM-IV-TR* for the diagnosis of autism; and may be found in other screening and diagnostic instruments for autism. These clinical clues are presented in Table 29.2.

ETIOLOGY

The specific causes of ASD are unknown. As is the case for many disorders, the interaction of a biological predisposition (e.g., genetic mutations) with environmental factors most likely underlies the etiology of ASD, but hypotheses regarding etiology along these lines continue to be investigated (Polleux & Lauder, 2004). Research on the genetic basis and neurobiology of ASD are two important fields that are illuminating possible contributions to the etiology of ASD, but research is still in the early stages.

Familial recurrence of autism is 100 times higher than in the general population. In addition, there are reported concordance rates of 70% to 90% in monozygotic twins, which is not found in dizygotic twins. In fact, most studies show near 0% concordance rates in dizygotic twins, which provides strong support for the genetic component of autism (Veenstra-VanderWeele & Cook, 2003). However, we are far from identifying a definitive etiology for autism.

Approximately 10% of cases of autism have comorbid genetic conditions, such as Fragile X, tuberous sclerosis, and Rett's Disorder (Polleux & Lauder, 2004). One-third of individuals with autism develop seizures, and over half have abnormal sleep patterns. Recently, researchers have developed a working hypothesis characterizing the cortex as having an imbalance of excitation and inhibition, which leads to hyperexcitability in the cortex (Polleux & Lauder, 2004). The imbalance in the cortex could have consequences "on the function of neuronal networks underlying perception and attention" (Polleux & Lauder, 2004, p. 304). Other neuroanatomical features that have been hypothesized to be linked to ASD are abnormal functioning of the brain stem, cerebellum, hippocampus, and amygdala. There is also evidence suggesting that the brain is enlarged in children with autism and that this enlargement occurs during the postnatal period, but there is no further enlargement in adulthood and studies show no differences in brain size in adulthood between adults with Autistic Disorder and typically developing adults. The exact specifications and implications of brain enlargement have yet to be determined.

Multiple genes are hypothesized to be linked to autism, but more research is needed and is currently ongoing to address this. Neurotransmitter abnormalities are also hypothesized to be implicated in autism. Serotonin and GABA are two that have the most evidence suggesting abnormalities in ASD. See Polleux and Lauder (2004) for a review of neurobiological evidence.

A major difficulty in drawing conclusions from the current research evidence from the neurobiology and neuroimaging fields is that the sample size of most studies is still small and is typically biased toward individuals with autism who are considered higher functioning. Thus, generalizing results from these studies is difficult. However, this is an important area of research, as Autistic Disorder is a neurobiological disorder and the genetics, specific brain areas affected, and other biological and neurological characteristics of Autistic Disorder need to be more fully understood. These same concepts hold true for determining biological etiologies or predispositions for Asperger's Disorder and PDD-NOS.

RELATED DEVELOPMENTAL DISORDERS: RETT'S DISORDER AND CHILDHOOD DISINTEGRATIVE DISORDER

Rett's Disorder

Rett's Disorder was discovered by Andreas Rett and (at a later time) Bengt Hagberg. Rett's Disorder is an X-linked, dominant disorder that occurs almost exclusively in females. It has been identified as the second leading disorder that causes MR in females (Down syndrome is the primary). Much is known about the genetic etiology of Rett's and the course of the disorder, which is described next. It is hoped that eventual understanding of this disorder will help with the understanding of related disorders.

The most prominent symptom in Rett's Disorder is stereotypic hand movements. These are often an early warning sign; that is, these types of stereotypies are often present before other associated symptoms of Rett's are noticed. For children with Rett's Disorder, the first 6 to 8 months of life are characterized by normal

development. About 50% of cases begin to show a delay in and/or deterioration of skills around 15 months of age. By age 3, a significant deterioration of development is noted, especially loss of language, increased stereotyped hand movements, and decreased interest in objects and social interaction. The deterioration of skills is rapid (i.e., after a 1-year period, severe to profound disabilities are noted). Additional motor difficulties and delays are also present. The characteristic stereotyped hand movements include hand-to-mouth movements, hand wringing, and hand clasping. By the early school years, the pace of deterioration has decreased and children are able to acquire and maintain skills. However, through adolescence there is a decrease in physical growth and problems with ambulation are noted; in addition, many children develop scoliosis (Van Acker, Loncola, & Van Acker, 2005).

In 1986, Hagberg and Witt-Engerstrom (Van Acker et al., 2005) provided a four-stage developmental model of Rett's Disorder, then called Rett syndrome. This model is generally accepted today. Between 6 and 18 months, Stage 1 begins. This stage is characterized by a noted deviation from normal development, particularly in motor abilities. Stage 2 begins between 1 year and 4 years of age. In this stage, a significant loss of previously acquired skills occurs; many symptoms are present that resemble those in individuals with autism. Thus, social, speech, cognitive, and motor skills are deteriorated or lost during this stage. The stereotypic hand movements tend to be continuous, without clear antecedents or triggers. Children with Rett's Disorder are diagnosed with severe to profound MR, and parents characterize their mood as labile and irritable. During this stage abnormal breathing patterns are noticed; 25% of children have clinically significant seizure activity and 75% present with sleep abnormalities. Stage 3 begins between ages 2 and 10. The symptoms characteristic of autism tend to decrease during this stage. In fact, an increase in social interaction, awareness of and interest in the environment, communication skills, and use and acquisition of functional skills occurs. Up to 80% of children have seizures and the motor difficulties become more prominent. Stage 4 begins when a child is 10 or older. During this stage motor deterioration continues, with decreased mobility and frequently need of assistance to move about (e.g., using a wheelchair). Individuals' cognitive functioning is stable, and social skills and attention improves. In addition, a decrease in seizure activity is observed.

Childhood Disintegrative Disorder

Childhood Disintegrative Disorder (CDD) is characterized by typical, age-appropriate development in the areas of language, social, cognition, and other skills, with a regression of skills occurring between the ages of 2 and 10. Some authors suggest that there may be a mild delay in development prior to age 2 (Kurita, Kita, & Miyake, 1992). However, the most common report of age of onset is between ages 3 and 5, with a mean of 3.4 years. With regard to differential diagnostic characteristics, in Autistic Disorder it is not uncommon to observe language regression; however, in CDD there are multiple areas of regression.

The onset of CDD is typically abrupt, with symptoms being noticed within days to weeks. Many cases report a psychosocial stress or medical event, but the etiology is unclear. Prevalence is estimated to be 1 to 9 cases per 100,000 children, according

to Fombonne (2002). Because this condition is so rare, it is likely that most clinicians have little familiarity with it, and thus the actual prevalence of CDD is uncertain. Similar to other developmental disabilities, CDD predominantly affects males. It is differentiated from Rett's Disorder by the longer period of normal development before regression begins.

For the most part, the regression and deterioration of skills typically plateaus. Most cases of CDD do not continue to regress to the point of mortality. However, Volkmar, Koenig, and State (2005) reviewed several case studies suggesting an increased mortality rate after a child is diagnosed with CDD. Most of the deaths are related to neuropathology and/or medical diseases. A comorbidity rate of seizures similar to that found in Autistic Disorder has also been suggested in CDD. However, there are no conclusive data suggesting a specific medical etiology for CDD nor a predictive relationship of CDD with other medical disorders or pathologies. The genetic mechanisms in CDD are also not understood, and further research needs to be conducted (Volkmar et al., 2005).

TREATMENT

The field of PDD has experienced notable advances in research and clinical intervention over the past 10 years. However, there is no one treatment that has been shown to be universally effective in the full remediation of ASD. Treatments for ASD are described as either comprehensive interventions that include elements of behavioral, educational, and psychotherapeutic approaches or specific interventions, most of which are behavioral or psychopharmacological, that focus on the treatment of specific behaviors.

In general, learning is quite challenging for individuals with Autistic Disorder, whose difficulties range from overselectivity (i.e., disproportionate attention to irrelevant aspects of a stimulus) to lack of generalization of skills. Most children with ASD tend to lack multiple foundational learning skills; thus, sensitivity to the sequence in typical development is important when designing interventions. It is important to consider the individual's age-equivalent or developmental level when providing interventions and determining expectations. This is important at all levels and ages of intervention. In a recent study by L. Turner et al. (2006), a relatively stable diagnosis was demonstrated in children diagnosed with ASD at age 2 and reassessed at age 9. Thus, it is important that children receive services throughout childhood.

In 2002, the National Institutes of Health sponsored a meeting on the current science of the treatment for children with ASD (Lord et al., 2005). There is no medical or biological treatment for the *core* features of ASD. However, there is medication that ameliorates some of the mood and behavioral problems associated with ASD. Given the heterogeneity of ASD, the specific language difficulties, and the comorbidity of mental retardation, it is not possible to say that a specific type and dosage of treatment is recommended. Interventions should begin as soon as the child is identified. Currently, behavioral interventions have the most empirical support (U.S. Public Health Service, 2001) for the treatment of ASD. Behavioral interventions for ASD have provided an impressive amount of empirical support. Applied behavior

analysis, cognitive-behavioral therapy, and psychopharmacology interventions are all discussed in turn. In addition, a selection of interventions specific to the social, communication, and behavioral impairments are also briefly discussed.

Applied Behavior Analysis

Applied behavior analysis (ABA) is a discipline that seeks to understand and improve human behavior and is grounded in principles of operant and respondent learning and conditioning. This approach has the strongest demonstrated research support for efficacy and has been widely acknowledged by independent reviews as a critical part of a child's treatment program (e.g., the report of the surgeon general; U.S. Public Health Service, 2001). There are guidelines for selecting qualified behavior analysts that provide interventions for individuals with ASD (Autism Special Interest Group of the Association for Behavioral Analysis, 2007). In addition, the Behavior Analyst Certification Board provides certification for behavior analysis practitioners.

In intervention programs that use ABA as a framework, operational definitions of behavior are utilized to objectively measure complex behavior patterns. Baseline data are collected on these behaviors, and interventions based on principles of behavior are implemented. Data collection occurs throughout the intervention. This process allows for the evaluation of the intervention on an ongoing basis, and modifications are made depending on the child's performance. Some of the instructional techniques used in ABA programs include discrete trial training, pivotal response training, incidental teaching, and generalization training.

An important component of ABA interventions for ASD is the use of functional behavioral assessments. Functional behavioral assessments assist in determining the functions of specific behaviors (e.g., self-stimulatory behavior, noncompliance, self-injurious behavior, aggressive behavior). A functional behavioral assessment may include interviews, direct behavioral observations, checklists, and manipulating the antecedents and consequences of a behavior. These methods assist in generating hypotheses of the function of a behavior and guide intervention. Common functions of disruptive behaviors are positive reinforcement, negative reinforcement, and nonsocial reinforcement, sometimes referred to as automatic reinforcement or sensory reinforcement (Iwata, Dorsey, Slifer, Bauman, & Richman, 1994; Reese, Richman, Belmont, & Morse, 2005).

Psychopharmacological Treatments

Psychopharmacological treatments for ASD target specific problematic behaviors. In addition, psychopharmacological treatment may be provided to treat comorbid disorders (such as anxiety or depression). A recent report of the American Psychological Association Working Group on Psychotropic Medications (2006) estimates that between 33% and 47% of individuals with ASD have been prescribed at least one psychotropic medication over the course of a 1-year period. The Working Group indicated that the evidence of the effectiveness of psychotropic medications is limited and variable, and that medication should be considered on an individual basis with careful consideration of the risks and benefits of the medication, expected

behavior change, and side effects. Recently, the Food and Drug Administration approved Risperdal (risperidone) for the treatment of severe irritability in children and adolescents with Autistic Disorder (U.S. Food and Drug Administration, 2006). Table 29.3 summarizes the information provided in the Working Group's report.

Cognitive-Behavioral Therapy

Cognitive-behavioral therapy (CBT) has garnered recent attention for treatment of anxiety for individuals with high-functioning autism and Asperger's Disorder. Sofronoff, Attwood, and Hinton (2005) implemented a 6-week brief CBT intervention for individuals with Asperger's Disorder and anxiety symptoms. These authors found that having the family (i.e., parents) involved led to the best outcomes (i.e., reduction in anxiety symptoms). However, these authors did rely on parent report to demonstrate the change in symptom frequency and severity, which is a limitation of the study.

Cardaciotto and Herbert (2004) modified a CBT intervention to treat social anxiety disorder (Social Phobia) for an individual with ASD who presented with fear of social situations, specifically assertiveness, talking in front of others, and initiating conversations with those he is familiar with. The authors provided 14 weeks of CBT. The outcome was successful and the client showed maintenance of gains. The recent research examining how to effectively modify traditional CBT interventions is a new and exciting area for the field of ASD.

Social Skills Training

There are many different social skills interventions for ASD. Research has shown that individuals with ASD do not learn social skills in the same way as typically developing children (Gresham, 1981). Developmental level and chronological age should be taken into consideration when teaching social skills. For example, social skills interventions during the early intervention years (i.e., 0 to 3 years old) focus on improving social behavior and interactions with an adult (e.g., parent). In the preschool years, the focus is on play skills and peer interactions. More advanced social skills are taught when a child reaches school age, such as initiation of play and conversations and developing friendships (Guralnick & Neville, 1997; Schreibman, 1996).

A review of social skills interventions for preschool-age children by Vaughn et al. (2003) revealed the following common features: prompting and rehearsal of target behaviors, play-related activities, free-play generalization, reinforcement of appropriate behaviors, modeling of specific skills, storytelling, direct instruction, and imitation of appropriate behaviors. The authors' review indicated that the interventions with the highest effect sizes were those that contained features of modeling, play-related activities, rehearsal and practice, and prompting. Based on their review of components of interventions that yielded improvements in social behavior, the authors suggest the following (pp. 12–13):

- Social skills programs that are embedded into regular class programs by teachers (Antia, Kreimeyer, & Eldredge, 1993; Ferentino, 1991; Fewell & Vadasy, 1989; Jenkins, Odom, & Speltz, 1989; Koenigs & Oppenheimer, 1985)

Table 29.3 Psychotropic Drugs Commonly Used in the Treatment of Symptoms of ASD

Drug Class	Specific Drug Studied	Target Behaviors/ Symptoms	Side Effects	Conclusions of APA Work Group
Stimulants	Methylphenidate (Ritalin)	Hyperactivity, inattention, impulsivity	Agitation, irritability, insomnia	48% to 50% experienced positive response; 18% experienced adverse effects.
Atypical antipsychotics	Risperidone*	Aggression, severe tantrum behavior	Drowsiness, fatigue, constipation, weight gain, may increase risk for diabetes and hyperlipidemia	Approximately 66% experienced improvement in symptoms. However, symptoms recur after medication is discontinued.
Selective serotonin reuptake inhibitors	Clomipramine, fluoxetine	Perseverative behaviors (e.g., compulsions, self-injury, stereotypies)		Limited evidence. Fluoxetine appears to be better tolerated than other SSRIs. Most studies are with adults; little is known about effects on children and adolescents.
Mood stabilizers	Lithium, divalproate	Explosive aggression, severe tantrum behavior		No conclusive evidence for children and adolescents. Only small studies have been conducted with adults, showing preliminary support.
Alpha agonists	Clonidine, guanfacine	Hyperactivity, aggression, severe tantrum behavior		Limited support in open-label, uncontrolled reports.

*Risperdone is now (as of October 6, 2006) FDA approved for the treatment of irritability associated with children and adolescents with Autistic Disorder (http://www.fda.gov/bbs/topics/NEWS/2006/NEW01485.html).

Adapted from *Psychopharmacological, Psychosocial, and Combined Interventions for Childhood Disorders: Evidence Base, Contextual Factors, and Future Directions,* by the American Psychological Association Working Group on Psychotropic Medications for Children and Adolescents, 2006, Washington, DC: Author.

- Interventions that include both instruction in social skills and behavioral contingencies for appropriate and inappropriate behaviors (LeBlanc & Matson, 1995; Matson, Fee, Coe, & Smith, 1991)
- Interventions that provide integrated or social interaction groups for children with and without disabilities (Jenkins et al., 1989)
- Interventions that train parents or peers as models to promote appropriate social behaviors (Dawson & Galpert, 1990; Strain, 1985)
- Interventions that involve an intensive social skills program (Rogers, Herbison, Lewis, Pantone, & Reiss, 1986)

Social skills interventions for individuals with ASD described as higher functioning have included group interventions (Howlin & Yates, 1999; Krasny, Williams, Provencal, & Ozonoff, 2003; Solomon, Goodlin-Jones, & Anders, 2004). Skills typically taught within the context of group interventions include awareness of emotions, face processing and nonverbal communication, theory of mind and perspective taking, conversations, problem solving, stress management, and beginning conversation skills (Solomon et al., 2004).

Communication

It has been observed that approximately 70% of children between the ages of 3 and 5 years with a disability have a delay in language development (Wetherby & Prizant, 1992). Thus, discovering new, innovative ways to teach individuals with ASD to communicate more effectively has been a priority in ASD treatment research. Most comprehensive treatment and educational programs for individuals with ASD include a communication acquisition component. Depending on the strengths and weaknesses of the individual, verbal language, nonverbal language, and social communication acquisition might be emphasized. Successful verbal language acquisition has been documented using interventions such as discrete trial training, incidental teaching, natural language teaching procedures, mand training (Peterson, 2004), and pivotal response training (Koegel, O'Dell, & Koegel, 1987).

The use of visual supports has been recommended and is frequently present in research studies and clinical settings. One such visual support is an activity schedule (McClannahan & Krantz, 1999). An activity schedule provides visual information (in the form of pictures, symbols, or text) to an individual with ASD. This information can be utilized to teach a skill (skills can range from daily living skills to academic skills, self-management skills, social skills) and to help individuals with ASD predict and plan a sequence of events (e.g., daily classroom activities). Research suggests that integrating multimedia components into teaching children with ASD shows initial positive results. However, the mechanism of multimedia effects is unknown. Future research should systematically examine whether and how the use of multimedia components is effective.

Even with available interventions, more than 50% of children with autism are nonverbal or mute (Sigman & Capps, 1997). Thus, language interventions are of

great importance to enable these children to use alternative means of communicating with others. Bondy and Frost (1994) developed the Picture Exchange Communication System (PECS) program for children with social communication deficits. This program is appealing to the ASD field for many reasons: (a) The PECS program requires simple motor movements; (b) PECS is portable and can be used across a wide variety of settings; (c) PECS is taught relatively quickly using behavioral principles of shaping, transfer of stimulus control, and differential reinforcement; (d) PECS facilitates child imitations of interactions with others; and (e) PECS has been demonstrated to facilitate verbal language (Charlop-Christy, Carpenter, Le, LeBlanc, & Kellet, 2002). The structured context and use of pictures in the PECS program is appealing because of the visual modality strength in children with ASD. The program enables a functional relation between the environment and the student to develop.

Behavior

Treatment for aggression is commonly requested from families of individuals with ASD. Because of the multiple functions that aggression can serve, no one specific treatment method is suggested. Behavioral and psychopharmacological treatments are typically used to treat aggression. Matson, Dixon, and Matson (2005) reviewed common behavioral interventions that have shown effectiveness for decreasing aggression, such as positive reinforcement, functional communication training (FCT; Carr & Durand, 1985), time-out, social skills training, and differential reinforcement procedures. The most commonly used interventions are FCT and differential reinforcement of other behavior. The authors emphasize the importance of conducting a functional analysis to understand factors influencing the behavior in order to treat aggression effectively.

Functional communication training (Durand, 1990) is an empirically validated procedure shown to be helpful in the management of challenging behaviors in individuals with ASD (Durand, 1990; Durand & Carr, 1991). This approach incorporates a functional behavior assessment to develop intervention plans to teach individuals with ASD to use alternative skills that decrease disruptive behaviors (Iwata et al., 1994). Thus, based on the functions of a behavior hypothesized from the functional behavior assessment, an individual is taught a more adaptive skill that serves the same function as the behavior. For example, if a child engaged in head-banging to escape a task, the child might be taught to use a Stop sign to signal a request to end a task. Treatment for the behavioral deficits in ASD continues to be an important area for future research.

FAMILY ISSUES

Having a child and/or family member with an ASD imposes a tremendous amount of stress on all family members. Family members may cope and manage stress differently. Because ASD is a lifelong disorder, there are challenges a family faces that vary from year to year, from coping with the diagnosis of an ASD to seeking appropriate educational treatment, dealing with adolescence issues, and changing

living situations when adult children become more challenging for parents. Families also appear to benefit from early intervention services for their children with ASD. The majority of parents indicated that early intervention services decreased the overall stress in the family (Glasberg, Martins, & Harris, 2005).

Lockshin, Gillis, and Romanczyk (2004) identify five general factors that shape family dynamics. The first is a need for continuous supervision of the child with ASD. This can limit the efficiency of completing other tasks. In addition, it requires family members to coordinate who is responsible for the child with ASD at specified times. Second, the inability or limited ability of the child with ASD to communicate the full range of his or her needs and wants can make the family feel that they are in a constant guessing game as to what they need to do to meet the child's needs. Third, atypical relationships with family members sometimes develop given the increased strain on each family member; this can lead to increased family conflict. Fourth, atypical relationships with peers can develop, in which the family members have limited peer and social support outside of the family. Fifth, atypical, difficult-to-manage behavior (e.g., aggression, rigidity, self-injury, lack of social boundaries) may require extraordinary resources for a family and increased stress if these behaviors are not under control in the family and public settings. Assisting families to identify resources in the community is important as a first step in decreasing family stress. It is then helpful to teach family members possible coping strategies. Helping families prioritize their needs so that all members are considered equally is important to decrease conflict and increase support within the family, as well as with friends and peers.

Research in the area of adolescents and adults with ASD is impoverished and is a critical area for growth. For the majority of individuals diagnosed with ASD in childhood, there is consensus that ASD is a lifelong disorder. However, several studies (Byrd, 2002; Mesibov, Schopler, Schaffer, & Michal, 1989; Piven, Harper, Palmer, & Arndt, 1996) have indicated that the core symptoms of ASD do improve with age. That is, the majority of individuals studied had noted improvements in communication, socialization, and behavior in adolescence and adulthood when compared to their functioning in these areas during childhood. Seltzer et al. (2003) found improvement in ADI-R scores in adolescence and adulthood, meaning that these individuals showed a reduction in the number of ASD-related symptoms. Despite these encouraging results, it remains clear that adolescents and adults with ASD continue to need extensive, but different types of, services.

SUMMARY

In the past 2 decades there have been substantial changes in diagnostic criteria and a broadening of the historically limited conceptualization of autism. Pervasive developmental disorders is the current *DSM-IV-TR* terminology which is used in parallel with the more recent and common umbrella term autism spectrum disorders. The ASD conceptualization emphasizes a focus on the social deficits that form the core deficit. There have been many advances in the field of ASD, and there are many exciting areas for future research. Increased availability and utility of technology, from functional magnetic resonance imaging to PDAs, will assist in measurement

that provides information that was not available to researchers 10 (even 5) years ago. Conceptualizing the pervasive developmental disorders as disorders that are on a continuous spectrum changes how we ask questions, think about these disorders, and define the population, and in the future will most likely eventually lead to a further delineation of specific subtypes of ASD.

In the area of intervention, although effective intervention procedures have been identified, more limited efficacy of the generalization of skills acquired is often observed. In addition, more detailed understanding of specific characteristics of individuals with ASD may guide choice of treatment for practitioners, parents, clinicians, therapists, and teachers. We do not yet understand the mediators that affect treatment outcome. There are well over 250 treatments available to individuals with ASD, with the vast majority not effective and wasteful of time and resources. But efficacy information often is overshadowed by marketing efforts by proponents of this vast array of treatments, and thus the treatment choice process is often stressful and difficult for parents and advocates for individuals with ASD. Demonstrating empirical evidence for available treatments and the ethical and responsible dissemination of the evidence is key to helping individuals with ASD and their families.

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CHAPTER 30

Parental Psychopathology and Its Relation to Child Psychopathology

Sherryl H. Goodman and Sarah R. Brand

Regardless of one's theoretical perspective, psychopathology in parents raises many concerns about children's psychological well-being. In this chapter, we focus on two of the most prevalent disorders in parents, depression in mothers and alcohol disorders in fathers, and examine the evidence for their effects on various aspects of children's psychological functioning. We also review the mechanisms or processes that best explain those outcomes: the mediators. Because not all children of parents with psychopathology develop psychopathology themselves, we also review the findings on the moderators: characteristics and circumstances that influence the strength of those associations. Finally, we suggest some areas for further research and some implications for clinical practice and policy. Throughout this chapter, we take a developmental psychopathology perspective.

Several sets of findings guided our decision to focus on depression in mothers and alcohol disorders in fathers. With regard to depression, it is common, especially among women of childbearing and child-rearing ages, severe and impairing, and recurrent or persistent. It is not surprising that its presence in mothers has fueled concern for its potential to disrupt aspects of caretaking known to be critical for healthy child development. Knowledge of the heritability of depression adds further concern for the children of depressed parents. More recently, emerging knowledge of the neuroendocrine correlates of depression during pregnancy has raised concern about possible adverse influences even on fetal development. Many of these issues also pertain to the study of paternal alcoholism and the effect on child development. Although mothers have been the dominant focus of high-risk studies, research on parental alcohol abuse has tended to focus more heavily on fathers, likely because of the greater prevalence of such disorders in men (Phares, 1992; Phares & Compas, 1992). Advances in genetics research have also shown a high genetic risk of alcoholism, especially in men passing on the genes to their sons. Also, although mothers remain the primary caregivers for children in the United States, fathers are becoming more involved and taking on greater caregivng roles. Thus, the impact of the environmental correlates of alcoholism in fathers on the development of psychopathology in children will only grow.

Data are consistent with the suggestion that depression in women may be more of a concern than depression in men. That is, although depression in both mothers and fathers has been shown to affect children's psychological functioning, a meta-analysis of 134 samples, with a total of over 60,000 parent-child dyads, showed that both internalizing and externalizing problems in children are more strongly associated with depression in mothers relative to depression in fathers (Connell & Goodman, 2002). On the other hand, the meta-analysis found that alcoholism and substance abuse disorders in mothers were more closely related to externalizing (but not internalizing) problems in children than were such disorders in fathers. This finding is contrary to our premise that paternal alcoholism would be more strongly linked to children's (especially boys') behavior problems, based on both genetic mechanisms of transmission and greater prevalence of alcoholism in men than women. However, the magnitude of the differences was small, and the meta-analysis findings may reflect the adverse influence of maternal substance use during pregnancy on prenatal development, which is later reflected in psychopathology in the children.

PREVALENCE

One reason to focus on depression in mothers is its prevalence in women of childbearing and child-rearing age. Depression is one of the most common psychiatric disorders in all adults. Among women, between 6% and 17% experience an episode of major depression at some point in their lifetime, a rate that is between 1.5 and 3 times higher than in men (Kessler, 2006). Moreover, depression is the leading cause of both economic and social disability in early to middle adulthood (Wilhelm, 2006), the primary childbearing and child-rearing years. The aspect of disability that has particular implications for depression in children of depressed mothers is depression's interference with quality of parenting, which has been the subject of a large body of research that will be reviewed later in this chapter and was the subject of a recent meta-analysis (Lovejoy, Graczyk, O'Hare, & Neuman, 2000).

Alcoholism is also highly prevalent in the U.S. population, with an estimated 8.45% (17.6 million) adult Americans meeting *Diagnostic and Statistical Manual of Mental Disorders* (*DSM-IV*; American Psychiatric Association, 1994) criteria for alcohol abuse or dependence during a 12-month period (Grant, Stinson, Dawson, Chou, Dufour, et al., 2004). As many as one in four children under 18 are exposed to alcohol abuse or dependency within their family (Grant, 2000). Because men show much higher rates of alcohol-related disorders than women, it is likely that a majority of those children are living in households with an alcoholic father.

THE NATURE OF THE DISORDERS

Researchers define clinically significant depression as either the set of symptoms, duration, and impairment by which an individual meets *DSM-IV* (American Psychiatric Association, 1994) criteria for a mood disorder or gets a high score on a self-report symptom rating scale of depression. The *DSM-IV* divides mood disorders into depressive disorders (sometimes referred to as unipolar depression) and Bipolar Disorders. Bipolar Disorder, which requires the presence of one or more

manic or hypomanic episodes, although a concern for its impact on children, has less often been the focus of studies of maternal depression and will not be discussed in this chapter.

The *DSM-IV* divides alcohol-related disorders into two separate categories: alcohol dependence and alcohol abuse. Like other substance abuse disorders in the *DSM-IV,* alcohol abuse requires fewer symptoms and therefore may be less severe than dependence. Alcohol abuse is diagnosed only once the absence of dependence has been established (American Psychiatric Association, 1994). We include studies of both disorders in this chapter.

In the research literature on alcoholism, the issue arises about how to best characterize alcoholics who also have antisocial personality traits. Data from the National Epidemiologic Survey on Alcohol and Related Conditions showed about 28% co-occurrence between personality disorders and alcohol disorders. Antisocial Personality Disorder occurred in 12.3% of alcoholics, both men and women (Grant, Stinson, Dawson, Chou, Raun, et al., 2004). When these traits were examined for their contribution to child outcomes independent of the influence of alcoholism, they were usually found to contribute more to negative child outcomes than the alcoholism. However, many studies do not assess for these traits or do not make it clear if fathers were excluded based on high antisocial personality characteristics. In this chapter, studies that also examine antisocial personality traits are noted. Caution is needed in interpreting findings from studies when measures of such traits were not included.

THEORIZED MECHANISMS/MEDIATORS

Goodman and Gotlib (1999) proposed a model to explain how depression in mothers might contribute to the development of psychopathology in children (see Figure 30.1). We use that model to structure our summary of the current status of evidence for the most likely mechanisms for the effects of maternal depression on children. We also begin with that model for examining evidence for various mechanisms of transmission in the effects of paternal alcoholism on children and modify the model in theory- and data-guided manners.

When reviewing the literature regarding the influence of paternal alcoholism on child development, some issues arose that are important for readers to keep in mind. First, in many cases, only sons of alcoholics were studied. This is most likely due to the strong findings of the genetic contribution to the development of alcoholism in men (Tyndale, 2004). Second, more often than in the parental depression literature, many studies of parental alcoholism failed to distinguish whether the alcoholism was in the father or the mother. We included such studies, but they clearly dilute the potential to inform the literature on paternal alcoholism specifically.

Biological Mechanisms for Effects of Depression in Mothers

Work on biological mechanisms that might explain the development of psychopathology in children of depressed mothers has focused on two lines of research. First, much research has centered on genetic influences, not only of depression as a disorder but also of vulnerabilities to depression and to other disorders. A

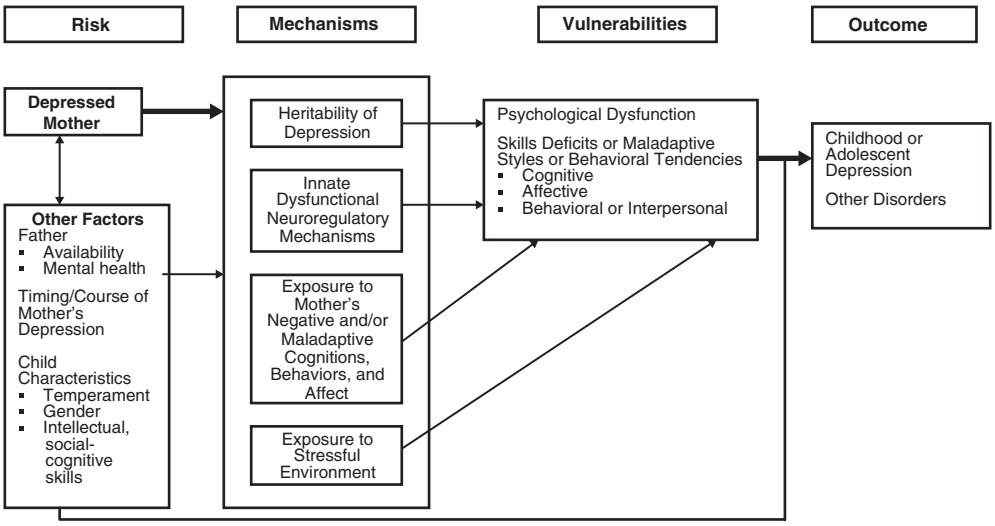

Figure 30.1 Integrative model for the transmission of risk to children of depressed mothers. *Source:* "Risk for Psychopathology in the Children of Depressed Mothers: A Developmental Model for Understanding Mechanisms of Transmission," by S. H. Goodman and I. H. Gotlib, 1999, *Psychological Review, 106,* pp. 458–490. Reprinted with permission.

second line of research has investigated the idea that fetal or postnatal exposures or experiences among children with depressed mothers might adversely influence brain development or hormonal functioning related to neuroregulatory mechanisms, thereby increasing risk for the later development of psychopathology. Support for both of those lines of research is reviewed in the following sections.

Genetic Influences

Research based on family studies, quantitative genetic studies (twin and adoption designs), and molecular genetic studies all provide strong support for heritability of depression in adults, although some evidence suggests that the specific genes related to risk for depression may differ for men compared to women (Caspi et al., 2003; Kendler, Gardner, Neale, & Prescott, 2001). Evidence for the heritability of depression in children and adolescents, however, is mixed. Based on family studies, strictly defined depression in children and adolescents is associated with increased rates of depression in relatives (Harrington, 1996; Harrington et al., 1997). Based on twin studies, heritability is higher in those whose levels of symptoms were below rather than above clinical cutoffs (Rende, Plomin, Reiss, & Hetherington, 1993), for depression in adolescents rather than in children (Eley & Stevenson, 1999; Murray & Sines, 1996; Scourfield et al., 2003; Silberg et al., 2001; Thapar & McGuffin, 1994), for parent-reported rather than child-reported symptoms (Eaves et al., 1997), and for parent-reported symptoms in girls relative to boys (Murray & Sines, 1996; Scourfield et al., 2003). In addition, because heritability of depression is probably not specific to depression, heritability likely contributes to risk for other disorders that are identified in higher rates in children of depressed mothers relative to controls (Moldin, 1999; Tsuang & Faraone, 1990). Thus, heritability likely contributes to

the risk for the development of psychopathology in children and adolescents with depressed mothers, although more research is needed to clarify the nature, extent, and specificity of the contribution.

An alternative to the notion that children with depressed mothers inherit a likelihood for depression per se is the idea that heritability contributes significantly to vulnerabilities to depression and other disorders, such as those proposed in the Goodman and Gotlib (1999) model. In particular, based on behavior genetics studies, high levels of heritability are found for behavioral inhibition and shyness (Cherny, Fulker, Corley, Plomin, & DeFries, 1994), low self-esteem (Loehlin & Nichols, 1976), neuroticism (Tellegen et al., 1988), sociability (Plomin et al., 1993), subjective well-being (Lykken & Tellegen, 1996), and expression of negative emotion (Plomin et al., 1993). This body of research suggests possible genetic transmission of affective, cognitive, and interpersonal vulnerabilities for the development of psychopathology.

Neuroregulatory Mechanisms

Biology may also play a role in the development of psychopathology in offspring of depressed mothers in that both prenatal and postnatal experiences might influence aspects of hormonal functioning and brain development, each of which may increase vulnerability to depression and other stress-related disorders in particular (Ashman & Dawson, 2002). Much of the evidence for this premise is based on research examining the role of early (neonatal) life stress in the development of psychopathology, emphasizing neuroregulatory processes and especially hypothalamic-pituitary-adrenocortical (HPA) axis functioning (Graham, Heim, Goodman, Miller, & Nemeroff, 1999). Typically measured in terms of abnormal cortisol responses to stress but sometimes as a baseline cortisol measure, HPA functioning has been associated with depression in adults (Ressler & Nemeroff, 2000) and may suggest vulnerability mechanisms for the development of depression and other disorders in children of depressed mothers. Stress-related physiological dysregulation in children may be a marker of children's sensitivity to stress, suggesting a particular pathway through which maternal depression during pregnancy and the postpartum may increase children's later development of depression (Sanchez, Ladd, & Plotsky, 2001).

Studies from Field's (2002) lab show that depression during pregnancy increases women's levels of cortisol. Other relevant findings are that pregnant women's levels of cortisol account for 50% of the variance in the fetus's levels of cortisol (Glover, 1999), that mothers' prenatal levels of cortisol predict newborns' cortisol levels (Lundy et al., 1999), and that number of months depressed during pregnancy predict preschool-age children's baseline cortisol levels (Ashman & Dawson, 2002). Elevated stress hormone levels, such as cortisol, in pregnant women are of concern because they have been associated with impaired neurodevelopment in the fetus (Wadhwa et al., 2002).

Postnatal exposure to stressors associated with maternal depression may also disrupt HPA processes (Ashman & Dawson, 2002). In particular, depressed mothers' insensitive or unresponsive behavior, discussed in a later section of this chapter,

may mediate the association between maternal depression and children's emotion regulation abilities, which in turn increases risk for the later development of psychopathology.

Along with disruptions in hormonal functioning, abnormal brain development (in frontal lobe activity) is another aspect of neuroregulatory functioning that may play a role in the development of psychopathology in children of depressed mothers. Researchers have been particularly interested in the abnormal EEG pattern of relatively greater baseline right (compared to left) frontal activation. This pattern is a stable individual difference at least by adolescence (Tomarken, Davidson, Wheeler, & Kinney, 1992) and has been associated with the tendency in children to experience withdrawal emotions such as sadness and fear (Davidson & Fox, 1989) and with depression in adolescents and adults (Davidson, Ekman, Saron, Senulis, & Friesen, 1990). Dawson and colleagues showed associations between frontal lobe activity and affective expressions in infants of depressed mothers (Dawson, Grofer Klinger, Panagiotides, Hill, & Spieker, 1992) and, further, that 3-year-olds' frontal brain activation mediated the relation between maternal depression and children's level of behavior problems (Dawson et al., 2003). That study was not designed to explain the relative (and likely interacting) influences of genetics and fetal and early postnatal experiences on the abnormal brain activation patterns. That is an especially important question given that the frontal lobe develops rapidly during the first 2 years of life (Chugani & Phelps, 1986), suggesting sensitivity to early adverse experiences such as exposure to a mother whose depression interferes with her providing the sensitive and responsive parenting required by infants. Also needed is more research on the extent to which the same or different predictors explain children's stress hormone functioning relative to their brain activation patterns. In 6-month-old infants, these two aspects of neuroregulatory functioning were found to be related; higher basal and stressor reactive cortisol levels were associated with relatively greater right EEG asymmetry (Buss et al., 2003).

Biological Mechanisms for Effects of Alcohol-Related Disorders in Fathers

Work on biological mechanisms that might explain the development of psychopathology in children of alcoholic fathers has focused primarily on genetics but also on electricortical arousal. Support for both of those lines of research is reviewed next.

Genetic Influences

Alcohol dependence is known to be highly heritable, with twin and adoption studies demonstrating that between 50% and 60% of phenotypic variations can be accounted for by a genetic component (Higuchi, Matsushita, & Kashima, 2006; McGue, 1999). Classic twin study estimates of the heritability of alcoholism in men are higher but also more variable (48% to 73%) than in women (51% to 65%; Tyndale, 2004). The variability is likely explained by differences in the classification of alcoholism subtypes. Cloninger and colleagues (Cloninger, Bohmann, & Sigvardsson, 1981) classified alcoholism into two subtypes associated with

variations in gender-specific heritability. Type I alcoholism (milieu-limited) is moderately heritable for both males and females, whereas Type II alcoholism (male-limited) is strongly associated with genetic factors in male offspring and is associated with an early age of onset and with elevated rates of externalizing behavior problems in fathers (Sigvardsson, Bohman, & Cloninger, 1996). Overall, the genetic contribution to alcohol dependence in males has been very well established (for review, see Higuchi et al., 2006) and is likely to be a particularly important mechanism in the transmission of alcoholism from fathers to sons. Genetic factors are also likely to bear primary responsibility for the association between paternal alcoholism and child Conduct Disorder (Haber, Jacob, & Heath, 2005).

Although traditional twin and adoption studies are particularly valuable for estimating heritability, they also play an essential role in revealing the relative contributions of shared and nonshared environment (Hopfer, Crowley, & Hewitt, 2003). Of particular interest from a psychosocial research perspective is altered family environments associated with paternal alcoholism and their role, independent of or in interaction with genes, in explaining the development of alcoholism (Jacob et al., 2003). In an effort to address this broader question, Jacob and colleagues used an offspring-of-twins design and assessed adult male monozygotic and dizygotic twins concordant or discordant for alcohol dependence and their biological offspring, examining both genetic and familial influences. Consistent with other genetics studies, offspring of alcoholic fathers were at a significantly higher risk for the development of an alcohol use disorder than were offspring of controls. Interestingly, however, the researchers found that the complete absence of family environmental correlates of alcoholism, even when high genetic risk is present, resulted in offspring alcoholism levels not significantly different from those of controls. If these findings are replicated, it could show that a genetic risk for alcoholism is not sufficient for offspring alcoholism to develop in the absence of significant environmental sequelae (Jacob et al., 2003). The relative contribution of heritability and environment may also vary depending on the particular outcome of interest and on developmental considerations. For example, in a large study of 16-year-old twins, shared environmental effects best explained drinking initiation, whereas genetic effects better explained frequency of drinking among subjects who had already started drinking (Viken, Kaprio, Koskenvuo, & Rose, 1999).

Despite the strong evidence for a genetic contribution to alcohol dependence in males, specific genes that alter the risk have been difficult to identify. This is partially due to the additional influence of environmental components as well as the understanding that multiple genes that have moderate effects are involved in alcoholism, and these different sets of genes may contribute to the disorder in different people (Higuchi et al., 2006). Thus more research is needed not only on the nature and extent of genetic contributions but also on the likely complex interplay between genes and environments.

Other Biological Mechanisms

Much of the research examining possible psychobiological risk factors for the development of alcoholism has focused on individual differences in electricortical

arousal, most often using the P300 responses. P300 is a positive event-related potential response wave that in itself is known to be highly heritable. The electricortical research has been focused on the P300 response because of its identification in working memory and attentional resource allocation and its reduced amplitude in recovering alcoholics (Windle & Tubman, 1999). To determine whether this reduced amplitude is a consequence of frequent inebriation or could be considered a risk factor for the development of alcoholism, children of alcoholics have been studied prior to their onset of drinking. Numerous such studies (see Windle & Tubman, 1999, for review) have found significant differences between children of alcoholics and children of nonalcoholics, with the children of alcoholics showing a reduced P300 response in visual task paradigms, which may generalize across the visual and auditory sensory modalities. Benegal and colleagues (Benegal, Jain, Subbukrishna, & Channabasavanna, 1995) took this a step further by showing that the P300 component was even smaller in children with multiple alcoholic relatives in comparison with children with only a single relative. These findings are consistent with the idea that reduced P300 amplitude is a risk factor for the development of an alcohol-related disorder.

Anecdotal reports from alcohol dependent children of alcoholics led researchers to consider one other possible biological mechanism for the development of alcoholism in children of alcoholic fathers. These narratives frequently describe children of alcoholics' abilities from a young age to consume large amounts of alcohol with few effects. Schuckit and colleagues (1996) examined the possible role of differences in alcohol tolerance by assessing time-to-peak alcohol concentration and the rate of disappearance of ethanol from the blood following the challenge in 453 sons of alcoholics and controls. Because of the confounding effect of fetal alcohol syndrome, the authors placed an emphasis on examining offspring of alcoholic fathers. There were no significant differences in blood alcohol concentration in the two groups, yet about 40% of the sons of alcoholics, but fewer than 10% of controls, reported low subjective feelings of intoxication (Schuckit & Smith, 1996). This finding has been replicated in many studies as well as confirmed by a meta-analysis (Pollock, 1992) and gives weight to the theory that sons of alcoholics possess less subjective (self-rated) sensitivity to alcohol effects than do controls. These findings have been interpreted to suggest that these individuals are more vulnerable to developing an alcohol-related disorder because they are less sensitive to alcohol effects and therefore find it more difficult to regulate their alcohol consumption because they are receiving less internal feedback. Thus, they may be more likely to consume more alcohol than those without a family history of alcoholism, which may contribute to their development of dependence or abuse (Pollock, 1992). The data do not allow us to conclude whether the findings are specific to sons or might generalize to daughters.

Quality of Care Mechanisms for the Effects of Maternal Depression

Social Learning Processes

Exposure to depressed mothers' negative affect, behavior, and cognitions is another potential mechanism for transmission of the risk of depression from parents

to offspring (see Figure 30.1). Through the social learning processes of modeling, observational learning, and reinforcement, children of depressed mothers may acquire cognitions, behaviors, and affect that resemble those of the depressed parent. These acquired depressotypic cognitions, behaviors, and affect may place the children at elevated risk for developing depression in particular, but also potentially other disorders.

Children have ample opportunities to be exposed to negative cognitions, behaviors, and affect associated with depression in their mothers. In terms of affect and behavior, a meta-analysis of 46 observation studies revealed that maternal depression is strongly associated with negative (hostile or coercive) parenting behavior, is moderately associated with parenting characterized as disengaged, and shows a small association with lower levels of positive behavior (Lovejoy et al., 2000). In terms of cognitions, the negatively biased cognitions associated with depression in adults (Gotlib, Gilboa, & Sommerfeld, 2000) emerge in parenting in depressed mothers' (a) more negative views of themselves as parents (Gelfand & Teti, 1990; Goodman, Sewell, Cooley, & Leavitt, 1993), (b) lower confidence in being able to positively influence their children (Kochanska, Radke-Yarrow, Kuczynski, & Friedman, 1987), and (c) more negative, critical perceptions of their children (Goodman, Adamson, Riniti, & Cole, 1994) relative to controls.

Researchers have linked these specific aspects of depression in mothers with the development of psychopathology in children and adolescents. Depressed mothers' negative, critical perceptions of their children are associated with children's self-blaming attributions for negative events and with lowered perceived self-worth (Goodman et al., 1994; Jaenicke et al., 1987; Kochanska et al., 1987; Radke-Yarrow, Belmont, Nottelman, & Bottomly, 1990). Children seem to internalize their depressed mother's negative views of them, thereby acquiring a cognitive vulnerability to depression (Beck, Rush, Shaw, & Emery, 1979). Depressed mothers' greater criticism and hostility (Frye & Garber, 2005; Webster-Stratton & Hammond, 1988) may also contribute to the emergence of coercive processes similar to those implicated primarily in the development of externalizing behavior problems but also of depression (Davis, Sheeber, & Hops, 2002; Hops, Sherman, & Biglan, 1990).

Inadequate Parenting

In addition to social learning processes, other mechanisms that may explain how qualities of parenting contribute to the development of problems in children focus on inadequate parenting (especially if early in life) as a stressor. Although not incompatible with social learning processes, these studies stem from a different tradition of research, emphasizing interpersonal processes such as the depressed mother being an inadequate social partner for the child and the mother being unable to meet the child's stage-salient social and emotional needs (Cicchetti & Toth, 1998). This inadequate parenting would, in turn, negatively affect the child's development of social and cognitive skills and beliefs and leave him or her vulnerable to the development of psychopathology.

In this line of research, inadequate parenting refers to the mother being unable to meet the needs associated with healthy psychological development specific to different phases of child development. Researchers have particularly emphasized

infancy given infants' dependence on their caregivers to meet their basic needs (Sroufe, Egeland, Carlson, & Collins, 2005). This research is consistent with the idea that depression in mothers is a stressor for their infants, and thus is related to the literature on HPA activity and EEG asymmetries reviewed earlier. Mothers may be stressors as a function of being unresponsive or inconsistent (unpredictable in their responsiveness) in relating to their infants, thereby interfering with infants' development of emotion regulation skills (Fox, 1994) and secure attachment relationships (Cicchetti, Rogosch, & Toth, 1998; Sroufe & Waters, 1977). Both of the latter are known to increase risk for the development of psychopathology in childhood or adolescence (Southam-Gerow & Kendall, 2002; Thompson, 2006).

Stress

In addition to the notion of parenting as a stressor, especially for infants, maternal depression increases children's exposure to stress in several other ways, which also may mediate associations between maternal depression and the development of psychopathology in offspring (Hammen, 2002). These include the stressfulness for children of the symptoms and episodic course of depression in their mothers (Compas, Langrock, Keller, Merchant, & Copeland, 2002), the chronic and episodic stressors that are often the context for depression (Monroe & Hadjiyannakis, 2002), and the stress-generating quality associated with depression (Hammen, 1991). Although these findings are consistent with the notion of stress as a mechanism, few studies have actually tested stress as a mediator in associations between depression in mothers and outcomes in children. One of the few such studies found support for the role of stressors as a mediator of the association between maternal depression and children's depression (Brennan, Hammen, Katz, & Le Brocque, 2002).

Quality of Care Mechanisms for the Effects of Alcoholism in Fathers

Because mothers are still the primary caregivers in most families, much of the research about alcoholism and parenting focuses specifically on the effects of maternal alcoholism or on familial alcoholism. Although we emphasize studies focusing solely on paternal alcoholism, we included studies involving familial alcoholism to illustrate the effect of alcohol-related disorders on quality of care given to the child.

Although there is a large range in quality of caregivng in families with an alcoholic parent, on average researchers find significantly poorer quality compared to controls. In the few studies involving observations of father-child interaction, paternal alcoholism was found to be associated with problematic parenting behaviors during unstructured free play with an infant, with alcoholic fathers showing higher negative affect, less sensitivity, lower positive engagement, and lower amount and quality of verbalizations relative to controls (Eiden, Edwards, & Leonard, 2002; Leonard & Eiden, 2002). Children of alcoholic parents are also exposed to significantly greater adverse circumstances. Similar to the studies of the role of stress as a mechanism in children of depressed mothers, one model in alcoholism studies is that the problems in family functioning associated with alcoholism in a parent

influence the expression of internalizing and externalizing problems among children of alcoholics by acting as a significant stressor for the child (Windle & Tubman, 1999). In homes where one or both parents are alcoholic, both child and parent reports reveal significantly poorer quality of home environment and more household dysfunction as well as exposure to more marital conflict, parent-child conflict, abuse (verbal, physical, and sexual), and domestic violence than in homes where neither parent is an alcoholic (Anda et al., 2002; Reich, Earls, & Powell, 1988). Among these adversities, much research has focused on the role of marital conflict as one of the mechanisms through which children of alcoholic parents are harmed. Marital conflict is related to ineffective parenting (including inconsistent discipline and psychological control), which in turn is associated with children's internalizing and externalizing problems (Cummings, Keller, & Davies, 2005). Thus parents' problematic drinking may be associated with children's behavior problems indirectly, through mediating variables such as marital conflict (Christensen & Bilenberg, 2000; Snow Jones, Miller, & Salkever, 1999).

Patterson and colleagues' (Patterson, Reid, & Dishion, 1992) well-established coercion model, which explains the relationship between parenting influences and childhood conduct disorders, is a useful theoretical model in that children's conduct problems increase the likelihood of alcohol or drug abuse. In this model, the most relevant force in determining a child's development of psychopathology is disturbances in the control dimension of parenting, especially a coercive parent-child interactional style, similar to the pattern that has been identified in studies of parenting in homes with an alcoholic parent. This style is marked by unclear behavioral expectations from the parent and lack of compliance by the child. The continuation of this coercive relational style results in inadequate parental monitoring, supervision, and discipline, which can lead to antisocial behavior in the child, which, in turn, increases the likelihood of the child's alcohol abuse (Patterson et al., 1992).

EVIDENCE FOR EFFECTS OF MATERNAL DEPRESSION AND PATERNAL ALCOHOLISM ON CHILDREN

Given the size and complexity of the literature, we begin with evidence for the effects of maternal depression and then separately review evidence of the effects of paternal alcoholism. Within each of these sections, we present data on effects on children's social, emotional, and cognitive development, which may be vulnerabilities for the later development of psychopathology, and then on psychopathology per se.

Evidence for Effects of Maternal Depression

Effects on Children's Social, Emotional, and Cognitive Development

Children of depressed mothers show problems in each of these aspects of functioning relative to controls. On social and interpersonal functioning, infants show less secure attachment relationships (van IJzendoorn, Goldberg, Kroonenberg, & Frenkel, 1992); preschool-age children engage in excessive compliance with their

depressed mothers (Zahn-Waxler, Radke-Yarrow, Wagner, & Chapman, 1992); kindergarten-age children are more often excluded by peers (Cummings et al., 2005); and sons, but not daughters, display more aggressive behavior during interactions with friends (note that the last effect was mediated by the children's exposure to interparental conflict; Hipwell, Murray, Ducournau, & Stein, 2005).

On emotional development, infants imitate their depressed mothers' negative affect (Field, 1994). Infants of depressed mothers show more negative affect and more self-directed regulatory behaviors (Field, 1992; Tronick & Gianino, 1986); toddlers show more dysregulated aggression and heightened emotionality (Zahn-Waxler, Cummings, Iannotti, & Radke-Yarrow, 1984); and adolescents (particularly girls) display more dysphoric and less happy affect (Hops et al., 1990). Infants interacting with their depressed mothers also engage in more gaze and head aversion, consistent with the idea that the infants were using self-regulatory behaviors to minimize the negative affect associated with maternal unresponsiveness (Pickens & Field, 1993).

Children of depressed mothers score lower on measures of intelligence and also have poorer academic performance overall (Anderson & Hammen, 1993; Hammen & Brennan, 2001; Hay & Kumar, 1995; Jaenicke et al., 1987). On other aspects of cognitive functioning, children of depressed mothers show early signs of cognitive vulnerability to depression, including being more likely than controls to blame themselves for negative outcomes, having a more negative attributional style, being less likely to recall positive self-descriptive adjectives, and having a lower self-concept (Garber & Martin, 2002).

Effects on the Development of Psychopathology

Beginning at the earliest ages when it is possible to reliably measure psychopathology in children, children and adolescents with depressed mothers show higher rates of depression relative to a variety of controls (Beardslee et al., 1988; Billings & Moos, 1985; Goodman et al., 1994; Lee & Gotlib, 1989; Malcarne, Hamilton, Ingram, & Taylor, 2000; Orvaschel, Walsh-Allis, & Ye, 1988; Weissman et al., 1984; Welner, Welner, McCrary, & Leonard, 1977). Rates of depression in the school-age children of depressed mothers vary from 20% to 41%, depending on the severity or impairment of the parent's depression, whether both parents are depressed, and a number of other sociodemographic variables (Gotlib & Goodman, 1999). Moreover, depression in offspring of depressed parents, compared with same-age offspring of nondepressed parents, has an earlier age of onset and longer duration and is associated with greater functional impairment and higher likelihood of recurrence (Hammen & Brennan, 2001; Hammen, Burge, Burney, & Adrian, 1990; Hammen, Shih, Altman, & Brennan, 2003; Keller et al., 1986; Lieb, Isensee, Hofler, Pfister, & Wittchen, 2002; Warner, Weissman, Fendrich, Wickramaratne, & Moreau, 1992).

Psychopathology in children with depressed mothers is not limited to depression. Researchers have also found higher rates of Social Phobia, Separation Anxiety Disorder and other anxiety disorders, attention deficit disorders, disruptive behavior disorders, overall levels of behavior problems, and poorer social functioning relative to controls (Anderson & Hammen, 1993; Biederman et al., 2001; Luoma et al., 2001;

Orvaschel et al., 1988; Weissman et al., 1984). In addition, levels of internalizing and externalizing behavior problems are higher in children with depressed mothers relative to controls (Brennan et al., 2002; Fergusson, Lynskey, & Horwood, 1993; Forehand, Brody, Slotkin, Fauber, & McCombs, 1988).

Evidence for Effects of Paternal Alcoholism

Effects on Children's Social, Emotional, and Cognitive Development

Similar to the findings from maternal depression, paternal alcoholism is associated with problematic patterns of father-infant attachment: infant avoidance and infant disorganization as assessed by the Strange Situation paradigm (Eiden et al., 2002; Leonard & Eiden, 2002; Leonard et al., 2000). Teenagers (18 years old) whose fathers were alcoholics rated their attachment to their fathers as significantly more negative than teenagers of nonalcoholic fathers (Cavell, Jones, Runyan, Constantin-Page, & Velasquez, 1993).

In terms of cognitive development, while there is strong empirical support for the severe cognitive deficiencies caused by maternal drinking during pregnancy (Steinhausen, Willms, & Spohr, 1993), the literature surrounding the impact of paternal alcoholism on a child's cognitive functioning is varied. Leonard and Eiden (2002) found no significant difference in the cognitive functioning of toddlers of alcoholic fathers, assessed at 12, 18, and 24 months, relative to toddlers in control families who were matched on maternal education, ethnicity, child sex, and marital status. However, children of alcoholics had poorer intellectual functioning than control children during the elementary school years, although still within the normal range (Poon, Ellis, Fitzgerald, & Zucker, 2000).

When Ozkaragoz, Satz, and Noble (1997) examined differences in neuropsychological functioning of sons of alcoholic fathers, recovering alcoholics, and social drinkers, they found that sons of active alcoholics performed worse on visuospatial, memory, and attention tasks as well as general intellectual functioning. However, no difference in neuropsychological functioning was found between the sons of the recovering alcoholics and those of social drinkers (Ozkaragoz et al., 1997). The latter finding is relevant to moderators, discussed later.

Effects on the Development of Psychopathology

Many studies have shown the link between paternal alcoholism and alcohol-related outcomes in children (Cavell et al., 1993; Chassin, Rogosch, & Barrera, 1991). Paternal alcoholism predicted initial alcohol and drug use in younger adolescents (Chassin et al., 1991) as well as greater alcohol involvement in older adolescents (Cavell et al., 1993). Overall, family alcoholism is associated with a threefold increase in risk for alcohol abuse in young adults (Schuckit & Smith, 1996).

Beyond an increased risk for developing an alcohol-related disorder, a family history of alcoholism put children at a higher risk for developing other psychopathologies, including Conduct Disorder, affective disorder, and Attention-Deficit/Hyperactivity Disorder (ADHD). Edwards, Eiden, and Leonard (2006) found that both male and female 18-month-old toddlers with alcoholic fathers had

higher levels of not only externalizing behaviors but also of internalizing behaviors relative to children of nonalcoholic fathers. Early adult (mean age 19) sons of alcoholic-dependent fathers were also significantly more likely to receive a diagnosis of Conduct Disorder than were matched controls from families without a history of alcoholism, indicating a diagnostic crossover in family transmission (Haber et al., 2005).

Many researchers have hypothesized that one potential pathway for the transmission of the risk of alcoholism from parent to child is the likelihood of early conduct problems among children of alcoholics, which has been linked to greater antisocial behavior in adolescence and adulthood, which, in turn, has been linked to a greater risk of substance abuse (Eiden, Leonard, & Morrisey, 2001). In a study examining toddler compliance at 18 and 24 months and comparing children of alcoholics and children of nonalcoholics, Eiden et al. showed that fathers' alcoholism was significantly associated with the development of compliance problems among boys, an effect more pronounced in the presence of maternal alcohol problems. These boys showed increased levels of resistance to parental requests (during a cleaning-up task) at 18 months and continued to show a higher level of resistance at 24 months. Contrary to expected results, daughters of alcoholic fathers showed the opposite pattern, even when the mother was also an alcoholic. The authors pointed out that the mothers with alcoholism were also more antisocial and aggressive than controls, and therefore the daughters could be displaying compulsive compliance. In general, frequencies of disruptive behavior problems in young children tend to decrease over time, most likely due to increases in self-regulatory capacities, which are developed during the preschool years. However, this decrease can be offset by the presence of paternal alcoholism, as Loukas and colleagues (Loukas, Zucker, Fitzgerald, & Krull, 2003) showed that paternal alcoholism was associated with higher levels of disruptive behavior problems across all time points.

MODERATORS

The processes underlying the developmental pathways that lead to psychological problems in children of depressed mothers or alcoholic fathers are likely to be affected by multiple moderating variables, which not only create individual pathways unique to each child but also more generally exacerbate or lessen the extent of risk associated with the psychopathology in the parent (Kraemer, Stice, Kazdin, Offord, & Kupfer, 2001). A transactional perspective suggests that through a process of continuous reciprocation, characteristics and traits of the child influence how the child experiences the environment, which, in turn, influences the child's characteristics and traits and so forth (Sameroff, 1975). Thus, from the perspective of transactional models, various factors may moderate the associations between psychopathology in a parent and children's development and account for some children functioning well and others developing psychopathology and other problems. These factors may include (a) qualities of the children, such as temperament, sex, and various cognitive variables; (b) the broader context of the children's environment, such as the presence and mental health of the other parent or another adult and the quality of the marital relationship; and (c) characteristics of the parent's psychopathology,

such as timing, course, and chronicity. Much heterogeneity has been noted in studies of children of depressed mothers and of alcoholic fathers. Moderating variables promise to explain much of that heterogeneity. Empirical support for each of these potential moderators is reviewed next, first for children of depressed mothers and then for children of alcoholic fathers.

Moderators of Associations between Maternal Depression and Children's Development

Qualities of the Children

The sex of the offspring of depressed mothers is one characteristic of the child that has been shown to moderate the association between mothers and limited aspects of children's development. Specifically, sex moderates associations between maternal depression and depression in adolescents (Sheeber, Davis, & Hops, 2002). Adolescent daughters of depressed mothers have more dysphoric and less happy affect than adolescent daughters of nondepressed mothers, whereas no significant difference in affect was found for adolescent boys of depressed mothers compared to sons of nondepressed mothers (Allgood-Merten, Lewinsohn, & Hops, 1990). Among the smaller body of support for sex differences among younger children, depressed mothers and their daughters, but not depressed mothers and their sons, have been shown to have synchronous bouts of depression (Radke-Yarrow, Nottelman, Belmont, & Welsch, 1993). Similarly, in community samples, levels of depression in mothers and daughters are interrelated, but levels of depression in mothers and sons are not related (Davies & Windle, 1997; Fergusson, Horwood, & Lynskey, 1995). Increased risk for depression in daughters compared to sons of depressed mothers is likely to be explained by the same mechanisms that explain risk for depression in adolescent girls compared to adolescent boys in general populations: gender-stereotyped socialization processes, especially early socialization of emotion expression, coping styles, and relationship orientation and possibly greater exposure of girls to family discord (Goodman & Tully, 2006; Sheeber et al., 2002).

Children's age is likely a complex moderator, perhaps most likely interacting with other moderators such as sex, as illustrated in Sheeber et al. (2002). In a meta-analysis, we found a negative correlation between children's age and effect size such that maternal depression was more strongly associated with younger children's internalizing and externalizing problems (Connell & Goodman, 2002). Age is also further complicated by the course of maternal depression in that older children are likely to have been more often exposed.

Researchers have studied children's cognitive abilities, such as IQ, and cognitive vulnerabilities, such as depressogenic attributions, as child qualities that may moderate the link between depression in mothers and children's psychological functioning. Children of depressed parents may be protected against adverse outcome concurrently and prospectively if they are more intelligent (Radke-Yarrow et al., 1990) or have better social-cognitive skills (Beardslee, Schultz, & Selman, 1987). Cognitive theories of depression suggest other potential moderating variables, such as children's perceptions and beliefs about their parents' depression, which may guide children's emotional and coping responses and, then, influence the impact of

the mother's depression on the children's development (Beardslee, 1989; Beardslee & Podorefsky, 1988; Compas et al., 2002; Garber & Martin, 2002; Klimes-Dougan & Bolger, 1998; Solantaus-Simula, Punamaki, & Beardslee, 2002).

Although studies have rarely directly tested the moderating role of temperament in associations between depression in mothers and their offspring, research suggests that children with difficult temperaments are more vulnerable to the effects of inadequate parenting, such as that shown by depressed mothers (Goldsmith, Buss, & Lemery, 1997). Other indirect support is that mothers' higher levels of depressive symptoms have been associated with their perceptions of more infant negativity (Pesonen, Raikkonen, Strandberg, Keltikangas-Jarvinen, & Jarvenpaa, 2004) and difficulty (Cutrona & Troutman, 1986; Edhborg, Seimyr, Lundh, & Widstrom, 2000) and with observational measures of infant difficulty (Cutrona & Troutman, 1986). Mothers of more difficult infants also perceive their parenting to be less efficacious, which in turn is linked to depression in mothers (Cutrona & Troutman, 1986; Porter & Hsu, 2003).

The Broader Context of the Children's Environment

Studies of depression are increasingly taking into account the broader context in which depression occurs (Keyes & Goodman, 2006). Researchers sensitive to the importance of context are beginning to find that fathers (their quantity and quality of involvement and their mental health) and marital conflict play important roles as moderators of associations between maternal depression and child outcomes (Goodman & Gotlib, 1999). Assortative mating, the tendency for adults with similar phenotypes to mate more frequently than expected by chance, is pertinent here. This type of mating practice can produce a double jeopardy situation for the children: Not only are the children inheriting a greater genetic predisposition toward psychopathology, but they are also being raised in a home environment that is likely to be highly compromised and have high levels of unpredictability and low levels of parental monitoring (Merikangas, Weissman, Prusoff, & John, 1988; Windle & Tubman, 1999). Assortative mating, in particular the selection of a mate who also has a mood disorder, is known to be a significant factor in mate selection for women with affective disorders (although not as much as for men; Matthews & Reus, 2001). In studies of maternal depression there is some evidence for the ameliorating effect of fathers' mental health (Thomas & Forehand, 1991), although other studies suggest a more complex model may be needed, such as one that takes into account transactional relationships over time (Hops et al., 1990). In addition, an adequate conceptualization of the role of fathers needs to take into account quality and quantity of involvement.

Marital discord is essential to consider in that it is a common contextual factor for depression in women and by itself is related to increased risk for psychopathology in children (Whisman, Weinstock, & Tolejko, 2006). Davies and Windle (1997) found that marital discord significantly mediated the association between histories of maternal depression symptoms and symptoms of depression in children at middle adolescence, but only in girls. Later, Davies and Dumenci (1999) found support for a pathway from marital distress through maternal depression (rather

than vice versa) to adolescents' depression. Moreover, particular conflict strategies that parents use (those involving depressive behaviors such as physical distress, withdrawal, sadness, and fear in contrast to destructive or constructive marital conflict strategies) mediate associations between maternal depression and children's depression (Du Rocher Schudlich & Cummings, 2003). These findings strongly support the importance of context and the need to develop and test models that consider its potentially moderating and mediating roles.

Parental Characteristics as Moderators

Timing and course of mothers' depression may also account for differences in risk among children of depressed mothers. The exact role of timing is likely to be complicated and may vary by child sex (Essex, Klein, Miech, & Smider, 2001). Timing and course are interrelated given that children who are exposed earlier in their lives are also likely to be exposed more often, not only to the depression but also to the stress correlates of depression, such as marital difficulty, all of which place the child at greater risk for psychopathology. Given changes in children's stage-salient needs, the timing of depression episodes in mothers may have implications for the development of specific vulnerabilities in their children. Finally, we know little about the effect on children of the unpredictable course of their mother's depression.

Moderators of Associations between Paternal Alcoholism and Children's Development

Qualities of the Children

As in studies of maternal depression, children's sex, age, and temperament have been studied as moderators of associations between paternal alcoholism and child outcomes. With regard to sex, the literature is limited by the majority of studies on the effect of paternal alcoholism on children having excluded daughters. Although it is commonly concluded that sons of alcoholics have a higher risk for the development of an alcohol-related disorder (Assaad et al., 2003), some studies have shown greater family history effects for women than for men (Sher, Walitzer, Wood, & Brent, 1991). Studies that included both sons and daughters of alcoholics have found support for a moderating role of child sex on the effect of paternal alcoholism. Paternal alcoholism had a greater effect on sons' relative to daughters' risk of developing ADHD depressive symptoms and substance use during adolescence (daughters of alcoholic mothers were more at risk for depression than sons; Chassin, Curran, Hussong, & Colder, 1996; Christensen & Bilenberg, 2000). These findings are consistent with both social-modeling theory, in that alcoholism in the parent of the same sex may have the most negative influences, and genetics in terms of male-specific inheritance.

In terms of age, Connell and Goodman's (2002) meta-analysis of the effect of parental psychopathology on children found that paternal alcoholism was more closely related to emotional and behavior problems in samples of older children, a finding opposite to that for maternal depression. This finding is difficult to interpret given the chronicity of alcoholism in that the older children were likely to have been

exposed for much or all of their lives; the finding may reflect cumulative effects of exposure.

As with maternal depression, we found few studies directly testing temperament as a moderator of the effect of parental alcoholism. In one study, among children of substance abusing fathers, those who had a difficult temperament, relative to others, were at higher risk for the development of a substance abuse disorder (Tarter, Kabene, Escallier, Laird, & Jacobs, 1990).

The Broader Context of the Children's Environment

Assortative mating is at least as relevant to studies of children of alcoholic fathers as with children of depressed mothers (Windle & Tubman, 1999). When studies have examined children raised in families with two alcoholic parents versus one or no alcoholic parents, children of two alcoholic parents were found to have far more behavior problems and earlier age of alcohol intoxication than children raised in families with only one alcoholic parent (McKenna & Pickens, 1983). In one study, for example, in households where both parents met criteria for current alcoholism or at least one parent had an antisocial alcoholic diagnosis, the children were more reactive and showed more externalizing behaviors relative to the children who only had one alcoholic parent or those with no parent with an antisocial alcoholic diagnosis (Wong, Zucker, Puttler, & Fitzgerald, 1999). As well as assortative mating with other alcoholics, partner pairings may occur between alcoholics and partners with another psychiatric disorder such as depression. Conversely, as with maternal depression, a child with an alcoholic father may benefit from an involved, mentally healthy mother. Edwards and colleagues (2006) found that 18-month-old toddlers with alcoholic fathers who had a secure infant-mother attachment at 12 months had fewer externalizing behavior problems at 24 months and fewer externalizing and internalizing problems at 36 months of age, showing a protective factor against the risk of paternal alcoholism.

Parental Characteristics as Moderators

Much of the variability in the literature on associations between paternal alcoholism and child outcomes could be due to authors either failing to identify or failing to test for differences within samples among types of alcoholics, frequency and quantity of drinking, presence of comorbid Antisocial Personality Disorder, and inclusion of recovering alcoholics. In one study of severity of alcoholism, Malone and colleagues (Malone, Iacono, & McGue, 2002) examined the effect of paternal alcoholism on child behavior using a measure of the maximum number of drinks the father had ever consumed in a 24-hour period. The authors suggest that individuals who consume a high number of alcoholic beverages may have developed a significant tolerance to the effects or possess an unusual insensitivity to the negative effects of alcohol. The maximum number of drinks consumed could be related to loss-of-control drinking, a distinguishing feature in Type 1 alcoholism using Cloninger's (1987) classification. Malone et al. found that a father's maximum number of drinks predicted children's externalizing disorders, especially Conduct Disorder and early substance use.

Other researchers studied the effect of severity or chronicity of parental alcoholism on child functioning by comparing children of active alcoholics to those of recovered alcoholics and social drinkers (DeLucia, Belz, & Chassin, 2001; Ozkaragoz et al., 1997). Ozkaragoz et al. compared children who had parents in these three groups and found that only children of active alcoholics showed neuropsychological deficits and reduced intellectual functioning compared to children of recovered alcoholics and social drinkers. On the other hand, when behavioral and emotional problems were compared, there were no group differences between active and recovered alcoholics' children, but these children had significantly more emotional problems than did the children of the social drinkers. DeLucia et al. found that adolescents of recovered alcoholics continued to show more emotional and behavioral problems than children of nonalcoholic parents. Because alcohol abuse and dependence are chronic conditions that can span decades, more studies are needed of the effects of chronicity.

Because many men with alcoholism also display antisocial personality traits, it is essential, though difficult, to tease apart the effect of paternal alcoholism from the effect of the antisocial traits on parental caregivng. For example, Eiden and Leonard (2000) found a significant impact of paternal alcoholism on fathers' aggravation toward their infants; however, this finding was found to be attributable to the antisocial behavior and depressive symptoms. In research on cognitive-intellectual functioning of children of alcoholic fathers, the children of antisocial alcoholics had significantly lower Full-Scale IQ scores (although still within the normal range) than children of nonantisocial alcoholics and controls (Poon et al., 2000). Further, the child's IQ was directly related to the severity of comorbid paternal pathology.

SUMMARY

The field of clinical psychology has increasingly recognized the importance of psychopathology in both mothers and fathers in relation to the development of psychopathology in children (Connell & Goodman, 2002; Phares & Compas, 1992). The goal of this chapter was to examine the effect of parental psychopathology on the development of psychopathology in children, specifically focusing on depression in mothers and alcohol disorders in fathers, as they are two of the most prevalent disorders in adults who are bearing and rearing children. This chapter adds to the emerging literature about the risks to the children, and while providing many answers also raises multiple questions needing to be addressed.

In bringing attention to this literature, we encourage researchers, clinicians, and public policy makers to consider the centrality of sex in all aspects of the issues raised here. First among those issues is the fact that depression is twice as common in women as men, whereas alcohol dependence and the often accompanying Antisocial Personality Disorder are at least twice as common in men as women. These epidemiological patterns suggest that important work needs to be done to balance efforts to minimize sex bias in the recognition, diagnosis, treatment, and study of these disorders, while at the same time recognizing sex-specific aspects of the disorders such as risk factors, symptom patterns, approaches to help seeking,

treatment options offered, barriers, and response to treatment. In this regard, as one example, sex specificity of genetic risks for depression (Kendler & Prescott, 1999) and alcohol abuse (Sigvardsson et al., 1996) are intriguing. Second, sex biases prevail in the studies of these high-risk children, as exemplified by the minimal attention to daughters of alcoholic fathers. Researchers may eventually conclude that sex specificity characterizes some risk and protective factors and pathways in the development of psychopathology in children of parents with psychopathology and identify others that characterize all children. An interesting likelihood is multifinality, the idea that similar risk factors, for example, uncontrollable life events, may increase the likelihood of different outcomes: depression emerging in women and alcohol abuse in men (Cicchetti & Rogosch, 1996). Third, even with the strong evidence of sex-specific prevalence patterns for depression and alcohol abuse in men and women, it is essential that models of the transmission of risk to the children take into account the fact that depression in mothers also affects the husbands and fathers, just as alcohol abuse in men affects the wives and mothers (El-Sheikh & Flanagan, 2001; Whisman et al., 2006). Fourth, researchers and practitioners need to be attentive to the emergence of changing epidemiological patterns such as the possibility that rates of depression worldwide are increasing more rapidly in men than in women (Klerman & Weissman, 1989), and women's rates of alcohol abuse, conversely, may increase worldwide as gender norms shift.

The newest work on identification of specific genes and other biological markers for psychiatric disorders offers promise for fresh insights into both alcoholism and depression research, especially if the work is developed in the context of biopsychosocial models and with sensitivity to developmental issues. Throughout the literature on maternal depression and paternal alcoholism, transactional models have emerged as a valuable tool in examining both the mediators for the effect of the parental psychopathology as well as evaluating potential moderators of risk. More research examining transactional effects has tremendous promise for revealing developmental pathways to disorder in children at risk and can benefit from increasing sophistication in longitudinal research design and approaches to data analysis. The findings reviewed here reveal the value of identifying the earliest emerging problems in children of parents with depression or alcohol abuse. Work based on the gateway hypothesis is useful in this regard (Kandel, Yamaguchi, & Klein, 2006). Both risk factor research and evidence-based preventive interventions will also benefit from the appreciation of interactions among risk factors as well as transactional patterns in pathways to disorder.

Research on the effect of paternal alcoholism on child development is not a new line, yet recently has not garnered as much attention as depression in mothers. By using the structure of the integrative model for the transmission of risk (Goodman & Gotlib, 1999; Figure 30.1), we have shown that many of the mechanisms and moderators found in the transmission of risk to children of depressed mothers also hold true for the transmission of risk to children of alcoholic fathers. The model bears strong resemblance to Windle and Tubman's (1999) model for the transmission of alcoholism, with several coinciding mechanisms and moderators, further suggesting that these two areas of research can learn from each other. The overlap between these two models and the empirical support reviewed here for common mechanisms

for the transmission of risk to children as well as moderators of the risk gives credibility to the potential for an overarching general model of the transmission of risk to children, regardless of the specific psychopathology of the parent. Such an approach has the benefit of recognizing common comorbidities with other disorders (including alcoholism with depression in both men and women), the tendency for risk factors to accumulate, and the essential role of context. While research will undoubtedly benefit from continued emphasis on each specific disorder, preventive mental health efforts may well also benefit from a focus on commonalities across disorders. An overarching model of risk would have practical screening, prevention, and treatment implications for children who are at risk for the development of psychopathology due to parental psychopathology. This model would facilitate the development of general approaches to prevention and treatment aimed at reducing the child's likelihood of the development of psychopathology and could be a cost-effective method of identifying and treating the children most at risk.

An essential message to clinical psychologists from the work reviewed in this chapter is that both maternal depression and paternal alcoholism confer not only specific risk for the transmission of the same disorder in offspring but also risk for the development of other forms of psychopathology, many of which show signs much earlier in development then when one typically first diagnoses depression or alcohol abuse. Although abstinence, if one could count on adherence, is an effective approach to the prevention of alcoholism in children of alcoholic fathers, there is no single such solution for children of depressed mothers. Neither does abstinence address the early emerging problems in youth or the other disorders for which children of alcoholic fathers are also at risk. At minimum, the findings in this chapter clearly support screening and prevention targeting risk factors for the development of the respective disorders, such as problems with emotion regulation abilities in both groups of offspring, a low response to alcoholism in children of alcoholic fathers, and cognitive vulnerabilities to depression in children of depressed mothers. Continued research on mechanisms holds promise for the design of specific effective interventions at the same time that ongoing work on moderators will help identify the individuals most likely to benefit from preventive interventions.

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CHAPTER 31

Problems in Infancy

Tiffany Field

The problems in infancy can be categorized as both the frustrations and challenges of the developmental milestones that occur and the problems they pose for the parent facilitating development and giving care along the way. This chapter reviews some of those problems under the headings of developmental milestones, developing emotions, behavior problems, and caregiving. Under each of these headings, the problems are given in an order that approximates their development or the order in which they may become concerns to parents.

DEVELOPMENTAL MILESTONES

Taste

Although most parents today realize that newborns can see and hear, many are surprised that they can also smell and taste and imitate facial expressions. Newborns are born with very sophisticated abilities to discriminate different tastes, probably because the amniotic fluid comprises all tastes, including sweet, salt, and bitter (Mennella, Jagnow, & Beauchamp, 2001). This has been observed in many studies based on newborns' facial expressions in response to different tastes. For example, disgust faces follow the baby tasting quinine and other bitter tastes, whereas contented faces follow a sweet taste (Liem & Mennella, 2002). Compared to older infants, however, newborns do not seem to reject salty and bitter substances, perhaps because those were familiar in utero. Slightly older infants seem to prefer novel tastes. For example, in one study, infants spent more time breastfeeding after their mothers ate garlic as compared to those infants whose mothers repeatedly consumed garlic. Clearly, though, the sweetest substances are preferred from the first day of life. In fact, sucrose has been used as an analgesic during painful procedures in the newborn nursery. Newborns who were given sucrose solutions to suck on during blood draws, for example, cried less following the procedure and had lower heart rate and more positive brain waves (Fernandez et al., 2003). Children

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who are introduced to new tastes and new foods in early childhood tend to enjoy a variety of different foods later, whereas those with a limited diet tend to have negative reactions to new foods.

Imitation

Until very recently people thought that imitation did not occur until around the end of the first year of life. This concept of imitation was advanced by the famous child psychologist Jean Piaget, who based his ideas on observations of his own two children. More recently several researchers have shown that even a newborn can imitate exaggerated facial expressions made by adults. These expressions include sticking out the tongue, an exaggerated smile, a surprised face, and a sad face. Of course, not all children show imitative behaviors as early as the newborn stage. In fact, some newborns have been described as "poker-faced" and merely stare inquiringly as you model these faces. In a study I conducted on newborn imitation, I first made sure that the newborn was awake and bright-eyed (Field, Guy, & Umbel, 1985). Then I held the baby in front of my face and bobbed up and down with a couple of knee bends to make sure the newborn's eyes were open, and then made a couple of tongue clicks to make sure the newborn was looking directly at my face. I then made an exaggerated facial expression, leaving it there until the infant looked away. I then repeated the action with the same or a different expression. When a baby was in a very alert state and in the mood for this game, we saw very interesting imitative behavior. Later, at around 3 months, mothers almost invariably imitate their baby's cooing sounds during face-to-face games, and the baby in turn imitates the mother. This game goes back and forth for many rounds. A little while later infants imitate gestures and fine motor behaviors. Although much of what infants show us they can do seems to be hardwired and occurs on some predetermined developmental schedule, imitation, like other skills, improves with experience (Learmonth, Lamberth, & Rovee-Collier, 2005).

Eye Contact

Eye contact is present as early as birth and is one of the most important features of social interactions (Farroni, Csibra, Simion, & Johnson, 2002). Without eye contact, it is extremely difficult to communicate with people. Infants and older children will sometimes turn off their eye contact or show gaze aversion when they are being overstimulated and need to process the information, much like taking a break from a conversation (Brooks & Meltzoff, 2002). However, excessive gaze aversion or the absence of eye contact early in infancy is often considered a sign of autism and should be further evaluated if this problem persists (Zwaigenbaum et al., 2005). A recent study of children ages 2 to 6 noted that children who averted their gaze from their parents in the first month of life (a) developed a maladaptive relationship in terms of their interactions by 2 years, (b) showed behavior problems and developmental delays for up to 6 years, and (c) had less favorable development during the preschool years.

Conversation Skills

Conversation skills start developing as early as birth, when babies have face-to-face interactions with their parents. These are particularly frequent and become exciting interactions at around 3 to 6 months, when babies not only have frequent eye contact but also smile, make a lot of funny faces, coo, and make funny sounds. Early conversations have been analyzed in detail by psychologists, who note that these conversations have many prespeech sounds (vowels and consonants), all the universal facial expressions (which occur as early as 5 months in utero), and many expressive hand gestures. When the timing of speech and pauses in speech and looking and looking away are studied, the baby's behavior is very much like an adult's. For example, babies look at their interaction partner when listening and look away when talking, as if to concentrate on what they are saying. Occasionally both speak at the same time, and occasionally there are interruptions, just as in adult conversation. But the turn-taking and the turn-taking signals are already fairly well developed.

Much of the infant's learning of these skills appears to come from imitation, which frequently happens during parent-infant interactions. Like a game, the parent imitates the infant and the infant imitates the parent. In fact, one of the easiest ways to get an infant to stop crying is to imitate that crying. The infant stops crying and pays close attention to the ridiculous-sounding adult. Another game parents often play is providing the words for the infant (Field, 1979). The mother says "I like your hat," and the baby coos; then the mother says "Oh, you like your hat too," as if providing the words for the baby. An important rule the parents learn immediately is to respect the infant's looking away. This indicates (as it does in adult conversations) that the infant is either bored, tired, or has had enough of that conversation. Repeating children's speech sounds and adding to their words helps as the infants become older. By toddlerhood children have conversations with the world, with themselves, and with their imaginary playmates. At that stage the silence at bedtime is a welcome relief.

Babbling

Babbling is the child's first attempt at conversation and consists of long strings of sounds formed with the lips and tongue, including *baba* and *dada* and arias of cooing that make the child seem musically talented (Goldstein, King, & West, 2003). These sounds are universal and gradually become differentiated into, say, American versus Japanese sounds as the child is further exposed to his or her unique speech environment. Importantly, babbling reinforces the parent's attempts to talk to the child. In the first conversations the parent says something, the child babbles, and the parent says "Oh, is that so?" as if the child has truly participated in the conversation.

Exploration

This is usually the precursor to playful behavior. The child explores objects to learn what the object can do and what the child can do to the object. One of the earliest forms of exploration is the newborn sucking on the mother's breast.

For the next 6 months everything gets explored by mouth, and much is learned (e.g., the shapes of things) in this way. The hands take over, and then exploration seems almost an obsession as infants make messes with their finger foods, throw objects on the ground for adults to retrieve, and take everything out of drawers and put them back in again (container behavior) as they explore and learn the properties of things. As exploration proceeds the child's behavior becomes more playful, repeating actions over and over, usually to the sounds and facial expressions of sheer joy. Stifling exploration should be avoided at all costs, as it would in turn stifle curiosity, creativity, and the entire learning process. Extremes of baby-proofing or childproofing the house can seriously interfere with exploration.

Handedness

Handedness, or right or left hand preference, develops very early in infancy and can be determined by placing a desirable object in front of the infant and determining which hand leads to grasp the object. The dominant hand will invariably be preferred for fine motor, dexterity activities (Hinojosa, Sheu, & Michel, 2003). However, a recent study highlights the important staging and guiding function of the nondominant hand from very early in infancy. Left-handedness is more frequent in twins (Derom, Theiry, Vlietinck, Loos, & Derom, 1996) than in single children and in males than in females.

The literature is very mixed about the association between handedness and problems. For example, in one study handedness and dyslexia (a learning disability) were significantly related, but handedness was only weakly associated with immune problems. In another study autoimmune disorders were more prevalent in left-handers. In a third study left-handers had nearly 3 times as many immune disorders as right-handers; the relatives of the left-handers suffered from the same conditions; there was a higher frequency of learning disabilities such as dyslexia, stuttering, and autism in the left-handers; and left-handers had migraine headaches more often (Eglinton & Annett, 1994). One of the theories for left-handedness, aside from the genetic theory, is that during fetal development testosterone (the predominant male hormone) slows the growth of the left hemisphere. When testosterone is elevated, it is more likely to produce left-handedness and, in extreme cases, to lead to alterations in the left hemisphere that may be responsible for learning disorders such as dyslexia. Excessive testosterone levels can also alter immune function, which can then lead to immune problems such as eczema, psoriasis, and asthma.

In earlier days attempts were made to shift children from being left-handed to being right-handed so that penmanship and scissors and virtually all hand-held devices (which are designed for right-handed people) would be more easily mastered. Today, parents and teachers do not attempt to alter handedness because of the undocumented theory that shifting handedness in itself may lead to learning disabilities.

Teething

Teething, or the breaking through of new teeth, can be uncomfortable for infants and distressful for parents. Parents' favorite treatment is probably rubbing the infant's

gums with a bit of liquor. Both the rubbing action and the liquor have analgesic effects.

Ear Infections

Ears are vulnerable sense organs because it is difficult to close them to unwanted noise. Ears are also a source of major discomfort to many young children because they experience earaches from inner and middle ear infections. Excessive pulling on the ears, irritability, and signs of difficulty hearing can mean another ear infection. After dozens of these infections and antibiotic treatments, pediatricians often refer the child for a myringotomy (Vaile, Williamson, Waddell, & Taylor, 2006). This procedure involves piercing holes in the eardrums to enable the fluid to drain and is considered a minor procedure, although anesthesia is typically required, and that is certainly not minor. Treatment is critical because data show that repeated, prolonged ear infections contribute to delays in language development and learning difficulties at school. For example, children with inner ear infections (otitis media) have significantly lower expressive language scores, suggesting impairments in language expression (Thomas, Simpson, Butler, & van der Voort, 2006). They also have deficits in processing auditory information and spelling skills and receive more special education services. Despite these problems, their academic achievement tests do not appear to suffer, and by later childhood their early language delays are no longer present.

Immunizations

Immunizations are thought to be critical for preventing disease in early childhood. The most common immunizations are for diphtheria, pertussis, and tetanus and are given at set intervals across the first few years of life (Cohen, 2002). Unfortunately, large numbers of children still are not receiving immunizations. For example, in a large U.S. midwestern city, a study recently revealed low immunization rates (31%), which were said to be due to cost, lack of insurance coverage, and long office visits. Not mentioned was the stressfulness of the immunization procedure on parents and children. Many studies have explored the effects of varying the conditions during immunizations to reduce the stress associated with them. In an attempt to reduce the fussiness and crying and elevated cortisol levels accompanying immunizations, researchers have varied the person accompanying the child for immunizations and varied different directions to the parents. For example, when parents accompany the child there is notably more distress in some studies, and in others there is notably less. That seems to depend on what the parent does during the immunization period. Reassurance, for example, reinforces crying, whereas distraction techniques are much more effective in reducing stress. Other calming interventions help, including music.

Crawling (or Creeping)

Crawling tends to occur at around 6 to 9 months, although some babies never crawl, and those babies sometimes walk at an earlier age (WHO Multicentre Growth

Reference Study Group, 2006). Every baby seems to have his or her own style of crawling, with some tending to crawl backward, some sideways, and some straight ahead. Some crawl on their hands and knees, while others do it on their hands with bended legs. Generally speaking, it seems an awkward way to get from one place to another, and it looks like hard work, so it is perhaps not surprising that there has to be something the baby really wants to reach for crawling to occur. A recent study was conducted on motor skill development in crawlers and noncrawlers during early infancy. The noncrawlers showed slower motor skill development, suggesting that those who do not crawl may develop motor skills more slowly, but crawling and other motor milestones do not seem to be related to later IQ (Capute, Shapiro, Palmer, Ross, & Wachtel, 1985).

A parent recently expressed concern that her child was retarded because she was not crawling at 9 months. After reminding the parent that some babies never crawl, I placed my keys on the carpet, and the baby proceeded to crawl directly to them. The mother, who typically carried her infant on her hip around the house for hours at a time, expressed amazement that all it took was placing her child on the floor with something desirable in view, and her child was no longer "retarded."

Walking

Children typically start walking at around 10 months of age and are pretty proficient by the time they are toddlers at around 16 to 18 months. Temperament differences affect the time of onset, so that one often sees children who are risk takers and exploratory starting earlier than children who are shy and slow to warm up. Although scientists have shown that walking can occur very early if it is practiced, there has been some concern that early walking may slow down the development of other skills, such as talking. Typically, children are focusing on one skill at a time, and those seem to be prewired developmentally to occur at specific times. Crawling and walking, of course, give a child more freedom to explore and get into trouble when the environment isn't childproof. Although walkers seem to be very effective in helping children get around, they are coming under increasing criticism because of their use as babysitters, because some serious falls have been reported, and because they delay the onset of walking (Burrows & Griffiths, 2002; Siegel & Burton, 1999). Hanging harnesses from door frames are very helpful for learning balance and walking movements. Whether walkers or jolly jumpers are used, all children seem to walk at about the same time. If walking is delayed, other functions may also be delayed. In one study, slow walking and slow talking were predictive of lower IQ scores and reading difficulties, but this was true only for children who started walking after 18 months, and only half those children later had low IQs and reading difficulties.

Terrible Twos

The 2-year-old stage has become so notorious for its difficulties that it is now classified as the terrible twos. This is the first of many autonomy-setting stages of the child's life, and it may be the worst (next to adolescence) because it is the first. Parents have come to totally enjoy their lovely child when all of a sudden they find

themselves faced with a little monster. Every other word is "no" and every other action is forbidden and accompanied by looks toward the parent of "I dare you." Language has emerged, and with that children act as if they're smarter than their parents. That might be true, as the child repeatedly outwits them. Because of the autonomy-establishing ways of the child at this age, it's important that parents not get into power struggles with their child over the child's behavior. For example, many children who have not been toilet trained by age 2 use toilet training as grounds for power struggles with their parents. Waiting until 2 virtually means waiting until at least 3, when autonomy issues are not as great. Thus, toilet training can dramatically exacerbate the problems of the terrible twos.

Aside from trying not to save important developmental milestones such as toilet training for this stage, parents are probably best advised not to get into verbal power struggles, but to simply move the child from the source of altercation (e.g., an electric socket or a restaurant where the child has resorted to temper tantrums) or to attempt to distract by activities that the child can clearly dominate. Parents' attempts to explain their "unpopular" position typically fall on deaf ears and accelerate the conflict. It is not surprising that this is an increasingly popular age for parents to send their children to school. There the children can have their terrible two struggles with teachers and other terrible-2-year-olds.

Autonomy

Autonomy is probably the primary goal of childhood. It is generally achieved by the end of childhood, just before the long struggle begins for adolescents to have complete autonomy. Teaching children how to take care of themselves, how to monitor and control their own behavior, how to relate to other people in a socially effective way, how to express their emotions, and how to achieve their potential is what has been called "the package of parenting." The developmental course of autonomy proceeds from the infant who cries at 6 months when you stifle his or her crawling attempts or remove a desired object, to the child who wants to walk unassisted, to the terrible twos when the child wants to have everything plus power over parents (LeCuyer-Maus & Houck, 2002). Developing all those skills—motor, cognitive, and social—are all in the service of achieving autonomy.

Problems that get in the way of developing autonomy include deficits in the child; temperament problems, such as inhibition and unwillingness to have commerce with the world; limited time and patience of parents, who might find it easier to do things for the child; and inability on the parents' part to allow the child to fail in trial-and-error learning. The frustration is felt on all sides, but ultimately the child will be autonomous and the parent will be proud. Even the terrible twos are a small price to pay. Actually, this is a time when parents can have fun, enjoying the slapstick humor of their child's clumsiness as he or she achieves the milestones of autonomy.

Play

Play and pretend play are critical to a child's development. Playfulness is measured by physical, social, and cognitive (mental) spontaneity, by joy and a sense of humor. Much of this develops in the child through experience playing, starting in preschool,

when a child's entire agenda is play, and tapering off as children get older and have less time for play. In one study on individual differences in young children's pretend play, early social pretend play was significantly related to the child's developing understanding of other people's feelings and beliefs (Keren, Feldman, Namdari-Weinbaum, Spitzer, & Tyano, 2005). These data provide strong support for the notion that early experience in pretend play is associated with mastery of relations between mental life and real life. Parents need to orchestrate some of the play of their children, inasmuch as the natural opportunities for play, such as neighborhood playgrounds, have become more rare in very busy and unsafe cities. Often all it takes is arranging for a child's friend to visit, gathering together the props, and getting the pretend play started. Once they get started, children are very good at pretend play.

Friends

Friendships develop at a very early stage in life, which we never knew until children started attending infant day care. In the nursery school my research group runs, we can no longer graduate infants at 12 months to the toddler nursery without also transferring their best friend with them. Parents told us very early on that they wanted their children to wait until they could move with their best friend, but it was only after studying this phenomenon that we appreciated that even infants experience anxiety and depression when separated from their best friends. In our study the infants who were transferred without their friends experienced sleep problems, eating problems (either more or less), toileting problems (constipation or diarrhea), and sometimes even illness (typically upper respiratory infections; Field, Vega-Lahr, & Jagadish, 1984). Thus, friendship starts at a very early age.

In our studies, when parents were absent for a meeting or for the birth of another child, the preschoolers who remained in day care with their friends were less depressed (Field & Reite, 1984). In the early stages, because of a child's shyness and less-developed social skills, parents may need to set the stage for developing relationships. This can be arranged through shared babysitting, inviting a peer along for special activities, and trying to facilitate cooperative play between the children by giving them the same opportunities to engage in the same activity and by having play materials in twos, ensuring that each child has a play object or a book, so that fighting is less likely to occur.

Sex Differences

Sex differences are present from birth. Research suggests that for most assessments, whether it is responding to social stimulation, discriminating faces and voices, or simple learning tasks, females are more skilled. This sexual advantage for girls persists until at least school age. Although it is widely known that boys are physically at a disadvantage at birth, often being smaller and having more perinatal complications, very little is known about the origins of boys' slower development. This is particularly a problem when parents compare their boy child to their girl child and conclude that the boy's slower language development, for example, is suggestive of developmental delay or retardation. Parents need to have more moderate expectations of their male children, particularly in the early preschool years.

Sometime later in children's schooling, female children lose their gains, often as they develop diffidence and experience more limited expectations for academic performance from adults.

There are, of course, many other sex differences that emerge, for example, in the expression of empathy and social interests. Although most people like to believe that these are socially determined, such as the introduction of dolls to girls and guns to boys, many of these preferences are apparent before the socializing experiences. For example, in a recent study on 12-month-old infants, girl infants presented with the choice of a toy airplane or a doll more often reached for the doll, while boy infants more often reached for the airplane (Servin, Bohlin, & Berlin, 1999). An anecdote highlights a similar response. When a teacher carried her baby into the preschool, the boy children rushed to the block corner and the girl children rushed to the teacher's baby, much to the chagrin of the teacher, who had tried very hard to encourage unisex activities and behavior in her preschool classroom.

Twins

Twins can come as a shock to any family, even though they seem to run in families. Senior family members try to prepare first-time parents for that possibility. Because ultrasound and other prenatal tests typically reveal twins sometime during pregnancy, most parents also have that time to prepare themselves. The downside is that you need two of everything and it is twice as much work, but the upside is that you can have twice the joy in half the time. Identical twins have quite amazing similarities as early as birth. For example, in a study we conducted on newborns, identical twins were almost identical in the amount of time they looked at faces and mimicked faces, whereas fraternal twins were very different in their facial expressions and their attentiveness. Fraternal twins are often perplexing because they seem no more alike than nontwins in the same family (Field, 1982). Despite their similar experiences at the same time in development, they can behave and develop very differently.

Intelligence Tests

Intelligence is still a hotly debated issue, particularly for infants and young children. Although there are now scales that are considered infant IQ tests, such as the Brazelton Scale and the Bayley Scales for Infant Development, these have very little stability over time (Lowman, Stone, & Cole, 2006). For example, some children score high early in infancy because of their precocious motor development, but then score poorly at a later age because of their limited language environment. The scales seem to focus more on motor skills in early infancy and language skills in early childhood. Also, scores are significantly affected by many environmental conditions, suggesting that performance is not strictly determined by genetic factors.

Bayley Scales

The Bayley Scales were created by Nancy Bayley in 1963 (and revised in 1993) to assess infants' mental and motor development (Lichtenberger, 2005). They can be given to children from about 3 months of age to about 4 years. Typically the

child is seated on the parent's lap while the Bayley Scales are given, and they last approximately 40 minutes. Developmental age-appropriate items are given, such as a red ring dangled in front of a 3-month-old infant's face, and pegs and peg boards are given at a slightly older toddler age. Motor skills are assessed by watching the child's natural motor behavior and offering more challenging tests at a later age, such as climbing stairs and walking on a rail. These scales yield a score, with an average at about 100. They are used simply to give the pediatrician or developmental professional a sense of how the child's mental and motor development are progressing and whether there are any areas that need work.

Prematurity

Infants who are born prematurely have a number of problems in infancy suggesting delayed growth and development (Vintzileos, Ananth, Smulian, Scorza, & Knuppel, 2002). Long-term follow-ups suggest that children born prematurely have more sleep-related problems and more accidents, inferior eye-hand coordination, delayed language development, more visual and hearing disorders, and more hospital care. Environmental factors account for more variation in their cognitive development than perinatal factors. Preterm children exhibit a self-righting tendency during early childhood, so that eventually environmental influence overshadows biological influences.

DEVELOPING EMOTIONS

Emotional Expressions

The necessary muscles for the universal emotional expressions are present very early in fetal development (3 months gestation). These universal expressions are happy, interest, surprise, sad, angry, fearful, and ashamed. Although the expressions seem exaggerated during early development, when children laugh without restraint as well as sometimes cry incessantly, the expressions become socialized with development, and we begin to see blends of expressions such as a combination of surprise and sadness when feelings are hurt. Some children display their emotions, as in the expression "wearing their heart on their sleeve," and others tend to be more "poker-faced" and show very little emotion. These extremes, sometimes labeled extraversion and introversion, appear in children as early as the newborn stage. In a study we conducted on newborns' imitations of facial expressions, about 20% of infants showed very exaggerated facial expressions and very little heart rate change, while the other 20% extreme showed emotional responses in their heart rate changes but were poker-faced (Field, 1982). Most children (like most adults) show their emotions not only on their face but in their physiological responses, for example in a racing heart, sweaty hands, or blushing.

Anger

Expressions of anger can be seen as early as the newborn period in crying outbursts by an uncomfortable newborn whose needs are not being met. The temper tantrums of a 2-year-old occur for the same reasons: discomfort and not getting what is wanted

or needed (Potegal & Davidson, 2003; Potegal, Kosorok, & Davidson, 2003). In the preschool years, transgressions against children either for their possessions or against their person, as in shoving, pushing, hitting, and kicking, sometimes cause anger at having been made uncomfortable or having their feelings hurt.

Anxiety

Children, like adults, have a host of stresses and anxieties that make their lives uncomfortable. Anxiety can cause sleep disturbances and illness as well as learning problems. The anxieties are different at different developmental stages. Infants and toddlers, for example, experience the stress of separation anxiety and stranger anxiety (Kochanska & Coy, 2002). Infants also experience frustration during developmental milestone periods such as crawling and walking, although anxiety is difficult to measure at those ages. Preschooler anxieties are more likely related to being hospitalized or starting school.

Children who are particularly prone to anxiety are those who are temperamentally shy and have difficulty making transitions. To help with the transitions, various comfort objects such as pacifiers at the younger age and blankies and stuffed animals at the older age are helpful (Donate-Bartfield & Passman, 2004). Other solutions are unique to the different types of anxiety. For example, for separation anxiety, which is typically related to the infant not realizing that the parents are going to come back, parents can reassure their infant that they are just going to get something and will return shortly. For stranger anxiety, a goodbye ritual helps, particularly if it is brief and upbeat (Field, 1995; Field, Gewirtz, et al., 1984). Sometimes play-acting these leave-takings with the child by using dolls and making the babysitter come and go in fantasy play helps. Allowing the child time to become familiar with the babysitter before the parents leave helps a great deal.

Most anxieties are normal and pass with the developmental stage, except for separation anxiety, which seems to persist. Most children do not particularly like their parents leaving them (until they reach adolescence and want to leave their parents). Separation anxiety in its extreme form, when the child will not leave the parent and develops symptoms such as headaches and stomach aches, may be a serious form of anxiety, labeled Separation Anxiety Disorder.

Massage therapy invariably helps. A 10-minute back rub on a daily basis invariably reduces anxiety (Field, 1995). This can be observed as reduced agitation and nervous behavior, and it can be measured as decreased stress hormones such as cortisol taken from saliva samples.

Stranger Anxiety

Stranger anxiety starts at approximately 9 months of age, after a rather extended period of infants loving to be held and played with by strangers. It is not clear why it emerges at this time, because even newborns have been shown to recognize the difference between their own mother and women who are strangers. In a study we conducted, newborns showed that they preferred their mothers by looking longer at them than at women who were strangers (Field, Cohen, Garcia, & Greenberg, 1984). This was surprising because they had had only 4 hours of contact with their

mothers. Equally surprising is their ability to identify their mother by her smell. Recognition of mother by her voice is probably less surprising since newborns would have had experience with the mother's voice in utero.

Nonetheless, even though infants recognize their mothers as early as the newborn stage, they do not appear to show stranger anxiety until approximately 9 months. This takes the form of fussing and showing fearful faces in the presence of the stranger, whether at a distance or when being handed to a stranger. This persists for anywhere between 3 months and a couple of years, in part depending on whether the infant remains at home with the mother or attends an infant nursery school. When an infant has experience with several adults, the infant shows less stranger fear and sometimes no fear. It is not clear what function the stranger fear serves. It certainly would not prevent kidnapping at this early helpless stage of life, and were it to happen any earlier it might even interfere with parents returning to work or hiring babysitters. Fortunately, it is a stage that disappears as quickly as it appears.

Separation Anxiety

Temporary separations from parents during hospitalizations of the parent or child or during parents' conference trips, and more permanent separation from peers associated with moves to other schools, are relatively stressful for young children (Field, 1984). In a study that we conducted on toddlers' and preschool children's responses to separation from the mother during the birth of another child, the children were agitated during the mother's hospitalization and then depressed after the mother's return (Field & Reite, 1984). During her hospitalization the children were irritable, hyperactive, and unable to sleep. When the mother returned the children showed depressed emotion, changes in eating and toileting, continuing sleep disturbances, and illnesses, including constipation and upper respiratory problems (all characteristic of adult depression). The children were clearly agitated by separation from the mother, even though they visited her at the hospital during this period and their father cared for them. The depression following her return from the hospital may have related to the mother's depression or exhaustion. This decreased animation of the mother, together with the arrival of a new sibling, seemed to alter the relationship the mother and child had previously experienced. Examples of the child's disturbance over this altered relationship were provided by the parents, who commented that their child "remained close" to them, "wanted to be rocked and held," "reverted to baby talk, whining, and screaming for attention," "destroyed his playroom," and "threatened to run a truck across the baby's head." In addition, increased fantasy play in the children was characterized by a number of scenes that involved aggression against the new sibling. For example, children built Lego constructions to "fall on the baby," "run over the baby," or "drive the baby to a high bridge and throw the baby in the water."

Some children did not seem to suffer separation stress as much as others. Those who seemed to suffer least were those who continued attending preschool while the mothers were hospitalized, as opposed to those who stayed at home with the grandmother or babysitter during the mother's hospitalization. Maintaining

the child's daily routine seemed to help buffer the effect of children's separation from their mothers.

In a subsequent study we observed children's responses to repeated separations from their mothers, who were attending conferences throughout the year (Field, 1991). Although we were concerned that there may be cumulative effects that made each separation worse than the last, we found that children became accustomed to these separations, and after a few of them were no longer showing the extreme behavioral and physiological changes or illness that accompanied the first and second separations.

Although most parents (and psychologists alike) had assumed that separation stress occurs only when parents are missing (because parents are supposedly the primary attachment figures), separation stress is also experienced over the loss of a teacher or favorite peer. Parents in our nursery school used to tell us that they wanted their infant or toddler to wait to graduate to the next class when their best friend was also ready to graduate. Sometimes parents actually transferred schools when their child's best friend transferred. In a recent study, preschool children who were transferring to new schools were observed during the 2 weeks prior to their leaving (Field, 1984). The children showed increased fantasy play, physical contact, negative statements, fussiness, hyperactivity, and illness, as well as changes in eating and sleeping patterns. Shortly after the children left, those who left the school showed less agitated behavior, but those who remained behind showed increased agitated behavior. Apparently the remaining children were being reminded of their lost friends by staying in the same environment. As in a divorced couple, the person remaining in the old house often has more difficulty because of reminders of the old relationship.

In another study we explored infants' and toddlers' reactions to being graduated to the next classroom with and without their best friends. Infants and toddlers who transferred without their best friend showed increased activity (Field, Vega-Lahr, et al., 1984), negative emotion, fussiness, and changes in eating and sleep patterns. Some children ate too much and some children ate too little; some children slept all the time and others slept not at all. Those who transferred with close friends did not experience the stressful effects of the transition period.

It is clear from these studies that children have difficulty with loss from a very early age, so that the loss of the old relationship with mother after a new sibling arrives or the loss of friends can be the most stressful. At those times, consistency in routine (e.g., remaining in school versus staying at home) can alleviate some of the stress. Having a familiar routine may help not only because it is familiar but also because it provides distraction and opportunities for the child to actively cope with the stress. Otherwise, parents simply need to ride it out and recognize that these are normal responses to loss and change.

Depression

Infants can have symptoms that mimic adult depression, including withdrawal and listlessness around other children, sad facial expressions, a monotone voice, and whining and complaining (Field, Gewirtz, et al., 1984). The child might also show

eating problems, such as overeating or undereating, and toileting regressions, difficulty going to sleep, frequent night wakings, and difficulty getting up in the morning. Symptoms of illness include respiratory problems, constipation, and diarrhea. Illness probably occurs because the immune system is negatively affected by high stress levels accompanying depression.

Empathy

Some form of empathy is apparent as early as birth, when newborns cry at the sound of other babies crying (Field, Hernandez-Reif, & Diego, 2006). A typical situation that is used to study empathy in toddlers and preschoolers is asking a mother to pretend she is crying. Almost always the child will show some signs of attempting to comfort the mother. However, some children, who, for example, have depressed mothers or have been prenatally exposed to drugs, show no empathy (Jones, Field, & Davalos, 2000). Some recent research in our lab suggests that there are different forms of nonempathetic behavior, including sad and angry responses. In the mother pretend-crying situation, for example, some preschoolers showed nonempathetic angry behavior, including kicking their crying mother or yelling at her, and other children showed nonempathetic sad behavior by simply ignoring the mother's crying.

Temperament

For most of history parents have blamed themselves for their children's temperaments. More recently considerable research suggests that infants come into the world with a temperament that is only then slightly modified by parents' temperament and things that happen in the environment. Parents will still blame themselves, but now they will do so for transmitting temperament traits through their genes rather than their childrearing. Temperament researchers have described babies, for example, as easy or difficult from the start, and they have based their conclusions on parents' reports and on observations of several characteristics they call temperament traits. These include the child's activity level, adaptability, distractibility, intensity, mood, persistence, regularity or rhythmicity, sensitivity, and weariness. Different temperament traits combine to determine whether a child is easy or difficult. In one of our studies, we noted that newborns in the first hours of life could be placed somewhere on a continuum between being extraverted and being introverted (Field & Greenberg, 1982). The extraverted newborns showed very expressive faces and were not very reactive when we measured heart rate. At the other extreme, the introverted newborns were poker-faced but showed considerable physiological reactivity as recorded by their heart rate. Most newborns (approximately 60%) fell in between these two extremes, being both facially expressive and physiologically reactive. When we looked at these behaviors in identical and nonidentical twins we found that there was more similarity in the personality types of identical than nonidentical twins, suggesting again that there was some genetic component to these personality profiles. Observing these infants over the first few years of life, we noted that if an extraverted infant was born to introverted parents, the extraverted infant moved toward the middle of the continuum, whereas if the extraverted infant

was born to extraverted parents, he or she more likely continued to be extraverted. Thus the interplay between genetics and environmental influences became clear.

Extraverted and introverted children have very different ways of interacting with the world. Extraverted children are more exploratory and risk taking and therefore likely to excel in development. On the other hand, if the parent is aware of the temperament traits of the introverted child and tries to match parenting strategies to that style, the child may end up being exploratory in spite of his or her introversion. Several researchers have tracked the development of temperament traits to determine whether they are stable over time and which traits seem to contribute to superior development. Dimensions of temperament that have been closely associated with less difficult responses to stress, fewer behavior problems, and lower stress include predictability, positive mood, adaptability, higher approach, lower intensity reaction, and lower responsiveness threshold (Huttunen & Nyman, 1982).

Because of the stability of temperament traits over development, psychiatrists have been concerned about the relation between those traits and behavioral syndromes in later childhood (Maziade, Cote, Thivierge, Boutin, & Bernier, 1989). In one study, for example, both high emotionality and low sociability predicted high scores on an anxiety/depression scale. In another study, high activity and low rhythmicity was associated with a high degree of psychopathology.

BEHAVIOR PROBLEMS

Crying

The most common causes of crying in infants and young children are fatigue, hunger, indigestion, soiling, wetness, and pain. The most common psychological reason is a parent's leaving. Very soon after birth, parents come to recognize the cry sounds of their infant. The frequency, intensity, and intonation of cry sounds are different for all of these problems. For example, fussy cries are less intense than hungry or pain-induced cries. Parents will often assure themselves that all of these problems are taken care of before letting the child cry it out. Sometimes when a child is overstimulated and, thus, overexhausted, crying serves as a kind of stimulus barrier to ward off all other sounds in the environment.

The parent may be able to stop the crying by rocking or walking or driving the infant around, but often the crying will start again when the infant is put to bed. This could become a pattern of crying for the parent to come and comfort (Räihä, Lehtonen, Huhtala, Saleva, & Korvenranta, 2002). The simplest method for solving this problem seems to be letting the infant cry it out. For the first night, the crying may persist for 20 to 30 minutes, and when there is no rescue from the parents, the infant finally falls asleep. The second night the crying may last only 10 minutes, and by the third night there usually is no crying. Getting up for every cry not only overstimulates and ultimately tires the infant but certainly tires the parent.

Colic

Colic is a syndrome of persistent crying in young infants. Colic is one of the most common problems in infancy (incidence ranges from 10% to 35%; Savino &

Oggero, 1996). Researchers disagree about the definition and clinical criteria for colic; some pediatricians even suggest that colic does not exist. This presents a very confusing picture to parents.

The word "colic" is derived from the Greek word *kolikos,* the adjective of *kolon,* meaning the large intestine. Colic also goes by the labels "infantile colic" and "paroxysmal fussing." Colic has been described as intermittent and unexplained crying during the first 3 months of life by infants who are otherwise well-fed and healthy. Colic usually starts during the second or third week of life and spontaneously disappears at around 3 months. The crying is generally intense and typically lasts from 30 minutes to 3 hours, often occurring in the late afternoon or evening, followed by difficulty falling asleep. The crying may also occur intermittently and last for several hours at any time of the day or night, although the crying episodes tend to recur at the same time of day.

In most cases, crying is considered normal infant behavior and can be treated with reassurance and counseling about appropriate parental responses. In a minority of cases, the crying pattern and other infant behaviors may suggest a gastrointestinal problem that may require treatment. Fussiness and crying are not easy to tolerate by most adults, which probably explains why colic is the most frequent problem presented to pediatricians. Although colic ends almost abruptly at 3 months, irritability and excessive crying have been known to recur during frustrating developmental milestones such as sitting, crawling, walking, and virtually any learning stage.

Some clinicians accept the definition of colic as full-force crying for at least 3 hours per day, for 3 or more days per week, for 3 weeks. Colic begins with a sudden onset of crying. The cry is often high pitched and accompanied by facial grimacing, suggesting that the infant is experiencing severe pain. The crying is accompanied by increased motor activity, flexion of the elbows, clenched fists, and hypertonicity (e.g., the knees are drawn up or the legs are stiff and extended), and the abdomen is tense and distended. The infant's eyes are either tightly closed or wide open, the back is arched, the feet are often cold, bowel sounds can be heard, infants may hold their breath for brief periods of time, and they may resist soothing.

Several factors may be involved in infant colic, including biological factors such as temperament and sleep-state organization and environmental factors, for example, overstimulation by parents. Biological factors include hypertonicity, gastrointestinal and allergic factors, and an immature central nervous system. Although gastrointestinal factors have typically been described for colic, serial radiographs suggest a normal amount of gas in the GI tracts of colic infants. Breast milk, cow's milk, and soy milk have also been implicated in colic. Digestion of milk and soy protein produces excessive intestinal gas and peristalsis, which can result in colic. However, colic occurs as commonly in breastfed as in formula-fed infants.

Feeding patterns have been implicated as a potential contributor to colic. Short interfeed intervals resulted in less crying. Infants who were carried more frequently did not cry at the usual 6-week crying peak, and the crying and fussing that tended to occur in the early evenings decreased. Overstimulation also contributes to colic. This became clear when infants who were hospitalized for colic cried less once they were hospitalized. Other overstimulating factors are excessive noise, bright lights, and irritating blankets or clothing. Colic may cease around 3 months because

infants can now use self-comforting techniques such as gaze aversion, thumb sucking, turning over, reaching, and grasping to avoid or remove the annoying stimulation. Having feeding, bathing, and sleeping routines is very effective. In addition, providing soothing stimulation such as massaging the infant, rocking or swinging, and playing classical music or white noise can also be extremely effective (Huhtala, Lehtonen, Heinonen, & Korvenranta, 2000). The most effective treatment is time, as colic usually disappears by 6 months (Crotteau, Wright, & Eglash, 2006).

Bedtime Problems

Bedtime is one of those quality times for parents and children. It needs to be a happy time even though children can fret or procrastinate. Having a bedtime ritual that the child looks forward to is often helpful. The teddy bear gets tucked in, the child hears a story, and then a back rub helps a child get to sleep. A recent study in our lab suggested that back rubs were more effective than rocking from early infancy through early toddlerhood (Field, Grizzle, et al., 1996). In the case of rocking, the child goes to sleep during the rocking but awakens as soon as he or she is carried to bed. In the case of back rubs, the child is awake during the back rub and goes to sleep in that position shortly thereafter.

Sleep Disturbances

Sleeplessness, in the form of difficulty falling asleep and/or nighttime waking, is a very common pediatric problem, affecting some 15% to 35% of infants and toddlers. Sleep problems as early as infancy may relate to physiological factors, temperament of the infant, and parents' responses to difficult sleep patterns. In recent years the rate of sleep disorders has reportedly increased, possibly because parents are increasingly less tolerant or more worried about sleep disturbances and how disruptive they are to the dual-career family, or possibly due to more sophisticated methods of monitoring sleep and knowledge about the development of sleep patterns. Various methods have been developed to study infant sleep, including measuring facial muscle activity, EEG, heart rate, and respiration. In addition, diaries are frequently kept by parents to record information on duration of sleep and crying and other activities.

Infants' wakeful time gradually increases from an average of 8 hours at birth to 12 hours at 6 months (Acebo et al., 2005). By the time the infant is 3 months old, sleep predominantly occurs during the nighttime hours, with the infant tending to remain awake during the later afternoon and early evening hours. By 6 months, the time awake has increased from a mean of 2 hours at 3 months to about 3.5 hours, and the longest waking period is about one-third of the total awake time. Although a diurnal (normal) sleep-wake pattern is established by this time and the infant is capable of sustaining sleep during the nighttime hours, many infants continue to wake during the night. At this time, the amount of time the infant takes to fall asleep increases, and parents begin to establish bedtime routines and the use of blankies and other self-comforting objects to facilitate earlier sleeping.

Once infants are put to bed, there is a large variation in the amount of time required for sleep onset, depending on age. For example, 2-month-olds are thought

to require approximately 28 minutes and 9-month-olds only 6 minutes to fall asleep. Night wakings occur in approximately 29% of infants during the third month, 17% by the sixth month, and 10% at the end of the first year. Once asleep, newborns typically sleep approximately 17 hours per day, and by the sixth month they are sleeping only approximately 14 hours (including naps), which remains the same across the first 2 years.

Temperament characteristics reported for sleep disturbances include low sensory threshold, difficulty establishing regular routines, and reluctance to move from one activity to another. Given that infants who are stimulated less seem to organize their sleep states sooner, overstimulation by parents may be disruptive to infant sleep. A study of healthy infants suggests that most children need something to assist them to fall asleep. In this study, regular use of a transitional object such as a blanket or toy was described for 44% of the children, bedtime sucking for 22%, and both for 14%, leaving only 20% of the children who fell asleep without assistance.

Parents use a number of comforting techniques to induce sleep in young infants, including swaddling, rocking, and sounds such as white noise or classical music. Others advocate letting the child cry or taking the child into the parents' bed or supplying toys. In addition, infants use their own body to self-comfort, including sucking on hands and rocking or head banging in a rhythmic way.

More formal procedures have been tried in studies giving parents instructions (Skuladottir, Thome, & Ramel, 2005). These have mostly involved having the parents find a consistent sleeping place, a regular bedtime, and a bedtime ritual. In one study parents were asked to try a quiet play routine before bed in which they used soothing and relaxing techniques such as massage and pleasant music. They were asked to refrain from using a light, changing diapers, and talking during their nighttime interactions. The majority of infants showed improvement during the first few weeks of treatment. Whenever parents are present at the time of sleep onset, there are more frequent night wakings.

We recently used massage therapy to help infants sleep (Field et al., 1996). The infants who experienced massage therapy compared to infants who were rocked cried less and had lower stress hormones (salivary cortisol levels), suggesting lower stress. After the massage versus the rocking sessions, the infants spent less time in an active awake state, suggesting that massage may be more effective than rocking for inducing sleep. Over the 6-week period the massage therapy infants gained more weight, showed greater improvement on emotionality, sociability, and soothability temperament dimensions, and had greater decreases in stress hormones.

Food Preferences

As early as birth infants appear to have food preferences. Studies on newborns show that they make different faces to different tastes, with sweet flavors being accompanied by happy faces and quinine or rotten egg flavors being accompanied by grimacing. This seems to happen whether the food is tasted or simply smelled. Of course, the amniotic fluid that engulfs the fetus contains all of the basic flavors, especially the more salty and bitter taste. Newborns will readily take glucose but tend to reject salty water. Taste preferences then shift across age. Very little is known

about young children's food texture preferences, although foods prepared for infants and young children tend to be pureed at first and only later have considerable texture. Their preferences for hot and cold foods is also relatively unknown.

Studies show that monozygotic (identical) twins have more similar food preferences than dizygotic (nonidentical) twins, suggesting a strong genetic determination of food preferences. Many children will follow parental likes and dislikes, particularly dislikes after seeing an adult grimace at the thought of eating a particular food. On the other hand, parents who expose their children to wide varieties of foods have children who are not fussy eaters and who are willing to try new foods.

Eating Problems

Eating problems can range from food refusals and minor conflicts with parents to eating disorders (Carruth & Skinner, 2000). The minor food refusals are sometimes just the child being finicky about food preferences, although very little is known about children's food preferences because all the nutritious foods are bottled in baby food form, with very little choice being given the children until they can select from their own lunch box or the school cafeteria menu. In addition to very little being known about taste preferences, very little is known about food textures that children prefer. In a recent study we were surprised to find that very young infants preferred chunky applesauce if they are given the choice before the designated age for chunky applesauce (Lundy et al., 1998). Infants also like vegetable flavors like broccoli and cauliflower (Mennella, Kennedy, & Beauchamp, 2006).

Other reasons for food refusals are the child's own power struggles that happen as early as the terrible twos, when children simply want to be autonomous and do what they want to do when they want to do it. These power struggles can continue into preschool and grade school, when children do not have the good models of their parents for food preferences but the fads of what other children are eating. At this time full lunch boxes might simply mean the child has eaten someone else's more desirable lunch.

Although playing with food is a good way to learn the art of eating at the infancy and toddler stage, by preschool the child should be learning appropriate restaurant eating habits. Around this time the child learns that mealtimes are a social time for the family, not a time for reading or watching TV. Unfortunately, because this is often the family's only time together, mealtimes can become times for conflict and conflict resolution. This can contribute to digestive problems and the association between food and anxiety or sadness. The association of food with joy is perhaps also not good, as it may come to be used as a way of solving problems and warding off depression.

Attention Seeking

Attention seeking is probably one of the most important motivating factors behind child behavior, second only to curiosity, exploration, and learning. From the time infants discover that they can make something happen, at around 3 months, when they can reach, grab on to something, and pull it toward them, to the time that they can crawl from one side of the room to the other, to the time they can walk by

holding on to furniture, they are discovering that they can do things. Typically the infant looks to whoever is in the near vicinity to see their reaction. The infant is innately a social being, and social interactions from the beginning of life involve getting attention and paying attention (Hobson, Patrick, Crandell, Garcia Perez, & Lee, 2004).

The problem is that as the world becomes busier and parents are juggling a thousand things, it is simply hard to pay attention to all the positive behaviors and all the accomplishments of the child. Doling out hugs and praise is time-consuming, can sometimes lead to dependency if given too often, and therefore is at odds with our cultural childrearing agenda, which is to facilitate autonomy (self-reliance). But children will inevitably get attention from someone by some behavior; often that is negative behavior, such as temper tantrums, interrupting, complaining, whining, breaking things, or having accidents. The child is impatient and wants attention when he or she wants it.

If possible, diverting the child's attention, mimicking the whining (babies stop crying when you imitate their crying), or attempting to ignore it and trying to engage the child in your activity (talking on the phone or laying out the lunches), will sometimes suffice. Escalations into full-blown temper tantrums, particularly when children are in public places such as restaurants, call for moving the child from the situation and having a time-out.

Clinging

Clinging is a fairly normal activity for young children when they are hesitant about new places and people. Clinging, or "holding on to the mother's apron strings," as they once called it, gives the child a sense of reassurance about the situation. Clinging is also fairly common during developmental milestone periods, for example, around 1 year, when the child is learning to walk and separation and stranger anxiety are occurring, and again during the time of the terrible twos. Situations during which children may cling are leave-takings at preschool and following special events like the birth of a sibling. This behavior occurs most frequently in temperamentally shy, slow-to-warm-up children who have difficulty making transitions to new situations.

Not showing your own anxiety about your child's anxious behavior will help. In addition, engaging children in play may help them adjust to the new situation. For example, when leaving children off at preschool, letting them have their stuffed animal or whatever makes them comfortable should help reduce clinging.

Thumb Sucking

Thumb sucking in newborns and young infants is considered a very sophisticated self-soothing behavior. Except for unusually shy children, most stop thumb sucking at around the preschool stage. After that, thumb sucking typically occurs only during times of stress and going to sleep. If this behavior continues into grade school, it is problematic because of peer rejection. In a study of 7-year-olds, for example, children viewed photos of boys and girls in thumb-sucking and non-thumb-sucking poses and rated the pictures (Friman, McPherson, Warzak, & Evans, 1993). The

results indicated that children in a thumb-sucking pose were rated as less intelligent, happy, attractive, likable, and fun and less desirable as a friend, playmate, seat mate, classmate, and neighbor than when they were in the non-thumb-sucking pose. Thus the risk of losing social acceptance is real at this stage. Typically thumb sucking subsides unless it is negatively reinforced. A parent's simple expression of concern that the child will be teased by his or her friends is enough warning for the child. Simply having distracters or substitute activities that are more interesting than thumb sucking can solve the problem.

Aggression

Aggression in young children is considered fairly normal. Usually it is their way of trying to get what they want, a toy or attention. Toddlers and preschoolers engage in pushing, shoving, hitting, kicking, yelling, and biting, and infants pull each other's hair and jab fingers in faces, which is not a problem until it is harmful to another child, a parent, or a teacher. Lots of the aggressive behaviors happen in the natural course of simply being active on the playground or in the classroom. Typically children pick themselves up or go on their way after being assaulted by another child in the course of rough-and-tumble play. Ultimately, with modeling from older children and adults, they learn to express their feelings verbally instead of physically (Tremblay et al., 2004).

Biting

Biting is perhaps the most difficult problem parents and teachers have with toddlers. Unfortunately, very little is known about how to prevent it because it is not a favorite topic for research. Generally children use biting when other behaviors do not manage to get them what they want from their playmates. Pushing another child often escalates into slapping, which escalates into biting. Biting attracts more attention from teachers because teeth marks are generally more alarming and potentially more dangerous than scrapes and bruises. Unfortunately, one child's biting often leads to an epidemic of biting, in which several children follow the lead and the problem becomes unmanageable. Biting occurs most often in the midmorning, randomly throughout the week, probably because children are more active in the midmorning, and most often in September, probably because this is a back-to-school period, which is a frustrating transition for children (Solomons & Elardo, 1989)

Generally teachers will pay additional attention to the biting child and try to predict when the child will bite. The teachers will typically intervene when they think the child is about to bite and attempt to distract the child with toys before the biting occurs. Other situations in which children might resort to biting are when they are extremely tired, frustrated, or overexcited. If children learn that they do not have to resort to biting to get what they want (teacher provides the wanted object before the biting occurs), they will probably no longer resort to biting. If the child appears to be compulsively biting, then more extreme measures may be necessary, such as providing a stuffed animal as a substitute for biting another child (it is less dangerous to bite the stuffed animal than another child). Another measure that

may be used is time-out. Usually a minute per year of the child's age is sufficient. Often children will see what they are missing during the time-out and will learn to control their behavior. Another effective measure is giving the child lemon juice. Most bites are minor and can be treated by washing, applying cold compresses, and comforting the child.

The advice of a child psychologist is that the best thing for an adult to do when one child bites another is to remain calm, take the child to a time-out area, tell the child "No biting" and that he or she should remain there until you return, and console the bitten child but without making a big deal of the injury. If the bite has broken the skin, administer a topical antibiotic and call a pediatrician. Then, without appearing that you are on the alert for another incident, be aware of signals that the child is working himself or herself up to a bite, move in, and either remove the child from the playing field or distract him or her. Research suggests that there is no connection between parenting style and biting, nor is biting indicative of psychological problems or predictive of later problems getting along with other children.

Because this behavior problem, more than virtually any other behavior problem, distresses parents, it needs special attention immediately. Parents become extremely emotional about this problem. For example, a thoracic surgeon parent was recently so hysterical about his child being bitten that he claimed the biting child could have pierced his child's lung (if he were being objective he would have remembered that he was a thoracic surgeon and knew very well that the rib cage stands outside the lung and would protect the lung from biting).

Head Banging

Head banging was once considered a pathological activity that only handicapped or emotionally disturbed children showed. However, recent studies on rhythmic behaviors of infants and toddlers suggest that many children engage in these rhythmic activities as part of normal development. Other behaviors are rocking, lip biting, and thumb sucking. They generally happen when the infant or toddler is otherwise unoccupied and appears to be bored. In this case, these rhythmic activities may be a source of substitute stimulation. At other times they occur when the child is extremely tired, and then they seem to pacify and quiet the child. Generally they will disappear after a short period of time, so they are probably best ignored. Children who show any of these behaviors excessively and are actually hurting themselves may need a professional evaluation (Berkson, 2002).

Stuttering

Approximately 8% of children stutter at some time, most usually between about 1.5 and 6 years of age, when the child is learning to talk, and later when the child begins school. Approximately 3 times as many boys as girls stutter, and stutterers are more often left-handed and have blond hair and blue eyes. Approximately 75% of children who stutter stop stuttering at around 12 or 13 years of age (Kalinowski, Saltuklaroglu, Dayalu, & Guntupalli, 2005).

Typically stutterers are delayed in their expressive language skills, not their receptive language skills, lending support to the hypothesis that language deficits

observed in stuttering children result at least in part from their attempts to simplify verbal responses as a means of coping with their stuttering. A recent study suggested that stutterers also show more nonspeech behavior during stuttered words than nonstutterers show during similar words (Bajaj, Hodson, & Westby, 2005). For example, they turn their head to the left, blink, and raise their upper lip excessively. Some evidence suggests a genetic predisposition for stuttering, and some researchers have found that the EEG of stutterers differs from those of nonstutterers, suggesting a disturbance of interhemispheric relationships. This may be a significant factor in the development of stuttering. Stressful events like the birth of siblings or transitions to new schools or new places can exaggerate the stuttering. The most difficult aspect of this problem is the embarrassment it causes the child and the delay it causes in language development.

Parents can help by not drawing attention to the stuttering, trying to ignore it, and certainly not speaking for the child. When children stutter, the tendency is to provide words for them rather than let the child complete the sentences. If the problem becomes so serious that the child cannot be understood, a speech therapist might be helpful. Often the therapist will work with the child to speak more slowly and to practice various breathing techniques that help the child become less stressed.

Temper Tantrums

Temper tantrums occur at frustrating stages in development. They are considered a mark of the child's attempts to become independent, for example around the terrible twos, when the child has power struggles with parents and because of limited language development has difficulty expressing his or her frustration except by tantrums. Temper tantrums typically arise from children's thwarted efforts to exercise mastery and autonomy. They occur more frequently in the active, determined child who has abundant energy. They may also occur in children who are hypersensitive to sounds, lights, and other stimulating features of the environment. Although these are considered normal behaviors, it is also clear that for a number of children temper tantrums turn into more difficult behavior problems (Leung & Fagan, 1991).

In one study, tantrums were reported in about 25% of children, most commonly at 2 to 5 years, becoming less common at 6 to 8 years. Children showing tantrums had a higher incidence of postnatal trauma. Parental overprotection and marital discord were stress factors in a greater number of boys than girls, and parental negligence was a significant stress factor for girls. Behavior problems that accompanied temper tantrums included thumb sucking, bedwetting, tics, head banging, and sleep disturbances. In another study, 52% of those who showed temper tantrums had multiple behavior problems. Factors independently associated with the tantrums included maternal depression and irritability, use of physical punishment, marital stress, child care provided exclusively by the mother, and poor child health (Potegal et al., 2003).

After having established that the child's temper tantrum is not simply related to discomfort, it's important to look at how the tantrum is being treated. Parenting practices that may encourage tantrums include inconsistency, unreasonable expectations, excessive strictness, overprotectiveness, and overindulgence. The child might also be bored, fatigued, hungry, or ill, thus reducing his or her tolerance for

frustration. In those situations it would obviously be important to distract the child or interest him or her in something less boring. Beyond these attempts, temper tantrums are probably best handled by ignoring the outburst and saving attention for the posttantrum, as well as helping the child learn to express negative feelings in less annoying ways.

Shyness

A significant body of research suggests that shyness is one of the strongest temperament traits in children (Stein, Chavira, & Jang, 2001). Research from our lab shows that even newborns show behaviors that look like shyness. Approximately 20% of newborns show shy, introverted behavior such as limited responses to social stimulation. We have called this being poker-faced (Field, 1982). At the other extreme, approximately 20% of children are extremely extraverted, or wear their heart on their sleeve. Approximately 60% of newborns and children lie somewhere in the middle of those extremes. In our study we found that when a child seemed to be born with a shy temperament but had extraverted parents, there was often a shift toward being somewhere in the middle. Thus, although this seems to be an inborn trait (particularly because we saw shyness more often being shared by identical than nonidentical twins), the environmental influences (such as shyness or no shyness in the parents) are also very strong (Gerhold, Laucht, Texdorf, Schmidt, & Esser, 2002).

Data also show that children who tend to be shy have unique physiological characteristics such as greater reactivity of heart rate and different brain waves. Sometimes they have very strong physiological responses to emotional situations, and their shyness or social withdrawal might be a way of providing a barrier to the excessive stimulation. If a child is socially withdrawn, there is often a tendency for people to respect the child's shyness and let the child take the initiative. Often the shy, slow-to-warm-up child will be most comfortable making his or her own overtures and contacts.

Researchers at Harvard found that children who were timid during infancy and as toddlers tended to be socially inhibited at age 7. They estimated that 15% of children, significantly more girls than boys, are born shy (Kagan, 2005), although they found that many outgrew their shyness.

Autism

Autism has a variety of genetic and nongenetic causes, although the etiology (development) is unknown in the majority of children. Boys are more frequently affected than girls, and autism first becomes apparent in the preschool years. It always affects sociability, communication, and the child's repertoire of activities and interests. Autism encompasses a broad range of severities and a variety of other signs of brain dysfunction. These include motor signs, notably stereotypes such as head banging and flapping of hands. Abnormal responses to a variety of sensory stimuli occur, and the children have very severe attention problems and problems showing their emotions. A significant number of autistic children experience epileptic seizures and have abnormal EEGs. Magnetic resonance imaging has revealed abnormalities

of brain development in some autistic children. The level of intelligence may range from profound mental deficiencies to giftedness. The pattern of cognitive skills is likely to be uneven, typically with better nonverbal than verbal skills (Mitchell et al., 2006). In the preschool years, autistic children have a language disorder. Verbal expression may range from total lack of language to being extremely verbose (some even have echolalia, which is endless talking). Comprehension and language use are invariably impaired. There are no specific drugs that can reduce the symptoms of attention disorder and seizures, but the autistic features can change with maturation and appropriate intervention. Communication skills and sociability remain deficient but improved for all but the most severely affected children. The children can be positively affected by early interventions focused on the development of appropriate social skills and meaningful communication.

Autism is a particularly difficult problem because the child is unresponsive. As early as infancy, autistic children are noted to arch their back when held, to make very little eye contact, and to simply dislike physical affection. At the preschool stage or as young children they prefer to play by themselves and often sit and do nothing, staring into space. Their difficulty staying on task and their autistic behaviors, such as flapping their hands and whirling around, spinning things, or rocking and banging their heads, is very disruptive behavior and often wins them no friends at school.

The number of autistic children has increased dramatically over the past several years, probably because screening techniques are more sensitive but less specific so that more children are labeled autistic (many of whom are not really autistic; Yeargin-Allsopp et al., 2003). Thus this is a "waste-basket" diagnosis, as some clinicians call it. Many schools are now providing special classes for autistic children, and some schools serve only autistic children. Statistically, approximately 1 in every 1,000 children is affected, with males being affected 4 times as often as females (Merrick, Kandel, & Morad, 2004). Generally the disorder is thought to be inherited. However, so many theories have been advanced about neurological and neurotransmitter causes that an entire journal is now devoted to autism. Medical conditions associated with autism include epilepsy, hearing deficits, speech and language impairments, and eyesight problems. A comprehensive medical workup is required in all cases of pervasive autistic symptomatology.

Children with autism have very little language and tend to repeat words, sometimes memorizing and repeating paragraphs of from magazines and in general speaking inappropriately. One study showed that using baby-talk-like language ("motherese") was more effective in gaining the attention of autistic children than using simple conversational tones. Many therapy attempts have focused on physical interactions such as wrapping the children in blankets and holding them to enhance their attentiveness. Although these children are also thought to have an aversion to touch, we were able to successfully perform a massage therapy study with them (Field et al., 1997). In our first study, the children who received massage therapy a couple of times a week for a month showed less off-task behavior, had fewer autistic mannerisms, and related more positively to their teachers, suggesting at least that they did not find touch aversive. In our second study the parents served as the therapists and the children had fewer sleep disturbances and less off-task

behavior in the classroom (Escalona, Field, Singer-Strunk, Cullen, & Hartshorn, 2001). This may be a technique that parents could use as a way of being physically more intimate with their typically rejecting autistic children and as a way of enhancing the children's attentiveness. In addition, we have used imitation therapy to help the parents and children interact. The parents imitate the child's behaviors, and the child becomes more social.

CAREGIVING

Touching

In a recent study on the touching behavior of mothers and fathers with their newborn infants, both mothers and fathers touched their infant for an equal amount of time and frequency, but mothers touched or stroked their infants more than they patted or moved them (Prodromidis et al., 1995). Fathers did not show a preference for either type of touching. Mothers preferentially touched their infant's hand and face more than the body, and fathers preferred to touch their infant's hand. The authors suggest that this is consistent with mothers' inclination to care for their infants and fathers' tendency to play with their infants. Another study conducted on touching premature newborns in intensive care suggested that touching occurs 63% of the observation time; 69% of the touching was treatment-oriented, 15% was caregiving-oriented, and 16% was simply comforting touch. The majority of touching was done by the nurses (Modrcin-Talbott, Harrison, Groer, & Younger, 2003).

Massage Therapy

Massage therapy has been noted to help infants and children in many ways. When we gave hospitalized premature infants three 15-minute massages a day for 10 days they gained 47% more weight, were hospitalized for 6 fewer days (at a hospital cost savings of approximately $10,000), and were more responsive to social stimulation (Field et al., 1986). Eight months later they still showed a growth advantage, and their performance on mental and motor developmental tests was superior (Field, Scafidi, & Schanberg, 1987). Massage therapy used with normal infants led to less irritability, more alertness, and better sleep patterns (Field, Grizzle, et al., 1996). Normal infants also gained weight following 5 weeks of massage. When compared to a group of infants who were rocked, the massaged infants were awake while they were being massaged and quickly went to sleep following the massage, whereas the rocked infants slept during the rocking but awakened as the rocking ended. This suggests that massage is a more effective way for parents to help their infants go to sleep.

Although the underlying mechanisms for the effectiveness of massage are not clear, massage seems to put infants and children in a more relaxed state. In that state their stress hormones (cortisol levels) are reduced and they are more alert. The reduction in stress hormones improves the functioning of the immune system.

Infant massage classes are being started in many places in the United States to teach parents how to massage their children. In at least one study we have shown that the person giving the massage benefits as much as the person receiving the massage (Field, Hernandez-Reif, Quintino, Schanberg, & Kung, 1998). In that

study we trained grandparent volunteers to massage babies, and after a month of massaging babies they showed several changes: fewer trips to the doctor, more social calls, fewer cups of coffee, less depression, and greater self-esteem. Although mothers are typically given the training, researchers have also trained fathers as massage therapists. For example, in one study Australian fathers were taught to give massages during bath time for 4 weeks after the infants were born (Scholz & Samuels, 1992). In another study by our group, by 3 months the infants were greeting their fathers with more eye contact, smiling, vocalizing, reaching, and orienting responses, and showed less avoidance behavior (Cullen, Field, Escalona, & Hartshorn, 2000). At this time the fathers who had been in the massage group also showed greater involvement with their infants. Thus, massage therapy has many benefits for both parents and their children.

Bathing

Bathing is typically one of those end-of-the-day activities when the parent has prime time with the child. From very early in infancy bath time is a favorite activity of the child. It becomes a family ritual and is intended not only for cleaning but also for relaxing the child at bedtime (Bryanton, Walsh, Barrett, & Gaudet, 2004). Children love the kicking and splashing, sliding under the water, exploring the taste of the shampoo, and even turning on the hot water without realizing it can burn. The most critical thing to remember is not only to have fun playing with the child and the bathtub toys but that the child should never be left unattended in the bathtub. The one inch of water in the bottom of the bathtub is the leading cause of drownings in the United States (Byard & Donald, 2004).

Swaddling

Swaddling, or the tight wrapping of babies (typically with a blanket), is done for a very limited period of time in the United States. Usually parents swaddle their babies to calm them down or help them go to sleep, and sometimes to make it easier to carry them (Franco et al., 2005). Although swaddling is more common in other cultures, such as American Indians wrapping infants tightly to carrying boards, it is discouraged in our country as a baby care practice because it is considered limiting to the development of hand-eye motor coordination.

Rocking

Rocking is a classic technique used by parents to calm their children and probably themselves as well (White-Traut, 2004). Recent studies suggest that pacifiers and massage are more effective at calming children and getting them to sleep, although rocking may very well serve the purpose of calming the parents. For that reason alone, its popularity may well continue.

Carriers

Carriers come in many varieties for various kinds of carrying positions. For example, in a recent study we observed that carrying the baby in a front-to-front

position led to infant sleep, whereas carrying the infant facing outward led to more wakeful and exploratory behavior (Field, Malphurs, Carraway, & Pelaez-Nogueras, 1996). The optimal carrier might allow for both positions. Being carried by the parent has a long history in evolution and is popular in many cultures, probably because it provides additional stimulation for the baby, enables the mother to closely monitor the baby, and lets her work unencumbered with hands free. This arrangement might also allow for less heat loss and the conservation of energy.

Music

Music has profound effects as early as birth. Studies on premature infants, for example, show that following repeated exposure to music, infants have reduced heart rate and deeper sleep and grow and develop better (Arnon et al., 2006; Standley, 1998; Standley & Moore, 1995). In a study on preschoolers, performance on cognitive tasks was superior following exposure to music. Much of these effects may be attributed to music's ability to relax infants and young children. When heart rate and blood pressure are recorded, for example, they are lower following music. When infants and children are distressed, they typically show greater EEG activity in the right frontal area of the brain (activity in this region is typically associated with negative emotions and mood states). Exposure to music can actually shift EEG activity from the right to the left side of the brain (Field et al., 1998).

Stimulation

Optimal parenting is often discussed in the context of providing the right amount of stimulation. This is particularly true for the infancy and toddler stages, when children need some help monitoring and regulating the amount of stimulation they receive. By virtue of having different temperaments, different children have different thresholds for the amount and kind of stimulation they desire. Matching the kind of stimulation to the child's personality is critical because either too much or too little stimulation will not enhance growth and development. Sometimes parents provide too little stimulation because they are tired, depressed, or preoccupied by their own problems. At other times, for example when children are unresponsive or slow in doing things, parents will be overstimulating and they try to get a response or an action.

Finding the optimal amount and kind of stimulation a child needs is one of the most difficult tasks of parenting. Probably the most effective approach is to take the cues from the child. When interacting with infants, it is easy to tell whether they are getting the kind of stimulation they want or need. Infants will avert their gaze, turn their head, arch their back, and, at the extreme, fuss and cry when they are either getting too little or too much stimulation. Slightly older children may go into a shell and ignore their parents' attempts to engage them. From very early infancy, children have conversation skills such as turn-taking and paying attention or ignoring. These skills can be used as signals that the parent is doing the right thing and the child is interested. Just like adults, children need breaks to not only process information from the conversation and formulate a response,

but also to take rests and have some quiet periods for thinking and imagining things.

Game Playing

Game playing punctuates early interactions between parents and their infants (Field, 1979). Infant games such as pat-a-cake and peek-a-boo appear with such frequency that they are universally recognized as infant games. The importance of early game playing has been underscored by a number of early interaction researchers, who suggest that game playing provides a context for learning one of the most important rules of conversation, that of turn-taking. Infant games are highly repetitive with simple and stereotyped roles for both participants. As such they exemplify the structure of conversational give and take or turn-taking. Even when the infant is yet too young to assume an active role, the structure of these repetitive games may enable the infant to assimilate the turn-taking nature of interaction.

Other important features of early games include the "contingency experience" they provide (i.e., learning to react to and innovate in changing situations) and the facilitation of positive emotion. An illustration of the contingent responsivity provided by a game can be seen in one of the most popular early games, called "Tell Me a Story." The words are provided by the parent, who treats the infants' vocalizations as if they too were words. The parent asks, "Do you want to tell me a story?" The infant coos, the parent responds, "Oh yeah? And then what happened?" The infant coos again and the parent replies, "Oh, that's funny." The infant contingently responds to the parent's words by cooing, and the parent contingently responds to infant cooing by further elaborations of the game.

The game "I'm Gonna Get You" is an illustration of the way in which games facilitate positive emotion. The parent repeatedly looms toward the infant, rubbing his or her head into the lap of the infant, tickling and jostling the infant, and accompanying each repetition with verbalizations such as "Ah, boom!" or "I'm gonna get you." These repetitious behaviors elicit infant smiles, vocalizations, and sometimes laughter. The parents contingently respond to these infant behaviors by playing more of the same game. The game ceases at the point at which it no longer elicits contented responses, and the parent moves on to another game.

Thus, early game playing provides an affective, contingency experience during which infants may learn important rules of communication such as turn-taking. When games are missing from the interaction or age-inappropriate games are played, the interaction appears to be badly timed, disorganized, or disturbed. When a mother plays age-inappropriate games or games that are too complex for a given age of infant, for example peek-a-boo with a 6-week-old infant, the interaction lacks positive emotion and is not sustained.

The early interactions of developmentally delayed, atypical, or failure-to-thrive infants and mothers feature very few games. These mothers do not appear to enjoy playing games and do not as frequently play games, particularly age-appropriate games, with their infants, probably because their infants are less responsive.

In one of our studies, the types and frequency of widely recognized infant games were observed during face-to-face interactions of normal and high-risk

infants and parents when the infants were 4 months old (Field, 1979). Approximately six different games were played for approximately one-third of the interaction time. Fathers played with normal male infants more often, and the high-risk infants and parents played games less frequently than the normal infants and parents.

Interaction Coaching

When parents, most typically mothers, are having difficulty interacting with their infants, they can be given interaction coaching, which helps modify the behavior of the parent, and, in turn, the behavior of the infant (Field, 1983b). Typically mothers and infants are seated in a face-to-face position, with the infant in an infant seat. The mothers are given instructions to slow down their behavior. For example, they slow down when they are asked to imitate their infants' behaviors and they provide more stimulation when they are asked to keep their infants' attention. Having to imitate the baby's behaviors slows down the mother's behavior simply because infant behaviors are slower than adults'. The infants usually delight at being imitated, and before long, mothers and infants are engaging in a back-and-forth imitation game. Another way that mothers learn to modify their behavior is by watching video feedback on their interactions. Mothers can usually see the immediate effects of their behaviors and will show change almost immediately.

Schedules

The importance of a schedule cannot be overrated. From the time of birth, newborns need to be on sleeping and feeding schedules so that they can organize their own internal clock and have sustained periods of sleeping and ultimately sustained periods of playfulness (Field, 1978). Having periods of consolidated sleep, including deep sleep, is critical for children to sustain their activities, attentiveness, and interest in their play. Without having a schedule when they are growing up, children can have serious difficulties organizing themselves and often will become hyperactive and develop learning disabilities.

I remember one of our laid back nursery school teachers who believed that every infant should have his or her own schedule, and she operated her infant nursery class on that premise. The infants were extremely fussy, often spitting up and not being able to focus their attention on any activities. Being on 16 different schedules, the four teachers also became exhausted and irritable. The teacher who replaced her ran the infant nursery like an army sergeant and had all the infants on the same schedule within a week. Parents were at first resistant but came to appreciate that their infants were much less irritable. The need for a schedule continues through childhood, which is why children typically do much better in a school setting than they do on weekends, when everyone is on a different schedule.

Quality Time

In today's world of dual-career families parents are concerned about snatching as much time as they can with their children (Broom, 1998). Often this leads to

children staying up late and sacrificing the quality time they spend in day care. Quality time, the time parents give their undivided attention to their children, is key—not quantity of time (Brooks-Gunn, Han, & Waldfogel, 2002).

Discipline

Affection and discipline are two of the most important aspects of parenting. Children growing up in the wild without parents become like wild animals because they are missing affection and discipline. In addition to needing affection for growth and development, children need discipline for their behavior to be organized and self-regulated (Socolar, Savage, Keyes-Elstein, & Evans, 2005). While some choices can be left to the child, for example, what the child wishes to wear, other matters that are critical to survival, such as not playing in the street and not being cruel to other children, require very firm limits. A parent should not ask a child if the child wants to hold the parent's hand while crossing the street; this is not a safe choice for a child to make. Having an intellectual discussion on the reasons why hurting someone is bad is also not effective. I recall a teacher in a model preschool quietly leaving the room after having been hurt by a flying block only to return several minutes later to have an intellectual discussion with the children about the "negative aspects of flying blocks." A better lesson would have been reacting normally to being hurt so the children would know the consequences of their actions.

Another problem with discipline is its frequent inconsistency. Recent reviews of the literature on behavior problems and on child abuse suggest that the most predictive factor for both is inconsistent discipline, such as parents being inconsistent from one time to the next or from one parent to the other (Field, 1983a). In societies where discipline is consistent within and across parents and between parents and teachers, such as in France and China, children have fewer behavior problems, and the incidence of child abuse is very low. Socially accepted forms of discipline are practiced by parents and teachers alike. Thus, the child is never seen driving the parent up the wall and the parent doesn't lose control.

A recent study revealed that verbal and physical forms of discipline are not substitutes but, instead, are commonly used together. Parents who yelled frequently were the ones most likely to hit frequently (Wade & Kendler, 2001). Respondents who were spanked (yelled at) frequently as children were more prone to frequently spank (yell at) their own children. Harsh parental discipline occurring early in life was associated with later aggression in children. The good news was that most people were able to break out of the cross-generational cycle of punitive childrearing, although physical punishment is still quite common. In a recent survey, 19% of mothers believed that there are times when it is appropriate to spank a child less than 1 year old, and 74% believed this about children 1 to 3 years old (Vittrup, Holden, & Buck, 2006); 42% reported that they had spanked their own child in the past week. Mothers believed more strongly in spanking for dangerous misbehaviors than for annoying ones. Other traditional disciplinary methods include rewards for good behavior and punishment for unacceptable behavior. The most common rewards are verbal praise, a smile, special attention or activities, or physical affection. Usual kinds of punishment include verbal disapproval, an unhappy look, ignoring

the behavior (extinction), or temporary isolation (time-out). Rewards are typically cited as being more effective than punishment.

Childproofing

Childproofing offers the opportunity to simplify your environment. It can even be fun to convert your living room to a playroom where the whole family can enjoy playing together. The obvious safety precautions are removing sharp objects, poisonous substances, plastic bags, and anything that can injure or anything small that can lead to choking or suffocation; placing guards on stairs, safety latches on cupboards, plugs on electric outlets, and locks on doors; and certainly locking up guns. For many years children do not know the difference between reality and fantasy, and for that reason alone guns should be under lock and key. Choking on foreign bodies is the cause of death for more than 300 children each year in the United States. A review of the records of 548 children (ages 4 months to 18 years) was undertaken to identify factors important in diagnosis. A wide variety of objects was recovered, the most common being peanuts, organic material, other nuts, popcorn, seeds, plastic objects, and pins (LeBlanc et al., 2006).

Airway obstruction from aspiration of a foreign body should be suspected in all infants and children who have swallowing or respiratory difficulties (Agran et al., 2003). If the patient is unable to clear the airway by coughing, the Heimlich maneuver should be attempted. Parents can help to prevent airway obstruction by keeping small objects away from infants and children and by teaching them to chew food thoroughly. Whenever the child is approaching a forbidden object or something dangerous, it is best to simply remove the child or distract him or her with a similar object or activity rather than saying no and expecting the child to leave the area. Saying no for most young children is simply a challenge and escalates into a struggle. Simply carrying or taking children by the hand and moving them to another area where they can be distracted is more effective (Glick, Greaves, Kronenfield, & Jackson, 1993).

Leave-takings

Leave-takings are increasingly common, given the increasing numbers of infants, toddlers, and preschoolers attending nursery schools. Research suggests that different children have different leave-taking styles. In one study, children who cried during the early morning separations showed either a slow approach or no response except looking at the mother when she returned from her day's work (Field, Gewirtz, et al., 1984). Those children who left their mothers quickly in the mornings typically continued playing and showed things to the mother when she returned in the afternoon. The mothers of children who cried during separations more often slipped out of the room. The authors of this study suggested that much of the separation and reunion behavior is determined by the child, with the mother merely being in a responding mode at these times, mainly waiting to receive and act on signals from her child. However, in a study we conducted, we thought that the parents were very influential in the leave-takings.

We studied the leave-taking and the reunion behaviors of the infants, toddlers, preschoolers, and their parents as the children were dropped off and picked up at their nursery school each day (Field, Gewirtz, et al., 1984). On arrival at their classroom, infants and toddlers related primarily to their parents, whereas preschoolers related to their teachers. Girls more frequently engaged in interaction with their teachers, and boys more frequently approached the children's play activities. Toddlers showed the most distress behavior during the parents' departures, and the toddlers' parents hovered about them and sneaked out of the room more frequently. When mothers dropped off the children, the children showed more attention-getting behavior and crying than when the children were dropped off by their fathers. Mothers engaged in more behaviors to distract the child and took a longer time to leave the classroom than fathers did. Giving a verbal explanation, distracting the child, lingering in the classroom, and sneaking out of the room were accompanied by more crying. We noted that some children had difficulty making transitions, and those children showed difficulty not only leaving their parents in the morning but leaving their classroom in the afternoon. In addition we noted that fathers were much more nonchalant in their behavior, and the children were less distressed when they were dropped off by their fathers. Thus, the rather business-like style of leave-taking by the fathers seems to work better if parents want to avoid the stress of seeing their children's distress.

Moving

Moving is one of the most stress-producing and growth-producing experiences in childhood. Whether the move is from classroom to classroom, neighborhood to neighborhood, city to city, or country to country, the source of stress is losing one's friends. Separation from peers leads to symptoms very much like those that occur following separation from parents. The child becomes agitated and then depressed. Sleep is most affected by moving, with night wakings and night terrors frequently happening. The child's behavior fluctuates between being aggressive and withdrawn, and like depression in adults, eating habits change (overeating or undereating). The children experience constipation or diarrhea, and sometimes they get sick because of immune problems. We have observed these changes in studies on moves to new classrooms and moves to new schools (Field, Gewirtz, et al., 1984; Field, Vega-Lahr, et al., 1984). Even as young as 1 year of age, we are having to move children's best friends with them when they graduate from infancy classes to toddler classes so that they will not experience these symptoms. The children left behind often experience more severe symptoms, perhaps because they are remaining in the same environment where they are reminded of their long-lost friends.

Maternal Depression

Mothers who are depressed or who have symptoms of depression such as flat affect (showing little emotion) and listless behavior tend to have a difficult time interacting and being responsive to their infants (McLearn, Minkovitz, Strobino, Marks, & Hou, 2006a, 2006b; Paulson, Dauber, & Leiferman, 2006). The infants in turn develop

flat affect and low activity levels. If the mother remains depressed over the first 6 months of life, the infant will begin to show growth and developmental delays by 1 year. By 3 years children of depressed mothers develop behavior problems such as depression and conduct problems. If mothers are simply depressed postpartum and the depression does not continue, the infants do not experience long-term effects.

Fathers and substitute caregivers such as infant nursery teachers can buffer some of these effects (Hossain et al., 1994; Pelaez-Nogueras, Field, Cigales, Gonzalez, & Clasky, 1994). Infants of depressed mothers often appear normal when interacting with their nondepressed fathers and teachers. Other effective interventions are the use of mood inductions with the mothers, such as music therapy and massage therapy (Jones & Field, 1999). Interaction coaching to help mothers be more active and responsive to their infants, such as playing peek-a-boo and imitation games, helps them improve their interactions. Teaching depressed mothers to massage their infants also helps improve the mothers' mood and reduce stress in the infants (Field, Grizzle, et al., 1996).

Fathers

In most studies people have found fathers to be more playful and stimulating and mothers to provide more caregiving and comforting. We found an exception to this when we studied fathers who were primary caregivers (Field, 1978). The stay-at-home fathers turned out to be more like mothers than fathers, and the mothers who were the breadwinners were more like the stereotypical father. We suggested that this meant that more caregiving is provided by the person who knows the infant better and more playfulness by the parent who knows the infant less well.

In a sense, the more stimulating, rough-and-tumble play of the fathers is complementary to the more soothing, comforting behavior of mothers. In a similar light, mothers tend to view their infants as being more difficult (perhaps because they are in a comforting role), and fathers are reported to enjoy their infants more (again because they engage in more play with them). Across the first several years, fathers are more involved in social interaction than in caregiving, and mothers continue to provide more of the caregiving. Although the differences between mothers and fathers has lessened over the years as more mothers have joined the workforce, some differences have continued, showing the mother providing more caregiving and the father more playful stimulation (St John, Cameron, & McVeigh, 2005).

Despite the differences in their parents' behaviors, infants and young children are noted to be equally attached to both parents. It may not appear that way, as infants and young children often protest their mothers leaving them at preschool and barely notice their fathers leaving (Field, Gewirtz, et al., 1984). This is probably because fathers matter-of-factly leave, saying they will see their child later, whereas mothers make a greater fuss about leaving their children and express more reassurance about their returning, which seems to draw out this fussing behavior. The children's ambivalence about leaving their mothers is also displayed upon her return, whereas infants greet fathers as they return with intense, brief displays of positive emotion.

Grandparents

Grandparents have a number of positive effects on child development (Harwood, 2001). Children are often more able to explore and try their skills around grandparents when not around their own parents. They seem to recognize a certain autonomy around grandparents, in contrast to their dependence around parents. This may be a two-sided process, as grandparents appear more relaxed and less overprotective around their grandchildren than parents do. They apparently also have less need to set limits than parents, as typically children seem more self-regulated in their presence. Part of this phenomenon may relate to grandparents giving children their undivided attention, whereas parents are often busy trying to juggle children and other activities at the same time. It could be that the child's more demanding and less compliant behavior in the parents' presence is attention-getting behavior. It could also be that the children sense greater demands being placed on them by their parents and higher expectations for performance. Combined experience with parents and grandparents, in any case, would appear to be optimal for any child.

Babysitters

Having a babysitter even for a newborn is a great break for parents and provides them needed time together. Finding someone who has experience, who loves taking care of babies, and who comes with good references from people you know is generally the best way to choose. On the first few occasions it is important to be there for a while until the babysitter becomes comfortable in the environment and familiar with the baby. Watching how the babysitter relates to your child and cares for your child in those few hours of getting acquainted is one way to develop confidence in the babysitter.

Day Care

Early day care helps develop a child's social and intellectual skills, but only if it is quality care. Penelope Leach, in her book *Children First,* and others have claimed that full-time day care can have harmful effects. However, their concerns relate to poor-quality day care (Geoffroy, Côté, Parent, & Seguin, 2006; Schwebel, Brezausek, & Belsky, 2006). In contrast, the average day care that most middle-class kids get is considered helpful for cognitive and social development, according to the National Research Council.

Quality care can be hard to define, but knowing what to look for in a day care center helps. State and federal governments have licensing requirements that ensure the provision of adequate space and safety features (e.g., smoke alarms and sprinkler jets). These requirements also ensure optimal teacher-student ratios and class sizes. For example, quality programs in Florida have no fewer than 1 teacher for every 4 infants, 1 teacher for every 6 toddlers, and 1 teacher for every 10 preschoolers in classrooms with no more than 20 infants, 30 toddlers, and 40 preschoolers. These figures, however, are meaningless if sufficient space is not provided (approximately 35 sq. ft. for indoor space and 50 sq. ft. for outdoor space per child).

Most parents know when they have discovered a quality day care setting. It will be immediately apparent that the teachers are nurturant and the children are happy. Children will be involved in creative activities and the environment will be organized, colorful, and rich with interesting materials and play opportunities. A class schedule with designated times for show and tell, snacks, naps, free play, and expressive arts, along with free time for fantasy play and gross motor play (play that uses muscles) is also an important part of quality day care. Outdoor play areas will include moving vehicles, sand play, and structures for climbing. Parents should ask questions about how long the teachers have been with the center (stability of the teachers is very important) and how much experience they have.

Some of the highest quality centers are found in university lab schools because they are partially subsidized by the universities. Some of the benefits of a laboratory school include active research and teacher training programs affiliated with their early childhood education and psychology departments. Religious centers (e.g., churches and synagogues), hospitals, corporate centers, and Montessori schools also provide quality day care. Specific listings are often provided by early childhood education departments at universities.

In a study on skills needed by preschool children for success in day care, teachers assigned the highest ratings to communication and independence. More specifically, these included children's ability to respond to their names, ask for help from an adult, stay out of restricted areas and respond to adults' questions. The teachers rated skills in the area of preacademics (including sitting cross-legged in a circle and cutting with scissors) the lowest.

In a study we conducted, the longer infants were in quality day care, the greater their emotional well-being and academic performance at grade school (Field, 1991). Similar data have been reported for quality day care in Sweden.

Pacifiers

Pacifiers are often viewed with disdain, and they are discouraged to the extent that premature infants in some hospitals require a physician's prescription for a pacifier. Yet documented and anecdotal reports show that sucking pacifiers helps calm infants during uncomfortable procedures, and toddlers are made more comfortable when adapting to unfamiliar situations (Stevens, Yamada, & Ohlsson, 2004). Pacifiers provide comfort, promote physiological tranquillity, and help in growth and development. In one study, we used pacifiers during painful hospital procedures (Field & Goldson, 1984). The infants who were allowed to suck on pacifiers cried significantly less, and their heart rate and respiration were slower. Pacifiers used during circumcisions reduced crying by 40%. It is not clear whether pacifier use reduces pain associated with invasive procedures or if the sucking is just incompatible with crying and therefore crying occurs less often. It is also possible that sucking on pacifiers releases pain-alleviating chemicals like serotonin. Sucking on pacifiers has also been used for weight gain, particularly with premature babies. In a study we conducted on premies, those who were allowed to use pacifiers gained weight more rapidly and were discharged from the hospital 6 days earlier than those not using pacifiers, at savings of approximately $3,000 per infant

(Field et al., 1982). These infants also performed better on a newborn assessment scale.

Given the positive effects of sucking, it is very fortunate that infants are inclined instinctively to engage in sucking behavior. That predisposition emerges very early in gestation, with fetuses as young as 5 months appearing to suck on their own hands in utero. During the first day of life, infants have been observed sucking as often as 30 minutes per hour. This does not appear to vary with the feeding experience. One survey found that 89% of parents introduced pacifiers to soothe their infants. Most studies in the literature concur that sucking is the most common method for soothing an infant. In another study pacifiers were compared with rocking. Pacifiers produced sleep states, and rocking produced alert states. Pacifiers also reduced heart rate levels significantly more than rocking. Thus pacifiers seem to be more soothing than rocking.

Sucking is also used for exploration and learning. Babies in one study sucked harder to hear their mother's voice than their father's voice, probably because they were more familiar with the vibrations of their mother's voice in utero (DeCasper & Prescott, 1984). Sucking is used in this and many studies to determine whether infants hear, see, make discriminations, and demonstrate sophisticated skills such as performing small math problems, discriminating different colors, and learning musical scales as early as the newborn period. Other investigators have noted how much learning occurs during infants' exploration by mouth. For example, infants are able to show that they can differentiate two different nipples (one with a knobby surface and one with a smooth surface) simply by having experienced sucking on the nipple and then looking longer at the picture of the nipple they had sucked, showing their preference for that familiar nipple (Hernandez-Reif, Field, & Diego, 2004).

Even if infants do not have nipples to suck on, they frequently suck on their hands or other objects. As early as 5 months gestation, fetuses explore their hands with their mouths. This persists through early infancy, when the mouth explores the hands, fist, and fingers. Similarly, for at least the first 6 months of life, everything the infant grasps goes into the mouth. In a recent court hearing I was asked how a 5-month-old infant could stuff a rather unpleasant-tasting alcohol wipe in his or her mouth. Invariably every 5-month-old I filmed for the hearing did exactly that. When babies have something more interesting to suck on, such as a nipple or rattle, they do not choose alcohol wipes. Infants prefer toys because of their interesting textures of softness, hardness, and firmness. Toys also give the infant information about features such as shape. Infants have individual preferences as well. Some infants enjoy sucking their soft toes, and others enjoy hard, firm plastic toys. Thus, sucking is a predominant activity during the first 6 months of life, just as walking is at 1 year, and it clearly provides a very basic learning function.

If sucking is so critical for reducing stress and facilitating growth and development, most especially cognitive (mental) development, and if sucking does not appear to have any negative effects, why should anyone want to eventually discourage the habit? One reason is that as a child develops, sucking on a pacifier or thumb may be incompatible with vocalizing and talking, just as it is incompatible with crying during the newborn stage. When infants become preschoolers they may

experience rejection by peers for engaging in "baby behavior." Thus pacifier use at preschool is probably not desirable. In one study pacifier use declined in a voluntary way between 3 and 24 months, suggesting that for most children, the attachment to a pacifier declines naturally. However, for others, sucking is a type of security blanket that becomes difficult to relinquish.

To prevent pacifier overuse parents should not get into the habit of popping pacifiers into their babies' mouth when the baby doesn't need pacifying. It is important to learn to tune in and appropriately respond to your baby's cues and prevent pacifier overuse. Older babies should be given a pacifier only for true pacifying, soothing, and calming.

As an aside, parents often ask whether sucking on a pacifier or thumb can mean that the child will need to wear braces. For most children this is not a problem. Pacifier and thumb sucking have very little effect on a child's oral-facial development and on the primary teeth and position of the jaw when it is done periodically (Adair, 2003). If the child continues to suck when the permanent teeth are coming in, pacifier or thumb sucking could cause movement of the teeth and bones, and orthodontic treatment might be needed.

Security Blankets

Security blankets have their place in infancy and toddlerhood, as they allow children to more securely leave their mother (father leave-takings are typically less difficult). However, beyond the toddler stage, security blankets may be a source of embarrassment for the children when their peers make fun of their security objects. Bringing toys to school may provoke less embarrassment, but that often leads to little wars over possession of each other's toys. For that reason most teachers prefer that children leave their toys at home.

Toilet Training

Toilet training is typically treated by pediatricians as an individual readiness phenomenon, although most children are toilet-trained (according to recent study) by 2.5 years, with physicians beginning to discuss toilet training with parents at around 12 to 18 months of age (Horn, Brenner, Rao, & Cheng, 2006). Although parents are initiating toilet training at later ages, around 3 years (Blum, Taubman, & Nemeth, 2004), waiting to start toilet training at age 2 is particularly problematic because the terrible twos involve power struggles between parents and children that make toilet training extremely difficult (Blum et al., 2004; Schum et al., 2001).

Placing a child in training pants and trying not to use diapers except during nighttime sleep is an effective introduction to toilet training. Buying children their own toilet seat gives them a sense of control over their own activity. Children seem to learn faster when they are attending preschool, probably because the older children model more mature toileting behavior. Taking children to the toilet frequently will give them the idea, whether at home or at school, that there are designated periods when all kids have bathroom time.

Flashcards

The flashcard method was designed by a "better baby" school in Philadelphia to teach children music, language, science, and virtually anything that can be taught on flashcards to infants and toddlers. The school has repeatedly made great claims that 2-year-olds can learn physics, violin, and the classification of insects, for example. This is done by repeated drilling with flashcards, typically at home by the parent. None of these claims, however, has been supported by research, and the directors of the school are reluctant to have their methods tested. It is also not clear why a parent would want their child to know all this by the time the child enters school, only to present the potential problem that the child will be rejected as "weird" by his or her peers. The child who is busily learning flashcard skills has very little time to learn social skills and how to get along with peers, which is certainly a more important priority at the preschool stage.

Television

The American Academy of Pediatrics recommends that children under the age of 2 not watch any television. Yet, several surveys indicate high numbers of infants routinely watching TV, including 17% of 0- to 11-month-old infants and 48% of 12- to 23-month-old toddlers in a large U.S. study, and even larger numbers in a Japanese study (30% of 3-month-olds and 90% of 12-month-olds). Exposure is also high, at an average 44 minutes per day among American 4-month-old infants, increasing to 62 minutes per day in 12-month-old infants; and again, there is higher exposure in Japan. In one study, 92% of exposure of 4-month-olds was to adult TV. Some view TV exposure as a cognitive learning experience, but others are concerned that infants are imitating TV models, both immediately and days later, remembering and mimicking the model's actors. They also are taking directions from TV. For example, in one study, 10- to 12-month-old infants who watched an actress respond to a toy with fear later avoided playing with it themselves, and they were more likely to appear worried and cry. In another study, food-related advertising on TV was associated with brand recognition in toddlers (Connor, 2006). Other studies suggest that television viewing in infants is associated with irregular sleep schedules (Thompson & Christakis, 2005) and with a risk of becoming overweight (Dennison, Erb, & Jenkins, 2002).

Children are first exposed to the audio and video stimulation of TV shortly after birth to the age of 2 months. They then begin to have an interest in and glance at or watch the TV screen at 4 to 5 months of age. When they are able to crawl at around 6 months, they approach and investigate the TV set. At 7 to 8 months they enjoy switching channels, and then they become interested in and speak to the on-screen characters at 10 to 11 months. At 12 to 14 months of age they have favorite TV characters and programs and enjoy TV as ordinary TV viewers. They move their bodies with the music, mimic the actors on-screen, smile, and ask questions of parents. They fix the information from TV in their memory, and they are pleased to see the things they like on TV. Even the fetus is affected by TV, showing heart rate acceleration or deceleration and fetal movement when the mother is watching

TV. Disruption of sleep patterns due to TV may also occur to fetuses. For some, though—for example, infants of depressed mothers—selected children's TV may provide more adequate stimulation.

Accidents

Approximately 80,000 children are killed and another 50,000 disabled by unnecessary accidents, according to the U.S. National Safe Kids Campaign. This makes accidents the number 1 killer of American children. Accidents are now the cause of more deaths in children than the next six most frequent causes combined. The leading causes of these accidents are traffic injury, drownings, burns, choking, poisoning, and falls.

Minor accidents such as scrapes and bruises are inevitable as infants, toddlers, and young preschool children learn to wend their way through the world. Both children and parents overreact to these minor injuries. Accident-prone behavior has been related to hyperactivity, clumsiness, and antisocial behavior in children and lack of parental discipline in the home. Injury-prone behavior is most common, for example, in active boys with irregular sleep and eating problems. Often, though, accidents happen to children who are simply adventuresome.

Solutions for accidents are making the environment safe, including using seat belts and child safety seats to avoid traffic injuries, teaching the infant how to swim, using pool detectors, never leaving an infant alone in the bathtub to avoid drownings, monitoring hot water in the bathtub to avoid burns, making certain that back burners are used on the stove and smoke detectors are in working order, and locking up all medications, liquor, and cleaning substances to avoid poisoning. Children's homes need to be childproofed at least until the early grade school years, including using electric socket covers and safety latches on cupboards and covers on sharp corners of furniture.

Automobile accidents are the number 1 killer of children (Rimsza, Schackner, Bowen, & Marshall, 2002). Until children are 5 years of age (or more than 45 pounds) they need to be in a car seat that protects them from head-on as well as side crashes and meets the U.S. federal motor vehicle safety standards. Every state has laws requiring that children younger than 5 must be strapped into car seats in cars in motion. Children under 5 are most commonly killed in cars. Children under 10 should not be allowed to cross streets without adult supervision. An immediate need following a car accident is, of course, an emergency rescue.

Death from *fires* is the second most common cause of death, with 75% of these being caused by smoke inhalation and 80% occurring in house fires (Istre, McCoy, Carlin, & McClain, 2002). Approximately 50% of house fires are caused by cigarettes. Smoke detectors should be installed in areas just outside bedrooms, and a fire extinguisher placed in the kitchen and on each floor of the house. Scalding is the most common cause of nonfatal burns; 20% of these result from tap water and 80% from spilled food. The average age of children with burns is 32 months. Hot water heaters should be regulated at 120 degrees Fahrenheit. Because coffee burns are most common, infants should not be seated on any adult's lap during a coffee break. For the treatment of burns, cold water should be dribbled on the burn,

sterile gauze squares held in place by gauze roll bandaging, and the doctor called or the child taken to the hospital. Other precautionary measures include not heating an infant's bottle in the microwave, dressing the child in fire-retardant sleep wear, and avoiding place mats and table cloths that can be pulled off the table.

The most common cause of *drowning* (the third most common cause of death to children) is leaving the child alone in the bathtub in a few inches of water in a minute's time. Even with swimming lessons, children less than 5 typically have less strength and coordination to swim out of danger, and their parents have a false sense of security from the swimming lessons, which often leads to minimal supervision. A safe rule of thumb is that any child under 10 should be supervised whenever near the water. An increasing percentage of drownings (44%) occur in nominally fenced-in pools in which the gate is open or the fence is in disrepair.

Approximately one-third of *poisonings* by medicine are due to children taking their grandparents' prescription drugs. Thus, not only your own home but the grandparents' home should be made child-safe. Cosmetics, shampoos, detergents, plant poisons, paint materials, and substances associated with automobiles need to be out of reach. Because different poisons require different actions—for example, for some it is important to induce vomiting, whereas others should not be treated that way—it is important to have a poison control center card and to call the poison control center, the doctor, or an emergency room.

An information card is also important in responding to *choking* situations. Parents should take cardiopulmonary resuscitation (CPR) classes in case artificial respiration is needed. And parents should certainly learn the Heimlich maneuver. *Miscellaneous dangers* in the home are stairs, windows, small objects, sharp objects, toilets, matches, tools, plastic bags, telephone cords, and, of course, guns. Most children do not know the difference between real and fantasy guns.

SUMMARY

The problems of infancy can be categorized as both the frustrations and challenges of the developmental milestones that occur and the problems they pose for the parents facilitating development and giving care along the way. The research on these problems is sparse and highlights the need for further studies on the problems of infancy.

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CHAPTER 32

Psychopharmacology as Practiced by Psychologists

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According to the *Oxford English Dictionary* (*OED*), the term "psychopharmacology" was first used in print by Mora in 1920 to refer to a yet to be developed science dealing with the behavioral effects of drugs (Macht & Mora, 1921). This meaning persists to the present, as indicated by the current *OED* definition: "Psychopharmacology: The branch of science concerned with the way drugs affect the mind and behavior." The mind, of course, cannot be directly assessed, but is studied indirectly by examining behavior, that is, what individuals say and do.

All drugs that affect behavior—broadly construed to include mood, cognitive status, and overt actions—are termed "psychoactive." Drugs that are prescribed with the purpose of improving behavior are called "psychotropic." All psychotropic drugs are psychoactive, but not all psychoactive drugs are psychotropic. The introduction of the first generally effective psychotropic drug, chlorpromazine (Thorazine), in the early 1950s revolutionized psychiatry and changed forever the practice of mental health.

Historically, only physicians prescribed psychotropic drugs. Over the past 20 years, however, the possibility of granting the privilege of prescribing psychotropic drugs to properly trained psychologists has become a highly visible, and hugely important, issue. The American Psychological Associations (APA) supports prescription privileges for psychologists, as do many state-level psychological associations. Two states recently enacted legislation allowing properly trained psychologists to prescribe these drugs, but there is continuing debate among psychologists as to whether or not prescription privileges are generally desirable. The issue is complex and we will examine it in a later section. But quite apart from the prescription privileges issue, psychologists have made and are making major contributions to the study of psychoactive drugs. We first examine some of those contributions.

PSYCHOLOGY AND THE STUDY OF DRUGS

Psychologists are expert in dividing the ongoing stream of behavior into meaningful units, in quantifying those units, in explaining why they occur, and in predicting and (sometimes) influencing future behavior. Their expertise in each of these areas has contributed to understanding the effects of psychoactive drugs.

Explanations of Behavior and Psychopharmacology: Mechanisms of Drug Action

Many important behaviors of humans and other animals are learned. That is, they occur through the processes of operant and classical conditioning. Operant conditioning is a form of learning in which the likelihood of occurrence of a particular response in a given situation is controlled primarily by its historical consequences in similar situations. If the consequences increase the likelihood of occurrence of the response, the process is termed reinforcement and the consequences are called reinforcers. If the consequences decrease the likelihood of occurrence of the response, the process is termed punishment and the consequences are called punishers.

Features of the environment that are present when consequences occur and subsequently exercise antecedent control over behavior are termed discriminative stimuli. Technically, a discriminative stimulus for reinforcement is a stimulus that evokes a response because in the past that kind of response has been more successful in producing reinforcement in the presence of that stimulus than in its absence. The classic example of operant conditioning involves Skinner's mildly food-deprived rats, which press a bar in the presence of a tone but not in its absence, because historically the bar-press response produced food (a reinforcer) in the presence of the tone but not in its absence. Regardless of whether or not the tone is present, substantial levels of bar pressing occur only when the rats are food-deprived. Food deprivation here is a motivational variable, specifically an establishing operation that increases the reinforcing capacity of food and increases the likelihood of occurrence of responses that historically produced food.

Unlike operant conditioning, classical (or respondent) conditioning is a form of learning in which behavior is controlled not by its consequences but by stimulus-stimulus pairings. Classical conditioning begins with a reflex, that is, an unlearned relation in which an unconditional stimulus elicits an unconditional response. By arranging conditions such that a previously neutral stimulus is repeatedly predictive of an unconditional stimulus, that neutral stimulus becomes established as a conditional stimulus, which reliably elicits a conditional response. The classic example involves Pavlov's mildly food-deprived dogs, in which food (unconditional stimulus) reflexively elicits salivation (unconditional response). By repeatedly presenting a sound (neutral stimulus) just before food (unconditional stimulus), the sound comes to elicit salivation (unconditional response) and is established as a conditional stimulus.

Understanding the processes of operant and classical conditioning provides a conceptual framework for examining behavioral mechanisms of drug action. "Mechanism of action" refers to a general drug effect at one level of analysis that produces (or at least, covaries with) another, usually more specific effect at another level of analysis. In lay terms, a drug's mechanism of action refers to how the drug "causes" or "works to produce" whatever effects are of interest. For example, chlorpromazine generally is effective in reducing what are referred to as the positive symptoms of Schizophrenia (hallucinations, thought disorders, bizarre behavior), which is one of the drug's most noteworthy effects (e.g., Baldessarini & Tarazi, 2006). The drug also produces adverse motor effects (rigidity, tremors,

slowed movement, restlessness). These effects, which can be observed directly, covary with events at another level of analysis, specifically, antagonism of the neurotransmitter actions of dopamine by competitively blocking dopamine receptors (Julien, 2001).

Early observations indicated that effective antipsychotic medications reduced dopaminergic activity in the brain and reduced positive symptoms of Schizophrenia. In addition, researchers observed that chronic exposure to high doses of drugs that increased dopaminergic activity, such as amphetamine, produced changes in behavior closely resembling those characteristic of Schizophrenia. These findings led to the speculation that Schizophrenia resulted from metabolic errors leading to either (a) overproduction of dopamine in the limbic system or cortex of the brain, or (b) production of an endogenous amphetamine-like compound. Similarly, the finding that drugs useful in treating depression increased monoaminergic activity suggested that depression results from a deficit in serotonergic and/or noradrenergic activity (Baldessarini, 2000). Studies of family resemblance suggested that genetics played some role in both Schizophrenia and depression, and it was soon widely believed that both conditions resulted from inherited diseases of the brain that produced abnormal neurochemical activity, which in turn produced abnormal behavior.

These early biological models of mental illness were crude, even primitive, and ran into problems from the start. For example, antidepressant drugs begin increasing serotonergic activity as soon as they are administered, but clinical benefits take weeks to appear. Also, no one has succeeded in demonstrating consistent neurochemical differences between people who are and are not diagnosed with depression or Schizophrenia. Finally, the role of inheritance in depression and Schizophrenia is rather weak, and the environment clearly plays a role in their genesis. How this occurs was not considered in the early models. In response to these problems, more sophisticated models have been proposed. They, too, are inadequate to explain mental illness (e.g., Baldessarini, 2000; Julien, 2001). Neuroscientists are making dramatic progress in understanding the human brain, however, and it is possible—perhaps even probable—than reasonably comprehensive, testable, and confirmed biological models of particular behavior disorders will soon appear.

Be that as it may, biological models developed to this point have had an enormous influence on how physicians, psychologists, and laypeople conceptualize mental illness. For many people, the signs (occurrence of undesired behaviors or absence of desired behavior) and symptoms (patient-reported affect and cognitive status) that lead to a clinical diagnosis are a manifestation of underlying pathophysiology. Effective treatment, regardless of its nature, corrects the underlying pathology, leading to alleviation of signs and symptoms. Under this model, drugs correct the pathology directly, whereas psychological interventions do so indirectly.

Although it has been criticized as not being supported by compelling evidence and focusing attention away from environmental variables that strongly affect the genesis, course, and treatment of mental illness (e.g., Malott, in press; Whitaker, 2002), this general view is widely held. Referred to variously as the medical, biological, or disease model of mental illness, this view at the extreme supports the notion that each condition listed in the *Diagnostic and Statistical Manual of Mental*

Disorders, fourth edition (*DSM-IV,* American Psychiatric Association, 1994) is a discrete disease with known signs and symptoms and a potentially specifiable neurochemical cause. This conception, however popular, is difficult to defend scientifically. There is, nonetheless, good evidence that particular disorders are fundamentally biological with strong genetic components, even if the specific structural and functional differences between people with and without the disorder cannot presently be specified precisely, if at all. For example, Russell Barkley, a psychologist well known for his research involving children with Attention-Deficit/Hyperactivity Disorder (ADHD), responded as follows when asked by a reporter for PBS's *Frontline* (2006), "So this [ADHD] is a biological disorder?"

> It's largely a biological disorder. It has many causes, but all of the known causes fall within the realm of neurology and genetics. We can rule out the social environment, such as bad parenting, intolerant teachers, the breakdown of the American family, a decline in family values, excess amount of TV viewing or video games. These have all been proposed as causes of ADHD. But there's no evidence that we can find that will substantiate them.
>
> All of the evidence that keeps turning up in our research points to genetics and neurology as being largely responsible for the excessive behavior and the poor self-control that we see in these children. So I think we can safely say that what we've learned in the past 10 years is that environmental causes of ADHD are not credible. They do not account for the substantial amount of scientific findings that exist on this disorder today.

Regardless of whether the drug in question is or is not a psychotropic medication, most of the interest in mechanisms of drug action has focused on the relation between neuropharmacological and behavioral events, and receptor models of drug action have proven invaluable in predicting and explaining drug effects. Some behavioral psychologists, however, have emphasized behavioral, not neuropharmacological, mechanisms of drug action. Behavioral mechanisms of drug action involve the stimulus properties of drugs in the context of operant and classical conditioning and the manner in which drugs modulate the actions of nondrug stimuli (Thompson, 1981; Thompson & Schuster, 1968). More specifically:

> By behavioral mechanism of drug action, we refer to a description of a drug's effect on a given behavioral system (locus) in terms of some more general set of environmental principles regulating behavior. Specifying the behavioral mechanism(s) responsible for an observed effect involves: (a) identifying the environmental variables which typically regulate the behavior in question, and (b) characterizing the manner in which the influence of these variables is altered by the drug. In some instances, the drug assumes the status of a behavioral variable per se, rather than modulating an existing environmental variable. (Thompson, 1981, p. 3)

In passing, it should be noted that the concept of behavioral mechanisms of drug action was developed and popularized by a psychologist, Travis Thompson, and his colleagues. This concept is abundantly supported by demonstrations that certain drugs inherently have, or can acquire through learning, the capacity to function as unconditional, conditional, discriminative, reinforcing, and punishing stimuli and as motivational operations (Poling & Byrne, 2000; Thompson & Pickens, 1971).

Especially significant is the demonstration that drug self-administration involves learned behavior that is maintained, in substantial part, by the short-term reinforcing effects of the drug in question (Laraway, Snycerski, Byrne, & Poling, 2000). Harmful patterns of self-administration, that is, drug abuse and drug addiction, are controlled by the same classes of variables that affect other kinds of excessive learned behavior, such as overeating. This conception of drug abuse suggests strategies for treatment, which often involve arranging the environment such that (a) the client's drug use is consistently monitored, (b) abstinence is consistently reinforced, (c) drug use results in loss of reinforcers, and (d) the density of nondrug reinforcers is sufficient to compete with the reinforcing effects of the drug (Higgins et al., 1991).

A study by Higgins et al. (1993) illustrates the potential value of this approach. They randomly assigned 38 cocaine-abusing clients to a 12-step counseling program or to a program that involved community-based reinforcement and vouchers, earned by producing drug-free urine samples, that could be exchanged for appropriate reinforcers. As part of the community-based reinforcement program, the clients attended behavior therapy sessions that involved employment counseling, assertiveness training, relaxation training, and counseling that targets participation in non-drug-related social and recreational activities. These therapy sessions were intended to increase drug users' participation in activities incompatible with drug use, which, given that they were sufficiently reinforcing, should compete effectively against drug-related activities.

They appeared to do so in many cases, in that only 5% of the clients in the 12-step program achieved 16 weeks of continuous abstinence, whereas 42% of clients in the behavioral program did so. Moreover, the former program appeared to be less acceptable to clients, insofar as only 11% of them completed 24 weeks of treatment, whereas 58% of those in the behavioral treatment did so.

Behavioral Mechanisms of Clinical Drug Effects

In principle, it should be possible to isolate the behavioral mechanisms involved in the beneficial effects produced by psychotropic drugs. For example, objects and events that previously were effective positive reinforcers (i.e., their delivery strengthened the behaviors that produced them) for a given person appear to lose their reinforcing capacity when she or he becomes depressed. Effective psychotropic medications might work, at least in part, by reestablishing reinforcing capacity, and they appear to do so.

With many disorders, however, no one has successfully categorized the characteristic changes in behavior in terms of behavioral processes. Consider, for example, how a person changes with the onset of Schizophrenia. Overt verbalizations become impoverished in terms of amount and content; they reflect delusional thoughts and auditory hallucinations. Anhedonia—a general lack of interest in once important objects and events (i.e., stimuli)—appears and self-care diminishes. Stimulus control, that is, the ability of objects and events to engender or suppress particular responses, clearly has changed, as has the capacity of certain objects and events to serve as reinforcers. But the generation and following of rules—verbal descriptions

of relations among stimuli and responses—also changes. Human behavior is unique in that it can be affected by descriptions of relations, such as the consequences of a particular response (e.g., "If you turn left at the second light, the diner will be two miles ahead on your right"), as well as by actual exposure to those relations. Delusions involve counterfactual rules (e.g., "They listen through the TV; if you turn it on they'll know what I'm thinking and hurt me"). It is unclear why people eventually diagnosed with Schizophrenia come to generate and follow counterfactual rules, or how psychotropic drugs reverse the process. The absence of viable, comprehensive behavioral models of mental illness makes it impossible to relate therapeutic drug effects to behavioral mechanisms of drug action. Moreover, the absence of such models appears to increase the appeal of biological models.

Nonetheless, researchers have had some success in relating the effects of psychotropic drugs to behavioral mechanisms of action. For example, Northup, Fusilier, Swanson, Roane, and Borrero (1997) examined the possibility that methylphenidate (Ritalin) functions as a motivational operation (MO) when administered to children diagnosed with ADHD. In brief, an MO is a stimulus that alters the reinforcing (or punishing) effectiveness of another stimulus. Establishing operations (EOs) increase reinforcing (or punishing) effectiveness; abolishing operations (AOs) decrease it (Laraway, Snycerski, Michael, & Poling, 2003).

Northup and his colleagues (1997) studied the behavior of three children diagnosed with ADHD who were receiving methylphenidate each day and attending a summer day treatment program. One of their activities at the camp was solving math problems. Solving a specified number of problems was reinforced with coupons that could be exchanged, depending on the color of the coupon selected, for edible items, other tangible items, activities, teacher attention, peer attention, and escape from academic programs. The effectiveness of each type of reinforcer was determined based on the number of coupons representing that class of reinforcer that a child selected.

The distribution of choices across reinforcer types was examined for each child when methylphenidate and placebo were administered. For all three children, methylphenidate changed the value of at least one type of reinforcer. For example, in the placebo condition, one boy's most preferred reinforcer type was edible items. At 20 mg methylphenidate, however, both activities and tangible nonfood items were preferred over food. Similar results were obtained with the other two children. In all three, activities functioned as more potent reinforcers following methylphenidate than following placebo. In contrast, edible items were more effective reinforcers following placebo than following methylphenidate. In technical terms, the drug served as an EO for activities as a reinforcer, and as an AO for food as a reinforcer.

The capacity of methylphenidate to act as an MO is important with respect to one of the potential side effects of the drug: its capacity to alter the effectiveness of behavioral interventions. That is, the effectiveness of behavior change programs that utilize reinforcement—and most behavioral interventions used to treat children diagnosed with ADHD do so—will differ in the presence and absence of methylphenidate if the drug alters the reinforcing capacity of the objects and events arranged to strengthen appropriate behavior. In addition, reinforcer assessment

conducted in the absence of the drug may yield information that is inaccurate in the presence of the drug, and vice versa. Clearly, the demonstration that methylphenidate acts as an MO in applied situations helps to explain how the drug may influence clinically significant behavior in children diagnosed with ADHD.

Methods for Assessing Drug Effects: Basic Research

Quantifying behavior is an important aspect of psychology. Many of the procedures developed by psychologists to quantify behavior and to examine the effects of nonpharmacological variables thereon were subsequently adopted for the study of drug effects. Substantial early impetus for doing so came from a 1956 conference sponsored by the New York Academy of Sciences. The conference, called "Techniques for the Study of the Behavioral Effects of Drugs," featured prominent behavioral psychologists, including Herrnstein, Miller, Morse, Sidman, and Skinner. They forcefully made the case that the methods characteristic of the experimental analysis of behavior were well suited for studying drugs.

The conference occurred shortly after chlorpromazine was introduced into psychiatric practice, and the remarkable success of the drug galvanized interest in basic and applied studies of its effects. Moreover, that success generated a heated quest to develop animal models that effectively screened other drugs for antipsychotic actions. Pickens (1977) reported that between 1917 and 1954 only 28 studies examining drug effects on learned behavior were published in English-language journals, whereas 274 such studies appeared from 1955 to 1963. The vast majority of these studies involved basic research. That is, their intent was to disclose the effects of particular drugs, the mechanisms responsible for those effects, and/or the variables that modulated them, not to help people directly.

Many early studies examined drug effects on learning and performance in mazes. Others were concerned with how drugs affected responding under various schedules of reinforcement. Research in both areas has continued to the present day, but scientists have developed many additional strategies for examining specific aspects of drug action. These strategies are discussed at length by Buccafusco (2001) and van Haaren (1993). Although most basic research has involved nonhuman subjects, a substantial number of studies with humans have appeared. Table 32.1, adapted from Snycerski, Laraway, and Poling (2000), provides examples of some of the behavioral domains examined in these studies and the procedures used to quantify those domains. Psychologists have made substantial contributions to the development and use of these procedures and thereby have played a quintessential role in basic psychopharmacological research.

Methods for Assessing Drug Effects: Applied Research

Psychotropic drugs have some special characteristics (e.g., slow onset of clinical effects) that must be considered in their evaluation. In addition, there are established conventions for clinical drug studies, such as the use of placebo-controlled and double-blind conditions, and determination of dose-response relations (see Gadow & Poling, 1988). Nonetheless, the value of a medication for treating a particular disorder can be determined via the same general tactics and strategies that are used to

Table 32.1 Procedures Commonly Used in Basic Psychopharmacological Research

Domain	Examples of Procedures
Arousal, mood, and activity	Profile of mood states Addiction research center inventory Visual analogue scale Beck Depression Inventory Electroencephalogram Skin temperature Skin resistance Heart rate
Perceptual performance	Critical frequency of fusion Signal detection
Cognitive performance	Repeated acquisition Delayed matching to sample Digit-symbol-substitution test Vigilance tasks
Motor performance	Reaction time Tapping rate Hand steadiness Pursuit rotor Driving simulations
Self-administration	Progressive-ratio schedules Concurrent schedules Discrete trials Single-access procedures
Motivation	Progressive-ratio schedules Contingent activity procedures
Drug discrimination	Two-choice procedures Three-choice procedures Stimulus equivalence procedures

From "Assessing Drug Effects in Humans: Basic Research" (pp. 111–140), by S. Snycerski, S. Laraway, and A. Poling, 2000, in A. Poling and T. Byrne (Eds.), *Introduction to Behavioral Pharmacology,* Reno, NV: Context Press. Adapted with Permission.

evaluate other behavior-change interventions (e.g., cognitive-behavioral therapy). There is no mystery to designing a clinical drug study: One determines whether any independent variable (including medication) affects a behavior of interest by comparing levels of behavior (the dependent variable) when the independent variable is and is not operative, or is operative at different levels (e.g., doses). Factors other than the independent variable that might affect the dependent variable (extraneous variables) are held constant across conditions or rendered inoperative. If appropriate data analysis reveals that levels of the dependent variable differ significantly when treatment is and is not present (or is present at different levels), it is logical to assume that behavior changed as a function of treatment, which is therefore deemed effective.

Many psychologists are truly expert in evaluating the effects of behavior-change interventions, and a substantial number of these experts have done extensive and excellent work in clinical drug assessment. As a case in point, consider Russell

Barkley (1998), who, as noted previously, is a leading ADHD researcher. For more than two decades, he has studied the pharmacological and nonpharmacological management of ADHD and translated the findings of his and others' research into practical, effective strategies for everyday use (e.g., Barkley, 2000). As the case of Barkley illustrates, psychologists can make a major contribution to clinical drug evaluation, even under circumstances in which they do not prescribe drugs themselves.

Several authors have proposed that the tactics and strategies of applied behavior analysis can be especially valuable in clinical psychopharmacology (e.g., Barlow & Hersen, 1984; Kollins, Ehrhardt, & Poling, 2000; Schroeder, 1985). In brief, this approach is underpinned by five beliefs (Poling & Ehrhardt, 1999; Poling & LeSage, 1995):

1. Intensive study of a few participants through the use of within-subject research designs (e.g., multiple-baseline, withdrawal) is a fruitful research strategy.
2. Behaviors targeted for change by a drug should be measured accurately and repeatedly in individual participants.
3. Variable data are best dealt with by isolating the responsible extraneous variables, not by statistical manipulations.
4. Visual (graphic) analysis of data for individual participants is a viable alternative, or adjunct, to statistical analysis.
5. Social validity data, evaluating the acceptability of the goals, methods, and outcomes of treatment to consumers, is desirable.

As Dermer and Hoch (1999) point out, it is common for authors of research methods texts to assert that the within-subject experiments characteristic of applied behavior analysis involve what Campbell and Stanley (1963) termed "quasi-experimental designs" with "low external validity." That is, their findings cannot be widely generalized. Certain between-subjects designs, in contrast, are described as producing results with much greater generalizability. The usual, and reasonable, argument is that between-subjects experiments characteristically involve the study of a clearly defined sample of subjects who are selected to be representative of a larger population with defined characteristics. If this is the case, then inferential statistics provide a basis for describing the true state of affairs in the population, given the results obtained in the sample selected for study.

Although this logic is sound, clinical drug evaluations that involve protected populations (e.g., children with autism) often are limited to small convenience samples; therefore, conventional between-subjects research designs (e.g., randomized between-groups) are inappropriate. Within-subjects designs, such as the multiple-baseline and withdrawal, appear to be the best alternative, even though their findings cannot be generalized with confidence to the population of ultimate concern (e.g., all children with autism). They should, however, extend to people who are very similar to the individuals who were studied. Whether or not this is true is determined by systematically replicating the study. Eventually, the range of conditions under which the drug is useful is established, and valuable prescription guidelines

can be established. Although some research in this vein has appeared, it seems that psychologists have had little success in convincing the medical profession that the strategy is useful. Should they eventually do so, there is potential to expand the literature in areas where further research is badly needed.

Everyday Drug Assessment

Psychotropic drugs are unique among medications in that they are used to deal with problems that are defined by, measured in terms of, and best understood at the level of overt behavior. The effects of these drugs on the biological processes responsible for the signs and symptoms characteristic of particular disorders are incompletely understood, and clinical effectiveness cannot be evaluated at this level of analysis. Therefore, clinical evaluations of psychotropic drugs must assess whether medication significantly improves some targeted aspect(s) of the participant's behavior without producing intolerable adverse reactions (side effects). To determine whether a pharmacological intervention is successful in treating a behavior disorder, three questions must be answered: (1) What were the desired effects of the medication? (2) Were those effects obtained? (3) Were there any significant adverse reactions? These are easy questions to ask but not to answer, whether in the context of formal research or in practical everyday drug evaluations (Kollins et al., 2000).

As Sprague and Werry (1971) pointed out 35 years ago, every prescription of a psychotropic medication is in essence an experiment in which the physician and other care providers hypothesize that administering a specific drug will produce a desired change in one or more aspects of a client's behavior. In the context of controlled research, the essential four features of a sound drug evaluation are that (1) treatment effects must be adequately measured, (2) medication must be administered according to the experimental regimen, (3) experimental conditions and their sequencing must allow observed changes in behavior to be attributed with confidence to the drug, and (4) data analysis must be adequate for detecting clinically important changes in client behavior. If these conditions are met, the evaluation is, in principle, sound.

These same characteristics are important in the practical evaluation of medications, although assessment procedures employed outside research settings rarely meet the rigorous standards advocated by scientists. When, for instance, a child diagnosed with ADHD receives methylphenidate for the first time, the child's parents, teachers, and physician are not interested in conducting a study worthy of presentation, but only in determining whether the drug is an appropriate treatment. An abundant scientific literature suggests that it is likely to be effective in a given individual who displays the behaviors that lead to a diagnosis of ADHD (restlessness, impulsivity, short attention span, excessive motor activity), but exceptions have been reported (some children show little or no positive response to the drug), and the most effective dose can vary. Therefore, the response of each young person with ADHD to methylphenidate needs to be monitored.

When competent adults contract for treatment by a physician, the evaluation process is relatively simple: The client and physician agree on treatment goals and a medication, and the treatment is successful, at least to some extent, if the

client deems it so and the physician agrees and also detects no serious side effects. Evaluation is complicated, however, when the client, by virtue of age or disability, is not able to decide whether to initiate or continue drug treatment. As Greiner (1958) pointed out soon after chlorpromazine was introduced:

> Sensible adult patients will usually balk when a drug is causing symptoms, but the very young and the very old are forced to take drugs, can't complain or stop on experiencing toxic symptoms, may not even connect them with the drug. The mentally deficient of any size or age cannot protect themselves either, and they also merit special care to avoid toxic doses.

The need for special care in assessing drug effects in people with mental retardation (and other developmental disabilities) is especially important in view of the lack of persuasive evidence of effectiveness in this population. In summarizing the most comprehensive recent review of psychopharmacology and mental retardation, Matson et al. (2000) concluded:

> Our findings were startling and consistent with Baumeister and Sevin's findings in 1990 [Baumeister and Sevin conducted a similarly comprehensive prior review]. Professionals continue to medicate individuals in their care despite a lack of a sound research base to support their treatment decisions. While a small amount of research does exist, it is seriously flawed from a research design perspective. The studies routinely lack appropriate experimental control, and lack reliable measures of drug effect, collateral behaviors, and side effects. Methodological controls are far short of psychological treatment in the field with little improvement over precious decades. This trend is extremely disappointing given the dangerous and restrictive nature of these treatment options.

Johnny Matson is one of a substantial number of psychologists whose conceptual and empirical work has major implications for the everyday use of psychotropic drugs. As regards drug effects in people with developmental disabilities, Matson et al. (2000) are not alone in their skepticism. For example, in reviewing studies examining the effects of antipsychotic medications for challenging behavior in people with intellectual disorder, Brylewski and Duggan (1999, p. 360) reported that they could find only three studies that could be classified as randomized controlled trials: "These trials provided no evidence as to whether antipsychotic medication does or does not help adults with [intellectual disorder] and challenging behavior."

Similar results have been found for persons receiving dual diagnoses of intellectual disability and Schizophrenia. In a review of studies that examined the effectiveness of antipsychotic medication in people dually diagnosed, Duggan and Brylewski (1999) found only one relevant randomized controlled trial, which was published over 40 years ago (i.e., Foote, 1958). These authors found that there is no empirical evidence that antipsychotic medications are, or are not, effective in individuals with developmental disabilities and Schizophrenia. Until methodologically sound trials are conducted, practitioners will have to rely on personal experience as well as extrapolations from studies with individuals with Schizophrenia and "normal" intelligence. These extrapolations may or may not be valid (Duggan & Brylewski, 1999).

Since 1949, when the "Boulder Conference on Graduate Education in Clinical Psychology" was held, the scientist-practitioner model has powerfully influenced clinical psychology, as well as other applied subdisciplines, such as school psychology and organizational behavior management. Three key provisions of this model, discussed in detail by others (e.g., Hayes, Barlow, & Nelson-Gray, 1999; Soldz & McCullough, 1999), are these:

1. Psychologists should use assessment and treatment procedures that have been scientifically validated. That is, whenever possible, applied psychologists are expected to stay abreast of the scientific literature relevant to the areas in which they conduct assessments and arrange treatments, and to use only those techniques supported by the results of methodologically rigorous research.
2. Psychologists should monitor the response of each individual who is assessed and treated to ensure that she or he actually benefits from these processes. Even empirically validated techniques may be ineffective or even harmful for some seemingly appropriate clients.
3. Psychologists should work with other health professions to improve the quality of care provided. This involves contributing the psychologists' expertise to health care teams and ensuring that those teams function in a manner consistent with the scientist-practitioner model.

We have contended elsewhere that these provisions, if widely upheld, would dramatically improve how psychotropic drugs are used in people with developmental disabilities (Poling, 1994; Poling & Ehrhardt, 1998; Poling, Laraway, Ehrhardt, Jennings, & Turner, 2004). Of course, the virtues of the scientist practitioner model are not limited to psychologists; psychiatrists and other physicians who rely on psychotropic medications of proven worth and monitor the effects of those medications carefully can provide excellent care with no involvement from psychologists. However, given that psychologists typically receive extensive training in behavioral assessment, we have advocated for a team approach in which well-trained psychologists and physicians work with other concerned individuals to provide high-quality drug treatments. Such treatments ensure that:

- The goals of treatment are clear, quantitative, and in the patient's best interest.
- The choice of a drug and dose for initially treating a particular patient are consistent with the best available scientific literature.
- Subsequent treatment decisions are made in view of meaningful information regarding the patient's response to the drug.
- Drug therapy is flexible and integrated with nonpharmacological interventions.
- Properly trained psychologists can do much to enhance the quality of drug treatment, regardless of whether or not they actually prescribe.

PRESCRIPTION PRIVILEGES FOR PSYCHOLOGISTS

Up to this point, we have endeavored to show that psychologists who do not prescribe drugs have made major contributions to many areas of psychopharmacology and potentially could do even more. One thing that we have not considered is the possibility that they could prescribe psychotropic drugs. Since 1990, 14 states have considered legislation to grant psychologists limited prescription privileges. Alaska, California, Florida, Georgia, Hawaii, Illinois, Missouri, Montana, Tennessee, and Texas rejected the legislation. In 2002, New Mexico became the first state allowing psychologists to prescribe drugs. It was followed in 2004 by Louisiana. Both states allow psychologists who meet certain criteria to prescribe psychotropic drugs independently. Too little time has passed to allow for meaningful evaluation of the effects of allowing psychologists to prescribe in New Mexico and Louisiana; whether or not psychologists should be allowed to prescribe drugs is a contentious issue, as it has been for the past 15 years.

A recent search of the PsychInfo database using "prescription privileges" as the keyword revealed 163 entries. Of them, 160 appeared from 1986 to 2006. Their distribution across years is shown in Figure 32.1. Clearly, there is a substantial literature relevant to whether or not psychologists should be allowed to prescribe drugs, and it is growing rapidly. This section summarizes that literature. In doing so, we provide an overview of past and current events relevant to psychologists' prescribing drugs and summarize arguments for and against their doing so. More detailed coverage is available elsewhere (e.g., Hayes & Heiby, 1998b; Lavoie & Barone, 2006; Sammons, Page, & Levant, 2003).

Historical Overview

In the United States, the Federal Food, Drug and Cosmetic Act, passed in 1938, determines which medications can be sold over the counter and which are available

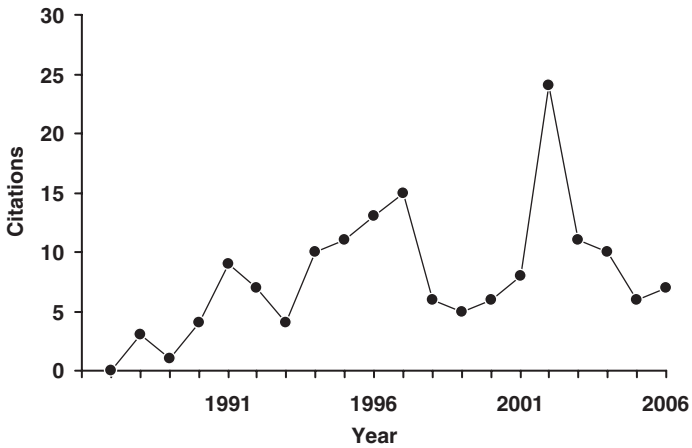

Figure 32.1 Number of publications listed in PsychInfo that used "prescription privileges" as a key word.

only by prescription (U.S. Food and Drug Administration, 2007). State laws, however, determine the professions that are allowed to prescribe, as well as (a) whether they can do so with or without physician supervision and (b) whether they can prescribe all classes of medication or only those relevant to their discipline. In every state, physicians can prescribe all medications independently. Other professions are more limited. Dentists, for example, can prescribe independently, but only those medications relevant to their practice (e.g., antibiotics, analgesics).

From its inception in the late nineteenth century until the onset of World War II, psychology was primarily an academic discipline concerned with understanding human behavior. Freudian psychoanalysis was the predominant treatment, and it was available to relatively few people. There were not sufficient psychiatrists available to serve the mental health needs of personnel returning from the war, and the Veterans Administration (VA) was severely taxed in trying to treat those people. In response, the VA dramatically extended the range of services they allowed psychologists to provide, including the delivery of psychotherapy under the supervision of psychiatrists (Lavoie & Fleet, 2002). Psychiatrists objected, claming that psychologists were not properly trained to provide psychotherapy, but an ever-increasing number of psychologists disagreed. In large part due to effective lobbying by APA and affiliated organizations, by the end of the 1950s psychologists were widely recognized as legitimate providers of clinical services. This recognition was codified in state licensure legislation, which specifies the range of activities in which properly trained and duly licensed (or, in some cases, certified) psychologists can legally engage. For example, in our home state of Michigan, Public Act 368 (Michigan Legislature, 2007):

> Defines the practice of psychology as the rendering to individuals, groups, organizations, or the public of services involving the application of principles, methods, and procedures of understanding, predicting, and influencing behavior for the purposes of the diagnosis, assessment related to diagnosis, prevention, amelioration, or treatment of mental or emotional disorders, disabilities or behavioral adjustment problems by means of psychotherapy, counseling, behavior modification, hypnosis, biofeedback techniques, psychological tests, or other verbal or behavioral means. The practice of psychology does not include the practice of medicine such as prescribing drugs, performing surgery, or administering electro-convulsive therapy.

Psychology flourished in the 1960s and 1970s and into the 1980s. Major conceptual advances, including those associated with behavior analysis and cognitive psychology, were accompanied by the development of effective interventions for various behavioral disorders. Cognitive-behavioral treatment for depression is an obvious case in point. Effective behavior-change interventions were developed and used not only by clinical psychologists, but by organizational and industrial, educational, and school psychologists as well. By 1980, several areas of psychology were well established and providing invaluable services to a wide range of individuals and organizations.

By that same year, the use of psychotropic drugs to treat a wide range of behavior disorders was firmly established. The success of chlorpromazine was

soon followed by the introduction of several other antipsychotic drugs, as well as antidepressants (monoamine oxidase inhibitors such as phenelzine [Nardil] and tranylcypromine [Parnate], tricyclics such as amitriptyline [Elavil, Endep] and imipramine [Tofranil]), and anxiolytics (primarily benzodiazepines such as diazepam [Valium]). The use of stimulants (such as methylphenidate [Ritalin]) to treat children diagnosed with hyperactivity (or Attention-Deficit Disorder) became more prevalent, and lithium became popular for the treatment of Bipolar Disorder. A bit later, in the early 1980s, the serotonin-specific reuptake inhibitors, beginning with fluoxetine (Prozac), were introduced for the treatment of depression. They soon became the most prescribed psychotropic medications.

Later still, a new group of antipsychotic drugs, labeled "atypical" because they produced fewer motor disturbances, was introduced. This group includes risperidone (Risperdal) and clozapine (Clozaril). A large number of studies in addition to the initial clinical trials required by the FDA prior to marketing have demonstrated that various psychotropic drugs are reasonably safe and at least somewhat effective (although the evidence in some cases is not overwhelming). The availability of effective psychotropic medications literally revolutionized psychiatry in terms of (a) increasing the effectiveness of available interventions, (b) fostering broad acceptance of a biological model of mental illness, and (c) generating widespread demand for the pharmacological management of behavior disorders.

More than 50 different drugs currently are FDA-approved as psychotropic medications. In 2005, over 10 million prescriptions were written in the United States for each of nine such drugs. The names of those drugs, the number of prescriptions written, and the overall ranking of each are shown in Table 32.2. These data make it clear that psychotropic medications play a central role in the current treatment of behavior disorders. As further evidence, a sample of practicing psychologists recently reported that 43% of their current patients were using psychotropic medication (VandenBos & Williams, 2000). Moreover, 94% of those psychologists indicated that they had consulted with physicians about changing the medications of joint clients.

As the pharmacological management of behavior disorders flourished, two major forces came into play that eventually led some psychologists to push for

Table 32.2 Psychotropic Drugs with >10 Million Prescriptions in 2005

Nonproprietary Name	Proprietary Name	Overall Rank	Number of Prescriptions
Alprazolam	Xanax	9	34,230,000
Sertraline	Zoloft	16	26,976,000
Fluoxetine	Prozac	29	21,403,000
Lorazepam	Ativan	30	19,002,000
Paroxetine	Paxil	38	18,889,000
Trazodone	Desyrel	48	14,505,000
Amitriptyline	Elavil	50	14,385,000
Diazepam	Valium	60	12,093,000
Bupropion	Wellbutrin	66	11,044,000

Source: The Top 300 Prescriptions for 2005 by Number of U.S. Prescriptions Dispensed, by RxList, 2005. Available from www.rxlist.com/top300/.

prescription privileges (Frank, Conti, & Goldman, 2005; Lavoie & Barone, 2006; Sanua, 1998). One, discussed previously, was conceptual, and took the form of growing acceptance among laypeople and health care providers alike of biological models of mental illness and of the logic and effectiveness of pharmacological management of those disorders.

A second was economic, leading ultimately to diminished job prospects for doctoral-level clinical psychologists. In brief, graduate schools had saturated the market with such psychologists, even as it was being demonstrated (e.g., Christiansen & Jacobson, 1994; Dawes, 1994) that non-doctoral-level professionals from a number of disciplines (e.g., psychology, nursing, social work) could provide effective, and relatively inexpensive, treatment for many psychological problems (Lavoie & Barone, 2006). As managed health care came to dominate the provision of mental health services, there was increased reliance on non-doctoral-level professionals and fewer lucrative positions for doctoral-level psychologists (Chiefetz & Salloway, 1984; Cummings, 1995). Concurrently, there was more money available for pharmacological treatments (Frank et al., 2005).

According to Frank et al. (2005), expenditures on psychotropic drugs in the United States in 1987, 1992, 1997, and 2001 were, in billions, $2.77, $3.83, $9.04, and $17.83, respectively. Expenditures on psychotropic drugs increased more than sixfold over this period, and the percentage of mental health spending devoted to them increased from 7.7% to 21% (Frank et al., 2005). Much of this money went to large pharmaceutical companies, which were growing enormously and exerting increasing—and sometimes harmful—influence on the medical profession (Antonuccio & Danton, 2003), but a substantial sum went to physicians who prescribed psychotropic drugs. Perhaps surprisingly, most of them were not psychiatrists, but general practitioners.

In what was in hindsight a predictable reaction to emerging financial and conceptual changes in the field of mental health, during the 1980s some influential psychologists and their allies in government began to advocate for prescription privileges. The APA, which currently has over 150,000 members, played a crucial role in the advocacy, with much of the impetus coming from members of Divisions 12 (Society of Clinical Psychology) and 42 (Psychologists in Independent Practice). Since 2000, a strong push for prescription privileges has come from Division 55, the American Society for the Advancement of Pharmacotherapy, which, according to the APA web site (APA, n.d., para 2), was created specifically "to enhance psychological treatments combined with psychopharmacological medications." Sanua (1998) details the contentious political history of the prescription movement within and outside APA.

As he relates, in 1981 APA convened the Psychologists' Use of Physical Intervention Committee to consider prescription privileges. The committee completed a report in 1981, with a revision in 1986 (see Sanua, 1998). Those reports did not specifically endorse prescription privileges, but instead suggested that psychologists should play a role in pharmacotherapy as dictated by consumer demand (Hayes, Walser, & Bach, 2002). In 1989, APA established an Ad Hoc Task Force on Psychopharmacology to study the issue of pharmacotherapy further; its report appeared in 1992 (APA, 1992). The report outlined three levels of training in

psychopharmacology, with the appropriate level determined by a psychologist's degree of involvement in the prescribing practice. Level 3, which has generated the most interest, is for psychologists who actually will prescribe. Level 2 is for collaborative practice, in which the psychologist would partner with a physician to determine the need for and monitor the effects of psychotropic drugs. Level 1 is for any psychologist providing mental health services and essentially provides an overview of psychopharmacology.

In 1995, in response to a request for support of a bill establishing prescription privileges in California, APA's Council of Representatives approved a resolution making prescription privileges for psychologists a major policy directive for the association. From this time forward, APA formally supported prescription privileges (Cullen and Newman, 1997). In 1996, the organization adopted a model prescription bill and training curriculum, and in 2000 it announced that APA-supplied insurance will cover prescription-related activities in any state where such activities are legal. Many APA-affiliated state organizations also have supported prescription privileges.

Many psychologists, however, have not. As Sanua (1998) relates, early on most APA members (and other psychologists) had little knowledge of or interest in the prescription privileges issue. Interest was galvanized by the APA's activities, by the 1985 introduction in Hawaii of legislation to study prescription privileges for psychologists, and by Congress directing the U.S. Department of Defense (DOD) in 1988 to establish a pilot project, called the Psychopharmacology Demonstration Project (PDP), to train psychologists to prescribe psychotropic medications. It became operational in 1991; 10 military psychologists completed training between 1991 and 1997 and were granted the right to prescribe psychotropic medications. Each received a total of 660 hours of academic training and 1,900 hours of clinical training.

By 1996, three national psychology associations, the Society for a Science of Clinical Psychology, the Council of University Directors of Clinical Psychology Programs, and the American Association of Applied and Preventative Psychology (AAAPP), had formally opposed prescription privileges for psychologists (Hayes & Heiby, 1996). For example, following a 1996 conference, AAAPP members generally agreed on three conclusions (Hayes & Heiby, 1998, p. vi):

1. The prescriptions privileges proposal is polarizing the discipline.
2. Psychology's resources should be focused on advocating the profession to managed care. Doing so will involve the development of treatment standards and the promotion of integrated care.
3. Psychologists wishing to obtain prescription privileges at this time are encouraged to obtained an advanced degree in nursing, a physician's assistant degree, or the like in order to develop more experience with the psychology/prescription privilege without first having to alter the discipline and its training focus.

Attendees also generally agreed that the push for prescription privileges was premature and that further reasoned discussion among psychologists should occur before any introduction of legislation seeking such privileges. This did not occur.

The move to secure prescription privileges continued unabated, and at present, psychologists wishing to obtain prescription privileges have options not available in 1996: They can obtain proper training and licensure and work in New Mexico or Louisiana. Whether or not this is good continues to be debated heatedly.

A recent meta-analysis of opinion data from 16 studies concerned with the prescription privileges debate (Walters, 2001) examined responses (agree, disagree) to three statements: (1) Properly trained psychologists should be allowed to prescribe psychotropic drugs; (2) APA should advocate in favor of prescription privileges for psychologists; and (3) I would personally seek prescription privileges should they become available. Collectively, these statements were taken to constitute the "prescription privilege agenda." Results indicated that "there was minimal consensus and a general split of opinion on the advisability of pursuing the prescription agenda" (p. 119). When data for all three statements were summed, there was no statistically significant difference in the number of respondents who agreed (52%; i.e., who supported the agenda) and the number who disagreed (35%; i.e., who opposed the agenda). Only statement 1 engendered significantly higher levels of agreement than disagreement.

The Department of Defense Psychopharmacology Demonstration Project

Good arguments can be made for and against prescription privileges, and data adequate for choosing between them are not currently available. As indicated, 10 psychologists were trained and allowed to prescribe drugs under DOD's PDP. Some information is available concerning the success of the program (Laskow & Grill, 2003; Newman, Phelps, Sammons, Dunivin, & Cullen, 2000). In general, the psychologists who participated demonstrated high levels of competency and achievement prior to entering the 2-year program. They primarily prescribed drugs in outpatient clinics, typically to deal with depressive disorders, anxiety disorders, or adjustment disorders. Psychiatrists who provided clinical supervision rated the quality of care provided by the psychologists as good to excellent and reported no adverse patient outcomes. Moreover, although some conflicts between psychologists and psychiatrists appeared early on, relations became generally harmonious; allowing psychologists to prescribe drugs did not degrade staff morale.

The General Accounting Office (GAO) estimated that the DOD would spend 7% more on its 10 prescribing psychologists than it would have spent to provide equivalent services using a traditional mix of psychiatrists and psychologists. The GAO noted that the PDP did not increase the wartime readiness of the DOD, primarily because psychotropic drugs are rarely a priority in combat, but that it might modestly enhance peacetime readiness. The PDP was successful, and a survey by Klusman (1998) indicated that most military primary care physicians, psychologists, and social workers strongly supported the program and the overall initiative to grant psychologists the authority to prescribe. Military psychiatrists, in contrast, were strongly negative regarding both the PDP and the overall initiative.

Save for demonstrating that psychologists can be trained to prescribe drugs in a safe and competent manner, the PDP's relevance for the prescription privileges debate at large appears to be minimal. A relatively small number of highly competent

people were trained, and they practiced in close association with psychiatrists in a highly structured setting. Opinions regarding the PDP's value differed across professions, and, more important, whatever virtues it possessed were not sufficient to lead to widespread DOD adoption of prescription privileges for psychologists. This outcome suggests an absence of social validity for the program. That is, one of its consumers—the DOD—was not satisfied with its goals, methods, or outcomes. Unfortunately, no data are available concerning the satisfaction of the military personnel who actually received medication at the hands of psychologists or, perhaps more important, the actual effectiveness of those medications in solving the mental health problems of those personnel.

Arguments for Granting Prescription Privileges

Advocates of prescription privileges make a number of solid arguments in support of their position. For example, Robert Sternberg (2003) wrote in the president's column of the *APA Monitor* that psychologists ought to be allowed prescription privileges because this would allow them to treat the whole person, provide full service, offer medications to patients who need them but are not presently being served, and reduce psychologists' dependence on medical practitioners. Moreover, he indicated that the success of the PDP program and the passage of a prescription authority bill in New Mexico demonstrate, respectively, the potential value and the political feasibility of obtaining prescription privileges. Other advocates have made similar points. In general, it appears that most of the arguments in favor of prescription privileges fall into two general categories.

1. Allowing Psychologists to Prescribe Drugs Will Help to Meet the Needs of People Not Currently Receiving Adequate Mental Health Services

This argument assumes that (a) psychotropic drugs are valuable treatments, (b) there are substantial numbers of people who currently need but do not have access to those drugs, and (c) allowing psychologists to prescribe would meet the needs of those people. Overall, a good case can be made for the value of psychotropic drugs, although they are not panaceas and, as discussed previously with respect to people with mental retardation, they historically have been misused in some populations. The relevant literature is gargantuan and cannot be addressed here. Perhaps a quote from the World Health Organization (WHO; 2005, p. 8) is sufficient:

> A large number of treatments are available for the pharmacological management of mental disorders. Many of these treatments have been shown to be effective in acute stages and in preventing relapses, but much remains unclear about their effectiveness in long-term treatment and in managing everyday mental disorders.

Despite the reservations expressed in the last sentence, the WHO considers some psychotropic drugs as "essential" medications (see Table 32.3), which should be cheaply and widely available worldwide.

Table 32.3 Psychotropic Drugs on the World Health Organization Model List of Essential Drugs

Drugs used in psychotic disorders	Chlorpromazine, fluphenazine, haloperidol
Drugs used in depressive disorders	Amitriptyline
Drugs used in bipolar disorders	Carbamazepine, lithium carbonate, valproic acid
Drugs used in generalized anxiety and sleep disorders	Diazepam
Drugs used in obsessive-compulsive disorders and panic attacks	Clomipramine

Source: Improving Access and Use of Psychotropic Medicines, by the World Health Organization, 2005, Geneva, Switzerland: Author.

Support for the view that there are many people with significant, untreated mental health needs comes from several sources. Among them is a 1999 report from the surgeon general (U.S. Department of Health and Human Services, 1999) indicating that fewer than one-third of all people in the United States with a diagnosable mental disorder receive treatment in a given year. Also commonly cited are data indicating that there is a shortage of psychiatrists, especially in rural areas.

Support for the view that allowing psychologists to prescribe drugs would meet currently unmet needs is conjecture. Presumably, at least some currently unmet mental health needs would respond to nonpharmacological interventions that could be, but are not, currently provided by psychologists. Why would allowing psychologists to prescribe substantially increase the number of people they serve or change patient demographics? Interestingly, a major argument leading to the acceptance of physician assistant as a profession was that physician assistants would provide medical care for rural and other historically underserved populations. In fact, very few (3%) of them do so (Lavoie & Barone, 2006). Complex interactions of sociocultural, geographic, and fiscal variables that influence patients as well as care providers determine who does and who does not receive treatment; creation of a new profession (psychologists who prescribe) will not alter these interactions.

Moreover, even if psychologists who prescribe will be able to increase access to mental health care, this does not mean that there is public demand for them to do so. Professionals from other disciplines (e.g., nurse practitioners, physician assistants), perhaps working in conjunction with psychologists, may well be in higher demand. Some critics suggest, in fact, that the push for prescription privileges has been driven primarily by financial considerations, as a limited number of influential psychologists work to secure continued access to a substantial share of the health care market, a share that is created by their lobbying efforts, not by legitimate need or public demand.

2. Allowing Psychologists to Prescribe Drugs Will Lead to High-Quality Care

This argument begins with the contention that it is possible to train psychologists to competently prescribe psychotropic medications. As noted previously, the success of the PDP generally supports this view. Compared to New Mexico and Louisiana,

however, the PDP training was more extensive. For example, the PDP required 660 hours of academic training and 1,900 hours of clinical training, whereas New Mexico requires 450 hours of academic training and 400 hours of clinical training (for a conditional prescribing certificate; further clinical training is required for a general prescription certificate).

Supporters contend that the specialized training required for psychologists to prescribe, coupled with the training they normally receive, is adequate to allow them to prescribe medications safely and effectively. As we have discussed previously, their training in behavioral assessment and commitment to the scientist-practitioner model could be especially useful in diagnosing behavior disorders, in selecting empirically validated interventions, and in monitoring the effects of psychotropic medications. In addition, competent psychologists with expertise in dealing with particular kinds of patients (e.g., people with developmental disabilities, elderly people, or children and adolescents) would be able to offer state-of-the-art nonpharmacological alternatives to psychotropic drugs when appropriate and to provide combined pharmacological and pharmacological interventions. Whether they would do so, or would instead come to rely on pharmacological interventions alone, is unclear. As time passes, data from Louisiana and New Mexico may bear on the issue.

Critics argue that medical training beyond that required to earn prescription privileges is required to use psychotropic medications safely. Expertise in monitoring the therapeutic and adverse effects in other medications is needed, given that many people treated with psychotropic medications receive other kinds of drugs.

If there is in fact a strong need for psychologists who prescribe drugs, and if they can be readily trained to do so without sacrificing other important activities, such as offering nonpharmacological treatments, then granting widespread prescription privileges for psychologists will benefit the profession and society at large.

Arguments against Granting Prescription Privileges

Like the advocates of prescription privileges, the critics have made a number of good arguments (e.g., Heiby, 1998; Lavoie & Barone, 2006). One common argument, mentioned previously, is that the proposed level of training is not adequate to allow psychologists to prescribe psychotropic drugs safely and effectively. Another argument, also mentioned previously, is that the push for prescription privileges is motivated primarily by potential financial gain for psychologists, which is not a legitimate reason for dramatically altering professional boundaries. Two other arguments, both very general, are discussed next.

1. There Is No Legitimate Need for Psychologists to Prescribe

This argument takes three forms. One rests on the simple contention that there is a strong and persistent demand for effective psychological interventions; competent and innovative clinical psychologists have more than enough to do without prescribing drugs (Hayes et al., 2002). Managed care has changed and will continue to change the role of doctorate psychologists, but the field of clinical psychology will remain viable for the foreseeable future. Rather than advocating for prescription

privileges, psychologists would be better served by advocating for enhanced acceptance of empirically validated psychological treatments of all mental disorders and by developing new and effective interventions.

A second form of the argument, alluded to in the preceding section, rests on the contention that the alleged demand for prescription privileges was created by a handful of influential psychologists, such as Pat DeLeon (e.g., DeLeon, Dunivin, & Newman, 2002; DeLeon & Wiggins, 1996) and their political allies, not by requests from mental health agencies, people with mental illness, or their advocates. In actuality, the argument goes, the demand for additional professionals who can prescribe is small and, even if prescription privileges are granted to psychologists, is likely to be filled from other professions (physician assistants, registered nurses). Services provided by doctoral-level psychologists are relatively expensive, and wherever possible managed care will seek to replace them with similar services obtained from less expensive professions.

A third form of the argument rests on the contention that psychotropic drugs are restrictive, largely ineffective treatments that are widely overused, at least in part due to questionable practices by the pharmaceutical industry (e.g., Malott, in press; Whitaker, 2002). Therefore, physicians should curb their willingness to prescribe such drugs and certainly no new profession should prescribe them.

All of these arguments have been countered. For example, many studies can be cited to show the clinical effectiveness of psychotropic drugs, including their value in managing conditions typically not responsive to psychological interventions alone (Schizophrenia). The practices of those who currently write most prescriptions for psychotropic drugs, general practitioners, are demonstrably far below optimal (Lavoie & Barone, 2006); therefore, psychologists who competently prescribe drugs can compete effectively against them. In addition, it is claimed, they can provide outreach to currently underserved patient groups. The truth of this contention, like that of all of the claims and counterclaims mentioned in this section, depends on one's source of information and how that information is interpreted.

2. Allowing Psychologists to Prescribe Drugs Will Harm the Profession

One part of this argument is the contention that the prescription privileges debate has eroded the discipline's cohesiveness and may lead to a split similar to that observed in the late 1980s, when the American Psychological Association, emphasizing basic science and the scientist-practitioner model, broke away from APA. Another part emphasizes dissolution of disciplinary boundaries, with a resulting erosion of focus and values. Allowing psychologists to prescribe drugs, the argument goes, would blur the distinction between psychology and medicine and create an unhealthy competition between the disciplines. Moreover, it would direct attention away from studying the variables historically emphasized by psychologists and on developing and employing new treatments based on those variables. Finally, it would shift emphasis away from advocating for and using psychological treatments. All of these happenings would be harmful for psychology.

Further harm would come in the form of altered training requirements. Although training to prescribe in New Mexico and Louisiana is at the postdoctoral level, widespread adoption of prescription privileges legislation might well create a demand for training at the graduate, or even undergraduate, level. Providing this training would seriously reduce opportunities for training in other important areas. Moreover, extensive training in psychopharmacology would be prohibitively expensive. Finally, allowing psychologists to prescribe drugs would open them to insidious influence from the pharmaceutical industry.

Advocates of prescription privileges have countered these arguments by contending that disciplines constantly evolve and, as we have discussed at some length, psychologists have long been involved in psychopharmacology, albeit not as direct treatment providers. For them, moving to the status of treatment providers is a small, logical, and beneficial step. They suggest that the necessary training can be provided at reasonable cost, without seriously compromising opportunities to develop competency in traditional areas. Psychologists competent in pharmacological and nonpharmacological treatments could provide a full range of services and would be in high demand. With careful planning, untoward influence from industry could be avoided (e.g., Antonuccio & Danton, 2003).

Our Position

In weighing arguments for and against prescription privileges, we find it noteworthy that most of the support comes from practice-based organizations, whereas most of the opposition comes from scientist-practitioner groups, with basic science organizations being largely indifferent (Hayes et al., 2002). Our view, expressed earlier, is that optimal use of psychotropic drugs demands acceptance of the scientist-practitioner perspective. The fact that those groups that are avowedly committed to this perspective argue against prescription privileges is noteworthy.

Although proponents have made a number of compelling arguments, our views parallel those of Lavoie and Barone (2006, p. 63), who end their extensive coverage of the prescription debate by noting, "There does not appear to be compelling evidence of the desirability of granting prescription privileges for psychologists." They go on to suggest that, instead of arguing for prescription privileges, psychologists would be better served by improving the dissemination of nonpharmacological services and working with general practitioners and psychiatrists to improve how psychotropic drugs are used. We agree on both counts.

In our view, it is particularly unfortunate that debate has focused almost entirely on psychologists being granted independent drug prescription privileges and has ignored the possibility of psychologists teaming with physicians and other medical personnel who are currently allowed to prescribe. As emphasized in the first section, many psychologists have skills that would allow them to play an invaluable role in assessing drug effects and in providing nonpharmacological alternatives and adjuncts to psychotropic drugs. Combining these skills with those of women and men specifically trained in medicine would go far in improving mental health care, without necessitating a dramatic change in the activities of either psychologists or medical doctors. Although psychologists have much to contribute to

psychopharmacology, our opinion at present, as it was in 1998 (Poling, Christian, & Ehrhardt, 1998), is that the move to obtain prescription privileges is premature, unwarranted, and unwise. Nonetheless, it is sure to continue.

SUMMARY

Psychologists have made (and are continuing to make) many significant contributions to the study of psychotropic and other psychoactive drugs. Their expertise in quantifying behavior has been invaluable in determining how drugs affect mood, cognitive status, and overt actions. Moreover, their conceptual models have been useful in explaining drug effects. Finally, their knowledge of the variables that control behavior has yielded effective strategies for treating drug abuse.

In recent years, some psychologists and their affiliates have argued that properly trained psychologists should be allowed to prescribe psychotropic medications, and New Mexico and Louisiana have recently allowed them to do so. Too little time has passed to allow the outcome of doing so to be determined. There are legitimate arguments for and against granting prescription privileges to psychologists; those arguments are summarized in the present chapter and elsewhere (e.g., Hayes & Heiby, 1998a; Lavoie & Barone, 2006; Sammons et al., 2003). After considering those arguments, our view is that psychologists can best serve mental health needs not by prescribing drugs, but by delivering nonpharmacological interventions and serving as members of multidisciplinary teams that arrange and evaluate pharmacological interventions.

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Job Description Summary – Clinical Psychologist

Job Title	Clinical Psychologist
Supervisor	Business Manager (Administration) and Clinical Supervisor
Location	Metropolitan SA
Direct Reports	N/A

JOB SUMMARY

The Clinical Psychologist is responsible for providing evidence-based treatment to members of the community**.** This includes services provided through a range of funding sources. The Clinical Psychologist will work autonomously in their duties, however, will be provided adequate supervision and support.

RESPONSIBILITIES AND DUTIES

Provide therapeutic sessions to referred clients using evidence based interventions.
Develop tailored treatment plans for referred clients using prior knowledge and research skills.
Collate handouts for referred clients for homework and therapeutic aids.
Complete electronic client notes and reports for sessions to an appropriate standard as outlined by APS Code of Ethics/Supervisor standards.
Write letters to other health professionals and external agencies to assist the client in continual access to health and other services.
Prepare for and attend weekly clinical supervision sessions as per AHPRA guidelines.
Attend professional development in accordance with AHPRA registration requirements.
Comply with subpoena or court appearance request on behalf of referred clients.
Attend case conferences for referred clients to assist in service provision.

JOB SPECIFICATIONS

Education	Masters of Clinical Psychology OR Doctorate in Psychology (Clinical) (Required) Endorsement by Clinical College
Knowledge	Knowledge of evidence based interventions, research, legal and ethical requirements, clinical skills, and referral sources.
Skills	Strong interpersonal skills, well organised, high level of verbal and written communication skills, well defined research and analytical skills. Strong problem solving skills, clinical skills based on experience, IT and social networking skills.
Ability	Ability to effectively communicate with people at all levels of the business. Ability to deal with sensitive and confidential information. Ability to work collaboratively with a wide range of stakeholders.

	Ability to listen and reflect on constructive feedback. Ability to be open to new ideas. Ability to critically analyse information. Ability to apply learned knowledge. Ability to remain calm under pressure. Ability to act in a professional manner.
Other	Experience using Microsoft Suite (Word, Excel and PowerPoint). Experience using electronic filing and medical software (e.g. MasterCare). Experience working with or a willingness to provide therapy to a range of client presentations. Must be flexible and able to take managerial direction as required. Willingness to engage in a team-based environment.
JOB CONTEXT	Indoors Office Business clothes Very minimal environmental or job hazards

Careers in Clinical Psychology

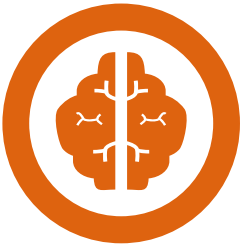
A career as a Clinical Psychologist is a popular, but competitive, route for Psychology graduates. The role involves working with clients who have a variety of mental health issues. Clinical Psychologists aim to improve the psychological wellbeing of their clients through various psychological interventions, methods and treatments.

For a full, in-depth description of the role of Clinical Psychologist, visit the role profiles at the following sites:

Prospects
www.prospects.ac.uk/job-profiles/clinical-psychologist

NHS Careers
www.healthcareers.nhs.uk/explore-roles/psychological-therapies/clinical-psychologist

British Psychological Society (BPS)
www.careers.bps.org.uk/area/clinical

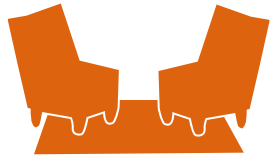


Qualifications

In order to work as a Clinical Psychologist you must be registered with the HCPC (Health and Care Professions Council). To be eligible to register you need to complete a **Doctorate in Clinical Psychology** (DClinPsy), or equivalent, approved by the HCPC. The Doctorate usually takes three years to complete and it can be **very** competitive to gain a place on the course. People often apply a number of times before being successful.

Applications are made through a central system (the Clearing House for Postgraduate Courses in Clinical Psychology) rather than directly to universities themselves, although the University of Hull and Queens University Belfast do accept direct applications. There is an annual cycle for applications, with the system opening to new applications in September and closing at the end of November, for candidates to commence their Doctoral training in the autumn the following year. Check online for the most up to date information and dates: **www.leeds.ac.uk/chpccp/index.html**

How to get a place on the Doctorate in Clinical Psychology



To be eligible for the Doctorate you will need Graduate Basis for Chartered Membership (GBC) through completing a BPS accredited undergraduate degree in Psychology or Psychology conversion course. Most providers look for a first class honours degree or an upper grading. You may also find that a Masters level qualification is helpful. Check directly with providers for their entry requirements and advice.

In addition to qualifications, Doctorate providers will be looking for relevant skills and experience in your application. In general, a minimum of 6 to 12 months of relevant clinical work experience is required to secure a place on the course. Some providers may expect even more than this – speak with them directly for their requirements and advice.

How to get relevant clinical experience

Roles which may help you gain the required experience to apply for the Doctorate in Clinical Psychology include:

- Assistant Psychologist
- Research Assistant
- Trainee Psychological Wellbeing Practitioner
- Mental Health Support Worker
- Healthcare Assistant or Health Care Support Worker on a relevant ward or department e.g. Psychiatric Unit, Community Mental Health Team
- Associate Mental Health Practitioner
- Social Work Assistant
- Community Support Worker

Paid roles and voluntary roles both count towards experience. Ideally you should aim for settings where you will be working with relevant client groups and gaining hands on experience in client facing work. You might find relevant roles in:

- The NHS
- Charitable or voluntary organisations e.g. Mind, Samaritans, Addaction
- Social Services
- The Prison Service
- Private healthcare settings

The Clearing House for Postgraduate Applications in Clinical Psychology has useful information about building up your experience, including links of sites which may be useful when searching for roles: **www.leeds.ac.uk/chpccp/entryexperience.html**

TOP TIP!

To find experience it can also be useful to put yourself in the shoes of a client who is searching for help – you may find it easier to discover local organisations who work with relevant client groups through a Google search such as 'help with an eating disorder in Lincoln' for example.

Other ways to boost your chances of getting a place

- The Clearing House for Postgraduate Courses in Clinical Psychology has produced a job description and person specification to help applicants understand what is being sought from trainees. It is worthwhile taking time to study this to ensure that you are building up relevant skills, knowledge and experience to be able to evidence that you meet the criteria on application: **www.leeds.ac.uk/chpccp/JobDescriptionPersonSpec.pdf**

- You may also benefit from joining and staying active within the BPS. The University is likely to pay for your membership in the third year of your course, but you might wish to continue with your membership and independently join the BPS Division of Clinical Psychology as a graduate member when you finish your studies.
- Network - remember that here at Lincoln we have qualified Clinical Psychologists within the School of Psychology who are involved with the Trent Doctoral programme, so it may be worthwhile to forge links with tutors and build your professional networks.
- Get good academic and clinical references. Work with referees to ensure that they will support your application.
- Get published! You can stand out from other applicants if you have evidence of peer reviewed publications in academic journals.
- Learn to drive to improve your chances of getting an assistant post where the ability to be geographically mobile is often important.

Making the most of your time during your undergraduate degree

If your overall goal is to become a Clinical Psychologist, you can start building your relevant experience throughout your undergraduate degree. This will you give you the best chance of securing the most sought after graduate roles (such as Assistant Psychologist or Research Assistant) prior to applying for the Doctorate.

The roles and organisations listed in the above section are mostly still applicable at undergraduate level. It would be worthwhile trying to get involved in roles in a volunteering capacity as early as possible, enabling you to build on your experience throughout the course.

You may find that volunteering is a good way to get into relevant organisations and the Students' Union Volunteering Team are available to help:
http://lincolnsu.com/work-ready/volunteer-opportunities

Organisations in and around Lincoln which may be able to offer you relevant experience include:

- Addaction
- Lincolnshire Action Trust
- NACRO
- Lincs shine
- LPFT – Lincolnshire Partnership Foundation Trust
- Richmond Fellowship
- Rethink
- Partnerships in Care
- The NHS
- The Prison Service

!

TOP TIP!

The NHS typically offers a variety of roles which are suitable for those looking to gain experience to lead up to training as a Clinical Psychologist. Use the NHS Jobs site **www.jobs.nhs.uk** as a database by searching for generic terms such as 'mental health' or 'psychology' rather than searching for specific job titles e.g. 'psychological wellbeing practitioner'. This will widen your field of results! To find roles which recent graduates are likely to be eligible for, consider filtering the results by pay band – looking at bands 2 – 4 is usually a good start.

Is a Career in Clinical Psychology right for me?

Given both the research intensive nature of the Doctorate in Clinical Psychology and the competitiveness of gaining a place on the programme, it is important to consider all of your options and you may wish to research alternative graduate career routes which have specialist options in mental health. Careers in areas such as occupational therapy, mental health nursing and social work might be of interest and the 'Options with Psychology' section on Prospects might provide a useful starting point for your research: **https://www.prospects.ac.uk/careers-advice/what-can-i-do-with-my-degree/psychology**

Useful websites

The British Psychological Society (BPS) Careers site
www.careers.bps.org.uk

Prospects
www.prospects.ac.uk/careers-advice/what-can-i-do-with-my-degree/psychology

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SCIENTIFIC AND SOCIAL CHALLENGES FOR CLINICAL PSYCHOLOGY

The article is an attempt to reflect on the current tasks faced by clinical psychology as a field of science and practice. These tasks stem from the constant need to improve the scientific level of this discipline and to solve problems connected with intensive social changes.We have considered the issues of globalization and the challenges it implies for clinical psychology, presenting selected theories of globalization and the patterns of reaction to these changes. We see evidence-based practice in psychology as a reflection of these transformations in science and clinical practice. In the article, we discuss the positive influence of this approach on the improvement of assessment practice standards and psychotherapy effectiveness, drawing attention to some negative consequences of universalization, particularly those that threaten human agency. Addressing and responding to these challenges requires close cooperation of researchers and practitioners, which was called for at the 1st National Clinical Psychology Conference in 2014.

Keywords: theory and practice of clinical psychology; globalization and culture; evidence-based practice; new challenges for assessment and therapy.

Introduction

The scientific level of clinical psychology is determined by creating a theory, adhering to methodological assumptions, and conducting modern research. Therefore, interrelations between theory and practice constantly constitute a subject for

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reflection and an area with new tasks to undertake. These reflections were the subject of the 1st National Clinical Psychology Conference after the political transformation in Poland. Analyzing the sources and nature of challenges to clinical psychology in the 21st century, we considered the key moments of its development that determined its theoretical basis and identity in the past fifty years.

In the present article, we reflect on the thesis that the most important problems regarding the individual's mental health and the mental condition of the population in many countries are related to the globalization processes taking place in various domains as well as to the preservation of natural and cultural identity. In science, globalization processes seem to be reflected in the ideas of evidence-based practice (EBP). In the early 1990s, Gordon Guyatt (1991, as cited in Spring, 2007) used the expression "evidence-based medicine" (EBM), pointing out the necessity of taking into account, above all, the results of scientific research rather than, as previously, the results of clinical research to justify the choice of the treatment procedure for a particular patient. The model of evidence-based psychological practice (EBPP) was officially recommended in 2005 by a team of experts of the American Psychological Association (APA, 2006). Since that time, just like in medicine (Rzepiński, 2013), evidence-based practice in psychology has been used in two different senses. First, it may refer to a certain way in which a psychologist engages in clinical practice in the areas of assessment and therapy. Activities in these areas can be evaluated as either meeting or failing to meet the criteria of evidence-based practice, with an indication of the degree to which the assessment-related and therapeutic decisions made by the psychologist possessing proper clinical skills and experience rely on the best results of scientific research, taking into account the values and preferences important for the patient (APA, 2006, p. 273). Second, it may refer to the general conception of evidence-based practice, in which assumptions are formulated and principles are established concerning the evaluation of assessment conceptualization reliability and in which the rules of determining the evidence value of various studies on psychotherapy effectiveness in specific mental disorders are formulated. In this perspective, EBP becomes a new, metaparadigmatic theory, which we recognize to be a manifestation of the trend towards the globalization of rules of evaluating the reliability and quality of assessment and therapeutic procedures in psychology, which has both positive and negative consequences for science as well as for clinical practice.

The ideas and assumptions underlying studies following the EBP approach, which were supposed to meet the criteria of the highest possible internal and external validity, evoked great hopes, but the ways in which they were used insti-

tutionally and individually in clinical practice brought not only positive outcomes. These studies were meant, above all, to improve the quality of medical services and contribute to the universalization of the standards of treating patients somatically ill and suffering from various mental disorders. In medicine and clinical psychology, many assessment and therapeutic procedures to follow in the case of various somatic diseases and mental disorders have been standardized thanks to EBP ideas. It turned out, however, that – just like in medicine – the phenomenon of universalization of assessment and therapeutic mental health services can lead to certain negative consequences for patients and for the effects of treatment.

In the current article we present clinical psychology as a science and social practice, taking into account the significance of EBP to the conceptualization of mental health. We not only focus on presenting the phenomenon of globalization as a social context influencing the mental health of individuals and social groups, but also attempt to point to the use of the evidence-based practice procedure in creating a universal model describing and explaining the determinants and mechanisms of mental health. Finally, we analyze the positive and negative consequences of translating the results of research in the EBP perspective into clinical practice, especially the consequences of the universalization of assessment and therapeutic procedures.

The theory and practice of clinical psychology

Like other disciplines of applied psychology, clinical psychology, is defined as a field of research and social practice directly related to achievements in various fields of scientific psychology (Lewicki, 1969; Sęk, 2000; Brzeziński, 2014a). It is currently assumed in the EBP perspective that the clinician is a researcher and a practitioner who translates psychological theories into practice – and the other way around, that his or her practice becomes a source of new research and conceptions explaining the determinants of mental health and effective therapy. The objective of clinical psychology is to describe mental health and disorders, to explain the intrapsychic mechanisms of health maintenance and disorders in accordance with the assumptions of psychological theories and the results of research on human nature, and to identify the determinants of these mechanisms in the context of knowledge about the significance of biological, psychological, and sociocultural factors. This knowledge is the basis for the clinical psychologist's assessment activity and the background for the formulation

of assumptions about effective procedures of psychological help in the form of preventive or psychotherapeutic interventions.

It is commonly believed that psychology as a science has a global character and – as opposed to extrascientific cognition, as emphasized by Brzeziński (2014a), citing Ajdukiewicz's propositions – is an intersubjectively communicable and verifiable discipline. Although the issues of knowledge accumulation methodology in psychology do not arouse much controversy, there is no agreement as to whether psychology is a universal science – namely, as to whether it explains human functioning above the sociocultural context. Some academics see it as a discipline accumulating knowledge that is highly supracultural and suprareligious (Łukaszewski, 2014), others view it as more local, derived from research conducted at a particular time and in a particular social and cultural context (Grzelak, 2014); still others regard it as a discipline that is, to various degrees, burdened with the cultural and national factor (Brzeziński, 2014a, 2014b). What is universal in psychology is research methodology, and what is more local is research on the mental health of individuals or local communities.

The issues of globalness and universality in clinical psychology are considered both at the descriptive level and at the level of explaining the sources of human experience and behavior. In psychopathology it is the universalistic approach that dominates, as it is pointed out that mental disorders – such as psychoses, personality disorders, or mood disorders – occur in the entire population. In every classification of mental disorders, for example in the current ICD-10 (WHO, 1997) and DSM-5 (APA, 2013), it is stressed that these systems have a universal and supracultural character. At the same time, these very classifications contain annexes in which specific mental disorders are presented that are strictly connected with a particular sociocultural environment and the beliefs found in it – for instance, koro anxiety disorder (the Chinese culture, Indonesia, Malaysia, Taiwan) or taijink yofusho (China, Japan, Korea; WHO, 1997, ICD-10).

At the level of explaining the determinants of mental health development, clinical psychology – just like developmental and personality psychology – is based on the fundamental assumption that this development is an outcome of the influence (interaction) of biological factors and specific sociocultural factors. In the light of the globalization and universalization of culture, the so far unquestionable thesis about sociocultural relativism in the understanding of mental health and disorders should be reconceptualized in psychology. In the radically universalistic biological model of the determinants of disorders, the influence of culture on mental health is considerably minimized. In contrast, the radically relativistic point of view posits that cultural influences are visible not only in

various symptoms of mental disorders but also in the development of pathomechanisms of disorders that can be shaped mainly by specific sociocultural communities. The two approaches gave rise to somewhat different research methodologies, procedures, and instruments (Stypuła, 2012). If we assume that globalization is the universalization of phenomena in various dimensions of reality, does this mean it abolishes the necessity of establishing the specific, individualized impact of social and cultural factors on the development of the individual's psychological structure? Or does the globalistic approach propose a new perspective in the understanding of the significance of sociocultural factors in the development of mental health? These questions are the subject of reflection below.

The cultural context and globalization processes – selected issues from the clinical psychologist's perspective

Globalization processes permeate various domains of the life of groups and individuals. They lead to the unification and interdependence of economic, political, and sociocultural processes on a global scale. These changes may have a positive and negative character. Positive phenomena include creating conditions for creative dynamism and mobility, opportunities for competition and cooperation as well as fast dissemination of new discoveries in the fields of science, technology, and information systems, the high speed of communication, and wide access to knowledge thanks to the constantly improving Internet. These changes stimulate the development of cognitive domains and some competencies in individuals (Friedman, 2000). The negative influences of globalizaton are analyzed, among other areas, in the theory of culture shock and its consequences (Nieman, 2011).

Globalization processes have also become an object of reflection in psychological publications. This can consist in a modification of the existing theories by introducing globalization phenomena into thinking about the patterns of human reactions to universalistic tendencies in economy, politics, ecology, language, culture, and individual self-regulatory development of personal identity and lifestyle, additionally characterized by a high potential for unification and obligation (cf. Bandura, 2001, 2002; Kowalik, 2015a, 2015b; Oleszkowicz & Senejko, 2013; Oleś, 2011; Salzman, 2001; Wosińska, 2007). Some authors propose new concepts and theories, making it possible to present the relationships between changes in the organization of social life and psychological reaction patterns. Kowalik (2015a) believes that new reciprocal influences should be revealed between macrosocial existence, which globalization amounts to, and individual

existence. For this purpose, he proposes a new concept of context, which, as opposed to environment, includes human actions – namely, the context of action such as work, education, family life, recreation, etc. A concept even more crucial and important for the clinical psychologist is contextualization. Its basic function is to give meaning to one's own behavior in terms of freedom of choice and in terms of maintaining the direction of one's activity in relation to the identified external macrosocial conditions (Kowalik, 2015a, p. 25). Such an understanding is consistent with the theory of self-regulating human agency as related to globalization, proposed by Bandura (2001).

A preliminary analysis of these phenomena can be based on the types of evaluations that people attribute to various aspects of globalization, but it is necessary to take positive, negative, as well as ambivalent evaluations into account (Oleszkowicz & Senejko, 2013; Łoś & Senejko, 2013). The attempt made below takes into account these evaluation and various behavioral consequences.

1. Positive evaluation and acceptance of globalization-related changes can take two forms: (a) the pattern of identification with globalization ideas accompanied by a rejection or denial of one's own cultural identity, and (b) the pattern of balancing globalization ideas with local and individual values.

1a. This pattern probably occurs in groups of people who are beneficiaries of positive changes in many life domains (prosperity, freedom, success, mobility, unrestricted enjoyment of neoliberal values). However, this positive pattern of reaction may be only apparent, since a weakening of social bonds may set in with time in families, groups of friends, and local communities. Excessive identification with the general systemic requirements of globalization can also lead to a decrease in the stable sense of personal identity and in the sense of personally created meaning in life. A phenomenon that can also occur among people involved in coping with the challenges of the globalizing social order is the self-exploration of personal resources and resignation from activities aimed at comprehensive development.

1b. The pattern of balancing, with self-regulating agency and with a tendency to resolve the conflict between universalization and the values of local culture. It is found in people who integrate globalistic values in their judgments with the values of the local sociocultural context. These are probably rational, realistic, pragmatic people, effective in achieving agentic and prosocial life goals. At the same time, they protect close relationships with family and friends as well as care for the development of personal emotional and coping resources and the sense of meaning in life.

The patterns of reacting positively to the universalization of life are not well known. This may stem from the fact that a vast majority of conceptions and studies in clinical psychology analyze negative consequences.

2. Negative evaluation of globalization-related changes can also take two forms: (a) the pattern of negation and rigid defense of local traditional values, and (b) the pattern of negation and struggle against globalistic tendencies.

2a. The pattern of *status quo* defense can take various forms of avoiding confrontation with globalization; this can be the cultivation of traditional rituals of family, religious, or social life with tendencies to exclude oneself from the changes taking place. These tendencies can be accompanied by processes of external exclusion. A consequence of this may be a sense of inadequacy or not being understood, a sense of harm, a sense of being under threat, as well as resentments, loneliness, etc.

2b. In the pattern of negation and struggle with globalistic tendencies, the dominant attitude will be rebellion as well as various forms of opposition and aggression. This pattern has been observed mainly in young people, socioeconomically disadvantaged and experiencing strong frustration and helplessness against the system (e.g., against injustice and harm). In the climate of neoliberal social changes and global market economy, there appears a large group of well-educated people who have no stable jobs, cannot do the kind of work they have been trained to do, and do not have access to state-guaranteed privileges, health or retirement insurance, etc. They are "precarians" (Standing, 2014), who experience a lack of work-related identity and a lack of belonging to a loyal community of employees. The lack of permanent employment and existential security leads to a variety of negative social and health-related consequences. The frustration of the activities one has engaged in evokes anger and aggression, a tendency to protest, and susceptibility to populist and nationalistic influenses.

As regards the thesis that globalization not only is significant to the problems faced by scholars but also influences clinical theory and practice via the EBP perspective, we referred to two examples. The first one is supposed to illustrate how the phenomenon of struggle with globalization processes is conceptualized in the cognitive-behavioral approach; the second one is meant to illustrate how the recommendations of EBP are used, EBP being a perspective promoting a certain ideal pattern of building psychological models that explain, for example, the determinants of mental health for resolving social problems.

This, however, requires an informed choice or creation of theoretical constructs within a psychological orientation – constructs concerning the determinants and mechanisms underlying the patterns of behavior in confrontation with

the demands of globalization. Such an attempt was made by the already mentioned Bandura (2001, 2002). He proposes that the cognitive-behavioral and self--regulating agency conceptions should embrace such a "changing face of psychology" (Bandura, 2001, p. 12) in the globalization era that would enhance the most humanistic tendency in human motivation and behavior, namely, the maintenance of agentic control over one's own life and action. In this connection, he analyzes educational activity, work, and health-related lifestyle. He mentions an increase in chance and incidental phenomena as well as the impact of the Internet and the erosion of intimacy as risk factors for developmental pathologies. He puts emphasis on self-regulation in coping with the stress involved in globalization, which promotes the formation and development of proactive tendencies and general resistance resources such as resiliency and positive emotions. This paradigmatic proposal requires further theoretical work to meet the conditions of EBP.

An attempt to develop a new model of psychological theory based on the research procedure derived from EBP was also proposed by a team of health psychologists from the United Kingdom (Michie et al., 2005). The procedure they applied was supposed to ensure a high internal and external validity of the model of individual's health determinants, arrived at through a consensus between theorists, researchers, and practitioners. The participants in the program were 18 renowned scholars investigating the mechanisms of health behavior change, a group of 13 people evaluating health care services, and a group of 30 practicing health psychologists. The group of theorists generated areas of knowledge about health behavior and theoretical constructs. Next, the group evaluated the interrelations between them and the importance of health behavior change for intervention issues. This conceptual work yielded 17 key constructs derived from 33 psychological theories. The interdisciplinary team and a group of practitioners evaluated the usefulness of these constructs in examinations consistent with EBP in health care. The group of health care theorists and researchers operationalized these constructs and together formulated questions for a questionnaire that served as the basis for pilot and standardization studies.

Challenges for clinical psychology in the areas of assessment and psychotherapy

The issues raised in the course of discussions at the conventions of the Polish Psychological Association (PPA) and at national scientific conferences in Poland focused on three important problems: (1) the low level of knowledge and pro-

fessional standards among psychologists using psychological tests and the low level of expectations regarding the psychometric parameters that psychological instruments should have (Paluchowski, 2010), (2) the low level of adherence to standards of conducting psychotherapy in accordance with the requirements of particular therapeutic schools, which leads to a situation in which, at best, various "signature" integrative therapies dominate in the Polish market (Szymańska, Dobrenko, Grzesiuk et al., 2014), and, as a result, (3) the need to appoint teams of experts to function at various institutions that would initiate research projects on assessment and therapy as well as promote professional and ethical standards for psychologists of various specialties (Brzeziński, 2014c; Cierpiałkowska & Sęk, 2015).

Three types of clinical assessment are usually distinguished in psychology: differential assessment, structural-functional assessment, and epigenetic assessment (Brzeziński & Kowalik, 2000; Sęk, 2000; Cierpiałkowska, 2007). Because there is not simply one clinical assessment, we cannot refer to one way of case conceptualization and, consequently, to the same principles and criteria of assessment reliability.

Generally, diagnosis has either a more or less descriptive character, in which case it presents the individual's functioning in various life domains in psychological or psychopathological terms (*case assessment*), or a more explanatory character, in which case it focuses on explaining, in accordance with the chosen paradigm in psychology, the salutogenic or pathogenic intrapsychic processes and mechanisms (*case formulation*) that sustain adaptive or maladaptive behaviors. Case conceptualization is the basis of therapy planning. As shown by Groenier, Pieters, Hulshof, Wilhelm, and Witteman (2008), clinical psychologists formulate explanatory diagnosis much less often than it seemed they did, particularly in the context of decision regarding the planned therapy. If they engage in such activity, they often do so in less complicated cases, in which empirical evidence on the effectiveness of various treatment programs is available (Groenier Pieters, Witteman, & Lehman, 2013). Therefore, the challenge for clinical psychology is to learn and establish the rules and ways of formulating explanatory diagnosis.

Studies on the reliability and validity of differential and structural-functional assessment, especially in the context of research instruments, were conducted before the emergence of the idea of EBP. Although assessment is something more than merely using techniques of measuring the patient's various traits and attributes, the greatest amount of attention in the model of evidence-based assessment is devoted to the validity, reliability and standardization of instruments.

When determining the directions of EBA development in adult psychopathology, Hunsley and Mash (2005) indicated, among other things, that the psychometric properties of measures are not absolute but vary depending on the clinical group, the context, and the aim of the study. For this reason, some of them are more reliable for screening purposes, others for assessment, and still others for treatment planning or monitoring (Soroko, 2016). At present, just like in medicine, thanks to studies conducted in accordance with the assumptions of EBA, specific techniques and strategies are recommended for assessing particular mental disorders, such as personality disorders (Widiger & Samuel, 2009), anxiety disorders (Antony & Rowa, 2005), posttraumatic stress disorder (Speroff et al., 2012), or alcohol addiction (Maisto & Connors, 2007). These procedures and instruments were useful until the introduction of the most recent DSM-5 classification (APA, 2013), which brought profound changes regarding the assessment criteria for some mental disorders. It turned out that the introduction of new categories of mental and behavioral disorders as well as the use of the hybrid dimensional-categorical model, for instance in the classification of personality disorders, requires a verification of assessment procedures and instruments (Cierpiałkowska & Soroko, 2014; Miller, Few, Lynam & McKillop, 2015). This is probably the greatest challenge facing EBA, especially as publications concerning the new version of ICD11 (iCAT, 2011), which is to appear soon, suggest that it is modeled on some solutions proposed in DSM-5 (APA, 2013). The dimensional approach in psychopathology poses a considerable challenge to scholars, which is to create sufficiently algorithmized and reliable differential assessment procedures. This is especially important as the dominant kind of patients is people with dual diagnosis, suffering from two or more mental disorders. Algorithmized operations involve at least a potential danger of ignoring individual patient characteristics not fitting into the algorithm.

A great challenge for EBA is to formulate the rules of conceptualizing paradigmatic structural-functional clinical assessment at the stage of qualifying a person for therapy and at various stages in the course of therapy. This assessment covers the process and effects of therapy, and its essence consists in continually repeating the evaluation of change in the patient – the significance of that change to the achievement of the effect that has been specified together in the contract. It amounts to continually evaluating the validity of successive interventions and the patient's responses to these interventions, which are evaluated as clinically significant or nonsignificant from the point of view of important therapeutic objectives, aimed at mental health. Therapeutic interventions are a kind of variant of the specific therapeutic strategy adopted during one session;

they constitute the entire therapeutic procedure followed in successive stages of the therapeutic process. While there are recommendations – including those inspired by EBP – regarding comprehensive models of the assessment procedure, which comprise successive stages of assessment, usually from the problem being reported by a person to checking the effectiveness of the psychological intervention after it has been completed, such as the integrative model by Fernández-Ballesteros and colleagues (2001) or the integrative and paradigmatic models by Ingram (2006), it is nevertheless difficult to find – apart from some attempts in the form of descriptions of therapeutic sessions – a model of the therapy assessment process in the context of specific interventions and intervention strategies.

Apart from data about the patient and the context of assessment, what is also taken into account in each type of assessment procedure is the clinician's reaction to the patient and his or her examination results (e.g., conscious and unconscious reactions) as well as the clinician's competence (e.g., in using the instruments) and characteristics (e.g., the preferred theoretical approach, cognitive structures, self-monitoring and self-control abilities; Trull & Prinstein, 2013; Cierpiałkowska & Soroko, 2015). Despite clinicians' belief in a significant positive influence of clinical experience on assessment validity, the results of meta-analyses show that the size of this effect ranges from $d = 0.12$ (Spengler et al., 2009) to $d = 0.15$ (Spengler & Pilipis, 2015). This is not a particularly strong effect in the context of other factors, whose significance to assessment validity is not fully known. It turned out that, despite the introduction of various changes and improvements to the training of clinical psychologists in recent years, high stability has been observed in both the size and the variability of this effect since 1999. The causes of this phenomenon may be different and are not very well known, since we do not know much about the process and the rules governing the formation of the empirical and theoretical model of the patient in the assessment psychologist's mind.

Current research indicates that what may be of special importance to assessment reliability is two groups of factors. The first group is factors connected with the clinician's ability to reflect on and monitor the course of the assessment process, enriched with the experience gained during diagnostic supervision, which broadens his or her self-awareness and provides instruments for self-control (Garb, 2010). Reflecting on the procedure will be particularly important in assessment based on the dimensional-categorical model of mental disorders (Barlow & Carl, 2014). The second group is factors concerning the time and level of the relationship between the clinician and the patient during assessment,

especially when assessment precedes decision on the program of therapy. The diagnostic relationship, which may become a therapeutic relationship (alliance), offers greater possibilities of reliable assessment because the patient provides more diverse – not only factual but also dynamic – information about the changing functioning in different life domains and in various social contexts (Tufekcioglu & Muran, 2015).

By the time the conception of EBP developed in the field of psychotherapy, results of studies on its effectiveness conducted in natural settings (*effectiveness* studies) and in laboratory conditions (*efficacy* studies) had already been known. Efficacy studies proved the effectiveness of cognitive-behavioral, psychodynamic, humanistic, and interpersonal psychotherapy in the treatment of various mental disorders. Finally, the controversy that had lasted for many years came to an end, and the verdict of the Dodo bird was: "Everybody has won, and all must have prizes" (as cited in Duncan, 2002, p. 12). But was that really the case? Fairly soon it turned out that therapists not only refused to use the therapeutic strategies described in handbook therapy based on sessions conducted during efficacy studies, but also pointed out their low usefulness for the patients they treated. Whereas the participants in therapy conducted in laboratory conditions were young, educated, and motivated patients with one mental disorder in acute condition, in natural conditions patients often had a dual diagnosis, were middle-aged, and did not have a particularly high motivation to participate in therapy (Nathan, Stuart, & Dolan, 2000; Lambert & Ogles, 2004).

Meta-analyses of many studies, including that by Lipsey and Wilson (2001), revealed that it is difficult to increase the effectiveness of therapy by manipulating specific healing factors, but it is possible to improve it by influencing common factors connected with the therapeutic relationship and alliance (Cooper, 2010). This finding triggered research on the influence of personality and situational factors on the level of therapeutic alliance both in the patient and in the therapist, which yielded many interesting results.

Today no one doubts that psychotherapy is an effective method of treating a variety of mental disorders. Yet, questions about its outcomes should be asked in a different way. For instance, is behavioral-cognitive therapy effective for a 50-year-old married woman with higher education, addicted to alcohol and also diagnosed with narcissistic personality disorder? It is becoming increasingly obvious to experts that various sources of information should be used in such a situation in order to find out if a particular treatment method is effective (Dozois, 2013). Experts of the EBP team of APA did not define the concept of evidence in psychological counseling and psychotherapy; therefore, experts and scholars

in many countries develop their own hierarchies of evidence value of studies on psychotherapy effectiveness in specific cases.

A hierarchy of various types of studies on effectiveness of therapeutic procedure according to their evidence value has been developed by experts of the Canadian Psychological Association (Dozois, Mikail, Bourgon et al., 2014). They arranged these types from the lowest to the highest evidence values (cf. Figure 1).

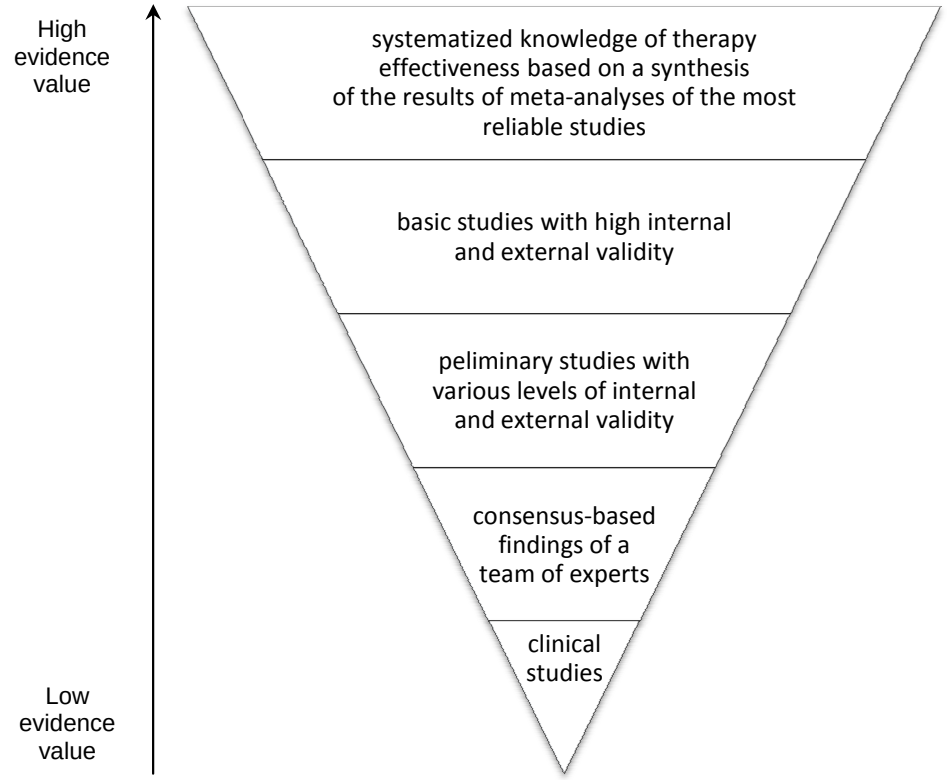

Figure 1. The hierarchy of evidence value of studies on psychotherapy effectiveness (based on Dozois, Mikail, Bourgon et al., 2014).

Barlow and Carl (2014) argue that studies with high evidence value have already become the basis for the creation of psychological procedures and strategies, particularly in cases of dual or triple diagnosis of mental disorders. The

verified effectiveness of these procedures contributed to the development of so-called transdiagnostic psychological intervention modules, which in turn are disseminated and attain the status of a standard in the treatment of mental disorders. It is predicted that the classic form of psychotherapy will dominate in nonpublic medicine and evidence-based psychological procedures will dominate in public medicine.

Knowledge derived from EBP research should have a significant impact on the assessment procedure and on the choice of the type of treatment to be recommended to patients with specific or multiple mental disorders, with patient preferences taken into account. At the same time, the therapist should monitor the effectiveness of therapy with regard to each individual patient and make changes to the therapeutic procedure when the expected positive effects are not achieved.

Conclusion

Summing up the main ideas of the article, we come to the conclusion that the challenges for clinical psychology concern both scientific and practical aspects. The scientific challenges are: to enhance the relations between theory, scientific research results, and clinical practice as well as the other way around, to constantly expand knowledge in accordance with the latest achievements in various areas of psychology, as well as to develop rules enabling the application of research results in practice. The challenge for clinicians is to prepare for cooperation in research teams solving problems of importance to individuals and social groups.

It more and more often happens that clinical psychologists also address the problems of the globalization era in the context of national and cultural movements. They conduct research on the meaning of global unification to the description and understanding of mental health and disorders. General patterns of positive and negative reactions to globalization-related changes in various domains of human life have been presented. Referring to the EBP perspective, scholars have proposed a paradigmatic approach to the individual's struggle with the globalization process and made attempts to apply its assumptions to the creation of health promotion conceptions.

The main aim of the EBP perspective is to increase the reliability of clinical assessment and the effectiveness of psychotherapy. In the field of clinical assessment, the results of research in this perspective has yielded many important findings making it possible to improve the psychometric properties of research instruments and the validity of differential diagnostic inference (case as-

sessment). What constitutes an enormous challenge is the creation of foundations for the assessment of mental disorders in the dimensional-categorical approach. Less has been done with regard to case formulation, the diagnosis indispensable for taking effective action in psychotherapy. Although the framework and principles of constructing an explanatory diagnosis have been defined in different schools of therapy, only learning the ways and rules of processing knowledge about the patient in the clinician's mind will make it possible to identify a more or less valid case formulation.

"Efficacy" research on the effectiveness of therapy, conducted in accordance with EBP recommendations, have not only yielded evidence that different therapeutic schools have similar effectiveness, but also led to the writing of numerous psychotherapy handbooks. The most effective therapeutic strategies and procedures have been presented in these handbooks, but therapists evaluated them as not very useful in practice. It turned out that patients selected for research on efficacy differ from the general population of patients. There is a shortage of studies and knowledge concerning patients resistant to standard treatment strategies in every school of psychotherapy as well as knowledge about the necessary conditions and rules of changing the therapeutic procedures. Another challenge for further research is to determine the significance of the healing factors common to different schools of psychotherapy, which are present and active to the greatest extent in the therapeutic relationship. Research consistent with the EBP perspective should answer the question of what psychological strategies should be used with regard to patients suffering from multiple mental disorders.

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HANDBOOK OF CLINICAL PSYCHOLOGY

Volume 2 Children and Adolescents

Edited by

MICHEL HERSEN
ALAN M. GROSS

John Wiley & Sons, Inc.

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Preface

HANDBOOK OF CLINICAL PSYCHOLOGY (VOLUMES 1 AND 2)

Over the past 3 decades the field of clinical psychology has periodically been reviewed in large tomes in multiauthor editions. Indeed, the senior editor of the present work has been involved in three such projects, published in 1983, 1991, and 1998. Since publication of the last of these works there have been major developments in clinical psychology, both with adults and with children and adolescents. Some of the most interesting innovations have appeared in the areas of ethics, legal issues, professional roles, cross-cultural psychology, psychoneuroimmunology, cognitive-behavioral treatment, psychopharmacology as practiced by clinical psychologists, political issues, geropsychology, parent training, pediatric psychology, assessment of child maltreatment and its remediation, and problems in infancy. The veritable explosion of new data has been fueled by the advent of new journals in highly specialized areas, in addition to the rapid dissemination of new ideas and data rapidly presented through electronic publication.

The plethora of data is distilled into 33 chapters for Volume 1 on adults and 32 chapters for Volume 2 on children. To enable the reader to easily traverse the two volumes we decided on a parallel structure. Therefore, each of the volumes is divided into six parts (Part I: General Issues, Part II: Theoretical Models, Part III: Research Contributions, Part IV: Diagnosis and Evaluation, Part V: Treatment, and Part VI: Special Issues). In both volumes a case illustration has been included in the section on Treatment. Authors here have been encouraged, to the extent possible, to use a relatively standard format: Identifying Features of the Client, Presenting Complaints, History, Assessment, Case Conceptualization, Course of Treatment and Assessment of Progress, Complicating Factors, Managed Care Considerations, Follow-up, Treatment Implications of the Case, and Recommendations to Clinicians and Students. In the volume on children and adolescents Developmental Factors and Parental Factors are added considerations that the authors have outlined.

These two volumes were conceived with graduate students and professionals in the field in mind. We believe that the material systematically presented herein is suitable both as a general reference work and separately as a major course text for clinical psychology with adults and clinical psychology applied to children. We, of course, eagerly anticipate the comments of both professors and students.

Many individuals have been involved in the development and fruition of this two-volume work. First and foremost we thank our contributors for taking time out

from their busy schedules to share their expertise with us. Second, we thank our friends at John Wiley and Sons for their appreciation of clinical psychology and the technical processing of this two-volume set. Third, once again we thank Carole Londeree for helping us to track all of the authors and manuscripts in her typical orderly and competent manner. Without her terrific help this could not have been accomplished. Finally, we thank Christopher Brown and Blake Kirschner for their wonderful technical work related to the indexes.

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PART I

General Issues

CHAPTER 1

Historical Perspectives

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In this chapter, we trace the historical development of the assessment and treatment of children's and adolescents' mental health issues. Although our overview is brief (see several sources for more detailed historical accounts: Doyle, 1974; Dubois, 1970; Linden & Linden, 1968; P. McReynolds, 1975), we provide some perspectives on contemporary evidence-based assessment and treatment. An examination of the historical factors in assessment and treatment is important for several reasons. First, it is important to understand that many of the contemporary issues in evidence-based practice have their origin in past practices. Second, it is important to realize that many contemporary issues are related to social or even political concerns that have their origin in the past. Third, the past has sometimes provided or even imposed a structure on assessment and treatment practices. It is important to understand this structure to understand contemporary models and the scope of psychological practices. Finally, it is important to focus on historical factors to introduce a variety of scholarly perspectives into the discussion of the issues surrounding evidence-based practice. We first review historical features of diagnosis and assessment and then turn our attention to child and adolescent therapy. However, the conceptual, theoretical, and practice issues in these domains overlap.

ASSESSMENT AND DIAGNOSIS: ANCIENT INFLUENCES

Most historical treatments of the assessment literature typically begin with a discussion of the work of Galton in England and Cattell in the United States (i.e., many books on assessment begin with this period; e.g., Sunberg, 1977). However, assessment has a much richer history, attesting to the assumption that many features of contemporary assessment actually date back to the beginnings of recorded history. L. V. McReynolds (1974) traced the historical antecedents of the current practices in assessment beginning with antiquity and extending to the second half of the twentieth century. Four phases were reviewed: antiquity, the medieval period and the Renaissance, the Age of Reason, and the period from Thomesius to Galton. We adopt this framework in this section of the chapter.

Antiquity

An examination of early assessment practices shows that there was a close interplay between the methods used and the cultural views held during that particular time.

This perspective is not unlike the contemporary views in the United States that led to the development of special education services for children (Kratochwill, Clements, & Kalymon, 2007), with its emphasis on fair assessment practices for handicapped children. It is possible that the first personality assessment procedure was based on *astrology,* and that the first psychological "test" was the horoscope. Although astrology can be regarded as invalid on scientific grounds, it did contribute to (a) the view that individual personalities represent the focus of assessment, (b) the belief that the psychological makeup of the individual is predetermined, and (c) the development of taxonomical (diagnostic) categories.

Another early assessment strategy involved physiognomy, the interpretation of an individual's character from body physique. Physiognomics, also a very limited assessment procedure, assumed a relatively fixed conception of personality but shared some methodological features with naturalistic observation, not unlike the naturalistic observations conducted using behavior modification procedures (Kazdin, 1978). L. V. McReynolds (1974) noted that the longest continued assessment technique with some claim to rationality and one that remains with us today is physiognomy. Thus, work by Mahl (1956) and Gleser, Gottshalk, and Springer (1961) on speech patterns; by Hall (1959), Eibl-Eibesfeldt (1971), and Haas (1972) on methodology of movements; of Izard (1971) and Ekman and associates (Ekman, 1973; Ekman, Friesen, & Ellsworth, 1972) on emotions and facial expressions; and of Hess and associates (Hess & Polt, 1960; Hess, Seltzer, & Schlien, 1965) on the relation of pupil size to affect can be related to earlier physiognomic conceptions (cf. L. V. McReynolds, 1974).

Developments in assessment during early times were not always limited to the area of personality assessment. For example, Civil Service examinations were used in ancient China for selection purposes. Dubois (1966, pp. 30–31) notes:

> The earliest development seems to have been a rudimentary form of proficiency testing. About the year 2200 B.C. the emperor of China is said to have examined his officials every third year. . . . A thousand years later in 1115 B.C., at the beginning of the Chan dynasty, formal examining procedures were established. Here the record is clear. Job sample tests were used requiring proficiency in the five basic arts: music, archery, horsemanship, writing, and arithmetic. . . . Knowledge of a sixth act was also required—skill in the rites and ceremonies of public and social life.

Medieval Period and the Renaissance

L. V. McReynolds (1974) notes that during this period, the acceptance of humeral psychology and physiognomic strategies of evaluating people were widespread. Generally, this period supported the recognition of the individual, and so we again see an example of cultural influences on assessment practices. In some respects this period set the occasion for what would later be a debate on research methodology, especially surrounding the use of group versus single-case research design in therapy research (Kratochwill & Levin, 1992).

Age of Reason

The Age of Reason covers the period from approximately the middle of the sixteenth century to the second half of the eighteenth. A major theme of this period was the focus on individual differences, as reflected in some important works on assessment: Huarte's *Tryal of Wits,* Wright's *Passions of the Minde,* and Thomesius's *New Discovery*. During this period, the recognition of individual differences prompted measurement so that an individual's sense of well-being could be more fully realized.

From Thomesius to Galton

A significant contribution to assessment during this period, particularly in the nineteenth century, was phrenology. Phrenology bears a similarity to physiognomy, but whereas physiognomy emphasized assessment of external body features such as facial and other characteristics, phrenology emphasized the assessment of the external formations of the skull. Phrenology assumed that mental functions were based on specific processes localized in certain areas of the brain and that the intensity or magnitude of these functions was indicated in the contours and external topography of the skull (L. V. McReynolds, 1974).

Four positive contributions of phrenology that have a resemblance to contemporary assessment practices or activities were identified (L. V. McReynolds, 1974). First, there was an emphasis on individual differences. Second, the assessment paradigm emphasized the notions of assessor and subject, the systematic collections of data during a single session, and written reports that usually included qualitative profiles. Third, the phrenological movement helped advance objectivity through blind assessment and rating scales. Fourth, phrenology contributed to the development of a primitive taxonomical system, which included affective faculties (e.g., propensities, sentiments) and intellectual faculties (e.g., perceptive, reflective). This line of reasoning was likely influential in later conceptualizations of diagnostic and classification systems.

Implications

This brief historical overview of ancient influences points out that many contemporary assessment practices have their roots deep in our past. Noteworthy is the fact that the work of the phrenologists (and later, Quetelet's work on psychological statistics) set the stage for the emergence of Galton's contributions and the subsequent more modern era in assessment. It is interesting to speculate how some of the ancient procedures might have set the stage for child diagnoses specifically. L. V. McReynolds (1974, pp. 524–525) raises an interesting point:

> We know that such techniques as chiromancy, metaposcopy, and phrenology are in principle all totally invalid, yet I suggest that in the hands of insightful and discerning practitioners they may, at least on occasion, have been more valid than we suppose, even if for different reasons than their users, much less their clients imagined.

ASSESSMENT AND DIAGNOSIS: NINETEENTH-CENTURY INFLUENCES

During the nineteenth century significant developments were taking place in Western Europe and the United States that would shape the future of psychological and educational assessment (cf. Carroll, 1978; Dubois, 1970; Laosa, 1977). Specifically, events were occurring in France, Germany, England, and the United States that were to have a profound influence on assessment practices in child therapy and education.

France

Attention to two movements occurred in France that made a significant impact on the history of testing and assessment in general and child assessment in particular (Maloney & Ward, 1976). One movement, pioneered by Berhheim, Liebault, Charcot, and Freud, was focused on a new view of deviant behavior. The influence of this movement was to take abnormal behavior out of the legal or moral realm with which it had been previously associated and cast it as a psychological or psychosocial problem. This focus prompted psychological assessment rather than moral or legal sanction, as had been common prior to this period.

Also noteworthy was the movement called "the science of education." Jacques Itard, a French physician, taught Victor, the "Wild Boy of Aveyron," various skills. Many of the procedures used in Itard's work were similar to later behavior modification procedures that emphasized environmental stimulus and response changes during intervention. Itard's contributions also provided a background for Binet's work on measurement of intelligence.

Esquivol's (1722–1840) work, represented in his book *Des Maladies Mentales,* was influential in that he distinguished between "emotional disorders" and "subaverage intellect." According to his views, subaverage intelligence consists of levels of individual performance: (a) those making cries only, (b) those using monosyllables, and (c) those using short phrases but not elaborate speech. Thus, here we see the basis for an early classification scheme that could organize human behavior.

Germany

Although some of the work in France emphasized individual differences in pathology and cognitive ability, German scientists perceived individual differences as a source of measurement error. A significant contribution to the individual differences theme is found in the "Maskelyne-Kinnebrook affair." The difference between Maskelyne (the astronomer) and Kinnebrook (the assistant) in their measurement of the timing of stellar transits was later analyzed by Bessel. Bessel concluded that different persons had different transit tracking times, and that when all astronomers were checked against one standard, individual error could be calculated; thus, a sort of "personal equation" was developed (cf. Boring, 1950).

Another significant influence on assessment came from Wundt, who set up a psychological laboratory in Leipzig to study such processes as reaction time, sensation, psychophysics, and association. This work, as well as the general work

occurring on measurement, was helpful to popularize the notion of measurement of differences between individuals. Some Americans who studied with Wundt were G. Stanley Hall and James McKeen Cattell. Both of these individuals were to have a large impact on future child psychological assessment.

England

The work of Charles Darwin was most influential in psychological and educational assessment, particularly through his theory of evolution presented in 1859 in *Origin of the Species*. Darwin's work emphasized that there were measurable and meaningful differences among members of each species. Galton, Darwin's half-cousin, was influential in applying evolutionary theory to humans. In his 1869 book, *Hereditary Genius,* Galton argued that genius has a tendency to run in families. He was greatly influenced by the Belgian statistician Quetelet (1770–1864), who was the first to apply the normal probability curve of Laplace and Gauss to human data. This work translated into the notion that nature's mistakes were represented as deviations from the average.

Several implications of this work for child assessment and treatment are noteworthy. First, Galton's system of classification represented a fundamental step toward the concept of standardized scores (Wiseman, 1967). Second, in the application of Quetelet's statistics, Galton demonstrated that many human variables, both physical and psychological, were distributed normally. This concept is a direct precursor to the notion of a norm and application of standardization (Laosa, 1977). Third, a major influence of this work was to establish that certain variables should be subjected to quantitative measurement. Galton's work was significant in that it encouraged other efforts in the area of measurement of individual differences in mental abilities that were considerably more sophisticated than previous efforts (Cooley & Lohnes, 1976). Finally, through the application of the normal curve, individual performance or standing could be classified as deviant or even as a mistake of nature. We know that although Galton was influenced by the phrenologists, he rejected this form of assessment. He noted in 1906, "Why capable observers should have come to such strange conclusions [can] be accounted for . . . most easily on the supposition of unconscious bias in collecting data" (quoted in Pearson, 1930, p. 577).

United States

Early work in the United States contributed to what was called the "mental testing" movement, a major part of clinical and school psychology. Cattell (1860–1944) was the first to use the term "mental test," and he is generally referred to as the father of mental testing (DuBois, 1970; Hunt, 1961). Cattell also introduced experimental psychology into the United States. A significant contribution to assessment was that he advocated testing in schools; he was also generally responsible for instigating mental testing in the United States (Boring, 1950).

In 1895, Cattell chaired the first American Psychological Association (APA) Committee on Mental and Physical Tests. Although Cattell made major changes in the nature of testing, his work was not accepted unconditionally. For example, Sharp (1899) published an article questioning the reliability of mental tests. Wissler

(1901) compared the reliability of some of Cattell's psychological measures with various measurement approaches from the physical sciences and concluded that tests used in Cattell's lab showed little correlation among themselves, did not relate to academic grades, and were unreliable (Maloney & Ward, 1976). Even Wundt was not supportive of Cattell's focus on mental measurements (Boring, 1950). Nevertheless, Cattell's work, as well as work in France, promoted the development of a movement called "differential psychology."

INTELLIGENCE TESTING MOVEMENT

Around the beginning of the twentieth century, assessment was given a new impetus through the development of differential psychology (Binet & Henri, 1895; Stern, 1900; Stern & Whipple, 1914). Stern and Whipple (1914) suggested that mental age be divided by chronological age to produce a "mental quotient," a procedure, with refinements, that evolved into the IQ concept (Laosa, 1977).

The work of Binet and his associates was quite influential, although not necessarily in the direction that Binet had envisioned or desired (Sarason, 1976; Wolf, 1973). Binet initially focused his efforts on the diagnosis of "mentally retarded" children around the late 1880s. At this time he was assisted by Theodore Simon, with whom he later worked in the development of the first formal measure of intellectual assessment for children (Wolf, 1973). Based on a study conducted for the Ministry of Public Instruction, he focused efforts on predicting which children would be unable to succeed in school (Resnick, 1982). Binet noted that performance on his scale had implications for diagnostic classification and education. Resnick (1982, p. 176) notes:

> A scale of thirty questions was developed, each of increasing difficulty. Idiots were those who could not go beyond the sixth item, and imbeciles were stymied after the twelfth. Morons were found able to deal with the first twenty-three questions. They were able to do the memory tests and arrange lines and weights in a series, but no more . . . the test . . . was designed as an examination to remove from the mainstream of schooling, and place in newly developed special classes for the retarded, those who would be unable to follow the normal prescribed curriculum. As such, it was a test for selection, removing from normal instruction those with the lowest level of ability. Binet argued, however, that the treatment the children would receive in the special classes would be more suited to their learning needs. The testing, therefore, was to promote more effective and appropriate instruction.

Interest in testing the abilities of children was at a high level during this time. This interest was prompted, in part, by the growing population of children in schools due to natural population growth and immigration and the fact that students began to stay in school longer (Chapman, Terman, & Movement, 1979). With the growing number of children in schools, it became clear that not all children could profit from regular instruction. The policies and procedures for diagnosis and assessment of children during this time set a direction that has only recently been changed with new federal regulations that has, as one focus, to reduce the emphasis on an IQ-achievement discrepancy to make a determination of disability status for children with learning problems. The new focus in assessment is called "response

to intervention" and involves determining the child's response to instruction as part of the diagnostic process (Kratochwill et al., 2007).

Several American psychologists promoted Binet's work. For example, Goddard published the first revision of the Binet scale, and Terman developed the Stanford-Binet. Thereafter, the Binet scale was used to identify children who were regarded as "backwards" or "feebleminded." In 1911 the Binet scale was being used in 71 of 84 cities that administered tests to identify feebleminded children. However, the Binet scale was also being used experimentally to screen out and turn back "retarded" immigrants (Knox, 1914, as cited in Widgor & Garner, 1982).

The Stanford version of the Binet-Simon scale was originally published in 1916 by Terman; it was revised by Terman and Merrill in 1937 and 1960 and renormed in 1972 and subsequently. This translation and revision of Binet's earlier work firmly established intelligence testing in schools and clinics throughout the United States (DuBois, 1970).

Development of Group Testing

The assessment movement was given a major thrust through the development of group tests during World War I. Many assessment efforts during this time reflected a pattern of procedures similar to that used by Binet (T. E. Newland, 1977). Ebbinghaus demonstrated the feasibility of group tests, and some American psychologists (e.g., Otis, 1918; Whipple, 1910) recognized that the Binet-Simon scale could be adapted for group testing. However, there were important differences. Whereas the Binet-type items typically required a definite answer provided by the child, group tests usually called for recognition of a correct answer among several alternatives (Carroll, 1978).

A committee of the APA chaired by Robert M. Yerkes developed the Army Alpha and Army Beta group tests. The Army Beta (a nonverbal group test) was designed so as not to discriminate against illiterates and individuals speaking foreign languages. Although the impact of this development was to create a new interest in and role for testing, a review of the tests used revealed that the source of many tests was increasingly used for nonmilitary purposes (T. E. Newland, 1977).

Following the war, many clinical psychologists who were involved in wartime testing sought employment in the civilian ranks, and many became involved in the schools. Resnick (1982, p. 183) notes:

> Aiding this movement was Philander P. Clarxton, U.S. Commissioner of education, who communicated to school superintendents throughout the country about the reserve of trained people that could be tapped for the needs of the schools. He wrote enthusiastically about the "unusual opportunity for city schools to obtain the services of competent men." Among the services that they could render was "discovering defective children and children of superior intelligence."

This movement, in part, facilitated the use of group intelligence tests in the public schools for purposes of diagnosis and classification. Many of these tests were administered to identify children who could not profit from regular instruction. Although some schools had made provisions for special children, the intelligence tests

formalized the decision-making process for these special services. Also, between 1919 and 1923, Terman introduced the National Intelligence Test for grades three to eight and the Terman Group Test, for grades seven to 12 and found that the schools were most receptive (Resnick, 1982). Resnick reported that the most important use of the tests was for placement of children into homogeneous groups:

> Sixty-four percent of the reporting cities used group intelligence tests for this purpose in elementary schools, 56% in junior high schools, and 41% in high schools. Enthusiasm for the use of testing systemwide for this purpose was at a high level. In 1923, Terman's group test for grades seven to thirteen sold more than a half-million copies. (pp. 184–185)

The stage for the rapid development of ability tests was also set by such psychologists as Spearman, Thorndike, and Thurstone and their respective theories of intelligence. For example, Spearman developed an elaborate theory of the organization of human abilities in which he concluded that all intellectual abilities have a common factor, *g*, and a number of specific factors, *s*, which relate uniquely to each presumed ability. Spearman's two-factor theory was the basis on which tests examining specific abilities (Edwards, 1971) rather than global scores were developed (Laosa, 1977).

Thorndike viewed intelligence as comprising a multitude of separate elements, each of which represented a specific ability. Intelligence was also perceived as having both hereditary and environmental components. Thurstone concluded that there were seven primary mental abilities (in contrast to Spearman's *s* and *g* factors) and developed the Primary Mental Abilities Test to measure each specific ability.

Intelligence tests gradually evolved into major diagnostic instruments throughout the world. Such instruments became an important diagnostic tool for identifying children with cognitive disabilities. However, not all countries accepted the use of these tests. For example, in the Soviet Union such tests were banned in 1936 by the Communist Party because they were considered methods that discriminated against the peasants and the working class in favor of the culturally advantaged (Sunberg, 1977; Wortis, 1960). As an alternative, diagnosis in the USSR was based primarily on neurophysiological evidence. The neurologist and psychophysiologist, rather than the clinical psychologist, were primarily engaged in diagnosing children with mental retardation (Dunn & Kirk, 1963).

Work in these areas, as well as other contributions prior to and during this period, led to diverse views on the nature of intelligence and its assessment. A major contribution to the testing movement was the development of the Wechsler intelligence scales. Psychologist David Wechsler developed the Wechsler Adult Intelligence Scale (WAIS) by including a group of subtests from WWI vintage that he found valuable in his work with adults. His criterion of "general adaptability" (Wechsler, 1975) was extended downward in the development of the Wechsler Intelligence Scale for Children (WISC) and the Wechsler Preschool and Primary Scale of Intelligence (WPPSI). The work of Wechsler contrasted with that of Binet. Whereas Wechsler's scales emerged from work with adults and were later developed for use with children, Binet's emerged from work with young children and later were developed for use with older children (T. E. Newland, 1977).

PERSONALITY ASSESSMENT MOVEMENT

While tests of cognitive ability were rapidly evolving during the early part of the twentieth century, tests of personality were in their infancy. Although such devices as the Woodworth Personal Data Sheet were used in the military during WWI, the personality assessment movement received increased attention through the development of projective techniques such as the Rorschach Inkblot Test and the Thematic Apperception Test (TAT).

World War II, like the first war, did much to set the stage for rapid proliferation of testing practices. Indeed, psychological testing combined with the military need for assessment was one of the primary factors leading to the development of clinical psychology as an independent specialty (Maloney & Ward, 1976).

During the period following WWII, testing practices developed dramatically. Most tests developed during this period were tied to an intrapsychic disease model or state-trait conceptualization of behavior (cf. W. Mischel, 1968). Psychoanalytic theory generally accelerated assessment procedures that would reveal unconscious processes. Assessment practices emphasized an "indirect-sign" paradigm. Assessment was indirect in that measurement of certain facets of behavior were disguised or hidden from the client (e.g., such as in the TAT). Moreover, within the context of the intrapsychic model, testing practices were said to predict certain states or traits. The clinician's task was to administer a battery of tests to a child and look for certain signs of traits or states. An example of this approach was represented in the work of Rappaport, Gill, and Schafer (1945). In their classic book, the authors demonstrated how a battery of tests (e.g., TAT, Rorschach, WAIS) could be used to diagnose deviant behavior within the intrapsychic model (in this case, the psychoanalytic model).

Similar to the sign approach was the "cookbook" method of assessment, which reached a zenith during the mid-1950s (cf. Meehl, 1956). An example of this approach is the Minnesota Multiphasic Personality Inventory (MMPI; Hathaway & McKinley, 1943). As these authors noted, one of the presumed advantages of the cookbook approach was that "it would stress representativeness of behavioral sampling, accuracy in recording and cataloguing data from research studies, and optional weighting of relevant variables and it would permit professional time and talent to be used economically" (p. 243).

Emergence of Behavior Modification and Assessment

Behavior modification (also referred to as behavior therapy) and the related assessment procedures associated with this model have made a tremendous impact on psychology and education (Kazdin, 1978; Kratochwill & Bijou, 1987). As some historical reviews illustrate (Hersen, 1976; Kazdin, 1978), behavior therapy represented a departure from traditional models of assessment and treatment of abnormal behavior, both psychological and educational. Although the history of behavior therapy cannot be traced along a single line, practice was characterized by diversity of viewpoints, a broad range of heterogeneous procedures with vastly different rationales, open debates over conceptual bases, methodological requirements,

and evidence of efficacy (Kazdin & Wilson, 1978). Some reports of behavioral treatment followed Watson and Raynor's (1920) work in conditioning of fear in a child, but a significant impetus to behavioral assessment and treatment is commonly traced to the publication in 1958 of Joseph Wolpe's reciprocal inhibition therapy.

Independent of Watson's and Wolpe's work was research in the psychology of learning, in both Russia and the United States. Particularly important in learning research was operant conditioning, which Skinner brought into focus in the late 1930s (e.g., Skinner, 1938). The evolution of operant work into experimental and applied behavior analysis had an extremely important influence on the development of behavior therapy and assessment practices in general.

Although behavior therapy and assessment evolved considerably over the years, some general characteristics represented unities within the heterogeneity of evolving practice:

1. Focus upon current rather than historical determinants of behavior;
2. Emphasis on overt behavior change as the main criterion by which treatment should be evaluated;
3. Specification of treatment in objective terms so as to make replication possible;
4. Reliance upon basic research in psychology as a source of hypotheses about treatment and specific therapy techniques; and
5. Specificity in defining, treating, and measuring the target problem in therapy. (Kazdin, 1978, p. 375)

With the advent of behavior modification and its proliferation, a new assessment role also developed, particularly for clinical child and school psychologists. Behavioral assessment emphasized *repeated measurement* of some target problem prior (baseline), during, and after (follow-up) the intervention. Hersen and Bellack (1976a) noted that the psychologist's expertise in theory and application of behavioral therapy techniques (e.g., classical and operant conditioning) also enabled both an *assessment* and a *treatment* role to emerge in psychiatric settings. This focus was also to occur as a basis for the scientist-practitioner model of psychological services (Hayes, Barlow, & Nelson-Gray, 1999) and represented a strong foundation for the evidence-based treatment movement (Kratochwill & Stoiber, 2002) and what is now called response to intervention in psychological practice in schools (Brown-Chidsey & Steege, 2006; Kratochwill et al., 2007). Thus, the psychologist in various settings (e.g., clinics, hospitals, schools) became involved in direct service rather than engaged in only testing and diagnosis, although this was slower to evolve in schools. Behavior modification provided the impetus for these new roles and has continued to move practice forward in both prevention and treatment.

Developments in behavioral assessment also influenced the field of personality assessment in general. In many respects, assessment has acted as a barometer for the thinking of personality theorists. For example, a barometer of change in views about assessment has been the evolution of the title of the journal

specifically devoted to assessment in professional psychology (Goldfried, 1976). The journal, founded in 1936, was initially entitled *Rorschach Research Exchange*. Other projective techniques came into existence in the assessment process, and by 1947 the journal title was changed to *Rorschach Research Exchange and Journal of Projective Techniques*. Gradually, the more objective personality assessment techniques (e.g., the MMPI) were being used, and in 1963 the title was again changed to *Journal of Projective Techniques and Personality Assessment*. Projective techniques continued to show disappointing research results, and in 1971 this may have prompted the journal's change to the title *Journal of Personality Assessment*.

Nevertheless, there remained some doubt as to whether the future direction of assessment would take a distinct behavioral orientation. Even in 1963, when the journal *Behavior, Research, and Therapy* made its appearance, the issue was raised as to whether there would be a large enough readership to justify its existence (Brady, 1976). However, as Hersen and Bellack (1976a) documented, the future looked very positive, as reflected in major journals inaugurated in the United States between 1968 and 1970 (e.g., *Journal of Applied Behavior Analysis, Behavior Therapy, Journal of Behavior Therapy and Experimental Psychiatry*). Moreover, several other behaviorally oriented journals emerged (e.g., *Cognitive Therapy and Research, Biofeedback and Self-control*), and specific journals devoted primarily to behavioral assessment (e.g., *Behavioral Assessment, Journal of Behavioral Assessment*) were formed, although they were eventually changed.

CONCEPTUAL MODELS OF HUMAN BEHAVIOR: IMPLICATIONS FOR CHILD THERAPY

An extraordinary amount of theory and research has been generated that has a bearing on child therapy (Morris & Kratochwill, in press). As a result, tremendous amounts of data have been accumulated concerning the origins, development, influences, and variations in human behavior. Nevertheless, the wealth of information has clearly not resulted in any integrated view of human performance. Indeed, the current state of knowledge generated from the various conceptual models has not only resulted in the lack of an integrated view of human functioning, but has yielded various conceptual positions that are diametrically opposed and has spawned debate in the evidence-based practice movement (Kratochwill & Shernoff, 2004).

Because our understanding of human behavior is influenced by basic assumptions concerning the "why" of behavior, assessment and treatment practices often become inextricably interwoven with the particular conceptual model of human functioning held by the psychologist. Different models, with their different perspectives on behavior, yield vastly different assessment approaches and data that are used in making decisions relative to assessment and intervention. In this section, we review some models of human behavior that influence contemporary psychological practices. The models reviewed include the medical or biogenetic model, intrapsychic disease model, psychoeducational process test-based model, and behavioral model. These various models have been discussed by others in the professional literature, and due to space limitations we are not able to discuss them in detail or cover other

models (see Kratochwill & Morris, 1993, for coverage of other models). The models differ in their conceptualization of deviant behavior and assessment procedures and devices (sometimes), as well as the nature of the intervention implemented. Each model is discussed in the context of various components and considerations in its use.

Medical or Biological Model

Components

The medical model is one of the oldest approaches guiding assessment and treatment. The medical model can be applied in either a literal or a metaphorical context (Phillips, Draguns, & Bartlett, 1975). We view the model in its literal sense. That is, abnormal biological systems can be traced to some underlying biological pathology which is then treated. For example, defective hearing (symptom) may be traced to some type of infection (the cause), which may be treated with antibiotics. The prevalence of medical problems in children is actually quite high (see Bear & Minke, 2006, for an overview of various problems). A variety of health problems may be found in children in the school setting, including those children who are chronically ill and those with nutritional disorders (undernutrition, obesity), hearing and visual disorders, dental problems, disorders of bones and joints, infectious disorders, respiratory disorders, allergic disorders, urinary disorders, and drug-related problems. It seems clear that a medical model is appropriate to deal with the diversity of medical problems in the schools.

The medical model is a disease-based model. The pathology is assumed to be within the individual, although the causes may be environmental. Some theorists consider biological deviations to be the necessary and sufficient factors in the development of the pathology; others claim that chemical or neurological anomalies are the necessary but not sufficient condition for pathogenesis. Here, environmental conditions may or may not catalyze a constitutional predisposition to pathology.

Considerations

Medical model procedures are clearly justifiable when there is no basis for assuming physiological change in the organism as a result of the sociocultural environment. Controversial practices characterize medical model procedures when they are used to interpret measures of learned behavior (e.g., various forms of disruptive behavior in children, academic skill deficits). Although genetic, developmental, neurological, and biochemical factors all undoubtedly influence behavior, in reality these factors are not discrete entities. They are interwoven with one another as well as with environmental factors.

Applications of the medical model influence assessment and treatment in various ways. Organic factors may not always be the cause of an observed medical or physical problem. There is growing recognition that psychological factors may affect a physical condition and that physical symptoms may have no known organic or physiological basis. In the past, various concepts such as “psychosomatic” or

"psychophysiological" have been used to describe the psychological basis for physical or somatic disorders. However, such perspectives may be of limited usefulness because they imply a simplistic relation between psychological factors and a distinct group of physical disorders when, in fact, there may be a complex interaction of biological, environmental, psychological, and social factors contributing to various physical disorders (Siegel, 1983). Long ago, Lipowski (1977, p. 234) noted:

> The concept of psychogenesis of organic disease . . . is no longer tenable and has given way to the multiplicity of all disease . . . the relative contribution of these factors [social and psychological] varies from disease to disease, from person to person, and from one episode of the same disease in the same person to another episode. . . . If the foregoing arguments are accepted then it becomes clear that to distinguish a class of disorders as "psychosomatic disorders" and to propound generalizations about psychosomatic patients is misleading and redundant. Concepts of single causes and cirilinear causal sequences for example, from psyche to soma and vise versa, are simplistic and obsolete.

The point here is that even in the treatment of physical disease, psychological factors may be involved. Exclusive reliance on medical (drug) interventions may bias treatment in the sense that psychological (or other) aspects of functioning may be ignored. Problems most often arise when behavioral measures that can be influenced by a variety of environmental circumstances are used to assess the potential organic origins of a perceived symptom. The more the individual differences observed on a behavioral measure are influenced by environmental factors, the more the measure has the potential of being biased. Such a circumstance may arise when the environmental factors that influence the measure differ across cultural groups.

Psychodynamic Model

Components

The psychodynamic model proposes that maladaptive behaviors are symptoms resulting from underlying processes analogous to disease in the literal sense. This model is sometimes labeled the medical model in psychological and psychoeducational practice. Because conceptualization and treatment of abnormal behavior initially resided largely within the domain of medicine, the medical model was extended to treatment of abnormal behavior, both medical and psychological. The historical developments of the model are not reviewed in detail here; instead, the reader is referred to several historical sources that discuss this approach (e.g., Alexander & Selesnick, 1968; Kraepelin, 1962).

The psychodynamic approach can be characterized by the following:

> (a) uses a number of procedures, (b) intended to tap various areas of psychological functioning, (c) both at a conscious and unconscious level, (d) using projective techniques as well as more objective and standardized tests, (e) in both cases, interpretation may rest on symbolic signs as well as scorable responses, (f) with the goal of describing individuals in personological rather than normative terms. (Korchin & Schuldberg, 1981, p. 1147)

As is evident in this characterization, the psychodynamic approach is aimed at providing a multifaceted description and inferences. This process is said to promote a unique and individual approach to child assessment.

The psychoanalytic model represents one example of the psychodynamic disease paradigm, as do many other dynamic models of human functioning. The dynamic approach to assessment of deviant behavior is best elucidated in the context of assumptions held about the internal dynamics of personality (W. Mischel, 1968). Traditionally, dynamic approaches inferred some underlying constructs that account for consistency in behavior. Assessment is viewed as a means of identifying some sign of these hypothetical constructs, which are of central importance in predicting behavior. This *indirect sign paradigm* in assessment (cf. W. Mischel, 1972, p. 319) includes the use of a large variety of projective tests (e.g., Rorschach, TAT, Human Figure Drawings, Sentence Completion Tests) as well as objective personality inventories (e.g., MMPI-A for adolescents, MMPI-II, California Psychological Inventory) that are still used in contemporary practice.

A second feature of the traditional psychodynamic approach is that it assumes that behavior will remain stable regardless of the specific environmental or situational context. In this regard, test content is of less concern and may even be disguised by making items ambiguous, as is true in projective testing (Goldfried & Linehan, 1977). Indeed, a particular response to a projective test is rarely examined in view of the overt qualities of the situation in which the test occurred, but rather is interpreted in the context of a complex theoretical structure.

Considerations

The dynamic approach to assessment can be criticized on several grounds. One problem is the preoccupation with historical events, often in the absence of any verifying data. The second criticism relates to the emphasis during assessment on the individual's presumed unconscious beliefs, attitudes, motivations, and so forth, as interpreted through projections. Third, behavior is assumed to be a consequence of internalized pathological features. This assumption ignores evidence showing that many behaviors are situationally specific.

The use of various psychodynamic indirect measurement procedures has direct implications for child assessment and treatment. These measures continue to be used in clinical practice despite data indicating their low predictive validity (cf. Hersen & Barlow, 1976). When these issues became important, Goldfried and Kent (1972) noted that although the interpretation of certain signs on the Bender-Gestalt test (Hutt & Briskin, 1960) had no empirical support (cf. Goldfried & Ingling, 1964; Hutt, 1968), the revised version of the Bender-Gestalt manual presumably discounted these research findings and still recommended the use of questionable interpretations. A rather extensive literature on the comprehensive (predictive) validity of indirect measurement techniques (T. Mischel, 1971; W. Mischel, 1968) suggested that the predictions made on the basis of self-reports were equal to or superior to those made on the basis of indirect measurement techniques that are interpreted and scored by clinical experts. These findings held true for a wide variety of content areas (cf. W. Mischel, 1972).

Perhaps the most important issue that has been raised over traditional dynamic assessment is its relation to treatment. A number of authors over the years have noted that there was little relation between traditional assessment and treatment (Bandura, 1969; Goldfried & Pomeranz, 1968; Kanfer & Phillips, 1970; Peterson, 1968; Stuart, 1970). Thus, although traditional dynamic assessment may lead to a diagnosis that may in turn lead to the recommendation of a particular treatment, diagnoses resulting from traditional assessment methods cannot accurately predict what particular treatment mode should be implemented (Ciminero, Calhoun, & Adams, 1977; Stuart, 1970).

Psychometric Test-Based Model

The psychometric test-based model bears similarity to the psychodynamic disease model in that underlying processes, specifically process deficits, are said to account for learning and behavior problems. In many respects, this model can be considered a part of the dynamic model; however, a psychometric approach is characterized by the use of a variety of individual and group tests to compare individuals along various trait or construct dimensions. In trait theory approaches, various personality structures are said to account for an individual's behavior (W. Mischel, 1968, 1974). Trait theorists disagreed on what traits explained certain patterns of behavior, but generally agreed that certain behaviors were consistent across time and settings and that these patterns are expressions or signs of underlying traits.

In contrast to the psychodynamic position, trait assessors traditionally placed a high premium on objective administration and scoring of tests. Attempts usually were made to establish formal reliability and validity of the various measures used. On empirical grounds, historically this statistical approach proved generally superior to the more clinical method in predicting behavior (cf. Korchin & Schuldberg, 1981), but questions have been raised over the manner in which the research reflects the reality of decision making in actual clinical practice.

Closely related to the psychometric approach is the psychoeducational process model used by many practicing school psychologists. The model can be considered analogous to the psychometric trait model in that assessment focuses on internal deficits, except its context is psychoeducational rather than personality or emotionally oriented. Because a variety of cognitive, perceptual, psycholinguistic, psychomotor, and neuropsychological processes or abilities have been cited as causes of children's academic failure, norm-referenced cognitive (e.g., WISC, McCarthy, Stanford-Binet), perceptual (Bender Visual Motor Gestalt Test, Developmental Test of Visual Perception, Developmental Test of Visual-Motor Integration), psycholinguistic (e.g., Illinois Test of Psycholinguistic Abilities), and psychomotor (e.g., Purdue Perceptual-Motor Survey) tests are used to assess these abilities.

Most of these assessment procedures follow a diagnostic-predescriptive approach. Ysseldyke and Mirkin (1982, p. 398) noted:

> All of the diagnostic-perspective approaches based on a process dysfunction viewpoint of the nature of exceptionality operate similarly. When students experience academic difficulties it is presumed that the difficulties are caused by inner process dysfunctions or

disorders. Tests are administered in an effort to identify the specific nature of the within-child disorder that is creating or contributing to learning difficulties. Disorders or deficits are test named (e.g., figure-ground deficiencies, auditory sequential memory deficits, body image problems, eye-hand coordination difficulties, visual association dysfunctions, and manual expression disorders). Specific interventions are developed to "cure" the underlying causative problems.

Considerations

There are several important implications that can be raised with regard to the assessment tactics used in the psychometric model. First, because norm-referenced devices are commonly used in the model, the clinician must assume that clients tested have background and acculturation comparable to those on whom the test was standardized (cf. Oakland & Matuszek, 1977). Yet the point was frequently raised that standardized tests are biased and unfair to individuals from cultural and socioeconomic minorities because they reflect predominantly White, middle-class values and do not reflect the experiences and the linguistic, cognitive, and other cultural values and styles of minority individuals (Laosa, 1977). For example, although the norms for some tests (e.g., some group achievement and aptitude tests, the 1972 Stanford-Binet, WISC-IV) were generally good, norming on other instruments (e.g., Illinois Test of Psycholinguistic Abilities [ITPA], Leiter International Performance Scale, Slosson Intelligence Test) was quite inadequate.

A second issue is that research examining components of reliability and validity on various process measures has not been optimistic (Salvia & Ysseldyke, 1978; Ysseldyke & Salvia, 1974). For example, several early reviews of research on the ITPA (e.g., Bateman, 1965; Buros, 1972; Sedlak & Weener, 1973) drew attention to these limitations. The magnitude of the problems of inadequate norming, inadequate or incomplete reliability data, or questionable validity was nicely represented in data presented by Salvia and Ysseldyke (1978). Clearly, the potential for biased assessment practice is high given the poor psychometric properties of these instruments.

Behavioral Model

Components

Technically, there is no one model of behavior therapy, and contemporary behavior therapy, despite commonalities, is characterized by a great deal of diversity.* Historically, the different approaches in behavior therapy include applied behavior analysis (e.g., Baer, Wolf, & Risley, 1968; Bijou, 1970), mediational stimulus-response model (e.g., Rachman, 1963; Wolpe, 1958), social learning theory (e.g., Bandura, 1969, 1977b), and cognitive-behavior modification (e.g., Meichenbaum, 1974, 1977; Mahoney, 1974; Mahoney & Arnkoff, 1978.)**

*A detailed account of the history of behavior modification can be found in Kazdin (1978).

**These approaches are only briefly reviewed here. The reader is referred to Kazdin and Wilson (1978) as well as original sources within each approach for a more detailed presentation. The reader is also referred to Kratochwill (1982) and Kratochwill and Bijou (1987).

Applied Behavior Analysis

This form of behavior therapy developed from the experimental analysis of behavior (cf. Day, 1976; Ferster & Skinner, 1957; Sidman, 1960; Skinner, 1945, 1953, 1957, 1969, 1974). It emphasized the analysis of the effects of independent events (variables) on the probability of specific behaviors (responses). Contemporary applied behavior analysis focuses on behaviors that are clinically or socially relevant (e.g., various social behaviors, learning disorders, mental retardation, social skills) and adheres to certain methodological criteria (e.g., experimental analysis, observer agreement on response measures, generalization of therapeutic effects).

Advocates of applied behavior analysis use a more restrictive sense of the term "behavior" than other areas represented in the field of behavior therapy: behavior as the overt activity of an organism. Internal feelings and cognitions are typically not considered a major focus for the techniques of therapy, research, and practice. However, it must be stressed that applied behavior analysis focuses on the behavior of an individual as a total functioning organism, although there is not always an attempt to observe, measure, and relate an organism's entire response taking place at one time (Bijou, 1976; Bijou & Baer, 1978).

Many intervention procedures associated with applied behavior analysis are derived from basic laboratory operant research (e.g., positive and negative reinforcement, punishment, time-out, response cost, shaping, fading stimulus control; see Bijou, 1976; Gelfand & Hartmann, 1975; Kazdin, 1980; Sulzer-Azaroff & Mayer, 1977). Assessment emphasizes the individual application of these procedures and a functional evaluation of their effectiveness (Bijou & Grimm, 1975; Emery & Marholin, 1977). Behavior analysis refers to the study of organism-environment interactions in terms of empirical concepts and laws for understanding, predicting, and controlling organism behavior and repeated measurement of well-defined and clearly observable responses (Bijou, 1976; Bijou, Peterson, & Ault, 1968; Bijou, Peterson, Harris, Allen, & Johnson, 1969).

Neobehavioristic Mediational Stimulus-Response Model

The noebehavioristic mediational stimulus-response (S-R) model is derived from the work of such learning theorists as Pavlov, Guthrie, Hull, Mower, and Miller (e.g., Eysenck, 1960, 1964; Rachman, 1963; Wolpe, 1958). These approaches are characterized by "the application of the principles of conditioning, especially classical conditioning and counter-conditioning to the treatment of abnormal behavior" (Kazdin & Wilson, 1978, p. 3). Although intervening variables and hypothetical constructs play a role in assessment and intervention, covert activities are most commonly defined in terms of a chain of S-R reactions, with cognitive formulations de-emphasized.

A number of treatment procedures such as counterconditioning and systematic desensitization have been used to treat anxiety reactions, phobic patterns, and other strong emotional disorders in children (Morris & Kratochwill, 1983; Morris, Kratochwill, Schoenfield, & Auster, in press). Systematic desensitization, based originally on the principle of reciprocal inhibition (Wolpe, 1958), has been

successfully used to treat a wide range of child and adult problem behaviors (see Morris & Kratochwill, 1983; Morris et al., in press). Assessment in the mediational S-R model relies on survey schedules (e.g., fear survey schedules), self-support data, and direct measures of client behavior (as in the use of behavioral avoidance tests).

Cognitive-Behavior Therapy

A unifying characteristic of the cognitive-behavior therapy approach was an emphasis on cognitive processes and private events as mediators of behavior change (Kendall, 1981b). The source of a client's problems were said to be related to the client's own interpretations and attributions of his or her behavior, thoughts, images, self-statements, and related processes (Kazdin & Wilson, 1978). Contemporary cognitive-behavior therapy emanates from Ellis's (1962) rational-emotive therapy, Beck's cognitive therapy, and Meichenbaum's self-instructional training. Treatment strategies are quite diverse and include such techniques as problem solving, stress inoculation, self-instructional training, coping skills training, language behavior therapy, thought stopping, and attribution therapy. These techniques represent procedures not generally addressed by other behavior therapy approaches (e.g., applied behavior analysis).

Assessment in cognitive-behavior therapy tended to be quite broad-based, taking into account many different dimensions of behavior. Yet there was still an emphasis on defining the nature of the target problem, whether overt or covert. In some cases, a more traditional functional analysis of behavior, which emphasizes a careful examination of environmental antecedents and consequents as related to a certain response repertoire, was recommended (e.g., Meichenbaum, 1977).

Some specific purposes for cognitive assessment were outlined by Kendall (1981a, pp. 3–4):

1. To study the relationships among covert phenomena and their relationship to patterns of behavior and expressions of emotion.
2. To study the role of covert processes in the development of distinct psychopathologies and the behavioral patterns associated with coping.
3. To confirm the effects of treatment.
4. To check studies where cognitive factors have either been manipulated or implicated in the effects of the manipulation.

Social Learning Theory

Social learning theory is based on the original work of Bandura and his associates (e.g., Bandura, 1969, 1971, 1977b; Bandura & Walters, 1963) and has evolved considerably over time. Bandura noted that in addition to outcome expectation, a person's sense of his or her ability to perform a certain behavior mediates performance. He referred to these latter expectations as efficacy expectations or self-efficacy, and suggested that they have important implications for treatment. Psychological treatment and methods were hypothesized to produce changes in a child's expectations of self-efficacy, as in the treatment of phobic behavior. Self-efficacy was said to determine the activation and maintenance of behavior strategies for coping with

anxiety-eliciting situations. Self-efficacy expectations were also said to be modified by different sources of psychological influence, including performance-based feedback (e.g., participant modeling), vicarious information (e.g., symbolic modeling), and physiological changes (e.g., traditional verbal psychotherapy; cf. Kazdin & Wilson, 1978). Intervention procedures such as symbolic modeling (e.g., Bandura, 1971), direct modeling (Bandura, 1977b; Rosenthal, 1976), and self-modeling (Brody & Brody, 1977) have been associated with the social learning theory approach. For example, modeling historically has been used to treat a variety of children's fears (e.g., animal fears, inanimate fears, dental and medical fears; Morris & Kratochwill, 1983; Morris et al., in press), socially maladjusted children (e.g., social withdrawal, aggression), distractibility, and severe deficiencies (e.g., autism, mental retardation) as well as a wide range of academic behaviors (cf. Zimmerman, 1977). In all these approaches, social learning theory stressed that human psychological functioning involved a reciprocal interaction between the individual's behavior and the environment in that a client is considered both the agent as well as the target of environmental influence, with assessment focusing on both dimensions of behavior.

Unifying Characteristics

Despite apparent diversity among the different areas within behavior therapy, several dimensions set it apart from traditional forms of psychological assessment and treatment, particularly the test-based psychometric models and psychodynamic models (Hartmann, Roper, & Bradford, 1979). Contemporary behavior therapy consists of the following characteristics:

1. A strong commitment to empirical evaluation of treatment and intervention techniques;
2. A general belief that therapeutic experiences must provide opportunities to learn adaptive or prosocial behavior;
3. Specification of treatment in operational and, hence, replicable terms, and
4. Evaluation of treatment effects through multiple-response modalities, with particular emphasis on overt behavior. (Kazdin & Hersen, 1980, p. 287)

Behavior therapy has become very diverse over time and includes a number of therapeutic strategies that were excluded from the field during the early years (e.g., rational-emotive therapy). Although these characteristics are tied to the therapeutic aspects of the behavioral approach, each can also be conceptually representative of the behavioral approach to child treatment (Bergan & Kratochwill, 1990).

HISTORICAL CONTEXT OF CHILD THERAPY

The Early Years

As we noted previously, although the study of psychological problems among adults can be traced back as far as the ancient Greeks (if not further), the study

of child psychopathology represents a relatively recent phenomenon in the history of psychological treatment. The most famous early example of such study is Jean Itard's examination of and attempt to educate the Wild Boy of Aveyron beginning in 1799. However, an increased focus on problems of childhood would have to follow a greater focus on childhood as a distinct period in human development and an increased interest in the nature of psychological problems in children. The emergence of compulsory education marks an important milestone in the development of such interest, in part due to the large number of children who struggled in school.

The study of children in the United States was largely founded by G. Stanley Hall, whose work spanned 1883 through 1918 (Davidson & Benjamin, 1987). Hall focused on problems with education, and although he was not primarily concerned with psychopathology, his work helped promote a better understanding of the problems of childhood.

Clifford Beers played an important role in reversing the attitude toward the treatment of the mentally ill generally, and later this focus was applied to children. Beers became clinically depressed and suicidal as a Yale University law student. After his hospitalization, he wrote *A Mind That Found Itself* (1908), a book describing the poor treatment he received while institutionalized. The success of the book helped inform the public of the terrible conditions in state hospitals. Beers would form a number of organizations, including the Connecticut Society for Mental Hygiene (in 1908) and the National Committee for Mental Hygiene (in 1909), which promoted better conditions in state hospitals, better treatment methods, and research on the prevention and treatment of psychopathologies. This increased concern for the treatment of persons with mental illness brought about "mental hygiene" programs in schools (Morris & Kratochwill, in press-b).

Lightner Witmer is credited as the first clinical psychologist, although he had a strong interest in schools and has an APA Division 16 (School Psychology) award named after him. Witmer established the first psychological clinic in the United States, at the University of Pennsylvania, in 1896. His first client was a 14-year-old boy who had difficulty spelling (Benjamin, 2005). In the first year of the clinic's operation, Witmer and his students saw a total of 24 children between the ages of 3 and 16 years, who had such presenting problems as "learning difficulties, speech problems, and possibly chorea, hydrocephalus, and hyperactivity" (P. McReynolds, 1996, p. 238). Witmer published the first journal devoted to clinical psychology, titled *The Psychological Clinic*, in 1907 (Routh, 1996; Witmer, 1907/1996). The focus on children with problems in schools would eventually lead to the development of a specialty of psychology that involved practice in schools.

In 1909, the Cook County Juvenile Psychopathic Institute in Chicago was formed under the leadership of William Healy. The institute staff worked directly with juvenile offenders and stressed an interdisciplinary approach to studying juvenile issues (e.g., psychiatrists, psychologists, and social workers worked together on particular cases, taking into account multiple causes and perspectives). Aided by Beers's National Committee for Mental Hygiene, numerous child guidance clinics also developed across the country over the next several decades, and by 1930 there were about 500 child guidance clinics in the United States (Kanner, 1948).

Through the introduction of dynamic psychiatry and psychoanalysis during the early twentieth century, specifically the work of Sigmund Freud in Vienna and Adolph Meyer in the United States, the psychodynamic approach maintained that the origins of behavior problems lay in the past experiences (typically, childhood) of the person (Kanner, 1948), thereby turning attention to childhood. For example, with adult patients, these experiences were explored retrospectively by the psychiatrist or psychoanalyst, who attempted to draw causal relationships between these past experiences and the patients' present behavior. During the early years of the dynamic psychiatry movement, children were usually not seen in treatment by psychiatrists (Kanner, 1948). Nevertheless, the retrospective search for the relationship between early childhood events and present functioning aroused sufficient interest among professionals that some began to acquaint themselves specifically with the behavior problems of children as well as with the dynamics that contributed to their difficulties (Morris & Kratochwill, in press-a). This psychodynamic interest in children was not formally realized in the literature until the publication in 1909 of Freud's (1909/1963) detailed case of "Little Hans." Interestingly, although Freud formulated his etiological theory of phobias on the basis of Hans's symptoms and experiences, he did not treat Hans directly; Hans's father treated him under Freud's direction and supervision (Morris & Kratochwill, 1983). Although Hans's problem was treated successfully, it was not until at least 15 to 20 years later that Freudian psychoanalytic child therapy came into existence. This was largely due to the contributions and adaptations of Freud's work for children by Melanie Klein, Freud's student, and his daughter, Anna Freud, and the subsequent publication beginning in 1945 of Anna Freud's multivolume edited book series (with Hans Hartmann and Ernst Kris), *The Psychoanalytic Study of the Child*. The changes made by these women, as well as the earlier psychoanalytic therapy work with children first initiated by Hermine Hug-Hellmuth (the third woman to join Freud's Vienna Psychoanalytic Society during the 1900s), made Freud's therapeutic approach very relevant to children and contributed to its increasing popularity in the twentieth century and later influence in the development of many other forms of child and adolescent psychotherapy (Benveniste, 1998).

Some of the major methodological changes that were made for children in Freud's psychoanalysis were the substitution of play activities for the technique of free association and the use of drawings and dreams to understand a child's problems. For example, through the use of the medium of play, both Klein and Anna Freud discovered that children were able to represent to the child analyst their inner conflicts and perceptions of important relationships in their lives, as well as portray in their play their unique feelings about, perceptions of, and concerns about their various pleasurable and traumatic experiences (Morris, Li, Lizardi-Sanchez, & Morris, 2002; Warshaw, 1997).

Morris and Kratochwill (in press-b) identified some additional developments that influenced the focus of child therapy. First, as we noted previously, the intelligence testing movement had a tremendous influence on the study of children; in particular, it became possible to learn the extent to which a particular child differed from the norm in cognitive ability. It also demonstrated clearly the diversity of children in

terms of their comprehension of classroom instruction (Kanner, 1948) and spawned a decade of debate on bias in testing and eventually, in the twenty-first century, federal regulations that would expand the focus of special services for children in schools (Kratochwill et al., in press).

Second, the formation of professional associations contributed to the increasing emphasis on the treatment of children. The first professional association was the Association of Medical Officers of American Institutions for Idiots and Feeble-minded Persons, founded in 1876, which evolved into the present-day American Association on Mental Retardation. Its first president was Edward Seguin. In 1892, another professional association was formed, the American Psychological Association, in which Witmer was a charter member. G. Stanley Hall was the first president of APA, and James Cattell was one of the first members of the Council (Board of Directors of APA) and the fourth president of APA. In 1922, the Council for Exceptional Children was formed; it consisted primarily of educators and other professionals, although parents were members too. The fourth early association was the American Orthopsychiatric Association, founded in 1924 and consisting primarily of psychiatrists, applied psychologists, and social workers, although educators as well as other professionals and parents were also members. Each of these groups encouraged the formulation and conduct of research with children having behavior and learning disorders, as well as the sharing of information regarding effective psychological and/or educational interventions.

Third, a treatment approach emerged that was initially much less popular than psychoanalytic child therapy. This approach was behaviorism and later became known as the behavior modification or behavior therapy movement (Kazdin, 1978; Morris, 1985). As we noted earlier, behavior modification emerged largely from the experimental psychology laboratory rather than from direct interaction with patients and was based on theories concerning how people and animals learn to behave through S-R learning and conditioning rather than through the conscious or unconscious thinking found in psychoanalytic writings. The two most famous behaviorists associated with this movement are John B. Watson (1913, 1919), often referred to as the "father of behaviorism," and B. F. Skinner (1938, 1953), the behavioral researcher and theorist who extended Watson's behaviorist views and developed a learning paradigm that Skinner referred to as "operant conditioning." The behavior modification and behavior therapy procedures that were derived from various learning theories were largely confined for many years to research settings. In fact, it was not until the mid-1960s to the late 1970s that these procedures began being applied on a regular basis in children's residential treatment settings, regular and special education classrooms, and outpatient mental health settings (Morris, 1985; Morris et al., 2002).

Developments in behavior modification set the stage for the evidence-based practice movement, when many of the procedures were implemented in research that offered strong support for effective outcomes, first with adults and eventually with children. Behaviorism also set the stage for prevention models in educational settings with particular emphasis on such models as positive behavior support, a movement with its roots in applied behavior analysis (Crone & Horner, 2003; Crone, Horner, & Hawken, 2004).

DEVELOPMENT OF THE EVIDENCE-BASED INTERVENTION AND PRACTICE MOVEMENT

As has been evident in the previous sections of this chapter, a number of different events shaped the future of psychological assessment and treatment of children. As the need for services for children and families increased, attention turned to the training of professionals in psychology. By the 1940s, considerable attention had been given to the need for a training model in clinical psychology. Up to this point, a great variety of training curricula existed in training programs, and training was often rushed and inadequate. Some programs produced statistically sophisticated therapists, and others focused on more service-oriented practice (Hayes et al., 1999). The issue had been present at least since 1918, when Leta Hollingworth argued that a training model was critical in the development of clinical psychology as a profession (Benjamin, 2005). However, there was little resolution until 1949, when 73 professionals met in Boulder, Colorado, for 15 days to discuss the training of clinical psychologists. The result became known as the scientist-practitioner model (or the Boulder model) and would profoundly influence professional psychology in subsequent years and, eventually, the evidence-based practice movement.

Training Psychologists: Advent of the Evidence-Based Practice Focus

The events leading up to the Boulder Conference were perhaps as rooted in social trends as they were in professional trends. By the 1940s, there was growing concern for mental health in the United States. The year 1944 saw written instruments and assessments used on close to 20 million persons (Reisman, 1976). World War I produced an abundance of psychological casualties for whom too few professionals could provide adequate care. In the wake of World War II, as professionals took steps to improve the quality of their care, the Veterans Administration officially declared clinical psychology a health care profession. The next major step was to improve and homogenize the training of clinical psychologists. Pressure to do so began to come from a variety of sources.

In 1941, the American Association for Applied Psychology (AAAP) endorsed a day-long conference for the development of a committee for improved training of clinical psychologists. The subsequent report, largely focused on training for the treatment of psychopathology, supported a 4-year PhD program in psychology: a year for systematic foundation in psychology, a year for psychometric and therapeutic principles and practice, a year for internship, and a final year for dissertation (Shakow, 1942). Reactions were largely positive, and the AAAP formed the Committee on Training in Clinical Psychology (CTCP). The integration of the AAAP with the APA in 1944 prevented prompt action, but the seeds were planted for the organization of professional training.

With concern for the welfare of returning soldiers, an increasing shortage of psychologists in the public, and concern over the quality of professional training, the VA and the U.S. Public Health Service (USPHS) began to commit funds for the training of clinical psychologists. In 1947, the VA and USPHS requested that the

APA assist in determining ways to train greater numbers of professional psychologists (Baker & Benjamin, 2000). The APA turned to the CTCP with four requests: recommend a clinical psychology training program, develop standards for institutions training clinical psychologists, visit and study these institutions, and maintain contact with other organizations invested in these problems (e.g., the American Orthopsychiatric Association, the National Committee for Mental Hygiene; APA, 1947). The chair of the CTCP, David Shakow, sent committee members a copy of a report titled "Graduate Internship Training in Psychology" (Shakow et al., 1945) and asked that it be critiqued. The Shakow Report, as it would become known, garnered significant support as the most comprehensive set of clinical psychology training recommendations to date.

In 1948 and 1949, the CTCP visited a number of training programs and awarded accreditation to 43 of them (APA, 1949). Although the committee was largely impressed with the efforts put forth by these programs, significant concerns were raised. First, it was contended that the focus of training was too narrow. There was certainly a social need for emphasis on psychiatric hospitals and the severely disturbed patients, but the committee argued that clinical psychologists should be trained to deal with other social issues. Second, many programs taught a variety of clinical techniques, placing limited emphasis on research methodology and theory—an issue that would become central to the Boulder Conference later that year (Baker & Benjamin, 2000).

When the Boulder Conference began, the adaptation of the scientist-practitioner model was not entirely agreed on. There was great concern that it would be difficult to train all graduate students in both research and practice. The role of research in training likely was the most difficult topic covered in the conference. However, by the end of the conference, agreement was reached among the large majority. In the most comprehensive report of the conference, titled *Training in Clinical Psychology,* Victor Raimy (1950, p. 23) reported, "The original chasm seemed to have largely disappeared and recognition of the importance of including research training in the preparation of all clinical psychologists was generally accepted." In fact, the decision to recommend training in research and practice did not require a compromise, but instead was nearly a unanimous consensus.

Recommendation for preparation in both areas rested on five major considerations. First, although most students would focus their career on either research or practice (not both), all students should be trained in both to encourage "cross fertilization and breadth of approach" (Raimy, 1950, p. 81) within the profession. That is, dual training would benefit both foci through collaboration; in addition, the study of research and practice would deter programs' tendencies to produce narrow thought processes and subsequent actions. Second, the extreme lack of dependable knowledge in the field demanded (and still demands) a need for research in clinical psychology. As Raimy reports, "Participants at the conference displayed considerable humility with respect to present techniques" (p. 80). Third, it was believed that students of clinical psychology should be capable of carrying out both roles of clinician and researcher. The number of applicants greatly surpassed the number of available positions in training programs; therefore, participants in the conference concluded that applicants should be considered for acceptance only if they showed

promise in both areas of study. Fourth, it was noted that researchers would have a better knowledge of important research topics if they were directly involved with the clinical process. Fifth, services provided by clinical practice could provide financial support for research causes. This has become even truer in the current age of managed care. Increasingly, health care systems rely on evidence-based practices for success in the marketplace (Hayes et al., 1999).

The following summarizes the roles of clinical psychologists as concluded by participants at the Boulder Conference:

> The basic needs of society for the services of clinical psychology are of two major kinds:
>
> a. Professional services to:
> (1) Individuals through corrective and remedial work as well as diagnostic and therapeutic practices
> (2) Groups and social institutions needing positive mental hygiene programs in the interest of better community health
> (3) Students in training, members of other professions, and the public through systematic education and the general dissemination of information
> b. Research contributions designed to:
> (1) Develop better understanding of human behavior
> (2) Improve the accuracy and reliability of diagnostic procedures
> (3) Develop more efficient methods of treatment
> (4) Develop methods of promoting mental hygiene and preventing maladjustment. (Raimy, 1950, pp. 20–21)

Though the decision to endorse the scientist-practitioner model was clear, participants recognized that there would be great difficulties involved in combining both roles. In this respect, participants in the Boulder Conference clearly anticipated the scientist-practitioner gap. A major criticism of the scientist-practitioner model developed at the Boulder Conference is its demonstrated failure to involve clinicians in research. The model has been much more universally adopted in research than in practice. The average PhD psychologist has published very few, if any, research articles.

Recent Developments in the Evidence-Based Intervention and Practice Movement

Despite great momentum in research in professional psychology, the evidence base for treatment of a variety of childhood disorders and problems lagged behind outcome research with adults. Nevertheless, as attention turned to mental health issues in children, psychologists began to review the literature on prevention and intervention (therapy) outcome research. To perform this task they needed guidelines to review the growing literature on prevention and intervention. Over the past decade criteria have been produced for developing evidence-based interventions and guidelines for how such interventions should be evaluated (see Barlow, 2004; Kratochwill & Stoiber, 2002). With pressure mounting from managed health care, governmental agencies, and professional organizations, a major task force on

evidence-based interventions was formed by the APA Task Force on Promotion and Dissemination of Psychological Procedures (1995). The report stimulated considerable debate. The major controversial themes were identified: (a) The focus of research studies identified as evidence-based in the adult literature were identified as behaviorally oriented; (b) the criteria used were judged as potentially biased (e.g., methodological and statistical decision rules in the literature reviews); (c) the use of intervention manuals to standardize interventions was deemed problematic (e.g., manuals remove flexibility and adaptation to individual or group needs in psychotherapy); and (d) the generalizability of findings was deemed limited (i.e., whether and how research studies can be generalized to actual clinical practice).

Early in the evidence-based intervention movement school psychology explored evidence-based interventions and their implications for the field. Division 16 of the APA, the Society for the Study of School Psychology, and the National Association of School Psychologists supported the development of a Task Force on Evidence-Based Interventions in Schools (now co-chaired by Thomas R. Kratochwill and Kimberly Hoagwood).

The initial purpose of the School Psychology Task Force was to examine and disseminate the knowledge based on what prevention and intervention programs or approaches for children, youth, and families demonstrate empirical support for application in the school and community and to facilitate strong research methodologies, technologies, and innovations. The Task Force worked cooperatively with the APA's Division of Clinical Child and Adolescent Psychology (Division 53).

Another major direction taken in the evidence-based practice movement has involved the development of practice guidelines (White & Kratochwill, 2005). Various practice guidelines have been constructed by a variety of professional groups: the Agency for Health Care Policy and Research (AHCPR), the American Psychiatric Association, the National Institutes of Health, and the Division of Clinical Psychology (Division 12) of the APA. The AHCPR has developed guidelines for treating and diagnosing depression in primary care. The guidelines are primarily aimed at physicians and the general public. Practice guidelines of the American Academy of Child and Adolescent Psychiatry have been evolving for some time and provide a variety of foci for child intervention, including anxiety disorders, conduct disorders, and Attention-Deficit/Hyperactivity Disorder. The Division 12 Task Force (Chambless & Ollendick, 2001) provided a preliminary set of practice guidelines based on their review of empirically validated psychosocial interventions, resulting in a proposed three-category system of efficacy: well-established interventions, probably efficacious interventions, and experimental interventions (those not yet established as at least probably efficacious).

Over the past decade the Committee on Science and Practice of the Society for Clinical Psychology of APA Division 12 developed a procedural and coding manual for identification of beneficial interventions (Weisz & Hawley, 1998). The manual provided guidelines for identification, review, and coding of studies of psychological interventions for behavioral, emotional, and adjustment problems and disorders. The manual was designed to (a) identify intervention outcome studies related to the effects of psychological interventions; (b) code studies according to the committee's criteria and provide related information on characteristics of interventions

in the studies that evaluate efficacy; and (c) make a determination on whether the intervention is beneficial. The procedural and coding manual extended the work of the Division 12 Task Force and the efforts of the Division 53 Task Force by clarifying what was meant by "empirically supported interventions" and expanded on a variety of published literature and the work of various committee members.

As various professional groups and researchers began to review the literature, narrative reviews and the application of meta-analysis to child psychotherapy outcome contributed to an evolving positive landscape for effective interventions (e.g., Lonigan, Elbert, & Johnson, 1998; Casey & Berman, 1985; Kazdin, Bass, Ayres, & Rodgers, 1990; Kazdin & Weisz, 2003; Weisz, Weiss, Alicke, & Klotz, 1987). Specifically, between 1952 and 1993, more than 300 research investigations of psychotherapy involving children between the ages of 2 and 18 years indicated that the intervention groups scored higher on various outcome measures than 76% to 81% of children in control groups (Casey & Berman, 1985; Kazdin, 2002). Hoagwood and Erwin (1997) conducted a 10-year research review and found that three types of interventions have empirical support: cognitive-behavior therapy, social skills training, and teacher consultation. Modest support for interventions delivered within school-based consultation occurred through traditional literature reviews (Kratochwill, Sheridan, & Van Someren, 1988; Mannino & Shore, 1975) and meta-analysis (Medway & Updyke, 1985; Sibley, 1986). Summaries of evidence-based treatments with children and adolescents can be found on a Division 53 website (www.effectivechildtherapy.org).

As the cost of treatment of adult and childhood disorders became clear, psychologists became vocal about the importance and effectiveness of prevention programs (e.g., Biglan, Mrazek, Carnine, & Flay, 2003; Coie et al., 1993; Kratochwill, Albers, & Shernoff, 2004; Nation et al., 2003; Tolan & Dodge, 2005; Weissberg, Kumpfer, & Seligman, 2003; Weisz, Sandler, Durlak, & Anton, 2005). Meta-analysis of prevention programs has also demonstrated positive outcomes (e.g., Durlak & Wells, 1997; Weisz, Jensen-Doss, & Hawley, 2006). To assist psychologists in implementing effective prevention programs, a website jointly sponsored by APA and the Society for Prevention Research presents information on over 100 reviews of prevention research on a variety of topics (www.oslc.org/spr/apa/summaries.html).

Tremendous advances have been made in the evidence-based practice area of child prevention and treatment. In recent years, several textbooks that emphasize evidence-based prevention and intervention for child and adolescent problems and disorders have appeared (see Table 1.1). The future looks very positive with regard to improving the quality of life of children and families who experience a wide range of academic and mental health concerns. To advance practice in the field, a number of issues have surfaced that have an important bearing on the future of this movement (see Kratochwill et al., in press). Although it is not possible to review these issues in great detail here, a sampling will convey the complexity of the tasks. To begin with, major concerns surround the research methods of establishing the evidence base in this area, with calls for expanding the methodologies of quantitative research and even the framework for empiricism of the movement (see Slife, Bradford, Wiggins, & Graham, 2005). Concerns have also been expressed about the

Table 1.1 Coverage and Number of Chapters of Specific Topics Addressed in Each Book on Evidence-Based Intervention

Topic	1[a]	2[b]	3[c]	4[d]	5[e]	6[f]
Treatment research methods	E	E	C	E	E	E
Developmental issues	NC	NC	C	E	E	E
Ethical issues	NC	NC	NC	E	NC	E
Assessment	NC	E	E	NC	E	E
Diversity	NC	NC	E	E	E	C
Future directions	E	E	E	E	E	C
Clinical disorders or topics						E
Adherence to medical regimens			1			E
Anxiety disorders	2	1	1	2	4	1
Attention Deficit Hyperactivity	2	1		1	1	1
Eating disorders		1		1		
Conduct Disorder, Oppositional Defiant Disorder						1
Depression	6	1	1	6	2	1
Depression	2	1		4	1	
Developmental disorders (e.g., language)		1				1
Encopresis			1			
Enuresis			1	1		
Habit disorders			1			
Obesity						
Obsessive-Compulsive Disorder				1	1	1
Pain management			1			
Pervasive developmental disorders (Autism)		1		2		1
Medical or physical conditions		1				1
Posttraumatic Stress Disorder					1	1
Prevention					4	1
Psychotic disorders		1				
School refusal						
Sleep problems			1			
Substance abuse		1				
Suicide		1				
Tourette's syndrome		1				
Somatic disorders						1
Child sexual abuse or Child maltreatment						1
Ethical and legal issues						1
Mental retardation and Intellectual disability						1
Psychopharmacology						1

Note: C = Mentioned but minimally; E = Emphasized; NC = Not covered.

[a] *Psychotherapy for Children and Adolescents: Evidence-Based Treatments and Case Examples,* by J. R. Weisz, 2004, Cambridge: Cambridge University Press.

[b] *What Works for Whom? A Critical Review of Treatments for Children and Adolescents*, by P. Fonagy, M. Target, D. Cottrell, J. Phillips, and Z. Kurtz, 2002, New York: Guilford Press.

[c] *Treatments that Work with Children: Empirically Supported Strategies for Managing Childhood Problems*, by E. R. Christophersen and S. L. Mortweet, 2003, Washington, DC: American Psychological Association.

[d] *Evidence-Based Psychotherapies for Children and Adolescents*, by A. E. Kazdin, & J. R. Weisz, (Eds.), 2003, New York: Guilford Press.

[e] *Handbook of Interventions that Work with Children and Adolescents: Prevention and Treatment,* by P. M. Barrett and T. H. Ollendick (Eds.), 2004, New York: Wiley.

[f] *The Practice of Child Therapy*, fourth edition, by R. J. Morris and T. R. Kratochwill, in press-a, Mawah, NJ: Erlbaum.

limitations of the focus of research, which traditionally has not given attention to the transportability of the prevention and intervention programs (e.g., Schoenwald & Hoagwood, 2001). Limitations of the sample of participants in evidence-based research and the need for cultural adaptations of interventions have also been noted (e.g., Bernal & Scharron-Del-Rio, 2001). In addition, limitations of outcome assessment strategies and, in particular, the arbitrary nature of the instruments used in evidence-based intervention research have been noted (see Blanton & Jaccard, 2006; Kazdin, 2006). Despite these challenges, major commitments to advance the quality of practice with a firm foundation of scientific support are in place (Frick, 2007; Kratochwill et al., in press). As the evidence-based movement progresses it will be important to consider the historical evidence and its importance in teaching us that diversity of viewpoints can best advance science and practice.

SUMMARY

Assessment and intervention for children and adolescents experiencing academic and behavior problems have come far from their ancient roots. In this chapter, we summarized the major trends and events that led to the current state of child and adolescent therapy. We began with a synopsis of progress prior to the nineteenth century, which set the stage for increasingly systematic investigations of human characteristics and behavior. This focus was exemplified by the work of Itard, Wundt, Galton, Cattell, and others. During the first half of the twentieth century, intelligence testing and personality assessment emerged as significant roles for psychologists. Methodological differentiation rapidly became an interest during World War I, and intelligence testing became important for the assessment of schoolchildren. In the second half of the twentieth century, we witnessed the rise of behavioral modification and assessment methods as greatly important in child and adolescent psychology. In particular, these assessment and intervention technologies set the stage for a scientific basis for therapy that would shape the evidence-based practice movement in psychology.

In addition to a discussion of trends and events, we described major conceptual models that have been developed and persist today, including the medical or biological model, the psychodynamic model, the psychometric test-based model, and the behavioral model. Implications for their use in current practices were discussed. We addressed the history of child therapy, from its most famous early roots to modern trends, including the emergence of the scientist-practitioner training model and the evidence-based practice movement. A brief discussion of major concerns surrounding evidence-based practice was provided. The future looks bright for the evidence-based practice movement as the complexities of selection, implementation, and sustainability of treatment technologies are being addressed in research and practice.

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CHAPTER 2

Clinical Training

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Clinical child and pediatric psychology might be considered adolescents in the field of professional psychology (Prinstein & Roberts, 2006). They have matured to the point of cohesive self-definitions that are accepted by the professional community and that guide training within the specialties. But these developments are relatively recent. Although interest in psychological treatment of youth dates back to the end of the nineteenth century, the need for specific training in child-related service provision was not vocalized until more than 50 years later, and formal initiatives to develop training standards and implement them consistently lagged even further behind. Fortunately, for both clinical child and pediatric psychology, adolescence has come with emergence of clear and complex identities that are grounded in the broad foundations of clinical psychology, integrated with related fields of human development and health psychology, and sensitive to the importance of context and diversity in addressing the needs of youth and their families. Adolescence has also come with increased autonomy and respect from the professional community, thanks to the passionate efforts of leaders in these areas. Thus, clinical child and pediatric psychology have become legitimate specialties within clinical psychology.

The current chapter focuses on issues involved in preparing aspiring psychologists for careers in clinical child psychology and pediatric psychology. We begin by highlighting some important foundational issues in general clinical psychology. We then discuss the emergence and definitions of clinical child and pediatric psychology and examine their relationship to other fields in psychology and associated disciplines. Next, we discuss training models and important components of clinical child and pediatric training at the graduate, internship, and postdoctoral levels. Finally, we address some of the current challenges in clinical psychology training, such as balancing broad and general training with more focused depth, defining and assessing professional competence, and identifying and addressing workforce issues.

GENERAL CLINICAL PSYCHOLOGY FOUNDATIONS

The birth of clinical psychology as a profession was really the birth of clinical *child* psychology, when, in 1896, Witmer established the first psychological clinic to treat youth with behavior and learning problems (Hothersall, 2004). Thus began

an era in which clinical psychology began to define itself, through journals devoted to clinically relevant research and practice (e.g., *Psychological Clinic; Journal of Consulting Psychology*, now published as the *Journal of Consulting and Clinical Psychology*), establishment of psychological testing and treatment clinics (e.g., the Juvenile Psychopathic Institute in Chicago and Judge Baker Foundation in Boston), a focus on training (e.g., increased availability of textbooks, university courses, and internships; development of training standards), and establishment of professional organizations such as the American Psychological Association (APA) Section on Clinical Psychology (Edelstein & Brasted, 1991; Hothersall, 2004).

Although some of the earliest work in clinical psychology focused on youth, various social forces, notably the need to evaluate large numbers of military personnel during World Wars I and II, moved substantial focus to adults. Nonetheless, as a profession, clinical psychology did not differentiate between child- and adult-focused professionals. Indeed, the 1949 Boulder Conference on graduate training in clinical psychology, on which many of our standards of graduate education are still based, emphasized the need to avoid specialization and to focus on broad training across many age levels (Raimy, 1950).

Over the last 110-plus years, the profession of clinical psychology has grown, matured, and become increasingly complex (Prinstein & Roberts, 2006). Clinical psychologists engage in activities ranging from basic and applied research to assessment and diagnosis, prevention and intervention, consultation, teaching and training, and public policy development, in settings as diverse as universities and research institutions, medical settings, schools, businesses, and government. They address the behavioral and emotional disorders, personality disorders, physical and mental disabilities, medical issues, and environmental stresses affecting individuals from infancy to late adulthood. Thus, it is not surprising that many leaders in the field (e.g., Roberts, 2006; Tuma, 1983) have questioned the wisdom, and even the ability, of continuing to pursue the type of generalist training espoused at the Boulder Conference. In essence, it is no longer feasible (if it ever was) for clinical psychologists to master the universe of relevant literature and skills required for professional competence as generalists. This has created an interesting dilemma for the profession. On the one hand, current standards for training, as operationalized by the field's accrediting body (APA Committee on Accreditation; CoA), still call for broad and general training in clinical psychology. On the other hand, providing such generalist training by itself is increasingly likely to produce graduates who are ill prepared to make positive contributions to the profession.

So how can clinical training address the challenge of providing broad foundational education in clinical psychology without drowning students in an ever-expanding sea of knowledge? There are many possible answers to this question, and given the complexity of the issues, multiple solutions almost certainly need to be implemented. But one first step in addressing the question is to acknowledge that general training in clinical psychology does not have the same meaning that it did in 1949 and to create training environments that retain the core aspects of clinical psychology while allowing focused experience in subfields or specialty areas. Interestingly, several leaders in the field (e.g., Roberts, 2006; Roberts & Sobel, 1999; Tuma, 1983) argue that is exactly what has happened in most clinical

programs, albeit somewhat covertly. Roberts and Sobel suggest that "current use of the term *clinical psychology* is today not as generally clinical as it might have been in earlier times. Today, clinical psychology typically represents only adult clinical psychology and only superficially represents a generalist clinical psychology" (p. 483). From this perspective, clinical (adult) psychology and clinical child psychology represent parallel paths to balancing the demands of breadth and focus. This *Handbook*, with separate volumes for clinical adult and clinical child psychology, overtly acknowledges the legitimacy of *broad and focused* training in adult- and child-focused clinical psychology and is an important step in advancing the field.

EMERGING CHILD CLINICAL AND PEDIATRIC IDENTITIES

During the 30 years following the Boulder Conference, several leaders in psychology education noted the existence of large numbers of unserved youth with behavioral and emotional problems or health-related issues, but a shortage of professionals with sufficient training to address the needs of these youth (Ross, 1959; Tuma, 1983; Wohlford, 1978; Wright, 1967). The fields of clinical child and pediatric psychology began to emerge, with professional organizations (e.g., APA Section on Clinical Child Psychology, now Division 53; APA Society of Pediatric Psychology, now Division 54), journals, and conferences. Graduate programs offering specialty training also started to emerge across the country, but there was little consistency in what comprised clinical child or pediatric training experiences (Roberts, Borden, Christiansen, & Lopez, 2005; Tuma, 1983). Indeed, when Tuma surveyed members of APA child, pediatric, and youth and family sections and divisions, respondents reported receiving a wide array of educational experiences to prepare them for working with youth and families, ranging from on-the-job or postgraduate training to elective coursework to more formal degree programs. Thus, leaders in clinical child and pediatric graduate training mounted several efforts to define these specialties, including training conferences, task forces, and journal sections devoted to training issues.

Hilton Head Conference

In 1985, the National Conference on Training Clinical Child Psychologists was held in Hilton Head, South Carolina. Although the need for a conference was first identified in 1977, it was almost 10 years before community need and professional training and funding priorities coalesced to allow the conference to proceed. By the time of the conference, researchers had established that the training community was falling far short of the need for well-trained clinical child psychologists, APA had recognized the need for child services and established a position in their central office to monitor child-related issues, and several child-related organizations were moving forward to establish training guidelines (Roberts, Erickson, & Tuma, 1985; Tuma, 1986). Thus, conference participants were motivated and well informed and were able to make significant progress toward addressing issues of clinical child specialty training. By the end of the 3-day conference, participants had drafted

21 recommendations, including 10 recommendations specific to graduate training, seven targeting internship, and four targeting postdoctoral training in clinical child psychology (see Table 2.1).

Interestingly, several of the graduate program recommendations focused on child-relevant training for *all* clinical psychology programs, not just those with child tracks or emphases. Specifically, the Hilton Head participants proposed that

Table 2.1 Hilton Head Recommendations: National Conference on Training Clinical Child Psychologists

Recommendations for All Clinical Training Programs
All clinical psychology programs should:
Require a course on the developmental bases of normal behavior, emphasizing a life span developmental approach.
Provide experiences with "normal" children and youth, as well as with other age groups, in diverse, "normal" settings to all graduate students.
Provide didactic and practicum training sufficient for minimal competencies in assessment, psychopathology, and intervention (e.g., treatment, consultation, prevention) with children, youth, and families.
Recommendations for Clinical Child Training Programs
Clinical child psychology programs should:
Provide didactic and practicum training sufficient for minimal competencies in assessment, psychopathology, and intervention (e.g., treatment, consultation, prevention) with adults.
Provide knowledge of and experience with children from different minority backgrounds.
Recommendations for Graduate Program Structure, Implementation, and Evaluation
APA's Committee on Accreditation should:
Ensure that accreditation site visitor teams reviewing programs that train clinical child psychologists include a member with expertise in clinical child psychology.
Clinical child psychology programs should:
Foster an appreciation of the broad and diverse contexts within which children develop and clinical child psychologists work (e.g., family, school, community).
Have at least one clinical child faculty member.
Use the Division 37 Task Force Guidelines for Training Psychologists to Work with Children, Youth and Families (Roberts, Erickson, & Tuma, 1985) as guiding principles for implementing training programs.
Endorse the Boulder scientist-practitioner model of training.
Recommendations for Internship-Level Training
Internship and graduate programs should communicate regarding:
Matching internship site characteristics with intern talents and interest and evaluating intern performance.
Development and ongoing evaluation of an individualized training plan for the intern's experience.
Child-oriented internships should:
Provide a minimum of two thirds of child-related activity training.
Within the unique constraints of the internship site, provide a broad base of experience with regard to diverse populations served (across age, gender, socioeconomic status, ethnic origin, presenting problems, theoretical orientation, and experiences).
Incorporate research into the internship by allowing time and providing role models.
Include broad-based and varied didactic and supervised training in assessment, intervention, case management, and consultation, at a minimum, including prevention if possible. Experiences should emphasize ethical considerations and evaluation of interventions.
The Section on Clinical Child Psychology should:
Create a standing committee on internship training to expand on these recommendations.

Recommendations for Postdoctoral and Continuing Education

All clinical child psychologists should:

Participate in continuing education to stay abreast of burgeoning knowledge in assessment, consultation, and prevention relevant to clinical child psychology.

The Section on Clinical Child Psychology should:

Establish a task force to create and publish criteria for continuing education offerings relevant to clinical child psychology.

Establish a task force to conduct a survey of current postgraduate programming in clinical child psychology and establish guidelines for such programming, including specification of levels and types of supervision.

Attempt to gain representation on appropriate governance structures of APA (e.g., Education and Training Committee, subcommittees on continuing education).

From *Proceedings: Conference on Training Clinical Child Psychologists*, by J. M. Tuma (Ed.), 1985, Baton Rouge, LA: Section on Clinical Child Psychology. Adapted with permission.

all clinical programs require a course on normal life span development and provide experience with normal individuals across the life span (i.e., children, youth, adults). They also proposed that all clinical programs provide coursework and experiential training to ensure minimal competencies in psychopathology, assessment, and intervention with children, youth, and families. These recommendations speak to the importance that conference participants placed on general clinical psychology training, regardless of age-related specialization. However, with the exception of required coursework in developmental bases of behavior (a requirement for accreditation), these recommendations have not been fully embraced by many clinical programs.

Recommendations focusing on clinical child psychology graduate training emphasized similar breadth, encouraging minimal competencies in work with adults, as well as with children from diverse backgrounds and in multiple contexts. Because clinical child training was so unstandardized at that time, the Hilton Head Conference also addressed issues of program structure and evaluation. In particular, they specified that programs providing clinical child training should have at least one clinical child faculty member and that accreditation teams evaluating these programs should have at least one member with expertise in clinical child psychology. These recommendations seem fairly obvious today, but illustrate the challenge of many emerging training models: that they may be implemented or judged by people without suitable backgrounds or appreciation for the important aspects of such programs. Finally, conference participants endorsed the work of other conferences and task forces, supporting the Boulder model of scientist-practitioner training, as well as the recommendations of the Division 37 Task Force for Training Psychologists to Work with Children, Youth, and Families (Roberts et al., 1985). Although the Division 37 recommendations were endorsed as flexible guidelines for program development and clinical child programs have evolved in diverse ways, the scientist-practitioner identity remains a core aspect of clinical child psychologist training.

Recommendations regarding internship and postdoctoral training were much less developed than those for graduate training but continued to stress the importance of diverse training across ages, client demographics and presenting problems, contexts, and professional duties (e.g., assessment, intervention, consultation, research). The committee's suggestions also emphasized the role of both internship

and postdoctoral fellowships as opportunities for continuing education in empirical, practical, and ethical aspects of clinical child work. Probably the most specific and unique recommendation to come out of this conference was that clinical child psychology internships should devote two thirds of the internship year to child-related experiences. Because few other specific recommendations for internship or postdoctoral training emerged from the conference, the Hilton Head conferees called for new task forces to explore clinical child internship and postdoctoral training more fully. These task forces, which would be charged with developing detailed recommendations for training experiences, emerged in the years following Hilton Head.

Post–Hilton Head: Addressing Internships and Postdoctoral Training

Between 1986 and 1988, the Section on Clinical Child Psychology commissioned two task forces to establish guidelines for clinical child psychology internships (Elbert, Abidin, Finch, Sigman, & Walker, 1988) and postdoctoral training (Ollendick, Drotar, Friedman, & Hodges, 1998). In general, these task forces reiterated the core training goals articulated at the Hilton Head Conference and subsequent internship training conferences (Belar et al., 1989). For example, they recommended that both internship and postdoctoral fellowships include opportunities to work with children, adolescents, and families across a wide range of ages (including adults), with a range of problems, from diverse backgrounds, and in diverse settings. As training experiences, both task forces also called for a range of educational activities, with internships including a combination of didactic seminars, clinical and research experiences, and supervision, and postdoctoral fellowships including approximately 50% clinical service delivery, 25% clinical research, and 25% supervision. Recommended professional activities included assessment, case conceptualization, intervention, prevention, interdisciplinary consultation, and research. The internship task force was fairly specific in the types of assessment experiences they recommended (e.g., intellectual, ability, and personality assessment, interviews, behavioral observation) and the importance of developing competencies in communicating assessment results and incorporating them into intervention planning. Both task forces emphasized that specific experiences could vary widely across training sites, and they noted the desirability of exposure to diverse treatment approaches, supervisors (including nonpsychologists), and professional experiences. They also noted the importance of addressing the increasingly complex ethical and legal issues that arise as trainees prepare for independent practice in a variety of contexts. For instance, the internship task force recommended that all interns gain awareness of the juvenile justice system, child legal rights and custody issues, and general issues in forensic psychology. In sum, these tasks forces paved the way for a sequence of training in clinical child psychology that was focused yet diverse and flexible.

Developing Pediatric Psychology Identity

At the same time that the profession was defining goals and training recommendations for clinical child psychology, psychologists who worked with children in

medical settings were also defining the roles and preparation of pediatric psychologists. These early efforts did not take place in formal training conferences or task forces, but in surveys of the field and articles published largely within the pediatric psychology community. Perhaps the most common theme of training in pediatric psychology, aside from its obvious focus on integrated attention to psychological and medical health, was its variability. For example, in the late 1960s and early 1970s, Routh (1970, 1972) described pediatric psychology practicum experiences offered by some medical school departments of pediatrics. However, as Routh's surveys suggested, clinical psychology students were not necessarily the ones who took advantage of these opportunities; pediatric psychology practicum training was just as often sought by students in school, developmental, counseling, or other areas of psychology. By the 1980s, pediatric psychology training had become aligned with clinical psychology (La Greca, Stone, & Swales, 1989; Routh & La Greca, 1990). Variability in training experiences was still evident, though. La Greca and colleagues (1989; Routh & La Greca, 1990) surveyed university graduate programs that had produced at least four graduates identified as pediatric psychologists by virtue of their membership in the Society of Pediatric Psychology. Of 41 programs surveyed, over 66% provided pediatric psychology practica, but fewer than 25% offered a course on the topic. Interestingly, only 17% of programs provided both didactic and applied pediatric experience, and 27% provided neither! Routh and La Greca noted that it is conceivable that students might gain experience in pediatric psychology during graduate training through their research and then complete more formal training during internship or postdoctoral training. This is consistent with results of an earlier survey (La Greca, Stone, Drotar, & Maddux, 1988) suggesting that graduate training in clinical and developmental psychology provided the ideal foundation for later specialization in pediatric psychology. Nonetheless, pediatric psychology training at the graduate level has continued to evolve, despite the lack of formal training guidelines. It was not until the twenty-first century that such guidelines were published (Spirito et al., 2003); we discuss these guidelines in "Curriculum Domains."

CURRENT DEFINITIONS

Because the fields of clinical child and pediatric psychology are evolving, so are their definitions, varying depending on when they were written and by whom. Although definitions of clinical psychology, clinical child psychology, and pediatric psychology abound in articles and texts, perhaps the most widely accepted definitions are those adopted by the relevant divisions of the APA. These are also typically the definitions that divisions of other organizations forward to APA's Commission for the Recognition of Specialties and Proficiencies in Professional Psychology (CRSPPP) when they seek specialty status.

Clinical Child Psychology

The current definition of clinical child psychology supported by APA Division 53 Clinical Child Psychology and approved by CRSPPP (2005) states:

> Clinical child psychology brings together the basic tenets of clinical psychology with a thorough background in child and family development. Clinical child psychologists conduct research and provide services aimed at understanding, preventing, and treating psychological, cognitive, emotional, developmental, behavior, and family problems of children across the age range from infancy through adolescence. Of particular importance to the Clinical Child Psychologist is an understanding of the basic psychological needs of children and how the family and other social contexts influence children's socioemotional adjustment, cognitive development, behavioral adaptation, and health status. There is an essential emphasis on a strong empirical research base recognizing the need for the documentation and further development of evidence-based assessments and treatments in clinical child and adolescent psychology.

Central to this definition are several themes. First, clinical child psychology's foundations are those of clinical psychology generally, including a focus on assessment, diagnosis, prediction, treatment, and prevention of psychopathology and adjustment problems, and an emphasis on scientifically based knowledge and application. Because the populations of interest range from infancy through adolescence, clinical child psychologists must have a thorough understanding of child and family development. Thus, the clinical child psychologist's view of pathology is based on a developmental psychopathology framework, understanding child maladjustment in the context of normal behavior and development (Rutter & Sroufe, 2000).

A second theme is the role of context in children's adjustment. Clinical child psychology explicitly emphasizes the notion that child adjustment and development do not occur in a vacuum, but in context. Understanding the relevant contexts is important for conceptualizing youths' emotional and behavioral problems, as well as for treating them effectively. To put it colloquially, contexts are usually part of the problem, and almost certainly part of the solution. Thus, for clinical child psychologists, treating children in context is both a luxury (that many clinical adult psychologists, who see individual clients during an office visit, do not have) and a necessity. As our brief definition implies, the family context has central importance in children's development and adjustment. This makes sense given that the family has typically been with the children since birth, observing and participating in their development over time and across multiple contexts. Thus, clinical child psychologists must have an understanding of family structures and systems, as well as of the personalities and issues of individuals who make up the family. How family members relate to one another regarding such issues as rules, boundaries, and values and how they share experiences and emotions influence the lessons that youth learn and carry into their relationships and roles throughout life (Minuchin, 1985). In addition, understanding family members' health and adjustment is helpful for contextualizing youth adjustment problems that may have a genetic basis (e.g., Moffitt, 2005).

Beyond the family, several other contexts are important. Schools, neighborhoods, peer groups, and religious and extracurricular organizations can have a large influence on social, emotional, cognitive, and behavioral adjustment, especially as children reach school age and adolescence and spend more time in school and peer settings (Lerner, 2006). Broader contextual factors, such as the child's culture of origin and racial/ethnic group, should also be considered (La Greca & Hughes, 1999;

Roberts, 1998). Ecological models, such as Bronfenbrenner's (Bronfenbrenner & Morris, 2006) model that considers the reciprocal interplay between a developing individual and the many environments in which he or she interacts as the determining factors in youth adjustment, are mainstays of clinical child approaches to understanding and intervening with children. Thus, for clinical child psychologists, it is routine to go beyond individual or family-focused assessment and treatment and to incorporate school, peer, and community settings in assessment, consultation, and intervention.

Finally, clinical child psychology endorses the view that psychological theory and practice are necessarily based on science, and further emphasizes that well-trained clinical child psychologists will be scientist-practitioners, integrating both science and application into their training and professional identity. The scientist-practitioner training model was endorsed at the Hilton Head Conference and continues to be considered a central aspect of clinical child psychology training (La Greca & Hughes, 1999; Tuma, 1985).

Pediatric Psychology

Pediatric psychology is defined as a subspecialty of clinical child psychology, building on clinical psychology and clinical child psychology foundations but integrating knowledge of child health issues (e.g., disease process, health promotion) and application in medical settings (Roberts, 2003; Spirito et al., 2003). Defining this field, the Society of Pediatric Psychology (APA Division 54) states:

> Society of Pediatric Psychology is dedicated to research and practice addressing the relationship between children's physical, cognitive, social, and emotional functioning and their physical well-being, including maintenance of health, promotion of positive health behaviors, and treatment of chronic or serious medical conditions. In recognition of the close interplay between children's psychological and physical health, SPP has developed liaisons with several pediatric societies and many pediatric psychologists have become experts in the behavioral and developmental aspects of acute and chronic diseases. (American Psychological Association, 2006c)

Integrating across these definitions, a picture emerges of pediatric psychology as *clinical child psychology plus*. The Society of Pediatric Psychology is rooted in clinical child psychology as opposed to health psychology, and as Spirito and colleagues (2003) note, pediatric psychology is frequently integrated with or synonymous with clinical child psychology, particularly at the graduate training stage. Pediatric psychologists share clinical child psychology's scientist-practitioner approach to understanding, assessing, and treating youth adjustment and a training focus that integrates clinical psychology foundations with a developmentally and contextually sensitive approach to assessment and treatment. Pediatric psychologists should be competent to provide general clinical child services to youth and families (Roberts & Steele, 2003; Spirito et al., 2003).

To these clinical child foundations, pediatric psychologists add didactic training and experience in issues specific to child health. Pediatric psychologists must have an understanding of disease process, including how illnesses and their treatments

affect psychological and behavioral functioning, as well as of disease prevention and health promotion (Spirito et al., 2003). Research and applied activities of pediatric psychologists focus, for example, on how a healthy lifestyle can prevent conditions such as obesity and diabetes, or how chronic illness affects youths' self-concept and peer relations (Roberts, 2003). Pediatric psychologists also must master skills required for effective consultation and liaison with physicians and other professionals who address child health, as they are almost always part of an interdisciplinary team. Because of the already large demands of graduate training in clinical psychology, pediatric psychology experience often occurs primarily at the internship and postdoctoral levels (Drotar, Palermo, & Ievers-Landis, 2003; Spirito et al., 2003). Indeed, a recent Division 53 survey of clinical child and pediatric graduate programs indicated that only 3 of 45 programs that provide child-relevant training have a formal pediatric psychology track (Division of Clinical Child and Adolescent Psychology, 2006).

Relationship to Other Areas of Professional Psychology

General Clinical Psychology

As noted earlier, the scope of clinical psychology has evolved since its description at the Boulder Conference; as the knowledge base has expanded, the notion of a generalist, competent across all ages and problem areas, has become increasingly unrealistic. As Roberts (2006) suggests, few clinical training programs actually provide generalist training, but instead focus on adults. Clinical child and pediatric psychology also provide focused training on a subset of the life span. However, the core aspects of clinical psychology, including a focus on understanding, assessing, preventing, and treating maladjustment in the context of established scientific methods and broader foundations in psychology, are integral parts of training in clinical child psychology. In addition, although clinical child and pediatric training focus on children and families, they also include educational background in adult development and psychopathology and the ability to work with parents and other adults who are part of the child's life. And beyond a focus on individual clients, clinical child and pediatric psychology emphasize systems and contexts and conducting research and intervention in interdisciplinary teams.

Although clinical child and pediatric psychology are more focused than generalist clinical psychology, they could be considered broader than many clinical (adult) programs (Roberts, 2006). In their discussion of clinical training in the twenty-first century, Snyder and Elliott (2005) maintain that most clinical (adult) psychology has been characterized by a focus on individuals' weaknesses, or psychopathology. They call for clinical psychology to broaden its horizons to encompass a focus on people's strengths and to consider not just person factors, but also environmental context. Interestingly, as several authors note in their responses to Snyder and Elliott's thought-provoking article, this seems to be exactly what clinical child and pediatric psychology have been doing for years (Kendall, 2005; Roberts, 2005). Thus, clinical child and pediatric psychology training models may provide a guide for clinical adult programs that seek to broaden their perspectives to encompass client strengths and contexts.

School Psychology

School psychology shares many similarities with clinical child and pediatric psychology, including a focus on assessment, intervention, prevention, and health promotion with youth and families, a scientist-practitioner approach, and sensitivity to contextual and diversity issues in addressing youth adjustment (Reynolds & Gutkin, 1999). School psychology, however, emphasizes educational and learning processes and environments. Relevant practice may extend to learners of all ages, including adults, and often takes place in an educational setting. Assessment is frequently focused on evaluation of students' intellectual, achievement, psychological, and behavioral functioning, particularly with regard to how these areas influence learning effectiveness. Common activities of school psychologists also include consultation with parents and educators, program development and evaluation of educational programs, and advocacy for education-relevant legislation. Thus, the day-to-day activities of applied clinical child, pediatric, and school psychologists may look quite similar: involving work with children and adolescents in their relevant contexts, on issues that affect their health and adjustment, using approaches that are supported by empirical literature.

Similarities are also evident in research and academic training contexts. Like research conducted by clinical child and pediatric psychologists, school psychologists' research may examine issues such as child emotional, behavioral, and developmental disorders and health; child adjustment and health promotion; and contextual factors such as families, peer groups, schools, culture, ethnicity, and economic disadvantage. Again, the major difference is one of relative emphasis, with school psychologists more likely to research areas relevant to learning or adjustment within the learning environment. Finally, as one of the three types of accredited programs in professional psychology (along with clinical and counseling psychology), school psychology graduate program curricula include foundational training in diverse bodies of knowledge within psychology (e.g., social, cognitive, and biological bases of behavior), research methods, normal and abnormal development, and professional ethics, as well as area-specific knowledge in assessment, intervention, and prevention (APA, 2005). Like clinical child and pediatric psychologists, school psychologists complete a predoctoral internship and supervised postdoctoral experience for licensure.

TRAINING ISSUES

Training in clinical child and pediatric psychology addresses multiple purposes and, as such, serves many masters. Academically, clinical child and pediatric psychology are usually (although not always) part of psychology departments, along with other subdisciplines such as social, cognitive, developmental, experimental, and quantitative psychology. Housed within liberal arts and sciences colleges or divisions, department priorities typically involve the traditional combination of research, graduate and undergraduate teaching, and service to the campus and academic community. Professional psychology programs are somewhat unique within departments in their additional priority of training graduate students for careers in

professional health service. Thus, accreditation, applied practicum and internship training, and preparation for licensure must be considered. Because accreditation provides standards for evaluating professional training programs, we discuss accreditation issues first, and then examine the context, goals, and curricula of clinical child and pediatric training programs.

Accreditation Issues

As with all professional psychology training, clinical child and pediatric psychology programs participate in the accreditation process. Accreditation is a process of voluntary self-regulation by the field, with a primary goal of protecting students and the public by ensuring that programs meet at least minimal standards of content and performance (APA, 2005). Clinical programs have been accredited since the mid-twentieth century, when the need for federally funded psychology training prompted development of a system and criteria for accreditation (Sheridan, Matarazzo, & Nelson, 1995). Since then, accreditation guidelines and procedures have undergone several revisions that reflect the evolving nature of professional psychology, and even as this chapter is being written the process is under review and modification (Schilling & Packard, 2005). General accreditation guidelines are beyond the scope of this chapter (interested readers can refer to the CoA's *Guidelines and Principles for Accreditation of Programs in Professional Psychology*, APA, 2005, as well as recent revisions to the *Policy Statements and Implementing Regulations*, APA, 2006a). Here, we focus on select issues specific to clinical child and pediatric psychology training.

As currently defined, the purpose of accreditation is to evaluate the extent to which a training program has achieved the goals and objectives of its stated training model, defined within the more broadly accepted principles of professional psychology (APA, 2005). In other words, programs must meet field-defined criteria, but beyond that are encouraged (or at least allowed) to articulate and pursue innovative training models and practices. As Roberts et al. (1998) noted, this flexibility paved the way for clinical child psychology programs to begin to define themselves more specifically than as just clinical programs and to more fully embrace the guidelines for child-focused training articulated at the Hilton Head and later conferences. As with many guidelines, standards, and rules, these accreditation guidelines are aspirational, and many would argue that they have been applied inconsistently or incompletely. This has been evident in the challenges that some clinical child programs have faced when seeking accreditation or reaccreditation. For example, Roberts (2006) notes that his clinical child training program includes formal training in adult psychopathology in order to satisfy the CoA, although he questions whether adult-focused programs receive the same level of scrutiny regarding their coverage of child psychopathology. Similar challenges likely affect other clinical child and pediatric programs, as well as the accreditation teams faced with the task of evaluating specific and varied applications of clinical psychology training goals.

Given the importance of accreditation for training programs and the time and expense involved in the process, it is easy for *Accreditation* (with a capital A)

to become reified in a way that leads both training programs and accreditation committee members to forget that the process is essentially intended to serve the field rather than to define it. Whereas accreditation should promote high standards within the field and facilitate innovation and development, it is the field itself that must self-define, innovate, and evolve. Thus, continued articulation of how the fields of clinical child and pediatric psychology define optimal graduate training remains crucial for influencing future development and application of accreditation guidelines.

Academic Context

As noted, most clinical programs are housed within departments of psychology. Within the department, however, the structure of clinical child and pediatric psychology training can range from a stand-alone academic training program to a track or emphasis within a more general clinical program, to simply providing informal opportunities to gain child-specific experience. Several surveys conducted over the past 30 years have documented the diverse structures and academic settings that characterize clinical child psychology training programs (Mannarino & Fischer, 1982; Roberts, 1992; Zeman, Nangle, & Sim, 1999), and this diversity remains true today. Currently, Divisions 53 and 54 provide an online directory of programs that report providing clinical child or pediatric psychology training (Division of Clinical Child and Adolescent Psychology, 2006). Of the 45 programs listed as having child or pediatric training, only three (6.7%) report having separate clinical child training programs. Over half of the programs (24; 53%) provide formal child or pediatric tracks or specialties, but 40% (18) report providing no formal track.

Decisions about the context in which to provide clinical child and pediatric training are likely complex, encompassing philosophical, historical, and practical considerations, and an understanding of how these issues influenced training program development at each institution is obviously beyond the scope of this chapter. However, the Division 53 data do provide some suggestions of how context relates to program issues such as size of faculty and student body, course offerings, and practicum opportunities. In examining these data, it is important to remember that they represent self-reported program characteristics, and respondents may have used different definitions and counting conventions. Also, we examine descriptive data without evaluating the statistical significance of patterns. With these caveats in mind, we note several interesting findings.

Table 2.2 presents data on characteristics of programs listed in the Division 53 directory as providing clinical child and pediatric psychology training in separate programs, formal tracks, or informal opportunities. Although the separate programs tended to have a somewhat smaller overall clinical faculty than programs with or without formal tracks (approximately five versus nine clinical faculty), their child-focused faculty averaged about one more faculty member than other programs (approximately five versus four clinical child or pediatric faculty). For programs with tracks or informal child psychology training opportunities, child-focused faculty represented approximately 40% of total faculty. Similarly, child-focused students in programs with child or pediatric tracks represented just over 40% of their overall

Table 2.2 Program Characteristics for Clinical Child or Adolescent and Pediatric Psychology Training Programs (2005)

	Type of Clinical Training Provided			
Characteristics	No Formal Track or Emphasis	C or P Emphasis	Formal C or P Track	Distinct C or P Program
N	15	7	18	3
Faculty	*M (SD)*	*M (SD)*	*M (SD)*	*M (SD)*
All clinical faculty	8.93 (3.06)	8.71 (2.29)	9.83 (2.28)	5.00 (2.65)
Child clinical faculty	2.80 (1.65)	3.71 (1.55)	3.81 (1.10)	5.00 (2.65)
Pediatric faculty	0.57 (0.78)	0.57 (0.93)	0.75 (0.67)	0.67 (1.15)
Students				
All clinical students	32.33 (10.11)	43.14 (10.16)	38.18 (12.90)	31.33 (5.51)
Clinical child students	7.33 (5.08)	12.86 (5.40)	15.50 (8.17)	31.33 (5.51)
Pediatric students	1.47 (2.03)	0.57 (0.98)	2.24 (3.09)	6.00 (10.39)
Course and Opportunities				
Child clinical courses	2.60 (1.24)	3.00 (1.29)	4.28 (1.41)	9.67 (4.73)
Pediatric courses	0.27 (0.46)	0.14 (0.38)	0.56 (0.78)	1.33 (2.31)
Child practica	2.43 (1.57)	7.86 (7.86)	5.78 (5.40)	7.00 (3.61)
Pediatric practica	0.73 (1.03)	2.29 (2.14)	1.88 (2.64)	1.00 (1.73)

Note: C = Child; P = Pediatric.

student body. However, programs without formal tracks may attract fewer child-focused students, as less than 30% of their student body comprised child or pediatric psychology students. Programs overall averaged 30 to 40 clinical graduate students, but for all but the separate clinical child or pediatric programs, child-focused students were in the minority.

Course requirements and practicum opportunities varied widely, both across and within program type. Separate clinical child or pediatric programs required substantially more child-focused courses than their counterparts with tracks or informal child psychology opportunities. On average, separate programs required 11 child clinical or pediatric courses, likely because they are designing courses for their child-focused students rather than teaching some courses for students from multiple (i.e., child and adult) tracks. In contrast, child track programs required an average of approximately four child clinical or pediatric courses, and informal opportunity programs required only three such courses. There are probably benefits and costs to both approaches. For example, providing general clinical courses to adult and child track students should expose all students to broad education about issues that arise across the life span and across diverse clinical subspecialties. Unfortunately, many general courses focus primarily on adults, with perhaps 1 to 2 weeks of child-relevant material, shortchanging both child- and adult-focused students. Child-focused courses offer maximum opportunity for students to learn about the content, process, and current issues in clinical child and pediatric psychology from faculty who are expert in these areas. However, explicit attention to helping students gain exposure to adult development and psychopathology is also important. Program type seems to have important implications for practicum opportunities

available to graduate students. Separate child or pediatric programs and programs with formal tracks reported having an average of eight practicum placements that focused on child or pediatric training. In contrast, programs without a formal track offered only about three child-focused practica.

Finally, across program type, clinical child training seems more extensive than pediatric psychology training. Of 24 programs that reported explicitly on the emphasis areas offered by their program or track (versus, for example, responding yes to the question about a child or pediatric track), only three reported having a pediatric track, and two of these programs also had a clinical child track. Overall, programs reported more clinical child than pediatric faculty, graduate students, courses, and practicum placements. Indeed, on average, pediatric psychology opportunities accounted for no more than 10% to 20% of child-related training opportunities. This is consistent with Spirito et al.'s (2003) statement that specialized training in pediatric psychology often does not occur until the internship or postdoctoral stage of training.

Training Goals

In preparing graduates for careers in the science and practice of clinical child and pediatric psychology, the field has emphasized several goals and competencies. For example, in the Division on Child, Family, and Youth Services (APA Division 37) training guidelines that were the basis of some of the work at the Hilton Head Conference, Roberts and colleagues (1985, p. 72) outlined four primary goals of training:

1. Training should provide at least minimal entry level skills for providing professional psychological services . . . on behalf of children, adolescents, and their mentors (e.g., parents and teachers) in a range of settings (e.g., home, school, institutions, and medical clinics). . . .
2. Training should provide the scientific basis for professional psychological applications.
3. Training should include specialized research contributing to an understanding of children, families, and psychological development, as well as in the evaluation of therapeutic interventions.
4. Child-, youth-, and family-related issues should be integrated into psychology-based courses at all levels, regardless of specialty designation.

More recently, La Greca and Hughes (1999, p. 441, Table 5) identified five competencies important for clinical child psychologists: (1) multicultural competencies; (2) delivery and evaluation of comprehensive and coordinated systems of care; (3) collaborative and interprofessional skills; (4) empirically supported assessment and treatments for promoting behavioral change in children, families, and other systems; and (5) entrepreneurial and supervisory skills. La Greca and Hughes argue that these competencies, reflecting the thinking of many educators in child-focused professional psychology, are increasingly important for effective clinical child practice in our society.

Separated by almost 15 years of developments in clinical child and pediatric psychology, these two sets of goals represent both foundational goals that are central to the definitions of clinical child and pediatric psychology as well as more progressive goals that reflect the evolution of professional psychology. In particular, the theme of empirically based assessment and intervention targeting youth in their environmental and cultural contexts has echoed throughout even the earliest definitions of the field. Likewise, the importance of an interdisciplinary perspective is one of the hallmarks of clinical child and pediatric psychology and a central goal for training and professional competence. Development of a scientist-practitioner identity is also fundamental and is reflected in the clinical child psychologist's research training as well as professional competence in evaluation of service delivery. A clinical scientist identity, based on the newer clinical science training model that emphasizes preparing students for careers that include generation, application, and dissemination of clinical science research (Academy of Psychological Clinical Science, 2007; McFall, 1991), would also be consistent with this goal.

Roberts et al.'s (1985) goal of infusing child-related issues into all clinical psychology curricula, regardless of specialty designation, underscores the importance of having a broad developmental base in all clinical psychology training. Assuming that clinical child and pediatric psychology training would of course attend to child issues, this goal seems more relevant to general or adult-focused clinical training programs. The importance of this goal for clinical child and pediatric training seems to lie primarily in its call for all clinical psychology training to view child issues as a legitimate part of broad and general training.

Some of La Greca and Hughes's (1999) competencies reflect newer or elaborated goals that have emerged in concert with changing issues and needs in society. Multicultural competence, although implicit in early definitions and goals emphasizing contextual views of child adjustment, is an explicit part of current-day competence for child psychologists. Increased attention to multicultural issues is especially important given the growing diversity of the U.S. population as well as priorities noted by APA, the CoA, and the National Institutes of Health. Finally, entrepreneurial and supervisory skills extend beyond the traditional purview of clinical psychology, but are important for preparing clinical child and pediatric psychologists for both applied and academic careers. In applied settings, these skills are important for helping clinical child psychologists function effectively as part of treatment teams that may include junior or in-training members and for maintaining fiscally viable practices. In academic settings, having faculty who have received formal training in supervision is clearly an essential component of providing high-quality practica. Entrepreneurial skills are becoming increasingly important for academics. As funding for higher education becomes tighter, universities are encouraging their faculty and departments to develop diverse sources of funding for their research and training. On the national level, psychologists are becoming increasingly active participants in legislative activity relevant to health care and education (e.g., through the APA Practice and Education Directorates, the Society for Research in Child Development's policy fellowships). To effectively influence the business and legislative sectors, convincing nonpsychologists of the importance of supporting

our efforts to address child health, psychologists must speak the language of these professionals.

TRAINING CONTENT: CURRICULUM DOMAINS

As with all professional psychology programs, clinical child and pediatric psychology curricula are shaped in part by accreditation requirements for clinical programs. However, as noted earlier, accreditation requirements are designed to allow individual training programs to define their goals and methods for attaining them. So how can clinical child and pediatric programs best integrate their training goals and accreditation requirements into a curriculum that prepares competent psychologists? As several investigators have noted, this is apparently not an easy task to do well; even after definitions and goals of clinical child and pediatric psychology were developed, training programs differed widely in their curricula and did not always meet minimal training goals for these specialty areas (Roberts et al., 1998; Zeman et al., 1999).

Recent advances in professional psychology have paved the way for both greater specificity of clinical child and pediatric curricula and increased flexibility in implementation. First, starting in 1996, changes to accreditation guidelines and procedures facilitated innovation in how professional psychology programs can articulate, meet, and evaluate their training goals (Roberts et al., 1998). Second, the field has begun moving from activity-based to competency-based models of training. This is reflected in accreditation guidelines and in recent writings and conferences on professional competencies (Kaslow et al., 2004; Sumerall, Lopez, & Oehlert, 2000).

Both clinical child and pediatric specialties have articulated curriculum recommendations that fit with these developments. In 1998, Roberts and colleagues outlined a model for training clinical child psychologists based on work begun by the National Institute of Mental Health work group on improving training for child and youth service providers (Wohlford, Myers, & Callan, 1993). Roberts et al.'s training model included 11 domains of training considered important for clinical child psychologists. Several years after the clinical child training model was published, the Society of Pediatric Psychology charged a task force with developing similar guidelines for training pediatric psychologists (Spirito et al., 2003). The resulting guidelines, adapted from the clinical child model, include 12 domains of training, with 11 overlapping with the clinical child model and the final domain specific to pediatric psychology.

Although coursework is a common way to meet curricular requirements, competence in many of the domains of training requires practical experience. The clinical child and pediatric training models acknowledge this, describing student training as a three-level sequence: *exposure* to the content and processes of clinical psychology through didactic training and observation, *experience* that involves mentored practice of the relevant skill, and *expertise* that is obtained through extensive exposure and experience and enables independent practice. By the end of graduate training, students should have had at least some exposure to and experience with each

domain, although they may not have expertise in any. Internship is an opportunity to gain more experience and develop expertise in some areas, and by the time students have completed postdoctoral training and are ready for licensure, they should have established sufficient expertise to support independent practice with a variety of youth and family issues. Next we outline the 12 recommended domains of clinical child and pediatric psychology training, from graduate through postdoctoral training, and offer comments about potential ways of providing didactic, applied, and research training in these domains.

Life Span Developmental Psychology

From a developmental psychopathology perspective, complete understanding of youths' adjustment problems requires an understanding of individual and contextual factors in normal development (van Eys & Dodge, 1999). At a minimum, coursework should expose students to current theory and research on life span development across social, emotional, cognitive, behavioral, and physical domains, as well as to the role of individual (e.g., sex, race) and contextual (e.g., family, peer, culture) factors in development. Practical experiences in assessment and intervention should involve youth from a variety of developmental levels and contexts, with particular emphasis on issues such as how developmental factors impact youth psychopathology and response to treatment. For pediatric psychology students, didactic and practical experiences should focus on issues relevant to child and adolescent health, such as the impact of disease process on cognitive development or the role of adolescent autonomy seeking on adherence to medical regimens. Student research should also incorporate developmental issues, addressing age, gender, and context in the specific research questions and methods, or at least conceptualizing the research from a developmentally informed perspective.

Beyond the minimum, several training programs integrate clinical child and developmental training more fully. In their survey of child-focused clinical programs, Zeman et al. (1999) found that programs characterizing themselves as clinical-developmental required more developmental courses than other programs, averaging just over five required developmental courses versus none or one course required by clinical child programs or clinical programs with a child emphasis. These programs also encourage student research mentored by developmental faculty, attendance at developmental conferences (e.g., Society for Research in Child Development, Society for Research on Adolescence), and publication in developmental as well as clinical journals. An example of a clinical child developmental program is the joint effort of the clinical and developmental programs at the University of Missouri–Columbia. Students admitted to the clinical child developmental program essentially complete requirements of both clinical and developmental training areas. Practically, this means that in addition to courses required for the clinical child program (including courses in normal development, developmental psychopathology, child assessment, child interventions, and family systems and interventions), joint program students take two additional courses in cognitive, social, and emotional development; explicitly incorporate developmental content into

thesis and dissertation research and comprehensive examination; and participate in both clinical and developmental faculty labs.

Life Span Developmental Psychopathology

With foundational knowledge of normal development, students can conceptualize deviations from typical or adaptive functioning. During graduate training, courses and seminars should expose students to abnormal behavior across the life span, as well as current research on trajectories of risk and resilience. Supervised practica with various forms of psychopathology in children, adolescents, and adults allow students to gain experience conceptualizing and diagnosing psychopathology. Again, a variety of settings, including youth in clinic, family, community, school, and medical settings, give graduate students experience in incorporating context into their understanding of youth psychopathology.

Child, Adolescent, and Family Assessment

Becoming a competent clinical child or pediatric psychologist requires a thorough understanding of assessment issues, including how to evaluate the empirical foundations and utility of a variety of assessment methods, how and when to use them, and how to interpret their results for both applied and research purposes. Relevant assessments for youth go beyond tests or measures that youth complete (e.g., intelligence tests) and include evaluations of youth functioning by others (e.g., caregiver reports of adaptive functioning, school reports of academic achievement), as well as evaluations of the systems in which youth function (e.g., parent-child and parent marital relations). Assessment also goes beyond standardized tests and includes clinical conceptualizations, functional analysis, and a general hypothesis-testing framework. Both clinical work and research benefit from a scientific approach to generating questions and using appropriate assessment to address them.

Students should develop an understanding of how developmental factors such as age, sex, ethnic background, and cultural conventions can influence assessment validity, and how to conduct multimethod assessments that can maximize the strengths and contributions of the multiple methods and reporters. Finally, students should obtain experience in interpreting and disseminating assessment results and in assessment reports for clinical purposes as well as in research reports. Roberts et al. (1998) suggest a minimum of two semesters of coursework and associated practica devoted to exposing students to all relevant assessment issues and methods and allowing them to gain experience with several assessments and develop expertise with at least a few. Overall, understanding the conceptual and psychometric aspects of assessment seems most critical to competence as a clinical psychologist. If students leave graduate training with expertise in how to formulate assessment questions, design developmentally and contextually sensitive assessment plans, and integrate results from multimethod assessment, as well as exposure to the mechanics of assessment, they should be able to quickly master specific assessment tools as needed.

Intervention Strategies

Training in interventions should reflect the core values of clinical child and pediatric psychology: empirically and contextually based treatment that is developmentally and multiculturally sensitive and interdisciplinary. This is a tall order, calling for treatment that is both standardized and individualized. To be effective at delivering the standardized aspects of treatment, clinicians in training must be familiar with empirically supported treatment techniques and protocols, including how, when, and with whom to use them. For example, during graduate training, students should be exposed to many of the treatments considered well-established or probably efficacious, as defined by APA Division 12 (Division of Clinical Psychology) and Division 53 (Society of Clinical Child and Adolescent Psychology) task forces (Chambless et al., 1996, 1998; Lonigan & Elbert, 1998), including parent training and multisystemic therapy for oppositional and conduct disorders, behavioral parent training and problem-solving skills training for Attention-Deficit/Hyperactivity Disorder, interpersonal and cognitive therapy for depression, cognitive-behavioral therapy and family anxiety management for generalized and separation anxiety disorders, exposure and relaxation therapies for anxiety disorders, and behavior therapy for childhood obesity. Although exposure to some of these treatments may be limited to didactic coverage in courses or readings, experience with at least some of the treatment protocols should occur during graduate school, with additional opportunities to develop experience and expertise coming during internship and postdoctoral training.

Clearly, the list of evidence-based treatments is incomplete, and even when such treatments exist, they frequently do not have demonstrated equal applicability across multiple ages, socioeconomic groups, cultural groups, or family constellations. Thus, it is also essential for students to gain experience in how to handle situations in which there is no prescribed, empirically supported treatment. Two general approaches are valuable here. First, familiarity with the developmental and clinical literatures on the relevant treatment moderators can help guide treatment decisions. For example, clinicians might use the literature on differences in effective parenting styles across socioeconomic or racial/ethnic groups (McLoyd, 1998) to tailor family treatment for a low-income ethnic minority family. Second, using their scientist-practitioner framework, students should become expert at evaluating their treatments as they deliver them. This way, even treatments that do not have established empirical support can be appraised empirically.

Research Methods and Systems Evaluations

As noted, clinical application from a scientist-practitioner perspective includes using the empirical literature to guide treatment decisions and using empirical evaluation methods in the treatment context to assess treatment process and outcome. More broadly, advancing the literature that informs treatment, prevention, and health promotion requires sound research. Thus, students in clinical child and pediatric psychology should become proficient as both consumers and producers of applied scientific knowledge. Background in methods of conducting scientifically

sound research and analyzing resulting data is an integral part of all professional psychology training. However, clinical child and pediatric training programs should ensure that methods that are especially relevant to understanding youth and families, such as longitudinal and prospective designs and hierarchical analyses that can evaluate child data in the context of schools or families, are included in child-relevant training. For pediatric psychologists, familiarity with research issues and methodologies common to medical settings is also important, including issues of health care utilization, clinical trials, and epidemiological and multisite methods. In addition, because youth work so often involves systems consultation and program development, expertise in systems and program evaluation is important for competent and marketable clinical child and pediatric psychologists. Consistent with a scientist-practitioner approach, practicum training should include opportunities to empirically evaluate treatment progress, using single-case design and other relevant methodologies. Finally, to be truly effective, research must be disseminated. Thus students should develop skills in scholarly publication and presentation and grant writing.

Although internship is acknowledged as primarily a clinical training year, continued opportunity to develop expertise in clinical research is an important part of a scientist-practitioner identity. In their recommendations for internship, Elbert et al. (1998) proposed that, at a minimum, interns be encouraged to systematically review the literature as they conceptualize cases and plan treatment. In addition, many scientist-practitioner and clinical science internships provide protected time for interns to conduct research, either independently or as part of ongoing research at the internship site. Some programs (e.g., Case Western Reserve University, Medical University of South Carolina, University of Mississippi Medical Center, Western Psychiatric Institute and Clinic) provide seminars in grant writing and funding for research and conference attendance, and may even require journal article or grant submission as part of the internship experience. Rather than distracting interns from their clinical duties, these research activities enrich interns' integration of science and application. Similarly, although a primary goal of many postdoctoral training fellowships is preparation for licensure, clinical research and development of grant writing skills can be effectively integrated into the training experience (Drotar, 1998; Ollendick et al., 1988). Often, the more research-oriented internship sites will also have postdoctoral positions in which trainees can continue the clinical and research work begun on internship.

Professional, Ethical, and Legal Issues

Working with children and adolescents comes with many unique professional, ethical, and legal issues. At the most basic level, minor youth are, in most cases, not legally in charge of their own lives. Thus, issues of who decides whether and how to pursue treatment, who receives the treatment (i.e., who is the client), and who provides the treatment must all be decided for the child. Parents are responsible for seeing that their children receive appropriate educational, medical, and psychological services, and graduate students must be aware of issues of parent informed consent, as well as how parent rights and responsibilities are influenced by divorce,

abuse, and neglect. Graduate students must also know how to define the client in family treatment or school or medical consultations. In addition to identifying the client, child and pediatric work can be associated with challenges in identifying the treatment agent. In the interdisciplinary settings in which child and pediatric psychologists often work, psychologists must negotiate and define their roles on the treatment team, along with educators, health professionals, and other social service providers. In all of these tasks, the child's role as an active (albeit legally nonresponsible) participant must be appreciated. Training should address how to involve children and adolescents in assenting to, planning, conducting, and evaluating assessment and interventions, as well as how to balance legal issues with children's rights to privacy.

The types of challenges that children and adolescents face require sensitivity to specific legal and ethical issues. For example, parent divorce and custody issues, abuse and neglect, and delinquency all require familiarity with the legal system. Clinical child and pediatric psychologists should be aware of local, state, and federal laws that define abuse (e.g., the types of physical injuries or conditions that qualify) and neglect (e.g., the age at which children can be left unattended), reporting requirements, and the process by which such cases are handled in the county. Knowledge of how child truancy, other status offenses, and criminal behavior are handled is helpful in treatment of child and family treatment. Particularly by the internship and postdoctoral levels, experience with child forensic issues such as custody evaluation and expert testimony is warranted (Elbert et al., 1988). In pediatric psychology settings, graduate students should gain experience in dealing with issues such as life-threatening and terminal illness, developmental and neurological conditions, and painful medical procedures. In all of these instances, dealing with the affected youth is just one aspect of psychological care; clinicians must help youth address the challenge in the context of their ongoing development, and also help parents, siblings, and perhaps friends cope with the implications of the illness or condition.

Issues of Diversity

Given that considering youth in context is a hallmark of clinical child and pediatric psychology training, attention to the context of diversity is a natural extension of this training. During graduate training, students should gain exposure to and experience with multiple aspects of diversity, including definitions of what constitutes diversity, the influence of values and prejudice on societal and personal attitudes and behavior, and the role of diversity factors in psychological adjustment and treatment. For example, diversity should be defined broadly, to include gender and age, as well as racial/ethnic, cultural, socioeconomic, religious, sexual orientation, disability, and family constellation factors. Graduate coursework and practicum experience should expose students to information on youth with as many of these backgrounds or experiences as possible, with advanced practicum, internship, and postdoctoral training experiences selected to broaden these experiences. Issues that are important to address include the role of diversity factors in definitions of psychopathology, risk factors for certain psychological or medical conditions (e.g., conduct disorders,

depression, heart disease), health care utilization and acceptability, and culturally sensitive assessment and treatment.

Class and supervision time should also be used to allow students to explore their own values and attitudes to become more effective treatment providers for diverse youth and families. Students should gain an appreciation for the balance between considering diversity factors and understanding the individual child or family. Although research may provide valuable information about diverse groups, this information may not generalize to individual cases. Additionally, a child's sex, disability status, and racial/ethnic group do not define him or her, but simply represent one aspect of the child's identity. Thus, students should remember to consider youth as individuals within the context of diversity. Finally, in addition to gaining and using information on diversity, students should use their research as an opportunity to add to the knowledge base on issues of diversity. Although psychology has made great strides in understanding male and female youth across ages and from diverse backgrounds, much remains to be done, and the scientist-practitioner and clinical science backgrounds of clinical child and pediatric psychology students places them in an ideal position to conduct clinically relevant and contextually sensitive research.

Role of Multiple Disciplines in Service Delivery Systems

As we've noted throughout this chapter, interdisciplinary work is another hallmark of clinical child and pediatric psychology. Much of working effectively within an interdisciplinary team is part of a professional issues skills set, involving mastery of how to fit into existing systems, how to divide and share tasks, and how to understand the language and approaches of multiple professions. As such, students must gain familiarity with other disciplines' approaches and contributions to child health promotion and experience in participating in interdisciplinary teams. This should include coursework in areas such as consultation, family and community psychology, and public health. Practicum experiences should allow students to observe and participate in a broad range of systems, including families, schools, social service agencies, intensive (e.g., inpatient or day) treatment programs, and medical settings. Internships are often an ideal place for trainees to gain a variety of interdisciplinary experiences, and by completion of postdoctoral training, clinical child psychologists should be able to work effectively with the full range of professionals involved in child health care.

Prevention, Family Support, and Health Promotion

A comprehensive approach to dealing with child and adolescent adjustment issues includes attending not just to diagnosed disorders, but also to factors that place youth at risk for disorder or maladjustment and factors that promote resilience and healthy adjustment. Thus, clinical child and pediatric psychologists should have a knowledge base in youth and family healthy adjustment and prevention science, as well as research and applied competence in prevention and health promotion. Such experiences can come from coursework in typical development and resilience, applied experiences in schools or primary care facilities, and interventions that focus on health

promotion issues such as parent-child relationship enhancement, social skills training, healthy eating or obesity prevention, and reduction of risky sexual behavior.

Social Issues Affecting Children, Adolescents, and Families

Many of the challenges to youth adjustment are the social-environmental issues that influence children and families. Clinical child and pediatric psychologists should be aware of the sorts of social issues particularly relevant to youth, both generally and in their home communities. For example, as many as 50% of children have been exposed, either directly or indirectly (e.g., through media reports), to traumatic events such as natural disasters or violence in their homes, schools, communities, or the world (Bell & Allwood, 2007). Thus, trauma exposure is a major public health issue and one that all clinical child and pediatric psychologists will almost certainly encounter. Another important public health concern is youth access to mental health services; as many as 20% of youth are estimated to be in need of psychological services, yet only about one third of these youth receive needed services (Department of Health and Human Services, 1999). For both of these issues, clinical child and pediatric psychologists can make positive contributions in several ways, including identifying youth in need of treatment, participating in prevention programs to reduce risk for maladjustment, and participating in child advocacy efforts that can impact both individual children's lives as well as broader public policy and legislation. Involvement in Society for Research in Child Development's Office for Policy and Communications, APA's Public Interest Directorate, or state or local government can be effective ways for clinical child and pediatric psychologists to leverage their influence.

Specialty Applied Experiences

Roberts's (1998) final recommendation for clinical child training guidelines is that trainees gain applied experience in specialty practice relevant to assessment, intervention, and consultation with children, adolescents, and families. Although this is part of most of the preceding recommendations, its reiteration here emphasizes the importance of providing trainees with child-specific experience versus simply assuming that skills will generalize from adult-focused experiences. Roberts suggests that sufficient experience includes a minimum of closely supervised child practica in at least two different settings, internships that are at least half child-focused, and postdoctoral experience that provides further depth in autonomous child-related practice. Interestingly, Roberts's recommendation regarding internship is less than the two thirds child focus recommended at the Hilton Head Conference.

For the Society of Pediatric Psychology Task Force (Spirito et al., 2003), this curriculum domain centers on the importance of consultation and liaison skill development. Again, this domain underscores recommendations regarding interdisciplinary practice made throughout the Task Force's document. However, in addition to the need for pediatric psychologists to consult with parents and other professionals on child assessment and treatment, they will often serve in a training role. Specifically, pediatric psychologists should be able to educate health care professionals about principles of learning and development, psychological health (of youth and

of health care professionals), and psychological research (Drotar et al., 2003; Spirito et al., 2003). This is particularly important because many psychological and behavioral issues are first raised with children's primary care physicians.

Disease Process and Medical Management

The final curriculum domain, specific to pediatric psychologists, notes the importance of understanding medical issues that they will be likely to encounter. Trainees should participate in coursework, seminars, supervision teams, and applied experiences that familiarize them with the phenomenology of relevant medical issues, as well as psychological issues related to the medical issue and its treatment (e.g., adverse behavioral effects of medical treatments). Spirito and colleagues (2003) suggest that pediatric psychology students complete practica and internship rotations in primary and specialty medical clinics in order to work directly with medical professionals and medical issues.

TRAINING OUTCOMES

There are undoubtedly many ways that successful outcomes for graduate students in clinical child and pediatric psychology can be defined. During graduate training, success can be evaluated by examining student retention, performance on program tasks (e.g., course grades, percentage of students passing comprehensive examinations), participation in professional activities such as conference presentations and publications, and more subjective issues such as personal and financial stability (e.g., the toll that graduate training takes on students' health, personal relationships, and financial debt). Postprogram outcome variables might include student program completion statistics (e.g., the likelihood that students will complete their graduate training and obtain relevant internships, the total debt load incurred to obtain the graduate degree), professional competence (e.g., number of graduates who attain licensure, diplomate status, or other demonstration of competent practice), and career tracks and success (e.g., the number of graduates who secure relevant jobs, research and mentoring productivity of graduates in academic positions, career satisfaction and longevity). Obviously, many of these outcomes are fairly subjective, multiply determined, and may not even be accessible to graduate programs. For example, the field is currently struggling with the best ways to define and assess professional competence (Kaslow et al., 2004; Roberts et al., 2005; Rodolfa, Kaslow, Stewart, Keilin, & Baker, 2005). Similarly, graduate programs are often unaware of students' personal issues unless they are clearly affecting professional functioning, and even if aware, are often in a poor position to evaluate the extent to which graduate program stresses versus other factors contributed to student health problems, relationship breakups, or decisions to leave a program or the field. Despite these limitations, however, training programs should routinely collect and evaluate data on student outcomes. At a minimum, these data are useful for the self-evaluation that programs can use to identify areas for growth (and that CoA requires). Beyond that, sharing these data with potential students and professional colleagues has the potential to improve training in the field as a whole.

For many years, the Council of University Directors of Clinical Psychology (CUDCP; the training council for scientist-practitioner and clinical science–oriented clinical doctoral programs) has supported a resolution strongly encouraging all member programs to make trainee outcome data available to the public. Programs collect and share information on program attrition, time to degree completion for graduates, and the percentage of students who obtain predoctoral internships. In addition to helping prospective graduate students evaluate training programs, these data have been useful in evaluating how clinical programs fare on these basic outcomes. For example, recent years have seen an increasing imbalance between the number of internship slots available through the Association of Psychology Postdoctoral and Internship Centers (APPIC; 2006) match system and the number of students applying for these slots, with only 77% of applicants being matched to internships on National Match Day in 2006. However, a survey of CUDCP programs indicated that 86% of students from CUDCP programs secured internships on the 2006 Match Day, and 92% of students ultimately obtained an internship through the match, internship clearinghouse, or other methods (Collins & Callahan, 2006). Although these data are not specific to clinical child or pediatric training, they do suggest that students trained in scientist-practitioner programs are faring better than the national norms.

Recently, the CoA has instituted a requirement that all accredited training programs make similar data available to the public (APA, 2006b). Specifically, beginning January 1, 2007, the CoA requires that training programs provide information on time to completion (including mean, median, and percentages of students completing the program in under 5, 6, 7, and more than 7 years), program costs (e.g., tuition, fees, any certain or possible adjustments to costs such as waivers, fellowships, or assistantships), internship placement (including APPIC match rates and overall placement rates through other methods, rates of paid and half-time internships), and attrition (i.e., the number and percentage of students who fail to complete the program for any reason). Additionally, beginning in 2008, programs will be required to report the number and percentage of graduates who have become licensed psychologists in the preceding decade.

Beyond program completion, graduates' career trajectories provide an index of training success. Given the scientist-practitioner emphasis of most clinical child and pediatric psychology programs, careers that integrate research and applied activity seem the ideal outcome, although the nature of this integration might range from jobs in academic departments or medical centers that combine research, clinical teaching and supervision, and perhaps some clinical practice, to more applied jobs in which practitioners provide evidence-based treatments, participate actively in continuing education, and perhaps provide clinical supervision to doctoral students. For clinical science programs, the desired integration is likely to emphasize research more heavily, with graduates being effective contributors to clinical science research and dissemination. Although they do not examine child-focused programs specifically, a few studies have examined clinical programs' success in placing graduates in academic department faculty positions (Ilardi, Rodriguez-Hanley, Roberts, & Seigel, 2000), graduates' research productivity once they begin their own academic careers (Roy, Roberts, & Stewart, 2006), and graduates' scores on

the Examination for Professional Practice in Psychology (Yu et al., 1997). These studies provide useful information that individual programs can use to compare themselves to national norms. Interestingly, although these outcome variables all tended to relate significantly to reputation rankings such as the one published by *U.S. News and World Report*, they offer a more objective and specific way to assess training outcomes.

FUTURE CHALLENGES

Although exciting advances have shaped the growth of clinical child and pediatric psychology over the past 50 years, important challenges lie ahead as the field emerges from its adolescence into its professional adulthood. In addition, current issues that confront clinical psychology as a whole may provide distinct opportunities for leaders in child and pediatric psychology to creatively solve problems and improve training experiences for students. These challenges are the focus of several recent articles and conferences (e.g., the special issue of *Clinical Psychology: Science and Practice* on the present and future of clinical psychology; Kendall, 2006). Here, we discuss a few of these issues, include the challenges of designing curricula that balance breadth and depth, defining what it means to be a competent clinical psychologist, addressing impairment among students and faculty, and responding to dynamic market forces that shape the job outlook for clinical psychology graduates.

Broad versus Specific Training

As we discussed earlier, a central issue in the growth of clinical child and pediatric psychology has been how best to balance general (thus far primarily adult-focused) training in psychology with specialized coursework and clinical experiences that focus on child or pediatric psychology. As the field evolves in the twenty-first century, it is critical to evaluate whether broad and general training, as it is currently defined, keeps clinical psychology training stagnant. Further, we foresee attaining an appropriate balance as becoming more difficult as the technology of the field advances and expands. Clinical research is reaching deeper into existing problems and stretching into novel areas. For example, the development and validation of new assessments, preventions, and treatments translates into additional skills and educational content that students must master to be competent psychologists upon graduation.

The question remains whether specialization at the level of clinical child or pediatric psychology is appropriate and beneficial to students. Or put another way, are clinical child and pediatric psychology more of a specialty than clinical adult psychology? To date the field has taken steps to define itself as such, and many leaders in the field have embraced this definition. Yet, clinical child and pediatric psychology are not the only fields to deal with this issue. Leading clinical gerontologists, health psychologists, and clinical neuroscientists also struggle with these same training decisions (Roberts, 2006). What is potentially lost in specialization's overemphasis in one area is the scope of clinical psychology as a whole. As currently implemented by many programs, specialized child clinical or pediatric tracks often take a life span developmental and systemic approach to psychology that ensures broad

training (Prinstein & Roberts, 2006; Roberts, 2006). One potential resolution to these specialization issues is to define core curricula as a set of skills that are common to all psychologists, such as training in scientific methods, data analytic techniques, and general clinical skills. In addition, training in core areas of psychology would provide content that takes a life span developmental and systemic approach. Then the focus on clinical adult or clinical child psychology would be considered at the same level in terms of specialization. This would alleviate the "adult clinical plus" model that permeates many training programs, in which students who are interested in working primarily with youth populations end up taking similar numbers of clinical classes plus additional classes or seek additional training opportunities with children (La Greca & Hughes, 1999; Roberts, 2006). Clearly, however, the significant challenge of this approach is to provide students with sufficient exposure to the burgeoning literature in important areas without extending an already lengthy training process.

Graduate students in clinical psychology fall on both sides of the argument. In a recent survey of over 1,000 graduate students in programs affiliated with the CUPCP, almost three quarters of clinical psychology graduate students report specializing within their graduate training, and over half of those who reported not specializing plan on doing so before the end of their graduate training (Luebbe, Green, & Malcolm, 2006). Further, just under 50% of those who reported specializing indicated that their specialty was a recognized or formal track within their program, and over 60% of students reported that their specialty area was a large influence on their choice of graduate programs. Interestingly, students reported specializing in a variety of ways beyond the types of populations (e.g., child) with whom they work. Students also consider themselves specializing in certain disorders, research areas, theoretical models of therapy, and specific skills such as statistics. These results suggest that graduate students want to and already are specializing during their graduate training. This makes a strong argument for officially recognizing certain specialties. Yet, these results also suggest that the word "specialty" has varied meanings, and students are finding opportunities to specialize within their current programs regardless of whether such specialties are officially recognized. Overall, designing curricula that provide broad but focused training remains a critical issue as clinical child and pediatric psychology grow.

Professional Competence

As we described, the field of clinical psychology, including its regulatory and accrediting bodies, is currently in flux as it moves away from an activity-based model of training to one that is competency-based. An ongoing challenge for training programs not only in clinical child and pediatric psychology but also clinical psychology as a whole is to continue to produce professionals who are competent to practice psychology and who are committed to a process of lifelong learning that will enable them to stay up-to-date with the best practices available. The development of training models and lists of core competencies in clinical child and pediatric psychology (e.g., La Greca & Hughes, 1999; Roberts et al., 1998; Spirito et al., 2003) are good first steps toward establishing competency-based training.

Still, several questions remain during this transition. First, how will competency be assessed? Second, *who* will be responsible for such assessments? Finally, and especially pertinent to clinical child and pediatric psychology, how will competency among specialties differ from one another regarding content of competencies, unique contexts in which competency is needed, sequence of competency development, and assessment of competency? In the future, leaders in clinical child and pediatric psychology will likely be charged not only with the task of developing novel techniques to assess competency across the span of a career, but also for conducting the necessary research showing that such assessments are reliable and valid indicators of competency (Roberts et al., 2005). For example, although the Examination for Professional Practice in Psychology provides a reliable assessment of knowledge, it does not assess other foundational competencies such as skills and attitudes. Several authors have proposed use of additional assessment techniques such as standardized patients, self-assessments, or "360 degree evaluations" (Kaslow, 2004 Roberts et al., 2005).

Recently, the field has been quite active in attempting to identify specific competencies that professional psychologists should attain across their development and exploring useful ways of assessing and documenting competence. The 2002 Competencies Conference initiated by APPIC identified several areas in which competencies should be established (Kaslow et al., 2004) and paved the way for ongoing efforts by APPIC, APA's Board of Educational Affairs, the Council of Chairs of Training Councils (a meta-organization representing the training councils in professional psychology as well as other organizations interested in training), and other training-related organizations. Hatcher and Lassiter (in press) describe the results of a multiyear practicum competencies workgroup spearheaded by the Association for Directors of Psychology Training Clinics with input from the Council of Chairs of Training Councils. With additional input from clinic and training program directors from all areas of professional psychology, the workgroup developed a document that outlines practicum competencies considered necessary at the beginning of practicum training, that develop over the course of graduate practicum, and that should be evident at the end of practicum training in order for students to be considered ready for internship. Although the document does not specify how these competencies should be assessed, training programs are already exploring creative ways of incorporating these competencies into supervisor evaluations and student self-assessments. The APA's Board of Educational Affairs is also currently active in identifying competency benchmarks for professional practice in areas such as assessment and diagnosis and multicultural competency. These efforts promise to make a substantial impact on the way professional psychologists evaluate their training efforts.

Because of the unique populations served and the varied contexts in which clinical child and pediatric psychologists work, new and specific ways to assess competency may be needed. For instance, ratings by peers or other professionals in schools, hospitals, or juvenile justice systems may complement more traditional competency assessments. In addition, as competency-based models become the norm for clinical child and pediatric training programs, it is imperative that the field as a whole also conduct an ongoing self-assessment of sorts. As the definition

of a clinical child or pediatric psychologist evolves, so too must the knowledge, skills, and attitudes that compose the core competencies of these professionals. Leaders in the field must be willing to adjust and strengthen current models of training by adding new domains of competency, modifying current domains, or removing some domains altogether.

Professional Impairment

Professional competence among practicing psychologists, faculty supervisors, and students helps both to build the public's trust and to ensure its safety. Detecting and intervening with impaired professionals is therefore a vital responsibility of all clinical psychologists. Empirical studies have begun to show that graduate student impairment, whether it be lack of competence or experienced personal distress, is prevalent (e.g., Vacha-Haase, Davenport, & Kerewsky, 2004). Yet on the whole, far less has been written about this population than about interns or licensed professionals (Schwartz-Mette, in press). Further, given the vulnerability of the population with whom child and pediatric psychologists primarily work, professional competence is even more critical for students and supervisors in these fields. Also, as previously mentioned, clinical child and pediatric psychologists often work with multiple individuals in a variety of systems. This puts these students, not to mention practicing professionals, in a unique position as they potentially interact with a wider spectrum of the public than many psychologists, and as such may have to be multicompetent.

While APA ethical codes stipulate that psychologists should monitor themselves and peers for signs of impairment, graduate students may be particularly unlikely to do so for reasons associated with their professional inexperience, added stress during training, growing pains as they transition from undergraduate experiences to being a publicly recognized professional, and reservations regarding identifying impaired peers for fear of negative social consequences (Schwartz-Mette, in press). Thus, detection and appropriate intervention with impaired students often falls on the shoulders of academic training programs. Unfortunately, although both the Council on Accreditation's *Guidelines and Practices* (APA, 2005) and APA's *Ethical Principles of Psychologists and Code of Conduct* (APA, 2002) discuss impairment, neither document provides recommendations for resolving these issues. As such, most training programs are left with creating program-specific guidelines for intervening in situations in which impairment occurs, and these guidelines vary widely from program to program.

Several authors (e.g., Schwartz- Mette, in press; Vacha-Haase et al., 2004) have provided recommendations for addressing professional impairment. These include improvement of self- and peer monitoring through education about self-care and transition-related stress in graduate school, increased monitoring by clinical supervisors of students, and creation of more sensitive and open cultures in training programs. Additionally, more research is needed in the field that examines the unique issues that arise when child clinical and pediatric psychologists are impaired (e.g., prevalence of inappropriate conduct with children). Finally, training programs will need to continue to take steps toward managing students with

impairments through informal (e.g., varied clinical training cases to improve competency with specific populations) and formal (e.g., required counseling for emotionally distressed students) channels.

Marketplace Developments

The exponential growth of clinical psychology over the past 100 years has not come about without some growing pains, and changing market forces have altered the job outlook for many graduates of clinical psychology training programs. In the future, clinical child and pediatric psychologists have reason to be cautious, but also optimistic. One factor that directly influences the job market for new psychologists is the ratio of qualified applicants to jobs. On the bright side, job opportunities are predicted to grow faster than average for practicing psychologists in the next 8 years, especially for clinical and school psychologists (U.S. Bureau of Labor Statistics, 2006). Despite this growth, training programs are unfortunately producing even more psychologists than there are adequate training opportunities (i.e., internships) and perhaps market demand. For example, the ratio of internship applicants to available slots decreased from 1999 to 2002 to levels close to 1:1. Yet, this ratio has risen sharply from 2003 to 2006 (APPIC, 2006). If this trend continues in a similar fashion, the market may become oversaturated with psychologists. Although the APA Office of Research collects data on employment of new graduates, no up-to-date formal workforce analysis exists. Fortunately, the Office of Research is currently pursuing such an analysis.

Positions for academic clinical child and pediatric psychologists also appear to be growing. Many departments are receiving funding for new faculty lines, and additional positions formerly occupied by retiring faculty are opening. More universities and colleges are creating interdisciplinary positions, such as developmental psychopathologist positions with dual responsibilities in clinical and developmental training areas, or have growing departments or programs in human development and family studies. Joint doctoral programs in clinical and developmental psychology, like that at the University of Missouri–Columbia described earlier, may help to expand students' marketability in the academic job market. One challenge that remains for academic psychologists deals with research funding. Unfortunately, grant funding is reliant on political factors, and despite funding levels remaining relatively stable from agencies like the National Institutes of Mental Health in the past 5 years, the competition has grown more intense and success rates for awards have decreased sharply. It is uncertain whether this trend will continue in the future.

Dynamics in the marketplace are leading many graduates of clinical training programs, including those in child and pediatric psychology, to redefine their professional roles or to enter nontraditional employment opportunities. Generally speaking, fewer clinical psychologists are working in traditional office settings seeing clients for the 50-minute hour, and this is especially true for clinical child and pediatric psychologists. In turn, training programs are challenged to keep pace with these changes by adjusting their curricula to provide relevant coursework and experiences that prepare their graduates to be competitive job applicants. However, within this trend there is reason for child and pediatric psychologists to be

optimistic. Specifically, students who are trained in an ecological framework should excel in opportunities to work in a variety of settings where child clinical or pediatric psychologists traditionally have not. This will require, however, that graduates of clinical child and pediatric training programs take the lead in developing partnerships with community agencies and institutions to develop such niche positions.

SUMMARY

Although adolescents in the field of professional psychology, the future of clinical child and pediatric psychology looks promising. Defining these fields has not been an easy task, given the variety of roles, responsibilities, and endeavors undertaken by clinical child and pediatric psychologists. Further, as these definitions continue to evolve, training programs will have to modify their curricula to provide state-of-the-art instruction for their students. In this chapter, we have outlined several training models and important components of clinical child and pediatric training at the graduate, internship, and postdoctoral levels. Special attention was paid to outcomes of training, including what a competent clinical child or pediatric psychologist might look like in areas as varied as assessment and diagnosis, multiculturalism, research methods, and ethics. Such curriculum modifications are not without their challenges, and we ended this chapter with a discussion of issues that will need to be addressed both in the near future and several years down the road. Overall, training in clinical child and pediatric psychology has made considerable strides to date, and many training programs produce competent, skilled psychologists. Yet, we think it is important that leaders in clinical child and pediatric psychology not be content with the progress made thus far, but work diligently and proactively to address the challenges facing these fields.

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CHAPTER 3

Professional Practice

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The practice of professional psychology has been in existence for roughly one century. Most readers will be quite aware of the roots of clinical psychology in the independent psychiatric practice of Sigmund Freud in the late nineteenth century. Perhaps less well known, but no less pivotal, was the establishment of the world's first psychological clinic by Lightner Witmer at the University of Pennsylvania in 1896 (which specialized in the provision of services to children and their families).

Professional psychology has witnessed considerable growth over the intervening decades. In the period between World War I and World War II, the number of psychologists in the United States increased from 300 to 3,000. Prior to WWI the vast majority of psychologists were academics teaching and conducting research in colleges and universities, yet by 1930 over one third were working in applied settings (Finch & Odoroff, 1939). This proportion remains essentially the same today. Recent data indicate that of the 179,000 psychologists employed in the United States in 2004 (U.S. Department of Labor, 2006), nearly 60,000 were engaged in full-time clinical work (with an additional 18,000 engaged in some form of part-time practice; Kohout & Wicherski, 2003).

WHAT DEFINES PROFESSIONAL PRACTICE?

Professional practice is a relatively broad term used to describe all professionals within the field of psychology, not just those who choose to practice independently. Researchers, academics, clinicians—all share certain things in common that allow them to be called professionals, including, but not limited to, a shared professional code of ethics, years of graduate education in research and treatment of psychological disorders, and a license or state certification to utilize knowledge to help the public. The focus on using and evaluating empirical methods helps shape the way professional psychologists approach their work, and this rigorous training is just one aspect of what sets professional psychology apart from other helping professions (Todd & Bohart, 2003).

Professional psychologists provide a diverse range of services and skills. As Blau (1983, p. 57) points out, psychologists

> conduct a wide variety of psychotherapeutic interventions, assess intellect, aptitude, personality, neuropsychological functioning, and marital adjustment—they design, direct,

> and monitor various prevention and intervention programs . . . work in many diverse medical settings utilizing uniquely psychological methods and tools that are required in health service, rarely work autonomously, and ordinarily function in psychology departments or psychology services in university clinics, hospitals, prisons, community mental health centers, agencies, courts, and treatment centers.

While many skills are shared by all professional psychologists, distinctions in preparatory work, professional roles, and skills are important to note. These differences begin in graduate training, as some psychologists are trained under the scientist-practitioner model (commonly known as the Boulder model), first outlined in 1949. This approach stresses the importance of academic research in combination with supervised clinical experience, resulting in professionals who are highly competent in the assessment, evaluation, and treatment of psychological disorders using empirical methods. As an alternative, the practitioner-scholar model places a heavy emphasis on clinical training instead of research. Although still requiring a doctoral degree, students trained under the Vail model have reduced academic research requirements and are often associated with separate schools of professional psychology rather than research universities. The two models typically provide professional psychologists with different degrees: A PhD or doctorate in philosophy is awarded upon completion of graduate work, successful defense of a research dissertation, and a year of internship at a clinical setting approved by the American Psychological Association (APA). This compares to a PhD, or doctorate in psychology, which is awarded after 4 to 6 years in a professional program where the emphasis is on clinical skills and training, a year of internship training, and research that tends to focus on more qualitative methods directly related to clinical practice. Both qualify under APA's (1987a) definition of professional psychologist: "Psychologists have a doctoral degree in psychology from an organized, sequential program in a regionally accredited university or professional school." Both models, despite their differences, allow for the ability to produce psychologists for independent practice. The Boulder model's training emphasis on the development of an evaluative attitude toward one's own and others' work is not antithetical to independent practice, and the delivery of services in the private sector by independent practitioners does not necessarily mean these psychologists are no longer critical scientists (Cantor & Moldawsky, 1985).

Professional practice is also shaped by what psychologists actually do on a day-to-day basis. The next logical question, then, is What distinguishes professional psychology from other professions? The APA (1987b, p. 2), in its Model Act for State Licensure of Psychologists, has outlined and given a definition for the role of professional psychologists, stating:

> The practice of psychology is defined as the observation, description, evaluation, interpretation, and modification of human behavior by the application of psychological principles, methods, and procedures, for the purpose of preventing or eliminating symptomatic, maladaptive, or undesired behavior and of enhancing interpersonal relationships, work and life adjustment, person effectiveness, behavioral health, and mental health. The practice of psychology includes, but is not limited to, psychological testing and the evaluation or assessment of personal characteristics, such as intelligence, personality, abilities, interests,

> aptitudes, and neuropsychological functioning; counseling, psychoanalysis, psychotherapy, hypnosis, biofeedback, and behavior analysis and therapy; diagnosis and treatment of mental and emotional disorder or disability, alcoholism and substance abuse, disorders of habit or conduct, as well as of the psychological aspects of physical illness, accident, injury, or disability; and psycho-educational evaluation, therapy, remediation, and consultation. Psychological services may be rendered to individuals, families, groups, and the public.

Although a better description of how psychologists are trained, along with what skills they possess, the definition provided thus far only scrapes the surface of the many different faces, forms, and functions a professional psychologist may represent. Differences in theoretical orientation, specialization in a specific disorder, and the type of job psychologists choose all reflect what duties they will perform, how they will go about them, and with whom they will work and collaborate in daily functioning. The purpose of the present chapter is to examine in closer detail one career path a psychologist may take: private practice.

INDEPENDENT PRIVATE PRACTICE IN PSYCHOLOGY

The idea of private practice in psychology is as old as the profession itself. For many, simply saying the words aloud conjures up the image of couches and clients recollecting dreams and early childhood experiences while a therapist sits back and attempts to make sense of it all. This romantic and glorified image of the practice of psychology is easily brought to mind, yet it could not be further from the truth. Today's private practitioner lives in a world dictated by managed health care, empirically supported treatments, and constant financial uncertainty. The lure of private practice may be what motivates many people to enter the field of psychology, yet fewer and fewer psychologists are ending up in private practice. One does not have to look far for a book, article, or speech that outlines the demise of private practice in psychology. Starting up and running a business, building a referral and client base, handling an increasingly large and difficult managed care system, and dealing with legal liabilities are generally not considered reasons that people look forward to starting a private practice. Yet the profession survives. With all its limitations and challenges, the benefits of a career in private practice remain. Professional practice will inevitably remain a viable career path for many psychologists, despite the ever-changing face and demands of the profession itself.

If private practice is not meant for everyone, the question becomes For whom is it meant? In a recent survey of over 1,600 psychologists, 2% reported working full time in independent private practice. This number rises to 5% when group practice settings are included (Wicherski & Kohout, 2005). These numbers show a slight decline from the 1999 Doctorate Employment Survey (6% in total were in some form of practice). Although numbers may be declining and the environment is becoming more challenging overall, numerous advantages of establishing a private practice still exist.

RELATIVE ADVANTAGES AND DISADVANTAGES OF PRIVATE PRACTICE

As with any profession, certain qualities and individual characteristics draw people to the field of psychology. As stated earlier, a career in private practice is not for everyone, but it certainly can be rewarding for those who enjoy the benefits it has to offer. One of the most salient rewards is the notion of being your own boss. Although the same freedoms may not exist today as were available years before, establishing an independent private practice means more control over your hours, clientele, and area of specialization. For many people, the idea of going to work each day and not having to face one or multiple supervisors is a truly appealing notion.

Being your own boss brings with it a certain level of flexibility and independence, along with more responsibility. Numerous decisions must be made, such as deciding where to establish a practice (urban versus rural), what the operating hours will be (flexible and spread out or a regular 9-to-5 workday), and what types of clients you are willing to see (specialization or generalization). In addition to these important decisions, an independent private practitioner also has full control over business decisions and directs how the practice will grow (or shrink, depending on one's business skills).

With this control, the practitioner's decisions play a large role in deciding how much he or she will earn that year. Once a strong client base has been established and clients are being seen regularly, the potential for earnings increases dramatically. A key point here is that independent practice allows for the potential to earn a large salary; the median starting salary of a psychologist in independent private practice is $50,000 (this is reduced to $45,000 when in group practice), which compares to a median starting salary of $45,000 for an assistant professor on a 9-month contract (Wicherski & Kohout, 2005). Earning substantially more than this requires a large investment of time and effort (both on a daily and a long-term basis). How much is earned also will depend on what services are provided, and whether you decide to accept insurance or have a sliding scale.

Having more power over decisions and the daily running of your practice can certainly be an advantage, but it brings with it a high degree of responsibility and personal liability. As primary decision maker, the practitioner must deal with all the consequences of those decisions. Many of the advantages listed previously also have the potential to become disadvantages if not handled correctly. Independence and autonomy may be great for some, but others may require a certain level of structure and consultation in order to succeed; thus, what may be advantageous for some may be a burden for others. If one's practice is badly located, one's hours are too few or constrained, and one's area of specialization is too focused, one may soon realize how challenging private practice can be.

Financially there exists a high level of uncertainty and risk in establishing a practice, as it will take some time to develop a strong client base. Many may choose to begin in another setting and slowly build up a private practice on the side, before fully committing to such a potentially risky endeavor. As the practitioner becomes responsible for all billing, collections, insurance, and employee decisions that must

be made (Barnett & Henshaw, 2003), he or she quickly realizes how important it is to be familiar with basic business principles—something many psychologists are not aware of or prepared for when entering this area. Some graduate schools are leading the way in providing courses that address the financial and business aspects of working in the field of psychology, covering aspects such as information and financial management, contract negotiation, and planning and budgeting for the future. It may be of great benefit for those interested in pursuing a career in private practice to take such business classes while in graduate school to be better prepared for the multiple financial and business responsibilities associated with working independently or as part of a private practice.

Others may experience a sense of professional isolation when practicing independently. Although this may be solved by deciding to enter into or form a group practice instead, the number of colleagues one interacts with is very limited in comparison to working in a larger institution, such as a university or medical setting. This also becomes a challenge when consultation on a case is necessary, as an independent practitioner may not have anyone to immediately turn to for advice. Such concerns may be addressed, in part, by maintaining involvement in a larger network of individuals in the same profession (state professional organizations, local treatment providers, fellow private practitioners in the area). Contact with these individuals may help ease isolation and provide a social outlet and support when necessary or desired.

When working independently it is important to remember that in many cases, one truly is working independent of others. This may require the psychologist to wear many hats, including those of accountant, public relations person, cleaning staff, insurance policy expert, receptionist, real estate agent, financial analyst, bill collector, and office manager (Vineberg, 2005). Access to clerical or support staff who can help with paperwork, correspondence, scheduling issues, and numerous other day-to-day operational tasks will affect how business is conducted in the office, as much of this work is necessary to keep a business running smoothly. All of these duties may be challenging by themselves, yet the larger burden may be the time spent ensuring that they are done correctly. Personal freedom may come at the cost of sacrificed time, and one may have to work long hours and through the weekend just to ensure that all necessary business details are covered. It may be important to remember here that these administrative duties are all above and beyond actually conducting therapy and performing one's duty as a clinical psychologist.

GROUP PRACTICE IN PSYCHOLOGY

If the disadvantages of independent private practice seem to outweigh the benefits, a viable alternative is joining or forming a group practice. Group practices can be small, composed of a few psychologists or other mental health professionals, or much larger organizations consisting of an interdisciplinary team of providers working collaboratively and sharing resources. In this manner, many of the potential downfalls of independent practice can be avoided, or at the very least reduced.

Joining a group practice allows one to experience many aspects of independent practice before actually committing to it. The daily exposure to aspects of running

a business, handling client records and paperwork, and other daily activities associated with work in a practice may provide a good first step in making an informed decision as to whether a future career in independent private practice is actually desirable. By joining a group practice and getting your foot in the door, it may be easier to see if this type of work is suitable and agreeable without having to commit large amounts of time and money to opening an independent practice, only to find out it is not the best professional setting for your needs.

As mentioned earlier, a rather large financial risk accompanies a solo venture into private practice. One of the main advantages to joining a group practice is that the financial risk, investment, and capital associated with beginning a practice is distributed among group members. This may be especially appealing given recent data showing that 74% of new psychologists in practice subfields are starting their professional life with a median debt level of $67,500 (Wicherski & Kohout, 2005). As high as this seems, the level of debt grows substantially higher for those graduating with a Clinical PsyD, with a median debt level of $90,000 and 79% of new graduates reporting some level of debt. Given these numbers, and knowing that it may take a substantial amount of time before a regular client base is established and income becomes steady, it is readily apparent why the idea of sharing financial burden and risk is appealing. When joining an already established group practice, a referral base may already exist, which eliminates a large degree of work and reduces the amount of stress associated with building a clientele from scratch. Along with defraying the initial start-up costs, it may be financially beneficial overall to be in group practice. Despite the fact that generated income may need to be split among members in the group practice, this is counterbalanced by having all members of the group split the cost of expenses generated in running the practice. Everything from rental and operating fees to paying office staff can be shared, whereas an independent practitioner must pay for all of these costs without help from others. To pay for many of these expenses, a solo practitioner needs to keep overhead costs at a minimum, and often will have to perform multiple roles within the office (e.g., receptionist, therapist, and custodian). As one author puts it:

> When you consider solo versus group practice, you eventually have to ask yourself whether you want to hand over a sizable portion of your revenue to cover expenses that are paid solely by you, whether you want to minimize these expenses somewhat and add more work and headaches to your already busy schedule, or whether you want to hand over a sizable portion of your revenue to others and let them take care of the headaches so you can focus your energy on being a clinician. (Habben, 2005, p. 101)

Working alongside others in group practice eliminates the disadvantage of professional isolation and provides other benefits as well. Colleagues may help in providing knowledge and expertise on a variety of issues, including how to run a successful practice, dealing with especially difficult cases, and providing resources for referrals in the community. Colleagues also provide friendship and may help with expanding one's social and professional network. Despite working alongside others, much of the freedom of independent practice still exists, as often one can still determine to a large degree the type and amount of patients seen.

Although it may appear as though group practice solves any and all disadvantages of independent private practice, some disadvantages remain. First and foremost, the level and degree of control over decisions being made are no longer solely in one person's hands in group practice. Decisions regarding the hiring and firing of staff, the fees charged to patients, questions about referrals and providers, and all other business decisions must now be made as a group and no longer individually. To some this may seem appealing; to others this loss of independence may be enough to turn them away from group practice.

Similar to independent practice, income from group practice may be variable. Although the potential for large earnings still exists, a lot depends on the overall strength and productivity of the practice as a whole, as well as your own work. Fluctuating levels of productivity will provide varying levels of pay across time, and this may cause a high degree of frustration in anyone who prefers a steady and stable paycheck.

Another disadvantage arises primarily when one is seeking another job, as working in a group practice does not carry with it the instant credibility and respect by association as working for a university may provide (Habben, 2005). Considering the low name recognition a group practice has outside of its own community, it may be hard to communicate to a future potential employer the exact nature of work conducted, the environment and conditions, and the skills needed to work successfully in the practice being left.

PREPARING FOR LIFE IN PRIVATE PRACTICE

Perhaps one of the most difficult aspects of beginning a private practice is the transition one must make in one's sense of identity, a transition from therapist to businessperson. As an independent therapist, you are providing and selling a service: therapy. Graduate school and internships do little to prepare the training psychologist for life in private practice, and it may be difficult at first to view therapy as a marketable service provided for a consumer. One of the first questions that must be asked is What do I have to offer? The service a clinical psychologist provides is a complicated blend of the therapist's personality, character, and previous experience; the theoretical or philosophical approach and orientation he or she espouses; and the techniques and procedures used that reflect this approach (Earle & Barnes, 1999).

CHALLENGES IN STARTING AND MAINTAINING A PRACTICE

One of the first concerns in establishing a private practice is the start-up cost, which will vary immensely depending on what approach is taken and what sort of practice is wanted. Deciding to practice as part of a group may help to defer some of the initial costs, the first of which may be choosing where the practice will be and renting adequate office space. Things to consider here are the size and cost of your office space, transportation to and from your office and how accessible it is to clients, and what clientele you may draw depending on where your practice is located. The decision of where to practice also carries with it the weight of deciding

where you would like to live—at least for some considerable time. Even though your skills and services may be transportable, clients are not, as they are unlikely to follow you to where you end up moving, whether that be across town or across the country.

Other initial costs include licensing fees, which will vary depending on one's state, and malpractice insurance, which will also vary depending on what coverage is desired. These fees relate to where you decide to practice, as requirements for licensing vary across the country. Substantial costs may be incurred in furnishing the office (everything from chairs to filing cabinets to children's furniture and toys), computers, assessment materials, business cards and office supplies, utilities, professional membership dues, taxes, and possibly advertising.

Of all initial start-up concerns, perhaps the most important is establishing a steady flow of referrals. This is especially difficult if one decides to enter private practice immediately after completing internship and before having a strong base of referral sources from which to draw clients. The need to begin forming professional relationships and contacts during the internship year or even earlier may be crucial to help lay the foundation for a later referral base, especially if the goal is to stay within that community. Forming ties with others in the community (schools, local pediatricians, other clinics) can help build reciprocal referral relationships and help make one's availability, services, and presence known in the area. Other mental health professionals may be vying for the same clients, making this a difficult process. Again, good business sense and a strong marketing strategy can make a substantial difference here in attracting new clients.

Numerous legal issues will inevitably arise as part of private practice, and it is imperative that all steps have been taken to prepare before the time comes. Good malpractice insurance is required, along with having a solid understanding of HIPAA regulations, licensing laws, and state regulations. Consultation with an attorney may be of great benefit as you establish your practice.

MANAGED CARE

The topic of managed care in the United States almost immediately invokes strong feelings among health care professionals, including practicing psychologists. Whether liked or not, it must be accepted at present as the medium through which a large proportion of services are delivered. The current system of managed care services can be seen positively as being a significant source of referrals for one's practice, offering clientele ranging from private direct pay to third-party payers, patients charged on a sliding scale, and pro bono cases. Unfortunately, the demand it places on both patients and providers creates an uneasy tension, ultimately changing the face of mental health treatment in ways that many psychologists find detrimental.

In a 1998 survey of almost 16,000 licensed psychologists, 4 out of 5 reported that managed care had a negative effect on their professional work, with independent practitioners and psychologists in medical settings reporting the most negative effects (Phelps, Eisman, & Kohout, 1998). Their concerns included how managed care is changing clinical practice, excess precertification requirements of managed care panels, decreases in income, and ethical dilemmas created by managed

care. The authors conclude that the "marketplace-driven demands for integrated services, large and diversified group practices, and greater accountability have created obstacles and declining market opportunities that threaten the very existence of traditional psychological practice" (p. 35). Another 1998 survey (Murphy, DeBernardo, & Shoemaker, 1998) of 442 members of APA's Division 42 (Division of Independent Practice) found similar results, with only 2% of respondents reporting that managed care practices had a positive impact on quality of care. Moreover, 80% indicated that caps on the number of sessions interfered with treatment, and 84% of respondents stated that managed care companies have control over aspects of patient care and treatment that clinicians should control. Most respondents indicated that managed care had led to inappropriate treatment, insufficient treatment, or both. More specifically, they believed that overall, the number of sessions had been reduced, flexibility and room for clinical judgment had decreased, premature termination had become common, there were too many restrictions on patients served, referrals for medication had increased, use of protocols for treatment and the use of treatments outside of their primary orientation had increased, and the amount of assessment time had decreased. The negative effect of managed care on assessment is also seen in a recent survey of over 400 psychologists, in which over half reported that choice of tests, the number of tests, and the time invested in testing have been negatively impacted (Rupert & Baird, 2004). Other aspects of managed care identified as being stressful on psychologists are external constraints on services, managed care reimbursement rates, and excessive paperwork (Rupert & Baird, 2004).

Various difficulties may arise when trying to balance the needs of managed health care and the needs of patients. At times, it may be difficult to know exactly who one is working for, client or corporation. One issue that arises in independent practice is the balancing of a patient's confidentiality and privacy with the need to keep third-party providers informed about the patient's progress and need for continued treatment. In the previously mentioned study, 75% of psychologists strongly believed that contact with managed care and utilization review compromises patient confidentiality (Murphy et al., 1998). Here, as in other areas of practice affected by managed care, there appears to be a struggle between what is required by managed care and what psychologists believe is the ethical course of action. In the opinion of one practitioner, "There is nothing private about private practice, at least when it is practiced with a managed care patient. Private practice is an oxymoron in the age of managed care" (Wolf, 1999, p. 114).

Previous work has suggested that the pressures of managed care on clinicians, such as loss of autonomy, the stress of dealing with ethical dilemmas, and feelings of frustration after having treatment recommendations denied, may eventually reduce morale and increase the potential for burnout among psychologists (Kisch, 1992; Wooley, 1993). Some smaller surveys of the impact of managed care in specific states further reveal how the pressures involved with managed care may be compromising effective and ethical practice. A 1998 survey of psychologists in New Jersey (Rothbaum, Bernstein, Haller, Phelps, & Kohout, 1998) found that those with high managed care caseloads reported an increase in paperwork, a decrease in the overall number of sessions, and a drop in morale. Perhaps more important,

they reported pressure to compromise quality of care and ethical principles under these circumstances. Higher proportions of managed care caseloads also have been associated with working longer hours, having more client contact, receiving less supervision, reporting more negative client behaviors, experiencing more stress, being less satisfied with income levels, and showing higher scores on emotional exhaustion (Rupert & Baird, 2004).

The implications of these findings serve to underscore the importance of preparing graduates interested in pursuing a career in private practice for the demands and ever-changing nature of managed health care in the area of mental health. Some of the major concerns and stressors indicated in the numerous surveys conducted over the past decade (i.e., dealing with paperwork, resolving ethical dilemmas related to disclosure of confidential information to third-party providers) are issues that are rarely, if ever, taught or discussed with graduate students. The increasing importance of documentation and record keeping in the field, due in large part to the increase in lawsuits and malpractice claims, also needs to be stressed to those preparing for private practice. What may be more difficult to convey is how difficult and stressful it can become for practicing professionals to stay up-to-date with their documentation and administrative duties, along with the substantial amount of time this may take. As this is generally not the reason why most practitioners go into private practice, the commitment to these side duties may become a large source of strain and frustration, and practitioners should be aware of and ready for this when entering practice.

THERAPIST BURNOUT AND IMPAIRMENT

Once in private practice, it may be difficult to ascertain exactly how much work and stress one can handle independently versus when help may be required. Multiple stressors exist in the practice of psychology, many of which can potentially cause serious distress or impairment if allowed to continue unchecked. Sherman and Thelen (1998) examined potential causes of distress and impairment for psychologists, comparing stress associated with life events and stress caused from work-related factors. The authors found that malpractice claims, changed work situations (e.g., different work responsibilities, major changes in working conditions or hours), inadequate time for all obligations, office politics, and conflicts in relationships with colleagues were all among the most distressing work-related factors in clinical practice. The greatest impairment at work was caused by malpractice claims and restrictions imposed by managed care companies. Questions have been raised as to whether psychologists can adequately perform their duties while they are distressed, and although this debate continues, research findings (Nathan, 1986) indicate high correlations between life distress and life impairment and work distress and work impairment. Distress was defined as "the interference in ability to practice therapy, which may be sparked by a variety of factors and results in a decline in therapeutic effectiveness" (p. 79). These data appear to indicate that many psychologists are not only experiencing high levels of distress, but their work, and therefore their patients, are also suffering as a consequence. The study also found that reducing the average number of client contact hours per week was not related to

improvements in work distress or impairment, and that working in a private practice (versus group practice) did not appear to be related to levels of work distress, impairment, or professional satisfaction. This information may be especially salient for young professionals who are just beginning their practice, as their level of stress is likely to be high due to the inherent challenges associated with beginning a private practice in psychology.

Another potential cause of professional impairment that may be difficult for the independent practitioner to become aware of is burnout, which has three defining features: (1) emotional exhaustion, (2) depersonalization (defined as a negative and cynical attitude about one's clients), and (3) decreased sense of personal accomplishment and/or an increased likelihood of evaluating oneself and one's work negatively (Rupert & Morgan, 2005). The consequences of burnout impact both the therapist and clients, often resulting in impaired delivery and quality of services and significant stress to all individuals involved. Although research on burnout among psychologists has provided mixed results, a consistent finding is less burnout and lower levels of stress among independent practitioners when compared to agency-employed psychologists (e.g., Ackerly, Burnell, Holder, & Kurdek, 1988; Farber, 1985; Hellman and Morrison, 1987; Raquepaw and Miller, 1989; Vredenburgh, Carlozzi, & Stein, 1999). Different reasons have been hypothesized for why this finding exists, including that independent practitioners have greater control over their work activities and environment, they have fewer bureaucratic responsibilities, and they often have more say in choosing which clients they see and thus potentially see less disturbed clients. These reasons may have been true at the time these studies were conducted, but the more recent emergence of managed health care has significantly changed the environment of independent practice by reducing the amount of control and autonomy practitioners have, along with increasing financial uncertainty, creating more ethical challenges, and increasing therapist workloads. Recent studies examining burnout report similar overall rates among practitioners in comparison to studies conducted prior to the advent of managed care (Rupert & Morgan, 2005). This seems to suggest that although the health care environment may be changing in numerous ways, it does not appear to be increasing levels of burnout among practicing professionals. This does not suggest that burnout and emotional exhaustion are not serious problems faced by psychologists, as roughly 44% of respondents in the Rupert and Morgan investigation scored within the high burnout range.

Emotional exhaustion is a critical first sign of burnout. Factors related to emotional exhaustion include total hours worked, administrative and paperwork hours, managed care client percentage, negative client behaviors, and overinvolvement with clients. It is imperative that professional practitioners make efforts to safeguard their own mental health in order that they may have the energy and cognitive resources to be of maximum benefit to their clients. Potential strategies to help reduce or prevent burnout include increased self-monitoring and awareness of one's own levels of stress and symptoms of burnout, increasing one's social support network and working relationships with colleagues in the field, attending professional development workshops and seminars on prevention of burnout, and becoming involved in state or nationwide professional organizations that may provide help or support in areas causing the greatest amount of stress.

ETHICAL AND LEGAL ISSUES IN INDEPENDENT PRACTICE

Conflict of Interest

Certain pressures exist in private practice that are unique to this setting, one of the most notable being the pressure to keep one's clients happy. This is important for a variety of reasons—not only therapeutically, but also from a business perspective. In other settings, a client's deciding to leave therapy may have detrimental effects on the progress of therapy and that client's recovery. The same is true for a client terminating therapy when seeing a private practitioner, but now there is an added financial aspect from the point of view of the therapist. An unhappy client who terminates has an immediate impact on the income of the practice, as well as having the potential to affect new referrals if that client chooses to make his or her unhappiness known to friends, family, and colleagues. The issue may be even more complex in the practice of clinical child psychology, as the client typically encompasses the family, which may include two parents (who may disagree on the course of therapy) and siblings, as well as the child or adolescent for whom the referral was initially made. As many child referrals come from school systems, the satisfaction of teachers and school counselors also may be relevant factors. Within larger organizations where financial and legal responsibilities are largely shouldered by others, the practitioner may feel more comfortable addressing clients directly than might a professional in solitary practice (Applebaum, 1992). The importance of this matter is also reflected in the kind of clients an independent practitioner chooses to accept. One survey has shown that approximately 30% of practicing psychologists have avoided taking on certain clients due to fear of being sued (Pope, Tabachnick, & Keith-Spiegel, 1987). In this case, not only is there a fear of having a client terminate prematurely with a potential loss of income, but the therapist is acting defensively and preemptively to protect his or her own interests for fear of being drawn into a lawsuit. Although such fear is understandable in some ways, this still leaves the potential client in the community without active treatment—an ethically questionable situation that may not be such a great concern for psychologists who are not acting in constant fear of litigation.

Fees

Another issue concerns decisions regarding appropriate charges for services. Managed health care and insurance complicates this matter even further, yet the basic need to set appropriate fees still exists and the practitioner is left with a difficult decision to make. With only a certain amount of time and space for clients, decisions must be made in respect to who will be seen. Will the therapist choose to see clients on a first-come-first-served basis, give preference to cases he or she finds most interesting, or accept only those who have the greatest ability to pay? Apart from client selection considerations, the dilemma about specific fees is an obvious one: Charge too little and the practice may not survive despite having many clients; charge too much and risk having too few clients to pay the bills at all. Some practitioners may consider charging clients on a sliding-scale system, where fees are

based on the client's income level. However, the problem of making enough income to keep the business afloat will continue if the majority of clients are being charged the minimal fee allowed. Many clients may still not be able to pay the lowest fee allotted, leaving the therapist to decide how much time can be spent doing pro bono work. Further, regulations exist restricting differential fees for clients covered by insurance compared to those who are not covered.

The complication of reimbursement by insurance raises ethical issues for many private practitioners. In a 1987 survey of 456 practicing APA psychologists, Pope et al. found that over half the respondents reported altering an insurance diagnosis to meet insurance criteria—in other words, committing insurance fraud—either rarely (26.5%) or more frequently (35.1%). A more recent report indicates that 49% of psychologists believed other psychologists were altering patient diagnoses to help receive reimbursement, and 59% believed others were altering diagnoses to help patients get reimbursed (Murphy et al., 1998). This illegal and unethical behavior may be occurring more frequently in private practice settings, where the need to retain clients is high and oversight is low.

Termination

Similar to the dilemma of trying to keep the client happy and remain in therapy, the decision about when to terminate may cause a certain ethical conflict between client and therapist. Even if treatment goals are set out at the beginning of therapy, whether they are defined as reducing symptoms or achieving more broad life goals, the therapist is put in the difficult position of having to willingly terminate not only a client, but also a source of income for the practice. This dilemma can actually work both ways, with some therapists terminating treatment early if the client is particularly difficult or unrewarding to work with, and others prolonging therapy with clients who may have achieved all they initially set out to when entering into therapy. Again, the issue of personal needs versus client needs enters into the decision-making process while in private practice, with the potential consequences of these decisions affecting client and therapist, both monetarily and emotionally.

Leaving Practice

Psychologists choose to change their professional positions for many reasons. Whether due to financial difficulties, taking a new position elsewhere, disputes with colleagues, or simply a desire to discontinue practicing psychology, many psychologists will at one point in their career face the reality of leaving their current position or workplace. This may be an especially difficult situation for a private practitioner, who faces multiple legal and ethical dilemmas when leaving private practice. Professional ethics codes, including that of the APA (2002), discuss the importance and obligation for clinicians not to abandon their clients. As Barnett (2001, p. 183) succinctly states, "Simply leaving a practice without addressing each patient's treatment needs, arranging for a smooth transition if transfer of care is in order, or preparing patients for termination would appear contrary to the intent of relevant case law and the ethics codes of all mental health professions." Steps must be taken to ensure that the client's needs have been fully documented, met, and

followed up on when a transition takes place between therapy providers. This may be especially important, not only ethically but legally, for clients who present with any sort of imminent crisis or risk. In all circumstances, it is best to anticipate and avoid potential problems with terminating or transferring clients rather than deal with the consequences of inappropriate action after the fact.

When working with other professionals, it is important to clearly outline legal responsibility prior to entering into the business relationship, seeking consultation and guidance on noncompetition clauses, record ownership, and the disposal of client records. Noncompetition clauses prohibit a departing psychologist from setting up another practice for approximately 1 to 3 years within a certain geographical area. This essentially protects a practice from a psychologist who "use[s] the practice's reputation, resources, and referral sources and then, after establishing a local reputation, leaves the practice and opens an office in the local area, taking along the patients the practice provided" (Barnett, 2001, p. 184). Record ownership generally remains with the practice and must be maintained there in accordance with state laws and ethical standards. If the client chooses to seek treatment elsewhere, copies of records can be sent to the new therapist, but the original client record with treatment reports and notes remain the property of the practice. Prior to such events, detailed plans should be made concerning the retention, storage, and potential destruction of records in accordance with state laws and regulations. Practitioners should pay special attention to the required retention times of different types of patient records, both for liability reasons and in the event a client chooses to return at a later time. This may be especially important with underage clients, whose files are required to be stored until 3 years past their age of majority. Knowledge of applicable laws and regulations is of utmost importance, as these may differ across states. Specific requirements now exist in some states outlining the proper course of action should a client's file need to be destroyed, including the notification of the former patient in writing that his or her records will be destroyed, the date of the planned record disposal, and how the record or a summary of it may be obtained prior to that date.

Rural Practice

One of the first things that one must consider in establishing a private practice is where the practice will be located. A basic distinction can be made between deciding to open a practice in an urban or a rural setting. When choosing the latter, a number of ethical issues arise specific to practicing psychology in a rural community. A recent study (Helbok, Marinelli, & Walls, 2006) found that psychologists in small towns or rural areas are significantly more likely to encounter and engage in multiple relationship behaviors and situations, have a higher degree of visibility (feeling as though they are therapists 24 hours a day, running into clients in the community, participating in events in which clients are also participating), are more likely to prepare their clients for chance encounters, and are more likely to learn information about the client from sources other than the client. A private practitioner choosing to open practice in this setting must be willing to be known as an individual and a member of the community and not just as a therapist. The alternative is to restrict one's social activities by avoiding local establishments, which may not even be a

viable option in very isolated communities. Even if avoidance is a possibility, the personal toll this may take on the psychologist should be considered; therapists should not strive to isolate themselves from the community in which they live. In rural settings, clients are likely to know more about a therapist's private life and daily activities outside of practice, which may be difficult for some practitioners to accept and should be considered before choosing to operate in such settings.

Aside from operating within a small community defined by geographical and population standards, psychologists may also practice within a small community of persons who share something in common, whether that be religion, ethnicity, personal disability, sexual orientation, or a special interest. Maintaining boundaries may be difficult enough in an urban setting, but being a member of a small and visible minority group within a rural community may present even greater difficulties to the practicing psychologist. Here, where personal interests and beliefs may become known to the community, the psychologist may need to be sensitive to integrating these in the therapy setting, where a client's beliefs may be contrary to the therapist's and potentially interfere with the therapeutic relationship. Difficulties exist in striking a balance between being accepted in the community as a person and maintaining professional boundaries.

Boundaries of Competence

As mentioned previously, the financial burdens related to private practice require a certain amount of income to be maintained to keep the practice operating. This pressure, in some cases, may tempt therapists to practice outside of their boundaries of competence. This presents a complicated ethical dilemma, one where more shades of gray may exist than clear-cut answers. The APA Ethics Code (APA, 2002, p. 1063) states, "Psychologists provide services, teach, and conduct research with populations and in areas only within their boundaries of competence, based on their education, training, supervised experience, consultation, study, or professional experience." No clear definition of competence is given in the Code; however, Knapp and VandeCreek (2003, p. 49) assert that the "general rule is that psychologists can ascertain if they have become proficient in a certain area of practice after submitting their work to external feedback," although they acknowledge that this becomes difficult for practitioners to achieve once they have left their doctoral program and are practicing independently. Complicating matters further, the Ethics Code allows for provisions to this rule, whereby psychologists can provide services with different populations, techniques, areas, or technologies as long they "undertake relevant education, training, supervised experience, consultation, or study" (p. 52).

Faced with these dilemmas, it may be especially difficult for independent practitioners to know exactly when they are practicing outside of their area of competence. A balance must be struck between keeping a steady flow of new clients, being open to accepting people of varied diversity and backgrounds with different presenting problems, and ensuring that the skills necessary to treat these individuals are present and do not stretch the therapist's competency boundaries too far so that treatment itself becomes unethical, potentially causing more harm than benefit. With little or no external feedback, the independent practitioner must be able to evaluate his or

her own competencies, abilities, and weaknesses and make firm decisions as to who can be seen and who would be better suited by being referred to another mental health professional.

Cultural Competency

A recent sample of professional psychologists (Hansen et al., 2006) revealed that a large proportion are practicing culturally sensitive therapy with their clients. Respecting a client's worldview and individuality, being aware of personal and societal biases (both positive and negative), establishing rapport in racially/ethnically sensitive ways, and considering the impact of race/ethnicity in diagnosis were all endorsed by at least 80% of those responding. Other areas suggest need for improvement: 50% of the sample rarely or never made efforts to improve their multicultural competencies (by creating a plan and seeking consultation or feedback), negotiating therapist-client language differences, using various resources to augment treatment (literature, translators, and indigenous healers), referring a client to a more qualified provider, and using racially/ethnically sensitive data-gathering techniques. The authors of the survey provide some hypotheses as to why these behaviors are not occurring more frequently, most notably that it may not necessarily be lack of knowledge, but rather that clinicians may be constrained by institutional or managed care mandates. Many of the behaviors listed as not being practiced would require substantial time and effort on the part of the practitioner to remedy, time and effort that may simply not be available given the stressful nature and demands of private practice. Alternatively, the authors suggest that it may be an issue of choice, as 51% of respondents indicated that they believe they were very or extremely multiculturally competent. Considering themselves experts, these practitioners may not believe they need to be any more sensitive or competent than they already are. This study further found a significant difference in what the psychologists practiced and their beliefs, indicating that they did not consistently behave in a manner congruent with what they believed is important for competent practice with a multicultural population (Hansen et al., 2006).

Given this knowledge, it is important for independent practitioners to ensure that they are continually taking steps to improve their cultural sensitivity and awareness rather than simply believing they are competent in this area. As supervision and consultation are often difficult to come by when practicing independently, practitioners should take steps to create a professional development plan that addresses any weaknesses that may detract from their ability to provide psychological services to minority groups in their community. Other behaviors that may help include "increasing the amount of culture-specific case consultations, reading relevant literature, developing a referral network of competent clinicians and translators, and learning about indigenous resources" (Hansen et al., 2006, p. 72).

SPECIAL CONSIDERATIONS WITH CHILD CLIENTS

One area of independent practice that merits further attention is that of child and adolescent populations. The number of children and adolescents with emotional and mental disorders ranges between 14% and 22% (Brandenburg, Friedman, & Silver,

1990), yet the number of private practitioners whose clinical focus is on children and adolescents remains low, and some evidence points to a national shortage of clinical child psychology specialists (Hanley, 1994; Tuma, 1989). Culbertson (1993, p. 119) states, "The United States continues to have serious deficiencies in the delivery of its mental health services for children. . . . An obvious problem is personnel shortage—there simply are not enough mental health professionals training to work with children." Overall, a need exists for greater numbers of practitioners and improved quality of mental health care for children, as the majority of the 8 million children in need of some form of mental health treatment will go unseen (Roberts, 1994).

Given this state of affairs, there is a pressing need to increase the number of graduates from clinical child psychology specialty programs. The APA defines clinical child psychology as

> a specialty of professional psychology which brings together the basic tenets of clinical psychology with a thorough background in child, adolescent and family development and developmental psychopathology. Clinical child and adolescent psychologists conduct scientific research and provide psychological services to infants, toddlers, children, and adolescents. The research and practices of Clinical Child Psychology are focused on understanding, preventing, diagnosing, and treating psychological, cognitive, emotional, developmental, behavioral, and family problems of children. Of particular importance to clinical child and adolescent psychologists is a scientific understanding of the basic psychological needs of children and adolescents and how the family and other social contexts influence socioemotional adjustment, cognitive development, behavioral adaptation, and health status of children and adolescents. . . . A defining feature of Clinical Child Psychology is the emphasis on understanding children from the perspective of both normal development and of psychopathology. Particular attention is given to specialty-specific skills for working with such diverse populations and an appreciation for those cultural, ethnic, and other diversity-related factors that must be considered in clinical practice. (APA Commission for the Recognition of Specialties and Proficiencies in Professional Psychology, 2005)

While general practitioners continue to provide a valuable service, we contend that to best meet the needs of today's youth, professionals working with substantive child or adolescent caseloads should be trained specialists.

As alluded to previously, presenting problems of children and adolescents often warrant contact with many individuals beyond the specified client. Academic performance problems and disruptive behavior in school settings are common presenting complaints. Competent assessment and intervention necessarily warrant involvement of parents as well as school personnel. Further, observations in school or home settings may be particularly advantageous. Unfortunately, independent practitioners may find it difficult to take the time for visits off-site, and equally problematic is the difficulty in obtaining reimbursement for such services. We encourage professionals to continue to lobby for appropriate funding for empirically supported assessment and intervention strategies to properly serve child and adolescent populations. In the meantime, professionals may elect to make consistent use of school-home notes and phone calls to relevant parties (see Kelly & McCain, 1995).

SUMMARY

Professional psychology has undergone significant advances since its inception. Research efforts have allowed us to continually improve our ability to accurately identify and provide treatment for those in need of services. The practice of professional psychology may be as varied as the number of individuals engaged in the pursuit. Although no single theoretical platform has been adopted, the prevailing sentiment is toward increased accountability and primary implementation of empirically supported strategies. Future efforts must be directed toward increasing the number of clinical child specialists, as the demand currently exceeds availability and the shortfall is expected to increase as the number of children and adolescents suffering from mental health problems continues to rise worldwide. In addition, we would be remiss if we did not highlight the pressing need for training in business management practices for those planning to undertake careers in independent practice. The many challenges and hours of hard work invested in a career in private practice can ultimately result in an exciting and rewarding career path, with the long-term benefit of professional independence and flexibility in one's work as a practice grows over time. Though professional psychologists are no doubt confronted with challenges on a regular basis, the vast majority report that the perceived benefits far exceed the disadvantages. For many of us, seeing just one child or adult improve, either through the efforts of our research or through direct service provision, makes all the years of training worthwhile.

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CHAPTER 4

Ethical and Legal Issues

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This chapter is divided into two sections. The first addresses the Ethical Standards and Guidelines of the American Psychological Association (APA) and its committees as these pertain to children and adolescents. It is governed by Table 4.1, which lists the APA documents. The second section focuses on the investigative and practice guidelines published by national organizations such as the American Prosecutors Research Institute (APRI), the American Professional Society on the Abuse of Children (APSAC), and the American Academy of Child and Adolescent Psychiatry (AACAP). Comparative research reports relevant to the interface of clinical and forensic child and adolescent psychology are cited in this section. This discussion may aid the psychologist seeking to determine when a guideline assists or may compromise professional practice.

ETHICAL STANDARDS AND GUIDELINES OF THE AMERICAN PSYCHOLOGICAL ASSOCIATION

Fifteen publications of the APA that contain statements of ethical standards and policy and practice guidelines have been reviewed. Table 4.1 identifies these documents and which of these speak to children, adolescents, and their families. For a psychologist seeking guidance regarding an ethics or practice guideline issue, the document's name provides some orientation. No one guideline is an exclusive source for ethical standards and principles on a particular topic. Hence, familiarity with them all is important. Legal standards vary from state to state and are found in state statutes and case law. Both of these topics are beyond the scope of this chapter. State resources may be particularly helpful in assisting psychologists to find and apply ethical guidelines to particular cases and issues. Some state psychological associations publish a compendium of laws affecting the practice of psychology.* State licensing boards and associations may provide, on request, a listing and copy of laws defining the practice of psychology in that state. The ethics chair of the state psychological association may be of help with the same issues and also in identifying attorneys who specialize in defending psychologists in grievance and malpractice actions. Malpractice insurance carriers have an interest in assisting psychologists to practice

See A Compendium of Laws Affecting the Practice of Psychology (1987) Connecticut Psychological Association.

Table 4.1 Ethical Principles, Standards, and Policies Published by the American Psychological Association

Document Name	Applies
Ethical Principles of Psychologists and Code of Conduct (APA, 2002c)	✓
Determination and Documentation of the Need for Practice Guidelines (APA, 2005a)	
Guidelines for Psychological Practice with Older Adults (APA, 2004b)	
Record Keeping Guidelines (APA, 1993)	
Guidelines for Child Custody Evaluations in Divorce Proceedings (APA, 1994)	✓
Guidelines for Psychological Evaluations in Child Protection Matters (APA, 1999)	✓
Guidelines for Psychotherapy with Lesbian, Gay, and Bisexual Clients (APA, 2000)	✓
Guidelines on Multicultural Education, Training, Research, Practice, and Organization Change for Psychologists (APA, 2002d)	✓
Criteria for Evaluating Treatment Guidelines (APA, 2002a)	
Criteria for Practice Guideline Development and Evaluation (APA, 2002b)	
Guidelines for Providers of Psychological Services to Ethnic, Linguistic, and Culturally Diverse Populations (APA, 1990)	✓
APA Resolution on Outpatient Civil Commitment (APA, 2004a)	✓
Training Grid Outlining Best Practices for Recovery and Improved Outcomes for People with Serious Mental Illness (APA, 2005c)	
Policy Statement on Evidence-Based Practice in Psychology (APA, 2005b)	✓
Specialty Guidelines for Forensic Psychologists (1991)	✓

Note: Applies = Applies to children, adolescents, and their families.

defensively and to avoid litigation. For practicing psychologists who are insured, contacting your carrier for recommendations and for standards and case law that affect your practice can be helpful. Sometimes an attorney will offer a legal opinion.

Ethical standards and guidelines are often revised and updated. There is no one location at the APA to access all of its guidelines and statements regarding ethical conduct and professional guidelines. This also is true of the APA web page. It can be challenging to become informed and to stay abreast of national and state laws and professional standards and guidelines that effect clinical and forensic practice.

1. The Ethical Principles of Psychologists and Code of Conduct

The Ethical Principles of Psychologists and Code of Conduct (APA, 2002c) is the major ethics guideline for psychologists in the United States. It does not contain subsections that specifically refer to adolescents and children. The general principles and ethical standards apply in general to professional activity regardless of client's age. For professionals serving adolescents and children, the principles and standards are clear. It is expected that the professional will have appropriate training and experience; will be knowledgeable about assessment instruments and procedures and their strengths and limitations; will avoid multiple relationships, sexual involvement, and questionable treatments and practices; and will obtain informed consent from authorized guardians and others for professional contact with minors.

The Ethics Code applies to the activities of psychologists in their scientific, educational, and professional roles, and includes the clinical, counseling, and school practice of psychology, research, teaching, trainee supervision, and professional activities of psychologists in the public sector. The Ethics Code applies to activities in a number of contexts, including in person, postal, telephone, and Internet communications. Psychologists are advised to consider the Ethics Code of the APA as well as applicable laws and psychology board regulations and decisions. Psychologists are urged to be alert to potential conflicts between the Ethics Code, their own conscience, and state and federal law, and to seek to resolve any such conflicts in a responsible manner.

The APA Ethics Code lists five general principles that are "aspirational" in nature, that is, which have the intent to inspire psychologists to the highest ethical ideals. These aspirational principles are not meant to be a basis for imposing sanctions. The principles include striving to benefit those with whom one works; establishing relationships of trust; promoting accuracy, honesty, and truthfulness in professional conduct; recognizing that fairness and justice entitle all persons access to the benefits of psychological practice; and respecting the dignity and worth of all people as well as their rights to privacy and confidentiality. Psychologists are urged to be mindful and respectful of individual differences based on age, sex, gender identity, race, ethnicity, culture, national origin, religion, sexual orientation, disability, language, and socioeconomic status and, with these sensitivities and knowledge in mind, to strive to eliminate bias from professional practice.

Ten ethical standards are defined in the Ethics Code. These are:

1. Resolving ethical issues
2. Competence
3. Human relations
4. Privacy and confidentiality
5. Advertising and other public statements
6. Record keeping and fees
7. Education and training
8. Research and publication
9. Assessment
10. Therapy

Each of these ethical standards has a number of subsections.

Ethical Standard 1

Psychologists are required to take reasonable steps to correct or minimize the misuse or misrepresentation of their work. They are expected to abide by the law and take steps to resolve conflicts when conflicts between the law, professional practice, and other ethical standards and regulations are perceived. This includes recognizing conflicts between ethics and organizational demands and resolving them, attempting informal resolution of ethical violations with colleagues, and reporting

ethical violations when professional practice has been found to cause harm or there is concern that it may do so. Psychologists are expected to cooperate with ethical investigations and proceedings, not file or encourage the filing of frivolous ethics complaints or without good cause discriminate against someone who has been the subject of an ethics complaint.

Ethical Standard 2

The ethical standard of competence requires psychologists to guide their professional practice based on the limitations of their education, training, supervised experience, consultation, study, and professional experience. Psychologists are required to obtain the training, experience, consultation, and/or supervision to ensure the competency of their professional practice according to the characteristics of the population they serve. They are expected to take "reasonable steps to ensure the competence of their work" and to protect those who receive their services. When working in a forensic role, psychologists are expected to become "reasonably familiar with the judicial or administrative rules governing their roles."

Psychologists are expected to make ongoing efforts to maintain and increase their competence, ensure that their professional practice is based on scientific principles and professional knowledge, delegate work appropriately and only when there is reason to expect it can be performed competently, and avoid the violation of ethical principles and licensure laws in such delegation.

Ethical Standard 3

Psychologists are expected to avoid unfair discrimination, sexual harassment, other forms of harassment, doing harm through professional practice, and conflicts of interest and multiple relationships. Multiple relationships that would "not be reasonably expected to cause impairment or risk exploitation or harm are not unethical." Distinguishing when a multiple relationship, a conflict of interest, and the potential for harm may or may not exist is not explained. Psychologists may want to seek peer and other consultations to clarify ambiguous situations. Psychologists are expected to avoid exploitive relationships, cooperate with other professionals, seek informed consent for their work, and clarify the implications of third-party requests for services that are court ordered or mandated. Also, the psychologist informs the recipient about the nature of the services and about limits of confidentiality. Psychologists who deliver services in an organizational setting are subject to a number of ethical standards that extend the requirements of informed consent, definition of client, issues of confidentiality, and the definition of organizational versus individual rights and responsibilities. Psychologists are expected to make reasonable efforts to plan for continuation of services when service delivery is interrupted.

Ethical Standard 4

Psychologists have an obligation to take reasonable precautions to protect confidential information developed during the course of professional practice. This includes discussing the limits of confidentiality with clients and patients, obtaining

permission before recording the voice or image of a client, minimizing intrusions on privacy through professional activity, and obtaining appropriate consent for the disclosure of confidential information. Provisions are made for the disclosure of confidential information without consent. This arises in situations that are mandated by law as needed to perform professional services, to obtain appropriate professional consultation, to protect those involved from harm, and to obtain payment. Disclosure is limited "to the minimum that is necessary to achieve the purpose." Psychologists are expected to disguise the person or organization when using confidential information for public purposes (publications, lectures, etc.) unless there is written consent or legal authorization.

Ethical Standard 5

In advertising and other public statements, psychologists are required to avoid false and deceptive statements both directly and by others whom they have engaged to represent them. Psychologists adhere to the Ethics Code in media presentations and should not solicit testimonials from current therapy clients or exploit the vulnerability of persons who may be actual or potential clients for other business ventures.

Ethical Standard 6

Psychologists are expected to maintain satisfactory documentation of their professional activities and the confidentiality of records, facilitate the appropriate transfer of records, and may not withhold records that are required for emergency treatment because payment has not been received. Psychologists are expected to reach payment and billing arrangements as soon as it is feasible in a professional relationship, state fees clearly, discuss service limitation that may result from limitations in financing as soon as is feasible, inform clients of the intended use of collection agencies or other measures to collect fees, and offer an opportunity for prompt payment. Barter in return for professional services is allowed if the barter is not clinically contraindicated and the arrangement is not exploitive.

Ethical Standard 7

In education and training activities, psychologists are required to uphold APA ethical standards. In the design and delivery of education and training programs, APA standards for licensure, certification, and other goals must be met; program content must be accurately described, including specification of training goals and program completion requirements. Course content must meet ethical standards for professionalism and scientific merit. Unethical demands are not to be made on students. When therapy participation is a program or course requirement, students are allowed to select therapy from practitioners unaffiliated with the program. Here, too, dual relationships are avoided.

Ethical Standard 8

Ethical standards should be maintained in research and publication. This includes obtaining informed consent when this is required and protecting research participants from participation and nonparticipation. Conditions are specified for dispensing with informed consent requirements. These include research that would not be expected to create distress or harm, that involves the study of normal phenomena, and that offers anonymous report forms that do not place participant information or a person at risk. Excessive and inappropriate financial or other rewards for research participation should be avoided. Deception in research should also be avoided, though it can be justified under some circumstances. Psychologists must provide research participants with a debriefing opportunity and information. When they become aware that the research has harmed a participant, reasonable steps must be taken to minimize the harm. Animal research should be conducted in compliance with federal, state, and local laws and regulations and also with professional standards. Psychologists do not plagiarize or fabricate research results. Psychologists must accept publication credit only when it is deserved for work done and contributions made. The position of principal authorship is based on contribution and not status.

Ethical Standard 9

Psychologists provide opinions on the psychological characteristics of people only after proper evaluation and define factors that limit the nature and extent of their conclusions and recommendations. Sources of information are defined. Psychologists use assessment instruments with known validity and reliability for the population evaluated. Language preference and other factors are considered in assessment instrument selection. Assessment instruments are administered, scored, and adapted according to instrument and research guidelines. Informed consent is obtained for assessment and also for the use of interpreter services. This information is conveyed in the report. Issues of confidentiality are maintained. Impact assessment of interpreter involvement is included in a discussion of data limitations. Situational, personal, linguistic, and cultural characteristics of clients as well as test-taking ability are considered along with data characteristics when considering data interpretation and validity. The norms, validity, reliability, and any other special qualifications for the use of assessment and scoring services are acknowledged. Psychologists retain responsibility for test selection, scoring, and interpretation when these aspects are delegated to other persons or to an automated service. Psychologists take reasonable steps to ensure that explanation of assessment results with appropriate qualifications occurs. Test security is maintained.

Ethical Standard 10

Psychologists obtain informed consent for therapy; provide information about the course of therapy, fees, third-party involvement, and limits of confidentiality; and provide the opportunity for questioning and receiving answers. They explain the potential risks of therapy and the availability of alternative treatments and disclose

therapy techniques that do not possess established results. When the therapist is a psychologist in training and is supervised, this is disclosed and the name of the supervisor is provided. When psychologists become involved in treating couples and families, the psychologist defines the professional relationship of each person, identifies which person is the client, and avoids role conflict, such as being a couples or family therapist and then a witness for one party in a divorce proceeding. Issues of confusion and conflict are also resolved in a timely fashion when deciding whether to offer services to persons receiving mental health services elsewhere. Psychologists do not provide treatment to persons with whom they have had sexual intimacy and do not have sexual intimacy with current therapy clients. Psychologists do not engage in sexual intimacy with former clients for 2 years after the termination of therapy. Even after the passage of 2 years, a psychologist may be challenged to demonstrate that a sexual involvement with a former client is consistent with the APA Ethics Code.

Therapy is terminated appropriately, such as when the service is no longer needed or is producing no benefit or when the client might be harmed by its continuation. A psychologist may terminate therapy when threatened. Psychologists provide pretermination counseling when feasible and also suggest alternative service providers as appropriate.

2. Determination and Documentation of the Need for Practice Guidelines

Determination and Documentation of the Need for Practice Guidelines (APA, 2005a) is an aspirational document in intent and references statements that are recommended for professional behavior and conduct but that are not mandatory and do not require an enforcement mechanism. Their purpose is to assist the psychologist to provide high-quality services. The Committee on Professional Practice and Standards identified a need for legal and regulatory, public benefit, and professional guidance. This document reflects the recognition that laws, court decisions, professional interaction with the legal system, and changes in regulatory and administrative systems may affect the practice of psychology. For example, *Daubert v. Merrill Dow Pharmaceuticals*, 509 U.S. 579 (1993), defines how expert opinions are to be delivered in court and evaluated there. Guidelines for Child Custody Evaluations in Divorce Proceedings (APA, 1994), Guidelines for Psychological Evaluations in Child Protection Matters (APA, 1999), and Specialty Guidelines for Forensic Psychologists (APA, 1991) are intended to assist the psychological practitioner when interacting with the legal system in the defined legal contexts. This document found that practice guidelines were needed to improve service delivery; avoid harm; meet the needs of emerging, underserved, and vulnerable client populations; and assist psychologists in responding to public policy initiatives. Psychologists may also seek guidelines for new areas of technology, for work in new contexts and emerging areas of practice with new approaches based on scientific advances, and with risk management issues.

3. The Guidelines for Psychological Practice with Older Adults

The Guidelines for Psychological Practice with Older Adults (APA, 2004b) does not apply.

4. Record Keeping Guidelines

Record Keeping Guidelines (APA, 1993) speak to an important area of psychological practice with children and adolescents. Formulated in 1993, these guidelines provide an introduction to documentation issues that receive specialized attention in later sections of this chapter, particularly in reference to forensic practice and documentation of investigatory procedures. Psychologists have many reasons for maintaining records, and the nature and extent of the documentation may vary according to the service performed. Records provide a history and offer a basis for case review, for planning, and for implementation of services. The General Guidelines for Providers of Psychological Services determined that psychologists hold widely varying rules on the wisdom of recording the content of the psychotherapeutic relationship. With respect to record keeping, a 50-state review of laws governing psychologists found substantial differences among the states on this issue. There are differing requirements in areas of professional practice between situations calling for assessment and those calling for psychotherapy.

Well-documented records may help psychologists in professional liability cases where the professional action of a psychologist is at issue.

Records include information about the nature, delivery, progress, and results of psychological services. Because of their nature, they can be reviewed and duplicated. Records of psychological services should minimally include identifying data, dates of services, types of services, fees, assessments, intervention plans, consultation, summary reports, testing reports, release of information, consent forms, and other supporting data. Psychologists are expected to maintain a reasonable degree of accurate, current, and pertinent records with sufficient detail to permit an understanding of the service by another psychologist.

Psychologists maintain the confidentiality of records, make reasonable efforts to protect against their misuse, develop reliable record keeping procedures, and strive to assure that record entries are readable and completed in a time that is proximate to the service delivery, if not simultaneous with it. Records may be maintained in a variety of formats and media. In the absence of laws and regulations to the contrary, these guidelines offer the following time periods for record maintenance:

- Complete records are maintained for a minimum of 3 years.
- Thereafter, a summary is maintained for an additional 12 years.
- If the client is a minor, the record period is extended until 3 years after the age of majority.
- A simple 7-year requirement for the retention of the complete record is preferred. The practitioner is urged to consult specialty guidelines in this regard.

5. Guidelines for Child Custody Evaluations in Divorce Proceedings

Guidelines for Child Custody Evaluations in Divorce Proceedings (APA, 1994) is one of two APA guidelines that address professional practice issues with children and their guardians and parents in legal contexts. Child custody and parenting issues arise in several legal contexts, including parental divorce, guardianship, abuse and neglect proceedings, and termination of parental rights. These guidelines were written specifically for the context of parental divorce, are aspirational in intent, are not mandatory or exhaustive, and have the stated purpose of promoting the efficiency of psychological expertise when conducting child custody evaluations. Psychologists can provide an important service to the courts and to children and families by providing "competent, objective, impartial information in assessing the best interests of the child." Psychologists performing child custody evaluations are expected to possess specialized competencies in the evaluation of adults, children, relationships, and parenting ability. Specific guidelines are discussed under three headings:

- Purpose of a Child Custody Evaluation
- Preparing for a Child Custody Evaluation
- Conducting a Child Custody Evaluation

There are 16 guidelines.

Guideline 1

The primary purpose of the evaluation is to assess the best psychological interests of the child.

Guideline 2

The child's interests and well-being are paramount.

Guideline 3

The focus of the evaluation is on parenting capacity, the psychological and developmental needs of the child, and the resulting fit. This involves focused assessment of parenting capacity in connection with the psychological characteristics and developmental needs of each involved child. Consequently, both adult and child assessments occur, and these include the assessment of the functional ability of each parent to meet the child's needs. An interactional evaluation between each adult and child also occurs.

Guideline 4

The role of the psychologist is that of a professional expert who strives to maintain an objective, impartial stance. The psychologist does not act as a judge or as an advocating attorney and remains impartial, balanced, and objective.

Guideline 5

The psychologist gains specialized competence. This includes competency in performing psychological assessments of children, adults, and families, knowledge of child and family development and child and family psychopathology, and knowledge of laws that govern divorce and custody decisions. This includes adherence to the Standards for Educational and Psychological Testing (APA, 1985). When specialized issues arise that are outside the evaluator's expertise, additional consultation and supervision should be sought.

Guideline 6

The psychologist is aware of personal and societal biases and engages in nondiscriminatory practices.

Guideline 7

The psychologist avoids multiple relationships.

Guideline 8

The scope of the evaluation is determined by the evaluator based on the nature of the referral questions. These may broaden or limit the scope of the evaluation from an entire family to a single family member. Sometimes psychologists are asked to review and critique the report of another mental health professional or serve as an expert in a specialized area of psychological knowledge.

Guideline 9

The psychologist obtains informed consent from all adult participants and, as appropriate, informs child participants. This entails informing each adult participant about the purpose, nature, and method of the evaluation, the source of the evaluation request, to whom the report will be sent, and who pays the fees. Information is given to children as is developmentally and otherwise appropriate.

Guideline 10

The psychologist informs participants about the limits of confidentiality and the disclosure of information. Participants are to be informed that they are consenting to disclosure of the evaluations' findings. It is recommended that the psychologist obtain a waiver of confidentiality from all adult participants.

Guideline 11

The psychologist uses multiple methods of data gathering: clinical interviews, observation, psychological assessments, and documentation from at least two sources about important facts. This can include the review of school, health care, and other kinds of records, and interviewing persons who know the family and who may contribute useful information. When external information is deemed significant,

psychologists are urged to corroborate it by an additional source whenever possible.

Guideline 12

The psychologist neither overinterprets nor inappropriately interprets clinical or assessment data. The psychologist is urged to seek convergent validity from multiple sources of information, including interviews and test data, and to cautiously and conservatively interpret evaluation results with awareness of reliability and validity restrictions.

Guideline 13

The psychologist does not give any opinion regarding the psychological functioning of any individual he or she has not personally evaluated.

Guideline 14

Recommendations, if any, are based on what is in the psychological interests of the child. It is noted that the profession itself has not reached consensus about whether psychologists ought to make recommendations about final custody determination to the courts. When the psychologist does make a custody recommendation, this should be based on sound psychological data and reflect the child's best interests in the case at hand.

Guideline 15

The psychologist clarifies financial arrangements.

Guideline 16

The psychologist maintains written records. Reference here is made to the APA Record Keeping Guidelines (APA, 1993) and relevant statutory guidelines. Further discussion of record keeping in the particular area of investigatory interviews is found in a later section of this chapter.

6. Guidelines for Psychological Evaluations in Child Protection Matters

Guidelines for Psychological Evaluations in Child Protection Matters (APA, 1999) is the second APA guideline to address professional practice issues with children. The legal context is litigation in child abuse and neglect cases. This guideline contains useful subsections of 17 guidelines, a glossary of terms, and a bibliography of major professional publications in the field. There is also a useful listing of national information resources, including the American Academy of Pediatrics, the American Bar Association Center on Children and the Law, and the Clearinghouse on Child Abuse and Neglect Information. These guidelines were developed for psychologists conducting psychological evaluations in child protection proceedings and have the goal of promoting "proficiency in using psychological expertise."

They do not address the issues of psychologists acting in other roles, such as when serving as a therapist for an individual or family. These guidelines build on those already discussed, including Ethical Principles of Psychologists in the Code of Conduct (APA, 1992), Record Keeping Guidelines (APA, 1993), and Standards for Educational and Psychological Testing (APA, 1985).

Three stages are described for state intervention in the affairs of the parents and children in child abuse and neglect matters. In the first stage, an investigation occurs following a report of suspected child abuse and neglect. In the second stage, the state may obtain custody of a child and a rehabilitation plan may be developed. When the child cannot be returned to the family, there may be a hearing for the final disposition of the case that can include termination of parental rights. States vary in statutory and case law that applies to these matters. Psychologists may be called on to evaluate the parties and "may function as agents of the court, of a child protection agency, or of the parents." In one or in all of these roles, the psychologist may be called on to evaluate the child, the parents, the parent and child relationship, or parenting capacity; to assess specialized issues such as the impact on the child of abuse and neglect; to recommend therapeutic interventions for the parents and children; and to make predictions about future harm if the child is reconciled with the parents and if parental rights are terminated. Psychologists are urged to develop the specialized knowledge and competencies required to do this work and seek consultation as needed. There are 17 guidelines.

Guideline 1

The primary purpose of the evaluation is to provide relevant, professionally sound results or opinions in matters where a child's health and welfare may have been and/or may in the future be harmed.

Guideline 2

In child protection cases, the child's interest and well-being are paramount.

Guideline 3

The evaluation addresses the particular psychological and developmental needs of the child and/or parents that are relevant to child protection issues such as physical issues, sexual abuse, neglect, and/or serious emotional harm.

Guideline 4

The role of the psychologist conducting the evaluation is that of a professional expert who strives to maintain an unbiased, objective stance.

Guideline 5

The serious consequences of psychological assessment in child protection matters place a heavy burden on psychologists.

Guideline 6

Psychologists gain specialized competence.

Guideline 7

Psychologists are aware of personal and societal biases and engage in nondiscriminatory practice.

Guideline 8

Psychologists avoid multiple relationships.

Guideline 9

On the basis of the nature of the referral questions, the scope of the evaluation is determined by the evaluator.

Guideline 10

Psychologists performing psychological evaluations in child protection matters obtain appropriate informed consent from all adult participants and, as appropriate, inform the child participant. Psychologists need to be particularly sensitive to informed consent issues.

Guideline 11

Psychologists inform participants about the disclosure of information and limits of confidentiality.

Guideline 12

Psychologists use multiple methods of data gathering.

Guideline 13

Psychologists neither overinterpret nor inappropriately interpret clinical or assessment data.

Guideline 14

Psychologists conducting a psychological evaluation in child protection matters provide an opinion regarding the psychological functioning of an individual only after conducting an evaluation of the individual adequate to support their statements or conclusions.

Guideline 15

Recommendations, if offered, are based on whether the child's health and welfare have been and/or may be seriously harmed.

Guideline 16

Psychologists clarify financial arrangements.

Guideline 17

Psychologists maintain appropriate records.

The glossary of terms is useful for psychologists when legal issues define the professional practice context and the evidentiary standards that will be applied to psychological data and opinion. Many of the definitions contained in the glossary were taken from Working with Courts in Child Protection (National Center on Child Abuse and Neglect, 1995). Selected terms are defined (author's selection and sequence):

- *Abuse, emotional:* Also referred to as psychological maltreatment; generally defined as a repeated pattern of behavior that conveys to children that they are worthless, unwanted, or only of value in meeting another's needs; may include serious threats of physical or psychological violence.
- *Abuse, physical:* Generally defined as the suffering of a child or substantial risk that a child will imminently suffer physical harm, inflicted nonaccidentally on him or her by his or her parents or caretakers.
- *Abuse, sexual (child):* Generally defined as contacts between a child and an adult or other persons significantly older or in a position of power or control over the child, where the child is being used for sexual stimulation of the adult or other person.
- *Beyond a reasonable doubt:* Highest standard of proof used in cases where the loss of liberty interests are at stake (e.g., incarceration or loss of life). Generally defined as the highest degree of support or level of certainty (90% to 95% chance).
- *Clear and convincing:* Intermediate standard of proof used in cases where significant liberty interests are at stake (e.g., loss of parental rights, civil commitment). Generally defined as a degree of support or level of certainty (75% chance).
- *Expert witness:* An individual who by reason of education or specialized experience possesses superior knowledge of a subject about which persons having no particular training are incapable of forming an accurate opinion or deducing correct conclusions. A witness who has been qualified as an expert will be allowed (through his or her answers to questions posed) to assist the jury in understanding complicated and technical subjects not within the understanding of the average layperson. Experts are also allowed to provide testimony based on hypothetical scenarios or information and opinions that are not specifically related to the parties in a particular legal action.
- *Fact witness:* Generally defined as an individual who, by being present, personally sees or perceives a thing; a beholder, spectator, or eyewitness. One who testifies to what he or she has seen, heard, or otherwise observed regarding a circumstance, event, or occurrence as it actually took place or a physical object or appearance as it usually exists or existed. Fact witnesses are generally

not allowed to offer opinion, address issues that they do not have personal knowledge of, or respond to hypothetical situations.

- *Family preservation or reunification:* The philosophical belief of social service agencies, established in law and policy, that children and families should be maintained together if the safety of the children can be ensured.
- *Guardian ad litem:* Generally defined as an adult appointed by the court to represent and make decisions for someone (such as a minor) legally incapable of doing so on his or her own in a civil legal proceeding.
- *Maltreatment:* Generally defined as actions that are abusive, neglectful, or otherwise threatening to a child's welfare. Commonly used as a general term for child abuse and neglect.
- *Neglect:* Generally defined as an act of omission, specifically the failure of a parent or other person legally responsible for a child's welfare to provide for the child's basic needs and proper level of care with respect to food, shelter, hygiene, medical attention, or supervision. There are two types of neglect:

    1. Emotional: Generally defined as the passive or passive-aggressive inattention to a child's emotional needs, nurturing, or emotional well-being. Also referred to as psychological unavailability to a child.
    2. Physical: Generally defined as a child suffering, or at substantial risk of imminently suffering, physical harm causing disfigurement, impairment of bodily functioning, or other serious physical injury as a result of conditions created by a parent or other person legally responsible for the child's welfare or by the failure of a parent or person legally responsible for a child's welfare to adequately supervise or protect him or her.

- *Preponderance of evidence:* Lowest of the three standards of proof, and the standard applied in most civil actions; generally defined as a "probable" degree of certainty (e.g., "more likely than not," or a 51% chance).

7. Guidelines for Psychotherapy with Lesbian, Gay, and Bisexual Clients

In general, the Guidelines for Psychotherapy with Lesbian, Gay, and Bisexual Clients (APA, 2000) may be thought to apply to older adolescent and adult populations and therefore not directly applicable to the evaluation and treatment of children and adolescents in clinical and forensic contexts. Professional practice, however, indicates otherwise. Lesbian, gay, and bisexual persons are involved with children as parents, guardians, and foster parents and may individually or as couples be subject to child protection action. Children with gender identity, sexual orientation, or child abuse and neglect issues, or some combination of these, are seen in clinical and forensic evaluation cases and are often identified for specialized care. These guidelines, therefore, though not specifically addressing children and adolescents, involve populations with whom practicing psychologists are likely to have professional contact. The stated goals of these guidelines are to provide

practitioners with a frame of reference for the treatment of lesbian, gay, and bisexual clients and to provide basic information and further references in the areas of assessment, intervention, identity, relationships, and the education and training of psychologists. The guidelines are organized into four sections: (1) attitudes toward homosexuality and bisexuality, (2) relationships and families, (3) issues of diversity, and (4) education. Some of the 16 guidelines are articulated.

Guideline 1

Psychologists understand that homosexuality and bisexuality are not indicative of mental illness.

Guideline 2

Psychologists are encouraged to recognize how their attitudes and knowledge about lesbian, gay, and bisexual issues may be relevant to assessment and treatment and seek consultation or make appropriate referrals when indicated.

Guideline 3

Psychologists recognize that families of lesbian, gay, and bisexual people may include people who are not legally or biologically related.

Guideline 4

Psychologists are encouraged to recognize the particular life issues or challenges that are related to multiple and often conflicting cultural norms, values, and beliefs that lesbian, gay, and bisexual members of racial and ethnic minorities face.

Guideline 5

Psychologists are encouraged to recognize the particular challenges that bisexual individuals experience.

Guideline 6

Psychologists strive to understand the special problems and risks that exist for lesbian, gay, and bisexual youth. Here it is noted that lesbian, gay, and bisexual adolescents face specialized risks, including higher rates of estrangement from their parents, rejection by their parents, and increased risks of homelessness, HIV infection, and becoming victims of violence and of attempting suicide. Gay, lesbian, and bisexual youth may have complications of adolescent development and may require specialized therapeutic strategies and a secure therapy context to address sexual orientation issues.

Guideline 7

Psychologists make reasonable efforts to familiarize themselves with relevant mental health, educational, and community resources for lesbian, gay, and bisexual people.

8. Guidelines on Multicultural Education, Training, Research, Practice, and Organizational Change for Psychologists

The increasing diversity of the American population underscores the importance of these guidelines. The four stated goals in the Guidelines on Multicultural Education, Training, Research, Practice, and Organizational Change for Psychologists (2002d) are to provide psychologists with a means of addressing multiculturalism with basic information to enhance ongoing education and broaden the purview of psychology as a profession. The guidelines address U.S. ethnic and racial minority groups as well as persons from biracial, multiethnic, and multiracial groups. The guidelines indicate that approximately one third of the American population, according to the U.S. Census Bureau 2001 data, belong to a minority group. While racial/ethnic diversity varies greatly by state, region, and urban versus rural environment, practicing psychologists are likely to encounter minority clients in their practice. This increases the "urgency for culturally responsive practices and services." Psychologists are urged to become conversant with terms of culture, race, and ethnicity. Six guidelines are defined. These encourage psychologists to develop a commitment to cultural awareness, to recognize the importance of multicultural sensitivity and responsiveness, to employ constructs of diversity in psychological education, to conduct culturally centered and ethical psychological research, to apply culturally appropriate skills in psychological practice, and to support culturally informed organizational development. In psychological assessment, this may mean determining the applicability of particular tests and measures to minority populations. For example, Grisso (1998) in "Forensic Evaluation of Juveniles" reports that no relation was found between competent versus incompetent status and gender and race. Grisso found that three causal factors were related to a clinician's conclusion about incompetence: severe mental disorder, cognitive and intellectual disabilities, and age-related characteristics. By contrast, the Wechsler Intelligence Scale for Children—Fourth Edition (Wechsler, 2004) discusses the importance of age, gender, regional, ethnic, and racial characteristics of standardization populations. The guidelines explain that cultural sensitivity means taking account of the client's cultural context, using culturally appropriate assessment tools, and using a broad array of interventions. Cases may involve unusual combinations of client characteristics with variable nationality, racial, ethnic, religious, regional, and sexual characteristics and orientations. Some persons may be in transition from one group to another. Others may be confused about their own identity and group memberships. Professional practice can lead to work with illegal aliens and requires special assistance with language issues. Critical judgment is recommended when using standardized assessment tools and methods. Knowledge of helping practices used in non-Western cultures and culture-specific therapies may be indicated for particular ethnic communities.

9. Criteria for Evaluating Treatment Guidelines

Criteria for Evaluating Treatment Guidelines (APA, 2002a) is aspirational in intent and was developed for the purpose of evaluating the guidelines of health care

organizations, government agencies, and professional associations. These pertain to specific recommendations about treatments to be offered to patients. They are, therefore, patient-focused rather than practitioner-focused as are practice guidelines. The purpose of treatment guidelines is to establish the most effective treatments available, translating knowledge into clinical practice. Treatment efficacy is defined as ascertaining the effects of a given intervention as compared to an alternative intervention or with no treatment, thereby establishing the scientific basis for determining treatment benefit.

Twenty criteria are listed for the evaluation of treatment guidelines. These describe the scientific method to be used in evaluating treatment efficacy. This requires establishing the clinical utility and cost effectiveness of treatment and that guideline panels be composed by persons with a broad range of expertise.

10. Criteria for Practice Guideline Development and Evaluation

Criteria for Practice Guideline Development and Evaluation (APA, 2002b) is intended to promote quality and consistency in practice guidelines and to produce a high quality of psychological services. These guidelines are aspirational and not intended to take precedence over professional judgment. Practice guidelines consist of recommendations to professionals concerning their conduct and issues in particular areas of psychological practice.

11. Guidelines for Providers of Psychological Services to Ethnic, Linguistic, and Culturally Diverse Populations

Guidelines for Providers of Psychological Services to Ethnic, Linguistic, and Culturally Diverse Populations (APA, 1990) represents general and aspirational principles for psychologists who work with ethnic, linguistic, and culturally diverse populations. Nine guidelines are articulated. These include the recognition of research and practice issues related to the population serviced, respect for the cultural structure of the community, use of the client's language, consideration of culture in designing intervention, elimination of discriminatory practice in service delivery, and documentation of relevant cultural factors affecting the family.

12. The APA Resolution on Outpatient Civil Commitment

The APA Resolution on Outpatient Civil Commitment (APA, 2004a) addresses the increasing use of this power in courts and civil commitment venues across the country. Although this resolution does not address children and adolescents specifically, it is well known that children and juveniles are often subject to this procedure. It recognizes that persons with serious mental illness and psychological disabilities can be subject to the imposition of involuntary mental health services, including hospitalization and incarceration. There is recognition of the controversy about the legitimacy of outpatient civil commitment (OCC), how and when it should be applied, and the professional practice standards, clinical methods, and level of proof involved.

The resolution recognizes that people possess behavioral disorders severe enough to reach the point of dangerousness, that these persons and persons with serious mental illness have a right to treatment, and that psychologists often play a role in providing the functional assessments and opinions that lead to OCC. There is recognition that some populations may have a greater vulnerability to the imposition of OCC. This APA resolution states, "It is never the role of any mental health practitioner, of any discipline, acting in the role of a caregiver, to make decisions that infringe upon a person's right to consent to services." Psychologists are encouraged to promote the development of evidence-based clinical methods for determining risk and dangerousness and determining competence to make specific judgments and decisions, promote rehabilitation from serious mental illness, research alternatives to involuntary commitment, and provide professional expertise and consultation to legal and judicial authorities to ensure that legal processes and decisions are appropriately informed by scientific and clinical considerations.

13. Training Grid Outlining Best Practices for Recovery and Improved Outcomes for People with Serious Mental Illness

The Training Grid Outlining Best Practices for Recovery and Improved Outcomes for People with Serious Mental Illness (APA, 2005c) is intended to assist providers in identifying the most effective interventions and resources for advanced clinical training. This document applies to adults with serious mental illness.

14. American Psychological Association Policy Statement on Evidence-Based Practice in Psychology

The APA Policy Statement on Evidence-Based Practice in Psychology (2005b) addresses the integration of the best available research with clinical expertise and patient characteristics. The purpose is to promote effective psychological practice and enhance public health. Best research evidence relies on the use of scientific method, beginning with clinical observation and leading to randomized clinical trials. Patient groups include child, adolescent, adult, older adult, couple, family, group, organization, community, and other populations that receive psychological services.

Researchers and practitioners are encouraged to combine their efforts to improve psychological practice. Psychologists are encouraged to use their clinical expertise to conduct assessments and develop diagnostic judgments, case formulations, and treatment plans; to evaluate treatment progress; to seek additional resources when indicated; and to have a cogent rationale for clinical strategies. Psychological services are considered to be most effective when they are responsive to patient characteristics and other factors such as cultural, family, environmental, values, and treatment expectations. Some effective treatments are directed to the patient's environment, such as parents, teachers, and caregivers.

15. Specialty Guidelines for Forensic Psychologists

Specialty Guidelines for Forensic Psychologists (APA, 1991) is designed to guide forensic psychologists in monitoring their professional conduct. The goal is to

improve the quality of forensic psychological services. The guidelines are aspirational. They apply to such diverse positions in forensic practice as clinical forensic examiners, psychologists employed by correctional and forensic mental health systems, researchers who offer direct testimony about the relevance of scientific data to a psycholegal issue, trial behavior consultants, and psychologists who provide professional opinions to the courts through amicus briefs or who appear as forensic experts before judicial, legislative, and administrative agencies. The guidelines amplify the Ethical Principles of Psychologists in the context of the practice of forensic psychology. The guidelines are designed to conform to state and federal laws. They specify the nature of desirable professional practice. They are not meant to apply to a psychologist who is not informed at the time of service delivery that the service would be used for forensic purposes.

Forensic psychologists have an obligation to provide services that are consistent with the highest standards of their profession. Forensic psychologists provide services only in areas of psychology in which they have specialized knowledge, skill, experience, and education. It is obligatory for forensic psychologists to inform the court about the boundaries of their competence, have a reasonable understanding of the legal and professional standards that govern psychological expertise in legal proceedings, and recognize that personal and professional relationships may be inconsistent with forensic involvement. Forensic psychologists explain their fees and their personal and professional activities that may produce a conflict of interest. They acknowledge areas of competence and limitations of competence. They do not provide services on the basis of contingent fees. Forensic psychologists are encouraged to offer a portion of their professional services on a pro bono or reduced fee basis.

Forensic psychologists inform prospective clients of their legal rights with respect to the forensic service and obtain informed consent before proceeding with the evaluation. An exception may be when the evaluation is court ordered. When the client is unwilling to proceed, the evaluation should be postponed and the psychologist should take steps to assist the client to receive legal advice regarding participation. When the client does not have the capacity to provide informed consent, the client's legal representative is informed, and if the legal representative objects to the evaluation, the forensic psychologist notifies the court that ordered the evaluation. The forensic psychologist has an obligation to inform legal authorities about any source of conflict and take reasonable steps to resolve it.

Forensic psychologists inform their clients about the limits of confidentiality. Information is released pursuant to statutory requirements, court order, or the consent of the client. Forensic psychologists document and are prepared to make available to the court all evidence that formed the basis for forensic services. They anticipate the detail and quality of documentation required by judicial scrutiny and recognize that this standard is higher than the standard for general clinical practice. Forensic psychologists also recognize that they have a special responsibility to provide the best documentation possible under the circumstances. It is understood that psychological evidence is subject to rules of discovery, disclosure, and confidentiality. The duties and obligations of forensic psychologists regarding documentation apply "from the moment they know or have a reasonable basis for knowing that their data and evidence derived from it are likely to enter into legally relevant decisions."

As an expert, the forensic psychologist actively seeks information that will test plausible rival hypotheses. Unless an individual is handling his or her representation pro se, a forensic psychologist does not provide forensic services to any party contemplating a legal proceeding prior to that person's representation by counsel. An exception to this rule is the delivery of emergency mental health services, when a failure to provide such services would pose a substantial risk of harm to the defendant or to others.

Forensic psychologists seek third-party data only with prior approval of the legal party or pursuant to a court order.

Forensic psychologists seek to obtain independent and personal verification of data relied on. An effort is made to minimize sole reliance on hearsay evidence when offering expert testimony. It is recognized that many forms of data used by forensic psychologists are hearsay; therefore, an attempt to corroborate critical data is required. When the status of the data is uncorroborated, this is acknowledged by the forensic psychologist. The forensic psychologist acknowledges information gathered by others and avoids offering information that does not directly bear on the legal purpose of the professional activity. Forensic psychologists avoid giving evidence about the psychological characteristics of a person they have not examined. When it is not possible to conduct a direct examination, the forensic psychologist explains limitations on the reliability and validity of professional products, evidence, or testimony. The forensic psychologist is mindful of the Federal Rules of Procedure, which restrict testimony by the expert based on statements made by the defendant in the course of the forensic examination.

Forensic psychologists maintain a high level of public and professional communication. This includes taking reasonable steps to correct misuse or misrepresentation of their professional services. Access to information is restricted to individuals with a legitimate professional interest in the data and steps are taken to ensure that test results are released to a qualified professional and in compliance with Principle 16 of the Standards for Educational and Psychological Testing.

A special responsibility for fairness and accuracy is a feature of forensic psychological practice. This includes a fair and accurate evaluation of the work and opinions of other experts and parties. Forensic psychologists testify fairly and avoid partisan distortion and misrepresentation and errors of commission and omission. Forensic psychologists disclose all sources of their information and which information was relied on in formulating a particular opinion. Forensic psychologists "are aware that their own professional observations, inferences, and conclusions must be distinguished from legal facts, opinions, and conclusions."

PRACTICE GUIDELINES FOR INVESTIGATIVE INTERVIEWING

The first section of this chapter reviewed the ethical standards and practice guidelines of the APA with special reference to clinical and forensic practice with children, adolescents, and their families. Now we turn to three of the more comprehensive guidelines of professional organizations which address investigative interviewing with alleged child and adolescent victims of sexual abuse. The APRI, the APSAC,

and the AACAP have formally addressed the topic of investigatory interviewing guidelines and standards.

1. The American Prosecutor's Research Institute

Clearly, the most comprehensive contribution on this topic is from the APRI. Its publication, *Investigation and Prosecution of Child Abuse*, third edition (APRI, 2004), is a multidisciplinary guide for attorneys, medical professionals, and behavioral scientists involved in the investigation and prosecution of child abuse. Chapter 2, "Investigation," is of particular relevance for psychologists practicing in both clinical and forensic settings. It presents a thorough and concentrated discussion of interviewing guidelines and validation principles. Because the sponsoring organization in this case is federal in scope and extends its training and practice guidelines to all of the states, the principles contained therein can be seen as offering national guidelines in a legal context.

Interviewing the Child

The stated purpose of this section is to assist multidisciplinary interviewers and investigators, including psychologists and prosecutors, to interview children properly and to analyze the evidentiary record for completeness, accuracy, and error. Techniques for the interviewing of preschool, grade school, and adolescent children are distinguished from those used for adults. The interviewer is expected to understand the field of child abuse and the developmental capacities of children of different ages. This includes the ability to judge the quality of prior interviews conducted with all witnesses. Interviewers are expected to be thoroughly trained and to develop a protocol for conducting interviews that reflect these guidelines and that can be defended in court. It is emphasized that comfort in talking with children is required. Investigators are urged to start the interview with no assumption about whether abuse occurred or about the identity of a perpetrator or how the child might feel about the perpetrator. A stance of neutrality about the allegation is required as it has the best chance at generating the greatest amount of reliable information. Interviewers are admonished not to coerce or manipulate children or to educate them about what may have happened or how they should feel. The author refers the reader to Ceci and Bruck (1995) and Poole and Lamb (1998), who provide detailed discussions of research on investigatory error and its impact on child narration, including autobiographical event narration.

Preparing for the Interview

Preparation for the interview requires thorough review of all known information on the case. This includes police and child protection reports, medical findings, and other materials. When possible, the person who made the report, the physician who examined the child, and the parent, guardian, and therapist who may have been treating the child should all be contacted. Interested parties are cautioned not to rehearse the child prior to the forensic or clinical interview. Gathering this information in advance is seen as strengthening the interviewer's understanding

of the nature of the case, or the child, and of the special issues that could affect the interview process. This discussion acknowledges that there is expert opinion that recommends "allegation-blind" interviews in which the interviewer knows nothing about the background of the child or of the allegations. This approach is recommended to neutralize interviewer expectation and to minimize leading and confirmatory errors. The work of Cantlin, Payne, and Erbaugh (1996) is referenced in this connection; they compared allegation-blind and allegation-informed structured interviews. Planning is recommended for situations in which a child may be handicapped or speaks a different language. This will necessarily involve training of the interpreter and/or a support person for the handicapped child to prepare him or her to participate in an investigative interview. Such persons should be instructed not to speak for the child and not to show reactions to what the child says. The author suggests that interpreters also be trained to give verbatim accounts and not to independently question the interviewee.

Interview Location and Circumstances

It is recommended that a neutral location be found for the investigative interview and the number of interviews be limited. Police department facilities are to be avoided as well as the presence of uniforms and weapons. The interview room should be free of distractions and furnished to help the child to feel welcome. For example, some interview rooms have child-size furniture. Anatomical drawings and dolls are not to be in view. The interview location should provide the child with a feeling of privacy. Providing a child with food during the interview should be avoided, as should the promise of a snack when the interview is over. Interviews should be scheduled at a time when the child is likely to be alert and with the least disruption to the child's daily routine.

Who Should Be Present During the Interview

It is firmly recommended that children be interviewed alone by a single adult interviewer. Only in rare circumstances should a parent of a very young child be allowed to be present during the investigative interview. Caretakers should be informed in advance that they will not be in the interview room. In rare circumstances, the sex of the interviewer may be of significance and may require rescheduling to make a child more comfortable. It is recognized that more than one interview may be necessary to receive a complete account from the child. All investigative interviews should be conducted by the same interviewer. Children should not be reinterviewed about information previously disclosed.

How to Record Information from an Interview

The APRI expresses some flexibility on this issue, recognizing that there are jurisdictions that require audiotaping or videotaping of forensic interviews with children and others that do not. The APRI notes that a videotape can be a powerful tool for demonstrating the behavior and statements of children both in the hands of the prosecution and in the hands of the defense. When taping is performed,

high-quality equipment is recommended. When taping is not used, interviews are to be accurately documented. The APRI indicates that "skilled interviewers are able to fulfill both functions [interviewing and note taking]." Lamb, Orbach, Sternberg, Hershkowitz, and Horowitz (2000) demonstrate that this is not the case and that even trained forensic interviewers produce significantly incomplete and flawed "verbatim notes." Therefore, for the sake of complete documentation, Lamb et al. recommend video- and audiotaping.

Conducting the Interview

Recommended interview protocols share common features. These begin with rapport building, the interview then progressing through a series of stages. At the outset, the interviewer should help the child to feel physically comfortable. For video- or audiotaping, the child should be positioned to facilitate that recording. Although the child should be asked, "Do you know why you're here?," he or she should not be told that the parent or guardian has already informed the interviewer about the interview purpose. The child should be canvassed about his or her understanding of the purpose of the interview and the role of the interviewer. Then the child should be instructed about a few rules. The first rule is that the child should say what he or she knows but not to guess. The second rule is that the child should correct the interviewer if he or she makes a mistake. The third rule is to inform the child that if the interviewer asks a question a second time the reason is not to cause the child to give a different answer. The fourth rule is that the child should be explicitly told that he or she can answer any question with "I don't know" or "I don't remember." The fifth rule is to ask the child to report when he or she does not understand a question. The interviewer should be mindful of the child's age, developmental capacities, and attention span as well as the child's capacity to remember instructions. Open-ended questions are encouraged as the best questioning method.

Canvassing the child's capacity to differentiate between the concepts of truth and lie and telling the truth versus telling a lie is treated as optional. It is noted that younger children have difficulty with these concepts and that this procedure is essentially an attempt to assess the child's competence to take an oath. The use of this procedure is apparently widespread; it is treated as an ordinary interviewing issue by Sattler (1988) and others (APRI, 2004; APSAC, 2002; Sparta & Koocher, 2006). White and Quinn (1988, p. 274) indicate that such canvassing of children can be intimidating, confusing, and coercive. They refer to this as the "truth-lie paradigm," categorizing it as coercive in all of its forms. They note that the labeling of statements as true or untrue is the function of the judicial system. The use of this technique "signals to the child that the interviewer believes there exists a particular truth to be found." Condie (2003, p. 222) sees such questioning of a child as a "fruitless endeavor" because neither a yes nor a no answer by the child has any predictive value for the child's ability to report events accurately. Truth-lie canvassing and developmental competency assessment do not access the core testimonial issue of the child's eyewitness memory for personal experience. As such, they provide no indication of the child's capacity and willingness to provide accurate autobiographical reports.

When competency questioning is undertaken, it can begin with testing the child's capacity to give accurate and inaccurate responses to fact questions in the present. It assesses the child's developmental abilities to name colors, count, count instances of events, tell the time of day, distinguish relative positions, and so on. Questioning about autobiographical events can provide an indication of the capacity for narrative recall. The author suggests that such questioning be directed at events that lend themselves to external corroboration. Sample topics are how and with whom the child came to the interview location, what the child ate that day, and major recent and remote events in the child's life. These can be accessed with such questions as "Tell me how you got here today"; "Who brought you here today?"; "What did you eat for breakfast today?"; "Tell me some things that you did yesterday"; and "Tell me about some of the best and worst things that ever happened to you."

The third stage is the abuse inquiry. The APRI guidelines are flexible on this issue. The discussion includes the optional use of anatomical drawings to orient the child to the subject of the inquiry. The APRI notes that some interviewers, however, postpone the introduction of body drawings for a later point in an interview that has produced an abuse report and use body drawings for body identification purposes. The discussion notes that when body drawings are introduced prior to an abuse report, this procedure is open to the attack that such pictures are suggestive in and of themselves. Poole and Lamb (1998) provide a detailed discussion of the pros and cons of the use of anatomical drawings and dolls for investigative interviewing purposes. They conclude with the recommendation that the use of such explicit stimuli be postponed to the identification and validation phases of an interview and that they not be used in advance of an abuse report by the child. They point to research that shows that the use of such props increases the percentage of false reports.

The APRI states that it is important to avoid asking children to simply repeat their prior report with a question like "I want to know what you told your mom about the babysitter." A recommended alternative is "Did you tell your mom about something that happened with the babysitter?" These examples are offered to illustrate attempts to access abuse reports. The author advises that both alternatives are objectionable as they access prior reports rather than prior experience. It may be a better practice to probe the issue of prior reports after concluding the abuse interview. This allows for the chronological recapitulation of experience and the assessment of the instant eyewitness report of the child. This method attempts to minimize the compounding effect of potential contextual influences and distortions. It also allows for a distinction between the child's report to the investigator and the disclosure process itself, which is a subject of separate inquiry. The forensic interviewer should seek "investigatory independence" (White & Quinn, 1988) and avoid direct and indirect alignment with prior interviewers. When an abuse report is obtained, it can be compared to prior abuse reports. If no abuse is reported or if abuse is denied, a decision can be made to probe the causes for inconsistency of reporting.

The APRI offers a table on a continuum of question types. These range from questions that are open-ended to those that are closed-ended. Answers to open-ended questions generate more confidence than those obtained from closed-ended questions. Question types identified are general, focused, multiple choice, yes/no, and leading. The opinion is firmly stated that the most reliable information comes in

response to open-ended questions. Interviewers are encouraged to avoid questions that sound accusatory, have erroneous content, and introduce elements of threat, coercion, and bribes. Putting pressure on a child to speak or to continue an interview is discouraged.

Possible Questions for the Sexual Abuse Inquiry

The APRI lists four pages of possible questions that fall into subcategories of general, penetration, erection or masturbation, ejaculation, nudity, oral contact, and pornography. There are then lists of questions about physical and sexual abuse, fears and reasons for secrecy, determining who was involved, where the abuse took place, and alternative hypothesis testing. Most of these questions are phrased in the affirmative and are therefore suitable for situations in which the possibility of abuse was previously indicated in the interview. Examples are:

- How did the touching start?
- What did his penis look like?
- What happened to his penis after he made you touch it? It is considered more appropriate in advance of an abuse report to ask questions in an open-ended interrogatory form to explore the possibility of experience that may or may not have an abusive character. Some APRI examples are:
- Have you had any touches you didn't like or that made you feel uncomfortable?
- Do you remember what you were wearing when you and . . . were playing?
- Did you see pictures/books/magazines/movies/videos at . . . ?
- What happens if you do something you're not supposed to?
- Have the police ever come to your house? Why did they come?
- Has anyone asked you to keep a secret? What was the secret?

Open-ended examples of follow-up questions are:

- Tell me about that.
- Tell me more about that.
- Do you remember more about that?

It is important to remember that requests for additional information can constitute a demand in the mind of the respondent and can cause the production of information and detail to please the questioner. Children and adolescents are rarely self-referred for investigative interviews, and most have had prior interviews by parents and others. The author advises that even general and open-ended requests for additional information be preceded by inquiry concerning the presence or absence of additional experience and knowledge. Poole and Lamb (1998) report that with additional questioning children add credible detail to both true and false reports. The risk of added detail to false reports is heightened when respondents have been exposed to misinformation in prior interviews. The well-informed investigative interview

strives to offer respondents the opportunity to drop misinformation and false reports if these exist and to pursue interviewing in the least suggestive manner.

Direct questions are seen as sometimes necessary, and a distinction is made between more and less suggestive direct questioning. The example "Did something happen to your [child's word for vagina]?" is considered superior to "Did your dad do something to your [child's word for vagina]?"

Anatomical Dolls

The APRI allows for the use of anatomical dolls for demonstration purposes after an abuse report has been received. The interviewer is encouraged to use dolls that are chosen for the child and for the perpetrator, and whenever possible these dolls should be appropriate for age, sex, and race. A list of inappropriate uses of dolls includes:

- Presenting the dolls to the child unclothed.
- Encouraging play with the dolls.
- Naming the sexual parts of the doll for the child.
- Probing the doll's genitals or breasts.
- Using the dolls to demonstrate sexual behavior to the child.
- Failing to elicit a verbal description of the abuse before presenting the dolls to the child.
- Incorrectly clarifying what the child demonstrated.

The APRI advises interviewers not to ask adolescents to name body parts at the beginning of an interview and to use an anatomical doll only when this can be done in a way that is not insulting. This author advises interviewers not to introduce drawings or dolls until a verbal abuse disclosure has been made and to restrict the use of drawings and dolls to clarify important points. This admonition reflects the author's recommendation for best practice methods.

Ending the Interview

It is recommended that the interview be ended after determining that the child is no longer able to provide information. This can be preceded by such questions as "Is there anything else you want to tell me today?" and "Is there anything you want to ask me?" It is recommended that the child be thanked for participating. After the meeting with the child, it may be appropriate to meet with the parents or caregivers to provide them with information. This can include scheduling follow-up interviews, making treatment recommendations, and giving information about the investigatory and court process. Interviewers are encouraged to develop a protocol that defines the stages of the interview, including introduction, rapport building, instructions, methodology for beginning the abuse inquiry, methodology for the abuse inquiry itself, and closure.

Assessing Validity in Sexual Abuse Cases

There are 13 subsections to the validity discussion that lead to a summary of indicators of report validity in sexual abuse cases. The purpose is to determine "whether a report of sexual abuse is valid." The 13 points describe the process of evaluating the available evidence. Evidence includes eyewitness statements, confessions, medical evidence, the child's statements, emotional and behavioral characteristics of the child as seen by others, and assessment of the validity of the abuse report. No factors are thought to conclusively demonstrate the validity or invalidity of an abuse report.

1. Alternative Explanations for the Child's Statement The investigator is required to consider alternative explanations throughout the investigation. This includes consideration of the possibility of deliberate falsehood, misinterpretation of innocent contacts, and coaching. The author adds ruling out professional error to this list of requirements. These possibilities are to be translated into active steps in document review and in investigative interviewing before the inquiry is ended.

2. Spontaneity Spontaneous statements from children are considered more reliable as they are unlikely to be the result of prompting. The evidentiary record should be reviewed to determine whether there is reliable indication of a spontaneous disclosure by the child and which parts of the child's narration in the investigative interview possess this characteristic.

3. Consistency Consistency of child statements in an interview and across interviews and statements is an essential feature of a validation determination. Consistency is tracked from the point of initial disclosure through all forensic interviews as well as the child's testimony in court. Consistency is tracked for core elements rather than for exact wording. The validation review should be alert to the issue of rote recital, and when this is found the record should be "scrutinized carefully for the possibility of coaching."

4. Sexual Knowledge or Behavior That Is Developmentally Unusual This refers to the acting out of adult forms of sexuality, including sexual talk, rather than age-appropriate touching and talk about body parts, including genitalia, that may occur out of curiosity and play. Friedrich (1992) identifies explicit advanced sexual behaviors that occur with low frequency and are considered indicators for sexual abuse. These include:

- A child putting his or her mouth on another's sexual parts.
- A child asking to engage in sexual acts.
- A child masturbating with an object.
- A child inserting an object into his or her own anus or vagina.
- A child imitating intercourse.
- A child making sexual sounds.

- French kissing by a child.
- A child talking about explicit sexual acts.
- A child undressing others.
- A child asking to watch explicit sexual television or movies.
- A child using sexual words.
- A child imitating sexual acts with dolls.

Schoentjes, Deboutte, and Friedrich (1999) confirmed these data in a study of 917 2- to 12-year-old Dutch-speaking children in Belgium. More recently, however, Drach, Wientzen, and Ricci (2001) studied the utility of sexual behavior problems as diagnostic indicators of sexual abuse in a sample of 247 children evaluated for sexual abuse at a forensic child abuse evaluation clinic. It was found that non-sexually abused children were just as likely to have high scores on the Child Sexual Behavior Inventory as sexually abused children. These results suggest that the occurrence of seemingly advanced sexual behavior in relatively young children should be viewed with caution as an indicator of sexual abuse. Both the absence and the presence of advanced sexual behavior was evident in children who had been and who had not been sexually abused. This nullified the use of advanced sexual behavior as an abuse indicator.

5. Developmentally Appropriate Language Reports about a child's disclosures and the child's interview language should be analyzed for its developmental characteristics. The interviewer is urged to adapt questions to the child's developmental language and to use the child's language during all phases of the interview. It is thought that children who have been coached or who have borrowed language from other sources for use in abuse reports may use language that is developmentally unexpected or unusual. This analysis should allow for children who have precocious language ability, who have been exposed to multiple interviews, and who may have acquired language from the offender.

6. Play and Gestures Indicative of Abuse A suspicion of abuse may have arisen because of the observation of sexualized play. Or, after an abuse disclosure, reports may arise about previously observed sexualized play. Particularly in younger children, such play may be seen as a reenactment of an advanced sexual exposure or experience.

7. Idiosyncratic Detail Children are considered less likely to have fabricated an abuse report that contains personalized, experiential detail. Sensory detail referencing taste, smell, touch, size, pressure, or pain, and containing mechanical descriptions is considered validating of the abuse report. Ceci and Bruck (1995), however, demonstrate that such detail can be abundantly produced by children in response to improper and suggestive interviewing. Nonetheless, an idiosyncratic report in the absence of evidence of improper questioning can be compelling. The APRI advises that this can occur when the abuse report contains "such description and detail that there can be little doubt that the child witnessed the sexual act." The child may have knowledge about a feature of the perpetrator's body such as a tattoo or knowledge

of taste and texture or visual exposure to sexual activity. The child may recount highly personalized details that lend credibility to the abuse report. It is stated that "a child's total inability to furnish details that demonstrate familiarity with the sex act forced upon the child may indicate the absence of abuse."

8. Content of the Statement The investigator is called on to consider whether the actions described make sense. This includes considering the possibility that innocent physical touches were misinterpreted as sexual and whether the progression of events fits known patterns of abuse. This can include identification of some features that have been found in validated abuse reports, including a description of multiple incidents over time, a progression of sexual activity that includes grooming, a request for secrecy, the use of power and authority to gain compliance, and attempts to cause a child to retract a complaint.

9. The Child's Manner and Emotional Response The APRI notes that a child's emotional presentation may give a sense of whether or not an abuse account is genuine, and the investigator is called on to note all of the child's verbal and nonverbal emotions. The advantage of videotaping to capture this information is acknowledged. Hesitancy, tone of voice, and facial and bodily expressions can offer clues about fearfulness, guilt, aversion, and distaste. It is noted that the absence of emotionality does not automatically invalidate an abuse report.

10. The Existence of a Motive to Fabricate This section states, "The assessment should consider what motives the child may have to fabricate the abuse or the motives others may have to coach a child to allege abuse." But the discussion of this topic offers little guidance about how to investigate this issue and how to translate hypotheses into investigatory procedures and questions. No specific validation procedures or questions are offered. We are cautioned that the existence of a motive is not the equivalence of a fabrication and to be alert to offenders who may attempt to confuse motive with misdeed. The author recommends that validation questions be adapted to the facts of the case and should reflect a knowledge of the complainant's personality and relationship characteristics, including his or her relationship with the accused. When the complainant's biography includes a history of lying or this issue otherwise arises (Benedek & Schetky, 1987), this issue should be explored. For younger children, the author suggests the use of questions such as:

- Is this something that really happened or is it make-believe?
- Are you saying this just to get [the accused] in trouble?
- Are you saying this because you're angry/mad at [the accused]?
- Does somebody else think this happened even though you don't?
- Are you saying this because you want to live with. . . ?

For older children, the author recommends such questions as:

- Are you saying this to get out of the house?
- Are you saying this because [a valued person] was not being nice to you or not paying enough attention to you?
- Are you saying this to cover up because you wanted to do it (have sex)?

Other authors provide some help in identifying motives to fabricate. Kanin (1994) investigated 45 consecutive, disposed, false rape allegations over a 9-year period in a small metropolitan community that constituted 41% of the total forcible rape cases in that jurisdiction ($n = 109$). Kanin determined that the allegations appeared to serve three major functions for the complainants: providing an alibi, seeking revenge, and obtaining sympathy and attention. The sample consisted of young adults. Retrospective comparison reports with college populations produced similar results. These motives may apply differently to child and younger adolescent populations. For example, providing an alibi for consensual sexual behavior may be more applicable in an adolescent population. Seeking revenge and obtaining sympathy and attention may apply equally to child and adolescent populations that have the developmental capacity to be motivated in these ways.

Benedek and Schetky (1987) investigated child sexual abuse reports in child custody and visitation disputes. They noted that adolescents may allege sexual abuse for secondary gain because they are angry at a parent, jealous of a parent's boyfriend, or want someone excluded from the home. For younger children, they found that a history of lying, the use of rote phrases, reports from the perspective of an adult rather than from the perspective of a child, and child reports delivered only in the presence of an accusing adult were characteristics of false reports. Ash and Gyer (1991) discuss biased reporting in parents undergoing child custody evaluation. They studied 196 court-ordered child custody evaluations and found that parents give symptomatic child reports that suit litigation purposes. Everson and Boat (1989) reviewed false allegations of sexual abuse by children and adolescents in reports given to child protection workers; 88 child protection respondents who had worked on 1,249 sexual abuse cases were in the sample. The children ranged in age from preschoolers to adolescents. Child protection workers reported fewer than 2% of cases involving children under the age of 6 and 8% of those involving adolescents as cases of false report. Subsequent retraction by the child was found to be the major reason for determining an allegation to be false. Children were found to have insufficient credibility when the report itself was improbable, the details were insufficient, there were inconsistencies in the report, or there was conflicting evidence. Everson and Boat caution that obvious evidence of coercion on a child to recant should not be considered a circumstance for the invalidation of a complaint.

Jones and McGraw (1987) reviewed all reports of suspected sexual abuse made to the Denver Department of Social Services in 1983 ($n = 576$). Each report was investigated by that department and designated as either "founded" or "unfounded." Two of the evaluation categories include fictitious reports by adults and fictitious reports by children. Fictitious reports by adults included deliberate falsifications, misperceptions, and confused interpretations of nonsexual events. Fictitious reports by children included those that were falsely made and involved deliberate

falsifications and misperceptions. Fictitious accounts by adults composed 5% of the sample and fictitious accounts from a child were 1% to 2%. Among the features of fictitious accounts by adults were no child reports despite careful and repeated interviewing, an ongoing custody and visitation dispute, abuse and neglect histories of alleging adults, and emotional disturbance in alleging adults that included two professionals. Among the fictitious child accounts were those from children who ranged in age from 3 to 10, and the context of all was a bitter custody or visitation dispute. The seven mothers were found to be psychiatrically disturbed. Reference is made to other studies that found a lack of detail or of accompanying emotion in fictitious reports. Reference is also made to children who had been sexually abused in the past and made an erroneous subsequent allegation of reabuse. Also noted was the absence of threats or coercive elements in the fictitious allegations.

Ceci and Bruck (1995) discuss false allegations of sexual abuse after reviewing some of the same literature. Discussing the Jones and McGraw (1987) study, they note that of the 576 cases reported that year in Denver, 23% were judged unfounded and in 24% of the cases no clear judgment could be made because of insufficient information. Six percent of the unfounded cases were considered to be deliberate or malicious and another 17% to have been wrong but made in good faith. Ceci and Bruck consider 23% of the reports to be properly categorized as false, of which 6% were deliberate lies and the other 17% baseless. Ceci and Bruck considered the Everson and Boat (1989) numbers to be underestimates. They referenced a study by Faller in 1991 that examined the reports of 136 children of divorced families referred to a project on abuse and neglect in which 35% of the reports were found to be untrue. It was noted that Faller broke down the 35% rate of unsubstantiated cases into three subgroups consisting of false allegations made (8.8%), false allegations reflecting dynamics besides divorce (11%), and false allegations arising in an atmosphere of divorce acrimony (14%). High rates of unsubstantiated sexual abuse reports are the norm, according to Besharov (1994), who studied child abuse and neglect substantiation rates in 30 states in 1986, 1987, and 1988, with rates ranging upwards of 77% in Montana in 1988 and as low as 15% in Alaska in 1986. Kuehnle and Sparta (2006) wrote of a marked nationwide decline during the 1990s of child sexual abuse reports made and in the percentage of reports that are substantiated. These statistics are a reminder to remain open-minded regarding the outcome of any particular investigation. They underscore the importance of a validation effort in every case.

Investigatory error, primarily in the form of improper questioning of alleged child victims, is the central topic of the Ceci and Bruck (1995) analysis. This topic is addressed by the APRI for the purpose of avoiding mistakes but not addressed for the purpose of identifying mistakes already made and estimating their impact. For impact assessment of professional error in the fabrication of abuse reports, we turn to other sources. Mantell (1988) based his analysis on several hundred child protection, family, and criminal court cases he evaluated. He indicates that professional error is one of several kinds of error causing false sexual abuse allegations. The impact is treated as potentially total such that an allegation can emerge from the questioning of a child without prior existence in the child's utterances or experience. The Ceci and Bruck (1995) analysis provides a full discussion of this topic and identifies

categories of error sources in investigatory interviewing. Their analysis focuses on the effects of several forms of suggestive questioning, including interviewer bias, repeated questioning, negative stereotyping, the emotional tone of the interview and the effects of peer pressure, of interviews by adults of high status, of misleading information, of failure to test an alternative hypothesis, and of questioning with the request for visual imaging. This is an analysis of the confounding impact of external influence on autobiographical memory.

The *Report on Scott County Investigations* (State of Minnesota, 1985) shows the effects of repeated questioning, cross-germination (witness contagion), and private motives and pressures in generating graphically detailed false reports by teenagers and young adults of homicide and sexual abuse. This began with the interview of a 12-year-old boy who provided detailed accounts of homicides in 1984, describing seven children being stabbed, mutilated, and shot. Other children of varying ages provided reports of ritualistic torture and murder. After state authorities determined that no homicides had occurred, they turned their focus to allegations of sexual abuse and pornography. The continued interviewing of the children led to an increasingly large list of accused citizens and ultimately to the children reporting abuse by their parents. Children as old as 21 and 18 as well as 12 provided such reports. In some cases, children were interviewed together and prompted to discuss allegations in group settings.

In some respects, the 1989 edition of the APRI manual provides a more detailed account of error sources to be assessed in the validation phase of the investigation. It calls for the child to be interviewed alone, out of the presence of other adults, to make sure that the child is not "overwhelmed" or "bombed" by adult interviewers, and to assess the impact of family and support persons in providing approval and disapproval to the child witness. The interviewer is told to ask simple, direct, open-ended questions, avoid leading questions, and make sure that the information the child is giving is really coming from the child and not from other sources, such as the interviewer. Answers need to be clarified regarding their reliability. A child is never to be pressured to talk or continue an interview. If a child thinks that he or she will get into trouble for talking, the interviewer should find out with whom and for what reason. The interviewer should try to find out if any threats, promises, or rewards were offered or given to the child for talking. If the child says that no abuse occurred, the interviewer should ask the child if there is someone who thinks that something did happen. Other explanations for any behavioral or medical data that may indicate abuse should be considered. Other explanations for the child's sexual knowledge and behavior should be considered and explored. The use of nonleading questions is identified as an important assurance for the analysis of the investigatory interview that the child is using his or her own language for abuse reports. Contamination of the child witness by family members is listed as a key error source to be ruled out. Witnesses who are likely to be in contact with the child or with each other are to be told not to question or rehearse the child, not to investigate the case on their own, and not to compare notes. Members of the child's family are to be interviewed because they may be able to provide direct and indirect corroboration for the abuse report and can also provide observations of behavioral, emotional, and physical indicators of abuse as well as information about the relationship between the child

and the suspect. Family members and others also should be interviewed to rule out their influence on the child to make an abuse report. Other witnesses, such as teachers, nurses, and police officers, are to be interviewed for their information. The APRI recommends a high level of care in the documentation of statements by these witnesses, including videotaping the interview.

11. The Child Corrects the Interviewer The APRI advises that children who correct the interviewer are listening attentively and are not judged to be overly suggestible.

12. and 13. Medical and Forensic Evidence Medical and forensic evidence are additional indicators for analysis. They are listed in the Summary of Indicators. Neither is behavioral in character, and thus they are not included in this discussion beyond noting that although both are infrequent they can be decisive in establishing that abuse occurred and in the identification of the perpetrator.

This section concludes with Table 4.2, a summary of the 13 validation indicators for sexual abuse. Table 4.2 offers a quick overview of the investigatory process and evidentiary record. It includes, as the final indicator, the nature of the alleged offender statement, if any, placing it last as an important abuse indicator. Construction of a case-specific evidentiary chart after completion of a thorough validation review offers a useful perspective on the case and can provide a basis for the development of a more thoroughly informed opinion about the strengths and weaknesses of the total evidentiary record. This can assist the development of an opinion about the completeness and reliability of the procedures used. Oftentimes, investigators are fact witnesses. They report what they have done and the results they have obtained but do not provide professional opinions about its meaning. Interviewers should remain sensitive to the nature of their role and not overstep their boundaries. Whether the interviewer considers the evidence to be strong or weak, it remains the domain of the trier of fact to determine whether or not abuse occurred. The author reminds interviewers of research by Poole and Lamb (1998) and Horner, Guyer, and Kalter (1992) that demonstrates that psychologists are not able to reliably distinguish between true and false reports of sexual abuse.

Table 4.2 Summary of Indicators of Report Validity in Sexual Abuse Cases (APRI)

1. Medical evidence
2. Forensic evidence
3. The child's statement
4. Statement spontaneity
5. Consistency—both internally in the statement and across statements
6. Developmentally unusual sexual knowledge/behavior
7. Idiosyncratic detail
8. Developmentally appropriate language
9. Play and gestures indicative of abuse
10. The child's manner and emotional responses
11. Lack of a motive to fabricate
12. The child corrects the interviewer
13. The offender's statement

2. The American Professional Society on the Abuse of Children

The APSAC publishes seven documents that address investigatory interviewing of alleged child abuse victims:

- Practice Guidelines: Code of Ethics (APSAC, 1997a)
- Practice Guidelines: Psychosocial Evaluation of Suspected Sexual Abuse in Children, second edition (APSAC, 1997b)
- Practice Guidelines: Psychosocial Evaluation of Suspected Psychological Maltreatment in Children and Adolescents (APSAC, 1995c)
- Practice Guidelines: Photographic Documentation of Child Abuse (APSAC, 1995b)
- Practice Guidelines: Use of Anatomical Dolls in Child Sexual Abuse Assessments (APSAC, 1995d)
- Practice Guidelines: Descriptive Terminology in Child Sexual Abuse Medical Evaluations (APSAC, 1995a)
- Practice Guidelines: Investigative Interviewing in Cases of Alleged Child Abuse (APSAC, 2002)

This discussion focuses on the last of the practice guidelines, Investigative Interviewing in Cases of Alleged Child Abuse (APSAC, 2003). The content of this guideline overlaps with many aspects of the APA and APRI guidelines. This guideline opens by reminding us that "inadequate and improper interviewing can lead to errors in decision-making about child safety and criminal prosecution." The guidelines are recommended for use in conjunction with the APSAC Guidelines for the Use of Anatomical Dolls and for the Psychosocial Evaluation of Suspected Sexual Abuse in Children. Aspects of those guidelines will be discussed as well.

The stated purpose of the investigative interview is to obtain as complete and accurate a report from an alleged victim as possible. It is noted that most investigative interviews are conducted by child protective service workers and law enforcement officers and that other mental health and medical professionals may also participate in or conduct such interviews. Interviewers and investigators are expected to develop specialized knowledge. Interviews should be timed to occur as close to the event in question as is feasible and with appropriate notification of parents and guardians. A neutral environment for the interview should be selected whenever possible. Electronic recording in the forms of videotaping and audiotaping are identified as the most comprehensive and accurate method of documentation. The number of interviews should be governed by the need to gather a complete and accurate report from the child. This can often be accomplished in one interview, although some children require more than one interview. However, multiple interviews also carry risks. They have been associated with memory errors, acquiescence to interviewer expectation, and child distress. Interviewing by one interviewer is the preferable practice. The presence of additional interviewers is allowed with the recommendation that a lead interviewer be designated. Closed-circuit TV and two-way mirrors can enable other professionals to monitor the interview while remaining

out of sight. Parents are not to be present during the investigative interview unless there is an exceptional circumstance. Siblings are to be interviewed separately.

Investigators are urged to practice cultural competence and to take cultural features into account during the interviewing process and also in review. Children with special needs may require accommodation.

The APSAC is concerned about interviewer bias and recommends that interviewers approach the interview with an open mind about what may have happened. The customary practice is to know the specifics of an allegation before the interview. Interviewers are reminded that the purpose of the interview is "not merely to confirm prior suspicions" but to "elicit information from the child." It is noted that in some jurisdictions there is a standard practice that the investigative interview begin with an interviewer who is blind to the allegations. Bias is seen as impairing the capacity of the interviewer to receive and to objectively interpret information and as a source for interviewer error. Negative statements about suspected abusers are one source of error and suggestion. A number of interview aids, such as anatomical dolls, anatomical drawings, and doll furniture, may be used, but the timing and manner of their use is important. Investigators are expected to have specialized knowledge about these aids. The Practice Guideline: Use of Anatomical Dolls in Child Sexual Abuse Assessments (APSAC, 1995d) essentially dismisses the concerns that the use of anatomical dolls leads to erroneous reports of sexual abuse and leaves the timing of their introduction to the judgment of the interviewer.

This position contrasts with that of Ceci and Bruck (1995), who in their own studies and the review of the literature found that with the use of anatomically detailed dolls many children made both verbal and nonverbal reports of genital touching and digital penetration though neither had occurred. Younger children were found to be more susceptible to such error. Ceci and Bruck recommend that the dolls not be used for diagnostic purposes, stating, "We have reviewed enough studies that demonstrate that this tool has the potential for serious misuse, including misdiagnosis" (p. 186). Poole and Lamb (1998) also provide a detailed discussion of the anatomical doll issue and of the research literature. They indicate that there are no standardized procedures for the use of anatomically detailed dolls in investigative interviewing and that they are not a psychological test for the diagnosis of abuse. Although the APSAC allows for them, Poole and Lamb note that the APSAC-defined preferred practice is for their use to identify body parts, to clarify previous statements, and for a demonstration by non- or low-verbal children after there is an indication of abuse activity. Sparta and Koocher (2006) appear to support the use of anatomically detailed dolls in investigative interviews without discussing the timing and manner of their use, while also noting that some researchers are concerned about the error rates associated with their use because of a reduction in the accuracy of children's reports.

The APSAC states that "research does not confirm that anatomical dolls are inherently too suggestive or sexually stimulating." The APSAC specifically recommends the use of anatomical dolls for several purposes: identification of body parts, as a visual aid for direct inquiry about the child's personal experience with private parts, and as a demonstration aid. The APSAC specifically recommends the use of dolls with preschool children, the age group in which Ceci and Bruck (1995)

found the highest error rates for children's abuse reports of genital touches in pediatric examinations that did not contain such touches. Melton, Petrila, Poythress, and Slobogin (1997, p. 476) conclude that anatomical dolls are "undoubtedly the most controversial evaluation technique." They note that there is a consensus that play with anatomically detailed dolls cannot be used as a test of whether or not child maltreatment has occurred. They note that a question remains about how the dolls can be used "even as demonstration aids to clarify a child's statements" (p. 477). Ceci and Bruck indicate that jurisdictions vary dramatically in legal decisions about the use of anatomical dolls, ranging from those that ban their use entirely to others that permit children to use the dolls in court to demonstrate their own sexual abuse.

In its practice guidelines, the APSAC distinguishes anatomical drawings from anatomical dolls and provides little guidance for how anatomical drawings might be used or about their limitations. They are identified as an interview aid that can assist the child to identify names for body parts and functions and also the location of possible sexual touching or physical injuries. The author recommends the best practice approach, which is to avoid issues of suggestive questioning and reserve the use of anatomical drawings and dolls for the confirmation phase of an interview following a demonstration that their use was needed after other avenues of explanation had been exhausted (Poole & Lamb, 1998).

The APSAC recommends careful preparation for the forensic interview by developing a list of specific topics and the hypothesis to be addressed. The interviewer should appear relaxed and nonthreatening. The interview itself should be paced to mesh with the child's developmental and temperamental characteristics, and the interviewer should proceed slowly and never pressure a child to respond to questions. Especially with younger children, the interviewer should be aware of signals indicating fatigue and loss of concentration as well as the need for a break.

In regard to questioning strategies, the APSAC recommends minimizing the introduction of information to the child that might be adopted by the child in his or her report. (This is the very principle underlying objections to the use of anatomical dolls and drawings and good and bad touch instruction prior to a child abuse inquiry and an abuse report.) The use of open-ended questions is recommended. Yes/no and multiple choice questions are allowed. Yet the interviewer is encouraged to focus open-ended questions to specific topics that the child has introduced by saying "Tell me more about. . . ." In a later section, specific, closed questions such as those that have yes/no and multiple-choice formats are recommended after open-ended questioning has concluded and appears to have exhausted the information that a child can provide. When the multiple-choice format is used, the example given is, "Was he in the house, in the yard, or someplace else?" This author notes that the forced-choice format of this formulation can be reduced by including the options "I don't know" and "I don't remember" as approved alternatives and then making sure that the child has the ability and willingness to use them. It is also essential that such questioning first be clarified as relevant. This can be achieved by asking "Do you know where he was when . . . ?"

The APSAC expresses uncertainty about how to define leading and suggestive questions. White and Quinn (White and Quinn, 1988, p. 271) define a leading question according to Black's Law Dictionary, "as one which instructs the witness

how to answer or puts into his mouth words to be echoed back." The suggestive question is defined as "presentation of an idea especially indirectly as through association of ideas, bringing before the mind for consideration in the nature of a hint" (p. 271). White and Quinn state that the question "Your daddy did put his finger in your vagina, didn't he?" is leading, whereas the question "Did your daddy put his finger in your vagina?" is suggestive. Others, such as the APRI, might classify the second alternative as a direct question. There would probably be unanimity about the fact that the question contains information descriptive of bodily touching and in this case of the digital penetration of the genitalia and that it is not an example of a general, open-ended question to which a child might provide a wholly spontaneous and independent response. There also can be no controversy about the fact that the question is formulated in the vocabulary of the interviewer, not the vocabulary of the child, and that its use poses the concern of the respondent acquiescing to the question with confirmation. Whenever possible, it is recommended to avoid these risks by asking questions that call on the child to independently express the substance of an idea descriptive of action and experience.

The APSAC describes the following components of investigative interviewing. It begins with the interviewer introducing himself or herself and the interviewer's role, followed by rapport building in which there is a discussion of neutral topics. Developmental screening can then occur to assess the child's ability to understand and respond to questions. The competency check may occur in jurisdictions that require an assessment of the child's understanding of the difference between the truth and a lie. This is distinguished from the assessment of the child's ability to provide information about events that are known to have occurred, such as a recent birthday party. The ground rules of the interview are then explained. The child should be told that the interview is about things that really happened, that the child will be asked many questions, and that the child cannot guess about answers and can say "I don't know" and "I don't remember." The child is encouraged to correct the interviewer when the interviewer makes a mistake. If the interviewer asks a question that is too hard or difficult for the child to talk about, the child should let the interviewer know.

The APSAC recommends that the topic of abuse be introduced in one of a number of ways:

- "I understand something may have happened to you. Tell me about it from beginning to end."
- "I understand you had to go to the doctor. Is there a reason you had to go?"

The author finds these questions problematic and recommends avoiding them. They can convey bias, expectation, and alignment of the interviewer with authority figures who may be close to the child and instructed the child about the purpose of the interview and what to say. They may convey to the child that the interviewer wants the child to repeat prior statements or the opinions of others. Getting started does not require reliance on methods that contain error and that may mislead the respondent. Alternative questions are:

- Do you know why your . . . brought you here to speak to me?
- What were you told about what we are going to do and talk about?
- How do you feel about being here?
- What would you like to do and talk about?

Situation-focused questioning is also encouraged, such as asking a child about bath time or toileting, about secrets, or about things the child doesn't like. The APSAC says that anatomical drawings and dolls are allowed at this time and are useful for the purpose of assessing the child's body knowledge, focusing the discussion on bodies and body experience, and giving the child permission to discuss sensitive topics such as private parts.

Children should be asked to provide details about concerning events. Here the APSAC recommends the use of open-ended narrative prompts, such as "Tell me more about that" and "Then what?" This actually constitutes an inversion of questioning sequence from more specific (body parts and touch experiences) to the open-ended or general questions. When abuse reports occur, the child should be encouraged to provide detailed information, contextual information, and information about abuse witnessing by other persons. The APSAC states that anatomical drawings and dolls can be useful during this phase of the interview for clarification purposes about what the child is attempting to describe. It appears that the APSAC is recommending the use of anatomical drawings and dolls for two purposes: first, to focus the child's attention on the topics of interest (to the interviewer), and second, to clarify with the child the report that has been received.

The psychologist may wonder how to start an abuse inquiry if anatomical dolls and drawings and good-touch/bad-touch clarification are not used. Opening an abuse inquiry is not difficult. Many children know why they have been brought to the interview. Often, asking about this alone is sufficient to elicit the child's report and to set the stage for the forensic interview. When the child says that he or she does not know the reason for the interview, the interviewer can inquire about who brought the child, what was discussed, and whether the person who brought the child said what the reason is. If this does not lead to mention of the forensic topic, the child can be generally canvassed about major life experiences and about relationships with people, and then ultimately about general classes of positive and negative and also body experience. This can include questioning about general safety rules when crossing the street, with hot things, with water, with dressing and undressing, with toileting, and with body touching.

The APSAC does not discuss validation procedures. The reader is referred to the APRI guidelines, Ceci and Bruck (1995), Poole and Lamb (1998), and White and Quinn (1988) for a detailed discussion of this issue.

3. Policy Statement from the American Academy of Child and Adolescent Psychiatry: Guidelines for the Clinical Evaluation of Child and Adolescent Sexual Abuse

The AACAP (1990) states that the purpose of a clinical evaluation in a child sexual abuse case is to determine whether abuse has occurred, if the child needs protection,

and if the child needs treatment for medical or emotional problems. The guidelines were developed to assist clinicians performing those evaluations. The AACAP sets high standards for the qualifications of evaluators. They should possess sound knowledge of child development, family dynamics related to sexual abuse, and the effects of sexual abuse on the child. They should be trained in diagnostic evaluation of both children and adults and should be comfortable testifying in court. If not a child and adolescent psychiatrist or psychologist, clinicians performing these evaluations should be under the supervision of professionals with these qualifications. Role differentiation should occur so that the child or adolescent therapist is not the evaluator.

The child should be seen for the minimum number of times necessary to conduct the evaluation, with cross-agency sharing of information to avoid duplication and the unnecessary interviewing of the child. Optimally, evaluation teams should be formed that integrate local police and reporting agencies. Multiple interviews may be viewed by the child as a demand for more information, and the concern is specific that this alone may encourage confabulation.

The evaluation should be conducted in a relaxed environment. Comprehensive history taking is important and should include developmental history, cognitive assessment, prior history of abuse and other traumas, relative medical history, behavioral change, parental history of abuse as children, and family attitudes toward sexuality and modesty. Prior psychiatric disorders in the child or parents should be identified, and allegiances by a child to parents should be assessed.

In intrafamilial abuse, the family history and perspective of each parent should be obtained. Stressors other than sexual abuse that might account for the child's symptoms need to be identified. This can include a psychiatric assessment of each parent, particularly if there is a concern about a false allegation. This concern can arise when it is determined that the allegation is coming from a parent rather than from a child, when the parents are engaged in a dispute, and when the child is a preschooler. False allegations are seen as arising from other situations as well, such as the misinterpretation of a child's statement and the behavior of adults. Adolescents may make a false allegation because of vindictiveness or to cover up their own sexuality, and children who experienced prior sexual abuse may misinterpret or mischaracterize the behavior of an adult.

In conducting a clinical evaluation, it is required that the clinician maintain emotional neutrality, approach the case with an open mind, and seek out the unique characteristics of each case. Clinicians are urged to avoid leading and coercive questions and techniques and to allow the child to tell the story in his or her own words. The need for external corroboration distinguishes the forensic evaluation from the usual clinical evaluation; this can include the need to review medical and school reports and prior psychiatric evaluations and talking with other persons.

A number of characteristics enhance a child's credibility. These include event descriptions in the child's own language and from the child's own point of view, spontaneity, an appropriate degree of anxiety, idiosyncratic or sensorimotor detail, consistency of allegation over time, behavioral changes consistent with abuse, absence of motivation or undue influence for fabrication, and corroborating evidence.

In regard to anatomically correct dolls, AACAP states that they may be used for eliciting the child's terminology for anatomical parts and for allowing a child who cannot tell or draw what happened to demonstrate what happened. The AACAP strongly advises that dolls not be used to instruct, coach, or lead the child or as a shortcut for a more comprehensive evaluation, and that in these assessments it is not necessary to use anatomically correct dolls at all. California has barred the admissibility of evidence obtained through the use of anatomically correct dolls.

By contrast, children's drawings are seen as a helpful assessment tool. This can include spontaneous drawings or requesting a child to draw a male or female, to draw what happened and where, or to draw a picture of the alleged offender. Videotaping is encouraged. Psychological testing may be helpful but is not seen in and of itself as a method for determining whether or not sexual abuse occurred.

SUMMARY

Nine of the 15 APA publications addressing ethical standards and guidelines apply to children and adolescents. Three of the guidelines speak specifically to clinical forensic practice. Aspirational in nature, these three guidelines have much in common with one another, with APA ethical principles, and with central features of the remaining guidelines. They all call for a high level of science-based professional practice based on the principles of thoroughness and fairness. Psychologists are expected to practice and testify within the limits of their knowledge and expertise, to communicate their results and opinions clearly, to understand their role and the law when practicing in a legal context, and to avoid dual relationships, bias, and error in their work. High standards for record keeping and for the documentation of clinical and forensic work are emphasized. The increasing importance of cultural competence is the focus in several statements. Evidence-based practice is the standard. Psychologists performing child custody evaluations are expected to possess specialized competencies for the evaluation of adults, children, relationships, and parenting ability. These skills and a matching knowledge base are also expected of psychologists performing court-ordered studies in child protection matters; they are additionally expected to have a solid understanding of the field of child abuse and neglect. Psychologists do not offer opinions about persons they have not examined or about issues for which they lack clear and verifiable evidence.

The APRI has articulated the most complete statement regarding the investigative interviewing of suspected child victims of abuse concentrating on sexual abuse. The strengths and weakness of the APRI guidelines were reviewed, along with those of the APSAC and the AACAP. Whereas the APA Guidelines for Psychological Evaluations in Child Protection Matters deals with general practice issues, the APRI guidelines offer specific instruction for the forensic interviewer on how to conduct and validate an investigative interview within the larger context of a multidisciplinary forensic investigation. The strengths and weaknesses in the APRI and APSAC guidelines were identified and discussed with the help of behavioral science research. Research-based recommendations were made for alternative strategies and methods. The APRI model for a comprehensive review of the evidentiary

record was presented and discussed. Unresolved and controversial issues and best practices were identified.

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CHAPTER 5

Professional Roles

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Given that this volume of the *Handbook of Clinical Psychology* focuses on children and adolescents, it seems fitting that a chapter be devoted to a discussion of the various professional roles assumed by psychologists who work with children and youth. While these psychologists are often referred to by various labels, such as clinical child psychologists and pediatric psychologists, professionals with other backgrounds, such as school psychologists, also work with children and adolescents.

Child-oriented psychologists vary in the nature of their professional training (clinical psychology, school psychology, developmental psychology, specialty-specific training in clinical child or pediatric psychology); theoretical orientation (behavioral, cognitive-behavioral, interpersonal, psychodynamic, humanistic); the degree to which they focus on research, clinical activities, or both; the nature of the professional activities they engage in (teaching, research, assessment, therapy, consultation, advocacy); the types of child or adolescent problems they work with (mental health problems, physical health problems, school-related difficulties); and the settings in which they work (universities, medical centers, children's hospitals, pediatric clinics, private practice, mental health centers, schools). Given the diversity of training, theoretical orientation, professional activities, types of problems addressed, and settings, it should not be surprising that these professionals often function in multiple roles and engage in a range of professional activities in the context of these roles.

This topic of professional roles is of special relevance given the rapid growth of clinical child and pediatric psychology in the past 25 years and the degree to which this growth has resulted in expanded role opportunities and, in some cases, significant changes in the nature of the activities engaged in by psychologists serving children and adolescents.

ROLES AND ROLE-RELATED ACTIVITIES OF CHILD-ORIENTED PSYCHOLOGISTS

Before embarking on the task of highlighting the professional roles assumed by child-oriented psychologists, it is important to attempt to define what is meant by the term "role" and differentiate this term from the professional activities associated with these roles.

Considering several dictionary definitions, the term role, as it is used here, generally refers to a *consistent pattern of behavior* that is expected of someone based on his or her status or position in society (Merriam-Webster's New Collegiate Dictionary, 11th ed.). From this perspective, functioning in the role of a clinical child or pediatric psychologist involves engaging in a pattern of professional behavior that is consistent with one's prior training as a specific type of child health service provider. At the simplest level, the term "professional role" as applied to a clinical child psychologist can be best thought of as a general *construct* used to summarize the range of interrelated specialty-specific behaviors that are engaged in as one trained in clinical child psychology carries out his or her professional activities.

Closely associated with the term role is the pattern of specialty-specific professional activities, engaged in by the psychologist by virtue of his or her professional training, that define him or her as a specific type of health care professional. As regards the various roles of child-oriented psychologists, some function in the role of clinical child psychologist, and others function in the role of pediatric psychologist. Still others function in the role of school or applied developmental psychologist. Both clinical child and pediatric psychologists may engage in somewhat similar activities to a greater or lesser degree. For example, both may engage in assessment, intervention, consultation, research, or various other professional activities with children and families. But, while there may be some similarity in terms of the general types of activities engaged in, the roles may differ significantly in the way these activities are carried out, the problems and clinical populations dealt with, and the settings in which these professional activities take place.

As but one example, the clinical child psychologist's activities might involve an interview and administering a well-chosen selection of psychological tests to assess for the presence of Attention-Deficit/Hyperactivity Disorder (ADHD) and possible comorbid conditions (e.g., learning disability, anxiety or depressive features), collaborating with a child psychiatrist in the treatment of the child's core ADHD symptoms and using cognitive-behavioral interventions to treat the child's diagnosed anxiety or depressive disorder, in a private practice or mental health clinic setting. In contrast, a pediatric psychologist's assessment activities might entail interviews with the child and parent, consultation with the child' physician and nursing staff, and behavioral observations at meal time to determine factors contributing to a 5-year-old's frequent vomiting and failure to eat. Treatment might involve behavioral interventions designed to teach the parent how to reward successive approximations to normal eating and avoid attending to statements of stomach upset and instances of vomiting. Other interventions might involve additional forms of reinforcement for increased caloric intake, as well as other contingencies following vomiting episodes, with such treatment taking place on a pediatric floor of a university-affiliated medical center. Thus, although both may engage in similar activities (e.g., assessment, consultation, intervention), the nature of the specific activities involved may differ depending on whether the clinician is functioning in the role of clinical child or pediatric psychologist.

It should also be emphasized that child-oriented psychologists may function in roles that represent a greater or lesser degree of specificity. For example, one clinical child psychologist may engage in a wide range of child-related professional

activities such as assessment, therapy, teaching, and research, and another may function more specifically in the role of child psychotherapist. Likewise, a pediatric psychologist may functional broadly in this professional role (e.g., working with children displaying both acute and chronic illnesses and engaging in a variety of child or adolescent consult-liaison activities) or may restrict his or her professional activities, as in the case of the pediatric psychologist whose work primarily involves conducting pretransplant evaluations with children and adolescents and working with children who have recently been transplanted (who may think of himself or herself as a pediatric transplant psychologist). Finally, whereas some clinical child or pediatric psychologists assume roles that involve varying levels of specialization, as described earlier, others simultaneously function in multiple professional roles (e.g., administrator *and* clinical child psychologist), each of which may vary in terms of breadth and complexity and relationship to one another.

Despite general agreement that there is an insufficient number of adequately trained child-oriented psychologists to meet the mental health needs of children, there is an increasingly large number of psychologists who, by virtue of their training and/or professional experience, currently define themselves as clinical child or pediatric psychologists and engage in a wide range of professional activities with children, youth, and families. In the sections to follow we provide an overview of the diverse range of professional activities and functions that are associated with these professional roles. This is followed by a brief commentary on the role of school psychologists, as they also often provide clinical services to children and adolescents.

CLINICAL CHILD AND ADOLESCENT PSYCHOLOGY

The area of clinical child and adolescent psychology represents a rapidly growing area of professional psychology. A capsule summary of the multiple roles and functions played by professionals working in this area can be seen in the ways this specialty has been self-defined.

The following definition of the specialty of clinical child psychology has been provided in the archival description of the specialty submitted to the Commission for the Recognition of Specialties and Proficiencies in Professional Psychology (1998), upon the formal recognition of the specialty by the American Psychological Association in 1998:

> Clinical Child Psychology is a specialty of professional psychology, which brings together the basic tenets of clinical psychology with a thorough background in child, adolescent and family development and developmental psychopathology. Clinical child and adolescent psychologists conduct scientific research and provide psychological services to infants, toddlers, children, and adolescents. The research and practices of Clinical Child Psychologists are focused on understanding, preventing, diagnosing, and treating psychological, cognitive, emotional, developmental, behavioral, and family problems of children. Of particular importance to clinical child and adolescent psychologists is a scientific understanding of the basic psychological needs of children and adolescents and how the family and other social contexts influence socioemotional adjustment, cognitive development, behavioral adaptation, and health status of children and adolescents. There is an essential

emphasis on a strong empirical research base recognizing the need for the documentation and further development of evidence-based assessments and treatments in clinical child and adolescent psychology.

Of special note in this definition is that the training of clinical child psychologists represents (or at least should represent) a blending of developmental psychology and clinical psychology in a framework that places emphasis on the developmental aspects of psychopathology, which has as its focus understanding the mechanisms of development and behavior change. Central to the focus on developmental psychopathology is the belief that the study of atypical development can add to the understanding of normal development and, conversely, that the methods and approaches used in normative developmental science can shed light on the etiology, course, and outcome of psychological disorders. It is this focus on training in child, adolescent, and family development *and* developmental psychopathology, along with the tenets of clinical psychology, that distinguishes the well-trained clinical child psychologist from the general clinical psychologists who happen to also work with children. It needs to be emphasized that clinical child psychology is more than just the knowledge of clinical psychology methods applied to children.

This definition also highlights the fact that clinical child psychologists work with a wide range of individuals, displaying a wide range of difficulties, through the application of a wide range of assessment and treatment methods. In the sections to follow we elaborate on these topics.

Populations Served

Clinical child psychologists work with children of all ages, ranging in age from infancy and childhood through the adolescent years, and their families. Nevertheless, children of some ages tend to be seen more frequently in clinical practice. For example, Tuma and Pratt (1982) found that the most common age groups seen were children between the ages of 5 and 10 (29%). The next most common age groups seen were adolescents 14 to 18 (23%), preadolescents 11 to 13 (22%), and preschoolers below 4 (13%). These child clinicians also reported seeing a significant number of adults, with 38% reporting working with patients over age 18.

Clinical child psychologists work with children who display a wide range of problems. Unlike the pediatric psychologist, who is likely to focus more on child and adolescent health-related problems, the clinical child psychologist is likely to focus more on mental health difficulties, although the distinction between the activities of clinical child and pediatric psychologists can be blurred on occasion. Problems that clinical child psychologists are often called on to assess and treat can range from easily managed problems such as bedwetting and soiling and helping parents manage a child with a difficult temperament to internalizing problems such as anxiety and depression and externalizing conditions such as Oppositional Defiant Disorder, Conduct Disorder, and ADHD. Clinical child psychologists also see children with learning disabilities, as well as those with severe psychological disorders such as mental retardation, pervasive developmental disorders, Schizophrenia, and pediatric Bipolar Disorder. Child clinicians are often called on to deal with problems

that may or may not be defined in terms of a specific diagnosis. Here, examples include working with children of divorce, those who have been vicariously exposed to marital abuse, those who have experienced physical or sexual abuse, or those who have been impacted by hurricanes or other natural disasters.

Practice Settings

For those clinical child psychologists engaging in applied clinical work, practice settings include private practice, children's residential treatment centers, mental health clinics, and public health agencies, schools, medical school clinics. Those whose primary role is university teaching (often in departments of psychology) may engage in part-time clinical activities in the university setting. As will be seen later, child clinicians who are more health-oriented often practice in children's hospitals or sometimes in private practice pediatric settings.

Some figures regarding clinical practice settings have been provided by Tuma and Pratt (1982) in their survey of 358 clinical child psychologists who were members of what is now the Society of Clinical Child and Adolescent Psychology. The data provided by this survey suggest employment in a wide variety of work settings, with most respondents working in more than one setting. Most reported working in medical school settings, primarily in departments of pediatrics and psychiatry (51%). The second largest work setting was private practice (44%). Other work settings included graduate departments of psychology (21%), schools (15%), mental health centers (11%), child guidance centers (9%), children's hospitals (9%), and general hospitals (5%). A much smaller number were employed in residential and state hospitals.

More recent survey findings regarding practice settings of 162 child-oriented psychologists (including clinical child and pediatric psychologists and school psychologists) have been provided by Cashel (2002). Of these clinicians, 29% reported working in a medical center or other hospital settings. Other settings included college or university (22%), school system (17%), outpatient clinic (11%), and community mental health center (1.9%). As in the Tuma and Pratt (1982) survey, many of the respondents in this study worked in multiple settings, with approximately 31% working exclusively in private practice.

Taken together, these findings suggest that, across time, the primary practice settings of clinical child psychologists appear to be medical schools and hospitals, private practice, and university settings where the child clinician likely also plays a major role in clinical training and supervision. In other settings the numbers are considerably smaller, with surprisingly few clinical child psychologists working in community mental health centers.

Theoretical Orientations

Of interest is the theoretical orientation of clinical child psychologists, especially as this may show changes over time. This issue is of relevance to clinical practice as approaches to both assessment and treatment are usually, to some degree, tied to the theoretical orientation of the clinician. And, to the extent that the primary theoretical orientation of clinicians changes over time, this is likely to have implications both

for the role functions of the clinician and for the type of training those aspiring to be clinical child psychologists will need to meet changing role functions.

Data relevant to the issue of theoretical orientation have been provided by the two surveys cited earlier. In the Tuma and Pratt (1982) survey, 28% of the 358 clinical child psychologists surveyed classified themselves as having a psychodynamic orientation, 25% described their orientation as behavioral, and 4% described their orientation as humanistic-existential; the remainder described their orientation as eclectic.

These findings regarding theoretical orientation can be contrasted with the findings 20 years later (Cashel, 2002). Here, the majority of child clinicians described themselves as cognitive-behavioral (52%), and an additional 7% described themselves as having a behavioral orientation. Thus, almost 60% of the respondents endorsed either a behavioral or cognitive-behavioral orientation. Respondents endorsing other orientations included psychodynamic (9%) and interpersonal (5%), with 25% labeling themselves eclectic.

These findings suggest that during the 20-year period between 1982 and 2002, the theoretical orientations of child clinicians have changed significantly, with a vastly increased number of clinical child psychologists now endorsing a behavioral or cognitive-behavioral theoretical framework (and presumably assessment and treatment methods based on this framework), with a concurrent reduction in clinicians claiming a psychodynamic orientation. These findings would seem to be consistent with the increasing focus on both evidence-based assessment (Kazdin, 2005; Mash & Hunsley, 2005) and evidence-based child treatment methods (Ollendick & King, 2004) that currently characterize clinical child psychology.

Role-Related Professional Activities

As suggested earlier, functioning in the role of clinical child psychologist can involve participating in an extensive array of professional activities that serve to define the specialty. Among the most prominent of these activities are assessment, treatment, and conducting research on a wide range of topics relevant to children and families. A sizable number of clinical child psychologists, especially those employed in academic settings, function as teacher and clinical supervisor, in addition to their involvement in research and perhaps part-time clinical work. A somewhat smaller but still significant number of clinical child psychologists also work in the area of prevention and engage in consultation activities. In the sections to follow we highlight the work of clinical child psychologists in a number of these areas and, where relevant, the ways these activities have changed as a function of the evolution of the specialty.

Assessment Activities

Assessment has long been a major professional activity of clinical child psychologists of various theoretical orientations, and for some practitioners, providing psychological assessments for children and adolescents is their primary professional activity. Indeed, assessment is one of the most frequently engaged in professional activities of psychologists working with children. Assessment can involve

open-ended interviews, the use of intelligence tests, and the assessment of academic achievement, adaptive behavior, visual-motor functioning, as well as the use of projective tests and various questionnaires to assess personality dynamics and what are thought to be enduring personality characteristics. Assessment can also involve the use of more structured interviews to assess specific symptoms of psychopathology, the use of objective behavior problem checklists designed to assess behavioral characteristics of the child, and approaches to assess overt behaviors and the factors that elicit (antecedents) and maintain (consequences) these behaviors in the natural environment. Also commonly used are measures designed to assess aspects of parenting stress and family environment as well as other variables thought to be relevant to the functioning of the child and his or her family. Given that assessment is essential to making an accurate diagnosis of childhood problems and is usually assumed to be an essential prerequisite to the development of an appropriate treatment plan, it is not surprising that clinical child psychologists have typically devoted a significant amount of time to assessment-related activities.

It is noteworthy that the nature and, to some extent, the *type* of involvement by clinical child psychologists in assessment activities has changed over the years. For example, in the Tuma and Pratt (1982) survey, the average amount of professional time spent in assessment activities with children, adolescents, parents, and families was 28%, as compared to 41% of their time being devoted to treatment-related activities. According to the authors, assessment and treatment were the two primary professional activities engaged in by these clinical child psychologists.

These authors also provided information regarding the nature of the assessment measures used by this sample of clinical child psychologists. Here, 83% reported using intelligence tests, 54% reported using the Rorschach, 53% the Thematic Apperception Test (TAT), and 60% figure drawings. A total of 53% indicated that they used the Bender Gestalt test, 51% indicated that they used achievement tests, 18% said they used the Minnesota Multiphasic Personality Inventory (MMPI), 17% reported use of sentence completion tests, and only 13% reported the use of behavioral assessment measures. As can be seen, apart from the use of intelligence and achievement tests, the child clinicians in this survey appeared to rely heavily on the use of projective tests of various types, to the exclusion of more behaviorally oriented assessment measures.

These assessment preferences can be compared to those found by Cashel (2002), where respondents reported spending 27.3% of their time conducting psychological assessments, supporting the view that assessment continues to be a major professional activity engaged in by child-oriented clinicians. When these respondents were asked to indicate their use of various assessment measures, the Wechsler Intelligence Scale for Children was, by far, the most frequently used measure. When asked to rank other assessment approaches by order of importance, the clinical interview was ranked as most important, followed by behavioral observations, behavior ratings, self-report inventories, and, finally, projective tests (Cashel, 2002).

These findings suggest that although some clinical child psychologists still use the Rorschach, the TAT, figure drawings, and other projective approaches, the perceived value of these measures has declined, while behaviorally focused assessment measures have become increasingly more popular and appear to be more highly

valued by today's clinical child psychologist. This trend parallels the increasing focus on evidence-based assessment, where the preference is not only for useful assessment methods that are supported by research findings but also for measures that are useful in monitoring treatment effectiveness (Kazdin, 2005). Not only has the focus on evidence-based practice likely had some impact on child assessment practice, but the nature of child assessment has also been impacted by managed care. Today there seems to be less of a focus on the use of standardized test batteries and more on assessment methods relevant to specific problem areas. In the Cashel (2002) survey, it was found that the utilization of many previously popular test measures such as the Rorschach and MMPI seems to have declined due to decreased reimbursement for such measures. Although assessment continues to be a major professional activity of clinical child psychologists, the nature of the assessment measures used by child clinicians today is increasingly moving away from subjective measures and projective techniques and toward more objective, behaviorally focused, and more empirically supported assessment measures.

Intervention Activities

Along with assessment, providing treatment for children, adolescents, and their families is one of the defining professional activities of clinical child psychologists. Indeed, many clinical child psychologists in private practice function primarily in the role of child and family therapist.

It can be recalled that Tuma and Pratt (1982) found that, on average, clinical child psychologists spent over 40% of their professional time engaging in treatment-related activities with children, adolescents, and families. The types of therapy engaged in included counseling with parents, play therapy, individual psychotherapy, behavior therapy, and family therapy. Not surprisingly, the type of therapy varied as a function of the child's age, with parent counseling, play therapy, and behavior therapy being used by 63%, 44%, and 36% of the therapists, respectively. With school-age children, play therapy (53%), family therapy (37%), and individual psychotherapy (35%) were the most widely used approaches, with fewer therapists endorsing parent counseling and behavior therapy (29% each). Individual psychotherapy (74%), family therapy (46%), and counseling with parents (32%) were the most widely used therapeutic approaches with adolescents.

Today, clinical child psychologists engage in a wide range of interventions with children and families. These continue to include many of the traditional approaches to child treatment described by Tuma and Pratt (1982; e.g., parent counseling, play therapy, individual psychotherapy, family therapy, behavior therapy) but with perhaps a much greater focus on behaviorally oriented parent training approaches and behavior management approaches applied in the school, community, and health care settings. Today, we also see the widespread use of more recently developed cognitive-behavioral approaches to treatment that have been found useful in dealing with children and adolescents displaying psychological difficulties such as anxiety, depression, anger management, and Obsessive-Compulsive Disorder (O'Donohue, Fisher, & Hayes, 2003), as well as multimodal interventions for children and adolescents exhibiting complex problems.

This movement toward the use of more behavioral and cognitive-behavioral child therapies appears due to two interrelated factors: the changing theoretical orientation of child-oriented clinicians and the coming of age of evidence-based treatments. As noted earlier, 25 years ago the two predominant theoretical orientations of clinical child psychologists were psychodynamic and behavioral, with somewhat more child clinicians endorsing the psychodynamic perspective and likely engaging in treatments that were in line with their orientation (Tuma & Pratt, 1982). More recent surveys suggest that almost 60% of child-oriented clinicians now define themselves as having cognitive-behavioral or behavioral orientations, with fewer than 10% identifying themselves with the psychodynamic perspective (Cashel, 2002). This represents a dramatic shift in theoretical perspective that, along with the increased focus on the development of evidence-based treatments, likely relates to the increased use of behaviorally related approaches to intervention that we see today.

Teaching and Training Activities

As was briefly noted earlier, surveys conducted 20 years apart by Tuma and Pratt (1982) and Cashel (2002) have both found that a sizable number of clinical child psychologists work in some type of academic setting. Both found that approximately 20% worked in college and university settings, often in graduate departments of psychology. Likewise, both found that many worked in medical school settings (51% in the Tuma and Pratt survey and 29% in the Cashel survey). Employment was usually in departments of pediatrics or psychiatry.

Clinical child psychologists employed in these settings are frequently involved in training of some type. In university departments of psychology, where the academic training of clinical child psychologists often takes place, teaching can take several forms. It can involve teaching basic psychology courses (e.g., introductory psychology, personality, abnormal psychology) or teaching academic courses relevant to specialty training (e.g., developmental psychology, developmental psychopathology, child assessment, child treatment, evidence-based practice). Teaching in this setting can also involve a supervisory role in the context of various clinical practica, where the primary focus is on helping the psychology trainees develop important conceptual and clinical skills. Importantly, teaching and mentoring may also take place in the context of the student's research training, which, ideally, should at some point involve research with clinical populations so the trainee can learn how clinical knowledge can inform research and how good clinical research can inform clinical practice.

For those clinical child (or pediatric) psychologists working in medical schools, teaching formal didactic courses may be less common than teaching seminars relevant to aspects of clinical practice and the clinical supervision of trainees (e.g., interns, postdoctoral fellows, psych techs). Not infrequently, psychologists working in this setting are asked to contribute to the behavioral science curriculum of medical students or to coteach courses such as "Introduction to Patient Evaluation," which are often designed to help medical students learn clinical interviewing and relationship-enhancement skills that will serve them well as professionals working with patients.

The teaching possibilities for clinical child psychologists are many. They are also important, as the quality of training that future clinical child psychology (or other) trainees receive is likely related not only to the trainees' ultimate level of professional competence but also to the quality of care that the trainees provide children and families upon completion of training.

Research Activities

Over the years, the predominant model of graduate training for clinical psychologists has been the Boulder or scientist-practitioner model (Raimy, 1950; Shakow, 1948). This model has also been repeatedly endorsed as the preferred model for training both clinical child and pediatric psychologists. As the basic premise of the scientist-practitioner model is that the clinical child psychologist should be trained as both a scientist and practitioner, it is not surprising that clinical child psychologists often adopt the roles of both researcher and practicing clinician. Although relatively few clinical child psychologists in private practice devote a significant amount of time to research, adoption of these dual roles is not at all uncommon for clinical child psychologists who work in academic or medical school settings, where they are likely to also function in the roles of teacher and supervisor.

The range of research topics addressed by clinical child psychologists is too broad to cover here but includes research related to the development and validation of assessment measures (e.g., structured and semi-structured interview measures, intelligence and achievement measures, measures of development and adaptive behavior, behavior problem checklists, personality questionnaires, computer-based assessment measures); research related to the development and documentation of the efficacy and effectiveness of various approaches to child treatment; research related to the etiology, correlates and developmental course of various types of child psychopathology; and research on factors that may place the child at risk for later psychopathology (e.g., genetic factors, difficult temperament, divorce, abuse), just to name a few areas.

One area that deserves special attention relates to efforts to identify empirically supported psychosocial treatments for children and adolescents. This work, which has a similar focus to the evidence-based practice movement in medicine (Sackett, Richardson, Rosenberg, & Haynes, 2000), resulted from the efforts of the Society of Clinical Psychology's (APA, Division 12) Task Force on Promotion and Dissemination of Psychological Procedures (Chambless, 1996; Chambless & Hollon, 1998).

This task force, which began its work in 1995, initially attempted to develop criteria for determining psychological and behavioral approaches to treatment that were well supported by research findings. Research support was to be determined by studies of treatment efficacy, with significant treatment-related change being documented in well-controlled research investigations, emphasizing internal validity. Using the criteria adopted by the task force, it was possible to categorize psychosocial treatments of various types into two efficacy-related categories: well-established treatments and probably efficacious treatments.

To qualify as a "well-established treatment," two adequately designed group studies (from different researchers), demonstrating the treatment to be significantly better than a pill placebo, psychological placebo, or other treatment or equivalent to an already established treatment were required. Alternatively, designation as a "well-established treatment" could result from a series of > 9 well-controlled, single-subject design studies that compared the experimental treatment to some other treatment. In both instances, the nature of the samples had to be adequately described and the treatment based on a treatment manual.

Treatments could qualify as "probably efficacious" based on (1) two well-controlled studies that demonstrated the treatment to be superior to a wait-list control group or (2) one or more studies that met criteria for "well-established treatments" (see previous) but conducted by the same researcher. Probably efficacious treatments could also be documented via a series (here > 3) of single-subject design studies by a single investigator that met the well-established treatment criteria. Again, treatment was to be based on a treatment manual and sample characteristics had to be clearly specified (Chambless & Hollon, 1998).

As Ollendick and King (2004) have noted, using criteria such as these, the task force was able to identify a total of 25 largely adult-focused treatments that were judged to be empirically supported to a greater or lesser degree (18 well-established treatments; 7 probably efficacious treatments). Of these, only three treatments for children were identified as well-established treatments, and one met criteria for a probably efficacious treatment. These included behaviorally oriented interventions for children with developmental delays, enuresis and encopresis, oppositional-defiant behavior (well-established treatments), and tics (probably efficacious).

Given that the initial focus of this task force was largely on adult-related treatments, the Society of Clinical Psychology and the Section on Clinical Child Psychology (Division 12; Section 1, presently Division 53 of the American Psychological Association), both subsequently developed task forces to determine empirical support for various approaches to child and adolescent treatments. The combined efforts of these two groups provided varying levels of evidence-based support for a number of additional child treatments (Lonigan, Elbert, & Johnson, 1998). Ollendick and King (2004) have also cited other attempts to delineate the degree of support available for child/adolescent treatments, using somewhat similar research criteria. Here, they cite the work of Nathan & Gorman (1998) and Roth, Fonagy, Parry, & Target (1996). Taken together, these efforts highlight a range of child treatments that have been judged as well-established or probably efficacious (Chambless & Ollendick, 2001; Ollendick & King, 2004). These treatments are listed in Tables 5.1 and 5.2.

While the information presented in Tables 5.1 and 5.2 highlights the significant advances that have been made in documenting the evidence base for a variety of approaches to child treatment, it is important to note that the treatments listed here are not likely to represent a final tally of effective psychosocial treatments for use with children and adolescents. For example, ongoing research related to the treatment of adolescent depression is providing increasing support for the role of interpersonal therapy as an evidence-based treatment for this disorder (Mufson,

Table 5.1 Examples of Well-Established Child Treatments

Problem or Disorder	Type of Treatment
Attention-Deficit/Hyperactivity Disorder	Behavioral parent training
	Classroom behavior modification
Anxiety disorders	None
Autism	None
Depression	None
Enuresis	Behavior modification
Encopresis	Behavior modification
Obsessive-Compulsive Disorder	None
Oppositional Defiant Disorder/ Conduct Disorder	Behavior parent training
	Functional family therapy
	Videotaped modeling
	Multisystemic therapy
Phobias	Graduated exposure
	Participant modeling
	Reinforced practice

From "Empirically Supported Treatments for Children and Adolescents: Advances toward Evidence-Based Practice" (p. 8), by T. H. Ollendick and N. J. King in *Handbook of Interventions that Work with Children and Adolescents: Prevention and Treatment,* P. M. Barrett and T. H. Ollendick (Eds.), 2004, Hoboken, NJ: Wiley. Adapted with permission.

Dorta, Moreau, & Weissman, 2004). Research related to the treatment of other types of child psychopathology is currently underway. As these research efforts continue, hopefully evidence will be found for the effectiveness of an even wider range of approaches that are useful in the treatment of child and adolescent problems.

Finally, although lagging behind research related to child treatment, the developing focus on evidence-based assessment (Mash & Hunsley, 2005; Kazdin, 2005) is also rapidly becoming an active area of research focus for clinical child psychologists.

Prevention Activities

Many psychologists focus on ways of treating problems already displayed by the child, adolescent, or family; the focus of other clinical child psychologists is on ways of preventing the initial development of childhood disorders (primary prevention) and/or on the early detection of problems so as to decrease the duration of the problem and minimize its effects (secondary prevention). A third form of prevention (tertiary prevention) is designed to minimize the impact of an already existing problem or disorder; in many ways, this can be equated with therapy.

Examples of primary prevention include programs designed to reduce teenage pregnancy and/or risky sexual behavior, programs designed to enhance seatbelt or car seat usage, and child-proofing homes to prevent unintentional injuries. Also included are programs designed to prevent substance abuse and family interventions to prevent child abuse in high-risk families.

Examples of secondary prevention include programs such as those designed to detect autism in very young children so that effective behavioral treatments can be used to teach social and language communication skills. Another example is the

Table 5.2 Examples of Probably Efficacious Child Treatments

Problem or Disorder	Type of Treatment
Attention-Deficit/Hyperactivity Disorder	Cognitive-behavior therapy (CBT)
Anxiety disorders	CBT
	CBT + family anxiety management
Autism	Contingency management
Depression	Behavioral self-control therapy
	Cognitive-behavioral coping skills
Enuresis	None
Encopresis	None
Obsessive-Compulsive Disorder	Exposure/response prevention
Oppositional Defiant Disorder/ Conduct Disorder	Anger control training with stress inoculation
	Anger coping therapy
	CBT
	Assertiveness training
	Delinquency prevention program
	Parent-child interaction therapy
	Problem-solving skills training
	Rational emotive therapy
	Time out + signal seat treatment
Phobias	Imaginal desensitization
	In vivo desensitization
	Live modeling
	Filmed modeling
	CBT

From "Empirically Supported Treatments for Children and Adolescents: Advances toward Evidence-Based Practice" (p. 8), by T. H. Ollendick and N. J. King, in *Handbook of Interventions that Work with Children and Adolescents: Prevention and Treatment*, P. M. Barrett and T. H. Ollendick (Eds.), 2004, Hoboken, NJ: Wiley. Adapted with permission.

Early Head Start program, which is designed to provide early education, health, and family support services for at-risk children up to age 3 from low-income families (Raikes & Love, 2002). Research has found that children benefit from this program in terms of increased cognitive development, more intellectually stimulating home environments, and the development of better paternal parenting skills (McNeil & Bernard, 2003).

Justifiably most often thought of as treatment, an excellent example of tertiary prevention is the use of parent-child interaction therapy (Brinkmeyer & Eyberg, 2003) to reduce disruptive behavior in preschool children with oppositional-defiant behavior, with the goal of preventing the long-term negative outcomes of adult antisocial behavior.

Work in the area of secondary prevention (early intervention) has been demonstrated to be of clear importance and to be of documented value in minimizing some of the negative impacts of childhood problems that, if left untreated, would likely result in negative long-term outcomes. Unfortunately, inadequate funding of primary prevention programs has left the potential value of many such programs unrealized (Boles, Mashunkashey, & Roberts, 2003). With increased levels of federal funding for the development and dissemination of such programs, work in the

area of primary prevention could become a more attractive professional activity for clinical child psychologists.

The Training of Clinical Child Psychologists

Over the years there has been a dramatic increase in the number of clinical child psychology training programs, which has been prompted, in part, by the shortage of a sufficient number of adequately trained psychologists to meet the mental health needs of children (Knitzer, 1982). In the 1976–1977 edition of the American Psychological Association's *Graduate Study in Psychology,* only eight graduate programs self-identified as offering specialty training in clinical child psychology (Johnson, 2003). A survey of graduate training programs in 1982 was able to identify only 15 doctoral programs that provided clinical child specialty training (Roberts, 1982). By 1995, the Directory of Graduate Programs in Clinical Child/Pediatric Psychology (Tarnowski & Simonian, 1995) listed more than 100 programs in the United States and Canada that *self-reported* offering training in clinical child or pediatric psychology, although many of these would likely not meet strict criteria for a formal training program. Indeed, it seems likely that the number of graduate training programs offering quality specialty training in clinical child psychology is well below this number.

Given the need for *adequately trained* clinicians to meet the mental health needs of children *and* the increasing number of programs offering training in clinical child and pediatric psychology, considerable attention has been given to developing training guidelines to ensure adequate preparation of those entering the field. Over the years, three efforts in this regard are especially noteworthy.

First was the work of a task force of the APA Division of Child Youth and Family Services (Division 37), which proposed initial general guidelines (academic coursework, research experiences, and applied training in assessment and therapy) for training psychologists for working with children, youth, and families (see Roberts, Erickson, & Tuma, 1985). This work was followed by the national conference on "Training Clinical Child Psychologists," held at Hilton Head, South Carolina, in May 1985. Here, participants, who were leaders in the fields of clinical child and pediatric psychology, endorsed the general recommendations of the Division 37 Task Force and a scientist-practitioner model of training that emphasized training clinical child psychologists to function both as scientist and clinician and agreed on a range of other recommendations for clinical child training at the graduate, internship, and postdoctoral levels (Johnson & Tuma, 1986; Tuma, 1985).

A more recent attempt to integrate and elaborate on the Hilton Head conference guidelines resulted from the efforts of a National Institute of Mental Health Center for Mental Health Issues Task Force, which held a subsequent training conference at the University of Kansas in 1993. The results of these efforts, detailed by Roberts et al. (1998), represent an elaborate and well-considered framework for the professional training of individuals desiring to work with children and families. It was recommended that trainees be exposed to a wide range of didactic and applied training experiences in (a) life span developmental psychology; (b) life span developmental psychopathology; (c) child, adolescent, and family assessment;

(d) intervention strategies; (e) professional, ethical, and legal issues pertaining to children, youth, and families; (f) research methods and approaches to system evaluation; (g) issues of diversity; (h) prevention, family support, and health promotion; (i) the role of multiple disciplines and service delivery systems; (j) social issues affecting children, youth, and families; and (k) specialized applied experiences in assessment, intervention, and consultation.

These recommendations indicated that clinical training in these areas should be designed to progress sequentially from simple exposure to the development of expertise in various areas, that they should involve structured research experiences relevant to the specialty, and that internship training should build on predoctoral training and provide a foundation for postdoctoral work. This framework appears to provide an excellent foundation for clinical child specialty training. Building on these guidelines, additional training recommendations related specifically to pediatric psychology training have been outlined by Spirito et al. (2003).

A significant issue in clinical child psychology training is that, at present, there are few quality assurance mechanisms to ensure that graduate programs purporting to offer specialty training do, in fact, offer training consistent with recommendations repeatedly endorsed by professional organizations representing the specialty. This is, in part, due to the fact that historically there has been no mechanism for the accreditation of specialty programs apart from clinical, counseling, and school psychology. Thus, although graduate programs in these areas are required to conduct self-studies and meet APA Committee on Accreditation standards to initially get or maintain accreditation, there are currently no provisions to ensure that clinical training in specialty tracks (where most of clinical child training takes place) is consistent with existing specialty training guidelines. It is clear that many existing clinical child psychology training programs provide quality graduate education in this area, as evidenced by a strong clinical child training faculty, a formal sequence of required clinical child training experiences (that are consistent with specialty training guidelines), and their having a strong track record of turning out successful clinical child psychologists. Quality assurance mechanisms are necessary to ensure that all programs purporting to offer clinical child training meet these standards.

Clinical Child Psychology: Present and Future

Clinical child psychology, along with pediatric psychology, has grown rapidly over the past several decades. Since the 1960s, when the first interest groups related to clinical child psychology were organized, the field has grown in the number of professionals identified with the area and in the number of graduate programs designed to train clinical child psychologists. Likewise, as noted, the area has developed clinical training guidelines in an initial attempt to ensure that those trained in the area are trained well.

After many years of being viewed as a subspecialty of clinical psychology, clinical child psychology has now been formally recognized as a separate specialty in professional psychology, and the primary organization representing the area, the Society of Clinical Child and Adolescent Psychology, has evolved from being a section of the APA Division of Clinical Psychology to being an APA division in

its own right, as has the Society of Pediatric Psychology. In the short time since its inception, research productivity of the members of the specialty has grown at an ever-increasing rate. Much of this research has been published by two journals maintained by these divisions, the *Journal of Clinical Child and Adolescent Psychology* and the *Journal of Pediatric Psychology*, which have grown from humble beginnings to currently enjoying the status of first-rate psychology journals. Cutting-edge research in the area of clinical child psychology is also regularly presented at the biannual Kansas Conference on Clinical Child and Adolescent Psychology, and current research in the area of pediatric psychology is presented at the biannual National Conference on Child Health Psychology.* The growth of the area is also highlighted by the fact that it is now possible for evidence of excellence in clinical child practice to be recognized through specialty board certification in clinical child psychology through the American Board of Clinical Child and Adolescent Psychology.

As suggested in the formal petition to the APA for specialty recognition in 1998, the area of clinical child psychology has indeed become a "well-developed, legitimate, and formally recognized area of clinical and research specialization, characterized by the development of an ever increasing body of specialized knowledge and a vibrant, diverse, and specialized area of practice" (Commission for the Recognition of Specialties and Proficiencies in Professional Psychology, 1998, p. 2). Given the rapid growth of the specialty during the past 20 to 25 years, questions arise as to where the specialty of clinical child psychology will go in the future. Here, change and advancement are likely on several fronts.

First, it seems likely that the evolving nature of the field will bring about significant changes in the practice of clinical child psychology in the years to come and result in new and different role functions for those identified with this area. As noted earlier, it appears that clinical child psychology has begun to experience a significant shift in the theoretical orientation of new clinical child psychologists coming into the field. There is clear movement toward an increasing number of practitioners with behavioral and cognitive-behavioral theoretical orientations and fewer clinical child psychologists with a predominantly psychodynamic orientation. Given this trend, in the future we will likely continue to see a de-emphasis on training in projective techniques and other more subjective assessment measures and an increased emphasis on more objective and evidence-based assessment measures. This is likely to be accompanied by a corresponding training emphasis on more evidence-based treatments for childhood disorders, which at this point seem to be interventions that are primarily behavioral or cognitive-behavioral in nature. These already evident trends in clinical training will likely also be reflected in the clinical practice of those clinical child psychologists entering the workforce.

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*The National Conference on Clinical Child and Adolescent Psychology was founded in October 1994 by Michael Roberts and colleagues in the Clinical Child Psychology program at the University of Kansas. The National Conference on Child Health Psychology (formerly the Florida Conference on Child Health Psychology) was founded in 1988 by James H. Johnson and Suzanne Bennett Johnson and their colleagues in the Department of Clinical and Health Psychology at the University of Florida.

It is also likely that the impact of managed care will continue to influence clinical practice in terms of assessment and treatment approaches as evidence-based assessment and treatment methods are likely to become increasingly favored for reimbursement.

Consistent with these projections, it is safe to predict that the next several years will see an increased focus on clinical child psychology research related to various aspects of evidence-based practice. This will likely include expanded efforts in researching evidence-based treatments for children and adolescents, including treatments other than those based on behavioral and/or cognitive-behavioral principles, and increased research focus on evidence-based assessment as well as research on the *integration* of evidence-based assessment and evidence-based treatment methods.

As regards child and adolescent treatment we can also expect to see a significant increase in research on the effectiveness of those child treatments that have been determined to be efficacious. It is clearly important to conduct research to document significant treatment effects in the research setting; it is also essential to document the effectiveness of these treatments in applied clinical settings. This research on effectiveness is essential and will likely become a major focus of child treatment research in the near future. Related to the issue of effectiveness, one can safely predict an increasing focus on finding ways of disseminating empirically supported treatments so they are likely to be used in clinical practice as well as researching ways of individualizing manualized treatments to ensure treatment fidelity while also making them more flexible and hence more palatable to practitioners (Kendall & Beidas, 2007). Continued advances in this area of research and the increased dissemination of evidence-based approaches to treatment are likely, over time, to have a significant impact on the role functions of clinical child practitioners in terms of their increasing reliance on empirically based methods of treatment. This would be a very positive outcome for children.

The changing roles of clinical child psychologists have also been highlighted by Prinstein and Roberts (2006) who have noted that, upon completing their graduate training, new clinical child psychologists are increasingly obtaining positions in new settings such as schools, primary care, and corporations that involve new role-related activities such as consulting, public policy, program evaluation, and the supervision of nonpsychology health care professionals. Regarding this trend, Prinstein and Roberts suggest that significant modifications of professional training will likely be needed to prepare clinical child psychologists for assuming these new employment opportunities and diverse professional roles in the future.

PEDIATRIC PSYCHOLOGY

Primary to the science and practice of pediatric psychology is an emphasis on health- and illness-related issues. More specifically, "pediatric psychology is a field of science and practice that addresses a range of physical, psychological, developmental, health, and illness issues affecting children and their families" (Roberts, Mitchell, & McNeal, 2003, p. 3). This often involves working with children, and their families, who are struggling with chronic (cystic fibrosis, asthma, diabetes) or acute

(enuresis, abdominal pain, headaches) health conditions. Compared to clinical child psychologists, pediatric psychologists are also more frequently involved in prevention and health promotion activities, such as promoting more healthful lifestyle behaviors and practices. Some psychologists who work with pediatric populations on health-related issues prefer to be referred to as child health psychologists. Those who differentiate between the two terms note that the term pediatric psychologist is more often associated with psychologists who focus on treatment of children with chronic health conditions, whereas child health psychologist often denotes a wider scope of practice, including work with acute and chronic health conditions, as well as general prevention and health promotion. However, for the purposes of this chapter, we use these terms interchangeably.

Practice Settings

There are a wide variety of settings in which pediatric psychologists practice their craft. The practice of pediatric psychology originated in inpatient medical centers dedicated to the treatment of disease and illness (Roberts et al., 2003). A recent survey of the members of the Society for Pediatric Psychology, Division 54 of the APA, found that pediatric psychology is still primarily a hospital-based field, with a majority (63%) of pediatric psychologists working in hospital settings (Opipari-Arrigan, Stark, & Drotar, 2006). Practice in inpatient settings often involves consultation services and treatment of children exhibiting less than optimal adjustment to illness or illness management behavior. Carter and colleagues (2003) completed a case-controlled study examining the nature of referrals to pediatric consultation and liaison service in 104 children. Reasons for referrals included problems in coping and adjustment (34.6%), medication and treatment noncompliance (30.7%), depression (29.8%), pain management (15.4%), anxiety (15.4%), illness exacerbation (14.4%), acute evaluation (13.4%), new medical diagnosis (13.4%), difficulties in parent coping (14.4%), ongoing treatment (12.5%), decision making regarding treatment (9.6%), differential diagnosis (9.6%), acting out (7.7%), general support (6.7%), and family conflict (6.5%), with many patients having multiple reasons for referral.

Hospital-based services may also involve consultation and clinical practice activities in emergency departments conducting assessments for depression or suicide risk, assessment and triage for victims of acute trauma or acute illness exacerbations, or assessment of children in acute psychosocial crisis. Medically based practice by pediatric psychologists also involves practice in medical outpatient clinics either independently or as part of multidisciplinary teams. The latter may include teams headed by pediatric subspecialties from endocrinology, pulmonology, gastroenterology, hematology, or rheumatology addressing health and coping issues related to diabetes, cystic fibrosis, asthma, inflammatory bowel disease, pediatric feeding aversion, sickle cell disease, juvenile rheumatoid arthritis, or other conditions. Pediatric psychologists also practice alongside health care professionals in primary care settings addressing acute health conditions (i.e., recurrent abdominal pain, headaches, toileting and soiling issues, sleep problems), as well as general emotional and behavior problems in children (Schroeder, 1979).

The science and practice of pediatric psychology is not limited to medical settings. Whether primarily involved in practice, training, research, or a combination of the three, pediatric psychologists also engage in role-related activities essential to the field in a variety of other settings. Whereas Opipari-Arrigan et al. (2006) found that most respondents worked in hospital settings (with about half in academic medical centers), a number of other practice settings were also noted. These included private practice (22%), academic psychology departments (5%), academic departments other than psychology (15%), mental health agencies (3%), and school systems (2%). Pediatric psychologists can also be found working or volunteering in camp settings designed for children suffering from chronic health conditions (Roberts et al., 2003). The following sections turn to the broad role and role-related activities engaged in by pediatric psychologists working in various settings.

Role-Related Professional Activities

As with clinical child psychologists, pediatric psychologists engage in a wide range of specialty-specific role-related activities. These include consultation, assessment, treatment, health promotion, promoting public health policy, training, and research. The ways pediatric psychologists carry out these activities in the context of their professional roles are detailed in the following sections.

Consultation Activities

As noted earlier, consultation is a critical role for pediatric psychologists, as our medical colleagues often represent the primary sources of patient referrals. The level of involvement between pediatric psychologists and medical colleagues during consultation may vary as a function of the case involved, the proximity of colleagues, or the presence of a formalized structure to support consultation. Consultation may be limited to brief discussions regarding general principles and strategies for dealing with children and families, provision of advice relevant to a specific case, brief involvement in assessment or development of a treatment plan, or active long-term involvement in patient treatment and case management. There are various models for the level of collaboration (Drotar, 1995; Roberts & Wright, 1982). For example, pediatric psychologists can function independently from medical colleagues (e.g., in their own outpatient clinic), providing assessment, diagnosis, and, if necessary, treatment of referred patients. Alternatively, they can function as part of a collaborative team in the assessment and treatment of children and families. This may involve working side by side with medical colleagues in multidisciplinary clinics, marked by active involvement and shared responsibility by team members leading to collaborative case conceptualization and treatment recommendations. It is within the rubric of consultations that pediatric psychologists often perform a number of functions, listed next.

Assessment Activities

As in all domains of applied clinical psychology, assessment is a central function of pediatric psychologists. In the ideal situation, most assessment leads to the

development of recommendations or potential treatment options to help address areas of concern. Assessment by pediatric psychologists commonly focuses on child and family behaviors, interaction patterns, coping strategies that impact adjustment to the child's medical condition, including the challenges associated with managing the child's illness, the implications for the long-term health of the child, and social and emotional sequelae of the health condition. Common strategies to gather information include (a) structured or nonstructured interviews with children and their parents; (b) completion of self-report or parent-report questions; (c) completion of questionnaires by schoolteachers or other professionals working with the child; (d) direct observations of family interactions patterns, child social skills, or completion of illness management tasks; and (e) self-monitoring of emotions, cognitions, or behaviors.

A couple of examples may help to illustrate the assessment process. Children with cystic fibrosis are often required to engage in a variety of illness management tasks, including increased caloric intake, airway clearance (chest physiotherapy and medications to loosen mucus in the lungs), pancreatic enzyme replacement therapy, and antibiotic medications to fight bronchial infections. Adherence to these treatment tasks, whether in children with cystic fibrosis or other illness conditions, is one of the most common and challenging referrals to pediatric psychologists. Accurate assessment of the actual adherence to various treatment tasks and barriers hindering adherence (including antecedents and consequences of engaging in or not engaging in certain behaviors) are critical to developing appropriate and effective intervention plans that ultimately can impact the child's long-term physical health and quality of life. Interviews (focusing on the child, parents, the medical staff, and possible school personnel), behavioral checklists and self-report questionnaires, and written logs can be critical to documenting adherence and critical barriers. In addition, over the past 20 years electronic monitoring devices have become an invaluable tool in providing more accurate assessment of adherence to various illness management tasks. These include metered dose inhalers and medication event monitoring systems that record the date and time of vial openings and closings, as well as chest physiotherapy assistance devices, such as the Flutter or the high-frequency chest compression vest, which vibrates to loosen mucus in the lungs and can track adherence to chest physiotherapy.

Assessment may also involve evaluation of child and family functioning in anticipation of challenging future medical procedures or health issues. A common example is pretransplant evaluation of children presenting for organ transplantation. Pretransplant evaluations assess a wide variety of critical strengths, potential barriers to graft maintenance, and risk factors that may impact a child's or family's ability to follow the pre- and posttransplant treatment regimen, as well as posttransplant adjustment and coping. These include, but are not limited to, an assessment of (a) adherence to treatment regimen (taking prescribed medications, adherence to dietary recommendations), (b) understanding of the transplant process; (c) emotional coping and adjustment of child and parents or legal guardians; (d) presence of oppositional or disruptive behavior; (e) academic or social functioning; (f) social, emotional, and logistical support available to the family; (g) family communication; (h) family's ability to continue to complete daily life management tasks (grocery

shopping, laundry, helping siblings with schoolwork, etc.); (i) family finances; and (j) other stressors such as additional family illness, potential job loss, and marital conflict. Accurate and reliable assessment from a multidisciplinary team is critical to establish a coherent picture of how these different areas ultimately can impact the child's and family's long-term health, quality of life, and maintenance of the graft.

Finally, assessment may involve psychoeducational assessment, including formalized measures such as the Wechsler Intelligence Scale for Children, the Stanford-Binet Intelligence Scale, or the Wechsler Individual Achievement Test to assess both cognitive functioning and academic achievement. Such testing may be advised to provide a baseline and track potential changes in cognitive functioning for children undergoing radiation or chemotherapy treatment for cancer or those who have experienced head trauma or other medical conditions that can lead to deterioration in cognitive functioning over time.

Treatment Activities

Treatment of children with acute or chronic health conditions is one of the most commonly recognized functions of pediatric psychologists. However, it is clear that treatment by pediatric psychologists (and clinical child psychologists), as well as others, does not occur in isolation. Prior to engaging in intervention efforts practitioners must travel a long road of didactic and applied clinical training to develop skills and competencies in evidence-based approaches to treatment. Moreover, prior to and throughout the course of treatment, assessment is essential to elucidate targets and pathways for change and to monitor ongoing progress. Research, either by the practitioner delivering the intervention or, more likely, by other psychologists and health care professionals, is necessary to provide data on best practices and evidence-based treatments to guide and inform intervention efforts. In fact, treatment-related research is becoming critically important, given the growing emphasis on evidence-based treatment in pediatric psychology. These treatments are highlighted by the ongoing series on empirically supported treatments in the *Journal of Pediatric Psychology*.

Potential child health-related difficulties that require intervention are numerous, but commonly involve coping and adjustment issues, adherence to treatment recommendations, or pain management. Problems of coping and adjustment may relate to death and dying, limits of daily functioning, family communication and conflict, school reentry, social interaction difficulties, or improving self-esteem. Adherence to treatment recommendations may involve increasing adherence to medication regimens, dietary recommendations (for cystic fibrosis, diabetes, inflammatory bowel disease, constipation), airway clearance (for cystic fibrosis), or increased physical activity (for obesity, Type 2 diabetes, juvenile rheumatoid arthritis). Pain management and coping with medical procedures may involve the use of diaphragmatic breathing, muscle relaxation, cognitive imagery or other active coping strategies to assist with injections, lumbar punctures, nausea associated with chemotherapy, or pain associated with sickle cell crisis or tumors. Treatment efforts often focus not only on the ill child, but also the parents and other members of the immediate and extended family, given the

bidirectional influences between the child and his or her immediate family members. As with other functions performed by pediatric psychologists, collaboration with other health care and school professionals in the child's environment is essential as effective treatment often requires intervention at multiple levels in the child's environment.

Teaching and Mentoring Activities

Many pediatric psychologists spend valuable time training and mentoring the next generation of pediatric psychologists in clinical service delivery, research methodology, grant writing, and professional practice. Mentoring may involve formal and/or informal interactions of students, interns, and postdoctoral fellows. Senior faculty can also play a valuable role in the mentorship of junior colleagues and trainees. As pediatric psychologists strive to promote the value of their services to their medical colleagues and hospital administrators, education of professionals in other health care disciplines can be invaluable. This can take the form of active involvement in teaching trainees from other health care disciplines or engaging in informal teaching interactions with more senior medical colleagues (Drotar, Spirito, & Stancin, 2003; Roberts et al., 2003). This topic is reviewed in more depth later in this chapter.

Health Promotion Activities

Pediatric psychologists are becoming increasingly involved in health promotion activities such as developing and implementing lifestyle interventions to improve dietary and physical activity habits, reducing risk for childhood injuries, reducing drug and alcohol abuse, reducing tobacco use, parent training for adolescent mothers to reduce the risk for child neglect or abuse, and osteoporosis prevention. For example, Stark and colleagues (Stark, Janicke, McGrath, Mackner, & Hommel, 2005) demonstrated the effectiveness of a behavior intervention designed to improve calcium intake and bone mineral content in children with juvenile rheumatoid arthritis and inflammatory bowel disease. More recently, pediatric psychologists have ventured into the public health domain and are working to help promote public health agendas through more broad-based community initiatives (Drotar et al., 2003). There are a variety of examples of quality research in this area. One excellent example is Planet Health (Gortmaker et al., 1999), a 2-year school-based intervention that has been demonstrated to have positive effects on the weight status of middle school youth (Gortmaker et al., 1999). Another obesity prevention program is the GEMS project (Girls' Health Enrichment Multisite Studies), which is an ongoing, multisite community-based program designed to identify and implement effective obesity prevention strategies for African American girls (Rochon et al., 2003). Finally, Levy and colleagues (Levy, Brugge, Peters, Clougherty, & Saddler, 2006) have recently reported on a community-based participatory research intervention that has demonstrated reductions in respiratory symptoms of pediatric asthmatics living in public housing. These types of programs demonstrate the positive impact that innovative prevention and health promotion efforts can have on children's long-term health and quality of life.

Research Activities

As greater demands are placed on the profession to provide evidence supporting the effectiveness of psychological interventions for health-related problems and as competition for the provision of services is growing from social workers and other medical professionals, research activities are taking on greater importance for pediatric psychologists. Many of our medical colleagues have recognized what pediatric psychologists have to offer as active members of a multidisciplinary team, and subsequently are very willing to collaborate on research endeavors related to pediatric and child health psychology. One of the most important areas for future collaboration is in the area of treatment outcome research, where there has been a growing emphasis on developing evidence-based assessment and treatment methods. There is also a greater push to promote translational research and practice, to improve access to effective mental health care in diverse populations and community settings (Glasgow, Lichtenstein, & Marcus, 2003; Strauman & Merrill, 2004).

A major focus of treatment outcome research is to enhance adherence to treatment recommendations in children struggling with chronic and acute health conditions such as diabetes, cystic fibrosis, and asthma. Another common theme of current research in pediatric psychology is the identification of mediators and moderators of positive coping and adjustment in children with chronic health conditions. Finally, pediatric psychologists are at the forefront of integrating modern technology to assist in data collection and distribution of treatments for children and families, including telehealth, accelerometers to track physical activity, metered dose inhalers to track medication use, and palm pilots to assist children with accurate self-monitoring of dietary intake or physical activity.

Pediatric Psychology Compared to Clinical Child Psychology

In many ways, the field of pediatric psychology can be considered a subspecialty of clinical child psychology. Strong training in clinical child psychology is essential, but not sufficient, to function as a pediatric psychologist. Skills in assessment and intervention and knowledge of the role of developmental processes and child psychopathology are critical for pediatric and clinical child psychologists. However, there are significant differences between these disciplines. Pediatric psychologists place a much greater emphasis on medically related issues and regularly work with children and adolescents experiencing acute and chronic health conditions. Pediatric psychologists spend more time working in medical settings and collaborating with medical professionals from a variety of disciplines. As a result, they must have a greater understanding of disease processes, medical management of illness, broader health systems issues, and the referral and consultation process than clinical child psychologists.

Frequently the time allotted to pediatric psychologists to complete assessments or intervention is less than that typically allocated to clinical child psychologists (Tuma & Grabert, 1983). Children often are admitted to inpatient units for only a few days, or weeks, at a time. Many families may drive great distances for expert pediatric care. Primary care settings, another common venue for pediatric

psychologists, are often fast-paced environments with a large number of patients, allowing for a limited amount of time with each family. In multidisciplinary clinics, often there is only a brief amount of time allotted to each member of the treatment team to work with a family. Moreover, whether in primary care settings, outpatient clinics, or hospital settings, pediatric psychologists may be forced to work with a child and/or family members in inpatient or exam rooms, with numerous interruptions. Given these limitations, brief, targeted, and flexible therapies that can address critical problematic behaviors or coping skills in a short period of time are often emphasized.

Training for Pediatric Psychology

As noted in the preceding sections, there are a variety of role-related professional activities and opportunities for practice open to pediatric psychologists. Not surprisingly, students pursuing a career in pediatric psychology require training to help them develop a blend of general or core skills similar to those developed through training in clinical child psychology, as well as specialized training related to the unique roles, settings, and contexts in which pediatric psychologists function. For many students, core training in child therapy and research skills comes first through coursework and in-house practicum training in graduate school. Fundamental therapy process skills and behavioral, cognitive-behavioral, interpersonal, and family systems intervention strategies form a foundation on which more specific training in pediatric psychology can build.

Regarding those elements seen as necessary for adequate training in this area, the Society of Pediatric Psychology's Task Force on Training of Pediatric Psychologists (Spirito et al., 2003) has outlined 11 domains of training that are believed to be necessary to develop the requisite knowledge and build the general and specific skills necessary for pursuing a career in pediatric psychology. These include training in the following: (a) life span developmental psychology; (b) life span developmental psychopathology; (c) child, adolescent, and family assessment; (d) child and family intervention strategies; (e) research methods and systems evaluation; (f) issues of diversity; (g) the role of multiple disciplines in service delivery systems; (h) prevention, family support, and health promotion; (i) social issues affecting children, adolescents, and families; (j) consultation and liaison roles; and (k) disease process and medical management.

One of the most salient aspects of training, unique to pediatric psychologists, is the need to have opportunities to work in interdisciplinary settings with professionals and trainees from other health professions (Spirito et al., 2003). Primary care and other outpatient clinical settings offer a fast-paced environment, where pediatric psychologists must think on their feet, remain flexible, build skills in effective but targeted brief therapy interventions, and learn to effectively communicate findings and impressions while also coordinating care with other health professionals. Given that most university academic settings have limited access to hospital-based training experiences, most in-depth hospital-based experiences usually occur during the internship and postdoctoral training experiences, although there are certainly exceptions to this rule.

Once again, the changing health care environment highlights the continuing need for additional treatment outcome research by pediatric psychologists to demonstrate the worth of our services to both patients and health care professionals. Conducting successful treatment outcome research in medical or community settings requires flexibility and teamwork, and knowledge of disease processes and medical systems, a deep respect for the issues and challenges presented to families of children struggling with acute and chronic health conditions, and research methodology considerations are critical. Almost all doctoral students in clinical psychology receive training in research methods and the evaluation of treatment outcomes. However, training students to conduct treatment outcome research studies on the scale required to support judgments of long-term efficacy and effectiveness necessitates more specialized training and experience. This may include, but are not limited to (a) training in participant recruitment and retention, (b) assessment methodology unique to specific health conditions, (c) longitudinal research design and corresponding data analytic approaches appropriate to such designs, (d) data management programs and protocol, (e) supervision and training of research team members and interventionists, (f) facilitating and maintaining collaboration with medical colleagues, (g) dissemination and translational research approaches, and (h) ethical considerations in the conduct of treatment outcome research.

Given the nature of the current health care environment with its increasing limitations on reimbursement for services provided by pediatric psychologists and the constraints on fiscal survival and success, "pediatric psychologists must be advocates not only for our services but also for the resources that support them" (Kronenberger, 2006, p. 648.). Thus, training in basic business management practices such as marketing of services, accounting, and financial management seem increasingly relevant to the ability to survive and, eventually, flourish as a profession. This may start with coursework during graduate training, but should be supplemented with seminars and hands-on practice during internship and postdoctoral training. In addition, as funding sources become more limited, training in grant writing will be beneficial for research and clinical pediatric psychologists. Again, these skills can be built through both didactic coursework and seminars, as well as hands-on experience assisting mentors with their applications or by submitting applications for their own National Research Service Awards during graduate and postdoctoral training.

Future Directions and Key Issues for Pediatric Psychologists

There are a number of key issues facing the field of pediatric psychology that will ultimately impact the direction of future practice as well as the viability of the field. Many of these issues were outlined by an expert panel of pediatric psychologists (Brown & Roberts, 2000), and some have been discussed previously in this chapter. The most important issue identified by respondents was related to potential threats to the viability of pediatric psychology. In response to this threat, there is a growing need to demonstrate the effectiveness of intervention protocols through controlled treatment outcome studies, a greater emphasis on interventions that improve health status as opposed to just coping with illness, and a greater need for

research examining the cost offset associated with psychological interventions. It will also be important for pediatric psychologists to assess alternative sources of funding and reimbursement (Drotar, 2004), as many of the services provided by pediatric psychologists, such as improving adherence, adjustment to chronic illness, consultation with medical colleagues, and pain management, are not technically in the categories of the *Diagnostic and Statistical Manual of Mental Disorders*. Increasing awareness of what pediatric psychologists can do in medical settings by marketing our services to medical professionals as well as patients may help ensure the viability of pediatric psychology as a field of practice (Bradford, 2004).

Many of the populations and issues researched by pediatric psychologists involve small numbers of patients in hospital settings. As such, future research, especially treatment outcome research, will require multisite investigations to enroll adequate numbers to provide sufficient power to test study hypotheses.

Most pediatric psychologists are congregated in large academic medical centers. As a result, many people have to travel long distances for pediatric care as they do not have access to pediatric psychology services in their home town (Kronenberger, 2006). Thus, a significant opportunity for future growth of pediatric psychology lies in outpatient, primary care pediatric settings. Development and behavioral problems often first present to primary care providers, who have been referred to as the gatekeepers to the health care system. Such a setting would seem ideal for pediatric psychologists to make significant contributions to the health status and quality of life of children and families and to demonstrate their usefulness and the benefit of their services to medical colleagues. The call for pediatric psychologists to build practices in primary care settings has been echoed over the past few decades, yet few psychologists have successfully accomplished this transition. Renewed efforts in this direction could provide big dividends to pediatric psychologists and may be vital for the future of our field.

New and emerging technologies provide pediatric psychologists with a variety of opportunities to expand their service net. Genetic testing, for example, has the potential to provide families with information on the risk of children developing life-threatening diseases, chronic conditions, and development irregularities (Patenaude, 2003). There are a variety of opportunities for research to ascertain the impact of knowledge of risk on adjustment and decision making, as well as the best ways to communicate information to parents and children (Tercyak, 2003). Another emerging technology is telehealth, which provides opportunities to bring treatment to children and families in underserved areas, but expertise and research will be required to determine effective strategies to intervene using this new medium (Harper, 2003).

Prevention and health promotion are areas that hold the potential for long-term health benefits and medical cost offsets. As noted earlier, there are numerous opportunities for pediatric psychologists to make valuable contributions in these areas. Included here is work in nutrition and exercise to impact weight status and physical fitness, new and more comprehensive efforts to reduce unintentional injuries, multimodal parent training programs to reduce abuse and neglect, and approaches to osteoporosis prevention, just to name a few. Historically, reimbursement for prevention activities has been limited; however, the growing emphasis on translation

and dissemination research and practice may provide more opportunities for pediatric psychologists to provide mental health services in underserved communities (Glasgow, Klesges, Dzewaltowski, Bull, & Estabrooks, 2004; Glasgow et al., 2003).

SCHOOL PSYCHOLOGY

Fagan and Wise (2000, p. 4) define a school psychologist as "a professional psychological practitioner whose general purpose is to bring a psychological perspective to bear on the problems of educators and the clients educators serve" and emphasize the "educational and psychological foundations" of the discipline. Unlike clinical child and pediatric psychology, school psychology is unique in terms of its focus on the functioning of individuals, groups, and systems in the school setting. As an area, school psychology also differs from clinical child and pediatric psychology in that the majority of practicing school psychologists have a master's or educational specialist degree rather than the doctorate, with the specialist degree being the most common. A National Association of School Psychologists membership renewal survey in 1999 found that 54% of school psychologists held a specialist degree, whereas only 26% held a doctoral degree (Fagan & Wise, 2000).

Not surprisingly, the most common setting for school psychologists is in schools, with approximately 80% of practitioners working in elementary or secondary schools. Private practice (3% to 5%) appears to be the second most common setting for school psychologists, and government agencies, community mental health centers, developmental disability centers, and universities are among some of the other work settings (Fagan & Wise, 2000). It should be noted that doctoral-level school psychologists frequently work with clinical populations outside of the school setting and engage in clinical activities similar to those performed by clinical child psychologists.

Role-Related Activities of School Psychologists

School psychologists engage in a variety of role-related professional activities, such as psychoeducational assessment, intervention, consultation, and, to a lesser degree, research and program evaluation, training, and supervision. These activities are not mutually exclusive as school psychologists often perform these activities concurrently.

Assessment Activities

School psychologists have historically spent much of their time conducting psychoeducational assessments. Indeed, Reschly (2000) has suggested that school psychologists spend approximately 50% to 55% of their time engaged in activities related to psychoeducational assessments. These assessments typically serve to gather information regarding the child's current difficulties to help plan interventions and/or provide a basis for special class placement or classroom accommodations. Presenting problems dealt with by school psychologists can relate to a range of possible areas, including academic, social, emotional, and behavioral functioning. Assessments can include multiple components, including review of school

records, classroom observation, and interviews with parents, teachers, and other persons who may be able to provide information related to the child's presenting difficulties. They can also include testing of cognitive ability, academic achievement, and perceptual and motor skills, as well as the administration of standardized rating scales measuring behavior problems, personality, and adaptive functioning (Fagan & Wise, 2000). It is noteworthy that the assessment activities of school psychologists are to some degree shifting away from an emphasis on psychoeducational testing toward more functional assessments designed to generate and evaluate interventions (Reschly, 2000).

Intervention Activities

School psychologists report spending approximately 20% of their time engaged in intervention-related activities, including planning, implementation, and evaluation of interventions (Reschly, 2000). These interventions can take various forms, including both individual and group counseling and planning and implementing classroom-based intervention. Individual counseling can focus on helping children learn skills and coping strategies to deal with a range of difficulties in the areas of academic, social, emotional, and behavioral functioning. Group counseling is often used to help children learn and use adaptive skills related to peer interactions, dealing with anger, and stressful situations such as divorce or death of a parent. Classroom-based interventions can involve collaborating with the child's teacher to arrange a particular seating arrangement, selecting an appropriate classroom placement, finding additional academic supports such as tutoring, and implementing behavior management programs in the classroom.

Consultation Activities

School psychologists are reported to devote approximately 23% of their time to consultation activities (Reschly, 2000). Consultation has been viewed as a collaborative, problem-solving endeavor between school psychologists and other personnel, often teachers, to help ensure that assessment-based recommendations are implemented (Fagan & Wise, 2000). Consultation can involve multiple components, including problem identification, assessment, and intervention planning, implementation, and evaluation. It has also been conceptualized as a way to help address the potential shortage of school psychologists in the future, given the potential to impact beyond the individual. For instance, consultation with a teacher can lead to interventions that have an impact on a whole classroom, and consultation on a systemic level may have the potential to effect positive change in an entire school district. Consultation has also been viewed as a way of promoting preventive strategies by providing children with supports, increasing their subjective sense of well-being, and building competencies (Meyers, Meyers, & Grogg, 2004).

Other Role-Related Activities

Unlike clinical child and pediatric psychologists, only a small percentage of school psychologists appear to be involved in conducting applied research and program

evaluation. Survey findings suggest that only about 2% of school psychologists engage in these activities (Reschly, 2000). This low number may be due in part to the fact that the large majority of school psychologists work in school settings, where research may not be supported (Fagan & Wise, 2000). Research that is conducted by school psychologists is likely most often conducted by PhD-level school psychologists who work in university settings.

Some school psychologists engage in activities related to the training of students, including teaching and supervision. School psychologists based in university settings likely engage in such roles to a significant degree, whereas school psychologists in other settings may engage in these activities on a part-time or adjunctive basis (Fagan & Wise, 2000). Other school psychologists function as administrators of psychological services or as supervisors of other school psychologists.

Factors Influencing Role-Related Activities

The amount of time school psychologists engage in various activities may be influenced by several factors. As mentioned earlier, laws and funding mechanisms can exert a significant influence. Historically, state and local education agencies have been most invested in having school psychologists conduct assessments to determine children's eligibility for special education services (Reschly, 2000).

The ratio of school psychologists to students can also influence the opportunity for school psychologists to engage in different roles. Generally, the more favorable the ratio, the more school psychologists are able to engage in activities beyond psychoeducational assessment, such as intervention and consultation (Curtis, Hunley, & Grier, 2002).

The level of training of the school psychologist is likely to be important, with doctoral-level practitioners generally being less involved in psychoeducational assessment activities and more involved in intervention, consultation, and applied research and program evaluation (Curtis et al., 2002). The availability of other providers in the setting, such as social workers and clinical psychologists, can influence the number of opportunities school psychologists have for intervention activities (Fagan & Wise, 2000).

Fagan (2002) noted that a consistent theme in the school psychology literature over the past several decades has been the call for school psychologists to expand their roles beyond assessment and become more involved in activities related to intervention and consultation. He notes that despite this consistent theme, research suggests that school psychologists continue to spend approximately half of their time involved in activities related to psychoeducational assessment, and report generally positive job satisfaction ratings.

The Expanding Role of the School Psychologist

Although the large majority of school psychologists work predominantly in school settings and spend a significant amount of time engaging in assessment-related activities, a significant number of doctoral-level school psychologists have expanded their range of clinical activities well beyond assessment and function in roles that

do not differ measurably from clinical child psychologists. School psychology programs are also now providing opportunities for students pursuing doctoral training in school psychology to obtain subspecialty training in the area of pediatric school psychology. Such training is designed to prepare school psychologists to engage in health-related activities such as serving as liaisons between schools and health care agencies to address the medical, educational, psychological, and community needs of children with or at risk for emotional and behavioral disabilities. As such, doctoral-level school psychologists are becoming increasingly similar to professionals in the area of pediatric psychology in terms of role functions.

SUMMARY

As can be seen from this overview, the specialties of clinical child and pediatric psychology have grown rapidly during the past 25 to 30 years. Indeed, in a recent paper, Prinstein and Roberts (2006) characterized these specialties as currently in their "professional adolescence" and as "grown up and striving for autonomy." As these two specialties have evolved, the roles of those working in these areas have become more and more complex, with child and pediatric psychologists assuming more varied role-related activities as health care providers and taking a prominent role in research focusing on the development of evidence-based practice. While primarily working in school-related settings, doctoral-level school psychologists also often engage in clinical activities that overlap significantly with the work of both clinical child and pediatric psychologists.

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PART II

Theoretical Models

CHAPTER 6

Psychodynamic

Sandra W. Russ

What is the future of psychodynamic psychotherapy with children? Although psychodynamic treatment has a rich theoretical and clinical underpinning, there is a relatively small empirical base for treatment effectiveness. Because of the nature of psychodynamic treatment, it has been difficult to carry out empirical studies. These studies are beginning to occur, but there is a long way to go to catch up with cognitive-behavioral approaches. The psychodynamic approach encompasses two categories of information important to the field of child psychotherapy. One is a number of psychological constructs that provide a way of understanding child development. Many of these constructs have empirical support. The second category is a wealth of psychotherapy techniques with children that have emerged from a psychodynamic understanding of the child. Of particular importance is the use of pretend play in therapy. This chapter reviews the psychological constructs and the treatment techniques. Empirical evidence is presented as much as possible, with guidelines for future research. Implications for treatment and prevention programs are discussed.

PSYCHODYNAMIC THERAPY

The psychodynamic approach focuses on the internal world of the child (Russ, 2006). What is the child thinking and feeling, and how does he or she experience other people and their environment? A developmental perspective is used to understand underlying cognitive, affective, and interpersonal processes (Shirk & Russell, 1996). The level of development of these processes and the interaction among them largely determines the child's behavior, relationships, and internal state. The psychodynamic framework is applied to understand these internal processes and childhood disorders. This psychodynamic understanding determines the specific intervention approach and techniques to be used with a particular child.

Historically, psychodynamic approaches evolved from psychoanalytic theory and therapy. As Fonagy and Moran (1990) pointed out, many forms of psychodynamic therapies are based on the psychoanalytic conceptualization of child development. Children's play was especially important in psychoanalytic approaches in that it was used as a vehicle for communication with the therapist. Also, the therapist actively worked to establish a positive relationship with the child (Freud,

1927). Anna Freud (1966) made significant contributions to psychoanalytic theory with her work on the ego and mechanisms of defense. Melanie Klein, in a different approach, developed early concepts in object relations theory and interpersonal relations (Tyson & Tyson, 1990). Klein also utilized active interpretation of children's play. A major goal of psychoanalytic and psychodynamic therapy is to return the child to normal developmental pathways (Freud, 1965; Shirk & Russell, 1996). Psychodynamic therapy differs from psychoanalytic therapy in that it has more focused treatment goals, is less frequent and intensive, and is more flexible in terms of types of interventions used and the integration of other theoretical perspectives and techniques (Fonagy & Moran, 1990; Russ, 1998; Tuma & Russ, 1993).

In most forms of psychodynamic therapy, the therapist and child meet individually once a week for a session of 45 to 50 minutes. The basic understanding between the therapist and the child is that the therapist is there to help the child express feelings and thoughts, understand causes of behavior, and form a relationship with the therapist (Freedheim & Russ, 1992). Goals of treatment are also discussed. Play is a major tool in the therapy. Traditionally, the child has usually structured the hour, chosen the topic to discuss, chosen the toys to play with, and has determined the pace of the therapy. But as therapy has become more short term, therapists are becoming more active and directive within the psychodynamic framework in focusing on specific topics and directing the play (Russ, 2004). Also, for most child therapists, parent guidance, parent therapy, and parent education and consultation with the school are essential components to the therapy. For a review of the practical issues that arise in psychodynamic therapy and for case presentations, see Chethik (2000), Kessler (1966, 1988), and Russ (2004).

How does change occur in psychodynamic approaches? Russ and Freedheim (2001) and Russ (2004, 2006) reviewed mechanisms of change in therapy most relevant to psychodynamic approaches. Some of the mechanisms of change are similar to those of other therapy approaches. Catharsis or the release of emotion and expression of feelings is thought to be therapeutic by a number of different schools of therapy (Axline, 1947; Freud, 1965). Expression of negative feelings in particular is important for many children (Moustakas, 1953). Helping children to know that they can safely express feelings through talk or play is a major task of the therapist. The therapist labels the feeling expressed by the child or puppet, accepts the feeling, and tries to understand the feeling. Words help put the feeling into a context for the child, thus making the feeling less overwhelming.

Another mechanism of change is a corrective emotional experience. The therapist accepts the child's feelings and thoughts. Often, the child's learned expectations are not met. The relationship between the therapist and child is especially important for a corrective emotional experience to occur. For example, a child expresses angry thoughts and feelings about his or her mother. The therapist, contrary to the child's expectations, is not angry or punishing; rather, the therapist is accepting of the feelings and works to understand the reasons for the anger. After a number of these therapeutic events, a corrective emotional experience occurs (Kessler, 1966). The automatic connection between angry thoughts or feelings and anxiety or guilt gradually decreases (or extinguishes). The child then becomes more comfortable with these feelings and thoughts. In children, expression of thoughts and feelings often

occurs through play. The therapist works with those expressions and, if possible, connects them to the child's daily life.

With some children, the therapist is directive and helps the child think about alternative ways of viewing a situation and generates problem-solving strategies. Role-playing and modeling of coping strategies are used by the therapist. Singer (1993) offers examples of modeling techniques during therapy. Although this mechanism of change is not often associated with psychodynamic therapy, it is frequently part of an intervention within a psychodynamic conceptualization of a case. Shapiro, Friedberg, and Bardenstein (2006) have proposed that life education also occurs in psychodynamic psychotherapy. Life education involves giving information about general human functioning. They pointed out that Anna Freud (1968) included educative components that taught adaptive skills.

The major mechanism of change associated with psychodynamic therapy is insight and the working-through process. It is this mechanism that differentiates psychodynamic therapy from other forms of interpersonal therapy. One goal of the therapist is to help the child reexperience major developmental conflicts or situational traumas in therapy. When underlying conflicts are a major issue, therapy has the goal of cognitive insight into origins of feelings and conflicts, causes of symptoms, and links between thoughts, feelings, and actions (Sandler, Kennedy, & Tyson, 1980; Shirk & Russell, 1996). Verbal labeling of unconscious impulses, conflicts, and causes of behavior helps lend higher order reasoning skills to understanding problems. Well-timed interpretations by the therapist attempt to make these links and to help the child understand the causes of his or her feelings and behavior. However, in many cases, especially with young children, cognitive insight does not occur. Rather, emotional reexperiencing, emotional working-through, and mastery of material occur and result in behavior change and changes in internal feeling states. This is an important mechanism of change and is often overlooked in the literature.

Messer and Warren (1995) stated that the goal of making the unconscious conscious needs to be modified in therapy with many children. In Erikson's (1963) concept of mastery, the child uses play to gain mastery over traumatic events and everyday conflicts. Resolving conflicts through play is part of normal child development. For normal children who are using play, cognitive insight into underlying issues does not usually occur. But the emotional processing that is occurring helps the child resolve the issues. This concept of emotional processing of negative events and traumas will be discussed further at a later point. Therapists who are using play in therapy may be helping the emotional processing occur without cognitive insight in many cases. Freedheim and Russ (1992) described the process of playing out conflict-laden content until the conflict and negative affect are resolved. The psychodynamic therapist helps guide the play, labels the feelings, describes the action, and makes interpretations to facilitate conflict resolution and the working-through process.

Another major mechanism of change with seriously emotionally disturbed children is the development of better object relations, which enables attachment and more stable day-to-day functioning. Many children have deficits in internal structures (such as internal representations) and processes that result in serious emotional

disorders. Deficits in self/other differentiation and object representations and object relations can result in serious problems with self-esteem regulation, reality testing, attachment, and impulse control. In these children, there are major deficits in underlying cognitive, affective, and interpersonal processes. Psychodynamically informed structure-building approaches are based on conceptualizations by Mahler (1968) and Kohut (1977). The therapist attempts to be seen as a caring, empathic, stable, and predictable figure. Development of good object relations is a major goal of therapy with these children. Gilpin (1976) stressed that the role of the therapist is to become an internalized object for the child. The therapist serves as a stable figure who helps the child develop these ego functions as much as possible. The relationship between the therapist and the child is probably the most important aspect of the therapy in helping this process to occur. The expression of empathy by the therapist for the child's experience is an essential technique that enables the child to develop.

Different forms of psychodynamic therapy emphasize different mechanisms of change. The two major forms are insight-oriented therapy and structure-building approaches. Shapiro et al. (2006, p. 117) list four objectives for psychodynamic therapy:

1. Increased self-understanding
2. Increased acceptance of feelings and wishes
3. Replacement of (unconscious) defense mechanisms with conscious coping strategies
4. Development of realistically complex and positive schemas for relationships between self and others

Which objectives and change processes are emphasized depends on the child's particular set of problems.

Insight-Oriented Therapy

The form of therapy most closely associated with the psychodynamic approach is insight-oriented therapy; it is most appropriate for the child with anxiety and internalized conflicts (Tuma & Russ, 1993). This approach is appropriate for children who have age-appropriate ego development, show evidence of internal conflicts, can trust adults, and, for young children, can use play effectively. Insight-oriented therapy is most appropriate for children with internalizing disorders involving anxiety or depression. Children with internalizing disorders often experience internal conflicts, have good development in major areas, and have good attachment and object relations. Goals of therapy are to resolve conflicts. The major mechanism of change is insight and working through. The therapist actively interprets thoughts and feelings in play and in talk about the child's life. As mentioned earlier, often cognitive insight does not occur. Rather, the child reexperiences difficult emotions and processes these feelings and develops a better understanding of them. Through the use of play and the therapist's reflection and interpretation of thoughts and

feelings, the child expresses "forbidden fantasy and feelings, works through and masters developmental problems, and resolves conflicts" (Freedheim & Russ, 1983, p. 982).

The technique of interpretation is important in facilitating insight and/or emotional reexperiencing. Shapiro et al. (2006) offer a good description of the various types of statements by the therapist that help bring new information to consciousness. They stated that interpretations are "causal explanations; they attempt to explain why the client feels, thinks, or behaves in some particular way" (p. 124).

Often, pretend play is a vehicle for the occurrence of insight and working through. The emotional resolution of conflict or trauma and negative feelings occurs in play. The play process has been thought of as a form of conflict resolution. For example, Waelder (1933) described the play process as one in which the child repeats an unpleasant experience over and over until it becomes manageable. Erikson (1963) presented the concept of mastery, in which the child uses play to gain mastery over traumatic events and everyday conflicts. During this process the therapist labels and interprets the play. The therapist will also tie the observations and interpretations to events in the child's life.

Insight and working through can be helpful for children with good internal resources who have experienced a major trauma. For example, Altschul (1988) described the use of psychoanalytic approaches in helping children to mourn the loss of a parent. Chethik (1989) discussed "focal therapy" as therapy that deals with "focal stress events" (p. 194) in the child's life. Chethik listed events such as death in the family, divorce, hospitalization, and illness as examples of specific stressors. Focal therapy focuses on the specific problem and is usually of short duration. It is a psychodynamic approach that uses insight and working through. Chethik viewed this approach as working best with children who have accomplished normal developmental tasks before the occurrence of stressful events.

Structure-Building Approaches

A second major form of psychodynamic therapy is the structure-building approach, which is used with children with deficits in object relations and interpersonal processes (Russ, 1998). For children with impaired object relations, self/other boundary disturbances, and problems distinguishing reality from fantasy, the therapist uses techniques that foster the development of object permanence, self/other differentiation, modulation of affect, and impulse control. The major mechanism of change is the development of these internal processes. The development of object relations is especially important. Mahler (1968) articulated the separation-individuation process and described the development of object constancy and object representations. Blank and Blank (1986) have pointed out that object relations plays a major role in the organization of other intrapsychic processes. Children with severely impaired object relations, such as borderline children and children with psychotic and characterological disorders, have early developmental problems in a variety of areas. Also, they tend to have severe dysfunction in the family and often a genetic predisposition.

In a structure-building approach, empathy on the part of the therapist is a much more important intervention than interpretation (Kohut & Wolfe, 1978). Kohut and

Wolfe discussed the failure of empathy from the parent that is the major issue in faulty parent-child interaction. Because of the frequency of this occurrence in the interaction between the child and parent, empathy from the therapist around the history of empathic failure becomes an important part of therapy. It is the empathy from the therapist that results in the therapist's being internalized and becoming a stable internal figure for the child. Chethik (1989) and Russ (2004) present cases of psychotherapy with borderline children. Often, help with problem solving and coping is used with these children as well. Therapy with these children is usually long term (1 to 2 years).

PSYCHODYNAMIC CONSTRUCTS

There are a number of psychodynamic constructs and hypotheses that are associated with psychodynamic theory. Research that investigates these constructs and hypotheses helps to build empirical support for psychodynamic theory. Although not directly investigating psychodynamic therapy, empirically supported constructs should guide therapy research in terns of what to focus on and what therapy techniques to emphasize. Westen (1998) reviewed the research literature and identified a number of psychodynamic constructs and propositions that have been supported. Much of the research has been carried out in cognitive, social, developmental, and personality psychology. Often, psychodynamic terminology is not used or referred to. Concepts are placed in a more contemporary cognitive-affective framework. Westen identified the following constructs and principles of psychodynamic theory that have received significant empirical support: unconscious processes, ambivalence and conflict, importance of childhood origins of personality and social dispositions, mental representations of self and others, and developmental dynamics.

Primary process thinking is another construct that has a large body of research support (Holt, 1977; Suler, 1980). Primary process thinking refers to drive-laden oral, aggressive, and libidinal material and illogical thinking related to that material (Holt, 1977). Holt described primary thought as an early, primitive system of thought that was drive-laden and not subject to rules of logic or oriented to reality. Affect and affect-laden cognition are a large component of primary process.

A large number of studies have found a relationship between access to primary process thinking and creativity, as psychoanalytic theory predicts, in adults and children (Russ, 1996; Suler, 1980). Russ has discussed these relationships within current cognitive-affective conceptualizations. The finding in a large number of studies that access to primary process thinking relates to measures of creativity supports the psychoanalytic theory that access to drive-laden thinking is important in the creative process. Repression of threatening drive-laden thoughts, images, and memories leads to a general intellectual restriction, which in turn constricts associative thinking. Access to a broad network of associations is important in most measures of creativity. Empirical support of this psychoanalytic theory is important in psychotherapy, where one goal is to help the child gain access to unconscious, repressed content so that the child can bring higher order problem-solving skills to daily problems and stressors.

Pretend play is a key resource for children. The important role of play in child development is not a concept that is unique to psychodynamic thinking. However, play has been a major technique in psychodynamic therapy and is viewed as having a central function in child development.

Pretend Play

Play has been a part of therapy with children since Melanie Klein and Anna Freud first began using play techniques in child psychotherapy in the 1930s. Currently, play is used in therapy from a variety of theoretical traditions. As of 1992, play in some form was used in child therapy by a majority of clinicians, as reported by Koocher and D'Angelo (1992).

In the child therapy literature, four broad functions of play emerge as important in therapy (Russ, 2004). First, play is a natural form of expression in children. Chethik (1989) refers to the language of play. Children use play to express feelings and thoughts. Chethik stated that play emerges from the child's internal life and reflects the child's internal world. Therefore, children use play to express affect and fantasy and, in therapy, to express troubling and conflict-laden feelings.

Second, the child uses the language of play to communicate with the therapist. By understanding these communications, the therapist develops a therapeutic relationship with the child (Chethik, 1989). The therapist actively labels, accepts, empathizes with, and interprets the play expressions, which, in turn, helps the child feel understood (Russ, 1995). These play expressions are important in enabling mechanisms of change to occur in therapy. For example, a corrective emotional experience can occur by the therapist accepting and understanding angry feelings expressed in play. Also, for many children, the feeling of empathy from the therapist around the play facilitates change in their internal representations and interpersonal functioning.

A third major function of play is as a vehicle for the occurrence of insight and working through. The emotional resolution of conflict or trauma and negative feelings is a major mechanism of change in psychodynamic child therapy. Children reexperience major developmental or situational trauma in therapy, and many of these conflicts are expressed in play. The play process itself has been thought of as a form of conflict resolution. During this process the therapist labels and interprets the play. The therapist will also tie the observations and interpretations to events in the child's life. Although there is controversy in the psychodynamic literature about how much interpretation to use (Freud, 1966; Klein, 1955), there is general agreement that working through and mastery are important mechanisms of change.

The whole area of emotion and how play helps process and regulate emotion is a key area for investigation. Shields and Cicchetti (1998) defined emotion regulation as the ability to modulate emotions and engage in an adaptive way with the environment. Mennin, Heimberg, Turk, and Fresco (2002) concluded that an emotion regulation perspective would have as goals of treatment to help individuals (a) become more comfortable with arousing emotional experience, (b) be more able to access and utilize emotional information in adaptive problem solving, and (c) be better able to modulate emotional experience and expression. Play in therapy

can aid with these three processes. The insight, emotional reexperiencing, and working-through process involve some additional components. For the child who is repressing or avoiding negative affect or thoughts and images that arouse conflict, play can help the expression of that content. Although some of the negative affect (e.g., anxiety) might be extinguished after repeated play, the emotional reexperiencing and working-through process involves more than extinction of emotion. It involves building a meaningful narrative around the emotion or trauma (Gaensbauer & Siegel, 1995). In addition, insight is applied to experiences and behaviors to which the insight pertains (Shapiro et al., 2006). Operationalizing the components of insight and working through and identifying how play facilitates this process is an important next step for research. It will also be important to identify when this type of process in therapy is important for a child, and when it is not necessary.

A fourth major function of play in therapy is to provide the child opportunities to practice with a variety of ideas, behaviors, interpersonal behaviors, and verbal expressions. Because play is occurring in a safe environment in a pretend world, with a permissive, nonjudgmental adult, the child can try out and rehearse a variety of expressions and behaviors without concern about real-life consequences. In some forms of play therapy, the therapist is quite directive in guiding the child to try new behaviors. Knell (1993) developed a cognitive-behavioral play therapy approach that actively uses modeling techniques and a variety of cognitive-behavioral techniques. Many of these can be used within a psychodynamic framework.

Although these functions of play occur in normal play situations, the therapist builds on these normal functions by enhancing the play experience. The therapist creates a safe environment, gives permission for play to occur, actively facilitates play, and labels the thoughts and feelings expressed. For the psychodynamic therapist, interpretation specifically aids conflict resolution.

For most children younger than 10 years in psychodynamic therapy, therapy is a mixture of play and talk. How play is used in psychotherapy depends on the child's ability to use play, developmental level, age, ability to verbalize, and the overall treatment approach and goals. A number of practical issues arise in using play in therapy. For suggestions about how to get the child started in using play, kinds of play materials, how much the therapist should engage in play, how much to interpret, and how much to set limits, see Chethik (1989), Singer, (1993), and Russ (2004).

RESEARCH WITH PSYCHODYNAMIC THERAPY

There has been a strong movement in the field of child psychotherapy to identify empirically supported treatments (Lonigan, Elbert, & Johnson, 1998). Most of the therapy outcome studies have been efficacy studies that are conducted under controlled conditions that involve random assignment, control groups, and single disorders. Effectiveness studies, on the other hand, are clinical utility studies that focus on treatment outcome in real-world environments. Empirically supported treatment reviews have focused on efficacy studies (Kazdin, 2000). In the child psychotherapy area, Kazdin concluded that the empirically supported treatments for children are few in number and consist mainly of cognitive-behavioral treatments.

Weisz and Weiss (1993) pointed out the low number of psychodynamic outcome studies in the literature. Psychodynamic treatments have not been investigated in the kind of controlled conditions necessary for efficacy studies. Criteria for efficacy studies include random assignment, specific child populations, use of treatment manuals, and use of multiple outcomes with blind raters.

Russ (2006) reviewed the research literature relevant to psychodynamic psychotherapy. Fonagy and Moran (1990) proposed that one reason for the paucity of research is that the rather global approach and broad goals of psychodynamic approaches do not lend themselves easily to carefully controlled outcome studies. Russ (1998) has proposed that the mechanisms of change are also broad and multiple: Therapy is individualized and the therapist is making decisions about intervention on a moment-to-moment basis. However, specificity in research is essential for the field to progress. An important conclusion from Weisz and Weiss (1993) was that studies showing positive results in psychotherapy research tend to be ones that focus on a specific problem, with careful planning of the intervention. In 1983, Freedheim and Russ stated, based on the literature to date, that therapy research needed to become very specific and investigate which specific interventions affect which cognitive, affective, and personality processes. Shirk and Russell (1996) developed a framework for conceptualizing intervention research that ties specific treatment processes to underlying cognitive, affective, and interpersonal processes within a developmental framework. They stress the importance of investigation of specific processes and change mechanisms. Psychodynamic conceptualizations such as the effect of conflict or of deficits in object relations should be testable within their framework.

There has been progress in the area of psychodynamic psychotherapy with children. Fonagy and Moran (1990) applied many of the current guidelines for psychotherapy outcome research to evaluating the effectiveness of child psychoanalysis. Their studies are models for how to investigate the efficacy of psychodynamic approaches. They have carried out different types of studies that are well suited to the psychoanalytic or psychodynamic approach. In one study (Moran & Fonagy, 1987), they used a time series analysis to study the 184 weeks of treatment of a diabetic teenager. Time series analysis investigates whether or not there is a time-bound relationship between events. A relationship was found between major themes in analysis and diabetic control. Moran and Fonagy concluded that the interpretation of conflicts in the treatment brought about an improvement in diabetic control. The improved control led to temporary increases in anxiety and guilt. The improved diabetic control appeared to increase the likelihood of manifest psychological symptomatology. This pattern fits the psychodynamic understanding of brittle diabetes.

In a second study, an inpatient program for diabetes was evaluated (Fonagy & Moran, 1990). Eleven patients received psychotherapy and medical supervision, while the comparison group received medical treatment with no psychotherapy. The analytic treatment was well defined and based on the psychoanalytic understanding of brittle diabetes as being caused by unconscious emotional factors influencing the significance of the disease or its treatment regimen. This leads to disregard for normal diabetic care. The goal of therapy was to make conscious the conflicts and

anxieties that were interwoven with the diabetes treatment regimen. The treated group showed significant improvement in diabetic control. None of the untreated group showed improvement. Improvement was maintained at 1-year follow-up. Fonagy and Moran stressed the importance and feasibility of systematic and specific intervention research that investigates changes in psychic structure, such as affect regulation and empathy, as a result of psychoanalysis.

Shirk and Russell (1996) reviewed a number of research programs and measures relevant to assessing change in psychodynamic psychotherapy. Wallerstein (1988) and Kernberg (1995) have coding systems for dimensions relevant to psychodynamic treatment. The Affect in Play Scale (Russ, 1993) can be used to measure change in play processes in psychodynamic therapy.

Muratori, Picchi, Bruni, Patarnello, and Romagnoli (2003) evaluated short- and long-term effects of time-limited psychodynamic psychotherapy for children with internalizing disorders. Fifty-eight outpatient children (6 to 10 years old) who met *DSM-IV* criteria for depressive or anxiety disorder were assigned to therapy or community services. Children were assessed at baseline, 6 months, and 2-year follow-up using the Children's Global Assessment Scale (C-GAS) and Child Behavior Checklist (CBCL). Group assignment was not random. Assignment to the treatment group occurred if there was an opening; if no opening was available, the child was referred to community services. There was no significant difference at baseline between the groups on major variables. Therapy consisted of 11 sessions that followed a protocol. There were five parent-child sessions, five sessions with the child alone, and a final parent-child session. There appear to be clear guidelines about therapist intervention in each session. The treatment approach was to point out the nature of the core conflictual theme and connect it to the child's symptoms and the representational world of the parent. In the child sessions, the therapist labeled feelings and connected mental content and symptoms. Play was seen as an important part of the therapy. Therapists were trained in this approach. Their interventions were videotaped and discussed weekly to monitor adherence to the protocol.

Results revealed major improvements in the treatment group on the C-GAS and CBCL. Improvements on the C-GAS were at 6 months, and CBCL findings were at 2-year follow-up. The changes in the internalizing syndrome on the CBCL at the follow-up only is evidence for the "sleeper effect," a delayed response to the treatment. Muratori et al. (2003) concluded that changes in the shared representational world brought about by the therapy resulted in changes in parent-child interaction only after a period of time. Also, the treatment group sought mental health services less than the comparison group during the 2-year period. This study followed many methodological guidelines for adequate research design described by Weisz and Weiss (1993). The sample was relatively homogeneous (anxious and depressed); a treatment protocol was followed; and therapists were trained and monitored to make sure they adhered to the intervention techniques. In addition, there was a theoretical rationale for why this approach should affect this syndrome. This is the type of rigorous research that needs to occur in the psychodynamic area.

Trowell et al. (2002) compared psychodynamic therapy with psychoeducational topic-focused discussions for 71 girls, 6 to 14 years old, who had experienced

sexual abuse. Children were randomly assigned to treatment groups and manualized treatment was used. Sessions were limited to a maximum of 30. Both treatment groups showed improvement, with the psychodynamic therapy group showing significantly more improvement on the "persistent avoidance of stimuli" measures on a PTSD scale. In a review, Ritvo (2006) pointed out that this study and the Fonagy and Moran (1990) study of children with diabetes both focused on children dealing with specific stress rather than with a clinical diagnosis. She stated that this focus on life stress is consistent with the emphasis that the psychodynamic approach places on adaptation and function rather than on clinical diagnosis per se.

We also need more studies that investigate specific mechanisms of change in therapy. A good example of the kind of study that needs to occur is that by Wiser and Goldfried (1998). Although this study was conducted with adults, it is a good model for the child area because it investigates specific types of interventions on specific processes. They investigated the effectiveness of therapeutic interventions on emotional experiencing. The expression of emotion in session has been thought to be important for different mechanisms of change to occur. An important question with adults and children is how to facilitate emotional experiencing in the session. Wiser and Goldfried coded interventions by therapists with anxious and/or depressed outpatient adults in either psychodynamic or cognitive-behavioral therapy. They found that interventions of reflections and acknowledgments, affiliative and noncontrolling interventions, and interventions highlighting nonspecific client content were associated with maintaining high emotional experiencing in both approaches (reflections were nonsignificant in the psychodynamic approach but may be significant with a larger sample size). Interestingly, no type of intervention was associated with a shift to higher emotional experiencing. Some interventions were related to shifts to low experiencing: those that were lengthy and those that were affiliative but moderately controlling. It is important to know which techniques to use to increase emotional experiencing and which to decrease emotional experiencing. Both types of interventions are therapeutic at different times with different populations. The Wiser and Goldfried study is an excellent example of the type of therapy process research that needs to be carried out with children.

Aspects of the relationship with the therapist are also important to investigate in psychodynamic therapy. Using meta-analytic procedures, Shirk and Karver (2003) reviewed associations between therapeutic relationship variables and treatment outcomes in child and adolescent therapy. Results were a modest association (.20), which was similar to that seen in adult studies. This relationship was found across diverse types and modes of child treatment.

Target (2002) reported on results from a retrospective chart study of treated children at the Anna Freud Center who were in psychodynamic or more intensive psychoanalytic treatment. Although this study was retrospective and there was no randomized design or comparison groups, the results are important and do provide guidance. Looking first at anxiety or depressive disorders, 72% of those treated for at least 6 months showed clinically significant improvement (Target & Fonagy, 1994). Phobic disorders were most likely to remit and depressive disorders least likely. Variables that identified children most likely to improve were higher IQ, younger age, longer treatment, good peer relations, poor overall adjustment of the

mother, the presence of anxiety in the mother, concurrent treatment of the mother, and absence of maternal antisocial behavior.

In the chart review of disruptive disorders, improvement rates were lower than for the internalizing disorders, with one third terminating therapy prematurely. Of those who stayed in treatment, 69% were no longer diagnosable upon termination. Predictors of improvement were the presence of an anxiety disorder, absence of other disorders, younger age, intensive treatment, longer treatment, maternal anxiety disorder, treatment of the mother, and foster care for the child. Target and Fonagy (1996) concluded that psychodynamic therapy was effective with disorders involving anxiety, even when the anxiety was coupled with disruptive disorders. A different picture emerged with the disruptive (externalizing) disorders. When therapy was effective, it was when intensive treatment was used. They reported that it was the oppositional disorders that were most likely to improve and conduct disorders the least likely.

It appears that before recommending psychodynamic therapy for externalizing disorders in which anxiety is not a major factor, several issues should be considered. Cognitive-behavioral approaches have been found to be very effective for this problem. For example, parent-child interaction therapy is very effective with Oppositional Defiant Disorder (Bodiford-McNeil, Hembree-Kigin, & Eyberg, 1996). If psychodynamic therapy is recommended, two things should be kept in mind and shared with the family: (1) There is no solid empirical support for psychodynamic therapy with externalizing disorders, and (2) the intensive therapy that would be attempted is costly.

The research studies that do exist in the area support the use of psychodynamic therapy for internalizing disorders where anxiety and conflict are important factors. Shapiro et al. (2006) concluded that psychodynamic therapy's focus on play, emotion-dominated thinking, and magical fantasies makes it appropriate for young children. They concluded that no treatment approach has been proven effective for anxious children under 7 (except for simple fears and phobias) or for depression below age 9. The play intervention research that exists also supports the use of a psychodynamic conceptualization for reducing anxiety and negative affect.

Play Intervention Studies

Russ (1995, 2004) reviewed studies investigating the effect of play interventions on specific types of child problems in specific populations. I call these studies play intervention because the focus is highly specific, usually involving only a few sessions with the child. These studies have focused on reducing anxiety in children in medical settings or dealing with separation anxiety. In two different studies, puppet play reduced anxiety in children facing medical procedures when compared with a control group (Cassell, 1965; Johnson & Stockdale, 1975). Rae, Worchel, Upchurch, Sanner, and Daniel (1989) found that children in a therapeutic play group showed significantly more reduction in self-reported hospital-related fears than children in three other treatment groups. This was a well-done study that controlled for time spent with an adult, verbal support, and play activity. They concluded that fantasy activity in the play resulted in fear reduction. Milos and

Reiss (1982) found that thematic play reduced separation anxiety in preschoolers when compared with a control group. Also, the quality of play ratings were significantly negatively related to posttest anxiety. High-quality play was defined as play that showed more separation themes and attempts to resolve conflicts. They concluded that their results supported the notion that play can reduce anxiety associated with psychological problems. The results are consistent with psychodynamic theory.

A well-designed study by Barnett (1984) investigated separation anxiety. Expanding on the work of Barnett and Storm (1981), Barnett found that free play reduced distress in children following a conflict situation the first day of school. Seventy-four preschool children were observed separating from their mothers and were rated anxious or nonanxious. These two groups were further divided into a play or no-play story-listening condition. For half of the play condition, play was solitary. For the other half, peers were present. The story condition was also split into solitary and peers-present segments. Play was rated by observers and categorized into types of play. Play significantly reduced anxiety in the high-anxious group. Anxiety was measured by the Palmer Sweat Index. There was no effect for low-anxious children. For the high-anxious children, solitary play was best in reducing anxiety. High-anxious children spent more time in fantasy play than did low-anxious children, who showed more functional and manipulative play. High-anxious children also engaged more in fantasy play when no other children were present. Barnett suggested that play was used to cope with a distressing situation. The findings supported her concept that it is not social play that is essential to conflict resolution, but rather imaginative play qualities that the child introduces into playful behavior.

The results of these studies suggest that play helps children deal with fears and reduce anxiety and that something about play itself is important and serves as a vehicle for change. Both the Milos and Reiss (1982) study and the Barnett (1984) study suggest that fantasy and make-believe are involved in the reduction of anxiety. The studies effectively controlled for the variable of an attentive adult. Results also suggest that children who are already good players are more able to use play opportunities to solve problems.

These research findings are consistent with psychodynamic theoretical and clinical literature suggesting that play therapy assists children with internal conflict resolution and mastery of internal issues, as well as with external traumas and stressful life events. As a result of this conflict resolution and problem solving, anxiety is reduced. Psychodynamic approaches also suggest the use of insight, conflict-resolution approaches for children whose fantasy skills are normally developed and who can use play in therapy.

Future research should focus on the mechanisms that account for the finding that play reduces anxiety. Conceptualizations and research from other theoretical frameworks could apply to play therapy. For example, Harris (2000) views play as helping the child construct a situation model that is revisable. Children go back and forth between an imagined world and reality and develop a new cognitive appraisal of the situation. Pennebaker's work is also relevant: Adults who develop a coherent and meaningful narrative of an event through an emotional

writing exercise have improved mental and physical health (Pennebaker & Graybeal, 2001). Children may be developing meaningful and coherent narratives in play therapy.

Future research should investigate techniques in therapy that facilitate play in therapy. Russ, Moore, and Pearson (2006) developed play intervention protocols that resulted in increased fantasy and imagination and affect in play in first- and second-grade children. In five 30-minute sessions with a play trainer, the child was instructed to play out various story scripts. In the affect play group, the child was asked to play out stories with different emotions and to express emotion. In the imagination play group, the child was asked to play out stories with a high fantasy content and high organization of the story. The instructions and prompts by the play trainer were standardized. The affect play group had higher scores on all play abilities when compared to a control group on the outcome play measure (Affect in Play Scale). The imagination play group had higher scores on the affect play variable. The results suggest that a brief standardized play intervention can improve play skills and increase emotional expression in play in psychotherapy. This technique could be helpful for children with constricted affect such as those with PTSD or depression. By having children make up various stories and directing them to express affect, with modeling and reinforcement by the therapist, children could increase affect expression quickly. This technique could be especially useful in time-limited therapy.

SHORT-TERM PSYCHODYNAMIC PSYCHOTHERAPY

Short-term psychotherapy (6 to 12 sessions) is a form of psychodynamic psychotherapy frequently used with children (Messer & Warren, 1995). The practical realities of HMOs and of clinical practice in general have led to briefer forms of treatment. Often, the time-limited nature of the therapy is by default, not by plan (Messer & Warren, 1995). The average number of sessions for children in outpatient therapy is 6 or fewer in private and clinical settings (Dulcan & Piercy, 1985).

There is little research or clinical theory about short-term therapy with children (Messer & Warren, 1995). A few research studies have shown that explicit time limits reduced the likelihood of premature termination (Parad & Parad, 1968), and that children in time-limited therapy showed as much improvement as those in long-term therapy (Smyrnios & Kirby, 1993). The time is right for the development of theoretically based short-term interventions for children. Messer and Warren suggest that the developmental approach utilized by psychodynamic theory provides a useful framework for short-term therapy. One can identify the developmental problems and obstacles involved in a particular case. They also stressed the use of play as a vehicle of change and, as Winnicott (1971) has said, of development. They suggest that the active interpretation of the meaning of play can help the child feel understood, which, in turn, can result in lifelong changes in self-perception and experience. In other words, the understanding of the metaphors in a child's play could give the child insight or an experience of empathy, or both. This lasting change could be accomplished in a short time.

As previously discussed, Chethik (1989) developed focal therapy to deal with specific stressful events in the child's life. Basic principles of psychodynamic therapy are applied in this short-term approach, with the basic mechanism of change being insight and working through. Chethik views this approach as working best with children who have accomplished normal developmental tasks before the stressful event occurred.

In general, brief forms of psychodynamic intervention are seen as more appropriate for the child who has accomplished the major developmental milestones. Proskauer (1969) stressed the child's ability to quickly develop a relationship with the therapist, good trusting ability, the existence of a focal dynamic issue, and flexible and adaptive defenses as criteria for short-term intervention. Messer and Warren (1995) concluded that children with less severe pathology are more responsive to brief intervention than children with chronic developmental problems. The Muratori et al. (2003) study that found that short-term intervention with anxious and depressed children was effective is consistent with this conclusion. The research and clinical literature suggest that internalizing disorders are most appropriate for brief psychodynamic intervention (Russ, 2004). The therapist is active, at times directive, and uses all mechanisms of change in the therapy. Insight and working through are essential, but modeling, rehearsal, and problem-solving strategies are also part of the therapy. Children with major deficits in object relations and those with early developmental problems need longer term structure-building approaches.

Shelby (2000) described the importance of using developmentally appropriate interventions in brief therapy with traumatized children. Working with traumatized children in Sarajevo, she used play and drawing. She described an experiential mastery technique in which children drew pictures of the thing that frightened them. Children were encouraged to verbalize their feelings about the drawing. They were also instructed to do anything they wanted to the drawing. Shelby described how this and other developmentally appropriate brief play interventions, although not empirically tested, helped a number of these traumatized children to integrate traumatic events and return to "normal developmental functioning" (p. 72).

Structured play techniques would be especially useful in short-term therapy. The MacArthur Story Stem Battery (MSSB), although designed as an assessment tool, can be used to structure the play situation. In an innovative approach, Kelsay (2002) used the MSSB to structure play therapy. The MSSB is a set of story beginnings (e.g., parents arguing over lost keys), and the child is asked to complete the story. The therapist can choose appropriate story stems tailored to the issues that the child is dealing with. This structured approach could move the therapy to central issues more quickly.

Structured play techniques have also been used with very young children. Gaensbauer and Siegel (1995) described structured play techniques with toddlers who have experienced traumatic events. They conceptualized the mechanisms of change when play is used as being similar to those in older children with PTSD. With these very young children, the therapist actively structures the play to recreate the traumatic event. Gaensbauer and Siegel outlined three purposes of structured play reenactment. First, play enables the child to organize the fragmented experiences into meaningful narratives. Second, the interpretive work by the therapist helps the

child understand the personal meanings of the trauma. Third, there is desensitization of the anxiety and fear and other negative emotions associated with the trauma. They stressed that the key element that enables the child to use play adaptively, rather than in a repetitive fashion, is the "degree to which affects can be brought to the surface so the child can identify them and integrate them in more adaptive ways" (p. 297).

RESEARCH GUIDELINES

Given the results of the research in the area, research guidelines suggested by Russ (2006) that would be the most fruitful and have the most immediate impact on the field would focus on the following:

- In young children, well-controlled efficacy and effectiveness studies with anxiety disorders, PTSD, depression, and focused problems involving anxiety and conflict should be carried out with psychodynamic therapy.
- Short-term interventions should be investigated and refined.
- Mechanisms of change in therapy, especially insight and emotional working through, are needed to identify mechanisms leading to anxiety reduction and reduction of negative affect.
- Play processes and techniques that facilitate play in therapy should be researched.

Demonstration of the effectiveness of structure-building approaches with seriously emotionally disturbed children is also necessary. Because the therapy is so long term and complex with these children, single-case designs would be especially appropriate.

SUMMARY

The psychodynamic approach focuses on the internal world of the child. A developmental perspective is used to understand underlying cognitive, affective, and interpersonal processes. A psychodynamic understanding determines the specific intervention approach and techniques to be used with a particular child. The mechanisms of change most associated with psychodynamic therapy are insight, emotional reexperiencing, and working through. A key question for future research is to identify the components of these mechanisms of change. In therapy, pretend play is frequently used as a vehicle for change. There are a number of constructs in psychodynamic theory that inform therapy. Empirical support for these constructs lends support for psychodynamic therapy. Outcome research on the effectiveness of psychodynamic therapy is sparse. A few studies do offer empirical support. Research in play intervention also suggests that the use of play in therapy can reduce anxiety. Future research should focus on the use of psychodynamic therapy with internalizing disorders in young children, play facilitation techniques, short-term therapy, and the components and role of the insight and working-through process.

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CHAPTER 7

Family Systems

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This chapter provides an overview of major contributions of family systems thinking to the field of clinical psychology. Consistent with the volume's other chapters, it attends to theory, research, assessment, and treatment issues. To provide additional substance for some of the central conceptual issues advanced in this chapter, a section on treatment considerations develops a clinical case example in an outline paralleling those of other chapters covering major treatment approaches. In so doing, we strive to clarify how clinicians working from a family systems approach might work with children and families in a manner quite distinct from, though often complementary to, interventionists working from psychodynamic, behavioral, or cognitive approaches.

What our field refers to as "family systems thinking" is actually a conceptual and philosophical framework that reflects a wide array of quite different approaches and influences—indeed, there *is* no single "systems" theory. What ties together this collection of rather disparate influences and approaches is the central, guiding notion that problem behavior exhibited by a child or an adult can never be understood devoid of its relational context. Further, from a systems view, the behavioral problems evident in any individual family member are perhaps best understood as manifestations of dysfunction within the broader family unit. From this basic tenet follow distinctive lines of inquiry in research studies, and approaches in clinical formulation, assessment, and treatment, that often differ substantially from paths followed in other theoretical traditions and frameworks.

We begin this chapter with a review of some of the major assumptions of systems approaches and a brief recounting of some of the historically important forms of influence in shaping this approach to psychopathology and its treatment. We summarize roles that empirical research has played in the family systems field. From this base, we outline how family systems principles might organize clinical work with a case referred for the treatment of ostensible problems in an individual family member. We close the chapter with some reflections on the value of systems-informed approaches in clinical work.

BASIC CONCEPTS AND PRINCIPLES

Historically, the systems theory developments that inspired and were infused by early family therapies presented exciting and revolutionary challenges to the

practices and theories of the times. Systems theory was held to differ from psychoanalytic approaches by emphasizing relational as opposed to intrapsychic processes, and by taking a holistic and multidirectional view of pathological behavior and its treatment (Bateson, Jackson, Haley, & Weakland, 1956; Lidz, Cornelison, Fleck, & Terry, 1957a, 1957b; Wynne, Ryckoff, Day, & Hirsch, 1958). But the sheer number of different thrusts within this emerging field and the complexity of some of the early theoretical formulations and writings meant that it would be many years before coherence in the field began to emerge. Accounts of "systems theory" and its pertinence to families and individual development often seek to trace the history of how principles governing physical systems came to be applied to social systems. In such accountings, special homage is often accorded the creative writings of von Bertanffly (1968). As Patricia Minuchin (1985) noted, however, the writings of Bateson (1972, 1979) and other family theoreticians more directly concerned with the functioning of human systems (e.g., Jackson, 1957; S. Minuchin, 1974; Watzlawick, Beavin, & Jackson, 1967) had a more formative impact on most practicing clinicians of the time. Among the most essential of systems tenets, outlined by P. Minuchin (1985) and others (e.g., Bornstein & Sawyer, 2006; Cox & Paley, 2003) are the following:

- Systems are organized wholes, and their constituting elements or subsystems are interdependent.
- Interconnected subsystems have their own integrity, are organized hierarchically, and are separated by boundaries.
- Patterns in a system are circular and not linear.
- Stable patterns are maintained over time through homeostatic processes.
- Open systems do adapt, change, reorganize, and develop.

Several important talking points follow from these central tenets. First, when clinicians consider any family system, they should find it possible to identify both a family group reality wherein patterns of organization governing the functioning of the overall unit operate as organized patterns in their own right, and a variety of subsystem realities that may or may not themselves mirror the same patterning and rules of the broader system. Most families are composed of multiple subsystems—among them, marital subsystems, parent-child subsystems, sibling subsystems, parent-grandparent and child-grandparent subsystems in multigenerational family systems, and so on. When clinicians and researchers assess family dynamics, they regularly detect logical interconnections among different subsystem levels; for example, high levels of distress in the marital subsystem typically coincide with evidence of impaired parenting of one or more children in the family (Erel & Burman, 1995). They also find that subsystem functioning both affects and is affected by broader systemic functioning, as when marital strife between a husband and wife within their dyadic marital subsystem comes to disrupt collaboration in effectively coparenting children at the family group level (Belsky, Crnic, & Gable, 1995; McHale, 1995). In other cases, such linkages are not as overt or easily detected, as when family members as a unit collude to obscure problematic

alignments or abuse within a particular dyadic subsystem in the family (Wynne, 1961).

Besides these interconnections between subsystems and the functioning of the family group, the functioning of the overall unit is also properly seen as constraining and organizing interactions at all other levels. For example, in Bowen's (1978; see also Friedman, 1992) conceptualization of the family's emotional system, family members are described as having developed emotional interdependencies to the point where the emotional system through which they are connected has evolved its own principles of organization. The resulting organizational structure then comes to influence the functioning of various individuals and dyads far more than any of them alter the functioning of the full system (S. Minuchin, 1974). For example, a variety of different interaction patterns and sequences (e.g., children diffusing conflict between parents; parents exhibiting preoccupation and overprotection of children) may be seen in families whose structure is guided by the covert rule that "family members must protect one another" (see Nichols & Schwartz, 1998); although changing any particular sequence would not alter this governing structure, altering the structure would reshape interaction patterns throughout the system.

Of course, if an interventionist was successful in meaningfully changing any given aspect of a family's entrenched pattern, this would have the effect of perturbing other patterns. Such temporary ripple effects and adjustments have the potential to affect more enduring change in the problematic family pattern—but only to the extent that the short-term adjustments could be sustained. Unfortunately, supporting enduring pattern changes is very difficult, in part because systems show marked propensities for preserving equilibrium. New patterns of behavior by a child or adult that challenge established family patterns and rules are often quickly countered by adjustments elsewhere in the system that are knowingly, or unconsciously, intended to reestablish the previously enduring status quo. Said differently, because everyone's behavior has become interdependent, the family members collectively collude to maintain their existing patterns. Moreover, repetitive patterns are especially unlikely to change if they have evolved to serve some important protective function for the system as a whole (Madanes, 1981; Selvini Palazzoli, Boscolo, Cecchin, & Prata, 1978), as when a child's misbehavior deflects attention away from a more serious rift in the parents' marital relationship.

The notion of circular causality in family systems (Bateson, 1979) is also central in family systems conceptualizations. Seen as a hallmark of systems thinking that stands in contrast to the linear thinking implicit in other approaches, the principle of circular causality is that the patterns families develop are actively maintained by all of the family's participant members. Such group and subsystem reciprocity is seen as largely responsible for the enduring nature of family patterns through time. As an example, extreme emotional overinvolvement by a child's mother, to the point where she interferes with that child's own capacities for autonomous emotional regulation, is often accompanied by pronounced disengagement by the child's father (in some families, the pattern is reversed, with intense father-child involvement co-occurring with maternal disengagement). Whether fathers disengage in response to intensive maternal involvement or whether maternal overinvolvement evolves as a response to inadequate father involvement is an immaterial question; all parties

(including the family's children) collude to help maintain system functioning by steadfastly keeping to the pattern that has evolved through a process of checks, balances, and recurring feedback loops.

Given these powerful corrective feedback loops self-regulating families and maintaining their enduring patterns, how do families ever change and evolve? Different theorists provide different perspectives, but most find consensus on certain points. First, all living systems do change in form (Speer, 1970). They do so in response to major and sustained perturbations either from within or from the outside that disrupt established patterns (Hoffman, 1981; Prigogine, 1973). The initial reactions of an intensively challenged system are to mobilize familiar modes of response. But as such responses fail to accommodate the new challenge, the system begins advancing previously untested, untried methods of adaptation until ultimately happening upon solutions relevant to addressing the challenge. During this period of duress, the system is in relative disarray. Indeed, it operates temporarily at a lower level of integration, as its previous modes of adaptation are abandoned in favor of newer and less refined adaptations. However, following this period of destructure and disintegration, newer structures and solutions effectively replace the old, becoming integrated into the system's patterning and modes of action. This process of disequilibration and reorganization (Block, 1982) typically eventuates in the emergence of a more complex and differentiated system, though devolution, as in the case of incestuous families, is also possible (Alexander, Sexton, & Robbins, 2002). The operating principle is that change is expectable and inevitable. In human systems this is largely because change is regularly initiated by individual development within the broader system.

Where, then, is the connection with von Bertanffly's general systems model? Extrapolating from the biological systems model, family theorists have often drawn connections to notions of systems as open, cybernetic, or closed (Bateson, 1972 1980; White, 1986, 1989). Open systems, von Bertanffly (1968) explains, have the intrinsic capacity for change, whereas change in closed (including cybernetic) systems occurs only in response to outside sources. Don Jackson (1957) was among the first to draw a parallel between closed information systems in states of homeostasis and the consistency of family disturbances and symptomatic behavior that resisted change. Appropriating the metaphor of homeostasis from general systems theory had appeal, as families stuck in recurrent maladaptive patterns could be viewed as being in a state of equilibrium, behavioral redundancy, and relationship balance. The homeostasis metaphor has remained a very influential one, though it is flawed in many ways. As Moyer (1994) has pointed out, whereas human systems are open rather than closed, theoretical concepts of cybernetics and homeostasis properly describe closed systems and hence are not truly applicable. For example, open (as opposed to cybernetic) systems are intrinsically capable of change, do possess the potential for free interchange of information, energy, or matter, and are composed of components that are simultaneously actors and acted upon (in a circular, rather than linear, dynamic).

Nonetheless, characterizations of disturbed family systems as closed, insular, and resistant (if not impermeable) to adaptive informational exchanges with the outside world have remained a recurring and robust metaphor in many clinical

writings. Certainly, though, not all theorists find it necessary to invoke open-closed and cybernetic metaphors when describing closure and insularity; Bowen (1978), for example, observed simply that family members withdraw from outside relationships and insulate themselves when emotional forces within the family are in opposition and internal anxiety and tension mount. Such closed or relatively closed-off family systems are usually seen as dysfunctional when contrasted with healthier open system functioning (Olsen, Sprenkle, & Russell, 1979; Satir, 1972), though open systems too can be plagued by serious problems, such as excessive and chaotic interaction, prolonged conflict, and ambiguity (Constantine, 1983).

We now turn to several of the most influential theoretical streams and concepts emanating from the related fields of family therapy and family psychology.

Historically Important Ideas and Lines of Influence: Family Therapy and Family Psychology

The view that children's behavioral problems are inextricable from their family group context and that they are most effectively addressed through family collective approaches has been active for nearly a century. Richmond (1917), echoing practices of the social work movement of her time, posited that child adjustment problems had to be understood in their family context, advocated work with family units, and emphasized the palliative role of family cohesion. Yet even as she wrote, common practice then as now was for children's mothers (and seldom fathers) to attend such family sessions. In clinical psychology, mainstream practice did not follow suit in assuming social work's collectivist stance through most of the first half of the twentieth century, even though positions developed by Sullivan, Horney, and Fromm emphasized the interpersonal nature of psychiatric disorders. The work of these pioneering figures, however, was not truly systemic in that treatment approaches seldom moved to involve family members; such inclusion was usually construed as a violation of client confidentiality.

During the 1950s, John E. Bell, Nathan Ackerman, and a handful of other clinicians broke from these long-standing traditions and began conducting conjoint therapy when seeing problem-referred children. This significant reconceptualization of work with child cases helped catalyze the gradual spread of family-level approaches and came to spark many generative debates and discussions among clinical psychologists. Herein were the gentle beginnings of a family therapy movement, guided by the clarion call that individual psychopathology signaled dysfunctional family systems. Distressed children were said to have accepted an assignment from the family to be its "identified patient" (IP), while the child's symptomatology came to be understood as serving a "messenger" function communicating dire problems at work in the broader family system. Yet even as the IP broadcast these messages, his or her problems were also playing some functional role in protecting the family and maintaining its balance (as when redirecting parents' attention away from their own strife and onto the child's problems). Moreover, this function often eluded the conscious awareness of some or all of the family's members. Based on this premise that child behavior problems were telegraphing broader system dysfunction, family practitioners argued that it was senseless to excise a child from his or her family

system to treat the child's behavior problems, and then return him or her to the very same context that had catalyzed the development and supported the maintenance of those problems to begin with. Rather, logic dictated attempting to effect change at the root source, alleviating the child's suffering by fostering meaningful change in the functioning of the child's family system.

There were many different pathways toward such ends charted by pioneers of the family therapy movement. Given space limitations, we highlight only some of the principal and most enduring contributions of various perspectives and underscore a few commonalities among approaches. However, we caution readers that by doing so we necessarily obfuscate much of the important detail and richness of theory differentiating the disparate approaches, and we recommend consulting original sources for fuller readings of theory exposition. As an orienting comment, it is probably fair to say that all of the major early schools of thought—including psychoanalytic, Bowenian, contextual, experiential, strategic, structural, communication, and behavioral therapies—shared a few things in common. All acknowledged mutual contributions by all family members to the problems they encountered, shared the therapeutic goals of helping family members assume greater responsibility and delimit blame, and aspired to clarify and enhance intrafamily communication patterns (Kaslow, 1982; Nichols & Schwartz, 1998). How the therapies sought to do so, what each took as their point of entry, how much each privileged the intrapsychic world or privileged external interpersonal relations, and how much emphasis each gave to the role of past as opposed to contemporary, here-and-now relationships—even insofar as differing in defining who actually constituted the family (and in multigenerational and extended kin frameworks, how far this reach needed to go)—varied greatly. Yet all approaches sought essentially similar ends: greater role flexibility and adaptability, greater clarity and specificity of communications, a more equitable balance of power among the family's coparenting figures, and promotion of greater individuality and differentiation of members within a cohesive family collective. Hence family cohesion, adaptability, and communication are some of the common themes tying together different approaches (Olson, Russell, & Sprenkle, 1980).

In most accounts Ackerman, who was a psychoanalyst and child psychiatrist by training, is hailed as the pioneering figure in the family therapy movement; his early publications, *The Unity of the Family* and *Family Diagnosis: An Approach to the Preschool Child,* are counted among the field's seminal works. Perhaps the most provocative conceptual advances early on in the family therapy movement, however, emanated from work with families of young men and women diagnosed with Schizophrenia. Among the innovative clinicians bucking a therapeutic tradition of segregating schizophrenic individuals from their families when providing treatment (and involving the family members only on a need-to-know basis) was Murray Bowen, who strove to treat the schizophrenic person's entire family unit. Bowen believed that the introduction of an outside person into disturbed relational systems had the potential to modify relationships within that system, but he emphasized the importance of the therapist's keeping the intensity of therapeutic focus within the family unit rather than siphoning it into an intense transference relationship. This stance, which broke with psychoanalytic convention, was one defining feature of the Bowenian approach. Bowen's theory described how individual

family members in troubled families experienced difficulty differentiating within a relatively closed-off emotional family system (in 1974, Bowen portrayed families with schizophrenic members as characterized by an "undifferentiated family ego mass"), and he was among the first family theorists to explicitly outline the role that multigenerational transmission played in family pathology. But perhaps the most central of all Bowen's many major contributions was his exposition of emotional triangles as the basic building blocks of families, and of how a third person helps to stabilize inherently unstable dyads. While similar ideas were developed by several other theorists, as outlined later, Bowen helped specify why it is that triangulation breeds stress; to the extent that one family member becomes responsible for or tries to change the relationship of two others, that person shoulders the stress for the others' relationship. Stress within emotional triangles thus came to be seen as a positional phenomenon; assuming responsibility overburdened younger, less differentiated family members, in particular, by trapping them in problematic relationships that they were ill equipped to handle (Friedman, 1992).

Elsewhere, Gregory Bateson and his colleagues in Palo Alto, California (among them Jay Haley, John Weakland, and later Don Jackson), had been developing a communication theory of Schizophrenia, from which surfaced the notions about family homeostasis discussed earlier. According to the group's double-bind theory, an individual's psychosis was understandable in the context of pathological family communication. When individuals receive contradictory messages or injunctions communicated on different (manifest content and metacommunication) levels, the theory went, they are inextricably trapped in contradictory, double-binding conundrums offering no clear channel for resolution or escape (Bateson et al., 1956). The Palo Alto group portrayed the cognitive distortions endemic to Schizophrenia as one adaptation to such insoluble double-binding communications. Despite its elegance and allure, double-bind theory as a theory for Schizophrenia did not stand up to clinical and empirical scrutiny and ultimately fell into disrepute. However, several innovative conceptualizations emanating from the group's work, including the coexistence and significance of multiple communication levels in family interactions and the notion that destructive relationship patterns endure as a result of self-regulating interactions, have had lasting impact.

As a family theory, many of the seminal ideas of the Palo Alto group, such as the double-bind notion, and Jackson's (1965) "quid pro quo," were limited in how much they could explain because they actually described dyads rather than full family systems. However, Haley developed an enduring interest in triads (which he later developed further in his writings about cross-generational coalitions and perverse triangles). This concept was influential in helping to bridge two of the more influential family systems approaches: strategic and structural family therapy (for greater historical perspective on this connection, see Nichols & Schwartz, 1998).

A third set of investigations of family dynamics in Schizophrenia that also had lasting impact in family thinking was that of the psychodynamic theorist Theodore Lidz and his colleagues (1957a, 1957b). Among Lidz's key contributions was his challenging of the blame burden that had historically been foisted exclusively upon mothers. Though he did not absolve mothers, Lidz did consider the potentially damaging role of fathering relationships in families, seeding a field now flourishing as

coparenting theory (McHale et al., 2002; McHale, 2007a). Focusing on what he termed role reciprocity, Lidz outlined ways in which parents of disturbed youth had failed to achieve mutually supportive, collaborative, and coordinated parenting stances within the family. In some cases, the problem was one of marital schism, whereby parents chronically undermined one another and competed for their children's loyalties. In other families, which he characterized as showing marital skew, one of the parents held inordinate sway in the family while the other passively gave way to the domineering parent. In Lidz's published cases it was typically the father who was the one who acceded, though the dynamic could also be flipped. In both kinds of family systems, children were caught in the middle, triangulated into the problematic interadult relations.

In many of these early initiatives, there was an implicit focus on nuclear family systems and problems that eventuated when power dynamics or family roles deviated from what were seen as optimal pathways. As will be detailed shortly, some of these presumptions have been critically evaluated by contemporary family scholars, including Hare-Mustin and Leupnitz. Yet even in advance of these critiques, not all theorists and clinicians were limiting the focus of their family conceptualizations and interventions to nuclear family units; in several approaches, present-day family dynamics were framed as temporal extensions of family lineages. These multigenerational approaches, in turn, laid important foundational groundwork for contemporary approaches that view family dynamics as embedded not just within extended and historical kin relations, but also within a broader network of systems that themselves impact the family's adaptations (Henggeler & Bourdin, 1990). One example of multigenerational thinking already mentioned is Bowen's work. In contrast with seemingly related psychodynamic approaches that had long maintained that people inherit, introject, and bring forward patterns from their historical past, Bowen's perspective on multigenerational transmission was that the past never left, and that families and family emotional systems press up against and interlock daily with prior generations. Friedman (1991) captured this position's essence well in observing that for Bowen, the intergenerational evolutionary flow had more power to format the structure of relationships than did the logic of their current connections. As should be apparent, true adoption of such a framework demands a thorough rethinking of the scope of work required to dislodge a family's symptomatic patterns and sustain newer forms of functioning.

Another seminal perspective on the historical embeddedness of contemporary child and family adjustment problems is Boszormeny-Nagy's contextual family therapy. As did Bowen, Nagy emphasized the continuing powerful influence of relations with families of origin, including parents' indebtedness to their origin families (innovatively cast as a ledger of owed balances and obligations). Many of these ideas were developed in Boszormeny-Nagy and Sparks's (1973) *Invisible Loyalties: Reciprocity in Intergenerational Family Therapy,* an influential work that illustrated the ways in which origin family dynamics reach into contemporary family processes even when parents are consciously unaware of or insist on having "disowned" them. Like Bowen, Nagy emphasized the unearthing of hidden ties and loyalties and the repair of unsolved problems. Unlike Bowen (who sent families home to do such work), Nagy gathered together various members of

the extended family network to address ruptured and strained relationships. The multigenerational perspective also helps to account for how it is that children in a contemporary family system come to be assigned (or to inherit) particular roles that reincarnate kindred roles that have existed in the family for generations. Attending only to the realities of the child's current-day nuclear family unit would not allow for detection of the source behind the historically entrenched family role the child has unwittingly stepped into and been expected to play.

Extended kin networks were also well attended to by the school of structural family therapy, though unlike the intergenerational approaches Salvador Minuchin's (1974) structural approach focused primarily on here-and-now family interaction and functioning. In the structural school of thought, the key core relational patterns in families are those revealed during family interactions. It would be inaccurate to imply that S. Minuchin did not give credence to the past. Rather, the structuralist perspective maintained that what is important about the past is reenacted in present transactions and is evident in current behavior. S. Minuchin, collaborating with colleagues Jay Haley (with whom he worked from 1967 to 1976), Braulio Montalvo, and Bernice Rossman, cultivated many of his seminal theses through work with low-income, inner-city families at the Child Guidance Clinic of Philadelphia; Haley later left to develop a distinctively different problem-solving therapy (see later discussion). Among the most influential core concepts contributed by structural family theory are those of family structure, boundaries, and subsystems, and in his therapeutic practice S. Minuchin worked to reconfigure nonadaptive family coalitions. He and his colleagues mapped family structure (the recurring and enduring patterns of interaction that had come to organize and structure daily family life) and then worked to foster change through techniques relying on the creation or amplification of challenges within here-and-now, ongoing family interactions. To be successful, they found it necessary to join and then work from within the family system, calling on action-oriented approaches as they worked directly with in-session enactments to help introduce, promote, and support change. By encouraging and supporting new patterns of behavior that introduced structural shifts in the family's hierarchical structure, alliance patterns, and boundary adherence, structural therapists sought to move families toward more adaptive functioning—with the identified child or adolescent client following suit. S. Minuchin and Haley each understood and respected family developmental issues as they worked to reconfigure nonadaptive family coalitions. The structural approach went further, however, to take stock of the family's entire relational field, including extended kin, other supportive figures, and helping professionals and institutions.

Perhaps most central to understanding S. Minuchin's theory are his views of hierarchy and boundaries. In structural theory, adaptive and healthy family systems are hierarchically organized, such that parents or parenting adults are clearly in charge as the system's executives. Families, as indicated earlier, can be composed of any number of different subsystems, including marital, coparenting, parent-child, and sibling subsystems. In well-functioning families, the adaptive emotional growth and development of children and other family members is contingent on the existence of appropriate boundaries between these subsystems. Conversely, when hierarchies break down, boundaries are violated, or children are triangulated

into adult-adult subsystems, symptomatic behavior is far more likely. S. Minuchin outlined a number of different ways in which children can be triangulated in family systems. In some cases, parents may compete for a child's loyalty and one may succeed in establishing a coalition with the child that excludes the other parent. In other cases, parents respond to distress in their relationship by deflecting their distress and anxiety to the needs or problems of a child. This solution provides short-term relief for the adults' relational distress and can artificially bond them together, as such detouring often results in psychosomatic or other emotional problems for the pathologized child. As did many other family theorists, S. Minuchin concurred that the identified patient's symptomatology played an important role in maintaining family homeostasis, keeping the family system together, and preventing it from fragmenting (S. Minuchin, Rosman, & Baker, 1978).

There has been a fair degree of clinical and research-based support for many of these basic suppositions (Andolfi, 1978; Buchanan, Maccoby, & Dornbusch, 1991; Camara & Resnick, 1989; Johnston, Campbell, & Mayes, 1985; S. Minuchin et al., 1978; S. Minuchin, Chamberlain, & Graubard, 1967; Rosenberg, 1978), and several integrative models have been derived in recent years using classic structural family therapy as their base. These include an attachment-based family therapy promulgated by Diamond, Siqueland, and Diamond (2003); a biobehavioral family model outlined by Wood, Klebba, and Miller (2000); a multidimensional family therapy propounded by Liddle (2000); and a multisystemic family therapy model, which explicitly takes into consideration other systems in which families are embedded, advanced by Henggeler, Schoenwald, Borduin, Rowland, and Cunningham (1998).

There have been other recent and major elaborations of practice promulgated by S. Minuchin's protégés at the Philadelphia Child and Family Therapy Training Center, expanding the basics of S. Minuchin's approach rather significantly. Earlier forms of structural family therapy emphasized a relatively rapid joining of the family by a strong and directive therapist who worked first to understand and accept the family's version of reality, and then intervened actively to alter here-and-now processes. By contrast, present-day practitioners often assess far beyond here-and-now family processes in an effort to more fully understand the family in its historical and larger community context and to take stock of potentially relevant individual biological, affective, and psychological processes of family members (Jones & Lindblad-Goldberg, 2002). They rely less on appealing to the therapist's expertise in establishing a hierarchical relationship with the family and fully engage families in planning and evaluation of assessment and treatment. They attend to the creation of relationships that will promote and nurture socioemotional competencies of family members, including emotion regulation and promotion of attachment.

Jones and Lindblad-Goldberg (2002) maintain that these shifts were natural ones as greater information than was available to S. Minuchin 30 years earlier came to the fore about the particular family and parenting processes that promote or constrain socioemotional development. They also point out that with the recent shifts in practice comes a more demanding accountability, from a focus on organizational functionality as the central treatment outcome to a focus on establishment of growth-promoting practices and strength of emotional connections. For the latter to succeed,

change will almost always be necessary at multiple system levels, including those beyond child and family at the child-family community interface. As Nichols and Schwartz (1998, p. 316) mused in commenting on the challenges of work with inner-city multiproblem families, "When between five to ten public agencies are involved in a family's life, and when the family is up against the crushing weight of poverty and racism, family therapy can seem quite puny. Recognizing limits has made therapists rethink their roles." We develop this systems-within-systems principle in the case example provided later in this chapter.

One other major effort from the early golden years of family therapy not yet discussed was Jay Haley's development of strategic family therapy. Following his work with the Palo Alto group and S. Minuchin, and influenced heavily by Milton Erickson's creative use of unorthodox therapeutic techniques and hypnosis to bring about rapid therapeutic change, Haley founded the Family Therapy Institute in 1976 and began offering brief problem-solving therapies. Recognizing that families in crisis sought rapid relief and seldom committed to participation in therapy for extended periods, Haley's time-compressed approach moved to initiate change in families not via insight and understanding but by the family carrying out directives prescribed by the therapist. Haley's goals were structural, and like S. Minuchin, he sought to address hierarchy and boundary problems that created and maintained problem sequences in families. Though his ideas owed in part to the cybernetic concept of positive feedback loops, Haley emphasized longer (and sometimes, much—month-long or more—longer) chains of sequence as well as chains that involved three, rather than two, people. Where he diverged from his earlier colleagues was in his calculated approach to effecting sequence and thereby system change, and in his view that family interactions could be understood as human struggles for control and power (Haley, 1976).

In strategic therapy, the therapist takes an authoritative and often manipulative stance designed to counter family resistance. Families are actively directed and maneuvered to undertake activities that amplify and exaggerate the troublesome symptoms that were actually serving as communication metaphors (see also Madanes, 1981, 1984). Strategic interventions are often subtle, usually covert, and frequently paradoxical (such as instructing families not to change and to exaggerate their disturbing symptoms). These kinds of interventions have been critiqued by many writers as both controversial and risky, not only because families inevitably deteriorated, at least temporarily, when following paradoxical directives, but also because the ultimate risk-benefit ratio and conditions for effective and safe use of paradoxical interventions were never empirically determined (Fruzzetti & Jacobson, 1991). Nonetheless, the strategic movement was one of the most active approaches on the scene during the second half of the twentieth century and gave rise to a number of creative variants (e.g., Selvini Palazzoli et al., 1978).

Prior to family therapy's embrace of multisystems thinking, most approaches had stayed squarely focused on family interiors, working with family systems as bounded entities unto themselves. As we've outlined, for some theorists this included the family's embeddedness in their multigenerational histories, and for others a lateral expansion to extended kin networks—but seldom did the views take stock of the broader organizational role of cultural attitudes and historical cohort.

This situation changed dramatically following searching critiques by Rachel Hare-Mustin (1978), Deborah Leupnitz (1988), and others who decried the gender bias inherent in the existing major schools of thought. For example, cybernetic theory, in repudiating unilateral control and power in systems (the components of which, instead, continually and circularly influenced one another in recurring feedback loops), implicitly absolved any particular system member of blame. When problems existed in a system, all of the system's components were involved and hence all could be said to bear equal responsibility. Such a posture, however, ignored the realities of the subjugation of women in a patriarchal society and effectively blamed victims in crimes against them. The implication that husbands and wives contributed equally to battering, for example, and bore equal responsibility for changing such problems reified cultural patriarchy.

Similarly, feminist writers drew attention to seldom examined historical realities driving one of the more common clinical presentations in two-parent families, in which mothers were cast as overinvolved and ineffectual and fathers as problematically underinvolved. Sadly, family therapy's seeming advance over psychodynamic theory in absolving mothers of primary blame for the child's ills (by factoring fathers into the family equation) had actually changed precious little. Mothers were still viewed negatively and pathologically, without recognition that they had been thrust by society into economically dependent, emotionally isolated, and hyperresponsible positions. Feminist therapists held that although a change of family circumstances was in order, it was not because incompetent mothers required help but because men's assumption of their responsibilities to their children would enable mothers to move out of the crazy-making situation that had been foisted upon them (McGoldrick, Anderson, & Walsh, 1989; Walters, Carter, Papp, & Silverstein, 1988).

In that systems approaches had introduced a revolutionary paradigm shift in the field of clinical psychology, it seems fitting that in closing this section we underscore how feminist theory in turn challenged existing systems beliefs. Feminist theory also had an impact on a final development highlighted here, which itself also deconstructed many of the unique expositions of earlier family theories. One critique of the early family systems work has been that its focus on cybernetics, systems, and patterns reflected a misguided search for grand theories that could explain human behavior, universal laws that could foster the understanding, diagnosis, repair, and resolution of most problems. This quest and its emphasis on the knowable and objectifiable were roundly criticized in postmodern critiques maintaining that there is no tangible reality, only the mental constructions of observers. Among the first theorists to broach this issue in family therapy were Paul Watzlawick (1984) and Lynn Hoffman (1985, 1988), who questioned the confidence and assuredness that had become such a central force in so many family therapies of the time. Collaborative approaches, such as the one introduced by Harlene Anderson (1993), followed soon thereafter, and took a position of *not* knowing in order to enable genuine conversations with families, and sticking firm to this stance rather than deftly shifting later into a position of expert. This approach, as did others, had its flaws, but it laid important groundwork for other critiques and new approaches, including narrative therapy, that were to follow.

In some ways, the essential tenets of narrative therapy threw into disarray all that family theories before it had promulgated. For decades, family therapy had stood in sharp contradistinction to psychodynamic approaches that viewed psychopathology as borne and bred within individuals; family therapy's basic tenet, as emphasized throughout this chapter, was that such problems were in interactions and relationships. But narrative theory eschewed this perspective as well, positing that problems were embedded in the stories that families lived by and that had come to organize their behavior. The focus shifted from the ways people behaved to the ways they constructed meaning. To move toward health, narrative approaches held, families needed to become aware of and discard these disempowering and self-defeating narratives they had been induced into, replacing them with alternate, health-promoting stories to govern their lives and future travels (Freedman & Combs, 1996; White, 1995). A major metaphor in such work was the externalization of problems: construing problems as forces threatening the family from the outside, rather than circulating within.

In many ways, narrative approaches incorporate very little of the systems thinking outlined throughout this chapter. Indeed, their social constructivist base deemphasizes dysfunctional family interactions and instead targets the malevolent influences of cultural biases and institutions. In so doing, narrative therapies provide a fresh, nonpathologizing view of families as capable and powerful. There is no question that their emphasis on collaboration and on expanding beyond families to frame families' problems infused creative new wisdom into the family field. It is less clear whether the accompanying wholesale rejection of systems theory by many narrative theorists was necessary. There remains relatively little evidence that any one system is inherently more or less helpful for aiding any particular problem, and there seem no inherent logical inconsistencies in retaining the valuable aspects of systems thinking while simultaneously attending to the humanizing, ethical, and empowering dimensions so celebrated in narrative frameworks (Eron & Lund, 1996). An ongoing challenge for those working in the field will be to evaluate when eclecticism in approach enhances work with families, and when it dilutes the power of particular approaches. Greater eclecticism and openness, greater collaboration with families, and refound respect for family competencies, worldviews, and inner wisdom have all enhanced modern practice, but in the end the approaches assumed by therapists are valuable only to the extent that they help effect real and meaningful change for families and family problems.

Family systems approaches and family therapy owe their place in the contemporary landscape of clinical psychology to the creativity and charismatic leadership of a relatively small group of brilliant individuals. This history has been well chronicled in a variety of sources, though this initial period of growth of the family therapy movement has been portrayed as a competitive and dogmatic one, guided by various leaders supported by groups of avid disciples emphasizing points of difference between the various approaches. As late as the 1980s, there had been relatively little integration across different approaches despite the similarities in many of the field's basic concepts. However, this circumstance has since changed. There has been a healthy and beneficial cross-pollination both across different schools of

thought within family systems approaches and in integrating important conceptualizations from other fields and disciplines, including developmental psychology and individual approaches to psychotherapy.

We have attempted to highlight some of the major areas of consensus among the various approaches. The majority of contemporary approaches view and treat families as systems, attend to forces that promote stability of behavior patterns and that delimit change, emphasize the triadic nature of human relationships while also recognizing the multiple systems in which families are embedded, and appreciate in one form or another the protective function of healthy boundaries not just for individuals and subsystems, but also for the family as a cohesive entity traveling through time together. Most schools concur that family groups function best when they are cohesive; freely, openly, and directly exchange information; and attend to the different developmental needs of different family members at changing points in the family life cycle. In addition, most perspectives view families as adaptive to the extent that they can show flexibility to adapt to shifting life circumstances, solve problems effectively, include adequate hierarchical structure, and support the individual autonomy and growth of all family members.

Beyond these similarities, there remain a variety of important differences in emphasis and scope. These include whether the proper focus of conceptualization and change efforts needs to be primarily on behavior and interactions, or whether it should take account of private inner experience of family members; the extent to which change efforts are enhanced by taking stock of individual character and personality dysfunction; and the extent to which family dynamics are seen as emanating from inner turmoil as opposed to being poor adaptations to the presses of an oppressive external world and culture. There are still disputes about whether the path to health will be charted through changes in behavior, action, and interaction or whether insight is required, and whether all family members must be a part of effective family solutions or whether it is possible to affect and sustain meaningful systemic change through work with individuals and subsystems. These things said, contemporary therapies rarely adhere tenaciously to theoretical dogma, and most now benefit from the convergence of ideas and rapprochement that began during the last decade of the twentieth century.

In an era of intense scrutiny on all forms of mental health intervention, cast in the name of accountability, family systems–based approaches to mental health and treatment have been called on to prove their merit to a widening external audience. In 1998, summarizing the current state of the field, Nichols and Schwartz anticipated this emerging challenge for family scholars and practitioners as they wrote "If family problems cause psychological problems, we should be able to demonstrate it; and perhaps more important, if we say we can resolve problems with family therapy, we'll probably have to prove it" (p. 498). The burgeoning field of family psychology, outlined later in the chapter, has largely shouldered the first part of this dual charge, while assumption of the second charge has also been formally taken on, though by a relatively smaller group of active researchers whose work has largely been informing recent national policy. These developments are summarized in the next section.

RESEARCH

If not already clear, from the days of the Palo Alto group's scholarly pursuit of communication theory, research has always been central to the understanding of family systems. Over the past quarter century, however, an interesting split of sorts has occurred. The growth of a family research field has been flourishing as empirical studies of families have benefited from advances in observational and recording capacities, digitalization, statistical modeling, and other scientific technologies and breakthroughs. Concurrently, a field bearing the name of family psychology has taken firm root. It is broad in scope and has been formally recognized as a new specialty within contemporary psychology. It is represented by a flagship journal, the *Journal of Family Psychology*, which debuted in 1986 and joined *Family Process* (an outlet that has been around since 1962, when Jay Haley served as its first editor) as the major source of breaking knowledge for professionals about family systems. Yet much of the scholarly work conducted under the flag of family psychology remains unfamiliar to active family practitioners.

The lack of dialogue is unfortunate, if understandable. James Alexander, who is among the small group of family psychologists whose work is well known to national policymakers, invoked Albert Einstein in a recent publication: "Not everything that can be counted counts, and not everything that counts can be counted" (Alexander et al., 2002, p. 17). Einstein's observation captures a rather interesting paradox in the family field: On the one hand, early family therapists took pains to insist that problem sequences in families are observable, identifiable, and deeply meaningful, and yet family therapy thinking has been replete with notions that would seem to defy objectification and standard measurement. Concepts such as triangulation, boundaries, hierarchy, differentiation, legacies, family projection processes, undifferentiated ego masses, rubber fences, and the like would seem to defy objective measurement. There have been both prevailing sentiments that any reductionist objectification of such inherently fluid family processes strips them of any true meaning, and contentions that empirical research is of little creative or additive value as it typically does little more than confirm established knowledge. Perhaps of greater concern, there is the fear that in objectifying complex structures and processes, researchers run the risk of obscuring and overlooking alternative interpretive frames. Indeed, from the postmodern perspective, everything is a construction and there is no objectifiable truth.

The discussion of operationalization and standardization has sometimes become most heated and strained when the topic shifts from the valid assessment of family processes to the assessment and efficacy of family therapies. Putting aside for the moment whether it is possible to evaluate important family group processes in meaningful ways (family psychologists have made a number of important strides in this arena, summarized later), a variety of writers have advanced the position that manualized treatments, insistence on adherence to treatment protocols, and evaluation of success in terms of statistical effect sizes undermines the creativity and fluidity that have been the hallmark of family therapy since its inception (Henry, 1998). This is certainly an extreme position, and not all practitioners share this concern that the empirical movement and move to more standardized treatment

regimens has a greater potential for harm than good (see, e.g., Jones & Lindblad-Goldberg, 2002).

Addressing this tension, Nichols and Schwartz (1998, p. 337) pointedly note that the research efforts of some of the major centers studying multiproblem families, including Henggeler's and Alexander's groups, Gerald Patterson's and Marion Forgatch's studies at the Oregon Social Learning Center, and major initiatives administered by Tolan and Gorman-Smith in Chicago and by Liddle and Szapocznik in Miami "provide credibility for family therapy within the larger mental health field, credibility that has been strained by the eccentric practices and unsubstantiated outcome claims" of earlier family models. Further, questioning the necessity and wisdom of casting blueprints as straitjackets, certain writers have decried the characteristically minimal dialectic between practicing family therapists and family therapy research groups. Nichols and Schwartz, for example, highlight both clinicians' ethical mandate to evaluate the efficacy of their interventions and the modern-day reality that family therapists must demonstrate the efficacy of their approaches to third-party payers. Echoing this latter point, Alexander and his colleagues (2002) noted that family-based approaches have been oddly absent from best practice lists generated by psychology's empirical treatments movement. They point out that this absence is out of keeping with accumulating empirical evidence for the effectiveness of family psychology interventions, evidence that has been gathered in research studies that have for the most part managed to sidestep the major criticisms typically levied against most other randomized clinical trial studies.

A comprehensive review of the efficacy literature generated by family psychology intervention scientists is beyond the scope of this chapter. Interested readers are referred to a comprehensive and readable volume published by the American Psychological Association, *Family Psychology: Science-Based Interventions*, edited by Liddle, Santisteban, Levant, and Bray (2002). It provides scholarly reviews and summaries of advances in family intervention research, including reviews of programs designed to enhance positive parenting practices through parent management training procedures, and other programs targeting parents' capacities for assuming responsibility (providing greater structure and closer monitoring), expressing warmth, and decreasing conflict and use of damaging disciplinary practices (e.g., Forgatch & Knutson, 2002; Liddle, 1995; Schoenwald & Henggeler, 2002; Szapocznik & Kurtines, 1989). Although initial work for some of these models was carried out in demonstration projects conducted in university settings, where conditions could be tightly controlled and titrated (e.g., Henggeler et al., 1986), there have since been concentrated efforts to render standardized family-based approaches within existing naturalistic contexts (homes, schools, community agencies, residential treatment facilities). An example of the more ambitious and intense work in this regard is multisystemic therapy provided within a family preservation model, introduced as a desirable alternative to outside-of-home placements of severely disturbed youth (Fraser, Nelson, & Rivard, 1997). In such work, outlined by Schoenwald and Henggeler (2002), anywhere from 2 to 15 hours of service per week is devoted to families over a concentrated 4- to 6-month period, with clinicians carrying low caseloads (4 to 6 families per clinician) but on call 24 hours 7 days a week.

Overall, empirically validated treatment approaches are informed by a systems perspective and by the contextual tradition of family therapy, but also focus broadly on behavioral, cognitive, and emotional transactions within the family and between the family and the influential social systems in which it resides. Such approaches generally see the focus of classic, pragmatic family therapies on intrafamilial interactions as insufficient to achieve and sustain meaningful gains with multiproblem youth. Most attend in one way or another to how children's intra-individual qualities (genetics, temperament) can challenge parenting systems and how social disadvantage, neighborhood, schools, and other contextual factors press on and interact with family processes. Many of these initiatives emphasize prevention and early identification of risk, and most are attentive to both cultural and ethic differences. Approaches vary a good deal with respect to how unabashedly they cast therapists into a role of expert educator, but all aim to affect some form of hierarchical change wherein parents assume greater responsibility for oversight and discipline of children while working to help promote greater family cohesion. Children themselves are often explicit targets of skill building as well, beyond the parenting and family work.

The other major focus of the family psychology field has been in providing evidence for ways families contribute in formative ways to the individual problems of their members. Such work often takes on a more decidedly linear focus than that endemic to cybernetic and circular perspectives, though researchers are often aware of this bias and episodically devote specific and concentrated energies to pursuing bidirectional, if not transactional, hypotheses. Again, a comprehensive review of this body of work is far beyond the scope of this chapter, but interested readers are referred to relevant overviews and contributions by P. Minuchin (1985), Cox and Paley (2003), McHale and Grolnick (2002), and Bornstein and Sawyer (2006). A special edition of the journal *Development and Psychopathology* (Davies & Cicchetti, 2004) includes a broad array of empirical papers presenting work from leading researchers involved in this area.

Much research in this tradition has been concerned with establishing and clarifying the nature of dynamic linkages between different family subsystems. Investigations of couple relationships, for example, reliably tie relationship functioning to both parenting behavior and child adjustment (e.g., Hetherington, 2006; Kaczynski, Lindahl, & Malik, 2006), as would be predicted by most theoretical frameworks in the family therapy field. Studies typically document child adjustment problems in the face of destructive marital conflict (Cummings & Davies, 1994), parenting disturbances in the face of marital distress (Erel & Burman, 1995), and child disturbances as a consequence of insecure parent-child attachments (Arend, Gove, & Sroufe, 1979; Speltz, DeKlyen, & Greenberg, 1999; Vondra, Shaw, & Swearingen, 2001; Weiss & Seed, 2002). Studies are often quite explicit in outlining the specific nature of dynamics in families, as when mothers in marriages devoid of intimacy are at greater risk for boundary violations and seductive behavior with children (Jacobvitz, Hazen, & Curran, 2004). They substantiate the cross-generational transmission of relationship patterns, as when parents who deny the significance of attachment-related experiences in their own origin families parent their infants in such a way that the infants likewise end up developing avoidant patterns of

attachment toward them (Steele, Steele, & Fonagy, 1996; Ward & Carlson, 1995; Zeanah, Benoit, & Barton, 1993). Such research studies are also attentive to context, as when outlining how changes in circumstances within or outside the family can alter risk trajectories (Egeland, Jacobvitz, & Sroufe, 1988).

Beginning in the mid-1990s, a new line of empirical investigations began appearing in the child and family development literature, explicitly assessing family group-level dynamics and the relationship of these whole family processes to children's adaptation (e.g., McHale & Cowan, 1996; McHale & Fivaz-Depeursinge, 1999). In particular, this work began paying explicit attention to specific coparenting dynamics in nuclear family systems, including hostility and competitiveness, alliance cohesion and support, and disconnection (Feinberg, 2003; McHale et al., 2002). A parallel line of inquiry also published during this time, fully systemic in nature, examined the development of family alliances from the prenatal period on through the toddler and preschool years (Favez et al., in press; Fivaz-Depeursinge & Corboz-Warnery, 1999). Among the most significant findings from these complementary lines of work are indications that early coparenting dynamics and family alliances can be forecast on the basis of both intrapsychic and interpersonal factors assessed during the pregnancy (McHale et al., 2004; von Klitzing, Simoni, & Burgin, 1999); coparenting processes are related to but also exert influences in the family that are independent of marital relationships and dynamics (Bearss & Eyberg, 1998; Katz & Low, 2004; McHale, 1995; Schoppe-Sullivan, Mangelsdorf, Frosch, & McHale, 2004); family alliance types consolidate early, are markedly stable through time, and affect children's social and emotional development in a host of important ways, even as early as the infant, toddler, and preschool years (Belsky, Putnam, & Crnic, 1996; Caldera & Lindsey, 2006; Fivaz-Depeursinge, Frascarolo, & Corboz-Warnery, 1996; Frosch, Mangelsdorf, & McHale, 2000; McHale, 2007b; McHale & Rasmussen, 1998); and family dynamics are sensitive to infant contributions from the very earliest months (Fivaz-Depeursinge, Favez, Lavanchy, de Noni, & Frascarolo, 2005; Fivaz-Depeursinge & Favez, 2006; McHale & Rotman, 2007). Related work has substantiated the protective effect of functional family hierarchies (Shaw, Criss, & Schonberg, 2004), and conversely, the destructive effects of family triangulation (Kerig, 1995).

These various lines of inquiry each, in their own way, have provided impetus for one form of intervention or another targeting family factors shown to increase risk. For example, if couple relationship problems before a baby's arrival are shown to foreshadow parenting and relationship problems afterward, then strengthening the couple relationship might be expected to help promote early parenting and couple adjustment (C. P. Cowan & Cowan, 1992; Shapiro & Gottman, 2005; Silliman, Stanley, Coffin, Markman, & Jordan, 2002). If maternal insensitivity places infants at risk for cultivating insecure attachments, interactive guidance and other forms of mother-infant psychotherapy targeting mothers' ability to read and respond warmly, contingently, and accurately to infants' signals may help mitigate such risk (McDonough, 1993; Robert-Tissot, Cramer, & Stern, 1996). If such reading is impeded or overwhelmed by maternal preoccupation with unresolved origin family issues, maternal representations may serve as an additional target for such interventions (Cramer, 1995; Fraiberg, 1980).

An interesting question from a systems perspective, of course, is whether long-term gains can be sustained if a single dyad in the family, be it husband-wife, mother-infant, or (as in several recent initiatives targeting father involvement; e.g., McBride & Mills, 1993) father-infant, is prioritized without parallel consideration of the ripple effects such interventions may cause throughout the family system (McHale, 2007a). Herein lies one crucial and ongoing challenge for systemically informed interventionists and clinical researchers of all sorts: establishing the extent to which an intervention with an individual or a dyad is sufficient to change a family's course, even if not all family members are the focus of intervention. Though the possibility of affecting meaningful systemic change without engaging all key players within the family would seem antithetical to most family systems theories, dyadic interventions are actually standard practice in the infant mental health field, as well as being an unfortunate reality of most community-based practice (P. A. Cowan, Cowan, Cohen, Pruett, & Pruett, in press). Hence, both basic and research efforts will continue to be crucial in helping to document the extent to which interventions seeking to effect enduring change show clinically meaningful results even without attending to all important relationships and networks in case formulations and intervention.

The vigorous dialogue about such issues that is active in family research circles exemplifies a level of inquiry and discourse healthy for all systems. For researchers, there is a never-ending revisiting and reworking of existing frameworks contingent on new data, even as certain bulwarks of the field (organizing effects of security of attachment, multirisk likelihood attendant to destructive marital conflict, and protective facets of supportive coparenting relations) provide necessary coherence. When there are episodic moves toward premature closure—some of the most vocal critiques of the movement toward empirically validated treatments have been along this line—there are corrective responses from within the research community. Episodically, someone can be counted on to stir the pot vigorously and threaten deconstruction of all held dear, as did Belsky, Campbell, Conn, and Moore, in one 1996 report failing to uncover significant stability in attachment security, either at the level of avoidance, security, and ambivalence (Mary Ainsworth's standard ABC classification rubric) or secure-insecure classifications. A more recent example involved Cook and Kenny's (2006) revisiting of data from a 1989 study by Cole and Jordan. Cole and Jordan had adapted Olson and colleagues' Cohesion and Adaptability scales on the Family Adaptability and Cohesion Evaluation Scale (FACES) instrument so that the scales' individual items reflected dyad-specific (mother-child, mother-father, father-child) rather than family-level processes. They had then asked each of the three family members to rate each of the three subsystems. Reanalyzing Cole and Jordan's data using advanced statistical analyses (EQS; Bentler, 1993), Cook and Kenny found that systematic variance at dyad subsystem levels was *rarely* significant after they took into account variance that was attributable to individuals within the family (mothers, fathers, or children). Taking the argument to its conceptual level, they speculated that in some circumstances what appears to be a systemic pattern can be as or more parsimoniously attributed to a single individual's influence, as when a father who does not like to be involved with people in general has difficulty being engaged with both his spouse and his children. Such a

problem, which when framed systemically would underscore a dialectic between dyadic (marital and parent-child subsystems), could actually be traced most fundamentally to the father's personality and habits. Any empirical finding like this is clearly delimited by the fragilities of instrumentation, yet such cyclical rekindling of old debates helps keep the field and its pursuits fresh, self-evaluative, and noncomplacent.

Perhaps somewhat surprisingly, despite the many, many volumes published on measurement issues in family research (e.g., Hofferth & Casper, 2006), assessments and evaluations of families in clinical practice seldom rely on instruments developed in laboratory research. In fact, most family assessments and evaluations do not rely on standardized instruments at all. In the next section, we summarize common practices and issues in family assessment and evaluation.

ASSESSMENT AND EVALUATION

As may not be surprising given the vast array of different approaches and areas of emphasis represented by different schools of thought, no standard or universally agreed upon set of assessment practices or techniques for evaluating families has ever emerged. Many students trained in family therapy approaches are introduced to some of the innovative ways of gathering information about families, such as completing detailed genograms (McGoldrick & Gerson, 1999)—sometimes indispensable in multigenerational formulations and interventions—and some are introduced to structured checklists or inventories (which help to organize and structure behavioral therapies). But most family therapists do not define formal assessment phases during which they systematically partake in standard assessment protocols or procedures. This is certainly not to imply that assessments are not conducted or are thought of as unimportant, but only that a cataloguing of different assessment tools in this section would badly misrepresent this facet of clinical work with families. Rather, we summarize here some of the major issues attended to by most systems therapists in the early stages of working with families.

Besides explicitly seeking information and clarity during family interviews about certain critical life events (e.g., domestic violence, sexual abuse, extramarital affairs, drug and alcohol abuse), most family systems approaches give special credence to the behavioral sequences and interactions that are revealed during early contacts with families. Such interactions are thought to reveal much of what is necessary to know about family communication patterns, boundaries, rules, and hierarchies. What family members do during assessments is as important as and sometimes more important than what they say, and hence very few family assessments take place without some direct observation of the family's interaction patterns. How early assessments are conducted, how directive therapists are during initial evaluations, and what the goals of early contacts are do differ from school to school, though there are some commonalities. First, most family therapists take pains during the early stages of work with families to listen to and take stock of the family's account of the presenting problem. During this stage, each family member is usually allowed a voice, and therapists acknowledge each person's perspective on the problem. This process is usually navigated in as open-ended a way as is

possible before additional detail is sought. The additional detail, when solicited, is geared to identify important details and to contextualize the problem with respect to time, place, and key players involved (including important individuals and systems outside of the immediate family). Information is also obtained about how the family has already tried to deal with the problem.

On the basis of these early assessments, interventionists are able to draw some preliminary hypotheses about structural and communication problems in families. From both session content and from observation of interaction sequences, the family hierarchy and the roles of different family members in the family's dynamics begin to become apparent. From observed family process patterns and sequences, the therapist is able to speculate about triangles, coalitions, and problematic boundaries within the system. This is particularly so when all of those principally involved in the maintenance of the problematic patterns are in attendance; understanding of family dynamics and communication patterns in families often shifts significantly when a sibling or live-in grandparent who had been absent during earlier family sessions subsequently attends a session and a different sequence pattern or alliance structure is revealed. Ultimately, however, an important aim for most therapies, not just structural approaches, is to develop an understanding of the invisible structures maintaining the referral problem.

Most therapies proceed from a general appreciation of the family life cycle and of normal family process. However, such understandings are often very global in scope and not always fully informed by contemporary developmental and family data. Most clinicians appreciate that the transition to new parenthood brings expectable strains that test the husband-wife dyad and increase risk for disengagement and/or depression by new mothers or fathers; that the early coparenting dynamic established in families often must reorganize as children move from infancy through the toddler and preschool years, since the behavioral challenges presented by mobile, verbal children introduce fresh new strain into the family system; and that adolescence requires parents to negotiate new relationships with their children to allow for greater autonomy and choice, without pulling the rug out from under them or affording too much freedom and not enough guidance. Less well understood are how normative developmental transitions are shaded by cultural expectations or by the origin family expectancies of the child's parents. For example, unlike most European and African American family systems, cosleeping and hand-feeding well through the preschool years is normative in many Asian heritage families. This cultural understanding is necessary to learn about during clinical assessments of Asian families, lest the family's parenting practices be cast as emotional overinvolvement. On a different front, cultural views of sexuality during early adolescence are also important to assess, as such views can be dramatically different in families where both parents became sexually active during their own early teens than they are in families where parents' sexual activity did not bloom until early adulthood. In research labs, developmental studies of family systems have uncovered revolutionary new data suggesting that even infants as young as 3 to 4 months are recruited into and become capable of actively shaping coparenting and family process (Fivaz-Depeursinge & Favez, 2006). Although very few families of infants this young come to the attention of family therapists because of ostensible problems in the

infant, assessing a family system without including infants would be as incomplete as omitting key members from assessments of families of older children.

Beyond the assessment of families' presenting problems and the patterns sustaining them, and the cultivation of an understanding of family adaptation apropos their position in the family life cycle (informed by relevant cultural and subcultural shading), a centrally important assessment task in most modern family therapy approaches is identifying contributory roles played by systems outside the family. Although this is most acutely so in assessments completed with multiproblem families (Henggeler et al., 1998), external systems do come into play for the majority of families with children and adolescents. However, though most assessments take stock of the family's interface with other systems, the extent to which such information is formally solicited during the early phase of family evaluations differs greatly from therapy to therapy. Typically, information about significant others in the family would be explicitly pursued during questioning in the earliest contacts with families, whereas information about outside systems (schools, the mental health system, the legal system, other social agencies) would be gathered as indicated as case material was shared by the family. More than occasionally, especially in cases where there is significant involvement by systems outside the family, some contact with these systems will ultimately be indicated, though in most approaches such contacts are more likely to take place later in the course of treatment than they are during the initial family assessment phase.

Finally, it is important to emphasize that although most family therapies do not systematically evaluate the psychological functioning of each individual family member, neither do they ignore developmental problems or psychopathology in individuals when it is present. Over the long history of the family field, there have been times when such individual problems were downplayed or cast as secondary and relatively inconsequential to the more significant relationship problems that entrapped the family, though this is seldom the case any more. As outlined earlier, therapy approaches now do consider the extent to which individual biological, affective, and psychological processes of family members are relevant (Jones & Lindblad-Goldberg, 2002), as such an understanding can help, rather than obfuscate, reasons why certain families get stuck.

To illustrate some of the major concepts described in the chapter, we now turn to a case conceptualized and treated from a family systems approach. In so doing, we underscore how systemic approaches help guide both the formulation of the problem and the path toward intervention chosen in a manner that might diverge substantially from courses charted by other major theoretical approaches in this volume.

Case Illustration

The case we have chosen to illustrate family systems intervention is one involving work with the Jackson family, a married heterosexual couple and their 14-year-old son, John, the family's identified patient. John's parents describe him as having shown precipitous deterioration in many areas of functioning, citing school failure (tardies, cuts, failing grades, warnings from administration), marijuana and

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alcohol use, and symptoms of depression (sleep disturbance, lethargy, lack of motivation, constriction of activities). For his part, John expresses anger and rejection of his mother and an alignment with his father in the marital breakdown presently occurring. John recently engaged in individual psychotherapy, though he also "forgot" several appointments. His parents were alarmed at the downward spiral in John's behavior and requested a psychiatric evaluation for medication.

John's father works as a business executive, and his mother has been a homemaker throughout the marriage. The father travels frequently and states that work has always been his primary involvement. Both parents agree that he is a good provider and concur that he has had only peripheral involvement in the domestic life of the family. Indeed, the coparenting partnership could be characterized as disconnected and skewed since John's birth. John's father took on very little of the hands-on, day-to-day work of parenting and almost always deferred to John's mother when child-related decisions had to be made. As individuals, the parents' personalities are also distinctively different. John's father impresses as low-key, intellectual, introverted, and conflict avoidant; when conflict escalates between him and his wife and/or between his wife and John, his response is typically to withdraw to his home office. By contrast, John's mother wears her emotions on her sleeve. She is extraverted and socially skilled, yet often impresses as dependent in her interpersonal style. The couple's long-term marriage (the first for both parents) became progressively less intimate over the years as the mother invested most of her attention in her parenting involvement with John, while the father invested in his successful career. Three years ago, the mother began an affair with a man in the community. She became increasingly preoccupied with this involvement, and as she did her parenting deteriorated. She became indulgent and less attuned and attentive to John's needs, and not surprisingly conflict between mother and son also escalated.

Presenting Complaints

The initial telephone contact was made by the father, who wanted to come in with his son to work on his son's acting-out at home. After some inquiry as to the family situation, the family therapist convinced the father that John's mother also needed to be included in the intake session. In the initial session, where all three family members were seen, the father and son sat close together on the therapist's couch. They exchanged frequent glances throughout the session, and on a few occasions finished one another's sentences. They both also engaged in verbal attacks of the mother, who remained quiet, self-recriminatory, and tearful during the session. The father reported that he had learned of his wife's affair only recently from a friend and had immediately confronted his wife, who confirmed its 3-year duration. The father described being furious, and announced his plans to separate and divorce. He had told his wife that she could not continue to live in the family's home, and she obliged by moving to a nearby apartment. The father said he has been torn because he cannot himself assume parental responsibility for John, given his work demands, whereas John's mother, whom John won't talk to, remains completely available. The father is unequivocal in asserting that he intends to separate from his wife, but he also understands that the son and his mother need to have a relationship. He acknowledges that he has discussed the affair with John and was appalled to have discovered that John had been colluding with his mother in keeping the affair a secret. This complicity has added to John's and John's father's shared rage and contempt for the mother.

During the session, the son reported, "My life used to be great, but now it sucks. . . . I hate my mother. She had an affair with this scumbag and she made me keep it a secret from my dad for 3 years. She used to bad-mouth my dad, but now I know she was lying. I can't trust anything she says. I wish my dad would divorce her so I wouldn't have to deal with her ever again." John is vehement in his insistence that he has no desire for contact with his mother.

John's mother expresses only remorse and seems accommodating of anything that might enable John's father to reconsider staying married. She states that she now recognizes that the main reason she pursued an affair was because of the increasing loneliness and frustration she felt in her attempts to rekindle intimacy with her husband. She says that she agreed to move out of the family home voluntarily (though she hopes it is temporary) and notes that since she moved out, John's father has made some changes so as to become more available to parent their son. She adds, however, that her husband is also calling on his extended family and friends to provide coverage when he is unavailable and laments that these individuals have also been quite disparaging of her to their son. She expresses concern that John has been unsupervised for the most part after school and reports that he has regularly been having friends over to the house to party. Currently, the mother comes to the home once or twice a week, at the request of the father (and over the objection of their son) to drive the son to school and to cook meals. She seems desperate to persuade her husband to consider reconciliation, fearing that separating will only add further stress to the son.

Both parents are extremely concerned about the son's functioning and agree that addressing his deterioration is the priority over sorting out their marital issues. Because the father seems resolute in his intent to separate and divorce, the stance the family therapist elects to take in closing the intake session is that the structure of their family needs to transition to more independent parenting of the son, consistent with a postseparation shared custody arrangement. The mother was clearly devastated by the therapist's recommendation and ongoing work was necessary to help her acknowledge that this was the only viable structure that would permit the couple to quickly begin addressing their son's significant dysfunction.

Family System Assessment

In keeping with contemporary structural family frameworks, John's current symptoms were presumed to derive from the mutual influences of both individual and family system factors, as well as factors external to the family system. On an individual level, John's adjustment was hampered by several individual vulnerabilities (including the attentional deficit, the current substance abuse, and the growing depression). Each of these factors would require ongoing assessment and monitoring because John's ability to respond positively to structural systemic interventions could potentially be affected adversely by these individual-level factors. For this reason, coordination with an effective individual psychotherapy and perhaps, at some phase of the treatment, a referral for psychiatric assessment may be indicated and pursued.

With respect to the context in which John's symptoms are developing, the marital subsystem had clearly deteriorated, perhaps irrevocably. The affair, triangulating a third party into the marital relationship, compromised the marital

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subsystem further, with the breach of trust, humiliation, and subsequent anger and contempt prompting the husband's unilateral decision to sever the marital partnership. Equally, the parental subsystem had been markedly compromised; the previous functionality of the mother's primary parental role, which had gradually deteriorated as she began neglecting domestic matters, took a further downward turn once the affair came to light because John began actively resisting her efforts to parent. While John looked to be intensely aligned with his father, the father remained substantially unavailable and was lacking in necessary parenting skills. Moreover, given the destructive conflict in the marital subsystem, the already disconnected coparenting subsystem became compromised even further.

The uncertain status of the Jackson's nuclear family given the current separation placed their family system in a family life cycle limbo. That is, it was far from clear whether the therapeutic aim should be working to restore greater functionality in the nuclear family system or working to restructure the Jacksons as a binuclear family (which would be the structural shift that would have to occur with a separation and divorce). The coparenting team's executive authority in structuring the family vis-à-vis the son had been undermined by multiple factors, including the mother's reliance on John as a confidant and the alignment John presently had with the father against John's mother. Both of these developments had disrupted the functional hierarchy in which the coparental subsystem trumped the parent-child dyadic subsystems.

Problems in the current coparenting alliance were numerous, dominated by the lack of parenting support between partners, leaving each to fly solo. For this family, the absence of a functional coparenting alliance was especially disastrous, given the father's historical absenteeism and lack of competence when he was in charge and John's newfound unwillingness to allow his mother to parent him. Successful coparenting was further compromised by ineffective child-focused communication and decision making, stemming from the adults' self-focus, hostility, and open conflict in the marriage. John's escalating symptomatology only placed added stress and pressure on this already compromised coparental subsystem. The lack of boundaries and structure (agreements and coordination about when each parent is to be parenting, if and when John's mother should and could be in the family home, who was responsible for coordinating with John's school, therapist, tutor) resulted in chaos, conflict, and further neglect of John's now considerable needs. From a family systems perspective, John's increasing symptomatology is readily cast as a response to the untenable role demanded of him in keeping his mother's affair a secret, and as a cry for help for a chaotic and disorganized family system.

This understanding of the family interior is insufficient, however. Effective family systems approaches also attend to systems outside the family currently (or anticipated to be) affecting the family system. For the Jackson family, the family therapist identified several such systems. First, there was the family's mental health care, specifically, an individual therapist whom John had been seeing. While consultation with outside therapists is often indicated in family work when such individuals are involved, it turned out to be especially pivotal in this case. There were some indications that the individual therapist may have unwittingly introduced some additional, complicating boundary problems that affected the family. By seeing John and his father conjointly but failing to involve John's mother at all in the treatment, she appeared to have unintentionally gotten entangled in the couple's marital conflict and reinforced John's alignment with his father. In their

conjoint meetings, it turned out, father and son had frequently raged to the therapist about John's mother, even as the mother's phone calls to the therapist had sometimes gone unanswered for as much as a week or more, by mother's report.

To assess the role being played by the mental health system, the family therapist obtained a waiver of confidentiality to connect with John's individual therapist. Such contact was in order both to facilitate assessment of the individual therapist's perspectives and goals and to permit the family therapist to deepen her understanding of the family system so that the interventionists could try to coordinate a treatment plan together that would be working toward parallel, appropriate goals. Similarly, given John's school problems it was important to initiate contact with a relevant liaison at the school. This was essential both to better understand the nature of John's current problems at school and to permit creation of a functional coordination of home-school support for him. Finally, given the strong inkling that John's father was preparing to initiate the process of divorce, potential legal implications had to be considered so that any future involvement with the legal system could circumvent the parents' involvement in an adversarial process certain to exacerbate the family's already high level of conflict.

Case Conceptualization

For over a decade, the Jackson family had sustained a reasonably stable and functional family adaptation, although it was one characterized by a compartmentalized and skewed parenting partnership in which John's mother carried out the active parenting (cementing a strong mother-son dyadic subsystem) as the father assumed an economic provider role (functioning only marginally in both marital and father-son dyadic subsystems). This once stable structure had gradually deteriorated over time, as avoidance and alienation in the marriage ultimately precipitated the mother's undertaking of an extramarital affair. As the mother's long-standing availability and quality of parenting the son deteriorated to the point of neglect, conflict increased between the son and his mother, while the father continued to avoid involvement. Keeping the affair secret from the father (seen by many family therapists as exemplifying a pathological coalition between mother and son) had infused additional toxicity into this triadic family system. The Jackson family's dynamics are a vivid example of family subsystems mutually influencing one another, both in contributing to the escalating dysfunction of the system and in calling out for an intervention that attended to each subsystem.

John's symptoms were both a functional cry for help for his family system and a reflection of marital and parental subsystem breakdowns. The family's hierarchy and many of its boundaries (mother-son, marital-parental) had disintegrated and now needed considerable support to restructure. The limbo state of the separation and divorce decision complicated matters by impairing the Jacksons' ability to functionally respond to the family life cycle crisis. The possibility of a high-conflict custody dispute necessitated structuring any intervention to protect the family and treatment process from being cast into the legal adversarial system. The professional systems outside the family were not adequately supporting the family. Coordination was poor (as there was no functional connection with the son's school), and the son's therapist was aligned with the father.

Every individual member and every dyadic relationship subsystem in this family was showing a significant need for intervention, and the deleterious impact of each subsystem on the other (affected through the family's mutual feedback

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loops) likewise needed to be interrupted. Given the multiple problems, a clear prioritization of need was required, and the therapist in this case chose to work from a directive, structurally based approach. The approach involved joining with the family quickly and building a collaborative working alliance by utilizing the parents' mutual concern for their son's welfare.

Complications

A number of complications became apparent only as the case progressed, introducing difficult decisions about how best to intervene. First, as already indicated, consultation with John's individual child therapist confirmed that her role in helping John process the emotional trauma of the family breakdown had been severely compromised by her unwitting but problematic entanglement in the marital conflict. Given the complications this triangulation introduced, the family therapist and John's therapist needed to process whether it would be in the Jacksons' best interests for the individual therapist to withdraw from the case and support John's engagement with a new individual therapist who specialized in high-conflict coparenting cases.

Second, the seeming inevitability of the family's move into a divorce transition led the therapist to incorporate proactive structural and psychoeducational interventions into the work. One focus of this work was on preempting new adversarial processes, inherent in the legal system, so as to keep them from infiltrating and further escalating the marital conflict and endangering the family and its members. From the outset, the family therapist also had to decide whether to advocate reconstitution of the nuclear family structure (which was the mother's clear desire) or to assist the family in restructuring to a binuclear family. Because the creation of a clear and functional structure would be the key to restoring adaptive functioning in this family, the family therapist decided to strongly advocate for disengaging the marital and parenting dyads. This stance, which prompted initial distress and resistance from John's mother, ultimately helped to enable the work of differentiating marital issues (which in a binuclear structure were not relevant targets for the therapy, except as they interfered with functional coparenting) from coparenting issues (which became the agreed-upon focus in helping the family to move toward greater functionality).

Finally, given the severity of the son's current dysfunction and the family's seeming inability to cooperate adaptively, the family therapist posed the possibility that the son may need to be placed outside the home. Individual factors such as the boy's substance use and the possibility of his school failure or dropout, complicated by the inadequate functional parental involvement and supervision, questioned whether continued in-home placement for the son would be a viable option. The out-of-home placement recommendation had not been considered by the parents, who were both resistant to the idea for multiple reasons. Their unified resistance was used to motivate all the family members to work on issues with focus and urgency.

Course of Treatment

The initial intervention occurred when the father contacted the family therapist asking to come in with the son, and the therapist insisted that the whole family come in together for the intake. The presenting problem was then reframed by the suggestion that resolution of John's symptoms needed to be addressed in the broader family system. The therapist highlighted the seriousness of the son's symptoms and used the family's immediate resistance to an outside-the-home

placement to motivate the parents to prioritize the provision of better structure and support to the son and stabilize the marital and parenting subsystems. This was handled by the therapist in a directive manner, and an explicit plan was worked out with the couple to implement a binuclear family structure. The specific plan called for a nesting residential arrangement, wherein the parents shared a single apartment where first one, and then the other, resided when not parenting John. John, however, stayed in the family home with whichever parent was actively parenting. This arrangement allowed for (a) a stabilization of the basic family structure—there was no time line imposed or immediate decision called for about separation and divorce—at the same time as (b) the marital and parenting subsystems were functionally separated. Treatment was then able to progress to structuring (by creating a schedule for) the parenting time, thereby assuring that appropriate responsibility and supervision would be occurring. The boundaries between spousal issues and parenting issues were functionally reinstituted, and John's parents and the therapist mutually agreed to move spousal issues to the back burner and to focus on parenting issues as primary.

Work with the parental subsystem became the primary thrust of therapy. Disengaging the couple's focus on the marriage and keeping their energies child-focused, with exchanges taking place principally over e-mail (augmented by occasional phone contacts and in family sessions), contained their conflict and made the coparenting relationship more manageable. Sessions often clarified when spousal relationship issues were compromising coparenting, and the parents were gradually able to attain better boundaries with regard to these issues. Conjoint work with the mother and son addressed the trauma induced by the affair and the secret, working on healing the relationship breach that had occurred so that the son could gradually accept more nurturance and parenting from his mother. In the beginning, parenting time was titrated between mother and son based on the boy's tolerance of being with her. There was more variability later, depending on advances and setbacks in the mother-son reunification work, and so titration continued to be monitored in an ongoing way by the family therapist. Work with John's father addressed his need to step up and assume greater parenting responsibility, to make himself more available, and to work on his deficient parenting skills. Coparenting during these early phases was coordinated by the family therapist, with explicit focus on ongoing information exchange and active decision making about the son.

As this work began, the conversations with the boy's individual therapist commenced. It became clear rather quickly that she was aligned with the father-son coalition and that her identification with the son's rejection of the mother was especially strong. She reluctantly acknowledged that she had probably alienated the mother and possibly also compromised her primary therapeutic role by conducting conjoint parent-child work with only the father and son. However, because she was steadfast in her conviction that she would almost certainly be unable to form a working alliance with the mother, she gradually acknowledged (and helped persuade the reluctant father) that it seemed best to have the son referred to another therapist. The family therapist assisted the family with a referral to a child therapist who specialized in high-conflict cases and established a collaborative team approach with this therapist from the beginning of their involvement. The family therapist also contacted the school counselor (who had not heard from either parent) and coordinated a meeting between relevant school staff and both parents. The school counselor and parents set up a plan

(*Continued*)

to support the son's functioning at school, which included structures for regular feedback from the school.

Progress was soon evident in multiple parameters of this family system's functioning. John's performance at school improved, and he gradually began working through his anger toward his mother, allowing her to reconnect as his primary parent. John's mother, who had ended the affair, redoubled her attentive parenting efforts and worked diligently to prove her commitment to preserve the marriage. John's father felt content with the moratorium on pursuing separation and divorce. He willingly accepted the parent coaching that the therapist offered and soon began looking forward to and enjoying his scheduled parenting time without John's mother present. Considerable work had to be done to help John's father support the healing of the mother-son relationship, despite his own continuing anger with her. The work of coparenting continued to be effectively coordinated by the family therapist; structures for communication were put into place, parenting issues that came up were discussed and decisions made in conjoint sessions with the parents, and written documentation of specific decisions made at these meetings were provided to John's parents to reinforce these structural shifts. Responsibility and accountability were supported throughout this process, from which a functional parental hierarchy came to be established with a transformation to a more equitable and shared parenting arrangement. This arrangement was established and documented in a formal parenting plan that evolved from the treatment, so that when the father made a decision to file for divorce 6 months after the work began, the family (despite some setbacks) was ultimately able to move collaboratively with only minimal legal-adversarial involvement to a shared custody arrangement. The family therapist–child therapist team continued to support the family during this transition.

Implications of the Case

John's individual symptoms were understood as occurring in the context of this family's triadic family process. Therefore, the goals of family systems treatment were broader than immediate resolution of the identified patient's symptoms. They were focused on addressing relational processes in the marital, coparental, and parent-child subsystems and in the extrafamilial system that was impacting the family. Helping the family to functionally restructure as they negotiated a transition from nuclear family to binuclear family was a critical focus of the treatment. Work on clarifying and reinforcing functional subsystem boundaries and reestablishing the hierarchy between parents and son remained a focus throughout the treatment. Given the destructive potential of the marital breakdown, structural disengagement of the marital couple (through the nesting residential arrangement and coordination of their coparenting) was an essential intervention in this case. Similarly, assessment of and direct intervention addressing extrafamilial involvement (with the son's individual therapist and relevant school personnel) was needed to create a collaborative professional context around the family system. Psychoeducation and recommendations about more collaborative divorce processes helped the parents resist the pulls from the legal adversarial family court system to escalate their spousal conflict. As the Jackson family moved through the divorce transition, the son benefited from a more balanced, competent, shared parenting arrangement, a marked change from the almost exclusive maternal involvement that had characterized his family experience prior to the beginning of treatment.

As illustrated in earlier sections of this chapter, there are myriad ways that family therapists of different schools might have chosen to approach and formulate this case, but there would also be a number of likely commonalities. For example, most systems-oriented clinicians would have sought to observe and assess the triadic or whole family process and to evaluate problems in the family's hierarchy and boundaries during initial sessions. Most also would have looked beyond the family's interior dynamics to understand the roles being played by interrelated systems—in this case, the school, legal, and mental health systems.

The therapeutic stance and the kinds of information gathered and privileged during this process would have varied from approach to approach, of course. The approach the family therapist took in working with the family was more old school than new age, in that a clear hierarchical arrangement installing the therapist as expert was set up from the outset to help stabilize the family quickly as a number of crucial decisions were made. The therapist was quite active in proposing an appropriate course of action, rather than coaxing the family to make these decisions themselves.

Different schools may very well have attended to other aspects of the case unaddressed in the work summarized here. For example, from certain frameworks, a spotlight may have been trained on John's father's financial and executive planning power and control in the family situation and on his mother's long-standing one-down position. Or focus may have been given to the cutoff of marital intimacy John's mother felt prior to embarking on the affair and/or to her empowerment as an individual (both absent in the case formulation) in the service of supporting and sustaining postdivorce parenting adaptation. The casting of out-of-home placement as an exterior threat to the family (which thereby mobilized their cohesiveness in combating the threat) resonated well with narrative approaches. Multigenerational theorists would have sought different insights to this case, perhaps unwrapping family members' collusion with the father in denigrating the mother to her son in case formulation.

Though there would almost certainly be differences in emphasis depending on the theoretical framework, the path toward recovery for the adolescent charted by helping to restructure the family, clarify boundaries, and strengthen the coparenting alliance was guided by a distinctively systems-driven approach. While remaining sensitive to the potential role played by intra-individual factors, the case exemplifies major ways in which therapists working from systems frameworks formulate and intervene in a manner differing in emphasis from those of other major traditions represented in this volume.

SUMMARY

Family therapy's revolutionary insight was that the problems of any given individual are a function of the whole family, and this systems viewpoint continues to guide virtually all contemporary family therapy practice. The field has undergone a dizzying number of changes in its 50-year history, and in at least certain schools of practice there have been some major breaks with long-standing traditions of focusing on interaction over cognition and on privileging relationships over individuals. Yet the hallmarks of systems approaches—thinking triadically about behavior problems, focusing on family process over content, understanding that

families are open systems embedded in extrafamilial systems—remain as fresh as ever. They will undoubtedly continue to have enduring impact and, when embraced, promise to infuse critically important insights into effective clinical practice.

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CHAPTER 8

Applied Behavior Analysis

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GENERAL OVERVIEW

The purpose of this chapter is to provide readers with basic information on applied behavior analysis (ABA). Important terms used throughout the chapter are defined first. Then, as context is highly relevant to ABA, a context for the work is provided. This context setting is accomplished through brief discussions of the philosophy and history associated with applied behavior analysis. Historical applications funnel to a discussion of the current uses of ABA. Finally, ethical considerations related to the use of ABA are covered.

Important Terms

Antecedent: A stimulus that acts as a signal or cue for future behavior either to repeat historically reinforced behavior or to avoid historically punished behavior (Baldwin & Baldwin, 2001).

Behavior: The organism's interaction with the environment in which there is a change in the environment due to an overt or covert action of the organism (Johnston & Pennypacker, 1993; Newman, Reeve, Reeve, & Ryan, 2003).

Conditioned stimulus: A stimulus that was formerly neutral that gains an ability to elicit respondent behavior through association with an unconditioned stimulus; was originally labeled a conditional stimulus (Newman et al., 2003).

Consequence: A stimulus that follows a behavior and results in a change in frequency of a behavior and/or in the probability of a behavior occurring and influences the significance of an antecedent stimulus as a cue for future behavior (Baldwin & Baldwin, 2001).

Differential reinforcement: A procedure in which one response is reinforced and another is extinguished; this process is involved in highly skilled and complex behaviors such as throwing a ball (Skinner, 1953).

Discrete trial training: A procedure used to teach new skills, in which there is a distinct block of time during which there is an instruction or prompt (an antecedent), a measure of emitted (or nonemitted) behavior, and consequential reinforcement (Newman et al., 2003).

Neutral stimulus: A stimulus that does not elicit a response without conditioning but can come to elicit a response after it is paired with an unconditioned stimulus and thus becomes a conditioned stimulus (Newman et al., 2003).

Operant conditioning: A procedure in which there is a behavior-environment interaction that changes the probability of a behavior (or the frequency of a behavior) occurring due to a consequence that contingently follows such behavior. This will also change the function of an antecedent consequence (Baldwin & Baldwin, 2001).

Punishment: A contingency between a behavior and a consequence that decreases the probability of that behavior occurring in the future under similar circumstances (Newman et al., 2003).

Reinforcement: A contingency between a behavior and a consequence that increases the probability of that behavior occurring in the future under similar circumstances (Baldwin & Baldwin, 2001; Leslie, 1996).

Respondent conditioning: Also known as classical conditioning or Pavlovian conditioning, a procedure in which a neutral stimulus is paired with an unconditioned stimulus and results in the neutral stimulus becoming a conditioned stimulus with the ability to elicit a response similar to that of the unconditioned stimulus with which it was paired (Newman et al., 2003).

Shaping: A process in which successively closer approximations, or variations, of a desired behavior are reinforced; learning occurs in a stepwise fashion and involves changing mastery criteria necessary to receive reinforcement so that more and more sophisticated behavior is required (Baldwin & Baldwin, 2001).

Target behavior: A behavior of interest in applied behavior analyses that can be increased or decreased depending on what is desired (Newman et al., 2003).

Unconditioned stimulus: A stimulus that elicits an innate or unlearned response, or a stimulus that leads to a response without teaching (reflex); originally called unconditional stimulus (Newman et al., 2003).

THE PHILOSOPHY OF APPLIED BEHAVIOR ANALYSIS

The Structure of Scientific Revolutions was published in 1962. In that seminal text, Thomas Kuhn wrote that it was impossible for a scientific community to conduct science without a set of foundational beliefs. Applied behavior analysis was still an emerging field in the early 1960s; however, the idea of communally shared beliefs has long been present in the philosophy underlying the work. As will become evident throughout this chapter, applied behavior analysis emphasizes not only recognition, but also explicit discussion of such a perspective.

The perspective on which applied behavior analysis is based is behaviorism. The term behaviorism can be used to refer to a philosophy of science or to a general approach to the scientific study of psychology (Lattal & Chase, 2003). At the most basic level, behaviorism refers to the idea that psychology can be explained through objective study, without making causal attributions to internal processes. There are

several varieties of behaviorism and definitions of behavior, but those who subscribe to behavior analytic work, often known as radical behaviorists, define behavior as anything that an integrated organism does in relation to context.

Defining Behavior

Defining behavior has been a key charge of many philosophers and psychologists throughout history. Aristotle attempted to understand behavior. Descartes did as well; he made a link between bodily movement and mechanical, as opposed to supernatural, causes. Descartes wrote that the soul and the body were two separate entities (i.e., he described an inherent dualism). He also proposed that some behaviors were caused by the soul, whereas other (less complex) behaviors were more mechanical, or determined by biological factors. Although his dualistic and mechanistic approach is quite different from much of the thought throughout behavior analysis, Descartes's work did set the stage for others, including Pavlov, to study reflexes (Leslie & O'Reilly, 1999). Classical conditioning principles as discovered and documented by Pavlov in the early 1900s play a crucial role in the historical development and the current conduct of applied behavior analysis. Therefore, Pavlov's work is elaborated further later in this chapter. For purposes of this philosophic discussion, it is sufficient to note that Pavlov's studies on salivation in dogs led to a link between biology and observable behavior.

Materialistic Approach

The emphasis on behavior as biological opened the door to an empirical analysis of materialistic philosophy. Materialism refers to the assumption that everything is composed of physical matter, and interactions between such material result in observable behavior. All supernatural explanations are rejected from a materialistic perspective (Cziko, 2000). Although early Greek philosophers, including Leucippus and Democritus, and early Western thinkers, such as Epicurus, proposed that behavior could be explained through the combination and separation of atoms, societal pressures about religion prevented materialism from taking hold until much later. There was some advancement of materialism in the seventeenth century when Hobbes wrote of the human body as a machine and LaMettrie and d'Holbach explained learning as resulting from changes in the brain. However, it was Newton's discovery of gravity and related physical principles that allowed for matter to be examined in a systematic manner (Cziko, 2000). Thorndike's studies on instrumental learning in animals, which will be elaborated on later, followed a materialistic approach and greatly advanced behaviorism.

Selectionism

Another, related advancement in philosophical thought and the study of behavior came when Darwin's work on evolution was published. Darwin's theory allowed for behavioral observations in biological contexts (Leslie & O'Reilly, 1999). The behavior analytic approach to studying psychological events parallels selectionism, the idea that characteristics of species evolve in response to success or effectiveness

(of survival and reproduction) in the environment. Behavior, like species development, is based on environmental contingencies; it is determined. Individual behavior and cultural behavior evolve in response to their effectiveness on and in context (Skinner, 1981). This emphasis on context, and behavior in interaction with context, is the reason it is so important to analyze and describe the philosophy underlying applied behavior analysis.

Contextualistic Approach

Contextualism refers to a philosophy of science that states that behavior is inseparable from the context in which it occurs. From a behavior analytic perspective, context can include the individual's learning history and internal states as well as the current physical environment. Contextualism emphasizes empiricism and experience. It is the opposite of mechanism, and involves the investigation of behavior at many levels that influence each other (Capaldi & Proctor, 1999). Contextualists believe that mechanistic attempts to study behavior lessen the quality of interpretation. Although early behaviorism was more mechanistic, and there are current behavioral works conducted from a mechanistic approach, there is wide consensus that ABA is contextualistic. One cannot perform ABA without careful consideration of context (Dougher & Hayes, 2000; Rosnow & Georgoudi, 1986). It is necessary to look at external factors that play a role in initiating and maintaining behavior.

Applied behavior analysis is built on a specific type of contextualism: functional contextualism (Biglan & Hayes, 1996). This term refers to a natural science that uses empirically based rules to predict and influence behavior with precision, scope, and depth (Biglan, 1995). From a behavior analytic perspective, prediction and influence are related to the function that is served by behavior. Related to that, causal variables are limited to those that are observable and manipulable. Given that internal variables do not meet those criteria, applied behavior analysis rejects the idea that a mind or soul causes behavior. Applied behavior analysis thus is monistic.

Pepper (1942) wrote that philosophical systems fall into one of four basic worldviews. Worldviews are characterized by root metaphors, which are how scientists attempt to understand the world, and by truth criteria, which are the measure by which analyses are evaluated. The root metaphor in functional contextualism is the act in and inseparable from its historical and current context, and the truth criterion is related to the function of the analysis. Truth depends on whether the goal is met (Biglan & Hayes, 1996). This is referred to as successful working, which was originally discussed in works on pragmatism by Pierce, Dewey, James, and Darwin. In fact, several behavior analysts have noted that contextualism and pragmatism occur symbiotically (e.g., Baer, Wolf, & Risley, 1968).

Pragmatism

Pragmatism emerged in the late nineteenth century as a mix of empiricism and psychology. More specifically, British empiricists, German Romantic idealists, Kant, Hegel, and others who worked in areas of philosophy, science, and education influenced pragmatism. Charles W. Pierce was the first in the United States to talk and

write about pragmatism. He talked about making ideas clear, which was related to using operational definitions and focusing on observable units of analyses as well as using successful working as a truth criterion. This is linked to behavior analysis in that the idea is to have correspondence between the goal and outcome after the analysis is complete. John Dewey and William James both expanded on Pierce's work. James took Pierce's work, which emphasized experimental (basic) work, and applied the philosophy to concerns about human behavior (Lattal & Laipple, 2003). James's work was part of an important shift from philosophy to the real world (Baum, 1994). Thus, our discussion shifts to a focus on the development and use of behavior analytic principles in applied settings.

A BRIEF HISTORY OF APPLIED BEHAVIOR ANALYSIS

Applied behavior analysis refers to the process of utilizing the principles of behavior in an effort to improve socially important behaviors (Baer et al., 1968). Principles of behavior were discovered through the use of carefully designed experiments, which have provided general information about respondent and operant conditioning as well as more specific knowledge about reinforcement, punishment, extinction, stimulus control, and their application. These principles are utilized in techniques such as shaping, chaining, token economies, and self-management—all of which were historically designed as methods to decrease maladaptive behavior with social relevance. Applied behavior analysis as we know it today began in the 1950s. At that time, researchers first applied the behavioral principles outlined by B. F. Skinner (1938) in his seminal work, *The Behavior of Organisms,* to clinical work with humans. However, it is important to also consider those figures and events that made early contributions to the development of behavioral principles.

Sechenov

Ivan Michailovich Sechenov (1829–1905) is considered the founder of Russian physiology (Koshtoyants, 1863/1965). At a time when the philosophical and psychological establishment emphasized mind-body duality, Sechenov took a monistic and materialistic approach to the study of mental processes, which he believed could be explained physiologically (Koshtoyants, 1863/1965). In his *Reflexes of the Brain,* Sechenov (1863/1965) proposed that all forms of behavior, including thoughts, emotions, and the holding of virtuous beliefs, were reflexes of the brain. Furthermore, he emphasized the importance of using strictly objective methods in the study of mental processes, and he stressed the significance of the role of the environment in an organism's behavior. Sechenov's ideas influenced subsequent Russian physiologists, most notably Ivan Pavlov (Koshtoyants, 1863/1965).

Pavlov

Ivan Petrovich Pavlov (1849–1936) shared Sechenov's materialistic approach, viewing psychological phenomena as physiological in origin. He also adhered to stringently objective methods in his study of higher nervous system activity (Pavlov, 1927/1928). Pavlov conducted his revolutionary research over the course

of 25 years, studying the gastric and salivary reflexes of dogs. His work on gastric reflexes earned him acclaim in physiology, not to mention the Nobel Prize, but it was his subsequent discovery and delineation of the principles of respondent, or classical, conditioning that made him a significant figure in psychology as well.

In the course of his research on gastric reflexes, Pavlov (1927/1928, p. 6) began to notice "psychic secretion" of saliva in his dogs. Specifically, the dogs began to salivate in the presence of external stimuli other than food (e.g., the experimenter who brought the food, the sound of the footsteps of said experimenter). Pavlov set about investigating this phenomenon in a strictly objective manner. So important to Pavlov was a precise experimental method in his studies of nervous system processes that a specially designed laboratory with a trench around the outside was constructed to ensure experimental control (Pavlov, 1927/1928). In his earliest experiments, Pavlov surgically transplanted the dog's salivary ducts to the outside of the cheek or under the chin for the purposes of observation and precise measurement. This surgical procedure made it possible to count the number of drops of saliva or measure the amount of colored fluid in a tube that was displaced by the dog's saliva. When food was presented to the dogs, they began to salivate. Pavlov considered the salivation response to food an unconditioned, or inborn, reflex that did not need to be trained, and food was considered an unconditioned stimulus. He then began preceding the presentation of food to the dogs with the sound of a beating metronome. After several trials in which the dogs heard the metronome and then received food, the sound of the metronome alone resulted in salivation. The metronome had become a conditioned stimulus, and the salivation in response to it was a conditioned response. Although at the time it was called a "conditioned reflex," today the term "reflex" is usually thought of in terms of the stimulus-response relationship. The conditioned response is a response that was previously an unlearned, or unconditioned, response to an unconditioned stimulus and is now occurring in response to a conditioned stimulus.

Pavlov (1927/1928) and his fellow researchers replicated these early experiments many times, conditioning a variety of stimuli to elicit salivation, including the buzzing of electric bells, the application of electric current, thermal stimuli, and the cessation of continuous metronome beating. In the process, Pavlov outlined several key principles:

- *Timing:* For a conditioned response to be established, the neutral stimulus must be presented such that it precedes or occurs at the same time as the unconditioned stimulus.
- *Extinction:* Repeated presentation of the conditioned stimulus without the unconditioned stimulus will result in a weakening of the conditioned response.
- *Differentiation, or discrimination:* Presentation of stimuli similar to the conditioned stimulus in the absence of the unconditioned stimulus will result in elicitation of the conditioned response only by the specific conditioned stimulus.
- *Generalization:* If discrimination training is not applied, the conditioned response may be elicited by stimuli that are similar to the conditioned stimulus even without direct pairing with the unconditioned stimulus.

Thorndike

At about the same time that Pavlov began investigating conditioned reflexes, Edward Lee Thorndike (1874–1949) was investigating other basic principles of learning in animals (Jonçich, 1968). Thorndike's work in animal, or comparative, psychology is considered part of the functionalist movement, a parent of behaviorism (O'Donnell, 1985). In his dissertation research at Columbia University, Thorndike conducted objective, methodical investigations of stimulus-response connections, or associations (Jonçich, 1968). The basic experiment consisted of placing a hungry kitten (and eventually hungry dogs and lonely chicks) in one of several puzzle boxes he designed, from which the animal could escape by actions such as pulling on the loop of a cord or pressing a lever (Thorndike, 1898). Food was placed visibly outside of the box (in the case of lonely chicks, other chicks were outside of the box), and Thorndike observed and recorded the animal's behavior and the time it took for the animal to escape the box. He plotted the time to escape over several trials on a graph and was one of the first researchers to capture evidence of learning in this way (Chance, 1999). Thorndike noted that, once the animal was successful, the time to escape generally consistently decreased with successive trials. He concluded that an association is formed between a certain situation and a certain response and that the resulting pleasure (obtaining food or company) served to strengthen this connection (Thorndike, 1898). In later works, Thorndike came to refer to this finding as the law of effect (Hearst, 1999).

Watson

If animal psychology can be thought of as a bridge between functionalism and behaviorism, John Broadus Watson (1878–1958) defiantly carried the flag of psychology across this bridge and planted it firmly on the other side. After 12 years of behavioral research with animals, Watson (1913) published "Psychology as the Behaviorist Views It," an article that called for a radically different approach to the science of psychology. He argued that psychology should abandon consciousness as its subject of study and forsake introspection as its method. Watson advocated psychology as an objective science, the goal of which would be the prediction and control of behavior through the study of observable responses to stimuli, using a precise and reproducible experimental method. The approach would be the same in studying animals and humans. Although Watson viewed the evaluation of the behavior of any species as important in its own right, he also criticized psychologists for failing to produce results that could be practically applied. Watson thus began the movement called behaviorism, which profoundly shaped psychological thought for much of the twentieth century (Bergmann, 1956).

Following his 1913 paper, Watson and others conducted behaviorist research that reflected a commitment to studying practical problems. His subsequent research on emotional responses as conditioned during childhood is a good example. In their famous experiment with 11-month-old Albert B., Watson and Rosalie Rayner (1920) endeavored to condition fear of an animal in the infant, to demonstrate that this conditioned emotional response could be transferred to

similar objects, to show that the response would persist over time, and to develop a method to remove the conditioned fear reaction. In the first three endeavors, they were successful. Prior to training, Albert showed no fear response to animals and objects presented to him, including a white rat, a rabbit, a dog, masks with hair, and cotton wool. He did exhibit a fear reaction in response to the noise made by a hammer being struck against a steel bar by an examiner standing behind him. To condition the fear response to a previously neutral animal, Watson and Rayner began trials in which the presentation of a white rat to Albert was paired with the striking of the hammer on the bar behind him. Within seven pairings of the two stimuli, the fear response had been completely conditioned, and presentation of the white rat alone was sufficient to elicit crying and escape behaviors in Albert. Upon further testing, the researchers determined that the conditioned fear response had transferred (or generalized) to a rabbit, a dog, a seal fur coat, cotton wool, Watson's hair, and a Santa Claus mask. Furthermore, the fear response to such objects was still present 1 month after the trials. Watson and Rayner planned to attempt the removal of the conditioned fear response in Albert, but the infant left the hospital where the experiments were conducted before any such attempts could be made.

Four years later, however, with assistance from Watson, Mary Cover Jones successfully unconditioned an existing fear response to white rabbits and similar objects in a 2-year, 10-month-old boy named Peter (Jones, 1924). To uncondition (or recondition) Peter's fear response, Jones first presented the rabbit to Peter on several occasions with other children in the room who eagerly interacted with the rabbit. Later she began pairing the presentation of the rabbit (successively closer, at first in a cage and then out of its cage) to Peter along with food and candy. By the end of the sessions, Peter showed no fear of the rabbit or similar objects. This was a significant early application of behavioral principles to a human problem. Through the work of B. F. Skinner, such applications would become an influential force in addressing countless human difficulties.

Skinner

Burrhus Frederick Skinner (1904–1990) graduated college with a major in English. After a year and a half as a struggling writer, he took a job in a bookstore in Greenwich Village in New York (Skinner, 1976). There he read Bertrand Russell's *Philosophy* (1925) and Watson's *Behaviorism* (1924/1930). Not long after, he applied and was accepted to graduate school in psychology at Harvard. In addition to psychology courses, Skinner also took classes in physiology under the department chair William Crozier. He studied Pavlov's work, and he began conducting research in the spirit of Pavlov and Watson's stimulus-response paradigm. Specifically, Skinner began investigating eating behavior in rats, which he postulated was a series of reflexes (Vargas, 2001). However, he soon observed that not all of the rats' behaviors were elicited by stimuli. Some behaviors were, instead, controlled by the stimuli that followed them. After 8 years of methodically investigating this phenomenon, Skinner (1938) published *The Behavior of Organisms,* in which he delineated the basic principles of behavior analysis.

In the studies summarized in *The Behavior of Organisms,* Skinner (1938) examined how the likelihood of a rat's emitting a target behavior could be changed by altering the consequences (in this case, the delivery of a food pellet) of that behavior. For these experiments, he constructed an "experimental box," which included a release chamber, a food tray, a waterspout, and a small brass lever that was connected to the food-dispensing magazine and to a recording device. Skinner used white rats because they were "cheap" (p. 47) and "amazingly stable" (p. 48). He chose lever pressing as the operant of study because it was an unconditioned response that occurred spontaneously in the rats from time to time before conditioning occurred. The food pellets he made contingent on responding were Purina Dog Chow.

In a typical experiment, a hungry rat would be placed behind the release door of the experimental box for 1 to 2 minutes before the experiment began (Skinner, 1938). This was done to control for any effects of the experimenter having just handled the rat. Once the door was opened, food would periodically be dispensed into the food tray from the magazine. The lever would not be connected to the magazine during this time, because it was first important for the rat to learn to respond to the tray only when it heard the magazine and not in the absence of the sound. Once this discriminative responding was established, the lever was connected and conditioning began. Any spontaneous lever pressing would now result in a food pellet being dispensed from the magazine into the tray. Lever pressing was recorded electrically in the form of a graph, with the total number of responses plotted over time. Because Skinner was interested in rate of responding, this was calculated from the slope of the lines created on the graphs. During initial conditioning, in general only four lever presses resulting in a food pellet being released into the tray were required for the rate of responding to increase dramatically. Skinner continued these experiments, delineating principles and demonstrating that the manipulation of reinforcement contingencies could powerfully shape behavior. He labeled these principles operant conditioning (Skinner, 1938).

Classical versus Operant Conditioning

Before continuing with the history of applied behavior analysis, it is important to note similarities and differences between classical, or respondent, conditioning as first described by Pavlov (1927/1928) and operant conditioning as described by Skinner, and to outline additional principles important in operant conditioning. Whereas respondent conditioning relies on antecedent stimuli predicting the presentation of another stimulus, operant conditioning relies on the consequent stimuli that follow a behavior. In other words, a consequent stimulus contingently following a behavior changes the future probability of the behavior recurring. Such a consequent stimulus is called a reinforcer.

In a situation in which a reinforcer is presented contingent on the occurrence of a particular behavior and subsequently increases the future probability of that behavior occurring, that reinforcer is referred to as a "positive reinforcer" (Newman et al., 2003). In a situation in which a reinforcer is removed contingent on the occurrence of a particular behavior and subsequently increases the future probability

of that behavior occurring, that reinforcer is referred to as a "negative reinforcer" (Newman et al., 2003). Note that in each case the reinforcers increase the future probability of a target behavior occurring. One factor that will increase the likelihood of a stimulus acting as a reinforcer is an organism's deprivation status (e.g., the longer an animal has been deprived of food, the greater the chance of food acting as a reinforcer, whereas a satiated animal's behavior may not be reinforced by food). Deprivation is an example of an establishing operation, which is a change in or characteristic of the environment that affects the power of stimuli to serve as antecedents and consequences that affect the initiation and maintenance of behavior (Newman et al., 2003).

In respondent conditioning, extinction refers to weakening of the conditioned response following presentation of the conditioned stimulus in the absence of the unconditioned stimulus (Pavlov, 1927/1928). In operant conditioning, extinction refers to a procedure or process in which the rate of a behavior of interest ultimately decreases. It involves the removal of the reinforcement contingency that produced operant conditioning, either by no longer presenting the reinforcing stimulus or by breaking the contingency that maintained the behavior of interest (Leslie, 1996, p. 57; Newman et al., 2003). In terms of behavior or process, extinction is an observed decline in the frequency of a response (Leslie, 1996).

Similar to the change in future probability of the occurrence of a target behavior (mentioned earlier in terms of reinforcement), there are also operant processes that decrease the likelihood of future behavior. As is the case with reinforcement, there are two types of punishment: positive and negative. Positive punishment is the contingent presentation of a stimulus following a particular behavior that subsequently decreases the future probability of that behavior occurring (Newman et al., 2003). Negative punishment is the contingent removal of a stimulus following a particular behavior that subsequently decreases the future probability of that behavior occurring (Newman et al., 2003). In instances of both reinforcement and punishment, positive and negative do not follow the popular connotations of "good" and "bad," but rather the presentation or removal of a stimulus, respectively. Reinforcement and punishment occur not in response to intentions, but in response to environmental contingencies that either increase or decrease the future probability of behavior, respectively (Newman et al., 2003).

The concepts of reinforcement and extinction are similar to Thorndike's instrumental behavior, but are spoken about in slightly different ways. Thorndike spoke in terms of the behavior as being influenced by pleasure or satisfaction (Leslie, 1996). Operant behavior, or more specifically, reinforcement, simply states that a reinforcer is a stimulus event that increases the probability of the behavior occurring in the future regardless of the organism's subjective evaluation (Skinner, 1953).

Operant behavior is generally thought of as voluntary behavior, and respondent behavior is thought of as involuntary behavior. Respondent behavior refers to a class of responses that are elicited by certain conditioned or unconditioned antecedent stimuli (Johnston & Pennypacker, 1993). Operants are a class of behaviors that may or may not be topographically similar and produce functionally similar consequences (Baldwin & Baldwin, 2001; Johnston & Pennypacker, 1993). The two classes of behavior are not mutually exclusive.

One way that these types of behavior overlap is that consequences can be classically conditioned. A primary, or unconditioned, reinforcer is one that is able to work effectively as a reinforcer without any prior learning or conditioning (Newman et al., 2003; Skinner, 1953). When a neutral consequence is paired with an established reinforcer, it becomes a reinforcer as well, that is then referred to as a conditioned, or secondary, reinforcer (Newman et al., 2003). One type of conditioned reinforcer is called a generalized reinforcer. A generalized reinforcer is a reinforcer that has been paired with more than one primary reinforcer and can often be exchanged for another reinforcer (Newman et al., 2003; Skinner, 1953). The benefit of a generalized reinforcer is that it has an effect on behavior despite the deprivation status of the organism. Examples of generalized reinforcers are tokens, successful control of our environment, attention, approval, affection, and the submissiveness of others (Skinner, 1953).

An area of additional overlap is that operant and respondent conditioning both affect antecedent control, albeit in different ways. An antecedent stimulus is said to have stimulus control when it reliably increases or decreases the probability of a response based on the organism's history, or experience (Baldwin & Baldwin, 2001; Leslie, 1996). Respondent conditioning involves an antecedent's ability to predict the presentation of an unconditioned stimulus. In operant conditioning, a behavior can be emitted in response to a particular antecedent stimulus but not another; these stimuli are called discriminative stimulus and S delta, respectively (Newman et al., 2003). Also in operant conditioning, one function of an antecedent is to set the occasion for an organism to repeat or avoid a past behavior that was met with reinforcement or punishment, respectively (Baldwin & Baldwin, 2001). In other words, in one situation the antecedent stimulus predicts the presence of an unconditioned stimulus (conditioned stimulus, or conditioned elicitor), and in the other case, it indicates the availability of a reinforcing or punishing consequence if a certain behavior is emitted.

Related to this issue, Skinner (1938) used the term "differentiation" to refer to operant processes, unlike Pavlov's definition, which referred to respondent responses. In *The Behavior of Organisms,* Skinner distinguished between differentiation and discrimination. Differentiation refers to distinction of the form of the response that will elicit the reinforcer from other responses in the same situation. Discrimination refers to the ability to distinguish which antecedent stimulus signals the availability of a reinforcer.

Skinner and many others continued basic laboratory research on behavioral principles (the experimental analysis of behavior) following the publication of *The Behavior of Organisms*. It became evident through such studies that reinforcement leads to maintenance of behavior, even when reinforcement is not continuous. In other words, a reinforcer does not have to immediately follow every response for behavior to be repeated (Leslie, 1996). Ferster and Skinner (1957) wrote about this phenomenon in *Schedules of Reinforcement*. The authors reported the results of comprehensive studies with pigeons, and they outlined the effect on behavior of various schedules of reinforcement. Specifically, they outlined fixed ratio (FR), fixed interval (FI), variable ratio (VR), and variable interval (VI) schedules:

- *FR:* A schedule of reinforcement in which a behavior is reinforced after a set number of responses.
- *FI:* A schedule of reinforcement in which a behavior is reinforced at the first response that occurs after a set interval of time elapses.
- *VR:* A schedule of reinforcement in which a behavior is reinforced after a variable number of responses occur that average around a value required for delivery of reinforcement.
- *VI:* A schedule of reinforcement in which a behavior is reinforced after a variable time interval elapses that averages around a value required for delivery of reinforcement.

Early Applications to Human Behavior

Ferster and Skinner's (1957) book as well as Keller and Schoenfeld's (1950) *Principles of Psychology,* which made Skinner's ideas more widely available, spurred continued *basic* research and necessitated the publishing of the *Journal of the Experimental Analysis of Behavior* (*JEAB*) in 1958 (Michael, 2004). However, it was Skinner's (1953) *Science and Human Behavior* that encouraged work in *applied* behavior analysis, or behavior modification. In this book, Skinner advocated the application of the science of behavior to human issues. He contended that human behavior could, in fact, be predicted and controlled by identifying and manipulating relevant environmental contingencies. With *Science and Human Behavior,* Skinner suggested how the basic principles outlined in *The Behavior of Organisms* apply to human individual behavior, to the behavior of groups, and to the behavior of organizations. Subsequent researchers in the 1950s and 1960s, including Skinner's students, were among the first to demonstrate that behavioral principles could indeed be applied to improving the human condition. Such effects were widely studied in work with children, adults, individuals with developmental disabilities, and individuals in mental health institutions.

During the early experiments in which Skinner outlined the importance of reinforcement, he discussed human analogues, but these plans did not come to fruition (Vargas, 2001). One very early demonstration of the principles of operant conditioning in a human was conducted by Paul R. Fuller (1949). Fuller precisely applied Skinner's methodology to operantly condition vertical right arm movements in an 18-year-old man with a severe developmental disability and limited mobility. Specifically, prior to reinforcement, he recorded the baseline rate of the participant's right-arm movements, then he reinforced the target movements with a sugar and milk solution until the rate of responding increased to desired levels, and finally he reestablished the baseline rate of responding using extinction. This experiment demonstrated that behavioral principles could indeed be applied to behavior change in humans. However, technically it was not applied behavior analysis because it did not improve a socially important behavior for the participant.

In one of the first studies that could be considered applied behavior analysis, graduate students of Skinner, Nathan Azrin and Ogden Lindsley (1956), increased cooperative behavior among children playing a game using operant conditioning

techniques, without giving any instructions to cooperate. The cooperative behaviors were reinforced using jellybeans. These behaviors increased when reinforced and decreased when reinforcement was no longer given. Free operant techniques were subsequently used to decrease a variety of problematic behaviors and to increase effective behaviors in children. These techniques involve altering contingencies involving free operants, which are behaviors that can occur at any time, without any specific cue, and often occur naturally (Newman et al., 2003). Early studies showed that the application of behavioral principles was effective in decreasing, for example, thumb sucking (Baer, 1962), tantrum behavior (Williams, 1959), and enuresis (Wickes, 1958). During this time, free operant techniques were also used with children to increase imitation (Baer & Sherman, 1964), walking (Harris, Johnston, Kelley, & Wolf, 1964), and appropriate verbal responses (Hart, Allen, Buell, Harris, & Wolf, 1964). Behavioral techniques were soon applied to the classroom setting, and researchers demonstrated that disruptive behaviors could be eliminated through extinction (ignoring such behaviors maintained by attention) and that appropriate classroom behaviors could be increased using positive reinforcement (Zimmerman & Zimmerman, 1962).

With adults, early studies in applied behavior analysis included increasing personal statements of opinion through positive reinforcement (Verplanck, 1955), reducing stuttering by making the presentation of an aversive stimulus (a loud tone) contingent upon it (Flanagan, Goldiamond, & Azrin, 1958), reducing the rate of tics in a musician by removing continuous music whenever a tic occurred (Barrett, 1962), and eliminating hysterical blindness by positively reinforcing use of visual cues (Brady & Lind, 1961). Techniques utilizing the principles of operant conditioning were also shown to be successful in treating obesity (Ferster, Nurnberger, & Levitt, 1962) and marital difficulties (Goldiamond, 1965).

Much of the early work in applied behavior analysis was conducted with individuals with developmental disabilities and with individuals in residential psychiatric facilities. Early studies successfully utilized the behavioral procedures of shaping, extinction, time-out (a period of removal from opportunities for reinforcement), and differential reinforcement to reduce problematic behaviors in children with autism (Wolf, Risley, & Mees, 1964) as well as positive reinforcement to increase the behavioral repertoires of these children (Ferster & DeMyer, 1962).

Of note, the use of differential reinforcement to reduce problematic behaviors often makes use of different intermittent reinforcement schedules. In procedures such as these, the problematic behaviors are often extinguished and other (desired) behaviors are reinforced (Leslie, 1996). There are five types of differential reinforcement schedules: differential reinforcement of other behavior (DRO), differential reinforcement of incompatible behavior (DRI), differential reinforcement of alternative behavior (DRA), differential reinforcement of low rates of responding (DRL), and differential reinforcement of high rates of responding (DRH; Leslie, 1996). In a DRO, any behavior other than the problematic behavior is reinforced. In a DRI, a particular behavior that is incompatible, or mutually exclusive, with the problem behavior that is reinforced. In a DRA, reinforcement is provided for alternative behavior to the problem behavior. In the DRL procedure, reinforcement is

delivered after a specific interval has passed in order to reduce the problem behavior by progressively lengthening the required interval. In opposition to this procedure, the DRH schedule delivers reinforcement only when a behavior is emitted rapidly (Baldwin & Baldwin, 2001; Leslie, 1996).

Additional early studies illustrated that parents and teachers of children with developmental disabilities such as autism and mental retardation could be trained to successfully utilize behavioral procedures to manage child behavior in the home and school settings (Bijou, Birnbrauer, Kidder, & Tague, 1966; Hawkins, Peterson, Schweid, & Bijou, 1966). In the school setting, the application of behavioral principles (including the use of secondary reinforcers such as stars or check marks that could be accumulated upon assignment completion and exchanged for other desired items) was shown to be effective, not only in eliminating inappropriate behaviors in these children, but also in facilitating learning of school subjects, including reading, writing, and arithmetic (Birnbrauer, Bijou, Wolf, & Kidder, 1965).

Behavioral principles as applied to work with residents in psychiatric institutions were effective in reducing a wide range of problematic behaviors in these settings, including vocalizations related to the experience of hallucinations (Lindsley, 1959), refusal to eat (Ayllon & Haughton, 1962), and hoarding behaviors (Ayllon, 1963). Once trained, nursing staff could effectively utilize techniques such as extinction and differential reinforcement to reduce problematic behaviors and increase adaptive behaviors in their patients, with researchers assuming a supervisory role (Ayllon & Michael, 1959).

Many of these institutional studies used various procedures to decrease or increase specific behaviors in one or two participants. Ayllon and Azrin (1965) endeavored to standardize and extend applications of behavior analysis to facilitate improvements in a wide array of behaviors in large numbers of adult patients in a psychiatric facility. Working with an entire ward population (up to 44 patients) at Anna State Hospital in Anna, Illinois, Ayllon and Azrin used a reinforcement program to increase participation and performance in ward jobs. These jobs included dietary assistant, cleaning assistant, laundry assistant, and recreational assistant. Because salient reinforcers for each patient tended to vary and many such reinforcers were impractical to deliver consistently and immediately following job participation (e.g., attention from the psychologist), the researchers used conditioned, or secondary, reinforcers in the form of specially designed tokens. Performance of each task was rewarded with a given number of tokens, which could then be exchanged in various amounts at designated times for a variety of desired reinforcers, for example, attention, privacy, and consumables. The program dramatically increased job participation and performance among patients, and it was quickly adapted for use by other professionals in facilities for children with mental retardation and juvenile offenders (Ayllon & Azrin, 1965). Programs such as this would become known as token economies (see Ayllon & Azrin, 1968).

Characteristics of Applied Behavior Analysis as Science

Several of the early endeavors in applied behavior analysis were published in *JEAB* and were collected in edited works such as *Case Studies in Behavior Modification*

(Ullmann & Krasner, 1965) and *Control of Human Behavior* (Ulrich, Stachnik, & Mabry, 1966). However, the body of applied research continued to grow and soon necessitated the publication of a counterpart to *JEAB*, the *Journal of Applied Behavior Analysis* (Risley, 2001). In the first issue in 1968, Baer et al. summarized dimensions relevant to studies in applied behavior analysis. In addition to the obvious characteristics of being "applied, behavioral, and analytic," the authors suggested that such studies should also be "technological, conceptually systematic, and effective" and "should display some generality" (p. 92). To be considered applied, studies should consider problems that are important to society and to the individual. As behavioral, the studies should precisely measure what an individual does and should include measures of reliability among those observing such behavior. Baer and colleagues suggested that in an analytic study, the experimenter must reliably demonstrate control over the behavior under study, such that the effect on behavior is replicable.

Two experimental designs that can illustrate such control are the reversal design and the multiple baseline design (Baer et al., 1968). In a reversal design, baseline levels of the target behavior are measured, then the variable of interest is applied and the target behavior is measured for change. If change occurs, the variable of interest is removed, and the target behavior is again measured. If the variable of interest was responsible for the behavior change, when it is removed the behavior change will no longer be observed. Finally, the variable of interest is reapplied, and the target behavior is measured to determine if the change recurs. If it does, there is evidence of behavioral control. When reversing a target behavior is not desired or impossible, the multiple baseline design can be used to demonstrate control (Baer et al., 1968). In this design, multiple target behaviors are identified and baseline levels are measured. A variable of interest is applied to one behavior and changes in that behavior are recorded, as well as the lack of change in the other behaviors. The variable of interest is then applied to another target behavior, and if it also changes while other behaviors go unchanged, there is evidence that this variable is successful in controlling behavior.

To be technological, Baer and colleagues (1968) suggested, the procedures used in studies in applied behavior analysis should be sufficiently described and should include specific details so that the study can be replicated. By conceptually systematic, the authors meant that such studies should include a discussion of the behavioral principles that are utilized in the procedure. With regard to effectiveness, the authors contended that studies in applied behavior analysis should strive for effects that are of sufficient size to be useful to the individual and those influenced by the individual's behavior. Finally, the authors indicated that such studies should aim for behavior change that has generality over time, across settings, and/or across similar behaviors.

Research including many of these dimensions of applied behavior analysis continued into the 1970s and 1980s, and, although work continued in educational and institutional settings, applied behavioral applications expanded into new arenas. Behavioral research provided benefits to community programs; for example, researchers used the principles of reinforcement (Kohlenberg & Phillips, 1973) and stimulus control (O'Neill, Blanck, & Joyner, 1980) to reduce littering and increase

proper waste disposal. In addition, behavioral techniques were successfully used to increase use of seat belts (Geller, Paterson, & Talbott, 1982). Behavioral principles were also applied successfully to improving health-related behaviors, including increasing medical appointment keeping (Friman, Finney, Rapoff, & Christophersen, 1985) and pursuit of dental care (Reiss & Bailey, 1982), as well as facilitating healthier dietary choices (Stark, Collins, Osnes, & Stokes, 1986). Early studies in the area of sports psychology indicated that application of behavioral principles could facilitate, for example, practice attendance among athletes (McKenzie & Rushall, 1974) as well as skills acquisition in sports such as football, tennis, and gymnastics (Allison & Ayllon, 1980). Studies of applied behavior analysis were also conducted in industrial settings. Applications in the business world were effective in changing a host of work-related behaviors, including reducing tardiness (Hermann, DeMontes, Domínguez, Montes, & Hopkins, 1973) and enhancing job performance (Pierce & Risley, 1974).

In the 20 years following the first publication of the *Journal of Applied Behavior Analysis,* research in behavior analysis and modification continued to proliferate, and several additional journals were developed to publish related work, including, *Behavior Therapy* in 1970, the *Journal of Organizational Behavior Management* in 1977, and the *Behavior Analyst* in 1978 (Miltenberger, 1997). In 1987, Baer, Wolf, and Risley again examined the dimensions of applied behavior analysis. They concluded that the components applied, behavioral, analytic, conceptual, technological, effective, and capable of generalization continued to be essential in studies of applied behavior analysis. They also acknowledged that the complexity of the definitions of these terms had increased. The authors suggested that studies after 20 years and into the future may include consideration of how socially important problems are identified, more complex experimental design to take into account contextual factors, consideration of the potential for methods to be adapted to additional settings, measures of social validity, and increased focus on interventions at both the individual and system levels.

CURRENT APPLICATIONS

To implement any psychological intervention effectively, it is imperative that the clinician and treatment team first conduct a comprehensive, detailed assessment. This assessment is done to best ascertain the nature of the problem and assist in subsequent treatment planning. Although many assessment procedures are based on a conceptualization of behavior as a manifestation of an underlying disorder, behavioral assessment assumes that an individual's behavior is context-dependent. Therefore, there is recognition that behavior may not be stable across various situations; thus, behavior should be assessed in a situation-dependent manner (Galassi & Perot, 1992). Behavioral assessment utilizes an ideographic approach, focusing on addressing contingencies that directly influence an *individual's* behavior. Given the need for detailed assessment that considers individual-specific context and contingencies, applied behavior analysis includes a procedure known as functional analysis to determine the form and function of relevant behavior.

Functional Analysis of Behavior

A crucial step in applied behavior analysis is conducting the functional assessment. A functional assessment allows the practitioner to identify and describe the problem behavior and the antecedents and consequences of that behavior in the context within which it occurs. The individual performing the behavior cannot be separated from current and historical contexts.

The objective of the functional assessment, or functional case conceptualization, is to obtain data that will be used to develop an effective treatment strategy that produces the desired outcome (S. C. Hayes, Nelson, & Jarrett, 1987). This is accomplished by identifying the purpose that behavior serves and hypothesizing, based on that information, which antecedents and consequences might be manipulated to decrease maladaptive behavior and optimize positive change.

Behavior analysts differentiate between functional assessment and functional analysis. Functional assessment refers to the procedures involved in clarifying antecedents and consequences; functional analysis refers to the direct manipulation of environmental contingencies in order to observe their impact on the target behavior (Gresham, Watson, & Skinner, 2001). In more complicated cases, functional analysis may include psychotherapy that focuses on reducing aversive (or appetitive) conditioning in addition to the direct altering of environmental factors. Proper functional analysis continues throughout treatment, and plans are modified if evaluation indicates that the treatment is not effective. Thorough analysis includes consideration of respondent and operant contingencies as well as their interactions (Wilson & Murrell, 2002).

Applied Behavior Analysis for Childhood Disorders

Although there are some concerns about the traditional use of behavior analysis, myriad complex behavioral issues have been dramatically improved through the application of applied behavior analytic procedures in home, school, and primary care settings. In addition, whereas pharmacological interventions have also been shown to be helpful with childhood disorders, research suggests that a combination of pharmacological and psychosocial interventions, particularly behavioral ones, can often maximize treatment gains (Hinshaw, Klein, & Abikoff, 2002).

Concerns, Common Obstacles, and Considerations

In addition to concerns about the use of traditional behavior analysis, there are common obstacles that occur during implementation. Therefore, several factors need to be considered in design and evaluation of applied behavior analyses. Of particular importance are generalizability, ease of implementation, and experimental design.

Pelham, Wheeler, and Chronis (1998) criticized behavioral interventions by pointing out that many treatment gains that are initially shown within the therapeutic milieu fail to generalize after the intervention is completed. Therefore, it is important to use interventions outside of a laboratory setting or therapeutic environment to increase the probability of the behavior generalizing to more natural settings. To do this, it is often necessary to train others to implement an intervention.

In child cases, it is important for involved parents and teachers to be motivated and able to carry out the ABA (Kazdin, 2000; Pelham et al., 1998). To enhance treatment success outside of the therapy setting, it is therefore important to identify all factors involved in enhancing and maintaining parents' and teachers' adherence to the treatment protocol. In addition, for functional relationships between independent and dependent variables to be established, it is imperative that interventions be implemented accurately and with integrity (Gresham, Gansle, & Noell, 1993). In other words, one wants to ensure that the individuals responsible for implementing the intervention are doing so correctly and consistently.

Many assessment techniques and interventions focus on the environmental contingencies regulating the child's behavior. Allen and Warzak (2000) emphasize the importance of also examining environmental contingencies that control parent, teacher, and clinician behavior. For example, many treatments are most successful when the parents are directly involved in implementing interventions for the child. Therefore, it is also important to examine the antecedents and consequences of parental behavior. The use of idiographic assessment increases the likelihood that appropriate contingencies will be identified. Often, environmental contingencies are directly related to the parent's adherence to treatment guidelines when implementing interventions (Allen & Warzak, 2000). It should be noted that contingencies regulating teachers' and clinicians' behavior should also be considered because they, too, can influence the effectiveness of the intervention.

Another factor related to effectiveness is strength of experimental design. Many behavioral studies have failed to operationally define the independent variable (such as the intervention) and to measure the accuracy with which it is implemented (Gresham et al., 1993). These flaws in experimental design can lead to failed attempts at using and evaluating behavior analytic interventions. To make scientific advances, the independent and dependent variables must be operationally defined. This is especially important in facilitating accurate replication (Gresham et al., 1993).

When generalizability, ease of implementation, and experimental design are considered, ABA evidences great success. Behavioral principles and applied behavior analytic techniques can be successfully used with children with Autistic Disorder and other pervasive developmental disabilities, Attention-Deficit/Hyperactivity Disorder (ADHD), and pediatric medical conditions such as elimination disorders (e.g., enuresis and encopresis), sleep problems, and a variety of habit disorders (trichotillomania, thumb sucking, and hair pulling). In addition, ABA is effective in treating children with Conduct Disorder and dealing with parenting problems that result from such symptomatology. It is also successfully used to address academic issues in classroom settings, with both children and teachers. The following discussion is meant to provide a sampling of some of the ways behavioral principles and the behavior analytic process, including functional assessment of problem behaviors, are currently applied to the treatment of various clinical problems across settings.

Pervasive Developmental Disorders

Applied behavior analysis facilitates the discovery and evaluation of interventions with individuals with developmental delays, such as autism spectrum disorders

and mental retardation (Frea & Vittimberga, 2000). In fact, a perusal of the current behavior analytic literature indicates that the majority of studies are being conducted with individuals with mental retardation or developmental delays (Carr, Coriaty, & Dozier, 2000; Reid, 1991).

Lovaas and colleagues (e.g., Lovaas, 1987) conducted early studies examining the impact of utilizing operant procedures to address behavioral problems in children with autism. This early work used discrete trial training to evoke and reinforce appropriate behaviors. Punishers were used to decrease inappropriate behaviors such as self-stimulatory behavior, self-injury, and incorrect vocalizations. A discrete trial is an intense, brief cycle of instruction between the instructor and the child that is used to facilitate the learning of new information and communication. One benefit of discrete trial training is that professional and trained paraprofessionals and family members can implement the procedures with the child (Smith, 2001). A typical discrete trial consists of five representative components: the cue (or discriminative stimulus), a prompt in which the instructor aids the child in responding, the actual response, a consequence, and an intertrial interval that is a brief pause before the next trial begins (Smith, 2001). In recent years, these discrete trial interventions have been modified to facilitate their use in classroom settings.

Lovaas's Young Autism Project at UCLA serves as the prototype of discrete trial training and is aimed at children under age 4. It is a time-intensive program that involves 40 hours per week of one-to-one training sessions for 3 years. It has shown success in helping children with developmental disabilities to acquire new information and new discrimination among cues.

Attention-Deficit/Hyperactivity Disorder

Direct contingency management, clinical behavior therapy, and self-management skills are three commonly used behavioral interventions for treatment of ADHD (Hinshaw et al., 2002). Contingency management procedures involve direct attempts to manipulate antecedent and reinforcement procedures to decrease problem behaviors and increase prosocial ones. Examples of contingency management procedures are time-out, in which the child is prevented from receiving reinforcement; token economy procedures; and regulating feedback from parents and/or teachers (Kronenberger & Meyer, 2001). Contingency management procedures are typically used in more professional settings, and treatment gains are often more pronounced than they are in the child's home environment (Pelham et al., 1998). Outpatient clinical behavior therapy is successful in helping parents learn to incorporate contingency management procedures in the home to facilitate treatment consistency and enhance treatment outcome (for a review, see Pelham et al., 1998).

Both direct contingency management and clinical behavior therapy place considerable demands on those implementing the programs. Self-management is a child-focused behavioral intervention for ADHD that enables the child to identify, monitor, and reinforce his or her own behaviors, thereby reducing demands on parents and teachers (Braswell & Bloomquist, 1991). Because self-management requires the child to focus on and monitor his or her own behaviors accurately in order to earn reinforcement, it not only reduces symptoms, but it also fosters

independence and personal control over behavior (Barry & Messer, 2003). Self-management can be implemented in a regular classroom or at home with some initial help from the teacher or parents, respectively. This technique is very effective at increasing on-task behaviors and improving academic performance while decreasing disruptive behaviors (Barry & Messer, 2003). Due to the self-monitoring and self-evaluation components, it is likely that self-management is more appropriate for older middle school, junior high, and high school students.

Behavioral Pediatrics

Behavior problems are often seen in primary care settings, and behavior pediatrics represents the merger between applied behavior analytic principles and pediatric medicine (Blum & Friman, 2000). Behavioral pediatrics is a broad area of study that focuses on pediatric behavior problems (such as sleeping dysfunction and elimination disorders) and health (Blum & Friman, 2000). Behavioral principles are used successfully to treat elimination disorders such as enuresis and encopresis. A number of behavioral interventions have been identified and studied to treat nocturnal enuresis in children, and the most well-known and accepted practice is the urine alarm ("bell and pad") and cleanliness training intervention (see Mowrer & Mowrer, 1938, as cited in Kronenberger & Meyer, 2001, for a description). Operant theorists suggest that the success of the bell and pad intervention among children can be attributed to negative reinforcement: specifically, the absence of the alarm and having to perform cleanliness training, which includes cleaning themselves and the toilet, changing their clothes, and changing the bed linens.

Applied behavior analysis is used with children who have a variety of other pediatric conditions, including asthma (e.g., Renne & Creer, 1976), diabetes (e.g., Epstein et al., 1981), feeding disorders (e.g., Coe et al., 1997), and other medical problems. Application of behavioral principles is also effective in increasing child and parent compliance with medical procedures and treatment (see Matthews, Spieth, & Christophersen, 1995). For example, in one study, contingency management was used to increase cooperative behavior in 3- to 6-year-old children during dental work (Allen & Stokes, 1987). Escape and attention were provided when children displayed nondisruptive behaviors during successively more invasive presentations of dentistry tools. Children were also rewarded with tangible reinforcers (i.e., stickers and toys) for remaining cooperative during increasing periods of time. Physiological arousal as well as disruptive behavior decreased from baseline (Allen & Stokes, 1987).

Conduct Disorder and Parenting Problems

Behavioral therapies based on operant learning have been used to address childhood conduct problems and to enhance the child's ability to formulate appropriate solutions to problems. Brestan and Eyberg (1998) reviewed current literature in hopes of identifying effective psychosocial treatments for Conduct Disorder, Oppositional Defiant Disorder, and associated behavior problems. Despite variability in methodological rigor, the more recent and well-designed studies were shown to be effective in decreasing negative behavior. One such treatment is

problem-solving skills training (PSST). This treatment, which is "probably efficacious," is often used to assist children in formulating prosocial responses. Games and actual social situations are used to decrease the likelihood of aggressive behaviors (Brestan & Eyberg, 1998). Recent modifications of PSST rely heavily on applied behavior analytic principles such as shaping appropriate behavior and extinguishing maladaptive responses (Kazdin, 2000).

Given the environmental contingencies that promote misconduct and possibly later antisocial behavior, PSST is often used in conjunction with parent management training (PMT; Weisz, 2004), a home-based intervention that is based on the premise that many maladaptive behaviors are due to dysfunctional parent-child interactions. Parents are taught underlying behavioral principles and skills to enhance the quality of interactions with their child, decrease problem behaviors, and facilitate adaptive behaviors. More specifically, parents are taught to implement reinforcement and punishment procedures and set appropriate rules and boundaries. In addition, the work focuses on enhancing problem-solving skills (Kazdin, 2000).

Parent-child interaction therapy (PCIT) is another evidence-based treatment designed to enhance interactions between parents and children and decrease parental stress (Eisenstadt, Eyberg, McNeil, Newcomb, & Funderburk, 1993). It is commonly used with preschool children and emphasizes teaching communication skills and behavioral principles to parents to implement in play situations with their children. The parents are given feedback on their use of the skills during the ongoing session. Initially, PCIT involves a child-directed play interaction in which the child leads the activities; this is followed by the parent-led interaction in which the parent utilizes appropriate communication and feedback to direct the play session. Finally, appropriate discipline is taught. Sessions use positive and negative reinforcement strategies to increase prosocial behaviors and decrease maladaptive ones (Eisenstadt et al., 1993).

Parent training programs often focus on enhanced consistency in efforts to maintain treatment gains. Most often, this work with youth and parents involves consistent application of reinforcement strategies and limited use of punishment procedures to facilitate behavior change (Thorpe & Olson, 1997). However, childhood behavior problems are characteristic of a number of childhood disorders and are often embedded within a problematic environmental context that maintains the child's problematic behaviors and makes lasting changes difficult. Kazdin and Whitley (2003) found that an added treatment component that specifically addressed parental stress improved overall therapeutic change when used with other models (such as PSST and PMT). Similarly, Weinberg (1999) found that training programs for parents of children with ADHD enhanced parental comprehension of behavior management principles and reduced overall stress related to parenting a child with ADHD.

Classroom Settings

In addition to substantial contributions to special education, applied behavior analysis is effectively used in general classroom settings. Historically, applied behavior analysis has focused on consequential procedures such as contingency management procedures using reinforcement and punishment. Due to ethical concerns, such as

excessive reliance on punitive or punishment procedures, applied behavior analysis has begun to shift focus and now also includes emphasis on antecedent control (Dunlap, Kern, & Worcester, 2001).

Token economies used in the classroom setting are well-established treatments for ADHD (Pelham et al., 1998). Whole-classroom token economies to manage disruptive classroom behavior may be more practical to implement than individualized token economies (Filcheck, McNeil, Greco, & Bernard, 2004). Token economies reward children for prosocial behaviors by awarding them with secondary reinforcers, such as tokens, that can be exchanged for a more powerful reinforcer. Occasionally, tokens can be taken away from the child contingent upon maladaptive behaviors. Removing tokens from the child is called response cost (Kronenberger & Meyer, 2001). Appropriate implementation is crucial to the success of such token economies.

Implementation of an intervention is an important matter in the classroom. A study by Greenwood, Terry, Arreaga-Mayer, and Finney (1992) looked at the possible moderating role of implementation factors in students' achievement in a classwide peer tutoring program. They instructed teachers to observe and record the students' successes and failures during periods when students were responsible for tutoring peers in spelling tasks. Students who were challenged by and mastered the material received the best implementation of the program. Interaction between the students was also seen to have an effect on student performance.

To enhance student performance, Lindsley (1992) expressed a hope for classroom settings to employ a behavioral precision teaching (PT) approach instead of the traditional lecturing approach. This method intends to provide more individualized, rapid progress, using a format in which the teacher acts more as a facilitator, and the students are responsible for their progress in terms of timing performance, counting, and daily charting (Lindsley, 1992). This approach involves the utilization of repeated practice with an error-correction procedure in place, along with aims expressed in terms of fluency and monitored with the use of standard celeration charts (McDowell & Keenan, 2001). The fluency factor is an important component in PT because it takes into account both accuracy and rate of responding, with the latter shown to be more sensitive to environmental changes than percentage correct (Lindsley, 1992). The application of rate of response and cumulative recording to the classroom setting comes from a free operant conditioning background (Lindsley, 1992). The use of PT in classrooms has allowed for exponential learning gains, such as seen in Morningside Academy with gains of two to three grade-level advances per year from 1980 to 1987 (Lindsley, 1992). (See Lindsley, 1992, for a brief review of important features involved in PT.)

The use of PT procedures provides gains in fluency and endurance, or attention span, in learned material (McDowell & Keenan, 2001). McDowell and Keenan reported a case study in which the use of PT procedures with a child diagnosed with ADHD resulted in an increase in correct responding, a decrease in incorrect responding, and an increase in on-task behavior. A return to baseline procedure prior to the acquisition of goals shows a decrease in performance, whereas a return to baseline after goals are acquired shows maintenance in rate of correct responding and on-task performance (Lindsley, 1992; McDowell & Keenan, 2001).

Concern with frequency of response can be seen in other classroom procedures as well. Programmed instruction is a technique based on reinforcement contingencies in which a correct answer is contingent upon a specified preceding behavior (Kritch & Bostow, 1998). In other words, instructional material is presented in a sequence that requires successively more correct responding. Typically, students are required to fill in the blanks in lessons as they work through these programs at their own pace. They are provided feedback and the correct responses immediately after providing an answer. Kritch and Bostow conducted a study that underscores the importance of frequency in responding in programmed instruction. The authors found that students in the high-density response requirement condition (i.e., the condition that demanded frequent response) performed better on tests than students who spent equivalent time on the task without any requirement that they respond. The participants in the high-density responding group were also better able to apply the knowledge than other groups.

Despite the widespread success of behavioral interventions in the classroom, Axelrod, Moyer, and Berry (1990) indicated that many teachers fail to use behavioral interventions. Teachers are rarely taught how to implement behavioral strategies prior to entering the classroom. Axelrod and colleagues suggest that behavior analysts should focus on facilitating the accurate use of behavioral principles by individuals other than behavior analysts. One way to address this problem is to teach behavior analysis to teachers; another solution is to train clients to monitor and adapt their own behavior.

Outpatient Therapy

As mentioned, applied behavior analysis is successful in treating myriad childhood behavioral disorders, pediatric medical disorders, and accompanying parent problems. There is a recent trend, often known as clinical behavior analysis, to rely on behavior analytic principles to address more complex psychological disorders. To best understand the emergence of clinical behavior analysis, a brief history of behavior therapy is required.

The Waves of Behavior Therapy

Traditionally, behavior therapy has shown a commitment to empiricism by examining overt behaviors and modifying environmental contingencies. The goal of early behavioral treatment was to reduce problematic behaviors and/or symptoms. Less overt psychological phenomena, such as thoughts and emotions, were of little interest to early behavior therapists. This lack of interest likely resulted from difficulty in manipulation, as behavior analysis (a radical behavioral perspective) never underestimated such factors. The focus on elimination of problematic behaviors and alleviation of symptoms represents *first-order* change processes: the direct targeting of symptoms and behaviors in clinical treatment with the goal of their alleviation and/or elimination. This first wave of behavior therapy was criticized, however, for not providing an adequate account of human language and cognition (Eifert & Forsyth, 2005).

The second wave of behavior therapies sought the integration of cognitive and behavioral models to facilitate treatment development and address criticisms in the literature. The application of direct change strategies to behavior and cognition has evidenced success in the reduction of psychological symptoms (S. C. Hayes, 2004). However, although research suggests that cognitive-behavior therapy (CBT) works with a variety of disorders, dismantling studies and meta-analyses suggest that researchers do not have a comprehensive understanding of *why* it works. It appears as if CBT might not work through its proposed mechanisms (e.g., Jacobson et al., 1996). Without an understanding of the mechanisms through which a treatment works, it is less likely that researchers and practitioners will be able to effectively reach the goals of behavior analysis: to predict and control (or influence) behavior.

Given the need to address language and cognition *and* have a treatment that is closely linked to examinable and manipulable explanatory mechanisms, many clinicians and applied researchers alike have made attempts to design and utilize behavior analytic treatments that address complex phenomena. These treatments, known as the *third-wave* behavior therapies, often focus on acceptance of private psychological events (e.g., thoughts and feelings) to facilitate behavior change. Further, recent therapies have incorporated spirituality, dialectics, Eastern wisdom traditions, and interpersonal functioning and values to broaden the scope of existing treatments. Consistent with behavior analysis, third-wave therapies rely on a philosophy of contextualism and pragmatism that emphasizes the function of the behavior (what purpose it serves and whether it results in positive or negative reinforcement) rather than its form (S. C. Hayes & Bissett, 2000).

These newer approaches encourage deeper analysis of private events in both assessment and treatment phases (S. C. Hayes & Brownstein, 1986). The third wave includes such treatments as dialectical behavior therapy (Linehan, 1993), mindfulness-based cognitive therapy (Segal, Williams, & Teasdale, 2002), integrative behavioral couples therapy (Jacobson, Christenson, Prince, Cordova, & Eldridge, 2000), functional analytic psychotherapy (Kohlenberg & Tsai, 2003), and acceptance and commitment therapy (S. C. Hayes, Strosahl, & Wilson, 1999). Third-wave behavior therapists may utilize first-order change techniques in their work. However, the ultimate and overarching goal of third-wave treatments is to facilitate *second-order* change. Second-order change refers to a shift in an individual's reactions and responses to his or her problems, regardless of whether the problems are decreased or eliminated. In other words, second-order change strategies target the processes of thinking and feeling rather than the content. This shift is achieved through the use of techniques that serve two primary purposes: (1) to alter the function (or meaning) of specific behaviors through the consideration of contextual factors and (2) to expand one's behavior repertoire so that, in the presence of distressing private events, one may flexibly choose to behave in a multitude of ways (S. C. Hayes, Strosahl, Bunting, Twohig, & Wilson, 2004.)

Overview of Acceptance and Commitment Therapy and Relational Frame Theory

Acceptance and commitment therapy (ACT; S. C. Hayes et al., 1999) has been influential in the emergence of third-wave behavior therapies. The developers describe

ACT as "a therapy approach that uses acceptance and mindfulness processes, and commitment and behavior change processes, to produce greater psychological flexibility" (S. C. Hayes et al., 2004, p. 13). The two halves of this description map onto the previously mentioned purposes of second-order change techniques. Work that increases a nonjudgmental awareness of private events and their relation to an individual in the present moment (i.e., acceptance and mindfulness processes) serves to facilitate a change in function of private events. Values assessment, behavioral contracting, behavior activation, and related techniques (i.e., commitment and behavior change processes) are used to expand behavior. Depending on their pragmatic utility in the ever-changing treatment context, didactics, paradox, metaphors, and experiential exercises are used to achieve the goal of psychological and behavioral flexibility.

Acceptance and commitment therapy is widely applied behavior analysis. Like other applied behavior analytic treatments, the work draws on the philosophy of functional contextualism and pragmatism as well as the principles of conditioning. Therapy begins with a functional assessment and includes ongoing functional analysis that identifies overt problem behaviors along with antecedents and consequences. In addition, the analyses include assessment of covert events, especially values and barriers to value-consistent behavior. These barriers may include variables in the external environment and/or conditioned aversives.

Proponents of the ACT model distinguish between direct and indirect conditioning. Direct conditioning (respondent and operant learning) results from personal experience or direct sensory perception of another's experience. In contrast, indirect conditioning emerges from verbal behavior. For example, a mother may become worried that her child will die of an obscure disease that she heard about on television, though neither she nor anyone she knows has the disease. When a problematic private event has been directly conditioned, such as when a child who has been bitten by a dog is scared in the presence of dogs, an ACT therapist will do exposure as it is traditionally done (e.g., in this case, a systematic desensitization, changing environmental contingencies, to the physical presence of dogs). When psychological distress is the result of indirect learning, exposure is labeled defusion, and acceptance and mindfulness processes are targeted in treatment. The intent of traditional exposure is to reduce elicitation, arousal, or avoidance in the presence of noxious stimuli. Defusion is intended to serve a different purpose: the development of wide and flexible behavior in response to noxious stimuli.

Most often, clients will present for treatment with a desire to eliminate problems. This is consistent with the traditional exposure paradigm. However, when given a choice, many will work for the possibility of a broader life. In those cases, defusion techniques are included. From an ACT perspective, clients are repeatedly asked to choose the therapeutic direction and goals. If the client chooses to work for the possibility of a broader life, successful therapy shifts the focus from a negative reinforcement paradigm to a positive reinforcement paradigm in which the emphasis is on utilizing values to create a rich, vital, and meaningful life. This emphasis on a rich life and on values can make the difficult work of treatment feel more dignified. Values, then, can be seen as powerful conditioned reinforcers. How such positive reinforcement develops with only the possibility of broader experience can be explained by the processes associated with human language and cognition.

Acceptance and commitment therapy is based on an underlying theory of human language and cognition called relational frame theory (RFT; S. C. Hayes, Barnes-Holmes, & Roche, 2001), which is an operant account of indirect learning processes. Using traditional behavioral principles, or the principles of direct conditioning, such as generalized operants (Baer et al., 1968) and positive and negative reinforcement, RFT explains how humans derive relationships between stimulus events that have never been directly conditioned and suggests how these processes may underlie psychopathology.

Consider the following interactions as an example of how direct conditioning is related to RFT. A mother smiles and asks her child, "What am I doing?" The child responds by saying "Smiling," and the mother says, "Yes, good job! I am smiling!" The mother then frowns, asks the same question, and responds similarly when her child says, "Frowning." Thus, the child has been directly trained and reinforced for correctly answering the question "What am I doing?" Although the content changes, a generalized response develops, one in which the child's behavior of providing the corresponding label has been positively reinforced through the use of praise. The mother then says, "Show me your smile," and the child responds with a smile. The child is reinforced for discriminating smiling from other facial expressions as well as performing the act. A few years later, through similar reinforcement strategies, the child is taught that the sound *smile* goes with the letters s-m-i-l-e. Without any additional training or reinforcement, a child with typical verbal capacities will know (or derive) that the facial expression is the same as the written word.

Words then become functionally equivalent to the stimuli they represent. This indirect conditioning ability, or relational conditioning, helps us balance short- and long-term contingencies, plan for the future, and categorize and sort information. However, when these relational processes become the primary source of stimulus control, these indirect conditioning principles can reduce our contact with actual environmental contingencies. Thus, our ability to react flexibly in situations is diminished, and this contributes to psychopathology. The ACT model suggests that psychological rigidity occurs as a by-product of language and verbal behavior. More specifically, humans tend to react to verbally constructed rules as though they are true rather than as what they are: verbally constructed rules, essentially thoughts. This phenomenon is referred to as *cognitive fusion* (S. C. Hayes et al., 1999). Further, humans also engage in *experiential avoidance,* which refers to the tendency to decrease, fix, or control the form or frequency of internal events, even when doing so contributes to psychological distress (S. C. Hayes, Wilson, Gifford, Follette, & Strosahl, 1996). Experiential avoidance may be adaptive in situations that are transient and time-limited. However, when it is primarily a means of addressing problems in daily living, it can contribute to decreased quality of life (Wilson & Murrell, 2004).

Returning briefly to how ACT attempts to increase quality of life, the model includes six primary and overlapping components: acceptance, defusion, contact with the present moment, self-as-context, valuing, and committed action. *Acceptance* refers to a nonjudgmental awareness of private events, along with the recognition that struggling with the content of those events is ineffective. *Defusion* work, as previously discussed, involves assisting clients in recognizing that words, thoughts,

feelings, and other such private events are not literal, real, or requiring of certain response. *Contact with the present moment* simply means that there is an ongoing effort by both the clinician and the client to remain focused on the here-and-now, as opposed to reliving the past or ruminating about the future. Although this is a process worked on in session, the skills ideally generalize to daily life. The aim of *self-as-context* is to differentiate between the content of psychological experiences and the context in which those experiences occur. In *valuing* work, clients are encouraged to identify what is important to them; therapists, along with clients, are expected to voice their values and make *committed action* to carrying out behaviors in accordance with those values.

Acceptance and Commitment Therapy with Children, Adolescents, and Families

There is growing empirical evidence to support the use of ACT with adults dealing with a variety of clinical disorders and behavioral problems (for a meta-analytic review, see S. C. Hayes, Luoma, Bond, Masuda, & Lillis, 2006). Researchers and clinicians are becoming interested in utilizing ACT for the treatment of childhood disorders as well as for the difficulties associated with parenting. Traditional ACT protocols are being modified to facilitate their applicability to children and adolescents. Such modifications in implementation include the use of more concrete language and multiple exemplars. Similarly, ACT techniques that are creative, interactive, and experiential rather than didactic or instructive are emphasized (Greco, Blackledge, Coyne, & Ehrenreich, 2005). Although treatment topics are chosen by the client, it is wise for the clinician to be aware of topics that are of typical relevance at various stages of development. For example, a clinician working with a 6-year-old should be prepared to discuss school entry difficulties. In addition, familiarity with games and television shows targeted toward young children is a plus. A clinician working with an adolescent may need to know more about current fashion trends and, more important, how the trends are related to burgeoning independence.

There are a few ACT treatment outcome studies that have been conducted with children, adolescents, and parents (for a review of current literature, see Murrell & Scherbarth, 2006). Case studies have shown promising results for youth with anorexia, anxiety, and chronic pain. One randomized clinical trial showed that ACT was more effective than medical treatment as usual for idiographic pain, and there are two ongoing randomized clinical trials that investigate the effectiveness of ACT in reducing delinquent behaviors. As with all child treatments, collateral information is often necessary while doing ACT, especially for functional analysis and other assessment. Information from parents is very important.

High levels of stress are often found among parents of children with psychological problems such as autism spectrum disorders and childhood aggression. From an ACT perspective, parenting deficits are thought to be associated with the parents' attempts to avoid painful emotions, thoughts, and experiences. Parents can make poor parenting choices to decrease their own negative psychological experiences. Consequently, this powerful motivation to avoid negative emotions

may actually prohibit the development of new, more adaptive parenting skills (Coyne & Wilson, 2004). Blackledge and Hayes (2006) utilized an ACT group therapy format to address depression and stress among parents of children with autism. Significant decreases among depression levels, psychiatric symptoms, and parental stress were found at a 3-month follow-up after the group ended. Treatment consistent with ACT principles has also been effective in working with parents of aggressive children. For example, in one case study, ACT treatment components were used to facilitate more value-consistent behavior on the part of the parent (Coyne & Wilson, 2004). As a result, child misbehavior and parental stress decreased after the 3-month treatment ended.

Murrell and Scherbarth (2006) encourage use of sophisticated research methodology to determine the effectiveness of ACT with children, adolescents, and families. Specifically, they point to the need for studies with larger sample sizes and controlled designs. To date, most of the ACT studies with children and adolescents involve case studies and uncontrolled group designs. To establish ACT as a credible and empirically based approach for use with children and adolescents, more rigorous research designs are needed.

ETHICAL CONSIDERATIONS

Utilizing a new, not yet empirically supported approach like ACT comes with ethical concerns. As with many topics in psychology, there are ethical concerns about the utilization of more traditional applied behavioral analysis as well. Common misperceptions of behavior analytic procedures, particularly negative reinforcement programs, may result in ABA having a negative connotation among the general public. Misunderstandings of terminology can be easily corrected. However, the majority of concerns about the use of aversive procedures are not easily allayed, because there is not a clear-cut line of right and wrong.

Punishment: To Use or Not to Use

There is a debate over the use of aversive procedures, such as punishment, and what reasonably justifies the use of such procedures (see Axelrod, 1990, for a description). Iwata (1988) advised that behavior analysts should make attempts to refrain from the use of aversive procedures to modify behavior, as there are several negative consequences that might come from that process. One such consequence is that those who implement aversive procedures also acquire aversive properties. This could decrease demand of their services. Another possible consequence is that the addressed problems will not benefit from the use of aversive procedures. This could leave behavior analysts in the negative position of developing punishment procedures for populations that exhibit the most extreme negative behaviors. Iwata notes that aversive procedures should be utilized only as a last resort; in other words, in addressing maladaptive behavior, preference should be given first to using positive reinforcement programs.

Iwata (1987) states that the attempt to extinguish behavior can be detrimental when there is a potential of an extinction burst, a phenomenon in which behavior

that is no longer reinforced increases in frequency, magnitude, and variability before it decreases (Newman et al., 2003). When the behavior in question is a life-threatening or harmful action, a temporary increase may be detrimental to the client. If unsuccessful attempts to extinguish behavior result in premature termination of the procedure, the behavior might inadvertently be reinforced at high levels (Iwata, 1987; Newman et al., 2003).

However, Iwata (1987) has also recognized the need for procedures such as punishment. Leslie (1996) suggests taking three steps when considering whether an aversive procedure should be used: First, define all involved terms; second, specify procedures to be considered; third, assess the possible consequences of the intervention, of the alternatives, and of taking no action. Leslie also stresses the need for an individualized assessment of the behavior and situation to examine important contextual aspects and the personal history of a client. General ethical principles may not be applicable because one solution will not be appropriate to all situations. There are multiple factors that should be considered in the decision of whether to use an aversive procedure. As mentioned, there is harm in failing to successfully implement an intervention. In addition, it is possible that behavioral effects such as aggression may emerge as a result of aversive procedure utilization (Leslie, 1996).

People often worry about the use of aversive procedures, yet they are already present in our natural environment (Axelrod, 1990; Leslie, 1996). The use of punishment allows for rapid response suppression (Leslie, 1996). When considering no intervention or alternatives to aversive procedures, Axelrod asks why one would want to waste a client's time with a procedure that is known to be less effective when there is a procedure available that is known to work. This is especially important when a problematic behavior could be life-threatening if allowed to continue, such as head banging or running into a busy street.

Sometimes Applied Behavior Analysis Is Painful

Sometimes punishment is necessary. Sometimes, even when punishment is not utilized, applied behavior analytic procedures can be painful. Treatment can be difficult. Another issue to consider is the balance between quality of life and the means of an intervention (L. J. Hayes & McCurry, 1990). One example is that of a cancer patient. If left untreated, the cancer may eventually be fatal. Yet while undergoing treatment, the side effects of a cure can cause an individual to be sick. The patient will need to come to a decision: to be sick now in order to feel better later, or to refuse treatment. An issue as complex as this would benefit from an empirical analysis of the phenomenon, which in turn carries its own ethical issues.

What to Do

Punishment, because it is easy and often effective, is reinforcing. Yet, there are clearly negative consequences related to its use. Some have suggested that an empirical understanding of topics such as punishment may allow us to reduce the use of such procedures. L. J. Hayes and McCurry (1990) remind us that aversive procedures serve a function for the greater good, but we cannot forget to contemplate

the ends versus the means of an intervention. Finally, the complexity of these issues highlights the need for an idiographic approach.

SUMMARY

Applied behavior analysis is an area of psychological research and treatment endeavoring to use behavioral principles in an effort to improve socially important human behavior (Baer et al., 1968). Philosophically, applied behavior analysis can be seen as a materialistic, deterministic, contextual, and functional approach to science. Historically influenced by Russian physiologists, American animal psychologists, and early behaviorism's call for psychology as an objective science, applied behavior analysis was realized in the 1950s and 1960s following the work of B. F. Skinner (1938, 1953) as researchers began using the principles of operant conditioning to predict and influence human problem behavior. These early researchers applied behavioral principles and techniques, manipulating the antecedents and consequences of behavior, in the successful elimination of undesirable behaviors and the development of adaptive behaviors in schoolchildren, adults, individuals with developmental disabilities, and individuals in mental health institutions. In the 1970s and 1980s applications of behavior analysis broadened to examine and improve community programs, health-related behaviors, athletic performance, and work-related behaviors. Today, applied behavior analysis continues to be a burgeoning area of research and to offer effective, empirically supported methods of psychological treatment useful in addressing a host of human difficulties, including problems associated with ADHD, Oppositional Defiant Disorder, parenting, elimination disorders, developmental delay, and serious mental illness in home, school, and institutional settings. With the emergence of third-wave behavior therapies, techniques rooted in applied behavior analysis are being successfully extended to work in outpatient settings with individuals and couples grappling with a host of psychological difficulties.

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CHAPTER 9

Cognitive-Behavioral Theory

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Cognitive-behavioral therapy (CBT) continues to evolve from its early roots in behavior therapy, and it is often represented in summaries of empirically supported interventions. Since the emergence of behavior therapy in the late 1950s and early 1960s, cognitive and behavioral theories have merged to produce a theoretically complex combination of therapeutic approaches known today as CBT, though some have continued to debate the merits of adding cognitive components to the traditional behavioral model (Beidel & Turner, 1986; Reitman & Drabman, 1997). After a period of development focusing primarily on adult psychopathology, recent years have witnessed the emergence of a wide range of CBT interventions and related research for children and adolescents. Dobson and Dozois (2001, p. 4) argue that CBT interventions for children are characterized by the following assumptions: " (a) Cognitive activity affects behavior, (b) cognitive activity may be monitored and altered, and (c) behavior change may be achieved through cognitive change."

One important facet of cognitive-behavioral theory is the assumption that "cognitive activity" and "behavior" are fundamentally different. Indeed, several authors have regarded cognitive activity as a subcategory of behavior (see Watson, 1924). In this chapter, we use the term "cognitive" to represent covert behavior, including thoughts and images. We use the term "overt behavior" when it is important to distinguish it from covert behavior. In addition to the three assumptions noted by Dobson and Dozois (2001), there are other key propositions within the cognitive-behavioral framework. Namely, the environment, overt behavior, and covert behavior all influence each other. Finally, Dobson and Dozois also distinguish cognitive-behavioral therapy from "pure" forms of behavior therapy and cognitive therapy. For example, in their view, applied behavior analysis for individuals with profound mental retardation and cognitive therapy focusing completely on memories of traumatic events would be viewed as theoretically distinct.

Kendall (2006, p. 7) regards the cognitive-behavioral framework as placing "greatest emphasis on the learning process and the influence of the models in the social environment, while underscoring the centrality of the individual's mediating/information processing style and emotional experiencing." This definition includes several key components, including (a) learning from direct experience, (b) social learning, and (c) cognitive and emotional mediation. Both Kendall and

Dobson and Dozois (2001) emphasize the mediational perspective. That is, in addition to environmental influences, a person's thoughts and feelings are believed to make a unique contribution to behavioral health.

The theoretical foundation of any form of therapy is regarded as critical by some authors who argue the merits of therapeutic techniques derived from a unifying theory. Hayes, Strosahl, and Wilson (1999, pp. 13–14), for example, distinguish between *theory* and *technique* by describing theory as "statements that have broad applicability" and technique as the "practical details" of therapy. The value of a well-developed theory is the ability to adapt techniques to new problems, develop new techniques, and add coherence to applied psychology (Hayes et al., 1999). Interestingly, most of the techniques used in CBT were derived from either behavioral theories or cognitive theories but not a unified "cognitive-behavioral" theory, and many have argued for a more flexible approach to case conceptualization that incorporates elements of many theoretical perspectives (A. Lazarus, 1996).

This chapter first provides brief descriptions of behavioral and cognitive theories, followed by a discussion of the continued integration of the two theories as they relate to children and adolescents. We close with a few comments about the future of the cognitive-behavioral framework.

BEHAVIORAL THEORY

Although there were many early contributors to the development of behavioral theory, John B. Watson's (1924) *Behaviorism* represented one of the first comprehensive efforts to describe behavioral theory. Elaborating on themes first developed in his "behaviorist manifesto" (Watson, 1913), Watson (1924) argues that behaviorism ought to focus on observable phenomena rather than consciousness. Observable responses were said to include what the organism "does," including overt actions such as speaking aloud and even covert actions such as speaking to oneself (i.e., thinking). Thus, cognitions are included in Watson's account of human behavior, and in his view, thinking is subvocal speech, observable, at least in principle, to the thinker. By contrast, psychodynamic theory held that many important motivations and explanations of human behavior were subconscious, neither observable nor available for further analysis (except indirectly through dream analysis or via unconscious responses to ambiguous or neutral stimuli, etc.). Watson also emphasizes the importance of stimuli (both environmental and physiological) that elicit observable responses.

In his early writings, Watson (1924) makes several important observations. First, he describes several kinds of responses that do not appear to require conditioning. For example, following Pavlov, he observed that responses to stimuli that are essential to the survival of the species (e.g., stimuli signaling danger or the availability of food) do not require elaborate conditioning histories. That being said, Watson also gathered data suggesting that these basic behavioral repertoires could be modified by experience. Ultimately, he concludes that much human behavior is learned via classical conditioning processes. Throughout his text, Watson consistently contrasts his view of learning with the prevailing view (in the 1920s and 1930s) that heredity alone accounts for most human behavior. Watson's work with an 11-month-old

infant, Albert, yielded one of the most potent examples of conditioning (Watson & Watson, 1921). Using classical conditioning procedures developed by Pavlov, Watson and Watson demonstrated that Albert showed no fear response (e.g., avoidance) to a white rat until experimenters repeatedly paired loud sounds (produced by striking a metal bar) with the introduction of the rat. After just a few pairings of the rat and sound, Albert began to show a fear response (i.e., crying and crawling away), even when the rat was presented in the absence of the sound. Albert's fear also transferred (or generalized) to similar stimuli, such as a rabbit. In other work, Watson and colleagues describe how one might use classical conditioning to alleviate fear, including (a) having other children handle the stimulus without fear and (b) deconditioning the fear by gradually presenting the feared stimulus without the accompanying aversive stimulus. These two techniques are today called participant modeling and in vivo desensitization, respectively, and both are counted among the handful of empirically supported interventions for anxiety problems (Ollendick & King, 1998). Watson and other early behaviorists went on to apply their conditioning model broadly to a wide range of human behaviors, including what they termed "emotional reactions" such as rage and love.

Over the course of the next few decades, B. F. Skinner continued to shape behavioral theory through extensive basic research and provocative theorizing about the implications of laboratory demonstrations of operant conditioning. Influenced by Thorndike's (1911) law of effect, Skinner gradually came to distinguish his work from the work of other behaviorists, including Watson. Whereas Watson tended to emphasize the stimuli that elicit responses, Skinner's work devotes greater attention to consequences of the behavioral response (see Skinner, 1953). One of the most important concepts to emerge from Thorndike's, and later, Skinner's work is the explicit recognition that under certain circumstances, humans and other organisms tend to repeat behaviors that are followed by particular events or consequences. Behavior can be strengthened through *positive reinforcement* by adding something (e.g., social, tangible, physical consequences) and can also be strengthened through *negative reinforcement,* typically by removing something aversive.

Skinner's theory also accounts for reductions in the strength of behavior. Specifically, *operant extinction* involves withholding the presentation of consequences maintaining a given behavior, whereas *negative punishment* involves reducing the frequency of behavior by the removal of a consequence following the performance of a behavior. *Positive punishment* involves weakening behavior by adding aversive stimuli contingent on the performance of a behavior. Skinner includes antecedents in his analyses of reinforcement contingencies, specifically focusing on antecedent stimuli (e.g., discriminative stimuli) that control behavioral responses because of a history of reinforcement in the presence of those stimuli. The three-term contingency describes the important relation that exists between *antecedent* stimuli, the *behavior,* and its *consequence* (i.e., the so-called A-B-Cs). Under certain circumstances, setting events that alter the discriminative functions of the antecedent stimuli are also included in the analysis (Iwata, Smith, & Michael, 2000). Ultimately, Skinner made many other important contributions to behavioral theory and applied behavior analysis, including the concepts of reinforcement schedules, shaping, fading, and functional analysis, just to name a few.

After an initial period of laboratory work with nonhumans, Skinner's students and colleagues rapidly expanded the application of learning principles utilizing laboratory methods to study, and influence, the behavior of children in child care, institutional, and educational settings (see Bijou & Baer, 1961, for an early example). Even today, the manipulation of antecedents and consequences is the basis for many empirically supported interventions for youth (Ollendick & King, 2000). Over the past few decades, applied behavior analysis has flourished, especially among individuals with developmental disabilities (see this volume's Chapter 8 for a more detailed account).

Like Watson, Skinner (1953, 1957) makes an explicit effort to address the role of cognition in his theory of human behavior. Skinner distinguishes between *private events* (e.g., thoughts, images, sensations, feelings) and *public events* (i.e., overt behavior), but does not afford private events a causal role independent of events (either contemporary or historical) occurring outside of the individual. For example, Skinner (1963, p. 953) once commented that "an adequate science of behavior must consider events taking place within the skin of the organism, not as physiological mediators of behavior, but as part of behavior itself." Put another way, Skinner considered both private events and public events to be forms of behavior that could serve as links in a "causal chain," ultimately determined by environmental factors.

In the late 1960s and early 1970s, Bandura initiated a series of studies examining the impact of social (or vicarious) learning, or what later became known as "modeling." Bandura (1974) agreed with Skinner's view that much human behavior was shaped by external consequences, but he suggests that learning would be exceptionally slow if humans were limited to learning via exposure to direct contingencies alone. Bandura (1973) argues that learning through observation is not only more efficient than direct contingency shaping; it also permits the individual to avoid aversive consequences. For example, in one of the landmark studies investigating observational learning, Bandura found that children who watched an aggressive model (i.e., live or video) demonstrated more aggressive acts with an inflatable doll than those who observed a nonaggressive model. Children observing the aggressive model also demonstrated more aggressive behavior than those who did not observe any model (see Bandura, 1973).

Although both Skinner and Bandura view verbal behavior as important to understanding behavior, Skinner (1957) emphasizes the role of the speaker and developed a fairly complex taxonomy of verbal stimuli that functioned as antecedents or consequences. Taking a different approach, Bandura places greater importance on cognitive variables (e.g., self-efficacy) and affords these variables a causal role in a wide range of human activity. Whether or not cognitions play a causal role in overt behavior continues to be debated (see Reitman & Drabman, 1997, on the relative merits of such debate).

COGNITIVE THEORY

Rational-emotive behavior therapy (REBT) was developed by Albert Ellis in the 1950s. First termed rational therapy, Ellis (1962) later adopted the name rational-emotive therapy (RET) to differentiate his theoretical approach from others sharing

the same name and to distance himself from the philosophical position known as rationalism. When fellow theorists and critics pointed out that many of his therapeutic techniques were essentially compatible with behavioral theory, Ellis (1999) incorporated behavior more explicitly, changing RET to rational-emotive behavior therapy. Long regarded as an inclusive therapeutic approach (A. Lazarus, 1989), REBT liberally incorporates elements of behavioral theory and practice, thus making it difficult to distinguish many of the various techniques used in REBT from behavior therapy.

The essence of Ellis's REBT is summarized by the acronym ABCDE (Dryden & Ellis, 2001; Ellis & Bernard, 1983). In this model, "A" represents an event that *activates* specific irrational thinking (e.g., a child gets a poor grade on an assignment). "B" represents *beliefs* that are irrational and illogical (e.g., "I can't stand to get a bad grade"). "C" represents the emotional and behavioral *consequences* of irrational and illogical beliefs (e.g., sadness and withdrawal). The focus of therapy, therefore, is to intervene at the level of beliefs (B), as opposed to the particular activating event or its consequences per se. In describing persons with psychological problems, Ellis (1989, p. 217) remarked, "Their emotional problems and their behavioral dysfunctions mainly stem from their grandiose demands on themselves, on others, and on environmental conditions." The role of the therapist, therefore, is to focus on the client's irrational thinking in order to produce a widespread change in emotional and behavioral functioning. The major efforts at producing cognitive and behavioral change occur in the final two steps of the model, where "D" represents the therapist *disputing* irrational thinking and beliefs (in conjunction with other REBT techniques) and "E" represents the healthier, more *effective,* state of emotional, behavioral, and cognitive functioning.

Although much of his theory places prime importance on a universally held irrational belief system, the REBT model reflects Ellis's (1962) view that all human existence consists of the primary domains of thought, emotion, and overt behavior. Ellis and Bernard (1983, p. 9) stated that people "almost never experience *pure* thoughts, feelings, or actions." Therefore, although Ellis suggests that cognitions can elicit behavioral and emotional consequences, he emphasizes that thoughts, emotions, and overt behavior are part of a holistic process in which no single domain determines activity in other domains.

Ellis contrasts REBT with humanistic models that emphasize the importance of unconditional support of the client as the primary goal in therapy. He argues that although the client should be valued, the dysfunctional thoughts and overt behavior of the client should be vigorously challenged (Ellis, 1962). In fact, Ellis is known for his often confrontational approach to therapy. Early critics of his theory questioned his forceful challenges to the client's irrational thinking, but Ellis argued that such efforts were necessary to maximize the effectiveness of therapy.

Aaron T. Beck is another highly influential figure in the history of CBT. Developing his theory independently of Ellis, A. Beck (1963) was one of the first researchers to offer a detailed account of the role of cognition in psychopathology. In one of his earliest works, he describes a pattern of cognitive distortions and themes in his adult patients with depression, and concludes that the actual source of his patients' depressed mood was distorted and irrational thinking (A. Beck, 1963). For example,

he observed that many patients perceived themselves "as incapable, incompetent, and helpless," regardless of their actual life circumstances (p. 327). In addition to having negative views of themselves, other common negative beliefs encompassed aspects of the future and the world. Collectively, these negative beliefs about the self, future, and world became known as the "cognitive triad" (DeRubeis, Tang, & Beck, 2001).

In his original formulation of the theory, A. Beck (1963) refutes the idea that mental health problems such as depression are driven by the affect experienced by the patient. Rather, he suggests that the affective, behavioral, and social facets of depression result from distorted thinking. Through interviews and extensive clinical experience with depressed patients, he began to notice certain patterns of cognitive distortions. Some of the most common forms of depressive cognitions were arbitrary inference, selective abstraction, overgeneralization, and exaggeration (A. Beck, 1963). For example, selective abstraction refers to the tendency for some persons to attend to only negative facets of a particular situation and neglect other details (A. Beck, Rush, Shaw, & Emery, 1979). In the case of depression, a person may recall a past conversation, focusing only on perceived subtle criticisms, while ignoring complimentary information.

A. Beck (A. Beck et al., 1979) has also described people with depression as shifting cognitively from "mature" thinking to more "primitive" thinking, which in turn leads to negative affective consequences. Mature thinking is characterized by reflection and insight that allows the individual to form a multidimensional, changeable, and relatively nonjudgmental self-concept. On the other hand, primitive thinking is absolute and global, leading the person to view themselves as unchangeable and fundamentally flawed (A. Beck et al., 1979). Early work by A. Beck (1963) focuses on depression, but he later suggests that cognitive theory could be applied to other disorders such as externalizing problems and anxiety (A. Beck, 1999; A. Beck, Emery, & Greenberg, 2005). Additionally, although cognitive theory focuses primarily on cognition, A. Beck and others have argued that the relation of cognition and affect may be complex (see Kohlenberg & Tsai, 1991). In any event, both Ellis and A. Beck were united in their hope that the explicit focus on cognitions would promote more efficient alleviation of negative affect (A. Beck et al., 1979).

A. Beck's (A. Beck et al., 1979) early work implies that there are different levels of belief, and his theory has evolved to incorporate these observations. For example, Judith Beck (1995) describes three levels of belief: automatic thoughts, intermediate beliefs, and core beliefs. *Automatic thoughts* are situation-specific and idiosyncratic (e.g., "I'm going to fail this test"). *Intermediate beliefs* are essentially rules or assumptions that a person applies to his or her life (e.g., "Things never work out for me"). Finally, *core beliefs* are both global and fixed (e.g., "I am a failure"). In this theory, automatic thoughts and intermediate beliefs are considered to be a reflection of a related core belief. Thus, the ultimate goal of treatment for a particular disorder is to modify irrational thinking at the level of the core belief, although treatment often begins with related automatic thoughts and intermediate beliefs.

Both Ellis and A. Beck have acknowledged the importance of reciprocal relations between cognition, the social environment, and overt behavior. For example,

if the host of a holiday party excuses herself to greet other guests, a person prone to depression and/or social anxiety may react with (a) the automatic thought "She thinks I'm boring," (b) the intermediate belief "Nobody wants to talk to a person like me," and (c) withdrawal from further social interaction. The withdrawal response may then prompt others to avoid the individual in that social situation. Consequently, the individual's intermediate belief (i.e., "Nobody wants to talk to me") may be supported by his or her subsequent experience at the party. Interestingly, the depressed or anxious individual is also likely to discount evidence contrary to the distorted hypotheses, reasoning "People only spoke to me because I was standing near the bar" and so forth. This sequence of events illustrates the reciprocal nature of individual cognition, overt behavior, and social conditioning that can maintain distorted thinking (A. Beck et al., 1979).

Although both Ellis and A. Beck postulate that irrational thinking maintains maladaptive behaviors, neither regards irrational thinking as the ultimate cause of psychopathology. Rather, both point to biological and environmental influences as important factors in the development of irrational thinking (A. Beck et al., 1979; Bernard & DiGiuseppe, 1989). According to A. Beck et al., a person's genetics and childhood environment may initiate a pattern of irrational thinking, which then leads to, and maintains, maladaptive behaviors and psychopathology. A. Beck suggests that most people lead life governed by rational thinking. However, irrational thoughts formed early in life could be triggered at some point by environmental stress, allowing for a shift from a pattern of rational thinking to irrational thinking.

The term cognitive-behavioral therapy is often used to refer specifically to the combination of A. Beck's cognitive techniques with other behavioral techniques. At other times, CBT is used as a generic term to refer to all manner of cognitive and behavior therapy blends. Although admitting many similarities, Ellis (1980; Ellis & Bernard, 1983) distinguishes between REBT and CBT. For example, Ellis states that CBT tends to focus more on cognitive change as it affects behavior, whereas REBT is primarily concerned with cognitive change as it leads to both emotional change and a broader change in one's philosophy of life, thus ostensibly affecting one's ability to cope with future problematic situations. Additionally, Ellis suggests that CBT focuses more on situation-specific cognitions and behaviors, and REBT addresses more global, irrational thinking and beliefs.

Most cognitive therapists began their work focused on problems confronted by adults in outpatient psychotherapy, so it is only recently that CBT has begun to formulate a coherent theoretical model to guide work with children. Ellis and Bernard (1983) were among the first to discuss the application of cognitive therapies to children and adolescents. Specifically, they noted that "although preferential RET is highly suitable for many bright adolescents, it may require too much philosophical analysis and more of an application of rigorous scientific method than many average youngsters, not to mention most younger children, are capable of fulfilling" (p. 8). Thus, children and many adolescents may benefit more from behavioral techniques and cognitive techniques that address situation-specific cognitions rather than belief systems.

A. Beck has written little on the subject of applying cognitive theory to children and adolescents. For example, a description of manualized cognitive therapy

treatment for adolescents who have attempted suicide seems more or less consistent with adult CBT as it progresses from the first session to termination, with no discussion of changes necessary to the therapy due to the developmental stage of adolescence (Henriques, Beck, & Brown, 2003). Recently, a few researchers have begun to examine the applicability of cognitive therapy to young children and adolescents. Specifically, Doherr, Reynolds, Wetherly, and Evans (2005) describe two studies that experimentally examined the ability of young children (5 to 7 years old) to complete cognitive tasks that would be required for cognitive therapy. Their findings indicate that, for the most part, the 5- to 7-year-olds possessed the cognitive abilities necessary to participate in cognitive therapy, although materials used for this therapy required alteration. For example, cognitive-behavioral therapy with adults often involves the use of a dysfunctional thought record in which clients record situations, automatic thoughts, and rational responses (J. Beck, 1995); some children may be able to talk about these concepts but would have difficulty writing them down independently.

At this time, the existing literature on adapting cognitive techniques for children is limited, and researchers and clinicians attempting to apply cognitive techniques to children rely on many theoretical and practical assumptions that were developed through research with adults (Derisley, 2004). On the other hand, many empirically supported interventions for youth (Ollendick & King, 2000) have been influenced by Ellis and A. Beck. Perhaps most important, changes in children's negative self-talk seem to mediate treatment outcome (Treadwell & Kendall, 1996). In the next section we present a brief review of CBT models developed more explicitly for children or that emerged based on work with child and adolescent populations.

FURTHER INTEGRATION OF BEHAVIORAL AND COGNITIVE THEORIES

While many theorists in the 1970s debated the strengths and weaknesses of the behavioral and cognitive models, some argued for an integrationist perspective including both theories. Cognitive therapists included many behavioral techniques, and some behaviorists focused more on covert behavior. These integrationist approaches ultimately had an important influence on the development of CBT interventions for children and adolescents.

Many of the integrated theories have been included under the broad label of "self-management" and include a variety of behavioral and cognitive approaches. Though sometimes referring to specific techniques, when the term self-management is used broadly, it implies that the person's behavior is partly controlled by external and internal stimuli and that the person is capable of manipulating those external and internal stimuli to achieve long-term goals (Rokke & Rehm, 2001). Thus, the derived self-management (or self-control) therapies focus on teaching learning and cognitive principles to clients so that they can apply them in their own lives. Self-management techniques therefore include monitoring one's own overt and covert behavior (i.e., self-monitoring), efforts to alter or change one's own external and internal antecedents to behavior (i.e., stimulus control), and the application of both external and internal consequences to one's own behavior (i.e., self-reward and

self-punishment; Mahoney & Arnkoff, 1978). Some self-management theorists place greater emphasis on adults (e.g., Goldfried, Decenteceo, & Weinberg, 1974); others focus more on children (Bandura, 1973; Meichenbaum, 1977). Theories of self-efficacy, problem solving, and self-instruction all fit within the self-management model.

Bandura (1997) first significantly influenced behavioral theory by adding social learning; he later continued to shape cognitive-behavioral theory with his self-efficacy model, which places primary importance on the beliefs of the individual. Although Bandura's writings are generally supportive of Skinner's behavioral theory, he suggests that earlier behavioral theories are incomplete, rather than inaccurate. For example, Bandura's research demonstrates that the consequences that follow a model's behavior play an important role in the behavior of the observer. That is, modeling is often more powerful when the model receives a reward, and Bandura (1977a) suggests that this increases the observers' expectations that reinforcers would likely follow their behavior as well. Indeed, basic research has shown that an individual's beliefs about a given reinforcement schedule could prove a more powerful predictor of behavior than the actual reinforcement schedule itself (Baron, Kaufman, & Stauber, 1969). As a result, social learning theory has increasingly emphasized the importance of cognition in predicting behavior. In particular, Bandura's recent work focuses on outcome expectations and self-efficacy. Outcome expectations are a person's predictions about future consequences, and self-efficacy describes the "beliefs in one's capabilities to organize and execute the courses of action required to produce given attainments" (Bandura, 1977b, p. 3). Bandura (1977b) describes four primary ways to develop self-efficacy; that is, self-efficacy develops when (1) mastery experiences result in the individual successfully obtaining a reinforcer, (2) social modeling involves observing others having success, (3) social persuasion is used to enhance the individuals' belief that they too can be successful, and (4) people are taught to read their own physical and emotional states and use these as a guide to success. Reflecting his greater emphasis on the role of cognitions in influencing behavior, Bandura (2004) now uses the term "social cognitive theory" to describe his theoretical model.

Cognitive problem solving is another model that emerged from the integrationist perspective. D'Zurilla and Goldfried (1971) present problem solving as the "self-directed cognitive-behavioral process by which a person attempts to identify or discover effective or adaptive solutions for specific problems encountered in everyday living" (D'Zurilla & Nezu, 2001, p. 212). D'Zurilla and Nezu draw largely from the work of R. Lazarus (1966; R. Lazarus & Folkman, 1984) in developing their "coping" or problem-solving model. In this model, stress is defined as a relationship between a person and the environment in which the person's coping skills are strained. Specific stressors include life's major problems as well as daily hassles. D'Zurilla and Nezu distinguish between two types of coping responses. *Problem-focused* coping responses are used to change something about a situation (e.g., communication style) and are best employed when the person has some control over the problem situation. On the other hand, *emotion-focused* coping responses are an attempt by the person to change his or her own emotional response

(e.g., relaxation techniques to decrease anxiety or anger) and are best used when the person does not have control over the problem situation.

Problem solving involves first identifying a situation requiring a coping response and then generating an adaptive cognitive coping response, a behavioral coping response, or both (D'Zurilla & Nezu, 2001). D'Zurilla and Nezu also distinguish between being *oriented* to a problem and then *solving* the problem. That is, before one can start generating effective solutions, one must (a) recognize that there is a problem, (b) attribute the problem to normal life rather than an internal stable deficiency, (c) see the problem as significant to one's well-being, (d) be willing to put time and effort into solving the problem, and (e) have high self-efficacy and an expectancy for change. Evidently, their problem-solving model was influenced by Bandura's work. Actually solving problems includes several components as well: (a) formulating the problem, (b) generating several possible solutions, (c) choosing the best solution, (d) implementing the solution, and (e) evaluating the solution's effectiveness. Interestingly, formulation of the problem might include cognitive and/or behavioral influences, and the solutions might be cognitive and/or behavioral in nature.

Meichenbaum (1977) offers yet another integrationist perspective, and he explicitly positioned his book *Cognitive-Behavior Modification* as an attempt to "bridge the gap" between behavior therapy and the cognitively focused therapies of Ellis and A. Beck. Meichenbaum also cites Luria (1959, 1961) and Vygotsky (1962) as important influences on his approach. Based on Luria and Vygotsky, Meichenbaum argues that the behavior of young children is first directed by their parents' verbal behavior. Children then begin to direct their own behavior through overt speech, and finally, inner speech is developed to direct their own behavior. Meichenbaum adds that once a child becomes proficient at a task, the inner speech for that task is no longer emitted. Thus, he hypothesized that maladaptive child behavior is partly the result of deficits in inner speech. For example, Meichenbaum theorizes that children with high rates of impulsive behavior have deficits in producing and effectively using inner speech. His theoretical focus on the progression of inner speech development was the impetus for the development of self-instruction training, which he applied to children with high rates of impulsive and hyperactive behavior. Self-instruction training assumes that children may have a deficit in that they do not produce enough effective inner speech to direct their behavior.

Meichenbaum was influenced by both social learning theory and the problem-solving model. Following from research suggesting that children imitate their parents' verbal reasoning (Maccoby, 1968), Meichenbaum extends Bandura's social learning theory (which focuses on modeling overt behaviors) to modeling inner speech, a form of covert behavior. Thus, self-instruction training first uses "cognitive modeling," in which adults model the use of problem-solving skills by talking aloud while performing a task. Using this technique, the adult verbally models defining the problem, focusing attention on task requirements, and providing self-praise and corrective feedback. The second step is for the child to perform the task while being instructed by the adult. The next steps involve the child imitating the adult with overt speech, then later with whispered speech, and finally with inner

speech. Meichenbaum (1977) also discusses integrating behavioral components, such as employing external reinforcers in addition to the use of self-praise.

Meichenbaum (1977) offers some important distinctions between his theory and those offered by Ellis and A. Beck. Specifically, he suggests that Ellis and A. Beck place a greater emphasis on the *presence* of maladaptive thoughts, and Meichenbaum places a greater emphasis on the *absence* of adaptive thoughts. He uses impulsive behavior as a prime example, arguing that it is not the presence of maladaptive thoughts that is problematic per se, but the absence of adaptive thoughts prior to impulsive behavior. Meichenbaum also suggests that some maladaptive thinking is normal, but that it is the degree of adaptive thinking that distinguishes normal functioning from abnormal functioning. Kendall and Hollon (1979) suggest that this distinction is the cognitive equivalent of behavioral excesses and behavioral deficits, and more recently, Kendall (2000) has used the terms "cognitive distortions" and "cognitive deficiencies" to explore this issue. Practically, this distinction suggests that for some problems maladaptive thoughts need to be directly challenged, whereas for other problems the focus of therapy should be on teaching more adaptive thoughts.

Meichenbaum's (1977, 1985) integrative theory also led to the development of stress inoculation training, another cognitive-behavioral approach to therapy. Stress inoculation training includes self-instruction training but incorporates additional components. The label "stress inoculation" is an analogy to medical inoculation, which builds antibodies to prevent diseases. In stress inoculation therapy, the concept is to build coping skills (i.e., "psychological antibodies") to prevent psychological problems (Meichenbaum, 1985). Meichenbaum gives credit to other investigators for independently developing similar ideas. For example, he cites Goldfried et al. (1974) as integrating Ellis's RET with behavioral theory. Overall, coping skill-based therapies involve teaching clients both more adaptive social behavior and cognitive coping skills. In the case of anxiety, a client is taught physiological relaxation (e.g., deep breathing and muscle relaxation) and adaptive self-statements. Clients first practice using these skills in a nonstressful situation (i.e., a therapy session) before deploying the skills in a more stressful situation.

Commenting on the diverse integrationist approaches to CBT that were emerging in the late 1970s, Mahoney and Arnkoff (1978) categorized cognitive-behavioral interventions as falling into three categories: cognitive restructuring, coping skills, and problem solving. The *cognitive restructuring* category includes the work of Ellis and A. Beck. In these therapies, maladaptive thinking is considered to be the primary cause of distress for the person. Cognitive restructuring therapies often use some behavioral techniques and carefully monitor the effects of treatment; however, the ultimate goal is to replace the person's maladaptive cognitions (internal problems) with more adaptive cognitions. The *coping skills* category includes Meichenbaum's stress inoculation therapy and other therapies that focus more on adapting to external challenges by increasing coping skills. Thus, though these therapies also address cognitions, successful therapy is determined by changes in behavioral responses rather than cognitive ones. Finally, the *problem-solving* category is perhaps the broadest in scope. These therapies focus on teaching the person a general strategy for identifying problems and generating solutions. Thus, the

focus for any given problem could be cognitive, behavioral, or both, and the range of possible solutions could include making changes to cognitions, overt behaviors, and/or the environment. Clearly, many empirically supported interventions with youth have been directly influenced by the work of cognitive-behavioral therapists with an integrationist perspective.

APPLYING THE COGNITIVE-BEHAVIORAL FRAMEWORK TO YOUTH

Cognitive-behavioral theory was largely developed by clinical researchers working with adult populations, and in recent years discussion has revolved around the extent to which the CBT framework applies to work with children and adolescents. Braswell and Kendall (2001) highlighted the importance of cognitive development in child CBT. Specifically, children's memory, attention, verbal expression, comprehension, and conceptual reasoning all might affect their response to treatment. Adaptation of CBT for children and adolescents would seem to minimally require that they be capable of sharing private events with the therapist and able to distinguish rational and irrational thoughts. To the extent that the child or adolescent is unable or unwilling to perform these tasks, the role that cognitive interventions such as cognitive restructuring can play in the therapy will be limited.

Another point of departure for adult and child CBT is that collaterals (e.g., parents and teachers) are unlikely to be directly involved in therapy for adults. Also, children rarely seek out therapy themselves, making the process of gaining their trust and engaging them in the therapeutic process especially challenging. Recognizing the importance of client motivation for treatment, Braswell and Kendall (2001) note that the manner in which CBT is presented must be altered to maximize the likelihood of positive therapy outcomes. Specifically, "a therapist must be able to teach in a playful manner and play in a way that teaches" (p. 250). To compensate for both limited cognitive capacity and motivation, Braswell and Kendall recommend greater emphasis on the incorporation of play (e.g., games, puppets), more role-plays, and the use of visual aids to supplement language-based aspects of therapy. For example, whereas adults often rate their fear on a 0 to 100 scale, in child CBT for anxiety children report on fears by choosing from an array of pictures (e.g., a row of faces depicting progressively greater fear). Some therapists use cartoon characters with thought bubbles as a tool in sharing thoughts.

Cognitive-behavioral therapy with children typically incorporates the four major components of behavioral skills training—modeling, instructions, rehearsal, and feedback (Miltenberger, 2004)—as well as manipulation of behavioral contingencies. The focus on behavioral skills training is to replace an undesirable behavior with a desirable behavior. That is, the goal is to teach the learner what to do in addition to what not to do. *Modeling* includes showing the learner how to perform a behavior, and examples include modeling behaviors for children (e.g., how to share) as well as modeling behaviors for their parents (e.g., how to use time-out). *Instructions* describe the desired behavior to the learner; examples include verbal instructions as well as written instructions. *Rehearsal* provides the learner with several opportunities to practice the desired behavior in front of the therapist.

Finally, the therapist provides *feedback* based on rehearsal, which includes praise and positively valenced critiques of the rehearsal. Thus, in the feedback component, behavioral contingencies are used to shape the learner's behavior, and the learner's behavior is ideally reinforced within the context of the role-play. Behavioral contingencies are employed outside of the therapy setting to increase generalization. Consistent with Bandura's (1974a) social learning theory, behavioral skills training increases self-efficacy through mastery experiences, social modeling, social persuasion, and sometimes through changing internal physical and emotional states.

Since their inception, behavioral and cognitive theories have drawn on other theoretical traditions. For example, the cognitive-behavioral framework is particularly compatible with *ecological systems theory* (Bronfenbrenner, 2005). The ecological systems perspective considers both characteristics of the individual and those of the family, school, and community; CBT is being increasingly used in schools (Mennuti, Freeman, & Christner, 2006). Importantly, both ecological systems theory and the cognitive-behavioral framework acknowledge that genetics and other biological influences play a role in the development of psychopathology. However, it is often difficult to determine the degree of bidirectional influence on behavior stemming from environmental and biological factors and, further, to distinguish normal behavioral variation from deviant developmental trajectories (Jensen, Knapp, & Mrazek, 2006).

In the following subsections, we discuss the application of cognitive-behavioral theory to specific problems for youth defined by the *Diagnostic and Statistical Manual of Mental Disorders,* fourth edition, text revision (*DSM-IV-TR;* American Psychiatric Association, 2000). Although our focus is on depression, anxiety, conduct problems, attention problems, and parenting stress, the cognitive-behavioral framework has also been successfully applied to many other problems affecting youth, including eating disorders and obesity (Wilfley, Passi, Cooperberg, & Stein, 2006), abuse (Deblinger, Behl, & Glickman, 2006), academic problems (Harris & Graham, 1996), and chronic pain (Masek, Russo, & Varni, 1984).

Depression

Cognitive-behavioral theorists have offered several accounts of potential etiological factors involved in youth depression (i.e., Major Depressive Disorder and Dysthymic Disorder). Lewinsohn (1974) hypothesizes that a lack of positive reinforcement could lead to depression. Some research has supported the notion that children with depression lack the social skills needed to obtain social reinforcement (Rudolph & Clark, 2001). Stressful life events (e.g., hospitalization, moving) have also been implicated in the development of depression among adolescents (Lewinsohn, Allen, Seeley, & Gotlib, 1999; Seligman, 1975). For example, in a study of over 1,500 adolescents, Lewinsohn et al. found that three or more stressful life events in the previous year appear to be the threshold for making predictions about the onset of a first depressive episode. Interestingly, recurrent depressive episodes could not be predicted using this threshold.

Consistent with a more cognitive (mediational) account of adolescent depression, experiencing several stressful life events could influence depression through

the development of a depressive attributional style (Seligman, 1975). Interestingly, negative attributional style does not predict depression when a high number of life stressors are present, but it does predict depression when there are lower levels of stress (Lewinsohn, Joiner, & Rohde, 2001). In a meta-analysis of 28 studies involving youth, depressive symptoms were correlated with internal (i.e., blame oneself), stable (i.e., not changeable), and global (i.e., applies to everything) attributions for *negative* life events and external, unstable, and specific attributions for *positive* life events (Gladstone & Kaslow, 1995). The influence of maladaptive thinking as a contributor to child and adolescent depression has also received support in the literature. Negative self-appraisals have been found to be predictive of depressive symptoms in children in sixth and seventh grade (Hoffman, Cole, Martin, Tram, & Seroczynski, 2000), but the study did not distinguish between cognitive distortions and potentially accurate negative self-statements. In another study, children with depression rated their own social status lower than peers rated them (Rudolph & Clark, 2001). Furthermore, McGrath and Repetti (2002) studied fourth- through sixth-grade students and found that depressive symptoms were predictive of later cognitive distortions; however, cognitive distortions were not predictive of an increase in depressive symptoms over 3 years. This study suggests that cognitive distortions may play a greater role in the maintenance of depression in children than in the development of depression.

The cognitive-behavioral framework allows for parental influence on the development of childhood depression. For example, recent research suggests that childhood depression is associated with family conflict (Pilowsky, Wickramaratne, Nomura, & Weissman, 2006). Also, though the study was correlational, parents of youth with depression appear to provide less praise and are more critical than parents of nondepressed youth (Puig-Antich et al., 1993). Meichenbaum's theory of cognitive modeling suggests a possible mechanism for the transmittal of negative self-appraisal. For example, a parent's critical statement "You are lazy" may later be covertly modeled "I am lazy."

Anxiety

As noted previously, Watson (1924) was among the first to examine how children's fears could be acquired through classical conditioning. According to Mowrer's (1939) two-factor theory, once acquired, fear-related behaviors can be maintained through negative reinforcement (e.g., through escape or avoidance behaviors). Following Bandura's social learning theory, Rachman (1991) notes that fears can develop both by watching other people engage in fearful behavior and by receiving other fear-provoking information through language. A. Beck's (A. Beck et al., 2005) theory of cognitive distortions was originally developed for depression but has been extended to anxiety disorders as well.

DSM-IV-TR anxiety-related problems that have been diagnosed among children and adolescents include Separation Anxiety Disorder, Panic, Agoraphobia, Specific Phobia, Social Phobia, Obsessive-Compulsive Disorder, Posttraumatic Stress Disorder, and Generalized Anxiety Disorder. With the exception of Separation Anxiety Disorder, the diagnostic criteria applied to adult anxiety problems and those applied

to child anxiety problems are nearly identical. Lang's (1968) tripartite model of anxiety is among the most influential of the cognitive-behavioral theories of anxiety and is composed of physiological (i.e., activation of the sympathetic nervous system), behavioral (e.g., fight-or-flight responses), and cognitive (i.e., fear-related speech) dimensions. More recently, a fourth dimension, emotional responding, has been explored (Southam-Gerow & Kendall, 2002). These four response dimensions are considered to be common to all of the anxiety disorders.

Research indicates that anxious children view ambiguous situations as more threatening than nonanxious children (Barrett, Rapee, Dadds, & Ryan, 1996). Barrett et al. also reported that parents play a significant role in children's anxiety. Specifically, like their children, parents of anxious children also interpret ambiguous situations as more threatening than do parents of nonanxious children. Furthermore, after anxious children had a chance to converse with their parents following an ambiguously threatening situation, the children's choice of avoidant behavior increased. The authors suggest that parents modeled anxious coping and reinforced anxious behavior by offering reassurance following avoidant responding.

Cognitive-behavioral approaches to child and adolescent anxiety have been designed to address concerns at each level of the model (Kendall & Suveg, 2006). For example, at the physiological level, children may learn relaxation techniques (e.g., diaphragmatic breathing, muscle relaxation, visual imagery). Cognitive rumination may be addressed by teaching children how to generate rational coping responses to cognitive distortions. At the behavioral level, the majority of treatment alternatives for anxiety continue to emphasize the role of exposure-based techniques such as systematic desensitization. Most recently, CBT researchers have begun to emphasize the role of the social environment in maintaining social anxiety. Thus, parents and teachers and other important influences (e.g., siblings, peers) on the child have been more aggressively recruited to participate in treatment. For example, in some cases, it has been suggested that to maximize the power of modeling and social reinforcement, parents must first be taught to manage their own anxiety before treatment for their child is initiated (Silverman & Kurtines, 1996).

Conduct Problems

As with internalizing problems, both environmental and cognitive variables are hypothesized to play a key role in the development of externalizing problems such as Conduct Disorder and Oppositional Defiant Disorder. Drawing both on Bandura's social learning theory and Skinner's operant model, Gerald Patterson's (Granic & Patterson, 2006) coercion model has been shown to account for significant variance in the development and maintenance of externalizing problems. According to the coercion model, parents may accidentally teach their children to exhibit externalizing problems. The cycle begins with parent demands for compliance. Children typically respond to requests for compliance with relatively mild coercive behavior (e.g., whining, aggression). For the coercive process to advance, parents then withdraw demands contingent on the child's aversive behaviors. Interestingly, both the child's behavior (via withdrawal of the demand) and the parent's permissive behavior (via the child's withdrawal of aversive behavior) are negatively reinforced. As

a result, the child becomes more likely to use coercive behavior in the future, and the parent is more likely to give in. The development of this pattern of behavior is deemed the "coercive cycle" or "coercion trap" and, if unabated, is hypothesized to lead to antisocial behavior.

Cognitions are believed by many to play an important role in externalizing problems. For example, in addition to studying anxiety in children, Barrett et al. (1996) also investigated oppositional behavior, with similar results. That is, consistent with Dodge (1986), children categorized in the oppositional group interpreted ambiguous situations as more threatening than did children in a control group. In fact, the differences observed in the oppositional group were even greater than the differences observed in the anxious group. Although the anxious and oppositional groups each viewed the situation as threatening, the groups differed in their choice of responses to the situation. As might be expected, the anxious group chose generally avoidant responses, and the oppositional group chose aggressive responses. As in the anxious group, parents of oppositional children viewed the ambiguous situation as more threatening than parents in the control group. Remarkably, parent-child discussions following the first exposure to the ambiguous stimulus also increased the likelihood of an aggressive child response.

Attention Problems

Kendall's (2000) distinction between cognitive distortions and cognitive deficiencies is relevant to both internalizing and externalizing problems; however, the distinction is perhaps most relevant to Attention-Deficit/Hyperactivity Disorder (ADHD). That is, researchers tend to characterize ADHD as a problem of cognitive deficiency rather than a problem of cognitive distortion, thus potentially rendering the disorder more difficult to treat (Barkley, 1997; Hinshaw, 2006). Indeed, cognitive-behavioral interventions for ADHD have generally been regarded as less effective than interventions for anxiety, depression, and conduct problems. Current state-of-the-art treatment for many children diagnosed with ADHD is a combination of stimulant medication and behavioral interventions in the classroom and at home (MTA Cooperative Group, 2004).

Despite their apparent utility for other problems affecting youth, CBT interventions for ADHD have received only inconsistent support (Pelham, Wheeler, & Chronis, 1998). On the other hand, some authors have claimed some success in shifting the control of behavior from external consequences to internal ones. As noted previously, Meichenbaum (1977) presented self-instructional training as one promising alternative. Some research has even supported using CBT procedures with preschoolers labeled impulsive, and treatment gains have sometimes generalized to the classroom (e.g., Bryant & Budd, 1982). However, little research has been conducted on self-instructional training as a treatment option for ADHD.

Because approximately 50% of children with ADHD also have significant social problems (Pelham & Bender, 1982), social skills training has become increasingly popular with this population (Hupp, Reitman, Northup, O'Callaghan, & LeBlanc, 2002; Pfiffner & McBurnett, 1997). Indeed, it is well known that effective social

skills training should incorporate elements of behavioral therapy such as teaching problem-solving skills in social contexts (Barkley, Edwards, Laneri, Fletcher, & Metevia, 2001; Fenstermacher, Olympia, & Sheridan, 2006). By and large, the effectiveness of such efforts in improving short-term outcomes for children diagnosed with ADHD is promising (see Pfiffner & McBurnett, 1997) but remains speculative over the long term.

Parenting and Parenting Stress

Cognitive-behavioral therapy has the potential to affect parents in at least four ways. First, CBT teaches parents how to more effectively manipulate antecedents and consequences associated with their child's problem behavior. Second, CBT often employs parents as models for desirable, adaptive behaviors. Third, CBT may address the parent's own psychopathology (e.g., depression, anger). Fourth, beyond simply addressing the need for some parents to obtain treatment of their own psychopathology, CBT has begun to focus more attention on the role of parenting stress. Specifically, researchers have become interested in how stress may influence parental perceptions of the child. The first three issues have been discussed in previous sections; we now examine how the cognitive-behavioral framework may be applied to parent perceptions of child behavior.

Cognitive-behavioral theorists have recently begun to examine the extent to which parenting behavior may be influenced by parental attributions about child behavior (Anastopolous & Farley, 2003; Sanders, Cann, & Markie-Dadds, 2003). In an experimental manipulation of parents' attributions, Slep and O'Leary (1998) randomly assigned 40 mothers to one of two conditions. In one condition, the mothers were given feedback that their child was primarily responsible for his or her own disruptive behavior (i.e., child-responsible group). In the other condition, parents were told that their child's disruptive behavior was developmentally appropriate (i.e., child-*not*-responsible group). This feedback was actually not based on anything other than random assignment, though parents were told it was based on a previous interview. During observations of a parent-child interaction that followed the bogus feedback, mothers in the child-responsible group demonstrated significantly more harsh parenting behaviors than parents in the child-not-responsible group. Thus, experimentally manipulated parenting attributions played a significant role in the parent's behavior. This line of research has now led to the development of child abuse prevention programs that incorporate components targeting parental attributions for change (Sanders et al., 2004).

SUMMARY

Cognitive-behavioral theory has been described as a big tent with room for many complementary (and some competing) theories. Cognitive-behavioral therapists can choose from an increasingly large number of intervention options, but there are at least three major unifying factors underlying CBT for children and adolescents. Although these factors are not uniformly characteristic of all research and theory in CBT, these general principles continue to offer guidance concerning the

future path that CBT research and practice may take. First, CBT for children should emphasize the development of a collaborative relationship between the therapist and the client (or family) as well as a focus on the response of the individual to treatment. Second, consistent with the view offered by Wolpe (1993) and other CBT founders, empirical data should serve as the basis for future modifications of CBT theory and practice (Reitman & Drabman, 1997). The development of manualized and, more recently, modularized treatment approaches are recent changes in practice that were driven by empirical data suggesting that the treatment outcomes could be enhanced with improvements in treatment integrity. Third, CBT therapists emphasize the importance of client activity outside of the therapy room, including homework exercises between sessions, self-monitoring, and a variety of other techniques that ensure that improvements are noted in session ultimately result in meaningful change (i.e., change that generalizes and can be maintained over time; see Stokes & Baer, 1977).

Despite the development of manualized treatments for children and adolescents, no two children, parents, or families that participate in CBT will have the same therapeutic experience. Moreover, as noted previously, treatment targets and developmental considerations may dictate the use of procedures that emphasize or de-emphasize the inclusion of cognitive variables. For some youth, overt behavioral problems may be the only focus of intervention, whereas other youth might be more likely to benefit from a therapeutic approach emphasizing cognitions. Also, as noted by Kendall (2000), cognitive-behavioral therapists may focus on cognitive distortions or cognitive deficiencies.

Nor is it presently necessary to make a hard choice between CBT approaches based on their overt and covert characteristics. Overt and covert approaches can certainly be used in a complementary fashion. For example, Mennuti, Christner, and Freeman (2006) recently enumerated common CBT procedures based on this overt-covert dimension. Drawing on their work, we have drafted a similar table (see Table 9.1) that more directly contrasts the overt behavioral technique with its covert counterpart. Though not comprehensive, this table provides a useful index of common cognitive techniques and nicely summarizes the therapeutic tools available to contemporary cognitive-behavioral therapists.

Table 9.1 Examples of Techniques Used in Cognitive-Behavioral Therapy

Overt Behavior Techniques	Covert Behavior Techniques
Self-monitoring	Thought record
Participant modeling	Cognitive modeling
Social skills training	Self-instructions
Competing response training	Rational responses
Positive activity scheduling	Increasing positive thoughts
Tokens and reinforcers	Self-praise
Role-playing	Mental rehearsal
Behavioral experiments	Socratic questioning
Progressive muscle relaxation	Visual imagery
Exposure to external stimuli	Exposure to internal stimuli

As can be seen in Table 9.1, monitoring can be employed to track either overt or covert behavior. Similarly, the covert counterpart to participant modeling is Meichenbaum's cognitive modeling. Social skills training is commonly used to overcome social deficits, and self-instruction training can be used to deal with cognitive deficits. Competing behavioral responses can be generated for overt or covert behavior (e.g., irrational thoughts). Increases in positive reinforcement theoretically can be achieved through positive activity scheduling (see Lewinsohn et al., 1999) or techniques aimed at increasing positive thoughts (see Henriques et al., 2003, for an example called the Hope Kit). Additionally, as seen at the bottom of Table 9.1, anxiety treatments may involve exposure to overt or covert (e.g., imaginal) stimuli. For example, a "worry chair" (e.g., making oneself worry for a set amount of time each day) is designed to expose a client with Generalized Anxiety Disorder to covert stimuli (e.g., worry), and interoceptive exposure (e.g., intentionally making oneself dizzy) is designed to produce covert exposure to panic symptoms (see Barlow, 2002, for a detailed account).

The last pairing in Table 9.1 hints at the newest theoretical wrinkle in the cognitive-behavioral framework. Specifically, Hayes (2004) describes "three waves" of behavior therapy. The first wave was heavily influenced by the work of Watson and Skinner. As shortcomings of the first wave became apparent, a second wave emerged. Influenced largely by Ellis and A. Beck, this second wave is sometimes referred to as the "cognitive revolution." Mahoney and Arnkoff (1978) summarized such innovations during this time as cognitive restructuring, coping skills, and problem solving as part of the behavior therapy tradition. The third wave of behavior therapy is thus characterized as a response to the shortcomings of the second wave. For example, Hayes (2004, p. 645) argues that it is "possible to alter the function of thoughts without first altering their form." Hayes goes on to present alternatives to directly attempting to modify cognitive distortions. These most recent additions to the CBT model are presently referred to as acceptance- or mindfulness-based therapies, and they incorporate principles borrowed from theology, Middle Eastern philosophy, and meditation.

One of the best known of the acceptance-based approaches is acceptance and commitment therapy (ACT; Hayes et al., 1999). In ACT, clients are instructed to learn to accept their cognitive distortions and emotional reactions without having to avoid or change them. Clients are taught to simply observe cognitions and, further, that the only cognitive change that is needed is to accept the fact that it is okay to have cognitive distortions and intense emotions. Similar therapies are mindfulness-based cognitive therapy (Teasdale et al., 2002) and dialectical behavior therapy (Linehan, 1993). Although acceptance- and mindfulness-based therapies are relatively new, early research appears encouraging (Hayes, Masuda, Bissett, Luoma, & Guerrero, 2004). Very little research has been devoted to ACT for youth, but a few case examples have yielded promising results (Heffner, Sperry, Eifert, & Detweiler, 2002; Wicksell, Dahl, Magnusson, & Olsson, 2005; see also Greco, Blackledge, Coyne, & Ehrenreich, 2005; Murrell, Coyne, & Wilson, 2004). As with other innovations within the cognitive-behavioral framework, ACT does not diminish the value of the other cognitive-behavioral techniques. Rather, it appears to offer yet another treatment alternative.

Throughout this chapter, CBT has been discussed primarily in the context of individual or single-family psychotherapy; however, cognitive and behavioral techniques hold significant promise for group therapy. In fact, CBT for adolescent depression may be enhanced by group social support (see Clarke, DeBar, & Lewinsohn, 2003). Because CBT is suitable for group administration it may also prove useful when incorporated into prevention programs. There are several forms of prevention: indicated, selective, universal, and health promotion models (Weisz, Sandler, Durlak, & Anton, 2005). Indicated prevention is intended to address the needs of individuals with significant symptoms that fall short of diagnosis. Selective prevention programs are for groups of individuals who share risk factors for developing problems (e.g., exposure to a traumatic event). Universal prevention programs are designed to limit the effects of potential risk factors for all youth (e.g., everyone in a school district), without specifically identifying individuals with certain risk factors. Finally, recent trends in the prevention literature point toward a health promotion focus that emphasizes building competencies within an entire population rather than focusing on risk reduction. An evidence base is quickly emerging for cognitive-behavioral prevention and health promotion programs at all four levels (Weisz et al., 2005).

In a recent symposium of former presidents of the Association for Behavioral and Cognitive Therapies (ABCT;* Albano, 2006), a number of topics relevant to the development of CBT were discussed, and two issues were central to this chapter and the future of CBT for children and adolescents. The first issue concerns core principles. Specifically, other than empiricism, are there any remaining core principles associated with CBT? If the answer is no, then theoretical considerations must always take a backseat to what works. Indeed, many early behavior therapists seemed to suggest that data should trump theory in behavior therapy, and that no aspect of the therapeutic model was to be regarded as sacred (see Reitman & Drabman, 1997). Nevertheless, in the symposium, Ollendick (2006) noted that he was initially drawn to the behavioral approach by something akin to core principles, specifically, its emphasis on functional assessment and idiographic case analysis. Wilson (2006) suggested that the validity of core theoretical principles should be subjected to rigorous analysis in dismantling studies, and he noted the substantive contributions made by Jacobson and Rachman in challenging the status quo via such an approach.

A second major issue discussed in the symposium concerned how best to go about exporting or disseminating behavioral work. Heimberg (2006) pointed out that many clients still do not benefit from cognitive-behavioral therapy and warned against complacency. Wilson (2006) suggested that efforts to disseminate existing treatment packages were unlikely to be successful without the elucidation of simpler, universal change principles underlying CBT. Interestingly, there were different ideas presented regarding how to best refine CBT. On the one hand, some viewed the future as requiring more sophisticated studies (with greater numbers of

*Reflecting a greater emphasis on cognitions, the association changed its name from Association for the Advancement of Behavior Therapy to Association for Behavioral and Cognitive Therapies in 2005.

participants) examining the mediators and moderators of treatment outcome; others thought more progress could be made by returning to idiographic approaches (to better understand change mechanisms at the level of the individual). Ultimately, this collection of ABCT past presidents suggested that much progress had been made on both theoretical and practical levels, yet there was broad consensus that much remained to be done.

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PART III

Research Contributions

CHAPTER 10

Statistical Considerations: Moderators and Mediators

Barbara Jandasek, Grayson N. Holmbeck, and Brigid M. Rose

In recent years, researchers who study pediatric populations have begun to posit complex theoretical models to explain developmental and psychological phenomena of interest. These models include longitudinal developmental pathways, risk and protective factors, and cognitive appraisal and coping processes (Thompson & Gustafson, 1996). This increase in theoretical complexity has necessitated an increase in terminological sophistication and the use of multivariate statistical strategies. Research based on mediational and moderational models has emerged as a critical method for testing competing theories about developmental pathways and other concepts central to clinical child and pediatric psychology. This chapter represents a compilation of the authors' previous works regarding mediation and moderation (Holmbeck, 1997, 2002; Rose, Holmbeck, Coakley, & Franks, 2004).

Specifically, the goal of this chapter is to provide an overview of mediators and moderators and their application to both research and practice in clinical child and pediatric psychology. First, definitions of each term and applied examples are provided, including more complex uses of these terms (e.g., moderated mediation, mediated moderation) and common definitional inconsistencies found in the literature. Next, statistical strategies used to evaluate mediator and moderator effects are reviewed (i.e., multiple regression and structural equation modeling techniques), including a presentation of strengths and limitations, common statistical inconsistencies, and applied statistical examples. Third, mediator and moderator effects in longitudinal research, intervention and outcome studies, and the study of risk and protective factors are discussed. Finally, we provide recommendations for applying mediational and moderational research to clinical practice.

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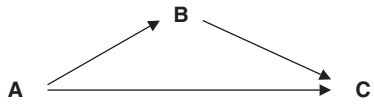

Figure 10.1 Mediated relationship among variables (A = Predictor; B = Mediator; C = Criterion/outcome).

DEFINITIONAL ISSUES

What Is a Mediator?

A mediator is an explanatory link in the relationship between two other variables (Figure 10.1; see Table 10.1 for definitions of all important terms). Often a mediator variable is conceptualized as the mechanism through which one variable (i.e., the predictor) influences another variable (i.e., the criterion; Baron & Kenny, 1986; Holmbeck, 1997, 2002). In other words, a mediator specifies how (or the mechanism by which) a given effect occurs (Baron & Kenny, 1986; James & Brett, 1984). Other terminology, such as *indirect effect* and *surrogate* or *intermediate endpoint effect,* is sometimes used interchangeably with mediation (MacKinnon, Lockwood, Hoffman, West, & Sheets, 2002), although these terms may not be conceptually identical (Holmbeck, 2002).

Suppose, hypothetically, that a researcher finds that parental intrusiveness is negatively associated with child adherence to a medical regimen. Although interesting, this finding does not tell us very much about processes that underlie the relationship between intrusiveness and adherence. By testing for mediational effects, a researcher can explore whether a third variable (e.g., child independence) might account for or explain the relationship between these variables. Continuing with the example, suppose that child independence mediates the

Table 10.1 Definition of Terms

Term	Definition
Mediator	The variable or mechanism by which a predictor influences an outcome variable (Figure 10.1).
Moderator	A variable that influences the strength or direction of a relationship between a predictor and an outcome (Figure 10.2).
Moderated mediation	A mediational model is significant only at certain levels of a moderator variable (Figure 10.3).
Mediated moderation	A moderational (or interaction) effect is mediated by another variable (Figure 10.4).
Protective factor	A variable that decreases the likelihood of a negative outcome under adverse conditions (Figure 10.7).
Resource factor	A variable that positively influences outcome, regardless of the presence of adversity (Figure 10.7).
Vulnerability factor	A variable that increases the likelihood of a poor outcome under adverse conditions (Figure 10.8).
Risk factor	A variable that negatively influences outcome regardless of the presence of adversity (Figure 10.8).

relationship between intrusiveness and adherence (more intrusive parenting → less child independence → less medical adherence). That is, parental intrusiveness impacts negatively on level of child independence, which in turn contributes to poor medical adherence (see Holmbeck, Shapera, & Hommeyer, 2002, for a similar example). In intervention studies, mediator effects may describe critical components of an intervention through which therapeutic change occurs (MacKinnon, Krull, & Lockwood, 2000, see later in chapter for more detail).

It is critical to note that a mediational model is, by its nature, a causal model. In other words, a mediational hypothesis is appropriate when one seeks to explain the causal process, or mechanism, by which an independent variable impacts a dependent variable (MacKinnon et al., 2000). Even if one's data do not permit causal conclusions (e.g., the data are cross-sectional and nonexperimental), the theory that underlies such a model is still inherently causal because it suggests a direction of influence.

One additional clarification about mediator variables is warranted. Typically, mediator mechanisms are proposed only after a significant association between the independent variable and the dependent variable has been fairly well established in the literature (i.e., indicating that there is, in fact, an effect to mediate). In other words, one does not typically begin study in a new research area by proposing mediational models. Such models are usually a natural extension of a well-established body of literature. Continuing with the earlier example, if one was interested in examining variables that mediate the link between parental intrusiveness and child medical adherence, one would want to be very confident that this link would be significant in one's own data set based on past research conducted on the variables.

What Is a Moderator?

Unlike a mediator, a moderator is a variable that influences the strength or the direction of a relationship between a predictor variable and a criterion variable (Figure 10.2). In other words, moderation describes an interaction effect whereby the impact of the independent variable on the dependent variable varies as a function of the level, or value, of the moderator variable. Moderator variables are frequently described as individual (e.g., sex, coping) or contextual (e.g., social support, family environment) characteristics (Grant et al., 2006). Suppose a researcher finds that familial stress is negatively associated with child psychological adjustment. Although this finding may be of interest to the researcher, it may be that the effect becomes more or less robust in the presence of other contextual variables. In fact, the researcher may develop specific theories about conditions that determine

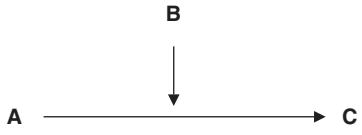

Figure 10.2 Moderated relationship among variables (A = Predictor; B = Moderator; C = Criterion/outcome).

the strength of the relationship between stress and adjustment. For example, the strength or the direction of the relationship between stress and adjustment may depend on the type of coping used by the family. That is, a significant association may emerge *only* when a child copes in a maladaptive manner. By testing coping style as a moderator of the relationship between stress and outcome, the researcher can specify certain conditions under which family stress predicts child adjustment. Such findings would likely have implications for future interventions.

Diagrammatic Distinction between Mediators and Moderators

Mediators and moderators can also be differentiated diagrammatically (see Figures 10.1 and 10.2; Baron & Kenny, 1986; J. Cohen & Cohen, 1983). A mediator (B; top of Figure 10.1) falls in the causal pathway between two variables (A and C in Figure 10.1; James & Brett, 1984). That is, if A is significantly associated with C, and A influences B, and B influences C, then B is a mediating variable between A and C. On the other hand, if A is expected to be related to C, but only under certain conditions of B, then B is a moderator variable. The moderator (B) can be drawn to indicate that it has an impact on the relationship between A and C (see Figure 10.2). Although some variables are more likely to be moderators than mediators (e.g., gender), some variables could serve either function—depending on the conceptual model under investigation (for examples where "coping strategies" were tested as both mediators and moderators in competing models, see Lewis & Kliewer, 1996; Sandler, Tein, & West, 1994). Moreover, both moderators and mediators can be specified within the same model (e.g., moderated mediation; James & Brett, 1984; for examples of this strategy, see Holmbeck, 1996; Simons, Lorenz, Wu, & Conger, 1993).

Applied Examples of Mediation and Moderation

An example will illustrate the distinction between moderated and mediated effects. This example is based on Kaczynski, Lindahl, Malik, and Laurenceau's (2006) study of marital conflict, parenting, and child adjustment. The authors hypothesized that maternal and paternal parenting would function as a mediator between marital functioning and adjustment difficulties (i.e., internalizing and externalizing symptoms) in children. In other words, marital conflict was expected to negatively impact the quality of parenting, which, in turn, would increase child maladjustment. Additionally, the authors tested a moderational effect: that parent gender would impact the nature of the relationship between marital conflict and disrupted parenting behavior. Specifically, this relationship was expected to be stronger for fathers than for mothers. Their findings fully supported the hypothesized mediational model. The moderational model was found to be valid as well, but only for families of boys.

As noted, the same variable can serve as either a mediator or a moderator (or both), depending on the research question. For example, Quittner (1992) provided a useful example of how a single construct (i.e., social support) could potentially serve as either a mediator *or* a moderator. She proposed two competing theoretical models in an attempt to study the role of social support in the relationship between parenting-related stress and psychological distress in parents of children

with hearing impairment and seizure disorder. She reasoned that parenting stress is more likely to have adverse effects on adjustment when parents have low levels of social support (i.e., social support is a moderator). On the other hand, it could also be that parenting stress undermines one's ability to garner social support, which in turn impacts on parental adjustment (i.e., social support is a mediator).

In her analyses, Quittner (1992) found that the relationship between parental stress and psychological distress did not vary as a function of level of social support. That is, social support did not *moderate* or alter the strength or direction of the relationship between parental stress and psychological distress. Having ruled out social support as a possible moderator, she tested social support as a mediator. Initially, she found that parenting stress (the predictor) was significantly associated with psychological distress (the outcome). Parenting stress was also a significant predictor of social support (mediator). Specifically, parents who experienced more stress tended to have fewer social contacts and to perceive themselves as less supported. Moreover, Quittner found a significant relationship between social support (the mediator) and psychological distress (the outcome), such that lower levels of social support were associated with higher distress ratings. When she tested the relationship between parental stress and psychological distress in the presence of this mediator, the strength of this previously significant relationship dropped significantly. She was able to conclude that the relationship between stress and psychological functioning was mediated by social support (parenting stress → social support → psychological distress) and that reduced social support may be one mechanism by which parental stress is linked with parental psychological distress.

Complex Models: Moderated Mediation and Mediated Moderation

Other types of theoretical models can be proposed that include both mediational and moderational effects. For example, with a moderated mediational model (Figure 10.3), a researcher can test whether a mediational model is significant only at certain levels of a moderator variable. Building on the earlier mediational example (i.e., parental intrusiveness → child independence → medical adherence), we might hypothesize that the entire mediational relationship will hold for only one gender and not the other. If this were found to be the case, this mediational model would be moderated by gender (i.e., moderated mediation). Similarly, mediated moderation is possible (Figure 10.4). For example, suppose that the association between intrusiveness and adherence is moderated by child gender (i.e., intrusiveness ×

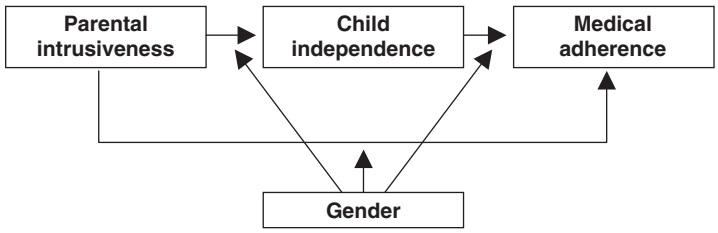

Figure 10.3 Moderated mediation.

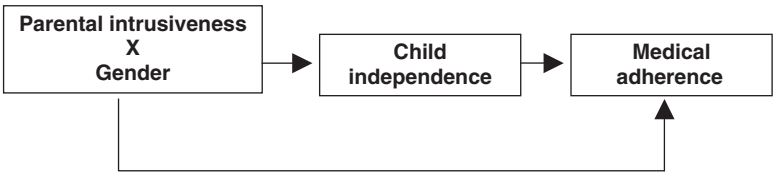

Figure 10.4 Mediated moderation.

gender → medical adherence; see later discussion regarding statistical methods for testing moderator effects as interaction terms). The investigator may then be interested in determining the mediational processes (e.g., child independence) that account for this significant moderational effect. One then tests whether the significant interaction (moderational) effect is mediated by child independence. Although statistically similar, it is important to note that moderated mediation and mediated moderation are not equivalent hypotheses when viewed conceptually. The former is based on the notion that an entire mediational model is significant only at certain levels of a moderator. The latter is based on the notion that a significant moderational effect is mediated by some variable of interest. The reader is referred to Muller and colleagues' (Mulleret, Judd, & Yzerbyt, 2005) article for a more detailed discussion regarding definitional issues and analytical procedures for mediated moderation and moderated mediation.

Common Definitional Inconsistencies

A lack of conceptual and statistical clarity in the study of mediated and moderated effects has become particularly prevalent in mental health literatures where investigators seek to examine factors that mediate or moderate associations between selected predictors and adjustment outcomes. In the child clinical and pediatric psychology literatures, for example, models of predictor-adjustment relationships have become quite complex (e.g., Grych & Fincham, 1990; Thompson, Gil, Burbach, Keith, & Kinney, 1993). Investigators working in these areas have found it necessary to invoke conceptual models that include mediated and moderated effects. Despite the appearance of several useful discussions of differences between mediated and moderated effects (e.g., Aldwin, 1994; Baron & Kenny, 1986; James & Brett, 1984), there continue to be inconsistencies in the use of these terms. More specifically, several types of definitional problems occur with some regularity: (a) vague or interchangeable use of the terms, (b) inconsistencies between terminology and the underlying conceptualization of the variables employed, and (c) a mismatch between written text and diagrammatic figures (see Holmbeck, 1997, for examples).

One concerning issue is that definitions used in the literature are not always consistent, in that they do not adhere to established statistical definitions (see Baron & Kenny, 1986) or the necessary A → B → C causal relationship. Although it is not the case that all investigators must adhere to the same definitions of all terms, it is likely that progress in the field will be hampered if the same term is used in different ways by different scholars. Research studies also often confuse or inappropriately

interchange the terms "mediation" and "moderation" (Holmbeck, 1997). Baron and Kenny provide some examples of this problem from the social psychology literature. For example, some research has referred to variables such as age of onset and socioeconomic status (SES) as mediators. However, it is likely that the effect described is moderational in nature. For example, variables such as age of onset and SES presumably dictate conditions under which a stressor is (or is not) associated with problematic outcomes (i.e., these variables are more likely to serve a moderational than a mediational role). Studies sometimes refer erroneously to an interaction effect as a mediation effect (Holmbeck, 1997). The opposite problem is also possible, where authors describe a moderational effect as mediation without making a clear case for how their variables could serve a mediational function. To be considered a mediational effect, the conceptual relationships between the variables of interest must adhere to a mediational model (A → B → C).

One potential cause for confusion regarding mediational models lies in the distinction between *temporal* and *causal* models (Holmbeck, 1997). For example, variables such as coping and appraisal are often misrepresented as mediators. This error is likely attributable to the use of models illustrating *temporal* pathways (see Thompson & Gustafson, 1996, for several examples of temporally oriented models). For example, one might provide a flowchart whereby a stressor is followed *in time* by various coping processes that are, in turn, followed by changes in psychosocial adjustment (i.e., stressor → coping → adjustment). Flowcharts like these are temporal (rather than mediational) in nature because they specify a time-based sequence of events rather than a causal sequence of events. For example, this type of model could illustrate how a stressor, a coping process, and changes in psychosocial adjustment are all placed *in time* relative to each other. The model becomes causal (and mediational) only if the investigator posits causal associations among the variables. The model just discussed would become mediational if the investigator hypothesized that the stressor actually impacted on or changed the way a person copes (e.g., the stressor causes a person to use more avoidance coping), which, in turn, impacted on subsequent levels of adjustment. Because researchers do not usually expect a stressor to influence or cause changes in the style with which a person copes, coping is not usually viewed as a mediational variable (although it *is* often viewed as a moderational variable).

Frequently, terminological, conceptual, and statistical errors are all present in the same study, such as when investigators conceptualize a variable as a moderator (e.g., coping strategies are hypothesized to serve a protective or buffering function), use the term mediator (rather than moderator) to describe the impact of the variable, provide a figure where the variable is presented as a mediator (rather than as a moderator), and conduct statistical analyses that test neither mediation nor moderation. When such a mismatch between terminology, theory, figures, and statistical analyses exists, findings become particularly difficult to interpret.

STATISTICAL STRATEGIES

Now that the conceptual differences between mediators and moderators have been clarified and their potential application within research has been addressed, the next

portion of the chapter is devoted to a more complete discussion of how mediational and moderational effects are tested statistically. For a more detailed overview of statistical methods and post-hoc probing (introduced later), as well as Statistical Package for the Social Sciences (SPSS) syntax, the reader is referred to relevant papers by Holmbeck (1997, 2002). This chapter focuses primarily on two types of statistical strategies: multiple regression (as suggested by Baron & Kenny, 1986, and as used by several investigators) and structural equation modeling (SEM; see Tabachnick & Fidell, 2007, for a relatively straightforward discussion; also see Bollen, 1989; Byrne, 1994; Hoyle, 1995; Jaccard & Wan, 1996; Joreskog and Yang, 1996; Mueller, 1996). Although SEM is often considered the preferred method because of the information it provides on the degree of fit for the entire model after controlling for measurement error (Peyrot, 1996), proper use of regression techniques can also provide meaningful tests of hypotheses. Moreover, for investigators working in areas of study where sample *N*s are often relatively small, such as pediatric psychology, use of regression techniques (as opposed to SEM) may be necessary because of power considerations (see Tabachnick & Fidell, 2007, for a discussion of sample size and SEM).

Statistical Approaches for Testing Mediation

This section addresses mediational analyses. Techniques utilizing both regression and SEM are presented, with a focus on the importance of evaluating the potency of a mediator. Criticisms of and alternatives to common approaches used in testing mediation are discussed, followed by an applied statistical example.

Regression Approach to Testing Mediated Effects

According to Baron and Kenny (1986), four conditions must be met for a variable to be considered a mediator: (1) The predictor, A, must be significantly associated with the hypothesized mediator, B (letters refer to variables in Figure 10.1); (2) the predictor, A, must be significantly associated with the dependent measure, C; (3) the mediator, B, must be significantly associated with the dependent variable, C, after controlling for the effect of A; and (4) the impact of the predictor, A, on the dependent measure, C, is less after controlling for the mediator, B. A corollary of the second condition is that there first has to be a significant relationship between the predictor and the dependent variable for a mediator to serve its mediating role. In other words, if A and C are not significantly associated, there is no significant effect to mediate. Such a bivariate association between A and C is not required in the case of moderated effects (nor is it required in the case on an indirect effect, as discussed later).

These four conditions can be tested with three multiple regression analyses (see Eckenrode, Rowe, Laird, & Brathwaite, 1995, for an example that includes figures as well as a complete explanation of this data analytic strategy). This strategy is similar to that employed when conducting a path analysis (J. Cohen & Cohen, 1983; Nie, Hull, Jenkins, Steinbrenner, & Bent, 1975). The significance of the A $\rightarrow$ B path (condition 1 above) is examined in the first regression, after controlling for any covariates. The significance of the A $\rightarrow$ C path (condition 2) is examined in

the second regression. Finally, A and C are employed as simultaneous predictors in the third equation, where C is the dependent variable. Baron and Kenny (1986) recommend using simultaneous entry (rather than hierarchical entry) in this third equation so that the effect of B on C is examined after controlling for A, and the effect of A on C is examined after controlling for B (borrowing from path analytic methodology; Nie et al., 1975). The significance of the B → C path in this third equation is a test of condition 3. The relative effect of A on C in this equation (when B is controlled), in comparison to the effect of A on C in the second equation (when B is not controlled), is the test of condition 4. Specifically, A should be less highly associated with C in the third equation than was the case in the second equation (see later discussion for a more detailed example). In other words, one is interested in determining whether the predictor → outcome effect is less after controlling for the mediator. The question that arises in this type of analysis is How much reduction in the total effect is necessary to claim the presence of mediation? The degree to which the effect is reduced is an indicator of the potency of the mediator. As Baron and Kenny discuss, it would be unusual in psychology for this A → C effect to be reduced from significance to zero.

In the past, some researchers have reported whether the predictor → outcome effect drops from significance (e.g., $p < .05$) to nonsignificance (e.g., $p > .05$) after the mediator is introduced into the model. This strategy is flawed, however, because a drop from significance to nonsignificance may occur, for example, when a regression coefficient drops from .28 to .27 but may not occur when the coefficient drops from .75 to .35 (Holmbeck, 2002). In other words, it is possible that significant mediation *has not* occurred when the test of the predictor → outcome effect drops from significance to nonsignificance after taking the mediator into account. On the other hand, it is also possible that significant mediation *has* occurred even when the statistical test of the predictor → outcome effect continues to be significant after taking the mediator into account. Clearly, a test is needed for the significance of this drop. This significance in the drop of the total effect on inclusion of the mediator in the model is computed using the Sobel test (Sobel, 1988).

Failure to test the significance of a mediated effect is likely to lead to false conclusions. One risks both false-positive and false-negative conclusions when one fails to test the significance of a mediated effect (Holmbeck, 2002). Suppose one tests the utility of a mediational model and seeks to determine whether the significance of the total effect drops to nonsignificance after the mediator is taken into account. Also suppose that this "drop to nonsignificance" criterion is used as the basis for whether significant mediation has taken place. If an initial total effect was just below the $p < .05$ threshold of significance (e.g., $p = .049$) and then dropped so that the significance level was now just above the $p < .05$ threshold (e.g., $p = .061$), one might conclude that significant mediation has occurred. Upon further analysis (using the strategy employed here), however, one may find that this represents a false-positive conclusion. On the other hand, if the original total effect was well under the significance threshold (e.g., $p = .001$) and remained under the threshold (e.g., $p = .040$) after accounting for the mediator, one might conclude that no mediation occurred (which would likely be a false-negative conclusion). Indeed, analyses using the Sobel equation may reveal that significant mediation

had occurred in this latter case. As noted earlier, the "drop to nonsignificance" criterion is flawed. It is also worth mentioning that there may be other reasons for false-negative conclusions (i.e., Type II errors). For example, there may be limited power to detect an effect, or the measures may be weak (e.g., they may have low reliability).

The reader is referred to an article by Sobel (1988, p. 56) that includes a complete explanation of how to apply the significance test and compute confidence intervals for the indirect effect (for empirical examples, see also Colder, Chassin, Stice, & Curran, 1997; Ireys, Werthamer-Larsson, Kolodner, & Gross, 1994; Lustig, Ireys, Sills, & Walsh, 1996). Holmbeck (2002) provides a detailed explanation, in addition to a computational example. Finally, an interactive website is available that conducts the Sobel test (with significance tests) if path coefficients and standard errors are entered (www.psych.ku.edu/preacher/sobel/sobel.htm).

Structural Equation Modeling Approach to Testing Mediated Effects

The SEM strategy is often preferred when the investigator has more than one measured variable for each of the constructs (or latent variables) assessed. Assuming there is a latent predictor variable (A), a hypothesized latent mediator variable (B), and a latent outcome variable (C), one would first assess the fit of the direct effect (A → C) model (Hoyle & Smith, 1994). Assuming an adequate fit, the investigator then tests the fit of the overall A → B → C model. Assuming the overall model provides an adequate fit, the A → B and B → C path coefficients are examined. At this point, the A → C, A → B, and B → C paths (as well as the A → B → C model) should all be significant (which is analogous to the regression strategy discussed earlier).

The final step in assessing whether there is a mediational effect is to assess the fit of the A → B → C model under two conditions (where the first condition is nested within the second): (1) when the A → C path is constrained to 0, and (2) when the A → C path is not constrained. One then examines whether the second model provides a significant improvement in fit over the first model. As noted earlier, improvement in fit is assessed with a significance test based on the difference between the two model chi-squares. If there is a mediational effect, the addition of the A → C path to the constrained model should not improve the fit. In other words, the previously significant A → C path is reduced to nonsignificance (i.e., it does not improve the fit of the model) when the mediator is taken into account (which is again analogous to the regression approach).

An additional consideration in using SEM to test for mediational effects is the important distinction between *indirect* and *mediated* effects. In this case, we define an indirect effect as a mediational effect where the A → C path is not significant (i.e., there is no effect to mediate, and only the A → B and B → C paths are significant). An example is used to highlight this distinction. Capaldi, Crosby, and Clark (1996) recently conducted a longitudinal study using structural equation modeling, where they concluded that the effect of aggression in the family of origin on aggression in young adult intimate relationships was mediated by the level of boys' antisocial behaviors during adolescence. On the other hand, Capaldi and her colleagues appear

to have found that the direct path between the predictor and the criterion was not significant even though the predictor-mediator and mediator-criterion paths were significant. Although there appears to be evidence for an indirect effect between predictor and criterion, the findings suggest that the mediator does not (and cannot) significantly account for the predictor-criterion relationship (because there was not a significant relationship between predictor and criterion in the first place; Hoyle & Smith, 1994). Thus, Capaldi et al.'s findings fit the criteria for an indirect effect but do not fit the criteria for a mediated effect (as they have been defined here). In the case of such an indirect effect, one must be conservative when discussing interpretations of links between predictor and criterion because one cannot claim that the predictor and the criterion are significantly associated.

Criticisms and Alternative Approaches for Testing Mediation

MacKinnon (MacKinnon et al., 2002; see David MacKinnon's website: www.public.asu.edu/~davidpm/ripl/mediate.htm) has argued recently that the regression approach to testing mediation may be overly conservative, with low power and inaccurate Type I error rates. In addition, MacKinnon and colleagues (2002, p. 87) criticize the Baron and Kenny (1986) approach, commenting that it does "not provide the full set of conditions for the strong inference of a causal effect of the independent variable on the dependent variable through the intervening variable." He and others are currently considering alternative methods in testing the significance of indirect effects (MacKinnon et al., 2002; Shrout & Bolger, 2002; also see David Kenny's website: http://davidakenny.net/cm/mediate.htm).

Some statisticians have argued that the Baron and Kenny (1986) approach, although appropriate in some cases, is overused (Spencer, Zanna, & Fong, 2005). These authors articulate several disadvantages of this approach, including that it (a) requires that the constructs of interest be easily measurable and that their measurement not interfere with the underlying processes (e.g., avoidance of priming effects); (b) relies on correlational, as opposed to causal, evidence; and (c) must adhere to statistical assumptions of multiple regression, which can, at times, be difficult to fulfill. Consideration of experimental designs to support mediational effects is suggested as an alternative to more traditional approaches, especially in cases where the constructs, or processes of interest, are easily manipulated.

Applied Statistical Example of Mediation

In a study of parental overprotection and adjustment in children with spina bifida, Holmbeck and colleagues (Holmbeck, Johnson, et al., 2002) found that parents of children with spina bifida tended to be more overprotective than parents of typically developing comparison children. The authors reasoned that group-related differences in the children's cognitive ability might explain or mediate the observed group differences in parenting style because developmental and intellectual delays might elicit more protective behavior from their parents. Specifically, it was hypothesized that level of cognitive functioning would mediate the relationship between group status and parental overprotective behavior (Figure 10.5).

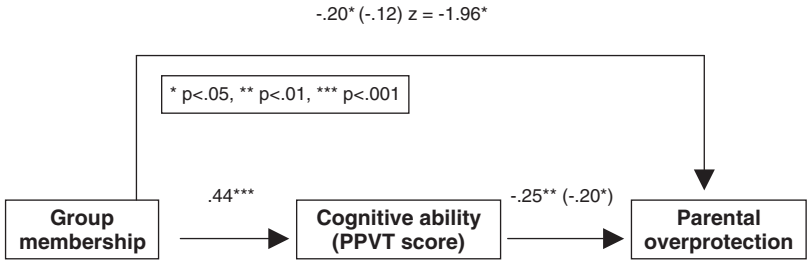

Figure 10.5 Mediational model: Cognitive ability as a mediator of the relationship between group status (spina bifida versus comparison sample) and parental overprotection.

As discussed previously, a mediational model is supported when four statistical criteria are met: (1) The predictor variable is significantly associated with the criterion outcome variable; (2) the predictor variable is significantly associated with the mediator; (3) the mediator is significantly associated with the outcome variable, after controlling for the predictor; and (4) the previously significant predictor → outcome relationship is significantly diminished when effects of the mediator are controlled (Barron & Kenny, 1986; Holmbeck, 1997). These four conditions can be tested with three regression equations (with steps 3 and 4 being tested with a single regression analysis; Holmbeck, 1997, 2002).

In the current example, the first step involved demonstrating that the predictor variable, group status (i.e., spina bifida versus comparison sample), was significantly associated with the outcome variable, overprotection. If the predictor and outcome variables were not found to be significantly associated, there would be no effect to mediate and further testing would not be necessary. In support of their hypothesis, Holmbeck and colleagues (Holmbeck et al., Holmbeck, Johnson, et al., 2002) found that group status significantly predicted parental overprotection, as measured by self-report questionnaires and observer ratings, so they proceeded to test condition 2 (the relationship between the predictor and the mediator). As expected, the second regression showed that the predictor variable (group status) was significantly associated with cognitive ability (as measured by the Peabody Picture Vocabulary Test—Revised; PPVT-R; Dunn & Dunn, 1981), thus supporting the second criterion for mediation.

Regarding the third and fourth mediational criteria, it was necessary to show that PPVT scores significantly predicted parental overprotection when group status was held constant (condition 3) and that the strength of the group status → overprotection effect dropped significantly when the mediator was controlled (condition 4). This was achieved by using simultaneous entry (rather than hierarchical entry) to enter group status and PPVT scores as predictors with parental overprotection as the dependent variable. The results of the analyses provided the information necessary to confirm the mediational effect. First, it was observed that the mediator (PPVT) was a significant predictor of overprotection, with group status controlled (condition 3). Second, in support of condition 4, it was shown that the relationship between group status and overprotection dropped significantly in this last analysis (when PPVT scores were controlled), as compared with the strength of the

relationship observed in the first regression model (when PPVT scores were not controlled). The significance of the drop was tested using Sobel's equation (Figure 10.5; Holmbeck, 2002; Sobel, 1988). This suggests that the mediator (PPVT score) accounted for a significant portion of the variance in the relationship between the predictor (group status) and the outcome (overprotection).

Statistical Approaches for Testing Moderation

Although statistical approaches used to test moderational effects primarily rely on both regression and SEM, this section focuses on regression. The use of various SEM techniques to evaluate moderation has demonstrated significant progress in the past several years, and a discussion of these approaches is, therefore, beyond the scope of this chapter. The reader should be aware that analytic strategies utilizing SEM may be advantageous when one has multiple indicators for the latent variables under investigation. Additionally, SEM analyses are generally less vulnerable to measurement error compared to those utilizing regression (Jaccard & Wan, 1995; Peyrot, 1996; Ping, 1996). For more information regarding SEM approaches used to test moderation, the reader is referred to articles by Moulder and Algina (2002) and Schumaker (2002).

Regression Approach to Testing Moderated Effects

Although the manner in which moderators are tested statistically varies somewhat depending on whether the predictor and/or moderator are continuous or dichotomous (Baron & Kenny, 1986; Mason Tu, & Cauce, 1996), the general strategy is the same regardless of the nature of the variables involved. As noted earlier, a moderator effect is an interaction effect. The preferred strategy is to use the variables in their continuous form (if they are not dichotomies) and employ multiple regression techniques (J. Cohen & Cohen, 1983; S. Cohen & Wills, 1985; Jaccard, Turrisi, & Wan, 1990; James & Brett, 1984; Mason et al., 1996).

The predictor and moderator main effects (and any covariates, if applicable) are entered into the regression equation first, followed by the interaction of the predictor and the moderator (e.g., Fuhrman & Holmbeck, 1995). Depending on the investigator's conceptual framework, the main effects can be entered in a hierarchical, stepwise, or simultaneous fashion (J. Cohen & Cohen, 1983). For example, in analyses involving marital conflict as a predictor and family structure as a moderator, marital conflict and family structure could be entered in any order or simultaneously. The interaction term is represented by the product of the two main effects (e.g., marital conflict $\times$ family structure) and "only becomes the interaction when its constituent elements are partialled" (J. Cohen & Cohen, 1983, p. 305; also see Aiken & West, 1991; Evans, 1991; Friedrich, 1982; Holmbeck, 1989). Thus, although the main effects may be entered in any order, they must be entered prior to the interaction term for the product of these two terms to represent the interaction when it enters the equation.

Given the manner in which the interaction is computed, the main effects (i.e., the predictor and the moderator) will be highly correlated with the interaction term, which can produce "ill-conditioning" error messages when using some statistical

software packages. To eliminate problematic multicollinearity effects between first-order terms (i.e., the independent variable and the moderator) and the higher order terms (i.e., the interaction terms), Aiken and West (1991) recommend that the independent variable and the moderator be "centered" prior to testing the significance of the interaction term. To center a variable, scores are put into deviation score form by simply subtracting the sample mean from all individuals' scores on the variable, thus producing a revised sample mean of zero. Such transformations have no impact on the level of significance of the interaction terms or the simple slopes of any plotted regression lines (see Holmbeck, 2002).

Post-Hoc Probing of Moderational Effects

Failure to conduct post-hoc probing with moderational effects may lead to false conclusions. When one tests for the presence of a moderational effect with multiple regression, one examines whether an interaction between two variables (one independent variable and a moderator) is a significant predictor of an outcome variable, after controlling for the effect of the two predictors. The presence of a significant interaction tells us that there is significant moderation (i.e., that the association between the predictor and the outcome is significantly different across levels of the moderator or that the association is conditional on values of the moderator), but tells us little about the specific conditions that dictate whether or not the predictor is significantly related to the outcome.

For example, if one was interested in whether the association between a parenting variable (e.g., father psychological control; Holmbeck, Shapera, et al., 2002) and an outcome (e.g., school grades) is moderated by group status (e.g., spina bifida versus an able-bodied comparison sample), one would test the interaction of psychological control and group as a predictor of school grades after controlling for the parenting and group main effects. If the interaction is significant, this tells us that the slope of the regression line (i.e., simple slope) that represents the association between parenting and grades for the spina bifida sample is significantly different from the slope for the comparison sample. Unfortunately, the significance of the interaction effect does *not* tell us whether either of the simple slopes is significantly different from zero. In other words, we do not know, based on the initial significant interaction effect, whether the relationship between parenting and grades is significant for the spina bifida sample, the comparison sample, or both samples. Post-hoc probing of the interaction effect via computation of the simple slopes with statistical tests is necessary to gain this information (see Holmbeck, 2002, for computational examples). Additionally, such information also facilitates the plotting of regression lines in figure form.

If an investigator is examining moderational effects by testing the significance of interaction terms, it is likely that the investigator has hypothesized previously that the impact of a predictor on an outcome is conditional on the level of a moderator variable. Suppose one has predicted that "A" will be related to "B" for males, but not for females. A significant "A $\times$ gender" interaction effect only tells you that A is related to B differentially as a function of gender; unfortunately, this statistical test does not answer the research question of interest. Only the post-hoc

probing procedure will tell you if A is significantly associated with B for males, but not for females. In the past, some investigators have merely plotted significant interaction effects and interpreted the significance of regression line slopes based on visual inspection, without conducting post-hoc probes. This strategy is likely to lead to false-positive results; one may be more likely to conclude that a slope is significantly different from zero based on eyeballing than via statistical tests. Thus, post-hoc probing is a critical step in the evaluation of a moderator effect.

One might also be tempted to employ post-hoc probing strategies that differ from those suggested here. For example, if one has isolated a significant interaction effect between a dichotomous variable and a continuous variable, one might choose to examine the bivariate correlation between the continuous predictor and the outcome at each level of the dichotomous moderator. Similarly, if one had found an interaction between two continuous variables, one might be tempted to examine the bivariate correlation between one of the continuous predictors and the outcome at high and low levels (usually based on a median split) of the other continuous variable.

Although this bivariate correlation approach is superior to doing no post-hoc probing, this strategy is less desirable for several reasons. First, the bivariate correlation strategy does not provide the investigator with a regression equation. Without such an equation, the plotting of findings is not a straightforward task. Second, when one generates regression line equations with slopes and intercepts, the slope is in the same metric as the outcome. Given the slope, one is able to determine the increase (or decrease) that will occur in the value of the outcome as a function of a 1 unit increase (or decrease) in the predictor (at a particular level of the moderator). Third, by computing the regression equations one can determine mathematically where the regression lines cross (see Aiken & West, 1991, pp. 23–24), which may be of practical or theoretical interest. Finally, the post-hoc strategy discussed here allows for greater flexibility in computing and plotting regression lines. In the case of an interaction between two continuous variables, one can use the + 1 *SD* convention *or* a variety of other values. When using the bivariate correlation strategy, one typically uses only the median split approach. An additional drawback of the median split strategy is that it yields a correlation for a fairly diverse subsample of participants (i.e., the association between the predictor and the outcome for all individuals above or below the median on the continuous moderator). The post-hoc strategy discussed in this chapter allows one to examine associations between predictor and outcome at any possible value of the moderator.

Applied Statistical Example of Moderation

The following example demonstrates how a moderational effect can be tested using multiple regression analyses. In a conference presentation drawn from the data set described earlier, the authors investigated the moderational influence of cognitive ability on the longitudinal relationship between early parental psychological control and adjustment in children with spina bifida 2 years later (Rose, Buck, Hagstrom, Skelnik, & Stafiej, 2002). Psychological control, or the use of guilt induction and coercive behavior as methods of influencing a child's behavior, has been shown to have detrimental effects on child adjustment in many developmental contexts.

However, it was hypothesized that psychological control might be less harmful in the context of spina bifida because children with greater physical and/or cognitive impairment tend to spend more time at home with their family and tend to be more dependent on their parents for assistance with basic needs. As a result, it is possible that they are less distressed by more intense parental involvement. Moreover, the authors predicted that intellectual ability (as approximated by PPVT score) would moderate the relationship between psychological control and adjustment, such that psychological control would be less strongly predictive of psychological maladjustment (as measured by the Children's Depression Inventory, CDI; Kovacs, 1992) in children who were more cognitively impaired. At the most complex level, it was hypothesized that associations between psychological control and adjustment would be the least strong among those with spina bifida *and* those with low IQ and most strong among those in an able-bodied comparison sample *and* those with high IQ. Thus, 3-way interactions (i.e., moderated moderation) among psychological control, group status, and PPVT scores were tested.

To demonstrate a moderational effect using multiple regression, the main effects and interaction effects of the predictor variables (i.e., group, psychological control, PPVT score) on the dependent variable (CDI score) were analyzed. Before the regression analyses were run, all continuous predictor variables (e.g., IQ) were centered around zero to prevent multicollinearity among the predictors and the interaction terms in the equation and to allow for proper testing of simple slopes. To do this, each predictor variable was transformed by subtracting the sample mean for that variable from all individual scores. Dichotomous variables (e.g., group) were recoded as 0 or 1. To allow for the testing of interaction terms, new variables were created consisting of all possible 2-way and 3-way products of the centered and/or dichotomous predictor variables. These new variables were then tested as interaction terms in the regression analyses.

Once these steps were complete, all predictor variables (i.e., group, psychological control, and PPVT) were entered in the first step of the regression equation to test for main effects. All possible 2-way products of the predictors (i.e., group $\times$ psychological control, group $\times$ PPVT, psychological control $\times$ PPVT) were entered on the second step. Finally, to test the 3-way interaction (in this case, group $\times$ psychological control $\times$ PPVT), all possible 3-way products of the predictors were entered in the third step.

In the current example, contrary to expectations, there were no significant 2-way interactions between PPVT and psychological control. In other words, PPVT did not appear to moderate the relationship between parental psychological control and child depression across the entire sample. The other 2-way interactions were also nonsignificant. However, there was a significant 3-way interaction (group $\times$ PPVT $\times$ psychological control), indicating that there was a significant difference in the interactions among the predictor variables (PPVT $\times$ psychological control) for the two groups. Once significant interactions were identified, simple regression slopes representing the predictor $\rightarrow$ outcome relationship at high and low levels of the two moderator variables were plotted and tested for significance.

Probing of the 3-way interaction revealed that the interaction between psychological control and PPVT in the prediction of depression varied by subsample.

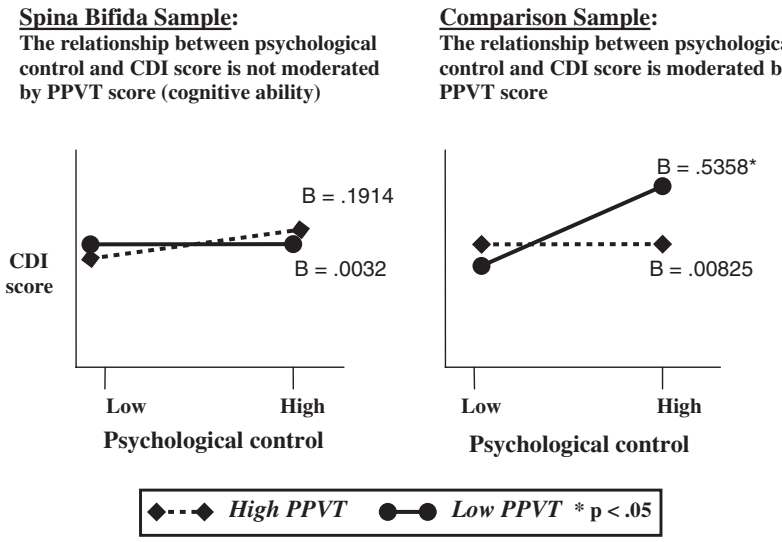

Figure 10.6 Three-way moderational model: In children with spina bifida there was no relationship between parental psychological control and child depressive symptoms. However, in the comparison sample, the relationship between psychological control and depressive symptoms was moderated by cognitive ability of the child (i.e., the relationship was significantly positive among children with relatively low PPVT-R scores).

Among children with spina bifida, the association between psychological control and depression was not significantly moderated by PPVT (Figure 10.6). In fact, psychological control and depression were not significantly related in this condition. However, among children in the comparison sample, the interaction between psychological control and PPVT was significant (Figure 10.6). Consistent with the hypotheses, PPVT scores appeared to influence or moderate the relationship between psychological control and depression in this group. On the other hand, and contrary to the hypotheses, tests of the simple slopes at two levels of the moderator (high PPVT and low PPVT) showed that there was a significant positive relationship between psychological control and depression among children with *low* PPVT scores. Among children with *higher* PPVT scores, the relationship between psychological control and depression was nonsignificant. Thus, it was concluded that the effects of psychological control on depression were moderated by condition status (i.e., psychological control was not associated with depression in children with spina bifida) and by level of intellectual functioning (among comparison children, higher intellectual ability appeared to protect against the effects of psychological control on depression).

Common Statistical Inconsistencies

In addition to the challenges already mentioned regarding inconsistent use of definitional terms in the literature, problems involving statistical testing of mediation and moderation also appear. These concerns fall into three general categories:

(1) use of data analytic procedures that fail to test for mediated and moderated effects when one of these effects is hypothesized, (2) a mismatch between terminology and statistical analyses (mediation is hypothesized, but moderation is tested), and (3) a lack of clarity in discussing implications of statistical results. These challenges often result in varying and discrepant interpretations of findings (see Holmbeck, 1997, for a detailed discussion and examples).

Inappropriate application of statistical strategies often occurs when a variable is included in a model that represents a response to another variable in the model. This issue is likely related to the use of temporal flowcharts that are subsequently interpreted as mediational models (see earlier discussion regarding diagrammatic inconsistencies). Variables such as coping strategies, cognitive appraisals, and causal attributions cannot exist in isolation; they exist only in relation to variables that have preceded them (e.g., marital conflict, a chronic illness). One cannot exhibit a coping strategy in response to marital conflict, for example, if there is no marital conflict in the first place. Some have also argued that such response variables are the mechanism through which the independent variable influences the dependent variable and are therefore best thought of as mediators. In many investigations, such variables are included in a box which is placed, connected by arrows, between antecedent (e.g., stress) and outcome (e.g., adjustment) variables.

Investigators who employ variables such as coping strategies and cognitive appraisals as mediators rarely provide a complete rationale for how these variables could serve a mediational function (as has been defined in this chapter). To do so, they would need to select a specific coping strategy (e.g., denial) and propose how such a coping strategy is expected to be employed with greater (or lesser) frequency when there are higher (or lower) levels of some illness parameter (e.g., severity of illness; Frese, 1986; A $\rightarrow$ B in Figure 10.1). They would also need to propose that higher (or lower) rates of certain maternal adjustment outcomes are expected when this particular coping strategy is employed with greater (or lesser) frequency (B $\rightarrow$ C in Figure 10.1). Finally, they would need to propose that the illness parameters are expected to be associated with the maternal adjustment outcomes (A $\rightarrow$ C in Figure 10.1).

RESEARCH APPLICATIONS IN CLINICAL CHILD PSYCHOLOGY

Now that mediators and moderators have been defined and differentiated and statistical approaches have been discussed, the next portion of this chapter focuses on the relevance and importance of mediator and moderator analyses in clinical child and pediatric psychology research. Interestingly, the study of mediators and moderators can be applied to a wide range of research questions. Specifically, mediational and moderational processes can be proposed in correlational and regression-oriented predictive utility studies, group differences research, longitudinal investigations, complex model testing (with structural equation modeling), studies of interventions including randomized clinical trials, and temporal relationship models in which "What precedes what?" questions are of interest. In this section, we discuss the application of mediation and moderation across three types of research

studies: longitudinal studies, risk and protective factor research, and intervention trials.

Longitudinal Research

Longitudinal mediational analyses allow researchers to test specific causal theories about time-ordered relationships among variables and the particular mechanism or pathway by which a relationship occurs. Such analyses differ from standard mediational analyses only insofar as they include variables gathered at different points in time. For example, information gathered using longitudinal data can advance what is known about the development of health behaviors, psychopathology, and problem behavior by allowing researchers to test the degree to which more immediate or proximal etiological factors can explain links between more remote or distal risk factors and adjustment, thereby contributing to intervention and prevention efforts (Weisz & Hawley, 2002). Furthermore, longitudinal analyses allow for a life span developmental perspective, which is helpful in understanding variability in adjustment trajectories (Cicchetti & Rogosch, 2002). In recent years, longitudinal mediational analyses have been used creatively to explain developmental pathways to resilience or maladjustment, beginning in early childhood and continuing through late adolescence (Masten et al., 1999). Knowledge of high-risk time points or critical transitional periods in child development may also facilitate the development of more effective treatments and interventions (a point discussed in more detail later; Weisz & Hawley, 2002).

Unlike research that is based solely on cross-sectional data, the use of mediational models with multiple data points enables the researcher to clarify the temporal sequencing of causal processes by ruling out bidirectional pathways (Holmbeck, 1997). Continuing with the mediational example discussed earlier, an investigator can propose a longitudinal hypothesis by arguing that high levels of parental intrusiveness are likely to be associated with *subsequent* decreases in child independence. The researcher might also hypothesize that low levels of child independence are likely to be associated with *subsequent* decreases in medical adherence. Longitudinal data would not only permit an examination of these hypotheses but would also allow the investigator to rule out the possibility that lower levels of child independence or lower levels of adherence are associated with *subsequent* increases in parental intrusiveness (an equally plausible scenario with a causal direction opposite from that proposed in the original model). Such hypotheses could not be tested with cross-sectional data.

At a minimum, two data points are needed to begin testing causal time-ordered mediational hypotheses (although three or more are preferred; Cole & Maxwell, 2003). To test the predictor → mediator link, initial Time 1 data are used for the predictor and Time 2 data are used for the mediator *after* controlling for Time 1 mediator data. More generally, partialing out the Time 1 dependent variable data from the Time 2 dependent variable data essentially converts the Time 2 dependent measure into a residual change variable. That is, a given participant's residual score represents the change in rank for this participant (relative to the other participants in the sample) on the dependent measure between Time 1 and Time 2. Similarly,

when examining the mediator → outcome link, initial Time 1 data are used for the mediator and Time 2 data are used for the criterion outcome, after controlling for the Time 1 data on the criterion outcome (Cole & Maxwell, 2003). One continues in this way when testing all pathways in the mediational model. It is important to note that if one has 3 waves of data, one could use Time 1 data for the predictor, Time 2 data for the mediator (controlling for Time 1 mediator data), and Time 3 data for the criterion outcome (controlling for Time 2 criterion outcome data).

Latent growth curve modeling is another statistical method by which to evaluate mediational processes using longitudinal data. Cheong, MacKinnon, and Khoo (2003) describe this type of mediational analysis as measuring two distinct parallel processes: (1) *growth* in the mediator and (2) concurrent *growth* in the outcome variable. This approach builds on the traditional A → B → C mediational model, in that the independent variable affects the growth (or trajectory) of the mediator, which in turn impacts the growth (or trajectory) of the outcome variable. In their study, Cheong and colleagues evaluated the effects of a prevention program geared toward reducing drug use and improving health behaviors among high school football players. Their results supported a longitudinal mediational model; exposure to the prevention program was found to increase perceived importance of team leaders over time, which, in turn, resulted in improvement in nutrition behaviors over time. Although this approach requires large sample sizes and multiple data points, it holds several advantages: It (a) allows for individual differences in development, (b) measures the impact of change in the mediator on *change* in the outcome, and (c) is able to model various trajectory shapes, including curvilinear trajectories (Cheong et al., 2003).

Longitudinal data can also be useful for testing moderator effects within a longitudinal or developmental framework. Using Cole and Maxwell's (2003) approach described earlier, it is necessary to control for earlier levels of the outcome variable when testing the relationship between a predictor (or a predictor-moderator interaction) and a later outcome. MacKinnon and Lockwood (2003) describe longitudinal moderational analyses that parallel Cheong and colleagues' (2003) approach to longitudinal mediation. These authors utilize growth modeling techniques to capture moderational effects on trajectories of change, or growth, over time. For example, one could test whether the impact of peer relationships on growth in deviant behavior during adolescence varies based on ethnicity.

Risk and Protective Factor Research

The study of risk and protective factors often implies mediational and moderational processes. Generally, research on risk and protective factors is focused on understanding the adjustment of youth who are exposed to varying levels of adversity. There is evidence that both contextual factors (e.g., SES, family-level functioning, peer relationships) and developmental variables (e.g., cognitive skills, autonomy development) can significantly influence outcomes for individuals living under adverse conditions, and thus serve a moderational role (Coaklay, Holmbeck, Friedman, Greenley, & Thill, 2002; Holmbeck et al., 2003; Holmes, Yu, & Frentz, 1999; Masten et al., 1999; Rutter, 1990). Risk and protective processes have been

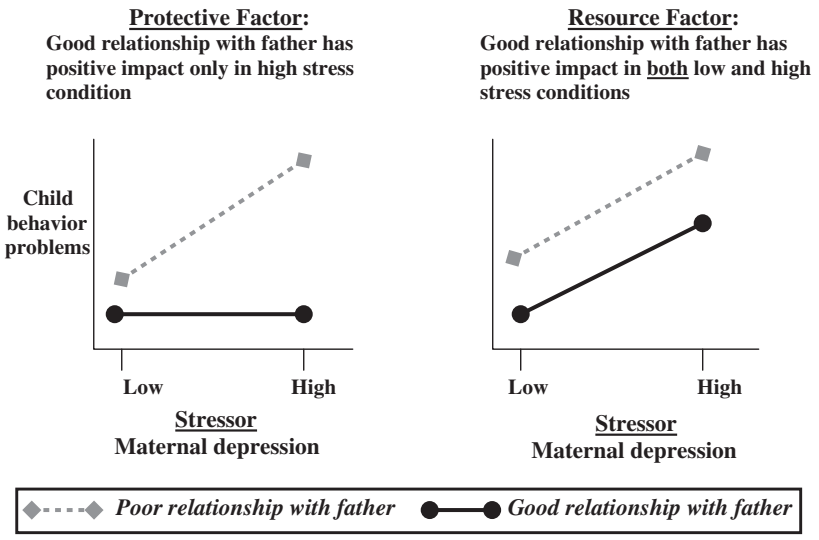

Figure 10.7 Factors associated with favorable outcomes: Protective and resource factors.

explored in the study of resilience, a term used with increasing frequency. Resilience refers to the process by which youth successfully navigate stressful situations or adversity and attain developmentally relevant competencies (Masten, 2001). Appropriate application of the terms commonly used to identify processes that influence the adjustment of youth exposed to adversity is necessary to promote terminological consistency. Thus, a brief description of several terms, and their relevance to moderational analyses, is presented using hypothetical examples (presented in Figures 10.7 and 10.8).

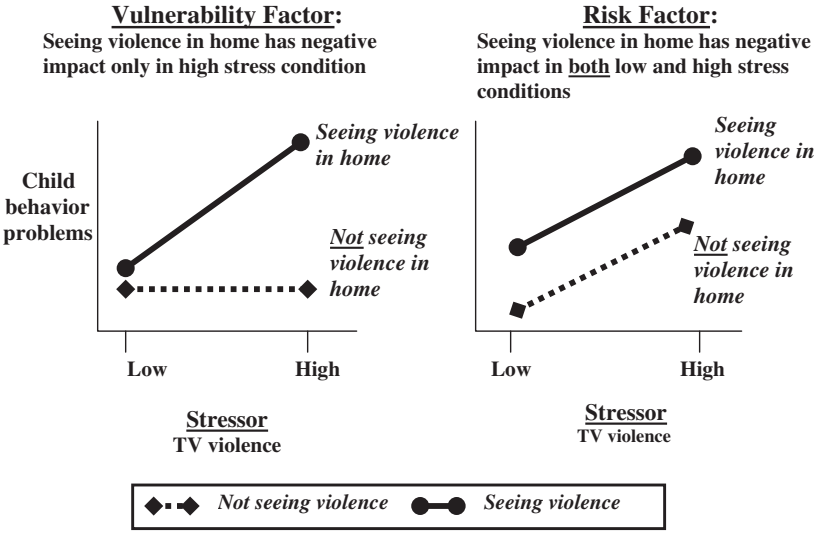

Figure 10.8 Factors associated with unfavorable outcomes: Vulnerability and risk factors.

Protective versus Resource Factors

A protective factor either ameliorates negative outcomes or promotes adaptive functioning. To isolate a true protective factor, however, there must be a particular stressor that influences the sample under investigation. The protective factor serves its protective role only in the context of adversity; a protective factor does not operate in low adversity conditions.

Protective factors are contrasted with resource factors. Specifically, a factor that has a positive impact on the sample *regardless* of the presence or absence of a stressor is a resource factor (Rutter, 1990). For example, if a positive father-child relationship reduces behavior problems only in children of depressed mothers, but has no impact for children of nondepressed mothers, then the father-child relationship would be conceptualized as a protective factor (Figure 10.7). However, if the positive father-child relationship reduces behavior problems in all children, regardless of mother's level of depression, then it would be conceptualized as a resource factor (Figure 10.7; Rutter, 1990). A model may also identify a positive father-child relationship as both a protective and a resource factor if it reduces behavior problems in children who have depressed mothers *more* than in children who have nondepressed mothers, but if it also produces a significant reduction in behavior problems for all children, regardless of level of maternal depression. It is important to note that a protective factor represents a moderational effect (see the statistically significant interaction effect in Figure 10.7), whereas a resource factor represents an additive effect. Statistically, a resource factor emerges as two main effects.

Risk versus Vulnerability Factors

Risk and vulnerability factors operate in much the same way as resource and protective factors, but in the opposite direction (Figure 10.8). A vulnerability factor is a moderator that increases the chances for maladaptive outcomes in the presence of adversity (Rutter, 1990). Similar to a protective factor, a vulnerability factor operates only in the context of adversity. By contrast, a variable that negatively influences outcome regardless of the presence or absence of adversity is a risk factor (Rutter, 1990). For example, witnessing violence in the home environment is conceptualized as a vulnerability factor if it only increases behavior problems in children who are also exposed to a stressor, such as viewing extensive violence on television (Figure 10.8). A vulnerability factor is a moderator and is demonstrated statistically with a significant interaction effect. Witnessing violence in the home can be conceptualized as a risk factor if it results in an increase in child behavior problems for all children, regardless of the amount of TV violence witnessed. As with resource factors, a risk factor represents an additive effect (i.e., two main effects). A model may also identify a factor as being both a risk and a vulnerability factor if it increases the chance of a maladaptive outcome in samples with and without exposure to a stressor, but increases the chances for maladaptive functioning significantly more in the sample with the stressor.

To summarize, if a factor significantly promotes or impairs the chances of attaining adaptive outcomes in the face of a stressor, then it operates via protective or

vulnerability mechanisms, respectively. In these cases, the factor serves a moderational role. However, if a factor significantly promotes or impairs the chance of attaining adaptive outcomes without differentiating between the presence or absence of a stressor, then it is conceptualized as operating via resource or risk mechanisms, respectively. Given that new research in the field has found that predictor → outcome effects may be moderated by developmental and contextual factors, there is an impetus for investigating these mechanisms at a more process-oriented level.

Intervention and Prevention Research

Research that involves a randomly assigned intervention as the predictor variable provides a particularly powerful design for drawing conclusions about causal mediational relationships (Clingempeel & Henggeler, 2002; Howe, Reiss, & Yuh, 2002). These types of models have three important strengths. First, significant mediational models that involve interventions inform us about the mechanisms through which treatments have their impact (Kraemer, Wilson, Fiarburn, & Agras, 2002; Rutter, Pickles, Murray, & Eaves, 2001; Weersing & Weisz, 2002). Simply put, with such models one is able to ask *how and why* an intervention or prevention program works (Kraemer et al., 2002; MacKinnon & Lockwood, 2003).

Second, as noted by Collins, Maccoby, Steinberg, Hetherington, and Bornstein (2000), if a manipulated variable (i.e., the randomly assigned intervention) is associated with change in the mediator, which is in turn associated with change in the outcome, there is significant support for the hypothesis that the mediator is a *causal* mechanism. One is more justified in invoking causal language when examining mediational models in which the predictor is manipulated than in mediational models in which no variables are directly manipulated.

Third, when one isolates a significant mediational process, one has learned that the mediator may play a role in the maintenance of the outcome (e.g., problem behavior). In this way, knowledge about mediational processes in the context of randomized clinical trials (RCTs) informs us about etiological theories of disorders (Kraemer et al., 2002). It is also important to note that when examining mediational processes in RCTs, one must demonstrate (via the research design and the timing of the data collections) that changes in the mediator *precede* changes in the treatment outcome (Kraemer, Stice, Kazdin, Offord, & Kupfer, 2001).

As an example of this strategy, Forgatch and DeGarmo (1999) examined the effectiveness of a parenting-training program for a large sample of divorcing mothers with sons. They also examined several parenting practices as mediators of the relationship between the intervention and child adjustment (see Figure 10.9 for an example of this type of model). Compared to mothers in the control sample, mothers in the intervention sample showed improvement in parenting practices. Improvements in parenting practices were linked with improvements in child adjustment (although, interestingly, the intervention did not have a direct effect on child outcome). As indicated in Figure 10.9, such models not only allow one to test potential mediators within an experimental design, but they also allow one to examine the differential utility of different mediators. In other words, one can determine which mediator best accounts for the effectiveness of a given treatment.

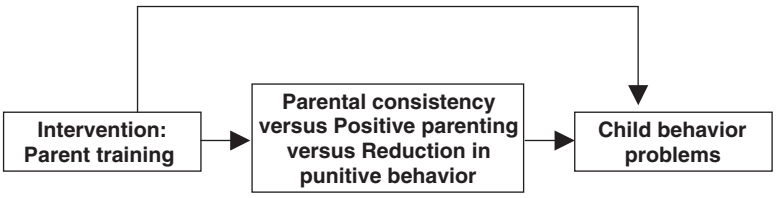

Figure 10.9 Mediators in intervention research: Parenting behaviors as mediators of the relationship between parent training (intervention) and child behavior (outcome).

At a more complex level of analysis, one could posit moderated mediational models. Such a model would not only suggest processes by which a treatment has an effect (i.e., a mediational process) but also which subsamples demonstrate the mediated treatment effects. Findings based on such complex models would provide important information that could be applied to modify the intervention so that it targets relevant mediational processes for specific subgroups of treatment participants.

Kraemer and colleagues (2002) have distinguished between moderators and mediators in the context of RCTs. According to these authors, a moderator is present prior to treatment and a mediator occurs during the treatment process. For example, a moderator might be a pretreatment characteristic of the individual (e.g., gender, IQ, initial severity of symptoms) or a contextual treatment variable (e.g., the setting of the treatment, of the level of exposure to the intervention). A mediator, on the other hand, is a variable that is expected to change as a result of treatment (e.g., parenting behaviors). In other words, while mediators inform *how* and *why* change occurs, moderators provide information regarding *for whom* the intervention or prevention is effective (MacKinnon & Lockwood, 2003). Given these definitions in the context of RCTs, Kraemer and colleagues maintain that the same variable cannot be both a moderator and a mediator (because of their differential temporal relationship to the treatment).

The ability of mediators and moderators to elucidate the impact of an intervention emphasizes the need to hypothesize and measure potential mediational and moderational effects. Unfortunately, although program effects are frequently reported, mediators and moderators of treatment often remain unspecified (MacKinnon & Lockwood, 2003; Weersing & Weisz, 2002). Measurement of potential mediators not only helps us to understand how and why an intervention program is successful, but also illuminates why certain programs may fail. In addition, evaluation of treatment mediators and moderators may facilitate informed modification of intervention efforts and improved cost-effectiveness of services (MacKinnon & Lockwood, 2003).

Research Findings and Future Directions

Grant and colleagues (2006) recently conducted a review of literature investigating mediational or moderational effects in the context of the relationship between stressors and child or adolescent adjustment. Their paper highlights several important

overall findings and challenges, as well as future directions for research incorporating mediation and moderation.

Examination of past moderational research revealed that demographic variables, such as age, sex, and ethnicity, were most frequently analyzed, with relatively few studies focusing on the role of cognition, competencies, and coping. Although age, sex, and ethnicity were not consistently significant across studies, significant findings did reveal some interesting patterns. For instance, it appears that for younger children, symptoms, as reported by parents, may be more strongly associated with stressors, whereas self-report may be more strongly related for adolescents. In terms of gender, girls may be more likely to develop internalizing symptoms in response to a stressor, whereas boys may be more prone to externalizing symptoms. Finally, associations between stressors and maladjustment may be stronger for White youth than for youth of other ethnicities. The moderational role of cognition was consistently supported, especially in the development of depressive symptoms. In terms of individual protective factors, academic competence appears to function as a buffer more consistently than social competence and coping. Finally, social support may protect against negative effects associated with exposure to violence and abuse.

The authors point to several challenges that permeate moderational research. First, inclusion of moderators in analyses frequently lacked theory. That is, analysis of potential moderators largely consisted of non-hypothesis-driven inclusion of demographic variables, with sparse research devoted to theory-based moderator variables, such as cognition. Second, measures and definitions of potential moderators were inconsistent, precluding generalization across findings. Finally, assessment of environmental moderators overrelied on use of child report, which can be problematic given possible common source variance and confounding with other child characteristics (e.g., competence).

In contrast to moderational studies, research studies investigating mediational effects were, for the most part, theory-driven. Although some evidence of statistical inconsistencies and use of inappropriate statistical analyses was found, sophistication of analyses appeared to increase over time. Across studies, family process (e.g., parenting behavior) emerged as a consistent mediator in the relationship between stressors and psychosocial adjustment. The authors noted that the majority of mediational research utilized multiple regression or SEM to analyze mediational effects. Weaknesses of mediational research included an overreliance on cross-sectional data and a failure to test competing mediational models.

This review indicates several future directions for mediation and moderation research in the area of clinical child and adolescent psychology. First, routine incorporation of conceptual models and theory grounded in developmental psychopathology would strengthen research pertaining to moderation. Second, utilization of consistent measures and definitions of moderator variables, such as coping, would promote cross-generalization across studies. Third, research would benefit from inclusion of other sources and methodologies in assessing environmental context beyond child report, such as observational measures and parent report. Fourth, longitudinal data are needed to explore mediational and moderational variables that impact on developmental phenomena. Fifth, recent criticisms of the multiple regression approach for analyzing mediation encourage exploration of newer

statistical methodology based on recent recommendations (e.g., Cheong et al., 2003; Cole & Maxwell, 2003; MacKinnon et al., 2002; Shrout & Bolger, 2002). Finally, mediation research should incorporate tests of competing mediational and integrative models investigating how family, child, and environmental variables operate in conjunction with each other (i.e., mediated moderation and moderated mediation).

IMPLICATIONS OF RESEARCH FOR CLINICAL PRACTICE

Now that the applications of mediational and moderational analyses in the field of research have been reviewed, it may be helpful to address briefly the relevance of this type of research to clinical practice. This chapter has reviewed the utility of mediational and moderational research for a variety of purposes: confirming or refuting theories about developmental pathways, identifying circumstances that promote or inhibit successful adaptation to chronic healthy or life stressors, and identifying specific mechanisms associated with positive outcomes following a therapeutic or clinical intervention. These accomplishments are important not only for the researcher, but also for the clinician seeking empirically validated prevention or intervention techniques.

For example, suppose that a clinician is treating a young child with a chronic illness (such as diabetes). The child lives with a single parent who is suffering from Major Depressive Disorder and is unable to provide consistent care. Mediational and moderational research can be of use to a clinician treating a child living under these circumstances. First, mediational research involving random assignment to intervention conditions can provide information about the relative efficacy of various treatments or combinations of treatments (e.g., medication, dietary changes, exercise, and all possible combinations of these). Mediational research can also clarify which specific aspects of traditionally accepted treatments are most responsible for observed changes, so that unnecessary components of the treatment may be eliminated. Mediational research such as this allows the clinician to make more informed decisions about treatment (e.g., which treatment is best, based on the means by which the treatment is shown to operate) and also provides information about critical points for intervention in a child's development.

Moderational research is also readily applicable to the clinical setting. First, it can clarify whether a treatment is equally effective when applied in the presence of various demographic, genetic, and environmental factors (e.g., child gender, family medical history, level of parental involvement). For example, perhaps a treatment found to be most effective in a sample of children in a rural setting is not very effective when applied to children living in an urban area. Or perhaps risk and protective factor research suggests that an intervention is beneficial only under particular conditions or circumstances. By considering these issues, moderational research fine-tunes the conclusions drawn from typical intervention studies. Through awareness of the variables that moderate the appropriateness of a given treatment, a clinician can tailor interventions to the individual rather than providing a "one size fits all" treatment.

SUMMARY

In this chapter, we provided definitions of mediated and moderated effects, reviewed statistical approaches, and discussed how such effects can be examined in longitudinal, intervention, and risk and protective factor research. The need for consistency in the use and analysis of mediation and moderation in the child clinical and pediatric psychology literatures was highlighted. It is recommended that care be taken in discussing these processes and that investigators be clear about what statistical approaches are appropriate for a given hypothesis. Because research in pediatric and child clinical psychology has important treatment, prevention, and public policy implications, appropriate modeling and statistical techniques are needed to move the field toward greater understanding. Additionally, this research could benefit from (a) greater inclusion of theory and complex models, (b) consistency in definitions and measures of mediational and moderational constructs, (c) integration of multiple methods and sources to assess environmental variables, (d) use of longitudinal data, (e) exploration of newer statistical methodologies, and (f) generation and testing of complex models.

To conclude, we provide a list of important concepts pertaining to mediational and moderational research. This list, which incorporates the major points of this chapter, is intended to serve as a guide and inspiration for future mediational and moderational research:

- If a researcher is interested in examining a relationship between two variables, the use of mediational and moderational models can lead to deeper and more comprehensive knowledge about the relationship by providing information about the conditions under which the two variables will be associated (moderation) and also about the intervening processes that help to explain the association (mediation).
- The same variable can either mediate *or* moderate a relationship between a predictor and an outcome, depending on the nature of the research question and the role the variable is thought to play in the relationship.
- Mediational models are usually proposed after an association between predictor and criterion variables is well established in the literature. If there is no established relationship, one may be better served by first examining associations among the predictor and criterion variables.
- Mediational models are causal models that illustrate a pathway of influence among variables. In this chapter, we distinguished between causal and temporal models.
- Risk, protective, vulnerability, and resource factors can all be identified and illustrated with the use of a moderational model.
- Mediational models are particularly informative and compelling when they are tested with longitudinal data and/or when they are tested in a randomized clinical treatment research design.
- The statistical strategies used to test mediational and moderational effects are distinct. The models differ both conceptually and statistically.

- In choosing a statistical approach for testing mediation and moderation, one should take into account the nature of one's data (e.g., sample size), as well as the strengths and weaknesses of each approach.
- Post-hoc probing is necessary to understand the nature of mediational and moderational effects.
- It is possible to propose complex theoretical models that include mediated moderation and moderated mediation.
- Mediational and moderational models are useful not only to researchers. Because they can provide information about developmental pathways and conditions affecting treatment efficacy, these models can be useful to clinicians as well.
- With the use of mediational and moderational models, researchers can test competing theories about relationships among variables of interest. By directly testing two or more alternative models, a researcher can determine statistically which theoretical model best captures or explains observed relations among the variables of interest.

As has been emphasized throughout this chapter, attention to mediation and moderation has great potential to enrich the field of clinical child and pediatric psychology. It is hoped that this chapter will encourage more widespread and accurate use of these concepts and techniques to enhance the scientific contribution of research.

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CHAPTER 11

Single-Case Research Designs

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Clinical child and adolescent psychology concerns itself with the assessment and treatment of abnormal behavior or psychopathology. Most branches of contemporary clinical psychology are firmly grounded in the scientific method or empirical approach to studying psychopathology and methods of its amelioration. This is affirmed by the adoption of the Boulder model of training by most clinical psychology doctoral training programs (though the development and growth of professional schools of psychology arguably have detracted from this emphasis), which espouses integration of scientific methods with clinical practice. For years, there has been concern about the disconnect between clinical psychology training in the scientist-practitioner model and real-world clinical practice (Hayes, Barlow, & Nelson-Gray, 1999). However, many argue that instead of abandoning training in research methods, the mandate remains to identify methods of bridging the gap between scientific endeavors and clinical practice through the identification of research methods and evidence-based practices applicable to clinical settings (Gelso, 2006; Hayes et al., 1999; Weisz, 2004; Weisz, Jensen-Doss, & Hawley, 2006).

Scientific inquiry assists in understanding the cause-effect relationships between various life events and situations and the clinical phenomena with which clients present. Such an understanding can be developed either via group or single-subject design methodologies, depending on the particular questions of interest and constraints present. In fact, though there are misunderstandings about each approach (e.g., Aeschleman, 1991) and some suggest or imply that one approach may be better than the other (Bailey & Burch, 2002; Gliner, Morgan, & Harmon, 2000), no one type of design is inherently better and each addresses important questions about clinical phenomena of interest.

Since the early days of the behavioral sciences there has been some debate regarding the most appropriate approach to scientific inquiry (Leary, 2001). Today most researchers in the behavioral sciences, including those in clinical child and adolescent psychology, take a nomothetic approach. This involves studying larger groups of individuals to identify general principles and broadly applicable generalizations (Blampied, 2001). Based on this perspective, using group designs (both between- and within-group designs) is most appropriate. They can assist with answering questions such as the percentage of individuals who respond a particular way given certain conditions (e.g., the percentage of youth with depressed mood

who respond to cognitive-behavior therapy) and the association between variables (e.g., child age or gender and treatment outcome). Others, however, have argued that the idiographic approach must be included to ensure that behavioral processes are understood at the level of the individual (for discussion, see Blampied, 2001; Leary, 2001). Idiographic research, therefore, focuses on describing, examining, and comparing the performance of an individual against his or her own performance at different times or in different situations, and thus typically involves single-case research designs. The current chapter focuses on the latter designs.

The primary goal of this chapter is to provide a discussion of single-case designs (SCDs). Specifically, an introduction to the various SCDs currently in use is described in terms of the processes involved in their use, as well as the benefits and weaknesses of each. Although some have argued that SCDs are most appropriate for testing the relation between particular variables of interest and targeted behavioral patterns because one can experimentally demonstrate a cause-effect relationship (e.g., Bailey & Burch, 2002), we intend to provide no judgment as to whether group or single-subject designs should be given greater value. In fact, the American Psychological Association's task force charged with determining guidelines for classifying empirical support for psychological treatments indicated that either can be an appropriate method for determining whether an intervention is efficacious (Chambless et al., 1996; Task Force on Promotion and Dissemination of Psychological Procedures, 1995), an issue that is discussed more thoroughly in a later section. Instead, our focus is on developing a better understanding of SCDs. We emphasize how SCDs are useful in clinically oriented research, but are also applicable to clinical practice.

CHARACTERISTICS OF SINGLE-CASE RESEARCH DESIGNS

When evaluated from a group design perspective, SCDs may be considered limited in scope and application, or even meaningless. For example, some have argued that SCDs are limited in terms of internal and external validity (Aeschleman, 1991). However, such concerns are grounded in the group design perspective of scientific inquiry, and taking such an approach may be misleading (Hillard, 1993). To appreciate the complexity and utility of the various single-subject designs, one needs to understand their fundamental premises and characteristics.

Repeated Observation of Phenomena of Interest

Single-case designs share in common repeated observation of the phenomena of interest across time under standard conditions (Holcombe, Wolery, & Gast, 1994). Specifically, measurement of the dependent variable occurs multiple times, across at least two separate conditions, which are usually referred to as *baseline* and *intervention*. Such a method of data collection is based on the perspective that determining whether the dependent variable differs from one condition to the next in a meaningful way requires multiple data points within each condition. Doing so allows the researcher to investigate whether desired trends in the data are evident (see "Visual Inspection" section) and protects against making assumptions about

differences that may be spurious or the result of natural variation in the dependent variable of interest.

Often in research, the dependent measure utilized is some form of observable behavioral phenomenon (e.g., aggressive behavior of a young child, out-of-seat behavior of a child with Attention-Deficit/Hyperactivity Disorder, duration of crying by a depressed adolescent). However, there is nothing inherent in SCDs that requires the dependent variable be of such form. End products (e.g., weekly weight loss for overweight children), scores on questionnaires (e.g., obtained scores on the Children's Depression Inventory), or subjective ratings (e.g., measurements of anxiety using the Subjective Units of Distress Scale) can be obtained repeatedly in SCD research, allowing for inferences regarding the impact of the independent variable on measures of outcome. Multiple examples abound regarding the use of various types of dependent measures in a single-subject research paradigm (e.g., Glicksohn, Gvirtsman, & Offer, 1997; Nugent, 1992).

In clinical practice, repeated observation can serve an important role in the therapeutic process. Specifically, if the clinician and/or client gathers data in a systematic and repeated manner, decisions can be made regarding whether to alter interventions when monitoring suggests a lack of progress. How one goes about making changes in clinical interventions will determine whether the practitioner adheres to methods that retain sufficient experimental rigor so as to empirically demonstrate causal relations between changes in the dependent and independent variables. However, because the goal of most clinical endeavors is not to demonstrate experimental control but to document change in the clinical phenomena of interest, such violations may be less important than when engaged in research practices.

Participants as Their Own Experimental Controls

Similar to what occurs in within-subjects group designs, participants serve as their own controls in SCDs. However, there is an avoidance of aggregating obtained data across cases with SCDs. Instead, they are based primarily on demonstrating experimental control through replication of the impact of an independent variable on a case-by-case basis. The main analytic process used to determine experimental control involves comparing an individual's behavior across experimental conditions, rather than comparing across individuals. This is viewed as the relevant comparison, because the real question is whether an individual's behavior has changed as a result of treatment relative to pretreatment behavioral presentation (Morgan & Morgan, 2001). For instance, a child psychologist is more concerned with whether depressive symptoms of an adolescent client have changed as a result of therapy relative to preintervention level of symptoms rather than whether the client's symptoms changed relative to a statistically derived group mean. Note that intrasubject comparison across experimental conditions as an empirical test of the effect of the independent variable on the symptoms of interest is different from completing a normative assessment (i.e., comparing a particular individual's performance or behavioral level to normative data) to determine if a person's current level of functioning is within normal limits. This latter practice is often an important issue clinically and has some relevance to determining whether changes observed are

clinically significant or meaningful (Atkins, Bedics, McGlinchey, & Beauchaine, 2005).

Given the focus on intrasubject variability as the primary issue of importance, SCDs are particularly relevant for psychotherapy research (Hayes et al., 1999; Hilliard, 1993; Lundervold & Bellwood, 2000; Morgan & Morgan, 2001). Typically, psychotherapy involves repeated interactions between the therapist and the client (e.g., child, adolescent, family) as a means of producing change. Further, there is usually a period of assessment in which the therapist learns about the client's presenting problems (analogous to baseline in research scenarios) and then treatment, during which the therapist creates conditions in an effort to produce change in the client's presenting problem. Like the SCD, psychotherapy as a process allows for comparison across conditions.

Replication of the Experimental Effect

Experimental replication of observed effects is one of the primary principles of scientific inquiry (Morgan & Morgan, 2001). Through replication, scientists are able to provide information about the reliability of observed effects, as well as to self-correct when initial findings are not consistently obtained. Unfortunately, replication studies are lacking in the behavioral sciences, including clinical child and adolescent psychology. This is likely due to a host of reasons, an important one being the energy and expense of replicating studies that rely on large groups of participants.

A hallmark of experimentally sound SCDs (i.e., those that allow for a determination of the functional relation between the independent variable and changes in dependent variables) is replication. That is, SCDs set as the standard repeated demonstration of the impact of the independent variable as the measure of experimental control. Support for the interpretation of the independent variable (e.g., training parents in methods of giving clear instructions to their children) as the cause of the change in the dependent variable (e.g., rate of child compliance to adult instructions) increases each time one demonstrates that the latter changes as a function of a change in the former. Because of this, each participant in single-subject research is repeatedly exposed to baseline and treatment conditions. This is in stark contrast to the most common group design used in psychological research, the between-groups design. In the between-groups design, the experimenter compares the aggregate level of the dependent variables across at least two groups: the comparison group, which is not exposed to the independent variable, and the treatment group, which is. In such a design, the experimenter rarely introduces, withdraws, and reintroduces the independent variable while obtaining measurement of the dependent variable across conditions. Instead, there is reliance on statistical analysis to determine whether the obtained difference would occur by chance versus as a result of the treatment.

Replication occurs across participants as well. The nomenclature describing SCDs implies that only one participant is involved (e.g., single-subject design, *n* of 1 designs). However, most research using these designs involves multiple participants (Horner et al., 2005; Leary, 2001), though there is no gold standard

for the number of individuals who should be included in a single study to allow for adequate assessment of intervention effects across individuals. Each time an intervention effect is observed with a different participant, confidence that the independent variable is responsible for the change increases. Thus, when there is a small effect, researchers strive to have more participants to increase confidence in interpretation of the source of the effect. The number of participants needed may be fewer if the effect is large.

There are instances in which repeatedly introducing and removing an intervention is impractical or unethical. For instance, one may implement an intervention focused on skills acquisition (e.g., teaching a child with an intellectual disability how to tie his shoes). In this instance, maintenance of the behavior is probable even if intervention is withdrawn because of the naturally rewarding properties of engaging in the behavior. In other words, once one has mastered the skill of tying one's shoes, the skill may not return to baseline levels when the intervention used to establish that skill is removed. When the intervention is designed to address a serious or threatening behavior (e.g., suicidal self-harm behavior), removing an effective intervention may be considered unethical. In these instances, researchers will rely on SCDs that more heavily focus on intersubject replication and, in some cases, cross-situational replication as a means of investigating experimental control of the independent variable on changes observed in the dependent variable (see "Multiple Baseline Designs" section).

Changing One Variable at a Time

Another essential tenet of SCDs is that only one variable should be altered at a time when moving from one phase of the experiment to the next (Barlow & Hersen, 1984; Hersen, 1982). For example, if a researcher is determining effective interventions for treating nocturnal enuresis, he or she would want to introduce only one intervention (e.g., the urine alarm) at the start of the treatment phase. If more than one variable is changed, it is impossible to determine which variable is producing change in the dependent variable. Using two strategies, such as decreasing evening fluid intake *and* introducing a particular behavioral intervention (e.g., the urine alarm) at the same time, would result in a situation in which it would be impossible to determine which produced any noted improvements that occur. Thus, unless the investigator is actually evaluating an intervention with multiple components (see later discussion), standard practice is to hold all variables constant except for one independent variable as the investigation moves from the baseline to the treatment phase.

Although the general guideline is to change only one variable at a time as one moves from baseline to treatment, as with any rule there are exceptions. Specifically, if a researcher is interested in the combined effect of several independent variables, as is often the case with many empirically supported interventions (Kazdin & Weisz, 2003; Mash & Barkley, 2006), it is acceptable to change more than one variable. For example, Freeman and Piazza (1998) were interested in the effects of a combined intervention for treating food refusal (i.e., stimulus fading, escape extinction, and reinforcement) and therefore introduced several strategies as a package intervention.

In this way, the authors were able to determine the impact of the combined treatment on the targeted behavior. It should be noted, however, that if investigators are using such an approach to experimentally demonstrate control over the dependent variable, all independent variables should be introduced simultaneously, rather than progressively. In this way, the package intervention essentially becomes one independent variable. Although by using a package treatment it is impossible to determine which component of the intervention is producing the change, or whether it is the components in concert, such an approach allows one to determine if the combined intervention impacts the dependent variable.

Utilizing a package intervention is becoming more common as psychotherapies become increasingly manualized (Drozd & Goldfried, 1996; Scaturo, 2005). Thus, at first glance, using SCDs to empirically test the effect of manualized interventions with different client populations or clinical settings seems reasonable. Theoretically this is true, but there are some challenges to testing manualized psychotherapies using SCDs. Specifically, manualized treatments typically involve multiple components that are applied in a sequential, versus simultaneous, fashion. For example, popular manuals to guide parent management training for young children with disruptive behavior disorders typically start by training parents in various positive parenting practices (e.g., use of labeled praise, token systems), followed by discipline strategies (e.g., time-out, response cost). Because of this, using relevant components of SCDs (e.g., repeated measurement) may be useful for documenting clinical outcome but may not be appropriate for empirically demonstrating cause-effect relations. Of course, if the package intervention is implemented such that all components are introduced simultaneously, one can systematically withdraw components of treatment over time to evaluate which components are essential (Kazdin, 2003; see also Jones, 1993).

Graphical Representation and Visual Analysis of Data

Data presentation and analysis differ markedly between SCDs and group designs. The most obvious difference is that data are presented for individual participants rather than as a summarized group aggregate. Additionally, information is typically presented in real-time, graphical representation, allowing the researcher to easily examine the data. Data are represented in the manner that most accurately captures the variable of interest (e.g., frequency per observation, duration of the behavior), typically with each data point representing one observation period (e.g., 1 day, one class period, one session).

Although data gathered using SCDs can be subjected to statistical analysis (for attention to this issue, see Parker et al., 2005; Todman & Dugard, 2001; see also Perone, 1999, for argument for experimental control versus statistical control), traditionally analysis involves visual inspection of data attending to level, trend, and variability within and across phases (Horner et al., 2005; Kazdin, 2003). *Level* refers to mean performance within a condition or phase; *trend* refers to the rate of increase or decrease in the hypothetical best-fit line of the variable being measured within a phase, or slope; *variability* refers to the amount that data vary around the mean or slope within a phase. Hayes and colleagues (1999) suggest that *course* is

also an important consideration when analyzing data gathered via SCDs. They define course as a more flexible method of investigating trend, allowing for a determination of whether data follow a linear or curvilinear pattern. Finally, latency to change in the dependent variable after changes in the independent variable is also important for understanding the impact of the latter on the former (Kazdin, 2003). Specifically, the closer in time between the introduction of the treatment and the change in the dependent variable, the more confident one is in the source of the change. By attending to each of these variables while visually inspecting data obtained, one can make determinations about the reliability and consistency of the intervention effect.

Compared against statistical analysis of data, visual inspection focuses much more heavily on clinical significance than statistical significance (Kazdin, 2003; Zhan & Ottenbacher, 2001). In clinical psychology, there has been increasing focus on clinical significance of intervention effectiveness versus simply statistical significance. Clinical significance refers to value or meaningful difference that an intervention has on a person's functioning or everyday life (Kazdin, 1999). Criteria have been suggested regarding how to determine clinical significance when using group designs (e.g., Jacobson, Roberts, Berns, & McClinchey, 1999; Jacobson & Truax, 1991), and some researchers argue that to determine clinical significance one must analyze the data at the level of the individual (Jacobson & Truax, 1991).

Single-case designs are well suited for attention to clinical significance of intervention effects. Because analysis relies on visual inspection, emphasis is placed on ensuring that there is a large effect present to say with confidence that the independent variable produced a meaningful change in the dependent variables. That is, to trip the belief operant that there is an effect present, single-subject researchers demand notable changes in dependent variables from baseline to intervention phases. Of course, for clinical purposes, there still may be valid reasons to complete norm-based assessment using standardized instruments to determine whether the change altered behaviors of interest so that they are now within typical or nonclinical limits (Hartmann, Roper, & Bradford, 1979).

VALIDITY AND SINGLE-CASE DESIGNS

Paramount to any research methodology is the ability to adequately address issues of internal and external validity. Without careful attention to these variables, obtained findings have little scientific meaning and thus are not helpful to the scientific or practitioner community. *Internal validity* refers to the extent to which one can say with confidence that a change in the dependent variable is the result of the independent variable. Multiple threats to internal validity exist and must be accounted for in order for research findings to be interpretable. A complete discussion of the factors that may threaten the internal validity of research findings is beyond the scope of this chapter. For a more in-depth discussion of these issues, readers are referred to Campbell and Stanley (1966), Scotti, Morris, and Cohen (2003), and Shadish, Cook, and Campbell (2001).

When using group designs, one of the primary methods for controlling for threats to internal validity is through random assignment of participants to experimental

groups (e.g., treatment, no treatment). Though some have argued for the use of randomization procedures in SCDs (e.g., Edgington, 2007), such a practice is not typical. Instead, SCDs adequately address threats to internal validity in several critical ways through other processes. For example, through use of repeated measurement of behavior across time, repeated demonstrations of the impact of independent variables, and intra- and interparticipant comparison, they control for threats such as maturation and history effects. Extraneous variables that threaten internal validity are assumed to be constant across baseline and treatment conditions, and thus any notable differences across the conditions can be attributed to the independent variable. How this is controlled for depends on the type of SCD used. Additionally, because data analysis involves frequent plotting and visual inspection of data, researchers using SCDs have the ability to observe whether aberrant or unusual patterns of behavior emerge during the course of an experiment and can go about searching for (and potentially controlling) the source of variability accounting for those patterns (Morgan & Morgan, 2001; Perone, 1999).

External validity relates to generality of findings and thus refers to the ability to extrapolate findings observed with participants from a particular study to a larger group of individuals who are similar on characteristics considered important. External validity of findings obtained using SCDs is enhanced with each replication of intervention effect across participants, settings, or conditions and/or different measures of the same independent variable (Horner et al., 2005). Thus, SCDs address external validity through two types of replication: direct and systematic replication (Sidman, 1960). Direct replication is accomplished by demonstrating the impact of the independent variable across participants with similar characteristics. As the number of replications of the impact increases across participants, the extent of generality is identified (McReynolds & Thompson, 1986). Systematic replication (i.e., varying some aspect of the original experimental conditions) is also used to address external validity. Thus, using SCDs to demonstrate replication of treatment effects across clinical disorders, settings, clinicians, and so forth allows for the assessment of the extent to which the findings are externally valid.

CHOOSING A BASELINE

With few exceptions, the experimenter utilizing an SCD initiates the process by completing a period of observations of the dependent variable as it occurs in the absence of the independent variable. Conventionally, this first phase is referred to as the *baseline* and is labeled with an "A" when representing the data on a graph. The data pattern that emerges during this phase of the research serves as the point of comparison for patterns of data during other (often intervention) conditions. Thus, careful attention to the patterns that develop during baseline data collection is important, with particular attention to level and trend (Bailey & Burch, 2002; Kazdin, 2003).

In basic research, investigators are typically concerned with establishing baseline responding under tightly controlled conditions that produce patterns of data with little variability (e.g., Perone, 1991; Sidman, 1960). In contrast, applied researchers usually collect baseline data during naturally occurring conditions

(Barlow & Hersen, 1984; Hersen, 1982). Rather than creating the baseline conditions, the applied researcher often simply documents the occurrence of the dependent variable in the typical environment. Thus, obtaining the level of control over the environment necessary to produce data with minimal fluctuation can be difficult. Further, applied researchers, because they are often working with human participants who may be experiencing significant distress, may not have the luxury of continuing baseline observations until a narrowly defined pattern of data emerges. As such, the definition of "stability" or "pattern" is typically viewed differently when in the applied context. This is not to suggest that applied researchers should not be vigilant in minimizing or eliminating variability in data patterns due to measurement errors or other extraneous variables that are controllable (and for which control is warranted). Rather, the point is that the acceptable amount of control, and thus the amount of variability, differs considerably across basic and applied research environments. The remaining discussion focuses on issues primarily relevant to applied research situations.

When conducting baseline observations, applied researchers and clinicians are interested in gathering a sample of the dependent variable sufficient to serve as a standard against which to compare patterns of data produced during other conditions. To date, there remains no definitive criterion for the right length of baseline. The most important consideration when determining whether one's baseline observation period is adequate is that of consistency in data patterns, which then allows one to predict future data patterns in the absence of any change to experimental conditions (Kazdin, 2003). Once a pattern of consistency in the desired direction has developed (e.g., the baseline pattern is increasing if one expects the independent variable to produce decreases), one can introduce the independent variable and obtain interpretable findings. It is conventional to consider a minimum of three data points as sufficient for determining whether consistency or pattern exists (Barlow & Hersen, 1984). However, one can easily find published examples of single-subject research baseline periods that include fewer or greater numbers of baseline observations. Thus, although most single-subject researchers strive for a minimum of three baseline data points, the ultimate criterion of being able to adequately assess the impact of the independent variable prevails.

Although various combinations and permutations could occur, several basic baseline patterns have been described (Barlow & Hersen, 1984; Hersen, 1982). Each has particular implications for the interpretation of the impact of the independent variable on the dependent variable. Figure 11.1 (top graph) depicts a stable baseline using a 6-point baseline observation period with hypothetical data. Here there is minimal variation with no significant increasing or decreasing trends across observations. Introducing the independent variable following this baseline period would result in unambiguous interpretation of its effect, easily determining whether there is no change (i.e., pattern remains the same during treatment), improvements (i.e., data patterns either increased or decreased, depending on the goal of the intervention), or worsening (i.e., again, data patterns either increased or decreased, depending on the goal of intervention).

The second general pattern of baseline is a worsening of the problem (i.e., increasing or decreasing trend). Take, for example, a situation in which a researcher

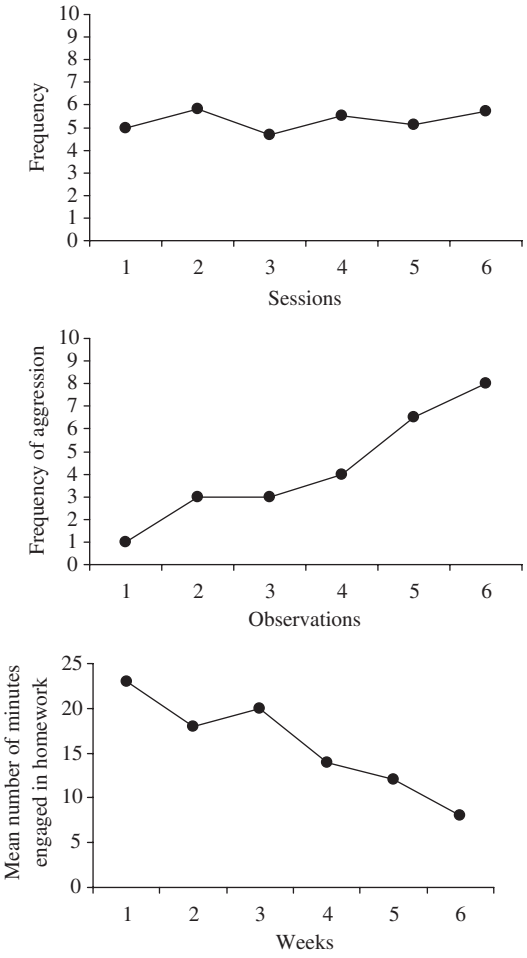

Figure 11.1 The stable baseline: hypothetical data demonstrating slight but negligible variation in data patterns (top graph). The increasing or decreasing baseline: hypothetical data on aggressive acts demonstrating an increasing baseline (middle graph) and hypothetical data on the number of minutes per evening spent engaging in homework (bottom graph), both representing worsening baseline trends.

is measuring the rate of aggressive acts displayed by a youth at recess (see Figure 11.1, middle graph). An increasing baseline pattern would document more frequent occurrence of the aggressive acts as baseline continued, suggesting that the behavior is worsening over time. Alternately, Figure 11.1 (bottom graph) depicts a decreasing trend in the mean number of minutes spent completing homework per evening across successive weeks, which is also indicative of a worsening of the problem during baseline. Both baselines are acceptable patterns for treatment comparison because meaningful interpretations of the impact of the independent variable can be made. Specifically, if the independent variable produces a reversed pattern of data (i.e., decrease in the number of aggressive acts or increase in minutes spent on homework), then the original pattern serves as an adequate comparison. An increasing or decreasing baseline trend becomes problematic, however, when

the independent variable does not produce a reversal in the direction of the behavior pattern. Specifically, such trends may mask whether an intervention has a detrimental effect, such as a worsening of the problem (Hersen, 1982). However, if there were a marked shift in the slope of the data (significantly steeper in either direction), one would be able to determine that the independent variable was having an adverse impact on the dependent measure.

In contrast to increasing or decreasing trends that represent a worsening in the dependent measure, the same data patterns that indicate improvements are troublesome as baseline patterns. Returning to the previous examples, one would not want a baseline that indicates a decreasing trend in aggressive acts or an increasing trend in number of minutes spent engaged in homework. These patterns are problematic as points of comparison because one could not determine whether continued improvements following the start of treatment are the result of naturally occurring factors (i.e., those that resulted in the initial improvement) or of the treatment.

The fourth pattern is the variable baseline (see Figure 11.2), which can be relatively common in applied research given difficulties in establishing tight experimental control over environmental variables. Although one could aggregate the data in a manner that minimizes the variability (e.g., by averaging the data collected over two sessions), this simply masks the variability and does not alter the basic pattern. Sidman (1960) recommended that, when a researcher is faced with such a pattern in baseline data, he or she should seek out and eliminate the extraneous factors that produce such variability. However, as has already been mentioned, this may be difficult or excessively time consuming for the applied researcher.

The utility of the variable baseline as a point of comparison depends on the impact of the independent variable. If the treatment produces a decrease in the variability of the data pattern *and* a change in the desired direction, then it is appropriate to conclude that the treatment had the desired impact. However, if variability is reduced, but the problem remains at unacceptable levels, then interpretation of the independent variable as useful would not be warranted. In this latter situation, although the dependent measure is impacted by the independent variable, the impact is not in the desired direction. Thus, one would not want to retain the treatment.

Another pattern of baseline is the variable-stable baseline. Here, the pattern is initially variable, but then becomes stable. Such a pattern may be achieved by

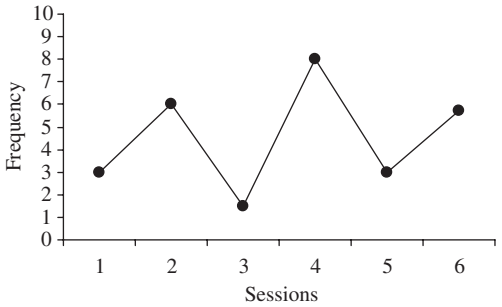

Figure 11.2 The variable baseline.

extending the baseline observation period for sufficient length so as to have the dependent measure come under the control of the baseline conditions. Such an approach might be warranted, for instance, if baseline conditions are novel to the participant. Because the independent variable is introduced following a period of stability, interpretation of its impact is the same as with the stable baseline: unambiguous. However, although perhaps the ideal scenario when initially obtaining variable data, practical and/or ethical constraints may limit the applied researcher's ability to extend the baseline to a sufficient length to establish stability (e.g., as in the case of a brief inpatient hospitalization, or when the dependent measure is severe self-injurious behavior).

The stable-variable baseline presents particular challenges as a point of comparison. Although unstable patterns of baseline data are not in and of themselves problematic, the stable-variable pattern suggests the influence of extraneous variables introduced after the initiation of baseline measurement. This may make it difficult to evaluate the impact of treatment as the extraneous variable may interfere with the independent variable. In these instances, attempts to seek out and control (or at least identify and describe) these outside influences are important.

SPECIFIC SINGLE-CASE DESIGNS

Now that defining characteristics of SCDs and issues related to evaluating baseline data patterns have been presented, the discussion is extended to encompass methods of introducing the independent variable to allow for appropriate comparisons. Specifically, defining features of commonly used SCDs are described.

Exploratory or Quasi-experimental Designs

There are times when it is not necessary or is impractical to utilize rigorous experimental SCDs when attempting to gather useful information about clinical phenomena of interest. Clinical psychology, as well as many other fields, is replete with many examples of interesting anecdotal descriptions of unique cases or systematic yet nonscientific observations of unusual phenomena. Gathering information using *exploratory* methods contributes to our understanding of phenomena of interest and often serves as the impetus for more in-depth and thorough analysis. In this section, we review two types of SCDs that allow for exploration of phenomena of interest without applying rigorous experimental criteria.

The Case Study

A case study is the "detailed study of a single individual, group, or event" (Leary, 2001, p. 321). The field of clinical psychology abounds with case studies of interesting, unique, and challenging clinical cases. In fact, early endeavors in this field to more rigorously analyze variables contributing to clinical phenomena relied heavily on the in-depth analysis of individual clients (Bailey & Burch, 2002). Even today, there are peer-reviewed journals devoted to publishing case studies based on psychotherapeutic work (e.g., *Clinical Case Studies*), a testament to the fact that there is much to be learned about the clinical phenomena experienced by our

clients and the experience of attempting to address those problems by presenting descriptive, typically qualitative summaries of observations and client reports.

Although case studies lack attention to systematic or replicable collection of information and are without controls to threats to validity, they have several beneficial characteristics (Kazdin, 2003). They serve as a source for information and ideas about clinical problems, as well as a source for developing therapy techniques. Further, they can be a beneficial way of studying phenomena of interest that are rare. A well-documented case study can also be used to challenge principles or practices that are considered well established by describing how they did not appear to apply to a given individual. Thus, though case studies are not experimental in nature, they play an important role in clinical psychology.

The Simple A-B Design

As was previously mentioned, the hallmarks of SCDs include repeated observation of the behavior of interest across time and systematic changes in independent variables. The least complex SCD utilized to accomplish these goals is the A-B design. With this design, the experimenter collects measures of the dependent variable during baseline and then again during the treatment condition. Thus, the effect of the independent variable is demonstrated once. The A-B design moves beyond the clinical case study in regard to experimental rigor because of its reliance on repeated and systematic collection of data on the dependent variable of interest.

However, in consideration with other SCDs, the simple A-B design lacks experimental rigor. Specifically, inclusion of only one phase of each of the components (i.e., baseline and treatment) decreases the ability to adequately assess experimental control or internal validity. Theoretically, it is possible that a change in an extraneous variable affecting the individual occurred concurrently with introduction of treatment. If or when this occurs, making inferences about the impact of the independent variable separate from the impact of the extraneous variable is impossible. Take a situation in which a clinician is working with an adolescent experiencing depressive symptoms. Suppose, after gathering baseline data on daily subjective assessment of mood, the clinician initiates behavioral activation strategies. Coinciding with this, the adolescent's family also initiates services with a psychiatrist, who prescribes a medication. If subjective ratings of mood change, the clinician will be unable to determine whether this change is due to the behavioral activation strategies, the medication, or a combination of both. Several other threats to internal validity (e.g., maturation, instrument decay, statistical regression to the mean) are also not well accounted for by the simple A-B design.

Despite limitations in the ability to make definitive conclusions about the causal role of independent variables using the A-B design, it can be useful given certain circumstances, and it offers advantages over the uncontrolled case study methodology. Specifically, for problems that have been resistant to change for significant periods of time, noticeable improvements observed during the B phase offer some support for the intervention as the causal mechanism. Further, the A-B design is appropriate when practical or ethical limitations decrease one's ability to remove the intervention as a means of demonstrating experimental control (e.g., treating

severe, life-threatening behavior). In these situations, the scientist-practitioner is at least gathering data to document whether change is occurring, regardless of whether effects can be fully ascribed to the specified intervention. Finally, for practitioners, reliance on A-B designs can be an important component of documenting that change occurred in a client's presenting problems, even if methodological weaknesses prevent making strong statements about the source of the change. Use of this design may be particularly valid for practitioners when they are implementing empirically supported interventions that have been previously evaluated via more rigorous methods and when one is primarily interested in documenting change rather than determining the source of change, a process referred to as Level II research (Hawkins & Matthews, 1999).

Classic Single-Case Withdrawal Designs

One of the most common categories of SCDs used in research involves systematically and repeatedly introducing and withdrawing an intervention and documenting the impact of those changes on the dependent variable of interest. This group of designs has been referred to as equivalent time-samples designs (Campbell & Stanley, 1966), withdrawal designs, and reversal designs (Kazdin, 2001). In this section, we describe common designs that are the result of permutations of the basic procedure of introducing and withdrawing intervention.

The A-B-A Design

The A-B-A design corrects the primary limitation of the A-B design by including a return to the original baseline conditions. This serves as a means of confirming experimental control over the dependent measure by allowing for repeated demonstration of experimental effect. If the dependent measure returns to a pattern that is similar to the original baseline, then stronger assumptions about the causal role of the independent variable can be made. In contrast to the A-B design, one is less concerned that some extraneous variables coincided with the introduction of the treatment phase. Although this may happen once, the possibility of another change in extraneous variables occurring at precisely the same time that the intervention is both introduced and withdrawn is so remote that it does not constitute a realistic threat to internal validity.

Although the A-B-A design corrects for problems of the simple A-B design, it has limitations. Most important for the clinical researcher, completing an investigation while the participant is exposed to baseline conditions is problematic. Let us return to the example of the adolescent client with depressive symptoms. Assume that an A-B-A design was used in which the B phase consisted of effective use of behavioral activation strategies. Ending the investigation while the participant is exposed to baseline conditions would be problematic if those conditions produced a return in data patterns similar to the original baseline. In this example, the client would be left to deal with his or her problems without the aid of intervention. Although perhaps not required to demonstrate experimental control, clinically and ethically it would seem prudent to reintroduce the intervention so as to address the presenting complaint.

The A-B-A-B Design

By introducing an additional treatment phase, the investigator creates the A-B-A-B design. This design controls for the limitation of the A-B-A design and is more methodologically rigorous due to the fact that three opportunities occur to compare the behavior patterns during baseline and treatment conditions (i.e., A to B, B to A, and A to B).

To illustrate the A-B-A-B design, consider Friman et al. (1999), who investigated the impact of an intervention for bedtime behavior problems exhibited by two children, ages 3 and 10 years. Both children exhibited bedtime problems in the form of calling out from their bedroom and leaving their room after their parents had put them to bed. Baseline data were collected across 15 days, during which time the parents were instructed to respond to the bedtime problems as they would typically. Treatment began on the 16th day and involved parents providing each child with a "bedtime pass" good for one free trip outside of the room past bedtime (e.g., get a drink of water, get another hug). Once the pass was used, it was relinquished to the parents, and no further trips were allowed. The pass was returned to the child on the subsequent night. During intervention, the parents were instructed to ignore all crying and to simply physically guide the child back into his room if he were to leave without the use of the pass.

Results of the experimental analysis are presented in Figure 11.3. Somewhat variable but increasing trends are noted in the dependent measure for both participants during the initial baseline. Following introduction of the intervention, a marked decrease in the targeted behavior occurred, to zero for the 10-year-old participant. Thus, it was demonstrated quite clearly with both participants that the behavior changed in the desired direction during the initial intervention phase. This is tempered slightly, however, with the 3-year-old participant. For him, as intervention continued the behavior problems began to increase again after the initial decrease. Reintroduction of the baseline conditions (i.e., withdrawal of the bedtime pass) resulted in the increased occurrence of the bedtime behavior problems, which decreased to zero for both participants upon the readministration of the intervention. Obtaining zero rates of behavior with the 3-year-old participant during the second intervention phase adds strength to the determination of experimental control given the initial variability in the first intervention phase.

The B-A-B Design

Although single-subject research typically starts with an initial baseline period, there may be instances in which this is not feasible. For example, perhaps the investigator is able to gather data only after some form of intervention is put into place. Or perhaps a participant's behavior problems are so severe that intervention is needed immediately. In either case, it is possible to use the B-A-B design. Although not as complete as the A-B-A-B design (because it does not allow for as many repeated comparisons of the experimental effect), it is more advantageous than the A-B-A design because it ends on a treatment phase (Hersen, 1982).

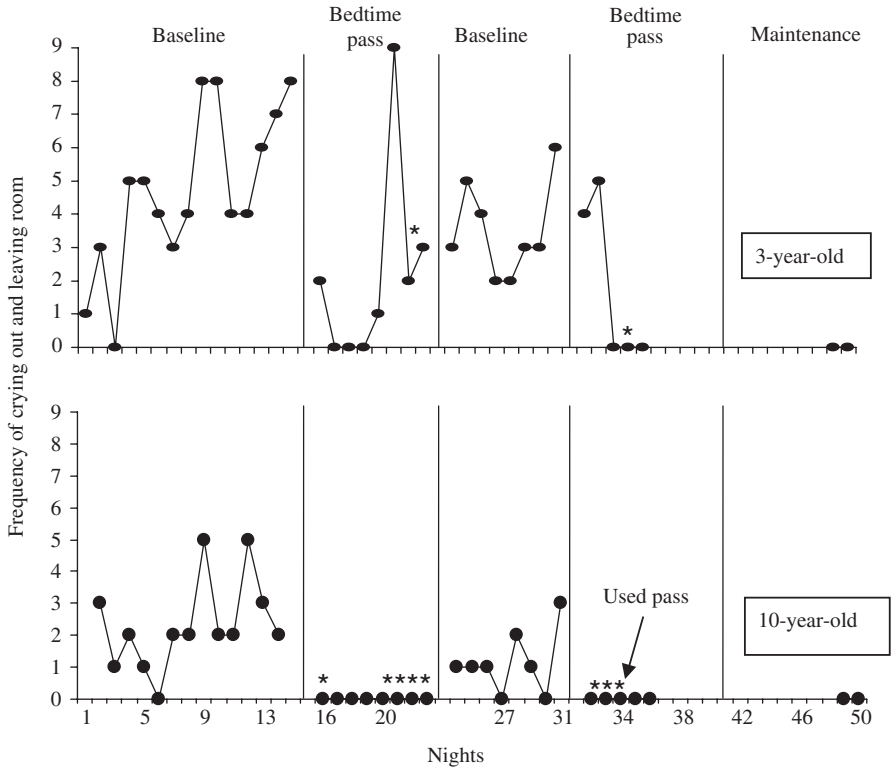

Figure 11.3 Nightly frequency of crying and leaving the room for the 3-year-old child (top) and the 10-year-old child (bottom). Asterisks indicate use of the pass. *Source:* "The Bedtime Pass: An Approach to Bedtime Crying and Leaving the Room," by P. C. Friman et al., 1999, *Archives of Pediatric and Adolescent Medicine, 153,* pp. 1027–1029, figure 1. Copyright 1999 by *Archives of Pediatric and Adolescent Medicine*. Reprinted with permission.

Variations of the Withdrawal Designs

The underlying strategy employed to demonstrate functional control using the various withdrawal designs (e.g., A-B-A, B-A-B)—the repeated introduction and withdrawal of the intervention—allows for various extensions of the basic designs. Thus, perhaps endless permutations or formations of the basic components of the designs can be created to address particular questions of interest. Several examples of such designs are considered next, although the list is not meant to be exhaustive.

A-B-A-B-A-B Design

The basic components of the A-B-A design can be extended, resulting in multiple withdrawals and administrations of the independent variable (e.g., the A-B-A-B-A-B design). Using such a design adds further support for the demonstration of functional control of the independent variable over the dependent variable. By demonstrating multiple times that the data patterns change in the expected direction when the treatment is added and withdrawn, the researcher can be more confident

that the effect is due to the particular treatment variables. Although this might not be necessary when the treatment effect is large (i.e., when there is a significant difference in the data pattern between baseline and treatment conditions), repeatedly demonstrating the desired effect may be necessary when the effect is small (i.e., when the difference between the two conditions is small). Observing minimal or moderate changes multiple times may provide more convincing evidence that the independent variable is producing the impact, above and beyond natural variability.

A-B-A-C-A-C Design

Introducing a condition in which the effects of a second independent variable are assessed, as with the A-B-A-C-A-C design, also can extend the basic A-B-A design. In such a design, the investigator compares the impact of treatment conditions B and C on the dependent variable. However, with such a design, one is not able to compare the *relative* effects of the different independent variables because they are confounded by the extraneous variable of time (Hersen, 1982). Further, order effects may confound the findings. Perhaps the conditions present in C have an impact only after the participant is exposed to the conditions present in B. With such a design, one is unable to determine whether this is the case, and thus there is a threat to the internal validity of the study. If completing intersubject comparisons, one can control for this by using a counterbalancing technique (i.e., participants 1 and 3 are exposed to the conditions in the following order: A-B-A-C-A-C, whereas participants 2 and 4 are exposed to them in the following order: A-C-A-B-A-B). By using counterbalancing, the researcher is able to determine whether the independent variables produce similar effects regardless of order of presentation. Thus, if all participants respond to condition C in similar ways regardless of when they were exposed to this condition, then one has greater confidence in the effect of C alone.

Interaction Designs

As stated earlier, single-subject research typically involves a change in only one variable at a time across conditions. This rule can be applied in a way that allows the investigator to evaluate the combined effects of multiple independent variables by using a withdrawal design. Specifically, by introducing a condition in which two independent variables are present, it becomes possible to assess the impact of the combined intervention relative to a single intervention. In some situations, a particular treatment may produce minimal impact or may produce an impact that is less than that which is desired. By extending the basic components of the A-B-A design, one can assess the impact of adding additional treatment components.

Assume, for example, that a researcher is interested in treating teen smoking. Figure 11.4 shows hypothetical data in this scenario. After establishing a baseline of the number of cigarettes smoked per day, the impact of a reinforcement-based treatment is evaluated. As is shown, although the number of cigarettes smoked decreases somewhat, the desired goal of zero cigarettes smoked is not achieved. Thus, the investigator adds an additional component to the intervention, the use of a nicotine patch. This results in a condition labeled “BC” because it involves

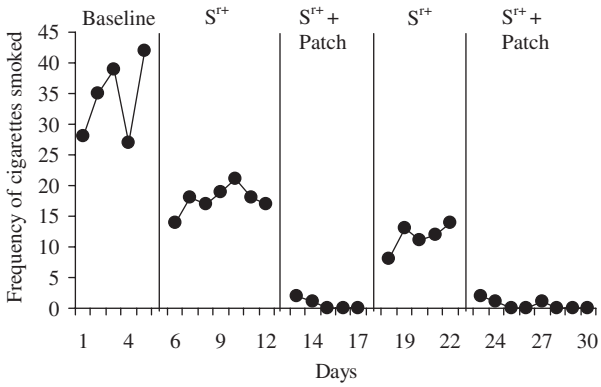

Figure 11.4 Number of cigarettes smoked per day. Hypothetical data of the assessment of interventions for teen smoking.

both components (i.e., reinforcement + nicotine patch). As shown in Figure 11.4, the combined intervention results in decreases in the dependent variable to zero. When the investigator withdraws the nicotine patch, the data patterns return to being similar to the original treatment phase. Finally, the combined intervention is reintroduced, resulting in the elimination of smoking. As illustrated, it is possible to demonstrate the impact of the combined intervention with such a design. Put another way, one is able to investigate the relative impact of condition C over condition B. The number of combinations is theoretically endless, potentially resulting in conditions such as BCD, BCDE, and so forth.

STAGE-PROCESS DESIGNS

Although the withdrawal designs have proven remarkably useful in scientific inquiry, partly because of the flexibility in applying the basic concepts of repeated observations and replication of experimental effect in multiple ways, there are situations in which their use is not appropriate. Withdrawal designs may be considered inappropriate due to a variety of reasons (e.g., ethical, practical). As mentioned earlier, withdrawal of a treatment may be unethical if the behavior problem is severe or life threatening. Also, it may not be possible to withdraw or remove certain therapeutic strategies. For example, once instructions have been given to a participant, it is impossible to remove them. Although the researcher could stop giving them in a particular condition, the fact that the instructions were given earlier could result in a change that is not removable. As another example, if the intervention results in behavior that is supported beyond the treatment conditions, then withdrawal of the intervention may have no impact on the behavior. For instance, if a clinician teaches someone to be appropriately assertive, the reactions of others may maintain use of assertive behavior regardless of whether the clinician removes the intervention. In this scenario, the lack of a return to baseline data patterns might suggest a lack of functional control of the independent variable. However, such is likely not the case; instead, other controlling variables continue to maintain the behavior patterns in the absence of the intervention. Because of these limitations, various other SCDs

are used. These include the multi-elements design, multiple baseline designs, and changing criterion designs.

The Multi-Elements Design

The multi-elements design (also referred to as the simultaneous treatment design or the alternating treatments design) differs from other SCDs in that multiple conditions (i.e., baseline and treatment conditions, two or more treatment conditions) are conducted in rapid succession, with the order of presentation typically determined through random selection, and compared against each other (Miltenberger, 2001). For example, perhaps baseline conditions are in effect on one day, treatment conditions the next, and so forth. Thus, unlike withdrawal designs, the effects of the different experimental conditions are evaluated across the same time frame. This helps eliminate the possibility of extraneous influences on the dependent variable during only one experimental condition. In other words, any extraneous variable is going to impact the dependent variable during all conditions because they are occurring essentially in conjunction. One is able to rule out the extraneous influence as the cause of any differences noted across conditions.

With the multi-elements design, often there are three phases: baseline, comparison (rapid alternation between two or more conditions), and the use of the effective intervention (Holcombe et al., 1994). In some situations, however, the baseline might not be necessary. This may be particularly true if one of the comparison conditions is a baseline condition. One treatment condition is judged to be superior if it produces data patterns in the expected direction at a level that is greater than other conditions. Another component of the multi-elements design is that an equal number of sessions of each condition should be conducted. To ensure discriminated responding across conditions, researchers often pair separate but salient stimuli with each condition. This design may be particularly useful if the investigator is comparing interventions that have an immediate effect, and when the dependent measure is particularly sensitive to changes in stimulus conditions (i.e., reversible; Holcombe et al., 1994).

The multi-elements design can be used to compare a variety of different variables of interest. For example, the design can be used to compare different assessment conditions, such as in functional analysis research (e.g., Anderson, Freeman, & Scotti, 1999). In addition, this design can be used to compare a treatment condition to baseline (e.g., Freeman & Piazza, 1998) or to compare multiple treatments (e.g., Kahng, Iwata, DeLeon, & Wallace, 2000). Such an approach allows the investigator to assess which intervention may prove to be the most effective in changing the targeted dependent variable. The benefit of such a design over other designs is the ability to make such treatment comparisons in a relatively short amount of time.

Multiple Baseline Designs

There are three types of multiple baseline designs: (1) multiple baseline across behaviors, (2) multiple baseline across persons, and (3) multiple baseline across settings (or situations; Hersen, 1982; Miltenberger, 2001). With the multiple baseline design across behaviors, the impact of an intervention across different behaviors

emitted by the same person is evaluated. As such, this is a within-subjects design. The intervention is applied sequentially to the different (presumably) independent behaviors. An example is evaluating the use of a particular reward strategy in affecting a child completing chores, cleaning his room, and completing his homework. The second design—multiple baseline across persons—involves the evaluation of the impact of a particular intervention across at least two individuals *matched* according to relevant variables who are presumed to be exposed to identical (or at least markedly similar) environments. For example, the investigator may compare an intervention across two students who attend math class with a particular teacher, one who attends in the morning and one who attends in the afternoon. Finally, with the multiple baseline across settings design, a particular intervention is applied sequentially to a single participant or group of participants across independent environments (e.g., home and school).

Technically, there must be at least two separate dimensions (i.e., behaviors, settings, or persons) present to utilize a multiple baseline design, although convention suggests a minimum of three or more. Multiple baseline designs are characterized by the presence of only two conditions: baseline and treatment. However, unlike the simple A-B design, treatment is introduced in such a way that one is able to evaluate experimental control of the independent variable. Specifically, the baseline condition is extended for increasing lengths of time as the intervention is introduced with the other dependent variables. Thus, these designs are particularly useful for studying irreversible effects because replication is achieved without withdrawal and reintroduction of the independent variable (Perone, 1991).

An assumption underlying the multiple baseline designs is that the dependent measures (i.e., behaviors, persons, settings) are functionally independent. If this is the case, a change in one dependent variable as a result of the independent variable should not produce changes in the other dependent variables. Treatment effects, therefore, are inferred when the dependent measure changes only when the intervention is applied to it. Variables still exposed to baseline conditions should show little to no change when the treatment is introduced with the other dependent variables. In this way, the multiple baseline designs are weaker than other designs (Barlow & Hersen, 1984; Hersen, 1982). Specifically, experimental control is inferred based on the comparison of nontreated dependent variables as compared to the treated variables, as well as the replication of the effect across behaviors, persons, or settings.

If the dependent variables are not functionally independent (i.e., treatment with one variable produces changes in the other variables), then the ability to establish experimental control is compromised. Determining a priori whether dependent variables are independent can be difficult, however. Take, for example, a situation in which the dependent measures are three problematic behaviors of a student in a classroom (i.e., talking out of turn, throwing spit wads, getting out of seat without permission). As an intervention, the investigator instructs the teacher to respond in a particular way to the behaviors (e.g., differential reinforcement of competing behavior) and evaluates the impact of the intervention using a multiple baseline design. The student may well learn to change each of the behaviors simply by exposure to the treatment for one behavior (e.g., differential reinforcement when the student

talks out of turn affects other behaviors). As this example illustrates, as an artifact of the lack of independence, intervention with one dependent variable may produce changes in the other variables, resulting in the *inability to demonstrate* experimental control. This is problematic because the intervention may be the controlling variable in this situation, but this cannot be demonstrated in a convincing manner.

Kazdin and Kopel (1975) provided three recommendations for addressing problems of dependence across dependent variables. First, they recommend selecting dependent measures that are as topographically distinct as possible as a means of increasing the likelihood that they are independent. However, such an approach still relies on successful guessing, as topographically distinct behaviors may still be functionally related. Second, they suggest that investigators utilize four or more baselines as compared to two or three. They argue that, by increasing the number of baselines, one is increasing the likelihood of selecting measures that are independent. However, as pointed out by Hersen (1982; Barlow & Hersen, 1984), the probability of interdependence may be enhanced with a larger number of dependent variables. Third, Kazdin and Kopel recommend that, when faced with the occurrence of dependence across variables, the investigator withdraw and then reintroduce the independent variable. By doing so, it may be possible to demonstrate adequate experimental control. Although the investigator originally may have selected the multiple baseline design to avoid using a withdrawal design, this may be necessary to adequately demonstrate control.

To a certain extent, there are competing clinical and research goals when considering multiple baseline designs. From a research standpoint, the goal is to identify target dependent variables that are less likely to be influenced by changes in other targeted dependent variables. However, clinically, one is often interested in designing interventions that not only address targeted behavior patterns, but are also likely to produce change that generalizes to other relevant behaviors or contexts. For instance, a therapist working with an adolescent with social anxiety is likely interested in implementing interventions that address anxious responding not only in the classroom setting, but also in the lunchroom, during extracurricular activities, and elsewhere. The clinical goal is to teach skills that will generalize to relevant contexts. Of course, whether generalization happens naturally (e.g., teaching use of relaxation skills in the classroom results in the person also using those skills in the lunchroom) or must be programmed explicitly (e.g., prompting the adolescent to use skills across settings) will be determined only as one progresses.

Nonconcurrent Multiple Baseline Design

The presumption of the multiple baseline designs is that the measurement of the different dependent variables occurs simultaneously. In this way, the designs control for threats to internal validity such as history effects. However, there may be situations in which it is particularly difficult to simultaneously observe multiple individuals who meet the specified criteria, thus limiting one's ability to utilize the multiple baseline across persons design. This may be particularly true in applied or clinical research (Hayes, 1985). In such situations, it may be possible to utilize the nonconcurrent multiple baseline design, originally described by Watson and Workman (1981).

With the nonconcurrent multiple baseline across persons design, each baseline length is predetermined (e.g., 3, 6, 9 days). Then, when a participant with the requisite features is available, he or she is exposed to a randomly selected baseline period. From this point, the methodology is conducted in the same manner as the simple A-B design: baseline observations are conducted for the predetermined length, followed by the application of treatment. In this way, baseline and treatment data are obtained from multiple participants, allowing for adequate between-persons comparisons. If a participant fails to display an acceptable baseline pattern, that individual is dropped from the investigation, although his or her eventual reaction to treatment may still be useful as a replication (Watson & Workman, 1981).

Some (e.g., Harris & Jenson, 1985a, 1985b) have argued that the nonconcurrent multiple baseline design across persons is equivalent to a series of A-B designs with replication. This nomenclature has been recommended because multiple baseline designs rely on simultaneous data collection as a means of demonstrating adequate control over threats to internal validity. As described earlier, by showing that the change in the dependent variable occurs only when the treatment is introduced, the investigator has confidence in the causal relationship. This is strengthened when data with several individuals are collected simultaneously because the design controls for history effects. Harris and Jenson argued that the nonconcurrent multiple baseline design does not control for such effects because the data are not collected across individuals simultaneously; thus, the individual is not exposed to the same environment.

By varying the length of baseline, however, the nonconcurrent multiple baseline design does control for threats to internal validity (Hayes, 1985; Mansell, 1982). Specifically, each time the investigator demonstrates that the dependent measure changes when the treatment is introduced, regardless of the length of baseline, the likelihood that an extraneous variable produced the change for each participant is greatly reduced. Further, by using increasing lengths of baseline, the researcher controls for the possibility that the exposure to baseline conditions naturally produces changes in the dependent measure. Thus, although generally considered one of the weaker SCDs, the nonconcurrent multiple baseline across persons design can serve as a useful methodology if other factors limit the ability to utilize a more stringent design.

Changing Criterion Design

The changing criterion design shares features with both the simple A-B design and the alternating treatments design (Hersen, 1982; Miltenberger, 2001). Specifically, this design is characterized by the presence of only one baseline and one treatment phase. However, what differentiates this design from the A-B design is that the treatment condition is defined by the sequential introduction of different performance goals. In other words, the treatment phase is applied until the targeted dependent variable achieves a specified level of performance. At that time, the goal (i.e., criterion) of performance is altered, and the intervention continues until the behavior again achieves the desired level. Changes in the criterion occur until the dependent measure is occurring at the desired terminal level. As such, the changing

criterion design is particularly well suited for situations in which the investigator is interested in evaluating shaping programs that are expected to result in increases or decreases in the dependent measure (e.g., decreased cigarette smoking, increased level of exercise; Hersen, 1982). Evaluation of the intervention as the causal agent occurs through two comparisons: between the occurrence of the dependent measure during baseline and during treatment, and between the occurrence of the dependent measure across the different levels of the intervention. If the dependent variable changes in the desired direction only when the criterion changes, then the investigator can have confidence in the controlling nature of the independent variable.

SINGLE-CASE DESIGNS AND EMPIRICALLY SUPPORTED TREATMENT

In 1993, the Division of Clinical Psychology of the American Psychological Association (APA) initiated efforts to define the criteria for determining whether treatments for clinical problems are adequately supported by empirical evidence. The task force charged with this assignment, now referred to as the Committee on Science and Practice (CSP), issued their first report in 1995 (Task Force, 1995), with updates offered in the following years by the committee (Chambless et al., 1996) and independently by members of the committee (Chambless & Hollon, 1998). The CSP provided guidelines or decision rules for defining empirically supported treatments (EST), as well as a list of psychotherapies meeting those criteria at the time. Establishing a process for determining whether an intervention approach is scientifically grounded, as well as a list of ESTs, has been viewed by many as important for promoting evidence-based practice within the fields of psychotherapy and clinical psychology (Herbert & Gaudiano, 2005), though others have questioned both the exercise and the criteria offered (for a summary, see Beutler, 1998).

The importance of evidence-based practice (a broader term used in other disciplines such as medicine and special education) has increased over the years and is now influenced by multiple sources. For instance, increasingly insurance companies are looking to providers to utilize therapies with known efficacy (Hayes et al., 1999; Morgan & Morgan, 2001). Additionally, certain federal initiatives and funding agencies are increasingly giving funding priority to proposed studies that utilize research methods to determine effectiveness of interventions (Horner et al., 2005; McDonald & O'Neill, 2003). Currently, many fields, including clinical psychology, tout between-group designs as the most scientifically rigorous method of determining effectiveness, with randomized clinical trials seen as the gold standard. However, SCDs can play an important role in determining empirical evidence for effectiveness.

The CSP (Task Force, 1995; Chambless et al., 1996) as well as modifications made since (Chambless & Hollon, 1998) provide guidelines for determining whether an intervention is "well established" or "probably efficacious." Initial CSP statements (Chambless et al., 1996; Task Force, 1995) included guidelines for determining whether an intervention was empirically supported using either group or single-subject designs. Regarding the latter, they recommended considering an intervention's efficacy to be well established if a large series of studies using SCDs

demonstrate intervention effects, and the intervention to be considered probably efficacious if a small series of studies supported the intervention. Chambless et al. specified that the series of studies needed to include nine or more participants total to be considered a large series.

Horner and colleagues (2005) provided greater specificity regarding how SCDs may be used to determine empirical evidence for interventions or practices. Though their discussion occurs in the context of educational practices, it has relevance for clinical child and adolescent psychology. Specifically, they provide five standards that must be met for SCDs to demonstrate evidence of efficacy for a practice (e.g., intervention, strategy, approach):

> (a) The practice is operationally defined; (b) the context in which the practice is to be tested is defined; (c) the practice is implemented with fidelity; (d) results from single subject research document the practice to be functionally related to change in dependent measures; and (e) the experimental effects are replicated across a sufficient number of studies, researchers, and participants to allow confidence in the findings. (pp. 175–176)

They elaborate on these standards, providing important information regarding suggested guidelines for ensuring that each is met. In particular, they suggest more rigorous criteria for the number of studies and participants using SCDs than the CSP guidelines. Specifically, they recommend that, for an intervention or practice to be considered efficacious, there be a minimum number of five studies using SCDs, that the studies be conducted by at least three different researchers or research teams, and that the number of participants included in those five studies total at least 20. Adherence to these criteria ensures that the effect of the independent variable on the dependent variable has been repeated multiple times by several researchers, increasing the internal and external validity of findings (Horner et al., 2005).

SUMMARY

An overview of the basic characteristics of various single-subject research designs has been provided, based in the context of clinical child and adolescent psychology. Single-case designs are distinguished from group design approaches to scientific inquiry by several features, including repeated observation of the dependent variable, replication of treatment effects, intrasubject and intersubject comparisons, visual analysis of individual participant data, and systematic manipulation of independent variables. Further, direct and systematic replications are used to establish the generality of the findings. Together, these features allow the investigator to guard against, or to detect when present, threats to internal and external validity. Thus, their use in psychological and psychotherapy research can allow investigators to answer questions of interest in a manner that meets the requirements of the scientific method (Hayes et al., 1999; Hilliard, 1993; Morgan & Morgan, 2001).

A particular strength of SCDs is their flexibility, allowing an investigator to change the purpose of the study when warranted as the investigation progresses (Barlow & Hersen, 1984; Holcombe et al., 1994). In fact, some have argued that the defining features of SCDs allow researchers to be more intimately involved with

their subject matter as compared to group designs (e.g., Perone, 1999). Assume, for example, that baseline data are variable. Rather than simply moving to the treatment phase in which the impact of a particular intervention is assessed, the investigator may wish to determine the source of the variability (Sidman, 1960). In this way, the investigation becomes an extended assessment that involves the identification of extraneous variables affecting the dependent variable. Once this is done, the researcher may wish to introduce a treatment (which, consequently, may be more effective now that sources of variability have been identified and can be accounted for).

Or perhaps the researcher was initially interested in the impact of a particular intervention on the dependent variable. For instance, perhaps a researcher is investigating the impact of time-out as an intervention for a noncompliant child using an A-B-A-B withdrawal design. Baseline data are collected on the rate of compliance per day. Then use of time-out as a consequence for noncompliance is introduced, which produces no change in the occurrence of the noncompliance. If this were a group design comparing the effects of time-out on noncompliance against either no treatment or an alternative treatment, the investigator would need to continue to utilize the exact procedures across all participants to assure experimental integrity, or consider the research unacceptable and stop the experiment altogether. With the SCDs, however, the investigator can extend the original lack of findings by evaluating another treatment (e.g., praise for compliance). Specifically, if there is no difference in the occurrence of the dependent variable between the initial baseline and treatment phases (A and B phases), it would be appropriate to introduce the second intervention (C phase) without returning to the original baseline conditions. If this produced an effect, withdrawal of the intervention would involve a return to the B phase. In other words, because data patterns do not differ between A and B, it is possible to use B as the comparison against patterns during the C phase. Thus, SCDs may be more economical than group designs in that they are malleable and changeable based on the data obtained.

In this way, single-subject designs are useful not only in the context of psychological research, but also in the context of demonstrating the effectiveness of clinical interventions (Hayes et al., 1999; Hilliard, 1993; Kazdin, 2003). Psychotherapy as a process involves, first, hypothesis building (i.e., developing reasonable explanations for the client's current conditions) and, second, hypothesis testing (i.e., assessing the effectiveness of an intervention expected to be effective based on the original hypothesis). Initial interventions may not be effective, and thus clinicians may experiment with others until improvements are noted. The flexibility of SCDs can allow clinicians to collect data and manipulate variables in a way that allows for adequate demonstration of therapeutic outcome as a result of specific interventions. Therefore, their use in psychotherapy can facilitate a clinician truly operating as a scientist-practitioner.

Multiple investigations have demonstrated that most graduates of clinical psychology programs, including those who come from scientist-practitioner programs, are involved in clinical endeavors rather than research pursuits (for review, see Gelso, 2006). As such, increased emphasis on training in empirical designs that have relevance for clinical settings seems prudent; in fact, cases for such an emphasis

have been made (Blampied, 1999; Hayes et al., 1999). As Hayes and colleagues noted, one of the major challenges in fostering greater connection between practice and scientific inquiry has been the lack of research designs appropriate for clinical settings. Single-case designs seem to have several distinct advantages over group designs in relation to conducting research in applied settings (e.g., applied at the individual level, more malleable so as to be responsive to information learned through the process); thus, they seem to have a greater probability of being incorporated into clinical practice.

Rigorously applying SCDs to clinical practice presents challenges, however. Specifically, as articulated at various places throughout this chapter, the requirements needed to experimentally demonstrate that changes in the dependent variable are the result of an intervention using SCDs can be at odds with the realities of clinical practice. A fundamental issue is identifying the variables (e.g., context, type of behavior, question being asked, population being served, resources available) for deciding which SCD is most appropriate for use in a particular clinical context for a specific purpose. For instance, if a treatment has already been established as evidence-based, then a simple A-B design may be appropriate as a mechanism of monitoring change. Further, reporting to the scientific community multiple applications of an intervention's effects using simple A-B designs will help support clinical application of the intervention that was originally tested in highly controlled conditions. On the other hand, because many interventions (including those identified as empirically supported according to APA criteria) have not been subjected to effectiveness trials or used with special populations (e.g., rare disorders, with individuals with comorbid disorders), use of SCDs provides an opportunity to establish effectiveness of the treatment in a clinical context and with variations of client characteristics that were not included in randomized clinical trials. One can envision a multistage process in which SCDs are initially used in clinical contexts to establish treatments as evidence-based; following this, treatments can be used more clinically in a single-case decision-making framework that meets certain purposes (e.g., case formulation, treatment design, monitoring progress) but that would not require withdrawal or multiple baseline designs, in part because the intervention had been established as effective in previous research.

The issue of how to successfully incorporate SCDs into clinical practice is complex, and full attention to it is beyond the scope of this chapter. We raise the issue here to encourage people to consider how to integrate SCDs into clinical practice, and how the pursuit of doing so is beneficial for not only clinicians and their clients (e.g., through documentation of change, better informed decision making) but also for the research community (e.g., by verifying previous outcomes with unique populations, by raising additional questions to be addressed through more comprehensive or precise research).

Single-case designs have critical underlying assumptions and basic characteristics that separate them from group designs, while still ensuring that data are collected and analyzed in an experimentally rigorous manner. Single-case designs are typically aligned with operant traditions of behavioral psychology, in particular Skinnerian radical behaviorism and the applied branch of this science, given ardent endorsements by critical leaders in this field (Sidman, 1960; Skinner, 1974).

However, as has been noted (e.g., Hayes, 1981), these designs are largely theory-free and can be useful for any investigator (whether a researcher or a practitioner) who gathers time-series data in a systematic fashion. Thus, regardless of theoretical premise of a given researcher or practitioner, SCDs may be appropriate for determining causal or functional relations between changes in the dependent variable and introduction of the independent variable. This is not to suggest that solid training in the philosophical and theoretical context in which these designs were developed is not important (Blampied, 1999), but instead to support their broad utility in the diverse field of clinical child psychology. Further, their utility is exemplified by the fact that there are frequently calls for more single-subject research in various fields of psychology and beyond (e.g., Cook, 1996; Hrycaiko & Martin, 1996; Lundervold & Bellwood, 2000). Not only are SCDs useful for explicit research purposes, but they can also be used to assist practitioners in determining the impact of their efforts. Because of this, understanding SCDs is important for both researchers and clinicians alike.

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CHAPTER 12

Personality in Childhood and Adolescence

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There has been a great deal of research on the personality characteristics of youth over the past 3 decades. In part, this blossoming of research has stemmed from the availability of models of general personality theories and the broadening of research within the adult literature on general personality. Much more research is being conducted on children and adolescents, and there has been a greater emphasis in better understanding the etiology, general structure, and stability of personality of youth and its connection to psychopathology. This area of study is somewhat controversial, as noted by the current nomenclature's lack of formal acknowledgment of personality and personality disorders in children and adolescence. However, our current nomenclature does suggest that personality disorders can be recognized in childhood and adolescence, and the research has been advancing to show that there are likely broad identifiable personality traits in children and adolescents.

The purpose of this chapter is to review aspects of research on early emerging individual differences during the years from infancy through late adolescence. First, we provide a general description of temperament and personality and discuss how the two have been conceptualized over time. In this section of the chapter we also discuss the potential links between temperament and personality and examine whether the two might be one and the same. Second, we describe what is known about the structure of personality in children and adolescents. We discuss four primary models of general personality developed over the past several decades and attempt to integrate the competing models. Third, we review the stability of childhood personality over time. We examine the data on personality models, showing the existence of both stability and change across major age groups. Fourth, and importantly, we explain some of the potential processes involved in continuity and change. Research examples of several processes are provided to shed light on this important issue. In this section of the chapter we also discuss Thomas and Chess's (e.g., Thomas & Chess, 1977; Thomas, Chess, & Birch, 1968; Thomas, Chess, Birch, Hertzig, & Korn, 1963) notion of the "goodness of fit" between the child's temperament and personality and his or her predominant environment. Fifth, we discuss childhood personality and its connection to both positive (adaptive) and negative (psychopathology) outcomes. This section of the chapter is important if

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we are to better understand how normal personality traits can lead to meaningful and fulfilling lives, and it also sheds light on how certain personality traits might be linked to psychopathology. Finally, we look at the *Diagnostic and Statistical Manual of Mental Disorders*, fourth edition, text revision (*DSM-IV;* American Psychiatric Association, 2000) model of pathology and its application to personality problems in youth. There is not an easy alliance between general models of personality and the *DSM-IV* categories of pathology, although researchers are beginning to bridge these gaps. The chapter concludes with suggestions for future research in this area, highlighting the exciting research on the near horizon for examining personality in children and adolescents as it relates to both positive and negative outcomes.

DESCRIPTION OF PERSONALITY IN CHILDREN AND ADOLESCENTS

Few would argue that children differ in their personalities. Some children are outgoing and exuberant, whereas others are more quiet and reserved. Some are irritable and emotionally labile; others show equanimity from an early age. Some are aggressive and others are gentle. These individual differences, although observed by parents, peers, teachers, and psychologists in everyday life, are not always easily described in research and clinical practice (Caspi & Shiner, in press; Shiner, 2006).

Traditionally, personality has been viewed as a stable and enduring set of characteristics that affects one's perceptions of oneself, others, and the world. The *DSM-IV* defines personality traits as "enduring patterns of perceiving, relating to, and thinking about the environment and oneself that are exhibited in a wide range of social and personal contexts" (American Psychiatric Association, 2000, p. 630). A personality disorder, in turn, is defined as an enduring pattern of inner experience and behavior that has deviated markedly from the expectations of the individual's culture, is pervasive and inflexible, has an onset in adolescence or adulthood, is stable over time, is evident across a broad range of personal and social situations, and leads to distress and impairment (p. 630).

Although there have been numerous studies of personality in adult samples, there has been limited research in the direct emergence or development of child personality and the concomitant problems that can stem from personality (Bernstein, Cohen, Velez, Schwab-Stone, Siever, & Shinsato, 1993). There is also little information on change or stability in childhood and adolescent personality. Part of the concern, when thinking about personality in children and adolescents is a general uneasiness that the identification of traits (or disorder) suggest little opportunity for change. Some scientists and clinicians believe that personality has not yet crystallized in children or adolescents; in this case, the existence of personality and/or personality disorder makes little conceptual sense. Others (e.g., Shapiro, 1990) question only whether general personality can be identified and personality disorders diagnosed before adolescence, when adultlike identity is thought to coalesce (Blos, 1967; Erikson, 1968). This is a nontrait and nondevelopmental approach because it does not consider the process by which, at each phase of development, an age-appropriate identity and personality exist and continue to form. However, researchers have noted

that ignoring the existence of a general blueprint for personality and the developmental line of the structuring of identity and personality means ignoring the ways personality development can occur across time but also how it might be positively or adversely affected at any age (Roberts & Wood, 2006; Shiner, 2006; Tackett, 2006).

BRIEF HISTORY OF THE STUDY OF PERSONALITY IN CHILDREN AND ADOLESCENTS

Individual differences in infants and children have focused on temperament and personality. The scientific study of early temperament can be traced to the seminal work of Thomas and Chess, who initiated the New York Longitudinal Study to study the significance of biologically based temperament in infancy and childhood (Thomas, Chess, Birch, Hertzig, & Korn, 1963). Thomas and Chess viewed temperament as a style of behavior and abilities. Their work challenged the way social development was studied at the time because they emphasized the two-way interaction of biology and environment in shaping children's outcomes. Prior to this view, many researchers and clinicians viewed child personality and outcome as being influenced almost exclusively by rearing environments. Thus, the work of Thomas and Chess was an important step in research because it suggested that there are preexisting personality styles from which individuals then operate.

Temperament has been defined differently by different researchers, and the link between temperament and personality is not fully understood (Frick, 2004). There are a variety of points of view on this topic: (a) that temperament is different from personality, (b) that temperament is linked to certain personality traits, and (c) that temperament is personality. Developmental and child psychologists traditionally describe stable and observable differences in young children by relying on temperamental constructs such as negative emotionality (Thomas & Chess, 1977) or emotionality (Buss & Plomin, 1984), sociability (Buss & Plomin, 1984) or surgency (Rothbart & Derryberry, 1981), task persistence (Thomas & Chess, 1977) or effortful control (Rothbart & Derryberry, 1981), and activity level (Buss & Plomin, 1984; Goldsmith & Campos, 1982; Putnam, Ellis, & Rothbart, 2001; Thomas & Chess, 1977). Researchers assume that temperament differences are expressions of neurobiological mechanisms that have a strong genetic basis (Mervielde, De Clercq, De Fruyt, & Van Leeuwen, 2005). In contrast, personality traits are mainly used to chart stable latent differences in adults, presumed to be partly influenced by temperament and interaction with the environment. According to this model, temperament can be considered a direct precursor to personality and, thus, is the earliest expression of individual difference.

Temperament is thought to be most directly observable during infancy and toddlerhood (Goldsmith et al., 1987) and to make up the entirety of personality during these early years (Caspi & Shiner, in press; Shiner & Caspi, 2003). It is thought that as children develop, early temperament traits develop into broader, more inclusive higher order personality traits as well as increasingly differentiated lower order traits (Buss & Finn, 1987; Caspi, Roberts, & Shiner, 2005; Caspi & Shiner, in press; Lahey, 2004; Rothbart & Ahadi, 1994; Shiner, 1998). Specifically, personality

develops as children progress through various cognitive and emotional stages that increasingly allow them to interact with, experience, and respond to the world in more complex ways (Caspi, 2000; Roberts & Wood, 2006; Rothbart & Ahadi, 1994). Thus, the structure of personality may change as children gain new skills (e.g., motor or language), gain the capacity to regulate emotions, and develop a sense of self (Roberst & Wood, 2006; Rothbart & Ahadi, 1994; Shiner, 1998, 2006; Shiner, Masten, & Tellegen, 2002). However, despite growing consensus that temperament traits make up the core of later personality, a clear understanding of the developmental relationship between temperament and personality is lacking (Halverson et al., 2003; Shiner & Caspi, 2003). This model is not embraced by all researchers, and the idea of temperament as "style" has been critiqued as vague and unnecessarily narrow. In fact, there are a variety of definitions of the term temperament (see Table 12.1), and some researchers claim that behavioral styles even early on are both biologically and environmentally influenced (see Frick, 2004; Plomin & Daniels, 1984), raising the question of whether the distinction between temperament and personality is necessary.

The resurgence of trait psychology, and especially the preponderance of the five-factor model (FFM; Digman, 1990), challenged both the viewpoint that personality could not be assessed in youth and that it could not also be considered a biologically based constellation of differences in youth. To elaborate on this point, personality psychologists have made several key and compelling arguments about the applicability of personality to children and adolescents. First, personality psychologists tend to agree that five broad dimensions—neuroticism, extraversion, openness, agreeableness, and conscientiousness—can be considered the basic dimensions underlying adult personality. Personality psychologists interested in the developmental antecedents of this FFM later showed that these five dimensions are also applicable to individual differences in children and adolescents (Digman, 1963; Digman & Inouye, 1986; Goldberg, 1993; John, Caspi, Robins, Moffitt, & Stouthamer-Loeber, 1994; Kohnstamm, Halverson, Mervielde, & Havill, 1998;

Table 12.1 Definitions of Temperament

Source	Definition
Buss (1995)	"Inherited personality traits that appear during the first two years of life and endure as basic components of personality." (p. 49)
Goldsmith (1996)	"Individual differences aspect of emotionality apparent early in development." (p. 221)
Kagan & Snidman (1991)	"Initial, inherited profiles that develop into different envelopes of psychological outcomes." (p. 856)
Rothbart & Derryberry (1981)	"Individual differences in reactivity and regulation assumed to have a constitutional basis." (p. 40)
Thomas & Chess (1977)	"The constellation of behaviors exhibited at any one age-period [and] are the result of all the influences, past and present, which shape and modify behaviors in a constantly evolving interactive process." (p. 171)

Note: Frick (2004) identified these definitions with the exception of Thomas and Chess earlier and also outlined specific dimensions of temperament.

Lamb, Chuang, Wessels, Broberg, & Hwang, 2002). In a key study, John et al. introduced the basic personality dimensions to developmental psychologists and demonstrated that the "little five" predict externalizing problem behavior in children.

McCrae and his colleagues (McCrae et al., 2000) cogently argued against the artificial distinction between temperamental constructs and personality traits, noting there are strong empirical and conceptual links between the domains of temperament and personality. The defining characteristics of temperament variables also apply to traits, including genetic basis, early observability, and pervasive impact on a range of behaviors (McCrae & Costa, 1997). In the past decade, behavior genetic studies have consistently documented the strong genetic basis of FFM traits (Jang, McCrae, Angleitner, Riemann, & Livesley, 1998), with heritability estimates ranging from .40 to .60 depending on the trait and method used for indexing the trait. McCrae and Costa (1996) conceptualized the FFM dimensions as basic tendencies shaping interactions with the environment, resulting in characteristic (mal)adaptations such as interests, values, and attitudes but also problem behavior and psychopathology. In another key study, Mervielde et al. (2005) compared the major models of temperament with the FFM and showed similarities and conceptual overlap between them. Frick (2004) commented that much of the work on temperament, personality, and psychopathology has been separated largely by discipline boundaries, which may account for differences in the terminology but not necessarily differences in the underlying concepts. Caspi et al. (2005) recently concluded that temperament and personality increasingly appear to be more alike than different.

Despite differences in terminology and perhaps conceptual differences, it has been suggested that the field of temperament and childhood personality needs a unifying framework to allow organization and integration of empirical findings and to facilitate communication among researchers within the field, as well as with researchers in clinical, developmental, and personality psychology subdisciplines (Frick, 2004; Halverson et al., 2003; Lonigan, Vasey, Phillips, & Hazen, 2004; Shiner & Caspi, 2003).

The popularity of the FFM in adult personality research has contributed to the growing body of literature documenting investigations of the five-factor structure in children and adolescents (e.g., Caspi & Shiner, in press; Digman, 1990; Graziano & Ward, 1992; John et al., 1994; Lamb et al., 2002; Martin, Wisenbaker, & Huttunen, 1994; Widiger & Trull, 2007). These studies tend to find evidence for the five-factor structure, although there is some evidence to suggest that openness to experience may not emerge until adolescence (Lamb et al., 2002). Halverson et al. (2003) in a novel study used a bottom-up approach to the structural debate by attempting to construct a cross-cultural, cross-age measure of childhood personality. These researchers collected parental descriptors of children (ages 3 to 12) across eight countries and, using a combination of rational (sorting done by focus groups) and factor analytic techniques, derived a somewhat new model of childhood personality. The resulting model consisted of a five-factor structure, although openness to experience was linked to only a single intellect scale, which reflects characteristics such as precociousness, intelligence, and speediness in comprehension. Overall, the

authors concluded that results of their effort produced a preliminary comprehensive taxonomy of childhood personality.

Shiner and Caspi (2003) also proposed a preliminary comprehensive taxonomy for childhood personality, which the authors recently extended (Caspi et al., 2005; Caspi & Shiner, 2006). They based their proposed taxonomy on a review of the literature on childhood and adolescent personality and also integrated existing and emerging work on the structure of personality. They initially defined a classification system consisting of four superordinate traits and 11 lower level traits, recently extending the taxonomy to include a fifth higher order trait (Caspi et al., 2005; Caspi & Shiner, 2006). The higher order traits mapped onto the five factors in the Big 5. They noted that questions remain over whether the fifth factor in the FFM, openness to experience, has a direct analogue in childhood (Shiner, 2006) or whether it emerges later in development (Caspi & Shiner, 2006). These two recent attempts to provide a preliminary classification system for childhood and adolescent personality virtually agreed on the relevant higher order traits. This convergence on the superordinate structure is particularly compelling given the different research approaches used to construct the two taxonomies.

Taken together, these studies provide a good starting point for discussing childhood personality within a comprehensive framework. Moreover, the framework put forth will allow for greater communication in the area of childhood personality, as well as linking areas of childhood and adult personality (Caspi & Shiner, 2006). These findings suggest that many researchers believe that the FFM should be chief in both child and adult research investigations, although we believe that it is also critical to reevaluate and revise theories as new evidence arises. Although the childhood measures have been developed from both bottom-up and top-down perspectives, we briefly review the adult models of personality and attempt to integrate the research on these models to better inform the models that have resulted for child and adolescent samples.

GENERAL MODELS OF PERSONALITY

Researchers (Watson, Kotov, & Gamez, 2006; Widiger & Trull, 2007) have noted that there are superordinate factors to the structure of human personality in adulthood, and the field is somewhat divided between two or three prominent models of personality. Watson et al. asked the question, If there is a stable factor structure for adult and child personality, why are there so many different models? These authors noted that despite a variety of models and some disagreement, these models are closely related or converge, and it has been shown that they can be easily integrated with each other (see Clark & Watson, 1999; Markon, Krueger, & Watson, 2005; Watson, Clark, & Harkness, 1994). We discuss each of these basic models of personality in turn. Then we show how the three major models can be integrated, drawing on the work of Clark and Watson (1999; see also Watson, Kotov, & Gamez, 2006).

The Big 5 Factors of Personality

The Big 5 factor model of personality developed out of a number of attempts to understand the lexicon or natural language of trait descriptors (John & Srivastava,

1999; McCrae et al., 2000). Factor analytic studies of these descriptors consistently revealed five broad factors: extraversion, agreeableness, conscientiousness, neuroticism, and openness to experience (or imagination, intellect, or culture). This structure has been shown to be robust, with the same five factors emerging in both self- and peer ratings (McCrae & Costa, 1987), in analyses of both children and adults (Digman, 1997), and across a wide variety of languages and cultures (McCrae & Costa, 1997).

The Big 3 Factors of Personality

The Big 3 factor structure of personality is based on a smaller set of superordinate factors: negative emotionality (paralleling neuroticism), positive emotionality (paralleling extraversion), and disinhibition versus constraint (paralleling conscientiousness; Clark & Watson, 1999; Markon et al., 2005). This structural format emerged from the seminal and pioneering work of Eysenck (Eysenck, 1997) and his colleagues. Eysenck originally created an influential two-factor model consisting of neuroticism and extraversion. Subsequent analyses led to the identification of a third broad dimension, labeled psychoticism. Although thus titled, this third dimension is better viewed as assessing individual differences in disinhibition versus constraint (Watson & Clark, 1993).

Other theorists subsequently posited similar three-factor models. For instance, Tellegen (1985) proposed a structure consisting of negative emotionality (paralleling neuroticism), positive emotionality (cf. extraversion), and constraint (which is strongly negatively correlated with psychoticism). Watson and Clark (1993) delineated a similar model, with factors named negative temperament, positive temperament, and disinhibition (versus constraint), respectively. In his reformulation of the California Psychological Inventory, Gough (1987) introduced the higher order "vectors" of self-realization, internality, and norm-favoring, which represent the low ends of neuroticism, extraversion, and psychoticism, respectively.

The Interpersonal Circumplex Model of Personality

Based on the work of Sullivan and Leary (Leary, 1957; Sullivan, 1953; Wiggins & Broughton, 1985; Wiggins & Pincus, 1989, 1994) proposed that personality was made up of octants: assured-dominant (PA), arrogant-calculating (BC), cold-hearted (DE), aloof-introverted (FG), unassured-submissive (HI), unassuming ingenuous (JK), warm-agreeable (LM), and gregarious-extraverted (NO). There exists considerable empirical literature on the correlates of the interpersonal circumplex model (e.g., Kiesler, 1996; Plutchik & Conte, 1997; Wiggins, Trapnell, & Phillips, 1988). Accordingly, the interpersonal circumplex model provides elements from a model of general personality and attempts to map the interpersonal circumplex onto Big 5 factor models that are generally well understood and are delineated in further detail next.

Integrating the Big 3, Big 5, and Interpersonal Circumplex Models of Personality

The accumulating data establish that the Big 5 model essentially represents an expanded and more differential version of the Big 3 (Markon et al., 2005; Rothbart,

Ahadi, Hershey, & Fisher, 2001; Watson, 2005). Most notably, the neuroticism and extraversion dimensions of the Big 5 essentially are equivalent to the negative emotionality and positive emotionality factors, respectively, of the Big 3. Thus, negative emotionality and positive emotionality are common to both models and constitute what Watson et al. (2006) refer to as the basic Big Two of personality. Moreover, the disinhibition versus constraint dimension of the Big 3 is a mixture of (low) conscientiousness and agreeableness. More specifically, according to Watson et al. (2006), disinhibited individuals tend to be impulsive, reckless, and irresponsible (i.e., low conscientiousness), as well as uncooperative, unsympathetic, and manipulative (i.e., low agreeableness). Finally, openness is strongly associated to the Big 3, although it does show a moderate positive association with positive emotionality (Digman, 1997; Markon et al., 2005). The interpersonal circumplex also maps onto the Big 5; many believe this personality framework taps two higher order dimensions, dominance and love. Dominance is most closely linked to extraversion and love is most closely linked to agreeableness and possibly openness (McCrae & Costa, 1989). Putting these data together, Clark and Watson (1999) noted that one can transform the Big 3 into the Big 5 by (a) decomposing disinhibition versus constraint into agreeableness and conscientiousness and (b) incorporating the additional dimension of openness. In addition the interpersonal circumplex can be subsumed under this broader umbrella of personality components and traits. Not surprisingly, of the somewhat varying models of personality in research, the FFM or Big 5 model of personality appears to subsume the others and thus is the most commonly used measure of personality traits.

It is important to note that some researchers have found flaws with this model. Specifically, although the Big 5 is the most common model, Block (1996, 2001) argued that the word "approach" might be more applicable than "model," since there is such an array of models that have not converged as of yet. He claimed that there are deficiencies in the FFM becoming the general perspective on personality, and pointed to the absence of an overall explanation for the five-factor approach. Others (Briggs, 1989) have voiced similar concerns, stating that there is no theoretical explanation for why these five dimensions were more coherent than other models' dimensions. On the other hand, no one has offered an explanation as to why this model is an invalid system of measure.

It is probably safe to say that the general consensus among researchers is that this model can be useful in describing the personality structure from which variant patterns can develop, thus forming the uniqueness of personality of an individual. Although it has shown reliable results when viewing adult differences, there have been limited findings with regard to children and the FFM. This could be for a number of reasons. For instance, there could be concerns about the reliability of self-reports and developmental and maturity levels, as well as diverging ideas about the age at which personality begins to develop from baseline attachments. In addition to these factors, reliability of parent reports of young children, or semantics, could have limited the degree to which research is conducted with children and the Big 5.

However limited, the FFM is increasingly credited with being one of the best currently available models for the study of personality, and research has been aiming to find a universality of personality as well as using it to track the development of

personality. Yamagata et al. (2006) conducted a study with 1,200 twins from Canada, Germany, and Japan to determine if the FFM identifies general personality structure that can be considered universal across cultures. These researchers found a significant congruence across genetic and environmental factors in relation to the FFM across all three countries, therefore providing a basic, universal genetic component in addition to the environmental factors involved in personality development.

A study by McCrae et al. (2002) examined changes in personality during adolescence, using the FFM to directly compare the adolescent results to the adult scores. They argue that adult personality measures can be meaningful when used with adolescent samples, as demonstrated by De Fruyt, Mervielde, Hoekstra, and Rolland (2000) and Parker and Stumpf (1998). Parker and Stumpf measured a sample of intellectually gifted adolescents and found that self- and parent reports were valid and reliable and yielded the same factor structure seen in adults. It is possible that psychologists underestimate the capacity of adolescents to relate to the adult personality items. Not only do adolescents understand the factors in the adult measure, but the factors seem to easily describe personality among adolescents when personality is evaluated by a variety of sources, including parents, teachers, and other adult observers. McCrae et al. (1999; see also McCrae et al., 2002) focused on the personality changes that occur during adolescence and used the FFM to examine each specific factor of neuroticism, extraversion, openness, agreeableness, and conscientiousness, predicting that adolescents would exhibit high neuroticism and extraversion and low agreeableness and conscientiousness. Although only moderate correlations were found with regard to high neuroticism, openness was also moderately elevated in adolescence. However, De Fruyt (De Fruyt et al., 2000) has noted that there is a need for continuity in measures. More specifically, it is possible that if there were a child version of the FFM the correlations may be stronger (or weaker). Either way, more research about the transition between childhood temperament and developmental maturity and later personality development should be examined not only with the FFM, but with other, equally strong measures of personality, such as the three, four, six or seven-factor models. We elaborate on this point next.

ASSESSING PERSONALITY AT A YOUNG AGE

A comprehensive assessment of age-specific indicators of traits is crucial to studying personality at a young age, especially in development. Different approaches have been adopted to assess FFM dimensions in children and adolescents, and as mentioned, work in this area is still in its infancy. Often, FFM measures—initially developed for adults—are downwardly extended and used to describe differences in younger age groups (see, e.g., studies using the Neuroticism Extraversion Openness Personality Inventory—Revised [NEO-PI-R] to assess adolescents' personality: De Clercq & De Fruyt, 2003; De Fruyt, Mervielde, Hoekstra, & Rolland, 2000; Goldberg, 2001; McCrae et al., 2002). Other studies adapted the phrasing of personality items for younger age groups (e.g., the junior version of Eysenck's Personality Questionnaire measure; Eysenck, 1963; Eysenck, Makaremi, & Barrett, 1994) to be more developmentally appropriate. However, it can be argued that these downwardly extended measures are probably not ideal for fine-grained

assessment of childhood and adolescent personality differences and especially not to index developmental change (De Clercq, De Fruyt, & Van Leeuwen, 2004). Therefore, according to De Clercq and De Fruyt an alternative approach that is more sensitive to subtle personality differences and developmental status at a variety of ages should be developed on the basis of the full range of personality differences observable prior to adulthood.

De Fruyt and his colleagues (De Fruyt et al., 2000; Slotboom, Havill, Pavlopoulos, & De Fruyt, 1998) have offered several suggestions for research and tool development in this area. Specifically, the lexical approach to personality description provides a convincing justification for the development of a comprehensive child and adolescent personality taxonomy. Mervielde and De Fruyt (1999) adopted this approach to construct such a taxonomy for classification of a large pool of parental personality descriptions in Flemish children between 6 and 13 years old (Kohnstamm et al., 1998). These authors subsequently developed the Hierarchical Personality Inventory for Children (HiPIC), representing the content of parental descriptions in short sentence items referring to concrete and observable behavior. The HiPIC can be considered a lexically based measure of the active parental vocabulary, in contrast to the NEO-PI-R, in which the facets are not derived empirically but are selected after a careful search of the adult personality literature. The HiPIC items span five broad domains: extraversion, benevolence, conscientiousness, emotional instability or neuroticism, and imagination. Some domain labels differ from the lexical adult Big 5 (Goldberg, 1993), although there is quite a bit of overlap in the content. It is beyond the scope of this chapter to cover comprehensively the measures used to assess the Big 5, but it should be noted that age-specific measures ought to undergo development and be tested. Researchers and clinicians alike are referred to Caspi and Shiner (2006) for a review of the existing measures, which might serve as a starting point for the development of newer measures with a developmentally sensitive theoretical framework. We comment on some important developmental factors that are likely to have an impact on personality development when considering the interaction between biology and environment and the two-way interplay that can lead to various outcomes for children and adolescents.

PERSONALITY DEVELOPMENT IN CHILDREN AND ADOLESCENCE

Developmental theories conceive a number of child and adolescent cognitive developments as well as puberty as important stages for social and personality development (Roberts & Wood, 2006), involving changing social interactions with parents and peers and increasing societal influences (e.g., Crick & Dodge, 1994; Roberts & Wood, 2006). Of course, we are increasingly aware that development occurs across the life span and that we are in the midst of development at any given age and developmental period. Nonetheless, the importance of adolescence as a key transitional phase has been acknowledged by a number of prominent theorists and researchers. Adolescence was acknowledged by Erikson (1950, 1968) as a critical stage in our development. Erikson considered adolescence a second individuation stage in which change is likely. Similarly, behavioral theories (Robin & Foster,

1989) emphasize that during adolescence, learning processes and contingencies are embedded in novel social networks and environments, including changing peer groups. In addition, organismic theories such as Piaget's (1983) theory of cognitive development and subsequent brain architecture suggest that newly acquired cognitive structures influence the way children and adolescents interact with their environment.

Personality theories such as Cloninger's (Cloninger, Svrakic, & Przybeck, 1993) theory of character development discern qualitatively different life strategies that an individual has to master before a more advanced developmental phase or level can be met. These theories emphasize developmental discontinuities, but it remains to be established whether these discontinuities reflect changes in basic tendencies or whether they are restricted to changing characteristic adaptations (McCrae & Costa, 1996; Shiner & Caspi, 2003). The demonstration of different forms of stability across the FFM trait hierarchy in childhood and adolescence would underscore McCrae and Costa's five-factor theory and therefore requires that the previously reviewed theories of personality development account not only for trait change but also for trait stability. We agree that there are some reasons to believe that personality would be stable across the life span; however, there are also compelling reasons with respect to maturation to expect both stability and change and that there would be a variety of outcomes for youth (both positive and negative) even if they manifested similar constellations of traits (e.g., extraversion and openness).

During the past several decades, research has uncovered considerable knowledge about the child's developing personality, including the emergence of a sense of identity, affect modulation, thinking style, and relationship with the external word (Mroczek & Little, 2006). With respect to the development of a sense of self, for example, it has been found that recognition of oneself by name in a mirror occurs by 3 years of age; a sense of shame, implying self-consciousness, emerges before 2 years of age (Lewis, 1993). Another example is empathy; a basic component of interpersonal functioning because of its role in the relationship between self and others, empathy develops in early childhood (Hoffman, 1977), evidencing clear signs by 2 years of age (see Salekin, 2006). These factors could also impact the development of personality in that youth might evaluate their basic personality characteristics and start to shape the various outcomes that result from their general personality makeup. Therefore, another component of developmental status and maturity—thinking style and the presence of a concrete operational reasoning system (Piaget, 1950)—is apparent by middle childhood and persists into adulthood. Differences in style of reasoning could impact the eventual outcome for youth. All these developmental considerations need a point of reference. That is, we need to know how stable personality is across childhood and adolescence and, more generally, across the life span.

Stability of Personality across Time: Continuity and Change

In the past decade, there has been a wealth of studies on personality continuity that were recently summarized in two meta-analyses of longitudinal data on differential (Roberts & DelVecchio, 2000) and mean-level (Roberts, Walton, & Viechtbauer,

2006) continuity. Differential continuity describes the degree to which the relative differences among individuals remain stable across time. Mean-level stability refers to the extent to which personality scores change over time. Longitudinal designs are required to investigate differential stability, looking at trait correlations across time. Mean-level stability can be studied with the use of longitudinal data (Roberts, Walton, Viechtbauer, 2006). Moreover, mean trait scores from cross-sectional age cohorts are useful for mean-level stability comparisons (McCrae et al., 1999).

Roberts and DelVecchio (2000) recently conducted a meta-analysis of differential stability and examined whether and when stability peaks during the life course, challenging Costa and McCrae's (1994; McCrae & Costa, 1984, 1990) claim that personality is "set like plaster" after the age of 30. They analyzed 3,217 test-retest correlation coefficients from 152 longitudinal studies and demonstrated an increase in stability from .31 during childhood to .54 in young adulthood, rising to .64 by age 30, with the highest stability of .74 observed between the ages of 50 and 70 years. These data suggest that personality is less stable during the preadult years; however, there was still a considerable amount of stability.

McCrae and colleagues (2002) examined both mean-level and differential changes during adolescence in self-descriptions using the NEO-PI-R, including an analysis of continuity at the level of the individual. These researchers found that mean-level personality scores for extraversion, agreeableness, and conscientiousness were stable between the ages of 12 and 18; however, neuroticism appeared to increase in girls and openness increased in boys and girls. A 4-year longitudinal study of intellectually gifted students showed a considerable degree of rank-order instability across the two assessment points. Stability coefficients for boys across a 4-year interval ranged from .31 (agreeableness) to .49 (conscientiousness) and for girls from .30 (neuroticism) to .63 (conscientiousness). Individual-level continuity analyses relying on the reliable change index (Jacobson & Truax, 1991) indicated that about 60% of the sample did not change over the 4-year interval for each of the FFM dimensions. A similar analysis for college-aged individuals (Robins, Fraley, Roberts, & Trzesniewski, 2001) showed that almost 80% were stable on each of the FFM dimensions, again suggesting that differential stability is somewhat lower during adolescence.

The meta-analysis of Roberts et al. (2006) on mean-level change during adolescence (10 to 18 years) showed a small increase for openness ($d = .23$) and a significant increase for social dominance ($d = .20$). No significant mean-level changes were reported for social vitality, agreeableness, and conscientiousness during adolescence. Studies examining stability of personality prototype classification using three personality profiles (resilient, undercontrolled, and overcontrolled children), provided partial evidence of ipsative stability (De Fruyt, Mervielde, & Van Leeuwen, 2002). However, according to De Fruyt et al., it was unclear whether the instability to prototype classification for some individuals should be attributed to real developmental change measurement error or the procedure to derive prototypes.

Taken together, these important studies and others (Funder, Parke, Tomlinson-Keasey, & Widaman, 1993; Graziano, 2003; Halverson, Kohnstamm, & Martin, 1994; Mroczek & Little, 2006) lead to two general conclusions. First, the level of

personality continuity in childhood and adolescence is higher than often expected (e.g., compared with young adulthood; Roberts et al., 2006). Second, personality change occurs further into adulthood than one might expect. Caspi and colleagues (2005) argued in this respect that personality trait development is not a continuity-versus-change proposition, but that continuity and change coexist. These authors concluded that the major challenge for developmental theories will be to account not only for trait continuity but also for trait changes. We return to this point later in the chapter when we cover contextual factors and what Thomas and Chess (1977) and recently Shiner (2006) have referred to as "goodness of fit."

PERSONALITY AND LIFE OUTCOMES

The importance of studying personality is that certain traits will be able to forecast certain outcomes for children and adolescents. Quite often in psychology and psychiatry, the fields have focused on the negative aspects of personality; this is particularly apparent in the *DSM-IV* and its successive versions, which have focused on personality disorder. Fortunately, there is also a movement in the field to focus on positive and healthy outcomes for children and adolescents. This section of the chapter highlights some previous work in the area of child and adolescent personality that has emphasized both positive and negative outcomes (Caspi & Bem, 1990; Caspi & Shiner, 2006; Rothbart & Bates, 1998; Shiner, 2006; Shiner & Caspi, 2003). Specifically, we focus on positive outcomes such as academic achievement, social competence with peers, rule-abiding conduct, the development of romantic relationships, and work competence in order to build on Shiner's and others work in this area (Funder et al., 1993; Masten et al., 1995; Shiner, 2006). With respect to negative outcomes, this section covers children's internalizing and externalizing behaviors. Taken together, the studies reviewed in this section point to two important conclusions: Children's traits are differentially linked with specific outcome, and these patterns are often replicated across studies, and there is convincing evidence for the consequential nature of children's individual differences or personality (Shiner, 2006). Typically, the prediction afforded by children's temperament and personality traits is modest in size. We take each of the Big 5 dimensions and tie each to positive and negative outcomes to build on Shiner's work in this area (2006). Although we do not examine personality disorders as conceptualized in the *DSM-IV,* we provide a road map for how this approach might allow for further study of *DSM-IV* disorders (see Table 12.2).

Extraversion: Negative and Positive Outcomes

It has been noted that children's differences in extraversion predict a complex set of outcomes (Shiner, 2006). There are some positive outcomes associated with extraversion. Shiner has noted that for some children extraversion is associated with positive and meaningful peer relations concurrently and across time into late adolescence and early adulthood. Moreover, extraversion predicts positive changes in social competence across time (Shiner, 2000; Shiner, Masten, & Roberts, 2003). Similarly, extraversion in early adolescence predicts growth in perceived social

Table 12.2 Outcomes Associated with Each Big 5 Factor Trait

Trait	Positive Outcome	Negative Outcome
Extraversion	Social competence Promotes good health	Antisocial behavior Callousness
Neuroticism	Conscience development Guilt when expected	Poor relationships Relationship conflict Relationship abuse Relationship dissolution Less competent parenting Risk for unemployment
Conscientiousness	School adjustment Educational achievement Occupational attainment Job performance	Obsessive
Agreeableness	Social competence Positive parenting Responsible parenting	
Openness	Exploring Friendliness Academic achievement	Exposure to risks

Note: This table is intended to be illustrative and not an exhaustive list of positive and negative outcomes.

support from early adolescence to late adolescence (Asendorpf & van Aken, 2003a, 2003b). Extraverted children are more likely than introverted children to have a life history of stable, high-quality romantic relationships by age 30 (Shiner, 2006; Shiner et al., 2003). Childhood extraversion is associated with lower levels of later internalizing symptoms (Chen et al., 2002).

Although being extraverted has some positive aspects to it, research has shown that some of the outcomes for those scoring high on extraversion measures are negative and may even point to certain types of pathology. For instance, a wide variety of studies have linked early signs of extraversion to heightened risk of externalizing symptoms and aggression (Chen et al., 2002; Frick & Scheffield-Morris, 2004; Huey & Weisz, 1997; Markey, Markey, & Tinsley, 2004). As Shiner (2006) indicated, the mechanisms for why extraverted children are more likely than introverted children to develop antisocial behavior are not yet clear; they may have stronger impulses that they must learn to regulate, or they may be more likely to engage in antisocial behavior with peers. Outcomes of shyness and inhibition essentially parallel those found with low extraversion, including lower social support (Newman, Caspi, Moffitt, & Silva, 1997) and higher rates of anxiety symptoms (Kagan, Snidman, Zentner, & Peterson, 1999). Socially inhibited and/or shy boys may experience more negative life outcomes than shy girls. According to Caspi and his colleagues (Caspi, Bem, & Elder, 1989; see also Gest, 1997), this might be because shyness is more socially sanctioned for girls than for boys.

Neuroticism: Negative and Positive Outcomes

Neuroticism and the broader negative emotionality trait are associated with negative outcomes in a variety of areas. Children high on negative emotionality are

at risk for a wide variety of social difficulties (see Eisenberg, Fabes, Guthrie, & Reiser, 2000; see also Shiner, 1998 and Muris, Schmidt, Merckelbach, & Schouten, 2001). Negative emotionality in children predicts both internalizing and externalizing psychopathology, suggesting a broad bandwidth of psychopathology (Lengua, 2002; Rothbart & Bates, 1998). Caspi and Shiner (2006; Shiner, 2006) have noted that it is important to consider the outcomes for anxiety and fear separately from the outcomes of irritability given that there is some heterogeneity in the neuroticism concept. According to these authors, this is important because in some cases, the lower order traits predict similar outcomes but may do so for quite different reasons. For example, they note that higher anxiety or fear and higher irritability both predict greater social difficulties (Asendorpf & van Aken, 2003a, 2003b; Eisenberg, Pidada, & Liew, 2001) and lower occupational attainment in adulthood (Caspi et al., 1989; Judge, Higgins, Thoresen, & Barrick, 1999). Yet, anxiety and fear may cause such difficulties as preventing children from fully participating in social and work spheres, whereas children's irritability may exert its effects through alienating other youths. Importantly, in other cases, these two components of negative emotionality may predict different outcomes altogether. For example, anxiety and fear are associated with greater risk for internalizing symptoms, and irritability with greater risk for externalizing symptoms (Rothbart & Bates, 1998; Shiner, 2006).

Although high negative emotionality spans a broad range of negative outcomes, it is equally important to recognize that higher fearfulness may promote positive outcomes for some children. For example, more fearful children experience greater guilt when they believe they have done something wrong, and this guilt proneness in turn appears to mediate the association between children's level of fear and their later compliance with rules (Kochanska, Gross, Linn, & Nichols, 2002). These findings are consistent with research documenting that childhood fearfulness protects against the development of externalizing behavior problems, including aggression (Raine, Reynolds, Venables, Mednick, & Farrington, 1998). These findings have also been linked to conscience development and the possibility of protecting against psychopathy-like traits in youth (Salekin, 2006). Like high extraversion, high fear predicts a complex mixture of both positive and negative outcomes.

Conscientiousness: Negative and Positive Outcomes

There is little to chart the negative outcomes for being conscientious. It might be that extreme forms of conscientiousness lead to obsessiveness and perhaps other forms of anxiety, but to our knowledge there are no data to support this supposition. Rather, scientific work on children's conscientiousness documents what might be considered somewhat obvious findings. That is, greater self-control appears to promote many good outcomes (Shiner, 2006). First, children's attention, self-control, and carefulness predict later academic achievement, even when controlling for earlier measurement of academic ability and IQ (Martin, Olejnik, & Gaddis, 1994; Shiner, 2000; Shiner et al., 2003). Childhood conscientiousness predicts positive changes in academic achievement from childhood to adulthood (Shiner, 2000; Shiner et al., 2003). These positive changes continue into adulthood, resulting in happiness and

well-being in other areas. Specifically, conscientiousness in adolescence forecasts adult career success in terms of both income and occupational status. Conscientiousness also predicts job satisfaction (Judge et al., 1999). Children's attention and self-control, regardless of the method of indexing these traits, also appear to promote the development of rule-abiding behavior rather than externalizing, antisocial behavior (Shiner, 2006). Put another way, children who are impulsive, inattentive, or careless are at risk for developing externalizing behaviors (Ackerman, Brown, & Izard, 2003; Olson, Schilling, & Bates, 1999; Shiner, 2000, 2006). Again, it is important to note that these links between early self-control and lower risk of externalizing problems may be mediated in part by more self-controlled children's tendencies to develop a stronger conscience (Kochanska & Knaack, 2003; Salekin, 2006). Finally, components of conscientiousness in childhood, including effortful attention, self-control, and carefulness, have predicted children's concurrent and later social competence with peers in several studies (Lamb et al., 2002; Hawley, 2006; Shiner, 2000).

Agreeableness: Negative and Positive Outcomes

Similar to conscientiousness, childhood agreeableness has few known negative outcomes, but instead points to many different positive life outcomes (Graziano & Eisenberg, 1997). With respect to negative outcomes, high compliance, an aspect of agreeableness, could be problematic for children and may be a particular risk factor for negative outcomes for girls (Pulkkinen, 2002). Although there are few data on the negative outcomes for agreeableness, it will be important for future work to address whether extremely high agreeableness poses adaptive risks for children.

Greater childhood agreeableness is associated with better peer relationships concurrently and across time (Graziano, Jensen-Campbell, & Finch, 1997; Shiner, 2000, 2006). However, some authors have noted that the name of this trait is somewhat misleading if it suggests that its importance is confined to interpersonal relationships. Childhood agreeableness is also linked to greater academic attainment and a reduced risk of externalizing behaviors and delinquency (Laursen, Pulkkinen, & Adams, 2002; Shiner, 2000, 2006; Shiner et al., 2003). In one longitudinal study, childhood agreeableness predicted positive work competence 20 years later, when childhood academic achievement and IQ were controlled (Shiner et al., 2003).

As noted previously, agreeableness has some heterogeneity to its lower order traits, and the components of this higher order trait include at least two major lower order traits: prosocial tendencies and antagonism (which includes aggressiveness). Findings for children's prosocial tendencies parallel those for agreeableness. For example, a stronger prosocial orientation in childhood predicts positive social adjustment and academic achievement. Moreover, and relatedly, a stronger prosocial orientation predicts fewer externalizing behaviors over time (Chen, Li, Li, Li, & Liu, 2000). Alternately, findings for antagonism and aggression typically are like those found for low agreeableness. More antagonistic, aggressive children are at risk for a number of negative outcomes, including academic failure, peer rejection, job difficulties and unemployment, and later conflictual romantic relationships (Kokko

& Pulkkinen, 2000; Moffitt & Caspi, 1998; Rubin, Bukowski, & Parker, 1998; Shiner, 2006). Boys who are chronically aggressive are at a heightened risk of serious delinquency and even violence (Nagin & Tremblay, 1999).

Thus, the traits underlying both conscientiousness and agreeableness promote positive adaptation in many different domains, and there are not a lot of data to suggest negative outcomes. As noted, these two higher order traits appear to share a common core of behavioral constraint or self-control, with conscientiousness focused on task-related behavior and agreeableness focused on interpersonal behavior, which may account for their concomitant positive outcomes. It is interesting to note, however, that the positive outcomes of conscientiousness are not restricted to the domains of achieving and striving. Nor should it be interpreted that the positive outcomes of agreeableness are limited to interpersonal domains. Rather, there is more likely a greater array of positive benefits that stem from each higher order construct. More detailed research on how highly conscientious and agreeable children approach a variety of situations will help shed light on why the two traits share such similar positive outcome profiles.

Openness to Experience: Negative and Positive Outcomes

Few studies have shown negative outcomes for the openness to experience construct. However, it is possible that being too open could leave one slightly more vulnerable. Research is needed in this area to determine if there are some negative aspects associated with extreme levels of openness. For the most part, research on the links between openness to experience and intellect shows positive outcomes. However, research in this area is more limited than for the other four higher order traits; yet, the findings generally are consistent. Openness to experience has been shown to have concurrent relations with academic achievement in samples of school-age children and young adolescents (Graziano et al., 1997; John et al., 1994; Measelle, Ablow, Cowan, & Cowan, 1998). Moreover, the size of the correlation varies from modest to strong, and this difference in magnitude could stem from the differences in trait descriptors for this dimension across indexes, with some emphasizing intellectual characteristics and others less broadly covering this aspect of the higher order trait (Shiner, 2006). Because openness and intellect in childhood are moderately associated with IQ or other measures of intellectual ability (Lamb et al., 2002), it is not clear to what extent these traits predict academic achievement; thus, these traits differ from the other Big 5 traits in that they may show more narrowly focused importance for adaptation (Caspi & Shiner, 2006; Shiner, 2006).

MODELS OF PERSONALITY AND THEIR CONNECTION TO PSYCHOPATHOLOGY

One of the questions that has fueled scholarly and popular interest in temperament is whether children with particular temperaments are at a greater risk for negative outcomes, including psychopathology. Much of the early interest in temperament traits was initiated by Thomas and Chess (1977), who suggested that early individual differences could set off a chain of transactions between the child and the

environment that could ultimately end with clinical disorders. The previous section highlighted general negative and positive outcomes associated with each of the Big 5 personality factors; this section focuses more specifically on the connection between the five factors and psychopathology.

Shiner and Caspi (2003) noted that although individual differences in personality shape individuals' adaptation over time, childhood personality also plays a key role in the development of psychopathology. Much of the current research on personality and psychopathology simply documents correlations between temperament or personality traits and aspects of psychopathology, without articulating how the two domains may be connected (Caspi & Shiner, 2006; Hoyle, 2000; Maziade et al., 1990; Rothbart & Bates, 1998). Shiner and Caspi (2003) and more recently Tackett (2006) present conceptual models of the possible associations between personality and psychopathology in childhood and adolescence. These childhood models have parallels to those developed by Clark, Watson, and Mineka (1994) and Widiger, Verheul, and van den Brink (1999). The models posit several different links to pathology. First, these authors all suggest that psychopathology may represent the extreme end of a continuously distributed personality trait or cluster of traits (this is also referred to as the spectrum model). Second, personality may set in motion processes that cause the development of psychopathology (referred to as the vulnerability model). Third, personality may protect against the development of psychopathology in the face of stress and adversity (referred to as the resilience model). Fourth, personality may influence the form and prognosis of a disorder, even if the personality trait is not a component or cause of the disorder (referred to as the pathoplastic model). Fifth, psychopathology may influence the course of personality development itself (referred to as the scar model; Shiner & Caspi, 2003; Tackett, 2006; see Table 12.3).

Over the past 2 decades, a growing body of research has attempted to explain the nature of these relationships (for reviews, see Krueger & Tackett, 2003; Widiger & Clark, 2000; Widiger et al., 1999), building on current research investigating the etiology and structure of both personality and psychopathology. However, this work has primarily focused on adult populations, despite the growing evidence of robust associations between personality traits and mental disorders in children and adolescents. Tackett (2006) has argued that to promote understanding and further investigation of the relationship between personality and psychopathology, existing work with children and adolescents must be integrated into research on adults to create a broader developmental picture.

Table 12.3 Models Depicting the Connection between Personality and Pathology

Model Name	Path (or Relation) to Pathology
Spectrum	Pathology is an extreme form of personality (viewed along a continuum).
Vulnerability	Personality sets in motion and is the cause of pathology.
Resilience	Personality may protect against or buffer against pathology.
Pathoplastic	Personality may influence the form of the disorder even if it is not the cause.
Scar	Pathology affects personality change.

Evidence for each of these models is limited in part because the study designs needed to detect the hypothesized relations are complex and time consuming. Nonetheless, there is some emerging research in this area. For instance, the one longitudinal study found that childhood antisocial behavior problems predicted an increase in neuroticism, or negative emotionality, in adulthood even after controlling for the level of neuroticism in childhood (Shiner et al., 2002); this model provides some support for the scar model, which suggests that childhood psychopathology may cause changes in personality later in life. However, without a measurement of personality before the onset of Axis I pathology, one cannot infer whether these changes in personality traits were a result of the Axis I disorder (Tackett, 2006).

There is also some evidence for the pathoplasty model. One area of study has concentrated on comorbidity between anxiety disorders and Conduct Disorder (CD) in children. This literature has largely found that disruptive children with high levels of inhibition or shyness, which would be related to the higher order factor of neuroticism, have less severe manifestations of CD and better prognosis compared to those with low levels of inhibition (Kerr, Tremblay, Pagani, & Vitaro, 1997; Walker et al., 1991). The bulk of research evidence to date appears to back the vulnerability and dimensional models.

The vulnerability model splits childhood pathology into externalizing and internalizing disorders. Several types of childhood and adolescent problem behaviors are typically categorized as externalizing pathology, such as Oppositional Defiant Disorder (ODD), Conduct Disorder, Attention-Deficit/Hyperactivity Disorder, and substance use. Disorders characterized by antisocial behaviors, such as ODD and CD, have received the greatest attention from the vulnerability perspective. In general, this literature has primarily identified conscientiousness and neuroticism as potentially relevant personality risk factors for the development of later psychopathology.

A number of studies have found that characteristics related to impulsivity (which has been related to low conscientiousness) were significantly related to antisocial behaviors throughout childhood (Hirshfeld et al., 1992; Raine et al., 1998; Tremblay, Pihl, Vitaro, & Dobkin, 1994) and adolescence (Lynam et al., 2000; Salekin, Lesitico, Trobst, Schrum, & Lochman, 2005). Some longitudinal studies that followed individuals into adulthood have also shown strong relationships between early characteristics of impulsivity and later antisocial behaviors. This finding has been substantiated by studies with measurements of impulsivity in adolescence (Sigvardsson, Bohman, & Cloninger, 1987; White, Bates, & Buyske, 2001), middle childhood (Farrington & West, 1993), and as early as 3 years of age (Caspi, 2000; Caspi, Moffitt, Newman, & Silva, 1996; Henry, Caspi, Moffitt, & Silva, 1996). In addition to the aforementioned findings related to low conscientiousness, early observations of negative emotions have been connected with antisocial behaviors in middle childhood (Renken, Egeland, Marvinney, Mangelsdorf, & Sroufe, 1989). Augmenting this work on general antisocial behavior, other lines of research have focused on relationships across time between personality traits and specific subtypes of antisocial behavior (Lynam, 1996; Moffitt, 1993; Moffitt, Caspi, Dickson, Silva, & Stanton, 1996; Salekin, 2006; White et al., 2001). Another strand of research

has measured personality traits related to the construct of psychopathy, such as callousness and impulsiveness, and found them to predict antisocial behavior across childhood (Lynam, 2006; Lynam et al., 2005; Salekin, 2006; Salekin & Frick, 2005; Salekin et al., 2005; Salekin, Neumann, Leistico, DiCicco, & Duros, 2004; Tremblay et al., 1994).

A larger body of work has demonstrated that measures of disinhibition predict externalizing pathology in early and middle childhood (Caspi, Henry, McGee, Moffitt, & Silva, 1995; Eisenberg et al., 2000 2004; Mun, Fitzgerald, Von Eye, Puttler, & Zucker, 2001; Rende, 1993; Rubin, Burgess, Dwyer, & Hastings, 2003; Silverman & Ragusa, 1992) and later into adolescence (Olson et al. 1999). Related to these results, other studies have reported that high levels of inhibition predicted lower levels of externalizing psychopathology in childhood and adolescence (Schwartz, Snidman, & Kagan, 1996; Sigvardsson et al., 1987; Tremblay et al., 1994).

Another major domain of disorders, typically referred to as internalizing pathology, consists primarily of problems with anxiety and depression. One personality construct that has received a great deal of attention regarding its connection with anxiety disorders is often called behavioral disinhibition (BI; Tackett notes that this is typically described as a temperament). In the context of personality, BI would represent a combination of high neuroticism and extraversion. Having said this, a substantial literature has reported a relation between BI and later internalizing problems (see Hirschfeld-Becker et al., 2003). Indices of BI in toddlerhood and early childhood predicted anxiety disorder in early middle childhood (Biederman et al., 1993, 1990; Hirschfeld et al., 1992; Rosenbaum et al., 1993) and adolescence (Hayward, Killen, Kraemer, & Taylor, 1998; Schwartz, Snidman, & Kagan, 1999).

A smaller set of studies has examined the links between personality characteristics and depression. These studies suggest that traits reflecting extreme inhibition in early childhood predict depression in adulthood and may show a significant relation with suicide attempts when combined with traits marking low conscientiousness (Caspi, 2000; Caspi et al., 1996). Another study reported that high levels of neuroticism and high levels of novelty seeking in adolescence predicted later suicidal behavior (Fergusson, Beautrais, & Horwood, 2003). Although the dimensional models are also very compelling, we do not cover them in as great of detail in this chapter. The essence of this model is that personality is pathology when the personality is extreme. One example of this is the work of Lynam (2006) and Salekin (2006), who have examined the extreme form of some personality traits that could in fact be psychopathy. They have found a potential connection between personality and externalizing disorders. The dominant model linking internalizing disorders to personality traits is the tripartite model, which has recently been applied to children (Clark & Watson, 1991; Clark et al., 1994).

In summary, the research on the connection between personality and pathology provides preliminary support for both vulnerability and spectrum approaches to conceptualizing the personality-psychopathology relationship in children and adolescents, with the vulnerability, or risk factor, approach receiving the greatest research attention. Specifically, neuroticism and conscientiousness have been implicated as particularly relevant in the development of later psychopathology. The

resilience or protective factor model also has support (Chiccetti & Luthar, 1990). There is less support for alternative conceptualizations, such as the scar and pathoplasty models, due to the lack of research. Although the spectrum model has not been a driving hypothesis to the extent that the vulnerability model has, it is difficult to disentangle evidence for these two approaches. Indeed, research supporting a vulnerability explanation for the personality-psychopathology relationship may also support a dimensional conceptualization, and these two approaches may ultimately work in concert to provide the best explanation. More direct tests of the scar, pathoplasty, and spectrum models will help elucidate a comprehensive approach to conceptualizing the personality-psychopathology relationship. Ultimately, we may find that all models are necessary to create such a comprehensive picture, possibly differing for various types of psychopathology or various individuals (Tackett, 2006).

CONNECTIONS BETWEEN BIG 5 AND THE *DIAGNOSTIC AND STATISTICAL MANUAL*

Some researchers have argued that the personality superfactors are too broad to capture all the interesting variation in children's personality, and distinctions at the level of more specific, lower order traits are necessary (McAdams, 1992; Tackett, 2006). Such is the case, for example, if using structural models of personality traits to distinguish between different personality disorders (Lynam & Widiger, 2001; Widiger, Trull, Clarkin, Sanderson, & Costa, 2002); recent work suggests that categorical personality disorders are distinguished by particular patterns of dimensional lower order personality traits, as shown earlier. The advantage of broad, higher order superfactors is their substantial bandwidth; the disadvantage is their low fidelity (Robins, John, & Caspi, 1994). There is much to learn from studying variation at different levels of the personality trait hierarchy, but this does not obviate the need to understand how different levels are associated hierarchically to each other.

The connection between the Big 5 and *DSM-IV* disorders is not all that well known. Research appears to be progressing along independent lines with little integration of the two (e.g., Brent, Zelenak, Buckstein, & Brown, 1990), similar to the problems with the research on temperament and personality (Bernstein, Cohen, Skodol, Bezirganian, & Brook, 1996). Although there is some connection between Axis I childhood pathology and the Big 5 (e.g., the connection between CD and neuroticism), much less has been conducted on the *DSM-IV* personality disorders themselves. Some have argued that the impact of *DSM* personality disorders on functioning can be as profound for a young person as for an adult (Kernberg, Weiner, & Bardenstein, 2000). For example, some preliminary research indicates that suicide is more likely for adolescents diagnosed with impulsive, dramatic, Avoidant or Dependent Personality Disorder (Brent et al., 1994), and suicidal behavior is more severe in older adolescents and young adults who have both Borderline Personality Disorder and depression (Friedman, 1982). Researchers should attempt to make connections between *DSM* disorders and the Big 5 and eventually integrate the models and then possibly tie the personality concepts to the *DSM-IV* Axis I disorders (see Table 12.4). Some significant strides have been made in this regard (see Krueger, 1999; Krueger, McGue, & Iacono, 2001; Tackett & Krueger, 2006).

Table 12.4 Connecting the Big 5 to Specific Childhood Disorders

Externalizing (high extraversion, low conscientiousness, and low agreeableness)	Internalizing (high neuroticism, low extraversion, high conscientiousness, low agreeableness)
Oppositional Defiant Disorder	Separation anxiety
Conduct Disorder	Depression
Attention-Deficit/Hyperactivity Disorder	Generalized anxiety
Substance abuse	Obsessive-Compulsive Disorder
	Eating disorders

Processes Underlying Continuity and Change in Personality Development

As noted in the preceding sections, children's early temperament and personality traits show both substantial continuity and change (Shiner, 2006). By the preschool years, individual differences are moderately stable. Children's individual differences moderately predict many aspects of development, including both positive and negative outcomes. Thus, early temperament and personality traits meaningfully predict later personality and life outcomes for some children (including both positive and negative outcomes, including psychopathology), but other children appear to undergo significant change over time. As Shiner points out, this finding regarding stability and change requires explanation: What is the process that underlies both the stability and the change? In part, the answer lies in contextual factors. This issue is hardly new, as Thomas and Chess (Thomas et al., 1963) raised the question of explaining continuity and change in their early theoretical work on temperament. Specifically, they argued that each early temperamental trait could yield a variety of subsequent outcomes, depending on the goodness of fit between the child's temperament and the context in which the child flourished, languished, or simply did okay. For example, some infants with a difficult temperament may continue to exhibit irritable, hard-to-manage behavior as children, whereas other infants with similarly challenging early temperaments may learn to manage their intense negative emotions if their parents can successfully adapt their parental style, and subsequently lead successful and happy lives. This means that a variety of intervening processes can play important roles in determining the eventual outcomes of children's early individual differences. These differences may promote personality continuity by shaping children's experience and engagement with their environment.

Caspi and Shiner (Caspi & Shiner, in press; Shiner & Caspi, 2003) compellingly cover the processes that facilitate stability. Briefly, these authors talk about elaboration, which could lead to further solidifying of a given personality style. According to these authors, the process of elaboration may involve at least six processes, thought to develop at different stages of development. The first of these processes starts in the first few months of life and includes learning processes and environmental elicitation. In early to middle childhood, after the emergence of necessary cognitive functions, the second set of processes—environmental construal and social comparison processes—may influence personality development.

Third, in late childhood and adolescence, environmental selection and manipulation emerge. These processes come online once the self-regulatory functions have developed. Shiner and Caspi describe these six processes through which endogenous biological and experiential learning sources of influence may serve to elaborate an initial disposition over time so that it increasingly organizes thought, emotion, and action. The theoretical models are in place; research is now needed on each of these processes in relation to different temperament and personality traits.

Other processes promoting continuity and change range from genetic or biological processes to more broad, contextual processes involving classrooms and neighborhoods (Kagan, 1998; Shiner, 2006). Genetic influences may play a role in shaping the continuity versus discontinuity of early individual differences. The MacArthur Longitudinal Twin Study (Saudino & Cherny, 2001) showed that for all of the traits except shyness, twins' stability in temperament—in terms of activity, task orientation, affect-extraversion, shyness at home, and behavioral inhibition in the lab—was derived from genetic influences; for shyness, both genetic and shared (familywide) environmental factors contributed to stability. On the other hand, sources of change varied for the traits; changes in shyness across time derived from both shared and nonshared (child-specific) environmental influences, whereas change in affect-extraversion was due to nonshared environmental influences only. These results fit well with other research on children and adults suggesting that, whereas personality stability appears to be due to genetic factors, change is due to environmental influences (Saudino & Plomin, 1996). Research in this area is preliminary. The Louisville Twin Study has found evidence for genetic influences on change in temperament in the first 4 years of life (Loehlin, 1992), but clearly, more work is needed to establish more definitively the genetic influences on change in children's individual differences, particularly focusing on both positive and negative outcomes.

Another factor that could be related to stability and change may have to do with the permutations of traits and other characteristics that accompany personality. Shiner (2006) has referred to these factors as intrapersonal factors. The point here is that the mix or makeup of children's characteristics is another source of stability and change. Allport's (1937, p. 48) classic definition of personality assumes the importance of intra-individual psychological organization: "Personality is the dynamic organization within the individual of those psychophysical systems that determine his unique adjustment to his environment." Researchers have recently attempted to understand how this "dynamic organization within the individual" may shape the stability and change in youths' personalities. These types of intra-individual characteristics have been investigated as potential moderators of personality stability and change and include other personality traits, IQ, and adaptation.

To elaborate on this point, Eisenberg and colleagues (2000) demonstrated that children's negative emotionality often moderates the links between aspects of conscientiousness and children's social behavior, such that children with poor attentional and behavioral control have poorer quality social functioning when they are also high on negative emotionality. In addition, poor self-control is particularly associated with later externalizing, antisocial behavior for children with higher levels of negative emotionality (Henry et al., 1996). Second, and importantly, children's

intelligence may moderate the stability and outcomes of their personalities. Brighter children may find it easier to handle challenging aspects of their personalities. For example, a longitudinal study of children from age 4 to age 10 demonstrated that IQ predicted the stability and change in behavioral inhibition or shyness. This study found that more intelligent children became increasingly less inhibited over the 6-year span (Asendorpf, 1994). Third, children's relative success or failure in important developmental domains (e.g., academic achievement, social competence with peers) may influence personality development. Research previously reviewed makes clear that childhood personality shapes later adaptation in work, school, and relationships (Shiner, 2006). It is true that the converse can occur, whereby life adaptation may influence children's typical ways of feeling, thinking, and behaving (Shiner & Masten, 2002; Shiner et al., 2002). Clearly, in the future it will be important to examine the patterning of children's individual differences in personality, IQ, and adaptation in order to understand the circumstances leading to stability versus change (Shiner, 2006).

Other factors impacting personality development are parenting, neighborhood context, schools, and the like. As noted previously, Thomas and Chess's (Thomas et al., 1963) early work on temperament highlighted the potential role of family environment in moderating the outcomes of children's early individual differences. This idea was at the core of their goodness-of-fit model of personality development. Recent research has shown several replicable patterns of interactions between childhood temperament and parenting in the prediction of a variety of outcomes (Bates & McFadyen-Ketchum, 2000; Gallagher, 2002; Hawley, 2003; Putnam, Sanson, & Rothbart, 2002). In a handful of studies, it has been found that parenting practices moderate the links between temperament and later externalizing problems. Specifically, when children are high on irritability or score low on agreeableness, several maternal parenting variables predict higher externalizing behaviors. These predictor variables include unskilled disciplining tactics, negativity toward the child, and a lack of monitoring and control over the child (Bates, Pettit, Dodge, & Ridge, 1998; Belsky, Hsieh, & Crnic, 1998; Hagekull & Bohlin, 2003; Rubin et al., 2003; Shiner, 2006; Stoolmiller, 2001). Some research has shown that parenting affects attachment, which impacts emotion regulation and developing extraversion (Sroufe, Carlson, Levy, & Egeland, 1999). Attachment was also found to predict neuroticism and openness (Hagekull & Bohlin, 2003). It appears that skillful, warm parenting may protect highly negative and hard-to-manage children from developing externalizing problems. These findings are similar to those noted by Dishion and Patterson (2006). Because it is possible that genetic factors can account for the covariance of children's temperaments and family variables, future work on goodness of fit and contextual factors should utilize genetically informative research designs (Caspi & Shiner, 2005; Shiner, 2006).

Another source of children's personality continuity and change involves the broader context in which children develop, including the neighborhoods in which they grow and the schools they attend. Children's school environments, peer relationships, and neighborhoods could all have important impacts on whether children's personalities remain stable and whether their personalities lead to either poor or good outcomes (Shiner, 2006). Although these broad contextual factors have

received less research attention than more proximal family processes, a number of recent studies have found evidence of moderating effects of children's contexts on their personality development (Chang, 2003; Dodge et al., 2003; Gazelle & Ladd, 2003). These studies suggest that peer relations, teacher warmth, and neighborhood warmth, safety and stability moderate the links between poor self-control and criminal involvement. Impulsive youth in better-off neighborhoods or who remain in school show less criminal behavior than similarly impulsive youth who live in poor neighborhoods or who drop out of school prematurely (Henry, Caspi, Moffitt, Harrington, & Silva, 1999; Lynam et al., 2000). Taken together, these studies provide clues that broader contexts are, in fact, important moderators and help chart the course of personality development.

CONCLUDING COMMENTS

Despite substantial progress in the decades since Thomas and Chess's (Thomas et al., 1963) classic work on temperament, some aspects of childhood personality development remain quite limited. Great strides have been made with respect to better understanding the connection between temperament and personality. In addition, the higher order structure of personality appears to be gaining further resolution. Yet, although these aspects of personality development have made remarkable progress, some of the same old debates continue to arise in personality research, particularly as it applies to children and adolescents. These include whether it is developmentally appropriate to look for and examine personality in children, and whether it should be assessed in child samples. Much of the concern has to do with whether personality is stable across the life span. Shiner and Caspi (2005) have argued that it is not an either/or argument with regard to stability, but rather both stability and change occur, and the process that underlies that change requires much greater understanding. This does not negate the importance of studying and using the concepts of personality with children and adolescents. The research cited in this chapter shows that not assessing personality means not understanding it. Along with these concerns, the connection between personality and psychopathology is not well understood. Importantly, Caspi and Shiner and Tackett (2006) have pointed the way to some important methods and research designs to advance our understanding of personality as it relates to positive and pathological outcomes in children and adolescents.

Although significant strides have been made in the area of personality in children and adolescents, this chapter signals the need for study in several other areas. The following are suggestions for potentially productive avenues of work on early personality structure that stem from the already important work conducted on children and adolescents. The research reviewed in this chapter leads to some natural conclusions and may provide a road map for where we need to go with personality investigations. These suggestions have in mind the goal of examining the long-term outcomes of childhood personality and the process shaping personality continuity and change from childhood to adulthood. One area of research that is needed is to continue to work on measures that are developmentally appropriate for children and adolescents. Although some work has been done in this area (De Fruyt et al.,

2006), efforts that focus on bottom-up approaches might be informative. As noted by Caspi et al. (2005), the field could benefit from continued creative measures of individual differences and an openness to the need to refine measures. Relatedly, another area of importance for research studies is to continue to examine the higher order structure of personality in children, particularly the structure of individual differences from the very earliest ages (infancy through to age 7 or 8). Shiner (2006) has noted that this is a particularly important area of study because the developmental changes during this period are rapid and sweeping. This chapter also signals the need for future research to specify lower order traits that can be identified in children. Many lower order traits examined in adults may prove to exist in children (e.g., integrity, talent, unconventionality).

Aside from structural issues, much more work is needed on the developmental mechanisms (or processes) at work. Although many of the findings on the long-term outcomes of children's personalities are robust and consistent, future work should examine the pathways linking personality and positive outcomes (Shiner & Masten, 2002). Scientists need to understand more about the mediating processes (Funder, 2001). As Shiner (2006) noted, many studies linking early personality with later outcomes span long periods of time. It could be very informative to also use more short-term longitudinal studies to provide an up-close glimpse of the processes at work. An avenue of research that needs work is the study of the genetic, intra-individual, family, neighborhood, and other contextual factors promoting personality continuity and change in children. Again, it will be important to explore the processes that actually mediate stability and change. Finally, much more research is needed to model the relation between personality and psychopathology. Caspi and Shiner (2005) and Tackett (2006) have provided explicit models for how these relations could be tested, and their guidelines offer a very good starting point.

Discoveries in all these areas may lead to interventions to help children with challenging temperaments or personality styles. Research is also likely to help us understand the features in children and adolescents that promote healthy development and well-being. Future work may serve to enrich understanding of how resilience emerges across development. Each of these avenues of research, and each of these research aims, will be very informative for intervention programs designed to help youth grow and flourish.

SUMMARY

This chapter reviewed research on the development of personality in children and adolescents. It covered the differences between the terms temperament and personality and examined the structure of personality in children and adolescents. An important aspect of studying personality is examining how stable it is across time. This chapter shows that personality is stable but that it also changes across time, even well into adulthood. This chapter also examined both the positive and negative outcomes associated with each of the Big Five dimensions. In addition, this chapter examined the connection between personality and psychopathology. Finally, the current chapter points to directions for future research that could shed light on the development of personality across the life span.

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CHAPTER 13

Treatment Research

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This chapter provides a broad overview of principles, practices, and issues related to planning, conducting, and evaluating treatment research with children. We begin by discussing relevant issues, considerations, and obstacles in conducting child research. These include developmental considerations, measurement issues, informed consent, and confidentiality. The ensuing sections recapitulate the evolving debate concerning clinical efficacy and clinical effectiveness, and the evolution of empirically supported treatments (ESTs) for children. Methodological approaches that address the selection and evaluation of outcome measures appropriate for children—including the clinical significance of findings—are highlighted afterward. The last two sections provide broad coverage of research designs frequently used in child research and discuss factors relevant to identifying a research sample, including sample size and power.

MEASUREMENT ISSUES AND DEVELOPMENTAL CONSIDERATIONS

Measurement Issues

Conducting research with children requires broad knowledge of measurement issues related to maturation, selection of dependent measures, and the degree to which informants (parents, teachers, observers) agree with one another when providing information about children's behavior. This is due to several factors. Children's behavior is qualitatively different from that of adults and thus requires special consideration with respect to measurement and observation. Extrapolating adult measures for use in child research—by changing wording, modifying administration procedures, and making other adjustments—frequently fails to capture the intended underlying construct. Childhood behavior problems typically entail a broad range of symptoms, and these difficulties can affect functioning in multiple situations and settings. Observations of core and secondary symptoms may consequently vary significantly depending on setting and task demands. For example, children with Attention-Deficit/Hyperactivity Disorder (ADHD) typically experience chronic behavioral and academic difficulties at home and at school, and there is generally good agreement among observers. In contrast, children with internalizing disorders are frequently underrecognized until daily functioning is sufficiently

impaired to render their suffering more transparent. Multiproblem and multisituation assessment and analysis are thus the rule rather than the exception in child research.

Finally, children's rapidly emerging and changing verbal and cognitive abilities require careful consideration for research studies in which children are asked questions about themselves, others, and internal (e.g., mood/affect, anxiety, concentration) or external (e.g., activity level, impulsiveness) states. This consideration is complicated by extant research demonstrating differences among raters (e.g., parents, teachers, and trained observers), across settings (e.g., home and school), and for different types of behavior problems, as discussed later in the chapter.

Developmental Considerations

Treatment research with children is an intriguing but complicated domain of clinical psychology due to developmental factors that play a key role in our understanding of and assessing maladaptive behavior. For example, many behaviors associated with or mistaken as maladjustment and emotional disturbance are relatively common during childhood. Estimated prevalence rates based on parent reports indicate that fears and worries (43%), temper tantrums (80%), bedwetting (17%), and restlessness (30%) occur frequently in 6- to 12-year-old children (Lapouse & Monk, 1959) but do not necessarily portend later psychological dysfunction. In a similar vein, most children exhibit stranger anxiety around 8 months of age that mirrors their ability to discriminate a familiar face from an unfamiliar face. Separation anxiety becomes evident shortly after stranger anxiety begins, characterized by distress and an inability to be readily comforted by others in the absence of the child's parents or primary care providers. This is a normal response pattern in very young children, whereas excessive anxiety concerning separation from major attachment figures in later years accompanied by other behaviors—such as reluctance to attend school, unrealistic worry that an untoward calamitous event will separate the child from his or her parents, and somatic complaints—may signal the onset of a clinical disorder. Oppositional behavior is a normal developmental phenomenon in 18- to 36-month-old children; however, its persistence and accompaniment by other behavior problems is considered maladaptive in later years.

Behavioral and emotional problems evident in less severe forms during different developmental stages frequently represent normal developmental progressions in children, whereas their persistence may predict developing psychopathology. Consider the heterotypic continuity of ADHD symptoms as an example. Symptoms of ADHD arise early in childhood; however, their presence does not necessarily portend a persistent pattern of ADHD beyond 3 years of age in an estimated 50% to 90% of children so characterized (Palfrey, Levine, Walker, & Sullivan, 1985). Continuation of early ADHD-like symptoms to 4 years of age, however, is highly predictive of clinical hyperactivity at 9 years of age (Campbell, 1990). Thus, the early onset, degree, and persistence of symptoms past 4 years of age is highly predictive of a clinical diagnosis (indicating continuing and worsening impairment) and continuing difficulties throughout adolescence and early adulthood.

Other problem behaviors show clear developmental trends and are not considered maladaptive until late childhood. Lying and destructiveness are common exemplars. An estimated 50% of boys and girls engage in lying by age 6, based on parent report. Frequency of lying decreases to approximately 25% and 13% by age 7 in boys and girls, respectively, and continues to follow a downward trend as a function of increasing age in most children. Destructive behavior shows a similar developmental trend, peaking at 3 and 5 years of age in girls and boys, respectively, with a clear downward trend for both sexes through age 13 (MacFarlane, Allen, & Honzik, 1954).

Some maladaptive behaviors remain stable over time in terms of their frequency, but change with respect to topography or form. The manifestation of childhood aggression is a good example. Threatening, pushing, and shoving in young children frequently evolve into verbal and physical assault in older children, who continue to manifest aggressive behavior (Dishion & Patterson, 2006).

The foregoing examples inform us that many problematic behaviors in children will diminish or remit over the course of normal development and may not be appropriate or high priorities for intervention unless the behavior is sufficiently impairing and more rapidly alleviated by clinical treatment. They also highlight the importance of understanding the typical frequency and topography with which specific behavioral and emotional problems occur in children, and how base rates change over the course of development. Failure to appreciate and control for these variables may inadvertently result in attributing change to treatment rather than maturation or other historical events. Study design must also plan for possible topographical changes in maladaptive behavior if outcome assessment spans several years (e.g., in some cross-sectional and longitudinal studies). This may necessitate using different age norms or even different measurement instruments, which introduces unwanted error into the protocol.

A different set of developmental considerations comes into play when research studies require investigators to assign children to groups based on the presence or absence of a clinical disorder. Structured or semi-structured clinical interviews, coupled with other instruments and measures, are characteristically used to render a clinical diagnosis for purposes of group assignment (for reviews, see Orvaschel, 2006; Rapport, Timko, & Wolfe, 2006). Skilled interviewers have little difficulty interviewing adult informants, whereas interviewing children about their emotional state can be extremely difficult, and at times impractical, depending on developmental factors. Children's emotional development significantly affects their ability to respond to clinical interview questions about themselves and how they feel. The ability to label emotions and talk to others about feelings develops by about 2 years of age, and children have a reasonably well-developed range of emotional displays and labels by school age. Children can report changes in positive emotions before negative ones, but most experience difficulty describing variability in negative emotions until late childhood or early adolescence. Conscious awareness of and the ability to report on emotions as a state or system is not fully developed until middle or late adolescence (Haviland-Jones, Gebelt, & Stapley, 1997).

MEASURING AND ASSESSING CHILDREN'S BEHAVIOR

Informant Ratings of Children's Behavior

Behavior checklists and rating scales play a prominent role in child research. They serve as an important source of information concerning a child's behavior in different settings, how others judge behavior, and the extent to which behavior deviates from age- and gender-related norms. Information gleaned from rating scales contributes to the diagnostic process, and several scales serve as treatment efficacy measures.

Standardized scales are available that provide broad (e.g., internalizing, externalizing behavior problems) and narrow (e.g., measures of particular clinical disorders or states such as depression) indices of behavior, as well as for particular constructs (e.g., self-esteem, perceived competence) and other types of functioning (e.g., classroom performance, adaptive behavior, peer perceptions). Incorporating rating scales in research necessitates examination of the scale's psychometric properties (i.e., whether the instrument provides valid and reliable information with respect to what it purports to measure), general knowledge concerning the degree to which different informants can be counted on to provide valid information about a child's behavior, and factors that influence informant ratings. A judicious review of an instrument's psychometric properties is a necessary first step in constructing a research protocol. This is accomplished by checking with the publisher or author of the instrument, conducting a literature search of the instrument's psychometric properties, and researching specialty texts that cover tests and rating scales (e.g., *The Fifteenth Mental Measurements Yearbook,* edited by Plake, Impara, & Spies, 2003).

Extant research examining the degree to which raters agree concerning the presence of behavior problems reveals several, relatively consistent trends. In general, informants who interact with the child in the same environment (e.g., parents) tend to show better agreement in their reports of behavior than those who interact with the child in different environments (e.g., parents versus teachers versus mental health workers; Achenbach, McConaughy, & Howell, 1987). Agreement between parents, however, is less than ideal, as highlighted in a recent meta-analytic review (Duhig, Renk, Epstein, & Phares, 2000). Correspondence between mother and father ratings of children's behavior problems varied depending on the category of problem behaviors examined. For example, mean correspondence between parents for internalizing behavior problems was .45 as opposed to .63 and .70 for externalizing and total behavior problem scores, respectively—perhaps because externalizing behavior problems are easier to judge and/or more consistent across situations or settings (Achenbach et al., 1987; Walker & Bracken, 1996). Although a correlation of .45 is considered a moderate level of agreement between raters, it nevertheless indicates that only 20% ($.45^2$) of the variability in one parent's ratings can be explained (predicted) by the variability in the other parent's ratings. Higher correspondence between parents was reported for adolescents than for younger children when examining both internalizing and externalizing behavior problems, and the family's socioeconomic status appears to exert small but significant effects

on parent ratings. Agreement among raters also varies for clinical and nonclinical samples, is more variable for particular types of behavior and emotional problems, and is unacceptably low when comparing children's self-ratings to adult ratings (Achenbach et al., 1987).

The foregoing summary indicates that parents and teachers can be relied on to provide reasonably reliable ratings of children's behavior in the context in which they observe children for broad indices of behavior problems such as internalizing and externalizing behavior problems. Diminished correspondence between informants is evident, however, for ratings of more discrete types of behavior problems, when informant ratings are based on different settings, and when comparing children's own ratings to those of adults. Investigators need to consider these issues during the research design stage and minimize error variance by utilizing a multitrait, multimethod approach. This approach recognizes the undesirable shared overlap (variance) among similar measures and raters of specific constructs and behaviors and informs the investigator how to maximize the unique contribution of measures relevant to the variables studied (cf. Burns & Haynes, 2006, for a review).

Measurement Limitations

Limitations common to most rating scales include their reliance on subjective judgments and multiple threats to internal validity. These include halo effects, response bias, intensity and immediacy effects, and rater expectation bias (Harris & Lahey, 1982; McClellen & Werry, 2000). The underlying assumptions that Likert rating formats reflect interval-level measurement (i.e., that the unit of measure between 2 and 4 is identical to the difference between 1 and 3, and that these behavioral units are consistent across scales) and that all behavioral and emotional problems should be equally weighted (count the same when endorsed at the same level) represent additional psychometric challenges to clinical rating scales. For example, two children could receive identical Conduct Disorder rating scale scores, yet be dramatically different with respect to behavioral disturbance. One child might receive four 2-point ratings (indicating the highest level of severity or frequency) for items such as lying, stealing, cursing, and truancy, whereas the other child receives four 2-point ratings for breaking and entering, rape, firearm violation, and physical aggressiveness. Their total scores would be identical, yet the second child's symptoms are clearly more serious than the former's. Other factors, such as the presence of psychopathology among informants, the informant's experience with children, and other demand characteristics, represent additional challenges to clinical child rating scales.

Several desirable psychometric properties have not been thoroughly established for many child rating scales. For example, scale validity is usually accomplished by demonstrating that scores derived from one instrument correlate with those derived from an already established scale of the same latent variable, or by demonstrating that children known to be highly active (e.g., ADHD) score significantly higher than normal peers on the scale. Most scales meet at least one of these two criteria. Extant research, however, reveals that even when scale scores are correlated, they may be unrelated to objective measures of the same trait (e.g., Rapoport, Abramson,

Alexander, & Lott, 1971; Stevens, Kupst, Suran, & Schulman, 1978). For example, when measured by step counter, nearly 64% of children rated as clinically hyperactive were less active than the most active child rated as being normal by the teacher (Tryon & Pinto, 1994). Collectively, these shortcomings limit the interpretability and usefulness of activity rating scales.

The diagnostic utility of most rating scales is unknown; however, this situation has improved considerably over the past several years. Four metrics address this concern: sensitivity, specificity, positive predictive power (PPP), and negative predictive power (NPP). Sensitivity and specificity indicate the proportion of the group with a target diagnosis who test positive and negative on a measure, respectively. These two indices are useful for examining the overall classification accuracy of rating scales and other instruments, but are not particularly valuable to clinicians who are unaware of a child's diagnostic standing prior to referral. The statistics most relevant for this purpose are PPP and NPP. As it applies to rating scale utility, PPP indicates the conditional probability that a child exceeding a rating scale cutoff score meets criteria for a particular diagnosis such as ADHD (i.e., the ratio of true positive cases to all test positives). In contrast, NPP indicates the conditional probability that a child who doesn't exceed an established cutoff score will not meet criteria for a particular clinical diagnosis (i.e., the ratio of true negative cases to all test negatives). High values (e.g., $>.80$) for all four indices are desirable.

Finally, it is becoming increasingly common for investigators to create specific measures for assessing outcome by borrowing from measures with established psychometric properties and reconfiguring them for the study. For example, items contained in adult rating scales may be reworded and rekeyed (i.e., changed from a 1–5 to a 1–3 metric scale) before administering them to children. Others borrow from different scales that reflect a similar latent construct (e.g., anxiety) and combine or select specific items thought to better reflect the construct. These well-intentioned efforts ignore the backbone of psychological measurement: psychometric theory. Altering a measure, no matter how strong its existing psychometric properties, requires the investigator to reestablish its psychometric properties with the intended population (e.g., children) before using it in research. The researcher must also consider relevant developmental phenomena when reconfiguring rating scales and other questionnaire instruments. For instance, young children are unable to describe variability in negative emotions until late childhood or early adolescence, and usually cannot reliably discriminate between behavioral descriptions on a 5-point frequency or severity scale.

Interview Measures of Children's Behavior

Conducting detailed clinical interviews with children and their caregivers is a common method for screening participants, determining group assignment, and obtaining study data for child research investigations. Structured and semi-structured clinical interviews provide unique information beyond the data gained from rating scales and currently represent the only assessment option that allows clinicians to probe for the onset, course, and duration of endorsed symptoms—a necessity for differential diagnosis. As a result, they are considered the gold standard for

assessment and diagnosis. The distinction between structured and semi-structured interviews lies in the degree of freedom granted to the clinician to stray from a given script and ask open-ended, probing questions in response to symptom endorsements made by the interviewee. As a general rule of thumb, semi-structured interviews evoke more detailed information concerning presenting symptomatology owing to the extensive probing and clarification permitted (e.g., by asking for examples) but require greater clinical acumen and training.

The development of structured and semi-structured interviews represents the recognition of both the questionable reliability and validity of unaided clinical interviews and the frequent disagreement between parent and child reports of symptom endorsement and severity. Structured and semi-structured clinical interviews result in decreased error compared to unstructured clinical interviews arising from both internal (e.g., differential training level of clinicians, clinician biases) and external (e.g., discrepancies between informants) sources. The interviews typically require between 1 and 2 hours to complete, depending on a range of factors. These include the clinician's experience, the informant's ability to remain focused and recall historical information, and the severity, range, and duration of presenting problems (Table 13.1). The time investment limits their practicality for repeated use (e.g., for assessing treatment effects). Financial investment varies significantly across interview schedules. Some are free and available online (e.g., Schedule for Affective Disorders and Schizophrenia for School-Age Children [K-SADS]), whereas others have an initial cost of $600 in addition to $2,000 fixed training costs (e.g., Child and Adolescent Psychiatric Assessment [CAPA]). All but one of the semi-structured interviews covers all major *DSM-IV* diagnoses for school-age children through age 18. Some of the semi-structured interviews provide separate versions for parents, children, and adolescents. None of the available semi-structured interviews includes teacher versions. Clinicians must use other instruments to obtain school-related information for purposes of establishing impairment across multiple settings.

Strong convergent and discriminant validity is reported for most of the available semi-structured interviews, and test-retest reliability suggests stability of diagnosis over 1 to 3 years in clinical samples (Pelham, Fabiano, & Massetti, 2005). High sensitivity and specificity are typically reported for the semi-structured interviews, with only limited information available pertaining to PPP and NPP. The lack of PPP and NPP metrics is due in part to the use of semi-structured clinical interviews as the gold standard from which the predictive power of other measures (e.g., rating scales) is established.

The CAPA appears to be the most extensively developed of the clinical interviews, but it requires up to 2 weeks of classroom instruction and an additional 1 to 2 weeks of practice to acquire the necessary certification (Angold & Costello, 2000). This training is estimated to cost $600, and there is an additional fixed cost of $2,000, which may limit widespread usage among clinicians, especially when schedules such as the K-SADS demonstrate adequate reliability and validity and are available online at no cost. The CAPA, however, may be superior to other available schedules due to several excellent features. The instrument provides for an intensity rating that varies by three symptom groupings: intrapsychic phenomena

Table 13.1 Clinical Child Diagnostic Interviews

Measures	Age Range	Time (min)	Test-Retest (Kappa)	Symptom History	Disorders Considered	Scoring Format	Instrument Cost ($)	Training Required
Structured Interviews								
Diagnostic Interview Schedule for Children IV (DISC-IV)	6–17 (P) 9–17 (C)	90–120 (P) 45–90 (C)	0.79 (P) 0.42 (C)	4 weeks/ 12 months	All major Dx	Y/N	150– 2,000	2–3 day training module
Diagnostic Interview for Children and Adolescents IV (DICA-IV)	6–17	60–120	NR (P) 0.32 (C) 0.59 (A)	4 weeks/ 12 months	All major Dx	Y/N	1,000	2–4 weeks
Children's Interview for Psychiatric Syndromes (CHiPS)	6–18	40	0.4	NR	All major Dx	Y/N	115	NDR
Semi-structured Interviews								
Schedule for Affective Disorders and Schizophrenia for School-Age Children (K-SADS)	6–17	30–90	0.63	6 months/ lifetime	All major Dx	0–3	Free online	CTR
Semi-structured Clinical Interview for Children and Adolescents (SCICA)	13–18	60–90	0.57 (AP Scale)	NR	Does not correspond with *DSM-IV*		110–295 25 for 50	NDR
Child and Adolescent Psychiatric Assessment (CAPA)	9–17	20–210 $M = 66$ (P) 22–150 $M = 59$ (C)	NR for ADHD 0.55 for CD	3 months	All major Dx	0–3	600 + 2,000 Fixed Costs	BA
Interview Schedule for Children and Adolescents (ISCA)	8–17	120–150 (P) 45–90 (C)	Between 0.64–1.0	NR	All major Dx	0–3		CTR

Notes: Properties are equivalent for parent and child/adolescent version unless otherwise indicated.
A = Adolescent; AP = Attention Problems; BA = Bachelor's level training; C = Child; CTR = Clinical training required; Dx = Diagnosis; NDR = No degree requirements; NR = Not reported; P = Parent.

such as worrying, qualitatively different symptoms such as psychosis, and conduct disturbances. Training and coding are based on a detailed glossary, and thoroughly investigated symptoms are matched to appropriate glossary definitions and levels of severity. Formal rules are provided for the use of screening, mandatory, and discretionary questions.

The Diagnostic Interview for Children and Adolescents, *DSM-IV* edition (DISC-IV), provides separate versions for children (ages 6 to 12) and adolescents (13 to 18) based on field testing of interview questions with different age, gender, and racial groups. Two to 4 weeks of training at a cost of approximately $1,000 are required to reach the desired level of competence. The duration of training is based on clinician experience and includes topics such as age-appropriate probe questions and maintaining a child's interest through techniques such as tone of voice and appropriate nonverbal gestures. Reliability estimates corroborate research findings indicating that children are less reliable reporters of externalizing symptoms but more reliable reporters on internalizing symptoms relative to their parents. Computer versions are available, but initial research suggests poor reliability compared to the standard interview format.

The K-SADS is currently the most widely used semi-structured clinical interview. The Present-Lifetime (PL) version collects information from the parent regarding both current symptomatology as well as symptomatology at its most frequent and severe levels in the past. A separate interview is conducted with the child, and a third pair of ratings is generated based on integration of the parent and child reports with historical information and other data (e.g., rating scales). The K-SADS-PL focuses on chronology, treatment, impairment, and severity of symptoms. The initial screening interview consists of 82 items covering all major *DSM-IV* diagnostic categories. Cutoff scores are used to determine the need to administer the in-depth supplementary sections available for each diagnostic category, thus shortening administration time by allowing the clinician to skip supplementary sections based on negative endorsement of key screening questions. Interrater reliability estimates for the K-SADS are among the highest of any of the semi-structured clinical interviews. Extensive knowledge of diagnostic and symptom subtleties is required owing to lack of formal training requirements, and clinicians must use their judgment to interject noncued verbal probes to clarify informant responses and elicit examples of problematic behavioral and emotional symptoms.

Observational Measures of Children's Behavior

Behavioral observation is a time-consuming but valuable approach for obtaining detailed information about children and the environments in which they learn, play, and interact with others. It refers to a process in which observers record motor, verbal, and interactive behaviors of children using carefully defined operational criteria as opposed to evaluative judgments (e.g., rating scales) or mechanically activated devices (e.g., movement monitors, physiological monitoring). As such, many consider it the sine qua non of child assessment. An advantage of using behavioral observations is that the technique is virtually unrestricted with respect

to context. Observations can be conducted in naturalistic settings (e.g., to record classroom, playground, or bus riding behavior), analogue settings (e.g., a clinic room furnished to resemble a classroom), or specialty clinics (e.g., to assess parent-child interactions as part of a treatment program for children with Conduct Disorder, to assist a diagnostic evaluation for autism).

Numerous observational coding schemas are available for research purposes. Some permit coding of a wide range of behaviors (e.g., parent-child interactions, general classroom deportment) and are best suited for particular settings (e.g., home or classroom observations). Others permit highly refined and detailed observation, recording, and measurement of more discrete types of behavior (e.g., gross motor activity). Assessing interobserver agreement is nearly always required for research studies and involves pretraining on the selected coding schema and arranging for multiple observers to code the targeted behaviors simultaneously while in the setting or based on taped recordings. Detailed information concerning how to select particular stimuli, responses, and specific recording techniques; how to operationalize particular types, forms, and classes of behavior; and how to calculate interobserver reliability is readily available in classic texts on child behavior assessment (Kazdin, 1982; Ollendick & Hersen, 1998).

Direct observations represent the gold standard for experimental and outcome research for many investigators, but their use in clinical assessment is limited by several factors. No two research teams or commercially available observation systems define behaviors in exactly the same way, and research indicates that differences in observational schema can produce widely discrepant results in collected data (see Kofler, Rapport, & Alderson, in press, for a meta-analytic review).

Commercially available observation systems shown in Table 13.2 are available for school-age children and typically require between 10 and 30 minutes of observation. Multiple observation days are required to produce representative and reliable data; however, some systems offer software versions of their product, allowing researchers to record behavior directly onto a personal digital assistant (PDA) or laptop computer.

Direct observations are commonly used in school and institutional settings that require a formal functional analysis to determine whether antecedent and consequential stimuli or events contribute to a child's maladaptive behavior. These specialized assessments are typically conducted by highly trained school psychologists and individuals specializing in applied behavior analysis.

Although direct observations by independent observers can provide more objective and valid data than any of the other assessment methods discussed here, their relatively high temporal cost, coupled with the general lack of norms, suggests that their usefulness for assessment and group assignment may be limited to situations where large discrepancies exist among informant reports.

An alternative and less costly procedure for obtaining information relevant to children's classroom functioning is to ask classroom teachers to answer specific questions about a child's day or to save work samples. For example, desk checks ("Is the child prepared for class?"), teacher records of verbally intrusive behavior, and the percentage of daily academic assignments completed correctly appear to be useful for discriminating between children with and without ADHD (Pelham

Table 13.2 Mechanical and Observational Assessment Tools

Instrument/ Distributor	Age Range	Recording Length	Norms	Software Available	Cost
Mechanical					
Actigraphs Ambulatory Monitoring MiniMitter MTI, Inc.	Any	22 days per 32 kilobytes of memory	N	Y	Starter: $1,000+ (with necessary software and reader interface); $500–$2,000 for each additional actigraph
Actometers[a] Model 108 Engineering Department Times Industries Waterbury, CT 06720	Any	Variable	N	Y[b]	*NR*
Pedometers (available at sporting goods stores) Stand-alone with data downloadable to PC	Any	Range: 99,999 steps (~5.25 miles) to 1,000 miles	N	N Y	$10–$40 $125–$400+
Direct Observations					
ADHD BCS Barkley, 1990	*NR*	15 min.	N	N	*NR*
AET-SSBD Sopris West	School age	15 min.	Y	N	Kit: $108 (includes all three parts of SSBD)
ADHD-SOC Checkmate Plus, Ltd.	School age	16 min.	N	N	Kit: $25
BASC-2 SOS AGS, Inc.	School age	15 min.	N	Y	25 forms @ $33
BOSS Harcourt Assessment	School age	15 min.	N	Y	Kit: $120
COC Abikoff, 1977/1980	School age	32 min.	N	N	NR
DOF ASEBA	5–14	10 min.	Y	Y	50 forms @ $25
SECOS Saudargas, 1997	Grades 1–5	20 min.	Y	N	
Noldus Observer Noldus Information Technology	Any	Variable	N	Y	Observer Basic 5.0 $1,795 Observer Video Pro 5.0 $5,850

Notes: AET-SSBD = Academic Engaged Time Code of the SSBD; BCS = Behavior Coding System; BOSS = Behavioral Observation of Students in Schools; COC = Classroom Observation Code; DOF = Direct Observation Form; NR = not reported; SECOS = State-Event Classroom Observation System; SOC = School Observation Code; SOS = Student Observation System.

[a]Many studies report either using the Kaulins & Willis actometers (no longer manufactured) or enlisting a jeweler to modify a self-winding wristwatch as described by Schulman & Reisman (1959).

[b]Eaton, McKeen, & Saudino (1996) provide SAS syntax for performing group-level data analysis based on actometer readings.

et al., 2005). These methods have the potential for objectivity that characterizes the independent observation methodologies but await critical evaluation to determine their utility for rendering diagnostic judgments at the individual level.

Potential Obstacles to Conducting Research with Children

Knowledge of regulatory statutes, the role and function of institutional review boards (IRBs), and general procedures for ensuring and obtaining informed consent is essential for conducting research with children. Detailed information concerning procedures and regulatory statutes related to studies involving children can be obtained from the Department of Health and Human Services website (www.nihtraining.com/ohsrsite/guidelines/guidelines.html). These regulations provide the minimum standards for protecting human subjects, particularly under the subpart D section, which includes detailed information concerning the involvement of children in research.

Regulations require the *assent* of the child or minor and the permission or *consent* of the parents or legally authorized guardian when children are involved in research. Assent is a child's affirmative agreement to participate in research. The standard for assent is the ability to understand, to some degree, the purpose of the research and what will happen if one participates.

Consent refers to the permission or agreement of parents or guardian to the participation of their child or ward in research. There are two forms of parental consent. *Passive* parental consent is a procedure that requires parents to respond only if they *do not* want their child to participate in a research project. *Active* parental consent requires the parent to return a signed consent form indicating whether or not they are willing to allow their child to participate. The latter method is recommended, although many researchers prefer the passive consent procedure because of the larger participant pool typically achieved. As a general rule for research involving minimal risk, it is sufficient to obtain informed consent from one parent.

Informed consent consists of three primary elements: knowledge, volition, and competency. Parents or guardians of children must be knowledgeable about all aspects of a study, which includes a thorough description of the facts, plausible risks, and potential sources of discomfort associated with an experimental study that may affect their decision to permit their child to participate. This information must be presented to the child's parents or legal guardian in an understandable fashion, followed by an opportunity for them to ask questions and clarify any issues that might not be thoroughly understood.

Volition refers to the process in which participants (or, in the case of minor children, their parents or guardian) agree to participate in a study free of coercion or threat. Subjects (or, in the case of minor children, their parents or guardian) must be able to decline or withdraw from participating at any time preceding or during an experiment without penalty, and should be informed of this provision (i.e., participation must be entirely voluntary).

Ensuring competency in child research typically entails the parents' or guardian's ability to render an educated decision and give consent for their child to participate based on thorough knowledge and understanding of the study.

A consent form signed by the parents or guardian of children indicates their willingness to have their child participate in the study based on an informed decision. These forms traditionally undergo formal review by an IRB or agency committee that evaluates the research proposal, consent procedures, and the form itself. Consent forms vary by design, but must contain a thorough description of the study, inherent and potential risks and benefits, procedures, and issues concerning confidentiality and how the study data may be used. Components of an informed consent form are described in Table 13.3.

Institutional review boards serve to assess the risks, possible benefits, and discomforts associated with a research project. Before beginning a project, an appropriate IRB committee must consider and formally approve all research proposals. University-based projects will typically be reviewed by the institution's IRB panel, and additional approvals may need to be obtained from other agency boards (e.g., hospitals, school systems, National Institute of Mental Health), depending on the selection of participants and whether the project is funded or sponsored by private or federal agencies. Many universities and research centers currently mandate

Table 13.3 Components of Informed Consent Forms

Section of the Form	Purpose and Contents
Overview	Presentation of the goals of the study, why this is conducted, who is responsible for the study and its execution.
Description of procedures	Clarification of the experimental conditions, assessment procedures, requirements of the subjects.
Risks and inconveniences	Statement of any physical and psychological risks and an estimate of their likelihood. Inconveniences and demands to be placed on the subjects (e.g., how many sessions, requests to do anything, contact at home).
Benefits	A statement of what the subjects can reasonably hope to gain from participation, including psychological, physical, and monetary benefits.
Costs and economic considerations	Charges to the subjects (e.g., in treatment) and payment (e.g., for participation or completing various forms).
Confidentiality	Assurances that the information is confidential and will only be seen by people who need to do so for the purposes of research (e.g., scoring and data analysis), procedures to assure confidentiality (e.g., removal of names from forms, storage of data). Also, caveats are included here if it is possible that sensitive information (e.g., psychiatric information, criminal activity) can be subpoenaed.
Alternative treatments	In an intervention study, alternatives available to the client before or during participation are outlined.
Voluntary participation	A statement that the subject is willing to participate and can say no now or later without penalty of any kind.
Questions and further information	A statement that the subject is encouraged to ask questions at any time and can contact an individual (or individuals) (listed by name and phone number) who is available for such contacts.
Signature lines	A place for the subject as well as the experimenter to sign.

Source: Clinical Psychotherapy: Developing and Identifying Effective Treatments, by A. E. Kazdin, 1988, New York: Pergamon Press. Reprinted with permission from Allyn & Bacon and the author.

that all persons involved in the conduct of research—from the investigators to the research assistants—take a specific training course and pass an exam concerning the ethical and appropriate conduct of research. Several of these are offered online (e.g., www6.miami.edu/citireg/), and results are forwarded to the host university's IRB as a precondition for conducting research.

CHILD RESEARCH SETTINGS

General Dimensions

A research study may be conducted in a laboratory (e.g., clinic, university laboratory) or an applied setting (e.g., school, home, hospital, community mental health center). The advantage of most laboratory settings is that they typically allow for maximal control over the experiment. Their most apparent disadvantage is that the nature of the setting is usually quite different from most real-life situations, which calls into question the generality of results. The advantage of applied settings is that they frequently represent real-life situations or environments, which allows for greater generalization of results. Their most obvious disadvantage is that they usually afford less control than a laboratory setting. Typically, one thinks of true experimental and quasi-experimental designs as taking place in a laboratory setting and observational designs as taking place in an applied setting. This is because the researcher attempts to exert a high degree of control in true and quasi-experimental designs, whereas few or no controls over extraneous variables are implemented in observational designs. True or quasi-experiments, however, can be conducted in applied settings, and observational studies can occur in a laboratory setting. Single-case research designs are used in both settings.

Gaining Access to Educational Settings

Conducting research in applied settings such as a school involves striking a balance between maintaining the requisite degree of scientific rigor and respecting the mission and structure of the educational setting. Research protocols must be planned such that disruption of the school or classroom routine is minimized. The investigator must be mindful that in the educational setting, classes (as opposed to individuals) are frequently the units of analysis. For protocols involving individual administration, pulling a child from some class periods (e.g., recreational reading) may be less disruptive than others (e.g., small group science activity) for the class as a whole and the child in particular. Moreover, schools may be especially protective of certain groups of students, such as children with developmental disabilities or behavioral/emotional problems. Conducting research in educational settings consequently requires careful planning and organization to maintain scientific rigor and accommodate the structure of the school. The ensuing discussion is particularly relevant for group studies in which a large sample of participants is required.

Subject recruitment is a necessary first step for most research projects, and schools are a logical source for studies involving children. The choice of school depends on a number of factors, which vary in degree of relevance depending on the focus of the investigation. Among the factors to consider are the socioeconomic

status of the community served by the school and the ethnic distribution of the school's population of students. For investigations involving multiple groups, the number of students enrolled in the school becomes an important consideration, in light of the power needs of the study's design to detect group differences.

Gaining access to research participants involves a stepwise progression to enhance the likelihood of a proposal's acceptance by educational institutions. Prior to contacting schools, it behooves the investigator to obtain endorsements from related agencies and the district superintendent of education—this smoothes the way to the engagement phase of decision making for the school principal. Subsequent contact with the principal should first be handled formally through an individually addressed letter that briefly explains the importance and nature of the study and mentions the endorsements received, followed by a telephone call approximately 1 week later to secure an appointment for a meeting.

Prior to contacting schools, the investigator must prepare to address any questions or concerns that may be forthcoming. Petosa and Goodman (1991) outline five phases of decision making that school officials undergo and discuss them in relation to decisions surrounding research proposals. In the *legitimacy phase,* the credibility of the investigator and the relevance of the study are of central concern. Representatives of the research project who contact the school should be prepared to address questions concerning the expertise and affiliation of the chief investigator, the source of funding for the project, and the project's relevance for the education and welfare of children or the community at large. In the second, *information-seeking phase,* the concerns of school officials involve the impact of the project on the day-to-day operation of the school. The investigator should consider the timing of the data collection with the school's schedule in mind. For example, the weeks preceding the winter holiday break and the end of the school year are often hectic for school personnel. Other times of the year, depending on the school system, are largely devoted to mandated standardized achievement testing. School officials will also wish to know the specific requirements of the project to gauge its impact on the school's mission and function. The investigator should be prepared to address questions pertaining to the measures to be administered, the class time required for participation in the project, the costs of participation accruing to the school (e.g., the extent of involvement of school staff), and the resources that will be available to the school (e.g., teacher training and curricular materials). During this phase, school officials will also be vigilant of any potential controversy that may arise due to the study's methodology, including parent objections to particular rating scales or survey questions and the potential for wasted classroom learning time.

In the ensuing *expression of limitations phase,* education officials are likely to express concerns over idiosyncratic situational factors that may pose obstacles to the school's participation in the project. The resourceful investigator should prepare to address concerns over issues of *fit* between the study's requirements and the school's structure, without undermining the integrity of the research project. If the investigator is successful, all parties enter the *engagement phase* of mutual problem solving, in which various ways of increasing the feasibility of the study are considered, such as finding the optimal dates and times and allocating physical space for data collection.

In the final, *commitment phase,* all parties make specific plans for the school's participation in the research project. Olds and Symons (1990) recommend presenting a *data collection participation form* for the principal's signature to secure the school's commitment. The initial start date for the study should be made far enough in advance to allow the school time to block the dates set aside for the data collection in its master calendar and to disseminate the information to parents and school staff. At least 1 week prior to the agreed-upon date for data collection, a follow-up reminder to the school should be made by telephone. At that time, information concerning any changes in the school's normal schedule or routine—such as assemblies or field trips—made subsequent to the meeting with the principal should be solicited.

The foregoing information and sequence of steps pertains primarily to schoolwide and classroom studies but is also applicable to conducting single-case research. Because single-case research impacts a classroom rather than a school, the key contact becomes the teacher and, as a courtesy, the principal, rather than the district superintendent. The decision-making process on the part of school personnel remains the same, however, and the investigator should follow the same suggestions for proactive planning and organization.

Additional considerations are warranted once project personnel are in the school to facilitate school support and cooperation. Gaining and maintaining the support of school personnel depend on the behavior and professionalism of research associates. These individuals need to be visible but unobtrusive, professional in appearance and demeanor, friendly and polite without exception, and respectful of school policies, procedures, and personnel. Research associates must be mindful of the school's mission to educate children in a safe environment and, from the outset, solicit information from school staff about such routine procedures as signing in and out upon entering and leaving the campus and whether identification badges are required. Most important, researchers should adhere to the original contract as agreed to by the school principal and refrain from such deviations as adding measures, collecting data on other dates, and attempting to deliver any intervention not previously agreed upon.

The research design and methodology are of prime concern to the investigator, but respect for the mission and structure of the educational setting is equally important. Demonstrating respect by minimizing disruption of daily operations increases the likelihood that the research proposal will be approved and ensures that the data collection process will run as smoothly as possible. Moreover, following through with promised benefits after the data have been collected represents a sound investment, as the school may be amenable to participation in future research projects. One means of accomplishing this is to volunteer to provide a teacher workshop that includes summarizing the study's results.

Clinic and Laboratory Investigations

Clinic- and laboratory-based investigations provide researchers with unique opportunities to manipulate carefully designed independent variables and observe their effects on dependent variables. These types of studies tend to have unparalleled

heuristic value and frequently test predictions stemming from extant models of child psychopathology. For example, a recent model of ADHD hypothesized a causal relationship between working memory and activity level in children (Rapport, Chung, Shore, & Isaacs, 2001; Rapport, Kofler, Alderson, & Raiker, in press) and challenged the notion that hyperactivity represents a ubiquitous deficit unrelated to task and setting demands (Porrino et al., 1983). Testing predictions stemming from the working memory model would be difficult to accomplish in a typical classroom for a variety of reasons. Measuring classroom activity level by utilizing actigraphs or behavioral observations does not represent a serious obstacle; however, introducing working memory tasks at varying degrees of difficulty in a counterbalanced manner to specific children and appropriate controls might prove impractical, if not unfeasible. Moreover, obtaining appropriate control over ambient noise levels (including ongoing student conversations or comments), lighting, chair type and placement, distractions, movement, and other setting characteristics is necessary to ensure that changes in the dependent variable (e.g., activity level) are due to differences in working memory demands (e.g., manipulating stimulus set size) rather than extraneous factors. The research clinic provides an ideal setting for these types of investigations.

Research clinics also enable investigators to scrutinize a vast array of personal and interpersonal processes by means of analogue investigations. These studies establish pseudo-situations that mimic important aspects of a child's life, either at home (e.g., asking parents to interact with their children following specific guidelines) or at school (e.g., having children attend a simulated classroom). Results of these studies often have important diagnostic and treatment implications. For example, observations of children with autism interacting with their parents in an analogue setting reveals a high degree of diagnostic sensitivity based on the child's reciprocal social interaction, play, stereotypic behaviors, gesturing, and communication (Filipek, Accardo, & Baranek, 1999). Teaching parents more adaptive ways to interact with their children in analogue settings has also shown exceptional promise for reducing Oppositional Defiant Disorder (Reid, Webster-Stratton, & Hammond, 2003).

CLINICAL EFFICACY, CLINICAL EFFECTIVENESS, AND EMPIRICALLY SUPPORTED THERAPIES

Overview

The terms *clinical efficacy* and *clinical effectiveness* emerged in response to calls for increased use of empirically validated treatments for individuals suffering from mental health problems. Debate evolved between research centers demonstrating that particular treatments resulted in significant client improvement following prescribed protocols, and the failure to find similar levels of improvement in applied settings such as community mental health centers. Hypothesized variables contributing to the lack of generalization of treatment effects across settings included differences in therapist training and orientation, use of manualized therapies to minimize therapist influences and maximize treatment integrity, and the extensive

oversight and supervision typically found in research centers. The essential concern was that treatments developed and tested in highly controlled settings were not enjoying the same level of success in applied settings.

Clinical Efficacy

The term clinical efficacy was coined to reflect the situation in which a particular treatment has proven useful and beneficial in treating psychological problems in children through controlled research. For example, Stein and colleagues (Stein, Jayxoc, & Kataoka, 2003) found that children treated with cognitive-behavior therapy for 10 weeks experienced significantly fewer symptoms associated with Posttraumatic Stress Disorder and depression, as well as improved psychosocial functioning and classroom behavior, relative to a 3-month wait-list control group in a randomized clinical trial.

Additional conditions are required to satisfy the recommended criteria for establishing clinical efficacy. The therapy must be significantly superior to no treatment, placebo, or an alternative treatment, or equivalent to a treatment already established as efficacious. These comparisons must be accomplished in the context of a randomized controlled trial, equivalent time-samples design, or controlled single-subject experimental design, and replicated by an independent research team.

Empirically Supported Treatments

Psychological therapies meeting clinical efficacy criteria are considered empirically supported treatments. Empirically supported treatments are psychological treatments proven effective for specific populations through controlled research (Chambless & Hollon, 1998). Meta-analytic reviews attest to the clinical efficacy of many therapies for children by computing and analyzing effect size estimates across studies. Thus, studies rather than subjects become the focus of analysis, and an overall effect size based on reviewed studies provides compelling evidence of the magnitude of the difference (in standard deviation units) between treatment and nontreatment groups. For those unfamiliar with meta-analysis, effect size estimates are calculated based on recommended formulas (Hedges, 1982) and involve subtracting treatment means from control group means and dividing by the pooled standard deviation of the two groups. The result is a number that reflects the difference between the two groups—for example, before and after therapy—in standard deviation units. Thus, an effect size of 1.2 indicates that the treated children are 1.2 standard deviations higher on a particular measure (e.g., in social competence) relative to the untreated group. Larger effect sizes are usually desirable and indicate greater mean differences in outcome measures between the treated and untreated or comparison groups.

Effect size metrics reported in meta-analyses can also be related back to a central clinical question. For example, Kofler et al. (in press) examined how often children with ADHD were on-task relative to control children in regular classroom environments. They used best case estimation to determine the magnitude of on-task deficits after accounting for significant moderators and obtained an effect size of 1.40. This effect size was translated onto the control group distribution to estimate

actual percentage rates, and revealed that children with ADHD were on-task an average of 74.7% compared to 87.9% for typically developing children.

Some of the published criteria for establishing a therapeutic intervention as an EST are dubious. For example, the superiority of the intervention must be demonstrated in at least two independent research settings *or* with a sample size of three or more subjects for single-subject design experiments also in at least two independent research settings. This means that a treatment can be considered clinically efficacious based on results from only six children—a rather weak case for suggesting that treatment effects are likely to generalize to other children with similar difficulties and in similar settings. The moniker *possibly efficacious* reflects the situation in which a single, well-controlled study supports the treatment's efficacy, highlighting the critical need for replication.

Clinical Effectiveness

In contrast to clinical efficacy, clinical effectiveness addresses whether the treatment is useful in applied clinical settings, that is, its clinical utility. The distinction between the two terms reflects the frequent failure to replicate treatment effects produced in highly controlled research environments—such as university- or hospital-based research clinics—in regular psychological service delivery systems such as community mental health centers and private practice settings. Variables likely to contribute to a treatment's lack of utility in applied settings, despite proving successful in controlled settings, are reviewed elsewhere (Chambless & Hollon, 1998; Kazdin, 1988). In response to these criticisms, several researchers have attempted to establish clinical effectiveness in applied settings; however, traditional design controls (e.g., random assignment to groups, nontherapeutic waiting-list groups) are certain obstacles to these efforts.

Relevant Issues

Determining whether a treatment or intervention is effective is not as straightforward as it may seem. Effectiveness varies significantly based on the type of measure (self-report, standardized rating scale, direct observation, clinical interview, objective measure such as an actigraph), the source of measurement (child self-ratings, teachers, parents, therapists), and the area of functioning assessed (e.g., social deportment versus academic functioning).

Many measures—and particularly rating scales—currently used in research studies have inadequate measurement units, insufficient psychometric properties, and lack age and sex norms. As an example, none of the available child activity level rating scales has a standard unit of measurement—that is, a meaningful unit of activity or movement that is equivalent within and across measures and raters. Accurately defining activity level is severely compromised as a result and leaves us in the uncomfortable position of describing differences and changes in children's activity level by referring back to the scales used to initially quantify the behavior. The moderate correlations reported between activity rating scales only partially mitigates this concern when contrasted with correlations between rating scales and objective measures of children's activity level. For example, correlations between activity

rating scales and actigraphs are generally between .32 and .58. These values indicate that 66% to 91% of the variability in activity rating scale scores is not linearly related to variability in actigraph scores in the same children measured at the same time. This finding probably reflects the fact that children's activity rating scales tend to reflect other aspects of behavior and not just activity level. The wording of most scale items and reliance on factor analytic scale construction methodology contribute to this phenomenon because descriptions of activity level may correlate highly, but usually reflect a broader range of behavior than just movement.

STATISTICALLY AND CLINICALLY SIGNIFICANT CHANGES

Overview

The importance of changes that occur following an intervention can be quantified in three ways. The first and most common approach is statistical significance testing, which examines group differences and estimates the reliability of change. Statistical significance testing calculates how certain we are that changes from pre- to posttest are robust—for example, not due to other factors such as measurement error. It does not tell us, however, about the magnitude of this change or its impact on children's lives. Effect size and correlational approaches can provide meaningful estimates of the magnitude of change, but these estimates do not necessarily have clinical or practical significance. Clinical significance refers to the practicality of an intervention—whether or not it makes a meaningful difference in the daily lives of children or those who interact with the child.

Intuitively, larger effects are more likely to be clinically relevant. The magnitude of effect, however, is not necessarily related to clinical significance. Kazdin (2004) has argued that an intervention can be clinically significant—or insignificant—with large, small, or no changes from pre- to posttest. As examples, he offers the possibility that a large but clinically insignificant effect may be obtained on a cognitive or laboratory task that has little to do with daily living. Likewise, a clinically significant finding of no change may occur when symptom deterioration was expected. The impetus of these examples is to demonstrate that a measure's ecological validity—its relevance to real life—must be considered when selecting measures used to test clinical significance.

Evaluating Clinical Significance

Several methods currently are used to evaluate clinical significance (Kazdin, 2004). Subjective evaluation methods rely on the judgments of the child or relevant others and ask whether they can discern meaningful improvement in their daily lives. Measures of social impact may also inform clinical significance, for example, whether there was a change in arrest record frequency, days worked, or missed school days. Absolute change methods are interested in determining whether a problem or symptom has been eliminated, or whether the child continues to meet diagnostic criteria for a psychiatric disorder.

The most common type of clinical significance testing is the *comparison method,* which asks whether members of the treatment group are distinguishable from

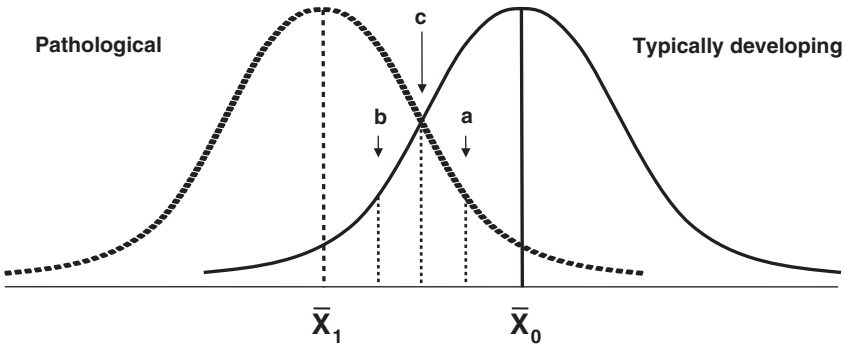

Figure 13.1 Clinical significance cutoff scores based on different criteria.

members of a typically developing (normal control) comparison group after treatment. Comparisons can be made to normative data or to a normal control group. Several commercially and freely available rating scales and direct observation systems provide standard scores based on normative data from nationwide community samples. Clinical significance may be demonstrated when a child changes from the clinical (usually 1.5 or 2 *SD* above the mean) to the normal (usually less than 1.5 *SD* above the mean) range following treatment. Comparison with rating scale norms is potentially problematic, however, due to different assessment conditions (Kazdin, 2004). For example, norms are typically based on a single test administration, whereas treatment outcomes are based on multiple administrations. Comparison with results from a concurrently collected normal control group is preferred.

Absolute change and two normal comparison, cutoff criterion methods are shown in Figure 13.1. X_0 represents the mean of the control group; X_1 represents the pathological group pretreatment mean. Three distinct cutoff scores have been proposed for determining clinical significance. The first (a) represents an absolute change method; the other two (b and c) estimate *normalization:* clinical significance in relation to a normal control group. Some argue that clinically significant improvement has occurred if a child's posttest functioning is (a) outside the range of the pathological group, defined as 2 standard deviations above the pathological control group pretreatment mean. Others argue that the child must fall (b) within the range of the normal population as defined by a score within 2 standard deviations of the normal group mean. A third proposal represents a compromise between these two options, defining normalization as occurring when a child is functioning (c) closer to the mean of the normal group than the pathological group.

In practice, children are classified as deteriorated, unchanged, improved, or normalized. The first three are determined by creating a confidence interval around the children's pretreatment score.* Children falling within the confidence interval are considered unchanged; those falling outside of it are labeled deteriorated

*Estimated true score is used if regression to the mean is considered a threat. See Speer (1992) for a discussion and simple method for determining whether regression to the mean is likely affecting posttreatment scores.

or improved. For example, children with a score of X_0 at pretreatment would be considered improved if they scored at or above cutoff score *a* in Figure 13.1. Children who improved and exceeded the predetermined cutoff score (*b* or *c*; see Figure 13.1) are considered normalized.

The difference between statistical and clinical significance is illustrated in the following example. Rapport, Denney, DuPaul, and Gardner (1994) evaluated the effectiveness of methylphenidate (Ritalin) on the classroom performance of children with ADHD. They used three outcome measures: teacher ratings of children's classroom deportment, academic efficiency scores (percentage of assigned class work completed correctly), and observed rates of on-task behavior in the classroom. Traditional significance tests revealed that all three outcome measures were significantly improved as a function of medication. Were these changes meaningful? Rapport and colleagues used the Jacobson and Truax (1991) model, correcting for regression to the mean as recommended by Speer (1992). As depicted in Figure 13.2, they found that between 50% and 78% of children were normalized, meaning that they were functioning within the normal control group range. Interestingly, although statistical significance was demonstrated for all three outcome variables, the rates of improvement or normalization were quite different. Most noticeable is the difference between teacher ratings of behavior, with 94% of ADHD children rated as improved or normalized, and actual classroom academic assignments completed correctly, with only 53% of ADHD children improved or normalized. Their findings also illustrate the importance of selecting relevant outcome

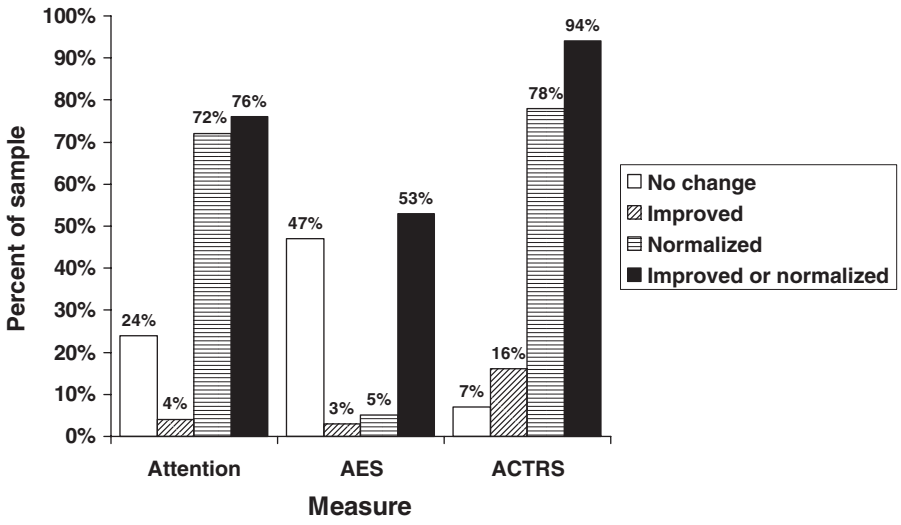

Figure 13.2 Clinical status of the group ($n = 76$) collapsed across methylphenidate dose conditions for three classroom measures. AES = Academic Efficiency Score; ACTRS = Abbreviated Conners Teacher Rating Scale. *Source:* "Attention Deficit Disorder and Methylphenidate: Normalization Rates, Clinical Effectiveness, and Response Prediction in 76 Children," by M. D. Rapport, C. Denney, G. J. DuPaul, and M. J. Gardner, 1994, *Journal of the American Academy of Child and Adolescent Psychiatry, 33*(6), pp. 882–893. Reprinted with permission.

measures. Teachers rated 94% of ADHD children as improving significantly when taking medication. When looking at their schoolwork, however, only 53% made actual improvements compared to their pretreatment performance. In most situations, when relevant outcome measures are used, statistical significance will be a necessary but not sufficient condition for clinical significance. Likewise, large effects may be more likely to be clinically significant effects. The presence of one or both of these conditions, however, does not demonstrate clinical significance.

In summary, clinical significance testing operates at the individual rather than group level. Beyond statistically reliable group differences, it is interested in the practical effects of an intervention—whether treatment makes a noticeable difference in the day-to-day lives of the clients we serve. Clinical significance can be defined subjectively by asking children about changes in their behavior, socially by comparing impact measures (e.g., school absences or test scores), or statistically using absolute and normative comparisons.

RESEARCH METHODS

Research *methodology* refers to the principles, practices, and procedures that direct research, whereas research *design* refers to the organized plan used to examine questions of interest. Both aspects represent important knowledge that guide the planning, implementation, and analysis of a research project.

Selection of Measures

Relationship between and among Variables

Selection of measures will depend on your specific research question as well as the level of understanding you wish to accomplish with respect to the phenomenon studied. For example, research may focus on examining *correlates,* or relationships between or among variables without consideration of time sequence or causality. Other studies may focus on identifying *risk factors,* or characteristics that precede and increase the likelihood of some event, outcome, or behavior pattern that may be modifiable. *Marker variables*—characteristics that precede and increase the likelihood of some event or behavior, but are not modifiable or have no effect on the outcome if modified—may be the focus of other investigations. Other measures will be more appropriate for questions concerning *treatment outcome* or *prevention* in children (for a detailed discussion, see Kazdin, 1999). Establishing causal factors, wherein one variable precedes and influences some other variable, represents the purest form of research if one embraces prediction of human behavior as the sine qua non of research.

Studying variables (*moderators*) that influence the direction, magnitude, and nature of a relationship between or among variables represents important avenues for clinical research, particularly when *protective factors* (a special case of moderator variables) can be identified that reduce the likelihood of an undesirable outcome. Researchers examine moderator variables when they suspect that the relationship among variables may differ because of the influence of some other variable. For example, if the relationship between type of aggressive behavior and a diagnosis of

Conduct Disorder is different for boys and girls, sex may be considered a moderator variable. Several moderating variables may be important to examine in research with children, including age, sex, ethnicity, height, weight, and motivation.

Once a causal relationship is established in the literature, researchers may wish to focus on explaining the process or mechanism by which the process evolves (*mediator variables* such as psychological or biological mechanisms). In this situation, a previously unconsidered variable is hypothesized to intercede between two related variables and reflects an indirect effect of one variable on another. For example, Rapport, Scanlan, and Denney (1999) found that the relationship between children's early attention problems and later scholastic achievement was mediated by differences in working memory.

In summary, a correlational relationship represents the most basic level of understanding between or among variables. A deeper level of understanding is achieved if it can be determined that one or more variables preceded the occurrence of other variables in time. To establish a causal linkage, investigators must provide evidence of a correlation and ordered temporal relationship between or among variables, complemented by three additional considerations: Other influences must be ruled out as possible explanations for the findings; the findings must be replicable across different samples from the same population; and a valid explanation concerning the mechanisms and processes through which the causal variables operate and are related to outcome variables must be explicated.

Types of Dependent Measures

The foregoing summary suggests that researchers need to consider the level of understanding they wish to achieve with respect to their research topic before selecting particular measures. Selection of measures will depend on a host of factors (e.g., availability of instruments or instrumentation, time constraints, costs) and can include a wide range of instruments and techniques. Popular exemplars include self-report inventories, behavior rating scales completed by others, and physiological instrumentation. Consultation of specialty texts on child assessment is advised for those interested in reviewing the broad range of instruments and techniques available (e.g., see Mash & Terdal, 1997; Hersen, 2006; and current volume).

Innovative measures can be created to assess discrete incidences (e.g., stereotypic movement, drooling) or categories of behavior (e.g., bizarre speech) but must meet expected standards of psychometric rigor (e.g., interrater reliability, validity). Computerized assessment is commonly used to assess important constructs such as vigilance (e.g., continuous performance tasks) and behavioral inhibition (e.g., stop-signal paradigm) in children. Incorporating current or archival indices of target behaviors or latent constructs reflects an additional method of data collection available to child researchers. For example, Fergusson and Horwood (1995) used arrest records as an index of later delinquency in children with early conduct problems and ADHD. Duration of hospital stay, school attendance, earned grades, completed academic assignments, and retention rates are other common examples of measures with high face validity used by researchers.

As a general rule of thumb, try to incorporate multiple indices of behavior that have strong psychometric properties. That is, select instruments, indices, or observations that provide valid and reliable measures of both the specific behaviors of interest as well as other potentially related areas of functioning. Doing so demonstrates that identified behavior problems or changes in behavior are not confined to a particular type of instrument or recording procedure and permits measurement of corresponding improvement in other domains related to the child's functioning. For example, designing a treatment outcome study to improve a child's academic performance in the classroom may result in corresponding improvement in the child's attention and classroom behavior. In such cases, teacher rating scales coupled with direct observational recording procedures that measure domains other than just academic performance are needed to document the broader spectrum of treatment gains related to the independent variable. Including collateral measures that reflect areas not expected to change (unless the researcher is expecting a generalized treatment effect) or undesirable change is necessary in some research protocols. The former provides evidence for discriminate validity (e.g., only the targeted area changes with the introduction of the independent variable), whereas the latter addresses the possibility of emergent symptoms or undesirable change due to the intervention.

Studies of specific clinical groups (e.g., ADHD, childhood depression), learning problems (e.g., academic deficits), or maladaptive behavior (e.g., peer aggression) may require measures different from those used to evaluate change due to intervention. For example, structured or semi-structured clinical interviews complemented by specific rating scales are traditionally used to define, classify, or describe children with a particular clinical disorder (i.e., serve as grouping variables), whereas other instruments, observations, or ratings may be used to measure change associated with intervention (i.e., serve as dependent variables). Therapy effects are strongest for outcome measures that match the problems targeted in treatment rather than through nonspecific or artifactual effects (Weisz, Weiss, Han, Granger, & Morton, 1995).

RESEARCH DESIGNS

Single-Subject Research Designs

Single-case research designs are valuable methodological tools used to evaluate different types of research questions involving individuals and groups. Consider some of the important scientific discoveries in psychology over the years. Single-case research was the principal paradigm in Wundt's (1832 to 1920) investigations of sensation and perception, Ebbinghaus's (1850 to 1909) studies of human memory, Pavlov's (1849 to 1936) classic experiments in respondent conditioning, and B. F. Skinner's (1904 to 1990) research in operant conditioning. These designs are particularly relevant to understanding children and the environments in which they live, where changes in everyday life may be of greater importance than obtaining a statistically significant change between groups (i.e., greater clinical relevance). Experimentation at the level of the individual case study may also provide greater insights with respect to understanding therapeutic change.

In contrast to the heavy reliance on established psychometric techniques and instruments required by between-group research, single-case methodology begins with identifying the focus of investigation (i.e., designating target behaviors) and proceeds to selecting potential strategies of assessment. When deciding on target behaviors, try to select observable behaviors and environment events as opposed to covert behaviors such as thoughts and ideas or hypothetical constructs such as anxiety and self-concept. Doing so will facilitate objective measurement and agreement between observers. Definitions should be written with absolute clarity to avoid ambiguity, and the boundaries of the defined target behavior must be clearly specified to minimize inference concerning whether a particular behavior qualifies as an occurrence or a nonoccurrence. On the surface, this may appear to be an easy task, but consider selecting classroom attentiveness ("on-task" in educational parlance) as a target behavior. Should off-task or inattentive behavior be recorded if a child is gazing up at the ceiling while working on a math assignment? Perhaps the child was thinking about the problem at hand or performing mental arithmetic. Are children permitted to ask peers for assistance when working on a problem, or allowed to leave their seat to sharpen a pencil during an academic work period? Does a momentary glance away from one's work count as off-task behavior? These and other definitional parameters must be decided and agreed upon, and preferably subjected to extensive pilot testing, prior to beginning formal observation and data collection. Once established, clear definitions of target behaviors will permit the recording of reliable baseline data and, in turn, serve as the traditional yardstick by which change is measured.

Assessing Behavior

Assessment of behavior can be accomplished in many different ways and will depend on what is being assessed and which method of recording best suits the needs of the researcher. The most commonly used methods of assessment include using frequency measures, classifying responses into discrete categories, counting the number of children or events, and measuring behavior based on discrete units of time.

Frequency counts are used when dealing with discrete behaviors that require a relatively constant time interval to perform. The first criterion enables observers to know when a designated behavior begins and ends, whereas the second permits recorded behavior to be treated as similar units for purposes of comparison. Consider the situation in which a researcher records the frequency with which a child speaks out of turn in a classroom. The child may blurt out an answer on one occasion and on another may turn and talk with a peer for 10 minutes or longer. Clearly, the two incidents are not comparable, and an alternative assessment method such as time interval recording should be considered. For situations in which frequency measures are taken for different periods of time (e.g., 20 minutes on day 1 and 40 minutes on day 2), calculate the frequency per minute or response rate by dividing the frequency of responses by the number of minutes observed each day. This metric, frequency per minute or response rate, will yield data that are comparable for different durations of observation.

Discrete categorization is used in situations in which behavior is best defined by categorical assignment. Commonly used categories include appropriate–not appropriate (e.g., social interactions between children), complete–not complete (e.g., classroom assignments), and discrete behaviors that form a functional (e.g., getting dressed in the morning, wherein each article of clothing counts as a discrete behavior) or correlated but unrelated (e.g., performing household chores) response chain.

Counting children is often used to assess the effectiveness of an intervention program. Examples include counting the number of children who perform a designated target behavior while on a school field trip, or calculating the percentage of children who complete their daily academic assignments in a classroom each morning. Contingencies can be introduced subsequently to reinforce daily academic assignment completion rates, for example, by scheduling in-class free time if at least 80% of the class meets the established criterion or for children who independently complete the assigned work.

Interval recording is used in situations in which the researcher wishes to obtain a representative sample of a target behavior during particular times of the day. Common examples of appropriate child behaviors include staying in one's seat during an academic work period, appropriate talk with peers, and paying attention. A block of time (e.g., a 30-minute daily or every other day observation period) is divided into a series of shorter intervals (e.g., 15-second observation blocks followed by 5-second recording blocks), and the target behavior is recorded as occurring or not occurring during each 15-second observation interval. This example would yield three observation blocks per minute and 90 intervals of recorded behavior per day. Variations of the interval recording method are common (e.g., time sampling) and might involve brief observations of a target behavior throughout the day rather than being confined to a single block of time.

Recording *duration* is another time-based method for observing behavior and is more appropriate for recording behaviors that are continuous rather than discrete acts. Common examples include observing ongoing social interactions between or among children or the total time required to complete an academic assignment. In these cases, the total duration or time interval that the behavior is performed serves as the dependent variable. An interesting but infrequently used variation of the duration method involves recording elapsed time before a particular behavior is performed (i.e., response latency). An example is to record how long a child takes to perform a particular behavior or chain of behaviors following adult instruction. Contingencies might be established subsequently based on the child's compliance within an increasingly shorter time interval over several days or weeks.

Design Types

The essence of single-subject research lies in its ability to demonstrate experimental control of an independent variable (IV) over one or more dependent variables (DV) by means of shrewd design and graphical illustration. Although statistical procedures exist that can be used to assess outcome effects associated with single-subject case studies, the more traditional means of demonstrating experimental control is to

provide a compelling visual (graphical) illustration that even a Doubting Thomas would acknowledge as evidence. To convince the scientific audience, an IV is introduced using a variety of design options such that its effects on the DV are systematically produced, reproduced, and/or eliminated, as in a carefully choreographed dance. Detailed descriptions of specific types of single-subject designs are provided in this volume (see Chapter 11).

Group Experimental Designs

Overview

Despite the numerous benefits associated with single-subject design methodology, the fact remains that an overwhelming majority of studies in psychology involve the comparison of groups, not individuals. Group designs are conventionally classified as true experimental designs, quasi-experimental designs, and observational designs, and may be used in a variety of contexts.

A true experimental design is a design in which the researcher manipulates one or more independent variables and measures one or more dependent variables. The researcher chooses what independent variables to manipulate, how they are manipulated (e.g., which levels to include), and what dependent variables to measure, based on the nature of the research question. For example, a researcher might be interested in determining what type of therapy works best for children with school phobia. Therefore, the research question has determined that the independent variable is *type of therapy* and the dependent variable is *school phobia*. Now, the researcher must decide whether any other independent or dependent variables should be included in the study and operationally define the independent variable(s) and the dependent variable(s).

Operationally defining independent variables refers to deciding what the levels should consist of, whereas operationally defining dependent variables refers to deciding exactly how to measure them. For example, the researcher needs to decide what levels of therapy to include (e.g., behavior therapy, cognitive therapy, cognitive-behavior therapy), whether or not to include a control group (condition), and how to measure school phobia. In the context of a between-subject design, a control group is a randomly assigned group that receives either no experimental treatment or a substitute for the experimental treatment. In the context of a within-subject design, a control condition is a condition (administered to all participants) in which either no experimental treatment or a substitute for the experimental treatment is administered. Control conditions (groups) are sometimes needed to control for nonspecific treatment effects (e.g., some children might believe that simply taking a pill helps because it's "medicine").

A quasi-experimental design is a design that is set up to emulate a true experimental design but includes one or more independent variables that cannot be manipulated by the researcher (often referred to as quasi-independent variables). These include variables such as sex, ethnicity, height, and weight, or clinical diagnoses such as ADHD or Reading Disorder. Another reason for the inability to manipulate certain variables may be due to ethical reasons such as substance abuse, smoking, exposure to harmful toxins, and cancer. The limitation to quasi-experimental

designs is the inability to express a causal relationship for quasi-independent variables.

Nonspecific Treatment Effects

Nonspecific treatment effects are any effects brought about by the experiment besides the treatment, such as being aware of what the experiment is about, contact with the experimenter, and discussing the experiment with other people. Nonspecific treatment effects are basically extraneous (nuisance) variables related to participants' perceptions concerning the experiment. Related to nonspecific treatment effects are placebo effects, in which participants improve simply because they believe that they are receiving treatment. If a control condition (group) is included, it is important that there are no other differences between the control condition and the experimental conditions except for an absence or substitute for the experimental treatment. Any other differences would introduce extraneous (nuisance) variables. Control conditions are necessary only when there is concern that nonspecific treatment effects may have an effect on the results. Nonspecific treatment effects are possible only if the participants' perceptions about the experiment can have an effect on the results. There are many situations in which participants' perceptions cannot have an effect on the results. For example, if a researcher is interested in studying what type of teaching method works best for teaching mathematics to third-grade children, a control condition consisting of teaching no mathematics would be benighted. Conversely, a child with social phobia may believe that taking a pill provides him with the necessary confidence to complete an in vivo exposure protocol.

There are three main classes of experimental designs: the between-subject design, the within-subject design, and the mixed-subject design, each with multiple variations possible.

Between-Subject Designs

The between-subject design is a design in which participants are randomly assigned to different treatment groups (levels of the independent variable) and each treatment group receives a different experimental condition. For example, in a single-factor experiment (an experiment with only one IV) examining two methods for teaching science to third-grade children (Technique 1 and Technique 2), half of the students would be randomly assigned to receive Technique 1 and half to receive Technique 2.

In a factorial experiment—an experiment with more than one independent variable—examining different methods for teaching science to third-grade children (Technique 1 and Technique 2) and the mode of presentation (teacher versus computer), one fourth of the students would be randomly assigned to receive Technique 1 via computer, one fourth to receive Technique 1 via teacher, one fourth to receive Technique 2 via computer, and one fourth to receive Technique 2 via teacher.

Advantages of the between-subject design as compared to the within-subject design include no carryover effects and no order or sequence effects. Disadvantages

of the between-subject design include the fact that many more participants are required than for the within-subject design, and the between-subject design yields less power than the within-subject design.

Ideally, there are an equal number of participants in each of the different treatment groups and assignment to the different treatment groups is random. Random assignment to treatment conditions simply means that each participant has an equal probability of being assigned to any given treatment group. Both ideals, equal number of participants per treatment group and random assignment to treatment groups, may not be possible for a given study. In the case of an equal number of participants per treatment group, the total number of participants may not be equally divisible among the various treatment groups or some participants may drop out of the study or miss the day that data are collected. In the case of random assignment of participants to the different treatment groups, this may not be possible. As in the example concerning methods of teaching science, it might be difficult to randomly assign half of each class to receive a different teaching method. The teacher cannot very well teach half of her class at a time, and even if he or she could, this would introduce the confound of which method is taught first. Both problems (unequal sample size and nonrandom assignment) can frequently be handled statistically. It is important, however, to take into account both unequal sample size and nonrandom assignment when analyzing data (i.e., one should not ignore them and analyze the data as if there were equal sample size and random assignment).

Another important issue is the loss of participants from the experiment, referred to as *attrition*. If the loss of participants is random (e.g., roughly an equal number of participants dropped out of each of the experimental groups), there is no problem and certain statistical techniques may be used to correct the situation. If, however, the loss of participants is not random but due to some aspect of particular treatment conditions (e.g., almost every participant who dropped out was in one particular treatment condition), then little can be done to salvage the experiment other than conducting a post-hoc test to determine whether attrition significantly biased one group relative to the other. For example, one may need to demonstrate that children with the most deviant scores did not drop out of one of the two groups receiving treatment, or that noncompleters were not significantly different on particular measures relative to completers.

As an additional note, the problems associated with randomly assigning some children within a given classroom to receive one level of the independent variable (treatment) and other children from the same classroom to receive other levels of the independent variable is not uncommon to research conducted in school settings (or even clinic or hospital settings). For example, it may not be possible for the teacher to teach some students using one teaching method and the other students using different teaching methods. When it is not possible to randomly assign children from the same class to different levels of the independent variable, an alternative is to randomly assign different classrooms to the different levels of the independent variable. This type of design is called a *hierarchical design* and requires special analysis. In a hierarchical design, the levels of at least one independent variable are nested under the levels of another independent variable, and the remaining independent variables are fully crossed. For example, if each level of IV 2 (e.g.,

Table 13.4 Hierarchical Design

	Technique 1		Technique 2
Classroom 1 Classroom 2		Classroom 3 Classroom 4	

classrooms) appears with only one level of IV 1 (e.g., teaching method), then IV 2 (classrooms) is said to be nested under IV 1 (teaching method). Examining two different methods for teaching science to third-grade children, two third-grade classes could be randomly assigned to receive Technique 1 and two third-grade classes could be randomly assigned to receive Technique 2. The design for this model is diagramed in Table 13.4.

The separate columns for Technique 1 and Technique 2 indicate that the model is not fully crossed, as classrooms (1, 2, 3, 4) are nested under techniques (1, 2). In this experiment, the independent variable *classroom* (classrooms 1, 2, 3, 4) is an extraneous (nuisance) variable. It is included in the design and analysis because it might have an effect on the dependent variable, and including it allows its effects to be isolated. If a hierarchical design is used, it is incorrect to analyze the data as if students were randomly assigned to each level of the independent variable (e.g., teaching method). There are additional complications that may arise when employing a hierarchical design, for example, when the levels of the nested variable (classroom) cannot be randomly assigned to the levels of the variable it is nested under (teaching method). Hierarchical designs are considered balanced if they satisfy two criteria. First, there must be an equal number of participants in each treatment combination (e.g., students in each class for each teaching method). Second, there must be an equal number of levels of the nested variable under each level of the other independent variable (e.g., two classes under each level of treatment method). If these criteria are not satisfied, the model is considered unbalanced and more complicated to analyze than balanced designs.

Within-Subject Designs

Every participant receives every treatment condition (levels of the independent variables) in a within-subject design. In the single-factor experiment on teaching techniques, every participant would receive Technique A and Technique B. In the factorial experiment involving teaching technique and mode of presentation, every participant would receive Technique A via computer, Technique A via teacher, Technique B via computer, and Technique B via teacher. A within-subject design may have children participating in the different levels of the independent variable simultaneously, or children may complete one level of the independent variable before participating in the next level. When participants must complete one level of the independent variable before participating in the next level, this is commonly referred to as a *crossover design*.

Within-subject designs present some special problems, including carryover effects, order effects, and sequence effects. Because participants receive every treatment condition, the effects of one treatment condition may carry over to the next

treatment condition. For example, once children are exposed to a particular reading method, they may be irreversibly changed. The investigator cannot simply cross them over into an alternative method and assume that nothing has been gained during the earlier experience. The order in which the treatment conditions are presented to the participants may also have an effect on the results. Finally, the sequence in which the treatment conditions are presented may have an effect. For example, in a study of perceived heaviness of objects, whether someone lifted a 10 pound object and then a 20 pound object or lifted a 20 pound object and then a 10 pound object would have an effect on the perceived heaviness of each object. Order and sequence effects may sound similar; however, order effects refer to the ordinal position in which the condition is presented in (e.g., first, second, third). Sequence effects, in contrast, have to do with which treatment follows which other treatment. As an example, consider the two sequences A—B—C and C—B—A. In both sequences, B has the same ordinal position (second). However, in the first case B follows A and in the second case B follows C; therefore, the sequence is different. Counterbalancing procedures are frequently used to control for problems related to carryover effects, order effects, and sequence effects.

Counterbalancing means that participants are exposed to the different treatment conditions in different orders. Ideally, one or more participants could be exposed to every possible order of treatment conditions. This is not possible with more than a few treatment conditions, as the number of possible orders increases rapidly (number of possible orders equals $n!$, such that three treatment conditions produces six possible orders: $3 \times 2 \times 1$). When the number of possible orders is too great, only a subset may be used. One method of obtaining a subset of the possible orders of treatment conditions is to simply randomly select a subset of the possible orders. This may pose a problem, however, because if the order or the sequence of treatment conditions has an effect on the dependent variable, random selection of treatment orders will not control for these effects.

Another method that not only controls for the order effect of treatment conditions but also allows for the order effect to be analyzed separately is the Latin square design. The Latin square design has as many orders represented as there are treatment conditions. Therefore, if there are four treatment conditions (Rx = treatment), four orders (indicated by uppercase letters A, B, C, D) would be represented as depicted in Table 13.5. Next, a random selection procedure is invoked to determine which of the four treatments corresponds to which letter (i.e., the order treatments will be administered). For example, if it was randomly determined that A = Treatment 3, B = Treatment 1, C = Treatment 4, and D = Treatment 2, then the matrix is designed as in Table 13.5.

Table 13.5 Simple Latin Square Design

A = Rx 3	B = Rx 1	C = Rx 4	D = Rx 2
B = Rx 1	C = Rx 4	D = Rx 2	A = Rx 3
C = Rx 4	D = Rx 2	A = Rx 3	B = Rx 1
D = Rx 2	A = Rx 3	B = Rx 1	C = Rx 4

Table 13.6 Balanced Latin Square Design

A = Rx 3	B = Rx 1	C = Rx 4	D = Rx 2
B = Rx 1	D = Rx 2	A = Rx 3	C = Rx 4
C = Rx 4	A = Rx 3	D = Rx 2	B = Rx 1
D = Rx 2	C = Rx 4	B = Rx 1	A = Rx 3

The Latin square design ensures that every treatment condition appears in every possible order (first, second, third, etc.). Therefore, if the order in which participants experience the different treatment conditions has an effect on the dependent variable, the Latin square design will cancel out that effect. The Latin square design also allows for the comparison of the different treatment orders (by including order as another independent variable) to determine whether there is a significant difference between the different orders. As mentioned earlier, in some situations both *order* and *sequence* may have an effect on the dependent variables. The balanced Latin square design controls for the effects of order and sequence effects and allows these differences to be compared. It ensures that treatment condition appears in every possible order (first, second, third, etc.) and that each treatment condition is preceded and followed by every other treatment condition exactly once (e.g., Rx 1 is preceded once by Rx 2, Rx 3, and Rx 4), as depicted in Table 13.6.

As a result, if order or sequence of treatment conditions affects the dependent variable, the balanced Latin square design will cancel out these effects. This design also allows for the comparison of the different treatment orders and sequences (taken together, not separately) to determine whether there is a significant difference between them. In Table 13.6, the rows indicate the order in which treatments will be assigned, and the columns indicate the sequence in which treatments will be received. As in the simple Latin square design, a random selection procedure is invoked to determine which treatment is assigned to which letter.

Mixed-Subject Designs

The mixed-subject design is a design in which one or more independent variables are between-subject variables (e.g., different participants randomly assigned to different levels of the independent variables) and one or more independent variables are within-subject variables (e.g., all participants receive all levels of the independent variables). For example, in the experiment involving teaching technique and mode of presentation, every participant would receive Technique A and Technique B (teaching technique is a within-subject variable), and half of the subjects would be taught by a computer and half would be taught by a teacher (mode of presentation is a between-subject variable). All of the issues discussed concerning between-subject designs and within-subject designs apply to mixed-subject designs.

Extraneous Variables

A major advantage of true experimental designs is that they are the only method that allow causal relationships among variables to be proven, which includes ruling out outside or extraneous influences. Thus, it is important to use a design in which

nothing else differs between the experimental groups except for the experimental conditions (levels of the independent variable). Any other differences besides the treatment conditions (called extraneous or nuisance variables) can call into question the causal inference.

Extraneous variables can sometimes affect all treatment conditions equally, either weakening or strengthening their effects. For example, in the study comparing teaching technique and mode of presentation for schoolchildren, if the treatment conditions were administered immediately following recess, it is possible that participants would be too wound up to pay attention, thus weakening the effects of all of the treatment conditions. When an extraneous variable varies systematically with the different treatment conditions, it is referred to as a *confounding variable* and poses an even greater danger to the interpretation of the results. When a confounding variable is present, any changes in the dependent variable cannot be attributed to the treatment condition with absolute certainty. Using our teacher study example, if everyone receiving Technique A via teacher had one teacher and everyone receiving Technique B via teacher had a different teacher, it could not be determined whether the different technique or the different teacher was responsible for any differences found in the dependent variable. Thus, it is important to keep everything constrained equally across the different treatment levels except for the treatment itself. This is not always possible, especially when conducting research in school and clinic settings. It does not necessarily invalidate the study to have extraneous or confounding variables present; however, it does limit the degree of causality that you can attribute to your treatment conditions.

Correlational Designs

In observational designs, the researcher simply measures variables as they occur naturally in the environment, which limits the degree to which causality can be inferred. Many observational studies, however, are conducted to examine whether a relationship exists between two or more variables or to predict certain variables (criterion variables) from other variables (predictor variables). Observing behavior and recording predictor and criterion variables is another method used in observational studies. For example, a researcher could observe children in the classroom and record the number of times they raise their hands to answer questions and the frequency of praise they receive from the teacher. Alternatively, observational studies may rely on self-report methods such as teacher or parent questionnaires that measure the variables of interest. Observational studies may also be conducted by obtaining ratings about children by people who know them, such as parents, teachers, or peers, or use combinations of these techniques.

Longitudinal and Cross-Sectional Designs

Overview

Longitudinal designs involve measuring participants repeatedly over an extended period of time, perhaps years or even decades. Cross-sectional designs and longitudinal designs are ways of obtaining different types of information. Cross-sectional

designs are best used for answering questions concerning how a treatment works at one point in time. Longitudinal designs are best used for answering questions concerning developmental change. For this reason, they can be especially informative when studying children because of their rapid developmental changes. Cross-sectional designs may be set up to assess different age groups simultaneously (e.g., children at ages 2, 6, 10, 14, 18), which would answer questions concerning possible age differences. However, it is possible that differences may exist between children of different age groups at a single point in time because they have different histories (cohort effects). Therefore, the children who are 2 years of age at the same point in time as other children who are 14 years of age might not display the same characteristics when they become 14 years of age. Longitudinal designs involve examining the long-term effects of some event or intervention. An additional advantage to longitudinal designs is that because information is collected with measures repeated at multiple time periods, error variance is reduced, often allowing for the detection of small behavior changes. Additionally, participants are compared to themselves at different points in time.

Potential Limitations

Disadvantages of longitudinal designs are mostly tied to the length of time required and the inherent cost associated with the study. Because the data are collected over long time periods, procedures and measures may become outdated and new procedures and measures may be developed. This leads to a major quandary: Should the outdated procedures and/or measures be continued so that differences may be compared across time, or should the new procedures and/or measures be adopted because they are better? One possible solution is to continue using the old procedures and/or measures to ensure accurate comparison and to adopt new procedures and/or measures as they are developed. Participant attrition is also a major problem with longitudinal designs because it is quite possible that the group that remains at the end of the study is not representative of those who dropped out along the way. Additionally, participant attrition increases the probability of a Type II error (failing to find a treatment effect when one exists) and decreases the generality of the results. Another problem is the potential confound between the effects of personal age and the effects of historical period (e.g., children growing up during a particular decade may be exposed to environmental or other events such as war that younger cohorts are not exposed to).

PARTICIPANT DEMOGRAPHICS AND SAMPLING

Identifying a Population of Interest

After formulating a research question and establishing a basic design, the next step is to consider how best to define and describe the population of interest. Review recent articles in credible journals to gain an understanding of how others have accomplished this task. For example, most research studies dealing with children contain basic sociodemographic information, such as children's age and grade (mean and *SD*), estimated level of intelligence, family socioeconomic status, sex, and ethnicity.

This information is included for other participants (e.g., parents) when relevant to the study (e.g., when studying parental attitudes toward medication compliance in children).

Detailed information is also included relevant to identifying, categorizing, or describing the population of interest. This may involve (a) a clear description of how a diagnosis is ascertained (e.g., using structured or semi-structured clinical interviews combined with using rating scale cutoff scores), (b) how a group is identified (e.g., a learning disability discrepancy formula), or (c) describing the characteristics of a select group of children (e.g., children attending a special education classroom). The use of diagnostic monikers (e.g., ADHD, Social Phobia) for identifying research samples is conventionally based on instruments with acceptable psychometric properties. This approach usually entails the administration of a structured or semi-structured clinical interview coupled with parent and teacher rating scales. Inclusion criteria for a particular diagnostic group is typically based on meeting *DSM-IV* criteria for the disorder in addition to exceeding an identified cutoff score, such as 2 standard deviations above the mean or the scale's identified range for clinical diagnosis. Exclusion criteria vary according to the research question posed. For example, a researcher studying children with learning disabilities may wish to exclude children with ADHD to determine how well an innovative intervention works with reading disabled children independent of ADHD. Conversely, a different research team investigating whether working memory deficits are unique to language-impaired children as opposed to a more generalized deficiency associated with psychiatric disability (e.g., common to many childhood disorders) may wish to include children with and without comorbid ADHD.

Alternative methods for identifying research samples are available, depending on the nature of the investigation. For example, children with poorly developed social skills who do not meet diagnostic criteria for a clinical disorder may nevertheless benefit from a social skills training program. Specific measures of social skill deficits, such as parent-teacher ratings and even direct observations, can be used if they reliably identify the sample.

Sampling

As a general guideline, researchers must sufficiently describe characteristics of their sample that might affect the generalizability of findings as discussed earlier and determine how best to obtain the research sample from the targeted population. This is particularly important for group designs but also applies to single-subject designs.

There are two classes of samples, *probability* and *nonprobability* samples. Probability samples are samples in which every member of the population has a known probability of being selected for inclusion, whereas nonprobability samples are samples in which the probability of being selected for inclusion is unknown.

Four well-known types of probability samples are *simple random samples, stratified random samples, systematic samples,* and *cluster samples*. A simple random sample is one in which every member of the population has an equal probability of being selected, whereas a stratified random sample is a random sample in which the proportion of certain characteristics in the population (e.g., race, ethnicity, culture,

sex, age, education, income, SES) are matched in the sample. For example, if sex is considered important to a study and the population under study consists of 58% boys and 42% girls, the sample reflects these same proportions. Samples may also be stratified on several different characteristics simultaneously. For the sample to be a stratified random sample, sampling would be random within each characteristic considered important to the study.

A systematic sample is a sample selected in a nonrandom fashion. For example, if 10% of the child population is to be included in the sample, then every 10th child would be chosen for inclusion.

A cluster sample is a sample in which clusters (groups) are randomly selected rather than individuals. For example, if researchers are interested in sampling grade school children, they would start with a list of grade schools and randomly select which schools to include. Within each school, you could include all of the students, randomly select students for inclusion, or randomly select additional clusters (e.g., grade levels).

The most commonly used types of nonprobability samples include the *convenience sample,* the *stratified convenience sample,* and the *snowball sample*. A convenience sample is a sample of participants that is convenient for the researcher to obtain (e.g., college undergraduates, hospital patients, child referrals to a university-based specialty clinic). A stratified convenience sample is the same as a stratified random sample, except that the participants are selected for convenience rather than randomly. A snowball sample is a sample that is created by having the initial participants (e.g., children's parents) suggest additional possible participants; these additional participants suggest additional possible participants, and so on. Because researchers do not typically have access to the entire population they wish to study, most studies use nonprobability samples.

Sample Size and Power

An important issue concerning samples is determining how many participants should be included. The main issue concerning sample size is that you want to recruit a sufficient number of participants to have a powerful test, but not more than you need, as this can be costly and time consuming. *Power* refers to the probability of finding a significant treatment effect when one truly exists (probability of rejecting a false null hypothesis). Power can be increased by setting alpha equal to 0.05 rather than 0.01, increasing the size of the treatment effect (increasing the between-condition or group variation), or reducing the error (reducing the within-condition or group variation). One method of reducing the within-condition variation (error) is to increase the number of participants. As sample size increases, within-group variation (error) decreases.

Convention generally holds that .80 is the minimum acceptable level of power. This means that if there truly is a treatment effect, your statistical test has an 80% probability of finding that treatment effect (rejecting the null hypothesis). Before you conduct an experiment, you should do a power analysis to determine how many subjects are needed to achieve (at least) 80% power. Conducting a power analysis requires the researcher to make several educated guesses concerning

the data (e.g., size of treatment effect and the population standard deviation). Specialty texts (see Cohen, 1988) and software (GPower: www.psycho.uni-duesseldorf.de/abteilungen/aap/gpower3/) for computing a power analysis are readily available. Researchers recognize that a certain degree of controversy exists concerning hypothesis testing, because with a large enough sample, even trivial treatment effects may be statistically significant. For this reason, researchers should specify the minimum interesting treatment effect (i.e., the minimum effect that would be of interest) to be found with 80% power before conducting a power analysis.

SUMMARY

Conducting research with children is a multifaceted enterprise that requires broad knowledge of research methods, research design, psychometric theory, statistics, and child development, coupled with a healthy dose of curiosity and tenacity. Textbooks and coursework are useful resources for novice researchers, but there is no substitution for working in an active research laboratory under the guidance of an accomplished mentor.

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PART IV

Diagnosis and Evaluation

CHAPTER 14

Assessment, Diagnosis, Nosology, and Taxonomy of Child and Adolescent Psychopathology

Thomas M. Achenbach

This chapter addresses a host of interrelated issues that are especially salient for those who work with children. (For brevity, I use "children" to include "adolescents.") These issues are also becoming increasingly salient for those who work with adults, so much so that they are shaping drafts of the forthcoming fifth edition of the American Psychiatric Association's *Diagnostic and Statistical Manual of Mental Disorders* (*DSM-V*). The issues are organized in terms of four interrelated aspects of what can be understood as continuous processes. These four aspects are assessment, diagnosis, nosology, and taxonomy.

In clinical services, assessment is often viewed as an initial step that then leads to diagnosis. Diagnosis, in turn, is shaped by nosologies (classifications of diseases) that specify the criteria for the disorders to be diagnosed. And taxonomies systematically represent similarities and differences among the phenomena to be distinguished from one another in terms of taxonomic constructs. Assessment data are essential for diagnosis, nosology, and taxonomy. However, the data sought and the procedures for seeking the data are shaped by diagnostic, nosological, and taxonomic concepts and criteria. Figure 14.1 illustrates the continuous process that includes assessment, diagnosis, nosology, and taxonomy. Although everyday clinical services focus primarily on assessment and diagnosis, these enterprises are ultimately structured according to nosological and taxonomic systems. Multiinformant, multicultural, and multitaxonomic perspectives are needed to take account of variations in the phenomena to be distinguished and in the models for representing the phenomena.

The clinical phenomena on which this chapter focuses include behavioral, emotional, social, and thought problems occurring between the ages of 18 months and 18 years with sufficient severity to warrant the term "psychopathology" and a likely need for professional help. The chapter does not focus on problems that result primarily from severe mental retardation or known physical abnormalities. The prospects for significant changes in the *DSM-V* argue for viewing existing *DSM* diagnostic criteria and categories as provisional and subject to change, despite their current influence on practice and reimbursement. Consequently, the *DSM* nosology

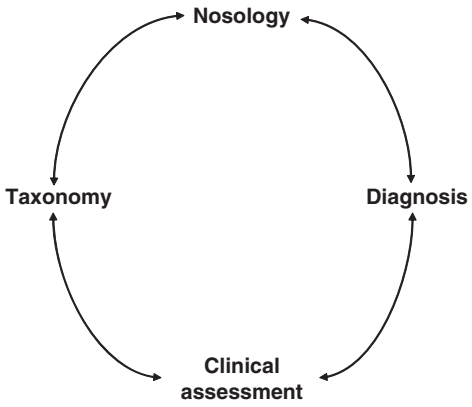

Figure 14.1 Schematic relations among assessment, diagnosis, nosology, and taxonomy.

is viewed as but one way to represent psychopathology rather than as embodying immutable truths concerning disorders and how to identify them. The *International Classification of Disease,* tenth edition (*ICD-10;* World Health Organization, 1992) provides a different nosology for representing psychopathology in terms of diagnostic categories, but it, too, is being revised. Continuing research on these and other ways of representing psychopathology is needed to optimize relations between clinical practice and advances in scientific findings.

CLINICAL ASSESSMENT

Clinical assessment refers to procedures for identifying the distinguishing features of problems, disorders, or cases, as well as relevant strengths. Many tests, questionnaires, inventories, interviews, and other procedures have been developed for the clinical assessment of children. Some assessment procedures, such as projective tests and unstructured clinical interviews, tend to be highly inferential. In using these assessment procedures, the practitioner makes inferences from responses to stimuli and questions in order to construct idiographic representations of the individual being assessed. Less inferential assessment procedures obtain responses whose meanings are more nomothetic. In other words, they are used to determine individuals' standing on scales or to place them in diagnostic categories, rather than to yield idiographic representations on the basis of the practitioner's interpretations. Assessment procedures also differ with respect to whether data obtained for individuals are judged in relation to norms versus nonnormed decision rules, such as those provided in the *DSM*.

Assessment procedures have long been used in clinical evaluations of children. However, the particular procedures, objectives, and types of conclusions have varied widely. In response to massive efforts to advance evidence-based treatment (EBT), Mash and Hunsley (2005) have argued for greater attention to what they call "evidence-based assessment" (EBA) as a necessary component of scientific support for clinical practice. Considering the flood of publications on EBT, it is

surprising how little attention has been paid to obtaining assessment data with which to choose the most appropriate treatment. Much of the EBT literature appears to assume that practitioners automatically know what should be treated and need only apply the appropriate EBT. The lack of explicit attention to systematically using assessment to determine what should be treated may reflect the fact that participants in EBT trials are often recruited on the basis of a priori definitions of the problems to be treated. Consequently, children are admitted to EBT trials if they are deemed to have the target disorder. However, criteria for recruiting participants for EBT trials may not be readily translatable into practical assessment procedures for helping clinicians decide which EBT to use with each client.

Mash and Hunsley (2005) have done a great service by calling attention to the failure of EBT to include generalizable assessment procedures. Nevertheless, their use of the term "evidence" may be somewhat ambiguous in "evidence-based assessment." In EBT, "evidence" appropriately refers to the informational basis for concluding that a particular kind of treatment is worthwhile. However, when applied to assessment, "evidence" is the *product* of the assessment process as well as the informational basis for deeming the assessment process to be worthwhile. In other words, assessment should *obtain evidence* on which to base choices of treatment, to judge the progress of treatment, and to evaluate the outcomes of treatment. Because assessment should serve these functions by empirically obtaining appropriate information about the case being assessed, I use EBA to stand for "empirically based assessment." This in no way weakens Mash and Hunsley's argument that EBT cannot stand alone without EBA. My own way of emphasizing their point is to observe that, "without EBA, EBT may be like a magnificent house with no foundation" (Achenbach, 2005, p. 547).

Obtaining Assessment Data on Children

Every individual differs from every other individual in countless ways. Consequently, no clinical assessment procedure can identify everything that is unique about each child and about the family members who are also important parts of each child's case. To focus on the subset of characteristics deemed most relevant to clinical decision making, assessment instruments are typically designed to obtain data according to particular diagnostic, nosological, and/or taxonomic models, as outlined in the following sections.

Diagnostically Based Assessment Instruments

Assessment instruments designed to make diagnoses include interviews consisting of questions about whether criteria for particular diagnoses are met. Several standardized interviews for making *DSM* diagnoses of children have been published. *ICD-10* diagnoses can also be made from some of the interviews, but most focus mainly on *DSM* diagnoses. Some diagnostic interviews, such as the Diagnostic Interview Schedule for Children (DISC; Shaffer, Fisher, Lucas, Dulcan, & Schwab-Stone, 2000), are highly structured in that they use precisely scripted questions to obtain yes-or-no answers as to whether each criterion is met for particular *DSM*

diagnostic categories. Other diagnostic interviews, such as the Schedule for Affective Disorders and Schizophrenia for School-Age Children (K-SADS; Ambrosini, 2000), are semi-structured in that interviewers are free to adapt the questions according to their clinical judgment.

In addition to ranging from highly structured to semi-structured, diagnostic interviews also vary in the degree to which conclusions are based only on the *respondent's* endorsement or denial of each *DSM* criterion versus being based on the *interviewer's judgment* of whether each criterion is met. The DISC is an example of a respondent-based interview, whereas the Child and Adolescent Psychiatric Assessment (Angold & Costello, 2000) is an example of an interviewer-based interview that requires interviewers to decide whether each *DSM* criterion is met.

When the first standardized *DSM* diagnostic interviews for children were developed in the 1980s, it was discovered that children's replies to interview questions often differed greatly from what parents and others reported about the children (Costello, Edelbrock, Kalas, Kessler, & Klaric, 1982; Herjanic & Reich, 1982). Consequently, parent versions of the interviews have been developed, and child versions have come to be used primarily with preadolescents and adolescents, rather than with younger children. However, even when restricted to preadolescents and adolescents, *DSM* diagnoses made from child interviews do not agree well with *DSM* diagnoses made from interviews with the children's parents. For example, in the massive Methods for Epidemiology of Child and Adolescent Disorders (MECA) Study of 1,283 9- to 18-year-olds, the mean kappa was .12 for agreement between diagnoses made from DISC interviews with children and their parents (P. Jensen et al., 1999). To provide some perspective on the meaning of the kappa statistic, Jacob Cohen (1960), the inventor of kappa, showed that the magnitude of kappa approximates the phi coefficient, which is the Pearson *r* scored from 2 $\times$ 2 tables (e.g., tables displaying relations between yes-versus-no categorizations of diagnoses made from child and parent interviews). Low agreement between parent and child reports of psychopathology is by no means limited to DISC interviews, as meta-analyses have yielded a mean *r* of only .25 between reports of child psychopathology by children and their parents, averaged across many instruments (Achenbach, McConaughy, & Howell, 1987).

In addition to interviews, another approach to assessing children in terms of *DSM* diagnoses is via questionnaires for rating *DSM* symptom criteria. An example is the ADHD Rating Scale IV, which consists of the *Diagnostic and Statistical Manual of Mental Disorders,* fourth edition (*DSM-IV;* American Psychiatric Association, 1994) symptom criteria for the inattentive and hyperactive impulsive types of Attention-Deficit/Hyperactivity Disorder (ADHD; DuPaul, Power, Anastopoulos, & Reid, 1998). Parent and teacher versions are available, on which each item is rated 0 to 3.

An example of a questionnaire for rating a broader range of *DSM* diagnostic categories is the Adolescent Symptom Inventory 4 (ASI-4; Gadow & Sprafkin, 1998). Parents and teachers rate symptoms for several *DSM-IV* diagnoses on 0 to 3 scales. A parallel self-rating questionnaire, the Youth Inventory-4 (YI-4; Gadow & Sprafkin, 1999), includes many of the same symptoms as the ASI-4.

Empirically Based Assessment Instruments

A key difference between the *DSM*-based instruments and what are here called empirically based instruments can be described in terms of their top-down versus bottom-up approaches. The *DSM*-based instruments are top-down in the sense that they start from the diagnostic categories and criteria that are listed in the *DSM*. The *DSM*-based assessment instruments are then constructed by wording the diagnostic criteria as questions that are either answered in interviews or are rated on questionnaires. The data sought with the *DSM*-based instruments are thus entirely shaped by the *DSM* nosology. Equally important, the conclusions drawn from assessment with these instruments are yes-versus-no judgments of whether criteria for particular *DSM* diagnoses are met.

By contrast, the empirically based instruments are bottom-up in the sense that their development starts with a large pool of assessment items that are chosen for their potential ability to identify children who are likely to need help for psychopathology. The items are pilot-tested by having them rated by appropriate respondents, such as parents, teachers, and children. Items that survive pilot testing are then used to assess large general population and clinical samples to determine how well they discriminate between children who have been referred for mental health services and demographically similar children who have not been referred for such services. Items that are found to be easily understood by the respondents, to be appropriately distributed, and to significantly discriminate between clinically referred and nonreferred children then become candidates for the scales of the assessment instruments. Items scored for large samples of children are subjected to statistical analyses, such as exploratory and confirmatory factor analyses (EFA and CFA), to identify sets (syndromes) of problems that tend to co-occur. The generalizability of the statistically derived syndromes is tested by performing similar analyses of items rated for children of each gender and different ages, by different informants, and in different cultures. These steps are outlined in Table 14.1.

Table 14.1 Developing Empirically Based Instruments for Assessing Child Psychopathology

1. Assemble pool of items that describe children's problems.
2. Formulate instructions and rating scales appropriate for the informants (e.g., parents, teachers, children).
3. Pilot-test and revise instruments based on informants' responses.
4. Use instruments to assess large clinical and epidemiological samples.
5. Do statistical analyses to identify syndromes of co-occurring problems.
6. Construct scales and profiles for scoring syndromes from the assessment forms.
7. Construct norms based on scores obtained by samples that are representative of relevant populations.
8. Test reliability and other psychometric properties.
9. Test validity in relation to external criteria.
10. Test replicability of syndrome structure in multiple cultures.

From *Multicultural Understanding of Child and Adolescent Psychopathology: Implications for Mental Health Assessment,* by T. M. Achenbach and L. A. Rescorla, 2007b, New York: Guilford Press. Adapted with permission.

Some of the steps required to develop empirically based instruments and scales for scoring them have been followed in the development of many instruments. For example, factor analyses were used in the development of some scales scored from the Behavior Assessment System for Children (Reynolds & Kamphaus, 2004). However, the results of the factor analyses were overridden when they contradicted the authors' judgments of what the scales and their constituent items should look like.

For another instrument, the Strengths and Difficulties Questionnaire (SDQ; Goodman, 1997), a priori scales were constructed and factor analyses were subsequently performed to see whether the obtained factors would match the scales (e.g., Dickey & Blumberg, 2004; Goodman, 2001; Koskelainen, Sourander, & Vauras, 2001; Muris, Meesters, Eijkelenboom, & Vincken, 2004; Rønning, Handegaard, Sourander, & Mørch, 2004). The factor analytic results varied considerably, perhaps owing at least partly to variations in methodology, samples, and criteria for determining whether the factors matched the SDQ scales.

Two families of widely used instruments have been developed according to steps like those outlined in Table 14.1. C. Keith Conners (1997, 2001) has developed a family of instruments primarily for assessing ADHD and related problems. The items comprising the instruments are rated on 0-1-2-3 scales by parents (Conners Parent Rating Scale—Revised; CPRS-R), teachers (Conners Teacher Rating Scale—Revised; CTRS-R), and youths (Conners-Wells Adolescent Self-Report Scale). The item pools and scales differ across the three instruments.

The CPRS-R and CTRS-R are scored for ages 3 to 17 and have been used much more widely than the youth form. The CPRS-R and CTRS-R both have factor-analytically derived scales designated as follows: Oppositional, Cognitive Problems/Inattention, Hyperactivity, Anxious-Shy, Perfectionism, and Social Problems. The CPRS-R is also scored on a scale designated as Psychosomatic, which cannot be scored from the CTRS-R because it lacks somatic items. Although the main scales were derived and then replicated by using factor analysis, items were eliminated if they lacked "conceptual coherence" with the factor on which they loaded (Conners, Sitarenios, Parker, & Epstein, 1998, p. 282). The CPRS-R and CTRS-R also have scales that paraphrase the *DSM-IV* symptom criteria for ADHD.

A family of instruments developed according to all the steps outlined in Table 14.1 includes the Child Behavior Checklist for Ages 6 to 18 (CBCL/6–18), Teacher's Report Form (TRF), and Youth Self-Report (YSR; Achenbach & Rescorla, 2001). Together with interview and observational instruments covering these ages, plus other instruments for ages 1.5 to 5, 18 to 59, and 60 to 90+, the CBCL/6–18, TRF, and YSR are components of the Achenbach System of Empirically Based Assessment (ASEBA; Achenbach & Rescorla, 2004; Rescorla & Achenbach, 2004). In addition to the factor-analytically derived scales, ASEBA instruments also have *DSM*-oriented scales comprising items identified by international experts as being very consistent with *DSM* diagnostic categories, as well as scales for assessing strengths. The problem items of most ASEBA instruments are rated 0 = not true, 1 = somewhat or sometimes true, and 2 = very true or often true.

As noted earlier, correlations between parents' and children's reports of child psychopathology are quite modest, as indicated by an *r* of only .25 in meta-analyses

(Achenbach et al., 1987). This low level of agreement means that parent reports and self-reports cannot be substituted for one another. Furthermore, agreement between adults who play different roles with respect to the children they rate is also modest, with meta-analyses yielding a mean *r* of only .28 between reports by parents versus teachers versus mental health workers versus observers. The *r*s averaged .60 between pairs of adults who play similar roles with respect to the children they rate (pairs of parents, teachers, mental health workers, observers). However, even these higher correlations do not indicate enough agreement to enable reports by any one informant to routinely serve as proxies for reports by other informants who play similar roles with respect to the children.

Although the meta-analytic findings were published in 1987, cross-informant agreement continues to be quite modest. As noted by De Los Reyes and Kazdin (2005, p. 483), the modest cross-informant agreement reported in the 1987 meta-analyses "has come to be one of the most robust findings in clinical child research: Different informants' (e.g., parents, children, teachers) ratings of social, emotional, or behavior problems in children are discrepant." Moreover, cross-informant discrepancies are not restricted to ratings of childhood problems. Meta-analyses of correlations between self-reports and others' reports of adult psychopathology have yielded mean *r*s of .43 for internalizing problems (depression, anxiety, somatic complaints) and .44 for externalizing problems (aggressive and rule-breaking behavior) when the probands and informants completed parallel instruments (Achenbach, Krukowski, Dumenci, & Ivanova, 2005). When probands and informants completed different instruments, the mean *r* was only .30. The fundamental fact is that agreement between different people's reports of psychopathology is simply too low to allow data from any single person to serve as a gold standard. Consequently, assessment, diagnosis, nosology, and taxonomy must all take account of the differences between the pictures obtained from different people.

The ASEBA instruments are designed to explicitly compare and contrast data from different informants. For preschoolers, the Child Behavior Checklist for Ages 1.5 to 5 (CBCL/1.5–5) is designed to be filled out by parent figures, and the Caregiver-Teacher Report Form (C-TRF) is designed to be filled out by day care providers and preschool teachers (Achenbach & Rescorla, 2000). These forms are scored on parallel syndrome scales that were derived from EFA and CFA of large general population and clinical samples. The scoring software provides profiles of scores and systematic comparisons of ratings by up to eight informants, as illustrated in Figures 14.2 and 14.3.

For school-age children, the scoring software compares scores from up to eight CBCL/6–18, TRF, and YSR forms. Furthermore, to take account of multicultural variations in norms, the scoring software enables users to display scale scores in relation to percentiles and *T* scores based on normative data obtained in many different societies (Achenbach & Rescorla, 2007a). For example, if a child lives in a society where problem scores tend to be relatively low, the user can elect to display the child's scale scores in relation to norms based on data from low-scoring societies. On the other hand, if a child comes from a low-scoring society but currently resides in a society where scores tend to be relatively high, the user can elect to display the child's scale scores in relation to norms for high-scoring

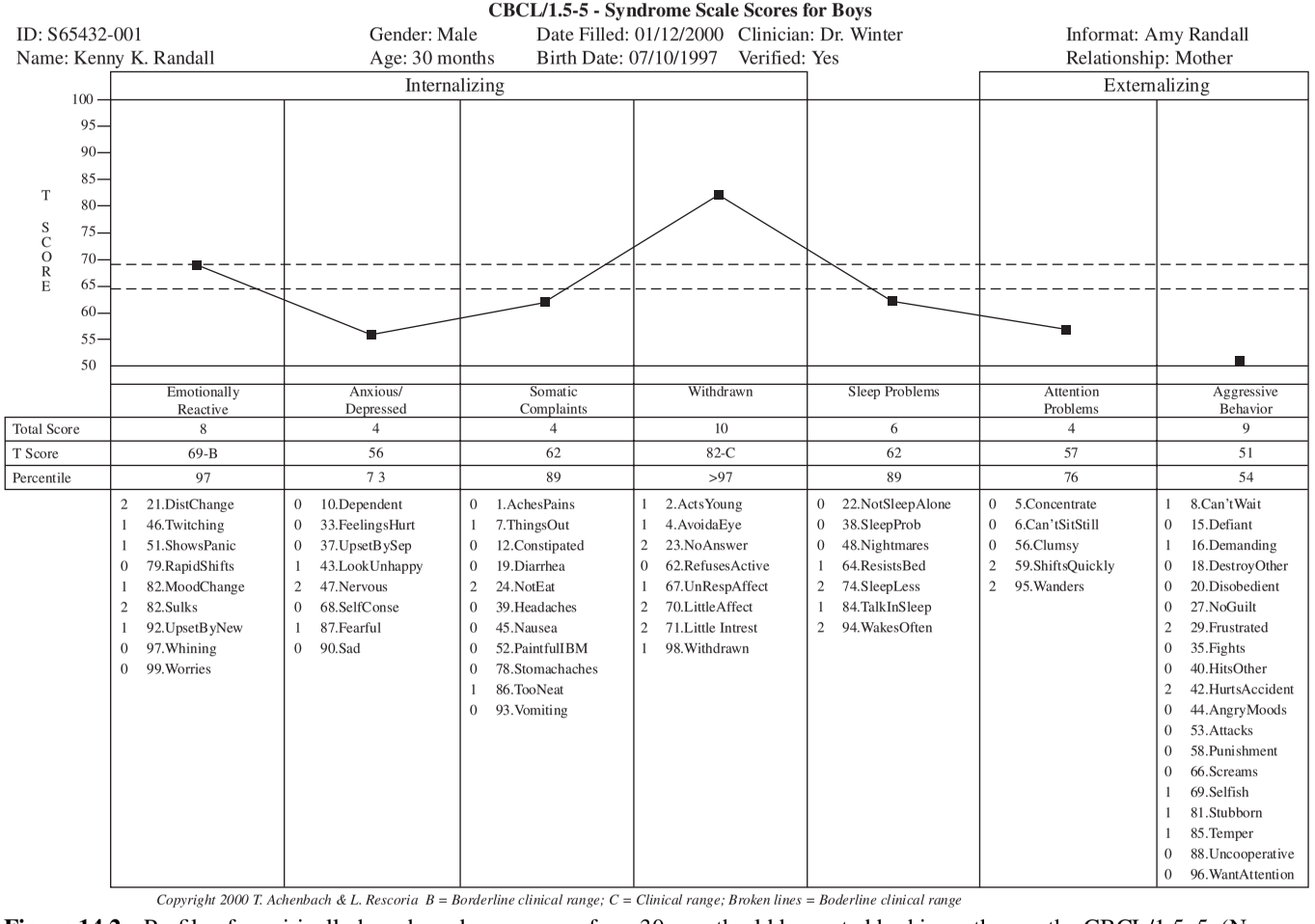

Figure 14.2 Profile of empirically based syndrome scores for a 30-month-old boy rated by his mother on the CBCL/1.5–5. (Names are fictitious.) *Source: Manual for the ASEBA Preschool Forms and Profiles,* by T. M. Achenbach and L. A. Rescorla, 2000, Burlington: University of Vermont, Research Center for Children, Youth, and Families. Reprinted with permission.

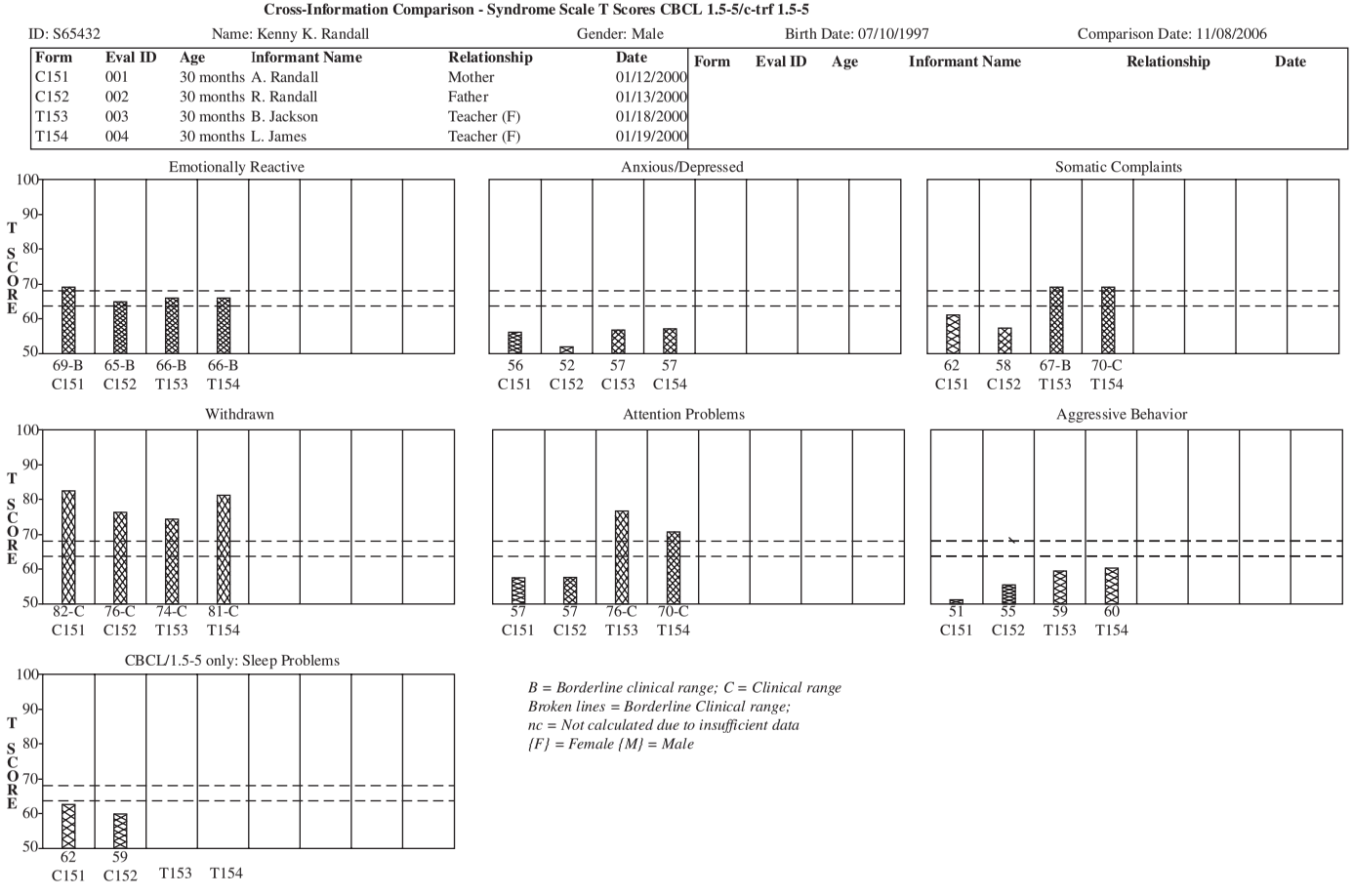

Figure 14.3 Cross-informant comparisons of syndrome scores for a 30-month-old boy rated on the CBCL/1.5–5 by his mother and father and on the C-TRF by two preschool teachers. (Names are fictitious.) *Source: Manual for the ASEBA Preschool Forms and Profiles,* by T. M. Achenbach and L. A. Rescorla, 2000, Burlington: University of Vermont, Research Center for Children, Youth, and Families. Reprinted with permission.

societies but can also display the child's scale scores in relation to norms for low-scoring societies like the child's home society. In this way, the user can determine whether the child's scores are clinically deviant according to both sets of norms, one set of norms, or neither set of norms. And, because the child's scale scores can be displayed for parent, teacher, and self-ratings, the user can compare scores obtained from each informant to whichever set of norms the user deems appropriate. Translations of ASEBA forms in more than 80 languages facilitate multicultural applications (Bérubé & Achenbach, 2008).

Obtaining Assessment Data on Parent Figures

Clinical assessment of children typically involves parent figures, including biological, adoptive, and stepparents. Parent figures may also include relatives, such as grandparents, aunts, uncles, and older siblings, and unrelated caregivers, such as foster parents, group home parents, the boyfriend or girlfriend of a child's parent, and child care workers. Parent figures (hereafter collectively designated as "parents") often instigate clinical assessment or seek it in response to urging by school, medical, welfare, or law enforcement personnel. Parents are key sources of information about children's functioning, are apt to affect children's problems and strengths, and are often pivotal in determining what will be done to help children.

Because child clinical cases typically need to be conceptualized in relation to parents, assessment of parents is a key component of the process of deciding how a child's problems should be viewed. When evaluating a child, mental health practitioners usually interview parents to learn about the background of the referral, to form impressions of the parents, and to discuss options. Parents may also be asked to complete instruments for assessing their child, such as *DSM*-based interviews, empirically based rating forms, and questionnaires focused on particular referral problems, such as ADHD, anxiety, or depression.

Because parents' reports and discrepancies between parents' reports are apt to be shaped by characteristics of the parents, it is helpful to illuminate these characteristics by assessing parents with instruments that parallel those for assessing their children. Although *DSM*-based interviews are typically too costly and intrusive to administer to parents, self-administered empirically based adult assessment instruments are available that parallel the ASEBA instruments for children (Achenbach & Rescorla, 2003).

The ASEBA adult instruments include parallel self- and other-report forms that tap adult versions of many of the strengths, problems, and scales that are scored from the child instruments, plus characteristics that are more specific to adults. To take account of differences between reports of adults' functioning by themselves and by others, the scoring software for the adult instruments can compare scores obtained from self-reports by each parent about himself or herself and reports by the parent's partner or someone else who knows the parent. Because the self- and other-report forms can each be completed in about 15 minutes at home, in a waiting room, or on the Internet, no professional time is needed to obtain the assessment data.

The profiles and comparisons of self- and other-reports can help the practitioner see relations between these different pictures of the parents' functioning and what the parents report about the child who is being evaluated. This approach enables the practitioner to efficiently document parental characteristics relevant to understanding children who are being assessed. The practitioner may also elect to show the parents the profiles obtained from the self- and other-reports to highlight similarities and differences in how their functioning is perceived. Viewing the profiles and cross-informant comparisons can teach parents about differences between perceptions of functioning and may motivate them to change their own behavior.

CLINICAL DIAGNOSIS

Since the publication of *DSM-III* (American Psychiatric Association, 1980), diagnosis has become an increasingly central focus of mental health services, training, research, and reimbursement. The introduction of explicit diagnostic criteria and fixed decision rules in *DSM-III* probably contributed to the ascendance of diagnosis. It was hoped that the change from inferential narratives in *DSM-II* (American Psychiatric Association, 1968) would produce more reliable diagnoses, etiological discoveries, and prescriptive treatments. Although the more explicit diagnostic criteria have led to much more detailed documentation of problems, the hopes for greater reliability of *DSM-III* diagnoses than achieved for *DSM-II* diagnoses were not realized for childhood disorders (Mattison, Cantwell, Russell, & Will, 1979; Mezzich, Mezzich, & Coffman, 1985). Furthermore, the hopes for etiological discoveries and for treatment specificity based on *DSM* diagnoses have not been realized either (Kupfer, First, & Regier, 2002; Rounsaville et al., 2002).

Although the term "diagnosis" may imply authoritative identification of a particular disorder, the Greek roots of *dia* = "apart" and *gignoskein* = "to know" pertain to distinguishing a particular condition from other conditions. In other words, diagnoses involve distinguishing among a variety of possibilities rather than simply identifying a condition whose identity can be understood without reference to similarities and differences between it and other conditions. Accordingly, the meaning of each possible diagnosis depends on how it is differentiated from other diagnoses within a system of diagnoses. The assessment data on which diagnoses are based must therefore ultimately be linked to a system that distinguishes among the phenomena that different diagnoses are supposed to represent. However, to clarify the relations between assessment and diagnosis and the nosological and taxonomic systems to which they should ultimately be linked, it is important to distinguish among the different meanings of diagnosis that sometimes confuse efforts to advance knowledge of child psychopathology. At least three meanings of diagnosis need to be distinguished, as outlined in the following sections.

Diagnostic Processes

Diagnostic processes consist of gathering data with which to make formal diagnoses and diagnostic formulations. Diagnostic processes can be thought of as constituting that portion of assessment that is directed toward making formal diagnoses and

diagnostic formulations. Although assessment thus includes diagnostic processes, it is not necessarily limited to diagnostic processes because it may include obtaining data on strengths, problems, and other variables that are not included in diagnostic criteria. However, assessment that is focused only on determining whether diagnostic criteria are met, such as *DSM*-based interviews, can be viewed as limited to diagnostic processes.

Formal Diagnoses

Formal diagnoses are made by determining the category of a nosology to which a case belongs. Diagnosis in this sense is what a leading psychiatric diagnostician called "the medical term for classification" (Guze, 1978, p. 53). When it is determined that a child meets criteria for a category of a nosology, such as *DSM* or *ICD,* this determination constitutes a formal diagnosis of the child, that is, a matching of the child's problems to a category of a nosology.

Diagnostic Formulations

Diagnostic formulations involve putting together various kinds of information about a case into a comprehensive picture of the case. For children, the information may include developmental history, family dynamics, physical liabilities, stressors, formal diagnoses, and prospects for change. In clinical practice, the diagnostic formulation may say more about the child's needs and the indications for particular treatments than formal diagnoses do.

To prevent undue reverence for diagnosis from clouding our thinking about relations among assessment, diagnosis, nosology, and taxonomy, it is important to be clear that the *DSM* and *ICD* consist of formal diagnoses but do not specify either diagnostic processes or diagnostic formulations. Although the *DSM* has been said to provide "operational definitions" for diagnoses (Rapoport & Ismond, 1996, p. 13), it does not in fact specify assessment operations (i.e., diagnostic processes) for most childhood disorders. About the only operationalized diagnostic criteria specified by *DSM-IV* are "an IQ of approximately 70 or below on an individually administered IQ test" (American Psychiatric Association, 2000, p. 49) as a criterion for mental retardation, plus low performance on relevant tests for learning disorders (Widiger & Clark, 2000). Thus, although diagnostic processes and formulations are certainly essential clinical activities, they are not within the purview of the *DSM* or *ICD* nosologies. Instead, these nosologies are classifications of formal diagnoses, as addressed in the following sections.

NOSOLOGICAL CATEGORIES

The *DSM* and *ICD* are nosologies, that is, systems for classifying diseases (called "disorders" in the *DSM*). They provide criteria for defining categories of disorders and for distinguishing particular disorders from other disorders. Since the publication of *DSM-III* (American Psychiatric Association, 1980), the *DSM* has included explicit criteria for each disorder and has specified fixed decision rules for determining whether a child has a particular disorder. Each criterion must be judged in a

yes-or-no fashion as either being met or not met. The *DSM* provides guidelines for describing some disorders as mild, moderate, or severe, but if the required number of criteria is not met, the disorder is deemed absent. Conversely, if the required number of criteria is met, the disorder is deemed present. The standard version of *ICD-10* (World Health Organization, 1992) does not provide such explicit diagnostic criteria or fixed rules for formal diagnoses, although a little-used research version of *ICD-10* does provide explicit criteria (World Health Organization, 1993).

Challenges of Comorbidity

Since the advent of explicit diagnostic criteria in *DSM-III,* numerous studies have found that many children qualify for multiple diagnoses. Termed "comorbidity," the tendency for disorders to co-occur could be very informative. For example, if many children diagnosed as having Disorder A are also diagnosed as having Disorder B, this could suggest that one of the disorders causes the other or that both disorders have a common cause. Comorbidity between Disorder A and Disorder B could also suggest that having one disorder makes a child more vulnerable to the other disorder, perhaps as a consequence of weakened resistance or of adverse effects of treatment for the initial disorder. However, because the *DSM* does not have well-validated markers for distinguishing each childhood disorder from each other disorder, it is also possible that apparent comorbidity between disorders reflects a lack of clear boundaries between them (Achenbach, 1991; Caron & Rutter, 1991). In other words, the diagnostic criteria for different nosological categories may not accurately represent truly different disorders. Nosologies may thus imply distinctions between disorders without evidence for true differences between the disorders.

As an example of a possible "distinction without a difference," two studies found that 96% of boys who met *DSM-III* criteria for Conduct Disorder (CD) also met criteria for Oppositional Defiant Disorder (ODD; Faraone, Biederman, Keenan, & Tsuang, 1991; Walker et al., 1991). Furthermore, field trials for the *Diagnostic and Statistical Manual of Mental Disorders,* third edition, revised (*DSM-III-R;* American Psychiatric Association, 1987) found that 84% of clinically referred children who met criteria for CD also met criteria for ODD (Spitzer, Davies, & Barkley, 1990). These findings certainly suggest that the nosological distinction between CD and ODD did not reflect two truly different disorders.

Unidirectional versus Bidirectional Comorbidity

Computing comorbidity in the other direction, that is, as the percentage of children diagnosed with ODD who were also diagnosed with CD, might yield a different picture. For example, if the proportion of children diagnosed with ODD were much greater than the proportion diagnosed with CD, the percentage of those diagnosed with ODD who had comorbid CD would be lower than the percentage of children diagnosed with CD who had comorbid ODD. Thus, if 200 children in a sample are diagnosed with ODD but only 100 are diagnosed with CD, the percentage of children who have ODD and also have CD could not exceed 50%. However, if all 100 of the children diagnosed with CD were among the 200 diagnosed with ODD, 100% of the children with CD would have comorbid ODD.

To avoid the misleading results of unidirectional computations of comorbidity rates, McConaughy and Achenbach (1994) recommended that comorbidity be computed as the mean of the comorbidities obtained by computing comorbidity in each of the two possible directions. Computing comorbidity bidirectionally in our hypothetical example would yield 100 CD/200 ODD = 50% + 100 ODD/100CD = 100%, for a bidirectional comorbidity of 50% + 100%/2 = 75%. Unfortunately, many studies of comorbidity fail to do this.

Although unidirectional computation of comorbidity makes it difficult to draw generalizable conclusions about precise rates of comorbidity, it is certainly clear that comorbidity rates have remained high even after *DSM* criteria were modified to reduce the amazingly high rates found for diagnoses such as CD and ODD (e.g., Angold, Costello, & Erkanli, 1999). In addition to high comorbidity between particular pairs of formal diagnoses, high comorbidity is also evident in findings that large proportions of children qualify for multiple diagnoses in community samples (e.g., Bird, Gould, & Staghezza, 1993) and clinical samples (e.g., A. Jensen & Weisz, 2002). In other words, children who have enough problems of a particular type to qualify for a diagnosis often have enough other kinds of problems to qualify for additional diagnoses as well.

Implications of Comorbidity

What are the implications of the high comorbidity rates found for formal diagnoses? As indicated by the 96% comorbidity found between CD and ODD in two studies, some comorbidity rates are so high that they raise questions about whether nosological distinctions between formal diagnostic categories accurately reflect true differences between disorders. If the different nosological categories do not in fact discriminate between different disorders, then comorbidity may be an artifact of having multiple categories for the same disorders.

The criteria for ODD and CD were changed from *DSM-III* to *DSM-III-R* to *DSM-IV* to reduce the overlap between these two diagnostic categories. However, it may still be an open question as to whether there is a meaningful difference between ODD, designated in *DSM-IV* as "[a] pattern of negativistic, hostile, and defiant behavior," and CD, designated as "[a] repetitive and persistent pattern of behavior in which the basic rights of others or major age-appropriate societal norms or rules are violated" (American Psychiatric Association, 2000, pp. 98, 102). Furthermore, the 15 criterial symptoms for CD are grouped into subcategories designated "aggression to people and animals," "destruction of property," "deceitfulness or theft," and "serious violations of rules." Because only 3 of the 15 criterial symptoms need to be judged present to warrant a diagnosis of CD, some children who qualify may have three aggressive symptoms, whereas others may have three rule-violating symptoms, and still others may have three deceitful symptoms or various combinations of aggressive, destructive, deceitful, and rule-violating symptoms. Consequently, diagnoses of CD may include very different patterns of behavior problems, some of which overlap considerably with ODD criteria, whereas others do not. Although changes in criteria since *DSM-III* may have reduced comorbidity between formal diagnoses of ODD and CD, overlaps between ODD kinds of problems and

particular subsets of CD problems may still artifactually contribute to apparent comorbidity.

Another implication of high comorbidity rates is that multiple diagnoses can greatly complicate use of treatments that are designed for particular disorders. For example, suppose that Treatment X has been found effective with children who were selected for having Disorder A. Does this mean that Treatment X should also be used for children who have Disorder B and Disorder C, as well as Disorder A? Even if the answer is yes, should additional Treatments Y and Z be used because they have found been effective with Disorders B and C?

If formal diagnoses of behavioral, emotional, social, and thought problems validly discriminate between disorders that are as distinct as physical diseases such as cancer, measles, and diphtheria, it would make sense to use multiple specific treatments for children who qualify for multiple diagnoses. However, because neither physical etiologies nor other physical abnormalities have been identified as underlying nosological categories for children's behavioral, emotional, social, or thought problems, treatments cannot be aimed at different physical abnormalities marked by different formal diagnoses. Instead, treatments are aimed at altering behaviors, feelings, and thoughts that may overlap among nosological categories. Consequently, research is needed not only on the effectiveness of specific treatments for specific diagnoses but also on the effectiveness of treatment for the many children who qualify for multiple diagnoses.

A related implication of comorbidity is that children who qualify for multiple diagnoses supposedly have multiple disorders. If such children in fact have multiple disorders, it could be hypothesized that they would have worse outcomes than children who have only one diagnosis. Doss and Weisz (2006) tested this hypothesis by analyzing the effects of initial severity on particular empirically based syndromes versus the effects of multiple syndrome deviance on treatment outcomes. They found that initial severity (high scores) on particular syndromes had much larger effects on outcomes than did deviance on multiple syndromes. Although these analyses did not directly test the effects of comorbidity among categorical diagnoses, such tests may be quite impractical, as discussed in the section on taxonomy.

New Directions in Nosology

A major thrust in planning for *DSM-V* involves possible quantification of at least some aspects of the diagnostic criteria (Helzer, Kraemer, Krueger, Wittchen & Regier, in press). Although quantified diagnostic criteria are often referred to as "dimensions," dimensions are not the only relevant forms of quantification. Consequently, I use the more comprehensive term "quantitative" to include dimensional as well as other quantitative approaches. Quantitative approaches have long been advanced for nosologies of personality disorders (Trull & Durrett, 2005), which primarily concern adult psychopathology. Quantitative approaches to nosologies of other adult disorders have also been proposed (Krueger & Markon, 2006). As will be evident in the section on taxonomic systems, quantitative approaches to child psychopathology have spawned extensive taxonomic research on psychopathology, including groupings compatible with existing categorical nosologies.

Although categorical and quantitative approaches are often assumed to be incompatible with one another, Kraemer (in press) has pointed out that each categorical diagnosis in the *DSM* nosology can be viewed as representing two (binary) values on a dimension. The two values are positive (the proband is deemed to have the disorder) versus negative (the proband is deemed free of the disorder). Most *DSM* diagnoses are specified in terms of a list of criterial symptoms and a cut point on the symptom list for distinguishing between positive and negative values (i.e., diagnosis present versus absent). The number of symptoms deemed present thus constitutes a score that ranges from 0 to *N* with *N* being the total number of symptoms on the list.

Kraemer (in press) argues that dichotomization is especially detrimental when, as is usually the case, there is no strong empirical basis for choosing cut points to discriminate between those who truly do versus do not have a particular disorder. To illustrate the detrimental effects of dichotomization, Kraemer cites studies of genetic influences on psychopathology, where unrealistically large samples are needed to detect genetic effects when the phenotypic characteristics are dichotomized rather than quantified. Kraemer also cites examples of research on treatment in which quantification of outcome measures revealed significant differences between the effects of different treatment conditions that were not detected when the same measures were analyzed dichotomously.

Although it has been argued that clinicians need categorical diagnoses (e.g., First, 2005), Kraemer (in press) provides examples of clinical diagnoses of hypertension, diabetes, and cancer that all involve quantitative specifications. She argues that the increasing importance of genetic, imaging, and biochemical data for research, training, and treatment related to psychopathology makes it essential to quantify aspects of the *DSM-V,* rather than deferring quantification to some unspecified future date, as was done when quantification of previous editions of the *DSM* was proposed. Preparations for the next edition of the *ICD* also include possibilities for quantification.

TAXONOMIC CONSTRUCTS

As discussed in the preceding section, nosologies are classifications of disorders. Like many classifications, they are constructed for the convenience of users. Accordingly, nosological categories may be constructed according to a variety of procedures and principles that are not necessarily consistent among all the categories. In a general medical nosology, for example, categories of bone fractures may be organized quite differently from categories of infectious diseases, cancers, and cardiac disorders.

Within the *DSM* and *ICD* nosologies, categories for disorders defined mainly in terms of behavioral and emotional problems are organized very differently from categories for mental retardation, learning disabilities, and dementias. Rather than being based directly on scientific principles or research findings, the *DSM* and *ICD* nosological categories were constructed mainly on the basis of negotiations among various stakeholders. A key consideration was whether the nosologies would be useful to clinicians, third-party payers, and those responsible for health care

records. Maintenance of continuity with previous editions of the nosologies was also an important objective.

The Nature of Taxonomies

Taxonomies are classification systems that are intended to capture intrinsic distinguishing features of the phenomena that are classified (Gove, 1971). Consequently, all taxonomies are classification systems, but classifications that do not apply scientific principles to distinctions among the classified phenomena are not taxonomies.

One view of taxonomies is that they should arrange entities into "natural categories on the basis of key features they share" (Millon, 1991, p. 246). Consistent with this view, it has been hoped that a taxonomy of psychopathology would "carve nature at its joints" (Kendell, 2002, p. 7). The goal of basing taxonomies on natural categories that carve nature at its joints is inspired largely by taxonomies of plant and animal species that are thought to distinguish among "natural kinds," each of which is inherently different from other natural kinds. However, no particular criteria have been found to consistently distinguish among species or even to provide an overarching definition of what "species" means. Consequently, biological species are now viewed as taxonomic constructs, rather than as embodying categories that are inherent in nature (Levin, 1979; Mayr, 1987). Considering that even taxonomies of biological species thus consist of constructs, current concepts of psychopathology are certainly unlikely to embody natural categories. Because it seems unrealistic to expect current concepts of psychopathology to yield taxonomies of natural kinds, Zachar (2000, p. 167) has argued that taxonomic constructs should be based on "patterns (of psychopathology) that can be identified with varying levels of reliability and validity."

If we acknowledge that natural taxonomic categories of psychopathology are unlikely to be discovered in the near future (if ever), it may be more realistic to consider multiple ways of taxonomically organizing data on psychopathology. This means that we should compare and contrast different ways of taxonomically organizing data on psychopathology. To avoid reifying any single set of taxonomic constructs, it may be helpful to view data on child psychopathology from multiple taxonomic perspectives. Considering the differences typically found between reports by different informants, it is also essential that taxonomic constructs take account of multi-informant variations. And because the patterning and prevalence of problems may differ among cultural groups, taxonomic constructs should take account of multicultural variations.

Nosologically Based Taxonomic Constructs

Although nosologies are classifications constructed for the convenience of users, it is assumed that each of their categories represents a particular type of disorder. In effect, each category represents a taxonomic construct that is defined in terms of the diagnostic criteria for the category. For example, the diagnostic criteria for CD define a taxonomic construct that is assumed to be different from the taxonomic constructs of ODD and ADHD. As illustrated in Figure 14.4, the approach to defining these taxonomic constructs can be described as top-down because it

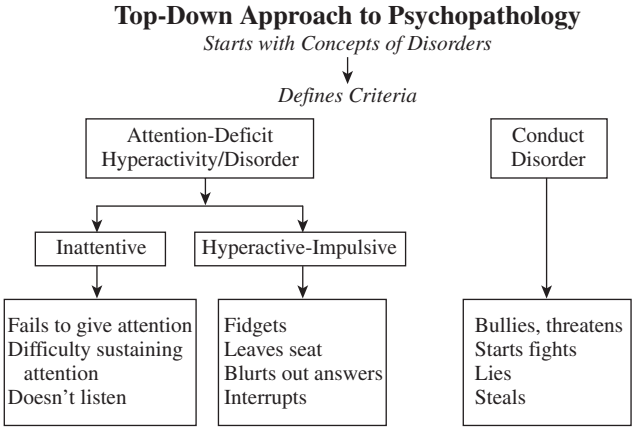

Figure 14.4 The top-down approach to defining taxonomic constructs. *Source: Multicultural Understanding of Child and Adolescent Psychopathology: Implications for Mental Health Assessment,* by T. M. Achenbach and L. A. Rescorla, 2007b, New York: Guilford Press. Reprinted with permission.

starts with a concept of a nosological category that is negotiated by the formulators of the nosology. The formulators of the nosology then work downward from the concept of the category to diagnostic criteria for determining which children fit the category.

Operationalizing Nosologically Based Criteria

Although diagnostic criteria have been made explicit in *DSM-III, DSM-III-R,* and *DSM-IV,* they are not specified in terms of operations for assessing individual children to determine which criteria are met. Instead, separate assessment procedures must be used to operationalize the diagnostic criteria to determine which children meet criteria for which diagnoses. Standardized diagnostic interviews, such as the DISC (Shaffer et al., 2000) and K-SADS (Ambrosini, 2000), and *DSM* rating forms, such as the ADHD Rating Scale-IV (DuPaul et al., 1998) and the YI-4 (Gadow & Sprafkin, 1999), operationalize criteria for those nosological categories that are based primarily on behavioral, emotional, social, and thought problems. However, each operationalization may yield different conclusions about which children meet criteria for which categories. For example, a mean kappa of only .03 was found for agreement between "definite" *DSM* diagnoses made from the DISC administered to mothers and their children versus the K-SADS administered to the same mothers and children (P. Cohen, O'Connor, Lewis, Velez, & Malachowski, 1987). When the diagnostic criteria were relaxed to include agreement between "possible" diagnoses, the mean kappa rose to only .14. Thus, even when criteria were operationalized by means of two of the most thoroughly researched and widely used interviews, the interviews failed to agree on their assignments of individual children to *DSM* categories. Furthermore, meta-analyses have yielded a mean kappa of only .15 between *DSM* diagnoses of children made from standardized interviews,

including the DISC and K-SADS, and diagnoses made from clinical evaluations of the children (Rettew, Doyle, Achenbach, Dumenci, & Ivanova, 2006).

DuPaul et al. (1998) reported point-biserial correlations averaging .31 between various combinations of *DSM* ADHD diagnoses and scores from the ADHD Rating Scale-IV completed by parents and teachers. When Gadow et al. (2002) selected the highest point-biserial correlations between 11- to 18-year-olds' self-ratings on YI-4 scales and *DSM* diagnoses in the youths' clinic charts, they found a mean correlation of .25 for diagnoses of ADHD, ODD, CD, depressive disorders, and anxiety disorders. Although not quite as low as the kappas, these modest correlations further indicate that major challenges remain for operationalizing the nosological taxonomic constructs.

Empirically Based Taxonomic Constructs

Empirically based assessment instruments were presented earlier in the chapter. Some of these instruments were explicitly designed to obtain data from which taxonomic constructs could be derived. For example, development of the CBCL/6–18, TRF, and YSR started with diverse assessment items that were then culled, revised, and augmented on the basis of extensive pilot testing. Subsequently, EFA and CFA were used to derive syndromes of co-occurring problems (Achenbach & Rescorla, 2001). The statistically derived syndromes serve as taxonomic constructs, where "constructs" are defined as "objects of thought constituted by the ordering or systematic uniting of experiential elements" (Gove, 1971, p. 489).

Although parent, teacher, and self-ratings differed with respect to some specific items, as well as with respect to the raters' perspectives, EFA and CFA identified eight similar syndromes in ratings by all three kinds of informants. The items common to the syndromes derived from the CBCL/6–18, TRF, and YSR define "cross-informant" syndrome constructs. Individuals' ratings of a particular child on the specific items composing the CBCL/6–18, TRF, and YSR versions of the syndromes operationalize the cross-informant syndrome constructs in ways that reflect variations in the raters' perceptions of the child. The cross-informant correlations between ratings on virtually all instruments, including the empirically based instruments, are inevitably limited by factors such as differences in the informants' roles (e.g., mother versus father versus math teacher versus history teacher versus self), knowledge and memory of the child's behavior, interpretations of assessment items, and candor. Because the empirically based approach derives syndromes statistically from data on large samples of real children, this approach to taxonomy can be described as working from the bottom up, as illustrated in Figure 14.5.

Relations between Nosologically Based and Empirically Based Taxonomic Constructs

Despite differences in their origins, some of the nosological categories include criterial symptoms analogous to the problem items comprising some of the empirically based syndromes. As an example, many of the criterial symptoms for ADHD have counterparts among the problem items that comprise the empirically based

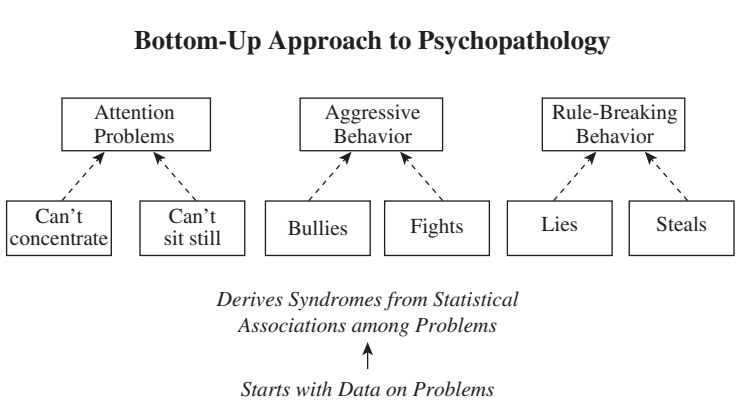

Figure 14.5 The bottom-up approach to defining taxonomic constructs. *Source: Multicultural Understanding of Child and Adolescent Psychopathology: Implications for Mental Health Assessment,* by T. M. Achenbach and L. A. Rescorla, 2007b, New York: Guilford Press. Reprinted with permission.

attention problems syndrome. On the other hand, some nosological categories group criterial symptoms differently than the empirically based syndromes group the corresponding problem items. For example, the nosological category of CD includes both overtly aggressive symptoms and nonaggressive rule-breaking symptoms. By contrast, factor analyses of the CBCL/6–18, TRF, and YSR, as well as of other instruments (Quay, 1986), have yielded separate syndromes of overtly aggressive versus rule-breaking behavior. Conversely, criterial symptoms for depression and anxiety are grouped in separate nosological categories, whereas factor analyses of the CBCL/6–18, TRF, and YSR, as well as other instruments (Achenbach & Rescorla, 2003), have yielded a syndrome that includes both depressive and anxiety problems.

Despite differences between the ways particular kinds of problems are partitioned, as well as differences between the categorical nosological criteria versus the quantification of items and syndromes, multiple studies have reported significant associations between *DSM* diagnoses and scores on empirically based syndromes (e.g., Bird, Gould, Rubio-Stipec, Staghezza, & Canino, 1991; Graetz, Sawyer, Hazell, Arney, & Baghurst, 2001; Steinhausen, Metzke, Meier, & Kannenberg, 1997; Verhulst, van der Ende, Ferdinand, & Kasius, 1997).

To facilitate cross-walks between nosological and empirically based approaches to taxonomy of child psychopathology, a top-down "expert judgment" approach has been used to construct *DSM*-oriented scales from the same pools of items that were factor analyzed to identify syndromes. This has been done for preschoolers, school-age children, adults, and the elderly (Achenbach, Dumenci, & Rescorla, 2003; Achenbach, Newhouse, & Rescorla, 2004; Achenbach & Rescorla, 2003). The *DSM*-oriented scales were constructed by having expert psychiatrists and psychologists from many cultural backgrounds identify problem items that they judged to be very consistent with particular *DSM-IV* categories. For nosological categories for which enough items were identified by a substantial majority of experts, the items were used to construct *DSM*-oriented scales. Like the empirically based

syndromes, the *DSM*-oriented scales are displayed on profiles in relation to age-, gender-, and informant-specific norms.

Multi-informant Perspectives

Previous sections have summarized findings of low to moderate levels of agreement between reports of psychopathology by different informants. Each informant may contribute clinically valuable information about a child's functioning and how it is perceived by the informant. The differences among the informants' reports are as valuable as the similarities for indicating the possible situational specificity versus cross-situational consistency of a child's problems. Differences among informants' reports may also reveal outlier informants whose perceptions of the child are idiosyncratic enough to warrant exploration of the informant's interactions with the child. Such findings may warrant interventions that focus on a particular informant or on the informant's interactions with the child.

To help clinicians efficiently document and compare reports by multiple informants, software is available that displays *DSM*-oriented scale scores, as well as syndrome scores obtained from any combination of up to eight forms (Achenbach & Rescorla, 2007a). As illustrated in Figure 14.6, bar graphs indicate standard scores (*T* scores) on each *DSM*-oriented scale in relation to norms for peers of the child's age and gender, as rated by different kinds of informants, such as parents, teachers, and children themselves. In each bar graph, the broken lines demarcate a borderline clinical range. Scores beneath the bottom broken line are in the normal range, whereas scores above the top broken line are in the clinical range. By looking at the bar graphs, the clinician can quickly identify clinically elevated scores and can also identify similarities and differences among informants' reports. If nosologically based taxonomic constructs are quantified, as exemplified by the *DSM*-oriented scales and as proposed for *DSM-V* (Helzer et al., in press), scores on such constructs can also be used for systematic cross-informant comparisons.

Multicultural Perspectives

Today's clinical caseloads include people from many different backgrounds. Native-born minority groups and immigrants from other regions and countries may need to be evaluated in relation to norms different from those of the majority groups in the societies where they are being clinically evaluated (Achenbach & Rescorla, 2007b). Before considering norms for different groups, it is important to determine whether particular taxonomic constructs are appropriate for representing psychopathology among different cultural groups. The multicultural generalizability of the statistically derived syndromes of the ASEBA instruments has been tested by performing CFA on large samples of parents' ratings of their children in 30 societies, teachers' ratings of their students in 20 societies, and youths' self-ratings in 23 societies (Ivanova, Achenbach, Dumenci, et al., 2007; Ivanova, Achenbach, Rescorla, Dumenci, Almqvist, Bathiche, et al., 2007; Ivanova, Achenbach, Rescorla, Dumenci, Almqvist, Bilenberg, et al., 2007). The societies differed in many ways from the United States, where the syndromes were originally derived. Examples of societies that differed in geographical region, language, ethnicity, religion, and

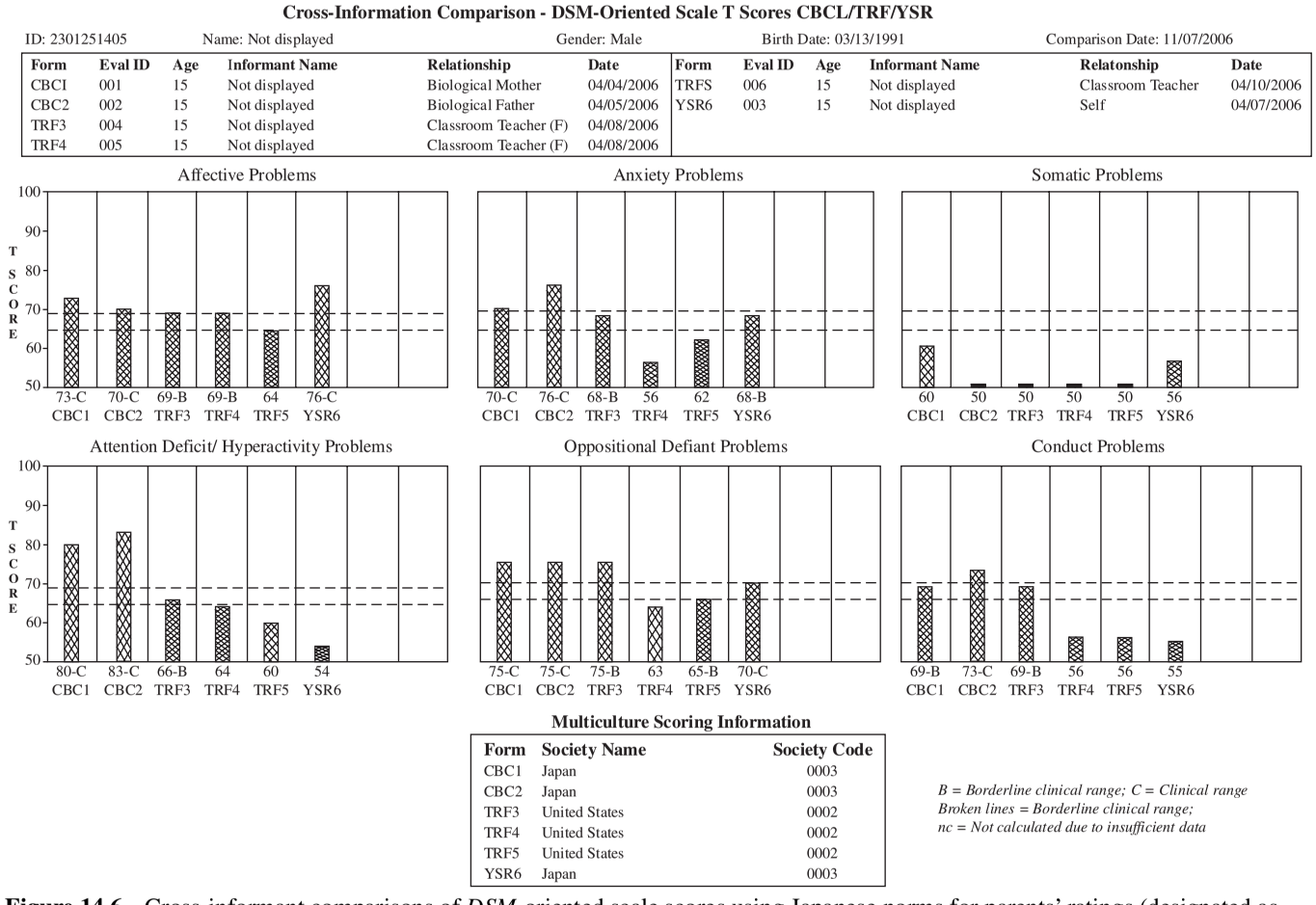

Figure 14.6 Cross-informant comparisons of *DSM*-oriented scale scores using Japanese norms for parents' ratings (designated as CBC) and youth's self-ratings (designated as YSR) and U.S. norms for teachers' ratings (designated as TRF). *Source: Multicultural Supplement to the Manual for the ASEBA School-Age Forms and Profiles,* by T. M. Achenbach and L. A. Rescorla, 2007a, Burlington: University of Vermont, Research Center for Children, Youth, and Families. Reprinted with permission.

political systems are Iran, China, Japan, Thailand, Korea, Ethiopia, Lebanon, Russia, and Turkey. Societies from the Caribbean and from western, eastern, northern, and southern Europe were also included.

To the surprise of the researchers, the syndromes derived primarily from U.S. samples were found to fit the data for all the other societies. The taxonomic constructs embodied by the statistically derived syndromes thus represent patterns of parent, teacher, and self-ratings of behavioral, emotional, social, and thought problems for at least this large array of very diverse societies. At this writing, no multicultural studies have tested taxonomic constructs embodied in the *DSM* or *ICD* nosologies for children's disorders.

The multicultural CFA support for the statistically derived syndromes justifies use of the syndromes for assessing children from many societies. However, this does not necessarily mean that the distributions of scale scores are similar enough to justify use of the same norms for different societies. The distributions of scale scores have been statistically compared for the United States and all the societies that were included in the CFA tests of the statistically derived syndromes (Rescorla et al., 2007, 2007a, 2007b). A great deal of overlap was found between the distributions of problem scores from all societies and the mean total problems scores from most societies were clustered within a narrow range. However, some societies had mean total problems scores that were low or high enough to warrant separate norms.

To take account of the relatively low scores, one set of norms was constructed by averaging the scores for societies whose mean total problems scores were more than 1 standard deviation below the mean of all the societies (called the "omnicultural mean"; Ellis & Kimmel, 1992). Another set of norms was constructed by averaging the scores for societies whose mean total problems scores were more than 1 standard deviation above the omnicultural mean. A third set of norms, for the middle scoring societies, consists of norms for U.S. samples, which are about in the middle of the middle group and which have been used in many clinical and research applications.

When evaluating individual children, clinicians can elect to have the children's syndrome and *DSM*-oriented scale scores displayed on profiles and cross-informant bar graphs in relation to norms for the low-scoring, middle-scoring, or high-scoring societies. The clinician can also choose to have percentiles and *T* scores from the low-, middle-, or high-scoring societies displayed. Equally important, the clinician can see how the child's scale scores compare with different sets of norms. This is especially useful for immigrant children and for children whose parents and/or teachers have very different cultural backgrounds.

As an example, for the cross-informant comparisons shown in Figure 14.6, the clinician elected to have the parents' CBCL/6–18 scale scores displayed in relation to norms for the high-scoring societies, because the parents had emigrated from a high-scoring society, maintained its customs, still spoke its language, and completed a translation of the CBCL/6–18 in that society's language. Because their child attended a mainstream school in the host society where TRF scores were in the middle-norm group, the clinician elected to have the TRF scale scores displayed in relation to the middle norms. Because YSR scores in the youth's home culture were in the high-norm group, the clinician elected to have the YSR scale scores displayed in relation to the high norms. However, because the host society's YSR scores

were in the middle-norm group, the clinician subsequently had the cross-informant comparisons displayed again with the youth's YSR scores displayed in relation to the middle norms. This enabled the clinician to see whether any scales would be clinically elevated that had not been clinically elevated in relation to the high norms. Note that all scales in the cross-informant comparisons are displayed in terms of *T* scores, but the user decides whether to display scores for each informant in relation to *T* scores based on the low-scoring, middle-scoring, or high-scoring societies.

Implications for Clinical Practice

Taxonomic constructs may seem irrelevant to clinical practice. However, they provide foundations for assessment, diagnosis, nosology, and treatment. Nosological categories such as ADHD, ODD, CD, Major Depressive Disorder, and Generalized Anxiety Disorder shape many aspects of clinical practice. Although such categories may seem to have an intrinsic reality, changes in the categories from one edition to another edition of a nosology, the lack of operational definitions, poor agreement among "gold standard" diagnostic procedures, and high comorbidity indicate that current nosological categories cannot be equated with categorically distinct disorders. Instead, the categories can be viewed as representing implicit taxonomic constructs. As such, they should be subjected to comparisons with alternative taxonomic constructs. For child psychopathology, statistically derived syndromes offer one alternative set of taxonomic constructs for which there is considerable empirical support. Options for scoring both statistically derived syndromes and *DSM*-oriented scales from the same sets of parent, teacher, and self-ratings can help clinicians view children's problems from both taxonomic perspectives. Multi-informant and multicultural comparisons of both kinds of scales further enhance opportunities for flexibly adapting clinical assessment to the needs of children from diverse backgrounds, as seen from different informants' perspectives.

SUMMARY

Assessment, diagnosis, nosology, and taxonomy can be understood as interdependent aspects of continuous processes. This chapter focused on these processes as they pertain to behavioral, emotional, social, and thought problems occurring between the ages of 18 months and 18 years with sufficient severity to warrant the term "psychopathology" and a likely need for professional help.

Clinical assessment refers to procedures for identifying the distinguishing features of problems, disorders, or cases, as well as relevant strengths. Some assessment instruments are designed to determine whether children meet criteria for *DSM* diagnoses, whereas other instruments assess broader spectra of functioning. Evidence- (or empirically) based assessment is necessary for appropriate applications of evidence-based treatment and other clinical services. Comprehensive clinical assessment of children should include assessment of parent figures.

Diagnosis has become a primary focus of clinical services since the *DSM-III* introduced explicit diagnostic criteria. Confusion often arises from failure to distinguish clearly among diagnostic processes, formal diagnoses, and diagnostic

formulations. Most relevant to nosology and taxonomy, formal diagnoses are made by determining how a case should be categorized.

Nosologies such as the *DSM* and the *ICD* are systems for classifying diseases (or disorders). Since the introduction of explicit diagnostic criteria by *DSM-III,* many children have been found to meet criteria for multiple diagnoses. Although comorbid diagnoses may provide useful information, the implications of comorbidity have been clouded by failure to compute it bidirectionally and by rates of comorbidity between some diagnoses that are so high as to cast doubt on whether they really represent different disorders. Quantitative approaches that are being advocated for *DSM-V* could help to alleviate problems arising from excess comorbidity among diagnoses as well as to bring diagnoses more in tune with advances in genetics, imaging, and biochemistry.

Taxonomic constructs may seem irrelevant to clinical practice, but they provide foundations for assessment, diagnosis, nosology, and treatment. The lack of known natural categories for psychopathology means that we should think in terms of multiple ways of taxonomically organizing data on psychopathology to avoid reifying any single set of taxonomic constructs. Nosological categories can be viewed as taxonomic constructs, although criteria for the constructs must be operationalized in a top-down process of developing assessment procedures such as diagnostic interviews. However, even the most thoroughly researched and widely used diagnostic interviews have yielded negligible agreement for child diagnoses made from other diagnostic interviews and from clinical evaluations.

Empirically based taxonomic constructs have been derived from statistical analyses of data obtained with empirically based assessment instruments. The generalizability of the syndromes derived from this bottom-up process has been supported by CFA of large samples from many societies.

There are important points of contact between nosologically based and empirically based taxonomic constructs. Both kinds of constructs have been operationalized via a common pool of assessment items. For both kinds of constructs, assessment data from multiple informants and normative data from multiple cultures are needed for clinical research and practice in the twenty-first century.

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CHAPTER 15

Diagnostic Interviewing

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It is well recognized by those involved in children's mental health that proper assessment and diagnosis of the diverse childhood psychopathologies are critical undertakings. The importance of these issues has been reflected in the numerous changes and developmental adaptations made to the *Diagnostic and Statistical Manual of Mental Disorders* (*DSM*). The criteria provided in the *DSM* are considered the basis for all major psychiatric diagnoses and constitute the necessary requirements for reimbursement or coverage by major health management organizations. Over the past 25 years, the *DSM* has undergone several changes specific to childhood disorders. In so doing, the *DSM* has begun to recognize that children are not miniature adults and that the identification of various disorders in childhood requires developmental adaptations (Ollendick & Hersen, 1983, 1989, 1998). For example, with the third revision of the *DSM,* a section specific to disorders that are usually first diagnosed in infancy, childhood, or adolescence was introduced. In addition, with each revision to the *DSM,* increasingly more diagnostic criteria that incorporate developmental considerations have been put forth. To illustrate, in the current *DSM-IV* (American Psychiatric Association, 1994, pp. 416–417), the diagnostic criteria for Social Phobia provide the following distinctions for children:

> (A) In children, there may be evidence of the capacity for age-appropriate social relationships with familiar people and the anxiety must occur in peer settings, not just in interactions with adults; (B) In children, the anxiety may be expressed by crying, tantrums, freezing, or shrinking from social situations with unfamiliar people; (C) In children, this feature (i.e., recognition of the fear as excessive or unreasonable) may be absent; and (F) In individuals under age 18 years, the duration is at least 6 months.

Similar modifications have been made to other psychiatric disorders to make them more applicable to children and adolescents for whom separate categories do not exist.

Increased attention to the issue of proper evaluation of child psychopathology has also been reflected in the actual role of children in the assessment and diagnostic process. In the past, parents were thought to be the best informants regarding

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a child's behavior and were, therefore, the only ones included in the assessment process (Loeber, Green, & Lahey, 1990; Rutter & Graham, 1968; Rutter, Tizard, & Whitmore, 1970; Williams, McGee, Anderson, & Silva, 1989). Any information collected from or reported by the child was considered secondary to the parents' reports. Reliance on parental report was primarily based on the assumption that children lacked the cognitive sophistication to respond appropriately in an interview format. For example, researchers reported concern that children would be inattentive to the assessment process, lack insight regarding their feelings, and be unable to accurately describe their own behaviors and symptoms (Edelbrock & Costello, 1990; Herjanic, Herjanic, Brown, & Wheatt, 1975; Schwab-Stone, Fallon, Briggs, & Crowther, 1994). Thus, the amount of credence placed on the child's report was minimal, and parents were considered the experts regarding the child's outward behaviors as well as their internal thoughts and feelings.

This view changed dramatically following seminal studies by Lapouse and Monk (1958), who included mothers and their children in the interview process, and Rutter, Graham, and colleagues (Rutter & Graham, 1968; Rutter et al., 1970; Rutter, Tizard, Yule, Graham, & Kingsley, 1977; Rutter, Tizard, Yule, Graham, & Whitmore, 1976), who designed and psychometrically validated the first structured interview procedures for children. The consideration of the child as a separate and valuable respondent by these researchers as well as their careful evaluation of the interview process paved the way for today's inclusion of multiple informants in clinical interview procedures. Currently, most researchers and clinicians consider a child assessment incomplete if information from numerous sources (e.g., teachers, parents, doctors), including the child, is not obtained (Ollendick & Hersen, 1984, 1993). Thus, children are now viewed as reliable, valid, and valuable informants regarding their subjective thoughts, behaviors, and emotions (Chambers et al., 1985; Edelbrock, Costello, Dulcan, Kalas, & Conover, 1985; Herjanic et al., 1975; Hodges, McKnew, Cytryn, Stern, & Kline, 1982; Kazdin, French, & Unis, 1983; Moretti, Fine, Haley, & Marriage, 1985; Verhulst, Althaus, & Berden, 1987). Consequently, companion diagnostic interviews for children and their parents have been developed (and revised) in line with the changes in the *DSM*. To demonstrate their psychometric soundness (e.g., reliability, validity, clinical utility), careful evaluations have been conducted with both the child and parent diagnostic interviews.

In general, "reliability refers to the consistency of measurements," whereas "the validity of a test refers to the extent to which a test measures what it is supposed to measure, and therefore the appropriateness with which inferences can be made on the basis of the test results" (Sattler, 1988, pp. 25, 30). Although a measure can be reliable while lacking validity, both of these psychometric features are necessary to proclaim an interview psychometrically sound.*

*The kappa statistic (i.e., κ) is commonly used in studies that assess reliability with dichotomous data. The kappa is a correlational statistic that corrects for chance and ranges from -1.00 (total disagreement) to 1.00 (pure agreement). Although the kappa statistic evaluates agreement beyond the level of chance, it is also sensitive to base rates, resulting in lower κs for less prevalent symptoms or disorders. Interpretations vary, but typically $\kappa s < .40$ are considered poor, $\kappa s = .40$ to .70 are considered acceptable, and $\kappa s > .70$ are considered good or excellent.

Important distinctions in the literature regarding reliability are those among different indices of interrater, test-retest, and multiple-informant agreement. *Interrater agreement* refers to the amount of consensus between two observers of the same child or of the same child's behaviors at the same point in time and in the same setting. With the diagnostic interviews, this is usually accomplished in one of two ways: live observation with simultaneous coding or video or audiotape recordings that are coded following the diagnostic interview. In general, interrater reliability tends to be high and is frequently higher than the other two types of reliability. *Test-retest reliability*, also called within-informant reliability, refers to the amount of consistency, or agreement, found regarding the same measure of behavior over time. In diagnostic interviews, this type of reliability customarily examines the level of symptom or diagnostic agreement *in the same informant* from one assessment time to another. *Multiple-informant agreement*, on the other hand, concerns the consensus of ratings made regarding a particular child *by different individuals* (e.g., parent-child, parent-teacher). Although discrepancies can affect all types of reliability, research findings suggest that more pronounced differences are found with multiple informants. That is, in contrast to generally acceptable levels of interrater and test-retest agreement, multiple-informant reliability studies have consistently yielded poor results. This has been so for child behavior rating scales as well as diagnostic interviews (cf. Achenbach, Dumenci, & Rescorla, 2002; Achenbach, McConaughy, & Howell, 1987; Grills & Ollendick, 2002).

In general, *validity* refers to how well a test measures what it is supposed to measure. Common types of validity examined include content, criterion-related, and construct validity. *Content validity* refers to the degree to which questions in the interview represent all aspects of the disorder and cannot be examined with traditional statistical procedures. Basically, it refers to the extent to which the diagnostic interview taps the various symptoms of a given disorder. *Criterion-related validity*, on the other hand, concerns the degree to which a measure like a diagnostic interview predicts an outcome on another measure such as adjustment, and is typically separated into concurrent (i.e., both measures obtained at the same time) and predictive (i.e., one measure obtained prior to the other) validity categories. Finally, *construct validity* measures how effectively the information obtained from the interview agrees with the theoretical construct being investigated (i.e., depression). Construct validity is typically determined via positive correlations with established measures of the construct under investigations (convergent validity) and nonsignificant associations with measures previously established to measure theoretically diverse constructs (divergent validity).

Establishing the validity of psychiatric diagnoses based on diagnostic interviews is frequently problematic because no gold standard exists with which to compare the findings. Consider the validity of diagnostic interviews and less structured and more traditional clinical interviews. Without a definitive comparison or standard, it cannot be determined whether it is the diagnostic interview or the clinical interview that is more accurate. Indeed, there are myriad factors that could influence either indicator, for example, the need for clinicians to obtain a diagnosis in order to receive insurance reimbursement or the inability of structured diagnostic interviews

to allow for additional inquiry of symptoms (see A. Jensen & Weisz, 2002, for further discussion of factors that could affect each). Moreover, very few studies have been conducted with regard to the validity of diagnostic interviews, and oftentimes estimates of validity are put together based on various methodological designs. For example, examination of content validity cannot be statistically evaluated, and yet the face validity of structured interviews is typically considered certain as items are derived from the *DSM* diagnostic criteria. A bit of circular reasoning is involved with such conclusions.

Criterion-related validity of the concurrent type is also difficult to determine. These findings are troubling as agreement between diagnoses generated by diagnostic interviews and real-world clinicians is often poor (cf. A. Jensen & Weisz, 2002), despite the use of varied methodologies and interviews. For example, A. Jensen and Weisz examined diagnostic interview diagnoses and diagnoses determined at community health clinics and found agreement not significantly better than chance for most individual disorders (e.g., Dysthymic Disorder) and broad categories of those disorders (e.g., mood disorders). Similarly, Lewczyk, Garland, Hurlburt, Gearity, and Hough (2003) found poor agreement with diagnostic interviews and diagnoses generated by a multidisciplinary team that included diverse mental health practitioners, including a child psychiatrist, a child psychologist, and a social worker. Overall, studies have shown poor validity when diagnoses obtained from diagnostic interviews are compared to clinician diagnoses at outpatient clinics (e.g., Ezpeleta, de la Osa, Doménech, Navarro, & Losilla, 1997), inpatient clinics (Pellegrino, Singh, & Carmanico, 1999; Vitello, Malone, Buschle, & Delaney, 1990), and on admission (Weinstein, Stone, Noam, & Grimes, 1989) and discharge (Aronen, Noam, & Weinstein, 1993; Welner, Reich, Herjanic, Jung, & Amado, 1987) from both of these settings.

Predictive validity has been demonstrated in only a few studies. For example, ratings of disruptive behavior disorders on diagnostic interviews and later problematic behaviors (e.g., delinquency, oppositional behavior) have been demonstrated (Jewell, Handwerk, Almquist, & Lucas, 2004). Furthermore, significantly better predictive validity (over a 6-month period) was found for child reports of Conduct Disorder on diagnostic interviews than for children who received diagnoses only from clinicians (and not reported by themselves) and a control group of children for whom no reports of externalizing problems were reported.

With regard to construct validity, diagnoses derived from diagnostic interviews are used to validate information obtained from rating scales, self-report instruments, and behavioral observations. In addition, some research has examined this process in reverse; that is, questionnaires and rating scales have been used as the criterion, and diagnoses obtained from diagnostic interviews have been examined in relation to these measures. For example, empirical research has confirmed the validity of diagnoses formulated using the child and parent versions of the Anxiety Disorders Interview Schedule (see later discussion) by showing that scores on child and parent rating scales converge in expected ways with diagnoses (e.g., Weems, Silverman, Saavedra, Pina, & Lumpkin, 1999; Wood, Piacentini, Bergman, McCracken, & Barrios, 2002). Wood et al. evaluated the validity of Anxiety Disorders Interview Schedule: Child and Parent Versions (ADIS-C/P) diagnoses of Social

Phobia, Separation Anxiety Disorder, Generalized Anxiety Disorder, and Panic Disorder in children and adolescents referred to an outpatient anxiety disorders clinic. In addition to youth and parents being administered their respective versions of the ADIS interviews, they were administered the Multidimensional Anxiety Scale for Children (MASC; March, Parker, Sullivan, Stallings, & Conners, 1997). Strong correspondence was found between ADIS-C/P diagnoses and empirically derived MASC factor scores corresponding to each of these diagnoses and disorders, with the exception of Generalized Anxiety Disorder. For example, MASC social anxiety factor scores, but no other factor scores, were significantly elevated for children meeting *DSM-IV* criteria for Social Phobia on the ADIS-C/P. Despite these positive findings, there is a need for research that evaluates the validity of diagnoses formulated with interview schedules beyond looking at diagnoses' convergence and divergence with questionnaire data (Jarrett, Wolff, & Ollendick, 2007). Ideally, other markers, including what have come to be known as endophenotypes, could be used for this purpose.

In addition to establishing the correct diagnosis, diagnostic interviews are also frequently used for case conceptualization and treatment planning purposes (Silverman & Ollendick, in press). Such purposes refer to the clinical utility of the instrument being used. Diagnoses themselves yield powerful information about treatment targets and potential treatment plans. For example, the anxiety and worry of Generalized Anxiety Disorder (GAD) is associated with one or more of the following six *DSM-IV* symptoms: being restless, being easily fatigued, having difficulty concentrating, being irritable, having muscle tension, and having difficulty falling or staying asleep. Any one or more of these problems (i.e., target symptoms) could lead to highly specific treatment plans designed to address them. This approach to prescriptive interventions is illustrated in the early work of Eisen and Silverman (1993, 1998) and is captured well in a recent book on modularized treatment by Chorpita (2007). In an early study, Eisen and Silverman (1993) showed that cognitive-behavior therapy was most effective for children with overanxious disorder (the *DSM-III-R* precursor to GAD) when they were matched with specific symptoms. For example, children with primary symptoms of worry and obsessions responded more favorably to cognitive therapy, whereas children with primary symptoms of somatic complaints responded more favorably to relaxation training aimed at dealing with physiological and somatic complaints. This early study was replicated in a second study by Eisen and Silverman (1998), in which these authors again showed that although both treatments were generally effective, only the prescriptive treatments produced sufficient improvements for the treated children to meet exacting positive end-state functioning criteria. Similar effects of matching were shown by Ollendick, Hagopian, and Huntzinger (1991) with separation anxious children and by Ollendick (1995) with adolescents with Panic Disorder and Agoraphobia. Thus, in these studies, treatments were shown to be maximally effective when they were closely matched with the symptom presentation of the individual child or adolescent. That is, specific diagnoses when supplemented with symptom profiles led to superior outcomes.

Likewise, diagnostic interviews have been used for treatment evaluation purposes. For example, in treatment outcome studies, 100% of youth necessarily meet

diagnostic criteria at pretreatment. In evaluating treatment outcome, of interest is the diagnostic recovery rate at posttreatment and follow-up; most studies report 60% to 80% of participants as recovered and no longer meeting diagnostic criteria at posttreatment and follow-up. As such, diagnostic interviews are extremely useful and provide one of the most rigorous ways of determining treatment outcomes and providing an evidence base for various treatment interventions (Ollendick, King, & Chorpita, 2006).

Finally, before proceeding to a review of specific diagnostic interviews, brief mention should be made of what has come to be called the treatment utility (a form of clinical utility) of measures such as diagnostic interviews. As we have noted, if one wishes to use the treatment that possesses the most research evidence for a given disorder, it is important to first have *confidence* (i.e., have reliable and valid information) that the youth or groups of youths with whom one is working are in fact accurately diagnosed with the disorder under study, rather than some other clinical disorder. Second, it is important to have confidence about the *specific* type(s) of disorder that the youth is diagnosed with. For example, with the anxiety disorders, it is crucial to know the exact disorder so that appropriate exposure tasks can be assigned in and out of the treatment sessions (e.g., expose the child to social evaluative situations if the child has a Social Phobia; to separation situations if the child has Separation Anxiety Disorder). However, as Nelson-Gray (2003) has pointed out, empirical verification of treatment utility is yet to be demonstrated. That is, it has not been empirically tested whether different outcomes result for children who have received diagnoses assigned with a diagnostic interview schedule and are given an intervention versus a group of children who have *not* been given a diagnostic interview schedule and receive the same treatment. Although such may be so, it seems to us that starting an intervention without this information would be premature.

As is evident, a host of issues surrounding diagnostic interviews exist, including their reliability, validity, and clinical utility. Although many questions remain about the psychometric properties of these interviews, much progress has been made. A number of reliable, valid, and clinically useful interviews have been developed, a topic to which we next turn our attention.

DIAGNOSTIC INTERVIEWS

Generally speaking, diagnostic interviews are considered to be a formal and full examination of a particular problem area or areas. There are several expectations imposed on these interviews: (a) They are psychometrically sound measurements, (b) they have specific rules regarding the content and order of the questions and topics being addressed (e.g., asking whether depressed mood is present prior to asking the possible effects of the depressed mood) as well as the manner of recording responses, and (c) some degree of structure is imposed with regard to final diagnostic decisions (Weiss, 1993). The degree of correspondence with these specifications appears to be directly related to the amount of structure imposed by the interview. Various descriptions have been applied to distinguish among diagnostic interviews, with the most common distinction pertaining to the

structure (semi- or highly structured) of the interview. Specifically, distinctions have been made among unstructured, semi-structured, and highly structured interviews.

Unstructured interviews give complete discretion to the interviewer as to what, when, and how questions will be asked and recorded. Although these interviews typically contain questions pertaining to the child's history (e.g., medical, psychological, developmental, academic, social), presenting problems (e.g., antecedents, behaviors, consequences), environmental contexts (e.g., home and family, school), and treatments (i.e., past and present), there are no set procedures used across clinicians. Semi-structured interviews also tend to allow leeway for the interviewer regarding the order in which questions are asked, the manner in which the questions are phrased, and the way responses are recorded. Emphasis is placed on obtaining consistent and reliable information; thus extensive training is generally required for administration of these interviews to ensure that clinical discretion will be applied judiciously. Highly structured interviews, on the other hand, are more restrictive in the amount of freedom allotted to the interviewer. With these interviews, it is generally expected that examiners ask all questions in the same manner and order, as well as record all responses in a prespecified manner. Clinical judgment is reduced with highly structured interviews, such that specific and/or extensive training is not required for their administration. In fact, highly structured interviews are commonly administered by laypersons (e.g., individuals without a formal degree in psychology, psychiatry, or social work), and several have been converted to computer-based formats (e.g., Diagnostic Interview Schedule for Children [DISC], Diagnostic Interview for Children and Adolescents [DICA]; see later discussion). Although highly structured formats allow for more confidence in the exactness of the interview's administration and perhaps more reliable findings, the rigidity of the interview may also make it seem impersonal, hinder the establishment of rapport, and interfere with reliability and validity by not providing the interviewee the opportunity to report all difficulties or to explore them in full depth. As a result, the use of highly structured interviews may result in unanswered questions for the clinician that might have been addressed in a less structured format (e.g., knowing a child feels sad most days does not answer questions of potential precipitants, etiological factors, responses by others in the child's environment).

In addition to differentiations based on interview structure, a variety of descriptors have been used in the literature (i.e., respondent-based versus interviewer-based, glossary-driven versus clinician-driven). Nevertheless, each of these distinctions concerns the degree of latitude allotted the interviewer, as well as the type of research with which the interview was designed to be used. Initially, highly structured interviews were used for epidemiological research, and semi-structured interviews were used in clinical research and randomized controlled trials. However, currently these distinctions seem less meaningful, with all research teams but that of the DISC considering their interviews semi-structured and some creating separate epidemiological and clinical interview versions (e.g., Schedule for Affective Disorders and Schizophrenia for School-Age Children [K-SADS]; see later discussion).

For the most part, the format of companion parent and child interviews is similar; each contains (a) an introductory section designed to help build rapport with the informant (e.g., demographics, school, free time activities) and elicit initial information regarding presenting problems; (b) symptom-specific modules (e.g., mood disorders) that usually begin with screener questions and determine whether to skip or continue with follow-up questions (i.e., regarding frequency, intensity, duration, and interference); and (c) the determination of diagnostic presence or absence based on algorithms or clinical judgments. Several of the most commonly utilized diagnostic interviews are discussed next, with general psychometric information provided. Overall, acceptable test-retest and interrater reliability coefficients have been reported for each of these interviews; however, findings for multiple-informant reliability have been more varied (cf. Grills & Ollendick, 2002), with little data published on the validity of these instruments.

The Anxiety Disorders Interview Schedule for Children for DSM-IV: Child and Parent Versions

The ADIS-CP for *DSM-IV* (Silverman & Albano, 1996; Silverman, Saavedra, & Pina, 2001) is a downward extension of the adult Anxiety Disorders Interview Schedule for *DSM-IV* (Brown, Di Nardo, & Barlow, 1994) and is most detailed in its coverage of the anxiety disorders, relative to the other interview schedules. The ADIS for *DSM-IV:* C/P and its previous *DSM-III* and *DSM-III-R* versions (Silverman, 1991) have been used most frequently in the youth anxiety disorders research literature, including randomized clinical trials. The ADIS for *DSM-IV:* C/P contains a series of modules that cover all the anxiety disorders described in *DSM-IV,* as well as sections that cover the most prevalent disorders of childhood and adolescence (e.g., Attention-Deficit/Hyperactivity Disorder [ADHD], depressive disorders) and screening questions for most other disorders (e.g., eating disorders, enuresis). Additional questions are included that allow interviewers to obtain information about the history of the problem as well as situational and youth cognitive factors influencing anxiety. In addition, the ADIS for *DSM-IV:* C/P contains clinician severity rating scales that assess for degree of impairment or interference in youth functioning associated with the specific anxiety disorder endorsed by the youth and parent, respectively. Specifically, based on the information obtained from the child and parent versions of the interview, interviewers assign the degree of distress and interference associated with each disorder (0 = None to 8 = Very severely disturbing/impairing) with respect to the youth's peer relationships, school work, family life, and personal distress. Each module in the interview also contains questions that allow interviewers to assign 0 to 8 ratings on the youth's fear and avoidance of diverse situations relevant to a particular disorder (e.g., Social Phobia). Similar to the adult ADIS, clinician severity ratings of 4 (definitely disturbing/impairing) or higher are viewed as clinical diagnoses, and those less than 4 are viewed as subclinical or subthreshold.

Acceptable test-retest reliability (7 to 14 days) estimates have been reported for the ADIS for *DSM-IV* Child (aged 7 to 16; $\kappa = .61$ to .80), Parent ($\kappa = .65$ to 1.00), and combined ($\kappa = .62$ to 1.00) diagnoses (Silverman et al., 2001). Likewise,

interrater agreement for earlier versions of the ADIS have generally fallen within the acceptable to excellent range for videotape ($\kappa = .45$ to .82; Rapee, Barrett, Dadds, & Evans, 1994) and live observer (.35 to 1.00; Silverman & Nelles, 1988) paradigms. In contrast, findings for multiple-informant reliability have been poor for both the most recent and earlier versions of the ADIS (Grills & Ollendick, 2003; Rapee et al., 1994).

Schedule for Affective Disorders and Schizophrenia for School-Age Children

There are currently several different forms of the K-SADS, all of which include corresponding parent and child (ages 6 to 18) versions. The three most current (*DSM-IV* applicable) and researched are Present state (P-IVR; Ambrosini & Dixon, 1996); Epidemiological (E-Version 5; Orvaschel, 1995); and Present/Lifetime (P/L; Kaufman et al., 1997). Each version of the K-SADS has undergone extensive revisions, which have been detailed elsewhere (see Ambrosini, 2000; Kaufman et al., 1997). Although the primary focus is affective disorders, these three K-SADS versions also assess for several additional disorders (see Table 15.1). The primary difference among these versions is the diagnostic time frame; that is, the K-SADS-P-IVR assesses disorders from the current episode (within the week preceding the interview) as well as from the past 12 months, whereas the K-SADS-E and P/L diagnose current and lifetime disorders (with a focus on the most severe past episode). In all cases, the interviews are conducted with the parent or primary guardian first, and the same interviewer then administers the corresponding version to the child. Information is gathered regarding the child's current symptoms, as well as from when the informant considered the symptoms worst during the specified time frame (i.e., past 12 months or lifetime). Discrepancies in reports of observable behaviors are addressed in a joint parent/child interview prior to the clinician's final ratings. Finally, based on all of the information obtained (e.g., parent, child, schools, charts), the clinician determines summary severity scores for each symptom and makes a diagnosis using accompanying criteria checklists. Research diagnostic criteria (RDC) are used whenever possible, with *DSM* diagnostic criteria used for those disorders not covered by RDC.

Using varied approaches (i.e., videotape, audiotape, live or joint observation), acceptable interrater reliability has been reported for the most recent (K-SADS-P IV, $\kappa = .80$ to 1.00) and earlier (K-SADS-P IIIR, $\kappa = .46$ to .91; K-SADS-E, $\kappa = .51$ to .77; Ambrosini, 2000; Ambrosini, Metz, Prabucki, & Lee, 1989) K-SADS versions. Likewise, test-retest studies have generally revealed adequate reliability coefficients for the K-SADS (P-IIIR, $\kappa = .46$ to .91; E, $\kappa = 1.00$ major depression past episode; P/L, .55 to 1.00). Despite these, agreement between parent and child reports on earlier versions of the KSADS, administered by the same interviewer, has varied greatly ($\kappa = –08$ to .96, Chambers et al., 1985: $\kappa = -.07$ to .52, Angold et al., 1987; $\kappa = .05$ to .66, Hodges, McKnew, Burbach, & Roebuck, 1987), and information on the most recent K-SADS is not currently available. In addition to these reliability estimates, predictive and construct validity have been demonstrated for the K-SADS interviews (see Ambrosini, 2000; Ambrosini et al., 1989).

Diagnostic Interview for Children and Adolescents

The DICA (Reich, 1998) is a semi-structured instrument, with parent and companion child (ages 6 to 12) and adolescent (ages 13 to 18) versions. The most recent revision of the DICA corresponds with the latest revisions to the *DSM* (*III-R, IV*) and is relatively unique in that it examines lifetime diagnoses (cf. Reich, 2000; Rourke & Reich, 2004). In addition to diagnostic sections (see Table 15.1), the DICA includes questions about psychosocial stressors, risk and protective factors, and perinatal, delivery, and early child development (parent interview only). Following completion of the interview, the interviewer pursues problematic areas which are then resolved by consultation with the DICA manual and/or discussion with more experienced clinicians or primary investigators. There are also more structured versions of the DICA, specifically a computerized version, which can be administered by a trained interviewer or completed by the informant alone.

Table 15.1 Summary of Diagnoses Possible with the Structured Interviews Reviewed

Diagnoses	K-SADS-P IV, -E, -P/L	ADIS-IV	DISC-IV	DICA
Major Depressive Disorder	Y	Y	Y	Y
Dysthymic Disorder	Y	Y	Y	Y
Bipolar Disorder	Y	N	N	N
Mania	Y	N	Y	Y
Hypomania	Y	N	Y	Y
Cyclothymia	Y	N	N	N
Anorexia Nervosa	Y	Y (screens for)	Y	Y
Bulimia Nervosa	Y	Y (screens for)	Y	Y
Enuresis	Y (E, P/L, only)	Y (parent version)	Y	N
Encopresis	Y (E, P/L, only)	N	Y	N
Tic disorders	Y (E, P/L, only)	N	Y	Y
Schizophrenia	Y	N	Y	N
Schizoaffective Disorder	Y	N	N	N
Generalized Anxiety Disorder	Y	Y	Y	Y
Separation Anxiety Disorder	Y	Y	Y	Y
Obsessive-Compulsive Disorder	Y	Y	Y	Y
Simple Phobia	Y	Y	Y	Y
Social Phobia	Y	Y	Y	Y
Agoraphobia	Y	Y	Y	N
Panic Disorder	Y	Y	Y	Y
Posttraumatic Stress Disorder	Y	Y	Y	Y
Oppositional Defiant Disorder	Y	Y(parent version)	Y	Y
Attention-Deficit/ Hyperactivity Disorder	Y	Y	Y	Y
Conduct Disorder	Y	Y(parent version)	Y	Y
Substance disorders	Y	Y	Y	Y
Personality disorders	Y	N	N	N

Generally acceptable 1-week test-retest reliability for the current DICA version has been reported for the adolescent ($\kappa = .59$ to .92) and child ($\kappa = .32$ to .65) versions. Examinations of earlier DICA versions revealed poor parent-child agreement for most diagnoses ($\kappa = .00$ to .58, Reich, Herjanic, Welner, & Gandhy, 1982; $\kappa =$ .00 to .35, Brunshaw & Szatmari, 1988; $\kappa = .11$ to .42, Sylvester, Hyde, & Reichler, 1987; $\kappa = -.05$ to .68, Boyle et al., 1993). In addition, at the symptom level, Herjanic and Reich (1982) found kappa coefficients ranging from .00 to .87; however, coefficients for only 16 of the 185 questions examined obtained acceptable status. A notable exception was found by Welner and colleagues (1987; $\kappa = .49$ to.80) through inclusion of probable cases of parent-child agreement (i.e., one symptom lacking or duration uncertain). Test-retest and multiple-informant agreement for the current DICA were not found.

The Diagnostic Interview Schedule for Children

The only interview considered highly structured, the DISC was originally designed for epidemiological surveys of children that were conducted by laypersons. The most recent revision (DISC-IV; Shaffer, Fisher, Lucas, Dulcan, & Schwab-Stone, 2000) contains criteria for disorders that can be diagnosed in children and adolescents using either the *DSM-IV* or *ICD-10* (see Table 15.1). Interviewers are trained to read questions exactly as written, with the majority of responses requested in a yes/no format. The DISC-IV inquires first about symptoms from the past 12 months. If the informant responds positively to this initial inquiry, information regarding symptoms from the past 4 weeks is obtained. The DISC is scored using a computer program (SAS) and algorithms that match *DSM-IV* (or *ICD-10*) criteria. In addition to the paper version, the DISC has computerized (C-DISC-4.0), Spanish, Present State, Teacher, Voice, Young Adult, and Quick versions (cf. Shaffer et al., 2000) and is considered the most highly researched and utilized of the parent/child companion interviews (A. Jensen & Weisz, 2002).

The DISC-IV diagnostic scales have evidenced generally acceptable test-retest reliability for parent ($\kappa = .43$ to .96) and combined parent/child ($\kappa = .48$ to .86) versions, with slightly less reliable findings for the child version ($\kappa = .25$ to .92, Fisher et al., 1997). Similar test-retest reliability rates were found for prior DISC versions in community and clinical samples (P. Jensen et al., 1995; Schwab-Stone et al., 1996). In addition, excellent interrater reliability has been reported for earlier versions of the DISC (Shaffer, Schwab-Stone, Fisher, & Cohen, 1993). Recent examinations of parent-child agreement have found poor reliability with the DISC-IV ($\kappa = -.08$ to .25, Crowley, Mikulich, Ehlers, Whitmore, & MacDonald, 2001; P. Jensen et al., 1999).

To summarize, several well-researched semi- or highly structured interviews exist for researchers and clinicians. As shown in Table 15.1, the majority of these interviews cover basic *DSM-IV* diagnoses for childhood disorders, with some focused more on specific areas than others (e.g., the ADIS and anxiety disorders versus the K-SADS and depressive disorders). In addition to procedural differences (e.g., level of structure, computer administration capability, same versus different interviewer

for parent and child components) among these interviews, several have also been adapted for use in multiple languages. For example, in addition to the standard English versions, there are currently Chinese (Ho et al., 2005), French (Breton, Bergeron, Valla, Berthiaume, & St. Georges, 1998), and Zhosa versions of the DISC, as well as a Greek version of the K-SADS (Kolaitis, Korpa, Kolvin, & Tsiantis, 2003). Therefore, although specific cases where the primary focus of study (e.g., epidemiological studies) or presenting problem (e.g., a disorder not assessed by all interviews, such as Bipolar Disorder) guides the choice of diagnostic interview, given the similarities among interviews, most often the clinician's or researcher's discretion or preference guides this decision. In fact, Hodges (1994) suggested that there is not one best interview, but that researchers and clinicians should determine which interview to use based on the sample and focus of their endeavor. Finally, it is important to note that the previously described interviews do not represent an exhaustive list of those available. Additional commonly utilized interviews are the Child Assessment Schedule (Hodges, Kline, Stern, Cytryn, & McKnew, 1982) and the Child and Adolescent Psychiatric Assessment (Angold, Prendergast, Cox, & Harrington, 1995), among others.

LINGERING ISSUES INVOLVED WITH DIAGNOSTIC INTERVIEWS

One of the most consistently raised difficulties involved in using parent-child diagnostic interviews concerns the discordant reports received. That is, symptom and diagnostic agreement within and across informants is generally poor—a finding that has led several researchers to examine potential factors that may influence multiple informants' reports and thus contribute understanding to the discrepancies found (cf. Grills & Ollendick, 2003). In general, these variables can be described in terms of whether they pertain to the interviewee, the interviewer, or the interview itself.

Beginning at the *interviewer* level, establishment of rapport is an important aspect of any clinical relationship. Although diagnostic interviews attempt to incorporate rapport-building strategies (e.g., beginning with general information), the lengthy interview format likely impedes rapport, in the traditional sense, from being fully developed (LaGreca & Stone, 1992; Verhulst et al., 1987). Individuals who have not established rapport with the interviewer may not feel comfortable disclosing diagnostic information, requesting clarification, or stopping the interview (Breton et al., 1995), thus potentially affecting the reliability and validity of the reports obtained. Unfortunately, this is likely often the case, as clinicians and researchers tend to use diagnostic interviews early in the assessment process to probe for areas of concern. In addition, despite requiring specific training in their administration, individual and across-site differences and errors likely remain. Furthermore, differences in clinician experience levels may affect interviews, as one with more experience may complete the interview in a more timely, fluid, and comprehensive manner. This latter issue would be particularly relevant for semi-structured diagnostic interviews (e.g., ADIS), as there is more clinical probing and leeway permitted with these instruments. An additional source of confounding can occur when the same interviewer administers both interviews. For instance, information

from the first interview may alter the interviewer's opinions and subsequently the style (nonverbal cues) and emphasis given to probing certain domains as well as the recording of responses in the second interview. With multiple informants, joint meetings may be held to resolve discrepant reports; however, the clinician cannot be sure that the final consensus accurately represents both informants' opinions or the concession of one of them (Angold et al., 1987; Sylvester et al., 1987).

Interview methodological problems are another area of concern for researchers. For example, it has been suggested that the sequence of disorder presentation in the interviews may affect informants' reports. Sylvester et al. (1987) noted that questions regarding externalizing disorders are asked first with the DICA and that this may influence parents' reports by focusing them on these outward behaviors. Thus, later questions about internalizing symptoms may be overshadowed by this earlier focus (Sylvester et al., 1987). This notion was recently supported by findings of order effects in the total number of symptoms endorsed, impairment ratings, and diagnostic criteria met for both parent and child reports on the DISC (P. Jensen et al., 1999). In contrast, the ADIS requests information on all anxiety-related symptoms early in the interview, with externalizing disorders probed later. Comparison of these measures and order effects may prove another area important for future studies. Similarly, parent-child concordance may be affected by differences in the informants' emotional state at the time of the interview. To illustrate, parents may overemphasize behavioral problems if the child was behaving especially poorly earlier that day or in the waiting room. They may also overemphasize behavioral problems if they themselves are having a difficult day or are under stress of one type or another. For children, poor reliability may be a reflection of their desire to quickly complete an interview (e.g., by saying no to all the probes) or impatience or boredom with the interview process (Edelbrock et al., 1985; P. Jensen et al., 1995; Reich, 2000). Potential differences concerning the degree of structure (semi- versus highly) imposed by interviews has also been reported. Whereas semi-structured interviews may lead to improved reliability over highly structured interviews (i.e., DISC) by allowing for more clinical probing (Verhulst et al., 107), highly structured interviews may limit confounding characteristics (e.g., interviewer experience, differences in probing across administrations). Thus, it is not surprising that studies conducted for the various interviews have revealed similar findings regardless of the degree of structure associated with the interview (Grills & Ollendick, 2003). It is important to note that these clinical interviews must be considered in the context of the diagnostic system on which they are based. If diagnostic criteria are not presented in a manner that allows for their adequate assessment, this will be reflected in the interviews as well. Furthermore, if the boundaries between normal and pathological child disorders are unclear, assessment strategies will also likely produce blurred and discrepant results. Thus, reliability and validity problems on multiple interviews may speak to the need for further alterations or amendments to the diagnostic system itself.

The most attention has clearly been devoted to *interviewee* (parent and child) characteristics proposed to affect agreement, including those particular to children and/or parents. For example, it is often presumed that younger children will evidence less agreement with their parents due to factors (e.g., verbal,

cognitive, memory, and attentional) related to their ability to understand and describe their feelings and behaviors (Breton et al., 1995; Edelbrock et al., 1985; Rapee et al., 1994; Schwab-Stone et al., 1994). Several researchers have indicated that structured interviews may require information that is too complex or beyond the cognitive capabilities of young children (Breton et al., 1995; Brunshaw & Szatmari, 1988; Edelbrock et al., 1985; Ezpeleta et al., 1997; Herjanic & Reich, 1982; Piacentini et al., 1993; Schwab-Stone et al., 1994; Valla, Bergeron, Berube, Gaudet, & St Georges, 1994; Welner et al., 1987; Young, O'Brien, Gutterman, & Cohen, 1987). Indeed, the majority of research studies have found improved ratings within (i.e., test-retest) and across (e.g., parent/child, clinician/child) informants for older children regardless of the interview used (Edelbrock, Costello, Dulcan, Conover, & Kalas, 1986; Edelbrock et al., 1985; Ezpeleta et al., 1997; Grills & Ollendick, 2002; P. Jensen et al., 1999; Rapee et al., 1994; Reich et al., 1982; Silverman & Eisen, 1992; Verhulst et al., 107). It appears that both the wording (e.g., length, vague concepts such as "most" and "fair," number of concepts included in the question) and types (e.g., yes/no versus open-ended, duration, age of onset, clustering, frequency, concrete information versus comparative judgments) of questions included in these interviews can be problematic. Concerns like these have been supported by studies revealing low overall understanding of concepts (16% to 56%) and time dimensions (24% to 30%) of DISC questions when children (ages 9 to 11) were asked to explain their answers (Breton et al., 1995). Likewise, Perez, Ezpeleta, Domenech, and de la Osa (1998) reported that the most reliable questions on the DICA-R were those containing no time concepts, clustering of symptoms, or impairment ratings and no peer comparisons or subjective judgments, externalizing (versus internalizing) content, and shorter duration (for younger children, but not adolescents). Although not all researchers have found support for age differences on the reliability of interviews (Angold et al., 1987; Boyle et al., 1993; Briggs-Gowan, Carter, & Schwab-Stone, 1996; Earls, Smith, Reich, & Jung, 1988; Perez et al., 1998; Thompson, Merritt, Keith, Murphy, & Johndrow, 1993), a higher minimal age for structured interviewing has been proposed by some (e.g., age 10+ for the DISC; Edelbrock et al., 1985; Schwab-Stone et al., 1994). Importantly, studies in this area also demonstrate aspects of potential revision for child structured interviews. For example, it may be helpful for interviewers to use multiple examples, visual aids, and explanations to ensure children's comprehension.

Interestingly, the child's age may also influence parent reports, as older children typically spend more time outside the home (e.g., with friends, in extracurricular activities), potentially decreasing parents' knowledge of their child's behaviors (Edelbrock et al., 1985). Although this notion has not been well-studied, it certainly remains an important consideration and further supports the need for both adolescent and parent reports.

Motives may also influence reliability. For example, symptoms may be overreported to gain services for a child that the parent feels are necessary to obtain. In addition, parent-child concordance may be disrupted if children intentionally hide information from their parents (e.g., drug use, truancy; Edelbrock et al., 1985), perhaps to avoid negative consequences (e.g., punishment, forced treatment)

resulting from disclosure. Alternatively, children may attempt to conceal emotional problems while in the presence of others, such as their peers or the clinician, to portray a positive image of themselves (DiBartolo, Albano, Barlow, & Heimberg, 1998; Grills & Ollendick, 2003; P. Jensen, Traylor, Xenakis, & Davis, 1988; Silverman & Rabian, 1995; Stavrakaki, Vargo, Roberts, & Boodoosingh, 1987). Similarly, parents may underreport symptoms in an attempt to conceal the child's behavioral problems (Crowley et al., 2001; Victor, Halverson, & Wampler, 1988), perhaps to prevent stigma or conceal family problems (e.g., marital conflicts). These social desirability concerns could then lead to the appearance of overreporting on the part of another informant. Social desirability may also influence concordance within or across informants if disagreeing with others or one's own earlier report is perceived as being viewed unfavorably.

Another commonly discussed area concerns the type of disorder on which the informants are reporting. That is, externalizing (e.g., hyperactivity, aggression), as compared to internalizing (e.g., anxiety, depression), symptoms are generally considered easier to observe and to report on by others, resulting in increased diagnostic agreement. Likewise, the severity of the disorder may influence reliability, perhaps through relation with more open dialogue about problems, more pronounced or observable symptoms, or less reliable answers (Cohen, O'Connor, Lewis, Velez, & Malachowski, 1987; Perez et al., 1998). Whereas findings with the various diagnostic interviews support the former proposal, such that increased reliability is often found for externalizing behaviors (Bidault et al., 1995; Chambers et al., 1985; Costello & Edelbrock, 1985; Edelbrock et al., 1986; Ezpeleta et al., 1997; Herjanic & Reich, 1982; Hodges, Gordon, & Lennon, 1990; P. Jensen et al., 1988; Kolko & Kazdin, 1993; Orvaschel, Weissman, Padian, & Lowe, 1981; Silverman & Eisen, 1992), the influence of disorder severity does not appear to be supported by multiple-informant reliability studies (e.g., Angold et al., 1987; Grills & Ollendick, 2003). Likewise, there remains debate as to whether parental psychopathology contributes to decreased interview reliability, perhaps due to greater sensitivity to their children's symptoms, misinterpreting of their child's behaviors, or projection of their own symptoms onto the child (Angold et al., 1987; Breslau, Davis, & Prabucki, 1988; Briggs-Gowan et al., 1996; Fergusson, Lynskey, & Horwood, 1993; P. Jensen et al., 1988; Kashani, Orvaschel, Burk, & Reid, 1985; Kolko & Kazdin, 1993; Miles et al., 1998; Moretti et al., 1985; Rapee et al., 1994). Poor family communication (Bidault-Russell et al., 1995) and family conflict or stress may also influence reliability. For example, children and parents from families characterized by high conflict have been found to evidence poorer agreement on the ADIS (Grills & Ollendick, 2003).

Although gender is also commonly examined, Klein (1991) has argued that there are no theoretical grounds on which to expect gender differences to occur. Indeed, findings for gender differences have been exceedingly mixed, and review of studies reporting them reveals generally few and small differences (cf. Grills & Ollendick, 2003). Furthermore, there remain myriad potential factors that have not yet been well examined for their influence on the reliability and validity of reports on clinical interviews (e.g., ethnicity, socioeconomic status). Although determining the role of these variables awaits future research, it seems likely that similar findings to

those already described will result. That is, despite some trends regarding particular interviewee characteristics, findings have not been particularly robust for any one area.

In sum, various interviewer, interviewee, and interview characteristics may influence the reliability and validity of clinical interviews. Future research should attempt to clarify the role of these characteristics to limit additional variance that may be introduced through them. The most robust findings appear to be related to children's comprehension of the interview itself.

Several procedures have been proposed to manage discordant findings with interviews: (a) basing diagnostic decisions on just one of the reports (Silverman et al., 2001) or only when both reports agree, (b) following specific guidelines for the consideration of each report (e.g., rules of evidence, P. Jensen et al., 1999; Shaffer et al., 2000), and (c) applying rules for how reports should be combined (e.g., ADIS Composite Diagnoses; Silverman & Albano, 1996; "or" rules; Bird, Gould, & Staghezza, 1992). Applying rules for combining reports appears most advantageous as this allows discrepancies to be interpreted as informative, not problematic (Boyle et al., 1996; Schwab-Stone et al., 1996). That is, parent and child reports should not be viewed as interchangeable, but complementary.

In light of these findings, the general consensus in the youth assessment area (Silverman & Ollendick, 2005, in press) is that parents and children should both be involved in the assessment of youth symptoms and diagnoses. One is not necessarily more right or wrong than the other. By carefully considering each source's information, there is increased likelihood that no child or adolescent is denied services (Comer & Kendall, 2004; Foley et al., 2005). De Los Reyes and Kazdin (2005), in a comprehensive review of the informant discrepancy literature in clinical child and adolescent psychology, further proposed that in addition to obtaining information of youths' problems from multiple informants, efforts should be made to collect information from these informants about their perceptions of why the youth is exhibiting these problems, as well as their perceptions of the desired treatment of the youth for these problems. De Los Reyes and Kazdin further discussed how gathering this type of information can serve as the basis for an assessment model that can guide research and clinical practice to help explain informant discrepancies in clinical child and adolescent research.

SUMMARY

A host of issues surrounding diagnostic interviews exist, including their reliability, validity, and clinical utility. Although many questions remain about the psychometric properties of these interviews, much progress has been made and a number of reliable, valid, and clinically useful interviews have been developed and are in wide use today. These interviews vary along a number of lines, including the amount and extent of structure embedded in them (unstructured, semi-structured, and highly structured). The semi-structured and the highly structured interviews enjoy the best psychometric properties. It is also evident that a number of complications can arise when using diagnostic interviews during the assessment process. For example, it is a common practice in the assessment of children to include multiple informants

since it is believed that adults alone cannot provide all of the information that is needed concerning the child's behavior. However, inclusion of multiple informants also creates a dilemma, as disagreements commonly occur. A number of potential variables related to the interviewer, the interviewee, and the interview itself have been considered in terms of their influence on the reliability, validity, and clinical utility of these instruments. However, few, if any, clear or consistent patterns have emerged. It seems likely that there is a complex interaction among the various factors that cannot be ignored and that will require additional research. Nonetheless, diagnostic interviews represent one of the most reliable, valid, and clinically useful instruments for diagnosis, treatment planning, and treatment evaluation. They are an indispensable part of our clinical armamentarium.

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CHAPTER 16

Intellectual Assessment

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The practice of estimating intellectual abilities dates back to the beginnings of scientific thought. Greek philosophers and modern-day psychologists alike have attempted to understand the nature of human intelligence and to devise methods by which to measure intellectual capabilities. These intellectual assessments often play important roles in diagnoses of learning disabilities, academic placement, occupational selection, and medical treatment plans. Additionally, intellectual assessments routinely have a significant role in the conceptualization of neurocognitive sequelae following traumatic injuries. Thus, the methods by which we estimate intellectual functioning, and the scientific integrity of these assessment tools, are critical to our practice as psychologists.

The goals of this chapter are to provide an overview of the history of intellectual assessment and common definitions of intelligence. We additionally review traditional intellectual assessment tools, as well as more recent additions to the intellectual assessment arsenal. We discuss the impact of intellectual assessment in the context of psychoeducational assessments and review legislation related to intellectual assessment in the context of educational settings. Finally, we provide two clinical case examples to reiterate the role of cognitive assessment in clinical practice.

HISTORY OF INTELLECTUAL ASSESSMENT

We begin our discussion of intellectual assessment with a review of historical perspectives. This vantage point is critical because little can be understood devoid of knowledge of its history.

Speculations into the nature of human intelligence can be traced back to the Greek philosopher Plato (ca. 428/427–348/347 B.C.E.) who referred to *reason* as the highest functioning part of the soul. Since that time, philosophers and scientists have engaged in dialogue pertaining to the nature of cognitive processing and intelligent behavior. Beginning in the sixteenth century, attempts were made to differentiate mental deficiencies from mental illness. Prominent philosophers and psychologists involved in this work include Jean Esquirol, Sir Francis Galton, Alfred Binet, and James Cattell.

Jean Etienne Esquirol

Jean Esquirol (1772–1840) was among the first scientists to differentiate mental deficiency, or "idiocy," from mental illness (Sattler, 2001). He regarded mental deficiency as a condition in which the intellectual facilities are not fully developed whereas mental illness involved the loss of functioning once possessed by the afflicted individual (Zusne, 1957). To differentiate these groups, Esquirol developed the first rudimentary test of cognitive functioning, in which he used a language criterion to determine mental retardation.

Sir Francis Galton

Sir Francis Galton (1822–1911) is commonly referred to as the founder of the testing movement (Kail & Pellegrino, 1985; Sattler, 2001). A diversified scholar, Galton's published work significantly contributed to the fields of geography, biology, psychology, and criminology. However, he is most highly recognized for his contributions to the field of behavioral genetics (Clayes, 2001). Greatly influenced by his cousin, Charles Darwin, Galton focused much of his work on the heritability of mental functioning. In his published work, Galton theorized that superior mental functioning was passed genetically from parent to offspring and used the bell-shape curve to describe differences in intellectual functioning (Sattler, 2001; Seligman, 2002). Additionally, Galton developed the statistical concepts of *regression to the mean* and *correlation,* which allowed for the empirical study of heritability of traits over time and among family members. His significant contributions to the fields of behavioral genetics and psychology paved the way for future statisticians and psychometricians, and his statistical concepts are still used as the gold standard in test construction and validation (Bynum, 2002).

Alfred Binet

While Esquirol developed the first crude measure of cognitive functioning, Alfred Binet (1857–1911) is credited with developing the first practical test of intelligence (Kail & Pellegrino, 1985; Sattler, 2001). In 1904, Binet was commissioned by the French government to develop an instrument that identified mentally retarded children in need of alternative education. In collaboration with his student, Theodore Simon, Binet created an intelligence test that consisted of 30 subtests measuring a broad range of cognitive abilities, including language skills, abstract reasoning, and memory functioning (Siegler, 1992). This test, commonly referred to as the 1905 Scale, compared the testing subject's performance to a chronological cohort deemed by their teachers to be of average intellectual functioning. The scale was revised in 1908 and adapted to English by Lewis Terman of Stanford University in 1916. Terman's adaptation, referred to as the Stanford-Binet, included culturally appropriate items and reported the child's level of performance in terms of an intelligence quotient (IQ) consisting of the child's mental age divided by chronological age multiplied by 100.

Although the Binet-Simon scale was created as a diagnostic instrument, Binet was equally interested in developing a theory of intelligence (Kail & Pellegrino,

1985). In fact, his definition of intelligence as consisting of complex cognitive processes including abstract reasoning, memory and comprehension greatly influenced the design of the Binet-Simon testing instrument. However, the practical demands of creating and maintaining the Binet-Simon scales overshadowed his pursuits in intelligence theory, thus marking a division in psychometrics between those scientists most interested in theory and those most interested in diagnostic measurement. The theory versus measurement discourse continues among psychologists.

James McKeen Cattell

James McKeen Cattell (1860–1944) was the first American psychologist to emphasize the study of individual differences and cognitive functioning (Kail & Pellegrino, 1985; Sattler, 2001). Cattell is often remembered for his work on theories of personality and bringing an empirical and scientific perspective to theories of personality. Rather than following in the footsteps of his predecessors and focusing on the specific *outcome* of his research, Cattell was more concerned with the *process* of research and shifting psychology to an empirical science based on observable behavior and subjected to statistical analysis. He believed that sensory, perceptual, and motor processes were the fundamental elements of cognitive functioning and, consequently, sought to measure these phenomena in the tests he designed. In his work, he attempted to use *mental tests,* a term he coined, to predict college success. Although the measures he used did not correlate well with educational achievement, Cattell was among the first scientists to empirically study cognitive functioning.

DEFINITIONS OF INTELLIGENCE

We now turn our attention to commonly employed definitions of intelligence. There are basically two main approaches utilized to understand and define intelligence. *Factor analytic theories* involve understanding intelligence by examining patterns of individual differences on assessments and then attempting to describe these patterns based on statistical principles. Theorists embracing this paradigm include Charles Spearman, Philip Vernon, Lewis Thurston, and Raymond Cattell.

Alternatively, *information-processing theories* describe the *components* of intelligent thought that translates external stimuli into an internal concept leading to an appropriate behavioral response. Theorists in this camp include psychologists such as Joseph Campione, Ann Brown and John Borkowski, and Robert Sternberg.

Factor Analytic Theories

Throughout the twentieth century, the factor analytic approach has dominated intelligence theory, research, and practice. Scientists adopting this approach attempt to understand intelligence by examining patterns of individual differences on tests measuring various cognitive skills such as vocabulary, general knowledge, visual-spatial processing, abstract reasoning, and arithmetic. These theories, also known as psychometric theories, statistically analyze test performance to identify the basic units, or *factors,* of intelligence. Although providing insight into the structure of intelligence, there is much debate regarding whether intelligence represents a

global, general ability or multiple abilities. The following section presents theories supporting a general factor of intelligence as well as those supporting a multifactor approach.

Spearman's Two-Factor Theory

Charles E. Spearman (1863–1945) was a prominent British psychologist best known for his statistical contributions to the study of intelligence (Kail & Pellegrino, 1985; Sattler, 2001). In his landmark 1904 publication, Spearman (1904a) introduced the statistical concept of factor analysis in an effort to measure factors associated with general intelligence. More specifically, he examined patterns of correlations among IQ subtest scores to look for consistency of performance with the reasoning that if intelligence is global, there would be a high degree of association among subtests. Once correcting for subtest attenuation (for more information, see Williams, Zimmerman, Zumbo, & Ross, 2003), Spearman found evidence in support of a two-factor theory of intelligence in which a general factor, or *g,* superseded more specific factors, or *s*. According to this theory, the *g-factor* accounts for complex cognitive processes such as deductive and analytic reasoning. It is available to the same individual across different intellectual acts and explains consistency in performance across these acts. In contrast, the *s-factors* account for more rudimentary mental processes, including processing speed, recognition, and recall. They are unique to the intelligent act and individuals can exhibit strengths and weaknesses across factors.

Vernon's Hierarchical Group Factor Theory

Heavily influenced by Spearman's work, Philip E. Vernon (1905–1987) adopted a factor analytic approach to intelligence. At the top of his hierarchical model was Spearman's *g,* which accounted for the largest source of variance in intelligence (Kail & Pellegrino, 1985; Sattler, 2001). Underlying *g* were two major group factors: verbal-educational ability (v:ed) and practical-spatial-mechanical abilities (k:m). These major group factors were subdivided into minor group factors, which were further decomposed into several more basic factors. For example, *k:m* subsumed more specific abilities such as spatial, mechanical, and physical abilities. Spatial ability, which was a minor group factor, could then be broken down into three specific factors: spatial orientation (i.e., ability to orient oneself in space relative to environmental markers), spatial relations (i.e., mental rotation of objects), and spatial visualization (i.e., complex mental manipulation of objects). Similar to Spearman's theory, Vernon believed that factors at the top of the hierarchy were more general abilities that affected a wider range of intelligent behaviors, and those at the bottom of the hierarchy involved skills unique to a specific act.

Thurstone's Theory of Primary Mental Abilities

Lewis L. Thurstone (1887–1955) was an established statistician, psychometrician, and intelligence theorist. Among his many contributions to the field of psychology was the development of a new method of factor analysis that allowed for the

exploration of multiple latent factors within a set of variables (Sattler, 2001). The use of this new analytic method challenged Spearman's theory of a unified intelligence in support of seven primary mental abilities: word fluency, verbal comprehension, spatial visualization, number facility, associative memory, reasoning, and perceptual speed (Thurstone, 1938). Moreover, Thurstone found that people with similar IQ scores often displayed vastly different profiles on the primary mental abilities, which he felt further substantiated his claim of multiple intelligence factors. Later studies conducted on a more heterogeneous group of people, however, found fewer discrepancies in primary mental abilities and, instead, supported the notion of a unitary intelligence. Thurstone conceded a statistical compromise that accounted for the presence of both a general factor and the seven primary mental abilities.

The Cattell-Horn Theory of Fluid and Crystallized Intelligence

Among many accomplishments, Raymond Bernard Cattell (1905–1998) is widely recognized for his contribution to intelligence theory. Along with his student, John Horn, Cattell proposed a model of intelligence describing general intelligence as a conglomeration of different abilities falling into two categories: fluid abilities (Gf) and crystallized abilities (Gc) (Horn & Cattell, 1966). Fluid abilities include novel problem-solving skills and the ability to think and act quickly. They are innate abilities that are free from education and acculturation. Examples of fluid abilities include spatial reasoning and symbolic classification. Fluid abilities have been shown to peak during adolescence and then gradually decline as the individual progresses into adulthood. They are more susceptible to brain injury and physiological degeneration than crystallized abilities. In contrast, crystallized abilities include acquired knowledge and skills that are the direct result of education, personal motivation, and culture. They develop throughout adulthood and are less susceptible to physiological degeneration. Examples of crystallized intelligence include general information and language skills.

Information Processing

Whereas the psychometric approach to intelligence focused on factor *structure,* the information-processing approach to intelligence is concerned with the *processes* involved in intelligent acts. Information processing theories elaborate on the structure of intelligence by describing the internal workings of intelligent thought and how it results in behavioral output. Rather than studying *factors,* information-processing theorists describe the *components* of intelligent thought that translates external stimuli into an internal concept and designates appropriate behavioral responses.

Campione, Brown, and Borkowski's Theory

In 1978, Joseph Campione and Ann Brown proposed an information-processing theory of intelligence that was later expanded on by John Borkowski (1985). The theory describes intelligence as consisting of both an architectural system, which describes the structure of cognition, and an executive system, which describes the processes involved in intellectual thought. Of particular importance to the model

were the executive skills of metacognition, which are skills used to monitor and regulate cognitive processes. The metacognitive skills utilized in intelligent thinking include (a) *planning* steps in problem solving, (b) *monitoring* effectiveness of the steps, (c) *revising* steps as needed, and (d) *evaluating* the efficacy of the problem-solving steps. An example of a metacognition is the selection of a rehearsal strategy to learn a word list or telephone number.

Sternberg's Triarchic Theory

Robert J. Sternberg (1985) proposed a triarchic theory of intelligence in which he divided intelligence into three different information-processing components. *Metacomponents* are higher order components that, much like Campione's metacognitive skills, are used to monitor and regulate cognitive processes. They involve the ability to recognize a problem as it arises, select a lower order strategy to correct the problem, monitor the efficacy of the strategy, and evaluate the outcome of the solution. *Performance components* are lower order components used to execute a solution to an identified problem, such as inferring the relationship between two objects. *Knowledge-acquisition components* are involved in learning new information and storing and retaining the information in long-term memory. Three knowledge-acquisition components involved in cognitive processing are (1) selective encoding, which involves distinguishing relevant from irrelevant information; (2) selective combinations, which is the process by which previously encoded information is combined in a new way that maximizes interconnectedness; and (3) selective comparison, in which the previously encoded and connected information is related to information stored in memory.

COMMONLY EMPLOYED MEASURES OF INTELLIGENCE

The impact of theory on the development of intellectual assessment tools cannot be overemphasized. Theories of intelligence have played an important role in the development of many of the traditional and more recent tools designed to measure intellectual abilities. Although many theorists focused mainly on the development and study of their unique theoretical assumptions regarding intelligence, their work has been instrumental to assessment tools that have been generated to measure intellectual capabilities. This section reviews traditional and more recent intellectual assessment tools, highlighting the theoretical basis of each measure where applicable.

Traditional Measures of Intelligence

Traditional measures of intellectual assessment include the Stanford-Binet Intelligence Scale, Wechsler Scales of Intelligence, and the Woodcock-Johnson Test of Cognitive Functioning.

Stanford-Binet

The Stanford-Binet Scale was developed to provide a global estimate of intellectual functioning in children. The most recent version, the Stanford-Binet Intelligence

Scales, 5th edition, was released in 1986 and incorporates into its arsenal of subtests a measure of working memory, which has been shown to relate to reading and math achievement (Johnson & D'Amato, 2004). The Stanford-Binet typically takes 45 to 75 minutes to administer and covers a wide age range (2 to 85 years).

The Full-Scale IQ score is based on the administration of 10 subtests and provides a global measure of cognitive ability (Johnson & D'Amato, 2004). The test battery includes two index scores: Verbal and Nonverbal IQ. The Verbal index requires the examinee to speak, read, and comprehend age-appropriate language. The Nonverbal index requires minimal language and, instead, emphasizes the use of fine motor coordination to maneuver toys and puzzle pieces and point to correct answers. The subtests on the Stanford-Binet measure concepts of fluid reasoning, knowledge, quantitative reasoning, visual-spatial processing, and working memory.

The Stanford-Binet has numerous strengths. For example, it has an expansive age range (2 to 85) and can be used in assessing a variety of disorders where verbal ability is limited (e.g., autism, hearing impairments, limited English-speaking skills). Internal consistency reliability is considered to be good, ranging from.95 to.98 for IQ scores; criterion validity using the individuals tested on the SB4 and SB5 was high ($r = .90$; Kush, 2004).

In spite of these strengths, the Stanford-Binet has been criticized for being racially and culturally biased, although efforts have been made to standardize recent editions on more heterogeneous populations. The Stanford-Binet was further criticized for its emphasis on speed, causing older adult test takers to be penalized for protracted test completion. As such, Binet's mental age norms were thought to be inadequate when applied to adult populations. Furthermore, historically, the Binet scales did not consider the deterioration of intellectual ability that coincides with aging.

Wechsler Intelligence Scales

David Wechsler is best known for his development of a widely used series of intelligence tests, including the Wechsler Preschool and Primary Scale of Intelligence, the Wechsler Intelligence Scales for Children, the Wechsler Adult Intelligence Scales, and the Wechsler Abbreviated Scale of Intelligence. Although a student of Spearman, Wechsler believed that the concept of general *g* was too narrow to adequately describe intellectual functioning. Rather, it was best described in terms of both verbal and nonverbal abilities. As a result, the Wechsler scales consist of several subsets designed to assess abilities within the domains of verbal and nonverbal intelligence, although an estimate of global cognitive functioning is additionally provided. The Wechsler scales have undergone numerous revisions and updates (approximately every 10 years) and are among the most widely used assessment instruments for intellectual functioning across the life span.

Strengths of the Wechsler scales include an adequate standardization sample and good satisfactory reliability and validity. The ability to estimate scores on other Wechsler scales is another advantage for both screening and reevaluation purposes (Keith, 2004).

In spite of efforts to improve on the original Binet scales, the first Wechsler scales were not without flaws. One criticism was the inaccuracy in population representation. The first Wechsler scale's normative sample was insufficient, comprising 1,081 Whites from the eastern United States (Kaplan & Saccuzzo, 2005). Not only was this sample underrepresentative of the country's various regions, but it was racially and ethnically restricted as well.

Woodcock Johnson Scales

The Woodcock Johnson scales are based on the Cattell-Horn-Carroll (CHC) theory of cognitive abilities, a derivative of Spearman's *g* (Kaufman & Lichtenberger, 2006). As previously mentioned, Spearman's theory states that a single *g* sufficiently describes cognitive abilities. This is in contrast to the Wechsler scales, which describe intelligence in terms of abilities across multiple domains, including verbal and performance capabilities.

The Woodcock Johnson scales consist of two separate batteries measuring cognitive ability and achievement. The WJ III cognitive battery consists of 10 subtests (e.g., Verbal Comprehension, Auditory Working Memory) and the extended form consisting of 10 additional subtests (e.g., Picture Recognition, Decision Speed). The cognitive battery provides two indices of general intelligence: a GIA score (general intellectual ability) and a BIA score (brief intellectual ability). The GIA is considered to be a highly reliable score, with the highest median reliability coefficient across ages of.97 on the standard battery (Sandoval, 2004). The BIA score is obtained by administering three subtests only and has a high reliability coefficient of.95 (Cizek, 2004).

The Woodcock Johnson scales have numerous strengths. Construct validity using factor analysis shows the 7-factor CHC cognitive model is more closely linked than any of the other six models frequently used in deriving cognitive batteries; however, more data on validity would be useful (Sandoval, 2004). The split-half method was used in computing reliability for a coefficient of.80 for 38 tests and greater than.90 for 11 tests (Sandoval, 2004). The WJ-III software, manuals, and test materials are impressively modern, but hand scoring is cumbersome. Another advantage of the WJ-III is that the user can administer only selected subtests that can be individually tailored to meet specific testing requirements of individual subjects without sacrificing validity. Extended subtests are available to acquire additional information (Sandoval, 2004).

Recent Measures of Intellectual Functioning

Recently, a number of psychologists have developed assessment tools in attempts to address shortcomings of more traditional measures. Specifically, these assessments have sought to demonstrate applicability across diverse populations, incorporate developmental and educational perspectives into the assessments, and provide a method to assess intellectual functioning among nonverbal and non-English-speaking clients. These assessments include the Kaufman Adolescent and Adult Intelligence Test, the Differential Abilities Scales, and the Test of Nonverbal Intelligence.

Kaufman Adolescent and Adult Intelligence Test

The Kaufman Adolescent and Adult Intelligence Test (KAIT), one of the newest measures of cognitive functioning, was derived from Horn and Cattell's theory of fluid (Gf) and crystallized (Gc) intelligence, Luria's definition of planning ability, and the formal operational stage of development in Piaget's theory (Kaufman & Lichtenberger, 2006). It is a standardized measure of old (crystallized) and new (fluid) learning which can be consolidated into the construct of general intelligence. The core battery consists of Definitions, Auditory Comprehension, Double Meanings, Rebus Learning, Logical Steps, and Mystery Codes (Kaufman & Lichtenberger, 2006).

The KAIT contains a Gf scale and a Gc scale. The Gf scale measures one's adaptability and flexibility when solving both verbal and nonverbal new problems. The Gc scale measures acquired facts, problem-solving ability, and concepts using stimuli dependent on schooling, acculturation, and verbal conceptual development for success, and a composite IQ score.

The strengths of the KAIT include its ease in administration, scoring, and manipulation (Flanagan, 2004). The KAIT yields high test-retest reliability coefficients, ranging from.87 to.97 (Flanagan, 2004). This test also exhibits high construct validity for Caucasian, African American, and Hispanic populations (Flanagan, 2004). It seems to be an adequate assessment of fluid ability and is stimulating to examinees (Flanagan, 2004).

A weakness of the KAIT is that the guiding theory of fluid and crystallized intelligence was included after the test had been standardized and, therefore, raises the question regarding the theoretical basis for the development of the test (Keith, 2004).

Differential Abilities Scale

The Differential Abilities Scale (DAS) was originally developed from the British Ability Scales (Aylward, 2004). The DAS consists of 17 cognitive and three achievement subtests and assesses individuals from 2 years, 6 months to 17 years, 11 months. The cognitive battery generates a General Conceptual Ability score, and the three achievement subtests yield measures of math, spelling, and reading. Internal consistency reliability coefficients of the DAS (based on item-response theory) ranged from.70 to.92 on individual subtests (Aylward, 2004). Correlations with the DAS and other cognitive measures are high (Reinehr, 2004).

The strengths of the DAS include the test's incorporation of a developmental and educational perspective and practicality in evaluating children suspected of having developmental delays, hearing or language difficulties, mild mental retardation, and learning disabilities (Aylward, 2004). The DAS includes special education categories in its normative sample. The chief limitation is the complexity in administration, scoring, and interpretation (Reinehr, 2004). The record form is a lengthy 22 pages and the summary page requires a 12-step process to complete (Reinehr, 2004). Additionally, the manual suggests that administration can be completed in less than $1\frac{1}{2}$ hours, yet this is difficult at best (Reinehr, 2004).

Test of Nonverbal Intelligence

The Test of Nonverbal Intelligence (TONI) was developed to evaluate aptitude intelligence, abstract reasoning, and problem solving in a nonverbal format. Nonverbal tests are useful to the mental health and education fields in that they allow for differential diagnoses, treatment, and placement recommendations. The TONI is brief, taking only 15 to 20 minutes to administer, and is a substitute intelligence measure for those who do not speak English or suffer from sensory deficits (Atlas, 2004). The TONI's Intelligence score was created from the Ravens Progressive Matrices Test (Atlas, 2004). It is used primarily in testing non-English-speaking, learning-disabled, emotionally disturbed, gifted, dyslexic, and ADHD populations (Atlas, 2004). Although the TONI is a popular instrument for nonverbal assessments, it has low concurrent validity, ranging from.53 to.63 (DeMauro, 2004).

PSYCHOEDUCATIONAL ISSUES RELATED TO INTELLIGENCE TESTING

There are numerous issues related to the use of intellectual assessments and testing in formal educational settings. Children are routinely placed in classrooms (either gifted classrooms or classrooms designed to provide children school-based interventions for learning difficulties) based on the outcomes of routine psychoeducational evaluations. Currently, approximately 5% of children attending public schools are diagnosed as having learning disabilities (LD), diagnoses that are routinely based on a discrepancy between intellectual abilities and academic achievement. Typically, these evaluations are conducted by clinical psychologists or school psychologists and frequently include assessment of intellectual functioning, along with achievement-based assessment measures. Intellectual assessments routinely serve as the benchmark for competence and capability.

All children suspected of having handicaps or disabilities impacting school-based performance must be identified and classified accordingly. A multidisciplinary team often is involved with this identification and typically includes a clinical and/or school psychologist, special education teacher with expertise in the area of the suspected disability, the regular education teacher, the parents, and sometimes the student. The team is formed to evaluate whether or not a student meets criteria for a disability using behavioral observations, a review of educational records, additional testing, and any other information necessary to determine eligibility for accommodations (McIntosh & Decker, 2005). Educational testing assesses the domains of intelligence and achievement, which have been established as substantially related (McIntosh & Decker, 2005). To evaluate learning disabilities, initial screening measures are administered to determine if the discrepancy between IQ and achievement meet state guidelines for qualification. To meet inclusion criteria for a learning disability, there must be a substantial difference between expected ability (estimated by intelligence tests) and actual academic performance (Gordan, Lewandowski, & Keiser, 1999). Thus, the role of accurate intellectual assessment is paramount in the determination of learning difficulties.

Outcomes from psychoeducational assessments often lead to specific accommodations for school-based interventions. The intent of these school-based interventions is to increase the likelihood that the student will be able to achieve academically at a level commensurate with his or her intellectual abilities. Obviously, the importance of the validity of the assessments utilized and the interventions subsequently employed is paramount. The significance of these assessments has led to the establishment of specific legislation aimed at increasing the likelihood that accommodations are employed and routinely monitored.

This section reviews the major legal issues related to utilization of intellectual assessments as a component of psychoeducational testing. Specifically, we provide an overview of 504 plans and individualized educational planning. In the general context of the discussion of intellectual assessment, this overview is important because intellectual abilities routinely serve as the starting point for assessment of learning disabilities (and, thus, psychoeducational accommodations).

504 Plan

Section 504 of the Rehabilitation Act of 1973 (34 CFR §104.3j; Rehabilitation Act, 1973) and the Americans with Disabilities Act (ADA) of 1990 (ADA; PL 101-336, 1990) were designed to protect the civil rights of students with disabilities and provide protection against discrimination for those with disabilities. Section 504 defines a disability as substantially limiting one or more major life activities, and this limitation is determined by comparing the individual's performance to the average performance of the general population (Smith, 2001). This definition includes the prevention of any form of discrimination against individuals who are otherwise qualified or capable with the exception of their disability limitations (Smith, 2001).

Section 504 applies to all entities that receive federal funding, but there is no federal funding attached to compliance with Section 504. The ADA applies to almost every entity in the United States, with the exception of churches and private clubs (Smith, 2001). If an individual is protected under the IDEA, that individual is also protected under Section 504. However, if an individual qualifies for services under Section 504, that individual may not necessarily qualify for services under IDEA (McIntosh & Decker, 2005). The basic requirements of Section 504 are nondiscrimination, equal access to free appropriate public education, and participation opportunity in the same academic and extracurricular activities as all other students (Smith, 2001). Section 504 plans may be implemented at any level of education (i.e., kindergarten through 12th grade, as well as at the collegiate level).

Under Section 504, "A qualified [college] student with a disability is one who is able to meet a program's admission, academic, and technical standards either with or without accommodation" (Thomas, 2000, p. 250). If a postsecondary institution receives federal aid of any sort, it is mandated to comply with Section 504. This compliance is verified by filing a document stating that the college does not discriminate based on disability, providing such a notice to recipients of the program, identifying an employee to coordinate efforts to comply, conducting an evaluation of programs, and adopting grievance procedures (Thomas, 2000).

During the preadmission process, colleges are prohibited from requesting information regarding disabilities; yet postadmission inquiries can be made confidentially to determine adequate accommodations (Thomas, 2000).

In determining whether a student is qualified under Section 504, any criteria that may adversely impact the admission of a student with a disability, such as first-year grades, must be excluded from admission requirements. Any admission test used to evaluate a student's performance must be designed to accommodate those with disabilities (Thomas, 2000). Accommodations are necessary only for students with appropriate documentation and should not result in the fundamental alteration of academic standards (Thomas, 2000). Once a student with a disability has been admitted and a review of his or her qualifications is conducted, assessment of that student's ability given the requested accommodations is necessary (Thomas, 2000). The caveat with accommodations is that they should not allow a student with a disability an unfair advantage over those students without disabilities (Thomas, 2000).

Individualized Education Plan

The IDEA mandates that all children, regardless of exceptionality, be entitled to free and appropriate public education in the least restrictive environment. Per the least restrictive environment clause, it is initially assumed that children with specific learning disabilities and/or behavior difficulties may be educated in the regular classroom with the help of supplementary aids and services. The IDEA further requires that each state develop and implement a plan for special education services, as outlined in an individualized education Program (Jacob & Hartshorne, 2003).

Typically, following a psychoeducational evaluation that includes an intellectual assessment, a multidisciplinary team is formed to develop and implement the individualized education program (IEP). A growing trend is including the student in his or her IEP meeting (Arndt, Konrad, & Test, 2006).

The IEP includes the student's present level of performance, defined as both academic achievement and functional performance. The annual goals are outlined to monitor progress in the general education curriculum, and the individualized education program must be reviewed, and if necessary revised, annually (Gartin & Murdick, 2005). Quarterly reports or other periodic reports are required to monitor a student's progress toward annual goals. To meet these goals, the IEP contains a description of the services to be offered and the extent to which the individual participates in general education with his or her peers. In compliance with state and district wide assessments, the IEP must include a statement regarding the appropriate accommodations to measure the student's academic achievement and functional performance. Finally, the IEP must specify a target start date for services to be provided.

CULTURAL ISSUES RELATED TO INTELLIGENCE TESTING

The number of children in the United States whose primary language is not English continues to grow; this trend has been apparent for the past two decades (Lam, 1993). This enormous growth of children in the United States whose primary language is not English and who vary in their rates of acculturation has occurred at a

time when public schools are placing increasing emphasis on the use of standardized tests and intellectual assessments to make placement decisions (Lam, 1993). In 1991, Torres reported that approximately 5 million students were inappropriately (because of their unique cultural background) administered standardized assessments in U.S. school systems. Obviously, that number has likely risen since 1991.

There are inherent problems related to the validity and reliability of intellectual assessments and standardized tests administered to individuals whose background differs from that of the normative population. As Lam (1993) discusses, legal cases aimed at challenging the validity of placement decisions based on standardized tests administered to non-English-speaking or diverse populations have led to mixed decisions.

Lam (1993) provides an excellent review of examiner and test-taker concerns related to administration of standardized tests to students whose primary language is not English. He describes five basic assumptions that clinicians should be mindful of when administering standardized assessments (e.g., the aforementioned intellectual assessments) to these children. First, the test-takers need to be able to follow the direction of the tests and have no linguistic barriers that limit their performance. Second, the test selected should have appropriate content and be consistent with the unique training the child has received prior to testing. In other words, the test administered (especially in the case of standardized achievement tests) should accurately reflect the child's exposure and past curricula. Third, the test-takers should have a previous history of taking standardized tests and some familiarity with this process in order for the results to be considered accurate reflections of their true abilities. Fourth, motivation plays an important factor in the validity of the results. A child who has a long-standing history of academic failure and/or difficulties attributable to cultural and language barriers may not be properly motivated to perform on standardized tests. The examiner should be aware of the impact of this lack of motivation on the interpretability of the results. Five, the examiner needs to be aware of any adverse reactions the test-taker may have to being assessed. Cultural barriers, language barriers, and past history of academic failures and struggles may lead to anxiety that impacts overall performance. While this type of anxiety can be found among the normative population, it has the potential to lead to greater difficulty among a non-English-speaking population. It is critically important that examiners take into account the test-taker's cultural background, specifically language barriers and previous educational curricula, when interpreting the results of standardized tests routinely used in the school setting. At a minimum, results should always be interpreted cautiously and judiciously.

CLINICAL APPLICATION

In an attempt to demonstrate the importance of intellectual assessment in the context of academic settings and medical treatment planning, two clinical case studies are reviewed next. The first case involves a 16-year-old, Caucasian American male referred for psychoeducational assessment following multiple academic failures at the college level. The second case involves a 10-year-old, African American male referred for academic difficulties following treatment for a pediatric brain tumor.

Case Illustration

C.R. was referred for a psychoeducational evaluation to assess for a specific learning disorder. According to his parents' report, he experienced difficulty learning foreign languages and failed several foreign-language classes in high school. C.R. reported difficulty remembering and pronouncing words and that spelling [was] always difficult for him. At the time of the referral, C.R. was required to take 2 years of foreign-language classes as a requirement for high school graduation. There was no history of psychiatric disorders or attention problems.

Cognitive testing revealed that C.R.'s overall ability fell within the average range, with little discrepancy between his verbal and nonverbal abilities. This pattern suggested that he performed equally well on verbal and performance-based tasks. On achievement testing, C.R. performed at a level commensurate with his academic history and estimated cognitive ability on tasks related to reading comprehension, pseudo-word decoding, numerical operations, math reasoning, and listening comprehension. However, his achievement scores for spelling and written expression were drastically lower than expected based on his educational background. Thus, he was generally achieving academically at a level consistent with his estimated cognitive functioning except for tasks involving written expression and word reading. This pattern suggested the presence of a learning disorder. More specifically, the discrepancy between his IQ and achievement test scores met criteria for the diagnosis of Disorder of Written Expression. Because academic curricula, including foreign languages and literature classes, rely heavily on written expression, C.R.'s difficulty in this domain was interpreted to negatively impact his academic performance in this content area and special classroom and curriculum accommodations were recommended to provide him with the optimal learning environment.

Case Illustration

C.T. was referred for a psychological evaluation by his mother for academic difficulties. At the time of testing, he was 5 years posttreatment (tumor resection, radiation, chemotherapy) for a pediatric brain tumor, with no signs of progressive disease. Although historically a good student, C.T. began to exhibit academic problems 2 years prior to the referral. More specifically, his mother reported that C.T. would "study for a test and know the material, but forget it" when he took the exam. This pattern resulted in a significant decline in his academic performance, with grades plummeting from primarily B's to mostly D's and F's. His mother reported that it appeared to take him longer to read than other children his age and that he had a hard time focusing while studying. C.T. indicated that he, also, felt it took him longer to read, but if given the time he could process the information. C.T. stated that he knew "the information, but when it [was] time for the test [he forgot] it all." To study, C.T. reported that he read the material several times and then tested himself on the material. Furthermore, he indicated that his difficulty seemed to be global and was not specific to one particular subject. He reported no sleep difficulties prior to a test

(*Continued*)

and no sudden change in appetite. He indicated that although he got nervous before a test (i.e., worried about not performing well, sweaty hands), he "never panicked" and did not feel that his nervousness interfered with his performance. Thus, his academic difficulties were not likely the result of test or performance anxiety. There was no reported family history of substance abuse, learning disorders, psychiatric disorders, or attention problems. He exhibited no significant behavior problems and denied the use of alcohol or drugs. Baseline testing (prior to diagnosis) indicated superior intellectual functioning.

C.T. was administered a comprehensive psychoeducational test battery to determine the nature of his recent academic declines. This battery included, among other instruments, the Wechsler Intelligence Scale for Children, Fourth Edition (WISC-IV) and the Wechsler Individual Achievement Test. Results of the WISC-IV indicated that C.T.'s Full-Scale IQ score was in the average range. Although consistent with scores of peers his age, his score was significantly below baseline testing and suggested a decline in intellectual functioning. Moreover, his verbal working memory index and processing speed fell within the low average range. Although these scores were not significantly lower than those of peers his age, the pattern showed a trend toward deficient memory functioning and processing speed in that the scores were two thirds of a standard deviation below average performance. The results from this achievement test suggested that C.T. performed at a level consistent with or better than his estimated cognitive ability on all academic tasks; thus, there was no evidence of a specific learning disorder.

Taken collectively, the pattern of results indicated a significant decline in cognitive functioning that was likely attributed to problems with memory and processing speed. This pattern is consistent with the cognitive late effects observed among pediatric cancer survivors. Thus, his academic difficulties were interpreted to be the result of cognitive late effects secondary to the disease process itself and/or aggressive treatment protocols (tumor resection, radiation). C.T. was diagnosed with Cognitive Disorder NOS and appropriate treatment recommendations were made.

The significance of intellectual assessment in both of these clinical cases is apparent. Accurate intellectual assessment assisted with the case conceptualization and, ultimately, the treatment planning for both clients. In the case of C.R., estimated intellectual abilities provided the basis to demonstrate a discrepancy between aptitude and achievement necessary for a diagnosis of Disorder of Written Expression and appropriate accommodations at school. In the case of C.T., declining intellectual functioning along with deficient memory and processing speed led to an efficacious treatment plan for treatment-related cognitive deficits and baseline to measure longitudinal changes postinjury.

SUMMARY

Intellectual assessment has a long-standing history in the disciplines of philosophy and psychology. Numerous theorists have developed theories of intelligence that have sought to best describe overall intellectual abilities of clients and patients.

Factor analytic theories attempt to understand intellectual functioning by examining patterns of individual differences on tests of cognitive skills. Information-processing theories of intelligence focus on the processes believed to be involved in intellectual activities by attempting to describe the internal workings of intelligent thought. Theories of intelligence have impacted both the definitions of intellectual functioning and the assessment tools utilized to measure intellectual abilities.

Intellectual assessments are clinically important for a variety of reasons. These measures routinely serve as the starting point of psychoeducational evaluations for learning disabilities. Baseline estimates of intellectual functioning are commonly utilized to demonstrate discrepancies between achievement (academic) and ability (intellectual prowess) necessary for the diagnosis of learning disabilities.

Intellectual assessments are also important in the classification of employees and in the context of military service. Assessment utilizing intelligence tests, specifically crystallized versus fluid intelligence, provides important diagnostic information for professionals working with patients who have suffered brain injuries and/or insults.

As a result of these multifaceted applications of intellectual assessments, it is important for the practitioner to remain abreast of novel and innovative approaches to assess intellectual abilities. It is equally important to remain mindful of the shortcomings of these assessment tools in light of their numerous ramifications, especially with respect to cultural biases that may be present. Finally, it is imperative that practitioners utilizing intellectual assessment tools accurately convey findings to clients and patients in such a manner that the information is easily understood.

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CHAPTER 17

Pediatric Neuropsychological Evaluation

Leslie D. Berkelhammer

"Left brained, right brained—I call that bird brained." Nearly 20 years ago, this was the opening sentence in the first lecture of my first neuropsychology professor, Ursula Kirk, EdD, at Teachers College of Columbia University. With characteristic dry wit, she encouraged us to think critically at a time when hemispheric specialization was all the rage. Neuropsychology is no longer the black box it once was and appreciation of brain-behavior relationships is more commonplace. Advances in our understanding of cognitive neuroscience, genetics, neuroimaging, and medical technology have enhanced neuropsychological research and clinical practice. Perhaps the challenge before today's researchers, clinicians, and students is best phrased by the late Ray Mulhern, PhD, of St. Jude Children's Research Hospital, who exhorted, "Don't think—know." Integration and consolidation of interdisciplinary findings in design research and clinical protocols is essential to advance the field and to gain recognition of the unique contributions of neuropsychologists working collaboratively with multidisciplinary colleagues in an academic medical setting.

DEFINITION OF A CLINICAL NEUROPSYCHOLOGIST

Clinical neuropsychology is a specialty formally recognized by the American Psychological Association, Canadian Psychological Association, and many other international bodies. In 1989, Division 40 of the APA first published the definition of a clinical neuropsychologist. The Houston Conference on Specialty Education and Training in Clinical Neuropsychology, held in 1997, formalized the programmatic and competency-based components of an integrated scientist-practitioner model for specialty training in clinical neuropsychology. The Houston Conference defined a clinical neuropsychologist as

> a professional psychologist trained in the science of brain-behavior relationships who specializes in the application of assessment and intervention principles based on the scientific study of human behavior across the life span as it relates to normal and abnormal functioning of the central nervous system.

In 2001, the National Academy of Neuropsychology updated the 1989 APA Division 40 definition with wording consistent with that of the Houston Conference. The

NAN definition further delineated the methods of a clinical neuropsychologist as one who

> uses psychological, neurological, cognitive, behavioral, and physiological principles, techniques and tests to evaluate patients' neurocognitive, behavioral, and emotional strengths and weaknesses and their relationship to normal and abnormal central nervous system functioning. The clinical neuropsychologist uses this information and information provided by other medical/health care providers to identify and diagnose neurobehavioral disorders, and plan and implement intervention strategies.

SCOPE OF PRACTICE

Some neuropsychologists are primarily clinicians or researchers; others combine these roles. Both those in academic medical centers and those in practice settings provide opportunities for training and mentorship. Professional organizations serve the interests of their constituents. The four primary professional organizations are the International Neuropsychological Society, the National Academy of Neuropsychology, Division 40 of the APA, and the Society for Neuroscience. A web-based, interactive database of programs offering training in clinical neuropsychology at the graduate, internship, and postdoctoral levels was recently added to the Division 40 website (www.apa.div40.org).

The Web has made access to information easier while also raising the bar for expectations of immediate responses and professional exposure. Many consumers of neuropsychological services now seek information on providers via the Internet that was previously based on a trusted referral. For professionals and future practitioners, opportunities for self-education are easily accessed on the Web, particularly for neuroanatomy tutorials and interactive brain atlases (e.g., www.digitalanatomist.com). Listservs such as the Pediatric Neuropsychology and Women in Neuropsychology listservs are useful resources for a wide audience, with discussions about particular cases, diseases and disorders, practice issues, and career development.

HYPOTHESIS-DRIVEN ASSESSMENT

Neuropsychological assessment is a hypothesis-driven assessment of higher brain functions resulting in an integrative analysis of findings in the context of a neurodevelopmental-systems approach and detailing recommendations for addressing the presenting problems as well as those that may be revealed in the course of the evaluation. Whether a neuropsychological test is conducted to profile the strengths and weaknesses of an individual or a population, the potential for changing how others view the presenting problem or research question is considerable. At an individual level, neuropsychological testing samples multiple cognitive domains and contextualizes findings within a neurodevelopmental approach. Indeed, skilled clinicians are able to translate test results into meaningful relevance for parents and educators such that the feedback session serves as a therapeutic intervention. Families and educators often change their perspective of the child and

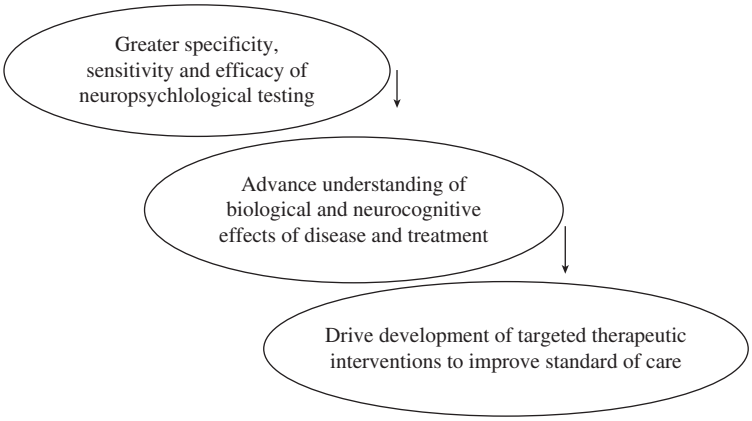

Figure 17.1 An integrative model of neuropsychological evaluation, treatment, and research.

reset expectations for parenting and teaching demands based on this more informed view. Further, the feedback session allows the child to gain a better understanding of his or her ability profile (Roman, 2004). Within and across groups, analysis of performance via empirical research affords opportunity to advance understanding of biological and neurocognitive effects of disease and treatment. Thus, neuropsychological data of individual and group performance drive development of targeted therapeutic interventions and have potential to improve standards of care (see Figure 17.1).

Pediatric or child neuropsychology is the evaluation, treatment, investigation, and teaching of brain-behavior relationships in children with a neurological, medical, or psychiatric disorder that negatively affects cognitive, psychological, and/or behavioral functioning. At its core, pediatric neuropsychology examines neurodevelopmental factors and the effects of disability, disease, and trauma in comparison to typical development and premorbid baseline abilities. In this chapter, I share with you some of my thoughts about child neuropsychological assessment. I describe the development of the discipline, distinguish pediatric neuropsychology from other child psychology specialties, and offer clinical tools for practice.

A NEURODEVELOPMENTAL APPROACH

The neuropsychologist approaches the testing session with an informed plan, based on a thorough review of the child's medical and school records, to test hypotheses regarding underlying neurological and developmental factors contributing to the child's presenting problem. It's not enough to simply document that the child's reading comprehension is at the 76th percentile compared to same-age peers. Rather, analysis of the subcomponents and reasons for the deficits as well as recommendations for accommodations and strategies for compensatory methods must be delineated within a brain-behavior and developmental-systems approach.

Bernstein (2000, p. 434) describes her neurodevelopmental model in terms of an interaction effect between the variables of brain, context, and development: "The

clinical assessment is not just a tally of skills made in a vacuum; it is a principled undertaking in the context of a theory of the organism." Neuropsychological testing is a dynamic (nonstatic) environment wherein the examiner, examinee, and examination are influenced by those same three factors (Bernstein, Kammerer, Prather, & Rey-Casserly, 1998).

Analogous to brain maturation and child developmental theory, the specialty of pediatric neuropsychology has evolved from infancy through childhood. The work of Bernstein and colleagues suggests the coming of age of child neuropsychological assessment. In essence, they have extended Kaplan's pioneering principle of the process approach inherent in clinical assessment of an individual to a more global view of the systems in which the particular child has lived, lives, and will live. Importantly, findings must take into account the four overlapping systems in which the child is affected and affects: medical, school, family, and community.

Levine (1991) has facilitated cross-talk among educators and more neuropsychologically based professionals interested in the interface of developmental pediatrics, neuropsychology, and pedagogy. He describes eight neurodevelopmental constructs (attention, temporal sequential ordering, spatial ordering, memory, language, neuromotor functions, social cognition, and higher order cognition) and their applications to specific curriculum objectives, instructional strategies, and accommodations for learning.

INDICATIONS FOR NEUROPSYCHOLOGICAL EVALUATION

A neuropsychological evaluation is indicated in cases in which brain-based impairment in thinking or behavior is suspected. Examples of relevant conditions include cognitive changes associated with medical conditions (e.g., childhood cancers, sickle cell disease, hematological disorders, HIV/AIDS, epilepsy) or medical treatments (e.g., brain surgery, radiation, chemotherapy, shunt placement, bone marrow transplantation, certain medicines) and brain injury, including stroke. Specific areas of difficulty may include language problems, memory difficulties, organizational problems, emotional difficulties, or behavior problems. Evaluation may be needed to characterize initial functioning and monitor change in children with potentially unstable conditions (e.g., children undergoing treatments with known cognitive risks, children at risk for stroke, children with a history of seizure), make a diagnosis, update academic or vocational plans, or assess for psychosocial problems related to a medical condition. A neuropsychological evaluation is particularly useful for tracking progress in rehabilitation after brain injury or neurological disease. Neuropsychological evaluations may confirm or clarify a diagnosis, identify changes in functioning since prior examinations, provide a profile of strengths and weaknesses to guide rehabilitation or educational or vocational services, identify beneficial compensatory strategies, or result in referrals to other specialists (e.g., educators, neurologists, psychologists, psychiatrists, social workers, vocational counselors).

SHARED VARIANCE AMONG CHILD PSYCHOLOGY SPECIALTIES

Neuropsychology is "the study, evaluation, and treatment of known and suspected brain disorders using the methods of psychology," according to the definition posted on the National Academy of Neuropsychology (NAN) website. By comparison, the American Psychological Association (APA) website posts definitions for clinical and pediatric psychology:

> The field of clinical psychology integrates science, theory, and practice to understand, predict, and alleviate maladjustment, disability, and discomfort as well as to promote human adaptation, adjustment, and personal development. Clinical Psychology focuses on the intellectual, emotional, biological, psychological, social, and behavioral aspects of human functioning across the life span, in varying cultures, and at all socioeconomic levels.

Finally, "Pediatric psychology is an interdisciplinary field addressing physical, cognitive, social, and emotional functioning and development as it relates to health and illness issues in children, adolescents, and their families."

Neuropsychological evaluations may include some of the same tests as those used to conduct a psychoeducational evaluation. However, a noticeable difference will be evident in the breadth and depth of skills measured and interpretation in relationship to brain processes. Neuropsychological assessment is not about the tests. It's about the dynamic testing environment and the theoretical approach of the professional who interprets the findings and ensures that a qualitative analysis of the child's performance is every bit as important as the quantitative results.

Commonalities and differences exist across specialties in child psychology. Pediatric neuropsychology incorporates elements of clinical, school, counseling, and organizational psychology. Pediatric neuropsychologists must be well versed in developmental psychopathology, knowledgeable about pedagogy and school systems, familiar with counseling principles, and informed about organizational behavior when rendering feedback and conducting cognitive rehabilitation to both the child and the family.

QUANTITATIVE AND QUALITATIVE DESCRIPTION OF GLOBAL AND SPECIFIC FUNCTIONS

A neuropsychological assessment provides quantitative and qualitative data on the domains of intelligence, attention, executive functions, language, learning and memory (verbal and nonverbal), visual-spatial, motor, behavioral, psychological, adaptive, and academic skills. However, select areas will be emphasized over others, based on the reasons for referral and data that emerge during the evaluation. The clinical interview is the foundation for the evaluation as it sets the tone for the patient and allows the examiner a period of focused data gathering on which to refine the preliminary testing. The Pediatric Neuropsychology Questionnaire (Baron, Fennell, & Voeller, 1995) is a particularly useful structured interview.

Components of the evaluation include the following: presenting problems and reasons for referral; relevant developmental, family, medical, educational, and psychosocial history; an integrative assessment and analysis of cognitive domains and skills within and across domains; comparison of the student's performance to same-age peers for normative comparison; description of the individual's relative strengths and weaknesses; specific recommendation for intervention; and prognosis, given all these considerations taken together.

Assessment of global cognitive abilities, or general *g*, should allow for comparison of verbal and nonverbal skills as well as analysis of specific subtests and the underlying skills presumed to be measured. Evaluation of attention and concentration abilities should compare auditory versus visual attentional skills, attention span, sustained and divided attention, and processing of simple versus complex information. Executive functions influence the expression within multiple domains and are particularly reliant on the connectivity of neural systems. Throughout the evaluation, task analysis of ability to prioritize, plan, sequence, analyze, reason, and make decisions should be prominent.

Language assessment should explore components such as expressive and receptive abilities, comprehension, verbal fluency, repetition, naming, speech production, conversational speech, organization of verbal output, and pragmatics (e.g., prosody, rambling speech, and nonverbal aspects of speech: eye contact, turn-taking, gestures, interpersonal space). Similarly, information processing should examine auditory versus visual abilities, response speed, and the relative pacing of simple versus complex information. Learning and memory assessment should also be multimodal and quantify functioning visual, verbal, auditory, tactile, and motor skills. There are several well-normed measures that allow the neuropsychologist to analyze which component of memory is problematic for the individual being examined: encoding versus retrieval. Recommendations should account for intra-individual performance in short- versus long-term memory and for simple versus complex information for both verbal and visual-spatial material.

Evaluation of current levels of academic achievement compared to peers is an integral part of the neuropsychological evaluation. Again, the goal should not be limited to mere reporting of ability levels in the core areas of mathematics, reading, spelling, and written expression. Instead, data-driven conclusions about the neuropsychological underpinnings of these expressed deficits should be offered. Recommendations to maximize strengths and minimize weaknesses should also be considered along the same lines. In sum, neuropsychological findings should inform opinions about the *goodness of fit* between the child and his or her learning environment.

Psychological, social-emotional, behavioral, and adaptive functions should be assessed. Recommendations for interventions and skill building should be grounded in an integrated understanding of the child's neuropsychological abilities. For example, the child with Attention-Deficit/Hyperactivity Disorder may require social skills training and cognitive rehabilitation to compensate for deficient frontal feedback loops that limit awareness of his or her effect on others. Temperament, or the biological bases of personality, seems to be underexamined by neuropsychologists, although there appears to be some common ground with child neuropsychology.

NEUROPSYCHOLOGICAL REPORT

The neuropsychological report is an important document. Often, there are multiple consumers of the report, each interpreting the information from the viewpoint of his or her respective disciplines (e.g., physician, educator, coach, and parent). Yet, each of these consumers relies on the information in the report to get a sense of the child's abilities under ideal conditions and to extrapolate to the situation in which the consumer of the report interacts with the child. Accordingly, the report should contain an integrated analysis of the child's cognitive abilities, academic achievement, and psychological functioning. The neuropsychologist should provide quantitative and qualitative data to elucidate multiple factors contributing to the presenting problems, including relevant neurological, medical, and developmental issues. Thus, the report is often a critical tool in decisions regarding school planning, educational program placement, medical care, and treatment monitoring.

By any measure, evaluation is an expensive procedure. Cognitive screening may be appropriate to prioritize those in need of a more comprehensive evaluation, estimate restricted functions, and generate epidemiological data. Regardless of the comprehensiveness of the testing, documentation of the child's strengths and weaknesses, comparison to age-level expectations, recommendations for instruction and curriculum, accommodations to optimize learning and development, and identification of children in need of additional services is expected.

COGNITIVE REHABILITATION

The emergence and development of the subspecialty of cognitive rehabilitation is a logical progression from the discipline's early emphasis on testing and test development (Sohlberg & Mateer, 2001). Cognitive rehabilitation is a subspecialty that employs functional activities to a systematic application of remediation strategies designed to ameliorate impaired brain functions after injury based on a thorough evaluation of the patient's acquired deficits and residual strengths. Treatment goals include strengthening premorbid adaptive skills and behaviors, establishing new patterns of cognition via compensatory strategies for impaired neurological systems, establishing new patterns via external compensatory supports, and adapting to reduced levels of ability. Learning therapy to improve acquired cognitive deficits by instruction and rehearsal of compensatory strategies with environmental modifications can have a profound difference on quality of life following injury or disease. Butler and colleagues (Butler & Copeland, 2002; Butler & Mulhern, 2005) have demonstrated the efficacy of cognitive rehabilitation in children treated for cancer. Moreover, evidence-based outcomes have been demonstrated on functional magnetic resonance imaging studies (Ogg, Zou, White, et al., 2002; Laatsch, Thulborn, Krisky, Shobat, & Sweeney, 2004).

FUTURE DIRECTIONS

As understanding of the biological basis of behavior and cognition become more important, and there are medical or biological treatments to be applied, the only

method of functionally monitoring such treatment is neuropsychological assessment. We need to be at the forefront of working with those who are devising treatments for diseases caused by both innate and acquired injuries to monitor functional outcomes of both biological and behavioral treatments.

—Elsa Shapiro

Historically, neuropsychologists have relied heavily on their expertise in assessment and test development to thoroughly document the cognitive and psychological strengths and weaknesses of the patient. Over time, demand for shorter test batteries, multidisciplinary treatment team involvement, and application of test findings to cognitive rehabilitation, forensics, and policy development has emerged. I have long posed a challenge to myself and colleagues with a two-word question: "So what?" That is, although quantification of an individual's neuropsychological profile in comparison to normally developing peers on standardized measures is important, relevance to the varied environments in which children operate is critical.

These are, indeed, exciting times for those who conduct neuropsychological assessment, treatment, and research and train the next generation of clinicians and researchers. Opportunities to incorporate neuropsychological theory and novel applications in forensic settings, day treatment programs, and automated environments push the envelope. Virtual reality applications to neuropsychological assessment and cognitive rehabilitation of acquired brain injury and neurological disorders are poised to further the field as we move beyond assessment and toward evidence-based outcomes (Schultheis, Himelstein, & Rizzo, 2002). Building on the well-received text, *Neuropsychological Evaluation of the Child*, Baron (2004) describes the growth and development of pediatric neuropsychology in the context of maturation of the field, development of cognitive rehabilitation models applied to a range of disorders, and increased cross-fertilization from allied health, psychology, neuroscience, and neuroradiology (Baron, in press).

I have included several clinical forms and a sample neuropsychological report in six appendixes. Appendix 1 is a disease-specific clinical intake form with content customized to known areas of cognitive vulnerability and contact information for the multidisciplinary team serving this group of patients. Appendix 2 is a test results summary grid provided by my colleague Jordi Bernabeu, PhD, of the La Fe Hospital in Valencia, Spain. He has found it helpful to explain test results to persons unfamiliar with specific tests using a common metric of z-scores with domains color-coded across multiple measures. I believe that we sometimes overestimate the readability of our reports. Even with tables embedded in the report to render scores into more meaningful functional categories, the consumer—multidisciplinary treatment team providers, parents, and educators—may not easily appreciate the psychometrics of reported data. Appendix 3 is a 24-hour preliminary feedback form on which test abbreviations and relevant scores can be entered by domain and categorical outcome. I have found this form helpful in rapidly communicating with the referral source. Appendix 4 is a preferred tests form, with measures grouped by domain and presented in a check box format to aid the examiner before and during the testing session. Given the proliferation of commercially available tests, the list is not exhaustive; it reflects my own selection biases. Appendix 5 is a recommendations form letter,

summarizing medical background and specific accommodations to be implemented in the school setting. Appendix 6 is a neuropsychological sample report.

SUMMARY

In this chapter, child neuropsychological evaluation was defined as a hypothesis-driven assessment of higher brain functions. Indications for and components of neuropsychological testing were defined. The importance of the neuropsychological report as a descriptive analysis of cognitive, psychological, and behavioral strengths and weaknesses, contextualized within a neurodevelopmental-systems framework, was explored. Professional qualifications, scope of practice, and differences between select child psychology specialties were described. Progress and future directions for the field of pediatric neuropsychology were discussed.

An integrated model of neuropsychological testing, treatment, and research was presented. Greater specificity, sensitivity, and efficacy of neuropsychological assessment was proposed as a means to advance understanding of the biological and neurocognitive effects of disease and treatment. In turn, these refinements were proposed as a catalyst to drive development of targeted therapeutic interventions and yield improvement in standards of care. As a specialty area of practice, research, and teaching, the field of neuropsychology has a bright future. Creative and well-trained neuropsychologists will, most certainly, further established lines of research and explore new applications with an increased reliance on emerging biotechnology.

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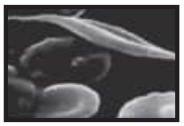
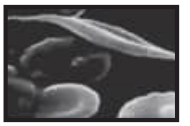
APPENDIX 1: CLINICAL REFERRAL FORM



CONFIDENTIAL
CLINICAL REFERRAL FORM
Sickle Cell Disease



Name: _____ **MRN #:** _____
Date of Testing: _____ **Date of Birth:** _____ **Age:** _____
Grade: _____ **School:** _____
Referring Physician: _____
SCD Genotype: ☐ HbSS ☐ HbSC ☐ HbSβ° thalassemia
☐ HbSβ± thalassemia ☐ HbSβ+ thalassemia

Therapy: ☐ None ☐ Chronic Transfusion ☐ Hydroxyurea ☐ Transplantation
Education: ☐ Retained _____ ☐ Sp. Ed. _____
Psychiatric: ☐ Diagnosis _____ ☐ Medication _____
☐ Psychotherapy _____ ☐ Pica ☐ Sleep Disorder
TCD: ☐ Normal ☐ Conditional ☐ Abnormal
MRI: ☐ Infarct _____ ☐ Other _____

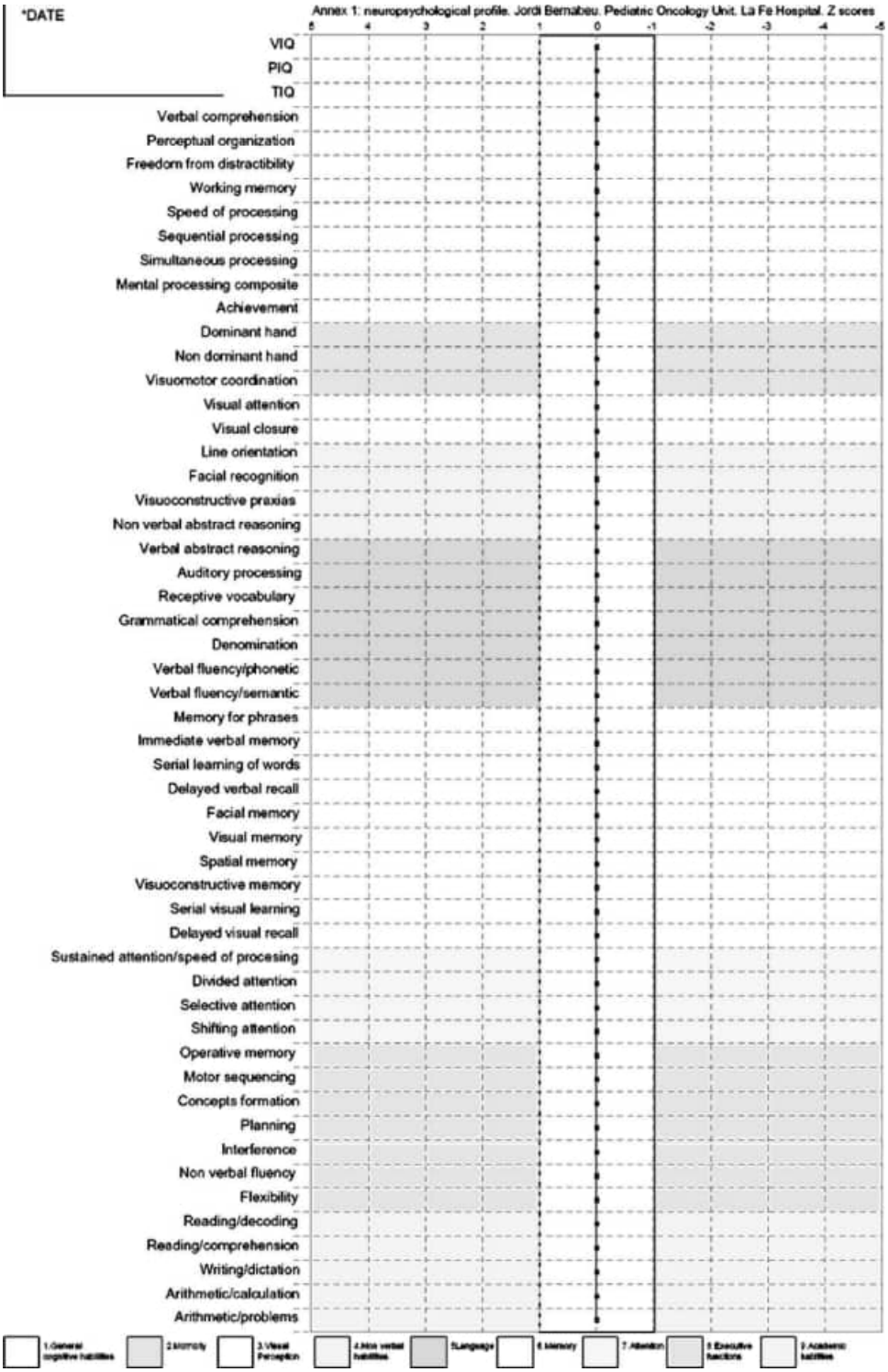
Presenting Problem(s) & Reason for Referral:

Evaluate:
☐ Intelligence
☐ Attention ☐ Processing Speed
☐ Executive Functions
☐ Working Memory ☐ Cognitive Flexibility ☐ Planning and Organization
☐ Fluency ☐ Task Initiation ☐ Task Persistence
☐ Learning and Memory
☐ Language
☐ Visual-Spatial
☐ Motor
☐ Psychosocial ☐ Behavioral / Adaptive ☐ Pain / Coping
☐ Academic Achievement
☐ Reading ☐ Math ☐ Writing ☐ Spelling

CONTACT INFORMATION FOR SCD TEAM

		Phone	Page
Neuropsychologist			
Nurse Case Managers	*Infant & Toddler (0–4 years)*		
	School-Age (5–13 years)		
	Teen (13–18 years)		
	Hydroxyurea (all ages)		
Educators			
Social Worker			
Physicians			

APPENDIX 2: TEST RESULTS SUMMARY GRID



APPENDIX 3: 24-HOUR PRELIMINARY FEEDBACK FORM

CONFIDENTIAL
PRELIMINARY RESULTS OF NEUROPSYCHOLOGICAL TESTING
Feedback provided within 24 hours as professional courtesy. Report to follow within 30 days.

Name: _____ **Medical Record #:** _____
Date of Birth: _____ **Date of Testing:** _____
Age: _____ **Grade:** _____
Examiner: _____ **Supervising Neuropsychologist:** _____
Diagnosis: _____ **Referring Physician:** _____
Reason for Referral:
_____
_____

	Impaired	Borderline	Below Average	Average	Above Average	Superior	Very Superior	Not Tested
Intelligence								
Attention & Executive								
Learning & Memory								
Language								
Visual-Spatial & Motor								
Psychosocial, Behavioral & Adaptive								
Academics								

IMPRESSIONS:

_____ _____
Neuropsychology Fellow Licensed Neuropsychologist

APPENDIX 4: PREFERRED TESTS FORM

CONFIDENTIAL
PREFERRED TESTS FORM

Please evaluate this patient's skills within the following domains and, if indicated, using the tests indicated. It is understood that clinical judgment may require changes to this plan.

NAME: _____ **MEDICAL RECORD #:** _____
DATE OF TESTING: _____ **DATE OF BIRTH:** _____
AGE: _____ **GRADE:** _____ **SCHOOL:** _____
DIAGNOSIS: _____ **MEDICATION:** _____
REFERRAL QUESTION: _____
BEHAVIORAL OBSERVATIONS: _____
_____
_____
_____
_____

☐ **Intellect and Global Cognitive**
- ☐ Bayley Scales of Infant and Toddler Development, Third Edition (Bayley-III)
- ☐ Bayley-III Screening Test
- ☐ Bracken Basic Concept Scale, Third Edition: Receptive (BBCS-3:R)
- ☐ Differential Abilities Scale (DAS)
- ☐ Kaufman Assessment Battery for Children, Second Edition (KABC-II)
- ☐ Kaufman Brief Intelligence Test, Second Edition (KBIT-2)
- ☐ McCarthy Scales of Children's Abilities
- ☐ McCarthy Screening Test (MST)
- ☐ Miller Assessment for Preschoolers (MAP)
- ☐ Mullen Scales of Early Learning
- ☐ NEPSY-II: A Developmental Neuropsychological Assessment, Second Edition (NEPSY-II)
- ☐ Raven's Progressive Matrices
- ☐ Stanford-Binet Intelligence Scales, Fifth Edition (SB-V)
- ☐ Universal Nonverbal Intelligence Test (UNIT)
- ☐ Wechsler Preschool and Primary Scale of Intelligence, Third Edition (WPPSI-III)
- ☐ Wechsler Intelligence Scale for Children, Fourth Edition (WISC-IV)
- ☐ Wechsler Intelligence Scale for Children, Fourth Edition, Integrated (WISC-IV-Integrated)
- ☐ Wechsler Abbreviated Scale of Intelligence (WASI)
- ☐ Wechsler Adult Intelligence Scale, Third Edition (WAIS-III)
- ☐ Woodcock-Johnson Tests of Cognitive Abilities, Third Edition (WJ-III COG)

☐ **Academic**
- ☐ Gray Oral Reading Test, Fourth Edition (GORT-4)
- ☐ Kaufman Test of Educational Achievement, Second Edition (KTEA-II)
- ☐ School Motivation and Learning Strategies Inventory (SMALSI)
- ☐ Wechsler Individual Achievement Test, Second Edition (WIAT-II)
- ☐ Woodcock-Johnson Tests of Achievement, Third Edition (WJ-III ACH)

☐ **Attention and Executive Function**
- ☐ Behavior Rating Inventory of Executive Function (BRIEF)
- ☐ Conners' Continuous Performance Test II, Version 5 (CPT II)
- ☐ Delis-Kaplan Executive Function System (D-KEFS)
- ☐ Clock Drawing
- ☐ Test of Variables of Attention (TOVA)
- ☐ Test of Everyday Attention in Children (TEA-Ch)
- ☐ Wisconsin Card Sort Test: Computer Version 4 (WCST: CV4)

☐ **Learning and Memory**
- ☐ California Verbal Learning Test, Children's Version (CVLT-C)
- ☐ California Verbal Learning Test, Second Edition (CVLT-II)
- ☐ Children's Memory Scale (CMS)
- ☐ Rey-Osterreith Complex Figure Test
- ☐ Wechsler Memory Scale, Third Edition (WMS-III)
- ☐ Wide Range Assessment of Memory and Learning, Second Edition (WRAML-2)

☐ **Language**
- ☐ Clinical Evaluation of Language Fundamentals, Fourth Edition (CELF-IV)
- ☐ Controlled Oral Word Association (COWA) / Verbal Fluency
- ☐ Expressive One-Word Picture Vocabulary Test, 2000 Edition (EOWPVT-2000)
- ☐ Comprehensive Test of Phonological Processing (CTOPP)
- ☐ Peabody Picture Vocabulary Test, Third Edition (PPVT-III)
- ☐ Receptive One-Word Picture Vocabulary Test (ROWVT-2000)
- ☐ Test of Written Language, Third Edition (TOWL-3)
- ☐ Writing Sample: ☐ Cookie Theft Test ☐ TOWL-3 ☐ WIAT-II
- ☐ Test of Word Reading Efficiency (TOWRE)

☐ **Visual-Spatial and Motor**
- ☐ Beery-Buktenica Developmental Test of Visual Motor Integration, Fifth Edition (Beery VMI)
- ☐ Finger Tapping Test
- ☐ Grooved Pegboard Test
- ☐ Hooper Visual Organization Test
- ☐ Judgment of Line Orientation, Short Form (JLO-S)
- ☐ Luria Sequential Hand Movements
- ☐ Wide Range Assessment of Visual-Motor Abilities (WRAVMA)

☐ **Psychosocial, Behavioral, and Adaptive**
- ☐ Adaptive Behavior Assessment System, Second Edition (ABAS-Second Edition)
- ☐ Behavior Assessment Scale for Children, Second Edition (BASC-2)
- ☐ Parenting Relationship Questionnaire (PRQ)
- ☐ Vineland Adaptive Behavior Scales, Interview Edition, Survey Form (VABS)
- ☐ Visual Analog Mood Scale (VAMS)
- ☐ PedsQL
- ☐ Pain Rating Scales (Faces)
- ☐ Sickle Cell Self-Efficacy Scale: Adolescent Version (SCSES-A)

APPENDIX 5: RECOMMENDATIONS FORM LETTER

[Letterhead]

[Director of Special Education]
[School or District]
[City, State Zip Code]

RE: **Patient Name & Medical Record Number**

Dear [Director of Special Education]:
[Full Name (Nickname)] completed a neuropsychological evaluation on [date(s)]. Recommendations and accommodations are summarized below. If you have any questions, please do not hesitate to contact me at [phone]. Respectfully,

[Signature & Title]

MEDICAL BACKGROUND
Diagnosis:
Treatment:
Vision Impairment ☐ Yes ☐ No
Hearing Impairment ☐ Yes ☐ No
Gross/Fine Motor Impairment ☐ Yes ☐ No
Speech Impairment ☐ Yes ☐ No
Assistive Device(s):

RECOMMENDATIONS
Due to the diagnosis, results of neuropsychological testing, and the educational impact of diagnosis and treatment, [Name] meets criteria to receive special education services and accommodations under the category [classification] as per [federal law #]. Specific recommendations are as follows:
☐ Regular classroom with accommodations
☐ Resource room for the following subjects: _____
☐ Self-contained classroom ☐ Full-time aide ☐ Part-time aide
☐ Abbreviated school day ☐ Part-time school attendance/part-time homebound

Instructional Strategies
☐ Review, preview, and summarize new material
☐ Specific, immediate, and frequent feedback
☐ Notebook color coded to textbook color
☐ Neuropage or other electronic prompting device
☐ Copy of material to be copied from book or board
☐ Copy of best notes in an anonymous manner
☐ Study guide
☐ Review assignment 1:1 to ensure understanding
☐ Modify assignment format (photo essay or skit)
☐ Other:

Classroom Modifications
☐ Preferential seating
☐ Second set of books
☐ Assignment book or Electronic organizer
☐ Abbreviated assignments, reduced homework
☐ Extended time to complete assignment
☐ Elevator or extra time to navigate hallway
☐ Color coded map of school
☐ Adaptive P.E.
☐ Other:

Testing and Grading Accommodations
☐ Extended time for testing
☐ Extended time to complete assignments
☐ Modify test format (word bank, multiple choice)
☐ Extra grade opportunities (redo items)
☐ Oral testing
☐ Portfolio-based assessment
☐ Quiet environment with few distractions to take tests
☐ Modified grading scale
☐ Read aloud/sign for internal instructions
☐ Read aloud/sign for test items
☐ Use of calculator
☐ Talking or electronic device with Braille display
☐ Word processor w/ or w/o talk-text technology
☐ Reads into auditory recorder and plays back
☐ Other:

APPENDIX 6: SAMPLE NEUROPSYCHOLOGICAL REPORT

CONFIDENTIAL
NEUROPSYCHOLOGICAL REPORT

KEISHA ST. JUDE
Dates of Testing: 7/14 and 27/2005
Dates of Feedback: 9/12/05 (mother, legal guardian)
9/21/05 (grandmother, durable power of attorney for health care decisions)
Date of Birth: 5/20/1988

REASON FOR EVALUATION: Keisha is a 17 year, 2 month old African American young woman with HbSS sickle cell anemia (SCA), status post cerebral vascular accident (CVA) at age 6 years old, who was referred by _____, MD for a comprehensive neuropsychological evaluation. A neuropsychological evaluation has not previously been conducted. The purpose of this neuropsychological evaluation was to evaluate her intelligence, attention, executive function, memory, language, behavioral, and psychosocial abilities for educational, vocational, and treatment planning.

BACKGROUND INFORMATION: Keisha had a documented CVA (stroke) in 1994 at age 6 with reported symptoms of slurred speech and weakness in her right hand and leg. Of note, Keisha and her mother reported a 2nd possible neurological event approximately 3 years ago which was previously not known to the treatment team prior to the clinical interview conducted as part of this neuropsychological evaluation. Symptoms of slurred speech, "eyes looking funny", weakness, and "not feeling good" were reported. Allegedly, the emergency room physicians in her home state determined that Keisha had an "allergic reaction" despite her mother reportedly stating that Keisha has a prior history of stroke secondary to SCA and denial of any known source for the "allergic reaction". It has been years since Keisha's last brain MRI and an updated MRI is pending.

Keisha's primary caretaker has been her maternal grandmother for most of Keisha's formative years. Keisha's biological mother, with whom she currently resides, remains her legal guardian but has awarded durable power of attorney for health care decisions to the grandmother. Keisha returned to Memphis to live with her mother and four younger half-siblings, reportedly, to remove her from school problems and the influence of negative peers as she had acquired a considerable history of multiple fights and association with delinquent peers.

Keisha and her mother denied basic knowledge of the effect of stroke on her cognitive and academic abilities. They also denied understanding of the importance of hydration and Desferal for iron overload due to chronic red blood cell transfusions. These medical and SCA management issues have been discussed at length on multiple occasions and by multiple providers with Keisha, her mother, and grandmother.

Keisha disclosed a prior episode of suicidal ideation with plan approximately one year ago (cutting wrists), which was previously unknown to the treatment team and grandmother. It was during a low point in her life at the time when she relocated to her current residence, thus being "made" to leave her boyfriend and grandmother. At the time, Keisha reportedly discussed her feelings with a trusted aunt who provided comfort and agreed not to tell anyone about Keisha's suicidal ideation with plan. Keisha denies current suicidal ideation or plan. She contracted to tell her grandmother or other responsible adult if she has suicidal ideation in the future. Grandmother was advised to ensure Keisha's immediate safety if suicidal ideation or plan were to become known.

There is a history of receiving special education services while in the 5th and 6th grades. However, Keisha and her mother have reportedly refused special education since the 7th grade. Keisha was retained in the 9th grade and is not currently receiving special education services in her current 9th grade placement. She enjoys gym and science. Keisha's career aspiration is to become a nurse.

BEHAVIORAL OBSERVATIONS: Keisha was accompanied by her mother to the first testing session and her grandmother to the next session. Initially, Keisha showed a defensive, passive response style and seemed somewhat despondent. However, she responded well to support with open communication and warmed to the examiner and the testing experience. On the second and third visits, Keisha smiled broadly upon greeting this examiner and seemed attentive and cooperative. Overall, Keisha presented as a very likeable young lady who demonstrated good effort on testing, although she lacked confidence in her performance and seemed to have limited ability to appreciate how her past choices have contributed to her current difficulties. Keisha tolerated participation well. Results are considered valid.

PROCEDURES:
Behavioral Observations
Clinical Interview with Keisha, mother, and grandmother
School consultation with two of Keisha's teachers
Wechsler Abbreviated Scale of Intelligence (WASI)
Tests of Variables of Attention (TOVA)
Wide Range Assessment of Memory and Learning-Second Edition (WRAML-2):
    Story and Design Memory subtests
Rey-Osterreith Complex Figure Test
Woodcock-Johnson Tests of Achievement-Third Edition (WJ-III):
    Word Attack, Reading Fluency, Calculation, and Spelling subtests
Wechsler Individual Achievement Test: Written Expression subtest
Behavior Assessment System for Children (BASC): Self-Report, Parent, and Teacher Forms
Childhood Depression Inventory: Short Form (CDI-S)
Feedback session with Keisha & her grandmother (9/12/05)
Feedback session with Keisha & her mother (9/14/05)

CPT CODE 96117, Neuropsychological Testing
Billable hours = 12 (7 hours testing; 5 hours scoring, interpretation, and documentation)

RESULTS:
Intelligence: Intellectual functioning fell within the lowest end of the *Borderline* range for verbal and *Impaired* range for nonverbal abilities, as measured on the Wechsler Abbreviated Scale of Intelligence (WASI) (VIQ=70, 2nd percentile; PIQ=54, ‹1st percentile). The WASI is an abbreviated test of general cognitive abilities, comprised of 4 subtests measuring verbal (i.e., word knowledge and verbal reasoning) and nonverbal (i.e., pattern analysis and nonverbal problem solving) skills compared to same-age peers. There was a significant difference between her obtained verbal and performance score. Therefore, the full scale IQ is not reported. The discrepancy is attributable to her T-score of 35 on the Similarities subtest as compared to T-scores of 20, 21, and 20 on the Vocabulary, Block Design, and Matrix Reasoning subtests, respectively. That is, the Similarities subtest requires associative reasoning to compare how two objects or concepts are alike. In contrast, on subtests requiring higher level language and conceptual reasoning skills, Keisha consistently performed three standard deviations below the mean. Qualitatively, Keisha's performance on a subtest of word knowledge was notable for her concrete responses and simple word associations. On a task of nonverbal problem solving, she demonstrated good task persistence, although an efficient trial and error approach lacking a systematic plan.

Attention and Executive Function: Keisha performed within the *Impaired* range on measures of attention and executive function. On a computerized test of visual attention, the Test of Variables of Attention (TOVA), Keisha demonstrated significant difficulty maintaining her attention after 5 to 6 minutes of a low-response (boring) task. Similarly, verbal and nonverbal subtests of the WASI requiring higher level conceptual reasoning were *Impaired* (i.e., Block Design=21 T-score, Similarities=35 T-score, and Matrix Reasoning=20 T-score). Further, Keisha's ability to analyze and copy a complex visual-spatial design indicated *Impaired* ability to integrate and synthesize the component parts into an integrated whole figure (Rey-Osterreith Complex Figure: Copy=9 of 36 raw score).

Learning and Memory: Learning and memory for language presented in context (i.e. 2 paragraph-length oral stories) was *Average* on the Wide Range Assessment of Memory and Learning, Second Edition (WRAML-2: Story Memory: Immediate recall=10 scaled score, Delayed Free Recall=9 scaled score). Cueing, using a multiple choice recognition format, was helpful (Delayed Recognition Recall=11 scaled score). Visual spatial learning and memory of simple to moderately complex designs was *Below Average* (WRAML-2: Design Memory= 7 scaled score). In contrast to her recall for verbal material, cueing was not helpful on the delayed visual-spatial recognition recall on which her performance declined to the *Borderline* range (WRAML-2: Delayed Recognition Recall=5 scaled score). It may be that Keisha's impaired sustained attention and decision-making abilities may have negatively influenced her demonstrated visual-spatial recall abilities. On a more complex task, copy and recall of the Rey-Osterreith Complex Figure was *Impaired* (Copy=9 raw score, Immediate recall=8.5 raw score, Delayed recall=7 raw score of 36 points, respectively). Despite several attempts at the copy task, she was unable to integrate the component parts. However, Keisha was able to retain the information initially acquired. This suggests that Keisha's deficiency in this area may be due to difficulty with visual-spatial analysis and integration rather than a memory problem.

Language: Possible expressive and receptive language problems were suggested. Qualitative analysis of Keisha's conversational speech was remarkable for sparse output, misarticulation, and word substitutions. For example, Keisha substituted the word "summer" for "day" in recalling the target phrase of the "hottest day in July". On several occasions, Keisha requested repetition of the target words. Her responses seemed to indicate that she did not hear the ending of select words presented in isolation. The possibility of an auditory processing deficit was discussed with Keisha who agreed that she often has difficulty understanding the speech and language of others. She agreed that her misunderstanding of language frequently results in miscommunication impacting socialization and school. Keisha reported that she sometimes studies material presented orally, only to get a poor test grade due to having learned erroneous information.

Academic: Tests of academic achievement were administered. Keisha's performance on both the Wechsler Individual Achievement Test, Second Edition (WIAT-II) and Woodcock-Johnson Tests of Achievement, Third Edition (WJ-III) fell within the *Impaired* range relative to same-age peers in reading, math, and spelling. Reading comprehension and speed on short passages was *Impaired* on the WIAT-II (Reading Comprehension=46 standard score, Reading Speed=4th quartile). The following subtest age- and grade-equivalent scores are reported for the WJ-III:

WJ-III Subtests	Age-Equivalent	Grade-Equivalent
Word Attack	7-7	2.1
Reading Fluency	10-1	4.7
Calculation	9-8	4.1
Spelling	13-3	7.1

Qualitatively, Keisha demonstrated some understanding of elementary rules of phonics, although she was inconsistent in her application of select rules. She was able to demonstrate accurate ability to read for understanding on simple sentences requiring basic reflective thought and

judgment of everyday events. It may be that residual effect of Keisha's stroke sometimes causes her to misinterpret the visual code essential for reading and application of rules of sound to symbol associations. Further, it may be that increased complexity of the task contributes to performance declines.

In terms of mathematical skills, Keisha was able to add, subtract, multiply, and divide single-digit problems. She was able to subtract two-digit numbers that did not require "borrowing", add a three-column straight addition problem, and sometimes multiply a two-digit number by a one-digit number. Keisha was observed to count on her fingers for single-digit arithmetic problems. She lacked automaticity for basic multiplication problems. For example, although Keisha was able to correctly mentally multiply 8×5, she required considerable time and effort to produce the correct response. Keisha remarked that she "goes through the times table" in her head to arrive at most answers on multiplication problems.

Written expression, although well below age- and grade-level expectations, was notable for the suggestion of functional writing skills. Keisha responded well to structure and support. Spontaneously, Keisha wrote the following to the scenario of the WIAT-II: "I am writing to the editor of the newspaper informing them why uniform is a bad opinion. First, everybody have on the same color everyday and the staff have a problem finding out who we are when we be in the cafeteria." Examiner prompting resulted in Keisha adding: "I think that uniforms are hot because most of the time there is no air on in the building and we can't be in the uniform all the time." Examiner encouragement yielded additional information: "A reason uniforms are good is because some kids can't afford cloths like everybody else. And they need to wear uniform because their parents can't afford those expenise cloths or shoes."

In clinical interview, Keisha remarked that it is often difficult for her to demonstrate what she knows and particularly difficult to elaborate on her thoughts in writing. Keisha agreed that she tends to exert less effort on academics because of repeated failure experiences and frustration with her lack of abilities. Moreover, attempts to work independently without adequate academic support for her efforts have not been positive and are, therefore, avoided. Keisha added that she enjoys reading, if the material is at her level.

Psychosocial and Behavioral: In clinical interview, Keisha acknowledged a prior history of suicidal ideation with plan (see Background Information section) but denies current suicidal ideation or plan and contracted to tell grandmother or another responsible adult if suicidal ideation or plan should return. Other significant psychosocial and behavioral concerns were reported, including current gang association and ongoing issues with anger control and poor interpersonal relations. Keisha acknowledged that she now has a reputation as a "fighter" and that it is difficult for her to change this expected pattern of behavior, even though she verbalized desire to change. Keisha reports that there have been times when she has been "provoked" into a fight and has not seen an alternative acceptable behavior. For example, a boy reportedly poured chocolate milk on her jacket. In another circumstance, Keisha fought to defend the honor of her stepfather subsequent to peer teasing. Of note, Keisha endorsed desire to "drop out" of school, which was new information to her mother and grandmother. Risk for school failure is assessed as extremely high.

School consultation with Keisha's English teacher was remarkable for her commitment to Keisha's success and her concern about Keisha's psychosocial problems. The teacher described that Keisha's behavior and attitude has improved following a recent intervention. That is, the teacher approached Keisha after she noted that Keisha seemed to be having difficulty with a female classmate. The teacher spoke privately with Keisha and presented her with a supportive challenge: "You're not going to let somebody else bring you down are you?" Since that time, the girls have not bothered each other in the classroom and Keisha's efforts in class seem considerably more appropriate.

The Behavior Assessment System for Children (BASC) was administered to Keisha, her grandmother, and her teacher. Responses of Keisha's grandmother, who has been her primary caretaker for many years, indicated clinically significant problems in several areas. Of note, Keisha's grandmother endorsed critical items of concern that Keisha sometimes says that she wants to kill herself, die, or wishes she were dead and often threatens to hurt others. The following T-score, percentile, and descriptive ratings are provided:

	T-Score	%ile	Rating
Conduct Problems	70	95	Mild
Anxiety	77	98	Mild
Depression	88	99	Moderate
Hyperactivity	66	93	At-Risk
Attention Problems	68	95	At-Risk
Withdrawal	64	91	At-Risk
Leadership	26	1	Mild
Social Skills	30	3	Mild
Externalizing Problems Composite	74	97	Mild
Internalizing Problems Composite	81	99	Moderate
Behavioral Symptoms Index	80	99	Moderate

Similarly, Keisha's English teacher indicated problems of clinical significance in the following areas:

	T-Score	%ile	Rating
Anxiety	77	99	Mild
Learning Problems	75	98	Mild
Withdrawal	81	99	Mild

In contrast, Keisha's responses on the BASC: Self-Report Form failed to indicate any problems of clinical significance and her response pattern fell within the *Caution* range, suggesting lack of forthrightness in her responses. However, Keisha did endorse critical items of "I can't seem to control what happens to me", "Nobody ever listens to me", and "I get into fights at school." Similarly, Keisha's responses to the Childhood Depression Inventory: Short Form (CDI-S) failed to indicate problems of significance. It may be that Keisha's responses underestimated her true feelings. Indeed, she cried during the initial feedback session and acknowledged that her mood is frequently sad, which she tends to mask with anger/socially-acting out behaviors and that she is frequently upset about her school failure. Keisha, her mother, and grandmother are in agreement that counseling, which has been strongly and repeatedly recommended by the SCA treatment team at the hospital, seems well indicated and may be helpful. However, the family has had difficulty following through on prior recommendations for counseling.

DSM-IV DIAGNOSES:

294.9 Cognitive Disorder NOS; status post cerebral vascular accident (CVA)
296.3x Major Depressive Disorder, Recurrent
312.82 Conduct Disorder, Adolescent-Onset Type

SUMMARY: Keisha is a 17 year, 2 month old African-American female with HbSS sickle cell anemia who is status post cerebral vascular accident (CVA) approximately eleven years ago. She was retained in the 9th grade and is currently receiving special education services in high school. Clinical interview revealed a possible second CVA, previously unknown, that was allegedly diagnosed as an "allergic reaction" by emergency room physicians in Arkansas approximately three years ago. An updated MRI is scheduled for 9/21/2006.

Results of this comprehensive neuropsychological evaluation revealed *Impaired* cognitive abilities (WASI: VIQ=70, 2nd percentile, *Borderline*, PIQ=54, 0.1 percentile, *Impaired*). Significant deficits in attention and executive function were indicated. Keisha was unable to sustain her attention for more than 6 minutes on a low-response (boring) task. On verbal and nonverbal tasks requiring high-level conceptual reasoning, Keisha's performance was deficient compared to same-age peers. Learning and memory for language presented in context, such as stories, was *Average*. However, visual-spatial learning and memory for simple to moderately complex designs was *Below Average* and dropped to *Impaired* with increased complexity of the material. Cueing was mildly helpful for verbal but not for nonverbal information. Possible deficits in expressive and receptive language functions were suggested. Academic achievement was approximately 8 years below age-level expectations in reading and math compared to approximately 4 years below age-level expectations in spelling.

Significant problems in psychosocial and behavioral functioning were reported by multiple informants at home, school, and with peers. Specifically, Keisha in addition to her mother, grandmother, and teachers described problems of clinical significance in the areas of anxiety, depression, hyperactivity, inattention, withdrawal, leadership, social skills, learning, and behavior. Ongoing problems of clinical significance were reported for physical aggression to peers and gang involvement. There is a reported history of suicidal ideation with active plan, approximately one year ago, which Keisha confided in a trusted aunt. Current suicidal ideation and/or plan were denied. Keisha contracted to immediately seek out a responsible adult if suicidal ideation and/or plan should occur in the future. Despite these difficulties, Keisha presents as a likeable young woman who seems distressed about her school failure yet lacks understanding as to how to alter her current maladaptive academic and psychosocial behaviors. Risk for school failure and drop out are assessed as extremely high. Keisha's knowledge and age-appropriate responsibility for medical management of her sickle cell anemia is well below expectations. Without significant intervention, the prognosis for this young lady is not positive.

RECOMMENDATIONS:

1. Continued consultation with the Director of the STAR Program at SJCRH, to further consider educational advocacy and vocational planning is strongly recommended. A highly structured academic program, with appropriate support and objectively rendered consequences, is strongly recommended. Teachers will need to become aware of Keisha's true cognitive abilities so as to structure expectations appropriately. It should be noted that Keisha seems to respond well to praise, reinforcement, and reward. Keisha should be encouraged to pursue her vocational aspirations at a level commensurate with her abilities and interests. Educators with the STAR Program were informed of the results of this evaluation.

2. Immediate need for counseling continues to be strongly recommended, particularly given newly disclosed report of a prior history of suicidal ideation with a plan. Current suicidal ideation and/or plan was denied. However, Keisha presented as tearful and fragile when discussing this history and her current psychological state. Continued assistance to the family in following through with this prior recommendation for counseling is of high priority. The LCSW was informed about these results. The family was informed and appeared to understand that if Keisha becomes suicidal in the future, they should take her to the emergency room to ensure safety. Keisha contracted for safety and agreed to inform her mother, grandmother, or other responsible adult if she were to become suicidal in the future. Keisha's current gang association and extremely high risk for school failure and "drop out" were discussed at length with Keisha and her mother and grandmother. This was allegedly new information to the mother and grandmother.

3. Recommendation for a speech/language evaluation is made. Of note, there were times when Keisha did not seem to hear the ending of select words, despite repetition. She reports and demonstrates considerable difficulty elaborating on her thoughts in both oral and written language. This relative weakness may be a reflection of her global cognitive abilities. However, additional evaluation seems warranted and may help elucidate findings and contribute to educational and vocational planning.

4. Educational recommendations for additional time on tests and assignments, reduced homework load, and ability to complete written assignments in alternative formats are suggested. For example, portfolio-based assessment and independent learning projects that allow a creative and hands-on demonstration of her knowledge should be encouraged.

5. Opportunities for appropriate and supervised peer socialization are strongly recommended. Participation in the Adolescent Support Group for teens with sickle cell anemia continues to be recommended. In the past, Keisha has declined participation in this group. Of note, in the interview, she expressed interest in attending the group. This information was communicated to the Teen Clinic Nurse Coordinator and Social Worker who co-lead the monthly Teen Group. Efforts will again be initiated to encourage Keisha to participate in the Teen Group. Also, Keisha's English teacher, Ms. Smith, described two positive peer socialization clubs available at school. She agreed to, again, encourage Keisha to consider participation.

6. Keisha, her mother, and grandmother were informed of the results of this neuropsychological evaluation and appeared to agree with and understand the findings. It was a pleasure to evaluate Keisha and to work collaboratively with her mother, grandmother, and school-based professionals. If there are additional questions, please do not hesitate to contact me at 901-495-5040.

_____

Leslie Berkelhammer, Ph.D.
Chief Neuropsychologist, Sickle Cell Program
St. Jude Children's Research Hospital

cc: Mother (legal guardian)
Grandmother (durable power of attorney for health care decisions)
Relevant providers on the sickle cell disease multidisciplinary treatment team

CHAPTER 18

Assessment of Psychopathology

Stefan E. Schulenberg, Jessica T. Kaster, Carrie Nassif, and
Erika K. Johnson-Jimenez

INTRODUCTION

Psychological assessment has long been a core aspect in the training and practice of applied psychologists, with trends over time characterized by remarkable consistency (Archer, Maruish, Imhof, & Piotrowski, 1991; Archer & Newsom, 2000; Camara, Nathan, & Puente, 2000; Cashel, 2002; Watkins, Campbell, Nieberding, & Hallmark, 1995). Applied psychology includes such fields as clinical psychology, counseling psychology, school psychology, and neuropsychology. Assessment comprises many areas (e.g., personality, behavior, intelligence) and is conducted in many professional contexts (e.g., inpatient, outpatient, forensic, medical, educational and occupational settings). For clinicians who work directly with consumers in applied settings, some spend a significant portion of professional time conducting assessments, with the remainder of time divided among other activities, such as therapy, research, teaching, supervision, consultation, and administration (Archer & Newsom, 2000; Cashel, 2002; Watkins et al., 1995). Assessment of personality and psychopathology is common, particularly among clinical psychologists (Camara et al., 2000). As for youth assessment, testing is a major aspect of psychological services that is experiencing swift expansion in measure development, with a concomitant challenge for clinicians to keep up with this growth (Geffken, Keeley, Kellison, Storch, & Rodrigue, 2006; Kamphaus, Petoskey, & Rowe, 2000).

The purpose of this chapter is to discuss issues in the assessment of personality in older children and adolescents, with particular regard for the evaluation of psychopathology. A comprehensive approach to youth clinical assessment is presented, including a discussion of the referral questions, informed consent and assent, rapport, clinical and collateral interviews and collateral records, test and measure selection, cognitive and achievement testing, self-reports via objective personality inventories, and behavioral rating scales (completed by youths, their parents, and their teachers) and observations. The controversy surrounding projective techniques is addressed, followed by a discussion of report writing and feedback. Potential complicating factors to conducting comprehensive assessments are outlined, as well as suggestions for developing assessment competency. The perspective of the authors is that of clinical psychology; however, the chapter has relevance to other

applied fields of psychology. Although a number of areas are discussed, the major focus is on how each area contributes to the understanding of psychopathology, and therefore greatest attention is paid to personality and behavioral assessment, where behavior is viewed as an external manifestation of personality.

THE IMPORTANCE OF IDENTIFYING PSYCHOPATHOLOGY IN OLDER CHILDREN AND ADOLESCENTS

Clinicians frequently are called on to assess disruptions in personality and behavior for a variety of referral questions, such as accurate diagnostic classification and assisting in psychological care and treatment monitoring. Given the variety of problems with which youths present, be they internalizing (depression, anxiety) or externalizing (Attention-Deficit/Hyperactivity Disorder or ADHD, Conduct Disorder), early intervention, accurate diagnosis, and congruence between consumer abilities and resources and treatment recommendations are critical. Kasen, Cohen, Skodol, Johnson, and Brook (1999) hypothesized that experiencing any disorder is disruptive to the general developmental process, but especially so with regard to social and interpersonal skills. A delay or lack of resolution of these developmental tasks would logically manifest in a higher risk of subsequent disturbed personality functioning. Additionally, Ramklint, von Knorring, von Knorring, and Ekselius (2003) found that the presence of a Major Depressive Disorder, disruptive disorder, and substance use disorder in childhood or adolescence each independently increases the odds of personality disorders in adulthood. Finally, youths with Conduct Disorder appear to be at increased risk for developing Antisocial Personality Disorder in adulthood, the connection of which is made in the *Diagnostic and Statistical Manual of Mental Disorders,* fourth edition, text revision (*DSM-IV-TR;* American Psychiatric Association, 2000; Kimonis & Frick, 2006).

Comprehensive assessment of psychopathology can aid the clinician in demarcating the line between adaptive and maladaptive functioning, accurately identifying problematic symptoms and behaviors, and developing a thorough and appropriate treatment strategy that is tailored to the unique attributes of the youth. Are the behaviors or symptoms extreme variations of otherwise normal development (and likely to be transient)? Or are the symptoms or behaviors no longer within the acceptable range and indicative of a potentially severe or chronic pathological syndrome that is causing impairment in functioning? Assessment is well suited to providing illumination to such important questions.

ASSESSING PSYCHOPATHOLOGY IN OLDER CHILDREN AND ADOLESCENTS

Clarifying the Referral Questions

Prior to beginning the assessment, it is necessary to clarify the referral questions and establish a collaborative working relationship with the consumer (Brenner, 2003; Harvey, 2006; Schroeder & Gordon, 2005; Snyder, Ritschel, Rand, & Berg, 2006). What is it that the referral source (e.g., parents or legal guardians) wishes to learn

about the youth's functioning? Working with the referral source throughout the evaluation is integral to conducting quality psychological assessments, in selecting the tests that are most appropriate to answering the referral questions effectively, and in tailoring the report to the referral questions (Brenner, 2003; Michaels, 2006; Snyder et al., 2006). With well-defined referral questions that take into account areas of difficulty and strength, clinicians can shape the evaluation and the report to the unique needs of the individual. Vague referral questions result in less precise evaluations, despite the efforts of clinicians to determine what problems may be evident. In many cases, the referral source will be a parent requesting assistance with better understanding a child's functioning or current problems. However, referral sources may also involve physicians, other mental health professionals, schools, or the courts.

Informed Consent and Assent

A variety of issues must be covered to obtain informed consent (American Psychological Association, 2002; Urbina, 2004). What purpose does the evaluation serve? How will the results be used? Who will receive a copy of the results? Will verbal feedback be available given the context of the assessment? Are the consumers apprised of the nature of confidentiality and the limits thereof? It is also important to determine the extent of the information requested to guard the youth's privacy. For instance, a clinical or forensic evaluation will contain more personal data than an evaluation conducted during a school-based assessment. When conducting assessments with older children and adolescents, informed consent must be obtained from the parent or legal guardian, and obtaining assent from the youth being tested is highly important to facilitate cooperation and build rapport.

Assent entails the explanation of the nature and purpose of the evaluation in language appropriate for the youth's developmental level. What is the youth's understanding as to why he or she is being assessed? Obtaining assent is an intricate and evolving issue, as it is unclear at what age a child may make an informed decision. Although offering a youth the opportunity to assent can be an affirming, rapport-building action on the part of the clinician, the question arises as to whether a youth with serious emotional problems referred by his or her parents should be able to refuse (Merrell, 2003). Depending on the maturity of the child, it is helpful to discuss the pros and cons of compliance, how the information will be used, the limits of confidentiality, and the like, offering youths the opportunity to have their questions answered.

Establishing Rapport

Through the clarification of the referral questions and by working with youths (assent) and their parents or legal guardians (informed consent), the chances that rapport will be healthy are enhanced significantly. The importance of finding ways to encourage youths and their parents to devote themselves throughout the assessment process cannot be overstated. Absence of rapport can negatively influence the evaluation to the extent that the results may be invalid, and it is necessary to prepare individuals and encourage them to be personally invested to do their best

on measures of ability and to respond frankly on personality instruments (Urbina, 2004). To build rapport with children, the examiner should be flexible and patient and possess an understanding of youth development; in addition, discussing topics familiar to youths, such as movies, sports, or computers, may be of assistance (Morrison & Anders, 1999).

As for adolescents, these youths often present difficulties for clinicians due to response styles characterized by resistance or denial (McCann, 1998). For instance, when adolescents are asked to complete the Minnesota Multiphasic Personality Inventory—Adolescent (MMPI-A) as a part of an assessment battery, they are essentially being asked to complete a long, exam-like questionnaire, which results in the possibility that they will not give their best effort (Archer & Krishnamurthy, 2002). Here we return to the issue of assent. It is important that the purpose of testing and the use of the results be explained (adolescents should be made aware that total confidentiality cannot be provided given their status as a minor), with opportunities for questions to be asked and answered, and with a clear indication that feedback of the results will be provided; such efforts demonstrate respect and create opportunities for youths to become involved (Archer, 2005; Archer & Krishnamurthy, 2002). It has been suggested that a firm, task-oriented approach may be helpful to solicit participation that will lead to meaningful results (Archer, 2005; Archer & Krishnamurthy, 2002). Other ways of increasing rapport include noting where restrooms and drinking fountains are, praising effort during testing, and offering adequate breaks (Archer & Krishnamurthy, 2002).

Conducting Interviews: Considering the Cultural Context

Adolescence is a time of transition, of expanding cognitive, physical, social, and personality development. The perceived and actual number of stressors experienced varies from youth to youth and is moderated by a diverse array of variables (e.g., coping skills, acculturative stress), and thus there may be a high level of variation when considering individual development.

For psychopathology to be accurately identified and understood, it must be assessed not only in the developmental context, but in the context of culture, gender, language, and socioeconomic status as well (Achenbach & Rescorla, 2007; Jones et al., 2001; Kuperminc & Brookmeyer, 2006; Leong, Levy, Gee, & Johnson, 2007). From culture to culture there may be different emphases on the importance of some personality traits being expressed over others, how youths should be socialized and values transferred, and how normal and unusual behaviors are defined. Thus, a behavior that is problematic in one cultural context may not be problematic in another. Clinicians also need to be aware that variations within cultures (from individual to individual) are just as significant as variations across cultures (Achenbach & Rescorla, 2007; Jones et al., 2001; Leong et al., 2007).

Cultural variables that clinicians should consider include acculturation, racial/ethnic identity, and minority status stress, all of which can significantly impact the results obtained and interpretations made in an assessment (Jones et al., 2001; Leong et al., 2007; Strack, Dunaway, & Schulenberg, in press). A variety of measures are available to assess these constructs; however, they are not used routinely

in clinical practice. Further research and developmental efforts must be undertaken with individuals of varied ages and backgrounds, and especially with youths, so that at some future point clinicians will be able to routinely and systematically measure these variables as part of a larger assessment battery when clinically indicated.

Another way that these variables may be examined is through clinical interviews. What values are important to the family? How strongly does the youth identify with his or her cultural background, and how does this affect his or her worldview? What communication style is most comfortable to the youth and his or her family? What is the family's perspective on mental illness, assessment, and mental health treatment? If there is a previous history of mental health assessment or intervention, this should be determined to assess the potential impact on the youth and his or her family. In the case of multicultural assessments, this may be of particular relevance given that some individuals may have had negative experiences that could influence the current assessment, and those with limited experience could be influenced by their expectations about what will happen (Allen, 2002; Strack et al., in press). These circumstances reflect on the importance of rapport and informed consent with parents or legal guardians and assent with youths.

Clinical Interviews: Selecting a Format

There are a variety of kinds of interviews that can be conducted, generally falling under the categories of unstructured, structured, and semi-structured. Unstructured interviews consist of open-ended questions that allow the interviewee to talk with little guidance from the examiner. Strengths of unstructured interviews are flexibility and facilitation of rapport (Schroeder & Gordon, 2005; Wiger & Huntley, 2002). Although follow-up information can be gathered based on what is learned during the interview, the content and quality of the information gathered, as well as the diagnoses assigned, may be substantially influenced by the training, experience, and perspective of the interviewer (Wiger & Huntley, 2002).

Structured interviews (increased exactitude, less flexibility) increase the psychometric properties of the data gathering process (Faul & Gross, 2006; Karg & Wiens, 2005; Wiger & Huntley, 2002). Questions are asked to determine the presence or absence of clinical symptoms and minimize the role of interviewer bias. Often there are versions for children and parents available. Structured interviews use the same questions across interviewees to maximize the chances that diagnostic areas are not overlooked, and they may help uncover problematic areas not identified in other sources (such as collateral records); however, although structured interviews tend to result in increased precision, the rigid format may interfere with rapport and the collecting of data not included on the interview form (Wiger & Huntley, 2002).

Semi-structured interviewing, a combination of the unstructured and structured approaches, is often used by clinicians to account for diagnostic precision and the uniqueness of the interviewee (Wiger & Huntley, 2002). It consists of a more guided approach, but also allows for following leads and inquiring further about information gathered. The interviewer decides which areas need to be addressed, such as school, home, friends, behavior, and history of the problem. While it is

generally the case that a semi-structured approach is useful in terms of collecting data and meeting the needs of the interviewee, clinicians themselves must be flexible in terms of their own style to account for important moderating factors (e.g., culture). There are numerous sources available for information on formal structured and semi-structured interviews (e.g., Loney & Frick, 2003; Orvaschel, 2006; Rourke & Reich, 2004; Shaffer, Fisher, & Lucas, 2004). The remainder of this discussion focuses on the standard clinical interview, in which a clinician probes a variety of symptom constellations and areas, using flexibility as necessary to evaluate for diagnoses and also to collect information relevant to understanding the youth's unique attributes. We begin with the parent or legal guardian interview.

Interviews with Parents or Legal Guardians

When conducting interviews, the clinician should be collecting data in such a way as to understand what events are occurring in the youth's life, when symptoms manifested themselves, whether there is an external cause evident, and whether symptoms wax and wane in accordance with certain events. Depth of understanding of contextual factors and how they intertwine with symptoms, coupled with an understanding of the youth's unique patterns of strength, will afford the clinician greater opportunities to discern whether a diagnosis is warranted, as well as to plan appropriate intervention, if needed.

When interviewing parents or legal guardians it is important that they feel comfortable and that the examiner is in tune with their concerns. It is ideal to interview both parents; however, this may not be possible given family dynamics (e.g., work schedule conflict, divorce). Interviews with parents or legal guardians offer a great deal of information about the youth's history and current functioning. In addition to providing information, parents can be brought on board early in the process, which can be beneficial when it is time to present and work on treatment interventions (Schroeder & Gordon, 2005).

Data to be gathered during an interview generally includes duration, frequency, and severity of youth behaviors and symptoms. Interviewing the child or adolescent independently, as well as together with parents or legal guardians, can yield information as well as family dynamics. Various symptom constellations should be addressed, whether or not they appear as part of the presenting complaint, to clarify the diagnostic picture. Other information to be gathered includes behavioral observations, developmental history (e.g., meeting of developmental milestones), family physical and mental health history, social history (e.g., friendships, relationships with siblings), losses in the family, and medical (e.g., injuries, medication allergies) and mental health (e.g., previous diagnoses, treatment) histories. Data should also be collected on the child's or adolescent's educational and cultural background, as well as occupational history in the case of some older adolescents. The interviewer can also inquire about parenting style, approach to discipline, the youth's behavior at home, and youth strengths. Focusing on these and related areas has been described to varying degrees in a number of sources (e.g., Merrell, 2003; Schroeder & Gordon, 2002 2005; Snyder et al., 2006). Finally, it is important to convey to parents the idea that positive action may be taken, and if additional data

are needed it is paramount to explain the rationale and methods consistent with the collaborative approach (Schroeder & Gordon, 2005).

Interviews with Older Children and Adolescents

With regard to conducting interviews with youths, behavioral observation and life history data are quite important to the overall quality of the assessment and should not be disregarded in favor of interviews with parents or legal guardians. Rather, a clinician may gather relevant information from parents or legal guardians and the youth being evaluated. Clinicians interviewing youths should expect to see variations based on such variables as development, education, and culture.

School-age children (ages 6 to 12) typically think in factual, literal, and rigid terms (they are able to consider some, but not all, eventualities) and are likely to participate little in discussions characterized as abstract or involving feelings and attitudes (Gudas & Sattler, 2006). Children may be concerned with how well they are doing on psychological tests and with pleasing the interviewer and/or deferring to the wishes of their parents, legal guardians, or other authority figures. Rapport building with youths is needed to account for such potential complicating factors.

Adolescents, in comparison to school-age children, think less concretely and are able to consider a substantially wider range of alternatives and to test hypotheses (Gudas & Sattler, 2006). They are able to report on symptoms and to consider elements of time and place; however, although they can think with increasing abstraction, problems with risk taking and invincibility may be evident (Gudas & Sattler, 2006). Their peer group tends to take on more influence in the determination of behavior, a role that typically was previously held by parents or legal guardians. Although this differentiation is often regarded as a part of developing into independent adulthood, it has specific implications during the interview. Adolescents tend to be more concerned about how they fit in and how they will be judged by others than with reporting how they actually experience things. They may be actively searching for an identity, including how they relate to ascribed gender and cultural roles. The adolescent interviewee may present at once as being genuinely both naive and blasé. Having an approachable yet frank presence can help foster a vitally important sense of acceptance and build rapport necessary to facilitate the interview.

As with older children, adolescents are capable of providing information useful to skilled clinicians. It is important for youths to know that their views are important to the clinician, and for the clinician such data are imperative in the process of developing and refining clinical hypotheses. In interviews of youths, especially adolescents, it is important to establish confidentiality and the limits thereof. With regard to interview content, it is important to assess such areas as drug and alcohol use, behaviors harmful to self or others, and sexual activity (Wiger & Huntley, 2002), in addition to the other areas outlined earlier.

Collateral Interviews

With appropriate consent, discussion with other relevant individuals, such as extended family members, teachers, and case managers, may provide important

contextual data about the youth's functioning across multiple settings. Information from as many sources as are reasonable will improve the ability of the interviewer to gain a comprehensive picture of the youth (Boggs, Griffin, & Gross, 2003). Such interview data may also be combined with behavior rating scales (discussed later) to enhance the clinical picture. For additional information on interviewing techniques and procedures as they relate to both clinical and collateral interviews, an array of sources is available (Groth-Marnat, 2003; Merrell, 2003; Morrison & Anders, 1999; Sattler & Hoge, 2006; Sharp, Reeves, & Gross, 2006; Wiger & Huntley, 2002).

Review of Collateral Records

The review of collateral records is of importance in improving the quality of psychological assessments. Are there certain medical or physical conditions that have a bearing on the current presentation? Are school records available that reveal academic performance or behavioral problems? In the case of adolescents in particular, if they are demonstrating behaviors that bring them into contact with the law, are police records available? Having access to collateral data is critical to understanding context, as a basis to facilitate discussion with the youth and his or her parents or legal guardians, and as a comparison with test data. Collateral records aid the clinician in establishing the youth's history, facilitating an individualized case conceptualization for the presenting complaints.

Selecting Tests and Measures for the Evaluation

There is a distinction between testing and assessment. A test is standardized, reliable, valid, and clinically useful (Hunsley, Lee, & Wood, 2003), whereas assessment involves the integration of data from multiple sources, such as tests (e.g., Wechsler Intelligence Scale for Children, fourth edition [WISC-IV]) and nontests (e.g., collateral records). Psychometric properties of a test may vary depending on the individual being tested (based on cultural or socioeconomic factors), the context (a test may be psychometrically sound for one clinical purpose but not another), and the composition of the test (tests with multiple subscales will have different psychometric properties for each scale). In a survey of psychologists who work with adolescents, Archer and Newsom (2000) learned of an array of factors relevant to instrument selection, including soundness of psychometric properties, relationship between the instrument and the referral question, the relevance of the generated data for treatment planning, provision of comprehensive information pertinent to psychopathology and diagnosis, and simplicity of scoring and administration.

As for a practical example of how tests may be selected, consider a relatively common referral question: "My teenage son is getting into fights in school, he's struggling academically, and he is having trouble paying attention to his teachers. What's wrong? Does he have ADHD?" An assessment battery may be composed of interviews with the youth and his parents; collateral interviews (e.g., with a teacher); a review of collateral records (e.g., report cards and other school records); intelligence and achievement testing (e.g., WISC-IV, Wechsler Individual Achievement Test, second edition [WIAT-II]); a continuous performance test to assess for difficulties with attention; self, parent, and teacher ratings (e.g., Behavior Assessment

System for Children, second edition [BASC-2]); and additional measures of functioning as needed (e.g., MMPI-A, Children's Depression Inventory [CDI]). Why such a comprehensive battery? Because ADHD is comorbid with learning disorders, externalizing behavior disorders such as Oppositional Defiant Disorder and Conduct Disorder, and internalizing disorders such as depressive and anxiety disorders (Sattler, Weyandt, & Willis, 2006). A thorough assessment facilitates a detailed diagnostic formulation. As a case in point, inattentiveness may be taken for ADHD, when in some cases it may be related to depression or anxiety. Aggression could be part of a larger constellation of symptoms meeting criteria for a diagnosis such as Conduct Disorder. A partial assessment may miss the mark.

Another consideration for clinicians when selecting a test is the degree of fit between the test and the language and cultural background of the respondent (Butcher & Pope, 2006; Flores & Obasi, 2003). This means determining that the norms, standardization, reliability, and validity are acceptable for the individual being evaluated and that the instrument is in the person's preferred language, all of which are necessary to minimize erroneous clinical interpretations (Flores & Obasi, 2003; Jones et al., 2001; Leong et al., 2007; Strack et al., in press). It is heartening that in recent years additional attention has been paid to multicultural issues in assessment (both within the United States and internationally), and this emphasis will likely continue; however, it is disheartening that empirical research continues to lag behind the need (Leong et al., 2007; Strack et al., in press; Wood, Garb, Lilienfeld, & Nezworski, 2002).

Cognitive and Achievement Testing: Clues to Personality Style?

In addition to clinical and collateral interviews and collateral records reviews, cognitive and achievement testing may reveal clues about personality functioning that may be evaluated using more systematic, standardized, and comprehensive methods. Of the intelligence tests used, the Wechsler scales are among the most commonly employed by clinicians and are quite common in adolescent and child batteries specifically (Archer & Newsom, 2000; Camara et al., 2000; Cashel, 2002; Kamphaus et al., 2000). The current edition of the Wechsler scales appropriate for older children and most adolescents is the WISC-IV (Wechsler, 2003).

Why mention a cognitive test in a chapter on assessment of psychopathology in older children and adolescents? As many clinicians know, behavioral observations are a heavily emphasized part of clinical assessments, and observing a youth's performance on the WISC-IV, and on achievement tests such as the WIAT-II (The Psychological Corporation, 2002), may reveal hypotheses about how the youth interacts with his or her world that may be examined subsequently through other methods. Looking at the WISC-IV specifically, a variety of tasks are asked of the youth, including those that tap performance-oriented abilities (e.g., Block Design) and verbal-oriented abilities (e.g., Comprehension). The administration of each subtest yields behavioral observation information for the clinician. For example, as noted by Sattler and Dumont (2004), during Block Design, where the task involves the examinee putting together blocks to match a series of designs, a number of hypotheses about performance may be generated. Is the youth deliberate or

impulsive? Does the youth persist in the task or give up easily? Fumbling too much may be indicative of anxiety, youths very slow in their work may be experiencing boredom, and youths who constantly check their work may be insecure (Sattler & Dumont, 2004).

Observations similar to these may be collected throughout a test such as the WISC-IV and also may be made in the area of achievement testing. In the case of the WIAT-II, what is the youth's attitude toward perceived success and perceived failure? Does the youth show resilience or quickly give up? How does the youth respond to the praising of effort? Such clinical observations and interpretive hypotheses are valued by many clinicians as a means to better understand the functioning of youths, but clinicians must exercise caution in relying on behavioral observations in isolation or otherwise placing too much emphasis on them. Intelligence and achievement tests are not standardized for these purposes, and interpreting clinical observations appropriately can be difficult, as there are many reasons why a youth may respond in a certain way in a certain context. Also, there is no guarantee that informal observations or interpretations relating to personality style will translate to the youth's typical environments. There is a danger for clinicians to form erroneous impressions early and stay with them. Multiple and appropriate standardized sources of data should be considered, and systematic evaluation of disconfirming data should be undertaken, prior to arriving at a diagnostic formulation.

Objective Comprehensive Measures of Psychopathology

Objective instruments of personality and psychopathology are typified by standardization and norms, as well as straightforward items, response format (e.g., true-false), and scoring (Segal & Coolidge, 2004; Urbina, 2004). As compared to specific domain measures (discussed later), comprehensive measures tend to be characterized by several hundred items and require added time to administer; however, they tend to have better norms and a wider range of applications and generate many scores (Hogan, 2007). Comprehensive measures are often desired in clinical assessments given the thorough data they provide across a range of domains. These data are generally useful for hypothesis testing and treatment planning. Commonly used comprehensive measures in youth assessments are the MMPI-A and the Millon Adolescent Clinical Inventory (MACI; Archer & Newsom, 2000; Cashel, 2002), both of which are considered to be effective tools (Groth-Marnat, 2003).

The Minnesota Multiphasic Personality Inventory—Adolescent

The MMPI-A (Butcher et al., 1992) is composed of 478 true-false items, is based on a large and diverse normative sample, and is used in a variety of settings to assess emotional functioning and/or refine clinical hypotheses in adolescents between the ages of 14 and 18 (Archer & Newsom, 2000; Baer & Rinaldo, 2004; Butcher & Pope, 2006; Graham, 2006). The normative sample consisted of 805 males and 815 females, and data on a clinical sample were also gathered (420 boys, 293 girls; Baer & Rinaldo, 2004). The MMPI-A contains a number of validity scales to detect random or unusual responding, and its many clinical (e.g., Depression, Psychasthenia, Psychopathic Deviate), content (e.g., School Problems, Conduct Problems), and

supplementary scales (e.g., Alcohol-Drug Problem Acknowledgment, Alcohol-Drug Problem Proneness) provide the clinician with a wealth of potentially useful data.

Raw scores are translated into *T* scores (uniform or linear translation, depending on the scale), and though there is no clear cutoff for scale elevations, scores as low as 60 to 64 possibly have interpretive significance (Baer & Rinaldo, 2004; Butcher & Pope, 2006; Graham, 2006). Interpretation is undertaken knowing the larger context of the adolescent's life history and presenting problems, proceeding with an examination of the MMPI-A's validity scales; if the profile is valid, the clinical scales are interpreted, with content and supplementary scales aiding in the refinement of clinical hypotheses (Baer & Rinaldo, 2004; Butcher & Pope, 2006). Thus, although the MMPI-A yields many data regarding an adolescent's personality functioning, clinicians must be aware of the psychometric properties of the many validity, clinical, content, and supplementary scales. For further information regarding developmental, psychometric, and interpretive considerations, numerous resources are available (e.g., Archer, 2005; Archer & Krishnamurthy, 2002; Archer, Krishnamurthy, & Stredny, 2007; Archer, Zoby, & Stredny, 2006; Butcher & Williams, 2000; Frick & Kamphaus, 2001).

There are a variety of positive aspects to using the MMPI-A, including provision of comprehensive information, useful adolescent-related material and norms, simplicity of administration, and research support with a wide variety of clinical populations (Archer & Newsom, 2000; Butcher & Pope, 2006). The measure is available in a standardized recorded version, it has been translated into Spanish with adolescent norms available for Spanish-speaking youths in the United States, and a variety of translations are available in languages such as Arabic, Chinese, Korean, and Hebrew, to name a few (Butcher & Pope, 2006).

Negative aspects of using the MMPI-A include the length of the measure and the test-like demands placed on the examinee, a demanding reading level for many adolescents (about seventh grade), time demands (administration, scoring, interpretation), and cost when considering the context of a managed care environment (Archer & Newsom, 2000). Additionally, Frick and Kamphaus (2001) criticized the MMPI-A's normative sample for undersampling Hispanic youths and 18-year-olds. Finally, although some multicultural studies have been conducted (e.g., Cashel, Rogers, Sewell, & Holliman, 1998; Gumbiner, 1998; Negy, Leal-Puente, Trainor, & Carlson, 1997), Baer and Rinaldo's (2004) review noted that in some cases there were no differences between racial/ethnic groups studied, but that when differences were found it was difficult to determine what the findings meant, or whether the differences were clinically meaningful. More multicultural research is needed (Gómez, Johnson, Davis, & Velásquez, 2000; Leong et al., 2007).

The Millon Adolescent Clinical Inventory

The MACI (Millon, Millon, & Davis, 1993) is a 160-item, true-false, self-report instrument based on the theory of personality and psychopathology put forth by Millon (Millon & Grossman, 2005; Tringone, 2002; Tringone, Millon, & Kamp, 2007). It is composed of 31 scales across four domains: Personality Patterns, Expressed Concerns, Clinical Syndromes, and Reliability and Modifying Indices (Millon &

Grossman, 2005; Tringone, 2002; Tringone et al., 2007). The MACI requires approximately a sixth-grade reading level and is useful for evaluating areas of concern, difficulties with personality styles, and psychopathology with adolescents ranging in age from 13 to 19 (Millon & Grossman, 2005; Tringone, 2002; Tringone et al., 2007). The measure has four separate norm tables (13- to 15-year-old males and females and 16- to 19-year-old males and females) seen in clinical settings, with the overall norm group consisting of over 1,000 adolescents from the United States and Canada (McCann, 2006; Tringone, 2002; Tringone et al., 2007).

Item development comprised a three-stage process: theoretical-substantive, internal-structural, and external-criterion (Millon & Grossman, 2005; Tringone, 2002; Tringone et al., 2007). The MACI utilizes base rate scores (BR), as opposed to *T* scores, where BR cutoff scores of 85 and 75 indicate trait prominence and presence of a trait, respectively (McCann, 2006; Millon & Grossman, 2005). Interpretation occurs in the context of the youth's history and involves systematic review and subsequent integration of the Modifying Indices (validity indicators), Personality Patterns (e.g., Inhibited, Unruly, Oppositional), Expressed Concerns (e.g., Sexual Discomfort, Social Insensitivity, Family Discord), and Clinical Syndromes (e.g., Substance-Abuse Proneness, Delinquent Predisposition, Anxious Feelings, Depressive Affect; McCann, 2006; Millon & Grossman, 2005; Tringone, 2002). The Expressed Concerns scales target areas that may be problematic to troubled youths, while elevations on the Personality Patterns and Clinical Syndromes scales signify psychopathology necessitating professional intervention (Millon & Grossman, 2005; Tringone, 2002). For more information on developmental, psychometric, and interpretive considerations, a variety of sources are available (e.g., McCann, 1999, 2006; Meagher, Grossman, & Millon, 2004; Pinto & Grilo, 2004; Tringone, 2002; Tringone et al., 2007).

The MACI may be useful in some clinical situations, such as when distortions in personality development appear to be evident, so long as such use is supported by the empirical literature and data are not used in isolation. Proponents of the MACI note that its advantages include its multistage construction, brevity, design for use with adolescent clinical populations (tapping both Axis I and Axis II symptoms), inclusion of Modifying Indices, and theoretical basis (Millon & Grossman, 2005; Tringone, 2002; Tringone et al., 2007). Noted disadvantages include a sixth-grade reading level being necessary (audiocassette administration is available for clients who speak English and Spanish), and more severe types of psychopathology such as thought and psychotic disorders, bipolar disorders, and paranoid ideation are not directly measured (McCann, 2006; Millon & Grossman, 2005; Tringone, 2002; Tringone et al., 2007). Moreover, few multicultural studies involving the MACI have appeared in the literature (for examples with Mexican American youths, see Blumentritt, Angle, & Brown, 2004; Blumentritt & Wilson VanVoorhis, 2004).

Objective Specific Domain Measures of Psychopathology

The MMPI-A and the MACI cast a broad net, assessing many different kinds of behavioral and emotional difficulties. Other measures assess specific constructs and therefore tend to have fewer items (faster administration times), yield fewer scores (perhaps a single score), and have norms and applications that are more limited

(Hogan, 2007). Specific domain instruments are often desirable to clinicians given their focus on a specific problem and the ease with which they may be combined with other data, such as comprehensive objective measures and interview data. Specific domain measures straddle many clinical areas, commonly depression and anxiety. Although a comprehensive review is not possible in a single chapter, we mention a few measures prior to discussing a recent constellation of specific domain measures that appears promising pending future research.

One of the most popular specific domain measures is the CDI (Kovacs, 1992). The CDI may be used with youths between the ages of 7 and 17. It includes 27 items of depressive symptoms with a choice of three response options (0, 1, or 2). Scores are summed, with higher scores considered to be more symptomatic. Raw scores are converted to *T* scores to yield a global score and five subscales: Negative Mood, Interpersonal Problems, Ineffectiveness, Anhedonia, and Negative Self-Esteem. The CDI is generally considered to be reliable (Campbell, 2006b; Merrell, 2003). However, given mixed discriminant validity support, and because it is more an indicator of distress as opposed to a diagnostic indicator, it should not be used in isolation (Fristad, Emery, & Beck, 1997; Schroeder & Gordon, 2002). This is excellent advice for any instrument, especially those of the specific domain variety. Sitarenios and Stein (2004) review the CDI's psychometric characteristics.

Additional self-report measures of depression include the Reynolds Child Depression Scale (RCDS; Reynolds, 1989) and the Reynolds Adolescent Depression Scale, second edition (RADS-2; Reynolds, 2002). The 30-item RCDS covers ages 8 to 13 years and is considered to thoroughly assess symptoms of depression, with good psychometric properties and adequate norms (Merrill, 2003; Reynolds, 2006; Schroeder & Gordon, 2002). The RADS-2 contains 30 items utilizing a 4-point response format and is used with youths between the ages of 11 and 20 (Campbell, 2006b; Reynolds, 2004, 2006). It yields *T* scores with a mean of 50 and a standard deviation of 10 and includes four factorially derived subscales in addition to the total score: Dysphoric Mood, Anhedonia/Negative Affect, Negative Self-Evaluation, and Somatic Complaints (Campbell, 2006b; Reynolds, 2004, 2006).

As for the measurement of anxiety, one self-report measure that has been receiving increasing attention is the Multidimensional Anxiety Scale for Children (MASC; March, 1997), which assesses anxiety in youths ages 8 to 19 years. The MASC's 39-item, 4-point response format yields *T* scores and is composed of a Total Anxiety score, Inconsistency Index, and Anxiety Disorder Index, as well as four factor scores: Physical Symptoms, Harm Avoidance, Social Anxiety, and Separation/Panic (Campbell, 2006a; Schroeder & Gordon, 2002). In reviews of the MASC, authors have noted the measure's reliability (particularly the internal consistency of the Total Anxiety score), convergent and discriminant validity support, and apparent utility in distinguishing youths with and without anxiety disorders (Campbell, 2006a; Schroeder & Gordon, 2002; Woodruff-Borden & Leyfer, 2006).

Beck Youth Inventories for Children and Adolescents, Second Edition

The Beck Youth Inventories for Children and Adolescents (BYI-II; Beck, Beck, Jolly, & Steer, 2005) are a recently available group of specific domain measures.

The original Beck Youth Inventories for Emotional and Social Impairment (Beck, Beck, & Jolly, 2001) were composed of five separate 20-item scales (Anxiety, Depression, Disruptive Behavior, Anger, and Self-Concept) that could be used separately or in combination with youths ages 7 to 14 (Beck et al., 2001; Bose-Deakins & Floyd, 2004; Steer, Kumar, Beck, & Beck, 2001). The scales appear to be internally consistent for clinical screening purposes, with growing evidence for convergent validity (Beck et al., 2001; Bose-Deakins & Floyd, 2004; Steer et al., 2001). However, Bose-Deakins and Floyd noted the need for additional discriminant validity data.

Given that the BYI-II was recently published, limited research is available beyond the manual at this time. The BYI-II continues the format of the original (second-grade reading level, five inventories of 20 questions each that may be used separately or in combination) with an expanded age range (7 to 18). Although initial psychometric properties appear promising, additional psychometric data are needed. Although the Anger, Anxiety, Depression, and Disruptive Behavior scales assess potentially problematic areas, it is noteworthy that the Self-Concept scale assesses positive attributes such as self-worth and may be one means of evaluating a youth's strengths. Considering that there are many specific domain measures available to assess personality and psychopathology in youths, it will be intriguing to see where these measures will eventually fit.

Behavior Rating Scales

Behavior rating scales are increasing in availability and popularity among clinicians (Archer & Newsom, 2000; Cashel, 2002; Kamphaus et al., 2000). They are advantageous in terms of their ease of use with a variety of respondents (youths, parents, teachers), cost effectiveness, and coverage of a wide variety of clinical domains such as ADHD and other disruptive behavior disorders (Elliot, Busse, & Gresham, 1993; Kamphaus et al., 2000; Piacentini, 1993). They are useful in providing data about low-frequency behaviors (e.g., assaults) that direct observation sessions may miss, assessing youths who are unwilling or unable to provide information about themselves, and accessing data in youths' typical environments, such as school and home settings, by those who know them well (Merrell, 2003). Potential disadvantages include their being subject to the perceptions of raters (e.g., response biases) and that they tend to focus on current functioning with little guidance as to etiology or intervention strategies (Elliot et al., 1993; Frick & Kamphaus, 2001; Merrell, 2003; Piacentini, 1993).

Rating scale systems frequently used by clinicians are the BASC, currently revised as the BASC-2, the Child Behavior Checklist (part of the Achenbach System of Empirically Based Assessment, or ASEBA), and the Conners Rating Scales (Archer & Newsom, 2000; Cashel, 2002; Kamphaus et al., 2000). For recent reviews of the ASEBA, see Achenbach (2007) and Achenbach and Rescorla (2006); for the Conners Rating Scales—Revised, see Kollins, Epstein, and Conners (2004). The rest of this discussion focuses on the most recently revised of these major rating systems, the BASC-2.

The BASC-2 (Reynolds & Kamphaus, 2004; see also Kamphaus, VanDeventer, Brueggemann, & Barry, 2007) contains separate rating forms for teachers, parents, and youths. The various 100- to 139-item Teacher Rating Scales (TRS) assess adaptive and problematic behaviors evidenced by the youth in the school setting, with the specific form used depending on the child's age (ages 2 to 5, 6 to 11, and 12 to 21). Clinical scales include Aggression, Anxiety, Atypicality, Depression, and Attention Problems. Adaptive scales include Adaptability and Functional Communication. The 134- to 160-item Parent Rating Scales (PRS; ages 2 to 5, 6 to 11, and 12 to 21) prompt parents to provide data on adaptive and problematic behaviors in the home and community. The 139- to 185-item Self-Report of Personality (SRP) forms provide information on the youth's perception of his or her functioning (child, ages 8 to 11; adolescent, ages 12 to 21; college, ages 18 to 25). Clinical scales include Attention Problems, Anxiety, Depression, Hyperactivity, Sense of Inadequacy, and Social Stress, and adaptive scales include Self-Reliance, Self-Esteem, Relations with Parents, and Interpersonal Problems. In general, there are many similarities across forms in terms of clinical and adaptive scales, item content, and composite scores, although there are some variations based on the age of the respondent and the rater (Kamphaus et al., 2007). Raw scores are converted to norm-referenced *T* scores to facilitate interpretation, and each of the sets of forms contains scales to evaluate the quality of obtained data (validity and response set indicators; Kamphaus et al., 2007). The BASC-2 also contains a Structured Developmental History (a data gathering tool) and a Student Observation System (a classroom observation method based on momentary time sampling).

With regard to how a rating system such as the BASC-2 may be used in a psychological evaluation, BASC-2 forms may be used to gather self-reports of youths (SRP), their parents or legal guardians (PRS), and their teachers (TRS). Such ratings generally provide psychometrically sound data to accompany clinical and collateral interviews. When feasible and depending on the clinical context, separate ratings may be sought from each parent, and if the youth has multiple teachers, more than one rater may be requested. Thus, rating scale data could be collected from three to five people (youth, one to two parents or legal guardians, one to two teachers) to get a better idea as to behaviors and symptoms that are considered problematic, to what degree, and in what context. Problems may be occurring at home and not in school, or vice versa; problems may be seen with one parent and not another, or with one teacher but not another. Such differences could be related to the clinical presentation or the environment, which has both diagnostic and treatment implications.

Behavioral Observations in Different Settings

As part of a multifaceted assessment, behavioral observations in the school, home, and other germane settings may be helpful in facilitating the development of a comprehensive picture of the context of the youth's behavior and how he or she interacts with others. Observations may take on a variety of forms, ranging from time-limited and informal to time-involved and intensely structured (Gudas & Sattler, 2006; Winsor, 2003). Although observations may be a rich source of data

that enhance the assessment, similar to interviews the quality of the data is largely dependent on the skill and objectivity of the observer and on the recording system that is employed (Urbina, 2004; Winsor, 2003).

The school is one environment where observations are typically conducted. The observation of older children (as opposed to adolescents, where observation tends to be more intrusive) in their school environment may be undertaken with the permission of the parents or legal guardians, the school, and teachers. With regard to aid in hypothesis generation, is the child inadvertently being reinforced for negative behaviors by the attention of a teacher? Are there alternative behavioral strategies that may be implemented by teachers to reinforce the child for more adaptive behaviors? Collecting behavioral observations may be helpful in certain cases, but they are somewhat limited by factors such as reactivity (behavior change in response to being evaluated) and the lack of ability to monitor internal affective states (Franz & Gross, 1998; Merrell, 2003; Tryon, 1998; Winsor, 2003). Direct observations in the schools are also substantially limited by the amount of time involved. Clinicians working under time constraints at community mental health centers, for example, may be hard-pressed to conduct observations of behaviors for this reason. For further information on direct observations, such as defining target behaviors, and the different forms of coding procedures (e.g., event recording, duration recording, time sampling) and observation systems, many references are available (e.g., Merrell, 2003; Miltenberger, 2001; Tryon, 1998; Winsor, 2003).

PROJECTIVE TECHNIQUES: A PSYCHOMETRIC CONTROVERSY

Projective techniques have been, and continue to be, an enduring aspect of the training and practice of the clinician (Camara et al., 2000; Hogan, 2007; Watkins et al., 1995), and they are commonly employed with children and adolescents (Archer & Newsom, 2000; Cashel, 2002). Projective techniques rely on vague stimuli and a free-response format. The idea is that with freedom to answer, the person's deep-seated personality characteristics will shape the response, revealing important information about underlying desires, motivations, perceptions, and so on (Hogan, 2007; Segal & Coolidge, 2004; Urbina, 2004). Examples of popular projective techniques are the Rorschach, the Thematic Apperception Test (TAT), the Children's Apperception Test, sentence completion forms, and projective drawings such as Human Figure Drawings, House-Tree-Person, and the Kinetic Family Drawing (Hogan, 2007; Urbina, 2004; Watkins et al., 1995).

On the Controversial Nature of Projective Techniques

Some have called into question the psychometric relevance of projective techniques to assessment and diagnosis, noting that they are not science but pseudoscience, and that in the rare cases where they are useful they do not appear to add much beyond other methods, such as interviews and objective inventories (Hunsley et al., 2003; Lilienfeld, Wood, & Garb, 2001). Another criticism centers on the lack of standardization among many of the techniques and the question of how well

clinicians adhere to procedures with the techniques that are standardized (Hunsley et al., 2003). Standardization is necessary to counter problems with subjectivity in administration, scoring, and interpretation. Despite such criticisms, projective techniques continue to be used with great frequency. Proponents argue that they are a source of insight beyond what objective inventories can provide, that they assess personality at a deeper level. Another justification is that they may be useful as a means to settle in to the assessment, perhaps as an "ice-breaker" (Hogan, 2007, p. 522).

Projective Drawings

Projective drawings may be popular for their rapport-building capabilities or as a means of facilitating discussion, because youths often enjoy such activities (Cashel, 2002; Hunsley et al., 2003). Some clinicians use projective drawings to get a sense of the youth. Others use specifics of the drawing, or signs, to make diagnostic inferences about personality and psychopathology (Lilienfeld et al., 2001). The validity of using signs is quite limited (Lally, 2001; Lilienfeld et al., 2001), and the diagnostic relevance of projective drawings to assessments has often been questioned (e.g., Motta, Little, & Tobin, 1993; Palmer et al., 2000; Stawar & Stawar, 1989). Some have argued that they should not be used in personality assessments given their psychometric and diagnostic shortcomings and considering the lack of scientific data on the variability of drawings by age, gender, and interest in the task (Hunsley et al., 2003; Motta et al., 1993). Such data are lacking with regard to multicultural variations in drawings as well.

With regard to formal scoring systems, some clinicians use them, but it does not seem to be a common practice. Structured scoring systems may fare better given the integration of multiple pieces of data (Lilienfeld et al., 2001). However, even the utility of one of the most recent attempts to establish a psychometrically sound, structured scoring system, the Draw-a-Person: Screening Procedure for Emotional Disturbance (Naglieri, McNeish, & Bardos, 1991), has been criticized as flawed, with poor validity and limited peer-reviewed, published data available (Lally, 2001; Motta et al., 1993). Others indicate that the psychometric success of such systems is "modest" (Hogan, 2007, p. 542), or holds "very limited promise" (Hunsley et al., 2003, p. 65). Ultimately, although some use projective drawings as a means of communicating with young clients and in establishing rapport, clinicians must exercise due caution given that the nature of the task pulls for impressionistic interpretation.

Sentence Completion Techniques

Sentence completion techniques also enjoy widespread use, as is evident from surveys of clinical assessment practices. They are not one technique, but rather a group of forms with similar formats, where the respondent completes a series of sentence fragments, or stems, the process of which is thought to reveal something about certain aspects of personality style (Hogan, 2007; Holaday, Smith, & Sherry, 2000; Rogers, Bishop, & Lane, 2003). Given the variety of sentence completion techniques available, clinicians may not be aware which form is best for which client,

the psychometric properties behind the form, or even the name of the form they are using (Holaday et al., 2000; Rogers et al., 2003). There is also some evidence that clinicians tend to use these measures informally, without recognized scoring procedures (Holaday et al., 2000). Interpreting sentence completion techniques through impressionistic methods, that is, based on clinical intuition, without a standardized and psychometrically sound scoring system, will leave clinicians susceptible to errors in decision making.

Storytelling Techniques

The Thematic Apperception Test (Murray, 1943) consists of 31 cards, where the goal is to solicit a narration from the respondent as to the content of the card. Through the narration the respondent is thought to reveal underlying attitudes, drives, and the like, and cards may be selected based on whether the respondent is male or female, a youth or an adult (Hunsley et al., 2003). A major problem with the TAT is standardization, as examiners select different sets of cards (different cards, different numbers of cards, and/or different orders of cards) for use with different examinees (Groth-Marnat, 2003; Hunsley et al., 2003; Lilienfeld et al., 2001). Each set of cards results in a different technique, and psychometric properties need to be established for each set. This property of the TAT also makes the evaluation of research difficult, as studies do not always report the combination of cards that were used. Although some scoring systems are available, there is currently no widely used, well-validated administration and scoring system for the TAT. Thus, those who use it in formal assessments risk misdiagnosis due to reliance on impressionistic interpretation (Hunsley et al., 2003; Lilienfeld et al., 2001). Finally, although a number of derivatives of the TAT have been developed for a variety of populations (e.g., the Children's Apperception Test; Bellak, 1954; see also Bellak & Abrams, 1997), the same caveats apply to the use of any narrative technique with regard to the importance of norms and standardized administration and scoring procedures in generating interpretations that are empirically supported as accurate and clinically useful.

The Rorschach

The Rorschach consists of 10 inkblots (one inkblot per card), and respondents reveal the way they perceive their world through their verbal responses to each card (Ritzler, 2005; Urbina, 2004). There are a variety of coding and scoring systems for the Rorschach; Exner's Comprehensive System (CS; Exner, 2001, 2003) is the most popular (Ritzler, 2005; Weiner, 2005). There are Rorschach proponents (e.g., Ritzler, 2005; Weiner, 2003, 2005), but there are those who raise serious psychometric criticisms. For example, it has been argued that the Rorschach's CS norms may result in erroneous clinical impressions (i.e., respondents may look like they have more pathology than they do or have psychopathology when they do not) and that there is a lack of data and inconsistent psychometric support for many of the CS scores (Garb, Wood, & Lilienfeld, 2005; Hunsley et al., 2003; Lilienfeld et al., 2001; Wood et al., 2002; Wood, Nezworski, Lilienfeld, & Garb, 2003). Psychometric data must be established for each Rorschach score, which is

quite a daunting task given the number of variables that make up the CS system. Psychometric properties will differ from variable to variable.

Some uses have been acknowledged even by critics of the method, as in the case of the Rorschach's apparent ability to assist in the evaluation of thought disorders (Garb et al., 2005; Hunsley et al., 2003; Wood et al., 2003). Beyond these rare points of consensus, however, the discussion is polarized, with impassioned debates characterized by point/counterpoint arguments appearing in various professional journals in recent years. The debate will likely continue for some time and is quite relevant to the assessment of older children and adolescents. It is not uncommon to find those who describe or otherwise support the use of the Rorschach with youths (e.g., Erdberg, 2007; Weiner, 2003, 2005). However, Wood et al. warn that the technique may overpathologize children, which may result in harm due to the possibility of inaccurate conclusions being drawn and a resulting incongruence of fit between the youth and recommended treatment options (Garb et al., 2005; Lilienfeld et al., 2001; Wood et al., 2003).

Projective Techniques: The Bottom Line

Clinicians are advised to carefully consider the arguments and data when selecting measures for formal clinical evaluations. Instruments incorporated into a psychological assessment battery should be selected with the referral questions in mind; be standardized and psychometrically sound with regard to norms, administration, scoring, and interpretive procedures; possess utility for the intended clinical purpose; and be appropriate for the context and the person being evaluated. Clinicians must make their own decisions whether to include projective techniques and be prepared to justify them based on these considerations. Lilienfeld et al. (2001) warn clinicians against the use of projective techniques, except in the rare cases where the merit of scoring and interpretation has been supported. For research-based guidelines on incorporating projective techniques into clinical practice, the reader is referred to Garb, Wood, Lilienfeld, and Nezworski (2002). As for the future of projective techniques, it is important for those who advocate their use to decide on standardized procedures (norms, administration, scoring, and interpretation) so that evaluation of their psychometric properties may take place through well-designed research studies (Hunsley et al., 2003).

THE PSYCHOLOGICAL REPORT: ENHANCING WRITTEN COMMUNICATION

In general, a good psychological report is one that explains the nature of the referral, documents observations of the person's behavior, lists the instruments and techniques used in the assessment and the dates employed, notes reliability and validity issues pertaining to the assessment, clearly and adequately explains results, offers diagnostic impressions, contains a summary and recommendations, and is signed with adequate credentials evident (Braaten, 2007; Brenner, 2003; Groth-Marnat & Horvath, 2006; Harvey, 1997, 2006; Koocher, 2005). The content of the report should be tailored to the individual (Groth-Marnat & Horvath, 2006), and the pros

and cons of diagnosis should be considered. When an integrated assessment supports the presence of a diagnosis, clinicians should make diagnostic distinctions that are appropriate, while also considering the diagnosis that is the least stigmatizing. A diagnosis may be helpful in fostering the implementation of services, as well as respecting the consumer's right to knowledge; however, less severe diagnoses may be associated with less stigmatization and emotional distress, and with youths there is the added danger of others reviewing the report, with certain elements being integrated into other documents and becoming part of the youth's records (Michaels, 2006). Although there are numerous variables to consider when writing a psychological report, a few important points are delineated here relating to inclusion of client strengths, improving readability, and using computerized interpretative narratives. For further general information on report writing, the reader is referred to Groth-Marnat (2003); Lichtenberger, Mather, Kaufman, and Kaufman (2004); and Sattler and Hoge (2006).

Inclusion of Strengths

Personal (e.g., good social skills) and environmental (e.g., solid social network) strengths should be highlighted, as often evaluations remain deficit-oriented and general (Brenner, 2003; Groth-Marnat & Horvath, 2006; Snyder et al., 2006). A separate section on strengths could be included in the psychological report (Groth-Marnat & Horvath, 2006; Snyder et al., 2006), or the material could be integrated into the narrative of the report and become a focal point of the feedback session. Focusing on an individual's assets is perhaps one means of alleviating stigmatization and emotional distress (Michaels, 2006), as well as empowering a youth to make adaptive and effective choices. Incorporating client strengths is a challenge given that it is more labor-intensive during the assessment and report writing phases; however, it results in a more thorough and appropriate case conceptualization (Snyder et al., 2006).

Improving Readability

Psychological reports need to be geared toward the consumer to be of utility (Brenner, 2003; Harvey, 2006). Reports are read by a variety of individuals with varied academic backgrounds, such as psychologists, educators, court employees, and, increasingly, clients and parents of youths (Groth-Marnat & Horvath, 2006; Harvey, 1997, 2006; Snyder et al., 2006). A major problem is that reports continue to be written at a level that is difficult for consumers to understand (Groth-Marnat & Horvath, 2006; Harvey, 1997, 2006), and use of technical terms may result in the misinterpretation of information, which is potentially harmful to the consumer (Michaels, 2006). To increase the readability of psychological reports, difficult and technical words, wordiness, passive verbs, and acronyms should be minimized (avoided if possible), sentences should be shortened, and subheadings employed (Brenner, 2003; Groth-Marnat & Horvath, 2006; Harvey, 1997, 2006). These efforts are critical to writing clear, consumer-oriented reports, although additional report writing time is required, which may be an added stressor for clinicians working in environments characterized by time constraints (Harvey, 2006).

Computerized Interpretive Narratives

Computerized interpretive narratives are often available with the computer scoring packages of personality instruments such as the MMPI-A and the MACI. These computerized narratives must be carefully tailored to the individual being assessed to combat interpretations and language that are hard to understand, lengthiness, and incorrect hypotheses and recommendations (Harvey, 2006). They should also be viewed as one source of data, confirmed by other sources (Michaels, 2006). Clinicians should be thorough consumers of the quality (e.g., validity) of the computerized interpretive packages that they use (Butcher, 2003; Garb, 2000; Snyder, 2000), as clinicians retain responsibility for the applicability of computerized interpretations and the content eventually included in the written psychological report (American Educational Research Association, American Psychological Association, & National Council on Measurement in Education, 1999; American Psychological Association, 2002; Association of State and Provincial Psychology Boards, 2001; Groth-Marnat & Horvath, 2006; Schulenberg & Yutrzenka, 2004).

THE FEEDBACK SESSION: ENHANCING ORAL COMMUNICATION

Conducting feedback sessions is important for cultivating relationships with consumers of assessment services (Brenner, 2003). Feedback sessions afford additional opportunities for clinicians to clarify hypotheses, refine the precision of their reports, work to ensure that the content of the report is understood, and emphasize the importance of following recommendations (Brenner, 2003; Harvey, 2006; Snyder et al., 2006). When communicating assessment results, clinicians are advised to tailor feedback to the educational level of the intended recipient and to be sensitive to cultural issues related to diagnosis and treatment. Communicating in clearly understandable language may be difficult, however, as many terms commonly used by professionals have different meanings from professional to professional, and therefore feedback provided to consumers may be substantially different based on the clinician providing the interpretations (and also by the consumers receiving feedback, who may be relying on their own definitions; Harvey, 2006).

Consumers should feel that they have been heard, that they are a part of the process, and that they have opportunities to pose questions, have concerns addressed, and comment on recommendations (Braaten, 2007; Brenner, 2003; Harvey, 2006; Snyder et al., 2006). Comments from parents or legal guardians and the youth being assessed can encourage the clinician to clarify the findings, particularly if there is a discrepancy between the test results and other data (e.g., collateral information). As for recommendations, they should be presented in a clear, straightforward, and convincing manner, relate to the referral questions, and take into account cultural variables, community resources, client strengths, and the ability of individuals (e.g., parents, teachers) to carry them out (Brenner, 2003; Gudas & Sattler, 2006; Harvey, 2006; Jones et al., 2001; Snyder et al., 2006). Explanation of recommendations is particularly important, as there may be a variety of barriers to implementing them (e.g., transportation, cultural differences; Geffken et al., 2006; Jones et al., 2001).

Potential barriers to following through with recommendations may be examined, and the clinician may problem-solve with consumers to minimize treatment nonadherence (see Geffken et al., 2006, for a discussion of recommendation follow-through).

Parents or legal guardians whose youths are undergoing psychological evaluations typically want to know if there is something "wrong," and they may experience anxiety at the prospect of the evaluation results and have questions about the testing and the implications of the findings. Parents or legal guardians may feel defensive regarding their youth's presenting problems (Lichtenberger et al., 2004), and some may need to be referred for treatment for their own psychological problems. The emotional responses of the parents or legal guardians should be addressed; however, the focus of the feedback session should remain on the youth (Braaten, 2007). Thus, clinicians must work to strike a delicate balance between providing information that is nonjudgmental and supportive, yet is accurate to the situation (Braaten, 2007; Lichtenberger et al., 2004). Starting the feedback session with positive comments about the youth, reiterating these positives throughout the evaluation, focusing on how parents or legal guardians can help their child, and conveying a sense of hope are generally useful strategies (Braaten, 2007).

A more complicated question is when and how to provide personality assessment feedback to the youths themselves. What age is appropriate for youths to hear this type of information? Could this information be detrimental? Young children who are able to read should not review their personality descriptions; however, it is appropriate to provide them with verbal feedback consistent with their developmental level (Lichtenberger et al., 2004). Older children and adolescents may be involved in feedback to the degree possible given a variety of variables, such as their developmental level, parental wishes, and the nature of the feedback. Focusing on a youth's strengths provides a critical starting point on which to build. Involving them in the assessment early on by building good rapport can be helpful later in involving them in the feedback session. Adolescents in particular may be interested in obtaining answers to their questions, and in some cases it is useful to provide feedback to the youth on an individual basis, as well as to the youth and his or her family together (Lichtenberger et al., 2004). Assessment and feedback should provide supportive and tactful illumination on areas of concern, identifying ways that youths may navigate their environments with greater effectiveness.

PSYCHOLOGICAL ASSESSMENT WITH YOUTHS: COMPLICATING FACTORS

Thus far we have presented a thorough model for the assessment of psychopathology in youths. Although the model presented is comprehensive, it should be noted that many clinicians work in different environments or under circumstances that complicate the thorough assessment of psychopathology. We now turn to a discussion of complicating factors: the forensic environment, mental retardation and developmental disabilities, and managed care.

The Forensic Environment

Increasingly, clinicians are conducting psychological assessments in juvenile forensic settings. To be effective working in the forensic environment, clinicians must be able to work with youths, families, attorneys, and legal officials, as well as understand the differences between clinical and forensic roles. Clinical roles tend to be supportive, whereas forensic roles tend to be more neutral. Additional intricacies relate to the referral questions, informed consent and assent, limitations of confidentiality, identifying who exactly is the client, establishing the role of the evaluator, and determining the content and length of the psychological report, as well as who will receive feedback and how feedback will be conducted (e.g., Ackerman, 2006; Butcher & Pope, 2006; Gudas & Sattler, 2006; Koocher, 2006). Is the client an attorney, a youth's parents, or a court? Does the referral relate to a specific legal issue, and if so, how is the assessment of personality relevant? The answers to such questions are important, as they have a bearing on how forensic interviews are conducted to a degree that requires specialized training (Gudas & Sattler, 2006).

With regard to the selection of measures appropriate to the clinical or forensic task at hand, specific guidelines are available (Butcher & Pope, 2006; Heilbrun, 1992), and clinicians should be prepared to justify their decisions (Gudas & Sattler, 2006; Koocher, 2006). In addition, given the forensic nature of the evaluation, the concern that adolescents may malinger is prominent (Butcher & Pope, 2006; McCann, 1998; Steffen & Ackerman, 1999). One means of countering this problem is by using objective personality inventories with validity indicators to aid in the identification of unusual response styles. As part of a comprehensive battery of assessment tools, the MMPI-A has some effectiveness in this regard (Butcher & Pope, 2006; McCann, 1998), and the MACI's utility is also viewed as promising (McCann, 1998, 2006). Thus, both the MMPI-A and the MACI are potentially useful in juvenile forensic settings (e.g., Archer et al., 2006; Butcher & Pope, 2006; McCann, 2006).

Working with Youths with Mental Retardation or Developmental Disabilities

Another issue that may arise is the assessment of psychopathology in youths with mental retardation or developmental disabilities (MR/DD). In an article focusing on working in a high-management group home setting with youths with MR/DD and who also demonstrated sexual offending behaviors, Schulenberg (2003) noted the impossibilities inherent in assessing these youths with self-report measures such as the MMPI-A and the MACI given the reading difficulties with this population. However, the potential assessment utility of having others complete standardized behavior rating scales in this context was noted.

Managed Care

In their review, Wood et al. (2002) noted the apparent impact of managed care on assessment practices, with psychologists facing increasing accountability to justify the utility, cost-effectiveness, and incremental validity of their services. In Cashel's (2002) survey, although the overall impact of managed care on test and technique

utilization was minimal, the largest and most noticeable declines in reported test usage were with measures such as the WISC, Rorschach, TAT, and MMPI-A. Similar findings were noted in Archer and Newsom's (2000) survey, with clinicians reporting less use of several measures as a result of managed care limitations, including the Rorschach, the TAT, the Wechsler scales, and the MMPI-A. For some clinicians, managed care may result in reduced activity in the area of psychological assessment, with fewer measures included in the battery (making batteries less comprehensive) and with diminished reimbursement (Archer & Newsom, 2000). With specific regard to the continued use of projective measures, although they continue to be popular among clinicians, it is possible that they may experience an eventual decline in the area of youth assessment. If such a decline were to occur, it could be due to the psychometric criticisms that have been raised, the increased use of behavior rating scales, and/or managed care (Archer & Newsom, 2000; Cashel, 2002; Groth-Marnat, 2003; Kamphaus et al., 2000).

SUMMARY

Delineating a specific standard for youth assessments is difficult given that they are conducted in a variety of environments (Palmiter, 2004). Our approach in this chapter was to paint a broad stroke with regard to major issues in the assessment of psychopathology in youths. In doing so a variety of issues were covered, beginning with the referral questions, informed consent and assent, rapport building, and measure selection, and continuing with clinical and collateral interviews, review of collateral records, and intelligence and achievement testing. The objective assessment of personality, as well as the psychometric controversy surrounding the use of projective techniques, was also discussed, and the potential utility of behavior ratings scales (administered to youths, their parents, and their teachers) and behavioral observations were underscored. Finally, issues in report writing and feedback were outlined, followed by complicating factors to the assessment process. Given space limitations, it was beyond our scope to provide an in-depth view of each area. For example, despite the emphasis on the importance of psychometric properties, a thorough treatment of each instrument discussed simply was not possible, nor was it possible to discuss all of the numerous measures available in a given area despite clinical popularity or psychometric support in certain circumstances.

To make sense of the wide array of instruments available, clinicians must be trained in test construction and psychometrics. They should be aware of the psychometric pros and cons of their tools and their applicability to the diagnostic question in the specific context of the assessment.

In addition to psychometrics, to competently conduct personality assessments with youths, clinicians must understand relevant developmental, environmental, personality, psychopathological, diagnostic, ethical, and multicultural variables. Assessments should be thorough, taking into account multiple sources of data through a variety of means, to generate hypotheses about a youth's functioning and to arrive at a precise and comprehensive case formulation. Thorough training in the administration, scoring, and interpretation of psychological tests, as well as strength identification, integration of findings, development of individualized

recommendations, and constructive methods of report writing and feedback provision is also necessary. These recommendations are consistent with other sources denoting standards for assessment competency (e.g., Krishnamurthy et al., 2004). For additional information on ethical and professional standards in assessment, see the publications by the American Educational Research Association, American Psychological Association, and National Council on Measurement in Education (1999); the American Psychological Association (1991, 1993, 2002); and the Association of State and Provincial Psychology Boards (2001). For those looking for additional information on psychometric properties of instruments useful in the personality and psychopathological assessment of older children and adolescents, beyond consulting the respective manuals, the *Mental Measurements Yearbook* (www.unl.edu/buros/), a comprehensive database for psychometric reviews of a variety of measures, is a useful place to begin. In addition, readers should consult the respective chapters, texts, and empirical articles that continue to increase in availability.

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CHAPTER 19

Behavioral Assessment

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One of the most important approaches to evaluating children and adolescents is behavioral assessment. No single definition of behavioral assessment exists, but the approach is often thought of as a problem-solving-oriented means of gathering information about current, observable behavior and what maintains behavior over time. Behavioral assessment is an idiographic approach involving detailed evaluations of specific behavior to develop a prescriptive treatment plan. A prescriptive treatment plan is one tailored to the individual needs of a given child. Although behavioral assessors derive from different traditions within psychology and other professions, the overarching goal of designing a customized treatment regimen for a particular child has remained largely unchanged over time (Mash & Terdal, 1997).

Behavioral assessment methods focus on *forms* of behavior that can be clearly defined and monitored. Examples include aggression in the form of slapping a classmate and fear in the form of avoiding a large dog. Frequency, duration, intensity, and other topographies of behavior may be assessed as well. Behavioral assessment methods also focus on *functions* of behavior, or antecedents and rewarding consequences that maintain behavior. Examples include teacher attention after slapping a classmate and reduction of unpleasant physical sensations after avoiding a large dog.

Behavioral assessment has been differentiated from global or traditional forms of assessment in several ways (Reitman, 2006). First, behavioral assessment emphasizes current, contextual factors in a child's life that are well defined and observable. Global forms of assessment such as intelligence and personality tests are designed to obtain information about general characteristics of a child such as problem-solving ability or psychological discomfort. In contrast, methods of behavioral assessment are designed to garner information about *molecular* characteristics of a child, such as length of attention when listening to a teacher command and amount of time spent in school.

Second, behavioral assessment emphasizes instability of behavior across settings. Global forms of assessment assume a more trait-like approach to behavior, wherein a child's cardinal characteristics, such as extraversion, will largely determine actions across settings. In contrast, behavioral assessors assume a child's behavior to be more situation-specific. A child may, for example, be relatively

gregarious and talkative with family members at home but somewhat inhibited and soft-spoken in peer-based social situations.

Third, behavioral assessment emphasizes multiple samplings of behavior to understand a child's actions. Global forms of assessment such as achievement tests are often given once to obtain information about general characteristics of a child, such as academic status. In contrast, behavioral assessors employ *ongoing* techniques, such as daily logbooks, to examine patterns of actions across settings and over time. In related fashion, behavioral assessment relies less on historical factors and more on present aspects of behavior and commonly involves multiple sources of information.

Fourth, behavioral assessment emphasizes direct, on-site methods such as behavioral observation and self-monitoring to gather information about specific aspects of a child's behavior. Global forms of assessment include more circuitous analyses of a child's behavior, such as a neuropsychological test in an office setting to indirectly assess brain dysfunction. In contrast, behavioral assessors actively evaluate a child's behavior in direct, natural environments such as home, school, playground, and social events.

Finally, behavioral assessment may be more appropriate for children with diverse ethnic backgrounds compared to traditional assessment. Because behavioral assessment emphasizes concrete and proximal influences on behavior, such as learning processes, the process may be freer of cultural bias compared to standardized tests. In addition, behavioral assessment techniques can be quite useful for identifying different behaviors across different cultural contexts (Castillo, Quintana, & Zamarripa, 2000; Suzuki, Alexander, Lin, & Duffy, 2006).

This chapter covers a central model of behavioral assessment, functional analysis, in addition to specific techniques endemic to the model. These techniques include interview, analogue assessment, behavioral observation, rating scales, self-monitoring, sociometric assessment, physiological assessment, and activity measures.

FUNCTIONAL ANALYSIS

A central model of behavioral assessment is functional analysis, defined as examining and manipulating antecedents and consequences of a behavior of interest to provide information about the relationship of these events to the behavior (McComas, Hoch, & Mace, 2000). Functional analysis is often linked to terms such as functional assessment and descriptive and experimental functional analysis. *Functional assessment* is a broader term that refers to developing general hypotheses about the function of problem behavior (Gresham & Lambros, 1998). A clinician addressing a child with temper tantrums in public, for example, might develop a hypothesis that the behavior is driven by attention and tangible rewards from parents. In this case, the child's parents may be inadvertently rewarding the tantrums by quickly attending and providing candy to quash the behavior.

Functional assessment typically consists of descriptive and experimental functional analysis. *Descriptive functional analysis* refers to indirect methods of gaining information about the relationship of a behavior to antecedent and consequent

events. Typical methods of assessment in descriptive functional analysis are interviews, rating scales, review of records, self-monitoring, and naturalistic observations. *Experimental functional analysis* refers to direct methods of gaining information about the relationship of a behavior to antecedent and consequent events. Experimental functional analysis involves standardized procedures to identify key maintaining variables of behavior and rule out competing explanations for why behavior continues to occur (Mace, 1994).

Functional analysis has been a key staple for assessing and treating developmental and disruptive behavior disorders in youth. Functional analysis is typically utilized for externalizing problems such as self-injurious behavior, aggression, tantrums, property destruction, and pica (Hanley, Iwata, & McCord, 2003). Functional analysts typically assess maintaining variables of behavior, such as sensory reinforcement, escape from aversive situations, attention, and tangible rewards. However, functional assessment may be broadened to include other contextual variables that trigger and reinforce behavior. Such assessment usually leads to effective treatments; such as training persons with developmental disorders to communicate desires more effectively so maladaptive behavior is not necessary (Durand & Merges, 2001).

Two examples of functional assessment using descriptive and experimental procedures are presented here. Field and colleagues (Field, Nash, Handwerk, & Friman, 2004) evaluated a 12-year-old male with disruptive behavior that included noncompliance and other misbehaviors in a residential group home. Noncompliance was defined as opposition to commands, rules, or stated expectations within the home. Descriptive functional analysis to garner hypotheses about the boy's behavior was conducted via teacher interviews, case file review, behavioral observations, and an interview with the boy. These data led the authors to hypothesize that attention was largely responsible for noncompliance. Specifically, attention typically followed noncompliance, and the boy often escalated noncompliance to force attention from others.

Experimental functional analysis then consisted of a reversal design that alternated two treatments. Standard treatment within the group home, which consisted of token economy and teacher interactions, was compared to modified treatment involving increased positive teacher interactions with the boy. Modified treatment led to a substantial decrease in noncompliance, thus supporting the original hypothesis that the misbehavior was reinforced by attention. In this case, the experimental functional analysis assessment strategy also consisted of effective intervention.

Kearney designed a functional assessment model for school refusal behavior, a behavior with many internalizing components. In this model, youths are thought to refuse school for one or more of several reasons: to avoid school-based stimuli that provoke negative affectivity, to escape aversive social and/or evaluative situations, to pursue attention from significant others, and to pursue tangible rewards outside of school. Descriptive functional analysis consists of a structured diagnostic interview, in-session behavioral observations, and a rating scale (School Refusal Assessment Scale—Revised) that measures relative influence of each of the four hypothesized functions of school refusal behavior (Kearney, 2002, 2006).

Procedures for experimental functional analysis include establishing parameters for a child's school day that best predict attendance and nonattendance (Kearney & Albano, 2007). If a child were suspected of refusing school to escape an aversive evaluative situation, such as tests, then the child's attendance could be evaluated on typical school days and days when the child is exempted from tests. Similarly, if a child were suspected of refusing school for tangible rewards outside of school, then attendance could be compared for typical school days and days when parents provide substantial tangible reinforcement for being in class. A functional assessment approach has been shown to be effective for predicting which prescriptive treatment strategy is best for youths who refuse school for different functions (Chorpita, Albano, Heimberg, & Barlow, 1996; Kearney, Pursell, & Alvarez, 2001; Kearney & Silverman, 1999).

The remaining portion of the chapter focuses on methods of behavioral assessment with children that may be used within a functional analytic model. These methods include interviews, analogue assessment, behavioral observation, rating scales, self-monitoring, sociometric assessment, physiological assessment, and activity measures.

BEHAVIORAL ASSESSMENT MEASURES

Interviews

Interviews with children and parents are the most common form of assessment used by mental health professionals. Although many professionals use unstructured interviews, clinicians and others who emphasize behavioral assessment tend to utilize structured interviews that are psychometrically stronger. In the area of behavioral assessment, structured interviews may be diagnostic or functional in nature. Structured diagnostic interviews are typically linked to *DSM* criteria for mental disorders (American Psychiatric Association, 2000). The interviews are designed to provide detailed information about the form of a child's mental disorder and help an assessor derive a formal diagnosis.

Diagnostic interviewing is discussed in more length in another chapter of this volume, but a brief list of measures is presented here:

- Anxiety Disorders Interview Schedule for DSM-IV: Child and Parent Versions (Silverman & Albano, 1996)
- Child and Adolescent Psychiatric Assessment (Angold & Costello, 2000)
- Child Assessment Schedule (Hodges, McKnew, Cytryn, Stern, & Kline, 1982)
- Children's Interview for Psychiatric Syndromes (Weller, Weller, Fristad, Rooney, & Schecter, 2000)
- Diagnostic Interview for Children and Adolescents (Welner, Reich, Herjanic, Jung, & Amado, 1987)
- Interview Schedule for Children (Kovacs, 1985)
- National Institute of Mental Health Diagnostic Interview for Children (Shaffer et al., 1996)

- Schedule for Affective Disorders and Schizophrenia for School-Age Children: Present and Lifetime Version (Kaufman et al., 1997)

Each of these interviews has demonstrated good reliability and validity. Though most cover externalizing disorders in youth in great detail, less attention has been paid historically to internalizing problems. However, the Anxiety Disorders Interview Schedule for DSM-IV: Child and Parent Versions (ADIS: C/P) has been recognized as the gold standard for assessing anxiety-related disorders in youth. In addition, the interview contains sections for other internalizing disorders, such as depression, as well as sections for externalizing, eating, developmental, and psychotic disorders. A section for school refusal behavior is also available and concentrates on yes/no questions regarding whether a child has problems attending school, whether a child is nervous when attending school, whether a child is upset about performing poorly in school, and whether a child has been medicated for nonattendance.

Structured diagnostic interviews have strengthened the child assessment process and are commonly used to steer treatment and evaluate outcome. Problems with these interviews have been noted, however. First, the interviews provide an overall view of a child's difficulty but may not provide specific information necessary for developing a detailed treatment plan. Second, the interviews are not particularly sensitive to developmental or cultural differences. An older adolescent with Attention-Deficit/Hyperactivity Disorder (ADHD), for example, is likely to present with different symptoms than a younger child with the disorder. In related fashion, a Hispanic child and a European American child may differ with respect to symptoms of Separation Anxiety Disorder. Third, the interviews are not generally designed for ongoing use or for multiple sources of information. A snapshot of a child's behavior is obtained, but behavioral changes that occur over time may not be examined (Kearney, 2005).

Interviews within a behavioral assessment paradigm may also be functional in nature and help address problems of structured diagnostic interviews. These interviews may be unstructured or structured, but more specifically concentrate on operationally defined forms of behavior and what maintains behavior over time. These interviews can be shaped in various ways to accommodate different developmental levels and can provide detailed information to help a clinician form a unique treatment plan.

Behavioral and functional interviews may be given to gather multiple sources of information; a sample excerpt with a parent is presented here. Contrast these questions with the general yes/no questions for the ADIS: C/P:

Therapist: What specifically does Brian do on the playground before school?

Parent: He just won't go into school.

Therapist: Okay, what does he say and do during that time?

Parent: He cries, clings to my leg, and sometimes tries to run away. When the teacher comes out to line everybody up for school, he runs behind me and holds tight.

Therapist: Does he do this every day?

Parent: No, not every day, but I notice it's worse on Mondays and Tuesdays or after a long break from school.

Therapist: How do you and the teacher react when Brian does this? What specifically do you and she do?

Parent: I try to comfort and reassure him as much as possible, giving him a lot of hugs even as the other kids are going into the school. The teacher tells Brian to come into the school but has to deal with the other kids, so she often leaves me and Brian behind.

Therapist: Does the extra attention and reassurance you give Brian help him go to class?

Parent: No, he just clings more and cries and I feel so bad that I just take him home.

Note that these questions are very specific about one episode of school refusal behavior. Questions surround the form of Brian's particular behaviors, when the behaviors occur, and what reinforcer (attention, in this case) might be maintaining the behavior. These kinds of questions can be asked of different people and at various times during the assessment and treatment process. The questions can also be tailored to the developmental level of a child. If Brian were an older child, for example, questions about his cognitions prior to entry into school might be particularly salient.

Behavioral and functional interviews are flexible enough to cover a wide range of material relevant to a particular case, especially the following topics:

- Specific behavioral, cognitive, and physiological aspects of a child's difficulty
- Effects of misbehavior on a child's and family's daily functioning
- Current life stressors
- Contextual or distal variables or risk factors that might influence a child's behavior, such as family dynamics, marital conflict or divorce, poor financial and social support, maltreatment, school-based threats, temperament, and physical and medical status
- Antecedents and consequences pertaining to specific episodes of misbehavior, such as sensory reinforcement, negative reinforcement, attention seeking, and pursuit of tangible rewards
- Child's skill levels in various areas, such as social, verbal, and academic
- Pertinent cultural variables such as language differences, normative behaviors for a particular culture, and perspectives on child misbehavior

Behavioral and functional interviews can be more structured as well. A good example is the Functional Diagnostic Profile modified for youths with selective mutism by Schill and colleagues (Schill, Kratochwill, & Gardner, 1996). The assessment protocol initially consisted of soliciting general information about a particular child with selective mutism to derive hypotheses about the function of the child's behavior. In this case, several hypotheses were formed. The child's mutism

was thought to be potentially related to several conditions: (a) escaping aversive requests, directives, or task demands; (b) avoiding attention from unfamiliar others; and (c) deriving attention from familiar others. Following this assessment process, the researchers engaged in analogue assessment to experimentally support these hypotheses and design individualized treatment that eventually led to increased verbalizations. A greater explication of analogue assessment is described next.

Analogue Assessment

Analogue assessment refers to behavioral observations typically conducted in a controlled setting, such as a therapist's office or research laboratory (Barkley, 1991). The observations are often done within a behavioral assessment paradigm to confirm hypotheses about the function of behavior. The premise behind analogue assessment is that a child's behavior in the contrived setting will mirror behavior in the child's natural environment. In the case presented by Schill and colleagues (1996), the child's mutism was explored under several conditions: allowing the child to escape a demand, asking the child to read before others, and providing positive reinforcers for speech such as play and parent praise. The child's responses under this set conditions helped the researchers confirm expectations about what maintained selective mutism.

A popular form of analogue assessment is a *behavioral assessment test* or *behavioral avoidance test,* in which a child role-plays a given situation or performs in some way before others. Behavioral assessment tests are commonly used to evaluate anxiety- and fear-based behavior in children. Among children with Social Phobia, for example, behavioral assessment tests may include conversing with others, writing or solving problems before others, speaking before others, reading a story aloud before others, and taking tests, such as subsections of an intelligence or achievement test (Beidel, 1991; Beidel, Turner, & Morris, 1999).

Behavioral assessment or avoidance tests may be used as well for youths who fear specific stimuli. A test in this situation might involve asking a child to approach a feared animal as close as possible, or spend as much time as possible in school. As a child approaches a feared stimulus, ratings of discomfort may be solicited and distance to the stimulus can be measured. Behavioral assessment or avoidance tests have been used to evaluate youth phobic of dogs, snakes, water, blood, heights, medical procedures, strangers, school-related events, and darkness, among other stimuli (Barrios & Hartmann, 1997).

Behavioral assessment tests have also been used to evaluate externalizing behavior problems in children. Children with ADHD, for example, have been subjected to controlled observations of their behavior in playrooms divided into grids to measure excess movement across the grids. Analogue classrooms with desks have also been used to measure attention problems, out-of-seat behavior, and inappropriate vocalizations. Analogue assessment procedures can help evaluate social skills, aggression, noncompliance, and academic behaviors such as completing worksheets (Barkley, 1991, 1997; Elliot & Gresham, 1991; Gable, Hendrickson, & Sasso, 1995; M. A. Roberts, 1990). Parent-adolescent conflict and interpersonal problem solving may be evaluated in this manner as well (Goldstein, 1999; Robin & Foster, 1989).

Traditional analogue assessments are common but inconsistent methods for evaluating children's behavior. The assessments have several advantages, including control of stimulus conditions, practicality, cost-effectiveness, and evaluation of multiple motor and other behaviors at one time. However, the assessments have variable test-retest reliability and construct validity because they have not been well standardized (Mori & Armendariz, 2001).

An increasingly popular form of analogue assessment among adults is *virtual reality assessment,* which involves placing a person in a digitally created environment that mimics a particular setting and allows for interaction within that setting. A person afraid of heights, for example, may interact in virtual scenes involving progressively higher places. The assessment can thus be conducted quite confidentially, can be manipulated to include many different contexts, and is amenable to examining different response systems at one time. Virtual reality assessment for children remains largely in development, with particular focuses on eating disorders, Social Phobia, ADHD, learning difficulties, and pain (Lannen, Brown, & Powell, 2002; Riva, Wiederhold, & Molinari, 1998; Rizzo et al., 2006).

Behavioral Observation

Analogue assessments are contrived observations often preferred because of their practicality. Direct naturalistic observations, however, represent the purest form of behavioral assessment and functional analysis. Analogue assessments often target a specific area of interest, such as approach toward a feared stimulus, whereas direct naturalistic observations usually involve a much wider range of target behaviors and contextual variables.

Behavioral observations typically concentrate on specific topographies of overt behaviors. The process typically begins by defining a behavior of interest in operational terms. Aggression, for example, may be operationally defined as a slap to the head of a classmate or verbal insult given to a sibling. In related fashion, aspects of a certain behavior may be observed and recorded. Behavior may be recorded with respect to intensity, duration, frequency, and latency. A rater may record the length of time between a teacher's command and a child's compliance. Behavior accuracy, such as number of mistakes on a spelling test, can be assessed as well.

Behavioral observations also include contextual variables or events that surround and influence behavior. Many researchers observe and code valence, which may involve classifying a behavior into positive, negative, neutral, and other categories. In addition, key antecedents and consequences to behavior are commonly observed for purposes of functional analysis. A child who slaps a classmate may do so with an angry facial expression (negative valence) immediately after a frustrating test (antecedent) and prior to a harsh teacher reprimand and attention from classmates (consequences; Watson & Steege, 2003).

Direct observation can involve simple systems of narrative recording as well as more complex research-based observational systems. Narrative recording involves simple written accounts of behavior as it transpires, though more sophisticated accounts may include recording topographies such as frequency and intensity. Raters

who are observing and interacting with a child simultaneously may find it difficult to maintain an ongoing log of transpired behavior, so interval recording is common. In this system, a rater may simply record the presence or absence of a behavior in blocks of time, such as every 20 seconds or 5 minutes (Skinner, Rhymer, & McDaniel, 2000).

Behavioral observations in research settings typically involve greater complexity. Many have been designed to evaluate youths with disruptive behavior disorders because of the overt nature of problems such as noncompliance. Some of these observational systems are highly contrived and qualify more as analogue assessments. These often involve analyses of free play, parent-directed play, and parent-directed chores (M. W. Roberts, 2001). Research-based observation systems for more naturally occurring behavioral phenomena have also been designed, however. These most prominently include systems for evaluating children's social and coping skills, play, and parent-adolescent communications (Anderson, Moore, Godfrey, & Fletcher-Flinn, 2004; Margolin et al., 1998; Pellegrini, 2001; Pretzlik & Sylva, 1999).

Whether simple or more complex behavioral observations are used, a key goal is to derive interrater reliability or consistency of ratings across two or more observers. At a basic level, interrater reliability can be calculated as the number of agreements divided by the number of agreements plus the number of disagreements times 100. Thus, 12 agreements and 3 disagreements would result in interrater reliability of 80%. At more complex levels, interrater reliability can be calculated in several ways, though a predominant way is Cohen's kappa.

Direct observations of behavior carry several strong advantages. Observations can supply a plethora of data regarding forms and functions of behavior and are often essential components of a detailed assessment plan for complex behaviors. In addition, observations in natural settings have high external validity. Observations can supply data about the molecular characteristics of a given behavior, such as a child's problem-solving method when completing a math worksheet or subtle changes in tone during parent-adolescent communications. Knowledge about these molecular characteristics is often invaluable for designing an individualized treatment plan.

A primary disadvantage to behavioral observation is that covert or low-frequency behaviors are not amenable to the process. Behavioral observation is not particularly useful for assessing internal mood states, visceral responses, or secretive behaviors. In addition, behavioral observations are subject to reactivity and other demand characteristics that could bias the assessment process. Observer drift, where raters deviate from the original operational definition of a targeted behavior, can reduce validity of the process as well. Finally, behavioral observations require substantial time, effort, training, and, ideally, multiple raters. As such, observations are not as practical or cost-effective as rating scales, which are described next.

Rating Scales

A particularly well-developed area of behavioral assessment for children is rating scales. *Rating scales* or questionnaires are instruments wherein children and

significant others endorse the presence or absence of various physiological, cognitive, and behavioral symptoms. Rating scales have several strong advantages, including ease of use, good reliability and validity, excellent coverage of many topics, sensitivity to treatment effects, standardization, and extension to multiple sources of information, such as parents and teachers. In addition, rating scales are good for assessing low-frequency behavior and behavior not amenable to direct observation, such as internalizing constructs of worry or sadness.

Rating scales also have several disadvantages, however. A key problem is rater bias, especially among parents and teachers, who may rate a child more negatively or positively than is actually the case. Measurement error is endemic to rating scales as well, and can come from variance in behavior across time as well as variable rater, setting, and instrument characteristics. In addition, rating scales have poor applicability to very young children and correlate inconsistently with real-life behavior. Developmental differences between children and adolescents are often not fully considered. Finally, rating scales provide summaries of perceptions of behavior rather than direct measures of actual behavior (Merrell, 2000; Myers & Winters, 2002a).

A full explication of all rating scales in all areas of child functioning is beyond the scope of this chapter, but a summary is presented here of the major rating scales available for children, parents, and teachers. Measures were chosen based on their psychometric strength, widespread use in the field, and applicability to designing an individualized treatment plan for a child. In addition, only the most prevalent areas of mental disorder among children and adolescents are covered here: anxiety, depression, ADHD and other disruptive behavior disorders, and developmental disorders. Primary measures in other areas are listed as well.

Anxiety Disorders

Anxiety disorders are the most common mental disorders in children, and research into the assessment and treatment of these problems has burgeoned in the past 2 decades. Because anxiety is largely an internalizing problem, the development and use of rating scales has become widespread in this area. Developers of anxiety rating scales must naturally contend with a phenomenon that changes quickly over time, but several scales appear to have excellent psychometric strength (Kearney & Bensaheb, 2007; Myers & Winters, 2002b).

Rating scales for anxiety disorders may be divided into traditional measures that assess broad swaths of anxiety and more recent measures that specifically target more narrow types of anxiety. Traditional child self-report measures include the Revised Children's Manifest Anxiety Scale (RCMAS), State-Trait Anxiety Inventory for Children (STAIC), and Fear Survey Schedule for Children—Revised (FSSC-R). The RCMAS measures physiological anxiety symptoms, worry and oversensitivity, and fear and concentration. The STAI-C assesses situation-specific and global anxiety. The FSSC-R contains subscales for fear of failure and criticism, injury and small animals, danger and death, medical topics, and a category of unknown fears (Ollendick, 1983; C. Reynolds & Richmond, 1985; Spielberger, 1973). Recent measures of general anxiety constructs include the Multidimensional Anxiety Scale

for Children, which measures social anxiety, separation anxiety and panic, physical anxiety, and harm avoidance, and the Screen for Child Anxiety Related Emotional Disorders, which measures somatic and panic symptoms, general and separation anxiety, and school and social phobias (Birmaher et al., 1999; March, 1997).

Though useful, traditional measures of anxiety are gradually becoming replaced by more specific measures that cover particular aspects of anxiety. With respect to social anxiety, for example, the Social Anxiety Scale for Children—Revised (SASC-R) and Social Phobia Anxiety Inventory for Children (SPAI-C) are most common (Beidel, Turner, & Morris, 1995; La Greca & Stone, 1993). These measures cover fear of negative evaluation and social avoidance and distress (SASC-R) as well as assertiveness and concerns about social encounters and public performance (SPAI-C).

With respect to Obsessive-Compulsive Disorder, the Leyton Obsessional Inventory—Child Version is a brief rating scale for compulsions, obsessions and incompleteness, and cleanliness (Bamber, Tamplin, Park, Kyte, & Goodyer, 2002). Finally, the Trauma Symptom Checklist for Children is useful for measuring children's reactions to trauma and the Child PTSD Symptom Scale is useful for measuring criteria necessary for a diagnosis of Posttraumatic Stress Disorder (Briere, 1996; Foa, Johnson, Feeny, & Treadwell, 2001).

Depression

Depression is a highly prevalent disorder among youth in general and adolescents in particular. The most robust and common child self-report measure of depression is the Children's Depression Inventory (CDI), which purportedly measures symptoms of negative mood, interpersonal problems, ineffectiveness, anhedonia, and negative self-esteem (Kovacs, 1992). Factor analysis of the measure, however, has yielded constructs for dysphoria-sadness, social self-esteem, and oppositional-misbehavior (D. Cole, Hoffman, Tram, & Maxwell, 2000). The CDI has been used extensively to assess youths with depression, to distinguish depressed youths from other populations, and to examine developmental, psychological, and ethnic differences in children and adolescents. Another common child self-report measure of depression is the Reynolds Child (and Adolescent) Depression Scales, which are based on *DSM* criteria (W. Reynolds & Mazza, 1998).

Internalizing problems of childhood, which include anxiety and depression, can also be measured via parent and teacher ratings. The most common measures for doing so are the Child Behavior Checklist (CBCL), Teacher Report Form (TRF), Conners Rating Scales—Revised (CRS-R), and Child Symptom Inventory 4 (CSI-4; Achenbach & Rescorla, 2001; Conners, 1997; Sprafkin, Gadow, Salisbury, Schneider, & Loney, 2002). The CBCL and TRF have an Anxious/Depressed subscale. The CRS-R has subscales for Anxious-Shy and Perfectionism, and the CSI-4 has subscales related to generalized, separation, and social anxiety.

Attention-Deficit/Hyperactivity Disorder

In contrast to anxiety and depressive problems, externalizing problems of childhood are more commonly rated by parents and teachers. A popular rating scale for

ADHD is the ADHD Rating Scale IV, which measures symptoms of hyperactivity-impulsivity and inattention in home and school settings. The scale is linked to *DSM-IV* criteria and can successfully distinguish youths with and without ADHD (DuPaul, Power, Anastopoulos, & Reid, 1998). Related measures include the ADHD Symptoms Rating Scale and Attention Deficit Disorders Evaluation Scale, second edition (Holland, Gimpel, & Merrell, 2001; McCarney, 1995). In addition, the Strengths and Weaknesses of ADHD Symptoms and Normal Behavior has been designed as a dimensional measure of ADHD symptoms. Items are presented in Likert-scale format so raters can judge the strength of certain symptoms such as inattention (Collett, Ohan, & Myers, 2003).

General rating scales such as the CBCL, TRF, and CRS-R are also commonly used to assess symptoms of ADHD. The CBCL and TRF have a subscale for Attention Problems, and the CRS-R has several ADHD subscales for Cognitive Problems/Inattention, Hyperactivity, and an ADHD index. The CSI-4 is useful for examining symptoms of ADHD across settings. The Conners ADHD/DSM-IV Scales may also be used to assess more specific symptoms of ADHD, such as trouble concentrating and difficulty playing quietly (Conners, 1999). Finally, the Behavior Assessment System for Children, second edition (BASC-2) allows for parent and teacher ratings of many different behaviors from the preschool period to adolescence. Relevant subscales regarding ADHD include Attention Problems and Hyperactivity (C. Reynolds & Kamphaus, 1998).

Other Disruptive Behavior Disorders

General scales for ADHD also contain subscales relevant to other disruptive behavior disorders such as Oppositional Defiant and Conduct Disorder. The CBCL and TRF contain subscales for Rule-breaking and Aggressive Behavior, the CRS-R contains an Oppositional subscale, the CSI-4 contains subscales for Oppositional Defiant and Conduct Disorder, and the BASC-2 contains subscales for Aggression and Conduct Problems.

Measures more specifically designed for oppositional and conduct problems have also been developed. The Eyberg Child Behavior Inventory (parent) and Sutter-Eyberg Student Behavior Inventory—Revised (teacher) evaluate the number of difficult behavior problems and the frequency with which they occur (Eyberg & Pincus, 1999). The ACQ Behavior Checklist is an amalgam of items from the CBCL, Conners parent scale, and Quay-Peterson Revised Behavior Problem Checklist that assesses aggressive and delinquent behavior (Achenbach, Howell, Quay, & Conners, 1991; Quay & Peterson, 1996). Finally, the Conduct Disorder Scale contains ratings and yes/no questions regarding key aspects of oppositional and conduct problems such as argumentativeness and cruelty to others (Gilliam, 2002).

Developmental Disorders

Rating scales for developmental disorders such as autism and mental retardation have also been widely developed in recent years. These scales are typically completed by clinicians or teachers. A popular one for autism is the Childhood Autism Rating Scale, which solicits ratings for difficulty relating to people, odd body

movements, intense reactions to change, and verbal and nonverbal communication (Schopler, Reichler, & Renner, 1988). In addition, the Autism Screening Instrument for Educational Planning 2 (ASIEP-2) assesses for sensory, relating, body concept, language, and social self-help behaviors. The Autism Behavior Checklist is part of the ASIEP-2 and can be completed by parents and teachers (Krug, Arick, & Almond, 1995). Other prominent measures include the Gilliam Asperger's Disorder Scale, Gilliam Autism Rating Scale, second edition, and the Social Communication Questionnaire (Gilliam, 2001, 2006; Rutter, Bailey, & Lord, 2003).

Clinician-administered rating scales are commonly used to assess adaptive behavior in youths with mental retardation. A predominant measure is the Vineland Adaptive Behavior Scales, second edition, which assesses communication, daily living skills, socialization, motor skills, and maladaptive behavior (Sparrow, Cicchetti, & Balla, 2005). In addition, the American Association on Mental Retardation's AAMR Adaptive Behavior Scale—School (second edition) includes domains of independent functioning, physical development, economic activity, language development, prevocational and vocational activity, numbers and time, self-direction, responsibility, socialization, and maladaptive behavior (Nihira, Leland, & Lambert, 1993). Other prominent measures in this area include the Adaptive Behavior Assessment System (second edition) and Scales of Independent Behavior—Revised (Bruininks, Woodcock, Weatherman, & Hill, 1996; Harrison & Oakland, 2003).

Other Disorders

A full explication of all rating scales for all other disorders and problems that apply to children and adolescents is beyond the scope of this chapter, but some prominent ones are listed here:

- Adolescent Substance Abuse Screening Inventory (Miller, 2001)
- Dyslexia Screening Instrument (Coon, Waguespack, & Polk, 1994)
- Eating Attitudes Test (Foreyt & Mikhail, 1997)
- Pediatric Pain Questionnaire (Varni, Thompson, & Hanson, 1987)
- Scales of Social Competence and School Adjustment (Walker & McConnell, 1995)

Self-Monitoring

Self-monitoring refers to observing and recording one's own behavior as well as antecedents and consequences to the behavior. At a basic level, self-monitoring involves asking a child to record whether a specific behavior occurred or did not occur during a certain time period. A child may, for example, record whether he or she fully completed homework during a prescribed evening time. Or a child may record the number of hours spent in school per day. In addition, a child may record events just prior to and following a behavior, such as parent prompts to do homework and playing with friends after school. Recordings are typically done in paper-and-pencil fashion but can be done via self-analysis of audiotaping or videotaping.

Self-monitoring is utilized within a behavioral assessment paradigm for several reasons. First, self-monitoring is intricately linked to self-regulation, as people more aware of their own behavior often become better at controlling the behavior. This is especially true of excess maladaptive behavior; the more one self-monitors and becomes aware of excess maladaptive behavior, the less one tends to engage in the behavior. Self-monitoring is often linked to self-regulation processes of goal setting, self-evaluation, and self-reinforcement to reduce maladaptive behavior (Agran & Wehmeyer, 2006).

A salient example is habit reversal training to treat Trichotillomania (compulsive hair pulling) in children. One component of habit reversal training for this population is awareness training in which a child is instructed to increase attention toward subtle movements that eventually lead to hair pulling. Awareness training may be supplemented with self-monitoring and other behavioral assessment measures to modify behavior associated with less conscious thought (Azrin, Nunn, & Frantz, 1980; Romaniuk, Miltenberger, & Deaver, 2003).

Second, self-monitoring is important for educating youths about a targeted behavior. For example, psychoeducation for anxious children includes a detailed explication of the three response systems of anxiety: physiological, cognitive, and behavioral. A clinician treating an anxious child may ask him or her to provide daily ratings of physiological distress as well as lists of problematic thoughts and avoided situations. These personal examples are then used to explain how one anxiety response system leads to another. In one case, for example, a child's aversive physiological symptoms may lead to worry about harm from the symptoms and subsequent avoidance of a certain situation (Kearney & Albano, 2007).

Third, self-monitoring is important for assessing treatment outcome. Global measures of behavior are not particularly sensitive to treatment changes over time, but measures such as daily ratings of distress, hours spent in school, frequency of aggressive acts, and words spoken audibly can provide much more information about patterns of behavior change across time (Kearney & Silverman, 1999). Finally, self-monitoring may be useful for assessing secretive behaviors and helps alleviate pressure on others, such as teachers, to supply data for behavioral analysis.

Basic self-monitoring, such as recording the presence or absence of a behavior, may be more appropriate for younger children, those with a singular, primary behavior problem, and those who need initial training in how to self-monitor behavior. More sophisticated self-monitoring strategies can involve recording multiple behaviors, providing ratings of severity and intensity of the behaviors, and rating internal sensations or mood states. Children and adolescents are generally considered to be the best reporters of their own internal mood states, so self-monitoring is particularly relevant to youths with anxiety and depressive disorders. However, self-monitoring has also been used to assess aggression, socially appropriate behavior, and academic tasks such as paying attention, staying seated, and completing work problems accurately (C. Cole, Marder, & McCann, 2000).

Formal methods of self-monitoring have been developed for anxious children (Beidel, Neal, & Lederer, 1991). These methods involve daily diaries in which children list anxiety-provoking situations, time and setting of the situations, and responses to the situations. Children may also be asked to endorse an illustration

that best represents how they felt in a given situation and to rate their level of avoidance, physical symptoms, and coping skills. Daily diaries such as these have adequate reliability, but a key problem is that children are not typically compliant with the task, nor do they tend to engage in the task for more than 2 weeks (Beidel et al., 1999). In addition, the quality and accuracy of self-monitoring data may be questionable given children's limited cognitive development, distractibility, and lack of cooperation.

To address these concerns, clinicians should carefully tailor self-monitoring to the individual needs of a given child, simplify the procedure as much as possible, provide detailed training on how to engage in self-monitoring, link incentives and negative consequences to compliance and noncompliance, and ask parents and others to closely supervise self-monitoring without unduly influencing a child's responses. In addition, child self-monitoring may be supplemented with reports from informed others such as parents and teachers to assess accuracy (C. Cole et al., 2000).

Sociometric Assessment

Sociometric assessment refers to measures of children's social relationships and standing among peers. Sociometric assessment is important for identifying children with poor social competence who may be at risk for emotional, behavioral, and academic problems. This assessment approach is also used to identify children with exceptional social skills and who have particularly good social standing within a peer group. In many cases, sociometric assessment is conducted in school settings such as classrooms or play areas. Procedures for sociometric assessment include peer nominations, rating scales, and rankings.

Peer nominations typically involve asking children to privately identify or sort peers or classmates along certain dimensions. In many cases, a simple listing is provided of names of children a child most and least likes to play with. Although younger children are often asked to nominate three classmates for a given question, older children and adolescents may provide longer lists. Peer nominations are usually scored by dividing the total number of nominations by the total number of raters.

From Chan and Mpofu (2001), the following questions are often asked as part of a peer nomination sociometric procedure:

- What three classmates do you like the best?
- What three classmates do you like the least?
- Which three people do you admire most in your class?
- Which three people do you respect most among your classmates?
- What three people would you wish to be like in your class?
- Which of your classmates try hard and get good grades?
- Which of your classmates do not try hard and get poor grades?
- Which of your classmates get angry much of the time?
- Which of your classmates is most cooperative?

- Which of your classmates is a good leader?
- Which of your classmates start fights or become disruptive in class?

Peer nominations have been used to classify children into social groups. Children who receive many positive peer nominations tend to be popular or socially accepted, and those who receive many negative peer nominations tend to be rejected. Neglected or isolated students generally receive no peer nominations of any sort, though controversial students are linked to many positive and negative peer nominations (Elksnin & Elksnin, 1997; Terry & Coie, 1991). In addition, peer nominations may be helpful for identifying children with Social Phobia who engage in avoidant self-isolation (Kearney, 2005).

Peer nominations have several advantages, including ease of use, portability, and linkage to social groupings that have good psychometric stability. Peer nominations also have disadvantages, however. First, only those children actually nominated by peers are available for analysis. As such, some students may be mistakenly classified as isolated and a full examination of a classroom's social dynamics may be impaired. Second, peer nominations may work less well in racially diverse classrooms where student groups may not readily interact (Chan & Mpofu, 2001).

An alternative to peer nominations is peer ratings, in which all children in a classroom or social setting rate all other peers in that setting. Ratings may be linked to the questions listed earlier for peer nominations, such as rating a particular child on a 1 to 5 Likert-type scale where 1 = not at all liked and 5 = very much liked. Ratings can also be obtained for traits such as cheerfulness, humorousness, and disruptiveness. In many cases, children are presented with a grid that lists names of relevant peers as well as spaces for ratings. Peer ratings are then averaged for each child. Peer sociometric ratings can also be supplemented by teacher ratings (Hintze, Stoner, & Bull, 2000).

A key advantage of peer ratings over peer nominations is that data for all children in a particular setting may be obtained. In addition, ratings are more amenable to psychometric analysis and have been shown to be reliable and valid measures. The dimensional aspect of peer ratings allows for a better way of assessing social acceptance of a particular child. A downside of peer ratings is they may be less useful than nominations for categorizing a child's social status. Also, younger children may have difficulty understanding a rating system; in this case, peer nominations and ratings may be integrated, or a simpler rating scale may be used (Chan & Mpofu, 2001).

Sociometric assessment also includes rankings whereby all children are ranked by all other children in a particular social setting. In this way, all children are included in the assessment process and ratings are unnecessary. Rankings may come from teachers and other knowledgeable adults as well. Rankings may involve comparison to a standard for a particular characteristic, such as likeability, or comparison against all other peers. Rankings could also be made for each possible peer pair, such as student A, B, C, and so on versus all other students. Unfortunately, rankings require substantial effort on the part of a clinician and raters, and so are rarely used except in conjunction with other sociometric techniques (Mpofu, Carney, & Lambert, 2006).

Physiological Assessment

Physiological assessment, sometimes referred to as *psychophysiological assessment,* includes measures designed to directly evaluate the relationship between physiological processes and thoughts, emotions, and behavior. Physiological assessment concentrates on motor and visceral responses that underscore higher order cognitive processes and behavior to increase the psychometric strength of other forms of behavioral assessment. Physiological assessment with children remains less well-developed than other forms of behavioral assessment, especially with respect to traditional measures such as electrocardiogram for changes in heart rate and electroencephalogram for changes in brain wave activity. With newer techniques such as functional magnetic resonance imaging, however, physiological assessment is rapidly becoming a dominant force.

Traditional physiological assessment primarily includes heart rate, blood pressure, and measures of skin conductance and resistance, perspiration, and muscle tension. Heart rate, blood pressure, and vasomotor activity (constriction and dilation of blood vessels) have been commonly measured via electrocardiogram, a device that records electrical activity of the heart. However, simple pulse rate and systolic/diastolic values are often obtained without formal use of an electrocardiogram (Chakrabarti & Stuart, 2005; King, 1994).

Skin conductance and resistance are measured via electrodermal devices; electrodes are placed on a person's palm, middle and index fingers, or wrist and forearm. An electrical signal is introduced and the device measures the length of time necessary for the signal to travel from one electrode to another. Greater conductance or less resistance to the signal generally indicates greater autonomic arousal. In related fashion, perspiration may be measured by the number of active sweat glands in a certain skin area, often the palm (palmar sweat index), by means of a plastic impression method. Muscle tension is typically assessed via electromyogram, a device that measures muscle contraction and atonia (Kohler & Schuschel, 1994; Latzman, Knutson, & Fowles, 2006; Tedroff, Knutson, & Soderberg, 2006).

These traditional physiological measures have been prominently used to study children with anxiety disorders, with variable results. Children with anxiety disorders generally show increased heart rate, perspiration, and muscle tension. However, these increases are not consistently found, nor are they always maintained over time (Beidel, 1991; Schniering, Hudson, & Rapee, 2000; Turner, Beidel, & Epstein, 1991). With respect to externalizing behavior problems, low resting and task-related electrodermal activity has been associated with psychopathy and conduct problems. Low resting heart rate and high heart rate reactivity have been associated with aggression and conduct problems (Lorber, 2004).

These traditional forms of physiological assessment are gradually giving way to more advanced techniques. These include electroencephalograph, computerized axial tomography, magnetic resonance imaging, functional magnetic resonance imaging, and positron emission tomography. The electroencephalograph traditionally involved multiple electrodes placed on a person's head to measure brain wave activity and evoked response potentials, but more contemporary devices involve elaborate and less intrusive head covers that contain high-density electrodes. Brain

wave measurement has been particularly salient with respect to child sleep problems, learning disorders, ADHD, and seizure disorder (Collins & Rourke, 2003; Kuntsi, McLoughlin, & Asherson, 2006; Tay et al., 2006).

Neuroimaging methods now produce computer-generated pictures of specific brain sections. Computerized axial tomography is used to assess structural abnormalities of the brain such as tumors or enlarged ventricles potentially related to mental disorder. Iodine is injected and brain tissue density is assessed by detecting the amount of radioactivity from X-rays that penetrate the tissue. Magnetic resonance imaging produces higher quality brain images without radiation by having a person lie in a large cylindrical magnet as radio waves are beamed through the magnetic field. Functional magnetic resonance imaging is a more advanced technique, whereby brain images are taken rapidly to assess metabolic changes in the brain.

Positron emission tomography is an invasive procedure whereby radioactive molecules are injected into the bloodstream and emit positrons that collide with electrons. The resulting light particles emitted from the skull are utilized by a computer to construct a full image of the brain. Neuroimaging techniques have received greater attention in the child literature, particularly with respect to major problems such as traumatic brain injury, Schizophrenia, autism, mood disorders, ADHD, and learning disorders (Ernst & Rumsey, 2000).

Physiological assessment with children may also include biochemical assessment, such as analysis of metabolites for neurotransmitters or toxicology examinations. Although much basic information about child behavior may be obtained from physiological assessment, key disadvantages include its prohibitive cost and need for technical expertise. In addition, physiological assessment has long been criticized for providing information not particularly useful for developing an individualized treatment plan. An exception may be biofeedback, where children are provided feedback regarding internal responses and trained to regulate them to help reduce problems such as hyperventilation and accelerated heart rate (Hirshberg, Chiu, & Frazier, 2005).

Other Areas

Behavioral assessment for children can consist of other measures as well. Children's activity levels, for example, can be measured via behavioral observation or rating scales, but objective *activity measures* include the following:

- Pedometer, a device that records steps taken while walking
- Accelerometer, an electronic device that measures accelerations produced by body movement
- Actometer, a modified watch that records frequency of movement

These measures are commonly used to assess youths with ADHD but can also be useful for evaluating inactivity from a developmental disorder or sedentary lifestyle (Rapport, Kofler, & Himmerich, 2006; Sirard & Pate, 2001).

Behavioral assessment can also include review of archival or record data. Case history files, academic and school attendance records, medical and psychiatric

reports, completed written or oral schoolwork, report cards, and audiotapes and videotapes can be useful sources of information. A particularly innovative use of archival data has involved analysis of home movies of youngsters very early in life to identify emerging patterns of developmental disorders such as autism (Werner & Dawson, 2005).

Behavioral assessment also includes methods of evaluating cognitions more specifically than the rating scales described earlier. *Thought listing,* for example, requires a child to maintain a written log of thoughts before, during, and after a critical event such as an oral presentation. Thought listing may be done in naturalistic settings such as a classroom, but many children have difficulty identifying or listing their thoughts in these situations. In research settings, therefore, thought listing is sometimes utilized as part of a behavioral assessment test where a child engages in a task while videotaped. The clinician and child can then review the videotape and the child can recall various cognitions during certain segments (Kendall & Chansky, 1991; Spence, Donovan, & Brechman-Toussaint, 1999).

An alternative method is a *think-aloud procedure,* in which a child is asked to verbalize thoughts and emotions before, during, and after a behavioral assessment test. Solicited thoughts and emotions can then be categorized with respect to content and valence, such as positive, negative, or neutral. Unfortunately, this procedure is highly subject to reactivity and the information received may not be useful for developing a treatment plan. As such, researchers have modified the think-aloud procedure to focus on simple ratings of distress or perceived threat as well as open-ended questions about a child's cognitions, the responses from which are then coded (Bogels & Zigterman, 2000; Chansky & Kendall, 1997).

SUMMARY

Behavioral assessment is a problem-solving-oriented means of gathering information about current, observable behavior and what maintains that behavior over time. Methods of behavioral assessment include interviews, analogue assessments, behavioral observation, rating scales, self-monitoring, sociometric assessment, physiological assessment, and activity and other measures. A central model of behavioral assessment is functional analysis, or examining and manipulating antecedents and consequences of a behavior to provide information about the relationship of these events to the behavior. Methods of behavioral assessment emphasize molecular aspects of behavior, situation-specific behavior, and ongoing sampling of overt and current behavior in multiple contexts and from multiple sources of information. Behavioral assessment is thus a key aspect of evaluating children with myriad behavior problems and mental disorders.

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PART V

Treatment

CHAPTER 20

Play Therapy

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Identifying early mental health interventions tailored to the maturational needs of children is a growing concern in the United States (New Freedom Commission on Mental Health, 2003). Increases in societal problems that directly involve and impact young children has focused greater attention on this issue. The most recent U.S. surgeon general's report on mental health described the shortage of appropriate services for children as a major health crisis and estimated that, although at least 1 in 10 of all children suffer from emotional and behavioral problems severe enough to impair normal functioning, fewer than half receive treatment (U.S. Public Health Service, 2000). Thus, providing families with greater access to interventions that are responsive to the distinct needs of young children is critical, not only to diminish unnecessary suffering, but to prevent the development of more serious impairment across the life span.

Play therapy is founded on the principle that children must be approached and understood from a developmental perspective. Children are not miniature adults, and just as their world is a world of concrete realities, so too their means of communicating their world is on a child's level through play (Landreth & Bratton, 1998). Play is to the child what verbalization is to the adult. Play provides a developmentally responsive means for expressing thoughts and feelings, exploring relationships, making sense of experiences, disclosing wishes, and developing coping strategies (Landreth, 2002).

Although play therapy has been a treatment of choice for children since the early 1900s, the formation of the Association for Play Therapy (APT) in 1982 established play therapy as a specialized treatment modality within the field of mental health. The Association's influence, along with an increase in university-based play therapy training programs, provided the momentum for the substantial growth of the field over the past 25 years. Today, play therapy is widely used among clinicians to successfully treat a range of emotional and behavioral problems (Bratton & Ray, 2000, 2002). According to APT's web site (2006), approximately 5,000 mental health professionals identify themselves as play therapists. The growing interest in the field is also evident in the more than 3,500 play therapy publications describing its use, rationale, and efficacy, the majority of which were produced after 1980 (Center for Play Therapy, 2006).

RATIONALE FOR PLAY THERAPY

Play has long been considered significant in children's lives. As early as the eighteenth century Rousseau (1762, 1930) acknowledged the importance of observing children's play as a means of understanding them; however, it was not until the early twentieth century that the symbolic function of play was recognized. Froebel (1903) proposed that children's play has a conscious and unconscious purpose and thus is meaningful and important. More recently, the value of play has been recognized by early childhood experts as the way children learn about themselves, others, and their world as they try to organize and make sense of their experiences. Frank (1982) suggested that play is the way children learn what no one can teach them.

Play has been used by child therapists since the early twentieth century to provide children with a medium for self-expression. Most children below the age of 11 lack a fully developed capacity for abstract thought, which is a prerequisite to meaningful verbal expression and understanding of complex issues, motives, and feelings (Piaget, 1962). The child's world is one of concrete realities; thus, children must be afforded a concrete means of expressing their perception of self, others, and the world. Children can more meaningfully express their inner world through the symbolic representation provided by toys, with toys and play materials serving as their language. Although children often use play to directly express their experiences, in play therapy it is the symbolic function of play that is most significant. Sometimes children's symbolization of their experiences is quite apparent; at other times the connection may seem rather remote. Regardless, children use symbolic play to develop mastery and a sense of control over their world as they reenact their experiences directly in the safety of the therapeutic milieu.

In play therapy, then, play is viewed as the vehicle for communication between child and therapist on the assumption that the child will use play materials to directly or symbolically act out feelings, thoughts, and experiences that the child is not able to express meaningfully through words alone (Axline, 1947; Kottman, 2001; Landreth, 2002; O'Connor, 2001; Schaefer, 2001). Play allows children to bridge the chasm between their experiences and understanding, thereby providing the means for insight, learning, problem solving, coping, and mastery. In play therapy, regardless of the reason for referral, the therapist has the opportunity to enter into and experience the child's world and actively deal with the issues that brought the child to therapy. Axline described this process as one in which the child has the opportunity to play out feelings as they emerge, getting them out in the open, and either learning to appropriately control them or abandon them.

DEVELOPMENT OF PLAY THERAPY

As early as the eighteenth century play was recognized as essential to children's healthy development, but it was not until the early twentieth century that play was introduced into a therapeutic setting. The use of play in therapy originated with the classical child analysis case of Little Hans by Sigmund Freud. Freud saw Hans only once for a brief visit and conducted treatment by advising Hans father of ways to respond, with suggestions based on the father's notes about Hans play.

Generally acknowledged as the originators of play therapy, Anna Freud (1928) and Melanie Klein (1932) used play as a substitute for verbalized free association in their efforts to apply analytic techniques to their work with children. David Levy's (1939) development of release play therapy, along with the structured approach of Gove Hambidge (1955), marked the next advance in the field. In these two approaches play materials were structured by the therapist to induce catharsis; this is in contrast to psychoanalytic methods, in which the analyst made no attempt to direct the child's play.

Virginia Axline (1947) was the first to apply nondirective therapeutic principles in play therapy based on her trust in children's capacity to resolve their own problems through their play. Her work and writings in the late 1940s and 1950s, including her accounting of play therapy with Dibs (Axline, 1964), heralded perhaps the most significant development in the field of play therapy and popularized play therapy as a psychotherapeutic treatment modality for children. Building on Axline's work, Haim Ginott (1961), Clark Moustakas (1953), Louise Guerney (1983), and Garry Landreth (1991, 2002) contributed significantly to the widespread acceptance and practice of what is now more commonly referred to as child-centered play therapy (CCPT). Landreth (1991, 2002) provides an in-depth description of the principles and procedures of CCPT.

The development of filial therapy by Bernard and Louise Guerney in the early 1960s marked a significant and innovative development in the field of play therapy. The Guerneys were the first to develop a model for training and supervising parents in client-centered play therapy methods to use with their own children (Guerney, 2000). Since the late 1980s the use of filial therapy by practitioners has increased tremendously. Guerney (2000) credited this growth, in part, to the efforts of Landreth (1991, 2002) and his protégés. Building on the work of the Guerneys, Landreth developed a more condensed, 10-session parent training format based on his experience that time and financial constraints often hindered parents' participation. Landreth and Bratton (2005) formalized the 10-session training format in a text, *Child Parent Relationship Therapy (CPRT): A 10-Session Filial Therapy Model.* Grounded in the principles and procedures of child-centered play therapy, the CPRT parent training model emphasizes a balance of didactic and supervision experiences in a 2-hour weekly support group format and requires parents to conduct weekly, videotaped play sessions at home. The CPRT protocol was recently manualized by Bratton, Landreth, Kellam, and Blackard (2006), allowing for replication of the model. Although originally conceived as a group model for training parents, filial therapy has also been successfully adapted for individual parents and couples (VanFleet, 1994) and for training teachers, mentors and other paraprofessionals who play a significant role in children's lives (L. Jones, Rhine, & Bratton, 2002; Morrison, 2006; White, Flynt, & Draper, 1997).

The field of play therapy grew dramatically during the 1980s and 1990s as various theorists, academicians, and practitioners developed specific play therapy approaches based on their theoretical views and personal experiences with children, including gestalt play therapy (Oaklander, 1994), Jungian play therapy (Allan, 1997), Adlerian play therapy (Kottman, 1995), developmental play therapy (Brody, 1997), cognitive-behavioral play therapy (Knell, 1997), ecosystemic play therapy

(O'Connor, 2000), and prescriptive play therapy (Schaefer, 2001), to name a few. The sheer number of contributors to the development of this discipline prevents mentioning each.

Play therapy has grown over its 100-year history to include a variety of theoretical schools of thought. Although they may differ in their approach, they all embrace the therapeutic and developmental properties of play: "to help [children] prevent or resolve psychosocial difficulties and achieve optimal growth and development" (APT, 2001, p. 20). It would be impossible to describe all the various theoretical approaches to play therapy within the scope of this chapter; thus, in the practice and case study sections, we focus on the principles and procedures of the play therapy approach with the longest history of use and the strongest base of research support, CCPT.

RESEARCH IN PLAY THERAPY

Identifying mental health interventions that are effective for young children is a critical need for the twenty-first century (U.S. Public Health Service, 2000). In spite of play therapy's growing popularity among clinicians, the scientific community has been slower to embrace this modality as an empirically sound intervention. Scientifically proving the effectiveness of any therapeutic intervention is essential to its widespread acceptance and use. Although the field of play therapy has a history of over 60 years of continuous research, historically play therapy studies have been hindered by inadequate research design, most notably the lack of a control group and small sample sizes. More current research has utilized well-designed methodologies and supports the use of play therapy as an effective treatment that is responsive to children's unique developmental needs.

Play therapy research dates back to the 1940s, demonstrating one of the longest histories of research of any psychological intervention. In 1942, Dulsky attempted to study the relationship between intellect and emotional problems. He inadvertently established the effect of nondirective play therapy, which was to improve social and emotional adjustments, yet no improvement was shown on intellect. Although the research demonstrated a positive effect, several flaws in the research are noted, including a lack of control or comparison group, and insufficient description of treatment and participants. Since Dulsky's 1942 study, an approximate count of play therapy research, excluding filial therapy research, includes 103 studies, of which 71 were published in professional journals and 32 remained in dissertation form. The majority of play therapy studies demonstrate some positive effect of play therapy on the participants.

An examination of the play therapy research focused on filial therapy modality reveals 37 studies using a pre-post control group design, 27 of which were published in professional journals. Of the 37 filial therapy outcome studies, 27 utilized the 10-session CPRT model (Landreth & Bratton, 2005), originally introduced by Landreth (1991). Of the 27 studies examining the effects of the 10-session model, 24 were conducted under the supervision of the CPRT authors to ensure treatment integrity. Videotaped and live supervision was used to assure adherence to the procedures originally developed by Landreth and recently formalized in a published treatment

manual (Bratton et al., 2006). A detailed summary of filial therapy research from 1967 to 2005 can be found in Landreth and Bratton. For the purpose of this chapter, we focus our discussion on research examining the effects of play therapy conducted directly by professionals.

Current Play Therapy Research

Since 1990, 36 (27 published) research studies on the impact of play therapy have been conducted. These recent studies have demonstrated the positive impact of play therapy on general behavioral problems (Raman & Kapur, 1999; Shashi, Kapur, & Subbakrishna, 1999), externalizing behavioral problems (Flahive & Ray, in press; Garza & Bratton, 2005; Karcher & Lewis, 2002; Kot, Landreth, & Giordano, 1998; Schumann, 2005), internalizing problems (Packman & Bratton, 2003), self-efficacy (Fall, Balvanz, Johnson, & Nelson, 1999), self-concept (Kot et al., 1998; Post, 1999), anxiety (Baggerly, 2004; Shen, 2002), depression (Baggerly, 2004), speech problems (Danger & Landreth, 2005), and diabetes treatment compliance (E. Jones & Landreth, 2002).

Meta-Analytic Research on Play Therapy

LeBlanc and Ritchie (1999) published the initial results of their meta-analysis of play therapy outcomes summarizing the results of 42 controlled studies, with an effect size of .66 standard deviation. The researchers further detailed their study in a later publication citing that benefits of play therapy appear to increase with the inclusion of parents and optimal treatment duration (LeBlanc & Ritchie, 2001). Using Cohen's (1988) guidelines for interpretation of effect size, .66 denotes a moderate treatment effect, similar to effect sizes (ES) found in other child psychotherapy meta-analyses (Casey & Berman, 1985, ES = .71; Weisz, Weiss, Han, Granger, & Morton, 1995, ES = .71).

Bratton, Ray, Rhine, and Jones (2005) conducted the largest meta-analysis on play therapy outcome research. This meta-analysis reviewed 180 documents that appeared to measure the effectiveness of play therapy from 1942 to 2000. Based on stringent criteria for inclusion—use of a controlled research design, sufficient data for computing effect size, and the identification by the author of a labeled "play therapy" intervention—93 studies were included in the final calculation of effect size. The overall effect size was calculated at .80 standard deviation, interpreted as a large effect, indicating that children receiving play therapy interventions performed .80 standard deviation above children who did not receive play therapy. LeBlanc and Ritchie (2001) and Bratton et al. (2005) included filial therapy research in their definitions of play therapy. Filial therapy is a parental intervention based on CCPT.

Bratton et al. (2005) coded specific characteristics of play therapy that impacted or had no impact on play therapy outcome. Effect sizes for humanistic (ES = .92) and nonhumanistic (ES = .71) play therapy interventions were considered to be effective regardless of theoretical approach. However, the effect size reported for the humanistic approach was in the large effect category, whereas the ES for the nonhumanistic approach was in the moderate category. This difference in effect may be attributed to a larger number of calculated humanistic studies (n = 73)

compared to nonhumanistic studies ($n = 12$). When play therapy was delivered by a parent (ES = 1.15), the effect size was much larger than when delivered by a mental health professional (ES = .72), indicating the importance of involving parents in treatment to increase the chances of a successful outcome. This finding was similar to the conclusions of LeBlanc and Ritchie (2001), who reported parent involvement as a predictor of play therapy outcome. Treatment duration was also a factor in the success of play therapy. Optimal treatment effects were obtained in 35 to 40 sessions. However, many studies with fewer than 14 sessions also produced medium and large effect sizes. Age and gender were not found to be significant factors from which to predict the effects of play therapy; play therapy appeared to be equally effective across age and gender. An effect size was not calculated for ethnicity due to the lack of specificity in the reporting of ethnicity in individual studies. In addressing presenting problems, the researchers encountered difficulty distinguishing specific diagnoses and symptoms due to the variation among the studies. However, 24 studies were coded as investigating internalizing problems; analysis revealed an effect size of .81. An effect size of .78 was calculated for the 17 studies coded as examining the effects of play therapy on externalizing problems. For the 16 studies that addressed a combination of internalizing and externalizing problems, the effect size was .93. These results indicated that play therapy had a moderate to large beneficial effect for internalizing, externalizing, and combined problem types.

Individual Study Research on Play Therapy

Ray (2006) highlighted specific empirical studies that support the use of CCPT with children with behavioral and emotional challenges. The following studies are the most recent supportive research on CCPT that were identified as adhering to current experimental rigor. Kot et al. (1998) found that among child witnesses of domestic violence between the ages of 4 and 10 years, children who participated in 12 CCPT sessions scored significantly higher on self-concept than children who did not receive play therapy. Their mothers further reported significantly fewer externalizing behavior problems and total behavioral problems than mothers of children who did not receive play therapy. Among Hispanic children identified by parents or teachers as demonstrating behavioral problems at school, parents reported that those children who participated in 15 CCPT sessions demonstrated significantly fewer externalizing problems than children who participated in a small group curriculum-based guidance program, with the greatest improvement in conduct problems (Garza & Bratton, 2005). The children who participated in CCPT also demonstrated moderate improvements in internalizing problems. Flahive and Ray (in press) found that, compared to the control group, after 10 sessions of a child-centered approach to sand tray therapy for preadolescents in upper elementary school, teachers reported that the students significantly improved in their internalizing and externalizing problem behaviors, and parents reported improvement in children's externalizing problem behaviors. In this study, the wait-list control group continued to experience increased behavioral problems, both internalizing and externalizing.

Fall et al. (1999) and Post (1999) found that children who participated in CCPT demonstrated statistically significantly greater self-efficacy and self-esteem over their peers who did not participate in play therapy. Shen (2002) explored the effect of CCPT on Chinese children who experienced a devastating earthquake in Taiwan. Children who participated in 10 group sessions of CCPT demonstrated a significant decrease in anxiety and suicide risk compared to children who did not participate in play therapy. E. Jones and Landreth (2002) examined the efficacy of 12 sessions of intensive individual play therapy sessions on anxiety levels and medical compliance of 30 insulin-dependent children, ages 7 to 11 years. Compared with the control group children who attended the same camp, the 15 experimental group children demonstrated significant improvement in medical compliance (giving shots, following diet, etc.), when measured immediately following camp and at a 3-month follow-up. These are just a few examples of recent research conducted on CCPT that support the validity of the intervention in helping to reduce childhood problems and improve emotional conditions.

Historical Play Therapy Studies

The focus on these exemplary play therapy research studies by no means devalues the significant contribution that many other historical play therapy studies have made to the field. A rich history of play therapy efficacy has been established in the areas of hospitalized children with symptoms of anxiety (Cassell, 1965; Clatworthy, 1981; Johnson & Stockdale, 1975; Rae, Worchel, Upchurch, Sanner, & Daniel, 1989), self-concept (Crow, 1990; Gould, 1980; House, 1970; Perez, 1987), social adjustment (Cox, 1953; Oualline, 1976; Pelham, 1972; Thombs & Muro, 1973), and behavioral difficulties (Brandt, 2001; Gaulden, 1975; Hannah, 1986; Quayle, 1991). Detailed accounts of these studies can be found in the compilation of play therapy research summarized in Bratton and Ray (2000).

PLAY THERAPY IN PRACTICE

As stated earlier, the scope of this chapter prevents a description of the practice of play therapy from all theoretical viewpoints. Although the various play therapy approaches may differ philosophically and technically, they are alike in their use of the therapeutic and developmental properties of play to help children benefit from treatment. In this section, we focus on the principles and procedures of CCPT. Of the many well-established play therapy schools of thought, CCPT has the longest history of use, the strongest research support, and, according to a survey of APT members (Lambert et al., 2005), the largest number of followers.

Role of Therapist and the Therapeutic Relationship

The goal of child-centered play therapy is to establish conditions so that the child can experience growth and integration. Raskin and Rogers (2005) postulated that if the therapist is successful in conveying genuineness, unconditional positive regard, and empathy, then the child will respond with a changed personality organization.

Child-centered play therapists believe that a child client's experience in the counseling relationship is the factor that is most meaningful and helpful in creating lasting, positive change.

Given the opportunity, children will play out their feelings and needs in a manner or process of expression that is similar for adults. Although the dynamics of expression and the vehicle for communication are different for children, the expressions (fear, satisfaction, anger, happiness, frustration, contentment) are similar to those of adults. Children may have considerable difficulty describing what they feel or how their experiences have affected them. If permitted, however, in the presence of a caring, sensitive, and empathetic adult, they will reveal inner feelings through the toys and materials they choose, what they do with and to the materials, and the stories they act out. Children's play is meaningful and significant to them, for through their play they extend themselves into areas they have difficulty entering verbally.

Because the child's world is a world of action and activity, play therapy provides the therapist with an opportunity to enter the child's world. The child is not restricted to discussing what happened; rather, the child lives out at the moment of play the past experience and associated feelings. If the reason the child was referred to the therapist is aggressive behavior, the medium of play gives the therapist an opportunity to experience the aggressive behavior firsthand as the child attempts to break toys or to shoot the therapist with a dart gun, while also allowing the child opportunities to learn self-control by responding with appropriate therapeutic limit-setting procedures.

Axline (1947) identified eight basic principles that guide the therapist in child-centered play therapy. These basic principles are consistent with a person-centered philosophy of working with children by emphasizing the primacy of the counseling relationship. These principles insist that the therapist (1) must develop a warm, friendly relationship with the child; (2) accepts the child unconditionally, without wishing the child were different in some way; (3) establishes a feeling of permissiveness in the relationship so that the child feels free to express himself or herself; (4) recognizes and reflects the feelings of the child to create understanding for the child; (5) respects the child's innate ability to solve his or her own problems and offers the child the opportunity to take responsibility for his or her own behavior; (6) does not attempt to direct the child's actions or conversation, but allows the child to lead the way; (7) recognizes the gradual nature of the child's process and does not try to rush counseling; and (8) establishes only those limitations that are necessary to anchor the child's experience in counseling to the world of reality.

Without the presence of play materials, the therapist could only talk with the child about the aggressive behavior the child exhibited yesterday or in the past weeks. In play therapy, whatever the reason for referral, the therapist has the opportunity to experience and actively deal with that problem in the immediacy of the child's experiencing.

How Play Therapy Facilitates Change in Children

Play therapy assists children's development by helping them learn to know and accept themselves. Most of what is learned in the play therapy relationship is not

cognitive learning, but experiential, intuitive learning about the self that occurs over the course of therapy. Landreth (2002) described the following learning experiences:

- *Children learn to respect themselves.* The play therapist maintains and communicates constant regard and respect for children regardless of their behavior, whether they are playing passively, acting out aggression, or being whiny and insisting on help with even the simplest tasks. Children sense the therapist's respect, feel respected, and, because of the absence of evaluation and an ever present acceptance, they internalize the respect. Once children have respect for themselves, they learn to respect others.
- *Children learn that their feelings are acceptable.* By playing out their feelings in the presence of an adult who understands and accepts even the intensity of their feelings, children learn that all of their feelings are acceptable. As children begin to experience that their feelings are acceptable, they begin to be more open in expressing them.
- *Children learn to express their feelings responsibly.* Once children have expressed their feelings openly and been accepted, those feelings lose their intensity and can more easily be controlled appropriately. This is a freeing process for children to experience in that they are free to go beyond those feelings.
- *Children learn to assume responsibility for themselves.* In the natural process of development, children strive toward independence and self-reliance but often are thwarted in their efforts by adults who, although well intentioned, take charge by doing things for them and thus deprive children of opportunities to experience how being responsible for themselves feels. In the play therapy relationship, the therapist believes in children's ability to be resourceful and so resists doing anything that would deprive children of the opportunity to discover their own strength. As the therapist allows children to struggle to do things for themselves, they discover what that responsibility feels like.
- *Children learn to be creative and resourceful in confronting problems.* When children are allowed to figure out things for themselves, to derive their own solutions to problems, and to complete their own tasks, their own creative resources are released. With increasing frequency, children will tackle their own problems and experience the satisfaction of doing things all by themselves. Although children may at first resist the opportunity to solve their own problems, the creative tendency of the self will come forth in response to the therapist's patience.
- *Children learn self-control and self-direction.* If no opportunities are available to experience being in control, learning self-control and self-direction is not possible. Although this principle seems simplistically obvious, the absence of opportunities like this in children's lives is conspicuous when one takes the time to carefully observe children's interactions with significant adults. Unlike most other adults in children's lives, the play therapist does not make decisions for children or try to control them either directly or subtly. Limits on children's

behavior in the playroom are verbalized in such a way that children are allowed to control their own behavior. Because control is not applied externally, children are able to make their own decisions.

- *Children gradually learn, at a feeling level, to accept themselves.* As children experience being accepted just as they are with no conditional expectations from the therapist, they gradually, and in sometimes imperceptible ways, begin to accept themselves as worthwhile. This is both a direct and an indirect process of communication and learning about self. Acceptance is an attitudinal message communicated verbally and nonverbally by all that the therapist is and does. The children first feel acceptance and then know it by being accepted nonjudgmentally for who they are, just as they are, with no desire that they be different. This increased self-acceptance is a major contributing factor to the development of a positive self-concept.
- *Children learn to make choices and to be responsible for their choices.* Life entails a never-ending series of choices. How can children learn how to make a choice if they are not afforded the opportunity of making choices, of being indecisive, of struggling, of wanting to avoid, of feeling incapable, of being anxious, and of feeling apprehensive that their choice will be unacceptable to others? Therefore, the therapist avoids making even simple choices for children.

Structuring the Relationship and Play Sessions

Beginning the Relationship

For most children, the therapeutic relationship is like no other they have experienced, one in which they are fully accepted and valued for who they are. The play therapist maintains and communicates a constant regard and respect for children. The structuring of the relationship begins with the initial interactions between therapist and child in the waiting room. The therapist's task is to give full attention to the child, communicating the child's importance. (*Note: Prior to the initial meeting with the child, the therapist has met with the parents without the child present to fully focus on the parents' needs and concerns, gather background history, and inform parents of what to expect.*)

In the initial meeting between therapist and child, the therapist should crouch down, make eye contact, and greet the child. Following a short introduction, the therapist can say, "We can go to the playroom now. Your mom (or the person who brought child) will wait here, so she will be here when we come back from the playroom." The play therapy relationship is further structured by how the therapist introduces the child to the playroom. Verbal communication should be kept to a minimum at this point. No amount of words will convey to the child that this is a safe place. The child learns this through experience. The therapist should choose words carefully to communicate to the child freedom, self-direction, and the parameters of the relationship. The therapist might say something such as "Sarah, this is our playroom, and this is a place where you can play with the toys in a lot of the ways you would like to." This statement conveys responsibility for direction, freeing the child. Boundaries on this freedom are conveyed by the words "in a lot of the ways,"

which in effect communicates limits on behavior. The playroom is not a place of complete freedom where children can do anything they want.

Playroom and Materials

Landreth (2002) suggested that the dimensions of a playroom be 12 by 15 feet. However, an open space of almost any size will suffice. In school buildings, we have used janitorial closets, old book rooms, and portable buildings for a playroom. If a therapist is very limited on space, he or she can develop a traveling playroom by maintaining a box or bag of materials that can be used in an office or conference room.

Because toys and materials are part of the communicative process for children, careful attention must be given to the selection of appropriate items. The rule is selection rather than accumulation. Play areas and playrooms containing an assortment of randomly acquired toys and materials often resemble a junk room and doom the play therapy process to failure. Toys and materials should be carefully selected for (a) the contribution they make to the accomplishment of the objective of play therapy and (b) the extent to which they are consistent with the rationale for play therapy. Not all play materials automatically encourage the expression of children's needs, feelings, and experiences. Because toys and materials are used by the child in the act of play to communicate a personal world to the counselor, toys should reflect the cultural experiences of children. Landreth (2002) suggested that toys and materials should be selected with the following goals in mind:

- Establishment of a positive relationship with the child.
- Expression of a wide range of feelings.
- Exploration of real life experiences.
- Testing of limits.
- Development of a positive image.
- Development of self-understanding.
- Opportunity to redirect behaviors unacceptable to others.

The following toys and materials are recommended because they can encourage a wide range of expressions and can be easily transported in a tote bag or stored out of the way in a corner or in a closet. They can be grouped into three broad categories:

1. Real-life toys, such as a doll (small size with soft body); a bendable doll family; a small cardboard box with rooms indicated by strips of tape or felt pen marker (could double as container for materials); doll house furniture (cost will probably necessitate purchase of plastic furniture and thus set limits on how it is used); a nursing bottle (plastic); two play dishes and cups (plastic or tin); spoons (avoid forks because of sharp points); a small car; a small airplane; and a telephone.

2. Acting-out or aggressive release toys, such as handcuffs (spring-release type without key because of possible embarrassment if child cannot get them off), a dart gun, toy soldiers (30-count size is sufficient because of the difficulty of picking up more than that), a pounding bench, and a rubber knife.
3. Toys for creative expression and emotional release, such as crayons (8-count size to avoid too many choices; break some in half and peel paper off to convey that it is not necessary to be neat and precise); newsprint; blunt scissors; pipe cleaners; popsicle sticks; Play-doh (place it in small plastic container to keep it pliable; the top of the cardboard Play-doh can may fray and allow air to enter can); hand puppets; a small plain mask (Lone Ranger type); a Nerf ball (a rubber ball bounces too much); Gumby (bendable, nondescript figure); scotch tape; nontoxic glue or paste; and an empty vegetable can (doubles as container for toy soldiers).

An inflatable punching toy (Bobo) is desirable if the room allows. Additional toys and materials can be added by counselors who have a permanent place for play therapy. Landreth (2002) provided a complete list of play materials in a fully equipped play therapy room. Highly structured materials and electronic toys should generally be avoided because they may interfere with rather than facilitate children's expressions. Because toys are the child's words, they should be in good condition.

Facilitative Responses

The following therapist skills are considered to be essential to the child-centered play therapy process. These are specific skills expected to be demonstrated in most play therapy sessions. The extent to which they will be used depends on the therapist's intuitive understanding of the child's needs in the immediacy of the session.

Nonverbal Skills

Play therapy is heavily reliant on nonverbal skills. Because play therapists believe that play is the language of children, the verbal world becomes less important in a play therapy session. Nonverbal skills are critical to any person-centered approach, but especially to play therapy. Nonverbal skills are demonstrated when the therapist leans toward the child and is physically directed toward the child at all times, showing keen interest in the child throughout the session. As with counseling adults, the therapist should strive to be congruent with how the child expresses himself or herself. The therapist should not only match the level of affect displayed by the child but should also convey a sense of genuineness. The skill of matching verbal response with nonverbal response is representative of the therapist's level of genuineness with the child. Mental health professionals who are new to play therapy often find this skill challenging, as they are not yet comfortable with responding to children's nonverbal play. Just as the therapist would not flatly respond, "You're happy that you got that to stick" (after the child looks over with a big smile after

finally getting the dart to stick on the wall after 10 unsuccessful attempts), the therapist should avoid being so animated that he or she overwhelms the child. Again, congruence and genuineness are the key.

Verbal Skills

How the therapist verbally responds to the child is almost as impactful as the words chosen. Effective delivery of responses includes a focus on succinct, interactive responses and rate of responses. Because play therapy is offered to young children and because play therapy recognizes the limited language ability of these children, the importance of short therapeutic responses is key. Lengthy responses quickly lose the child's interest, confuse the child, and often convey a lack of understanding on the part of the therapist. Therapists match their level of interaction to that of the child. If the child is quiet and reserved, the play therapist will slow his or her responses; if the child is highly interactive and talkative, the play therapist will match this level of energy with an increased number of responses.

Child-centered play therapy verbal responses can be structured into categories that help facilitate growth in the child. The following are several relevant categories of verbal responses. Therapist responses should reflect the developmental level of the child; responses to a 3-year-old and an 8-year-old should be different. The examples that follow are appropriate for most young children.

Reflecting Nonverbal Play Behavior: Tracking The therapist tracks or reflects the nonverbal content of the child's play behavior simply by stating what he or she observes. Reflecting the child's nonverbal behavior allows the child to know that the therapist is interested and accepting of the child. It also helps the therapist immerse himself or herself into the child's world. Examples of tracking behavior are "Hmm, you're not sure what you want to play with" (as a child walks around, briefly touching several different toys) or "You decided to play with that" (as the child picks up the plastic horse and puts in the sandbox).

Reflecting Content Reflecting verbal content in play therapy is identical to reflecting content in talk therapy. To reflect content, the play therapist paraphrases the verbal interaction of the child. Reflecting content validates the children's perceptions of their experience and helps to clarify their understanding of themselves (Landreth, 2002). An example of reflecting content is "You know a lot about dinosaurs" (after the child shares detailed information on dinosaurs).

Reflecting Feeling Reflecting feeling is the verbal response to emotions expressed by children in play therapy. The reflection of feeling can sometimes be threatening to a child and should be presented carefully. Reflecting feeling helps a child become aware of emotions, thereby leading to the appropriate acceptance and expression of such emotions. Examples of reflecting feeling are "You are really angry with that guy" (after the child throws the big elephant across the room, saying, "He stepped on everyone. I hate him") and "You're frustrated that it's not working out the way you wanted" (when the child tries unsuccessfully to cut a heart design out of paper

and then crumples it up). As stated previously, therapist nonverbals and intonation of voice are as important, if not more so, as the words used.

Facilitating Decision Making and Returning Responsibility One of the play therapist's goals is to help children experience a sense of their own capability and to take responsibility for their expression of capability. Responses that facilitate decision making or return responsibility help children experience themselves as able and empowered. Examples of responses that facilitate decision making or return responsibility are "In here, you can decide what color you want it to be" or "In here, that's up to you" (when the child wants to draw a picture and asks, "What color of paper should I use?") and "Show me what you want me to do" (when, without making an attempt, the child points to her wet painting on the easel and asks, "Can you take this off for me?"). The latter response allows the therapist to see if the child already knows how to use the clips holding the paper to the easel or is quite capable of doing so once she tries. Many children come to play therapy with little experience of figuring things out for themselves. In their experience, adults have all the answers, so there is no point in their trying. The rule of thumb in play therapy is to never do for a child what the child can do for himself or herself (Landreth, 2002).

Facilitating Creativity and Spontaneity Helping children experience their own sense of creativity and freedom is another benefit of play therapy. Acceptance and encouragement of creativity sends a message to children that they are unique and special in their own way. Experiencing the freedom of expression allows them to develop flexibility in thought and action. Examples of responses that facilitate creativity or spontaneity are "You can make whatever you want" (when the child picks up craft material sticks and asks, "What do I make with these?") and "You've got a lot of ideas about want you want to make today" (when the child moves from making one craft project to another in play session).

Esteem Building and Encouraging Encouraging children to feel better about themselves and their abilities is a constant objective for the play therapist. The use of esteem-building statements works to help children experience themselves as capable. Examples of esteem-building and encouraging responses are "You're trying lots of different ideas. You're not giving up" (when the child has trouble getting the glue bottle open and tries several different solutions) and "You did it" or "You kept at it until you found a way" (when the child tries and tries to reach the top shelf, and after a few attempts, succeeds).

Facilitating Relationship Because the therapeutic relationship serves as a model for all intimate relationships, the play therapist seeks to respond to any attempt by the child to address the relationship. Relational responses help the child learn effective communication patterns and express the therapist's care for the child. Examples of responses that facilitate the relationship are "You're wondering what I think about that" (when the child is drawing a picture and stops to look up at the therapist but says nothing) and "You want us to look just alike" (when the child puts a hat on

the therapist and says, "Now yours is like mine") and "You're mad at me, because I wouldn't do what you wanted" (after the therapist declines to eat the sand given by the child, and the child responds, "You're no fun, I don't want to come here anymore"). Relationship responses should always include a reference to the child and a reference to oneself as therapist.

Limit Setting In an attempt to provide an environment for a child that allows self-direction and self-responsibility, minimal limits are established in child-centered play therapy. The goal is to help the child move toward the ability to self-limit. Typically, limits are set when children attempt to damage themselves, another person, and certain expensive or irreplaceable toys or if their behavior impedes counselor acceptance.

Landreth (2002) proposed a specific method for setting limits in play therapy. This method has been widely adopted by play therapists, regardless of theoretical orientation, as the initial response to setting a limit in the playroom. The A-C-T model of limit-setting is Acknowledge the feeling, Communicate the limit, and Target an alternative. In this model, the play therapist first recognizes and addresses the child's feelings in the moment: "You're really excited about your project." Then the therapist sets a short, concrete, definitive limit: "But it's not for stapling to the walls." Finally, the therapist provides an alternative to the action: "You can tape it to the walls." When children have directed energy in the moment, it is important to provide them an alternative for that energy so that they do not feel the need to act on impulse. Although there are other methods for setting limits, the A-C-T model is short, direct, and works effectively.

Involving Parents as Partners in the Therapeutic Process

Prior to the first play session, the play therapist should meet with the parents without the child present. The content and length of the initial parent meeting will vary with the therapist's theoretical and personal approach to involving parents in their child's therapy and the importance placed on intake assessment. The therapist's primary task during this session is to form a connection with the parents, focusing entirely on the parents' needs and concerns and conveying to them their importance in their child's therapy. The initial parent consultation provides the play therapist with the opportunity to inform parents what they need to know about play therapy and what to expect during the process. During this initial session, the therapist will explain the play therapy process to the parents, including the description of the playroom, how the therapist responds to the child, and how the process is helpful to the child. The therapist will also want to inquire about special considerations for the child that would impact the therapeutic process. For example, some young children have a history of difficulty separating from their parents. It is helpful to know that information prior to the child's first visit for play therapy so the therapist can discuss possible options with the parents and coach them on how to respond if the child clings to them.

The therapist will also take this opportunity to convey the importance of confidentiality. Although, in most states, children do not have the legal right to

confidentiality, the play therapist emphasizes the need for children to experience confidentiality of specific actions and words in the playroom. In play therapy, it is expected that parents will not observe their children's play so that the child can feel free to fully express thoughts, feelings, and actions. In the first consultation, the therapist will also review the importance of attendance, indicating that sporadic participation in play therapy interferes with progress.

Finally, the play therapist highlights the need for a final session if the parents decide to prematurely terminate therapy. Often, when parents decide to terminate therapy without the support of the therapist, the process is abrupt and disconcerting to the child. A final session allows children to experience a positive end to a relationship that has been important to them. Play therapists approach the initial consultation with an attitude of relational alignment. For parents to invest in the therapeutic process, they need to feel that the play therapist is understanding and accepting of their plight. An initial consultation is not a time for education and instruction, which can quickly become overwhelming and feel judgmental to the parents.

The play therapy process encourages consistent contact with parents on an individual basis for subsequent consultations. Meeting with parents on a regular basis (at least monthly) from 30 minutes to an hour allows the play therapist to develop a relationship with the parents that will, consequently, support the child's progress. The length and frequency of parent meetings is determined by the unique needs of each family. In continuing consultations, therapists typically check with the parents about new events at home and school. To keep parents involved in their child's play therapy and help them develop a better understanding of their child, the play therapist will discuss play themes exhibited by the child during play and discuss the relevance to the child's home and school life. For example, a therapist may share that the child in play therapy seems to exhibit a significant need for power and control and responds well to choice giving by the therapist. After helping the parents better understand the child, a therapist may then teach a short concept. Using the example just given, the play therapist may follow the explanation of the theme with teaching how choice giving works and then practicing with the parents. Common skills for play therapists to teach parents are choice giving, encouragement, reflection of feeling, problem-solving techniques, returning responsibility to the child, and limit setting (Bratton et al., 2006).

Facilitating consistent consultations with parents encourages continuation of the child's therapy, improves in parenting skills, and fosters better systemic relationships within the therapeutic context. For some children who are experiencing significant difficulties at school, similar consultations with an invested teacher can be helpful and provide greater consistency in how significant adults respond to the child.

In addition, the use of CCPT principles, including involving parents more fully in their child's therapy through filial therapy training, can impact the family system by encouraging the adults in the system to make use of CCPT's relational principles, thereby creating a safer, more enhancing environment for children. The scope of this chapter prevents a detailed description of filial therapy procedures. Landreth and Bratton (2005) provide a comprehensive description of their 10-session filial therapy model based on CCPT principles and skills.

Group Play Therapy Applications

The skills used in child-centered individual play therapy are the same skills used in child-centered group play therapy. As in individual play therapy, a child-centered play therapist recognizes the group members' ability to develop their own potential to move in a positive direction (Landreth & Sweeney, 1999). Ginott (1961) recognized a child's "social hunger" as a need to interact and develop relationships with other children. Group play therapy feeds this social hunger by placing children together in an environment that provides for freedom of expression and understanding of multiple worldviews. A child's appropriateness for a referral to group play therapy is dependent on developmental readiness and presenting concerns (Landreth & Sweeney, 1999). White and Flynt (1999) advocated the use of child-centered group play therapy due to its focus on developmental appropriateness, variation of function (applicable for preventive, remedial, or crisis intervention counseling), vicarious learning and cathartic effect, positive socialization, and growth through insight.

The practicality of using child-centered group play therapy is limited by space and scheduling. In response to this limitation, the play therapist may be able to serve only two to three children in a group setting. This number allows for comfortable movement in the space, while still providing the benefits of a social environment.

Multicultural Applications

Play therapy, including filial therapy, has been used as an effective treatment modality for children across cultures (Bratton et al., 2005; Landreth & Bratton, 2005). Because of the acceptance of the child as a holistic organism, negating any preconceived labels or designations, child-centered play therapists have the opportunity to serve all children while embracing their cultural and ethnic backgrounds. As the play therapist seeks to fully know and understand each child, it is imperative that the therapist recognizes the significance of the child's culture and is open to learning about the child's and family's unique social and cultural environment. Person-centered philosophy is sometimes criticized for its individual focus that potentially ignores the contextual nature of some cultures (Kirschenbaum, 2004). However, a well-trained child-centered play therapist will recognize the impact of culture and embrace the culture as well as embracing the individual. Because toys and materials are used by the child to communicate a personal world to the counselor, toys should reflect the cultural experiences of the child.

Therapist Training

Mental health professionals who practice play therapy should have specific training in child development, as well as didactic training and supervised experience in play therapy theories and procedures. Children are not miniature adults. Because talk therapy may not be developmentally appropriate, they require an alternative form of communication. Thus, those who provide therapy for children require specialized training. All too often, graduate students receive no training to counsel children and then are assigned to an internship setting that requires them to work exclusively with

children. Many universities do not have specific play therapy training; however, play therapy training and supervision is widely available through conferences and workshops. To find training near you, check with the Association for Play Therapy (www.a4pt.org) and the Center for Play Therapy (www.centerforplaytherapy.com).

Case Illustration

The following case provides a glimpse into the process of play therapy with a young child with a history of multiple interpersonal traumas. The child participated in play therapy and filial therapy with the first author for a total of 55 sessions over a 20-month period using CCPT procedures presented in the practice section. My initial objective was to establish a therapeutic relationship in which the child would feel secure and safe enough to explore all his experiences, including those that in the past had been too threatening to admit to awareness. Axline (1947) described the optimal play therapy climate as "good growing ground," where the child can fully experience all aspects of self and realize potential for growth. Specifically, this case illustration shows the child's movement toward trusting primary adults in his life, integration of his early experiences into his self-concept, and increased levels of self-acceptance and self-regulation. Regular parent consultation and filial therapy training were important components of treatment. As part of the filial therapy training, parents participated in 16 supervised play sessions with their child during the treatment period. Identifying information has been changed to protect the child and his family.

Identifying Features of the Client

Dom, an approximately 6-year-old male of Chinese descent, was referred to play therapy by his adoptive parents, both of whom are of Anglo descent. He was adopted at the age of approximately 4 years of age. Prior to then, he lived at an orphanage in China. According to sketchy records, he was abandoned at the orphanage when he was approximately 1 year old by a neighbor and given the name Dom.

At the time of intake, Dom had recently decided he wanted to be called Andy. He was living in the southwestern region of the United States with his adoptive parents, Carol and Tom; a 14-year-old stepsister, Ann (biological child of parents); and a 7-year-old stepbrother, Chris, also adopted from China when Andy was $5^1/_2$ years of age. His mother was 45 years old and employed as a teacher in the private school that Andy and Chris attended. Andy's father, also 45 years old, was employed as a sales professional and traveled several days a week. The parents shared similar values and traditional Protestant beliefs. Both were actively involved in their children's extracurricular activities.

Presenting Complaints

During a phone consultation with Andy's mother, she reported concerns about his mood swings, violent outbursts, running away, stealing (at home), and refusing to let her leave home without him. Secondary concerns included eating and sleeping difficulties, gender identity issues, and possible sexual abuse from when he lived in the orphanage. She further disclosed that Andy had been diagnosed by the school psychologist with Reactive Attachment Disorder and that he was receiving special education services for a learning disability and speech

and language delays. Carol's stress was obvious in her voice. I immediately scheduled her and her husband for an intake and initial parent consultation, emphasizing the importance of both of them attending. I allotted 2 hours to permit sufficient time to focus on Carol's and Tom's needs as parents of this child and to help them feel supported, while also allowing them time to tell Andy's story, a story that appeared to be rather complicated for such a short life.

Although both Tom and Carol attended the initial intake session, the following information, unless otherwise specified, was reported by Andy's mother. Intake was conducted 1 week after Andy's sixth birthday, during the summer after he had completed kindergarten. Carol began by relaying her concerns almost verbatim to her phone report, with Tom nodding but not adding anything. When asked what they enjoyed most about Andy, Carol described Andy as charming, adding, "He is very affectionate and funny and loves to cuddle." However, she quickly refocused on her concern about his episodes of sudden violent outbursts that she described as "coming out of nowhere." She reported that Andy purposely damaged things during his violent outbursts, adding that he had put holes in the wall and broken windows in their home. She reported that when he was in a "rage," she physically restrained him until he calmed down, adding that Andy generally wanted her to continue to hold him and "cuddle" after an episode.

Carol expressed a high level of frustration about not being able to leave the house without Andy. When she tried, he would chase the car until she came back. At the time of intake, she had resigned to not leaving him except at school and to occasionally try to run an errand after he was asleep. Because Carol was employed as a reading specialist at Andy's school, administrators allowed him to come to her classroom when his teacher determined that he needed his mother. Both boys came to her classroom after school while she completed her workday. When I inquired about when she had any time to herself, she joked, "Only in the bathroom . . . and even then, not always!" Although she did not complain during the intake, my impression of Carol was that she was highly stressed in her role as parent and feeling resentful.

Carol and Tom both expressed their alarm that Andy had lately begun stealing from them. When confronted, he would first deny that he took anything, and then immediately tell Carol where she "might find it." Andy's speech delay, possible learning problems, and eating and sleeping problems were briefly discussed, but those issues were not at the top of their list of current concerns.

When asked about Andy's adjustment to his adoption experience and his new family, Carol stated that he had adjusted well to their family, and Tom agreed. She reported that he refused to discuss that he was adopted or that he had lived anywhere other than with them, adding, "Andy tells people that he was born in my [Carol's] tummy." Andy acknowledged the difference in his skin color by telling Carol that she needed to "get a tan like mine." Carol found this humorous and endearing. She believed that he had blocked out all his early memories from the orphanage. Tom said little during the intake, other than that he agreed with Carol's report. Tom acknowledged that Andy is more of a problem for Carol than for him and that Andy was much closer to Carol than to him. Tom acknowledged that because he traveled several days a week, he had much less responsibility for caretaking of Andy and Chris. Tom reported that although Andy seemed to obey him better than Carol, Andy would not stay with him without Carol present.

(*Continued*)

Tom explained that Carol was generally the one who restrained Andy when he was out of control, because he was afraid that he would hurt Andy.

Socially, Andy was described by his parents and teacher as "acting younger than his age." He enjoyed playing with a few younger children; however, his preference was to play at home with his mother. The decision had been made that Andy would repeat kindergarten in the fall with the same teacher.

Carol's and Tom's responses on the Child Behavior Checklist (CBCL) administered at intake supported their worries about Andy's behavior at home and his performance at school. Carol's scores on the CBCL (reported in Table 20.1) fell in the clinical range in several areas. Tom's responses on the CBCL revealed a pattern of concern similar to Carol's, but showed a lesser degree of concern. A psychiatric referral for Andy was briefly discussed, but Tom and Carol were adamantly opposed to medication as a treatment option.

Table 20.1 Carol's Scores on the Child Behavior Checklist

Elements	*5 months Prior	July Pretreatment	December 5 month	May 9 month	August 12 month	December 16 month	April 20 month
Total problems	64C	65C	61B	59	59	60	58
Externalizing problems	69C	72C	67C	63B	62B	62B	60B
Syndrome scales							
Aggressive behavior	71C	72C	69B	65B	63	64	62
Rule-breaking behavior	70C	73C	65B	60	60	65B	62
Thought problems	66B	65B	58	57	58	61	58
Social problems	70C	68B	60	63	60	61	59
DSM-oriented scales							
Conduct problems	72C	73C	66B	64	63	64	63
Oppositional defiant	68B	71C	64	62	61	60	58
Affective problems	70C	70C	65B	64	61	61	60
Competency scale scores							
School	27C	24C	27C	29C	35B	30C	31B

Note: Only CBCL scales that were elevated at pretreatment intake are included.
B = Borderline clinical range; C = Clinical range.
*Administration by school psychologist 5 months prior to intake.

Carol and Tom seemed to recognize that Andy's behavior stemmed from his early experiences and genuinely wanted to understand how to help him. During this initial consultation, I believed that I had forged a strong connection with Andy's parents and that they would be active partners in the therapeutic process. We discussed what they could expect during the process of play therapy and the important role that they would play in Andy's treatment. I reviewed *What Parents and Children Need to Know about Play Therapy* (Center for Play Therapy, 1998) and gave them a copy to take home. Because Andy was genuinely remorseful

after he calmed down after his violent outbursts, and Carol had already been responding by saying, "I know you don't like it when you feel this way," we mutually decided that the next time he had an episode, she would take that opportunity to explain about coming to play therapy and how it could help Andy. Because of Carol's worries regarding Andy's inability to separate from her, we decided that the three of us would meet together for Andy's first introduction to the playroom and me.

Developmental History

Very little is known about Andy's developmental history prior to adoption at age 4. His birth date and exact age were unknown, and his adoptive parents had no pictures of him until he was about 3 years old. Sketchy orphanage records revealed that when Andy was a small baby, his birth mother abruptly abandoned him with a neighbor under the pretense of running an errand, and that the neighbor kept him for an unspecified length of time prior to leaving him at the orphanage at what officials guessed to be around the age of 1 year. Andy was initially abandoned at the orphanage with no information, and these brief details were later reported by the neighbor. The orphanage staff guessed at his age based on this limited information and gave him a birth date of July 1.

Orphanage records stated that Andy never talked but that he was a very affectionate and loving baby and child. His developmental milestones were not recorded, but he was mostly toilet-trained at the time of adoption. His gross motor skills and coordination were reported by his adoptive parents to be above average. When Carol and Tom first met Andy in the orphanage they noticed that he appeared cross-eyed but that he was able to focus when they came close to him. Andy had just turned 4 years of age in July when Carol and Tom brought him from China to join their family in the United States. A checkup by the family pediatrician determined that Andy's eye condition was so severe that he could see very little and only from very close up. After being in the United States a little over 1 week, he underwent major eye surgery. Carol recalled the week directly after surgery as very stressful for Andy and for her. As part of his eye treatment, Andy wore glasses with a very strong correction.

Within a month of his arrival, 4-year-old Andy was enrolled in the prekindergarten program at Carol's school. He was described as "extremely affectionate, clingy, and shy" and as "not wanting to be away from Carol. . . ever!" At the age of 5, Andy was enrolled in kindergarten at the same school. During his second year with Carol and Tom, Andy began exhibiting some anger outbursts and uncontrollable crying, but not to the current extent. Because of his immaturity, Andy was scheduled to repeat kindergarten. As noted earlier, he had recently requested that everyone call him Andy, saying that Dom was a stupid name (his official American name was Andy Dom Smith).

Parental Factors and Family Dynamics

Tom and Carol had been married for 18 years at the time of intake and reported satisfaction with their marriage and lifestyle. Neither reported any remarkable history. Neither set of grandparents lived close, so the family had little contact with them other than an annual summer vacation at the beach with Carol's parents. Tom and Carol reported no more than the usual problems in rearing their

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14-year-old daughter, Ann, whom they described as loving but stubborn and strong-willed as a child. Both parents stated that they felt closer to Andy than to Chris (who had been recently diagnosed with Fetal Alcohol Syndrome), adding that Ann and Andy also enjoyed a close relationship. Ann appeared to assume a significant amount of responsibility in helping with Andy and his brother, including discipline. Family church activities were important, although more so to Carol. Carol was the primary disciplinarian; however, she reported, "When Tom speaks the children tend to listen to him more than they do to me." She also noted that when Tom got involved in discipline it was when "he has had enough," typically when Andy and Chris are fighting and their behavior and emotions are out of control. Andy tended to be the catalyst for most fights and enjoyed being "the boss." The brothers reportedly played together fairly well because Chris would generally go along with what Andy wanted to do.

In later consultations with Carol, she disclosed feeling overwhelmed with the role of primary caretaking, and that she did not feel supported by Tom. Carol's self-report was supported by her scores on the Parenting Stress Index (PSI; Abidin, 1983), reported in Table 20.2.

Table 20.2 Parent Stress Index: Percentile Rankings for Carol's Scores

Elements	July Pretreatment (%)	December 5 month (%)	May 9 month (%)	August 12 month (%)	December 16 month (%)	May 20 month (%)
Total stress	85	83	75	50	65	55
Child domain	96	93	80	55	75	60
Adaptability	99+	98	85	55	80	70
Reinforces parent	90	80	80	55	80	55
Demandingness	85	90	65	65	75	55
Mood	85	85	60	50	60	60
Acceptability	85	90	50	50	60	40
Parent domain						
Isolation	90	85	80	70	80	70
Attachment	85	80	75	65	75	50
Spouse	85	85	80	70	80	70

Note: Scores at or above 85% are considered high. Only those scales that were elevated at time of intake are included.

Assessment

Information was gathered through interviews, informal observation, and formal assessment. Andy presented as a much smaller than average (by American norms) 6-year-old male who had just completed public school kindergarten. Formal assessment by the school district 5 months prior to his referral for play therapy confirmed significant language delays and revealed cognitive delays in several areas. However, the school psychologist determined that intelligence test results could not be considered reliable due to his significant receptive and expressive language delay. The CBCL (Achenbach, 2001) was also administered as part of the school's assessment battery. Carol's scores indicated clinical levels of concern regarding Andy's externalizing behaviors. In addition, Andy had been diagnosed by the school psychologist with Reactive Attachment Disorder (American Psychiatric Association, 1994). Carol and Tom's description of Andy's current behavior and early history were consistent with diagnostic criteria for this disorder, particularly Andy's ambivalent and contradictory approach/avoidance response to Carol and, to a lesser degree, his excessive familiarity with relative

strangers. It made sense that Andy's early history placed him at risk for difficulties in developing appropriate and stable attachments.

I administered the CBCL (Achenbach, 2001) and the PSI (Abidin, 1983) to Tom and Carol. Carol's elevated scores on the CBCL were consistent with her self-report at intake and similar to, although slightly higher than, her report at the time of the school administration 5 months prior. Carol reported clinical levels of concern in Externalizing and Total Problems, School Competency, and several subscales. Her scores are shown in Table 20.1. Tom's scores on the CBCL reflected a similar pattern of behavioral concern, but at consistently lower levels than Carol. Carol's scores on the PSI were higher than what I expected based on her initial verbal report and, as shown in Table 20.2, indicate high levels of Total Stress and high levels of stress related to the Child Domain. Tom did not complete the PSI during intake. The CBCL and PSI were administered approximately every 4 months for the duration of play therapy to provide a measure of treatment outcome. Although both parents were asked to complete both instruments at each point of measurement, only Carol complied. Tables 20.1 and 20.2 show CBCL and PSI results at time of assessment, along with results from repeated measurements of the PSI and CBCL over the treatment period.

Informal assessment supported the school's findings regarding Andy's significant expressive and receptive language deficits. Immature social-emotional behavior was noted by parents and confirmed through informal observation over the first several weeks of therapy. Consistent with parent report, Andy's motor skills and coordination were observed to be very well developed for his age. Because he was receiving special education services for a learning disability and speech and language delays, and because the school recommended retesting after 1 year, no further testing was done at time of intake.

Case Conceptualization

My understanding of Andy and his treatment needs was based more on what I believe all children need rather than on the facts presented during intake. Andy needed to experience a secure and consistent relationship, both in play therapy and within his family, in which he could fully experience all aspects of self and become the person he was meant to be. That does not mean that I view all children the same, nor does that mean that the background history and information presented by Carol and Tom were not useful. It provided me with a context for understanding Andy's experiences played out in therapy as well as an understanding of the child's system, including sources of support to tap into and elements within the system that were likely hindering Andy's optimal growth.

Andy was displaying many of the symptoms one would expect from a child with his history: hypervigilance to noise (exacerbated by his inability to see much during the first 4 years of his life), difficulty going to sleep and staying asleep, eating difficulties (refusing to eat much, but wanting to save the food for later), inappropriate affection with relative strangers, and incessant fear that his adoptive mother would leave him. Andy's early experiences were characterized by a series of abandonment and loss of primary caregivers—first his mother, then the neighbor who cared for him for several months, then caregivers that came and left in the orphanage—and the loss of his cultural identity and the only home Andy could remember in his young life. That these experiences likely occurred

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during the time in his development when separation anxiety was at its peak confirmed for Andy that his fears were real: his significant caregivers (mother, neighbor) left and did not come back. These early traumatic events likely compounded the impact of his hospitalization shortly after he arrived in the United States and his brief but traumatic separation from his new mother.

Based on Andy's behavior as reported by his parents, and consistent with my experiences of Andy as we began our relationship, I guessed that his early attachment experiences had been inconsistent; sometimes his needs had been responded to and his cries had been soothed, and other times they had not. He seemed to view the world as sometimes safe and secure and sometimes unpredictable and scary. These inconsistent early experiences with primary caregivers seemed to have resulted in two internalized and contradictory messages: (1) "I cannot depend on you. I must not be worthy of being loved and cared for. I have to take care of myself. I don't need you," and (2) "I need you, but I don't trust you. I cannot let you out of my sight, because you might not come back when I need you. I have to make you like me so you will not want to leave me." Thus, Andy approached relationships with ambiguity, which was very puzzling to Tom, and even more so to Carol, who felt rejected and confused when Andy pushed her away. I knew that helping Carol to understand that Andy's contradictory approach/avoidance response to her was "normal" given his history would be a top priority as we began therapy.

I believed that a significant factor in Andy's progress would revolve around helping Carol and Tom better understand how Andy's previous experiences impacted his current functioning. As they began to see the relationship between his early experiences, his unmet needs, and the resulting behavior that he chose to get his needs met, I believed that they would be able to accept him more fully. Several events from Andy's early experiences seemed particularly salient to his current functioning, as evidenced by his repeatedly playing out related themes. Because Andy was able to symbolically, and at times quite literally, express his world to me through play therapy, I was able to help Carol and Tom better understand Andy's needs and how they could meet them. This type of collaborative feedback proved particularly helpful to Carol.

A wise mentor once told me that if I focus too much on the problem I will lose sight of the child. With a case as complicated as this one proved to be, I had to be vigilant to not let that happen. My primary objective would be to first establish a consistent relationship in which Andy would feel secure and safe enough to explore all his experiences, including those in the past that were too threatening to admit to awareness. I would need to be patient in developing this kind of relationship with Andy, whose experience had taught him that relationships were not consistent, safe, or secure. It was my job to provide him with a corrective experience, an experience in which he would come to perceive me as an adult who was trustworthy, consistent, and predictable, one who would not leave him prematurely. An additional objective was for Andy to experience our relationship as unconditional, one in which he did not have to earn my approval or acceptance by impressing me or being overtly affectionate (his mode of connecting when we first began our relationship). My hope was that as Andy experienced unconditional acceptance in our relationship, he would begin to alter his self-perception of "damaged . . . no good," not worthy of love, to one of "lovable" and worthy of positive regard. Through these experiences, I hoped that Andy would

form new perceptions of his world as safe and predictable. I believed that by providing Andy with this kind of experience he would be ready for the most critical therapeutic task of all, to experience this kind of relationship with his adoptive parents. I believed that as Andy grew to allow himself to fully experience his parents' love for him, he would no longer feel the need to push them away for fear that they would eventually leave him.

Andy's current family situation was a significant strength, a strength that I would build on during his therapy. Carol and Tom seemed amenable to participating in additional parent or family therapy services that might be recommended as an adjunct to play therapy. Although there were parenting strategies that I believed would help Carol and Tom to respond more effectively to Andy's needs, their interactions did not appear to be contributing to Andy's difficulties in any significant way. As is true for most parents and caregivers of children who have experienced trauma, I believed that Tom and Carol would benefit from direction and support regarding how to talk to Andy about his early experiences, particularly his complicated adoption story. Andy's refusal to stay with Tom was an issue that we would focus on to provide Carol with some relief and hopefully reduce her level of stress related to parenting Andy.

Helping Andy's parents learn how to establish a healthy attachment with their son would be critical to his healing; therefore, I planned to incorporate CPRT into our overall treatment plan. Additional goals that would be addressed through CPRT were (a) fostering greater understanding of Andy and his experiences, (b) normalizing Carol and Tom's understanding and response to Andy's behavior problems, (c) enhancing parent-child attunement through creating mutually enjoyable and developmentally responsive interactions, and (d) strengthening parenting skills and confidence in responding to Andy's unique and challenging needs. However, my first priority in our parent consults was to address several safety issues around Andy's violent outbursts and to help Carol become more attuned to what precipitated the events as well as develop strategies that she could implement to prevent or de-escalate the situation. One strategy that I expected would be useful was helping Carol identify times throughout the day when she could initiate holding Andy close in a loving and playful manner, rather than waiting for him to get his "holding" needs met during the aftermath of his "rages."

Course of Treatment and Assessment of Progress

Andy's mother brought him to therapy and participated in regular parent consultation. Andy attended 55 play therapy sessions, approximately 45 minutes in length, over a 20-month period. I saw Andy weekly for the first 15 weeks of treatment and after that an average of 3 times per month with the exception of summer. I met with Carol every 2 to 3 weeks to provide support and help her develop a better understanding of Andy by describing themes I observed in his play, the connection between his play and his early experiences, and how these themes were relevant to his behavior at home and school. I also used this opportunity to introduce brief CPRT parenting skills that I believed Carol could implement successfully, including reflecting feelings (particularly anger), limit setting, and choice giving (Bratton et al., 2006). Carol competed a weekly parent report form during each session while Andy was in play therapy

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to keep me informed of any changes or significant events at home or school over the past week, as well as to rate Andy's behavior and mood compared to the week before. This was a very effective communication tool for Carol and me, providing an efficient means for us to stay connected on the weeks that we didn't formally meet.

As therapy progressed, I suggested many strategies that Carol and Tom could implement at home to promote the work being done in play therapy. They also participated in intensive filial therapy training using the CPRT protocol (Bratton et al., 2006), meeting 2 times per week over 6 weeks in the summer, when they both had breaks from teaching. During that time Andy participated in 10 supervised play sessions with his father, and Carol conducted play sessions with Chris. In the final phase of counseling, Carol participated in 6 additional weeks of supervised filial play sessions with Andy.

The playroom where Andy and I met was a fully equipped playroom similar to Landreth's (2002) description. I made sure there were baby dolls and doll figures with varying skin tones, including dolls representative of Andy's ethnicity. Andy was immediately drawn to a Native American baby doll with thick, short black hair that he initially decided was a girl. As Carol was not aware of any particular toys that Andy had played with in the orphanage, the only other culturally responsive toys included were musical instruments, food items, and eating utensils.

During a brief meeting with Andy and his mother prior to his first play therapy session, I introduced Andy to the playroom, saying, "Andy, this is the playroom where we will come each week. In here you can play with the toys in lots of the ways that you'd like," and then I allowed him time to explore. I bent down to be on his level and inquired if there was anything he wanted to ask or know more about. When he shook his head, I asked him if his mother had told him why they were here today. He informed me (with very limited English), as he looked at his mother, "Get mad and break things." I responded to let him know that lots of kids come here for the same reason, and added that this was a special place where he could play and talk about things that bothered him. Andy looked up at me and smiled, and although he didn't let go of his mother's hand, I believed that in those brief moments we had made a tentative connection. I ended the initial meeting by loaning Andy and Carol a child's storybook on play therapy to read together to help Andy feel more secure about coming to play therapy. I followed up with a phone call to Carol to discuss her concern that Andy would not separate from her and go to the playroom with me in the upcoming week and to formulate a plan for how we would respond if that occurred.

Andy was sitting in his mother's lap when I greeted him in the waiting room to take him to the playroom for our first session. I noted that he seemed a little anxious. I leaned down and made eye contact and said, "Hi, Andy. Hi, Carol. Andy, remember I showed you and your mom the playroom last week? It's time for us to go back to the playroom and play" (I slowly stood up). As if on cue (and following the instructions I had given her during intake), Carol stated, "Andy, I'll be right here waiting for you when you get through playing!" He smiled up at me, but then hesitated and looked at his mother. I quickly added, "You can choose: Should we run to the playroom or walk?" His face lit up, and he jumped up and said "Run!" as he took off down the hall with me running behind him trying to

catch up. That started a weekly ritual of Andy running to beat me to the playroom. Although I gave Andy the choice of running or walking to the playroom to take his focus off separating from his mother, I was surprised he went so easily. Based on Carol's report of his extreme difficulty leaving her, I had prepared her for the eventuality of coming to the playroom with us for at least the first week. (As Andy and I ran to the playroom, I was thinking to myself that this child has more resiliency than his mother gives him credit for!) As we entered the playroom, I reminded Andy, "You can play with the toys in lots of the ways you'd like to. We will have 45 minutes to play, and then it will be time to go back to the waiting room where your mom is."

Andy's first four sessions were characterized by his need to win my approval and affection by impressing me, playing close to me, and by overt demonstrations of affection, which included touching me, patting my head, and putting his face right up next to mine. Based on his early history, it was not a surprise that he acted overly familiar with me. I knew that no amount of words would convince Andy that I already accepted him without condition—that he did not need to win my acceptance and approval. He would have to experience my acceptance firsthand before he could come to see himself as acceptable. During the fourth session, he used the flashlight to connect with me in a playful way by turning off the light in the room, then coming close to me and shining the flashlight so we could see each other's faces, saying, "You see me; I see you." I responded in kind, "I see you; you see me." He enjoyed repeating this over and over, often moving his face closer and farther away and giggling, but without trying to touch me. He seemed to have discovered a safe and fun way to connect with me.

Andy's expressive language was very underdeveloped and his speech was often difficult to understand. Many of his interactions with me consisted of one- to two-word utterances. Among his favorites were "Hey!"; "No good"; "Bad"; "What?"; "What that?"; and later, an exclaimed "Yes!" to show pride in his accomplishments. Nevertheless, he managed to communicate his world to me quite eloquently through his play. As might be expected from a traumatized child, Andy was initially hypervigilant to every noise he heard in and outside of the playroom. These disruptions made it difficult for him to focus. It was no wonder that learning was so difficult for him!

In the beginning phase of developing our relationship, Andy worked hard to gain my approval by showing me he was strong and could do difficult things, including hammering nails, lifting the sandbox and other heavy objects in the playroom, and climbing to the top of the shelves. After he successfully completed these tasks, he would often make a "muscle" with his arm to show me how strong he was. One of the first ways that he chose to impress me with his strength and ability was hammering nails into the log stump in the playroom. Because the log and hammer were next to where I usually sat, this also allowed him to remain very close to me. During our third session, as he stood by me and looked around the room, he kept glancing over at the log and hammer. I reflected, "Looks like you're wondering about that." Andy looked at me and said, "Dangerous" (obviously a response he had heard from adults). I reflected his concern, "You think maybe that would be dangerous to play with," and he shook his head no, as if to say he did not think it was dangerous. I responded, "You're not sure.

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Well, in here it's up to you to decide if you want to play with that," letting him know that he was free to decide what to play with, just as adult clients are free to talk about what they want to talk about. Andy looked at me and looked at the log wistfully, letting me know he really would like to play with it. I reflected, "Looks like you're thinking you might like to try it out. There are some safety glasses on the hook." Andy immediately picked up the hammer and a nail and exclaimed a word I did not understand, but his actions and affect clearly communicated what he was trying to say. I replied, "You decided you want to hammer!" This brief interaction revealed Andy's struggle between his internal motivation and desire, wanting to hammer, and the external message he had received from adults that hammers are dangerous. Because I was able to respond to his internal world, he became more focused in his play and his anxiety was reduced. After he carefully hammered the first nail in, he looked at me and exclaimed, "Look!" I leaned over to look closely at his work and reflected his obvious excitement: "You wanted me to see. You got it all the way in!" He proceeded to hammer in several more nails with strength and precision and looked up at me and showed me the muscle in his arm, to which I replied, "You're really strong." He nodded his head in agreement. Then I understood better what the hammering was all about and added, "It took a lot of strength to get those nails all the way in." Although hammering initially served as an activity that Andy chose to gain my admiration, later this activity served other important functions, including demonstrating competency and expression of anger. *(A note of caution: because children can accidentally injure themselves with a hammer, I always sit close by when a child chooses this activity.)*

The fifth session marked a shift in Andy's play. He was now feeling safe enough to leave my side and explore the playroom. He delighted in his discovery of a western-type hat that had a whistle attached by a string and proceeded to spend a considerable amount of time exploring the toys on the shelves and happily blowing the whistle. He noticed the baby bottle for the first time and filled it with water from the sink, sucking contentedly on it for approximately the last third of our session while he continued to wear the hat. When I told Andy that we had 5 minutes left to play, he went to the sandbox, climbed in, lay back against the edge, and sucked on the bottle with a glazed look on his face. When I announced that our time was up, he shook his head with the bottle still in his mouth, letting me know that he wasn't ready to leave. I was patient and restated the limit as I leaned down to his level and made eye contact and reflected his desire to stay longer: "Andy, I know that you'd like to stay longer, but our time is up for today. You can drink some more from the bottle next week." I repeated similar statements three more times, letting him know I understood it was hard to leave when he was enjoying himself. Still he shook his head and remained in the sandbox. Ordinarily, I would tell my students that this is a time when you need to be more firm in your tone of voice, but I sensed that there was something different about this experience. I realized in that moment that I wished that Andy didn't have to leave either. I understood that the sucking was serving a very important function for him, but that it was also important that he experience the consistency and predictability of our time together. I bent down even closer to him and said, "Andy, I wish we didn't have to leave either, but we do. It's time for us to go back to the waiting room where Mom is." Focusing my responses on our relationship, rather than on Andy's behavior, seemed to be what he needed—a

lesson that I shared with Carol in our next parent consultation. This brief but powerful interaction with Andy reminded me how important it is for me to trust my experience with children—that when I do, they let me know what they need from me.

Our next several sessions together were characterized by Andy's fascination with the whistle and the bottle (although his sucking was less intense), while continuing to show me how strong and competent he was at hammering. Blowing the whistle seemed to be serving an important expressive function for Andy. During session 7, as he happily blew to his own beat while he played, he suddenly stopped and looked at me, saying "Take whistle home?" Then he added, "Please . . . I bring back next time." For a brief moment as I looked into his pleading eyes, I almost gave in; however, I knew that this, too, was part of the consistency and predictability of the play therapy experience that Andy needed so desperately to experience. Instead, I let him know that I understood how much he wished that he could take it home and explained the playroom rule: "All the toys are for staying here in the playroom, but it will be here for you to play with next week." He seemed to be considering the rule, but stopped playing with the whistle. (Because of Andy's delay in language development, I sometimes wondered how much he understood.) A few minutes later I told Andy that we had 5 minutes left (a time when he often introduced significant play themes or play sequences). He began to paint for the first time, painting all the colors on top of each other to make a blob of brownish-black. The following brief interaction seemed to reveal Andy's inner conflict between his own wishes and needs and trying to please others, especially "Mommy."

Andy: Mommy be mad (as he began to use his hands to smear the paint).

S: You're worried that Mommy will be mad.

Andy: Messy (as he went to the sink and washed the paint off hurriedly).

S: You're worried that Mom wouldn't like it if you got messy. But before I could get the words out, he began to take his shoes off and looked over at me to see what I thought (as if to ask my approval).

S: Looks like you're not sure if that's okay. In here you can decide if you want to take your shoes off or leave them on. (Andy takes both shoes off.) You decided you wanted them off. (I responded to his internal desire.)

Andy went back to the paints and picked up the paint brush and looked at me and said something I did not understand as he proceeded to paint one hand and then the other, then sat down and began to paint the bottom of his feet, enjoying the feel of the brush (much like the sensorimotor play of a younger child).

Andy: Tickles (a big smile on his face).

S: You like how that feels.

S: Andy, we have 1 minute left to play, then it's time to wash up to go home. (I was careful to make sure that we washed all the paint off his feet and hands before we went back to the waiting room to prevent potential guilt feelings that Andy might experience if Carol reacted adversely to a "messy Andy.")

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In our next parent consultation, I reminded Carol of the importance of Andy's wearing old clothes to play therapy so that he could get messy if he needed to, and we discussed what making "messes" might symbolize for him.

For the next 2 sessions, Andy repeated this ambivalent play theme of "Mommy be mad" and "Not supposed to get messy" and the obvious joy of getting messy, smearing paint on the paper and painting his hands and feet. My responses followed a similar pattern, reflecting that he was concerned about what his mom would think, but focusing on his enjoyment of the experience and always making sure he was returned to his mother mostly clean. Andy's play seemed to reflect his continued internal struggle between perceived external demands of pleasing others (hypervigilance), including an awareness that his "out-of-control" behavior was unacceptable ("not okay to be messy"), and his internal need to express all aspects of self, even the "messy and chaotic."

Andy reveled in his newfound freedom in the playroom. As he experienced the safety and freedom to fully express himself, his play became more focused and meaningful. It was during this phase of play therapy that he began to display more of his conflicted and difficult feelings, most notably his sense of being damaged or "no good." He would inspect all the toys, and any that had the slightest flaw or did not seem to work properly were deemed "no good" and thrown away. During this period, he would sometimes mumble "No good" up to two dozen times in a session.

During our parent consult in week 9, Carol announced that Andy's "stealing" was now happening at school as well as at home. It was obvious from Andy's play ("Mommy be mad") that he perceived that Carol was angry and upset with him, although she denied it initially. On the surface Carol was trying to appear understanding, but underneath her façade she was embarrassed and upset with Andy. Her response seemed exacerbated by the fact that she taught at the same school and was concerned about what her colleagues would think. I helped her understand that Andy was picking up on her underlying feelings, even though she was not directly expressing them to him. We again discussed Andy's hypersensitivity to nonverbals. This problem was significantly impacting Carol's acceptance of Andy; thus we needed to actively address possible solutions.

My first priority was to help Carol and others to reframe how they viewed Andy's behavior. It seemed obvious that Andy's stealing, as described by Carol, was not malicious. Thus I asked Carol to use the term "taking things that belong to others" instead of "stealing" when talking to Andy (and others) about his behavior. Although I could have made guesses about several possible underlying reasons why Andy was taking things that did not belong to him, the explanation that seemed to help Carol and others normalize Andy's actions was that this behavior could best be understood as a function of a significant developmental delay (one of many regressed behaviors he demonstrated). I explained to Carol that her description of the events and Andy's response when confronted were consistent with what one would expect from much younger children who simply see something they want or need and take it. I reminded her of a toddler's worldview: "If I want it, it's mine!" She agreed that if Andy were 2 years old she would not see his behavior as stealing. Carol and I devised a plan for how to respond to Andy at home and at school if and when he initiated this behavior again. Carol and I talked with Andy's teacher at school to discuss the plan, and Carol also

elicited school administrators' help. We all approached this as if Andy's behavior was a function of development and that it was our job as adults to help further his development by helping him learn the concept of ownership and people's related feelings about their property. Andy's parents and the school were instrumental in helping Andy move through this phase over the next several weeks. This is just one example of many instances in which they proved to be incredible allies throughout the therapeutic process!

Andy's play during the next several weeks could best be described as disorganized and chaotic. This relatively short phase was marked by the most intensely negative affect during our time together. Themes of "bad boy" and "no good" persisted. He began to actively deal with the issue of stealing. For the next 4 weeks he was focused on trying to figure out a way to take the whistle from the playroom. Each week he would retrieve the hat with whistle from the shelf, take it behind the puppet theater, cut the string to get the whistle off, and then try to figure out how to take it out without my knowing. He was generally agitated and often unfocused in his play once he had the whistle cut off. The first time he hid the whistle under the beanbag chair behind the puppet theater. He kept asking me how much time we had left and for the first time wanted to leave early. After I stated that there were 5 minutes left, he went behind the theater and put the whistle in his pocket and then came over to me and pulled it out and showed me "his whistle" that he had "brought from home." I responded that it looked just like the one we had in the playroom. At first he tried to convince me that it was his and then said, "Just kidding," and put it back on the shelf. I let him know that I understood how much he liked the whistle and wished that he had one like it.

During the next 2 sessions his level of agitation was at an all-time high. In each session, Andy immediately took the hat and whistle behind the theater, cut the whistle off, and stuffed it in his pocket. His play was highly disorganized and he was clearly distressed. (I was having difficulty being with him in this experience and found myself wanting to rescue him. I had to work very hard to not let my need to protect Andy from these difficult feelings interfere with his need to work through them in his own way. I found myself needing to seek peer supervision over the next few weeks to process my own feelings and avoid imposing my needs onto Andy.) During session 13, with the whistle already pocketed, Andy spent 15 minutes throwing toys, repeating, "No good, no good." I reflected, "Nothing seems to be going right today." He seemed to calm down for a moment, and with less intensity than before, he hit himself in the head (with his hand) and said, "Stupid." I went over to him and reflected, "Andy, I know you're upset about something, but you're not for hitting." He paused and looked up at me and quietly said, "I want to go." I knew it was important to maintain the structure and consistency of our time together, so I held firm and replied with all the empathy I was feeling for this little boy, "Andy, I know that you're having a hard time in the playroom today, but we have 20 minutes left and then it will be time to go." He insisted, "I don't want to play anymore." I replied softly, "That's up to you. . . . You can choose to play or not to play in the 20 minutes that we have left." He did not try to leave, but every few minutes he looked at the clock and asked me how much time we had left. When I told him we had 1 minute left, he went straight to the door with obvious relief. At this point, he still had the whistle in his pocket. As he got to the door, I gently responded, "Andy, I think maybe you put the whistle in your pocket

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when you were playing with it earlier. Remember, the toys are for staying in the playroom." First he shook his head no. "Maybe you should check to make sure." He put his hand in his pocket and said, "I forgot," and pulled out the whistle and handed to me. As I put it back on the shelf I reminded him that "it will be right here for you next week." (I tried to remember each week to check the playroom in advance to make sure I kept my promise.) The next session followed a similar pattern; however, Andy was noticeably less agitated, and when it was time to leave he pulled the whistle out of his pocket without any prompting and said, "Oh yeah," and handed it to me matter-of-factly.

The change in Andy's affect in our next session was remarkable. He was very excited to see me and ran straight to find the hat and whistle and put it on his head. It was the first time in 5 weeks that he had not cut the string and hidden the whistle. He put the hat on and marched around the room happily blowing the whistle (very free), all the while sucking on the baby bottle for the first time since he became preoccupied with the whistle. He spent the majority of the session with the whistle in one hand and the bottle in the other, much of the time marching to a beat of suck-suck—whistle-whistle. Whereas his play over the past 4 weeks was constricted, agitated, and chaotic, his play this week was spontaneous and joyful. It was during the last 5 minutes of this session that Andy first introduced the "spinning game," his name for a game he devised that involved his directing me to spin him around while he sat in the swivel chair that I normally sat in. There were several variations of this game, but they all had in common a theme of mother-baby play. Little did I know just how important this game would become!

This tumultuous phase in Andy's play seemed to represent his struggle for self-regulation as he internalized the concept that the whistle was not his and that although he wanted it, he could now choose to play with it and give it back (just as outside of the session, he was struggling with understanding this concept on a larger scale). The combined impact of his being able to play out his struggles in the safety of the playroom with me as a supportive witness to his expression of difficult feelings, and his parents' and teacher's changed attitude and response to his behavior at home and school resulted in an almost complete halt in the stealing behavior. Carol reported that Andy was no longer taking things at school and that he had mostly stopped this behavior at home. Her scores on the CBCL and PSI completed after session 15 supported her verbal report of Andy's improvement in this area as well as in the frequency and intensity of his anger outbursts.

Andy's play over the next 10 sessions was characterized by greater spontaneity in self-expression and increased positive affect. For a child that had begun therapy being so hypersensitive to what he needed to be in order to win others' approval, this movement was just what I had hoped for. During this time, he found the baby dolphin and chose it to become a central figure in his play. He often carried "Baby" around while he played or gave it to me to hold. He determined that the dolphin was male and was especially gentle with it. Andy continued the spinning game, which became increasingly more intimate. He would often lie in the chair and suck from the bottle, gazing at me and sometimes giggling while he directed me to spin him around and around.

It was also during this time that he introduced his favorite game, "hide-and-find." With the exception of a few breaks, this was a game that Andy delighted in

up until our termination, sometimes spending over half of our session in this activity. From the first time he introduced this activity, with "Close eyes, I hide," I sensed that this game was very important to him. I wanted to play my part just the way he needed me to; thus I inquired, "How will I know when you're ready?" Andy said, as if it were obvious, "Count to 10," and after I did, "Find me!" he yelled. I slowly began to look for him. (I was wondering to myself, Does he need me to find him easily or take a long time? Because of his abandonment issues, I believed that how I responded to my task of finding him was vital.) In a soft voice, I asked, "Are you going to be easy to find or hard to find?" Andy paused a minute and said, "Hard!" So I began a long search around the room, amplifying my movements and efforts to find him, often exclaiming variations of "I bet you're over here. . . . Hmm, not here . . . You hid really well. . . . I wonder where you are?" When he would giggle, I would reflect, "I hear something. You must be over there." Each time we played this out, no matter whether he chose to be easy to find or hard to find, he always gave me clues to let me know when he was ready to be found. We were in synch with each other; I knew when and how he needed me to respond. Again, Andy had chosen a playful and safe way to deepen our relationship and work through his early experiences of what I guessed to be an inconsistent response to his nurturing needs. He was learning that I could be trusted to consistently respond to what he needed. Through this experience and his experience of being with a loving family, I hoped that Andy would begin to see the world as secure and predictable—a place where he could depend on adults to keep him safe and respond appropriately to his needs.

Andy was clearly working on attachment and trust issues that I believed could be more effectively addressed by involving Carol and Tom in his play therapy. We discussed starting CPRT training in the near future so that Andy could play out his attachment needs with his parents. We agreed that Andy was making good progress, including a reduction in his violent outbursts, a marked decrease in stealing, and fewer sleep problems. We tentatively discussed ending therapy by the end of the school year.

During this time, Carol reported that Andy was showing more interest in what he was like as a baby. When he saw babies on television, he would ask, "Did I look like that?" When he saw pictures of babies, he would ask, "Was that me?" Once again, I recommended that Carol help Andy make an adoption scrapbook to help him understand his adoption story, and we practiced how she could explain events to him in appropriate words (suggestions that Carol continued to resist; it was, after all, a difficult story to explain). However, as a result of our discussions, Carol and Tom came up with a remarkable plan. They had arranged for a 17-year-old exchange student from China to live with them for the next academic school year. Because of Andy's increased interest in adoption issues and Tom's work schedule which prevented him from starting the CPRT and filial training, we decided to continue play therapy until the end of the school year and begin CPRT the first week in June, when his schedule allowed him to travel less.

Sessions 25 through 28 marked the introduction of another significant play theme, although I had no idea what it meant at the time. Andy's play was highly

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organized and purposeful. He clearly had a plan in mind. He began to use the dress-up clothes trunk as an important part of his play, first by taking everything out and trying on most of the clothes to see how he looked. He would often dress up both of us so that we looked similar and exclaim, "I girl like you!" I was a bit puzzled by what this meant for Andy, but he took great delight in this activity. However, what puzzled me most was that he ended this play sequence each week with a ritual of putting the lit flashlight in the trunk, along with the baby dolphin, and closing the lid. Before he left the playroom, he would open the lid to make sure the light was still on. He seemed to gain a great deal of satisfaction from his play during these weeks. Although at this point I did not comprehend the meaning his play held, I conveyed my understanding that it was important to him.

By this time, school was out and Andy had successfully completed his second year of kindergarten. Carol's scores on the CBCL and PSI (Table 20.1 and 20.2) reflected her weekly reports and comments regarding Andy's improved behavior, most notably in the areas of sleep, stealing, and anger outbursts.

In keeping with our plan, Carol and Tom began intensive CPRT training twice a week, following the treatment protocol outlined by Bratton et al. (2006). We had 6 weeks before they and their family would be going on vacation. During that time I did not see Andy in play therapy, because he was coming to the clinic twice a week to have play sessions with his dad as part of the CPRT and filial therapy training. Carol conducted her play sessions with Andy's stepbrother, Chris. Carol and Tom and I decided that this was a good opportunity to focus on helping Andy develop a closer bond with Tom, in hopes of relieving Carol of her full-time parenting role. Andy played out some of the same themes with his dad that he played with me, particularly doctoring the baby and sucking from the baby bottle. I had prepared Tom for the possibility that Andy might want to suck from the bottle, but Tom struggled to accept that behavior from his now almost 7-year-old son. He understood that it was important to Andy and that he should be accepting, but as he said, "That's easier said than done." We role-played how Tom could respond in a way that would convey his acceptance and encourage Andy to involve him in his play, but what seemed to help most was asking Tom to remember when Anne (his 14-year-old biological daughter) was a baby and how he had enjoyed holding her close and giving her a bottle. Andy had not had the opportunity to have that experience with him or Carol (and maybe not with anyone).

Both Tom and Carol were motivated students and became fairly proficient in demonstrating the CCPT play skills taught in CPRT. Each session began with a review of skills, role-play, and questions, immediately followed by Tom and Carol taking turns conducting play sessions with their respective child of focus, while I supervised through the two-way mirror. After both parents had conducted a play session, I met briefly with them to give feedback and to process their play sessions. I focused primarily on helping them better understand what Andy and Chris were communicating through their play. The children were supervised by clinic staff when they were not participating in play sessions. During this time, both Tom and Carol reported that this was the least stressful time since they had adopted both boys. They agreed to continue their weekly play sessions while they were on their monthlong vacation at the beach. Because Andy seemed to be actively addressing his attachment needs, we decided that I would see

Andy for at least one more month after they returned to allow for appropriate termination of our relationship.

Andy and I resumed play therapy 2 weeks before he would start first grade and the week before Dong, the exchange student, arrived from China. I had not seen Andy in 5 weeks and had not had a play session with him in 11 weeks. Carol completed the fourth measurement of the CBCL and PSI to assess progress over the summer months, when the only treatment provided had been CPRT (and going to the beach for a month!).

Andy was very excited to see me and pulled me down the hall running. He ran ahead and when I entered the room, he was inside the trunk covered with clothes. He delighted in jumping up and yelling, "Boo!" Andy's play seemed to pick up where he had left off, with a fascination for the dress-up clothes and the trunk. With 5 minutes left, I guessed that he had something important in mind as he blocked the door and taped it so "we can't get out." He quickly threw the rest of the clothes from the trunk and climbed inside, trying several different positions to see how he could make himself fit. Then he climbed in again and curled up in a distinctly fetal position and told me he wanted me to close the lid. I clarified that he would let me know when I should open it. He told me he would knock, and he demonstrated that he would knock three times. He replayed this scenario three more times, staying inside only a few seconds each time. He obviously had something else in mind, as he very purposefully retrieved the flashlight from its spot on the shelf to take it with him inside the trunk. He turned on the light and curled into a little ball and told me to shut the lid "all the way." This time he stayed for almost a minute, although it seemed like several minutes to me. I glanced at my watch and noted that we had 1 minute left and bent down and whispered into the trunk, "Andy, we have 1 minute left." He immediately knocked on the lid to let me know he was ready to come out and ran over to get the baby dolphin and the baby bottle of juice. He carefully laid them inside the trunk on a soft piece of fabric, along with the lit flashlight and shut the lid. Again, I had no idea what all of this meant for Andy, only that he was working hard and had a definite plan in mind, one that he first conceived of 12 weeks ago!

Following this session, Carol and I met for our regular consultation. She looked more relaxed than I had ever seen her. She discussed Dong's arrival from China in the coming week and how excited Andy was that he and Dong would be sharing a room. Then Carol said, "I just remembered what I wanted to tell you. Andy said the strangest thing to me this week. He asked me if he had a flashlight when he was in his mommy's tummy." (By now, Andy was able to acknowledge that he came from someone else's tummy—not Carol's.) Carol disclosed that Andy had continued to be preoccupied with babies over the summer. He loved to watch shows on cable television that showed babies being born and persistently asked, "Was that me?" They had recently purchased *The Miracle of Life,* which was now Andy's favorite movie. Now, Andy's flashlight-in-the-trunk play made perfect sense: He was trying to make sense of where he came from, quite literally! Carol's scores on the fourth measure of the CBCL and PSI in August reflected her self-report of "how relaxing and uneventful" the summer had been. The CPRT filial training, 1 month at the beach, and continuing the home play sessions seemed to be just what the doctor ordered for this family.

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"I got new big brother!" Andy exclaimed when I greeted him the next week. He was obviously excited, and much of his play revolved around his "new big brother." He played inside of the trunk again with the baby dolphin and the flashlight, but with less intensity. He did not ask me to close the lid. This was the end of Andy's important in-the-trunk play until our closing session. In the next 5 sessions, his play was primarily focused on issues related to attachment and mastery. His play showed increased levels of reciprocity and sharing. He continued to suck the bottle, but with much less intensity (his sucking need seemed to have decreased in response to the play sessions with his mom and dad over the summer). We spent extended periods of time playing the spinning game, which had become increasingly more intimate, sometimes with the light off and sometimes with Andy directing me to shine the flashlight under my chin or his so we could see each other. He delighted in inventing several variations of baby-like games in which he directed me to touch and count his fingers and toes while I spun him around. During this time, he also began to express negative feelings toward me directly. One day I was 15 minutes late for a session and he wouldn't look at me as we walked to the playroom (no running today!). When we entered the playroom, he looked at me and then kicked the inflatable bop bag. I reflected, "Seems like you might be mad that I was late. It's okay to tell me that you're mad." Andy shook his head slightly as if to say he was not mad. It seemed important to help him understand that it is okay for people to get mad at each other and that our relationship could recover. I bent down to Andy and gently said, "You were expecting me to be here today when I was supposed to, and I wasn't. I'm sorry" (something I should have said earlier).

Dong's arrival seemed an impetus for Andy to more directly address issues around the orphanage, his adoption, and cultural identity. Themes of attachment, trust, and mothers and babies (often looking for each other) continued over the next 15 weeks. His play became more complex and imaginative. During this time he often had difficulty leaving the playroom. Carol and I discussed the importance of Andy's play themes and his need for the safety of the play therapy relationship to play out and express these confusing experiences. We decided he should continue in play therapy.

Session 36 marked Andy's first attempts to play out what seemed to be experiences that were directly related to his abandonment by his birth mother and his orphanage experience. The baby dolphin had long been a favorite and a meaningful toy for him. I already sensed that he used the dolphin to represent himself. (Andy had previously identified it as a baby and as a male during in-the-trunk play.) He took the baby dolphin and for the first time buried it deep in the sand. He abruptly left it there and began to hammer nails in the log for the first time in several weeks. I saw this behavior as a "play disruption," a sudden shift from play that seems to hold specific meaning for the child in response to anxiety or discomfort (Findling, Bratton, & Hensen, 2006). Dealing with traumatic material can be retraumatizing; thus it was important for Andy to approach this at his own pace. Later he went back to the sandbox and looked at me and said, shaking his head, "Don't know where he is." I simply reflected, "Not sure where he is," not knowing who it was that didn't know where the baby was. I waited patiently to see if a story would unfold. Andy found the large dolphin (which I interpreted as a mother or caregiver figure) and a few other caregiver-figure animals (a seal, a walrus, and a polar bear) off the shelf and placed them carefully in the

sandbox. He again abruptly went back to hammering (an activity that he had long ago mastered). After I announced that we had 5 minutes left, Andy went to the sandbox and hurriedly moved the "mother" animals all around on top of the sand as if they were all looking for the baby, then had the large dolphin dig down and find the baby and exclaim, "There he is!" He then took the little dolphin and made it flip up in the air as if it were jumping out of water. I reflected, "It looks like he's happy that they found him." (Inside, I was struggling with staying with Andy's expression of wanting his mother to find him and knowing that that will not happen. My need to try to protect him from being hurt had surfaced again.) I wondered if Andy's play represented wishful thinking that he could go back and find his mom (or remembered "mother figure"), like Dong would go back to his family in China at the end of the year.

From the beginning of Dong's stay Carol and I had discussed the importance of helping Andy understand that Dong would be going back to China at the end of the school year. Because of Andy's thematic play, I made a note to further discuss these issues with Carol, along with adding more facts to his adoption story. I also found a child's book on adoption that more closely mirrored Andy's own experience than the books that Carol had been able to find. *We Adopted You, Benjamen Koo* (Girard, 1989) became a favorite of Andy's; he could relate to the boy, whose skin color was different from his parents', who did not know his real birthday or name, and who would probably "never know his birth mother."

Using animals rather than people figures to make sense of his adoption experience seemed to provide Andy with the safety and distance he needed. On two occasions he initiated brief play with mother dolls with two different skin colors, but both times he abandoned the play. Over the next several weeks Andy was preoccupied with babies and mothers. In our clinic playrooms, we have several sets of animal families grouped together. He readily identified the "babies" and separated them from the mother-figures, devising various scenarios of what happened next. Often he would place the babies across the room from the mothers, sometimes out of sight behind the puppet theater. My responses mirrored his actions; sometimes the babies could not find the mothers and sometimes the mothers could not find the babies. In Andy's play there were always more babies than mothers, which seemed to reflect his orphanage experience. Another theme that emerged was that the baby dolphin began to receive better care than the other babies; the others often were thrown down from the shelf, but the baby dolphin was always gently handed to me to hold. (I took that as a positive sign of self-nurturing, and that Andy was internalizing messages from his adoption story that Carol and he had been working on. I knew that the story conveyed the message, "We chose you out of all the children," and other messages that conveyed that he was "special.") Andy returned to burying the baby dolphin deeply in the sand, but with a different story line this time. For the first time he identified the large dolphin as a mother and then handed it to me as if I knew what I should do.

S: What am I supposed to do? (soft voice).

Andy: Look for baby.

S: Is he going to be hard to find or easy to find?

Andy: Hard (I started to look in the sandbox).

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Andy: No—everywhere, not here. (Now, I understood, I was supposed to search for the baby dolphin the way I searched for Andy when we played hide-and-find.)

S: I may need some help if you want me to find him. (After I had spent a long time looking for the baby and not getting a sense of what Andy needed from me. I was not sure who I represented—birth mother, orphanage caregiver, or maybe even Carol; neither did I understand if he wanted me to find the baby or not, so I was careful to let him fully direct this play sequence. Andy never did direct me to find the baby, but he seemed to enjoy that I was methodically looking all over the playroom.)

S: I'm looking everywhere I can think to look, but I can't find him. . . . I wonder where he is? (Andy looked at me and shrugged his shoulders.)

S: I don't know where he is either, but I'm going to keep looking. . . . I'm not giving up.

Andy seemed content that I (as the mother dolphin) looked everywhere and never gave up. He also seemed content or resigned to the fact that in this session the mother never found the baby. Andy seemed to be integrating his previous experiences into a positive memory of being wanted and special, even if his mother had left him. Just before we left the playroom each week, he was always careful to return the dolphin to its place on the shelf. It seemed important to him that the baby was left in its safe and predictable spot. In subsequent sessions Andy's mother–baby animal play continued, but with less focus on the mother dolphin. He no longer buried the baby dolphin and began to include the polar bear, seal, and walrus as more central figures in his play. Although he occasionally continued to place the mother dolphin in the sandbox, the other figures played a more active part in his play. These figures seemed to represent his current family, with the polar bear taking on an increasingly nurturing role. Often, he would make a "home" just for the polar bear and baby dolphin, and the other animals were left out. Consistent with Carol's report, although Andy would now stay with Tom while she went out for a brief period of time, he still preferred to spend time just with her. Although we were making progress, Andy was not at the point with Carol that he had fully internalized the important attachment messages "When you leave me, I know you will come back" and "I can count on you to meet my needs." Andy was clearly working on attachment issues and making sense of his adoption experience; thus Carol and I decided to continue play therapy and reevaluate monthly.

During this phase of therapy, hide-and-find and the spinning game were central to Andy's play activity. He continued to fill the bottle with juice each week, but often simply drank from it more than he sucked. He would often fill it and leave it on the shelf (in case he needed it). He explored the limits of our relationship by purposefully, yet playfully, choosing to break limits that neither hurt the room nor me, such as shooting me in the back with the dart gun and spraying me with water. The only new play that he introduced during this time was related to making sense of his cultural identity. He introduced two puppets, "Shake Your Bootie Guy," a ferocious-looking dragon who spoke English, and "Donga," a cute little girl puppet with a big smile that made baby-like noises that sounded like the Chinese equivalent of "ga-ga-goo-goo." From Carol's report, Andy did not know any Chinese-language words other than a few words he had learned from Dong. Andy incorporated Shake Your Bootie Guy and Donga primarily into our

hide-and-find play, varying how he used them. I took it as a positive sign that both puppets were important to him. Toward the end of his therapy, Andy began to choose a similar, but friendlier-looking version of Shake Your Bootie Guy to sometimes represent Donga. I thought this might be a sign that he was beginning to integrate his two cultures.

The fifth measure of treatment outcome was administered after session 42, at the end of the fall semester. Carol reported overall improvement in Andy's behavior, although school continued to place additional stress on Andy, which placed additional stress on Carol. Her scores on the CBCL and the PSI reflected Andy's school stress, as well as her stress related to parenting a more stressed Andy. Although he continued to be much improved from when he started play therapy, his scores were slightly more elevated than they had been prior to the beginning of this school year. Carol and I decided that Andy would continue in play therapy. I encouraged Carol to resume her home play sessions with Andy, at least during the holidays.

Sessions 43 to 45 marked a brief but tumultuous shift in Andy's play, with a marked increase in expression of negative affect. He was clearly feeling conflicted about something. His anger was expressed more directly toward me, although I suspected this was transference of his feelings toward his mother. On two occasions he stated, "I'm gonna make a big mess" and proceeded to follow through with his threat, purposefully knocking toys off the shelves that he knew would not break. (I thought to myself that this was progress: Andy seldom expressed his anger directly and in such a controlled manner.) I reflected that he seemed angry and that he was showing me how mad he was. The second week that this happened, I wondered aloud, "Seems like maybe you're still mad about something." Andy exclaimed, "Yeah—you're mean!" He then proceeded to hammer the nails really hard for several minutes until he seemed spent, and I reflected, "It feels good to hit it really hard when you're mad." Andy spent the rest of the session focusing on our relationship by playing hide-and-find in the dark with the flashlight. During our last 5 minutes, he initiated the spinning game with a new twist; he filled the bottle with juice, took his glasses off, and pushed the chair so it was touching my legs. Then he showed me he how he wanted me to just barely move the chair back and forth (it felt like rocking). He lay across the chair and contentedly sucked while he gazed into my eyes with a glazed look. It was a very intimate moment, the kind he needed to be having with Carol and Tom. I made a mental note to talk with Carol about taking a break from play therapy and beginning filial therapy in the next couple of weeks.

When Andy and I returned to the reception area, Carol was insistent that she needed to talk to me before they left. I knew by her tone of voice that it must be important, and so did Andy. I bent down to let Andy know that his mom and I would be back in 15 minutes (that was all the time I had before another appointment) and that he would stay with Jane (the receptionist) while we were gone. Andy grabbed his mom around the waist and stated adamantly that he didn't want her to go and was not going to let her go. Although on the surface his words and actions conveyed one message, I had the strongest sense that the real issue was that he did not want me to hear what his mother was planning to say, that he believed that what she had to say would negatively impact my view of him. I did something I have never done before; I knelt down beside Andy and whispered so that only he could hear, "Andy,

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there's nothing your mom could say to me that would ever change how I feel about you." Although, in general, I believe that my actions with children speak much louder than my words, I sensed that Andy needed to hear this. I knew that he experienced my acceptance of him as unconditional, but his past experiences told him that things can (and will) change. He looked at me for a few moments as if he was deciding if I was telling the truth, then smiled at me as if to say "Okay" and went over to watch the cartoon playing in the waiting room. Our relationship had passed a test. I was reminded once again of the importance of trusting my intuitive sense of what children need from me.

Andy's academic performance and behavior at school were bothering Carol, and her frustration was carrying over to home, where she felt unsupported. Although Andy seemed to be experiencing more problems at school than he had in a while, Carol's extreme reaction was my main concern. Her escalating level of stress was an indication that I needed to shift my focus to strengthening the relationship between Carol and Andy and at the same time provide Carol with some much-needed parental support. Carol was initially unsure about taking a break from play therapy to focus on filial therapy, but I emphasized to her that right now Andy needed her more than he needed me. Coincidentally, I also sensed at this moment that Carol needed me more than Andy did. We decided that she would begin filial play sessions with Andy in 2 weeks during our regularly scheduled appointment time.

At the beginning of our next session, I explained to Andy that he and I would be taking a break from playing together: "You and I will come to our playroom two more times, and after that, you and Mom will get to play together in the family playroom." (I always use a different playroom for parent-child play sessions if I am concurrently seeing the child in play therapy.) In Andy's next two sessions with me, he showed me that he was secure and confident in our relationship. The spinning game and sucking on the baby bottle were his primary play activities. Instead of just counting fingers and toes, Andy taught me how to sign "I love you" as I spun him around.

For the next 4 weeks, I coached and supervised Carol's play sessions with Andy. I spent time supporting her worries and concerns about Andy and about her being a good enough mother for him. We actively worked on increasing her enjoyment and acceptance of Andy for who he is, school difficulties and all. Andy enjoyed seeing me at the clinic, but was eager to play with his mom. Carol reported feeling much less stressed, and we both agreed that Andy's behavior had improved markedly by the end of the 4 weeks. A return to filial play sessions proved to be just what Carol and Andy needed. With the end of the school year 5 weeks away, Carol and I decided that it would be a good time to end treatment, at least for now.

Planning for the ending of therapy is important for all children, but even more so for children who have had multiple experiences of abandonment and loss. I knew that how I approached the ending of our relationship would be critical so that Andy would not perceive me as yet another important adult (and female) who was leaving him. I explained to Carol that I would meet with her and Andy briefly before Andy's play session in the following week to talk with both of them about termination. I explained to Carol that I would be letting Andy know why I believed that he no longer needed to come, and that I would ask Andy and Carol what they thought about ending our sessions. I encouraged Carol to say to Andy

what she had reported to me about the improvements she was seeing. Because Andy and I had such a long-standing relationship, I decided that 4 weeks would be an appropriate termination phase. In our meeting with Andy, Carol did a remarkable job of conveying to Andy all the positive changes that she had seen over the past several months. Although Andy did not say much, he seemed to understand that we would be ending soon. Marking the weeks on a calendar I gave him, I told Andy he would come to the clinic four more times, and that each week he could decide if he wanted to play with his mom or with me. Because he had been playing most recently with his mother, it seemed important to give him the option of play therapy with me or filial play sessions with his mom.

For the first of our termination sessions, Andy decided to have a filial play session with his mother, but afterward he announced, "All the next times, just you and me play." His play over the last three sessions was primarily constructive and showed his mastery over the playroom. He didn't introduce any new play and seemed to be replaying our time together, almost like fast-forwarding a favorite home movie. He briefly played out some of his more destructive, confused, and chaotic play, but without the previous intensity. Mostly, he enjoyed repeatedly playing his favored hide-and-find and spinning game, while making sure he had some time for hammering the log in our last moments together.

When I went to meet Andy in the waiting room for our last time together he showed me the cupcake he had brought. He whispered in my ear that he needed a lighter and showed me the candle he had in his pocket. (I later found out that he wanted Carol to let him bring a lighter and, understandably, she refused.) It seemed very important to Andy, so I retrieved a lighter from my office. He directed me to light the candle on the cupcake and sing "Happy Birthday," after which he blew out the candle and said, "Do it again!" Andy's birthday was 2 months away, but he seemed to have found a perfect way to celebrate our ending and at the same time, his beginning. We repeated the scenario 12 times, with Andy sometimes joining in the singing. He looked into my face with a huge smile the entire time I sang to him. After the 12th round of "Happy Birthday," he said that we could now eat the cupcake and carefully cut it in half with a play knife from the kitchen.

"Spinning game!" Andy announced, which I had long ago learned meant he would sit in the swivel chair and that I would sit in the smaller chair. After a few rounds of playing our traditional spinning game, touching fingers, counting, and signing, he went to get the flashlight and announced with a huge grin, "Hide and find!" At that point in our relationship, we needed few words to communicate meaningfully with each other. Andy spent most of the rest of the session hiding in all the places he had hid over the months we had been together, starting with the oven in the play stove that he was no longer able to fit in and ending with the dress-up clothes trunk that had been such an important part of his play. With each new hiding place he would giggle as he told me without my prompting whether he was going to be "easy to find" or "hard to find."

After I announced that there were 5 minutes left in the session, he came back and sat down by me and began to tell me about going to his grandmother's beach house for the summer (it was very unusual for Andy to sit and talk!). He let me know that he is going to have two birthday parties, one here with his friends and his "real" birthday party at his grandmother's house. Then Andy looked up

(*Continued*)

at me and said, “I won’t see you.” I assured him that although we wouldn’t see each other for a while, he could always call me. I wrote down the clinic phone number and gave it to him. Then he looked up at me for a few moments and said quietly, “You might need to call me.” I leaned over and quietly replied, “Sounds like you’re worried about us not seeing each other. Anytime you need me or want to come back here, you can call or tell your mom.” Then Andy responded with a big smile, “I know! I give you my phone number,” as he wrote the number and gave it to me. I smiled in response and exclaimed, “Now we both know how to call each other if we need to.” Andy seemed very satisfied that he had figured this out; it was important for him to know that although I would not physically be with him, I was not abandoning him, that I would be available if he needed me. Andy then jumped up and ran behind the puppet theater and yelled, “Guess where I am?” (without turning out the lights). I stood up to begin our usual ritual of hide-and-find, prepared to search all around the room for him, when after just a few moments (not waiting for me to find him, as he had always done in the past), he jumped from behind the puppet theater, threw his arms wide in the air, and with a huge smile on his beautiful face, exclaimed, “HERE I AM!” I could not have summed it up any better! I responded by throwing my arms in the air and exclaimed, “Yes, THERE YOU ARE!”

I am always amazed by how children know what they need to bring closure to the therapeutic relationship if they have proper notice. Just thinking about our last moment together brings a huge smile to my face. Andy was free; he no longer needed me. He was well on his way to fully becoming all that he was meant to be.

In the following week, Carol and I met for a final consultation without Andy to bring closure to our relationship, to review Andy’s progress, and to devise a plan that would support Andy’s continued progress without therapy. We briefly reviewed Carol’s scores on the CBCL and PSI and discussed Andy’s progress since the beginning of treatment. Andy’s posttreatment CBCL scores indicated normal functioning in 8 of the 10 areas that were initially elevated, with an average mean decrease of 8.5 across all areas. Carol seemed almost surprised at her initial scores on the CBCL and how much differently she now viewed Andy’s behavior. Tears came to her eyes (for the first time in our 20 months together) when we reviewed the PSI outcomes and she realized how much her stress related to parenting Andy had been affecting their relationship. She finally seemed to understand that her view of Andy had a considerable influence on Andy’s view of himself. We discussed, among other things, his school placement for next year, strategies to minimize Andy’s stress related to school, involvement in extracurricular sports that he excelled in, and continuing special family activities at home, including reading Andy’s adoption story, “Chinese Night,” and weekly filial play sessions. Carol’s recent experience with Andy in their play sessions had been particularly rewarding for her, and she seemed highly motivated to continue. But I suspected that without support, she might have difficulty following through once the stress of the new school year began. We discussed the importance of follow-up and agreed to talk after Andy had started second grade in approximately 6 months (or sooner, if Carol saw a need).

Complicating Factors

Children’s ability to benefit fully from treatment is dependent on their caregivers and other systemic factors. When parents are unwilling or unable to commit to

participating in the therapeutic process or when children remain in an environment that is unsafe or is contributing significantly to the presenting issue, therapy is compromised. In this particular case, neither of these scenarios was an issue. Quite the contrary, the systemic factors related to this case were highly facilitative of Andy's growth in therapy. I noted no significant complications to the successful outcome of Andy's treatment.

Managed Care Considerations

In this particular case, managed care considerations were not an issue. Andy was seen at a community clinic with a sliding-scale fee. For those families that rely on managed care systems, educating third-party payors about the length of time of treatment needed for children with a history of interpersonal trauma is critical to ensure that these children receive the help they need. It is also imperative that the value and necessity of working with parents as part of a child's therapy be acknowledged and reimbursed by third-party payors. Although cases such as the one presented are complicated and require longer term therapy, play therapy research shows this method is effective in only a few sessions (Bratton et al., 2005). More research is needed to examine the efficacy of play therapy with specific populations and to determine optimal length of treatment relative to presenting concern.

Follow-Up Treatment

As a matter of practice, with the termination of all child clients I encourage parents to report back in around 4 to 6 months on how their child is progressing. At the time of this writing, 6 months had passed since termination of this case. During a follow-up phone consultation, Carol reported that school continued to be a challenge for Andy, but he was now better able to cope with school-related stress. His extracurricular involvement in tae kwon do and basketball seemed to be an effective outlet and a source of feeling competent. Andy excelled in both sports and particularly enjoyed his family's cheering him on in basketball. He was receiving educational support services in speech and language, but was on grade level in math and science. Although Andy continued to have occasional anger outbursts, the intensity and duration were greatly reduced. He had been sleeping in his own room for the past 9 months, and it had been well over a year since he had exhibited any stealing at home or school. Thus, according to Carol, there was no need for additional counseling at the time of follow-up. I followed up on the suggestions I had given her in our final consultation. She confessed that since school had started she had not been consistent with any of the family activities we had discussed, but acknowledged their importance and committed to setting aside 30 minutes each week for an activity of Andy's choosing. I encouraged Carol to call if any concerns arose and discussed the possibility that Andy might need additional counseling at certain points in his development.

Implications and Recommendations for Practice

Andy was an amazing child. In the face of overwhelming circumstances, he used play to fully experience all aspects of his life, both positive and negative, and assimilate those experiences into a more integrated self. What a privilege it was to be with him on this journey! Andy illustrates children's remarkable resiliency

(*Continued*)

and ability to master difficult situations if allowed to do so at their own pace and in their own way. Play provided the way for Andy to express, make sense of, and develop mastery over his confusing experiences. I could never have devised a treatment plan that was as brilliant as Andy's internal plan for what he needed in order to heal and overcome his early experiences.

Just as children have an innate drive for mastery and self-actualization, they also possess a protective tendency to suppress that which is painful until they are strong enough to allow difficult and confusing memories into their experience. Thus, the nonthreatening nature of CCPT was particularly appropriate to allow Andy to choose how, when, and to what to degree he would address his worries. Creating a safe, accepting, and respectful relationship provided Andy with the "good growing ground" (Axline, 1947) he needed to support his movement toward fully becoming the person he was meant to be. With no specific direction from me, he actively used his play to confront difficult issues in his life, including abandonment, loss, self-identity (including cultural identity), adoption, attachment, and belonging. Children who have experienced trauma are often triggered by what would be normal events for most children. Through my understanding of what Andy communicated to me in his play, I was able to help his parents and teachers identify potentially retraumatizing events (bedtime, separation, doctor visits) and devise strategies for altering or minimizing their impact on him.

Through the process of play therapy, Andy discovered new dimensions of himself that resulted in a revised self-image and more congruent and satisfying behavior. Outwardly, he demonstrated marked improvement in targeted behaviors, most notably self-regulation and a secure attachment to his parents, particularly to Carol. Andy was blessed to have a home with such remarkable parents as Carol and Tom. In my view, an invested parent is one of the most important tools in play therapy, or any child therapy. Their active participation in almost anything I suggested and willingness to implement changes were critical factors in Andy's growth. Their involvement in CPRT served as both curative and protective factors. It strengthened their confidence in their parenting and gave them a sense of empowerment as they successfully learned to utilize therapeutic play skills with Andy and saw what a difference they were making. Tom and Carol ended therapy armed with the confidence and skills they needed to continue to help Andy reach his potential. Providing parents with the skills and experiences to build a closer parent-child relationship is an essential element of therapy for a child like Andy, whose early experiences disrupted the formation of a strong bond with a caregiver.

For myself, my experiences with Andy on his complicated journey of growth and healing forced me to confront my own issues around wanting to protect children from difficult and overwhelming experiences. Andy did not need me to protect him from his feelings. He needed me to facilitate and witness the full expression of all of his experiences, both positive and negative. Peer consultations during some of the more difficult periods in Andy's therapy helped ensure that my needs were not interfering with Andy's need to fully convey all aspects of himself. Because children so readily express painful events and related emotions through the symbolism of play, play therapists may be more vulnerable to their own unresolved issues surfacing in the context of the child's playing out difficult experiences.

This case has shown how using a child-centered play therapy approach allows child therapists to join children at their level. The toys and materials provide children with a developmentally sensitive and personal means of expressing feelings, experiences, thoughts, and wishes in a way that child therapists can understand and respond to. The implication of this case is that play therapy is an effective and responsive treatment for young children. Furthermore, this case illustrates the value of involving parents as partners in the play therapy process for optimal treatment effects.

SUMMARY

Play therapy has grown over its over 100-year history to include a variety of treatment methodologies and theoretical schools of thought that embrace the therapeutic powers of play. Play therapy is based on developmental principles, and thus provides, through play, developmentally appropriate means of expression and communication for children. Enjoying a rich history of literature and research dating back to 1942, play therapy is the longest standing modality of psychotherapy developed specifically for children. Through early studies and recent meta-analyses, play therapy demonstrates a base of evidence to support its effectiveness.

This chapter has focused specifically on the principles and procedures of child-centered play therapy. In practice, CCPT offers a clear method of treatment, in which the relationship between child and therapist is the catalyst for all therapeutic change. The child's experience within the counseling relationship is the factor that is most meaningful and helpful in creating lasting, positive change. Hence, the therapist works toward creating an environment of acceptance and understanding that helps the child feel safe to fully express thoughts and feelings, moving toward more positive functioning. Following guiding CCPT principles (Axline, 1947; Landreth, 2002), the therapist develops a warm relationship with the child, accepts the child unconditionally, allows permissiveness within the relationship, reflects the child's feelings, respects the child's ability to solve problems, does not attempt to direct the child's actions, recognizes the gradual nature of the child's process, and establishes minimal limits.

For further effectiveness in play therapy, the play therapist is trained to work within the system that supports the child. Working with parents on understanding and responding to their children's needs is critical to the progress of play therapy. Parent consultation is one method of alignment with parents so that the therapist makes consistent intimate contact with the parents to sustain their commitment to their child's play therapy and to offer brief opportunities for education. Filial therapy is a more in-depth approach and provides extensive education in play therapy procedures and group support to parents so that they are empowered to use therapeutic play skills with their child.

This chapter has served to provide an introduction to the modality of play therapy for child therapists. The rationale and history of play therapy, as well as the research in the field, presented a base for its therapeutic use. The practice of CCPT was enumerated through descriptions of the therapist role, objectives, structure, therapist responses and attitudes, and applications of play therapy concepts. The case study

offered a concrete example of how play therapy is applied in a real-life setting, working with a child with a complicated presentation. Play therapy serves as a developmentally responsive intervention for children, allowing therapists to speak the language of the child. Skill in using play therapy is an essential tool for mental health professionals who work with children. Therapeutic play allows children to express themselves fully and at their own pace with the assurance that they will be understood and accepted. Through this experience, children learn to express, accept, respect, and take responsibility for themselves and their feelings, which leads to creative insight, learning, problem solving, coping, and mastery.

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CHAPTER 21

Behavior Modification

Raymond G. Miltenberger

Behavior modification is the field of psychology devoted to understanding and changing human behavior. In behavior modification, assessment procedures are used to record the problem behavior and identify the factors that contribute to its occurrence, and intervention procedures are used to help people change their behavior. Behavior modification procedures are based on basic principles of behavior discovered in experimental research with laboratory animals and humans over the past 70 years (e.g., Skinner, 1938; Ullmann & Krasner, 1965; Ulrich, Stachnik, & Mabry, 1966). As behavior modification is currently practiced, it is synonymous with the field of applied behavior analysis (Baer, Wolf, & Risley, 1968; Miltenberger, 2004).

In behavior modification there is an emphasis on observable behaviors, both behavioral excesses and behavioral deficits. Objective definition and measurement of behavior is a hallmark of behavior modification. Behavior change is accomplished by changing environmental events occurring at the time of the behavior, antecedents that evoke the behavior, and consequences that maintain the behavior. Behavior modification is largely based on operant conditioning principles and, to a lesser extent, on respondent conditioning principles (Miltenberger, 2004). Behavior modification has been successful in helping people of all ages and ability levels change a wide variety of socially significant behaviors in a wide variety of natural contexts (J. Carr & Austin, 2001; Gross & Drabman, 2005; Hersen & Rosqvist, 2005; Sugai & Horner, 2005; Vollmer et al., 2000).

This chapter provides an overview of the field of behavior modification, starting with issues in assessment and research design, followed by a review of basic behavioral principles that form the foundation of behavior modification procedures. Next, procedures to establish new behaviors are reviewed, the functional approach to assessing and changing behavior is presented, and various behavioral interventions are described.

USING DATA IN BEHAVIOR MODIFICATION

Observing, recording, and displaying data to evaluate treatment and make treatment decisions are important activities that help define behavior modification.

Observing and Recording Behavior

In behavior modification emphasis is placed on objective measurement of behavior through direct observation procedures. A number of steps are involved in the process of recording behavior (Miltenberger, 2004).

Defining the Behavior

The first step in the process of recording behavior is to define the target behaviors in objective terms. Target behaviors are chosen based on their social importance (Baer et al., 1968; Wolf, 1978) and may include problematic behaviors to be decreased (behavioral excesses) and desirable behaviors to be increased (behavioral deficits). Target behaviors are defined using action verbs so that two independent observers can record their occurrence reliably.

Logistics of Recording

Once the target behaviors are defined, the next steps are to choose an observer and the time and place of observation. The observer may be the person engaging in the behavior (self-monitoring) or a care provider (e.g., parent, teacher, staff member), therapist, or researcher. The observation period will be the time and place in which the behavior is most problematic or in which the behavior has the highest likelihood of occurrence.

Recording Method

The next step in the process of recording behavior is to decide on the use of event recording or a sampling procedure. With event recording, every instance of the behavior is recorded during the observation period. The four dimensions of behavior that can be measured are frequency (number of times the behavior occurred), duration (the time from onset to offset of the behavior), intensity (a measure of the force of the behavior), and latency (the time from some stimulus to the onset of the behavior). One or more dimensions are chosen for recording based on their relevance to the target behavior.

Sampling procedures, including interval and time sample recording, are not designed to record every instance of the behavior but rather to provide an estimate of the occurrence of the behavior. In *interval recording,* the observation period is divided into smaller consecutive intervals of time and the behavior is recorded as occurring or not in each interval. *Time sample recording* involves recording the occurrence of the behavior in brief intervals of time separated by longer intervals of time without observation. The outcome of interval or time sample recording is reported as the percentage of observation intervals in which the behavior occurred.

Recording Instrument

The final step is to choose a recording instrument. The most common recording instrument is a data sheet prepared for the particular client and target behaviors in question. The data sheet, designed to be convenient and easy to use, provides a

space for recording each instance of the behavior as it is observed. Other recording instruments are counters or timers, stopwatches, PDAs or other hand-held devices, and laptop computers, to name a few. The recording instrument is designed to be practical, to minimize disruption of ongoing activities during behavior recording, and to facilitate immediate recording of the behavior as it occurs.

Interobserver Agreement

Behavior recording procedures are designed to produce information that is as reliable as possible. One index of reliability commonly used in behavior modification is interobserver agreement. Interobserver agreement is the degree to which two independent observers agree on the occurrence of the behavior during the same observation period. Interobserver agreement is a necessary component of research in behavior modification, although it is used less often in clinical applications of behavior modification. The reader is referred to Miltenberger (2004), Bailey and Burch (2002), and J. Cooper, Heron, and Heward (2007) for details on calculating interobserver agreement.

Indirect Measures

In behavior modification, emphasis is placed on direct observation of behavior. However, in some instances, indirect measures of behavior, including product measures and retrospective reports, are used. Product measures may be used when the target behavior produces permanent changes in the environment (products) that can be measured. Examples of product recording are the number of assignments completed in an educational setting, the number of units assembled in a factory setting, and the number of hairs pulled by a person diagnosed with Trichotillomania. Retrospective reports about the occurrence of behavior can be gathered through interviews or questionnaires. Retrospective reports of behavior are not used as the sole assessment method but are typically used during the initial phase of assessment to plan direct observation procedures.

Data Display and Research Design

In behavior modification, the target behavior is recorded before, during, and after intervention is implemented. By recording before the intervention, a baseline level of the behavior is established and the effects of the intervention can be assessed by looking at changes in the behavior. A graph is the tool used in behavior modification to identify changes in the behavior from baseline to intervention. In a graph, the level of the behavior is plotted on the vertical axis (y axis) and time is plotted on the horizontal axis (x axis) so that changes in behavior over time can be easily discerned.

In most clinical applications of behavior modification the target behavior is recorded and displayed on a graph in two phases: baseline and intervention. However, when research is conducted in behavior modification, more sophisticated graphs are used that demonstrate a functional relationship between environmental

events (the intervention) and the target behavior. To demonstrate a functional relationship between intervention and the target behavior the researcher must:

- Reliably record the target behavior.
- Show that the target behavior changes when (and only when) the intervention is implemented.
- Provide a replication of the effect. In essence, the intervention is implemented two or more times and the behavior changes each time the intervention is implemented, but only when the intervention is implemented.

A number of research designs are used in behavior modification to demonstrate a functional relationship. These include the ABAB reversal design; multiple baseline designs across behaviors, participants, or settings; alternative treatment design; and changing criterion design. See Miltenberger (2004), Bailey and Burch (2002), or J. Cooper et al. (2007) for descriptions of these behavior modification research designs.

BASIC PRINCIPLES

Behavior modification procedures are based on a core set of basic behavioral principles that describe functional relationships between specific types of environmental events and behavior. These basic behavioral principles are reinforcement, extinction, punishment, stimulus control, and respondent conditioning.

Reinforcement

Reinforcement is a fundamental principle that underlies many behavior modification procedures. First studied extensively by Skinner (1938) and Ferster and Skinner (1957), reinforcement is defined as the process in which the consequence of a behavior strengthens the future probability of the behavior. Understanding the principle of reinforcement is important for understanding the causes of behavior and for strengthening appropriate behavior. There are two forms of reinforcement: positive reinforcement, in which behavior is strengthened when it results in the delivery of a positive reinforcer, and negative reinforcement, in which behavior is strengthened when it results in escape or avoidance of aversive stimuli. The consequences of the behavior involved in both positive and negative reinforcement may consist of changes in the physical environment or the behavior of others.

Reinforcement is influenced by a number of factors, including immediacy, contingency, establishing operations, individual differences, and magnitude (see Miltenberger, 2004, for discussion of these factors). One important factor that is increasingly utilized in behavior modification procedures is the establishing operation (e.g., Vollmer & Iwata, 1991). An establishing operation is an environmental event that makes a reinforcer more potent and evokes the behavior that produces that reinforcer (Michael, 1982).

Extinction

Extinction occurs when the reinforcing consequence maintaining a behavior is no longer present following the behavior, and as a result, the behavior decreases and eventually stops occurring. For behavior maintained by positive reinforcement, extinction occurs when the positive reinforcer no longer follows the behavior. For behavior maintained by negative reinforcement, extinction occurs when the individual is no longer able to escape or avoid the aversive stimulus following the behavior. In each case, when the behavior is no longer reinforced, it will stop occurring.

One phenomenon frequently seen during extinction is an extinction burst (Lerman & Iwata, 1995). In an extinction burst, there is a temporary increase in the behavior or the occurrence of novel or emotional behaviors when the target behavior no longer produces the reinforcer. The extinction burst is a transient phenomenon, and the target behavior will eventually stop occurring if the reinforcer continues to be withheld following the behavior. Understanding the extinction burst is important for the successful use of extinction in clinical practice.

Punishment

Punishment occurs when the consequence that follows a behavior decreases the future probability of the behavior. There are two variations of punishment. In positive punishment the behavior is followed by the addition of an aversive stimulus (also called a punisher); in negative punishment, the behavior is followed by the removal of a positive reinforcer. In both cases, the behavior is weakened. As with reinforcement, punishment occurs naturally in our contact with the physical environment and in our interactions with others. Some behavior modification procedures are based on the principle of punishment (e.g., time-out and response cost). The same factors that influence reinforcement influence punishment: immediacy, contingency, establishing operations, magnitude, and individual differences (Miltenberger, 2004). See Lerman and Vorndran (2002) for an in-depth discussion of punishment.

Stimulus Control

Although reinforcement, extinction, and punishment are the basic principles that determine whether a behavior will continue to occur, the effects of these operations are relatively situation-specific. Reinforcement strengthens behavior in situations that are similar to the ones in which the behavior was reinforced in the past. Likewise, extinction and punishment weaken behavior in situations that are similar to the ones in which the behavior was subjected to extinction or punishment procedures in the past. The increased probability that behavior will occur in specific situations (in the presence of specific stimuli) is referred to as stimulus control. A behavior is under the stimulus control of a specific stimulus when it is more likely to occur when that stimulus is present.

Stimulus control is developed through stimulus discrimination training. In stimulus discrimination training (also called discrimination training), the behavior is reinforced in the presence of one stimulus (a discriminative stimulus or S^D) and is not reinforced in the presence of other stimuli. As a result, when the S^D is present

in the future, the behavior is more likely to occur. The S^D has stimulus control over the behavior. The S^D will continue to have stimulus control over the behavior as long as the behavior continues to be reinforced in its presence.

Although discrimination training makes the behavior more likely to occur in the presence of the S^D, the behavior is also likely to occur in the presence of stimuli that are similar to the S^D. The more similar a stimulus is to the S^D, the more likely the behavior is to occur in its presence (Guttman & Kalish, 1956; Lalli, Mace, Livezey, & Kates, 1998). The occurrence of behavior in the presence of stimuli that are similar to the S^D is called generalization.

Respondent Conditioning

Reinforcement, extinction, punishment, and stimulus control are all operant conditioning principles. In operant conditioning, behavior is controlled by its consequences (reinforcement strengthens behavior, and extinction or punishment weaken behavior). Although operant behavior comes under the stimulus control of antecedent stimuli (the S^D), stimulus control is developed through the differential consequences that are present when the S^D is present. Whereas operant behavior is controlled by its consequences, respondent behavior is controlled by its antecedents. Respondent behavior is elicited by a prior stimulus, either an unconditioned stimulus or a conditioned stimulus.

Respondent conditioning occurs when a neutral stimulus is paired with an unconditioned stimulus (US). The US elicits an unconditioned response (UR) and, as a result of its pairing with the US, the neutral stimulus comes to elicit a response similar to the UR. The neutral stimulus is then called a conditioned stimulus, and the response it elicits is called a conditioned response. Anxiety, anger, and inappropriate sexual arousal are examples of conditioned responses that may be targets of intervention in behavior modification procedures.

Behavior modification procedures, the actual techniques used to help people change their behavior, are based on the basic principles just described. Behavior modification procedures are used to establish new behaviors, to identify the variables contributing to the occurrence of behaviors, and to help people increase or decrease existing behaviors.

PROCEDURES TO ESTABLISH NEW BEHAVIORS

When a desirable behavior is not currently in a person's repertoire, a number of procedures can be used to help the person acquire the behavior. These procedures include shaping, prompting and fading, chaining, and behavioral skills training procedures.

Shaping

Shaping is the differential reinforcement of successive approximations of a target behavior. Shaping is used to help an individual engage in a new behavior (e.g., a child saying a word for the first time), a new dimension of an existing behavior (e.g., a patient speaking with increasing voice volume), or a previously exhibited

behavior that is no longer occurring (e.g., a stroke patient walking again; Jackson & Wallace, 1974; Miltenberger, 2004; G. W. O'Neill & Gardner, 1983). To use shaping, the target behavior is defined and then a starting behavior is chosen. The starting behavior (or first approximation), a behavior that already occurs at least occasionally, is reinforced a number of times until it occurs consistently. Then it is no longer reinforced (extinction), and the next approximation to the target behavior is reinforced instead. Once this behavior occurs a number of times, it is no longer reinforced, and a closer approximation to the target behavior is then reinforced. This process continues until the individual engages in the target behavior. Once the target behavior is exhibited it is reinforced consistently to strengthen it and maintain its occurrence over time.

Shaping, a procedure that came from experimental research with laboratory animals (Skinner, 1938, 1951), has been used extensively in animal training (Pryor, 1985) and with humans to help individuals acquire a number of socially significant target behaviors in areas such as education, developmental disabilities, rehabilitation, and sports (Hagopian & Thompson, 1999; Horner, 1971; G. W. O'Neill & Gardner, 1983; Scott, Scott, & Goldwater, 1997). Shaping is used when more efficient behavioral acquisition strategies (e.g., prompting, instructions, modeling) are not possible.

Prompting and Fading

Prompting is used when a desired behavior is not currently occurring or not occurring in the correct circumstances (in the presence of the S^D). Prompting is a procedure in which an antecedent stimulus is delivered to evoke the behavior in the presence of the S^D so that the behavior can get reinforced. Fading is the gradual elimination of the prompt once the behavior is occurring in the presence of the S^D. The prompt gets the correct behavior to occur at the right time, and fading eliminates the prompt to get the behavior under the stimulus control of the S^D.

Prompts can include the behavior of another person (response prompt) or a change in an antecedent stimulus (stimulus prompt). There are four types of response prompts: verbal, gestural, modeling, and physical. A *verbal prompt* is when the verbal behavior of another person is used to get the correct behavior to occur in the presence of the S^D. A *gestural prompt* is when a physical movement or gesture of another person gets the correct behavior to occur. A *modeling prompt* is when a demonstration of the behavior gets the correct response to occur. A *physical prompt* is when another person physically assists or guides a person to engage in the correct behavior. Physical prompting typically consists of hand-over-hand guidance.

When prompting is used to evoke the correct behavior, the trainer first presents the S^D and, if a response does not occur immediately, the trainer provides the prompt. When the prompted response occurs, the trainer provides a reinforcer.

Consider the case of a training program to teach a child with autism to hit a baseball off a tee as part of a recreational program. The S^D is the ball on the tee on home plate on a baseball field in front of the child while she holds the bat. In the presence of the S^D, hitting the ball is the correct response and praise from the

coach is the reinforcer. A verbal prompt might be to say "Hit the ball" while the child is standing in position holding the bat. If the child hits the ball after receiving the prompt, the behavior can be reinforced. If the child does not hit the ball after receiving the verbal prompt, a gestural prompt such as pointing to the ball on the tee can be delivered. If the child does not engage in the correct behavior following the gestural prompt, a modeling prompt can be delivered. A modeling prompt might consist of hitting the ball off the tee yourself as the child watches. If the modeling prompt does not evoke the correct behavior, then a physical prompt can be used. A physical prompt might involve holding the child's hands on the bat and swinging the bat with the child to hit the ball. The use of hand-over-hand guidance in this way would result in the correct response and praise from the coach.

In the baseball example, once the child is hitting the ball correctly with the coach's physical guidance, the coach will fade the physical prompt. Over consecutive trials, the coach will provide less and less assistance (e.g., he will help the child start to swing and let her finish with no assistance) until the child swings the bat to hit the ball without any further assistance. Once prompts are no longer needed, the behavior is under the control of the S^D (the ball on the tee). See J. Cooper et al. (2007), Miltenberger (2004), and Sulzer-Azaroff and Mayer (1991) for more information on the use of prompting and fading.

Chaining

A chaining procedure is used to teach the learner to engage in a chain of behaviors. A behavioral chain (also called a stimulus-response chain) consists of a number of behaviors, each consisting of an S^D and a response, that occur together in a sequence. Each response in the chain creates the S^D for the next response in the chain. For example, consider the behavior of getting a drink from a water fountain. There are at least three behaviors in the chain: pushing the handle for water, bending over the stream of water, and drinking the water. In this example, the water fountain is the S^D. The first response is turning on the water. This behavior creates the second S^D (the water stream). The second response is bending over the water stream. This behavior creates the third S^D (the water stream in proximity to or touching your mouth). The third response is drinking the water.

Before using a chaining procedure to teach a chain of behaviors, you must conduct a task analysis. A task analysis identifies every S^D and response in the chain of behaviors (e.g., Bellamy, Horner, & Inman, 1979). Careful observation of someone engaging in the chain of behaviors is used to develop the task analysis.

Three procedures are used to teach a chain of behaviors: forward chaining, backward chaining, and total task presentation (Miltenberger, 2004).

Forward Chaining

In forward chaining you use prompting and fading to teach the first component in the chain. After the learner engages in the first response when the first S^D is presented without any prompts, prompting and fading are used to teach the first two components. At this time, the learner will engage in the first response when the first S^D is presented. When the first response creates the second S^D, the learner is

prompted to engage in the second response. Once prompts are faded, the learner can now engage in the first two responses in the chain without prompts. Then prompting and fading are used to teach the first three components, and so on until all of the component behaviors occur together.

Backward Chaining

In backward chaining, the last S^D is presented and prompting and fading are used to get the last response to occur. Once the last response is under the stimulus control of the last S^D, the next to last S^D is presented and the next to last response is prompted. Once the next to last response occurs, it creates the last S^D and the learner will then engage in the last response. After the last two behaviors in the chain occur without prompts, the third to last S^D is presented and prompting and fading are used until the last three responses occur without prompts. This process continues until the learner engages in the whole chain of behaviors without prompts.

Total Task Presentation

In total task presentation, you present the first S^D and physically guide the learner through the entire chain of behaviors from start to finish. Over trials, you fade the physical guidance to shadowing until the physical prompt is eliminated. This fading method, in which physical prompting is faded to shadowing and eventually eliminated, is called graduated guidance. Total task presentation is typically used with more capable learners, and forward or backward chaining is used with learners with significant disabilities.

Behavioral Skills Training

Behavioral skills training (BST) is one final strategy for helping a learner acquire new skills. Behavioral skills training consists of four training components: instructions, modeling, rehearsal, and feedback (e.g., Himle & Miltenberger, 2004). With instructions and modeling, the skills are described and demonstrated for the learner. Rehearsal is an opportunity for the learner to practice the skills in a simulation or role-play situation. Finally, feedback consists of praise for correct performance of the skills and correction or further instruction when the skill is not performed correctly. Typically, BST is used for behaviors that can be demonstrated and practiced in a simulation or role-play format. Various skills have been taught with BST procedures, including social skills, parenting skills, job-related skills, child safety skills, and clinical interviewing skills (Bakken, Miltenberger, & Schauss, 1993; Himle, Miltenberger, Flessner, & Gatheridge, 2004; Johnson et al., 2005; Miltenberger & Fuqua, 1985).

To conduct BST, you first define the target behaviors and all of the situations in which the behaviors are to be performed. In teaching abduction prevention skills, for example, the target behavior consists of saying no, running away, and telling an adult when faced with an abduction lure. All of the different ways an adult might try to lure a child would then be identified and incorporated into training (e.g., offering the child candy or a toy to leave, asking for assistance, invoking

authority). Once the target behaviors and S^Ds (situations) are identified, training begins with instructions and modeling. The trainer describes the behavior in detail and the situations in which the behavior is to be performed. The trainer then models the behavior in the context of a role-play of one of the relevant situations.

After receiving instructions and modeling, the learner then has the opportunity to rehearse the behavior. The trainer role-plays the same situation that was just modeled and the child practices the skill in response to the situation. For example, the trainer might tell a child, "Pretend I am a stranger that walks up to you in your front yard and asks you to help me look for my puppy down the street. Show me what you would do if this happened." The trainer then role-plays the situation, delivers the lure, and observes the child's response. Immediately after the child rehearses the behavior, the trainer provides feedback. The trainer provides descriptive praise for any aspect of the behavior the child performed correctly and then tells the child how the behavior could be improved (if needed). The trainer then immediately conducts the same role-play and provides the child the opportunity to rehearse the behavior again after receiving feedback. Once the child demonstrates the correct behavior in the role-play, other situations are incorporated into role-plays so that the full range of situations and responses is trained.

PROCEDURES TO INCREASE OR DECREASE EXISTING BEHAVIORS

A number of behavior modification procedures are used to help individuals decrease undesirable behaviors and increase desirable behaviors. The functional approach to assessment and intervention starts with a functional assessment to identify the antecedents and consequences influencing the behavior (R. E. O'Neill et al., 1997). Functional interventions, designed to address the antecedents and consequences of the behavior, include extinction, differential reinforcement, and antecedent control procedures (Miltenberger, 2004). Changing the antecedents and consequences of the behavior and promoting functionally equivalent alternative behavior alter the conditions that influence the behavior, resulting in the greatest likelihood of change in the behavior.

Functional Assessment

Functional assessment is a process of gathering information on the target behavior and the antecedents and consequences that are functionally related to the occurrence of the behavior (Lennox & Miltenberger, 1989). The outcome of a functional assessment is a hypothesis about the reinforcing consequences maintaining the behavior and the antecedents that evoke the behavior (R. E. O'Neill et al., 1997).

There are four categories of reinforcing consequences that could be maintaining a target behavior: social positive reinforcement (e.g., attention, tangible, or activity reinforcers provided by another person), automatic positive reinforcement (some form of sensory stimulation arising directly from the behavior), social negative

reinforcement (escape or avoidance of aversive activities or interactions mediated by another person), and automatic negative reinforcement (escape from aversive stimulation such as pain, discomfort, negative emotional responding; Iwata, Vollmer, Zarcone, & Rodgers, 1993). A functional assessment produces information that enables you to identify the type (or types) of reinforcement maintaining the problem behavior.

Antecedents that are identified in a functional assessment are the events, people, and circumstances that are reliably associated with the occurrence of the behavior. Antecedents can include S^Ds that occur immediately prior to the behavior and establishing operations (EO) whose onset may be proximal or more distal to the occurrence of the behavior. Identifying the S^D and EO associated with the behavior allows you to alter these antecedents to influence the problem behavior and alternative behaviors (Smith & Iwata, 1997).

There are three methods for conducting a functional assessment: indirect assessment, direct assessment, and functional analysis.

Indirect Assessment

An indirect assessment involves interviews, questionnaires, or rating scales in which individuals who have frequent contact with the client (informants) provide information from their recall of antecedents and consequences associated with the behavior (Lennox & Miltenberger, 1989; R. E. O'Neill et al., 1997). R. E. O'Neill et al., for example, developed the Functional Analysis Interview Format to gather detailed functional assessment information. The advantage of an indirect functional assessment is that it is easy and convenient to administer and does not take much time. The limitation is that it does not involve direct observation of the behavior as it occurs but instead relies on the informant's recall of events.

Direct Observation Assessment

A direct observation functional assessment (also called ABC recording) involves direct observation and recording of the behavior and its antecedents and consequences as the behavior occurs (Lennox & Miltenberger, 1989; R. E. O'Neill et al., 1997). Observations occur across a number of instances of the behavior until patterns in the antecedents and consequences are evident in the data. ABC recording can involve descriptive recording or the use of a checklist. In descriptive ABC recording, the observer writes down a description of antecedents, behavior, and consequences as they occur. To be of most value, the descriptions must be detailed and objective. In the checklist method, the observer has a checklist of possible target behaviors, antecedents, and consequences and places a checkmark in the appropriate column to designate the particular antecedents and consequences that were observed at the time of the behavior. The advantage of ABC recording is that the antecedents and consequences are observed as they occur, which results in objective (and likely more accurate) recording of the events. The limitation is that it takes more time to conduct multiple observations of the individual in the natural context (group home, school, etc.).

Functional Analysis

A functional analysis involves direct observation of the target behavior as antecedents and consequences are systematically manipulated to identify their influence on the behavior (e.g., E. G. Carr & Durand, 1985; Iwata, Dorsey, Slifer, Bauman, & Richman, 1982). For example, you could provide attention following each instance of the target behavior to see if the behavior increased, thus demonstrating that attention functioned as a reinforcer. Likewise, you could provide escape from demands contingent on the behavior or tangible reinforcers contingent on the behavior to see whether these events functioned as reinforcers for the behavior.

A functional analysis might be used to test a hypothesis, in which case one reinforcer might be manipulated, or a functional analysis might test a number of possible reinforcers. For example, to test the hypothesis that attention is reinforcing the target behavior, a functional analysis would alternate conditions in which attention is provided contingent on the behavior and conditions in which attention is withheld contingent on the behavior (e.g., Arndorfer, Miltenberger, Woster, Rortvedt, & Gaffaney, 1994). If attention is a reinforcer, the behavior will increase in the first condition relative to the second. To test a number of reinforcers, you might alternate conditions in which attention, escape, or tangible reinforcers are made contingent on the behavior. An alone condition would also be used to see if the behavior persisted in the absence of social consequences (e.g., Iwata et al., 1982). The advantage of a functional analysis is that, because antecedents and consequences are manipulated and resulting changes in the target behavior are recorded, a functional relationship between the antecedents and consequences and the behavior is demonstrated. The limitation of a functional analysis is that it takes substantial time and expertise to conduct and requires temporarily reinforcing the undesirable behavior.

Once a functional assessment is conducted and hypotheses about the antecedents and consequences of the behavior have been developed, functional interventions (described later) are chosen based on the hypotheses.

Extinction

Extinction, described earlier, is a basic principle of behavior. Extinction is also a behavioral procedure when it is used to help an individual eliminate an undesirable behavior (Lerman & Iwata, 1996). To use extinction, you must first conduct a functional assessment to identify the reinforcer maintaining the problem behavior. Once the reinforcer is identified, extinction involves eliminating the reinforcer after each instance of the behavior. If the behavior no longer gets reinforced, it will cease to occur. However, the successful use of extinction in clinical practice requires careful consideration of a number of questions (Miltenberger, 2004):

- Have you identified the reinforcer for the behavior? You cannot assume a particular reinforcer is maintaining a problem behavior; you must identify the reinforcer from a careful functional assessment. Sometimes, multiple reinforcers are maintaining a problem behavior in different contexts.

- Can you eliminate the reinforcer? Eliminating the reinforcer can be a problem when the behavior is reinforced by other people and you do not have influence over those individuals or when the behavior is automatically reinforced.
- Is extinction safe to use? In some cases of aggressive or self-injurious behavior, if the parent or teacher no longer responds to the behavior, it might put individuals at risk for harm. Even though the adult's response to the behavior may be reinforcing the behavior (i.e., attention), during extinction it may be necessary for the adult to respond to interrupt the behavior and assure the individual's safety. In such cases, the reinforcing consequence may be attenuated if it cannot be totally eliminated.
- Will individuals using extinction be able to deal effectively with an extinction burst? In most cases when extinction is used, the problem behavior temporarily worsens (increases in frequency, duration, or intensity) before it decreases. The parents, teachers, or others using extinction must be alerted to the probability of an extinction burst and be prepared to deal with it effectively. If they are not prepared, they might accidentally reinforce a more extreme problem behavior when it emerges during an extinction burst.
- Can consistency be maintained over time with all people? For extinction to be effective, it must be implemented each time the problem behavior occurs. If some instances of the behavior are reinforced only at some times by some individuals, then the behavior will likely persist and possibly worsen due to intermittent reinforcement.

When extinction is used clinically, it must be used in the context of other functional interventions as well. Not only should the problem behavior result in extinction, but differential reinforcement should be used to strengthen more desirable behavior. In addition, antecedent conditions should be altered to prevent the occurrence of the problem and evoke desired behaviors while using extinction.

Differential Reinforcement

Differential reinforcement involves the use of reinforcement to increase desirable behavior and extinction to decrease the occurrence of the problem behavior. There are three variations of differential reinforcement: differential reinforcement of alternative behavior (DRA), differential reinforcement of other behavior (DRO), and differential reinforcement of low rates of behavior (DRL; Miltenberger, 2004).

Differential Reinforcement of Alternative Behavior

In DRA, reinforcement is provided for a desirable alternative behavior, and extinction is used for the problem behavior. The goal of DRA is for the desirable behavior to increase in frequency and replace the problem behavior as it is decreased through extinction. For DRA to be effective it is important to reinforce the desirable behavior each time it occurs while withholding reinforcement for every instance of the problem behavior. In addition, the reinforcer chosen for the desirable behavior should be a known reinforcer. When possible, it is best to reinforce the desirable behavior with the same reinforcer that was maintaining the problem behavior prior

to the use of extinction. For example, if the problem behavior was maintained by attention, attention should be provided as a reinforcer for the alternative behavior. In this way, the desirable alternative behavior is functionally equivalent to the problem behavior because it produces the same outcome that used to produce the problem behavior (e.g., E. G. Carr, 1988).

Differential reinforcement of alternative behavior may involve positive reinforcement or negative reinforcement. In the case of positive reinforcement involving attention, attention no longer follows the problem behavior and instead is delivered following each occurrence of a desirable alternative behavior (Durand & Carr, 1991, 1992). In the case of negative reinforcement involving escape from aversive activities, escape is no longer contingent on the problem behavior and is delivered instead contingent on the occurrence of a desirable behavior (Allen & Stokes, 1987; Marcus & Vollmer, 1995). Steege et al. (1990) used DRA to decrease self-injurious behavior that was maintained by escape from grooming activities. In this case, a child with intellectual disabilities bit his wrist while an adult was brushing the child's teeth and the biting was reinforced by the termination of tooth brushing. When DRA was used, wrist biting no longer resulted in termination of tooth brushing (extinction). Instead, the child was prompted to press a switch to activate a tape-recorded message asking for a break. When this alternative behavior occurred, it resulted in a break (termination of tooth brushing for a brief period of time). When pushing the button to activate the taped message was reinforced with escape but wrist biting was not, the child quit biting his wrist and engaged in the alternative behavior instead.

Differential Reinforcement of Other Behavior

Differential reinforcement of other behavior is a procedure that utilizes extinction for the problem behavior and reinforcement for the absence of the problem behavior. If the problem behavior no longer produces a reinforcer and the reinforcer can be acquired only for the absence of the behavior, the behavior will decrease. Before using DRO you must first identify the reinforcer maintaining the problem behavior. Then you identify the reinforcer to be delivered contingent on the absence of the problem behavior. In most cases, the reinforcer delivered for the absence of the problem behavior will be the reinforcer that was maintaining the problem. Next, you identify the interval for the delivery of the reinforcer. The interval will be based on the average time between occurrences of the problem behavior in baseline. For example, if a problem behavior occurs on average 3 times per minute, the average time between responses is 20 seconds and the DRO interval should be set at 20 seconds or less.

To use DRO you provide the reinforcer after each interval of time in which the problem behavior does not occur. If the problem behavior does occur at any time, the reinforcer is withheld and the interval is reset. Again, the problem behavior must be absent for the entire interval for the reinforcer to be delivered. Once the problem behavior begins to decrease in frequency and the reinforcer is being delivered for its absence in the majority of intervals, the interval is gradually lengthened. As the interval is lengthened, the problem behavior must be absent for longer periods of

time before the reinforcer is delivered. The goal when lengthening the DRO interval is to end up with an interval length that is practical for the parent or teacher to implement. Although it will require the complete attention of the caregiver to deliver the reinforcer after every 20-second interval with the absence of the problem behavior, once the interval is lengthened to a more manageable time (such as 30 minutes) the caregiver will be able to resume normal activities while implementing DRO (e.g., Cowdery, Iwata, & Pace, 1990; Iwata, Pace, Cowdery, & Miltenberger, 1994; Vollmer & Iwata, 1992).

Differential Reinforcement of Low Rates of Behavior

Differential reinforcement of low rates of behavior is a procedure in which a reinforcer is delivered if a problem behavior happens fewer than a specified number of times in an observation period. The goal is to provide a reinforcer for a lower rate of the behavior. Consider the case of a child with Attention-Deficit/Hyperactivity Disorder who gets up from the dinner table on average more than 10 times per meal. In a DRL procedure, the parent might arrange a contingency in which the child gets dessert (a reinforcer) only if she gets up from the table fewer than 5 times during the meal. Once the child is successful, the number of acceptable responses can be lowered to a more desired level. When using DRL, it is important to provide the child with a way to keep track of the number of responses that have occurred. In this case, the parent might bring to the table a chart with five boxes and place an X in a box each time the behavior of getting up from the table occurs. The child would have to have at least one blank box at the end of the meal to receive the reinforcer.

In another form of DRL, a reinforcer is delivered only if x amount of time has elapsed since the last response. In this form of DRL, the timing of responses is important. As the interresponse time increases, the response rate decreases. An example of this form of DRL is from a study by Lennox, Miltenberger, and Donnelly (1987). The authors worked with adults with mental retardation who engaged in rapid eating at meals. To decrease the rate of eating to make the individuals more socially acceptable, the authors used a DRL procedure in which a bite of food was allowed in the mouth only if 15 seconds had elapsed since the previous bite of food. If 15 seconds had not elapsed since the previous bite, the authors blocked the participants from putting the bite in their mouth. In this way, the time between bites increased to more than 15 seconds (thus, eating rate decreased).

Antecedent Control

Antecedent control procedures involve the manipulation of S^Ds, EOs, or response effort to make the occurrence of a problem behavior less likely or evoke the occurrence of desirable behaviors (Miltenberger, 2004).

To use antecedent control to increase desirable behavior you can present the S^D or an EO for the behavior or decrease response effort for the behavior. Presenting an S^D for a behavior makes the behavior more likely to occur because it has been reinforced in the presence of the S^D in the past. An example might be to send a student to a quiet room with textbooks on a desk to increase the likelihood of studying. Presenting an EO makes a behavior more probable because it increases

the reinforcing value of the outcome of the behavior (Vollmer & Iwata, 1991). In the case of a training program with a child with autism in which small edible reinforcers are used (bits of favorite cereal), the trainer can run the training program just before meals, when the child has not eaten in a few hours, as a way to make the edible reinforcers more potent. The trainer can also make sure the child does not have access to the favorite cereal until the training session as a way to make the cereal more reinforcing. Decreasing response effort for a behavior makes the behavior more probable than a functionally equivalent alternative behavior that requires more effort (e.g., Horner & Day, 1991). When a child with tantrum behavior maintained by attention is taught how to ask for attention, the child is more likely to ask for attention if asking takes less effort than engaging in the tantrum behavior and produces an equal amount of attention.

To use antecedent control to decrease a problem behavior, you can remove an S^D or EO for the behavior or increase response effort for the behavior. A behavior is less likely to occur in the absence of the S^D, so removing the S^D is one way to make the behavior less likely. If watching television and playing computer games interfere with doing homework, then turning off the television and computer removes the S^Ds for these competing behaviors to make them less likely (and make doing homework more likely). If the EO for a particular reinforcer is removed, the reinforcer is no longer potent and the behavior producing that reinforcer will be less likely. In the case of the child whose tantrums are reinforced by parent attention, frequent noncontingent attention from the parent will make the tantrums less likely because the reinforcing value of parental attention is decreased through satiation (e.g., Vollmer, Iwata, Zarcone, Smith, & Mazaleski, 1993). Noncontingent reinforcement can make behavior maintained by various forms of reinforcement (attention escape, tangibles) less likely by altering the EO through satiation (Wilder & Carr, 1998). Increasing response effort for a problem behavior makes the behavior less likely because the individual will be more likely to engage in a less effortful, functionally equivalent alternative behavior. For example, when a student with multiple disabilities in a special education classroom slapped students near him, the behavior was reinforced by teacher attention (the teacher came over to him, expressed concern, and explained why the behavior was inappropriate). In this case, the student was moved to a location in the classroom where other children were out of reach. Because it now required more effort to slap another student, the student was more likely to engage in alternative behavior (e.g., raising his hand) for teacher attention.

To use antecedent control procedures successfully, you must identify relevant antecedents through a functional assessment and alter the antecedents that are functionally related to the target behaviors. Numerous studies have demonstrated the effectiveness of antecedent control procedures for problem behaviors maintained by positive reinforcement and negative reinforcement (Luiselli, 2006).

Punishment

In behavior modification problem behaviors are addressed first through a functional approach to assessment and intervention. After conducting a functional assessment,

extinction, differential reinforcement, and antecedent control procedures are utilized, often with success. In cases where problem behaviors are not successfully treated with these functional interventions, punishment may be considered (Iwata, 1988; Miltenberger, 2004). The most commonly used punishment procedures are based on the principle of negative punishment and include time-out and response cost. Although positive punishment procedures involving the delivery of aversive stimulation (such as the use of contingent shock) have been used in the past, such procedures are rarely if ever used in contemporary behavior modification.

Time-out

Time-out from positive reinforcement ("time-out" for short) is a procedure in which, contingent on a problem behavior, a child is removed from a reinforcing environment for a brief period of time. Time-out is a negative punishment procedure because the child is removed from all sources of positive reinforcement contingent on the problem behavior and the behavior decreases. The child is typically escorted to a nearby room or an area devoid of any access to reinforcers. Time-out has been shown to be a successful intervention for a range of childhood problem behaviors (e.g., Mathews, Friman, Barone, Ross, & Christophersen, 1987; Porterfield, Herbert-Jackson, & Risley, 1976; Rortvedt & Miltenberger, 1994).

For time-out to be effective, a number of factors must be considered (Miltenberger, 2004):

- First, the environment from which the child is removed (the time-in environment) must be reinforcing (must contain positively reinforcing activities, interactions, and/or stimuli). If the time-in environment is not reinforcing, time-out will not function as negative punishment.
- Second, time-out must be initiated immediately contingent on the problem behavior.
- Third, time-out must be brief (a few minutes in duration) to provide minimal disruption of ongoing educational or other important activities.
- Fourth, attention or other reinforcers must not be available in the time-out area. If the time-out room or area is reinforcing, time-out will not be effective.
- Fifth, the child should not get any attention while being taken to time-out. Contingent on the problem behavior the parent should simply state the problem behavior and that the child has to go to time-out and proceed to escort the child to time-out without any further discussion.
- Sixth, the child should not escape from time-out until released by the parent when time-out is over.
- Seventh, the time-out room or area must be safe.

Response Cost

Response cost is a procedure in which, contingent on the occurrence of the problem behavior, a specified amount of a reinforcer is removed to decrease the behavior.

Losing the opportunity to have dessert contingent on problem behavior at supper, or loss of a portion of an allowance contingent on the occurrence of a problem behavior are examples of response cost that might be used by parents. In each case, the loss of the reinforcer makes the problem behavior less likely in the future. Response cost procedures are used frequently by governments and other organizations (e.g., speeding tickets, fines for parking in handicapped parking spaces, fees for bounced checks, late fees from credit card companies). Response cost procedures are used effectively for a variety of child behavior problems as well. For example, Little and Kelley (1989) use response cost to decrease noncompliance to parental requests; Long, Miltenberger, and Rapp (1999) use response cost to decrease thumb sucking; and Rapport, Murphy, and Bailey (1982) used response cost to decrease hyperactive behavior in children.

For response cost to be effective, the loss of the reinforcer (or a statement that the reinforcer will be lost) should occur immediately contingent on each instance of the problem behavior. The magnitude of the reinforcer loss should be sufficient to function as a punisher. Finally, ethical issues must be considered such that the loss of the reinforcer does not harm the child in any way or violate the child's rights (e.g., loss of meals, for example, is not an acceptable form of response cost).

Promoting Generalization

In behavior modification, generalization is defined as the occurrence of target behavior outside of the treatment or training setting in the natural environment. For example, if a child learns social skills in treatment sessions with the school psychologist, generalization occurs when the social skills are used with peers at school or in other natural settings. A number of strategies can be used to promote generalization (e.g., Stokes & Baer, 1977; Stokes & Osnes, 1989):

- Provide reinforcement for the target behavior when it occurs in the natural environment. In the social skills example, the school psychologist could provide praise whenever she observes the child engaging in good social skills with peers.
- Teach skills that will contact natural contingencies of reinforcement. The child should be taught the kind of social skills likely to generate a reinforcing response from peers.
- Modify natural contingencies of reinforcement. The psychologist might talk to the student's teachers and have them provide praise for the student's appropriate use of social skills.
- Incorporate a variety of relevant stimulus situations in training so that the child has practiced and received reinforcement for engaging in the correct behavior in a wide range of situations he might encounter in the natural environment.
- Incorporate stimuli from the natural environment into the training situation (common stimuli). An example would be to have some same-age peers in the social skills training sessions so the student practices with the kinds of peers he is likely to encounter in school.

- Teach self-generated mediators of generalization. This strategy might include teaching the student some simple self-instructions that he could use in social situations to cue himself to use the social skills that he had learned in training sessions.

Another form of generalization occurs when a target behavior that decreased in treatment sessions also decreases in the natural environment. To promote a generalized reduction in problem behavior, it is important to reinforce a functionally equivalent alternative behavior so that the desirable behavior takes the place of the problem and produces the same type and amount of reinforcement that the problem behavior did before treatment (E. G. Carr, 1988). It is also important to maintain extinction or punishment contingencies across situations and over time in case the problem behavior occurs again for some reason (e.g., spontaneous recovery).

Habit Reversal

Habit reversal is a procedure developed by Azrin and Nunn (1973) as a treatment for nervous habits (hair pulling, thumb sucking, nail biting, skin picking, and other repetitive self-directed behaviors) and motor and vocal tics. Habit reversal procedures were also applied successfully to stuttering by Azrin and Nunn (1974). Since the original studies describing these procedures, numerous studies have demonstrated the effectiveness of the procedures or streamlined versions of the procedures (Miltenberger, Fuqua, & Woods, 1998).

Habit reversal is a multicomponent procedure with the following components: (a) motivational procedures involving review of all the inconveniences, embarrassment, or social and personal problems caused by the habit behavior; (b) awareness training in which the client describes the habit behaviors and learns to detect each instance of the habit as it occurs or as it is about to occur; (c) competing response training in which the client learns to engage in a physically incompatible motor response contingent on the occurrence of the habit behavior or in anticipation of the habit behavior; and (d) social support procedures in which a parent or other adult helps the child control the habit behavior by prompting the child to use the competing response when necessary, praising the child for using the competing response at the appropriate time (DRA), and praising the child when the habit behavior is not occurring (DRO).

Habit reversal procedures are taught to the child with the parent present in a few treatment sessions. Booster sessions are then used as needed to help the child continue to use the procedures successfully over time (Rapp, Miltenberger, Long, Elliott, & Lumley, 1998; Romaniuk, Miltenberger, & Deaver, 2003). Although habit reversal has been used successfully for child habit behaviors, research has shown that it may not be effective for younger children (Long, Miltenberger, & Rapp, 1999; Rapp, Miltenberger, Galensky, Roberts, & Ellingson, 1999) or individuals with disabilities (Long, Miltenberger, Ellingson, & Ott, 1999).

Token Economy

A token economy involves the systematic application of conditioned reinforcers called tokens to promote and maintain desirable behaviors (Ayllon & Azrin, 1968). In a token economy, tokens are delivered for desirable behavior and later exchanged for backup reinforcers. Tokens maintain their effectiveness as conditioned reinforcers because they are paired with already established backup reinforcers. The following steps are involved in developing and implementing a token economy (Miltenberger, 2004):

1. Identify and define the target behaviors you want to strengthen through the use of the token economy.
2. Identify the objects that will used as tokens. Tokens must be able to be delivered immediately to the client following the desirable behavior. They can consist of small objects such as poker chips, pennies, or marbles that can be accumulated by the client. They can also include points, checkmarks, hole punches in a card, or anything else that can be delivered and accumulated easily.
3. Identify the backup reinforcers. Because tokens maintain their power as conditioned reinforcers by being paired with backup reinforcers, it is important to choose items or activities that are known to be powerful reinforcers for use in the token economy. Furthermore, it is best to have a variety of backup reinforcers available to help avoid satiation.
4. Identify the number of tokens that will be delivered for each of the target behaviors. Typically, more tokens are delivered for more difficult or more important target behaviors.
5. Identify the exchange rate for backup reinforcers. You need to determine how many tokens are required to exchange for each backup reinforcer. Larger or more valued reinforcers will require more tokens.
6. Identify a time and place for exchanging tokens for backup reinforcers. Clients will be receiving and accumulating tokens each day for engaging in desirable behavior. Unless a time and place is set for exchanging tokens, the clients may seek to purchase backup reinforcers numerous times daily, thus disrupting ongoing activities.

To use a token economy effectively, it must be planned in advance, taking into consideration each of these steps. In addition, staff must be trained and supervised to carry out the program successfully. Finally, it is important to develop the token economy in such a way that the tokens cannot be acquired in any way other than by engaging in the target behaviors and that the backup reinforcers cannot be acquired in any way other than with tokens. If backup reinforcers are freely available, individuals will not work to earn tokens to exchange for them.

Behavioral Contracts

A behavioral contract is a written agreement specifying a desired behavior and a consequence for the behavior (O'Banion & Whaley, 1981). By writing a behavioral contract, a person is making a commitment to engage in a specified behavior in a specified time frame and arranging a consequence as a reinforcer for the behavior. Typically, a contract manager implements the contract with the person seeking behavior change. It is the contract manager's job to verify the occurrence of the behavior and carry out the consequence specified in the contract. For example, a student might agree to a behavioral contract with a school counselor in which the student commits to being on time for school each day for a week. When agreeing to the contract, the student gives the counselor five personal checks for $10 each. Each day the student gets to school on time, she gets one of the checks back. If she is late, the check for that day is sent to charity. Knowing that she will avoid the loss of $10 by arriving on time helps motivate the student to engage in the behavior.

The essential components of a behavioral contract include (a) a clear description of the target behavior to be engaged in or avoided, (b) the time frame in which the target behavior (or its absence) should occur, (c) how the target behavior will be measured, (d) the consequence for engaging in (or not engaging in) the target behavior, and (e) signatures of the individual seeking behavior change and the contract manager (Miltenberger, 2004).

Case Illustration

Christy was a 25-month-old girl referred by her mother for feeding difficulties. She was seen by a psychology graduate student under the supervision of a faculty member in clinical psychology.

Identifying Features of the Client
Christy was 25 months of age at the time of referral. She lived at home with her parents and 5-year-old brother. Due to her feeding problems, she had a gastric tube and received supplemental feedings each night while she was asleep.

Presenting Complaints
The problem for which her mother sought help was a feeding problem involving food refusal. Christy refused to accept bites of food presented to her on a spoon by closing her mouth, turning her head, placing her hands in front of her mouth, and trying to get away from the table. Another problem was gagging and food expulsion, which occurred to a lesser extent than food refusal. Gagging was most likely to occur when coarser foods were eaten or larger bites were attempted. Christy refused most foods except baby foods, pureed foods, soft foods such as ice cream and pudding, and shakes.

History and Developmental Factors
Christy's mother had an uncomplicated pregnancy and delivery. Within 10 days after birth, during which time Christy cried, grimaced, and vomited frequently during bottle feeding, she was diagnosed with gastro-esophageal reflux. She received a medication but the problem continued for the next 4 months. Because Christy did not gain weight at the appropriate rate in the first 6 months of life,

a gastric tube was implanted and she received supplemental feedings while asleep in bed. Her medication was also changed and the reflux problem abated. However, Christy exhibited food refusal problems when her mother started feeding her baby food and then table food.

At the time of referral, Christy was developmentally normal in all areas except oral-motor development, where she was slightly delayed. However, she did have the ability to chew and swallow food cut up into small pieces. She was at the 30th percentile for height and the 2nd percentile for weight.

Parental Factors

Both parents were present in the home and committed to solving Christy's feeding problem. Her mother was at home with her while her father worked for the first 2 years of her life, so her mother was the one who usually fed Christy. Christy started day care 2 months before the referral.

Assessment

Assessment was conducted through interviews with the mother, direct observation of meal times by the graduate student therapist, and behavioral recording by her mother. The behaviors recorded during direct observation were (a) number of bites presented, (b) number of bites accepted, (c) type of food, and (d) amount of food consumed. Functional assessment information was gathered through the interview and through direct observation to determine the types of food refused and parental responses to food refusal.

Case Conceptualization

It appeared from history that food refusal was initially negatively reinforced by escape from aversive stimulation arising from reflux. It also appeared that food refusal was reinforced through avoidance of gagging. At the time of referral, both positive and negative reinforcement appeared to play a role in maintaining food refusal. Christy escaped from eating foods that were more coarse or required more chewing by engaging in food refusal behaviors. In addition, when she refused to accept bites of food, her mother provided substantial coaxing and other attention in an attempt to get her to eat. Furthermore, when she refused to eat food being served at the meal, her mother gave her preferred foods instead so that she was consuming some calories. Therefore, the food refusal appeared to be maintained in part by attention and the receipt of preferred foods. Gagging was probably reinforced in part by attention, preferred food, and escape as her mother expressed great concern each time she gagged, let her stop eating, and let her have access to the pureed foods or baby foods instead.

Course of Treatment and Assessment of Progress

Treatment consisted of (a) a shaping program, (b) differential reinforcement involving the use of attention and bites of preferred food for accepting bites of nonpreferred food, and (c) extinction in which attention and preferred foods were no longer given as a consequence for food refusal.

In the shaping procedure, Christy was asked to eat very small portions of nonpreferred foods at the beginning of treatment, and the portion size was gradually increased across days and weeks. In the differential reinforcement procedure, each time she ate the nonpreferred food that was presented to her,

(*Continued*)

she received her preferred food. She also received enthusiastic praise for each bite of nonpreferred food she accepted. In the extinction component of treatment her mom and dad were taught to withhold their attention for any refusal behaviors or gagging. They were also taught not to provide preferred food following food refusal behaviors.

The graduate student therapist went to the family home and modeled the intervention procedures at a few mealtimes and coached the mom and dad as they used the procedures in subsequent meals. Once the parents were successfully using the procedures and stated that they were comfortable with the procedures, home visits were terminated and office visits were continued to discuss progress and troubleshoot any problems. Initially, mealtimes were videotaped to collect data on Christy's refusals and gagging. When it was evident that refusals were declining greatly and the amount eaten was increasing, videotaping was no longer conducted. Christy's mother continued to record type and amount of food consumed and number of refusals.

Complicating Factors

There were no complicating factors making assessment or intervention more difficult in this case.

Managed Care Considerations

The intensity of the intervention, conducted at supper time in the home a few times per week, would be difficult to carry out in a typical managed care environment. Because a graduate student receiving practicum training was involved in the case, the home visits and increased frequency of sessions could be accomplished.

Follow-up Treatment

Christy continued to make improvements over the course of 3 months, engaging in far fewer food refusal behaviors and eating larger quantities of food. Her weight increased into the 10th percentile after 3 months of treatment. Although she continued to be fed through the gastric tube each evening, the amount of supplementation was gradually decreased.

Implications and Recommendations to Clinicians and Students

This case illustrates how a functional approach to assessment and intervention can effectively address a serious child behavior problem. The intervention procedures were chosen based on the results of the functional assessment, which identified the antecedents and consequences maintaining the problem behavior. The antecedents and consequences were then addressed in treatment to change the variables contributing to the problem. Clinicians and students are encouraged to utilize a functional case conceptualization and a functional approach to treatment as best practice when using behavior modification.

SUMMARY

Behavior modification is a field of clinical psychology that emphasizes objective measurement of behavior before and after intervention to document the effects of the

intervention. Intervention involves the application of behavioral principles to help people change socially significant behaviors. Behavior modification procedures are aimed at changing aspects of the physical or social environment to bring about changes in behavior. Behavior modification procedures are used to help people develop new behaviors (to overcome behavior deficits) and to help people stop engaging in undesirable behavior (to decrease behavioral excesses).

When using behavior modification procedures to help people address behavior problems, a functional approach to assessment and intervention is utilized. A functional assessment is conducted first to identify the variables contributing to the problem behavior. Interventions are then developed based on the results of the functional assessment information. Interventions seek to alter the antecedents and consequences that contribute to the problem behavior while promoting functionally equivalent alternative behavior to replace the problem behavior. Behavior modification procedures are successful when they produce desired behavior changes that generalize to the client's everyday environment.

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CHAPTER 22

Parent Training

Mark W. Roberts

Surely parents were given advice about how to raise their difficult children long before the introduction of parent training to the child clinical literature in the 1960s. McLoughlin (1982) cites explicit American efforts dating back to 1815. What is surprising is that professional therapists attempted to treat emotionally disturbed children during the first 60 years of the twentieth century without involving parents! This chapter traces the therapeutic use of parents to socialize defiant, aggressive children from its humble beginnings at midcentury to the relatively robust protocols of today. The entire parent training literature is too vast to be integrated into a single book, let alone a chapter, so a specific focus must be adopted. First, only articles regarding overt problems of conduct are considered. Specifically excluded are literatures on teaching parents to promote skills in children with disabilities (e.g., S. L. Harris, Wolchick, & Milch, 1982), to facilitate normal adaptive skills such as toileting (e.g., Azrin & Besalel, 1979), or to treat internalizing disorders (e.g., Barmish & Kendall, 2005) or covert problems (e.g., stealing, vandalism). Second, the chapter is limited to children in the preschool and middle childhood periods (roughly 2 to 12 years of age), eliminating publications addressing infants and adolescents. Third, only parent training based on social learning theory is considered (Miller & Prinz, 1990), excluding parent training based on humanistic, dynamic, or family systems theory (Rinn & Markle, 1977). Fourth, the use of parent training as a primary prevention tool (Sanders, 1999) or a secondary prevention strategy (Conduct Problems Prevention Research Group, 2004) is beyond the scope of this chapter. The chapter does presume to accurately represent parent training studies that treat problems of overt social aggression in 2- to 12-year-old children from a social learning perspective.

The chapter is organized into six sections in which the current status and limitations of knowledge are reviewed. A brief history of parent training, its purpose and origins, is presented, which includes references to major review articles and identification and discussion of six ongoing research domains. The theoretical basis for parent training is examined from a historical perspective with an eye toward case conceptualization and treatment selection. The various methods of teaching skills to parents are compared in terms of effectiveness and efficiency. Current parent

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training curricula are reviewed, with a focus on treatments for noncompliance and aggression plus brief discussions of negative vocalizations and situational issues. A case illustration and articulation of future challenges conclude the chapter.

A BRIEF HISTORY OF PARENT TRAINING

The Early Years: 1959 to 1973

Children disobey parents and teachers; they fight with other children; they scream and yell at adults who attempt to discipline them or deny their requests; they argue with "uncooperative" parents, siblings, and peers; and they talk rudely to those who attempt to thwart their interests. Essentially, the presenting problems of oppositional and disruptive children are problems of social interaction, including parent-child, teacher-child, sibling-child, and peer-child interaction. It is to the credit of the small set of clinical child researchers who launched parent training research in the 1960s that they recognized the inextricable relationship between child social aggression and the social contexts in which it flourishes.

A case study by Williams (1959) is often cited as the first published account of parent training, in which the parents of a 21-month-old were instructed to ignore his bedtime screaming. The recommended extinction procedure did successfully covary with decreased tantrum duration at bedtime. Boardman's (1962) florid account of a conduct disordered 6-year-old is often cited as the second documented parent training case. "Rusty" defied his mother, left home and school without permission, set fires, and destroyed property. Boardman met with the family just once, but followed up with several phone calls. Room time-outs and logical consequences were prescribed. Rusty reacted with violent resistance, running away, and continued defiance until an intense spanking episode appeared to suppress his coercive efforts. Positive reinforcement for cooperation had also been recommended by Boardman, but there was no evidence of its use. Boardman's treatment plan elicited concerned and scholarly comments from Bandura (1962), who noted the limitations of punishment-only strategies and recommended the systematic use of positive reinforcement. Intriguingly, the 1962 Boardman and Bandura exchange foreshadowed some of the basic questions faced every day by clinicians, only some of which have been resolved by research. Can we treat social aggression by reinforcing socially appropriate alternative responses or periods of inhibition? Must we actively teach alternative responses to replace coercion, or might such skills already reside within the youth's repertoire? How might we alter natural contingencies among siblings and peers to support alternatives to social aggression, given the effort and nonreinforcement that often accompanies "social skills"? If discipline is necessary to block reinforcers for aggression, how do we advise parents to manage the child's aversive reaction to the recommended discipline routine?

Neither Williams nor Boardman articulated the case for parent training. The case studies, single-case designs, and uncontrolled group studies that followed in the next decade, however, did offer explicit rationales for using "mothers as behavior therapists" (Wahler, Winkel, Peterson, & Morrison, 1965, p. 113) to "reprogram the social environment" (Patterson, McNeal, Hawkins, & Phelps, 1967, p. 181).

Review articles of published studies from the 1960s and early 1970s summarized the many reasons advanced in these vanguard publications for abandoning individual psychotherapy with socially aggressive youth and replacing it with the new therapeutic modality, parent training. Three good reviews from this period are available: Berkowitz and Graziano (1972), Johnson and Katz (1973), and O'Dell (1974). Each reviewer looked at the same set of studies a bit differently, but induced three rather similar themes.

First, all reviewers noted that traditional therapy (i.e., relationship-based, insight-oriented) occupies a minor amount of time in the life of a child (e.g., 1 hour a week) and is performed in an unusual situation (i.e., an office playroom with an unfamiliar adult). Even if successful, therapeutic gains might not generalize to home, school, or neighborhood settings where the social aggression was manifest, let alone persist in those natural settings. In contrast, parents interact almost continuously with the young child referred for treatment. Generalization of treatment effects to the home setting and across time could be directly programmed by providing parents with guidance on how to perform the therapy themselves at home. The parent would assume the primary therapy role, while the licensed clinician would assume a consulting role. Essentially, the first theme culled by reviewers of the period was one of quantitative advantage and setting relevance for parents over traditional therapists.

Second, Berkowitz and Graziano (1972) and O'Dell (1974), and to a lesser extent, Johnson and Katz (1973), inferred a theoretical rationale for parent training from the initial spate of publications. Because parents are intrinsically involved in the socialization of their own children, and because referred children present as socialization failures (i.e., significant aggression and defiance beyond the age of 4 years), parents with socially aggressive children are likely to be unwitting contributors. Typical parenting errors might include providing attention for misbehavior; succumbing to the relentless drone of crying, arguing, or sultry mood displays by providing previously denied objects or activities; allowing natural reinforcers for aggression to operate by ignoring sibling fighting; or modeling angry, defiant responses with one's adult partner. Whatever the parenting mistakes might be, the therapeutic implications of the new theoretical position were clear. A psychopathogenic home environment could overwhelm any therapeutic change a clinician might achieve in a clinic playroom, since the pattern of child management in the home had not been addressed and would continue to elicit and reinforce social aggression. The extreme case of the new theoretical perspective was that targeting social aggression without changing the behavior of parents would simply fail.

Third, all three reviews called for improved research methodology. Published studies at that time often failed to embrace basic research procedures needed for replication and to establish internal validity. These basics include pretreatment measures of child deviance, objective definitions of parenting skills and the techniques of teaching those skills, measurements of parent acquisition and use of therapeutic skills, measurements of child reactions to changed parent behavior, and adequate experimental controls. The early work clearly set the stage for the very productive science of the next 3 decades, which addressed methodological shortcomings, proposed theoretical models, and expanded the research questions.

Patterson, Wahler, and Hanf

Three early entries into this literature warrant special recognition: Gerald Patterson at the University of Oregon (and later the Oregon Social Learning Center, hereafter OSLC), Robert Wahler at the University of Tennessee, and Constance Hanf at the University of Oregon Medical Center. Each contributed substantially to measurement, treatment, and theory during a sensitive period in the development of parent training based on social learning. A new measurement strategy often leads the way in science, and Patterson, Hanf, and Wahler did just that. Each developed microbehavioral observational systems to record parent-child event sequences. Although Patterson and Wahler focused on home settings under general conditions (Reid, 1978; Wahler, House, & Stambaugh, 1976) and Hanf on clinic settings under specific simulated conditions (Hanf, 1968), each coding system could detect frequencies of child and parent behavior, as well as conditional probabilities that revealed the stimulus control and contingent relationships among the behavioral events. Verifiable, observational data were used to confirm the presence of child deviance, identify likely parenting mistakes, provide treatment outcome data, and contribute to theory development.

Patterson and his colleagues defined 29 behavioral codes that sampled an alternating sequence of child and family member interaction in a general home context. Wahler's observational system is similar. He defined 24 child and parent codes, but used an interval-sampling approach, rather than the event-sequence approach pioneered by Patterson. The Patterson and Wahler observational systems provided data that served as the foundation of theoretical advancements, including the measurement of coercive processes (Patterson, 1982; Patterson, Reid, & Dishion, 1992; Wahler & Dumas, 1987). In contrast, Hanf defined nine standard laboratory situations derived from parent interviews about the conditions associated with preschool children's misbehavior (e.g., a parent denies a child's request). Unlike Patterson and Wahler, Hanf's codes did not apply to all measurement situations, and only some codes operationalized an event sequence (e.g., child compliance given a parent's command). Nevertheless, Hanf's observational system directly influenced the coding systems constructed by Forehand (Forehand & McMahon, 1981, pp. 183–218) at the University of Georgia, Eyberg (Eyberg, Bessmer, Newcomb, Edwards, & Robinson, 1994) at the University of Florida, and Webster-Stratton at the University of Washington, who adopted Eyberg's Dyadic Parent-Child Interaction Coding System. By moving the field of child clinical psychology toward observational methods and away from a reliance on interviews, symptom checklists, and personality measures, Patterson, Wahler, and Hanf created the primary tool needed for the development of parent training as a legitimate alternative to traditional psychotherapy.

Hanf and Patterson also developed influential treatment protocols (Eyberg, 1992), based in part on the earlier work of Wahler. In 1965, Wahler and his colleagues trained two parents of oppositional children to use differential attention, and one parent to use differential attention and a room time-out, based on the time-out procedure first defined by Wolf, Risely, and Mees (1964). Wahler (1969a) subsequently published one of the most widely cited empirical demonstrations of

the combined use of social reinforcement and time-out by parents to reduce oppositional behavior. Concurrently, Patterson and his colleagues (Patterson, Reid, Jones, & Conger, 1975) were developing a general set of learning-based strategies to use with parents of socially aggressive 3- to 14-year-old youth. The OSLC introduces parents to social learning principles; specific misbehaviors and prosocial activities are then defined and tracked by the parent in the home; some combination of token reinforcement, token fines, and room time-outs are tailored to the individual child to modify behavior and replace the parenting errors presumed to maintain child misbehavior; the token economy is then extended to address classroom misbehavior as needed; finally, the token economy is gradually replaced by more typical contingencies (e.g., an allowance system). Hanf's (Hanf, 1969; Hanf & Kling, 1973) treatment protocol targeted a narrower developmental range, 2 to 7 years, and focused on noncompliance. It was not Hanf's (Hanf, 1968) intent to focus exclusively on noncompliance. Indeed, she attempted to construct clinic simulations to measure and treat the full range of parent-reported child misbehaviors for this age group. Unfortunately, her outcome measures were limited to noncompliance and parent-child play quality, to the exclusion of other social aggression targets like fighting, tantrums, and negative talk. Hanf's treatment protocol for noncompliance recommended that parents first improve the quality of play with their child in order to enhance the mother's "social desirability" (Hanf & Kling, 1973, p. 7). Parents are taught to discriminate and positively attend to the child's age-appropriate play, to inhibit instructions and questions, and to ignore minor misbehavior. Hanf's first treatment stage is commonly referred to as "the child's game" by the Forehand group (McMahon & Forehand, 2003, *Helping the Noncompliant Child* [HNC]) and "child-directed interaction" by the Eyberg group (Eyberg, 1988, parent-child interaction therapy [PCIT]). Hanf's second stage of therapy taught parents to issue direct commands, praise compliance efforts, warn the child given noncompliance, and use a chair time-out system for continued noncompliance. Hanf's second treatment stage is referred to as "the parent's game" in HNC projects and "parent-directed interaction" in the PCIT tradition. It would be very difficult to find a parent training publication with preadolescent externalizing disordered children that did not include some or all of the Wahler, Patterson, or Hanf treatment protocols. Arguably, Patterson, Wahler, and Hanf, in conjunction with their prominent students (e.g., Chamberlain, Dumas, Eyberg, Forehand, and Webster-Stratton), and the students of their students, have produced much of the knowledge on social learning–based parent training reported in this chapter. Joining these direct descendants is Matthew Sanders (1999) of the University of Queensland in Australia and his many students reporting on the Triple P-Positive Parenting Program.

Discoveries and Reactions

Parent training has been reviewed many times since the early 1970s in peer-reviewed journals, book chapters, and monographs. Journal reviews focusing exclusively on social learning–based parent training are instructive of the maturation of the field across the past 3 decades. The scholarly reader is directed to the work

chronologically: Moreland, Schwebel, Beck, and Wells (1982), Griest and Wells (1983), Miller and Prinz (1990), Graziano and Diament (1992), Taylor and Biglan (1998), Brestan and Eyberg (1998), and Nixon (2002). In some cases, parent training reviews are embedded within the larger topic of treatments for antisocial children (Dumas, 1989; Estrada & Pinsof, 1995; Kazdin, 1987, 1997; Kazdin & Weisz, 1998). Journal reviews have also included more focused aspects of the parent training literature, such as noncompliance (Houlihan, Sloane, Jones, & Patten, 1992), father involvement in parent training (Coplin & Houts, 1991), and applications with Attention-Deficit/Hyperactivity Disorder youth (Chronis, Chacko, Fabiano, Wymbs, & Pelham, 2004). Noteworthy book chapters include Forehand's (1977) initial review of the noncompliance literature; O'Dell's (1985) efforts to clarify the different methods of teaching skills to parents; theoretical contributions by Eyberg (Foote, Eyberg, & Schuhmann, 1988), Wahler and Dumas (1987), and Patterson, Dishion, and Chamberlain (1993); and a relatively recent comprehensive parent training chapter by McMahon (1999). Monographs have also contributed to organizing the field, from those presenting PCIT and HNC traditions to clinicians (Hembree-Kigin & McNeil, 1995; McMahon & Forehand, 2003) to more general parenting applications (Briesmeister & Schaefer, 1998; Dangel & Polster, 1984; Mash, Handy, & Hamerlynck, 1976). The OSLC contributions are presented in the five-volume series, *A Social Learning Approach to Family Intervention* (Patterson, 1982; Patterson, Reid, & Dishion, 1992; Patterson et al., 1975; Reid, 1978), which spans 1975 to 1994, plus a more recent collection of invited chapters in Reid, Patterson, and Snyder (2002).

Consolidating the parent training literature to capture its essential themes and historical transitions is a daunting task. McMahon (1999) conceptualized three stages. The first stage witnessed the development of parent training as a viable model of intervention for socially aggressive children, which corresponds nicely to this chapter's previous discussion of the "Early Years: 1959 to 1973." McMahon's second stage was stimulated by Forehand and Atkeson's (1977) clarion call for research to evaluate and promote generalization of parent training effects from clinic to other relevant settings (e.g., home, community), to untreated siblings, to untreated child misbehaviors, and across time. Part of the promise of parent training, relative to individual psychotherapy, was exactly these generalizations. It was hoped that once the parent understood social learning principles and had gained some degree of reinforcement (and associated self-confidence) for the successful application of new skills, the parent would continue to use effective child management skills in the home (promoting generalization from clinic to home and across time) and to apply principles and specific skills to new problems (generalization across responses) and with other family members (sibling generalization). McMahon's third stage emphasized the development and empirical evaluation of methods to enhance and expand basic prescriptive parent training to address newly identified child problems (e.g., cognitive distortions), relevant contexts (e.g., unsupervised postschool periods), and barriers to successful treatment (e.g., maternal depression, marital discord). See the review on enhancements to parent training by Miller and Prinz (1990) and the earlier call to reformulate the field to "behavioral family therapy" (Griest & Wells, 1983).

A complementary model to McMahon's (1999) three stages is offered next. Six distinct and unending waves of research can be conceptualized, each a reaction to discoveries made at different points in time, and each wave a contributor to progress in the others.

Wave I

The initial spate of reviews (e.g., Berkowitz & Graziano, 1972) challenged the field to empirically test the effectiveness of parent training approaches to externalizing disorders. Clear progress has been made. Social learning theory–based parent training now meets APA's Division 12 criteria as an "empirically supported treatment" (Brestan & Eyberg, 1998; Chambless & Ollendick, 2001; Kazdin & Weisz, 1998). Meta-analytic reviews of parent training effects, contrasted primarily with wait-list control conditions, have reported average effect sizes of .86 (Serketich & Dumas, 1996) and .56 (Weisz, Weiss, Han, Granger, & Morton, 1995). Barlow and Stewart-Brown (2000) reported effect sizes from .20 to .68 for observational data following group-based parenting programs (e.g., Webster-Stratton, 1981a). Far less is known about how well social learning–based parent training compares to alternative family-based treatments, such as Parent Effectiveness Training (Gordon, 1970), family systems theory (Minuchin, 1974), or Systematic Training for Effective Parenting (Dinkmeyer & McKay, 1973, 1976). These counseling approaches to parent training were positively reviewed during the 1970s (Reisinger, Ora, & Frangia, 1976; Rinn & Markle, 1977; Tavormina, 1974, 1975), although more recent social learning-oriented reviewers have found little empirical support for such models (McMahon, 1999; Miller & Prinz, 1990). A meta-analytic review of Parent Effectiveness Training (Cedar & Levant, 1990), however, arrived at a different conclusion. Treatment comparisons appear to be hampered by disagreements over participant inclusion criteria and outcome measures. When overt misbehaviors are measured (Patterson, Chamberlain, & Reid, 1982; Wells & Egan, 1988), social learning programs have usually outperformed the alternative models, but not always (Bernal, Klinnert, & Schultz, 1980). A second ongoing task of Wave I research is dismantling the major treatment programs to determine the active components to successful treatment (e.g., Roberts, 1985).

Wave II

Generality of treatment effects constitutes the second ongoing research domain, corresponding to McMahon's second stage. Much knowledge has been gained (e.g., Sanders & James, 1983) since Forehand and Atkeson (1977) articulated the issues. Generalization of parenting skills from clinic to home settings has been documented multiple times by all major research programs (HNC: e.g., Peed, Roberts, & Forehand, 1977; PCIT: e.g., McNeil, Eyberg, Eisenstadt, Newcomb, & Funderburk, 1991; the group discussion videotape modeling [GDVM] protocol developed by Webster-Stratton: e.g., Webster-Stratton, Kolpacoff, & Hollinsworth, 1988; Triple P: e.g., Sanders & Christensen, 1985; and OSLC: e.g., Patterson, 1974). It is reasonable that home data would confirm setting generalization effects, since clinic work is designed explicitly to reprogram the home environment, giving parents new

methods to manage child misbehavior. Generalization to school settings, however, has quite understandably yielded mixed results, since the teacher is not involved in the parent training, nor are relevant classroom conditions targeted by parent trainers (e.g., independent seat work, peer conflict on the playground). As early as 1969 Wahler (1969b) demonstrated independence of the home and school settings, which necessitated specific interventions designed for the school. Similarly, HNC researchers have not found generality to school settings, despite positive home setting outcomes (Breiner & Forehand, 1981). In contrast, PCIT scientists, using a very similar Hanf-based protocol, have found positive generalizations to untreated school settings (McNeil et al., 1991). Durability of treatment effects has also been repeatedly demonstrated with young noncompliant children (McMahon, Wells, & Kotler, 2006, pp. 162–172) for periods up to 14 years (Long, Forehand, Wierson, & Morgan, 1994). Wahler's (1980) data, however, documents the clear failure to maintain treatment gains over even brief periods when treatment samples include impoverished or isolated mothers. Some evidence of generality to untreated siblings has been detected by various groups (e.g., Eyberg, Boggs, & Algina, 1995; Humphreys, Forehand, McMahon, & Roberts, 1978), which is reasonable, given shared conditions and treatment procedures when both children are in the same developmental period. In contrast, response generalization seems less likely, since untargeted child misbehaviors are likely to involve different psychosocial conditions (e.g., a sibling conflict versus a parental instruction), different functions (e.g., attention seeking versus task avoidance), and different topographies (e.g., hit versus argue). Nevertheless, it is possible that response classes exist to the degree that conditions and functions overlap, despite differences in response topography (Wahler, 1975). Unfortunately, such overlaps are rare (Sanders & James, 1983), although a few have been reported (Wells, Forehand, & Griest, 1980). It appears that parent trainers are usually required to program change across a child's set of presenting problems, rather than focusing on a narrow band (e.g., noncompliance).

Wave III

Techniques for training parents effectively and efficiently have drawn the attention of many researchers, especially O'Dell (1985) at the University of Mississippi and Webster-Stratton (e.g., 1994) at the University of Washington. Given the centrality of parent training techniques to this chapter, research representative of Wave III is reviewed later in some depth.

Wave IV

Treatment failures and dropout data led Wahler (1980) and his colleagues to study the adverse impact of social disadvantage and social isolation on parenting practices. Specifically, inattention, inadequate monitoring, and indiscriminate responding characterize the child management patterns of poor, uneducated, socially isolated women. Wahler and Dumas (1987, 1989) hypothesized that distal setting events (e.g., an aversive interchange with a relative) disrupt parent attention, monitoring, and recognition of the stimulus events actually associated with coercive child behavior, especially for disadvantaged parents. Subsequently, Wahler, Cartor,

Fleischman, and Lambert (1993) developed and evaluated synthesis training, a supplement to basic parent training that has the look and feel of traditional insight-based psychotherapy, centered on each mother's child care experiences and associated extrafamilial events. Synthesis training combined with basic parent training was found to improve child behavior and reduce indiscriminate parenting, whereas basic parent training alone was ineffective with these socially disadvantaged and isolated parents. Teaching complex parenting skills to multistressed, uneducated parents remains a critical and unsolved problem in the literature.

Wave V

Parent pathology has been found to increase risk for inadequate parenting and poor responses to parent training interventions (Griest & Wells, 1983, pp. 42–46; McMahon, 1999, pp. 168–170; McMahon et al., 2006, pp. 149–152; Miller & Prinz, 1990, pp. 294–295). Specifically, parental depression has been associated with attenuated parental monitoring and inaccurate judgments of child behavior, poorer treatment outcomes, and dropout. Similarly, marital distress may disrupt the parent training process. Meta-analyses performed by Erel and Burman (1995) and by Krishnakumar and Buehler (2000) reported significant average correlations between marital distress and child misbehavior. Although distressed couples may profit from parent training to treat externalizing disorders (Forehand, Griest, Wells, & McMahon, 1982), follow-up data reported by Dadds, Schwartz, and Sanders (1987) indicated that marital distress adversely impacts treatment durability. This fifth research wave implies that interventions addressing parental depression and marital discord should be screened for, and treated if necessary, to render parent training effective. Fortunately, empirically supported treatments exist for both adult depression and marital distress (Chambless & Ollendick, 2001). In addition to depression and marital discord, the OSLC group (e.g., Patterson et al., 1992, pp. 102–112) and developmental psychopathologists (e.g., Frick & Loney, 2002) have repeatedly noted the empirical association between parents' antisocial behavior and child externalizing disorders. It is not difficult to understand the adverse effects of an antisocial parent who models angry, aggressive reactions to normal frustrators and interacts coercively with both society and the developing child. Moreover, antisocial adults may be court-ordered to treatment, bringing little motivation to accommodate to therapist recommendations. Parental substance abuse can also disrupt parenting and appears to be exacerbated by disruptive children, creating an event chain rendering both parent substance abuse and child misbehavior more likely (Pelham & Lang, 1993). Unlike parental depression and marital distress, there are no robust outpatient treatment protocols for Antisocial Personality Disorder or substance abuse, leaving the parent trainer with substantive challenges. For the reader interested in the extensive work on parent pathology and child developmental problems, see the recent volume edited by McMahon and Peters (2002).

Wave VI

Variables that promote or degrade parent engagement in the parent training process are now recognized as quite important. See discussions by Miller and Prinz

1990, pp. 299–300), McMahon (1999, pp. 163–164), and Patterson et al. (1993, pp. 65–67) regarding the relatively small and emerging literature on the therapeutic alliance. Although the child's misbehavior is the precipitating factor for the clinic referral, the parent is the mechanism of change, and hence the individual with whom a therapeutic alliance is central. Chamberlain, Patterson, Reid, Kavanagh, and Forgatch (1984) found that high rates of parent refusal statements ("I can't" and "I won't") during the first two treatment sessions are linked to dropout and are associated with maternal depression, paternal social disadvantage, and high base rates of antisocial child behavior. Erstwhile parent trainers inherently place many demands on parents in sessions (e.g., to role-play a new skill) and between sessions (e.g., to use time-out for noncompliance), some of which are very difficult as a result of complexity, child resistance, and/or associated parent emotionality. Parents may have even attempted some variation of a therapist-recommended routine at some point in the past and met with blatant failure. Consequently, many parents argue with the therapist and/or passively avoid therapeutic activities between sessions, to the great consternation of therapists. The OSLC group has consistently called for therapist support and supervision (e.g., Patterson, 1985) to mitigate the potentially adverse effects of nonresponsive parents on therapists.

Balancing traditional supportive therapeutic processes (e.g., active listening skills, reframing, empathic communications, and genuine regard for family members) with prescriptive parent training is an important goal for all therapists. Balancing, however, does not imply abrogating one's professional responsibility to confront difficult problems for the sake of parent comfort, even if that confrontation requires complex and demanding interventions. Flexibility is recommended. Tailoring interventions to accommodate the idiosyncratic issues of a specific family is always appropriate, given that the recommended procedures are theoretically reasonable and the outcome can be monitored. For example, a therapist might opt for a relatively slow shaping process with an unusual but salient positive reinforcement menu, despite an available, efficient, and socially valid standard treatment strategy. Clearly, much is unknown about the dialectical process between prescriptive therapies like parent training and needed therapeutic alliance building.

THEORETICAL FOUNDATIONS OF PARENT TRAINING

Operant Conditioning

The continuing strength of parent training protocols rests on operant principles. Early clinical interventions were attempts to apply learning principles to the treatment of child misbehavior. Patterson (1982, p. 2) views the seminal publications by Skinner (1953), Bandura and Walters (1963), and Ullmann and Krasner's (1965) edited collection of *Case Studies in Behavior Modification* as setting the stage for applied behavior analysis with parents in a therapist role. Early efforts to positively reinforce prosocial child behavior with parent attention or token reinforcers, to extinguish minor misbehaviors via parent ignoring, and to punish aggression and noncompliance with time-outs, fines, or privilege losses are all operant techniques. After a decade of data collection and clinical trials, the OSLC group articulated a

series of conceptual breakthroughs. First, they defined socially aggressive behavior as aversive and contingent (Patterson, 1982, p. 12). An aggressive act is aversive if interacting agents (parents, teachers, siblings, or peers [PTSP]) report it as unpleasant and work to actively escape from it or avoid it. An aggressive act is contingent if its conditional probability is greater that its base rate probability (i.e., demonstrating it is not displayed at random). The OSLC data convinced most scientists that socially aggressive child behavior could coerce interacting agents (PTSP) into changing their own behavior in exchange for temporary cessation of the child's aggression. A "reinforcement trap" (Patterson, 1982, p. 145) was hypothesized to exist, in which the child coerces positive or negative reinforcement from PTSP with aggressive behavior, while PTSP is negatively reinforced by aggression termination. Patterson's widely accepted coercion theory is, therefore, based on rather simple operant processes, but its implications for the psychopathology of childhood social aggression are profound:

1. Aggression is likely to be stable across time, since the child and PTSP are both reinforced, and resistant to extinction, since reinforcement schedules are likely to be intermittent.
2. The child and/or PTSP will escalate to even greater intensities of aggression if either party withholds expected reinforcers (i.e., an extinction burst).

No mechanisms beyond positive reinforcement, negative reinforcement, schedules of reinforcement, and extinction are necessary to appreciate implications 1 and 2. Treatment implications of coercion theory are equally important:

3. Parents and teachers must block reinforcers coerced by social aggression; treatments that do not do so will likely fail.
4. Because most siblings and peers are not sufficiently mature to assume any responsibility to change on behalf of the target child, therapists must teach parents and teachers to guide and protect siblings and peers if the child's aggression is to be successfully treated; otherwise, siblings and peers will continue to unwittingly reinforce child aggression.
5. Parental ignoring will be an ineffectual treatment for child coercion, which is negatively reinforced by parent withdrawal of events aversive to the child (e.g., instructions, denials, discipline efforts), since ignoring allows the negative reinforcement process to operate.
6. Implications 1 through 5 render the successful treatment of socially aggressive children very difficult, especially for middle childhood youth who have experienced thousands of reinforced coercive trials over the course of many years and who can violently oppose parent and teacher efforts to adjust.

Important assessment and case conceptualization strategies are also implicated by coercion theory. Assessments should identify the idiosyncratic set of stimulus events that discriminate the occasion for coercion, as well as guide hypotheses about the parenting errors that are likely to reinforce coercion. The specific stimulus

events (e.g., parent "Stop X" instructions) and settings in which they occur (e.g., bedtime, unsupervised sibling play) must be known to construct a comprehensive yet family-specific treatment plan. Because aggressive behavior is partly defined by its contingent quality, coercion theory implies there will be a set of 1 though K events, presented by PTSP, each of which are discriminative for coerced reinforcement. Figure 22.1 is offered as a heuristic device for the practitioner. Common interpersonal events that elicit social aggression are included in the first two columns of the figure. Any given referred youth will display some subset of the many possible event sequences displayed in the figure. Knowing the relevant operant sequences, however, is not an adequate assessment. See the recent chapter by McMahon and Frick (2005) for a full discussion of evidence-based assessment procedures for externalizing disordered youth.

Figure 22.1 requires elaboration. It is based entirely on coercion theory as presented by Patterson and his many colleagues (Patterson, 1982; Patterson et al., 1992, 1993) and Hanf's (1968) identification of common situations provoking child misbehavior. The figure attempts to characterize typical event sequences consisting of

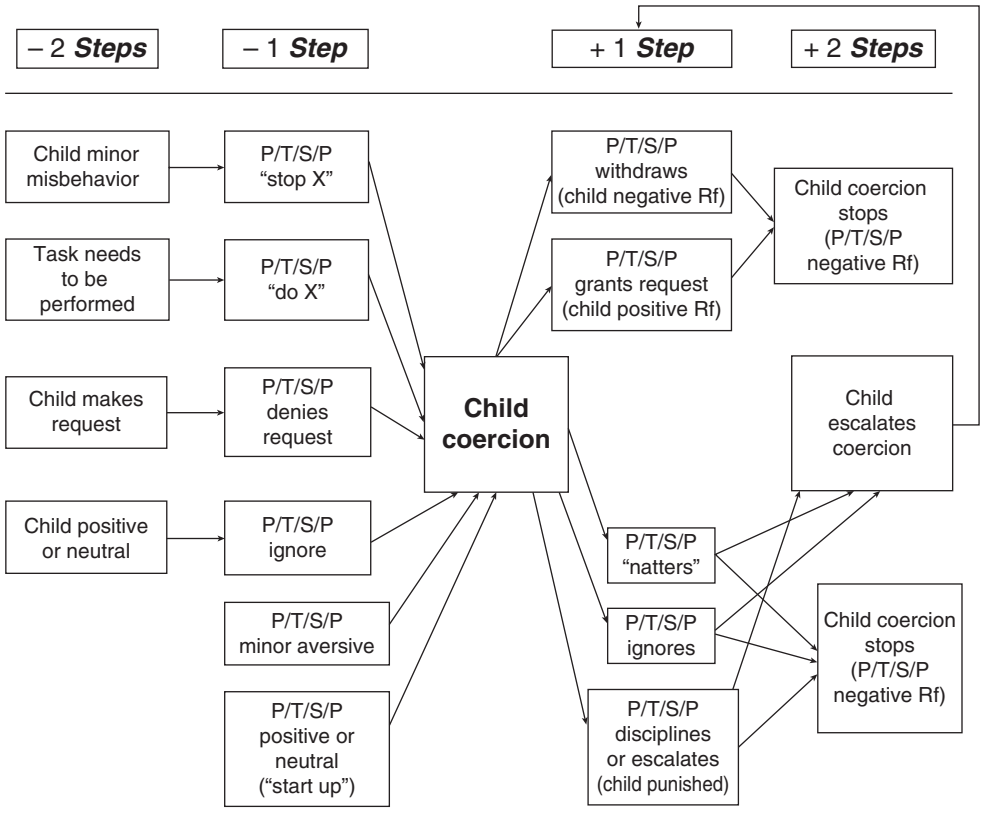

Figure 22.1 Coercive event sequences based on the work of Patterson, other OSLC scientists, and Hanf, designed as a heuristic device for clinicians to conceptualize the specific clinical presentations of referred families. P/T/S/P = Parent/Teacher/Sibling/Peer; Rf = Reinforcement.

six antecedent events ("–1 Step") and five consequent events ("+1 Step"), anchored in the middle by child coercion. Child coercion is defined as overt, contingent, and aversive behavior that is most commonly expressed by the subjects of this chapter as fight, noncomply, negative verbalization, and/or tantrum. Up to two behavioral events that precede coercion ("−2 Steps" & "−1 Step") and two that follow coercion ("+1 Step" & "+2 Steps") are needed to conceptualize the operant mechanisms. At "−2 Steps" an identifiable behavior or condition is present (e.g., child teases his sibling) that increases the likelihood of the "−1 Step" event (e.g., parent instructs child to inhibit teasing). Note that a "−2 Steps" condition has not been identified for all "−1 Step" events. For example, a host of different events probably precede "P/T/S/P minor aversive" (e.g., sibling teases the target child) that are difficult to characterize. Similarly, what precedes the positive or neutral behavior of family members that sometimes evokes child coercion (known as "Start Up"; Patterson et al., 1992, pp. 47–54) is currently unknown. Child coercion is likely to follow "−1 Step" events, especially in clinic-referred youth. At "+1 Step" the child may be reinforced, ignored, or punished for coercion in an indiscriminant manner, resulting in the hypothesized intermittent reinforcement schedule.

The concept of "nattering," introduced by John Reid (Patterson, 1982, p. 69), may be unfamiliar to some and requires discussion. Nattering is interpreted as a hostile, verbal outburst by PTSP (e.g., threatening gestures, angry tone of voice, criticisms, stop instructions, empty threats) that fails to discriminate subsequent discipline but may provide positive reinforcement in the form of contingent attention and certainly provides a hostile verbal model. Figure 22.1 supposes that on some occasions, nattering is followed by child cessation ("+2 Steps"), negatively reinforcing the natterer, but more often than not, nattering simply prolongs the aversive interchange and sets the occasion for escalated intensities of social aggression (Patterson, 1982, pp. 67–84; Patterson et al., 1992, pp. 39–54). The model hypothesized in Figure 22.1 also captures the paradoxical concept of punishment acceleration (Patterson, 1976), in that referred children are much more likely than normal controls to react to discipline efforts ("+1 Step") with escalated intensities of coercion ("+2 Steps"), throwing PTSP back to "+1 Step" options, where the PTSP may finally relent and provide reinforcement, attempt to ignore, or continue the struggle by escalating the intensity of punishment or by nattering.

Patterson's (1976) concept of the aggressive child as both "victim and architect" of his or her own environment is germane within the operant framework. The model provided in Figure 22.1 implicates both PTSP and the child. Certainly the operant model implies that PTSP lack skills. These important socialization agents are reinforcing social aggression on an indiscriminant, intermittent basis. To that extent, the child is a "victim" of dysfunctional child management. On the other hand, the operant model exposes the child's unintended role as "architect" of the very environment that sustains his or her psychopathology. First, aggressive child actions coerce their own reinforcement from the many unwitting interactive agents who seek short-term relief from the child's aversiveness via misguided child management strategies. Second, to avoid the aversive child coercion altogether, PTSP are likely to inhibit presenting "–1 Step" events, thus abrogating their normal socialization roles.

Social Learning Theory

Most parent trainers ascribe to the theoretical framework provided by social learning theory (Bandura, 1977), rather than relying on the more narrow principles of operant learning. Indeed, Patterson's published series is entitled *A Social Learning Approach to Family Intervention*. Social learning theory embraces the full range of biopsychosocial variables that interact continuously during the child's development. Consequently, social learning theory provides a better theoretical home for evidence-based parent trainers than the more limited operant model. For example, it is presumed that biological-based temperaments impact parenting (e.g., Wahler & Dumas, 1987), that models of antisocial behavior promote coercion (Frick & Loney, 2002), that modeling and discussing child management skills can result in parent knowledge and performance acquisitions (e.g., Webster-Stratton, 1994), that social-cognitive processes influence overt social behavior in middle childhood (e.g., Dodge, 1980), and that a family's cultural background is an important consideration in parent training (Forehand & Kotchick, 1996). Each of these biopsychosocial processes are well beyond the confines of operant mechanisms.

Developmental psychopathology is consistent with social learning theory. The relevant examples of this are the child-onset (i.e., "early starter") Conduct Disorder model, nicely summarized in McMahon et al. (2006, pp. 147–152), the Patterson stage model (e.g., Patterson et al., 1992, pp. 115–141), and Frick's (2004) recent review of the developmental psychopathology of severe antisocial behavior. Child-onset Conduct Disorder models are based on the continuous interaction of biopsychosocial variables from a developmental perspective across the changing social contexts of the child's cultural group from infancy to adolescence. Of particular relevance to clinicians are two classes of findings from the child-onset developmental model. First, inept parental discipline and inadequate parental monitoring have been repeatedly found to increase the risk of social aggression (Patterson et al., 1993). Consequently, parent trainers must invariably evaluate and improve parent discipline and monitoring. Second, social (e.g., Dodge, 1980) and academic (e.g., Hinshaw, 1992) skill deficits are very likely to emerge in externalizing disordered youth by middle childhood (Patterson et al., 1992, pp. 120–131). Consequently, parent trainers should assess sibling and peer skills and classroom function and, if needed, intervene or refer to address the identified deficits.

The empirical and theoretical work of Wahler and Dumas with impoverished, socially isolated mothers deserves special comment here. It is consistent with the cognitive elements of social learning theory (Wahler & Dumas, 1987, p. 616) and adds to our theoretical understanding of seemingly intractable dysfunctional parenting. Specifically, Wahler and Dumas (1989) argue that multistressed parents (e.g., impoverished, socially isolated, depressed, abusive) operate with trait-like response classes, such that setting events irrelevant to parent-child interaction adversely impact temporally distal parent monitoring, judgments, and responses to children. Coercion theory cannot explain the apparent overgeneralization of parent responding from a child-irrelevant context (e.g., friend, employer, spouse) to child encounters. Moreover, this parental failure to attend to the differences in social context and respond adaptively may be a significant cognitive barrier to parent

training with multistressed parents. All social learning theory–based parent training programs attempt to teach parents to discriminate specific child actions, track those actions in the home, and respond in the prescribed manner. In opposition to these well-intended recommendations, multistressed parents appear to operate with macro-, trait-like cognitive strategies that link seemingly disparate social encounters. It is currently unknown if synthesis training, a very cognitive, insight-based therapeutic process, will reduce this barrier to treatment success.

Attachment Theory

Eyberg and her colleagues (Foote et al., 1998) have oriented the child clinical field to the special importance of parental sensitivity to child signals. Parental sensitivity and the establishment of synchronized turn taking have long been found to facilitate secure attachments and promote general socialization (Maccoby & Martin, 1983, pp. 26–37). Developmental psychologists have repeatedly found specific positive associations between mother-child reciprocity and successful socialization (e.g., Kochanska, 1997; Stayton, Hogan, & Ainsworth, 1971). In contrast, parents of aggressive children are likely to be indiscriminate and aversive (e.g., Dumas & Wahler, 1985) in comparison to normal control parents. Eyberg's child-directed interaction (in Hembree-Kigin & McNeil, 1995) and McMahon and Forehand's (2003) child's game are treatment components in noncompliance programs that actively teach the parent to be sensitive to age-appropriate child signals during free-play conditions. Eyberg argues that by enhancing the responsiveness of parents to appropriate child signals, therapists will strengthen the parent-child relationship in a manner analogous to the development of secure attachments in normal infants. Whaler and Dumas would support this position, since child-directed interaction skills teach the parent to discriminate and attend to age-appropriate child behavior. Even operant devotees would applaud the parental attention to "good" behavior found in child-directed interaction. Because teaching parent responsiveness is a fairly widespread clinical practice, the theoretical linkage to normal child development is important. Indeed, child-directed interaction looks like the nurturance component of Baumrind's (1967) classic conceptualization of the successful authoritative parent, who combines firm control with nurturance and communication skills. A second intriguing linkage is to traditional humanistic-based play therapy (Axline, 1969), with the major procedural exception, of course, that parents are trained to perform the therapist role.

Summary of Theoretical Perspectives

Parent trainers earn their daily living with operant strategies. Coercive processes must be minimized if the child is to gain a normal socialization path. The much broader shoulders of social learning theory, however, are embraced to consider the wider range of biopsychosocial variables needed to assess, conceptualize, and treat externalizing disorders. Finally, attachment theory, stripped of its psychodynamic overtones, provides a basis for the common goal of strengthening the parent-child relationship.

TECHNIQUES FOR TRAINING PARENTS

Training techniques can be dimensionalized in terms of therapist time and degree of contact. At the least intrusive, least expensive level, therapists can impart skills didactically via discussion, handouts, and programmed reading materials, all of which can be used in group formats to add even more efficiency. Modeling is usually conceptualized as more intrusive than didactic approaches, especially in vivo modeling, since it requires an actor in addition to the therapist. Videotaped modeling, however, does not require an actor and is very compatible with group training (e.g., Webster-Stratton, 1994). Because discussion, written materials, and modeling are passively received by parents, the clinician remains uncertain if acquisition has occurred. In contrast, role-playing and guided practice are more intrusive, active, and informative strategies, since the parent must perform overtly. In role-play formats the parent responds to an actor (or therapist) simulating child behavior. During guided practice the therapist provides immediate feedback to the parent while the parent interacts with his or her own child. Guided practice provides direct evidence of both parent acquisition and child reaction to the recommended protocol. Finally, techniques to prompt and motivate home use of newly acquired skills include specific homework assignments (accompanied by written materials for reference), visually posted behavioral tracking systems, and phone contacts.

Psychologists tailor the set of training techniques to the child's presenting problems, developmental level, social-cultural circumstance, and treatment protocol. For example, introducing a home token system for use with aggressive grade-school siblings would likely require discussion, written materials, a home tracking system, modeling, and role-playing. In addition to the parent, both siblings would be actively involved in all facets of the training to gain their input regarding the reinforcement menu, as well as their awareness of the new rules, the reasons behind the rules, and all the changed contingencies. In contrast, introducing differential attention skills to the parents of a defiant 3-year-old would emphasize guided practice using a bug-in-the-ear during free-play conditions. Moreover, the 3-year-old would not be involved in discussion with the therapist about goals, rationales, and choices.

Psychologists also adjust their training techniques, based on parent performance. For example, role-playing might reveal a parent tendency to repeat instructions quickly, despite verbal feedback. Consequently, the therapist might revert to several modeling trials before attempting a second try at the more demanding role-play routine. Conversely, efficiencies can be obtained by sampling a parent's performance following minimal training, for example, brief discussion. If the advice is specific and meaningful, a parent might adjust his or her behavior immediately and with little difficulty (e.g., "Ignore whichever child is crying and play with the other sibling"). The data will inform the clinician if this 3-second intervention worked!

A brief and representative review of empirical findings is offered.

Didactic Strategies

Simple advice giving in the absence of other techniques has been described (Boardman, 1962; Holland, 1969), but not studied with adequate experimental control.

Obviously, verbal instructions are an inherent component of all other training strategies (i.e., "Do X under Y conditions for Z reasons"). The therapist's interpersonal skill in communicating these instructions is likely an important component of engaging (or failing to engage) the parent in the therapeutic process, represented by Wave VI of ongoing research in parent training discussed earlier.

Written materials can be a sufficient intervention under limited circumstances. Clark et al. (1977) used a multiple-baseline design with observational data to test the effects of a written manual of advice for parents on how to shop successfully with preadolescent children. Procedures included providing information about shopping, differential attention, token reinforcement for rule following, and token loss for misbehavior. Child misbehavior and ineffective parenting declined as a function of reading the manual (Green, Clark, & Risley, 1977). Similarly, McMahon and Forehand (1978) used a brochure to instruct parents on a mealtime routine, consisting of differential attention, instruction giving, and brief room time-outs. Again, a multiple-baseline design indicated that children reduced their overt misbehavior and parents displayed new skills. Sloane, Endo, Hawkes, and Jenson (1991) found that written self-instructional materials alone helped most parents provoke some improvement in their child's compliance.

Together, these three multiple-baseline projects reveal the potential contribution of a well-written manual or handout for changing overt parent behavior and subsequent child adjustment. Each project was limited, however, by the use of recruited subjects. Connell, Sanders, and Markie-Dadds (1997) evaluated the effects of written materials on parents of oppositional children relative to a wait-list control group. The positive results, however, were limited to improved questionnaire data. When parents have been randomly assigned to reading-only conditions and compared to those experiencing more active training techniques, reading-only is usually less effective on performance measures (Eyberg & Matarazzo, 1980; Flanagan, Adams, & Forehand, 1979; Nay, 1975; O'Dell et al., 1982), but can be equivalent (O'Dell, Krug, Patterson, & Faustman, 1980; O'Dell, Mahoney, Horton, & Turner, 1979). See McMahon and Forehand (1980) for an early review of self-instructional studies and McMahon and Forehand (1981) for a discussion of the many parameters (e.g., parent reading level) that should be considered when using or researching written instruction techniques.

Modeling

Modeling specific skills has a long history (Mash & Terdal, 1973) and a sound theoretical basis in social learning theory (Bandura, 1977). Several years ago Webster-Stratton (1981b) developed a training protocol in which videotape modeling is the primary technique. Parents in small groups (8 or 9) view vignettes of parents displaying specific skills with children in a variety of normal social contexts. After viewing each vignette (about 2 minutes), a therapist prompts discussion of the critical events, helps the group discuss how to apply the skill at home, and assigns homework. A programmatic series of publications has largely supported the efficacy of Webster-Stratton's GDVM protocol. Performance measures of parental responsiveness during play contexts improved relative to wait-list controls

posttreatment (Webster-Stratton, 1981a) and were maintained at 1-year follow-up (Webster-Stratton, 1982). Early conclusions were attenuated, however, by the use of recruited subjects and nonspecific play observation conditions in a clinic (i.e., a condition not likely to provoke coercion). More recently, the GDVM protocol was found to be as effective as an individually administered videotape (IVM) modeling procedure and a group discussion protocol, relative to a wait-list condition (Webster-Stratton et al., 1988). Importantly, participants in these later projects were referred children who met a questionnaire criterion for deviance, and observational data were collected in the home. Consumer satisfaction with GDVM and group discussion protocols has been rated higher than for IVM, suggesting the importance of the therapist contribution (Webster-Stratton, 1989); moreover, parents found that GDVM led to skills that were easier to implement at 1-year follow-up than either IVM or group discussion alone. Therapist consultation appears to improve the effectiveness of IVM (Webster-Stratton, 1990). Because highly stressed families (cf. Wahler & Dumas, 1987) were less likely to profit from IVM (Webster-Stratton, 1992), GDVM can be augmented by parent communication, problem solving, and self-control components (Webster-Stratton, 1994), a nice example of McMahon's Stage 3 in parent training research. The enhanced program yielded increased benefits in consumer satisfaction, child pro-social knowledge, and problem-solving and communication skills not targeted by GDVM.

Recently, Foster and Roberts (2007) replicated the effectiveness of videotape modeling combined with didactics to improve parent skills in both play and clinic task conditions, but also replicated the need for therapist monitoring of initial parent use of time-out (Roberts & Powers, 1990). In addition, technique comparison studies have generally favored more active techniques, which include modeling, over didactic strategies and wait-list on performance measures (Eyberg & Matarazzo, 1980; Flanagan et al., 1979; Nay, 1975), although O'Dell et al. (1982) found that written instructions compared well with videotape modeling and a more intrusive procedure, live modeling plus role-playing.

In summary, modeling, like written instructions, has been demonstrated to be an important technique for parent trainers. Modeling is invariably more effective than no training and competes well with alternative techniques. The effectiveness of Webster-Stratton's GDVM protocol is truly impressive. The opportunity to discuss the modeled event sequence with other parents under the watchful eye of a therapist appears to be an important component of GDVM. If videotapes are self-administered, a therapist consultant is recommended, especially to assist the parent in initial use of potentially emotion-provoking routines such as time-out.

Role-Playing and Guided Practice

Role-playing has three distinct advantages over all other techniques: flexibility, control, and the opportunity to present retrials immediately. A little imagination and a few props allow psychologists to tailor role-play analogues to family-specific problem conditions (e.g., bedtime, shopping, car trips, sibling aggression, toilet training). For example, the trainer might say, "Let's pretend this doll represents your child and these four chairs are the family car. When the doll starts fighting

with his pretend brother (another doll), you practice stopping the 'car' just like you saw me do it a moment ago." There is no limit to creative simulations. Second, the therapist has complete control over the doll, enabling task difficulty to be modulated to accommodate the parent's acquisition rate. Third, immediate retrials of exactly the same stimulus array can be presented to practice a skill or to correct an error. For example, given an error, the parent trainer can discuss the error, the targeted response, the reason for the targeted response, and then immediately present the task again. No other training technique allows this intensive, individualized practice. Written materials and videotape models are inflexible; guided practice puts the trainer at the mercy of the child, who often exhibits nongraduated task difficulties, will not pause for discussion, and may not give a second chance at the same misbehavior. Only in vivo modeling with an actor matches the flexibility and control of role-playing, but lacks the retrial component, since performance is not required. Consequently, parent trainers often provide live models followed by a role-play to teach specific event sequences.

Several experiments (Flanagan et al., 1979; Nay, 1975; O'Dell et al., 1982; Webster-Stratton, 1984) have found that live modeling plus role-playing outperformed didactics and no training on performance measures, but sometimes, equivalent acquisition rates are found (O'Dell et al., 1979, 1980). Powers and Roberts (1995) contrasted a standard training protocol, toy cleanup, with an augmented simulation procedure designed to enhance generalization. Parents who rehearsed skills to mastery via modeling and role-playing for wake-up/dressing, mealtime, and cleanup/bedtime conditions performed better during unstructured clinic and home conditions than parents trained solely under toy cleanup conditions, suggesting that multiple, content-valid exemplars should be incorporated into modeling and role-play regimens.

If the parent is provided prompts and feedback during interaction with his or her child, the training technique is considered "guided practice." A bug-in-the-ear device is often used to allow the therapist in an observation room to talk to a parent in an adjacent clinic room without the child's awareness. The "bug" is a small radio receiver worn by the parent like a hearing aid; the radio signal is transmitted locally to the clinic room via an antenna. Hanf (1969) introduced the procedure into the literature, which is now a staple of HNC and PCIT parent training programs that target 2- to 7-year-old disruptive and oppositional children. Note that guided practice loses its applicability in middle childhood, as youth become savvy to one-way windows and clinic simulations lack credibility. The training advantages to guided practice are obvious: immediate social reinforcement, corrections for errors of commission, and prompts for errors of omission. Therapist support via the bug can be gradually faded as the parent displays mastery, allowing the parent to practice with the child under more home-like conditions. Because the target child is providing the behavior display, programming for setting generality is maximized. Unlike all other training techniques, a psychologist using guided practice has the opportunity to observe the child's reaction to the recommended changes in parent behavior, which allows the psychologist to make procedural adjustments as needed. Because young defiant children are especially prone to react to discipline with coercion (Roberts & Powers, 1988), a phenomenon consistent with

theory and data (e.g., Patterson, 1982; Roberts, 1982b), it is particularly important to monitor initial time-outs with overtly defiant children (Roberts & Powers, 1990). Guided practice provides that mechanism. Only one thorough test of this position, however, is available. Webster-Stratton (1984) randomly assigned recruited parents to individual therapy, which made use of live modeling, role-playing, and guided practice with a "bug," to GDVM and a no-treatment condition. Both treatment groups performed equally well posttreatment and better than wait-list controls; no behavioral differences were present at 1-year follow-up. Neither treatment group, however, manifested a significant decrease in child noncompliance ratios, which is inconsistent with data from HNC and PCIT outcome studies. It is hypothesized that guided practice fills an important niche in the tools of the parent trainer, but no group comparison data currently support that contention.

Homework, Posted Tracking Systems, and Phone Contacts

To facilitate transfer of training from clinic to home settings, specific homework is routinely assigned (e.g., "Use the compliance program every time you give an instruction at home this week"). In addition, some form of record keeping system is provided to parents to monitor progress at home. These participant-observation devices are usually posted somewhere central (e.g., the refrigerator), thereby serving a prompting role in addition to their use as treatment outcome data. If a therapist anticipates parent difficulty in implementing a recommended child management routine (e.g., initial time-outs, a bedtime program), phone calls can be scheduled to provide support and guidance. Parents are invariably invited to contact the therapist during working hours if problems arise at home and can be provided after-hours emergency contact systems if needed.

Although homework, participant observation, and phone support are well integrated into parent training programs, only participant observation has been studied systematically. The parent daily report (PDR; Chamberlain & Reid, 1987) is the best example. Parents are contacted by phone and report on the occurrence or nonoccurrence of specific misbehaviors during the prior 24 hours. Test-retest reliability of PDR data has been adequate in clinic samples ($r = .60$) and excellent in recruited samples ($r = .82$); interparental agreement has been modest ($r = .56$), but given differential daily interaction with the child, would not be expected to be a sensitive index of accuracy. Importantly, PDR data have converged well with total aversive behavior rates collected by professional observers in three samples ($.46 < r < .69$). Moreover, normative data are available, indicating an average of 5.3 misbehaviors per day ($SD = 3.1$) in 4- to 10-year-olds. Unfortunately, the PDR system is more of a research tool than a practical clinical instrument at this point, and does not serve the prompting role of posted tracking systems. Most clinicians cannot make daily phone calls, even for brief periods, and most clinicians want daily frequency counts rather than binary data. Parent-collected daily frequency counts of targeted sibling aggression (Olson & Roberts, 1987) have yielded good evidence of internal consistency (i.e., the odd/even day correlation for sibling aggression was $r = .88$), but nothing is known about the accuracy or concurrent validity of such parent-collected frequency data.

CHILD MANAGEMENT SKILLS

The core curricula of all modern social learning–based parenting programs were established by the early work of Patterson, Wahler, and Hanf. Six generic parenting skills can be identified. First, parents are introduced to the social learning principles that underlie the acquisition and maintenance of externalizing disorders (see Figure 22.1) and the use of operant, cognitive, and developmental principles to treat the child's coercive patterns. Patterson and Gullion's (1968) *Living with Children,* a programmed textbook, served as the prototype for all future efforts (e.g., McMahon, Forehand, & Griest, 1981; Patterson, 1975) to help parents understand why their child misbehaves and how they could be part of the solution. Second, parents are taught to define, observe, and record the presence of targeted misbehaviors and prosocial accomplishments in the home setting. Participant observation provides more than assessment and treatment outcome data; it enhances the parent's awareness of child acts that will become discriminative for implementing therapist-recommended interventions and could help multistressed parents with the cognitive appraisal problems identified by Wahler and Dumas (discussed earlier). Third, parents are trained to attend to age-appropriate child behavior while ignoring coercive behavior maintained by parent attention. There are two distinct and different purposes behind differential attention skills. First, clinic-referred parents commonly provide attention to misbehavior in the form of reprimands, "Don't" instructions, and/or nattering in general (e.g., Patterson, 1976), which at worst positively reinforces coercion and at least models arguing. Teaching parents to withhold attention for specific misbehaviors can effectively extinguish some coercive acts, such as negative verbalizations (Herbert & Baer, 1972), tantrums during time-out administrations (Hobbs & Forehand, 1975), and defiant talk during compliance training (Roberts & Hatzenbuehler, 1981). Note, however, that differential attention can make some misbehaviors worse (Herbert et al., 1973). Moreover, differential attention does not play a substantive role in treating noncompliance (Roberts, 1985), since noncompliance appears to be reinforced by task avoidance rather than misguided parental attention. For an early discussion of differential attention, see Sajwaj and Dillon (1977), and for a modern procedural description of the effective use of differential attention, see McMahon and Forehand (2003, pp. 106–129). The second purpose of differential attention skills is to improve the parent-child relationship via responsiveness during play. Eyberg and her many students (e.g., Foote et al., (1998) emphasize parental responsiveness during play contexts as a mechanism to enhance the parent-child bond, as discussed earlier under attachment theory. Fourth, because noncompliance is a foundational problem in externalizing disorders (Loeber & Schmaling, 1985; Patterson et al., 1992), parenting programs invariably teach parents to use verbal instructions that provoke child attention and provide precise information. Verbal instructions that specify a specific motor task, followed by a 5-second pause, elevate compliance ratios of 3- to 7-year-olds relative to experimental control subjects (Roberts, McMahon, Forehand, & Humphreys, 1978). In addition, eye contact has been experimentally shown to increase compliance ratios in middle childhood (Hamlet, Axelrod, & Kuerschner, 1984), a finding supported by a group comparison project with 3- to 7-year-olds (Hudson & Blane, 1985). Specifically,

nonclinic parents used more eye contact than parents of referred children. In addition, Hudson and Blane found that instructions issued in close proximity, at the child's eye level, while orienting toward the relevant objects and using a pleasant tone of voice were more likely in parents of normal children than clinic-referred children. Wruble, Sheeber, Sorensen, Boggs, and Eyberg (1991) demonstrated that direct commands (e.g., "Come here") were more effective than indirect commands (e.g., "Would you come here?") in a recruited sample of 3- to 5-year-olds. These data are consistent with the common recommendation to use an imperative grammatical form of verbal instruction rather than the interrogative form in which choice is implied. Fifth, most parenting programs introduce some form of discipline to block the reinforcers previously coerced by the child. Time-outs, token fines, privilege losses, or work chores are routinely recommended for noncompliance and aggression, if not for other targets. Sixth, token reinforcement systems are introduced to provide a mechanism to deliver activity and material reinforcers for prosocial behavior on the assumption that social reinforcers (e.g., praise) are relatively weak compared to coerced reinforcers (e.g., task avoidance), especially for referred middle childhood youth (e.g., Patterson et al., 1975; Wahler, 1969a). Successful applications of token systems are not limited to middle childhood, however, and can be used successfully with children as young as 3 years (Christophersen & Mortweet, 2001, pp. 223–239).

Noncompliance Programs

The classic Hanf protocol for teaching noncompliant 2- to 7-year-olds to obey parent instructions has been repeatedly evaluated and come through 35 years of empirical scrutiny relatively unscathed. Figure 22.2 presents the often displayed familiar event sequence (e.g., McMahon & Forehand, 2003, p. 131). Parents are first taught to present effective instructions as discussed earlier. If the child makes any motoric

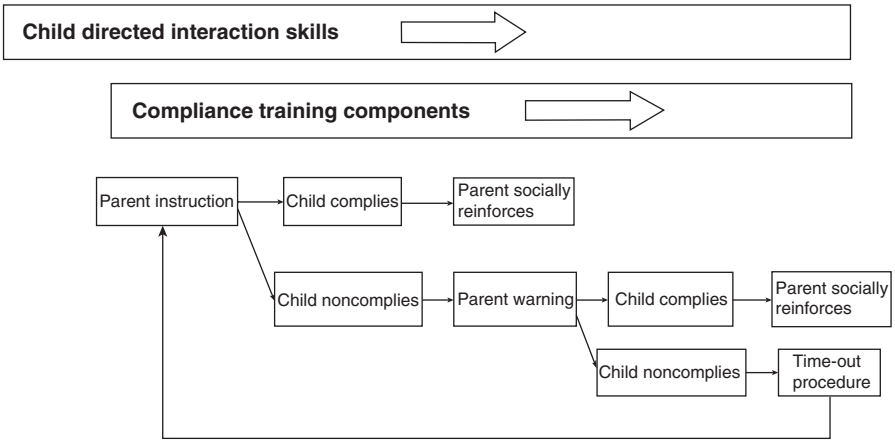

Figure 22.2 Recommended compliance training protocol for 2- to 7-year-olds based on the sequential strategy of current HNC and PCIT programs.

effort to comply within 5 seconds, social reinforcement is provided. If the child makes no effort within 5 seconds of the termination of the instruction, a warning is issued. Child effort within 5 seconds of the warning is socially reinforced, whereas a lack of effort results in a time-out administration. Upon completion of time-out, the original instruction is re-presented and the process is recycled as needed until the child complies. It sounds simple, but there are endless complexities. Fortunately, some empirical guidance is available.

Because the parent's game and cleanup conditions frequently provide the context for teaching compliance-eliciting skills in both HCN and PCIT traditions, the many successful demonstrations of compliance acquisition focus on child initiations to perform discrete acts (e.g., "Put the toy on the shelf"). Less is known about motivating children to display sustained effort to complete a substantial task (e.g., "Clean your room") or to cease an ongoing response (e.g., "Stop yelling"). The ability to sustain effort in the absence of an adult requires complex self-regulatory skills that emerge gradually (e.g., Maccoby & Martin, 1983, pp. 35–37), highlighting the need for parents to discriminate noncompliance from normal developmental errors. For example, parents might instruct an 8-year-old to clean his or her room, but not a 3-year-old for whom parent proximity, modeling, and repeated instructions would be needed. A different set of issues involves "Don't" instructions. We know that even 2-year-olds can inhibit actions given an instruction (Minton, Kagan, & Levine, 1971), but the immediate cessation is less relevant than the resultant frequency of the targeted misbehavior. For example, does the child's base rate of teasing his sibling decline, given compliance to the adult instruction "Stop teasing"? That is currently an empirical question, addressed at the level of the individual child. For physical aggression, however, some general knowledge exists. Jones, Sloane, and Roberts (1992) found clear evidence that parent "Stop fighting" instructions can function like nattering, socially reinforcing misbehavior and allowing coerced reinforcers from siblings to operate. Because the child stops fighting briefly, the parent is negatively reinforced for issuing "Stop fighting" instructions, as predicted by coercion theory (Figure 22.1), despite an unchanged or worsened fight frequency. Similarly, Kendall, Nay, and Jeffers (1975) observed that adolescent peer aggression in a residential facility increased when an instruction or warning procedure preceded time-out. Therefore, parent trainers are advised to consider the type of instruction used by the parent and adjust the treatment plan according to developmental variables and empirical evidence. Using a noncompliance protocol to reduce physical aggression is contraindicated.

Hanf recommended a 5-second interval as the window for child initiation of effort to instructions and warnings, which intriguingly has some observational support (Wruble et al., 1991), despite its clearly arbitrary roots. If the child does display effort during the 5-second window, social reinforcement is recommended. Interestingly, social reinforcement is not necessary for compliance acquisition or maintenance (Roberts, 1985; Roberts, Hatzenbuehler, & Bean, 1981) in overtly noncompliant referred preschoolers, failing to augment instruction and time-out effects. Noncompliance appears to be negatively reinforced by task avoidance rather than misguided parental attention, and compliance acquisition, at least in

previously noncompliant 2- to 7-year-olds, appears to be negatively reinforced by active avoidance of time-out. Compliance acquisition in normally developing toddlers may be quite a different process, however, and involve such variables as infant temperament and parental responsivity, as discussed earlier. Hopefully, all children, including previously noncompliant ones, eventually internalize a standard of conduct to obey legitimate authority and, subsequently, regulate their own behavior rather than obey simply to avoid discipline. Nothing in these empirical findings and theoretical speculations indicates that parents should not be taught to praise child compliance. Theoretically, every opportunity to respond positively to prosocial child behavior is an opportunity to promote the parent-child relationship. The high consumer satisfaction (McMahon, Tiedemann, Forehand, & Griest, 1984) and social validity (Forehand, Wells, & Griest, 1980) of HNC protocols suggest that the Hanf two-stage protocol, which includes social reinforcement for compliance and mastery skills in child-directed interaction (PCIT) or the child's game (HNC), is thoroughly justified. Consequently, child-directed interaction is included in Figure 22.2 as the first step in compliance training. Nevertheless, the scholarly skeptic is not yet convinced that parent social reinforcement of child compliance is important, given the current limited set of dependent variables. See Forehand (1986) for a more thorough look at this issue, and the ensuing rejoinder by Lutzker, Touchette, and Campbell (1988) and the Forehand reply (1988).

Warnings to obey instructions (or be sent to time-out) have proven useful when the instruction is to initiate a discrete task (Roberts, 1982a). The warning component has reduced the number of time-outs needed to elicit compliance criterion responding and has not attenuated compliance acquisition rates relative to no-warning control conditions. The concern has always been that warnings will function like nattering, which appears not to be the case for instructions to initiate behavior. As noted earlier, however, fighting is not well served by "Don't" instructions and warnings, suggesting that the type of instruction interacts with the utility of warnings. If parent trainers are trying to manage a misbehavior other than noncompliance (e.g., tease, yell, dangerous climbing) with "Don't" instructions, the base rate of the misbehavior should be used to evaluate the protocol, rather than compliance to the instruction or warning.

The time-out component of Hanf-based programs has been repeatedly shown to be a critical treatment component (Roberts et al., 1981), yet one plagued with difficulty (Roberts, 1982b, 1984). Time-out procedures are complex. Early reviews (K. R. Harris, 1985; MacDonough & Forehand, 1973) explicated the many parameters of time-out (e.g., duration, location, release rules), some of which are now understood. Decades ago Leitenberg (1965) queried whether time-out is an aversive condition. It is. Overtly noncompliant children will fiercely resist initial time-outs (Roberts, 1982b), and successfully treated referred children will persistently comply to avoid them (Roberts, 1985). The more noncompliant the child at baseline, the more intense the time-out resistance (Roberts & Powers, 1988, table 2). If parent trainers allow children to successfully escape from time-out, compliance acquisition is significantly degraded (Bean & Roberts, 1981; Roberts & Powers, 1990). Moreover, contingency awareness does not mitigate the defiant child's behavioral insistence on testing the limits of the newly imposed contingencies (Roberts, 1984).

Fortunately, the spanking component of Hanf's original time-out maintenance procedure proved unnecessary (Day & Roberts, 1982; Roberts, 1988). Brief room time-outs maintained by a barrier suppress chair time-out escape efforts just as well as spanking, while avoiding corporal punishment and modeling aggression. That finding does not imply that putting a barrier or door between the parent and an angry child is a pleasant process. For just that reason, Roberts and Powers (1990) and Foster and Roberts (2007) have called for clinicians to monitor initial time-outs, support parents and children through the process, and adjust the procedures as needed. A legitimate and unanswered question focuses on time-out location. Why not use the prototype room time-out (Wolf et al., 1964) and avoid the escape problems of chair time-outs? The early work of Wahler and the OSLC group did just that. The argument voiced by HNC and PCIT programs is that 2- to 7-year-olds are better served by chair time-out conditions that can be visually monitored for safety and readily adapted to community settings. This argument, however, is rational rather than empirical. For a recent discussion of time-out parameters, see Roberts (2005).

Aggression Programs

The OSLC database has yielded some important empirical insights into the interrelationships of aggressive boys, their siblings, and their mothers. All parties appear to elicit, model, and reinforce physical aggression, yielding about one fight per hour, with the most likely victim a male sibling (Patterson, 1984, 1986). Years ago O'Leary, O'Leary, and Becker (1967) established the basic intervention for sibling aggression: immediate chair time-out combined with reinforcement for cooperative sibling interaction. An important distinction between treatments for noncompliance and aggression was anticipated by the O'Learys. The noncompliance protocol promotes its prosocial replacement skill (i.e., compliance) via an instruction and two reinforcement mechanisms: time-out avoidance and social reinforcement. In contrast, suppressing aggression with time-out or other strategies (e.g., token fines) does not directly provoke an acceptable replacement skill. Indeed, siblings might actually avoid each other, as was observed by O'Leary et al. The immense social skill training literature is designed specifically to teach a repertoire of nonaggressive strategies to solve myriad interpersonal sibling and peer conflicts (see McMahon et al., 2006, pp. 178–186), and, although well beyond the scope of this chapter, needs attention from parent trainers working with sibling aggression. The social skill assessment and training issues are complex and unresolved. For example, does the child know how to "share," or "take turns," or "ask nicely," or "take 'no' for an answer," or "be verbally assertive," given a specific and complex social condition (Roberts, Arnold, & Mangum, 1992)? What is the best way to teach those skills (Thomas & Roberts, 2006), let alone motivate performance of those skills in relatively unmonitored home settings?

Immediate time-out is effective at suppressing sibling aggression in 2- to 7-year-olds (Allison & Allison, 1971; Jones et al., 1992; O'Leary et al., 1967; Olson & Roberts, 1987). Referred to as a "standing rule" by HNC theorists (McMahon & Forehand, 2003, pp. 150–155), both children are immediately sent to

time-out for fighting, effectively blocking all reinforcers that might be coerced by the aggressive act and simultaneously calming down both children via the quiet release contingency inherent in time-out routines. Following time-out, parents are taught to discuss the event-sequence leading up to the fight to enhance awareness of the conflict situation and alternative solutions. Does the parent trainer also need to program reinforcement for aggression inhibition or prosocial alternatives? Leitenberg, Burchard, Burchard, Fuller, and Lysaght (1977) have done so successfully with recruited families. Moreover, there is a huge literature on peer aggression that indicates success with various combinations of discipline, reinforcement, and skill building (e.g., Fehrenbach & Thelen, 1982). Therefore, a comprehensive and sequential approach to sibling aggression is recommended and is displayed in Figure 22.3: reinforcement for inhibition, immediate time-out, and skill building, in that order. Each component can be added as needed, based on parent records of sibling aggression at home. Developmental limitations, however, preclude most 2- and 3-year-olds from the skill-building and reinforcement components, given cognitive and attentional limitations. Aggressive siblings under age 4 might profit from immediate material reinforcement for cooperative play if a therapist could be inserted into the home (O'Leary et al., 1967), but such a routine seems impractical for parents and would lack social validity. Reducing sibling aggression without an immediate time-out seems especially unlikely in children under age 4. It is true that one can find theoretical arguments and some empirical support from developmental psychology (e.g., Brody & Stoneman, 1987) and family systems therapy (e.g., Dinkmeyer & McKay, 1973) to ignore sibling aggression. Others vehemently disagree (Bennett, 1990; Ihinger, 1975), noting that a quiet tyranny of the mighty could emerge, consistent with coercion theory (Figure 22.1). It is difficult to imagine how a delayed reinforcer for inhibition and/or skill building role-played in a clinic could override the immediately coerced reinforcers for sibling aggression in children under 4. In contrast, middle childhood youth (7 to 12 years old) might be effectively discouraged from sibling aggression by reinforced inhibition and/or social skill building alone (see the peer aggression literature reviewed by

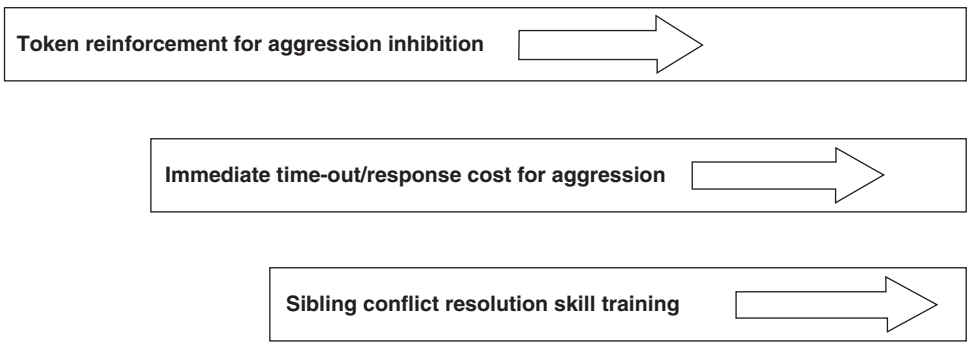

Figure 22.3 Recommended sibling aggression protocol for 2- to 12-year-olds based on a sequential strategy. Immediate time-out or response cost is needed if reinforcing inhibition is insufficient; skill training is needed if the combined reinforcement and time-out/response cost programs are insufficient.

McMahon et al., 2006, pp. 178–186). If sibling aggression still persists, then immediate time-outs, token fines, privilege losses, or work chores are indicated.

Other Programs and Situational Variables

Unlike noncompliance and aggression, which are motoric actions, tantrums and negative verbalizations are both vocal, suggesting extinction processes might be successful. If the function of coercive vocalizations is to provoke adult attention (e.g., nattering), differential attention procedures can effectively extinguish verbal coercion (e.g., Herbert & Baer, 1972). If the function is to terminate time-out conditions, a quiet release rule is effective (Hobbs & Forehand, 1975). If the coercive verbalizations covary with instruction presentations, simply persisting with the instruction routine can extinguish the negative verbal output despite socially reinforcing compliance in the presence of negative talk (Roberts & Hatzenbuehler, 1981). Therefore, teaching parents to persist with the stimulus conditions (Figure 22.1, "–1 Step") that provoke the negative vocal outburst is just as important as ignoring negative vocalizations (Figure 22.1, "+1 Step") in other contexts (e.g., sibling tease, disagree, bossy talk). Extinction, however, can be an aversive routine for many adults, given an extinction burst, and is sometimes impossible to implement (e.g., the child may climb on parent and scream). Teaching the parent to assume a physical posture and activity (e.g., turn one's back, leave the area, stand up) that makes ignoring viable is one option. Some unignorable emotionally aroused preschoolers, however, may need a time-out to successfully structure the extinction procedure for the parent. Only the quiet contingency of the time-out needs to be met to terminate time-out conditions for unignorable tantrums. In middle childhood, specific negative verbalizations (e.g., cursing) might be targeted within a larger token system by reinforcing inhibition and fining specified negative talk with tokens, backed up by activity or material reinforcers. Contingent chores (Fischer & Nehs, 1978) have also been used to suppress negative verbalizations in middle childhood.

Three social contexts require accommodations and are briefly discussed: bedtime, community, and car. Defiance, aggression, and tantrums at bedtime can befuddle the parent trainer if standard daytime treatments are applied. Time-out on a chair in the living room may function like "time-in"; differential attention may lead to reinforcing the child for refusing to go to bed, since staying up is ignored and attention for remaining in bed cannot compete with unrestricted access to the common rooms; token reinforcers for successful bedtime behavior are often delayed until morning. Traditional routines targeting bedtime crying, such as ignoring and graduated extinction (i.e., a gradual and systematic increase in parent response latency to crying), are effective (Kuhn & Elliot, 2003) but aversive to parents and presume that the child will remain in bed. Repeated "walk-backs" to bed or ultimately a shut door may prove necessary. Edwards and Christophersen (1994) recommend a sequential strategy, in which daytime self-quieting is targeted prior to attempting a bedtime extinction procedure (which requires children to calm themselves). Daytime ignoring and/or time-outs with quiet contingencies for tantrum and negative verbalizations provide practice opportunities for successful self-quieting. Recently,

Burke, Kuhn, and Peterson (2004) tested a clever positive reinforcement system (a prize under the pillow, left by the Sleep Fairy) that warrants replication with an oppositional sample. See Christophersen and Mortweet (2001, pp. 99–122) for a recent review of the psychopathology and empirically supported interventions for managing sleep problems.

Car and community situations often fail to provide the rich array of play options found at home (e.g., a long car trip, watching mother shop, sitting through big sister's piano recital), increasing the chance of coercion to provoke reinforcers. Further, community settings may be discriminative for misbehavior, since some parents actively mollify the child in public to avoid the embarrassment of child tantrums. For those two reasons, planned activity training combined with modified time-out routines are recommended. Sanders and Dadds (1982) helped parents identify and promote age-appropriate activities in specific community settings (e.g., quiet toys and games for the car; reinforced shopping participation; outdoor toys for a teenage sister's soccer match) that significantly enhanced generalization of parent training effects to community settings. Time-out locations can be defined in any specific, low-traffic area, backed up by the two private locations found in the community: bathrooms and the parent's car (Christophersen & Rapoff, 1979, p. 371; McMahon & Forehand, 2003, pp. 155–157). Sibling aggression in the car or seat belt refusal may necessitate stopping the car in a safe location and declaring time-out until all children are sitting quietly.

Some children may not adjust to community or car settings despite good planned positive activities and appropriately modified time-out routines. Consequently, parent trainers should consider inserting more powerful token systems tailored to the community setting (e.g., Barnard, Christophersen, & Wolf, 1977) to reinforce rule following and participation in planned positive activities. In addition, parent trainers can shape community skills by initially assigning brief, repetitive exposures under rich reinforcement conditions combined with simulation practice in the clinic. Data collected by the parent in the target setting can be used to adjust the treatment protocol.

Case Illustration

Ms. Smith and her three children, ages 7, 5, and 2, were referred to the clinic by their pediatrician for evaluation of conduct problems and possible therapy. During the standard interview, Ms. Smith reported that the 5-year-old, Zachary, was her main concern. Zach had been asked to leave a day care facility that summer for repeatedly fighting with peers and was now "in trouble" in kindergarten. Ms. Smith reported that Zach had always been "difficult" to raise. He argued, disobeyed, tantrummed, and was even aggressive with her when he didn't get his way, his requests were denied, or when she attempted to spank him or send him to his room. Zach was very bossy with his 2-year-old brother, which led to angry outbursts and aggression. In contrast, the 7-year-old sister had been "very easy" to raise and seemed to disregard the two boys. Ms. Smith voiced worry that her 2-year-old was learning "bad habits"

from Zach. Questionnaire data from the Child Behavior Checklist (Achenbach & Rescorla, 2000) and the Eyberg Child Behavior Inventory (Eyberg & Pincus, 1999) both yielded elevated scores on Social Aggression scales for Zach, but not for either sibling. The Slosson Intelligence Test Revised 3 (Slosson, 2002) was administered to screen for delays in Zach's cognitive/verbal development. He was normal (SS = 92). Maternal depression was also screened and fortunately absent (Beck Depression Inventory II = 9, "minimal"; Beck, Steer, & Brown, 1996). Since Ms. Smith was a single parent, a marital adjustment scale was not administered, but an instrument like the Marital Satisfaction Inventory R (Snyder, 1997) is recommended to screen for marital conflict. Because neither depression nor marital conflict was present, referrals to address these issues were not needed. A phone call to Zach's kindergarten teacher revealed disruptive behavior and occasional aggression, but no teacher interest in assistance from an outside professional at that time. The school had its own system for managing misconduct (an escalating response-cost system). His teacher did request (and was eventually mailed) a copy of his psychological evaluation, findings, and recommendations.

Home record data collected by the mother during the 2-week evaluation period yielded 3.6 fights per day (usually with the 2-year-old brother) and over 5 tantrums per day. In addition, Zach ran away from the house twice when confronted by his mother over conflict with his brother, provoking necessary but distressing "chase" episodes. Clinic observations under child's game conditions showed a responsive mother (64% positive or neutral reactivity, given a signal from Zach or the 2-year-old), but a very high rate of questions plus instructions (8.3 per minute) and a virtual absence of positive maternal signals (praise + descriptions + imitations = 0.2 per minute). The 7-year-old was not available for the observational measures since she was in school when the family attended clinic sessions. Under cleanup task analogue conditions, Zach's compliance ratio was 22%. Ms. Smith, however, repeated her instructions so quickly that her beta instruction ratio was almost 75% (i.e., repeated + vague instructions divided by the total number of instructions issued); her praise ratio was 0%. A compliance test was administered to ensure that Zach's defiance was not simply an artifact of maternal instruction style. It was not; his compliance ratio, given well-presented instructions to pick up toys, was 3%. He also became quite angry during the compliance test, yielding cry and back-talk ratios over 50%.

Given the interview, questionnaire data, home record data, clinic observations, and school information, Zach met criteria for Oppositional Defiant Disorder. Using Figure 22.1, it could be seen that several paths were relevant to help conceptualize the case. Coercion was very likely, given maternal or sibling instructions to "Do X" or "Stop Y," as well as denials of Zach's requests. Fortunately, there was no evidence that minor aversive events (e.g., disagree with siblings), ignoring, or positive or neutral events set the occasion for Zach's misbehavior. Zach played alone quite well. Discipline efforts, however, were ineffectual and provoked escalating intensities of coercion (e.g., he had physically hurt his mother on more than one occasion when she attempted to guide him to his room). Common parenting errors (i.e., Figure 22.1, "+1 Step"), although not observed, presumably reinforced Zach's coercion on a partial reinforcement schedule. Social learning–based parent training was recommended.

(*Continued*)

In addition, Zach's school progress was to be monitored to ensure that programs implemented in that setting were successful.

Child-directed interaction skills were targeted first. Within three sessions Ms. Smith became rather skillful at giving attention to age-appropriate play when interacting with Zach and his 2-year-old brother in the clinic playroom. Her responsiveness index remained high (over 70%), while the quality of her responses became more positive (praise + descriptions + imitations exceeded 4 per minute). It was difficult for her to learn to inhibit question asking, since she seemed to want to "teach" skills via this route, but she eventually adopted a more child-centered strategy, limiting questions plus instructions to under one per minute. She also had difficulty learning to ignore the boys when they became bossy, angry, or disruptive with each other, but finally learned to turn her back and play with the other sibling or simply model quiet toy play until one or both siblings began to play more appropriately. Ms. Smith was asked to set aside some morning time (e.g., 15 to 20 minutes) to use child-centered play with her 2-year-old and a similar time after school with all three children. In addition, she was instructed to look for opportunities throughout the day to socially reinforce child initiations with her new listening skills. Handouts on responsive play skills and the principles of shaping and positive reinforcement were provided for review and future discussion.

Compliance training, which began during the third treatment session, was more difficult. During modeling and role-playing of instruction, social reinforcement, warning, and time-out skills, Zach displayed interest in his mother's reactions to the doll, but when queried about why the doll went to time-out, Zach refused to talk to the therapist. During the initial guided practice with Zach, his mother, wearing a bug-in-the-ear, met with immediate defiance and time-out resistance. Zach refused to go to the time-out chair and physically fought his mother when she guided him to the backup room. He banged on the door and yelled angrily for the entire 60-second period; when escorted back to time-out for his second opportunity at chair time-out conditions, he repeated the entire sequence of escape, a furious 1-minute room backup, and subsequent replacement on the time-out chair. During his third experience with chair time-out conditions, he inhibited escape for about 5 seconds before arising and throwing the chair against the wall. He was immediately escorted to the backup room for a third and final time; there to remain until a 5-second quiet rule was met. Ten minutes later he finally extinguished his crying and yelling sufficiently for the therapist to instruct the mother to open the door and say, "Since you're quiet, you can come out now." Zach immediately reinitiated screaming and refused to leave the time-out room. Pouting after initial time-outs is not unusual, but Zach was extreme. The therapist had to enter the playroom, prop the time-out backup room door open (because Zach kept slamming it), and prompt the mother to play positively with the 2-year-old out of Zach's visual field. This process also allowed the therapist to provide social support and compassion for the mother beyond what could be done over the radio device. Zach was ignored. Four minutes later he emerged from the time-out room. Rather than re-present the disobeyed instruction immediately (as in Figure 22.2), the therapist and mother continued in a responsive play mode with the younger sibling and Zach until Zach began to play and talk normally. The therapist then left the room and resumed compliance training via the bug. Zach immediately disobeyed the same cleanup instruction and was sent to

time-out again. The entire coercive cycle repeated itself, although greatly toned down. Zach remained on the time-out chair after just one 60-second trip to the backup room and became quiet within 3 minutes. He again refused to leave time-out when told he could, but when the mother began playing with his little brother, Zach left the chair and began playing (the therapist did not have to enter the playroom this time). Once Zach began playing, the therapist instructed the mother to say (via the bug), "Okay Zach, we still need to put the stuffed animals back in the toy box." Zach started yelling, but when given the warning, he picked up one toy despite continued verbal protests. His mother praised his efforts and continued to use the basic Figure 22.2 system under guidance via the bug. Zach met the criterion of 10 compliant responses in a row (despite the necessity of three warnings, which he obeyed). Ms. Smith was instructed to give him a big hug and inform Zach that he could play the way he wanted to now. Responsive play was resumed until both boys were playing and talking appropriately. The 2-year-old was then issued similar, but very simple motor tasks (e.g., "Come here"; "Take this (toy)"; or "Put it (location)," always accompanied by close proximity, eye contact, and a gesture. The 2-year-old met compliance criterion with only one time-out and virtually no time-out resistance. A record card was sent home to monitor chair time-outs for disobeyed warnings and needed room backups (using the child's bedroom). Because Zach had been extremely resistant to time-out, phone contacts were scheduled during the next few days and until the mother was confident she could administer the Figure 22.2 program. Handouts on the compliance program and the principles of extinction and punishment were sent home and reviewed at a later session.

During the fifth clinic visit, the home record data indicated that Zach had required from one to four chair time-outs per day for disobeyed warnings and room backup procedures about half the time. His mother did report that Zach often obeyed warnings, which pleased her, and the younger brother obeyed almost all the time if she "made it simple" as instructed. The mother had clearly gained some confidence in the compliance routine, but felt helpless about the boys' fighting. A no-fighting rule (Figure 22.3) was instituted in addition to reviewing responsive play and compliance protocols. Zach sat beside his mother while the therapist sent two dolls to time-out for fighting over a toy. An effort was made to discuss the conditions, mistake, and optional responses, but Zach would have none of it. Ms. Smith, however, role-played the post-time-out effort at incidental training regarding the conflict conditions (e.g., brother 1 had taken brother 2's toy), the mistake (i.e., fighting), and the optional responses (e.g., using an assertive instruction or seeking maternal help). During the responsive play sample, Zach pushed his brother intentionally; the mother was prompted to implement the no-fight rule, marching Zach immediately to the time-out chair while labeling the contingency in an authoritative manner (i.e., "Since you hit your brother, you must go to time-out"). Zach actually stayed on the time-out chair, but refused to talk to the mother during the incidental learning procedure after time-out. There was no more fighting during that clinic session. Time-out for fighting was added to that week's home record card and monitored thereafter. A token system was also initiated. Zach was shown spaces on the home card where stickers would be entered for remembering his no-fight rule; he could earn up to two on school

(*Continued*)

days and three on the weekends. Both Zach and his brother were shown the clinic store where stickers could buy prizes the next week. His mother was instructed to insert stickers for specified periods of aggression inhibition and lavish praise for rule following.

During the sixth through eighth clinic sessions home cards were reviewed, stickers were rewarded with clinic prizes, responsive play and clinic task conditions were observed, and rule protocols were reviewed. Zach's little brother always received a prize, despite not participating in the token system. A set of planned positive activities and adaptations for time-out, if needed, were modeled and role-played for shopping and car situations. Both boys watched with great interest as Ms. Smith role-played with the dolls. Shopping and car opportunities were monitored on the home record card and discussed with Ms. Smith during follow-up sessions. During the eighth session an allowance system was established for Zach and his older sister to substitute for the token system. Both older siblings earned a set amount of money each week (10 cents per day for Zach) for completing at least one household chore each day. In addition, Zach was fined 5 cents for every time-out, which had dropped to about four per week at that point. The token reinforcement system for aggression inhibition was then discontinued. Because fighting rates had declined dramatically, sibling conflict resolution skill training (Figure 22.3) was not considered necessary. Zach was, however, still having trouble in school. A school psychologist had been called in by the principal to address continued classroom disruption and occasional fighting. Following phone consultation and one lengthy meeting, a program was jointly evolved that included a token reinforcement system for rule following and classroom productivity, token fines and privilege losses for fighting, and a weekly social skill session with two other boys and two "peer helpers" led by the school psychologist.

After eight treatment sessions, two follow-up sessions, and 3 months, Zach's home data finally looked normal for a 5-year-old (i.e., 1 to 2 time-outs per week, no room backups), but school problems persisted. Long-term follow-up was planned to provide continued support and educational services to the mother, as well as consultation services for the school district as needed.

FUTURE CHALLENGES

Current parent training treatment modalities have limitations. In particular, effective discipline procedures for middle childhood have not been identified. Theoretically (Figure 22.1), the coerced reinforcers of violence must be blocked to make progress with oppositional and conduct disordered grade-schoolers. Unfortunately, parent trainers lack an effective procedure to enforce time-outs, token fines, privilege losses, or work chores with those 8- to 12-year-olds who fiercely resist discipline efforts. Their 2- to 7-year-old counterparts can be safely escorted to time-out as needed. In contrast, older children can intensify the level of violence toward the parent, thwart the implementation of the discipline, coerce denied reinforcers, and/or run out of the house! Increasing the level of care to psychiatric facilities, therapeutic foster care, community residential programs, or juvenile detention centers is the current solution, but it is not satisfactory.

Similarly, some middle childhood referrals continue to display unacceptably high rates of aggression and defiance, despite the establishment of a seemingly good

set of treatment procedures. Frick (2004) has identified a subset of externalizing disordered children he calls "callous-unemotional" who may not respond to family-based interventions. Specifically, family dysfunction may be less important than temperamental and physiological differences for this minority of violent clinic referrals, suggesting a need for supplemental treatment modalities. Unfortunately, there are no specific and effective interventions for instilling guilt for transgressions, acceptance of responsibility for mistakes, empathy for victims, or identification with mainstream social activities. These important self-regulatory processes emerge gradually in normal children during socialization. Parent trainers often observe phenomena that suggest improvement in these important covert processes by many of our child clients, but not all. The discussion of "child effects" is not new in our field (Lytton, 1990), nor resolved (Dodge, 1990; Wahler, 1990). There is, however, no doubt that some young males seem untreatable. Relentless and remorseless appear apt descriptors to this author. Psychiatric medication and higher levels of care appear to be their current destination.

SUMMARY

Parent training is based on the recognition that externalizing disorders of childhood are rooted in coercive patterns of social interaction. Intentional, aversive child behavior is very likely to coerce parents, teachers, siblings, and peers into unwittingly reinforcing common referral symptoms, such as disobedience, fighting, negative talk, and tantrums. Parent training evolved as a new treatment modality to address the reinforcement traps perpetuating the child's social aggression. The pioneering work of Patterson, Wahler, and Hanf led the way toward the sophisticated theoretical models, assessment strategies, and treatment protocols of today. Social learning–based parent training is now widely recognized as an empirically supported treatment. Parents can learn child management skills, use those skills at home, and reduce socially aggressive child behavior. Improvements observed in clinic settings generalize to home settings and persist over time. Basic child management skills include knowledge and use of social learning principles, observation, differential attention, responsiveness, instruction giving, discipline, and token systems. Basic methods of teaching skills to parents include didactic strategies, modeling, role-playing, guided practice, and homework. Both parent group and individual family approaches have been effective.

Problems not initially considered became evident as clinical scientists pursued the dropouts and treatment failures from parent training programs: poverty, social isolation, parent pathology, and therapeutic disengagement head the list. Moreover, there are some child presentations we still cannot manage on an outpatient family therapy basis (e.g., violence in middle childhood), let alone successfully treat. Parent training is a work in progress, as witnessed by the ongoing waves of clinical research discussed in the chapter. Nevertheless, routine defiance, aggression, negative talk, and tantrums, especially for oppositional and disruptive children under 8 years of age, can be successfully treated at home and in other relevant situations via parent training. Under those conditions, parent training is the empirically supported treatment of choice.

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CHAPTER 23

Cognitive-Behavioral Treatment

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If you have ever observed or listened closely to children in clinical situations, at school, or in a playground, you have most likely heard at least a few of the following:

- "She said no when I asked to sit at her lunch table. I'll never have any friends."
- "I can't do this math problem—I'm a total idiot!"
- "If I don't get this homework done perfectly, it just shows how stupid I am."
- "I'll never hit the ball, not even if I try a million times."

For a century or more, we have known that children aren't little adults (Kingery et al., 2006). However, like adults, they too experience cognitions that reflect irrationality or overgeneralization or catastrophic thinking. They may predict that small upsets in peer relationships will continue "forever"; they may interpret spelling errors as total failure; and they may describe normative fluctuations in mood as making life utterly impossible and overwhelming. Such patterns of thinking, in concert with individual personality characteristics and contextual factors, interact to determine both adaptive and maladaptive psychological functioning. Although prevalence rates vary with methodologies used, recent epidemiological estimates have found that roughly 15% to 22% of youngsters will suffer from psychological issues and their related functional impairment during childhood and adolescence (Costello, 1989; McCracken, 1992).

Cognitive-behavioral models posit that, like adults, children and adolescents have faulty beliefs and automatic thoughts that negatively impact their behaviors and contribute to the development and maintenance of psychopathology. Cognitions refer to beliefs, schemas, attributions, perceptions, or mental representations of events. Cognitive-behavioral theory is phenomenological in that it assumes that behavioral responses are a function of an individual's perception of events, rather than the events themselves. In other words, we imbue our worlds with meaning and respond to them in terms of these meanings. Attributions we make about the world may be more or less accurate and, similarly, more or less helpful in directing our behavior.

Recent approaches argue that cognitions may function as "rules." They suggest that because the behavioral responses associated with such thoughts are thus

rule-governed, they may be especially intractable (Hayes, Strosahl, & Wilson, 1999). Very often, we don't have control over environmental contextual factors that maintain maladaptive behaviors. However, we do have access to thoughts and emotions: We can monitor internal psychological events, notice how they appear to be connected to behavioral responses, evaluate them, and, if we choose, engage in attempts to change them. Changes in thoughts are linked with changes in behavior, and thus, cognitive-behavioral theorists ascribe them causal status. As such, cognitive-behavioral approaches contend, our thoughts may be ideal targets for treatment.

Cognitive-behavioral therapy (CBT) has enjoyed a wealth of empirical attention over the past few decades. In the past 10 years or so, these approaches have been adapted for use with children and adolescents. Although originally developed for use with adults, child-focused CBTs are now numerous and have been investigated for youth of varying ages and across a variety of disorders. Whereas early efforts in treatment development involved downward extensions of adult models of CBT (e.g., based on the work of Ellis and Beck), newer approaches reflect more sophisticated thinking about the developmental nature of child psychopathology, representing more bottom-up innovations (e.g., Kendall & Hedtke, 2006; Kendall, Kane, Howard, & Sigueland, 1990). Interventions have been designed in more contextually and developmentally sensitive ways, thus expanding the utility of CBTs in broader domains and with more diverse populations. Recent advances include applications with very young children (Freeman, Garcia, Coyne, Leonard, & Compton, 2007), family-based CBT models (Barrett, Dadds, & Rapee, 1996; Freeman, Garcia, Fucci, Karitani, Miller, & Leonard, 2003), and those incorporating acceptance- and mindfulness-based approaches (Semple, Reid, & Miller, 2005), to mention just a few. Moreover, CBT has been adapted for use across multiple settings, including primary care, behavioral health, and with severely emotionally disturbed youth; for multiple purposes, including prevention and intervention efforts; and with diverse racial-ethnic populations. This chapter provides an overview of contemporary cognitive-behavioral treatments with children, discusses current research in CBT efficacy, with particular focus on childhood depression and anxiety, and illustrates an innovative use of family-based CBT for young children through a case study depicting intervention for a young child with Obsessive-Compulsive Disorder (OCD).

OVERVIEW

Cognitive-behavioral therapy is based on principles of operant and respondent conditioning. Consistent with a social learning perspective, CBT models assume that an individual's behaviors are a function of antecedent and consequent control. Contingencies may be directly experienced or observed. However, perceived environmental contingencies, in concert with individual characteristics such as biological predispositions, temperament, or personality, as well as particular contexts, interact to influence behavioral responses and dynamically developing repertoires. Cognitive-behavioral therapy holds that although distorted or irrational thoughts alone are not sufficient to precipitate psychological difficulties, they are a major

contributor to maladaptive behavioral repertoires. As such, unrealistic or unhelpful thoughts and emotions are primary treatment targets in cognitive-behavioral approaches. Clinicians strive to help clients modify or reframe irrational thinking to effect desired changes in behavior. Thus, CBT dually targets client thoughts and behaviors to ameliorate symptoms and improve functioning. This is accomplished through a collaborative relationship between clinician and client that is at times didactic and at times experiential. Often, clinicians use Socratic questioning to call client attention to maladaptive thinking patterns and to challenge the logic of such cognitions. Treatment tends to be structured, goal-oriented, and time-limited and may be delivered in an individual or group format.

Although there is variation across different treatment manuals and clinical issues, CBT consists of three fundamental components: psychoeducation, cognitive restructuring, and behavioral experiments. *Psychoeducation* is perhaps the most didactic element of CBT and serves as an important first step in treatment. Children are educated about the role that cognitions play in their emotions and behavior. Clinicians help children recognize and monitor their response patterns and appraisals of problematic situations. By educating children about how their response patterns work, they may exhibit greater facility in applying their knowledge and new skills after treatment concludes and across different situations. Though psychoeducation is the first step in treatment, the education process continues for the duration of a given intervention.

Cognitive restructuring is the central cognitive feature of CBT and was originally defined by Albert Ellis (1962) and Aaron Beck (1976). Cognitive restructuring requires that children strive to do two things: first, identify their erroneous or distorted thoughts or beliefs, and second, dispute them. This "rational disputation" is the hallmark of this therapeutic approach. Contemporary conceptualizations of cognitive restructuring include a variety of techniques meant to help children and adolescents reframe or challenge their own beliefs, automatic thoughts, and schemas. Techniques include, but are not limited to, perspective taking, problem solving, replacing negative self-evaluations with positive self-statements, revising distorted views of social interactions and relationships, as well as more experiential exercises such as guided imagery and relaxation. Although these strategies may differ somewhat among CBTs, their overarching goal is to assist individuals in modifying the content of their thoughts, reducing their believability, and thus supporting the eventual development of more functionally useful behavioral responses.

Another core component of CBT is a set of skill-building techniques that strive to help children develop behavioral competencies for coping with difficult or feared situations. In this component of treatment, children are asked to try out a variety of responses, including testing their faulty appraisals, automatic thoughts, and beliefs through behavioral experiments. Two examples of this set of techniques are exposure with response prevention (EX/RP) in the treatment of OCD and behavioral activation in the treatment of depression. Exposure with response prevention consists of exposing children to feared stimuli, then preventing their ritualized response. In so doing, children experience their discomfort rise and fall and learn that the situations they have avoided are not catastrophic after all. Behavioral experiments can either be *in vivo*, whereby children practice engaging in the

behavior, or *imaginal,* whereby they visualize the behavior and its consequences. In practicing this technique, children learn that they need not engage in rituals to reduce their anxiety and that they may use more adaptive coping skills to address their fears. This technique is most commonly found in treatment for OCD, although exposure is used more broadly across numerous child and adolescent anxiety disorders.

Behavioral activation involves pleasant event scheduling, which assists depressed children and adolescents in more frequently engaging in behaviors that will allow them to experience rewarding consequences. Engaging in behaviors that are either inherently reinforcing or that bring children in contact with naturally occurring rewards helps children to refute their beliefs of hopelessness and the intractability of their negative moods. It may also create behavioral momentum by increasing the density of reinforcers experienced.

Although each of these components is present in some degree in cognitive-behavioral treatments, some approaches rely more heavily on more traditional behavioral techniques (e.g., treatment for child and adolescent OCD), and others use more explicitly cognitive tools (e.g., treatment for depression or Generalized Anxiety Disorder) that emphasize the modification of thoughts, beliefs, and appraisal processes. The degree to which each of these strategies is used is in part a function of the childhood disorder targeted and in part a function of the age of the child or adolescent in treatment. Cognitive-behavioral therapy also incorporates a number of other techniques, such as problem solving and skills training, in the treatment of adolescent depression.

DEVELOPMENTAL CONSIDERATIONS

When using cognitive-behavioral therapy with children and adolescents, it is critical to evaluate the developmental stage of the client. Careful evaluation of the cognitive and emotional functioning of a given child will help guide the selection of therapeutic techniques and strategies: Children 8 and younger may have great difficulty with tactics such as thought monitoring and rational disputation, whereas older children might benefit greatly from these exercises.

Similarly, parents may be helpful as architects of their young children's attempts at behavior change, given that young children are far more dependent on parents for structure and guidance. The purpose of enlisting parents in treatment is twofold: First, they can ensure that behavioral changes and homework are occurring at home; second, the clinician can intervene in any parent behaviors that may inadvertently exacerbate the child's problems. In contrast, adolescents may benefit more if given greater autonomy to effect behavioral changes at their own pace, rather than as dictated by parents.

It is also crucial to consider the contextual factors that may impact a child's or adolescent's behavioral presentation. These include home, school, and peer environments. Depending on a child's age and interests, the relative importance of these factors may vary over time. For adolescents, peer relationships may stand out as a critical treatment issue. Young children, on the other hand, may be more influenced by their parents. Additionally, youngsters' functioning may differ across each of

these environments; thus, using multiple informants (e.g., child, teacher, parent) to describe behavior in each of these domains is recommended.

Finally, a developmental perspective is important in helping to determine whether behaviors are timely and normative or reflective of psychopathology. One example of this is the topography of normal fears across childhood and adolescence. Fear of monsters under the bed is quite normative (if distressing) in preschool-age children; however, such a fear in a teenager might represent a hallucinatory event. Conversely, teens are often concerned with violence and unfairness in the world at large, but a 6-year-old overly concerned with terrorism might merit assessment for anxiety or trauma.

EMPIRICAL SUPPORT FOR COGNITIVE-BEHAVIORAL THERAPY WITH YOUTH

Cognitive-behavioral treatments have been used with a variety of child and adolescent disorders, including anxiety disorders (Albano & Kendall, 2002; Berman, Weems, Silverman, & Kurtines, 2000; Flannery-Schroeder, Choudhury, & Kendall, 2005), OCD (Franklin et al., 1998; J. M. March, Mulle, & Herbel, 1994; Piacentini, 1999; Pediatric OCD Treatment Study [POTS], 2004), depression (Compton, Burns, Egger, & Robertson, 2002; Crisp, Gudmundsen, & Shirk, 2006), eating disorders (Lock, 2005; G. T. Wilson & Sysko, 2006), Attention-Deficit/Hyperactivity Disorder (ADHD; Hinshaw, 2005), conduct problems (Southam-Gerow, 2003), substance abuse (Kaminer & Slesnick, 2006; Kaminer & Waldron, 2006), and trauma (Cohen, 2005; Cohen, Mannarino, & Knudsen, 2005). Gains following CBT appear robust over time (Albano & Kendall, 2002; Dadds et al., 1999; Flannery-Schroeder et al., 2005). A thorough review of the treatment outcome literature is beyond the scope of this chapter; however, in the following we describe the relative efficacy of CBT across childhood and adolescent depression and anxiety and briefly describe applications with each, as well as recent innovations in their use.

Cognitive-Behavioral Therapy for Child Depression

Data suggest that prevalence rates of Major Depressive Disorder (MDD) among children and adolescents are increasing (Cross-National Collaborative Group, 1993; Kessler et al., 1994; Kessler & Walters, 1998). Moreover, suicide is the third leading cause of death in children ages 10 to 19 (Centers for Disease Control and Prevention, 2004). Yet, in an overwhelming majority of youngsters, MDD goes undiagnosed for a year or more, and few receive treatment even by the end of adolescence (Jensen, 2006; Kessler, Avenevoli, & Merikangas, 2001; Leaf et al., 1996). Thus, the need for accurate detection, diagnosis, and effective treatment of depression during childhood is critical. Applications of cognitive-behavioral treatments constitute a strong attempt to meet that need.

Cognitive models of MDD highlight the role of maladaptive beliefs in development and maintenance of depression, although there is some variability across conceptualizations. Recent theoretical work has stressed the importance of evaluating cognitive vulnerability in context and has relied on diathesis-stress

conceptualizations. As originally described by Beck, Rush, Shaw, and Emory (1979), depressed individuals are thought to have systematic biases in thinking that lead to idiosyncratic interpretations of events. Additionally, individuals who are vulnerable to depression appear to have schemas that guide information processing and may potentiate negative self-attributions that are at the heart of depression. Finally, cognitive models characterize vulnerability to depression in terms of the negative cognitive triad: negative perceptions of the self, the world, and one's future. All these increase an individual's risk for depression. Depression has also been explained in terms of Seligman's (1975) model of learned helplessness and as a function of negative attributional style (Abramson, Seligman, & Teasdale, 1978). This latter term refers to the tendency to attribute poor outcomes to internal, global, and stable factors and to ascribe positive outcomes to external, specific, fleeting, or transient factors. Some data suggest that links between attributional style and depression get stronger as children age (Nolen-Hoeksema, Girgus, & Seligman, 1992) and with the increasing sophistication of abstract thinking (Turner & Cole, 1994).

Research on CBT for adolescent depression has found it to be a promising treatment, although not without shortcomings (Birmaher et al., 1996; Brent, Birmaher, Holder, Johnson, & Kolko, 1995; Kolko, Brent, Baugher, Bridge, & Birmaher, 2000). Very recently, the most comprehensive study ever conducted on the efficacy of CBT for adolescent depression was completed (Jensen, 2006). The Treatment of Adolescent Depression Study (TADS) was an incredibly sophisticated, 6-year-long, multisite, randomized controlled trial comparing cognitive-behavioral therapy alone, fluoxetine + CBT, fluoxetine alone, and a placebo control in a sample of 439 adolescents with moderate to severe levels of depression (March et al., 2004). Results indicated that the combined CBT + fluoxetine treatment led to statistically significant reductions in depression compared to placebo. Data also revealed that neither fluoxetine alone nor CBT alone fared better than placebo, although fluoxetine alone had a slight advantage over CBT (March et al., 2004). However, other studies have shown that CBT and CBT + antidepressant medication were similarly effective in symptom reduction (Melvin et al., 2006). Severity of depression, comorbid anxiety or dysthymia, and family factors, such as parent psychopathology, parent conflict, low socioeconomic status, and exposure to stressors, predict attenuated treatment gains (Birmaher et al., 1996; March et al., 2004). In the TADS trial, age, expectations about treatment, severity, complexity, and chronicity also contributed to poorer outcomes (Brent, 2006; Curry et al., 2006). Further work is needed to better elucidate the efficacy of CBT and to develop more effective interventions. However, initial data are positive.

A meta-analysis of published studies on internalizing disorders indicated the following components as key features of CBT for depression in children: cognitive restructuring, problem-solving skills, decision-making skills, relaxation training, coping strategies, and skill generalization strategies (Compton et al., 2002). These combined strategies get at the main goals of CBT, which are to challenge and change thoughts, and then change behaviors. Recent data examining mediators of treatment response suggest that CBT effects treatment outcome though reducing cognitive distortions (Kolko et al., 2000). However, such data are scarce, and more research is needed (Chorpita, Brown, & Barlow, 1998; Kazdin, 2004).

The treatment of depression via CBT is a common practice for older children and adolescents, although randomized clinical trials are far more rare in younger children. However, one innovative cognitive approach to childhood depression is contextual emotion regulation therapy (CERT; Kovacs et al., 2006). The model on which CERT is based focuses on the emotion regulatory aspects of depression. In particular, youngsters who are unable to regulate their distress or dysphoria are at increased risk of developing symptoms of depression. Thus, treatment focuses on teaching children coping skills to reframe their experiences of compromised mood in terms of coping with accumulating stressors in their daily lives. A small, open trial in 15 children ages 7 to 12 years found that 53% of completers had full remission of dysthymia, and at 1-year follow-up this number grew to 92% (Kovacs et al., 2006).

Cognitive-Behavioral Therapy with Anxiety Disorders

Recent epidemiological data suggest that between 12% and 20% of children and adolescents experience clinically significant anxiety disorders (Achenbach, Howell, McConaughy, & Stanger, 1995; Velting, Setzer, & Albano, 2004). Although anxiety may change in presentation, it tends to be stable over time and if left untreated results in significant functional impairment in social, academic, and vocational domains (Gosch, Flannery-Schroeder, Mauro, & Compton, 2006; Velting et al., 2004). As was the case for depression in childhood, only a small proportion of children who suffer anxiety receive adequate assessment and treatment services. Thus, making effective treatments available is essential.

Anxiety has been conceptualized in terms of respondent, operant, and social learning principles. With regard to respondent conditioning models, the association of a neutral stimulus repeatedly paired with an unconditioned stimulus may lead to an unconditioned anxiety response. Unpleasant internal states, in addition to observable environmental stimuli, can also come to have elicitative properties (Gosch et al., 2006; Watson & Raynor, 1920). Mowrer (1960) argued that a respondent conditioning paradigm was not sufficient in and of itself to explain anxiety. He proposed a two-factor theory, which relies on both respondent and operant models. Operant models suggest that avoidance behaviors arise and are maintained by the reduction of fear and/or positive reinforcement by significant others, such as parents (Gosch et al., 2006; Skinner, 1969). Social learning models of anxiety hold that individuals may begin to doubt their self-efficacy in anxiety-provoking situations and thus learn avoidant repertoires vicariously, through observing others (Bandura, 1977). Cognitive models of anxiety, such as Rachman's (1991) neoconditioning theory, stress the importance of verbal constructions of events.

The use of cognitive-behavioral approaches for the treatment of childhood anxiety disorders has been well-studied. Empirical data suggest the efficacy of CBT across a broad range of anxiety and anxiety-related disorders, including simple phobias, Social Phobia, Generalized Anxiety Disorder, Agoraphobia, and Separation Anxiety Disorder (Albano & Kendall, 2002; Berman et al., 2000; Flannery-Schroeder et al., 2005). Findings are positive, and largely consistent, although some

data suggest that children and adolescents who struggle with anxiety and comorbid depression may be less likely to complete or benefit from treatment. For example, in one study of exposure-based treatment for anxiety in children ages 6 to 17, results indicated that children in the treatment failure group were significantly more likely to have comorbid depression (Berman et al., 2000).

With regard to maintenance of treatment gains in anxious children and adolescents, CBT has been found to have significant effects at long-term follow-up (Albano & Kendall, 2002; Flannery-Schroeder et al., 2005). For example, one investigation of how children fared several years out from termination of treatment indicated maintenance of gains from CBT for anxiety at an average of 7.4-year follow-up (Kendall, Safford, Flannery-Schroeder, & Webb, 2004). This may indicate that the skills and patterns learned during CBT for childhood anxiety remain with the child throughout subsequent years and even into different developmental stages. Cognitive-behavioral therapy has been found to be equally efficacious at 1-year follow-up, regardless of whether it was delivered in an individual or group format (Flannery-Schroeder et al., 2005). Delivery of CBT in a group format provides an efficacious yet cost-effective treatment.

Cognitive-behavioral therapy has also been found to be an effective tool for preventing the development of anxiety in at-risk children. A study by Dadds and colleagues (Dadds, Spence, Holland, Barrett, & Laurens, 1997) examined the effects of a school-based cognitive-behavioral intervention program for anxiety disorders in at-risk children ages 7 to 14. Treatment focused on the use of coping strategies in group-based exposure exercises and included child management skills sessions for parents. Results indicate that children in the intervention group had significantly lower rates of anxiety disorders at 6-month follow-up. Intervention also reduced the rates of anxiety in those children who had mild to moderate levels of anxiety at onset. At 2-year follow-up, participants in the intervention group reported significantly lower rates of anxiety when compared to the monitoring group (Dadds et al., 1997).

Cognitive-behavioral approaches to anxiety disorders in children and adolescents contain the same core components common to CBT for other issues. Generally speaking, these include psychoeducation, coping skills training, exposure tasks, and contingency management procedures (Gosch et al., 2006). As in CBT for depression, psychoeducation aims to educate the children and their parents about the cognitive, affective, and physiological components of anxiety. Child behavioral responses to feared situations are typically conceptualized in two ways. Attempts to avoid feared situations or stimuli are described to children as "avoidant coping" and are discouraged. More appropriate replacement behaviors, classed as "coping, or bravery-based responses," are encouraged (Kendall, 1990; Pincus, Eyberg, & Choate, 2005). One well-researched treatment manual targeting younger children uses the analogy of a "scaredy cat" versus a "coping cat" to assist children in understanding and engaging in treatment (Coping Cat; Kendall, 1990).

Cognitive restructuring for child anxiety focuses on teaching children to test their hypotheses about feared situations by engaging in approach rather than avoidance behaviors. For example, socially anxious children often come into treatment

believing they "cannot" tolerate their fear in social situations, or with certainty that they will do "something stupid," peers will laugh at them, and they will feel greatly embarrassed. Cognitive restructuring then revolves around approaching these thoughts as hypotheses and testing them via actually placing children in feared social situations and asking children to self-monitor their fear-based cognitions and check the degree of truth of these cognitions. Children may also be coached to use self-instructions, or positive self-statements, as an additional strategy to support the development and generalization of approach-based responses.

Exposure-based treatment components involve approaching feared situations in a hierarchical manner. Children practice "behavioral experiments," which consist of approaching feared stimuli so that they may watch their level of anxiety rise and then fall until habituation is reached. Children then progress through treatment until mastery is reached at each step; in other words, they have learned and effectively used their approach-based, rather than avoidance-based, behaviors.

Many manualized treatments for childhood anxiety have been developed in recent years and reflect innovative uses of CBT for a spectrum of anxiety disorders. For example, the Coping Cat Program (Kendall, 1990) is widely used and has versions for use with adolescents and school-age children. Other novel uses of cognitive principles are Pincus and colleagues (2005) adaptation of parent-child interaction therapy (PCIT), a well-established behavioral training program typically used with parents of young oppositional children, for use with young separation-anxious children. Two treatment phases in PCIT target the development of a warm and responsive relationship between children and caregivers and then focus on imparting appropriate behavior management techniques. To make it more relevant to separation-anxious children, Pincus and colleagues added a "bravery directed interaction" module in which parents are taught how to encourage approach behaviors and set limits around avoidance. This component specifically targets parent cognitions that may impede their encouragement of their children's approach attempts (Pincus et al., 2005). Other innovative uses include recent adaptations of CBT for groups of young separation-anxious children in the context of a summer camp Ehrenreich, Goldstein, Wright, & Barlow, under review) and for young children with OCD (Freeman et al., 2003).

Cognitive-Behavioral Therapy for Obsessive-Compulsive Disorder

Recent data suggest that CBT is a treatment of choice for childhood OCD (Pediatric OCD Treatment Study, 2004). When treating adults and older children, CBT with EX/RP is considered the first-line intervention (Franklin et al., 1998; J. M. March et al., 1994; Piacentini, 1999). Only recently has the efficacy of CBT, and in particular, EX/RP, been carefully reviewed and studied in children and adolescents (J. March, 1995; Piacentini, 1999). The publication of the Pediatric OCD Treatment Study, a multisite, randomized controlled trial, reported that CBT and CBT plus augmentative pharmacotherapy with serotonin reuptake inhibitors were efficacious in treating pediatric OCD. Although CBTs have been developed and effectively used for older children and adolescents, there has been growing interest in adapting this approach to younger children.

Until very recently, there were no cognitive-behavioral treatments available for children 8 and younger with OCD. However, Freeman et al. (2007) formulated a manualized CBT with EX/RP for early childhood OCD in children ages 4 to 8. Initial results from a small randomized clinical trial indicated that this treatment resulted in more positive gains for children in a family-based cognitive-behavioral treatment condition compared to those in a family-based relaxation treatment (Freeman et al., 2007).

Cognitive-Behavioral Therapy for Young Children with Obsessive-Compulsive Disorder

Onset of OCD has been documented in children as young as 3 years and may be more common in younger children than previously believed (Hollingsworth, Tanguay, Grossman, & Pabst, 1980; Swedo, Rapoport, Leonard, Lenane, & Cheslow, 1989). Although the presentation of OCD is similar across child and adult populations, the developmental context of young children complicates how cognitive-behavioral treatment may best be adapted. When using CBT with young children, it is important to consider (a) their developmental level and (b) their dependence on and involvement in their family system. Failure to address these issues would constitute significant treatment obstacles for the use of CBT with younger populations.

Developmental Issues

It is important to consider developmental factors in treating early childhood-onset OCD for numerous reasons. First, cognitive-behavioral treatments for anxiety and related disorders in children often involve the use of client-generated exposure hierarchies both in and between sessions. Young children may be unable to accurately collaborate in creating an exposure hierarchy due to their more limited skills in recognizing and articulating the level of distress or discomfort elicited by particular stimuli. This may lead clinicians or parents who assist with between-session homework assignments to implement exposure hierarchies that are too easy or too difficult. It is also important to note that early-onset OCD cases are more likely to have comorbid tic disorders and "just-right" obsessions (Geller, Biederman, Griffin, Jones, & Lefkowitz, 1996; Geller et al., 1998; Pauls, Alsobrook, Goodman, Rasmussen, & Leckman, 1995). It is difficult for children of any age to describe "just right" urges, but it is considerably harder for very young children to describe these urges. In a related vein, compulsive behavior in young children is not always perceived as ego-dystonic, but rather, as "just what kids do." Thus, it is important to include the observations of parents in developing an appropriate hierarchy. Additionally, it is useful to revisit the hierarchy in a systematic way during each session, to continually assess its level of appropriateness, and to provide opportunities to incorporate new information as young children progress through the treatment.

When treating young children, it is essential to use simple, concrete, child-friendly descriptions of symptoms or exposure tasks. For example, in J. S. March and Mulle's (1998) cognitive-behavioral treatment for older children and teens with

OCD, the disorder is described in psychobiological terms. With young children, it may be more useful to conceptualize OCD symptoms as a "hiccup of the brain." Younger children may be asked to label OCD symptoms with a silly name or to draw a picture of OCD as a "worry monster." Active participation in exposure may be described as "bossing back the worry monster" (e.g., J. S. March & Mulle, 1998, p. 152). Where possible, relying more on experiential tools in treatment is helpful. For example, active modeling of approach behaviors by clinicians and parents is particularly helpful for younger children.

When considering a plan for assessment during cognitive-behavioral treatment, it is helpful to evaluate symptoms and functional impairment not just at the outset of treatment, but throughout treatment, at termination, and, if possible, at follow-up check-in appointments. Systematic assessment assists in accurately and comprehensively describing treatment gains, as well as possible plateaus and exacerbations of symptoms. Before engaging in treatment, it is helpful to perform a structured clinical interview to assess the breadth of possible symptom areas given the high rate of comorbidity in childhood anxiety disorders, but especially due to higher rates and different patterns in young children relative to older children (Geller et al., 1996, 1998; Pauls et al., 1995). In particular, OCD is often comorbid with other anxiety disorders, mood disorders, tic disorders, and ADHD. With older children, the Anxiety Disorders Interview Schedule (both parent and child versions) is useful.

Family Contextual Issues

Consideration of family contextual factors that may impact how young children will respond to cognitive-behavior therapy is also critical. Young children are far more dependent on their parents for assistance and support than are older children. Parents of younger children are more likely to inadvertently reinforce or even actively accommodate their child's rituals (Lenane, 1989, 1991; Pollack & Carter, 1999; Steketee, 1997). Young children are particularly likely to involve family members in rituals, especially in the form of reassurance seeking (Lenane, 1989; Leonard et al., 1989; Rettew, Swedo, Leonard, Lenane, & Rapoport, 1992). Moreover, parents of younger children may have greater difficulty tolerating the distress that OCD symptoms precipitate in their children (Freeman et al., 2003). Thus, parents are often trained as coaches or cotherapists in CBT approaches with younger children. However, children who experience early childhood-onset OCD are far more likely to have parents who may struggle with their own OCD symptoms, other anxiety disorders, or depression (Nestadt et al., 2000; Pauls et al., 1995; Rosario-Campos et al., 2005). This is complicated by the fact that very young children rely on parents for guidance and direction in remembering and practicing exposure skills at home between sessions. This constellation of factors may lead to significant disruption of family routines and emotional distress as families progress through exposure-based treatment.

Parents struggling with their own compromised mental health may be less willing or able to actively support their children in treatment. In addition, they may have more difficulty tolerating their own or their child's negative affect during

challenging exposure exercises (Freeman et al., 2003). They may also have difficulty modeling adaptive, approach-oriented coping strategies. Studies of child anxiety have shown an effect of parental anxiety on child outcome (Cobham, Dadds, & Spence, 1998). Whether parents use avoidant- or approach-oriented coping strategies with their own anxiety issues, they may be at risk of undermining treatment, whether intentionally or not. This may occur through overt resistance during sessions, noncompliance with between-session homework tasks, or potentially, more subtle indicators such as providing distraction for children during EX/RP or "bailing out" of EX/RP tasks before habituation occurs.

Family accommodation has been associated with a host of additional negative factors, such as less developed problem solving, higher rates of depression in family members, and more expressed anger (Amir, Freshman, & Foa, 2000; Przeworski et al., 1999). Because EX/RP requires children to resist engaging in rituals, such a treatment approach often causes a great deal of distress and disruption of the family. This, unfortunately, is a typical pattern in families of individuals with OCD and may be especially problematic in families of younger children. Thus, effective use of behavior management strategies is critical to support the effective use of exposure and ritual prevention, especially with regard to between-session homework.

It is not uncommon for parents to disagree about the nature or intensity of symptoms or about their engagement in treatment. Thus, clinicians should carefully consider viewpoints of both parents, in addition to children's own report of their symptoms. It is fairly common to receive discrepant reports about an individual child's symptoms across parents and the anxious child. In particular, children may report more distress than parents, although given young children's less developed ability to appreciate and express complex or intense emotions, the reverse may be true here.

Because of these complex issues, the direct involvement of families in treatment is essential. Direct involvement in planning, exposure, coaching, and psychoeducation may improve the child's treatment outcome, adherence, and motivation. Research has indicated that the addition of a developmentally appropriate family component to CBT leads to more positive outcomes (Waters, Barrett, & March, 2001; Wood, Piacentini, Southam-Gerow, Chu, & Sigman, 2006). This may be especially true for younger children with OCD.

The cognitive-behavioral treatment illustrated in this chapter reflects current clinical treatment outcome research investigating family-based CBT for young children (ages 4 to 8) with OCD. This treatment explicitly addresses young children's cognitive and emotional sophistication, the role of parents or caregivers, and common comorbid conditions (externalizing and internalizing disorders). To date, a preliminary clinical trial has concluded, and a larger multisite replication and extension is under way (e.g., Freeman et al., 2003). Initial results are promising. The following case study illustrates the adaptation of cognitive-behavioral treatment for a young child struggling with OCD and his family. It also describes the use of new advances in cognitive-behavioral treatments, including acceptance and mindfulness techniques.

Case Illustration

Identifying Features of Client

Max was a 5-year-old Latino boy presenting to an outpatient child anxiety treatment center that is part of a Department of Child and Family Psychiatry at a large public hospital in an urban setting. Max was brought in for an initial evaluation by his mother and was accompanied by his older sister, Nina, who was 9, and his infant brother, Carlos, who was 7 months old. Max appeared slightly tall and underweight for his age. He made limited eye contact with his interviewers. He was neatly groomed, although his clothing fit poorly and seemed to be several sizes too large. He was extremely polite during the interview, often replying "Yes, sir" and "No, sir" to the interviewer. He fidgeted and shifted in his chair throughout the evaluation and appeared to make some subtle facial grimaces, lifted his eyebrows, and moved his eyes upward and from side to side. Max spoke quickly and quietly during the interview and at times seemed distracted from what he was saying. This tendency was so pronounced that the interviewer prompted him several times to finish what he had started to say.

Max spoke English quite clearly, without accent, although the family spoke primarily Spanish in the home. Max's mother spoke some English, but often relied on Max or his older sister to translate the interviewer's questions and comments. The family emigrated from the Dominican Republic 2 years ago, when Max was 3. They moved to the United States to join other members of their extended family who live and work here. Max's father worked for an upholstery cleaning company. His mother stayed at home with the children and also cared for Max's maternal grandmother, who was infirm. The family rented a small second-floor apartment in a predominantly Latino neighborhood.

Presenting Complaints

Max's mother and Max himself served as informants. Max was referred for evaluation by his family physician after a visit during which his mother complained that he was often noncompliant and was very anxious about getting sick. Max's mother described him as "stubborn," always wanting things his own way, even if it bothered other members of the family. Her primary complaint was that Max spent a great deal of time in the bathroom, often using up all the soap, shampoo, and toilet paper so that there was none left for the rest of the family. Often, there was no hot water left for showers or baths. This was a particular problem due to the family's limited budget. Max also insisted that his mother wash his toys when he felt they were dirty, and he grew quite distressed if they were not "clean enough" or if she refused to wash them.

In addition to this, Max also was a "picky eater." Max's mother reported that she often cajoled and scolded him to eat his meals, but typically, he would take only a few bites. Max described feeling that there was something "wrong" with his food. According to his mother, he tended to get particularly distressed if different types of food "touched" or got mixed together on the plate. Max's mother also described incidents during which Max would come down to breakfast and refuse to eat anything, and she reported that he would not say why. When asked directly about this by the interviewer, Max simply sat silently. Finally, Max reportedly tended to hoard a number of items that did not appear to have any sentimental value, notably, used tissues, rocks, and twigs. Max threw tantrums if he was prevented

from "doing things his way," according to his mother. He also sometimes hit or screamed at his older sister if she tried to "boss him," according to Max.

The amount of time Max spent washing, eating meals, refusing to speak, and hoarding objects had grown over time, and at the time of evaluation, constituted a significant proportion of the day. Max often missed opportunities to play with his older sister and with neighborhood children due to his involvement in ritualized behavior. In addition, his distress had also grown over time. At first, he did not appear to notice how disruptive his symptoms were and simply thought of them as "what I do." However, over the past several weeks, he had become more aware of them, concurrent with their increasing intensity. According to his mother, he described having more "bad days" than good days, and he sometimes cried in frustration at feeling that he "had to" continue engaging in rituals.

Max's mother reported that he was having difficulties at school, in addition to his struggles at home. According to her, his teacher was concerned that he had difficulty sitting still in class, completing tasks and art projects, and practicing holding and using pencils correctly. He also had some difficulty making friends and tended to play by himself on the playground. Occasionally, he got into physical fights with the other children, although he often denied this and would not tell his mother any details about the altercations.

Max's father reportedly did not think there was anything wrong with Max. According to Max's mother, he viewed Max as very similar to how he himself was as a child. Max's father felt strongly that Max's behavior was a "family issue" and that the family should not be seeking special help from another doctor. He believed that Max would "grow out of it" and was not supportive of Max's initial evaluation visit.

Developmental Factors

A number of factors were considered essential to adapting a cognitive-behavioral treatment for Max and his family. These included, first, assessment of Max's rituals as developmentally normative versus clinically significant. Second, it was important to determine Max's ability to recognize and express his anxiety in such a way as to contribute to building an accurate, workable fear hierarchy.

Parental Factors

Given Max's young age and relative dependence on his mother for guidance and limit setting, family accommodation of his rituals constituted a significant obstacle to treatment. In Max's case, there was already some evidence of his mother's acquiescence to his requests that she assist in his rituals. Moreover, other family members, especially his older sister, were considered unwitting participants in helping maintain his OCD-related behaviors.

In addition, several other familial factors stood out as potentially critical factors in using CBT effectively. First, his parents disagreed about the nature of Max's symptoms. His mother appeared concerned, at least enough to follow the advice given her by her family physician. However, Max's father saw his symptoms as more of an adjustment issue that Max would outgrow. Thus, the viewpoints of both parents, in concert with Max's own, were considered. In light of the parents' disagreement, the clinician decided to carefully observe Max's level of anxiety during sessions and to foster his ability to communicate how he felt throughout the treatment. Additionally, psychoeducation provided during CBT

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helped to create more consistency in how Max and his parents understood his symptoms, his level of distress, and the functional impairment the symptoms created.

Given that Max often grew oppositional when limits were set on his compulsive behavior, his family may have been involved in subtle or overt accommodation of these symptoms. Again, educating the parents with regard to the nature of Max's noncompliance or distress around thwarted rituals was central to treatment. Highlighting the parents' distress when observing their son's discomfort when he was prevented from engaging in rituals enhanced parent treatment compliance. It was also useful to enlist the help of his older sister to prevent her inadvertent accommodation of his compulsions.

Finally, because Max and his family were Dominican, and perhaps more important, were recent immigrants dependent, at least in part, on an extended family network, the therapist took great care in assessing cultural differences in how young children were treated in both Max's immediate and extended family. It was also important to accurately conceptualize the degree to which the family had acculturated. In less acculturated Latino families, children may be given less autonomy than might be expected in the majority culture in the United States. However, there is such diversity in Latino groups that it is important that clinicians do not make assumptions about cultural differences (Szalacha et al., 2003). Both of these related factors impacted Max and his family's ability to understand, implement, and benefit from cognitive-behavioral treatment. Max's clinicians made every attempt to elicit and welcome discussion of cultural differences and to structure CBT in a culture-specific, culturally acceptable way. Failure to do so might have resulted in treatment resistance or even dropout.

History

Max reached all developmental milestones on time. As an infant, he had difficulties with transitions and novelty, often becoming overwhelmed when experiencing new situations, toys, or people. He stayed at home with his mother until the year of his initial assessment, when he began kindergarten at a local public school. At the beginning of the school year, Max had separation difficulties and often cried inconsolably when his mother dropped him off at school. He often sought reassurance about when his mother would pick him up from school and where he should wait for her when she did so.

Max had engaged in ritualistic behavior since he was 4, but this became more pronounced upon his entry into preschool. At age 5, one of Max's more involved rituals involved complicated sequences of events, which he requested his mother to perform, at bedtime. For example, Max required his mother to lay out his pajamas in a particular way on the bed and help him take off his play clothes in a particular order. After his pajamas were put on in a circumscribed sequence, Max brushed his teeth until they "felt" clean, which often went on for more than 5 minutes. Following that, he required his mother to line up his animals, tuck him in "just so," read three bedtime stories, and kiss him twice on each cheek. She then had to say "I love you" and "Don't let the bedbugs bite." Failure to perform any of these behaviors or doing them out of sequence caused Max great distress. If that happened, Max often cried until they could repeat the sequence correctly. Repeating morning routines resulted in lateness to school as well as general irritability toward his teacher and peers.

Most recently, Max had become concerned with "germs" and often complained to his mother that things were dirty and need to be cleaned. He made frequent attempts to have his mother wash his toys. She acquiesced because this placated him. Max was also very worried about toileting and spent an excessive amount of time cleaning himself after using the bathroom. This often resulted in dry, irritated skin and a great deal of discomfort. Other areas of concern for Max were that his food may be contaminated and that he felt compelled to keep nonsentimental items. His collection of used tissues, for example, constituted two large black garbage bags in his room. He became very distressed when any family member touched or moved the bags.

In general, the frequency and intensity of Max's rituals were exacerbated by stressful events. In addition, Max appeared to tic more during symptom exacerbations. Onset of his symptoms was gradual, rather than acute. As a toddler, he hated sticky or greasy substances on his hands and typically asked his mother or sister to wipe or wash them. He would tantrum until they did so.

Max did not have a history of significant medical problems. His mother had discussed Max's "picky eating" with the family physician, who reported that Max was "on the thin side" but that his weight and growth were within normal limits. Max's mother reported that she suffered from "nervous spells" and that Max's father was "moody." No other history of mental health issues was reported.

Assessment

Because Max was 5 and the Anxiety Disorders Interview Schedule for *DSM-IV* has not been normed in young children, a broader structured clinical interview, the Schedule for Affective Disorders and Schizophrenia for School-Age Children—Present and Lifetime Version (K-SADS-P/L; Chambers et al., 1985; Kaufman et al., 1997) was used to establish his diagnostic status. The K-SADS-P/L is a semi-structured, clinician-rated interview that yields *DSM-IV* diagnoses across the Axis I domains. The interview is administered to the parents (or primary caretakers) regarding the child, and to children 6 years and older. The K-SADS is routinely used to assess psychiatric diagnoses in children as young as 5 years (Hirshfeld-Becker et al., 2004; Youngstrom, Gracious, Danielson, Findling, & Calabrese, 2003).

The Children's Yale-Brown Obsessive Compulsive Scale (CY-BOCS), a 10-item semi-structured clinician-rated interview, was administered to assess OCD symptoms and severity at baseline, at midtreatment, at termination, and at all follow-up check-ins. The CY-BOCS rates obsessions and compulsions on 5-point scales on five dimensions (time, interference, distress, resistance, control) and yields a total obsessions score (0 to 20), a total compulsion score (0 to 20), and a combined total score (0 to 40). This scale has adequate psychometric properties and, more important, has been shown to be sensitive to change in both behavioral therapy and pharmacotherapy treatment studies. The CY-BOCS has been successfully used with children as young as 6 years (J. S. March & Leonard, 1998).

Because Max demonstrated some facial tics, the presence of a tic disorder was assessed with the Yale Global Tic Severity Scale (Leckman et al., 1989). This is a clinician-rated scale used to assess severity of motor or phonic tics, as well as related impairment. Motor and phonic tics are each rated on a 6-point scale

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regarding number of tics present, frequency, intensity, complexity, and interference. Motor and phonic tic scores may range from 0 to 25, and the combined total score ranges from 0 to 50. There is also an impairment score that provides a score from 0 to 50 that reflects overall burden due to tics. This instrument has demonstrated excellent psychometric properties (Leckman et al., 1989).

In addition to these measures, more general indexes of functioning in both home and school environments were sought. The Child Behavior Checklist Parent and Teacher Report forms (CBCL; Achenbach & Rescorla, 2001) is a widely used scale with well-established psychometric properties used to assess an array of behavioral problems in children ages 1.5 to 18 years.

Max's clinician was interested in both the impact of treatment and the rate at which Max's behaviors changed. The clinician also wanted to provide feedback to Max's mother regarding her son's progress, in the hopes that when treatment hit a rough patch, the feedback would be a motivating factor for the family. Thus, symptoms were assessed at several points throughout the treatment. To assess symptoms at midpoint, termination, and follow-up appointments, the CY-BOCS and CBCL were used, as well as unstructured interviews with Max's mother and teachers. General indexes of functional impairment were used, including Max's self-reported distress and level of satisfaction with school, his peer relationships, and his interactions with his family.

It is important to note that the assessment measures just described have not been used extensively with minority groups. However, they were considered the most appropriate measures available for use in this case. Thus, the scores ascribed to Max from these instruments were interpreted with some caution and were reported to Max's family and teachers with the appropriate caveats.

Case Conceptualization

Max is a 5-year-old Dominican male who, after participating in an extensive assessment procedure, was diagnosed with clinically significant OCD and secondary Tic Disorder. Although the externalizing disorders ADHD and Oppositional Defiant Disorder were considered, Max's presentation did not warrant assignment of these diagnoses.

Obsessive-Compulsive Disorder in children, as well as in adults, is characterized by recurrent, upsetting obsessions and repetitive, purposeful compulsive behaviors. Max's obsessions involved contamination fears: excessive and irrational worry about getting severely ill through exposure to germs and microbes, as well as contaminating others. He exhibited a number of compulsions related to these fears, including excessive hand washing, showering, and wiping after using the toilet. Max also had concerns about germs or poison in his food, although these worries included a general sense that "something was wrong" with his food when different food items touched on his plate. His compulsion regarding this was to refuse to eat or request a new plate of food items that did not touch. Max also felt compelled to hoard items that had no sentimental value and did not constitute items that would be considered developmentally normative to save. At school, Max reported that when he made pictures or held his pencil, it did not feel "right." Thus, he avoided coloring and writing altogether. If that was impossible, he would continue to make new pictures until he created one that was "perfect."

Max exhibited a number of motor tics, particularly facial grimaces, eyebrow movements, and shifting his eyes from side to side. Occasionally he did exhibit mild vocal tics, including clearing his throat and sniffing. However, the tics did not tend to co-occur and waxed and waned over time and across situations. They grew more pronounced in stressful situations or when Max felt vulnerable. The frequency and intensity of these tics was mild to moderate, and often Max did not realize that he was engaging in them. His teacher and the other children had begun to notice them at school; however, they did not constitute a significant source of distress for him.

Although Max was described by his mother as oppositional and noncompliant, those behaviors typically occurred in the context of ritual prevention. Rather than reflecting clinically significant oppositionality, they were more a manifestation of Max's frustration and anxiety when he felt compelled to engage in ritualized behavior. For example, Max's refusal to talk, although suggestive of oppositionality, actually represented his compulsive attempt to "protect" his family from contamination by germs Max breathed out. Max was also assessed for ADHD, in part because of a high level of observed activity as well as some symptoms typically attributed to inattention. However, interviews suggested that these were more reflective of his being distracted by obsessive fears. Given the high rate of comorbidity of OCD, ADHD, and Tourette's Disorder, Max would be watched over the course of treatment to remain vigilant for any changes in his symptom presentation.

Max's family had certain strengths on which to draw as they progressed through treatment. Namely, they were part of an extended family network that may be helpful in providing social support. Additionally, the fact that Max's mother sought the help of her family physician, and followed that advice to seek the services of a clinical psychologist suggested that she recognized a problem in her child's behavior and was amenable, at least initially, to participating in treatment.

Max's family was also characterized by what might constitute significant obstacles to treatment. These included many of the developmental factors outlined earlier, including Max's young age, his dependence on his parents, the family's potential difficulty in understanding the rationale of treatment, and issues with emotion regulation around his symptoms. Parental issues, including frank accommodation of Max's symptoms, his mother's level of anxiety and potential difficulty tolerating her own and Max's negative affect, and her lack of support from her husband, were complicating factors.

Course of Treatment and Assessment of Progress

Given Max's symptom presentation, in concert with developmental and familial issues outlined earlier, it was decided that Max would benefit from a family-based cognitive-behavioral approach, specifically adapted to a child of his age. This treatment was developed by researchers at Rhode Island Hospital/Brown Medical School in the Pediatric Anxiety Research Clinic (Freeman et al., 2003). The treatment was designed specifically to treat OCD in young children and, as such, does not directly target Tic Disorder. Treatment of choice for clinically significant tics would be habit reversal. However, given that Max's tics were not as pronounced or distressing as his symptoms of OCD, treatment was designed in a sequential way to address his OCD first.

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The family-based cognitive-behavioral treatment was designed to address the unique needs of young children with OCD. In particular, it attends to young children's cognitive and socioemotional issues, the critical role of the family and their beliefs and behaviors around OCD symptoms, and the potential need for training in appropriate behavior management skills. Therapeutic material is tailored to younger children by including simple, concrete, child-friendly metaphors and experiential exercises. A great deal of attention is paid to young children's relatively less sophisticated emotion recognition and expression skills. Thus, a "child tools" section of each therapy session includes strategies to help children understand the treatment rationale and develop skill in acknowledging and expressing their anxiety along a continuum. The treatment also incorporates basic behavioral management strategies that parents may utilize in many situations, including, but not limited to, the child's OCD symptoms. These include overt modeling of approach behaviors, as well as psychoeducation regarding accommodation of symptoms. As an additional measure, the treatment deals with parents' and children's thoughts and feelings about and within their relationship and considers the role these may play in parent responses to child symptoms. Thus, both child and parent behaviors are targets of treatment.

Treatment consists of five main components, which are described next. This particular treatment was designed and its feasibility and efficacy compared to a family-based relaxation treatment (Freeman et al., 2007). It comprises 14 sessions. The first two sessions are with the parent(s) alone and generally last 90 minutes. The remaining 10 sessions include both parents and child and last approximately 60 minutes. Follow-up sessions are added as needed to address any residual OCD symptoms or symptom exacerbations that may occur over time. Because this is a manualized treatment, there is a suggested structure in the sequence of presentation of each element. However, as with most CBT manualized treatments, flexible use tailored to each individual client's needs is encouraged, with the constraint that all treatment components are eventually addressed.

Psychoeducation

During this portion of the treatment, the family is given education about the nature of OCD. Psychoeducation occurs during the first two sessions for the parents and the third for the child, according to the manual. However, in practice, it is also woven throughout sessions as needed. The main goals of this treatment component are to describe OCD as a neurobiological disorder such that it is understandable to the family. Oftentimes, parents who do not understand the nature of a psychological illness such as OCD are likely to blame their child or interpret symptoms as oppositionality or stubbornness. Similarly, they may be likely to require their child to "just cut it out." Providing detailed information conceptualizing OCD as a brain-based behavioral disorder helps to redirect parent change strategies in ways that may be more useful for their child. Thus, a second goal of this therapeutic component is to correct misattributions about OCD.

Children with OCD may blame themselves for their symptoms. They may also see their symptoms as part of themselves and view them as impossible to change. A core goal of psychoeducation is to help children view OCD symptoms as separate from themselves. This is accomplished through a number of strategies, including asking children to draw a picture of their OCD, asking

them to think in terms of "bossing back" (see J. S. March & Mulle, 1998) their OCD instead of allowing the OCD to "boss" them, and drawing a map of what parts of their life are under their own control and what parts are under the control of the OCD. The overarching goal of therapy is to increase the domains of the children's life in which they, and not OCD, are in charge.

During this portion of the treatment, it is of utmost importance to describe to parents what they can expect. Treatment is designed to facilitate the child to participate in exposure and response prevention in a graduated way both within and outside of session. Parents are asked to become part of the treatment team and, as such, act as coaches for their child throughout the course of therapy. Because exposure and response prevention are inherently uncomfortable and stressful for children and parents, therapists must take care to address this issue in an honest and realistic way. Parents should be told that the treatment is often stressful and may disrupt family routines. If this is not carefully discussed, parents may have difficulty accepting their child's or their own affect during exposure exercises. This can impede the progress of therapy, lead to setbacks, and, ultimately, to dropout.

The following is a sample clinical transcript of how the therapist approached Max's parents:

Therapist: First and foremost, it is important for you to know how exposure works and what it may be like for your family to go through it. Helping Max participate in exposure will be hard, although we will start with easier exercises, and then move on to harder ones. It will likely be stressful for you as parents to watch Max go through the treatment, and watch him feel anxious. He may get scared, sad, or angry. Those are normal responses, and come from being prevented from engaging in rituals, which he now uses to make himself feel better. Even though he will feel uncomfortable, and it may be hard for you to watch and participate in that, it is important that he continue the treatment, but also that he does his exposure exercises until his level of distress goes down. Through doing this, he will learn that whatever anxiety and distress he feels is simply his OCD bossing him around. And if we as a team help him put his foot down and stand up to his OCD, he will learn that he doesn't have to do his rituals. Eventually, our goal is to help him get more of his life back, so that he can do more of the fun things he wants to do, without being disrupted by his OCD symptoms. So if it gets difficult for you, try to keep that in mind if you feel your resolve weakening.

To increase both parent and child participation, it is important to make the process of psychoeducation as simple and interesting as possible. This not only assists therapists in building rapport with the family, but also sets the tone for therapy. Presenting the information in several modalities and using visual imagery, metaphors, and concrete examples make it more memorable.

The following is an example of how a therapist described the nature of OCD to Max:

Therapist: Sometimes we all get thoughts that pop into our heads over and over again, like when a song gets stuck in your head. The problem is that sometimes those thoughts are *worries* and they scare or upset us. When someone

(*Continued*)

gets lots of worries that keep popping into their head over and over again, and they feel scared or upset a lot, we might call these thoughts (for example) "The Worry Monster." To try to make those worries stop, a person might try doing some different things. Like, if someone felt dirty, they might wash their hands, or if they felt like they said something wrong, they might need to repeat it or apologize for it. These behaviors are called (for example) "re-do's." The problem is that you feel better for a second, but then you worry again and you have to do these things again and again. (Coyne, McVey-Noble, Freeman, & Garcia, 2005, p. 5)

To describe to Max how OCD might impair his functioning, the therapist relied on a comical, child-friendly metaphor:

Therapist: Imagine you bought a hermit crab and it was really cute and cool to play with. Every day you fed it and changed its water and hung out with it. That would be fun, right? Then imagine that it started to grow really big, and it started to follow you everywhere. Sometimes it even ate your lunch and your favorite dessert, and it annoyed you when you were playing with your friends and wouldn't let you watch your favorite TV shows, and it made a lot of noise when you were trying to go to sleep at night, and maybe even tried to sleep in your bed, which would squish you. Pretty silly, but not much fun, right? Then, imagine that it started being a really bratty crab and giving you little pinches unless you paid attention to it and gave it snacks all the time. Not nice. That's what happens when those worries take over and start being the boss of you. So, let's make you the boss of you and your worries! (Coyne et al., 2005, p. 2)

Parent Education

In addition to educating parents and children about the nature of OCD symptoms and distinguishing those symptoms from normative developmental rituals and any comorbid symptoms of other anxiety disorders, therapists also teach parents skills for behavior management. Although parents may enter treatment with their children at various levels of knowledge and performance of appropriate behavior management strategies, it is important to help parents strengthen this skill set and adapt it to their child's OCD symptoms. Tools addressed in this program include differential attention, modeling, and scaffolding. These skills are taught in session and practiced in the home. Therapists are trained to troubleshoot incorrect or inconsistent use of these skills and to help foster better usage, especially with regard to the child's OCD symptoms.

Differential attention refers to removing attention to a child's attempts to resist or refuse to engage in exposure or homework exercises, while attending to the child's approach behaviors. Use of this tool reflects the hypothesis that parent attention is an important motivational factor in CBT for early-onset OCD, as previous research has shown that family members often accommodate children's symptoms.

As previously discussed, parents learn to model approach behaviors for their children. Reasons for teaching this particular skill come from a social learning perspective that assumes that the parents' modeling of approach behaviors will help motivate and support their children in engaging in exposure and exposure-based homework during the course of treatment.

Finally, scaffolding (or coaching) involves parents' assessment (guided by the therapist) of their child's progress through treatment and encouragement of their

child to engage in increasingly difficult exposures. It involves empathic encouragement and includes eliciting the child's feelings, use of a brainstorming approach to collaboratively come up with a way to approach rather than avoid the stimulus, choosing a method of approach, and implementation of that method. Clinicians may teach parents by example or in a more didactic way. Approaching the potentially hesitant child in this way helps to engender and maintain the child's positive momentum and may prevent treatment resistance or fallbacks. It also assists the child in learning emotional and behavioral regulation and the internalization of these responses. This is thought to promote generalization of gains made across settings and time.

The following is a sample of how the therapist addressed the brainstorming component of scaffolding with Max's parents:

Therapist: It is often helpful to brainstorm with your child reasons *why* he should do EX/RP. You might remind him about the hermit crab metaphor [see previous example], and say that OCD doesn't go away simply if you ignore it—it just gets bigger! Another way is to use the idea of "bossing back" OCD. You might ask Max who he wants to be the boss—himself, or mean old Mr. Worry.

Externalizing Obsessive-Compulsive Disorder and Exposure/Response Prevention

Exposure and response prevention is a core component of cognitive-behavioral treatment for OCD. This involves graduated exposure to feared stimuli and prevention of discomfort-reducing responses until habituation is reached. Generalization of exposure practice and response prevention is facilitated by the use of a token reward system implemented by parents. To that end, both parents and children learn to externalize, or separate, OCD symptoms from themselves. For young children, this is best accomplished through having them "boss back" OCD. Children also learn how to use a "fear thermometer" to help them report gradations in their experience of discomfort, whether to structure an anxiety hierarchy or to communicate rising and falling levels of anxiety during exposure. Parents are encouraged to use rewards to encourage children to practice EX/RP at home between sessions.

The following is an example of how a clinician educated Max about the use of a fear thermometer:

Therapist: Do you know how a thermometer works? (Child nods.) Well, my feelings thermometer works exactly the same way. When you don't feel scared at all, the red stuff stays down at the bottom. If you are sort of scared, the red is somewhere near the middle. If you are really, really scared, the red is all the way at the top. Can you think of how scared you might be if you were sitting on your couch, watching your favorite show, eating popcorn? Can you show me where the red might be on my thermometer? (Give child a chance to point out fear level, typically at the bottom.) What about if you were really really scared of heights, and you had to parachute out of an airplane? Where would the red be then? (Give child a chance to point out fear level, usually near the top.) Well, we are going to use this thermometer to figure out how much your OCD bugs you when you don't do your re-dos (compulsions).

(*Continued*)

Family Process Components

Family accommodation of symptoms is addressed very specifically via this component of treatment. This is often a process that continues throughout the course of therapy, as parent insight and ability to discriminate between accommodating and nonaccommodating responses grows. Sometimes parents are angry with or hostile toward their children, given that child symptoms have the potential to significantly disrupt family routines. In a related vein, parents may have great difficulty tolerating their own distress when watching their children struggle with symptoms. Thus, parents are encouraged to discuss these emotions, and in turn, to engage in more adaptive problem solving with regard to their child's symptoms. Finally, in this treatment component, therapists may highlight for parents their own anxious interpretations and behaviors about their child's symptoms.

The following reflects a conversation between the clinician and Max's parents regarding their feelings about his OCD:

Therapist: What is it like for you when you see Max struggling to wash his hands over and over?

Parent: Well, it's really hard. It breaks my heart to see him so anxious. I'd do anything to make him feel better.

Therapist: What are some things you do when you see him like that?

Parent: I try to tell him it's okay, but that doesn't help at all. I plan on extra time getting anywhere, because I know that Max will have to spend some time washing and checking his hands. I buy really gentle soap so that his hands don't get so chapped. I even write notes to his teachers when he is having a bad day, so that they will be more understanding and let him spend extra time in the bathroom, and he won't get in trouble.

Therapist: That sounds like it's pretty distressing for you as well, and that you try to do a lot of things to give Max the space he needs to do his rituals.

Parent: It is hard for me, but it's definitely harder for Max. I do what I can to help.

Therapist: get how important it is for you to help Max. I want you to consider this: What if Max had a medical disorder, like diabetes, and you had to teach him to manage his sugar intake?

Parent: Okay.

Therapist: What if he threw a really big fit one day, because he really wanted a piece of chocolate cake? Say his sugar was fairly high already that day.

Parent: Well, he couldn't have it. I wouldn't let him have it.

Therapist: But what if he got really upset?

Parent: Well, it just wouldn't be an option—it would make him sick.

Therapist: But it would be hard to say no, wouldn't it?

Parent: Sure, but I would know I was doing what was best for him.

Therapist: OCD is exactly like that. If Max is allowed to do his rituals, then he will likely keep having his OCD symptoms. If you help him learn to resist them, even though it will be hard, you will be doing what is best for him.

Contingency Management

In addition to the parenting tools described earlier, therapists also address potential (or actual) resistance to engaging in EX/RP by teaching parents appropriate limit setting. Sometimes young children develop compensatory strategies to cope with their OCD, such as oppositional behavior when required to approach a feared stimulus. If this is the case, clinicians may impart to parents how to effectively use response-cost procedures to address child resistance or token systems to encourage EX/RP practice. Once parents learn this component, their skills in this area may generalize to other domains of their children's behavior that are not specific to their OCD.

Mindfulness, Acceptance, and Values

What have been called "third wave" behavior therapies have developed technologies for addressing thoughts and emotions in a slightly different way than traditionally defined cognitive-behavior therapy. Acceptance and commitment therapy (ACT) is one such treatment paradigm. It may be classed as a cognitive-behavioral therapy in that it addresses the role of cognition in behavior. Thus, individuals progressing through the therapy learn to attend to their internal psychological experiences and the meanings with which they imbue external events. However, whereas traditional cognitive therapies conceptualize maladaptive behavior as the product of faulty cognition and therefore strive to change the content of such mentation, ACT stresses mindful awareness and acceptance of unpleasant or unwanted thoughts. As such, individuals are taught in an experiential way that they may simply notice and detach from their thoughts without evaluation or response. Doing so permits them to engage in valued, personally meaningful activities. Accruing evidence suggests that ACT may work through different mechanisms than CBT (Hayes, 2005).

Core components of ACT include mindfulness or awareness and valuing. Incorporating these components of ACT into CBT approaches may help to address motivational issues in treatment. Mindfulness has been defined in similar ways across the literature and generally means moment-to-moment awareness of one's experience—including thoughts, feelings, and physiological responses—without judgment. With regard to parents, mindful acceptance may help parents identify without response their thoughts and feelings about their child. This may minimize parent reactivity to their child's expression of strong emotion when rituals are prevented, and thus may help parents resist accommodation. For children, learning to be mindfully aware of their own thoughts and feelings in a detached way may augment their approach attempts during exposure exercises. Moreover, it may facilitate children's attending more fully to their actual experience of exposure, rather than their "stories" or expectations about how it will go. In this way, they may be more sensitive to actual environmental contingencies, and thus may benefit from exposure more fully. Finally, teaching mindfulness skills to children may help inoculate against mental rituals or "white knuckling" and engender more willingness to engage in exposure. This may also reduce the possibility of parents' coercion of the child into the exposure.

Here is an example of how mindfulness was used to increase the efficacy of exposure with Max:

(*Continued*)

Therapist: I am going to show you a way to be what we call "mindful" when you are doing your exposure work. We're going to practice being scientists. Do you know what scientists do?

Max: They find out stuff, I guess.

Therapist: How do you think they do that?

Max: I don't know. Maybe they watch stuff to see what happens?

Therapist: That's exactly right. Scientists watch stuff and see what happens. But here's the trick—they watch stuff without trying to change it.

Max: I don't get it.

Therapist: Think of it this way. When you eat an ice cream cone, what do you think about?

Max: Nothing, I guess. I just eat it.

Therapist: Yup. And you probably notice that it's cold, and that it's sweet, and maybe a little melty if it's hot out, and that the cone is crunchy, like a cracker, but sweet too. Well, that's like what a scientist does—noticing stuff, without trying to change it. That's what we call being mindful. Can we try to practice that right now?

Max: I guess so.

Therapist: Okay. Close your eyes. . . get comfy in your chair. . . notice how your body feels. . . where it touches the chair, where your feet touch the floor. . . notice all the sounds you hear. . . see how many different ones you can notice. . . see if you can notice how warm or cool the room is. . . see if you can tell what color the light is on your eyelids. . . see if you can notice your breathing, gently, in and out. . . notice that the air is cool when you breathe it in, and warmer as you breathe it out. Now notice what you are thinking. . . if you are seeing pictures in your head, see if you can print them on billboards on a highway. . . imagine that each thought or picture in your head is on a billboard. . . and just look at it until it passes and a new image pops in.

The therapist practiced this skill with Max in the context of his neutral thoughts, feelings, and bodily sensations, and then progressed onto distressing thoughts and images in the context of exposure exercises. Because mindful awareness constitutes exposure to feared psychological events (called *defusion* in the ACT literature), it is helpful to encourage children to practice it in a graduated way, just as they would with more traditional modes of exposure.

Mindfulness and acceptance are components that have been used in a number of different cognitive-behavioral therapies, including those targeting reduction of stress, depression, and anxiety (Hayes, 2005; Hoppes, 2006; Levitt & Karekla, 2005; Roemer, Salters-Pedneault, & Orsillo, 2006). *Valuing* is a component unique to ACT. It refers to activities and ideals that are meaningful and vital to an individual (K. G. Wilson & Murrell, 2004). Values are not goals; rather, they are directions in which to move. As such, engaging in a valued activity is inherently reinforcing, and self-renewing, since values cannot be reached, but rather, serve to direct behavior. Clinicians may harness valued action in treatment to help motivate parents and their children to engage and actively participate in treatment. For example, if parents feel that helping children engage in EX/RP is consistent with the value of being a good parent, this may inspire better treatment adherence, as well as dignify their often difficult work in and outside of session.

The following is a sample of how the clinician worked with Max's mother to identify her values with regard to Max:

Therapist: (After a brief, experiential centering exercise, eyes closed) I want you to imagine Max 13 years from now, at his high school graduation. There are many people at the ceremony, graduates, their families and friends. You see Max from where you are sitting. . . . As part of the ceremony, there is an announcement that one of the graduates will give a speech. There is silence, and you see Max slowly rise and walk toward the podium. When he reaches it, he clears his throat, and looks for you in the crowd. Notice his face, his eyes, the way he looks at you. He smiles. Imagine that he begins to speak about his struggle with OCD, and your role in that struggle. I want you to take a few minutes and imagine where you would like for him to be at that moment in his life, and how you would like him to describe your role in his struggle. (Pause.)

Mother: I want him to be happy, to be successful, to have a good life. I want to fight for him. I want him to say I fought for him, and that I helped him. That I didn't give up on him.

Therapist: (Pause) Is that what is most important to you?

Mother: Yes. More than anything. I want to protect him.

Therapist: What if our work here, even when it is at its hardest, is the best way to protect Max? What if it is a step toward Max having a successful, good life? What if this is one way you can fight for him, right now, by helping him to do his exposure work?

Mother: I would do it.

Therapist: Even when it is hard for him, and for you. . . and it will be, if it is to work. . . I want you to keep that picture of him in your mind, and in your heart. That will be our goal—to do the work here, now, that will help Max get to that place in the future, where he is living the life he wants, and doing the things he most cares about. You will help him get there.

Complicating Factors

Many factors that may complicate a family's progress through treatment have been detailed in previous sections. Two broad areas—developmental issues and parenting issues—were identified. Given Max's young age, his developmental stage and skills in emotion recognition and expression, care should be taken by his clinician to ensure that encouragement of these skills is woven into treatment. Similarly, given his developmental level, Max is more embedded within his family and dependent on his parents than an older child might be. This highlights the importance of motivating parents to become involved and to maintain their involvement in Max's treatment. One obvious difficulty is Max's parents' contrasting views of his OCD symptoms. His mother views them as impairing and emblematic of a pathological process; his father views them as a phase that he will outgrow. A clinician working with this family would be well advised to help build consensus between the parents, as well as to be vigilant for any parent behaviors that might undermine Max's treatment.

Parents may have difficulty tolerating their child's distress as the family progresses through treatment. This may lead to parent accommodation of child symptoms in the name of reducing child distress and minimizing disruption to

(*Continued*)

the family. Clinicians should offer support for parents in this area and take opportunities to notice, empathize with, and normalize perceived parent stress. Additional treatment components, such as motivational interviewing and acceptance and mindfulness-based techniques may be helpful with regard to facilitating treatment adherence in both parents and children, although data on these approaches are limited (Barmish & Kendall, 2005; Gance-Cleveland, 2005; Semple et al., 2005).

Max's Dominican culture may also prove to be a complicating factor in treatment, especially if the clinician is unaware of (or worse, insensitive to) important cultural differences that may impact treatment. Unfortunately, there is very little research involving anxiety in Hispanic and Latino cultures, and less in its particular subgroups (Varela, Vernberg, & Sanchez-Sosa, 2004). There is some evidence, however, that Hispanic Americans of Dominican descent may frame their troubling anxiety symptoms using idioms such as *nervios,* which may reflect a more benign, culturally sanctioned conceptualization (Salman, Diamond, Jusino, Sanchez-LaCay, & Liebowitz, 1997). This could suggest that Dominican families may view mental health issues, and anxiety in particular, as stigmatizing or shameful. This may result in a family's attempts to minimize or conceal anxiety symptoms and their related interference in family function (Salman et al., 1997). This view may be more or less pronounced, depending on differences in the level of acculturation within a particular family and its members. In Max's case, it may in part explain his father's minimization of his son's OCD symptoms.

When faced with families with racial and ethnic origins different from their own, clinicians should remain open and maintain the stance that clients are the experts on their own culture, degree of acculturation, and culturally accepted practices with regard to mental health issues. If issues of shame or stigma are identified, it may be useful to address these with the family in a nonevaluative way. It is important for clinicians not to make assumptions, but rather, to approach those from diverse cultural or ethnic groups with openness and respect throughout the course of treatment.

Treatment resistance constitutes another broad obstacle to treatment progress. For example, one hindrance to cognitive-behavioral treatment for young children with OCD is failure to complete homework tasks between sessions. To address this issue, clinicians should plan and troubleshoot homework assignments with great care. Some parents are more likely than others to comply with homework assignments. Reasons for variability in compliance include level of family disorganization, time constraints, parent accommodation and unwillingness to tolerate distress or disruption of family routines, poor planning, overzealousness, and failure to make treatment a priority. Clinicians should spend time assessing for these issues and making as detailed a homework plan as possible. It is also helpful if the development of such a plan is undertaken collaboratively, with compromises on the part of both family and clinician. Clinical judgment must be wisely used in terms of weighing issues of family skill, commitment to treatment, and logistical problems against the necessity of the child's engaging in sustainable, slightly difficult EX/RP exercises throughout the week. It is also useful to elicit information from the family about what worked and what did not, in the service of better planning future homework assignments.

Within-session resistance, whether from parents or children, may also be a problem. Parent tools, including differential attention, positive verbal praise, scaffolding, use of a token system, and limit setting techniques, have been

incorporated into the treatment to address child resistance. Clinician attention to family process issues, such as parents' emotions and attributions about their child's symptoms, parents' role on the treatment team, and how parents cope with their own anxiety around treatment, is essential to treatment success.

Children, young children in particular, may have difficulty generalizing the skills that they learn in session to their lives more broadly, especially as time passes after treatment termination. Between-session homework addresses this issue, to some degree. As Max's family gained mastery over particular rituals, the therapist encouraged them to practice in a wide variety of settings, from home to school, restaurants, movie theaters, and elsewhere. The family was also encouraged to try impromptu practice; in other words, whenever Max's symptoms crept up, his parents were asked to remind him to use his skills to "boss back" his OCD. The therapist also educated the family with regard to the tendency of OCD symptoms to wax and wane, or shift in topography, so that they might be watchful and ready to use their skills as needed. The package of skills taught in therapy is conceptualized as a life change—one that the family must continue to practice as needed should symptoms return in times of stress or transition. Finally, the therapist highlighted major developmental events in Max's life—transition to first grade, increasing demands in school, puberty—that might herald periods when Max might be vulnerable to symptom recurrence or difficulty implementing his EX/RP skills.

Managed Care Considerations

Cognitive-behavioral treatments for children and adolescents are oftentimes constrained by managed care. Psychological practice undertaken in a managed care setting must address careful measurement of client progress and outcomes to justify continued reimbursement by insurance providers. There are many downsides to this situation, including reliance on "quick and dirty" assessment methods that may not capture the full complexity of a family's struggle, the fact that families that do not progress quickly may exhaust their allotted mental health services, and that some families have less access to needed mental health services because they cannot afford insurance. Some may view the increased accountability of health care providers, emphasis on developing more effective treatments, and greater emphasis on careful tracking and measurement of meaningful outcomes as positive consequences of managed care (Fonagy, Target, Cottrell, Phillips, & Kurtz, 2002). For Max's treatment, the clinician included multiple methods of measuring treatment gains, as detailed earlier.

Follow-Up Treatment

Follow-up visits were useful to continue to assess Max's symptom picture. In addition, booster sessions may be offered as his symptoms wax and wane or change in topography over time. Just as it is important to meet Max at his developmental level, it is also important to consider the demands of developmental transitions: progressing from kindergarten to first grade, the beginning of each school year, the onset of puberty, the increasing autonomy that comes with adolescence, and so on. The option of follow-up treatment is especially critical at these junctures, as families may need additional help tailoring their use of EX/RP skills in more developmentally sensitive ways.

(*Continued*)

Implications of the Case

The case described in this chapter illustrates that cognitive-behavioral therapy can indeed be adapted to younger populations. However, such adaptations must take into account issues such as the phenomenology of different mental health issues in young children, their increased reliance on and embeddedness in the family, and levels of cognitive and socioemotional sophistication.

Despite the increasingly recognized need, there still exists a gap in the treatment literature on the application of CBT to young children with OCD. The majority of the published treatment studies do not include children under the age of 7 or 8, nor do they address early developmental considerations. Though current clinical trials are working to adapt CBT with EX/RP for children ages 4 to 8, there still exists the need to address the unique developmental factors for this younger age group and its possible moderators of treatment. It is for this reason that treatment formulation for young children must be approached by considering all of these factors as well as any other developmental factors unique to a particular child. Comprehensive assessment of a child's developmental attainment and stage will allow for the augmentation of current CBT approaches that have been found to be highly successful in treating OCD.

Recommendations to Clinicians and Students

Increasingly, CBT approaches for children also address parent issues, such as stress, deficits in skills acquisition or performance of appropriate behavior management techniques, and parents' own struggles with mental health issues (Barmish & Kendall, 2005; Barrett et al., 1996; Chronis, Chacko, Fabiano, Wymbs, & Pelham, 2004). Given the increasing diversity of the U.S. population, coupled with disparities in access to and utilization of mental health services to minority populations, it is essential for clinicians to keep apprised of empirical research regarding the use of CBT with these groups. Additionally, the increasing use of acceptance and mindfulness-based techniques and motivational interviewing may open up new ways to improve and broaden outcomes of more traditional CBT approaches. Finally, although we as a field have accrued a good deal of evidence detailing the efficacy of CBT, we need to turn our focus to the mechanisms of treatment, as well as mediators and moderators of treatment outcome (Kazdin & Nock, 2003). Without investigation in this area, we will not be able to make continued advances in the broad goals of helping to ameliorate mental health concerns so that individuals may live more effective, meaningful lives.

SUMMARY

Cognitive-behavioral treatment approaches have been successfully developed and used with a variety of psychological issues in children, adolescents, and their parents. Although empirical data suggest that CBTs are efficacious in many domains and across many disorders, there is still room for growth and refinement in how they are conceived, operationalized, applied, and evaluated. Recent developments include applications with diverse groups, in diverse settings, using different media (e.g., virtual reality, computer-based programs, use of palm devices), and with younger children. In addition, clinical treatment researchers have been increasing their emphasis on mechanisms of action in treatment and mediators and moderators

of treatment outcome. Taken together, these new directions promise continued sophistication, efficacy, and, hopefully, effectiveness in CBTs as the empirical literature continues to advance.

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CHAPTER 24

Primary Care Behavioral Pediatrics

Patrick C. Friman

Children are not born civilized and socialized. They do not emerge from the womb predisposed to use the toilet appropriately, sleep through the night on their own, wait their turn, sit still or be quiet when stillness or quiet are called for, or to exhibit any of the myriad behavioral requirements of civilized society. Unfortunately for parents, teachers and other caretakers, some children will resist most, and most children will resist some, critical aspects of the socialization and education processes in this culture, and a vast number and array of child behavior problems are the result (e.g., American Academy of Pediatrics, 2001; Christophersen, 1982; Christophersen & Rapoff, 1979; Costello & Shugart, 1992; Earls, 1980). For example, nutritional and maturational health is predicated on food preferences that include the major food groups, and yet many children, apparently satisfied with milk and pureed food, resist parental attempts to introduce new tastes and textures into the daily diet. Adaptive child performance during the day is dependent on receipt of adequate sleep at night, yet many children resist parental efforts to impose a reasonable bedtime. Most parents, preschools, and many day care programs require full toilet training during the third year of life, yet many children resist parental training efforts. Success in most life situations requires a reasonable amount of instructional control, yet many children resist following important adult instructions. There are many other examples, and they generally emerge in situations that require child adherence to or conformance with the family, school, or societal standards or requirements. There are also a large number of child problems that do not involve child resistance so much as child inability to perform, maintain, reduce, or cease important behavior. Although all of these problems pose risks for the child and family, they mostly represent skill deficits rather than pathologies, especially in young children (e.g., Christophersen, 1982, 1988; Friman, 2002; Earls, 1980; also see Horwitz, 2002). An obstacle for mental health practitioners interested in addressing these problems is that they are much more likely to surface initially in primary medical care (PMC) than in mental health service (MHS) settings (e.g., American Academy of Pediatrics, 2001; Costello, 1988; Costello, Burns, et al., 1988; Costello, Edelbrock, et al., 1988; Costello & Shugart, 1992; Dulcan et al., 1990; Goldberg, Roghmann, McInerny, & Burke, 1984; Hickson, Altemeier, & O'Connor, 1983; Horwitz, Leaf, Leventhal, Forsyth, & Speechley, 1992; Lavigne et al., 1993; Starfield et al., 1980).

Although the skills children need to overcome these problems can emerge through vicarious experiences, most skills have to be trained directly by parents or caregivers. Unfortunately, there is an implicit assumption in this culture that those who provide the training should be able to do so without benefit of MHS. Although there is substantially more cultural permission to seek assistance with child behavior problems in PMC settings, as well as a much greater likelihood of parents doing so, the structure of typical PMC visits imposes substantial limits on the assistance actually made available (e.g., Finney et al., 1990; Wissow et al., 2002). Mental health service is vouchsafed by the culture only when child behavior problems reflect significant deviations from cultural norms (e.g., it meets criteria for a psychiatric diagnosis; American Psychiatric Association, 1994); correspondingly, costs accrued from obtaining MHS for such problems are usually reimbursable through third-party payers. Some assistance for problems that are not considered clinically significant is available outside of MHS and PMC (e.g., websites, books, television shows, lay counseling), which is fortunate for parents because they appear to be mostly on their own when training children whose behavior problems are not considered clinically significant. Unfortunately, the quality of that assistance is hard to determine and impossible to guarantee to any meaningful degree, and reimbursement for it is not available from third-party payers. In addition to that assistance, there are multiple child training methods that are an integral part of the parenting process in this country, and virtually all parents have direct access to them. Prominent examples include praising, scolding, yelling, warning, restrictions, and rewards (e.g., Blum, Williams, Friman, & Christophersen, 1995; Peters, 1998; Regalado, Sereen, Inkelas, Wissow, & Halfon, 2004) and a range of methods collectively referred to as time-out (Friman, 2005c; Friman & Finney, 2003). How well the nonprofessional assistance and freely available training methods work can be estimated through inspection of epidemiological surveys of children whose behavioral concerns meet criteria for diagnostic classification; presumably these are children whose behavior has been exposed to some combination of the assistance and the methods. Major surveys indicate that between 11% and 25% of children in this country have behavioral problems that meet criteria for a psychiatric diagnosis (cf. American Academy of Pediatrics, 2001; Costello, 1988; Costello, Burns, et al., 1988; Costello, Edelbrock, et al., 1988; Costello & Shugart, 1992; Dulcan et al., 1990; Goldberg et al., 1984; Hickson et al., 1983; Lavigne et al., 1993; Starfield et al., 1980). These data suggest that a substantial portion of child behavior problems, although not initially deviant from cultural norms, devolve to something much more serious and that a very large number of parents need more help addressing the problems than is made available to them outside of professional contexts.

The two most logical sources of professional help are PMC (e.g., pediatricians) and MHS (e.g., psychologists). The former source is utilized by virtually all parents, although mostly for medical care. Nonetheless, child behavior problems are much more likely to be initially presented in PMC than MHS. Unfortunately, service seeking for children with behavior problems presenting in PMC is restricted by three challenges: limited time, training, and reimbursement. Regarding limited time, early time-and-motion studies showed that the length of the average visit in

PMC for children was between 8 and 15 minutes, with only a small portion of that time devoted to behavior (Christophersen, 1982; Reisinger & Bires, 1980); more current research shows that these early findings have not changed appreciably in recent times (Cooper, Valleley, Polaha, Begeny, & Evans, 2005). Regarding limited training, the curriculum for physicians planning to work in pediatric settings has historically included very little training for assessing and treating behavior problems (e.g., Friedman, Phillips, & Parrish, 1983; Glasscock, O'Brien, Friman, Christophersen, & MacLean, 1989), although the American Academy of Pediatrics is attempting to remedy this problem by requiring a rotation in developmental-behavioral pediatrics (2001; Hagan, 2001). The average rotation, however, is only a month long, which is certainly better than no training at all but is hardly sufficient to develop the expertise necessary to identify and manage the range and number of behavior problems that present in PMC. Regarding reimbursement, physician provision of behavior assessment and treatment involves services that managed care companies are increasingly reluctant to cover (Kelleher, Scholle, Feldman, & Nace, 1999), resulting in non-revenue-generation time by PMC providers and/or limits on services for child behavior problems supplied by PMC providers.

In MHS, providers typically have both the time and the training to competently assess and treat child behavior problems, and they have a service code and nosological system that can lead more effectively to reimbursement for related services than the system used in PMC. Unfortunately, whereas very few parents have reservations about seeking advice about child behavior problems from providers in PMC, many parents have substantial reservations about seeking advice from providers in MHS. Among the reasons for these reservations are the routine use of psychopathology in the conceptualization of child behavior problems, the related stigma associated with the failure of tactics available outside professional contexts to address the problems, the common reliance on strong medication in treatment, high costs, and fear of permanent labeling by third-party payers. To phrase this point in blunt colloquial terms, most parents are loath to see a shrink about their children's behavior problems unless those problems have become severe (e.g., Friman, 2002; Leinwand, 2006).

As a result of the limited services in PMC and limited service seeking in MHS, a large gap in behavioral health care for children has opened up in this country. In that gap is a large number and array of child behavior problems. By definition, these are addressed only with tactics made available outside of professional contexts, and, as indicated by data from epidemiological research, a large portion worsen over time (cf. American Academy of Pediatrics, 2001; Costello, 1988; Costello, Burns, et al., 1988; Costello, Edelbrock, et al., 1988; Costello & Shugart, 1992; Dulcan et al., 1990; Goldberg et al., 1984; Hickson et al., 1983; Lavigne et al., 1993; Starfield et al., 1980). A professional movement focused on addressing this mounting problem involves primary care behavioral pediatrics (PCBP). Primary care behavioral pediatrics does not currently have a widely accepted technical definition, and yet a seemingly noncontroversial one can be abstracted from various descriptive papers. Specifically, PCBP involves behavioral health care that employs applications drawn from the behavioral and child developmental sciences to

address child behavior problems that populate the intersection between child PMC and MHS. Although the problems that present in PCBP can involve secondary (curative) or tertiary (rehabilitative) features, PCBP upholds the long-standing tradition in pediatric medicine of emphasizing prevention over cure or rehabilitation. In the words of Stanford Friedman (1975, p. 515), an early architect of the general field of behavioral pediatrics, "Curative and rehabilitative orientation [is] always second best to preventing the disease or defect in the first place."

DESCRIPTION OF PRIMARY CARE BEHAVIORAL PEDIATRICS TREATMENT

Essentially pragmatic, PCBP treatment is much more concerned with problem resolution than with explication of hypothetical underlying psychogenic causes or constructing elaborate theories about such causes. Additionally, PCBP is concerned mostly with problems that have not devolved sufficiently to fit major diagnostic categories, such as suicidal behavior, Major Depressive Disorder, Conduct Disorder, Bipolar Disorder, Posttraumatic Stress Disorder, or psychosis. These major disorders and conditions are the ones most likely to generate complex theorizing about mental illness, and they are generally outside the domain of PCBP. Nonetheless, it is important that PCBP practitioners be able to recognize major disorders so that they can refer clients to professionals who do specialize in the appropriate evaluation and treatment. The therapy provided for those problems that remain in PCBP typically involves two intersected kinds of therapy: supportive health education and prescriptive behavioral treatment.

Supportive Health Education

Supportive health education involves provision of information about the appropriate classification and developmental, learning, and prognostic context of presenting problems. As an example, there is a widespread cultural ignorance and misunderstanding of the etiology of elimination disorders, and belief in characterological and/or psychopathological defects as major contributing factors has historically been widespread (Friman, 1986, 2002). Relatedly, incontinence is a major contributor to child abuse (Helfer & Kempe, 1976; Kempe, Helfer, & Krugman, 1997; Schmitt, 1987). To address this problem, PCBP supplies health education about elimination processes, ordered and disordered, summarized nicely by the term "demystification" first offered by Levine (1982) and now used by multiple others (e.g., Christophersen & Friman, 2004; Friman, 1986; Friman & Jones, 1998, 2005). As another example, supportive health education often involves the reassurance that many child behaviors reported in primary care are actually developmentally normal and will resolve with the tincture of time. For example, common problems such as thumb sucking, periodic separation anxiety, and crying at bedtime in 3-year-olds, although often of concern to parents and disruptive of the family (Earls, 1980; Costello & Shugart, 1992; Christophersen, 1982, 1988; Christophersen & Rapoff, 1979), are usually benign and not deviant from a developmental perspective. If such

problems perpetuate beyond certain age limits, substantially worsen, or begin to cause health concerns, supportive health education is insufficient, and prescriptive behavioral treatment recommendations are provided.

Prescriptive Behavioral Treatment

If a behavior problem does not resolve following the provision of supportive health education and does not involve a major psychiatric condition (e.g., major depression) at the boundary of PCBP, practitioners recommend a series of therapeutic steps for the parents and child to follow at home (or school). The therapeutic recommendations typically emphasize procedure over process, and most procedures used are derived from the more pragmatic parts of the behavioral sciences, particularly those focused on learning and development (Blum & Friman, 2000; Christophersen, 1982; Friman, 2005a; Friman & Blum, 2003; Friman & Piazza, in press). Practitioners of PCBP tend to interpret behavior problems not as representative of pathologies per se, but as representative of skill deficits whose signature behaviors are currently aggravating health and development and that could devolve to actual pathologies if not remedied. Treatments that work are valued for their own sake, and their importance is not diminished because they are at odds with this or that theory. Efficiency, effectiveness, and acceptance are valued over and above theoretical consistency, precision, and scope.

Treatment, although predominantly verbal and thus consistent topographically with traditional child psychotherapy, differs from it in at least two important ways. First and perhaps most fundamental, parents (or primary caregivers) rather than children are often the direct recipients of treatment (i.e., supportive health education and the recommendations that make up the prescriptive behavioral treatment regimens). Children are, of course, the ultimate recipients of PCBP treatment regardless of its form, and they are often present during its delivery and even participate in its preparation. But the fundamental vehicle for PCBP therapy is educational and/or prescriptive advice that pertains to the parent portion of parent-child interactions. Although the comparison is not perfect, it may be helpful to view PCBP therapy as a specialized form of parent training. The critical point, however, is that although child problems are the reason for PCBP therapy, parents are the proximal recipients of the therapeutic advice pertaining to those problems.

Second, because of limitations on time and an emphasis on procedure in pediatric settings, PCBP treatments are often brief and protocol-driven. In this respect they differ substantially from the process-based, time-intensive interventions that characterize traditional child psychotherapy. However, PCBP therapy is consonant with the increasing emphasis on empirically supported treatment and manualized practice in the contemporary practice of psychotherapy (e.g., Christophersen & Mortweet, 2001). The therapeutic armamentarium of the PCBP therapist includes a variety of procedures, each with abundant empirical support, including (but not limited to) time-out, contingency management, home-school notes, simple token economies, and various empirically supported procedures for simple habits, chronic incontinence, bedtime struggles, feeding problems, mild depression, and most of the anxiety disorders.

WHO ARE THE PRACTITIONERS?

Because of its preventive emphasis, clinical expertise in the major psychiatric problems of childhood is not a prerequisite for practitioners in the field. In some respects, such expertise could even handicap practitioners by predisposing them to interpret routine child problems as evidence of a latent or occult pathology (Friman, 2002). For example, hair play, twirling, and pulling in toddlers, although potentially problematic and certainly important enough to address in a behavioral pediatric visit, is not necessarily reflective of psychopathology or indicative of true Trichotillomania (Drews & Friman, 2003; Friman & Hove, 1987). Thumb sucking in preschoolers is more likely a benign source of self-soothing than a malignant sign of oral fixation or regressive personality disorder (Friman, Byrd, & Oksol, 2001). Soiling in young children is much more likely to result from constipation than from psychic mechanisms such as resentment, regression, or anal fixation (Field & Friman, 2006; Friman, 2003). Daytime wetting in first graders is more likely due to incomplete training than insufficient personality (Christophersen & Friman, 2004). Nocturnal wetting is almost always due to developmental variables (e.g., reduced functional bladder capacity) and only rarely to psychological variables suggestive of pathology (Friman, 1986, 1995, in press-a; Friman & Jones, 1998, 2005). There are many other examples.

Because highly specialized psychiatric expertise is unnecessary for effective practice, and because the child problems that occupy the intersection between MHS and PMC are diverse, the field of PCBP is professionally eclectic. Thus primary care physicians (e.g., pediatricians, family practitioners) can specialize in PCBP just as readily as pediatric, school, and clinical child psychologists, and there are specialized training programs for various types of professionals. One essential qualification for all PCBP practitioners is some knowledge of the biological variables that are functionally related to child behavior problems. The primary reason this specialized knowledge is needed is that several behavioral problems that respond well to PCBP intervention involve significant medical (i.e., biological) dimensions. The overall field includes the evaluation and treatment of at least three categories of problems: (1) routine behavior problems, (2) behavior problems with significant medical dimensions, and (3) medical problems with significant behavioral dimensions. Although a fourth category can plausibly be described, one that emphasizes the interaction between medical and behavioral issues more than is typical of categories 2 and 3 (e.g., Blum & Friman, 2000; Friman, 2005a; Friman & Piazza, in press), this chapter focuses only on these three categories, presenting a descriptive PCBP analysis of and therapeutic approach to a representative problem from each area.

COMMON BEHAVIOR PROBLEMS IN PRIMARY CARE SETTINGS

As indicated, a substantial portion of children seen in PMC have symptoms that meet criteria for a behavioral or emotional disorder (e.g., American Academy of Pediatrics, 2001; Costello, 1988; Costello, Burns, et al., 1988; Costello, Edelbrock, et al., 1988; Costello & Shugart, 1992; Dulcan et al., 1990; Goldberg et al., 1984;

Hickson et al., 1983; Lavigne et al., 1993; Starfield et al., 1980), and a much larger percentage exhibit behaviors or emotions that cause their parents concern and/or cause some functional impairment for the child but do not meet criteria for a disorder (American Academy of Pediatrics, 2001; Costello & Shugart, 1992; Earls, 1980; Christophersen, 1982, 1988; Friman, 2002). The problems parents report to PMC providers vary with the age and developmental level of their children. During infancy, parents are most likely to report concerns about their children's excessive crying and problems initiating and maintaining sleep. During the preschool years, parents are most likely to report concerns about their children's oppositional behavior, toileting practices (especially resistance), limited attention span, overly selective eating, and fears or worries (Earls, 1980; Lavigne et al., 1993). During the school years, parents continue to report concerns about problems first encountered during the preschool years that have persisted, as well as new concerns about academics, school behavior, and peer problems (Kanoy & Schroeder, 1985).

Many of these behavior problems will ultimately resolve without the direct intervention of a professional. For this reason, PMC providers often initially describe presenting problems as fitting within the range of normal expectations and recommend that parents let their children grow out of it. The "tincture of time" approach, however, does little to address the substantial stress that behavior problems can place on a family while they are occurring, a fact that is highlighted by the increased risk of child abuse that occurs in association with many problem behaviors (Helfer & Kempe, 1976; Kempe et al., 1997; Schmitt, 1987). Furthermore, children do not always grow out of their behavior problems, and, in most cases, it is very difficult to distinguish those who will from those who will not. Without treatment of problems that persist, children can have serious consequences later in life. For example, untreated, persistent oppositional behavior can devolve into major conduct problems that require extraordinary therapeutic interventions for remission (Caspi & Moffitt, 1995). Unresolved toileting problems can lead to serious medical problems, such as megacolon, urinary tract infection, and the unstable bladder of childhood (Christophersen & Friman, 2004; Friman, 1986, 1995; Friman & Christophersen, 1986; Friman & Jones, 1998). Untreated bedtime problems can lead to habitually disrupted sleep patterns, family discord, and child maltreatment (Blampied & France, 1993; Blum & Carey, 1996; Edwards & Christophersen, 1994; Ferber, 1985). Unresolved school problems can lead to incomplete education, truancy, and school failure, which, in turn, are instrumental in the development of delinquency, drug use, and ultimately criminal behaviors (Friman, 2000). There are many other examples; collectively, they underscore the benefit of brief, problem-specific, effective treatment for children whose behavior problems have persuaded parents to discuss those problems in PMC, even if they involve low-intensity, high-frequency problems such as those just mentioned. The next section discusses one example in detail.

Bedtime Problems

Teaching children to go to bed, go to sleep, and stay asleep throughout the night is difficult for many families in this culture (Blampied & France, 1993; Blum

& Carey, 1996; Edwards & Christophersen, 1994; Ferber, 1985; Friman, 2005c). A representative survey indicated that at least 30% of families contend with this problem three or more nights a week (Lozoff, Wolf, & Davis, 1985). The difficulties reported by parents include bedtime struggles such as resistance to going to bed, fussing and crying while in bed, and night waking with fussing, crying, and unauthorized departures from the bedroom. These problems are important because they disrupt parents' and their children's sleep, which, in turn, decreases the subsequent capacity of both to execute emotional self-control (Teitelbaum, 1977). Although PCBP advice about sleep is needed from the age of 3 months on, its importance is greatly amplified if bedtime struggles continue after 6 months of age. One intervention for child bedtime problems employed by many pediatric providers (and often parents acting on their own) involves medication. Although sometimes effective in the short term, all soporific medication results in unwanted side effects, and sleep-related benefits are usually lost when the medication is withdrawn. Additional information on medication for bedtime problems is available in published reviews of the research on them (e.g., Edwards & Christophersen, 1994).

Treatment for Bedtime Problems

The cardinal component of the most effective behavioral interventions for bedtime problems involves one of the first documented and most frequently used in behavioral psychology: extinction. As children develop sleep habits, they often learn to associate specific environmental factors with self-quieting and the induction of sleep. Misinformed parental efforts to help children sleep often result in problematic sleep associations that mitigate the process of falling asleep. For example, for most children sleep induction is enhanced by the presence of a parent to sooth and cuddle them and thereby ease the transition from wakefulness to sleep. These parental activities can contribute functional status to a constellation of stimuli that influence children's behavior in bed. With the parent present, rapid sleep induction typically occurs. Unfortunately, when the parent is absent at bedtime, the child is left without the stimulus that is most powerfully associated with sleep. The children's response to parental absence typically involves prolonged and intensive crying that resembles an extinction burst (Blampied & France, 1993; Ferber, 1985; Edwards & Christophersen, 1994; Friman, 2005b). This response motivates most parents to intervene either by soothing or disciplining their child, which, unfortunately, usually worsens the problem. Soothing responses to crying can reinforce the crying; disciplinary responses to crying often provokes more crying; and both responses limit the child's learning of self-quieting skills (Blampied & France, 1994; Edwards & Christophersen, 1994; Ferber, 1985; Friman, 2005b; Lozoff et al., 1985). Not surprisingly, after failed attempts to solve the problem themselves, parents whose children exhibit bedtime problems often seek professional advice. The four procedures most likely to be prescribed by PCBP practitioners are described next.

Complete Extinction This approach to bedtime problems involves no visits by the parent to the child's bedroom after the child has gone to bed. In effect, the child

is left to cry it out. Generally, complete extinction works more rapidly than other approaches, but it presents problems that mitigate its overall effectiveness (Adams & Rickert, 1989; Edwards & Christophersen, 1994; Rickert & Johnson, 1988). For example, the child's crying can be highly aversive during the first nights of implementation. If the family lives in an apartment complex, or the treatment is implemented during a season when windows are open, the crying and screaming can draw attention from the neighbors, with predictably problematic consequences. Additionally, extended crying and screaming differentially affects parents. Discord is possible (probable) when crying is substantially more aversive for one parent than the other (Adams & Rickert, 1989; Ferber, 1985). Thus, complete extinction is a straightforward behavioral approach to bedtime problems in children but has limited social validity. In an attempt to improve social validity, sleep researchers have developed other, multicomponent methods that employ extinction but decrease its intensity and aversiveness for parents through the inclusion of other procedures.

Graduated Extinction This procedure involves advising parents to ignore bedtime problem behavior for specific time intervals that gradually increase. The optimal length of the intervals has not been established empirically, but expert advice (Ferber, 1985) and related research studies (e.g., Adams & Rickert, 1989) recommend beginning with a 5-minute interval on the first episode, 10 at the second, and 15 minutes for subsequent episodes on night 1. These intervals increase over the course of a week, ending with 35 minutes for the first episode on night 7, 40 for the second, and 45 minutes for all subsequent episodes and nights. Although children can tantrum for longer than 45 minutes at night, published data (Adam & Rickert, 1989; see also Edwards & Christophersen, 1994) and a large amount of clinical experience described by Ferber (1985) suggests that very few do.

An important question about graduated extinction is why it works. At first glance, it would seem to be the perfect procedure for teaching children to cry gradually longer and longer, culminating in a reliable 45-minute bout reproducible throughout the night. This, however, is not what happens. No research has yet addressed why, and the following explanation is speculation. The procedure is perhaps more appropriately labeled a differential reinforcement schedule than scheduled extinction. The parents visit the room only if their child is crying; thus, visits are contingent. However, increasing the response requirement to 45 minutes of crying may stretch the schedule so much that the reinforcing effects of sleep supersede the reinforcing effects of parental visitation.

Positive Routines This procedure involves a hybrid of extinction and a rewarding bedtime ritual. In the procedure, parents determine when they would like their child to go to sleep and when the child typically falls asleep. Beginning shortly before the time the child typically falls asleep, parents engage the child in several quiet activities lasting no longer than 20 minutes total. During the activities, the parents issue easily followed instructions and richly supply rewarding responses for compliance. In this respect, the procedure resembles a widely studied procedure for increasing instructional control called behavioral momentum (Mace et al., 1988). In positive

routines, as in behavioral momentum, the child is given a series of instructions with a high probability of compliance, followed by a terminal instruction whose probability of compliance is low. In this case, the terminal instruction is "Now stay in bed and go to sleep," or something equivalent. If at any time after the completion of the routines and the terminal instruction the child leaves the bed, the parent places the child back in bed, telling him or her that the routine is over and it is time for bed. Crying and verbalizations are ignored. At specified intervals (e.g., 1 week) the parents move the positive routine back in time 5 to 10 minutes. They continue this backward movement until they arrive at the bedtime they prefer for their child. For example, the child may typically fall asleep at 9:15, and the parent's preferred bedtime might be 8:15. Thus, it could take between 6 and 8 weeks for parents to arrive at the preferred bedtime. Experimental comparison of the positive routines procedure with scheduled extinction showed that both produced substantially improved bedtime behavior for children, but that the parents using positive routines reported significantly improved marital relations, suggesting a more socially valid procedure (Adams & Rickert, 1989).

The Bedtime Pass The bedtime pass program involves (a) requiring that children get into bed; (b) providing them with a small, laminated note card exchangeable for one free trip out of the bedroom or one visit to them by the parent after being put to bed in order to have a small request satisfied (e.g., for a drink, hug, or visit to the bathroom); (c) surrender of the pass after it is used; and (d) extinction thereafter. In the initial study, the program eliminated the high rates of crying out, calling out, and coming out of the bedroom after bedtime exhibited by two children, ages 3 and 10 years. Additionally, these successful results were achieved without an accompanying extinction burst during initial intervention periods, and a large group of sample parents rated the intervention as more acceptable than extinction alone (Friman et al., 1999). The pass program was subsequently replicated in a single-subject analysis of four 3-year-old children (Freeman, 2006) and a randomized trial involving nineteen 3- to 6-year-old children (Moore, Friman, Fruzetti, & MacAleese, 2007).

Two plausible possibilities for how the pass achieves its effects have been proposed. First, the pass program is analogous to differential reinforcement of alternative behavior (with use of the pass as an alternative to crying, calling, or coming out from the bedroom and the parents' response as reinforcement), which is a well-established method for reducing disruptive behavior (Vollmer & Iwata, 1992). Second, the use of the pass involves the reinforcement of a request made by the child. A small line of recent research shows that reinforcing appropriate behavior in disruptive individuals and subsequently placing disruption on a course of extinction can reduce the disruption without producing an extinction burst (e.g., Bowman, Fisher, Thompson, & Piazza, 1997).

This brief discussion of child bedtime problems and their treatment is by no means complete. It merely describes a PCBP analysis and four representative PCBP interventions that are the most frequently used and have the most empirical support. There are other interventions used by parents (and prescribed by some professionals) but usually avoided in PCBP (e.g., medication, the family bed; see Friman,

2005b, for description) as well as others that almost certainly will become part of the PCBP armamentarium if they generate sufficient empirical support. Examples include the sleep fairy intervention, a variation on the tooth fairy (Burke, Kuhn, & Peterson, 2004), and the teaching of self-quieting skills during the day to promote a better transition to bed at night (Edwards, 1993; Edwards & Christophersen, 1994).

The general premise of this section is that child bedtime concerns represent a very important and common presenting complaint in PMC, one that is not appropriate for conventional medical evaluation and not optimally responsive to conventional medical treatment (e.g., medication). The vast majority of bedtime concerns fit within the category of common behavioral problems: They are high-frequency, low-intensity presenting complaints. As with most common behavior problems, bedtime problems have a fundamental behavioral and learning basis; thus, not surprisingly, they respond well to behavioral- and learning-based evaluation and treatment. Yet conventional pediatric training does not emphasize this type of evaluation and treatment, and even if it did, most pediatricians would not have sufficient time during the average visit to accomplish it (cf. Christophersen, 1982).

As indicated, PCBP research has contributed several behavioral- and learning-based treatments for bedtime problems; however, the research agenda is far from complete, especially in light of the extent of the problems in this culture. For example, little research on the actual function of bedtime problems has been reported and, although the assumption that the primary function is attention seeking has served the field well, there may be other functions. Additionally, although the treatment procedures described here extend the clinical effectiveness of PCBP practitioners, new procedures, especially those with high social validity, are needed. More generally, new effective and socially valid treatments are needed for all of the common behavior problems presenting in pediatric settings, and PCBP is uniquely well positioned to address that need.

BEHAVIOR PROBLEMS WITH SIGNIFICANT MEDICAL DIMENSIONS

A fundamental assumption of PCBP is that behavior occurs mostly as a function of environmental circumstances, but this assumption does not disavow the influence of medical (i.e., physiological) variables. Rather, many physiological variables are indeed seen as determinative, but are themselves the result of environmental behavior contingencies that occurred in a phylogenetic context (Skinner, 1966). Physiological variables often play an initiating role in behavioral problems that present in PMC. For example, an immune response can lead to a rash, a rash to scratching, scratching to tissue abrasions, and tissue abrasions to scratching unrelated to the rash. Child stomach pain brought on by physiological variables (e.g., flu virus) can lead to missed school. While the child is home from school, the child's behavior is influenced by reinforcers involving avoidance of schoolwork as well as contact with sympathetic responses from caregivers. These influences can, in turn, result in complaints of stomach pain that do not involve physiological variables, a condition sometimes referred to as recurrent abdominal pain (Finney, Lemanek, Cataldo, Katz, & Fuqua, 1989). Hair loss caused by medical problems

(e.g., alopecia areata) can lead to social avoidance, the extended duration of which can lead to a deterioration of social skills that can, in turn, lead to social avoidance even after hair has grown back. There are many other examples; the one elaborated upon here involves constipation, a physiological variable that can cause common toileting problems, the most serious of which is retentive encopresis.

Retentive Encopresis

Descriptive Features

Definition Encopresis, a common presenting complaint in pediatrics (3% to 5% of all referrals), is a disorder in which children either voluntarily or involuntarily pass feces into or onto an inappropriate location, usually their clothing (Friman & Jones, 1998; Wright, 1973; Wright, Schaefer, & Solomons, 1979). Encopresis is not diagnosed if the problem is exclusively due to an anatomic or neurologic abnormality that prevents continence. The current criteria from *DSM-IV* (American Psychiatric Association, 1994) are (a) inappropriate passage of feces at least once a month for at least 3 months, (b) chronological or developmentally equivalent age of 4 years, and (c) not due exclusively to the direct physiological effects of a substance (e.g., laxatives) or a general medical condition, except through a mechanism involving constipation. The *DSM-IV* distinguishes two subtypes of encopresis, one with constipation and overflow incontinence (i.e., retentive encopresis) and one without these symptoms (i.e., nonretentive encopresis). For reasons that will be discussed later, this section focuses only on encopresis with constipation.

Relevant Physiology The large intestine, or colon, is the distal end of the alimentary tract, which is sequentially composed of the esophagus, stomach, small intestine, and colon. A thorough review of the colonic system is beyond the scope of this chapter (for more thorough reviews, see Weinstock & Clouse, 1987, or Whitehead & Schuster, 1985). Some rudimentary description of the system, however, is necessary to understand the physiology that supplies the logic of effective treatment. The colon is a tubular organ with a muscular wall. It connects the small intestine to the rectum and anus. It has three primary functions: fluid absorption, storage, and evacuation. Extended storage and planned evacuation are the defining features of fecal continence. Movement of waste through the colon is achieved through muscular contractions called peristalsis, which produce a wave-like motion of the colon walls. As the waste moves through the colon moisture is absorbed from it, creating semisolid feces. Movement through the colon is potentiated by a variety of external events that instigate muscular contractions in the colonic wall. For example, eating a meal increases colonic contractions (referred to as the gastrocolonic reflex), and moving about will have a similar effect (referred to as the orthocolonic reflex).

Usually the rectum, a hollow receptacle at the distal end of the large colon, contains little or no feces, but prior to defecation muscular contractions in the colonic wall propel feces into the rectum. This results in distension, which stimulates sensory receptors in the rectal mucosa and in the muscles of the pelvic floor, resulting in relaxation of the internal sphincter, which facilitates defecation. This process is involuntary, but the child can constrict the anal canal and inhibit

defecation by contracting the external anal sphincter and the functionally related puborectalis muscle. When the urge to defecate is suppressed this way, the rectum accommodates the retained stool through the adaptive pliance of its structure and terminates the reflex relaxation of the internal sphincter. The urge gradually decays, and some of the fecal matter in the rectum is returned to the descending colon by retroperistalsis.

Etiology Between 80% and 95% of encopresis cases can be traced to a primary causal variable: constipation (Hatch, 1988; Levine, 1975, 1982; Wright et al., 1979). Although definitions for constipation vary, children who frequently go two or more days without a bowel movement are probably prone to constipation. A common complaint by the parents of encopretic children is that they deliberately soil their clothing (Wright et al., 1979), but this accusation is usually false (Levine, 1982). The primary cause of soiling is fecal retention (constipation). In most cases, retention is not caused by characterological or psychopathological problems (Friman, Mathews, Finney, & Christophersen, 1988; Gabel, Hegedus, Wald, Chandra, & Chaponis, 1986). Retention is usually the result of a constellation of factors, many of which are beyond a child's immediate control (Levine, 1982). These factors include a constitutional predisposition (i.e., slow gastrointestinal transit time), diet, insufficient leverage for passage of hard stools, and occasional or frequent painful passage of hard stools, resulting in negative reinforcement for holding stools (Christophersen & Rapoff, 1983). Although retention is frequently attributed to sexual abuse, an association between fecal retention and accidents has not been established (e.g., Mellon, Whiteside, & Friedrich, 2006). For some children, especially those with extreme constipation and/or treatment failure, there is an increased threshold of awareness of rectal distension, a possibly weak internal sphincter, and/or a tendency to contract the external sphincter during the act of defecation (Meunier, Marechal, & De Beaujeu, 1979; Wald, Chandra, Chiponis, & Gabel, 1986). The combined effect of all these factors is a lowered probability of voluntary stool passage and a heightened probability of fecal retention.

Chronic fecal retention results in fecal impaction, which results in enlargement of the colon. Colon enlargement results in decreased motility of the bowel system and, occasionally, in involuntary passage of large stools and frequent soiling due to seepage of soft fecal matter. The seepage is often referred to as paradoxical diarrhea because the children retain large masses of stool and thus are functionally constipated, but their colon allows passage of soft stool around the mass, which results in diarrhea (Christophersen & Rapoff, 1983; Levine, 1982).

That fecal impaction is related to retentive encopresis has been established by several investigators, primary among whom are Davidson (1958), Levine (1975), and Wright (1975). All independently reported that 80% of their patients had fecal impaction accompanying fecal incontinence at the first clinic visit. Subsequent to his 1975 report, Levine and his colleagues developed a simple clinical procedure to identify fecal impaction from an X-ray of the lower abdomen (Barr, Levine, Wilkinson, & Mulvihill, 1979). As a result of the improved diagnostic method, Levine revised his initial 80% estimate of fecal impaction's coexistence with fecal incontinence to 90% (Christophersen & Rapoff, 1979).

Differential Diagnosis There are rare anatomic and neurologic problems that can lead to fecal retention and soiling. Anatomic problems include a variety of malformations and locations of the anus that are detectable on physical exam and require medical management (Hatch, 1988). Hirschsprung's disease or congenital aganglionosis is a disorder in which the nerves that control the muscles in the wall of the colon or in an entire section of the colon are absent, causing severe constipation. Its incidence is approximately 1 in 25,000, and it usually causes severe symptoms in infancy (Levine, 1975). Thus the clinical presentation itself should prevent the astute clinician from mistaking one for the other. The possible exception is ultrashort segment Hirschsprung's disease, which has a more subtle clinical picture. However, the existence of this condition is controversial, and, even if it does exist, proper collaboration between pediatrician and behavior analyst should ensure timely diagnosis.

Evaluation

The initial PCBP evaluation should begin with a physical examination that usually includes a routine check of history, abdominal palpation, rectal examination, and sometimes an X-ray of the abdomen to determine the extent of fecal impaction. A barium enema is rarely necessary unless features of the exam suggest Hirschsprung's disease. The evaluation should also include a behavior assessment that begins with a behavioral interview that involves critical questions related to constipation. These include asking whether there is ever a long period between bowel movements, whether bowel movements are atypically large (stop up the toilet), whether fecal matter ever has an unusually foul odor, whether fecal matter is ever hard, difficult, or painful to pass, and whether the child ever complains of not being able to feel the movement or make it to the toilet on time. An additional question that pertains more to treatment history than to pathogenesis is whether the child ever hides soiled underwear. Affirmative answers to any or all of these questions are highly suggestive of retentive encopresis, and hiding underwear indicates a history that includes some form of punishment.

The encopresis evaluation is usually the first step in treatment because it always includes a substantial amount of supportive health education. Encopresis is not well understood outside of the medical community, and the child's parents are likely to be under the influence of the characterological and psychopathological interpretations that are prevalent in Western culture (e.g., Friman, 2002; Horwitz, 2002). The parents' interpretation of the condition is also likely to influence how the children view their problem. Thus, the encopresis evaluation can actually begin treatment by providing accurate information that demystifies the problem. Last, the evaluation should include questions about diet and timing of meals. Low-fiber diets and irregular meals can be contributing factors in encopresis.

Treatment

During the past 15 years, several descriptive and controlled experimental studies have supported a multicomponent approach to treatment of chronic retentive encopresis partly derived from the pioneering work of Davidson (1958), Christophersen

and Rainey (1976), Levine (1975), and Wright (1975). As indicated earlier, the first component can be addressed within the evaluation. Specifically, the entire elimination process, including its disordered manifestations, should be demystified (Christophersen & Rapoff, 1983; Levine, 1982). Generally this means providing information about bowel dynamics and the relationship of the problem to constipation (Levine, 1982). Second, if there is a fecal impaction it should be removed with enemas and/or laxatives (Christophersen & Rapoff, 1983; Levine, 1982; O'Brien, Ross, & Christophersen, 1986). Third, the child should sit on the toilet for about 5 minutes one or two times a day (O'Brien et al., 1986; Wright, 1975). Fourth, the parents should promote proper toileting with encouragement and not with coercion. Additionally, they should not reserve all of their praise and affection for proper elimination; a child should be praised just for sitting on the toilet (Christophersen & Rapoff, 1983; Levine, 1982; Wright, 1975). Fifth, a stool softener such as mineral oil (Davidson, 1958) or glycerin suppositories (O'Brien et al., 1986; Wright & Walker, 1977) should be used to ease the passage of hard stools. Sixth, dietary fiber should be increased in the child's diet (Houts, Mellon, & Whelan, 1988; O'Brien et al., 1986). Seventh, to increase and maintain motility in the child's colon, the child's activity levels and fluid intake should be increased (Levine, 1982). Eighth, during toileting episodes the child's feet should be on a flat surface. Foot placement is crucial to the Valsalva maneuver (grunting push necessary to produce a bowel movement; Levine, 1982; O'Brien et al., 1986). And ninth, the child should be rewarded for all bowel movements in the toilet (Christophersen & Rainey, 1976; Levine, 1982; O'Brien et al., 1986; Wright & Walker, 1977).

The literature on this approach (or variations thereof) has progressed sufficiently to lead to group trials. For example, in a study of 58 children with encopresis, 60% were completely continent after 5 months, and those who did not achieve full continence averaged a 90% decrease in accidents (Lowery, Srour, Whitehead, & Schuster, 1985). There are multiple other examples that document similar levels of success (e.g., Stark et al., 1997). However, not all children succeed with the conventional approach; for these children, augmentative methods have been developed. These methods developed through study of behaviors associated with treatment failure (Stark, Spirito, Lewis, & Hart, 1990). Incorporating behavior management methods relevant to the behaviors, teaching parents to use them, and delivering treatment in a group format resulted in an 83% decrease in accidents in 18 treatment-resistant children with encopresis, and the results were maintained or even improved at 6-month follow-up (Stark, Owens-Stively, Spirito, Lewis, & Guevremont, 1990).

The general premise of this section—that medical variables can influence or cause behavior problems—is not controversial, even in sciences such as behavior analysis that are devoted to the study of environmental cause (e.g., Friman & Piazza, in press). Constipation is one such variable, and it can cause behavior problems ranging from minor toileting resistance to encopresis, and there are many others (e.g., anorexia due to gastroesophogeal reflux, restricted activities due to pain). Because of the medical component of these problems, there are frequently serious health consequences for unsuccessful treatment that compound any

psychological or behavioral components. For example, extreme fecal retention can be life-threatening, and even routine cases seriously decrease social standing and increase social distancing in affected children. Because of the behavioral component of these problems, a solely medical intervention is insufficient for effective treatment. Needed are treatments that deliver or aid the delivery of the medical components of treatment while addressing the behavioral components. Thus, PCBP is an appropriate context. Additionally, although there are many types of behavior problems that stem from physiological influences, for the medical dimensions the most frequently occurring problem is some form of noncompliance with a treatment regimen. As an example, cooperation with prescribed treatment for encopresis is so necessary for success that instructional control training is frequently either a component of treatment or a prerequisite for it (Friman, 2003). Furthermore, although reasonable progress has been made in improving pediatric compliance, encopresis remains one of the most chronic problems in all of pediatric medicine.

MEDICAL PROBLEMS WITH SIGNIFICANT BEHAVIORAL DIMENSIONS

The immediately preceding section discussed the influence of medical variables on child behavior problems, with emphasis on constipation and its influence on toileting resistance and retentive encopresis, problems that are frequently seen in PMC. This section discusses the influence of behavior on medical variables with emphasis on the behavioral treatment of physiologically based behavior problems that are also frequently seen in PCBP. For decades health-based sciences have demonstrated relationships between child behavior and health. For example, eating nutritious food and engaging in modest exercise can improve child cardiovascular health. Obtaining sufficient sleep can help sustain immunological responses, emotional resiliency, and adaptability. Maintaining adequate personal hygiene can decrease child susceptibility to infectious disease. There are many other examples. More recently, PCBP has demonstrated a variety of healthful outcomes from changes in behavior (e.g., Finney, Miller, & Adler, 1993; Friman & Christophersen, 1986; Irwin, Cataldo, Matheny, & Peterson, 1992; Stark et al., 1993). A review of this literature is beyond the scope of this chapter, and thus the focus here is on biofeedback, a treatment involving the manipulation of behavioral variables to improve health; treatment of nocturnal enuresis is offered as the primary example.

Nocturnal Enuresis

Biofeedback Perspective

Biofeedback involves the use of electrical or electromechanical equipment to measure and increase the salience of stimuli associated with pertinent physiologic processes and training patients to discriminate and control them in order to improve their own health. The penultimate goal of biofeedback is to train individuals to alter the physiologic processes in healthful directions, and the ultimate goal is to train them to do so without biofeedback (Culbert, Kajander, & Reaney, 1996; Friman, in press-b). Most biofeedback-type treatments require sophisticated instrumentation

and specialized training and thus may not be incorporated readily into primary care practices (e.g., anorectal manometry combined with electromyography for treatment of fecal incontinence). However, nocturnal enuresis, a physiologically based behavioral problem that is one of the most frequent presenting behavioral complaints in PMC, is highly responsive to urine alarm treatment, a minimally technical, uncomplicated form of biofeedback that is readily used in PMC (Christophersen & Friman, 2004; Friman, 1986, 1995, in press-a; Friman & Jones, 1998, 2005). In fact, urine alarm treatment is the most empirically supported treatment for nocturnal enuresis, by a wide margin, despite more than 50 years of research devoted to discovery of effective physiologically based treatments (e.g., medication). There are at least two medications (DDAVP and imipramine) that can improve enuresis, but with significantly smaller magnitude and duration of effects than the urine alarm. This section discusses descriptive features of nocturnal enuresis as well as its evaluation and treatment.

Descriptive Features

Definition The current criteria for enuresis (nocturnal and diurnal) from *DSM-IV* (American Psychiatric Association, 1994) are (a) repeated urination into bed or clothing; (b) at least two occurrences per week for at least 3 months or a sufficient number of occurrences to cause clinically significant distress; (c) chronological age of 5 or, for children with developmental delays, a mental age of at least 5; and (d) not due exclusively to the direct effects of a substance (e.g., diuretics) or a general medical condition (e.g., diabetes). There are three subtypes of enuresis: nocturnal only, diurnal only, and mixed nocturnal and diurnal. There are two courses: The primary course includes children who have never established continence, and the secondary course involves children who, after establishing continence, resume having accidents. This section primarily discusses nocturnal enuresis (although some discussion of diurnal enuresis is supplied), which is estimated to occur in as many as 20% of first-grade children (Christophersen & Friman, 2004; Friman, 1986, 1995, in press-a; Friman & Jones, 1998, 2005).

Relevant Physiology The bladder (detrusor) is an elastic hollow organ with a muscular wall. Its shape resembles an upside-down balloon with a long narrow neck; it has two, primarily mechanical functions: storage and release of urine (Vincent, 1974). Extended storage and volitional release are the defining properties of urinary continence. In infancy, distension of the bladder leads to contraction of the bladder and automatic (nonvolitional) evacuation of urine. As children mature, the capacity of the central nervous system to inhibit bladder contraction increases, typically coinciding with the development of continence in early childhood (Berk & Friman, 1990; Koff, 1995).

The components of the urogenital system that are under volitional control and that are necessary for continence skills are the muscles of the pelvic floor. Except during imminent or actual urination, these muscles remain in a state of tonus or involuntary partial contraction, which maintains the bladder neck in an elevated and closed position (Vincent, 1974). Even after initiation of urination has begun, contraction of the pelvic floor muscles can abruptly raise the bladder neck and terminate urination.

But for children with nocturnal enuresis, these urinary inhibitory responses are either not present or are sporadic (Friman, 1995; Friman & Jones, 1998; Houts, 1991; Mellon, Scott, Haynes, Schmidt, & Houts, 1997).

Etiology Although nocturnal enuresis has a strong genetic basis, its exact cause is unknown. For decades there has been a sustained effort to link it to causal psychopathology, but contemporary research (Friman, Handwerk, Swearer, McGinnis, & Warzak, 1998), as well as several reviews of older research (Friman, 1986, 1995, 2002; Friman & Jones, 1998), suggest that most enuretic children do not exhibit clinically significant psychopathology, and when they do, it is more likely to be an outcome than a cause of nocturnal enuresis. Physiologically oriented studies of nocturnal enuresis suggest that some affected children may have difficulty concentrating their urine during the night and, thus, produce more urine nocturnally than their nonenuretic peers (Lackgren, Neveus, & Stenberg, 1997; Rittig, Knudsen, Norgaard, Pedersen, & Djurhuus, 1989). The overall importance of this factor, however, is controversial because the proportion of enuretic children with urine concentration problems may be small (Eggert & Kuhn, 1995). Finally, nocturnal enuresis may be viewed as merely a deficit in the skills necessary to prevent urination while asleep (Houts, 1991).

Evaluation

As with encopresis, the initial stage of an enuresis evaluation should include a physical examination. Once a thorough history has been obtained and preliminary information about enuresis has been shared with the parents and child, PCBP goes no further until all possible pathophysiological causes of enuresis have been ruled out. There are several such causes, some of which are common (e.g., urinary tract infection) and some rare (e.g., diabetes), and they must be ruled out before treatment begins. There are several other elements necessary for a full evaluation of enuresis, and these are well documented in multiple other papers (e.g., Christophersen & Friman, 2004; Friman, 1986, 1995; Friman & Jones, 1998; in press-a).

Treatment

The two most common treatments for nocturnal enuresis are the urine alarm and the medications DDAVP and imipramine. These medications can provide symptomatic relief; approximately 25% to 40% of children will be dry most nights when taking them. However, when the medications are stopped, the enuresis usually returns (Moffatt, 1997). The urine alarm is a moisture-sensitive switching system, the closing of which (when the child wets the bed) rings the alarm. Repeated pairing of awakening by the alarm with episodes of wetting is consistently described in the literature as the single most effective treatment for enuresis (Christophersen & Friman, 2004; Friman, 1986, 1995; Friman & Jones, 1998, in press-a). Its success rate is higher (approximately 75%) and its relapse rate lower (approximately 41%) than any other drug- or skill-based treatment.

The urine alarm is a simple form of biofeedback treatment because its primary function is to provide feedback for a physiological event, urination, which in

enuretic episodes occurs beneath awareness. The feedback (ringing of the alarm) increases the salience of the urination and, thus, aids in the ultimate establishment of urinary self-control. The mechanism by which the alarm improves enuresis, however, is still unknown. Changes in secretion of hormones that affect the ability to concentrate urine (Friman, 1995; Friman & Jones, 1998; Houts, 1991) or alterations in the brain's inhibition of bladder contraction are at least theoretically possible, but have not been investigated. Early conceptualizations described the mechanism as classical conditioning, with the alarm as the unconditioned stimulus, bladder distention the conditioned stimulus, and waking the conditioned response, but most children successfully treated with the alarm learn to sleep through the night without wetting (or waking). Subsequent conceptualizations emphasized a negative reinforcement paradigm in which the children repeatedly awakened by the alarm avoided it by urinating in the toilet (initially) and (ultimately) holding their urine until a more convenient time. The current prevailing account involves a combination of the two previous accounts with classical conditioning of pelvic floor muscles and operant conditioning of volitional behaviors related to continence (Houts, 1991). The key distinguishing feature of the current account is that children are not necessarily trained to awaken to the alarm; most merely learn to engage their urinary inhibition system even if they are asleep, a skill that would be difficult to teach without the sensory feedback provided by the alarm.

Alarm-based treatment produces cure slowly, and during the first few weeks of alarm use, waking, if it occurs at all, occurs only after a complete voiding. Recent research using the size of the urine stain on the soiled sheets as the dependent measure showed that prior to accident-free nights, the stain grew increasingly smaller on successive nights, suggesting a graduated process of continence attainment (Ruckstuhl & Friman, 2003). In other words, the feedback properties of the alarm gradually but inexorably strengthened the skills necessary to avoid it. The core skill involves contraction of the pelvic floor muscles, resulting in sustained elevation of the bladder neck, resulting in the cessation or, preferably, prevention of urination. There is also the possibility of increased sensory awareness of urinary need and waking to urinate, but this is a less likely outcome and actually inferior to sustained, accident-free sleep throughout the night.

However, increasing sensory awareness of urinary need prior to daytime accidents is also a key component in the most empirically supported treatment for diurnal enuresis. Only two studies are available (very little research has been conducted on diurnal enuresis), and the first used a much simpler conceptualization (Halliday, Meadow, & Berg, 1987). Specifically, this early study merely suggested that the alarm served as a reminder to urinate. A subsequent study utilizing the biofeedback conceptualization was conducted with a young girl who was initially unresponsive to urinary urge and onset, but who rapidly became responsive with use of the alarm (Friman & Vollmer, 1995). The decreasing latency between alarm onset and appropriate response was characteristic of learning curves resulting from alarm-based treatment for nocturnal enuresis and biofeedback treatments in general.

As indicated earlier, most biofeedback treatments are much more technically complex than the urine alarm, and they are used for a broad range of physiologically

based behavioral concerns that often initially present in PMC. Among the physiological processes that can be monitored are muscle tension, skin temperature, respiratory rate, blood pressure, and skin moisture (perspiration). Biofeedback devices sensitive to these processes have been used in treatment of a wide variety of disorders, including headaches, other varieties of chronic pain, asthma, bruxism, anxiety disorders, sleep disorders, and dysfunction of the autonomic nervous system (Culbert et al., 1996). Additionally, evidence is mounting that biofeedback can generate operant responses that, in turn, lead to control over physiologic processes long thought to be outside of volition, such as skin temperature and blood pressure. For example, verbally based awareness-enhancement methods have been shown to alter the level of mediators of the immune system in saliva (Olness, Culbert, & Uden, 1989) and to decrease the recurrence of chronic mouth ulcers (Andrews & Hall, 1990). Collectively, the large body of research documenting the effectiveness of the urine alarm, along with the even larger literature on the effectiveness of biofeedback treatment for a broad range of medical conditions, underscores the research and clinical potential represented by the influence of behavioral variables on physiology.

Case Illustration: A Routine Case of Bedtime Problems

Identifying Features of the Client

Tom was an 8-year-old Caucasian boy who lived at home with his natural parents, his 12-year-old brother, Bill, and his 3-year-old sister, Suzie. His medical, developmental, psychiatric, and educational histories were unremarkable. He attended a local parochial school and was in the third grade. According to his teachers and his parents his grades were excellent and his behavior was very good. For example, the school used a demerit system and he had received only one demerit at the midyear point and that was for misplacing an assignment. At school, he had several good friends, and during his initial clinic session he identified three of them as best friends. When asked what he liked about his best friends, he mentioned that they were nice to him, funny, and shared similar interests.

Presenting Complaint

Tom's presenting complaint included disruptive emotional behavior after bedtime. Approximately 4 weeks prior to the initial visit, Tom began waking up 2 to 3 hours after bedtime and crying out for his parents or coming into their room to summon them if they did not heed his call. When asked about the nature of his distress, he typically used a range of complaints, including nightmares, upset stomach, noises in his bedroom, and worry over school the following day. In response, the parents either allowed him to sleep in their room until the following morning or took him back to his room, at which point one parent would lie with him until he fell asleep. These responses to his distress actually appeared to worsen rather than improve his nighttime concerns, and by the time of the initial appointment, he had begun exhibiting distress at or even prior to bedtime, often refusing to go to bed without being accompanied by a parent.

(*Continued*)

Parental Factors

The parents described their marriage as good and their relationship with all three of their children as loving and supportive. They did exhibit a difference of opinion about how to manage Tom's bedtime concerns. The father thought that too much attention was being paid to these concerns, whereas the mother thought Tom needed the extra attention. Due to Tom's distress, his mother's concern, and the conflict between the parents, the mother presented the problems to Tom's pediatrician, who initially recommended that the parents ignore Tom's nocturnal distress. The parents found this advice impossible to follow because of Tom's extraordinary emotional response, and subsequently the pediatrician recommended they seek services from my office.

During the interview, the parents revealed several potentiating factors, the most significant of which appeared to be a recent move from a small home in a well-established neighborhood to a large home in a new suburb. Tom had shared a bedroom with his sister in his previous home that was very close to their parents' room. He had his own room in the new home, and it was on the opposite end of the second floor from the parents' room. In addition, Tom's mother disclosed that she was an anxious person who had had trouble with bedtime, sleep onset, and nightmares as a child. She also reported that several of her relatives, including her own mother, had histories of anxiety problems. Thus she empathized with her son and said it was very hard for her to ignore him when he was distressed at night. According to Dad, Mom was overly solicitous with Tom not only at bedtime but in response to any form of distress, physical or emotional. Mom reported feeling that Dad was possibly too emotionally distant when it came to addressing Tom's distress and felt that she needed to do extra to supplement Dad's presumed insufficient response. Thus she was the parent targeted by Tom at night.

History

A typical day interview revealed that bedtime was at 8:30 and included a well-established bedtime routine involving hygiene (e.g., bath), dressing (e.g., pajamas), stories, hugs, and lights out. Immediately prior to the initial episode, the bedtime routine had been nonproblematic. However, the parents reported an early history of resistance to bedtime, resulting in periodic stays by Tom in their room. Although their memory of these events was somewhat hazy, they reported that the peak incidence occurred around the age of 3. Additionally, Mom reported that Tom had been mildly colicky. Neither of Tom's siblings had exhibited bedtime problems, and both had their own room and their own routine, which they followed regularly. However, the intensity of Tom's distress at night had begun distressing his siblings at the time of the referral, especially his younger sister. The parents were fearful that she would begin to mimic his responses in order to obtain their attention.

The remainder of the typical day interview revealed little of clinical significance. For example, Tom's morning routine was regularly completed without disruption or instances of significant countercontrol. There was no resistance to the seatbelt on the car ride to school. Upon returning home from school he routinely had a snack, did about 15 minutes of homework, and then played with friends in the neighborhood or his siblings. The family evening meal was a sit-down version, and there were no reported instances of significant misbehavior during it. He

watched approximately 1 hour of television a day and was allowed to play video games for 1/2 hour.

However, he had no household responsibilities that were regularly enforced. For example, the parents had requested that he make his bed on a daily basis and also bring his dishes to the sink after the evening meal, but, on average, he made his bed once a week and it usually had to be redone by his mother. He also rarely brought his dishes to the sink. The parents reported that bedroom time-out was their primary disciplinary method but that they employed it rarely and could not remember a recent instance. As the parents described expectations and discipline, their tone took on a confessional note and they seemed somewhat embarrassed by how easy they had made life for Tom. Nonetheless, behavior screening via the parent version of the Child Behavior Checklist (Achenbach, 1991) did not reveal any significant externalizing problems; the only elevations were in the Anxiety subscale.

Case Conceptualization

My clinical impression of this case was that it involved a form of separation anxiety brought on by problematic adjustment to a recent move, a related dramatically altered bedroom configuration, and nocturnal separation between child and parent. Tom was probably predisposed to anxiety and worry as a function of family history stemming from mother's own history and symptoms that, in turn, had their own family history leading back through a grandparent. The difference of opinion between parents about how best to approach Tom's distress and the limited extent to which household expectations of him had been enforced and discipline had been imposed may have increased his susceptibility to clinically significant experiential avoidance problems such as separation anxiety (Friman, in press-c; Hayes, Wilson, Gifford, Follette, & Strosahl, 1996). In other words, he may not have been prepared to manage the sudden onset of bedtime distress because he had had little experience addressing daytime distress. The progression of his overt distress that was occurring an hour or two after bedtime prior to the referral to distress occurring at, and increasingly before, bedtime at the time of the referral suggests that Tom's condition was worsening and that he was at risk for more generalized problems.

Course of Treatment

The treatment prescribed involved the bedtime pass procedure. In addition, Tom was taught relaxation exercises, to visualize pleasant settings and events when distressed, and controlled breathing. In the bedtime pass portion of treatment, Tom was given a bedtime pass after he had completed his bedtime routine and was in bed. He was informed that he could use the pass to summon his parents to address a simple request or to seek them out for the same purpose. Once the request had been satisfied, he would surrender the pass and spend the remainder of the evening in bed alone. An incentive was added to the program, allowing Tom to turn in unused passes for use in a simple reward program. In that program, Tom identified something that he wanted his parents to purchase for him and that they agreed was appropriate (a video game). Tom and his parents drew a picture of the item using a dot-to-dot format. For each unused pass, Tom was allowed to connect one dot, and when the dots were completely connected, the parents purchased the item for him.

(*Continued*)

Estimate of Progress

Immediately upon implementation of the program, Tom completed three nights without calling out or coming out of his bedroom, thus earning three dots on his dot-to-dot incentive program. On the fourth night, a strong storm moved through the area and Tom called out for his parents and subsequently joined the entire family in the basement because of a tornado warning. Because of the extraordinary nature of this night, he was allowed to keep his pass and turn it in for a dot. However, the next night he again called out for his parents and used his pass even though the weather was calm. Subsequently, he stopped calling out or coming out and earned 12 continuous days of nonpass use. On the 13th night he had a bad dream and called out for his parents and used his pass. In the 3 weeks since that night, he has not used the pass, has earned his reward, and the parents asked if they could discontinue the pass program and transfer the dot-to-dot reward program to other accomplishments.

Complicating Factors

The only complicating factor in this case involved the disagreement between the parents over how to manage Tom's nocturnal distress, and it was resolved easily with the pass program. Tom's enthusiastic response to the pass program and to the exercises he was taught seemed to reassure the parents that they had pursued the right course of action.

Managed Care Considerations

Primary managed care considerations in cases such as this one involve third-party reimbursement. There were no concerns here, however, because Tom met criteria for a covered condition, specifically Adjustment Disorder with Anxiety Features. Only three visits were needed to resolve the problem, and although a fourth was scheduled, it was canceled by the parents by telephone because there was no apparent need. Therefore, the diagnosable condition along with the short number of visits rendered any managed care concerns moot. After the third and final treatment session, follow-up was accomplished by telephone just prior to the scheduled fourth visit, and during the telephone call, the parents expressed no concern. Tom also participated in the call and expressed his excitement about mastering bedtime and also earning his video game.

Implications of the Case

This case represents a classic example of PCBP treatment for routine behavior problems. The problem initially surfaced in PMC and failed a course of treatment provided there. Subsequently, the PMC provider referred the case to a PCBP clinic, where it was quickly classified and effectively treated with just three clinic visits. Additionally, a generalized anxiety condition did not occur, and thus the brief course of treatment could be characterized as preventive. This last point is important because I suspect Tom was at risk for more generalized conditions given his history and the spread of symptoms that had occurred prior to the first visit. Thus, this case represents the fundamental goal of treatment in primary care behavioral pediatrics: prevention.

RECOMMENDATIONS TO CLINICIANS AND STUDENTS

My primary recommendation is to consider following a similar approach to problems of a routine sort that are referred to practitioners by PMC providers. Specifically, routine problems initially presenting in PMC, as was the case with this one, do not necessarily require a comprehensive psychiatric evaluation before treatment is pursued. A simple behavioral screening in conjunction with a good, verbally based diagnostic interview can direct the clinician to prescribe treatment for the vast majority of referral problems or more evaluation for those problems that appear to involve clinically significant dimensions that extend beyond the initial description of the presenting complaint. In the majority of cases where treatment is pursued, the treatment itself becomes a form of assessment because response to it directs the next course of action. In situations where the problem resolves, such as this one, no further clinical care is needed. In situations where it does not, additional assessment and treatment revision are needed.

A second recommendation is to consider modifying treatment protocols in accord with the presentation of the case, even if an empirically supported treatment is used. The safest way to do this is to include all of the components of the empirically supported treatment and add other components in accord with presumed clinical benefit. In this case, the bedtime pass program was used in a fashion consistent with its description in the two experimental small *n* studies (Freeman, 2006; Friman et al., 1999) and randomized clinical trial (Moore et al., 1997) mentioned in the section on common behavior problems, and it was augmented by visualization, relaxation, and a breathing exercise, along with an incentive system. Whether these augmentations added to the active ingredients in treatment or merely enhanced its social validity isn't possible to determine from a case report. Nonetheless, the added components did appear to please Tom as well as his parents and may possibly have made the entire treatment protocol more acceptable. Whether this can be demonstrated empirically is a question for future research.

SUMMARY

The three domains of care—common behavior problems, behavior problems with significant medical dimensions, and medical problems with significant behavioral dimensions—the representative problems used to illustrate them, and the case illustration were supplied to provide an overview of PCBP. The main purpose of this chapter was to present PCBP as a partial solution to problems occupying the gap in behavioral health care between PMC and MHS for children with behavior problems. Because many more providers are needed, another purpose of the chapter was to increase interest in PCBP. Generally, the field favors those whose orientation to practice is guided by science more than art, whose claim to expertise is predicated on empiricism more than clinical or ex cathedra authority, and whose methods are typified more by procedure than process. Thus, there are similarities between PCBP providers and those in some branches of psychology (e.g., behavior therapy, applied behavior analysis, pediatric psychology) but not in others (e.g., psychoanalysis, existentialist psychology, human potential psychology).

Although many of the behavior problems addressed in PCBP are clinically unremarkable, some actually are, resemble, or if unresolved can lead to clinically significant conditions, and thus they require some level of professional assessment prior to intervention. For example, most bedtime problems are functional, but they can chronically resist lay attempts at resolution, resulting in multiple other problems that put not only children at increased risk, but also the parents and family. Furthermore, a small percentage of problems actually involve biologically disordered sleep processes, and these need to be identified and referred to appropriate specialists. As another example, most cases involving child incontinence are also functional, but they can easily be under- or overinterpreted and correspondingly under- or overtreated, resulting in serious health consequences for affected children. Furthermore, a small percentage of cases actually involve pathological processes, and these need to be identified early and referred to appropriate specialists. There are many other examples, generally involving behavior problems resulting from a mismatch between situational requirements for child exhibition of adaptive behavior and/or inhibition of maladaptive behavior and child behavioral skill levels (Blum & Friman, 2000; Christophersen, 1982; Friman, 2005a; Friman & Blum, 2003; Friman & Piazza, in press).

The PCBP perspective on these problems is that most do not involve pathology, many will resolve with time and health education, some will require prescriptive treatment, and all can benefit from some level of professional assessment to determine which is which. As indicated, the problems usually emerge as a function of the friction between child preference and/or skill level and the requirements inherent in socialization and education processes. If unresolved, the problems can devolve into bona fide pathological conditions, but the PCBP perspective is that little scientific or clinical benefit is gained from the assumption that the presence of a behavior problem necessarily occurs as a function of an underlying psychopathology (e.g., Friman, 2002; also see Horwitz, 2002). This perspective, unfortunately, appears to be at odds with the vast majority of psychology and psychiatric literature on child behavior problems, which is focused almost entirely on detection of psychopathology with minimal regard for detection of child health. In fact, very few clinical assessment instruments are even designed to detect healthy child behavior. Behavior assessment instruments used in the research on and practice of child behavior problems ask questions about symptoms or behavior problems, and the typical intent is to determine whether a given child has significantly more symptoms or problems than the children of a similar age in the group used to norm the instruments. The de facto definition of child behavioral health within clinical child psychology and psychiatry appears to involve a composite of symptoms and problems that are below a threshold established for clinical significance, not a composite of healthful behaviors. In other words, appropriate functioning is conventionally viewed as the absence of illness, not necessarily the presence of health.

Because it is much more concerned with prevention than cure or rehabilitation, PCBP provides a health-based alternative to the psychopathological context that pervades mainstream clinical child psychology and psychiatry (Friman, 2002; Horwitz, 2002). This is not to say that PCBP disputes or disregards the possibility of child psychopathology. Standard practice in PCBP involves referral to appropriate

specialists when presenting problems are deemed to be beyond the bounds of primary care or when they prove resistant to interventions attempted with PCBP. This referral practice is as much a part of PCBP as it is any other branch of pediatrics. The essential point here, however, is that the conventional assumption in PCBP is one of child behavioral health until proven otherwise.

Perhaps the best way to view the PCBP approaches to behavior problems is as a first line of defense. To use a medical metaphor, the typical intent is to strengthen children's immunity against threats to behavioral health. The problems that pose such threats typically involve mild to moderate child behavior problems, most of which are responsive to changes in practices by parents, other caregivers, and teachers. Although some of the problems may meet diagnostic criteria for clinical conditions (e.g., enuresis, encopresis, simple phobias, Oppositional Defiant Disorder), the problems are usually in their early stages and are much more responsive to changes in teaching or training practices than problems that have been chronic for years. Thus, even though PCBP provides treatment, the context of care is still characterized as preventive rather than curative or rehabilitative. And, as the provision of PCBP services widens, the gap in behavioral health care for children in this country could correspondingly narrow.

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CHAPTER 25

Peer Intervention

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Peers are being used as intervention agents with increasing frequency. No doubt, this growth is largely attributable to the many advantages of peer interventions, perhaps the most prominent of which are convenience and cost-effectiveness (Covert & Wangberg, 1992; Fowler, 1988; Topping & Ehly, 1998). Though most often used with elementary-school-age children and adolescents, these interventions have also targeted preschoolers (e.g., Carey, 1997; Kohler, Strain, Hoyson, & Jamieson, 1997) and adults (e.g., Moore, 2005). The wide scope of peer interventions is evidenced not only in their application across developmental levels, but also in the diverse range of targeted clinical problem areas and disorders. By no means an exhaustive list, peers have been used in interventions aimed at improving the social and communication skills of children with autism (e.g., Kohler et al., 1997), reducing medical fears (e.g., Elkins & Roberts, 1986), decreasing delinquency and aggression (e.g., Becker, Hall, Ursic, Jain, & Calhoun, 2004), resolving conflicts (e.g., Johnson & Johnson, 1996), promoting health behaviors (e.g., Agha & Rossem, 2004), and preventing suicide (e.g., Bagley, Tse, & Hoi-Wah, 1994).

This chapter provides an overview of peer interventions as they have been used with children and adolescents. Given the variety of populations, targets, and intervention formats, we begin the chapter with a review of recent categorization efforts and the delineation of specific intervention types. This is followed by coverage of more general issues in intervention implementation. Among these are important considerations regarding the peer interveners themselves, which include selection, training, and often overlooked concerns over the relative benefits and risks of participation. Continued discussion of general implementation issues, including advantages, disadvantages, ethical considerations, and evaluation, is followed by a review of applications targeting specific clinical problem areas. The chapter closes with an illustrative case example.

PEER INTERVENTION FORMS

The frequent and widespread use of peer interventions has given rise to a number of attempts to define and categorize the various approaches. One area of agreement is that such interventions are a method of delivery rather than a specific treatment technique (e.g., Shiner, 1999). Their common element is that a child, rather than a

therapist or teacher, implements the intervention (Shiner, 1999). Of course, adult involvement is still needed, but should not include continual prompting of the child's intervention efforts (Fowler, 1988). Common roles for adults are training the interveners, supervising the intervention, maintaining systems to monitor behavior, and arranging for rewards (Fowler, 1988). Some intervention formats may require greater adult supervision to ensure appropriate implementation, and younger children may require more adult support than older children or adolescents. Even with such adult involvement, however, peer interventions are "designed to be by and for young people; they themselves largely determine what is relevant in terms of information and how it is to be delivered" (Backett-Milburn & Wilson, 2000, p. 94).

Researchers have categorized past efforts in a number of different ways. Distinctions have been made between methods emphasizing skill development and those focusing on knowledge acquisition (Topping & Ehly, 1998). Shiner (1999) distinguishes between the constructs of peer development (i.e., a focus on the development of the interveners) and delivery (i.e., the provision of services). Perhaps the most widely discussed and useful distinctions are based on the roles of the children involved. That is, the child can either intervene directly or indirectly and either be the intervener or the target.

Peer interventions can be direct or indirect (Kalfus, 1984; Odom & Strain, 1984). In indirect approaches, an adult implements the intervention and the child is used in a supplementary role (e.g., as a model, employing group reinforcement contingencies). For instance, to reduce fears of medical procedures, children observed the responses of another child going through the same procedures (Elkins & Roberts, 1986; Stokes & Kennedy, 1980). In direct approaches, the peer implements the intervention and works to directly influence the target child's behavior. For example, Krantz, Ramsland, and McClannahan (1989) had one child with autism prompt others with the disorder to engage in conversations about sports. Most peer interventions appear to fall into the latter category. In the review of specific approaches that follows, we therefore include several direct approaches (e.g., education, mentoring, and counseling) and only one indirect approach (i.e., modeling).

The second distinction focuses on the rigidity of roles and whether a child can be both the intervener and the target or can serve only one of those functions (e.g., Fowler, 1988; Johnson & Johnson, 1995). Fowler describes three basic models characterizing the relationship between the target and the intervening child. In the "subject as client" model, the target child receives an intervention implemented by another child who generally exhibits desired behaviors (e.g., good social skills and teacher compliance) and is well-liked by peers. As such, one child or a small group of children are selected, by adults, to perform the role of intervener. In the "subject alternating as client and therapist" model, all group members have the opportunity to be both therapist and client and to switch between these roles. Members of the group may differ in their intervention needs, and some children may not be in need of intervention. Specific criteria may be used to choose the child who will serve as intervener on a particular day. For instance, the child may be required to exhibit a minimum level of a particular behavior. In the "subject as therapist" model, the target child serves as the intervener rather than the "client." This encourages the

child to focus on the positive behaviors that he or she is rewarding in another child and may result in new positive interaction patterns between the child and his or her peers. Similarly, Johnson and Johnson described two models of the roles in peer counseling interventions. In the "cadre approach," a small number of children are trained to serve as counselors to aid other children in their problem-solving efforts. In the "student body approach," the entire student population is trained in intervention strategies (e.g., problem solving, conflict resolution) and has the opportunity to serve as counselors. These authors suggest that an advantage to this approach is that all children learn new strategies and can apply them outside of a formal intervention.

Having discussed common elements and distinguishing features, we now turn to a review of specific peer intervention types, including modeling, prompting and reinforcement, initiation, tutoring, education, mentoring, and counseling.

Peer Modeling

In peer modeling approaches, one or more competent children are used to exemplify the desired behaviors (Schunk, 1998). Three features are required: one or more models, one or more observers (learners), and a task to be performed (Schunk, 1998). Generally, a target child is paired with a more competent peer model whose behavior can be imitated. A common distinction is between mastery and coping models. The mastery model engages competently and skillfully in the desired behavior. In contrast, the coping model still needs to improve and learn to overcome problems. Thus, through observing a peer who is a mastery model, the target child can see what the desired behavior should look like. By observing a coping model, the target child can learn how to improve performance and handle difficulties or failures, in addition to the possibility of being comforted by the realization that other children face such challenges. Research has shown that both coping and mastery models can be effective in changing children's behavior (e.g., Schunk and Hansen, 1985). Regardless of whether a mastery or a coping model is used, the model and behavior should be selected carefully to reduce the chance of incorrect modeling or inability to learn the behavior (Schunk, 1998).

Another variant commonly used in social skills interventions is the "peer pairing" approach (e.g., Bierman & Furman, 1984; Prinz, Blechman, & Dumas, 1994). Target children are paired with a more competent peer, who receives no training and very little guidance. For instance, Prinz and colleagues used teacher ratings to select aggressive and nonaggressive socially competent children. These two types of children were then paired and given instructions (e.g., "Find out something that made your partner laugh this week") for performing role-plays. The socially competent peer always enacted the role-play first in an attempt to model appropriate behaviors; however, this peer was not given any prior instructions.

The effectiveness of modeling interventions depends on characteristics of the target child, the model, and the behavior to be learned. The modeled behavior is more likely to be followed if the model is the same sex as the target child and is perceived to have high power and the target child has high self-efficacy for and places importance on performing the action (Bandura, Ross, & Ross, 1963;

Grusec, 1992). When behaviors are rare or difficult to elicit or the model may have an unpredictable response (e.g., coping with medical procedures), video modeling of the desired behavior may be preferable (e.g., Gilbert et al., 1982). Observation of the model may be sufficient for simple behaviors, but further intervention is often required for more complicated actions (Schunk, 1998). When modeling alone is insufficient, the target child will need to practice the observed behavior and receive feedback regarding performance.

Modeling can be used to teach a variety of behaviors, including cognitive strategies, social skills, and coping methods. In modeling cognitive strategies, other children can be asked to verbalize their thought processes in solving problems. For example, Schunk and Hanson (1989) used two types of coping models to exhibit cognitive processes involved in learning math skills. The coping-emotive model initially made negative statements (e.g., exhibiting low self-esteem or the perceived difficulty of the task) and then positive coping statements (e.g., regarding the use of concentration or working hard), and the coping-alone model verbalized only positive coping statements. Peers have also been used to model social skills, such as conversational skills (e.g., Kohler et al., 1997) and play behaviors (e.g., Pierce & Schreibman, 1995). As part of a larger intervention, Pierce and Schreibman taught children to model social skills, including complex play behaviors and appropriate verbalizations during play in an attempt to improve the social behavior of children with autism. Others have used children to model emotional coping with stressful or anxiety-provoking events. For instance, Elkins and Roberts (1986) had children observe the emotional coping responses (e.g., talking about and successfully coping with fear) of a peer model undergoing an operation in an attempt to reduce fear of medical procedures.

With its versatility and potential for teaching novel responses, modeling has been included in interventions targeting quite a variety of clinical problems, such as coping with medical fears (e.g., Elkins & Roberts, 1986; Gilbert et al., 1982; Stokes & Kennedy, 1980), ameliorating academic difficulties (e.g., Schunk, Hanson, & Cox, 1987), and improving language skills (e.g., Kohler et al., 1997) and affective behavior in children with autism (e.g., Gena, Couloura, & Kymissis, 2005). In a particularly interesting application, Gena et al. used video modeling to help preschool-age children with autism better regulate their emotional expressions in reaction to situations eliciting the emotions of sympathy, appreciation, and disapproval. Videotapes exhibited age- and sex-matched peers displaying appropriate verbal and facial responses for each situation. Targeted children evidenced 100% accuracy in emotional responses within 10 sessions of video modeling.

Peer Prompting and Reinforcement

Peers are often used to prompt target children to engage in particular behaviors or to reinforce the appropriate use of behaviors. Prompting includes instruction and reinforcement for appropriately following the instruction (Odom & Strain, 1984). This strategy has been used to increase autistic children's language skills (e.g., Krantz et al., 1989), improve autistic children's social skills (e.g., Odom & Strain, 1986), and improve healthy eating behaviors in obese children (Foster, Wadden,

& Brownell, 1985). For example, Foster and colleagues conducted an intervention to improve health-related behaviors of obese children using behavioral change strategies that included a peer intervention component. In this component, eighth graders were trained to reinforce the eating behaviors of selected second- through fifth-grade students. A few times a week, children met with their peer intervener, who examined their lunchbox and provided verbal praise and stickers based on food choices. When compared to control children, those receiving the intervention lost significantly more weight and evidenced greater improvement in self-concept.

Peer Initiation

In peer initiation interventions, other children are used to initiate or maintain social interactions, such as play or conversation, with the target child (Carey, 1997; Guevremont, MacMillan, Shawchuck, & Hansen, 1989; Odom & Strain, 1984). It is sometimes difficult to ensure consistent implementation by the peer initiators, and adult prompts or rewards may be beneficial (Carey, 1997; Fowler, 1988). All in all, peer initiation interventions are relatively straightforward, and even preschoolers are able to successfully implement them (Carey, 1997).

Peer initiation has been included as a component of social skills training studies targeting children with autism (e.g., Goldstein, Kaczmarek, Pennington, & Shafer, 1992; Sasso, Mundschenck, Melloy, & Casey, 1998) and behaviorally disordered or developmentally delayed preschoolers (Goldstein & Wickstrom, 1986). Goldstein and Wickstrom (1992) taught two nonhandicapped peers to intervene with three target children with behavioral problems and developmental delays. The peer interveners prompted social interaction through actions such as establishing eye contact or a joint focus, responding to the target child's verbalizations, and describing play behaviors. The teacher used rewards (tokens that could be traded in for stickers or toys) and prompts (e.g., suggested new strategies) to ensure the cooperation of the interveners. All three target children exhibited and maintained improvements in the frequency of social interactions.

Peer Tutoring

Kalfus (1984) describes peer tutoring as needing to include any two or more of the following: instructions to the student, prompts for correct responding, praise, corrective feedback, and ignoring of specific student behaviors. Peer tutors are generally thought of as individuals who focus on ameliorating academic problems, and tutoring programs can take a structured or unstructured format (Kalfus, 1984). Structured programs provide formal training and clearly delineate who is the student and who is the tutor, whereas unstructured programs do not. The independence conferred on the tutor can vary across programs (Topping & Ehly, 1998). Tutors may be provided with specific materials or a detailed curriculum. In contrast, other tutors may be expected to design their own materials and curriculum content.

These programs have been applied effectively at the classroom (see Greenwood, Maheady, & Delquadri, 2002, for a review) and individual (Calhoon, 2005; DuPaul, Ervin, Hook, & McGoey, 1998) levels. Though specific academic skills

are generally targeted, broader improvements in classroom behavior and on-task performance are often seen. In one example, DuPaul and colleagues used a classroomwide peer tutoring program to help improve the mathematics, spelling, and reading skills of target children diagnosed with Attention-Deficit/Hyperactivity Disorder (ADHD). Children were paired with each individual taking on the role of tutor. The tutor was given academic materials, including correct answers so that he or she could provide praise and reinforcement to the target child. Targeted ADHD children exhibited improvements in their on-task behavior as well as specific academic skills. In an example of a more structured program, Wright and Cleary (2006) implemented a cross-age reading skills tutoring intervention in which one child served exclusively as the tutor and the other as student. The tutor received four sessions of training that included appropriate behavior for tutors, the use of praise and other forms of encouragement, how to manage problem behavior, and instruction in the specific tutoring activity. Overall, children exhibited an increase in reading fluency of one word per week.

Peer Education

Peer education programs serve as a method of dispersing important or sensitive information throughout the peer group (Topping & Ehly, 1998). Children can, and already do, provide information to each other in a shared language (Mathie & Ford, 1998; Topping, 1996). Unlike teachers and other adults, peers can impart knowledge without being perceived as someone with authority. Children and adolescents may be unlikely to speak freely with individuals whom they know in other roles (e.g., teacher, administrator) that carry the weight of authority. Formalizing previously informal methods of knowledge transmission serves to increase the accuracy and credibility of the information communicated (Mathie & Ford, 1998; Topping, 1996; Topping & Ehly, 1998).

Peer education programs have been used to address substance abuse (Botvin, Baker, Filazzola, & Botvin, 1990), bullying (Salmivalli, 2001), and the prevention of HIV/AIDS (Agha & Rossem, 2004) and suicide (Sandoval, Davis, & Wilson, 1987). These educational programs can take a variety of forms. Some are comprehensive and ongoing, whereas others are focused and brief. An example of the former is a bullying prevention program described by Salmivalli. Students gave a schoolwide presentation (including information about bullying, participants' roles in bullying, and dramatic skits demonstrating bullying) and led small group discussions in classes (focusing on classroom climate, bullying problems, and how students could reduce bullying). During a weeklong period, the topic was also integrated in several school activities. Examples included coverage of conflict negotiation topics in the school news and related posters displayed in art classes. Following the intervention, reductions were found in self-reported victimization across all grades and peer-reported bullying for seventh-grade girls. In contrast, an HIV/AIDS prevention program with Zambian adolescents was conducted in single sessions lasting less than 2 hours (Agha & Rossem, 2004). Adolescents attended one discussion-oriented session focused primarily on condom use and abstinence.

The educators provided factual information, performed skits, and provided the adolescents with informational pamphlets. Adolescents exhibited improvements in their beliefs regarding abstinence and condom use.

Peer Mentoring

Unlike other peer interventions where similarity in age is generally preferred, peer mentoring is often characterized by an age discrepancy (Topping & Ehly, 1998). In this relationship, the mentor exhibits more experience, skill, or knowledge in a particular area and fills the role of passing this information on to the target child. Sometimes, peer mentors and their mentees are chosen from different schools in an effort to smooth future school transitions. For instance, in one mentoring program, youth in a secondary school were chosen as mentors for primary school students (Dearden, 1998). The mentors served as resources during the transition to the secondary school.

Mentoring interventions have also been used in efforts to scaffold appropriate behaviors (e.g., Fair, Vandermaas-Peeler, Beaudry, & Dew, 2005), improve school connectedness (e.g., Karcher, 2005), and decrease violent or delinquent behavior (e.g., Becker et al., 2004). In a cross-age mentoring program, high school students served as mentors to fourth- and fifth-grade students (Karcher, 2005). Mentors and mentees were given the opportunity to select their pairs, and the mentees included several children whom teachers had identified as being at high risk for academic failure. Meetings between the dyads occurred in a group setting twice a week for 2 hours. The mentoring program positively improved school and parent connectedness in mentees. In the Caught in the Crossfire intervention, Becker and colleagues employed young adults from backgrounds similar to that of target adolescents as peer mentors and provided them with training in counseling skills, anger management, cultural competency, and resource identification. Target adolescents were individuals who had been hospitalized as the result of a violent injury. For many target adolescents, these young adults served as role models for up to a year following the initial hospitalization. Mentees were less likely to be arrested or have any criminal involvement when compared to youth receiving no intervention, but there was no difference between the groups in injury and death rates.

Peer Counseling

Despite their nonprofessional status, peer counselors serve functions similar to those of professional counselors (Topping & Ehly, 1998). They offer help to other children and adolescents by listening, providing empathy, and using problem-solving skills (D'Andrea & Salovey, 1983; Topping & Ehly, 1998). These services can be offered in groups or individually, in person or over the phone, and can involve the provision of information and resources rather than advice giving.

Most often, peer counselors address typically occurring stressors and life issues, such as conflict resolution, relationship problems, and school transitions (Bell, Coleman, Anderson, Whelan, & Wilder, 2000; Johnson & Johnson, 1995; Morey, Miller, Rosen, & Fulton, 1989). For instance, in one high school peer counseling

program, the issues most frequently discussed were relationship problems (Morey et al., 1989). Another peer counseling program addressed conflict resolution (Bell et al., 2000). Teachers and students selected a small group of sixth through eighth graders to serve as peer mediators for the school, which included students in the first through eighth grades. When conflicts arose, teachers had the option of referring children to these counselors or to the principal. Over a 6-week period, 32 of the 34 conflicts mediated by peers were deemed successful; that is, the mediator and disputants came to a written agreement and the conflict was not referred to the principal for follow-up.

Individuals presenting with more serious concerns (e.g., depression and suicide) have also been served by peer counselors (e.g., Bagley et al., 1994; Boehm & Campbell, 1996). When difficult or complicated problems arise, peer counselors should have a method for referring the counselee for professional services or other appropriate resources (Leader, 1996; Topping & Ehly, 1998). This is often the case with hotlines developed for adolescents. Callers to such hotlines present with concerns ranging from family and peer relationship problems to sexuality, academic difficulties, suicide, and drug and alcohol use (Boehm, Schondel, Marlowe, & Manke-Mitchell, 1999). An example is Teen Line, a counseling hotline staffed by adolescents trained to handle a variety of calls, including those from suicidal individuals, and provide appropriate resources as needed (Boehm & Campbell, 1996). Additional supports for the peer counselors included an available adult and the ability to trace the calls of suicidal individuals.

GENERAL ISSUES AND CONSIDERATIONS

Identifying and Selecting Peer Interveners

To start, it is important to delineate who constitutes a "peer." In a very broad definition, peers can include everyone from close friends to virtual strangers involved in a shared activity (Coleman & Hendry, 1990). In peer interventions, peers are often described as nonprofessionals who are similar to the target child in age, status, and knowledge (Ehly & Vasquez, 1998; Leader, 1996). Though using age as the sole factor in establishing similarity is a rather common practice, Shiner (1999) points to other identity characteristics, such as ethnicity, sexual orientation, social class, and sex that may be more salient features.

In addition to ensuring similarity, there are a variety of additional criteria and methods for selecting peer interveners. Many times, children are chosen because they exhibit exemplary behaviors (Fowler, 1988). Similarly, there is evidence that children who are highly respected and socially accepted can have a more significant influence on behavior change (Hamburg, 1989). Exhibiting appropriate behaviors and being socially accepted, however, are not absolute requirements. Kalfus (1984) points out that a variety of children (e.g., mentally retarded, learning disabled, juvenile delinquents) have served as effective peer agents in past interventions. Additional selection criteria might include an interest in performing the role and a consistent ability to follow adult instructions (Fowler, 1988; Kalfus, 1984; Strain, 1981). Others include a specific grade-point average, a certain time commitment, or

enrollment in a course designed to provide appropriate skills and training (Covert & Wangberg, 1992).

Although some criteria can be broadly applied, others are specific to the characteristics of the intervention (e.g., fixed versus flexible roles), the amount of adult supervision provided, or the clinical problem being addressed (Fowler, 1988). For instance, good verbal and other communication skills may be desirable in verbally based interventions, but may be unnecessary for a modeling intervention (Hamburg, 1989; Kalfus, 1984). Counseling interventions may require adequate problem-solving abilities and social competence (Leader, 1996), and regular school attendance is an important element for school-based interventions (Fowler, 1988; Odom & Strain, 1984; Strain, 1981).

Selection methods vary considerably as well. The first component of the selection process is choosing potential interveners from a broader population; this has been done in a number of ways. Bell and colleagues (2000) had students choose a pool of potential interveners from which teachers or school administrators then selected appropriate individuals. In another intervention, teachers or school staff provided recommendations, and the potential interveners were given the ultimate decision regarding participation (Kim, McLeod, Rader, & Johnston, 1992). Chapman (1998) suggests using sociometric nominations of social acceptance as a method of identification to ensure that interveners are drawn from various subgroups of the population. Greater diversity of interveners in terms of race, cultural background, interests, personality, or other characteristics improves the chances that a target child will have someone with whom he or she can identify (Chapman, 1998; Covert & Wangberg, 1992). Sociometric measures of acceptance have been the major selection method for social skills interventions (e.g., Fantuzzo et al., 1988).

Selection often includes an effort to determine whether or not the peers meet recommended criteria. Generally, the potential peer interveners and others (e.g., teachers) are included in this process in order to assess the ability to intervene, intellectual and emotional capabilities, commitment, tolerance, and empathy (Chapman, 1998; Leader, 1996). Chapman recommends including the intervention recipients, the potential interveners, school or community personnel, and the program coordinators in this step. Leader described use of a multiple-stage process for selecting adolescents to serve as counselors for a teen hotline. Potential peer counselors were asked to complete a questionnaire and were individually interviewed by a clinician. This was followed by a group interview and an orientation meeting with their parents.

Training Peer Interveners

Once selected, interveners need sufficient instruction to adequately implement the intervention. As mentioned, the selection process often elicits interveners with specific personality characteristics or skills. Although this expertise should be respected, these individuals will benefit from further training (Leader, 1996). The format, duration, and components of training vary across interventions. Training may include lectures on important topics, modeling of techniques, role-playing, pre- and postassessments of skills, and continued monitoring and supervision (e.g., Covert & Wangberg, 1992; Fowler, 1988; Leader, 1996; Martin, Martin, & Barrett, 1987).

A number of peer interventions include instruction on similar topics and skills, such as communication and listening skills; delineation of responsibilities, requirements, and rules; confidentiality; and enumeration of additional community resources or referral sources (Bishop, 2003; Covert & Wangberg, 1992; Fowler, 1988). Bishop recommended that certain core skills (e.g., active listening, empathy) be taught to train individuals in supporting others, regardless of the specific intervention to be used. Of course, most interventions will also necessitate the addition of more tailored instruction topics. For example, training for an adolescent suicide prevention counseling program included topics such as depression and characteristics of individuals who are suicidal (Martin et al., 1987).

The depth and breadth of training needed will depend on the type of intervention. Little or no training might be needed to teach a child to model appropriate behaviors, whereas in-depth training may be needed for peer education or peer counseling formats. In one school intervention, for example, training for peer mediators required sessions held once every 2 weeks over a 2-month period (Bell et al., 2000). To ensure the maintenance of training benefits, two booster sessions were held. For another intervention, adolescent hotline counselors participated in a 13-week training program that included two weekly sessions focusing on listening skills and adolescent psychology (Leader, 1996). Training also involved the observation of others fielding phone calls and participation in monthly supervision sessions.

Studies examining training intensity needs for peer counseling interventions have been inconclusive. For example, Guttman (1987) randomly assigned peer counselors to either a high-training or minimal-training condition. Counselors in the high-training condition participated in a yearlong training and practicum, whereas those in the minimal-training condition participated in three workshops, each 4 hours long. Both groups exhibited improvements; the only group difference was that the minimal-training group used a greater number of statements identifying the group or group process. In another study, Cooker and Cherchia (1976) found that adolescents were able to competently facilitate adolescent groups on drug use after only 8 hours of training. Although there is not agreement regarding the optimal duration or intensity of treatment, there is consensus that supervision or adult monitoring should continue beyond the constraints of formal training (e.g., Covert & Wangberg, 1992; Fowler, 1988; Leader, 1996).

Considering the Peer Interveners Themselves

In addition to the benefits afforded to target children, peer interventions have been shown to have positive effects on the interveners such as improvements in classroom behavior, grade point average (Garner, Martin, & Martin, 1989), mental health (Silver, Coupey, Bauman, Doctors, & Boeck, 1992), and self-esteem (Kim et al., 1992). Indeed, Hamburg (1989, p. 9) suggests that "the long-term effects of peer counseling may be more important for the peer leaders than for the youngsters counseled. By being useful to others, adolescents can build a more durable basis for self-esteem."

Some gains may come from participating in training and gaining the skills necessary to intervene. In an intervention described by Silver and colleagues (1992),

inner-city adolescents who participated in a 3-month counselor training and accepted positions as peer counselors exhibited improvements in their mental health. The training, however, did not promote such improvements in all of the adolescents. Those who declined counseling positions or who were not judged to be adequately skilled following training did not exhibit similar gains. In another peer counseling intervention, Kim et al. (1992) reported that trainees in their intervention evidenced improvements in empathic responding and personal growth (i.e., communication skills, emotional control, helping others, and interpersonal relations).

Unfortunately, there is an absence of research examining the potential risks for peer interveners. In the absence of empirical findings, we are left to speculation. For instance, Fowler (1988) suggests the possibility that the target child could retaliate against the peer implementing treatment. Perhaps worthy of some exploration is the potential for risks associated with increased deviant peer affiliation. In well-publicized research, Dishion and his colleagues have demonstrated iatrogenic effects for group interventions that bring together at-risk youth (e.g., Dishion, McCord, & Poulin, 1999). They reviewed evidence from two experiments suggesting that youth participating in group interventions had increased negative behaviors (e.g., smoking, delinquent behavior) in short- and long-term follow-ups. Dishion and colleagues suggest that "deviancy training" may be occurring within adolescent groups and interfering with intervention effects. That is, deviant peers may socialize each other to engage in antisocial behavior. Though it seems plausible that such processes could operate in peer interventions, there are unique elements that are likely to afford some protection. For instance, peer interveners typically pass through selection procedures and are often chosen based on their social competence. As such, they may not be as prone to deviancy training.

Advantages and Disadvantages of Peer Interventions

Using youth as interveners has several advantages relative to more traditional adult-mediated approaches (e.g., teacher-implemented educational strategies, psychotherapy). Their convenience and cost-effectiveness enhances their applicability to school settings (Covert & Wangberg, 1992; Fowler, 1988; Topping & Ehly, 1998). In this way, they have the potential of expanding service provision to larger numbers of children and establishing links between at-risk children and needed professional services (Henriksen, 1991). They also allow adults to focus efforts on other activities and interventions (Kalfus, 1984; Topping & Ehly, 1998). Even if not always meeting the same quality standards as adult-mediated approaches, peers have the advantage of being able to implement without delay (Greenwood, Carta, & Kamps, 1990; Topping & Ehly, 1998). They are in a unique social position that allows them to observe behaviors that may go unnoticed by adults and to effect change on other children's behavior (Carey, 1997; Fowler, 1988). More immediate feedback and consequences are also possible. Another advantage is the facilitation of maintenance and generalization. Typically, the intervention is carried out in the target child's natural environment; thus, the peer intervener, as well as other relevant stimuli, remain in place after treatment and serve as cues for appropriate responding in the future (Stokes & Osnes, 1989).

Given the social milieu in which children operate, peers may be especially important for influencing and changing behavior. This may be particularly true during adolescence, when greater amounts of time are spent with peers, who become a primary source of social support. Also important to consider is that such peer influence can be positive or negative. Well before adolescence, for example, children can unintentionally contribute to unwanted behaviors. Fowler (1988) suggests four main ways in which children reinforce negative behaviors. First, children positively reinforce the negative behavior. For instance, a child who uses aggression to resolve a conflict may be subsequently accepted by the peer group. Second, peers negatively reinforce the behavior. In this case, a child who is being teased might hit the bully, resulting in the cessation of taunting. Third, alternative positive behaviors are ignored or inappropriately reinforced, resulting in their eventual diminishment. Thus, a child may attempt to enter group play appropriately (e.g., asking nicely to play, attempting to take turns with a toy) but be denied access to the group. Over time, the child may cease to use these appropriate behaviors if they do not result in the desired outcome. Fourth, positive behaviors or social attempts are met with negative peer responses. In this case, children may respond to an appropriate social request, such as asking to share a toy, with aggression or by making negative comments (e.g., "I don't want to play with you"; "I don't like you").

Including peers in intervention has other advantages. For instance, having another child model appropriate behavior can be more beneficial than using an adult model (Schunk, 1998). The peer model is likely to be perceived as more similar to the target child and is less likely to demonstrate a perfect performance. As such, the target child may better identify with the child model and perceive himself or herself as able to mimic the model's behavior. In addition, children might be more likely to use services provided by their peers than more traditional services (Covert & Wangberg, 1992). For example, the use of traditional counseling services may carry a greater stigma than meeting with a school's peer counselor. For some topics, youth may be perceived as more expert than adults and better able to understand the child's or adolescent's experience (Mathie & Ford, 1998). When attempting to educate adolescents about sexual behavior or social decision making, for instance, it may be difficult to separate an adult from his or her other roles (e.g., a teacher), increasing discomfort and decreasing willingness to openly engage in discussions. Another adolescent may be perceived as having a better understanding of the youth's real-life struggles with these topics.

Despite these advantages, there are some limitations. Increased adult involvement can undermine efficiency and cost-effectiveness and might suggest that the intervention is no longer truly peer-mediated (Fowler, 1988; Kalfus, 1984). Indeed, the many advantages of using peers notwithstanding, Kalfus suggests that adults may be more efficient and effective in producing behavior change. For example, an adult may be better able to teach new skills or effectively implement rewards. Another disadvantage of using peers relates to the need for informed consent and confidentiality (Ehly & Vasquez, 1998). The peer intervener might need to be privy to sensitive information regarding the target child's behavior and need for treatment. As discussed earlier, there may also be risks for the peer intervener. Finally, as illustrated in our case example, these interventions are often implemented in

schools and other nonclinic settings, and their feasibility for clinicians not already working within such environments may be an important concern.

Ethical Considerations

The use of peers as intervention agents raises some unique ethical considerations. One of the major issues is maintaining client confidentiality. The peer intervener may need to be made aware that treatment is occurring and learn private information. The ability of the intervener to understand and protect confidentiality should be carefully considered in the selection process. Such concerns may be greater in some forms of intervention than others. For instance, peer education interventions rely primarily on the dispensation of information and may not include disclosure of personal information. In contrast, peer counseling interventions often involve discussion of sensitive, private matters that would require counselors to have an understanding of and respect for confidentiality. Furthermore, peer counselors should be trained on the limits of confidentiality and when it may be appropriate to seek adult support or intervention (Ehly & Vasquez, 1998).

A second major concern pertains to informed consent. In some interventions, target children may not need to know that they are participating in an intervention. In fact, the intervention may be more successful if the target child does not realize another child has been instructed to help them. For example, in a social skills intervention, the target child may be naive to the fact that other children are being instructed to initiate social interactions (e.g., Middleton, Zollinger, & Keene, 1986). The degree to which the intervener participates voluntarily is also a significant ethical concern. Individuals selected as peer interveners should have the option of whether or not to participate (Greenwood et al., 1990). Some children may feel more comfortable in this role than others. Given the responsibilities placed on the peer intervener, he or she should be informed of all aspects of the role in the intervention and be allowed the choice of whether or not to be involved.

Evaluating Peer Interventions

A variety of approaches have been used to evaluate peer interventions, ranging from assessments of the interveners' abilities to reports of treatment satisfaction and objective measures of behavioral changes (Covert & Wangberg, 1992). Fowler (1988) recommends the use of a number of different methods to assess intervention effects, including consumer satisfaction surveys, daily assessments of effectiveness, assessment of generalization and maintenance of behavior changes, and assessment of behavioral changes in the intervener. Compliance with treatment protocols and the accuracy with which youth implement the intervention or reward systems should also be examined. Although the evaluation of peer interventions is important for many reasons (e.g., determining behavior change, funding, and gaining continued commitment from schools), the quality of program evaluations varies dramatically, and a number of studies do not use any form of objective evaluation (Covert & Wangberg, 1992).

In spite of such variation in outcome assessment quality, there is a fair degree of empirical support for the effectiveness of peer interventions. As indicated in

our review thus far, peer interventions have been effective in addressing a wide range of presenting concerns. Published reviews of two of the major forms of peer intervention have also been favorable. In a review of peer education programs, Mellanby, Rees, and Tripp (2000) reported that, in general, peer-mediated programs were at least as effective as adult-led programs in improving knowledge and inducing behavioral changes. Johnson and Johnson (1996) completed a review of school counseling programs evaluating the knowledge gained from training as well as outcomes at the individual and school level. After conflict resolution training, students were found to have made improvements in their management of conflicts and use of resolution skills. In peer mediation programs, high levels (i.e., over 80%) of successful resolutions were generally obtained, although some programs had lower rates (e.g., 58% to 74%) and not all programs resulted in changes to the overall school climate.

CLINICAL APPLICATIONS

Not surprisingly, with their flexibility and many other advantages, peer interventions have been applied to many different clinical problem areas. Our review of peer intervention forms included many example applications. In this section, we provide a selective review highlighting the targeted problem areas.

Communication and Social Skills Deficits

Related interventions for communication and social skills deficits generally take the form of peer modeling, peer social initiations, or peer prompting and reinforcement. Children targeted have included those with autism, developmental delays, and cognitive deficits, as well as those evidencing social withdrawal. In an intervention aimed at improving social and communication skills in children with autism, normally developing preschoolers were used as models of appropriate play behavior (Kohler et al., 1997). Target children's social interactions with other children increased during the peer intervention phase and were maintained at follow-up. Peer modeling, however, is not always successful with autistic children. For instance, Charlop and Walsh (1986) used time delay and modeling procedures to facilitate affection responses in autistic children. The children were quick to learn the behaviors through the time delay technique, but peer modeling was ineffective. The authors suggest that the loosely structured environment of the modeling intervention may have contributed to its failure.

In a unique application, Krantz and colleagues (1989) trained an adolescent with autism to prompt three other autistic children in an effort to increase conversational abilities. Through a series of role-play sessions, the adolescent prompter was trained to talk about sports and ask five questions that could be applied to a variety of topics (e.g., "Do you like basketball/baseball?"). The target children evidenced improved conversational skills in dyadic and group conversations. Moreover, the skills generalized to new situations and were maintained at a 1-month follow-up. Other evidence suggests that peer initiation interventions may be most successful when the target child is paired with one child instead of multiple children. In a study

by Sasso and colleagues (1998), children with developmental delays and autism showed greater improvement in social behaviors when paired than in a similar intervention conducted in triads.

Socially withdrawn children have also been the focus of such interventions. Guevremont and colleagues (1989) trained classmates to initiate and maintain interactions with two withdrawn children. Significant improvements in each child's social behavior were observed during and following the intervention. Similar success was observed in an intervention for four blind, socially withdrawn elementary school children (Sisson, Van Hasselt, Hersen, & Strain, 1985). Peers were used to initiate social behaviors in the target children, and increases in social behaviors were maintained at a 4-month follow-up assessment. In a treatment for socially neglected children, Middleton and colleagues (1986) selected accepted or popular peers to act as interveners and provided them with weekly instruction on behaviors to use with the target children (e.g., greeting or sitting next to at lunch). Also included were instruction and modeling components in which the peer modeled proper behavioral responses and offered suggestions for coping with unwanted behaviors. Socially neglected children who received treatment showed increases in peer nominations during posttest assessments.

Academic Problems

Tutoring is the most common form of peer intervention used to address problems of an academic nature, although other approaches (e.g., mentoring, modeling) have been used as well. Intervention targets have included specific academic skills, such as writing, spelling, and reading (e.g., Braaksma, Rijlaarsdam, & van den Bergh, 2002; James, Charlton, Leo, & Indoe, 1991), as well as school behaviors, such as work skills and attention to task (e.g., DuPaul et al., 1998). Formats of these interventions have also varied in terms of who is selected to intervene and the rigidity of roles. Competent peers may be used in the fixed role of intervener (e.g., Topping, Peter, Stephen, & Whale, 2004) or children may alternate between being the target child and the intervener (e.g., Kohler, Schwartz, Cross, & Fowler, 1989).

Several studies have suggested that children learn academic skills from peer models better when the model's competency level is similar to their own (e.g., Schunk et al., 1987). For instance, young teens were assessed on their ability to learn argumentative writing in three different conditions (Braaksma et al., 2002). Two conditions involved observing another child who was either a good or a weak model. The final condition was a control condition in which the children had to write short arguments. Individuals learned better from the model that was more similar to them in competence. Similarly, children had greater self-efficacy for learning and actually learned math skills better when observing a peer of similar competence gradually learn the skill than when observing a child who had already mastered the skill (Schunk et al., 1987). Notably, however, repeated observations of a mastery model resulted in similar effects.

Other interventions have been aimed at improving general academic skills. Kohler and colleagues (1989) selected three low-achieving students to receive a peer intervention aimed at improving independent work skills. Rather than assume

solely the client role, these children and their classmates alternated roles daily. Regardless of the role assumed, the target children exhibited improvements in work skills (i.e., on-task behavior); however, task completion and accuracy did not improve.

Bullying, Aggression, and Disruptive Behavior

Peer programs aimed at bullying, aggression, and disruptive behavior have taken on a number of formats, including counseling, mentoring, mediation, and education. In his suggested strategies for preventing school violence, Stephens (1997) encourages the implementation of peer counseling and mediation programs. In his words:

> Students represent some of the best agents for promoting and maintaining a safe and secure campus. An effective peer counseling program can head off many problems before they reach explosive and violent levels. In addition, the students who are trained as counselors become wonderfully influential resources for nonviolent problem solving. (p. 74)

Bullying has been spotlighted recently, and a number of programs focus on providing support services to victims. For example, Boulton (2005) developed one-on-one peer counseling programs for victims in two secondary schools in the United Kingdom. When students were asked about their previous and future use of these programs, 22% stated that they had previously used the program and the majority said they would definitely (37%) or maybe (60%) use the program if needed in the future. In choosing the people they were most likely to go to for help with bullying, students chose close friends over the peer counselor, but the peer counselor was rated higher than a teacher or sibling. There was no difference in the likelihood of seeking help for bullying between the peer counselors and school counselor or parents.

Other programs focus on the aggressors or individuals who exhibit disruptive behaviors. For example, the Caught in the Crossfire program involved peer mentoring in a community setting for adolescents who sustained violent injuries (Becker et al., 2004). This program was developed to reduce the likelihood of reinjury or future involvement in delinquent activities. Recipients were less likely to be arrested or have any criminal involvement when compared to individuals who did not receive the intervention. Rather than target individuals at high risk for violence, other programs use a broader approach by intervening schoolwide. For instance, sixth-grade students received a violence prevention program taught by a teacher with or without a peer leader (Orpinas, Parcel, McAlister, & Frankowski, 1995). Individuals who received the teacher plus peer leader intervention exhibited greater attitude changes toward violent responses.

Peer Mediation and Conflict Resolution

Although in some ways similar to peer counseling, peer mediation is a more focused intervention strategy aimed at resolving conflicts between two individuals. In a schoolwide implementation of a peer mediation program, Bell et al. (2000) observed important changes in student conduct. Thirty students in the sixth through eighth

grades were trained to be peer mediators for the entire school population of first through eighth graders. Teachers were given the option of referring students to the program or to the principal. During the 6-week study period, 34 mediations were conducted, and the percentage of suspensions and fighting decreased. Johnson and Johnson (1995) evaluated the effectiveness of conflict resolution training using both student body and cadre approaches. Overall, the results indicated that conflict resolution strategies and problem solving can be learned by the students and used in real-world settings.

Suicide and Depression

Suicide and depression interventions are generally in the form of prevention and detection programs or counseling programs. Some interventions encompass a two-tiered system of primary (curriculum-based prevention program) and secondary (peer counseling) prevention components (Sandoval et al., 1987). Further, a number of peer counseling programs address depression and suicide prevention but do not focus exclusively on these issues (e.g., Boehm et al., 1999). For example, one teen counseling hotline received 441 calls pertaining to depression or suicide over a 5-year period (Boehm & Campbell, 1996). Researchers in Hong Kong developed a schoolwide suicide prevention program that was designed to address concerns with available alternatives (Bagley et al., 1994). The Health Intervention Training—Mutual Aid Network promotes positive social support among students using a "student-body approach" in which all students serve as both intervener and client.

Health Behaviors

Health behaviors have been addressed mainly through peer education programs focusing on topics such as sexual behaviors (Agha & Rossem, 2004), AIDS and HIV prevention (Backett-Milburn & Wilson, 2000), substance abuse (Botvin et al., 1990), and asthma education (Shah et al., 2006). A review of studies comparing youth-led and adult-led health education programs indicated that participants in youth-led programs evinced knowledge gains, attitude changes, and behavioral outcomes that were as positive, or more so, as in adult-led programs (Mellanby et al., 2000).

Backett-Milburn and Wilson (2000) described an HIV/AIDS peer education program in which educators worked together in groups. With this enhanced flexibility, the peer educators could capitalize on individual strengths. In addition, while some members led the groups, others were able to work behind the scenes to plan sessions or search for information and resources. In another intervention focused on sexual health and HIV prevention, peer educators conducted single sessions with groups of Zambian adolescents (Agha & Rossem, 2004). The trained educators used dramatic skits to depict sexual behavior models. At follow-up, the intervention group showed significant improvements in their approval of condom use and intent to use condoms. At the 6-month follow-up, intervention adolescents were more likely to think that it was normal for both women and men to suggest using a condom.

Coping with Illness or Medical Procedures

Acknowledging the negative impact of chronic illness on children's social relations and development, coping interventions have also targeted peer interactions (Harbeck-Weber, Fisher, & Dittner, 2003). For instance, Greco, Shroff Pendley, McDonnell, and Reeves (2001) included the best friends of adolescents with diabetes in a group intervention. In this informal peer intervention, the friends learned diabetes knowledge and ways to better support the adolescent with diabetes. A novel intervention approach, STARBRIGHT, capitalizes on computer technology in a peer intervention for hospitalized children (Holden, Bearison, Rode, Rosenberg, & Fishman, 1999). The STARBRIGHT program connects hospitalized children (ages 9 through 19) through an online community. Less formal than many other peer interventions, it allows children to gain support from others experiencing similar conditions. Children using STARBRIGHT reported less intense pain and reduced anxiety compared to those receiving standard pediatric care alone (Holden et al., 1999).

More conventional modeling approaches have been frequently used to help children with medical fears (e.g., Elkins & Roberts, 1986; Melamed & Siegel, 1975). In a normative population of children, modeling was examined by Elkins and Roberts to see whether medical fears could be reduced. Third- and fifth-grade children were randomly assigned to view one of four videotapes. Peer models were used in two videotapes (either one child alone or two children talking together), an adult was used in another videotape, and the last film was a control video in which a child models coping but outside the hospital setting. Children high in medical anxiety reported reduction in their anxiety after viewing all three medically related films; a similar effect was not seen in children with low medical anxiety. Similar methods have been used with children who have significant medical fears or who are uncooperative with medical treatments. For instance, as part of a larger intervention, highly uncooperative children observed peers undergoing a dental procedure and being reinforced for appropriate behavior just prior to their own treatment (Stokes & Kennedy, 1980). Children's uncooperative behaviors (e.g., body movements, complaints, crying) were reduced following the intervention.

Case Illustration

Identifying Features of the Client

Bobby D. is a 7-year-old boy who was diagnosed with Autistic Disorder at age 3. Bobby attends first grade at a public elementary school and engages in an intensive applied behavioral analysis program for children with autism and other developmental delays approximately 20 hours per week. Currently, Bobby lives with both parents and two siblings (4-year-old sister and 9-year-old brother).

Presenting Complaints

Bobby currently divides his weekdays between a structured applied behavioral analysis program and a regular classroom with the support of a behavioral specialist. He also receives individual speech and language therapy 3 days a week. All of these services are coordinated by his school as part of his individual education plan. At home, Bobby's parents receive

additional support from a behavior specialist contracted through a community agency who provides extra instruction on activities such as communication skills (e.g., asking for objects) and daily living skills (e.g., mealtime behaviors).

More recently, Mr. and Mrs. D. sought some adjunctive intervention to help Bobby improve his social behavior. On the advice of school personnel, they contacted a clinical psychologist who was in private practice and had frequently consulted with Bobby's school on cases involving autism spectrum disorders. Although Bobby's social behavior (e.g., eye contact and use of gestures) had improved, it was still of major concern. He is relatively high-functioning and has adequate communication skills, as he is able to speak in simple sentences and occasionally uses nonverbal gestures or sign language to communicate. According to his most recent speech and language evaluation, Bobby has mildly delayed expressive language and moderately delayed receptive language. At the present time, Mr. and Mrs. D. would like to see improvements in Bobby's ability to appropriately respond to and engage with his peers.

Developmental Factors

Like many children with this disorder, Bobby exhibited developmental delays in the communication and social domains. Though he started to walk and crawl on time, his first spoken words came at about 27 months. His parents also noted his frequent resistance to eye contact, lack of response to his name, and failure to reciprocate social initiations. Mrs. D. reported a normal pregnancy and on-time delivery without medical complications. Bobby was a relatively healthy child and had no major illnesses, injuries, or hospitalizations.

Parental Factors

Bobby lives in an intact, two-parent household in which both parents work full time. As such, Bobby spends several hours a day with a behavioral specialist while his parents are at work. Mr. D. has some college education and Mrs. D. completed high school. Both Mr. and Mrs. D. are involved in Bobby's treatment and work with his behavior specialist to provide Bobby with discrete trial training to address daily living skills such as appropriate mealtime and bedtime behaviors.

Bobby's behavior and special needs contribute to Mr. and Mrs. D.'s daily stress. At times their marriage has become strained due to differing opinions regarding Bobby's care. Another concern is that the additional attention given to Bobby will have detrimental effects on their other children. Mr. and Mrs. D. appear to be coping well with this stress and are involved in a support group for parents with children with autism.

History

Prior to intervention efforts starting at age 3, Bobby had very limited language abilities. He exhibited severe delays in both expressive and receptive language. As previously mentioned, Bobby's language skills have improved and he now evidences only mild to moderate delays. At times, Bobby uses jargon and echolalic speech, but he can communicate using simple sentences.

Bobby has also exhibited a number of delays in the acquisition of social skills. When initially diagnosed, Bobby rarely made eye contact and made very few

(*Continued*)

attempts at social referencing. For instance, he would occasionally grab one of his parent's hands and pull him or her toward a requested item. However, he never made attempts to show his parents completed activities or gain their assistance when injured. When around his siblings or peers, Bobby rarely made any attempts to engage them and often ignored other children's social bids. For instance, in preschool, he would sometimes push other children away when they approached him to engage in a joint activity. At this time, Bobby has made improvements in his eye contact such that he consistently makes eye contact with familiar individuals (e.g., parents, teachers, siblings, familiar peers). He still has difficulty in his peer interactions and does not spontaneously engage in reciprocal play behaviors.

Assessment

The consulting psychologist conducted a clinical interview with Bobby's parents. In addition, Mr. and Mrs. D. each completed the Achenbach Child Behavior Checklist for ages 6 to 18 (CBCL) and his teacher, Ms. Baker, completed the Teacher Report Form (TRF) and Social Skills Rating System (SSRS). Bobby was also observed in his classroom and on the school playground to assess for the frequency and quality of interactions with peers. Based on information obtained from the parental interview and observations of Bobby's behavior, the psychologist completed the Childhood Autism Rating Scale (CARS), a clinician-completed scale assessing abilities or domains that are frequently impaired in children with autism.

On the CBCL, Mr. and Mrs. D.'s responses both resulted in clinical elevations on the Social Problems, Attention Problems, Externalizing, and Total Problems subscales. Mrs. D.'s responses also resulted in an elevation on the Withdrawn/Depressed subscale. On the TRF, Ms. Baker's responses resulted in clinical elevations on the Social Problems, Attention Problems, Aggressive Behavior, Externalizing, and Total Problems subscales. Overall, Ms. Baker endorsed a greater severity of symptoms than either Mr. or Mrs. D. Her SSRS endorsements indicated that Bobby's social skills were well below that of more normative comparison children, and his overall conduct was somewhat more problematic. Frequently occurring behaviors included temper tantrums, whereas behaviors such as making friends easily and giving compliments to peers were endorsed as never occurring.

Behavioral observations in both the classroom and on the playground revealed that Bobby rarely engaged in social interactions with his peers. During a 20-minute play period, Bobby was observed to interact with his peers for less than 5 minutes. All interactions were initiated by peers, but Bobby often ignored such initiations. On two occasions, Bobby was observed to push other children away. In the classroom, he ignored other children even during a group task. While three other children in his group interacted and worked together to complete a puzzle, Bobby was observed to stare off into space and rock in his chair.

On the CARS, Bobby received an overall score that placed him in the mild autism range. Domains in which he exhibited the most significant delays included his relations to others, interactions with objects, nonverbal communication, and emotional responding.

Case Conceptualization

A major characteristic of autism is impairment in social functioning, including difficulties initiating or maintaining social interactions, impaired social relationships, and a lack of reciprocity or sharing of social or emotional experiences. Similar to other children with autism, Bobby has difficulty engaging in social interactions with peers and often refuses or ignores other children's social bids. Bobby exhibits other behaviors characteristic of autism, such as rocking, spinning, and repetitive play behaviors. Unlike other children his age, Bobby's play activities are generally limited to playing with trains or railroad-related toys and pictures. When looking at pictures of trains or playing with trains, he often becomes immersed and has difficulty being redirected to other activities. At this time, the major treatment concern is to increase the frequency of Bobby's social interactions with his peers. Improvements in social and other domains are difficult to achieve, and progress is incremental.

Course of Treatment and Assessment of Progress

The major goals of treatment were to improve Bobby's responses to other children's social initiations. The treatment approach was based on that used by Goldstein and colleagues (1992). Based on Ms. Baker's report, three socially competent peers were selected from Bobby's classroom to facilitate the intervention. The psychologist obtained Mr. and Mrs. D.'s permission to involve these children in intervention efforts. Prior to the start of the intervention, the psychologist discussed the responsibilities and behaviors required with each peer intervener and their parents. Parental consent and child assent were obtained. The parents were informed of the nature of the intervention and the need for confidentiality. The children were told that they had been selected to play with Bobby, but were not explicitly informed of the treatment aspect of this play to minimize responsibility for maintaining confidentiality. Bobby was not informed that he was the target of an intervention. The children were trained in separate steps to use several strategies to promote social interaction. Consistent with Goldstein and colleagues, the steps included (a) establishing joint attention by moving into Bobby's visual space, (b) saying "Bobby," (c) repeating "Bobby" and physically tapping him if he remained unresponsive, (d) starting a conversation and listening to see if Bobby responded, and (e) continuing to speak to Bobby. Each peer had individual play sessions with Bobby in which these steps were practiced. To gain peer intervener compliance, rewards were given following each play session for following the steps and engaging in conversations with Bobby.

Bobby's progress during these sessions was monitored by a teacher or teacher's aide who recorded the number of steps that were required to gain Bobby's attention and whether or not he engaged in conversation with the peer. To assess generalization, the psychologist observed Bobby on a biweekly basis in his regular classroom setting or on the playground. As with the baseline observations, the frequency of Bobby's interactions with peers was tallied. Over a 4-week period, Bobby exhibited substantial improvement in his responses to the three peers during the one-on-one play sessions. However, these improvements failed to generalize to other, less structured settings.

(*Continued*)

Complicating Factors

Although Bobby's behavior improved in the one-on-one play sessions, it failed to generalize to other situations, such as unstructured play with other peers. During her observations, the psychologist noticed that Bobby's responses to peer initiations had not improved and that peers rarely attempted to initiate such interactions. Even those peers who engaged in the one-on-one play sessions were rarely observed to attempt social interactions outside of these sessions. To address these concerns, the peer prompters were instructed to also initiate social interactions outside of the individual sessions. Teachers took note of these initiations and rewarded interveners for attempts outside of the individual sessions. Over the next 6-week period, Bobby's peers engaged in more frequent initiation attempts. Further, Bobby's responses to these attempts improved such that the frequency of ignoring initiations decreased dramatically and aggressive behavior (e.g., pushing children away) was eliminated completely.

Managed Care Considerations

Receiving compensation for services provided in schools is often a challenge. In this case, only the time spent with the parents in the clinic may be billable according to most managed care guidelines. The time spent training and supervising the peer interveners, as well as coordinating and monitoring the ongoing intervention, would probably not be covered by most insurance companies. Travel to and from the school would likely be an additional unrecoverable expense. Being directly contracted by the school as a consultant is a preferable arrangement in cases such as the one presented here.

Follow-Up Treatment

Once the frequency of Bobby's social interactions and the quality of his responding had improved to reasonable levels, the active peer intervention was terminated. Continued observation and teacher report indicated that Bobby was engaging in more frequent social interactions with his classmates and rarely ignored other children's social bids. Subsequently, a follow-up treatment was started that consisted of attempts to use peers to model more complicated and extended social interactions.

Implications of the Case

Bobby's case highlights several concerns regarding the implementation of peer interventions. In this case, some difficulty in achieving generalization was encountered. Although the peer interveners initiated interactions with Bobby during structured play sessions, they failed to initiate in other situations in which they were not directly reinforced. This emphasizes the need for continued monitoring of progress and supervision of peer interveners. Bobby's situation also illustrates some of the challenges in obtaining consent and maintaining confidentiality. Given the young age of the children involved in this intervention, they were not fully apprised of the therapeutic nature of their involvement. Further, Bobby was not made aware that he was being targeted in these social initiations.

Recommendations to Clinicians and Students

Despite their many advantages and seeming simplicity, peer interventions pose some significant challenges. Given issues relating to reimbursement and

availability of peers to intervene, a majority of these interventions take place in a school setting. Even those interventions in community settings occur outside of a typical clinic setting. Added demands include the time required for training the peer interveners, consulting with teachers, and providing ongoing supervision and monitoring. In combination with the unique concerns regarding informed consent and confidentiality, the feasibility of this intervention form is an issue for most practitioners. Such interventions may be best suited to those already working within the school setting.

SUMMARY

With their many inherent advantages, peer interventions have grown in popularity and have been used in efforts to address a wide variety of clinical problem areas and disorders. For many reasons, peer interventions may be preferable to adult-led treatments. Due to their similar social status, children may be perceived as more accessible and can be convenient for implementing treatment. Information may be more readily received when coming from one's peer, and the stigmas associated with treatment seeking may be lessened. The convenience and cost-effectiveness of peer interventions enhances their applicability to school settings, thus potentially expanding service provision to large numbers of children. Using peers as treatment agents and implementation in the natural environment also facilitates maintenance and generalization.

With these advantages come some considerations and potential concerns. Though often touted for their convenience and cost-effectiveness, peer interventions are not as straightforward as they may seem. Clinicians must carefully weigh the pros and cons relative to more traditional adult-mediated approaches. Setting is one consideration, as these interventions are most frequently implemented in the child's natural environment and seem most amenable to school environments. A variety of particular intervention forms are available, and one most suitable to the target child and presenting problem must be chosen. Implementation requires a carefully considered approach to the selection, training, and supervision of the peer interveners, as well as sensitivity to the unique ethical challenges posed by peer interventions. Securing the cooperation of others in the child's environment, such as teachers and other school personnel, and obtaining quality outcome data are additional considerations.

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PART VI

Special Issues

CHAPTER 26

Cultural Issues

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Culture is the lens through which we view children and adolescents. Culture provides the frames we use to label, categorize, and make sense of childhood development and behaviors. It defines our relationship to children, what is considered normal and desired child behavior, and what is considered pathological. Culture indisputably impacts childhood psychopathology. How disorders are expressed and communicated, whether help is sought for children, treatment processes and desired outcomes are all influenced by culture. Psychologists and other mental health professionals must have a thorough understanding of culture and the ways it impacts child and adolescent mental health, psychopathology, service utilization, assessment, and treatment (American Psychological Association [APA], 2003; Constantine & Sue, 2005; Cuellar & Paniagua, 2000).

In this chapter, we first provide an overview of major sociocultural changes occurring globally and in the United States that have implications for children. With this contextual frame we highlight the need for cultural competence in the assessment and treatment of children. Next we define culture and describe its complexity, outlining the ways that it influences the status of children, child development, socialization processes, family and social relationships, and beliefs about child and adolescent psychopathology. We introduce key cultural concepts, such as acculturation, ethnic identity, and worldview, and provide suggestions for working with culture at each phase of assessment and treatment. Problems related to culture, cultural adaptations to treatment, and unique populations are also discussed.

A CHANGING MULTICULTURAL CONTEXT

Global Perspectives

Worldwide increases in population mobility and economic globalization are bringing together people from all cultures at unprecedented rates (Global Commission on International Migration [GCIM], 2005; Population Resource Center [PRC], 2005; N. Kaufman & Rizzini, 2002). Children and adolescents, accounting for 39% of the world's population, are increasingly growing up in a multicultural context (GCIM, 2005). Economic globalization and infrastructure development have resulted in increased opportunities for living, working, and studying in other countries. There is an overall increase in voluntary migration, usually from rural to urban areas

and from developing countries to economically developed countries (GCIM, 2005; PRC, 2005). There is also an increase in forced migration, resulting in 25 million children uprooted from their home due to natural disasters, ethnic conflict, and war (United Nations High Commission for Refugees, 2001). While the importance of the extended family network remains high on a global level, population mobility breaks up traditional kinship networks and strains family and community supports.

Currently, the global population exceeds 6 billion, and over 1 billion children will be born in the next decade (PRC, 2005). Most of these children will be children of color, many will be poor, and most will live in developing countries (GCIM, 2005; PRC, 2005). Critical challenges facing these children include poverty, violations of human rights (e.g., child trafficking, involvement in armed conflict), and social justice concerns, such as the widening gap between rich and poor and between developed and developing countries (GCIM, 2005; PRC, 2005). A lack of sociopolitical power, racism, environmental threats, health risk behaviors, and global population pressures are other critical concerns facing today's children (GCIM, 2005; N. Kaufman & Rizinni, 2002; Lerner, Fisher, & Weinberg, 2000; PRC, 2005). While population rates are on the rise, natural resources are dwindling, and cultural ideology (based on ethnicity, religion, or social class) often frames the argument over resources and supplies the justification for violent conflict. Trends in population growth and mobility are expected to continue, resulting in more opportunities and demands for intercultural contact and communication (Martin & Nakayama, 2005).

Cultural Diversification of the United States

The United States is becoming an increasingly diverse, multicultural society. Children today are more likely to belong to an ethnic minority group (Hernandez, 2004, 2006). As illustrated in Figure 26.1, ethnic minority children are quickly becoming the new statistical majority. Immigrant families and children account for the greatest growth in the ethnic minority population, and about 84% come from countries in Latin America and Asia (Hernandez, 2004). Current estimates indicate that approximately 35 million foreign-born persons live in the United States, with about 1 in 5 children having at least one foreign-born parent (Hernandez, 2004; PRC, 2005). The top countries of origin are Mexico (30.7%), the Philippines (4.5%), India (4%), Mainland China (3.4%), and Vietnam (3%), with all other countries combined accounting for 45.6% (PRC, 2005).

Implications for Children

The changing multicultural context in the United States and worldwide has implications for how children and adolescents will interface with educational, legal, occupational, and health and mental health care systems (Hernandez, 2004; GCIM, 2005). These institutions were designed from a particular cultural perspective (European American) and, at least historically, were for the benefit of a population with that same cultural perspective (Marsella & Yamada, 2000). Educational, economic, health, and mental health disparities are well documented (U.S. Department of Health and Human Services, 2001). Compared to European Americans, ethnic minorities in the United States face lower educational status, higher unemployment,

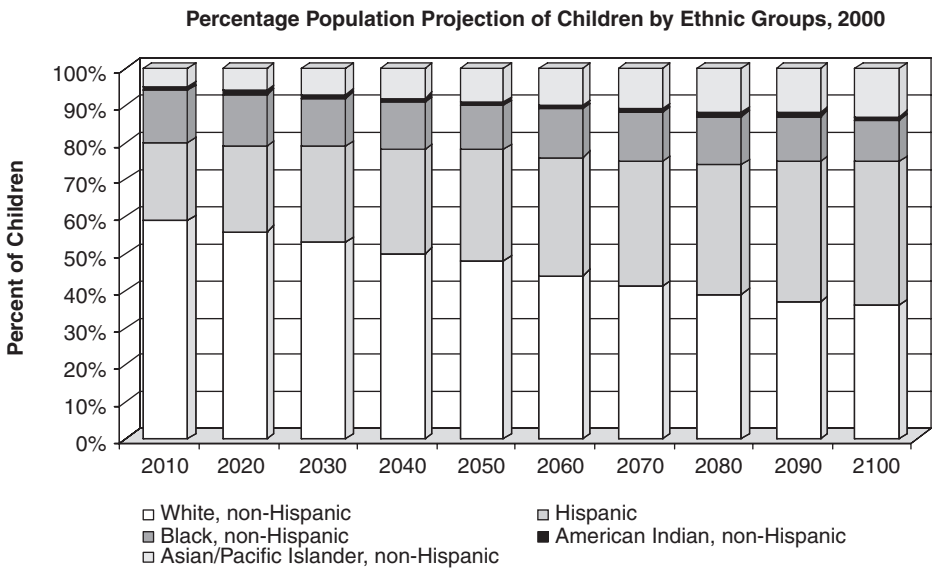

Figure 26.1 Changing ethnic profile of children in the United States. *Source:* "Demographic Change and Life Circumstances of Immigrant Families," by D. J. Hernandez, April 2004, *Future of Children*, pp. 17–47. Reprinted with permission.

higher rates of poverty, a higher risk of health and mental health problems, and increased rates of victimization and incarceration (Cuellar & Paniagua, 2000). Children living in immigrant families have higher poverty rates, are more likely to have poorly educated parents, and are more likely to live in linguistically isolated households than native-born families (Hernandez, 2004).

Despite the cultural and linguistic diversification of the United States, ethnic minority and immigrant populations continue to underutilize outpatient professional psychological services. Those who do seek help are more likely to be misdiagnosed, receive inappropriate treatment, drop out of treatment early, and receive fewer positive benefits overall compared to European Americans (Kurasaki, Sue, Chun, & Gee, 2000; Sue, Chun, & Gee, 1995; Sue, Zane, & Young, 1994). Psychologists have an ethical imperative to understand these disparities and to work toward clinical, scientific, and educational excellence in the applicability and effectiveness of their approaches (APA, 2003; Whol & Aponte, 2000).

Cultural Competence

Cultural competence is a product of general, cross-cultural, and culture-specific competencies that allow for effective functioning in the context of cultural differences (APA, 2003). It applies to individual and organizational practices and spans the areas of education, research, clinical training, and practice in psychology (APA, 2003; Constantine & Sue, 2005; D. Sue, Arredondo, & McDavis, 1992). Operationally, cultural competence is often described as three interrelated components of awareness, knowledge, and skills (Arredondo et al., 1996; D. Sue et al., 1992).

Awareness

Clinician *awareness* is commonly viewed as a prerequisite for the development of cultural competence. This requires clinicians to become aware of themselves as cultural beings. Clinicians are encouraged to intentionally bring their attention to the role of culture in their own life and to consider how it impacts their own values, beliefs, and behaviors, including biases and stereotypes about diverse children and families (APA, 2003). Skills in self-reflection can be developed and awareness enhanced through written reflection assignments, such as journal writing or reaction papers, case studies, small experiential exercises, group discussions, and coaching (Hong, Garcia, & Soriano, 2000; Martin & Nakayama, 2005).

Clinical self-awareness must extend to the profession. It is crucial to understand the sociocultural and historical context in which mainstream (i.e., European and American) theories of psychological assessment and treatment emerged (Pedersen, 2004). Although individuals of European or North American ancestry account for only one sixth of the world's population, a Western worldview and cultural perspective dominates modern psychological therapies and practices (Marsella & Yamada, 2000). Psychologists are gaining increased awareness of the cultural embeddedness of their approaches to defining, assessing, and treating childhood psychological problems.

Knowledge

Beyond awareness, clinicians must enhance their *knowledge* and understanding of culture and how it impacts children, adolescents, and the assessment and treatment process. This includes general cultural concepts applicable across many cultural groups and culture-specific information related to a particular cultural group (APA, 2003; Constantine & Sue, 2005). General knowledge includes a clear understanding of the nature of culture and how it impacts the expression of pathology, beliefs, and attitudes about childhood problems, expectations for the assessment and treatment process, and the clinician's relationship with the family.

Specific knowledge about one's *own culture* and *other cultures* is also necessary. Clinicians can enhance their cultural knowledge by reading literature, attending continuing education conferences or workshops, watching films, and reading professional books and journals as well as popular books or magazines. Experiential-based activities, such as visits to ethnic neighborhoods, international travel, second-language learning, interviews with diverse others, volunteer work, and practicum experiences, are additional ways of enhancing one's knowledge of other cultural groups.

Clinicians should undertake particular efforts to educate themselves about the major cultural groups with whom they are likely to come in contact. Additionally, when children or their family members come from cultural backgrounds with which the clinician is unfamiliar, clinicians should learn about relevant culture-specific group norms, including, but not limited to, sociopolitical history or history of oppression, cultural beliefs and values, communication styles, and attitudes toward mental health and illness (Pedersen, 2004; D. Sue & Sue, 2003). Although

it is important to be knowledgeable about group norms, there is vast within-group variability.

Skills

Clinician *skills* for cultural competence include reflective thinking, establishing rapport, bridging cultural differences, intercultural communication, and accurate and culturally appropriate assessment, diagnosis, conceptualization, and treatment (APA, 2003; Arredondo et al., 1996; Constantine & Sue, 2005; Hays, 2001). In the following sections, we provide information to further clinician awareness and knowledge about the nature of culture and how it impacts the lives of children and adolescents.

THE NATURE AND COMPLEXITY OF CULTURE

A better understanding of culture and its role in shaping beliefs and behaviors is essential for clinicians working in a multicultural environment. Although there are well over 150 definitions of culture, most describe culture as being learned and shared among a group of people and passed on through the generations (Kluckholm & Kroeber, 1952). Culture includes external factors, such as style of dress, language, food, traditions, kinship structures, and social patterns, and behavioral norms, including norms for expression of psychological problems (Draguns, 2000; Paniagua, 2000). Culture can be considered a macro-template for predicting and explaining child and adolescent behavior. However, this view of culture is deceivingly simple. Culture is quite complex and has a number of features that present challenges in the clinical setting. A failure to understand the complexity of culture can result in an over- or underemphasis on culture, stereotyping, misdiagnosis, misuse or misinterpretation of tests, or culturally incompatible treatment (Canino, Canino, & Arroyo, 1998; Canino & Spurlock, 2000).

Culture Is Pervasive and Subjective

Culture has a pervasive influence in the lives of children. It encompasses a wide range of human behaviors and activities and ways of seeing, understanding, and interacting with the world. It influences our sense of time and time consciousness, relational styles, patterns of communication, sense of personal space, family structures, worldview, and ways of thinking, categorizing, and problem solving. Culture includes overt aspects, which are driven by internal features, such as beliefs, attitudes, norms, and values, that are largely unobservable and subjective. These are the unwritten rules of culture. An iceberg analogy is often used to illustrate that the observable aspects of culture are just the tip of the iceberg. Not only are most cultural factors unobservable, but they are also out of our awareness. Like fish in water we go about our daily lives, behaving and interpreting from a particular cultural frame without the awareness that we are doing so. Intentionally bringing culture into one's awareness (and keeping it there) can help clinicians recognize cultural influences and biases (Pedersen, 2000).

Culture Is Multidimensional and Dynamic

Culture is multidimensional, and children and families may describe their culture in relation to one or more dimensions or sociocultural reference groups, such as sociorace, ethnicity, tribal or clan affiliation, language group, religion, socioeconomic status, age or generational status, gender, sexual orientation, disability, geographic or regional location, nationality, and political affiliation (D. Sue & Sue, 2003; Hays, 2001). Although these groups have common features that are shared with others, the importance of different dimensions of culture (and therefore their relative importance in assessment and treatment) is in a state of flux as children develop and move in and out of various cultural milieus.

What cultural dimensions are salient will vary from child to child and family to family across time and contexts. The importance of various cultural factors and identity dimensions may change as a function of developmental stage. For example, ethnic identity and sexual orientation may be more relevant for adolescents than for young children. Relevance may also be spurred by an event, such as experiencing an act of racism. Many ethnic minority children and adolescents report that they were unaware of their race until they were called a racist name. Other events that require a child or adolescent to recognize and adapt to another culture, such as moving to a new country, a new school, or even a different state or region, can highlight the importance of different aspects of culture and identity.

Culture Is Uniquely Expressed

Cultural influences interact with individual and environmental factors to manifest idiographically in each child. The relative importance of different dimensions will vary based on a child's learning history, life experiences, and context. For example, in the United States, where there is a history of slavery, segregation, and oppression and no shortage of racial discrimination today, "being Black" is often a salient feature in determining the worldview and identity of African American children, adolescents, and their parents (Helms, 1990). In other countries, religion, social class, or tribal affiliation may be more contextually salient. Status as a member of the majority or minority may also influence a child or family's worldview or identity. Sexual identity may be more salient for gay and lesbian youth than heterosexual youth, and religious identity may be more salient for Muslim youth growing up in a majority Christian environment. It is important to note that cultural contexts change as well, bringing to the forefront different dimensions. For example, prior to the Belgian colonization of Rwanda, one's ethnicity (e.g., Hutu or Tutsi) was largely irrelevant, but from 1993 to 1994 it became the basis for who would kill or be killed.

Culture Is Interactive

Children and their families are involved in multiple cultural systems and interactions. Children and youth move in and out of different cultural contexts and social environments in their day-to-day lives (Akiba & Garcia-Coll, 2004; Bronfenbrenner, 1979, 1989; Cuellar, 2000). As illustrated in Figure 26.2, culture

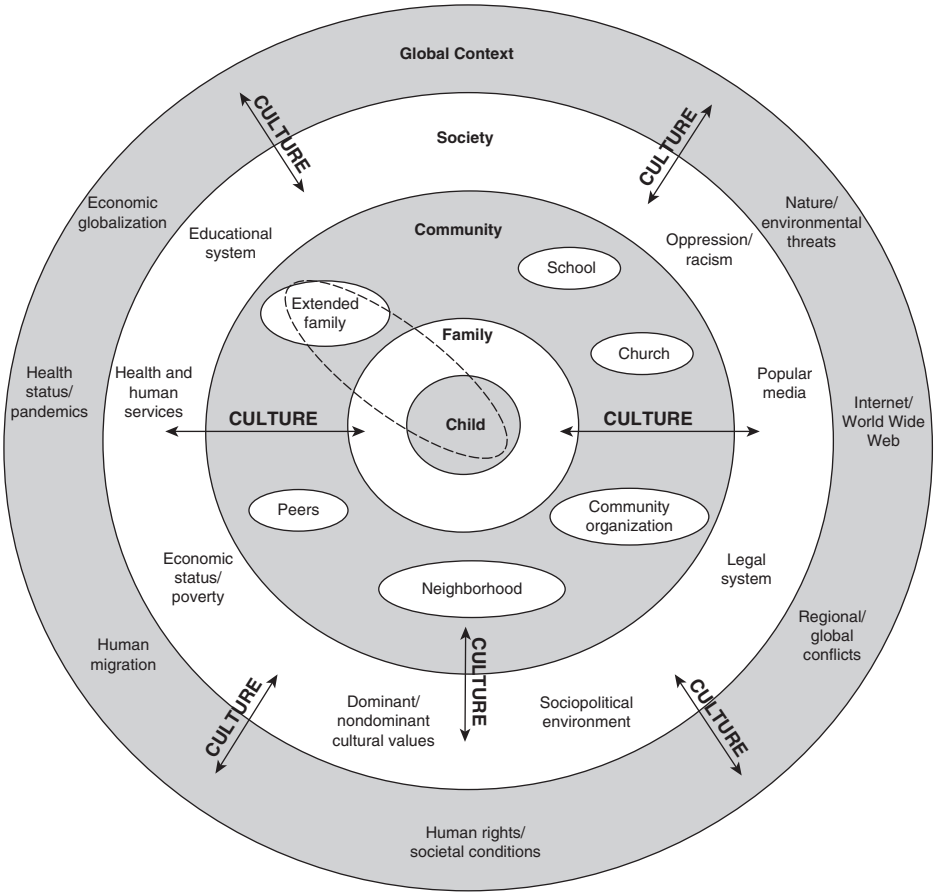

Figure 26.2 Interactive and reciprocal influence of culture across ecological contexts.

has a multilevel and reciprocal influence across children's contexts, at the proximal, microsystem level as well as at the more distal, macrosocial level. Family and community cultures interact with larger societal cultural systems and global influences. There may be drastic differences in communication, accepted norms, rules, and behaviors in the various microsystems of home, community, and school, or between the child's community and the larger society. Clinicians should consider how culture impacts the child's functioning and fit within his or her range of daily experiences. This includes the cultures of the home, school, neighborhood, and community and how they interact with each other (Bronfenbrenner, 1979, 1989).

Macrosystem factors include globalization, sociopolitical atmosphere (e.g., human rights, degree of democracy and freedom), socioeconomic conditions (e.g., poverty, educational opportunities), environmental threats related to industrialization and urbanization (pollution, safety concerns, overcrowding, noise, connection to natural environment), and overall health status (epidemics, pandemics, health risks). Of key importance is that children and adolescents are engaged in a reciprocal relationship with culture. The ecological model of child development shows

that children play an active role in defining, interpreting, and shaping their cultural environment (Brislin, 2000; Bronfenbrenner, 1979, 1989).

THE IMPACT OF CULTURE ON CHILDREN

Cross-cultural research in psychology, anthropology, and sociology has revealed much about cultural variation in the role and status of children, approaches to education and socialization, cultural and social norms for development, family structure, and social relationships (Brislin, 2000; Schmidt, 2006; Tseng, 2003). Clinicians should be aware of these differences, as they have important implications for assessment and treatment.

Role and Status of Children

Around the world there is considerable cultural variation in children's roles and status, influencing how they are viewed and treated (Brislin, 2000; Petersen, Steinmetz, & Wilson, 2005a; Tseng, 2003). For example, children and adolescents living in more individualistic cultures, such as in the United States and Western Europe, may be viewed as separate, equal individuals deserving of respect. They may have considerable choice in their day-to-day lives and are frequently given a voice in family and community matters. In the context of the one-child policy in China, it has been suggested that Chinese children are likely to be pampered and doted on by parents and grandparents. However, their status as only children in a collectivist society also brings an expectation for filial piety, increased responsibilities for future caretaking, and intense pressure to succeed (Feng, 2002; Tseng, 2003). Other cultures view children as property or as essentially providing a service. In some parts of the world, for example in New Guinea, children as young as 8 may be responsible for growing, harvesting, and cooking food for themselves and their younger siblings.

The responsibilities, role, and status of children can influence risk and protective factors that may enhance or reduce their physical and psychological health. This has implications for a child's overall development; the supports, services, and resources afforded them; and their potential to be neglected or abused (Tseng, 2003). Gender, class, and cultural issues tend to interact in this regard, often conferring male and class privilege though reduced social status and fewer economic and educational opportunities for young girls worldwide (United Nations Children's Fund, 2004).

Education and Socialization

Cultural factors shape the ways that children and adolescents are educated, socialized, enculturated, and disciplined. Major differences in child-rearing practices are found cross-culturally in terms of who is involved in education and socialization of children and also in the methods used to teach children (Brislin, 2000; Petersen et al., 2005a). In a Western setting, teaching styles favor a direct approach emphasizing written history, direct verbal communication, and linear and dialectical thinking. Critical thinking and Socratic questioning are often encouraged, and the uniqueness of each individual is valued. Teaching in the United States, in particular,

is considered child-centered, with the responsibility for the child's learning placed with the parent or teacher. In other cultures, approaches to teaching, and often parenting, tend to place more responsibility on the child and have high expectations for obedience and respect for authority (Petersen, Steinmetz, & Wilson, 2005b). Blending in and working for the good of the whole is valued, and personal distinction is frowned upon. Asking questions of persons in authority (teachers or parents) is desirable behavior in one context, whereas in another it brings ridicule and shame.

Cultural differences exist in tolerance for misbehavior and in approaches to punishment (e.g., short or long term, mild or severe) and types of punishment (e.g., physical; suspension of social, material, or activity rewards; emotional response such as shame or embarrassment, or withdrawal of love; Brislin, 2000; Peterson, Steinmetz, & Wilson, 2005c; Tseng, 2003). Families may have different approaches for boys and girls, or for children of different ages. Discipline may be used consistently or haphazardly (Tseng, 2003). In the United States, African American parents often have a more authoritarian approach to child rearing than European American parents. American Indians, on the other hand, may value a natural unfolding of development (Brislin, 2000). Misunderstanding of the cultural norms for parenting may lead one to incorrectly jump to assumptions about abuse or neglect with these and other populations.

Sense of Self, Family, and Social Relationships

How one views the self and an emphasis on different aspects of the self is a fundamental concern to multicultural assessment. The idea of the unique, autonomous, and bounded self and the American ideal of rugged individualism are in contrast with the view of the majority of the world's population, which is more interdependent or sociocentric in nature. In these cultures, the boundaries of the child's sense of self extend to include siblings, parents, grandparents, aunts, uncles, and other extended family members; members of the community, ethnic group or tribe, and nation; ancestors; even the solar system and the universe (Brislin, 2000). The nature of interpersonal relationships that children and adolescents share with others varies from culture to culture, and the differences in closeness of relationships and the kinship structures can be drastic. American ethnic minority groups and immigrants are more likely to have important extended family structures and obligations than Euro-American children and families. Immigrant children are more likely to live with extended family members (Hernandez, 2004). Clinicians should pay careful attention to flexible family roles and assumptions related to the closeness and the importance of significant relationships of children. Ideas about dependence, enmeshment, and differentiation of the self should be examined critically (Hays, 2001; Tseng, 2003).

Normative (and Valued) Development

Definitions about what is healthy and desired child and adolescent development and behavior vary culturally. Different modes of behaving (passive versus active), communication (oral versus written, verbal versus nonverbal), relating to others

(independent versus interdependent), and acquisition of different skills (e.g., motor versus language skills) vary culturally (Brislin, 2000). So, too, do the ways that development is shaped by a child's family and society and how poor (or abnormal) development is tolerated and handled. Cultural differences in expectations and valued processes and outcomes are commonly seen between individualist cultures and collectivist cultures (Triandis, 1995). It is useful for clinicians to keep in mind that worldwide, the majority of cultures (including all major ethnic minority groups in the United States) are rooted in collectivism. Individualistic cultures tend to value independence in behavior and relationships, competitiveness, and autonomy. Egalitarianism and fairness are also valued, and individuality and uniqueness are encouraged. On the other hand, traditional collectivist cultures tend to stress an interdependent sense of self, cooperative behaviors, and dependence, and value social responsibility over personal interests.

In collectivist cultures, consideration of children's attachments may be expanded to include attachment and bonding relationships with extended family and other important caregivers. Developmentally, 2- to 4-year-old children in a Eurocentric culture may be encouraged toward autonomy, whereas other cultures may view too much independence as problematic and thus place a greater value on shame. In some sociocultural contexts, child initiative may be expressed as aggression, whereas other cultures may value a "being" over "doing" orientation toward activity (Tseng, 2003). The age at entering adulthood also varies. Instead of moving toward independence, some cultures may emphasize adolescents' understanding and fulfilling their family or societal roles. Gender expectations and roles will interact with both age and cultural differences, and these roles may be more or less defined and rigid.

Variation in Psychopathology

How problems of a psychological nature are defined is determined by culture, and behaviors can be understood only within their cultural context. Culture impacts the incidence of various problems of childhood, symptom expression, help seeking, and treatment process (Aponte & Johnson, 2000; Draguns, 2000). Culture may directly or indirectly contribute to pathology, shaping the clinical picture as a whole or at level of symptom manifestation (Jenkins, 1996).

Some challenges to clinicians include cultural issues related to mind-body dualism, somatic presentation, the role of spirituality, ideas of selfhood, and categories of emotion and related semantics (Angel & Williams, 2000; Draguns, 2000; Good & Good, 1986). This separation of mind and body and the associated symptoms stems from Western dualism and runs contrary to holistic views of the mind and body that dominate non-Western thinking. How one views the self and emphasis on different aspects of the self will impact the cognitive and affective aspects of disorders and their associated meanings. This has implications for the prominence or absence of symptoms and how they are expressed. Statements assessing depression, for example, emphasize the self ("I feel sad"), whereas individuals from non-Western cultures may be more apt to express dysphoria in terms of their relations to others, for example, concerns about disappointing or embarrassing other

individuals (Manson, 1995). Symptoms associated with anxiety and the content of phobias also vary greatly across cultures.

Variations in the language of emotion and differential emphasis on particular emotions also present difficulties. *Indeterminacy of meaning* refers to problems that arise in the semantic translation of concepts and symptoms and the assumptions made about emotional categories (Good & Good, 1986). For example, Western-based diagnostic systems (e.g., the *Diagnostic and Statistical Manual of Mental Disorders* [*DSM*]) may construe guilt, shame, and sinfulness as generally having the same meaning, whereas other cultures may see important differences. In other instances, symptoms or disorders that are conceptually distinct (e.g., anxiety and depression) according to Western nosology may be presented as fused (e.g., feeling "sad and bothered"; Good & Good, 1986).

Another issue of concern for clinicians is how to respond to somatic and spiritual or religious presentations of problems. Somatic symptoms play a more central role in both the experience and expression of mental disorders worldwide. Somatic distress is common in non-Western populations and in American ethnic minority groups (Kleinman, 1987). Similarly, spiritual and religious factors may play a greater role in understanding culturally diverse populations. Spiritual concerns may include concerns related to religious identity, loss of faith, religious beliefs that a child's disorder is a sign from God (blessing or punishment), problems of the soul, angered ancestors or God(s), concerns related to being cursed though witchcraft or sorcery, or problems interacting within a religious community. These symptoms may be puzzling to Western clinicians and are not well integrated into Western conceptions of mental illness. Spiritual or religious beliefs may also come into play as cultural strengths, coping resources, paths to treatment, or the means of treatment.

Cultural Strengths and Resources

Thinking about the complexity of culture and the many ways that it can impact children and their families, it is easy to see how clinicians may become overwhelmed with all that cultural competence requires of them. Culture can also be a great help to clinicians, allowing them to tap natural strengths and coping resources of children and families. Identifying family cultural values and resources can help the clinicians develop treatment plans that are acceptable to the family and also work with their strengths and supports (Hill, 1993). Some of the cultural strengths and values of the major ethnocultural groups in the United States are shown in Table 26.1.

CULTURAL CONCEPTS IN ASSESSMENT AND TREATMENT

Responding to culture effectively in the clinical setting requires that we unpackage culture into its many parts (or cultural factors). Defining and assessing specific cultural concepts can allow for a deeper understanding of culture as it impacts children within and across cultural groups. Included among these, and essential for accurate assessment and treatment, are etic and emic approaches, enculturation, acculturation, cultural identity, worldview, and explanatory models.

Table 26.1 Cultural Strengths and Resources of U.S. Ethnic Minority Groups

African Americans	American Indians	Asian Americans	Latinos/as
Strong desire to achieve in education	Success in learning important cultural values of tribe	Family expects high academic performance from children; Peer support for academics	Academic success supported by family and community
Strong work ethic	Strong work ethic	Intense loyalty to employer	Vigorous work ethic
Flexible family roles; Extended kinship and supports; Connection to other African Americans	Diverse family structures; Communal values of sharing within group and extended family	Family roles are hierarchical and interdependent	Value marriage, importance of immediate, extended family, and friends
Strong spiritual and religious orientation	Sense of connection between nature and self	Strong sense of connection among family members	Importance of interpersonal relationships and sensitivity

Etic and Emic Approaches

Western European and American psychological research and practice traditionally relied on a nomothetic or etic approach aimed at discovering universals (Lewis-Fernandez & Kleinman, 1995). An idiographic, or emic, approach emphasizes the importance of the local cultural context in shaping phenomena (Munroe & Munroe, 1980). Kleinman (1977, 1980) offers a conceptual distinction between the etic construct of "disease/disorder" and the emic construct of "illness." Disorder refers to the malfunctioning of biological or psychological processes. Illness is a cultural construct that refers to the expression and experience of the disorder. It includes the personal and social meanings of the disorder and how it impacts the lives of children and their families (Kleinman, 1977, 1980). Use of the two perspectives is considered essential to adequate assessment and treatment planning (Lewis-Fernandez & Kleinman, 1995). The right balance between etic and emic approaches largely depends on the child's and family's level of enculturation and acculturation.

Enculturation

Enculturation refers to the process of socialization into a particular cultural, ethnic, or social group (Aponte & Johnson, 2000; Brislin, 2000). Enculturation occurs directly and indirectly through teaching, coaching, reinforcing various behaviors, social role modeling, media, and other environmental and contextual influences. Enculturation of children is an ongoing process and is impacted by the parents' enculturation history, ethnic identity, social class, and current living context. Cultural knowledge, awareness, pride, and feelings of connectedness in one's ethnic or cultural group (i.e., a strong ethnic or cultural identity) are possible outcomes of enculturation. Successful enculturation also results in skills for interacting effectively

within one's cultural group (Aponte & Johnson, 2000). In some cases, children and adolescents will not be well enculturated. Children of refugees and immigrants may reject parental attempts at enculturation, whereas other children are not exposed to their cultural heritage due to parental and/or societal pressure to assimilate into the majority cultural group. Enculturation processes and outcomes interact with acculturation processes that occur as a result of contact with other cultures.

Acculturation

Acculturation Process

Acculturation refers to changes in values, beliefs, and behaviors that occur as a result of sustained contact with a second culture (Berry, 1997; Berry & Kim, 1988). Due to the increase in cultural contact worldwide, the concept of acculturation has garnered increased attention and application over the past few decades. It is considered crucial in the experiences of refugees, immigrants, indigenous groups, ethnic minorities, and sojourners, such as international students and business travelers (Aponte & Johnson, 2000; Berry, 1997; Roysircar, 2004). Additionally, attention has recently been given to "the other side of acculturation," referring to acculturation processes occurring in dominant majority cultures as a result of immigration and other population trends that cause increased intercultural interactions.

Whereas acculturation generally refers to changes resulting from group-level encounters between two or more cultures, *psychological acculturation* refers to psychological changes occurring at the individual level (Berry, 1998). Psychological acculturation is of key importance in assessing culturally diverse children because it involves changes across a number of domains, including language, cognitive style, attitudes, style of relating, and identity (Berry, 1986).

Acculturative Stress

In the context of acculturation, children and families may experience *acculturative stress.* This refers to mildly pathological and disruptive behaviors and experiences that are commonly generated during acculturation. Symptoms of acculturative stress may include family conflict, academic or behavioral problems, depression, anxiety, physical complaints, anger, identity confusion, and substance abuse (Berry, Kim, Minde, & Mok, 1987). Several group and individual factors interact to impact the amount of acculturative stress experienced by children and families. These include (a) macrosocial influences of the child's environment, such as legal constraints, discrimination, degree of tolerance for cultural diversity; (b) factors associated with the background of the acculturating family, such as worldview, social and familial network structure, cultural distance from American society; and (c) individual factors such as age, sex, language proficiency, personality, coping skills, and cultural knowledge (Aponte & Johnson, 2000; Berry, 1986, 1997). Two other factors affecting acculturative stress are (d) phase of acculturation and (e) acculturation strategy (Berry, 1986, 1997, 1998). Phase of acculturation refers to the pattern of adjustment over time (Berry, 1986). Different patterns of responding to the demands of acculturation have been referred to as acculturation strategies or modes (Berry, 1997).

Modes of Acculturation

Of the many factors impacting acculturation problems in children and families, the mode of acculturation may be the most important in predicting overall adjustment outcomes. Four different modes are assimilation, separation, marginalization, and integration. The modes of acculturation account for the relative degree of cultural contact and participation by families and the amount of value placed on such actions (Berry, 1997). As seen in Table 26.2, those with an *assimilation* strategy try to disengage completely from their culture of origin in hopes of being completely absorbed and accepted into the dominant host culture (LaFromboise, Coleman, & Gerton, 1993). Assimilated children may operate effectively within the new social, cultural, and academic environment, but they may sacrifice their sense of identity, their cultural or familial supports, and groundedness in their culture of origin. Families and children using the *separation* strategy will segregate themselves from mainstream American society and establish relationships primarily with others from their own ethnic or cultural group. Often living in ethnic pockets or neighborhoods, these families may garner much needed social support. However, as these children and their parents face demands to speak English, perform academically, access

Table 26.2 Acculturation Modes and Approaches to Working with Children and Families

Mode of Acculturation	*Positive* Contact, Affiliation, and Effectiveness with *Dominant/Majority* Culture	*Lack of* Contact, Affiliation, and Effectiveness with *Dominant/Majority* Culture
Positive Contact, Affiliation, and Effectiveness with *Own Culture*	**Bicultural** Children and families have an integrated identity; skills to interact across multiple, cultural settings; ability to access services and benefits from heritage culture and majority culture. Consider bicultural approaches; adapt mainstream approaches; integrate culturally related or traditional approaches; use culture as a resource.	**Separated** Children and families are supported in their community but separated from the dominant culture. Conflicts arise in negotiating school, legal, health, or other systems. Mistrust related to racism, oppression, or cultural differences may exist. Consider culturally related and traditional approaches; be aware of biases and miscommunication.
Lack of Contact, Affiliation, and Effectiveness with *Own Culture*	**Assimilated** Children and families function reasonably well in dominant society and systems. May lack cultural identity and traditional supports; internalized racism or inability to fully assimilate may cause conflict. Consider mainstream approaches and culturally adapted approaches.	**Marginalized** Children and families lack effectiveness in both cultures. Without resources, natural coping strengths, and the ability to access help, these families are likely to suffer from multiple hardships, including psychological problems. Consider bicultural approaches.

resources, and/or negotiate school, legal, or health care systems, they are likely to experience high levels of acculturative stress (Berry, 1997; LaFromboise et al., 1993). *Marginalized* families will experience the highest levels of acculturative stress, as they lack the behavioral repertoire needed to interact with members of their own culture and with the dominant society. They have the greatest risk of psychological maladjustment (Berry, 1997; Berry & Kim, 1988).

Children and families with a certain degree of integration into the majority culture will experience the fewest problems (Berry, 1997). The *bicultural* strategy suggests that children can acquire the skills and psychological flexibility to function effectively in both their culture of origin and the majority culture. Achievement of bicultural effectiveness serves as both a coping strategy for and a protective factor against acculturative stress (Berry, 1997, 1998; LaFromboise et al., 1993). In practice, bicultural effectiveness may involve the techniques of *code switching* (completely shifting behavior depending on the context) and *culture blending* (merging aspects of two or more cultures; Garcia Coll & Magnuson, 1997; Szapocznik, Santisteban, Kurtines, Perez-Vidal, & Hervis, 1984).

Acculturation modes are not dichotomous, and families can be at the extremes or in the middle on either dimension. Moreover, children and their families usually do not undergo acculturation at the same rate or approach it in the same way. For example, among immigrant and refugee families, children and youth may tend toward assimilation, while their parents, fearing loss of their traditional heritage, may tend toward separation.

Cultural, Ethnic, and Racial Identity

Cultural identity is a multidimensional construct that results from a dynamic combination of different aspects of social identity, such as ethnic, racial, and gender identity (Aponte & Johnson, 2000; Schmidt, 2006). Theory and research related to cultural, ethnic, and racial identity are prominent contributions to the field of multicultural psychology (Atkinson, Morten, & Sue, 1998). One's cultural identity is usually evolving, but at any point, it represents an outcome of one's age and developmental stage, processes of enculturation and acculturation, and immediate and historical contextual situations. A strong or well-developed sense of one's ethnic or cultural identity is considered to be a protective factor, or developmental asset, among youth (Phinney, 1990).

Numerous models of cultural, ethnic, and racial identity have been developed for various groups, for example, Black racial identity development (Cross, 1971, 1995; Helms, 1990); minority identity development (Atkinson et al., 1998); person of color identity development (Helms, 1995); homosexual identity development (Cass, 1979); biracial identity development (Root, 1990); and White racial identity development (Helms, 1990, 1995). Most of the models describe a process of development from a lack of awareness of oneself as a cultural being and the implications of culture (or race) to an integrated and balanced identity. Along the way, individuals may experience dissonance and confusion about cultural differences (often spurred by an event), and at other times may develop a strong sense of cultural identity and pride coupled with complete immersion into their own cultural heritage

and background. In the more advanced stages of identity development, individuals incorporate many identity dimensions and reference groups into their overall self-concept. This includes the ability to respond objectively to members of the dominant cultural group. Integrated identity awareness often extends to advocacy or social action to benefit culturally marginalized or oppressed groups. Although these models often use a stage process to describe identity development, development does not always occur in a linear fashion. Children may have characteristics of more than one stage at any time (Helms, 1995).

Worldview

Worldview refers to psychological and value orientations that influence how people think, behave, and interpret and experience situations, including the assessment and treatment process. Several models draw attention to different aspects or conceptualizations of worldview. Kluckhohn and Strodtbeck's (1952) orientation model (as cited in Mio, Barker-Hackett, & Tumambing, 2006) is based on four dimensions: (1) orientation to time (e.g., past, present, future), time flexibility, and time consciousness; (2) relationship to nature (e.g., harmony or control over nature); (3) view of the self and social relations (e.g., independent or interdependent); and (4) activity orientation (e.g., focus on being or doing). Other concepts of worldview highlight the two dimensions of locus of control and locus of responsibility, suggesting that unique worldviews emerge based on their combination (D. Sue, 1978). A strong belief in fate among parents, or feeling incapable of controlling or changing the behavior of their children or their circumstances, would have major implications for the assessment and treatment process. Other commonly discussed worldviews are collectivism and individualism (Triandis, 1995). Clients from individualistic cultures tend to place the individual's rights and needs before that of the group, whereas clients from collectivistic backgrounds emphasize the rights and gains of the group over the individual. Another concept of worldview, the minority worldview, highlights one's status as a minority (e.g., ethnic, gender, sexual orientation) and related experiences of oppression as influencing worldview (Mio et al., 2006). These aspects of worldview may be extremely important in guiding the behaviors of the client and the clinician.

Explanatory Models

Explanatory models are beliefs about an episode of illness that give meaning to the illness and convey how the illness is understood or conceptualized by the child and the child's family. These beliefs play a key role in understanding the illness experience of the child and family, their treatment choices, patterns of mental health service utilization, and treatment outcomes (Kleinman, 1980; Pelto & Pelto, 1996; Weiss & Kleinman, 1988). Explanatory models contain information about the cause of the illness, its onset, the probable course of the illness, its severity, and level of impairment. Eliciting an explanatory model will provide useful information about fears associated with the illness, the main problems it has caused, and, importantly, what kind of treatment the family is expecting and what they hope the outcomes will be (Chrisman & Johnson, 1996; Kleinman, 1980).

CULTURE AND ASSESSMENT

The influential nature of assessments and their results can dramatically affect the lives of children and their families. All clinical interactions are to some extent intercultural. Therefore, it is safe to assume that culture is always operating in *some* capacity in *every* case, and the need to address culture when conducting assessments is always applicable. When interpreting test results, making diagnoses, and conceptualizing cases clinicians should undertake a cultural formulation of the client. The *DSM-IV* (Appendix I) offers clinicians an outline for things to keep in mind while composing the cultural formulation. This formulation, in addition to other assessment procedures, will assist the examiner in understanding the relative importance of culture and the extent to which procedures, instruments, and interpretations need to be adjusted. In this section, we discuss the cultural formulation along with other cultural aspects of assessment, such as assessing the context, identifying biases of the examiner and the tests, interviewing the family, selecting and administering tests, adapting tests, assessing the child and family's culture, and interpreting the results.

Identify Cultural Bias in Assessment and Testing

The *DSM-IV* Outline for Cultural Formulation stresses that clinicians must identify cultural elements of the relationship between the client and the clinician (APA, 1994). Bias in testing and assessment may be related to the examiner, the approach to assessment, and the testing tools and procedures. Determining which questions to ask and to explore, deciding which instruments to administer, and interpreting the results are all influenced by unique, culture-bound beliefs. Clinician self-awareness is key to accurately observing and interpreting the verbal and nonverbal behaviors of clients. For instance, the Western idea that a high level of eye contact is acceptable and preferred may be contrary to that in other cultures, and a lack of eye contact may be misinterpreted as a negative characteristic. Sullivan (1954, as cited in Takushi & Uomoto, 2001) states that clinicians should continually ask themselves about alternative explanations for behaviors (e.g., "Could this mean something different from what readily comes to mind?" or "What might the client mean by that?").

The cultural formulation also states that clinicians must identify cultural differences between themselves and their clients that may influence the therapeutic relationship. This suggests that clinicians need to identify their personal biases, stereotypes, and assumptions about diverse children and families. Clinicians should be aware of within-group variability. Each family has its own cultural traditions and belief systems and each child should be thought of as coming from his or her own unique amalgam of cultural influences. Failure to account for within-group variability and a reliance on stereotypes or group norms can lead to misdiagnosis or inappropriate treatment (Canino & Spurlock, 2000). It is impossible to learn all of the details of every cultural system, but it is the clinician's responsibility to understand the limits of his or her own cultural knowledge and seek out additional knowledge as needed.

In addition to clinician biases, many standardized tests are biased. In general, assessments favor European American and middle-class children, while disadvantaging ethnic minority children (Kaufman & Lichtenberger, 2002). If interpreted incorrectly, results can be particularly harmful to children in school settings. To provide a culturally responsive assessment to diverse children, the following steps are suggested:

- Research the cultural context of the referral question.
- Conduct a culturally informative and sensitive clinical interview.
- Select and administer tests in a culturally responsive way.
- Use the cultural formulation to guide assessment and interpretation.

Research the Cultural Context of the Referral Question

It is essential that the child's behaviors be considered in multiple contexts (i.e., within the microsystems of home, peer network, school). This gives the examiner information regarding the development of the child's behaviors as well as the responses of parents, siblings, teachers, and extended family. Working with the culturally diverse child, clinicians should try to understand any discrepancies regarding how the behavior is viewed across contexts. It is possible for a behavior to be interpreted as extremely disruptive, offensive, and maladaptive within one context; however, the same behavior may be considered adaptive and culturally appropriate within the child's family system or community. During this process, it is important to consider not only maladaptive behaviors, but also the child's strengths.

The *DSM-IV* Outline for Cultural Formulation states that the clinician must gather information regarding the cultural factors related to the psychosocial environment and levels of functioning for each client. Important cultural and historical information may be gathered from records prior to testing. For instance, knowing the regions a child has lived in prior to the current context, the origin of the child's family and ancestors, and the child's first language could be helpful for better understanding the nature of the question, the family's cultural perspective, and whether an interpreter will be needed. If possible, the examiner should obtain information regarding the client's socioeconomic status, family history, language, and acculturation experiences before the interview. While cultural information gathered prior to the assessment can be helpful as a starting point, a culturally sensitive and informative interview is essential for understanding the unique perspective of each child and family.

Conduct a Culturally Informative Clinical Interview

This portion of the assessment is critical for obtaining culturally valid information. During a parental interview there are a number of goals to be considered. The examiner should first offer information regarding the process of the assessment and answer any questions to help ease possible anxiety or cultural discrepancies. The examiner should be aware that it is considered offensive in some cultures to ask

personal questions and should therefore address this possibility in the beginning. Perhaps a cultural broker could be used to facilitate this process in a more culturally responsive manner. The examiner should explain not only confidentiality and its limitations but also the purpose of the questioning. It should be clear to the parents that the intent of the examiner is not to offend but to gather information to gain a full picture of the child's environment. However, the examiner should also explain that if anything is asked that is deemed too personal by the parents, they may choose not to disclose, and they should understand that the examiner means no disrespect.

It is essential to gain the parents' trust to obtain all the information needed regarding the child's history and current functioning. The clinician may also want to ask about any previous experiences with mental health and let the client's parents explain their frustrations or satisfactory experiences. This gives the examiner a better picture of the level of trust the parents have in mental health providers in general and any cultural misunderstandings that may have occurred in the past. In the case of school or court referrals, parents may feel that an evaluation insinuates that there is a deficit or something abnormal with their child; if the "maladaptive" behaviors being assessed are viewed as adaptive within their own culture, the parents may believe an assessment is unnecessary. This may result in defensiveness, and parents may withhold information they consider evidence of such a deficit. It is the examiner's responsibility to inform the parents of the objective and neutral nature of the assessment.

Next, the examiner should obtain a familial and developmental history of the child. It should also be noted what languages are spoken within the home as this may be a factor in language acquisition and development. The examiner should inquire about any incidences reported in the records, such as previous illnesses or developmental milestones, and what the parents' explanatory models are regarding these occurrences. Information to be considered by the clinician includes the following:

- The child's cultural context within all settings.
- Year or time period each developmental stage occurred within the child's life and the correlating events or cultural beliefs.
- Client's language usage and comfort speaking the clinician's language (may require an interpreter or referral).
- Cultural strengths the client may gain from membership in a particular cultural group.
- Cultural information apparent in how the parents and family interact with the child.

Assess Child's Enculturation, Acculturation, and Ethnic Identity

In addition to the issues presented by the clients, psychologists must also keep in mind generational history (how many family members in the country, circumstances of the move, etc.); citizenship or residency status (e.g., number of years in the country); fluency in standard English or other language; extent of family

support or disintegration of family; availability of community resources; level of education; change in social status; and work history and level of stress related to acculturation (APA, 2003). According to the *DSM-IV* Outline for Cultural Formation, psychologists must explore these issues and their influence in the progression of treatment. Hays (2001) suggests the following questions for inquiring about the client's enculturation and acculturation:

- On what terms did you or your family arrive here (refugees, students, etc.)?
- How long have you lived here?
- What language did you first learn to speak? What is spoken in your home? What language do you prefer?
- What is your cultural (or ethnic) heritage or background?
- What is your involvement with _____ (Americans and American culture, own ethnic culture)?
- Would you mind telling me about your religious upbringing? Do you have a religious (or spiritual) practice now?
- What is it like to grow up as a boy/girl in your culture and family?
- Would it be okay if I ask you about your family?
- How is your family's economic situation now? Are there ways it impacts your family life or child's problem?
- Are there ways in which discrimination (prejudice, racism) impacts your family or your child's problem?
- Are there ways in which your (or your child's) disability contributes to the problem?

Assess the Family's Explanatory Model

Assessing the explanatory model of the child and family will help the clinician understand the child's and the family's perspectives on the problem. The model will reveal important cultural and/or personal beliefs about the child's problem that have implications for assessing and treating the child. Research suggests that the family's explanatory model may be just as important as, if not more important than the child's in determining help seeking. Kleinman (1980, p. 106) suggests several questions for exploring explanatory models:

- What do you call your child's problem? What name does it have?
- What do you think caused your child's problem?
- Why do you think it started when it did?
- What does your child's illness do to him or her? How does it work?
- How severe is it? Will it have a short or long course?
- What do you fear most about your child's problem?

- What are the problems your child's illness has caused? What is the impact on the family?
- What kind of help (assessment/treatment) do you think your child needs? What are the most important results you hope to get from treatment?

These questions may be asked during the clinical interview, but keep in mind communication norms and sensitivities. Rather than barrage the family with direct questions, questions can be incorporated more naturally into a conversation about the family's concerns. Young children in particular may not have a complete explanatory model; some parts may be missing, vague, or fused together. The beliefs will change over time, and the child's and parents' views may be different from each other. Depending on the age, developmental stage, and cultural background of the child, consideration of the child's perspective may be more or less important in the family's eyes. The clinician needs to consider the family structure when making efforts to elicit or communicate the child's perspective. For example, having the child vocalize a difference or disagreement with a parent, either alone or in front of the parent, could cause a loss of face, shame, or embarrassment for the parent. Moreover, giving equal weight to a child's or adolescent's explanatory model may threaten the parent's authority and jeopardize the examiner's credibility.

Select Appropriate Test Instruments

Clinicians must understand and be knowledgeable about the reliability, validity, standardization, and appropriate use of any procedures or instruments used in an assessment (APA, 2002). The APA Ethics Code urges psychologists to "use assessment instruments whose validity and reliability have been established for use with members of the population tested" (p. 13). The Ethics Code states that "when such validity or reliability has not been established, psychologists must describe the strengths and limitations of test results and interpretation" (p. 13). It is the responsibility of the examiner to understand the instrument's cultural limitations and adapt procedures and interpretations of results as needed.

The number of school-age children coming from minority backgrounds is increasing rapidly; however, assessment instruments normed on nonmajority populations are sparse. Padilla (2001) states that using a test with individuals of cultural or linguistic groups who were not included in the standardization group calls into question the reliability and validity of the results. Instruments appropriate for use with children from diverse cultural backgrounds and varying linguistic abilities are not commonly available in educational and psychological settings. According to Torres (1991, as cited in Padilla, 2001), studies have shown that approximately 5 million students are inappropriately tested each year by standardized assessment instruments. This is of great concern, particularly in special education, where standardized testing is mandatory to give a child a diagnosis and secure appropriate support. Students' results may be deflated, resulting in inappropriate academic labeling and placement, such as being retained in a grade or placed on a low-expectation education track (Padilla, 2001).

Suggestions for Making Standardized Tests More Culturally Responsive

Although no test can be considered culture-free, there are some tests considered to be culturally responsive in testing with minorities. Some of the instruments were developed with a particular ethnicity in mind; there are also many available for use with larger diverse populations. For instance, the Wechsler Intelligence Scale for Children III, Woodcock-Johnson Psychoeducational Battery—Revised, Child Behavior Checklist, and the Kaufman Assessment Battery for Children are all considered appropriate for use with children from diverse populations (Canino & Spurlock, 2000).

In an attempt to lessen the cultural bias toward diverse populations, many clinicians choose to alter the administration of the instrument. For instance, when giving timed tests, it is important to understand that the emphasis in Western culture on quick responding as an indicator of intelligence is culture-based. Some cultures (e.g., Asian) may emphasize careful consideration of a question before responding and may further consider a careful, unrushed, and thoughtful response to be a sign of intelligence (Gopaul-McNicol & Thomas-Presswood, 1998). As a result, some clinicians eliminate (or extend) the time limits for some subtests or items (Canino & Spurlock, 2000).

Gopaul-McNicol and Thomas-Presswood (1998) advocate altering vocabulary portions of instruments to allow the child to use the vocabulary item within a sentence (Canino & Spurlock, 2000). This allows the child to contextualize the word and may give the examiner a better picture of the true knowledge the child has regarding the vocabulary word. Another method used to alter testing administration and minimize cultural bias is to advocate the use of paper and pencil to work out math problems. This not only allows the examiner to see the process the examinee uses but may also help the examinee in the task. It may be beneficial for the examiner to model the required behavior prior to giving actual test items. Doing so may clarify directions that were misunderstood due to a language deficiency or cultural misunderstanding. It also gives the child the opportunity to ask questions regarding the task and minimizes the possibility that the child's behaviors during testing are misinterpreted as low levels of ability rather than a lack of clarity (Paniagua, 1994). Adaptations should be made with awareness and knowledge of the child's acculturation level and language usage and fluency. Some strategies for culturally responsive testing include:

- When using an interpreter or cultural broker, set up a meeting prior to the assessment to discuss the process and possible cultural concerns.
- Explore all possible venues to explain the client's responses, scores, and behaviors during testing.
- If possible, current scores should be compared relative to the client's prior performance instead of against others' scores or the norm.
- Avoid the use of standardized personality tests for diagnostic purposes with clients of minority identities.

Interpretation and Case Conceptualization

The clinician's overall interpretation of the client's situation stems from his or her own way of perceiving the client's presentation of problems. This evaluation is guided by the clinician's professional orientation, sociocultural background, cultural awareness, and experiences. It is the clinician who tailors the interviewing process, test selection and administration, interpretation and perception, and ultimately the understanding and reporting of the problem. Therefore, it is essential that clinicians compose a thorough case conceptualization that takes culture into account and is alert to possible biases. It is through this culturally relativistic prism that test scores and behaviors should be interpreted.

CULTURE AND TREATMENT

Culturally responsive treatment can be developed through collaborative negotiation between the clinician and the family (Hays, 2001). How and what cultural material should be integrated vary with each child and family. According to the *DSM-IV* Outline for Cultural Formation, information about the client and the client's cultural background should be organized into an overall approach for treatment that links the conceptualization to the treatment. Culture frames beliefs of the clinician and of the family about the best route to recovery and the desired result. Therapeutic theories, orientations, and approaches have their own cultural roots and reflect a particular worldview and philosophy (Pedersen, 2004). Whether the goal of therapy is to work for a reduction of psychological symptoms or for an acceptance of them is a value-laden question (Tseng, 2003).

Ideally the treatment plan will flow from two sources: the clinician's conceptualization (based on theoretical orientation and comprehensive assessment) and the perspective and goals of the child and family. Cultural factors should be considered at every phase in the treatment process.

Next we consider culture across the phases of treatment, from before treatment begins to building rapport, establishing roles and boundaries, and planning and adapting treatments.

Establish a Relationship

First, clinicians should take the necessary time to establish rapport, respect, and credibility with the family (Hays, 2001; Hong et al., 2000; Okazaki, 2000). This asks the clinician to develop an appropriate level of intimacy with the child and other family members. What level is appropriate and the expected degree of formality will vary based on cultural, gender, and age-related factors, as well as the nature of the problem. Clinicians should consider the use of titles, differences in communication (e.g., language usage and fluency, paralanguage), and social interaction (formal versus informal, gender roles). With clients from patriarchal and authoritarian cultures, it will be important to pay special attention to establishing respect and credibility with the father; in other cultures it could be a maternal grandmother who deserves special consideration. Necessary family support for treatment may hinge, to some extent, on the support, approval, and cooperation of these individuals. With

other groups, such as African Americans, efforts to connect equally with all family members, including the child and other siblings, will be valued.

Therapists should also appreciate that some families will have a need to contextualize the clinician, or otherwise understand something about the clinician or the clinician's background (Hays, 2001). To build trust and engender respect and rapport, a clinician may engage in small talk, use appropriate self-disclosure, and adapt language style by avoiding jargon and using the language of the family members (e.g., idioms, ways of describing) when possible. Providing concrete help to the family or otherwise quickly addressing a family concern can also build trust. Clinicians should be mindful of personal reactions or biases related to race or culture that may result in mistrust, suspicion, hostility, ambivalence, overreliance or overfriendliness, excessive guilt, anger, or denial of ethnicity on the part of the family or the clinician. All of these interfere with the therapeutic relationship (Vaughn, 2004; Wohl, 2000). In all cases, a clinician's skills in conveying warmth, openness, and respect can go a long way in building a working alliance with the child and family members. A certain degree of humility and appropriate use of humor are also considered to be essential to building intercultural relationships (Hays, 2001).

Understand and Address Treatment Barriers

Understanding and addressing obstacles to treatment barriers is essential to equitable service provision. Ethnic and cultural minorities and low-income and other disenfranchised groups face a number of barriers in accessing and benefiting from professional psychological treatment. These barriers include a lack of affordability, accessibility, acceptability, and appropriateness in current services. Efforts to address the psychological needs of children and adolescents should be particularly attuned to culture-based beliefs about psychological problems, such as stigmas and beliefs regarding from whom and what type of help is needed. A lack of familiarity with mental health services or counseling, cultural mistrust due to racism, or previous negative experiences with mental health services are additional factors that prevent psychological help seeking. Instead, families may seek informal help from family and friends; from religious, spiritual, or other community supports; or from medical providers. Psychological services may be sought only as a last resort, when problems have escalated or the family is in a state of crisis. Moreover, service may be sought due to outside pressures (e.g., legal or school referral) rather than family choice. Once help is sought, clients may experience communication barriers or institutional or individual racism and treatment that may be culturally incongruent with their values and beliefs.

The nature and extent of these treatment barriers will need to be assessed and addressed for each child and family. In general, clinicians should be aware of common barriers within their treatment context (location, service population, type of service). Culturally competent responses to such barriers might include sliding fee scales; transportation vouchers; flexible operating hours; community-based or school-based service provision; flexible treatment modalities, such as group, family, and culturally adapted approaches; and culturally sensitive and knowledgeable clinicians.

Primary and secondary prevention approaches to child and adolescent problems should be considered for their nonstigmatizing nature. Likewise, strengths-based approaches that view culture as a resource and aim to build protective factors and resilience in children and adolescents are encouraged (Hays, 2001; Lerner, Brentano, Dowling, & Anderson, 2002). Coordination of services and collaboration among stakeholders, such as community participation in planning and evaluation, can also aid in successful programming.

Determine the Family's Familiarity with the Treatment Setting

It is important to determine, early on, the child's and family's familiarity with psychotherapy and expectations for treatment. This may include discussing their previous experiences with mental health services or other interactions with the health care system. Personal and social meanings and beliefs about seeking psychological help should also be explored. There may be a lack of familiarity with the concept and practice of Western counseling and psychotherapy. Even when psychological treatment is sought, families may continue to have a certain level of cultural mistrust and apprehension entering the therapy setting. Certain routine intake procedures, such as signing consent and confidentiality forms, may increase distrust, especially if the need for such forms is not well explained (Hays, 2001). Additionally, the typical therapy format relies on a direct style of communication to identify and address personal problems in the presence of a professional (the psychologist or clinician), who is usually a stranger. This format may be very foreign to families coming from cultures that rely on familiar social and religious supports, such as community leaders, village elders, family members, priests, religious leaders, or shamans for addressing such problems. Counselors and therapists who lack cultural knowledge can exacerbate the problems and feelings of unease.

Conduct Pretherapy

Families that are unfamiliar with the therapy setting (e.g., ethnic minorities, immigrants, rural populations, and lower income populations) may benefit from pretherapy, which refers to time spent orienting the child and family to the therapist and the treatment setting. The family should be invited into the therapist's culture (Hays, 2001). Pretherapy may include added time for rapport building, education about what to expect in therapy, knowledge about the therapist's credentials and approach, and demystification of the therapy process. At this time, the role of the therapist will be clarified and so will the general expectations of the child and family members. Some cultures do not have a history of keeping regular, scheduled appointments that begin and end at a certain time. Additionally, children and families with no transportation or a lack of child care for siblings may have difficulty keeping appointments.

In many traditional approaches to helping, payment or reciprocity for service is not required on a regular basis, but is offered as a donation after treatment. Expectations regarding payment should be handled in a delicate and sensitive manner. Procedures related to confidentiality should be given careful attention, as parents with strong authoritarian family roles may expect that the clinician will automatically

share all information with them (Tseng, 2003). The child's parents will need to be carefully informed of limits to confidentiality, such as child abuse reporting laws. The time spent in pretherapy can also be used to further establish credibility and rapport and to gather cultural information.

Gather Cultural Information

Next, attempts are made to learn about and understand the cultural perspectives of the child and family by crossing into the client's culture (Hays, 2001). That is, the clinician should learn about the child's cultural context and background. The client and the client's family are good sources for relevant cultural information. However, it is not appropriate to rely solely on clients, and the clinician should seek information about a child's cultural background and group-related norms outside of the therapy session. There are a number of excellent books and book chapters, scientific and practice-related journals, and Web-based resources that provide information specific to different cultural and minority groups. Important background information includes sociopolitical and historical events, history of oppression, worldview, communication and language issues, cultural variation in symptom expression, stigmas and related help-seeking patterns, cultural strengths, values, and supports.

In learning about cultural norms and values, clinicians should attempt to suspend judgment about a client's cultural beliefs and remain mindful of their own, culture-based reactions to areas of difference (Martin & Nakayama, 2005). In addition to learning about cultural norms and values, identifying the strengths of children and families and mobilizing them as a resource in treatment is one of the most important skills for effective practice with diverse clients (Hays, 2001).

Clarify Clinician Roles and Boundaries

Working in a multicultural context, careful attention should be given to clarifying the function and roles of the therapist, determining the nature of the therapeutic relationship, and defining the boundaries of the relationship (Constantine & Sue, 2005; Tseng, 2003). The role of clinicians and the nature of the therapist-client relationship should be viewed from a cultural perspective. Practitioners from multicultural, feminist, and humanistic approaches, for example, may encourage enhanced authenticity and genuineness in the client-therapist relationship. Boundaries are not as rigidly defined, but continuous in nature, with shifts readily occurring in areas such as self-disclosure, giving and receiving gifts, multiple roles, bartering, and nonsexual touch and other expressions of care. Indeed, when grounded in one's theoretical approach and well thought out in each case, relaxed boundaries in some of these areas are considered culturally competent service provision. Although research findings are inconclusive, practitioners suggest that appropriately timed self-disclosure by the clinician can engender feelings of trust and respect and facilitate the development of rapport with youth and with some cultural groups. Acceptable practices vary greatly across different treatment settings and populations.

Proponents of multicultural approaches also call on psychologists to take on multiple helping roles beyond traditional individual therapy (Atkinson et al., 1998).

With a host of concerns related to diverse clients' environment or external context, a case management approach is often called for to assess and treat children with awareness of their familial, cultural, and ecological contexts. A task for the clinician may be building parents' skills for successfully advocating for their children and accessing needed resources and services. Other clinician roles may include conducting groups, intervening in systems, serving as a consultant or cultural broker, and implementing community-based preventive interventions (Atkinson et al., 1998).

Involve the Family

Working with the family members of children and adolescents is a common practice generally, and in the context of cultural difference can be a great asset (Wilson, Kohn, & Lee, 2000). Family members, including parents and siblings, grandparents, aunts and uncles, and close others in the child's family network can be valuable resources of collateral and/or corroborating information about the occurrence of various behaviors and how they are interpreted. Family members can provide a base for assessing cultural perspectives and for determining the extent to which various behaviors or other psychological manifestations are culturally acceptable. Family members will also be knowledgeable about culturally appropriate coping choices and ways of responding. It is important that clinicians be aware of culturally diverse family hierarchies, rules for relating among family members, and taboos. For example, it may be inappropriate for a family member to mention the cause of a parent's death (Tseng, 2003).

Address Communication Differences

Even when clinicians and their clients are able to communicate in the same language, cultures vary greatly in their communication styles (whether they are direct or indirect, high or low context), communication patterns, and norms and rules for verbal and nonverbal expressions (Martin & Nakayama, 2005). Clinicians should pay attention to the language used by themselves and the children and families with whom they work. Efforts should be made to use simple and clear language and to minimize the use of psychological jargon, idioms, and colloquialisms. The process of communication and rules for formality and nonverbal communication, such as eye contact, gestures, and body movements, vary widely. Clinicians should clarify the intended meaning of any expression with which they are unfamiliar. An incorrect interpretation can result in minor misunderstandings or serious mistakes, such as misdiagnosis. It is important for clinicians to know that a client who says "Yes, yes" does not necessarily mean "I agree with you." How symptoms are expressed is also a matter of culture, and expressions may be culturally stylized (Tseng, 2003).

Direct communication about emotions or discussions of certain other matters may be taboo, or considered very private. If an indirect style is used, the clinician may need to adjust his or her communication style and seek consultation on specific indirect methods, such as making use of proverbs or stories that reflect cultural wisdom. Clinicians should proceed with caution and determine cultural

norms for discussing family matters, sexual or religious practices, money, suicide, and ancestors (alive or dead). Individuals with a collectivist background may consider it improper to discuss family members or family matters, because to do so would bring great shame to the family. Clinicians should explain their need for the information and ask permission first (e.g., "May I ask you about your family?"). Invite clients to share and then pay attention for signs (nonverbal and verbal) that you might have overstepped cultural boundaries (Pedersen, 1997).

Work with Interpreters

Children and their families should have access to treatment in a language that they completely understand and in which they can comfortably express themselves (APA, 2003; Constantine & Sue, 2005). With 19% of families in the United States speaking a language other than English in the home, mental health practitioners will increasingly be providing services to individuals with limited English proficiency (LEP). In all cases in which children, youth, or their families are considered LEP or prefer speaking in their native language, clinicians should arrange for the services of a professional interpreter. Under Title VI of the 1964 Civil Rights Act, institutions and service providers that receive federal funds are required to provide such interpretation services. Professional interpreters are trained to provide direct, simultaneous, or sequential interpretation in a professional and ethical manner. They often have had training in confidentiality, dual roles, and medical or psychological terminology.

Interpretation is provided verbatim and in the first person. Typically, interpreters will sit beside the client and family. During the session, an interpreter often takes a rather technical role, serving as the conduit for communication between the clinician and the patient. Eye contact, to the extent that it occurs, will largely remain between the clinician and the child and family. If the interpreter needs to interject or further explain something, he or she may enter the conversation by saying "The interpreter would like to say. . . . " This may be done to clarify a communication, explain a concept, or provide important cultural information. If trained professionals are not available, for example in rural locations, telephone interpreters can be arranged. Some things to consider when selecting an interpreter are sex and ethnic background, as these may influence client disclosure and trust. Despite their level of training, brief pre- and posttherapy sessions with interpreters can provide a chance to discuss therapeutic concepts that may be difficult to translate, such as relaxation training. Presessions may help prepare the interpreter for a potentially difficult (e.g., crisis) situation.

Use a Cultural Consultant

Cultural brokers may be useful in accessing cultural knowledge or advice. Who is a cultural broker? This role is rather loosely defined, but may include a professional (e.g., anthropologist, sociologist, psychologist, or other mental health professional) with experience or training in a cultural population. A cultural broker could be a person with knowledge and experience in the culture or community in question, such as a community leader, spiritual person, pastor, or other traditional helper.

Cultural brokers can provide background, group-level cultural information, or case-specific consultation. Interpreters can also serve as cultural brokers and have the benefit of providing this service as treatment sessions unfold and directly afterward. Interpreters serving as cultural brokers may be able to further elaborate or explain culturally embedded concepts and recognize and head off cultural clashes. For example, the interpreter may inform the clinician that, if a concept is literally translated, it will offend the client. Cultural brokers may even assist the therapist in coming up with a culturally acceptable way to address a sensitive issue, such as suicidality.

Negotiate a Common Explanatory Model

In selecting and developing a treatment plan, clinicians should begin with what the family presents (their explanatory model) and make appropriate adjustments.

After assessing the family's explanatory model, clinicians can compare it to their own (Chrisman & Johnson, 1996; Lewis-Fernandez & Kleinman, 1995). Studies suggest that, more often than not, the models of clinicians differ from their patients' (Cohen, Tripp-Reimer, Smith, Sorofman, & Lively, 1994; Kleinman, 1980). Disparate beliefs, left unexplored, may result in poor patient-practitioner communication and lead to misdiagnosis, inappropriate treatment, treatment nonadherence, early attrition, and reduced treatment outcomes (Chrisman & Johnson, 1996; Kleinman, 1980; Wohl, 2000). The following steps are suggested to negotiate a common model with children and their families.

- Start with what the family presents.
- Share your own model, using client language when possible.
- Bridge the models by emphasizing similarities and discussing differences.
- Make modifications that are culturally salient for the family.
- Agree on a common model and a treatment plan that emphasizes strengths.
- Continue to monitor and modify as needed.

First, the clinician should summarize the family's view of the problem and expectations and goals for treatment. This will convey an accurate understanding and highlight misunderstandings that the clinician has. Next, the clinician can share his or her conceptualization of the problem and suggested treatment. Next, clinicians should make efforts to engage the family in a collaborative approach to goal setting and treatment planning. Even when there are drastic differences in practitioner and family explanatory models, clinicians can emphasize similarities and work toward shared goals. To the extent that the clinician has built credibility and trust, families may be more receptive to new ideas or methods proposed by the therapist. Reframing or psychoeducational approaches can be used to present treatments in culturally palatable ways. Strengths-based approaches are also suggested, and because they are congruent with cultural and family values may be more likely to engender family support (Hays, 2001).

Adjust the Therapeutic Frame

Efforts to make treatments culturally relevant and meaningful often involve making adjustments to the therapeutic frame. Although some families may not be familiar with formal, Western-based psychological treatments, families all over the world are familiar with the idea of talking to or otherwise seeking help from a caring and concerned person. Keeping this basic similarity in mind, clinicians can make a variety of technical and theoretical adjustments, ranging from minor adjustments to mainstream approaches to drastic changes in the foci, goals, and process of treatment so that treatments are developed largely from cultural (emic) approaches (Koss-Chiono, 2000).

Adaptations should be made with consideration of the family's cultural perspective and available resources and services. Simple adjustments in mainstream approaches can lay the groundwork for effective services. For example, although many therapists are taught not to give advice, families from diverse backgrounds may seek out the therapist's guidance in various dilemmas (Tseng, 2003). Clinicians who can provide concrete help to families in solving problems are adjusting the relational frame to meet the family's expectations and needs.

Other therapeutic adjustments may come into play as treatment foci, goals, and processes are selected. Goals should be culturally congruent and contextually relevant. Which behaviors are targeted for change should be considered in light of the child's multiple contexts. For example, working to increase assertiveness with a Chinese adolescent may be helpful in the school environment, but problematic at home. These issues should be explored thoroughly before treatments are pursued.

Adapt Treatments

Clinicians consider a number of factors when selecting the treatment mode and format, and culture should be among these. Culturally sensitive and adapted treatment programs may garner increased patient support and satisfaction.

Level and Mode of Intervention

Family approaches to treatment are often suggested for ethnic minority and culturally diverse youth, as are small groups and peer-based and systems-level interventions (Aponte & Bracco, 2000; Wilson et al., 2000). Community approaches are considered preferable to individual approaches in meeting the needs of ethnic minorities. Typically, they emphasize ecological validity, a multiproblem focus, commitment to interagency collaboration, employment of bicultural and bilingual staff, development of cultural competency, strengthening social support networks, and use of prevention and consultation strategies (Aponte & Bracco, 2000; Cuellar, 2000).

Culturally Related and Culture-Specific Approaches

Treatments range widely in terms of intensity, focus, modality, and the degree to which cultural material is incorporated. Many clinicians choose to adapt standard European-based strategies such as behavioral, cognitive, interpersonal, or family

systems, by incorporating cultural values, strengths, and resources into therapeutic goals and techniques (Hays, 2001). Examples of adjustments in a cognitive-behavioral treatment might include using proverbs, religious verses, or other cultural metaphors to help counter negative thoughts, and using family to help support home-based exposures. Culture-based coping, such as meditation or prayer, could be integrated into a stress reduction program, and other cultural rituals related to healing may be included in an overall treatment plan. For example "having a sweat" (i.e., in a sweat lodge) may be recommended alongside other treatment recommendations for an American Indian adolescent living on an Indian reservation. Alternative approaches to treatment include narrative, testimonial, and bibliotherapy, as well as the use of creative, expressive, and somatically based therapies (Hays, 2001).

Culture-specific approaches are based on specific cultural frameworks. Most of these interventions provide cultural education based on traditional values, and then use these values as the underpinnings of the treatment. For example, clinicians may use concepts of *familismo, simpatia,* and *respeto* as themes when working with Latino/a children and adolescents, and themes related to collective responsibility, self-determination, or spirituality when working with African American youth. Culture-adapted and culture-specific treatments may make use of ethnically similar role models, discussion of cultural attitudes regarding risk behaviors, ethnic identity development, and issues related to racism and discrimination (Belgrave et al., 1994). Traditional cultural customs, stories, songs, and other cultural materials may be used, and treatment should be in accessible locations where families can draw support from community leaders or organizations. Some culture-specific approaches make use of a "rites of passage" paradigm to facilitate, formally recognize, and celebrate the physical and emotional development of youth. Although there has been little controlled research on the therapeutic efficacy of these interventions, cross-sectional research has indicated that a grounding in traditional cultural values is negatively associated with substance abuse, anxiety, hostility, juvenile crime, and depression (Belgrave et al., 1994).

Two culturally related treatments are cuento therapy for Latinos/as (Constantino, Malgady, & Rogler, 1986) and Afrocentric "rights of passage" programs for African American youth (Brookins, 1996; P. Hill, 1992). Cuento therapy draws on the themes in Hispanic cultural stories of heroes, heroines, and moral messages. Afrocentric programs have youth undertake a series of behavioral tasks and related rituals that signify growth, maturity, and a commitment to self and community. These programs aim to educate and empower adolescents and enhance their life skills, cultural identity, self-esteem, and academic performance. One obvious concern is the practicality and costs of designing culture-specific treatments. Moreover, to make treatment decisions based on the ethnic background of the child and family may be considered an ethnic slur and a violation of ethical and clinical rules of practice (Wohl, 2000). Instead, multiple cultural factors will overlap and manifest idiographically, accounting for vast diversity within ethnic groups.

Work with Traditional Healers

Children and families may benefit from traditional approaches, such as in the case of culture-bound syndromes or when families are highly *separated* from mainstream

society. Clinicians should develop awareness and understanding of indigenous healing and helping methods so that they can better understand their client's cultural perspective and provide referrals (Koss-Chioino, 2000). Clinicians should recognize that there are several parallels between the functions of traditional healers and those of psychologists. Traditional healers vary greatly in type, level of training, and approaches to treatment, and include shamans, priests or pastors, imams, herbalists, curanderos, and spiritualists. Traditional and indigenous healing approaches are gaining increased attention, and it is suggested that clinicians learn about practicing local healers and develop respectful relationships with them (Koss-Chioino, 2000).

Select a Focus for Treatment

The focus of treatment may or may not be culturally related. Clinicians will assess and treat children referred for common child and adolescent problems involving conduct, social relations, mood, mental retardation or other developmental delays, learning problems, and adjustment difficulties. Although there may be phenomenological variation in the presentation of these disorders, if the presenting problem is directly related to or influenced by cultural factors, then a more direct focus on cultural issues is needed. Some common treatment foci that are culturally related are school adjustment, acculturation-related problems, and identity concerns.

School Adjustment

Children from low socioeconomic backgrounds, ethnic minorities, immigrants, and refugees all face formidable challenges in the school system. The school system in the United Sates, for example, has a pedagogical philosophy valuing written expression, direct verbal communication, student-centered approaches, competitiveness, and individual achievement. To the extent that children live in households or communities with different cultural values, or have a history of schooling in another system, adjusting to school can be quite a challenge. Language problems, cultural differences in learning style or behavioral norms, discrimination, and the inability of parents to help with schoolwork are challenges faced by immigrant and ethnic minority youth.

Among refugees and immigrants, language issues, difficulty making American friends, and differences in educational systems are some of the acculturative problems reported (Vinokurov, Trickett, & Birman, 2002). Other problems are discrimination by peers, such as name-calling and teasing (Bemak & Greenberg, 1994). A lack of previous access to formal education can make the adjustment even more difficult for some immigrant and refugee youth. Self-efficacy will be important for adjustment in the school setting, with some studies suggesting an impact on academic skills and performance, computer skills, and interpersonal skills. A thorough assessment will be needed to determine the appropriate type, target, and level of intervention (e.g., a system- or organizational-level change or child behavior change).

Acculturation Problems

The seriousness of symptoms and the degree of psychological difficulty experienced by children and families in cultural conflict will vary considerably. Berry (1997) breaks symptom severity into three levels of distress: (1) mild to moderate cultural conflict, (2) acculturative stress, and (3) psychopathology. Mild to moderate cultural conflict is explained by the need to learn new ways of behaving (Ward, 1997). These conflicts arise as certain behaviors are unlearned or shed, while being replaced with a new behavioral repertoire. If behavioral shifts are not easily made, the second level of difficulty, acculturative stress, will occur (Berry, 1997). When major conflicts are experienced, a psychopathology perspective is helpful. Children and their family members may be in crisis or otherwise suffering from incapacitating levels of depression, anxiety, and/or other mental disorders needing serious attention (Berry, 1997).

Parent-Child Acculturation Gap

Children of immigrants and refugees often acculturate more rapidly than their parents and older family members. They may adopt new behaviors and values that are incongruent with their parents' expectations. Refugee and immigrant youth may become ashamed of their parents, reject their culture of origin, and rapidly assimilate to America's youth-oriented society. Parents are often fearful that their children will forget their cultural values, and they may become upset as their adolescents adopt new styles of dress, wear makeup, or engage in unacceptable social behaviors, such as dating and attending parties (Sourander, 2003).

Children may learn English before their parents do, and power shifts may occur as children take on new roles interpreting for the family. At the same time, parents may lose their status and effectiveness in the new culture (Tousignant et al., 1999). These shifts in power dynamics can be quite upsetting for children and adults, especially those coming from more traditional, patriarchal families or collectivist cultures. Family conflict may result, which can easily escalate to child or domestic abuse situations that did not exist prior to migration (Bemak & Chung, 2000; Colic-Peisker & Walker, 2003).

Bicultural Effectiveness

In working with mild to moderate acculturation concerns, clinicians may employ strategies for culture learning and social skills acquisition. A bicultural (or multicultural) approach will reinforce the value and importance of maintaining effectiveness in the culture of origin and the new culture. Some suggestions for addressing the acculturation gap between parents and children include the following:

- Reframe family conflict as acculturation differences that normally occur.
- Identify cultural similarities and differences.
- Teach families about the process of cultural adjustment and how it can play out in families.

- Examine the cultural values and demands faced by the child outside the home (e.g., in school, with peers).
- Help youth understand the value in their heritage culture.
- Learn skills needed to be effective in different cultural environments (i.e., bicultural competence).

Bicultural effectiveness programs have been developed to reduce intercultural conflict among Cuban American youth and their parents (Szapocznik et al., 1984) and to reduce substance abuse among American Indian adolescents (Schinke et al., 1988).

Identity Concerns

Questions related to one's personal and social identity, such as "What kind of person am I? Do others like me? Do I belong to a group? What is my ethnic/racial or other cultural group? How do others see me as a part of this group?" and "How do I see myself as a part of this group?" are common beginning with middle-age children and may be especially so for ethnic and cultural minority youth (Schmidt, 2006). Efforts to nurture a positive social and cultural identity can be integrated into standard treatment approaches. Effort to enhance a child's awareness of and sense of belonging to his or her cultural group may include bibliotherapy (including cultural stories, myths, proverbs), attendance at cultural events, cultural and personal values exploration, identification of cultural strengths and resources, learning about the cultural or sociopolitical history of one's cultural group, and appreciation of or participation in creative or expressive arts and artistic traditions of one's group. Approaches that work toward an integrated (e.g., bicultural or multicultural) identity offer maximal benefits across settings.

Mistakes and Modifications

Once the treatment focus and techniques are agreed upon, clinicians need to consider cultural factors and influences throughout treatment. During the entire process, clinicians should make efforts to maintain therapeutic rapport, monitor the child's and family's progress and commitment, be mindful of mistakes that are made, and make modifications as necessary. Maintaining a good working alliance with children and credibility with families requires ongoing attention and effort. Clinician roles, boundaries, and style of interaction may need adjusting as treatment and the level of familiarity increases. Latino families, for example, may expect the clinician to shift from an interactional style favoring *formaliso* to one of *personalismo* (Hays, 2001). Cultural impasses may occur as new material comes up and/or new techniques or interventions are discussed or implemented.

Multicultural psychologists point out that mistakes will be made in treating the culturally diverse. Successfully recognizing and responding to cultural missteps is an important clinical skill (Pedersen, 1997, 2000). Awareness of changes in relating or treatment nonadherence may indicate a potential miscommunication. Mistakes may be addressed directly or indirectly (as culturally appropriate) and

modifications made in the technique or approach. Respect, humility, and humor are considered among the most important cross-cultural skills and can be quite effective in maintaining relationships in spite of mistakes (Hays, 2001; Pederson 1997, 2000).

Termination

Cultural expectations of the relationship may impact termination. Families may not understand such an artificial termination and may feel a need to maintain the relationship. Clinicians may help terminating children and families by linking them with community supports and also by scheduling spaced-out sessions and booster sessions or otherwise inviting families to return for additional treatment (Tseng, 2003).

SPECIAL POPULATIONS AND EMERGING TOPICS

Children Living in Poverty

Children and adolescents growing up in poverty are affected on multiple levels by compounding factors. High poverty often goes hand in hand with lack of education or poor education; lack of good nutrition and health care; unsafe, high-crime neighborhoods; lack of supervision; few social supports; poor environmental conditions; and uprooted cultural values (Vargas & Koss-Chiono, 1992). The background of children with low socioeconomic status emphasizes survival. These children's behaviors could be the direct result of living without proper care or sustenance, lack of formal education, and an insecure environment. Clinicians need to keep in mind the influence their own worldview may have when interpreting the behaviors of children from such groups. Flexible clinician roles, case management approaches to treatment, and community-based approaches may be helpful.

Immigrant and Refugee Children and Youth

Challenges faced by immigrant and refugee children and youth include changes in daily living, language, behavior, education system, and socialization practices. Clinicians should facilitate new culture acquisition and also encourage retaining cultural ties (see skills under "Bicultural Effectiveness"). Immigration status, legal or illegal, will have implications for the degree of stress experienced and service available. Immigrant youth who do not have legal status or who have undocumented family members face challenges accessing health and educational resources and are vulnerable to legal retribution, unfair and unsafe labor practices, family separation, and added stress related to their uncertain status and need for secrecy.

Refugee children's adjustment problems are underscored by multiple losses, family separation, refugee camp experiences, and trauma (Bemak & Greenberg, 1994). Resettling refugee children and adolescents report depression, irritability, withdrawal, low self-image, isolation, estrangement, loneliness, a loss of power, feeling out of place, sleep problems, feeling that their old values are useless, and a sense of being devalued (Sourander, 2003). Refugee and immigrant services

organizations provide some services for refugees and immigrants, such as language classes, assistance in finding housing and enrolling children in school, accessing health care services, and limited economic support. Clinicians should consider a case management approach to service provision and bicultural effectiveness.

Ethnic Minority Children and Adolescents

Ethnic minority children and adolescents face discrimination, racism, and prejudice related to their minority status (Porter, 2000). Stress associated with racism is one factor accounting for higher rates of stress-related health problems in ethnic minority groups. Overall, ethnic minority youth are at increased risk for developing emotional and behavioral problems. Issues facing ethnic minority youth include physically threatening environments, conflicting messages about appropriate identity development, economic hardship, and limited access to supportive services that may place many ethnic youth at a significant disadvantage when compared to their White counterparts (Hoberman, 1992). Experiences of oppression and racism further impede their healthy psychosocial adjustment and development. Signs of maladjustment associated with these many contextual factors include substance abuse, mood disturbances, academic underachievement, premature sexuality and childbearing, delinquency, underemployment, and instability of family life (Barbarin, 1993).

Biracial Children

Children of biracial or multiethnic backgrounds may have unique concerns related to their personal and social identity. A child's physical appearance may impact his or her self-identity and also influence parental identification. Models of biracial or multiracial identity development can help clinicians get a sense of where biracial children are in their identity development (Root, 1990). For biracial children, cultural conflicts regarding values, family roles, and parenting may arise in multiple contexts. If parents and families successfully negotiate these cultural differences, problems may not arise, but conflict in these areas could result in inconsistency, confusion, and family stress. Depending on the parents' origins, the child could be attempting to understand and negotiate multiple cultural influences. The clinician must assess the child's identification with each culture and the impact this may have across contexts.

Internationally Adopted Children

Rates of international adoption in the United States have increased dramatically in the past decade (Friedlander et al., 2000). More and more children are growing up with parents and siblings from a different language and cultural background from themselves and, importantly, who look different from themselves. These children may face teasing, name-calling, and discrimination, just as ethnic minority children do. White parents may be ill equipped to understand and/or address their adopted child's experiences with racism or struggles with racial or ethnic identity.

Families may also experience conflict with other family members or the community regarding their decision to adopt an ethnically different child.

Children with a history of institutionalization (e.g., living in an orphanage) face health problems, delays in cognitive or socioemotional development, sensorimotor integration difficulties, and problems with attachment and bonding. Additionally, adoptive parents of these children may be dealing with unresolved fertility issues, disappointments related to the child, or problems related to family integration. As young children mature, issues related to ethnic and cultural identity may emerge to an even greater extent. Family activities and routines that promote bonding and relationships among all family members can be encouraged. Families may need assistance resolving cultural conflicts and skills training on how to become a bicultural (or multicultural) family (Friedlander et al., 2000).

Gay, Lesbian, Bisexual, and Transgender Youth

When dealing with children and adolescents who identify themselves as gay, lesbian, bisexual, or transgender (GLBT), the clinician must be aware of not only the client's current functioning within that identity but also the past experiences associated with it. Worldwide, GLBT youth face discrimination and social marginalization related to their sexual or gender orientation. Models of identity development indicate that GLBT individuals may undergo a series of stages of confusion, identification, and acceptance of their sexual orientation (Cass, 1979). When this occurs, how it occurs, and the outcomes will vary from person to person. Individual and contextual factors certainly come into play when assessing each client's experience of this process. In an atmosphere of family support and openness, the client is psychologically equipped to handle outside pressures or discrimination. However, in an environment with strong family, religious, or cultural sanctions against homosexuals, children and adolescents may have to face these societal pressures alone.

Given these factors, it is important to give GLBT youth a safe and open environment to express their experiences, frustrations, and ideas. The worldview of GLBT individuals is one that has developed alongside experiences of discrimination, fear, and possibly danger (Herek, 1995, 2000). It is important for the clinician to recognize the multidimensional nature of the client and address GLBT issues as they relate to the presenting problem.

NEW DIRECTIONS IN CULTURALLY RESPONSIVE TREATMENT

Consistent with an increased global interest in positive psychology and cultural competence, new directions in child and adolescent approaches to treatment view children and culture as resources (Lerner et al., 2002). Positive youth development (PYD) supports the notion that cultivating social, emotional, and cognitive competencies in children and adolescents can head off, or at least ameliorate some emotional and behavioral problems (Larson, 2000). Programs employing PYD promote

resilience by helping children and youth to better understand their environment and act in ways that promote healthy development (Dworkin, Larson, & Hansen, 2003; Lerner et al., 2002).

These programs may be particularly appropriate for addressing the mental health needs of culturally diverse youth (Johnson & Johnson-Pynn, 2007). They offer a broad-based approach that is widely culturally acceptable and accessible (i.e., often school- or community-based). Nurturing cultural strengths and community connections is integral to most PYD programs. Program outcomes, such as social competencies, positive bonds with people and institutions, positive self-regard, self-efficacy, positive ethnic identity, and social justice values, have obvious benefits for culturally diverse youth (Catalano, Berglund, Ryan, Lonczak, & Hawkins, 2002; Lerner, 2002).

SUMMARY

In this chapter, we have described the nature and functioning of culture as it impacts children and families and the process of assessment and treatment. First we discussed important contextual factors, such as the changing demographic landscape in the United States and the increasing multicultural nature of society, that highlight the need for cultural competence in child assessment and treatment. We discussed the complex, multifaceted, subjective, and dynamic aspects of culture that must be understood in order to fully appreciate culture's influence and intricacies. Next we outlined some of the ways that culture interacts with other factors to affect child development, socialization, education, values, and norms. We also described how culture results in variations in mental heath functioning and psychopathology in children, shaping symptom expression and communication, help seeking, illness beliefs and expectations, and desired outcomes for assessment and treatment.

Next, we provided an overview of key cultural constructs to consider in assessment and treatment. Acculturation, ethnic identity, worldview, and explanatory models were defined, as was their potential to influence the assessment and treatment process. We then discussed how culture may come into play at each phase in the assessment and treatment process. We integrated suggestions for attending to culture into each section and provided practical ideas, such as how to build a satisfactory relationship with clients, how to ask clients about their culture, how to adapt one's communication style, how to incorporate cultural strengths into treatment, and how to work with interpreters and cultural brokers. Culture-related problems, such as acculturative stress and identity concerns, were discussed, as were culturally responsive interventions.

Finally, we called attention to the unique concerns of certain groups of children, including refugee and immigrant children, gay and lesbian youth, bicultural children, ethnic minorities, children living in poverty, and internationally adopted children. In closing, we point to the promise of positive youth development programs as culturally responsive approaches to meeting the mental health needs of significant numbers of culturally diverse children and adolescents. Although this chapter covered many aspects important for understanding and working with clients from a variety of cultures, it is impossible to sufficiently address cultural issues in

assessment and treatment in a single chapter. Clinicians working in the context of cultural differences are encouraged to make ongoing efforts to enhance their cultural competence in assessing and treating children and adolescents.

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CHAPTER 27

Divorce and Custody

Shannon M. Greene, Kate Sullivan, and Edward R. Anderson

Divorce is a common transition for growing numbers of parents and children in the United States. Each year over a million American children experience their parents' divorce (U.S. Census Bureau, 2000). As part of the divorce, families encounter the legal system as they attempt to make plans for how children will be raised in the new family structure. Clinical psychologists may become involved in several ways, such as by providing mediation and custody evaluations that influence the way family life will proceed after divorce. Thus, clinical psychologists may benefit from information on divorce and custody issues for these litigating families.

This chapter provides an overview of divorce and custody issues relevant for clinical psychologists. It introduces a general model of divorce adjustment, followed by a review of critical factors affecting child outcome for divorcing families. Then it presents three areas that affect postdivorce legal decisions: guiding legal principles, custody and visitation arrangements, and child support. Finally, it focuses on mediation and custody evaluations as mechanisms for assisting families through the divorce transition.

A GENERAL MODEL OF DIVORCE ADJUSTMENT

Perhaps the most widely accepted theoretical model of divorce involves a process perspective. Specifically, divorce is viewed not as a single isolated event, but as a complex chain of marital transitions and family reorganizations that alter roles and relationships and affect individual and family adjustment (Hetherington, Bridges, & Insabella, 1998). Prior to divorce, conflict within the family may increase, accompanied by periods of physical separation between spouses and attempts at reconciliation. Divorce itself may lead to additional disruptions in daily routines, roles, activities, and social relationships, including continued shifts in living arrangements and visitation, such as physical separation, accompanied by changes in contact between the child and the nonresidential parent. Even after the divorce occurs, changes may continue, brought about by parental efforts to establish new partnerships, including dating and nonmarital cohabitation. These changes present children and

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families with adaptive challenges. The response to these challenges is influenced by previous functioning and experiences and subsequent stresses and resources (Amato, 2000; Hetherington et al., 1998).

Converging research indicates that parenting practices mediate many of the contextual changes that occur during and after divorce (e.g., Simons & Associates, 1996). Thus, divorce precipitates changes in family life that have the potential to disrupt effective parenting practices. These practices include monitoring children's whereabouts, providing consistent discipline, using constructive problem-solving strategies, and engaging in positive involvement. Disruption in parenting subsequently increases the likelihood of children's exhibiting behavioral and emotional difficulties.

Empirical sources suggest there are four key factors that have implications for how children and families adapt: economic hardship, interparental conflict, parental distress, and cumulative transitions and related life stress. These factors are related (e.g., parents experience distress when conflict is high and economic resources are low), but each has potential challenges and implications for child adjustment.

Economic Hardship

With divorce, families often encounter dramatic reductions in income, with per capita declines averaging between 13% and 35% in national samples (R. Peterson, 1996). Economic pressure may mean that families are unable to meet material needs, fall behind on debt payments, and have to cut back on everyday expenses to live within available means (Conger, Ge, Elder, Lorenz, & Simons, 1994). Economic hardship, when combined with high levels of life stress, is associated with increased psychological distress and decreased parenting skills. Not surprisingly, some parents become distracted and emotionally or physically unavailable to their children (Simons et al., 1996). Such disruptions in parenting lead to problematic behavior in children (Forgatch, Patterson, & Ray, 1996). With sufficient time, stress usually abates, although families with lower income may continue to experience disruptive events. The increase in economic hardship following divorce is a potent factor in the emergence of problematic outcomes for children and thus a key reason why courts attempt to determine and enforce child custody payments.

Interparental Conflict

Research indicates that the frequency and intensity of conflict, as well as more overt forms of conflict (e.g., verbal and physical aggression), are related to problematic adjustment in children, regardless of marital status of the parents (Amato, 1999). In fact, interparental conflict has been shown to be a better predictor of child maladjustment than changes in the parents' marital status (Davies, Harold, Goeke-Morey, & Cummings, 2002). Interparental conflict appears to increase children's emotional distress, disrupt peer relations and academic performance, and lead to physical health difficulties. Children who have a family history of exposure to interparental conflict (particularly if conflict is unresolved) show increasing sensitivity or reactivity over time, with reactions ranging from fear and sadness to anger and aggressiveness. Children may be particularly sensitive when conflict concerns

child-related issues such as child rearing and custody arrangements. Conflict over these topics may cause children to feel shameful, to blame themselves for the divorce, or to fear becoming involved in the conflict (Grych, Harold, & Miles, 2003). Some theorize that exposure to highly conflictual situations undermines children's sense of emotional security, as evidenced by emotional and behavioral problems (Davies et al., 2002). Others show that high levels of conflict disrupt parenting practices, so that mothers show more negativity and inconsistent discipline, and fathers become less involved with their children, in contrast to counterparts in low-conflict marriages (Gonzales, Pitts, Hill, & Roosa, 2000).

Parental Distress

Divorce is one of the most stressful experiences that an individual may experience. Many individuals experience a variety of adjustment problems, such as increased risk of psychopathology, higher incidence of motor vehicle accidents, elevated drinking and drug use, alcoholism, suicide, and even death (see review by Amato, 2000). More recently separated and more emotionally attached divorcing individuals have poorer immune functioning, greater levels of depression, and other health problems (Kiecolt-Glaser, McGuire, Robles, & Glaser, 2002). For some, divorce may bring about positive effects, such as increased personal autonomy, overall happiness, social involvement, and career development (Hetherington & Kelly, 2002).

Adults who experience difficulties in adapting to stressful postdivorce changes may have problematic relationships with their children (Tein, Sandler, & Zautra, 2000). These problematic relationships interfere with adjustment in children (e.g., Martinez & Forgatch, 2002). In the context of the divorced family, parental adjustment assumes a critical role, given that child adjustment typically follows from how well adults in the household fare. Children's own adjustment difficulties also contribute to the custodial parent's level of stress and quality of parenting (Forgatch et al., 1996). So processes that operate within the parent-child relationship appear to contribute to the long-term adjustment of each family member.

Cumulative Transitions and Related Life Stress

Adjustment problems also arise from transitions that surround relationship dissolutions and consequent repartnering in both formal and informal unions (e.g., Martinez & Forgatch, 2002). After divorce, most adults enter into new relationships; one study found that at the time of the divorce filing, half of the parents already had begun to date, and over one quarter were in a serious romantic relationship (Anderson et al., 2004). Within 10 years of separation, about two thirds of families with two or more children have experienced a nonmarital cohabitation (Bramlett & Mosher, 2002). Parents who remarried had dated an average of three to five people prior to meeting the person they would eventually remarry (Montgomery, Anderson, Hetherington, & Clingempeel, 1992). These relationships introduce additional transitions and reorganizations within the family (Raley & Wildsmith, 2004). Echoing findings discussed earlier, the effect of repartnering transitions on child adjustment is mediated by parenting practices (DeGarmo & Forgatch, 1997).

Short-Term and Long-Term Adjustment to Divorce

Children's adaptation to divorce depends on the particular combination of stress and resources in their lives (Amato, 1993). As stresses begin to decrease and resources become more plentiful, many children who had exhibited problematic adjustment will improve. However, problems with adjustment may persist when stressors such as financial hardship, interparental conflict, and disrupted parenting continue.

POSTDIVORCE DECISIONS

Divorce calls for a restructuring of parental rights and responsibilities along two broad areas relating to child rearing: custody and visitation, and child support. Before discussing each area, we present information on the overarching or guiding legal principles in decisions related to child custody.

Guiding Legal Principles in Child Custody Decision Making

Divorce law and the determination of child custody are established through state statutes. The most prevalent criterion for determining placement of children after divorce is the Best Interest of the Child standard, which emerged in the 1960s and 1970s as an alternative to the Tender Years Doctrine, an earlier standard favoring placement with mothers. The Tender Years Doctrine itself supplanted a custody preference for fathers, when children were considered legal property. Most jurisdictions now explicitly state a gender-neutral preference for custody decisions (see O'Donohue & Bradley, 1999, for more details).

The Best Interest of the Child standard set forth a list of factors to be used in determining child placement after divorce, although states vary in their relative consideration and weighting of these factors. Judges too have great latitude in applying the standard for making determinations in custody cases. Buehler and Gerard (1995) provide a compilation of all Best Interest factors used throughout the United States (see Table 27.1).

Before considering Table 27.1 in detail, note that a critical distinction should be made with respect to legal custody versus physical custody. Legal custody refers to the relative level of continuing responsibility and involvement each parent will have in decisions about the child, ranging from joint, where custody is shared, to sole, where custody is assigned to one parent. Physical custody refers to the daily residential and supervisory arrangements for the child. In practice, these two decisions are independent; for example, parents may be awarded joint legal custody, but one parent may receive physical custody (i.e., the child resides primarily within that parent's household and sees the other parent at established times). Or there may be some form of joint physical custody awarded. Although less clearly defined than sole physical custody (Buchanan & Williams, 2006), a minimum criterion in use for joint physical custody is that children spend at least 25% of the time with each parent (Bauserman, 2002).

With respect to Table 27.1, factors 1 to 5 correspond to the criteria outlined in the Uniform Marriage and Divorce Act (1970); the remaining factors consist of

Table 27.1 "Best Interest of the Child" Factors

Criteria Described in the Uniform Marriage and Divorce Act (1970)

1. Wishes of the child's parent or parents as to his/her custody
2. Wishes of the child as to his/her custodian
3. Interaction and interrelationship of the child with his/her parents, his/her siblings, and any other persons who may significantly affect the child's best interest
4. Child's adjustment to his/her home, school, community
5. Mental and physical health of all individuals involved

Criteria Related to the Primary Caretaker Standard

6. If one parent, both parents, or neither parent has provided primary care of the child
7. Capacity and interest of each parent to provide for the emotional, social, moral, material, and educational needs of the child
8. Capacity and disposition of each party to provide the child with food, clothing, medical care, and other material needs

Criteria Related to Safety and Well-being of Children

9. Evidence of domestic violence, abuse, or neglect
10. Evidence that substance use affects the child's well-being
11. Moral fitness of the parents

Criteria Related to the Cooperative Parent Factor

12. Which parent is more likely to allow the child frequent and continuing contact with the noncustodial parent
13. Nature and extent of coercion or duress used by a parent in obtaining an agreement regarding custody
14. Ability of both parties to cooperate and make decisions jointly
15. Child support noncompliance

Other Criteria

16. Out-of-state residence
17. Length of time the child has lived in a stable, satisfactory environment and the desirability of maintaining continuity
18. Permanence of custodial home
19. Nature and amount of contact with both parents
20. Love and affection existing between the child and each parent
21. Age and sex of the child
22. Parents' respective home environments
23. Physical, emotional, mental, religious, and social needs of the child
24. Health, safety, and welfare of the child
25. Child's cultural background

Source: "Divorce Law in the United States: A Focus on Child Custody," by C. Buehler and J. M. Gerard, 1995, *Family Relations, 44,* pp. 439–458.

outgrowths or restatements of these basic criteria. It is noteworthy that the first two factors involve the direct wishes of the parties involved, although the preference for the child's wishes typically is reserved for older children. Standards are emerging for representing these and other child interests in custody cases (e.g., Ackerman, 2006; American Bar Association [ABA], 2003; American Law Institute, 2002; American Psychological Association [APA], 2002; Elrod, 2003).

Factors 6 to 8 in Table 27.1 address the preference for placing the child with the primary caretaker, the individual who has provided the daily pattern of care.

In principle, this preference appears determinative and relatively easy to establish (Buehler & Gerard, 1995). In actual practice, however, it may lead to inequitable physical custody arrangements, such as granting sole physical custody to a parent who has performed only marginally more than the other with respect to caretaking. An alternative standard has evolved, called the approximation rule, with custody arrangements attempting to resemble what had been in place prior to the divorce. Emery (1999) has advocated for this preference because it combines the best aspects of the primary caretaker and joint custody standards. The determinate standard of time spent caring for the child is used to establish the custody arrangement, thereby accommodating a wide variety of potential predivorce parenting situations.

Factors 9 to 10 in Table 27.1, domestic violence and substance abuse, arguably may override other factors as they directly impact the safety of the child (Buehler & Gerard, 1995). Most states prohibit custody for a parent with verified domestic violence. Factor 11 mentions "moral fitness," a vague term often equated with sexual promiscuity. Seven states, for example, outlaw cohabitation between unmarried partners, which may indirectly affect awards of child custody.

Factors 12 to 15 in Table 27.1 relate to the "cooperative or friendly parent," such that parents who act to restrict access to the child in the absence of compelling reasons of safety may face greater challenges in obtaining custody and visitation. Relatedly, efforts to restrict access (along with other behaviors such as efforts to "brainwash" the child from wanting to have a relationship with the other parent and unfounded accusations of abuse) have been the focus of a controversial proposed psychiatric disorder for understanding children who become resistant to contact with the noncustodial parent, called Parental Alienation Syndrome (Gardner, 1998). More recent views, however, have recast the situation of the child who is resistant to visitation more broadly, now called the *alienated child;* sources of resistance now include the behaviors of both parents, along with developmental preferences and reactions to the divorce. In support of this reformulation, results of two cross-sectional studies show child resistance to visitation arising from multiple determinants, such as prolonged litigation, parental lack of warmth, and the child's own emotional and social problems. Johnston (2005) has integrated this work into Figure 27.1.

Finally, as listed in Table 27.1, Buehler and Gerard (1995) provide factors relating to contact and continuity. Out-of-state residence, for example, is likely to affect practical access to the child. Clearly, the information in Table 27.1 shows that no single approach will fit all potential situations. In the case of incarcerated fathers, for example, judges must weigh the desire of the parties to maintain contact, along with a consideration of whether the visit itself may be difficult for the child, or whether written or phone contact may be more appropriate (Menno, 2003). Moreover, issues such as the nature of the offense, length of prison stay, and past history may affect the form that custody will take.

Thus, in practice, there is wide latitude given to the courts in determining the nature of custody and visitation arrangements. Buehler and Gerard (1995) propose a presumption of favoring parental agreements, such that parents forward their preference for custody and visitation to the court, which the court will honor. In the absence of an established parental agreement, these authors propose a presumption

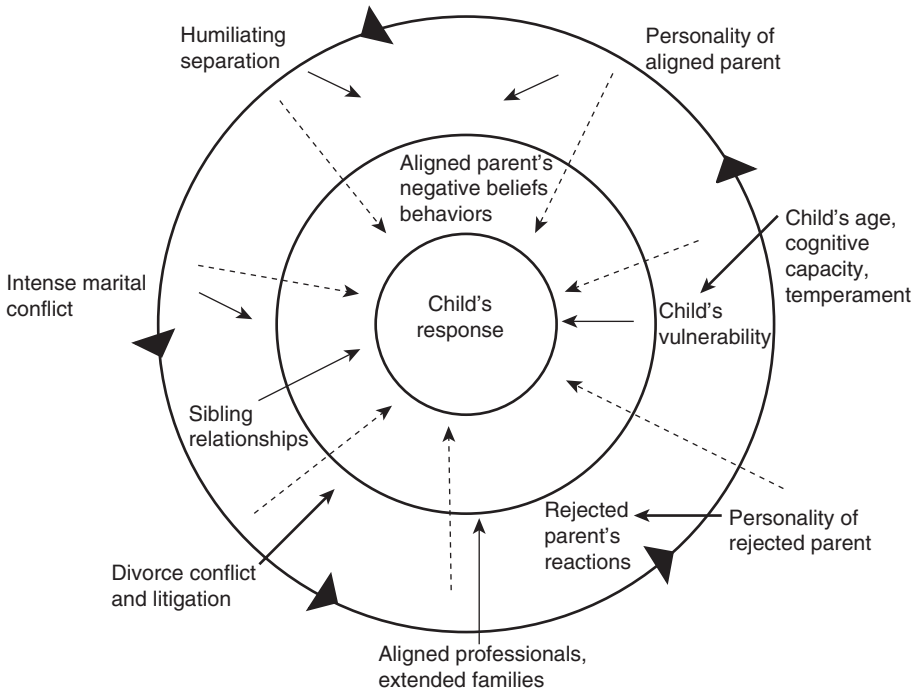

Figure 27.1 A model of factors predicting child alienation. *Source:* "Children of Divorce Who Reject a Parent and Refuse Visitation: Recent Research and Social Policy Implications for the Alienated Child," by J. R. Johnston, 2005, *Family Life Quarterly, 38*(4), pp. 757–775. Reprinted with permission.

of shared legal custody, sole physical custody to the primary caretaker, and substantial access by the nonresidential parent.

Custody and Visitation Arrangements

In this section, we consider differences across custody arrangements identified in previous research, issues related to maintaining contact with the nonresidential parent, and special considerations that impact custody and visitation arrangements.

Common Practices and Comparisons across Custody Arrangements

Following divorce, up to 80% of children are in the sole physical custody of their mother (Emery, 1999). The prevalence of sole father custody, however, is increasing rapidly, with fathers more likely to be awarded custody when children are adolescent (Maccoby & Mnookin, 1992). Buchanan, Maccoby, and Dornbusch (1996) note that mother and father sole custody arrangements were similar on a wide variety of dimensions. Differences were noted in emotional closeness and monitoring, which favored mother custody households; this can be explained in part by the lower rate of behavior problems for children in mother custody households. It also should be noted that fathers essentially may be taking on more challenging situations to begin with, given that they gain custody under more difficult family circumstances than

mothers (Emery, 1999). Sole custodial father families are less likely to be granted child support awards, and nonresidential mothers are less likely to comply with the child support awards given (Meyer & Garasky, 2003).

In response to social and economic changes beginning in the 1970s, there has been an increased preference for parents to share custody, with the hope of maximizing involvement of both parents in their children's lives. There has been a call for current legal practices to recognize the child's need to have both parents actively involved in his or her life (Menno, 2003). The movement toward recognizing the importance of both parents has coincided with the tendency for states to deregulate custody arrangements, allowing parents to establish individualized arrangements (Buehler & Gerard, 1995). Additionally, developmental research has stressed the role that both parents play in children's lives as well as the need for custody arrangements to be flexible (J. B. Kelly & Lamb, 2000). When awarded, joint custody is more common for children in middle childhood and for families with higher income (Cancian & Meyer, 1998; Maccoby & Mnookin, 1992).

As opposed to comparisons across different kinds of sole custody homes, meta-analytic evidence suggests that children in joint custody arrangements fare better across a variety of indexes; in fact, they showed comparable levels of adjustment with children from never-divorced households (Bauserman, 2002). Better adjustment appears to be accounted for by greater closeness between the child and both parents. In addition, interparental conflict is lower in joint custody homes. Finally, children from joint custody families were more likely to see their father weekly, have more overnight stays, and benefit from more child support (Seltzer, 1998). However, the families that enter into joint custody arrangements may themselves be different from those that enter into sole custody arrangements; as described earlier, states likely award joint custody to families that demonstrate that they can get along well enough to manage the close contact and coordination between parents that such arrangements may require. So better outcomes may be due, in part, to selection effects.

What factors are associated with acquiring sole custody? Several studies have shown that parents are more likely to receive custody when they are the plaintiffs and have higher income and education levels compared to their spouse, but not when they have prior marriages, children from prior marriages, or marital infidelity (Cancian & Meyer, 1998; Fox & Kelly, 1995; R. F. Kelly, Redenbach, & Rinaman, 2005; Seltzer, 1998). Fathers are more likely to receive sole custody if ending a long-term marriage; they are less likely to receive custody when they are unemployed or owe child support arrears or when the family is eligible for welfare support (Cancian & Meyer, 1998; Fox & Kelly, 1995). Mothers are more likely to receive custody when they are older, when the oldest child in the family is female, or when there are high levels of conflict (Cancian & Meyers, 1998; Fox & Kelly, 1995; R. F. Kelly, et al., 2005). Joint custody is more likely when all the children are male, both parents work and own homes (Cancian & Meyer, 1998), and when fathers are more educated (R. F. Kelly et al., 2005).

In addition to sole and joint custody, split custody is available to families with more than one child. In this individualized custody arrangement, some of the children may live with the father while other children may live with the mother (Ellis, 2000).

Continuing Contact with the Nonresidential Parent

A critical issue with sole custody arrangements concerns continued contact with the nonresidential parent, typically fathers, given prevailing custody arrangements. Adults who experienced parental divorce in childhood report a sense of loss due to the lack of contact with their father (Laumann-Billings & Emery, 2000) and a wish to have been more involved with him while growing up (Fabricius & Hall, 2000). Some level of contact is necessary for closeness (Fabricius, 2003), especially for younger children (Leventhal, Kelman, Galatzer-Levy, & Hraus, 1999; Pruett & Johnston, 2004), though contact alone is not always a sufficient determinant for relationship quality (Berscheid & Peplau, 1983; Buchanan & Williams, 2006).

When fathers were able to maintain strong emotional ties with their nonresidential children, their children had greater psychological and behavioral adjustment during adolescence (Buchanan et al., 1996; McLanahan & Sandefur, 1994; Seltzer, 1994; Whiteside & Becker, 2000). Sheer contact with the nonresidential father may not be directly tied to children's adjustment (Amato & Gilbreth, 1999; Hetherington & Kelly, 2002; Marsiglio, Amato, Day, & Lamb, 2000; McLanahan & Sandefur, 1994; Seltzer, 1994; Whiteside & Becker, 2000), although its absence is certainly reflected in other ways, such as adult children's sense of loss mentioned earlier. Rather than a direct effect on children's adjustment, the impact of contact between the noncustodial parent and the child on the child's well-being may depend on other factors, most notably the amount of conflict between the divorced parents (e.g., Buchanan & Williams, 2006; Lamb et al., 1997). High interparental conflict may undermine the closeness between the nonresidential father and child, or involve children directly in marital disputes. Relatedly, conflictual parents are less likely to carry out visitation plans, even when court-imposed (Pruett, Williams, Insabella, & Little, 2003; Whiteside & Becker, 2000). Conflict over visitation is one of the primary reasons for the courts to determine that supervised visits should occur. In a multisite study of supervised visitation, over three-quarters of the families reported that anger toward the other parent was the primary reason for the court determination of supervised visitation (Pearson & Thoennes, 2000). Those most satisfied with supervised visits were the families who had reported fighting at drop-off and pickup times.

Special Considerations

There is a considerable controversy over whether to encourage or discourage overnight stays when children are infants and toddlers. Much of this controversy concerns a study by Solomon and George (1999a, 1999b) in which overnight stays were linked to problems in infant-mother attachment. However, infant adjustment to the overnight stays could be promoted with parental cooperation and maternal support. Results highlight the need to consider more than simply the child's age when making decisions about overnight contact.

There also is legal recognition that high-conflict custody cases represent a substantial risk factor for the involved children and families (High-Conflict Custody Cases, 2001). The Family Law section of the ABA, in conjunction with the Johnson

Foundation (2001), developed standards of practice to recognize and address the challenges that these children and families face.

Given high rates of geographic mobility in the United States, parental relocation following separation and divorce is increasingly problematic as joint custody has become the preferred custody arrangement (J. B. Kelly & Lamb, 2003). Joint physical and/or legal custody assumes that both parents maintain geographically close residences to enable contact between children and parents. Relocation by one or both parents makes this difficult to maintain (Austin, 2000). Relocation of one parent also may reduce financial support and lead to a more distant relationship between that parent and the child (Braver, Ellman, & Fabricius, 2003). Relocation may depend on the children's own developmental status (J. B. Kelly & Lamb, 2003) or, as Warshak (2000) suggests, on unique considerations that require a case-by-case analysis.

Child Support

Though the emotional upheaval that typifies divorce is most commonly at the forefront of issues for families experiencing this transition, financial problems also present challenges to child well-being. Single-parent families are more likely to be poor (Bartfeld, 2000). Thus, children have a greater chance of living in poverty after divorce.

Public policy in the United States has increasingly sought to address this issue by establishing the financial responsibility of both the custodial and the noncustodial parent for their children. As early as the nineteenth century, public policy began to enforce parental financial responsibility for dependent children. More sweeping changes to child support policy appeared during the twentieth century, coinciding with the dramatic increase in separation and divorce in American families. Beginning with the Social Security Act of 1935, government support was granted to women and their dependent children based on need, specifically in cases where one parent was not providing any financial support. Additional policies since then have addressed the need for parents to assume greater parental responsibility for their children, such as the Uniform Interstate Family Support Act in 1992 and the Personal Responsibility and Work Opportunity Reconciliation Act in 1996. Currently, child support is regulated by the Child Support Enforcement Program, a partnership of federal, state, and local authorities to collect child support (U.S. Department of Health and Human Services, 2005). Current policy efforts to enforce child support have yielded benefits: Children are less likely to live in poverty (Edin, 1995), their families are less reliant on government support, and their parents take on greater financial responsibility for their children (Altman, 2003).

More generally, child support represents an important resource on which children rely (Amato & Gilbreth, 1999). Increases in their standard of living likely improve children's health and educational attainment (Furstenberg, Morgan, & Allison, 1987; King, 1994; McLanahan, Seltzer, Hanson, & Thompson, 1994). There is evidence, for example, that child support shows positive effects on children's school achievement (Argys, Peters, Brooks-Gunn, & Smith, 1998; Knox, 1996). A meta-analysis of 14 studies shows child support related to greater academic success and fewer externalizing problems in children (Amato & Gilbreth, 1999).

Developmental gains from child support may be more important for younger children (King & Soboleski, 2006) and for involuntary types of payments (Argys et al., 1998; Hernandez, Beller, & Graham, 1995).

Though the general consensus favors child support enforcement, establishing guidelines for child support awards is a more complex issue. Before legislative changes in the 1980s, child support was typically awarded on a case-by-case basis (Rogers, 1999). However, policy changes with the Family Support Act sought to ensure a more equitable system that made use of economic data to estimate the financial costs associated with raising children (Beld & Biernant, 2003). Within these federal guidelines, individual states were free to establish their own formulas; the more common formulas are described next. More detailed reviews of these and other formulas can be found elsewhere (Beld & Biernant, 2003; Blumberg, 1999; Morgan, 2004).

> *Income Shares.* Developed by the U.S. Department of Health and Human Services, the basis for this formula is to provide children with the same proportion of financial resources and support that would have been available to them had their parents remained married.
>
> *Percentage of Obligor Income.* Developed by the state of Wisconsin, this formula takes into account the parent's income, the amount of time the child spends with each parent, and whether the parent is supporting other children. Formulas may differ based on the complexity of the family arrangement, although the basic support guidelines assume that the noncustodial parent is responsible for providing 17% of gross income for 1 child, 25% for 2, 29% for 3, 31% for 4, and 34% for 5 or more children.
>
> *Cost Shares.* The cost shares formula takes into account the costs of raising children, as well as tax benefits associated with dependent children.
>
> *Hybrid Models.* Evolving out of some of the limitations with previous formulas, various hybrid models have emerged, such as one proposed by the ALI [American Law Institute] that allows for greater discretion in deciding costs associated with living arrangements. (Bartlett, 2001; Czapanskiy, 2001; Harris, 2001)

As with any public policy, the various formulas for determining child support have come under fire. Some charge that states do not raise award standards often enough to fairly reflect the actual cost of raising children, given federal legislation that requires updates only every 4 years (Venohr & Griffith, 2005; Venohr & Williams, 1999). Others criticize specifics of the formulas, charging that the income shares and the percentage of obligor income models allow custodial parents to achieve a higher standard of living than nonresidential parents (Rogers, 1999). In fact, formulas such as income shares were designed to protect the primary custodial parent and residential children from a precipitous drop in standard of living after the divorce, although Braver and colleagues (2003) counter that the drop may not be as drastic as once thought. Criticism also is levied because many of the formulas assume that one parent will have sole custody, rather than acknowledging the growing numbers of shared custody arrangements and situations in which visitation and residence vary over time (Melli & Brown, 1994). In fact, child support awards typically do not factor in costs of rearing children for the nonresidential parent until time spent in that household approaches the level of joint custody. Fabricius and Braver (2003) provide evidence for a linear relationship between household

expenses and amount of contact, suggesting that nonresidential expenses are being underestimated. Findings by Garfinkel, McLanahan, and Wallerstein (2004) differ, perhaps because of sample characteristics. The use of a nationally representative, multi-informant sample may be required to address this issue further.

There also is disagreement with whether child support should end when children achieve majority status or after receiving additional post–high school education and training. Current formulas and legislation typically do not consider expenses incurred after children turn 18 years old. New state laws are beginning to address this issue, although there is evidence that some divorced parents are voluntarily providing equal proportions of support to their college-age children (Fabricius, Braver, & Deneau, 2003).

Finally, difficulties surround international child support enforcement. There is currently no global legislation regarding child support, although the United States has reached agreement with several nations on a case-by-case basis (Duncan, 2004). Efforts have been made to develop international policy, but there is no obligation to adhere to the legislation.

A recent report published by the U.S. Census Bureau provides a report card of sorts on child support awards, actual receipts, and enforcement, based on the Current Population Survey reports from 1994 to 2004 (Grall, 2006). It showed that the proportion of children living in poverty after divorce has dropped, from one third in 1993 to less than one fourth in 2003. However, only half of custodial parents had received all their child support award due; the average amount of child support in arrears was $5,100, for a combined total of $37 billion past due. Many custodial parents acknowledge receiving noncash support, such as gifts and coverage of expenses, and over half of child support awards included health insurance provisions from the nonresidential parent.

What would help the parents without primary custody maintain their child support payments? With respect to policy, Freeman and Waldfogel (2001) found that compliance with child support was greatest when paired with stronger enforcement laws. Additionally, greater compliance occurs when child support awards explicitly state the amount to be paid (J. Peterson & Nord, 1990). With respect to individual considerations, greater compliance is linked to access to children, as measured by actual contact (Argys, Peters, & Waldman, 2001; Seltzer, Schaeffer, & Charng, 1989) or geographic proximity (Teachman, 1991). Greater compliance occurs when the award is perceived as fair (Lin, 2000) and when mediation has been used (Emery, Matthews, & Kitzmann, 1994). Greater compliance also occurs when noncustodial parents believe they have some say in the child's life and how the money will be used (Braver et al., 1993; Weiss & Willis, 1985). Perceptions of control influence the likelihood of child support payments indirectly through visitation. Payment of child support is more likely among nonresidential parents who see their child frequently (Seltzer et al., 1989).

Other factors also may decrease the likelihood of child support receipt. Noncompliance is associated with more marginalized groups, such as the less educated and the poor (J. Peterson & Nord, 1990). It also is associated with families with greater interparental conflict (Hetherington & Stanley-Hagen, 1999) and domestic violence, although victims of domestic violence are less likely to seek and receive

child support awards (Menard & Turetsky, 1999). Though recent policy efforts have been made to ensure the safety of those seeking child support, greater advances need to be made in securing the safety of victims of domestic abuse in seeking child support.

Finally, there are few established policies or laws that dictate how child support should be handled among divorces involving stepparents. Few states have laws requiring continuing support from the stepparent following divorce (Hans, 2002; Fine & Fine, 1992).

ASSISTING FAMILIES WITH THE DIVORCE TRANSITION

When assisting parents with the divorce transition, clinical psychologists may make use of various strategies, some of which place them in an advisory role to the family or legal community. This section discusses two such strategies: mediation and custody evaluation.

Mediation

Reflective of dissatisfaction with litigation, mediation has evolved as an alternative. Mediation is essentially cooperative in nature, having originated from the field of negotiation. Thus, mediation is based on the fundamental premise of self-determination: Each parent makes a voluntary effort to reach mutually agreed upon decisions about a parenting plan.

Mediators are typically attorneys, psychologists, or social workers who assist with identifying needs and interests and facilitating an agreement between the parents. The format used is goal-oriented and short term, usually lasting from 1 to 5 sessions. The mediator acts as an advocate for both parents in attempting to create a mutually agreeable solution. Emery and Wyer (1987) outline four aims for mediation, otherwise known as the four "Cs": to reduce *conflict,* to increase *cooperation,* to allow individuals *control* over key decisions affecting their lives, and to reduce *costs* associated with divorce and litigation.

There has been rapid growth in the use of mediation in divorce over the past 2 decades, with some states mandating an attempt at mediation in custody disputes. When mandated, parents must attend one or more meetings with a mediator in an attempt to reach an agreement, but they are not required to reach an agreement. Additionally, some states have implemented programs to educate parents about the impact of divorce and related transitions on children's well-being.

Typically, mediators seek to generate a mutually agreed upon, workable coparenting plan that defines the respective boundaries of each parent and, in doing so, minimizes the need for future contact. The process of mediation helps to model how the coparenting relationship should operate in future interactions, with communication that is less emotional and more businesslike in tone. A distinct advantage with mediation is the evolution of a detailed, uniquely tailored parenting plan. Such plans may generate greater satisfaction, thereby yielding better adherence.

Reviewers of related research have concluded that mediation appears to have moderate success (J. B. Kelly, 1996; Saposnek, 1998). Benefits are cost savings incurred when litigation can be avoided, improved parental satisfaction, and increased compliance for mediated agreements compared to those reached through adjudication. Perhaps because of the increases in parental satisfaction, parents tend to report lower levels of conflict at 1 year after the divorce, although such results are less consistently seen over the long term. Mediation does not appear to garner benefits to parent and child psychological adjustment. Also, court-mandated mediation appears to be less successful at producing an agreed upon parenting plan. Finally, mediation may be less appropriate in situations involving domestic violence and child abuse, where there are substantial discrepancies in bargaining power, or when parents are severely disturbed.

Custody Evaluation

When collaborative efforts such as mediation fail, families may turn to the courts for adjudication. Maccoby and Mnookin (1992) note that about 50% of families settle out of court, with custody uncontested; about 30% contest custody but settle through attorney negotiations out of court; another 11% resolve custody issues with mediation; the remaining are given mandatory custody evaluations consistent with the local courts, with about half of those settling out of court after the evaluation is conducted. Thus, approximately 4% of the sample proceeded to trial, and about half of those cases settled during the course of the trial. Only 1.5% of the sample completed a custody trial and received a final determination by the judge. The authors conclude that about 25% of the sample had entrenched legal battles over their children. These families were not distinguished from the larger sample with respect to race, education, and income; discriminating factors were children younger than 3 and mothers not engaged in paid employment.

Custody evaluations have become a routine part of litigation, either sponsored by the parties themselves or initiated by the courts. The custody evaluation comprises an assessment of the child, the parents, and the family by a psychologist. Because the stakes are high—an evaluation carries weight in determining parental rights relative to the child—custody evaluations are often difficult and demanding work. Adding to the difficulty are evaluations that involve accusations of child abuse, negligence, and family violence. Either parent may retain the psychologist, but generally most psychologists prefer to be retained by the court or jointly by the attorneys of both sides (Ackerman & Ackerman, 1997).

Although state statutes differ, generally the evaluation assesses "parenting capacity" and its fit with the psychological and developmental needs of the child: (a) parental capacity of the prospective custodians: their knowledge, skills, and abilities; (b) psychological functioning and developmental needs of the child (can also include the wishes of the child in some cases); and (c) functional ability of each parent to meet the child's needs. Thus the assessment includes the mental and physical health of all family members involved; the child's adjustment at home, school, and community settings; and the interaction of the child with parents and siblings. The APA (1994) directs evaluations to focus on three issues: (1) the needs of the

child, (2) the abilities and capacities of the parents, and (3) the best fit between the two.

Custody evaluations can be extensive, as the psychologist conducts clinical interviews with each parent and child (and any cohabiting romantic partner), reviews all relevant documents, conducts parent-child observations, conducts psychological testing on parents and children, and talks with teachers, attorneys, and therapists. The total time to conduct a custody evaluation (excluding court testimony time) ranges from 18 to 28 hours, and fees average $3,000 (Bow & Quinnell, 2001). As such, custody evaluations are often a critical part of the divorce proceedings, and may have a pivotal effect on the determination of the so-called ultimate issues of a case: custody and visitation rights.

In this legal setting, the custody evaluator is a professional expert who is called on to inform and advise the court in a balanced and impartial way, in contrast to a judge, who renders a final decision, or an advocating attorney, who represents the interest of one party. More recently, this role of imputed impartiality has come under increased scrutiny. Tippins and Wittmann (2005) state that custody evaluations are most valuable (and potentially least biased) when focused on facts and observations rather than clinical inference or judgments. The authors propose a four-level model for directing clinical involvement in custody cases, with the levels reflecting the degree of clinical inference involved (see Figure 27.2). At Level I, the lowest level of clinical inference, custody evaluators make observations about actual behaviors and test outcomes and processes occurring as part of the evaluation. Examples the authors provide include "She hung her head low and was often tearful" and "There is an elevation on scale two of the MMPI-2 [Minnesota Multiphasic Personality Inventory-2]."

At Level II, the custody evaluator begins to make simple inferences about Level I data. Examples the authors provide include "The father's style of parenting is very authoritarian" and "The mother's blue mood, the child's report of her chronic sadness, and her MMPI-2 elevations suggest that she is moderately depressed." In contrast, at Level III, the evaluator makes reference to custody-specific constructs, such as "parenting capacity" and "potential psychological risks of primary custody with this parent." The authors provide this example: "Multiple days away from the mother, this child's primary psychological parent, will be emotionally stressful." Finally, Level IV comprises the evaluator's custody-related recommendations about the ultimate issues, such as a proposed access schedule and physical custody. The authors direct evaluators to be circumspect in their evaluation, charging that specific empirical support may be lacking in making recommendations at Level IV. They and others (Karras & Berry, 1985; Melton, Petriala, Poythress, & Slobogin, 1997) recommend that custody evaluators avoid making specific recommendations about what a court "should" do. Ironically, specific recommendations are often viewed as what is most valuable about custody evaluations (Bow & Quinnell, 2004). Some argue instead that custody evaluators should provide recommendations, but that judges should view them as advisory in nature (Stahl, 2005).

If custody evaluators cannot be impartial, they should attempt to withdraw from a case, or at a minimum provide information to the court about factors that affect their impartiality. Custody evaluators also should avoid multiple relationships. In

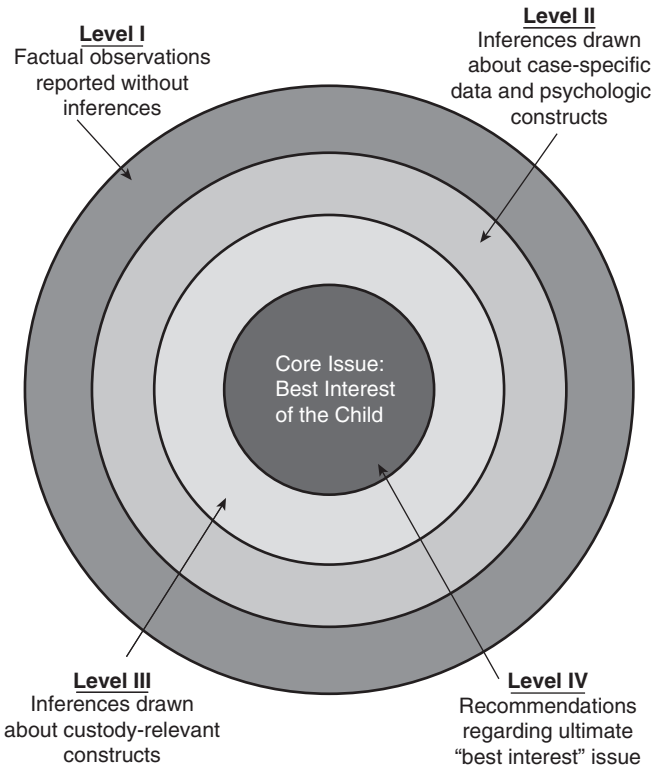

Figure 27.2 A model for directing clinical involvement in custody cases. *Source:* "Empirical and Ethical Problems with Custody Recommendations: A Call for Clinical Humility and Judicial Vigilance," by T. M. Tippins and J. P. Wittmann, 2005, *Family Court Review, 43*(2), pp. 193–222. Reprinted with permission.

cases where they have served in a therapeutic role, they should avoid providing custody evaluations. They may testify as a witness, although recommendations are generally to decline such a role unless ordered by the court to do so. During the custody evaluation, they should refrain from accepting any of the involved parties as therapy clients, and even after the evaluation is complete, they are urged to be cautious in agreeing to treat former participants of the custody evaluation.

Custody evaluators should have a thorough understanding of ethical guidelines for conducting custody evaluations. For example, APA guidelines specify that the custody evaluator obtain informed consent from each adult participant, including identifying the assessment tools and techniques that will be used. Informed consent also specifies fees, who will pay, and the limits of confidentiality, including what information will be disclosed. In conducting the evaluation, the psychologist should refrain from inappropriately interpreting the data and should use multiple methods of data collection.

The APA standards do not address many of the issues involved in child custody evaluations, and thus are considered aspirational in nature. Additionally, custody evaluations have a high risk for malpractice suits and ethical complaints (Kirkland &

Kirkland, 2001). Thus, the psychologist should follow closely the ethical guidelines and other risk management procedures. See Craig (2005) for other guidelines that may apply for custody evaluations.

Because of the nature of the work, custody evaluators should have competency in psychological assessment of adults, children, and families. They should be knowledgeable in child development, family development, adult and child psychopathology, and research on children and divorce. Further, they should be familiar with applicable legal standards, especially for their state or jurisdiction. Their work should reflect acceptable clinical and scientific methods. Their evaluation should be based on multiple methods, with important facts or opinions based on at least two sources of information.

Additionally, the psychologist should use standard and commonly used assessment techniques in custody evaluations. Measures that have been frequently used can be found in Bow and Quinnell (2001) and Hagen and Castagna (2001). The measure most frequently used by psychologists in evaluating child custody issues is the MMPI. This test is often used to assess parenting capacity and emotional adjustment of each party. It also allows for detecting impression management, or the degree to which individuals are consciously inflating or concealing various aspects of themselves. This test has been normed on child custody litigants (e.g., Strong, Greene, Hoppe, Johnston, & Olesin, 1999) and has been used to identify parents who attempt to alienate the child from the other parent (Siegel & Langford, 1998). Craig (2005) recommends that evidence from the MMPI should be explored further as part of the workup. However, Bradley (2004) warns of potential cultural biases in this instrument and others. The Millon Clinical Multiaxial Inventory (MCMI) is being used increasingly in child custody evaluations. There is some controversy over the use of the MCMI because it lacks established norms for differentiating clinical from nonclinical respondents (Butcher & Miller, 1999) and there are issues with scale interpretation. Craig recommends caution when using the MCMI for custody evaluations. Hysjulien, Wood, and Benjamin (1994) report the use of various instruments, including the Achenbach Child Behavior Checklist (McConaughy & Achenbach, 1996), FACES III (Edmon, Cole, & Howard, 1990), and the Parent Attachment Structured Interview (Roll, Lockwood, & Roll, 1981). Bradley recommends the Child Perception of Parent and Parent Awareness Skills Survey (Bricklin, 1995) and the Parental Discipline Techniques Self Report Instrument (Gardner, 1997).

SUMMARY

Parental divorce has the potential to create considerable risk for children, but there is large variation in how children respond to the experience. Key risk factors for maladjustment after divorce are economic hardship, interparental conflict, parental distress, and cumulative transitions and related life stress. Clinical psychologists may provide mediation or custody evaluation, both of which require a sophisticated knowledge of legal, ethical, and clinical issues. The work is painstaking and difficult, but offers help to families attempting to make plans for how children will be raised in the new family structure.

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CHAPTER 28

Child Maltreatment

Christine Wekerle, Harriet L. MacMillan, Eman Leung, and Ellen Jamieson

> It was a sunny day in a popular vacation spot, crowded with children and adults in the pool area, some swimming, some lying about on chairs, and some in conversation. A mom, lying face down, was oblivious to the world. A dad, sitting in a nearby chair, seemed quite uncomfortable. Perhaps he did not favor the heat. His daughter, about 12 years old, was restlessly standing in front of him. She approached and sat on his knee, moving from side to side. Finally, she got up, turned around and straddled him. The frankness of that straddle, his delay in responding to this movement, and my audible response of shock put this man in motion and he roughly pushed her off him. The child's face registered no emotional reaction, just a watchful blankness. The child left her dad and came in the pool beside me. I tried to make friendly conversation—"Hi." "The pool water is really warm!" "Do you like to swim?" The total lack of response from the child in any dimension indicating registering the prior event or registering my communication efforts led me to entertain several hypotheses: non-English speaker, hearing-impaired, cognitively delayed, or fearful of connecting to another (the first three could later be ruled out). After a little while, the father said words to the mother. He called his daughter. Child and father walked away together, leaving the mother behind who had remained frozen in her ideal tanning position.

Child maltreatment disturbs your peace. If it is clear, it sets you in motion. If it is unclear, it may nag, percolate, and unsettle you in the background workings of your mind. You struggle because the pieces of information do not align well and, likely, the child is not talking. For the safety of the child, you move carefully through ruling out more hypotheses than is typical. In child clinics, the assessment question may come as broadly as "What is going on?" and in person, your colleague seems at a loss. The clinician may observe or have reported such behaviors as public self-stimulation, "daydreaming" during a conversation, back arching from a light touch, aggression or emotional overreactivity at the slightest provocation, or selective mutism. As hyperactivity is a nonspecific symptom, relevant to Attention-Deficit/Hyperactivity Disorder, genetic syndrome disorders, children of newly clinically depressed mothers, and so on, maltreatment leaves scant symptom

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profiles. This is due to the possibility of maltreatment affecting every developmental domain—difficulties sustaining attention and learning, behavioral hyperactivity and aggression, depression, anxious clinging to or overreaching with friendly adults, nonnormative fears, and bizarre or nonnormative behaviors—potentially layered on top of preexisting problems.

Childhood maltreatment is no disorder. It is the result of adult action serving adult purposes or stemming from adult deficits. People who interact regularly with maltreated children may note that things seem "off." Clinical examples include a kindergarten child repeatedly referring to himself as "having the devil inside him," and teachers describing a child sibling pair as "weird" given their unusual behaviors (e.g., licking a classmate's hand) and another child as "Dr. Jekyll/Mr. Hyde" given his extreme behavioral and affective changes. For other children, their adaptation may not include individual signs, but may take on compulsive caregiving of younger siblings, sublimating their own distress. Other children may find avenues for more positive development by becoming essentially members of another family, be it via other family members, school and work relations, or in their neighborhood. Finally, for some children, symptomatology is not evident or may be delayed or very restricted in its expression, even in the context of high-impact trauma (e.g., witnessing one parent's murder by the other parent) or chronic maltreatment. Regardless of symptom expression, we have child protection laws acknowledging that it is not the child's responsibility to fend off violence from adults.

Professionals wrestle with the legal requirement to report suspected maltreatment and the ethical requirement to intervene and document concerns. Documentation is one way to establish a pattern of concerns, even if the initial report is not investigated or substantiated by child protective services (CPS). With clear injury patterns, indications of patterned behavior, and verbal disclosures, the clarity of reasonable *suspicion* of abuse, neglect, or risk increases. Professional hesitation regarding reporting and quantifying confidently "reasonable suspicion" in less than clear situations has reasonable grounds. These include concern about civil litigation challenging "good faith" reporting; the fact that the short-term and long-term child effects of CPS involvement have received little research in its 30 years of existence; the concern that the only CPS service offered will be investigation; and the puzzlement, given current clinic wait-lists, that most children are returned to the home within 6 months. Further, there is a community threshold for reportable abuse or neglect within the practical reality of limited resources.

Erring on the side of caution is a prudent approach when considering that U.S. figures of mortality from child abuse are likely underestimates (Herman-Giddens et al., 1999). Using North Carolina medical examiner data over a 10-year period, nearly 60% of fatal cases were recorded as due to battery, not child abuse specifically. The probable assailant was male (66%) and a biological parent (63%). It is generally accepted that much serious abuse and neglect go undetected by formal systems. The World Health Organization (WHO) estimates that 57,000 children die yearly from maltreatment fatalities (Krug, Mercy, Dahlberg, & Zwi, 2002). In the United States, falling is the most common reason given initially, with the child being found unresponsive as the next most prevalent (Collins & Nichols, 1999).

Most deaths occur when the mother and her unrelated boyfriend live in the home (Stiffman et al., 2002). Another large category is neglect deaths from starvation, exposure to heat or cold, and failure to safeguard. As a risk factor for childhood death, any protection system involvement for neglect or physical and/or sexual abuse dramatically increases mortality, especially for children under 1 year of age (Sabotta & Davis, 1992).

When maltreatment is repeated, we can only acknowledge all of our system failures, from public monitoring to front-line professionals (community recreation, faith-based institutions, education, health, substance abuse, mental health, hospital emergency, police, domestic violence shelters, etc.) to child protective services. Part of our citizenry social contract is to provide a safety net for expectant and at-risk parents, and government is the ultimate partner in providing leadership in this.

Most United Nations member states have committed to an action plan to uphold the child's human rights to be protected from abuse and to have the basic necessities of life provided. The U.S. surgeon general's "Healthy People 2010" targets specific reductions in rates of child abuse and related fatalities and research goals of understanding the magnitude of new cases and evaluating the impact of existing interventions. Child maltreatment is properly regarded as an issue of human rights, public health, mental and physical illness prevention, and economics. The remainder of this chapter addresses the definition and scope of the problem of maltreatment, using brief case descriptions to illustrate these. An overview of theoretical models and current knowledge on outcomes is presented. Finally, current gold standards in intervention are highlighted. We end with thoughts on future directions for research and practice and provide a summary for this chapter.

DEFINITION AND SCOPE OF THE PROBLEM OF CHILD MALTREATMENT

Barber and Sibert (2000, p. 743) note, "The initial recognition of the possibility of abuse is in some ways the most important step . . . and one should never walk alone" in the child protection process. Thus, the first step is to have maltreatment in mind as an active possible hypothesis to rule out, as well as to know local applicable laws and key consultation experts. U.S. figures (Federal Fiscal Year, October 2004–September 2005) place the annual substantiated rate of abuse or neglect at about 12.10/1,000 children. Most (75%) are first-time victims. The recurrence rate for substantiated abuse over 6 months is about 6% (U.S. Department of Health and Human Services [USDHHS], 2007). Canadian rates of entries into child protection, which may reflect substantiated high risk as well as substantiated discrete events, are about 22/1,000 children (Canadian Incidence Study; Trocmé et al., 2005). Consistent over recent years, neglect is the most common form presenting to CPS (62.8%), with psychological abuse the least frequent in the United States (7.1%) and sexual abuse the least frequent in Canada (3%). With the exception of sexual abuse, parents are the vast majority of perpetrators, with mothers somewhat higher than fathers (83.4% abuse by a parent acting alone or with another person; USDHHS, 2007). Dramatic gender differences exist only with sexual abuse, the majority

being female victims and the majority of perpetrators being male. Younger children show the highest victimization rate (16.1/1,000 children, birth to age 3), with decreasing rates across ages to teens. However, substantial levels of maltreatment do occur in adolescence (12–15 year-olds: rate of 10.2/1,000; 16–17 year-olds: rate of 6.2/1,000; USDHHS, 2007). It should be noted that victims in CPS have an estimated 6-month maltreatment recurrent rate of 94.6%, as determined by the 2006 Child and Family Services Reviews (USDHHS, 2007). Further, community survey data suggest that physical abuse is more than 40 times more prevalent and sexual abuse is more than 15 times more prevalent than official rates (Wekerle, Miller, Wolfe, & Spindel, 2006).

Neglect

Neglect is challenging to define as it covers a wide range of failure-to-care events. Neglect can be willful disregard of child needs or more passive acts of omission. Neglect examples include abandonment or desertion; failure to provide a home or a safe, sanitary shelter; supervisory neglect; neglect of personal hygiene; refusal or delay in providing normative health care (e.g., specific concern visits, such as ear infection, maintaining recommended immunization schedule, routine dental care); nutritional neglect; failure to meet standards for education (e.g., high absenteeism); and failure to meet special needs (e.g., failure to treat asthma from secondary smoke, failure to treat learning disorders with educational support). Exposure to physical hazards (e.g., exposed wiring, broken glass, nonsecured and loaded guns, illegal drugs, dangerous or unhygienic pets, asking child to perform dangerous activities, driving with child while intoxicated) is another area of concern. Neglect is a chronic condition with high recidivism, experiencing the greatest rate of placement in out-of-home care. In the United States, parents are perpetrators in 86.6% of cases. Neglect is more prevalent among younger caregivers, households with a primary male caregiver, and families with low socioeconomic status (Trocmé et al., 2005).

The U.S. case-level statistics indicate that neglect accounts for 60% of children receiving clinical services (USDHHS, 2007). The rate of cases of neglect reported to CPS is also around 60%, or 7.4/1,000 children (USDHHS, 2007). The most common form of investigated neglect is "failure to supervise leading to physical harm," where the child suffered or was at substantial risk of suffering physical harm because of the caregiver's failure to monitor and protect the child adequately. This form of neglect constitutes 37% of the substantiated forms of maltreatment (3.06/1,000 children) among new reports to CPS (Trocmé et al., 2005).

Research indicates that caregiver substance abuse is strongly linked to neglect (Takayam, Wolfe, & Coulter, 1998; Wekerle & Wall, 2002; Yampolskaya & Banks, 2006). Among a range of caregiver vulnerability factors, caregiver substance abuse was the single best predictor of neglect, although the composite of vulnerabilities was the best overall predictor (Wekerle, Wall, Leung, & Trocmé, 2006) after controlling for other child and parent characteristics. Other significant single predictors were socioeconomic disadvantage, caregiver history of childhood maltreatment, and caregiver lack of social support. Recent caregiver violent partnership interacted with total caregiver vulnerability, indicating an exacerbating impact, more

than doubling the likelihood of a substantiated versus nonsubstantiated neglect finding. Addressing caregiver vulnerabilities early on, when the family first comes to the attention of CPS, seems a critical move.

Some definitions of neglect also include emotional neglect (a marked indifference to children's need for affection, attention, and emotional support), as well as exposure to chronic or extreme spousal abuse or intimate partner violence (IPV). These subtypes can be summarized as involving three major forms of neglect: (1) lack of emotional support, affection, or both; (2) lack of protection from family conflict, violence, or both; and (3) lack of protection from community violence (Wekerle et al., 2006). A recent epidemiological survey of randomly selected teens on the active caseload of an urban center found that the teens reported higher prevalence for emotional neglect items than physical neglect items, as measured by the Childhood Trauma Questionnaire (Bernstein & Fink, 1998). Item examples include often or sometimes "did not have enough to eat" (37%) and often or sometimes "parent too drunk/high to care for them" (27%; Wekerle et al., 2006).

Case Illustration

Robert, age 4, was observed playing on the roof of his house by his neighbors, who called the child protective agency. When the social worker investigated the home situation, she found Robert and two other siblings, Nancy, age 5, and Alice, age 2, living in unsafe and filthy conditions in the care of their mother, age 30, and her boyfriend, who was the children's stepfather. The mother, Carol, reported that her partner, Frank, had a cocaine habit and used much of what she received from social assistance to buy drugs. He was physically and verbally abusive to her and threatened to hurt the children if she told anyone about the abuse that she was experiencing. They had been together for 1 year; previously, Carol had been with another man, the father of two of the three children, who had also been abusive to her. Carol had grown up in a home where she had been sexually abused by a maternal uncle; when she disclosed this to her mother, she was not believed, and so left home at the age of 15 years. She also witnessed violence between her parents. Carol dropped out of school in grade 10 and had hopes of eventually finishing her education, but was struggling to first look after the children. She acknowledged that she occasionally left the children locked in a room together while she went out to buy groceries; it was during one of these occasions that Robert had climbed out the window onto the roof.

Physical Abuse

According to the World Health Organization (WHO, 2006), physical abuse is defined as the intentional use of physical force against a child that results in, or has a high likelihood of resulting in, harm to the child's health, survival, development, or dignity. This includes exposure to such behaviors as hitting, beating, kicking, shaking, biting, strangling, scalding, burning, poisoning, and suffocating. Some physical violence against children in the home occurs in the context of parental discipline attempts in response to normative child behaviors such as constant crying,

whining, toilet-training issues, defiance, and sibling fighting. Closed-fisted strikes, striking with a hard object (belt, board, bat), and throwing the child into objects (furniture, wall) are extremes on a continuum of corporal punishment. Public debate continues about whether spanking constitutes a form of maltreatment; however, there is growing consensus that certain types of spanking—for example, of a teenager, or spanking resulting in injury, such as a bruise—is abusive. Multiple bruising is more consistent with abuse, with slap marks being common; evidence of severe physical injury describes a minority of substantiated physical abuse cases. In the United States, 16.6% of cases were physical abuse (about 2/1,000 children; USDHHS, 2007). In Canada, the rate of substantiated physical abuse (or deemed at high risk of) was 5/1,000 children: 41% were hit with a hand; 21% shaken, pushed, or thrown; 19% were hit with objects; and 8% were punched, kicked, or bitten (Trocmé et al., 2005).

The U.S. Developmental Victimization Survey (Finkelhor, Ormrod, Turner, & Hamby, 2005) found that over half of community children (age 2–17) surveyed experienced a physical assault in the past year. Poly-victimization (four or more different types of events) best described 18% of surveyed children (Finkelhor, Ormrod, & Turner, 2007). One state population survey found physical abuse in the form of kicking, slapping, or punching was most common among teenagers (Theodore et al., 2005). Also, shaking of children less than 2 years of age was reported by caregivers more often than previously recognized. It was estimated that for every child who sustains a serious medical injury as a result of shaking, an estimated 150 children may have been shaken yet were undetected. These authors noted that comparing their survey rate to those from the National Child Abuse and Neglect Data System places physical abuse at more than 40 times higher than the official state-level statistic. Bruising is suspicious for abuse when the marks occur are recent, when the bruising is over nonbony areas (ears, abdomen, anterior chest), and when bruise patterns capture the image of an object (e.g., a hand). Considerable force is needed to cause a bone fracture; this type of abuse is seen mostly in children 6 months and under (Barber & Sibert, 2000). Although most bruising is the result of physically abusive discipline, some actions are more sadistic and planful.

One type of physical abuse more likely to present in hospital emergency rooms and to hospital-based abuse and neglect specialty teams is Munchausen by proxy syndrome (MBPS), in which the caregiver (typically the mother) purposefully makes the child ill (e.g., induces vomiting, administers poison), fabricates the medical complaint and/or psychological symptoms, or subjects the child to medical testing repeatedly, including invasive procedures. Moving from expert to expert increases the amount of medical testing. Fabricated and induced illness is also a diagnostic term (Craft & Hall, 2004), as is fictitious disorder by proxy (*DSM-IV-TR;* American Psychological Association, 2004). These conditions are considered uncommon, although their incidence is unknown.

Although there is no clear understanding of parental motivation in these cases, it has been suggested that such behaviors result from a desire to be recognized as a caring parent, to gain positive adult attention from medical professionals, to prevent a partnership breakup, to avoid criminal culpability, and for secondary economic gain. Others may witness the illness crisis but not the initiation of the

medical event, yet the MBPS caregiver is present consistently. Girls and boys seem equally targeted. Younger children tend to be the focus, although MBPS can span childhood and include participation of a teenager in self-harm. The symptoms cease or decrease when the child is separated from the MBPS caregiver. In a study on male MPBS caregivers, perpetration of physical abuse (e.g., bone fractures) or sexual abuse emerged in the background data collection (Meadow, 1998). Fathers often themselves had Munchausen syndrome or a somatoform or factitious disorder; no father was in regular employment or had been in long-term employment, and about a third of fathers were receiving a disability allowance. Unlike MBPS mothers, fathers were noted by hospital staff as overbearing, aggressive, and quick-tempered. Other situations more likely to occur in MBPS families included unusual pet deaths, home fires, and formal complaints filed by a parent about the hospital.

In a British study that used covert video recordings of suspected MBPS in a hospital ($N = 39$ children; median age 9 months), evidence of abuse (suffocation, poisoning, arm breaking) was captured in the majority of cases. Risk of abuse extended to other children in these families. About 30% of siblings had died suddenly and unexpectedly (most were previously categorized as sudden infant deaths); abuse was documented in a further 36% (Southall, Plunkett, Banks, Falkov, & Samuels, 1997; see article for case examples of sequential observation notes from recordings). Compared with controls, the abused children were less likely to have been born prematurely and more likely to present with bleeding from the nose or mouth, and more of the abusive parents were diagnosed as having personality disorders. These parents were not deemed psychotic; they were observed to attack their children in the absence of challenging child behavioral conditions (e.g., when child was asleep). Maternal allegations of their own maltreatment or the presence of domestic violence were common. The likelihood of a problematic childhood is strengthened by the finding that over 80% of the mothers in one study showed insecure adult attachment of a dismissing or disorganized type, reflecting an undervaluing of close relationships and a frightened and frightening caregiver (Adshead & Bluglass, 2005). Although only a minority of these mothers scored in the personality disorder range, about half showed clinically elevated scores, mostly on Cluster B personality problems (*DSM-IV-TR;* American Psychiatric Association, 2000). Given the planful and persistent nature of this behavior and the harm to the children, the potential role of antisocial and dissociative parental tendencies and subdisorder level of clinical issues need to be studied.

In one follow-up study of maternal MBPS ($n = 17$) with coordinated intervention for the individual (offending) mother and family, the majority of children were functioning reasonably well in their biological family homes (Berg & Jones, 1999). Parents who persistently denied abuse and culpability or had a severe parental personality disorder were not considered suitable for treatment; in these cases, alternative care (fostering, adoption) was maintained. Despite the severity of the initial abuse, in other cases family reunification was achieved and maintained in the context of treatment with attachment-based psychotherapy, cognitive-behavior therapy, and systemic family therapy, along with in-depth assessments of child outcomes, intensive liaison with key professionals local to the family, and involvement of other family members and friends in the treatment process. One of the

13 treatment cases re-presented with MSBP. As these children ranged from 2.5 to 12 years at the 2-year follow-up mark, what remains unclear is the impact of developmental level on outcome and recidivism assessment. Full assessment of sibling outcomes and longer term follow-up are important research goals. For instance, when all siblings are considered, reabuse rates of 20% have been reported (Davis et al., 1998).

It should be noted that criminal and clinical approaches to MPBS are distinct, with intentionality difficult to confirm and more central to legal objectives (Pankratz, 2006). It is important to consider in MSBP, and indeed in all forms of maltreatment, that positive, appropriate, and affectionate parent behavior is likely to be observed or reported. Observational studies of parenting have clearly shown that aversive parenting and positive affect are independent (Solomon & George, 1999). In the vast majority of substantiated abuse cases, parents do declare that they love their child. Affective switching and reactivity are prominent in problematic personality styles, as well as in personality, mood, and anxiety disorders. Careful attention to context is critical in understanding the range of parental behaviors historically and currently demonstrated. Erring on the side of caution for child safety is a reasonable professional stance.

Case Illustration

Susan, age 9 years, was noted by her teacher to wear long sweaters even in summer. She presented as subdued and rarely interacted with the other children; in class, she seemed distracted at times. During an outdoor track and field day, when the children were in gym clothes with short sleeves, the teacher noted that Susan had two or three bruises brown to red in color on each forearm. When the teacher met with Susan and asked her about the bruises, Susan said that she had fallen off her bicycle. In fact, the injuries were not consistent with such an explanation; they fit the pattern of grab marks. The teacher was alert to the fact that these physical marks did not fit with a fall from a bicycle and contacted the local child protection agency.

Psychological and Emotional Abuse

According to the WHO (2006), emotional abuse involves both isolated incidents and a pattern of failure over time on the part of the caregiver to provide a developmentally appropriate and supportive environment. Acts in this category may have a high probability of damaging the child's physical or mental health, or physical, mental, spiritual, moral, or social development. Acts include the restriction of movement; a pattern of belittling, blaming, threatening, frightening, discriminating against, or ridiculing; and other nonphysical forms of rejection or hostile treatment. In the United States, as a single type, the psychological abuse rate was .9/1,000 children in 2004 (USDHHS, 2005). In Canada, the rate of substantiated emotional abuse was 3/1,000 children in 2003 (Trocmé et al., 2005). Emotional abuse is considered to be an undercurrent to other types of maltreatment.

Case Illustration

Tiffany was a 15-year-old waiting to go to court for her sexual abuse victimization by her biological father. As part of a victim impact statement, she spoke of the severe impact of his emotional abuse and how it was connected to the sexual abuse. She recalled that only two hugs in her life from her dad were nonsexualized. There was an unwritten rule that you were not allowed to close or lock the bathroom door when you were using it. She was given many household jobs, living with a single dad. When she failed to comply according to his view, she was required to eat her dinner from the dog bowl. When younger, she was required to kneel in a corner for long periods of time as a parental sanction. Her father played her compliance off her brother's, referring to who was his "best child" that day. Tiffany viewed her father as part of her biology, but never as a dad.

Exposure to Intimate Partner Violence

One type of child maltreatment that has been recognized only recently is exposure to IPV. Empirical studies examining this type of abuse first appeared in the 1980s (Kitzmann, Gaylord, Holt, & Kenny, 2003). The WHO defines IPV as

> any behaviour within an intimate relationship that causes physical, sexual or psychological harm to those in that relationship. It includes acts of physical aggression (slapping, hitting, kicking or beating), psychological abuse (intimidation, constant belittling or humiliation), forced sexual intercourse or any other controlling behaviour (isolating a person from family and friends, monitoring their movements and restricting access to information or assistance. (WHO Intimate Partner Violence and Alcohol Fact Sheet, page 1)

Most *reported* IPV is committed by men against women (Krug et al., 2002). Some surveys suggest that similar proportions of men and women experience IPV; however, abused women experience more physical and emotional impairment in relation to IPV than men (Archer, 2000; Tjaden & Thoennes, 2000). Violence against men by women and between same-sex partners is important for future research to consider at greater depth.

Two recent reviews underscore the broad range of health problems experienced by children exposed to IPV (Bair-Merritt, Blackstone, & Feudtner, 2006; Kitzmann et al., 2003). In a meta-analysis of 118 original studies, Kitzmann and colleagues examined the psychosocial effects of IPV; child outcomes included psychological, social, emotional and behavioral, and academic measures. About 63% of children who witnessed interparental violence were "doing more poorly" than the average child without such exposure. The authors emphasize that effect size did not depend on outcome type. Children who witnessed violence had levels of adjustment problems that were similar to those exposed to physical abuse. A second review that included 22 research articles examined the physical health outcomes of IPV exposure (Bair-Merritt et al., 2006). Exposure to IPV among children was associated with increased risk-taking behavior, such as alcohol abuse and risky sexual activities, during adolescence and adulthood.

There is substantial overlap between exposure to IPV and other types of maltreatment. In a review focusing on the co-occurrence of these exposures, Osofsky (2003) referred to several studies concluding that in homes where a woman is physically abused, 60% to 75% of children have also experienced physical abuse. Children residing with a mother exposed to physical abuse are at increased risk of sexual abuse, either by the mother's partner or a perpetrator outside the home (McCloskey, Figueredo, & Koss, 1995). Osofsky suggests that the co-occurrence of child maltreatment and IPV is important "from a risk and protective factors framework" since it may be more difficult to ensure adequate safety and protection in these families. As a general rule for all maltreatment concerns, it is essential when one subtype of child maltreatment is identified to consider the co-occurrence of other types of abuse and neglect. Research is clear on the predominance of severe male-to-female battering as opposed to female violence. However, child abuse and neglect is concerned with the negative impact on the child, irrespective of the sex of the primary perpetrator.

Case Illustration

Tyrone was working full time while his wife, Jan, was at home with their five young children. When Tyrone came home from work, he completely took over all child care activities. Jan was always finding fault with Tyrone and put him down in front of the children, calling him "fat," "useless," and "a loser," often within a string of expletives. The children, ranging in age from preschooler to preteen, were typically present during fights that often escalated to Jan's threatening to kill Tyrone and slapping, hitting, scratching, and pulling his hair. On several occasions she gave him a black eye, which was noted at his place of work. Tyrone always gave excuses for his marks, such as that he fell over something or the kids got carried away in rough-housing. The children would reenact these scenarios at school, causing concern among the teachers, and the youngest child would use inappropriate language, including swear words and "sex." Generally, Jan presented well outside the home for short durations, confining most of her explosiveness and inappropriate behaviors to the household, although she had a history of suicidal gestures. She alternated rage with a hopeless, helpless depression and would lock herself in her room for days. She had a history of suicidal gestures and had threatened self-harm in front of her children. She had no friends or social life outside of her family and had cut off all contact with her own and Tyrone's family. Eventually, Tyrone's boss confronted Tyrone on the repeated patterns of his injuries and harassing phone calls from Jan to his place of work. Tyrone's boss initiated intervention for Tyrone, which led to other system involvement.

Sexual Abuse

Sexual abuse is defined by the WHO as the involvement of a child in sexual activity that he or she does not fully comprehend, is unable to give informed consent to, or for which the child is not developmentally prepared, or else that violates the laws or social taboos of the society. Children can be sexually abused by both

adults and other children who, by virtue of their age or stage of development, are in a position of responsibility, trust, or power over the victim. Authors of a U.S. community survey study noted that the rate of childhood sexual abuse generated from their self-report data is more than 15 times higher than the official state-level statistic (Theodore et al., 2005). For 2004, 1.2/1,000 children was the official sexual abuse rate (USDHHS, 2005), keeping the same rate over 5 years. Three-quarters of perpetrators were friends or neighbors. Victims are predominantly in the age 12 and up category, although 9% are in the 4- to 7-year category and 2% are in the 3 and under category. In Canada, for 2003, the rate was less than 1/1,000 children (Trocmé et al., 2005).

Case Illustration

Scott, age 9, lived with his mother, Susan, who worked part time as a teacher; his father, Brant, a physician; and his younger brother, Stephen, age 7. The family got along well together with no major stressors, although Brant worked long hours. Scott was an excellent student attending grade 3 and loved all sports. The couple had been married 5 years before Scott was born; both children were planned pregnancies. The family did many activities together; one night a week, Susan attended an art class and Brant worked in the local walk-in clinic. The parents arranged to have a 16-year-old son of close friends, Bruce, as a babysitter. This arrangement appeared to work well for about 6 weeks, until Scott became increasingly resistant to his mother's attending art class. On two occasions, he said that he felt ill and expressed tearfully that he wanted her to stay home. When Susan questioned Scott about his reluctance at her going out, Scott disclosed that on two occasions, Bruce had made him "suck his penis." Bruce had told Scott that this was the latest thing at school, and all teenagers were doing it. Scott became tearful while telling his mother what had happened; he expressed confusion and not wanting to do this, but Bruce was someone that he and all the other kids looked up to, so he thought it must be okay.

MALTREATMENT-ASSOCIATED MALADJUSTMENT

Childhood maltreatment is not a psychiatric disorder but an exposure. It is associated with a range of impairment in areas of basic functioning (eating, sleep), cognition (attention, memory, learning, academic achievement), affective style (modulation, regulation, mood disorders), motivation, and relationships. The challenges in managing tendencies toward both avoidance and overreactivity can create secondary problems in somatization, substance abuse, behavioral risk taking, and aggression toward oneself and others. These problems can persist, yielding implications for partnering and parenting.

The cycle of violence is a clinical reality, particularly when IPV is included with abuse and neglect, but, empirically, is of modest predictive significance in many studies. Across types of maladjustment, a parent's history of maltreatment no more than doubled his or her risk of abusive parenting. This suggests that parental psychiatric status and socioeconomic disadvantage are more weighty, proximal predictors,

compared to a parent's own childhood history (Hurley, Chiodo, Leschied, & Whitehead, 2003; Renner & Slack, 2006; Sidebotham & Heron, 2006). In reviewing generational physical abuse (parent and offspring), most studies fell short methodologically, with the strongest study estimating a relative risk of 12.6 (Ertem, Leventhal, & Dobbs, 2000). In a longitudinal study that involved 224 former male victims of sexual abuse, who were identified based on a nationwide search of clinical and social service case records, 26 had subsequently committed sexual offenses, in most cases involving children outside their family (Salter et al., 2003). Thus, it would seem appropriate to consider a cycle of dysfunction and disadvantage, in addition to a cycle of violence.

Maltreatment is linked with lower school achievement (Wekerle & Wolfe, 2003), which has been accounted for by such factors as school engagement and social competence (e.g., Shonk & Cicchetti, 2001). One prospective, longitudinal study of substantiated abused and neglected children found that only 22% were deemed resilient in adulthood (e.g., no period of homelessness, consistent employment, and no juvenile or adult arrests), which significantly differed from matched controls (McGloin & Widom, 2001). As such, maltreatment may be one of the most modifiable barriers to living a "normal" life, including freedom from mental illness (DeBellis, 2001).

Psychiatric Disorders Associated with Childhood Maltreatment

Mood, anxiety, substance use, eating, and personality disorders in youth will be considered. Research in childhood maltreatment is uneven; most studies have focused on the areas of sexual and physical abuse, and on adults, mostly females. To provide an overview of the associated risk of the considered disorders, odds ratios have been compiled into a table on adult female functioning (Table 28.1).

Mood Disorders

Depressive symptoms have commonly been associated with childhood maltreatment (Danielson, de Arellano, Kilpatrick, Saunders, & Resnick, 2005; Gover, 2004; Meyerson, Long, Miranda, & Marx, 2002). As can be seen from Table 28.1, depression and anxiety problems far outstrip other maladjustment for female survivors of childhood maltreatment. In a community-based longitudinal study, individuals with a history of child abuse and neglect reported a two- to threefold increased likelihood of Dysthymic Disorder and Major Depressive Disorder during adolescence and young adulthood, as compared to nonabused individuals, even after statistically controlling for certain confounding variables (Brown, Cohen, Johnson, & Smailes, 1999). In a meta-analysis of child sexual abuse, victims exhibited significantly greater social withdrawal and internalizing symptoms, such as depressed mood, anhedonia (i.e., lack of pleasure in previously reinforcing activities, such as leisure, sex), and feelings of worthlessness and guilt (Kendall-Tackett et al., 1993). Other common emotional effects include guilt (i.e., feeling responsible for the abuse), helplessness and hopelessness, sleep disturbance, appetite disturbance, and low self-esteem. According to the *DSM-IV-TR* (APA, 2004), the symptoms of depression for children and adolescents are the same as those for adults. However, it is suggested that different symptoms may be more prominent at different ages.

Table 28.1 Adult (Female) Maladjustment Based on Childhood History of Maltreatment

Diagnoses/problems	Physical Abuse	Sexual Abuse	Emotional Abuse	Child Neglect
Anxiety	39.4 (33.8–45.1)	42.9 (35.3–50.4)		
Major Depressive Disorder	23.3 (18.2–28.3)	28.5 (20.8–36.2)	2.7 (2.3–3.2)	4.1 (1.9–8.8)
Alcohol abuse	9.9 (7.0–12.8)	10.3 (5.9–14.8)		1.65
Other substance use disorders	4.1 (2.7–5.5)	6.6 (2.5–10.7)		1.7 (1.5–2.0)
Posttraumatic Stress Disorder	1.03 (1.00–1.06)	10.2 (7.1–14.5)	5.67 (2.74–11.74)	3.5 (1.5–7.9)
Antisocial personality	4.1 (2.5–5.7)	4.7 (2.2–7.2)		1.32 (0.6–2.88)
Criminality	1.09 (1.01–1.19)	1.14 (1.02–1.27)	b = 2.06	1.7 (1.2– 2.3)
Suicidal ideation	2.77 (1.26–6.12)			
Suicide attempt	1.64 (0.63–4.28)		b = 0.57	3.03 (1.11–8.33)

Note: Odds ratios and confidence intervals are presented unless otherwise specified. *Source:* H. A. Bergen, G. Martin, A. S. Richardson, S. Allison, & L. Roeger (2003); A. C. Boudewyn & J. H. Liem (1995); K. Brewer-Smyth, A. W. Burgess, & J. Shults (2004); J. Brown, P. Cohen, J. G. Johnson, & E. M. Smailes (1999); D. P. Chapman, C. L. Whitfield, V. J. Felitti, S. R. Dube, V. J. Edwards, & R. F. Anda (2003); E. Evans, K. Hawton, & K. Rodham (2005); C. Evren, S. Kural, & D. Cakmak (2005); J. G. Johnson, P. Cohen, J. Brown, E. M. Smailes, & D. P. Bernstein (1999); P. R. Joyce, J. M. McKenzie, S. E. Luty, R. T. Mulder, J. D. Carter, P. F. Sullivan, et al. (2003); S. J. Kaplan, D. P. Covitz, S. Salzinger, F. Mandel, M. Weiner, & V. Labruna (1999); D. G. Kilpatrick, K. J. Ruggiero, R. Acierno, B. E. Saunders, H. S. Resnick, C. L. Best, et al. (2000); D. G. Kilpatrick, K. J. Ruggiero, R. Acierno, B. E. Saunders, H. S. Resnick, & C. L. Best (2003); J. B. Kingree, D. Phan, & M. Thomson (2003); D. E. Nelson, G. K. Higginson, & J. A. Grant-Worley (1995); C. S. Widom (1999).

For example, somatic complaints and irritable mood are especially common during childhood and adolescence. Sleep and appetite disturbances in children and adolescents, as in adults, may vary, with some patients oversleeping and/or overeating (hypersomnia and hyperphagia, respectively), while others present with difficulty falling or staying asleep and/or having no appetite.

Anxiety Disorders

General symptoms of anxiety, nightmares, inappropriate fears of certain places, and clinging to parents are commonly found among maltreated children (Giardino & Giardino, 2002; Kendall-Tackett et al., 1993). Child inpatients with histories of exposure to abuse were described by their parents as being significantly more afraid of being left alone with others, exhibiting more suspicion, and getting more upset when touched, as compared to nonabused inpatients (Kolko, Moser, & Weldy, 1988). Based on self-report, sexually abused children were significantly more anxious than both clinical and normal controls (Mannarino, Cohen, & Gregor, 1989). Physical abuse has been shown to increase the likelihood for any anxiety disorder, as well as for Generalized Anxiety Disorder (GAD), indicating excessive worries (Flisher et al., 1997).

The initial response to single-event trauma may be hyperarousal, notably, hyperactivity, disorganization, and disruption in maintaining routines. However, as trauma is repeated, the adaptation response can become more complicated by dissociation, affective disturbance, and prolonged gaps in historical memory. With

a single-event trauma, there is an emphasis on pathological fear and erroneous associations (people, object) with danger (Tolin & Foa, 2002). Most of the focus has been on the Posttraumatic Stress Disorder (PTSD) diagnosis, although subclinical symptom levels are important as they have been linked to significant functional impairment in youth (Carrion, Weems, Ray, & Reiss, 2002; Putnam, 1998). From age 8 to 10 years, PTSD reactions are more in line with adult diagnostic criteria, with females more likely to show PTSD-like responses and males to show aggression responses (Dyregrov & Yule, 2006).

Beyond the presence of a traumatic stressor, there are three main symptom clusters, with the *DSM* diagnostic requirement that criteria be met in each of the clusters. The three-cluster symptom model consists of (1) reexperiencing the trauma (e.g., flashbacks, nightmares); (2) avoidance of trauma-related cues and emotional numbing; and (3) problems with arousal (e.g., sleep onset, maintenance; exaggerated or diminished startle response). Some of these symptoms may manifest in other ways, including somatic symptoms (e.g., stomachache, headache to support escape or avoidance), self-destructiveness (e.g., cutting to support release from numbing), and hostility (e.g., lashing out under hyperarousal; sexual play as trauma-based reenactment).

In a clinical sample (most had multiple traumas and prior child protective services involvement), PTSD was commonly comorbid with other anxiety and mood disorders (Carrion et al., 2002). The most intense PTSD symptoms were irritability and anger, distressing dreams, and feelings of detachment from others. The most frequent symptoms were avoidance of trauma-related thoughts, feelings, and cues; distressing recollections; inability to recall important aspects of the trauma; distressing dreams; and difficulty concentrating. Full diagnosis for PTSD was predicted by severe and frequent hypervigilance, exaggerated startle, and detachment from others. The most common symptom cluster was reexperiencing or intrusions about the trauma. Clearly this symptom pattern of sleep and concentration difficulties must challenge the PTSD child and youth in school.

Similarly, in a study of randomly selected teens involved with the child protection system, reexperiencing was among the most common self-reported PTSD symptoms ("bad dreams or nightmares"; "remembering things that happened that I didn't like"). As is consistent with developmental level and abstract thinking about a shortened future, teens reported feeling a vague sense of impending doom ("feeling afraid something bad might happen"; Wekerle et al., 2006). In a study of teen inpatients, emotional abuse was the only consistent predictor of PTSD symptoms across all three clusters when considering all types of abuse and neglect simultaneously. Sexual abuse history also predicted the reexperiencing symptom cluster (Sullivan, Fehon, Andrew-Hyman, Lipschitz, & Grilo, 2006). Taken together, PTSD symptomatology may be as important to consider as the issue of meeting full criteria for the disorder. Given that the likelihood of meeting all three cluster criteria is greater in older children, current *DSM* criteria for PTSD may not be sensitive to the developmental manifestation in younger children. Symptomatology of PTSD has been shown empirically to mediate maltreatment and negative outcome (e.g., Risser, Hetzel-Riggin, Thomsen, & McCanne, 2006; Wekerle et al., 2001); therefore, targeting PTSD symptomatology is a fruitful intervention avenue.

Research has suggested that those with PTSD tend to have lengthier histories of maltreatment and/or broader exposure to violence (Deblinger, Steer, & Lippman, 1999; Linning & Kearney, 2004; Wood, Foy, Layne, Pynoos, & James, 2002). This may be why sexual abuse tends to predict PTSD diagnosis better than physical abuse, if both are present. Symptoms of PTSD may be associated with behavior inhibition problems (Ruchkin, Schwab-Stone, Koposov, Vermeiren, & Steiner, 2002). One study of youth from shelter facilities found that PTSD was comorbid with other mood or anxiety disorders, most notably dysthymia, major depression, GAD, panic with and without agoraphobia, and specific phobias (Linning & Kearney, 2004). In studies of adult PTSD, the majority of PTSD diagnoses occurred among females and in the context of other diagnoses, notably depression, where PTSD contributes to functional impairment across social, physical, emotional, and general health domains (Cloitre, Koenen, Gratz, & Jakupcak, 2002). The adult literature is investigating the tenet that chronic maltreatment is more likely to manifest as PTSD "personality" or complex PTSD, with characterological symptoms (impulsivity, aggression, self-destructiveness), pervasive despair and hopelessness, somatization symptoms, transient dissociative episodes, and difficulties in maintaining close relationships. In this way, PTSD has become a central organizing construct for personal and interpersonal style for the maltreated individual.

Overall, PTSD may be a relatively underassessed condition, whereas depression or anxiety may be noted rather than the more complex symptom picture. However, given that studies suggest reexperiencing is a frequent outcome, sleep patterns and distressing or intrusive dreams and thoughts should be considered in the context of coping with maltreatment. Further, with adolescence and young adulthood, the issue of personality involvement is raised.

Behavioral Problems and Disorders

Children exposed to maltreatment, especially those who are physically abused, show problems with aggression (Joshi, Daniolos, & Salpekar, 2004a, 2004b; Kendall-Tackett et al., 1993), and it is a problem that continues to emerge in adulthood (see Table 28.2); Also, see Botash and Ricci (2006). Social competence is further challenged by the tendency for aggression to co-occur with withdrawing behaviors in maltreated samples (e.g., Rogosch & Cicchetti, 1994). Physically abused youth show greater sensitivity to anger displays and threat cues (Pollak & Kistler, 2002, Pollak & Tolley-Schell, 2003) and more hostile attributions toward others (Dodge, Pettit, Bates, & Valente, 1995). During adolescence, physical abuse victims are more likely to engage in delinquency and risk-taking behaviors (Giardino & Giardino, 2002). Prevalence studies of disruptive behavior disorders among physically abused youth, as compared to those who are nonabused, reveal high rates of Conduct Disorder (CD) and Oppositional Defiant Disorder (Flisher et al., 1997; Pelcovitz et al., 2000; Pelcovitz, Kaplan, Goldenberg, Mandel, Lehane, & Guarrera, 1994). Both physically abused males and females are 9 times as likely to meet criteria for CD, even after controlling for potential mediating factors (e.g., parental psychopathology, lack of family cohesion, and lack of parental support; Kaplan et al., 1998). However, other research suggests that *females,* rather than

Table 28.2 Signs of Maltreatment

Type of Abuse	Signs
Physical Abuse	*Bruising of soft tissue (e.g., head, neck, trunk, arms):* → Pattern bruising in the shape of objects or hand marks. → Imprint of large, multiple, clustered bruises; bruises at different stages of healing (multiple color bruising). *Burns:* → Normally occur on hands, feet, and genitalia. → Burn patterns: • Inflicted burns are symmetrical when forced. • Accidental burns are irregularly shaped. → Cigarette burns: • Inflicted cigarette burns appear 7 to 10 mm round and have deep central crater. • Accidental cigarette burns appear oval in shape. → Scald burn (with hot liquid) is a common cause of childhood burns, both accidental and inflicted: • Accidental burns with hot liquids are irregularly shaped, with patches and splash marks. • Immersion burn injury occurs when a child is dunked in hot liquid; it can leave a stocking- or glove-like burn pattern found on the hands or feet. *Bites:* → Human bites cover a larger surface area than animal bites and tend to tear or crush (3 cm or greater space between canines indicates the bite was most likely by an adult). *Other:* → Inadequately explained bone fractures (X-ray evidence of history of multiple fractures). → Brain injuries from blunt head trauma: • Shaken baby syndrome: cerebral contusions, subdural hemorrhages, injury to cervical spinal cord, rib fractures; may present with seizures, decreased consciousness, respiratory difficulty, irritability, lethargy, vomiting, and sleep apnea.
Emotional Abuse	Unexplained physical symptoms/complaints with no medical basis. Sleep disturbances. Extreme dependence. Inability to trust. Feelings of shame and guilt. Child's own sense of his or her potential is limited. Child fails to thrive. Withdrawal. Poor self-esteem.
Neglect	Child's inability to adequately grow and develop to or above the 3rd percentile in height and weight, with no organic reason. Malnutrition, poor growth, consistent hunger, distended stomach. Untidy appearance, inappropriate dress, poor hygiene. Pale or listless appearance. Repeated or unattended lice. Recurring and unexplained patterns of minor injuries. Aches, pains, migraine, gastrointestinal problems.

(*continued*)

Table 28.2 (*Continued*)

Type of Abuse	Signs
Sexual abuse	Trauma to genital or anal area: →Abrasions or bruising of the genitalia. →Abdominal, genital, or rectal pain. →Genital or rectal discharge; unexplained sexually transmitted infection. Findings sometimes seen following sexual abuse but also other causes: History and other investigation is important in diagnosing abuse. → Frequent urinary tract or yeast infections.
Exposure to Intimate Partner Violence	Increased risk of physical harm or injury due to proximity to an act of family violence. Headaches. Stomach pains (e.g., ulcers). Difficulty sleeping/insomnia. Difficulty with memory. Hearing and articulation problems. Anxiety-related increases of physical issues (asthma, tics, eczema, epileptic fits, enuresis/encopresis).

Note: This table is an expanded version of Cheng, Munn, Jack, & MacMillan, 2006 based on American Academy of Pediatrics (n.d.); American Humane Association (2006); E. D. Bariciak, A. C. Plint, I. Gaboury, & S. Bennett (2003); J. Bays & D. Chadwick (1993); W. Bernet, W. Ayres, J. E. Dunne, E. Benedek, G. A. Bernstein, E. Bryant, et al. (1997); S. E. Chaney (2000); C. Doyle (1997); K. M. Drach, J. Wientzen, & L. R. Ricci (2001); J. Emans, E. Woods, N. Flagg, & A. Freeman (May 1987); E. Evans, K. Hawton, & K. Rodham (2005); J. Fantuzzo, R. Boruch, A. Beriama, M. Atkins, & S. Marcus (1997); J. W. Fantuzzo & C. U. Lindquist (1989); M. E. Herman-Giddens, G. Brown, S. Verbiest, P. J. Carlson, E. G. Hooten, E. Howell, et al. (1999); V. M. Herrera & L. McCloskey (2003); G. Hornor (2005); N. Kellogg & The Committee on Child Abuse & Neglect (2005, August); W. J. King, M. MacKay, & A. Sirnick (2003); J. R. Kolbo, E. H. Blakely, & D. Engleman (1996); L. Kos & T. Shwayder (2006); G. W. Lambie (2005); S. Maguire, M. K. Mann, J. Sibert, & A. Kemp (2005); B. W. Mayer & P. Burns (2000); S. S. Mudd & J. S. Findlay (2004); K. Nimkink & P. K. Kleinman (2001); *Ontario Association of Children's Aid Societies* (n.d.); V. J. Palusci, E. O. Cox, T. A. Cyrus, S. W. Heartwell, F. E. Vandervort, & E. S. Pott (1999); M. D. Peck & D. Priolo-Kapel (2002); C. Powell (2003); A. Slep & R. Heyman (2006, August); R. Wright, H. Mitchell, C. M. Visness, S. Cohen, J. Stout, R. Evans, et al. (2004, April).

males, are at high risk for CD when physically abused (MacMillan et al., 1999) or when experiencing multiple forms of maltreatment (Romano, Zoccolillo, & Paquette, 2006). Similarly, E. Nelson and colleagues (2002) found an increased risk for the development of CD and Antisocial Personality Disorder among women who report histories of childhood sexual abuse. When compared to physical abuse and exposure to marital violence, childhood sexual abuse appeared to be the strongest predictor of self-reported violent and nonviolent criminal behavior among females (Herrera & McCloskey, 2003). Epidemiological work suggests that type of maltreatment exposure is not specific in explaining the increased risk of conduct problems (MacMillan et al., 2001).

Substance Use Disorders

There has been greater recognition of the overlap between substance abuse and maltreatment, both in terms of childhood history and in perpetrators of maltreatment (Wekerle & Wall, 2002). Both physical and sexual abuse are linked with a greater

likelihood of early-onset smoking, drinking, illicit drug use, and injection drug use and nonmedical prescription drug abuse compared to nonmaltreated children and adolescents (De Marco & Phelan, 2004; Kessler et al., 1996; National Clearinghouse on Child Abuse and Neglect, 2004; Raghavan & Kingston, 2006; Vermeiren, Schwab-Stone, Deboutte, Leckman, & Ruchkin, 2003). In a study using a nationally representative probability sample, investigators found that having a history of physical or sexual abuse doubled the risk of alcohol, marijuana, or other hard drug abuse or dependence in the past year. The victimizations reported by this sample occurred prior to nonexperimental drug use, suggesting that victimization is a risk factor for later abuse or dependence (Kilpatrick et al., 2000). Further, sexual abuse history directly predicts both earlier age of substance abuse and higher reported rates of lifetime traumatic events among adults, raising the issue of an interplay between substance use and revictimization (Raghavan & Kingston, 2006).

Drug and alcohol use may function as a coping mechanism for increased stress brought on by childhood abuse. Alcohol and other illegal drug use may reduce symptoms of hyperarousal and unpleasant emotions and produce emotional numbing or euphoria (Cohen, Mannarino, Zhitova, & Capone, 2003; Kilpatrick et al., 2000). The frequent comorbidity between PTSD and substance use is consistent with a self-medication hypothesis (Cohen et al., 2003; Stewart & Israeli, 2002).

Youth with a history of maltreatment may enter service systems through a number of doors. Aarons et al. (2001) found high rates of lifetime prevalence of substance use disorders (SUDs) among youth from child welfare (19%), mental health (24%), and juvenile justice (41%) systems, whereas comparable community rates are less than 10%. Although child welfare youth seemed to be less vulnerable than those from other entry points, this study did not investigate the association between child maltreatment across service site and SUDs. This study is important, however, as child welfare youth are not often included as a population for epidemiological study.

Eating Disorders

Elevated risk for eating disorders or weight problems among teens reporting sexual abuse or physical neglect has been noted, above and beyond the risk from parental psychopathology, childhood eating disorders, and difficult temperaments (Johnson, Cohen, Kasen, & Brook, 2002). Sexually abused teens also appear to report significantly higher levels of distress about their weight more often than nonabused adolescents (Wonderlich et al., 2000). Women diagnosed with Bulimia Nervosa and those who have recovered for at least 1 year reported significantly higher levels of physical and psychological abuse (Rorty, Yeager, & Rossotto, 1994).

Personality Disorders

An interesting prelude to personality research is the finding based on story retelling tasks that maltreated children display from an early age (e.g., 3.5 years) affective splitting, and that greater negative emotionality is evidenced thereafter. Most nonmaltreated individuals would develop a positivity bias, being able to see the positives in themselves, others, and situations (Ayoub et al., 2006). Thus, it is not a surprise that childhood abuse has been linked with an increased risk of personality

disorders (Battle et al., 2004). Specifically, documented and self-reported physical abuse is significantly associated with Antisocial Personality Disorder, even after controlling for previous sexual abuse, neglect, and characteristics of other personality disorders. When controlling for these mediating factors, childhood sexual abuse was significantly associated with Borderline Personality Disorder (Johnson, Cohen, Brown, Smailes, & Bernstein, 1999). Borderline Personality Disorder has been frequently cited as the personality disorder most commonly associated with sexual abuse (Battle et al., 2004; Zanarini et al., 1997).

Suicidal and Self-Harm Behaviors

A history of childhood maltreatment is a significant risk factor for suicide (Giardino & Giardino, 2002; Wozencraft, Wagner, & Pellegrin, 1991). For instance, a Turkish study of teenagers revealed that the rate of suicide attempt for those with a history of childhood maltreatment was 7.6 times higher than for those without such a history (Zoroglu et al., 2003). Autopsy studies of youth have found an association between physical abuse and suicide (Kaplan, Pelcovitz, Salzinger, Mendel, & Wiener, 1997; Riggs, Alario, & McHorney, 1990). These findings are consistent with prospective longitudinal community studies (Brown et al., 1999; Johnson et al., 2002). Johnson and colleagues (2002) found an increased risk of suicide attempts in early adulthood with a history of maltreatment, even after adjusting for demographic characteristics and psychiatric symptoms and disorders in childhood and early adolescence. It is suggested that interpersonal skills may be the linking factor, with the maltreated individual experiencing greater social isolation and/or antagonistic interactions with others, which in turn adds risk for suicidal behavior. A study reviewing the risk factors for youth suicide reported that youth with a history of childhood sexual abuse had an increased risk for suicide up to about 12 times that of youth with no reported history of abuse (Beautrais, 2000). Individuals with histories of sexual abuse who report suicidal ideation tended to be older at the time of abuse and to have remained in their home after abuse was detected (Wozencraft et al., 1991). The relationship between childhood sexual abuse and suicidality was greatly reduced, but not eliminated, after controlling for a wide range of potentially confounding factors (Fergusson & Mullen, 1999). Thus, the increased risk for suicide from maltreatment may be partly accounted for by other factors.

A significant portion of the empirical literature demonstrating the association between maltreatment and *nonsuicidal* self-injurious behavior has focused on adults (e.g., van der Kolk, Perry, & Herman, 1991; Wiederman, Sansone, & Sansone, 1999). For example, 62% of subjects who self-mutilate reported being sexually or physically abused at a young age (Favazza & Conterio, 1989). However, a number of studies specifically focusing on childhood maltreatment and adolescent self-injurious behavior exist as well. In one inpatient adolescent sample, those in the self-mutilation group reported histories of physical and sexual abuse more often than those in the control group. However, when all types of maltreatment were controlled simultaneously, only sexual abuse and emotional neglect appeared to be significant predictors of self-mutilation (Lipschitz et al., 1999). Similarly, more sexually abused adolescents reported deliberate self-harm, as compared to the

non-sexually abused adolescents. In this study, 30% of the abused teenagers reported deliberately harming themselves five or more times (Martin, Bergen, Richardson, Roeger, & Allison, 2004). Zoroglu and colleagues (2003) found a 27-fold increase in self-mutilation in adolescents reporting a history of maltreatment compared to those who did not. It is advised that clinicians working with maltreated populations be mindful of, and regularly assess, potential risk for self-injurious behavior.

Dissociation

Dissociation is considered a psychological avoidance strategy, or an unconscious attempt to become emotionally numb via altering one's level of conscious self-awareness. To accomplish an emotionally disconnected state, one's emotions, thoughts, and behavioral experiences are separated from one another (Giardino & Giardino, 2002). In children and adolescents, dissociation can take the form of a trance-like state (e.g., abruptly distracted, sudden change in head orientation in midconversation) or manifest as a sudden change in behavior (e.g., one moment cooperative and the next moment oppositional or aggressive). Typically, activities during the dissociative episode are not remembered during nondissociative states. Dissociation may manifest as functional amnesias for sets of behaviors, a sense of depersonalization or derealization (e.g., problems with tracking of own activities in real time; sense of detachment from others; sense of being outside of oneself; repressing pain sensations in parts of the body), and experiences in identity alterations (e.g., true self versus projected, public self; chameleon-like personality with different interactants, pretending to be someone else; Putnam, 1996). Dissociation is thought to be a key contributor to the maltreated child's fragmented sense of self. As early as the preschool years, physically and sexually abused children exhibit signs of dissociation, although this is not as apparent for neglected children (e.g., Macfie, Cicchetti, & Toth, 2001). There is evidence for differential relationships between dissociation and physical and sexual abuse (Joshi et al., 2004a, 2004b; Giardino & Giardino, 2002). Using a longitudinal, prospective design, Kisiel and Lyons (2001) evaluated the prevalence of dissociation among physically and sexually abused youth using self-report and parent report questionnaires. Whereas sexually abused children reported significantly higher levels of dissociation compared to nonabused subjects, physically abused children did not. However, parental report of child dissociative symptoms indicated a significant association with both physical and sexual abuse histories. Dissociation may be more likely when maltreatment involves physical violence and multiple perpetrators (Trickett, Noll, Reiffman, & Putnam, 2001).

Asymptomatic Victims

Maltreatment covers a wide range of behaviors, from single events to chronic exposure. Within types there is also substantial variability; for example, fondling is the most frequently substantiated form of sexual abuse, as compared to penetration acts (Trocmé et al., 2005). The impact of maltreatment can manifest in myriad ways, suggesting that there is no single symptom profile associated with maltreatment (Kendall-Tackett et al., 1993).

A meta-analysis of sexual abuse studies found that between 21% and 49% of the subjects appeared asymptomatic at the time of assessment (Kendall-Tackett et al., 1993). McLeer and colleagues (McLeer, Deblinger, Atkins, Foa, & Rolphe, 1988) suggest that psychiatric sequelae can develop well after the time of the actual abuse (McLeer et al., 1988). Such a "sleeper effect" (Fergusson & Mullen, 1999) may emerge at developmentally sensitive times; for example, in teen dating, maltreatment-relevant threat cues may be heightened (e.g., physical proximity, physical contact, push-pull affect, sexual response; Wekerle et al., 2001). Maltreatment characteristics predict symptomatology, such as child age at onset, the type of abuse, the frequency and severity of abuse, the relationship between the victim and the perpetrator, the family environment, and family psychiatric history (Boney-McCoy & Finkelhor, 1995; Flisher et al., 1997; National Clearinghouse on Child Abuse and Neglect, 2004). The likelihood of having adjustment problems after experiencing abuse increases if the abuse is severe, violent, and lengthy in duration, and if limited family support is available (Boney-McCoy & Finkelhor, 1995; Fergusson & Mullen, 1999). Elements such as coping style, attachment status, and perceptions of self-identification as abused may influence distress and adjustment levels (Wekerle & Wolfe, 2003; Wekerle et al., 2001). For assessment of adjustment and trauma-related issues, see Cohen, Mannarino, Murray, and Igelman (2006), Kolko (1996), Wekerle et al. (2006), and V. Wolfe (in press).

Continued research on child and adolescent adjustment postmaltreatment is needed; the majority of work to date has been conducted on adults, mostly with retrospective reporting on maltreatment, with some important twin and prospective longitudinal studies. As Table 28.1 demonstrates, the importance of childhood maltreatment is underscored when considering adult disorders and impairment. As with adults, a similar pattern in impairment domains is shown for youth: substance abuse, antisociality and aggression, anxiety, depression, PTSD, and suicidality. We are forced to acknowledge that it is hard for many abused individuals to cope and to stay alive.

THEORETICAL MODELS OF MALTREATMENT-ASSOCIATED MALADJUSTMENT

Social Learning and Attachment Theories

Caregiver vulnerabilities of various kinds have been linked to compromised parenting, for example, anxiety and depression, substance abuse, aggression and antisociality, social support, and poor physical health (e.g., Egeland, Yates, Appleyard, & van Dulmen, 2002; Eiden & Leonard, 2000; Lyons-Ruth, Lyubchik, Wolfe, & Bronfman, 2002; MacMillan & Munn, 2001; Pine et al., 2005; Rogosch, Cicchetti, & Toth, 2004; Shea, Walsh, MacMillan, & Steiner, 2005). Developmental research shows that inadequately responsive parenting increases children's risk for problems with future relationships, managing emotions, self-efficacy, and violence (Dube et al., 2001; Kim & Cicchetti, 2003; National Institute of Child Health and Human Development Early Child Care Research Network, 2004).

According to social learning theory, children form a schema for how stress is regulated in a close relationship through their experience of interacting with their caregivers. Longitudinal research has shown, for example, that maternal depression typically precedes child aggression and hyperactivity, as well as affective disturbances (Elgar, Curtis, McGrath, Waschbusch, & Stewart, 2003). Early abusive relationships inform relationship schema, wherein maladaptive stress–maladaptive coping is learned via modeling and experience with reinforcement schedules. For example, attack can end the stress from the interactant by causing harm and shutting down conflict or interaction, promoting the interactant to acquiesce, appease, and remove himself or herself from the immediate environment. Maltreated individuals may maintain developed dysfunctional coping partly as the result of their dysfunctional relationship strategy.

According to attachment theory, the end goal of the early caregiving relationship is to protect us from harm and enhance our exploratory behavior. The confidence in the protection that subsequently develops serves as both the source for stress regulation and a model of stress regulation to be internalized. This internal model is carried into future situations as a central means to regulate stress (Goldberg, 2001). An abusive relationship turns the source of predictable comfort and first choice for support into a source of fear, confusion, and hesitancy, thereby compromising the development and practice of stress-regulating strategy (Lyons-Ruth & Jacobvitz, 1999). Thus, from both social learning and attachment perspectives, relationship functioning represents a key ongoing area of adaptational challenge for maltreated children (Rutter & Sroufe, 2000). This heightened vulnerability has been tied, theoretically, to holding a conceptualization of relationships that are organized along power abusive, victim-victimizer dimensions (e.g., Crittenden & Ainsworth, 1989; Crittenden & Claussen, 2002), as attachment theory (Bowlby, 1970, 1972, 1980, 1982) advances that both sides of a relationship are learned.

Maltreated victims may, by implication, have a more developed (and more generally applied) conceptualization of their victimizer. Defensive processes (e.g., including perceptual information that is consistent with one's relationship schema and excluding perceptual information that is inconsistent with the relationship schema) are thought to interfere with being open to new input, thereby increasing the likelihood of schema maintenance. Maltreated teens would be expected to select partners who would contribute to the maintenance of distressed relationships (Crittenden & Ainsworth, 1989).

Much recent evidence has accumulated to show that maltreated youth, relative to their nonmaltreated counterparts, exhibit indicators of self-system dysfunction, including poorer self-representations, self-esteem, and ego control (e.g., Cicchetti & Rogosch, 1997; Toth, Cicchetti, Macfie, & Emde, 1997). Such disorganization has been linked with higher trauma-related symptoms, specifically dissociation, in adolescents who had childhood histories of maltreatment and other trauma (Carlson, 1998). The PTSD trauma symptomatology is also functionally related to relationship disturbance, including adolescent dating violence (Wolfe, 1999; Wekerle et al., 2001). Dissociation, for example, may contribute to asynchrony in and abrupt withdrawal from interacting. Hypervigilance may lead to an agitated, highly reactive response in interactions.

Developmental Traumatology Theory

According to the theory of developmental traumatology, early exposure to the intense traumatic stress of maltreatment, compounded by the underdevelopment of an optimally functioning stress-regulating system, will result in an overreactive stress system (Cicchetti & Walker, 2001; DeBellis, 2001; DeBellis & Putnam, 1994). Such a long-term effect of maltreatment experience on stress and adverse health outcomes has been linked to the impairment of the hypothalamic-pituitary-adrenal axis, the psychobiological stress system that is sensitive to the effect of early environment. In turn, this compromises other stress-related systems, including the immune, neurotransmitter, and sympathetic nervous systems, and may be associated with structural changes to the brain (DeBellis, 2001). Nowhere is this convergence of malfunctions across difference stress systems more apparent than in PTSD.

In the developmental traumatology model, PTSD symptomatology is seen as the key mediator of negative outcomes. Such symptoms may persist and overwhelm youth who are disadvantaged by their stress response mechanisms, thereby interfering with their sense of self-continuity and ability to cope, leading to maladjustment (Bradley, 2000; Chandler & Lalonde, 1998). Due to maltreatment-related cues directly leading to aversive experience, escape-avoidance mechanisms may be reinforced. Substance abuse and self-harming may be efforts to self-medicate in response to PTSD symptoms via reducing negative affect, numbness, dissociation, tension, and hypervigilance or enhancing positive affect. However, progression to more harmful behaviors (e.g., abuse of cocaine, opiates) may exacerbate affect problems and increase victimization risk (e.g., Ginzler, Cochran, Comenech-Rodriguez, Cauce, & Whitbeck, 2003).

Personality Trait Model

Personality research posits that people differ on a set of underlying hypothetical dimensions, and these represent stable dispositions that are inherent in the individual, manifest early in life, and generalize across a variety of situations. Research on maltreating caregivers has highlighted the role of the caregiver's personality in limiting his or her ability to cope with stress associated with parenting, thus predisposing the individual to abusive behavior (Marziali, Damianakis, & Trocmé, 2003). To further understand the role of the caregiver's personality in the perpetration of child maltreatment and spousal violence, the model suggests that certain personality traits are a risk factor for child and partner abuse, operating singly or in conjunction with other risk factors such as substance abuse (Flett & Hewitt, 2002). In addition, it has been shown that personality may serve as a mediator in the relation between the individual's own history of childhood maltreatment and the perpetration of partner violence (Ornduff, Kelsey, & O'Leary, 2001). This indicates that the development of personality is shaped by the abusive experience during early childhood and consolidated into a stable behavioral and cognitive tendency that individuals bring to close relationships, perpetuating the cycle of violence and abuse (see also Pickering, Farmer, & McGuffin, 2004).

It has also been suggested that childhood maltreatment and individual personality traits contribute synergistically to adverse behavioral and health outcomes. For

example, Blackson, Tarter, and Mezzich (1996) showed that the adverse effect of parental discipline by substance-abusing fathers on their child's behavioral adjustment is exacerbated by the child's difficult temperament, an indicator of poor affect regulation. Similarly, Trocki and Caetano (2003) have shown that children who were exposed to family violence were more likely to develop depressive symptoms and abuse substances during adulthood if they also have a difficult temperament. Child response to trauma may be moderated by preexisting psychopathology, attributional or coping style, and temperament (Feiring, Taska, & Lewis, 2002). The mechanism underlying such interaction between personality and maltreatment can be conceptualized along gene-environment interaction effects. Because of its effect on the monoamine (e.g., serotonin, norepinephrine, and dopamine) level at the prefrontal cortex, different polymorphisms of genes associated with the monoamine system have been linked to individual differences in temperamental affect regulation, measured in terms of maternal report as well as behavioral observation (Benjamin, Ebstein, & Belmaker, 2002). In a now classic study, Caspi and colleagues (2002) found that male children who were maltreated were more likely to exhibit externalizing behavior if they had the low monoamine oxidase A activity genotype. Similarly, Kaufman et al. (2004) demonstrated that maltreated children who receive low social support and carry the short allele of the serotonin transporter gene promoter polymorphism were twice as likely to develop depression, even compared to their nonmaltreated counterparts who carry the same polymorphism. Maltreated children who carry the short allele but receive high social support had only minimal increase in the depression score, indicating the buffering effect of a positive environment. Recently, in validating the gene-environment interaction effect observed among maltreated children, Kim-Cohen and colleagues (2006) examined two different types of gene-environment correlation that may have potentially confounded the interaction effect: (1) The abusive parent may have transmitted to the child both an adverse environment and a genetic susceptibility for behavioral maladjustment, and (2) children who carry the risk genotype may behave in ways that are more challenging to an underskilled, stress-vulnerable parent. To address these issues, the researchers tested for the former by controlling for maternal antisocial personality and for the latter by assessing whether exposure to child maltreatment is predicted by the child's genetic polymorphism. Results indicate that even after controlling for the two types of gene-environment correlation, the interaction effect remains significant, indicating that children who carry the polymorphism that calls for low monoamine oxidase A activity were more vulnerable to the exposure to physical abuse.

One point to consider in reviewing these studies of gene-environment interaction in maltreated samples is the potential difference in maltreatment experience even between genetically identical twins. The shared environment may not be shared in a chronically chaotic and dysfunctional household. Such family dynamics as role reversals (e.g., parentification), scapegoating (e.g., targeting one child), and changing household rules with changing caregivers, adults, and siblings may lead to very different daily realities, as well as different maltreatment events, when considered in detail. Further, rather than the actual event, it may be the perception of the experience of maltreatment that differs, even when children are similarly maltreated

by the same perpetrator. For example, one child may view himself or herself as a victim, whereas the other embraces spirituality, such as forgiveness of the perpetrator, seeing a greater value in individual suffering, believing he or she has a special relationship with God, and so on. In our clinical experience, one mother identified that God had allowed such substantial abuse when she was a child since he knew she was "strong enough" to "take it" and survive. In research, youth differed in their acknowledgment of exposure to physical abuse even when endorsing physical abuse items with clear injury outcomes (Wekerle et al., 2001). Capturing the maltreatment event, the perception of the maltreatment, and the "theory" or story (or lack thereof) created to understand the maltreatment may be important aspects of a shared environment.

SCREENING FOR CHILD MALTREATMENT

During the 1980s and 1990s, attempts were made to screen individuals to determine risk for committing child maltreatment (Brayden, MacLean, Bonfiglio, & Altemeier, 1993; Caldwell, Bogat, & Davidson, 1988; Center on Child Abuse Prevention Research, 1996; Kotelchuck, 1982; McCurdy, 1995). One study examined the association between staff assessment of a participant's risk for committing abuse based on personal and familial characteristics such as economic difficulties, and scores on the Child Abuse Potential Inventory, a self-report measure (McCurdy, 1995). The predictive validity of this strategy for subsequent abuse, however, was not determined. Two investigations followed families prospectively and measured risk of future maltreatment (Brayden et al., 1993; Center on Child Abuse Prevention Research, 1996). Such risk approaches have shown only limited ability to predict future maltreatment. For example, the interview used by Brayden and colleagues had a positive predictive value of 6.6% and a sensitivity of 55.6% for physical abuse. The main factor that precludes use of child maltreatment screening approaches continues to be the unacceptably high false-positive rate (Caldwell et al., 1988; Kotelchuck, 1982); individuals are identified as potentially at risk for abusing a child and yet do not go on to do so. Certain child, parental, and social factors are important to consider in any assessment of child maltreatment, including evaluations of parents or caregivers to determine their ability to care for a child (often referred to as a parenting capacity assessment).

Table 28.2 outlines physical and other signs linked with an increased risk for occurrence of child maltreatment. Although the presence of these variables increases the risk of maltreatment, it is important to recognize that abuse or neglect can occur without any risk indicators present. At the same time, many individuals and families with one or more of these indicators present are not experiencing child maltreatment. Adverse events for children in the home tend to cluster together (multiple forms of maltreatment, parental alcohol and drug abuse, IPV, parental marital discord, divorce or separation, parental criminality; Dong et al., 2004) and may also be reflected in the community context (e.g., safe, affordable, low crime, high in parenting resource areas).

There is a strong tendency among both researchers and child protection agencies to focus exclusively on mothers when considering issues of child neglect. Yet

some research suggests that the involvement of a father (biological or the mother's partner) protects against abuse and neglect (Biller & Solomon, 1986; Dubowitz et al., 2000; Egeland, Jacobvitz, & Sroufe, 1988; Quinton, Rutter, & Liddle, 1984; Turcotte, Dubeau, Bolté, & Paquette, 2001). Other research, however, found that male partners were rarely a significant source of support to neglecting mothers (Polansky, Chamlers, Buttenwieser, & Williams, 1981). The research suggests that an analysis of the family type leads to a better understanding of the problem of neglect (Mayer, Dufour, Lavergne, Girard, & Trocmé, 2003). This underscores both the challenge and the importance of conducting a careful history with all family members. The assessment of the child in multiple locales (at home during family routine times, such as meals; at school; in service centers) and in multiple contexts (separating and returning to parent, response to strange adults, bonding with teacher and school staff) may be helpful. It is essential that all forms of maltreatment be considered as simultaneous working hypotheses for the clinician in order to capture possible polyvictimization (Finkelhor et al., 2007).

Another issue that has arisen more recently is screening women for exposure to IPV. Citing the prevalence, consequences, and costs associated with IPV (Campbell et al., 2002; Ulrich et al., 2003), some organizations advocate screening as a key aspect of identification and response; the American Medical Association (1992), the Family Violence Prevention Fund, and the American Academy of Pediatrics, among others, have recommended universal screening to their members (Bair-Merritt, Mollen, Yau, & Fein, 2006). Taket and colleagues (2003, p. 674) have asserted that "routine inquiry is the only way to increase the proportion of women who disclose abuse and who may benefit from intervention." These recommendations disregard the fact that no evidence exists regarding the possible harm associated with universal screening.

Those relying on evidence-based approaches favor caution. Systematic reviews by the Canadian Task Force on Preventive Health Care, the U.S. Preventive Services Task Force, and Ramsay and colleagues (MacMillan & Wathen, 2003; Nelson, Nygren, McInerney, & Klein, 2004; Ramsay, Richardson, Carter, Davidson, & Feder, 2002; U.S. Preventive Services Task Force, 2004; Wathen & MacMillan, 2003) have concluded that there is insufficient evidence to recommend universal screening for IPV. Efforts have been made to assist health care providers to use the best available evidence to address the issue of IPV with their clients (Ferris, 2004; Liebschutz, Frayne, & Saxe, 2003; MacMillan & Wathen, 2003a, 2003b; Rhodes & Levinson, 2003). The key question, however, is not whether to screen universally; the issue is the lack of evidence for the effectiveness of interventions for IPV. Identification is only the first step.

ASSESSMENT OF CHILD MALTREATMENT

In most regions in North America, it is the responsibility of child protection agencies to carry out investigative interviews with children and family members when child maltreatment is suspected. Although there is no evidence to support general screening of children for child maltreatment, it is important to include questions about exposure to the four main types of child abuse and neglect (physical, sexual,

and emotional abuse and neglect) in any diagnostic assessment of a child that involves emotional or behavioral problems (some clinicians separate exposure to IPV as a type of maltreatment, while others include it as a form of emotional abuse).

Clinicians must ask clients directly whether they have been abused. A history of victimization form that focuses on the behavioral aspects of maltreatment may be an appropriate prelude to the clinical interview for age-appropriate children. Examples of appropriate forms are the Juvenile Victimization Questionnaire (Hamby & Finkelhor, 2004), Childhood Trauma Questionnaire (Bernstein & Fink, 1998), and Childhood Experience of Violence Questionnaire (Walsh, MacMillan, Trocmé, Dudziuk, & Boyle, 2000). The use of behavioral referents and the avoidance of labels are critical and need to be contextualized (e.g., discipline, relationships). Such paper-and-pencil or computer-administered questionnaires may be helpful because they do not require individuals to self-label and conceptualize their experiences as abuse or neglect. An initial family interview is a good time to explain the goals, perceptions of concerns, expected procedures, and ethical issues (i.e., harm to self, harm to others, child abuse and neglect) of assessment. In maltreating families, it is important to understand who in the family is the person in power so that the clinician can support child empowerment. Going directly to the child may lead to more conflict for the child if the perception is that a child broke an adult's rule. Research shows that most children do not tell.

Children will most typically indicate their maltreatment behaviorally rather than in terms of verbal disclosures. Children will not respond with disclosures readily to the caregiver unless there is a "talking" and warm relationship with a reliable caregiver. For example, in one case, a child did not disclose sexual abuse when directly queried by an older sibling witnessing the child's distress after an incident had occurred with a neighbor. In another example, three male siblings did not disclose bizarre punishment (i.e., hanging a child upside down over an apartment balcony to elicit compliance) by their mother's boyfriend. In this case, the mother herself reported in a family session with the boyfriend present that she was physically abusive to her children (e.g., punching child in the face, knocking out child's teeth). Also present in the family assessment session was the maternal grandmother, who declared a "no tears" rule for this extended family. In all the discussions of violence, no distress was expressed by any family member. Thus, it is important for the clinician to maintain openness to disclosures of violence from a variety of family members and in a variety of presentational forms.

In the event of a disclosure, the maltreatment utterances and clinician questions and responses should be recorded verbatim in writing, in real time, and dated and signed. Clinicians need to be prepared to relay their concerns in terms of observed and reported behavior or signs, the sequential actions leading to the discovery, incongruence and improbabilities in explaining physical signs (see Table 28.2), the presence of risk indicators, as well as the basic facts (alleged perpetrator name, relationship, address, child level of fear of current danger or risk of victimization, knowledge of other victims and/or contact with other children, whether others were told about the abuse or neglect). The person who receives the information directly should act on the report or suspicion forthwith and should not engage the alleged

perpetrator or other caregiver in the report. Such limits to confidentiality should be clearly conveyed prior to clinical intervention.

All clinicians involved in the care of children should be alert to entertaining a hypothesis of maltreatment. A general principle to assist in identifying abuse injuries is when the history is not consistent with the physical findings. Even when the client is an adult or a couple, the best interests of the child remain the first priority. As a helping professional, the clinician has entered into the social contract that makes protection of children a priority over family privacy and unification or adult treatment goals. Discipline-specific codes of ethics and legal statutes exist to ensure child health and safety. The duty to report overrides most privileged information, including with police and religious authorities. Beyond familiarity with applicable laws, clinicians may consult local CPS to discuss the situation and obtain guidance anonymously. Abuse or neglect does not have to be proven for a report to be made; "reasonable" or "probable" grounds for suspicion are required. With the best interests of the child in mind, reports are made in "good faith," and malicious reporting is rare. Failure to report carries with it penalties such as fines, disciplinary action from the professional licensing body, civil suits, and prosecution by authorities. For further details, see texts by Wekerle et al. (2006), Reese & Ludwig (2001), and an article by Sirotnak, Grigsby, and Krugman (2004).

A helpful approach to understanding the context of symptomatology is to query caregivers and youths about a typical day. This allows the clinician to understand who does what generally and how the day proceeds, including levels of child self-care and sibling responsibilities, eating times, hygiene, affection displays, and discipline routines. It may be helpful to compare the child's perspective with that of siblings and caregivers. Examples of broad questions that provide opportunities to ask about child maltreatment are: What things do you worry about? What happens in your family when someone gets in trouble? Has anyone ever touched you in a way that made you feel uncomfortable or upset? Has anyone ever made you do something sexually you did not want to do? It is generally not appropriate to ask detailed questions about a disclosure of maltreatment because in many jurisdictions, a child protection agency worker needs to obtain the disclosure. Children should not be subjected to investigative interviews by clinicians without specialized training in this area. Maltreatment victims hope for an empathic response, reassuring them that telling was the right thing to do, allowing them to express their feelings and thoughts about the disclosure; the clinician should convey positive messages but not promises that cannot be kept. Helpful messages include conveying that the children did not do anything to provoke or deserve their maltreatment, that abuse and neglect are against the law, that you are sorry that they have had to go through this or suffer so, that they are not alone, and that other children have experienced it. The child should be monitored and supported postdisclosure with a known professional (i.e., CPS staff, family doctor). In cases where abuse is suspected but not substantiated by CPS investigation, the risk of subsequent reports and substantiation is high. As with many clinical phenomena, a good predictor of future behavior is past behavior.

TREATMENT AND PREVENTION OF CHILD MALTREATMENT

There are several promising and effective interventions on different aspects of maltreatment effects and violence prevention (for a review of the evidence base and available interventions, see J. Cohen et al., 2006). Recent infant-parent psychotherapy reports indicate success in altering the insecure attachment status of infants (about 66% were abused and/or neglected in the first year of life), as compared to low-income normative controls. There were substantial increases in secure attachment, and disorganized attachment was prevalent among the treatment-as-usual community maltreated infants (Cicchetti, Rogosch, & Toth, 2006). Certain interventions appear to hold promise for neglected children and families: Group play training and resilient peer-mediated treatment for withdrawn preschoolers may improve social behavior; therapeutic day treatment may improve self-esteem; and multisystemic therapy and parent training may improve parent-child relations, reduce parenting problems, and decrease family social problems (National Clearinghouse on Family Violence, 2006).

Currently, the two most researched and supported interventions are home visitation models (HVM) for younger children and trauma-focused cognitive-behavioral therapy (TF-CBT) for preschool and school-age children and adolescents. Research studies are under way to assess effectiveness in community settings, including child protection populations. Further research is needed in terms of entry into these interventions. Both HVM and TF-CBT are very promising approaches to support child protection services work in the area of referring families whom CPS does not serve but who are reported to the system, if families are interested and deemed appropriate. For example, a dating violence prevention program, the Youth Relationships Project, was found to significantly reduce the self-reported trauma symptomatology and dating violence among CPS and at-risk youth over a 2-year postintervention, as compared to youth receiving typical agency service (Wolfe et al., 1996).

Home Visitation Models

The prevalence and consequences of maltreatment have prompted many efforts in primary prevention in the past 2 decades. The U.S. Advisory Board on Child Abuse and Neglect identified child maltreatment as a national emergency in 1991 and promoted an expansion of Hawaii Healthy Start, a program of paraprofessional home visitation that had shown some evidence of effectiveness in nonrandomized trials (Duggan et al., 1999). Healthy Families America (Daro & Harding, 1999) was modeled after Hawaii Healthy Start and disseminated nationally. Currently, Health Families America programs are running in 430 locations in the United States and Canada.

In the past 10 years, randomized controlled trials have been conducted on Hawaii Healthy Start and Healthy Families America in New York State, San Diego, and Alaska. The trials have found significant effects on some domains of self-reported parenting behaviors (including abusive behaviors) and beliefs. The Alaskan study showed positive effects on some aspects of the child's environment, child

development, and parent-reported child behavior at age 2, but the San Diego trial showed that the effect on child development was not sustained. The programs showed no effect on official reports of child abuse or neglect, childhood injuries, or other objective measures that are proxies for maltreatment (Olds, 2006). In 2004, the Washington State Institute for Public Policy estimated that Healthy Families America has incurred a $4,500 (U.S.) per family loss in investment.

The 1991 home visitation recommendations of the Advisory Board were based on evidence from research on the Nurse-Family Partnership Program (NFP; Olds, Henderson, Chamberlin, & Tatelbaum, 1986), a substantively different program from that offered by Hawaii Healthy Start and Healthy Families America. The NFP is a program of prenatal and infancy home visiting by nurses (not paraprofessionals) for low-income mothers having their first child. Nurses follow detailed visit-by-visit guidelines in their efforts to (a) help women improve the outcomes of pregnancy by promoting healthy prenatal behaviors, (b) improve child health and development by promoting parents' competent care of their children, and (c) enhance parents' life-course development by encouraging pregnancy planning and parents' education and work. The nurses help families make use of needed health and human services and attempt to involve other family members and friends in the pregnancy, birth, and early care of the child. The NFP has a schedule of visits beginning before the end of the second trimester in pregnancy, as follows: weekly for the first four visits, every 2 weeks for the remainder of the pregnancy, every week for 6 weeks in the postpartum period, every 2 weeks until the infant is 21 months old, then once a month for the next 3 months until the child is 2 years old.

The NFP has been tested in three randomized controlled trials over 3 decades. The first trial was conducted in Elmira, New York ($N = 400$, 89% White). The second, conducted in Memphis, Tennessee ($N = 1139$, 92% Black), was designed to determine if the results from Elmira could be replicated when a health department delivered the NFP and when it served low-income Blacks in an urban area (Kitzman, Cole, Yoos, & Olds, 1997). The Denver trial ($N = 735$, 45% Hispanic) was designed to determine whether the NFP might be delivered effectively by paraprofessionals who shared many of the characteristics of the families they visited (Olds et al., 2002). The Denver trial also included a randomly assigned group visited by nurses and a control group. At randomization and in subsequent assessments, treatment and control groups were essentially equivalent in terms of demographics, and high rates of sample retention were maintained in each trial.

In each of the trials, women were randomized to receive either home visiting during pregnancy and in the first 2 years of their children's lives or comparison services that consisted of some combination of free transportation for prenatal and well child care and sensory and developmental screening and referral of the child for potential problems. All three studies employed a variety of data sources, including maternal interviews, developmental tests of the child, and medical and administrative data. The randomized controlled trials fell somewhere between efficacy and effectiveness trials, in that the samples had few restrictions, nearly entire populations of low-income first-time mothers were registered in the communities where the trials were conducted, and the interventions were conducted in collaboration with local agencies (Olds, 2002).

In Elmira, the intervention reduced rates of state-verified reports of child abuse and neglect. A 15-year follow-up showed that this effect was increased: the nurse-visited group had 48% fewer verified reports of maltreatment than did the control group, and among mothers who were at higher risk, 77% fewer verified reports compared to control group counterparts (Olds et al., 1997). However, rates of maltreatment found in official records in Memphis and Denver were too low in the first 2 years of children's lives to be reliable outcomes. Rates in Elmira were about 3 times higher than in Memphis and Denver, and 6 to 7 times greater for high-risk families. Child protection service records typically pick up only a fraction of actual maltreatment; therefore, the results in Elmira may be an underestimate of the effect. As well, a surveillance bias could have been operating in the study; maltreatment could be detected at a greater rate in nurse-visited families because the nurses are mandated reporters.

Because of the low rates of substantiated reports of maltreatment in Memphis, the research used childhood injuries reported in the medical record as a proxy (Kitzman et al., 1997). Program effects on injuries were similar to the pattern in Elmira; program effects were enhanced among children born to mothers with low psychological resources. In Denver, the health care delivery system made it impossible to trace children's utilization patterns reliably, so infants' early emotional expressions (which are associated with child abuse and neglect) were measured. Nurse-visited 6-month-olds born to mothers with low psychological resources had higher rates of observed emotional vitality and lower rates of emotional vulnerability than their control group counterparts. As the children matured, the program produced effects on language and executive functioning consistent with better prenatal and caregivng environments.

The Elmira follow-up showed that the program effect on child maltreatment decreased as the level of IPV increased, although the program had no impact on the incidence of IPV in this trial. The moderating effect of IPV was specific to child maltreatment and did not attenuate program effects on any other outcomes (Eckenrode et al., 2000).

It is noteworthy that nurse-visited women in the Denver trial reported significantly less IPV from partners during the 6-month interval prior to the 4-year interview than those in the control group (6.9% versus 13.6%, $p = .05$; Olds et al., 2004). Future research should be focused on this interplay between IPV and child maltreatment and efforts made to enhance IPV-effective components for these families.

The best evidence for prevention of physical abuse and neglect in high-risk groups comes from the nurse home visitation program developed by Olds and colleagues in 2007. MacMillan and colleagues examined the effectiveness of an intensive program of nurse home visitation in preventing recidivism of physical abuse and neglect among CPS-involved families referred because of at least one child in the family experiencing either physical abuse or neglect. Families were randomly assigned to a control group, which received CPS standard service such as risk assessment, parenting education, or referral, or the intervention group, which was provided standard CPS service plus a 2-year program of nurse home visitation. Both the control and the intervention group were followed up annually

over 3 years. Results showed no statistically significant reduction in recurrence of either child physical abuse or neglect; the recidivism rate for both intervention and control groups was high. The only promising finding was that among those families whose involvement with child protection lasted less than 3 months, there was a reduction in recurrence of physical abuse but not neglect. This suggests the possibility that early intervention with families involved with CPS might lead to less recurrence of child physical abuse. While the NFP has been shown to be effective in preventing maltreatment and associated outcomes, the trials by Olds and colleagues and MacMillan et al. (2005) have substantial differences. The NFP (Olds et al., 2007) targeted high-risk first-time mothers and began prenatally, whereas the study by MacMillan and colleagues (2005) involved a much broader range of families. In the NFP, the nurses assisted mothers prenatally, and then after the birth of the infant worked closely with mothers in helping them understand the needs of their infants. Although the original rationale for assessing a program of nurse home visitation with a sample of families involved with the CPS was based on the positive findings of the NFP, it was structured very differently from the NFP. The trial was designed to evaluate an intervention that paralleled the in-home services being offered by CPS in general, but at a more intensive level than was available in the community. The negative findings, though disappointing, emphasize that randomized, controlled clinical trials in the CPS population should be a priority for research. Disseminating findings from strong methodological studies should be a priority for CPS practice and policy, regardless of whether the outcome was positive or negative. Further, there remains an urgent need for rigorous evaluation of the effectiveness of parenting programs that are specifically designed to treat physical abuse and neglect, either independently or as part of broader packages of care. Studies need to measure outcome broadly to assess effects on child abuse and neglect per the legal definitions (CPS reports) as well as objective measures of parenting (e.g., home behavioral observations). A recent systematic review indicates treatment effectiveness of some programs on associated aspects of parenting; currently, the evidence on reduced maltreatment rates is limited for treatment compared to prevention with at-risk samples (Barlow, Johnston, Kendrick, Polnay, & Stewart-Brown, 2006).

Clearly, the effects seen in the NFP were not observed in the paraprofessional visiting model of Hawaii Healthy Start and Healthy Families America. The Denver trial of the NFP showed that nurses produce a larger and broader range of beneficial effects in, among others, infant caregivng and language development versus paraprofessionals. However, a one-size-fits-all model is not effective. The NFP model showed substantial limitations when interparental violence was present; the positive findings in reduction of child maltreatment were not maintained under such conditions (Eckenrode et al., 2000). This may be the best option for high-risk, young mothers prenatally or beginning their parenting career. Program implementation and fidelity are critical. In 1996, the U.S. Justice Department recommended replication of the NFP in community settings. This led to the creation of the NFP National Service, which develops and maintains procedures for ensuring faithful reproduction of the program's essential elements as it is replicated in new communities. The NFP is currently operating in over 170 sites, serving 20,000 families per

year in 250 counties in the United States. The Washington State Institute and the RAND Corporation estimate that the NFP saves $17,000 (U.S.) for every family served.

Trauma-Focused Cognitive Behavior Therapy

When the diagnosis of PTSD was first formulated in 1980, it was not believed to be relevant to children and adolescents. Child stress reactions were considered short-lived (Masten, Best, & Garmezy, 1990). The empirical basis for such views, however, consisted mostly of reports from parents and teachers, who may underreport child distress (Handford, Mayes, Mattison, & Humphrey, 1986). As the assessment moved to asking children and adolescents themselves about their thoughts, feelings, memories, and emotions following trauma, debilitating and long-lasting PTSD symptoms were noted (Dalgleish, Meiser-Stedman, & Smith, 2005). Pediatric PTSD symptomatology is consistent with children behaviorally enacting their emotional reaction to trauma (Dyregrov & Yule, 2006). For example, it was observed that young children show less emotional numbing and have more problems in reporting avoidance; they require rather complex cognitive introspection.

Consequently, the need for a thorough assessment following exposure to trauma is crucial to the implementation of age-appropriate and developmentally sensitive interventions to capture the intense feelings of shock, threat, terror, horror, or helplessness. In a recent review (Strand, Sarmiento, & Pasquale, 2005), assessment and screening tools currently available to researchers and practitioners were divided into four categories in terms of how trauma is constructed: those that screen for (1) both a history of exposure to traumatic events and the presence of symptoms of trauma, (2) only a history of exposure, (3) symptoms of PTSD or dissociation, and (4) symptoms other than or in addition to PTSD. Alternately, Perrin, Smith, and Yule (2000) categorize these instruments into four different modes: (1) *DSM*-criteria-based semi-structured interview, (2) parental interview, (3) unstructured child interview, and (4) child self-report. In evaluating these instruments, Strand et al. argued that most instruments are not designed to be age-specific, raising questions about the influence of age on outcome. Specifically, only 8 of the 33 measures available are designed for children 10 years or younger, and fewer of these are geared for the very young child. In addition, Perrin et al. point out that since many of the *DSM* PTSD criteria are not developmentally sensitive, existing semi-structured interviews may have only limited usefulness for young children. Further, because children are better reporters of internalizing symptoms and parents are more accurate reporters of externalizing or behavioral symptoms (Greenwald & Rubin, 1999), only instruments that collect information from both sources can provide a balanced account. Very few self-report measures were developed specifically to assess child PTSD. Instead, the majority of these instruments are adapted from adult questionnaires with little normative child data available (see McNally, 1991; Nader, 1997). Clearly, there is a need to develop instruments that measure more developmentally sensitive child PTSD.

The treatment, however, has centered on Trauma-Focused Cognitive Behavior Therapy (TF-CBT), which can be used to target trauma deriving from all types

of maltreatment, although it was initially developed for sexually abused children (Berliner, 2005; Cohen & Mannarino, 1996, 1998; Cohen, Mannarino, & Deblinger, 2006; Cohen, Mannarino, & Knudsen, 2005; Deblinger, Stauffer, & Steer, 2001; Stovall-McClough & Dozier, 2004). A systematic review of outcome studies on sexually abused children concludes that some outcomes have been more successfully demonstrated than others (e.g., decreases in child depression, improvements in parenting practices), and some outcomes have yet to be assessed. Overall, more controlled outcomes studies on various populations are encouraged (Macdonald, Higgins, & Ramchandani, 2006).

Treatment proceeds along 12 to 16 weekly sessions. In addition to the recently published manual (Cohen et al., 2006), free Web-based training is available (www.musc.edu/tfcbt/). TF-CBT has demonstrated efficacy for multiply traumatized children, maintaining treatment gains at 1-year follow-up (Cohen, Deblinger, Mannarino, & Steer, 2004; Deblinger, Mannarino, Cohen, & Steer, 2006). While not direct targets of treatment, parental PTSD and depression symptoms have been reduced with TF-CBT, further enhancing the likelihood of adaptive parental support. As is the case with most protocols, where additional intervention is deemed necessary, referral is provided. This is seen as particularly important when parental psychiatric issues are interfering with parenting capacity.

A key issue with TF-CBT is to engage a nonoffending caregiver as an active agent of change, providing the daily continuity needed to support the child's new learning (relaxation, leisure skills) and ability to talk openly about the maltreatment (emotions vocabulary, assertive communication) and to give the child confidence in the caregiver as a protector (caregiver affect management, parent training for child behavior management). Further, relaxation skills can be widely applied by the child and supported by the parent to address sleep, hyperarousal, and somatic issues. For the child, graduated exposure to the traumatizing maltreatment, addressing the phobic response to trauma memories and cues, is addressed in the creation of a narrative. This achieves the goal of decoupling maltreatment cues from debilitating anxiety. The child's story of the maltreatment can be in pictorial, audio, written, or typed form and is developed over time and at the child's pace. The practice of the cognitive components, addressing cognitive errors (e.g., dwelling on the worst possibilities, self-blame for maltreatment) via thought-stopping, positive self-talk, and practice applying the cognitive triangle (linking feeling, thoughts, behaviors), is achieved in therapist-child discussions during rereading and reworking of the narrative in a manner consistent with the CBT gains. Opportunities for corrective relationship experiences are available in individual sessions with the therapist and ongoing supportive work with respect to parenting. In this way, TF-CBT addresses the common relationship issues in maltreatment: impaired trust, social incompetence, poor relationship problem-solving skills, problems in sustaining relationships, and hypersensitivity in interactions. Another key maltreatment issue, affect regulation, is addressed via skill development in self-soothing, distress tolerance, and managing negative affect (sadness, fear, anger, anxiety). Risk behaviors (sexualized, aggressive, unsafe) are addressed in skill-building components.

The flow of TF-CBT sessions moves from assessment to assessment feedback, parallel child and adult individual sessions, and conjoint child-parent sessions.

During assessment, the central aspect of TF-CBT (i.e., the trauma narrative) as a skill is evaluated and practiced in a nonthreatening context. Children share their feelings, thoughts, and bodily sensations about a positive experience or favorite activity. The TF-CBT components form the acronym PRACTICE, reflecting the specific session goals across time: Psychoeducation and parenting skills, Relaxation, Affective modulation, Cognitive coping and processing, Trauma narrative, In vivo mastery of trauma reminders, Conjoint parent-child sessions, Enhancing future safety and development. Although caregiver involvement is optimal, significant PTSD symptom improvement has been documented with child-only work. Clinicians are advised to utilize their clinical judgment in determining the initial focus of treatment, which may need to be crisis management, reinstitution of the child's routine, and child behavior stabilization. Clinicians should assess for fragile states, as creating the trauma narrative could worsen any suicidal ideation or substance abuse. An alternative treatment strategy based on dialectical behavior therapy (Linehan, 1993) that targets suicidal risk and other multiproblem situations, particularly for adolescents, developed by Miller, Rathus, and Linehan (2007), is described in Wekerle et al. (2006).

SUMMARY

Child maltreatment is a serious challenge to adaptation and development across domains and across the life span. Children who experience multiple and/or chronic exposure to maltreatment events adapt less well to normative challenges in life. When maltreatment effects are evident, development is compromised as misses in early skills weaken the foundation for more complex skill attainment. In particular, maltreatment appears to carry an additional cognitive load for the individual such that academic performance is compromised, information integration is slower, affective splitting is more likely, emotion-laden and self-referential communication is more difficult, and negative biases in emotionality, attributions, and behavior (harm to self, harm to others) are more likely. Particular disorder vulnerabilities occur in the domains of mood, anxiety, substance abuse, aggression, eating, and personality. Further, survival is challenged with suicidal ideation and attempts, and suicide is a real possibility. Relationship functioning and self-preservation are consistent themes, with greater insecurities and disorganization in attachment, more problematic peer relationships, and greater likelihood of dating and intimate partner violence. Consistent with theoretical formulations, PTSD symptomatology appears to be central, at least for female victims. TF-CBT is a first line of treatment for those with impairing levels of symptomatology. The bulk of the data on maltreatment, though, continues to urge us toward violence prevention and early intervention. The NFP for first-time socially disadvantaged mothers shows the best evidence for prevention of maltreatment and associated outcomes such as injuries. However, once the line of abuse and official system involvement has been crossed, developing and testing approaches for parents continue to be pressing research questions.

Abuse and neglect of children are entirely unnecessary adult actions that can leave much damage to address in their wake. Even so, most maltreated individuals do not

go on to maltreat their partners and children. Many maltreated individuals achieve high functionality in some domains, but may continue to struggle in others (e.g., romantic relationships). Given the social contract that children's human rights be ensured and protected, research must continue to be productive in understanding the full experience of maltreatment for the victim, support the broad uptake of effective programs while continuing to rigorously test them in their field settings, and thank those maltreated individuals and their families for participating in research to intervene, treat, and, most of all, prevent abuse and neglect.

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CHAPTER 29

Autism Spectrum Disorders and Related Developmental Disabilities

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This chapter addresses the pervasive developmental disorders (PDD) included in the *Diagnostic and Statistical Manual of Mental Disorders*, fourth edition, text revision (*DSM-IV-TR;* American Psychiatric Association, 2000). These disorders include Autistic Disorder (autism), Asperger's Disorder, Pervasive Developmental Disorder—Not Otherwise Specified (PDD-NOS), Rett's Disorder, and Childhood Disintegrative Disorder. Given the sheer amount of information currently available on all of the PDDs, we highlight aspects of the disorders that are particularly pertinent to clinical psychologists.

This chapter first addresses terminology related to the PDDs. In particular, the increased attention to specific PDDs by diverse groups, such as consumers, educators, legislators, and basic researchers, has produced an imprecise umbrella term, autism spectrum disorder (ASD). The lack of clarity in terminology has produced, and continues to produce, significant confusion and miscommunication. Second, we address the three factors common across Autistic Disorder, Asperger's Disorder, and PDD-NOS: social impairment, language and communication impairments, and restricted, repetitive, and/or stereotyped behaviors. This discussion is important because these three areas vary greatly within and across each of these disorders. Issues related to assessment and diagnosis are discussed next. This section introduces the reader to commonly used assessments as well as the issues surrounding diagnostic evaluations for individuals with an ASD. A discussion of intellectual functioning and mental retardation follows. Comorbidity, trends in prevalence, early intervention, and etiology of ASD are covered next. Rett's Disorder and Childhood Disintegrative Disorder are discussed afterward, as these disorders are rare in occurrence and have distinct differences from the other three disorders. Next, treatment for the core symptoms of the PDDs is discussed. A brief section on family issues is last.

TERMINOLOGY

A review of the terminology over the past decade shows that the term pervasive developmental disorder as defined by the *DSM-IV-TR* (American Psychiatric Association, 2000) is often used synonymously with autism spectrum disorder.

Unfortunately, many studies applying either term to describe participants, samples, or population fail to specifically denote which disorders, if any, are included in these categories. Moreover, there is no standard operational definition of ASD, as the term is not in the *DSM-IV-TR,* nor have other attempts to adequately and consistently define this term been made. For consistency purposes, we suggest conceptualizing ASD as a spectrum including Autistic Disorder, Asperger's Disorder, and PDD-NOS. When referring to ASD, Rett's Disorder and Childhood Disintegrative Disorder are not included, mostly due to the low occurrence of each disorder and their specific course.

An additional problem is the use of descriptive labels for presumed subcategories of Autistic Disorder that also have yet to be adequately defined (Volkmar & Klin, 2005). For example, the term "high-functioning autism" is widely used yet has no reference in the *DSM-IV-TR* (American Psychiatric Association, 2000). The term typically refers to individuals with a diagnosis of Autistic Disorder who have adequate verbal language abilities and an average or above average level of cognitive ability (Baron-Cohen, O'Riordan, Stone, Jones, & Plaisted, 1999; Escalante-Mead, Minshew, & Sweeney, 2003; Klin & Jones, 2006). Individuals described as having high-functioning autism present clinically similar to individuals with Asperger's Disorder. However, researchers debate whether there are meaningful differences between individuals diagnosed with Asperger's Disorder and those described as having high-functioning autism (Lincoln, Courchesne, Kilman, Elmasian, & Allen, 1998; Miller & Ozonoff, 2000). This is an important area that has begun to receive more attention, as it might have implications for the study of etiology, response to treatment, type of treatment recommended, and prognosis. Using the term high-functioning autism as a specifier that is not always consistently defined creates potential diagnostic overlap. This can lead to problems when conducting research and when attempting to diagnose an individual with a PDD using the current *DSM* classification system.

Clearly, more precise definitions for Autistic Disorder, Asperger's Disorder, and PDD-NOS are needed. An examination of the *DSM-IV-TR* (American Psychiatric Association, 2000) diagnostic criteria for the PDDs might be helpful in understanding how similar these disorders are to each other. Table 29.1 lists the major diagnostic criteria in the areas of social skills, communication, and behavior for each PDD. Notice that the social skills and behavior criteria are the same for both Autistic Disorder and Asperger's Disorder. Under the PDD-NOS category, the criteria define a less severe impairment than in Autistic Disorder and Asperger's Disorder for social and behavior areas. Given the overlap of symptoms and characteristics of these disorders, the next sections describe the main impairments (social, communication, and behavioral) in turn.

As indicated in Table 29.1, describing the characteristics in the social, communication, and behavioral domains of Autistic Disorder, Asperger's Disorder, and PDD-NOS involves discussion of many similar and overlapping characteristics. We describe these characteristics under the term ASD, given the current trend in the literature. However, characteristics and symptoms specific to a particular disorder are noted.

Table 29.1 Major *DSM-IV TR* Criteria for Pervasive Developmental Disorders

Impairments	Autistic Disorder	Asperger's Disorder	PDD-NOS	Rett's Disorder	Childhood Disintegrative Disorder
Social	Qualitative impairment in social interaction (at least two specific impairments needed)	Same as Autistic Disorder	"A severe and pervasive impairment in development of reciprocal social interaction" (p. 84)	Loss of social engagement early in the course (although often social interaction develops later)	Clinically significant loss in social skills; abnormalities of functioning in qualitative impairments in social interaction
Communication	Qualitative impairment in communication (at least one specific impairment needed)	No clinically significant general delay in language	Impairment in verbal or nonverbal communication skills	Severely impaired expressive and receptive language development (after a period of normal development)	Clinically significant loss in expressive/receptive language; abnormalities of functioning in qualitative impairments in communication
Behavior	Restricted, repetitive, and stereotyped patterns of behavior, interests, and activities (at least one specific impairment needed)	Same as Autistic Disorder	Presence of stereotyped behavior, interests, and activities	Loss of previously acquired purposeful hand skills between 5 to 30 months of age with subsequent development of stereotyped hand movements	Clinically significant loss in adaptive behavior/motor skills; abnormalities of functioning in restricted, repetitive, and stereotyped patterns of behavior, interests, and activities, including motor stereotypes and mannerisms

Adapted from *Diagnostic and Statistical Manual of Mental Disorders*, Fourth Edition, Revised, by the American Psychiatric Association, 2000, Washington DC: Author.

SOCIAL IMPAIRMENT

The impairment in social skills, social interactions, social reciprocity, and social competence in individuals with Autistic Disorder, Asperger's Disorder, or PDD-NOS is central to the diagnosis. In this section we describe the specific impairments associated with each of the disorders in turn.

Autistic Disorder

Leo Kanner (1971) identified social impairment as the primary hallmark of autism (his term was "early infantile autism"). That is, the children's "inability to relate themselves in the ordinary way to people and situations from the beginning of life" (p. 140). Indeed, developmental research supports this claim. For example, infants show preferential attention toward social stimuli (i.e., faces) over other types of stimuli within the first few months of life (Sigman, Dijamco, Gratier, & Rozga, 2004), with initial social development occurring between the infant and caregiver. As children grow, social behavior begins to become increasingly differentiated; children develop peer relationships, pro-social skills, and an increasing capacity for self-regulation (Singer, 1996). In children with Autistic Disorder, however, retrospective studies suggest that aberrant social development starts within the first few months after birth (Goin & Myers, 2004; Osterling, Dawson, & Munson, 2002).

One profound social impairment in individuals with Autistic Disorder is a lack of eye contact. Unlike children with mental retardation or other developmental disabilities (but not a PDD), children with Autistic Disorder may not engage in consistent, predictable eye contact with parents, caretakers, and others (Carter, Davis, Klin, & Volkmar, 2005). Similarly, individuals with Autistic Disorder do not readily initiate other social behavior involving eye contact, such as joint attention, which in typically developing children develops in infancy (Mundy & Burnette, 2005).

Social impairment is not only considered to be the core deficit, but is also complex, as it varies with chronological age and developmental level. Considered a primary deficit, social impairment has been implicated in the aberrant development of other areas, such as language abilities, emotion regulation, and behavioral patterns of rigidity, stereotypies, or self-stimulatory behavior (Bachevalier & Merjanian, 1994; Happe & Frith, 1995). For example, some children have difficulty using nonverbal cues to modulate their social interaction (e.g., continuing a conversation when the listener is looking at his or her watch repeatedly and yawning; Lord, 1990).

Typically, social competence is markedly below the cognitive ability of an individual with Autistic Disorder (and more broadly ASD). For example, a 7-year-old child with Autistic Disorder with cognitive ability in the mental retardation range may neither display eye contact nor respond to the greeting of another person, whereas a same-age child with Autistic Disorder but of average intelligence may display partial eye contact and may have a limited repertoire in responding to a greeting. Both children show deficits in social interaction and social skills when compared to their typically developing peers.

Social skills vary by age as well. By adulthood individuals may develop some social skills that are typical in childhood, while other social skills important in

adulthood remain deficient (e.g., relating to others on an intimate level). In addition, some individuals will have more diverse and advanced social abilities than others.

Play skills are also impaired (Kasari, Freeman, & Paparella, 2006). By the end of 2 years of age, most typically developing children demonstrate functional and symbolic play skills. The play skills of children with Autistic Disorder are severely diminished compared to typically developing children. Children with Autistic Disorder may seek out play items but use them in a self-stimulatory manner. Most children with Autistic Disorder show difficulty using a toy or object in a more social, playful manner (Thomasa & Smith, 2004). A preference to play alone and with only a few specific toys is usually observed, whereas the typically developing child would more likely prefer to play with others, explore a variety of toys, and engage in other social interactions with his or her environment.

Deficits in expressive social communication (e.g., gestures) and engaging in eye contact, joint attention, and social reciprocity (Lord, 1990) are observed. It is also the case that abilities to initiate social behavior and understand the rules governing social interaction are impaired (Wing, 1981). Further, individuals with Autistic Disorder are typically not responsive when others are in distress (Travis, Sigman, & Ruskin, 2001). There may also be difficulty with perception of affective content (Hobson & Lee, 1989). One study showed that adolescents with Asperger's Disorder are significantly impaired in their ability to detect the emotional content of videotaped scenes that lack speech content cues as compared to typically developing adolescents (Koning & Magill-Evans, 2001). A series of studies using the Vineland Adaptive Behavior Scales, which is a norm-referenced assessment, found that socialization deficits are greater than expected based on overall developmental level (Sparrow, Balla, & Cichetti, 1984; Volkmar et al., 1987). Additionally, these studies demonstrated that children with Autistic Disorder exhibit significantly lower socialization scores than children with mental retardation.

Decoding of emotions from facial expressions is also impaired relative to typically developing children. It appears that individuals with ASD do not rely on facial expressions for emotional information. In addition, they may have difficulty relying on other situational cues to determine a person's emotional state (Egan, Brown, Goonan, Goonan, & Celano, 1998).

Asperger's Disorder

Individuals with Asperger's Disorder may exhibit all or many of the social impairments just described. The socioemotional and social communicative deficits in Asperger's Disorder are primary deficits. Individuals with Asperger's Disorder usually have significant difficulties identifying ("reading") and understanding social cues. It appears that individuals with Asperger's can learn cues and facial expressions to correctly identify emotions, but do not show generalization of these skills to spontaneous use (Grossman, Klin, Carter, & Volkmar, 2000; Serra, Minderaa, van Geert, & Jackson, 1999). Egan et al. (1998) found that individuals with ASD were unable to use prosody (tone of voice) and facial expression to identify emotions. A more recent study by Lidner and Rosén (2006) supported these results. These authors found that children with Asperger's Disorder are able to utilize verbal content

to assist them with identifying emotions. Grossman et al. found that this is not the case for individuals with Autistic Disorder; individuals with Autistic Disorder are unable to compensate using verbal ability to identify emotions.

Individuals with Asperger's Disorder typically desire friendships and want to make connections with others. However, they lack social skills to do so in a meaningful manner. Because individuals with Asperger's Disorder are intellectually typical with regard to their cognitive and verbal language abilities, they are aware that they are "socially different" from others. These individuals, therefore, are at a higher risk for developing depression or anxiety as a result, especially in adolescence and adulthood. Because of the compromised ability of individuals with Asperger's Disorder to understand and interpret social behavior appropriately, forming sustainable relationships is a challenge

Pervasive Developmental Disorder—Not Otherwise Specified

Individuals with PDD-NOS share many of the characteristics of both Autistic Disorder and Asperger's Disorder. However, the severity of this impairment may not be to the same degree. Thus, a separate section that covers PDD-NOS is redundant, even though this is not a reflection of a less important diagnosis.

COMMUNICATION

Children with Autistic Disorder either develop verbal language later than typically developing children or do not develop functional language at all. Fifty percent of children with Autistic Disorder do not develop spoken communication (Sigman & Capps, 1997). For individuals who do develop spoken language, abnormal features are typically present (Sheinkopf, Mundy, Oller, & Steffens, 2000; Sigman & Capps, 1997). For example, Sheinkopf et al. noted that preschool children with Autistic Disorder showed a rate of verbalization similar to normally developing children for their developmental level. However, the types of verbalizations and vocal quality were noticeably abnormal (e.g., growling, squeals).

Echolalia is a characteristic of speech patterns in approximately 75% of individuals with Autistic Disorder (Prizant, 1983). Echolalia is defined as the repetition of the speech of another person or oneself and is often present in typically developing infants. Echolalia can be immediate (i.e., repeating a word, phrase, or sentence immediately after it is spoken by someone else), delayed (i.e., repeating speech heard in the past), or mitigated (e.g., imperfectly repeating speech heard in the past). Echolalia is not unique to Autistic Disorder, as it is an abnormal speech quality observed in other psychiatric disorders and sometimes as a result of brain damage (Sigman & Capps, 1997). There is some question in the field as to whether there is a communicative function of echolalia for individuals with ASD (Tager-Flusberg, Paul, & Lord, 2005). Some researchers suggest that children with ASD use echolalia to engage in a conversation because they do not understand how to use speech to communicate (Loveland & Tunali-Kotoski, 2005).

Prosody is impaired in individuals with ASD. Prosody is the rhythm of spoken language; it helps the speaker add specific meaning to a phrase or sentence. For

example, when asking a question, most individuals raise their voice slightly at the end of the sentence. This helps the conversational partner understand that the intent of the sentence is to ask a question. Individuals with ASD typically have abnormal intonation and use of rhythm and stress in speech production. Other characteristic speech abnormalities are pronoun reversals, articulation difficulties, and idiosyncratic use of speech, including the development and use of neologisms (Paul, Augustyn, Klin, & Volkmar, 2005). By age 5 or 6, most children with Autistic Disorder still show a delay in spoken language. If speech is present by this age, it is typically an indicator of better prognosis for the child (Tager-Flusberg et al., 2005).

In typically developing children who have not acquired speech, for example, due to a specific speech delay, the use of nonverbal communication is usually intact. A child who is deaf, for instance, is able to acquire nonverbal forms of communication (e.g., including sign language or other communication methods) and will compensate through the use of physical gestures, eye contact, pointing, and body language to communicate with others. For individuals with Autistic Disorder, these areas of nonverbal communication are either absent or noticeably impaired (Buffington, Krantz, McClannahan, & Poulson, 1998; Kasari, Sigman, Mundy, & Yirmiya, 1990; Tager-Flusberg et al., 2005).

Communicative intent is the planned purpose of a specific communication interaction. There are many purposes or goals for communicating with others; for instance, communicating that one is feeling sad, hungry, or that one would like to read a book located in an unreachable location, are all different yet important reasons to communicate to another person. Communicating with another person is categorized as either declarative (e.g., asking a friend what he received for his birthday) or regulatory (e.g., asking a friend to pass a piece of birthday cake; Landa, 2005; also known in behavior analysis terms as tact and mand, respectively). One of the significant characteristics of communicative intent in individuals with Autistic Disorder is that the intent is typically toward gaining access to an object or activity, not an interaction or conversation with a person. Thus children will use communicative gestures to gain access to an object, but not use such gestures in a social way.

There has recently been much attention devoted to joint attention deficits in ASD. Joint attention is a form of communicative intent that typically develops in infancy. Joint attention involves gestures such as pointing and use of eye contact that serve to focus another person's attention on an object or event for the purpose of shared enjoyment (Landa, 2005). Joint attention begins to emerge between 9 and 18 months (Bakeman & Adamson, 1984). Deficits in joint attention in children with ASD are noticed at an early age and have been well documented in the literature. These deficits include failure to orient to social stimuli, speak sounds, follow eye gaze, and follow pointing in conjunction with eye gaze (Leekam, Hunnisett, & Moore, 1998; Mundy, Sigman, & Kasari, 1994). Children with ASD also exhibit deficits in the initiation of joint attention, including a lack of showing objects of interest and failure to use declarative pointing and referential looking (MacDonald et al., 2006). Deficits in responding to joint attention appear to be less severe than the initiation to joint attention (MacDonald et al., 2006).

Interestingly, it does not appear that joint attention skills are delayed per se, as they sometimes fail to develop after more advanced social skills, such as nonverbal

pointing to request an item (Mundy et al., 1994). MacDonald et al. (2006) reported relatively no differences in the use of joint attention behaviors between 2-, 3-, and 4-year-olds with autism. Carpenter, Pennington, and Rogers (2002) reported that the development of joint attention skills in children with autism does not follow the same pattern of development of these skills in typically developing children. These authors suggested that there might be a direct relationship between language ability and joint attention skills. Joint attention has also been shown to be related to social development (Schertz & Odom, 2004) with regard to emotion regulation and the ability to develop connections with other individuals, given that joint attention is related to the ability to share enjoyment and other forms of affect with others (Schertz & Odom, 2004; Wetherby, Prizant, & Schuler, 2000).

Most communication is used in the context of a social interaction. Given that individuals with ASD have pronounced difficulties in the area of social interaction, it follows that these individuals also have difficulties in aspects of communication that require interaction with others. Presuppositional skills represent the ability to produce the necessary quality and quantity of communication given different contexts. This skill emerges in preschool-age children and improves as the child's working memory, social cognitive skills, and metacognitive and linguistic skills develop. Discourse management, which typically develops by age 4, includes conversational skills, such as turn taking, starting and ending conversations, and staying on the topic of conversation (Landa, 2005).

Pragmatic language skills include the ability to adapt communication to listeners and situations. Although typical children become adept at pragmatic language skills as early as the preschool years, individuals with ASD show a wide variety of skill deficits in this area. Specific pragmatic language deficits include asking inappropriate questions (that may result in embarrassment or discomfort to the conversational partner), inappropriately interrupting others, inability to start or end conversations, and perseverating on a topic of interest. When neologisms are used in conversation, confusion may occur, especially when the individual with ASD does not acknowledge that the conversational partner does not understand the neologism. Individuals with ASD have difficulty understanding sarcasm, jokes, and puns. They tend to interpret the meaning of a joke, for example, at face value, or literally (Emerich, Creaghead, Grether, Murray, & Grasha, 2003; Van Bourgondien & Mesibov, 1987).

To converse with another person requires numerous skills in addition to the ability to produce and understand spoken language. A high percentage of children with Autistic Disorder do not acquire verbal skills but do use augmentative or alternative ways to communicate. Thus, these children should be encouraged to engage in conversations with others (Loveland & Tunali-Kotoski, 2005).

Asperger's Disorder

Characteristics similar to those just described are present in Asperger's Disorder. Although there is no delay in the acquisition of language in Asperger's Disorder, the way individuals with this disorder communicate to others, particularly in the

social context, is noticeably impaired. For instance, the topic of conversation with someone with Asperger's Disorder may be narrow in scope and idiosyncratic. The individual will typically attempt to keep the conversation focused on a specific topic of interest to him or her, be indifferent to requests to change topics, have considerable difficulty taking turns in the conversation (i.e., social reciprocity), and have difficulty ending the conversation.

Receptive language difficulties include making inferences from others' nonverbal behavior and with certain elements of speech. For example, perspective taking can be difficult, as well as recognizing indirect social cues. Other elements of receptive language that are difficult for individuals with Asperger's Disorder are nonverbal (e.g., eye contact, facial expressions) and paralinguistic (e.g., tone of voice, prosody) cues and discriminating the meaning of a phrase, joke, or pun (i.e., they might have difficulty distinguishing something that has a literal versus a sarcastic meaning). Individuals with Asperger's Disorder tend to have a pedantic style of speech and prefer factual information.

BEHAVIOR

The broad category of restricted and repetitive behaviors in individuals with Autistic Disorder and Asperger's Disorder includes self-stimulatory behavior (e.g., repetitive finger movements, gazing at lights, rocking, toe-walking), stereotypical motor behavior (e.g., posturing), preoccupation with objects and activities, insistence on sameness or inflexibility in routines, and restrictive interest in objects or activities (South, Ozonoff, & McMahon, 2005; M. Turner, 1999). Children with ASD tend to have a narrow range of interests and activities, whereas adults have a somewhat wider range of interests but continue to be more resistant to change. Depending on routine and familiarity, the rigid or stereotyped thinking processes characteristic of ASD make abstraction and comprehension difficult. These clusters of behaviors are not unique to ASD. In fact, they may be seen in individuals with psychiatric conditions (e.g., Obsessive-Compulsive Disorder) and neurological conditions (e.g., Tourette's Disorder).

Recent studies are beginning to identify types or categories of atypical behaviors in ASD (Bishop, Richler, & Lord, 2006; Carcani-Rathwell, Rabe-Hasketh, & Santosh, 2006). In the current literature, the atypical behaviors within the category of restricted and repetitive behaviors are classified as either "lower order" or "higher order" behaviors. Lower order behaviors include stereotypical motor movements (e.g., hand and finger mannerisms) and preoccupations with parts of objects. Higher order behaviors include circumscribed interests, adherence to routines, and resistance to change (Bishop et al., 2006). In addition, Cuccaro et al. (2003) and Bishop et al. reported that younger children more readily engage in lower order behaviors (i.e., sensory-motor behaviors), whereas older children with ASD engage in higher order behaviors (i.e., insistence on sameness).

Bishop et al. (2006) point out that identifying specific atypical behaviors with different subtypes of ASD may be helpful in locating underlying genetic components and pathology among the ASDs. For instance, M. Turner (1999) and Lord, Rutter, and Le Couteur (1994) found that certain atypical behaviors (e.g., repetitive

sensory-motor behaviors) were associated with lower levels of adaptive functioning as measured by the Vineland Adaptive Behavior Scales (Sparrow et al., 1984). However, both studies indicated that other atypical behaviors, such as difficulty with or resistance to change, were not associated with adaptive functioning. In addition, there have been advances in research linking repetitive behaviors to specific neurobiological regions (Lewis & Bodfish, 1998; Pierce & Courchesne, 2001) as well as chromosomal regions (Alarcón, Cantor, Liu, Gilliam, & Geschwind, 2001; Cuccaro et al., 2003). Certainly, though more research needs to be conducted, the field is coming to understand the behavioral aspects of ASD much better now than in the past.

Disruptive behaviors are common in children and adolescents with ASD. Common aggressive behaviors observed and reported in individuals with ASD and other developmental disabilities include physical aggression (e.g., hitting, kicking, biting, shoving, pinching), verbal aggression (e.g., yelling, calling names), and destructive behavior (e.g., throwing objects, breaking objects of others). Assessment and interventions for aggression are discussed in the treatment section of this chapter. Other types of disruptive behaviors commonly observed in children with ASD are sleeping and eating problems.

It has been noted that individuals with ASD have significant feeding and sleeping issues. These two problems are serious given the impact they have on parental, family, and child functioning. However, these problems are not part of the *DSM-IV-TR* diagnostic criteria. In the context of this chapter, feeding problems are maladaptive feeding behaviors not due to a medical condition and/or without a medical explanation. They include aversion to food, refusal to eat, gagging or choking responses to food, and restrictive eating patterns (i.e., eating only certain very specific types of food).

Ahearn, Castine, Nault, and Green (2001) suggest that feeding problems in children with ASD may be categorized under the restrictive patterns of interests and activities characteristic of ASD. The *DSM-IV-TR* (American Psychiatric Association, 2000) includes a diagnostic category of feeding problems, which includes failure to thrive. Many children with ASD eat enough to not meet this criterion; however, their restrictive eating patterns might lead to significant nutritional deficiencies that will cause developmental problems later in life. Ledford and Gast (2006) reported that even though children with ASD may not be considered at risk for an eating disorder, the long-term health-related consequences of having a restricted diet should be taken into consideration and treatment of feeding problems in young children with ASD should be a priority.

Ledford and Gast (2006) conducted a comprehensive review of the literature on descriptive studies of feeding problems in ASD and also on interventions for feeding problems. Based on their review, these authors concluded that 46% to 89% of children with ASD have feeding problems. The authors reviewed nine intervention studies, all of which demonstrated positive outcomes of the interventions. It should be noted that the interventions were behavioral and included differential reinforcement, stimulus fading, escape extinction, and appetite manipulation (simultaneous and sequential presentation of a nonpreferred food with a preferred food).

Children with ASD have significantly more sleep problems than typically developing children (Hoffman, Sweeney, Gilliam, & Lopez-Wagner, 2006). Parents reported that these sleep problems include bedtime resistance, parasomnias, night wakings, sleep anxiety, sleep disordered breathing, and sleep-onset delay. However, children with autism are not observed to be sleepier during the daytime than children who do not have sleep problems. Sleep problems tend to decrease as children age, according to the study by Hoffman et al.

Empirically supported treatment of sleep difficulties in ASD has not been documented. However, traditional behavioral methods for sleep intervention do show some promise for children with ASD. It is necessary to understand the etiology of sleep disorders in ASD to provide more efficacious treatments for these individuals. Many parents of children with ASD report that their children have sleep difficulties that exceed those of typically developing children.

Hoffman et al. (2006) surveyed parents of children with ASD and parents of typically developing children and found that children with ASD have significantly more sleep problems in the following areas: (a) bedtime resistance, (b) sleep-onset delay, (c) sleep duration, (d) sleep anxiety, (e) night wakings, (f) parasomnia, and (g) sleep disordered breathing. These authors also found that as children with ASD age, the severity of some sleep problems decreases. A previous paper by Hoffman et al. (2005) demonstrated a positive correlation between sleep difficulties and autism symptomatology. Research is needed to evaluate sleep interventions for children with ASD and the impact that these interventions might have on learning, family and parental stress, and other symptoms of ASD.

ASSESSMENT AND DIAGNOSIS

Autistic Disorder is typically diagnosed before the age of 3. It is quite common for a diagnosis to be given around the age of 18 to 24 months (Robins, Fein, Barton, & Green, 2001). However, the diagnosis of young children with ASD can be quite difficult. There are no reliable biological markers for ASD; thus diagnosis relies on the use of instruments designed specifically for the diagnosis of ASD. These include the Autism Diagnostic Observation Schedule—Generic (ADOS-G; Lord, Rutter, DiLavore, & Risi, 2000), the Autism Diagnostic Interview—Revised (ADI-R; Rutter, LeCouteur, & Lord, 1995), and the Childhood Autism Rating Scale (CARS; Schopler, Reichler, & Renner, 1980). A recent study by Ventola et al. (2006) indicated that the use of the ADOS-G, CARS, and clinical judgment based on the *DSM-IV* (American Psychiatric Association, 1994) criteria demonstrated strong agreement with each other. Even though these results are encouraging, as they suggest that clinicians might be able to use different assessment instruments, more research needs to be conducted to determine which method for diagnosing ASD has the highest sensitivity and specificity.

Autism spectrum disorders are heterogeneous, and each individual with an ASD will present differently in certain areas. In fact, a diagnosis of Autistic Disorder is sometimes more straightforward than a diagnosis of high-functioning autism, Asperger's Disorder, or PDD-NOS. This is because some of the characteristics of these other disorders are more subtle and may not appear to be severe enough

according to a measurement instrument. Distinguishing between the boundaries of PDD-NOS, Asperger's Disorder, and high-functioning Autistic Disorder remains a challenge to both researchers and clinicians. As more research is conducted on diagnostic instruments, the symptom clusters and developmental trajectories specific to different ASDs should be illuminated.

The psychological assessment of an individual with ASD should include, but not be limited to, an understanding of the individual's current developmental level, his or her verbal and nonverbal strengths and weaknesses, and adaptive functioning (i.e., daily living skills). Because children with ASD have deficits in other areas than those listed in the *DSM-IV-TR,* the use of instruments that assess cognitive, language, motor, and adaptive skills are important to include in a diagnostic assessment battery. Standardized assessment instruments that can be utilized to assess standard intellectual functioning include the Wechsler Scales (the specific scale depends on the age of the individual); to assess mental processing and acquired language, the Kaufman Autism Behavior Checklist II; to assess developmental abilities, the Differential Ability Scales; and for children with severely limited language abilities, the Leiter-R. For developmental assessment, the Mullen Scales measure nonverbal, language, and motor skills. The Vineland Adaptive Behavior Scales measure adaptive functioning. Speech, language, and communication assessments should evaluate basic speech and language skills (including receptive and expressive language, prosody, paralinguistics, metalinguistics, and narrative skills) as well as social communication skills (such as joint attention, reciprocal social and communicative interactions, and self-regulation).

Importantly, when providing assessment, consultation, or intervention services to an individual with ASD, there will be numerous other professionals with whom a clinical psychologist is likely to interact for a variety of reasons (i.e., assessment, treatment, referral). Professionals range from early intervention service coordinators, special education teachers, and certified behavior analysts to a variety of family members, vocational skills training professionals, and psychiatrists. Of course, this is a short list, as the number and type of professionals depend on a host of factors, such as the client's age and level of functioning and the services available.

Above all, it is important to understand the child's functional skills, that is, skills the child can use on a day-to-day basis to communicate, interact socially, and accomplish tasks. Clinically it may not be important for a person with ASD to be able to do complex math problems consistent with his or her chronological age, but it may be far more important for that person to tell time on a digital clock in order to participate in an environment where schedules must be kept.

INTELLECTUAL FUNCTIONING AND MENTAL RETARDATION

Mental retardation (MR) is diagnosed when an individual's cognitive ability is assessed as an IQ of 70 or less, with adaptive skills deficits, and the age of onset is before 18 years (American Psychiatric Association, 2000). Given the criteria required for a diagnosis of Autistic Disorder, the criteria for impaired adaptive skills and age of onset overlaps. Thus, the critical aspect of diagnostic importance

for MR is strictly the cognitive skills deficit (Edelson, 2006). For years, reports of a rate of 70% to 80% of children with Autistic Disorder with comorbid MR have been suggested (Bohman, Bohman, Björck, & Sjöholm, 1983; Chung, Luk, & Lee, 1990; Gillberg & Coleman, 2000). A recent report suggests that between 40% and 55% of children with Autistic Disorder have comorbid MR (Chakrabarti & Fombonne, 2001).

Edelson (2006) examined retroactively the statistics in previous reports and studies to ascertain the nature of support for the claim of the incidence of MR in Autistic Disorder. Edelson provided an excellent and critical review of over 215 studies. For articles in which the methodology for determining MR was described (only 15.7%), the incidence of MR was 71.89%. Interestingly, Edelson found that 157 of the 215 articles reported the diagnosis of MR with autism, but did not provide a methodology for diagnosing MR.

Instruments used to assess cognitive ability are limited because when assessing the cognitive ability of children with ASD, poor motivation in particular may significantly attenuate performance. Given the variability of behavior in some children with ASD, it is difficult to say whether additional factors (e.g., attention, language comprehension, unfamiliarity with structured interactions) negatively influence performance on cognitive tests. Ritvo and Ritvo (2006) insist that whether a child with autism has MR does not really matter; what matters is whether the child can carry out daily activities, including academic ones. These authors suggest that the focus of treatment should be on social and communication deficits, not on cognitive and academic deficits. Future prevalence surveys should include the whole spectrum of Autistic Disorder and other ASDs to ascertain a more accurate estimate of the prevalence of MR (Freeman & Van Dyke, 2006).

COMORBIDITY OF PSYCHIATRIC DISORDERS

Comorbidity refers to the co-occurrence of two disorders where the occurrence of one disorder does not exclude the occurrence of the other (Krishnan et al., 2002). Over the past 10 years, comorbidity issues in ASD have appeared increasingly in the literature, especially with regard to anxiety disorders in ASD.

A recent study by Leyfer et al. (2006) piloted an instrument called the Autism Comorbidity Interview—Present and Lifetime Version (ACI-PL), which was modified from the Schedule for Affective Disorders and Schizophrenia for School-Age Children (K-SADS; Chambers et al., 1985). The ACI-PL is a semi-structured interview and is similar in format and scoring to the K-SADS. In the two pilot samples, the ACI-PL demonstrated adequate specificity, sensitivity, and reliability for three *DSM* diagnoses: (1) Major Depressive Disorder, (2) Obsessive-Compulsive Disorder, and (3) Attention-Deficit Hyperactivity Disorder. This study found that the most frequent disorder comorbid with Autistic Disorder was Specific Phobia, with a comorbidity rate of 44%. Obsessive-Compulsive Disorder was the second most common (37%), and the most common type of compulsion was performing a ritual in a certain way. The full diagnosis of Attention-Deficit/Hyperactivity Disorder (ADHD) is the third most common comorbid disorder (31%). Interestingly, 55% of individuals with Autistic Disorder also had subsyndromal ADHD. The authors

note that the inattentive type was more common in individuals with Autistic Disorder (whereas the hyperactive type is most common in the typically developing population). Ten percent of individuals with Autistic Disorder had experienced at least one episode of depression.

Overall, this study found that approximately 30% of the pilot sample had on average two comorbid diagnoses along with Autistic Disorder. The pilot sample tended to represent children with Autistic Disorder who would be described as high functioning, as 67.71% of children tested had Full-Scale IQ scores greater than 70, with a mean age of 9 years.

Comorbidity is an important area in ASD, given that the diagnostic criteria for ASD appear to be broadening (despite no change in the *DSM-IV-TR;* American Psychiatric Association, 2000) and the field is beginning to learn more about these comorbid disorders. The data Leyfer et al. (2006) provided should be accepted cautiously, given that this was a pilot study. However, the results do confirm an increased comorbidity of ASD with other *DSM* diagnoses reported by other investigators (Gillot, Furniss, & Walter, 2001; Kim, Szatmari, Bryson, Streiner, & Wilson, 2000; Towbin, Pradella, Gorrindo, Pine, & Leibenluft, 2005).

CURRENT TRENDS IN THE PREVALENCE OF AUTISTIC SPECTRUM DISORDERS

Historically, autism has been considered a very rare disorder. The *DSM-IV-TR* (American Psychiatric Association, 2000) reports the prevalence of Autistic Disorder to be 5 per 10,000, ranging from 2 to 20 per 10, 000. At the time of publication of the *DSM-IV-TR,* prevalence data were lacking for Asperger's Disorder, PDD-NOS, Rett's Disorder, and Childhood Disintegrative Disorder. Over the past 6 years, since the latest revision to the *DSM,* epidemiological research has been published providing new estimates of the prevalence, not only of Autistic Disorder, but also for Asperger's Disorder, PDD-NOS, and ASD (Fombonne, 2005). Thus, information about the prevalence of ASD in the *DSM-IV-TR* is no longer accurate.

In a recent review of prevalence studies from 1966 through 2004, Fombonne (2005) suggested that the current prevalence of Autistic Disorder is 13 per 10,000, of PDD-NOS is 21 per 10,000, of Asperger's Disorder is 2.6 per 10,000, with a prevalence of 60 per 10,000 for all PDDs. Croen, Grether, Hoogstrate, and Selvin (2002) conducted a study that investigated the change in prevalence of ASD in California from 1987 to 1994. The prevalence of ASD rose from 5.8 per 10,000 in 1987 to 14.9 per 10,000 in 1994. Interestingly, the prevalence of mental retardation showed the opposite trend; in 1987, the prevalence was 28.8 per 10,000 compared to 19.5 per 10,000 in 1994.

The male-to-female ratio of Asperger's Disorder, Autistic Disorder, and PDD-NOS is 4:1. Notably, when a female is diagnosed with Autistic Disorder, she typically has a more severe form (i.e., lower cognitive abilities, increased behavioral problems, and more severe social and communicative impairments).

Questions loom as to why there has been an increase in the prevalence of ASD. Croen et al. (2002) propose that factors accounting for the increase in prevalence rate may include improvements in identification of ASD and changes in the criteria

for a diagnosis of ASD in the *DSM-IV-TR* (American Psychiatric Association, 2000) compared to the *DSM-III* (American Psychiatric Association, 1980). Children diagnosed with Autistic Disorder in the *DSM-IV-TR* may represent a broader phenotype than if they were diagnosed using the *DSM-III*. Even though the precise answer to this question is not available, it is the case that we now have assessment tools to identify ASD at a younger age and more accurately, increased awareness and knowledge in communities and among practitioners, and increased services available for ASD (Croen et al., 2002; Fombonne, 2005). Regardless of the reasons for the increase in prevalence of ASD, early identification of the disorder is crucial for early intervention services to begin (Romanczyk et al., 2005).

EARLY IDENTIFICATION

Early identification and early intervention are necessary to provide positive outcomes for children with autism. Of the 3.75 million infants born each year in the United States, 20% are at risk for developmental disabilities due to pre- and postnatal factors (Haber, 1991).

Two federal laws that mandate service provision to children in the early intervention years (birth to 3 years) are Public Law 99-457 and Public Law 105-17. Public Law 99-457 was enacted in 1986 to provide early intervention services for at-risk infants and children (Haber, 1991). In addition, Public Law 105-17 allows eligible infants, toddlers, and children with disabilities to receive both appropriate and free early intervention services (birth to 35 months) and public education (starting at 36 months). Public awareness of the importance of early intervention services for children with autism has also increased. Several studies have shown both the short-term and long-term benefits for children receiving appropriate early intervention services (Anderson & Romanczyk, 1999; Guralnick, 1996; Lovaas, 1987; Smith, Eikeseth, Klevstrand, & Lovaas, 1997). These services are required to be free and it is mandatory for all states to follow the statutes in both public laws. Early identification, which leads to early intervention, is extremely important for individuals with ASD. Numerous studies have demonstrated impressive gains in language, cognition, and social development when individuals received early intervention services (Bondy & Frost, 1995; Harris & Handleman, 2000; National Research Council, 2001; L. Turner, Stone, Pozdol, & Coonrod, 2006).

The importance of early identification and intervention has spurred the development and validation of many screening measures. Given research on early signs of ASD, a diagnosis of ASD can be made before 24 months of age. Screening instruments are not substitutes for a diagnostic assessment, but do allow for the higher probability of identification of children who may be at risk for a developmental disability, such as ASD, in order to receive a more comprehensive diagnostic evaluation. Multiple screening measures are available, including the Checklist for Autism in Toddlers (Baird et al., 2000), the Modified Checklist for Autism in Toddlers (Robins et al., 2001), the Social Communication Questionnaire (Rutter, Bailey, Berument, Lord, & Pickles, 2003), and the Screening for Autism in 2-Year-Olds (Stone & Ousley, 1997).

Table 29.2 Clinical Clues for Autism

Delay or absence of spoken language
Looks through people; not aware of others
Not responsive to other people's facial expressions or feelings
Lack of pretend play; little or no imagination
Does not show typical interest in or play near peers purposefully
Lack of turn taking
Unable to share pleasure
Qualitative impairment in nonverbal communication
Not pointing at an object to direct another person to look at it
Lack of gaze monitoring
Lack of initiation of activity or social play
Unusual or repetitive hand and finger mannerisms
Unusual reactions, or lack of reaction, to sensory stimuli

Excerpted from *Clinical Practice Guideline for Autism/Pervasive Developmental Disorders: Assessment and Intervention for Young Children Ages 0–3 Years* (Guideline Technical Report. no. 4217), Table III-4, by the New York State Department of Health, Early Intervention Program, 1999, Albany, New York: Author.

The New York State Department of Health published *The Clinical Practice Guideline for Autism/Pervasive Developmental Disorders* in 1999. There are three versions of the guideline: *The Guideline Technical Report, Quick Reference Guide,* and *Report of the Recommendations.* Common to each version are basic recommendations for assessment and intervention methods for children with autism or other PDD. Within the assessment content area is a table listing 13 clinical clues recommended for the purpose of screening children for autism or other PDD. These clinical clues were taken from empirical studies that listed diagnostic criteria for autism according to the *DSM-III-R* and *DSM-IV-TR.* Each clue has been validated by empirical studies; has been shown to have a sensitivity or specificity of greater than 50%; is relevant to the *DSM-IV-TR* for the diagnosis of autism; and may be found in other screening and diagnostic instruments for autism. These clinical clues are presented in Table 29.2.

ETIOLOGY

The specific causes of ASD are unknown. As is the case for many disorders, the interaction of a biological predisposition (e.g., genetic mutations) with environmental factors most likely underlies the etiology of ASD, but hypotheses regarding etiology along these lines continue to be investigated (Polleux & Lauder, 2004). Research on the genetic basis and neurobiology of ASD are two important fields that are illuminating possible contributions to the etiology of ASD, but research is still in the early stages.

Familial recurrence of autism is 100 times higher than in the general population. In addition, there are reported concordance rates of 70% to 90% in monozygotic twins, which is not found in dizygotic twins. In fact, most studies show near 0% concordance rates in dizygotic twins, which provides strong support for the genetic component of autism (Veenstra-VanderWeele & Cook, 2003). However, we are far from identifying a definitive etiology for autism.

Approximately 10% of cases of autism have comorbid genetic conditions, such as Fragile X, tuberous sclerosis, and Rett's Disorder (Polleux & Lauder, 2004). One-third of individuals with autism develop seizures, and over half have abnormal sleep patterns. Recently, researchers have developed a working hypothesis characterizing the cortex as having an imbalance of excitation and inhibition, which leads to hyperexcitability in the cortex (Polleux & Lauder, 2004). The imbalance in the cortex could have consequences "on the function of neuronal networks underlying perception and attention" (Polleux & Lauder, 2004, p. 304). Other neuroanatomical features that have been hypothesized to be linked to ASD are abnormal functioning of the brain stem, cerebellum, hippocampus, and amygdala. There is also evidence suggesting that the brain is enlarged in children with autism and that this enlargement occurs during the postnatal period, but there is no further enlargement in adulthood and studies show no differences in brain size in adulthood between adults with Autistic Disorder and typically developing adults. The exact specifications and implications of brain enlargement have yet to be determined.

Multiple genes are hypothesized to be linked to autism, but more research is needed and is currently ongoing to address this. Neurotransmitter abnormalities are also hypothesized to be implicated in autism. Serotonin and GABA are two that have the most evidence suggesting abnormalities in ASD. See Polleux and Lauder (2004) for a review of neurobiological evidence.

A major difficulty in drawing conclusions from the current research evidence from the neurobiology and neuroimaging fields is that the sample size of most studies is still small and is typically biased toward individuals with autism who are considered higher functioning. Thus, generalizing results from these studies is difficult. However, this is an important area of research, as Autistic Disorder is a neurobiological disorder and the genetics, specific brain areas affected, and other biological and neurological characteristics of Autistic Disorder need to be more fully understood. These same concepts hold true for determining biological etiologies or predispositions for Asperger's Disorder and PDD-NOS.

RELATED DEVELOPMENTAL DISORDERS: RETT'S DISORDER AND CHILDHOOD DISINTEGRATIVE DISORDER

Rett's Disorder

Rett's Disorder was discovered by Andreas Rett and (at a later time) Bengt Hagberg. Rett's Disorder is an X-linked, dominant disorder that occurs almost exclusively in females. It has been identified as the second leading disorder that causes MR in females (Down syndrome is the primary). Much is known about the genetic etiology of Rett's and the course of the disorder, which is described next. It is hoped that eventual understanding of this disorder will help with the understanding of related disorders.

The most prominent symptom in Rett's Disorder is stereotypic hand movements. These are often an early warning sign; that is, these types of stereotypies are often present before other associated symptoms of Rett's are noticed. For children with Rett's Disorder, the first 6 to 8 months of life are characterized by normal

development. About 50% of cases begin to show a delay in and/or deterioration of skills around 15 months of age. By age 3, a significant deterioration of development is noted, especially loss of language, increased stereotyped hand movements, and decreased interest in objects and social interaction. The deterioration of skills is rapid (i.e., after a 1-year period, severe to profound disabilities are noted). Additional motor difficulties and delays are also present. The characteristic stereotyped hand movements include hand-to-mouth movements, hand wringing, and hand clasping. By the early school years, the pace of deterioration has decreased and children are able to acquire and maintain skills. However, through adolescence there is a decrease in physical growth and problems with ambulation are noted; in addition, many children develop scoliosis (Van Acker, Loncola, & Van Acker, 2005).

In 1986, Hagberg and Witt-Engerstrom (Van Acker et al., 2005) provided a four-stage developmental model of Rett's Disorder, then called Rett syndrome. This model is generally accepted today. Between 6 and 18 months, Stage 1 begins. This stage is characterized by a noted deviation from normal development, particularly in motor abilities. Stage 2 begins between 1 year and 4 years of age. In this stage, a significant loss of previously acquired skills occurs; many symptoms are present that resemble those in individuals with autism. Thus, social, speech, cognitive, and motor skills are deteriorated or lost during this stage. The stereotypic hand movements tend to be continuous, without clear antecedents or triggers. Children with Rett's Disorder are diagnosed with severe to profound MR, and parents characterize their mood as labile and irritable. During this stage abnormal breathing patterns are noticed; 25% of children have clinically significant seizure activity and 75% present with sleep abnormalities. Stage 3 begins between ages 2 and 10. The symptoms characteristic of autism tend to decrease during this stage. In fact, an increase in social interaction, awareness of and interest in the environment, communication skills, and use and acquisition of functional skills occurs. Up to 80% of children have seizures and the motor difficulties become more prominent. Stage 4 begins when a child is 10 or older. During this stage motor deterioration continues, with decreased mobility and frequently need of assistance to move about (e.g., using a wheelchair). Individuals' cognitive functioning is stable, and social skills and attention improves. In addition, a decrease in seizure activity is observed.

Childhood Disintegrative Disorder

Childhood Disintegrative Disorder (CDD) is characterized by typical, age-appropriate development in the areas of language, social, cognition, and other skills, with a regression of skills occurring between the ages of 2 and 10. Some authors suggest that there may be a mild delay in development prior to age 2 (Kurita, Kita, & Miyake, 1992). However, the most common report of age of onset is between ages 3 and 5, with a mean of 3.4 years. With regard to differential diagnostic characteristics, in Autistic Disorder it is not uncommon to observe language regression; however, in CDD there are multiple areas of regression.

The onset of CDD is typically abrupt, with symptoms being noticed within days to weeks. Many cases report a psychosocial stress or medical event, but the etiology is unclear. Prevalence is estimated to be 1 to 9 cases per 100,000 children, according

to Fombonne (2002). Because this condition is so rare, it is likely that most clinicians have little familiarity with it, and thus the actual prevalence of CDD is uncertain. Similar to other developmental disabilities, CDD predominantly affects males. It is differentiated from Rett's Disorder by the longer period of normal development before regression begins.

For the most part, the regression and deterioration of skills typically plateaus. Most cases of CDD do not continue to regress to the point of mortality. However, Volkmar, Koenig, and State (2005) reviewed several case studies suggesting an increased mortality rate after a child is diagnosed with CDD. Most of the deaths are related to neuropathology and/or medical diseases. A comorbidity rate of seizures similar to that found in Autistic Disorder has also been suggested in CDD. However, there are no conclusive data suggesting a specific medical etiology for CDD nor a predictive relationship of CDD with other medical disorders or pathologies. The genetic mechanisms in CDD are also not understood, and further research needs to be conducted (Volkmar et al., 2005).

TREATMENT

The field of PDD has experienced notable advances in research and clinical intervention over the past 10 years. However, there is no one treatment that has been shown to be universally effective in the full remediation of ASD. Treatments for ASD are described as either comprehensive interventions that include elements of behavioral, educational, and psychotherapeutic approaches or specific interventions, most of which are behavioral or psychopharmacological, that focus on the treatment of specific behaviors.

In general, learning is quite challenging for individuals with Autistic Disorder, whose difficulties range from overselectivity (i.e., disproportionate attention to irrelevant aspects of a stimulus) to lack of generalization of skills. Most children with ASD tend to lack multiple foundational learning skills; thus, sensitivity to the sequence in typical development is important when designing interventions. It is important to consider the individual's age-equivalent or developmental level when providing interventions and determining expectations. This is important at all levels and ages of intervention. In a recent study by L. Turner et al. (2006), a relatively stable diagnosis was demonstrated in children diagnosed with ASD at age 2 and reassessed at age 9. Thus, it is important that children receive services throughout childhood.

In 2002, the National Institutes of Health sponsored a meeting on the current science of the treatment for children with ASD (Lord et al., 2005). There is no medical or biological treatment for the *core* features of ASD. However, there is medication that ameliorates some of the mood and behavioral problems associated with ASD. Given the heterogeneity of ASD, the specific language difficulties, and the comorbidity of mental retardation, it is not possible to say that a specific type and dosage of treatment is recommended. Interventions should begin as soon as the child is identified. Currently, behavioral interventions have the most empirical support (U.S. Public Health Service, 2001) for the treatment of ASD. Behavioral interventions for ASD have provided an impressive amount of empirical support. Applied behavior

analysis, cognitive-behavioral therapy, and psychopharmacology interventions are all discussed in turn. In addition, a selection of interventions specific to the social, communication, and behavioral impairments are also briefly discussed.

Applied Behavior Analysis

Applied behavior analysis (ABA) is a discipline that seeks to understand and improve human behavior and is grounded in principles of operant and respondent learning and conditioning. This approach has the strongest demonstrated research support for efficacy and has been widely acknowledged by independent reviews as a critical part of a child's treatment program (e.g., the report of the surgeon general; U.S. Public Health Service, 2001). There are guidelines for selecting qualified behavior analysts that provide interventions for individuals with ASD (Autism Special Interest Group of the Association for Behavioral Analysis, 2007). In addition, the Behavior Analyst Certification Board provides certification for behavior analysis practitioners.

In intervention programs that use ABA as a framework, operational definitions of behavior are utilized to objectively measure complex behavior patterns. Baseline data are collected on these behaviors, and interventions based on principles of behavior are implemented. Data collection occurs throughout the intervention. This process allows for the evaluation of the intervention on an ongoing basis, and modifications are made depending on the child's performance. Some of the instructional techniques used in ABA programs include discrete trial training, pivotal response training, incidental teaching, and generalization training.

An important component of ABA interventions for ASD is the use of functional behavioral assessments. Functional behavioral assessments assist in determining the functions of specific behaviors (e.g., self-stimulatory behavior, noncompliance, self-injurious behavior, aggressive behavior). A functional behavioral assessment may include interviews, direct behavioral observations, checklists, and manipulating the antecedents and consequences of a behavior. These methods assist in generating hypotheses of the function of a behavior and guide intervention. Common functions of disruptive behaviors are positive reinforcement, negative reinforcement, and nonsocial reinforcement, sometimes referred to as automatic reinforcement or sensory reinforcement (Iwata, Dorsey, Slifer, Bauman, & Richman, 1994; Reese, Richman, Belmont, & Morse, 2005).

Psychopharmacological Treatments

Psychopharmacological treatments for ASD target specific problematic behaviors. In addition, psychopharmacological treatment may be provided to treat comorbid disorders (such as anxiety or depression). A recent report of the American Psychological Association Working Group on Psychotropic Medications (2006) estimates that between 33% and 47% of individuals with ASD have been prescribed at least one psychotropic medication over the course of a 1-year period. The Working Group indicated that the evidence of the effectiveness of psychotropic medications is limited and variable, and that medication should be considered on an individual basis with careful consideration of the risks and benefits of the medication, expected

behavior change, and side effects. Recently, the Food and Drug Administration approved Risperdal (risperidone) for the treatment of severe irritability in children and adolescents with Autistic Disorder (U.S. Food and Drug Administration, 2006). Table 29.3 summarizes the information provided in the Working Group's report.

Cognitive-Behavioral Therapy

Cognitive-behavioral therapy (CBT) has garnered recent attention for treatment of anxiety for individuals with high-functioning autism and Asperger's Disorder. Sofronoff, Attwood, and Hinton (2005) implemented a 6-week brief CBT intervention for individuals with Asperger's Disorder and anxiety symptoms. These authors found that having the family (i.e., parents) involved led to the best outcomes (i.e., reduction in anxiety symptoms). However, these authors did rely on parent report to demonstrate the change in symptom frequency and severity, which is a limitation of the study.

Cardaciotto and Herbert (2004) modified a CBT intervention to treat social anxiety disorder (Social Phobia) for an individual with ASD who presented with fear of social situations, specifically assertiveness, talking in front of others, and initiating conversations with those he is familiar with. The authors provided 14 weeks of CBT. The outcome was successful and the client showed maintenance of gains. The recent research examining how to effectively modify traditional CBT interventions is a new and exciting area for the field of ASD.

Social Skills Training

There are many different social skills interventions for ASD. Research has shown that individuals with ASD do not learn social skills in the same way as typically developing children (Gresham, 1981). Developmental level and chronological age should be taken into consideration when teaching social skills. For example, social skills interventions during the early intervention years (i.e., 0 to 3 years old) focus on improving social behavior and interactions with an adult (e.g., parent). In the preschool years, the focus is on play skills and peer interactions. More advanced social skills are taught when a child reaches school age, such as initiation of play and conversations and developing friendships (Guralnick & Neville, 1997; Schreibman, 1996).

A review of social skills interventions for preschool-age children by Vaughn et al. (2003) revealed the following common features: prompting and rehearsal of target behaviors, play-related activities, free-play generalization, reinforcement of appropriate behaviors, modeling of specific skills, storytelling, direct instruction, and imitation of appropriate behaviors. The authors' review indicated that the interventions with the highest effect sizes were those that contained features of modeling, play-related activities, rehearsal and practice, and prompting. Based on their review of components of interventions that yielded improvements in social behavior, the authors suggest the following (pp. 12–13):

- Social skills programs that are embedded into regular class programs by teachers (Antia, Kreimeyer, & Eldredge, 1993; Ferentino, 1991; Fewell & Vadasy, 1989; Jenkins, Odom, & Speltz, 1989; Koenigs & Oppenheimer, 1985)

Table 29.3 Psychotropic Drugs Commonly Used in the Treatment of Symptoms of ASD

Drug Class	Specific Drug Studied	Target Behaviors/ Symptoms	Side Effects	Conclusions of APA Work Group
Stimulants	Methylphenidate (Ritalin)	Hyperactivity, inattention, impulsivity	Agitation, irritability, insomnia	48% to 50% experienced positive response; 18% experienced adverse effects.
Atypical antipsychotics	Risperidone*	Aggression, severe tantrum behavior	Drowsiness, fatigue, constipation, weight gain, may increase risk for diabetes and hyperlipidemia	Approximately 66% experienced improvement in symptoms. However, symptoms recur after medication is discontinued.
Selective serotonin reuptake inhibitors	Clomipramine, fluoxetine	Perseverative behaviors (e.g., compulsions, self-injury, stereotypies)		Limited evidence. Fluoxetine appears to be better tolerated than other SSRIs. Most studies are with adults; little is known about effects on children and adolescents.
Mood stabilizers	Lithium, divalproate	Explosive aggression, severe tantrum behavior		No conclusive evidence for children and adolescents. Only small studies have been conducted with adults, showing preliminary support.
Alpha agonists	Clonidine, guanfacine	Hyperactivity, aggression, severe tantrum behavior		Limited support in open-label, uncontrolled reports.

*Risperdone is now (as of October 6, 2006) FDA approved for the treatment of irritability associated with children and adolescents with Autistic Disorder (http://www.fda.gov/bbs/topics/NEWS/2006/NEW01485.html).

Adapted from *Psychopharmacological, Psychosocial, and Combined Interventions for Childhood Disorders: Evidence Base, Contextual Factors, and Future Directions,* by the American Psychological Association Working Group on Psychotropic Medications for Children and Adolescents, 2006, Washington, DC: Author.

- Interventions that include both instruction in social skills and behavioral contingencies for appropriate and inappropriate behaviors (LeBlanc & Matson, 1995; Matson, Fee, Coe, & Smith, 1991)
- Interventions that provide integrated or social interaction groups for children with and without disabilities (Jenkins et al., 1989)
- Interventions that train parents or peers as models to promote appropriate social behaviors (Dawson & Galpert, 1990; Strain, 1985)
- Interventions that involve an intensive social skills program (Rogers, Herbison, Lewis, Pantone, & Reiss, 1986)

Social skills interventions for individuals with ASD described as higher functioning have included group interventions (Howlin & Yates, 1999; Krasny, Williams, Provencal, & Ozonoff, 2003; Solomon, Goodlin-Jones, & Anders, 2004). Skills typically taught within the context of group interventions include awareness of emotions, face processing and nonverbal communication, theory of mind and perspective taking, conversations, problem solving, stress management, and beginning conversation skills (Solomon et al., 2004).

Communication

It has been observed that approximately 70% of children between the ages of 3 and 5 years with a disability have a delay in language development (Wetherby & Prizant, 1992). Thus, discovering new, innovative ways to teach individuals with ASD to communicate more effectively has been a priority in ASD treatment research. Most comprehensive treatment and educational programs for individuals with ASD include a communication acquisition component. Depending on the strengths and weaknesses of the individual, verbal language, nonverbal language, and social communication acquisition might be emphasized. Successful verbal language acquisition has been documented using interventions such as discrete trial training, incidental teaching, natural language teaching procedures, mand training (Peterson, 2004), and pivotal response training (Koegel, O'Dell, & Koegel, 1987).

The use of visual supports has been recommended and is frequently present in research studies and clinical settings. One such visual support is an activity schedule (McClannahan & Krantz, 1999). An activity schedule provides visual information (in the form of pictures, symbols, or text) to an individual with ASD. This information can be utilized to teach a skill (skills can range from daily living skills to academic skills, self-management skills, social skills) and to help individuals with ASD predict and plan a sequence of events (e.g., daily classroom activities). Research suggests that integrating multimedia components into teaching children with ASD shows initial positive results. However, the mechanism of multimedia effects is unknown. Future research should systematically examine whether and how the use of multimedia components is effective.

Even with available interventions, more than 50% of children with autism are nonverbal or mute (Sigman & Capps, 1997). Thus, language interventions are of

great importance to enable these children to use alternative means of communicating with others. Bondy and Frost (1994) developed the Picture Exchange Communication System (PECS) program for children with social communication deficits. This program is appealing to the ASD field for many reasons: (a) The PECS program requires simple motor movements; (b) PECS is portable and can be used across a wide variety of settings; (c) PECS is taught relatively quickly using behavioral principles of shaping, transfer of stimulus control, and differential reinforcement; (d) PECS facilitates child imitations of interactions with others; and (e) PECS has been demonstrated to facilitate verbal language (Charlop-Christy, Carpenter, Le, LeBlanc, & Kellet, 2002). The structured context and use of pictures in the PECS program is appealing because of the visual modality strength in children with ASD. The program enables a functional relation between the environment and the student to develop.

Behavior

Treatment for aggression is commonly requested from families of individuals with ASD. Because of the multiple functions that aggression can serve, no one specific treatment method is suggested. Behavioral and psychopharmacological treatments are typically used to treat aggression. Matson, Dixon, and Matson (2005) reviewed common behavioral interventions that have shown effectiveness for decreasing aggression, such as positive reinforcement, functional communication training (FCT; Carr & Durand, 1985), time-out, social skills training, and differential reinforcement procedures. The most commonly used interventions are FCT and differential reinforcement of other behavior. The authors emphasize the importance of conducting a functional analysis to understand factors influencing the behavior in order to treat aggression effectively.

Functional communication training (Durand, 1990) is an empirically validated procedure shown to be helpful in the management of challenging behaviors in individuals with ASD (Durand, 1990; Durand & Carr, 1991). This approach incorporates a functional behavior assessment to develop intervention plans to teach individuals with ASD to use alternative skills that decrease disruptive behaviors (Iwata et al., 1994). Thus, based on the functions of a behavior hypothesized from the functional behavior assessment, an individual is taught a more adaptive skill that serves the same function as the behavior. For example, if a child engaged in head-banging to escape a task, the child might be taught to use a Stop sign to signal a request to end a task. Treatment for the behavioral deficits in ASD continues to be an important area for future research.

FAMILY ISSUES

Having a child and/or family member with an ASD imposes a tremendous amount of stress on all family members. Family members may cope and manage stress differently. Because ASD is a lifelong disorder, there are challenges a family faces that vary from year to year, from coping with the diagnosis of an ASD to seeking appropriate educational treatment, dealing with adolescence issues, and changing

living situations when adult children become more challenging for parents. Families also appear to benefit from early intervention services for their children with ASD. The majority of parents indicated that early intervention services decreased the overall stress in the family (Glasberg, Martins, & Harris, 2005).

Lockshin, Gillis, and Romanczyk (2004) identify five general factors that shape family dynamics. The first is a need for continuous supervision of the child with ASD. This can limit the efficiency of completing other tasks. In addition, it requires family members to coordinate who is responsible for the child with ASD at specified times. Second, the inability or limited ability of the child with ASD to communicate the full range of his or her needs and wants can make the family feel that they are in a constant guessing game as to what they need to do to meet the child's needs. Third, atypical relationships with family members sometimes develop given the increased strain on each family member; this can lead to increased family conflict. Fourth, atypical relationships with peers can develop, in which the family members have limited peer and social support outside of the family. Fifth, atypical, difficult-to-manage behavior (e.g., aggression, rigidity, self-injury, lack of social boundaries) may require extraordinary resources for a family and increased stress if these behaviors are not under control in the family and public settings. Assisting families to identify resources in the community is important as a first step in decreasing family stress. It is then helpful to teach family members possible coping strategies. Helping families prioritize their needs so that all members are considered equally is important to decrease conflict and increase support within the family, as well as with friends and peers.

Research in the area of adolescents and adults with ASD is impoverished and is a critical area for growth. For the majority of individuals diagnosed with ASD in childhood, there is consensus that ASD is a lifelong disorder. However, several studies (Byrd, 2002; Mesibov, Schopler, Schaffer, & Michal, 1989; Piven, Harper, Palmer, & Arndt, 1996) have indicated that the core symptoms of ASD do improve with age. That is, the majority of individuals studied had noted improvements in communication, socialization, and behavior in adolescence and adulthood when compared to their functioning in these areas during childhood. Seltzer et al. (2003) found improvement in ADI-R scores in adolescence and adulthood, meaning that these individuals showed a reduction in the number of ASD-related symptoms. Despite these encouraging results, it remains clear that adolescents and adults with ASD continue to need extensive, but different types of, services.

SUMMARY

In the past 2 decades there have been substantial changes in diagnostic criteria and a broadening of the historically limited conceptualization of autism. Pervasive developmental disorders is the current *DSM-IV-TR* terminology which is used in parallel with the more recent and common umbrella term autism spectrum disorders. The ASD conceptualization emphasizes a focus on the social deficits that form the core deficit. There have been many advances in the field of ASD, and there are many exciting areas for future research. Increased availability and utility of technology, from functional magnetic resonance imaging to PDAs, will assist in measurement

that provides information that was not available to researchers 10 (even 5) years ago. Conceptualizing the pervasive developmental disorders as disorders that are on a continuous spectrum changes how we ask questions, think about these disorders, and define the population, and in the future will most likely eventually lead to a further delineation of specific subtypes of ASD.

In the area of intervention, although effective intervention procedures have been identified, more limited efficacy of the generalization of skills acquired is often observed. In addition, more detailed understanding of specific characteristics of individuals with ASD may guide choice of treatment for practitioners, parents, clinicians, therapists, and teachers. We do not yet understand the mediators that affect treatment outcome. There are well over 250 treatments available to individuals with ASD, with the vast majority not effective and wasteful of time and resources. But efficacy information often is overshadowed by marketing efforts by proponents of this vast array of treatments, and thus the treatment choice process is often stressful and difficult for parents and advocates for individuals with ASD. Demonstrating empirical evidence for available treatments and the ethical and responsible dissemination of the evidence is key to helping individuals with ASD and their families.

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CHAPTER 30

Parental Psychopathology and Its Relation to Child Psychopathology

Sherryl H. Goodman and Sarah R. Brand

Regardless of one's theoretical perspective, psychopathology in parents raises many concerns about children's psychological well-being. In this chapter, we focus on two of the most prevalent disorders in parents, depression in mothers and alcohol disorders in fathers, and examine the evidence for their effects on various aspects of children's psychological functioning. We also review the mechanisms or processes that best explain those outcomes: the mediators. Because not all children of parents with psychopathology develop psychopathology themselves, we also review the findings on the moderators: characteristics and circumstances that influence the strength of those associations. Finally, we suggest some areas for further research and some implications for clinical practice and policy. Throughout this chapter, we take a developmental psychopathology perspective.

Several sets of findings guided our decision to focus on depression in mothers and alcohol disorders in fathers. With regard to depression, it is common, especially among women of childbearing and child-rearing ages, severe and impairing, and recurrent or persistent. It is not surprising that its presence in mothers has fueled concern for its potential to disrupt aspects of caretaking known to be critical for healthy child development. Knowledge of the heritability of depression adds further concern for the children of depressed parents. More recently, emerging knowledge of the neuroendocrine correlates of depression during pregnancy has raised concern about possible adverse influences even on fetal development. Many of these issues also pertain to the study of paternal alcoholism and the effect on child development. Although mothers have been the dominant focus of high-risk studies, research on parental alcohol abuse has tended to focus more heavily on fathers, likely because of the greater prevalence of such disorders in men (Phares, 1992; Phares & Compas, 1992). Advances in genetics research have also shown a high genetic risk of alcoholism, especially in men passing on the genes to their sons. Also, although mothers remain the primary caregivers for children in the United States, fathers are becoming more involved and taking on greater caregivng roles. Thus, the impact of the environmental correlates of alcoholism in fathers on the development of psychopathology in children will only grow.

Data are consistent with the suggestion that depression in women may be more of a concern than depression in men. That is, although depression in both mothers and fathers has been shown to affect children's psychological functioning, a meta-analysis of 134 samples, with a total of over 60,000 parent-child dyads, showed that both internalizing and externalizing problems in children are more strongly associated with depression in mothers relative to depression in fathers (Connell & Goodman, 2002). On the other hand, the meta-analysis found that alcoholism and substance abuse disorders in mothers were more closely related to externalizing (but not internalizing) problems in children than were such disorders in fathers. This finding is contrary to our premise that paternal alcoholism would be more strongly linked to children's (especially boys') behavior problems, based on both genetic mechanisms of transmission and greater prevalence of alcoholism in men than women. However, the magnitude of the differences was small, and the meta-analysis findings may reflect the adverse influence of maternal substance use during pregnancy on prenatal development, which is later reflected in psychopathology in the children.

PREVALENCE

One reason to focus on depression in mothers is its prevalence in women of childbearing and child-rearing age. Depression is one of the most common psychiatric disorders in all adults. Among women, between 6% and 17% experience an episode of major depression at some point in their lifetime, a rate that is between 1.5 and 3 times higher than in men (Kessler, 2006). Moreover, depression is the leading cause of both economic and social disability in early to middle adulthood (Wilhelm, 2006), the primary childbearing and child-rearing years. The aspect of disability that has particular implications for depression in children of depressed mothers is depression's interference with quality of parenting, which has been the subject of a large body of research that will be reviewed later in this chapter and was the subject of a recent meta-analysis (Lovejoy, Graczyk, O'Hare, & Neuman, 2000).

Alcoholism is also highly prevalent in the U.S. population, with an estimated 8.45% (17.6 million) adult Americans meeting *Diagnostic and Statistical Manual of Mental Disorders* (*DSM-IV*; American Psychiatric Association, 1994) criteria for alcohol abuse or dependence during a 12-month period (Grant, Stinson, Dawson, Chou, Dufour, et al., 2004). As many as one in four children under 18 are exposed to alcohol abuse or dependency within their family (Grant, 2000). Because men show much higher rates of alcohol-related disorders than women, it is likely that a majority of those children are living in households with an alcoholic father.

THE NATURE OF THE DISORDERS

Researchers define clinically significant depression as either the set of symptoms, duration, and impairment by which an individual meets *DSM-IV* (American Psychiatric Association, 1994) criteria for a mood disorder or gets a high score on a self-report symptom rating scale of depression. The *DSM-IV* divides mood disorders into depressive disorders (sometimes referred to as unipolar depression) and Bipolar Disorders. Bipolar Disorder, which requires the presence of one or more

manic or hypomanic episodes, although a concern for its impact on children, has less often been the focus of studies of maternal depression and will not be discussed in this chapter.

The *DSM-IV* divides alcohol-related disorders into two separate categories: alcohol dependence and alcohol abuse. Like other substance abuse disorders in the *DSM-IV,* alcohol abuse requires fewer symptoms and therefore may be less severe than dependence. Alcohol abuse is diagnosed only once the absence of dependence has been established (American Psychiatric Association, 1994). We include studies of both disorders in this chapter.

In the research literature on alcoholism, the issue arises about how to best characterize alcoholics who also have antisocial personality traits. Data from the National Epidemiologic Survey on Alcohol and Related Conditions showed about 28% co-occurrence between personality disorders and alcohol disorders. Antisocial Personality Disorder occurred in 12.3% of alcoholics, both men and women (Grant, Stinson, Dawson, Chou, Raun, et al., 2004). When these traits were examined for their contribution to child outcomes independent of the influence of alcoholism, they were usually found to contribute more to negative child outcomes than the alcoholism. However, many studies do not assess for these traits or do not make it clear if fathers were excluded based on high antisocial personality characteristics. In this chapter, studies that also examine antisocial personality traits are noted. Caution is needed in interpreting findings from studies when measures of such traits were not included.

THEORIZED MECHANISMS/MEDIATORS

Goodman and Gotlib (1999) proposed a model to explain how depression in mothers might contribute to the development of psychopathology in children (see Figure 30.1). We use that model to structure our summary of the current status of evidence for the most likely mechanisms for the effects of maternal depression on children. We also begin with that model for examining evidence for various mechanisms of transmission in the effects of paternal alcoholism on children and modify the model in theory- and data-guided manners.

When reviewing the literature regarding the influence of paternal alcoholism on child development, some issues arose that are important for readers to keep in mind. First, in many cases, only sons of alcoholics were studied. This is most likely due to the strong findings of the genetic contribution to the development of alcoholism in men (Tyndale, 2004). Second, more often than in the parental depression literature, many studies of parental alcoholism failed to distinguish whether the alcoholism was in the father or the mother. We included such studies, but they clearly dilute the potential to inform the literature on paternal alcoholism specifically.

Biological Mechanisms for Effects of Depression in Mothers

Work on biological mechanisms that might explain the development of psychopathology in children of depressed mothers has focused on two lines of research. First, much research has centered on genetic influences, not only of depression as a disorder but also of vulnerabilities to depression and to other disorders. A

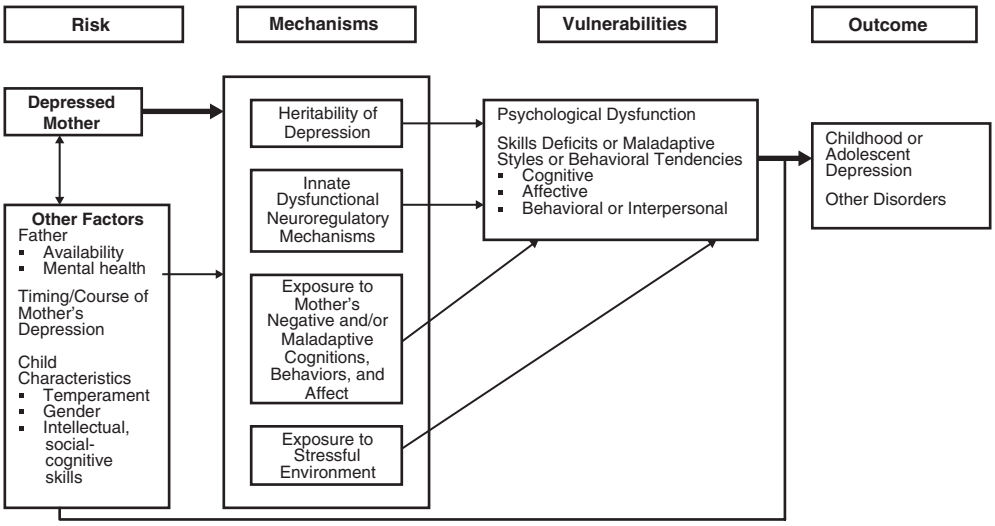

Figure 30.1 Integrative model for the transmission of risk to children of depressed mothers. *Source:* "Risk for Psychopathology in the Children of Depressed Mothers: A Developmental Model for Understanding Mechanisms of Transmission," by S. H. Goodman and I. H. Gotlib, 1999, *Psychological Review, 106,* pp. 458–490. Reprinted with permission.

second line of research has investigated the idea that fetal or postnatal exposures or experiences among children with depressed mothers might adversely influence brain development or hormonal functioning related to neuroregulatory mechanisms, thereby increasing risk for the later development of psychopathology. Support for both of those lines of research is reviewed in the following sections.

Genetic Influences

Research based on family studies, quantitative genetic studies (twin and adoption designs), and molecular genetic studies all provide strong support for heritability of depression in adults, although some evidence suggests that the specific genes related to risk for depression may differ for men compared to women (Caspi et al., 2003; Kendler, Gardner, Neale, & Prescott, 2001). Evidence for the heritability of depression in children and adolescents, however, is mixed. Based on family studies, strictly defined depression in children and adolescents is associated with increased rates of depression in relatives (Harrington, 1996; Harrington et al., 1997). Based on twin studies, heritability is higher in those whose levels of symptoms were below rather than above clinical cutoffs (Rende, Plomin, Reiss, & Hetherington, 1993), for depression in adolescents rather than in children (Eley & Stevenson, 1999; Murray & Sines, 1996; Scourfield et al., 2003; Silberg et al., 2001; Thapar & McGuffin, 1994), for parent-reported rather than child-reported symptoms (Eaves et al., 1997), and for parent-reported symptoms in girls relative to boys (Murray & Sines, 1996; Scourfield et al., 2003). In addition, because heritability of depression is probably not specific to depression, heritability likely contributes to risk for other disorders that are identified in higher rates in children of depressed mothers relative to controls (Moldin, 1999; Tsuang & Faraone, 1990). Thus, heritability likely contributes to

the risk for the development of psychopathology in children and adolescents with depressed mothers, although more research is needed to clarify the nature, extent, and specificity of the contribution.

An alternative to the notion that children with depressed mothers inherit a likelihood for depression per se is the idea that heritability contributes significantly to vulnerabilities to depression and other disorders, such as those proposed in the Goodman and Gotlib (1999) model. In particular, based on behavior genetics studies, high levels of heritability are found for behavioral inhibition and shyness (Cherny, Fulker, Corley, Plomin, & DeFries, 1994), low self-esteem (Loehlin & Nichols, 1976), neuroticism (Tellegen et al., 1988), sociability (Plomin et al., 1993), subjective well-being (Lykken & Tellegen, 1996), and expression of negative emotion (Plomin et al., 1993). This body of research suggests possible genetic transmission of affective, cognitive, and interpersonal vulnerabilities for the development of psychopathology.

Neuroregulatory Mechanisms

Biology may also play a role in the development of psychopathology in offspring of depressed mothers in that both prenatal and postnatal experiences might influence aspects of hormonal functioning and brain development, each of which may increase vulnerability to depression and other stress-related disorders in particular (Ashman & Dawson, 2002). Much of the evidence for this premise is based on research examining the role of early (neonatal) life stress in the development of psychopathology, emphasizing neuroregulatory processes and especially hypothalamic-pituitary-adrenocortical (HPA) axis functioning (Graham, Heim, Goodman, Miller, & Nemeroff, 1999). Typically measured in terms of abnormal cortisol responses to stress but sometimes as a baseline cortisol measure, HPA functioning has been associated with depression in adults (Ressler & Nemeroff, 2000) and may suggest vulnerability mechanisms for the development of depression and other disorders in children of depressed mothers. Stress-related physiological dysregulation in children may be a marker of children's sensitivity to stress, suggesting a particular pathway through which maternal depression during pregnancy and the postpartum may increase children's later development of depression (Sanchez, Ladd, & Plotsky, 2001).

Studies from Field's (2002) lab show that depression during pregnancy increases women's levels of cortisol. Other relevant findings are that pregnant women's levels of cortisol account for 50% of the variance in the fetus's levels of cortisol (Glover, 1999), that mothers' prenatal levels of cortisol predict newborns' cortisol levels (Lundy et al., 1999), and that number of months depressed during pregnancy predict preschool-age children's baseline cortisol levels (Ashman & Dawson, 2002). Elevated stress hormone levels, such as cortisol, in pregnant women are of concern because they have been associated with impaired neurodevelopment in the fetus (Wadhwa et al., 2002).

Postnatal exposure to stressors associated with maternal depression may also disrupt HPA processes (Ashman & Dawson, 2002). In particular, depressed mothers' insensitive or unresponsive behavior, discussed in a later section of this chapter,

may mediate the association between maternal depression and children's emotion regulation abilities, which in turn increases risk for the later development of psychopathology.

Along with disruptions in hormonal functioning, abnormal brain development (in frontal lobe activity) is another aspect of neuroregulatory functioning that may play a role in the development of psychopathology in children of depressed mothers. Researchers have been particularly interested in the abnormal EEG pattern of relatively greater baseline right (compared to left) frontal activation. This pattern is a stable individual difference at least by adolescence (Tomarken, Davidson, Wheeler, & Kinney, 1992) and has been associated with the tendency in children to experience withdrawal emotions such as sadness and fear (Davidson & Fox, 1989) and with depression in adolescents and adults (Davidson, Ekman, Saron, Senulis, & Friesen, 1990). Dawson and colleagues showed associations between frontal lobe activity and affective expressions in infants of depressed mothers (Dawson, Grofer Klinger, Panagiotides, Hill, & Spieker, 1992) and, further, that 3-year-olds' frontal brain activation mediated the relation between maternal depression and children's level of behavior problems (Dawson et al., 2003). That study was not designed to explain the relative (and likely interacting) influences of genetics and fetal and early postnatal experiences on the abnormal brain activation patterns. That is an especially important question given that the frontal lobe develops rapidly during the first 2 years of life (Chugani & Phelps, 1986), suggesting sensitivity to early adverse experiences such as exposure to a mother whose depression interferes with her providing the sensitive and responsive parenting required by infants. Also needed is more research on the extent to which the same or different predictors explain children's stress hormone functioning relative to their brain activation patterns. In 6-month-old infants, these two aspects of neuroregulatory functioning were found to be related; higher basal and stressor reactive cortisol levels were associated with relatively greater right EEG asymmetry (Buss et al., 2003).

Biological Mechanisms for Effects of Alcohol-Related Disorders in Fathers

Work on biological mechanisms that might explain the development of psychopathology in children of alcoholic fathers has focused primarily on genetics but also on electricortical arousal. Support for both of those lines of research is reviewed next.

Genetic Influences

Alcohol dependence is known to be highly heritable, with twin and adoption studies demonstrating that between 50% and 60% of phenotypic variations can be accounted for by a genetic component (Higuchi, Matsushita, & Kashima, 2006; McGue, 1999). Classic twin study estimates of the heritability of alcoholism in men are higher but also more variable (48% to 73%) than in women (51% to 65%; Tyndale, 2004). The variability is likely explained by differences in the classification of alcoholism subtypes. Cloninger and colleagues (Cloninger, Bohmann, & Sigvardsson, 1981) classified alcoholism into two subtypes associated with

variations in gender-specific heritability. Type I alcoholism (milieu-limited) is moderately heritable for both males and females, whereas Type II alcoholism (male-limited) is strongly associated with genetic factors in male offspring and is associated with an early age of onset and with elevated rates of externalizing behavior problems in fathers (Sigvardsson, Bohman, & Cloninger, 1996). Overall, the genetic contribution to alcohol dependence in males has been very well established (for review, see Higuchi et al., 2006) and is likely to be a particularly important mechanism in the transmission of alcoholism from fathers to sons. Genetic factors are also likely to bear primary responsibility for the association between paternal alcoholism and child Conduct Disorder (Haber, Jacob, & Heath, 2005).

Although traditional twin and adoption studies are particularly valuable for estimating heritability, they also play an essential role in revealing the relative contributions of shared and nonshared environment (Hopfer, Crowley, & Hewitt, 2003). Of particular interest from a psychosocial research perspective is altered family environments associated with paternal alcoholism and their role, independent of or in interaction with genes, in explaining the development of alcoholism (Jacob et al., 2003). In an effort to address this broader question, Jacob and colleagues used an offspring-of-twins design and assessed adult male monozygotic and dizygotic twins concordant or discordant for alcohol dependence and their biological offspring, examining both genetic and familial influences. Consistent with other genetics studies, offspring of alcoholic fathers were at a significantly higher risk for the development of an alcohol use disorder than were offspring of controls. Interestingly, however, the researchers found that the complete absence of family environmental correlates of alcoholism, even when high genetic risk is present, resulted in offspring alcoholism levels not significantly different from those of controls. If these findings are replicated, it could show that a genetic risk for alcoholism is not sufficient for offspring alcoholism to develop in the absence of significant environmental sequelae (Jacob et al., 2003). The relative contribution of heritability and environment may also vary depending on the particular outcome of interest and on developmental considerations. For example, in a large study of 16-year-old twins, shared environmental effects best explained drinking initiation, whereas genetic effects better explained frequency of drinking among subjects who had already started drinking (Viken, Kaprio, Koskenvuo, & Rose, 1999).

Despite the strong evidence for a genetic contribution to alcohol dependence in males, specific genes that alter the risk have been difficult to identify. This is partially due to the additional influence of environmental components as well as the understanding that multiple genes that have moderate effects are involved in alcoholism, and these different sets of genes may contribute to the disorder in different people (Higuchi et al., 2006). Thus more research is needed not only on the nature and extent of genetic contributions but also on the likely complex interplay between genes and environments.

Other Biological Mechanisms

Much of the research examining possible psychobiological risk factors for the development of alcoholism has focused on individual differences in electricortical

arousal, most often using the P300 responses. P300 is a positive event-related potential response wave that in itself is known to be highly heritable. The electricortical research has been focused on the P300 response because of its identification in working memory and attentional resource allocation and its reduced amplitude in recovering alcoholics (Windle & Tubman, 1999). To determine whether this reduced amplitude is a consequence of frequent inebriation or could be considered a risk factor for the development of alcoholism, children of alcoholics have been studied prior to their onset of drinking. Numerous such studies (see Windle & Tubman, 1999, for review) have found significant differences between children of alcoholics and children of nonalcoholics, with the children of alcoholics showing a reduced P300 response in visual task paradigms, which may generalize across the visual and auditory sensory modalities. Benegal and colleagues (Benegal, Jain, Subbukrishna, & Channabasavanna, 1995) took this a step further by showing that the P300 component was even smaller in children with multiple alcoholic relatives in comparison with children with only a single relative. These findings are consistent with the idea that reduced P300 amplitude is a risk factor for the development of an alcohol-related disorder.

Anecdotal reports from alcohol dependent children of alcoholics led researchers to consider one other possible biological mechanism for the development of alcoholism in children of alcoholic fathers. These narratives frequently describe children of alcoholics' abilities from a young age to consume large amounts of alcohol with few effects. Schuckit and colleagues (1996) examined the possible role of differences in alcohol tolerance by assessing time-to-peak alcohol concentration and the rate of disappearance of ethanol from the blood following the challenge in 453 sons of alcoholics and controls. Because of the confounding effect of fetal alcohol syndrome, the authors placed an emphasis on examining offspring of alcoholic fathers. There were no significant differences in blood alcohol concentration in the two groups, yet about 40% of the sons of alcoholics, but fewer than 10% of controls, reported low subjective feelings of intoxication (Schuckit & Smith, 1996). This finding has been replicated in many studies as well as confirmed by a meta-analysis (Pollock, 1992) and gives weight to the theory that sons of alcoholics possess less subjective (self-rated) sensitivity to alcohol effects than do controls. These findings have been interpreted to suggest that these individuals are more vulnerable to developing an alcohol-related disorder because they are less sensitive to alcohol effects and therefore find it more difficult to regulate their alcohol consumption because they are receiving less internal feedback. Thus, they may be more likely to consume more alcohol than those without a family history of alcoholism, which may contribute to their development of dependence or abuse (Pollock, 1992). The data do not allow us to conclude whether the findings are specific to sons or might generalize to daughters.

Quality of Care Mechanisms for the Effects of Maternal Depression

Social Learning Processes

Exposure to depressed mothers' negative affect, behavior, and cognitions is another potential mechanism for transmission of the risk of depression from parents

to offspring (see Figure 30.1). Through the social learning processes of modeling, observational learning, and reinforcement, children of depressed mothers may acquire cognitions, behaviors, and affect that resemble those of the depressed parent. These acquired depressotypic cognitions, behaviors, and affect may place the children at elevated risk for developing depression in particular, but also potentially other disorders.

Children have ample opportunities to be exposed to negative cognitions, behaviors, and affect associated with depression in their mothers. In terms of affect and behavior, a meta-analysis of 46 observation studies revealed that maternal depression is strongly associated with negative (hostile or coercive) parenting behavior, is moderately associated with parenting characterized as disengaged, and shows a small association with lower levels of positive behavior (Lovejoy et al., 2000). In terms of cognitions, the negatively biased cognitions associated with depression in adults (Gotlib, Gilboa, & Sommerfeld, 2000) emerge in parenting in depressed mothers' (a) more negative views of themselves as parents (Gelfand & Teti, 1990; Goodman, Sewell, Cooley, & Leavitt, 1993), (b) lower confidence in being able to positively influence their children (Kochanska, Radke-Yarrow, Kuczynski, & Friedman, 1987), and (c) more negative, critical perceptions of their children (Goodman, Adamson, Riniti, & Cole, 1994) relative to controls.

Researchers have linked these specific aspects of depression in mothers with the development of psychopathology in children and adolescents. Depressed mothers' negative, critical perceptions of their children are associated with children's self-blaming attributions for negative events and with lowered perceived self-worth (Goodman et al., 1994; Jaenicke et al., 1987; Kochanska et al., 1987; Radke-Yarrow, Belmont, Nottelman, & Bottomly, 1990). Children seem to internalize their depressed mother's negative views of them, thereby acquiring a cognitive vulnerability to depression (Beck, Rush, Shaw, & Emery, 1979). Depressed mothers' greater criticism and hostility (Frye & Garber, 2005; Webster-Stratton & Hammond, 1988) may also contribute to the emergence of coercive processes similar to those implicated primarily in the development of externalizing behavior problems but also of depression (Davis, Sheeber, & Hops, 2002; Hops, Sherman, & Biglan, 1990).

Inadequate Parenting

In addition to social learning processes, other mechanisms that may explain how qualities of parenting contribute to the development of problems in children focus on inadequate parenting (especially if early in life) as a stressor. Although not incompatible with social learning processes, these studies stem from a different tradition of research, emphasizing interpersonal processes such as the depressed mother being an inadequate social partner for the child and the mother being unable to meet the child's stage-salient social and emotional needs (Cicchetti & Toth, 1998). This inadequate parenting would, in turn, negatively affect the child's development of social and cognitive skills and beliefs and leave him or her vulnerable to the development of psychopathology.

In this line of research, inadequate parenting refers to the mother being unable to meet the needs associated with healthy psychological development specific to different phases of child development. Researchers have particularly emphasized

infancy given infants' dependence on their caregivers to meet their basic needs (Sroufe, Egeland, Carlson, & Collins, 2005). This research is consistent with the idea that depression in mothers is a stressor for their infants, and thus is related to the literature on HPA activity and EEG asymmetries reviewed earlier. Mothers may be stressors as a function of being unresponsive or inconsistent (unpredictable in their responsiveness) in relating to their infants, thereby interfering with infants' development of emotion regulation skills (Fox, 1994) and secure attachment relationships (Cicchetti, Rogosch, & Toth, 1998; Sroufe & Waters, 1977). Both of the latter are known to increase risk for the development of psychopathology in childhood or adolescence (Southam-Gerow & Kendall, 2002; Thompson, 2006).

Stress

In addition to the notion of parenting as a stressor, especially for infants, maternal depression increases children's exposure to stress in several other ways, which also may mediate associations between maternal depression and the development of psychopathology in offspring (Hammen, 2002). These include the stressfulness for children of the symptoms and episodic course of depression in their mothers (Compas, Langrock, Keller, Merchant, & Copeland, 2002), the chronic and episodic stressors that are often the context for depression (Monroe & Hadjiyannakis, 2002), and the stress-generating quality associated with depression (Hammen, 1991). Although these findings are consistent with the notion of stress as a mechanism, few studies have actually tested stress as a mediator in associations between depression in mothers and outcomes in children. One of the few such studies found support for the role of stressors as a mediator of the association between maternal depression and children's depression (Brennan, Hammen, Katz, & Le Brocque, 2002).

Quality of Care Mechanisms for the Effects of Alcoholism in Fathers

Because mothers are still the primary caregivers in most families, much of the research about alcoholism and parenting focuses specifically on the effects of maternal alcoholism or on familial alcoholism. Although we emphasize studies focusing solely on paternal alcoholism, we included studies involving familial alcoholism to illustrate the effect of alcohol-related disorders on quality of care given to the child.

Although there is a large range in quality of caregivng in families with an alcoholic parent, on average researchers find significantly poorer quality compared to controls. In the few studies involving observations of father-child interaction, paternal alcoholism was found to be associated with problematic parenting behaviors during unstructured free play with an infant, with alcoholic fathers showing higher negative affect, less sensitivity, lower positive engagement, and lower amount and quality of verbalizations relative to controls (Eiden, Edwards, & Leonard, 2002; Leonard & Eiden, 2002). Children of alcoholic parents are also exposed to significantly greater adverse circumstances. Similar to the studies of the role of stress as a mechanism in children of depressed mothers, one model in alcoholism studies is that the problems in family functioning associated with alcoholism in a parent

influence the expression of internalizing and externalizing problems among children of alcoholics by acting as a significant stressor for the child (Windle & Tubman, 1999). In homes where one or both parents are alcoholic, both child and parent reports reveal significantly poorer quality of home environment and more household dysfunction as well as exposure to more marital conflict, parent-child conflict, abuse (verbal, physical, and sexual), and domestic violence than in homes where neither parent is an alcoholic (Anda et al., 2002; Reich, Earls, & Powell, 1988). Among these adversities, much research has focused on the role of marital conflict as one of the mechanisms through which children of alcoholic parents are harmed. Marital conflict is related to ineffective parenting (including inconsistent discipline and psychological control), which in turn is associated with children's internalizing and externalizing problems (Cummings, Keller, & Davies, 2005). Thus parents' problematic drinking may be associated with children's behavior problems indirectly, through mediating variables such as marital conflict (Christensen & Bilenberg, 2000; Snow Jones, Miller, & Salkever, 1999).

Patterson and colleagues' (Patterson, Reid, & Dishion, 1992) well-established coercion model, which explains the relationship between parenting influences and childhood conduct disorders, is a useful theoretical model in that children's conduct problems increase the likelihood of alcohol or drug abuse. In this model, the most relevant force in determining a child's development of psychopathology is disturbances in the control dimension of parenting, especially a coercive parent-child interactional style, similar to the pattern that has been identified in studies of parenting in homes with an alcoholic parent. This style is marked by unclear behavioral expectations from the parent and lack of compliance by the child. The continuation of this coercive relational style results in inadequate parental monitoring, supervision, and discipline, which can lead to antisocial behavior in the child, which, in turn, increases the likelihood of the child's alcohol abuse (Patterson et al., 1992).

EVIDENCE FOR EFFECTS OF MATERNAL DEPRESSION AND PATERNAL ALCOHOLISM ON CHILDREN

Given the size and complexity of the literature, we begin with evidence for the effects of maternal depression and then separately review evidence of the effects of paternal alcoholism. Within each of these sections, we present data on effects on children's social, emotional, and cognitive development, which may be vulnerabilities for the later development of psychopathology, and then on psychopathology per se.

Evidence for Effects of Maternal Depression

Effects on Children's Social, Emotional, and Cognitive Development

Children of depressed mothers show problems in each of these aspects of functioning relative to controls. On social and interpersonal functioning, infants show less secure attachment relationships (van IJzendoorn, Goldberg, Kroonenberg, & Frenkel, 1992); preschool-age children engage in excessive compliance with their

depressed mothers (Zahn-Waxler, Radke-Yarrow, Wagner, & Chapman, 1992); kindergarten-age children are more often excluded by peers (Cummings et al., 2005); and sons, but not daughters, display more aggressive behavior during interactions with friends (note that the last effect was mediated by the children's exposure to interparental conflict; Hipwell, Murray, Ducournau, & Stein, 2005).

On emotional development, infants imitate their depressed mothers' negative affect (Field, 1994). Infants of depressed mothers show more negative affect and more self-directed regulatory behaviors (Field, 1992; Tronick & Gianino, 1986); toddlers show more dysregulated aggression and heightened emotionality (Zahn-Waxler, Cummings, Iannotti, & Radke-Yarrow, 1984); and adolescents (particularly girls) display more dysphoric and less happy affect (Hops et al., 1990). Infants interacting with their depressed mothers also engage in more gaze and head aversion, consistent with the idea that the infants were using self-regulatory behaviors to minimize the negative affect associated with maternal unresponsiveness (Pickens & Field, 1993).

Children of depressed mothers score lower on measures of intelligence and also have poorer academic performance overall (Anderson & Hammen, 1993; Hammen & Brennan, 2001; Hay & Kumar, 1995; Jaenicke et al., 1987). On other aspects of cognitive functioning, children of depressed mothers show early signs of cognitive vulnerability to depression, including being more likely than controls to blame themselves for negative outcomes, having a more negative attributional style, being less likely to recall positive self-descriptive adjectives, and having a lower self-concept (Garber & Martin, 2002).

Effects on the Development of Psychopathology

Beginning at the earliest ages when it is possible to reliably measure psychopathology in children, children and adolescents with depressed mothers show higher rates of depression relative to a variety of controls (Beardslee et al., 1988; Billings & Moos, 1985; Goodman et al., 1994; Lee & Gotlib, 1989; Malcarne, Hamilton, Ingram, & Taylor, 2000; Orvaschel, Walsh-Allis, & Ye, 1988; Weissman et al., 1984; Welner, Welner, McCrary, & Leonard, 1977). Rates of depression in the school-age children of depressed mothers vary from 20% to 41%, depending on the severity or impairment of the parent's depression, whether both parents are depressed, and a number of other sociodemographic variables (Gotlib & Goodman, 1999). Moreover, depression in offspring of depressed parents, compared with same-age offspring of nondepressed parents, has an earlier age of onset and longer duration and is associated with greater functional impairment and higher likelihood of recurrence (Hammen & Brennan, 2001; Hammen, Burge, Burney, & Adrian, 1990; Hammen, Shih, Altman, & Brennan, 2003; Keller et al., 1986; Lieb, Isensee, Hofler, Pfister, & Wittchen, 2002; Warner, Weissman, Fendrich, Wickramaratne, & Moreau, 1992).

Psychopathology in children with depressed mothers is not limited to depression. Researchers have also found higher rates of Social Phobia, Separation Anxiety Disorder and other anxiety disorders, attention deficit disorders, disruptive behavior disorders, overall levels of behavior problems, and poorer social functioning relative to controls (Anderson & Hammen, 1993; Biederman et al., 2001; Luoma et al., 2001;

Orvaschel et al., 1988; Weissman et al., 1984). In addition, levels of internalizing and externalizing behavior problems are higher in children with depressed mothers relative to controls (Brennan et al., 2002; Fergusson, Lynskey, & Horwood, 1993; Forehand, Brody, Slotkin, Fauber, & McCombs, 1988).

Evidence for Effects of Paternal Alcoholism

Effects on Children's Social, Emotional, and Cognitive Development

Similar to the findings from maternal depression, paternal alcoholism is associated with problematic patterns of father-infant attachment: infant avoidance and infant disorganization as assessed by the Strange Situation paradigm (Eiden et al., 2002; Leonard & Eiden, 2002; Leonard et al., 2000). Teenagers (18 years old) whose fathers were alcoholics rated their attachment to their fathers as significantly more negative than teenagers of nonalcoholic fathers (Cavell, Jones, Runyan, Constantin-Page, & Velasquez, 1993).

In terms of cognitive development, while there is strong empirical support for the severe cognitive deficiencies caused by maternal drinking during pregnancy (Steinhausen, Willms, & Spohr, 1993), the literature surrounding the impact of paternal alcoholism on a child's cognitive functioning is varied. Leonard and Eiden (2002) found no significant difference in the cognitive functioning of toddlers of alcoholic fathers, assessed at 12, 18, and 24 months, relative to toddlers in control families who were matched on maternal education, ethnicity, child sex, and marital status. However, children of alcoholics had poorer intellectual functioning than control children during the elementary school years, although still within the normal range (Poon, Ellis, Fitzgerald, & Zucker, 2000).

When Ozkaragoz, Satz, and Noble (1997) examined differences in neuropsychological functioning of sons of alcoholic fathers, recovering alcoholics, and social drinkers, they found that sons of active alcoholics performed worse on visuospatial, memory, and attention tasks as well as general intellectual functioning. However, no difference in neuropsychological functioning was found between the sons of the recovering alcoholics and those of social drinkers (Ozkaragoz et al., 1997). The latter finding is relevant to moderators, discussed later.

Effects on the Development of Psychopathology

Many studies have shown the link between paternal alcoholism and alcohol-related outcomes in children (Cavell et al., 1993; Chassin, Rogosch, & Barrera, 1991). Paternal alcoholism predicted initial alcohol and drug use in younger adolescents (Chassin et al., 1991) as well as greater alcohol involvement in older adolescents (Cavell et al., 1993). Overall, family alcoholism is associated with a threefold increase in risk for alcohol abuse in young adults (Schuckit & Smith, 1996).

Beyond an increased risk for developing an alcohol-related disorder, a family history of alcoholism put children at a higher risk for developing other psychopathologies, including Conduct Disorder, affective disorder, and Attention-Deficit/Hyperactivity Disorder (ADHD). Edwards, Eiden, and Leonard (2006) found that both male and female 18-month-old toddlers with alcoholic fathers had

higher levels of not only externalizing behaviors but also of internalizing behaviors relative to children of nonalcoholic fathers. Early adult (mean age 19) sons of alcoholic-dependent fathers were also significantly more likely to receive a diagnosis of Conduct Disorder than were matched controls from families without a history of alcoholism, indicating a diagnostic crossover in family transmission (Haber et al., 2005).

Many researchers have hypothesized that one potential pathway for the transmission of the risk of alcoholism from parent to child is the likelihood of early conduct problems among children of alcoholics, which has been linked to greater antisocial behavior in adolescence and adulthood, which, in turn, has been linked to a greater risk of substance abuse (Eiden, Leonard, & Morrisey, 2001). In a study examining toddler compliance at 18 and 24 months and comparing children of alcoholics and children of nonalcoholics, Eiden et al. showed that fathers' alcoholism was significantly associated with the development of compliance problems among boys, an effect more pronounced in the presence of maternal alcohol problems. These boys showed increased levels of resistance to parental requests (during a cleaning-up task) at 18 months and continued to show a higher level of resistance at 24 months. Contrary to expected results, daughters of alcoholic fathers showed the opposite pattern, even when the mother was also an alcoholic. The authors pointed out that the mothers with alcoholism were also more antisocial and aggressive than controls, and therefore the daughters could be displaying compulsive compliance. In general, frequencies of disruptive behavior problems in young children tend to decrease over time, most likely due to increases in self-regulatory capacities, which are developed during the preschool years. However, this decrease can be offset by the presence of paternal alcoholism, as Loukas and colleagues (Loukas, Zucker, Fitzgerald, & Krull, 2003) showed that paternal alcoholism was associated with higher levels of disruptive behavior problems across all time points.

MODERATORS

The processes underlying the developmental pathways that lead to psychological problems in children of depressed mothers or alcoholic fathers are likely to be affected by multiple moderating variables, which not only create individual pathways unique to each child but also more generally exacerbate or lessen the extent of risk associated with the psychopathology in the parent (Kraemer, Stice, Kazdin, Offord, & Kupfer, 2001). A transactional perspective suggests that through a process of continuous reciprocation, characteristics and traits of the child influence how the child experiences the environment, which, in turn, influences the child's characteristics and traits and so forth (Sameroff, 1975). Thus, from the perspective of transactional models, various factors may moderate the associations between psychopathology in a parent and children's development and account for some children functioning well and others developing psychopathology and other problems. These factors may include (a) qualities of the children, such as temperament, sex, and various cognitive variables; (b) the broader context of the children's environment, such as the presence and mental health of the other parent or another adult and the quality of the marital relationship; and (c) characteristics of the parent's psychopathology,

such as timing, course, and chronicity. Much heterogeneity has been noted in studies of children of depressed mothers and of alcoholic fathers. Moderating variables promise to explain much of that heterogeneity. Empirical support for each of these potential moderators is reviewed next, first for children of depressed mothers and then for children of alcoholic fathers.

Moderators of Associations between Maternal Depression and Children's Development

Qualities of the Children

The sex of the offspring of depressed mothers is one characteristic of the child that has been shown to moderate the association between mothers and limited aspects of children's development. Specifically, sex moderates associations between maternal depression and depression in adolescents (Sheeber, Davis, & Hops, 2002). Adolescent daughters of depressed mothers have more dysphoric and less happy affect than adolescent daughters of nondepressed mothers, whereas no significant difference in affect was found for adolescent boys of depressed mothers compared to sons of nondepressed mothers (Allgood-Merten, Lewinsohn, & Hops, 1990). Among the smaller body of support for sex differences among younger children, depressed mothers and their daughters, but not depressed mothers and their sons, have been shown to have synchronous bouts of depression (Radke-Yarrow, Nottelman, Belmont, & Welsch, 1993). Similarly, in community samples, levels of depression in mothers and daughters are interrelated, but levels of depression in mothers and sons are not related (Davies & Windle, 1997; Fergusson, Horwood, & Lynskey, 1995). Increased risk for depression in daughters compared to sons of depressed mothers is likely to be explained by the same mechanisms that explain risk for depression in adolescent girls compared to adolescent boys in general populations: gender-stereotyped socialization processes, especially early socialization of emotion expression, coping styles, and relationship orientation and possibly greater exposure of girls to family discord (Goodman & Tully, 2006; Sheeber et al., 2002).

Children's age is likely a complex moderator, perhaps most likely interacting with other moderators such as sex, as illustrated in Sheeber et al. (2002). In a meta-analysis, we found a negative correlation between children's age and effect size such that maternal depression was more strongly associated with younger children's internalizing and externalizing problems (Connell & Goodman, 2002). Age is also further complicated by the course of maternal depression in that older children are likely to have been more often exposed.

Researchers have studied children's cognitive abilities, such as IQ, and cognitive vulnerabilities, such as depressogenic attributions, as child qualities that may moderate the link between depression in mothers and children's psychological functioning. Children of depressed parents may be protected against adverse outcome concurrently and prospectively if they are more intelligent (Radke-Yarrow et al., 1990) or have better social-cognitive skills (Beardslee, Schultz, & Selman, 1987). Cognitive theories of depression suggest other potential moderating variables, such as children's perceptions and beliefs about their parents' depression, which may guide children's emotional and coping responses and, then, influence the impact of

the mother's depression on the children's development (Beardslee, 1989; Beardslee & Podorefsky, 1988; Compas et al., 2002; Garber & Martin, 2002; Klimes-Dougan & Bolger, 1998; Solantaus-Simula, Punamaki, & Beardslee, 2002).

Although studies have rarely directly tested the moderating role of temperament in associations between depression in mothers and their offspring, research suggests that children with difficult temperaments are more vulnerable to the effects of inadequate parenting, such as that shown by depressed mothers (Goldsmith, Buss, & Lemery, 1997). Other indirect support is that mothers' higher levels of depressive symptoms have been associated with their perceptions of more infant negativity (Pesonen, Raikkonen, Strandberg, Keltikangas-Jarvinen, & Jarvenpaa, 2004) and difficulty (Cutrona & Troutman, 1986; Edhborg, Seimyr, Lundh, & Widstrom, 2000) and with observational measures of infant difficulty (Cutrona & Troutman, 1986). Mothers of more difficult infants also perceive their parenting to be less efficacious, which in turn is linked to depression in mothers (Cutrona & Troutman, 1986; Porter & Hsu, 2003).

The Broader Context of the Children's Environment

Studies of depression are increasingly taking into account the broader context in which depression occurs (Keyes & Goodman, 2006). Researchers sensitive to the importance of context are beginning to find that fathers (their quantity and quality of involvement and their mental health) and marital conflict play important roles as moderators of associations between maternal depression and child outcomes (Goodman & Gotlib, 1999). Assortative mating, the tendency for adults with similar phenotypes to mate more frequently than expected by chance, is pertinent here. This type of mating practice can produce a double jeopardy situation for the children: Not only are the children inheriting a greater genetic predisposition toward psychopathology, but they are also being raised in a home environment that is likely to be highly compromised and have high levels of unpredictability and low levels of parental monitoring (Merikangas, Weissman, Prusoff, & John, 1988; Windle & Tubman, 1999). Assortative mating, in particular the selection of a mate who also has a mood disorder, is known to be a significant factor in mate selection for women with affective disorders (although not as much as for men; Matthews & Reus, 2001). In studies of maternal depression there is some evidence for the ameliorating effect of fathers' mental health (Thomas & Forehand, 1991), although other studies suggest a more complex model may be needed, such as one that takes into account transactional relationships over time (Hops et al., 1990). In addition, an adequate conceptualization of the role of fathers needs to take into account quality and quantity of involvement.

Marital discord is essential to consider in that it is a common contextual factor for depression in women and by itself is related to increased risk for psychopathology in children (Whisman, Weinstock, & Tolejko, 2006). Davies and Windle (1997) found that marital discord significantly mediated the association between histories of maternal depression symptoms and symptoms of depression in children at middle adolescence, but only in girls. Later, Davies and Dumenci (1999) found support for a pathway from marital distress through maternal depression (rather

than vice versa) to adolescents' depression. Moreover, particular conflict strategies that parents use (those involving depressive behaviors such as physical distress, withdrawal, sadness, and fear in contrast to destructive or constructive marital conflict strategies) mediate associations between maternal depression and children's depression (Du Rocher Schudlich & Cummings, 2003). These findings strongly support the importance of context and the need to develop and test models that consider its potentially moderating and mediating roles.

Parental Characteristics as Moderators

Timing and course of mothers' depression may also account for differences in risk among children of depressed mothers. The exact role of timing is likely to be complicated and may vary by child sex (Essex, Klein, Miech, & Smider, 2001). Timing and course are interrelated given that children who are exposed earlier in their lives are also likely to be exposed more often, not only to the depression but also to the stress correlates of depression, such as marital difficulty, all of which place the child at greater risk for psychopathology. Given changes in children's stage-salient needs, the timing of depression episodes in mothers may have implications for the development of specific vulnerabilities in their children. Finally, we know little about the effect on children of the unpredictable course of their mother's depression.

Moderators of Associations between Paternal Alcoholism and Children's Development

Qualities of the Children

As in studies of maternal depression, children's sex, age, and temperament have been studied as moderators of associations between paternal alcoholism and child outcomes. With regard to sex, the literature is limited by the majority of studies on the effect of paternal alcoholism on children having excluded daughters. Although it is commonly concluded that sons of alcoholics have a higher risk for the development of an alcohol-related disorder (Assaad et al., 2003), some studies have shown greater family history effects for women than for men (Sher, Walitzer, Wood, & Brent, 1991). Studies that included both sons and daughters of alcoholics have found support for a moderating role of child sex on the effect of paternal alcoholism. Paternal alcoholism had a greater effect on sons' relative to daughters' risk of developing ADHD depressive symptoms and substance use during adolescence (daughters of alcoholic mothers were more at risk for depression than sons; Chassin, Curran, Hussong, & Colder, 1996; Christensen & Bilenberg, 2000). These findings are consistent with both social-modeling theory, in that alcoholism in the parent of the same sex may have the most negative influences, and genetics in terms of male-specific inheritance.

In terms of age, Connell and Goodman's (2002) meta-analysis of the effect of parental psychopathology on children found that paternal alcoholism was more closely related to emotional and behavior problems in samples of older children, a finding opposite to that for maternal depression. This finding is difficult to interpret given the chronicity of alcoholism in that the older children were likely to have been

exposed for much or all of their lives; the finding may reflect cumulative effects of exposure.

As with maternal depression, we found few studies directly testing temperament as a moderator of the effect of parental alcoholism. In one study, among children of substance abusing fathers, those who had a difficult temperament, relative to others, were at higher risk for the development of a substance abuse disorder (Tarter, Kabene, Escallier, Laird, & Jacobs, 1990).

The Broader Context of the Children's Environment

Assortative mating is at least as relevant to studies of children of alcoholic fathers as with children of depressed mothers (Windle & Tubman, 1999). When studies have examined children raised in families with two alcoholic parents versus one or no alcoholic parents, children of two alcoholic parents were found to have far more behavior problems and earlier age of alcohol intoxication than children raised in families with only one alcoholic parent (McKenna & Pickens, 1983). In one study, for example, in households where both parents met criteria for current alcoholism or at least one parent had an antisocial alcoholic diagnosis, the children were more reactive and showed more externalizing behaviors relative to the children who only had one alcoholic parent or those with no parent with an antisocial alcoholic diagnosis (Wong, Zucker, Puttler, & Fitzgerald, 1999). As well as assortative mating with other alcoholics, partner pairings may occur between alcoholics and partners with another psychiatric disorder such as depression. Conversely, as with maternal depression, a child with an alcoholic father may benefit from an involved, mentally healthy mother. Edwards and colleagues (2006) found that 18-month-old toddlers with alcoholic fathers who had a secure infant-mother attachment at 12 months had fewer externalizing behavior problems at 24 months and fewer externalizing and internalizing problems at 36 months of age, showing a protective factor against the risk of paternal alcoholism.

Parental Characteristics as Moderators

Much of the variability in the literature on associations between paternal alcoholism and child outcomes could be due to authors either failing to identify or failing to test for differences within samples among types of alcoholics, frequency and quantity of drinking, presence of comorbid Antisocial Personality Disorder, and inclusion of recovering alcoholics. In one study of severity of alcoholism, Malone and colleagues (Malone, Iacono, & McGue, 2002) examined the effect of paternal alcoholism on child behavior using a measure of the maximum number of drinks the father had ever consumed in a 24-hour period. The authors suggest that individuals who consume a high number of alcoholic beverages may have developed a significant tolerance to the effects or possess an unusual insensitivity to the negative effects of alcohol. The maximum number of drinks consumed could be related to loss-of-control drinking, a distinguishing feature in Type 1 alcoholism using Cloninger's (1987) classification. Malone et al. found that a father's maximum number of drinks predicted children's externalizing disorders, especially Conduct Disorder and early substance use.

Other researchers studied the effect of severity or chronicity of parental alcoholism on child functioning by comparing children of active alcoholics to those of recovered alcoholics and social drinkers (DeLucia, Belz, & Chassin, 2001; Ozkaragoz et al., 1997). Ozkaragoz et al. compared children who had parents in these three groups and found that only children of active alcoholics showed neuropsychological deficits and reduced intellectual functioning compared to children of recovered alcoholics and social drinkers. On the other hand, when behavioral and emotional problems were compared, there were no group differences between active and recovered alcoholics' children, but these children had significantly more emotional problems than did the children of the social drinkers. DeLucia et al. found that adolescents of recovered alcoholics continued to show more emotional and behavioral problems than children of nonalcoholic parents. Because alcohol abuse and dependence are chronic conditions that can span decades, more studies are needed of the effects of chronicity.

Because many men with alcoholism also display antisocial personality traits, it is essential, though difficult, to tease apart the effect of paternal alcoholism from the effect of the antisocial traits on parental caregivng. For example, Eiden and Leonard (2000) found a significant impact of paternal alcoholism on fathers' aggravation toward their infants; however, this finding was found to be attributable to the antisocial behavior and depressive symptoms. In research on cognitive-intellectual functioning of children of alcoholic fathers, the children of antisocial alcoholics had significantly lower Full-Scale IQ scores (although still within the normal range) than children of nonantisocial alcoholics and controls (Poon et al., 2000). Further, the child's IQ was directly related to the severity of comorbid paternal pathology.

SUMMARY

The field of clinical psychology has increasingly recognized the importance of psychopathology in both mothers and fathers in relation to the development of psychopathology in children (Connell & Goodman, 2002; Phares & Compas, 1992). The goal of this chapter was to examine the effect of parental psychopathology on the development of psychopathology in children, specifically focusing on depression in mothers and alcohol disorders in fathers, as they are two of the most prevalent disorders in adults who are bearing and rearing children. This chapter adds to the emerging literature about the risks to the children, and while providing many answers also raises multiple questions needing to be addressed.

In bringing attention to this literature, we encourage researchers, clinicians, and public policy makers to consider the centrality of sex in all aspects of the issues raised here. First among those issues is the fact that depression is twice as common in women as men, whereas alcohol dependence and the often accompanying Antisocial Personality Disorder are at least twice as common in men as women. These epidemiological patterns suggest that important work needs to be done to balance efforts to minimize sex bias in the recognition, diagnosis, treatment, and study of these disorders, while at the same time recognizing sex-specific aspects of the disorders such as risk factors, symptom patterns, approaches to help seeking,

treatment options offered, barriers, and response to treatment. In this regard, as one example, sex specificity of genetic risks for depression (Kendler & Prescott, 1999) and alcohol abuse (Sigvardsson et al., 1996) are intriguing. Second, sex biases prevail in the studies of these high-risk children, as exemplified by the minimal attention to daughters of alcoholic fathers. Researchers may eventually conclude that sex specificity characterizes some risk and protective factors and pathways in the development of psychopathology in children of parents with psychopathology and identify others that characterize all children. An interesting likelihood is multifinality, the idea that similar risk factors, for example, uncontrollable life events, may increase the likelihood of different outcomes: depression emerging in women and alcohol abuse in men (Cicchetti & Rogosch, 1996). Third, even with the strong evidence of sex-specific prevalence patterns for depression and alcohol abuse in men and women, it is essential that models of the transmission of risk to the children take into account the fact that depression in mothers also affects the husbands and fathers, just as alcohol abuse in men affects the wives and mothers (El-Sheikh & Flanagan, 2001; Whisman et al., 2006). Fourth, researchers and practitioners need to be attentive to the emergence of changing epidemiological patterns such as the possibility that rates of depression worldwide are increasing more rapidly in men than in women (Klerman & Weissman, 1989), and women's rates of alcohol abuse, conversely, may increase worldwide as gender norms shift.

The newest work on identification of specific genes and other biological markers for psychiatric disorders offers promise for fresh insights into both alcoholism and depression research, especially if the work is developed in the context of biopsychosocial models and with sensitivity to developmental issues. Throughout the literature on maternal depression and paternal alcoholism, transactional models have emerged as a valuable tool in examining both the mediators for the effect of the parental psychopathology as well as evaluating potential moderators of risk. More research examining transactional effects has tremendous promise for revealing developmental pathways to disorder in children at risk and can benefit from increasing sophistication in longitudinal research design and approaches to data analysis. The findings reviewed here reveal the value of identifying the earliest emerging problems in children of parents with depression or alcohol abuse. Work based on the gateway hypothesis is useful in this regard (Kandel, Yamaguchi, & Klein, 2006). Both risk factor research and evidence-based preventive interventions will also benefit from the appreciation of interactions among risk factors as well as transactional patterns in pathways to disorder.

Research on the effect of paternal alcoholism on child development is not a new line, yet recently has not garnered as much attention as depression in mothers. By using the structure of the integrative model for the transmission of risk (Goodman & Gotlib, 1999; Figure 30.1), we have shown that many of the mechanisms and moderators found in the transmission of risk to children of depressed mothers also hold true for the transmission of risk to children of alcoholic fathers. The model bears strong resemblance to Windle and Tubman's (1999) model for the transmission of alcoholism, with several coinciding mechanisms and moderators, further suggesting that these two areas of research can learn from each other. The overlap between these two models and the empirical support reviewed here for common mechanisms

for the transmission of risk to children as well as moderators of the risk gives credibility to the potential for an overarching general model of the transmission of risk to children, regardless of the specific psychopathology of the parent. Such an approach has the benefit of recognizing common comorbidities with other disorders (including alcoholism with depression in both men and women), the tendency for risk factors to accumulate, and the essential role of context. While research will undoubtedly benefit from continued emphasis on each specific disorder, preventive mental health efforts may well also benefit from a focus on commonalities across disorders. An overarching model of risk would have practical screening, prevention, and treatment implications for children who are at risk for the development of psychopathology due to parental psychopathology. This model would facilitate the development of general approaches to prevention and treatment aimed at reducing the child's likelihood of the development of psychopathology and could be a cost-effective method of identifying and treating the children most at risk.

An essential message to clinical psychologists from the work reviewed in this chapter is that both maternal depression and paternal alcoholism confer not only specific risk for the transmission of the same disorder in offspring but also risk for the development of other forms of psychopathology, many of which show signs much earlier in development then when one typically first diagnoses depression or alcohol abuse. Although abstinence, if one could count on adherence, is an effective approach to the prevention of alcoholism in children of alcoholic fathers, there is no single such solution for children of depressed mothers. Neither does abstinence address the early emerging problems in youth or the other disorders for which children of alcoholic fathers are also at risk. At minimum, the findings in this chapter clearly support screening and prevention targeting risk factors for the development of the respective disorders, such as problems with emotion regulation abilities in both groups of offspring, a low response to alcoholism in children of alcoholic fathers, and cognitive vulnerabilities to depression in children of depressed mothers. Continued research on mechanisms holds promise for the design of specific effective interventions at the same time that ongoing work on moderators will help identify the individuals most likely to benefit from preventive interventions.

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CHAPTER 31

Problems in Infancy

Tiffany Field

The problems in infancy can be categorized as both the frustrations and challenges of the developmental milestones that occur and the problems they pose for the parent facilitating development and giving care along the way. This chapter reviews some of those problems under the headings of developmental milestones, developing emotions, behavior problems, and caregiving. Under each of these headings, the problems are given in an order that approximates their development or the order in which they may become concerns to parents.

DEVELOPMENTAL MILESTONES

Taste

Although most parents today realize that newborns can see and hear, many are surprised that they can also smell and taste and imitate facial expressions. Newborns are born with very sophisticated abilities to discriminate different tastes, probably because the amniotic fluid comprises all tastes, including sweet, salt, and bitter (Mennella, Jagnow, & Beauchamp, 2001). This has been observed in many studies based on newborns' facial expressions in response to different tastes. For example, disgust faces follow the baby tasting quinine and other bitter tastes, whereas contented faces follow a sweet taste (Liem & Mennella, 2002). Compared to older infants, however, newborns do not seem to reject salty and bitter substances, perhaps because those were familiar in utero. Slightly older infants seem to prefer novel tastes. For example, in one study, infants spent more time breastfeeding after their mothers ate garlic as compared to those infants whose mothers repeatedly consumed garlic. Clearly, though, the sweetest substances are preferred from the first day of life. In fact, sucrose has been used as an analgesic during painful procedures in the newborn nursery. Newborns who were given sucrose solutions to suck on during blood draws, for example, cried less following the procedure and had lower heart rate and more positive brain waves (Fernandez et al., 2003). Children

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who are introduced to new tastes and new foods in early childhood tend to enjoy a variety of different foods later, whereas those with a limited diet tend to have negative reactions to new foods.

Imitation

Until very recently people thought that imitation did not occur until around the end of the first year of life. This concept of imitation was advanced by the famous child psychologist Jean Piaget, who based his ideas on observations of his own two children. More recently several researchers have shown that even a newborn can imitate exaggerated facial expressions made by adults. These expressions include sticking out the tongue, an exaggerated smile, a surprised face, and a sad face. Of course, not all children show imitative behaviors as early as the newborn stage. In fact, some newborns have been described as "poker-faced" and merely stare inquiringly as you model these faces. In a study I conducted on newborn imitation, I first made sure that the newborn was awake and bright-eyed (Field, Guy, & Umbel, 1985). Then I held the baby in front of my face and bobbed up and down with a couple of knee bends to make sure the newborn's eyes were open, and then made a couple of tongue clicks to make sure the newborn was looking directly at my face. I then made an exaggerated facial expression, leaving it there until the infant looked away. I then repeated the action with the same or a different expression. When a baby was in a very alert state and in the mood for this game, we saw very interesting imitative behavior. Later, at around 3 months, mothers almost invariably imitate their baby's cooing sounds during face-to-face games, and the baby in turn imitates the mother. This game goes back and forth for many rounds. A little while later infants imitate gestures and fine motor behaviors. Although much of what infants show us they can do seems to be hardwired and occurs on some predetermined developmental schedule, imitation, like other skills, improves with experience (Learmonth, Lamberth, & Rovee-Collier, 2005).

Eye Contact

Eye contact is present as early as birth and is one of the most important features of social interactions (Farroni, Csibra, Simion, & Johnson, 2002). Without eye contact, it is extremely difficult to communicate with people. Infants and older children will sometimes turn off their eye contact or show gaze aversion when they are being overstimulated and need to process the information, much like taking a break from a conversation (Brooks & Meltzoff, 2002). However, excessive gaze aversion or the absence of eye contact early in infancy is often considered a sign of autism and should be further evaluated if this problem persists (Zwaigenbaum et al., 2005). A recent study of children ages 2 to 6 noted that children who averted their gaze from their parents in the first month of life (a) developed a maladaptive relationship in terms of their interactions by 2 years, (b) showed behavior problems and developmental delays for up to 6 years, and (c) had less favorable development during the preschool years.

Conversation Skills

Conversation skills start developing as early as birth, when babies have face-to-face interactions with their parents. These are particularly frequent and become exciting interactions at around 3 to 6 months, when babies not only have frequent eye contact but also smile, make a lot of funny faces, coo, and make funny sounds. Early conversations have been analyzed in detail by psychologists, who note that these conversations have many prespeech sounds (vowels and consonants), all the universal facial expressions (which occur as early as 5 months in utero), and many expressive hand gestures. When the timing of speech and pauses in speech and looking and looking away are studied, the baby's behavior is very much like an adult's. For example, babies look at their interaction partner when listening and look away when talking, as if to concentrate on what they are saying. Occasionally both speak at the same time, and occasionally there are interruptions, just as in adult conversation. But the turn-taking and the turn-taking signals are already fairly well developed.

Much of the infant's learning of these skills appears to come from imitation, which frequently happens during parent-infant interactions. Like a game, the parent imitates the infant and the infant imitates the parent. In fact, one of the easiest ways to get an infant to stop crying is to imitate that crying. The infant stops crying and pays close attention to the ridiculous-sounding adult. Another game parents often play is providing the words for the infant (Field, 1979). The mother says "I like your hat," and the baby coos; then the mother says "Oh, you like your hat too," as if providing the words for the baby. An important rule the parents learn immediately is to respect the infant's looking away. This indicates (as it does in adult conversations) that the infant is either bored, tired, or has had enough of that conversation. Repeating children's speech sounds and adding to their words helps as the infants become older. By toddlerhood children have conversations with the world, with themselves, and with their imaginary playmates. At that stage the silence at bedtime is a welcome relief.

Babbling

Babbling is the child's first attempt at conversation and consists of long strings of sounds formed with the lips and tongue, including *baba* and *dada* and arias of cooing that make the child seem musically talented (Goldstein, King, & West, 2003). These sounds are universal and gradually become differentiated into, say, American versus Japanese sounds as the child is further exposed to his or her unique speech environment. Importantly, babbling reinforces the parent's attempts to talk to the child. In the first conversations the parent says something, the child babbles, and the parent says "Oh, is that so?" as if the child has truly participated in the conversation.

Exploration

This is usually the precursor to playful behavior. The child explores objects to learn what the object can do and what the child can do to the object. One of the earliest forms of exploration is the newborn sucking on the mother's breast.

For the next 6 months everything gets explored by mouth, and much is learned (e.g., the shapes of things) in this way. The hands take over, and then exploration seems almost an obsession as infants make messes with their finger foods, throw objects on the ground for adults to retrieve, and take everything out of drawers and put them back in again (container behavior) as they explore and learn the properties of things. As exploration proceeds the child's behavior becomes more playful, repeating actions over and over, usually to the sounds and facial expressions of sheer joy. Stifling exploration should be avoided at all costs, as it would in turn stifle curiosity, creativity, and the entire learning process. Extremes of baby-proofing or childproofing the house can seriously interfere with exploration.

Handedness

Handedness, or right or left hand preference, develops very early in infancy and can be determined by placing a desirable object in front of the infant and determining which hand leads to grasp the object. The dominant hand will invariably be preferred for fine motor, dexterity activities (Hinojosa, Sheu, & Michel, 2003). However, a recent study highlights the important staging and guiding function of the nondominant hand from very early in infancy. Left-handedness is more frequent in twins (Derom, Theiry, Vlietinck, Loos, & Derom, 1996) than in single children and in males than in females.

The literature is very mixed about the association between handedness and problems. For example, in one study handedness and dyslexia (a learning disability) were significantly related, but handedness was only weakly associated with immune problems. In another study autoimmune disorders were more prevalent in left-handers. In a third study left-handers had nearly 3 times as many immune disorders as right-handers; the relatives of the left-handers suffered from the same conditions; there was a higher frequency of learning disabilities such as dyslexia, stuttering, and autism in the left-handers; and left-handers had migraine headaches more often (Eglinton & Annett, 1994). One of the theories for left-handedness, aside from the genetic theory, is that during fetal development testosterone (the predominant male hormone) slows the growth of the left hemisphere. When testosterone is elevated, it is more likely to produce left-handedness and, in extreme cases, to lead to alterations in the left hemisphere that may be responsible for learning disorders such as dyslexia. Excessive testosterone levels can also alter immune function, which can then lead to immune problems such as eczema, psoriasis, and asthma.

In earlier days attempts were made to shift children from being left-handed to being right-handed so that penmanship and scissors and virtually all hand-held devices (which are designed for right-handed people) would be more easily mastered. Today, parents and teachers do not attempt to alter handedness because of the undocumented theory that shifting handedness in itself may lead to learning disabilities.

Teething

Teething, or the breaking through of new teeth, can be uncomfortable for infants and distressful for parents. Parents' favorite treatment is probably rubbing the infant's

gums with a bit of liquor. Both the rubbing action and the liquor have analgesic effects.

Ear Infections

Ears are vulnerable sense organs because it is difficult to close them to unwanted noise. Ears are also a source of major discomfort to many young children because they experience earaches from inner and middle ear infections. Excessive pulling on the ears, irritability, and signs of difficulty hearing can mean another ear infection. After dozens of these infections and antibiotic treatments, pediatricians often refer the child for a myringotomy (Vaile, Williamson, Waddell, & Taylor, 2006). This procedure involves piercing holes in the eardrums to enable the fluid to drain and is considered a minor procedure, although anesthesia is typically required, and that is certainly not minor. Treatment is critical because data show that repeated, prolonged ear infections contribute to delays in language development and learning difficulties at school. For example, children with inner ear infections (otitis media) have significantly lower expressive language scores, suggesting impairments in language expression (Thomas, Simpson, Butler, & van der Voort, 2006). They also have deficits in processing auditory information and spelling skills and receive more special education services. Despite these problems, their academic achievement tests do not appear to suffer, and by later childhood their early language delays are no longer present.

Immunizations

Immunizations are thought to be critical for preventing disease in early childhood. The most common immunizations are for diphtheria, pertussis, and tetanus and are given at set intervals across the first few years of life (Cohen, 2002). Unfortunately, large numbers of children still are not receiving immunizations. For example, in a large U.S. midwestern city, a study recently revealed low immunization rates (31%), which were said to be due to cost, lack of insurance coverage, and long office visits. Not mentioned was the stressfulness of the immunization procedure on parents and children. Many studies have explored the effects of varying the conditions during immunizations to reduce the stress associated with them. In an attempt to reduce the fussiness and crying and elevated cortisol levels accompanying immunizations, researchers have varied the person accompanying the child for immunizations and varied different directions to the parents. For example, when parents accompany the child there is notably more distress in some studies, and in others there is notably less. That seems to depend on what the parent does during the immunization period. Reassurance, for example, reinforces crying, whereas distraction techniques are much more effective in reducing stress. Other calming interventions help, including music.

Crawling (or Creeping)

Crawling tends to occur at around 6 to 9 months, although some babies never crawl, and those babies sometimes walk at an earlier age (WHO Multicentre Growth

Reference Study Group, 2006). Every baby seems to have his or her own style of crawling, with some tending to crawl backward, some sideways, and some straight ahead. Some crawl on their hands and knees, while others do it on their hands with bended legs. Generally speaking, it seems an awkward way to get from one place to another, and it looks like hard work, so it is perhaps not surprising that there has to be something the baby really wants to reach for crawling to occur. A recent study was conducted on motor skill development in crawlers and noncrawlers during early infancy. The noncrawlers showed slower motor skill development, suggesting that those who do not crawl may develop motor skills more slowly, but crawling and other motor milestones do not seem to be related to later IQ (Capute, Shapiro, Palmer, Ross, & Wachtel, 1985).

A parent recently expressed concern that her child was retarded because she was not crawling at 9 months. After reminding the parent that some babies never crawl, I placed my keys on the carpet, and the baby proceeded to crawl directly to them. The mother, who typically carried her infant on her hip around the house for hours at a time, expressed amazement that all it took was placing her child on the floor with something desirable in view, and her child was no longer "retarded."

Walking

Children typically start walking at around 10 months of age and are pretty proficient by the time they are toddlers at around 16 to 18 months. Temperament differences affect the time of onset, so that one often sees children who are risk takers and exploratory starting earlier than children who are shy and slow to warm up. Although scientists have shown that walking can occur very early if it is practiced, there has been some concern that early walking may slow down the development of other skills, such as talking. Typically, children are focusing on one skill at a time, and those seem to be prewired developmentally to occur at specific times. Crawling and walking, of course, give a child more freedom to explore and get into trouble when the environment isn't childproof. Although walkers seem to be very effective in helping children get around, they are coming under increasing criticism because of their use as babysitters, because some serious falls have been reported, and because they delay the onset of walking (Burrows & Griffiths, 2002; Siegel & Burton, 1999). Hanging harnesses from door frames are very helpful for learning balance and walking movements. Whether walkers or jolly jumpers are used, all children seem to walk at about the same time. If walking is delayed, other functions may also be delayed. In one study, slow walking and slow talking were predictive of lower IQ scores and reading difficulties, but this was true only for children who started walking after 18 months, and only half those children later had low IQs and reading difficulties.

Terrible Twos

The 2-year-old stage has become so notorious for its difficulties that it is now classified as the terrible twos. This is the first of many autonomy-setting stages of the child's life, and it may be the worst (next to adolescence) because it is the first. Parents have come to totally enjoy their lovely child when all of a sudden they find

themselves faced with a little monster. Every other word is "no" and every other action is forbidden and accompanied by looks toward the parent of "I dare you." Language has emerged, and with that children act as if they're smarter than their parents. That might be true, as the child repeatedly outwits them. Because of the autonomy-establishing ways of the child at this age, it's important that parents not get into power struggles with their child over the child's behavior. For example, many children who have not been toilet trained by age 2 use toilet training as grounds for power struggles with their parents. Waiting until 2 virtually means waiting until at least 3, when autonomy issues are not as great. Thus, toilet training can dramatically exacerbate the problems of the terrible twos.

Aside from trying not to save important developmental milestones such as toilet training for this stage, parents are probably best advised not to get into verbal power struggles, but to simply move the child from the source of altercation (e.g., an electric socket or a restaurant where the child has resorted to temper tantrums) or to attempt to distract by activities that the child can clearly dominate. Parents' attempts to explain their "unpopular" position typically fall on deaf ears and accelerate the conflict. It is not surprising that this is an increasingly popular age for parents to send their children to school. There the children can have their terrible two struggles with teachers and other terrible-2-year-olds.

Autonomy

Autonomy is probably the primary goal of childhood. It is generally achieved by the end of childhood, just before the long struggle begins for adolescents to have complete autonomy. Teaching children how to take care of themselves, how to monitor and control their own behavior, how to relate to other people in a socially effective way, how to express their emotions, and how to achieve their potential is what has been called "the package of parenting." The developmental course of autonomy proceeds from the infant who cries at 6 months when you stifle his or her crawling attempts or remove a desired object, to the child who wants to walk unassisted, to the terrible twos when the child wants to have everything plus power over parents (LeCuyer-Maus & Houck, 2002). Developing all those skills—motor, cognitive, and social—are all in the service of achieving autonomy.

Problems that get in the way of developing autonomy include deficits in the child; temperament problems, such as inhibition and unwillingness to have commerce with the world; limited time and patience of parents, who might find it easier to do things for the child; and inability on the parents' part to allow the child to fail in trial-and-error learning. The frustration is felt on all sides, but ultimately the child will be autonomous and the parent will be proud. Even the terrible twos are a small price to pay. Actually, this is a time when parents can have fun, enjoying the slapstick humor of their child's clumsiness as he or she achieves the milestones of autonomy.

Play

Play and pretend play are critical to a child's development. Playfulness is measured by physical, social, and cognitive (mental) spontaneity, by joy and a sense of humor. Much of this develops in the child through experience playing, starting in preschool,

when a child's entire agenda is play, and tapering off as children get older and have less time for play. In one study on individual differences in young children's pretend play, early social pretend play was significantly related to the child's developing understanding of other people's feelings and beliefs (Keren, Feldman, Namdari-Weinbaum, Spitzer, & Tyano, 2005). These data provide strong support for the notion that early experience in pretend play is associated with mastery of relations between mental life and real life. Parents need to orchestrate some of the play of their children, inasmuch as the natural opportunities for play, such as neighborhood playgrounds, have become more rare in very busy and unsafe cities. Often all it takes is arranging for a child's friend to visit, gathering together the props, and getting the pretend play started. Once they get started, children are very good at pretend play.

Friends

Friendships develop at a very early stage in life, which we never knew until children started attending infant day care. In the nursery school my research group runs, we can no longer graduate infants at 12 months to the toddler nursery without also transferring their best friend with them. Parents told us very early on that they wanted their children to wait until they could move with their best friend, but it was only after studying this phenomenon that we appreciated that even infants experience anxiety and depression when separated from their best friends. In our study the infants who were transferred without their friends experienced sleep problems, eating problems (either more or less), toileting problems (constipation or diarrhea), and sometimes even illness (typically upper respiratory infections; Field, Vega-Lahr, & Jagadish, 1984). Thus, friendship starts at a very early age.

In our studies, when parents were absent for a meeting or for the birth of another child, the preschoolers who remained in day care with their friends were less depressed (Field & Reite, 1984). In the early stages, because of a child's shyness and less-developed social skills, parents may need to set the stage for developing relationships. This can be arranged through shared babysitting, inviting a peer along for special activities, and trying to facilitate cooperative play between the children by giving them the same opportunities to engage in the same activity and by having play materials in twos, ensuring that each child has a play object or a book, so that fighting is less likely to occur.

Sex Differences

Sex differences are present from birth. Research suggests that for most assessments, whether it is responding to social stimulation, discriminating faces and voices, or simple learning tasks, females are more skilled. This sexual advantage for girls persists until at least school age. Although it is widely known that boys are physically at a disadvantage at birth, often being smaller and having more perinatal complications, very little is known about the origins of boys' slower development. This is particularly a problem when parents compare their boy child to their girl child and conclude that the boy's slower language development, for example, is suggestive of developmental delay or retardation. Parents need to have more moderate expectations of their male children, particularly in the early preschool years.

Sometime later in children's schooling, female children lose their gains, often as they develop diffidence and experience more limited expectations for academic performance from adults.

There are, of course, many other sex differences that emerge, for example, in the expression of empathy and social interests. Although most people like to believe that these are socially determined, such as the introduction of dolls to girls and guns to boys, many of these preferences are apparent before the socializing experiences. For example, in a recent study on 12-month-old infants, girl infants presented with the choice of a toy airplane or a doll more often reached for the doll, while boy infants more often reached for the airplane (Servin, Bohlin, & Berlin, 1999). An anecdote highlights a similar response. When a teacher carried her baby into the preschool, the boy children rushed to the block corner and the girl children rushed to the teacher's baby, much to the chagrin of the teacher, who had tried very hard to encourage unisex activities and behavior in her preschool classroom.

Twins

Twins can come as a shock to any family, even though they seem to run in families. Senior family members try to prepare first-time parents for that possibility. Because ultrasound and other prenatal tests typically reveal twins sometime during pregnancy, most parents also have that time to prepare themselves. The downside is that you need two of everything and it is twice as much work, but the upside is that you can have twice the joy in half the time. Identical twins have quite amazing similarities as early as birth. For example, in a study we conducted on newborns, identical twins were almost identical in the amount of time they looked at faces and mimicked faces, whereas fraternal twins were very different in their facial expressions and their attentiveness. Fraternal twins are often perplexing because they seem no more alike than nontwins in the same family (Field, 1982). Despite their similar experiences at the same time in development, they can behave and develop very differently.

Intelligence Tests

Intelligence is still a hotly debated issue, particularly for infants and young children. Although there are now scales that are considered infant IQ tests, such as the Brazelton Scale and the Bayley Scales for Infant Development, these have very little stability over time (Lowman, Stone, & Cole, 2006). For example, some children score high early in infancy because of their precocious motor development, but then score poorly at a later age because of their limited language environment. The scales seem to focus more on motor skills in early infancy and language skills in early childhood. Also, scores are significantly affected by many environmental conditions, suggesting that performance is not strictly determined by genetic factors.

Bayley Scales

The Bayley Scales were created by Nancy Bayley in 1963 (and revised in 1993) to assess infants' mental and motor development (Lichtenberger, 2005). They can be given to children from about 3 months of age to about 4 years. Typically the

child is seated on the parent's lap while the Bayley Scales are given, and they last approximately 40 minutes. Developmental age-appropriate items are given, such as a red ring dangled in front of a 3-month-old infant's face, and pegs and peg boards are given at a slightly older toddler age. Motor skills are assessed by watching the child's natural motor behavior and offering more challenging tests at a later age, such as climbing stairs and walking on a rail. These scales yield a score, with an average at about 100. They are used simply to give the pediatrician or developmental professional a sense of how the child's mental and motor development are progressing and whether there are any areas that need work.

Prematurity

Infants who are born prematurely have a number of problems in infancy suggesting delayed growth and development (Vintzileos, Ananth, Smulian, Scorza, & Knuppel, 2002). Long-term follow-ups suggest that children born prematurely have more sleep-related problems and more accidents, inferior eye-hand coordination, delayed language development, more visual and hearing disorders, and more hospital care. Environmental factors account for more variation in their cognitive development than perinatal factors. Preterm children exhibit a self-righting tendency during early childhood, so that eventually environmental influence overshadows biological influences.

DEVELOPING EMOTIONS

Emotional Expressions

The necessary muscles for the universal emotional expressions are present very early in fetal development (3 months gestation). These universal expressions are happy, interest, surprise, sad, angry, fearful, and ashamed. Although the expressions seem exaggerated during early development, when children laugh without restraint as well as sometimes cry incessantly, the expressions become socialized with development, and we begin to see blends of expressions such as a combination of surprise and sadness when feelings are hurt. Some children display their emotions, as in the expression "wearing their heart on their sleeve," and others tend to be more "poker-faced" and show very little emotion. These extremes, sometimes labeled extraversion and introversion, appear in children as early as the newborn stage. In a study we conducted on newborns' imitations of facial expressions, about 20% of infants showed very exaggerated facial expressions and very little heart rate change, while the other 20% extreme showed emotional responses in their heart rate changes but were poker-faced (Field, 1982). Most children (like most adults) show their emotions not only on their face but in their physiological responses, for example in a racing heart, sweaty hands, or blushing.

Anger

Expressions of anger can be seen as early as the newborn period in crying outbursts by an uncomfortable newborn whose needs are not being met. The temper tantrums of a 2-year-old occur for the same reasons: discomfort and not getting what is wanted

or needed (Potegal & Davidson, 2003; Potegal, Kosorok, & Davidson, 2003). In the preschool years, transgressions against children either for their possessions or against their person, as in shoving, pushing, hitting, and kicking, sometimes cause anger at having been made uncomfortable or having their feelings hurt.

Anxiety

Children, like adults, have a host of stresses and anxieties that make their lives uncomfortable. Anxiety can cause sleep disturbances and illness as well as learning problems. The anxieties are different at different developmental stages. Infants and toddlers, for example, experience the stress of separation anxiety and stranger anxiety (Kochanska & Coy, 2002). Infants also experience frustration during developmental milestone periods such as crawling and walking, although anxiety is difficult to measure at those ages. Preschooler anxieties are more likely related to being hospitalized or starting school.

Children who are particularly prone to anxiety are those who are temperamentally shy and have difficulty making transitions. To help with the transitions, various comfort objects such as pacifiers at the younger age and blankies and stuffed animals at the older age are helpful (Donate-Bartfield & Passman, 2004). Other solutions are unique to the different types of anxiety. For example, for separation anxiety, which is typically related to the infant not realizing that the parents are going to come back, parents can reassure their infant that they are just going to get something and will return shortly. For stranger anxiety, a goodbye ritual helps, particularly if it is brief and upbeat (Field, 1995; Field, Gewirtz, et al., 1984). Sometimes play-acting these leave-takings with the child by using dolls and making the babysitter come and go in fantasy play helps. Allowing the child time to become familiar with the babysitter before the parents leave helps a great deal.

Most anxieties are normal and pass with the developmental stage, except for separation anxiety, which seems to persist. Most children do not particularly like their parents leaving them (until they reach adolescence and want to leave their parents). Separation anxiety in its extreme form, when the child will not leave the parent and develops symptoms such as headaches and stomach aches, may be a serious form of anxiety, labeled Separation Anxiety Disorder.

Massage therapy invariably helps. A 10-minute back rub on a daily basis invariably reduces anxiety (Field, 1995). This can be observed as reduced agitation and nervous behavior, and it can be measured as decreased stress hormones such as cortisol taken from saliva samples.

Stranger Anxiety

Stranger anxiety starts at approximately 9 months of age, after a rather extended period of infants loving to be held and played with by strangers. It is not clear why it emerges at this time, because even newborns have been shown to recognize the difference between their own mother and women who are strangers. In a study we conducted, newborns showed that they preferred their mothers by looking longer at them than at women who were strangers (Field, Cohen, Garcia, & Greenberg, 1984). This was surprising because they had had only 4 hours of contact with their

mothers. Equally surprising is their ability to identify their mother by her smell. Recognition of mother by her voice is probably less surprising since newborns would have had experience with the mother's voice in utero.

Nonetheless, even though infants recognize their mothers as early as the newborn stage, they do not appear to show stranger anxiety until approximately 9 months. This takes the form of fussing and showing fearful faces in the presence of the stranger, whether at a distance or when being handed to a stranger. This persists for anywhere between 3 months and a couple of years, in part depending on whether the infant remains at home with the mother or attends an infant nursery school. When an infant has experience with several adults, the infant shows less stranger fear and sometimes no fear. It is not clear what function the stranger fear serves. It certainly would not prevent kidnapping at this early helpless stage of life, and were it to happen any earlier it might even interfere with parents returning to work or hiring babysitters. Fortunately, it is a stage that disappears as quickly as it appears.

Separation Anxiety

Temporary separations from parents during hospitalizations of the parent or child or during parents' conference trips, and more permanent separation from peers associated with moves to other schools, are relatively stressful for young children (Field, 1984). In a study that we conducted on toddlers' and preschool children's responses to separation from the mother during the birth of another child, the children were agitated during the mother's hospitalization and then depressed after the mother's return (Field & Reite, 1984). During her hospitalization the children were irritable, hyperactive, and unable to sleep. When the mother returned the children showed depressed emotion, changes in eating and toileting, continuing sleep disturbances, and illnesses, including constipation and upper respiratory problems (all characteristic of adult depression). The children were clearly agitated by separation from the mother, even though they visited her at the hospital during this period and their father cared for them. The depression following her return from the hospital may have related to the mother's depression or exhaustion. This decreased animation of the mother, together with the arrival of a new sibling, seemed to alter the relationship the mother and child had previously experienced. Examples of the child's disturbance over this altered relationship were provided by the parents, who commented that their child "remained close" to them, "wanted to be rocked and held," "reverted to baby talk, whining, and screaming for attention," "destroyed his playroom," and "threatened to run a truck across the baby's head." In addition, increased fantasy play in the children was characterized by a number of scenes that involved aggression against the new sibling. For example, children built Lego constructions to "fall on the baby," "run over the baby," or "drive the baby to a high bridge and throw the baby in the water."

Some children did not seem to suffer separation stress as much as others. Those who seemed to suffer least were those who continued attending preschool while the mothers were hospitalized, as opposed to those who stayed at home with the grandmother or babysitter during the mother's hospitalization. Maintaining

the child's daily routine seemed to help buffer the effect of children's separation from their mothers.

In a subsequent study we observed children's responses to repeated separations from their mothers, who were attending conferences throughout the year (Field, 1991). Although we were concerned that there may be cumulative effects that made each separation worse than the last, we found that children became accustomed to these separations, and after a few of them were no longer showing the extreme behavioral and physiological changes or illness that accompanied the first and second separations.

Although most parents (and psychologists alike) had assumed that separation stress occurs only when parents are missing (because parents are supposedly the primary attachment figures), separation stress is also experienced over the loss of a teacher or favorite peer. Parents in our nursery school used to tell us that they wanted their infant or toddler to wait to graduate to the next class when their best friend was also ready to graduate. Sometimes parents actually transferred schools when their child's best friend transferred. In a recent study, preschool children who were transferring to new schools were observed during the 2 weeks prior to their leaving (Field, 1984). The children showed increased fantasy play, physical contact, negative statements, fussiness, hyperactivity, and illness, as well as changes in eating and sleeping patterns. Shortly after the children left, those who left the school showed less agitated behavior, but those who remained behind showed increased agitated behavior. Apparently the remaining children were being reminded of their lost friends by staying in the same environment. As in a divorced couple, the person remaining in the old house often has more difficulty because of reminders of the old relationship.

In another study we explored infants' and toddlers' reactions to being graduated to the next classroom with and without their best friends. Infants and toddlers who transferred without their best friend showed increased activity (Field, Vega-Lahr, et al., 1984), negative emotion, fussiness, and changes in eating and sleep patterns. Some children ate too much and some children ate too little; some children slept all the time and others slept not at all. Those who transferred with close friends did not experience the stressful effects of the transition period.

It is clear from these studies that children have difficulty with loss from a very early age, so that the loss of the old relationship with mother after a new sibling arrives or the loss of friends can be the most stressful. At those times, consistency in routine (e.g., remaining in school versus staying at home) can alleviate some of the stress. Having a familiar routine may help not only because it is familiar but also because it provides distraction and opportunities for the child to actively cope with the stress. Otherwise, parents simply need to ride it out and recognize that these are normal responses to loss and change.

Depression

Infants can have symptoms that mimic adult depression, including withdrawal and listlessness around other children, sad facial expressions, a monotone voice, and whining and complaining (Field, Gewirtz, et al., 1984). The child might also show

eating problems, such as overeating or undereating, and toileting regressions, difficulty going to sleep, frequent night wakings, and difficulty getting up in the morning. Symptoms of illness include respiratory problems, constipation, and diarrhea. Illness probably occurs because the immune system is negatively affected by high stress levels accompanying depression.

Empathy

Some form of empathy is apparent as early as birth, when newborns cry at the sound of other babies crying (Field, Hernandez-Reif, & Diego, 2006). A typical situation that is used to study empathy in toddlers and preschoolers is asking a mother to pretend she is crying. Almost always the child will show some signs of attempting to comfort the mother. However, some children, who, for example, have depressed mothers or have been prenatally exposed to drugs, show no empathy (Jones, Field, & Davalos, 2000). Some recent research in our lab suggests that there are different forms of nonempathetic behavior, including sad and angry responses. In the mother pretend-crying situation, for example, some preschoolers showed nonempathetic angry behavior, including kicking their crying mother or yelling at her, and other children showed nonempathetic sad behavior by simply ignoring the mother's crying.

Temperament

For most of history parents have blamed themselves for their children's temperaments. More recently considerable research suggests that infants come into the world with a temperament that is only then slightly modified by parents' temperament and things that happen in the environment. Parents will still blame themselves, but now they will do so for transmitting temperament traits through their genes rather than their childrearing. Temperament researchers have described babies, for example, as easy or difficult from the start, and they have based their conclusions on parents' reports and on observations of several characteristics they call temperament traits. These include the child's activity level, adaptability, distractibility, intensity, mood, persistence, regularity or rhythmicity, sensitivity, and weariness. Different temperament traits combine to determine whether a child is easy or difficult. In one of our studies, we noted that newborns in the first hours of life could be placed somewhere on a continuum between being extraverted and being introverted (Field & Greenberg, 1982). The extraverted newborns showed very expressive faces and were not very reactive when we measured heart rate. At the other extreme, the introverted newborns were poker-faced but showed considerable physiological reactivity as recorded by their heart rate. Most newborns (approximately 60%) fell in between these two extremes, being both facially expressive and physiologically reactive. When we looked at these behaviors in identical and nonidentical twins we found that there was more similarity in the personality types of identical than nonidentical twins, suggesting again that there was some genetic component to these personality profiles. Observing these infants over the first few years of life, we noted that if an extraverted infant was born to introverted parents, the extraverted infant moved toward the middle of the continuum, whereas if the extraverted infant

was born to extraverted parents, he or she more likely continued to be extraverted. Thus the interplay between genetics and environmental influences became clear.

Extraverted and introverted children have very different ways of interacting with the world. Extraverted children are more exploratory and risk taking and therefore likely to excel in development. On the other hand, if the parent is aware of the temperament traits of the introverted child and tries to match parenting strategies to that style, the child may end up being exploratory in spite of his or her introversion. Several researchers have tracked the development of temperament traits to determine whether they are stable over time and which traits seem to contribute to superior development. Dimensions of temperament that have been closely associated with less difficult responses to stress, fewer behavior problems, and lower stress include predictability, positive mood, adaptability, higher approach, lower intensity reaction, and lower responsiveness threshold (Huttunen & Nyman, 1982).

Because of the stability of temperament traits over development, psychiatrists have been concerned about the relation between those traits and behavioral syndromes in later childhood (Maziade, Cote, Thivierge, Boutin, & Bernier, 1989). In one study, for example, both high emotionality and low sociability predicted high scores on an anxiety/depression scale. In another study, high activity and low rhythmicity was associated with a high degree of psychopathology.

BEHAVIOR PROBLEMS

Crying

The most common causes of crying in infants and young children are fatigue, hunger, indigestion, soiling, wetness, and pain. The most common psychological reason is a parent's leaving. Very soon after birth, parents come to recognize the cry sounds of their infant. The frequency, intensity, and intonation of cry sounds are different for all of these problems. For example, fussy cries are less intense than hungry or pain-induced cries. Parents will often assure themselves that all of these problems are taken care of before letting the child cry it out. Sometimes when a child is overstimulated and, thus, overexhausted, crying serves as a kind of stimulus barrier to ward off all other sounds in the environment.

The parent may be able to stop the crying by rocking or walking or driving the infant around, but often the crying will start again when the infant is put to bed. This could become a pattern of crying for the parent to come and comfort (Räihä, Lehtonen, Huhtala, Saleva, & Korvenranta, 2002). The simplest method for solving this problem seems to be letting the infant cry it out. For the first night, the crying may persist for 20 to 30 minutes, and when there is no rescue from the parents, the infant finally falls asleep. The second night the crying may last only 10 minutes, and by the third night there usually is no crying. Getting up for every cry not only overstimulates and ultimately tires the infant but certainly tires the parent.

Colic

Colic is a syndrome of persistent crying in young infants. Colic is one of the most common problems in infancy (incidence ranges from 10% to 35%; Savino &

Oggero, 1996). Researchers disagree about the definition and clinical criteria for colic; some pediatricians even suggest that colic does not exist. This presents a very confusing picture to parents.

The word "colic" is derived from the Greek word *kolikos,* the adjective of *kolon,* meaning the large intestine. Colic also goes by the labels "infantile colic" and "paroxysmal fussing." Colic has been described as intermittent and unexplained crying during the first 3 months of life by infants who are otherwise well-fed and healthy. Colic usually starts during the second or third week of life and spontaneously disappears at around 3 months. The crying is generally intense and typically lasts from 30 minutes to 3 hours, often occurring in the late afternoon or evening, followed by difficulty falling asleep. The crying may also occur intermittently and last for several hours at any time of the day or night, although the crying episodes tend to recur at the same time of day.

In most cases, crying is considered normal infant behavior and can be treated with reassurance and counseling about appropriate parental responses. In a minority of cases, the crying pattern and other infant behaviors may suggest a gastrointestinal problem that may require treatment. Fussiness and crying are not easy to tolerate by most adults, which probably explains why colic is the most frequent problem presented to pediatricians. Although colic ends almost abruptly at 3 months, irritability and excessive crying have been known to recur during frustrating developmental milestones such as sitting, crawling, walking, and virtually any learning stage.

Some clinicians accept the definition of colic as full-force crying for at least 3 hours per day, for 3 or more days per week, for 3 weeks. Colic begins with a sudden onset of crying. The cry is often high pitched and accompanied by facial grimacing, suggesting that the infant is experiencing severe pain. The crying is accompanied by increased motor activity, flexion of the elbows, clenched fists, and hypertonicity (e.g., the knees are drawn up or the legs are stiff and extended), and the abdomen is tense and distended. The infant's eyes are either tightly closed or wide open, the back is arched, the feet are often cold, bowel sounds can be heard, infants may hold their breath for brief periods of time, and they may resist soothing.

Several factors may be involved in infant colic, including biological factors such as temperament and sleep-state organization and environmental factors, for example, overstimulation by parents. Biological factors include hypertonicity, gastrointestinal and allergic factors, and an immature central nervous system. Although gastrointestinal factors have typically been described for colic, serial radiographs suggest a normal amount of gas in the GI tracts of colic infants. Breast milk, cow's milk, and soy milk have also been implicated in colic. Digestion of milk and soy protein produces excessive intestinal gas and peristalsis, which can result in colic. However, colic occurs as commonly in breastfed as in formula-fed infants.

Feeding patterns have been implicated as a potential contributor to colic. Short interfeed intervals resulted in less crying. Infants who were carried more frequently did not cry at the usual 6-week crying peak, and the crying and fussing that tended to occur in the early evenings decreased. Overstimulation also contributes to colic. This became clear when infants who were hospitalized for colic cried less once they were hospitalized. Other overstimulating factors are excessive noise, bright lights, and irritating blankets or clothing. Colic may cease around 3 months because

infants can now use self-comforting techniques such as gaze aversion, thumb sucking, turning over, reaching, and grasping to avoid or remove the annoying stimulation. Having feeding, bathing, and sleeping routines is very effective. In addition, providing soothing stimulation such as massaging the infant, rocking or swinging, and playing classical music or white noise can also be extremely effective (Huhtala, Lehtonen, Heinonen, & Korvenranta, 2000). The most effective treatment is time, as colic usually disappears by 6 months (Crotteau, Wright, & Eglash, 2006).

Bedtime Problems

Bedtime is one of those quality times for parents and children. It needs to be a happy time even though children can fret or procrastinate. Having a bedtime ritual that the child looks forward to is often helpful. The teddy bear gets tucked in, the child hears a story, and then a back rub helps a child get to sleep. A recent study in our lab suggested that back rubs were more effective than rocking from early infancy through early toddlerhood (Field, Grizzle, et al., 1996). In the case of rocking, the child goes to sleep during the rocking but awakens as soon as he or she is carried to bed. In the case of back rubs, the child is awake during the back rub and goes to sleep in that position shortly thereafter.

Sleep Disturbances

Sleeplessness, in the form of difficulty falling asleep and/or nighttime waking, is a very common pediatric problem, affecting some 15% to 35% of infants and toddlers. Sleep problems as early as infancy may relate to physiological factors, temperament of the infant, and parents' responses to difficult sleep patterns. In recent years the rate of sleep disorders has reportedly increased, possibly because parents are increasingly less tolerant or more worried about sleep disturbances and how disruptive they are to the dual-career family, or possibly due to more sophisticated methods of monitoring sleep and knowledge about the development of sleep patterns. Various methods have been developed to study infant sleep, including measuring facial muscle activity, EEG, heart rate, and respiration. In addition, diaries are frequently kept by parents to record information on duration of sleep and crying and other activities.

Infants' wakeful time gradually increases from an average of 8 hours at birth to 12 hours at 6 months (Acebo et al., 2005). By the time the infant is 3 months old, sleep predominantly occurs during the nighttime hours, with the infant tending to remain awake during the later afternoon and early evening hours. By 6 months, the time awake has increased from a mean of 2 hours at 3 months to about 3.5 hours, and the longest waking period is about one-third of the total awake time. Although a diurnal (normal) sleep-wake pattern is established by this time and the infant is capable of sustaining sleep during the nighttime hours, many infants continue to wake during the night. At this time, the amount of time the infant takes to fall asleep increases, and parents begin to establish bedtime routines and the use of blankies and other self-comforting objects to facilitate earlier sleeping.

Once infants are put to bed, there is a large variation in the amount of time required for sleep onset, depending on age. For example, 2-month-olds are thought

to require approximately 28 minutes and 9-month-olds only 6 minutes to fall asleep. Night wakings occur in approximately 29% of infants during the third month, 17% by the sixth month, and 10% at the end of the first year. Once asleep, newborns typically sleep approximately 17 hours per day, and by the sixth month they are sleeping only approximately 14 hours (including naps), which remains the same across the first 2 years.

Temperament characteristics reported for sleep disturbances include low sensory threshold, difficulty establishing regular routines, and reluctance to move from one activity to another. Given that infants who are stimulated less seem to organize their sleep states sooner, overstimulation by parents may be disruptive to infant sleep. A study of healthy infants suggests that most children need something to assist them to fall asleep. In this study, regular use of a transitional object such as a blanket or toy was described for 44% of the children, bedtime sucking for 22%, and both for 14%, leaving only 20% of the children who fell asleep without assistance.

Parents use a number of comforting techniques to induce sleep in young infants, including swaddling, rocking, and sounds such as white noise or classical music. Others advocate letting the child cry or taking the child into the parents' bed or supplying toys. In addition, infants use their own body to self-comfort, including sucking on hands and rocking or head banging in a rhythmic way.

More formal procedures have been tried in studies giving parents instructions (Skuladottir, Thome, & Ramel, 2005). These have mostly involved having the parents find a consistent sleeping place, a regular bedtime, and a bedtime ritual. In one study parents were asked to try a quiet play routine before bed in which they used soothing and relaxing techniques such as massage and pleasant music. They were asked to refrain from using a light, changing diapers, and talking during their nighttime interactions. The majority of infants showed improvement during the first few weeks of treatment. Whenever parents are present at the time of sleep onset, there are more frequent night wakings.

We recently used massage therapy to help infants sleep (Field et al., 1996). The infants who experienced massage therapy compared to infants who were rocked cried less and had lower stress hormones (salivary cortisol levels), suggesting lower stress. After the massage versus the rocking sessions, the infants spent less time in an active awake state, suggesting that massage may be more effective than rocking for inducing sleep. Over the 6-week period the massage therapy infants gained more weight, showed greater improvement on emotionality, sociability, and soothability temperament dimensions, and had greater decreases in stress hormones.

Food Preferences

As early as birth infants appear to have food preferences. Studies on newborns show that they make different faces to different tastes, with sweet flavors being accompanied by happy faces and quinine or rotten egg flavors being accompanied by grimacing. This seems to happen whether the food is tasted or simply smelled. Of course, the amniotic fluid that engulfs the fetus contains all of the basic flavors, especially the more salty and bitter taste. Newborns will readily take glucose but tend to reject salty water. Taste preferences then shift across age. Very little is known

about young children's food texture preferences, although foods prepared for infants and young children tend to be pureed at first and only later have considerable texture. Their preferences for hot and cold foods is also relatively unknown.

Studies show that monozygotic (identical) twins have more similar food preferences than dizygotic (nonidentical) twins, suggesting a strong genetic determination of food preferences. Many children will follow parental likes and dislikes, particularly dislikes after seeing an adult grimace at the thought of eating a particular food. On the other hand, parents who expose their children to wide varieties of foods have children who are not fussy eaters and who are willing to try new foods.

Eating Problems

Eating problems can range from food refusals and minor conflicts with parents to eating disorders (Carruth & Skinner, 2000). The minor food refusals are sometimes just the child being finicky about food preferences, although very little is known about children's food preferences because all the nutritious foods are bottled in baby food form, with very little choice being given the children until they can select from their own lunch box or the school cafeteria menu. In addition to very little being known about taste preferences, very little is known about food textures that children prefer. In a recent study we were surprised to find that very young infants preferred chunky applesauce if they are given the choice before the designated age for chunky applesauce (Lundy et al., 1998). Infants also like vegetable flavors like broccoli and cauliflower (Mennella, Kennedy, & Beauchamp, 2006).

Other reasons for food refusals are the child's own power struggles that happen as early as the terrible twos, when children simply want to be autonomous and do what they want to do when they want to do it. These power struggles can continue into preschool and grade school, when children do not have the good models of their parents for food preferences but the fads of what other children are eating. At this time full lunch boxes might simply mean the child has eaten someone else's more desirable lunch.

Although playing with food is a good way to learn the art of eating at the infancy and toddler stage, by preschool the child should be learning appropriate restaurant eating habits. Around this time the child learns that mealtimes are a social time for the family, not a time for reading or watching TV. Unfortunately, because this is often the family's only time together, mealtimes can become times for conflict and conflict resolution. This can contribute to digestive problems and the association between food and anxiety or sadness. The association of food with joy is perhaps also not good, as it may come to be used as a way of solving problems and warding off depression.

Attention Seeking

Attention seeking is probably one of the most important motivating factors behind child behavior, second only to curiosity, exploration, and learning. From the time infants discover that they can make something happen, at around 3 months, when they can reach, grab on to something, and pull it toward them, to the time that they can crawl from one side of the room to the other, to the time they can walk by

holding on to furniture, they are discovering that they can do things. Typically the infant looks to whoever is in the near vicinity to see their reaction. The infant is innately a social being, and social interactions from the beginning of life involve getting attention and paying attention (Hobson, Patrick, Crandell, Garcia Perez, & Lee, 2004).

The problem is that as the world becomes busier and parents are juggling a thousand things, it is simply hard to pay attention to all the positive behaviors and all the accomplishments of the child. Doling out hugs and praise is time-consuming, can sometimes lead to dependency if given too often, and therefore is at odds with our cultural childrearing agenda, which is to facilitate autonomy (self-reliance). But children will inevitably get attention from someone by some behavior; often that is negative behavior, such as temper tantrums, interrupting, complaining, whining, breaking things, or having accidents. The child is impatient and wants attention when he or she wants it.

If possible, diverting the child's attention, mimicking the whining (babies stop crying when you imitate their crying), or attempting to ignore it and trying to engage the child in your activity (talking on the phone or laying out the lunches), will sometimes suffice. Escalations into full-blown temper tantrums, particularly when children are in public places such as restaurants, call for moving the child from the situation and having a time-out.

Clinging

Clinging is a fairly normal activity for young children when they are hesitant about new places and people. Clinging, or "holding on to the mother's apron strings," as they once called it, gives the child a sense of reassurance about the situation. Clinging is also fairly common during developmental milestone periods, for example, around 1 year, when the child is learning to walk and separation and stranger anxiety are occurring, and again during the time of the terrible twos. Situations during which children may cling are leave-takings at preschool and following special events like the birth of a sibling. This behavior occurs most frequently in temperamentally shy, slow-to-warm-up children who have difficulty making transitions to new situations.

Not showing your own anxiety about your child's anxious behavior will help. In addition, engaging children in play may help them adjust to the new situation. For example, when leaving children off at preschool, letting them have their stuffed animal or whatever makes them comfortable should help reduce clinging.

Thumb Sucking

Thumb sucking in newborns and young infants is considered a very sophisticated self-soothing behavior. Except for unusually shy children, most stop thumb sucking at around the preschool stage. After that, thumb sucking typically occurs only during times of stress and going to sleep. If this behavior continues into grade school, it is problematic because of peer rejection. In a study of 7-year-olds, for example, children viewed photos of boys and girls in thumb-sucking and non-thumb-sucking poses and rated the pictures (Friman, McPherson, Warzak, & Evans, 1993). The

results indicated that children in a thumb-sucking pose were rated as less intelligent, happy, attractive, likable, and fun and less desirable as a friend, playmate, seat mate, classmate, and neighbor than when they were in the non-thumb-sucking pose. Thus the risk of losing social acceptance is real at this stage. Typically thumb sucking subsides unless it is negatively reinforced. A parent's simple expression of concern that the child will be teased by his or her friends is enough warning for the child. Simply having distracters or substitute activities that are more interesting than thumb sucking can solve the problem.

Aggression

Aggression in young children is considered fairly normal. Usually it is their way of trying to get what they want, a toy or attention. Toddlers and preschoolers engage in pushing, shoving, hitting, kicking, yelling, and biting, and infants pull each other's hair and jab fingers in faces, which is not a problem until it is harmful to another child, a parent, or a teacher. Lots of the aggressive behaviors happen in the natural course of simply being active on the playground or in the classroom. Typically children pick themselves up or go on their way after being assaulted by another child in the course of rough-and-tumble play. Ultimately, with modeling from older children and adults, they learn to express their feelings verbally instead of physically (Tremblay et al., 2004).

Biting

Biting is perhaps the most difficult problem parents and teachers have with toddlers. Unfortunately, very little is known about how to prevent it because it is not a favorite topic for research. Generally children use biting when other behaviors do not manage to get them what they want from their playmates. Pushing another child often escalates into slapping, which escalates into biting. Biting attracts more attention from teachers because teeth marks are generally more alarming and potentially more dangerous than scrapes and bruises. Unfortunately, one child's biting often leads to an epidemic of biting, in which several children follow the lead and the problem becomes unmanageable. Biting occurs most often in the midmorning, randomly throughout the week, probably because children are more active in the midmorning, and most often in September, probably because this is a back-to-school period, which is a frustrating transition for children (Solomons & Elardo, 1989)

Generally teachers will pay additional attention to the biting child and try to predict when the child will bite. The teachers will typically intervene when they think the child is about to bite and attempt to distract the child with toys before the biting occurs. Other situations in which children might resort to biting are when they are extremely tired, frustrated, or overexcited. If children learn that they do not have to resort to biting to get what they want (teacher provides the wanted object before the biting occurs), they will probably no longer resort to biting. If the child appears to be compulsively biting, then more extreme measures may be necessary, such as providing a stuffed animal as a substitute for biting another child (it is less dangerous to bite the stuffed animal than another child). Another measure that

may be used is time-out. Usually a minute per year of the child's age is sufficient. Often children will see what they are missing during the time-out and will learn to control their behavior. Another effective measure is giving the child lemon juice. Most bites are minor and can be treated by washing, applying cold compresses, and comforting the child.

The advice of a child psychologist is that the best thing for an adult to do when one child bites another is to remain calm, take the child to a time-out area, tell the child "No biting" and that he or she should remain there until you return, and console the bitten child but without making a big deal of the injury. If the bite has broken the skin, administer a topical antibiotic and call a pediatrician. Then, without appearing that you are on the alert for another incident, be aware of signals that the child is working himself or herself up to a bite, move in, and either remove the child from the playing field or distract him or her. Research suggests that there is no connection between parenting style and biting, nor is biting indicative of psychological problems or predictive of later problems getting along with other children.

Because this behavior problem, more than virtually any other behavior problem, distresses parents, it needs special attention immediately. Parents become extremely emotional about this problem. For example, a thoracic surgeon parent was recently so hysterical about his child being bitten that he claimed the biting child could have pierced his child's lung (if he were being objective he would have remembered that he was a thoracic surgeon and knew very well that the rib cage stands outside the lung and would protect the lung from biting).

Head Banging

Head banging was once considered a pathological activity that only handicapped or emotionally disturbed children showed. However, recent studies on rhythmic behaviors of infants and toddlers suggest that many children engage in these rhythmic activities as part of normal development. Other behaviors are rocking, lip biting, and thumb sucking. They generally happen when the infant or toddler is otherwise unoccupied and appears to be bored. In this case, these rhythmic activities may be a source of substitute stimulation. At other times they occur when the child is extremely tired, and then they seem to pacify and quiet the child. Generally they will disappear after a short period of time, so they are probably best ignored. Children who show any of these behaviors excessively and are actually hurting themselves may need a professional evaluation (Berkson, 2002).

Stuttering

Approximately 8% of children stutter at some time, most usually between about 1.5 and 6 years of age, when the child is learning to talk, and later when the child begins school. Approximately 3 times as many boys as girls stutter, and stutterers are more often left-handed and have blond hair and blue eyes. Approximately 75% of children who stutter stop stuttering at around 12 or 13 years of age (Kalinowski, Saltuklaroglu, Dayalu, & Guntupalli, 2005).

Typically stutterers are delayed in their expressive language skills, not their receptive language skills, lending support to the hypothesis that language deficits

observed in stuttering children result at least in part from their attempts to simplify verbal responses as a means of coping with their stuttering. A recent study suggested that stutterers also show more nonspeech behavior during stuttered words than nonstutterers show during similar words (Bajaj, Hodson, & Westby, 2005). For example, they turn their head to the left, blink, and raise their upper lip excessively. Some evidence suggests a genetic predisposition for stuttering, and some researchers have found that the EEG of stutterers differs from those of nonstutterers, suggesting a disturbance of interhemispheric relationships. This may be a significant factor in the development of stuttering. Stressful events like the birth of siblings or transitions to new schools or new places can exaggerate the stuttering. The most difficult aspect of this problem is the embarrassment it causes the child and the delay it causes in language development.

Parents can help by not drawing attention to the stuttering, trying to ignore it, and certainly not speaking for the child. When children stutter, the tendency is to provide words for them rather than let the child complete the sentences. If the problem becomes so serious that the child cannot be understood, a speech therapist might be helpful. Often the therapist will work with the child to speak more slowly and to practice various breathing techniques that help the child become less stressed.

Temper Tantrums

Temper tantrums occur at frustrating stages in development. They are considered a mark of the child's attempts to become independent, for example around the terrible twos, when the child has power struggles with parents and because of limited language development has difficulty expressing his or her frustration except by tantrums. Temper tantrums typically arise from children's thwarted efforts to exercise mastery and autonomy. They occur more frequently in the active, determined child who has abundant energy. They may also occur in children who are hypersensitive to sounds, lights, and other stimulating features of the environment. Although these are considered normal behaviors, it is also clear that for a number of children temper tantrums turn into more difficult behavior problems (Leung & Fagan, 1991).

In one study, tantrums were reported in about 25% of children, most commonly at 2 to 5 years, becoming less common at 6 to 8 years. Children showing tantrums had a higher incidence of postnatal trauma. Parental overprotection and marital discord were stress factors in a greater number of boys than girls, and parental negligence was a significant stress factor for girls. Behavior problems that accompanied temper tantrums included thumb sucking, bedwetting, tics, head banging, and sleep disturbances. In another study, 52% of those who showed temper tantrums had multiple behavior problems. Factors independently associated with the tantrums included maternal depression and irritability, use of physical punishment, marital stress, child care provided exclusively by the mother, and poor child health (Potegal et al., 2003).

After having established that the child's temper tantrum is not simply related to discomfort, it's important to look at how the tantrum is being treated. Parenting practices that may encourage tantrums include inconsistency, unreasonable expectations, excessive strictness, overprotectiveness, and overindulgence. The child might also be bored, fatigued, hungry, or ill, thus reducing his or her tolerance for

frustration. In those situations it would obviously be important to distract the child or interest him or her in something less boring. Beyond these attempts, temper tantrums are probably best handled by ignoring the outburst and saving attention for the posttantrum, as well as helping the child learn to express negative feelings in less annoying ways.

Shyness

A significant body of research suggests that shyness is one of the strongest temperament traits in children (Stein, Chavira, & Jang, 2001). Research from our lab shows that even newborns show behaviors that look like shyness. Approximately 20% of newborns show shy, introverted behavior such as limited responses to social stimulation. We have called this being poker-faced (Field, 1982). At the other extreme, approximately 20% of children are extremely extraverted, or wear their heart on their sleeve. Approximately 60% of newborns and children lie somewhere in the middle of those extremes. In our study we found that when a child seemed to be born with a shy temperament but had extraverted parents, there was often a shift toward being somewhere in the middle. Thus, although this seems to be an inborn trait (particularly because we saw shyness more often being shared by identical than nonidentical twins), the environmental influences (such as shyness or no shyness in the parents) are also very strong (Gerhold, Laucht, Texdorf, Schmidt, & Esser, 2002).

Data also show that children who tend to be shy have unique physiological characteristics such as greater reactivity of heart rate and different brain waves. Sometimes they have very strong physiological responses to emotional situations, and their shyness or social withdrawal might be a way of providing a barrier to the excessive stimulation. If a child is socially withdrawn, there is often a tendency for people to respect the child's shyness and let the child take the initiative. Often the shy, slow-to-warm-up child will be most comfortable making his or her own overtures and contacts.

Researchers at Harvard found that children who were timid during infancy and as toddlers tended to be socially inhibited at age 7. They estimated that 15% of children, significantly more girls than boys, are born shy (Kagan, 2005), although they found that many outgrew their shyness.

Autism

Autism has a variety of genetic and nongenetic causes, although the etiology (development) is unknown in the majority of children. Boys are more frequently affected than girls, and autism first becomes apparent in the preschool years. It always affects sociability, communication, and the child's repertoire of activities and interests. Autism encompasses a broad range of severities and a variety of other signs of brain dysfunction. These include motor signs, notably stereotypes such as head banging and flapping of hands. Abnormal responses to a variety of sensory stimuli occur, and the children have very severe attention problems and problems showing their emotions. A significant number of autistic children experience epileptic seizures and have abnormal EEGs. Magnetic resonance imaging has revealed abnormalities

of brain development in some autistic children. The level of intelligence may range from profound mental deficiencies to giftedness. The pattern of cognitive skills is likely to be uneven, typically with better nonverbal than verbal skills (Mitchell et al., 2006). In the preschool years, autistic children have a language disorder. Verbal expression may range from total lack of language to being extremely verbose (some even have echolalia, which is endless talking). Comprehension and language use are invariably impaired. There are no specific drugs that can reduce the symptoms of attention disorder and seizures, but the autistic features can change with maturation and appropriate intervention. Communication skills and sociability remain deficient but improved for all but the most severely affected children. The children can be positively affected by early interventions focused on the development of appropriate social skills and meaningful communication.

Autism is a particularly difficult problem because the child is unresponsive. As early as infancy, autistic children are noted to arch their back when held, to make very little eye contact, and to simply dislike physical affection. At the preschool stage or as young children they prefer to play by themselves and often sit and do nothing, staring into space. Their difficulty staying on task and their autistic behaviors, such as flapping their hands and whirling around, spinning things, or rocking and banging their heads, is very disruptive behavior and often wins them no friends at school.

The number of autistic children has increased dramatically over the past several years, probably because screening techniques are more sensitive but less specific so that more children are labeled autistic (many of whom are not really autistic; Yeargin-Allsopp et al., 2003). Thus this is a "waste-basket" diagnosis, as some clinicians call it. Many schools are now providing special classes for autistic children, and some schools serve only autistic children. Statistically, approximately 1 in every 1,000 children is affected, with males being affected 4 times as often as females (Merrick, Kandel, & Morad, 2004). Generally the disorder is thought to be inherited. However, so many theories have been advanced about neurological and neurotransmitter causes that an entire journal is now devoted to autism. Medical conditions associated with autism include epilepsy, hearing deficits, speech and language impairments, and eyesight problems. A comprehensive medical workup is required in all cases of pervasive autistic symptomatology.

Children with autism have very little language and tend to repeat words, sometimes memorizing and repeating paragraphs of from magazines and in general speaking inappropriately. One study showed that using baby-talk-like language ("motherese") was more effective in gaining the attention of autistic children than using simple conversational tones. Many therapy attempts have focused on physical interactions such as wrapping the children in blankets and holding them to enhance their attentiveness. Although these children are also thought to have an aversion to touch, we were able to successfully perform a massage therapy study with them (Field et al., 1997). In our first study, the children who received massage therapy a couple of times a week for a month showed less off-task behavior, had fewer autistic mannerisms, and related more positively to their teachers, suggesting at least that they did not find touch aversive. In our second study the parents served as the therapists and the children had fewer sleep disturbances and less off-task

behavior in the classroom (Escalona, Field, Singer-Strunk, Cullen, & Hartshorn, 2001). This may be a technique that parents could use as a way of being physically more intimate with their typically rejecting autistic children and as a way of enhancing the children's attentiveness. In addition, we have used imitation therapy to help the parents and children interact. The parents imitate the child's behaviors, and the child becomes more social.

CAREGIVING

Touching

In a recent study on the touching behavior of mothers and fathers with their newborn infants, both mothers and fathers touched their infant for an equal amount of time and frequency, but mothers touched or stroked their infants more than they patted or moved them (Prodromidis et al., 1995). Fathers did not show a preference for either type of touching. Mothers preferentially touched their infant's hand and face more than the body, and fathers preferred to touch their infant's hand. The authors suggest that this is consistent with mothers' inclination to care for their infants and fathers' tendency to play with their infants. Another study conducted on touching premature newborns in intensive care suggested that touching occurs 63% of the observation time; 69% of the touching was treatment-oriented, 15% was caregiving-oriented, and 16% was simply comforting touch. The majority of touching was done by the nurses (Modrcin-Talbott, Harrison, Groer, & Younger, 2003).

Massage Therapy

Massage therapy has been noted to help infants and children in many ways. When we gave hospitalized premature infants three 15-minute massages a day for 10 days they gained 47% more weight, were hospitalized for 6 fewer days (at a hospital cost savings of approximately $10,000), and were more responsive to social stimulation (Field et al., 1986). Eight months later they still showed a growth advantage, and their performance on mental and motor developmental tests was superior (Field, Scafidi, & Schanberg, 1987). Massage therapy used with normal infants led to less irritability, more alertness, and better sleep patterns (Field, Grizzle, et al., 1996). Normal infants also gained weight following 5 weeks of massage. When compared to a group of infants who were rocked, the massaged infants were awake while they were being massaged and quickly went to sleep following the massage, whereas the rocked infants slept during the rocking but awakened as the rocking ended. This suggests that massage is a more effective way for parents to help their infants go to sleep.

Although the underlying mechanisms for the effectiveness of massage are not clear, massage seems to put infants and children in a more relaxed state. In that state their stress hormones (cortisol levels) are reduced and they are more alert. The reduction in stress hormones improves the functioning of the immune system.

Infant massage classes are being started in many places in the United States to teach parents how to massage their children. In at least one study we have shown that the person giving the massage benefits as much as the person receiving the massage (Field, Hernandez-Reif, Quintino, Schanberg, & Kung, 1998). In that

study we trained grandparent volunteers to massage babies, and after a month of massaging babies they showed several changes: fewer trips to the doctor, more social calls, fewer cups of coffee, less depression, and greater self-esteem. Although mothers are typically given the training, researchers have also trained fathers as massage therapists. For example, in one study Australian fathers were taught to give massages during bath time for 4 weeks after the infants were born (Scholz & Samuels, 1992). In another study by our group, by 3 months the infants were greeting their fathers with more eye contact, smiling, vocalizing, reaching, and orienting responses, and showed less avoidance behavior (Cullen, Field, Escalona, & Hartshorn, 2000). At this time the fathers who had been in the massage group also showed greater involvement with their infants. Thus, massage therapy has many benefits for both parents and their children.

Bathing

Bathing is typically one of those end-of-the-day activities when the parent has prime time with the child. From very early in infancy bath time is a favorite activity of the child. It becomes a family ritual and is intended not only for cleaning but also for relaxing the child at bedtime (Bryanton, Walsh, Barrett, & Gaudet, 2004). Children love the kicking and splashing, sliding under the water, exploring the taste of the shampoo, and even turning on the hot water without realizing it can burn. The most critical thing to remember is not only to have fun playing with the child and the bathtub toys but that the child should never be left unattended in the bathtub. The one inch of water in the bottom of the bathtub is the leading cause of drownings in the United States (Byard & Donald, 2004).

Swaddling

Swaddling, or the tight wrapping of babies (typically with a blanket), is done for a very limited period of time in the United States. Usually parents swaddle their babies to calm them down or help them go to sleep, and sometimes to make it easier to carry them (Franco et al., 2005). Although swaddling is more common in other cultures, such as American Indians wrapping infants tightly to carrying boards, it is discouraged in our country as a baby care practice because it is considered limiting to the development of hand-eye motor coordination.

Rocking

Rocking is a classic technique used by parents to calm their children and probably themselves as well (White-Traut, 2004). Recent studies suggest that pacifiers and massage are more effective at calming children and getting them to sleep, although rocking may very well serve the purpose of calming the parents. For that reason alone, its popularity may well continue.

Carriers

Carriers come in many varieties for various kinds of carrying positions. For example, in a recent study we observed that carrying the baby in a front-to-front

position led to infant sleep, whereas carrying the infant facing outward led to more wakeful and exploratory behavior (Field, Malphurs, Carraway, & Pelaez-Nogueras, 1996). The optimal carrier might allow for both positions. Being carried by the parent has a long history in evolution and is popular in many cultures, probably because it provides additional stimulation for the baby, enables the mother to closely monitor the baby, and lets her work unencumbered with hands free. This arrangement might also allow for less heat loss and the conservation of energy.

Music

Music has profound effects as early as birth. Studies on premature infants, for example, show that following repeated exposure to music, infants have reduced heart rate and deeper sleep and grow and develop better (Arnon et al., 2006; Standley, 1998; Standley & Moore, 1995). In a study on preschoolers, performance on cognitive tasks was superior following exposure to music. Much of these effects may be attributed to music's ability to relax infants and young children. When heart rate and blood pressure are recorded, for example, they are lower following music. When infants and children are distressed, they typically show greater EEG activity in the right frontal area of the brain (activity in this region is typically associated with negative emotions and mood states). Exposure to music can actually shift EEG activity from the right to the left side of the brain (Field et al., 1998).

Stimulation

Optimal parenting is often discussed in the context of providing the right amount of stimulation. This is particularly true for the infancy and toddler stages, when children need some help monitoring and regulating the amount of stimulation they receive. By virtue of having different temperaments, different children have different thresholds for the amount and kind of stimulation they desire. Matching the kind of stimulation to the child's personality is critical because either too much or too little stimulation will not enhance growth and development. Sometimes parents provide too little stimulation because they are tired, depressed, or preoccupied by their own problems. At other times, for example when children are unresponsive or slow in doing things, parents will be overstimulating and they try to get a response or an action.

Finding the optimal amount and kind of stimulation a child needs is one of the most difficult tasks of parenting. Probably the most effective approach is to take the cues from the child. When interacting with infants, it is easy to tell whether they are getting the kind of stimulation they want or need. Infants will avert their gaze, turn their head, arch their back, and, at the extreme, fuss and cry when they are either getting too little or too much stimulation. Slightly older children may go into a shell and ignore their parents' attempts to engage them. From very early infancy, children have conversation skills such as turn-taking and paying attention or ignoring. These skills can be used as signals that the parent is doing the right thing and the child is interested. Just like adults, children need breaks to not only process information from the conversation and formulate a response,

but also to take rests and have some quiet periods for thinking and imagining things.

Game Playing

Game playing punctuates early interactions between parents and their infants (Field, 1979). Infant games such as pat-a-cake and peek-a-boo appear with such frequency that they are universally recognized as infant games. The importance of early game playing has been underscored by a number of early interaction researchers, who suggest that game playing provides a context for learning one of the most important rules of conversation, that of turn-taking. Infant games are highly repetitive with simple and stereotyped roles for both participants. As such they exemplify the structure of conversational give and take or turn-taking. Even when the infant is yet too young to assume an active role, the structure of these repetitive games may enable the infant to assimilate the turn-taking nature of interaction.

Other important features of early games include the "contingency experience" they provide (i.e., learning to react to and innovate in changing situations) and the facilitation of positive emotion. An illustration of the contingent responsivity provided by a game can be seen in one of the most popular early games, called "Tell Me a Story." The words are provided by the parent, who treats the infants' vocalizations as if they too were words. The parent asks, "Do you want to tell me a story?" The infant coos, the parent responds, "Oh yeah? And then what happened?" The infant coos again and the parent replies, "Oh, that's funny." The infant contingently responds to the parent's words by cooing, and the parent contingently responds to infant cooing by further elaborations of the game.

The game "I'm Gonna Get You" is an illustration of the way in which games facilitate positive emotion. The parent repeatedly looms toward the infant, rubbing his or her head into the lap of the infant, tickling and jostling the infant, and accompanying each repetition with verbalizations such as "Ah, boom!" or "I'm gonna get you." These repetitious behaviors elicit infant smiles, vocalizations, and sometimes laughter. The parents contingently respond to these infant behaviors by playing more of the same game. The game ceases at the point at which it no longer elicits contented responses, and the parent moves on to another game.

Thus, early game playing provides an affective, contingency experience during which infants may learn important rules of communication such as turn-taking. When games are missing from the interaction or age-inappropriate games are played, the interaction appears to be badly timed, disorganized, or disturbed. When a mother plays age-inappropriate games or games that are too complex for a given age of infant, for example peek-a-boo with a 6-week-old infant, the interaction lacks positive emotion and is not sustained.

The early interactions of developmentally delayed, atypical, or failure-to-thrive infants and mothers feature very few games. These mothers do not appear to enjoy playing games and do not as frequently play games, particularly age-appropriate games, with their infants, probably because their infants are less responsive.

In one of our studies, the types and frequency of widely recognized infant games were observed during face-to-face interactions of normal and high-risk

infants and parents when the infants were 4 months old (Field, 1979). Approximately six different games were played for approximately one-third of the interaction time. Fathers played with normal male infants more often, and the high-risk infants and parents played games less frequently than the normal infants and parents.

Interaction Coaching

When parents, most typically mothers, are having difficulty interacting with their infants, they can be given interaction coaching, which helps modify the behavior of the parent, and, in turn, the behavior of the infant (Field, 1983b). Typically mothers and infants are seated in a face-to-face position, with the infant in an infant seat. The mothers are given instructions to slow down their behavior. For example, they slow down when they are asked to imitate their infants' behaviors and they provide more stimulation when they are asked to keep their infants' attention. Having to imitate the baby's behaviors slows down the mother's behavior simply because infant behaviors are slower than adults'. The infants usually delight at being imitated, and before long, mothers and infants are engaging in a back-and-forth imitation game. Another way that mothers learn to modify their behavior is by watching video feedback on their interactions. Mothers can usually see the immediate effects of their behaviors and will show change almost immediately.

Schedules

The importance of a schedule cannot be overrated. From the time of birth, newborns need to be on sleeping and feeding schedules so that they can organize their own internal clock and have sustained periods of sleeping and ultimately sustained periods of playfulness (Field, 1978). Having periods of consolidated sleep, including deep sleep, is critical for children to sustain their activities, attentiveness, and interest in their play. Without having a schedule when they are growing up, children can have serious difficulties organizing themselves and often will become hyperactive and develop learning disabilities.

I remember one of our laid back nursery school teachers who believed that every infant should have his or her own schedule, and she operated her infant nursery class on that premise. The infants were extremely fussy, often spitting up and not being able to focus their attention on any activities. Being on 16 different schedules, the four teachers also became exhausted and irritable. The teacher who replaced her ran the infant nursery like an army sergeant and had all the infants on the same schedule within a week. Parents were at first resistant but came to appreciate that their infants were much less irritable. The need for a schedule continues through childhood, which is why children typically do much better in a school setting than they do on weekends, when everyone is on a different schedule.

Quality Time

In today's world of dual-career families parents are concerned about snatching as much time as they can with their children (Broom, 1998). Often this leads to

children staying up late and sacrificing the quality time they spend in day care. Quality time, the time parents give their undivided attention to their children, is key—not quantity of time (Brooks-Gunn, Han, & Waldfogel, 2002).

Discipline

Affection and discipline are two of the most important aspects of parenting. Children growing up in the wild without parents become like wild animals because they are missing affection and discipline. In addition to needing affection for growth and development, children need discipline for their behavior to be organized and self-regulated (Socolar, Savage, Keyes-Elstein, & Evans, 2005). While some choices can be left to the child, for example, what the child wishes to wear, other matters that are critical to survival, such as not playing in the street and not being cruel to other children, require very firm limits. A parent should not ask a child if the child wants to hold the parent's hand while crossing the street; this is not a safe choice for a child to make. Having an intellectual discussion on the reasons why hurting someone is bad is also not effective. I recall a teacher in a model preschool quietly leaving the room after having been hurt by a flying block only to return several minutes later to have an intellectual discussion with the children about the "negative aspects of flying blocks." A better lesson would have been reacting normally to being hurt so the children would know the consequences of their actions.

Another problem with discipline is its frequent inconsistency. Recent reviews of the literature on behavior problems and on child abuse suggest that the most predictive factor for both is inconsistent discipline, such as parents being inconsistent from one time to the next or from one parent to the other (Field, 1983a). In societies where discipline is consistent within and across parents and between parents and teachers, such as in France and China, children have fewer behavior problems, and the incidence of child abuse is very low. Socially accepted forms of discipline are practiced by parents and teachers alike. Thus, the child is never seen driving the parent up the wall and the parent doesn't lose control.

A recent study revealed that verbal and physical forms of discipline are not substitutes but, instead, are commonly used together. Parents who yelled frequently were the ones most likely to hit frequently (Wade & Kendler, 2001). Respondents who were spanked (yelled at) frequently as children were more prone to frequently spank (yell at) their own children. Harsh parental discipline occurring early in life was associated with later aggression in children. The good news was that most people were able to break out of the cross-generational cycle of punitive childrearing, although physical punishment is still quite common. In a recent survey, 19% of mothers believed that there are times when it is appropriate to spank a child less than 1 year old, and 74% believed this about children 1 to 3 years old (Vittrup, Holden, & Buck, 2006); 42% reported that they had spanked their own child in the past week. Mothers believed more strongly in spanking for dangerous misbehaviors than for annoying ones. Other traditional disciplinary methods include rewards for good behavior and punishment for unacceptable behavior. The most common rewards are verbal praise, a smile, special attention or activities, or physical affection. Usual kinds of punishment include verbal disapproval, an unhappy look, ignoring

the behavior (extinction), or temporary isolation (time-out). Rewards are typically cited as being more effective than punishment.

Childproofing

Childproofing offers the opportunity to simplify your environment. It can even be fun to convert your living room to a playroom where the whole family can enjoy playing together. The obvious safety precautions are removing sharp objects, poisonous substances, plastic bags, and anything that can injure or anything small that can lead to choking or suffocation; placing guards on stairs, safety latches on cupboards, plugs on electric outlets, and locks on doors; and certainly locking up guns. For many years children do not know the difference between reality and fantasy, and for that reason alone guns should be under lock and key. Choking on foreign bodies is the cause of death for more than 300 children each year in the United States. A review of the records of 548 children (ages 4 months to 18 years) was undertaken to identify factors important in diagnosis. A wide variety of objects was recovered, the most common being peanuts, organic material, other nuts, popcorn, seeds, plastic objects, and pins (LeBlanc et al., 2006).

Airway obstruction from aspiration of a foreign body should be suspected in all infants and children who have swallowing or respiratory difficulties (Agran et al., 2003). If the patient is unable to clear the airway by coughing, the Heimlich maneuver should be attempted. Parents can help to prevent airway obstruction by keeping small objects away from infants and children and by teaching them to chew food thoroughly. Whenever the child is approaching a forbidden object or something dangerous, it is best to simply remove the child or distract him or her with a similar object or activity rather than saying no and expecting the child to leave the area. Saying no for most young children is simply a challenge and escalates into a struggle. Simply carrying or taking children by the hand and moving them to another area where they can be distracted is more effective (Glick, Greaves, Kronenfield, & Jackson, 1993).

Leave-takings

Leave-takings are increasingly common, given the increasing numbers of infants, toddlers, and preschoolers attending nursery schools. Research suggests that different children have different leave-taking styles. In one study, children who cried during the early morning separations showed either a slow approach or no response except looking at the mother when she returned from her day's work (Field, Gewirtz, et al., 1984). Those children who left their mothers quickly in the mornings typically continued playing and showed things to the mother when she returned in the afternoon. The mothers of children who cried during separations more often slipped out of the room. The authors of this study suggested that much of the separation and reunion behavior is determined by the child, with the mother merely being in a responding mode at these times, mainly waiting to receive and act on signals from her child. However, in a study we conducted, we thought that the parents were very influential in the leave-takings.

We studied the leave-taking and the reunion behaviors of the infants, toddlers, preschoolers, and their parents as the children were dropped off and picked up at their nursery school each day (Field, Gewirtz, et al., 1984). On arrival at their classroom, infants and toddlers related primarily to their parents, whereas preschoolers related to their teachers. Girls more frequently engaged in interaction with their teachers, and boys more frequently approached the children's play activities. Toddlers showed the most distress behavior during the parents' departures, and the toddlers' parents hovered about them and sneaked out of the room more frequently. When mothers dropped off the children, the children showed more attention-getting behavior and crying than when the children were dropped off by their fathers. Mothers engaged in more behaviors to distract the child and took a longer time to leave the classroom than fathers did. Giving a verbal explanation, distracting the child, lingering in the classroom, and sneaking out of the room were accompanied by more crying. We noted that some children had difficulty making transitions, and those children showed difficulty not only leaving their parents in the morning but leaving their classroom in the afternoon. In addition we noted that fathers were much more nonchalant in their behavior, and the children were less distressed when they were dropped off by their fathers. Thus, the rather business-like style of leave-taking by the fathers seems to work better if parents want to avoid the stress of seeing their children's distress.

Moving

Moving is one of the most stress-producing and growth-producing experiences in childhood. Whether the move is from classroom to classroom, neighborhood to neighborhood, city to city, or country to country, the source of stress is losing one's friends. Separation from peers leads to symptoms very much like those that occur following separation from parents. The child becomes agitated and then depressed. Sleep is most affected by moving, with night wakings and night terrors frequently happening. The child's behavior fluctuates between being aggressive and withdrawn, and like depression in adults, eating habits change (overeating or undereating). The children experience constipation or diarrhea, and sometimes they get sick because of immune problems. We have observed these changes in studies on moves to new classrooms and moves to new schools (Field, Gewirtz, et al., 1984; Field, Vega-Lahr, et al., 1984). Even as young as 1 year of age, we are having to move children's best friends with them when they graduate from infancy classes to toddler classes so that they will not experience these symptoms. The children left behind often experience more severe symptoms, perhaps because they are remaining in the same environment where they are reminded of their long-lost friends.

Maternal Depression

Mothers who are depressed or who have symptoms of depression such as flat affect (showing little emotion) and listless behavior tend to have a difficult time interacting and being responsive to their infants (McLearn, Minkovitz, Strobino, Marks, & Hou, 2006a, 2006b; Paulson, Dauber, & Leiferman, 2006). The infants in turn develop

flat affect and low activity levels. If the mother remains depressed over the first 6 months of life, the infant will begin to show growth and developmental delays by 1 year. By 3 years children of depressed mothers develop behavior problems such as depression and conduct problems. If mothers are simply depressed postpartum and the depression does not continue, the infants do not experience long-term effects.

Fathers and substitute caregivers such as infant nursery teachers can buffer some of these effects (Hossain et al., 1994; Pelaez-Nogueras, Field, Cigales, Gonzalez, & Clasky, 1994). Infants of depressed mothers often appear normal when interacting with their nondepressed fathers and teachers. Other effective interventions are the use of mood inductions with the mothers, such as music therapy and massage therapy (Jones & Field, 1999). Interaction coaching to help mothers be more active and responsive to their infants, such as playing peek-a-boo and imitation games, helps them improve their interactions. Teaching depressed mothers to massage their infants also helps improve the mothers' mood and reduce stress in the infants (Field, Grizzle, et al., 1996).

Fathers

In most studies people have found fathers to be more playful and stimulating and mothers to provide more caregiving and comforting. We found an exception to this when we studied fathers who were primary caregivers (Field, 1978). The stay-at-home fathers turned out to be more like mothers than fathers, and the mothers who were the breadwinners were more like the stereotypical father. We suggested that this meant that more caregiving is provided by the person who knows the infant better and more playfulness by the parent who knows the infant less well.

In a sense, the more stimulating, rough-and-tumble play of the fathers is complementary to the more soothing, comforting behavior of mothers. In a similar light, mothers tend to view their infants as being more difficult (perhaps because they are in a comforting role), and fathers are reported to enjoy their infants more (again because they engage in more play with them). Across the first several years, fathers are more involved in social interaction than in caregiving, and mothers continue to provide more of the caregiving. Although the differences between mothers and fathers has lessened over the years as more mothers have joined the workforce, some differences have continued, showing the mother providing more caregiving and the father more playful stimulation (St John, Cameron, & McVeigh, 2005).

Despite the differences in their parents' behaviors, infants and young children are noted to be equally attached to both parents. It may not appear that way, as infants and young children often protest their mothers leaving them at preschool and barely notice their fathers leaving (Field, Gewirtz, et al., 1984). This is probably because fathers matter-of-factly leave, saying they will see their child later, whereas mothers make a greater fuss about leaving their children and express more reassurance about their returning, which seems to draw out this fussing behavior. The children's ambivalence about leaving their mothers is also displayed upon her return, whereas infants greet fathers as they return with intense, brief displays of positive emotion.

Grandparents

Grandparents have a number of positive effects on child development (Harwood, 2001). Children are often more able to explore and try their skills around grandparents when not around their own parents. They seem to recognize a certain autonomy around grandparents, in contrast to their dependence around parents. This may be a two-sided process, as grandparents appear more relaxed and less overprotective around their grandchildren than parents do. They apparently also have less need to set limits than parents, as typically children seem more self-regulated in their presence. Part of this phenomenon may relate to grandparents giving children their undivided attention, whereas parents are often busy trying to juggle children and other activities at the same time. It could be that the child's more demanding and less compliant behavior in the parents' presence is attention-getting behavior. It could also be that the children sense greater demands being placed on them by their parents and higher expectations for performance. Combined experience with parents and grandparents, in any case, would appear to be optimal for any child.

Babysitters

Having a babysitter even for a newborn is a great break for parents and provides them needed time together. Finding someone who has experience, who loves taking care of babies, and who comes with good references from people you know is generally the best way to choose. On the first few occasions it is important to be there for a while until the babysitter becomes comfortable in the environment and familiar with the baby. Watching how the babysitter relates to your child and cares for your child in those few hours of getting acquainted is one way to develop confidence in the babysitter.

Day Care

Early day care helps develop a child's social and intellectual skills, but only if it is quality care. Penelope Leach, in her book *Children First,* and others have claimed that full-time day care can have harmful effects. However, their concerns relate to poor-quality day care (Geoffroy, Côté, Parent, & Seguin, 2006; Schwebel, Brezausek, & Belsky, 2006). In contrast, the average day care that most middle-class kids get is considered helpful for cognitive and social development, according to the National Research Council.

Quality care can be hard to define, but knowing what to look for in a day care center helps. State and federal governments have licensing requirements that ensure the provision of adequate space and safety features (e.g., smoke alarms and sprinkler jets). These requirements also ensure optimal teacher-student ratios and class sizes. For example, quality programs in Florida have no fewer than 1 teacher for every 4 infants, 1 teacher for every 6 toddlers, and 1 teacher for every 10 preschoolers in classrooms with no more than 20 infants, 30 toddlers, and 40 preschoolers. These figures, however, are meaningless if sufficient space is not provided (approximately 35 sq. ft. for indoor space and 50 sq. ft. for outdoor space per child).

Most parents know when they have discovered a quality day care setting. It will be immediately apparent that the teachers are nurturant and the children are happy. Children will be involved in creative activities and the environment will be organized, colorful, and rich with interesting materials and play opportunities. A class schedule with designated times for show and tell, snacks, naps, free play, and expressive arts, along with free time for fantasy play and gross motor play (play that uses muscles) is also an important part of quality day care. Outdoor play areas will include moving vehicles, sand play, and structures for climbing. Parents should ask questions about how long the teachers have been with the center (stability of the teachers is very important) and how much experience they have.

Some of the highest quality centers are found in university lab schools because they are partially subsidized by the universities. Some of the benefits of a laboratory school include active research and teacher training programs affiliated with their early childhood education and psychology departments. Religious centers (e.g., churches and synagogues), hospitals, corporate centers, and Montessori schools also provide quality day care. Specific listings are often provided by early childhood education departments at universities.

In a study on skills needed by preschool children for success in day care, teachers assigned the highest ratings to communication and independence. More specifically, these included children's ability to respond to their names, ask for help from an adult, stay out of restricted areas and respond to adults' questions. The teachers rated skills in the area of preacademics (including sitting cross-legged in a circle and cutting with scissors) the lowest.

In a study we conducted, the longer infants were in quality day care, the greater their emotional well-being and academic performance at grade school (Field, 1991). Similar data have been reported for quality day care in Sweden.

Pacifiers

Pacifiers are often viewed with disdain, and they are discouraged to the extent that premature infants in some hospitals require a physician's prescription for a pacifier. Yet documented and anecdotal reports show that sucking pacifiers helps calm infants during uncomfortable procedures, and toddlers are made more comfortable when adapting to unfamiliar situations (Stevens, Yamada, & Ohlsson, 2004). Pacifiers provide comfort, promote physiological tranquillity, and help in growth and development. In one study, we used pacifiers during painful hospital procedures (Field & Goldson, 1984). The infants who were allowed to suck on pacifiers cried significantly less, and their heart rate and respiration were slower. Pacifiers used during circumcisions reduced crying by 40%. It is not clear whether pacifier use reduces pain associated with invasive procedures or if the sucking is just incompatible with crying and therefore crying occurs less often. It is also possible that sucking on pacifiers releases pain-alleviating chemicals like serotonin. Sucking on pacifiers has also been used for weight gain, particularly with premature babies. In a study we conducted on premies, those who were allowed to use pacifiers gained weight more rapidly and were discharged from the hospital 6 days earlier than those not using pacifiers, at savings of approximately $3,000 per infant

(Field et al., 1982). These infants also performed better on a newborn assessment scale.

Given the positive effects of sucking, it is very fortunate that infants are inclined instinctively to engage in sucking behavior. That predisposition emerges very early in gestation, with fetuses as young as 5 months appearing to suck on their own hands in utero. During the first day of life, infants have been observed sucking as often as 30 minutes per hour. This does not appear to vary with the feeding experience. One survey found that 89% of parents introduced pacifiers to soothe their infants. Most studies in the literature concur that sucking is the most common method for soothing an infant. In another study pacifiers were compared with rocking. Pacifiers produced sleep states, and rocking produced alert states. Pacifiers also reduced heart rate levels significantly more than rocking. Thus pacifiers seem to be more soothing than rocking.

Sucking is also used for exploration and learning. Babies in one study sucked harder to hear their mother's voice than their father's voice, probably because they were more familiar with the vibrations of their mother's voice in utero (DeCasper & Prescott, 1984). Sucking is used in this and many studies to determine whether infants hear, see, make discriminations, and demonstrate sophisticated skills such as performing small math problems, discriminating different colors, and learning musical scales as early as the newborn period. Other investigators have noted how much learning occurs during infants' exploration by mouth. For example, infants are able to show that they can differentiate two different nipples (one with a knobby surface and one with a smooth surface) simply by having experienced sucking on the nipple and then looking longer at the picture of the nipple they had sucked, showing their preference for that familiar nipple (Hernandez-Reif, Field, & Diego, 2004).

Even if infants do not have nipples to suck on, they frequently suck on their hands or other objects. As early as 5 months gestation, fetuses explore their hands with their mouths. This persists through early infancy, when the mouth explores the hands, fist, and fingers. Similarly, for at least the first 6 months of life, everything the infant grasps goes into the mouth. In a recent court hearing I was asked how a 5-month-old infant could stuff a rather unpleasant-tasting alcohol wipe in his or her mouth. Invariably every 5-month-old I filmed for the hearing did exactly that. When babies have something more interesting to suck on, such as a nipple or rattle, they do not choose alcohol wipes. Infants prefer toys because of their interesting textures of softness, hardness, and firmness. Toys also give the infant information about features such as shape. Infants have individual preferences as well. Some infants enjoy sucking their soft toes, and others enjoy hard, firm plastic toys. Thus, sucking is a predominant activity during the first 6 months of life, just as walking is at 1 year, and it clearly provides a very basic learning function.

If sucking is so critical for reducing stress and facilitating growth and development, most especially cognitive (mental) development, and if sucking does not appear to have any negative effects, why should anyone want to eventually discourage the habit? One reason is that as a child develops, sucking on a pacifier or thumb may be incompatible with vocalizing and talking, just as it is incompatible with crying during the newborn stage. When infants become preschoolers they may

experience rejection by peers for engaging in "baby behavior." Thus pacifier use at preschool is probably not desirable. In one study pacifier use declined in a voluntary way between 3 and 24 months, suggesting that for most children, the attachment to a pacifier declines naturally. However, for others, sucking is a type of security blanket that becomes difficult to relinquish.

To prevent pacifier overuse parents should not get into the habit of popping pacifiers into their babies' mouth when the baby doesn't need pacifying. It is important to learn to tune in and appropriately respond to your baby's cues and prevent pacifier overuse. Older babies should be given a pacifier only for true pacifying, soothing, and calming.

As an aside, parents often ask whether sucking on a pacifier or thumb can mean that the child will need to wear braces. For most children this is not a problem. Pacifier and thumb sucking have very little effect on a child's oral-facial development and on the primary teeth and position of the jaw when it is done periodically (Adair, 2003). If the child continues to suck when the permanent teeth are coming in, pacifier or thumb sucking could cause movement of the teeth and bones, and orthodontic treatment might be needed.

Security Blankets

Security blankets have their place in infancy and toddlerhood, as they allow children to more securely leave their mother (father leave-takings are typically less difficult). However, beyond the toddler stage, security blankets may be a source of embarrassment for the children when their peers make fun of their security objects. Bringing toys to school may provoke less embarrassment, but that often leads to little wars over possession of each other's toys. For that reason most teachers prefer that children leave their toys at home.

Toilet Training

Toilet training is typically treated by pediatricians as an individual readiness phenomenon, although most children are toilet-trained (according to recent study) by 2.5 years, with physicians beginning to discuss toilet training with parents at around 12 to 18 months of age (Horn, Brenner, Rao, & Cheng, 2006). Although parents are initiating toilet training at later ages, around 3 years (Blum, Taubman, & Nemeth, 2004), waiting to start toilet training at age 2 is particularly problematic because the terrible twos involve power struggles between parents and children that make toilet training extremely difficult (Blum et al., 2004; Schum et al., 2001).

Placing a child in training pants and trying not to use diapers except during nighttime sleep is an effective introduction to toilet training. Buying children their own toilet seat gives them a sense of control over their own activity. Children seem to learn faster when they are attending preschool, probably because the older children model more mature toileting behavior. Taking children to the toilet frequently will give them the idea, whether at home or at school, that there are designated periods when all kids have bathroom time.

Flashcards

The flashcard method was designed by a "better baby" school in Philadelphia to teach children music, language, science, and virtually anything that can be taught on flashcards to infants and toddlers. The school has repeatedly made great claims that 2-year-olds can learn physics, violin, and the classification of insects, for example. This is done by repeated drilling with flashcards, typically at home by the parent. None of these claims, however, has been supported by research, and the directors of the school are reluctant to have their methods tested. It is also not clear why a parent would want their child to know all this by the time the child enters school, only to present the potential problem that the child will be rejected as "weird" by his or her peers. The child who is busily learning flashcard skills has very little time to learn social skills and how to get along with peers, which is certainly a more important priority at the preschool stage.

Television

The American Academy of Pediatrics recommends that children under the age of 2 not watch any television. Yet, several surveys indicate high numbers of infants routinely watching TV, including 17% of 0- to 11-month-old infants and 48% of 12- to 23-month-old toddlers in a large U.S. study, and even larger numbers in a Japanese study (30% of 3-month-olds and 90% of 12-month-olds). Exposure is also high, at an average 44 minutes per day among American 4-month-old infants, increasing to 62 minutes per day in 12-month-old infants; and again, there is higher exposure in Japan. In one study, 92% of exposure of 4-month-olds was to adult TV. Some view TV exposure as a cognitive learning experience, but others are concerned that infants are imitating TV models, both immediately and days later, remembering and mimicking the model's actors. They also are taking directions from TV. For example, in one study, 10- to 12-month-old infants who watched an actress respond to a toy with fear later avoided playing with it themselves, and they were more likely to appear worried and cry. In another study, food-related advertising on TV was associated with brand recognition in toddlers (Connor, 2006). Other studies suggest that television viewing in infants is associated with irregular sleep schedules (Thompson & Christakis, 2005) and with a risk of becoming overweight (Dennison, Erb, & Jenkins, 2002).

Children are first exposed to the audio and video stimulation of TV shortly after birth to the age of 2 months. They then begin to have an interest in and glance at or watch the TV screen at 4 to 5 months of age. When they are able to crawl at around 6 months, they approach and investigate the TV set. At 7 to 8 months they enjoy switching channels, and then they become interested in and speak to the on-screen characters at 10 to 11 months. At 12 to 14 months of age they have favorite TV characters and programs and enjoy TV as ordinary TV viewers. They move their bodies with the music, mimic the actors on-screen, smile, and ask questions of parents. They fix the information from TV in their memory, and they are pleased to see the things they like on TV. Even the fetus is affected by TV, showing heart rate acceleration or deceleration and fetal movement when the mother is watching

TV. Disruption of sleep patterns due to TV may also occur to fetuses. For some, though—for example, infants of depressed mothers—selected children's TV may provide more adequate stimulation.

Accidents

Approximately 80,000 children are killed and another 50,000 disabled by unnecessary accidents, according to the U.S. National Safe Kids Campaign. This makes accidents the number 1 killer of American children. Accidents are now the cause of more deaths in children than the next six most frequent causes combined. The leading causes of these accidents are traffic injury, drownings, burns, choking, poisoning, and falls.

Minor accidents such as scrapes and bruises are inevitable as infants, toddlers, and young preschool children learn to wend their way through the world. Both children and parents overreact to these minor injuries. Accident-prone behavior has been related to hyperactivity, clumsiness, and antisocial behavior in children and lack of parental discipline in the home. Injury-prone behavior is most common, for example, in active boys with irregular sleep and eating problems. Often, though, accidents happen to children who are simply adventuresome.

Solutions for accidents are making the environment safe, including using seat belts and child safety seats to avoid traffic injuries, teaching the infant how to swim, using pool detectors, never leaving an infant alone in the bathtub to avoid drownings, monitoring hot water in the bathtub to avoid burns, making certain that back burners are used on the stove and smoke detectors are in working order, and locking up all medications, liquor, and cleaning substances to avoid poisoning. Children's homes need to be childproofed at least until the early grade school years, including using electric socket covers and safety latches on cupboards and covers on sharp corners of furniture.

Automobile accidents are the number 1 killer of children (Rimsza, Schackner, Bowen, & Marshall, 2002). Until children are 5 years of age (or more than 45 pounds) they need to be in a car seat that protects them from head-on as well as side crashes and meets the U.S. federal motor vehicle safety standards. Every state has laws requiring that children younger than 5 must be strapped into car seats in cars in motion. Children under 5 are most commonly killed in cars. Children under 10 should not be allowed to cross streets without adult supervision. An immediate need following a car accident is, of course, an emergency rescue.

Death from *fires* is the second most common cause of death, with 75% of these being caused by smoke inhalation and 80% occurring in house fires (Istre, McCoy, Carlin, & McClain, 2002). Approximately 50% of house fires are caused by cigarettes. Smoke detectors should be installed in areas just outside bedrooms, and a fire extinguisher placed in the kitchen and on each floor of the house. Scalding is the most common cause of nonfatal burns; 20% of these result from tap water and 80% from spilled food. The average age of children with burns is 32 months. Hot water heaters should be regulated at 120 degrees Fahrenheit. Because coffee burns are most common, infants should not be seated on any adult's lap during a coffee break. For the treatment of burns, cold water should be dribbled on the burn,

sterile gauze squares held in place by gauze roll bandaging, and the doctor called or the child taken to the hospital. Other precautionary measures include not heating an infant's bottle in the microwave, dressing the child in fire-retardant sleep wear, and avoiding place mats and table cloths that can be pulled off the table.

The most common cause of *drowning* (the third most common cause of death to children) is leaving the child alone in the bathtub in a few inches of water in a minute's time. Even with swimming lessons, children less than 5 typically have less strength and coordination to swim out of danger, and their parents have a false sense of security from the swimming lessons, which often leads to minimal supervision. A safe rule of thumb is that any child under 10 should be supervised whenever near the water. An increasing percentage of drownings (44%) occur in nominally fenced-in pools in which the gate is open or the fence is in disrepair.

Approximately one-third of *poisonings* by medicine are due to children taking their grandparents' prescription drugs. Thus, not only your own home but the grandparents' home should be made child-safe. Cosmetics, shampoos, detergents, plant poisons, paint materials, and substances associated with automobiles need to be out of reach. Because different poisons require different actions—for example, for some it is important to induce vomiting, whereas others should not be treated that way—it is important to have a poison control center card and to call the poison control center, the doctor, or an emergency room.

An information card is also important in responding to *choking* situations. Parents should take cardiopulmonary resuscitation (CPR) classes in case artificial respiration is needed. And parents should certainly learn the Heimlich maneuver. *Miscellaneous dangers* in the home are stairs, windows, small objects, sharp objects, toilets, matches, tools, plastic bags, telephone cords, and, of course, guns. Most children do not know the difference between real and fantasy guns.

SUMMARY

The problems of infancy can be categorized as both the frustrations and challenges of the developmental milestones that occur and the problems they pose for the parents facilitating development and giving care along the way. The research on these problems is sparse and highlights the need for further studies on the problems of infancy.

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CHAPTER 32

Psychopharmacology as Practiced by Psychologists

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According to the *Oxford English Dictionary* (*OED*), the term "psychopharmacology" was first used in print by Mora in 1920 to refer to a yet to be developed science dealing with the behavioral effects of drugs (Macht & Mora, 1921). This meaning persists to the present, as indicated by the current *OED* definition: "Psychopharmacology: The branch of science concerned with the way drugs affect the mind and behavior." The mind, of course, cannot be directly assessed, but is studied indirectly by examining behavior, that is, what individuals say and do.

All drugs that affect behavior—broadly construed to include mood, cognitive status, and overt actions—are termed "psychoactive." Drugs that are prescribed with the purpose of improving behavior are called "psychotropic." All psychotropic drugs are psychoactive, but not all psychoactive drugs are psychotropic. The introduction of the first generally effective psychotropic drug, chlorpromazine (Thorazine), in the early 1950s revolutionized psychiatry and changed forever the practice of mental health.

Historically, only physicians prescribed psychotropic drugs. Over the past 20 years, however, the possibility of granting the privilege of prescribing psychotropic drugs to properly trained psychologists has become a highly visible, and hugely important, issue. The American Psychological Associations (APA) supports prescription privileges for psychologists, as do many state-level psychological associations. Two states recently enacted legislation allowing properly trained psychologists to prescribe these drugs, but there is continuing debate among psychologists as to whether or not prescription privileges are generally desirable. The issue is complex and we will examine it in a later section. But quite apart from the prescription privileges issue, psychologists have made and are making major contributions to the study of psychoactive drugs. We first examine some of those contributions.

PSYCHOLOGY AND THE STUDY OF DRUGS

Psychologists are expert in dividing the ongoing stream of behavior into meaningful units, in quantifying those units, in explaining why they occur, and in predicting and (sometimes) influencing future behavior. Their expertise in each of these areas has contributed to understanding the effects of psychoactive drugs.

Explanations of Behavior and Psychopharmacology: Mechanisms of Drug Action

Many important behaviors of humans and other animals are learned. That is, they occur through the processes of operant and classical conditioning. Operant conditioning is a form of learning in which the likelihood of occurrence of a particular response in a given situation is controlled primarily by its historical consequences in similar situations. If the consequences increase the likelihood of occurrence of the response, the process is termed reinforcement and the consequences are called reinforcers. If the consequences decrease the likelihood of occurrence of the response, the process is termed punishment and the consequences are called punishers.

Features of the environment that are present when consequences occur and subsequently exercise antecedent control over behavior are termed discriminative stimuli. Technically, a discriminative stimulus for reinforcement is a stimulus that evokes a response because in the past that kind of response has been more successful in producing reinforcement in the presence of that stimulus than in its absence. The classic example of operant conditioning involves Skinner's mildly food-deprived rats, which press a bar in the presence of a tone but not in its absence, because historically the bar-press response produced food (a reinforcer) in the presence of the tone but not in its absence. Regardless of whether or not the tone is present, substantial levels of bar pressing occur only when the rats are food-deprived. Food deprivation here is a motivational variable, specifically an establishing operation that increases the reinforcing capacity of food and increases the likelihood of occurrence of responses that historically produced food.

Unlike operant conditioning, classical (or respondent) conditioning is a form of learning in which behavior is controlled not by its consequences but by stimulus-stimulus pairings. Classical conditioning begins with a reflex, that is, an unlearned relation in which an unconditional stimulus elicits an unconditional response. By arranging conditions such that a previously neutral stimulus is repeatedly predictive of an unconditional stimulus, that neutral stimulus becomes established as a conditional stimulus, which reliably elicits a conditional response. The classic example involves Pavlov's mildly food-deprived dogs, in which food (unconditional stimulus) reflexively elicits salivation (unconditional response). By repeatedly presenting a sound (neutral stimulus) just before food (unconditional stimulus), the sound comes to elicit salivation (unconditional response) and is established as a conditional stimulus.

Understanding the processes of operant and classical conditioning provides a conceptual framework for examining behavioral mechanisms of drug action. "Mechanism of action" refers to a general drug effect at one level of analysis that produces (or at least, covaries with) another, usually more specific effect at another level of analysis. In lay terms, a drug's mechanism of action refers to how the drug "causes" or "works to produce" whatever effects are of interest. For example, chlorpromazine generally is effective in reducing what are referred to as the positive symptoms of Schizophrenia (hallucinations, thought disorders, bizarre behavior), which is one of the drug's most noteworthy effects (e.g., Baldessarini & Tarazi, 2006). The drug also produces adverse motor effects (rigidity, tremors,

slowed movement, restlessness). These effects, which can be observed directly, covary with events at another level of analysis, specifically, antagonism of the neurotransmitter actions of dopamine by competitively blocking dopamine receptors (Julien, 2001).

Early observations indicated that effective antipsychotic medications reduced dopaminergic activity in the brain and reduced positive symptoms of Schizophrenia. In addition, researchers observed that chronic exposure to high doses of drugs that increased dopaminergic activity, such as amphetamine, produced changes in behavior closely resembling those characteristic of Schizophrenia. These findings led to the speculation that Schizophrenia resulted from metabolic errors leading to either (a) overproduction of dopamine in the limbic system or cortex of the brain, or (b) production of an endogenous amphetamine-like compound. Similarly, the finding that drugs useful in treating depression increased monoaminergic activity suggested that depression results from a deficit in serotonergic and/or noradrenergic activity (Baldessarini, 2000). Studies of family resemblance suggested that genetics played some role in both Schizophrenia and depression, and it was soon widely believed that both conditions resulted from inherited diseases of the brain that produced abnormal neurochemical activity, which in turn produced abnormal behavior.

These early biological models of mental illness were crude, even primitive, and ran into problems from the start. For example, antidepressant drugs begin increasing serotonergic activity as soon as they are administered, but clinical benefits take weeks to appear. Also, no one has succeeded in demonstrating consistent neurochemical differences between people who are and are not diagnosed with depression or Schizophrenia. Finally, the role of inheritance in depression and Schizophrenia is rather weak, and the environment clearly plays a role in their genesis. How this occurs was not considered in the early models. In response to these problems, more sophisticated models have been proposed. They, too, are inadequate to explain mental illness (e.g., Baldessarini, 2000; Julien, 2001). Neuroscientists are making dramatic progress in understanding the human brain, however, and it is possible—perhaps even probable—than reasonably comprehensive, testable, and confirmed biological models of particular behavior disorders will soon appear.

Be that as it may, biological models developed to this point have had an enormous influence on how physicians, psychologists, and laypeople conceptualize mental illness. For many people, the signs (occurrence of undesired behaviors or absence of desired behavior) and symptoms (patient-reported affect and cognitive status) that lead to a clinical diagnosis are a manifestation of underlying pathophysiology. Effective treatment, regardless of its nature, corrects the underlying pathology, leading to alleviation of signs and symptoms. Under this model, drugs correct the pathology directly, whereas psychological interventions do so indirectly.

Although it has been criticized as not being supported by compelling evidence and focusing attention away from environmental variables that strongly affect the genesis, course, and treatment of mental illness (e.g., Malott, in press; Whitaker, 2002), this general view is widely held. Referred to variously as the medical, biological, or disease model of mental illness, this view at the extreme supports the notion that each condition listed in the *Diagnostic and Statistical Manual of Mental*

Disorders, fourth edition (*DSM-IV,* American Psychiatric Association, 1994) is a discrete disease with known signs and symptoms and a potentially specifiable neurochemical cause. This conception, however popular, is difficult to defend scientifically. There is, nonetheless, good evidence that particular disorders are fundamentally biological with strong genetic components, even if the specific structural and functional differences between people with and without the disorder cannot presently be specified precisely, if at all. For example, Russell Barkley, a psychologist well known for his research involving children with Attention-Deficit/Hyperactivity Disorder (ADHD), responded as follows when asked by a reporter for PBS's *Frontline* (2006), "So this [ADHD] is a biological disorder?"

> It's largely a biological disorder. It has many causes, but all of the known causes fall within the realm of neurology and genetics. We can rule out the social environment, such as bad parenting, intolerant teachers, the breakdown of the American family, a decline in family values, excess amount of TV viewing or video games. These have all been proposed as causes of ADHD. But there's no evidence that we can find that will substantiate them.
>
> All of the evidence that keeps turning up in our research points to genetics and neurology as being largely responsible for the excessive behavior and the poor self-control that we see in these children. So I think we can safely say that what we've learned in the past 10 years is that environmental causes of ADHD are not credible. They do not account for the substantial amount of scientific findings that exist on this disorder today.

Regardless of whether the drug in question is or is not a psychotropic medication, most of the interest in mechanisms of drug action has focused on the relation between neuropharmacological and behavioral events, and receptor models of drug action have proven invaluable in predicting and explaining drug effects. Some behavioral psychologists, however, have emphasized behavioral, not neuropharmacological, mechanisms of drug action. Behavioral mechanisms of drug action involve the stimulus properties of drugs in the context of operant and classical conditioning and the manner in which drugs modulate the actions of nondrug stimuli (Thompson, 1981; Thompson & Schuster, 1968). More specifically:

> By behavioral mechanism of drug action, we refer to a description of a drug's effect on a given behavioral system (locus) in terms of some more general set of environmental principles regulating behavior. Specifying the behavioral mechanism(s) responsible for an observed effect involves: (a) identifying the environmental variables which typically regulate the behavior in question, and (b) characterizing the manner in which the influence of these variables is altered by the drug. In some instances, the drug assumes the status of a behavioral variable per se, rather than modulating an existing environmental variable. (Thompson, 1981, p. 3)

In passing, it should be noted that the concept of behavioral mechanisms of drug action was developed and popularized by a psychologist, Travis Thompson, and his colleagues. This concept is abundantly supported by demonstrations that certain drugs inherently have, or can acquire through learning, the capacity to function as unconditional, conditional, discriminative, reinforcing, and punishing stimuli and as motivational operations (Poling & Byrne, 2000; Thompson & Pickens, 1971).

Especially significant is the demonstration that drug self-administration involves learned behavior that is maintained, in substantial part, by the short-term reinforcing effects of the drug in question (Laraway, Snycerski, Byrne, & Poling, 2000). Harmful patterns of self-administration, that is, drug abuse and drug addiction, are controlled by the same classes of variables that affect other kinds of excessive learned behavior, such as overeating. This conception of drug abuse suggests strategies for treatment, which often involve arranging the environment such that (a) the client's drug use is consistently monitored, (b) abstinence is consistently reinforced, (c) drug use results in loss of reinforcers, and (d) the density of nondrug reinforcers is sufficient to compete with the reinforcing effects of the drug (Higgins et al., 1991).

A study by Higgins et al. (1993) illustrates the potential value of this approach. They randomly assigned 38 cocaine-abusing clients to a 12-step counseling program or to a program that involved community-based reinforcement and vouchers, earned by producing drug-free urine samples, that could be exchanged for appropriate reinforcers. As part of the community-based reinforcement program, the clients attended behavior therapy sessions that involved employment counseling, assertiveness training, relaxation training, and counseling that targets participation in non-drug-related social and recreational activities. These therapy sessions were intended to increase drug users' participation in activities incompatible with drug use, which, given that they were sufficiently reinforcing, should compete effectively against drug-related activities.

They appeared to do so in many cases, in that only 5% of the clients in the 12-step program achieved 16 weeks of continuous abstinence, whereas 42% of clients in the behavioral program did so. Moreover, the former program appeared to be less acceptable to clients, insofar as only 11% of them completed 24 weeks of treatment, whereas 58% of those in the behavioral treatment did so.

Behavioral Mechanisms of Clinical Drug Effects

In principle, it should be possible to isolate the behavioral mechanisms involved in the beneficial effects produced by psychotropic drugs. For example, objects and events that previously were effective positive reinforcers (i.e., their delivery strengthened the behaviors that produced them) for a given person appear to lose their reinforcing capacity when she or he becomes depressed. Effective psychotropic medications might work, at least in part, by reestablishing reinforcing capacity, and they appear to do so.

With many disorders, however, no one has successfully categorized the characteristic changes in behavior in terms of behavioral processes. Consider, for example, how a person changes with the onset of Schizophrenia. Overt verbalizations become impoverished in terms of amount and content; they reflect delusional thoughts and auditory hallucinations. Anhedonia—a general lack of interest in once important objects and events (i.e., stimuli)—appears and self-care diminishes. Stimulus control, that is, the ability of objects and events to engender or suppress particular responses, clearly has changed, as has the capacity of certain objects and events to serve as reinforcers. But the generation and following of rules—verbal descriptions

of relations among stimuli and responses—also changes. Human behavior is unique in that it can be affected by descriptions of relations, such as the consequences of a particular response (e.g., "If you turn left at the second light, the diner will be two miles ahead on your right"), as well as by actual exposure to those relations. Delusions involve counterfactual rules (e.g., "They listen through the TV; if you turn it on they'll know what I'm thinking and hurt me"). It is unclear why people eventually diagnosed with Schizophrenia come to generate and follow counterfactual rules, or how psychotropic drugs reverse the process. The absence of viable, comprehensive behavioral models of mental illness makes it impossible to relate therapeutic drug effects to behavioral mechanisms of drug action. Moreover, the absence of such models appears to increase the appeal of biological models.

Nonetheless, researchers have had some success in relating the effects of psychotropic drugs to behavioral mechanisms of action. For example, Northup, Fusilier, Swanson, Roane, and Borrero (1997) examined the possibility that methylphenidate (Ritalin) functions as a motivational operation (MO) when administered to children diagnosed with ADHD. In brief, an MO is a stimulus that alters the reinforcing (or punishing) effectiveness of another stimulus. Establishing operations (EOs) increase reinforcing (or punishing) effectiveness; abolishing operations (AOs) decrease it (Laraway, Snycerski, Michael, & Poling, 2003).

Northup and his colleagues (1997) studied the behavior of three children diagnosed with ADHD who were receiving methylphenidate each day and attending a summer day treatment program. One of their activities at the camp was solving math problems. Solving a specified number of problems was reinforced with coupons that could be exchanged, depending on the color of the coupon selected, for edible items, other tangible items, activities, teacher attention, peer attention, and escape from academic programs. The effectiveness of each type of reinforcer was determined based on the number of coupons representing that class of reinforcer that a child selected.

The distribution of choices across reinforcer types was examined for each child when methylphenidate and placebo were administered. For all three children, methylphenidate changed the value of at least one type of reinforcer. For example, in the placebo condition, one boy's most preferred reinforcer type was edible items. At 20 mg methylphenidate, however, both activities and tangible nonfood items were preferred over food. Similar results were obtained with the other two children. In all three, activities functioned as more potent reinforcers following methylphenidate than following placebo. In contrast, edible items were more effective reinforcers following placebo than following methylphenidate. In technical terms, the drug served as an EO for activities as a reinforcer, and as an AO for food as a reinforcer.

The capacity of methylphenidate to act as an MO is important with respect to one of the potential side effects of the drug: its capacity to alter the effectiveness of behavioral interventions. That is, the effectiveness of behavior change programs that utilize reinforcement—and most behavioral interventions used to treat children diagnosed with ADHD do so—will differ in the presence and absence of methylphenidate if the drug alters the reinforcing capacity of the objects and events arranged to strengthen appropriate behavior. In addition, reinforcer assessment

conducted in the absence of the drug may yield information that is inaccurate in the presence of the drug, and vice versa. Clearly, the demonstration that methylphenidate acts as an MO in applied situations helps to explain how the drug may influence clinically significant behavior in children diagnosed with ADHD.

Methods for Assessing Drug Effects: Basic Research

Quantifying behavior is an important aspect of psychology. Many of the procedures developed by psychologists to quantify behavior and to examine the effects of nonpharmacological variables thereon were subsequently adopted for the study of drug effects. Substantial early impetus for doing so came from a 1956 conference sponsored by the New York Academy of Sciences. The conference, called "Techniques for the Study of the Behavioral Effects of Drugs," featured prominent behavioral psychologists, including Herrnstein, Miller, Morse, Sidman, and Skinner. They forcefully made the case that the methods characteristic of the experimental analysis of behavior were well suited for studying drugs.

The conference occurred shortly after chlorpromazine was introduced into psychiatric practice, and the remarkable success of the drug galvanized interest in basic and applied studies of its effects. Moreover, that success generated a heated quest to develop animal models that effectively screened other drugs for antipsychotic actions. Pickens (1977) reported that between 1917 and 1954 only 28 studies examining drug effects on learned behavior were published in English-language journals, whereas 274 such studies appeared from 1955 to 1963. The vast majority of these studies involved basic research. That is, their intent was to disclose the effects of particular drugs, the mechanisms responsible for those effects, and/or the variables that modulated them, not to help people directly.

Many early studies examined drug effects on learning and performance in mazes. Others were concerned with how drugs affected responding under various schedules of reinforcement. Research in both areas has continued to the present day, but scientists have developed many additional strategies for examining specific aspects of drug action. These strategies are discussed at length by Buccafusco (2001) and van Haaren (1993). Although most basic research has involved nonhuman subjects, a substantial number of studies with humans have appeared. Table 32.1, adapted from Snycerski, Laraway, and Poling (2000), provides examples of some of the behavioral domains examined in these studies and the procedures used to quantify those domains. Psychologists have made substantial contributions to the development and use of these procedures and thereby have played a quintessential role in basic psychopharmacological research.

Methods for Assessing Drug Effects: Applied Research

Psychotropic drugs have some special characteristics (e.g., slow onset of clinical effects) that must be considered in their evaluation. In addition, there are established conventions for clinical drug studies, such as the use of placebo-controlled and double-blind conditions, and determination of dose-response relations (see Gadow & Poling, 1988). Nonetheless, the value of a medication for treating a particular disorder can be determined via the same general tactics and strategies that are used to

Table 32.1 Procedures Commonly Used in Basic Psychopharmacological Research

Domain	Examples of Procedures
Arousal, mood, and activity	Profile of mood states Addiction research center inventory Visual analogue scale Beck Depression Inventory Electroencephalogram Skin temperature Skin resistance Heart rate
Perceptual performance	Critical frequency of fusion Signal detection
Cognitive performance	Repeated acquisition Delayed matching to sample Digit-symbol-substitution test Vigilance tasks
Motor performance	Reaction time Tapping rate Hand steadiness Pursuit rotor Driving simulations
Self-administration	Progressive-ratio schedules Concurrent schedules Discrete trials Single-access procedures
Motivation	Progressive-ratio schedules Contingent activity procedures
Drug discrimination	Two-choice procedures Three-choice procedures Stimulus equivalence procedures

From "Assessing Drug Effects in Humans: Basic Research" (pp. 111–140), by S. Snycerski, S. Laraway, and A. Poling, 2000, in A. Poling and T. Byrne (Eds.), *Introduction to Behavioral Pharmacology,* Reno, NV: Context Press. Adapted with Permission.

evaluate other behavior-change interventions (e.g., cognitive-behavioral therapy). There is no mystery to designing a clinical drug study: One determines whether any independent variable (including medication) affects a behavior of interest by comparing levels of behavior (the dependent variable) when the independent variable is and is not operative, or is operative at different levels (e.g., doses). Factors other than the independent variable that might affect the dependent variable (extraneous variables) are held constant across conditions or rendered inoperative. If appropriate data analysis reveals that levels of the dependent variable differ significantly when treatment is and is not present (or is present at different levels), it is logical to assume that behavior changed as a function of treatment, which is therefore deemed effective.

Many psychologists are truly expert in evaluating the effects of behavior-change interventions, and a substantial number of these experts have done extensive and excellent work in clinical drug assessment. As a case in point, consider Russell

Barkley (1998), who, as noted previously, is a leading ADHD researcher. For more than two decades, he has studied the pharmacological and nonpharmacological management of ADHD and translated the findings of his and others' research into practical, effective strategies for everyday use (e.g., Barkley, 2000). As the case of Barkley illustrates, psychologists can make a major contribution to clinical drug evaluation, even under circumstances in which they do not prescribe drugs themselves.

Several authors have proposed that the tactics and strategies of applied behavior analysis can be especially valuable in clinical psychopharmacology (e.g., Barlow & Hersen, 1984; Kollins, Ehrhardt, & Poling, 2000; Schroeder, 1985). In brief, this approach is underpinned by five beliefs (Poling & Ehrhardt, 1999; Poling & LeSage, 1995):

1. Intensive study of a few participants through the use of within-subject research designs (e.g., multiple-baseline, withdrawal) is a fruitful research strategy.
2. Behaviors targeted for change by a drug should be measured accurately and repeatedly in individual participants.
3. Variable data are best dealt with by isolating the responsible extraneous variables, not by statistical manipulations.
4. Visual (graphic) analysis of data for individual participants is a viable alternative, or adjunct, to statistical analysis.
5. Social validity data, evaluating the acceptability of the goals, methods, and outcomes of treatment to consumers, is desirable.

As Dermer and Hoch (1999) point out, it is common for authors of research methods texts to assert that the within-subject experiments characteristic of applied behavior analysis involve what Campbell and Stanley (1963) termed "quasi-experimental designs" with "low external validity." That is, their findings cannot be widely generalized. Certain between-subjects designs, in contrast, are described as producing results with much greater generalizability. The usual, and reasonable, argument is that between-subjects experiments characteristically involve the study of a clearly defined sample of subjects who are selected to be representative of a larger population with defined characteristics. If this is the case, then inferential statistics provide a basis for describing the true state of affairs in the population, given the results obtained in the sample selected for study.

Although this logic is sound, clinical drug evaluations that involve protected populations (e.g., children with autism) often are limited to small convenience samples; therefore, conventional between-subjects research designs (e.g., randomized between-groups) are inappropriate. Within-subjects designs, such as the multiple-baseline and withdrawal, appear to be the best alternative, even though their findings cannot be generalized with confidence to the population of ultimate concern (e.g., all children with autism). They should, however, extend to people who are very similar to the individuals who were studied. Whether or not this is true is determined by systematically replicating the study. Eventually, the range of conditions under which the drug is useful is established, and valuable prescription guidelines

can be established. Although some research in this vein has appeared, it seems that psychologists have had little success in convincing the medical profession that the strategy is useful. Should they eventually do so, there is potential to expand the literature in areas where further research is badly needed.

Everyday Drug Assessment

Psychotropic drugs are unique among medications in that they are used to deal with problems that are defined by, measured in terms of, and best understood at the level of overt behavior. The effects of these drugs on the biological processes responsible for the signs and symptoms characteristic of particular disorders are incompletely understood, and clinical effectiveness cannot be evaluated at this level of analysis. Therefore, clinical evaluations of psychotropic drugs must assess whether medication significantly improves some targeted aspect(s) of the participant's behavior without producing intolerable adverse reactions (side effects). To determine whether a pharmacological intervention is successful in treating a behavior disorder, three questions must be answered: (1) What were the desired effects of the medication? (2) Were those effects obtained? (3) Were there any significant adverse reactions? These are easy questions to ask but not to answer, whether in the context of formal research or in practical everyday drug evaluations (Kollins et al., 2000).

As Sprague and Werry (1971) pointed out 35 years ago, every prescription of a psychotropic medication is in essence an experiment in which the physician and other care providers hypothesize that administering a specific drug will produce a desired change in one or more aspects of a client's behavior. In the context of controlled research, the essential four features of a sound drug evaluation are that (1) treatment effects must be adequately measured, (2) medication must be administered according to the experimental regimen, (3) experimental conditions and their sequencing must allow observed changes in behavior to be attributed with confidence to the drug, and (4) data analysis must be adequate for detecting clinically important changes in client behavior. If these conditions are met, the evaluation is, in principle, sound.

These same characteristics are important in the practical evaluation of medications, although assessment procedures employed outside research settings rarely meet the rigorous standards advocated by scientists. When, for instance, a child diagnosed with ADHD receives methylphenidate for the first time, the child's parents, teachers, and physician are not interested in conducting a study worthy of presentation, but only in determining whether the drug is an appropriate treatment. An abundant scientific literature suggests that it is likely to be effective in a given individual who displays the behaviors that lead to a diagnosis of ADHD (restlessness, impulsivity, short attention span, excessive motor activity), but exceptions have been reported (some children show little or no positive response to the drug), and the most effective dose can vary. Therefore, the response of each young person with ADHD to methylphenidate needs to be monitored.

When competent adults contract for treatment by a physician, the evaluation process is relatively simple: The client and physician agree on treatment goals and a medication, and the treatment is successful, at least to some extent, if the

client deems it so and the physician agrees and also detects no serious side effects. Evaluation is complicated, however, when the client, by virtue of age or disability, is not able to decide whether to initiate or continue drug treatment. As Greiner (1958) pointed out soon after chlorpromazine was introduced:

> Sensible adult patients will usually balk when a drug is causing symptoms, but the very young and the very old are forced to take drugs, can't complain or stop on experiencing toxic symptoms, may not even connect them with the drug. The mentally deficient of any size or age cannot protect themselves either, and they also merit special care to avoid toxic doses.

The need for special care in assessing drug effects in people with mental retardation (and other developmental disabilities) is especially important in view of the lack of persuasive evidence of effectiveness in this population. In summarizing the most comprehensive recent review of psychopharmacology and mental retardation, Matson et al. (2000) concluded:

> Our findings were startling and consistent with Baumeister and Sevin's findings in 1990 [Baumeister and Sevin conducted a similarly comprehensive prior review]. Professionals continue to medicate individuals in their care despite a lack of a sound research base to support their treatment decisions. While a small amount of research does exist, it is seriously flawed from a research design perspective. The studies routinely lack appropriate experimental control, and lack reliable measures of drug effect, collateral behaviors, and side effects. Methodological controls are far short of psychological treatment in the field with little improvement over precious decades. This trend is extremely disappointing given the dangerous and restrictive nature of these treatment options.

Johnny Matson is one of a substantial number of psychologists whose conceptual and empirical work has major implications for the everyday use of psychotropic drugs. As regards drug effects in people with developmental disabilities, Matson et al. (2000) are not alone in their skepticism. For example, in reviewing studies examining the effects of antipsychotic medications for challenging behavior in people with intellectual disorder, Brylewski and Duggan (1999, p. 360) reported that they could find only three studies that could be classified as randomized controlled trials: "These trials provided no evidence as to whether antipsychotic medication does or does not help adults with [intellectual disorder] and challenging behavior."

Similar results have been found for persons receiving dual diagnoses of intellectual disability and Schizophrenia. In a review of studies that examined the effectiveness of antipsychotic medication in people dually diagnosed, Duggan and Brylewski (1999) found only one relevant randomized controlled trial, which was published over 40 years ago (i.e., Foote, 1958). These authors found that there is no empirical evidence that antipsychotic medications are, or are not, effective in individuals with developmental disabilities and Schizophrenia. Until methodologically sound trials are conducted, practitioners will have to rely on personal experience as well as extrapolations from studies with individuals with Schizophrenia and "normal" intelligence. These extrapolations may or may not be valid (Duggan & Brylewski, 1999).

Since 1949, when the "Boulder Conference on Graduate Education in Clinical Psychology" was held, the scientist-practitioner model has powerfully influenced clinical psychology, as well as other applied subdisciplines, such as school psychology and organizational behavior management. Three key provisions of this model, discussed in detail by others (e.g., Hayes, Barlow, & Nelson-Gray, 1999; Soldz & McCullough, 1999), are these:

1. Psychologists should use assessment and treatment procedures that have been scientifically validated. That is, whenever possible, applied psychologists are expected to stay abreast of the scientific literature relevant to the areas in which they conduct assessments and arrange treatments, and to use only those techniques supported by the results of methodologically rigorous research.
2. Psychologists should monitor the response of each individual who is assessed and treated to ensure that she or he actually benefits from these processes. Even empirically validated techniques may be ineffective or even harmful for some seemingly appropriate clients.
3. Psychologists should work with other health professions to improve the quality of care provided. This involves contributing the psychologists' expertise to health care teams and ensuring that those teams function in a manner consistent with the scientist-practitioner model.

We have contended elsewhere that these provisions, if widely upheld, would dramatically improve how psychotropic drugs are used in people with developmental disabilities (Poling, 1994; Poling & Ehrhardt, 1998; Poling, Laraway, Ehrhardt, Jennings, & Turner, 2004). Of course, the virtues of the scientist practitioner model are not limited to psychologists; psychiatrists and other physicians who rely on psychotropic medications of proven worth and monitor the effects of those medications carefully can provide excellent care with no involvement from psychologists. However, given that psychologists typically receive extensive training in behavioral assessment, we have advocated for a team approach in which well-trained psychologists and physicians work with other concerned individuals to provide high-quality drug treatments. Such treatments ensure that:

- The goals of treatment are clear, quantitative, and in the patient's best interest.
- The choice of a drug and dose for initially treating a particular patient are consistent with the best available scientific literature.
- Subsequent treatment decisions are made in view of meaningful information regarding the patient's response to the drug.
- Drug therapy is flexible and integrated with nonpharmacological interventions.
- Properly trained psychologists can do much to enhance the quality of drug treatment, regardless of whether or not they actually prescribe.

PRESCRIPTION PRIVILEGES FOR PSYCHOLOGISTS

Up to this point, we have endeavored to show that psychologists who do not prescribe drugs have made major contributions to many areas of psychopharmacology and potentially could do even more. One thing that we have not considered is the possibility that they could prescribe psychotropic drugs. Since 1990, 14 states have considered legislation to grant psychologists limited prescription privileges. Alaska, California, Florida, Georgia, Hawaii, Illinois, Missouri, Montana, Tennessee, and Texas rejected the legislation. In 2002, New Mexico became the first state allowing psychologists to prescribe drugs. It was followed in 2004 by Louisiana. Both states allow psychologists who meet certain criteria to prescribe psychotropic drugs independently. Too little time has passed to allow for meaningful evaluation of the effects of allowing psychologists to prescribe in New Mexico and Louisiana; whether or not psychologists should be allowed to prescribe drugs is a contentious issue, as it has been for the past 15 years.

A recent search of the PsychInfo database using "prescription privileges" as the keyword revealed 163 entries. Of them, 160 appeared from 1986 to 2006. Their distribution across years is shown in Figure 32.1. Clearly, there is a substantial literature relevant to whether or not psychologists should be allowed to prescribe drugs, and it is growing rapidly. This section summarizes that literature. In doing so, we provide an overview of past and current events relevant to psychologists' prescribing drugs and summarize arguments for and against their doing so. More detailed coverage is available elsewhere (e.g., Hayes & Heiby, 1998b; Lavoie & Barone, 2006; Sammons, Page, & Levant, 2003).

Historical Overview

In the United States, the Federal Food, Drug and Cosmetic Act, passed in 1938, determines which medications can be sold over the counter and which are available

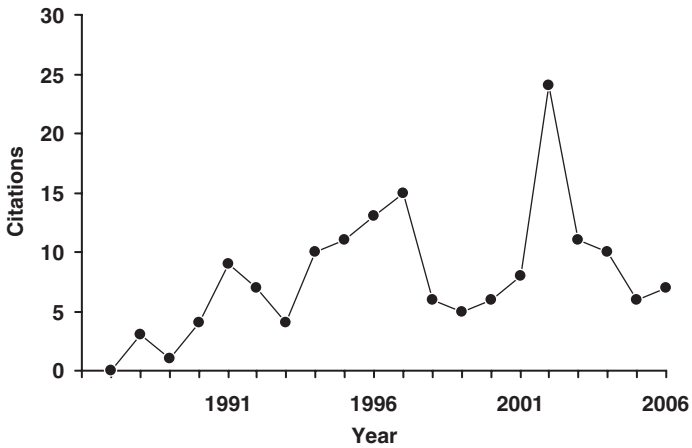

Figure 32.1 Number of publications listed in PsychInfo that used "prescription privileges" as a key word.

only by prescription (U.S. Food and Drug Administration, 2007). State laws, however, determine the professions that are allowed to prescribe, as well as (a) whether they can do so with or without physician supervision and (b) whether they can prescribe all classes of medication or only those relevant to their discipline. In every state, physicians can prescribe all medications independently. Other professions are more limited. Dentists, for example, can prescribe independently, but only those medications relevant to their practice (e.g., antibiotics, analgesics).

From its inception in the late nineteenth century until the onset of World War II, psychology was primarily an academic discipline concerned with understanding human behavior. Freudian psychoanalysis was the predominant treatment, and it was available to relatively few people. There were not sufficient psychiatrists available to serve the mental health needs of personnel returning from the war, and the Veterans Administration (VA) was severely taxed in trying to treat those people. In response, the VA dramatically extended the range of services they allowed psychologists to provide, including the delivery of psychotherapy under the supervision of psychiatrists (Lavoie & Fleet, 2002). Psychiatrists objected, claming that psychologists were not properly trained to provide psychotherapy, but an ever-increasing number of psychologists disagreed. In large part due to effective lobbying by APA and affiliated organizations, by the end of the 1950s psychologists were widely recognized as legitimate providers of clinical services. This recognition was codified in state licensure legislation, which specifies the range of activities in which properly trained and duly licensed (or, in some cases, certified) psychologists can legally engage. For example, in our home state of Michigan, Public Act 368 (Michigan Legislature, 2007):

> Defines the practice of psychology as the rendering to individuals, groups, organizations, or the public of services involving the application of principles, methods, and procedures of understanding, predicting, and influencing behavior for the purposes of the diagnosis, assessment related to diagnosis, prevention, amelioration, or treatment of mental or emotional disorders, disabilities or behavioral adjustment problems by means of psychotherapy, counseling, behavior modification, hypnosis, biofeedback techniques, psychological tests, or other verbal or behavioral means. The practice of psychology does not include the practice of medicine such as prescribing drugs, performing surgery, or administering electro-convulsive therapy.

Psychology flourished in the 1960s and 1970s and into the 1980s. Major conceptual advances, including those associated with behavior analysis and cognitive psychology, were accompanied by the development of effective interventions for various behavioral disorders. Cognitive-behavioral treatment for depression is an obvious case in point. Effective behavior-change interventions were developed and used not only by clinical psychologists, but by organizational and industrial, educational, and school psychologists as well. By 1980, several areas of psychology were well established and providing invaluable services to a wide range of individuals and organizations.

By that same year, the use of psychotropic drugs to treat a wide range of behavior disorders was firmly established. The success of chlorpromazine was

soon followed by the introduction of several other antipsychotic drugs, as well as antidepressants (monoamine oxidase inhibitors such as phenelzine [Nardil] and tranylcypromine [Parnate], tricyclics such as amitriptyline [Elavil, Endep] and imipramine [Tofranil]), and anxiolytics (primarily benzodiazepines such as diazepam [Valium]). The use of stimulants (such as methylphenidate [Ritalin]) to treat children diagnosed with hyperactivity (or Attention-Deficit Disorder) became more prevalent, and lithium became popular for the treatment of Bipolar Disorder. A bit later, in the early 1980s, the serotonin-specific reuptake inhibitors, beginning with fluoxetine (Prozac), were introduced for the treatment of depression. They soon became the most prescribed psychotropic medications.

Later still, a new group of antipsychotic drugs, labeled "atypical" because they produced fewer motor disturbances, was introduced. This group includes risperidone (Risperdal) and clozapine (Clozaril). A large number of studies in addition to the initial clinical trials required by the FDA prior to marketing have demonstrated that various psychotropic drugs are reasonably safe and at least somewhat effective (although the evidence in some cases is not overwhelming). The availability of effective psychotropic medications literally revolutionized psychiatry in terms of (a) increasing the effectiveness of available interventions, (b) fostering broad acceptance of a biological model of mental illness, and (c) generating widespread demand for the pharmacological management of behavior disorders.

More than 50 different drugs currently are FDA-approved as psychotropic medications. In 2005, over 10 million prescriptions were written in the United States for each of nine such drugs. The names of those drugs, the number of prescriptions written, and the overall ranking of each are shown in Table 32.2. These data make it clear that psychotropic medications play a central role in the current treatment of behavior disorders. As further evidence, a sample of practicing psychologists recently reported that 43% of their current patients were using psychotropic medication (VandenBos & Williams, 2000). Moreover, 94% of those psychologists indicated that they had consulted with physicians about changing the medications of joint clients.

As the pharmacological management of behavior disorders flourished, two major forces came into play that eventually led some psychologists to push for

Table 32.2 Psychotropic Drugs with >10 Million Prescriptions in 2005

Nonproprietary Name	Proprietary Name	Overall Rank	Number of Prescriptions
Alprazolam	Xanax	9	34,230,000
Sertraline	Zoloft	16	26,976,000
Fluoxetine	Prozac	29	21,403,000
Lorazepam	Ativan	30	19,002,000
Paroxetine	Paxil	38	18,889,000
Trazodone	Desyrel	48	14,505,000
Amitriptyline	Elavil	50	14,385,000
Diazepam	Valium	60	12,093,000
Bupropion	Wellbutrin	66	11,044,000

Source: The Top 300 Prescriptions for 2005 by Number of U.S. Prescriptions Dispensed, by RxList, 2005. Available from www.rxlist.com/top300/.

prescription privileges (Frank, Conti, & Goldman, 2005; Lavoie & Barone, 2006; Sanua, 1998). One, discussed previously, was conceptual, and took the form of growing acceptance among laypeople and health care providers alike of biological models of mental illness and of the logic and effectiveness of pharmacological management of those disorders.

A second was economic, leading ultimately to diminished job prospects for doctoral-level clinical psychologists. In brief, graduate schools had saturated the market with such psychologists, even as it was being demonstrated (e.g., Christiansen & Jacobson, 1994; Dawes, 1994) that non-doctoral-level professionals from a number of disciplines (e.g., psychology, nursing, social work) could provide effective, and relatively inexpensive, treatment for many psychological problems (Lavoie & Barone, 2006). As managed health care came to dominate the provision of mental health services, there was increased reliance on non-doctoral-level professionals and fewer lucrative positions for doctoral-level psychologists (Chiefetz & Salloway, 1984; Cummings, 1995). Concurrently, there was more money available for pharmacological treatments (Frank et al., 2005).

According to Frank et al. (2005), expenditures on psychotropic drugs in the United States in 1987, 1992, 1997, and 2001 were, in billions, $2.77, $3.83, $9.04, and $17.83, respectively. Expenditures on psychotropic drugs increased more than sixfold over this period, and the percentage of mental health spending devoted to them increased from 7.7% to 21% (Frank et al., 2005). Much of this money went to large pharmaceutical companies, which were growing enormously and exerting increasing—and sometimes harmful—influence on the medical profession (Antonuccio & Danton, 2003), but a substantial sum went to physicians who prescribed psychotropic drugs. Perhaps surprisingly, most of them were not psychiatrists, but general practitioners.

In what was in hindsight a predictable reaction to emerging financial and conceptual changes in the field of mental health, during the 1980s some influential psychologists and their allies in government began to advocate for prescription privileges. The APA, which currently has over 150,000 members, played a crucial role in the advocacy, with much of the impetus coming from members of Divisions 12 (Society of Clinical Psychology) and 42 (Psychologists in Independent Practice). Since 2000, a strong push for prescription privileges has come from Division 55, the American Society for the Advancement of Pharmacotherapy, which, according to the APA web site (APA, n.d., para 2), was created specifically "to enhance psychological treatments combined with psychopharmacological medications." Sanua (1998) details the contentious political history of the prescription movement within and outside APA.

As he relates, in 1981 APA convened the Psychologists' Use of Physical Intervention Committee to consider prescription privileges. The committee completed a report in 1981, with a revision in 1986 (see Sanua, 1998). Those reports did not specifically endorse prescription privileges, but instead suggested that psychologists should play a role in pharmacotherapy as dictated by consumer demand (Hayes, Walser, & Bach, 2002). In 1989, APA established an Ad Hoc Task Force on Psychopharmacology to study the issue of pharmacotherapy further; its report appeared in 1992 (APA, 1992). The report outlined three levels of training in

psychopharmacology, with the appropriate level determined by a psychologist's degree of involvement in the prescribing practice. Level 3, which has generated the most interest, is for psychologists who actually will prescribe. Level 2 is for collaborative practice, in which the psychologist would partner with a physician to determine the need for and monitor the effects of psychotropic drugs. Level 1 is for any psychologist providing mental health services and essentially provides an overview of psychopharmacology.

In 1995, in response to a request for support of a bill establishing prescription privileges in California, APA's Council of Representatives approved a resolution making prescription privileges for psychologists a major policy directive for the association. From this time forward, APA formally supported prescription privileges (Cullen and Newman, 1997). In 1996, the organization adopted a model prescription bill and training curriculum, and in 2000 it announced that APA-supplied insurance will cover prescription-related activities in any state where such activities are legal. Many APA-affiliated state organizations also have supported prescription privileges.

Many psychologists, however, have not. As Sanua (1998) relates, early on most APA members (and other psychologists) had little knowledge of or interest in the prescription privileges issue. Interest was galvanized by the APA's activities, by the 1985 introduction in Hawaii of legislation to study prescription privileges for psychologists, and by Congress directing the U.S. Department of Defense (DOD) in 1988 to establish a pilot project, called the Psychopharmacology Demonstration Project (PDP), to train psychologists to prescribe psychotropic medications. It became operational in 1991; 10 military psychologists completed training between 1991 and 1997 and were granted the right to prescribe psychotropic medications. Each received a total of 660 hours of academic training and 1,900 hours of clinical training.

By 1996, three national psychology associations, the Society for a Science of Clinical Psychology, the Council of University Directors of Clinical Psychology Programs, and the American Association of Applied and Preventative Psychology (AAAPP), had formally opposed prescription privileges for psychologists (Hayes & Heiby, 1996). For example, following a 1996 conference, AAAPP members generally agreed on three conclusions (Hayes & Heiby, 1998, p. vi):

1. The prescriptions privileges proposal is polarizing the discipline.
2. Psychology's resources should be focused on advocating the profession to managed care. Doing so will involve the development of treatment standards and the promotion of integrated care.
3. Psychologists wishing to obtain prescription privileges at this time are encouraged to obtained an advanced degree in nursing, a physician's assistant degree, or the like in order to develop more experience with the psychology/prescription privilege without first having to alter the discipline and its training focus.

Attendees also generally agreed that the push for prescription privileges was premature and that further reasoned discussion among psychologists should occur before any introduction of legislation seeking such privileges. This did not occur.

The move to secure prescription privileges continued unabated, and at present, psychologists wishing to obtain prescription privileges have options not available in 1996: They can obtain proper training and licensure and work in New Mexico or Louisiana. Whether or not this is good continues to be debated heatedly.

A recent meta-analysis of opinion data from 16 studies concerned with the prescription privileges debate (Walters, 2001) examined responses (agree, disagree) to three statements: (1) Properly trained psychologists should be allowed to prescribe psychotropic drugs; (2) APA should advocate in favor of prescription privileges for psychologists; and (3) I would personally seek prescription privileges should they become available. Collectively, these statements were taken to constitute the "prescription privilege agenda." Results indicated that "there was minimal consensus and a general split of opinion on the advisability of pursuing the prescription agenda" (p. 119). When data for all three statements were summed, there was no statistically significant difference in the number of respondents who agreed (52%; i.e., who supported the agenda) and the number who disagreed (35%; i.e., who opposed the agenda). Only statement 1 engendered significantly higher levels of agreement than disagreement.

The Department of Defense Psychopharmacology Demonstration Project

Good arguments can be made for and against prescription privileges, and data adequate for choosing between them are not currently available. As indicated, 10 psychologists were trained and allowed to prescribe drugs under DOD's PDP. Some information is available concerning the success of the program (Laskow & Grill, 2003; Newman, Phelps, Sammons, Dunivin, & Cullen, 2000). In general, the psychologists who participated demonstrated high levels of competency and achievement prior to entering the 2-year program. They primarily prescribed drugs in outpatient clinics, typically to deal with depressive disorders, anxiety disorders, or adjustment disorders. Psychiatrists who provided clinical supervision rated the quality of care provided by the psychologists as good to excellent and reported no adverse patient outcomes. Moreover, although some conflicts between psychologists and psychiatrists appeared early on, relations became generally harmonious; allowing psychologists to prescribe drugs did not degrade staff morale.

The General Accounting Office (GAO) estimated that the DOD would spend 7% more on its 10 prescribing psychologists than it would have spent to provide equivalent services using a traditional mix of psychiatrists and psychologists. The GAO noted that the PDP did not increase the wartime readiness of the DOD, primarily because psychotropic drugs are rarely a priority in combat, but that it might modestly enhance peacetime readiness. The PDP was successful, and a survey by Klusman (1998) indicated that most military primary care physicians, psychologists, and social workers strongly supported the program and the overall initiative to grant psychologists the authority to prescribe. Military psychiatrists, in contrast, were strongly negative regarding both the PDP and the overall initiative.

Save for demonstrating that psychologists can be trained to prescribe drugs in a safe and competent manner, the PDP's relevance for the prescription privileges debate at large appears to be minimal. A relatively small number of highly competent

people were trained, and they practiced in close association with psychiatrists in a highly structured setting. Opinions regarding the PDP's value differed across professions, and, more important, whatever virtues it possessed were not sufficient to lead to widespread DOD adoption of prescription privileges for psychologists. This outcome suggests an absence of social validity for the program. That is, one of its consumers—the DOD—was not satisfied with its goals, methods, or outcomes. Unfortunately, no data are available concerning the satisfaction of the military personnel who actually received medication at the hands of psychologists or, perhaps more important, the actual effectiveness of those medications in solving the mental health problems of those personnel.

Arguments for Granting Prescription Privileges

Advocates of prescription privileges make a number of solid arguments in support of their position. For example, Robert Sternberg (2003) wrote in the president's column of the *APA Monitor* that psychologists ought to be allowed prescription privileges because this would allow them to treat the whole person, provide full service, offer medications to patients who need them but are not presently being served, and reduce psychologists' dependence on medical practitioners. Moreover, he indicated that the success of the PDP program and the passage of a prescription authority bill in New Mexico demonstrate, respectively, the potential value and the political feasibility of obtaining prescription privileges. Other advocates have made similar points. In general, it appears that most of the arguments in favor of prescription privileges fall into two general categories.

1. Allowing Psychologists to Prescribe Drugs Will Help to Meet the Needs of People Not Currently Receiving Adequate Mental Health Services

This argument assumes that (a) psychotropic drugs are valuable treatments, (b) there are substantial numbers of people who currently need but do not have access to those drugs, and (c) allowing psychologists to prescribe would meet the needs of those people. Overall, a good case can be made for the value of psychotropic drugs, although they are not panaceas and, as discussed previously with respect to people with mental retardation, they historically have been misused in some populations. The relevant literature is gargantuan and cannot be addressed here. Perhaps a quote from the World Health Organization (WHO; 2005, p. 8) is sufficient:

> A large number of treatments are available for the pharmacological management of mental disorders. Many of these treatments have been shown to be effective in acute stages and in preventing relapses, but much remains unclear about their effectiveness in long-term treatment and in managing everyday mental disorders.

Despite the reservations expressed in the last sentence, the WHO considers some psychotropic drugs as "essential" medications (see Table 32.3), which should be cheaply and widely available worldwide.

Table 32.3 Psychotropic Drugs on the World Health Organization Model List of Essential Drugs

Drugs used in psychotic disorders	Chlorpromazine, fluphenazine, haloperidol
Drugs used in depressive disorders	Amitriptyline
Drugs used in bipolar disorders	Carbamazepine, lithium carbonate, valproic acid
Drugs used in generalized anxiety and sleep disorders	Diazepam
Drugs used in obsessive-compulsive disorders and panic attacks	Clomipramine

Source: Improving Access and Use of Psychotropic Medicines, by the World Health Organization, 2005, Geneva, Switzerland: Author.

Support for the view that there are many people with significant, untreated mental health needs comes from several sources. Among them is a 1999 report from the surgeon general (U.S. Department of Health and Human Services, 1999) indicating that fewer than one-third of all people in the United States with a diagnosable mental disorder receive treatment in a given year. Also commonly cited are data indicating that there is a shortage of psychiatrists, especially in rural areas.

Support for the view that allowing psychologists to prescribe drugs would meet currently unmet needs is conjecture. Presumably, at least some currently unmet mental health needs would respond to nonpharmacological interventions that could be, but are not, currently provided by psychologists. Why would allowing psychologists to prescribe substantially increase the number of people they serve or change patient demographics? Interestingly, a major argument leading to the acceptance of physician assistant as a profession was that physician assistants would provide medical care for rural and other historically underserved populations. In fact, very few (3%) of them do so (Lavoie & Barone, 2006). Complex interactions of sociocultural, geographic, and fiscal variables that influence patients as well as care providers determine who does and who does not receive treatment; creation of a new profession (psychologists who prescribe) will not alter these interactions.

Moreover, even if psychologists who prescribe will be able to increase access to mental health care, this does not mean that there is public demand for them to do so. Professionals from other disciplines (e.g., nurse practitioners, physician assistants), perhaps working in conjunction with psychologists, may well be in higher demand. Some critics suggest, in fact, that the push for prescription privileges has been driven primarily by financial considerations, as a limited number of influential psychologists work to secure continued access to a substantial share of the health care market, a share that is created by their lobbying efforts, not by legitimate need or public demand.

2. Allowing Psychologists to Prescribe Drugs Will Lead to High-Quality Care

This argument begins with the contention that it is possible to train psychologists to competently prescribe psychotropic medications. As noted previously, the success of the PDP generally supports this view. Compared to New Mexico and Louisiana,

however, the PDP training was more extensive. For example, the PDP required 660 hours of academic training and 1,900 hours of clinical training, whereas New Mexico requires 450 hours of academic training and 400 hours of clinical training (for a conditional prescribing certificate; further clinical training is required for a general prescription certificate).

Supporters contend that the specialized training required for psychologists to prescribe, coupled with the training they normally receive, is adequate to allow them to prescribe medications safely and effectively. As we have discussed previously, their training in behavioral assessment and commitment to the scientist-practitioner model could be especially useful in diagnosing behavior disorders, in selecting empirically validated interventions, and in monitoring the effects of psychotropic medications. In addition, competent psychologists with expertise in dealing with particular kinds of patients (e.g., people with developmental disabilities, elderly people, or children and adolescents) would be able to offer state-of-the-art nonpharmacological alternatives to psychotropic drugs when appropriate and to provide combined pharmacological and pharmacological interventions. Whether they would do so, or would instead come to rely on pharmacological interventions alone, is unclear. As time passes, data from Louisiana and New Mexico may bear on the issue.

Critics argue that medical training beyond that required to earn prescription privileges is required to use psychotropic medications safely. Expertise in monitoring the therapeutic and adverse effects in other medications is needed, given that many people treated with psychotropic medications receive other kinds of drugs.

If there is in fact a strong need for psychologists who prescribe drugs, and if they can be readily trained to do so without sacrificing other important activities, such as offering nonpharmacological treatments, then granting widespread prescription privileges for psychologists will benefit the profession and society at large.

Arguments against Granting Prescription Privileges

Like the advocates of prescription privileges, the critics have made a number of good arguments (e.g., Heiby, 1998; Lavoie & Barone, 2006). One common argument, mentioned previously, is that the proposed level of training is not adequate to allow psychologists to prescribe psychotropic drugs safely and effectively. Another argument, also mentioned previously, is that the push for prescription privileges is motivated primarily by potential financial gain for psychologists, which is not a legitimate reason for dramatically altering professional boundaries. Two other arguments, both very general, are discussed next.

1. There Is No Legitimate Need for Psychologists to Prescribe

This argument takes three forms. One rests on the simple contention that there is a strong and persistent demand for effective psychological interventions; competent and innovative clinical psychologists have more than enough to do without prescribing drugs (Hayes et al., 2002). Managed care has changed and will continue to change the role of doctorate psychologists, but the field of clinical psychology will remain viable for the foreseeable future. Rather than advocating for prescription

privileges, psychologists would be better served by advocating for enhanced acceptance of empirically validated psychological treatments of all mental disorders and by developing new and effective interventions.

A second form of the argument, alluded to in the preceding section, rests on the contention that the alleged demand for prescription privileges was created by a handful of influential psychologists, such as Pat DeLeon (e.g., DeLeon, Dunivin, & Newman, 2002; DeLeon & Wiggins, 1996) and their political allies, not by requests from mental health agencies, people with mental illness, or their advocates. In actuality, the argument goes, the demand for additional professionals who can prescribe is small and, even if prescription privileges are granted to psychologists, is likely to be filled from other professions (physician assistants, registered nurses). Services provided by doctoral-level psychologists are relatively expensive, and wherever possible managed care will seek to replace them with similar services obtained from less expensive professions.

A third form of the argument rests on the contention that psychotropic drugs are restrictive, largely ineffective treatments that are widely overused, at least in part due to questionable practices by the pharmaceutical industry (e.g., Malott, in press; Whitaker, 2002). Therefore, physicians should curb their willingness to prescribe such drugs and certainly no new profession should prescribe them.

All of these arguments have been countered. For example, many studies can be cited to show the clinical effectiveness of psychotropic drugs, including their value in managing conditions typically not responsive to psychological interventions alone (Schizophrenia). The practices of those who currently write most prescriptions for psychotropic drugs, general practitioners, are demonstrably far below optimal (Lavoie & Barone, 2006); therefore, psychologists who competently prescribe drugs can compete effectively against them. In addition, it is claimed, they can provide outreach to currently underserved patient groups. The truth of this contention, like that of all of the claims and counterclaims mentioned in this section, depends on one's source of information and how that information is interpreted.

2. Allowing Psychologists to Prescribe Drugs Will Harm the Profession

One part of this argument is the contention that the prescription privileges debate has eroded the discipline's cohesiveness and may lead to a split similar to that observed in the late 1980s, when the American Psychological Association, emphasizing basic science and the scientist-practitioner model, broke away from APA. Another part emphasizes dissolution of disciplinary boundaries, with a resulting erosion of focus and values. Allowing psychologists to prescribe drugs, the argument goes, would blur the distinction between psychology and medicine and create an unhealthy competition between the disciplines. Moreover, it would direct attention away from studying the variables historically emphasized by psychologists and on developing and employing new treatments based on those variables. Finally, it would shift emphasis away from advocating for and using psychological treatments. All of these happenings would be harmful for psychology.

Further harm would come in the form of altered training requirements. Although training to prescribe in New Mexico and Louisiana is at the postdoctoral level, widespread adoption of prescription privileges legislation might well create a demand for training at the graduate, or even undergraduate, level. Providing this training would seriously reduce opportunities for training in other important areas. Moreover, extensive training in psychopharmacology would be prohibitively expensive. Finally, allowing psychologists to prescribe drugs would open them to insidious influence from the pharmaceutical industry.

Advocates of prescription privileges have countered these arguments by contending that disciplines constantly evolve and, as we have discussed at some length, psychologists have long been involved in psychopharmacology, albeit not as direct treatment providers. For them, moving to the status of treatment providers is a small, logical, and beneficial step. They suggest that the necessary training can be provided at reasonable cost, without seriously compromising opportunities to develop competency in traditional areas. Psychologists competent in pharmacological and nonpharmacological treatments could provide a full range of services and would be in high demand. With careful planning, untoward influence from industry could be avoided (e.g., Antonuccio & Danton, 2003).

Our Position

In weighing arguments for and against prescription privileges, we find it noteworthy that most of the support comes from practice-based organizations, whereas most of the opposition comes from scientist-practitioner groups, with basic science organizations being largely indifferent (Hayes et al., 2002). Our view, expressed earlier, is that optimal use of psychotropic drugs demands acceptance of the scientist-practitioner perspective. The fact that those groups that are avowedly committed to this perspective argue against prescription privileges is noteworthy.

Although proponents have made a number of compelling arguments, our views parallel those of Lavoie and Barone (2006, p. 63), who end their extensive coverage of the prescription debate by noting, "There does not appear to be compelling evidence of the desirability of granting prescription privileges for psychologists." They go on to suggest that, instead of arguing for prescription privileges, psychologists would be better served by improving the dissemination of nonpharmacological services and working with general practitioners and psychiatrists to improve how psychotropic drugs are used. We agree on both counts.

In our view, it is particularly unfortunate that debate has focused almost entirely on psychologists being granted independent drug prescription privileges and has ignored the possibility of psychologists teaming with physicians and other medical personnel who are currently allowed to prescribe. As emphasized in the first section, many psychologists have skills that would allow them to play an invaluable role in assessing drug effects and in providing nonpharmacological alternatives and adjuncts to psychotropic drugs. Combining these skills with those of women and men specifically trained in medicine would go far in improving mental health care, without necessitating a dramatic change in the activities of either psychologists or medical doctors. Although psychologists have much to contribute to

psychopharmacology, our opinion at present, as it was in 1998 (Poling, Christian, & Ehrhardt, 1998), is that the move to obtain prescription privileges is premature, unwarranted, and unwise. Nonetheless, it is sure to continue.

SUMMARY

Psychologists have made (and are continuing to make) many significant contributions to the study of psychotropic and other psychoactive drugs. Their expertise in quantifying behavior has been invaluable in determining how drugs affect mood, cognitive status, and overt actions. Moreover, their conceptual models have been useful in explaining drug effects. Finally, their knowledge of the variables that control behavior has yielded effective strategies for treating drug abuse.

In recent years, some psychologists and their affiliates have argued that properly trained psychologists should be allowed to prescribe psychotropic medications, and New Mexico and Louisiana have recently allowed them to do so. Too little time has passed to allow the outcome of doing so to be determined. There are legitimate arguments for and against granting prescription privileges to psychologists; those arguments are summarized in the present chapter and elsewhere (e.g., Hayes & Heiby, 1998a; Lavoie & Barone, 2006; Sammons et al., 2003). After considering those arguments, our view is that psychologists can best serve mental health needs not by prescribing drugs, but by delivering nonpharmacological interventions and serving as members of multidisciplinary teams that arrange and evaluate pharmacological interventions.

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Roles and Responsibilities of Clinical and Forensic Psychologist

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Abstract

Clinical psychology is the branch of psychology concerned with the assessment and treatment of mental illness, abnormal behavior, psychiatric problems and emotional disturbance and it includes scientific study and application of psychology for the purpose of Understanding, preventing and Relieving physiological based distress or dis function and to promote subjective well-being and personal development.

American Psychological Association defines clinical psychology as a clinical discipline that involves the assessment, diagnostic, treatment, prevention and consultative service to the patients.

Categories of clinical and Experimental Psychology include: Social

Psychology, Forensic Psychology, Biopsychology, Cognitive Psychology and Counseling Psychology. This field is first begun in 1896. The first psychological clinic is at the University of Pennsylvania by Lighter Witmer. Advances in clinical health psychology are to include number of physical disorders and diseases. Psychologists contribute directly to the prevention and treatment of among other diseases, Diabetes, hypertension and aids

Keywords: Psychology• wellbeing• Clinical Psychologist• Forensic Psychologist

Characteristics of Clinical Psychology includes: Emphasis on science, Emphasis on maladjustment, Emphasis on individual, Emphasis on helping.

Parameters to define Practice in Clinical Psychology

Procedures: Assessment, Intervention, Consultation, Research, Problems/Issues, Populations.

Roles and Responsibilities of Clinical Psychologist

Clinical psychologists meet with clients to identify problems such as emotional, mental and behavioural in their lives. Through observation and by using specific methods the psychologist will diagnose any existing or potential disorders and they are able to understand and they know how to treat people suffering from psychological problems

Psychologist do not engage knowingly in behaviour that is harassing to the person to whom they interact in their work based on such factor such as age, gender.

Goals of Clinical Psychologist

They help people, frequently through talk therapy resolve a variety of emotional, behavioural and mental health problems. To meet this goal, they first determine a person's condition and how serious it is by asking some questions and sometimes providing psychological tests.

Advantages of Being Clinical Psychologist

Opportunity to work with new people every day.

High hearing potential.

Flexible work schedules.

The reward of helping people overcomes their challenges.

Disadvantages of Being Clinical Psychologist

Constantly drumming up new business

Having to set up your own practice

Dealing with billing issues.

Having to set up your own practice

Dealing with clients can be stressful and draining.

Forensic Psychology:

Is the Subfield of psychology, which analyze to compare identity and interpret physical evidence of psychological knowledge and methods to both criminal legal and civil cases.

Clinical Forensic Psychology: It is similar to clinical psychology. Clients here are not only suffering from some type of mental problems, but their issues are of important of legal decision making as well. The main scope of forensic psychologist is the specific set of events or interactions.

Roles of Forensic Psychologist

Forensic Psychologist will often conduct their own research, Apply psychiatry to the law They may study different types of criminals have, Forensic psychologist may involve interviewing criminals along with their loved ones and victims. Jury selection, Expert witness, Consultation with Lawyers, Insanity assessment, Custody assessment, Competency assessment, Lethality assessment, Law enforcement screening Researcher.

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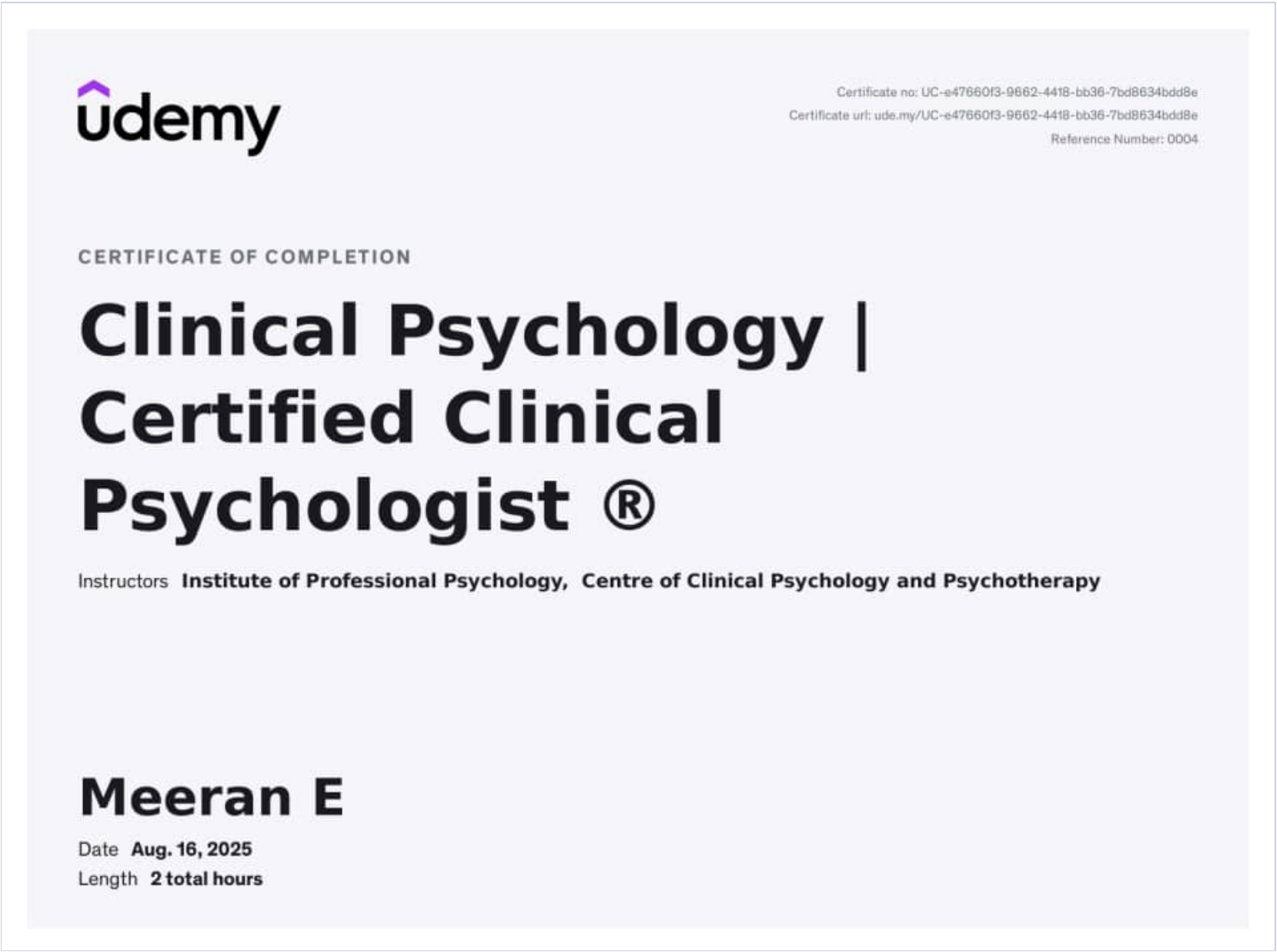

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